

index.html

```
1 <!-- About System Application Tab -->
2 <style>
3   .about-sys-wrapper {
4     padding: 0.75rem;
5     color: var(--text-color, #f8fafc);
6   }
7   .about-sys-top-header {
8     background: var(--surface-2);
9     border: 1px solid var(--border-color);
10    border-radius: var(--radius-md, 8px);
11    padding: 0.75rem 1rem;
12    margin-bottom: 0.75rem;
13  }
14  .about-sys-progress-track {
15    background: var(--surface-2);
16    border-radius: 4px;
17    height: 8px;
18    overflow: hidden;
19    min-width: 100px;
20  }
21  .about-sys-progress-bar {
22    height: 100%;
23    border-radius: 4px;
24    transition: width 0.3s ease;
25  }
26  .about-sys-badge-ok {
27    background: var(--status-success-bg);
28    color: var(--status-success-text);
29    border: 1px solid var(--status-success-border);
30    font-size: 0.72rem;
31    padding: 2px 6px;
32    border-radius: var(--radius-xs, 4px);
33    font-weight: 600;
34  }
35  .about-sys-badge-warn {
36    background: var(--status-warning-bg);
```

```
37 |     color: var(--status-warning-text);
38 |     border: 1px solid var(--status-warning-border);
39 |     font-size: 0.72rem;
40 |     padding: 2px 6px;
41 |     border-radius: var(--radius-xs, 4px);
42 |     font-weight: 600;
43 | }
44 | .about-sys-tree-node {
45 |     margin-bottom: 6px;
46 |     background: var(--surface-2, #1e293b);
47 |     border: 1px solid var(--border-color, #334155);
48 |     border-radius: 6px;
49 |     overflow: hidden;
50 | }
51 | .about-sys-tree-title {
52 |     padding: 6px 10px;
53 |     font-size: 0.82rem;
54 |     font-weight: 600;
55 |     cursor: pointer;
56 |     background: transparent;
57 |     display: flex;
58 |     align-items: center;
59 |     justify-content: space-between;
60 |     color: var(--text-color, #f8fafc);
61 |     transition: background 0.12s;
62 | }
63 | .about-sys-tree-title:hover {
64 |     background: rgba(56, 189, 248, 0.08);
65 | }
66 | .about-sys-tree-body {
67 |     padding: 8px 12px;
68 |     font-size: 0.80rem;
69 |     border-top: 1px solid var(--border-color, #334155);
70 |     background: rgba(15, 23, 42, 0.45);
71 |     display: grid;
72 |     grid-template-columns: repeat(auto-fill, minmax(360px, 1fr
73 | ));
74 |     gap: 4px 20px;
```

```
74 | }
75 | .about-sys-prop-row {
76 |     display: grid;
77 |     grid-template-columns: max-content 1fr;
78 |     align-items: baseline;
79 |     gap: 6px;
80 |     padding: 4px 8px;
81 |     border-bottom: 1px solid rgba(255, 255, 255, 0.04);
82 |     border-radius: 4px;
83 |     transition: background 0.12s ease;
84 | }
85 | .about-sys-prop-row:hover {
86 |     background: rgba(56, 189, 248, 0.05);
87 | }
88 | .about-sys-prop-row:last-child {
89 |     border-bottom: 1px solid rgba(255, 255, 255, 0.04);
90 | }
91 | .about-sys-prop-key {
92 |     color: var(--text-muted, #94a3b8);
93 |     font-weight: 500;
94 |     font-size: 0.80rem;
95 |     user-select: text;
96 |     white-space: nowrap;
97 | }
98 | .about-sys-prop-key::after {
99 |     content: ":";
100 |     margin-left: 1px;
101 |     color: var(--text-muted, #64748b);
102 | }
103 | .about-sys-prop-val {
104 |     color: var(--text-color, #f8fafc);
105 |     font-weight: 500;
106 |     font-size: 0.82rem;
107 |     word-break: break-word;
108 |     user-select: text;
109 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
110 |     Consolas, monospace;
110 | }
```

```
111 | .about-sys-spec-table {
112 |     width: 100%;
113 |     border-collapse: separate;
114 |     border-spacing: 0;
115 | }
116 | .about-sys-spec-table td {
117 |     padding: 10px 16px;
118 |     border-bottom: 1px solid var(--border-subtle, rgba(51, 65,
      | 85, 0.35));
119 |     vertical-align: middle;
120 |     line-height: 1.5;
121 | }
122 | .about-sys-spec-table tr:hover td {
123 |     background: rgba(56, 189, 248, 0.04);
124 | }
125 | .about-sys-spec-table tr:last-child td {
126 |     border-bottom: none;
127 | }
128 | .about-sys-spec-key {
129 |     color: var(--text-muted, #94a3b8);
130 |     width: 230px;
131 |     min-width: 210px;
132 |     max-width: 260px;
133 |     white-space: nowrap;
134 |     font-weight: 500;
135 |     font-size: 0.82rem;
136 |     font-family: var(--font-sans, system-ui, -apple-system, sa
      | ns-serif);
137 |     vertical-align: middle;
138 | }
139 | .about-sys-spec-key i {
140 |     margin-right: 8px;
141 |     font-size: 0.92rem;
142 |     vertical-align: -1px;
143 | }
144 | .about-sys-spec-val {
145 |     color: var(--text-color, #f8fafc);
146 |     font-weight: 500;
```



```
147     font-size: 0.82rem;
148     font-family: var(--font-mono, monospace);
149     vertical-align: middle;
150     word-break: break-word;
151 }
152 .about-ai-copilot-banner {
153     background: linear-gradient(135deg, rgba(15, 23, 42, 0.98)
0%, rgba(20, 35, 60, 0.95) 50%, rgba(15, 23, 42, 0.98) 100%);
154     border: 1px solid rgba(56, 189, 248, 0.45) !important;
155     border-radius: 10px;
156     box-shadow: 0 0 22px rgba(56, 189, 248, 0.15), inset 0 1px
0 rgba(255, 255, 255, 0.1);
157     position: relative;
158     overflow: hidden;
159 }
160 .about-ai-pulse-icon {
161     width: 34px;
162     height: 34px;
163     border-radius: 8px;
164     background: rgba(56, 189, 248, 0.12);
165     border: 1px solid rgba(56, 189, 248, 0.35);
166     display: flex;
167     align-items: center;
168     justify-content: center;
169 }
170 .about-ai-stages-box {
171     background: rgba(10, 15, 29, 0.7);
172     max-height: 240px;
173     overflow-y: auto;
174 }
175 .about-ai-stage-row {
176     padding: 3px 6px;
177     border-radius: 4px;
178     font-size: 0.74rem;
179     line-height: 1.35;
180 }
181 .about-ai-stage-row:hover {
182     background: rgba(255, 255, 255, 0.04);
```

```
183 | }
184 | .anomaly-pill {
185 |     font-size: 0.72rem;
186 |     padding: 2px 8px;
187 |     border-radius: 4px;
188 |     background: rgba(239, 68, 68, 0.12);
189 |     border: 1px solid rgba(239, 68, 68, 0.4);
190 |     color: #f87171;
191 |     font-weight: 500;
192 | }
193 | .anomaly-pill.warning {
194 |     background: rgba(234, 179, 8, 0.12);
195 |     border-color: rgba(234, 179, 8, 0.4);
196 |     color: #fde047;
197 | }
198 |
199 | /* Стили переключателя подвкладок */
200 | .about-subtabs-nav {
201 |     background: var(--surface-2);
202 |     border: 1px solid var(--border-color);
203 |     border-radius: var(--radius-md, 8px);
204 |     padding: 0.35rem 0.5rem;
205 | }
206 | .about-subtab-btn {
207 |     transition: all 0.18s ease;
208 |     font-size: 0.78rem;
209 | }
210 | .about-subtab-btn.active {
211 |     background: #0ea5e9 !important;
212 |     color: #0f172a !important;
213 |     border-color: #0ea5e9 !important;
214 |     font-weight: 700;
215 | }
216 | .diag-badge-critical {
217 |     background: rgba(239, 68, 68, 0.2);
218 |     color: #f87171;
219 |     border: 1px solid rgba(239, 68, 68, 0.4);
220 |     font-size: 0.70rem;
```

```
221 | padding: 2px 6px;
222 | border-radius: 4px;
223 | font-weight: 600;
224 | }
225 | .diag-badge-warning {
226 |   background: rgba(234, 179, 8, 0.2);
227 |   color: #fde047;
228 |   border: 1px solid rgba(234, 179, 8, 0.4);
229 |   font-size: 0.70rem;
230 |   padding: 2px 6px;
231 |   border-radius: 4px;
232 |   font-weight: 600;
233 | }
234 | .diag-badge-normal {
235 |   background: rgba(34, 197, 94, 0.15);
236 |   color: #4ade80;
237 |   border: 1px solid rgba(34, 197, 94, 0.3);
238 |   font-size: 0.70rem;
239 |   padding: 2px 6px;
240 |   border-radius: 4px;
241 |   font-weight: 500;
242 | }
243 | .diag-panel {
244 |   background: var(--surface-2, #1e293b);
245 |   border: 1px solid var(--border-color, #334155);
246 |   border-radius: 8px;
247 |   overflow: hidden;
248 | }
249 | .diag-panel-header {
250 |   background: rgba(15, 23, 42, 0.6);
251 |   padding: 8px 14px;
252 |   border-bottom: 1px solid var(--border-color, #334155);
253 |   font-weight: 600;
254 |   font-size: 0.85rem;
255 | }
256 | .diag-meter-card {
257 |   background: var(--surface-2);
258 |   border: 1px solid var(--border-color);
```

```

259     border-radius: var(--radius-md, 8px);
260     padding: 0.75rem 1rem;
261 }
262 </style>
263
264 <div class="about-sys-wrapper">
265     <!-- Top Navigation / Refresh & Status Bar -->
266     <div class="about-sys-top-header d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3">
267         <div class="d-flex align-items-center gap-2 flex-wrap">
268             <span class="fs-5">i</span>
269             <h6 class="mb-0 fw-bold" style="font-size: 1rem; color: #38bdf8;">0 Система (System Specification & Telemetry)</h6>
270             <span class="badge rounded-pill bg-info-subtle text-info border border-info px-2.5 py-1" style="font-size: 0.72rem;" id="about-sys-live-badge">
271                 • Live Host Telemetry
272             </span>
273             <span class="badge rounded-pill bg-success-subtle text-success border border-success px-2.5 py-1" style="font-size: 0.72rem;" id="about-sys-cache-badge" title="Состояние кеша вкладки">
274                 □ Cache: Ready
275             </span>
276             <span class="badge rounded-pill bg-secondary-subtle text-secondary border border-secondary px-2.5 py-1" style="font-size: 0.72rem;" id="about-sys-privilege-badge">
277                 □ Privilege Check
278             </span>
279         </div>
280         <div class="d-flex align-items-center gap-2 flex-wrap">
281             <!-- Button to Open AI Diagnostics Modal -->
282             <button class="btn btn-sm btn-info text-dark fw-bold rounded-pill px-3 py-1 shadow-sm d-flex align-items-center gap-1.5"
283                 id="btn-open-about-ai-modal"
284                 type="button"
285                 data-bs-toggle="modal"

```

```

286 |         data-bs-target="#aboutAiDiagnosticsModal"
287 |         title="Открыть AI Copilot: Диагностика сенсоров
    | и оборудования">
288 |         <i class="bi bi-robot"></i>
289 |         <span>AI Диагностика</span>
290 |         <span class="badge bg-dark text-info border border-inf
    | o font-monospace ms-1" id="about-ai-health-badge-btn">Health:
    | 100/100</span>
291 |     </button>
292 |     <button class="btn btn-sm btn-outline-info rounded-pill
    | px-3 py-1" id="btn-about-sys-refresh" title="Обновить всю инфо
    | рмацию">
293 |         <i class="bi bi-arrow-clockwise me-1"></i> Обновить
294 |     </button>
295 |     <button class="btn btn-sm btn-outline-warning rounded-pi
    | ll px-2.5 py-1" id="btn-about-sys-clear-cache" title="Очистить
    | кеш вкладки и обновить">
296 |         <i class="bi bi-trash3 me-1"></i> Очистить кеш
297 |     </button>
298 |     <button class="btn btn-sm btn-outline-secondary rounded-
    | pill px-2.5 py-1" id="btn-about-sys-live-toggle" title="Приост
    | ановить или возобновить поток">
299 |         <i class="bi bi-pause-fill me-1" id="icon-about-sys-li
    | ve"></i> <span id="txt-about-sys-live">Пауза</span>
300 |     </button>
301 | </div>
302 | </div>
303 |
304 | <!-- AI Copilot Diagnostics Modal Dialog -->
305 | <div class="modal fade" id="aboutAiDiagnosticsModal" tabinde
    | x="-1" aria-labelledby="aboutAiDiagnosticsModalLabel" aria-hid
    | den="true">
306 |     <div class="modal-dialog modal-dialog-centered modal-dialo
    | g-scrollable modal-fullscreen-xl-down" style="max-width: 95vw;
    | width: 95vw; height: 92vh; margin: 1.5vh auto;">
307 |         <div class="modal-content bg-dark text-white border bord
    | er-info border-opacity-40 shadow-lg" style="height: 100%; back
    | ground: linear-gradient(135deg, rgba(13, 20, 36, 0.98) 0%, rgb

```

```

a(18, 30, 52, 0.98) 100%);">
308
309         <div class="modal-header border-secondary border-opaci
ty-25 pb-2 px-3">
310             <div class="d-flex align-items-center justify-conten
t-between w-100 flex-wrap gap-2">
311                 <div class="d-flex align-items-center gap-2.5">
312                     <div class="about-ai-pulse-icon">
313                         <i class="bi bi-robot text-info fs-5"></i>
314                     </div>
315                     <div>
316                         <div class="d-flex align-items-center gap-2 fl
ex-wrap">
317                             <h6 class="modal-title fw-bold text-white mb
-0" id="aboutAiDiagnosticsModallabel" style="font-size: 0.95re
m;">
318                                 AI Copilot: Анализ производительности и уз
ких мест
319                             </h6>
320                             <span class="badge bg-success-subtle text-su
ccess border border-success font-monospace" id="about-ai-healt
h-badge">Health: 100/100</span>
321                         </div>
322                         <div class="small text-muted" style="font-size
: 0.72rem;">Интеллектуальный диагностический аудит хоста и обо
рудования</div>
323                     </div>
324                 </div>
325
326             <div class="d-flex align-items-center gap-2">
327                 <span class="badge bg-dark text-info border bord
er-info border-opacity-30 px-2.5 py-1 font-monospace" id="abou
t-ai-engine-name" style="font-size: 0.72rem;">
328                     <i class="bi bi-cpu me-1"></i>Unified AI / Heu
ristic
329                 </span>
330                 <button class="btn btn-sm btn-info text-dark fw-
bold rounded-pill px-3 py-1 shadow-sm d-flex align-items-cente

```

```

    r gap-1.5" id="btn-about-ai-rescan" title="Выполнить глубокий
пересканирующий аудит с вызовом AI">
331         <i class="bi bi-lightning-charge-fill" id="ic-
about-ai-rescan"></i>
332         <span id="txt-about-ai-rescan">Rescan</span>
333     </button>
334     <button type="button" class="btn-close btn-close
-white ms-1" data-bs-dismiss="modal" aria-label="Заккрыть"></bu
tton>
335
336 </div>
337 </div>
338
339 <div class="modal-body p-3" id="about-ai-copilot-card"
style="overflow-y: auto; max-height: 80vh;">
340
341 <!-- Live Polling & Active Telemetry Feed Banner (Sh
ows what is being collected right now) -->
342 <div class="p-2.5 mb-2.5 rounded border border-info
border-opacity-30 bg-black bg-opacity-40 d-flex flex-column ga
p-2" id="about-ai-live-feed-banner">
343     <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-2">
344         <div class="d-flex align-items-center gap-2">
345             <span class="spinner-grow spinner-grow-sm text
-info" id="about-ai-live-pulse" role="status" style="width: 0.
75rem; height: 0.75rem;"></span>
346             <strong class="text-info small" style="font-si
ze: 0.80rem;">Текущий сбор сенсоров:</strong>
347             <span class="text-light small font-monospace"
id="about-ai-live-status-text">Опрос датчиков и системных инте
рфейсов хоста...</span>
348         </div>
349         <div class="d-flex align-items-center gap-1.5 fl
ex-wrap font-monospace" style="font-size: 0.70rem;">
350             <span class="badge bg-secondary bg-opacity-40
text-light border border-secondary" id="chip-sensor-lhm"> LHM Sensors</span>

```

```

351 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-wmi"> WMI
| Cores</span>
352 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-ram"> RA
| M & Procs</span>
353 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-disks">
| Disks & SMART</span>
354 |         <span class="badge bg-secondary bg-opacity-40
| text-light border border-secondary" id="chip-sensor-net"> Ne
| twork & OS</span>
355 |     </div>
356 | </div>
357 | <!-- Live metrics snapshot ticker -->
358 | <div class="small text-secondary font-monospace d-
| flex align-items-center gap-3 flex-wrap border-top border-seco
| ndary border-opacity-20 pt-1.5" id="about-ai-live-metrics-bar"
| style="font-size: 0.72rem;">
359 |     <span><strong class="text-muted">CPU:</strong> <
| span class="text-info fw-semibold" id="live-tick-cpu">--</span
| ></span>
360 |     <span><strong class="text-muted">Max Temp:</stro
| ng> <span class="text-warning fw-semibold" id="live-tick-temp"
| >--</span></span>
361 |     <span><strong class="text-muted">RAM:</strong> <
| span class="text-success fw-semibold" id="live-tick-ram">--</s
| pan></span>
362 |     <span><strong class="text-muted">Disks:</strong>
| <span class="text-light fw-semibold" id="live-tick-disks">--<
| /span></span>
363 |     <span><strong class="text-muted">Top Process:</s
| trong> <span class="text-danger fw-semibold" id="live-tick-top
| -proc">--</span></span>
364 | </div>
365 | </div>
366 |
367 | <!-- Execution Stages / Steps Inspector -->

```



```

368         <div class="mb-2.5" id="about-ai-stages-section">
369             <div class="d-flex align-items-center justify-cont
ent-between text-muted small mb-1.5" style="font-size: 0.72rem
; ">
370                 <span class="text-uppercase tracking-wider fw-se
mibold text-info"><i class="bi bi-activity me-1"></i>Цепочка п
ошаговой AI-диагностики</span>
371                 <button class="btn btn-link btn-sm p-0 text-info
text-decoration-none" type="button" id="btn-about-ai-toggle-s
tages" style="font-size: 0.70rem;">
372                     <span id="txt-about-ai-toggle-stages">Свернуть
этапы</span> <i class="bi bi-chevron-up ms-0.5" id="ic-about-
ai-toggle-stages"></i>
373                 </button>
374             </div>
375             <!-- Stepwise Progress Bar -->
376             <div class="progress mb-2" style="height: 4px; bac
kground-color: rgba(255,255,255,0.08);">
377                 <div class="progress-bar bg-info progress-bar-st
riped progress-bar-animated" role="progressbar" id="about-ai-s
tep-progress" style="width: 0%;"></div>
378             </div>
379             <div class="about-ai-stages-box p-2 rounded border
border-secondary border-opacity-25" id="about-ai-stages-conta
iner">
380                 <div class="small text-muted text-center py-2" i
d="about-ai-stages-empty">Ожидание запуска пошагового аудита..
.</div>
381             </div>
382         </div>
383
384         <!-- 4 Functional Domain Cards Grid -->
385         <div class="mt-2.5" id="about-ai-domain-cards-sectio
n">
386             <div class="row g-2" id="about-ai-groups-container
"></div>
387         </div>
388

```

```
389 |         <!-- Anomalies Pills Container -->
390 |         <div id="about-ai-anomalies-container" class="mt-2 d
    | -flex flex-wrap gap-1.5"></div>
391 |
392 |         <!-- Error Message Alert (if present) -->
393 |         <div class="mt-2 p-2 rounded bg-danger bg-opacity-10
    | border border-danger border-opacity-40 d-none" id="about-ai-e
    | rror-section">
394 |             <div class="d-flex align-items-center gap-2 text-d
    | anger small">
395 |                 <i class="bi bi-exclamation-triangle-fill"></i>
396 |                 <strong>Ошибка выполнения AI-диагностики:</stron
    | g>
397 |             </div>
398 |             <div class="small text-light font-monospace mt-1 p
    | s-4" id="about-ai-error-text" style="font-size: 0.72rem; word-
    | break: break-word;"></div>
399 |         </div>
400 |
401 |         <!-- AI Final Synthesis / Executive Summary Output -
    | ->
402 |         <div class="mt-2.5 p-3 rounded bg-black bg-opacity-5
    | 0 border border-info border-opacity-30 shadow-sm" id="about-ai
    | -synthesis-card">
403 |             <div class="d-flex align-items-center justify-cont
    | ent-between mb-1.5 pb-1 border-bottom border-secondary border-
    | opacity-25">
404 |                 <span class="small text-info fw-bold font-monosp
    | ace text-uppercase" style="font-size: 0.75rem;">
405 |                     <i class="bi bi-award-fill me-1"></i>Итоговый
    | синтез и вердикт инженера
406 |                 </span>
407 |                 <span class="badge bg-secondary-subtle text-ligh
    | t font-monospace" id="about-ai-synthesis-status" style="font-s
    | ize: 0.68rem;">Готов к запуску</span>
408 |             </div>
409 |             <div class="small text-light" id="about-ai-summary
    | -text" style="line-height: 1.55; white-space: pre-line;">
```

```
410 |             Нажмите «Rescan» для запуска пошаговой AI-диагно
| стики сенсоров по 4 доменам (CPU/GPU, Память, Диски, Сеть).
411 |         </div>
412 |         <div class="mt-2 pt-2 border-top border-secondary
| border-opacity-20 d-none" id="about-ai-actions-box">
413 |             <div class="text-warning small fw-semibold mb-1"
| style="font-size: 0.72rem;">
414 |                 <i class="bi bi-exclamation-circle-fill me-1">
| </i>Приоритетные действия для администратора:
415 |             </div>
416 |             <ul class="list-unstyled mb-0 small text-light p
| s-2" id="about-ai-actions-list" style="font-size: 0.73rem;"></
| ul>
417 |         </div>
418 |     </div>
419 |
420 |     <!-- Collapsible Live Telemetry Payload Inspector (W
| hat was collected) -->
421 |     <div class="mt-2" id="about-ai-snapshot-section">
422 |         <div class="d-flex align-items-center justify-cont
| ent-between flex-wrap gap-1">
423 |             <button class="btn btn-sm btn-link p-0 text-info
| text-decoration-none d-flex align-items-center gap-1.5 font-m
| onospace" type="button" id="btn-about-ai-toggle-snapshot" styl
| e="font-size: 0.74rem;">
424 |                 <i class="bi bi-cpu-fill text-info"></i>
425 |                 <span><strong>Срез собираемых сенсоров и сырых
| данных (Live Telemetry Snapshot)</strong></span>
426 |                 <i class="bi bi-chevron-down text-muted ms-1"
| id="ic-about-ai-snapshot"></i>
427 |             </button>
428 |             <button class="btn btn-dark btn-sm py-0 px-2 tex
| t-secondary border border-secondary-subtle d-flex align-items-
| center gap-1" type="button" id="btn-about-ai-copy-snapshot" st
| yle="font-size: 0.68rem;" title="Скопировать JSON данных">
429 |                 <i class="bi bi-clipboard"></i>
430 |                 <span id="txt-about-ai-copy-snapshot">Копирова
| ть JSON</span>
```

```
431         </button>
432     </div>
433     <div id="about-ai-snapshot-collapse" class="d-none
mt-1.5 p-2 bg-black bg-opacity-75 border border-info border-o
capacity-30 rounded">
434         <pre class="m-0 p-1.5 text-info font-monospace"
id="about-ai-snapshot-text" style="font-size: 0.70rem; line-he
ight: 1.35; max-height: 250px; overflow-y: auto; white-space:
pre-wrap; word-break: break-word;">Ожидание сбора телеметрии..
.</pre>
435     </div>
436 </div>
437
438     <!-- Collapsible System Instruction Inspector -->
439     <div class="mt-2" id="about-ai-instruction-section"
style="display: none;">
440         <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-1">
441             <button class="btn btn-sm btn-link p-0 text-info
text-decoration-none d-flex align-items-center gap-1.5 font-m
onospace" type="button" id="btn-about-ai-toggle-instruction" s
tyle="font-size: 0.74rem;">
442                 <i class="bi bi-shield-lock-fill text-info"></
i>
443                 <span><strong>Системная инструкция модели</str
ong></span>
444                 <i class="bi bi-chevron-down text-muted ms-1"
id="ic-about-ai-instruction"></i>
445             </button>
446             <button class="btn btn-dark btn-sm py-0 px-2 tex
t-secondary border border-secondary-subtle d-flex align-items-
center gap-1" type="button" id="btn-about-ai-copy-instruction"
style="font-size: 0.68rem;" title="Скопировать системную инст
рукцию">
447                 <i class="bi bi-clipboard"></i>
448                 <span id="txt-about-ai-copy-instruction">Копир
овать</span>
449             </button>
```

```
450         </div>
451         <div id="about-ai-instruction-collapse" class="d-n
one mt-1.5 p-2 bg-black bg-opacity-75 border border-info borde
r-opacity-30 rounded">
452             <pre class="m-0 p-1.5 text-info font-monospace"
id="about-ai-instruction-text" style="font-size: 0.70rem; line
-height: 1.35; max-height: 200px; overflow-y: auto; white-spac
e: pre-wrap; word-break: break-word;"></pre>
453         </div>
454     </div>
455
456     <!-- Collapsible Sent Prompt Inspector -->
457     <div class="mt-2" id="about-ai-prompt-section" style
="display: none;">
458         <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-1">
459             <button class="btn btn-sm btn-link p-0 text-warn
ing text-decoration-none d-flex align-items-center gap-1.5 fon
t-monospace" type="button" id="btn-about-ai-toggle-prompt" sty
le="font-size: 0.74rem;">
460                 <i class="bi bi-terminal-fill text-warning"></
i>
461                 <span><strong>Сформированный промпт модели</st
rong></span>
462                 <i class="bi bi-chevron-down text-muted ms-1"
id="ic-about-ai-prompt"></i>
463             </button>
464             <button class="btn btn-dark btn-sm py-0 px-2 tex
t-secondary border border-secondary-subtle d-flex align-items-
center gap-1" type="button" id="btn-about-ai-copy-prompt" styl
e="font-size: 0.68rem;" title="Скопировать отправленный промпт
">
465                 <i class="bi bi-clipboard"></i>
466                 <span id="txt-about-ai-copy-prompt">Копировать
</span>
467             </button>
468         </div>
469         <div id="about-ai-prompt-collapse" class="d-none m
```

```

t-1.5 p-2 bg-black bg-opacity-75 border border-warning border-
opacity-30 rounded">
470         <pre class="m-0 p-1.5 text-warning font-monospac
e" id="about-ai-prompt-text" style="font-size: 0.70rem; line-h
height: 1.35; max-height: 220px; overflow-y: auto; white-space:
pre-wrap; word-break: break-word;"></pre>
471     </div>
472 </div>
473
474     <!-- Collapsible Raw Response / Diagnostic Data Insp
ector -->
475     <div class="mt-2" id="about-ai-row-section" style="d
isplay: none;">
476         <div class="d-flex align-items-center justify-cont
ent-between flex-wrap gap-1">
477             <button class="btn btn-sm btn-link p-0 text-info
text-decoration-none d-flex align-items-center gap-1.5 font-m
onospace" type="button" id="btn-about-ai-toggle-row" style="fo
nt-size: 0.74rem;">
478                 <i class="bi bi-code-slash text-info"></i>
479                 <span><strong>Сырой ответ модели / Данные</str
ong></span>
480                 <i class="bi bi-chevron-down text-muted ms-1"
id="ic-about-ai-row"></i>
481             </button>
482             <button class="btn btn-dark btn-sm py-0 px-2 tex
t-secondary border border-secondary-subtle d-flex align-items-
center gap-1" type="button" id="btn-about-ai-copy-row" style="
font-size: 0.68rem;" title="Скопировать ответ">
483                 <i class="bi bi-clipboard"></i>
484                 <span id="txt-about-ai-copy-row">Копировать</s
pan>
485             </button>
486         </div>
487         <div id="about-ai-row-collapse" class="d-none mt-1
.5 p-2 bg-black bg-opacity-75 border border-info border-opacit
y-30 rounded">
488             <pre class="m-0 p-1.5 text-light font-monospace"

```

```

    id="about-ai-raw-text" style="font-size: 0.70rem; line-height
: 1.35; max-height: 220px; overflow-y: auto; white-space: pre-
wrap; word-break: break-word;"></pre>
489         </div>
490     </div>
491 </div>
492
493     <div class="modal-footer border-secondary border-opaci
ty-25 py-2 px-3 d-flex justify-content-between align-items-cen
ter">
494         <div class="small text-muted" style="font-size: 0.72
rem;">
495             <i class="bi bi-shield-check text-info me-1"></i>S
ystem Diagnostics & AI Sensors Audit
496         </div>
497         <button type="button" class="btn btn-sm btn-outline-
secondary rounded-pill px-3" data-bs-dismiss="modal">
498             Закрыть
499         </button>
500     </div>
501
502 </div>
503 </div>
504 </div>
505
506
507 <!-- Top 4 Summary KPI Cards (Vital Overview) -->
508 <div class="row g-2 mb-3">
509     <!-- Card 1: Operating System & Host -->
510     <div class="col-12 col-sm-6 col-lg-3">
511         <div class="about-sys-card">
512             <div class="about-sys-label">
513                 <span>Платформа & ОС</span>
514                 <i class="bi bi-windows text-info"></i>
515             </div>
516             <div class="about-sys-value text-info text-truncate mt
-1" id="about-kpi-os-title">Windows 11 (AMD64)</div>
517             <div class="about-sys-sub text-truncate d-flex align-i

```

```

tems-center justify-content-between">
518     <span id="about-kpi-os-host">Host: DELL-VOSTRO</span
>
519     <span class="badge bg-dark border border-secondary t
ext-info py-0 px-1" id="about-spec-uptime">Uptime: --</span>
520     </div>
521 </div>
522 </div>
523
524 <!-- Card 2: Security & Defender -->
525 <div class="col-12 col-sm-6 col-lg-3">
526     <div class="about-sys-card">
527         <div class="about-sys-label">
528             <span>Безопасность системы</span>
529             <i class="bi bi-shield-check text-success"></i>
530         </div>
531         <div class="about-sys-value text-success text-truncate
mt-1" id="about-kpi-sec-title">Active & Protected</div>
532         <div class="about-sys-sub text-truncate" id="about-kpi
-sec-sub">Firewall: ON | UAC: ON</div>
533     </div>
534 </div>
535
536 <!-- Card 3: System Protection & Restore -->
537 <div class="col-12 col-sm-6 col-lg-3">
538     <div class="about-sys-card">
539         <div class="about-sys-label">
540             <span>Точки восстановления</span>
541             <i class="bi bi-shield-lock text-warning"></i>
542         </div>
543         <div class="about-sys-value text-warning text-truncate
mt-1" id="about-kpi-prot-title">3 Checkpoints</div>
544         <div class="about-sys-sub text-truncate" id="about-kpi
-prot-sub">Protection: Active</div>
545     </div>
546 </div>
547
548 <!-- Card 4: System Storage (C:) -->

```



```
549 |     <div class="col-12 col-sm-6 col-lg-3">
550 |         <div class="about-sys-card">
551 |             <div class="about-sys-label">
552 |                 <span>Системный накопитель (C:)</span>
553 |                 <i class="bi bi-hdd-fill text-primary"></i>
554 |             </div>
555 |             <div class="about-sys-value text-white text-truncate m
| t-1" id="about-kpi-stor-title">1014.8 GB Free</div>
556 |             <div class="about-sys-sub text-info text-truncate" id=
| "about-kpi-stor-clean">Cleanable: ~190 MB</div>
557 |         </div>
558 |     </div>
559 | </div>
560 |
561 | <!-- Live Resource Telemetry Bar -->
562 | <div class="row g-2 mb-3">
563 |     <!-- CPU Utilization -->
564 |     <div class="col-12 col-sm-6 col-lg-3">
565 |         <div class="about-sys-card">
566 |             <div class="about-sys-label">
567 |                 <span>Загрузка CPU</span>
568 |                 <span id="about-telemetry-cpu-cores" class="small te
| xt-muted">6 физ. / 12 Поточков</span>
569 |             </div>
570 |             <div class="d-flex align-items-baseline justify-conten
| t-between mt-1">
571 |                 <div class="about-sys-value text-info" id="about-tel
| emetry-cpu-val">22.2%</div>
572 |                 <span class="small text-muted" id="about-telemetry-c
| pu-freq">2900 MHz</span>
573 |             </div>
574 |             <div class="about-sys-progress-track mt-1.5">
575 |                 <div class="about-sys-progress-bar bg-info" id="abou
| t-telemetry-cpu-bar" style="width: 22%;"></div>
576 |             </div>
577 |         </div>
578 |     </div>
579 |
```

```
580 | <!-- Memory RAM -->
581 | <div class="col-12 col-sm-6 col-lg-3">
582 |   <div class="about-sys-card">
583 |     <div class="about-sys-label">
584 |       <span>Память (RAM)</span>
585 |       <span id="about-telemetry-ram-sub" class="small text
| -muted">58.0% занято (13.1 GB свободно)</span>
586 |     </div>
587 |     <div class="about-sys-value text-warning text-truncate
| mt-1" id="about-telemetry-ram-val">18.7 / 31.8 GB</div>
588 |     <div class="about-sys-progress-track mt-1.5">
589 |       <div class="about-sys-progress-bar bg-warning" id="a
| bout-telemetry-ram-bar" style="width: 58%;"></div>
590 |     </div>
591 |   </div>
592 | </div>
593 |
594 | <!-- GPU Accelerator -->
595 | <div class="col-12 col-sm-6 col-lg-3">
596 |   <div class="about-sys-card">
597 |     <div class="about-sys-label">
598 |       <span>GPU Ускоритель</span>
599 |       <span id="about-telemetry-gpu-badge" class="badge bg
| -success-subtle text-success border border-success" style="fon
| t-size: 0.68rem;">CUDA + DirectML</span>
600 |     </div>
601 |     <div class="about-sys-value text-danger text-truncate
| mt-1" id="about-telemetry-gpu-name">GeForce GT 710</div>
602 |     <div class="about-sys-sub text-truncate" id="about-tel
| emetry-gpu-vram">VRAM: 2.0 GB | Нагрузка: 0%</div>
603 |   </div>
604 | </div>
605 |
606 | <!-- Disk I/O Rate -->
607 | <div class="col-12 col-sm-6 col-lg-3">
608 |   <div class="about-sys-card">
609 |     <div class="about-sys-label">
610 |       <span>Диск (C:) I/O</span>
```

```

611         <i class="bi bi-speedometer2 text-info"></i>
612     </div>
613     <div class="about-sys-value text-white text-truncate m
t-1" id="about-telemetry-disk-val">0.00 MB/s</div>
614     <div class="about-sys-sub text-truncate" id="about-tel
emetry-disk-rates">Чтение: 0 КБ/с | Запись: 0 КБ/с</div>
615 </div>
616 </div>
617 </div>
618
619
620 <!-- Reorganized Dual Master Panels: Hardware & Platform Spe
cs vs User Environment & Security -->
621 <div class="row g-2 mb-3">
622     <!-- Left: Hardware & Platform Specification -->
623     <div class="col-12 col-lg-6">
624         <div class="about-sys-panel h-100">
625             <div class="about-sys-panel-header">
626                 <div class="d-flex align-items-center gap-2">
627                     <i class="bi bi-cpu-fill text-info"></i>
628                     <span class="text-white">Конфигурация оборудования
и ОС</span>
629                 </div>
630                 <span class="badge bg-dark border border-secondary t
ext-info" id="about-ident-os-build">Windows 11</span>
631             </div>
632             <table class="about-sys-spec-table">
633                 <tbody>
634                     <tr>
635                         <td class="about-sys-spec-key"><i class="bi bi-c
pu me-1.5 text-info"></i>Процессор (CPU)</td>
636                         <td class="about-sys-spec-val" id="about-spec-cp
u">Intel Processor</td>
637                     </tr>
638                     <tr>
639                         <td class="about-sys-spec-key"><i class="bi bi-m
emory me-1.5 text-warning"></i>Установленная память</td>
640                         <td class="about-sys-spec-val" id="about-spec-ra

```

```

m">31.8 GB RAM</td>
641         </tr>
642         <tr>
643             <td class="about-sys-spec-key"><i class="bi bi-d
isplay me-1.5 text-primary"></i>Мониторы и дисплеи</td>
644             <td class="about-sys-spec-val text-truncate" id=
"about-spec-monitors">1920x1080 @ 60Hz</td>
645         </tr>
646         <tr>
647             <td class="about-sys-spec-key"><i class="bi bi-l
ightning-charge me-1.5 text-success"></i>Схема питания</td>
648             <td class="about-sys-spec-val" id="about-spec-po
wer">Balanced (Сбалансированная)</td>
649         </tr>
650         <tr>
651             <td class="about-sys-spec-key"><i class="bi bi-c
alendar-check me-1.5 text-warning"></i>Дата установки ОС</td>
652             <td class="about-sys-spec-val" id="about-ident-i
nstall-date">Не определена</td>
653         </tr>
654         <tr>
655             <td class="about-sys-spec-key"><i class="bi bi-a
rrow-repeat me-1.5 text-info"></i>Служба обновлений</td>
656             <td class="about-sys-spec-val" id="about-spec-up
date">Up to date</td>
657         </tr>
658     </tbody>
659 </table>
660 </div>
661 </div>
662
663 <!-- Right: User Environment, Regional & Security -->
664 <div class="col-12 col-lg-6">
665     <div class="about-sys-panel h-100">
666         <div class="about-sys-panel-header">
667             <div class="d-flex align-items-center gap-2">
668                 <i class="bi bi-person-badge-fill text-success"></
i>

```

```

669 |         <span class="text-white">Окружение пользователя и
      | безопасность</span>
670 |     </div>
671 |     <span class="small text-muted font-monospace" id="ab
      | out-ident-time-badge">Локальное время: --:--:--</span>
672 | </div>
673 | <table class="about-sys-сpec-table">
674 |     <tbody>
675 |         <tr>
676 |             <td class="about-sys-сpec-key"><i class="bi bi-p
      | erson-circle me-1.5 text-success"></i>Пользователь и права</td
      | >
677 |             <td class="about-sys-сpec-val">
678 |                 <span class="fw-bold text-white me-1" id="abou
      | t-ident-username">Пользователь</span>
679 |                 <span class="text-muted small" id="about-ident
      | -privilege">(Права)</span>
680 |             </td>
681 |         </tr>
682 |         <tr>
683 |             <td class="about-sys-сpec-key"><i class="bi bi-t
      | ranslate me-1.5 text-warning"></i>Язык и локали</td>
684 |             <td class="about-sys-сpec-val">
685 |                 <span id="about-ident-language">Русский</span>
686 |                 <div class="small text-muted font-monospace" i
      | d="about-ident-locales">User: ru-RU | Sys: ru-RU</div>
687 |             </td>
688 |         </tr>
689 |         <tr>
690 |             <td class="about-sys-сpec-key"><i class="bi bi-c
      | lock me-1.5 text-primary"></i>Часовой пояс и кодировка</td>
691 |             <td class="about-sys-сpec-val">
692 |                 <span id="about-ident-timezone">UTC+03:00</spa
      | n>
693 |                 <div class="small text-muted font-monospace" i
      | d="about-ident-codepage">Кодировка: UTF-8</div>
694 |             </td>
695 |         </tr>

```

```

696         <tr>
697             <td class="about-sys-spec-key"><i class="bi bi-c
loud-check me-1.5 text-info"></i>Офис и OneDrive</td>
698             <td class="about-sys-spec-val">
699                 <span>MS Office: <strong id="about-ident-ms-of
fice" class="text-light font-monospace">Не установлен</strong>
</span>
700                 <div class="small text-muted font-monospace">O
neDrive: <span id="about-ident-onedrive" class="text-light">Не
настроено</span></div>
701             </td>
702         </tr>
703         <tr>
704             <td class="about-sys-spec-key"><i class="bi bi-a
rchive me-1.5 text-info"></i>Резервные копии</td>
705             <td class="about-sys-spec-val" id="about-ident-b
ackup">Нет записей</td>
706         </tr>
707         <tr>
708             <td class="about-sys-spec-key"><i class="bi bi-s
hield-shaded me-1.5 text-success"></i>Брандмауэр Windows</td>
709             <td class="about-sys-spec-val">
710                 Domain: <span id="about-sec-fw-domain" class="
text-success fw-bold">Active</span> |
711                 Private: <span id="about-sec-fw-private" class
="text-success fw-bold">Active</span> |
712                 Public: <span id="about-sec-fw-public" class="
text-success fw-bold">Active</span>
713             </td>
714         </tr>
715         <tr>
716             <td class="about-sys-spec-key"><i class="bi bi-s
hield-lock me-1.5 text-success"></i>Защитник и UAC</td>
717             <td class="about-sys-spec-val">
718                 Defender: <span id="about-sec-defender" class=
"text-success fw-bold">Enabled</span> |
719                 Realtime: <span id="about-sec-realtime" class=
"text-success fw-bold">Enabled</span> |

```

```

720 |             UAC: <span id="about-sec-uac" class="text-succ
      | ess fw-bold">Enabled</span>
721 |         </td>
722 |     </tr>
723 | </tbody>
724 | </table>
725 | </div>
726 | </div>
727 | </div>
728 |
729 | <!-- Dual Panel: LibreHardwareMonitor Sensors + Storage Volu
      | mes & Partitions -->
730 | <div class="row g-2 mb-3">
731 |     <!-- Left: Sensors LibreHardwareMonitor -->
732 |     <div class="col-12 col-lg-5">
733 |         <div class="about-sys-panel h-100">
734 |             <div class="about-sys-panel-header">
735 |                 <div class="d-flex align-items-center gap-2">
736 |                     <i class="bi bi-activity text-danger"></i>
737 |                     <span class="text-white">Сенсоры LibreHardwareMoni
      | tor</span>
738 |                 </div>
739 |                 <span class="badge bg-success-subtle text-success bo
      | rder border-success" id="about-sensors-count-badge">LHM Active
      | </span>
740 |             </div>
741 |             <div class="p-2 border-bottom border-secondary-subtle
      | d-flex align-items-center justify-content-between flex-wrap ga
      | p-1">
742 |                 <!-- Filter Buttons -->
743 |                 <div class="btn-group btn-group-sm" role="group" id=
      | "about-sensor-filter-group">
744 |                     <button type="button" class="btn btn-outline-info
      | active py-0.5 px-2" data-sensor-cat="all" style="font-size: 0.
      | 72rem;">Все</button>
745 |                     <button type="button" class="btn btn-outline-secon
      | dary py-0.5 px-2" data-sensor-cat="voltage" style="font-size:
      | 0.72rem;">Вольтаж</button>

```

```

746         <button type="button" class="btn btn-outline-secon
dary py-0.5 px-2" data-sensor-cat="clock" style="font-size: 0.
72rem;">Частоты</button>
747         <button type="button" class="btn btn-outline-secon
dary py-0.5 px-2" data-sensor-cat="temperature" style="font-si
ze: 0.72rem;">Температуры</button>
748         <button type="button" class="btn btn-outline-secon
dary py-0.5 px-2" data-sensor-cat="power" style="font-size: 0.
72rem;">Нагрузка</button>
749     </div>
750     <input type="text" class="form-control form-control-
sm" id="about-sensor-search" placeholder="Фильтр..." style="ma
x-width: 130px; font-size: 0.72rem; height: 26px;">
751 </div>
752 <div class="p-2" id="about-sensors-container" style="m
ax-height: 400px; overflow-y: auto;">
753     <div class="text-center py-4 text-muted small">Опрос
сенсоров оборудования...</div>
754 </div>
755 </div>
756 </div>
757
758 <!-- Right: Storage Volumes & Partitions -->
759 <div class="col-12 col-lg-7">
760     <div class="about-sys-panel h-100">
761         <div class="about-sys-panel-header">
762             <div class="d-flex align-items-center gap-2">
763                 <i class="bi bi-hdd-stack-fill text-primary"></i>
764                 <span class="text-white">Storage Volumes & Partiti
ons</span>
765             </div>
766             <span class="badge bg-secondary" id="about-disks-cou
nt">-- Drives</span>
767         </div>
768         <div class="table-responsive" style="max-height: 440px
; overflow-y: auto;">
769             <table class="table table-dark table-hover table-sm
mb-0" style="font-size: 0.8rem; font-family: ui-monospace, SFM

```



```

ono-Regular, Menlo, Monaco, Consolas, monospace;">
770         <thead>
771             <tr class="text-secondary border-secondary">
772                 <th style="width: 12%;">Drive</th>
773                 <th style="width: 14%;">FS</th>
774                 <th style="width: 14%;">Total</th>
775                 <th style="width: 14%;">Used</th>
776                 <th style="width: 14%;">Free</th>
777                 <th style="width: 20%;">Usage</th>
778                 <th style="width: 12%;">SMART</th>
779             </tr>
780         </thead>
781         <tbody id="about-sys-disks-tbody">
782             <!-- Injected dynamically -->
783             <tr>
784                 <td colspan="7" class="text-center py-3 text-m
uted">Загрузка разделов дисков...</td>
785             </tr>
786         </tbody>
787     </table>
788 </div>
789 </div>
790 </div>
791 <!-- 4. Износ накопителей и батареи (Storage & Battery Healt
h) -->
792 <div class="card shadow-sm border-secondary-subtle mb-3" sty
le="background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
0%); border-radius: 10px;">
793     <div class="card-body p-3">
794         <div class="d-flex flex-wrap align-items-center justify-
content-between gap-2">
795             <div class="d-flex align-items-center gap-2">
796                 <div class="fs-4">□ </div>
797             <div>
798                 <h5 class="mb-0 fw-bold text-white">4. Износ накоп
ителей и батареи (Storage & Battery Health)</h5>
799                 <small class="text-muted">Мониторинг ресурса SSD/N
VMe, файловых систем, SMART и деградации аккумулятора</small>

```

```

800         </div>
801     </div>
802
803     <div class="d-flex align-items-center gap-2 flex-wrap"
804 >
805         <div class="form-check form-switch mb-0 small text-m
806 uted d-flex align-items-center gap-1">
807             <input class="form-check-input" type="checkbox" id
808 ="diag-wear-auto-refresh">
809             <label class="form-check-label small text-nowrap"
810 for="diag-wear-auto-refresh">Авто (10c)</label>
811         </div>
812         <button class="btn btn-sm btn-outline-info rounded-p
813 ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
814 "btn-diag-wear-refresh">
815             <i class="bi bi-arrow-clockwise"></i> Обновить
816         </button>
817     </div>
818 </div>
819 </div>
820
821 <div class="row g-2 mb-3">
822     <!-- Disks Health & Wear Panel -->
823     <div class="col-12 col-xl-8 col-lg-8">
824         <div class="diag-panel h-100">
825             <div class="diag-panel-header d-flex justify-content-b
826 etween align-items-center">
827                 <div class="d-flex align-items-center gap-2">
828                     <i class="bi bi-hdd-fill text-primary"></i>
829                     <span class="text-white">Здоровье накопителей SSD/
830 NVMe и износ (Wear & SMART)</span>
831                 </div>
832                 <span class="badge bg-secondary" id="diag-wear-disks
833 -count">0 дисков</span>
834             </div>
835             <div class="table-responsive" style="max-height: 480px
836 ; overflow-y: auto;">

```

```

828 |         <table class="table table-dark table-hover table-sm
      | mb-0" style="font-size: 0.82rem; font-family: ui-monospace, SF
      | Mono-Regular, Menlo, Monaco, Consolas, monospace;">
829 |             <thead>
830 |                 <tr class="text-secondary border-secondary">
831 |                     <th>Физический накопитель</th>
832 |                     <th>Тип / Шина</th>
833 |                     <th>Разделы</th>
834 |                     <th style="text-align: right;">Объем</th>
835 |                     <th style="text-align: right;">Свободно</th>
836 |                     <th>Наработка / Старт</th>
837 |                     <th style="text-align: right;">Запись / Чтение
      | </th>
838 |                     <th style="width: 130px;">Здоровье / Износ</th>
      | >
839 |                         <th>Статус</th>
840 |                 </tr>
841 |             </thead>
842 |             <tbody id="diag-wear-disks-tbody">
843 |                 <tr><td colspan="9" class="text-center py-4 text
      | -muted">Опрос контроллеров накопителей...</td></tr>
844 |             </tbody>
845 |         </table>
846 |     </div>
847 | </div>
848 | </div>
849 |
850 | <!-- Battery Deep Telemetry -->
851 | <div class="col-12 col-xl-4 col-lg-4">
852 |     <div class="diag-panel h-100">
853 |         <div class="diag-panel-header d-flex justify-content-b
      | etween align-items-center">
854 |             <div class="d-flex align-items-center gap-2">
855 |                 <i class="bi bi-battery-charging text-warning"></i>
      | >
856 |                 <span class="text-white">Деградация и состояние ба
      | тарей (Battery Degradation)</span>
857 |             </div>

```

```

858 |         <span class="badge bg-dark border border-secondary t
ext-info" id="diag-battery-source-badge">AC Mains</span>
859 |     </div>
860 |     <div class="p-3" id="diag-battery-details-container">
861 |         <div class="text-center py-4 text-muted small">
862 |             <i class="bi bi-plug-fill fs-4 text-info d-block m
b-1"></i>
863 |             Стационарный компьютер – питание напрямую от элект
росети
864 |         </div>
865 |     </div>
866 | </div>
867 | </div>
868 | </div>
869 |
870 | <!-- Full Hardware Specification Tree Panel (Hardware Tree)
-->
871 | <div class="about-sys-panel">
872 |     <div class="about-sys-panel-header">
873 |         <div class="d-flex align-items-center gap-2">
874 |             <i class="bi bi-diagram-3-fill text-info"></i>
875 |             <span class="text-white">Полная спецификация оборудова
ния</span>
876 |             <span class="badge bg-secondary" id="about-sys-node-co
unt">Hardware Tree</span>
877 |         </div>
878 |         <div class="d-flex align-items-center gap-2">
879 |             <button class="btn btn-sm btn-outline-secondary rounde
d-pill px-2.5 py-0.5" style="font-size: 0.72rem;" id="btn-abou
t-sys-expand-all">
880 |                 <i class="bi bi-arrows-expand me-1"></i> Развернуть
все
881 |             </button>
882 |             <button class="btn btn-sm btn-outline-secondary rounde
d-pill px-2.5 py-0.5" style="font-size: 0.72rem;" id="btn-abou
t-sys-collapse-all">
883 |                 <i class="bi bi-arrows-collapse me-1"></i> Свернуть
все

```

```
884 |         </button>
885 |         <input type="text" class="form-control form-control-sm
      | " id="about-sys-search"
886 |             placeholder="Поиск компонентов..." style="max-w
      | idth: 220px; font-size: 0.76rem; height: 26px;">
887 |     </div>
888 | </div>
889 |     <div class="p-3" id="about-sys-tree-container" style="max-
      | height: min(85vh, 1200px); overflow-y: auto; transition: max-h
      | eight 0.25s ease-in-out;">
890 |         <div class="text-center py-4 small text-muted">
891 |             <div class="spinner-border spinner-border-sm text-info
      | mb-2" role="status"></div>
892 |             <div>Загрузка спецификации оборудования...</div>
893 |         </div>
894 |     </div>
895 | </div>
896 |
897 | </div> <!-- /.about-sys-wrapper -->
898 |
899 |
900 |
```

main.js

```
1 // =====
  =====
2 // Process Name: About System Tab Web Controller
3 // =====
  =====
4 // Description:
5 //   Client-side JavaScript controller for the About System ta
  b.
6 //   Provides real-time host telemetry, system identity & loca
  le parameters,
7 //   AIDA64-like hardware tree, LibreHardwareMonitor sensors,
  live top processes,
8 //   disk volumes and security status.
9 //
10 //   КЕШИРОВАНИЕ (IndexedDB + Memory):
11 //   - Stale-While-Revalidate для всех данных
12 //   - Cache-First для hardware спец (меняется редко)
13 //   - Network-First для live telemetry (всегда свежие)
14 //   - Prefetching при активации вкладки
15 //
16 // File: main.js
17 // Package: src.api.webgui.about_system_tab
18 // Author: hypo69
19 // Copyright: © 2026 hypo69
20 // =====
  =====
21
22 (function () {
23   let hardwareData = [];
24   let currentSensors = [];
25   let currentProcesses = [];
26   let currentDisks = [];
27   let activeSensorFilter = 'all';
28   let isLiveActive = true;
29   let liveIntervalId = null;
30   let wearAutoRefreshTimer = null;
```

```

31 | let isUpdating = false;
32 | let isAiRunning = false;
33 |
34 | // Кеш-стратегия и TTL
35 | const CACHE_STRATEGY = {
36 |     HARDWARE_SPEC: { ttl: 30 * 60 * 1000, strategy: 'cache-first' }, // 30 мин, редко меняется
37 |     SYSTEM_SUMMARY: { ttl: 5 * 1000, strategy: 'stale-while-revalidate' }, // 5 сек
38 |     SENSORS: { ttl: 3 * 1000, strategy: 'network-first' }, // 3 сек, всегда свежие
39 |     CONTROL_STATUS: { ttl: 60 * 1000, strategy: 'stale-while-revalidate' }, // 1 мин
40 |     BACKUP_STATUS: { ttl: 5 * 60 * 1000, strategy: 'cache-first' }, // 5 мин
41 |     STORAGE_BATTERY: { ttl: 10 * 1000, strategy: 'stale-while-revalidate' } // 10 сек
42 | };
43 |
44 | const CACHE_STORE = 'api_cache';
45 | const CACHE_TAG = 'about-system-tab';
46 |
47 | /**
48 |  * Универсальная функция для работы с кешем
49 |  * Поддерживает стратегии: cache-first, network-first, stale-while-revalidate
50 |  */
51 | async function cachedFetch(url, cacheKey, options = {}) {
52 |     const { ttl = 60000, strategy = 'stale-while-revalidate', forceNetwork = false } = options;
53 |     const cache = window.browseCache;
54 |
55 |     if (!cache) {
56 |         // Fallback: кеш недоступен, прямой запрос
57 |         return await apiFetch(url);
58 |     }
59 |
60 |     // Cache-First: сначала кеш, потом сеть (для редко меняющи

```

```

хся данных)
61     if (strategy === 'cache-first' && !forceNetwork) {
62         const cached = await cache.get(CACHE_STORE, cacheKey);
63         if (cached) {
64             console.log(`[Cache] HIT (cache-first): ${cacheKey}`);
65             return cached;
66         }
67     }
68
69     // Stale-While-Revalidate: показываем кеш, обновляем в фон
е
70     if (strategy === 'stale-while-revalidate' && !forceNetwork
) {
71         const cached = await cache.get(CACHE_STORE, cacheKey);
72         if (cached) {
73             console.log(`[Cache] HIT (stale-while-revalidate): ${c
acheKey}, updating in background...`);
74             // Запускаем обновление в фоне
75             apiFetch(url)
76                 .then(fresh => cache.set(CACHE_STORE, cacheKey, fres
h, { ttl, tags: [CACHE_TAG] })))
77                 .catch(err => console.warn(`[Cache] Background updat
e failed for ${cacheKey}:`, err));
78             return cached;
79         }
80     }
81
82     // Network-First или первый запрос: сначала сеть, потом ке
ш
83     try {
84         const fresh = await apiFetch(url);
85         // Сохраняем в кеш
86         await cache.set(CACHE_STORE, cacheKey, fresh, { ttl, tag
s: [CACHE_TAG] });
87         console.log(`[Cache] MISS → stored: ${cacheKey}`);
88         return fresh;
89     } catch (err) {
90         // Если сеть недоступна, пытаемся вернуть устаревший кеш

```



```

91 |         const cached = await cache.get(CACHE_STORE, cacheKey);
92 |         if (cached) {
93 |             console.warn(`[Cache] Network failed, returning stale
cache: ${cacheKey}`);
94 |             return cached;
95 |         }
96 |         throw err;
97 |     }
98 | }
99 |
100 | /**
101 |  * Очистка кеша вкладки (при необходимости)
102 |  */
103 | async function clearTabCache() {
104 |     const cache = window.browserCache;
105 |     if (cache) {
106 |         await cache.invalidateByTag(CACHE_STORE, CACHE_TAG);
107 |         console.log('[Cache] Tab cache cleared');
108 |     }
109 | }
110 |
111 | async function apiFetch(url, options = {}) {
112 |     const opts = { ...options };
113 |     if (opts.body && typeof opts.body === 'string') {
114 |         opts.headers = {
115 |             'Content-Type': 'application/json',
116 |             ...(opts.headers || {})
117 |         };
118 |     }
119 |     if (window.api && typeof window.api.fetch === 'function')
{
120 |         return await window.api.fetch(url, opts);
121 |     }
122 |     const res = await fetch(url, opts);
123 |     if (!res.ok) {
124 |         let errMsg = `HTTP ${res.status}`;
125 |         try {
126 |             const errJson = await res.json();

```

```

127         if (errJson && errJson.detail) {
128             errMsg += ` (${typeof errJson.detail === 'object' ?
JSON.stringify(errJson.detail) : errJson.detail})`;
129         }
130     } catch (_) {}
131     throw new Error(errMsg);
132 }
133 return await res.json();
134 }
135
136 function setText(id, text) {
137     const el = document.getElementById(id);
138     if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
139 }
140
141 function setHtml(id, html) {
142     const el = document.getElementById(id);
143     if (el) el.innerHTML = html;
144 }
145
146 async function initAboutSystemTab() {
147     console.log('[AboutSystemTab] Initializing tab controller.
...');
148     bindEvents();
149     bindAIEvents();
150     updateLocalClock();
151
152     // Проверка доступности кеша
153     await updateCacheStatus();
154
155     // Prefetch: предзагрузка критичных данных для мгновенного
отображения
156     const prefetchPromises = [
157         fetchSystemSummary().catch(err => console.warn('[AboutSy
stemTab] Quick summary error:', err)),
158         fetchHardwareSpec().catch(err => console.warn('[AboutSys
temTab] Hardware spec error:', err))

```

```

159 | ];
160 |
161 | // Ждем критичные данные перед отображением
162 | await Promise.allSettled(prefetchPromises);
163 |
164 | // Фоновая загрузка менее критичных данных
165 | fetchHardwareSensors().catch(err => console.warn('[AboutSystemTab] Sensors error:', err));
166 | fetchBackupStatus().catch(err => console.warn('[AboutSystemTab] Backup status error:', err));
167 | fetchSystemControlStatus().catch(err => console.warn('[AboutSystemTab] Control status error:', err));
168 | fetchStorageBatteryWear().catch(err => console.warn('[AboutSystemTab] Storage & Battery wear error:', err));
169 |
170 | // AI-диагностика запускается только пользователем по кнопке Rescan / Запуск
171 | // Start live telemetry ticker
172 | startLiveStream();
173 | }
174 | window.initAboutSystemTab = initAboutSystemTab;
175 |
176 | /**
177 |  * Обновление статуса кеша в интерфейсе
178 |  */
179 | async function updateCacheStatus() {
180 |     const badge = document.getElementById('about-sys-cache-badge');
181 |     if (!badge) return;
182 |
183 |     const cache = window.browserCache;
184 |     if (!cache) {
185 |         badge.textContent = '❌ Cache: Unavailable';
186 |         badge.className = 'badge rounded-pill bg-warning-subtle text-warning border border-warning px-2.5 py-1';
187 |         badge.title = 'Кеш недоступен, используется прямой запрос к API';
188 |         return;

```

```
189 |     }
190 |
191 |     const ready = await cache.ready();
192 |     if (ready) {
193 |         const stats = await cache.getDetailedStats();
194 |         const hitRate = stats.metrics.hits + stats.metrics.misses > 0
195 |             ? Math.round((stats.metrics.hits / (stats.metrics.hits
196 | + stats.metrics.misses)) * 100)
197 |             : 0;
198 |         badge.textContent = `📊 Cache: ${hitRate}% hit`;
199 |         badge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-2.5 py-1';
200 |         badge.title = `Кеш активен\nПопаданий: ${stats.metrics.hits}\nПромашов: ${stats.metrics.misses}\nВ памяти: ${stats.memoryCacheSize} записей`;
201 |     } else {
202 |         badge.textContent = '🔴 Cache: Error';
203 |         badge.className = 'badge rounded-pill bg-danger-subtle text-danger border border-danger px-2.5 py-1';
204 |         badge.title = 'Ошибка инициализации кеша';
205 |     }
206 | }
207 |
208 | function updateLocalClock() {
209 |     const clockEl = document.getElementById('about-ident-time-badge');
210 |     if (clockEl) {
211 |         const now = new Date();
212 |         clockEl.textContent = `Локальное время: ${now.toLocaleTimeString()}`;
213 |     }
214 | }
215 |
216 | function startLiveStream() {
217 |     if (window.registerTabPoller) {
218 |         window.registerTabPoller('tab-about-system', async () =>
```

```

    {
219         updateLocalClock();
220         if (isLiveActive && !isUpdating) {
221             await pollLiveTelemetry();
222             if (activeSubtab !== 'subtab-overview') {
223                 await refreshActiveSubtab();
224             }
225         }
226         // Обновляем статус кеша каждые 3 секунды
227         await updateCacheStatus();
228     }, 3000, { immediate: true });
229 } else {
230     if (liveIntervalId) clearInterval(liveIntervalId);
231     liveIntervalId = setInterval(async () => {
232         updateLocalClock();
233         if (isLiveActive && !isUpdating) {
234             await pollLiveTelemetry();
235             if (activeSubtab !== 'subtab-overview') {
236                 await refreshActiveSubtab();
237             }
238         }
239         await updateCacheStatus();
240     }, 3000);
241 }
242 }
243
244 function bindEvents() {
245     const btnRefresh = document.getElementById('btn-about-sys-
refresh');
246     if (btnRefresh) {
247         btnRefresh.onclick = async () => {
248             btnRefresh.disabled = true;
249             const icon = btnRefresh.querySelector('i');
250             if (icon) icon.classList.add('spin-animation');
251             await refreshAllData();
252             if (icon) icon.classList.remove('spin-animation');
253             btnRefresh.disabled = false;
254         };

```

```
255 |     }
256 |
257 |     const btnClearCache = document.getElementById('btn-about-s
ys-clear-cache');
258 |     if (btnClearCache) {
259 |         btnClearCache.onclick = async () => {
260 |             btnClearCache.disabled = true;
261 |             const icon = btnClearCache.querySelector('i');
262 |             if (icon) icon.classList.add('spin-animation');
263 |
264 |             // Очищаем кеш вкладки
265 |             await clearTabCache();
266 |
267 |             // Принудительно обновляем все данные из сети
268 |             await refreshAllData(true);
269 |
270 |             if (icon) icon.classList.remove('spin-animation');
271 |             btnClearCache.disabled = false;
272 |
273 |             if (window.showToast) {
274 |                 window.showToast('Кеш вкладки очищен, данные обновле
ны', 'success');
275 |             }
276 |         };
277 |     }
278 |
279 |     const btnLiveToggle = document.getElementById('btn-about-s
ys-live-toggle');
280 |     if (btnLiveToggle) {
281 |         btnLiveToggle.onclick = () => {
282 |             isLiveActive = !isLiveActive;
283 |             if (window.setTabPollerEnabled) {
284 |                 window.setTabPollerEnabled('tab-about-system_default
', isLiveActive);
285 |             }
286 |             const icon = document.getElementById('icon-about-sys-l
ive');
287 |             const txt = document.getElementById('txt-about-sys-liv
```

```

    e');
288     const liveBadge = document.getElementById('about-sys-1
live-badge');
289     if (isLiveActive) {
290         if (icon) icon.className = 'bi bi-pause-fill me-1';
291         if (txt) txt.textContent = 'Пауза';
292         if (liveBadge) {
293             liveBadge.className = 'badge rounded-pill bg-info-
subtle text-info border border-info px-2.5 py-1';
294             liveBadge.textContent = '● Live Host Telemetry';
295         }
296     } else {
297         if (icon) icon.className = 'bi bi-play-fill me-1';
298         if (txt) txt.textContent = 'Возобновить';
299         if (liveBadge) {
300             liveBadge.className = 'badge rounded-pill bg-warni
ng-subtle text-warning border border-warning px-2.5 py-1';
301             liveBadge.textContent = '□ Stream Paused';
302         }
303     }
304 };
305 }
306
307     const btnExpandAll = document.getElementById('btn-about-sy
s-expand-all');
308     if (btnExpandAll) {
309         btnExpandAll.onclick = () => {
310             document.querySelectorAll('.about-sys-tree-body').forE
ach(b => b.classList.remove('d-none'));
311             document.querySelectorAll('.about-sys-chevron').forEac
h(c => c.textContent = '▲');
312             adjustHardwareTreeHeight();
313         };
314     }
315
316     const btnCollapseAll = document.getElementById('btn-about-
sys-collapse-all');
317     if (btnCollapseAll) {

```

```
318 |         btnCollapseAll.onclick = () => {
319 |             document.querySelectorAll('.about-sys-tree-body').forEach(b => b.classList.add('d-none'));
320 |             document.querySelectorAll('.about-sys-chevron').forEach(c => c.textContent = '▼');
321 |             adjustHardwareTreeHeight();
322 |         };
323 |     }
324 |
325 |     window.addEventListener('resize', () => adjustHardwareTreeHeight());
326 |
327 |     const treeSearch = document.getElementById('about-sys-search');
328 |     if (treeSearch) {
329 |         treeSearch.oninput = () => {
330 |             renderHardwareTree(hardwareData, treeSearch.value.trim().toLowerCase());
331 |         };
332 |     }
333 |
334 |     const sensorSearch = document.getElementById('about-sensor-search');
335 |     if (sensorSearch) {
336 |         sensorSearch.oninput = () => {
337 |             renderSensorsList(currentSensors);
338 |         };
339 |     }
340 |
341 |     const procSearch = document.getElementById('about-proc-search');
342 |     if (procSearch) {
343 |         procSearch.oninput = () => {
344 |             renderProcessesTable(currentProcesses);
345 |         };
346 |     }
347 |
348 |     // Sensor category filter buttons
```



```

349     const filterGroup = document.getElementById('about-sensor-
filter-group');
350     if (filterGroup) {
351         filterGroup.querySelectorAll('button').forEach(btn => {
352             btn.onclick = () => {
353                 filterGroup.querySelectorAll('button').forEach(b =>
{
354                     b.classList.remove('active', 'btn-outline-info');
355                     b.classList.add('btn-outline-secondary');
356                 });
357                 btn.classList.add('active', 'btn-outline-info');
358                 btn.classList.remove('btn-outline-secondary');
359                 activeSensorFilter = btn.getAttribute('data-sensor-c
at') || 'all';
360                 renderSensorsList(currentSensors);
361             };
362         });
363     }
364
365     const btnWearRefresh = document.getElementById('btn-diag-w
ear-refresh');
366     if (btnWearRefresh) {
367         btnWearRefresh.onclick = () => fetchStorageBatteryWear(t
rue);
368     }
369
370     const wearAutoSwitch = document.getElementById('diag-wear-
auto-refresh');
371     if (wearAutoSwitch) {
372         wearAutoSwitch.onchange = (e) => {
373             if (e.target.checked) {
374                 wearAutoRefreshTimer = setInterval(() => fetchStorag
eBatteryWear(true), 10000);
375             } else if (wearAutoRefreshTimer) {
376                 clearInterval(wearAutoRefreshTimer);
377                 wearAutoRefreshTimer = null;
378             }
379         };

```

```

380     }
381 }
382
383 async function refreshAllData(forceNetwork = false) {
384     isUpdating = true;
385     try {
386         if (forceNetwork) {
387             // Принудительная загрузка из сети (игнорируем кеш)
388             await Promise.allSettled([
389                 apiFetch('/api/v1/system/summary?process_limit=25').
390 then(data => {
391                 if (window.browserCache) {
392                     window.browserCache.set(CACHE_STORE, 'system_sum
393 mary', data, {
394                         ttl: CACHE_STRATEGY.SYSTEM_SUMMARY.ttl,
395                         tags: [CACHE_TAG]
396                     });
397                 }
398                 return fetchSystemSummary();
399             }),
400                 apiFetch('/api/system-control/status').then(data =>
401 {
402                 if (window.browserCache) {
403                     window.browserCache.set(CACHE_STORE, 'system_con
404 trol_status', data, {
405                         ttl: CACHE_STRATEGY.CONTROL_STATUS.ttl,
406                         tags: [CACHE_TAG]
407                     });
408                 }
409                 return fetchSystemControlStatus();
410             }),
411                 apiFetch('/api/v1/system/hardware').then(data => {
412                     if (window.browserCache) {
413                         window.browserCache.set(CACHE_STORE, 'hardware_s
414 pec', data, {
415                             ttl: CACHE_STRATEGY.HARDWARE_SPEC.ttl,
416                             tags: [CACHE_TAG]
417                         });

```

```

413         }
414         return fetchHardwareSpec();
415     }},
416     apiFetch('/api/v1/system/sensors').then(data => {
417         if (window.browserCache) {
418             window.browserCache.set(CACHE_STORE, 'hardware_s
419 ensors', data, {
420                 ttl: CACHE_STRATEGY.SENSORS.ttl,
421                 tags: [CACHE_TAG]
422             });
423         }
424         return fetchHardwareSensors();
425     }},
426     apiFetch('/api/v1/windows-backup/health').then(data
427 => {
428         if (window.browserCache) {
429             window.browserCache.set(CACHE_STORE, 'backup_sta
430 tus', data, {
431                 ttl: CACHE_STRATEGY.BACKUP_STATUS.ttl,
432                 tags: [CACHE_TAG]
433             });
434         }
435         return fetchBackupStatus();
436     }},
437     apiFetch('/api/v1/system/diagnostics/storage-battery
438 ').then(data => {
439         if (window.browserCache) {
440             window.browserCache.set(CACHE_STORE, 'storage_ba
441 ttery_wear', data, {
442                 ttl: CACHE_STRATEGY.STORAGE_BATTERY.ttl,
443                 tags: [CACHE_TAG]
444             });
445         }
446         return fetchStorageBatteryWear(true);
447     })
448 ]);
449 } else {
450     await Promise.allSettled([

```

```

446         fetchSystemSummary(),
447         fetchSystemControlStatus(),
448         fetchHardwareSpec(),
449         fetchHardwareSensors(),
450         fetchBackupStatus(),
451         fetchStorageBatteryWear()
452     ]);
453 }
454 } finally {
455     isUpdating = false;
456 }
457 }
458
459 async function fetchBackupStatus() {
460     try {
461         const config = CACHE_STRATEGY.BACKUP_STATUS;
462         const data = await cachedFetch(
463             '/api/v1/windows-backup/health',
464             'backup_status',
465             { ttl: config.ttl, strategy: config.strategy }
466         );
467         if (!data) return;
468
469         const cfg = data.file_history?.config;
470         const storage = data.storage_audit;
471
472         let lastTimeStr = 'Нет записей';
473         if (cfg && cfg.last_backup_time) {
474             try {
475                 const d = new Date(cfg.last_backup_time);
476                 if (!isNaN(d.getTime())) {
477                     lastTimeStr = d.toLocaleDateString() + ' ' + d.toL
478 ocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
479                 }
480             } catch (_) {}
481             } else if (storage && storage.sample_versions && storage
482 .sample_versions.length > 0) {
483                 const latestSample = storage.sample_versions[0];

```

```
482         if (latestSample && latestSample.version_timestamp) {
483             try {
484                 const d = new Date(latestSample.version_timestamp)
485                 ;
486                 if (!isNaN(d.getTime())) {
487                     lastTimeStr = d.toLocaleDateString() + ' ' + d.toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
488                 } catch (_) {}
489             }
490         }
491
492         let storageStr = '';
493         if (storage && storage.target_exists && storage.free_space_gb !== null && storage.free_space_gb !== undefined) {
494             const target = storage.target_path || 'Диск';
495             const freeGb = typeof storage.free_space_gb === 'number' ? storage.free_space_gb.toFixed(1) : storage.free_space_gb;
496             storageStr = `${target} (${freeGb} GB своб.)`;
497         } else if (cfg && (cfg.target_drive_letter || cfg.target_url)) {
498             storageStr = cfg.target_drive_letter || cfg.target_url
499             ;
500         } else {
501             storageStr = 'Хранилище не найдено';
502         }
503
504         const backupSummary = `${lastTimeStr} | ${storageStr}`;
505         const backupEl = document.getElementById('about-ident-backup');
506         if (backupEl) {
507             backupEl.textContent = backupSummary;
508             backupEl.title = `Последний бэкап: ${lastTimeStr}\nХранилище: ${storageStr}\nСлужба File History: ${data.file_history?.service_status || '-'}\nHealth Score: ${data.health_score ? '---'}/100`;
509         }
510     } catch (e) {
```

```

510     console.warn('[AboutSystemTab] fetchBackupStatus warning
    :', e);
511     setText('about-ident-backup', 'Не настроено');
512 }
513 }
514
515 async function fetchStorageBatteryWear(forceNetwork = false)
    {
516     try {
517         const config = CACHE_STRATEGY.STORAGE_BATTERY;
518         const data = await cachedFetch(
519             '/api/v1/system/diagnostics/storage-battery',
520             'storage_battery_wear',
521             { ttl: config.ttl, strategy: config.strategy, forceNet
work }
522         );
523         if (!data) return;
524
525         // Disks Wear & SMART
526         const disksTbody = document.getElementById('diag-wear-di
sks-tbody');
527         if (disksTbody && data.disks_wear) {
528             setText('diag-wear-disks-count', `${data.disks_wear.le
ngth} дисков`);
529             if (data.disks_wear.length === 0) {
530                 disksTbody.innerHTML = '<tr><td colspan="9" class="t
ext-center py-4 text-muted">Физических накопителей не обнаруже
но</td></tr>';
531             } else {
532                 const formatBytesLocal = (bytes) => {
533                     if (bytes == null || bytes === 0) return '0 B';
534                     const k = 1024;
535                     const sizes = ['B', 'KB', 'MB', 'GB', 'TB', 'PB'];
536                     const i = Math.floor(Math.log(bytes) / Math.log(k)
);
537                     return `${parseFloat((bytes / Math.pow(k, i)).toFi
xed(1)))} ${sizes[i]}`;
538                 };

```

```

539
540         const formatPohLocal = (hours) => {
541             if (hours == null || hours <= 0) return null;
542             const days = Math.floor(hours / 24);
543             const years = (hours / (24 * 365.25)).toFixed(1);
544             if (days >= 365) return `${hours.toLocaleString()}
ч (≈ ${years} г)`;
545             if (days >= 1) return `${hours.toLocaleString()} ч
(${days} д)`;
546             return `${hours.toLocaleString()} ч`;
547         };
548
549         disksTbody.innerHTML = data.disks_wear.map(d => {
550             const diskName = d.name || d.model || d.device_id
|| 'Физический диск';
551             const devId = d.device_id ? `<span class="badge bg
-secondary-subtle text-light border border-secondary me-1.5">${
escapeHtml(d.device_id)}</span>` : '';
552             const isSsd = (d.media_type && d.media_type.toUppe
rCase().includes('SSD')) || (d.bus_type && d.bus_type.toUpperC
ase().includes('NVME'));
553             const diskIcon = isSsd ? 'bi-device-ssd text-info'
: 'bi-hdd text-primary';
554             const typeBus = [d.media_type, d.bus_type].filter(
Boolean).join(' / ') || 'Storage';
555             const partsBadge = d.partitions && d.partitions !=
= '_'
556             ? `<span class="badge bg-primary-subtle text-inf
o border border-info-subtle font-monospace">${escapeHtml(d.par
titions)}</span>`
557             : `<span class="text-muted">—</span>`;
558             const freeGbStr = d.free_gb != null ? `${d.free_gb
} GB` : `<span class="text-muted">—</span>`;
559             const healthPct = d.health_pct != null ? d.health_
pct : 100;
560             const healthColor = healthPct >= 80 ? 'success' :
(healthPct >= 50 ? 'warning' : 'danger');
561

```

```

562         // Нарботка и дата первого включения
563         const pohFormatted = formatPohLocal(d.power_on_hou
rs);
564         const pohHtml = pohFormatted
565             ? `
```



```

587         </div>
588     </td>
589     <td class="text-muted font-monospace"><span clas
s="badge bg-dark border border-secondary text-light">${escapeH
tml(typeBus)}</span></td>
590     <td>${partsBadge}</td>
591     <td style="text-align: right;" class="text-light
font-monospace">${d.total_gb} GB</td>
592     <td style="text-align: right;" class="text-succe
ss font-monospace">${freeGbStr}</td>
593     <td>${pohHtml}</td>
594     <td style="text-align: right;">${ioHtml}</td>
595     <td>
596         <div class="d-flex align-items-center gap-1.5"
>
597             <div class="about-sys-progress-track flex-gr
ow-1" style="height: 6px; min-width: 55px;">
598                 <div class="about-sys-progress-bar bg-{$he
althColor}" style="width: ${healthPct}%;"></div>
599             </div>
600             <span class="small font-monospace text-{$hea
lthColor} ms-1">${healthPct}%</span>
601         </div>
602     </td>
603     <td><span class="badge bg-{$healthColor}-subtle
text-{$healthColor} border border-{$healthColor}">${escapeHtml
(d.status || 'OK')}</span></td>
604 </tr>
605 `;}).join('');
606     }
607 }
608
609 // Battery Degradation
610 const batContainer = document.getElementById('diag-batte
ry-details-container');
611 const batSourceBadge = document.getElementById('diag-bat
tery-source-badge');
612 if (batContainer && data.battery_wear) {

```

```

613         const b = data.battery_wear;
614         if (batSourceBadge) {
615             batSourceBadge.textContent = b.power_source || 'AC M
ains';
616         }
617         if (!b.has_battery) {
618             batContainer.innerHTML = `
619                 <div class="text-center py-4 text-muted small">
620                     <i class="bi bi-plug-fill fs-4 text-info d-block
mb-1"></i>
621                         Стационарный компьютер – питание напрямую от эле
ктросети
622                 </div>
623                 `;
624         } else {
625             batContainer.innerHTML = `
626                 <table class="about-sys-spec-table">
627                     <tbody>
628                         <tr>
629                             <td class="about-sys-spec-key"><i class="bi
bi-battery-charging me-1.5 text-warning"></i>Уровень заряда</t
d>
630                             <td class="about-sys-spec-val fw-bold text-w
hite">${b.percent}% (${b.is_charging ? 'Заряжается' : 'Разряд'
})</td>
631                         </tr>
632                         <tr>
633                             <td class="about-sys-spec-key"><i class="bi
bi-shield-shaded me-1.5 text-info"></i>Заводская емкость</td>
634                             <td class="about-sys-spec-val font-monospace
">${b.design_capacity_mwh ? b.design_capacity_mwh + ' mWh' : '
-- '}</td>
635                         </tr>
636                         <tr>
637                             <td class="about-sys-spec-key"><i class="bi
bi-battery-full me-1.5 text-success"></i>Текущая емкость</td>
638                             <td class="about-sys-spec-val font-monospace
">${b.full_charge_capacity_mwh ? b.full_charge_capacity_mwh +

```

```

    ' mWh' : '--'}</td>
639 |         </tr>
640 |         <tr>
641 |             <td class="about-sys-spec-key"><i class="bi
bi-heart-pulse me-1.5 text-danger"></i>Деградация (Износ)</td>
642 |             <td class="about-sys-spec-val ${b.wear_level
_pct > 20 ? 'text-danger fw-bold' : 'text-success'}">
643 |                 ${b.wear_level_pct}% износа
644 |             </td>
645 |         </tr>
646 |     </tbody>
647 | </table>
648 |     `;
649 |     }
650 | }
651 | } catch (err) {
652 |     console.warn('[AboutSystemTab] fetchStorageBatteryWear e
rror:', err);
653 | }
654 | }
655 |
656 | async function pollLiveTelemetry() {
657 |     isUpdating = true;
658 |     try {
659 |         await Promise.allSettled([
660 |             fetchSystemSummary(true),
661 |             fetchHardwareSensors(true)
662 |         ]);
663 |     } finally {
664 |         isUpdating = false;
665 |     }
666 | }
667 |
668 | async function fetchSystemSummary(isLightPoll = false) {
669 |     try {
670 |         const config = CACHE_STRATEGY.SYSTEM_SUMMARY;
671 |         const snap = await cachedFetch(
672 |             '/api/v1/system/summary?process_limit=25',

```

```

673         'system_summary',
674         { ttl: config.ttl, strategy: isLightPoll ? 'stale-while-revalidate' : 'network-first' }
675     );
676     if (!snap) return;
677
678     // 1. Identity & Locale Block
679     const host = snap.hostname || 'DELL-VOSTRO';
680     const user = snap.username || 'onela';
681     const lang = snap.system_language || 'Русский (Россия) [ru-RU]';
682     const userLoc = snap.user_locale || 'ru-RU';
683     const sysLoc = snap.system_locale || 'ru-RU';
684     const tz = snap.timezone || 'UTC+03:00';
685     const cp = snap.codepage || 'UTF-8 (ACP: 65001)';
686     const inputs = Array.isArray(snap.input_languages) && snap.input_languages.length > 0
687         ? snap.input_languages.join(', ')
688         : 'Русский (RU), English (US),  (IL)';
689     const osBuild = snap.os_build ? `${snap.os_name || 'Windows 11'} (Build ${snap.os_build})` : (snap.os_name || 'Windows 11');
690
691     setText('about-ident-hostname', host);
692     setText('about-ident-domain', `Workgroup / Host: ${host}`);
693     setText('about-ident-username', user);
694     setText('about-ident-language', lang);
695     setText('about-ident-locales', `Локали: User: ${userLoc} | Sys: ${sysLoc}`);
696     setText('about-ident-timezone', tz);
697     setText('about-ident-codepage', `Кодировка: ${cp}`);
698     setText('about-ident-inputs', inputs);
699     setText('about-ident-os-build', osBuild);
700     setText('about-ident-install-date', snap.os_install_date || 'Не определена');
701
702     // Top KPI Card 1: Operating System

```

```

703 |         setText('about-kpi-os-title', `${snap.os_name || 'Window
s 11'} (${snap.cpu?.architecture || 'AMD64'})`);
704 |         setText('about-kpi-os-host', `Host: ${host}`);
705 |
706 |         // 2. Telemetry Live Bar
707 |         if (snap.cpu) {
708 |             const cpuPct = snap.cpu.total_percent || 0;
709 |             setText('about-telemetry-cpu-val', `${cpuPct.toFixed(1
)}}%`);
710 |             const cpuBar = document.getElementById('about-telemetr
y-cpu-bar');
711 |             if (cpuBar) {
712 |                 cpuBar.style.width = `${Math.min(100, Math.max(0, cp
uPct))}%`;
713 |                 cpuBar.className = cpuPct > 85 ? 'about-sys-progress
-bar bg-danger' : (cpuPct > 60 ? 'about-sys-progress-bar bg-wa
rning' : 'about-sys-progress-bar bg-info');
714 |             }
715 |             const freqTxt = snap.cpu.frequency_mhz ? `${snap.cpu.f
requency_mhz} MHz` : '';
716 |             setText('about-telemetry-cpu-freq', freqTxt);
717 |             const cores = snap.cpu.physical_cores || 6;
718 |             const threads = snap.cpu.logical_cores || 12;
719 |             setText('about-telemetry-cpu-cores', `${cores} физ. /
${threads} Поток`);
720 |
721 |             // Specification Table CPU
722 |             setText('about-spec-cpu', `${snap.cpu.model || 'Intel
Processor'} (${threads} logical cores)`);
723 |         }
724 |
725 |         if (snap.memory) {
726 |             const total = Number(snap.memory.total_gb || 0).toFixe
d(1);
727 |             const used = Number(snap.memory.used_gb || 0).toFixed(
1);
728 |             const avail = Number(snap.memory.available_gb || 0).to
Fixed(1);

```

```

729         const pct = snap.memory.percent || 0;
730
731         setText('about-telemetry-ram-val', `${used} / ${total}
GB`);
732         setText('about-telemetry-ram-sub', `${pct}% занято (${
avail} GB свободно)`);
733         const ramBar = document.getElementById('about-telemetr
y-ram-bar');
734         if (ramBar) {
735             ramBar.style.width = `${Math.min(100, Math.max(0, pc
t))}%`;
736             ramBar.className = pct > 85 ? 'about-sys-progress-ba
r bg-danger' : (pct > 65 ? 'about-sys-progress-bar bg-warning'
: 'about-sys-progress-bar bg-info');
737         }
738
739         // Specification Table RAM (Hardware installed capacit
y)
740         setText('about-spec-ram', `${total} GB RAM (${avail} G
B свободно)`);
741     }
742
743     if (Array.isArray(snap.gpus) && snap.gpus.length > 0) {
744         const gpu = snap.gpus[0];
745         setText('about-telemetry-gpu-name', gpu.name || 'GPU')
;
746         const vramTxt = gpu.memory_total_gb ? `VRAM: ${Number(
gpu.memory_total_gb).toFixed(1)} GB` : 'VRAM: N/A';
747         const loadTxt = gpu.load_percent !== null && gpu.load_
percent !== undefined ? ` | Нагрузка: ${gpu.load_percent}%` :
'';
748         setText('about-telemetry-gpu-vram', `${vramTxt}${loadT
xt}`);
749
750         const badgeGpu = document.getElementById('about-teleme
try-gpu-badge');
751         if (badgeGpu) {
752             const caps = [];

```

```
753 |         if (gpu.has_cuda) caps.push('CUDA');
754 |         if (gpu.has_directml) caps.push('DirectML');
755 |         badgeGpu.textContent = caps.length > 0 ? caps.join('
| + ') : 'Active GPU';
756 |     }
757 | }
758 |
759 |     if (snap.disk_io) {
760 |         const rKbs = Math.round((snap.disk_io.read_bytes_per_s
| ec || 0) / 1024);
761 |         const wKbs = Math.round((snap.disk_io.write_bytes_per_
| sec || 0) / 1024);
762 |         const totalMb = (((snap.disk_io.read_bytes_per_sec ||
| 0) + (snap.disk_io.write_bytes_per_sec || 0)) / (1024 * 1024))
| .toFixed(2);
763 |         setText('about-telemetry-disk-val', `${totalMb} MB/s`
| ;
764 |         setText('about-telemetry-disk-rates', `Чтение: ${rKbs}
| КБ/с | Запись: ${wKbs} КБ/с`);
765 |     }
766 |
767 |     // Specification Table general
768 |     setText('about-spec-host', host);
769 |     setText('about-spec-os', osBuild);
770 |     if (snap.uptime_seconds) {
771 |         const sec = snap.uptime_seconds;
772 |         const h = Math.floor(sec / 3600);
773 |         const m = Math.floor((sec % 3600) / 60);
774 |         setText('about-spec-uptime', `Uptime: ${h}h ${m}m`);
775 |     }
776 |
777 |     // 3. Disks Table
778 |     if (Array.isArray(snap.disks) && snap.disks.length > 0)
| {
779 |         currentDisks = snap.disks;
780 |         renderDisksTable(snap.disks);
781 |
782 |         // Update Top KPI Storage C:
```

```

783 |         const cDrive = snap.disks.find(d => (d.device || '').t
      | oUpperCase().startsWith('C')) || snap.disks[0];
784 |         if (cDrive) {
785 |             const freeGb = Number(cDrive.free_gb || 0).toFixed(1
      | );
786 |             const totalGb = Number(cDrive.total_gb || 0).toFixed
      | (1);
787 |             setText('about-kpi-stor-title', `${freeGb} GB Free /
      | ${totalGb} GB`);
788 |         }
789 |     }
790 |
791 |     // 4. Monitors & Displays Block
792 |     if (Array.isArray(snap.monitors) && snap.monitors.length
      | > 0) {
793 |         const monShorts = snap.monitors.map(m => {
794 |             const prim = m.is_primary ? ' [Основной]' : '';
795 |             return `${m.name || 'Monitor'} (${m.width}x${m.heigh
      | t}@${m.frequency_hz}Hz${prim})`;
796 |         });
797 |         setText('about-ident-monitors', monShorts.join(', '));
798 |         setText('about-spec-monitors', monShorts.join(', '));
799 |     }
800 |
801 |     // 5. Windows Updates Block
802 |     if (snap.updates) {
803 |         const kbCount = snap.updates.installed_kb_count ? ` ($
      | {snap.updates.installed_kb_count} KBs)` : '';
804 |         const updTxt = `${snap.updates.status || 'Up to date'}
      | ${kbCount}`;
805 |         setText('about-ident-updates', updTxt);
806 |         setText('about-spec-update', updTxt);
807 |     }
808 |
809 |     // 5.1 MS Office Block
810 |     if (snap.office) {
811 |         const offTxt = snap.office.status || (snap.office.inst
      | alled ? `${snap.office.product_name || 'MS Office'} (${snap.of

```



```

    fice.version || '')` : 'Не установлен');
812     setText('about-ident-ms-office', offTxt);
813   } else {
814     setText('about-ident-ms-office', 'Не установлен');
815   }
816
817   // 5.2 OneDrive Storage Block
818   if (snap.onedrive) {
819     const odTxt = snap.onedrive.status || (snap.onedrive.i
nstalled ? `${snap.onedrive.free_gb || 0} GB своб.` : 'Не наст
роено');
820     setText('about-ident-onedrive', odTxt);
821   } else {
822     setText('about-ident-onedrive', 'Не настроено');
823   }
824
825   // 6. Processes Table
826   if (Array.isArray(snap.top_processes)) {
827     currentProcesses = snap.top_processes;
828     renderProcessesTable(snap.top_processes);
829   }
830
831   } catch (e) {
832     console.warn('[AboutSystemTab] fetchSystemSummary warnin
g:', e);
833   }
834 }
835
836 async function fetchSystemControlStatus() {
837   try {
838     const config = CACHE_STRATEGY.CONTROL_STATUS;
839     const res = await cachedFetch(
840       '/api/system-control/status',
841       'system_control_status',
842       { ttl: config.ttl, strategy: config.strategy }
843     );
844     if (!res) return;
845

```

```

846     const isElevated = res.is_elevated;
847     const privBadge = document.getElementById('about-sys-privilege-badge');
848     if (privBadge) {
849         if (isElevated) {
850             privBadge.className = 'badge rounded-pill bg-success -subtle text-success border border-success px-2.5 py-1';
851             privBadge.textContent = 'Administrator Access';
852         } else {
853             privBadge.className = 'badge rounded-pill bg-secondary-subtle text-secondary border border-secondary px-2.5 py-1';
854             privBadge.textContent = 'Standard User Mode';
855         }
856     }
857     setText('about-ident-privilege', isElevated ? 'Права: Администратор (Full Access)' : 'Права: Стандартный пользователь');
858
859     // Security Details
860     const sec = res.security || {};
861     const rest = res.restore || {};
862     const disk = res.disk || {};
863     const pwr = res.power || {};
864     const upd = res.update || {};
865
866     // KPI 2: Security & Defender
867     const defActive = sec.defender_enabled !== false;
868     setText('about-kpi-sec-title', defActive ? 'Active & Protected' : 'Attention Required');
869     const fwOk = sec.firewall_overall_enabled !== false;
870     const uacOk = sec.uac_enabled !== false;
871     setText('about-kpi-sec-sub', `Firewall: ${fwOk ? 'ON' : 'OFF'} | UAC: ${uacOk ? 'ON' : 'OFF'}`);
872
873     // KPI 3: System Protection
874     const countPoints = rest.restore_points_count || 0;
875     setText('about-kpi-prot-title', `${countPoints} Checkpoints`);

```

```

876 |         setText('about-kpi-prot-sub', `Protection: ${rest.system
|_protection_enabled ? 'Active' : 'Disabled'}`);
877 |
878 |         // KPI 4: Cleanable estimate & Total Disk Size
879 |         const cleanMb = disk.cleanup_estimate?.total_cleanable_m
|b || 150;
880 |         const cDrive = currentDisks.find(d => (d.device || '').t
|oUpperCase().startsWith('C')) || currentDisks[0];
881 |         const totalGbTxt = cDrive ? ` | Bcero: ${Number(cDrive.t
|otal_gb || 0).toFixed(1)} GB` : '';
882 |         setText('about-kpi-stor-clean', `Cleanable: ~${cleanMb}
|MB${totalGbTxt}`);
883 |
884 |         // Security table rows
885 |         const setSecStatus = (id, active, activeText = 'Enabled'
|, inactiveText = 'Disabled') => {
886 |             const el = document.getElementById(id);
887 |             if (!el) return;
888 |             const stateText = active ? activeText : inactiveText;
889 |             el.setAttribute('data-status', stateText);
890 |             el.className = active ? 'text-success fw-bold' : 'text
|-danger fw-bold';
891 |             el.textContent = stateText;
892 |         };
893 |         setSecStatus('about-sec-defender', defActive);
894 |         setSecStatus('about-sec-realtime', sec.realtime_protecti
|on_enabled !== false);
895 |         setSecStatus('about-sec-fw-domain', sec.firewall_profile
|s?.Domain, 'Active', 'Disabled');
896 |         setSecStatus('about-sec-fw-private', sec.firewall_profil
|es?.Private, 'Active', 'Disabled');
897 |         setSecStatus('about-sec-fw-public', sec.firewall_profile
|s?.Public, 'Active', 'Disabled');
898 |         setSecStatus('about-sec-uac', uacOk);
899 |
900 |         // Power Scheme & Updates
901 |         if (pwr.active_plan_name) setText('about-spec-power', pw
|r.active_plan_name);

```

```

902         if (upd.status) {
903             const kbCount = upd.recent_hotfixes_count ? ` (${upd.r
recent_hotfixes_count} KBs installed)` : '';
904             const updTxt = `${upd.status}${kbCount}`;
905             setText('about-spec-update', updTxt);
906             setText('about-ident-updates', updTxt);
907         }
908
909     } catch (e) {
910         console.warn('[AboutSystemTab] fetchSystemControlStatus
warning:', e);
911     }
912 }
913
914 async function fetchHardwareSensors(isPoll = false) {
915     try {
916         const config = CACHE_STRATEGY.SENSORS;
917         const sensors = await cachedFetch(
918             '/api/v1/system/sensors',
919             'hardware_sensors',
920             { ttl: config.ttl, strategy: config.strategy }
921         );
922         if (Array.isArray(sensors)) {
923             currentSensors = sensors;
924             const badge = document.getElementById('about-sensors-c
ount-badge');
925             if (badge) badge.textContent = `LHM Active (${sensors.
length} шт.)`;
926             renderSensorsList(sensors);
927         }
928     } catch (e) {
929         console.warn('[AboutSystemTab] fetchHardwareSensors warn
ing:', e);
930     }
931 }
932
933 function bindAIEvents() {
934     const btnRescan = document.getElementById('btn-about-ai-re

```

```
scan');
935     if (btnRescan) {
936         btnRescan.onclick = () => runAIDiagnostics(true);
937     }
938
939     const btnToggleStages = document.getElementById('btn-about
-ai-toggle-stages');
940     const stagesBox = document.getElementById('about-ai-stages
-container');
941     const icStages = document.getElementById('ic-about-ai-togg
le-stages');
942     const txtStages = document.getElementById('txt-about-ai-to
ggle-stages');
943     if (btnToggleStages && stagesBox) {
944         btnToggleStages.onclick = () => {
945             const isHidden = stagesBox.classList.toggle('d-none');
946             if (icStages) {
947                 icStages.className = isHidden ? 'bi bi-chevron-down
ms-0.5' : 'bi bi-chevron-up ms-0.5';
948             }
949             if (txtStages) {
950                 txtStages.textContent = isHidden ? 'Показать этапы'
: 'Свернуть этапы';
951             }
952         };
953     }
954
955     const btnToggleInstruction = document.getElementById('btn-
about-ai-toggle-instruction');
956     const instructionCollapse = document.getElementById('about
-ai-instruction-collapse');
957     const icInstruction = document.getElementById('ic-about-ai
-instruction');
958     if (btnToggleInstruction && instructionCollapse) {
959         btnToggleInstruction.onclick = () => {
960             const isShown = !instructionCollapse.classList.toggle(
'd-none');
961             if (icInstruction) {
```

```

962         icInstruction.className = isShown ? 'bi bi-chevron-u
p text-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
963     }
964     };
965 }
966
967     const btnCopyInstruction = document.getElementById('btn-ab
out-ai-copy-instruction');
968     if (btnCopyInstruction) {
969         btnCopyInstruction.onclick = () => {
970             const text = document.getElementById('about-ai-instruc
tion-text')?.textContent || '';
971             navigator.clipboard.writeText(text).then(() => {
972                 const s = document.getElementById('txt-about-ai-copy
-instruction');
973                 if (s) s.textContent = 'Скопировано!';
974                 setTimeout(() => { if (s) s.textContent = 'Копироват
ь'; }, 1800);
975             });
976         };
977     }
978
979     const btnTogglePrompt = document.getElementById('btn-about
-ai-toggle-prompt');
980     const promptCollapse = document.getElementById('about-ai-p
rompt-collapse');
981     const icPrompt = document.getElementById('ic-about-ai-prom
pt');
982     if (btnTogglePrompt && promptCollapse) {
983         btnTogglePrompt.onclick = () => {
984             const isShown = !promptCollapse.classList.toggle('d-no
ne');
985             if (icPrompt) {
986                 icPrompt.className = isShown ? 'bi bi-chevron-up tex
t-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
987             }
988         };
989     }

```

```
990
991     const btnCopyPrompt = document.getElementById('btn-about-a
i-copy-prompt');
992     if (btnCopyPrompt) {
993         btnCopyPrompt.onclick = () => {
994             const text = document.getElementById('about-ai-prompt-
text')?.textContent || '';
995             navigator.clipboard.writeText(text).then(() => {
996                 const s = document.getElementById('txt-about-ai-copy
-prompt');
997                 if (s) s.textContent = 'Скопировано!';
998                 setTimeout(() => { if (s) s.textContent = 'Копироват
ь'; }, 1800);
999                 });
1000         };
1001     }
1002
1003     const btnToggleRaw = document.getElementById('btn-about-ai
-toggle-row');
1004     const rawCollapse = document.getElementById('about-ai-row-
collapse');
1005     const icRaw = document.getElementById('ic-about-ai-row');
1006     if (btnToggleRaw && rawCollapse) {
1007         btnToggleRaw.onclick = () => {
1008             const isShown = !rawCollapse.classList.toggle('d-none'
);
1009             if (icRaw) {
1010                 icRaw.className = isShown ? 'bi bi-chevron-up text-m
uted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1011             }
1012         };
1013     }
1014
1015     const btnCopyRaw = document.getElementById('btn-about-ai-c
opy-row');
1016     if (btnCopyRaw) {
1017         btnCopyRaw.onclick = () => {
1018             const text = document.getElementById('about-ai-row-tex
```

```

1019         navigator.clipboard.writeText(text).then(() => {
1020             const s = document.getElementById('txt-about-ai-copy
-raw');
1021             if (s) s.textContent = 'Скопировано!';
1022             setTimeout(() => { if (s) s.textContent = 'Копироват
ь'; }, 1800);
1023         });
1024     };
1025 }
1026
1027     const btnToggleSnapshot = document.getElementById('btn-abo
ut-ai-toggle-snapshot');
1028     const snapshotCollapse = document.getElementById('about-ai
-snapshot-collapse');
1029     const icSnapshot = document.getElementById('ic-about-ai-sn
apshot');
1030     if (btnToggleSnapshot && snapshotCollapse) {
1031         btnToggleSnapshot.onclick = () => {
1032             const isShown = !snapshotCollapse.classList.toggle('d-
none');
1033             if (icSnapshot) {
1034                 icSnapshot.className = isShown ? 'bi bi-chevron-up t
ext-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1035             }
1036         };
1037     }
1038
1039     const btnCopySnapshot = document.getElementById('btn-about
-ai-copy-snapshot');
1040     if (btnCopySnapshot) {
1041         btnCopySnapshot.onclick = () => {
1042             const text = document.getElementById('about-ai-snapsho
t-text')?.textContent || '';
1043             navigator.clipboard.writeText(text).then(() => {
1044                 const s = document.getElementById('txt-about-ai-copy
-snapshot');
1045                 if (s) s.textContent = 'Скопировано!';

```



```
1046 |         setTimeout(() => { if (s) s.textContent = 'Копироват  
    | б JSON'; }, 1800);  
1047 |     });  
1048 |     };  
1049 |     }  
1050 | }  
1051 |  
1052 | function renderAIDiagnosticStages(stages, isCompleted = true  
    | ) {  
1053 |     const container = document.getElementById('about-ai-stages  
    | -container');  
1054 |     if (!container) return;  
1055 |     if (!Array.isArray(stages) || stages.length === 0) {  
1056 |         container.innerHTML = '<div class="small text-muted text  
    | -center py-2">Этапы пошагового аудита не запущены</div>';  
1057 |         return;  
1058 |     }  
1059 |  
1060 |     container.innerHTML = stages.map((st, idx) => {  
1061 |         const isLatest = idx === stages.length - 1 && !isComple  
    | ted;  
1062 |         const icon = isLatest  
1063 |             ? '<div class="spinner-grow spinner-grow-sm text-info"  
    | style="width: 0.6rem; height: 0.6rem;" role="status"></div>  
1064 |             : (st.stage === 'error' || st.stage === 'warning')  
1065 |             ? '<span class="text-warning small fw-bold" style="fon  
    | t-size: 0.72rem;">⚠ </span>'  
1066 |             : '<span class="text-success small fw-bold" style="fon  
    | t-size: 0.72rem;">✅ </span>';  
1067 |         const textClass = isLatest ? 'text-light fw-semibold' :  
    | 'text-muted';  
1068 |         return `  
1069 |             <div class="about-ai-stage-row ${textClass}">  
1070 |                 <div class="d-flex align-items-center gap-2">  
1071 |                     <span>${icon}</span>  
1072 |                     <span>${escapeHtml(st.message || st.title || 'Выпо  
    | лнение этапа...')}</span>  
1073 |                 </div>
```

```

1074 |         ${st.details ? `<div class="text-secondary ps-3 font
      | -monospace" style="font-size: 0.68rem; word-break: break-all;"
      | >␣ ${escapeHtml(st.details)}</div>` : ''}
1075 |         </div>
1076 |         `;
1077 |     }).join('');
1078 | }
1079 |
1080 | function renderAIDomainCards(groupsResults) {
1081 |     const container = document.getElementById('about-ai-groups
      | -container');
1082 |     if (!container) return;
1083 |
1084 |     container.innerHTML = groupsResults.map(g => {
1085 |         const isPending = g.status === 'pending';
1086 |         const isAnalyzing = g.status === 'analyzing';
1087 |         const isCrit = g.status === 'critical';
1088 |         const isWarn = g.status === 'warning';
1089 |
1090 |         const statusBadge = isPending
1091 |             ? '<span class="badge bg-secondary bg-opacity-25 text-
      | muted border border-secondary border-opacity-25"><i class="bi
      | bi-hourglass me-1"></i>Ожидание</span>'
1092 |             : isAnalyzing
1093 |             ? '<span class="badge bg-info bg-opacity-20 text-info
      | border border-info border-opacity-50"><span class="spinner-bor
      | der spinner-border-sm me-1" style="width:0.6rem;height:0.6rem;
      | "></span>Анализ LLM...</span>'
1094 |             : isCrit
1095 |             ? '<span class="badge bg-danger text-white"><i class="
      | bi bi-exclamation-octagon-fill me-1"></i>Критично</span>'
1096 |             : isWarn
1097 |             ? '<span class="badge bg-warning text-dark"><i class="
      | bi bi-exclamation-triangle-fill me-1"></i>Внимание</span>'
1098 |             : '<span class="badge bg-success-subtle text-success b
      | order border-success"><i class="bi bi-check-circle-fill me-1">
      | </i>В норме</span>';
1099 |

```

```

1100     const cardBorder = isCrit
1101       ? 'border-danger border-opacity-75 shadow-sm'
1102       : isWarn
1103       ? 'border-warning border-opacity-50 shadow-sm'
1104       : isAnalyzing
1105       ? 'border-info border-opacity-60 shadow-sm'
1106       : 'border-secondary border-opacity-30';
1107
1108     const metricsList = g.key_metrics && Object.keys(g.key_m
metrics).length > 0
1109     ? `<div class="mt-2 pt-1.5 border-top border-secondary
border-opacity-20 d-flex flex-wrap gap-1.5 font-monospace" st
yle="font-size: 0.70rem;">
1110         ${Object.entries(g.key_metrics).map(([k, v]) => `<
span class="badge bg-dark bg-opacity-75 border border-secondar
y text-light px-2 py-1"><strong class="text-info">${escapeHtml
(k)}:</strong> <span class="text-white">${escapeHtml(String(v)
)}</span></span>`).join('')}
1111         </div>`
1112     : '';
1113
1114     return `
1115         <div class="col-12 col-md-6">
1116             <div class="p-2.5 rounded bg-black bg-opacity-40 bor
der ${cardBorder} h-100 d-flex flex-column justify-content-bet
ween transition-all" id="domain-card-${g.group_id}">
1117                 <div>
1118                     <div class="d-flex align-items-center justify-co
ntent-between mb-1.5">
1119                         <div class="d-flex align-items-center gap-1.5"
>
1120                             <i class="bi ${escapeHtml(g.icon || 'bi-cpu'
)} text-info fs-6"></i>
1121                             <strong class="text-light" style="font-size:
0.82rem;">${escapeHtml(g.title)}</strong>
1122                         </div>
1123                         ${statusBadge}
1124                     </div>

```

```

1125 |         <div class="small text-light mt-1" style="font-s
      | size: 0.73rem; line-height: 1.45; white-space: pre-line;">
1126 |             ${escapeHtml(g.summary || g.description || 'Ож
      | идание запуска группы...')}
1127 |         </div>
1128 |     </div>
1129 |     ${metricsList}
1130 | </div>
1131 | </div>
1132 | `;
1133 | }).join('');
1134 | }
1135 |
1136 | function renderAIAnomalies(anomalies) {
1137 |     const container = document.getElementById('about-ai-anomal
      | ies-container');
1138 |     if (!container) return;
1139 |     if (!Array.isArray(anomalies) || anomalies.length === 0) {
1140 |         container.innerHTML = '<span class="badge bg-success-sub
      | tle text-success border border-success px-2.5 py-1" style="fon
      | t-size: 0.72rem;"><i class="bi bi-check-circle me-1"></i> Аном
      | алий в подсистемах оборудования не обнаружено</span>';
1141 |         return;
1142 |     }
1143 |
1144 |     container.innerHTML = anomalies.map(a => {
1145 |         const isCrit = a.severity === 'critical';
1146 |         return `<span class="anomaly-pill ${isCrit ? 'critical'
      | : 'warning'}">🔍 [${escapeHtml(a.subsystem || 'SYS')}] ${escape
      | Html(a.title || '')}: ${escapeHtml(a.description || '')}</span
      | >`;
1147 |     }).join('');
1148 | }
1149 |
1150 | async function runAIDiagnostics(isManualRescan = false) {
1151 |     if (isAiRunning) return;
1152 |     isAiRunning = true;
1153 |

```

```
1154 // Устанавливаем флаг processing для немедленного переключ
      ения вкладок
1155 if (window.setTabProcessing) {
1156     window.setTabProcessing('tab-about-system', true);
1157 }
1158
1159 const btnRescan = document.getElementById('btn-about-ai-re
      scan');
1160 const txtRescan = document.getElementById('txt-about-ai-re
      scan');
1161 const icRescan = document.getElementById('ic-about-ai-resc
      an');
1162 const badgeHealth = document.getElementById('about-ai-heal
      th-badge');
1163 const badgeHealthBtn = document.getElementById('about-ai-h
      ealth-badge-btn');
1164 const engineTag = document.getElementById('about-ai-engine
      -name');
1165 const summaryEl = document.getElementById('about-ai-summar
      y-text');
1166 const errorSec = document.getElementById('about-ai-error-s
      ection');
1167 const errorText = document.getElementById('about-ai-error-
      text');
1168 const progressBar = document.getElementById('about-ai-step
      -progress');
1169 const synthesisStatus = document.getElementById('about-ai-
      synthesis-status');
1170 const actionsBox = document.getElementById('about-ai-actio
      ns-box');
1171 const actionsList = document.getElementById('about-ai-acti
      ons-list');
1172
1173 const liveStatusText = document.getElementById('about-ai-l
      ive-status-text');
1174 const chipLhm = document.getElementById('chip-sensor-lhm')
      ;
1175 const chipWmi = document.getElementById('chip-sensor-wmi')
```

```

;
1176     const chipRam = document.getElementById('chip-sensor-ram')
;
1177     const chipDisks = document.getElementById('chip-sensor-dis
ks');
1178     const chipNet = document.getElementById('chip-sensor-net')
;
1179     const snapshotText = document.getElementById('about-ai-sna
pshot-text');
1180     const tickCpu = document.getElementById('live-tick-cpu');
1181     const tickTemp = document.getElementById('live-tick-temp')
;
1182     const tickRam = document.getElementById('live-tick-ram');
1183     const tickDisks = document.getElementById('live-tick-disks
');
1184     const tickTopProc = document.getElementById('live-tick-top
-proc');
1185
1186     if (btnRescan) btnRescan.disabled = true;
1187     if (icRescan) icRescan.className = 'spinner-border spinner
-border-sm text-dark';
1188     if (txtRescan) txtRescan.textContent = 'Диагностика...';
1189
1190     if (errorSec) errorSec.classList.add('d-none');
1191     if (actionsBox) actionsBox.classList.add('d-none');
1192     if (progressBar) progressBar.style.width = '5%';
1193
1194     if (liveStatusText) liveStatusText.textContent = 'Опрос ап
паратных датчиков хоста (LibreHardwareMonitor, WMI, psutil, SM
ART)...';
1195
1196     // Reset chips to active polling state
1197     [chipLhm, chipWmi, chipRam, chipDisks, chipNet].forEach(c
=> {
1198         if (c) c.className = 'badge bg-secondary bg-opacity-40 t
ext-light border border-secondary';
1199     });
1200

```

```
1201     if (badgeHealth) {
1202         badgeHealth.textContent = 'Health: Поэтапный опрос...';
1203         badgeHealth.className = 'badge bg-warning-subtle text-wa
rning border border-warning font-monospace';
1204     }
1205     if (badgeHealthBtn) {
1206         badgeHealthBtn.textContent = 'Health: Опрос...';
1207         badgeHealthBtn.className = 'badge bg-warning-subtle text
-warning border border-warning font-monospace ms-1';
1208     }
1209
1210     const stagesLog = [
1211     {
1212         stage: 'init',
1213         title: 'Сбор телеметрии',
1214         message: '□ Получение среза телеметрии и сенсоров хос
та...',
1215         details: 'Опрос LibreHardwareMonitor, WMI счетчиков и
SystemSnapshot',
1216     }
1217 ];
1218     renderAIDiagnosticStages(stagesLog, false);
1219
1220     try {
1221         // 1. Получение списка подготовленных 4 групп телеметрии
1222         const groups = await apiFetch('/api/v1/system/diagnose/g
roups');
1223         if (!Array.isArray(groups) || groups.length === 0) {
1224             throw new Error('Не удалось получить список диагностич
еских групп');
1225         }
1226
1227         if (snapshotText) {
1228             snapshotText.textContent = JSON.stringify(groups, null
, 2);
1229         }
1230
1231         // Обновление живой информационной плашки собранных метр
```

```
иК
1232     const compGroup = groups.find(g => g.group_id === 'compu
te_thermals');
1233     if (compGroup && compGroup.payload) {
1234         const cpu = compGroup.payload.cpu || {};
1235         if (tickCpu) tickCpu.textContent = `${cpu.model || 'CP
U'} (${cpu.load_percent ?? 0}%)`;
1236         const temps = compGroup.payload.sensors?.temperatures_
celsius || {};
1237         const tempVals = Object.values(temps);
1238         const maxT = tempVals.length > 0 ? Math.max(...tempVal
s) : null;
1239         if (tickTemp) tickTemp.textContent = maxT !== null ? `
${maxT}°C` : 'В норме';
1240     }
1241
1242     const memGroup = groups.find(g => g.group_id === 'memory
_processes');
1243     if (memGroup && memGroup.payload) {
1244         const ram = memGroup.payload.ram || {};
1245         if (tickRam) tickRam.textContent = `${ram.used_gb ?? 0
} / ${ram.total_gb ?? 0} GB (${ram.used_percent ?? 0}%)`;
1246         const procs = memGroup.payload.top_active_processes ||
[];
1247         if (tickTopProc) tickTopProc.textContent = procs.lengt
h > 0 ? `${procs[0].name} (${procs[0].cpu_percent ?? 0}%)` : '
Нет активных';
1248     }
1249
1250     const diskGroup = groups.find(g => g.group_id === 'stora
ge_smart');
1251     if (diskGroup && diskGroup.payload) {
1252         const parts = diskGroup.payload.storage_partitions ||
[];
1253         if (tickDisks) tickDisks.textContent = parts.map(p =>
`${p.mountpoint || p.device || 'Vol'}: ${p.free_gb ?? 0} GB св
.`).join(', ') || 'OK';
1254     }
```



```

1255
1256 // Отрисовка начальных 4 карточек в режиме ожидания с уж
1257 е доступными собранными метриками
1258 const cardsState = groups.map(g => {
1259   const keyMetrics = {};
1260   if (g.group_id === 'compute_thermals') {
1261     const cpu = g.payload?.cpu || {};
1262     keyMetrics['cpu_load'] = `${cpu.load_percent ?? 0}%`
1263   };
1264   keyMetrics['cpu_model'] = cpu.model || 'CPU';
1265   const temps = g.payload?.sensors?.temperatures_celsius || {};
1266   const maxT = Object.values(temps).length > 0 ? Math.
1267 max(...Object.values(temps)) : null;
1268   keyMetrics['max_temp'] = maxT !== null ? `${maxT}°C`
1269   : 'В норме';
1270   } else if (g.group_id === 'memory_processes') {
1271     const ram = g.payload?.ram || {};
1272     keyMetrics['ram_used'] = `${ram.used_gb ?? 0} / ${ra
1273 m.total_gb ?? 0} GB (${ram.used_percent ?? 0}%)`;
1274     const procs = g.payload?.top_active_processes || [];
1275     keyMetrics['top_process'] = procs.length > 0 ? `${pr
1276 ocs[0].name} (${procs[0].cpu_percent ?? 0}%)` : 'Нет';
1277   } else if (g.group_id === 'storage_smart') {
1278     const parts = g.payload?.storage_partitions || [];
1279     const maxUsed = parts.length > 0 ? Math.max(...parts
1280 .map(p => p.used_percent || 0)) : 0;
1281     keyMetrics['max_volume_fill'] = `${maxUsed}%`;
1282     keyMetrics['volumes_count'] = parts.length;
1283   } else if (g.group_id === 'system_network') {
1284     const updates = g.payload?.system_updates || {};
1285     keyMetrics['reboot_pending'] = updates.reboot_pendin
1286 g ? 'Да' : 'Нет';
1287     keyMetrics['update_status'] = updates.status || 'Up
1288 to date';
1289   }
1290 }
1291
1292 return {

```

```
1283         group_id: g.group_id,
1284         title: g.title,
1285         icon: g.icon,
1286         status: 'pending',
1287         summary: g.description,
1288         key_metrics: keyMetrics,
1289         payload: g.payload,
1290     };
1291 });
1292 renderAIDomainCards(cardsState);
1293
1294     const completedGroups = [];
1295
1296     // 2. Последовательный вызов каждой группы (цепочка Запр
1297     ос -> Ответ -> Отображение)
1298     for (let i = 0; i < groups.length; i++) {
1299         const group = groups[i];
1300         const stepNum = i + 1;
1301         const totalSteps = groups.length;
1302         const pct = Math.round((stepNum / totalSteps) * 85);
1303         if (progressBar) progressBar.style.width = `${pct}%`;
1304
1305         // Подсветка активного чипа собираемого домена
1306         if (group.group_id === 'compute_thermals') {
1307             if (chipLhm) chipLhm.className = 'badge bg-info text
1308             -dark border border-info fw-bold';
1309             if (chipWmi) chipWmi.className = 'badge bg-info text
1310             -dark border border-info fw-bold';
1311         } else if (group.group_id === 'memory_processes') {
1312             if (chipRam) chipRam.className = 'badge bg-info text
1313             -dark border border-info fw-bold';
1314         } else if (group.group_id === 'storage_smart') {
1315             if (chipDisks) chipDisks.className = 'badge bg-info
1316             text-dark border border-info fw-bold';
1317         } else if (group.group_id === 'system_network') {
1318             if (chipNet) chipNet.className = 'badge bg-info text
1319             -dark border border-info fw-bold';
1320         }
1321     }
1322 }
```

```
1315 |
1316 |         if (liveStatusText) {
1317 |             liveStatusText.textContent = `🔄 Анализ [${stepNum}/${
1318 | ${totalSteps}]: ${group.title} (оценка сенсоров и метрик)...`;
1319 |         }
1320 |
1321 |         // Обновляем статус карточки на "Анализ"
1322 |         cardsState[i].status = 'analyzing';
1323 |         cardsState[i].summary = `🔄 Модель выполняет целевой а
1324 | нализ телеметрии группы ${stepNum}/${totalSteps}...`;
1325 |         renderAIDomainCards(cardsState);
1326 |
1327 |         stagesLog.push({
1328 |             stage: 'group_run',
1329 |             title: `Группа ${stepNum}/${totalSteps}`,
1330 |             message: `🔄 [${stepNum}/${totalSteps}] Запрос модел
1331 | и: ${group.title}...`,
1332 |             details: group.description,
1333 |         });
1334 |         renderAIDiagnosticStages(stagesLog, false);
1335 |
1336 |         // Запрос к AI модели по данной группе
1337 |         const res = await apiFetch('/api/v1/system/diagnose/gr
1338 | oup', {
1339 |             method: 'POST',
1340 |             body: JSON.stringify({
1341 |                 group_id: group.group_id,
1342 |                 title: group.title,
1343 |                 payload: group.payload,
1344 |             }),
1345 |         });
1346 |
1347 |         if (res) {
1348 |             cardsState[i] = res;
1349 |             completedGroups.push(res);
1350 |         } else {
1351 |             cardsState[i].status = 'warning';
1352 |             cardsState[i].summary = 'Ответ не получен, использую
```

```

    тся базовые метрики.';
1349         completedGroups.push(cardsState[i]);
1350     }
1351
1352     // Отмечаем чип как завершённый (зелёный)
1353     if (group.group_id === 'compute_thermals') {
1354         if (chipLhm) chipLhm.className = 'badge bg-success-s
ubtle text-success border border-success';
1355         if (chipWmi) chipWmi.className = 'badge bg-success-s
ubtle text-success border border-success';
1356     } else if (group.group_id === 'memory_processes') {
1357         if (chipRam) chipRam.className = 'badge bg-success-s
ubtle text-success border border-success';
1358     } else if (group.group_id === 'storage_smart') {
1359         if (chipDisks) chipDisks.className = 'badge bg-succe
ss-subtle text-success border border-success';
1360     } else if (group.group_id === 'system_network') {
1361         if (chipNet) chipNet.className = 'badge bg-success-s
ubtle text-success border border-success';
1362     }
1363
1364     // Немедленно обновляем красивую карточку на экране!
1365     renderAIDomainCards(cardsState);
1366
1367     stagesLog.push({
1368         stage: 'group_done',
1369         title: `Группа ${stepNum} завершена`,
1370         message: `▢ [${stepNum}/${totalSteps}] ${group.title
}: ${cardsState[i].status.toUpperCase()}`,
1371         details: cardsState[i].summary.substring(0, 120) + (
cardsState[i].summary.length > 120 ? '...' : ''),
1372     });
1373     renderAIDiagnosticStages(stagesLog, false);
1374 }
1375
1376 if (liveStatusText) {
1377     liveStatusText.textContent = '▢ Поэтапный опрос и ана
лиз всех доменов завершён.';

```

```

1378     }
1379
1380     // 3. Финальный синтез итогового вердикта
1381     if (progressBar) progressBar.style.width = '95%';
1382     stagesLog.push({
1383         stage: 'synthesis',
1384         title: 'Финальный синтез',
1385         message: '□ Формирование итогового заключения и расче
т Health Score...',
1386         details: 'Агрегация заключений всех 4 доменов',
1387     });
1388     renderAIDiagnosticStages(stagesLog, false);
1389
1390     const synthesis = await apiFetch('/api/v1/system/diagnos
e/synthesize', {
1391         method: 'POST',
1392         body: JSON.stringify({ groups: completedGroups }),
1393     });
1394
1395     if (progressBar) progressBar.style.width = '100%';
1396
1397     // Обновление итогового Health Badge
1398     const score = Number(synthesis.health_score || 100);
1399     const healthClass = score >= 80 ? 'bg-success-subtle tex
t-success border border-success' :
1400     score >= 60 ? 'bg-warning-subtle text-warning border b
order-warning' :
1401     'bg-danger-subtle text-danger border border-danger';
1402
1403     if (badgeHealth) {
1404         badgeHealth.textContent = `Health: ${score}/100`;
1405         badgeHealth.className = `badge font-monospace ${health
Class}`;
1406     }
1407     if (badgeHealthBtn) {
1408         badgeHealthBtn.textContent = `Health: ${score}/100`;
1409         badgeHealthBtn.className = `badge font-monospace ms-1
${healthClass}`;

```

```
1410     }
1411
1412     if (engineTag) {
1413         engineTag.innerHTML = `<i class="bi bi-cpu me-1"></i>${
1414             escapeHtml(synthesis.ai_model_used || 'Heuristic Engine')}`;
1415     }
1416
1417     if (synthesisStatus) {
1418         synthesisStatus.textContent = `${synthesis.status_label} (${score}/100)`;
1419         synthesisStatus.className = `badge font-monospace ${
1420             score >= 80 ? 'bg-success text-white' :
1421             score >= 60 ? 'bg-warning text-dark' :
1422             'bg-danger text-white'
1423         }`;
1424     }
1425
1426     if (summaryEl) {
1427         summaryEl.textContent = synthesis.executive_summary ||
1428         'Анализ завершён.';
1429     }
1430
1431     // Отрисовка приоритетных действий
1432     if (Array.isArray(synthesis.critical_actions) && synthesis.critical_actions.length > 0) {
1433         if (actionsBox && actionsList) {
1434             actionsBox.classList.remove('d-none');
1435             actionsList.innerHTML = synthesis.critical_actions.map(act => `<li class="mb-1"><i class="bi bi-arrow-right-short text-warning me-1"></i>${escapeHtml(act)}</li>`).join('');
1436         }
1437     }
1438
1439     stagesLog.push({
1440         stage: 'done',
1441         title: 'Аудит завершен',
1442         message: `□ Поэтапная AI-диагностика успешно завершена (Health: ${score}/100)`,
```

```
1441         details: 'Все 4 функциональные группы оценены моделью'
1442     ,
1443     });
1444     renderAIDiagnosticStages(stagesLog, true);
1445 } catch (e) {
1446     console.error('[AboutSystemTab] Grouped AI Diagnosis err
1447 or:', e);
1448     if (summaryEl) summaryEl.textContent = 'Ошибка выполнени
1449 я поэтапной AI-диагностики: ' + e.message;
1450     if (badgeHealth) {
1451         badgeHealth.textContent = 'Health: Error';
1452         badgeHealth.className = 'badge bg-danger-subtle text-d
1453 anger border border-danger font-monospace';
1454     }
1455     if (badgeHealthBtn) {
1456         badgeHealthBtn.textContent = 'Health: Error';
1457         badgeHealthBtn.className = 'badge bg-danger-subtle tex
1458 t-danger border border-danger font-monospace ms-1';
1459     }
1460     if (errorSec) {
1461         errorSec.classList.remove('d-none');
1462         if (errorText) errorText.textContent = e.message;
1463     }
1464     stagesLog.push({
1465         stage: 'error',
1466         title: 'Ошибка',
1467         message: `❌ Ошибка выполнения: ${e.message}`,
1468         details: 'Проверьте доступность API-сервера',
1469     });
1470     renderAIDiagnosticStages(stagesLog, true);
1471 } finally {
1472     isAiRunning = false;
1473     // Сбрасываем флаг processing после завершения
1474     if (window.resetTabProcessing) {
1475         window.resetTabProcessing('tab-about-system');
1476     }
1477 }
```

```
1474
1475     if (btnRescan) btnRescan.disabled = false;
1476     if (icRescan) icRescan.className = 'bi bi-lightning-char
ge-fill';
1477     if (txtRescan) txtRescan.textContent = 'Rescan';
1478 }
1479 }
1480
1481
1482 async function fetchHardwareSpec() {
1483     const container = document.getElementById('about-sys-tree-
container');
1484     const badgeCount = document.getElementById('about-sys-node
-count');
1485     if (!container) return;
1486
1487     try {
1488         const config = CACHE_STRATEGY.HARDWARE_SPEC;
1489         const nodes = await cachedFetch(
1490             '/api/v1/system/hardware',
1491             'hardware_spec',
1492             { ttl: config.ttl, strategy: config.strategy }
1493         );
1494         hardwareData = Array.isArray(nodes) ? nodes : [];
1495
1496         if (badgeCount) {
1497             badgeCount.textContent = `${hardwareData.length} кате
рий`;
1498         }
1499
1500         const searchInput = document.getElementById('about-sys-s
earch');
1501         const filter = searchInput ? searchInput.value.trim().to
LowerCase() : '';
1502         renderHardwareTree(hardwareData, filter);
1503     } catch (e) {
1504         console.error('[AboutSystemTab] Failed to fetch hardware
spec:', e);
    }
```



```
1505     container.innerHTML = `<div class="text-center py-4 text
    -danger small">Ошибка загрузки оборудования: ${escapeHtml(e.me
    ssage)}</div>`;
1506   }
1507 }
1508
1509 function renderDisksTable(disks) {
1510   const tbody = document.getElementById('about-sys-disks-tbo
    dy');
1511   const countBadge = document.getElementById('about-disks-co
    unt');
1512   if (!tbody) return;
1513
1514   if (!Array.isArray(disks) || disks.length === 0) {
1515     tbody.innerHTML = '<tr><td colspan="7" class="text-cen
    te
    r py-3 text-muted">Дисковые разделы не обнаружены</td></tr>';
1516     return;
1517   }
1518
1519   if (countBadge) countBadge.textContent = `${disks.length}
    Drives`;
1520
1521   tbody.innerHTML = disks.map(d => {
1522     const pct = Math.min(100, Math.max(0, d.percent || 0));
1523     let barClass = 'bg-info';
1524     if (pct > 88) barClass = 'bg-danger';
1525     else if (pct > 70) barClass = 'bg-warning';
1526
1527     const devStr = String(d.device || d.mountpoint || '');
1528     const volStr = String(d.volume_name || '');
1529     const isGoogleDrive = d.is_virtual || d.drive_type === '
    Google Drive' || volStr.toLowerCase().includes('google') || vo
    lStr.includes('@') || devStr.startsWith('G:') || devStr.starts
    With('J:');
1530
1531     let driveIcon = 'bi-hdd-fill text-primary';
1532     let driveBadge = '';
1533     let driveSubtitle = '';
```

```

1534 |         let fsBadge = ``;
1535 |         let smartBadge = `

```

```

1559         <i class="bi ${driveIcon} me-1.5 fs-6"></i>
1560         <div>
1561             <div class="d-flex align-items-center">
1562                 <strong>${escapeHtml(devStr || 'C:\\')}</str
ong>
1563                 ${driveBadge}
1564             </div>
1565             ${driveSubtitle}
1566         </div>
1567     </div>
1568 </td>
1569 <td>${fsBadge}</td>
1570 <td class="font-monospace">${Number(d.total_gb || 0)
.toFixed(1)} GB</td>
1571 <td class="text-light font-monospace">${Number(d.use
d_gb || 0).toFixed(1)} GB</td>
1572 <td class="text-info fw-semibold font-monospace">${N
umber(d.free_gb || 0).toFixed(1)} GB</td>
1573 <td>
1574     <div class="d-flex align-items-center gap-2">
1575         <div class="about-sys-progress-track flex-grow-1
" style="height: 6px;">
1576             <div class="about-sys-progress-bar ${barClass}
" style="width: ${pct}%;"></div>
1577         </div>
1578         <span class="small font-monospace" style="min-wi
dth: 42px; text-align: right;">${pct.toFixed(1)}%</span>
1579     </div>
1580 </td>
1581 <td>${smartBadge}</td>
1582 </tr>
1583 `;
1584 }).join('');
1585 }
1586
1587 function renderSensorsList(sensors) {
1588     const container = document.getElementById('about-sensors-c
ontainer');

```

```
1589     if (!container) return;
1590
1591     if (!Array.isArray(sensors) || sensors.length === 0) {
1592         container.innerHTML = '<div class="text-center py-4 text
- muted small">Нет доступных сенсоров LHM</div>';
1593         return;
1594     }
1595
1596     const searchInput = document.getElementById('about-sensor-
search');
1597     const query = searchInput ? searchInput.value.trim().toLow
erCase() : '';
1598
1599     const filtered = sensors.filter(s => {
1600         if (activeSensorFilter !== 'all') {
1601             const cat = (s.category || '').toLowerCase();
1602             if (activeSensorFilter === 'temperature' && !cat.inclu
des('temp')) return false;
1603             if (activeSensorFilter === 'voltage' && !cat.includes(
'volt')) return false;
1604             if (activeSensorFilter === 'clock' && !cat.includes('c
lock') && !cat.includes('freq')) return false;
1605             if (activeSensorFilter === 'power' && !cat.includes('p
ower') && !cat.includes('load')) return false;
1606         }
1607         if (query) {
1608             return (s.name || '').toLowerCase().includes(query) ||
(s.category || '').toLowerCase().includes(query);
1609         }
1610         return true;
1611     });
1612
1613     if (filtered.length === 0) {
1614         container.innerHTML = '<div class="text-center py-3 text
- muted small">Сенсоры по заданному фильтру не найдены</div>';
1615         return;
1616     }
1617
```

```

1618     container.innerHTML = filtered.map(s => {
1619         let icon = 'bi bi-thermometer-half text-danger';
1620         let valColor = 'text-info';
1621         const cat = (s.category || '').toLowerCase();
1622
1623         if (cat.includes('volt')) {
1624             icon = 'bi bi-lightning-charge text-warning';
1625             valColor = 'text-warning';
1626         } else if (cat.includes('clock') || cat.includes('freq'))
1627     ) {
1628         icon = 'bi bi-speedometer text-info';
1629         valColor = 'text-info';
1630     } else if (cat.includes('power')) {
1631         icon = 'bi bi-plug text-success';
1632         valColor = 'text-success';
1633     } else if (cat.includes('load')) {
1634         icon = 'bi bi-percent text-primary';
1635         valColor = 'text-light';
1636     }
1637
1638     return `
1639         <div class="d-flex align-items-center justify-content-
1640         between py-1 px-2 border-bottom border-secondary-subtle">
1641             <div class="d-flex align-items-center gap-2 text-tru
1642             ncate" style="max-width: 68%;">
1643                 <i class="${icon}" style="font-size: 0.85rem;"></i
1644             >
1645                 <span class="small text-truncate" title="${escapeH
1646                 tml(s.name)}">${escapeHtml(s.name)}</span>
1647             </div>
1648                 <div class="sensor-badge-val ${valColor}">${s.value}
1649                 ${escapeHtml(s.unit || '')}</div>
1650             </div>
1651         `;
1652     }).join('');
1653 }
1654
1655 function renderProcessesTable(procs) {

```

```
1650     const tbody = document.getElementById('about-sys-procs-tbo
dy');
1651     if (!tbody) return;
1652
1653     if (!Array.isArray(procs) || procs.length === 0) {
1654         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
ter py-4 text-muted">Процессы не найдены</td></tr>';
1655         return;
1656     }
1657
1658     const searchInput = document.getElementById('about-proc-se
arch');
1659     const query = searchInput ? searchInput.value.trim().toLow
erCase() : '';
1660
1661     const filtered = procs.filter(p => {
1662         if (!query) return true;
1663         return String(p.pid).includes(query) ||
1664             (p.name || '').toLowerCase().includes(query) ||
1665             (p.username || '').toLowerCase().includes(query);
1666     });
1667
1668     if (filtered.length === 0) {
1669         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
ter py-4 text-muted">Процессы по запросу не найдены</td></tr>';
1670         return;
1671     }
1672
1673     tbody.innerHTML = filtered.slice(0, 30).map(p => {
1674         const cpuVal = Number(p.cpu_percent || 0).toFixed(1);
1675         const memVal = Number(p.memory_mb || 0).toFixed(1);
1676         const isHigh = p.cpu_percent > 10;
1677
1678         return `
1679         <tr>
1680             <td class="text-muted">${p.pid}</td>
1681             <td class="text-white fw-semibold text-truncate" sty
le="max-width: 160px;" title="${escapeHtml(p.name)}">
```

```

1682         ${escapeHtml(p.name)}
1683     </td>
1684     <td><span class="badge bg-dark border border-seconda
ry text-secondary" style="font-size: 0.68rem;">${escapeHtml(p.
status || 'running')}</span></td>
1685     <td class="${isHigh ? 'text-danger fw-bold' : 'text-
info'}">${cpuVal}%</td>
1686     <td class="text-light">${memVal} MB</td>
1687     <td class="text-secondary">${p.num_threads || 1}</td
>
1688     <td class="text-muted small text-truncate" style="ma
x-width: 140px;" title="${escapeHtml(p.username || '')}">
1689         ${escapeHtml(p.username || '-')}
1690     </td>
1691 </tr>
1692 `;
1693 }).join('');
1694 }
1695
1696 /**
1697  * Динамический расчет высоты контейнера дерева спецификации
оборудования
1698  * в зависимости от количества категорий и развернутых строк
1699  */
1700 function adjustHardwareTreeHeight() {
1701     const container = document.getElementById('about-sys-tree-
container');
1702     if (!container) return;
1703
1704     const visibleNodes = container.querySelectorAll('.about-sy
s-tree-node');
1705     if (visibleNodes.length === 0) {
1706         container.style.maxHeight = '300px';
1707         return;
1708     }
1709
1710     let totalEstimatedHeight = 32;
1711     visibleNodes.forEach(node => {

```

```

1712     totalEstimatedHeight += 38;
1713     const body = node.querySelector('.about-sys-tree-body');
1714     if (body && !body.classList.contains('d-none')) {
1715         const rows = body.querySelectorAll('.about-sys-prop-ro
w');
1716         const gridRowCount = Math.ceil(rows.length / 2);
1717         totalEstimatedHeight += gridRowCount * 28 + 16;
1718     }
1719 });
1720
1721     const vhLimit = Math.floor(window.innerHeight * 0.85);
1722     const dynamicMaxHeight = Math.max(500, Math.min(totalEstim
atedHeight + 20, Math.max(900, vhLimit)));
1723     container.style.maxHeight = `${dynamicMaxHeight}px`;
1724 }
1725
1726 function renderHardwareTree(nodes, filter) {
1727     const container = document.getElementById('about-sys-tree-
container');
1728     if (!container) return;
1729
1730     if (!Array.isArray(nodes) || nodes.length === 0) {
1731         container.innerHTML = '<div class="text-center py-4 text
-muted small">Оборудование не обнаружено или опрашивается...</
div>';
1732         adjustHardwareTreeHeight();
1733         return;
1734     }
1735
1736     const filtered = nodes.filter(node => {
1737         if (!filter) return true;
1738         const catMatch = (node.category || '').toLowerCase().inc
ludes(filter);
1739         const nameMatch = (node.name || '').toLowerCase().includ
es(filter);
1740         const propsMatch = Object.entries(node.properties || {}).
some(
1741             ([k, v]) => k.toLowerCase().includes(filter) || String

```



```

    (v).toLowerCase().includes(filter)
1742     );
1743     return catMatch || nameMatch || propsMatch;
1744 });
1745
1746     if (filtered.length === 0) {
1747         container.innerHTML = `
```

```

1771         <span class="d-flex align-items-center gap-2">
1772             <i class="${iconClass} text-info"></i>
1773             <strong>${escapeHtml(node.category)}:</strong>
1774             <span class="text-white">${escapeHtml(node.name
|| ' ')}</span>
1775         </span>
1776         <span class="about-sys-chevron" id="about-chevron-
${idx}" style="font-size: 0.72rem; color: #38bdf8;">▼</span>
1777     </div>
1778     <div class="about-sys-tree-body" id="about-node-body
-${idx}">
1779         ${propsHtml}
1780     </div>
1781 </div>
1782 `;
1783 }).join('');
1784
1785     adjustHardwareTreeHeight();
1786 }
1787
1788     window.toggleNode = function(idx) {
1789         const body = document.getElementById(`about-node-body-${id
x}`);
1790         const chevron = document.getElementById(`about-chevron-${i
dx}`);
1791         if (body) {
1792             body.classList.toggle('d-none');
1793             if (chevron) {
1794                 chevron.textContent = body.classList.contains('d-none'
) ? '▼' : '▲';
1795             }
1796             adjustHardwareTreeHeight();
1797         }
1798     };
1799
1800     function getNodeIcon(category) {
1801         const cat = (category || '').toLowerCase();
1802         if (cat.includes('processor') || cat.includes('cpu')) retu

```

```
rn 'bi bi-cpu';
1803     if (cat.includes('module') || cat.includes('планка')) retu
rn 'bi bi-memory';
1804     if (cat.includes('memory') || cat.includes('ram')) return
'bi bi-sd-card';
1805     if (cat.includes('system') || cat.includes('os')) return '
bi bi-laptop';
1806     if (cat.includes('motherboard') || cat.includes('mainboard
')) return 'bi bi-motherboard';
1807     if (cat.includes('display') || cat.includes('gpu') || cat.
includes('video') || cat.includes('graphics')) return 'bi bi-g
pu-card';
1808     if (cat.includes('monitor') || cat.includes('screen') || c
at.includes('экран')) return 'bi bi-display';
1809     if (cat.includes('physical') || cat.includes('диск')) retu
rn 'bi bi-hdd-fill';
1810     if (cat.includes('disk') || cat.includes('storage') || cat
.includes('drive') || cat.includes('volume')) return 'bi bi-hd
d-stack';
1811     if (cat.includes('network') || cat.includes('adapter') ||
cat.includes('ethernet') || cat.includes('wi-fi') || cat.inclu
des('сеть')) return 'bi bi-ethernet';
1812     if (cat.includes('audio') || cat.includes('sound') || cat.
includes('звук')) return 'bi bi-volume-up-fill';
1813     if (cat.includes('usb') || cat.includes('контроллер')) ret
urn 'bi bi-usb-symbol';
1814     if (cat.includes('update') || cat.includes('servicing')) r
eturn 'bi bi-arrow-repeat';
1815     return 'bi bi-gear-fill';
1816 }
1817
1818 function escapeHtml(str) {
1819     if (str === null || str === undefined) return '';
1820     return String(str)
1821         .replace(/&/g, '&')
1822         .replace(/</g, '&lt;')
1823         .replace(/>/g, '&gt;')
1824         .replace(/"/g, '&quot;');
```

```
1825 |         .replace(/'/g, '&#039;');
1826 |     }
1827 |
1828 |     // Self-trigger if container is already in DOM
1829 |     if (document.getElementById('about-sys-wrapper') || document
1830 |         .getElementById('tab-about-system')) {
1831 |         if (document.readyState === 'loading') {
1832 |             document.addEventListener('DOMContentLoaded', () => init
1833 | AboutSystemTab());
1834 |         } else {
1835 |             setTimeout(() => initAboutSystemTab(), 10);
1836 |         }
1837 |     }
1838 | }));
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title>AI Assistant - Admin</title>
8   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
9   t/css/bootstrap.min.css" rel="stylesheet">
10  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
11  0.0/font/bootstrap-icons.css" rel="stylesheet">
12  <link href="/html/css/main.css?v=20260925_v1" rel="styleshee
13  t">
14  <script src="/html/js/toast.js?v=20260925_v1"></script>
15  <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
16  8next.min.js"></script>
17 </head>
18 <body>
19
20 <!-- Password Protection Modal -->
21 <div class="modal fade" id="passwordModal" tabindex="-1" aria-
22 labelledby="passwordModalLabel" aria-hidden="true" data-bs-bac
23 kdrop="static" data-bs-keyboard="false">
24   <div class="modal-dialog modal-dialog-centered">
25     <div class="modal-content bg-dark text-white border-second
26     ary">
27       <div class="modal-header border-secondary">
28         <h5 class="modal-title" id="passwordModalLabel"><i cla
29         ss="bi bi-lock-fill"></i> Вход в админку</h5>
30         <button type="button" class="btn-close btn-close-white
31         " data-bs-dismiss="modal" aria-label="Close" id="closePassword
32         Modal"></button>
33       </div>
34       <div class="modal-body p-4">
35         <p class="text-muted">Введите пароль для доступа к адм
36         инистраторскому интерфейсу:</p>
```

```

25         <div class="mb-3">
26             <input type="password" class="form-control bg-dark t
ext-white border-secondary" id="admin-password" placeholder="•
....." autocomplete="current-password" data-lpignore="true" da
ta-lp-ignore="true" autofocus>
27         </div>
28         <div id="password-error" class="alert alert-danger d-n
one">Неверный пароль</div>
29         <button type="button" class="btn btn-primary w-100" id
="login-btn">Войти</button>
30     </div>
31 </div>
32 </div>
33 </div>
34
35 <div class="container-fluid mt-2" id="admin-interface">
36
37     <!-- Navigation Bar -->
38     <div class="d-flex align-items-center justify-content-betwee
n mb-3 main-nav-container rounded px-2 py-1">
39         <ul class="nav nav-tabs flex-nowrap flex-grow-1 border-0 m
e-2 align-items-center" id="mainTabs">
40             <!-- Кнопка: 0 системе (сверху и первая) -->
41             <li class="nav-item">
42                 <button class="nav-link d-flex align-items-center gap-
1 active" data-tab="tab-about-system" data-bs-target="#tab-abo
ut-system" type="button" data-i18n="tabs.aboutSystem">
43                     <i class="bi bi-info-circle-fill text-info me-1"></i
>
44                     <span>0 системе</span>
45                 </button>
46             </li>
47
48             <!-- Разделитель -->
49             <li class="nav-item d-flex align-items-center">
50                 <div class="nav-group-divider"></div>
51             </li>
52

```

```

53 |     <!-- Группа: Диалог и аудио -->
54 |     <li class="nav-item dropdown">
55 |         <button class="nav-link dropdown-toggle active d-flex
| align-items-center gap-1" type="button" id="dialogTabsDropdown
| ">
56 |             <i class="bi bi-chat-dots-fill text-primary me-1"></
| i>
57 |             <span data-i18n="tabs.groupDialog">Диалог и аудио</s
| pan>
58 |         </button>
59 |         <div class="dropdown-menu shadow border-secondary-subt
| le">
60 |             <button class="dropdown-item active d-flex align-ite
| ms-center gap-2" data-tab="tab-chat" data-bs-target="#tab-chat
| " type="button" data-i18n="tabs.chat">
61 |                 □ Чат
62 |             </button>
63 |             <button class="dropdown-item d-flex align-items-cent
| er gap-2" data-tab="tab-voice" data-bs-target="#tab-voice" typ
| e="button" data-i18n="tabs.voice">
64 |                 □ Голос и диаризация
65 |             </button>
66 |             <button class="dropdown-item d-flex align-items-cent
| er gap-2" data-tab="tab-tts" data-bs-target="#tab-tts" type="b
| utton" data-i18n="tabs.tts">
67 |                 □ Настройка TTS
68 |             </button>
69 |         </div>
70 |     </li>
71 |
72 |     <!-- Разделитель -->
73 |     <li class="nav-item d-flex align-items-center">
74 |         <div class="nav-group-divider"></div>
75 |     </li>
76 |
77 |     <!-- Группа: ИИ и Знания -->
78 |     <li class="nav-item dropdown">
79 |         <button class="nav-link dropdown-toggle d-flex align-i

```

```

    tems-center gap-1" type="button" id="aiTabsDropdown">
80         <i class="bi bi-cpu-fill text-info me-1"></i>
81         <span data-i18n="tabs.groupAi">ИИ и Знания</span>
82     </button>
83     <div class="dropdown-menu shadow border-secondary-subt
le">
84         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-rag" data-bs-target="#tab-rag" type="b
utton" data-i18n="tabs.rag">
85             □ RAG
86         </button>
87         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-pixelrag" data-bs-target="#tab-pixelra
g" type="button">
88             □ PixelRAG
89         </button>
90         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-models" data-bs-target="#tab-models" t
ype="button" data-i18n="tabs.models">
91             □ Модели и API
92         </button>
93         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-agents" data-bs-target="#tab-agents" t
ype="button" data-i18n="tabs.agents">
94             □ Агенты ИИ
95         </button>
96         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-skills" data-bs-target="#tab-skills" t
ype="button" data-i18n="tabs.skills">
97             □ Навыки
98         </button>
99         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-mcp" data-bs-target="#tab-mcp" type="b
utton" data-i18n="tabs.mcp">
100             □ MCP Серверы
101         </button>
102     </div>
103 </li>

```



```

104
105     <!-- Разделитель -->
106     <li class="nav-item d-flex align-items-center">
107         <div class="nav-group-divider"></div>
108     </li>
109
110     <!-- Группа: Плагины -->
111     <li class="nav-item dropdown" id="pluginsNavDropdownWrap
per">
112         <button class="nav-link dropdown-toggle d-flex align-i
tems-center gap-1" type="button" id="pluginsTabsDropdown">
113             <i class="bi bi-puzzle-fill text-warning me-1"></i>
114             <span data-i18n="tabs.groupPlugins">Плагины</span>
115         </button>
116         <div class="dropdown-menu shadow border-secondary-subt
le" id="nav-plugins-dropdown-menu">
117             <div id="nav-enabled-plugins-list" class="nav-enable
d-plugins-list">
118                 <!-- Dynamic enabled plugins will be injected here
-->
119             </div>
120             <hr class="dropdown-divider my-1">
121             <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-plugins" data-bs-target="#tab-plugins"
type="button" data-i18n="tabs.managePlugins">
122                 □ Все плагины
123             </button>
124         </div>
125     </li>
126
127     <!-- Разделитель -->
128     <li class="nav-item d-flex align-items-center">
129         <div class="nav-group-divider"></div>
130     </li>
131
132     <!-- Группа: Приложения (/apps) -->
133     <li class="nav-item dropdown">
134         <button class="nav-link dropdown-toggle d-flex align-i

```

```

    items-center gap-1" type="button" id="appsTabsDropdown">
135         <i class="bi bi-grid-fill text-danger me-1"></i>
136         <span data-i18n="tabs.groupApps">Приложения</span>
137     </button>
138     <div class="dropdown-menu shadow border-secondary-subt
le">
139         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-about-system" data-bs-target="#tab-abo
ut-system" type="button" data-i18n="tabs.aboutSystem">
140             i O системе
141         </button>
142         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-trading" data-bs-target="#tab-trading"
type="button" data-i18n="tabs.trading">
143             □ Trading Terminal
144         </button>
145         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-network" data-bs-target="#tab-network"
type="button" data-i18n="tabs.network">
146             □ Network Terminal
147         </button>
148         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-system-inspector" data-bs-target="#tab
-system-inspector" type="button" data-i18n="tabs.systemInspect
or">
149             □ System Inspector
150         </button>
151         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-chat" data-bs-target="#tab-chat" type=
"button" data-i18n="tabs.chat">
152             □ Чат
153         </button>
154         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-scenarios" data-bs-target="#tab-scenar
ios" type="button" data-i18n="tabs.scenarios">
155             □ □ Чат и сценарии
156         </button>
157         <button class="dropdown-item d-flex align-items-cent

```

```

er gap-2" data-tab="tab-user-assistant" data-bs-target="#tab-u
ser-assistant" type="button">
158 |         □ User Assistant
159 |     </button>
160 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-gcloud" data-bs-target="#tab-gcloud" t
ype="button">
161 |         □ Google Cloud Monitor
162 |     </button>
163 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-website-monitor" data-bs-target="#tab-
website-monitor" type="button">
164 |         □ Website Intelligence
165 |     </button>
166 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-cloudflared" data-bs-target="#tab-clou
dflared" type="button">
167 |         □ Cloudflared Monitor
168 |     </button>
169 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-windows-backup" data-bs-target="#tab-w
indows-backup" type="button">
170 |         □ Windows Backup
171 |     </button>
172 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-startup-auditor" data-bs-target="#tab-
startup-auditor" type="button">
173 |         □ Windows Startup Auditor
174 |     </button>
175 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-process-leaks" data-bs-target="#tab-pr
ocess-leaks" type="button">
176 |         □ 1. Утечки и микрофризы
177 |     </button>
178 |     <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-forensics" data-bs-target="#tab-forens
ics" type="button">
179 |         □ 2. Поведенческая форензика

```

```

180         </button>
181         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-throttling" data-bs-target="#tab-throt
tling" type="button">
182             □ 3. Качество ядра и Троттлинг
183         </button>
184         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-storage-wear" data-bs-target="#tab-sto
rage-wear" type="button">
185             □ 4. Износ дисков и батареи
186         </button>
187         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-peripherals" data-bs-target="#tab-peri
pherals" type="button">
188             □ 5. Сеть и периферия
189         </button>
190         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-system-logs" data-bs-target="#tab-syst
em-logs" type="button" data-i18n="tabs.systemLogs">
191             □ Логи операционной системы
192         </button>
193     </div>
194 </li>
195
196     <!-- Разделитель -->
197     <li class="nav-item d-flex align-items-center">
198         <div class="nav-group-divider"></div>
199     </li>
200
201     <!-- Группа: Администрирование -->
202     <li class="nav-item dropdown">
203         <button class="nav-link dropdown-toggle d-flex align-i
tems-center gap-1" type="button" id="adminTabsDropdown">
204             <i class="bi bi-gear-wide-connected text-success me-
1"></i>
205             <span data-i18n="tabs.groupSystem">Администрирование
</span>
206         </button>

```

```
207 |         <div class="dropdown-menu shadow border-secondary-subtle"
|         le">
208 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-admin" data-bs-target="#tab-admin" type="button" data-i18n="tabs.admin">
209 |                 □ Управление
210 |             </button>
211 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-users" data-bs-target="#tab-users" type="button" data-i18n="tabs.users">
212 |                 □ Внутренние пользователи
213 |             </button>
214 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-windows-users" data-bs-target="#tab-windows-users" type="button">
215 |                 □ Пользователи Windows
216 |             </button>
217 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-google-accounts" data-bs-target="#tab-google-accounts" type="button" data-i18n="tabs.googleAccounts">
218 |                 □ Google аккаунты
219 |             </button>
220 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-logs" data-bs-target="#tab-logs" type="button" data-i18n="tabs.logs">
221 |                 □ Системные логи
222 |             </button>
223 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-instructions" data-bs-target="#tab-instructions" type="button" data-i18n="tabs.instructions">
224 |                 □ Системные инструкции
225 |             </button>
226 |             <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-help" data-bs-target="#tab-help" type="button" data-i18n="tabs.help">
227 |                 □ Справочник
228 |             </button>
```

```

229         </div>
230     </li>
231 </ul>
232
233 <!-- Auth Container -->
234 <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto pb-1" id="nav-auth-container">
235     <!-- Active Model & Search Badges -->
236     <span id="chat-model-badge" class="badge rounded-pill al
ign-items-center gap-1 px-3 py-2 shadow-sm border" style="font
-size: 0.8rem; display: none; cursor: pointer;" title="Текущая
модель ИИ">
237         □ Загрузка...
238     </span>
239     <span id="chat-search-badge" class="badge rounded-pill a
lign-items-center gap-1 px-3 py-2 shadow-sm border" style="fon
t-size: 0.8rem; display: none;" title="Провайдер поиска">
240     </span>
241
242 <!-- Google OAuth Register / Sign-in Button (Shown when
not logged in) -->
243 <button class="btn btn-sm btn-light text-dark fw-semibol
d d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-
pill btn-google-login" id="nav-google-btn" title="Войти или за
регистрироваться через Google" data-i18n-title="auth.googleSig
nInTitle" style="border: 1px solid rgba(0,0,0,0.15);">
244     <svg width="18" height="18" viewBox="0 0 24 24"><path
fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h
5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.
74 3.28-8.09z"/><path fill="#34A853" d="M12 23c2.97 0 5.46-.98
7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-
1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/><path fill
="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.
35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.
85-2.22.81-.63z"/><path fill="#EA4335" d="M12 5.38c1.62 0 3.06
.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.4
7 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/></svg>
245     <span data-i18n="auth.googleSignUp">Регистрация через

```

```

Google</span>
246     </button>
247
248     <!-- Authenticated User Menu Dropdown (Shown when logged
in) -->
249     <div class="dropdown d-none" id="user-menu-dropdown-wrap
per">
250         <button class="btn btn-sm btn-outline-secondary d-flex
align-items-center gap-2 px-2 py-1 rounded-pill dropdown-togg
le" id="user-settings-btn" type="button" data-bs-toggle="dropd
own" aria-expanded="false" title="Пользовательское меню">
251             <img id="nav-user-avatar" src="" class="rounded-circ
le d-none" width="22" height="22" style="object-fit: cover;">
252             <i class="bi bi-person-circle" id="nav-user-icon"></i>
253             <span id="nav-user-label" class="small fw-semibold"
data-i18n="userSettings.button">Пользователь</span>
254         </button>
255         <ul class="dropdown-menu dropdown-menu-end shadow bord
er-secondary-subtle py-1" aria-labelledby="user-settings-btn"
style="min-width: 220px;">
256             <li class="px-3 py-2 border-bottom border-secondary-
subtle">
257                 <div class="fw-bold text-truncate" id="user-menu-d
isplay-name">Пользователь</div>
258                 <div class="small text-muted text-truncate" id="us
er-menu-display-email" style="font-size: 0.75rem;">user@gmail.
com</div>
259             </li>
260             <li><h6 class="dropdown-header small text-uppercase
text-muted fw-semibold mt-1">Сервисы</h6></li>
261             <li>
262                 <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-mail">
263                     <i class="bi bi-envelope-check text-primary"></i>
264                     <span>Почта</span>
265                 </button>

```

```

266         </li>
267         <li>
268             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-docs">
269                 <i class="bi bi-file-earmark-text text-info"></i>
>
270                 <span>Google Docs</span>
271             </button>
272         </li>
273         <li>
274             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-sheets">
275                 <i class="bi bi-file-earmark-spreadsheet text-su
ccess"></i>
276                 <span>Google Таблицы</span>
277             </button>
278         </li>
279         <li>
280             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-news">
281                 <i class="bi bi-newspaper text-warning"></i>
282                 <span>Новости</span>
283             </button>
284         </li>
285         <li>
286             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-facebook">
287                 <i class="bi bi-facebook text-primary"></i>
288                 <span>Facebook</span>
289             </button>
290         </li>
291         <li><hr class="dropdown-divider my-1"></li>
292         <li>
293             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" data-bs-toggle="modal" data-bs-
target="#userSettingsModal">
294                 <i class="bi bi-gear-fill text-secondary"></i>
295                 <span data-i18n="userSettings.title">Настройки п

```



```

    рофия</span>
296         </button>
297     </li>
298     <li>
299         <button class="dropdown-item d-flex align-items-center gap-2 py-1 text-danger btn-user-logout" type="button">
300             <i class="bi bi-box-arrow-right"></i>
301             <span>Выйти</span>
302         </button>
303     </li>
304 </ul>
305 </div>
306
307
308     <!-- Mode Switcher: Test Computer (/tc) -->
309     <a href="/tc" class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5 px-2.5 py-1 rounded-pill shadow-sm"
310         title="Перейти в режим Test Computer (/tc)">
311         <i class="bi bi-display"></i>
312         <span class="small fw-semibold">Test Computer</span>
313     </a>
314
315     <!-- Quick Settings Gear Button -->
316     <button class="btn btn-sm btn-outline-secondary d-flex align-items-center justify-content-center p-0 rounded-circle" id="user-settings-gear-btn" data-bs-toggle="modal" data-bs-target="#userSettingsModal" title="Настройки" style="width: 32px; height: 32px;" data-i18n-title="userSettings.title">
317         <i class="bi bi-gear-fill"></i>
318     </button>
319 </div>
320
321 <!-- Main Tab Content Container -->
322 <div class="tab-content" id="mainTabContent">
323     <!-- Chat Tab -->
324     <div id="tab-chat" class="tab-pane fade show active"></div>
325 >

```

```
325 |
326 | <!-- Plugins Tab -->
327 | <div id="tab-plugins" class="tab-pane fade"></div>
328 |
329 | <!-- Admin Tab -->
330 | <div id="tab-admin" class="tab-pane fade"></div>
331 |
332 | <!-- Users Tab -->
333 | <div id="tab-users" class="tab-pane fade"></div>
334 |
335 | <!-- Windows Users Tab -->
336 | <div id="tab-windows-users" class="tab-pane fade"></div>
337 |
338 | <!-- Google Accounts Tab -->
339 | <div id="tab-google-accounts" class="tab-pane fade"></div>
340 |
341 | <!-- RAG Tab -->
342 | <div id="tab-rag" class="tab-pane fade"></div>
343 |
344 | <!-- PixelRAG Tab -->
345 | <div id="tab-pixelrag" class="tab-pane fade"></div>
346 |
347 | <!-- Telegram RAG Tab -->
348 | <div id="tab-telegram-rag" class="tab-pane fade"></div>
349 |
350 | <!-- Smart News Tab -->
351 | <div id="tab-news" class="tab-pane fade"></div>
352 |
353 | <!-- Voice Diarization Tab -->
354 | <div id="tab-voice" class="tab-pane fade"></div>
355 |
356 | <!-- Models Tab -->
357 | <div id="tab-models" class="tab-pane fade"></div>
358 |
359 | <!-- Agents Tab -->
360 | <div id="tab-agents" class="tab-pane fade"></div>
361 |
362 | <!-- Search Tab -->
```

```
363 | <div id="tab-search" class="tab-pane fade"></div>
364 |
365 | <!-- TTS Tab -->
366 | <div id="tab-tts" class="tab-pane fade"></div>
367 |
368 | <!-- Sources Tab -->
369 | <div id="tab-sources" class="tab-pane fade"></div>
370 |
371 | <!-- Logs Tab -->
372 | <div id="tab-logs" class="tab-pane fade"></div>
373 |
374 | <!-- Help Tab -->
375 | <div id="tab-help" class="tab-pane fade"></div>
376 |
377 | <!-- Instructions Tab -->
378 | <div id="tab-instructions" class="tab-pane fade"></div>
379 |
380 | <!-- Skills Tab -->
381 | <div id="tab-skills" class="tab-pane fade"></div>
382 |
383 | <!-- MCP Servers Tab -->
384 | <div id="tab-mcp" class="tab-pane fade"></div>
385 |
386 | <!-- Trading Terminal Tab -->
387 | <div id="tab-trading" class="tab-pane fade"></div>
388 |
389 | <!-- Network Terminal Tab -->
390 | <div id="tab-network" class="tab-pane fade"></div>
391 |
392 | <!-- System Inspector Tab -->
393 | <div id="tab-system-inspector" class="tab-pane fade"></div>
394 | >
395 | <!-- About System Tab -->
396 | <div id="tab-about-system" class="tab-pane fade"></div>
397 |
398 | <!-- Observability & Diagnostics Tab -->
399 | <div id="tab-observability" class="tab-pane fade"></div>
```

```
400
401 <!-- Windows System Administrator Tab -->
402 <div id="tab-windows-admin" class="tab-pane fade"></div>
403
404 <!-- Cloudflared Monitor Tab -->
405 <div id="tab-cloudflared" class="tab-pane fade"></div>
406
407 <!-- User Assistant Tab -->
408 <div id="tab-user-assistant" class="tab-pane fade"></div>
409
410 <!-- Google Cloud Monitor Tab -->
411 <div id="tab-gcloud" class="tab-pane fade"></div>
412
413 <!-- Website Intelligence Monitor Tab -->
414 <div id="tab-website-monitor" class="tab-pane fade"></div>
415
416 <!-- System Control Center Tab -->
417 <div id="tab-system-control" class="tab-pane fade"></div>
418
419 <!-- System Log Viewer Tab -->
420 <div id="tab-system-logs" class="tab-pane fade"></div>
421
422 <!-- Software Audit Tab -->
423 <div id="tab-software-audit" class="tab-pane fade"></div>
424
425 <!-- Registry Viewer Tab -->
426 <div id="tab-registry-viewer" class="tab-pane fade"></div>
427
428 <!-- Windows Backup Manager Tab -->
429 <div id="tab-windows-backup" class="tab-pane fade"></div>
430
431 <!-- Startup Auditor Tab -->
432 <div id="tab-startup-auditor" class="tab-pane fade"></div>
433
434 <!-- Scenarios Tab -->
435 <div id="tab-scenarios" class="tab-pane fade"></div>
436
437 <!-- Wikipedia Research Lab Tab -->
```

```
438 |     <div id="tab-wikipedia-research" class="tab-pane fade"></div>
439 |
440 |     <!-- User Directories and Storage Tab -->
441 |     <div id="tab-user-directories" class="tab-pane fade"></div>
442 |
443 |     <!-- Diagnostic and Forensics Tabs -->
444 |     <div id="tab-process-leaks" class="tab-pane fade"></div>
445 |     <div id="tab-forensics" class="tab-pane fade"></div>
446 |     <div id="tab-throttling" class="tab-pane fade"></div>
447 |     <div id="tab-storage-wear" class="tab-pane fade"></div>
448 |     <div id="tab-peripherals" class="tab-pane fade"></div>
449 | </div>
450 | </div>
451 |
452 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
453 | /js/bootstrap.bundle.min.js"></script>
454 | <script src="/html/js/window-api.js?v=20260926_v2"></script>
455 | <script src="/html/js/appConfigEditor.js?v=20260926_v2"></scri
456 | pt>
457 | <script src="/html/js/chatService.js?v=20260926_v2"></script>
458 |
459 | <!-- Admin Main - Modular Architecture -->
460 | <script type="module" src="/html/admin/main-refactored.js?v=20
461 | 260926_v2"></script>
462 |
463 | <!-- Optimization Patches -->
464 | <script type="module" src="/html/admin/lazy-init-patch.js?v=20
465 | 260926_v2"></script>
466 | <script type="module" src="/html/users_tab/debounce-patch.js?v
467 | =20260926_v2"></script>
468 |
469 | <!-- HELP Modal -->
470 | <div class="modal fade modal-help" id="help-modal" tabindex="-
471 | 1" aria-labelledby="help-modal-label" aria-hidden="true">
472 |     <div class="modal-dialog modal-dialog-scrollable">
473 |         <div class="modal-content">
```

```
468     <div class="modal-header">
469         <h5 class="modal-title" id="help-modal-label" data-i18
n="helpModal.title">Помощь</h5>
470         <button type="button" class="btn-close" data-bs-dismis
s="modal" aria-label="Close"></button>
471     </div>
472     <div class="modal-body" id="help-modal-content">
473         <p data-i18n="common.loading">Загрузка...</p>
474     </div>
475     <div class="modal-footer">
476         <button type="button" class="btn btn-secondary" data-b
s-dismiss="modal" data-i18n="helpModal.close">Закрыть</button>
477     </div>
478 </div>
479 </div>
480 </div>
481
482 <!-- Prompt Preview Modal -->
483 <div class="modal fade" id="prompt-preview-modal" tabindex="-1
" aria-labelledby="prompt-preview-modal-label" aria-hidden="tr
ue">
484     <div class="modal-dialog modal-lg">
485         <div class="modal-content">
486             <div class="modal-header">
487                 <h5 class="modal-title" id="prompt-preview-modal-label
">Предпросмотр промта</h5>
488                 <button type="button" class="btn-close" data-bs-dismis
s="modal" aria-label="Close"></button>
489             </div>
490             <div class="modal-body" id="prompt-preview-content">
491                 <p>Загрузка...</p>
492             </div>
493             <div class="modal-footer">
494                 <button type="button" class="btn btn-secondary" data-b
s-dismiss="modal">Закрыть</button>
495             </div>
496         </div>
497     </div>
```

```

498 | </div>
499 |
500 | <!-- Chat Logic & RAG Architecture Modal -->
501 | <div class="modal fade" id="chat-logic-modal" tabindex="-1" ar
    | ia-labelledby="chat-logic-modal-label" aria-hidden="true">
502 |   <div class="modal-dialog modal-xl modal-dialog-scrollable">
503 |     <div class="modal-content bg-dark text-white border-second
    | ary">
504 |       <div class="modal-header border-secondary d-flex justify
    | -content-between align-items-center">
505 |         <h5 class="modal-title d-flex align-items-center gap-2
    | " id="chat-logic-modal-label">
506 |           <i class="bi bi-diagram-3-fill text-info"></i> □ Ар
    | хитектура и алгоритм диспетчеризации ответов чата
507 |         </h5>
508 |         <div class="d-flex align-items-center gap-2">
509 |           <a href="/html/assets/chat_and_rag_logic.svg" target
    | ="_blank" class="btn btn-sm btn-outline-info" title="Открыть S
    | VG в новой вкладке">
510 |             <i class="bi bi-arrows-fullscreen"></i> SVG
511 |           </a>
512 |           <button type="button" class="btn-close btn-close-whi
    | te" data-bs-dismiss="modal" aria-label="Close"></button>
513 |         </div>
514 |       </div>
515 |       <div class="modal-body p-3">
516 |         <!-- Tabs for Schema vs Documentation -->
517 |         <ul class="nav nav-pills mb-3" id="chatLogicTabs" role
    | ="tablist">
518 |           <li class="nav-item" role="presentation">
519 |             <button class="nav-link active" id="chat-logic-svg
    | -tab" data-bs-toggle="pill" data-bs-target="#chat-logic-svg-pa
    | ne" type="button" role="tab">□ Интерактивная блок-схема</butt
    | on>
520 |           </li>
521 |           <li class="nav-item" role="presentation">
522 |             <button class="nav-link" id="chat-logic-doc-tab" d
    | ata-bs-toggle="pill" data-bs-target="#chat-logic-doc-pane" typ

```

```

e="button" role="tab">Описание алгоритма и сценариев</button>
523     </li>
524 </ul>
525
526     <div class="tab-content" id="chatLogicTabContent">
527         <!-- SVG Diagram Tab -->
528         <div class="tab-pane fade show active text-center" id="chat-logic-svg-pane" role="tabpanel">
529             <div class="p-2 border border-secondary rounded bg-black d-flex justify-content-center align-items-center" style="overflow-x: auto; min-height: 520px;">
530                 
531             </div>
532         </div>
533
534         <!-- Documentation Text Tab -->
535         <div class="tab-pane fade" id="chat-logic-doc-pane" role="tabpanel">
536             <div class="p-3 bg-black rounded border border-secondary text-light small">
537                 <h6 class="text-info fw-bold mb-2">Режимы работы (config.json &rarr; rag.mode)</h6>
538                 <ul class="mb-3">
539                     <li><strong class="text-success">rag+model (Гибридный / Основной):</strong> Сначала проверяется локальная медиатека. Если фильм найден – LLM красиво рассказывает о нём со ссылкой на наличие в медиатеке. Если фильм отсутствует – LLM обращается к веб-поиску (MCP/Grounding) и рассказывает о фильме с кнопкой загрузки <code>&lt;film&gt;</code>.</li>
540                     <li><strong class="text-warning">rag (Строгий локальный):</strong> Работает только по базе <code>media.db</code>. При отсутствии тайтла выдаёт чёткое <em>«В локальной базе не найдено»</em> без ложных подстановок.</li>
541                     <li><strong class="text-primary">model (Без базы):</strong> Прямой ответ LLM и веб-поиск без обращения к дис

```



```

кам.</li>
542         </ul>
543
544         <h6 class="text-info fw-bold mb-2">□ Сценарии о
бработки</h6>
545         <div class="table-responsive mb-3">
546             <table class="table table-dark table-bordered
table-sm small">
547                 <thead>
548                     <tr class="table-secondary text-dark">
549                         <th>Сценарий</th>
550                         <th>Пример</th>
551                         <th>Алгоритм</th>
552                         <th>Результат</th>
553                     </tr>
554                 </thead>
555                 <tbody>
556                     <tr>
557                         <td class="fw-bold text-success">Фильм е
сть в базе</td>
558                         <td><em>«Расскажи о фильме А зори здесь
тихие»</em></td>
559                         <td>Точный поиск в <code>media.db</code>
(Score 1.0) / FAISS &ge; 0.60 &rarr; контекст в LLM</td>
560                         <td>LLM сообщает, что фильм <strong>дост
упен в медиатеке</strong>, описывает сюжет и тегирует <code>&l
t;film&gt;</code>.</td>
561                     </tr>
562                     <tr>
563                         <td class="fw-bold text-danger">Фильма н
ет в базе</td>
564                         <td><em>«Расскажи о сериале Сваты»</em><
/td>
565                         <td>Локальный RAG даёт 0 совпадений &rar
r; Fallback на Web Search (MCP)</td>
566                         <td>LLM рассказывает о сериале, указывае
т, что его <strong>нет на дисках</strong>, тегирует <code>&lt;
film&gt;</code> для торрентов.</td>

```

```

567 |         </tr>
568 |         <tr>
569 |             <td class="fw-bold text-info">Команда за
пуска</td>
570 |             <td><em>«Включи А зори здесь тихие»</em>
</td>
571 |             <td>Проверка пути на диске &rarr; WebSoc
ket <code>play_file_by_path</code></td>
572 |             <td>Мгновенный старт в плеере (Direct Pl
ay).</td>
573 |         </tr>
574 |         <tr>
575 |             <td class="fw-bold text-primary">Подборк
а / Настроение</td>
576 |             <td><em>«Найди боевик на вечер»</em></td>
>
577 |             <td>Извлечение подходящих жанров &rarr;
LLM формирует диалоговую подборку</td>
578 |             <td>Живая рекомендация с обоснованием «п
очему стоит посмотреть».</td>
579 |         </tr>
580 |     </tbody>
581 | </table>
582 | </div>
583 |
584 |     <h6 class="text-info fw-bold mb-1">□ Порог коси
нусного сходства (Cosine Similarity)</h6>
585 |     <p class="text-muted mb-0">Для исключения ложных
срабатываний порог в <code>rag_search_tool</code> установлен
на <strong>0.60</strong>. Запросы с более низким сходством счи
таются отсутствующими в локальной базе и корректно перенаправл
яются в интернет-поиск.</p>
586 |     </div>
587 | </div>
588 | </div>
589 | </div>
590 | <div class="modal-footer border-secondary">
591 |     <button type="button" class="btn btn-sm btn-secondary"

```

```

    data-bs-dismiss="modal">Заккрыть</button>
592     </div>
593   </div>
594 </div>
595 </div>
596
597 <!-- User Settings Modal -->
598 <div class="modal fade" id="userSettingsModal" tabindex="-1" a
    ria-labelledby="userSettingsModalLabel" aria-hidden="true">
599   <div class="modal-dialog modal-dialog-centered">
600     <div class="modal-content shadow border-secondary-subtle">
601       <div class="modal-header border-secondary-subtle">
602         <h5 class="modal-title d-flex align-items-center gap-2
    " id="userSettingsModalLabel">
603           <i class="bi bi-person-gear text-primary"></i>
604           <span data-i18n="userSettings.title">Пользовательски
    е настройки</span>
605         </h5>
606         <button type="button" class="btn-close" data-bs-dismis
    s="modal" aria-label="Close"></button>
607       </div>
608       <div class="modal-body p-3">
609         <!-- User Account & Google OAuth Card -->
610         <div class="card bg-body-tertiary border-secondary-sub
    tle mb-3 p-3" id="user-settings-auth-card">
611           <div class="d-flex align-items-center gap-3">
612             <img id="user-settings-avatar" src="" class="round
    ed-circle border border-2 border-primary" width="48" height="4
    8" style="object-fit: cover;">
613             <div class="flex-grow-1 min-w-0">
614               <div class="d-flex align-items-center gap-2">
615                 <strong id="user-settings-name" class="text-tr
   uncate">Пользователь</strong>
616                 <span id="user-settings-role-badge" class="bad
    ge bg-primary">User</span>
617               </div>
618               <div id="user-settings-email" class="small text-
    muted text-truncate">email@gmail.com</div>

```

```

619 |         <div id="user-settings-google-badge" class="small mt-1 text-muted">
620 |             <i class="bi bi-google text-danger me-1"></i>Google OAuth
621 |         </div>
622 |     </div>
623 |     <button type="button" class="btn btn-outline-danger btn-sm btn-user-logout" title="Выйти">
624 |         <i class="bi bi-box-arrow-right"></i>
625 |     </button>
626 | </div>
627 | </div>
628 |
629 | <!-- Guest / Connect Google OAuth Card -->
630 | <div class="card bg-body-tertiary border-secondary-subtle mb-3 p-3 text-center d-none" id="user-settings-guest-card">
631 |     <div class="text-muted small mb-2">
632 |         <i class="bi bi-shield-lock me-1"></i> Авторизация и синхронизация аккаунта
633 |     </div>
634 |     <button type="button" class="btn btn-outline-danger btn-sm w-100 d-flex align-items-center justify-content-center gap-2 btn-google-login">
635 |         <i class="bi bi-google"></i>
636 |         <span>Войти через Google (OAuth)</span>
637 |     </button>
638 |     <div class="small text-muted mt-2" style="font-size: 0.75rem;">
639 |         Синхронизация с Google Drive, Docs, Calendar и Contacts
640 |     </div>
641 | </div>
642 |
643 | <!-- Language Selection -->
644 | <div class="mb-3">
645 |     <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-lang-selector">

```

```

646         <i class="bi bi-translate text-info"></i>
647         <span data-i18n="userSettings.language">Язык интер
фейса</span>
648     </label>
649     <select class="form-select lang-selector" id="modal-
lang-selector">
650         <option value="ru">Русский</option>
651         <option value="en">English</option>
652         <option value="he">□ □ □ □ □ </option>
653     </select>
654 </div>
655
656 <!-- Theme Selection -->
657 <div class="mb-2">
658     <label class="form-label fw-semibold small d-flex al
ign-items-center gap-2" for="modal-theme-selector">
659         <i class="bi bi-palette text-warning"></i>
660         <span data-i18n="userSettings.theme">Тема оформлен
ия</span>
661     </label>
662     <select class="form-select theme-selector" id="modal
-theme-selector">
663         <option value="light" data-i18n="userSettings.them
eLight">□ Светлая</option>
664         <option value="dark" data-i18n="userSettings.theme
Dark">□ Тёмная</option>
665         <option value="system" data-i18n="userSettings.the
meSystem">□ Системная</option>
666     </select>
667 </div>
668 </div>
669 <div class="modal-footer border-secondary-subtle py-2">
670     <button type="button" class="btn btn-secondary btn-sm"
data-bs-dismiss="modal" data-i18n="userSettings.close">Закрыт
ь</button>
671 </div>
672 </div>
673 </div>

```

```
674 | </div>
675 |
676 | </body>
677 | </html>
```

lazy-init-patch.js

```
1 /**
2  * Lazy Init Patch - Патч для переключения на ленивую загрузку
3  * вкладок
4  *
5  * Этот скрипт перехватывает инициализацию main.js и использует
6  * LazyTabLoader вместо стандартной загрузки всех вкладок сразу
7  */
8 import {
9   initLazyTabLoading,
10  enhanceTabSwitching,
11  restoreLastTab
12 } from './optimization-init.js';
13 import { autoPatchTab } from '../js/tab-debounce-auto-patch.js';
14 import TabStatePersistence from '../js/tab-state-persistence.js';
15 import TabSwitchOptimizer from '../js/tab-switch-optimizer.js';
16
17 console.log('[LazyInitPatch] Patching admin interface initialization...');
18
19 // Инициализируем persistence для админки
20 const tabPersistence = new TabStatePersistence('admin');
21 window.tabPersistence = tabPersistence;
22
23 // Ждем когда main.js загрузится и инициализирует базовые элементы
24 window.addEventListener('load', async () => {
25   console.log('[LazyInitPatch] Window load event, initializing lazy loading...');
26
27   // Ждем немного чтобы main.js выполнил свои скрипты
```

```
28 |   await new Promise(resolve => setTimeout(resolve, 100));
29 |
30 |   // Если есть оригинальная функция switchTab, оборачиваем ее
31 |   if (window.switchTab && typeof window.switchTab === 'function') {
32 |     const originalSwitchTab = window.switchTab;
33 |
34 |     // Применяем оптимизацию переключения
35 |     const switchOptimizer = new TabSwitchOptimizer();
36 |     switchOptimizer.enableDOMCaching();
37 |
38 |     // Оборачиваем в enhanced версию с lazy loading
39 |     let enhancedSwitchTab = enhanceTabSwitching(originalSwitch
40 | Tab);
41 |     // Оборачиваем в optimizer версию для отслеживания произво
42 | дительности
43 |     const optimizedSwitchTab = switchOptimizer.optimizeSwitchi
44 | ng(enhancedSwitchTab);
45 |
46 |     window.switchTab = async function(targetId) {
47 |       // Вызываем оптимизированную функцию
48 |       await optimizedSwitchTab.call(this, targetId);
49 |
50 |       // Сохраняем активную вкладку
51 |       if (targetId) {
52 |         const cleanId = targetId.startsWith('#') ? targetId.sl
53 | ice(1) : targetId;
54 |         tabPersistence.saveActiveTab(cleanId);
55 |       }
56 |     };
57 |
58 |     // Инициализируем ленивую загрузку с определением всех вклад
59 | ок
```



```
59 | const allTabDefinitions = [  
60 |     // Основные вкладки (Диалог и аудио)  
61 |     { tabName: 'chat', htmlUrl: `/html/chat/index.html`, jsUrl  
62 |       : `/html/chat/main.js` },  
63 |     { tabName: 'voice', htmlUrl: `/html/voice_tab/index.html`,  
64 |       jsUrl: `/html/voice_tab/main.js` },  
65 |     { tabName: 'tts', htmlUrl: `/html/tts_tab/index.html`, jsU  
66 |       rl: `/html/tts_tab/main.js` },  
67 |  
68 |     // ИИ и Знания  
69 |     { tabName: 'rag', htmlUrl: `/html/rag_tab/index.html`, jsU  
70 |       rl: `/html/rag_tab/main.js` },  
71 |     { tabName: 'models', htmlUrl: `/html/models_tab/index.html  
72 |       `, jsUrl: `/html/models_tab/main.js` },  
73 |     { tabName: 'agents', htmlUrl: `/html/agents_tab/index.html  
74 |       `, jsUrl: `/html/agents_tab/main.js` },  
75 |     { tabName: 'skills', htmlUrl: `/html/skills_tab/index.html  
76 |       `, jsUrl: `/html/skills_tab/main.js` },  
77 |     { tabName: 'mcp', htmlUrl: `/html/mcp_tab/index.html`, jsU  
78 |       rl: `/html/mcp_tab/main.js` },  
79 |     { tabName: 'observability', htmlUrl: `/html/system_inspect  
80 |       or_tab/index.html?v=20260926_v1`, jsUrl: `/html/system_inspect  
      or_tab/main.js?v=20260926_v1` },  
81 |     { tabName: 'telegram-rag', htmlUrl: `/html/telegram_rag_ta  
82 |       b/index.html`, jsUrl: `/html/telegram_rag_tab/main.js` },  
83 |     { tabName: 'sources', htmlUrl: `/html/sources_tab/index.ht  
84 |       ml`, jsUrl: `/html/sources_tab/main.js` },  
85 |     { tabName: 'search', htmlUrl: `/html/search_tab/index.html  
86 |       `, jsUrl: `/html/search_tab/main.js` },  
87 |  
88 |     // Система и Администратор  
89 |     { tabName: 'admin', htmlUrl: `/html/admin_tab/index.html`,  
90 |       jsUrl: `/html/admin_tab/main.js` },  
91 |     { tabName: 'users', htmlUrl: `/html/users_tab/index.html`,  
92 |       jsUrl: `/html/users_tab/main.js` },  
93 |     { tabName: 'plugins', htmlUrl: `/html/plugins_tab/index.ht  
94 |       ml`, jsUrl: `/html/plugins_tab/main.js` },  
95 |     { tabName: 'google-accounts', htmlUrl: `/html/google_accou
```

```
nts_tab/index.html`, jsUrl: `/html/google_accounts_tab/main.js` },
81 { tabName: 'instructions', htmlUrl: `/html/instructions_tab/index.html`, jsUrl: `/html/instructions_tab/main.js` },
82 { tabName: 'news', htmlUrl: `/html/news_tab/index.html`, jsUrl: `/html/news_tab/main.js` },
83 { tabName: 'logs', htmlUrl: `/html/logs/index.html`, jsUrl: `/html/logs/main.js` },
84 { tabName: 'help', htmlUrl: `/html/help/index.html`, jsUrl: `/html/help/main.js` },
85
86 // Микроприложения (с привязкой к ID приложения в конфиге
  профиля)
87 { tabName: 'trading', appId: 'trading_terminal', htmlUrl:
  `/html/trading_tab/index.html`, jsUrl: `/html/trading_tab/main
  .js` },
88 { tabName: 'network', appId: 'network_terminal', htmlUrl:
  `/html/network_tab/index.html`, jsUrl: `/html/network_tab/main
  .js` },
89 { tabName: 'system-inspector', appId: 'system_inspector',
  htmlUrl: `/html/system_inspector_tab/index.html?v=20260926_v1`,
  jsUrl: `/html/system_inspector_tab/main.js?v=20260926_v1` },
90 { tabName: 'about-system', appId: 'about_system', htmlUrl:
  `/html/about_system_tab/index.html?v=20260926_v1`, jsUrl: `/h
  tml/about_system_tab/main.js?v=20260926_v1` },
91 { tabName: 'windows-admin', appId: 'windows_sysadmin', htm
  lUrl: `/html/windows_admin_tab/index.html`, jsUrl: `/html/wind
  ows_admin_tab/main.js` },
92 { tabName: 'cloudflared', appId: 'cloudflared_monitor', ht
  mlUrl: `/html/cloudflared_tab/index.html`, jsUrl: `/html/cloud
  flared_tab/main.js` },
93 { tabName: 'google-desktop', appId: 'google_user_desktop',
  htmlUrl: `/html/google_desktop_tab/index.html?v=20260928_v1`,
  jsUrl: `/html/google_desktop_tab/main.js?v=20260928_v1` },
94 { tabName: 'user-assistant', appId: 'user_assistant', html
  Url: `/html/user_assistant_tab/index.html`, jsUrl: `/html/user
  _assistant_tab/main.js` },
95 { tabName: 'gcloud', appId: 'gcloud_monitor', htmlUrl: `/h
```

```
    tml/gcloud_tab/index.html`, jsUrl: `/html/gcloud_tab/main.js`
  },
96    { tabName: 'website-monitor', appId: 'website_monitor', ht
    mlUrl: `/html/website_monitor_tab/index.html`, jsUrl: `/html/w
    ebsite_monitor_tab/main.js` },
97    { tabName: 'system-control', appId: 'system_control_center
    ', htmlUrl: `/html/system_control_tab/index.html?v=20260924_v1
    `, jsUrl: `/html/system_control_tab/main.js?v=20260924_v1` },
98    { tabName: 'system-logs', appId: 'system_log_viewer', html
    Url: `/html/system_logs_tab/index.html`, jsUrl: `/html/system_
    logs_tab/main.js` },
99    { tabName: 'software-audit', appId: 'software_audit', html
    Url: `/html/software_audit_tab/index.html`, jsUrl: `/html/soft
    ware_audit_tab/main.js` },
100   { tabName: 'registry-viewer', appId: 'registry_viewer', ht
    mlUrl: `/html/registry_viewer_tab/index.html`, jsUrl: `/html/r
    egistry_viewer_tab/main.js` },
101   { tabName: 'startup-auditor', appId: 'windows_startup_audi
    tor', htmlUrl: `/html/startup_auditor_tab/index.html`, jsUrl:
    `/html/startup_auditor_tab/main.js` },
102   { tabName: 'windows-backup', appId: 'windows_backup_manage
    r', htmlUrl: `/html/windows_backup_tab/index.html?v=20260924_v
    2`, jsUrl: `/html/windows_backup_tab/main.js?v=20260924_v2` },
103   { tabName: 'wikipedia-research', appId: 'wikipedia_researc
    h', htmlUrl: `/html/wikipedia_research_tab/index.html`, jsUrl:
    `/html/wikipedia_research_tab/main.js` },
104   { tabName: 'process-leaks', appId: 'process_leaks', htmlUr
    l: `/html/process_leaks_tab/index.html?v=20260924_v1`, jsUrl:
    `/html/process_leaks_tab/main.js?v=20260924_v1` },
105   { tabName: 'forensics', appId: 'forensics', htmlUrl: `/htm
    l/forensics_tab/index.html?v=20260924_v1`, jsUrl: `/html/foren
    sics_tab/main.js?v=20260924_v1` },
106   { tabName: 'throttling', appId: 'throttling', htmlUrl: `/h
    tml/throttling_tab/index.html?v=20260924_v1`, jsUrl: `/html/th
    rottling_tab/main.js?v=20260924_v1` },
107   { tabName: 'storage-wear', appId: 'storage_wear', htmlUrl:
    `/html/storage_wear_tab/index.html?v=20260924_v1`, jsUrl: `/h
    tml/storage_wear_tab/main.js?v=20260924_v1` },
```

```
108     { tabName: 'peripherals', appId: 'peripherals', htmlUrl: `
    /html/peripherals_tab/index.html?v=20260924_v1`, jsUrl: `/html
    /peripherals_tab/main.js?v=20260924_v1` },
109 ];
110
111   try {
112     // Получаем статус приложений для активного сценария (Run-
    Dashboard, Run-TC, su)
113     let appsStatus = window.appsStatusMap || null;
114     if (!appsStatus) {
115       try {
116         const res = await (window.api ? window.api.fetch('/api
    /apps/status') : fetch('/api/apps/status')).then(r => r.json())
    );
117         appsStatus = res?.apps || null;
118       } catch (err) {
119         console.warn('[LazyInitPatch] Could not fetch apps sta
    tus for profile filtering:', err);
120       }
121     }
122
123     // Фильтруем вкладки: загружаем ТОЛЬКО те, которые разреше
    ны в активном профиле конфига
124     const activeTabDefinitions = allTabDefinitions.filter(tab
    => {
125       if (tab.appId && appsStatus) {
126         const appInfo = appsStatus[tab.appId] || Object.values
    (appsStatus).find(a => a.id === tab.appId || a.tab === `tab-${
    tab.tabName}`);
127         if (appInfo && appInfo.enabled === false) {
128           return false; // Отключено в активном сценарии/профи
    ле
129         }
130       }
131       return true;
132     });
133
134     // Инициализируем систему ленивой загрузки только для разр
```

```
ешенных в профиле вкладок
135     await initLazyTabLoading(activeTabDefinitions);
136
137     console.log(`[LazyInitPatch] Lazy loading initialized: ${a
ctiveTabDefinitions.length} tabs registered on-demand.`);
138
139     // Восстанавливаем и загружаем ТОЛЬКО одну активную вкладк
y
140     const restored = await restoreLastTab('tab-about-system');
141     console.log(`[LazyInitPatch] Initial active tab loaded: ${
restored}`);
142
143   } catch (error) {
144     console.error('[LazyInitPatch] Error during initialization
:', error);
145   }
146 });
147
148 console.log('[LazyInitPatch] Patch loaded successfully');
149
150 export {};
151
```

main-refactored.js

```
1 /**
2  * Admin Interface Main - Оркестратор
3  *
4  * Главный модуль для инициализации административного интерфей
5  * са.
6  * Все реальная реализация находится в модулях, этот файл толь
7  * ко оркестрирует.
8  *
9  * Модули:
10 * - auth-handler.js: Аутентификация и проверка пароля
11 * - tab-manager.js: Управление вкладками и навигацией
12 * - ui-handler.js: Модали, уведомления, помощь
13 * - init-interface.js: Инициализация компонентов
14 * - apps-sync.js: Синхронизация приложений
15 */
16
17 // =====
18 // ИМПОРТ МОДУЛЕЙ
19 // =====
20
21 import { setupAuthHandlers, verifyPassword } from './modules/a
22  uth-handler.js';
23 import { setupTabManagement, onTabSwitched } from './modules/t
24   ab-manager.js';
25 import { setupUIHandlers, showHelpModal, showNotification, sho
26   wChatLogicModal } from './modules/ui-handler.js';
27 import {
28   initializeInterface,
29   setupGlobalFunctions,
30   setupLanguageSelector,
31   setupThemeSelector
32 } from './modules/init-interface.js';
33 import { syncApplicationsVisibility } from './modules/apps-syn
34   c.js';
```

```

29
30 // =====
   =====
31 // ГЛОБАЛЬНЫЕ ПЕРЕМЕННЫЕ
32 // =====
   =====
33
34 let tabLoader = null;
35 let apiFetcher = null;
36
37 // =====
   =====
38 // ГЛАВНАЯ ФУНКЦИЯ ИНИЦИАЛИЗАЦИИ
39 // =====
   =====
40
41 async function initAdminInterface() {
42   console.log(' [AdminInterface] Starting initialization...'
43   );
44   try {
45     // 1. Инициализируем компоненты интерфейса
46     console.log('[AdminInterface] Step 1: Initializing core co
47     mponents...');
48     await initializeInterface();
49     setupGlobalFunctions();
50     setupLanguageSelector();
51     setupThemeSelector();
52
53     // 2. Регистрируем глобальные функции для UI
54     console.log('[AdminInterface] Step 2: Setting up UI handle
55     rs...');
56     registerGlobalFunctions();
57     setupUIHandlers();
58     setupAuthHandlers();
59
60     // 3. Инициализируем управление вкладками
61     console.log('[AdminInterface] Step 3: Setting up tab manag

```

```

    ement...');
60     setupTabManagement();
61
62     // 4. Синхронизируем видимость приложений
63     console.log('[AdminInterface] Step 4: Syncing applications
    visibility...');
64     await syncApplicationsVisibility();
65
66     // 5. Инициализируем оптимизации (ленивая загрузка, кеширо
    вание и т.д.)
67     console.log('[AdminInterface] Step 5: Initializing optimiz
    ations...');
68     await initializeOptimizations();
69
70     // 6. Инициализируем API клиент
71     console.log('[AdminInterface] Step 6: Setting up API clien
    t...');
72     setupAPIClient();
73
74     console.log('□ [AdminInterface] Initialization complete!')
    ;
75
76     } catch (error) {
77         console.error('□ [AdminInterface] Initialization error:',
            error);
78         showNotification('Ошибка инициализации интерфейса: ' + err
            or.message, 'danger');
79     }
80 }
81
82 // =====
    =====
83 // РЕГИСТРАЦИЯ ГЛОБАЛЬНЫХ ФУНКЦИЙ
84 // =====
    =====
85
86 function registerGlobalFunctions() {
87     window.switchTab = switchTab;

```



```

88 | window.onTabSwitched = onTabSwitched;
89 | window.showHelpModal = showHelpModal;
90 | window.showNotification = showNotification;
91 | window.showChatLogicModal = showChatLogicModal;
92 | window.verifyPassword = verifyPassword;
93 | }
94 |
95 | // =====
96 | // УПРАВЛЕНИЕ ВКЛАДКАМИ
97 | // =====
98 |
99 | /**
100 |  * Переключить на вкладку
101 |  */
102 | async function switchTab(targetId) {
103 |     if (!targetId) return;
104 |
105 |     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
106 |     : targetId;
107 |     const tabName = cleanId.replace(/^tab-/, '');
108 |     console.log(`[AdminInterface] Switching to tab: ${tabName}`)
109 |     ;
110 |     // Используем LazyTabLoader если доступен
111 |     if (tabLoader && typeof tabLoader.isLoading === 'function') {
112 |         if (!tabLoader.isLoading(tabName)) {
113 |             console.log(`[AdminInterface] Loading tab: ${tabName}`);
114 |             await tabLoader.loadTab(tabName);
115 |         }
116 |     }
117 |
118 |     // Обновляем визуальное состояние
119 |     updateTabUI(cleanId);
120 |
121 |     // Вызываем callback

```

```
122 |   if (typeof window.onTabSwitched === 'function') {
123 |       window.onTabSwitched(cleanId);
124 |   }
125 |
126 |   console.log(`[AdminInterface] Tab switched: ${tabName}`);
127 | }
128 |
129 | /**
130 |  * Обновить визуальное состояние UI
131 |  */
132 | function updateTabUI(cleanId) {
133 |   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
134 |     cleanId}`;
135 |
136 |   // Скрываем все вкладки
137 |   document.querySelectorAll('#mainTabContent > .tab-pane, body
138 |     > .container-fluid > .tab-content > .tab-pane, [role="tabpanel
139 |     l"]').forEach(tab => {
140 |     tab.classList.remove('active', 'show');
141 |     tab.setAttribute('aria-hidden', 'true');
142 |   });
143 |
144 |   // Показываем целевую вкладку
145 |   const targetTab = document.getElementById(tabId) || document
146 |     .getElementById(cleanId);
147 |   if (targetTab) {
148 |     targetTab.classList.add('active', 'show');
149 |     targetTab.setAttribute('aria-hidden', 'false');
150 |   }
151 |
152 |   // Обновляем состояние элементов меню
153 |   document.querySelectorAll('#mainTabs .list-group-item, #main
154 |     Tabs .dropdown-item, #mainTabs [data-tab], #mainTabs [data-bs-
155 |     target]').forEach(item => {
156 |     const itemTarget = item.getAttribute('data-tab') || item.g
157 |     etAttribute('data-bs-target')?.replace('#', '');
158 |     if (itemTarget === tabId || itemTarget === cleanId) {
159 |       item.classList.add('active');
```

```

153     item.setAttribute('aria-selected', 'true');
154 } else {
155     item.classList.remove('active');
156     item.setAttribute('aria-selected', 'false');
157 }
158 });
159
160 // Обновляем активность родительских кнопок dropdown-toggle
161 document.querySelectorAll('#mainTabs .dropdown').forEach(dro
pdown => {
162     const toggle = dropdown.querySelector('.dropdown-toggle');
163     const hasActiveChild = dropdown.querySelector('.dropdown-i
tem.active, .list-group-item.active');
164     if (toggle) {
165         if (hasActiveChild) {
166             toggle.classList.add('active');
167         } else {
168             toggle.classList.remove('active');
169         }
170     }
171 });
172 }
173
174 // =====
=====
175 // ИНИЦИАЛИЗАЦИЯ ОПТИМИЗАЦИЙ
176 // =====
=====
177
178 async function initializeOptimizations() {
179     // Оптимизации автоматически инициализируются в lazy-init-pa
tch.js
180     // Здесь просто сохраняем ссылки для последующего использова
ния
181
182     if (window.optimizationModule) {
183         tabLoader = window.optimizationModule.tabLoader;
184         apiFetcher = window.optimizationModule.apiFetcher;

```

```

185     console.log('[AdminInterface] Optimizations initialized');
186 } else {
187     console.warn('[AdminInterface] Optimization module not found');
188 }
189 }
190
191 // =====
192 // ИНИЦИАЛИЗАЦИЯ API КЛИЕНТА
193 // =====
194
195 function setupAPIClient() {
196     if (!window.api) {
197         console.warn('[AdminInterface] API client not found, skipping setup');
198         return;
199     }
200
201     // API клиент уже подключен в main.js (исходный), здесь могут быть
202     // дополнительные настройки
203     console.log('[AdminInterface] API client ready');
204 }
205
206 // =====
207 // ОБРАБОТКА СОБЫТИЯ LOAD
208 // =====
209
210 if (document.readyState === 'loading') {
211     document.addEventListener('DOMContentLoaded', initAdminInterface);
212 } else {
213     initAdminInterface();
214 }

```

```
215 |
216 | console.log(' [AdminInterface] Module loaded and ready');
217 |
218 | // =====
    | =====
219 | // ЭКСПОРТЫ (для модульного использования)
220 | // =====
    | =====
221 |
222 | export {
223 |     initAdminInterface,
224 |     switchTab,
225 |     setupAuthHandlers,
226 |     setupTabManagement,
227 |     setupUIHandlers,
228 |     syncApplicationsVisibility,
229 |     initializeInterface,
230 |     setupGlobalFunctions
231 | };
232 |
```

main.js

```
1 // Admin Interface Main JS
2 import { initI18n, switchLang, applyTranslations } from '../js/i18n.js';
3 import { initTheme, setTheme, getThemeMode, getResolvedTheme } from '../js/theme.js';
4 import { initUserSettings, refreshUserProfile } from '../js/userSettings.js';
5
6 // Make switchLang and theme functions available globally
7 window.switchLang = switchLang;
8 window.applyTranslations = applyTranslations;
9 window.setTheme = setTheme;
10 window.getThemeMode = getThemeMode;
11 window.getResolvedTheme = getResolvedTheme;
12 window.initUserSettings = initUserSettings;
13 window.refreshUserProfile = refreshUserProfile;
14
15 // Helper to open plugin from top navbar dropdown
16 window.openPluginFromDropdown = function(pluginName) {
17   if (!pluginName) return;
18   document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((m) => {
19     m.classList.remove('show');
20     m.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
21   });
22
23   const normalized = pluginName.toLowerCase().replace(/-/g, '_');
24
25   if (normalized === 'telegram_channel_rag' || normalized === 'telegram_rag') {
26     const pane = document.getElementById('tab-telegram-rag') | document.getElementById('tab-telegram_rag');
27     if (pane && typeof window.switchTab === 'function') {
28       window.switchTab('tab-telegram-rag');
```

```
29 |     return;
30 | }
31 | }
32 |
33 | if (normalized === 'news_feed' || normalized === 'news' || normalized === 'smart_news' || normalized === 'smart_feed_news') {
34 |     const pane = document.getElementById('tab-news');
35 |     if (pane && typeof window.switchTab === 'function') {
36 |         window.switchTab('tab-news');
37 |         return;
38 |     }
39 | }
40 |
41 | if (typeof window.switchTab === 'function') {
42 |     window.switchTab('tab-plugins');
43 | }
44 |
45 | const select = () => {
46 |     if (typeof window.selectPlugin === 'function') {
47 |         window.selectPlugin(pluginName);
48 |     }
49 | };
50 |
51 | select();
52 | setTimeout(select, 150);
53 | setTimeout(select, 400);
54 | };
55 |
56 | // API module
57 | window.api = {
58 |     async fetch(url, options = {}) {
59 |         const response = await fetch(url, options);
60 |         if (!response.ok) {
61 |             let msg = response.statusText;
62 |             try {
63 |                 const data = await response.json();
64 |                 if (data && data.detail) {
```

```

65         if (typeof data.detail === 'string') {
66             msg = data.detail;
67         } else if (Array.isArray(data.detail)) {
68             msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
69         } else {
70             msg = JSON.stringify(data.detail);
71         }
72     }
73     } catch {}
74     throw new Error(`${response.status} ${msg}`);
75 }
76 return response.json();
77 },
78
79 // User management API
80 users: {
81     async list(params = {}) {
82         const q = new URLSearchParams(params).toString();
83         return window.api.fetch(`/api/admin/users${q ? '?' + q :
''}`);
84     },
85
86     async get(userId) {
87         return window.api.fetch(`/api/admin/users/${userId}`);
88     },
89
90     async create(data) {
91         return window.api.fetch('/api/admin/users', {
92             method: 'POST',
93             headers: { 'Content-Type': 'application/json' },
94             body: JSON.stringify(data)
95         });
96     },
97
98     async update(userId, data) {
99         return window.api.fetch(`/api/admin/users/${userId}`, {
100             method: 'PUT',

```



```
101         headers: { 'Content-Type': 'application/json' },
102         body: JSON.stringify(data)
103     });
104 },
105
106     async setPassword(userId, password) {
107         return window.api.fetch(`/api/admin/users/${userId}/pass
word`, {
108             method: 'POST',
109             headers: { 'Content-Type': 'application/json' },
110             body: JSON.stringify({ password })
111         });
112     },
113
114     async toggleActive(userId) {
115         return window.api.fetch(`/api/admin/users/${userId}/togg
le-active`, {
116             method: 'POST'
117         });
118     },
119
120     async toggleRole(userId) {
121         return window.api.fetch(`/api/admin/users/${userId}/togg
le-role`, {
122             method: 'POST'
123         });
124     },
125
126     async delete(userId) {
127         return window.api.fetch(`/api/admin/users/${userId}`, {
128             method: 'DELETE'
129         });
130     }
131 },
132
133 // Skills management API
134 skills: {
135     async list(params = {}) {
```

```
136 |         const q = new URLSearchParams(params).toString();
137 |         return window.api.fetch(`/api/admin/skills${q ? '?' + q
    | : ''}`);
138 |     },
139 |     async get(name) {
140 |         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
    | mponent(name))}`);
141 |     },
142 |     async create(data) {
143 |         return window.api.fetch('/api/admin/skills', {
144 |             method: 'POST',
145 |             headers: { 'Content-Type': 'application/json' },
146 |             body: JSON.stringify(data)
147 |         });
148 |     },
149 |     async update(name, data) {
150 |         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
    | mponent(name))}`, {
151 |             method: 'PUT',
152 |             headers: { 'Content-Type': 'application/json' },
153 |             body: JSON.stringify(data)
154 |         });
155 |     },
156 |     async delete(name) {
157 |         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
    | mponent(name))}`, {
158 |             method: 'DELETE'
159 |         });
160 |     },
161 |     async package(name) {
162 |         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
    | mponent(name))}/package`, {
163 |             method: 'POST'
164 |         });
165 |     }
166 | },
167 |
168 | // MCP Servers management API
```

```
169 | mcp: {
170 |     async list() {
171 |         return window.api.fetch('/api/admin/mcp/servers');
172 |     },
173 |     async get(id) {
174 |         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`);
175 |     },
176 |     async create(data) {
177 |         return window.api.fetch('/api/admin/mcp/servers', {
178 |             method: 'POST',
179 |             headers: { 'Content-Type': 'application/json' },
180 |             body: JSON.stringify(data)
181 |         });
182 |     },
183 |     async update(id, data) {
184 |         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
185 |             method: 'PUT',
186 |             headers: { 'Content-Type': 'application/json' },
187 |             body: JSON.stringify(data)
188 |         });
189 |     },
190 |     async delete(id) {
191 |         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
192 |             method: 'DELETE'
193 |         });
194 |     },
195 |     async toggle(id) {
196 |         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/toggle`, {
197 |             method: 'POST'
198 |         });
199 |     },
200 |     async test(id) {
201 |         return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/test`, {
```

```

202         method: 'POST'
203     });
204 },
205 async testAdhoc(data) {
206     return window.api.fetch('/api/admin/mcp/test', {
207         method: 'POST',
208         headers: { 'Content-Type': 'application/json' },
209         body: JSON.stringify(data)
210     });
211 },
212 async getTools() {
213     return window.api.fetch('/api/admin/mcp/tools');
214 }
215 },
216
217 // Google Workspace Accounts Pool API
218 googleAccounts: {
219     async list() {
220         return window.api.fetch('/api/admin/google-accounts');
221     },
222     async get(name) {
223         return window.api.fetch(`/api/admin/google-accounts/${encodeURIComponent(name)}`);
224     },
225     async create(data) {
226         return window.api.fetch('/api/admin/google-accounts', {
227             method: 'POST',
228             headers: { 'Content-Type': 'application/json' },
229             body: JSON.stringify(data)
230         });
231     },
232     async upload(formData) {
233         const response = await fetch('/api/admin/google-accounts/upload', {
234             method: 'POST',
235             body: formData
236         });
237         if (!response.ok) {

```

```
238 |         let msg = response.statusText;
239 |         try {
240 |             const data = await response.json();
241 |             if (data && data.detail) msg = data.detail;
242 |         } catch {}
243 |         throw new Error(`${response.status} ${msg}`);
244 |     }
245 |     return response.json();
246 | },
247 | async setDefault(name) {
248 |     return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/default`, {
249 |         method: 'POST'
250 |     });
251 | },
252 | async resetStatus(name) {
253 |     return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/reset-status`, {
254 |         method: 'POST'
255 |     });
256 | },
257 | async test(name) {
258 |     return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/test`, {
259 |         method: 'POST'
260 |     });
261 | },
262 | async delete(name) {
263 |     return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}`, {
264 |         method: 'DELETE'
265 |     });
266 | }
267 | }
268 | };
269 |
270 | // HELP content
271 | // Password protection state
```

```
272 let isPasswordProtected = false;
273 let hasEnteredPassword = false;
274
275 // Tab Switch Handler
276 function onTabSwitched(targetId) {
277     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
278     : targetId;
279     if (cleanId === 'tab-chat') {
280         const msgInput = document.getElementById('message-input');
281         if (msgInput) msgInput.focus();
282     } else if (cleanId === 'tab-voice' && typeof window.initVoiceTab === 'function') {
283         console.log('[AdminInterface] Switching to voice tab...');
284         window.initVoiceTab();
285     } else if (cleanId === 'tab-tts' && typeof window.initTtsTab === 'function') {
286         console.log('[AdminInterface] Switching to tts tab...');
287         window.initTtsTab();
288     } else if (cleanId === 'tab-plugins' && typeof window.initPluginsTab === 'function') {
289         console.log('[AdminInterface] Switching to plugins tab...');
290         window.initPluginsTab();
291     } else if (cleanId === 'tab-admin' && typeof window.initAdminTab === 'function') {
292         console.log('[AdminInterface] Switching to admin tab...');
293         window.initAdminTab();
294     } else if (cleanId === 'tab-users' && typeof window.initUsersTab === 'function') {
295         console.log('[AdminInterface] Switching to users tab...');
296         window.initUsersTab();
297     } else if (cleanId === 'tab-windows-users' && typeof window.initWindowsUsersTab === 'function') {
298         console.log('[AdminInterface] Switching to Windows users tab...');
299         window.initWindowsUsersTab();
300     } else if (cleanId === 'tab-user-directories' && typeof window.initUserDirectoriesTab === 'function') {
```

```
300     console.log('[AdminInterface] Switching to user directorie
s tab...');
301     window.initUserDirectoriesTab();
302 } else if (cleanId === 'tab-google-accounts' && typeof windo
w.initGoogleAccountsTab === 'function') {
303     console.log('[AdminInterface] Switching to google accounts
tab...');
304     window.initGoogleAccountsTab();
305 } else if (cleanId === 'tab-instructions' && typeof window.i
nitInstructionsTab === 'function') {
306     console.log('[AdminInterface] Switching to instructions ta
b...');
307     window.initInstructionsTab();
308 } else if (cleanId === 'tab-skills' && typeof window.initSki
llsTab === 'function') {
309     console.log('[AdminInterface] Switching to skills tab...')
;
310     window.initSkillsTab();
311 } else if (cleanId === 'tab-mcp' && typeof window.initMcpTab
=== 'function') {
312     console.log('[AdminInterface] Switching to MCP tab...');
313     window.initMcpTab();
314 } else if (cleanId === 'tab-rag' && typeof window.initRagTab
=== 'function') {
315     console.log('[AdminInterface] Switching to RAG tab...');
316     window.initRagTab();
317 } else if (cleanId === 'tab-pixelrag' && typeof window.initP
ixelRagTab === 'function') {
318     console.log('[AdminInterface] Switching to PixelRAG tab...
');
319     window.initPixelRagTab();
320 } else if ((cleanId === 'tab-telegram-rag' || cleanId === 't
ab-telegram_rag') && typeof window.initTelegramRagTab === 'fun
ction') {
321     console.log('[AdminInterface] Switching to Telegram RAG ta
b...');
322     window.initTelegramRagTab();
323 } else if (cleanId === 'tab-search' && typeof window.initSea
```

```
rchTab === 'function') {
324     console.log('[AdminInterface] Switching to search tab...')
    ;
325     window.initSearchTab();
326 } else if (cleanId === 'tab-models' && typeof window.initModelsTab === 'function') {
327     console.log('[AdminInterface] Switching to models tab...')
    ;
328     window.initModelsTab();
329 } else if (cleanId === 'tab-agents' && typeof window.initAgentsTab === 'function') {
330     console.log('[AdminInterface] Switching to agents tab...')
    ;
331     window.initAgentsTab();
332 } else if (cleanId === 'tab-sources' && typeof window.initSourcesTab === 'function') {
333     console.log('[AdminInterface] Switching to sources tab...')
    );
334     window.initSourcesTab();
335 } else if (cleanId === 'tab-logs' && typeof window.initLogsTab === 'function') {
336     console.log('[AdminInterface] Switching to logs tab...');
337     window.initLogsTab();
338 } else if (cleanId === 'tab-trading' && typeof window.initTradingTab === 'function') {
339     console.log('[AdminInterface] Switching to trading tab...')
    );
340     window.initTradingTab();
341 } else if (cleanId === 'tab-network' && typeof window.initNetworkTab === 'function') {
342     console.log('[AdminInterface] Switching to network tab...')
    );
343     window.initNetworkTab();
344 } else if (cleanId === 'tab-system-inspector' && typeof window.initSystemInspectorTab === 'function') {
345     console.log('[AdminInterface] Switching to system inspector tab...');
346     window.initSystemInspectorTab();
```



```
347 | } else if (cleanId === 'tab-about-system' && typeof window.i
    | nitAboutSystemTab === 'function') {
348 |     console.log('[AdminInterface] Switching to about system ta
    | b...');
349 |     window.initAboutSystemTab();
350 | } else if (cleanId === 'tab-windows-admin' && typeof window.
    | initWindowsAdminTab === 'function') {
351 |     console.log('[AdminInterface] Switching to windows admin t
    | ab...');
352 |     window.initWindowsAdminTab();
353 | } else if (cleanId === 'tab-cloudflared' && typeof window.in
    | itCloudflaredTab === 'function') {
354 |     console.log('[AdminInterface] Switching to cloudflared tab
    | ...');
355 |     window.initCloudflaredTab();
356 | } else if (cleanId === 'tab-google-desktop' && typeof window
    | .initGoogleDesktopTab === 'function') {
357 |     console.log('[AdminInterface] Switching to google desktop
    | tab...');
358 |     window.initGoogleDesktopTab();
359 | } else if (cleanId === 'tab-user-assistant' && typeof window
    | .initUserAssistantTab === 'function') {
360 |     console.log('[AdminInterface] Switching to user assistant
    | tab...');
361 |     window.initUserAssistantTab();
362 | } else if (cleanId === 'tab-gcloud' && typeof window.initGCl
    | oudTab === 'function') {
363 |     console.log('[AdminInterface] Switching to gcloud tab...')
    | ;
364 |     window.initGCloudTab();
365 | } else if (cleanId === 'tab-website-monitor' && typeof windo
    | w.initWebsiteMonitorTab === 'function') {
366 |     console.log('[AdminInterface] Switching to website monitor
    | tab...');
367 |     window.initWebsiteMonitorTab();
368 | } else if (cleanId === 'tab-system-control' && typeof window
    | .initSystemControlTab === 'function') {
369 |     console.log('[AdminInterface] Switching to system control
```

```

    tab...');
370     window.initSystemControlTab();
371   } else if (cleanId === 'tab-news' && typeof window.initNewsTab === 'function') {
372     console.log('[AdminInterface] Switching to news tab...');
373     window.initNewsTab();
374   } else if (cleanId === 'tab-windows-backup' && typeof window.initWindowsBackupTab === 'function') {
375     console.log('[AdminInterface] Switching to windows backup tab...');
376     window.initWindowsBackupTab();
377   } else if (cleanId === 'tab-help' && typeof window.initHelpTab === 'function') {
378     console.log('[AdminInterface] Switching to help tab...');
379     window.initHelpTab();
380   }
381 }
382
383 // Programmatic tab switcher
384 function switchTab(targetId) {
385   if (!targetId) return;
386   let cleanId = targetId.startsWith('#') ? targetId.slice(1) : targetId;
387   if (cleanId === 'observability' || cleanId === 'tab-observability') {
388     cleanId = 'tab-system-inspector';
389   }
390   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${cleanId}`;
391
392   // 1. Update active state on dropdown items & toggles
393   document.querySelectorAll('#mainTabs .dropdown-item').forEach((item) => {
394     const itemTarget = item.getAttribute('data-tab') || item.getAttribute('data-bs-target')?.replace('#', '');
395     if (itemTarget === tabId || itemTarget === cleanId) {
396       item.classList.add('active');
397     } else {

```

```

398     item.classList.remove('active');
399   }
400 });
401
402 document.querySelectorAll('#mainTabs .dropdown').forEach((dr
opdown) => {
403   const toggle = dropdown.querySelector('.dropdown-toggle');
404   const hasActiveChild = dropdown.querySelector('.dropdown-i
tem.active');
405   if (toggle) {
406     if (hasActiveChild) {
407       toggle.classList.add('active');
408     } else {
409       toggle.classList.remove('active');
410     }
411   }
412 });
413
414 // 2. Switch tab-pane (scoped to top-level container to pres
erve nested subtabs)
415 const mainTabContent = document.getElementById('mainTabConte
nt');
416 if (mainTabContent) {
417   Array.from(mainTabContent.children).forEach((pane) => {
418     if (pane.classList.contains('tab-pane')) {
419       pane.classList.remove('show', 'active');
420     }
421   });
422 } else {
423   document.querySelectorAll('#mainTabContent > .tab-pane, bo
dy > .container-fluid > .tab-content > .tab-pane').forEach((pa
ne) => {
424     pane.classList.remove('show', 'active');
425   });
426 }
427 const targetPane = document.getElementById(tabId) || documen
t.getElementById(cleanId);
428 if (targetPane) {

```

```
429     targetPane.classList.add('show', 'active');
430 }
431
432 // 3. Notify lifecycle callback
433 onTabSwitched(tabId);
434 }
435 window.switchTab = switchTab;
436 window.switchToTab = switchTab;
437
438 // Setup dropdowns navigation
439 function setupDropdownTabs() {
440     const mainTabs = document.getElementById('mainTabs');
441     if (!mainTabs) return;
442
443     // 1. Dropdown Toggle Buttons
444     mainTabs.querySelectorAll('.dropdown-toggle').forEach((btn)
=> {
445         btn.onclick = (e) => {
446             e.preventDefault();
447             e.stopPropagation();
448
449             const dropdown = btn.closest('.dropdown');
450             const menu = dropdown?.querySelector('.dropdown-menu');
451             const isAlreadyOpen = menu?.classList.contains('show');
452
453             // Close all dropdowns
454             document.querySelectorAll('#mainTabs .dropdown-menu.show
').forEach((m) => {
455                 m.classList.remove('show');
456                 m.closest('.dropdown')?.querySelector('.dropdown-toggl
e')?.classList.remove('show');
457             });
458
459             // Toggle clicked dropdown
460             if (!isAlreadyOpen && menu) {
461                 menu.classList.add('show');
462                 btn.classList.add('show');
463             }
```

```

464     };
465   });
466
467   // 2. Dropdown Item Buttons (Tab Switchers & Action Buttons)
468   mainTabs.querySelectorAll('.dropdown-item').forEach((item) =
469   > {
470     item.onclick = (e) => {
471       const targetId = item.getAttribute('data-tab') || item.g
472       etAttribute('data-bs-target')?.replace('#', '');
473       const pluginName = item.getAttribute('data-plugin');
474
475       // Close dropdown menu
476       item.closest('.dropdown-menu')?.classList.remove('show')
477       ;
478       item.closest('.dropdown')?.querySelector('.dropdown-togg
479       le')?.classList.remove('show');
480
481       if (pluginName && typeof window.openPluginFromDropdown =
482       == 'function') {
483         e.preventDefault();
484         e.stopPropagation();
485         window.openPluginFromDropdown(pluginName);
486       } else if (targetId) {
487         e.preventDefault();
488         e.stopPropagation();
489         switchTab(targetId);
490       }
491     };
492   });
493
494   // 3. Document Click to Close Dropdowns
495   if (!document.body.dataset.dropdownOutsideBound) {
496     document.body.dataset.dropdownOutsideBound = 'true';
497     document.addEventListener('click', (e) => {
498       if (!e.target.closest('#mainTabs .dropdown')) {
499         document.querySelectorAll('#mainTabs .dropdown-menu.sh
500         ow').forEach((menu) => {
501           menu.classList.remove('show');

```

```

496 |         menu.closest('.dropdown')?.querySelector('.dropdown-
| toggle')?.classList.remove('show');
497 |     });
498 | }
499 | });
500 | }
501 | }
502 |
503 | async function startAdmin() {
504 |     console.log('Admin interface initializing...');
505 |     try {
506 |         initTheme();
507 |         setupDropdownTabs();
508 |         const el = document.getElementById('admin-interface');
509 |         if (el) el.style.display = 'block';
510 |
511 |         if (isPasswordProtected) {
512 |             showPasswordModal();
513 |         } else {
514 |             await initInterface();
515 |         }
516 |         console.log('Admin interface ready');
517 |     } catch (err) {
518 |         console.error('Error during startAdmin initialization:', e
| rr);
519 |         const el = document.getElementById('admin-interface');
520 |         if (el) el.style.display = 'block';
521 |     }
522 | }
523 |
524 | if (document.readyState === 'loading') {
525 |     document.addEventListener('DOMContentLoaded', () => {
526 |         setupDropdownTabs();
527 |         startAdmin();
528 |     });
529 | } else {
530 |     setupDropdownTabs();
531 |     startAdmin();

```

```

532 }
533
534 async function initInterface() {
535     // Initialize theme
536     initTheme();
537
538     // Initialize i18n
539     const savedLang = localStorage.getItem('app_language') || 'ru';
540     await initI18n(savedLang);
541
542     // Setup language selector
543     document.querySelectorAll('.lang-selector').forEach((sel) => {
544         sel.value = savedLang;
545         sel.addEventListener('change', (e) => {
546             switchLang(e.target.value);
547         });
548     });
549
550     // Initialize User Settings & Google OAuth
551     await initUserSettings();
552
553
554     // Initialize HELP content
555     initHelpContent();
556
557     const cb = Date.now();
558
559     // Синхронизация видимости приложений (/apps) перед загрузкой вкладок
560     let appsMap = {};
561     try {
562         let appsData = null;
563         try {
564             appsData = await window.api.fetch('/api/apps/status');
565         } catch {
566             appsData = await window.api.fetch('/api/admin/apps/status');

```

```

s');
567     }
568     if (appsData && appsData.apps) {
569         appsMap = appsData.apps;
570         window.appsStatusMap = appsData.apps;
571         syncAppsTabsVisibility(appsData.apps);
572     }
573 } catch (err) {
574     console.warn('Ошибка синхронизации видимости приложений (/
apps):', err);
575 }
576
577 // Загрузка базовых вкладок администратора
578 const coreTabLoads = [
579     loadTabContent('chat', `/html/chat/index.html?v=${cb}`, `/
html/chat/main.js?v=${cb}`),
580     loadTabContent('plugins', `/html/plugins_tab/index.html?v=
${cb}`, `/html/plugins_tab/main.js?v=${cb}`),
581     loadTabContent('admin', `/html/admin_tab/index.html?v=${cb
}`, `/html/admin_tab/main.js?v=${cb}`),
582     loadTabContent('users', `/html/users_tab/index.html?v=${cb
}`, `/html/users_tab/main.js?v=${cb}`),
583     loadTabContent('windows-users', `/html/windows_users_tab/i
ndex.html?v=${cb}`, `/html/windows_users_tab/main.js?v=${cb}`)
,
584     loadTabContent('user-directories', `/html/user_directories
_tab/index.html?v=${cb}`, `/html/user_directories_tab/main.js?
v=${cb}`),
585     loadTabContent('google-accounts', `/html/google_accounts_t
ab/index.html?v=${cb}`, `/html/google_accounts_tab/main.js?v=$
{cb}`),
586     loadTabContent('instructions', `/html/instructions_tab/ind
ex.html?v=${cb}`, `/html/instructions_tab/main.js?v=${cb}`),
587     loadTabContent('rag', `/html/rag_tab/index.html?v=${cb}`,
`/html/rag_tab/main.js?v=${cb}`),
588     loadTabContent('pixelrag', `/html/pixelrag_tab/index.html?
v=${cb}`, `/html/pixelrag_tab/main.js?v=${cb}`),
589     loadTabContent('telegram-rag', `/html/telegram_rag_tab/ind

```



```
ex.html?v=${cb}` , `/html/telegram_rag_tab/main.js?v=${cb}`),
590   loadTabContent('voice', `/html/voice_tab/index.html?v=${cb}
` , `/html/voice_tab/main.js?v=${cb}`),
591   loadTabContent('models', `/html/models_tab/index.html?v=${
cb}` , `/html/models_tab/main.js?v=${cb}`),
592   loadTabContent('agents', `/html/agents_tab/index.html?v=${
cb}` , `/html/agents_tab/main.js?v=${cb}`),
593   loadTabContent('search', `/html/search_tab/index.html?v=${
cb}` , `/html/search_tab/main.js?v=${cb}`),
594   loadTabContent('tts', `/html/tts_tab/index.html?v=${cb}` ,
`/html/tts_tab/main.js?v=${cb}`),
595   loadTabContent('sources', `/html/sources_tab/index.html?v=
${cb}` , `/html/sources_tab/main.js?v=${cb}`),
596   loadTabContent('skills', `/html/skills_tab/index.html?v=${
cb}` , `/html/skills_tab/main.js?v=${cb}`),
597   loadTabContent('mcp', `/html/mcp_tab/index.html?v=${cb}` ,
`/html/mcp_tab/main.js?v=${cb}`),
598   loadTabContent('news', `/html/news_tab/index.html?v=${cb}`
, `/html/news_tab/main.js?v=${cb}`),
599   loadTabContent('logs', `/html/logs/index.html?v=${cb}` , `
/html/logs/main.js?v=${cb}`),
600   loadTabContent('help', `/html/help/index.html?v=${cb}` , `
/html/help/main.js?v=${cb}`),
601 ];
602
603 // Определение и фильтрация вкладок микроприложений (/apps)
604 const appTabDefs = [
605   { id: 'about_system', tab: 'about-system', html: '/html/ab
out_system_tab/index.html?v=20260926_v2', js: '/html/about_sys
tem_tab/main.js?v=20260926_v2' },
606   { id: 'trading_terminal', tab: 'trading', html: '/html/tra
ding_tab/index.html', js: '/html/trading_tab/main.js' },
607   { id: 'network_terminal', tab: 'network', html: '/html/net
work_tab/index.html', js: '/html/network_tab/main.js' },
608   { id: 'system_inspector', tab: 'system-inspector', html: '
/html/system_inspector_tab/index.html?v=20260926_v2', js: '/ht
ml/system_inspector_tab/main.js?v=20260926_v2' },
609   { id: 'chat', tab: 'chat', html: '/html/chat/index.html',
```

```

js: '/html/chat/main.js' },
610   { id: 'scenarios', tab: 'scenarios', html: '/html/scenario
s_tab/index.html', js: '/html/scenarios_tab/main.js' },
611   { id: 'user_assistant', tab: 'user-assistant', html: '/htm
l/user_assistant_tab/index.html', js: '/html/user_assistant_ta
b/main.js' },
612   { id: 'gcloud_monitor', tab: 'gcloud', html: '/html/gcloud
_tab/index.html', js: '/html/gcloud_tab/main.js' },
613   { id: 'website_monitor', tab: 'website-monitor', html: '/h
tml/website_monitor_tab/index.html', js: '/html/website_monito
r_tab/main.js' },
614   { id: 'cloudflared_monitor', tab: 'cloudflared', html: '/h
tml/cloudflared_tab/index.html', js: '/html/cloudflared_tab/ma
in.js' },
615   { id: 'google_user_desktop', tab: 'google-desktop', html:
'/html/google_desktop_tab/index.html?v=20260928_v1', js: '/htm
l/google_desktop_tab/main.js?v=20260928_v1' },
616   { id: 'process_leaks', tab: 'process-leaks', html: '/html/
process_leaks_tab/index.html?v=20260924_v1', js: '/html/proces
s_leaks_tab/main.js?v=20260924_v1' },
617   { id: 'forensics', tab: 'forensics', html: '/html/forensic
s_tab/index.html?v=20260924_v1', js: '/html/forensics_tab/main
.js?v=20260924_v1' },
618   { id: 'throttling', tab: 'throttling', html: '/html/thrott
ling_tab/index.html?v=20260924_v1', js: '/html/throttling_tab/
main.js?v=20260924_v1' },
619   { id: 'storage_wear', tab: 'storage-wear', html: '/html/st
orage_wear_tab/index.html?v=20260924_v1', js: '/html/storage_w
ear_tab/main.js?v=20260924_v1' },
620   { id: 'peripherals', tab: 'peripherals', html: '/html/peri
peripherals_tab/index.html?v=20260924_v1', js: '/html/peripherals_
tab/main.js?v=20260924_v1' },
621 ];
622
623 const appTabLoads = appTabDefs
624   .filter(def => {
625     const appInfo = appsMap[def.id] || Object.values(appsMap
).find(a => a.tab === `tab-${def.tab}` || a.id === def.id || a

```

```

        .folder === def.id);
626     return appInfo ? appInfo.enabled : true;
627   })
628   .map(def => loadTabContent(def.tab, `${def.html}?v=${cb}`,
    `${def.js}?v=${cb}`));
629
630   await Promise.all([...coreTabLoads, ...appTabLoads]);
631
632   // Синхронизация видимости вкладок плагинов
633   try {
634     const pluginsData = await window.api.fetch('/api/admin/plu
gins');
635     if (window.syncPluginTabsVisibility && pluginsData && plug
insData.plugins) {
636       window.syncPluginTabsVisibility(pluginsData.plugins);
637     }
638   } catch (err) {
639     console.error('Ошибка синхронизации видимости плагинов:',
err);
640   }
641
642   setupDropdownTabs();
643 }
644
645 function syncAppsTabsVisibility(appsMap) {
646   if (!appsMap || typeof appsMap !== 'object') return;
647   const dropdown = document.getElementById('appsTabsDropdown')
;
648   if (!dropdown) return;
649   const dropdownMenu = dropdown.nextElementSibling || dropdown
.closest('.dropdown')?.querySelector('.dropdown-menu');
650   if (!dropdownMenu) return;
651
652   const appButtons = dropdownMenu.querySelectorAll('button[dat
a-tab]');
653   let visibleCount = 0;
654
655   appButtons.forEach((btn) => {

```

```

656 |     const dataTab = btn.getAttribute('data-tab') || btn.getAtt
| ribute('data-bs-target')?.replace('#', '');
657 |     const cleanId = dataTab ? dataTab.replace(/^tab-/, '') : '
| ';
658 |     const appInfo = Object.values(appsMap).find(a => a.tab ===
| dataTab || a.id === cleanId || a.folder === cleanId);
659 |     const isEnabled = appInfo ? appInfo.enabled : true;
660 |
661 |     if (isEnabled) {
662 |         btn.style.display = '';
663 |         btn.classList.remove('d-none');
664 |         visibleCount++;
665 |     } else {
666 |         btn.style.display = 'none';
667 |         btn.classList.add('d-none');
668 |         const pane = document.getElementById(dataTab);
669 |         if (pane) {
670 |             pane.style.display = 'none';
671 |             pane.classList.remove('show', 'active');
672 |         }
673 |         if (btn.classList.contains('active')) {
674 |             btn.classList.remove('active');
675 |             if (typeof window.switchTab === 'function') {
676 |                 window.switchTab('tab-chat');
677 |             }
678 |         }
679 |     }
680 | });
681 |
682 | const dropdownWrapper = dropdown.closest('li.nav-item');
683 | if (dropdownWrapper) {
684 |     const prevDivider = dropdownWrapper.previousElementSibling
| ;
685 |     if (visibleCount === 0) {
686 |         dropdownWrapper.style.display = 'none';
687 |         dropdownWrapper.classList.add('d-none');
688 |         if (prevDivider && prevDivider.querySelector('.nav-group
| -divider')) {

```

```

689     prevDivider.style.display = 'none';
690     prevDivider.classList.add('d-none');
691 }
692 } else {
693     dropdownWrapper.style.display = '';
694     dropdownWrapper.classList.remove('d-none');
695     if (prevDivider && prevDivider.querySelector('.nav-group
    -divider')) {
696         prevDivider.style.display = '';
697         prevDivider.classList.remove('d-none');
698     }
699 }
700 }
701 }
702 window.syncAppsTabsVisibility = syncAppsTabsVisibility;
703
704 function showPasswordModal() {
705     const modal = new bootstrap.Modal(document.getElementById('p
    asswordModal'));
706     modal.show();
707
708     const loginBtn = document.getElementById('login-btn');
709     const passwordInput = document.getElementById('admin-passwor
    d');
710     const passwordError = document.getElementById('password-erro
    r');
711
712     passwordInput?.addEventListener('keypress', (e) => {
713         if (e.key === 'Enter') verifyPassword();
714     });
715
716     loginBtn?.addEventListener('click', verifyPassword);
717 }
718
719 async function verifyPassword() {
720     const passwordInput = document.getElementById('admin-passwor
    d');
721     const passwordError = document.getElementById('password-erro

```

```

    r');
722     const password = passwordInput?.value;
723     if (!password) return;
724
725     const formData = new FormData();
726     formData.append('password', password);
727
728     try {
729         const res = await fetch('/admin', {
730             method: 'POST',
731             body: formData
732         });
733         if (res.ok) {
734             hasEnteredPassword = true;
735             passwordError?.classList.add('d-none');
736             const modalElement = document.getElementById('passwordMo
dal');
737             const modal = bootstrap.Modal.getInstance(modalElement);
738             modal?.hide();
739             document.getElementById('admin-interface').style.display
= 'block';
740             await initInterface();
741         } else {
742             throw new Error('Invalid password');
743         }
744     } catch {
745         passwordError?.classList.remove('d-none');
746         if (passwordError) {
747             passwordError.textContent = 'Неверный пароль';
748         }
749         if (passwordInput) {
750             passwordInput.value = '';
751             passwordInput.focus();
752         }
753     }
754 }
755
756 async function loadTabContent(tabName, url, jsOverrideSrc) {

```

```
757 |   try {
758 |     const response = await fetch(url);
759 |     if (!response.ok) throw new Error(`HTTP ${response.status}
    | `);
760 |     const html = await response.text();
761 |     const container = document.getElementById(`tab-${tabName}`
    | );
762 |     if (!container) return;
763 |     container.innerHTML = html;
764 |
765 |     // Load JS for the tab
766 |     const cacheBuster = Date.now();
767 |     const scriptSrc = jsOverrideSrc || `/html/${tabName}/main.
    | js?v=${cacheBuster}`;
768 |
769 |     await new Promise((resolve) => {
770 |       const script = document.createElement('script');
771 |       if (tabName === 'instructions' || tabName === 'rag') {
772 |         script.type = 'module';
773 |       }
774 |       script.src = scriptSrc;
775 |       script.onload = () => {
776 |         console.log(`[Admin] Loaded JS for ${tabName}`);
777 |         resolve();
778 |       };
779 |       script.onerror = (err) => {
780 |         console.warn(`[Admin] Note: No JS loaded for ${tabName}
    | } from ${scriptSrc}`);
781 |         resolve();
782 |       };
783 |       document.body.appendChild(script);
784 |     });
785 |
786 |     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
    | .toUpperCase());
787 |     const initFuncName = 'init' + camelName.charAt(0).toUpperC
    | ase() + camelName.slice(1) + 'Tab';
788 |     if (typeof window[initFuncName] === 'function') {
```

```

789     console.log(`[Admin] Calling ${initFuncName} for ${tabName}
me}`);
790     try {
791         await window[initFuncName]();
792     } catch (err) {
793         console.error(`[Admin] Error executing ${initFuncName}
:`, err);
794     }
795 }
796 applyTranslations();
797 } catch (e) {
798     console.error(`[Admin] Error loading tab ${tabName}:`, e);
799     const container = document.getElementById(`tab-${tabName}`
);
800     if (container) {
801         container.innerHTML = `<div class="alert alert-danger">О
шибка загрузки: ${e.message}</div>`;
802     }
803 }
804 }
805
806 // Initialize HELP content
807 function initHelpContent() {
808     window.HELP_CONTENT = window.HELP_CONTENT || {};
809     const defaults = {
810         'overview': `<h4>□ Обзор проекта</h4><p>ai-breadboard – и
нтегрированная среда для работы с AI, RAG и системным управлен
ием.</p>`,
811         'google_oauth': `<h4>□ Google OAuth и аккаунты</h4><p>Инте
грация с Gmail, Drive, Sheets и Docs через OAuth 2.0 и Servic
e Accounts.</p>`,
812         'ai_models': `<h4>□ ИИ Провайдеры и Модели</h4><p>Облачны
е (Gemini, OpenAI, Groq) и локальные (Ollama, Foundry, DirectM
L) модели.</p>`,
813         'gdrive_sync': `<h4>□ Google Drive Sync</h4><p>Автоматичес
кая синхронизация баз данных и файлов с Google Drive.</p>`,
814         'rag_knowledge': `<h4>□ База знаний и RAG</h4><p>Загрузка
и поиск по вашим документам (PDF, Word, TXT, CSV, JSON).</p>`

```



```

,
815     'rag': `

#### ▣ RAG-индекс (Векторный поиск)</h4> 816 <p><strong>1. База RAG:</strong> Индексирует системные базы зн аний, системные инструкции и документы.</p> 817 <p><strong>2. Загрузка документов:</strong> Позволяет загрузит ь внешние <code>.json</code>, <code>.txt</code>, <code>.md</co de>, <code>.pdf</code> файлы или сканировать директории напрям ую в RAG-индекс.</p> 818 <p><strong>3. Чат-RAG:</strong> Индексирует историю сохраненны х диалогов и ответов ассистента.</p>`, 819 'voice_tts': `▣ Голос и Озвучка</h4><p>Голосовой ввод (Whisper/WebSpeech) и синтез речи (Edge TTS).</p>`, 820 'plugins_skills': `▣ Плагины, Навыки и MCP</h4><p>Tel egram-бот, IFTTT, распознавание счетов и расширение через MCP. </p>`, 821 'storage_disks': `▣ Диски и Хранилище</h4><p>Сканиров ание накопителей, проверка целостности и консолидация дублей.< /p>`, 822 'troubleshooting': `▣ Решение проблем (FAQ)</h4><p>Отв еты на частые вопросы и устранение ошибок подключения.</p>` 823 }; 824 window.HELP_CONTENT = Object.assign(defaults, window.HELP_CO NTENT); 825 } 826 827 function showHelpModal(key) { 828 const content = window.HELP_CONTENT[key] || '<p>Информация н е найдена</p>'; 829 document.getElementById('help-modal-content').innerHTML = co ntent; 830 const modal = new bootstrap.Modal(document.getElementById('h elp-modal')); 831 modal.show(); 832 } 833 834 function showChatLogicModal() { 835 const modalEl = document.getElementById('chat-logic-modal'); 836 if (modalEl) {


```

```
837     const modal = new bootstrap.Modal(modalEl);
838     modal.show();
839 }
840 }
841 window.showChatLogicModal = showChatLogicModal;
842 window.showHelpModal = showHelpModal;
843
844 // Уведомления
845 function showNotification(message, type = 'info') {
846     const notification = document.createElement('div');
847     notification.className = `alert alert-${type} position-fixed
    top-0 end-0 m-3`;
848     notification.style.zIndex = '9999';
849     notification.style.maxWidth = '400px';
850     notification.textContent = message;
851     document.body.appendChild(notification);
852
853     setTimeout(() => {
854         notification.remove();
855     }, 5000);
856 }
857
858 async function initAdminTab() {
859     const modelSelect = document.getElementById('admin-model-select');
860     const saveBtn = document.getElementById('btn-admin-save-model');
861
862     if (!modelSelect || !saveBtn) return;
863
864     // 1. Очищаем селект
865     modelSelect.innerHTML = '';
866
867     let modelsGrouped = {};
868     // 2. Загружаем доступные модели
869     try {
870         const modelsData = await window.api.fetch('/api/chat/models');
```

```

871     modelsGrouped = modelsData.models || {};
872     if (Array.isArray(modelsGrouped)) {
873         modelsGrouped = { 'gemini': modelsGrouped };
874     }
875 } catch (err) {
876     console.error('Ошибка загрузки моделей:', err);
877     showNotification('Ошибка загрузки моделей AI: ' + err.mess
age, 'danger');
878 }
879
880 // 3. Заполняем селект моделями с иерархией optgroup
881 const providerMeta = {
882     'gemini': { label: '☐ Google Gemini', order: 1 },
883     'agy': { label: '☐ Google Antigravity (AGY)', order: 2 },
884     'foundry': { label: '☐ Microsoft Foundry', order: 3 },
885     'ollama': { label: '☐ Ollama (Local)', order: 4 },
886     'onnx': { label: '☐ Microsoft ONNX / Olive', order: 5 }
887 };
888
889 const providers = Object.keys(modelsGrouped).sort((a, b) =>
{
890     const oA = providerMeta[a]?.order ?? 99;
891     const oB = providerMeta[b]?.order ?? 99;
892     return oA - oB;
893 });
894
895 let totalModels = 0;
896 providers.forEach(p => {
897     const list = modelsGrouped[p] || [];
898     if (!Array.isArray(list) || list.length === 0) return;
899
900     const optgroup = document.createElement('optgroup');
901     optgroup.label = providerMeta[p]?.label || `☐ ${p.toUpper
Case()}`;
902
903     list.forEach(m => {
904         totalModels++;
905         const option = document.createElement('option');

```

```
906 |     option.value = m;
907 |     option.textContent = m;
908 |     optgroup.appendChild(option);
909 | });
910 |
911 |     modelSelect.appendChild(optgroup);
912 | });
913 |
914 | if (totalModels === 0) {
915 |     const option = document.createElement('option');
916 |     option.value = '';
917 |     option.textContent = 'Нет доступных моделей';
918 |     modelSelect.appendChild(option);
919 |     saveBtn.disabled = true;
920 | } else {
921 |     saveBtn.disabled = false;
922 | }
923 |
924 | // 4. Загружаем текущие настройки пользователя (выбранную мо
    дель)
925 | try {
926 |     const settingsData = await window.api.fetch('/auth/settings');
927 |     const savedModel = settingsData && settingsData.model ? settingsData.model : '';
928 |     if (savedModel) {
929 |         modelSelect.value = savedModel;
930 |     }
931 |     if (typeof window.updateChatBadges === 'function') {
932 |         window.updateChatBadges(savedModel);
933 |     }
934 | } catch (err) {
935 |     console.error('Ошибка загрузки настроек AI пользователя:', err);
936 | }
937 |
938 | // 5. Навешиваем обработчик сохранения
939 | saveBtn.onclick = async () => {
```

```
940 |     const selectedModel = modelSelect.value;
941 |     saveBtn.disabled = true;
942 |     const originalText = saveBtn.textContent;
943 |     saveBtn.textContent = 'Сохранение...';
944 |
945 |     try {
946 |         await window.api.fetch('/auth/settings', {
947 |             method: 'POST',
948 |             headers: { 'Content-Type': 'application/json' },
949 |             body: JSON.stringify({ model: selectedModel })
950 |         });
951 |         showNotification('Модель успешно обновлена на: ' + selectedModel, 'success');
952 |
953 |         // Обновляем бейджи модели в реальном времени на странице
954 |         window.activeModelName = selectedModel;
955 |         if (typeof window.updateChatBadges === 'function') {
956 |             window.updateChatBadges(selectedModel);
957 |         } else {
958 |             const badges = document.querySelectorAll('#chat-model-badge, #chat-popup-model-badge');
959 |             badges.forEach(badge => {
960 |                 badge.textContent = selectedModel;
961 |                 badge.style.display = 'inline-block';
962 |             });
963 |         }
964 |     } catch (err) {
965 |         console.error('Ошибка сохранения модели:', err);
966 |         showNotification('Ошибка сохранения: ' + err.message, 'danger');
967 |     } finally {
968 |         saveBtn.disabled = false;
969 |         saveBtn.textContent = originalText;
970 |     }
971 | };
972 | }
973 |
```

```
974 | window.initAdminTab = initAdminTab;
```

apps-sync.js

```
1 /**
2  * Apps Sync Module - Синхронизация видимости приложений
3  */
4
5 export async function syncApplicationsVisibility() {
6   console.log('[AppSync] Syncing application visibility...');
7
8   let appsMap = {};
9
10  try {
11    let appsData = null;
12    try {
13      appsData = await window.api.fetch('/api/apps/status');
14    } catch {
15      appsData = await window.api.fetch('/api/admin/apps/status');
16    }
17
18    if (appsData && appsData.apps) {
19      appsMap = appsData.apps;
20      window.appsStatusMap = appsData.apps;
21      syncAppsTabsVisibility(appsData.apps);
22    }
23  } catch (error) {
24    console.warn('[AppSync] Failed to sync apps visibility:',
25      error);
26  }
27  return appsMap;
28 }
29
30 export function syncAppsTabsVisibility(appsMap) {
31   if (!appsMap || typeof appsMap !== 'object') return;
32
33   const dropdown = document.getElementById('appsTabsDropdown')
34   ;
```

```

34 |   if (!dropdown) return;
35 |
36 |   const dropdownMenu = dropdown.nextElementSibling ||
37 |       dropdown.closest('.dropdown')?.querySelector(
38 |   '.dropdown-menu');
39 |   if (!dropdownMenu) return;
40 |
41 |   const appButtons = dropdownMenu.querySelectorAll('button[dat
42 |   a-tab]');
43 |   let visibleCount = 0;
44 |
45 |   appButtons.forEach((btn) => {
46 |       const dataTab = btn.getAttribute('data-tab') ||
47 |       btn.getAttribute('data-bs-target')?.replace
48 |   ('#', '');
49 |       const cleanId = dataTab ? dataTab.replace(/^tab-/, '') : '
50 |       '
51 |       const appInfo = Object.values(appsMap).find(
52 |       a => a.tab === dataTab || a.id === cleanId || a.folder =
53 |   == cleanId
54 |   );
55 |       const isEnabled = appInfo ? appInfo.enabled : true;
56 |
57 |       if (isEnabled) {
58 |           btn.style.display = '';
59 |           btn.classList.remove('d-none');
60 |           visibleCount++;
61 |       } else {
62 |           btn.style.display = 'none';
63 |           btn.classList.add('d-none');
64 |
65 |           const pane = document.getElementById(dataTab);
66 |           if (pane) {
67 |               pane.style.display = 'none';
68 |               pane.classList.remove('show', 'active');
69 |           }
70 |
71 |           if (btn.classList.contains('active')) {

```



```

67 |         btn.classList.remove('active');
68 |         if (typeof window.switchTab === 'function') {
69 |             window.switchTab('tab-chat');
70 |         }
71 |     }
72 | }
73 | });
74 |
75 | // Hide/show apps dropdown if all apps are disabled
76 | const dropdownWrapper = dropdown.closest('li.nav-item');
77 | if (dropdownWrapper) {
78 |     const prevDivider = dropdownWrapper.previousElementSibling
79 |     ;
80 |     if (visibleCount === 0) {
81 |         dropdownWrapper.style.display = 'none';
82 |         if (prevDivider) prevDivider.style.display = 'none';
83 |     } else {
84 |         dropdownWrapper.style.display = '';
85 |         if (prevDivider) prevDivider.style.display = '';
86 |     }
87 | }
88 |
89 | console.log(`[AppSync] Synced apps visibility: ${visibleCount} visible`);
90 | }
91 |
92 | export function getAppsMap() {
93 |     return window.appsStatusMap || {};
94 | }
95 |
96 |

```

auth-handler.js

```
1 /**
2  * Auth Handler Module - Обработка аутентификации и пароля
3  */
4
5 export function setupAuthHandlers() {
6   setupPasswordModal();
7   setupPasswordVerification();
8 }
9
10 function setupPasswordModal() {
11   const modal = document.getElementById('passwordModal');
12   const loginBtn = document.getElementById('login-btn');
13   const closeBtn = document.getElementById('closePasswordModal');
14   const adminPassword = document.getElementById('admin-password');
15
16   if (loginBtn) {
17     loginBtn.addEventListener('click', verifyPassword);
18   }
19
20   if (adminPassword) {
21     adminPassword.addEventListener('keypress', (e) => {
22       if (e.key === 'Enter') {
23         verifyPassword();
24       }
25     });
26   }
27
28   if (closeBtn) {
29     closeBtn.addEventListener('click', () => {
30       if (modal) {
31         const m = bootstrap.Modal.getInstance(modal);
32         if (m) m.hide();
33       }
34     });
35   }
36 }
```

```
35 | }
36 |
37 | // Show modal on load if not authenticated
38 | setTimeout(() => {
39 |     const isAuthenticated = localStorage.getItem('admin_authen
    ticated');
40 |     if (!isAuthenticated) {
41 |         showPasswordModal();
42 |     }
43 | }, 500);
44 | }
45 |
46 | function setupPasswordVerification() {
47 |     // Проверка аутентификации при загрузке
48 |     const isAuthenticated = localStorage.getItem('admin_authenti
    cated');
49 |     const adminPassword = document.getElementById('admin-passwor
    d');
50 |
51 |     if (isAuthenticated && adminPassword) {
52 |         adminPassword.value = '';
53 |         localStorage.removeItem('admin_authenticated');
54 |     }
55 | }
56 |
57 | async function verifyPassword() {
58 |     const passwordInput = document.getElementById('admin-passwor
    d');
59 |     const passwordError = document.getElementById('password-erro
    r');
60 |     const modal = document.getElementById('passwordModal');
61 |
62 |     if (!passwordInput.value) {
63 |         if (passwordError) {
64 |             passwordError.classList.remove('d-none');
65 |             passwordError.textContent = 'Пароль не может быть пустым
    ';
66 |         }
```

```
67 |     return;
68 | }
69 |
70 | try {
71 |     const response = await fetch('/api/admin/verify-password',
    | {
72 |         method: 'POST',
73 |         headers: { 'Content-Type': 'application/json' },
74 |         body: JSON.stringify({ password: passwordInput.value })
75 |     });
76 |
77 |     if (response.ok) {
78 |         // Пароль верен
79 |         if (modal) {
80 |             const m = bootstrap.Modal.getInstance(modal);
81 |             if (m) m.hide();
82 |         }
83 |         localStorage.setItem('admin_authenticated', 'true');
84 |         passwordInput.value = '';
85 |         if (passwordError) passwordError.classList.add('d-none')
    | ;
86 |     } else {
87 |         if (passwordError) {
88 |             passwordError.classList.remove('d-none');
89 |             passwordError.textContent = 'Неверный пароль';
90 |         }
91 |         passwordInput.value = '';
92 |         passwordInput.focus();
93 |     }
94 | } catch (error) {
95 |     console.error('Password verification error:', error);
96 |     if (passwordError) {
97 |         passwordError.classList.remove('d-none');
98 |         passwordError.textContent = 'Ошибка проверки пароля';
99 |     }
100 | }
101 | }
102 |
```

```
103 function showPasswordModal() {
104     const modal = document.getElementById('passwordModal');
105     if (modal) {
106         const m = new bootstrap.Modal(modal, { backdrop: 'static',
107         keyboard: false });
108         m.show();
109         const passwordInput = document.getElementById('admin-passw
110         ord');
111         if (passwordInput) {
112             passwordInput.focus();
113         }
114     }
115 }
116 export { verifyPassword, showPasswordModal };
```

init-interface.js

```
1 /**
2  * Interface Initialization Module - Инициализация компонентов
   | интерфейса
3  */
4
5 import { initI18n, switchLang, applyTranslations } from '../..
   | /js/i18n.js';
6 import { initTheme, setTheme, getThemeMode, getResolvedTheme }
   | from '../.. /js/theme.js';
7 import { initUserSettings, refreshUserProfile } from '../.. /js
   | /userSettings.js';
8
9 export async function initializeInterface() {
10   console.log('[AdminInterface] Initializing core components..
   | .');
11
12   try {
13     // Initialize theme
14     initTheme();
15
16     // Initialize i18n
17     await initI18n();
18
19     // Initialize user settings
20     await initUserSettings();
21
22     // Apply translations
23     applyTranslations();
24
25     // Обновляем бейджи активной модели и поиска
26     if (typeof window.updateChatBadges === 'function') {
27       window.updateChatBadges();
28     }
29
30     console.log('[AdminInterface] Core components initialized'
   | );
   }
```

```

31 |   } catch (error) {
32 |       console.error('[AdminInterface] Initialization error:', error);
33 |       throw error;
34 |   }
35 | }
36 |
37 | export function setupGlobalFunctions() {
38 |     // Make functions available globally
39 |     window.switchLang = switchLang;
40 |     window.applyTranslations = applyTranslations;
41 |     window.setTheme = setTheme;
42 |     window.getThemeMode = getThemeMode;
43 |     window.getResolvedTheme = getResolvedTheme;
44 |     window.initUserSettings = initUserSettings;
45 |     window.refreshUserProfile = refreshUserProfile;
46 |
47 |     // Help and modals
48 |     window.showHelpModal = window.showHelpModal || (() => {});
49 |     window.showChatLogicModal = window.showChatLogicModal || (()
    => {});
50 |     window.showNotification = window.showNotification || (() =>
    {});
51 |
52 |     // API Fallback
53 |     window.api = window.api || {};
54 |     if (!window.api.fetch) {
55 |         window.api.fetch = async function(url, options = {}) {
56 |             const response = await fetch(url, options);
57 |             if (!response.ok) {
58 |                 let msg = response.statusText;
59 |                 try {
60 |                     const data = await response.json();
61 |                     if (data && data.detail) {
62 |                         if (typeof data.detail === 'string') msg = data.de
tail;
63 |                         else if (Array.isArray(data.detail)) msg = data.de
tail.map(d => d.msg || JSON.stringify(d)).join(', ');

```

```

64         else msg = JSON.stringify(data.detail);
65     } else if (data && data.message) {
66         msg = data.message;
67     }
68     } catch {}
69     throw new Error(`${response.status} ${msg}`);
70 }
71 return response.json();
72 };
73 }
74
75 console.log('[AdminInterface] Global functions registered');
76 }
77
78 export function setupLanguageSelector() {
79     const langSelectors = document.querySelectorAll('.lang-selector');
80     const savedLang = localStorage.getItem('language') || 'ru';
81
82     langSelectors.forEach((selector) => {
83         selector.value = savedLang;
84         selector.addEventListener('change', (e) => {
85             switchLang(e.target.value);
86         });
87     });
88
89     console.log('[AdminInterface] Language selector setup');
90 }
91
92 export function setupThemeSelector() {
93     const themeSelectors = document.querySelectorAll('.theme-selector');
94     const savedTheme = getThemeMode();
95
96     themeSelectors.forEach((selector) => {
97         selector.value = savedTheme;
98         selector.addEventListener('change', (e) => {
99             setTheme(e.target.value);

```



```
100 |     });
101 | });
102 |
103 | console.log('[AdminInterface] Theme selector setup');
104 | }
105 |
106 | export { initI18n, switchLang, applyTranslations };
107 | export { initTheme, setTheme, getThemeMode, getResolvedTheme }
    | ;
108 | export { initUserSettings, refreshUserProfile };
109 |
```

tab-manager.js

```
1  /**
2   * Tab Manager Module - Управление вкладками и навигацией
3   */
4
5  export function setupTabManagement() {
6      setupDropdownTabs();
7      setupTabEvents();
8  }
9
10 function setupDropdownTabs() {
11     const mainTabs = document.getElementById('mainTabs');
12     if (!mainTabs) return;
13
14     const dropdowns = mainTabs.querySelectorAll('.dropdown-toggle');
15
16     dropdowns.forEach((dropdown) => {
17         dropdown.addEventListener('click', (e) => {
18             e.preventDefault();
19             e.stopPropagation();
20
21             const parentDropdown = dropdown.closest('.dropdown');
22             const menu = dropdown.nextElementSibling || parentDropdown?.querySelector('.dropdown-menu');
23             const isAlreadyOpen = menu?.classList.contains('show');
24
25             // Close other dropdowns
26             document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((otherMenu) => {
27                 otherMenu.classList.remove('show');
28                 otherMenu.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
29             });
30
31             // Toggle current dropdown
32             if (!isAlreadyOpen && menu) {
```

```

33 |         dropdown.classList.add('show');
34 |         menu.classList.add('show');
35 |     }
36 | });
37 | });
38 |
39 | // Handle all clickable navigation items inside mainTabs (ta
bs and dropdown items)
40 | mainTabs.querySelectorAll('.dropdown-item, .list-group-item,
[data-tab], [data-bs-target], [data-plugin]').forEach((item)
=> {
41 |     item.addEventListener('click', (e) => {
42 |         const pluginName = item.getAttribute('data-plugin');
43 |         const targetId = item.getAttribute('data-tab') ||
44 |             item.getAttribute('data-bs-target')?.rep
lace('#', '');
45 |
46 |         // Close parent dropdown menu
47 |         const menu = item.closest('.dropdown-menu');
48 |         if (menu) {
49 |             menu.classList.remove('show');
50 |             menu.closest('.dropdown')?.querySelector('.dropdown-to
GGLE')?.classList.remove('show');
51 |         }
52 |
53 |         if (pluginName && typeof window.openPluginFromDropdown =
== 'function') {
54 |             e.preventDefault();
55 |             e.stopPropagation();
56 |             window.openPluginFromDropdown(pluginName);
57 |         } else if (targetId) {
58 |             e.preventDefault();
59 |             e.stopPropagation();
60 |             if (typeof window.switchTab === 'function') {
61 |                 window.switchTab(targetId);
62 |             }
63 |         }
64 |     });

```

```

65 | });
66 |
67 | // Close dropdowns on outside click
68 | document.addEventListener('click', (e) => {
69 |     if (!mainTabs.contains(e.target)) {
70 |         dropdowns.forEach((dropdown) => {
71 |             dropdown.classList.remove('show');
72 |             const menu = dropdown.nextElementSibling;
73 |             if (menu) menu.classList.remove('show');
74 |         });
75 |     }
76 | });
77 | }
78 |
79 | function setupTabEvents() {
80 |     // Tab content click handlers
81 |     const tabPane = document.getElementById('mainTabsContent');
82 |     if (tabPane) {
83 |         tabPane.addEventListener('click', (e) => {
84 |             const target = e.target.closest('[data-tab]');
85 |             if (target) {
86 |                 const tabId = target.getAttribute('data-tab');
87 |                 if (tabId) {
88 |                     window.switchTab(tabId);
89 |                 }
90 |             }
91 |         });
92 |     }
93 | }
94 |
95 | export function onTabSwitched(targetId) {
96 |     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
97 |     : targetId;
98 |
99 |     if (cleanId === 'tab-chat') {
100 |         if (typeof window.initChatTab === 'function') {
101 |             window.initChatTab();

```

```
102 |   }  
103 | }  
104 |  
105 | export { setupDropdownTabs, setupTabEvents };  
106 |
```

ui-handler.js

```
1 /**
2  * UI Handler Module - Работа с модалями, уведомлениями и помо
  щью
3  */
4
5 let helpContent = {};
6
7 export function setupUIHandlers() {
8   initHelpContent();
9   setupModalHandlers();
10 }
11
12 function initHelpContent() {
13   helpContent = {
14     'overview': `

#### □ Обзор проекта</h4><p>ai-breadboard – и нтегрированная среда для работы с AI, RAG и системным управлен ием.</p>`, 15 'google_oauth': `□ Google OAuth и аккаунты</h4><p>Инт еграция с Gmail, Drive, Sheets и Docs через OAuth 2.0 и Servic e Accounts.</p>`, 16 'ai_models': `□ ИИ Провайдеры и Модели</h4><p>Облачны е (Gemini, OpenAI, Groq) и локальные (Ollama, Foundry, DirectM L) модели.</p>`, 17 'gdrive_sync': `□ Google Drive Sync</h4><p>Автоматичес кая синхронизация баз данных и файлов с Google Drive.</p>`, 18 'rag_knowledge': `□ База знаний и RAG</h4><p>Загрузка и поиск по вашим документам (PDF, Word, TXT, CSV, JSON).</p>` , 19 'rag': `□ RAG-индекс (Векторный поиск)</h4> 20 <p><strong>1. База RAG:</strong> Индексирует системные базы зн аний, системные инструкции и документы.</p> 21 <p><strong>2. Загрузка документов:</strong> Позволяет загрузит ь внешние <code>.json</code>, <code>.txt</code>, <code>.md</co de>, <code>.pdf</code> файлы или сканировать директории напрям ую в RAG-индекс.</p> 22 <p><strong>3. Чат-RAG:</strong> Индексирует историю сохраненны


```

```

    х диалогов и ответов ассистента.</p>`,
23     'voice_tts': `<h4>□ Голос и Озвучка</h4><p>Голосовой ввод
    (Whisper/WebSpeech) и синтез речи (Edge TTS).</p>`,
24     'plugins_skills': `<h4>□ Плагины, Навыки и MCP</h4><p>Tel
    egram-бот, IFTTT, распознавание счетов и расширение через MCP.
    </p>`,
25     'storage_disks': `<h4>□ Диски и Хранилище</h4><p>Сканиров
    ание накопителей, проверка целостности и консолидация дублей.<
    /p>`,
26     'troubleshooting': `<h4>□ Решение проблем (FAQ)</h4><p>Отв
    еты на частые вопросы и устранение ошибок подключения.</p>`
27   };
28
29   window.HELP_CONTENT = helpContent;
30 }
31
32 function setupModalHandlers() {
33   // Help modal
34   const helpBtn = document.getElementById('help-btn');
35   if (helpBtn) {
36     helpBtn.addEventListener('click', () => {
37       showHelpModal('overview');
38     });
39   }
40
41   // Chat logic modal
42   const chatLogicBtn = document.getElementById('chat-logic-btn
    ');
43   if (chatLogicBtn) {
44     chatLogicBtn.addEventListener('click', () => {
45       showChatLogicModal();
46     });
47   }
48 }
49
50 export function showHelpModal(key) {
51   const content = helpContent[key] || '<p>Информация не найден
    а</p>';

```

```
52 |   const contentDiv = document.getElementById('help-modal-conte
    | nt');
53 |   if (contentDiv) {
54 |     contentDiv.innerHTML = content;
55 |   }
56 |
57 |   const modal = document.getElementById('help-modal');
58 |   if (modal) {
59 |     const m = new bootstrap.Modal(modal);
60 |     m.show();
61 |   }
62 | }
63 |
64 | export function showChatLogicModal() {
65 |   const modalEl = document.getElementById('chat-logic-modal');
66 |   if (modalEl) {
67 |     const modal = new bootstrap.Modal(modalEl);
68 |     modal.show();
69 |   }
70 | }
71 |
72 | export function showNotification(message, type = 'info') {
73 |   const notification = document.createElement('div');
74 |   notification.className = `alert alert-${type} position-fixed
    | top-0 end-0 m-3`;
75 |   notification.style.zIndex = '9999';
76 |   notification.style.maxWidth = '400px';
77 |   notification.textContent = message;
78 |   document.body.appendChild(notification);
79 |
80 |   setTimeout(() => {
81 |     notification.remove();
82 |   }, 5000);
83 | }
84 |
85 | export { initHelpContent, setupModalHandlers };
86 |
```


optimization-init.js

```
1  /**
2   * Optimization Init - Интеграция систем оптимизации в админку
3   *
4   * Этот скрипт инициализирует:
5   * - Ленивую загрузку вкладок
6   * - Кеширование API
7   * - Дебаунсинг запросов
8   * - Сохранение состояния вкладок
9   */
10
11 import LazyTabLoader from '../js/tab-loader.js';
12 import { APICache, APIFetcher } from '../js/api-cache.js';
13 import { RequestManager } from '../js/request-manager.js';
14
15 // =====
16 // 1. ИНИЦИАЛИЗАЦИЯ СИСТЕМ ОПТИМИЗАЦИИ
17 // =====
18
19 const tabLoader = new LazyTabLoader();
20 const apiCache = new APICache();
21 const apiFetcher = new APIFetcher(apiCache);
22 const requestManager = new RequestManager();
23
24 // Экспортируем в глобальный скоп для доступа из других модуле
25 // й
26 window.optimizationModule = {
27   tabLoader,
28   apiCache,
29   apiFetcher,
30   requestManager,
31   stats: {
32     tabsLoaded: new Set(),
33     apiCallsUsed: 0,
34     cachedCallsUsed: 0,
```

```

34     requestsDebounced: 0
35   }
36 };
37
38 console.log('[Optimization] System initialized');
39
40 // =====
41 // 2. ПЕРЕХВАТ API.FETCH ОБЕРТКА
42 // =====
43
44 // Сохраняем оригинальный fetch
45 const originalApiFetch = window.api?.fetch;
46
47 // Переопределяем window.api.fetch для использования кеша
48 if (window.api && originalApiFetch) {
49   window.api.fetch = async function(url, options = {}) {
50     const method = (options.method || 'GET').toUpperCase();
51     const isGetRequest = method === 'GET' || method === 'HEAD'
52     ;
53     // Для GET запросов проверяем кеш
54     if (isGetRequest) {
55       const cached = apiCache.get(url);
56       if (cached) {
57         window.optimizationModule.stats.cachedCallsUsed++;
58         console.log(`[Optimization] API call from cache: ${method} ${url}`);
59         return cached;
60       }
61     }
62
63     // Делаем реальный запрос
64     window.optimizationModule.stats.apiCallsUsed++;
65     const result = await originalApiFetch.call(this, url, options);
66

```

```

67 // Кешируем для GET запросов (5 минут TTL)
68 if (isGetRequest) {
69     apiCache.set(url, result, 5 * 60 * 1000); // 5 минут
70 } else {
71     // Для POST/PUT/DELETE инвалидируем релевантный кеш
72     const basePath = url.split('?')[0];
73
74     // Умная инвалидация - очищаем кеш связанных GET запросо
    в
75     if (method === 'POST' || method === 'PUT' || method ===
    'DELETE') {
76         // Инвалидируем основной путь и все его вариации
77         const pattern = new RegExp(`${basePath.replace(/\/$/,
    '')}.*`);
78         const invalidatedCount = apiCache.invalidate(pattern);
79         if (invalidatedCount > 0) {
80             console.log(`[Optimization] Invalidated ${invalidate
    dCount} cache entries for ${basePath}`);
81         }
82
83         // Для пользователей - инвалидируем список пользовател
    ей
84         if (basePath.includes('/users/')) {
85             apiCache.invalidate(/\/api\/admin\/users/);
86             apiCache.invalidate(/\/api\/users/);
87         }
88
89         // Для конфигурации - инвалидируем конфиги
90         if (basePath.includes('/config/')) {
91             apiCache.invalidate(/\/api\/config/);
92             apiCache.invalidate(/\/api\/admin\/config/);
93         }
94
95         // Для плагинов - инвалидируем список плагинов
96         if (basePath.includes('/plugins/')) {
97             apiCache.invalidate(/\/api\/plugins/);
98             apiCache.invalidate(/\/api\/admin\/plugins/);
99         }

```

```

100
101     // Для модели - инвалидируем список моделей
102     if (basePath.includes('/models') || basePath.includes(
103         '/chat')) {
104         apiCache.invalidate(/\/api\/chat\/models/);
105         apiCache.invalidate(/\/api\/models/);
106     }
107 }
108
109     return result;
110 };
111 }
112
113 // =====
114 // 3. ПЕРЕДЕЛКА ИНИЦИАЛИЗАЦИИ ВКЛАДОК (LAZY LOADING)
115 // =====
116
117 export async function initLazyTabLoading(tabDefinitions) {
118     console.log('[Optimization] Initializing lazy tab loading (o
119 n-demand mode)...');
120
121     // Регистрируем все вкладки в системе ленивой загрузки
122     tabDefinitions.forEach(({ tabName, htmlUrl, jsUrl }) => {
123         tabLoader.registerTab(tabName, htmlUrl, jsUrl);
124     });
125
126     // Перехватываем loadTab чтобы применять автопатч
127     const originalLoadTab = tabLoader.loadTab.bind(tabLoader);
128     tabLoader.loadTab = async function(tabName) {
129         const result = await originalLoadTab(tabName);
130
131         // Применяем дебаунс автопатч к загруженной вкладке
132         if (result && window.TabDebounceAutoPatch) {
133             setTimeout(() => {
134                 try {

```

```

134         window.TabDebounceAutoPatch.autoPatchTab(tabName, {
135             searchEnabled: true,
136             autoSaveEnabled: true,
137             refreshEnabled: true
138         });
139     } catch (e) {
140         console.warn(`[Optimization] Failed to apply debounc
141         e patch to ${tabName}:`, e);
142     }
143     }, 100);
144 }
145     return result;
146 };
147
148     const allowedTabNames = tabDefinitions.map(def => def.tabNam
149     e);
150     console.log(`[Optimization] Registered ${allowedTabNames.len
151     gth} tabs for on-demand loading.`);
152
153     return { registeredTabs: allowedTabNames };
154 }
155
156 // =====
157 // 4. ИНТЕГРАЦИЯ С ПЕРЕКЛЮЧАТЕЛЕМ ВКЛАДОК
158 // =====
159
160 export function enhanceTabSwitching(originalSwitchTab) {
161     return async function switchTabOptimized(targetId) {
162         if (!targetId) return;
163
164         const cleanId = targetId.startsWith('#') ? targetId.slice(
165         1) : targetId;
166         const tabName = cleanId.replace(/^tab-/, '');
167
168         console.log(`[Optimization] Switching to tab: ${tabName}`)

```

```

165 | ;
166 |
167 |     // Если вкладка еще не загружена - загружаем ее по требова
    нию
168 |     if (!tabLoader.isLoaded(tabName)) {
169 |         console.log(`[Optimization] Tab not loaded yet, loading
on-demand: ${tabName}`);
170 |         await tabLoader.loadTab(tabName);
171 |         window.optimizationModule.stats.tabsLoaded.add(tabName);
172 |     }
173 |
174 |     // Вызываем оригинальную функцию переключения
175 |     if (typeof originalSwitchTab === 'function') {
176 |         originalSwitchTab.call(this, targetId);
177 |     }
178 |
179 |     // Сохраняем состояние вкладки в localStorage
180 |     localStorage.setItem('admin:lastActiveTab', cleanId);
181 | };
182 | }
183 |
184 | // =====
    =====
185 | // 5. ВОССТАНОВЛЕНИЕ ПОСЛЕДНЕЙ ВКЛАДКИ
186 | // =====
    =====
187 |
188 | export async function restoreLastTab(fallbackTab = 'tab-about-
system') {
189 |     // Проверяем хэш в URL
190 |     const hash = location.hash.replace('#', '');
191 |     let initialTab = hash || null;
192 |
193 |     // Если в хэше ничего нет, проверяем persistence
194 |     if (!initialTab) {
195 |         if (window.tabPersistence && typeof window.tabPersistence.
getLastActiveTab === 'function') {
196 |             initialTab = window.tabPersistence.getLastActiveTab();

```

```

197     } else {
198         initialTab = localStorage.getItem('admin:lastActiveTab')
199     };
200 }
201
202 const targetTabId = (initialTab && document.getElementById(i
initialTab.startsWith('tab-') ? initialTab : `tab-${initialTab}
`))
203     ? (initialTab.startsWith('tab-') ? initialTab : `tab-${ini
tialTab}`)
204     : fallbackTab;
205
206 const tabName = targetTabId.replace(/^tab-/, '');
207 console.log(`[Optimization] Loading strictly initial active
tab: ${tabName}`);
208
209 if (!tabLoader.isLoaded(tabName)) {
210     await tabLoader.loadTab(tabName);
211     window.optimizationModule.stats.tabsLoaded.add(tabName);
212 }
213
214 if (typeof window.switchTab === 'function') {
215     window.switchTab(targetTabId);
216 }
217 return targetTabId;
218 }
219
220 // =====
221 // 6. УТИЛИТЫ ДЛЯ ДЕБАУНСИНГА И БАТЧИНГА
222 // =====
223
224 export function createSearchDebounce(searchFn, delay = 500) {
225     return (query) => {
226         window.optimizationModule.stats.requestsDebounced++;
227         return requestManager.debounce('search-query', () => searc

```

```

    hFn(query), delay);
228   };
229 }
230
231 export function createAutoSaveDebounce(saveFn, delay = 1000) {
232   return (data) => {
233     window.optimizationModule.stats.requestsDebounced++;
234     return requestManager.debounce('autosave-data', () => save
    Fn(data), delay);
235   };
236 }
237
238 // =====
    =====
239 // 7. ФУНКЦИИ ОТЛАДКИ И СТАТИСТИКИ
240 // =====
    =====
241
242 export function getOptimizationStats() {
243   return {
244     ...window.optimizationModule.stats,
245     cacheStats: apiCache.getStats(),
246     requestManagerStats: requestManager.getStats(),
247     tabsPreloaded: Array.from(window.optimizationModule.stats.
    tabsLoaded)
248   };
249 }
250
251 export function printOptimizationStats() {
252   const stats = getOptimizationStats();
253   console.group('□ Optimization Statistics');
254   console.log('API Calls:', stats.apiCallsUsed);
255   console.log('Cached Calls:', stats.cachedCallsUsed);
256   console.log('Cache Hit Rate:', (stats.cachedCallsUsed / (sta
    ts.apiCallsUsed + stats.cachedCallsUsed) * 100).toFixed(1) + '
    %');
257   console.log('Requests Debounced:', stats.requestsDebounced);
258   console.log('Tabs Preloaded:', stats.tabsPreloaded);

```



```
259 | console.log('Cache Entries:', stats.cacheStats.size);
260 | console.groupEnd();
261 | }
262 |
263 | // Делаем доступной из консоли для отладки
264 | window.printOptimizationStats = printOptimizationStats;
265 | window.getOptimizationStats = getOptimizationStats;
266 |
267 | // Выводим статистику каждые 30 секунд (опционально, для отлад
    | ки)
268 | // Раскомментировать в production если нужна мониторинг
269 | // setInterval(() => {
270 | //   console.clear();
271 | //   printOptimizationStats();
272 | // }, 30000);
273 |
274 | export default {
275 |   initLazyTabLoading,
276 |   enhanceTabSwitching,
277 |   restoreLastTab,
278 |   createSearchDebounce,
279 |   createAutoSaveDebounce,
280 |   getOptimizationStats,
281 |   printOptimizationStats,
282 |   tabLoader,
283 |   apiCache,
284 |   apiFetcher,
285 |   requestManager
286 | };
287 |
```

sync-api.js

```
1 /**
2  * Google Drive Sync API Client
3  * Клиент для работы с REST API синхронизации
4  */
5
6 class SyncApiClient {
7   constructor(baseUrl = '/api/admin/sync') {
8     this.baseUrl = baseUrl;
9     this.timeout = 10000;
10  }
11
12  /**
13   * Выполнить GET запрос
14   */
15   async get(endpoint) {
16     try {
17       const response = await fetch(`${this.baseUrl}${endpoint}
18     `, {
19       method: 'GET',
20       headers: { 'Content-Type': 'application/json' }
21     });
22
23     if (!response.ok) {
24       throw new Error(`API Error: ${response.status}`);
25     }
26
27     return await response.json();
28   } catch (error) {
29     console.error(`[SyncApiClient] GET ${endpoint}:`, error)
30   };
31   throw error;
32 }
33
34 /**
35  * Выполнить POST запрос
```

```

35 |     */
36 |     async post(endpoint, data = {}) {
37 |         try {
38 |             const response = await fetch(`${this.baseUrl}${endpoint}
|         `, {
39 |                 method: 'POST',
40 |                 headers: { 'Content-Type': 'application/json' },
41 |                 body: JSON.stringify(data)
42 |             });
43 |
44 |             if (!response.ok) {
45 |                 throw new Error(`API Error: ${response.status}`);
46 |             }
47 |
48 |             return await response.json();
49 |         } catch (error) {
50 |             console.error(`[SyncApiClient] POST ${endpoint}:`, error
|         );
51 |             throw error;
52 |         }
53 |     }
54 |
55 |     /**
56 |      * Получить статус синхронизации
57 |      */
58 |     async getStatus() {
59 |         return this.get('/status');
60 |     }
61 |
62 |     /**
63 |      * Получить статистику
64 |      */
65 |     async getStats() {
66 |         return this.get('/stats');
67 |     }
68 |
69 |     /**
70 |      * Получить информацию о Google Drive

```

```
71 |     */
72 |     async getDriveInfo() {
73 |         return this.get('/drive-info');
74 |     }
75 |
76 |     /**
77 |      * Получить конфигурацию
78 |      */
79 |     async getConfig() {
80 |         return this.get('/config');
81 |     }
82 |
83 |     /**
84 |      * Запустить планировщик
85 |      */
86 |     async startScheduler(interval = 6) {
87 |         return this.post('/start', { sync_interval_hours: interval
88 |     });
89 |     }
90 |
91 |     /**
92 |      * Остановить планировщик
93 |      */
94 |     async stopScheduler() {
95 |         return this.post('/stop');
96 |     }
97 |
98 |     /**
99 |      * Синхронизировать
100 |     */
101 |     async syncNow(syncType = 'all') {
102 |         return this.post('/sync-now', { sync_type: syncType });
103 |     }
104 |
105 |     /**
106 |      * Проверить подключение
107 |      */
108 |     async testConnection() {
```

```
108 |     return this.post('/test-connection');
109 | }
110 |
111 | /**
112 |  * Обновить конфигурацию
113 |  */
114 | async updateConfig(config) {
115 |     return this.post('/config', config);
116 | }
117 | }
118 |
```

sync-tab.js

```
1 /**
2  * Google Drive Sync Management Tab
3  * Управление синхронизацией Google Drive через админ-панель
4  */
5
6 class SyncTab {
7   constructor() {
8     this.initialized = false;
9     this.messageTimeout = null;
10  }
11
12  /**
13   * Инициализация вкладки
14   */
15  async init() {
16    if (this.initialized) return;
17
18    console.log('[SyncTab] Initializing...');
19
20    // Загрузка HTML
21    await this.loadHTML();
22
23    // Установка обработчиков
24    this.attachEventListeners();
25
26    // Загрузка данных
27    await this.updateStatus();
28    await this.updateStats();
29
30    // Установка автообновления
31    if (window.registerTabPoller) {
32      window.registerTabPoller('tab-sync', () => this.updateSt
33        atus(), 5000, { pollerId: 'sync_status', immediate: false });
34      window.registerTabPoller('tab-sync', () => this.updateSt
35        ats(), 30000, { pollerId: 'sync_stats', immediate: false });
36    } else {
```

```

35     setInterval(() => this.updateStatus(), 5000);
36     setInterval(() => this.updateStats(), 30000);
37 }
38
39 this.initialized = true;
40 console.log('[SyncTab] Initialized');
41 }
42
43 /**
44  * Загрузить HTML компонента
45  */
46 async loadHTML() {
47     const container = document.getElementById('tab-sync');
48     if (!container) {
49         console.warn('[SyncTab] Container not found');
50         return;
51     }
52
53     const html = `
54         <div class="sync-panel">
55             <!-- Заголовок -->
56             <div class="mb-4">
57                 <h4 class="d-flex align-items-center gap-2">
58                     <i class="bi bi-cloud-upload-fill text-primary"></i>
59                     <span>Google Drive Synchronization</span>
60                 </h4>
61                 <p class="text-muted mb-0">Manage data synchronizati
62 on and storage migration</p>
63             </div>
64
65             <!-- Статус -->
66             <div class="card border-secondary-subtle mb-3">
67                 <div class="card-header bg-body-tertiary border-seco
68 ndary-subtle">
69                     <h6 class="mb-0">□ Synchronization Status</h6>
70                 </div>
71                 <div class="card-body">

```

```

70         <div class="row g-3">
71             <div class="col-md-3">
72                 <div class="ps-3 border-start border-primary-s
ubtle">
73                     <small class="text-muted">Scheduler</small>
74                     <div>
75                         <span id="scheduler-status" class="badge b
g-danger">Inactive</span>
76                     </div>
77                 </div>
78             </div>
79             <div class="col-md-3">
80                 <div class="ps-3 border-start border-primary-s
ubtle">
81                     <small class="text-muted">Last Sync</small>
82                     <div id="last-sync-time" class="small">Never
</div>
83                 </div>
84             </div>
85             <div class="col-md-3">
86                 <div class="ps-3 border-start border-primary-s
ubtle">
87                     <small class="text-muted">Interval</small>
88                     <div id="sync-interval" class="small">6 hour
s</div>
89                 </div>
90             </div>
91             <div class="col-md-3">
92                 <div class="ps-3 border-start border-primary-s
ubtle">
93                     <small class="text-muted">Drive Folder</smal
l>
94                     <div>
95                         <a id="drive-url" href="#" target="_blank"
class="small text-decoration-none">
96                             Open
97                         </a>
98                     </div>

```



```
99         </div>
100     </div>
101 </div>
102 </div>
103 </div>
104
105 <!-- Статистика -->
106 <div class="card border-secondary-subtle mb-3">
107     <div class="card-header bg-body-tertiary border-sec-
108 ndary-subtle">
109         <h6 class="mb-0"> Storage Statistics</h6>
110     </div>
111     <div class="card-body">
112         <div class="row g-3">
113             <div class="col-md-3 text-center">
114                 <div class="fs-5 fw-bold text-info" id="total-
115 size">-</div>
116                 <small class="text-muted">Total Size</small>
117             </div>
118             <div class="col-md-3 text-center">
119                 <div class="fs-5 fw-bold text-success" id="fil-
120 es-count">-</div>
121                 <small class="text-muted">Files</small>
122             </div>
123             <div class="col-md-3 text-center">
124                 <div class="fs-5 fw-bold text-warning" id="fol-
125 ders-count">-</div>
126                 <small class="text-muted">Folders</small>
127             </div>
128             <div class="col-md-3 text-center">
129                 <div class="fs-5 fw-bold text-danger" id="sync-
130 -errors">0</div>
131                 <small class="text-muted">Errors</small>
132             </div>
133         </div>
134     </div>
135 </div>
```

```

132 |         <!-- Управление -->
133 |         <div class="card border-secondary-subtle mb-3">
134 |             <div class="card-header bg-body-tertiary border-seco
ndary-subtle">
135 |                 <h6 class="mb-0">□ Synchronization Controls</h6>
136 |             </div>
137 |             <div class="card-body">
138 |                 <div class="mb-3">
139 |                     <label class="form-label">Sync Interval (hours)<
/label>
140 |                     <div class="input-group mb-2">
141 |                         <input type="number" id="sync-interval-input"
min="1" max="24" value="6" class="form-control">
142 |                         <button class="btn btn-outline-secondary" oncl
ick="window.syncTab.updateInterval()">
143 |                             Update
144 |                         </button>
145 |                     </div>
146 |                 </div>
147 |
148 |                 <div class="d-flex gap-2 flex-wrap">
149 |                     <button id="start-scheduler-btn" class="btn btn-
success btn-sm" onclick="window.syncTab.startScheduler()">
150 |                         <i class="bi bi-play-fill"></i> Start Schedule
r
151 |                     </button>
152 |                     <button id="stop-scheduler-btn" class="btn btn-d
anger btn-sm" disabled onclick="window.syncTab.stopScheduler()
">
153 |                         <i class="bi bi-stop-fill"></i> Stop Scheduler
154 |                     </button>
155 |                     <button class="btn btn-primary btn-sm" onclick="
window.syncTab.showSyncOptions()">
156 |                         <i class="bi bi-arrow-repeat"></i> Sync Now
157 |                     </button>
158 |                     <button class="btn btn-info btn-sm" onclick="win
dow.syncTab.testConnection()">
159 |                         <i class="bi bi-link-45deg"></i> Test Connecti

```



```

187 |             <input class="form-check-input" type="radio"
      | name="sync-type" value="logs" id="sync-logs">
188 |             <label class="form-check-label" for="sync-lo
      | gs">
189 |                 <strong>Logs Only</strong> - Application l
      | ogs
190 |             </label>
191 |         </div>
192 |         <div class="form-check mb-2">
193 |             <input class="form-check-input" type="radio"
      | name="sync-type" value="secrets" id="sync-secrets">
194 |             <label class="form-check-label" for="sync-se
      | crets">
195 |                 <strong>Secrets Only</strong> - Configurat
      | ion and keys
196 |             </label>
197 |         </div>
198 |     </div>
199 |     <div class="modal-footer border-secondary">
200 |         <button type="button" class="btn btn-secondary
      | btn-sm" data-bs-dismiss="modal">
201 |             Cancel
202 |         </button>
203 |         <button type="button" class="btn btn-primary b
      | tn-sm" onclick="window.syncTab.performSync()">
204 |             Sync Selected
205 |         </button>
206 |     </div>
207 | </div>
208 | </div>
209 | </div>
210 |
211 | <!-- Сообщения -->
212 | <div id="sync-messages" class="mb-3"></div>
213 |
214 | <!-- Лог активности -->
215 | <div class="card border-secondary-subtle">
216 |     <div class="card-header bg-body-tertiary border-seco

```

```

ndary-subtle">
217         <h6 class="mb-0"> Recent Activity</h6>
218     </div>
219     <div class="card-body" style="max-height: 300px; ove
rflow-y: auto;">
220         <div id="activity-log">
221             <p class="text-muted mb-0">No recent activity</p
>
222         </div>
223     </div>
224 </div>
225 </div>
226 `;
227
228     container.innerHTML = html;
229 }
230
231 /**
232  * Присоединить обработчики событий
233  */
234 attachEventListeners() {
235     // Обработка модального окна
236     const modal = new bootstrap.Modal(document.getElementById(
'sync-options-modal'), {
237         backdrop: 'static'
238     });
239
240     window.syncTab = this; // Для доступа из HTML
241 }
242
243 /**
244  * Обновить статус синхронизации
245  */
246 async updateStatus() {
247     try {
248         const response = await fetch('/api/admin/sync/status');
249         if (!response.ok) throw new Error('Failed to fetch statu
s');

```

```
250 |
251 |     const data = await response.json();
252 |
253 |     // Обновить статус
254 |     const badge = document.getElementById('scheduler-status'
    | );
255 |     badge.textContent = data.is_running ? 'Running' : 'Inact
    | ive';
256 |     badge.className = data.is_running
257 |         ? 'badge bg-success'
258 |         : 'badge bg-danger';
259 |
260 |     // Обновить кнопки
261 |     document.getElementById('start-scheduler-btn').disabled
    | = data.is_running;
262 |     document.getElementById('stop-scheduler-btn').disabled =
    | !data.is_running;
263 |
264 |     // Обновить время
265 |     document.getElementById('last-sync-time').textContent =
    | data.last_sync_time || 'Never';
266 |     document.getElementById('sync-interval').textContent = `
    | ${data.sync_interval_hours} hours`;
267 |
268 |     // Обновить Google Drive папку
269 |     try {
270 |         const driveResponse = await fetch('/api/admin/sync/dri
    | ve-info');
271 |         if (driveResponse.ok) {
272 |             const driveData = await driveResponse.json();
273 |             const link = document.getElementById('drive-url');
274 |             link.href = driveData.folder_url;
275 |         }
276 |     } catch (e) {
277 |         console.warn('[SyncTab] Failed to fetch drive info');
278 |     }
279 | } catch (error) {
280 |     console.error('[SyncTab] Error updating status:', error)
```

```

    ;
281     }
282   }
283
284   /**
285    * Обновить статистику
286    */
287   async updateStats() {
288     try {
289       const response = await fetch('/api/admin/sync/stats');
290       if (!response.ok) throw new Error('Failed to fetch stats
291       ');
292       const data = await response.json();
293
294       document.getElementById('total-size').textContent =
295         `${data.total_size_mb.toFixed(1)} MB`;
296       document.getElementById('files-count').textContent = dat
297       a.files_count;
298       document.getElementById('folders-count').textContent = d
299       ata.folders_count;
300       document.getElementById('sync-errors').textContent = dat
301       a.sync_errors;
302     } catch (error) {
303       console.error('[SyncTab] Error updating stats:', error);
304     }
305   }
306
307   /**
308    * Запустить планировщик
309    */
310   async startScheduler() {
311     try {
312       const interval = parseInt(document.getElementById('sync-
313       interval-input').value);
314       const response = await fetch('/api/admin/sync/start', {
315         method: 'POST',
316         headers: { 'Content-Type': 'application/json' },

```

```

313         body: JSON.stringify({ sync_interval_hours: interval }
314     );
315
316     const data = await response.json();
317     this.showMessage(data.message || 'Scheduler started', 's
318 uccess');
319     await this.updateStatus();
320     } catch (error) {
321         this.showMessage('Error: ' + error.message, 'danger');
322     }
323
324     /**
325      * Остановить планировщик
326      */
327     async stopScheduler() {
328         try {
329             const response = await fetch('/api/admin/sync/stop', {
330                 method: 'POST'
331             });
332
333             const data = await response.json();
334             this.showMessage(data.message || 'Scheduler stopped', 's
335 uccess');
336             await this.updateStatus();
337         } catch (error) {
338             this.showMessage('Error: ' + error.message, 'danger');
339         }
340
341         /**
342          * Проверить подключение
343          */
344         async testConnection() {
345             try {
346                 const response = await fetch('/api/admin/sync/test-conne
347 ction', {

```



```
347         method: 'POST'
348     });
349
350     const data = await response.json();
351     const type = data.connected ? 'success' : 'warning';
352     this.showMessage(data.message || 'Connection test comple
ted', type);
353 } catch (error) {
354     this.showMessage('Error: ' + error.message, 'danger');
355 }
356 }
357
358 /**
359  * Показать опции синхронизации
360  */
361 showSyncOptions() {
362     const modal = bootstrap.Modal.getInstance(document.getElem
entById('sync-options-modal')) ||
363     new bootstrap.Modal(document.getElementById('sync-option
s-modal'));
364     modal.show();
365 }
366
367 /**
368  * Выполнить синхронизацию
369  */
370 async performSync() {
371     const syncType = document.querySelector('input[name="sync-
type"]:checked').value || 'all';
372
373     try {
374         const response = await fetch('/api/admin/sync/sync-now',
{
375             method: 'POST',
376             headers: { 'Content-Type': 'application/json' },
377             body: JSON.stringify({ sync_type: syncType })
378         });
379
```

```
380     const data = await response.json();
381     this.showMessage(data.message || 'Sync started', 'success');
382     this.addActivityLog(`Started ${syncType} synchronization`);
383
384     // Скрыть модаль
385     const modal = bootstrap.Modal.getInstance(document.getElementById('sync-options-modal'));
386     if (modal) modal.hide();
387   } catch (error) {
388     this.showMessage('Error: ' + error.message, 'danger');
389   }
390 }
391
392 /**
393  * Обновить интервал
394  */
395 async updateInterval() {
396   const interval = parseInt(document.getElementById('sync-interval-input').value);
397
398   try {
399     const response = await fetch('/api/admin/sync/config', {
400       method: 'POST',
401       headers: { 'Content-Type': 'application/json' },
402       body: JSON.stringify({ sync_interval_hours: interval })
403     });
404
405     this.showMessage(`Interval updated to ${interval} hours`, 'success');
406     await this.updateStatus();
407   } catch (error) {
408     this.showMessage('Error: ' + error.message, 'danger');
409   }
410 }
411
```

```

412 | /**
413 |  * Показать сообщение
414 |  */
415 | showMessage(text, type = 'info') {
416 |     const container = document.getElementById('sync-messages')
417 |     ;
418 |     if (!container) return;
419 |     const alert = document.createElement('div');
420 |     alert.className = `alert alert-${type} alert-dismissible fade show`;
421 |     alert.innerHTML = `
422 |         ${text}
423 |         <button type="button" class="btn-close" data-bs-dismiss=
424 | "alert"></button>
425 |     `;
426 |     container.appendChild(alert);
427 |
428 |     clearTimeout(this.messageTimeout);
429 |     this.messageTimeout = setTimeout(() => {
430 |         alert.remove();
431 |     }, 5000);
432 | }
433 |
434 | /**
435 |  * Добавить в лог активности
436 |  */
437 | addActivityLog(text) {
438 |     const log = document.getElementById('activity-log');
439 |     if (!log) return;
440 |
441 |     const item = document.createElement('div');
442 |     item.className = 'small mb-2 pb-2 border-bottom';
443 |
444 |     const now = new Date().toLocaleTimeString();
445 |     item.innerHTML = `<span class="text-muted">${now}</span> -
    ${text}`;

```

```
446 |
447 |     if (log.textContent.includes('No recent activity')) {
448 |         log.innerHTML = '';
449 |     }
450 |
451 |     log.insertBefore(item, log.firstChild);
452 | }
453 | }
454 |
455 | // Инициализация при загрузке документа
456 | document.addEventListener('DOMContentLoaded', async () => {
457 |     window.syncTab = new SyncTab();
458 |
459 |     // Инициализировать когда вкладка станет видимой
460 |     const tab = document.getElementById('tab-sync');
461 |     if (tab) {
462 |         // Проверить если вкладка уже активна
463 |         const tabPane = tab.closest('.tab-pane');
464 |         if (tabPane && tabPane.classList.contains('active')) {
465 |             await window.syncTab.init();
466 |         } else if (tabPane) {
467 |             // Инициализировать при активации вкладки
468 |             tabPane.addEventListener('shown.bs.tab', () => {
469 |                 window.syncTab.init();
470 |             });
471 |         }
472 |     }
473 | });
474 |
```

index.html

```
1 <!-- System Admin Tab -->
2 <div class="container-fluid py-2">
3   <div class="alert alert-dark d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3 shadow-sm border-secondary">
4     <div class="d-flex align-items-center gap-2">
5       <span class="fs-4">⏏ </span>
6       <div>
7         <h5 class="mb-0 fw-bold">Центральное управление систем
ой</h5>
8         <p class="mb-0 small text-muted">Мониторинг ядра, системные сервисы, провайдеры моделей и глобальное администрирование</p>
9       </div>
10    </div>
11    <div class="d-flex gap-2 flex-wrap">
12      <button class="btn btn-warning btn-sm fw-bold d-flex align-items-center gap-1 shadow-sm" onclick="switchToTab('helpdesk')" id="btn-switch-to-helpdesk">
13        <i class="bi bi-headset"></i> Режим Helpdesk
14      </button>
15      <button class="btn btn-outline-info btn-sm" onclick="rescanStorageDrives()">
16        <i class="bi bi-hdd-network"></i> Пересканировать диски
17      </button>
18      <button class="btn btn-outline-primary btn-sm" onclick="actualizeAiModels()">
19        <i class="bi bi-cpu"></i> Обновить модели ИИ
20      </button>
21    </div>
22  </div>
23
24 <!-- System Health & Status Cards -->
25 <div class="row g-3 mb-3">
26   <div class="col-md-3 col-sm-6">
```

```
27 |     <div class="card shadow-sm border-secondary-subtle h-100
    | ">
28 |         <div class="card-body p-3">
29 |             <div class="text-muted small text-uppercase fw-semib
    | old mb-1">Сервер / Режим</div>
30 |             <div class="d-flex align-items-center justify-conten
    | t-between">
31 |                 <h5 class="mb-0 fw-bold text-primary" id="sys-serv
    | er-mode">FastAPI (DEV)</h5>
32 |                 <span class="badge bg-success-subtle text-success"
    | >Online</span>
33 |             </div>
34 |             <small class="text-muted" id="sys-server-host">0.0.0
    | .0:8000</small>
35 |         </div>
36 |     </div>
37 | </div>
38 |
39 | <div class="col-md-3 col-sm-6">
40 |     <div class="card shadow-sm border-secondary-subtle h-100
    | ">
41 |         <div class="card-body p-3">
42 |             <div class="text-muted small text-uppercase fw-semib
    | old mb-1">AI Провайдеры</div>
43 |             <div class="d-flex align-items-center justify-conten
    | t-between">
44 |                 <h5 class="mb-0 fw-bold text-info" id="sys-ai-prov
    | iders">Gemini / AGY / Foundry</h5>
45 |                 <i class="bi bi-robot fs-4 text-info"></i>
46 |             </div>
47 |             <small class="text-muted">Актуализированный пул</sma
    | ll>
48 |         </div>
49 |     </div>
50 | </div>
51 |
52 | <div class="col-md-3 col-sm-6">
53 |     <div class="card shadow-sm border-secondary-subtle h-100
```

```

">
54         <div class="card-body p-3">
55             <div class="text-muted small text-uppercase fw-semib
old mb-1">Подключено дисков</div>
56             <div class="d-flex align-items-center justify-conten
t-between">
57                 <h5 class="mb-0 fw-bold text-warning" id="sys-driv
es-count">Загрузка...</h5>
58                 <i class="bi bi-device-hdd fs-4 text-warning"></i>
59             </div>
60             <small class="text-muted" id="sys-drives-list">CONNE
CTED_DRIVES</small>
61         </div>
62     </div>
63 </div>
64
65     <div class="col-md-3 col-sm-6">
66         <div class="card shadow-sm border-secondary-subtle h-100
">
67             <div class="card-body p-3">
68                 <div class="text-muted small text-uppercase fw-semib
old mb-1">Подключаемые плагины</div>
69                 <div class="d-flex align-items-center justify-conten
t-between">
70                     <h5 class="mb-0 fw-bold text-success" id="sys-plug
ins-count">Загрузка...</h5>
71                     <a href="javascript:void(0)" onclick="switchToTab(
'plugins')" class="badge bg-primary text-decoration-none">Упра
вление </a>
72                 </div>
73                 <small class="text-muted" id="sys-plugins-active-cou
nt">Реестр модулей</small>
74             </div>
75         </div>
76     </div>
77 </div>
78
79     <!-- Fast Navigation Hub to Core Subsystems -->

```

```
80 | <h6 class="fw-bold mb-3 text-uppercase small text-muted"><i
    | class="bi bi-grid-fill me-1"></i> Системные подсистемы</h6>
81 | <div class="row g-3 mb-4">
82 |     <div class="col-lg-4 col-md-6">
83 |         <div class="card shadow-sm border-secondary-subtle h-100
    | p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
    | ab('plugins')">
84 |             <div class="d-flex align-items-center gap-3">
85 |                 <span class="fs-1">☐ </span>
86 |                 <div>
87 |                     <h6 class="fw-bold mb-1">Подключатель плагинов (Hu
    | b)</h6>
88 |                     <p class="small text-muted mb-0">Управление модуля
    | ми (web_search, rag, langchain_media и др.), включение/отключе
    | ние и их органы настройки.</p>
89 |                 </div>
90 |             </div>
91 |         </div>
92 |     </div>
93 |
94 |     <div class="col-lg-4 col-md-6">
95 |         <div class="card shadow-sm border-secondary-subtle h-100
    | p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
    | ab('users')">
96 |             <div class="d-flex align-items-center gap-3">
97 |                 <span class="fs-1">☐ </span>
98 |                 <div>
99 |                     <h6 class="fw-bold mb-1">Управление пользователями
    | </h6>
100 |                     <p class="small text-muted mb-0">Создание аккаунто
    | в, назначение ролей (admin/user), пароли, активность и права д
    | оступа.</p>
101 |                 </div>
102 |             </div>
103 |         </div>
104 |     </div>
105 |
106 |     <div class="col-lg-4 col-md-6">
```



```
107 |         <div class="card shadow-sm border-secondary-subtle h-100
      |         p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('models')">
108 |             <div class="d-flex align-items-center gap-3">
109 |                 <span class="fs-1">📄 </span>
110 |                 <div>
111 |                     <h6 class="fw-bold mb-1">Модели и API-ключи</h6>
112 |                     <p class="small text-muted mb-0">Конфигурация Google Gemini, Antigravity, Ollama и AI Foundry, пул ротации ключей.</p>
113 |                 </div>
114 |             </div>
115 |         </div>
116 |     </div>
117 |
118 |     <div class="col-lg-4 col-md-6">
119 |         <div class="card shadow-sm border-secondary-subtle h-100
      |         p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('instructions')">
120 |             <div class="d-flex align-items-center gap-3">
121 |                 <span class="fs-1">📄 </span>
122 |                 <div>
123 |                     <h6 class="fw-bold mb-1">Системные инструкции</h6>
124 |                     <p class="small text-muted mb-0">Редактирование системных промптов чата, стилей диктора и версионирование инструкций.</p>
125 |                 </div>
126 |             </div>
127 |         </div>
128 |     </div>
129 |
130 |     <div class="col-lg-4 col-md-6">
131 |         <div class="card shadow-sm border-secondary-subtle h-100
      |         p-3 hover-shadow" style="cursor: pointer;" onclick="switchToTab('logs')">
132 |             <div class="d-flex align-items-center gap-3">
133 |                 <span class="fs-1">📄 </span>
134 |                 <div>
```

```

135 |         <h6 class="fw-bold mb-1">Системные логи</h6>
136 |         <p class="small text-muted mb-0">Просмотр журнала
    | событий uvicorn, ошибок FastAPI, истории запросов и аудита.</p>
    | >
137 |     </div>
138 | </div>
139 | </div>
140 | </div>
141 |
142 | <div class="col-lg-4 col-md-6">
143 |     <div class="card shadow-sm border-secondary-subtle h-100
    | p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
    | ab('agents')">
144 |         <div class="d-flex align-items-center gap-3">
145 |             <span class="fs-1">🔍 </span>
146 |             <div>
147 |                 <h6 class="fw-bold mb-1">Агенты ИИ</h6>
148 |                 <p class="small text-muted mb-0">Конфигурация авто
    | номных ReAct агентов, тулов поиска и параметров инференса.</p>
149 |             </div>
150 |         </div>
151 |     </div>
152 | </div>
153 |
154 | <div class="col-lg-4 col-md-6">
155 |     <div class="card shadow-sm border-secondary-subtle h-100
    | p-3 hover-shadow" style="cursor: pointer;" onclick="switchToT
    | ab('models'); setTimeout(() => { const btn = document.getEleme
    | ntById('pill-google-accounts-tab'); if (btn) btn.click(); }, 1
    | 50);">
156 |         <div class="d-flex align-items-center gap-3">
157 |             <span class="fs-1">🔍 </span>
158 |             <div>
159 |                 <h6 class="fw-bold mb-1">Google Workspace & Аккаун
    | ты</h6>
160 |                 <p class="small text-muted mb-0">Пул аккаунтов OAu
    | th 2.0 и Service Account для Gmail, Drive, Sheets и Docs.</p>
161 |             </div>

```

```

162         </div>
163     </div>
164 </div>
165
166     <div class="col-lg-4 col-md-6">
167         <div class="card shadow-sm border-secondary-subtle h-100
168         p-3 hover-shadow border-warning-subtle" style="cursor: pointe
169         r;" onclick="switchToTab('helpdesk')">
170             <div class="d-flex align-items-center gap-3">
171                 <span class="fs-1">🔍 </span>
172                 <div>
173                     <h6 class="fw-bold mb-1 text-warning">Helpdesk & С
174                     лужба поддержки</h6>
175                     <p class="small text-muted mb-0">Управление заявка
176                     ми, живой чат с пользователями, шаблоны быстрых ответов и метр
177                     ики SLA.</p>
178                 </div>
179             </div>
180         </div>
181
182     <div class="col-lg-4 col-md-6">
183         <div class="card shadow-sm border-secondary-subtle h-100
184         p-3 hover-shadow border-primary-subtle" style="cursor: pointe
185         r;" onclick="switchToTab('observability')">
186             <div class="d-flex align-items-center gap-3">
187                 <span class="fs-1">🔍 </span>
188                 <div>
189                     <h6 class="fw-bold mb-1 text-primary">Observabilit
190                     y & Диагностика</h6>
191                     <p class="small text-muted mb-0">Мониторинг ресурс
192                     ов хоста в реальном времени, дерево оборудования и эвристическ
193                     ий AI-аудит.</p>
194                 </div>
195             </div>
196         </div>
197     </div>
198 </div>

```

```

190     <div class="col-lg-4 col-md-6">
191         <div class="card shadow-sm border-secondary-subtle h-100
p-3 hover-shadow border-info-subtle" style="cursor: pointer;"
onClick="document.getElementById('scheduler-management-card')
.scrollIntoView({behavior: 'smooth'})">
192             <div class="d-flex align-items-center gap-3">
193                 <span class="fs-1">□ </span>
194                 <div>
195                     <h6 class="fw-bold mb-1 text-info">Фоновый планиро
вщик (Scheduler)</h6>
196                     <p class="small text-muted mb-0">24ч переиндексац
ия RAG коллекций, 5м мониторинг почты Gmail и автообновление.</
p>
197                 </div>
198             </div>
199         </div>
200     </div>
201 </div>
202
203 <!-- Background Schedulers Management Section -->
204 <div class="card shadow-sm border-secondary-subtle mb-3" id=
"scheduler-management-card">
205     <div class="card-header bg-body-tertiary d-flex flex-wrap
justify-content-between align-items-center py-2 gap-2">
206         <div class="d-flex align-items-center gap-2">
207             <span class="fw-bold"><i class="bi bi-clock-history te
xt-info me-2"></i>□ Фоновые периодические задачи (Schedulers)<
/span>
208             <span id="sched-main-badge" class="badge bg-success-su
btle text-success">Активен</span>
209         </div>
210         <div class="d-flex gap-2">
211             <button class="btn btn-outline-info btn-sm" onclick="r
efreshSchedulersConfig()">
212                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
статус
213             </button>
214             <button class="btn btn-primary btn-sm" onclick="saveSc

```

```

hedulersConfig()">
215         <i class="bi bi-check2-circle me-1"></i> Сохранить н
астройки
216     </button>
217 </div>
218 </div>
219 <div class="card-body p-3">
220     <!-- Master toggle -->
221     <div class="form-check form-switch mb-3">
222         <input class="form-check-input" type="checkbox" id="sc
hed-master-switch" checked>
223         <label class="form-check-label fw-bold" for="sched-mas
ter-switch">Включить глобальный фоновый планировщик (Backgroun
dScheduler)</label>
224         <div class="small text-muted">При отключении все фонов
ые интервальные циклы будут остановлены.</div>
225     </div>
226
227     <div class="row g-3">
228         <!-- RAG 24h Re-indexing Task -->
229         <div class="col-md-6">
230             <div class="card bg-body-tertiary border-secondary-s
ubtle h-100 p-3">
231                 <div class="d-flex justify-content-between align-i
tems-center mb-2">
232                     <span class="fw-bold text-primary"><i class="bi
bi-cpu me-1"></i> □ Переиндексация RAG (RAG Re-indexing)</span
>
233                     <div class="form-check form-switch m-0">
234                         <input class="form-check-input" type="checkbox
" id="sched-rag-enabled" checked>
235                     </div>
236                 </div>
237                 <p class="small text-muted mb-2">Автоматический фо
новый пересчет и обновление коллекций векторных/TF-IDF индексо
в.</p>
238                 <div class="mb-2">
239                     <label class="form-label small fw-semibold mb-1"

```

```

    for="sched-rag-interval">Интервал запуска (в часах):</label>
240         <div class="input-group input-group-sm">
241             <input type="number" class="form-control" id="
sched-rag-interval" value="24" min="0.1" step="0.5">
242             <span class="input-group-text">час(ов)</span>
243         </div>
244     </div>
245     <div class="d-flex justify-content-between align-i
tems-center small text-muted mt-auto pt-2 border-top border-se
condary-subtle">
246         <span>Посл. запуск: <strong id="sched-rag-last-r
un">Никогда</strong></span>
247         <button class="btn btn-outline-primary btn-sm py
-0 px-2" onclick="triggerSchedulerJob('rag')" title="Запустить
переиндексацию сейчас">
248             <i class="bi bi-play-fill"></i> Запустить сейч
ас
249         </button>
250     </div>
251 </div>
252 </div>
253
254 <!-- Gmail 5m Check Task -->
255 <div class="col-md-6">
256     <div class="card bg-body-tertiary border-secondary-s
ubtle h-100 p-3">
257         <div class="d-flex justify-content-between align-i
tems-center mb-2">
258             <span class="fw-bold text-success"><i class="bi
bi-envelope-check me-1"></i> □ Проверка Gmail (Email Check)</s
pan>
259             <div class="form-check form-switch m-0">
260                 <input class="form-check-input" type="checkbox
" id="sched-email-enabled" checked>
261             </div>
262         </div>
263         <p class="small text-muted mb-2">Опрос ящиков Goog
le Workspace на наличие новых непрочитанных сообщений.</p>

```

```

264 |         <div class="mb-2">
265 |             <label class="form-label small fw-semibold mb-1"
    | for="sched-email-interval">Интервал опроса (в минутах):</label>
266 |             <div class="input-group input-group-sm">
267 |                 <input type="number" class="form-control" id="
    | sched-email-interval" value="5" min="1" step="1">
268 |                 <span class="input-group-text">мин.</span>
269 |             </div>
270 |         </div>
271 |         <div class="d-flex justify-content-between align-items-center small text-muted mt-auto pt-2 border-top border-secondary-subtle">
272 |             <span>Посл. запуск: <strong id="sched-email-last-run">Никогда</strong></span>
273 |             <button class="btn btn-outline-success btn-sm py-0 px-2" onclick="triggerSchedulerJob('email')" title="Проверить почту сейчас">
274 |                 <i class="bi bi-play-fill"></i> Запустить сейчас
    |
275 |             </button>
276 |         </div>
277 |     </div>
278 | </div>
279 | </div>
280 | </div>
281 | </div>
282 |
283 | <!-- Storage & Disks Management Section -->
284 | <div class="card shadow-sm border-secondary-subtle mb-3">
285 |     <div class="card-header bg-body-tertiary d-flex flex-wrap justify-content-between align-items-center py-2 gap-2">
286 |         <span class="fw-bold"><i class="bi bi-hdd-network text-warning me-2"></i> ☐ Управление дисками и накопителями</span>
287 |         <button class="btn btn-outline-info btn-sm" onclick="rescanStorageDrives()">
288 |             <i class="bi bi-arrow-clockwise me-1"></i> Пересканировать диски

```

```

289 |         </button>
290 |     </div>
291 |     <div class="card-body p-3">
292 |         <div id="sys-drives-detailed-container" class="row g-3">
293 |             <div class="col-12 text-muted small">Загрузка информац
|ии о накопителях...</div>
294 |         </div>
295 |     </div>
296 | </div>
297 |
298 | <!-- User Telemetry & Ngrok Tunnel Section -->
299 | <div class="card shadow-sm border-secondary-subtle mb-4">
300 |     <div class="card-header bg-body-tertiary d-flex flex-wrap
|justify-content-between align-items-center py-2 gap-2">
301 |         <div class="d-flex align-items-center gap-2">
302 |             <span class="fw-bold"><i class="bi bi-graph-up-arrow t
|ext-info me-2"></i> □ Активность пользователей и туннель Ngrok
|</span>
303 |             <span id="ngrok-status-badge" class="badge bg-secondar
|y-subtle text-secondary">Туннель: проверка...</span>
304 |         </div>
305 |         <div class="d-flex gap-2">
306 |             <button class="btn btn-outline-primary btn-sm" onclick
|="refreshTelemetryStats()">
307 |                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
|статистику
308 |             </button>
309 |         </div>
310 |     </div>
311 |     <div class="card-body p-3">
312 |         <!-- Tunnel info alert -->
313 |         <div id="ngrok-info-alert" class="alert alert-secondary
|py-2 px-3 small d-flex flex-wrap align-items-center justify-co
|ntent-between gap-2 mb-3">
314 |             <div class="d-flex align-items-center gap-2">
315 |                 <i class="bi bi-broadcast fs-5 text-primary"></i>
316 |                 <div>
317 |                     <span class="fw-semibold">Ngrok шлюз приема статис

```



```
тики:</span>
318         <span id="ngrok-public-url-text" class="text-prim
ry font-monospace ms-1">Ожидание запуска...</span>
319     </div>
320 </div>
321     <small class="text-muted" id="ngrok-hint-text">Скрипт:
<code>.\launchers\Run-Ngrok.ps1</code></small>
322 </div>
323
324 <!-- Telemetry Stats Cards -->
325 <div class="row g-3 mb-3">
326     <div class="col-md-4">
327         <div class="card bg-body-tertiary border-0 p-3 h-100
">
328             <div class="small text-muted text-uppercase fw-sem
ibold mb-1">Событий за 30 дней</div>
329             <h4 class="mb-0 fw-bold text-primary" id="telemetr
y-total-events">0</h4>
330             <small class="text-muted" id="telemetry-active-use
rs">0 пользователей</small>
331         </div>
332     </div>
333     <div class="col-md-4">
334         <div class="card bg-body-tertiary border-0 p-3 h-100
">
335             <div class="small text-muted text-uppercase fw-sem
ibold mb-1">Популярные вкладки</div>
336             <div id="telemetry-tabs-breakdown" class="small te
xt-muted">
337                 Загрузка...
338             </div>
339         </div>
340     </div>
341     <div class="col-md-4">
342         <div class="card bg-body-tertiary border-0 p-3 h-100
">
343             <div class="small text-muted text-uppercase fw-sem
ibold mb-1">Топ кликов и действий</div>
```

```

344 |         <div id="telemetry-clicks-breakdown" class="small
      | text-muted">
345 |             Загрузка...
346 |         </div>
347 |     </div>
348 | </div>
349 | </div>
350 |
351 |     <!-- Recent Telemetry Log Table -->
352 |     <h6 class="fw-bold small text-muted text-uppercase mb-2"
      | ><i class="bi bi-clock-history me-1"></i> Последние действия з
      | арегистрированных пользователей</h6>
353 |     <div class="table-responsive">
354 |         <table class="table table-sm table-hover align-middle
      | mb-0" style="font-size: 0.85rem;">
355 |             <thead class="table-light">
356 |                 <tr>
357 |                     <th scope="col">Время</th>
358 |                     <th scope="col">Пользователь</th>
359 |                     <th scope="col">Тип</th>
360 |                     <th scope="col">Вкладка</th>
361 |                     <th scope="col">Действие / Элемент</th>
362 |                     <th scope="col">Длительность</th>
363 |                 </tr>
364 |             </thead>
365 |             <tbody id="telemetry-events-tbody">
366 |                 <tr>
367 |                     <td colspan="6" class="text-center text-muted py
      | -3">Загрузка журнала активности...</td>
368 |                 </tr>
369 |             </tbody>
370 |         </table>
371 |     </div>
372 | </div>
373 | </div>
374 | </div>

```

main.js

```
1 // System Admin Tab – Central Management Logic
2
3 'use strict';
4
5 async function initSystemAdminTab() {
6   console.log('Инициализация панели системного управления...')
7   ;
8   await refreshSystemDashboard();
9   await refreshSchedulersConfig();
10  await refreshTelemetryStats();
11 }
12
13 async function refreshSystemDashboard() {
14   try {
15     // 1. Загрузка дисков
16     const drivesData = await window.api.fetch('/api/control/re
17 scan', { method: 'GET' }).catch(() => ({ drives: [], details:
18 [] }));
19     const drives = Array.isArray(drivesData.drives) ? drivesDa
20 ta.drives : [];
21     const details = Array.isArray(drivesData.details) ? drives
22 Data.details : [];
23     const drivesCountEl = document.getElementById('sys-drives-
24 count');
25     const drivesListEl = document.getElementById('sys-drives-l
26 ist');
27     if (drivesCountEl) drivesCountEl.textContent = `${drives.l
28 ength} диск(ов)`;
29     if (drivesListEl) drivesListEl.textContent = drives.join('
30 , ') || 'Диски не обнаружены';
31
32
33     const detailedContainer = document.getElementById('sys-dri
34 ves-detailed-container');
35     if (detailedContainer) {
36       if (details.length === 0 && drives.length === 0) {
37         detailedContainer.innerHTML = '<div class="col-12 text
```

```

    -muted small">Накопители не обнаружены или доступ ограничен.</
div>';
27     } else if (details.length > 0) {
28         detailedContainer.innerHTML = details.map(d => {
29             const percent = d.percent || 0;
30             const barClass = percent > 90 ? 'bg-danger' : percen
t > 75 ? 'bg-warning' : 'bg-primary';
31             return `
32                 <div class="col-lg-4 col-md-6">
33                     <div class="card bg-body-tertiary border-seconda
ry-subtle h-100 p-3">
34                         <div class="d-flex justify-content-between ali
gn-items-center mb-2">
35                             <span class="fw-bold text-primary"><i class=
"bi bi-hdd me-1"></i>${d.device || d.mountpoint}</span>
36                             <span class="badge bg-secondary-subtle text-
body">${d.fstype || 'NTFS'}</span>
37                         </div>
38                         <div class="d-flex justify-content-between sma
ll text-muted mb-1">
39                             <span>Занято: <strong>${d.used_gb} GB</stron
g></span>
40                             <span>Свободно: <strong>${d.free_gb} GB</str
ong></span>
41                         </div>
42                         <div class="progress mb-2" style="height: 8px;
">
43                             <div class="progress-bar ${barClass}" role="
progressbar" style="width: ${percent}%;" aria-valuenow="${perc
ent}" aria-valuemin="0" aria-valuemax="100"></div>
44                         </div>
45                         <div class="d-flex justify-content-between sma
ll text-muted">
46                             <span>Всего: ${d.total_gb} GB</span>
47                             <span class="fw-semibold">${percent}%</span>
48                         </div>
49                     </div>
50                 </div>

```

```

51 |         `;
52 |     }).join('');
53 | } else {
54 |     detailedContainer.innerHTML = drives.map(drv => `
55 |         <div class="col-lg-3 col-md-4 col-sm-6">
56 |             <div class="card bg-body-tertiary border-secondary
57 | -subtle p-2 text-center">
58 |                 <span class="fw-bold text-primary"><i class="bi
59 | bi-hdd me-1"></i>${drv}</span>
60 |             </div>
61 |         </div>
62 |     `).join('');
63 | }
64 |
65 | // 2. Загрузка статуса плагинов
66 | const pluginsData = await window.api.fetch('/api/admin/plu
67 | gins').catch(() => ({ plugins: [] }));
68 | const plugins = pluginsData.plugins || [];
69 | const pluginsCountEl = document.getElementById('sys-plugin
70 | s-count');
71 | const pluginsActiveEl = document.getElementById('sys-plugi
72 | ns-active-count');
73 | const activeCount = plugins.filter(p => p.enabled).length;
74 | if (pluginsCountEl) pluginsCountEl.textContent = `${plugin
75 | s.length} модулей`;
76 | if (pluginsActiveEl) pluginsActiveEl.textContent = `${acti
77 | veCount} активно из ${plugins.length}`;
78 | } catch (err) {
79 |     console.error('Ошибка загрузки системного дашборда:', err)
80 |     ;
81 | }
82 | }
83 |
84 | async function rescanStorageDrives() {
85 |     try {
86 |         if (typeof showNotification === 'function') showNotificati
87 | on('Пересканирование накопителей ОС...', 'info');

```

```

80     const result = await window.api.fetch('/api/control/rescan
    ', { method: 'GET' });
81     const drivesList = Array.isArray(result.drives) ? result.d
    rives.join(', ') : 'OK';
82     if (typeof showNotification === 'function') showNotificati
    on(`Диски обновлены: ${drivesList}`, 'success');
83     await refreshSystemDashboard();
84 } catch (e) {
85     if (typeof showNotification === 'function') showNotificati
    on(`Ошибка: ${e.message}`, 'danger');
86 }
87 }
88
89 async function actualizeAiModels() {
90     try {
91         if (typeof showNotification === 'function') showNotificati
    on('Актуализация пула моделей ИИ...', 'info');
92         await window.api.fetch('/api/keys/actualize-all', { method
    : 'POST' }).catch(() => {});
93         if (typeof showNotification === 'function') showNotificati
    on('Модели успешно синхронизированы', 'success');
94         await refreshSystemDashboard();
95     } catch (e) {
96         if (typeof showNotification === 'function') showNotificati
    on(`Ошибка: ${e.message}`, 'danger');
97     }
98 }
99
100 async function refreshTelemetryStats() {
101     try {
102         const res = await fetch('/api/telemetry/stats?days=30');
103         if (!res.ok) return;
104         const stats = await res.json();
105
106         // 1. Ngrok Tunnel Status
107         const tunnel = stats.tunnel_status || {};
108         const badgeEl = document.getElementById('ngrok-status-badg
    e');

```

```
109 |     const urlTextEl = document.getElementById('ngrok-public-ur
    | l-text');
110 |     const alertEl = document.getElementById('ngrok-info-alert'
    | );
111 |
112 |     if (tunnel.is_active && tunnel.public_url) {
113 |         if (badgeEl) {
114 |             badgeEl.className = 'badge bg-success-subtle text-succ
    | ess';
115 |             badgeEl.textContent = 'Туннель: Онлайн';
116 |         }
117 |         if (urlTextEl) {
118 |             urlTextEl.innerHTML = `
```

```
users');
135     if (totalEl) totalEl.textContent = (stats.total_events ||
    0).toLocaleString();
136     if (usersEl) usersEl.textContent = `${stats.active_users_c
    ount || 0} пользователей`;
137
138     // 3. Tab views
139     const tabsEl = document.getElementById('telemetry-tabs-bre
    akdown');
140     if (tabsEl) {
141         const tabEntries = Object.entries(stats.tab_views || {})
    ;
142         if (tabEntries.length === 0) {
143             tabsEl.innerHTML = '<span class="text-muted">Нет данны
    x</span>';
144         } else {
145             tabsEl.innerHTML = tabEntries.slice(0, 4).map(([tab, i
    nfo]) => `
146                 <div class="d-flex justify-content-between align-ite
    ms-center mb-1">
147                     <span class="badge bg-secondary-subtle text-body">
    ${tab}</span>
148                     <span class="fw-semibold">${info.count} раз</span>
149                 </div>
150             `).join('');
151         }
152     }
153
154     // 4. Click actions
155     const clicksEl = document.getElementById('telemetry-clicks
    -breakdown');
156     if (clicksEl) {
157         const clickEntries = Object.entries(stats.top_clicks ||
    {});
158         if (clickEntries.length === 0) {
159             clicksEl.innerHTML = '<span class="text-muted">Нет кли
    ков</span>';
160         } else {
```



```

161 |         clicksEl.innerHTML = clickEntries.slice(0, 4).map(([ta
| rget, info]) => `
162 |             <div class="d-flex justify-content-between align-ite
| ms-center mb-1 text-truncate" title="${target}">
163 |                 <span class="text-truncate me-1" style="max-width:
| 140px;">${info.action || target}</span>
164 |                 <span class="badge bg-primary-subtle text-primary"
| >${info.count}</span>
165 |             </div>
166 |         `).join('');
167 |     }
168 | }
169 |
170 | // 5. Recent events table
171 | const tbody = document.getElementById('telemetry-events-tb
| ody');
172 | if (tbody) {
173 |     const recent = stats.recent_events || [];
174 |     if (recent.length === 0) {
175 |         tbody.innerHTML = '<tr><td colspan="6" class="text-cen
| ter text-muted py-3">Пока нет зарегистрированных событий актив
| ности.</td></tr>';
176 |     } else {
177 |         tbody.innerHTML = recent.map(ev => {
178 |             const dt = ev.timestamp ? new Date(ev.timestamp).tol
| ocaleTimeString() : '-';
179 |             const userName = ev.name || ev.email || `ID ${ev.use
| r_id}`;
180 |             const typeBadge = ev.event_type === 'tab_view' ? 'bg
| -info-subtle text-info' : ev.event_type === 'click' ? 'bg-prim
| ary-subtle text-primary' : 'bg-secondary-subtle text-secondary
| ';
181 |             const durStr = ev.duration_ms > 0 ? `${(ev.duration_
| ms / 1000).toFixed(1)}s` : '-';
182 |             return `
183 |                 <tr>
184 |                     <td class="text-muted font-monospace small">${dt
| }</td>

```

```

185         <td class="fw-semibold">${userName}</td>
186         <td><span class="badge ${typeBadge}">${ev.event_
type || 'action'}</span></td>
187         <td><span class="badge bg-light text-dark border
">${ev.tab_name || '-'}</span></td>
188         <td class="text-truncate" style="max-width: 250p
x;" title="${ev.target_element || ''}">${ev.action || ev.targe
t_element || '-'}</td>
189         <td class="text-muted">${durStr}</td>
190     </tr>
191     `;
192     }).join('');
193     }
194     }
195     } catch (e) {
196         console.debug('Failed to load telemetry stats:', e);
197     }
198 }
199
200 function switchToTab(tabName) {
201     if (typeof window.switchTab === 'function') {
202         window.switchTab(tabName);
203         return;
204     }
205     const cleanName = tabName.replace(/^tab-/ , '');
206     const triggerEl = document.querySelector(`[data-bs-target="#
tab-${cleanName}"]`, [data-tab="${cleanName}"]`);
207     if (triggerEl && typeof bootstrap !== 'undefined' && bootstr
ap.Tab) {
208         const tab = new bootstrap.Tab(triggerEl);
209         tab.show();
210     }
211 }
212
213 // Schedulers Management Logic
214 async function refreshSchedulersConfig() {
215     try {
216         const res = await window.api.fetch('/api/admin/schedulers'

```

```

    );
217     if (!res || !res.config) return;
218
219     const cfg = res.config;
220     const status = res.status || {};
221
222     const masterEl = document.getElementById('sched-master-swi
tch');
223     const ragEnabledEl = document.getElementById('sched-rag-en
abled');
224     const ragIntervalEl = document.getElementById('sched-rag-i
nterval');
225     const emailEnabledEl = document.getElementById('sched-emai
l-enabled');
226     const emailIntervalEl = document.getElementById('sched-ema
il-interval');
227     const badgeEl = document.getElementById('sched-main-badge'
);
228     const ragLastRunEl = document.getElementById('sched-rag-la
st-run');
229     const emailLastRunEl = document.getElementById('sched-emai
l-last-run');
230
231     if (masterEl) masterEl.checked = cfg.enabled !== false;
232     if (ragEnabledEl) ragEnabledEl.checked = cfg.rag_reindex?.
enabled !== false;
233     if (ragIntervalEl) ragIntervalEl.value = cfg.rag_reindex?.
interval_hours || 24;
234
235     if (emailEnabledEl) emailEnabledEl.checked = cfg.email_che
ck?.enabled !== false;
236     if (emailIntervalEl) emailIntervalEl.value = cfg.email_che
ck?.interval_minutes || 5;
237
238     if (badgeEl) {
239         if (status.running && cfg.enabled) {
240             badgeEl.className = 'badge bg-success-subtle text-succ
ess';

```

```
241 |         badgeEl.textContent = 'Активен (Работает)';
242 |     } else {
243 |         badgeEl.className = 'badge bg-secondary-subtle text-se
condary';
244 |         badgeEl.textContent = 'Отключен';
245 |     }
246 | }
247 |
248 | if (ragLastRunEl) {
249 |     ragLastRunEl.textContent = status.last_rag_run ? new Dat
e(status.last_rag_run).toLocaleString() : 'Никогда';
250 | }
251 | if (emailLastRunEl) {
252 |     emailLastRunEl.textContent = status.last_email_run ? new
Date(status.last_email_run).toLocaleString() : 'Никогда';
253 | }
254 | } catch (err) {
255 |     console.error('Ошибка загрузки настроек планировщика:', er
r);
256 | }
257 | }
258 |
259 | async function saveSchedulersConfig() {
260 |     try {
261 |         const masterEl = document.getElementById('sched-master-swi
tch');
262 |         const ragEnabledEl = document.getElementById('sched-rag-en
abled');
263 |         const ragIntervalEl = document.getElementById('sched-rag-i
nterval');
264 |         const emailEnabledEl = document.getElementById('sched-ema
il-enabled');
265 |         const emailIntervalEl = document.getElementById('sched-ema
il-interval');
266 |
267 |         const payload = {
268 |             enabled: masterEl ? masterEl.checked : true,
269 |             rag_reindex: {
```

```
270 |         enabled: ragEnabledEl ? ragEnabledEl.checked : true,
271 |         interval_hours: parseFloat(ragIntervalEl ? ragInterval
    | El.value : 24) || 24
272 |     },
273 |     email_check: {
274 |         enabled: emailEnabledEl ? emailEnabledEl.checked : tru
    | e,
275 |         interval_minutes: parseFloat(emailIntervalEl ? emailIn
    | tervalEl.value : 5) || 5
276 |     }
277 | };
278 |
279 | if (typeof showNotification === 'function') showNotificati
    | on('Сохранение конфигурации планировщика...', 'info');
280 |
281 | const res = await window.api.fetch('/api/admin/schedulers'
    | , {
282 |     method: 'POST',
283 |     headers: { 'Content-Type': 'application/json' },
284 |     body: JSON.stringify(payload)
285 | });
286 |
287 | if (typeof showNotification === 'function') {
288 |     showNotification('Настройки планировщика успешно сохране
    | ны и применены!', 'success');
289 | }
290 | await refreshSchedulersConfig();
291 | } catch (err) {
292 |     if (typeof showNotification === 'function') {
293 |         showNotification(`Ошибка сохранения: ${err.message}`, 'd
    | anger');
294 |     }
295 | }
296 | }
297 |
298 | async function triggerSchedulerJob(jobName) {
299 |     try {
300 |         if (typeof showNotification === 'function') {
```

```
301     showNotification(`Запуск фоновой задачи '${jobName}'...`  
    , 'info');  
302   }  
303  
304   const res = await window.api.fetch(`/api/admin/schedulers/  
trigger/${jobName}`, {  
305     method: 'POST'  
306   });  
307  
308   if (jobName === 'rag') {  
309     const count = res.result?.updated_count ?? 0;  
310     if (typeof showNotification === 'function') {  
311       showNotification(`Переиндексация RAG завершена! Обновл  
ено коллекций: ${count}`, 'success');  
312     }  
313   } else if (jobName === 'email') {  
314     const count = res.count ?? 0;  
315     if (typeof showNotification === 'function') {  
316       showNotification(`Проверка почты завершена! Найдено не  
прочитанных писем: ${count}`, 'success');  
317     }  
318   }  
319  
320   await refreshSchedulersConfig();  
321 } catch (err) {  
322   if (typeof showNotification === 'function') {  
323     showNotification(`Ошибка запуска задачи: ${err.message}`  
    , 'danger');  
324   }  
325 }  
326 }  
327  
328 window.initAdminTab = initSystemAdminTab;  
329 window.rescanStorageDrives = rescanStorageDrives;  
330 window.actualizeAiModels = actualizeAiModels;  
331 window.switchToTab = switchToTab;  
332 window.refreshTelemetryStats = refreshTelemetryStats;  
333 window.refreshSchedulersConfig = refreshSchedulersConfig;
```

```
334 | window.saveSchedulersConfig = saveSchedulersConfig;  
335 | window.triggerSchedulerJob = triggerSchedulerJob;  
336 |  
337 |
```

index.html

```
1 <!-- AI Agents Management & Builder Tab -->
2 <style>
3   /* High contrast & modern dark styling for AI Agents Tab */
4   .agent-card {
5     background: #0f172a !important;
6     border: 1px solid #334155 !important;
7     border-radius: 12px;
8     transition: transform 0.15s ease, border-color 0.15s ease,
9     box-shadow 0.15s ease;
10  }
11  .agent-card:hover {
12    border-color: #475569 !important;
13    box-shadow: 0 6px 20px rgba(0, 0, 0, 0.45);
14  }
15  .agent-card .card-header {
16    background: #1e293b !important;
17    border-bottom: 1px solid #334155 !important;
18    color: #ffffff !important;
19    border-top-left-radius: 11px;
20    border-top-right-radius: 11px;
21  }
22  .agent-card .agent-desc {
23    color: #cbd5e1 !important;
24  }
25  .agent-spec-box {
26    background: #020617 !important;
27    border: 1px solid #334155 !important;
28    border-radius: 8px;
29    padding: 0.6rem 0.8rem;
30  }
31  .agent-spec-box .spec-label {
32    color: #cbd5e1 !important;
33    font-weight: 600;
34    font-size: 0.82rem;
35  }
36  .agent-spec-box .spec-value-badge {
```



```
36 |     background-color: #2563eb !important;
37 |     color: #ffffff !important;
38 |     font-weight: 600;
39 |     font-size: 0.78rem;
40 | }
41 | .agent-spec-box .spec-tech-val {
42 |     color: #38bdf8 !important;
43 |     font-weight: 700;
44 |     font-size: 0.85rem;
45 | }
46 | .tool-badge {
47 |     background: #1e293b !important;
48 |     border: 1px solid #475569 !important;
49 |     color: #f8fafc !important;
50 |     font-size: 0.75rem;
51 |     font-weight: 500;
52 |     border-radius: 6px;
53 |     padding: 0.25rem 0.5rem;
54 |     display: inline-flex;
55 |     align-items: center;
56 |     gap: 0.25rem;
57 | }
58 | .agent-card .btn-test-sandbox {
59 |     color: #fbbf24 !important;
60 |     border: 1.5px solid #f59e0b !important;
61 |     font-weight: 600;
62 |     font-size: 0.82rem;
63 |     background: transparent;
64 | }
65 | .agent-card .btn-test-sandbox:hover {
66 |     background: #f59e0b !important;
67 |     color: #0f172a !important;
68 | }
69 | .agent-card .btn-edit-agent {
70 |     color: #38bdf8 !important;
71 |     border: 1.5px solid #0284c7 !important;
72 |     font-weight: 600;
73 |     background: transparent;
```

```
74 | }
75 | .agent-card .btn-edit-agent:hover {
76 |     background: #0284c7 !important;
77 |     color: #ffffff !important;
78 | }
79 | .agent-card .btn-delete-agent {
80 |     color: #f87171 !important;
81 |     border: 1.5px solid #dc2626 !important;
82 |     font-weight: 600;
83 |     background: transparent;
84 | }
85 | .agent-card .btn-delete-agent:hover {
86 |     background: #dc2626 !important;
87 |     color: #ffffff !important;
88 | }
89 | #agents-filter-group .btn {
90 |     background: #1e293b;
91 |     border: 1px solid #475569;
92 |     color: #f1f5f9;
93 |     font-weight: 600;
94 |     font-size: 0.82rem;
95 | }
96 | #agents-filter-group .btn:hover {
97 |     background: #334155;
98 |     color: #ffffff;
99 |     border-color: #64748b;
100 | }
101 | #agents-filter-group .btn.active {
102 |     background: #2563eb !important;
103 |     border-color: #3b82f6 !important;
104 |     color: #ffffff !important;
105 |     box-shadow: 0 0 8px rgba(37, 99, 235, 0.4);
106 | }
107 | #btn-refresh-agents {
108 |     background: #1e293b;
109 |     border: 1px solid #475569;
110 |     color: #f1f5f9;
111 |     font-weight: 600;
```

```

112     font-size: 0.82rem;
113 }
114 #btn-refresh-agents:hover {
115     background: #334155;
116     color: #ffffff;
117     border-color: #64748b;
118 }
119 </style>
120
121 <div class="container-fluid py-3">
122
123     <!-- Header Alert & Quick Actions -->
124     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 p-3 rounded-3 shadow-sm text-white" style="background: linear-gradient(135deg, #0f172a 0%, #1e1b4b 100%); border: 1px solid #334155;">
125         <div>
126             <h5 class="mb-1 fw-bold text-white d-flex align-items-center gap-2">
127                 <span class="fs-4">🔮 </span> Управление и создание ИИ-Агентов
128             </h5>
129             <p class="mb-0 small" style="color: #cbd5e1;">
130                 Конфигурирование специализированных агентов, назначение моделей из пула, подключение инструментов и генерация через AI.
131             </p>
132         </div>
133         <div class="d-flex align-items-center gap-2 flex-wrap">
134             <button class="btn btn-sm btn-outline-info rounded-pill px-3 shadow-sm fw-semibold" id="btn-open-ai-builder" style="border-width: 1.5px;">
135                 <i class="bi bi-magic me-1"></i> 🔮 Создать с помощью AI
136             </button>
137             <button class="btn btn-sm btn-primary rounded-pill px-3 shadow-sm fw-semibold" id="btn-open-create-agent">
138                 <i class="bi bi-plus-lg me-1"></i> Добавить вручную

```

```
139 |         </button>
140 |     </div>
141 | </div>
142 |
143 | <!-- Collapsible Architecture & SVG Scheme Block -->
144 | <div class="card mb-4 shadow-sm text-white" style="background-color: #0f172a !important; border: 1px solid #334155 !important; border-radius: 10px;">
145 |     <div class="card-body p-3">
146 |         <div class="d-flex align-items-center justify-content-between">
147 |             <div class="d-flex align-items-center gap-2">
148 |                 <i class="bi bi-diagram-3-fill text-info fs-5"></i>
149 |                 <span class="fw-bold" style="color: #f1f5f9;">Интерактивная справка: архитектура работы и создания агентов</span>
150 |             </div>
151 |             <button class="btn btn-sm btn-link text-decoration-none p-0 fw-semibold" style="color: #38bdf8;" type="button" data-bs-toggle="collapse" data-bs-target="#agentsArchCollapse" aria-expanded="false" aria-controls="agentsArchCollapse">
152 |                 <i class="bi bi-chevron-down"></i> Развернуть / Скрыть схему
153 |             </button>
154 |         </div>
155 |         <div class="collapse" id="agentsArchCollapse">
156 |             <div class="mt-3 pt-3" style="border-top: 1px solid rgba(255, 255, 255, 0.1);">
157 |                 <div class="text-center p-2 rounded mb-3" style="background: #020617; border: 1px solid rgba(255, 255, 255, 0.08);">
158 |                     
159 |                 </div>
160 |                 <div class="row g-3 small">
161 |                     <div class="col-md-3">
162 |                         <div class="p-3 rounded h-100" style="background
```

```

: #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
163         <strong class="d-block mb-1" style="color: #38bdf8; font-size: 0.88rem;">1. Пул моделей (Model Pool)</strong>
>
164         Каждый агент может использовать независимую модель: Google Gemini (с ротацией ключей), Antigravity AGY, Microsoft Foundry или Ollama.
165     </div>
166 </div>
167     <div class="col-md-3">
168         <div class="p-3 rounded h-100" style="background: #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
169             <strong class="d-block mb-1" style="color: #4ade80; font-size: 0.88rem;">2. Каталог инструментов (Tools)</strong>
170             Набор готовых модулей: интернет-поиск (MCP / Google Grounding), RAG база знаний, Python Code Execution, чтение файлов и синтез речи TTS.
171         </div>
172     </div>
173     <div class="col-md-3">
174         <div class="p-3 rounded h-100" style="background: #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
175             <strong class="d-block mb-1" style="color: #fbfb24; font-size: 0.88rem;">3. AI-Архитектор промптов</strong>
176             Модель из пула анализирует вашу цель и генерирует структурированную роль, правила ReAct и список инструментов.
177         </div>
178     </div>
179     <div class="col-md-3">
180         <div class="p-3 rounded h-100" style="background: #1e293b; border: 1px solid #334155; color: #f1f5f9; line-height: 1.55;">
181             <strong class="d-block mb-1" style="color: #f8

```

```

7171; font-size: 0.88rem;">4. Песочница (Sandbox)</strong>
182     Тестирование поведения агента, отслеживание це
почки рассуждений (Thought & Action) и замер задержек ответа д
о боевого использования.
183         </div>
184     </div>
185 </div>
186 </div>
187 </div>
188 </div>
189 </div>
190
191 <!-- Filters and Status Toolbar -->
192 <div class="d-flex justify-content-between align-items-cente
r mb-3 flex-wrap gap-2">
193     <div class="btn-group btn-group-sm" role="group" id="agent
s-filter-group">
194         <button type="button" class="btn active" data-filter="al
1">Все (<span id="count-all">0</span>)</button>
195         <button type="button" class="btn" data-filter="active">А
ктивные (<span id="count-active">0</span>)</button>
196         <button type="button" class="btn" data-filter="system">C
истемные (<span id="count-system">0</span>)</button>
197         <button type="button" class="btn" data-filter="custom">П
ользовательские (<span id="count-custom">0</span>)</button>
198     </div>
199     <div class="d-flex align-items-center gap-2">
200         <button class="btn btn-sm rounded-pill px-3" id="btn-ref
resh-agents" title="Обновить список">
201             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
202         </button>
203     </div>
204 </div>
205
206 <!-- Agents Grid -->
207 <div class="row g-3" id="agents-grid">
208     <div class="col-12 text-center py-5" style="color: #cbd5e1
; ">

```

```

209     <div class="spinner-border text-primary mb-2" role="stat
| us"></div>
210     <div class="fw-semibold">Загрузка агентов и пула моделей
| ...</div>
211     </div>
212 </div>
213
214 </div>
215
216 <!-- =====
| ===== -->
217 <!-- MODAL: AI Agent Builder (Генерация через LLM)
|         -->
218 <!-- =====
| ===== -->
219 <div class="modal fade" id="modal-ai-builder" tabindex="-1" ar
| ia-labelledby="modalAiBuilderLabel" aria-hidden="true">
220     <div class="modal-dialog modal-lg modal-dialog-centered">
221         <div class="modal-content text-white" style="background-co
| lor: #0f172a !important; border: 1px solid #334155 !important;
| border-radius: 12px;">
222             <div class="modal-header" style="background-color: #1e29
| 3b !important; border-bottom: 1px solid #334155 !important;">
223                 <h5 class="modal-title fw-bold" style="color: #38bdf8;
| " id="modalAiBuilderLabel">
224                     <i class="bi bi-magic me-2"></i>AI Agent Builder – К
| онструктор агентов
225                 </h5>
226                 <button type="button" class="btn-close btn-close-white
| " data-bs-dismiss="modal" aria-label="Close"></button>
227             </div>
228             <div class="modal-body p-4">
229                 <p class="small mb-3" style="color: #cbd5e1; line-heig
| ht: 1.5;">
230                     Опишите задачу агента своими словами. Выбранная моде
| ль из пула проанализирует цель, подберёт необходимые инструмен
| ты и сформирует профессиональный системный промпт.
231                 </p>

```

```
232 |
233 |         <div class="row g-3 mb-3">
234 |             <div class="col-md-6">
235 |                 <label class="form-label small fw-semibold" style=
"color: #f1f5f9;">Провайдер модели-архитектора:</label>
236 |                 <select class="form-select form-select-sm" style="
background-color: #020617 !important; color: #f8fafc !important; border: 1px solid #475569 !important;" id="ai-builder-provider">
237 |                     <option value="gemini">Google Gemini</option>
238 |                     <option value="gemini_cli">Google Gemini CLI</option>
239 |                     <option value="agy">Google Antigravity (AGY)</option>
240 |                     <option value="foundry">Microsoft Foundry (Local)
</option>
241 |                     <option value="ollama">Ollama (Local)</option>
242 |                 </select>
243 |             </div>
244 |             <div class="col-md-6">
245 |                 <label class="form-label small fw-semibold" style=
"color: #f1f5f9;">Модель из пула:</label>
246 |                 <select class="form-select form-select-sm" style="
background-color: #020617 !important; color: #f8fafc !important; border: 1px solid #475569 !important;" id="ai-builder-model">
247 |                     <!-- Заполняется динамически -->
248 |                 </select>
249 |             </div>
250 |         </div>
251 |
252 |         <div class="mb-3">
253 |             <label class="form-label small fw-semibold" style="color: #f1f5f9;">Опишите задачу агента:</label>
254 |             <textarea class="form-control" style="background-color: #020617 !important; color: #f8fafc !important; border: 1px solid #475569 !important;" id="ai-builder-prompt" rows="4" placeholder="Например: Создай агента для автономного поиска субт
```


итров к фильмам, проверки их таймингов и сопоставления с медиа
текой..."></textarea>

255 </div>

256

257 <div class="d-flex justify-content-end mb-3">

258 <button class="btn btn-info rounded-pill px-4 fw-bol
d" id="btn-run-ai-generate" style="background-color: #0284c7 !
important; border-color: #0284c7 !important; color: #ffffff !i
mportant;">

259 <i class="bi bi-stars me-1"></i> Сгенерировать спе
цификацию

260 </button>

261 </div>

262

263 <!-- Result Preview Section -->

264 <div id="ai-builder-result" class="d-none pt-3" style=
"border-top: 1px solid #334155;">

265 <h6 class="fw-bold mb-2" style="color: #4ade80;"><i
class="bi bi-check2-circle me-1"></i> Спецификация сформирован
а:</h6>

266 <div class="p-3 rounded mb-3 small font-monospace" s
tyle="background: #020617; border: 1px solid #334155; color: #
f8fafc;" id="ai-builder-preview-box">

267 <!-- Превью JSON / данных -->

268 </div>

269 <div class="d-flex justify-content-end gap-2">

270 <button class="btn btn-outline-secondary btn-sm ro
unded-pill px-3" data-bs-dismiss="modal" style="color: #cbd5e1
; border-color: #475569;">Отмена</button>

271 <button class="btn btn-success btn-sm rounded-pill
px-3 fw-semibold" id="btn-apply-ai-generated">

272 <i class="bi bi-pencil-square me-1"></i> Открыть
в редакторе агента

273 </button>

274 </div>

275 </div>

276 </div>

277 </div>

```

278 |     </div>
279 | </div>
280 |
281 | <!-- =====
| ===== -->
282 | <!-- MODAL: Agent Editor (Создание и Редактирование)
|         -->
283 | <!-- =====
| ===== -->
284 | <div class="modal fade" id="modal-agent-editor" tabindex="-1"
| aria-labelledby="modalAgentEditorLabel" aria-hidden="true">
285 |     <div class="modal-dialog modal-xl modal-dialog-centered moda
| l-dialog-scrollable">
286 |         <div class="modal-content text-white" style="background-co
| lor: #0f172a !important; border: 1px solid #334155 !important;
| border-radius: 12px;">
287 |             <div class="modal-header" style="background-color: #1e29
| 3b !important; border-bottom: 1px solid #334155 !important;">
288 |                 <h5 class="modal-title fw-bold text-white" id="modalAg
| entEditorLabel">
289 |                     <i class="bi bi-robot me-2" style="color: #38bdf8;">
| </i>Настройка агента
290 |                 </h5>
291 |                 <button type="button" class="btn-close btn-close-white
| " data-bs-dismiss="modal" aria-label="Close"></button>
292 |             </div>
293 |             <div class="modal-body p-4">
294 |                 <form id="agent-form">
295 |                     <input type="hidden" id="agent-edit-mode" value="cre
| ate">
296 |
297 |                     <div class="row g-3 mb-3">
298 |                         <div class="col-md-4">
299 |                             <label class="form-label small fw-semibold" styl
| e="color: #f1f5f9;">Уникальный ID агента:</label>
300 |                             <input type="text" class="form-control form-cont
| rol-sm font-monospace" style="background-color: #020617 !impor
| tant; color: #f8fafc !important; border: 1px solid #475569 !im

```

```

    portant;" id="agent-id" placeholder="my_custom_agent" required
    >
301         </div>
302         <div class="col-md-5">
303             <label class="form-label small fw-semibold" styl
e="color: #f1f5f9;">Отображаемое имя:</label>
304             <input type="text" class="form-control form-cont
rol-sm" style="background-color: #020617 !important; color: #f
8fafc !important; border: 1px solid #475569 !important;" id="a
gent-name" placeholder="Название агента" required>
305         </div>
306         <div class="col-md-3">
307             <label class="form-label small fw-semibold" styl
e="color: #f1f5f9;">Статус активности:</label>
308             <div class="form-check form-switch mt-1">
309                 <input class="form-check-input" type="checkbox"
" id="agent-enabled" checked>
310                 <label class="form-check-label small fw-medium
" style="color: #e2e8f0;" for="agent-enabled">Агент активен</l
abel>
311             </div>
312         </div>
313     </div>
314
315     <div class="mb-3">
316         <label class="form-label small fw-semibold" style=
"color: #f1f5f9;">Краткое описание роли:</label>
317         <input type="text" class="form-control form-contro
l-sm" style="background-color: #020617 !important; color: #f8f
af8 !important; border: 1px solid #475569 !important;" id="age
nt-desc" placeholder="Какую задачу решает данный агент">
318     </div>
319
320     <!-- Model & Provider Selection from Pool -->
321     <div class="card mb-3" style="background: #0b0f19 !i
mportant; border: 1px solid #334155 !important; border-radius:
8px;">
322         <div class="card-header py-2" style="background: #

```

```

1e293b !important; border-bottom: 1px solid #334155 !important
; ">
323         <span class="fw-bold small" style="color: #38bdf
8; "><i class="bi bi-cpu me-1"></i> Привязка к модели из пула</
span>
324     </div>
325     <div class="card-body p-3">
326         <div class="row g-3">
327             <div class="col-md-6">
328                 <label class="form-label small fw-semibold"
style="color: #f1f5f9; ">Провайдер ИИ:</label>
329                 <select class="form-select form-select-sm" s
tyle="background-color: #020617 !important; color: #f8fafc !im
portant; border: 1px solid #475569 !important;" id="agent-prov
ider">
330                     <option value="gemini">Google Gemini</opti
on>
331                     <option value="gemini_cli">Google Gemini C
LI</option>
332                     <option value="agy">Google Antigravity (AG
Y)</option>
333                     <option value="foundry">Microsoft Foundry<
/option>
334                     <option value="ollama">Ollama (Local)</opt
ion>
335                 </select>
336             </div>
337             <div class="col-md-6">
338                 <label class="form-label small fw-semibold"
style="color: #f1f5f9; ">Модель:</label>
339                 <select class="form-select form-select-sm" s
tyle="background-color: #020617 !important; color: #f8fafc !im
portant; border: 1px solid #475569 !important;" id="agent-mode
l">
340                     <!-- Заполняется динамически -->
341                 </select>
342             </div>
343         </div>

```

```
344 |
345 |         <div class="row g-3 mt-1">
346 |             <div class="col-md-4">
347 |                 <label class="form-label small fw-semibold d
348 | -flex justify-content-between" style="color: #f1f5f9;">
349 |                     <span>Температура:</span> <span id="agent-
350 | temp-val" class="fw-bold" style="color: #38bdf8;">0.3</span>
351 |                 </label>
352 |                 <input type="range" class="form-range" id="a
353 | gent-temperature" min="0" max="1" step="0.05" value="0.3">
354 |             </div>
355 |             <div class="col-md-4">
356 |                 <label class="form-label small fw-semibold"
357 | style="color: #f1f5f9;">Макс. шагов ReAct (Max Steps):</label>
358 |                 <input type="number" class="form-control for
359 | m-control-sm" style="background-color: #020617 !important; col
360 | or: #f8fafc !important; border: 1px solid #475569 !important;"
361 | id="agent-max-steps" min="1" max="50" value="15">
362 |             </div>
363 |             <div class="col-md-4">
364 |                 <label class="form-label small fw-semibold"
365 | style="color: #f1f5f9;">Таймаут (сек):</label>
366 |                 <input type="number" class="form-control for
367 | m-control-sm" style="background-color: #020617 !important; col
368 | or: #f8fafc !important; border: 1px solid #475569 !important;"
369 | id="agent-timeout" min="5" max="300" value="60">
370 |             </div>
371 |         </div>
372 |     </div>
373 |
374 |     <!-- Tools Selection Matrix -->
375 |     <div class="card mb-3" style="background: #0b0f19 !i
376 | mportant; border: 1px solid #334155 !important; border-radius:
377 | 8px;">
378 |         <div class="card-header py-2 d-flex justify-conten
379 | t-between align-items-center" style="background: #1e293b !impo
380 | rtant; border-bottom: 1px solid #334155 !important;">
```

```

367 |         <span class="fw-bold small" style="color: #4ade8
    | 0;"><i class="bi bi-tools me-1"></i> Подключенные инструменты
    | (Capabilities)</span>
368 |         <span class="badge" style="background: #334155;
    | color: #f8fafc; font-weight: 600;" id="agent-tools-count">0 вы
    | брано</span>
369 |     </div>
370 |     <div class="card-body p-3">
371 |         <div class="row g-2" id="agent-tools-matrix">
372 |             <!-- Чекбоксы инструментов загружаются динамич
    | ески -->
373 |         </div>
374 |     </div>
375 | </div>
376 |
377 |     <!-- System Instruction Editor -->
378 |     <div class="mb-3">
379 |         <div class="d-flex justify-content-between align-i
    | tems-center mb-1">
380 |             <label class="form-label small fw-semibold mb-0"
    | style="color: #f1f5f9;">Системная инструкция (System Prompt):
    | </label>
381 |             <div class="btn-group btn-group-sm">
382 |                 <button type="button" class="btn btn-outline-s
    | econdary btn-sm py-0" style="color: #cbd5e1; border-color: #47
    | 5569;" id="btn-insert-react-template">Шаблон ReAct</button>
383 |                 <button type="button" class="btn btn-outline-s
    | econdary btn-sm py-0" style="color: #cbd5e1; border-color: #47
    | 5569;" id="btn-insert-json-template">Шаблон JSON-ответа</butto
    | n>
384 |             </div>
385 |         </div>
386 |         <textarea class="form-control font-monospace" styl
    | e="background-color: #020617 !important; color: #f8fafc !impor
    | tant; border: 1px solid #475569 !important; font-size: 0.85rem
    | ; line-height: 1.45;" id="agent-system-prompt" rows="7" placeh
    | older="Опишите поведение агента, правила вызова инструментов и
    | формат ответа..." spellcheck="false" required></textarea>

```

```

387         </div>
388
389     </form>
390 </div>
391     <div class="modal-footer" style="background-color: #1e29
3b !important; border-top: 1px solid #334155 !important;">
392         <button type="button" class="btn btn-outline-secondary
rounded-pill px-3" data-bs-dismiss="modal" style="color: #cbd
5e1; border-color: #475569;">Отмена</button>
393         <button type="button" class="btn btn-primary rounded-p
ill px-4 fw-bold" id="btn-save-agent">
394         <i class="bi bi-check-circle me-1"></i> Сохранить аг
ента
395     </button>
396 </div>
397 </div>
398 </div>
399 </div>
400
401 <!-- =====
===== -->
402 <!-- MODAL: Agent Sandbox (Песочница тестирования)
-->
403 <!-- =====
===== -->
404 <div class="modal fade" id="modal-agent-sandbox" tabindex="-1"
aria-labelledby="modalAgentSandboxLabel" aria-hidden="true">
405     <div class="modal-dialog modal-lg modal-dialog-centered moda
l-dialog-scrollable">
406         <div class="modal-content text-white" style="background-co
lor: #0f172a !important; border: 1px solid #334155 !important;
border-radius: 12px;">
407             <div class="modal-header" style="background-color: #1e29
3b !important; border-bottom: 1px solid #334155 !important;">
408                 <h5 class="modal-title fw-bold" style="color: #fbbf24;
" id="modalAgentSandboxLabel">
409                 <i class="bi bi-play-circle me-2"></i> Песочница т
естирования агента

```

```
410         </h5>
411         <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
412     </div>
413     <div class="modal-body p-4">
414
415         <!-- Active Agent Summary in Sandbox -->
416         <div class="d-flex align-items-center justify-content-
between p-2 px-3 rounded mb-3" style="background: #020617; bor
der: 1px solid #334155;">
417             <div>
418                 <strong id="sandbox-agent-name" class="text-white
fs-6">--</strong>
419                 <span class="badge ms-2" style="background-color:
#2563eb; color: #ffffff; font-weight: 600;" id="sandbox-agent-
model">--</span>
420             </div>
421             <span class="small fw-medium" style="color: #cbd5e1;
" id="sandbox-agent-tools-summary">Инструментов: 0</span>
422         </div>
423
424         <!-- Chat / Trace Container -->
425         <div class="rounded p-3 mb-3" style="background: #0206
17; border: 1px solid #334155; min-height: 240px; max-height:
380px; overflow-y: auto;" id="sandbox-chat-window">
426             <div class="text-center small py-4" style="color: #9
4a3b8;" id="sandbox-placeholder">
427                 <i class="bi bi-chat-dots fs-3 d-block mb-1"></i>
428                 Введите тестовый запрос ниже для проверки логики а
гента и цепочки вызовов инструментов.
429             </div>
430             <div id="sandbox-messages-list"></div>
431         </div>
432
433         <!-- Input Box -->
434         <div class="input-group">
435             <input type="text" class="form-control" style="backg
round-color: #020617 !important; color: #f8fafc !important; bo
```



```
border: 1px solid #475569 !important;" id="sandbox-input-msg" placeholder="Например: Найди фильм Интерстеллар и проверь торренты...">
```

```
436         <button class="btn btn-warning fw-bold px-4" id="btn-sandbox-send" style="background-color: #f59e0b; border-color: #f59e0b; color: #0f172a;">
```

```
437             <i class="bi bi-send-fill me-1"></i> Тест
```

```
438         </button>
```

```
439     </div>
```

```
440
```

```
441 </div>
```

```
442     <div class="modal-footer" style="background-color: #1e293b !important; border-top: 1px solid #334155 !important;">
```

```
443         <button type="button" class="btn btn-outline-secondary rounded-pill px-3" data-bs-dismiss="modal" style="color: #cbd5e1; border-color: #475569;">Заккрыть песочницу</button>
```

```
444     </div>
```

```
445 </div>
```

```
446 </div>
```

```
447 </div>
```

```
448
```

main.js

```
1 /**
2  * =====
3  * Process Name: AI Agents Management and Builder Interface
4  * =====
5  * Description:
6  *   Client-side controller for managing AI agent registry, model pool binding,
7  *   automated specification generation, and execution sandbox testing.
8  *
9  * File: main.js
10 * Project: Mediteka
11 * Module: AgentsTab
12 * Author: hypo69
13 * Copyright: © 2026 hypo69
14 * =====
15 */
16
17 'use strict';
18
19 /**
20 * Global reactive state for the AI Agents tab.
21 */
22 const _state = {
23   agents: [],
24   tools: [],
25   providers: {},
26   currentFilter: 'all',
27   activeSandboxAgent: null,
28   lastGeneratedSpec: null,
29   isEventsBound: false
30 };
31
```

```

32 | /**
33 |  * Initializes the AI Agents management tab.
34 |  * Called on tab switch from the main admin interface.
35 |  *
36 |  * @returns {Promise<void>}
37 |  */
38 | async function initAgentsTab() {
39 |     _bindStaticEvents();
40 |     await _loadAllData();
41 | }
42 |
43 | window.initAgentsTab = initAgentsTab;
44 |
45 | /**
46 |  * Executes an HTTP JSON request using window.api or standard
47 |  * fetch.
48 |  *
49 |  * @param {string} endpoint - Target API endpoint path.
50 |  * @param {RequestInit} [options={}] - Optional fetch configur
51 |  * ation options.
52 |  * @returns {Promise<any>} Parsed JSON response.
53 |  * @throws {Error} Thrown if response status is not OK.
54 |  */
55 | async function _fetchJson(endpoint, options = {}) {
56 |     if (window.api && typeof window.api.fetch === 'function') {
57 |         return await window.api.fetch(endpoint, options);
58 |     }
59 |     const response = await fetch(endpoint, options);
60 |     if (!response.ok) {
61 |         throw new Error(`HTTP ${response.status}: ${response.statu
62 | sText}`);
63 |     }
64 |     return await response.json();
65 | }
66 |
67 | /**
68 |  * Loads agents, capabilities catalog, and model providers fro
69 |  * m the backend API.

```

```

66 | *
67 | * @returns {Promise<void>}
68 | */
69 | async function _loadAllData() {
70 |     try {
71 |         const [agentsRes, toolsRes, providersRes] = await Promise.
all([
72 |             _fetchJson('/api/agents'),
73 |             _fetchJson('/api/agents/tools'),
74 |             _fetchJson('/api/agents/providers')
75 |         ]);
76 |
77 |         _state.agents = Array.isArray(agentsRes) ? agentsRes : [];
78 |         _state.tools = Array.isArray(toolsRes) ? toolsRes : [];
79 |         _state.providers = providersRes ?? {};
80 |
81 |         _renderCounters();
82 |         _renderGrid();
83 |         _populateToolsMatrix();
84 |         _updateModelDropdown('agent-provider', 'agent-model');
85 |         _updateModelDropdown('ai-builder-provider', 'ai-builder-mo
del');
86 |     } catch (err) {
87 |         console.error('[AgentsTab] Failed to load data:', err);
88 |         _showToast(`Failed to load data: ${err.message}`, 'danger'
);
89 |         const grid = document.getElementById('agents-grid');
90 |         if (grid) {
91 |             grid.innerHTML = `
92 |                 <div class="col-12 text-center py-5 text-danger">
93 |                     <i class="bi bi-exclamation-triangle fs-1 d-block mb
-2"></i>
94 |                     <h5>Не удалось загрузить данные агентов</h5>
95 |                     <p class="small text-secondary">${err.message}</p>
96 |                     <button class="btn btn-sm btn-outline-primary rounde
d-pill px-3" onclick="window.initAgentsTab()">Повторить попытку</button>
97 |                 </div>

```

```

98     `;
99     }
100 }
101 }
102
103 /**
104  * Binds static DOM event listeners for buttons, filters, and
105  * modal templates.
106  * @returns {void}
107  */
108 function _bindStaticEvents() {
109     if (_state.isEventsBound) return;
110     _state.isEventsBound = true;
111
112     // Refresh agents list
113     const refreshBtn = document.getElementById('btn-refresh-agents');
114     if (refreshBtn) {
115         refreshBtn.onclick = async () => {
116             refreshBtn.disabled = true;
117             await _loadAllData();
118             refreshBtn.disabled = false;
119             _showToast('Список агентов обновлен', 'success');
120         };
121     }
122
123     // Filter toolbar buttons
124     const filterGroup = document.getElementById('agents-filter-group');
125     if (filterGroup) {
126         filterGroup.querySelectorAll('button').forEach((btn) => {
127             btn.onclick = () => {
128                 filterGroup.querySelectorAll('button').forEach((b) =>
129                     b.classList.remove('active'));
130                 btn.classList.add('active');
131                 _state.currentFilter = btn.getAttribute('data-filter')
132                     ?? 'all';

```

```
131 |         _renderGrid();
132 |     };
133 | });
134 | }
135 |
136 | // Modal trigger buttons
137 | document.getElementById('btn-open-create-agent')?.addEventListener(
138 |     'click', () => _openEditor(null));
139 | document.getElementById('btn-open-ai-builder')?.addEventListener(
140 |     'click', _openAiBuilder);
141 |
142 | // Provider change handlers
143 | document.getElementById('agent-provider')?.addEventListener(
144 |     'change', () => {
145 |         _updateModelDropdown('agent-provider', 'agent-model');
146 |     });
147 | document.getElementById('ai-builder-provider')?.addEventListener(
148 |     'change', () => {
149 |         _updateModelDropdown('ai-builder-provider', 'ai-builder-model');
150 |     });
151 |
152 | // Temperature slider synchronization
153 | const tempSlider = document.getElementById('agent-temperature');
154 | const tempVal = document.getElementById('agent-temp-val');
155 | if (tempSlider && tempVal) {
156 |     tempSlider.oninput = () => {
157 |         tempVal.textContent = tempSlider.value;
158 |     };
159 | }
160 |
161 | // Action buttons
162 | document.getElementById('btn-save-agent')?.addEventListener(
163 |     'click', _handleSaveAgent);
164 | document.getElementById('btn-run-ai-generate')?.addEventListener(
165 |     'click', _handleRunAiGenerate);
166 |
```

```
161 | // Apply AI Generated specification to editor
162 | document.getElementById('btn-apply-ai-generated')?.addEventListener('click', () => {
163 |     if (!_state.lastGeneratedSpec) return;
164 |     const builderModalEl = document.getElementById('modal-ai-builder');
165 |     bootstrap.Modal.getInstance(builderModalEl)?.hide();
166 |     _openEditor(null, _state.lastGeneratedSpec);
167 | });
168 |
169 | // System Prompt templates
170 | document.getElementById('btn-insert-react-template')?.addEventListener('click', () => {
171 |     const promptInput = document.getElementById('agent-system-prompt');
172 |     if (promptInput) {
173 |         promptInput.value =
174 |             'Ты автономный ReAct-агент Mediteka.\n\n' +
175 |             'ПРИНЦИП РАБОТЫ (Thought -> Action -> Observation):\n'
176 |             +
177 |             '1. Тщательно анализируй вопрос пользователя (Thought).\n' +
178 |             '2. Выбирай необходимый инструмент из доступных (Action) с корректными параметрами.\n' +
179 |             '3. Анализируй результат выполнения инструмента (Observation).\n' +
180 |             '4. Сформируй итоговый структурированный ответ пользователю.\n\n' +
181 |             'ФОРМАТ ОТВЕТА:\n' +
182 |             '- Используй Markdown с красивым форматированием (списки, жирный шрифт, эмодзи).\n' +
183 |             '- Не выводите технические ошибки напрямую, объясняй суть решения.';
184 |     };
185 | });
186 | document.getElementById('btn-insert-json-template')?.addEventListener('click', () => {
```

```
187     const promptInput = document.getElementById('agent-system-
prompt');
188     if (promptInput) {
189         promptInput.value =
190             'Ты специализированный аналитический агент Mediteka.\n
\n' +
191             'Твоя задача: структурировать полученные данные и всег
да отвечать строго валидным JSON-объектом без лишнего обрамлен
ия и markdown-блоков.\n\n' +
192             'Пример структуры:\n{\n  "status": "success",\n  "item
s": [],\n  "summary": "краткое резюме"\n}';
193     }
194 });
195
196 // Sandbox inputs
197 const sandboxSendBtn = document.getElementById('btn-sandbox-
send');
198 const sandboxInput = document.getElementById('sandbox-input-
msg');
199 if (sandboxSendBtn && sandboxInput) {
200     sandboxSendBtn.onclick = _handleRunSandboxTest;
201     sandboxInput.onkeypress = (e) => {
202         if (e.key === 'Enter') _handleRunSandboxTest();
203     };
204 }
205 }
206
207 /**
208  * Updates filter counter badges on top toolbar buttons.
209  *
210  * @returns {void}
211  */
212 function _renderCounters() {
213     const allCount = _state.agents.length;
214     const activeCount = _state.agents.filter((a) => a.enabled).l
ength;
215     const systemCount = _state.agents.filter((a) => a.is_system)
.length;
```



```

216 |   const customCount = _state.agents.filter((a) => !a.is_system
    | ).length;
217 |
218 |   document.getElementById('count-all')?.replaceChildren(docume
    | nt.createTextNode(String(allCount)));
219 |   document.getElementById('count-active')?.replaceChildren(doc
    | ument.createTextNode(String(activeCount)));
220 |   document.getElementById('count-system')?.replaceChildren(doc
    | ument.createTextNode(String(systemCount)));
221 |   document.getElementById('count-custom')?.replaceChildren(doc
    | ument.createTextNode(String(customCount)));
222 | }
223 |
224 | /**
225 |  * Renders the responsive agent cards grid based on current fi
    | lter state.
226 |  *
227 |  * @returns {void}
228 |  */
229 | function _renderGrid() {
230 |   const grid = document.getElementById('agents-grid');
231 |   if (!grid) return;
232 |
233 |   const filter = _state.currentFilter;
234 |   let items = _state.agents;
235 |
236 |   if (filter === 'active') items = items.filter((a) => a.enabl
    | ed);
237 |   else if (filter === 'system') items = items.filter((a) => a.
    | is_system);
238 |   else if (filter === 'custom') items = items.filter((a) => !a
    | .is_system);
239 |
240 |   if (items.length === 0) {
241 |     grid.innerHTML = `
242 |       <div class="col-12 text-center py-5" style="color: #cbd5
    | e1;">
243 |         <i class="bi bi-robot fs-1 d-block mb-2" style="color:

```

```

#38bdf8;"></i>
244     <h5 class="text-white">Агенты не найдены</h5>
245     <p class="small" style="color: #94a3b8;">Создайте ново
го агента с помощью AI или вручную</p>
246   </div>
247   `;
248   return;
249 }
250
251 grid.innerHTML = items
252   .map((agent) => {
253     const isSystem = Boolean(agent.is_system);
254     const isEnabled = Boolean(agent.enabled);
255     const providerName = _getProviderDisplayName(agent.provi
der);
256
257     const toolsBadges = (agent.tools ?? [])
258       .map((toolId) => {
259         const tool = _state.tools.find((t) => t.id === toolI
d);
260         const icon = tool?.icon ?? '⬜';
261         const name = tool?.name ?? toolId;
262         return `<span class="badge tool-badge me-1 mb-1 font
-monospace">${icon} <span>${name}</span></span>`;
263       })
264       .join('');
265
266     return `
267       <div class="col-xl-4 col-lg-6">
268         <div class="card h-100 agent-card shadow-sm" style="
${!isEnabled ? 'opacity: 0.65;' : ''}">
269           <div class="card-header d-flex justify-content-bet
ween align-items-center py-2 px-3">
270             <div class="d-flex align-items-center gap-2">
271               <span class="fs-5">${isSystem ? '⬜' : '⬜'}</s
pan>
272               <strong class="text-white fs-6 fw-bold">${agen
t.name}</strong>

```

```
273 |         ${
274 |             isSystem
275 |             ? '<span class="badge fw-bold" style="font
| -size: 0.7rem; background-color: #0284c7 !important; color: #f
| fffff !important;">SYSTEM</span>'
276 |             : '<span class="badge fw-bold" style="font
| -size: 0.7rem; background-color: #d97706 !important; color: #f
| fffff !important;">CUSTOM</span>'
277 |         }
278 |     </div>
279 |     <div class="form-check form-switch m-0" title="В
| ключить / Выключить">
280 |         <input class="form-check-input agent-toggle-sw
| itch" type="checkbox" data-agent-id="${agent.id}" ${isEnabled
| ? 'checked' : ''}>
281 |     </div>
282 | </div>
283 | <div class="card-body p-3 d-flex flex-column">
284 |     <p class="small mb-3 flex-grow-1 agent-desc" sty
| le="min-height: 40px; color: #cbd5e1; line-height: 1.45;">${ag
| ent.description || 'Описание отсутствует'}</p>
285 |
286 |     <div class="agent-spec-box mb-3 small">
287 |         <div class="d-flex justify-content-between ali
| gn-items-center mb-1 pb-1" style="border-bottom: 1px solid rgb
| a(255, 255, 255, 0.08);">
288 |             <span class="spec-label">Провайдер &amp; Мод
| ель:</span>
289 |             <span class="badge spec-value-badge text-tru
| ncate" style="max-width: 190px;">${providerName}: ${agent.mode
| l}</span>
290 |         </div>
291 |         <div class="d-flex justify-content-between ali
| gn-items-center">
292 |             <span class="spec-label">Температура / Шаги:
| </span>
293 |             <span class="spec-tech-val font-monospace">T
| : ${agent.temperature} <span style="color: #64748b;">|</span>
```

```

Max: ${agent.max_steps}</span>
294     </div>
295     </div>
296
297     <div class="mb-3">
298         <div class="small fw-semibold mb-2" style="col
or: #f1f5f9;">Инструменты (${(agent.tools ?? []).length}):</di
v>
299         <div class="d-flex flex-wrap">${toolsBadges ||
'<span class="small" style="color: #94a3b8;">Без внешних инст
рументов</span>'}</div>
300     </div>
301
302     <div class="mt-auto d-flex justify-content-betwe
en gap-2 pt-2" style="border-top: 1px solid #334155;">
303         <button class="btn btn-sm rounded-pill px-3 bt
n-test-sandbox" data-agent-id="${agent.id}">
304             <i class="bi bi-play-fill"></i> Тест в Sandb
ox
305         </button>
306         <div class="d-flex gap-1">
307             <button class="btn btn-sm rounded-pill px-2
btn-edit-agent" data-agent-id="${agent.id}" title="Редактирова
ть">
308                 <i class="bi bi-pencil-fill"></i>
309             </button>
310             ${
311                 !isSystem
312                 ? `
313                 <button class="btn btn-sm rounded-pill px-
2 btn-delete-agent" data-agent-id="${agent.id}" title="Удалить
">
314                     <i class="bi bi-trash-fill"></i>
315                 </button>
316                 `
317                 : `
318                 `
319             </div>

```

```

320         </div>
321     </div>
322 </div>
323 </div>
324 `;
325 })
326 .join('');
327
328 // Attach card event listeners
329 grid.querySelectorAll('.agent-toggle-switch').forEach((sw) =
330 > {
331     sw.onchange = async () => {
332         const agentId = sw.getAttribute('data-agent-id');
333         if (agentId) await _toggleAgentState(agentId, sw.checked
334 );
335     };
336 });
337
338 grid.querySelectorAll('.btn-test-sandbox').forEach((btn) =>
339 {
340     btn.onclick = () => {
341         const agentId = btn.getAttribute('data-agent-id');
342         if (agentId) _openSandbox(agentId);
343     };
344 });
345
346 grid.querySelectorAll('.btn-edit-agent').forEach((btn) => {
347     btn.onclick = () => {
348         const agentId = btn.getAttribute('data-agent-id');
349         if (agentId) _openEditor(agentId);
350     };
351 });
352
353 grid.querySelectorAll('.btn-delete-agent').forEach((btn) =>
354 {
355     btn.onclick = () => {
356         const agentId = btn.getAttribute('data-agent-id');
357         if (agentId) _deleteAgent(agentId);

```

```

354     };
355   });
356 }
357
358 /**
359  * Toggles an agent's enabled state and persists it to the server.
360  *
361  * @param {string} agentId - Target agent unique identifier.
362  * @param {boolean} enabled - New enabled state.
363  * @returns {Promise<void>}
364  */
365 async function _toggleAgentState(agentId, enabled) {
366   const agent = _state.agents.find((a) => a.id === agentId);
367   if (!agent) return;
368
369   agent.enabled = enabled;
370   try {
371     await _fetchJson(`/api/agents/${agentId}`, {
372       method: 'PUT',
373       headers: { 'Content-Type': 'application/json' },
374       body: JSON.stringify(agent)
375     });
376     _renderCounters();
377     _renderGrid();
378     _showToast(`Агент "${agent.name}" ${enabled ? 'включен' :
379 'отключен'}`, 'success');
380   } catch (err) {
381     console.error('[AgentsTab] Error toggling agent:', err);
382     _showToast(`Ошибка: ${err.message}`, 'danger');
383     await _loadAllData();
384   }
385 }
386
387 /**
388  * Populates capability checkboxes inside the editor modal.
389  *
390  * @returns {void}

```

```

390 | */
391 | function _populateToolsMatrix() {
392 |     const container = document.getElementById('agent-tools-matri
    x');
393 |     if (!container) return;
394 |
395 |     container.innerHTML = _state.tools
396 |         .map(
397 |             (tool) => `
398 |                 <div class="col-md-6">
399 |                     <div class="p-3 rounded h-100" style="background: #1e2
    93b; border: 1px solid #334155;">
400 |                         <div class="form-check form-switch m-0">
401 |                             <input class="form-check-input tool-checkbox" type
    ="checkbox" value="${tool.id}" id="tool-chk-${tool.id}">
402 |                             <label class="form-check-label small text-white fw
    -bold" for="tool-chk-${tool.id}">
403 |                                 ${tool.icon} ${tool.name}
404 |                             </label>
405 |                         </div>
406 |                         <div class="small mt-1" style="color: #cbd5e1; font-
    size: 0.8rem; line-height: 1.4;">${tool.description}</div>
407 |                     </div>
408 |                 </div>
409 |             `
410 |         )
411 |         .join('');
412 |
413 |     container.querySelectorAll('.tool-checkbox').forEach((chk) =
    > {
414 |         chk.onchange = _updateSelectedToolsCounter;
415 |     });
416 | }
417 |
418 | /**
419 |  * Updates selected capabilities counter badge in the editor m
    odal.
420 |  *

```

```

421 | * @returns {void}
422 | */
423 | function _updateSelectedToolsCounter() {
424 |     const count = document.querySelectorAll('#agent-tools-matrix
    .tool-checkbox:checked').length;
425 |     const badge = document.getElementById('agent-tools-count');
426 |     if (badge) badge.textContent = `${count} выбрано`;
427 | }
428 |
429 | /**
430 |  * Updates model selection dropdown options when provider sele
    ction changes.
431 |  *
432 |  * @param {string} providerSelectId - ID of provider select el
    ement.
433 |  * @param {string} modelSelectId - ID of model select element.
434 |  * @param {string} [selectedModel=''] - Model ID to select by
    default.
435 |  * @returns {void}
436 |  */
437 | function _updateModelDropdown(providerSelectId, modelSelectId,
    selectedModel = '') {
438 |     const pSel = document.getElementById(providerSelectId);
439 |     const mSel = document.getElementById(modelSelectId);
440 |     if (!pSel || !mSel) return;
441 |
442 |     const prov = _state.providers[pSel.value];
443 |     const models = prov?.models ?? [];
444 |
445 |     mSel.innerHTML = models.map((m) => `<option value="${m.id}">
    ${m.name || m.id}</option>`).join('');
446 |
447 |     if (selectedModel && models.some((m) => m.id === selectedMod
    el)) {
448 |         mSel.value = selectedModel;
449 |     } else if (prov?.default_model) {
450 |         mSel.value = prov.default_model;
451 |     }

```



```

452 }
453
454 /**
455  * Opens agent editor modal for creating or updating an agent.
456  *
457  * @param {string|null} [agentId=null] - Agent ID to edit, or
  null for create.
458  * @param {object|null} [prefillData=null] - Pre-filled data f
  rom AI builder.
459  * @returns {void}
460  */
461 function _openEditor(agentId = null, prefillData = null) {
462   const modalEl = document.getElementById('modal-agent-editor'
  );
463   if (!modalEl) return;
464
465   const modeInput = document.getElementById('agent-edit-mode')
  ;
466   const idInput = document.getElementById('agent-id');
467   const nameInput = document.getElementById('agent-name');
468   const descInput = document.getElementById('agent-desc');
469   const enabledChk = document.getElementById('agent-enabled');
470   const providerSel = document.getElementById('agent-provider'
  );
471   const tempSlider = document.getElementById('agent-temperatur
  e');
472   const tempVal = document.getElementById('agent-temp-val');
473   const maxStepsInput = document.getElementById('agent-max-ste
  ps');
474   const timeoutInput = document.getElementById('agent-timeout'
  );
475   const promptTa = document.getElementById('agent-system-promp
  t');
476
477   // Reset tools checkboxes
478   document.querySelectorAll('#agent-tools-matrix .tool-checkbo
  x').forEach((chk) => {
479     chk.checked = false;

```

```
480 | });
481 |
482 | if (agentId) {
483 |     const agent = _state.agents.find((a) => a.id === agentId);
484 |     if (!agent) return;
485 |
486 |     if (modeInput) modeInput.value = 'edit';
487 |     if (idInput) {
488 |         idInput.value = agent.id;
489 |         idInput.disabled = true;
490 |     }
491 |     if (nameInput) nameInput.value = agent.name;
492 |     if (descInput) descInput.value = agent.description ?? '';
493 |     if (enabledChk) enabledChk.checked = Boolean(agent.enabled
    | );
494 |     if (providerSel) providerSel.value = agent.provider ?? 'ge
    | mini';
495 |
496 |     _updateModelDropdown('agent-provider', 'agent-model', agen
    | t.model);
497 |
498 |     if (tempSlider) {
499 |         tempSlider.value = agent.temperature ?? 0.3;
500 |         if (tempVal) tempVal.textContent = tempSlider.value;
501 |     }
502 |     if (maxStepsInput) maxStepsInput.value = agent.max_steps ?
    | ? 15;
503 |     if (timeoutInput) timeoutInput.value = agent.timeout_secon
    | ds ?? 60;
504 |     if (promptTa) promptTa.value = agent.system_prompt ?? '';
505 |
506 |     (agent.tools ?? []).forEach((tId) => {
507 |         const chk = document.getElementById(`tool-chk-${tId}`);
508 |         if (chk) chk.checked = true;
509 |     });
510 | } else if (prefillData) {
511 |     if (modeInput) modeInput.value = 'create';
512 |     if (idInput) {
```

```
513 |         idInput.value = (prefillData.name || 'agent').toLowerCase().replace(/^[^a-z0-9_]/g, '_').slice(0, 25);
514 |         idInput.disabled = false;
515 |     }
516 |     if (nameInput) nameInput.value = prefillData.name ?? 'Новый ИИ-Агент';
517 |     if (descInput) descInput.value = prefillData.description ?? '';
518 |     if (enabledChk) enabledChk.checked = true;
519 |     if (providerSel) providerSel.value = prefillData.provider ?? 'gemini';
520 |
521 |     _updateModelDropdown('agent-provider', 'agent-model', prefillData.model);
522 |
523 |     if (tempSlider) {
524 |         tempSlider.value = prefillData.temperature ?? 0.3;
525 |         if (tempVal) tempVal.textContent = tempSlider.value;
526 |     }
527 |     if (maxStepsInput) maxStepsInput.value = prefillData.max_steps ?? 15;
528 |     if (timeoutInput) timeoutInput.value = 60;
529 |     if (promptTa) promptTa.value = prefillData.system_prompt ?? '';
530 |
531 |     (prefillData.recommended_tools ?? []).forEach((tId) => {
532 |         const chk = document.getElementById(`tool-chk-${tId}`);
533 |         if (chk) chk.checked = true;
534 |     });
535 | } else {
536 |     if (modeInput) modeInput.value = 'create';
537 |     if (idInput) {
538 |         idInput.value = '';
539 |         idInput.disabled = false;
540 |     }
541 |     if (nameInput) nameInput.value = '';
542 |     if (descInput) descInput.value = '';
543 |     if (enabledChk) enabledChk.checked = true;
```

```

544     if (providerSel) providerSel.value = 'gemini';
545
546     _updateModelDropdown('agent-provider', 'agent-model');
547
548     if (tempSlider) {
549         tempSlider.value = 0.3;
550         if (tempVal) tempVal.textContent = '0.3';
551     }
552     if (maxStepsInput) maxStepsInput.value = 15;
553     if (timeoutInput) timeoutInput.value = 60;
554     if (promptTa) promptTa.value = '';
555 }
556
557 _updateSelectedToolsCounter();
558 new bootstrap.Modal(modalEl).show();
559 }
560
561 /**
562  * Saves agent configuration (POST create or PUT update).
563  *
564  * @returns {Promise<void>}
565  */
566 async function _handleSaveAgent() {
567     const mode = document.getElementById('agent-edit-mode')?.value ?? 'create';
568     const id = document.getElementById('agent-id')?.value.trim();
569     const name = document.getElementById('agent-name')?.value.trim();
570     const desc = document.getElementById('agent-desc')?.value.trim();
571     const enabled = document.getElementById('agent-enabled')?.checked ?? true;
572     const provider = document.getElementById('agent-provider')?.value ?? 'gemini';
573     const model = document.getElementById('agent-model')?.value ?? 'gemini-2.5-flash';
574     const temperature = parseFloat(document.getElementById('agen

```

```
t-temperature')?.value ?? '0.3');
575 |   const max_steps = parseInt(document.getElementById('agent-max-steps')?.value ?? '15', 10);
576 |   const timeout_seconds = parseInt(document.getElementById('agent-timeout')?.value ?? '60', 10);
577 |   const system_prompt = document.getElementById('agent-system-prompt')?.value.trim() ?? '';
578 |
579 |   if (!id || !name) {
580 |     _showToast('ID и Название агента обязательны для заполнения', 'warning');
581 |     return;
582 |   }
583 |
584 |   const selectedTools = [];
585 |   document.querySelectorAll('#agent-tools-matrix .tool-checkbox:checked').forEach((chk) => {
586 |     selectedTools.push(chk.value);
587 |   });
588 |
589 |   const payload = {
590 |     id,
591 |     name,
592 |     description: desc,
593 |     enabled,
594 |     provider,
595 |     model,
596 |     temperature,
597 |     max_steps,
598 |     timeout_seconds,
599 |     tools: selectedTools,
600 |     system_prompt
601 |   };
602 |
603 |   const saveBtn = document.getElementById('btn-save-agent');
604 |   if (saveBtn) saveBtn.disabled = true;
605 |
606 |   try {
```

```

607     if (mode === 'create') {
608         await _fetchJson('/api/agents', {
609             method: 'POST',
610             headers: { 'Content-Type': 'application/json' },
611             body: JSON.stringify(payload)
612         });
613         _showToast(`Агент "${name}" успешно создан!`, 'success')
        ;
614     } else {
615         await _fetchJson(`/api/agents/${id}`, {
616             method: 'PUT',
617             headers: { 'Content-Type': 'application/json' },
618             body: JSON.stringify(payload)
619         });
620         _showToast(`Агент "${name}" успешно обновлен!`, 'success
        ');
621     }
622
623     const modalEl = document.getElementById('modal-agent-edito
        r');
624     bootstrap.Modal.getInstance(modalEl)?.hide();
625     await _loadAllData();
626 } catch (err) {
627     console.error('[AgentsTab] Error saving agent:', err);
628     _showToast(`Ошибка сохранения: ${err.message}`, 'danger');
629 } finally {
630     if (saveBtn) saveBtn.disabled = false;
631 }
632 }
633
634 /**
635  * Deletes an existing custom agent after confirmation.
636  *
637  * @param {string} agentId - Target agent unique identifier.
638  * @returns {Promise<void>}
639  */
640 async function _deleteAgent(agentId) {
641     const agent = _state.agents.find((a) => a.id === agentId);

```

```

642 |   if (!agent) return;
643 |
644 |   if (!confirm(`Вы действительно хотите удалить агента "${agent.name}" (${agent.id})?`)) {
645 |       return;
646 |   }
647 |
648 |   try {
649 |       await _fetchJson(`/api/agents/${agentId}`, { method: 'DELETE' });
650 |       _showToast(`Агент "${agent.name}" удален`, 'success');
651 |       await _loadAllData();
652 |   } catch (err) {
653 |       console.error('[AgentsTab] Error deleting agent:', err);
654 |       _showToast(`Ошибка удаления: ${err.message}`, 'danger');
655 |   }
656 | }
657 |
658 | /**
659 |  * Opens AI prompt architect builder modal.
660 |  *
661 |  * @returns {void}
662 |  */
663 | function _openAiBuilder() {
664 |     const modalEl = document.getElementById('modal-ai-builder');
665 |     if (!modalEl) return;
666 |
667 |     const promptInput = document.getElementById('ai-builder-prompt');
668 |     const resultBox = document.getElementById('ai-builder-result');
669 |     if (promptInput) promptInput.value = '';
670 |     if (resultBox) resultBox.classList.add('d-none');
671 |     _state.lastGeneratedSpec = null;
672 |
673 |     new bootstrap.Modal(modalEl).show();
674 | }
675 |

```

```

676 /**
677  * Generates an agent specification using selected LLM archite
    ct model.
678  *
679  * @returns {Promise<void>}
680  */
681 async function _handleRunAiGenerate() {
682     const taskDesc = document.getElementById('ai-builder-prompt'
    )?.value.trim();
683     const provider = document.getElementById('ai-builder-provide
    r')?.value ?? 'gemini';
684     const model = document.getElementById('ai-builder-model')?.v
    alue ?? 'gemini-2.5-flash';
685
686     if (!taskDesc) {
687         _showToast('Пожалуйста, опишите задачу агента', 'warning')
    ;
688         return;
689     }
690
691     const runBtn = document.getElementById('btn-run-ai-generate'
    );
692     if (runBtn) {
693         runBtn.disabled = true;
694         runBtn.innerHTML = '<span class="spinner-border spinner-bo
    rder-sm me-2"></span>Анализ задачи и генерация...';
695     }
696
697     try {
698         const res = await _fetchJson('/api/agents/generate-prompt'
    , {
699             method: 'POST',
700             headers: { 'Content-Type': 'application/json' },
701             body: JSON.stringify({ task_description: taskDesc, provi
    der, model })
702         });
703
704         if (res?.data) {

```



```

705 |     _state.lastGeneratedSpec = { ...res.data, provider, mode
    | 1 };
706 |     const resultBox = document.getElementById('ai-builder-re
    | sult');
707 |     const previewBox = document.getElementById('ai-builder-p
    | review-box');
708 |     if (resultBox && previewBox) {
709 |         resultBox.classList.remove('d-none');
710 |         previewBox.innerHTML = `
711 |             <div><strong style="color: #38bdf8;">Название:</stro
    | ng> <span class="text-white fw-bold">${res.data.name}</span></
    | div>
712 |             <div class="mt-1"><strong style="color: #cbd5e1;">Оп
    | исание:</strong> <span style="color: #f1f5f9;">${res.data.desc
    | ription}</span></div>
713 |             <div class="mt-1"><strong style="color: #4ade80;">Ин
    | струменты:</strong> <span style="color: #f8fafc;">${(res.data.
    | recommended_tools ?? []).join(', ') || 'нет'}</span></div>
714 |             <div class="mt-2 p-2 rounded" style="background: #0f
    | 172a; border: 1px solid #334155; color: #f8fafc; white-space:
    | pre-wrap; font-size: 0.85rem; line-height: 1.45;">${res.data.s
    | ystem_prompt}</div>
715 |         `;
716 |     }
717 |     _showToast('Спецификация агента успешно сгенерирована!',
    | 'success');
718 | }
719 | } catch (err) {
720 |     console.error('[AgentsTab] AI Generator error:', err);
721 |     _showToast(`Ошибка генерации: ${err.message}`, 'danger');
722 | } finally {
723 |     if (runBtn) {
724 |         runBtn.disabled = false;
725 |         runBtn.innerHTML = '<i class="bi bi-stars me-1"></i> Cre
    | нерировать спецификацию';
726 |     }
727 | }
728 | }

```

```
729 |
730 | /**
731 |  * Opens sandbox modal for interactive agent testing.
732 |  *
733 |  * @param {string} agentId - Target agent unique identifier.
734 |  * @returns {void}
735 |  */
736 | function _openSandbox(agentId) {
737 |     const agent = _state.agents.find((a) => a.id === agentId);
738 |     if (!agent) return;
739 |
740 |     _state.activeSandboxAgent = agent;
741 |     const modalEl = document.getElementById('modal-agent-sandbox
742 | ');
743 |     if (!modalEl) return;
744 |
745 |     document.getElementById('sandbox-agent-name')?.replaceChildr
746 | en(document.createTextNode(agent.name));
747 |     const modelBadge = document.getElementById('sandbox-agent-mo
748 | del');
749 |     if (modelBadge) modelBadge.textContent = `${_getProviderDisp
750 | layName(agent.provider)}: ${agent.model}`;
751 |
752 |     const toolsSummary = document.getElementById('sandbox-agent-
753 | tools-summary');
754 |     if (toolsSummary) toolsSummary.textContent = `Инструментов:
755 | ${((agent.tools ?? []).length)} `;
756 |
757 |     const msgsList = document.getElementById('sandbox-messages-l
758 | ist');
759 |     const placeholder = document.getElementById('sandbox-placeho
760 | lder');
761 |     if (msgsList) msgsList.innerHTML = '';
762 |     if (placeholder) placeholder.style.display = 'block';
763 |
764 |     new bootstrap.Modal(modalEl).show();
765 | }
```

```

759 /**
760  * Executes a test query against the active sandbox agent and
    renders execution trace.
761  *
762  * @returns {Promise<void>}
763  */
764 async function _handleRunSandboxTest() {
765     const agent = _state.activeSandboxAgent;
766     const input = document.getElementById('sandbox-input-msg');
767     const sendBtn = document.getElementById('btn-sandbox-send');
768     const msgsList = document.getElementById('sandbox-messages-l
    ist');
769     const placeholder = document.getElementById('sandbox-placeho
    lder');
770
771     if (!agent || !input || !msgsList) return;
772     const query = input.value.trim();
773     if (!query) return;
774
775     if (placeholder) placeholder.style.display = 'none';
776
777     // Append user message
778     const userMsgEl = document.createElement('div');
779     userMsgEl.className = 'd-flex justify-content-end mb-3';
780     userMsgEl.innerHTML = `
781         <div class="p-2 px-3 rounded shadow-sm text-white small" s
    tyle="background-color: #2563eb; max-width: 80%; font-weight:
    500;">
782             <strong>Вы:</strong> ${query}
783         </div>
784     `;
785     msgsList.appendChild(userMsgEl);
786
787     input.value = '';
788     if (sendBtn) sendBtn.disabled = true;
789
790     // Append response placeholder with loader
791     const botMsgEl = document.createElement('div');

```

```

792 | botMsgEl.className = 'd-flex flex-column mb-3';
793 | botMsgEl.innerHTML = `
794 |     <div class="p-3 rounded small" style="background: #0f172a;
      |     border: 1px solid #334155; color: #f8fafc;">
795 |         <div class="d-flex align-items-center gap-2 mb-2" style=
      |         "color: #fbbf24; font-weight: 600;">
796 |             <span class="spinner-border spinner-border-sm"></span>
797 |             <span>Агент выполняет рассуждение и вызовы инструменто
      | в...</span>
798 |         </div>
799 |         <div class="steps-trace small font-monospace" style="col
      | or: #cbd5e1;"></div>
800 |     </div>
801 | `;
802 | msgsList.appendChild(botMsgEl);
803 | msgsList.scrollTop = msgsList.scrollHeight;
804 |
805 | try {
806 |     const res = await _fetchJson('/api/agents/test', {
807 |         method: 'POST',
808 |         headers: { 'Content-Type': 'application/json' },
809 |         body: JSON.stringify({ agent_id: agent.id, test_message:
      | query })
810 |     });
811 |
812 |     const stepsHtml = (res.steps ?? [])
813 |         .map((s) => {
814 |             let icon = '❏';
815 |             if (s.type === 'tool_init') icon = '🔧';
816 |             else if (s.type === 'action') icon = '👉';
817 |             else if (s.type === 'finish') icon = '✅';
818 |             else if (s.type === 'error') icon = '❌';
819 |
820 |             return `<div class="mb-1">${icon} <span style="color:
      | #38bdf8; font-weight: bold;">[Шаг ${s.step}]</span> <span styl
      | e="color: #e2e8f0;">${s.content}</span></div>`;
821 |         })
822 |         .join('');

```

```

823
824     botMsgEl.innerHTML = `
825         <div class="p-3 rounded small" style="background: #0f172
a; border: 1px solid #334155; color: #f8fafc;">
826             <div class="d-flex justify-content-between align-items
-center mb-2 pb-1" style="border-bottom: 1px solid #334155;">
827                 <strong style="color: #38bdf8; font-size: 0.95rem;">
❑ ${agent.name}</strong>
828                 <span class="badge" style="background: #334155; colo
r: #f8fafc; font-weight: 600;">${res.duration_ms ?? 0} мс</spa
n>
829             </div>
830
831             <div class="mb-2" style="white-space: pre-wrap; color:
#f8fafc; line-height: 1.5;">${res.response || 'Пустой ответ'}
</div>
832
833             <div class="p-2 rounded mt-2 small font-monospace" sty
le="background: #020617; border: 1px solid #334155; font-size:
0.8rem; line-height: 1.45;">
834                 <div class="fw-bold mb-1" style="color: #cbd5e1;">Тр
ассировка выполнения (ReAct Trace):</div>
835                 ${stepsHtml}
836             </div>
837         </div>
838     `;
839 } catch (err) {
840     botMsgEl.innerHTML = `
841         <div class="p-3 rounded small" style="background: #450a0
a; border: 1px solid #dc2626; color: #fecaca;">
842             <strong style="color: #ef4444;">Ошибка выполнения:</st
rong> ${err.message}
843         </div>
844     `;
845 } finally {
846     if (sendBtn) sendBtn.disabled = false;
847     const chatWin = document.getElementById('sandbox-chat-wind
ow');

```

```

848     if (chatWin) chatWin.scrollTop = chatWin.scrollHeight;
849 }
850 }
851
852 /**
853  * Returns human-readable display name for provider identifier
854  *
855  * @param {string} provKey - Provider key (e.g. 'gemini', 'agy
856  *                            ', 'foundry', 'ollama').
857  * @returns {string} Provider display label.
858  */
859 function _getProviderDisplayName(provKey) {
860     return _state.providers[provKey]?.name ?? provKey;
861 }
862
863 /**
864  * Displays floating feedback notification toast.
865  *
866  * @param {string} message - Notification text message.
867  * @param {string} [type='info'] - Bootstrap color variant ('i
868  *                                nfo', 'success', 'warning', 'danger').
869  * @returns {void}
870  */
871 function _showToast(message, type = 'info') {
872     if (typeof window.showModelsNotification === 'function') {
873         window.showModelsNotification(message, type);
874         return;
875     }
876     const alertDiv = document.createElement('div');
877     alertDiv.className = `alert alert-${type} position-fixed bot
878     tom-0 end-0 m-3 shadow`;
879     alertDiv.style.zIndex = '9999';
880     alertDiv.textContent = message;
881     document.body.appendChild(alertDiv);
882     setTimeout(() => alertDiv.remove(), 3500);
883 }
884
885

```

index.html

```
1 <!-- AI Inference Benchmark Tab -->
2 <style>
3   .bm-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .bm-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.6rem 0.85rem;
11    height: 100%;
12    transition: border-color 0.15s ease, box-shadow 0.15s ease
13  ;
14  }
15  .bm-card:hover {
16    border-color: #38bdf8 !important;
17    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
18  }
19  .bm-card-label {
20    font-size: 0.7rem;
21    text-transform: uppercase;
22    letter-spacing: 0.05em;
23    color: #94a3b8;
24    font-weight: 600;
25    margin-bottom: 0.25rem;
26  }
27  .bm-card-value {
28    font-size: 1.4rem;
29    font-weight: 700;
30    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
31    Consolas, monospace;
32    color: #38bdf8;
33  }
34  .bm-panel {
35    background: #0f172a !important;
36    border: 1px solid #334155 !important;
```

```
35 |     border-radius: 8px;
36 |     margin-bottom: 0.75rem;
37 |     overflow: hidden;
38 | }
39 | .bm-panel-header {
40 |     background: #1e293b !important;
41 |     border-bottom: 1px solid #334155 !important;
42 |     padding: 0.45rem 0.85rem;
43 |     font-weight: 600;
44 |     font-size: 0.85rem;
45 |     color: #38bdf8;
46 |     display: flex;
47 |     align-items: center;
48 |     justify-content: space-between;
49 | }
50 | .bm-table {
51 |     margin-bottom: 0;
52 |     font-size: 0.8rem;
53 |     width: 100%;
54 | }
55 | .bm-table th {
56 |     background: #1e293b;
57 |     color: #94a3b8;
58 |     font-size: 0.72rem;
59 |     text-transform: uppercase;
60 |     padding: 0.45rem 0.6rem;
61 |     border-bottom: 1px solid #334155;
62 | }
63 | .bm-table td {
64 |     border-bottom: 1px solid #1e293b;
65 |     padding: 0.45rem 0.6rem;
66 |     vertical-align: middle;
67 | }
68 | </style>
69 |
70 | <div class="bm-container">
71 |     <!-- Заголовок и панель параметров запуска -->
72 |     <div class="bm-panel mb-3">
```



```
73     <div class="bm-panel-header">
74         <span><i class="bi bi-speedometer2 me-1"></i> Параметры
запуска AI Inference Benchmark</span>
75         <span class="badge bg-primary-subtle text-primary border
border-primary-subtle" id="bm-status-badge">Готов</span>
76     </div>
77     <div class="p-3">
78         <div class="row g-2 align-items-end">
79             <div class="col-md-3 col-sm-6">
80                 <label class="form-label small text-muted mb-1">Пров
айдер</label>
81                 <select class="form-select form-select-sm bg-dark te
xt-white border-secondary" id="bm-provider-select">
82                     <option value="gemini" selected>Google Gemini API<
/option>
83                     <option value="ollama">Ollama (Local)</option>
84                     <option value="foundry">Microsoft Foundry Local</o
ption>
85                     <option value="onnx">ONNX DirectML / NPU</option>
86                 </select>
87             </div>
88             <div class="col-md-3 col-sm-6">
89                 <label class="form-label small text-muted mb-1">Моде
ль</label>
90                 <input type="text" class="form-control form-control-
sm bg-dark text-white border-secondary" id="bm-model-input" va
lue="gemini-2.5-flash">
91             </div>
92             <div class="col-md-2 col-sm-6">
93                 <label class="form-label small text-muted mb-1">Max
Tokens</label>
94                 <input type="number" class="form-control form-contro
l-sm bg-dark text-white border-secondary" id="bm-tokens-input"
value="150" min="20" max="2000">
95             </div>
96             <div class="col-md-2 col-sm-6">
97                 <label class="form-label small text-muted mb-1">Temp
erature</label>
```

```

98 |         <input type="number" class="form-control form-control-sm bg-dark text-white border-secondary" id="bm-temp-input" value="0.7" min="0" max="1" step="0.1">
99 |     </div>
100 |     <div class="col-md-2 col-sm-12">
101 |         <button class="btn btn-sm btn-primary w-100 fw-semibold d-flex align-items-center justify-content-center gap-1.5" id="bm-run-btn">
102 |             <i class="bi bi-play-fill fs-6"></i> Запустить тест
103 |         </button>
104 |     </div>
105 | </div>
106 | <div class="mt-2">
107 |     <label class="form-label small text-muted mb-1">Тестовый промпт</label>
108 |     <textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="bm-prompt-input" rows="2">Напиши краткий рассказ о квантовых вычислениях на 100 слов.</textarea>
109 | </div>
110 | </div>
111 | </div>
112 |
113 | <!-- Карточки последних результатов -->
114 | <div class="row g-2 mb-3">
115 |     <div class="col-md-3 col-6">
116 |         <div class="bm-card">
117 |             <div class="bm-card-label"><i class="bi bi-stopwatch text-info me-1"></i> Time to First Token (TTFT)</div>
118 |             <div class="bm-card-value" id="bm-card-ttft">— ms</div>
119 |         </div>
120 |     </div>
121 |     <div class="col-md-3 col-6">
122 |         <div class="bm-card">
123 |             <div class="bm-card-label"><i class="bi bi-lightning-charge text-warning me-1"></i> Скорость (TPS)</div>

```

```

124 |         <div class="bm-card-value text-success" id="bm-card-tp
      | s">- t/s</div>
125 |     </div>
126 | </div>
127 | <div class="col-md-3 col-6">
128 |     <div class="bm-card">
129 |         <div class="bm-card-label"><i class="bi bi-hourglass-s
      | plit text-primary me-1"></i> Общее время (Latency)</div>
130 |         <div class="bm-card-value" id="bm-card-total">- ms</di
      | v>
131 |     </div>
132 | </div>
133 | <div class="col-md-3 col-6">
134 |     <div class="bm-card">
135 |         <div class="bm-card-label"><i class="bi bi-chat-left-t
      | ext text-secondary me-1"></i> Сгенерировано токенов</div>
136 |         <div class="bm-card-value text-white" id="bm-card-toke
      | ns">-</div>
137 |     </div>
138 | </div>
139 | </div>
140 |
141 | <!-- Таблица истории замеров -->
142 | <div class="bm-panel">
143 |     <div class="bm-panel-header">
144 |         <span><i class="bi bi-clock-history me-1"></i> История б
      | енчмарков инференса</span>
145 |         <button class="btn btn-xs btn-outline-secondary py-0 px-
      | 2 text-white border-secondary" id="bm-refresh-history-btn" sty
      | le="font-size: 0.75rem;">
146 |             <i class="bi bi-arrow-clockwise"></i> Обновить
147 |         </button>
148 |     </div>
149 |     <div class="table-responsive">
150 |         <table class="table table-dark table-hover bm-table" id=
      | "bm-history-table">
151 |             <thead>
152 |                 <tr>

```

```
153 |         <th>Время</th>
154 |         <th>Провайдер</th>
155 |         <th>Модель</th>
156 |         <th>TTFT</th>
157 |         <th>Скорость (TPS)</th>
158 |         <th>Общее время</th>
159 |         <th>Токены (Prompt / Gen)</th>
160 |         <th>Статус</th>
161 |     </tr>
162 | </thead>
163 | <tbody id="bm-history-tbody">
164 |     <tr>
165 |         <td colspan="8" class="text-center text-muted py-3
166 |         ">История бенчмарков пуста. Нажмите "Запустить тест".</td>
167 |     </tr>
168 | </tbody>
169 | </table>
170 | </div>
171 </div>
172
```

main.js

```
1 /**
2  * ai_benchmark_tab/main.js – Контроллер вкладки AI Inference
  Benchmark
3  */
4
5 export async function initAiBenchmarkTab() {
6   const runBtn = document.getElementById('bm-run-btn');
7   const refreshBtn = document.getElementById('bm-refresh-histo
  ry-btn');
8   const providerSelect = document.getElementById('bm-provider-
  select');
9   const modelInput = document.getElementById('bm-model-input')
  ;
10  const tokensInput = document.getElementById('bm-tokens-input
  ');
11  const tempInput = document.getElementById('bm-temp-input');
12  const promptInput = document.getElementById('bm-prompt-input
  ');
13  const statusBadge = document.getElementById('bm-status-badge
  ');
14
15  // Автозаполнение модели при смене провайдера
16  if (providerSelect) {
17    providerSelect.addEventListener('change', () => {
18      const p = providerSelect.value;
19      if (p === 'gemini') modelInput.value = 'gemini-2.5-flash
  ';
20      else if (p === 'ollama') modelInput.value = 'llama3.2:la
  test';
21      else if (p === 'foundry') modelInput.value = 'phi-3.5-mi
  ni';
22      else if (p === 'onnx') modelInput.value = 'directml-phi3
  ';
23    });
24  }
25
```

```
26 | // Запуск бенчмарка
27 | if (runBtn) {
28 |     runBtn.addEventListener('click', async () => {
29 |         runBtn.disabled = true;
30 |         runBtn.innerHTML = '<span class="spinner-border spinner-
border-sm me-1"></span> Тестирование...';
31 |         if (statusBadge) {
32 |             statusBadge.className = 'badge bg-warning-subtle text-
warning border border-warning-subtle';
33 |             statusBadge.textContent = 'Выполняется замер...';
34 |         }
35 |
36 |         try {
37 |             const payload = {
38 |                 provider: providerSelect ? providerSelect.value : 'g
emini',
39 |                 model_name: modelInput ? modelInput.value : 'gemini-
2.5-flash',
40 |                 prompt: promptInput ? promptInput.value : 'Тестовый
запрос',
41 |                 max_tokens: tokensInput ? parseInt(tokensInput.value
, 10) || 150 : 150,
42 |                 temperature: tempInput ? parseFloat(tempInput.value)
|| 0.7 : 0.7,
43 |             };
44 |
45 |             const res = await fetch('/api/windows/benchmark/ai', {
46 |                 method: 'POST',
47 |                 headers: { 'Content-Type': 'application/json' },
48 |                 body: JSON.stringify(payload),
49 |             });
50 |
51 |             const data = await res.json();
52 |             if (res.ok && data.success) {
53 |                 updateCards(data);
54 |                 if (statusBadge) {
55 |                     statusBadge.className = 'badge bg-success-subtle t
ext-success border border-success-subtle';
```

```
56 |         statusBadge.textContent = 'Успешно завершен';
57 |     }
58 |     if (window.toast) {
59 |         window.toast.success('Бенчмарк завершен', `Скорост
60 | б: ${data.tokens_per_second} t/s | TTFT: ${data.ttft_ms} ms`);
61 |     }
62 |     } else {
63 |         if (statusBadge) {
64 |             statusBadge.className = 'badge bg-danger-subtle te
65 | xt-danger border border-danger-subtle';
66 |             statusBadge.textContent = 'Ошибка';
67 |         }
68 |         if (window.toast) {
69 |             window.toast.error('Ошибка бенчмарка', data.error_
70 | message || 'Не удалось выполнить замер');
71 |         }
72 |     } catch (err) {
73 |         if (statusBadge) {
74 |             statusBadge.className = 'badge bg-danger-subtle text
75 | -danger border border-danger-subtle';
76 |             statusBadge.textContent = 'Ошибка сети';
77 |         }
78 |         if (window.toast) {
79 |             window.toast.error('Сбой запроса', err.message);
80 |         }
81 |     } finally {
82 |         runBtn.disabled = false;
83 |         runBtn.innerHTML = '<i class="bi bi-play-fill fs-6"></
84 | i> Запустить тест';
85 |         await loadHistory();
86 |     }
87 | });
88 | }

89 | // Обновление истории
90 | if (refreshBtn) {
91 |     refreshBtn.addEventListener('click', loadHistory);
```

```
89 | }
90 |
91 | // Загрузка начальной истории
92 | await loadHistory();
93 | }
94 |
95 | function updateCards(data) {
96 |   const ttftEl = document.getElementById('bm-card-ttft');
97 |   const tpsEl = document.getElementById('bm-card-tps');
98 |   const totalEl = document.getElementById('bm-card-total');
99 |   const tokensEl = document.getElementById('bm-card-tokens');
100 |
101 |   if (ttftEl) ttftEl.textContent = `${data.ttft_ms} ms`;
102 |   if (tpsEl) tpsEl.textContent = `${data.tokens_per_second} t/
    | s`;
103 |   if (totalEl) totalEl.textContent = `${data.total_time_ms} ms
    | `;
104 |   if (tokensEl) tokensEl.textContent = `${data.prompt_tokens}
    | / ${data.completion_tokens}`;
105 | }
106 |
107 | async function loadHistory() {
108 |   const tbody = document.getElementById('bm-history-tbody');
109 |   if (!tbody) return;
110 |
111 |   try {
112 |     const res = await fetch('/api/windows/benchmark/ai/history
    | ');
113 |     const items = await res.json();
114 |
115 |     if (!Array.isArray(items) || items.length === 0) {
116 |       tbody.innerHTML = '<tr><td colspan="8" class="text-cente
    | r text-muted py-3">История бенчмарков пуста. Нажмите "Запустит
    | ь тест".</td></tr>';
117 |       return;
118 |     }
119 |
120 |     tbody.innerHTML = items.slice().reverse().map(item => {
```



```

121     const timeStr = item.timestamp ? new Date(item.timestamp
    * 1000).toLocaleTimeString() : '-';
122     const statusBadge = item.success
123     ? '<span class="badge bg-success-subtle text-success">
    OK</span>'
124     : '<span class="badge bg-danger-subtle text-danger" ti
    tle="${item.error_message || ''}">Ошибка</span>`;
125
126     return `
127         <tr>
128             <td><span class="text-muted small">${timeStr}</span>
    </td>
129             <td><span class="badge bg-secondary-subtle text-info
    ">${item.provider}</span></td>
130             <td class="fw-semibold text-white">${item.model_name
    }</td>
131             <td><span class="text-info">${item.ttft_ms} ms</span
    ></td>
132             <td><span class="text-success fw-bold">${item.tokens
    _per_second} t/s</span></td>
133             <td>${item.total_time_ms} ms</td>
134             <td><span class="text-muted small">${item.prompt_tok
    ens} in / ${item.completion_tokens} out</span></td>
135             <td>${statusBadge}</td>
136         </tr>
137     `;
138     })).join('');
139 } catch (err) {
140     tbody.innerHTML = `<tr><td colspan="8" class="text-danger
    text-center py-2">Ошибка загрузки истории: ${err.message}</td>
    </tr>`;
141 }
142 }
143
144 // Автозапуск при инициализации вкладки
145 if (document.readyState === 'loading') {
146     document.addEventListener('DOMContentLoaded', initAiBenchmar
    kTab);

```

```
147 | } else {  
148 |     initAiBenchmarkTab();  
149 | }  
150 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title>AI Breadboard – Приложения (/apps)</title>
8   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
9   t/css/bootstrap.min.css" rel="stylesheet">
10  <link href="https://cdn.jsdelivrivr.net/npm/bootstrap-icons@1.1
11  0.0/font/bootstrap-icons.css" rel="stylesheet">
12  <link href="/html/css/main.css?v=20260926_v4" rel="styleshee
13  t">
14  <script src="https://cdn.jsdelivrivr.net/npm/chart.js"></script
15  >
16  <script src="/html/js/toast.js?v=20260926_v4"></script>
17  <script src="https://cdn.jsdelivrivr.net/npm/i18next@23.11.5/i1
18  8next.min.js"></script>
19 </head>
20 <body>
21
22 <div class="container-fluid mt-1 px-2" id="apps-interface">
23
24   <!-- Navigation Bar (/tc, /apps) -->
25   <div class="d-flex align-items-center justify-content-between mb-2 main-nav-container rounded px-2 py-1 flex-wrap gap-2">
26     <!-- Левая часть: кнопка меню, брендинг и индикатор активн
27     ого раздела -->
28     <div class="d-flex align-items-center gap-2">
29       <!-- Кнопка открытия левого выдвижного меню -->
30       <button class="btn btn-sm btn-primary d-flex align-items
31       -center gap-1.5 px-2 py-1 rounded shadow-sm"
32         id="apps-drawer-toggle-btn"
33         type="button"
34         data-bs-toggle="offcanvas"
35         data-bs-target="#appsSideNavOffcanvas">
```

```

28         aria-controls="appsSideNavOffcanvas"
29         title="Открыть меню разделов"
30         style="font-size: 0.82rem; font-weight: 500;">
31         <i class="bi bi-list fs-6"></i>
32         <span class="d-none d-md-inline">Меню</span>
33     </button>
34
35     <!-- Брендинг -->
36     <span class="fw-bold d-flex align-items-center gap-1.5 text-
ext-primary fs-6 user-select-none" style="letter-spacing: 0.3p
x;">
37         <i class="bi bi-cpu-fill fs-5"></i>
38         <span class="d-none d-sm-inline">Test Computer</span>
39     </span>
40     <span id="active-tab-title-badge" class="badge bg-body-s
econdary text-secondary-emphasis border rounded-pill px-2 py-1
small fw-normal d-flex align-items-center gap-1" title="Текущ
ий активный раздел">
41         i 0 Системе
42     </span>
43 </div>
44
45 <!-- Top Menu Buttons – генерируются из tc_menu_config.js
n -->
46 <div class="d-flex align-items-center gap-1 flex-wrap"></d
iv>
47
48 <!-- Controls (Model Dropdown, Language, Theme, Switcher)
-->
49 <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto">
50     <!-- Active AI Model Dropdown -->
51     <div class="dropdown">
52         <button class="btn btn-sm btn-outline-primary d-flex a
lign-items-center gap-1.5 py-1 px-2 rounded dropdown-toggle"
53             type="button"
54             id="apps-model-dropdown-btn"
55             data-bs-toggle="dropdown">

```

```

56 |             aria-expanded="false"
57 |             title="Выберите активную AI-модель">
58 |         <i class="bi bi-cpu-fill text-info"></i>
59 |         <span class="text-white fw-semibold" id="apps-model-
| text">AI: Загрузка...</span>
60 |     </button>
61 |     <ul class="dropdown-menu dropdown-menu-dark" aria-labe
| lledby="apps-model-dropdown-btn" id="apps-model-dropdown-menu"
| >
62 |         <li><span class="dropdown-item-text text-muted small
| px-2">Загрузка моделей...</span></li>
63 |     </ul>
64 | </div>
65 |
66 | <!-- Language Selector -->
67 | <select class="form-select form-select-sm bg-dark text-w
| hite border-secondary lang-selector" id="apps-lang-selector" s
| tyle="width: 100px;">
68 |     <option value="ru">Русский</option>
69 |     <option value="en">English</option>
70 |     <option value="he">□ □ □ □ □ </option>
71 | </select>
72 |
73 | <!-- Theme Selector -->
74 | <select class="form-select form-select-sm bg-dark text-w
| hite border-secondary theme-selector" id="apps-theme-selector"
| style="width: 105px;">
75 |     <option value="dark">□ Тёмная</option>
76 |     <option value="light">□ Светлая</option>
77 |     <option value="system">□ Система</option>
78 | </select>
79 |
80 | <!-- Mode Switcher (Test Computer / Admin Panel) -->
81 | <div class="btn-group btn-group-sm shadow-sm" role="grou
| p" aria-label="Режим работы">
82 |     <a href="/tc" class="btn btn-primary active d-flex ali
| gn-items-center gap-1 py-1 px-2.5" title="Режим Test Computer
| (/tc)">

```

```

83 |         <i class="bi bi-display"></i>
84 |         <span class="small fw-semibold">Test Computer</span>
85 |     </a>
86 |     <a href="/admin" class="btn btn-outline-secondary d-fl
    | ex align-items-center gap-1 py-1 px-2.5" title="Открыть полную
    | панель управления">
87 |         <i class="bi bi-shield-lock-fill"></i>
88 |         <span class="small d-none d-md-inline">Админка</span>
    | >
89 |     </a>
90 | </div>
91 | </div>
92 | </div>
93 |
94 | <!-- Плавающий боковой ярлычок (вкладка) слева -->
95 | <button class="left-drawer-tab-trigger btn shadow"
96 |     type="button"
97 |     data-bs-toggle="offcanvas"
98 |     data-bs-target="#appsSideNavOffcanvas"
99 |     aria-controls="appsSideNavOffcanvas"
100 |     title="Открыть меню навигации">
101 |     <i class="bi bi-list fs-6"></i>
102 |     <span class="left-drawer-tab-text">МЕНЮ</span>
103 | </button>
104 |
105 | <!-- Левая выдвижная вкладка / панель навигации (Offcanvas)
    | -->
106 | <div class="offcanvas offcanvas-start left-drawer-nav" tabin
    | dex="-1" id="appsSideNavOffcanvas" aria-labelledby="appsSideNa
    | vOffcanvasLabel">
107 |     <div class="offcanvas-header border-bottom py-2 px-3">
108 |         <h5 class="offcanvas-title d-flex align-items-center gap
    | -2 fs-6 fw-bold" id="appsSideNavOffcanvasLabel">
109 |             <i class="bi bi-grid-fill text-primary"></i>
110 |             <span>Приложения (/apps)</span>
111 |         </h5>
112 |         <button type="button" class="btn-close text-reset" data-
    | bs-dismiss="offcanvas" aria-label="Закрыть"></button>

```

```
113     </div>
114     <div class="offcanvas-body p-3 d-flex flex-column justify-
content-between" style="overflow-y: auto;">
115         <div class="drawer-sections-wrapper">
116             <div class="nav-drawer-group mb-3">
117                 <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
118                     <i class="bi bi-grid-fill text-primary"></i>
119                     <span>Разделы Test Computer</span>
120                 </div>
121                 <div class="list-group list-group-flush rounded bord
er" id="appsNavTabs"></div>
122             </div>
123         </div>
124
125         <!-- Footer Drawer -->
126         <div class="nav-drawer-footer border-top pt-2 mt-auto">
127             <button class="btn btn-sm btn-outline-secondary w-100
d-flex align-items-center gap-2 py-2 rounded" id="menu-editor-
btn" type="button">
128                 <i class="bi bi-gear-fill"></i>
129                 <span class="fw-medium">Редактор меню</span>
130             </button>
131         </div>
132     </div>
133 </div>
134
135 <!-- Menu Editor Modal -->
136 <div class="modal fade" id="menuEditorModal" tabindex="-1" a
ria-labelledby="menuEditorModallLabel" aria-hidden="true">
137     <div class="modal-dialog modal-xl modal-dialog-centered">
138         <div class="modal-content bg-dark text-white">
139             <div class="modal-header border-secondary">
140                 <h5 class="modal-title fw-bold" id="menuEditorModall
abel">
141                     <i class="bi bi-gear-fill text-primary"></i> Редак
тор меню
142                 </h5>
```

```

143         <button type="button" class="btn-close btn-close-whi
te" data-bs-dismiss="modal" aria-label="Заккрыть"></button>
144     </div>
145     <div class="modal-body">
146         <div class="alert alert-info small mb-3">
147             <i class="bi bi-info-circle"></i> Настройте видимо
сть и порядок элементов меню. Используйте переключатели для вы
бора позиции (сверху/снизу/скрыть) и слайдеры для изменения по
рядка элементов.
148         </div>
149
150         <div class="menu-editor-section">
151             <h6 class="text-primary fw-bold mb-3">
152                 <i class="bi bi-list-ul"></i> Все элементы меню
153             </h6>
154             <div class="list-group list-group-flush" id="allMe
nuEditor">
155                 <div class="text-muted small p-3 text-center">За
грузка...</div>
156             </div>
157         </div>
158     </div>
159     <div class="modal-footer border-secondary">
160         <button type="button" class="btn btn-secondary" data
-bs-dismiss="modal">Отмена</button>
161         <button type="button" class="btn btn-primary" id="sa
veMenuConfig">
162             <i class="bi bi-save"></i> Сохранить
163         </button>
164     </div>
165 </div>
166 </div>
167 </div>
168
169 <!-- Main Tab Content Container (Apps Only) -->
170 <div class="tab-content" id="mainTabContent">
171     <!-- About System Tab (Default Primary) -->
172     <div id="tab-about-system" class="tab-pane fade show activ

```



```
172 | e"></div>
173 |
174 | <!-- Chat Tab -->
175 | <div id="tab-chat" class="tab-pane fade"></div>
176 |
177 | <!-- Scenarios Tab -->
178 | <div id="tab-scenarios" class="tab-pane fade"></div>
179 |
180 | <!-- Network Terminal Tab -->
181 | <div id="tab-network" class="tab-pane fade"></div>
182 |
183 | <!-- System Inspector Tab -->
184 | <div id="tab-system-inspector" class="tab-pane fade"></div>
185 | >
186 | <!-- Windows System Administrator Tab -->
187 | <div id="tab-windows-admin" class="tab-pane fade"></div>
188 |
189 | <!-- System Control Center Tab -->
190 | <div id="tab-system-control" class="tab-pane fade"></div>
191 |
192 | <!-- System Log Viewer Tab -->
193 | <div id="tab-system-logs" class="tab-pane fade"></div>
194 |
195 | <!-- Software Audit Tab -->
196 | <div id="tab-software-audit" class="tab-pane fade"></div>
197 |
198 | <!-- Registry Viewer Tab -->
199 | <div id="tab-registry-viewer" class="tab-pane fade"></div>
200 |
201 | <!-- Microsoft Defender Security Tab -->
202 | <div id="tab-defender" class="tab-pane fade"></div>
203 |
204 | <!-- Startup Auditor Tab -->
205 | <div id="tab-startup-auditor" class="tab-pane fade"></div>
206 |
207 | <!-- Windows Backup & Libraries Manager Tab -->
208 | <div id="tab-windows-backup" class="tab-pane fade"></div>
```

```
209 |
210 | <!-- Hardware & Sensors Monitor Tab -->
211 | <div id="tab-hardware-monitor" class="tab-pane fade"></div>
    | >
212 |
213 | <!-- Cloudflared Monitor Tab -->
214 | <div id="tab-cloudflared" class="tab-pane fade"></div>
215 |
216 | <!-- Google User Desktop Tab -->
217 | <div id="tab-google-desktop" class="tab-pane fade"></div>
218 |
219 | <!-- Google Cloud Monitor Tab -->
220 | <div id="tab-gcloud" class="tab-pane fade"></div>
221 |
222 | <!-- Website Intelligence Monitor Tab -->
223 | <div id="tab-website-monitor" class="tab-pane fade"></div>
224 |
225 | <!-- Exchange Trading Terminal Tab -->
226 | <div id="tab-trading" class="tab-pane fade"></div>
227 |
228 | <!-- User Assistant Tab -->
229 | <div id="tab-user-assistant" class="tab-pane fade"></div>
230 |
231 | <!-- Helpdesk Tab -->
232 | <div id="tab-helpdesk" class="tab-pane fade"></div>
233 |
234 | <!-- Wikipedia Research Lab Tab -->
235 | <div id="tab-wikipedia-research" class="tab-pane fade"></div>
    | iv>
236 |
237 | <!-- Auto-Logging & CSV Management Tab -->
238 | <div id="tab-autolog" class="tab-pane fade"></div>
239 |
240 | <!-- AI Software Transparency Scanner Tab -->
241 | <div id="tab-software-transparency" class="tab-pane fade">
    | </div>
242 |
243 | <!-- User Directories and Storage Tab -->
```

```
244 | <div id="tab-user-directories" class="tab-pane fade"></div
    | >
245 |
246 | <!-- Ninite Auto-Updater Tab -->
247 | <div id="tab-ninite-updater" class="tab-pane fade"></div>
248 |
249 | <!-- File Recovery (R-Studio) Tab -->
250 | <div id="tab-file-recovery" class="tab-pane fade"></div>
251 |
252 | <!-- File History AI Search Tab -->
253 | <div id="tab-file-history-search" class="tab-pane fade"></
    | div>
254 |
255 | <!-- Telemetry History & Visualization Tab -->
256 | <div id="tab-telemetry-history" class="tab-pane fade"></di
    | v>
257 |
258 | <!-- Telemetry Research Tab -->
259 | <div id="tab-telemetry-research" class="tab-pane fade"></d
    | iv>
260 |
261 | <!-- Process Leak Hunter Tab -->
262 | <div id="tab-process-leaks" class="tab-pane fade"></div>
263 |
264 | <!-- Behavioral Forensics Tab -->
265 | <div id="tab-forensics" class="tab-pane fade"></div>
266 |
267 | <!-- Kernel Quality & Throttling Tab -->
268 | <div id="tab-throttling" class="tab-pane fade"></div>
269 |
270 | <!-- Storage & Battery Wear Tab -->
271 | <div id="tab-storage-wear" class="tab-pane fade"></div>
272 |
273 | <!-- Peripherals & Wireless Network Tab -->
274 | <div id="tab-peripherals" class="tab-pane fade"></div>
275 |
276 | <!-- RAG Tab -->
277 | <div id="tab-rag" class="tab-pane fade"></div>
```

```
278 |
279 |     <!-- PixelRAG Tab -->
280 |     <div id="tab-pixelrag" class="tab-pane fade"></div>
281 |
282 |     <!-- Models & API Tab -->
283 |     <div id="tab-models" class="tab-pane fade"></div>
284 |
285 |     <!-- AI Agents Tab -->
286 |     <div id="tab-agents" class="tab-pane fade"></div>
287 |
288 |     <!-- Skills Tab -->
289 |     <div id="tab-skills" class="tab-pane fade"></div>
290 |
291 |     <!-- MCP Servers Tab -->
292 |     <div id="tab-mcp" class="tab-pane fade"></div>
293 |
294 |     <!-- AI Inference Benchmark Tab -->
295 |     <div id="tab-ai-benchmark" class="tab-pane fade"></div>
296 | </div>
297 </div>
298 |
299 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    | /js/bootstrap.bundle.min.js"></script>
300 <script src="/html/js/chatService.js?v=20260926_v4"></script>
301 <script src="/html/js/appConfigEditor.js?v=20260926_v4"></scri
    | pt>
302 <script type="module" src="/html/apps/main.js?v=20260926_v4"><
    | /script>
303 |
304 </body>
305 </html>
306 |
```

main.js

```
1 /**
2  * apps/main.js – оркестратор страницы /tc (tc.ps1)
3  * Архитектура: UI_ARCHITECTURE.md
4  */
5
6 import { setupGlobalApi, setupThemeAndLang } from './modules/i
nit-interface.js';
7 import { fetchAppsStatus, updateModelDropdown } from './module
s/status-manager.js';
8 import { APP_TAB_DEFS } from './modules/tabs-config.js';
9 import { applyTranslations } from '../js/i18n.js';
10 import { switchTab, loadTab, setupTabClicks } from '../js/tab-
core.js';
11
12 // Глобальный экспорт
13 window.switchTab = switchTab;
14 window.switchToTab = switchTab;
15
16 // Карта: tabId → пути (из tabs-config, не генерируем)
17 const TAB_PATHS = Object.fromEntries(
18   APP_TAB_DEFS.map(d => [d.tabId, { html: d.html, js: d.js }])
19 );
20
21 // Lazy-загрузка: вкладка грузится только при первом открытии
22 const loadedTabs = new Set();
23
24 async function lazyLoad(tabId) {
25   const normId = tabId.startsWith('tab-') ? tabId : `tab-${tab
Id}`;
26   if (loadedTabs.has(normId)) return;
27   const paths = TAB_PATHS[normId];
28   if (!paths) return;
29   loadedTabs.add(normId);
30   const v = Date.now();
31   const name = normId.replace(/^tab-/, '');
32   await loadTab(name, `${paths.html}?v=${v}`, `${paths.js}?v=$
```

```

    {v}`);
33   applyTranslations();
34 }
35
36 const _switchTab = switchTab;
37 export async function switchAppTab(tabId) {
38   if (!tabId) return;
39   const normId = tabId.startsWith('tab-') ? tabId : `tab-${tab
  Id}`;
40   await lazyLoad(normId);
41   _switchTab(normId);
42 }
43 window.switchTab = switchAppTab;
44 window.switchToTab = switchAppTab;
45
46 // — МЕНЮ —————
  _____
47
48 /**
49  * Создает DOM-элемент иконки для кнопок и редактора (поддержи
  вает bi-*, SVG и эмодзи)
50  * @param {string} iconStr
51  * @param {string} defaultIcon
52  * @returns {HTMLElement}
53  */
54 export function createIconElement(iconStr, defaultIcon = '❏ ')
  {
55   const icon = (iconStr || defaultIcon).trim();
56   if (/^bi-[a-z0-9-]+$/i.test(icon) || icon.startsWith('bi-') |
  | icon.startsWith('bi ')) {
57     const i = document.createElement('i');
58     const iconClass = icon.startsWith('bi ') ? icon : `bi ${ic
  on}`;
59     i.className = iconClass;
60     return i;
61   }
62   if (icon.startsWith('<svg') || icon.startsWith('<i ')) {
63     const span = document.createElement('span');

```

```

64     span.innerHTML = icon;
65     return span;
66 }
67 const span = document.createElement('span');
68 span.textContent = icon;
69 return span;
70 }
71
72 /**
73  * Обработчик прямого запуска программ (например R-Studio)
74  * @param {Object} item
75  * @param {HTMLElement} [iconEl]
76  */
77 async function executeLaunchItem(item, iconEl = null) {
78     const origContent = iconEl ? iconEl.innerHTML : '';
79     if (iconEl) iconEl.innerHTML = '⏏';
80     try {
81         const res = await fetch('/api/recovery/launch', { method:
82         'POST' });
83         const data = await res.json();
84         if (res.ok && data.success) {
85             if (window.toast) {
86                 window.toast.success('R-Studio запущена', data.message
87                 || 'Программа восстановления файлов запущена');
88             }
89         } else {
90             if (window.toast) {
91                 window.toast.error('Ошибка запуска', data.detail || 'Н
92                 е удалось запустить программу');
93             }
94         }
95     } catch (err) {
96         if (window.toast) {
97             window.toast.error('Ошибка связи', err.message);
98         }
99     } finally {
100         if (iconEl) iconEl.innerHTML = origContent;
101     }

```

```
99 | }
100 |
101 | function getMenuTarget() {
102 |     const p = location.pathname.toLowerCase();
103 |     if (p.startsWith('/su') || location.search.includes('target=
    su')) {
104 |         return 'su';
105 |     }
106 |     return 'tc';
107 | }
108 |
109 | async function buildMenu(appsMap = {}, customCfg = null) {
110 |     let cfg = customCfg;
111 |     const target = getMenuTarget();
112 |     if (!cfg) {
113 |         try {
114 |             const r = await fetch(`/api/menu/config?target=${target}
    &t=${Date.now()}`);
115 |             if (r.ok) {
116 |                 cfg = await r.json();
117 |             }
118 |         } catch (e) {
119 |             console.warn(`Не удалось загрузить /api/menu/config?target=${target}, пробуем статический файл`, e);
120 |         }
121 |     }
122 |
123 |     if (!cfg) {
124 |         try {
125 |             const staticFile = target === 'su' ? 'su_menu_config.json' : 'tc_menu_config.json';
126 |             const r = await fetch(`/html/config_menus/${staticFile}
    ?t=${Date.now()}`);
127 |             if (r.ok) {
128 |                 cfg = await r.json();
129 |             }
130 |         } catch (e) {
131 |             console.error('Ошибка загрузки статического файла конфи
```



```

    урации меню', e);
132     }
133   }
134
135   if (!cfg || !cfg.menu) return null;
136
137   const enabled = item => appsMap[item.id]?.enabled !== false;
138   const sorted = arr => [...arr]
139     .filter(x => x.visible !== false && enabled(x))
140     .sort((a, b) => (a.order || 0) - (b.order || 0));
141
142   // Верхнее меню
143   const topEl = document.querySelector('.main-nav-container .d
    -flex.gap-1.flex-wrap');
144   if (topEl) {
145     topEl.innerHTML = '';
146     sorted(cfg.menu.topButtons || []).forEach(item => {
147       const btn = document.createElement('button');
148       btn.className = 'btn btn-sm btn-outline-primary d-flex a
    lign-items-center gap-1.5 py-1 px-2 rounded';
149       btn.type = 'button';
150       btn.title = item.label;
151
152       const iconEl = createIconElement(item.icon, 'bi-app');
153       const labelSpan = document.createElement('span');
154       labelSpan.className = 'd-none d-sm-inline';
155       labelSpan.textContent = item.label;
156       if (item.i18n) labelSpan.dataset.i18n = item.i18n;
157
158       btn.append(iconEl, labelSpan);
159
160       if (item.id === 'file_recovery' || item.action === 'laun
    ch') {
161         btn.dataset.action = 'launch';
162         btn.onclick = async (e) => {
163           e.preventDefault();
164           e.stopPropagation();
165           await executeLaunchItem(item, iconEl);

```

```
166 |         };
167 |     } else {
168 |         btn.dataset.tab = item.tab;
169 |     }
170 |
171 |     topEl.appendChild(btn);
172 | });
173 | }
174 |
175 | // Боковое меню
176 | const navEl = document.getElementById('appsNavTabs');
177 | if (navEl) {
178 |     navEl.innerHTML = '';
179 |     sorted(cfg.menu.sidebarItems || []).forEach(item => {
180 |         const btn = document.createElement('button');
181 |         btn.className = 'list-group-item list-group-item-action
182 | d-flex align-items-center gap-2 py-2 px-3';
183 |         btn.type = 'button';
184 |
185 |         const iconEl = createIconElement(item.icon, ' ');
186 |         iconEl.classList.add('fs-6');
187 |         const label = document.createElement('span');
188 |         label.className = 'fw-medium';
189 |         label.textContent = item.label;
190 |         if (item.i18n) label.dataset.i18n = item.i18n;
191 |
192 |         btn.append(iconEl, label);
193 |
194 |         if (item.id === 'file_recovery' || item.action === 'laun
195 | ch') {
196 |             btn.dataset.action = 'launch';
197 |             btn.title = 'Запуск программы восстановления файлов (R
198 | -Studio)';
199 |             btn.onclick = async (e) => {
200 |                 e.preventDefault();
201 |                 e.stopPropagation();
202 |                 await executeLaunchItem(item, iconEl);
203 |             };
204 |         }
205 |     });
206 | }
```

```

201     } else {
202         btn.dataset.tab = item.tab;
203     }
204
205     navEl.appendChild(btn);
206 });
207 }
208
209 // Обновление активного состояния кнопок вкладок
210 const activePane = document.querySelector('#mainTabContent .
tab-pane.active') || document.querySelector('.tab-pane.active'
);
211 const activeTabId = activePane ? activePane.id : null;
212 if (activeTabId) {
213     document.querySelectorAll('[data-tab]').forEach(btn => {
214         btn.classList.toggle('active', btn.dataset.tab === activ
eTabId);
215     });
216 }
217
218 applyTranslations();
219 return cfg;
220 }
221
222 export async function loadRequiredTabs(tabs = []) {
223     for (const tab of tabs) {
224         await lazyLoad(tab);
225     }
226 }
227
228 // — РЕДАКТОР МЕНЮ —————
229
230 function initMenuEditor(cfg, appsMap = {}) {
231     const editorBtn = document.getElementById('menu-editor-btn')
;
232     const modal = document.getElementById('menuEditorModal');
233     const list = document.getElementById('allMenuEditor');

```

```
234 | const saveBtn = document.getElementById('saveMenuConfig');
235 | if (!editorBtn || !modal || !list || !cfg) return;
236 |
237 | let currentItems = [];
238 | let draggedIdx = null;
239 |
240 | function collectInitialItems() {
241 |     const knownIds = new Set();
242 |     const items = [];
243 |
244 |     (cfg.menu.topButtons || []).forEach(x => {
245 |         knownIds.add(x.id);
246 |         items.push({ ...x, position: x.visible === false ? 'hidden' : 'top' });
247 |     });
248 |
249 |     (cfg.menu.sidebarItems || []).forEach(x => {
250 |         if (!knownIds.has(x.id)) {
251 |             knownIds.add(x.id);
252 |             items.push({ ...x, position: x.visible === false ? 'hidden' : 'bottom' });
253 |         }
254 |     });
255 |
256 |     // Автоподгрузка незарегистрированных вкладок из APP_TAB_DEFS со статусом Скрыть (hidden)
257 |     APP_TAB_DEFS.forEach(def => {
258 |         if (!knownIds.has(def.id)) {
259 |             items.push({
260 |                 id: def.id,
261 |                 label: def.label || def.id.replace(/_/g, ' ').replace(/\\b\\w/g, c => c.toUpperCase()),
262 |                 icon: def.icon || 'bi-app',
263 |                 tab: def.tabId || `tab-${def.tab}`,
264 |                 order: items.length + 1,
265 |                 visible: false,
266 |                 position: 'hidden'
267 |             });

```

```

268         knownIds.add(def.id);
269     }
270 });
271
272     return items;
273 }
274
275 function renderList() {
276     list.innerHTML = '';
277     const total = currentItems.length;
278
279     currentItems.forEach((item, idx) => {
280         const safeId = (item.id || `item_${idx}`).replace(/[^a-zA-Z0-9_-]/g, '');
281         const div = document.createElement('div');
282         div.className = 'menu-editor-item d-flex align-items-center gap-2 p-2 mb-2 rounded border border-secondary-subtle';
283         div.dataset.id = item.id;
284         div.dataset.index = idx;
285         div.draggable = true;
286
287         // 1. Ручка перетаскивания (Drag Handle)
288         const dragHandle = document.createElement('div');
289         dragHandle.className = 'drag-handle-container text-muted px-1';
290         dragHandle.title = 'Перетащите для изменения порядка';
291         dragHandle.innerHTML = '<i class="bi bi-grip-vertical fs-5"></i>';
292
293         // 2. Порядковый номер
294         const orderBadge = document.createElement('span');
295         orderBadge.className = 'badge bg-secondary-subtle text-light border px-2 py-1';
296         orderBadge.textContent = `#${idx + 1}`;
297         orderBadge.style.minWidth = '38px';
298         orderBadge.style.textAlign = 'center';
299
300         // 3. Иконка элемента

```

```

301 |     const iconContainer = document.createElement('div');
302 |     iconContainer.className = 'item-icon d-flex align-items-
    | center justify-content-center px-1 text-center';
303 |     iconContainer.style.minWidth = '30px';
304 |     const iconEl = createIconElement(item.icon, ' ');
305 |     iconContainer.appendChild(iconEl);
306 |
307 |     // 4. Информация об элементе (название и вкладка)
308 |     const info = document.createElement('div');
309 |     info.className = 'item-info flex-grow-1 min-w-0 px-1';
310 |     const labelDiv = document.createElement('div');
311 |     labelDiv.className = 'fw-semibold text-truncate small';
312 |     labelDiv.textContent = item.label;
313 |     const tabDiv = document.createElement('div');
314 |     tabDiv.className = 'text-muted text-truncate font-monosp
    | ace';
315 |     tabDiv.style.fontSize = '0.75rem';
316 |     tabDiv.textContent = item.tab;
317 |     info.append(labelDiv, tabDiv);
318 |
319 |     // 4. Слайдер-перетаскиватель порядка и кнопки перемещен
    | ия
320 |     const orderCtrl = document.createElement('div');
321 |     orderCtrl.className = 'd-flex align-items-center gap-1 m
    | e-2';
322 |     orderCtrl.style.width = '160px';
323 |     orderCtrl.style.flexShrink = '0';
324 |
325 |     const btnUp = document.createElement('button');
326 |     btnUp.type = 'button';
327 |     btnUp.className = 'btn btn-sm btn-outline-secondary p-0
    | px-1';
328 |     btnUp.title = 'Переместить выше';
329 |     btnUp.disabled = idx === 0;
330 |     btnUp.innerHTML = '<i class="bi bi-chevron-up"></i>';
331 |     btnUp.addEventListener('click', () => {
332 |         if (idx > 0) {
333 |             const [moved] = currentItems.splice(idx, 1);

```

```

334 |         currentItems.splice(idk - 1, 0, moved);
335 |         renderList();
336 |     }
337 | });
338 |
339 |     const slider = document.createElement('input');
340 |     slider.type = 'range';
341 |     slider.className = 'form-range order-slider flex-grow-1
    | m-0';
342 |     slider.min = '1';
343 |     slider.max = String(total);
344 |     slider.value = String(idk + 1);
345 |     slider.title = `Слайдер порядка: ${idk + 1} из ${total}`
    | ;
346 |
347 |     slider.addEventListener('input', (e) => {
348 |         const targetIdk = parseInt(e.target.value, 10) - 1;
349 |         if (targetIdk !== idk && targetIdk >= 0 && targetIdk <
    | total) {
350 |             const [moved] = currentItems.splice(idk, 1);
351 |             currentItems.splice(targetIdk, 0, moved);
352 |             renderList();
353 |         }
354 |     });
355 |
356 |     const btnDown = document.createElement('button');
357 |     btnDown.type = 'button';
358 |     btnDown.className = 'btn btn-sm btn-outline-secondary p-
    | 0 px-1';
359 |     btnDown.title = 'Переместить ниже';
360 |     btnDown.disabled = idk === total - 1;
361 |     btnDown.innerHTML = '<i class="bi bi-chevron-down"></i>'
    | ;
362 |     btnDown.addEventListener('click', () => {
363 |         if (idk < total - 1) {
364 |             const [moved] = currentItems.splice(idk, 1);
365 |             currentItems.splice(idk + 1, 0, moved);
366 |             renderList();

```

```

367         }
368     });
369
370     orderCtrl.append(btnUp, slider, btnDown);
371
372     // 5. Переключатели позиции (Сверху / Слева / Скрыть)
373     const group = document.createElement('div');
374     group.className = 'btn-group btn-group-sm flex-shrink-0'
375     ;
376     group.setAttribute('role', 'group');
377     [['top', 'Сверху', 'btn-outline-primary'], ['bottom', 'С
378     лева', 'btn-outline-secondary'], ['hidden', 'Скрыть', 'btn-out
379     line-danger']]
380     .forEach(([val, text, cls]) => {
381         const inp = document.createElement('input');
382         inp.type = 'radio'; inp.className = 'btn-check';
383         inp.name = `pos-${safeId}`; inp.id = `pos-${val}-${s
384         afeId}`; inp.value = val;
385         inp.checked = item.position === val;
386         inp.addEventListener('change', () => {
387             item.position = val;
388             item.visible = val !== 'hidden';
389         });
390         const lbl = document.createElement('label');
391         lbl.className = `btn ${cls}`; lbl.setAttribute('for'
392         , inp.id); lbl.textContent = text;
393         group.append(inp, lbl);
394     });
395
396     // 6. Drag & Drop события
397     div.addEventListener('dragstart', (e) => {
398         draggedIdx = idx;
399         div.classList.add('dragging');
400         e.dataTransfer.effectAllowed = 'move';
401         e.dataTransfer.setData('text/plain', String(idx));
402     });

```



```
400 |         div.addEventListener('dragend', () => {
401 |             div.classList.remove('dragging');
402 |             document.querySelectorAll('.menu-editor-item').forEach
403 |             (el => el.classList.remove('drag-over'));
404 |         });
405 |
406 |         div.addEventListener('dragover', (e) => {
407 |             e.preventDefault();
408 |             e.dataTransfer.dropEffect = 'move';
409 |             div.classList.add('drag-over');
410 |         });
411 |
412 |         div.addEventListener('dragleave', () => {
413 |             div.classList.remove('drag-over');
414 |         });
415 |
416 |         div.addEventListener('drop', (e) => {
417 |             e.preventDefault();
418 |             div.classList.remove('drag-over');
419 |             if (draggedIdx !== null && draggedIdx !== idx) {
420 |                 const [moved] = currentItems.splice(draggedIdx, 1);
421 |                 currentItems.splice(idx, 0, moved);
422 |                 draggedIdx = null;
423 |                 renderList();
424 |             }
425 |         });
426 |
427 |         div.append(dragHandle, orderBadge, info, orderCtrl, group);
428 |         list.appendChild(div);
429 |     });
430 |
431 |     return currentItems;
432 | }
433 |
434 | editorBtn.addEventListener('click', () => {
435 |     currentItems = collectInitialItems();
436 |     renderList();
```

```

436     new bootstrap.Modal(modal).show();
437 });
438
439 saveBtn.addEventListener('click', async () => {
440     cfg.menu.topButtons = currentItems
441         .filter(x => x.position === 'top')
442         .map(({ position, ...x }, idx) => ({ ...x, visible: true
, order: idx + 1 }));
443
444     cfg.menu.sidebarItems = currentItems
445         .filter(x => x.position !== 'top')
446         .map(({ position, ...x }, idx) => ({ ...x, visible: posi
tion === 'bottom', order: idx + 1 }));
447
448     try {
449         const target = getMenuTarget();
450         const r = await fetch(`/api/menu/config?target=${target}
`, {
451             method: 'POST',
452             headers: { 'Content-Type': 'application/json' },
453             body: JSON.stringify(cfg)
454         });
455
456         if (!r.ok) {
457             throw new Error(r.statusText || `HTTP ${r.status}`);
458         }
459
460         // Немедленно переопределяем меню и структуру вкладок в
интерфейсе
461         await buildMenu(appsMap, cfg);
462
463         const modalInstance = bootstrap.Modal.getInstance(modal)
;
464         if (modalInstance) {
465             modalInstance.hide();
466         }
467
468         // Проверяем, активна ли ещё текущая вкладка, если нет –

```

```

    переключаемся на первую доступную
469     const currentTabEl = document.querySelector('.tab-pane.a
ctive');
470     const activeTabId = currentTabEl ? currentTabEl.id : '';
471     const isStillInMenu = document.querySelector(`[data-tab=
"${activeTabId}"]`);
472     if (!isStillInMenu) {
473         const firstVisible = document.querySelector('[data-tab
]');
474         if (firstVisible?.dataset.tab) {
475             await switchAppTab(firstVisible.dataset.tab);
476         }
477     }
478     } catch (e) {
479         window.showToast?.('Ошибка при сохранении меню: ' + e.me
ssage, 'danger') || alert('Ошибка при сохранении меню: ' + e.m
essage);
480     }
481 });
482 }
483
484 // — INIT —————
485
486 async function init() {
487     setupGlobalApi();
488     await setupThemeAndLang();
489
490     if (location.pathname.startsWith('/tc'))
491         document.title = 'AI Breadboard – Test Computer (/tc)';
492
493     // Единый обработчик кликов
494     setupTabClicks();
495
496     // Статус + выпадающий список моделей
497     const statusData = await fetchAppsStatus();
498     window.appsStatusMap = statusData?.apps || {};
499     await updateModelDropdown();

```

```
500 | const appsMap = statusData?.apps || {};
501 |
502 | // Меню из конфига
503 | const cfg = await buildMenu(appsMap).catch(() => null);
504 | initMenuEditor(cfg, appsMap);
505 |
506 | applyTranslations();
507 |
508 | // Активировать вкладку по hash, defaultTab из конфига или п
    | ервую доступную
509 | const hash = location.hash.replace('#', '');
510 | const defaultTab = cfg?.settings?.defaultTab;
511 | const first = document.querySelector('#appsNavTabs [data-tab
    | ]') || document.querySelector('[data-tab]');
512 | const initialTab = hash || (defaultTab && document.querySele
    | ctor('[data-tab="${defaultTab}"]') ? defaultTab : first?.datas
    | et.tab || 'tab-chat');
513 | await switchAppTab(initialTab);
514 | }
515 |
516 | document.readyState === 'loading'
517 |   ? document.addEventListener('DOMContentLoaded', init)
518 |   : init();
519 |
```

init-interface.js

```
1 /**
2  * apps/modules/init-interface.js – Инициализация тем, языка и
   глобального окружения /apps
3  */
4
5 import { initI18n, switchLang, applyTranslations } from '../..
   /js/i18n.js';
6 import { initTheme, setTheme, getThemeMode, getResolvedTheme }
   from '../.. /js/theme.js';
7
8 export function setupGlobalApi() {
9   window.switchLang = switchLang;
10  window.applyTranslations = applyTranslations;
11  window.setTheme = setTheme;
12  window.getThemeMode = getThemeMode;
13  window.getResolvedTheme = getResolvedTheme;
14
15  window.api = window.api || {
16    async fetch(url, options = {}) {
17      const response = await fetch(url, options);
18      if (!response.ok) {
19        let msg = response.statusText;
20        try {
21          const data = await response.json();
22          if (data && data.detail) {
23            if (typeof data.detail === 'string') {
24              msg = data.detail;
25            } else if (Array.isArray(data.detail)) {
26              msg = data.detail.map(d => d.msg || JSON.stringi
27 fy(d)).join(', ');
28            } else {
29              msg = JSON.stringify(data.detail);
30            }
31          } catch {}
32          throw new Error(`${response.status} ${msg}`);
```

```

33     }
34     return response.json();
35 }
36 };
37
38 window.appsStatusMap = window.appsStatusMap || {};
39 }
40
41 export async function setupThemeAndLang() {
42     initTheme();
43
44     const savedLang = localStorage.getItem('app_language') || 'ru';
45     await initI18n(savedLang);
46
47     const langSel = document.getElementById('apps-lang-selector');
48     if (langSel) {
49         langSel.value = savedLang;
50         langSel.addEventListener('change', (e) => {
51             switchLang(e.target.value);
52         });
53     }
54
55     const themeSel = document.getElementById('apps-theme-selector');
56     if (themeSel) {
57         themeSel.value = getThemeMode();
58         themeSel.addEventListener('change', (e) => {
59             setTheme(e.target.value);
60         });
61     }
62 }
63

```

status-manager.js

```
1 /**
2  * apps/modules/status-manager.js – Управление состоянием прил
  | ожений и бейджем/дропдауном модели
3  */
4
5 /**
6  * Получение текущей активной модели из настроек пользователя
  | и активного профиля
7  * @returns {Promise<{provider: string, model: string, display
  | : string, configFile: string}>}
8  */
9 export async function fetchActiveModelInfo() {
10   const isTcRoute = window.location.pathname.startsWith('/tc')
  | ;
11   const query = isTcRoute ? '?profile=tc' : '';
12
13   let provider = '';
14   let model = '';
15   let configFile = '';
16
17   // 1. Проверяем настройки пользователя
18   try {
19     const sResp = await fetch('/auth/settings');
20     if (sResp.ok) {
21       const sData = await sResp.json();
22       if (sData && sData.model) {
23         let raw = sData.model.trim();
24         if (raw.includes(':')) {
25           const parts = raw.split(':');
26           provider = parts[0].toUpperCase();
27           model = parts.slice(1).join(':');
28         } else if (raw.startsWith('agy-')) {
29           provider = 'AGY';
30           model = raw;
31         } else if (raw.toLowerCase().includes('gemini')) {
32           provider = 'GEMINI';
```

```

33         model = raw;
34     } else {
35         provider = 'AI';
36         model = raw;
37     }
38 }
39 }
40 } catch (err) {
41     console.warn('[AppsHub] Could not fetch user settings:', err);
42 }
43
44 // 2. Если модель не получена из профиля, запрашиваем эндпоинт активной модели
45 if (!model) {
46     try {
47         const resp = await fetch(`/api/chat/active-model${query}`);
48         if (resp.ok) {
49             const data = await resp.json();
50             if (data && data.model) {
51                 provider = data.provider || provider || 'AI';
52                 model = data.model;
53                 configFile = data.config_file || '';
54             }
55         }
56     } catch (err) {
57         console.warn('[AppsHub] Could not fetch active model from /api/chat/active-model:', err);
58     }
59 }
60
61 // Очищаем префикс провайдера из имени модели, если он продублирован
62 if (provider && model && model.toLowerCase().startsWith(`${provider.toLowerCase()}:`)) {
63     model = model.substring(provider.length + 1);
64 }

```



```

65
66   const display = (provider && model && model !== 'default')
67     ? `${provider}: ${model}`
68     : (model && model !== 'default')
69       ? model
70       : (provider ? provider : 'AI: Не определена');
71
72   return { provider, model, display, configFile };
73 }
74
75 /**
76  * Получение списка доступных моделей из /api/chat/models
77  * @returns {Promise<Array<{provider: string, model: string, fullId: string, label: string}>>}}
78  */
79 export async function fetchAvailableModels() {
80   try {
81     const resp = await fetch('/api/chat/models');
82     if (!resp.ok) return [];
83
84     const data = await resp.json();
85     const grouped = data.models || {};
86     const result = [];
87
88     for (const [prov, mList] of Object.entries(grouped)) {
89       if (!Array.isArray(mList)) continue;
90       const provUpper = prov.toUpperCase();
91
92       for (const m of mList) {
93         let cleanName = m;
94         if (cleanName.startsWith(`${prov}:`)) cleanName = cleanName.substring(prov.length + 1);
95         else if (cleanName.startsWith('gemini_cli:')) cleanName = cleanName.substring(11);
96         else if (cleanName.startsWith('agy-')) cleanName = cleanName.substring(4);
97         else if (cleanName.startsWith('foundry:')) cleanName = cleanName.substring(8);

```

```
98 |         else if (cleanName.startsWith('ollama:')) cleanName =
| cleanName.substring(7);
99 |         else if (cleanName.startsWith('onnx:')) cleanName = cl
| eanName.substring(5);
100 |
101 |         result.push({
102 |             provider: provUpper,
103 |             model: cleanName,
104 |             fullId: m.includes(':') || m.startsWith('agy-') ? m
| : `${prov}:${m}`,
105 |             label: `${provUpper}: ${cleanName}`
106 |         });
107 |     }
108 | }
109 |
110 |     return result;
111 | } catch (err) {
112 |     console.warn('[AppsHub] Could not fetch available models f
| rom /api/chat/models:', err);
113 |     return [];
114 | }
115 | }
116 |
117 | /**
118 |  * Обновляет выпадающий список и текст кнопки активной модели
119 |  */
120 | export async function updateModelDropdown() {
121 |     const dropdownBtn = document.getElementById('apps-model-drop
| down-btn');
122 |     const dropdownMenu = document.getElementById('apps-model-dro
| pdown-menu');
123 |     const modelText = document.getElementById('apps-model-text')
| ;
124 |
125 |     if (!dropdownBtn && !modelText) return;
126 |
127 |     // 1. Получаем и отображаем текущую активную модель
128 |     const activeInfo = await fetchActiveModelInfo();
```

```
129 |   if (modelText) {
130 |       modelText.textContent = activeInfo.display;
131 |   }
132 |   if (dropdownBtn) {
133 |       const fileLabel = activeInfo.configFile ? `, Профиль: ${activeInfo.configFile}` : '';
134 |       dropdownBtn.title = `Используемая модель: ${activeInfo.model || 'не определена'} (Провайдер: ${activeInfo.provider || 'AI'})${fileLabel}`;
135 |   }
136 |
137 |   if (!dropdownMenu) return;
138 |
139 |   // 2. Загружаем доступные модели для меню
140 |   try {
141 |       const models = await fetchAvailableModels();
142 |
143 |       if (models.length === 0) {
144 |           dropdownMenu.innerHTML = '<li><span class="dropdown-item-text text-muted small px-2">Нет доступных моделей</span></li>';
145 |           return;
146 |       }
147 |
148 |       dropdownMenu.innerHTML = '';
149 |
150 |       // Группируем по провайдерам
151 |       let currentProvider = '';
152 |       models.forEach(({ provider, model, fullId, label }) => {
153 |           if (provider !== currentProvider) {
154 |               if (currentProvider !== '') {
155 |                   const divider = document.createElement('li');
156 |                   divider.innerHTML = '<hr class="dropdown-divider border-secondary my-1">';
157 |                   dropdownMenu.appendChild(divider);
158 |               }
159 |               currentProvider = provider;
160 |               const header = document.createElement('li');
```

```

161 |         header.innerHTML = `

###### </i><stro ng>${model}</strong>` 173 | : `


```

```

193 |         modelText.textContent = `${provider}: ${model}`;
194 |     }
195 |     if (window.toast && typeof window.toast.success ==
    | = 'function') {
196 |         window.toast.success('Модель обновлена', `Активн
    | ая модель: ${provider}: ${model}`);
197 |     }
198 |     // Перерисовываем список для обновления галочки
199 |     await updateModelDropdown();
200 | }
201 | } catch (saveErr) {
202 |     console.error('[AppsHub] Error saving selected model
    | :', saveErr);
203 | }
204 | });
205 |
206 |     li.appendChild(a);
207 |     dropdownMenu.appendChild(li);
208 | });
209 | } catch (err) {
210 |     console.error('[AppsHub] Failed to build model dropdown:',
    | err);
211 |     dropdownMenu.innerHTML = '<li><span class="dropdown-item-t
    | ext text-muted small px-2">Ошибка загрузки моделей</span></li>
    | ';
212 | }
213 | }
214 |
215 | /**
216 |  * Получение статуса приложений и AI-конфига
217 |  */
218 | export async function fetchAppsStatus() {
219 |     const isTcRoute = window.location.pathname.startsWith('/tc')
    | ;
220 |     const query = isTcRoute ? '?profile=tc' : '';
221 |
222 |     let statusData = null;
223 |     const urls = [

```

```
224 |     `/api/apps/status${query}`,
225 |     `/api/admin/apps/status${query}`,
226 |     `/apps/status${query}`
227 | ];
228 |
229 | for (const url of urls) {
230 |     try {
231 |         const resp = await fetch(url);
232 |         if (resp.ok) {
233 |             statusData = await resp.json();
234 |             if (statusData && (statusData.apps || statusData.ai))
235 |             {
236 |                 break;
237 |             }
238 |         } catch {}
239 |     }
240 |
241 |     if (statusData && statusData.apps) {
242 |         window.appsStatusMap = statusData.apps;
243 |     }
244 |
245 |     return statusData;
246 | }
247 |
248 | /**
249 |  * Алиас для совместимости с бейджами
250 |  */
251 | export async function updateModelBadge(statusData) {
252 |     return updateModelDropdown();
253 | }
254 |
```

tab-manager.js

```
1 // Реэкспорт для обратной совместимости
2 export const switchTab = (id) => window.switchTab?.(id);
3 export const setupNavTabs = () => {};
4 export const loadTabContent = async () => {};
5
```

tabs-config.js

```
1 /**
2  * apps/modules/tabs-config.js – Реестр определений вкладок для
  | интерфейса /apps
3  */
4
5 export const APP_TAB_DEFS = [
6   { id: 'about_system', label: 'О Системе', icon: 'i', tab: 'a
  | bout-system', tabId: 'tab-about-system', html: '/html/about_sy
  | stem_tab/index.html?v=20260926_v1', js: '/html/about_system_ta
  | b/main.js?v=20260926_v1' },
7   { id: 'scenarios', label: 'Чат и сценарии', icon: '☐ ', tab:
  | 'scenarios', tabId: 'tab-scenarios', html: '/html/scenarios_t
  | ab/index.html?v=20260925_v2', js: '/html/scenarios_tab/main.js
  | ?v=20260925_v2' },
8   { id: 'chat', label: 'Чат', icon: '☐ ', tab: 'chat', tabId:
  | 'tab-chat', html: '/html/chat/index.html?v=20260923_v5', js: '
  | /html/chat/main.js?v=20260923_v5' },
9   { id: 'network_terminal', label: 'Сетевой терминал', icon: '
  | ☐ ', tab: 'network', tabId: 'tab-network', html: '/html/networ
  | k_tab/index.html', js: '/html/network_tab/main.js' },
10  { id: 'system_inspector', label: 'Потребление ресурсов', ico
  | n: 'bi-graph-up-arrow', tab: 'system-inspector', tabId: 'tab-s
  | ystem-inspector', html: '/html/system_inspector_tab/index.html
  | ?v=20260926_v1', js: '/html/system_inspector_tab/main.js?v=202
  | 60926_v1' },
11  { id: 'windows_sysadmin', label: 'Windows Sysadmin', icon: '
  | bi-server', tab: 'windows-admin', tabId: 'tab-windows-admin',
  | html: '/html/windows_admin_tab/index.html', js: '/html/windows
  | _admin_tab/main.js' },
12  { id: 'system_control_center', label: 'System Control', icon
  | : 'bi-tools', tab: 'system-control', tabId: 'tab-system-contro
  | l', html: '/html/system_control_tab/index.html?v=20260925_v1',
  | js: '/html/system_control_tab/main.js?v=20260925_v1' },
13  { id: 'system_log_viewer', label: 'Системные логи', icon: '☐
  | ☐ ', tab: 'system-logs', tabId: 'tab-system-logs', html: '/html
  | /system_logs_tab/index.html', js: '/html/system_logs_tab/main.
```



```
js' },
14 { id: 'software_audit', label: 'Аудит программ', icon: '☐ ',
    tab: 'software-audit', tabId: 'tab-software-audit', html: '/html/software_audit_tab/index.html', js: '/html/software_audit_tab/main.js' },
15 { id: 'registry_viewer', label: 'Registry Viewer', icon: '☐ ',
    tab: 'registry-viewer', tabId: 'tab-registry-viewer', html: '/html/registry_viewer_tab/index.html', js: '/html/registry_viewer_tab/main.js' },
16 { id: 'windows_defender', label: 'Defender Security', icon: 'bi-shield-lock', tab: 'defender', tabId: 'tab-defender', html: '/html/defender_tab/index.html', js: '/html/defender_tab/main.js' },
17 { id: 'windows_startup_auditor', label: 'Автозапуск программ', icon: '☐ ', tab: 'startup-auditor', tabId: 'tab-startup-auditor', html: '/html/startup_auditor_tab/index.html', js: '/html/startup_auditor_tab/main.js' },
18 { id: 'windows_backup_manager', label: 'Windows Backup', icon: '☐ ', tab: 'windows-backup', tabId: 'tab-windows-backup', html: '/html/windows_backup_tab/index.html?v=20260924_v2', js: '/html/windows_backup_tab/main.js?v=20260924_v2' },
19 { id: 'hardware_monitor', label: 'Мониторинг железа', icon: 'bi-cpu', tab: 'hardware-monitor', tabId: 'tab-hardware-monitor', html: '/html/hardware_monitor_tab/index.html', js: '/html/hardware_monitor_tab/main.js' },
20 { id: 'cloudflared_monitor', label: 'Cloudflared', icon: '☐ ', tab: 'cloudflared', tabId: 'tab-cloudflared', html: '/html/cloudflared_tab/index.html', js: '/html/cloudflared_tab/main.js' },
21 { id: 'google_user_desktop', label: 'Google User Desktop', icon: '☐ ', tab: 'google-desktop', tabId: 'tab-google-desktop', html: '/html/google_desktop_tab/index.html?v=20260928_v1', js: '/html/google_desktop_tab/main.js?v=20260928_v1' },
22 { id: 'gcloud_monitor', label: 'Google Cloud', icon: '☐ ', tab: 'gcloud', tabId: 'tab-gcloud', html: '/html/gcloud_tab/index.html', js: '/html/gcloud_tab/main.js' },
23 { id: 'website_monitor', label: 'Website Intelligence', icon: '☐ ', tab: 'website-monitor', tabId: 'tab-website-monitor',
```

```
html: '/html/website_monitor_tab/index.html', js: '/html/websi
te_monitor_tab/main.js' },
24 { id: 'trading_terminal', label: 'Trading Terminal', icon: '
', tab: 'trading', tabId: 'tab-trading', html: '/html/tradin
g_tab/index.html', js: '/html/trading_tab/main.js' },
25 { id: 'user_assistant', label: 'User Assistant', icon: ' ',
tab: 'user-assistant', tabId: 'tab-user-assistant', html: '/h
tml/user_assistant_tab/index.html', js: '/html/user_assistant_
tab/main.js' },
26 { id: 'helpdesk', label: 'Helpdesk', icon: ' ', tab: 'helpd
esk', tabId: 'tab-helpdesk', html: '/html/helpdesk_tab/index.h
tml', js: '/html/helpdesk_tab/main.js' },
27 { id: 'wikipedia_research', label: 'Wikipedia Research', ico
n: ' ', tab: 'wikipedia-research', tabId: 'tab-wikipedia-rese
arch', html: '/html/wikipedia_research_tab/index.html', js: '/
html/wikipedia_research_tab/main.js' },
28 { id: 'autolog_manager', label: 'Автолог', icon: ' ', tab:
'autolog', tabId: 'tab-autolog', html: '/html/autolog_tab/inde
x.html?v=20260925_v2', js: '/html/autolog_tab/main.js?v=202609
25_v2' },
29 { id: 'software_transparency_scanner', label: 'Transparency
Scanner', icon: ' ', tab: 'software-transparency', tabId: 'ta
b-software-transparency', html: '/html/software_transparency_t
ab/index.html?v=20260926_v2', js: '/html/software_transparency
_tab/main.js?v=20260926_v2' },
30 { id: 'user_directories', label: 'Пользовательские папки', i
con: ' ', tab: 'user-directories', tabId: 'tab-user-directori
es', html: '/html/user_directories_tab/index.html', js: '/html
/user_directories_tab/main.js' },
31 { id: 'ninite_updater', label: 'Ninite Updater', icon: ' ',
tab: 'ninite-updater', tabId: 'tab-ninite-updater', html: '/h
tml/ninite_updater_tab/index.html?v=20260924_v2', js: '/html/n
inite_updater_tab/main.js?v=20260924_v2' },
32 { id: 'file_recovery', label: 'Восстановление файлов', icon:
' ', tab: 'file-recovery', tabId: 'tab-file-recovery', html:
'/html/file_recovery_tab/index.html', js: '/html/file_recover
y_tab/main.js' },
33 { id: 'file_history_ai_search', label: 'Поиск истории файлов
```

```
AI', icon: '🔍', tab: 'file-history-search', tabId: 'tab-file-  
-history-search', html: '/html/file_history_ai_search_tab/inde  
x.html?v=20260928_v1', js: '/html/file_history_ai_search_tab/m  
ain.js?v=20260928_v1' },  
34 { id: 'telemetry_history', label: 'История телеметрии', icon  
: 'bi-graph-up', tab: 'telemetry-history', tabId: 'tab-telemet  
ry-history', html: '/html/telemetry_history_tab/index.html?v=2  
0260924_v1', js: '/html/telemetry_history_tab/main.js?v=202609  
24_v1' },  
35 { id: 'telemetry_research', label: 'Анализ телеметрии', icon  
: '🔍', tab: 'telemetry-research', tabId: 'tab-telemetry-resea  
rch', html: '/html/telemetry_research_tab/index.html?v=2026092  
4_v1', js: '/html/telemetry_research_tab/main.js?v=20260924_v1  
' },  
36 { id: 'process_leaks', label: 'Утечки процессов', icon: '🔍',  
tab: 'process-leaks', tabId: 'tab-process-leaks', html: '/htm  
l/process_leaks_tab/index.html?v=20260924_v1', js: '/html/proc  
ess_leaks_tab/main.js?v=20260924_v1' },  
37 { id: 'forensics', label: 'Форензика', icon: '🔍', tab: 'for  
ensics', tabId: 'tab-forensics', html: '/html/forensics_tab/in  
dex.html?v=20260924_v1', js: '/html/forensics_tab/main.js?v=20  
260924_v1' },  
38 { id: 'throttling', label: 'Троттлинг', icon: '🔍', tab: 'th  
rottling', tabId: 'tab-throttling', html: '/html/throttling_ta  
b/index.html?v=20260924_v1', js: '/html/throttling_tab/main.js  
?v=20260924_v1' },  
39 { id: 'storage_wear', label: 'Износ дисков', icon: '🔍', tab  
: 'storage-wear', tabId: 'tab-storage-wear', html: '/html/stor  
age_wear_tab/index.html?v=20260924_v1', js: '/html/storage_wea  
r_tab/main.js?v=20260924_v1' },  
40 { id: 'peripherals', label: 'Периферия', icon: '🔍', tab: 'p  
eripherals', tabId: 'tab-peripherals', html: '/html/peripheral  
s_tab/index.html?v=20260924_v1', js: '/html/peripherals_tab/ma  
in.js?v=20260924_v1' },  
41 { id: 'rag', label: 'RAG Знания', icon: '🔍', tab: 'rag', ta  
bId: 'tab-rag', html: '/html/rag_tab/index.html', js: '/html/r  
ag_tab/main.js' },  
42 { id: 'pixelrag', label: 'PixelRAG', icon: '🔍', tab: 'pixel
```

```
rag', tabId: 'tab-pixelrag', html: '/html/pixelrag_tab/index.html', js: '/html/pixelrag_tab/main.js' },
43 { id: 'models', label: 'Модели ИИ', icon: '🤖', tab: 'models', tabId: 'tab-models', html: '/html/models_tab/index.html', js: '/html/models_tab/main.js' },
44 { id: 'agents', label: 'Агенты ИИ', icon: '🤖', tab: 'agents', tabId: 'tab-agents', html: '/html/agents_tab/index.html', js: '/html/agents_tab/main.js' },
45 { id: 'skills', label: 'Навыки ИИ', icon: '🤖', tab: 'skills', tabId: 'tab-skills', html: '/html/skills_tab/index.html', js: '/html/skills_tab/main.js' },
46 { id: 'mcp', label: 'MCP Серверы', icon: '🤖', tab: 'mcp', tabId: 'tab-mcp', html: '/html/mcp_tab/index.html', js: '/html/mcp_tab/main.js' },
47 { id: 'ai_benchmark', label: 'AI Benchmark', icon: '🤖', tab: 'ai-benchmark', tabId: 'tab-ai-benchmark', html: '/html/ai_benchmark_tab/index.html?v=20260926_v1', js: '/html/ai_benchmark_tab/main.js?v=20260926_v1' },
48 ];
49
50 export const TC_EXCLUDES = new Set([
51   'cloudflared_monitor',
52   'gcloud_monitor',
53   'website_monitor',
54   'user_assistant',
55   'wikipedia_research',
56   'trading_terminal',
57   'helpdesk'
58 ]);
59
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Applications Auto-Logging and Telemetry Sensors
5 Web UI Tab
6 |=====
7 |=====
8 Description:
9 |    Модульный интерфейс управления периодическим автологгирова
10 |нием приложений,
11 |    сенсорами телеметрии оборудования, инспекции и скачивания
12 |CSV-логов.
13 |
14 |File: index.html
15 |Project: AI-Breadboard
16 |Module: WebInterface.AutoLogTab
17 |Author: hypo69
18 |Copyright: © 2026 hypo69
19 |=====
20 |=====
21 -->
22 <div class="container-fluid p-2" id="autolog-tab-container">
23 |
24 |    <!-- Compact Header & Summary Banner -->
25 |    <div class="card shadow-sm border-secondary-subtle mb-3" sty
26 |le="background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
27 |0%); border-radius: 10px;">
28 |        <div class="card-body p-3">
29 |            <div class="d-flex flex-wrap align-items-center justify-
30 |content-between gap-2 mb-3">
31 |                <div class="d-flex align-items-center gap-2">
32 |                    <div class="fs-3">❏ </div>
33 |                    <div>
34 |                        <h5 class="mb-0 fw-bold text-white">Управление авт
35 |ологгированием и сенсорами телеметрии</h5>
36 |                        <small class="text-muted">Единый конфигурационный
```

профиль телеметрии и периодический сбор в CSV (`%APPDATA%/AI-Breadboard/apps/logs`)

```
27         </div>
28     </div>
29
30     <div class="d-flex align-items-center gap-2 flex-wrap"
31 >
32         <span id="autolog-engine-badge" class="badge bg-secondary d-flex align-items-center gap-1.5 px-3 py-2 fs-6">
33             <span class="spinner-grow spinner-grow-sm" role="status" aria-hidden="true" style="width: 0.75rem; height: 0.75rem;"></span>
34             <span id="autolog-engine-badge-text">Проверка статуса...</span>
35         </span>
36
37         <button id="btn-autolog-poll-all" class="btn btn-sm btn-outline-info d-flex align-items-center gap-1 shadow-sm" title="Немедленно выполнить опрос всех логов">
38             <i class="bi bi-arrow-repeat"></i>
39             <span>Опросить все сейчас</span>
40         </button>
41
42         <button id="btn-autolog-toggle-engine" class="btn btn-sm btn-outline-success d-flex align-items-center gap-1 shadow-sm" title="Запустить / Остановить движок">
43             <i class="bi bi-power"></i>
44             <span id="btn-autolog-toggle-text">Запустить</span>
45         </button>
46
47         <button id="btn-autolog-save-config" class="btn btn-sm btn-primary d-flex align-items-center gap-1 shadow-sm" title="Сохранить настройки в конфигурационный файл">
48             <i class="bi bi-floppy-fill"></i>
49             <span>Сохранить настройки</span>
50         </button>
51     </div>
```

```
51 |         </div>
52 |
53 |         <!-- Quick Metrics Counters -->
54 |         <div class="row g-2">
55 |             <div class="col-6 col-md-3">
56 |                 <div class="p-2 rounded border border-secondary" sty
| le="background: rgba(15, 23, 42, 0.6);">
57 |                     <div class="text-muted small text-uppercase" style
| ="font-size: 0.7rem;">Глобальное автологгирование</div>
58 |                     <div class="d-flex align-items-center justify-cont
| ent-between mt-1">
59 |                         <span id="autolog-master-status-text" class="fw-
| bold text-info">Включено</span>
60 |                         <div class="form-check form-switch mb-0">
61 |                             <input class="form-check-input" type="checkbox
| " role="switch" id="switch-enable-autolog" checked>
62 |                         </div>
63 |                     </div>
64 |                 </div>
65 |             </div>
66 |
67 |             <div class="col-6 col-md-3">
68 |                 <div class="p-2 rounded border border-secondary" sty
| le="background: rgba(15, 23, 42, 0.6);">
69 |                     <div class="text-muted small text-uppercase" style
| ="font-size: 0.7rem;">Активных фоновых задач</div>
70 |                     <div class="fw-bold text-white fs-5 mt-1" id="metr
| ic-active-tasks">0</div>
71 |                 </div>
72 |             </div>
73 |
74 |             <div class="col-6 col-md-3">
75 |                 <div class="p-2 rounded border border-secondary" sty
| le="background: rgba(15, 23, 42, 0.6);">
76 |                     <div class="text-muted small text-uppercase" style
| ="font-size: 0.7rem;">Сенсоров телеметрии</div>
77 |                     <div class="fw-bold text-success fs-5 mt-1" id="me
| tric-active-sensors">0</div>
```

```

78         </div>
79     </div>
80
81     <div class="col-6 col-md-3">
82         <div class="p-2 rounded border border-secondary" sty
83         le="background: rgba(15, 23, 42, 0.6);">
84             <div class="text-muted small text-uppercase" style
85             ="font-size: 0.7rem;">Конфигурационный файл</div>
86             <div class="fw-bold text-warning fs-6 mt-1 text-tr
87             uncate" id="metric-active-config-file">~autolog_sensors.json</
88             div>
89         </div>
90     </div>
91
92     <!-- Main View Navigation Pills -->
93     <ul class="nav nav-tabs border-secondary mb-3" id="autologTa
94     bSections" role="tablist">
95         <li class="nav-item" role="presentation">
96             <button class="nav-link active d-flex align-items-center
97             gap-1.5 py-1.5 px-3" id="subtab-loggers-btn" data-bs-toggle="
98             tab" data-bs-target="#subtab-loggers" type="button" role="tab"
99             >
100                 <i class="bi bi-sliders2-vertical text-info"></i>
101                 <span>Логгеры приложений</span>
102             </button>
103         </li>
104         <li class="nav-item" role="presentation">
105             <button class="nav-link d-flex align-items-center gap-1.
106             5 py-1.5 px-3" id="subtab-sensors-btn" data-bs-toggle="tab" da
107             ta-bs-target="#subtab-sensors" type="button" role="tab">
108                 <i class="bi bi-cpu text-warning"></i>
109                 <span>Сенсоры телеметрии</span>
110                 <span class="badge bg-secondary ms-1" id="badge-sensor
111                 s-count">0</span>

```



```

105         </button>
106     </li>
107     <li class="nav-item" role="presentation">
108         <button class="nav-link d-flex align-items-center gap-1.
109 5 py-1.5 px-3" id="subtab-telemetry-service-btn" data-bs-toggl
110 e="tab" data-bs-target="#subtab-telemetry-service" type="butto
111 n" role="tab">
112         <i class="bi bi-activity text-success"></i>
113         <span>Служба телеметрии (ai-telemetry)</span>
114         <span class="badge bg-secondary ms-1" id="badge-teleme
115 try-status">Проверка...</span>
116     </button>
117 </li>
118 </ul>
119
120 <!-- Tab Content Panes -->
121 <div class="tab-content" id="autologTabSectionsContent">
122
123     <!-- Section 1: Loggers Table & Frequency Configuration --
124     >
125     <div class="tab-pane fade show active" id="subtab-loggers"
126 role="tabpanel">
127         <div class="card shadow-sm border-secondary-subtle">
128             <div class="card-header bg-dark d-flex flex-wrap justi
129 fy-content-between align-items-center gap-2 py-2 px-3 border-s
130 econdary">
131                 <div class="d-flex align-items-center gap-2">
132                     <span class="fw-semibold text-white small">Логгеры
133 приложений и периодичность</span>
134                     <span class="badge bg-secondary" id="badge-total-l
135 oggers">0</span>
136                 </div>
137
138                 <div class="d-flex align-items-center gap-2">
139                     <div class="input-group input-group-sm" style="max
140 -width: 250px;">
141                         <span class="input-group-text bg-dark text-muted
142 border-secondary"><i class="bi bi-search"></i></span>

```

```

131 |         <input type="text" id="input-search-loggers" cla
      | ss="form-control bg-dark text-white border-secondary" placehol
      | der="Поиск приложения...">
132 |     </div>
133 |     <button id="btn-refresh-loggers" class="btn btn-sm
      | btn-outline-secondary" title="Обновить список">
134 |         <i class="bi bi-arrow-clockwise"></i>
135 |     </button>
136 | </div>
137 | </div>
138 |
139 |     <div class="table-responsive" style="max-height: 520px
      | ;">
140 |         <table class="table table-dark table-hover table-bor
      | dered align-middle mb-0 small" id="table-loggers">
141 |             <thead class="table-secondary text-nowrap sticky-t
      | op">
142 |                 <tr>
143 |                     <th style="width: 45px;" class="text-center">В
      | кл</th>
144 |                     <th style="min-width: 180px;">Приложение / Лог
      | гер</th>
145 |                     <th style="min-width: 200px;">Интервал опроса<
      | /th>
146 |                     <th style="width: 100px;" class="text-center">
      | Опросов</th>
147 |                     <th style="min-width: 170px;">Последний опрос<
      | /th>
148 |                     <th style="width: 170px;" class="text-center">
      | Действия</th>
149 |                 </tr>
150 |             </thead>
151 |             <tbody id="tbody-loggers">
152 |                 <tr>
153 |                     <td colspan="6" class="text-center text-muted
      | py-4">
154 |                         <div class="spinner-border spinner-border-sm
      | text-info me-2" role="status"></div>

```

```

155 |             Загрузка списка логов...
156 |         </td>
157 |     </tr>
158 | </tbody>
159 | </table>
160 | </div>
161 |
162 |     <div class="card-footer bg-dark border-secondary p-2 d
    | -flex justify-content-between align-items-center flex-wrap gap
    | -2">
163 |         <small class="text-muted">
164 |             <i class="bi bi-info-circle me-1"></i>
165 |             Изменения сохраняются в единый конфигурационный фа
    | йл и мгновенно перезапускают фоновые воркеры.
166 |         </small>
167 |         <button id="btn-autolog-save-config-bottom" class="b
    | tn btn-sm btn-primary d-flex align-items-center gap-1 shadow-s
    | m">
168 |             <i class="bi bi-floppy-fill"></i>
169 |             <span>Сохранить изменения</span>
170 |         </button>
171 |     </div>
172 | </div>
173 | </div>
174 |
175 | <!-- Section 2: Telemetry Sensors Visual Editor -->
176 | <div class="tab-pane fade" id="subtab-sensors" role="tabpa
    | nel">
177 |     <div class="card shadow-sm border-secondary-subtle mb-3"
    | >
178 |         <div class="card-header bg-dark d-flex flex-wrap justi
    | fy-content-between align-items-center gap-2 py-2 px-3 border-s
    | econdary">
179 |             <div class="d-flex align-items-center gap-2">
180 |                 <span class="fw-semibold text-white small">□ Глоба
    | льные параметры сбора телеметрии</span>
181 |             </div>
182 |         </div>

```

```
183 |         <div class="card-body bg-dark text-white p-3">
184 |             <div class="row g-3">
185 |                 <div class="col-12 col-md-4">
186 |                     <div class="form-check form-switch mb-2">
187 |                         <input class="form-check-input" type="checkbox"
188 | " role="switch" id="opt-collect-inventory" checked>
189 |                         <label class="form-check-label small" for="opt-
190 | -collect-inventory">Инвентаризация оборудования (WMI/Win32)</l
191 | abel>
192 |                     </div>
193 |                     <div class="form-check form-switch">
194 |                         <input class="form-check-input" type="checkbox"
195 | " role="switch" id="opt-collect-serials" checked>
196 |                         <label class="form-check-label small" for="opt-
197 | -collect-serials">Сбор серийных номеров накопителей/CPU</label
198 | >
199 |                     </div>
200 |                 </div>
201 |                 <div class="col-12 col-md-4">
202 |                     <div class="form-check form-switch mb-2">
203 |                         <input class="form-check-input" type="checkbox"
204 | " role="switch" id="opt-collect-file-events" checked>
205 |                         <label class="form-check-label small" for="opt-
206 | -collect-file-events">Мониторинг файловых событий</label>
207 |                     </div>
208 |                     <div class="d-flex align-items-center gap-2">
209 |                         <label class="small text-muted text-nowrap mb-
210 | 0" for="opt-max-file-size">Макс. лог (MB):</label>
211 |                         <input type="number" id="opt-max-file-size" cl
212 | ass="form-control form-control-sm bg-dark text-white border-se
213 | condary" style="max-width: 100px;" value="50" min="1" max="100
214 | 0">
215 |                     </div>
216 |                 </div>
217 |             </div>
218 |             <div class="col-12 col-md-4">
219 |                 <label class="small text-muted mb-1" for="opt-wa
220 | tch-dirs">Отслеживаемые директории (через запятую):</label>
221 |                 <input type="text" id="opt-watch-dirs" class="fo
```

```

rm-control form-control-sm bg-dark text-white border-secondary
" value="C:\Users\">
208     </div>
209     </div>
210     </div>
211     </div>
212
213     <div class="card shadow-sm border-secondary-subtle">
214         <div class="card-header bg-dark d-flex flex-wrap justi
fy-content-between align-items-center gap-2 py-2 px-3 border-s
econdary">
215             <div class="d-flex align-items-center gap-2">
216                 <span class="fw-semibold text-white small">☐ Сенс
оры оборудования и метрики телеметрии</span>
217             </div>
218             <button id="btn-refresh-sensors" class="btn btn-sm b
tn-outline-secondary" title="Обновить сенсоры">
219                 <i class="bi bi-arrow-clockwise"></i>
220             </button>
221         </div>
222
223         <div class="card-body p-3 bg-dark">
224             <div class="row g-3" id="sensors-cards-container">
225                 <div class="col-12 text-center text-muted py-4">
226                     <div class="spinner-border spinner-border-sm tex
t-info me-2" role="status"></div>
227                     Загрузка конфигурации сенсоров...
228                 </div>
229             </div>
230         </div>
231
232         <div class="card-footer bg-dark border-secondary p-2 d
-flex justify-content-between align-items-center flex-wrap gap
-2">
233             <small class="text-muted">
234                 <i class="bi bi-info-circle me-1"></i>
235                 Нажмите на бейджи метрик для включения/выключения
отдельных параметров сбора.

```

[illegible]

```
262 |         </button>
263 |         <button id="btn-telemetry-restart" class="btn btn-
    | sm btn-outline-warning d-flex align-items-center gap-1 shadow-
    | sm" title="Перезапустить процесс">
264 |             <i class="bi bi-arrow-clockwise"></i>
265 |             <span>Перезапуск</span>
266 |         </button>
267 |     </div>
268 | </div>
269 |
270 | <div class="card-body bg-dark text-white p-3">
271 |     <div class="row g-3">
272 |         <div class="col-12 col-md-3">
273 |             <div class="p-2 rounded border border-secondary"
    | style="background: rgba(15, 23, 42, 0.6);">
274 |                 <div class="text-muted small text-uppercase" s
    | tyle="font-size: 0.7rem;">Активные процессы</div>
275 |                 <div class="fw-bold text-info fs-6 mt-1 text-t
    | runcate" id="svc-info-pids">—</div>
276 |             </div>
277 |         </div>
278 |         <div class="col-12 col-md-3">
279 |             <div class="p-2 rounded border border-secondary"
    | style="background: rgba(15, 23, 42, 0.6);">
280 |                 <div class="text-muted small text-uppercase" s
    | tyle="font-size: 0.7rem;">Потребление памяти (RAM)</div>
281 |                 <div class="fw-bold text-success fs-6 mt-1" id
    | ="svc-info-memory">— МБ</div>
282 |             </div>
283 |         </div>
284 |         <div class="col-12 col-md-3">
285 |             <div class="p-2 rounded border border-secondary"
    | style="background: rgba(15, 23, 42, 0.6);">
286 |                 <div class="text-muted small text-uppercase" s
    | tyle="font-size: 0.7rem;">Снимков в базе SQLite</div>
287 |                 <div class="fw-bold text-warning fs-6 mt-1" id
    | ="svc-info-snapshots">—</div>
288 |             </div>
```

```
289 |         </div>
290 |         <div class="col-12 col-md-3">
291 |             <div class="p-2 rounded border border-secondary"
| style="background: rgba(15, 23, 42, 0.6);">
292 |                 <div class="text-muted small text-uppercase" s
| tyle="font-size: 0.7rem;">Windows Task Scheduler</div>
293 |                 <div class="fw-bold fs-6 mt-1" id="svc-info-sc
| heduler">--</div>
294 |             </div>
295 |         </div>
296 |     </div>
297 |
298 |     <div class="d-flex flex-wrap gap-2 mt-3 pt-2 border-
| top border-secondary align-items-center">
299 |         <span class="small text-muted">Планировщик заданий
| :</span>
300 |         <button id="btn-telemetry-install-task" class="btn
| btn-sm btn-outline-info d-flex align-items-center gap-1 shado
| w-sm" title="Зарегистрировать в Task Scheduler с флагом WakeTo
| Run">
301 |             <i class="bi bi-calendar-plus"></i>
302 |             <span>Установить в Task Scheduler (WakeToRun)</s
| pan>
303 |         </button>
304 |         <button id="btn-telemetry-uninstall-task" class="b
| tn btn-sm btn-outline-secondary d-flex align-items-center gap-
| 1 shadow-sm" title="Удалить из Task Scheduler">
305 |             <i class="bi bi-trash"></i>
306 |             <span>Удалить из Task Scheduler</span>
307 |         </button>
308 |     </div>
309 | </div>
310 | </div>
311 |
312 | <!-- Mode & Intervals Configuration Card -->
313 | <div class="card shadow-sm border-secondary-subtle">
314 |     <div class="card-header bg-dark d-flex flex-wrap justi
| fy-content-between align-items-center gap-2 py-2 px-3 border-s
```



```

econdary">
315     <div class="d-flex align-items-center gap-2">
316         <span class="fw-semibold text-white small">□ Конфи
    гурация режимов и периодичности сбора</span>
317     </div>
318 </div>
319
320 <div class="card-body bg-dark text-white p-3">
321     <div class="row g-4">
322         <!-- Режим сбора -->
323         <div class="col-12 col-md-4 border-end border-seco
    ndary">
324             <label class="form-label fw-bold small text-info
    text-uppercase">Режим работы службы</label>
325             <div class="d-flex flex-column gap-2 mt-1">
326                 <div class="form-check">
327                     <input class="form-check-input" type="radio"
    name="telemetryModeRadio" id="mode-minimal" value="minimal">
328                     <label class="form-check-label small" for="m
    ode-minimal">
329                         <span class="fw-bold text-success">Minimal
    (Ультралегковесный)</span>
330                         <div class="text-muted" style="font-size:
    0.75rem;">Только базовая телеметрия (CPU, RAM, GPU, диски, сет
    ь, топ процессов). 28 мс замер, ~120 МБ RAM, 0% CPU. Лишние би
    блиотеки отключены.</div>
331                     </label>
332                 </div>
333                 <div class="form-check mt-2">
334                     <input class="form-check-input" type="radio"
    name="telemetryModeRadio" id="mode-hybrid" value="hybrid" che
    cked>
335                     <label class="form-check-label small" for="m
    ode-hybrid">
336                         <span class="fw-bold text-warning">Hybrid
    (Адаптивный гибридный)</span>
337                     <div class="text-muted" style="font-size:
    0.75rem;">Постоянный ультрабыстрый минимальный сбор + периодич

```

еский опрос ресурсоемких сенсоров (LHM, SMART, сеть) с заданным интервалом.</div>

338 </label>

339 </div>

340 <div class="form-check mt-2">

341 <input class="form-check-input" type="radio"
name="telemetryModeRadio" id="mode-full" value="full">

342 <label class="form-check-label small" for="mode-full">

343 Full (Полный стек)

344 <div class="text-muted" style="font-size: 0.75rem;">Опрос всех подключенных аппаратных сенсоров, LHM и с
пидтестов на каждом тике.</div>

345 </label>

346 </div>

347 </div>

348 </div>

349

350 <!-- Интервалы опроса -->

351 <div class="col-12 col-md-4 border-end border-secondary">

352 <label class="form-label fw-bold small text-info text-uppercase">Интервалы опроса (секунды)</label>

353

354 <div class="mb-3">

355 <label for="cfg-fast-interval" class="form-label small mb-1">

356 Быстрый сбор базовых метрик: 5с

357 </label>

358 <div class="input-group input-group-sm">

359 <input type="number" class="form-control bg-dark text-white border-secondary" id="cfg-fast-interval" value="5.0" min="1" max="60" step="0.5">

360 секунд

361 </div>

```

362 |         <small class="text-muted" style="font-size: 0.
      | 72rem;">Частота выборки CPU, RAM, дисков, сети и топа процессо
      | в.</small>
363 |     </div>
364 |
365 |     <div class="mb-3">
366 |         <label for="cfg-heavy-interval" class="form-la
      | bel small mb-1">
367 |             Периодический тяжелый опрос: <span id="val-h
      | eavy-interval" class="fw-bold text-warning">60с</span>
368 |         </label>
369 |         <div class="input-group input-group-sm">
370 |             <input type="number" class="form-control bg-
      | dark text-white border-secondary" id="cfg-heavy-interval" valu
      | e="60.0" min="10" max="3600" step="5">
371 |             <span class="input-group-text bg-dark text-m
      | uted border-secondary">секунд</span>
372 |         </div>
373 |         <small class="text-muted" style="font-size: 0.
      | 72rem;">Применяется в режиме Hybrid для опроса тяжелых сенсоро
      | в.</small>
374 |     </div>
375 |
376 |     <div>
377 |         <label for="cfg-top-processes" class="form-lab
      | el small mb-1">Топ активных процессов:</label>
378 |         <select class="form-select form-select-sm bg-d
      | ark text-white border-secondary" id="cfg-top-processes">
379 |             <option value="5">Топ 5 процессов</option>
380 |             <option value="10" selected>Топ 10 процессов
      | (рекомендуется)</option>
381 |             <option value="20">Топ 20 процессов</option>
382 |         </select>
383 |     </div>
384 | </div>
385 |
386 | <!-- Тяжелые сенсоры в гибридном цикле -->
387 | <div class="col-12 col-md-4">

```

```
388         <label class="form-label fw-bold small text-info
      text-uppercase">Тяжелые сенсоры (Hybrid опрос)</label>
389         <div class="d-flex flex-column gap-2 mt-1">
390             <div class="form-check form-switch">
391                 <input class="form-check-input" type="checkbox"
      role="switch" id="h-sensor-lhm" checked>
392                 <label class="form-check-label small" for="h
      -sensor-lhm">
393                     Датчики LibreHardwareMonitor
394                     <div class="text-muted" style="font-size:
      0.72rem;">Температуры VRM/чипсета, обороты кулеров, вольтаж. <
      /div>
395                 </label>
396             </div>
397             <div class="form-check form-switch mt-1">
398                 <input class="form-check-input" type="checkbox"
      role="switch" id="h-sensor-smart" checked>
399                 <label class="form-check-label small" for="h
      -sensor-smart">
400                     SMART и износ накопителей
401                     <div class="text-muted" style="font-size:
      0.72rem;">Оценка ресурса SSD и SMART-атрибутов дисков.</div>
402                 </label>
403             </div>
404             <div class="form-check form-switch mt-1">
405                 <input class="form-check-input" type="checkbox"
      role="switch" id="h-sensor-ping" checked>
406                 <label class="form-check-label small" for="h
      -sensor-ping">
407                     Сетевой пинг и задержки
408                     <div class="text-muted" style="font-size:
      0.72rem;">Пинг до сетевого шлюза и DNS.</div>
409                 </label>
410             </div>
411             <div class="form-check form-switch mt-1">
412                 <input class="form-check-input" type="checkbox"
      role="switch" id="h-sensor-inventory">
413                 <label class="form-check-label small" for="h
```

```

    -sensor-inventory">
414         Инвентаризация WMI
415         <div class="text-muted" style="font-size:
0.72rem;">Сбор серийных номеров и спецификаций оборудования.</
div>
416         </label>
417     </div>
418 </div>
419 </div>
420 </div>
421 </div>
422
423     <div class="card-footer bg-dark border-secondary p-2 d
-flex justify-content-between align-items-center flex-wrap gap
-2">
424         <small class="text-muted">
425             <i class="bi bi-info-circle me-1"></i>
426             Конфигурация сохраняется в <code>apps/windows/tele
metry/config.json</code> и немедленно применяется службой.
427         </small>
428         <button id="btn-save-telemetry-config" class="btn bt
n-sm btn-primary d-flex align-items-center gap-1 shadow-sm">
429             <i class="bi bi-floppy-fill"></i>
430             <span>Сохранить настройки телеметрии</span>
431         </button>
432     </div>
433 </div>
434 </div>
435
436 </div>
437
438 <!-- Notification Toast Container -->
439 <div class="toast-container position-fixed bottom-0 end-0 p-
3" style="z-index: 1090;">
440     <div id="autolog-toast" class="toast align-items-center te
xt-white bg-dark border border-secondary shadow-lg" role="aler
t" aria-live="assertive" aria-atomic="true">
441         <div class="d-flex">

```

```
442 |         <div class="toast-body d-flex align-items-center gap-2  
    | " id="autolog-toast-body">  
443 |             <i class="bi bi-info-circle-fill text-info" id="auto  
    | log-toast-icon"></i>  
444 |             <span id="autolog-toast-msg">Сообщение</span>  
445 |         </div>  
446 |         <button type="button" class="btn-close btn-close-white  
    | me-2 m-auto" data-bs-dismiss="toast" aria-label="Закреть"></b  
    | utton>  
447 |     </div>  
448 | </div>  
449 | </div>  
450 |  
451 | </div>  
452 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Auto-Logging and Telemetry Sensors Interface
  Controller
4  * =====
5  * Description:
6  *   Модульный контроллер вкладки управления автологгированием
  и телеметрией в /tc.
7  *   Обеспечивает опрос REST API (/api/autolog), сохранение об
  ъединенной конфигурации,
8  *   графическое редактирование сенсоров, метрик, JSON-редакто
  р и выгрузку CSV-файлов.
9  *
10 * File: main.js
11 * Project: AI-Breadboard
12 * Module: WebInterface.AutoLogTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17
18 (function () {
19   'use strict';
20
21   console.log(' [AutoLogTab] Initializing Auto-Logging & Sen
  sors Controller...');
22
23   // Preset intervals for quick selection
24   const INTERVAL_PRESETS = [
25     { label: '5 сек (5 seconds)', value: '5 seconds' },
26     { label: '10 сек (10 seconds)', value: '10 seconds' },
27     { label: '30 сек (30 seconds)', value: '30 seconds' },
28     { label: '1 мин (1 minute)', value: '1 minute' },
```

```

29     { label: '5 мин (5 minutes)', value: '5 minutes' },
30     { label: '10 мин (10 minutes)', value: '10 minutes' },
31     { label: '30 мин (30 minutes)', value: '30 minutes' },
32     { label: '1 час (1 hour)', value: '1 hour' },
33     { label: '6 часов (6 hours)', value: '6 hours' },
34     { label: '24 часа (24 hours)', value: '24 hours' },
35     { label: '1 день (1 day)', value: '1 day' },
36     { label: '7 дней (7 days)', value: '7 days' },
37 ];
38
39 // App readable metadata (icons and titles)
40 const APP_META = {
41     system_inspector: { icon: '🔍', title: 'System Inspector (
Потребление ресурсов)' },
42     hardware_monitor: { icon: '🔧', title: 'Hardware Sensors (Д
атчики оборудования)' },
43     librehardwaremonitor: { icon: '📡', title: 'LibreHardwareM
onitor API' },
44     website_monitor: { icon: '🌐', title: 'Website Intelligenc
e & Heartbeat' },
45     gcloud_monitor: { icon: '☁️', title: 'Google Cloud Observab
ility' },
46     cloudflared_monitor: { icon: '🌐', title: 'Cloudflared Tun
nel Supervisor' },
47     windows_sysadmin: { icon: '🖥️', title: 'Windows Sysadmin Se
rvices' },
48     windows_defender: { icon: '🛡️', title: 'Windows Defender S
ecurity' },
49     windows_startup_auditor: { icon: '🔍', title: 'Windows Sta
rtup Auditor' },
50     windows_backup_manager: { icon: '💾', title: 'Windows Back
up & Restore Points' },
51     trading_terminal: { icon: '💻', title: 'Trading Terminal C
onnection' },
52     user_assistant: { icon: '👤', title: 'Personal User Assist
ant' },
53     helpdesk: { icon: '🎫', title: 'Helpdesk Tickets Queue' },
54     registry_viewer: { icon: '📁', title: 'Windows Registry Hi

```



```

    ves' },
55     software_audit: { icon: '🔍', title: 'Software Transparenc
y Audit' },
56     autolog_manager: { icon: '📝', title: 'Auto-Logging Manage
r' },
57 };
58
59 // Sensor metadata
60 const SENSOR_META = {
61     cpu: { icon: '🖥️', title: 'Процессор (CPU)', allMetrics: [
'temperature', 'load', 'clocks', 'voltage', 'power' ] },
62     gpu: { icon: '🎮', title: 'Видеокарта (GPU)', allMetrics:
['temperature', 'load', 'memory', 'power', 'fan', 'clocks' ] },
63     ram: { icon: '💾', title: 'Оперативная память (RAM)', allM
etrics: ['usage', 'swap', 'available', 'total' ] },
64     disk: { icon: '💿', title: 'Дисковая подсистема (Disk)', a
llMetrics: ['usage', 'io', 'read_bytes', 'write_bytes', 'queue
_length' ] },
65     network: { icon: '🌐', title: 'Сетевой интерфейс (Network)
', allMetrics: ['throughput', 'connections', 'bytes_sent', 'by
tes_recv', 'errors' ] },
66     sensors: { icon: '🌡️', title: 'Сенсоры LHM / WMI', allMetr
ics: ['temperature', 'fan', 'voltage', 'power', 'control' ] },
67     internet: { icon: '🌐', title: 'Интернет-канал (Speed/Ping
)', allMetrics: ['ping', 'download', 'upload', 'dns', 'jitter'
] },
68     storage: { icon: '💾', title: 'Здоровье дисков (S.M.A.R.T.
)', allMetrics: ['smart_attributes', 'temperature', 'wear_leve
l', 'health_status' ] },
69     device_flapping: { icon: '🌀', title: 'Дребезг устройств (
Flapping)', allMetrics: ['connect_events', 'disconnect_events'
, 'flapping_count' ] },
70 };
71
72 // State
73 let _configData = null;
74 let _statusData = null;
75 let _toastInstance = null;

```

```

76 |
77 | function showToast(message, type = 'info') {
78 |     const toastEl = document.getElementById('autolog-toast');
79 |     if (!toastEl) return;
80 |     const msgEl = document.getElementById('autolog-toast-msg')
    | ;
81 |     const iconEl = document.getElementById('autolog-toast-icon
    | ');
82 |
83 |     if (msgEl) msgEl.textContent = message;
84 |
85 |     if (iconEl) {
86 |         iconEl.className = 'bi me-1 ';
87 |         if (type === 'success') {
88 |             iconEl.className += 'bi-check-circle-fill text-success
    | ';
89 |         } else if (type === 'error' || type === 'danger') {
90 |             iconEl.className += 'bi-exclamation-octagon-fill text-
    | danger';
91 |         } else if (type === 'warning') {
92 |             iconEl.className += 'bi-exclamation-triangle-fill text
    | -warning';
93 |         } else {
94 |             iconEl.className += 'bi-info-circle-fill text-info';
95 |         }
96 |     }
97 |
98 |     if (!_toastInstance && window.bootstrap?.Toast) {
99 |         _toastInstance = new window.bootstrap.Toast(toastEl, { d
    | elay: 4000 });
100 |     }
101 |     if (_toastInstance) {
102 |         _toastInstance.show();
103 |     }
104 | }
105 |
106 | // --- API Fetch Helpers ---
107 | async function apiGet(endpoint) {

```

```
108 |     const res = await fetch(endpoint);
109 |     if (!res.ok) {
110 |         throw new Error(`HTTP ${res.status}: ${res.statusText}`)
111 |     };
112 |     return await res.json();
113 | }
114 |
115 | async function apiPost(endpoint, body = {}) {
116 |     const res = await fetch(endpoint, {
117 |         method: 'POST',
118 |         headers: { 'Content-Type': 'application/json' },
119 |         body: JSON.stringify(body),
120 |     });
121 |     if (!res.ok) {
122 |         const err = await res.json().catch(() => ({}));
123 |         throw new Error(err.detail || `HTTP ${res.status}: ${res
124 | .statusText}`);
125 |     }
126 |     return await res.json();
127 | }
128 |
129 | async function apiDelete(endpoint) {
130 |     const res = await fetch(endpoint, { method: 'DELETE' });
131 |     if (!res.ok) {
132 |         const err = await res.json().catch(() => ({}));
133 |         throw new Error(err.detail || `HTTP ${res.status}: ${res
134 | .statusText}`);
135 |     }
136 |     return await res.json();
137 | }
138 |
139 | // --- Load and Render Status & Config ---
140 | async function loadStatusAndConfig() {
141 |     try {
142 |         const [status, config] = await Promise.all([
143 |             apiGet('/api/autolog/status'),
144 |             apiGet('/api/autolog/config'),
```

```
143 |     });
144 |
145 |     _statusData = status;
146 |     _configData = config;
147 |
148 |     renderHeaderStatus();
149 |     renderLoggersTable();
150 |     renderSensorsCards();
151 |     renderTelemetryOptions();
152 | } catch (err) {
153 |     console.error('[AutoLogTab] Ошибка загрузки статуса/конф
игурации:', err);
154 |     showToast(`Ошибка загрузки данных: ${err.message}`, 'dan
ger');
155 | }
156 | }
157 |
158 | function renderHeaderStatus() {
159 |     if (!_statusData || !_configData) return;
160 |
161 |     const isRunning = _statusData.running || _configData.is_ru
nning;
162 |     const isAutologEnabled = _configData.enable_autolog;
163 |
164 |     // Badge
165 |     const badge = document.getElementById('autolog-engine-badg
e');
166 |     const badgeText = document.getElementById('autolog-engine-
badge-text');
167 |     if (badge && badgeText) {
168 |         if (isRunning) {
169 |             badge.className = 'badge bg-success d-flex align-items
-center gap-1.5 px-3 py-2 fs-6 shadow-sm';
170 |             badgeText.textContent = 'Движок активен';
171 |         } else {
172 |             badge.className = 'badge bg-secondary d-flex align-ite
ms-center gap-1.5 px-3 py-2 fs-6 shadow-sm';
173 |             badgeText.textContent = isAutologEnabled ? 'Движок ост
```

```
    ановлен' : 'Автологгирование выключено';
174     }
175   }
176
177   // Toggle button text
178   const btnToggleText = document.getElementById('btn-autolog
-toggle-text');
179   const btnToggle = document.getElementById('btn-autolog-tog
gle-engine');
180   if (btnToggleText && btnToggle) {
181     if (isRunning) {
182       btnToggleText.textContent = 'Остановить';
183       btnToggle.className = 'btn btn-sm btn-outline-danger d
-flex align-items-center gap-1 shadow-sm';
184     } else {
185       btnToggleText.textContent = 'Запустить';
186       btnToggle.className = 'btn btn-sm btn-outline-success
d-flex align-items-center gap-1 shadow-sm';
187     }
188   }
189
190   // Counters
191   const activeTasksEl = document.getElementById('metric-acti
ve-tasks');
192   if (activeTasksEl) {
193     activeTasksEl.textContent = _statusData.active_tasks_cou
nt ?? 0;
194   }
195
196   const sensorsCount = Object.keys(_configData.sensors || {}).length;
197   const activeSensorsCount = Object.values(_configData.senso
rs || {}).filter(s => s.enabled).length;
198   const metricSensorsEl = document.getElementById('metric-ac
tive-sensors');
199   if (metricSensorsEl) {
200     metricSensorsEl.textContent = `${activeSensorsCount} / ${
{sensorsCount}`;
```

```
201     }
202     const badgeSensorsEl = document.getElementById('badge-sens
ors-count');
203     if (badgeSensorsEl) {
204         badgeSensorsEl.textContent = `${activeSensorsCount}`;
205     }
206
207     const cfgFileEl = document.getElementById('metric-active-c
onfig-file');
208     if (cfgFileEl) {
209         cfgFileEl.textContent = _configData.config_file || '~aut
olog_sensors.json';
210     }
211
212     // Master Switch
213     const masterSwitch = document.getElementById('switch-enabl
e-autolog');
214     const masterText = document.getElementById('autolog-master
-status-text');
215     if (masterSwitch) {
216         masterSwitch.checked = Boolean(isAutologEnabled);
217     }
218     if (masterText) {
219         masterText.textContent = isAutologEnabled ? 'Включено' :
'Выключено';
220         masterText.className = isAutologEnabled ? 'fw-bold text-
info' : 'fw-bold text-muted';
221     }
222 }
223
224 // --- Render Loggers Table ---
225 function renderLoggersTable() {
226     const tbody = document.getElementById('tbody-loggers');
227     if (!tbody || !_configData) return;
228
229     const loggers = _configData.loggers || {};
230     const loggerKeys = Object.keys(loggers);
231
```

```

232     const badgeTotal = document.getElementById('badge-total-loggers');
233     if (badgeTotal) badgeTotal.textContent = loggerKeys.length;
234
235     if (loggerKeys.length === 0) {
236         tbody.innerHTML = `
237             <tr>
238                 <td colspan="6" class="text-center text-muted py-4">
Логгеры не найдены в конфигурации</td>
239             </tr>
240         `;
241         return;
242     }
243
244     let html = '';
245     for (const name of loggerKeys) {
246         const item = loggers[name];
247         const meta = APP_META[name] || { icon: '🔍', title: name };
248         const isEnabled = Boolean(item.enabled);
249         const intervalVal = item.interval || '1 minute';
250         const pollCount = item.poll_count ?? 0;
251         const lastPoll = item.last_poll
252             ? new Date(item.last_poll).toLocaleTimeString([], { hour: '2-digit', minute: '2-digit', second: '2-digit' })
253             : '<span class="text-muted">—</span>';
254
255         // Presets options HTML
256         let optionsHtml = '';
257         let isCustom = true;
258         for (const preset of INTERVAL_PRESETS) {
259             const selected = preset.value.toLowerCase() === intervalVal.toLowerCase() ? 'selected' : '';
260             if (selected) isCustom = false;
261             optionsHtml += `<option value="${preset.value}" ${selected}>${preset.label}</option>`;
262         }

```

```

263         if (isCustom) {
264             optionsHtml += `<option value="${intervalVal}" selecte
d>Пользовательский (${intervalVal})</option>`;
265         }
266
267         html += `
268             <tr data-logger-name="${name}">
269                 <td class="text-center">
270                     <div class="form-check form-switch d-inline-block
mb-0">
271                         <input class="form-check-input logger-enable-swi
tch" type="checkbox" role="switch"
272                             data-app="${name}" ${isEnabled ? 'checked
' : ''}>
273                     </div>
274                 </td>
275                 <td>
276                     <div class="d-flex align-items-center gap-2">
277                         <span class="fs-5">${meta.icon}</span>
278                         <div>
279                             <div class="fw-semibold text-white">${meta.tit
le}</div>
280                             <code class="text-secondary small">${name}</co
de>
281                         </div>
282                     </div>
283                 </td>
284                 <td>
285                     <div class="d-flex align-items-center gap-1">
286                         <select class="form-select form-select-sm bg-dar
k text-white border-secondary select-logger-interval"
287                             data-app="${name}" style="min-width: 170
px;">
288                             ${optionsHtml}
289                         </select>
290                     </div>
291                 </td>
292                 <td class="text-center fw-mono">

```



```

293 |         <span class="badge bg-secondary">${pollCount}</spa
n>
294 |     </td>
295 |     <td>
296 |         <small class="text-light">${lastPoll}</small>
297 |     </td>
298 |     <td class="text-center">
299 |         <button class="btn btn-sm btn-outline-info btn-pol
l-single d-inline-flex align-items-center gap-1"
300 |             data-app="${name}" title="Опросить немедле
нно">
301 |             <i class="bi bi-play-fill"></i>
302 |             <span>Опросить</span>
303 |         </button>
304 |     </td>
305 | </tr>
306 | `;
307 | }
308 |
309 | tbody.innerHTML = html;
310 | bindLoggersTableEvents();
311 | }
312 |
313 | function bindLoggersTableEvents() {
314 |     const tbody = document.getElementById('tbody-loggers');
315 |     if (!tbody) return;
316 |
317 |     // Single poller trigger
318 |     tbody.querySelectorAll('.btn-poll-single').forEach((btn) =
> {
319 |         btn.addEventListener('click', async (e) => {
320 |             const appName = btn.getAttribute('data-app');
321 |             if (!appName) return;
322 |
323 |             btn.disabled = true;
324 |             btn.innerHTML = '<span class="spinner-border spinner-b
order-sm" role="status"></span>';
325 |

```

```

326         try {
327             await apiPost(`/api/autolog/poll/${appName}`);
328             showToast(`Опрос логгера '${appName}' успешно выполн
ен`, 'success');
329             await loadStatusAndConfig();
330         } catch (err) {
331             showToast(`Ошибка опроса '${appName}': ${err.message
}`, 'danger');
332         } finally {
333             btn.disabled = false;
334             btn.innerHTML = '<i class="bi bi-play-fill"></i> <sp
an>Опросить</span>';
335         }
336     });
337 });
338
339 // Logger switch change
340 tbody.querySelectorAll('.logger-enable-switch').forEach((s
w) => {
341     sw.addEventListener('change', (e) => {
342         const appName = sw.getAttribute('data-app');
343         if (_configData?.loggers?.[appName]) {
344             _configData.loggers[appName].enabled = sw.checked;
345         }
346     });
347 });
348
349 // Logger interval change
350 tbody.querySelectorAll('.select-logger-interval').forEach(
(sel) => {
351     sel.addEventListener('change', (e) => {
352         const appName = sel.getAttribute('data-app');
353         if (_configData?.loggers?.[appName]) {
354             _configData.loggers[appName].interval = sel.value;
355         }
356     });
357 });
358 }

```

```

359
360 // --- Render Telemetry Sensors Cards ---
361 function renderSensorsCards() {
362     const container = document.getElementById('sensors-cards-c
ontainer');
363     if (!container || !_configData) return;
364
365     const sensors = _configData.sensors || {};
366     const sensorKeys = Object.keys(sensors);
367
368     if (sensorKeys.length === 0) {
369         container.innerHTML = `
370         <div class="col-12 text-center text-muted py-4">Сенсор
ы телеметрии не найдены в конфигурации</div>
371         `;
372         return;
373     }
374
375     let html = '';
376     for (const sensorName of sensorKeys) {
377         const cfg = sensors[sensorName] || {};
378         const meta = SENSOR_META[sensorName] || { icon: '⚙', ti
tle: sensorName, allMetrics: cfg.metrics || [] };
379         const isEnabled = Boolean(cfg.enabled);
380         const intervalSec = cfg.interval_seconds ?? 5.0;
381         const activeMetrics = new Set(cfg.metrics || []);
382
383         // Combine default/known metrics with active metrics
384         const allKnownMetrics = Array.from(new Set([...(meta.all
Metrics || []), ...activeMetrics]));
385
386         let metricsHtml = '';
387         for (const m of allKnownMetrics) {
388             const isActive = activeMetrics.has(m);
389             metricsHtml += `
390             <button type="button" class="btn btn-sm ${isActive ?
'btn-primary' : 'btn-outline-secondary'} py-0 px-2 rounded-pi
ll metric-pill-btn"

```

```

391 |             data-sensor="${sensorName}" data-metric="${m
| }" style="font-size: 0.75rem;">
392 |             ${isActive ? '⏏ ' : '+ '}${m}
393 |         </button>
394 |     `;
395 | }
396 |
397 |     html += `
398 |         <div class="col-12 col-md-6 col-lg-4" data-sensor-card
| ="${sensorName}">
399 |             <div class="card bg-black border ${isEnabled ? 'bord
| er-secondary' : 'border-secondary-subtle opacity-75'} h-100 p-
| 3 shadow-sm rounded-3">
400 |                 <div class="d-flex align-items-center justify-cont
| ent-between mb-2">
401 |                     <div class="d-flex align-items-center gap-2">
402 |                         <span class="fs-4">${meta.icon}</span>
403 |                         <div>
404 |                             <h6 class="mb-0 text-white fw-bold">${meta.t
| itle}</h6>
405 |                             <code class="text-secondary small">${sensorN
| ame}</code>
406 |                         </div>
407 |                     </div>
408 |                     <div class="form-check form-switch mb-0">
409 |                         <input class="form-check-input sensor-enable-s
| witch" type="checkbox" role="switch"
410 |                             data-sensor="${sensorName}" ${isEnabled
| ? 'checked' : ''}>
411 |                     </div>
412 |                 </div>
413 |
414 |                 <div class="d-flex align-items-center gap-2 my-2">
415 |                     <label class="small text-muted text-nowrap mb-0"
| >Интервал (сек):</label>
416 |                     <input type="number" class="form-control form-co
| ntrol-sm bg-dark text-white border-secondary sensor-interval-i
| nput"

```

```

417 |             data-sensor="${sensorName}" value="${inte
| rvalSec}" min="0.1" step="0.5" style="max-width: 90px;">
418 |         </div>
419 |
420 |         <div class="mt-2">
421 |             <div class="small text-muted mb-1">Собираемые ме
| трики:</div>
422 |             <div class="d-flex flex-wrap gap-1">
423 |                 ${metricsHtml || '<span class="text-muted smal
| l">Метрики не заданы</span>'}
424 |             </div>
425 |         </div>
426 |     </div>
427 | </div>
428 | `;
429 | }
430 |
431 |     container.innerHTML = html;
432 |     bindSensorsEvents();
433 | }
434 |
435 | function bindSensorsEvents() {
436 |     const container = document.getElementById('sensors-cards-c
| ontainer');
437 |     if (!container) return;
438 |
439 |     // Sensor enabled switch
440 |     container.querySelectorAll('.sensor-enable-switch').forEac
| h((sw) => {
441 |         sw.addEventListener('change', () => {
442 |             const sName = sw.getAttribute('data-sensor');
443 |             if (_configData?.sensors?.[sName]) {
444 |                 _configData.sensors[sName].enabled = sw.checked;
445 |                 const card = container.querySelector(`[data-sensor-c
| ard="${sName}"] .card`);
446 |                 if (card) {
447 |                     if (sw.checked) {
448 |                         card.classList.remove('opacity-75', 'border-seco

```

```

    ndary-subtle');
449         card.classList.add('border-secondary');
450     } else {
451         card.classList.add('opacity-75', 'border-seconda
ry-subtle');
452         card.classList.remove('border-secondary');
453     }
454 }
455 renderHeaderStatus();
456 }
457 });
458 });
459
460 // Sensor interval input
461 container.querySelectorAll('.sensor-interval-input').forEa
ch((inp) => {
462     inp.addEventListener('input', () => {
463         const sName = inp.getAttribute('data-sensor');
464         const val = parseFloat(inp.value);
465         if (_configData?.sensors?.[sName] && !isNaN(val)) {
466             _configData.sensors[sName].interval_seconds = val;
467         }
468     });
469 });
470
471 // Metric toggle pill
472 container.querySelectorAll('.metric-pill-btn').forEach((bt
n) => {
473     btn.addEventListener('click', () => {
474         const sName = btn.getAttribute('data-sensor');
475         const metric = btn.getAttribute('data-metric');
476         if (!_configData?.sensors?.[sName]) return;
477
478         let metrics = _configData.sensors[sName].metrics || []
;
479         if (metrics.includes(metric)) {
480             metrics = metrics.filter(m => m !== metric);
481             btn.className = 'btn btn-sm btn-outline-secondary py

```

```

    -0 px-2 rounded-pill metric-pill-btn';
482         btn.textContent = '+' + metric;
483     } else {
484         metrics.push(metric);
485         btn.className = 'btn btn-sm btn-primary py-0 px-2 rounded-pill metric-pill-btn';
486         btn.textContent = '□ ' + metric;
487     }
488     _configData.sensors[sName].metrics = metrics;
489 });
490 });
491 }
492
493 function renderTelemetryOptions() {
494     if (!_configData) return;
495     const opts = _configData.telemetry_options || {};
496
497     const optInv = document.getElementById('opt-collect-inventory');
498     const optSer = document.getElementById('opt-collect-serials');
499     const optFiles = document.getElementById('opt-collect-file-events');
500     const optMax = document.getElementById('opt-max-file-size');
501     const optDirs = document.getElementById('opt-watch-dirs');
502
503     if (optInv) optInv.checked = opts.collect_hardware_inventory ?? true;
504     if (optSer) optSer.checked = opts.collect_serial_numbers ? true;
505     if (optFiles) optFiles.checked = opts.collect_file_events ?? true;
506     if (optMax) optMax.value = opts.max_file_size_mb ?? 50;
507     if (optDirs) optDirs.value = (opts.watch_directories || ['C:\\Users\\']).join(', ');
508 }
509

```

```

510 function collectTelemetryOptions() {
511     const optInv = document.getElementById('opt-collect-invent
    ory');
512     const optSer = document.getElementById('opt-collect-serial
    s');
513     const optFiles = document.getElementById('opt-collect-file
    -events');
514     const optMax = document.getElementById('opt-max-file-size'
    );
515     const optDirs = document.getElementById('opt-watch-dirs');
516
517     const dirsRaw = optDirs ? optDirs.value.split(',').map(s =
    > s.trim()).filter(Boolean) : ['C:\\Users\\'];
518
519     return {
520         collect_hardware_inventory: optInv ? optInv.checked : tr
    ue,
521         collect_serial_numbers: optSer ? optSer.checked : true,
522         collect_file_events: optFiles ? optFiles.checked : true,
523         max_file_size_mb: optMax ? parseInt(optMax.value, 10) ||
    50 : 50,
524         watch_directories: dirsRaw,
525     };
526 }
527
528 // --- Save Config via API ---
529 async function saveFullConfiguration() {
530     if (!_configData) return;
531
532     const masterSwitch = document.getElementById('switch-enabl
    e-autolog');
533     const enableAutolog = masterSwitch ? masterSwitch.checked
    : true;
534
535     // Collect loggers
536     const loggersPayload = {};
537     for (const [name, item] of Object.entries(_configData.logg
    ers || {})) {

```



```
538 |     loggersPayload[name] = {
539 |         interval: item.interval || '1 minute',
540 |         enabled: Boolean(item.enabled),
541 |     };
542 | }
543 |
544 | // Collect sensors
545 | const sensorsPayload = {};
546 | for (const [name, item] of Object.entries(_configData.sens
ors || {})) {
547 |     sensorsPayload[name] = {
548 |         enabled: Boolean(item.enabled),
549 |         interval_seconds: item.interval_seconds ?? 5.0,
550 |         metrics: item.metrics || [],
551 |     };
552 | }
553 |
554 | const telemetryOptions = collectTelemetryOptions();
555 |
556 | const payload = {
557 |     enable_autolog: enableAutolog,
558 |     default_interval: _configData.default_interval || '1 min
ute',
559 |     loggers: loggersPayload,
560 |     sensors: sensorsPayload,
561 |     telemetry_options: telemetryOptions,
562 | };
563 |
564 | try {
565 |     showToast('Сохранение конфигурации...', 'info');
566 |     const res = await apiPost('/api/autolog/config', payload
);
567 |     showToast(res.message || 'Конфигурация успешно сохранена
!', 'success');
568 |     await loadStatusAndConfig();
569 | } catch (err) {
570 |     console.error('[AutoLogTab] Ошибка сохранения:', err);
571 |     showToast(`Ошибка сохранения: ${err.message}`, 'danger')
```

```

572     }
573 }
574
575 // --- Telemetry Service Logic (ai-telemetry.exe) ---
576 let _telemetryStatusData = null;
577
578 async function loadTelemetryServiceStatusAndConfig() {
579     try {
580         const [statusRes, configRes] = await Promise.all([
581             apiGet('/api/autolog/telemetry/status'),
582             apiGet('/api/autolog/telemetry/config'),
583         ]);
584         _telemetryStatusData = statusRes;
585
586         // Обновление карточки живого статуса
587         const badgeNav = document.getElementById('badge-telemetry-status');
588         const badgeCard = document.getElementById('svc-status-badge');
589         const pidsEl = document.getElementById('svc-info-pids');
590         const memEl = document.getElementById('svc-info-memory');
591         const snapEl = document.getElementById('svc-info-snapshots');
592         const schedEl = document.getElementById('svc-info-scheduler');
593
594         const isRunning = Boolean(statusRes.is_running);
595         if (badgeNav) {
596             badgeNav.textContent = isRunning ? 'Online' : 'Offline';
597             badgeNav.className = isRunning ? 'badge bg-success ms-1' : 'badge bg-secondary ms-1';
598         }
599         if (badgeCard) {
600             badgeCard.textContent = isRunning ? 'Активен (ai-telemetry.exe)' : 'Остановлен';

```

```
601 |         badgeCard.className = isRunning ? 'badge bg-success' :
    | 'badge bg-secondary';
602 |     }
603 |
604 |     if (pidsEl) {
605 |         if (isRunning && statusRes.processes && statusRes.proc
    | esses.length > 0) {
606 |             const list = statusRes.processes.map(p => `${p.name}
    | (PID: ${p.pid})`).join(', ');
607 |             pidsEl.textContent = list;
608 |             pidsEl.title = list;
609 |         } else {
610 |             pidsEl.textContent = 'Не запущен';
611 |         }
612 |     }
613 |
614 |     if (memEl) {
615 |         memEl.textContent = isRunning ? `${statusRes.total_mem
    | ory_mb} МБ` : '0 МБ';
616 |     }
617 |
618 |     if (snapEl) {
619 |         snapEl.textContent = `${statusRes.snapshots_count} сн
    | мков`;
620 |         if (statusRes.latest_timestamp) {
621 |             snapEl.title = `Последний: ${statusRes.latest_timest
    | amp}`;
622 |         }
623 |     }
624 |
625 |     if (schedEl) {
626 |         const sched = statusRes.task_scheduler || {};
627 |         if (sched.installed) {
628 |             schedEl.innerHTML = `<span class="text-success">Уста
    | новлено (${sched.state})</span> <small class="text-muted" styl
    | e="font-size:0.7rem;">[Wake: ${sched.wake_to_run}]</small>`;
629 |         } else {
630 |             schedEl.innerHTML = '<span class="text-muted">Не уст
```

```
        ановлено</span>';
631     }
632 }
633
634     // Заполнение полей формы конфигурации
635     const mode = (configRes.mode || statusRes.mode || 'hybrid').toLowerCase();
636     const radioMode = document.querySelector(`input[name="telemetryModeRadio"][value="${mode}"]`);
637     if (radioMode) radioMode.checked = true;
638
639     const fastInt = configRes.interval_seconds || statusRes.interval_seconds || 5.0;
640     const inputFast = document.getElementById('cfg-fast-interval');
641     const valFast = document.getElementById('val-fast-interval');
642     if (inputFast) inputFast.value = fastInt;
643     if (valFast) valFast.textContent = `${fastInt}c`;
644
645     const heavyInt = configRes.heavy_interval_seconds || statusRes.heavy_interval_seconds || 60.0;
646     const inputHeavy = document.getElementById('cfg-heavy-interval');
647     const valHeavy = document.getElementById('val-heavy-interval');
648     if (inputHeavy) inputHeavy.value = heavyInt;
649     if (valHeavy) valHeavy.textContent = `${heavyInt}c`;
650
651     const topProcs = configRes.top_processes || statusRes.top_processes || 10;
652     const selTop = document.getElementById('cfg-top-processes');
653     if (selTop) selTop.value = String(topProcs);
654
655     const hColls = configRes.heavy_collectors || statusRes.heavy_collectors || {};
656     const swLhm = document.getElementById('h-sensor-lhm');
```

```
657 |         if (swLhm) swLhm.checked = Boolean(hColls.hardware_senso
|rs ?? true);
658 |         const swSmart = document.getElementById('h-sensor-smart'
|);
659 |         if (swSmart) swSmart.checked = Boolean(hColls.storage_sm
|art ?? true);
660 |         const swPing = document.getElementById('h-sensor-ping');
661 |         if (swPing) swPing.checked = Boolean(hColls.network_ping
| ?? true);
662 |         const swInv = document.getElementById('h-sensor-inventor
|y');
663 |         if (swInv) swInv.checked = Boolean(hColls.inventory_wmi
| ?? false);
664 |
665 |     } catch (err) {
666 |         console.warn('[AutoLogTab] Ошибка загрузки статуса служб
|ы телеметрии:', err);
667 |     }
668 | }
669 |
670 | async function saveTelemetryConfiguration() {
671 |     try {
672 |         const selectedMode = document.querySelector('input[name=
|"telemetryModeRadio"]:checked')?.value || 'hybrid';
673 |         const fastInt = parseFloat(document.getElementById('cfg-
|fast-interval')?.value || '5.0');
674 |         const heavyInt = parseFloat(document.getElementById('cfg
|-heavy-interval')?.value || '60.0');
675 |         const topProcs = parseInt(document.getElementById('cfg-t
|op-processes')?.value || '10', 10);
676 |
677 |         const hColls = {
678 |             hardware_sensors: document.getElementById('h-sensor-lh
|m')?.checked ?? true,
679 |             storage_smart: document.getElementById('h-sensor-smart
|')?.checked ?? true,
680 |             network_ping: document.getElementById('h-sensor-ping')
|?.checked ?? true,
```

```
681 |         inventory_wmi: document.getElementById('h-sensor-inven
| tory')?.checked ?? false,
682 |     };
683 |
684 |     const payload = {
685 |         mode: selectedMode,
686 |         interval_seconds: fastInt,
687 |         heavy_interval_seconds: heavyInt,
688 |         top_processes: topProcs,
689 |         heavy_collectors: hColls,
690 |     };
691 |
692 |     showToast('Сохранение параметров телеметрии...', 'info')
| ;
693 |     const res = await apiPost('/api/autolog/telemetry/config
| ', payload);
694 |     showToast(res.message || 'Параметры телеметрии успешно с
| охранены!', 'success');
695 |     await loadTelemetryServiceStatusAndConfig();
696 | } catch (err) {
697 |     showToast(`Ошибка сохранения телеметрии: ${err.message}`
| , 'danger');
698 | }
699 | }
700 |
701 | async function controlTelemetryService(action) {
702 |     try {
703 |         showToast(`Выполняется: ${action}...`, 'info');
704 |         const res = await apiPost('/api/autolog/telemetry/contro
| l', { action });
705 |         showToast(`Действие '${action}' успешно выполнено!`, 'su
| ccess');
706 |         setTimeout(loadTelemetryServiceStatusAndConfig, 1000);
707 |     } catch (err) {
708 |         showToast(`Ошибка: ${err.message}`, 'danger');
709 |     }
710 | }
711 |
```

```
712 | // --- Setup Master Event Listeners ---
713 | function setupEventListeners() {
714 |     // Save buttons
715 |     const btnSaveHeader = document.getElementById('btn-autolog
-s-save-config');
716 |     const btnSaveBottom = document.getElementById('btn-autolog
-s-save-config-bottom');
717 |     const btnSaveSensors = document.getElementById('btn-save-s
ensors-config');
718 |     if (btnSaveHeader) btnSaveHeader.addEventListener('click',
saveFullConfiguration);
719 |     if (btnSaveBottom) btnSaveBottom.addEventListener('click',
saveFullConfiguration);
720 |     if (btnSaveSensors) btnSaveSensors.addEventListener('click
', saveFullConfiguration);
721 |
722 |     // Telemetry service controls & save
723 |     const btnSaveTelemetry = document.getElementById('btn-save
-telemetry-config');
724 |     if (btnSaveTelemetry) btnSaveTelemetry.addEventListener('c
lick', saveTelemetryConfiguration);
725 |
726 |     const btnTelStart = document.getElementById('btn-telemetry
-start');
727 |     if (btnTelStart) btnTelStart.addEventListener('click', ()
=> controlTelemetryService('start'));
728 |
729 |     const btnTelStop = document.getElementById('btn-telemetry-
stop');
730 |     if (btnTelStop) btnTelStop.addEventListener('click', () =>
controlTelemetryService('stop'));
731 |
732 |     const btnTelRestart = document.getElementById('btn-telemet
ry-restart');
733 |     if (btnTelRestart) btnTelRestart.addEventListener('click',
() => controlTelemetryService('restart'));
734 |
735 |     const btnTelInstallTask = document.getElementById('btn-tel
```

```

    telemetry-install-task');
736     if (btnTelInstallTask) btnTelInstallTask.addEventListener(
    'click', () => controlTelemetryService('install-task'));
737
738     const btnTelUninstallTask = document.getElementById('btn-t
    telemetry-uninstall-task');
739     if (btnTelUninstallTask) btnTelUninstallTask.addEventListe
    ner('click', () => controlTelemetryService('uninstall-task'));
740
741     // Dynamic label sync for intervals
742     const inputFast = document.getElementById('cfg-fast-interv
    al');
743     if (inputFast) {
744         inputFast.addEventListener('input', () => {
745             const valFast = document.getElementById('val-fast-inte
    rval');
746             if (valFast) valFast.textContent = `${inputFast.value}
    c`;
747         });
748     }
749     const inputHeavy = document.getElementById('cfg-heavy-inte
    rval');
750     if (inputHeavy) {
751         inputHeavy.addEventListener('input', () => {
752             const valHeavy = document.getElementById('val-heavy-in
    terval');
753             if (valHeavy) valHeavy.textContent = `${inputHeavy.val
    ue}c`;
754         });
755     }
756
757     // Toggle Engine button
758     const btnToggle = document.getElementById('btn-autolog-tog
    gle-engine');
759     if (btnToggle) {
760         btnToggle.addEventListener('click', async () => {
761             btnToggle.disabled = true;
762             try {

```



```
763 |         const isRunning = _statusData?.running || _configData?.is_running;
764 |         if (isRunning) {
765 |             await apiPost('/api/autolog/stop');
766 |             showToast('AutoLogEngine успешно остановлен', 'info');
767 |         } else {
768 |             await apiPost('/api/autolog/start');
769 |             showToast('AutoLogEngine успешно запущен', 'success');
770 |         }
771 |         await loadStatusAndConfig();
772 |     } catch (err) {
773 |         showToast(`Ошибка: ${err.message}`, 'danger');
774 |     } finally {
775 |         btnToggle.disabled = false;
776 |     }
777 | });
778 | }
779 |
780 | // Poll All button
781 | const btnPollAll = document.getElementById('btn-autolog-poll-all');
782 | if (btnPollAll) {
783 |     btnPollAll.addEventListener('click', async () => {
784 |         btnPollAll.disabled = true;
785 |         btnPollAll.innerHTML = '<span class="spinner-border spinner-border-sm" role="status"></span> Опрос...';
786 |         try {
787 |             const res = await apiPost('/api/autolog/poll-all');
788 |             showToast(`Опрос завершен: успешно ${res.successful_count} из ${res.total_polled}`, 'success');
789 |             await loadStatusAndConfig();
790 |         } catch (err) {
791 |             showToast(`Ошибка разового опроса: ${err.message}`, 'danger');
792 |         } finally {
793 |             btnPollAll.disabled = false;
```

```

794 |         btnPollAll.innerHTML = '<i class="bi bi-arrow-repeat
    | "></i> <span>Опросить все сейчас</span>';
795 |     }
796 | });
797 | }
798 |
799 | // Refresh Loggers & Sensors
800 | const btnRefLoggers = document.getElementById('btn-refresh
    | -loggers');
801 | if (btnRefLoggers) btnRefLoggers.addEventListener('click',
    | loadStatusAndConfig);
802 | const btnRefSensors = document.getElementById('btn-refresh
    | -sensors');
803 | if (btnRefSensors) btnRefSensors.addEventListener('click',
    | loadStatusAndConfig);
804 |
805 | // Вкладка службы телеметрии
806 | const btnTabTelemetry = document.getElementById('subtab-te
    | lemetry-service-btn');
807 | if (btnTabTelemetry) {
808 |     btnTabTelemetry.addEventListener('click', loadTelemetryS
    | erviceStatusAndConfig);
809 | }
810 |
811 | // Search Loggers filter
812 | const inputSearch = document.getElementById('input-search-
    | loggers');
813 | if (inputSearch) {
814 |     inputSearch.addEventListener('input', () => {
815 |         const q = inputSearch.value.toLowerCase().trim();
816 |         const rows = document.querySelectorAll('#tbody-loggers
    | tr[data-logger-name]');
817 |         rows.forEach(r => {
818 |             const name = r.getAttribute('data-logger-name') || '
    | ';
819 |             const text = r.textContent.toLowerCase();
820 |             r.style.display = (name.includes(q) || text.includes
    | (q)) ? '' : 'none';

```

```
821 |         });
822 |     });
823 | }
824 | }
825 |
826 | // --- Initialization ---
827 | function initAll() {
828 |     setupEventListeners();
829 |     loadStatusAndConfig();
830 |     loadTelemetryServiceStatusAndConfig();
831 |     setInterval(loadTelemetryServiceStatusAndConfig, 10000);
832 | }
833 |
834 | document.addEventListener('DOMContentLoaded', initAll);
835 |
836 | // Also initialize immediately if DOM is already ready
837 | if (document.readyState === 'interactive' || document.readyState === 'complete') {
838 |     initAll();
839 | }
840 | })();
841 |
```

index.html

```
1 <div class="chat-container d-flex flex-row h-100 w-100 overflow-
  w-hidden gap-2" style="min-height: 80vh; height: calc(100vh -
  65px);">
2   <!-- Sessions Sidebar -->
3   <aside class="chat-sidebar d-flex flex-column flex-shrink-0
  p-2 border rounded" id="chat-sidebar" style="width: 250px; min
  -width: 250px; max-width: 250px; height: 100%;">
4     <div class="d-flex align-items-center justify-content-betw
  een mb-2 flex-shrink-0">
5       <button class="btn btn-primary btn-sm w-100 d-flex align
  -items-center justify-content-center gap-2" id="chat-new-sessi
  on-btn" type="button" title="Начать новый диалог">
6         <i class="bi bi-plus-lg"></i>
7         <span>Новый чат</span>
8       </button>
9     </div>
10
11   <!-- Sessions List -->
12   <div class="chat-sessions-list flex-grow-1 overflow-auto d
  -flex flex-column gap-1 my-1" id="chat-sessions-list" style="m
  in-height: 0;">
13     <!-- Session items injected via JS -->
14   </div>
15
16   <!-- Sidebar Footer -->
17   <div class="pt-2 border-top border-secondary-subtle d-flex
  align-items-center justify-content-between flex-shrink-0">
18     <div class="d-flex align-items-center gap-2 flex-grow-1"
  >
19       <small class="text-muted" id="chat-sessions-count">0 ч
  атов</small>
20       <span class="badge bg-secondary-subtle text-secondary
  cache-badge" id="chat-cache-badge" title="Кеш: попадания/всего
  " style="cursor: help; font-size: 0.7rem;">
21       <i class="bi bi-lightning-charge"></i> <span id="cha
  t-cache-ratio">-</span>
```

```

22         </span>
23     </div>
24     <div class="d-flex align-items-center gap-1">
25         <button class="btn btn-sm btn-link text-warning text-d
ecoration-none p-0" id="chat-clear-cache-btn" type="button" ti
tle="Очистить кеш">
26             <i class="bi bi-arrow-clockwise"></i>
27         </button>
28         <button class="btn btn-sm btn-link text-danger text-de
coration-none p-0" id="chat-clear-all-sessions-btn" type="butt
on" title="Удалить все чаты">
29             <i class="bi bi-trash3"></i>
30         </button>
31     </div>
32 </div>
33 </aside>
34
35 <!-- Main Chat Conversation Area -->
36 <main class="chat-main-area d-flex flex-column flex-grow-1 h
-100 overflow-hidden" style="min-width: 0;">
37     <!-- Top Session Header Bar -->
38     <div class="chat-header-bar d-flex align-items-center just
ify-content-between p-1 px-2 mb-2 rounded border flex-shrink-0
">
39         <div class="d-flex align-items-center gap-2 flex-grow-1
overflow-hidden">
40             <button class="btn btn-sm btn-outline-secondary py-0 p
x-2" id="chat-sidebar-toggle-btn" type="button" title="Показат
ь/скрыть панель чатов">
41                 <i class="bi bi-layout-sidebar"></i>
42             </button>
43             <span class="fw-semibold text-truncate" id="chat-curre
nt-title" title="Текущий диалог">Новый чат</span>
44             <button class="btn btn-sm btn-link text-muted p-0" id=
"chat-rename-current-btn" type="button" title="Переименовать д
иалог">
45                 <i class="bi bi-pencil-square"></i>
46             </button>

```

```
47 |         </div>
48 |
49 |         <div class="d-flex align-items-center gap-1 flex-shrink-
50 | 0">
51 |             <button class="btn btn-sm btn-outline-secondary py-0 p
52 | x-2" id="chat-clear-btn" type="button" title="Очистить текущий
53 | диалог">
54 |                 <i class="bi bi-eraser me-1"></i><span class="d-none
55 | d-sm-inline">Очистить</span>
56 |             </button>
57 |         </div>
58 |     </div>
59 |
60 |     <!-- Chat Messages Window -->
61 |     <div class="chat-window flex-grow-1 mb-2 overflow-auto" id
62 | ="chat-window"></div>
63 |
64 |     <!-- Execution Status Indicator -->
65 |     <div id="chat-status-indicator" class="d-none small text-m
66 | uted mb-2 d-flex align-items-center gap-2 flex-shrink-0">
67 |         <span class="spinner-border spinner-border-sm text-prima
68 | ry" role="status" aria-hidden="true"></span>
69 |         <span id="chat-status-text">Обработка запроса...</span>
70 |     </div>
71 |
72 |     <!-- Query Input & Settings Bar -->
73 |     <div class="d-flex flex-wrap align-items-center gap-2 flex
74 | -shrink-0">
75 |         <!-- RAG Search Toggle Switch -->
76 |         <div class="form-check form-switch d-flex align-items-ce
77 | nter gap-1 m-0">
78 |             <input class="form-check-input mt-0" type="checkbox" r
79 | ole="switch" id="chat-rag-toggle" checked title="Использовать
80 | базу знаний (RAG)">
81 |             <label class="form-check-label small fw-semibold text-
82 | muted text-nowrap user-select-none" for="chat-rag-toggle" titl
83 | e="Использовать базу знаний (RAG)">
84 |                 <i class="bi bi-database-check me-1"></i>RAG
```

```
72 |         </label>
73 |     </div>
74 |
75 |     <!-- TOP_K Setting -->
76 |     <div class="d-flex align-items-center gap-1">
77 |         <label for="chat-top-k" class="small fw-semibold text-
78 | muted text-nowrap">TOP_K:</label>
79 |         <input type="number" class="form-control form-control-
80 | sm" id="chat-top-k" value="3" min="1" max="50" style="width: 7
81 | 5px;" title="Количество результатов (TOP_K)">
82 |     </div>
83 |
84 |     <!-- MIN_SCORE Setting -->
85 |     <div class="d-flex align-items-center gap-1">
86 |         <label for="chat-min-score" class="small fw-semibold t
87 | ext-muted text-nowrap">MIN_SCORE:</label>
88 |         <input type="number" class="form-control form-control-
89 | sm" id="chat-min-score" value="0.45" min="0.0" max="1.0" step=
90 | "0.05" style="width: 85px;" title="Минимальный порог (MIN_SCOR
91 | E)">
92 |     </div>
93 |
94 |     <!-- Output Mode (Text only / Voice only / Text and Voic
95 | e) -->
96 |     <div class="d-flex align-items-center gap-1">
97 |         <label for="chat-output-mode" class="small fw-semibold
98 | text-muted text-nowrap user-select-none" title="Режим вывода
99 | ответа">
100 |             <i class="bi bi-sliders me-1"></i>Режим:
101 |         </label>
102 |         <select class="form-select form-select-sm" id="chat-ou
103 | tput-mode" style="width: auto; min-width: 145px;" title="Режим
104 | вывода ответа">
105 |             <option value="text_only" selected>☐ Только текст</
106 | option>
107 |             <option value="text_and_voice">☐ +☐ Текст и голос</
108 | option>
109 |             <option value="voice_only">☐ Только голос</option>
```

```
96 |         </select>
97 |     </div>
98 |
99 |     <!-- Message Input and Send Button -->
100 |     <div class="input-group flex-grow-1">
101 |         <input type="text" class="form-control" id="message-in
102 | put" placeholder="Введите сообщение..." autocomplete="off" dat
103 | a-lpignore="true" data-1p-ignore="true">
104 |         <button class="btn btn-primary" id="send-button" type=
105 | "button">Отправить</button>
106 |     </div>
107 | </main>
108 | </div>
```


main.js

```
1 // — CHAT.JS —————  
2  
3 import { cachedApiFetch, clearCacheByTag } from '../js/api-cache.js';  
4  
5 const STORAGE_KEY_SESSIONS = 'breadboard_chat_sessions_v2';  
6 const STORAGE_KEY_ACTIVE_SESSION_ID = 'breadboard_active_session_id';  
7 const STORAGE_KEY_SIDEBAR_COLLAPSED = 'breadboard_chat_sidebar_collapsed';  
8 const LEGACY_STORAGE_KEY = 'breadboard_chat_history';  
9 const CACHE_TAG = 'chat';  
10  
11 let sessions = [];  
12 let activeSessionId = null;  
13  
14 /**  
15  * Generates a clean, human-friendly title based on the first  
16  * prompt (ChatGPT-style).  
17  */  
18 function generateSessionTitle(prompt) {  
19   if (!prompt || typeof prompt !== 'string') return 'Новый чат';  
20  
21   let clean = prompt  
22     .replace(/^```\s\S]*?```/g, '')  
23     .replace(/<[^>]*>/g, '')  
24     .replace(/[*_#`[\]()]/g, '')  
25     .trim();  
26  
27   if (!clean) return 'Новый чат';  
28  
29   // Take the first line or first sentence  
30   const firstLine = clean.split('\n')[0].trim();  
31   const sentenceMatch = firstLine.match(/^(.+)?(?:[.!?]|$)/);
```

```
31 | let text = sentenceMatch ? sentenceMatch[1].trim() : firstLi
    | ne;
32 |
33 | if (text.length > 36) {
34 |     text = text.substring(0, 36).trim();
35 |     const lastSpace = text.lastIndexOf(' ');
36 |     if (lastSpace > 15) {
37 |         text = text.substring(0, lastSpace);
38 |     }
39 |     text += '...';
40 | }
41 |
42 | // Capitalize first character
43 | return text.charAt(0).toUpperCase() + text.slice(1);
44 | }
45 |
46 | let syncDebounceTimer = null;
47 |
48 | function saveLocalSessionsOnly() {
49 |     try {
50 |         localStorage.setItem(STORAGE_KEY_SESSIONS, JSON.stringify(
    | sessions));
51 |         if (activeSessionId) {
52 |             localStorage.setItem(STORAGE_KEY_ACTIVE_SESSION_ID, acti
    | veSessionId);
53 |         }
54 |     } catch (err) {
55 |         console.warn('Failed to save chat sessions to localStorage
    | :', err);
56 |     }
57 | }
58 |
59 | function saveSessions(syncToServer = true) {
60 |     saveLocalSessionsOnly();
61 |     if (syncToServer) {
62 |         debounceSyncActiveSession();
63 |     }
64 | }
```

```
65 |
66 | function debounceSyncActiveSession() {
67 |   if (syncDebounceTimer) {
68 |     clearTimeout(syncDebounceTimer);
69 |   }
70 |   syncDebounceTimer = setTimeout(() => {
71 |     const active = getActiveSession();
72 |     if (active) {
73 |       syncSessionToServer(active);
74 |     }
75 |   }, 500);
76 | }
77 |
78 | async function syncSessionToServer(session) {
79 |   if (!session || !session.id) return;
80 |   try {
81 |     await fetch('/api/chat/sessions', {
82 |       method: 'POST',
83 |       headers: { 'Content-Type': 'application/json' },
84 |       body: JSON.stringify({
85 |         id: session.id,
86 |         title: session.title || 'Новый чат',
87 |         isCustomTitle: Boolean(session.isCustomTitle),
88 |         createdAt: session.createdAt || Date.now(),
89 |         updatedAt: session.updatedAt || Date.now(),
90 |         messages: session.messages || [],
91 |         chatHistory: session.chatHistory || []
92 |       })
93 |     });
94 |   } catch (err) {
95 |     console.debug('Background session sync to server skipped o
96 | r failed:', err);
97 |   }
98 |
99 | async function deleteSessionFromServer(sessionId) {
100 |   if (!sessionId) return;
101 |   try {
```

```

102 |     await fetch(`/api/chat/sessions/${encodeURIComponent(sessionId)}`, {
103 |       method: 'DELETE'
104 |     });
105 |   } catch (err) {
106 |     console.debug('Failed to delete session on server:', err);
107 |   }
108 | }
109 |
110 | async function clearAllSessionsFromServer() {
111 |   try {
112 |     await fetch('/api/chat/sessions', {
113 |       method: 'DELETE'
114 |     });
115 |   } catch (err) {
116 |     console.debug('Failed to clear sessions on server:', err);
117 |   }
118 | }
119 |
120 | async function syncSessionsWithServer() {
121 |   try {
122 |     const res = await cachedApiFetch('/api/chat/sessions', {
123 |       tags: [CACHE_TAG, 'chat-sessions']
124 |     });
125 |     if (!res.ok) return;
126 |     const data = await res.json();
127 |     const serverSessions = data.sessions || [];
128 |
129 |     // If we have local sessions and server is empty, or server
130 |     // has fewer/older sessions, bulk sync
131 |     if (sessions.length > 0 && serverSessions.length === 0) {
132 |       const syncRes = await fetch('/api/chat/sessions/sync', {
133 |         method: 'POST',
134 |         headers: { 'Content-Type': 'application/json' },
135 |         body: JSON.stringify({ sessions: sessions })
136 |       });
137 |       if (syncRes.ok) {
138 |         const syncData = await syncRes.json();

```

```

138         if (syncData.sessions && syncData.sessions.length > 0)
139         {
140             sessions = syncData.sessions;
141         }
142     } else if (serverSessions.length > 0) {
143         // Merge server sessions with any unique local sessions
144         const serverMap = new Map();
145         serverSessions.forEach((s) => serverMap.set(s.id, s));
146
147         const localOnly = [];
148         sessions.forEach((ls) => {
149             if (!serverMap.has(ls.id)) {
150                 localOnly.push(ls);
151             } else {
152                 // If local has newer updatedAt, prefer local and sync
153                 // to server
154                 const ss = serverMap.get(ls.id);
155                 if ((ls.updatedAt || 0) > (ss.updatedAt || 0)) {
156                     serverMap.set(ls.id, ls);
157                     syncSessionToServer(ls);
158                 }
159             }
160         });
161
162         if (localOnly.length > 0) {
163             // Push local only to server
164             await fetch('/api/chat/sessions/sync', {
165                 method: 'POST',
166                 headers: { 'Content-Type': 'application/json' },
167                 body: JSON.stringify({ sessions: localOnly })
168             });
169         }
170
171         sessions = Array.from(serverMap.values()).sort((a, b) =>
172             (b.updatedAt || 0) - (a.updatedAt || 0));
173     }

```

```
173 |     if (sessions.length === 0) {
174 |         createNewSession('Новый чат', true);
175 |         return;
176 |     }
177 |
178 |     const savedActiveId = localStorage.getItem(STORAGE_KEY_ACTIVE_SESSION_ID);
179 |     if (savedActiveId && sessions.some((s) => s.id === savedActiveId)) {
180 |         activeSessionId = savedActiveId;
181 |     } else {
182 |         activeSessionId = sessions[0].id;
183 |     }
184 |
185 |     saveLocalSessionsOnly();
186 |     renderSessionsList();
187 |     renderActiveSessionMessages();
188 | } catch (err) {
189 |     console.debug('Failed to sync sessions with server (working in offline/local mode):', err);
190 | }
191 | }
192 |
193 | function loadSessions() {
194 |     try {
195 |         const raw = localStorage.getItem(STORAGE_KEY_SESSIONS);
196 |         if (raw) {
197 |             const parsed = JSON.parse(raw);
198 |             if (Array.isArray(parsed) && parsed.length > 0) {
199 |                 sessions = parsed;
200 |                 const savedActiveId = localStorage.getItem(STORAGE_KEY_ACTIVE_SESSION_ID);
201 |                 if (savedActiveId && sessions.some((s) => s.id === savedActiveId)) {
202 |                     activeSessionId = savedActiveId;
203 |                 } else {
204 |                     activeSessionId = sessions[0].id;
205 |                 }

```

```
206 |         syncSessionsWithServer();
207 |         return;
208 |     }
209 | }
210 |
211 | // Migrate from legacy single-session chat history if present
212 | const legacyRaw = localStorage.getItem(LEGACY_STORAGE_KEY);
213 | ;
214 | if (legacyRaw) {
215 |     try {
216 |         const legacyLog = JSON.parse(legacyRaw);
217 |         if (Array.isArray(legacyLog) && legacyLog.length > 0)
218 |         {
219 |             const firstUserMsg = legacyLog.find((m) => m.sender
220 |             === 'user');
221 |             const title = firstUserMsg ? generateSessionTitle(firstUserMsg.text) : 'Предыдущий диалог';
222 |             const legacyHistory = [];
223 |             for (const item of legacyLog) {
224 |                 legacyHistory.push({
225 |                     role: item.sender === 'user' ? 'user' : 'model',
226 |                     parts: [item.text || item.voiceText || '']
227 |                 });
228 |             }
229 |
230 |             const legacySession = {
231 |                 id: 'session_' + Date.now(),
232 |                 title: title,
233 |                 isCustomTitle: false,
234 |                 createdAt: Date.now(),
235 |                 updatedAt: Date.now(),
236 |                 messages: legacyLog,
237 |                 chatHistory: legacyHistory
238 |             };
239 |             sessions = [legacySession];
240 |             activeSessionId = legacySession.id;
241 |             saveSessions();
242 |         }
243 |     }
244 | }
```

```
239 |         syncSessionsWithServer();
240 |         return;
241 |     }
242 | } catch (e) {
243 |     console.warn('Failed to migrate legacy chat history:',
    | e);
244 | }
245 | }
246 |
247 | // Default clean initial session and attempt server sync
248 | createNewSession('Новый чат', true);
249 | syncSessionsWithServer();
250 | } catch (err) {
251 |     console.warn('Failed to load chat sessions:', err);
252 |     sessions = [];
253 |     createNewSession('Новый чат', true);
254 |     syncSessionsWithServer();
255 | }
256 | }
257 |
258 | function getActiveSession() {
259 |     if (!activeSessionId || sessions.length === 0) {
260 |         return createNewSession('Новый чат', true);
261 |     }
262 |     let session = sessions.find((s) => s.id === activeSessionId)
    | ;
263 |     if (!session) {
264 |         session = sessions[0];
265 |         activeSessionId = session.id;
266 |     }
267 |     return session;
268 | }
269 |
270 | function createNewSession(title = 'Новый чат', makeActive = tr
    | ue) {
271 |     const newSession = {
272 |         id: 'session_' + Date.now() + '_' + Math.random().toString
    | (36).substring(2, 7),
```



```
273 |     title: title,
274 |     isCustomTitle: false,
275 |     createdAt: Date.now(),
276 |     updatedAt: Date.now(),
277 |     messages: [],
278 |     chatHistory: []
279 | };
280 |
281 | sessions.unshift(newSession);
282 | if (makeActive) {
283 |     activeSessionId = newSession.id;
284 | }
285 | saveSessions();
286 | renderSessionsList();
287 | renderActiveSessionMessages();
288 | return newSession;
289 | }
290 |
291 | function switchSession(sessionId) {
292 |     if (sessionId === activeSessionId) return;
293 |     const session = sessions.find((s) => s.id === sessionId);
294 |     if (!session) return;
295 |
296 |     activeSessionId = sessionId;
297 |     saveSessions(false);
298 |     renderSessionsList();
299 |     renderActiveSessionMessages();
300 |
301 |     const msgInput = document.getElementById('message-input');
302 |     if (msgInput) msgInput.focus();
303 | }
304 |
305 | function renameSession(sessionId, newTitle = null) {
306 |     const session = sessions.find((s) => s.id === sessionId);
307 |     if (!session) return;
308 |
309 |     let title = newTitle;
310 |     if (title === null) {
```

```
311     title = prompt('Введите новое название диалога:', session.  
title);  
312 }  
313  
314 if (title && title.trim()) {  
315     session.title = title.trim();  
316     session.isCustomTitle = true;  
317     session.updatedAt = Date.now();  
318     saveSessions();  
319     renderSessionsList();  
320     if (session.id === activeSessionId) {  
321         updateCurrentSessionHeader();  
322     }  
323 }  
324 }  
325  
326 function deleteSession(sessionId) {  
327     const index = sessions.findIndex((s) => s.id === sessionId);  
328     if (index === -1) return;  
329  
330     sessions.splice(index, 1);  
331     if (sessions.length === 0) {  
332         createNewSession('Новый чат', true);  
333     } else if (activeSessionId === sessionId) {  
334         activeSessionId = sessions[0].id;  
335         renderActiveSessionMessages();  
336     }  
337  
338     saveSessions(false);  
339     deleteSessionFromServer(sessionId);  
340     renderSessionsList();  
341 }  
342  
343 function clearCurrentSession() {  
344     const session = getActiveSession();  
345     if (!session) return;  
346  
347     session.messages = [];
```

```
348 session.chatHistory = [];
349 session.title = 'Новый чат';
350 session.isCustomTitle = false;
351 session.updatedAt = Date.now();
352
353 saveSessions();
354 renderSessionsList();
355 renderActiveSessionMessages();
356 }
357
358 function clearAllSessions() {
359     if (!confirm('Вы уверены, что хотите удалить ВСЕ сохраненные
        | диалоги?')) return;
360
361     sessions = [];
362     try {
363         localStorage.removeItem(STORAGE_KEY_SESSIONS);
364         localStorage.removeItem(STORAGE_KEY_ACTIVE_SESSION_ID);
365         localStorage.removeItem(LEGACY_STORAGE_KEY);
366     } catch (e) {}
367
368     // Clear cache for chat sessions
369     clearCacheByTag(CACHE_TAG).catch(err => console.debug('Faile
        | d to clear chat cache:', err));
370
371     clearAllSessionsFromServer();
372     createNewSession('Новый чат', true);
373 }
374
375
376 function updateCurrentSessionHeader() {
377     const titleEl = document.getElementById('chat-current-title'
        | );
378     const session = getActiveSession();
379     if (titleEl && session) {
380         titleEl.textContent = session.title || 'Новый чат';
381         titleEl.title = session.title || 'Новый чат';
382     }
```

```

383 }
384
385 function renderSessionsList() {
386     const listEl = document.getElementById('chat-sessions-list')
387     ;
388     const countEl = document.getElementById('chat-sessions-count
389     ');
390     if (!listEl) return;
391     listEl.innerHTML = '';
392     if (countEl) {
393         countEl.textContent = `${sessions.length} ${sessions.lengt
394         h === 1 ? 'чат' : (sessions.length < 5 ? 'чата' : 'чатов')}`;
395     }
396     updateCurrentSessionHeader();
397     sessions.forEach((session) => {
398         const item = document.createElement('div');
399         const isActive = session.id === activeSessionId;
400         item.className = 'chat-session-item d-flex align-items-cen
401         ter justify-content-between p-1 px-2 rounded mb-1' + (isActive
402         ? ' active' : '');
403         item.dataset.sessionId = session.id;
404         item.innerHTML = `
405             <div class="chat-session-title d-flex align-items-center
406             gap-1 text-truncate flex-grow-1 me-1" title="${session.title}
407             ">
408                 <i class="bi bi-chat-text ${isActive ? 'text-primary'
409                 : 'text-muted'} flex-shrink-0"></i>
410                 <span class="text-truncate">${session.title}</span>
411             </div>
412             <div class="chat-session-actions d-flex align-items-cent
413             er gap-1 flex-shrink-0">
414                 <button class="btn btn-sm btn-link text-muted p-0 btn-
415                 rename" title="Переименовать" type="button">
416                 <i class="bi bi-pencil" style="font-size: 0.8rem;"><

```

```

    /i>
411         </button>
412         <button class="btn btn-sm btn-link text-danger p-0 btn
    -delete" title="Удалить" type="button">
413             <i class="bi bi-trash" style="font-size: 0.8rem;"></
    i>
414         </button>
415     </div>
416     `;
417
418     item.addEventListener('click', (e) => {
419         if (e.target.closest('.btn-rename') || e.target.closest(
    '.btn-delete')) return;
420         switchSession(session.id);
421     });
422
423     const renameBtn = item.querySelector('.btn-rename');
424     if (renameBtn) {
425         renameBtn.addEventListener('click', (e) => {
426             e.stopPropagation();
427             renameSession(session.id);
428         });
429     }
430
431     const deleteBtn = item.querySelector('.btn-delete');
432     if (deleteBtn) {
433         deleteBtn.addEventListener('click', (e) => {
434             e.stopPropagation();
435             if (confirm(`Удалить диалог "${session.title}"?`)) {
436                 deleteSession(session.id);
437             }
438         });
439     }
440
441     listEl.appendChild(item);
442 });
443 }
444

```

```
445 function renderActiveSessionMessages() {
446   const chatWindow = document.getElementById('chat-window');
447   if (!chatWindow) return;
448
449   chatWindow.innerHTML = '';
450   const session = getActiveSession();
451
452   if (session && session.messages && session.messages.length >
    0) {
453     for (const msg of session.messages) {
454       addMessageToDom(msg.sender, msg.text, msg.voiceText, msg
        .timeStr, msg.mode);
455     }
456   } else {
457     const welcome = document.createElement('div');
458     welcome.className = 'message bot-message';
459     welcome.innerHTML = '<strong>AI Assistant</strong>: Добро
        пожаловать! Задайте любой вопрос в поле ввода.';
460     chatWindow.appendChild(welcome);
461   }
462
463   chatWindow.scrollTop = chatWindow.scrollHeight;
464   updateCurrentSessionHeader();
465 }
466
467 /**
468  * Update cache status badge
469  */
470 function updateChatCacheStatus() {
471   const badge = document.getElementById('chat-cache-badge');
472   const ratio = document.getElementById('chat-cache-ratio');
473   if (!badge || !ratio) return;
474
475   try {
476     const stats = window.apiCacheStats || { hits: 0, total: 0
    };
477     const hitRate = stats.total > 0 ? Math.round((stats.hits /
    stats.total) * 100) : 0;
```

```
478     ratio.textContent = `${hitRate}%`;
479
480     // Color coding based on hit rate
481     badge.className = 'badge cache-badge';
482     if (hitRate >= 70) {
483         badge.classList.add('bg-success-subtle', 'text-success')
484     };
485     } else if (hitRate >= 40) {
486         badge.classList.add('bg-warning-subtle', 'text-warning')
487     };
488     } else {
489         badge.classList.add('bg-secondary-subtle', 'text-seconda
490 ry');
491     }
492 } catch (err) {
493     console.debug('Failed to update cache status:', err);
494 }
495
496 /**
497  * Clear chat cache
498  */
499 async function clearChatCache() {
500     try {
501         await clearCacheByTag(CACHE_TAG);
502         updateChatCacheStatus();
503         console.log('Chat cache cleared');
504     } catch (err) {
505         console.error('Failed to clear chat cache:', err);
506     }
507 }
508
509 function initChatTab() {
510     loadSessions();
511     renderSessionsList();
512     renderActiveSessionMessages();
513     // Update cache status
```

```
513 |   updateChatCacheStatus();
514 |
515 |   // Bind Sidebar Toggle
516 |   const sidebar = document.getElementById('chat-sidebar');
517 |   const sidebarToggleBtn = document.getElementById('chat-sideb
  ar-toggle-btn');
518 |   if (sidebar && sidebarToggleBtn) {
519 |     const isCollapsed = localStorage.getItem(STORAGE_KEY_SIDE
  BAR_COLLAPSED) === 'true';
520 |     if (isCollapsed) {
521 |       sidebar.classList.add('collapsed');
522 |       sidebar.style.display = 'none';
523 |     }
524 |
525 |     sidebarToggleBtn.addEventListener('click', () => {
526 |       const isCurrentlyCollapsed = sidebar.classList.toggle('c
  ollapsed');
527 |       sidebar.style.display = isCurrentlyCollapsed ? 'none' :
  'flex';
528 |       localStorage.setItem(STORAGE_KEY_SIDE BAR_COLLAPSED, isCu
  rrentlyCollapsed);
529 |     });
530 |   }
531 |
532 |   // Bind New Session Button
533 |   const newSessionBtn = document.getElementById('chat-new-sess
  ion-btn');
534 |   if (newSessionBtn) {
535 |     newSessionBtn.addEventListener('click', () => {
536 |       createNewSession('Новый чат', true);
537 |       const msgInput = document.getElementById('message-input'
  );
538 |       if (msgInput) msgInput.focus();
539 |     });
540 |   }
541 |
542 |   // Bind Clear Current / Clear All buttons
543 |   const clearBtn = document.getElementById('chat-clear-btn');
```



```
544 |   if (clearBtn) {
545 |       clearBtn.addEventListener('click', () => {
546 |           if (confirm('Очистить сообщения текущего диалога?')) {
547 |               clearCurrentSession();
548 |           }
549 |       });
550 |   }
551 |
552 |   const clearAllBtn = document.getElementById('chat-clear-all-
sessions-btn');
553 |   if (clearAllBtn) {
554 |       clearAllBtn.addEventListener('click', () => {
555 |           clearAllSessions();
556 |       });
557 |   }
558 |
559 |   const clearCacheBtn = document.getElementById('chat-clear-ca
che-btn');
560 |   if (clearCacheBtn) {
561 |       clearCacheBtn.addEventListener('click', async () => {
562 |           await clearChatCache();
563 |           // Force refresh sessions
564 |           await syncSessionsWithServer();
565 |           renderSessionsList();
566 |       });
567 |   }
568 |
569 |   const renameCurrentBtn = document.getElementById('chat-renam
e-current-btn');
570 |   if (renameCurrentBtn) {
571 |       renameCurrentBtn.addEventListener('click', () => {
572 |           if (activeSessionId) {
573 |               renameSession(activeSessionId);
574 |           }
575 |       });
576 |   }
577 |
578 |   // Expose sendMessage globally for user menu actions
```

```
579 | window.sendMessage = sendMessage;
580 |
581 | // Bind Input and Controls
582 | const sendBtn = document.getElementById('send-button');
583 | const msgInput = document.getElementById('message-input');
584 | const ragToggle = document.getElementById('chat-rag-toggle')
    | ;
585 | const topKInput = document.getElementById('chat-top-k');
586 | const minScoreInput = document.getElementById('chat-min-score');
587 |
588 | const updateRagInputsState = (enabled) => {
589 |   if (topKInput) {
590 |     topKInput.disabled = !enabled;
591 |     topKInput.style.opacity = enabled ? '1' : '0.5';
592 |   }
593 |   if (minScoreInput) {
594 |     minScoreInput.disabled = !enabled;
595 |     minScoreInput.style.opacity = enabled ? '1' : '0.5';
596 |   }
597 | };
598 |
599 | if (ragToggle) {
600 |   const savedRagPref = localStorage.getItem('chat_rag_enabled');
601 |   if (savedRagPref !== null) {
602 |     ragToggle.checked = savedRagPref === 'true';
603 |     updateRagInputsState(ragToggle.checked);
604 |   }
605 |   fetch('/auth/settings', { cache: 'no-cache' })
606 |     .then((res) => (res.ok ? res.json() : null))
607 |     .then((settings) => {
608 |       if (settings && typeof settings.rag_enabled !== 'undefined') {
609 |         const isEnabled = settings.rag_enabled !== 0;
610 |         ragToggle.checked = isEnabled;
611 |         localStorage.setItem('chat_rag_enabled', isEnabled);
612 |         updateRagInputsState(isEnabled);
```

```

613         }
614     })
615     .catch(() => {});
616
617     ragToggle.addEventListener('change', async (e) => {
618         const isEnabled = e.target.checked;
619         localStorage.setItem('chat_rag_enabled', isEnabled);
620         updateRagInputsState(isEnabled);
621         try {
622             await fetch('/auth/settings', {
623                 method: 'POST',
624                 headers: { 'Content-Type': 'application/json' },
625                 body: JSON.stringify({ rag_enabled: isEnabled ? 1 :
626 | 0 })
627             });
628         } catch (err) {
629             console.error('Error saving RAG setting:', err);
630         }
631     });
632
633     const modeSelect = document.getElementById('chat-output-mode
634 | ');
635     if (modeSelect) {
636         const savedMode = localStorage.getItem('chat_output_mode')
637 | ;
638         if (savedMode) {
639             modeSelect.value = savedMode;
640         } else {
641             modeSelect.value = 'text_only';
642         }
643         modeSelect.addEventListener('change', (e) => {
644             localStorage.setItem('chat_output_mode', e.target.value)
645 | ;
646             if (e.target.value === 'text_only' && window.chatService
647 | ?.stop) {
648                 window.chatService.stop();
649             }

```

```

646     });
647 }
648
649 if (sendBtn) {
650     sendBtn.addEventListener('click', sendMessage);
651 }
652 if (msgInput) {
653     msgInput.addEventListener('keypress', (e) => {
654         if (e.key === 'Enter') {
655             sendMessage();
656         }
657     });
658 }
659
660 // Check for pending external chat payload (from Chrome exte
nsion or URL parameters)
661 try {
662     const pending = sessionStorage.getItem('pending_breadboard
_chat');
663     if (pending) {
664         const payload = JSON.parse(pending);
665         sessionStorage.removeItem('pending_breadboard_chat');
666         setTimeout(() => handleExternalChatPrompt(payload), 300)
;
667     } else {
668         const urlParams = new URLSearchParams(window.location.se
arch);
669         const promptParam = urlParams.get('prompt') || urlParams
.get('message');
670         if (promptParam) {
671             setTimeout(() => {
672                 handleExternalChatPrompt({
673                     prompt: promptParam,
674                     autoSend: urlParams.get('autosend') !== 'false'
675                 });
676             }, 300);
677         }
678     }

```

```
679 | } catch (e) {
680 |     console.error('Error checking pending chat payload:', e);
681 | }
682 | }
683 |
684 | function handleExternalChatPrompt(payload) {
685 |     if (!payload || !payload.prompt) return;
686 |
687 |     const chatTabBtn = document.querySelector('[data-bs-target="
#tab-chat"]');
688 |     if (chatTabBtn && window.bootstrap?.Tab) {
689 |         bootstrap.Tab.getOrCreateInstance(chatTabBtn).show();
690 |     }
691 |
692 |     const msgInput = document.getElementById('message-input');
693 |     if (msgInput) {
694 |         msgInput.value = payload.prompt;
695 |         if (payload.autoSend !== false) {
696 |             setTimeout(() => {
697 |                 sendMessage();
698 |             }, 150);
699 |         }
700 |     }
701 | }
702 |
703 | function handleRemoteTranscript(text) {
704 |     if (!text || !text.trim()) return;
705 |     const cleanText = text.trim();
706 |
707 |     // Ensure active session exists
708 |     const session = getActiveSession();
709 |     if (!session) return;
710 |
711 |     const userTimeStr = new Date().toLocaleTimeString();
712 |
713 |     // Clear welcome placeholder if this is the first message
714 |     const chatWin = document.getElementById('chat-window');
715 |     if (chatWin && session.messages.length === 0) {
```

```

716     chatWin.innerHTML = '';
717 }
718
719 // Add message to DOM and save to active session without tri
ggering AI model generation
720 addMessageToDom('user', cleanText, null, userTimeStr);
721 session.messages.push({ sender: 'user', text: cleanText, tim
eStr: userTimeStr });
722
723 if ((!session.title || session.title === 'Новый чат' || sess
ion.title === 'Предыдущий диалог') && !session.isCustomTitle)
{
724     session.title = generateSessionTitle(cleanText);
725     renderSessionsList();
726 }
727
728 session.updatedAt = Date.now();
729 saveSessions();
730
731 if (chatWin) {
732     chatWin.scrollTop = chatWin.scrollHeight;
733 }
734 }
735
736 if (typeof window !== 'undefined') {
737     window.initChatTab = initChatTab;
738     window.handleExternalChatPrompt = handleExternalChatPrompt;
739     window.handleRemoteTranscript = handleRemoteTranscript;
740     window.clearChatHistory = clearCurrentSession;
741     window.createNewSession = createNewSession;
742     window.switchSession = switchSession;
743     window.renameSession = renameSession;
744     window.deleteSession = deleteSession;
745
746     if (typeof chrome !== 'undefined' && chrome.runtime?.onMessa
ge) {
747         chrome.runtime.onMessage.addListener((msg) => {
748             if (msg && msg.action === 'breadboard_external_chat_prom

```

```

    pt') {
749         handleExternalChatPrompt(msg);
750     }
751 });
752 }
753 }
754
755 function parseContentToHtml(text) {
756     if (!text) return '';
757
758     let cleaned = text.trim();
759     const codeBlockRegex = /^```(?:html|xml)?\s*([\s\S]*?)\s*```$/i;
760     const match = cleaned.match(codeBlockRegex);
761     if (match) {
762         cleaned = match[1];
763     }
764
765     if (window.marked && typeof window.marked.parse === 'function') {
766         return window.marked.parse(cleaned);
767     }
768
769     return cleaned
770         .replace(/\*\*(.*?)\*\*/g, '<strong>$1</strong>')
771         .replace(/\*(.*?)\*/g, '<em>$1</em>')
772         .replace(/\`([^\`]+)\`/g, '<code>$1</code>')
773         .replace(/\n/g, '<br>');
774 }
775
776 function addMessageToDom(sender, text, voiceText = null, times
tr = null, outputMode = null) {
777     const win = document.getElementById('chat-window');
778     if (!win) return null;
779
780     const el = document.createElement('div');
781     el.className = 'message ' + (sender === 'user' ? 'user-messa
ge' : 'bot-message');

```

```

782
783     const displayText = parseContentToHtml(text);
784     const formattedTime = timeStr || new Date().toLocaleTimeString();
785
786     if (sender === 'user') {
787         el.innerHTML = `Вы (${formattedTime}): <div class="message-content">${displayText}</div>`;
788     } else {
789         const modeBadge = outputMode === 'voice_only' ? '<span class="badge bg-secondary-subtle text-secondary me-1"><i class="bi bi-volume-up"></i> Голос</span>' : '';
790         el.innerHTML = `
791             <div class="d-flex align-items-center justify-content-between mb-1">
792                 <div><strong>AI Assistant</strong> ${modeBadge}<span class="text-muted">(${formattedTime})</span></div>
793                 <button class="btn btn-sm btn-outline-secondary py-0 px-2 btn-speak-message" title="Озвучить ответ" type="button">
794                     <i class="bi bi-volume-up"></i>
795                 </button>
796             </div>
797             <div class="message-content">${displayText || (voiceText ? `<em><i class="bi bi-soundwave me-1"></i>${parseContentToHtml(voiceText)}</em>` : '')}</div>
798         `;
799         const speakBtn = el.querySelector('.btn-speak-message');
800         if (speakBtn) {
801             speakBtn.addEventListener('click', () => {
802                 if (window.chatService?.speak) {
803                     window.chatService.speak(voiceText || text);
804                 }
805             });
806         }
807     }
808
809     win.appendChild(el);
810     win.scrollTop = win.scrollHeight;

```



```
811 |   return el;
812 | }
813 |
814 | function setStatus(statusText, isVisible = true) {
815 |   const indicator = document.getElementById('chat-status-indicator');
816 |   const textEl = document.getElementById('chat-status-text');
817 |   if (!indicator || !textEl) return;
818 |
819 |   if (isVisible) {
820 |     textEl.textContent = statusText || 'Обработка запроса...';
821 |     indicator.classList.remove('d-none');
822 |   } else {
823 |     indicator.classList.add('d-none');
824 |   }
825 | }
826 |
827 | async function sendMessage() {
828 |   const input = document.getElementById('message-input');
829 |   const btn = document.getElementById('send-button');
830 |   const topKInput = document.getElementById('chat-top-k');
831 |   const minScoreInput = document.getElementById('chat-min-score');
832 |   const modeSelect = document.getElementById('chat-output-mode');
833 |   const ragToggle = document.getElementById('chat-rag-toggle');
834 |
835 |   if (!input || !btn) return;
836 |
837 |   if (!window.activeModelName) {
838 |     input.disabled = true;
839 |     btn.disabled = true;
840 |     input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
841 |     input.classList.add('is-invalid');
842 |     return;
843 |   }
```

```
844
845   const msg = input.value.trim();
846   if (!msg) return;
847
848   const session = getActiveSession();
849   if (!session) return;
850
851   const topK = topKInput ? parseInt(topKInput.value, 10) || 3
      : 3;
852   const minScore = minScoreInput ? parseFloat(minScoreInput.value) || 0.45 : 0.45;
853   const outputMode = modeSelect ? modeSelect.value : 'text_only';
854   const shouldSpeak = outputMode === 'voice_only' || outputMode === 'text_and_voice';
855   const isRagEnabled = ragToggle ? ragToggle.checked : true;
856
857   input.value = '';
858   input.disabled = true;
859   btn.disabled = true;
860
861   setStatus('Отправка запроса...', true);
862   const userTimeStr = new Date().toLocaleTimeString();
863
864   // Clear welcome placeholder if this is the first message
865   const chatWin = document.getElementById('chat-window');
866   if (chatWin && session.messages.length === 0) {
867     chatWin.innerHTML = '';
868   }
869
870   addMessageToDom('user', msg, null, userTimeStr);
871   session.messages.push({ sender: 'user', text: msg, timeStr: userTimeStr });
872
873   // Auto-name the session based on the first query if not customized
874   if ((!session.title || session.title === 'Новый чат' || session.title === 'Предыдущий диалог') && !session.isCustomTitle)
```

```

    {
875     session.title = generateSessionTitle(msg);
876     renderSessionsList();
877 }
878
879 saveSessions();
880
881 const win = document.getElementById('chat-window');
882 const botMessageEl = document.createElement('div');
883 botMessageEl.className = 'message bot-message';
884 const timeStr = new Date().toLocaleTimeString();
885 const modeBadge = outputMode === 'voice_only' ? '<span class
="badge bg-secondary-subtle text-secondary me-1"><i class="bi
bi-volume-up"></i> Голос</span>' : '';
886 botMessageEl.innerHTML = `
887     <div class="d-flex align-items-center justify-content-betw
888     een mb-1">
889         <div><strong>AI Assistant</strong> ${modeBadge}<span cla
890         ss="text-muted">(${timeStr})</span></div>
891         <button class="btn btn-sm btn-outline-secondary py-0 px-
892         2 btn-speak-message d-none" title="Озвучить ответ" type="butto
893         n">
894             <i class="bi bi-volume-up"></i>
895             </button>
896         </div>
897         <div class="bot-text"><span class="text-muted">□ Ожидание
898         ответа...</span></div>
899     `;
900     if (win) {
901         win.appendChild(botMessageEl);
902         win.scrollTop = win.scrollHeight;
903     }
904     const textDiv = botMessageEl.querySelector('.bot-text');
905     const speakBtn = botMessageEl.querySelector('.btn-speak-mess
906     age');
907     let fullReply = '';
908     let started = false;

```

```
904 |
905 | try {
906 |     const generationConfig = {
907 |         top_k: topK,
908 |         min_score: minScore,
909 |         threshold: minScore,
910 |         rag_enabled: isRagEnabled,
911 |         output_mode: outputMode,
912 |         tts_enabled: shouldSpeak,
913 |         model: window.activeModelName || undefined,
914 |         search_engine: window.activeSearchEngine || undefined
915 |     };
916 |
917 |     let rafId = null;
918 |     const scheduleRender = () => {
919 |         if (rafId) return;
920 |         rafId = requestAnimationFrame(() => {
921 |             rafId = null;
922 |             if (textDiv) {
923 |                 textDiv.innerHTML = parseContentToHtml(fullReply);
924 |             }
925 |             if (win) win.scrollTop = win.scrollHeight;
926 |         });
927 |     };
928 |
929 |     const replyObj = await window.chatService.sendChatMessage(
930 |         msg,
931 |         (chunk, status) => {
932 |             if (status) {
933 |                 setStatus(status, true);
934 |             }
935 |             if (chunk) {
936 |                 if (!started) {
937 |                     if (textDiv) textDiv.textContent = '';
938 |                     started = true;
939 |                 }
940 |                 fullReply += chunk;
941 |                 scheduleRender();
```

```
942 |     }
943 | },
944 | session.chatHistory || [],
945 | generationConfig
946 | );
947 |
948 | if (rafId) {
949 |     cancelAnimationFrame(rafId);
950 |     rafId = null;
951 | }
952 |
953 | const finalReply = replyObj.text || fullReply;
954 | if (textDiv) {
955 |     if (outputMode === 'voice_only' && !finalReply.trim() &&
956 | replyObj.voice) {
957 |         textDiv.innerHTML = `<em><i class="bi bi-soundwave me-
958 | 1"></i>${parseContentToHtml(replyObj.voice)}</em>`;
959 |     } else {
960 |         textDiv.innerHTML = parseContentToHtml(finalReply);
961 |     }
962 | }
963 |
964 | if (speakBtn) {
965 |     speakBtn.classList.remove('d-none');
966 |     speakBtn.addEventListener('click', () => {
967 |         if (window.chatService?.speak) {
968 |             window.chatService.speak(replyObj || finalReply);
969 |         }
970 |     });
971 | }
972 |
973 | // Auto-speak response if mode is voice_only or text_and_v
974 | oice
975 | if (shouldSpeak && window.chatService?.speak) {
976 |     window.chatService.speak(replyObj || finalReply);
977 | }
978 |
979 | if (!session.chatHistory) session.chatHistory = [];
```

```
977 |     session.chatHistory.push({ role: 'user', parts: [msg] });
978 |     session.chatHistory.push({ role: 'model', parts: [finalReply || replyObj.voice || ''] });
979 |
980 |     session.messages.push({
981 |         sender: 'bot',
982 |         text: finalReply,
983 |         voiceText: replyObj.voice || null,
984 |         timeStr: timeStr,
985 |         mode: outputMode
986 |     });
987 |     session.updatedAt = Date.now();
988 |     saveSessions();
989 |
990 |     setStatus('', false);
991 | } catch (err) {
992 |     const errText = err instanceof Error ? err.message : typeof err === 'object' && err !== null ? JSON.stringify(err) : String(err);
993 |     if (textDiv) {
994 |         textDiv.innerHTML = `Ошибка: ${errText}</span>`;
995 |     }
996 |     setStatus('Ошибка исполнения', true);
997 |     setTimeout(() => setStatus('', false), 4000);
998 | } finally {
999 |     if (window.activeModelName) {
1000 |         input.disabled = false;
1001 |         btn.disabled = false;
1002 |         input.placeholder = 'Введите сообщение...';
1003 |         input.classList.remove('is-invalid');
1004 |         input.focus();
1005 |     } else {
1006 |         input.disabled = true;
1007 |         btn.disabled = true;
1008 |         input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
1009 |         input.classList.add('is-invalid');
```

```
1010 |     }
1011 |     if (win) win.scrollTop = win.scrollHeight;
1012 | }
1013 | }
1014 |
1015 |
```

index.html

```
1 <!-- Cloudflare Tunnel Monitor Application Tab -->
2 <style>
3   .cf-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .cf-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    min-height: 58px;
15    gap: 0.2rem;
16  }
17  .cf-label {
18    font-size: 0.68rem;
19    text-transform: uppercase;
20    letter-spacing: 0.05em;
21    color: #94a3b8;
22    line-height: 1.1;
23  }
24  .cf-value {
25    font-size: 1.25rem;
26    font-weight: 700;
27    color: #f8fafc;
28    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
29    Consolas, monospace;
30    line-height: 1.15;
31  }
32  .cf-panel {
33    background: #0f172a !important;
34    border: 1px solid #334155 !important;
35    border-radius: 8px;
36    overflow: hidden;
```



```

36 | }
37 | .cf-panel-header {
38 |     background: #1e293b !important;
39 |     border-bottom: 1px solid #334155 !important;
40 |     padding: 0.45rem 0.85rem;
41 |     font-weight: 600;
42 |     font-size: 0.82rem;
43 |     color: #38bdf8;
44 |     display: flex;
45 |     align-items: center;
46 |     justify-content: space-between;
47 | }
48 | .cf-log-terminal {
49 |     background: #020617;
50 |     color: #cbd5e1;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
52 |     font-size: 0.78rem;
53 |     line-height: 1.45;
54 |     padding: 0.5rem 0.75rem;
55 |     border-radius: 6px;
56 |     max-height: 380px;
57 |     overflow-y: auto;
58 | }
59 | </style>
60 |
61 | <div class="cf-container">
62 |     <!-- Header Alert & Status (Compact Strip) -->
63 |     <div class="d-flex justify-content-between align-items-center
    flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style
    ="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%
    ); border: 1px solid #334155;">
64 |         <div class="d-flex align-items-center gap-2 flex-wrap">
65 |             <span class="fs-5">❏ </span>
66 |             <span class="fw-bold text-white small" style="font-size:
    0.92rem;">Cloudflare Tunnel Monitor</span>
67 |             <span class="badge bg-dark border border-secondary text-
    info fw-normal" style="font-size: 0.7rem;">Diagnostics</span>

```

```

68 |     <span class="text-secondary small d-none d-lg-inline" st
| yle="font-size: 0.76rem;">| Служба Cloudflare Tunnel, доступно
| сть URL & логи</span>
69 |     </div>
70 |     <div class="d-flex align-items-center gap-1.5 flex-wrap">
71 |         <span class="badge rounded-pill bg-success-subtle text-s
| uccess border border-success px-2 py-1" style="font-size: 0.72
| rem;" id="cf-daemon-badge">
72 |             • АКТИВЕН
73 |         </span>
74 |         <button class="btn btn-xs btn-outline-info rounded-pill
| px-2.5 py-1" style="font-size: 0.75rem;" id="btn-cf-refresh">
75 |             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
76 |         </button>
77 |         <button class="btn btn-xs btn-outline-warning rounded-pi
| ll px-2 py-1" style="font-size: 0.75rem;" id="btn-cf-config">
78 |             <i class="bi bi-gear-fill me-1"></i> Настройки
79 |         </button>
80 |         <button class="btn btn-xs btn-primary rounded-pill px-2.
| 5 py-1" style="font-size: 0.75rem;" id="btn-cf-test-endpoint">
81 |             <i class="bi bi-broadcast me-1"></i> Ingress
82 |         </button>
83 |     </div>
84 | </div>
85 |
86 | <!-- Top Metrics Bar (Compact Density Grid) -->
87 | <div class="row g-2 mb-2">
88 |     <div class="col-6 col-md-3">
89 |         <div class="cf-card">
90 |             <span class="cf-label">Публичный Ingress URL</span>
91 |             <div class="d-flex align-items-baseline justify-conten
| t-between mt-1">
92 |                 <span class="cf-value text-info text-truncate" id="c
| f-ingress-url" style="font-size: 0.88rem;">https://kino.davidk
| a.net</span>
93 |             </div>
94 |         </div>
95 |     </div>

```

```
96 |     <div class="col-6 col-md-3">
97 |         <div class="cf-card">
98 |             <span class="cf-label">Доступность эндпоинта</span>
99 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
100 |                 <span class="cf-value text-success" style="font-size: 0.95rem;" id="cf-endpoint-status">200 OK</span>
101 |             </div>
102 |         </div>
103 |     </div>
104 |     <div class="col-6 col-md-3">
105 |         <div class="cf-card">
106 |             <span class="cf-label">CPU / Память демона</span>
107 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
108 |                 <span class="cf-value" style="font-size: 0.95rem;" id="cf-process-usage">0.5% / 45 MB</span>
109 |             </div>
110 |         </div>
111 |     </div>
112 |     <div class="col-6 col-md-3">
113 |         <div class="cf-card">
114 |             <span class="cf-label">Health Score</span>
115 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
116 |                 <span class="cf-value text-success" id="cf-health-score">100/100</span>
117 |             </div>
118 |         </div>
119 |     </div>
120 | </div>
121 |
122 | <div class="row g-3">
123 |     <!-- Tunnel Logs Tail Panel -->
124 |     <div class="col-lg-7">
125 |         <div class="cf-panel h-100">
126 |             <div class="cf-panel-header">
127 |                 <span><i class="bi bi-terminal me-1"></i> Журнал clo
```

```

udflared.log</span>
128     <span class="badge bg-secondary" id="cf-log-count-ba
dge">50 строк</span>
129     </div>
130     <div class="p-3">
131         <div class="cf-log-terminal" id="cf-log-terminal-out
put">
132             Загрузка журналов...
133         </div>
134     </div>
135 </div>
136 </div>
137
138 <!-- AI Diagnostics & Recommendations Panel -->
139 <div class="col-lg-5">
140     <div class="cf-panel h-100">
141         <div class="cf-panel-header">
142             <span><i class="bi bi-heart-pulse-fill text-danger m
e-1"></i> Диагностика & Рекомендации</span>
143             <button class="btn btn-sm btn-outline-info py-0 px-2
" id="btn-cf-run-diagnostics">Диагностировать</button>
144         </div>
145         <div class="p-3">
146             <div class="card bg-dark border-secondary p-3 mb-3">
147                 <h6 class="fw-bold text-info mb-1" id="cf-diag-sum
mary-title">Статус оптимален</h6>
148                 <p class="small text-muted mb-0" id="cf-diag-summa
ry-desc">Все туннельные соединения стабильны, аномалий не обна
ружено.</p>
149             </div>
150             <h6 class="small text-uppercase text-muted fw-bold m
b-2">Рекомендации:</h6>
151             <ul class="list-group list-group-flush bg-transparen
t small" id="cf-recommendations-list">
152                 <li class="list-group-item bg-transparent text-lig
ht border-secondary py-1 px-0">• Поддерживайте актуальную верс
ию cloudflared</li>
153                 <li class="list-group-item bg-transparent text-lig

```

```
ht border-secondary py-1 px-0">• Проверьте сертификат SSL дом  
ена</li>
```

```
154|         </ul>
```

```
155|     </div>
```

```
156| </div>
```

```
157| </div>
```

```
158| </div>
```

```
159| </div>
```

```
160|
```

main.js

```
1 // =====
  =====
2 // Process Name: Cloudflared Monitor Web Tab Module
3 // =====
  =====
4 // Description:
5 //   Client-side JavaScript controller for the Cloudflare Tunnel Monitor
6 //   administrative web interface tab.
7 //
8 // File: main.js
9 // Project: ai-breadboard
10 // Package: src.api.webinterface.cloudflared_tab
11 // Author: hypo69
12 // Copyright: © 2026 hypo69
13 // =====
  =====
14
15 (function() {
16   let isCfInitialized = false;
17
18   async function fetchTunnelStatus() {
19     try {
20       const res = await fetch('/api/cloudflared/status');
21       if (!res.ok) return;
22       const data = await res.json();
23
24       const badge = document.getElementById('cf-daemon-badge')
25       ;
26       const ingressEl = document.getElementById('cf-ingress-url');
27       const endpointEl = document.getElementById('cf-endpoint-status');
28       const usageEl = document.getElementById('cf-process-usage');
29       const healthEl = document.getElementById('cf-health-score');
```

```

    e');
29
30     if (badge) {
31         badge.className = data.daemon_running ? 'badge rounded
-pill bg-success-subtle text-success border border-success px-
3 py-2' : 'badge rounded-pill bg-danger-subtle text-danger bor
der border-danger px-3 py-2';
32         badge.innerText = data.daemon_running ? '● Туннель акт
ивен' : '● Туннель остановлен';
33     }
34
35     if (ingressEl) ingressEl.innerText = data.public_url ||
'https://kino.davidka.net';
36     if (endpointEl) {
37         endpointEl.innerText = data.endpoint_reachable ? `Дост
упен (${data.latency_ms || 0} ms)` : 'Недоступен';
38         endpointEl.className = data.endpoint_reachable ? 'cf-v
alue text-success' : 'cf-value text-danger';
39     }
40     if (usageEl) {
41         usageEl.innerText = `${data.cpu_percent || 0}% / ${dat
a.memory_mb || 0} MB`;
42     }
43     if (healthEl) {
44         healthEl.innerText = `${data.health_score || 100}/100`
;
45     }
46 } catch (e) {
47     console.error('[CloudflaredTab] Failed to fetch status:'
, e);
48 }
49 }
50
51 async function fetchLogs() {
52     try {
53         const res = await fetch('/api/cloudflared/logs?limit=50'
);
54         if (!res.ok) return;

```

```

55 |     const data = await res.json();
56 |     const logs = data.logs || [];
57 |
58 |     const terminal = document.getElementById('cf-log-termina
l-output');
59 |     const countBadge = document.getElementById('cf-log-count
-badge');
60 |     if (countBadge) countBadge.innerText = `${logs.length} c
трок`;
61 |
62 |     if (terminal) {
63 |         if (logs.length === 0) {
64 |             terminal.innerText = 'Журнал cloudflared пуст.';
65 |             return;
66 |         }
67 |         terminal.innerHTML = logs.map(l => {
68 |             const color = l.level === 'error' ? '#f87171' : (l.l
evel === 'warn' ? '#fbbf24' : '#94a3b8');
69 |             return `<div style="color: ${color};">[${l.time} | '
'|] [${(l.level || 'INFO').toUpperCase()}] ${l.msg || l.messag
e || ''}</div>`;
70 |         }).join('');
71 |     }
72 | } catch (e) {
73 |     console.error('[CloudflaredTab] Failed to fetch logs:',
e);
74 | }
75 | }
76 |
77 | async function runDiagnostics() {
78 |     try {
79 |         const res = await fetch('/api/cloudflared/diagnostic');
80 |         if (!res.ok) return;
81 |         const report = await res.json();
82 |
83 |         const titleEl = document.getElementById('cf-diag-summary
-title');
84 |         const descEl = document.getElementById('cf-diag-summary-

```



```

    desc');
85     const recList = document.getElementById('cf-recommendati
ons-list');
86
87     if (titleEl) titleEl.innerText = `Здоровье: ${report.sta
tus || 'OK'} (${report.health_score || 100}/100)`;
88     if (descEl) descEl.innerText = report.summary || 'Все ту
ннельные соединения стабильны.';
89
90     if (recList && Array.isArray(report.recommendations)) {
91         recList.innerHTML = report.recommendations.map(r => `
92             <li class="list-group-item bg-transparent text-light
border-secondary py-1 px-0">• ${r}</li>
93         `).join('');
94     }
95     } catch (e) {
96         console.error('[CloudflaredTab] Diagnostic error:', e);
97     }
98 }
99
100 function initCloudflaredTab() {
101     console.log('[CloudflaredTab] Initializing Cloudflared Mon
itor tab...');
102     fetchTunnelStatus();
103     fetchLogs();
104
105     if (!isCfInitialized) {
106         const refreshBtn = document.getElementById('btn-cf-refre
sh');
107         const configBtn = document.getElementById('btn-cf-config
');
108         const diagBtn = document.getElementById('btn-cf-run-diag
nostics');
109         const testBtn = document.getElementById('btn-cf-test-end
point');
110
111         if (refreshBtn) {
112             refreshBtn.onclick = () => {

```

```

113         fetchTunnelStatus();
114         fetchLogs();
115     };
116 }
117
118 if (diagBtn) diagBtn.onclick = runDiagnostics;
119
120 if (testBtn) {
121     testBtn.onclick = async () => {
122         testBtn.disabled = true;
123         testBtn.innerHTML = '<span class="spinner-border spi
124 nner-border-sm me-1"></span> Проверка...';
125         try {
126             const res = await fetch('/api/cloudflared/test-end
127 point');
128             const data = await res.json();
129             const msg = `Результат проверки Ingress:\nДоступен
130 : ${data.reachable ? 'ДА' : 'НЕТ'}\nКод ответа: ${data.status_
131 code || 'N/A'}\nЗадержка: ${data.response_time_ms || 0} ms`;
132             window.showToast?.(msg, data.reachable ? 'success'
133 : 'warning') || alert(msg);
134         } catch (e) {
135             window.showToast?.('Ошибка проверки эндпоинта: ' +
136 e.message, 'danger') || alert('Ошибка проверки эндпоинта: ' +
137 e.message);
138         } finally {
139             testBtn.disabled = false;
140             testBtn.innerHTML = '<i class="bi bi-broadcast me-
141 1"></i> Проверить Ingress';
142             fetchTunnelStatus();
143         }
144     };
145 }
146
147 if (configBtn) {
148     configBtn.onclick = () => {
149         if (typeof window.openAppConfigModal === 'function')
150         {

```

```
142 |         window.openAppConfigModal('cloudflared_monitor', '
    | Cloudflare Tunnel Monitor');
143 |     }
144 | };
145 | }
146 |
147 |     isCfInitialized = true;
148 | }
149 | }
150 |
151 | window.initCloudflaredTab = initCloudflaredTab;
152 | ))();
153 |
```

index.html

```
1 <!-- Link CosmicPlayer custom style -->
2 <link rel="stylesheet" href="/html/cosmicplayer/styles.css">
3 <!-- HLS.js library for streaming M3U8 inside custom HTML5 vid
   eo player -->
4 <script src="https://cdn.jsdelivr.net/npm/hls.js@latest"></scr
   ipt>
5
6 <div class="cosmic-player-tab">
7     <!-- Sidebar: Video URL list / Library / Settings -->
8     <aside class="cosmic-sidebar">
9         <div class="logo">
10             <span class="logo-icon">❏ </span>
11             <h2>CosmicPlayer</h2>
12         </div>
13
14         <!-- Paste link panel -->
15         <div class="url-input-section">
16             <h3>Вставить ссылку</h3>
17             <form id="stream-form">
18                 <input type="url" id="stream-url" placeholder=
19                 "Ссылка (YouTube, VK, RuTube, Reza...)" required>
20                 <button type="submit" class="btn btn-glow">Заг
21                 рузить видео</button>
22             </form>
23         </div>
24
25         <!-- Supported sources badges -->
26         <div class="sources-panel">
27             <h4>Совместимые источники</h4>
28             <div class="sources-grid">
29                 <span class="badge" data-provider="youtube">Yo
30                 uTube</span>
31                 <span class="badge" data-provider="vk">VK Vide
32                 o</span>
33                 <span class="badge" data-provider="rutube">RuT
34                 ube</span>
```

```
30 |         <span class="badge" data-provider="hdrezka">HD
    | Rezka</span>
31 |         <span class="badge" data-provider="seasonvar">
    | SeasonVar</span>
32 |         <span class="badge" data-provider="kinogo">Kin
    | ogo</span>
33 |         <span class="badge" data-provider="direct">Dir
    | ect HLS/MP4</span>
34 |     </div>
35 | </div>
36 |
37 | <!-- History List -->
38 | <div class="history-section">
39 |     <div class="section-header">
40 |         <h4>История воспроизведения</h4>
41 |         <button id="clear-history" class="btn-text">Оч
    | истить</button>
42 |     </div>
43 |     <div id="history-list" class="history-list">
44 |         <div class="empty-state">Нет недавно проигранн
    | ых видео</div>
45 |     </div>
46 | </div>
47 | </aside>
48 |
49 | <!-- Main Content Area: Player and Description -->
50 | <main class="main-content">
51 |     <!-- Movie Screen and Ambience -->
52 |     <div class="player-container" id="player-container">
53 |         <!-- Background Ambient Glow -->
54 |         <div class="ambient-glow" id="ambient-glow"></div>
55 |
56 |         <!-- Display Iframe (for embeds) -->
57 |         <iframe id="embed-player" class="player-element hi
    | dden" allowfullscreen allow="autoplay; encrypted-media"></ifra
    | me>
58 |
59 |         <!-- Custom HTML5 Player (for direct video feeds)
```

```

-->
60         <div id="html5-player-wrapper" class="player-eleme
nt hidden">
61             <video id="html5-video" class="html5-video-el"
></video>
62             <!-- Custom Player Skin -->
63             <div class="custom-controls">
64                 <div class="progress-container" id="progre
ss-container">
65                     <div class="progress-bar" id="progress
-bar">
66                         <div class="progress-hover"></div>
67                     </div>
68                 </div>
69                 <div class="controls-row">
70                     <div class="controls-left">
71                         <button class="ctrl-btn" id="play-
pause-btn">▶ </button>
72                         <button class="ctrl-btn" id="stop-
btn">■</button>
73                         <div class="time-display">
74                             <span id="current-time">0:00</
span> / <span id="duration-time">0:00</span>
75                         </div>
76                     </div>
77                     <div class="controls-right">
78                         <div class="volume-container">
79                             <button class="ctrl-btn" id="m
ute-btn">□ </button>
80                             <input type="range" id="volume
-slider" min="0" max="1" step="0.05" value="1">
81                         </div>
82                         <select id="speed-select" class="s
peed-dropdown">
83                             <option value="0.5">0.5x</opti
on>
84                             <option value="1" selected>1.0
x</option>

```

```

85 |         <option value="1.5">1.5x</opti
| on>
86 |         <option value="2">2.0x</option
| >
87 |     </select>
88 |     <button class="ctrl-btn" id="fulls
| creen-btn">⏮ </button>
89 |         </div>
90 |     </div>
91 | </div>
92 | </div>
93 |
94 | <!-- Initial Empty State -->
95 | <div id="player-placeholder" class="player-placeho
| lder">
96 |     <div class="placeholder-content">
97 |         <span class="play-icon">⏮ </span>
98 |         <h3>Видео не загружено</h3>
99 |         <p>Вставьте ссылку на видео или выберите о
| дин из демонстрационных источников слева.</p>
100 |     </div>
101 | </div>
102 | </div>
103 |
104 | <!-- Current Video Information -->
105 | <div class="video-info-card glass-card">
106 |     <div class="info-header">
107 |         <span class="provider-tag" id="current-provide
| r-tag">Offline</span>
108 |         <h3 id="video-title">Добро пожаловать в Cosmic
| Player</h3>
109 |     </div>
110 |     <div class="info-body">
111 |         <p id="video-source-url">Пожалуйста, введите U
| RL видео для воспроизведения.</p>
112 |     </div>
113 | </div>
114 | </main>

```

115 | </div>
116 |

main.js

```
1 window.initCosmicplayerTab = function() {
2     console.log("CosmicPlayer tab initialized");
3
4     // Elements
5     const streamForm = document.getElementById('stream-form');
6     const streamUrlInput = document.getElementById('stream-url
7     ');
8     const embedPlayer = document.getElementById('embed-player'
9     );
10    const html5PlayerWrapper = document.getElementById('html5-
11    player-wrapper');
12    const html5Video = document.getElementById('html5-video');
13    const playerPlaceholder = document.getElementById('player-
14    placeholder');
15    const videoTitle = document.getElementById('video-title');
16    const videoSourceUrl = document.getElementById('video-sour
17    ce-url');
18    const currentProviderTag = document.getElementById('curren
19    t-provider-tag');
20    const historyList = document.getElementById('history-list'
21    );
22    const clearHistoryBtn = document.getElementById('clear-his
23    tory');
24    const badges = document.querySelectorAll('.cosmic-player-t
25    ab .badge');
26    const ambientGlow = document.getElementById('ambient-glow'
27    );
28
29    // Custom controls
30    const playPauseBtn = document.getElementById('play-pause-b
31    tn');
32    const stopBtn = document.getElementById('stop-btn');
33    const currentTimeDisplay = document.getElementById('curren
34    t-time');
35    const durationTimeDisplay = document.getElementById('durat
36    ion-time');
```

```

24     const progressContainer = document.getElementById('progress-container');
25     const progressBar = document.getElementById('progress-bar');
26     const progressHover = document.querySelector('.progress-hover');
27     const muteBtn = document.getElementById('mute-btn');
28     const volumeSlider = document.getElementById('volume-slider');
29     const speedSelect = document.getElementById('speed-select');
30     const fullscreenBtn = document.getElementById('fullscreen-btn');
31
32     let hlsInstance = null;
33     let history = JSON.parse(localStorage.getItem('cosmic_player_history') || '[]');
34
35     // Dynamic ambient glow colors per provider
36     const providerColors = {
37         youtube: 'radial-gradient(circle, rgba(255, 0, 0, 0.25) 0%, transparent 60%)',
38         vk: 'radial-gradient(circle, rgba(74, 118, 168, 0.25) 0%, transparent 60%)',
39         rutube: 'radial-gradient(circle, rgba(235, 95, 30, 0.25) 0%, transparent 60%)',
40         hdrezka: 'radial-gradient(circle, rgba(242, 197, 17, 0.2) 0%, transparent 60%)',
41         seasonvar: 'radial-gradient(circle, rgba(0, 242, 254, 0.2) 0%, transparent 60%)',
42         kinogo: 'radial-gradient(circle, rgba(155, 81, 224, 0.2) 0%, transparent 60%)',
43         direct: 'radial-gradient(circle, rgba(121, 40, 202, 0.25) 0%, transparent 60%)',
44         offline: 'radial-gradient(circle, rgba(255, 255, 255, 0.05) 0%, transparent 60%)'
45     };
46

```

```

47 | // Load history on start
48 | renderHistory();
49 |
50 | // Form submission
51 | streamForm.addEventListener('submit', (e) => {
52 |     e.preventDefault();
53 |     const url = streamUrlInput.value.trim();
54 |     if (url) {
55 |         loadVideo(url);
56 |     }
57 | });
58 |
59 | // Clear history
60 | clearHistoryBtn.addEventListener('click', () => {
61 |     history = [];
62 |     localStorage.setItem('cosmic_player_history', JSON.str
63 | ingify(history));
64 |     renderHistory();
65 | });
66 |
67 | // Router and url parser
68 | function loadVideo(url, title = null) {
69 |     resetPlayers();
70 |
71 |     const providerInfo = parseVideoUrl(url);
72 |     const resolvedTitle = title || providerInfo.defaultTit
73 | le;
74 |
75 |     highlightBadge(providerInfo.provider);
76 |
77 |     // Update info panel
78 |     videoTitle.textContent = resolvedTitle;
79 |     videoSourceUrl.textContent = url;
80 |     currentProviderTag.textContent = providerInfo.provider
81 | ;
82 |
83 |     // Update ambient glow color
84 |     ambientGlow.style.background = providerColors[provider

```

```

    Info.provider] || providerColors.direct;
82
83     // Route to the correct player engine
84     if (providerInfo.type === 'embed') {
85         embedPlayer.src = providerInfo.embedUrl;
86         embedPlayer.classList.remove('hidden');
87     } else if (providerInfo.type === 'html5') {
88         initHTML5Player(url);
89         html5PlayerWrapper.classList.remove('hidden');
90     } else if (providerInfo.type === 'iframe_fallback') {
91         embedPlayer.src = url;
92         embedPlayer.classList.remove('hidden');
93     }
94
95     // Add to history
96     addToHistory(url, resolvedTitle, providerInfo.provider
    );
97 }
98
99 function resetPlayers() {
100     embedPlayer.classList.add('hidden');
101     embedPlayer.src = '';
102     html5PlayerWrapper.classList.add('hidden');
103     playerPlaceholder.classList.add('hidden');
104
105     html5Video.pause();
106     html5Video.src = '';
107
108     if (hlsInstance) {
109         hlsInstance.destroy();
110         hlsInstance = null;
111     }
112 }
113
114 // Video URL Parser
115 function parseVideoUrl(url) {
116     // YouTube Matcher
117     const ytReg = /(?:youtube\.com\/(?:[^\./]+\./+\/|(?:(v|e

```

```

    (? : mbed ) ? ) \ / | . * [ ? & ] v = ) | youtu \ . be \ / ) ( [ ^ " & ? \ \ / \ s ] { 11 } ) / ;
118     const ytMatch = url.match(ytReg);
119     if (ytMatch) {
120         return {
121             provider: 'youtube',
122             type: 'embed',
123             embedUrl: `https://www.youtube.com/embed/${ytM
atch[1]}?autoplay=1&rel=0`,
124             defaultTitle: `YouTube Video (${ytMatch[1]})`
125         };
126     }
127
128     // VK Matcher
129     const vkReg = /vk \. com \ / video ( - ? \ d + ) _ ( \ d + ) / ;
130     const vkMatch = url.match(vkReg);
131     const vkEmbedReg = /oid = ( - ? \ d + ) & id = ( \ d + ) ( ? : & hash = ( [ a - f
0 - 9 ] + ) ) ? / ;
132     const vkEmbedMatch = url.match(vkEmbedReg);
133
134     if (vkMatch) {
135         return {
136             provider: 'vk',
137             type: 'embed',
138             embedUrl: `https://vk.com/video_ext.php?oid=${{
vkMatch[1]}&id=${vkMatch[2]}&autoplay=1`,
139             defaultTitle: `VK Video (${vkMatch[1]})_${vkMat
ch[2]})`
140         };
141     } else if (vkEmbedMatch) {
142         const hashParam = vkEmbedMatch[3] ? `&hash=${vkEmb
edMatch[3]}` : '';
143         return {
144             provider: 'vk',
145             type: 'embed',
146             embedUrl: `https://vk.com/video_ext.php?oid=${{
vkEmbedMatch[1]}&id=${vkEmbedMatch[2]}${hashParam}&autoplay=1`
147             ,
            defaultTitle: `VK Embed Video`

```

```
148         };
149     }
150
151     // RuTube Matcher
152     const rutubeReg = /rutube\.ru\/(?:video|play\/embed)\/
153     ([a-f0-9]{32})/;
154     const rutubeMatch = url.match(rutubeReg);
155     if (rutubeMatch) {
156         return {
157             provider: 'rutube',
158             type: 'embed',
159             embedUrl: `https://rutube.ru/play/embed/${rutu
160 beMatch[1]}?autoplay=1`,
161             defaultTitle: `RuTube Video`
162         };
163     }
164
165     // Direct Stream (MP4/HLS)
166     const isHLS = url.includes('.m3u8');
167     const isMP4 = url.includes('.mp4');
168     if (isHLS || isMP4) {
169         return {
170             provider: 'direct',
171             type: 'html5',
172             defaultTitle: isHLS ? 'Прямой HLS поток' : 'Пр
173 ямой MP4 файл'
174         };
175     }
176
177     // SeasonVar
178     if (url.includes('seasonvar.ru')) {
179         return {
180             provider: 'seasonvar',
181             type: 'iframe_fallback',
182             defaultTitle: 'СезонВар плеер'
183         };
184     }
185 }
```

```

183 |         // HDRezka
184 |         if (url.includes('rezka.ag') || url.includes('hdrezka'
    |     )) {
185 |             return {
186 |                 provider: 'hdrezka',
187 |                 type: 'iframe_fallback',
188 |                 defaultTitle: 'Резка плеер'
189 |             };
190 |         }
191 |
192 |         // Kinogo
193 |         if (url.includes('kinogo')) {
194 |             return {
195 |                 provider: 'kinogo',
196 |                 type: 'iframe_fallback',
197 |                 defaultTitle: 'Киного плеер'
198 |             };
199 |         }
200 |
201 |         // Default fallback
202 |         return {
203 |             provider: 'direct',
204 |             type: 'html5',
205 |             defaultTitle: 'Прямой видеоисточник'
206 |         };
207 |     }
208 |
209 |     function initHTML5Player(url) {
210 |         if (url.includes('.m3u8')) {
211 |             if (typeof Hls !== 'undefined' && Hls.isSupported(
    |     )) {
212 |                 hlsInstance = new Hls();
213 |                 hlsInstance.loadSource(url);
214 |                 hlsInstance.attachMedia(html5Video);
215 |                 hlsInstance.on(Hls.Events.MANIFEST_PARSED, ()
    |     => {
216 |                     html5Video.play();
217 |                 });

```

```
218         } else if (html5Video.canPlayType('application/vnd
    |.apple.mpegurl')) {
219             html5Video.src = url;
220             html5Video.play();
221         }
222     } else {
223         html5Video.src = url;
224         html5Video.play();
225     }
226
227     playPauseBtn.textContent = '▶';
228 }
229
230 // Video events
231 html5Video.addEventListener('timeupdate', () => {
232     const current = html5Video.currentTime;
233     const total = html5Video.duration || 0;
234
235     if (total > 0) {
236         const pct = (current / total) * 100;
237         progressBar.style.width = `${pct}%`;
238     }
239
240     currentTimeDisplay.textContent = formatTime(current);
241     durationTimeDisplay.textContent = formatTime(total);
242 });
243
244 html5Video.addEventListener('loadedmetadata', () => {
245     durationTimeDisplay.textContent = formatTime(html5Vide
    |o.duration);
246 });
247
248 playPauseBtn.addEventListener('click', () => {
249     if (html5Video.paused) {
250         html5Video.play();
251         playPauseBtn.textContent = '▶';
252     } else {
253         html5Video.pause();
```



```
254         playPauseBtn.textContent = '▶';
255     }
256 });
257
258 stopBtn.addEventListener('click', () => {
259     html5Video.pause();
260     html5Video.currentTime = 0;
261     playPauseBtn.textContent = '▶';
262 });
263
264 volumeSlider.addEventListener('input', (e) => {
265     const vol = e.target.value;
266     html5Video.volume = vol;
267     if (vol === '0') {
268         muteBtn.textContent = '🔇';
269     } else {
270         muteBtn.textContent = '🔊';
271     }
272 });
273
274 muteBtn.addEventListener('click', () => {
275     if (html5Video.muted) {
276         html5Video.muted = false;
277         volumeSlider.value = html5Video.volume;
278         muteBtn.textContent = '🔊';
279     } else {
280         html5Video.muted = true;
281         volumeSlider.value = 0;
282         muteBtn.textContent = '🔇';
283     }
284 });
285
286 speedSelect.addEventListener('change', (e) => {
287     html5Video.playbackRate = parseFloat(e.target.value);
288 });
289
290 progressContainer.addEventListener('click', (e) => {
291     const rect = progressContainer.getBoundingClientRect()
```

```

    ;
292     const clickX = e.clientX - rect.left;
293     const width = rect.width;
294     const pct = clickX / width;
295
296     html5Video.currentTime = pct * html5Video.duration;
297 });
298
299 progressContainer.addEventListener('mousemove', (e) => {
300     const rect = progressContainer.getBoundingClientRect()
    ;
301     const hoverX = e.clientX - rect.left;
302     const pct = hoverX / rect.width;
303     progressHover.style.width = `${pct * 100}%`;
304 });
305
306 progressContainer.addEventListener('mouseleave', () => {
307     progressHover.style.width = '0%';
308 });
309
310 fullscreenBtn.addEventListener('click', () => {
311     if (!document.fullscreenElement) {
312         html5PlayerWrapper.requestFullscreen().catch(err =
    > {
313             console.error(`Error enabling fullscreen: ${er
    r.message}`);
314         });
315     } else {
316         document.exitFullscreen();
317     }
318 });
319
320 function formatTime(sec) {
321     if (isNaN(sec)) return '0:00';
322     const mins = Math.floor(sec / 60);
323     const secs = Math.floor(sec % 60);
324     return `${mins}:${secs < 10 ? '0' : ''}${secs}`;
325 }

```

```

326
327     function highlightBadge(provider) {
328         badges.forEach(badge => {
329             if (badge.getAttribute('data-provider') === provid
er) {
330                 badge.classList.add('active');
331             } else {
332                 badge.classList.remove('active');
333             }
334         });
335     }
336
337     function addToHistory(url, title, provider) {
338         history = history.filter(item => item.url !== url);
339
340         history.unshift({
341             url,
342             title,
343             provider,
344             timestamp: new Date().toLocaleTimeString([], { hou
r: '2-digit', minute: '2-digit' })
345         });
346
347         if (history.length > 8) history.pop();
348
349         localStorage.setItem('cosmic_player_history', JSON.str
ingify(history));
350         renderHistory();
351     }
352
353     function renderHistory() {
354         historyList.innerHTML = '';
355         if (history.length === 0) {
356             historyList.innerHTML = '<div class="empty-state">
Нет недавно проигранных видео</div>';
357             return;
358         }
359

```

```

360 |         history.forEach(item => {
361 |             const el = document.createElement('div');
362 |             el.className = 'history-item';
363 |             el.innerHTML = `
364 |                 <div class="history-title" title="${item.title
| }">${item.title}</div>
365 |                 <div class="history-meta">
366 |                     <span>${item.provider.toUpperCase()}</span>
| >
367 |                     <span>${item.timestamp}</span>
368 |                 </div>
369 |             `;
370 |             el.addEventListener('click', () => {
371 |                 streamUrlInput.value = item.url;
372 |                 loadVideo(item.url, item.title);
373 |             });
374 |             historyList.appendChild(el);
375 |         });
376 |     }
377 |
378 |     badges.forEach(badge => {
379 |         badge.addEventListener('click', () => {
380 |             const provider = badge.getAttribute('data-provider
| ');
381 |             let demoUrl = '';
382 |
383 |             switch (provider) {
384 |                 case 'youtube':
385 |                     demoUrl = 'https://www.youtube.com/watch?v
| =aqz-KE-bpKQ';
386 |                     break;
387 |                 case 'vk':
388 |                     demoUrl = 'https://vk.com/video-22822305_4
| 56239018';
389 |                     break;
390 |                 case 'rutube':
391 |                     demoUrl = 'https://rutube.ru/video/e7cfcb8
| cb4310d54026fb4bd56e828d1/';

```

```
392 |             break;
393 |         case 'direct':
394 |             demoUrl = 'https://test-streams.mux.dev/x3
| 6xhzz/x36xhzz.m3u8';
395 |             break;
396 |         case 'seasonvar':
397 |             demoUrl = 'http://seasonvar.ru';
398 |             break;
399 |         case 'hdrezka':
400 |             demoUrl = 'https://rezka.ag';
401 |             break;
402 |         case 'kinogo':
403 |             demoUrl = 'https://kinogo.biz';
404 |             break;
405 |     }
406 |
407 |     if (demoUrl) {
408 |         streamUrlInput.value = demoUrl;
409 |         loadVideo(demoUrl);
410 |     }
411 | });
412 | });
413 | };
414 |
```

styles.css

```
1 | .cosmic-player-tab {
2 |     --primary: #7928ca;
3 |     --primary-glow: #ff007a;
4 |     --primary-gradient: linear-gradient(135deg, #7928ca 0%, #f
    f007a 100%);
5 |     --panel-border: rgba(255, 255, 255, 0.08);
6 |     --bg-dark: #0a0714;
7 |     --text-muted: #a0aec0;
8 |     --transition: all 0.25s cubic-bezier(0.25, 0.8, 0.25, 1);
9 |
10 |     display: flex;
11 |     width: 100%;
12 |     height: calc(100vh - 80px);
13 |     background: rgba(20, 16, 36, 0.6);
14 |     backdrop-filter: blur(25px);
15 |     -webkit-backdrop-filter: blur(25px);
16 |     border: 1px solid var(--panel-border);
17 |     border-radius: 12px;
18 |     overflow: hidden;
19 |     margin-top: 6px;
20 | }
21 |
22 | .cosmic-sidebar {
23 |     width: 260px;
24 |     background: rgba(14, 10, 26, 0.75);
25 |     border-right: 1px solid var(--panel-border);
26 |     padding: 16px 14px;
27 |     display: flex;
28 |     flex-direction: column;
29 |     gap: 14px;
30 |     overflow-y: auto;
31 | }
32 |
33 | .cosmic-sidebar .logo {
34 |     display: flex;
35 |     align-items: center;
```

```
36 |     gap: 8px;
37 | }
38 |
39 | .cosmic-sidebar .logo h2 {
40 |     font-size: 1.1rem;
41 |     font-weight: 700;
42 |     margin: 0;
43 |     background: var(--primary-gradient);
44 |     -webkit-background-clip: text;
45 |     -webkit-text-fill-color: transparent;
46 | }
47 |
48 | .cosmic-sidebar .logo-icon {
49 |     font-size: 1.3rem;
50 |     filter: drop-shadow(0 0 6px rgba(255, 0, 122, 0.5));
51 | }
52 |
53 | .cosmic-player-tab .url-input-section h3,
54 | .cosmic-player-tab .sources-panel h4,
55 | .cosmic-player-tab .history-section h4 {
56 |     font-size: 0.72rem;
57 |     font-weight: 600;
58 |     text-transform: uppercase;
59 |     letter-spacing: 0.6px;
60 |     color: var(--text-muted);
61 |     margin-bottom: 8px;
62 | }
63 |
64 | #stream-form {
65 |     display: flex;
66 |     flex-direction: column;
67 |     gap: 10px;
68 | }
69 |
70 | #stream-url {
71 |     background: rgba(0, 0, 0, 0.4);
72 |     border: 1px solid var(--panel-border);
73 |     border-radius: 8px;
```

```
74 | padding: 10px 12px;
75 | color: #fff;
76 | font-size: 0.85rem;
77 | outline: none;
78 | transition: var(--transition);
79 | }
80 |
81 | #stream-url:focus {
82 |     border-color: var(--primary-glow);
83 |     box-shadow: 0 0 10px rgba(255, 0, 122, 0.2);
84 | }
85 |
86 | .btn-glow {
87 |     background: var(--primary-gradient);
88 |     color: white;
89 |     border: none;
90 |     border-radius: 8px;
91 |     padding: 10px 15px;
92 |     font-size: 0.85rem;
93 |     font-weight: 600;
94 |     cursor: pointer;
95 |     transition: var(--transition);
96 | }
97 |
98 | .btn-glow:hover {
99 |     transform: translateY(-1px);
100 |     box-shadow: 0 4px 15px rgba(255, 0, 122, 0.4);
101 | }
102 |
103 | .sources-grid {
104 |     display: flex;
105 |     flex-wrap: wrap;
106 |     gap: 6px;
107 | }
108 |
109 | .cosmic-player-tab .badge {
110 |     background: rgba(255, 255, 255, 0.08);
111 |     border: 1px solid var(--panel-border);
```



```
112 |     border-radius: 6px;
113 |     padding: 5px 10px;
114 |     font-size: 0.7rem;
115 |     font-weight: 500;
116 |     color: #e6edf3;
117 |     transition: var(--transition);
118 |     cursor: pointer;
119 | }
120 |
121 | .cosmic-player-tab .badge:hover,
122 | .cosmic-player-tab .badge.active {
123 |     color: white;
124 |     background: rgba(121, 40, 202, 0.2);
125 |     border-color: var(--primary);
126 | }
127 |
128 | .history-section {
129 |     flex-grow: 1;
130 |     display: flex;
131 |     flex-direction: column;
132 | }
133 |
134 | .section-header {
135 |     display: flex;
136 |     justify-content: space-between;
137 |     align-items: center;
138 |     margin-bottom: 10px;
139 | }
140 |
141 | .btn-text {
142 |     background: none;
143 |     border: none;
144 |     color: var(--text-muted);
145 |     font-size: 0.75rem;
146 |     cursor: pointer;
147 | }
148 |
149 | .history-list {
```

```
150 |     display: flex;
151 |     flex-direction: column;
152 |     gap: 6px;
153 |     overflow-y: auto;
154 |     max-height: 200px;
155 | }
156 |
157 | .history-item {
158 |     background: rgba(255, 255, 255, 0.01);
159 |     border: 1px solid var(--panel-border);
160 |     border-radius: 8px;
161 |     padding: 8px 12px;
162 |     cursor: pointer;
163 |     display: flex;
164 |     flex-direction: column;
165 |     gap: 2px;
166 |     transition: var(--transition);
167 | }
168 |
169 | .history-item:hover {
170 |     background: rgba(255, 255, 255, 0.04);
171 | }
172 |
173 | .history-title {
174 |     font-size: 0.8rem;
175 |     font-weight: 500;
176 |     white-space: nowrap;
177 |     overflow: hidden;
178 |     text-overflow: ellipsis;
179 | }
180 |
181 | .history-meta {
182 |     display: flex;
183 |     justify-content: space-between;
184 |     font-size: 0.65rem;
185 |     color: var(--text-muted);
186 | }
187 |
```

```
188 /* Main Panel Layout */
189 .main-content {
190     flex-grow: 1;
191     display: flex;
192     flex-direction: column;
193     padding: 14px;
194     gap: 12px;
195     background: rgba(10, 7, 20, 0.2);
196 }
197
198 .player-container {
199     position: relative;
200     width: 100%;
201     height: 60%;
202     min-height: 380px;
203     background: #000;
204     border-radius: 15px;
205     overflow: hidden;
206     border: 1px solid var(--panel-border);
207 }
208
209 .ambient-glow {
210     position: absolute;
211     top: -20%;
212     left: -20%;
213     width: 140%;
214     height: 140%;
215     background: radial-gradient(circle, rgba(121, 40, 202, 0.2
216     ) 0%, transparent 60%);
217     filter: blur(50px);
218     z-index: 0;
219     pointer-events: none;
220     transition: background 1s ease;
221 }
222
223 .player-element {
224     position: relative;
225     width: 100%;
```

```
225 |     height: 100%;
226 |     border: none;
227 |     z-index: 2;
228 | }
229 |
230 | .hidden {
231 |     display: none !important;
232 | }
233 |
234 | .html5-video-el {
235 |     width: 100%;
236 |     height: 100%;
237 |     object-fit: contain;
238 | }
239 |
240 | #html5-player-wrapper {
241 |     position: relative;
242 | }
243 |
244 | .custom-controls {
245 |     position: absolute;
246 |     bottom: 0;
247 |     left: 0;
248 |     right: 0;
249 |     background: linear-gradient(to top, rgba(0, 0, 0, 0.9) 0%,
    | rgba(0, 0, 0, 0.4) 70%, transparent 100%);
250 |     padding: 15px 20px;
251 |     display: flex;
252 |     flex-direction: column;
253 |     gap: 10px;
254 |     opacity: 0;
255 |     transition: opacity 0.3s ease;
256 |     z-index: 10;
257 | }
258 |
259 | #html5-player-wrapper:hover .custom-controls {
260 |     opacity: 1;
261 | }
```

```
262 |
263 | .progress-container {
264 |     height: 4px;
265 |     background: rgba(255, 255, 255, 0.2);
266 |     border-radius: 2px;
267 |     cursor: pointer;
268 |     position: relative;
269 | }
270 |
271 | .progress-bar {
272 |     height: 100%;
273 |     background: var(--primary-gradient);
274 |     width: 0%;
275 |     border-radius: 2px;
276 | }
277 |
278 | .progress-hover {
279 |     position: absolute;
280 |     top: 0;
281 |     left: 0;
282 |     height: 100%;
283 |     background: rgba(255, 255, 255, 0.15);
284 |     width: 0%;
285 |     pointer-events: none;
286 | }
287 |
288 | .controls-row {
289 |     display: flex;
290 |     justify-content: space-between;
291 |     align-items: center;
292 | }
293 |
294 | .controls-left, .controls-right {
295 |     display: flex;
296 |     align-items: center;
297 |     gap: 12px;
298 | }
299 |
```

```
300 | .ctrl-btn {
301 |     background: none;
302 |     border: none;
303 |     color: white;
304 |     font-size: 1rem;
305 |     cursor: pointer;
306 | }
307 |
308 | .time-display {
309 |     font-size: 0.75rem;
310 |     color: var(--text-muted);
311 | }
312 |
313 | .volume-container {
314 |     display: flex;
315 |     align-items: center;
316 |     gap: 6px;
317 | }
318 |
319 | #volume-slider {
320 |     width: 60px;
321 |     height: 3px;
322 |     background: rgba(255, 255, 255, 0.2);
323 |     outline: none;
324 |     accent-color: var(--primary-glow);
325 |     cursor: pointer;
326 | }
327 |
328 | .speed-dropdown {
329 |     background: rgba(255, 255, 255, 0.1);
330 |     border: 1px solid rgba(255, 255, 255, 0.2);
331 |     color: white;
332 |     border-radius: 4px;
333 |     padding: 2px 6px;
334 |     font-size: 0.75rem;
335 |     outline: none;
336 |     cursor: pointer;
337 | }
```

```
338 |
339 | .player-placeholder {
340 |     position: absolute;
341 |     top: 0;
342 |     left: 0;
343 |     width: 100%;
344 |     height: 100%;
345 |     display: flex;
346 |     justify-content: center;
347 |     align-items: center;
348 |     z-index: 1;
349 |     background: rgba(10, 7, 20, 0.95);
350 | }
351 |
352 | .placeholder-content {
353 |     text-align: center;
354 |     max-width: 320px;
355 | }
356 |
357 | .placeholder-content .play-icon {
358 |     font-size: 3rem;
359 |     display: block;
360 |     margin-bottom: 12px;
361 | }
362 |
363 | .placeholder-content h3 {
364 |     font-size: 1.15rem;
365 |     font-weight: 600;
366 |     margin-bottom: 8px;
367 | }
368 |
369 | .placeholder-content p {
370 |     color: var(--text-muted);
371 |     font-size: 0.8rem;
372 |     line-height: 1.4;
373 | }
374 |
375 | .glass-card {
```

```
376 |     background: rgba(255, 255, 255, 0.02);
377 |     border: 1px solid var(--panel-border);
378 |     border-radius: 12px;
379 |     padding: 18px;
380 | }
381 |
382 | .video-info-card {
383 |     flex-grow: 1;
384 |     display: flex;
385 |     flex-direction: column;
386 |     justify-content: center;
387 |     gap: 8px;
388 | }
389 |
390 | .info-header {
391 |     display: flex;
392 |     align-items: center;
393 |     gap: 12px;
394 | }
395 |
396 | .provider-tag {
397 |     background: var(--primary-gradient);
398 |     color: white;
399 |     padding: 3px 10px;
400 |     border-radius: 15px;
401 |     font-size: 0.65rem;
402 |     font-weight: 600;
403 |     text-transform: uppercase;
404 | }
405 |
406 | #video-title {
407 |     font-size: 1.1rem;
408 |     font-weight: 600;
409 |     margin: 0;
410 | }
411 |
412 | #video-source-url {
413 |     font-size: 0.8rem;
```



```
414 |     color: var(--text-muted);
415 |     word-break: break-all;
416 |     margin: 0;
417 | }
418 |
419 | .empty-state {
420 |     color: var(--text-muted);
421 |     font-size: 0.75rem;
422 |     text-align: center;
423 |     padding: 15px;
424 |     border: 1px dashed var(--panel-border);
425 |     border-radius: 8px;
426 | }
427 |
```

base.css

```
1  /* =====
   |
2  BASE STYLES & SCROLLBARS – AI Breadboard WebGUI
3  Базовые стили страницы, скроллбары и контейнеры вкладок
4  =====
   |
   | ===== */
5
6  /* Базовый шрифт – 14px для компактного вида */
7  html {
8      font-size: 14px;
9      height: 100%;
10 }
11
12 body {
13     display: flex;
14     flex-direction: column;
15     height: 100%;
16     min-height: 100vh;
17     background: var(--bg-color);
18     color: var(--text-color);
19     font-size: 0.9375rem; /* ≈ 13.125px */
20     line-height: 1.5;
21     transition: background-color 0.25s ease, color 0.25s ease;
22 }
23
24 /* =====
   |
25 СКРОЛЛБАРЫ
26 =====
   |
   | ===== */
27 ::-webkit-scrollbar { width: 6px; height: 6px; }
28 ::-webkit-scrollbar-track { background: var(--scrollbar-track); border-radius: 3px; }
29 ::-webkit-scrollbar-thumb { background: var(--scrollbar-thumb); border-radius: 3px; }
30 ::-webkit-scrollbar-thumb:hover { opacity: 0.8; }
```

```
31 |
32 | /* =====
33 |     ВКЛАДКИ (TAB PANES)
34 |     ===== */
35 | .tab-content {
36 |     flex-grow: 1;
37 |     display: flex;
38 |     flex-direction: column;
39 | }
40 |
41 | .tab-pane {
42 |     min-height: auto;
43 |     padding-bottom: 16px;
44 | }
45 |
46 | .tab-pane.active {
47 |     display: block;
48 | }
49 |
50 | /* =====
51 |     АДМИН-ИНТЕРФЕЙС
52 |     ===== */
53 | /* Стили для #admin-interface внутри .container-fluid (как в i
54 |     ndex.html) */
55 | #admin-interface,
56 | .container-fluid#admin-interface {
57 |     display: flex;
58 |     flex-direction: column;
59 |     flex-grow: 1;
60 |     height: 100%;
61 |     overflow-y: auto;
62 | }
63 | #admin-interface .tab-content,
```

```
64 | .container-fluid#admin-interface .tab-content {
65 |     flex-grow: 1;
66 |     overflow-y: auto;
67 | }
68 |
69 | #admin-interface .tab-pane,
70 | .container-fluid#admin-interface .tab-pane {
71 |     flex-grow: 1;
72 |     overflow-y: auto;
73 |     overflow-x: hidden;
74 | }
75 |
76 | #admin-interface .tab-pane.active,
77 | .container-fluid#admin-interface .tab-pane.active {
78 |     display: block;
79 | }
80 |
81 | /* Скроллбары для админки */
82 | #admin-interface ::-webkit-scrollbar {
83 |     width: 8px;
84 |     height: 8px;
85 | }
86 |
87 | #admin-interface ::-webkit-scrollbar-track {
88 |     background: var(--scrollbar-track);
89 |     border-radius: 4px;
90 | }
91 |
92 | #admin-interface ::-webkit-scrollbar-thumb {
93 |     background: var(--scrollbar-thumb);
94 |     border-radius: 4px;
95 | }
96 |
97 | #admin-interface ::-webkit-scrollbar-thumb:hover {
98 |     opacity: 0.85;
99 | }
100 |
```

chat.css

```
1  /* =====  
2  CHAT – AI Breadboard WebGUI  
3  Стили диалогового окна, сайдбара сессий, сообщений и дебагг  
   ера  
4  ===== */  
5  
6  /* — Основной макет вкладки чата —  
   — */  
7  #tab-chat {  
8    height: calc(100vh - 55px);  
9    max-height: calc(100vh - 55px);  
10   min-height: 80vh;  
11   padding-bottom: 0;  
12   overflow: hidden;  
13 }  
14  
15 #tab-chat.active {  
16   display: flex !important;  
17   flex-direction: column;  
18   height: calc(100vh - 55px);  
19   max-height: calc(100vh - 55px);  
20   min-height: 80vh;  
21 }  
22  
23 .chat-container {  
24   display: flex;  
25   flex-direction: row;  
26   height: 100%;  
27   min-height: 0;  
28   gap: 8px;  
29   position: relative;  
30 }  
31  
32 /* — Сайдбар чата (история диалогов) —
```

```
_____* /
33 .chat-sidebar {
34   width: 220px;
35   min-width: 220px;
36   max-width: 220px;
37   display: flex;
38   flex-direction: column;
39   background: var(--surface-1);
40   border: 1px solid var(--border-color);
41   border-radius: 6px;
42   padding: 6px;
43   transition: margin-left 0.22s ease, opacity 0.18s ease;
44   height: 100%;
45   flex-shrink: 0;
46 }
47
48 .chat-sidebar.collapsed {
49   margin-left: -232px;
50   opacity: 0;
51   pointer-events: none;
52   display: none !important;
53 }
54
55 .chat-sessions-list {
56   flex: 1 1 auto;
57   overflow-y: auto;
58   display: flex;
59   flex-direction: column;
60   gap: 2px;
61   margin-top: 4px;
62   margin-bottom: 4px;
63   padding-right: 2px;
64 }
65
66 .chat-session-item {
67   display: flex;
68   align-items: center;
69   justify-content: space-between;
```

```
70 | padding: 5px 7px;
71 | border-radius: 5px;
72 | cursor: pointer;
73 | color: var(--text-color);
74 | font-size: 0.8rem;
75 | border: 1px solid transparent;
76 | transition: background 0.12s ease, border-color 0.12s ease;
77 | user-select: none;
78 | }
79 |
80 | .chat-session-item:hover {
81 |   background: rgba(128, 128, 128, 0.1);
82 | }
83 |
84 | .chat-session-item.active {
85 |   background: rgba(37, 99, 235, 0.1);
86 |   border-color: rgba(37, 99, 235, 0.25);
87 |   font-weight: 500;
88 | }
89 |
90 | [data-theme="dark"] .chat-session-item.active {
91 |   background: rgba(88, 166, 255, 0.1);
92 |   border-color: rgba(88, 166, 255, 0.2);
93 | }
94 |
95 | .chat-session-title {
96 |   white-space: nowrap;
97 |   overflow: hidden;
98 |   text-overflow: ellipsis;
99 |   flex: 1 1 auto;
100 |   margin-right: 3px;
101 | }
102 |
103 | .chat-session-actions {
104 |   display: none;
105 |   align-items: center;
106 |   gap: 2px;
107 |   flex-shrink: 0;
```

```
108 }
109
110 .chat-session-item:hover .chat-session-actions,
111 .chat-session-item.active .chat-session-actions {
112     display: flex;
113 }
114
115 .chat-session-action-btn {
116     background: transparent;
117     border: none;
118     padding: 0 2px;
119     color: var(--text-muted);
120     border-radius: 3px;
121     cursor: pointer;
122     line-height: 1;
123     font-size: 0.75rem;
124 }
125
126 .chat-session-action-btn:hover {
127     color: var(--text-color);
128     background: rgba(128, 128, 128, 0.18);
129 }
130
131 /* — Главная область чата —————
    ————— */
132 .chat-main-area {
133     flex: 1 1 auto;
134     min-width: 0;
135     display: flex;
136     flex-direction: column;
137     height: 100%;
138 }
139
140 /* Хедер-бар чата */
141 .chat-header-bar {
142     display: flex;
143     align-items: center;
144     justify-content: space-between;
```



```
145 | padding: 3px 8px;
146 | margin-bottom: 5px;
147 | border-radius: 5px;
148 | background: var(--surface-1);
149 | border: 1px solid var(--border-color);
150 | font-size: 0.82rem;
151 | flex-shrink: 0;
152 | }
153 |
154 | /* Окно чата */
155 | .chat-window {
156 |   height: calc(100vh - 260px);
157 |   min-height: 160px;
158 |   overflow-y: auto;
159 |   border: 1px solid var(--border-color);
160 |   border-radius: 5px;
161 |   padding: 8px 10px;
162 |   background: var(--surface-1);
163 | }
164 |
165 | #tab-chat .chat-window {
166 |   height: auto;
167 |   flex: 1 1 auto;
168 |   min-height: calc(80vh - 120px);
169 |   max-height: calc(100vh - 140px);
170 |   overflow-y: auto;
171 |   margin-bottom: 5px !important;
172 | }
173 |
174 | #tab-chat .media-controls { flex-shrink: 0; margin-bottom: 5px !important; }
175 | #tab-chat #chat-debugger-toolbar { flex-shrink: 0; margin-bottom: 5px !important; }
176 | #tab-chat .input-group { flex-shrink: 0; margin-bottom: 2px !important; }
177 |
178 | /* — Сообщения —————
    | ————— */
```

```
179 | .message {
180 |     margin-bottom: 5px;
181 |     padding: 5px 10px;
182 |     border-radius: 5px;
183 |     font-size: 1.16rem;
184 |     line-height: 1.55;
185 | }
186 |
187 | /* — Поле ввода чата —————
    | ————— */
188 | #message-input,
189 | .chat-main-area #message-input,
190 | #chat-popup-input,
191 | #scenario-chat-input {
192 |     font-size: 1.15rem;
193 | }
194 |
195 | .user-message {
196 |     background: var(--chat-user-bg);
197 |     color: var(--chat-user-text);
198 |     text-align: right;
199 | }
200 |
201 | .bot-message {
202 |     background: var(--chat-bot-bg);
203 |     color: var(--chat-bot-text);
204 | }
205 |
206 | /* Чипы сообщений в чате */
207 | .chat-window .badge,
208 | .message .badge,
209 | .bot-message .badge,
210 | .user-message .badge {
211 |     font-size: 0.7rem;
212 |     font-weight: 500;
213 | }
214 |
215 | .chat-window .badge.bg-secondary,
```

```
216 | .chat-window .badge.bg-primary,
217 | .chat-window .badge.bg-danger,
218 | .chat-window .badge.bg-success,
219 | .chat-window .badge.bg-dark,
220 | .message .badge.bg-secondary,
221 | .message .badge.bg-primary,
222 | .message .badge.bg-danger,
223 | .message .badge.bg-success,
224 | .message .badge.bg-dark {
225 |     color: #ffffff !important;
226 | }
227 |
228 | /* — Дебаггер-тулбар чата —————
    | ————— */
229 | #chat-debugger-toolbar {
230 |     background-color: var(--surface-2);
231 |     border: 1px solid var(--border-color);
232 |     border-radius: 5px;
233 |     font-size: 0.8rem;
234 | }
235 |
236 | [data-theme="light"] #chat-debugger-toolbar {
237 |     background-color: var(--surface-2) !important;
238 |     border-color: var(--border-color) !important;
239 | }
240 |
241 | [data-theme="light"] #chat-debugger-toolbar select {
242 |     background-color: var(--surface-1) !important;
243 |     color: var(--text-color) !important;
244 |     border-color: var(--border-color) !important;
245 | }
246 |
```

components.css

```
1  /* =====
   |
2  COMPONENTS — AI Breadboard WebGUI
3  Базовые UI-компоненты: карточки, формы, кнопки, модалки, та
   | блицы, алерты, бейджи
4  =====
   | ===== */
5
6  /* =====
   |
7  КАРТОЧКИ
8  =====
   | ===== */
9  .card {
10   background-color: var(--card-bg);
11   border: 1px solid var(--border-color);
12   color: var(--card-text);
13   transition: border-color 0.2s, background-color 0.2s;
14   border-radius: 6px;
15 }
16
17 .card-header {
18   background-color: var(--surface-2);
19   border-bottom: 1px solid var(--border-color);
20   color: var(--text-color);
21   padding: 0.375rem 0.75rem;
22   font-size: 0.85rem;
23   font-weight: 600;
24 }
25
26 .card-body {
27   color: var(--card-text);
28   padding: 0.75rem;
29 }
30
31 .card-footer {
```

```
32 | background-color: var(--surface-2);
33 | border-top: 1px solid var(--border-color);
34 | padding: 0.375rem 0.75rem;
35 | }
36 |
37 | .card.bg-light {
38 |   background-color: var(--surface-2) !important;
39 |   color: var(--text-color);
40 | }
41 |
42 | .card.bg-dark > .card-header,
43 | .modal-content.bg-dark > .modal-header,
44 | .modal-content.bg-dark > .modal-footer,
45 | .agent-card .card-header {
46 |   background-color: var(--surface-2) !important;
47 |   border-color: var(--border-color) !important;
48 |   color: var(--text-color) !important;
49 | }
50 |
51 | [data-theme="light"] .card.bg-dark,
52 | [data-theme="light"] .modal-content.bg-dark,
53 | [data-theme="light"] .alert-dark {
54 |   background-color: var(--surface-1) !important;
55 |   color: var(--text-color) !important;
56 |   border-color: var(--border-color) !important;
57 | }
58 |
59 | [data-theme="light"] .card.bg-dark > .card-header,
60 | [data-theme="light"] .modal-content.bg-dark > .modal-header,
61 | [data-theme="light"] .modal-content.bg-dark > .modal-footer {
62 |   background-color: var(--surface-2) !important;
63 |   color: var(--text-color) !important;
64 |   border-color: var(--border-color) !important;
65 | }
66 |
67 | [data-theme="light"] .bg-black {
68 |   background-color: var(--bg-color) !important;
69 |   color: var(--text-color) !important;
```

```

70 }
71
72 [data-theme="light"] .text-light {
73     color: var(--text-color) !important;
74 }
75
76 [data-theme="light"] .btn-close-white {
77     filter: none !important;
78 }
79
80 /* =====
81
82     ФОРМЫ И ИНПУТЫ
83
84     */
85 .form-control,
86 .form-select {
87     background-color: var(--surface-1);
88     border: 1px solid var(--border-color);
89     color: var(--text-color);
90     font-size: 0.85rem;
91     padding: 0.3rem 0.5rem;
92     transition: border-color 0.15s, box-shadow 0.15s;
93     border-radius: 5px;
94 }
95
96 .form-control:focus,
97 .form-select:focus {
98     background-color: var(--surface-1);
99     border-color: var(--nav-active);
100     color: var(--text-color);
101     box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.18);
102     outline: none;
103 }
104
105 [data-theme="dark"] .form-control:focus,
106 [data-theme="dark"] .form-select:focus {
107     box-shadow: 0 0 0 2px rgba(88, 166, 255, 0.2);

```

```
106 }
107
108 .form-control::placeholder {
109     color: var(--text-muted);
110     opacity: 0.8;
111 }
112
113 .form-control-sm,
114 .form-select-sm {
115     font-size: 0.8rem;
116     padding: 0.22rem 0.4rem;
117 }
118
119 .form-label {
120     font-size: 0.82rem;
121     margin-bottom: 0.3rem;
122     color: var(--text-color);
123 }
124
125 .input-group-text {
126     background-color: var(--surface-2);
127     border: 1px solid var(--border-color);
128     color: var(--text-color);
129     font-size: 0.85rem;
130     padding: 0.3rem 0.55rem;
131 }
132
133 /* =====
134     КНОПКИ
135     ===== */
136 .btn {
137     font-size: 0.82rem;
138     font-weight: 500;
139     transition: background-color 0.15s, border-color 0.15s, box-
140     shadow 0.15s;
141     border-radius: 5px;
```

```
141 | }
142 |
143 | .btn-sm {
144 |     padding: 0.2rem 0.45rem;
145 |     font-size: 0.78rem;
146 | }
147 |
148 | .btn-lg {
149 |     font-size: 0.9rem;
150 |     padding: 0.45rem 0.9rem;
151 | }
152 |
153 | .btn-primary {
154 |     background-color: var(--nav-active);
155 |     border-color: var(--nav-active);
156 |     color: #ffffff;
157 | }
158 |
159 | .btn-primary:hover {
160 |     background-color: #1d4ed8;
161 |     border-color: #1d4ed8;
162 |     color: #ffffff;
163 | }
164 |
165 | [data-theme="dark"] .btn-primary {
166 |     background-color: #1f6feb;
167 |     border-color: #1f6feb;
168 | }
169 |
170 | [data-theme="dark"] .btn-primary:hover {
171 |     background-color: #388bfd;
172 |     border-color: #388bfd;
173 | }
174 |
175 | .btn-outline-secondary {
176 |     background-color: transparent;
177 |     border-color: var(--border-color);
178 |     color: var(--nav-text);
```



```
179 | }
180 |
181 | .btn-outline-secondary:hover {
182 |     background-color: var(--nav-hover-bg);
183 |     border-color: var(--border-color);
184 |     color: var(--text-color);
185 | }
186 |
187 | /* =====
188 |
189 |     МОДАЛЬНЫЕ ОКНА
190 |
191 | ===== */
192 |
193 | .modal-content {
194 |     background-color: var(--modal-bg);
195 |     border: 1px solid var(--modal-border);
196 |     color: var(--modal-text);
197 |     border-radius: 8px;
198 |     box-shadow: var(--shadow-lg);
199 | }
200 |
201 | .modal-header {
202 |     border-bottom: 1px solid var(--border-color);
203 |     padding: 0.6rem 0.9rem;
204 |     background-color: var(--surface-2);
205 |     border-radius: 8px 8px 0 0;
206 | }
207 |
208 | .modal-title {
209 |     font-size: 0.9rem;
210 |     font-weight: 600;
211 | }
212 |
213 | .modal-body {
214 |     padding: 0.75rem 0.9rem;
215 |     font-size: 0.85rem;
216 | }
```

```

215 | .modal-footer {
216 |     border-top: 1px solid var(--border-color);
217 |     padding: 0.5rem 0.9rem;
218 |     background-color: var(--surface-2);
219 |     border-radius: 0 0 8px 8px;
220 | }
221 |
222 | /* Модальное окно помощи */
223 | .modal-help .modal-dialog { max-width: 600px; }
224 | .modal-help .modal-content {
225 |     border-radius: 8px;
226 |     background: var(--modal-bg);
227 |     color: var(--modal-text);
228 |     border: 1px solid var(--modal-border);
229 | }
230 | .modal-help .modal-header {
231 |     border-bottom: 1px solid var(--border-color);
232 |     background: var(--surface-2);
233 | }
234 | .modal-help .modal-body { font-size: 0.85rem; line-height: 1.6
    | ; }
235 | .modal-help .modal-footer {
236 |     border-top: 1px solid var(--border-color);
237 |     background: var(--surface-2);
238 | }
239 |
240 | /* =====
    | =====
241 |     ТАБЛИЦЫ
242 |     =====
    | ===== */
243 | .table {
244 |     --bs-table-color: var(--text-color);
245 |     --bs-table-bg: transparent;
246 |     --bs-table-border-color: var(--border-color);
247 |     font-size: 0.82rem;
248 | }
249 |

```

```
250 | .table th,
251 | .table td {
252 |     padding: 0.4rem 0.6rem;
253 |     vertical-align: middle;
254 | }
255 |
256 | .table-hover tbody tr:hover {
257 |     background-color: var(--nav-hover-bg);
258 | }
259 |
260 | .table-dark {
261 |     --bs-table-bg: var(--surface-2);
262 |     --bs-table-color: var(--text-color);
263 | }
264 |
265 | [data-theme="dark"] .table-light,
266 | [data-theme="dark"] thead.table-light,
267 | [data-theme="dark"] thead.table-light th {
268 |     background-color: var(--surface-2) !important;
269 |     color: var(--text-color) !important;
270 |     border-color: var(--border-color) !important;
271 | }
272 |
273 | [data-theme="dark"] .table {
274 |     color: var(--text-color) !important;
275 |     border-color: var(--border-color) !important;
276 | }
277 |
278 | .table-responsive-adaptive {
279 |     height: auto;
280 |     min-height: 120px;
281 |     max-height: clamp(320px, 58vh, 720px);
282 |     overflow-y: auto;
283 |     overflow-x: auto;
284 |     scrollbar-width: thin;
285 | }
286 |
287 | .table-responsive-adaptive thead th,
```

```
288 | .table-sticky-header thead th {
289 |     position: sticky;
290 |     top: 0;
291 |     z-index: 2;
292 |     background-color: var(--surface-2);
293 | }
294 |
295 | /* =====
296 |     АЛЕРТЫ — адаптивные для обеих тем
297 |     ===== */
298 | .alert {
299 |     font-size: 0.83rem;
300 |     padding: 0.5rem 0.75rem;
301 |     border-radius: 6px;
302 |     border-width: 1px;
303 |     border-style: solid;
304 | }
305 |
306 | .alert-success {
307 |     background-color: var(--alert-success-bg);
308 |     border-color: var(--alert-success-border);
309 |     color: var(--alert-success-text);
310 | }
311 |
312 | .alert-danger {
313 |     background-color: var(--alert-danger-bg);
314 |     border-color: var(--alert-danger-border);
315 |     color: var(--alert-danger-text);
316 | }
317 |
318 | .alert-warning {
319 |     background-color: var(--alert-warning-bg);
320 |     border-color: var(--alert-warning-border);
321 |     color: var(--alert-warning-text);
322 | }
323 |
```

```
324 | .alert-info {
325 |   background-color: var(--alert-info-bg);
326 |   border-color: var(--alert-info-border);
327 |   color: var(--alert-info-text);
328 | }
329 |
330 | .alert:not([class*="alert-"]) {
331 |   background-color: var(--surface-1);
332 |   border-color: var(--border-color);
333 |   color: var(--text-color);
334 | }
335 |
336 | /* =====
337 |    БЕЙДЖИ
338 |    =====
339 |    */
340 | .badge {
341 |   font-size: 0.72rem;
342 |   font-weight: 500;
343 |   padding: 0.22em 0.5em;
344 |   letter-spacing: 0.02em;
345 | }
346 |
347 | .badge:not([class*="bg-"]):not([class*="badge-"]) {
348 |   background-color: var(--border-color);
349 |   color: var(--text-color);
350 | }
351 |
352 | .badge.bg-light,
353 | .badge.badge-light {
354 |   background-color: var(--border-subtle) !important;
355 |   color: var(--text-color) !important;
356 | }
357 |
358 | .badge.bg-secondary,
359 | .badge.bg-primary,
```

```
360 | .badge.bg-success,
361 | .badge.bg-dark,
362 | .badge-secondary,
363 | .badge-primary,
364 | .badge-danger,
365 | .badge-success,
366 | .badge-dark,
367 | .film-inline-badge {
368 |   color: #ffffff !important;
369 | }
370 |
371 | .badge.bg-warning,
372 | .badge.badge-warning {
373 |   color: #1a1a1a !important;
374 |   font-weight: 600;
375 | }
376 |
377 | .badge.bg-info,
378 | .badge.badge-info {
379 |   color: #042743 !important;
380 |   font-weight: 600;
381 | }
382 |
383 | /* Бейджи модели/поиска в навбаре */
384 | #chat-model-badge,
385 | #chat-popup-model-badge,
386 | #chat-search-badge,
387 | #chat-popup-search-badge {
388 |   background-color: var(--surface-2) !important;
389 |   border: 1px solid var(--border-color) !important;
390 |   color: var(--text-color) !important;
391 |   font-weight: 600;
392 |   font-size: 0.75rem;
393 | }
394 |
395 | [data-theme="dark"] #chat-model-badge,
396 | [data-theme="dark"] #chat-popup-model-badge {
397 |   color: var(--nav-active) !important;
```

```
398 | border-color: var(--border-color) !important;
399 | }
400 |
401 | [data-theme="dark"] #chat-search-badge,
402 | [data-theme="dark"] #chat-popup-search-badge {
403 |     color: #38bdf8 !important;
404 |     border-color: var(--border-color) !important;
405 | }
406 |
407 | /* Разделители в выпадающих меню */
408 | .dropdown-divider {
409 |     border-color: var(--border-color);
410 |     opacity: 0.6;
411 |     margin: 3px 0;
412 | }
413 |
414 | /* =====
415 |     СИСТЕМА ПОМОЩИ
416 |     ===== */
417 | .help-icon {
418 |     display: inline-flex;
419 |     align-items: center;
420 |     justify-content: center;
421 |     width: 16px;
422 |     height: 16px;
423 |     border-radius: 50%;
424 |     background: var(--nav-active);
425 |     color: white;
426 |     font-size: 11px;
427 |     font-weight: bold;
428 |     cursor: help;
429 |     margin-left: 3px;
430 |     vertical-align: middle;
431 |     opacity: 0.8;
432 |     transition: opacity 0.15s;
433 | }
```

```
434 |
435 | .help-icon:hover { opacity: 1; }
436 |
437 | .help-popover {
438 |     max-width: 320px;
439 |     font-size: 0.8rem;
440 |     background: var(--modal-bg);
441 |     color: var(--modal-text);
442 |     border: 1px solid var(--border-color);
443 | }
444 |
445 | /* =====
446 |
447 |     ИНТЕРАКТИВНЫЕ ЧИПЫ (ФИЛЬМЫ)
448 |
449 | ===== */
450 |
451 | .film-inline-badge {
452 |     background-color: #2563eb !important;
453 |     color: #ffffff !important;
454 |     font-weight: 500;
455 |     border: 1px solid rgba(255, 255, 255, 0.2) !important;
456 |     box-shadow: 0 1px 2px rgba(0, 0, 0, 0.2);
457 | }
458 |
459 | .film-inline-badge:hover {
460 |     background-color: #1d4ed8 !important;
461 |     color: #ffffff !important;
462 | }
463 |
464 | .film-interactive-chip {
465 |     display: inline-flex;
466 |     align-items: center;
467 |     gap: 3px;
468 |     background: linear-gradient(135deg, #1e3a8a 0%, #2563eb 100%
469 | );
470 |     color: #ffffff !important;
471 |     font-size: 0.85em;
472 |     font-weight: 500;
```



```
469 | padding: 2px 4px 2px 7px;
470 | border-radius: 14px;
471 | border: 1px solid rgba(255, 255, 255, 0.2);
472 | box-shadow: 0 1px 3px rgba(0, 0, 0, 0.18);
473 | margin: 2px 2px;
474 | vertical-align: middle;
475 | transition: all 0.18s ease;
476 | }
477 |
478 | .film-interactive-chip:hover {
479 |   background: linear-gradient(135deg, #1d4ed8 0%, #3b82f6 100%
480 |   );
481 |   box-shadow: 0 2px 6px rgba(37, 99, 235, 0.35);
482 |   transform: translateY(-1px);
483 | }
484 |
485 | .film-chip-title {
486 |   cursor: pointer;
487 |   color: #ffffff !important;
488 |   text-decoration: none;
489 |   font-weight: 600;
490 |   font-size: 0.82em;
491 | }
492 |
493 | .film-chip-title:hover { text-decoration: underline; }
494 |
495 | .film-chip-btn {
496 |   background: rgba(255, 255, 255, 0.15);
497 |   border: none;
498 |   color: #ffffff;
499 |   width: 20px;
500 |   height: 20px;
501 |   border-radius: 50%;
502 |   display: inline-flex;
503 |   align-items: center;
504 |   justify-content: center;
505 |   font-size: 0.7rem;
506 |   padding: 0;
```

```
506 | cursor: pointer;
507 | transition: all 0.12s ease;
508 | }
509 |
510 | .film-chip-btn:hover { background: rgba(255, 255, 255, 0.35
    | ); transform: scale(1.1); }
511 | .film-chip-play:hover { background: #10b981; }
512 | .film-chip-youtube:hover{ background: #ef4444; }
513 | .film-chip-online:hover { background: #06b6d4; }
514 | .film-chip-torrent:hover{ background: #f59e0b; color: #111827;
    | }
515 |
516 | [data-theme="dark"] .film-interactive-chip {
517 |   background: linear-gradient(135deg, #1e293b 0%, #334155 100%
    | );
518 |   border: 1px solid rgba(148, 163, 184, 0.25);
519 | }
520 |
521 | [data-theme="dark"] .film-interactive-chip:hover {
522 |   background: linear-gradient(135deg, #334155 0%, #475569 100%
    | );
523 |   box-shadow: 0 2px 6px rgba(0, 0, 0, 0.45);
524 | }
525 |
526 | /* =====
    | =====
527 |   PROMPT DUMP (КОНСОЛЬ АДМИНИСТРАТОРА)
528 |   =====
    | ===== */
529 | .prompt-dump-details {
530 |   margin-bottom: 8px;
531 |   font-size: 0.8em;
532 |   background: #0d1117 !important;
533 |   padding: 8px;
534 |   border-radius: 5px;
535 |   border: 1px solid #30363d !important;
536 |   box-shadow: inset 0 1px 3px rgba(0, 0, 0, 0.7);
537 | }
```

```
538 |
539 | .prompt-dump-details summary {
540 |     cursor: pointer;
541 |     color: #e3b341 !important;
542 |     font-weight: bold;
543 |     user-select: none;
544 |     font-size: 0.85em;
545 | }
546 |
547 | .prompt-dump-details pre {
548 |     white-space: pre-wrap;
549 |     margin-top: 8px;
550 |     color: #00ff66 !important;
551 |     font-family: 'Consolas', 'Courier New', Courier, monospace;
552 |     font-size: 0.9em;
553 |     line-height: 1.4;
554 |     max-height: 280px;
555 |     overflow-y: auto;
556 | }
557 |
558 | /* =====
559 |
560 | MENU EDITOR (DRAG AND DROP)
561 |
562 | ===== */
563 |
564 | .menu-editor-item {
565 |     display: flex;
566 |     align-items: center;
567 |     gap: 10px;
568 |     padding: 10px 12px;
569 |     background: var(--surface-1);
570 |     border: 1px solid var(--border-color);
571 |     border-radius: 6px;
572 |     margin-bottom: 8px;
573 |     transition: all 0.15s ease;
574 | }
575 |
576 | .menu-editor-item:hover {
```

```
574 | border-color: var(--nav-active);
575 | background: var(--surface-2);
576 | }
577 |
578 | .menu-editor-item:active {
579 |   cursor: grabbing;
580 | }
581 |
582 | .menu-editor-item.dragging {
583 |   opacity: 0.5;
584 |   background: var(--nav-active);
585 |   border-color: var(--nav-active);
586 | }
587 |
588 | .menu-editor-item .drag-handle-container {
589 |   cursor: grab;
590 |   color: var(--text-muted);
591 |   font-size: 1.2rem;
592 |   display: flex;
593 |   align-items: center;
594 | }
595 |
596 | .menu-editor-item .drag-handle-container:active {
597 |   cursor: grabbing;
598 | }
599 |
600 | .menu-editor-item .item-icon {
601 |   font-size: 1.3rem;
602 |   min-width: 28px;
603 |   text-align: center;
604 | }
605 |
606 | .menu-editor-item .item-info {
607 |   flex: 1;
608 |   min-width: 0;
609 | }
610 |
611 | .menu-editor-item .item-label {
```

```
612 | font-size: 0.88rem;
613 | font-weight: 500;
614 | white-space: nowrap;
615 | overflow: hidden;
616 | text-overflow: ellipsis;
617 | }
618 |
619 | .menu-editor-item .item-tab {
620 |   font-size: 0.72rem;
621 |   color: var(--text-muted);
622 |   font-family: monospace;
623 | }
624 |
625 | .menu-editor-item .editor-controls {
626 |   display: flex;
627 |   flex-direction: column;
628 |   gap: 6px;
629 |   min-width: 180px;
630 | }
631 |
632 | .menu-editor-item .editor-visibility,
633 | .menu-editor-item .editor-order {
634 |   display: flex;
635 |   align-items: center;
636 |   gap: 6px;
637 | }
638 |
639 | .menu-editor-item .editor-visibility .visibility-label,
640 | .menu-editor-item .editor-order .order-label {
641 |   font-size: 0.75rem;
642 |   color: var(--text-muted);
643 |   white-space: nowrap;
644 | }
645 |
646 | .menu-editor-item .editor-visibility .visibility-select {
647 |   flex: 1;
648 |   font-size: 0.78rem;
649 |   padding: 2px 6px;
```

```
650 }
651
652 .menu-editor-item .editor-order .order-slider {
653     flex: 1;
654     height: 6px;
655     padding: 0;
656 }
657
658 .menu-editor-item .editor-order .order-value {
659     font-size: 0.75rem;
660     color: var(--text-muted);
661     min-width: 24px;
662     text-align: right;
663 }
664
665 .menu-editor-section {
666     margin-bottom: 20px;
667 }
668
669 .menu-editor-section h6 {
670     font-size: 0.8rem;
671     text-transform: uppercase;
672     letter-spacing: 0.5px;
673     margin-bottom: 12px;
674     color: var(--text-muted);
675 }
676
677 .menu-editor-section h6 i {
678     margin-right: 6px;
679 }
680
681 #menuEditorModal .modal-body {
682     max-height: 70vh;
683     overflow-y: auto;
684 }
685
686 /* =====
```

```
687 | FLOATING TOAST NOTIFICATIONS (ВСПЛЫВАЮЩИЕ ОКНА)
688 |
689 | ===== */
690 | .floating-toast-container {
691 |     position: fixed;
692 |     top: 18px;
693 |     right: 18px;
694 |     z-index: 99999;
695 |     display: flex;
696 |     flex-direction: column;
697 |     gap: 10px;
698 |     max-width: 420px;
699 |     width: calc(100vw - 36px);
700 |     pointer-events: none;
701 | }
702 |
703 | .floating-toast {
704 |     pointer-events: auto;
705 |     background-color: var(--surface-1, #1e293b);
706 |     color: var(--text-color, #f8fafc);
707 |     border: 1px solid var(--border-color, #334155);
708 |     border-left-width: 4px;
709 |     border-radius: 8px;
710 |     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.4), 0 8px 10px
711 | -6px rgba(0, 0, 0, 0.3);
712 |     padding: 12px 14px;
713 |     opacity: 0;
714 |     transform: translateX(40px) scale(0.96);
715 |     transition: opacity 0.25s cubic-bezier(0.16, 1, 0.3, 1), tra
716 | nsform 0.25s cubic-bezier(0.16, 1, 0.3, 1);
717 |     overflow: hidden;
718 |     position: relative;
719 | }
720 |
721 | .floating-toast.show {
722 |     opacity: 1;
723 |     transform: translateX(0) scale(1);
724 | }
```

```
722 |
723 | .floating-toast.hide {
724 |   opacity: 0;
725 |   transform: translateX(40px) scale(0.92);
726 | }
727 |
728 | .floating-toast-success {
729 |   border-left-color: #22c55e !important;
730 | }
731 |
732 | .floating-toast-danger {
733 |   border-left-color: #ef4444 !important;
734 | }
735 |
736 | .floating-toast-warning {
737 |   border-left-color: #f59e0b !important;
738 | }
739 |
740 | .floating-toast-info {
741 |   border-left-color: #3b82f6 !important;
742 | }
743 |
744 | .floating-toast-header {
745 |   display: flex;
746 |   align-items: center;
747 |   gap: 8px;
748 |   margin-bottom: 6px;
749 | }
750 |
751 | .floating-toast-icon {
752 |   font-size: 1.1rem;
753 |   line-height: 1;
754 | }
755 |
756 | .floating-toast-title {
757 |   font-size: 0.88rem;
758 |   font-weight: 600;
759 |   flex: 1;
```



```
760 |   color: var(--text-color, #f8fafc);
761 | }
762 |
763 | .floating-toast-close {
764 |   background: transparent;
765 |   border: none;
766 |   color: var(--text-muted, #94a3b8);
767 |   font-size: 1.25rem;
768 |   line-height: 1;
769 |   padding: 0 4px;
770 |   cursor: pointer;
771 |   transition: color 0.15s ease;
772 | }
773 |
774 | .floating-toast-close:hover {
775 |   color: var(--text-color, #ffffff);
776 | }
777 |
778 | .floating-toast-body {
779 |   font-size: 0.84rem;
780 |   line-height: 1.45;
781 |   color: var(--card-text, #cbd5e1);
782 |   word-break: break-word;
783 | }
784 |
785 | .floating-toast-progress {
786 |   position: absolute;
787 |   bottom: 0;
788 |   left: 0;
789 |   right: 0;
790 |   height: 3px;
791 |   background: rgba(255, 255, 255, 0.08);
792 | }
793 |
794 | .floating-toast-progress-bar {
795 |   height: 100%;
796 |   width: 100%;
797 | }
```

```

798
799 /* =====
      =====
800     КАСКАДНАЯ ДИЗАЙН-СИСТЕМА (UNIFIED CASCADING UI SYSTEM)
801     =====
      ===== */
802
803 /* — 1. KPI & Метрические карточки —————
      ————— */
804 .ui-kpi-card {
805     background: var(--kpi-bg, var(--surface-1)) !important;
806     border: 1px solid var(--kpi-border, var(--border-color)) !im
portant;
807     border-radius: var(--radius-md, 8px);
808     padding: var(--kpi-padding, 0.65rem 0.9rem);
809     min-height: var(--kpi-min-height, 72px);
810     display: flex;
811     flex-direction: column;
812     justify-content: space-between;
813     height: 100%;
814     box-sizing: border-box;
815     transition: border-color 0.15s ease, box-shadow 0.15s ease,
transform 0.15s ease;
816 }
817
818 .ui-kpi-card:hover {
819     border-color: var(--kpi-accent, var(--nav-active)) !importan
t;
820     box-shadow: var(--shadow-sm, 0 2px 8px rgba(0, 0, 0, 0.15));
821 }
822
823 .ui-kpi-label {
824     font-size: var(--kpi-label-size, 0.72rem);
825     text-transform: uppercase;
826     letter-spacing: 0.05em;
827     font-weight: 600;
828     color: var(--text-muted, #8b949e);
829     display: flex;

```

```
830 | justify-content: space-between;
831 | align-items: center;
832 | line-height: 1.2;
833 | }
834 |
835 | .ui-kpi-value {
836 |   font-size: var(--kpi-val-size, 1.25rem);
837 |   font-weight: 700;
838 |   font-family: var(--font-mono, monospace);
839 |   color: var(--text-color, #f8fafc);
840 |   line-height: 1.2;
841 | }
842 |
843 | .ui-kpi-sub {
844 |   font-size: var(--kpi-sub-size, 0.72rem);
845 |   color: var(--text-muted, #8b949e);
846 |   line-height: 1.2;
847 | }
848 |
849 | /* Модификаторы состояний KPI карточек через каскад */
850 | .ui-kpi-card.is-success {
851 |   --kpi-border: var(--status-success-border, #22c55e);
852 |   --kpi-accent: var(--status-success-border, #22c55e);
853 | }
854 | .ui-kpi-card.is-danger {
855 |   --kpi-border: var(--status-danger-border, #ef4444);
856 |   --kpi-accent: var(--status-danger-border, #ef4444);
857 | }
858 | .ui-kpi-card.is-warning {
859 |   --kpi-border: var(--status-warning-border, #eab308);
860 |   --kpi-accent: var(--status-warning-border, #eab308);
861 | }
862 | .ui-kpi-card.is-info {
863 |   --kpi-border: var(--status-info-border, #0284c7);
864 |   --kpi-accent: var(--status-info-border, #0284c7);
865 | }
866 |
867 | /* — 2. Модульные заголовки и Hero-баннеры —————
```

```

      _____ */
868 .ui-module-header {
869     background: var(--surface-2) !important;
870     border: 1px solid var(--border-color) !important;
871     border-radius: var(--radius-md, 8px);
872     padding: 0.6rem 1rem;
873     margin-bottom: 0.75rem;
874     display: flex;
875     align-items: center;
876     justify-content: space-between;
877     flex-wrap: wrap;
878     gap: 0.5rem;
879 }
880
881 .ui-module-title {
882     font-size: 0.95rem;
883     font-weight: 700;
884     color: var(--text-color);
885     display: flex;
886     align-items: center;
887     gap: 0.5rem;
888     margin-bottom: 0;
889 }
890
891 .ui-module-desc {
892     font-size: 0.78rem;
893     color: var(--text-muted);
894     margin-bottom: 0;
895 }
896
897 /* — 3. Панели и контейнеры таблиц _____
      _____ */
898 .ui-panel {
899     background: var(--panel-bg, var(--surface-1)) !important;
900     border: 1px solid var(--panel-border, var(--border-color)) !
important;
901     border-radius: var(--radius-md, 8px);
902     overflow: hidden;

```

```
903 | margin-bottom: 0.75rem;
904 | box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
905 | }
906 |
907 | .ui-panel-header {
908 |   background: var(--panel-header-bg, var(--surface-2)) !important;
909 |   border-bottom: 1px solid var(--border-color) !important;
910 |   padding: 0.5rem 0.9rem;
911 |   font-size: 0.85rem;
912 |   font-weight: 600;
913 |   color: var(--text-color);
914 |   display: flex;
915 |   align-items: center;
916 |   justify-content: space-between;
917 |   flex-wrap: wrap;
918 |   gap: 0.5rem;
919 | }
920 |
921 | .ui-panel-body {
922 |   padding: 0.75rem;
923 | }
924 |
925 | .ui-panel-footer {
926 |   background: var(--surface-2) !important;
927 |   border-top: 1px solid var(--border-color) !important;
928 |   padding: 0.4rem 0.75rem;
929 |   font-size: 0.78rem;
930 | }
931 |
932 | /* — 4. Унифицированные суб-навигационные переключатели (Sub-
933 |   Nav Tabs) — */
934 | .ui-subnav {
935 |   display: flex;
936 |   align-items: center;
937 |   gap: 0.4rem;
938 |   border-bottom: 1px solid var(--border-color);
939 |   margin-bottom: 0.75rem;
```

```
939 | padding-bottom: 0.35rem;
940 | overflow-x: auto;
941 | flex-wrap: nowrap;
942 | }
943 |
944 | .ui-subnav-btn {
945 |   background: transparent;
946 |   border: 1px solid transparent;
947 |   color: var(--text-muted);
948 |   font-size: var(--subnav-font-size, 0.82rem);
949 |   font-weight: 500;
950 |   padding: var(--subnav-btn-padding, 0.35rem 0.75rem);
951 |   border-radius: var(--radius-sm, 6px);
952 |   cursor: pointer;
953 |   white-space: nowrap;
954 |   display: inline-flex;
955 |   align-items: center;
956 |   gap: 0.4rem;
957 |   transition: all 0.15s ease;
958 | }
959 |
960 | .ui-subnav-btn:hover {
961 |   background: var(--surface-2);
962 |   color: var(--text-color);
963 | }
964 |
965 | .ui-subnav-btn.active {
966 |   background: var(--surface-2);
967 |   border-color: var(--nav-active);
968 |   color: var(--nav-active);
969 |   font-weight: 600;
970 | }
971 |
972 | /* — 5. Унифицированные статусные бейджи —————
973 |    |————— */
974 | .ui-badge {
975 |   font-size: 0.72rem;
976 |   font-weight: 600;
```

```
976 | padding: 0.2rem 0.55rem;
977 | border-radius: var(--radius-xs, 4px);
978 | display: inline-flex;
979 | align-items: center;
980 | gap: 0.3rem;
981 | line-height: 1.2;
982 | }
983 |
984 | .ui-badge-success,
985 | .status-active,
986 | .status-enabled,
987 | .status-on,
988 | .status-running,
989 | .status-online,
990 | .status-connected,
991 | .status-true,
992 | .badge-status-on,
993 | .badge-status-active,
994 | .badge-status-enabled,
995 | .badge-status-running,
996 | .badge-status-online,
997 | .badge-status-true,
998 | .is-status-positive,
999 | [data-status="active"],
1000 | [data-status="enabled"],
1001 | [data-status="on"],
1002 | [data-status="running"],
1003 | [data-status="online"],
1004 | [data-status="true"],
1005 | [data-status="включен"],
1006 | [data-status="включена"],
1007 | [data-status="включено"],
1008 | [data-status="активен"],
1009 | [data-status="активна"],
1010 | [data-status="активно"],
1011 | [data-status="работает"],
1012 | [data-status="подключен"],
1013 | [data-state="active"],
```

```
1014 [data-state="enabled"],
1015 [data-state="on"],
1016 [data-state="running"],
1017 [data-state="online"],
1018 [data-state="true"],
1019 [data-state="включен"],
1020 [data-state="включена"],
1021 [data-state="включено"],
1022 [data-state="активен"],
1023 [data-state="активна"],
1024 [data-state="активно"],
1025 [data-state="работает"],
1026 [data-state="подключен"] {
1027     background-color: var(--status-success-bg) !important;
1028     color: var(--status-success-text) !important;
1029     border: 1px solid var(--status-success-border) !important;
1030 }
1031
1032 .ui-badge-danger,
1033 .status-inactive,
1034 .status-disabled,
1035 .status-off,
1036 .status-stopped,
1037 .status-offline,
1038 .status-danger,
1039 .status-error,
1040 .status-disconnected,
1041 .status-false,
1042 .badge-status-off,
1043 .badge-status-inactive,
1044 .badge-status-disabled,
1045 .badge-status-stopped,
1046 .badge-status-offline,
1047 .badge-status-false,
1048 .is-status-negative,
1049 [data-status="inactive"],
1050 [data-status="disabled"],
1051 [data-status="off"],
```



```
1052 [data-status="stopped"],
1053 [data-status="offline"],
1054 [data-status="false"],
1055 [data-status="error"],
1056 [data-status="выключен"],
1057 [data-status="выключена"],
1058 [data-status="выключено"],
1059 [data-status="неактивен"],
1060 [data-status="неактивна"],
1061 [data-status="неактивно"],
1062 [data-status="остановлен"],
1063 [data-status="остановлена"],
1064 [data-status="отключен"],
1065 [data-status="ошибка"],
1066 [data-state="inactive"],
1067 [data-state="disabled"],
1068 [data-state="off"],
1069 [data-state="stopped"],
1070 [data-state="offline"],
1071 [data-state="false"],
1072 [data-state="error"],
1073 [data-state="выключен"],
1074 [data-state="выключена"],
1075 [data-state="выключено"],
1076 [data-state="неактивен"],
1077 [data-state="неактивна"],
1078 [data-state="неактивно"],
1079 [data-state="остановлен"],
1080 [data-state="остановлена"],
1081 [data-state="отключен"],
1082 [data-state="ошибка"] {
1083     background-color: var(--status-danger-bg) !important;
1084     color: var(--status-danger-text) !important;
1085     border: 1px solid var(--status-danger-border) !important;
1086 }
1087
1088 .ui-badge-warning {
1089     background: var(--status-warning-bg);
```

```
1090 | color: var(--status-warning-text);
1091 | border: 1px solid var(--status-warning-border);
1092 | }
1093 |
1094 | .ui-badge-info {
1095 |     background: var(--status-info-bg);
1096 |     color: var(--status-info-text);
1097 |     border: 1px solid var(--status-info-border);
1098 | }
1099 |
1100 | .ui-badge-neutral {
1101 |     background: var(--surface-2);
1102 |     color: var(--text-muted);
1103 |     border: 1px solid var(--border-color);
1104 | }
1105 |
1106 | /* Точечный индикатор состояния (Status Dot) */
1107 | .status-dot,
1108 | .status-indicator {
1109 |     display: inline-block;
1110 |     width: 8px;
1111 |     height: 8px;
1112 |     border-radius: 50%;
1113 |     vertical-align: middle;
1114 |     margin-right: 4px;
1115 | }
1116 |
1117 | .status-dot.active,
1118 | .status-dot.enabled,
1119 | .status-dot.on,
1120 | .status-dot.running,
1121 | .status-dot.online,
1122 | .status-dot.positive,
1123 | .status-indicator.active,
1124 | .status-indicator.enabled,
1125 | .status-indicator.on,
1126 | .status-indicator.running,
1127 | .status-indicator.online,
```

```

1128 | .status-indicator.positive {
1129 |     background-color: #22c55e !important;
1130 |     box-shadow: 0 0 6px rgba(34, 197, 94, 0.6);
1131 | }
1132 |
1133 | .status-dot.inactive,
1134 | .status-dot.disabled,
1135 | .status-dot.off,
1136 | .status-dot.stopped,
1137 | .status-dot.offline,
1138 | .status-dot.negative,
1139 | .status-indicator.inactive,
1140 | .status-indicator.disabled,
1141 | .status-indicator.off,
1142 | .status-indicator.stopped,
1143 | .status-indicator.offline,
1144 | .status-indicator.negative {
1145 |     background-color: #ef4444 !important;
1146 |     box-shadow: 0 0 6px rgba(239, 68, 68, 0.6);
1147 | }
1148 |
1149 |
1150 | /* — 6. Унифицированная таблица спецификаций (Ключ - Значение
      | ) ————— */
1151 | .ui-spec-table,
1152 | .about-sys-spec-table,
1153 | .sysctl-spec-table,
1154 | .def-spec-table,
1155 | .winadmin-spec-table {
1156 |     width: 100%;
1157 |     margin-bottom: 0;
1158 |     font-size: 0.82rem;
1159 |     border-collapse: separate;
1160 |     border-spacing: 0;
1161 | }
1162 |
1163 | .ui-spec-table td,
1164 | .about-sys-spec-table td,

```

```
1165 | .sysctl-spec-table td,
1166 | .def-spec-table td,
1167 | .winadmin-spec-table td {
1168 |     padding: 9px 14px;
1169 |     border-bottom: 1px solid var(--border-subtle, rgba(51, 65, 8
        | 5, 0.35));
1170 |     vertical-align: middle;
1171 |     line-height: 1.45;
1172 | }
1173 |
1174 | .ui-spec-table tr:hover td,
1175 | .about-sys-spec-table tr:hover td,
1176 | .sysctl-spec-table tr:hover td,
1177 | .def-spec-table tr:hover td,
1178 | .winadmin-spec-table tr:hover td {
1179 |     background: rgba(56, 189, 248, 0.04);
1180 | }
1181 |
1182 | .ui-spec-table tr:last-child td,
1183 | .about-sys-spec-table tr:last-child td,
1184 | .sysctl-spec-table tr:last-child td,
1185 | .def-spec-table tr:last-child td,
1186 | .winadmin-spec-table tr:last-child td {
1187 |     border-bottom: none;
1188 | }
1189 |
1190 | .ui-spec-key,
1191 | .about-sys-spec-key,
1192 | .sysctl-spec-key,
1193 | .def-spec-key,
1194 | .winadmin-spec-key {
1195 |     color: var(--text-muted, #94a3b8);
1196 |     width: 230px;
1197 |     min-width: 210px;
1198 |     max-width: 260px;
1199 |     white-space: nowrap;
1200 |     font-weight: 500;
1201 |     font-family: var(--font-sans, system-ui, -apple-system, sans
```

```

    -serif);
1202     vertical-align: middle;
1203 }
1204
1205 .ui-spec-key i,
1206 .about-sys-spec-key i,
1207 .sysctl-spec-key i,
1208 .def-spec-key i,
1209 .winadmin-spec-key i {
1210     margin-right: 8px;
1211     font-size: 0.92rem;
1212     vertical-align: -1px;
1213 }
1214
1215 .ui-spec-val,
1216 .about-sys-spec-val,
1217 .sysctl-spec-val,
1218 .def-spec-val,
1219 .winadmin-spec-val {
1220     color: var(--text-color, #f8fafc);
1221     font-weight: 500;
1222     font-family: var(--font-mono, monospace);
1223     vertical-align: middle;
1224     word-break: break-word;
1225 }
1226
1227 /* — 7. Контекстные каскадные темы (Тематические переопределе
    ния) ————— */
1228 .theme-defender {
1229     --kpi-min-height: 64px;
1230     --kpi-padding: 0.5rem 0.75rem;
1231     --kpi-accent: #22c55e;
1232 }
1233
1234 .theme-sysadmin {
1235     --kpi-min-height: 68px;
1236     --kpi-padding: 0.55rem 0.85rem;
1237 }

```

```
1238 |
1239 | .theme-inspector {
1240 |   --kpi-min-height: 80px;
1241 |   --kpi-val-size: 1.35rem;
1242 | }
1243 |
1244 | .theme-sysctl {
1245 |   --kpi-min-height: 72px;
1246 | }
1247 |
1248 | /* — 8. Алиасы совместимости (Backward-compatible Aliases) —
      | _____ */
1249 | .about-sys-card,
1250 | .def-card,
1251 | .sysctl-card,
1252 | .scc-card,
1253 | .winadmin-card,
1254 | .sys-card {
1255 |   composes: ui-kpi-card;
1256 |   background: var(--kpi-bg, var(--surface-1)) !important;
1257 |   border: 1px solid var(--kpi-border, var(--border-color)) !im
      | portant;
1258 |   border-radius: var(--radius-md, 8px);
1259 |   padding: var(--kpi-padding, 0.65rem 0.9rem);
1260 |   min-height: var(--kpi-min-height, 72px);
1261 |   display: flex;
1262 |   flex-direction: column;
1263 |   justify-content: space-between;
1264 |   height: 100%;
1265 |   box-sizing: border-box;
1266 |   transition: border-color 0.15s ease, box-shadow 0.15s ease;
1267 | }
1268 |
1269 | .about-sys-card:hover,
1270 | .def-card:hover,
1271 | .sysctl-card:hover,
1272 | .scc-card:hover,
1273 | .winadmin-card:hover,
```

```
1274 | .sys-card:hover {
1275 |   border-color: var(--kpi-accent, var(--nav-active)) !important;
1276 | }
1277 |
1278 | .about-sys-label,
1279 | .def-card-label,
1280 | .sysctl-label,
1281 | .scc-label,
1282 | .winadmin-label,
1283 | .sys-label {
1284 |   font-size: var(--kpi-label-size, 0.72rem);
1285 |   text-transform: uppercase;
1286 |   letter-spacing: 0.05em;
1287 |   font-weight: 600;
1288 |   color: var(--text-muted, #8b949e);
1289 |   display: flex;
1290 |   justify-content: space-between;
1291 |   align-items: center;
1292 |   line-height: 1.2;
1293 | }
1294 |
1295 | .about-sys-value,
1296 | .def-card-value,
1297 | .sysctl-value,
1298 | .scc-value,
1299 | .winadmin-value,
1300 | .sys-value {
1301 |   font-size: var(--kpi-val-size, 1.25rem);
1302 |   font-weight: 700;
1303 |   font-family: var(--font-mono, monospace);
1304 |   color: var(--text-color, #f8fafc);
1305 |   line-height: 1.2;
1306 | }
1307 |
1308 | .about-sys-sub,
1309 | .def-card-sub,
1310 | .sysctl-sub,
```

```
1311 | .scc-sub,
1312 | .winadmin-sub,
1313 | .sys-sub {
1314 |     font-size: var(--kpi-sub-size, 0.72rem);
1315 |     color: var(--text-muted, #8b949e);
1316 |     line-height: 1.2;
1317 | }
1318 |
1319 | .about-sys-panel,
1320 | .def-panel,
1321 | .sysctl-panel,
1322 | .winadmin-panel,
1323 | .sys-panel {
1324 |     background: var(--panel-bg, var(--surface-1)) !important;
1325 |     border: 1px solid var(--panel-border, var(--border-color)) !
1326 | important;
1327 |     border-radius: var(--radius-md, 8px);
1328 |     overflow: hidden;
1329 |     margin-bottom: 0.75rem;
1330 |     box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
1331 | }
1332 |
1333 | .about-sys-panel-header,
1334 | .def-panel-header,
1335 | .sysctl-panel-header,
1336 | .winadmin-panel-header,
1337 | .sys-panel-header {
1338 |     background: var(--panel-header-bg, var(--surface-2)) !import
1339 | ant;
1340 |     border-bottom: 1px solid var(--border-color) !important;
1341 |     padding: 0.5rem 0.9rem;
1342 |     font-size: 0.85rem;
1343 |     font-weight: 600;
1344 |     color: var(--text-color);
1345 |     display: flex;
1346 |     align-items: center;
1347 |     justify-content: space-between;
1348 |     flex-wrap: wrap;
```



```
1347 | gap: 0.5rem;
1348 | }
1349 |
1350 |
1351 |
1352 |
```

main.css

```
1  /* =====
   |
2  MAIN STYLESHEET MANIFEST – AI Breadboard WebGUI
3  Модульная точка входа стилей веб-интерфейса
4  ===== */
5
6  /* 1. Дизайн-токены и переменные тем (светлая/тёмная) */
7  @import "variables.css";
8
9  /* 2. Базовые стили страницы, скроллбары и контейнеры вкладок
   | */
10 @import "base.css";
11
12 /* 3. Навигационная панель, вкладки, выпадающие меню и переключатели */
13 @import "nav.css";
14
15 /* 4. Общие UI-компоненты (карточки, формы, кнопки, модалки, таблицы, бейджи, чипы) */
16 @import "components.css";
17
18 /* 5. Чат, сообщения, сайдбар истории диалогов и дебаггер */
19 @import "chat.css";
20
21 /* 6. Управление торрентами, поиск релизов, карточки трекеров */
22 @import "torrents.css";
23
24 /* 7. Адаптивные переопределения и совместимость тем с Bootstrap */
25 @import "overrides.css";
26
```

nav.css

```
1  /* =====
   |=====
2  NAVIGATION – AI Breadboard WebGUI
3  Главная панель навигации, вкладки, выпадающие меню и элемен
   |ты управления
4  =====
   |===== */
5
6  /* — Главная панель навигации —————
   |————— */
7  .main-nav-container {
8      background-color: var(--nav-bg);
9      border-bottom: 1px solid var(--border-color);
10     min-height: 34px;
11     position: relative;
12     z-index: 1020;
13     overflow: visible !important;
14     box-shadow: var(--shadow-xs);
15 }
16
17 .nav-tabs {
18     border-bottom: none;
19     background-color: transparent;
20     flex-wrap: nowrap;
21     overflow: visible !important;
22 }
23
24 .nav-tabs .nav-item {
25     flex-shrink: 0;
26     position: relative;
27 }
28
29 .nav-tabs .nav-item.dropdown {
30     position: relative;
31 }
32
```

```
33 /* Nav-links — компактные */
34 .nav-tabs .nav-link {
35     color: var(--nav-text);
36     border: none;
37     border-bottom: 2px solid transparent;
38     background: transparent;
39     padding: 5px 10px;
40     font-size: 0.82rem;
41     font-weight: 500;
42     white-space: nowrap;
43     transition: color 0.15s, border-color 0.15s, background-color 0.15s;
44     border-radius: 4px 4px 0 0;
45 }
46
47 .nav-tabs .nav-link:hover {
48     color: var(--nav-active);
49     background-color: var(--nav-hover-bg);
50     border-bottom-color: transparent;
51 }
52
53 .nav-tabs .nav-link.active {
54     color: var(--nav-active);
55     background: transparent;
56     border-bottom-color: var(--nav-active);
57 }
58
59 /* — Выпадающие меню навбара ————— */
60 .nav-tabs .dropdown-menu {
61     position: absolute !important;
62     top: 100% !important;
63     left: 0 !important;
64     margin-top: 3px !important;
65     z-index: 9999 !important;
66     min-width: 180px;
67     display: none !important;
68     padding: 4px;
```

```
69 | border-radius: 8px;
70 | background-color: var(--surface-1);
71 | border: 1px solid var(--border-color);
72 | box-shadow: var(--shadow-md) !important;
73 | }
74 |
75 | .nav-tabs .dropdown-menu.show {
76 |   display: block !important;
77 |   visibility: visible !important;
78 |   opacity: 1 !important;
79 | }
80 |
81 | .nav-tabs .dropdown-item {
82 |   width: 100%;
83 |   border: none;
84 |   background: transparent;
85 |   text-align: left;
86 |   padding: 5px 10px;
87 |   border-radius: 5px;
88 |   font-size: 0.82rem;
89 |   font-weight: 500;
90 |   cursor: pointer;
91 |   color: var(--text-color);
92 |   transition: background-color 0.12s ease, color 0.12s ease;
93 | }
94 |
95 | .nav-tabs .dropdown-item:hover,
96 | .nav-tabs .dropdown-item:focus {
97 |   background-color: var(--nav-hover-bg);
98 |   color: var(--nav-active);
99 | }
100 |
101 | .nav-tabs .dropdown-item.active {
102 |   background-color: rgba(37, 99, 235, 0.1);
103 |   color: var(--nav-active);
104 |   font-weight: 600;
105 | }
106 |
```

```
107 [data-theme="dark"] .nav-tabs .dropdown-item.active {
108   background-color: rgba(88, 166, 255, 0.12);
109   color: var(--nav-active);
110 }
111
112 /* — Разделители и метки групп —————
    |————— */
113 .nav-group-divider {
114   width: 1px;
115   height: 16px;
116   background-color: var(--divider-color);
117   margin: 0 6px;
118   opacity: 0.7;
119   align-self: center;
120   flex-shrink: 0;
121 }
122
123 .nav-group-label {
124   display: inline-flex;
125   align-items: center;
126   font-size: 0.68rem;
127   font-weight: 600;
128   text-transform: uppercase;
129   letter-spacing: 0.5px;
130   color: var(--text-muted);
131   padding: 2px 6px;
132   background-color: var(--surface-2);
133   border: 1px solid var(--border-color);
134   border-radius: 5px;
135   user-select: none;
136 }
137
138 /* — Выпадающие меню Bootstrap (глобально) —————
    |————— */
139 [data-theme="light"] .dropdown-menu {
140   background-color: var(--surface-1);
141   border-color: var(--border-color);
142   box-shadow: var(--shadow-md);
```

```
143 }
144
145 [data-theme="light"] .dropdown-item {
146     color: var(--text-color);
147     border-radius: 4px;
148     font-size: 0.82rem;
149 }
150
151 [data-theme="light"] .dropdown-item:hover,
152 [data-theme="light"] .dropdown-item:focus {
153     background-color: var(--nav-hover-bg);
154     color: var(--nav-active);
155 }
156
157 [data-theme="light"] .dropdown-item.active {
158     background-color: #2563eb;
159     color: #ffffff;
160 }
161
162 [data-theme="dark"] .dropdown-menu {
163     background-color: var(--surface-1);
164     border-color: var(--border-color);
165     box-shadow: var(--shadow-md);
166 }
167
168 [data-theme="dark"] .dropdown-item {
169     color: var(--text-color);
170     border-radius: 4px;
171     font-size: 0.82rem;
172 }
173
174 [data-theme="dark"] .dropdown-item:hover,
175 [data-theme="dark"] .dropdown-item:focus {
176     background-color: var(--nav-hover-bg);
177     color: var(--nav-active);
178 }
179
180 [data-theme="dark"] .dropdown-item.active {
```

```
181 | background-color: #1f6feb;
182 | color: #ffffff;
183 | }
184 |
185 | /* — Кнопки и контролы навбара —————
186 | |————— */
187 | .main-nav-container .lang-selector {
188 |   background-color: var(--nav-bg);
189 |   border-color: var(--border-color);
190 |   color: var(--nav-text);
191 |   font-size: 0.8rem;
192 | }
193 | #nav-google-btn {
194 |   background: #ffffff;
195 |   color: #1f2937 !important;
196 |   font-size: 0.78rem;
197 |   font-weight: 600;
198 |   border-radius: 20px;
199 |   border: 1px solid rgba(0, 0, 0, 0.12);
200 |   padding: 3px 10px;
201 |   transition: all 0.18s ease;
202 | }
203 |
204 | #nav-google-btn:hover {
205 |   background: #f8fafc;
206 |   box-shadow: var(--shadow-sm);
207 |   transform: translateY(-1px);
208 | }
209 |
210 | [data-theme="dark"] #nav-google-btn {
211 |   background: var(--surface-2);
212 |   color: var(--text-color) !important;
213 |   border-color: var(--border-color);
214 | }
215 |
216 | #user-settings-btn {
217 |   font-size: 0.8rem;
```



```
218 | font-weight: 500;
219 | border-radius: 20px;
220 | color: var(--nav-text);
221 | border-color: var(--border-color);
222 | transition: all 0.18s ease;
223 | padding: 3px 10px;
224 | }
225 |
226 | #user-settings-btn:hover {
227 |   background-color: var(--nav-hover-bg);
228 |   color: var(--nav-active);
229 |   border-color: var(--nav-active);
230 | }
231 |
232 | #user-settings-gear-btn,
233 | #theme-toggle-btn {
234 |   color: var(--nav-text);
235 |   border-color: var(--border-color);
236 |   transition: all 0.18s ease;
237 |   width: 28px;
238 |   height: 28px;
239 |   padding: 0;
240 |   display: inline-flex;
241 |   align-items: center;
242 |   justify-content: center;
243 |   border-radius: 50%;
244 |   font-size: 0.85rem;
245 | }
246 |
247 | #user-settings-gear-btn:hover {
248 |   background-color: var(--nav-hover-bg);
249 |   color: var(--nav-active);
250 |   border-color: var(--nav-active);
251 |   transform: rotate(30deg);
252 | }
253 |
254 | #theme-toggle-btn:hover {
255 |   background-color: var(--nav-hover-bg);
```

```
256 | color: var(--nav-active);
257 | border-color: var(--nav-active);
258 | }
259 |
260 | /* — Переключатель темы (Dropdown / иконки) —————
    | ————— */
261 | #theme-dropdown-container .dropdown-menu {
262 |   min-width: 140px;
263 |   padding: 3px;
264 | }
265 |
266 | .theme-check-icon {
267 |   font-size: 0.85rem;
268 | }
269 |
270 | .theme-check-icon.d-none {
271 |   display: none !important;
272 | }
273 |
274 | /* — Выдвижной боковой ярлычок (Левая вкладка-триггер) —————
    | ————— */
275 | .left-drawer-tab-trigger {
276 |   position: fixed;
277 |   top: 48%;
278 |   left: 0;
279 |   transform: translateY(-50%);
280 |   z-index: 1040;
281 |   background-color: var(--surface-1);
282 |   border: 1px solid var(--border-color);
283 |   border-left: none;
284 |   border-radius: 0 8px 8px 0;
285 |   padding: 10px 6px;
286 |   display: flex;
287 |   flex-direction: column;
288 |   align-items: center;
289 |   gap: 6px;
290 |   cursor: pointer;
291 |   color: var(--nav-text);
```

```
292 | box-shadow: var(--shadow-md);
293 | transition: all 0.2s ease;
294 | user-select: none;
295 | }
296 |
297 | .left-drawer-tab-trigger:hover {
298 |   background-color: var(--nav-hover-bg);
299 |   color: var(--nav-active);
300 |   padding-right: 10px;
301 |   box-shadow: var(--shadow-lg);
302 | }
303 |
304 | .left-drawer-tab-text {
305 |   writing-mode: vertical-rl;
306 |   text-orientation: mixed;
307 |   font-size: 0.68rem;
308 |   font-weight: 700;
309 |   letter-spacing: 2px;
310 |   text-transform: uppercase;
311 | }
312 |
313 | /* — Левая выдвижная панель навигации (Offcanvas Drawer) —
    | _____ */
314 | .left-drawer-nav {
315 |   width: 320px !important;
316 |   background-color: var(--surface-1);
317 |   color: var(--text-color);
318 |   border-right: 1px solid var(--border-color);
319 |   box-shadow: var(--shadow-lg);
320 |   z-index: 1060;
321 | }
322 |
323 | .left-drawer-nav .offcanvas-header {
324 |   background-color: var(--surface-2);
325 |   border-bottom: 1px solid var(--border-color);
326 | }
327 |
328 | .nav-drawer-group-title {
```

```
329 | font-size: 0.72rem;
330 | font-weight: 700;
331 | text-transform: uppercase;
332 | letter-spacing: 0.8px;
333 | color: var(--text-muted);
334 | padding: 0 4px;
335 | }
336 |
337 | .left-drawer-nav .list-group-item {
338 |   background-color: var(--surface-1);
339 |   border-color: var(--border-color);
340 |   color: var(--text-color);
341 |   font-size: 0.85rem;
342 |   transition: background-color 0.15s ease, color 0.15s ease, t
ransform 0.1s ease;
343 |   cursor: pointer;
344 |   border-radius: 6px;
345 |   margin-bottom: 3px;
346 |   border: 1px solid transparent;
347 | }
348 |
349 | .left-drawer-nav .list-group-item:hover {
350 |   background-color: var(--nav-hover-bg);
351 |   color: var(--nav-active);
352 |   transform: translateX(2px);
353 | }
354 |
355 | .left-drawer-nav .list-group-item.active {
356 |   background-color: #2563eb;
357 |   border-color: #2563eb;
358 |   color: #ffffff;
359 |   font-weight: 600;
360 | }
361 |
362 | [data-theme="dark"] .left-drawer-nav .list-group-item.active {
363 |   background-color: #1f6feb;
364 |   border-color: #1f6feb;
365 |   color: #ffffff;
```

```
366 | }
367 |
368 | .left-drawer-nav .list-group-item.active:hover {
369 |     background-color: #1d4ed8;
370 | }
371 |
372 | [data-theme="dark"] .left-drawer-nav .list-group-item.active:h
    | over {
373 |     background-color: #388bfd;
374 | }
375 |
376 |
377 |
```

overrides.css

```
1  /* =====  
2  THEME OVERRIDES & COMPATIBILITY – AI Breadboard WebGUI  
3  Адаптивные переопределения Bootstrap и специфичных компонен  
тов для тем оформления  
4  ===== */  
5  
6  /* — Light Theme – Переопределения Bootstrap и общих утилит –  
   ===== */  
7  [data-theme="light"] {  
8      /* Инпуты, к которым JS или Bootstrap добавил bg-dark */  
9      input.bg-dark,  
10     textarea.bg-dark,  
11     select.bg-dark,  
12     .form-control.bg-dark,  
13     .form-select.bg-dark {  
14         background-color: var(--surface-1) !important;  
15         color: var(--text-color) !important;  
16         border-color: var(--border-color) !important;  
17     }  
18  
19     /* div/section с bg-dark (не кнопки и не бейджи) */  
20     div.bg-dark,  
21     section.bg-dark,  
22     aside.bg-dark,  
23     article.bg-dark {  
24         background-color: var(--surface-1) !important;  
25         color: var(--text-color) !important;  
26     }  
27  
28     /* Bootstrap bg-dark / bg-black */  
29     .bg-dark:not(.btn):not(.badge):not(.navbar) {  
30         background-color: var(--surface-1) !important;  
31         color: var(--text-color) !important;  
32         border-color: var(--border-color) !important;
```

```
33 | }
34 |
35 | .bg-black:not(.btn) {
36 |     background-color: var(--bg-color) !important;
37 |     color: var(--text-color) !important;
38 | }
39 |
40 | /* text-white / text-light в карточках и панелях */
41 | .card .text-white,
42 | .card .text-light,
43 | [class*="-panel"] .text-white,
44 | [class*="-panel"] .text-light,
45 | [class*="-card"] .text-white,
46 | [class*="-card"] .text-light,
47 | .tab-pane .text-white,
48 | .tab-pane .text-light,
49 | .tab-pane .text-white:not(.btn):not(.badge),
50 | .card .text-white:not(.btn):not(.badge) {
51 |     color: var(--text-color) !important;
52 | }
53 |
54 | /* Бейджи bg-dark → нейтральный цвет */
55 | .badge.bg-dark {
56 |     background-color: var(--border-color) !important;
57 |     color: var(--text-color) !important;
58 | }
59 |
60 | /* pre/code без явного светлого фона (кроме prompt-dump) */
61 | pre:not(.prompt-dump-details pre) {
62 |     background-color: var(--surface-2) !important;
63 |     color: #1e2328 !important;
64 |     border: 1px solid var(--border-color);
65 |     border-radius: 5px;
66 | }
67 |
68 | code.bg-dark {
69 |     background-color: var(--surface-2) !important;
70 |     color: #1e2328 !important;
```

```
71     border: 1px solid var(--border-color);
72 }
73
74 /* Прогресс-бар: трек */
75 .progress:not([style]),
76 .progress {
77     background-color: var(--border-color) !important;
78 }
79
80 /* — Системный инспектор, трейдинг, сеть и дерево компонент
ОВ — */
81 .sys-card,
82 .sys-panel,
83 .sys-panel-header,
84 .sys-tree-node,
85 .sys-tree-body,
86 .sys-proc-table th {
87     background-color: var(--surface-1) !important;
88     border-color: var(--border-color) !important;
89     color: var(--text-color) !important;
90 }
91
92 .sys-panel-header {
93     background-color: var(--surface-2) !important;
94     color: var(--nav-active) !important;
95 }
96
97 .sys-tree-title {
98     color: var(--text-color) !important;
99     background: transparent !important;
100 }
101
102 .sys-tree-body {
103     background-color: var(--bg-color) !important;
104 }
105
106 .sys-proc-table th {
107     background-color: var(--surface-2) !important;
```



```
108     color: var(--text-muted) !important;
109 }
110
111 .sys-value {
112     color: var(--text-color) !important;
113 }
114
115 /* — Карточки агентов, моделей, плагинов — */
116 .agent-card,
117 .model-card,
118 .plugin-card,
119 [class$="-card"]:not(.btn):not(.badge) {
120     background-color: var(--surface-1) !important;
121     border-color: var(--border-color) !important;
122     color: var(--text-color) !important;
123 }
124
125 /* — card-header с тёмным фоном — */
126 .card-header.bg-dark,
127 .card-header[style*="background"],
128 .card-header[style*="#0"],
129 .card-header[style*="#1"] {
130     background-color: var(--surface-2) !important;
131     color: var(--text-color) !important;
132     border-color: var(--border-color) !important;
133 }
134
135 /* — Логи (textarea/div с моноширинным шрифтом) — */
136 #media-log,
137 [id$="-log"]:not(canvas),
138 .log-output {
139     background-color: #f1f3f5 !important;
140     color: #1e2328 !important;
141     border-color: var(--border-color) !important;
142 }
143
144 /* — Stripe-полосы в таблицах — */
145 .table-striped > tbody > tr:nth-of-type(odd) {
```

```
146 |     --bs-table-bg-type: var(--surface-2);
147 | }
148 |
149 | /* — border-secondary — */
150 | .border-secondary {
151 |     border-color: var(--border-color) !important;
152 | }
153 | }
154 |
155 | /* — Dark Theme – Повышенный контраст и доминантность описани
    | я процессов — */
156 | [data-theme="dark"] .def-desc-text,
157 | [data-theme="dark"] .process-desc,
158 | [data-theme="dark"] .table-dark td.text-secondary,
159 | [data-theme="dark"] .table td.text-secondary,
160 | .theme-defender td.text-secondary,
161 | .theme-defender .text-secondary {
162 |     color: #cbd5e1 !important; /* Высококонтрастный доминантный
    | оттенок (Slate 300) для идеальной читаемости */
163 | }
164 |
165 |
```

torrents.css

```
1  /* =====
   |=====
2  TORRENTS & MEDIA – AI Breadboard WebGUI
3  Стили управления торрентами, поиска релизов и логов медиате
   |ки
4  =====
   |===== */
5
6  /* — Таблица и статусы торрентов =====
   |===== */
7  #torrents-table th[data-col] {
8      cursor: pointer;
9      user-select: none;
10 }
11
12 .state-missingFiles { color: #dc3545; font-weight: 600; }
13 .state-uploading,
14 .state-seeding      { color: #16a34a; }
15 .state-downloading  { color: #2563eb; }
16 .state-checkingUP,
17 .state-checkingDL   { color: #ea580c; }
18
19 .progress            { height: 5px; }
20 .path-cell           { font-size: 0.78rem; color: var(--text-mu
   |ted); }
21 .path-edit           { display: none; }
22 .path-edit.active    { display: flex; }
23
24 /* — Панель отсутствующих файлов =====
   |===== */
25 #missing-panel       { border-left: 2px solid #dc3545; }
26 .missing-row         { cursor: pointer; }
27 .missing-row:hover   { background: rgba(255, 200, 0, 0.08); }
28 .missing-row.selected { background: rgba(37, 99, 235, 0.1); }
29
30 /* — Лог медиатеки =====
```

```

| _____ */
31 #media-log {
32     font-size: 0.78rem;
33     font-family: 'Consolas', monospace;
34     background: #0d1117;
35     color: #d4d4d4;
36     border-radius: 5px;
37     padding: 8px;
38     height: 160px;
39     overflow-y: auto;
40     white-space: pre-wrap;
41 }
42
43 .dup-badge      { font-size: 0.72rem; }
44 .integrity-ok   { color: #16a34a; }
45 .integrity-warn { color: #dc3545; }
46
47 /* =====
| =====
48     ТОРРЕНТ-ПОИСК
49     =====
| ===== */
50 .torrent-card {
51     border: 1px solid var(--border-color);
52     border-radius: 6px;
53     background-color: var(--surface-1);
54     padding: 10px 12px;
55     margin-bottom: 8px;
56     box-shadow: var(--shadow-xs);
57     transition: transform 0.15s, box-shadow 0.15s;
58     color: var(--card-text);
59 }
60
61 .torrent-card:hover {
62     transform: translateY(-1px);
63     box-shadow: var(--shadow-sm);
64 }
65

```

```
66 | .torrent-badge {
67 |   display: inline-block;
68 |   padding: 0.2em 0.5em;
69 |   font-size: 0.72rem;
70 |   font-weight: 600;
71 |   border-radius: 50rem;
72 |   margin-right: 3px;
73 |   margin-bottom: 3px;
74 | }
75 |
76 | .badge-rutracker { background-color: #6e7a8a; color: #fff; }
77 | .badge-nnmclub    { background-color: #16a34a; color: #fff; }
78 | .badge-seeds      { background-color: #22c55e; color: #fff; }
79 | .badge-peers      { background-color: #6b7280; color: #fff; }
80 | .badge-size       { background-color: #2563eb; color: #fff; }
81 |
82 | .torrent-title {
83 |   font-size: 0.88rem;
84 |   font-weight: 600;
85 |   margin-bottom: 4px;
86 |   color: var(--text-color);
87 | }
88 |
89 | .torrent-title a {
90 |   color: var(--nav-active);
91 |   text-decoration: none;
92 | }
93 |
94 | .torrent-title a:hover {
95 |   text-decoration: underline;
96 | }
97 |
98 | .torrent-desc {
99 |   font-size: 0.8rem;
100 |   color: var(--text-muted);
101 |   margin-bottom: 6px;
102 | }
103 |
```

variables.css

```
1  /* =====  
2  THEME VARIABLES — AI Breadboard WebGUI  
3  Переменные оформления и дизайн-токены  
4  ===== */  
5  
6  /* — Светлая тема (по умолчанию) —  
7  ===== */  
8  :root,  
9  [data-theme="light"] {  
10   /* Фоны (иерархия поверхностей) */  
11   --bg-color:          #f2f4f7;    /* страница */  
12   --surface-1:         #ffffff;    /* карточки, панели */  
13   --surface-2:         #f7f8fa;    /* хедеры карточек, тулбары */  
14   --card-bg:           #ffffff;  
15   --card-text:         #1e2328;  
16   --nav-bg:            #ffffff;  
17  
18   /* Текст */  
19   --text-color:        #1e2328;  
20   --text-muted:        #6b7280;  
21   --nav-text:          #4b5563;  
22   --nav-active:        #2563eb;  
23   --nav-hover-bg:      #eff2f6;  
24  
25   /* Линии и границы */  
26   --border-color:      #e4e6eb;  
27   --border-subtle:     #eef0f3;  
28   --divider-color:     #dde1e7;  
29   --modal-border:      #e4e6eb;  
30  
31   /* Тени (лёгкие) */  
32   --shadow-xs:         0 1px 2px rgba(0, 0, 0, 0.05);  
33   --shadow-sm:         0 1px 4px rgba(0, 0, 0, 0.07);  
34   --shadow-md:         0 4px 12px rgba(0, 0, 0, 0.09);
```

```
34 | --shadow-lg: 0 8px 24px rgba(0, 0, 0, 0.11);
35 |
36 | /* Чат */
37 | --chat-user-bg: #2563eb;
38 | --chat-user-text: #ffffff;
39 | --chat-bot-bg: #f0f2f5;
40 | --chat-bot-text: #1e2328;
41 |
42 | /* Скроллбар */
43 | --scrollbar-track: #f0f2f5;
44 | --scrollbar-thumb: #ced4da;
45 |
46 | /* Модальные окна */
47 | --modal-bg: #ffffff;
48 | --modal-text: #1e2328;
49 |
50 | /* Алерты – светлая тема */
51 | --alert-success-bg: #f0fdf4;
52 | --alert-success-border: #86efac;
53 | --alert-success-text: #166534;
54 | --alert-danger-bg: #fff1f2;
55 | --alert-danger-border: #fca5a5;
56 | --alert-danger-text: #991b1b;
57 | --alert-warning-bg: #fffbeb;
58 | --alert-warning-border: #fcd34d;
59 | --alert-warning-text: #92400e;
60 | --alert-info-bg: #eff6ff;
61 | --alert-info-border: #93c5fd;
62 | --alert-info-text: #1e40af;
63 |
64 | /* Геометрия и радиусы */
65 | --radius-xs: 4px;
66 | --radius-sm: 6px;
67 | --radius-md: 8px;
68 | --radius-lg: 12px;
69 |
70 | /* Типографика */
71 | --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
```

```
Helvetica Neue", Arial, sans-serif;
72  --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
    nsolas, "Courier New", monospace;
73
74  /* Каскадные токены KPI-карточек по умолчанию */
75  --kpi-bg:          var(--surface-1);
76  --kpi-border:      var(--border-color);
77  --kpi-accent:      var(--nav-active);
78  --kpi-min-height:  72px;
79  --kpi-padding:     0.65rem 0.9rem;
80  --kpi-val-size:    1.25rem;
81  --kpi-label-size:  0.72rem;
82  --kpi-sub-size:    0.72rem;
83
84  /* Каскадные токены панелей */
85  --panel-bg:        var(--surface-1);
86  --panel-border:    var(--border-color);
87  --panel-header-bg: var(--surface-2);
88
89  /* Каскадные токены суб-навигации */
90  --subnav-btn-padding: 0.35rem 0.75rem;
91  --subnav-font-size:  0.82rem;
92
93  /* Статусные цвета */
94  --status-success-bg:    #dcfce7;
95  --status-success-text:  #15803d;
96  --status-success-border: #86efac;
97
98  --status-danger-bg:     #fee2e2;
99  --status-danger-text:   #b91c1c;
100  --status-danger-border: #fca5a5;
101
102  --status-warning-bg:    #fef3c7;
103  --status-warning-text:  #b45309;
104  --status-warning-border: #fde047;
105
106  --status-info-bg:       #e0f2fe;
107  --status-info-text:     #0369a1;
```



```
108 | --status-info-border:    #7dd3fc;
109 | }
110 |
111 | /* — Тёмная тема —————
    | ————— */
112 | [data-theme="dark"] {
113 |     /* Фоны (иерархия поверхностей) */
114 |     --bg-color:           #0d1117;    /* страница – GitHub dark */
115 |     --surface-1:          #161b22;    /* карточки */
116 |     --surface-2:          #1c2128;    /* хедеры карточек */
117 |     --card-bg:            #161b22;
118 |     --card-text:          #c9d1d9;
119 |     --nav-bg:             #161b22;
120 |
121 |     /* Текст */
122 |     --text-color:         #c9d1d9;
123 |     --text-muted:         #8b949e;
124 |     --nav-text:           #8b949e;
125 |     --nav-active:         #58a6ff;
126 |     --nav-hover-bg:      #21262d;
127 |
128 |     /* Линии и границы */
129 |     --border-color:       #30363d;
130 |     --border-subtle:      #21262d;
131 |     --divider-color:      #30363d;
132 |     --modal-border:       #30363d;
133 |
134 |     /* Тени (глубже) */
135 |     --shadow-xs:  0 1px 2px rgba(0, 0, 0, 0.3);
136 |     --shadow-sm:  0 1px 4px rgba(0, 0, 0, 0.35);
137 |     --shadow-md:  0 4px 12px rgba(0, 0, 0, 0.45);
138 |     --shadow-lg:  0 8px 24px rgba(0, 0, 0, 0.55);
139 |
140 |     /* Чат */
141 |     --chat-user-bg:      #1f6feb;
142 |     --chat-user-text:    #ffffff;
143 |     --chat-bot-bg:       #21262d;
144 |     --chat-bot-text:     #c9d1d9;
```

```
145 |
146 | /* Скроллбар */
147 | --scrollbar-track: #0d1117;
148 | --scrollbar-thumb: #484f58;
149 |
150 | /* Модальные окна */
151 | --modal-bg: #161b22;
152 | --modal-text: #c9d1d9;
153 |
154 | /* Алерты – тёмная тема */
155 | --alert-success-bg: #132c1e;
156 | --alert-success-border: #2ea043;
157 | --alert-success-text: #7ee787;
158 | --alert-danger-bg: #2d1218;
159 | --alert-danger-border: #da3633;
160 | --alert-danger-text: #ff7b72;
161 | --alert-warning-bg: #2d2008;
162 | --alert-warning-border: #d29922;
163 | --alert-warning-text: #e3b341;
164 | --alert-info-bg: #0d1f38;
165 | --alert-info-border: #1f6feb;
166 | --alert-info-text: #79c0ff;
167 |
168 | /* Геометрия и радиусы */
169 | --radius-xs: 4px;
170 | --radius-sm: 6px;
171 | --radius-md: 8px;
172 | --radius-lg: 12px;
173 |
174 | /* Типографика */
175 | --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
176 | Helvetica Neue", Arial, sans-serif;
177 | --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
178 | nsolas, "Courier New", monospace;
179 |
180 | /* Каскадные токены KPI-карточек */
181 | --kpi-bg: var(--surface-1);
182 | --kpi-border: var(--border-color);
```

```
181 | --kpi-accent:      var(--nav-active);
182 | --kpi-min-height: 72px;
183 | --kpi-padding:     0.65rem 0.9rem;
184 | --kpi-val-size:    1.25rem;
185 | --kpi-label-size:  0.72rem;
186 | --kpi-sub-size:    0.72rem;
187 |
188 | /* Каскадные токены панелей */
189 | --panel-bg:        var(--surface-1);
190 | --panel-border:    var(--border-color);
191 | --panel-header-bg: var(--surface-2);
192 |
193 | /* Каскадные токены суб-навигации */
194 | --subnav-btn-padding: 0.35rem 0.75rem;
195 | --subnav-font-size:  0.82rem;
196 |
197 | /* Статусные цвета (темная тема) */
198 | --status-success-bg:    #14532d;
199 | --status-success-text:  #86efac;
200 | --status-success-border: #22c55e;
201 |
202 | --status-danger-bg:     #7f1d1d;
203 | --status-danger-text:   #fca5a5;
204 | --status-danger-border: #ef4444;
205 |
206 | --status-warning-bg:    #78350f;
207 | --status-warning-text:  #fde047;
208 | --status-warning-border: #eab308;
209 |
210 | --status-info-bg:       #0c4a6e;
211 | --status-info-text:     #7dd3fc;
212 | --status-info-border:   #0284c7;
213 | }
214 |
```

index.html

```
1 <!-- Microsoft Defender & AI Security Diagnostic Center Tab -->
2 <style>
3   .def-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .def-badge-active { background: var(--status-success-bg); color: var(--status-success-text); border: 1px solid var(--status-success-border); }
7   .def-badge-disabled { background: var(--status-danger-bg); color: var(--status-danger-text); border: 1px solid var(--status-danger-border); }
8   .def-badge-audit { background: var(--status-warning-bg); color: var(--status-warning-text); border: 1px solid var(--status-warning-border); }
9   .def-badge-warn { background: var(--status-warning-bg); color: var(--status-warning-text); border: 1px solid var(--status-warning-border); }
10  .def-badge-critical { background: var(--status-danger-bg); color: var(--status-danger-text); border: 1px solid var(--status-danger-border); font-weight: bold; }
11  /* Доминантный цвет описания процессов и эшелонов в темной теме */
12  .def-desc-text,
13  .theme-defender td.text-secondary,
14  .theme-defender .table td.text-secondary {
15    color: #cbd5e1 !important;
16  }
17 </style>
18
19 <div class="def-container theme-defender">
20   <!-- Top Stat Cards -->
21   <div class="row g-2 mb-2">
22     <div class="col-6 col-md-3 col-xl">
23       <div class="def-card">
24         <div class="def-card-label"><span>Индекс защиты</span>
```

```

    </div>
25     <div class="def-card-value" id="def-score-val">--/100<
    /div>
26     <div class="def-card-sub" id="def-score-sub">Загрузка.
    ..</div>
27     </div>
28 </div>
29 <div class="col-6 col-md-3 col-xl">
30     <div class="def-card">
31         <div class="def-card-label"><span>Real-time Shield</sp
    an> </div>
32         <div class="def-card-value text-success" id="def-realt
    ime-val">--</div>
33         <div class="def-card-sub" id="def-realtime-sub">MsMpEn
    g.exe</div>
34     </div>
35 </div>
36 <div class="col-6 col-md-3 col-xl">
37     <div class="def-card">
38         <div class="def-card-label"><span>Cloud & Tamper</span
    > </div>
39         <div class="def-card-value" id="def-cloud-val">--</div
    >
40         <div class="def-card-sub" id="def-cloud-sub">MAPS Prot
    ection</div>
41     </div>
42 </div>
43 <div class="col-6 col-md-3 col-xl">
44     <div class="def-card">
45         <div class="def-card-label"><span>PUA & Ransomware</sp
    an> </div>
46         <div class="def-card-value" id="def-pua-cfa-val">--</d
    iv>
47         <div class="def-card-sub" id="def-pua-cfa-sub">CFA Fol
    der Guard</div>
48     </div>
49 </div>
50 <div class="col-6 col-md-3 col-xl">

```

```

51     <div class="def-card">
52         <div class="def-card-label"><span>Исключения</span> <
53     /div>
54         <div class="def-card-value text-warning" id="def-exclu
55     sions-val">0</div>
56         <div class="def-card-sub" id="def-exclusions-sub">0 по
57     дозрительных</div>
58     </div>
59     </div>
60     <div class="col-6 col-md-3 col-xl">
61         <div class="def-card">
62             <div class="def-card-label"><span>Активные угрозы</spa
63             n> </div>
64             <div class="def-card-value text-danger" id="def-threat
65             s-val">0</div>
66             <div class="def-card-sub" id="def-threats-sub">Каранти
67             н: 0</div>
68         </div>
69     </div>
70 </div>
71 <!-- Action Bar -->
72 <div class="def-panel mb-2 p-2 d-flex align-items-center jus
73     tify-content-between flex-wrap gap-2">
74     <div class="d-flex align-items-center gap-2 flex-wrap">
75         <button class="btn btn-sm btn-primary d-flex align-items
76         -center gap-1" id="btn-def-quick-scan">
77             <i class="bi bi-play-circle-fill"></i> <span>Быстрое с
78             канирование</span>
79         </button>
80         <button class="btn btn-sm btn-outline-primary d-flex ali
81         gn-items-center gap-1" id="btn-def-full-scan">
82             <i class="bi bi-shield-shaded"></i> <span>Полное скани
83             рование</span>
84         </button>
85         <button class="btn btn-sm btn-outline-info d-flex align-
86         items-center gap-1" id="btn-def-update-sigs">
87             <i class="bi bi-cloud-arrow-down-fill"></i> <span>Обно

```

```

    вить сигнатуры</span>
77     </button>
78     <button class="btn btn-sm btn-outline-warning d-flex ali
    gn-items-center gap-1" id="btn-def-ai-diag">
79         <i class="bi bi-cpu-fill"></i> <span>AI Диагностика</s
    pan>
80     </button>
81 </div>
82 <div class="d-flex align-items-center gap-2">
83     <span class="badge bg-dark border border-secondary" id="
    def-engine-ver-badge">Движок: --</span>
84     <span class="badge bg-dark border border-secondary" id="
    def-sigs-ver-badge">Базы: --</span>
85     <button class="btn btn-sm btn-outline-secondary" id="btn
    -def-refresh" title="Обновить">
86         <i class="bi bi-arrow-clockwise"></i>
87     </button>
88 </div>
89 </div>
90
91 <!-- Scan Execution Progress / Notification Alert -->
92 <div id="def-alert-box" class="d-none alert alert-info py-2
    px-3 mb-2 small"></div>
93
94 <!-- Sub Navigation Tabs -->
95 <ul class="nav nav-tabs mb-2 border-secondary" id="defSubTab
    s" role="tablist">
96     <li class="nav-item" role="presentation">
97         <button class="nav-link active py-1 px-3 small" id="def-
    tab-overview-btn" data-bs-toggle="tab" data-bs-target="#def-pa
    ne-overview" type="button">🔍 Обзор и Службы</button>
98     </li>
99     <li class="nav-item" role="presentation">
100         <button class="nav-link py-1 px-3 small" id="def-tab-asr
    -btn" data-bs-toggle="tab" data-bs-target="#def-pane-asr" type
    ="button">📋 Правила ASR</button>
101     </li>
102     <li class="nav-item" role="presentation">

```

```
103 |         <button class="nav-link py-1 px-3 small" id="def-tab-cfa
      | -btn" data-bs-toggle="tab" data-bs-target="#def-pane-cfa" type
      | ="button"> Ransomware / CFA</button>
104 |     </li>
105 |     <li class="nav-item" role="presentation">
106 |         <button class="nav-link py-1 px-3 small" id="def-tab-exc
      | lusions-btn" data-bs-toggle="tab" data-bs-target="#def-pane-ex
      | clusions" type="button"> Искключения</button>
107 |     </li>
108 |     <li class="nav-item" role="presentation">
109 |         <button class="nav-link py-1 px-3 small" id="def-tab-thr
      | eats-btn" data-bs-toggle="tab" data-bs-target="#def-pane-threa
      | ts" type="button"> Угрозы и Карантин</button>
110 |     </li>
111 |     <li class="nav-item" role="presentation">
112 |         <button class="nav-link py-1 px-3 small" id="def-tab-pro
      | cs-btn" data-bs-toggle="tab" data-bs-target="#def-pane-procs"
      | type="button"> Дерево Процессов / Fileless</button>
113 |     </li>
114 |     <li class="nav-item" role="presentation">
115 |         <button class="nav-link py-1 px-3 small" id="def-tab-eve
      | nts-btn" data-bs-toggle="tab" data-bs-target="#def-pane-events
      | " type="button"> Журнал Событий</button>
116 |     </li>
117 |     <li class="nav-item" role="presentation">
118 |         <button class="nav-link py-1 px-3 small text-warning fw-
      | semibold" id="def-tab-diag-btn" data-bs-toggle="tab" data-bs-t
      | arget="#def-pane-diag" type="button"> AI Диагностика</button
      | >
119 |     </li>
120 | </ul>
121 |
122 | <!-- Tab Content Panes -->
123 | <div class="tab-content" id="defTabPanes">
124 |
125 |     <!-- 1. Overview & Services Pane -->
126 |     <div class="tab-pane fade show active" id="def-pane-overvi
      | ew" role="tabpanel">
```



```

127 |     <div class="row g-2">
128 |         <div class="col-md-7">
129 |             <div class="def-panel h-100">
130 |                 <div class="def-panel-header">
131 |                     <span>Эшелоны Защиты Microsoft Defender</span>
132 |                 </div>
133 |                 <div class="table-responsive p-0">
134 |                     <table class="table table-dark table-sm table-ho
ver mb-0 align-middle small">
135 |                         <thead>
136 |                             <tr class="text-secondary">
137 |                                 <th>Компонент</th>
138 |                                 <th class="text-center">Статус</th>
139 |                                 <th>Описание</th>
140 |                             </tr>
141 |                         </thead>
142 |                         <tbody id="def-shields-tbody">
143 |                             <tr><td colspan="3" class="text-center text-
muted py-3">Загрузка данных...</td></tr>
144 |                             </tbody>
145 |                         </table>
146 |                     </div>
147 |                 </div>
148 |             </div>
149 |             <div class="col-md-5">
150 |                 <div class="def-panel h-100">
151 |                     <div class="def-panel-header">
152 |                         <span>Системные Службы и Процессы</span>
153 |                     </div>
154 |                     <div class="table-responsive p-0">
155 |                         <table class="table table-dark table-sm table-ho
ver mb-0 align-middle small">
156 |                             <thead>
157 |                                 <tr class="text-secondary">
158 |                                     <th>Процесс</th>
159 |                                     <th class="text-center">Статус</th>
160 |                                     <th class="text-end">PID</th>
161 |                                     <th class="text-end">Память</th>

```

```

162         </tr>
163     </thead>
164     <tbody id="def-services-tbody">
165         <tr><td colspan="4" class="text-center text-
166 muted py-3">Загрузка данных...</td></tr>
167     </tbody>
168 </table>
169 </div>
170 </div>
171 </div>
172 </div>
173
174 <!-- 2. ASR Rules Pane -->
175 <div class="tab-pane fade" id="def-pane-asr" role="tabpane
176 l">
177     <div class="def-panel">
178         <div class="def-panel-header">
179             <span>Правила Attack Surface Reduction (ASR)</span>
180             <span class="badge bg-secondary" id="def-asr-count-b
181 adge">Всего: 0</span>
182         </div>
183         <div class="table-responsive p-0">
184             <table class="table table-dark table-sm table-hover
185 mb-0 align-middle small">
186                 <thead>
187                     <tr class="text-secondary">
188                         <th>Правило ASR</th>
189                         <th>Категория</th>
190                         <th class="text-center">Состояние</th>
191                         <th>Рекомендация</th>
192                         <th>GUID</th>
193                     </tr>
194                 </thead>
195                 <tbody id="def-asr-tbody">
196                     <tr><td colspan="5" class="text-center text-mute
197 d py-3">Загрузка правил ASR...</td></tr>
198                 </tbody>

```

```

195         </table>
196     </div>
197 </div>
198 </div>
199
200 <!-- 3. CFA (Ransomware) Pane -->
201 <div class="tab-pane fade" id="def-pane-cfa" role="tabpanel
202     1">
203     <div class="row g-2">
204         <div class="col-md-6">
205             <div class="def-panel h-100">
206                 <div class="def-panel-header">
207                     <span>Защищенные Каталоги (Protected Folders)</s
208                 pan>
209                 </div>
210                 <div class="p-2" id="def-cfa-folders-list">
211                     <div class="text-muted small">Загрузка...</div>
212                 </div>
213             </div>
214             <div class="col-md-6">
215                 <div class="def-panel h-100">
216                     <div class="def-panel-header">
217                         <span>Доверенные Приложения (Allowed Apps)</span
218                     >
219                     </div>
220                     <div class="p-2" id="def-cfa-apps-list">
221                         <div class="text-muted small">Загрузка...</div>
222                     </div>
223                 </div>
224             </div>
225         </div>
226     </div>
227 <!-- 4. Exclusions Security Audit Pane -->
228 <div class="tab-pane fade" id="def-pane-exclusions" role="
229     tabpanel">
230     <div class="def-panel">

```

```

229 |         <div class="def-panel-header">
230 |             <span>Аудит Исключений из Проверки</span>
231 |             <span class="badge bg-warning text-dark" id="def-exc
-summary-badge">0 исключений</span>
232 |         </div>
233 |         <div class="p-2 mb-1 bg-dark text-muted small border-b
ottom border-secondary" id="def-exc-recommendation-box">
234 |             Анализ исключений...
235 |         </div>
236 |         <div class="table-responsive p-0">
237 |             <table class="table table-dark table-sm table-hover
mb-0 align-middle small">
238 |                 <thead>
239 |                     <tr class="text-secondary">
240 |                         <th>Тип</th>
241 |                         <th>Значение</th>
242 |                         <th class="text-center">Уровень Риска</th>
243 |                         <th>Обоснование</th>
244 |                     </tr>
245 |                 </thead>
246 |                 <tbody id="def-exclusions-tbody">
247 |                     <tr><td colspan="4" class="text-center text-mute
d py-3">Загрузка исключений...</td></tr>
248 |                 </tbody>
249 |             </table>
250 |         </div>
251 |     </div>
252 | </div>
253 |
254 | <!-- 5. Threats History Pane -->
255 | <div class="tab-pane fade" id="def-pane-threats" role="tab
panel">
256 |     <div class="def-panel">
257 |         <div class="def-panel-header">
258 |             <span>Журнал Обнаружения Угроз и Карантин</span>
259 |             <span class="badge bg-danger" id="def-threats-count-
badge">0 записей</span>
260 |         </div>

```

```

261         <div class="table-responsive p-0">
262             <table class="table table-dark table-sm table-hover
mb-0 align-middle small">
263                 <thead>
264                     <tr class="text-secondary">
265                         <th>Угроза</th>
266                         <th>Категория</th>
267                         <th class="text-center">Опасность</th>
268                         <th>Статус</th>
269                         <th>Время</th>
270                         <th>Ресурсы</th>
271                     </tr>
272                 </thead>
273                 <tbody id="def-threats-tbody">
274                     <tr><td colspan="6" class="text-center text-mute
d py-3">Угроз не обнаружено.</td></tr>
275                 </tbody>
276             </table>
277         </div>
278     </div>
279 </div>
280
281 <!-- 6. Process Trees / Fileless Activity Pane -->
282 <div class="tab-pane fade" id="def-pane-procs" role="tabpa
nel">
283     <div class="def-panel">
284         <div class="def-panel-header">
285             <span>Инспекция Цепочек Процессов и Индикаторы Filel
ess-атак</span>
286             <span class="badge bg-secondary" id="def-procs-count
-badge">0 аномалий</span>
287         </div>
288         <div class="table-responsive p-0">
289             <table class="table table-dark table-sm table-hover
mb-0 align-middle small">
290                 <thead>
291                     <tr class="text-secondary">
292                         <th>Родительский Процесс</th>

```

```

293 |         <th>Дочерний Процесс</th>
294 |         <th>PID</th>
295 |         <th>Опасность</th>
296 |         <th>Причина срабатывания</th>
297 |     </tr>
298 | </thead>
299 | <tbody id="def-procs-tbody">
300 |     <tr><td colspan="5" class="text-center text-mute
    d py-3">Подозрительных цепочек процессов не обнаружено.</td></
    tr>
301 |     </tbody>
302 | </table>
303 | </div>
304 | </div>
305 | </div>
306 |
307 | <!-- 7. Operational Events Pane -->
308 | <div class="tab-pane fade" id="def-pane-events" role="tabp
    anel">
309 |     <div class="def-panel">
310 |         <div class="def-panel-header">
311 |             <span>Журнал Microsoft-Windows-Windows Defender/Oper
    ational</span>
312 |             <span class="badge bg-secondary" id="def-events-coun
    t-badge">0 событий</span>
313 |         </div>
314 |         <div class="table-responsive p-0" style="max-height: 4
    80px; overflow-y: auto;">
315 |             <table class="table table-dark table-sm table-hover
    mb-0 align-middle small">
316 |                 <thead>
317 |                     <tr class="text-secondary">
318 |                         <th>Event ID</th>
319 |                         <th>Время</th>
320 |                         <th>Уровень</th>
321 |                         <th>Категория</th>
322 |                         <th>Сообщение</th>
323 |                     </tr>

```

```

324 |         </thead>
325 |         <tbody id="def-events-tbody">
326 |             <tr><td colspan="5" class="text-center text-mute
| d py-3">Загрузка журнала событий...</td></tr>
327 |             </tbody>
328 |         </table>
329 |     </div>
330 | </div>
331 | </div>
332 |
333 | <!-- 8. AI Diagnostics Pane -->
334 | <div class="tab-pane fade" id="def-pane-diag" role="tabpane
| el">
335 |     <div class="row g-2">
336 |         <div class="col-md-4">
337 |             <div class="def-panel h-100 p-3 text-center d-flex f
| lex-column justify-content-center">
338 |                 <div class="text-secondary small mb-1">Security Po
| sture Score</div>
339 |                 <div class="display-4 fw-bold text-success" id="de
| f-diag-score-display">--</div>
340 |                 <div class="fw-semibold mt-2" id="def-diag-summary
| -display">Анализ состояния...</div>
341 |             </div>
342 |         </div>
343 |         <div class="col-md-8">
344 |             <div class="def-panel h-100 p-3">
345 |                 <h6 class="text-warning fw-bold mb-2">⚠ Рекоменда
| ции AI по Усилению Защиты:</h6>
346 |                 <div id="def-diag-recs-container" class="small">
347 |                     <div class="text-muted">Загрузка рекомендаций...
| </div>
348 |                 </div>
349 |                 <hr class="border-secondary my-2">
350 |                 <h6 class="text-danger fw-bold mb-2">⚠ Критические
| уязвимости и риски:</h6>
351 |                 <div id="def-diag-criticals-container" class="small
| l">

```

```
352 |           <div class="text-muted">Нет критических уязвимос  
    | теЙ.</div>  
353 |           </div>  
354 |           </div>  
355 |           </div>  
356 |           </div>  
357 |       </div>  
358 |  
359 |   </div>  
360 </div>  
361 |
```


main.js

```
1 // Microsoft Defender Security Center Frontend Controller
2 (function() {
3   'use strict';
4
5   let defenderData = null;
6
7   function showAlert(msg, isError = false) {
8     const box = document.getElementById('def-alert-box');
9     if (!box) return;
10    box.className = `alert py-2 px-3 mb-2 small ${isError ? 'a
    lert-danger' : 'alert-info'}`;
11    box.textContent = msg;
12    box.classList.remove('d-none');
13    setTimeout(() => {
14      box.classList.add('d-none');
15    }, 8000);
16  }
17
18  async function fetchJSON(url, options = {}) {
19    if (window.api && typeof window.api.fetch === 'function')
20    {
21      return await window.api.fetch(url, options);
22    }
23    const res = await fetch(url, options);
24    if (!res.ok) {
25      throw new Error(`HTTP ${res.status}: ${res.statusText}`)
26    };
27    return await res.json();
28  }
29
30  async function loadDefenderStatus() {
31    try {
32      const status = await fetchJSON('/api/v1/defender/status'
33    );
34      defenderData = status;
```

```

33
34     // Badges in header
35     const engBadge = document.getElementById('def-engine-ver
-badge');
36     if (engBadge) engBadge.textContent = `Движок: ${status.e
ngine_version || 'N/A'}`;
37     const sigsBadge = document.getElementById('def-sigs-ver-
badge');
38     if (sigsBadge) sigsBadge.textContent = `Базы: ${status.a
ntivirus_signature_version || 'N/A'}`;
39
40     // Stat Cards
41     const rtVal = document.getElementById('def-realtime-val'
);
42     if (rtVal) {
43         rtVal.textContent = status.real_time_protection_enable
d ? 'ВКЛ' : 'ВЫКЛ';
44         rtVal.className = `def-card-value ${status.real_time_p
rotection_enabled ? 'text-success' : 'text-danger'}`;
45     }
46
47     const cloudVal = document.getElementById('def-cloud-val'
);
48     if (cloudVal) {
49         cloudVal.textContent = status.cloud_protection_enabled
? 'АКТИВНА' : 'ВЫКЛ';
50         cloudVal.className = `def-card-value ${status.cloud_pr
otection_enabled ? 'text-success' : 'text-warning'}`;
51     }
52
53     const puaCfaVal = document.getElementById('def-pua-cfa-v
al');
54     if (puaCfaVal) {
55         const puaText = status.pua_protection_enabled ? 'PUA:
' : 'PUA: ';
56         const cfaText = status.controlled_folder_access_enable
d ? 'CFA: ' : 'CFA: ';
57         puaCfaVal.textContent = `${puaText} | ${cfaText}`;

```

```
58 |         puaCfaVal.className = 'def-card-value small text-info'
    | ;
59 |     }
60 |
61 |     // Shields Table
62 |     const shieldsTbody = document.getElementById('def-shield
    | s-tbody');
63 |     if (shieldsTbody) {
64 |         const shields = [
65 |             { name: 'Real-time Protection', enabled: status.real
    | _time_protection_enabled, desc: 'Постоянный перехват и проверк
    | а файлов при создании и обращении' },
66 |             { name: 'Cloud Protection (MAPS)', enabled: status.c
    | loud_protection_enabled, desc: `Мгновенная блокировка новых уг
    | роз через облачный ИИ Microsoft (${status.cloud_block_level})`
    | },
67 |             { name: 'Behavior Monitoring', enabled: status.behav
    | ior_monitor_enabled, desc: 'Анализ поведения запущенных процес
    | сов и цепочек активности' },
68 |             { name: 'IOAV Protection', enabled: status.ioav_prot
    | ection_enabled, desc: 'Проверка файлов, загружаемых из браузер
    | а, почты и мессенджеров' },
69 |             { name: 'AMSI Script Scanning', enabled: status.scri
    | pt_scanning_enabled, desc: 'Инспекция скриптов PowerShell, VBS
    | cript, JavaScript до их выполнения' },
70 |             { name: 'Tamper Protection', enabled: status.tamper_
    | protection_enabled, desc: 'Защита настроек антивируса от отклю
    | чения вредоносным ПО' },
71 |             { name: 'PUA Protection', enabled: status.pua_prot
    | ection_enabled, desc: 'Блокировка нежелательного ПО, рекламных и
    | нсталляторов и майнеров' },
72 |             { name: 'Controlled Folder Access', enabled: status.
    | controlled_folder_access_enabled, desc: 'Защита папок Документ
    | ы/Рабочий стол от шифрования Ransomware' },
73 |             { name: 'Network Protection', enabled: status.networ
    | k_protection_enabled, desc: 'Блокировка сетевых соединений с в
    | редоносными доменами и IP' },
74 |         ];
```

```

75 |
76 |     shieldsTbody.innerHTML = shields.map(s => `
77 |         <tr>
78 |             <td class="fw-semibold text-light">${s.name}</td>
79 |             <td class="text-center">
80 |                 <span class="badge ${s.enabled ? 'def-badge-acti
| ve' : 'def-badge-disabled'}">
81 |                     ${s.enabled ? 'ВКЛЮЧЕНО' : 'ОТКЛЮЧЕНО'}
82 |                 </span>
83 |             </td>
84 |             <td class="def-desc-text">${s.desc}</td>
85 |         </tr>
86 |     `).join('');
87 | }
88 |
89 | // Services Table
90 | const servicesTbody = document.getElementById('def-servi
| ces-tbody');
91 | if (servicesTbody && status.services) {
92 |     servicesTbody.innerHTML = status.services.map(svc => `
93 |         <tr>
94 |             <td>
95 |                 <div class="fw-semibold text-light">${svc.name}<
| /div>
96 |                 <div class="def-desc-text" style="font-size: 0.7
| 2rem;">${svc.display_name}</div>
97 |             </td>
98 |             <td class="text-center">
99 |                 <span class="badge ${svc.running ? 'bg-success'
| : 'bg-secondary'}">
100 |                     ${svc.running ? 'АКТИВЕН' : 'СТОП'}
101 |                 </span>
102 |             </td>
103 |             <td class="text-end text-secondary">${svc.pid || '
| -'}</td>
104 |             <td class="text-end font-monospace">${svc.running
| ? svc.memory_mb + ' MB' : '-'}</td>
105 |         </tr>

```

```

106 |         `).join('');
107 |     }
108 |
109 |     } catch (err) {
110 |         console.error('[DefenderTab] Error loading status:', err
111 | );
112 |         showAlert(`Ошибка загрузки статуса Defender: ${err.messa
113 | ge}`, true);
114 |     }
115 | }
116 |
117 | async function loadASRRules() {
118 |     try {
119 |         const rules = await fetchJSON('/api/v1/defender/asr');
120 |         const tbody = document.getElementById('def-asr-tbody');
121 |         const badge = document.getElementById('def-asr-count-bad
122 | ge');
123 |         if (badge) badge.textContent = `Bcero: ${rules.length}`;
124 |
125 |         if (tbody) {
126 |             tbody.innerHTML = rules.map(r => {
127 |                 let stateBadge = '<span class="badge bg-secondary">H
128 | е задано</span>';
129 |                 if (r.state === 'enabled') stateBadge = '<span class
130 | ="badge def-badge-active">Блокировка (Block)</span>';
131 |                 else if (r.state === 'audit') stateBadge = '<span cl
132 | ass="badge def-badge-audit">Аудит (Audit)</span>';
133 |                 else if (r.state === 'warn') stateBadge = '<span cla
134 | ss="badge def-badge-warn">Предупреждение</span>';
135 |                 else if (r.state === 'disabled') stateBadge = '<span
136 | class="badge def-badge-disabled">Отключено</span>';
137 |
138 |                 return `
139 |                     <tr>
140 |                         <td>
141 |                             <div class="fw-semibold text-light">${r.name}<
142 | /div>
143 |                             <div class="def-desc-text" style="font-size: 0

```

```

        .72rem;">${r.description}</div>
135         </td>
136         <td><span class="badge bg-dark border border-sec
ondary">${r.category}</span></td>
137         <td class="text-center">${stateBadge}</td>
138         <td class="def-desc-text small">${r.recommendati
on || '-'}</td>
139         <td class="text-secondary font-monospace" style=
"font-size: 0.68rem;">${r.guid}</td>
140     </tr>
141     `;
142     }).join('');
143     }
144     } catch (err) {
145         console.error('[DefenderTab] Error loading ASR rules:',
err);
146     }
147 }
148
149 async function loadCFA() {
150     try {
151         const cfa = await fetchJSON('/api/v1/defender/cfa');
152         const foldersList = document.getElementById('def-cfa-fol
ders-list');
153         const appsList = document.getElementById('def-cfa-apps-l
ist');
154
155         if (foldersList) {
156             if (cfa.protected_folders && cfa.protected_folders.len
gth > 0) {
157                 foldersList.innerHTML = `
158                 <ul class="list-group list-group-flush bg-transpar
ent">
159                     ${cfa.protected_folders.map(f => `
160                     <li class="list-group-item bg-transparent text
-light border-secondary py-1 px-0 font-monospace small">
161                         □ ${f}
162                     </li>

```

```

163         `).join(''))}
164     </ul>
165     `;
166     } else {
167         foldersList.innerHTML = '<div class="text-muted small
168 l">Нет настроенных защищенных папок.</div>';
169     }
170
171     if (appsList) {
172         if (cfa.allowed_applications && cfa.allowed_applicatio
173 ns.length > 0) {
174             appsList.innerHTML = `
175                 <ul class="list-group list-group-flush bg-transpar
176 ent">
177                     ${cfa.allowed_applications.map(a => `
178                         <li class="list-group-item bg-transparent text
179 -light border-secondary py-1 px-0 font-monospace small">
180                             □ ${a}
181                         </li>
182                     `).join(''))}
183                 </ul>
184             `;
185         } else {
186             appsList.innerHTML = '<div class="text-muted small">
187 Список доверенных приложений пуст.</div>';
188         }
189     }
190
191     async function loadExclusions() {
192         try {
193             const rep = await fetchJSON('/api/v1/defender/exclusions
194 ');

```

```
195     const countVal = document.getElementById('def-exclusions
| -val');
196     const countSub = document.getElementById('def-exclusions
| -sub');
197     const summaryBadge = document.getElementById('def-exc-su
| mmary-badge');
198     const recBox = document.getElementById('def-exc-recommen
| dation-box');
199     const tbody = document.getElementById('def-exclusions-tb
| ody');
200
201     if (countVal) countVal.textContent = rep.total_exclusion
| s;
202     if (countSub) countSub.textContent = `${rep.suspicious_c
| ount} подозрительных`;
203     if (summaryBadge) summaryBadge.textContent = `${rep.tota
| l_exclusions} исключений (${rep.suspicious_count} риск)`;
204     if (recBox) recBox.textContent = rep.summary_recommenda
| tion;
205
206     const allItems = [...rep.path_exclusions, ...rep.extensi
| on_exclusions, ...rep.process_exclusions];
207
208     if (tbody) {
209         if (allItems.length === 0) {
210             tbody.innerHTML = '<tr><td colspan="4" class="text-c
| enter text-muted py-3">Исключения отсутствуют. Защита работает
| без пропусков.</td></tr>';
211         } else {
212             tbody.innerHTML = allItems.map(itm => {
213                 let riskBadge = '<span class="badge bg-success">Бе
| зопасно</span>';
214                 if (itm.risk_level === 'critical') riskBadge = '<s
| pan class="badge def-badge-critical">КРИТИЧЕСКИЙ</span>';
215                 else if (itm.risk_level === 'high') riskBadge = '<
| span class="badge def-badge-disabled">ВЫСОКИЙ</span>';
216                 else if (itm.risk_level === 'medium') riskBadge =
| '<span class="badge def-badge-warn">СРЕДНИЙ</span>';
```



```

217 |
218 |         return `
219 |             <tr>
220 |                 <td><span class="badge bg-dark border border-s
221 | econdary">${itm.type.toUpperCase()}</span></td>
222 |                 <td class="font-monospace text-light">${itm.va
223 | lue}</td>
224 |                 <td class="text-center">${riskBadge}</td>
225 |                 <td class="text-secondary small">${itm.risk_re
226 | ason}</td>
227 |             </tr>
228 |         `;
229 |     }).join('');
230 | }
231 | }
232 | }
233 |
234 | async function loadThreats() {
235 |     try {
236 |         const threats = await fetchJSON('/api/v1/defender/threat
237 | s?limit=50');
238 |         const countBadge = document.getElementById('def-threats-
239 | count-badge');
240 |         const countVal = document.getElementById('def-threats-va
241 | l');
242 |         const countSub = document.getElementById('def-threats-su
243 | b');
244 |         const tbody = document.getElementById('def-threats-tbody
245 | ');
246 |
247 |         if (countBadge) countBadge.textContent = `${threats.leng
248 | th} записей`;
249 |         if (countVal) countVal.textContent = threats.length;
250 |     } catch (err) {
251 |         console.error('[DefenderTab] Error loading exclusions:',
252 | err);
253 |     }
254 | }

```

```

245     const quarantined = threats.filter(t => t.status.toLowerCase
Case().includes('quarantine')).length;
246     if (countSub) countSub.textContent = `Карантин: ${quaran
tined}`;
247
248     if (tbody) {
249         if (threats.length === 0) {
250             tbody.innerHTML = '<tr><td colspan="6" class="text-c
enter text-muted py-3">Активных и архивных угроз не зафиксиров
ано.</td></tr>';
251         } else {
252             tbody.innerHTML = threats.map(t => `
253                 <tr>
254                     <td class="fw-semibold text-danger">${t.threat_n
ame}</td>
255                     <td><span class="badge bg-dark border border-sec
ondary">${t.category}</span></td>
256                     <td class="text-center"><span class="badge bg-wa
rning text-dark">${t.severity.toUpperCase()}</span></td>
257                     <td><span class="badge bg-success">${t.status}</
span></td>
258                     <td class="text-secondary small">${t.last_detect
ion_time || t.initial_detection_time || '-'}</td>
259                     <td class="text-secondary font-monospace small">
${(t.resources || []).join(', ')}</td>
260                 </tr>
261             `).join('');
262         }
263     }
264 } catch (err) {
265     console.error('[DefenderTab] Error loading threats:', er
r);
266 }
267 }
268
269 async function loadProcesses() {
270     try {
271         const chains = await fetchJSON('/api/v1/defender/process

```

```

    -tree');
272     const badge = document.getElementById('def-procs-count-b
    adge');
273     const tbody = document.getElementById('def-procs-tbody')
    ;
274
275     if (badge) badge.textContent = `${chains.length} аномали
    й`;
276
277     if (tbody) {
278         if (chains.length === 0) {
279             tbody.innerHTML = '<tr><td colspan="5" class="text-c
    enter text-muted py-3">Подозрительных связей процессов и filel
    ess индикаторов не обнаружено.</td></tr>';
280         } else {
281             tbody.innerHTML = chains.map(c => `
282                 <tr>
283                     <td class="font-monospace text-warning">${c.pare
    nt_process} (PID: ${c.parent_pid || '-'})</td>
284                     <td class="font-monospace text-danger fw-semibol
    d">${c.child_process}</td>
285                     <td class="text-center text-secondary">${c.child
    _pid || '-'}</td>
286                     <td class="text-center"><span class="badge def-b
    adge-critical">${c.severity.toUpperCase()}</span></td>
287                     <td class="text-secondary small">
288                         <div>${c.reason}</div>
289                         ${c.command_line ? `<div class="font-monospace
    text-muted mt-1" style="font-size: 0.68rem;">cmd: ${c.command
    _line}</div>` : ''}
290                     </td>
291                 </tr>
292             `).join('');
293         }
294     }
295     } catch (err) {
296         console.error('[DefenderTab] Error loading process chain
    s:', err);

```

```

297 |     }
298 | }
299 |
300 | async function loadEvents() {
301 |     try {
302 |         const events = await fetchJSON('/api/v1/defender/events?
limit=50');
303 |         const badge = document.getElementById('def-events-count-
badge');
304 |         const tbody = document.getElementById('def-events-tbody'
);
305 |
306 |         if (badge) badge.textContent = `${events.length} событий
`;
307 |
308 |         if (tbody) {
309 |             if (events.length === 0) {
310 |                 tbody.innerHTML = '<tr><td colspan="5" class="text-c
enter text-muted py-3">Событий в журнале Defender Operational
не обнаружено.</td></tr>';
311 |             } else {
312 |                 tbody.innerHTML = events.map(e => `
313 |                     <tr>
314 |                         <td class="font-monospace fw-semibold text-info"
>${e.event_id}</td>
315 |                         <td class="text-secondary small font-monospace">
${e.timestamp}</td>
316 |                         <td><span class="badge ${e.level.toLowerCase().i
ncludes('err') || e.level.toLowerCase().includes('crit') ? 'bg
-danger' : (e.level.toLowerCase().includes('warn') ? 'bg-warni
ng text-dark' : 'bg-secondary')}>${e.level}</span></td>
317 |                         <td><span class="badge bg-dark border border-sec
ondary">${e.category}</span></td>
318 |                         <td class="text-light small">${e.message}</td>
319 |                     </tr>
320 |                 `).join('');
321 |             }
322 |         }

```

```
323     } catch (err) {
324         console.error('[DefenderTab] Error loading events:', err
    );
325     }
326 }
327
328 async function loadDiagnostics() {
329     try {
330         const rep = await fetchJSON('/api/v1/defender/diagnostic
    s');
331
332         const scoreVal = document.getElementById('def-score-val'
    );
333         const scoreSub = document.getElementById('def-score-sub'
    );
334         const diagScoreDisplay = document.getElementById('def-di
    ag-score-display');
335         const diagSummaryDisplay = document.getElementById('def-
    diag-summary-display');
336         const recsContainer = document.getElementById('def-diag-
    recs-container');
337         const critsContainer = document.getElementById('def-diag
    -criticals-container');
338
339         const scoreColor = rep.security_score >= 80 ? 'text-succ
    ess' : (rep.security_score >= 50 ? 'text-warning' : 'text-dang
    er');
340
341         if (scoreVal) {
342             scoreVal.textContent = `${rep.security_score}/100`;
343             scoreVal.className = `def-card-value ${scoreColor}`;
344         }
345         if (scoreSub) scoreSub.textContent = rep.status_summary;
346
347         if (diagScoreDisplay) {
348             diagScoreDisplay.textContent = `${rep.security_score}/
    100`;
349             diagScoreDisplay.className = `display-4 fw-bold ${scor
```

```

    eColor}`;
350     }
351     if (diagSummaryDisplay) diagSummaryDisplay.textContent =
    rep.status_summary;
352
353     if (recsContainer) {
354         if (rep.recommendations && rep.recommendations.length
    > 0) {
355             recsContainer.innerHTML = `
356                 <ol class="ps-3 mb-0">
357                     ${rep.recommendations.map(r => `<li class="py-1
    text-light">${r}</li>`).join('')}
358                 </ol>
359             `;
360         } else {
361             recsContainer.innerHTML = '<div class="text-success"
    >Все рекомендации соблюдены. Защита оптимальна.</div>';
362         }
363     }
364
365     if (critsContainer) {
366         if (rep.critical_findings && rep.critical_findings.len
    gth > 0) {
367             critsContainer.innerHTML = `
368                 <ul class="ps-3 mb-0 text-danger">
369                     ${rep.critical_findings.map(c => `<li class="py-
    1">${c}</li>`).join('')}
370                 </ul>
371             `;
372         } else {
373             critsContainer.innerHTML = '<div class="text-muted">
    Критических уязвимостей не зафиксировано.</div>';
374         }
375     }
376
377     } catch (err) {
378         console.error('[DefenderTab] Error loading diagnostics:'
    , err);

```

```
379 |     }
380 | }
381 |
382 | function setupActions() {
383 |     const btnQuick = document.getElementById('btn-def-quick-sc
384 | an');
385 |     if (btnQuick) {
386 |         btnQuick.onclick = async () => {
387 |             try {
388 |                 showAlert('Запуск быстрого сканирования Defender...'
389 | );
390 |                 const res = await fetchJSON('/api/v1/defender/scan',
391 | {
392 |                 method: 'POST',
393 |                 headers: { 'Content-Type': 'application/json' },
394 |                 body: JSON.stringify({ scan_type: 'quick' }),
395 |             });
396 |                 showAlert(`Быстрое сканирование: ${res.message}`);
397 |                 await loadDefenderStatus();
398 |             } catch (err) {
399 |                 showAlert(`Ошибка запуска сканирования: ${err.messag
400 | e}`, true);
401 |             }
402 |         };
403 |     }
404 |
405 |     const btnFull = document.getElementById('btn-def-full-scan
406 | ');
407 |     if (btnFull) {
408 |         btnFull.onclick = async () => {
409 |             try {
410 |                 showAlert('Запуск полного сканирования системы (фоно
411 | вый процесс Defender)...');
412 |                 const res = await fetchJSON('/api/v1/defender/scan',
413 | {
414 |                 method: 'POST',
415 |                 headers: { 'Content-Type': 'application/json' },
416 |                 body: JSON.stringify({ scan_type: 'full' }),
```

```
410         });
411         showAlert(`Полное сканирование: ${res.message}`);
412     } catch (err) {
413         showAlert(`Ошибка полного сканирования: ${err.messag
414 e}`, true);
415     }
416 };
417 }
418 const btnUpdate = document.getElementById('btn-def-update-
419 sigs');
420 if (btnUpdate) {
421     btnUpdate.onclick = async () => {
422         try {
423             showAlert('Обновление баз сигнатур Defender...');
424             const res = await fetchJSON('/api/v1/defender/update
425 -signatures', { method: 'POST' });
426             showAlert(`Обновление сигнатур: ${res.message}`);
427             await loadDefenderStatus();
428         } catch (err) {
429             showAlert(`Ошибка обновления баз: ${err.message}`, t
430 rue);
431         }
432     };
433 }
434 const btnRefresh = document.getElementById('btn-def-refres
435 h');
436 if (btnRefresh) {
437     btnRefresh.onclick = async () => {
438         showAlert('Обновление данных Defender...');
439         await refreshAll();
440     };
441 }
442 const btnAiDiag = document.getElementById('btn-def-ai-diag
443 ');
444 if (btnAiDiag) {
```



```
442     btnAiDiag.onclick = () => {
443         const diagTabBtn = document.getElementById('def-tab-di
ag-btn');
444         if (diagTabBtn) {
445             const tab = new bootstrap.Tab(diagTabBtn);
446             tab.show();
447         }
448     };
449 }
450 }
451
452 async function refreshAll() {
453     await Promise.all([
454         loadDefenderStatus(),
455         loadASRRules(),
456         loadCFA(),
457         loadExclusions(),
458         loadThreats(),
459         loadProcesses(),
460         loadEvents(),
461         loadDiagnostics(),
462     ]);
463 }
464
465 window.initDefenderTab = async function() {
466     console.log('[DefenderTab] Initializing Defender Security
Center Tab...');
467     setupActions();
468     await refreshAll();
469 };
470
471 // Auto initialize if container exists
472 if (document.readyState === 'loading') {
473     document.addEventListener('DOMContentLoaded', () => {
474         if (document.getElementById('def-shields-tbody')) {
475             window.initDefenderTab();
476         }
477     });
478 }
```

```
478 |     } else {  
479 |         if (document.getElementById('def-shields-tbody')) {  
480 |             window.initDefenderTab();  
481 |         }  
482 |     }  
483 |  
484 | })();  
485 |
```

index.html

```
1 <!-- File History AI Search Tab -->
2 <style>
3   .fh-container {
4     padding: 0.75rem 1rem;
5     max-width: 1400px;
6     margin: 0 auto;
7   }
8   .fh-card {
9     background: var(--bs-body-bg, #1e293b);
10    border: 1px solid var(--bs-border-color, #334155);
11    border-radius: 12px;
12    padding: 1.25rem;
13    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
14  }
15  .fh-header {
16    background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
17  0%);
18    border: 1px solid #334155;
19    border-radius: 12px;
20    padding: 1rem 1.5rem;
21    color: #f8fafc;
22  }
23  .fh-stat-card {
24    background: rgba(30, 41, 59, 0.7);
25    border: 1px solid var(--bs-border-color, #334155);
26    border-radius: 10px;
27    padding: 0.85rem 1.1rem;
28    transition: transform 0.15s ease, border-color 0.15s ease;
29  }
30  .fh-stat-card:hover {
31    transform: translateY(-2px);
32    border-color: #3b82f6;
33  }
34  .fh-stat-val {
35    font-size: 1.4rem;
36    font-weight: 700;
```

```
36     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,  
    Consolas, monospace;  
37 }  
38 .fh-search-box {  
39     background: var(--bs-dark, #0f172a);  
40     border: 1px solid #3b82f6;  
41     border-radius: 12px;  
42     padding: 0.75rem 1.25rem;  
43 }  
44 .fh-result-card {  
45     border-left: 4px solid #3b82f6;  
46     background: var(--bs-body-bg, #1e293b);  
47     border-top: 1px solid var(--bs-border-color, #334155);  
48     border-right: 1px solid var(--bs-border-color, #334155);  
49     border-bottom: 1px solid var(--bs-border-color, #334155);  
50     border-radius: 8px;  
51     padding: 1rem;  
52     margin-bottom: 0.75rem;  
53     transition: background-color 0.15s ease;  
54 }  
55 .fh-result-card:hover {  
56     background: rgba(59, 130, 246, 0.05);  
57 }  
58 .fh-score-badge {  
59     font-size: 0.82rem;  
60     font-weight: 600;  
61     font-family: monospace;  
62 }  
63 .fh-snippet-box {  
64     background: rgba(0, 0, 0, 0.2);  
65     border-radius: 6px;  
66     padding: 0.6rem 0.85rem;  
67     font-family: SFMono-Regular, Menlo, Monaco, Consolas, mono  
    space;  
68     font-size: 0.85rem;  
69     color: var(--bs-body-color, #cbd5e1);  
70     white-space: pre-wrap;  
71     word-break: break-all;
```

```
72     max-height: 120px;
73     overflow-y: auto;
74 }
75 </style>
76
77 <div class="fh-container">
78     <!-- Верхняя информационная панель -->
79     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 fh-header">
80         <div class="d-flex align-items-center gap-3">
81             <div class="bg-primary bg-opacity-25 p-2.5 rounded-3 text-primary border border-primary border-opacity-25">
82                 <i class="bi bi-clock-history fs-3"></i>
83             </div>
84             <div>
85                 <h5 class="mb-0 fw-bold d-flex align-items-center gap-2">
86                     Поиск по истории файлов Windows (File History AI Search)
87                     <span class="badge bg-primary bg-opacity-75 fs-7">RAG v1.0</span>
88                 </h5>
89                 <span class="text-secondary small">Семантический векторный поиск по Windows File History, Recent LNK, Activity DB и файловой системе</span>
90             </div>
91         </div>
92         <div class="d-flex align-items-center gap-2 flex-wrap">
93             <button class="btn btn-sm btn-outline-light rounded-pill px-3" id="btn-fh-refresh-status" type="button">
94                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить статус
95             </button>
96             <button class="btn btn-sm btn-success rounded-pill px-3" id="btn-fh-reindex" type="button">
97                 <i class="bi bi-cpu me-1"></i> Перестроить RAG
98             </button>
99         </div>
```

```
100 | </div>
101 |
102 | <!-- Карточки метрик и статусов -->
103 | <div class="row g-3 mb-3">
104 |     <div class="col-12 col-sm-6 col-md-3">
105 |         <div class="fh-stat-card">
106 |             <div class="text-secondary small text-uppercase fw-sem
107 | ibold mb-1">Всего в RAG-индексе</div>
108 |             <div class="d-flex align-items-center justify-content-
109 | between">
110 |                 <div class="fh-stat-val text-primary" id="fh-stat-do
111 | cs-count">0</div>
112 |                 <i class="bi bi-database-check fs-4 text-primary opa
113 | city-50"></i>
114 |             </div>
115 |             <div class="small text-muted mt-1" id="fh-stat-last-in
116 | dex">Индексация не проводилась</div>
117 |         </div>
118 |     </div>
119 |     <div class="col-12 col-sm-6 col-md-3">
120 |         <div class="fh-stat-card">
121 |             <div class="text-secondary small text-uppercase fw-sem
122 | ibold mb-1">Фоновый планировщик</div>
123 |             <div class="d-flex align-items-center justify-content-
124 | between">
125 |                 <div class="fh-stat-val" id="fh-stat-scheduler-stat
126 | us">0<div class="fh-stat-val" id="fh-stat-scheduler-status">Остановлен</div>
127 |                 <i class="bi bi-arrow-repeat fs-4 text-warning opaci
128 | ty-50" id="fh-scheduler-icon"></i>
129 |             </div>
130 |             <div class="small text-muted mt-1" id="fh-stat-schedul
131 | er-interval">Интервал: 15 мин</div>
132 |         </div>
133 |     </div>
134 | </div>
```

```
128 |         <div class="text-secondary small text-uppercase fw-sem
| ibold mb-1">Файл FAISS индекса</div>
129 |         <div class="d-flex align-items-center justify-content-
| between">
130 |             <div class="fh-stat-val fs-6 text-truncate" id="fh-s
| tat-faiss-file" style="max-width: 180px;">file_history.faiss</
| div>
131 |             <i class="bi bi-file-earmark-binary fs-4 text-info o
| pacity-50"></i>
132 |         </div>
133 |         <div class="small text-muted mt-1">Векторы Gemini Embe
| ddings</div>
134 |     </div>
135 | </div>
136 |
137 |     <div class="col-12 col-sm-6 col-md-3">
138 |         <div class="fh-stat-card">
139 |             <div class="text-secondary small text-uppercase fw-sem
| ibold mb-1">Управление демоном</div>
140 |             <div class="d-flex align-items-center gap-2 mt-1">
141 |                 <button class="btn btn-sm btn-outline-success w-50 p
| y-1" id="btn-fh-start-scheduler">
142 |                     <i class="bi bi-play-fill me-1"></i>Старт
143 |                 </button>
144 |                 <button class="btn btn-sm btn-outline-danger w-50 py
| -1" id="btn-fh-stop-scheduler">
145 |                     <i class="bi bi-stop-fill me-1"></i>Стоп
146 |                 </button>
147 |             </div>
148 |             <div class="small text-muted mt-1 text-center">Автообн
| овление каждые 15м</div>
149 |         </div>
150 |     </div>
151 | </div>
152 |
153 | <!-- Блок Семантического Поиска -->
154 | <div class="fh-card mb-4">
155 |     <h6 class="fw-bold mb-3 d-flex align-items-center gap-2">
```

```

156 |         <i class="bi bi-search text-primary"></i>
157 |         Быстрый семантический поиск по истории файлов
158 |     </h6>
159 |
160 |     <div class="row g-2 mb-3">
161 |         <div class="col-12 col-md-7">
162 |             <div class="input-group">
163 |                 <span class="input-group-text bg-body-tertiary borde
164 |                 <i class="bi bi-search text-muted"></i>
165 |                 </span>
166 |                 <input type="text" class="form-control border-start-
167 |                 0" id="fh-search-input" placeholder="Введите поисковый запрос
168 |                 (напр., 'финансовый отчет', 'скрипт бэкапа', 'документ Word').
169 |                 ..">
170 |                 <button class="btn btn-primary px-4 fw-semibold" id=
171 |                 "btn-fh-search" type="button">
172 |                 <i class="bi bi-arrow-right-circle me-1"></i> Найд
173 |                 и
174 |                 </button>
175 |             </div>
176 |             <div class="col-6 col-md-3">
177 |                 <select class="form-select" id="fh-filter-source">
178 |                     <option value="">Все источники (All Sources)</option
179 |                     <option value="file_history_xml">Windows File Histor
180 |                     y (XML)</option>
181 |                     <option value="recent_lnk">Recent Files (LNK)</optio
182 |                     n>
183 |                     <option value="activity_db">Activity History (DB)</o
184 |                     ption>
185 |                     <option value="fs_scan">Сканер модификаций ФС</optio
186 |                     n>
187 |                 </select>
188 |             </div>
189 |             <div class="col-6 col-md-2">
190 |                 <select class="form-select" id="fh-filter-topk">

```



```

183         <option value="5" selected>Топ 5</option>
184         <option value="10">Топ 10</option>
185         <option value="20">Топ 20</option>
186         <option value="50">Топ 50</option>
187     </select>
188 </div>
189 </div>
190
191 <!-- Индикатор загрузки поиска -->
192 <div class="text-center py-4 d-none" id="fh-search-spinner
">
193     <div class="spinner-border text-primary" role="status">
194         <span class="visually-hidden">Выполняется RAG поиск...
</span>
195     </div>
196     <div class="small text-muted mt-2">Выполняется векторный
семантический поиск по FAISS...</div>
197 </div>
198
199 <!-- Контейнер Результатов Поиска -->
200 <div id="fh-search-results-container">
201     <div class="text-center text-muted py-4 border rounded-3
bg-body-tertiary bg-opacity-50">
202         <i class="bi bi-search fs-2 d-block mb-1 text-secondar
y opacity-50"></i>
203         <span>Введите ключевые слова или естественный запрос в
ыше для семантического поиска по истории файлов.</span>
204     </div>
205 </div>
206 </div>
207
208 <!-- Блок Обнаруженных Элементов Истории (Телеметрия) -->
209 <div class="fh-card">
210     <div class="d-flex justify-content-between align-items-cen
ter mb-3 flex-wrap gap-2">
211         <h6 class="fw-bold mb-0 d-flex align-items-center gap-2"
>
212         <i class="bi bi-list-stars text-info"></i>

```

```

213 |         Телеметрия истории файлов Windows (Сканер системы)
214 |     </h6>
215 |     <button class="btn btn-sm btn-outline-info rounded-pill
    | px-3" id="btn-fh-scan-now">
216 |         <i class="bi bi-arrow-repeat me-1"></i> Запустить скан
    | ирование ОС
217 |     </button>
218 | </div>
219 |
220 | <div class="table-responsive">
221 |     <table class="table table-hover align-middle mb-0" id="f
    | h-items-table">
222 |         <thead class="table-light">
223 |             <tr>
224 |                 <th scope="col" style="width: 40px;">#</th>
225 |                 <th scope="col">Файл</th>
226 |                 <th scope="col">Источник</th>
227 |                 <th scope="col">Событие</th>
228 |                 <th scope="col">Время</th>
229 |                 <th scope="col">Размер</th>
230 |                 <th scope="col" class="text-end">Действия</th>
231 |             </tr>
232 |         </thead>
233 |         <tbody id="fh-items-tbody">
234 |             <tr>
235 |                 <td colspan="7" class="text-center py-4 text-muted
    | ">
236 |                     Нажмите «Запустить сканирование ОС» для получени
    | я свежей истории файлов Windows.
237 |                 </td>
238 |             </tr>
239 |         </tbody>
240 |     </table>
241 | </div>
242 | </div>
243 | </div>
244 |

```

main.js

```
1 /**
2  * file_history_ai_search_tab/main.js – Логика веб-интерфейса
  поиска по истории файлов Windows через RAG
3  */
4
5 (function () {
6   'use strict';
7
8   let currentItems = [];
9
10  /**
11   * Инициализация событий и обработчиков вкладки
12   */
13   function initFileHistoryTab() {
14     const btnSearch = document.getElementById('btn-fh-search')
15     ;
16     const searchInput = document.getElementById('fh-search-input');
17     const btnRefreshStatus = document.getElementById('btn-fh-refresh-status');
18     const btnReindex = document.getElementById('btn-fh-reindex');
19     const btnScanNow = document.getElementById('btn-fh-scan-now');
20     const btnStartSched = document.getElementById('btn-fh-start-scheduler');
21     const btnStopSched = document.getElementById('btn-fh-stop-scheduler');
22
23     if (btnSearch) {
24       btnSearch.addEventListener('click', executeSearch);
25     }
26
27     if (searchInput) {
28       searchInput.addEventListener('keydown', (e) => {
29         if (e.key === 'Enter') {
```

```
29         executeSearch();
30     }
31     });
32 }
33
34 if (btnRefreshStatus) {
35     btnRefreshStatus.addEventListener('click', loadStatus);
36 }
37
38 if (btnReindex) {
39     btnReindex.addEventListener('click', triggerReindex);
40 }
41
42 if (btnScanNow) {
43     btnScanNow.addEventListener('click', triggerScan);
44 }
45
46 if (btnStartSched) {
47     btnStartSched.addEventListener('click', () => toggleScheduler('start'));
48 }
49
50 if (btnStopSched) {
51     btnStopSched.addEventListener('click', () => toggleScheduler('stop'));
52 }
53
54 // Первоначальная загрузка статуса
55 loadStatus();
56 }
57
58 /**
59  * Получение текущего статуса RAG индекса и планировщика
60  */
61 async function loadStatus() {
62     try {
63         const resp = await fetch('/api/windows/file-history/status');
64     }
```

```

64 |         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
65 |         const data = await resp.json();
66 |
67 |         const elCount = document.getElementById('fh-stat-docs-co
unt');
68 |         const elLastIdx = document.getElementById('fh-stat-last-
index');
69 |         const elSchedStat = document.getElementById('fh-stat-sch
eduler-status');
70 |         const elSchedIcon = document.getElementById('fh-schedule
r-icon');
71 |         const elSchedInt = document.getElementById('fh-stat-sche
duler-interval');
72 |         const elFaissFile = document.getElementById('fh-stat-fai
ss-file');
73 |
74 |         if (elCount) elCount.textContent = data.total_indexed_do
cuments || 0;
75 |         if (elLastIdx) {
76 |             elLastIdx.textContent = data.last_indexed_at
77 |                 ? `Обновлено: ${new Date(data.last_indexed_at).toLoc
aleString('ru-RU')}`
78 |                 : 'Индексация не проводилась';
79 |         }
80 |
81 |         if (elSchedStat) {
82 |             if (data.scheduler_running) {
83 |                 elSchedStat.textContent = 'Активен';
84 |                 elSchedStat.className = 'fh-stat-val text-success';
85 |                 if (elSchedIcon) elSchedIcon.className = 'bi bi-arro
w-repeat fs-4 text-success opacity-75 spin';
86 |             } else {
87 |                 elSchedStat.textContent = 'Остановлен';
88 |                 elSchedStat.className = 'fh-stat-val text-warning';
89 |                 if (elSchedIcon) elSchedIcon.className = 'bi bi-arro
w-repeat fs-4 text-warning opacity-50';
90 |             }
91 |         }

```

```

92 |
93 |     if (elSchedInt) {
94 |         elSchedInt.textContent = `Интервал: ${data.update_inte
rval_minutes || 15} мин`;
95 |     }
96 |
97 |     if (elFaissFile && data.index_file_path) {
98 |         const parts = data.index_file_path.split(/[\\\/]/);
99 |         elFaissFile.textContent = parts[parts.length - 1] || d
ata.index_file_path;
100 |         elFaissFile.title = data.index_file_path;
101 |     }
102 | } catch (err) {
103 |     console.error('[FileHistoryUI] Ошибка загрузки статуса:'
, err);
104 | }
105 | }
106 |
107 | /**
108 |  * Принудительный запуск сканирования истории файлов Windows
109 |  */
110 | async function triggerScan() {
111 |     const tbody = document.getElementById('fh-items-tbody');
112 |     if (tbody) {
113 |         tbody.innerHTML = `
114 |             <tr>
115 |                 <td colspan="7" class="text-center py-4">
116 |                     <div class="spinner-border spinner-border-sm text-
info me-2" role="status"></div>
117 |                     <span>Сканирование источников истории файлов Windo
ws...</span>
118 |                 </td>
119 |             </tr>`;
120 |     }
121 |
122 |     try {
123 |         const resp = await fetch('/api/windows/file-history/scan
');

```

```

124         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
125         currentItems = await resp.json();
126         renderItemTable(currentItems);
127         if (window.showToast) {
128             window.showToast(`Сканирование завершено: найдено ${currentItems.length} элементов`, 'info');
129         }
130     } catch (err) {
131         console.error('[FileHistoryUI] Ошибка сканирования:', err);
132         if (tbody) {
133             tbody.innerHTML = `
134                 <tr>
135                     <td colspan="7" class="text-center py-4 text-danger">
136                         <i class="bi bi-exclamation-triangle me-1"></i>
137                         Ошибка при сканировании: ${err.message}
138                     </td>
139                 </tr>`;
140         }
141     }
142
143     /**
144     * Отрисовка таблицы сохраненных элементов телеметрии
145     */
146     function renderItemTable(items) {
147         const tbody = document.getElementById('fh-items-tbody');
148         if (!tbody) return;
149
150         if (!items || items.length === 0) {
151             tbody.innerHTML = `
152                 <tr>
153                     <td colspan="7" class="text-center py-4 text-muted">
154                         Элементы истории файлов не обнаружены.
155                     </td>
156                 </tr>`;
157             return;

```

```

158     }
159
160     let html = '';
161     items.slice(0, 100).forEach((item, idx) => {
162         const badgeSource = getSourceBadge(item.source_type);
163         const formattedSize = formatBytes(item.file_size);
164         const timeStr = item.timestamp ? new Date(item.timestamp
165         ).toLocaleString('ru-RU') : '-';
166
167         html += `
168             <tr>
169                 <td class="text-muted small">${idx + 1}</td>
170                 <td>
171                     <div class="fw-semibold text-truncate" style="max-
172                     width: 280px;" title="${escapeHtml(item.file_name)}">
173                         <i class="bi bi-file-earmark-text me-1 text-prim
174                         ary"></i>${escapeHtml(item.file_name)}
175                     </div>
176                     <div class="small text-muted text-truncate" style=
177                     "max-width: 320px;" title="${escapeHtml(item.file_path)}">
178                         ${escapeHtml(item.file_path)}
179                     </div>
180                 </td>
181                 <td>${badgeSource}</td>
182                 <td><span class="badge bg-secondary bg-opacity-50 te
183                 xt-light">${escapeHtml(item.event_type)}</span></td>
184                 <td class="small text-muted">${timeStr}</td>
185                 <td class="small font-monospace">${formattedSize}</t
186                 d>
187                 <td class="text-end">
188                     <button class="btn btn-sm btn-outline-secondary py
189                     -0 px-2 btn-copy-path" data-path="${escapeHtml(item.file_path)
190                     }" title="Копировать путь">
191                         <i class="bi bi-clipboard"></i>
192                     </button>
193                 </td>
194             </tr>`;
195     });

```



```
188
189     tbody.innerHTML = html;
190
191     // Привязка копирования пути
192     tbody.querySelectorAll('.btn-copy-path').forEach((btn) =>
193     {
194         btn.addEventListener('click', (e) => {
195             const path = e.currentTarget.getAttribute('data-path')
196             ;
197             if (path) {
198                 navigator.clipboard.writeText(path);
199                 if (window.showToast) window.showToast('Путь скопиро
200 ван в буфер обмена', 'success');
201             }
202         });
203     });
204
205     /**
206     * Пересчет и векторизация RAG-индекса
207     */
208     async function triggerReindex() {
209         const btnReindex = document.getElementById('btn-fh-reindex
210 ');
211         if (btnReindex) btnReindex.disabled = true;
212
213         if (window.showToast) {
214             window.showToast('Запущен процесс сканирования и векторн
215 ой RAG-индексации...', 'info');
216         }
217
218         try {
219             const resp = await fetch('/api/windows/file-history/inde
220 x', { method: 'POST' });
221             if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
222             const data = await resp.json();
223             if (window.showToast) {
224                 window.showToast(`RAG-индекс перестроен: проиндексиров
```

```

    ано `${data.indexed_count} элементов!`, 'success');
220     }
221     loadStatus();
222   } catch (err) {
223     console.error('[FileHistoryUI] Ошибка перестроения RAG: '
    , err);
224     if (window.showToast) {
225       window.showToast(`Ошибка индексации RAG: ${err.message}
    ` , 'error');
226     }
227   } finally {
228     if (btnReindex) btnReindex.disabled = false;
229   }
230 }
231
232 /**
233  * Выполнение семантического поиска
234  */
235 async function executeSearch() {
236   const searchInput = document.getElementById('fh-search-inp
    ut');
237   const filterSource = document.getElementById('fh-filter-so
    urce');
238   const filterTopk = document.getElementById('fh-filter-topk
    ');
239   const spinner = document.getElementById('fh-search-spinner
    ');
240   const container = document.getElementById('fh-search-resul
    ts-container');
241
242   const query = searchInput ? searchInput.value.trim() : '';
243   if (!query) {
244     if (window.showToast) window.showToast('Введите поиско
    вый запрос', 'warning');
245     return;
246   }
247
248   const topK = filterTopk ? parseInt(filterTopk.value, 10) :

```

```

5;
249     const sourceType = filterSource && filterSource.value ? fi
    lterSource.value : null;
250
251     if (spinner) spinner.classList.remove('d-none');
252     if (container) container.innerHTML = '';
253
254     try {
255         const resp = await fetch('/api/windows/file-history/sear
    ch', {
256             method: 'POST',
257             headers: { 'Content-Type': 'application/json' },
258             body: JSON.stringify({
259                 query: query,
260                 top_k: topK,
261                 source_type: sourceType,
262             }),
263         });
264
265         if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
266         const data = await resp.json();
267         renderSearchResults(data);
268     } catch (err) {
269         console.error('[FileHistoryUI] Ошибка поиска:', err);
270         if (container) {
271             container.innerHTML = `
272                 <div class="alert alert-danger mb-0">
273                     <i class="bi bi-exclamation-octagon me-1"></i> Оши
    бка семантического поиска: ${err.message}
274                 </div>`;
275         }
276     } finally {
277         if (spinner) spinner.classList.add('d-none');
278     }
279 }
280
281 /**
282  * Отрисовка карточек результатов семантического поиска

```

```

283     */
284     function renderSearchResults(data) {
285         const container = document.getElementById('fh-search-resul
ts-container');
286         if (!container) return;
287
288         if (!data.results || data.results.length === 0) {
289             container.innerHTML = `
290                 <div class="text-center text-muted py-4 border rounded
-3">
291                     <i class="bi bi-emoji-frown fs-3 d-block mb-1 text-w
arning opacity-50"></i>
292                     <span>Совпадений по запросу «<strong>${escapeHtml(da
ta.query)}</strong>» не найдено.</span><br>
293                     <small class="text-secondary">Попробуйте изменить за
прос или нажать «Перестроить RAG» для обновления индекса.</small>
294                     </div>`;
295             return;
296         }
297
298         let html = `
299             <div class="d-flex justify-content-between align-items-c
enter mb-2 px-1">
300                 <span class="small text-muted">
301                     Найдено результатов: <strong>${data.total_found}</st
rong> (Время выполнения: <strong>${data.execution_time_ms} мс<
/strong>)
302                 </span>
303             </div>`;
304
305         data.results.forEach((item, idx) => {
306             const badgeSource = getSourceBadge(item.source_type);
307             const scorePct = Math.round(item.score * 100);
308             const badgeScoreClass = item.score > 0.6 ? 'bg-success'
: item.score > 0.3 ? 'bg-info' : 'bg-secondary';
309             const timeStr = item.timestamp ? new Date(item.timestamp
).toLocaleString('ru-RU') : '-';

```

```

310 |
311 |     html += `
312 |         <div class="fh-result-card">
313 |             <div class="d-flex justify-content-between align-items-start gap-2 mb-1">
314 |                 <div class="d-flex align-items-center gap-2">
315 |                     <span class="badge rounded-circle bg-body-tertiary text-muted p-2 fw-bold me-1" style="width: 26px; height: 26px; display: inline-flex; align-items: center; justify-content: center;">
316 |                         ${idx + 1}
317 |                     </span>
318 |                     <h6 class="mb-0 fw-bold text-truncate" style="max-width: 450px;" title="${escapeHtml(item.file_name)}">
319 |                         <i class="bi bi-file-earmark-code me-1 text-primary"></i>${escapeHtml(item.file_name)}
320 |                     </h6>
321 |                     ${badgeSource}
322 |                 </div>
323 |                 <div class="d-flex align-items-center gap-2">
324 |                     <span class="badge ${badgeScoreClass} fh-score-badge" title="Косинусное сходство / Релевантность">
325 |                         Score: ${item.score}
326 |                     </span>
327 |                     <button class="btn btn-sm btn-outline-secondary py-0 px-2 btn-copy-path" data-path="${escapeHtml(item.file_path)}" title="Скопировать путь">
328 |                         <i class="bi bi-clipboard"></i>
329 |                     </button>
330 |                 </div>
331 |             </div>
332 |
333 |             <div class="small text-muted text-truncate mb-2 font-monospace">
334 |                 <i class="bi bi-folder2-open me-1 text-secondary"></i>${escapeHtml(item.file_path)}
335 |                 <span class="ms-2 text-secondary">| Время: ${timeStr}</span>

```

```

336         </div>
337
338         <div class="fh-snippet-box">
339             ${escapeHtml(item.snippet)}
340         </div>
341     </div>`;
342 });
343
344     container.innerHTML = html;
345
346     // Привязка копирования в результатах поиска
347     container.querySelectorAll('.btn-copy-path').forEach((btn)
348     => {
349         btn.addEventListener('click', (e) => {
350             const path = e.currentTarget.getAttribute('data-path')
351             ;
352             if (path) {
353                 navigator.clipboard.writeText(path);
354                 if (window.showToast) window.showToast('Путь скопиро
355 ван в буфер обмена', 'success');
356             }
357         });
358     });
359
360     /**
361     * Управление фоновым планировщиком
362     */
363     async function toggleScheduler(action) {
364         const url = `/api/windows/file-history/scheduler/${action}
365 `;
366         try {
367             const resp = await fetch(url, { method: 'POST' });
368             if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
369             const data = await resp.json();
370             if (window.showToast) {
371                 window.showToast(
372                     action === 'start' ? 'Фоновый планировщик RAG запуще

```

```

н' : 'Фоновый планировщик остановлен',
370     action === 'start' ? 'success' : 'info'
371 );
372 }
373 loadStatus();
374 } catch (err) {
375     console.error('[FileHistoryUI] Ошибка управления планиро
вщиком:', err);
376     if (window.showToast) window.showToast(`Ошибка: ${err.me
ssage}`, 'error');
377 }
378 }
379
380 /**
381  * Получение бейджа источника
382  */
383 function getSourceBadge(sourceType) {
384     switch (sourceType) {
385         case 'file_history_xml':
386             return '<span class="badge bg-primary bg-opacity-75"><
i class="bi bi-file-earmark-xml me-1"></i>File History XML</spa
n>';
387         case 'recent_lnk':
388             return '<span class="badge bg-info bg-opacity-75"><i c
lass="bi bi-link-45deg me-1"></i>Recent LNK</span>';
389         case 'activity_db':
390             return '<span class="badge bg-warning text-dark bg-opa
city-75"><i class="bi bi-database me-1"></i>Activity DB</span>
';
391         case 'fs_scan':
392             return '<span class="badge bg-success bg-opacity-75"><
i class="bi bi-hdd-network me-1"></i>FS Scan</span>';
393         default:
394             return '<span class="badge bg-secondary">${escapeHtml(
sourceType || 'unknown')}</span>';
395     }
396 }
397

```

```

398 | /**
399 |  * Форматирование байт
400 |  */
401 | function formatBytes(bytes, decimals = 1) {
402 |     if (!bytes || bytes === 0) return '0 B';
403 |     const k = 1024;
404 |     const dm = decimals < 0 ? 0 : decimals;
405 |     const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
406 |     const i = Math.floor(Math.log(bytes) / Math.log(k));
407 |     return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
    | ' ' + sizes[i];
408 | }
409 |
410 | /**
411 |  * Экранирование спецсимволов HTML
412 |  */
413 | function escapeHtml(str) {
414 |     if (!str) return '';
415 |     return String(str)
416 |         .replace(/&/g, '&#039;');
417 |         .replace(/</g, '&#039;');
418 |         .replace(/>/g, '&#039;');
419 |         .replace(/"/g, '&#039;');
420 |         .replace(/'/g, '&#039;');
421 | }
422 |
423 | // Запуск при загрузке DOM
424 | if (document.readyState === 'loading') {
425 |     document.addEventListener('DOMContentLoaded', initFileHist
    | oryTab);
426 | } else {
427 |     initFileHistoryTab();
428 | }
429 | })();
430 |

```


index.html

```
1 <!-- File Recovery (R-Studio Technician Portable) Tab -->
2 <style>
3   .recovery-container {
4     padding: 0.75rem 1rem;
5     max-width: 1200px;
6     margin: 0 auto;
7   }
8   .recovery-card {
9     background: #0f172a !important;
10    border: 1px solid #334155 !important;
11    border-radius: 10px;
12    padding: 1.25rem;
13    color: #f8fafc;
14    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.2);
15  }
16  .recovery-header {
17    background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
18    0%);
19    border: 1px solid #334155;
20    border-radius: 10px;
21    padding: 0.85rem 1.25rem;
22  }
23  .recovery-stat-box {
24    background: #1e293b;
25    border: 1px solid #334155;
26    border-radius: 8px;
27    padding: 0.75rem 1rem;
28  }
29  .recovery-stat-title {
30    font-size: 0.75rem;
31    text-transform: uppercase;
32    letter-spacing: 0.05em;
33    color: #94a3b8;
34  }
35  .recovery-stat-val {
36    font-size: 1.1rem;
```

```

36     font-weight: 600;
37     color: #f8fafc;
38     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
39 }
40 .recovery-feature-item {
41     display: flex;
42     align-items: flex-start;
43     gap: 0.5rem;
44     margin-bottom: 0.5rem;
45     font-size: 0.85rem;
46     color: #cbd5e1;
47 }
48 </style>
49
50 <div class="recovery-container">
51     <!-- Верхняя информационная панель -->
52     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 recovery-header text-white">
53         <div class="d-flex align-items-center gap-2.5">
54             <span class="fs-4">❏ </span>
55             <div>
56                 <h5 class="mb-0 fw-bold fs-6">Восстановление удаленных
    файлов</h5>
57                 <span class="text-secondary small">Запуск профессионал
    ьного комплекса R-Studio Technician Portable</span>
58             </div>
59         </div>
60         <div class="d-flex align-items-center gap-2">
61             <span class="badge rounded-pill bg-dark border border-se
    condary text-info px-2.5 py-1.5" id="recovery-status-badge">
62                 Проверка утилиты...
63             </span>
64             <button class="btn btn-sm btn-outline-info rounded-pill
    px-3" id="btn-recovery-refresh" type="button">
65                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить ст
    атус
66             </button>

```

```

67     </div>
68 </div>
69
70 <div class="row g-3 mb-3">
71     <!-- Левая колонка: Запуск и возможности -->
72     <div class="col-12 col-lg-7">
73         <div class="recovery-card h-100 d-flex flex-column justify-content-between">
74             <div>
75                 <h6 class="fw-bold text-info mb-3 d-flex align-items-center gap-2">
76                     <i class="bi bi-arrow-counterclockwise"></i> Восстановление данных с накопителей
77                 </h6>
78
79                 <p class="text-light small mb-3">
80                     <strong>R-Studio Technician</strong> – мощный комплекс для восстановления удалённых, повреждённых и отформатированных файлов с локальных и сетевых дисков, RAID-массивов, флеш-накопителей и виртуальных дисков.
81                 </p>
82
83                 <div class="recovery-feature-item">
84                     <i class="bi bi-check2-circle text-success fs-5"></i>
85                     <div>
86                         <strong>Поддерживаемые файловые системы:</strong>
87                         > NTFS, NTFS5, ReFS, FAT12/16/32, exFAT, HFS/HFS+, APFS, Ext2/Ext3/Ext4 FS, UFS.
88                     </div>
89                 </div>
90                 <div class="recovery-feature-item">
91                     <i class="bi bi-check2-circle text-success fs-5"></i>
92                     <div>
93                         <strong>Сценарии восстановления:</strong> Случайное удаление (Shift+Del), форматирование разделов, повреждение

```

MBR/GPT, сбой питания, вирусные атаки, сбойные сектора.

```
94         </div>
95     </div>
96
97     <div class="recovery-feature-item">
98         <i class="bi bi-check2-circle text-success fs-5"><
99         /i>
100         <div>
101             <strong>Режим работы:</strong> Portable (не треб
102             ует установки, запускается напрямую из <code>\bin\R-Studio_Tec
103             hnician_9.5.191810_Portable\R-Studio_9.5.exe</code>).
104         </div>
105     </div>
106
107     <div class="mt-4">
108         <!-- Большая главная кнопка запуска -->
109         <button class="btn btn-primary btn-lg w-100 py-3 fw-
110         bold d-flex align-items-center justify-content-center gap-2 sh
111         adow" id="btn-recovery-launch" type="button">
112             <i class="bi bi-play-circle-fill fs-4"></i> Восста
113             новить удаленные файлы (Запустить R-Studio)
114         </button>
115         <div id="recovery-alert-box" class="mt-3" style="dis
116         play: none;"></div>
117     </div>
118 </div>
119
120 <!-- Правая колонка: Информация об исполняемом файле -->
121 <div class="col-12 col-lg-5">
122     <div class="recovery-card h-100 d-flex flex-column justi
123     fy-content-between">
124         <div>
125             <h6 class="fw-bold text-info mb-3 d-flex align-items
126             -center gap-2">
127                 <i class="bi bi-info-circle-fill"></i> Информация
128                 о пакете утилиты
```

```
121 |         </h6>
122 |
123 |         <div class="row g-2 mb-3">
124 |             <div class="col-12">
125 |                 <div class="recovery-stat-box">
126 |                     <div class="recovery-stat-title">Исполняемый ф
айл</div>
127 |                         <div class="recovery-stat-val text-truncate" i
d="recovery-file-status">Проверка...</div>
128 |                         <div class="text-secondary small mt-1 text-bre
ак font-monospace" id="recovery-file-path" style="font-size: 0
.75rem;">
129 |                             \bin\R-Studio_Technician_9.5.191810_Portable
\R-Studio_9.5.exe
130 |                         </div>
131 |                     </div>
132 |                 </div>
133 |
134 |             <div class="col-12">
135 |                 <div class="recovery-stat-box">
136 |                     <div class="recovery-stat-title">Версия и сбор
ка</div>
137 |                         <div class="recovery-stat-val text-info" id="r
ecovery-version">R-Studio Technician 9.5</div>
138 |                         <div class="text-secondary small mt-1">Сборка:
191810 (Portable x64/x86)</div>
139 |                     </div>
140 |                 </div>
141 |
142 |             <div class="col-12">
143 |                 <div class="recovery-stat-box">
144 |                     <div class="recovery-stat-title">Размер файла
/ Дата модификации</div>
145 |                         <div class="recovery-stat-val text-truncate" i
d="recovery-file-size">--</div>
146 |                         <div class="text-secondary small mt-1" id="rec
overy-file-date">--</div>
147 |                     </div>
```

```
148 |         </div>
149 |     </div>
150 | </div>
151 |
152 |     <div class="p-2 rounded bg-dark border border-secondary
    | y text-secondary small">
153 |         <i class="bi bi-shield-check text-success me-1"></i>
    | Приложение запускается с повышенными привилегиями Windows для
    | доступа к секторам физических дисков.
154 |     </div>
155 | </div>
156 | </div>
157 | </div>
158 | </div>
159 |
```

main.js

```
1 /**
2  * file_recovery_tab/main.js – логика вкладки «Восстановить удаленные файлы» (R-Studio Technician Portable)
3  */
4
5 (function () {
6     'use strict';
7
8     function formatBytes(bytes) {
9         if (!bytes || bytes <= 0) return '0 B';
10        const units = ['B', 'KB', 'MB', 'GB', 'TB'];
11        const i = Math.floor(Math.log(bytes) / Math.log(1024));
12        return (bytes / Math.pow(1024, i)).toFixed(2) + ' ' + units[i];
13    }
14
15    function showAlert(msg, type = 'info') {
16        const box = document.getElementById('recovery-alert-box');
17        if (!box) return;
18        box.className = `alert alert-${type} alert-dismissible fade show small py-2 px-3`;
19        box.innerHTML = `
20            <div class="d-flex align-items-center justify-content-between">
21                <div>${msg}</div>
22                <button type="button" class="btn-close py-2" data-bs-dismiss="alert" aria-label="Close"></button>
23            </div>
24        `;
25        box.style.display = 'block';
26    }
27
28    async function loadRecoveryStatus() {
29        const badge = document.getElementById('recovery-status-badge');
30        const fileStatus = document.getElementById('recovery-file-
```

```
status');
31   const filePath = document.getElementById('recovery-file-pa
th');
32   const fileSize = document.getElementById('recovery-file-si
ze');
33   const fileDate = document.getElementById('recovery-file-da
te');
34
35   try {
36     if (badge) badge.textContent = 'Обновление...';
37     const res = await fetch(`/api/recovery/status?t=${Date.now()}`);
38     if (!res.ok) throw new Error(`HTTP ${res.status}`);
39
40     const data = await res.json();
41     const tool = data.tool || {};
42
43     if (tool.exists) {
44       if (badge) {
45         badge.className = 'badge rounded-pill bg-success-sub
tle border border-success text-success px-2.5 py-1.5';
46         badge.textContent = 'Готов к запуску';
47       }
48       if (fileStatus) {
49         fileStatus.className = 'recovery-stat-val text-succe
ss text-truncate';
50         fileStatus.textContent = 'Файл найден';
51       }
52       if (filePath) filePath.textContent = tool.path || tool
.relative_path || '';
53       if (fileSize) fileSize.textContent = formatBytes(tool
.size_bytes);
54       if (fileDate) fileDate.textContent = `Изменён: ${tool
.modified_time || '--'}`;
55     } else {
56       if (badge) {
57         badge.className = 'badge rounded-pill bg-warning-sub
tle border border-warning text-warning px-2.5 py-1.5';
```



```

58 |         badge.textContent = 'Исполняемый файл не найден';
59 |     }
60 |     if (fileStatus) {
61 |         fileStatus.className = 'recovery-stat-val text-warni
ng text-truncate';
62 |         fileStatus.textContent = 'Не найден в bin';
63 |     }
64 |     if (filePath) filePath.textContent = tool.relative_pat
h || '';
65 |     if (fileSize) fileSize.textContent = '0 B';
66 |     if (fileDate) fileDate.textContent = '--';
67 |     }
68 | } catch (err) {
69 |     console.error('Ошибка загрузки статуса утилиты восстано
ления:', err);
70 |     if (badge) {
71 |         badge.className = 'badge rounded-pill bg-danger-subtle
border border-danger text-danger px-2.5 py-1.5';
72 |         badge.textContent = 'Ошибка связи';
73 |     }
74 | }
75 | }
76 |
77 | async function launchRecoveryTool() {
78 |     const btn = document.getElementById('btn-recovery-launch')
;
79 |     if (!btn) return;
80 |
81 |     const originalContent = btn.innerHTML;
82 |     btn.disabled = true;
83 |     btn.innerHTML = `<span class="spinner-border spinner-borde
r-sm me-2" role="status" aria-hidden="true"></span>Запуск R-St
udio...`;
84 |
85 |     try {
86 |         const res = await fetch('/api/recovery/launch', {
87 |             method: 'POST',
88 |             headers: { 'Content-Type': 'application/json' },

```

```

89 |     });
90 |
91 |     const data = await res.json();
92 |
93 |     if (res.ok && data.success) {
94 |         showAlert(`<strong>Успех:</strong> ${data.message} || '
Программа R-Studio успешно запущена!'}`, 'success');
95 |         if (window.toast) {
96 |             window.toast.success('R-Studio запущена', data.messa
ge || 'Окно программы открывается...');
97 |         }
98 |     } else {
99 |         const err = data.detail || 'Не удалось запустить R-Stu
dio';
100 |        showAlert(`<strong>Ошибка:</strong> ${err}`, 'danger')
;
101 |        if (window.toast) {
102 |            window.toast.error('Ошибка запуска', err);
103 |        }
104 |    }
105 | } catch (err) {
106 |     console.error('Ошибка вызова API запуска:', err);
107 |     showAlert(`<strong>Сетевая ошибка:</strong> ${err.messag
e}`, 'danger');
108 |     if (window.toast) {
109 |         window.toast.error('Сетевая ошибка', err.message);
110 |     }
111 | } finally {
112 |     btn.disabled = false;
113 |     btn.innerHTML = originalContent;
114 | }
115 | }
116 |
117 | function initListeners() {
118 |     const refreshBtn = document.getElementById('btn-recovery-r
efresh');
119 |     if (refreshBtn) {
120 |         refreshBtn.onclick = () => loadRecoveryStatus();

```

```
121     }
122
123     const launchBtn = document.getElementById('btn-recovery-la
    unch');
124     if (launchBtn) {
125         launchBtn.onclick = () => launchRecoveryTool();
126     }
127 }
128
129 function initFileRecoveryTab() {
130     initListeners();
131     loadRecoveryStatus();
132 }
133
134 // Экспорт для жизненного цикла tab-core.js
135 window.initFile_recoveryTab = initFileRecoveryTab;
136 window.initFilerecoveryTab = initFileRecoveryTab;
137 window.initFileRecoveryTab = initFileRecoveryTab;
138 window.initRecoveryTab = initFileRecoveryTab;
139
140 // Автозапуск при самостоятельной загрузке
141 if (document.readyState === 'loading') {
142     document.addEventListener('DOMContentLoaded', initFileReco
    veryTab);
143 } else {
144     initFileRecoveryTab();
145 }
146 }));
147
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Behavioral Forensics Tab
6 =====
7
8 Description:
9     Вкладка форензики активности пользователя, активного окна
10    в фокусе,
11    времени простоя (Idle Time), сенсоров конфиденциальности (
12    Camera/Mic Radar)
13    и истории запуска приложений UserAssist.
14
15 File: index.html
16 Project: AI-Breadboard
17 Module: WebInterface.ForensicsTab
18 Author: hypo69
19 Copyright: © 2026 hypo69
20 =====
21
22 -->
23 <style>
24     .forensics-tab-wrapper {
25         padding: 0.75rem;
26         color: var(--text-color, #f8fafc);
27     }
28     .diag-panel {
29         background: var(--surface-2, #1e293b);
30         border: 1px solid var(--border-color, #334155);
31         border-radius: 8px;
32         overflow: hidden;
33     }
34     .diag-panel-header {
35         background: rgba(15, 23, 42, 0.6);
36         padding: 8px 14px;
37         border-bottom: 1px solid var(--border-color, #334155);
```

```

32     font-weight: 600;
33     font-size: 0.85rem;
34 }
35 .about-ai-pulse-icon {
36     width: 42px;
37     height: 42px;
38     border-radius: 8px;
39     display: flex;
40     align-items: center;
41     justify-content: center;
42     border: 1px solid rgba(56, 189, 248, 0.3);
43 }
44 </style>
45
46 <div class="forensics-tab-wrapper">
47     <!-- Top Header & Status -->
48     <div class="card shadow-sm border-secondary-subtle mb-3" sty
49 le="background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
50 0%); border-radius: 10px;">
51         <div class="card-body p-3">
52             <div class="d-flex flex-wrap align-items-center justify-
53 content-between gap-2">
54                 <div class="d-flex align-items-center gap-2">
55                     <div class="fs-4">□ </div>
56                     <div>
57                         <h5 class="mb-0 fw-bold text-white">2. Поведенческ
58 ая форензика активности</h5>
59                         <small class="text-muted">Телеметрия активного фок
60 уса, время бездействия, аудит доступа к камере/микрофону и рее
61 стр UserAssist</small>
62                     </div>
63                 </div>
64             </div>
65
66             <div class="d-flex align-items-center gap-2 flex-wrap"
67 >
68                 <div class="form-check form-switch mb-0 small text-m
69 uted d-flex align-items-center gap-1">
70                     <input class="form-check-input" type="checkbox" id

```

```

    ="diag-forensics-auto-refresh">
62         <label class="form-check-label small text-nowrap"
    for="diag-forensics-auto-refresh">Авто (3с)</label>
63     </div>
64     <button class="btn btn-sm btn-outline-info rounded-p
    ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
    "btn-diag-forensics-refresh">
65         <i class="bi bi-arrow-clockwise"></i> Обновить
66     </button>
67 </div>
68 </div>
69 </div>
70 </div>
71
72 <div class="row g-2 mb-3">
73     <!-- Active Foreground Window Card -->
74     <div class="col-12 col-lg-6">
75         <div class="diag-panel h-100">
76             <div class="diag-panel-header d-flex justify-content-b
    etween align-items-center">
77                 <div class="d-flex align-items-center gap-2">
78                     <i class="bi bi-window-stack text-primary"></i>
79                     <span class="text-white">Текущее активное окно в ф
    окусе (Foreground Window)</span>
80                 </div>
81                 <span class="badge bg-success-subtle text-success bo
    rder border-success" id="diag-forensics-live-badge">Live Focus
    </span>
82             </div>
83             <div class="p-3">
84                 <div class="d-flex align-items-center gap-3 mb-2">
85                     <div class="about-ai-pulse-icon" style="background
    : rgba(14, 165, 233, 0.15); border-color: rgba(14, 165, 233, 0
    .4);">
86                         <i class="bi bi-cursor-fill text-info fs-5"></i>
87                     </div>
88                     <div class="flex-grow-1 overflow-hidden">
89                         <div class="fw-bold text-white text-truncate" id

```

```

    ="diag-fg-title" style="font-size: 0.95rem;">Рабочий стол Wind
ows</div>
90         <div class="small text-muted font-monospace" id=
"diag-fg-proc">Процесс: explorer.exe (PID: 0)</div>
91         </div>
92     </div>
93     <div class="border-top border-secondary-subtle pt-2
mt-2 d-flex justify-content-between align-items-center flex-wr
ap gap-2">
94         <div>
95             <span class="text-muted small">Время бездействия
пользователя (Idle Time):</span>
96             <strong class="text-warning font-monospace ms-1"
id="diag-user-idle-time">0.0 сек</strong>
97         </div>
98         <span class="badge bg-dark border border-secondary
text-info" id="diag-user-status-badge">Пользователь активен</
span>
99     </div>
100 </div>
101 </div>
102 </div>
103
104 <!-- Privacy Sensors (Camera & Mic Radar) -->
105 <div class="col-12 col-lg-6">
106     <div class="diag-panel h-100">
107         <div class="diag-panel-header d-flex justify-content-b
etween align-items-center">
108             <div class="d-flex align-items-center gap-2">
109                 <i class="bi bi-camera-video-fill text-danger"></i
>
110                 <span class="text-white">Контроль доступа к камере
и микрофону (Privacy Radar)</span>
111             </div>
112             <span class="badge bg-dark border border-secondary t
ext-info">ConsentStore</span>
113         </div>
114         <div class="p-3">

```

```

115         <div class="row g-2">
116             <div class="col-6">
117                 <div class="p-2 rounded bg-black bg-opacity-30 b
order border-secondary-subtle">
118                     <div class="small text-muted mb-1"><i class="b
i bi-camera-video me-1 text-danger"></i>Камера:</div>
119                     <div id="diag-cam-access-list" class="small te
xt-light" style="max-height: 90px; overflow-y: auto;">
120                         <span class="text-muted">Нет активных обраще
ний</span>
121                     </div>
122                 </div>
123             </div>
124             <div class="col-6">
125                 <div class="p-2 rounded bg-black bg-opacity-30 b
order border-secondary-subtle">
126                     <div class="small text-muted mb-1"><i class="b
i bi-mic me-1 text-info"></i>Микрофон:</div>
127                     <div id="diag-mic-access-list" class="small te
xt-light" style="max-height: 90px; overflow-y: auto;">
128                         <span class="text-muted">Нет активных обраще
ний</span>
129                     </div>
130                 </div>
131             </div>
132         </div>
133     </div>
134 </div>
135 </div>
136 </div>
137
138 <!-- UserAssist Application Run History -->
139 <div class="diag-panel">
140     <div class="diag-panel-header d-flex justify-content-betwe
en align-items-center">
141         <div class="d-flex align-items-center gap-2">
142             <i class="bi bi-clock-history text-info"></i>
143             <span class="text-white">Форензика запуска программ и

```



```

    время удержания фокуса (UserAssist ROT13)</span>
144     </div>
145     <span class="badge bg-secondary" id="diag-userassist-cou
    nt">0 записей</span>
146   </div>
147   <div class="table-responsive" style="max-height: 480px; ov
    erflow-y: auto;">
148     <table class="table table-dark table-hover table-sm mb-0
    " style="font-size: 0.80rem; font-family: ui-monospace, SFMono
    -Regular, Menlo, Monaco, Consolas, monospace;">
149       <thead>
150         <tr class="text-secondary border-secondary">
151           <th>Приложение / Программа</th>
152           <th style="width: 140px; text-align: right;">Счетч
    ик запусков</th>
153           <th style="width: 180px; text-align: right;">Время
    в фокусе</th>
154           <th>Полный путь / Идентификатор</th>
155         </tr>
156       </thead>
157       <tbody id="diag-userassist-tbody">
158         <tr><td colspan="4" class="text-center py-4 text-mut
    ed">Загрузка истории активности UserAssist...</td></tr>
159       </tbody>
160     </table>
161   </div>
162 </div>
163 </div>
164

```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows Behavioral Forensics Logic
4  * =====
5  * Description:
6  *   Клиентский контроллер вкладки форензики активности:
7  *   активное окно, время бездействия пользователя, Privacy Ra
8  *   dar и UserAssist.
9  *
10 * File: main.js
11 * Project: AI-Breadboard
12 * Module: WebInterface.ForensicsTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17 (function () {
18   'use strict';
19
20   let autoRefreshTimer = null;
21
22   function escapeHtml(str) {
23     if (str === null || str === undefined) return '';
24     return String(str)
25       .replace(/&/g, '&amp;')
26       .replace(/</g, '&lt;')
27       .replace(/>/g, '&gt;')
28       .replace(/"/g, '&quot;')
29       .replace(/'/g, '&#039;');
30   }
31
32   function setText(id, text) {
```

```

33     const el = document.getElementById(id);
34     if (el) el.textContent = text !== null && text !== undefined ? String(text) : '--';
35 }
36
37 async function apiFetch(url) {
38     if (window.api && typeof window.api.fetch === 'function')
39     {
40         return await window.api.fetch(url);
41     }
42     const res = await fetch(url);
43     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.statusText}`);
44     return await res.json();
45 }
46
47 async function fetchForensicsActivity() {
48     try {
49         const data = await apiFetch('/api/v1/system/diagnostics/forensics');
50         if (!data) return;
51
52         if (data.foreground_window) {
53             setText('diag-fg-title', data.foreground_window.title || 'Рабочий стол Windows');
54             setText('diag-fg-proc', `Процесс: ${data.foreground_window.process_name || 'unknown'} (PID: ${data.foreground_window.pid || 0})`);
55         }
56
57         setText('diag-user-idle-time', `${data.user_idle_seconds.toFixed(1)} сек`);
58
59         const statusBadge = document.getElementById('diag-user-status-badge');
60         if (statusBadge) {
61             if (data.user_idle_seconds > 300) {
62                 statusBadge.className = 'badge bg-warning-subtle text-warning border border-warning';
63             }
64         }
65     }
66 }

```

```

61 |         statusBadge.textContent = 'Пользователь отошел (>5 м
    | ин)';
62 |     } else {
63 |         statusBadge.className = 'badge bg-success-subtle tex
    | t-success border border-success';
64 |         statusBadge.textContent = 'Пользователь за ПК';
65 |     }
66 | }
67 |
68 | // Camera Apps
69 | const camContainer = document.getElementById('diag-cam-a
    | ccess-list');
70 | if (camContainer) {
71 |     if (data.camera_active_apps && data.camera_active_apps
    | .length > 0) {
72 |         camContainer.innerHTML = data.camera_active_apps.map
    | (a => `
73 |             <div class="d-flex align-items-center justify-cont
    | ent-between py-1 border-bottom border-secondary-subtle">
74 |                 <span class="text-truncate" style="max-width: 14
    | 0px;">${escapeHtml(a.app_name)}</span>
75 |                 <span class="badge ${a.is_active_now ? 'bg-dange
    | r' : 'bg-secondary'}" style="font-size: 0.65rem;">
76 |                     ${a.is_active_now ? 'Захват камеры' : 'В досту
    | пе'}
77 |                 </span>
78 |             </div>
79 |         `).join('');
80 |     } else {
81 |         camContainer.innerHTML = '<span class="text-muted sm
    | all">Нет активных обращений к камере</span>';
82 |     }
83 | }
84 |
85 | // Mic Apps
86 | const micContainer = document.getElementById('diag-mic-a
    | ccess-list');
87 | if (micContainer) {

```

```

88         if (data.microphone_active_apps && data.microphone_active_apps.length > 0) {
89             micContainer.innerHTML = data.microphone_active_apps
90             .map(a => `
91                 <div class="d-flex align-items-center justify-content-between py-1 border-bottom border-secondary-subtle">
92                     <span class="text-truncate" style="max-width: 140px;">${escapeHtml(a.app_name)}</span>
93                     <span class="badge ${a.is_active_now ? 'bg-danger' : 'bg-secondary'}" style="font-size: 0.65rem;">
94                         ${a.is_active_now ? 'Запись звука' : 'В доступе'}
95                     </span>
96                 </div>
97             `).join('');
98         } else {
99             micContainer.innerHTML = '<span class="text-muted small">Нет активных обращений к микрофону</span>';
100         }
101
102         // UserAssist Apps Table
103         const uaTbody = document.getElementById('diag-userassist-tbody');
104         if (uaTbody && data.userassist_top_apps) {
105             setText('diag-userassist-count', `${data.userassist_top_apps.length} программ`);
106             if (data.userassist_top_apps.length === 0) {
107                 uaTbody.innerHTML = '<tr><td colspan="4" class="text-center py-4 text-muted">Данных UserAssist в реестре не обнаружено</td></tr>';
108             } else {
109                 uaTbody.innerHTML = data.userassist_top_apps.map(a => `
110                     <tr>
111                         <td class="fw-bold text-white"><i class="bi bi-app text-info me-1.5"></i>${escapeHtml(a.name)}</td>
112                         <td style="text-align: right;" class="text-info

```

```

font-monospace">${a.run_count} paz</td>
113 |         <td style="text-align: right;" class="text-warni
font-monospace">${a.focus_formatted}</td>
114 |         <td class="small text-muted text-truncate" style
="max-width: 280px;" title="${escapeHtml(a.path)}">${escapeHtm
l(a.path)}</td>
115 |     </tr>
116 |     `).join('');
117 |     }
118 |     }
119 |     } catch (err) {
120 |         console.warn('[ForensicsTab] fetchForensicsActivity erro
r:', err);
121 |     }
122 | }
123 |
124 | function bindEvents() {
125 |     const btnRefresh = document.getElementById('btn-diag-foren
sics-refresh');
126 |     if (btnRefresh) {
127 |         btnRefresh.onclick = () => fetchForensicsActivity();
128 |     }
129 |
130 |     const autoSwitch = document.getElementById('diag-forensics
-auto-refresh');
131 |     if (autoSwitch) {
132 |         autoSwitch.onchange = (e) => {
133 |             if (e.target.checked) {
134 |                 autoRefreshTimer = setInterval(fetchForensicsActivit
y, 3000);
135 |             } else if (autoRefreshTimer) {
136 |                 clearInterval(autoRefreshTimer);
137 |                 autoRefreshTimer = null;
138 |             }
139 |         };
140 |     }
141 | }
142 |

```

```
143 |   async function init() {
144 |       bindEvents();
145 |       await fetchForensicsActivity();
146 |   }
147 |
148 |   if (document.readyState === 'loading') {
149 |       document.addEventListener('DOMContentLoaded', init);
150 |   } else {
151 |       setTimeout(init, 10);
152 |   }
153 | })();
154 |
```

index.html

```
1 <!-- Google Cloud Monitor Application Tab -->
2 <style>
3   .gcloud-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .gcloud-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    min-height: 58px;
15    gap: 0.2rem;
16  }
17  .gcloud-label {
18    font-size: 0.68rem;
19    text-transform: uppercase;
20    letter-spacing: 0.05em;
21    color: #94a3b8;
22    line-height: 1.1;
23  }
24  .gcloud-value {
25    font-size: 1.25rem;
26    font-weight: 700;
27    color: #f8fafc;
28    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
29    Consolas, monospace;
30    line-height: 1.15;
31  }
32  .gcloud-panel {
33    background: #0f172a !important;
34    border: 1px solid #334155 !important;
35    border-radius: 8px;
36    overflow: hidden;
```



```

36 | }
37 | .gcloud-panel-header {
38 |     background: #1e293b !important;
39 |     border-bottom: 1px solid #334155 !important;
40 |     padding: 0.45rem 0.85rem;
41 |     font-weight: 600;
42 |     font-size: 0.82rem;
43 |     color: #38bdf8;
44 |     display: flex;
45 |     align-items: center;
46 |     justify-content: space-between;
47 | }
48 | .gcloud-log-terminal {
49 |     background: #020617;
50 |     color: #cbd5e1;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
52 |     font-size: 0.78rem;
53 |     line-height: 1.45;
54 |     padding: 0.5rem 0.75rem;
55 |     border-radius: 6px;
56 |     max-height: 400px;
57 |     overflow-y: auto;
58 | }
59 | </style>
60 |
61 | <div class="gcloud-container">
62 |     <!-- Header Alert & Status (Compact Strip) -->
63 |     <div class="d-flex justify-content-between align-items-center
    flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style
    ="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%
    ); border: 1px solid #334155;">
64 |         <div class="d-flex align-items-center gap-2 flex-wrap">
65 |             <span class="fs-5">❏ </span>
66 |             <span class="fw-bold text-white small" style="font-size:
    0.92rem;">Google Cloud Monitor</span>
67 |             <span class="badge bg-dark border border-secondary text-
    info fw-normal" style="font-size: 0.7rem;">Observability</span>

```

```

>
68     <span class="text-secondary small d-none d-lg-inline" st
yle="font-size: 0.76rem;">| Cloud Logging, метрики Cloud Monit
oring & IAM</span>
69     </div>
70     <div class="d-flex align-items-center gap-1.5 flex-wrap">
71         <span class="badge rounded-pill bg-success-subtle text-s
uccess border border-success px-2 py-1" style="font-size: 0.72
rem;" id="gcloud-status-badge">
72             • GCP Подключен
73         </span>
74         <button class="btn btn-xs btn-outline-info rounded-pill
px-2.5 py-1" style="font-size: 0.75rem;" id="btn-gcloud-refres
h">
75             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
76         </button>
77         <button class="btn btn-xs btn-outline-warning rounded-pi
ll px-2 py-1" style="font-size: 0.75rem;" id="btn-gcloud-confi
g">
78             <i class="bi bi-gear-fill me-1"></i> Настройки
79         </button>
80     </div>
81 </div>
82
83 <!-- Top Metrics Bar (Compact Density Grid) -->
84 <div class="row g-2 mb-2">
85     <div class="col-6 col-md-3">
86         <div class="gcloud-card">
87             <span class="gcloud-label">Проект GCP</span>
88             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
89                 <span class="gcloud-value text-info text-truncate" i
d="gcloud-project-id">-</span>
90             </div>
91         </div>
92     </div>
93     <div class="col-6 col-md-3">
94         <div class="gcloud-card">

```

```
95 |         <span class="gcloud-label">Тип авторизации</span>
96 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
97 |             <span class="gcloud-value text-light" style="font-size: 1.05rem;" id="gcloud-auth-type">-</span>
98 |         </div>
99 |     </div>
100 | </div>
101 | <div class="col-6 col-md-3">
102 |     <div class="gcloud-card">
103 |         <span class="gcloud-label">Служебный аккаунт</span>
104 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
105 |             <span class="gcloud-value text-secondary text-truncate" style="font-size: 0.85rem;" id="gcloud-client-email">-</span>
106 |         </div>
107 |     </div>
108 | </div>
109 | <div class="col-6 col-md-3">
110 |     <div class="gcloud-card">
111 |         <span class="gcloud-label">Здоровье инфраструктуры</span>
112 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
113 |             <span class="gcloud-value text-success" id="gcloud-health-score">100/100</span>
114 |         </div>
115 |     </div>
116 | </div>
117 | </div>
118 |
119 | <!-- Main Content Panels Grid -->
120 | <div class="row g-3">
121 |     <!-- Cloud Logging Terminal Panel -->
122 |     <div class="col-lg-7">
123 |         <div class="gcloud-panel h-100">
124 |             <div class="gcloud-panel-header">
```

```

125 |         <span>☐ Cloud Logging (Последние события)</span>
126 |         <span class="badge bg-secondary" id="gcloud-logs-count-badge">0 строк</span>
127 |     </div>
128 |     <div class="p-2">
129 |         <div class="gcloud-log-terminal" id="gcloud-log-terminal-output">
130 |             Загрузка журналов Cloud Logging...
131 |         </div>
132 |     </div>
133 | </div>
134 | </div>
135 |
136 | <!-- AI Diagnostics & Recommendations Panel -->
137 | <div class="col-lg-5">
138 |     <div class="gcloud-panel h-100">
139 |         <div class="gcloud-panel-header">
140 |             <span>☐ AI-Диагностика и рекомендации</span>
141 |             <span class="badge bg-info text-dark">Gemini Diagnostics</span>
142 |         </div>
143 |         <div class="p-3">
144 |             <h6 class="fw-bold text-info mb-1" id="gcloud-diag-summary-title">Анализ телеметрии...</h6>
145 |             <p class="small text-light mb-3" id="gcloud-diag-summary-desc">Загрузка диагностического заключения...</p>
146 |             <div class="fw-semibold text-secondary small text-uppercase mb-2">Рекомендованные действия:</div>
147 |             <ul class="list-group list-group-flush small" id="gcloud-recommendations-list">
148 |                 <li class="list-group-item bg-transparent text-light border-secondary py-1 px-0">• Ожидание данных телеметрии...
149 |             </li>
150 |             </ul>
151 |         </div>
152 |     </div>
153 | </div>

```

154 | </div>
155 |

main.js

```
1 // =====
  =====
2 // Process Name: Google Cloud Monitor Web Tab Module
3 // =====
  =====
4 // Description:
5 //   Client-side JavaScript controller for the Google Cloud Mo
  nitor
6 //   administrative web interface tab.
7 //
8 // File: main.js
9 // Project: ai-breadboard
10 // Package: src.api.webinterface.gcloud_tab
11 // Author: hypo69
12 // Copyright: © 2026 hypo69
13 // =====
  =====
14
15 (function() {
16   let isGcloudInitialized = false;
17
18   async function fetchStatus() {
19     try {
20       const res = await fetch('/api/gcloud/status');
21       if (!res.ok) return;
22       const data = await res.json();
23
24       const badge = document.getElementById('gcloud-status-bad
  ge');
25       const projectEl = document.getElementById('gcloud-projec
  t-id');
26       const authTypeEl = document.getElementById('gcloud-auth-
  type');
27       const emailEl = document.getElementById('gcloud-client-e
  mail');
28
```

```
29 |         if (badge) {
30 |             badge.className = data.authenticated ? 'badge rounded-
| pill bg-success-subtle text-success border border-success px-3
| py-2' : 'badge rounded-pill bg-warning-subtle text-warning bo
| rder border-warning px-3 py-2';
31 |             badge.innerText = data.authenticated ? (data.is_mock ?
| '● Mock Режим' : '● GCP Подключен') : '● Не авторизован';
32 |         }
33 |
34 |         if (projectEl) projectEl.innerText = data.project_id ||
| 'default-project';
35 |         if (authTypeEl) authTypeEl.innerText = (data.auth_type |
| | 'service_account').toUpperCase();
36 |         if (emailEl) emailEl.innerText = data.client_email || 'l
| ocal-service-account';
37 |     } catch (e) {
38 |         console.error('[GCloudTab] Failed to fetch status:', e);
39 |     }
40 | }
41 |
42 | async function fetchLogs() {
43 |     try {
44 |         const res = await fetch('/api/gcloud/logs?limit=40');
45 |         if (!res.ok) return;
46 |         const logs = await res.json();
47 |
48 |         const terminal = document.getElementById('gcloud-log-ter
| minal-output');
49 |         const countBadge = document.getElementById('gcloud-logs-
| count-badge');
50 |         if (countBadge) countBadge.innerText = `${logs.length} c
| трок`;
51 |
52 |         if (terminal) {
53 |             if (!Array.isArray(logs) || logs.length === 0) {
54 |                 terminal.innerText = 'Журнал Cloud Logging пуст.';
55 |                 return;
56 |             }
```

```

57 |         terminal.innerHTML = logs.map(l => {
58 |             const sev = (l.severity || 'INFO').toUpperCase();
59 |             const color = sev === 'ERROR' || sev === 'CRITICAL'
60 | ? '#f87171' : (sev === 'WARNING' ? '#fbbf24' : '#94a3b8');
61 |             return `<div style="color: ${color};">[${l.timestamp
62 | || ' '}] [${sev}] ${l.resource_type ? '[' + l.resource_type +
63 | ']' ' : ''}${l.message || ''}</div>`;
64 |         }).join('');
65 |     }
66 | } catch (e) {
67 |     console.error('[GCloudTab] Failed to fetch logs:', e);
68 | }
69 |
70 | async function runDiagnostics() {
71 |     try {
72 |         const res = await fetch('/api/gcloud/diagnostic');
73 |         if (!res.ok) return;
74 |         const report = await res.json();
75 |
76 |         const scoreEl = document.getElementById('gcloud-health-s
77 | core');
78 |         const titleEl = document.getElementById('gcloud-diag-sum
79 | mary-title');
80 |         const descEl = document.getElementById('gcloud-diag-summ
81 | ary-desc');
82 |         const recList = document.getElementById('gcloud-recommen
83 | dations-list');
84 |
85 |         if (scoreEl) scoreEl.innerText = `${report.health_score
86 | || 100}/100`;
87 |         if (titleEl) titleEl.innerText = `Статус: ${report.overa
88 | ll_status || 'HEALTHY'}`;
89 |         if (descEl) descEl.innerText = report.summary || 'Все се
90 | рвисы Google Cloud функционируют в штатном режиме.';
91 |
92 |         if (recList && Array.isArray(report.recommendations)) {
93 |             recList.innerHTML = report.recommendations.map(r => `

```



```

85 |         <li class="list-group-item bg-transparent text-light
      border-secondary py-1 px-0">• ${r}</li>
86 |     `).join('');
87 |     }
88 | } catch (e) {
89 |     console.error('[GCloudTab] Diagnostic error:', e);
90 | }
91 | }
92 |
93 | function initGCloudTab() {
94 |     fetchStatus();
95 |     fetchLogs();
96 |     runDiagnostics();
97 |
98 |     if (!isGcloudInitialized) {
99 |         const refreshBtn = document.getElementById('btn-gcloud-r
      efresh');
100 |         if (refreshBtn) {
101 |             refreshBtn.addEventListener('click', () => {
102 |                 fetchStatus();
103 |                 fetchLogs();
104 |                 runDiagnostics();
105 |             });
106 |         }
107 |
108 |         const configBtn = document.getElementById('btn-gcloud-co
      nfig');
109 |         if (configBtn) {
110 |             configBtn.addEventListener('click', () => {
111 |                 if (typeof window.openAppConfigModal === 'function')
      {
112 |                     window.openAppConfigModal('gcloud_monitor');
113 |                 }
114 |             });
115 |         }
116 |         isGcloudInitialized = true;
117 |     }
118 | }

```

```
119 |  
120 |     window.initGCloudTab = initGCloudTab;  
121 | })();  
122 |
```

index.html

```
1 <!-- Google Workspace Accounts Pool Management Tab -->
2 <div class="container-fluid py-2" id="google-accounts-tab-root
  ">
3   <!-- Top Banner & Stats -->
4   <div class="d-flex flex-wrap align-items-center justify-cont
  ent-between gap-3 mb-3 p-3 bg-body-tertiary border rounded sha
  dow-sm">
5     <div>
6       <h5 class="mb-1 text-primary d-flex align-items-center g
  ap-2">
7         <i class="bi bi-google text-danger"></i>
8         <span>Управление аккаунтами Google Workspace</span>
9       </h5>
10      <p class="mb-0 text-muted small">Пул учетных записей (OA
  uth 2.0 и Service Accounts) для интеграции с Gmail, Google Dri
  ve, Sheets и Docs.</p>
11    </div>
12    <div class="d-flex gap-2">
13      <button class="btn btn-outline-primary btn-sm d-flex ali
  gn-items-center gap-1 rounded-pill px-3" id="tab-btn-refresh-g
  accounts" title="Обновить список">
14        <i class="bi bi-arrow-clockwise"></i> Обновить
15      </button>
16    </div>
17  </div>
18
19  <!-- Summary Metric Cards -->
20  <div class="row g-3 mb-3">
21    <div class="col-6 col-md-3">
22      <div class="card bg-body-tertiary border h-100 shadow-sm
  ">
23        <div class="card-body p-3 d-flex align-items-center ju
  stify-content-between">
24          <div>
25            <div class="text-muted small">Всего аккаунтов</div>
  >
```

```

26 |         <h4 class="mb-0 fw-bold" id="gstat-total-accounts"
    | >0</h4>
27 |     </div>
28 |     <div class="bg-secondary bg-opacity-25 rounded p-2 t
    | ext-primary">
29 |         <i class="bi bi-collection-fill fs-4"></i>
30 |     </div>
31 | </div>
32 | </div>
33 | </div>
34 | <div class="col-6 col-md-3">
35 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
36 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
37 |             <div>
38 |                 <div class="text-muted small">Активные</div>
39 |                 <h4 class="mb-0 text-success fw-bold" id="gstat-ac
    | tive-accounts">0</h4>
40 |             </div>
41 |             <div class="bg-success bg-opacity-25 rounded p-2 tex
    | t-success">
42 |                 <i class="bi bi-check-circle-fill fs-4"></i>
43 |             </div>
44 |         </div>
45 |     </div>
46 | </div>
47 | <div class="col-6 col-md-3">
48 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
49 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
50 |             <div>
51 |                 <div class="text-muted small">OAuth 2.0 (User)</di
    | v>
52 |                 <h4 class="mb-0 text-info fw-bold" id="gstat-oauth
    | -accounts">0</h4>
53 |             </div>

```

```

54 |         <div class="bg-info bg-opacity-25 rounded p-2 text-i
    | nfo">
55 |             <i class="bi bi-person-badge-fill fs-4"></i>
56 |         </div>
57 |     </div>
58 | </div>
59 | </div>
60 | <div class="col-6 col-md-3">
61 |     <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
62 |         <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
63 |             <div>
64 |                 <div class="text-muted small">Service Accounts</di
    | v>
65 |                 <h4 class="mb-0 text-warning fw-bold" id="gstat-sa
    | -accounts">0</h4>
66 |             </div>
67 |             <div class="bg-warning bg-opacity-25 rounded p-2 tex
    | t-warning">
68 |                 <i class="bi bi-cpu-fill fs-4"></i>
69 |             </div>
70 |         </div>
71 |     </div>
72 | </div>
73 | </div>
74 |
75 | <!-- Info Alert -->
76 | <div class="alert alert-info border-0 small shadow-sm rounde
    | d-3 mb-4" role="alert">
77 |     <div class="d-flex align-items-center gap-2 mb-1">
78 |         <i class="bi bi-info-circle-fill text-info fs-5"></i>
79 |         <strong class="text-info">Автономная ротация и приоритет
    | ы аккаунтов</strong>
80 |     </div>
81 |     <p class="mb-1 text-secondary">
82 |         Агенты и инструменты AI (<code>gmail_search</code>, <cod
    | e>gmail_create_draft</code>, <code>gdrive_list_files</code>, <

```

`gsheets_read`, `gsheets_append`) могут явно вызывать нужный аккаунт через параметр `account_name="work"`, либо автоматически использовать основной аккаунт по умолчанию (*Default*).

83

84

>

85 Учетные записи и токены авторизации хранятся в `src/secrets/` в изолированном защищенном виде.

86

87

88

89

>

90 <!-- Add Account Card -->

91

>

92

93

94

######

95 *Добавить Google аккаунт*

96

97 <button type="button" class="btn btn-outline-info btn-sm py-0 px-2 rounded-pill d-flex align-items-center gap-1" data-bs-toggle="modal" data-bs-target="#googleCredsHelpModal" style="font-size: 0.75rem;">

98 *Инструкция*

99

100

101

102 <form id="tab-form-add-gaccount" onsubmit="event.preventDefault();">

103

104 <label class="form-label small text-muted fw-bold">Имя аккаунта (ID): *</label>

105 <input type="text" class="form-control form-cont

```

    rol-sm" id="tab-new-gacc-name" placeholder="Например: work, pe
    rsonal, bot" autocomplete="off" required>
106         <div class="form-text small text-secondary">Уник
    альный системный идентификатор для вызова в агентах.</div>
107     </div>
108
109     <div class="mb-3">
110         <label class="form-label small text-muted fw-bol
    d">Тип аутентификации:</label>
111         <select class="form-select form-select-sm" id="t
    ab-new-gacc-type">
112             <option value="oauth2">OAuth 2.0 Client (User
    Login / Desktop App)</option>
113             <option value="service_account">Service Accoun
    t Key (Server JSON)</option>
114         </select>
115     </div>
116
117     <div class="mb-3">
118         <label class="form-label small text-muted fw-bol
    d">Email (опционально):</label>
119         <input type="email" class="form-control form-con
    trol-sm" id="tab-new-gacc-email" placeholder="user@gmail.com и
    ли sa@project.iam.gserviceaccount.com" autocomplete="off">
120     </div>
121
122     <!-- Credentials Input Mode Toggle -->
123     <div class="mb-3">
124         <label class="form-label small text-muted fw-bol
    d d-flex justify-content-between align-items-center">
125             <span>Учетные данные (Credentials JSON): <span
    class="text-danger">*</span></span>
126             <span class="badge bg-info text-dark" role="bu
    tton" data-bs-toggle="modal" data-bs-target="#googleCredsHelpM
    odal" style="cursor: pointer; font-size: 0.72rem;">
127                 <i class="bi bi-question-circle-fill me-1"><
    /i>Как получить JSON?
128             </span>

```

```
129         </label>
130         <ul class="nav nav-pills nav-fill mb-2" id="tab-
gacc-input-mode-tabs" role="tablist">
131             <li class="nav-item" role="presentation">
132                 <button class="nav-link active py-1 small" i
d="tab-gacc-mode-file-tab" data-bs-toggle="pill" data-bs-targe
t="#tab-gacc-mode-file" type="button" role="tab"> Загрузить
файл</button>
133             </li>
134             <li class="nav-item" role="presentation">
135                 <button class="nav-link py-1 small" id="tab-
gacc-mode-json-tab" data-bs-toggle="pill" data-bs-target="#tab
-gacc-mode-json" type="button" role="tab"> Вставить JSON</bu
tton>
136             </li>
137         </ul>
138
139         <div class="tab-content" id="tab-gacc-input-mode
-tabContent">
140             <div class="tab-pane fade show active" id="tab
-gacc-mode-file" role="tabpanel">
141                 <input class="form-control form-control-sm"
type="file" id="tab-new-gacc-file" accept=".json,application/j
son">
142                 <div class="form-text small text-secondary">
Загрузите <code>credentials.json</code> или <code>service_accou
nt.json</code> из <a href="https://console.cloud.google.com/a
pis/credentials" target="_blank" rel="noopener noreferrer" cla
ss="text-info text-decoration-none">Google Cloud Console <i cl
ass="bi bi-box-arrow-up-right ms-1" style="font-size: 0.75rem;
"></i></a>.</div>
143             </div>
144             <div class="tab-pane fade" id="tab-gacc-mode-j
son" role="tabpanel">
145                 <textarea class="form-control font-monospace
form-control-sm" id="tab-new-gacc-json" rows="4" placeholder=
'{"installed": {"client_id": "...", "client_secret": "..."}}'>
</textarea>
```



```

146         </div>
147     </div>
148 </div>
149
150     <div class="form-check mb-4 mt-2">
151         <input class="form-check-input" type="checkbox"
152         id="tab-new-gacc-default" style="cursor: pointer;">
153         <label class="form-check-label small fw-semibold
154         " for="tab-new-gacc-default" style="cursor: pointer;">
155             Сделать аккаунтом по умолчанию (Default)
156         </label>
157     </div>
158
159     <button type="button" class="btn btn-primary btn-s
160     m w-100 rounded-pill fw-bold" id="tab-btn-add-gacc">
161         <i class="bi bi-plus-circle me-1"></i> Сохранить
162         Google аккаунт
163     </button>
164 </form>
165 </div>
166 </div>
167 </div>
168
169 <!-- Accounts Table -->
170 <div class="col-lg-7">
171     <div class="card shadow-sm border mb-3">
172         <div class="card-header bg-transparent border-bottom p
173         y-2 d-flex justify-content-between align-items-center">
174             <h6 class="fw-bold mb-0"><i class="bi bi-list-check
175             me-1"></i> Зарегистрированные аккаунты</h6>
176             <span class="badge bg-secondary" id="tab-gaccounts-c
177             ount-badge">0</span>
178         </div>
179         <div class="card-body p-0">
180             <div class="table-responsive">
181                 <table class="table table-hover align-middle mb-0"
182                 id="tab-gaccounts-table">
183                     <thead class="table-light small text-uppercase">

```

```

176 |         <tr>
177 |             <th>Аккаунт</th>
178 |             <th style="width: 130px;" class="text-center
    | ">Тип</th>
179 |             <th style="width: 100px;" class="text-center
    | ">Статус</th>
180 |             <th style="width: 110px;" class="text-center
    | ">Токен</th>
181 |             <th style="width: 180px;" class="text-center
    | ">Действия</th>
182 |         </tr>
183 |     </thead>
184 |     <tbody id="tab-gaccounts-list-body">
185 |         <tr>
186 |             <td colspan="5" class="text-center py-4 text
    | -muted">Загрузка аккаунтов...</td>
187 |         </tr>
188 |     </tbody>
189 | </table>
190 | </div>
191 | </div>
192 | </div>
193 |
194 | <!-- Test & Diagnostics Output Card -->
195 | <div class="card bg-body-tertiary border shadow-sm" id="
    | tab-gaccount-test-card" style="display: none;">
196 |     <div class="card-header bg-transparent border-bottom d
    | -flex justify-content-between align-items-center py-2">
197 |         <span class="small fw-bold text-info"><i class="bi b
    | i-terminal me-1"></i> Результат проверки доступа</span>
198 |         <button class="btn btn-sm btn-outline-secondary py-0
    | px-2" type="button" onclick="document.getElementById('tab-gac
    | count-test-card').style.display='none'">❏ </button>
199 |     </div>
200 |     <div class="card-body p-3 font-monospace small bg-body
    | " id="tab-gaccount-test-body" style="max-height: 200px; overfl
    | ow-y: auto; white-space: pre-wrap;">
201 |     </div>

```

```

202     </div>
203 </div>
204 </div>
205
206 <!-- Google Credentials Instruction Modal -->
207 <div class="modal fade" id="googleCredsHelpModal" tabindex="
-1" aria-labelledby="googleCredsHelpModalLabel" aria-hidden="t
rue">
208     <div class="modal-dialog modal-lg modal-dialog-centered mo
dal-dialog-scrollable">
209         <div class="modal-content shadow-lg">
210             <div class="modal-header">
211                 <h5 class="modal-title d-flex align-items-center gap
-2" id="googleCredsHelpModalLabel">
212                     <i class="bi bi-google text-danger"></i> Как получ
ить и загрузить Credentials JSON
213                 </h5>
214                 <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Close"></button>
215             </div>
216             <div class="modal-body p-4">
217                 <ul class="nav nav-tabs mb-3" id="helpModalTabs" rol
e="tablist">
218                     <li class="nav-item" role="presentation">
219                         <button class="nav-link active" id="help-oauth-t
ab" data-bs-toggle="tab" data-bs-target="#help-oauth-pane" typ
e="button" role="tab">
220                             <i class="bi bi-person-check-fill text-primary
me-1"></i> 1. OAuth 2.0 Client (Личный вход)
221                         </button>
222                     </li>
223                     <li class="nav-item" role="presentation">
224                         <button class="nav-link" id="help-sa-tab" data-b
s-toggle="tab" data-bs-target="#help-sa-pane" type="button" ro
le="tab">
225                             <i class="bi bi-robot text-warning me-1"></i>
2. Service Account (Сервисный ключ)
226                         </button>

```

```

227         </li>
228     </ul>
229
230     <div class="tab-content">
231         <!-- OAuth 2.0 Pane -->
232         <div class="tab-pane fade show active" id="help-oa
233         uth-pane" role="tabpanel">
234             <p class="text-secondary small">
235                 Используется для работы от имени вашего личног
236                 о или рабочего аккаунта Google (чтение/отправка почты в Gmail,
237                 доступ к Google Drive, редактирование Google Sheets/Docs).
238             </p>
239             <ol class="list-group list-group-numbered list-g
240             roup-flush small">
241                 <li class="list-group-item">
242                     Перейдите в консоль управления Google: <br>
243                     <a href="https://console.cloud.google.com/ap
244                     is/credentials" target="_blank" rel="noopener noreferrer" clas
245                     s="btn btn-sm btn-outline-info mt-1">
246                         <i class="bi bi-box-arrow-up-right me-1"><
247                         /i> Google Cloud Console □ Credentials
248                     </a>
249                 </li>
250                 <li class="list-group-item">
251                     Если проект еще не создан, выберите существу
252                     ющий или создайте новый проект (кнопка в верхней панели).
253                 </li>
254                 <li class="list-group-item">
255                     В разделе <strong>OAuth consent screen</stro
256                     ng> (Экран согласия OAuth):
257                     <ul class="mt-1 text-secondary">
258                         <li>Выберите тип: <code>External</code> (В
259                         нешний) или <code>Internal</code>.</li>
260                         <li>Укажите название приложения (например,
261                         <code>AI Breadboard</code>) и ваш контактный email.</li>
262                         <li>В разделе <em>Test users</em> добавьте
263                         ваш Google email (если статус приложения Testing).</li>
264                     </ul>

```

```

253 |         </li>
254 |         <li class="list-group-item">
255 |             В разделе <strong>Credentials</strong> нажми
| те <code>+ CREATE CREDENTIALS</code> □ выберите <strong>OAuth
| client ID</strong>.
256 |         </li>
257 |         <li class="list-group-item">
258 |             В поле <em>Application type</em> выберите <s
| trong>Desktop app</strong> (Приложение для ПК) и задайте имя (
| например <code>Breadboard-Desktop</code>), затем нажмите <stro
| ng>Create</strong>.
259 |         </li>
260 |         <li class="list-group-item">
261 |             В появившемся окне нажмите <i class="bi bi-d
| ownload text-info"></i> <strong>DOWNLOAD JSON</strong> (или ск
| ачайте файл в таблице OAuth 2.0 Client IDs).
262 |         </li>
263 |         <li class="list-group-item">
264 |             Включите необходимые API в <a href="https://
| console.cloud.google.com/apis/dashboard" target="_blank" rel="
| noopener noreferrer" class="text-info">Enabled APIs & Services
| <i class="bi bi-box-arrow-up-right"></i></a>:
265 |                 <div class="mt-1 d-flex flex-wrap gap-1">
266 |                     <span class="badge bg-secondary">Gmail API
| </span>
267 |                     <span class="badge bg-secondary">Google Dr
| ive API</span>
268 |                     <span class="badge bg-secondary">Google Sh
| eets API</span>
269 |                     <span class="badge bg-secondary">Google Do
| cs API</span>
270 |                 </div>
271 |             </li>
272 |             <li class="list-group-item">
273 |                 В этой карточке выберите <strong>Загрузить ф
| айл</strong> и укажите скачанный <code>credentials.json</code>
| (или вставьте его содержимое во вкладку <strong>Вставить JSON
| </strong>), введите ID аккаунта и нажмите <strong>Сохранить Go

```

```

    ogle аккаунт</strong>.
274         </li>
275     </ol>
276 </div>
277
278     <!-- Service Account Pane -->
279     <div class="tab-pane fade" id="help-sa-pane" role=
"tabpanel">
280         <p class="text-secondary small">
281             Используется для серверных / автономных фоновы
х задач без необходимости интерактивного логина в браузере.
282         </p>
283         <ol class="list-group list-group-numbered list-g
roup-flush small">
284             <li class="list-group-item">
285                 Перейдите в раздел сервисных аккаунтов: <br>
286                 <a href="https://console.cloud.google.com/ia
m-admin/serviceaccounts" target="_blank" rel="noopener norefer
rer" class="btn btn-sm btn-outline-warning mt-1">
287                     <i class="bi bi-box-arrow-up-right me-1"><
/i> Google Cloud Console □ Service Accounts
288                 </a>
289             </li>
290             <li class="list-group-item">
291                 Нажмите <code>+ CREATE SERVICE ACCOUNT</code>
>, введите имя (например <code>ai-agent-sa</code>) и нажмите <
strong>Done</strong>.
292             </li>
293             <li class="list-group-item">
294                 Кликните по созданному сервисному аккаунту и
перейдите во вкладку <strong>Keys</strong> (Ключи).
295             </li>
296             <li class="list-group-item">
297                 Нажмите <strong>ADD KEY</strong> □ <strong>C
reate new key</strong> □ выберите тип <strong>JSON</strong> □
нажмите <strong>Create</strong>.
298             </li>
299             <li class="list-group-item">

```

```
300 |             Файл ключа (<code>project-id-*.json</code>)
      | скачается на ваш компьютер.
301 |         </li>
302 |         <li class="list-group-item">
303 |             В форме выберите тип <em>Service Account Key
      | </em>, загрузите JSON файл и сохраните аккаунт.
304 |         </li>
305 |     </ol>
306 | </div>
307 | </div>
308 | </div>
309 | <div class="modal-footer">
310 |     <button type="button" class="btn btn-secondary btn-s
      | m rounded-pill px-4" data-bs-dismiss="modal">Понятно</button>
311 | </div>
312 | </div>
313 | </div>
314 | </div>
315 | </div>
316 |
317 |
```

main.js

```
1 // Google Workspace Accounts Tab JavaScript Logic
2
3 function escapeHtml(text) {
4     if (!text) return '';
5     return String(text)
6         .replace(/&/g, '&amp;')
7         .replace(/</g, '&lt;')
8         .replace(/>/g, '&gt;')
9         .replace(/"/g, '&quot;')
10        .replace(/'/g, '&#039;');
11 }
12
13 function showGAccountsNotification(message, type = 'info') {
14     const notification = document.createElement('div');
15     notification.className = `alert alert-${type} position-fixed
16         top-0 end-0 m-3 shadow-lg`;
17     notification.style.zIndex = '9999';
18     notification.style.maxWidth = '400px';
19     notification.innerHTML = message;
20     document.body.appendChild(notification);
21
22     setTimeout(() => {
23         notification.remove();
24     }, 4000);
25 }
26
27 async function refreshTabGoogleAccountsList() {
28     const container = document.getElementById('tab-gaccounts-list-body');
29     if (!container) return;
30     try {
31         container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted"><span class="spinner-border spinner-border-sm me-2 text-primary"></span>Загрузка аккаунтов...</td></tr>';
32     }
```



```

32     const data = await window.api.googleAccounts.list();
33     const accounts = data.accounts || [];
34
35     // Update stats
36     const totalEl = document.getElementById('gstat-total-accounts');
37     const activeEl = document.getElementById('gstat-active-accounts');
38     const oauthEl = document.getElementById('gstat-oauth-accounts');
39     const saEl = document.getElementById('gstat-sa-accounts');
40     const countBadge = document.getElementById('tab-gaccounts-count-badge');
41
42     if (totalEl) totalEl.textContent = accounts.length;
43     if (activeEl) activeEl.textContent = accounts.filter(a =>
44     a.status === 'active').length;
45     if (oauthEl) oauthEl.textContent = accounts.filter(a => a.
46     type === 'oauth2').length;
47     if (saEl) saEl.textContent = accounts.filter(a => a.type =
48     == 'service_account').length;
49     if (countBadge) countBadge.textContent = `${accounts.length} акк.`;
50
51     if (accounts.length === 0) {
52         container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Список аккаунтов пуст. Загрузите credentials.json или service_account.json.</td></tr>';
53         return;
54     }
55
56     container.innerHTML = '';
57     accounts.forEach(acc => {
58         const row = document.createElement('tr');
59
60         // 1. Account Name + Default Badge + Email
61         const tdName = document.createElement('td');
62         tdName.innerHTML = `

```

```

60 |         <div class="d-flex align-items-center gap-2">
61 |             <strong>${escapeHtml(acc.name)}</strong>
62 |             ${acc.is_default ? '<span class="badge bg-warning te
| xt-dark"><i class="bi bi-star-fill"></i> Default</span>' : ''}
63 |         </div>
64 |         ${acc.email ? `<div class="small text-muted font-monos
| pace">${escapeHtml(acc.email)}</div>` : ''}
65 |         `;
66 |         row.appendChild(tdName);
67 |
68 |         // 2. Type
69 |         const tdType = document.createElement('td');
70 |         tdType.className = 'text-center';
71 |         const isOAuth = acc.type === 'oauth2';
72 |         tdType.innerHTML = isOAuth
73 |             ? '<span class="badge bg-primary-subtle text-primary b
| order border-primary-subtle"><i class="bi bi-person me-1"></i>
| OAuth 2.0</span>'
74 |             : '<span class="badge bg-info-subtle text-info border
| border-info-subtle"><i class="bi bi-cpu me-1"></i>Service Acco
| unt</span>';
75 |         row.appendChild(tdType);
76 |
77 |         // 3. Status / Quota
78 |         const tdStatus = document.createElement('td');
79 |         tdStatus.className = 'text-center';
80 |         const isActive = acc.status === 'active';
81 |         if (isActive) {
82 |             tdStatus.innerHTML = '<span class="badge bg-success">A
| ктивен</span>';
83 |         } else if (acc.status === 'exhausted') {
84 |             tdStatus.innerHTML = '<span class="badge bg-danger" ti
| tle="Исчерпан суточный лимит квоты">Лимит</span>';
85 |         } else {
86 |             tdStatus.innerHTML = `<span class="badge bg-secondary"
| >${escapeHtml(acc.status)}</span>`;
87 |         }
88 |         row.appendChild(tdStatus);

```

```
89 |
90 |     // 4. Token / Credentials Status
91 |     const tdToken = document.createElement('td');
92 |     tdToken.className = 'text-center';
93 |     if (acc.has_token || acc.type === 'service_account') {
94 |         tdToken.innerHTML = '<span class="badge bg-success-subtle text-success border border-success-subtle"><i class="bi bi-shield-check me-1"></i>Готов</span>';
95 |     } else {
96 |         tdToken.innerHTML = '<span class="badge bg-warning-subtle text-warning border border-warning-subtle"><i class="bi bi-key me-1"></i>Нужен токен</span>';
97 |     }
98 |     row.appendChild(tdToken);
99 |
100 |    // 5. Actions
101 |    const tdActions = document.createElement('td');
102 |    tdActions.className = 'text-center';
103 |
104 |    // Set Default button
105 |    if (!acc.is_default) {
106 |        const btnDefault = document.createElement('button');
107 |        btnDefault.className = 'btn btn-xs btn-outline-warning btn-sm me-1';
108 |        btnDefault.title = 'Сделать аккаунтом по умолчанию';
109 |        btnDefault.innerHTML = '<i class="bi bi-star"></i>';
110 |        btnDefault.onclick = () => setTabGoogleAccountDefault(acc.name);
111 |        tdActions.appendChild(btnDefault);
112 |    }
113 |
114 |    // Reset status button
115 |    if (acc.status === 'exhausted') {
116 |        const btnReset = document.createElement('button');
117 |        btnReset.className = 'btn btn-xs btn-outline-info btn-sm me-1';
118 |        btnReset.title = 'Сбросить статус исчерпания';
119 |        btnReset.innerHTML = '<i class="bi bi-arrow-repeat"></i>';
```

```

    i>';
120     btnReset.onclick = () => resetTabGoogleAccountStatus(a
cc.name);
121     tdActions.appendChild(btnReset);
122     }
123
124     // Test button
125     const btnTest = document.createElement('button');
126     btnTest.className = 'btn btn-xs btn-outline-info btn-sm
me-1';
127     btnTest.title = 'Проверить доступ и авторизацию';
128     btnTest.innerHTML = '<i class="bi bi-play-circle"></i> Т
ест';
129     btnTest.onclick = () => testTabGoogleAccount(acc.name);
130     tdActions.appendChild(btnTest);
131
132     // Delete button
133     const btnDelete = document.createElement('button');
134     btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
135     btnDelete.title = 'Удалить аккаунт из пула';
136     btnDelete.innerHTML = '<i class="bi bi-trash"></i>';
137     btnDelete.onclick = () => deleteTabGoogleAccount(acc.nam
e);
138     tdActions.appendChild(btnDelete);
139
140     row.appendChild(tdActions);
141     container.appendChild(row);
142     });
143
144     } catch (err) {
145         console.error('Ошибка загрузки аккаунтов Google Workspace:
', err);
146         container.innerHTML = `<tr><td colspan="5" class="text-cen
ter py-4 text-danger">Ошибка: ${escapeHtml(err.message)}</td><
/tr>`;
147     }
148 }

```

```
149 |
150 | async function handleAddTabGoogleAccount() {
151 |     const nameInput = document.getElementById('tab-new-gacc-name
    | ');
152 |     const typeSelect = document.getElementById('tab-new-gacc-typ
    | e');
153 |     const emailInput = document.getElementById('tab-new-gacc-ema
    | il');
154 |     const fileInput = document.getElementById('tab-new-gacc-file
    | ');
155 |     const jsonTextarea = document.getElementById('tab-new-gacc-j
    | son');
156 |     const defaultCheckbox = document.getElementById('tab-new-gac
    | c-default');
157 |     const addBtn = document.getElementById('tab-btn-add-gacc');
158 |
159 |     if (!nameInput || !addBtn) return;
160 |
161 |     const accountName = nameInput.value.trim();
162 |     const accountType = typeSelect ? typeSelect.value : 'oauth2'
    | ;
163 |     const email = emailInput ? emailInput.value.trim() : '';
164 |     const setAsDefault = defaultCheckbox ? defaultCheckbox.check
    | ed : false;
165 |
166 |     if (!accountName) {
167 |         showGAccountsNotification('Введите имя аккаунта (например:
    | work, personal)', 'warning');
168 |         nameInput.focus();
169 |         return;
170 |     }
171 |
172 |     const file = fileInput && fileInput.files && fileInput.files
    | [0];
173 |     const jsonContent = jsonTextarea ? jsonTextarea.value.trim()
    | : '';
174 |
175 |     if (!file && !jsonContent) {
```

```
176 |     showGAccountsNotification('Загрузите файл credentials.json
    | / service_account.json или вставьте JSON', 'warning');
177 |     return;
178 | }
179 |
180 | addBtn.disabled = true;
181 | const originalText = addBtn.innerHTML;
182 | addBtn.innerHTML = '<span class="spinner-border spinner-bord
    | er-sm me-1"></span> Сохранение...';
183 |
184 | try {
185 |     if (file) {
186 |         const formData = new FormData();
187 |         formData.append('account_name', accountName);
188 |         formData.append('account_type', accountType);
189 |         if (email) formData.append('email', email);
190 |         formData.append('set_as_default', setAsDefault ? 'true'
    | : 'false');
191 |         formData.append('file', file);
192 |
193 |         await window.api.googleAccounts.upload(formData);
194 |     } else {
195 |         let parsedJson = null;
196 |         try {
197 |             parsedJson = JSON.parse(jsonContent);
198 |         } catch (e) {
199 |             throw new Error('Некорректный JSON в поле учетных данн
    | ых');
200 |         }
201 |
202 |         await window.api.googleAccounts.create({
203 |             account_name: accountName,
204 |             account_type: accountType,
205 |             email: email || undefined,
206 |             credentials_dict: parsedJson,
207 |             set_as_default: setAsDefault
208 |         });
209 |     }
```

```
210
211     showGAccountsNotification(`Аккаунт Google "${accountName}"
    успешно сохранен в пул`, 'success');
212     nameInput.value = '';
213     if (emailInput) emailInput.value = '';
214     if (fileInput) fileInput.value = '';
215     if (jsonTextarea) jsonTextarea.value = '';
216     if (defaultCheckbox) defaultCheckbox.checked = false;
217
218     await refreshTabGoogleAccountsList();
219   } catch (err) {
220     console.error('Ошибка сохранения аккаунта Google:', err);
221     showGAccountsNotification('Ошибка сохранения: ' + err.mess
    age, 'danger');
222   } finally {
223     addBtn.disabled = false;
224     addBtn.innerHTML = originalText;
225   }
226 }
227
228 async function setTabGoogleAccountDefault(name) {
229   try {
230     await window.api.googleAccounts.setDefault(name);
231     showGAccountsNotification(`Аккаунт "${name}" назначен по у
    молчанию`, 'success');
232     await refreshTabGoogleAccountsList();
233   } catch (err) {
234     console.error('Ошибка установки аккаунта по умолчанию:', e
    rr);
235     showGAccountsNotification('Ошибка: ' + err.message, 'dange
    r');
236   }
237 }
238
239 async function resetTabGoogleAccountStatus(name) {
240   try {
241     await window.api.googleAccounts.resetStatus(name);
242     showGAccountsNotification(`Статус аккаунта "${name}" сброс
```

```

    ен в активный`, 'success');
243     await refreshTabGoogleAccountsList();
244   } catch (err) {
245     console.error('Ошибка сброса статуса аккаунта:', err);
246     showGAccountsNotification('Ошибка сброса: ' + err.message,
    'danger');
247   }
248 }
249
250 async function testTabGoogleAccount(name) {
251   const testCard = document.getElementById('tab-gaccount-test-
card');
252   const testBody = document.getElementById('tab-gaccount-test-
body');
253
254   if (testCard && testBody) {
255     testCard.style.display = 'block';
256     testBody.innerHTML = `

```



```

272     `${res.scopes && res.scopes.length > 0 ? `<div class="t
ext-muted mt-2"><strong>Доступные Scopes:</strong><br>${res.sc
opes.map(s => '• ' + escapeHtml(s)).join('<br>')}}</div>` : ''}
273     `;
274     }
275   } catch (err) {
276     if (testBody) {
277       testBody.innerHTML = `<span class="text-danger"><i class
="bi bi-x-circle me-1"></i>Ошибка проверки: ${escapeHtml(err.m
essage)}</span>`;
278     }
279   }
280 }
281
282 async function deleteTabGoogleAccount(name) {
283   if (!confirm(`Вы уверены, что хотите удалить аккаунт "${name}
}" из пула Google Workspace?`)) return;
284   try {
285     await window.api.googleAccounts.delete(name);
286     showGAccountsNotification(`Аккаунт "${name}" успешно удале
н`, 'success');
287     await refreshTabGoogleAccountsList();
288   } catch (err) {
289     console.error('Ошибка удаления аккаунта Google:', err);
290     showGAccountsNotification('Ошибка удаления: ' + err.messag
e, 'danger');
291   }
292 }
293
294 async function initGoogleAccountsTab() {
295   const refreshBtn = document.getElementById('tab-btn-refresh-
gaccounts');
296   const addBtn = document.getElementById('tab-btn-add-gacc');
297
298   if (refreshBtn) {
299     refreshBtn.onclick = async () => {
300       refreshBtn.disabled = true;
301       const orig = refreshBtn.innerHTML;

```

```
302 |         refreshBtn.innerHTML = '<span class="spinner-border spin  
|ner-border-sm me-1"></span> Обновление...';  
303 |         try {  
304 |             await refreshTabGoogleAccountsList();  
305 |             showGAccountsNotification('Список Google аккаунтов обн  
|овлен', 'success');  
306 |         } finally {  
307 |             refreshBtn.disabled = false;  
308 |             refreshBtn.innerHTML = orig;  
309 |         }  
310 |     };  
311 | }  
312 |  
313 | if (addBtn) {  
314 |     addBtn.onclick = handleAddTabGoogleAccount;  
315 | }  
316 |  
317 | await refreshTabGoogleAccountsList();  
318 | }  
319 |  
320 | window.initGoogleAccountsTab = initGoogleAccountsTab;  
321 | window.refreshTabGoogleAccountsList = refreshTabGoogleAccounts  
|List;  
322 |
```

index.html

```
1 <!-- Google User Desktop Web Tab Module -->
2 <style>
3   .gdesktop-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .gdesktop-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.5rem 0.85rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    min-height: 64px;
15    gap: 0.2rem;
16  }
17  .gdesktop-label {
18    font-size: 0.7rem;
19    text-transform: uppercase;
20    letter-spacing: 0.05em;
21    color: #94a3b8;
22    line-height: 1.1;
23  }
24  .gdesktop-value {
25    font-size: 1.3rem;
26    font-weight: 700;
27    color: #f8fafc;
28    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
29    Consolas, monospace;
30    line-height: 1.15;
31  }
32  .gdesktop-panel {
33    background: #0f172a !important;
34    border: 1px solid #334155 !important;
35    border-radius: 8px;
36    overflow: hidden;
```

```
36 | }
37 | .gdesktop-panel-header {
38 |     background: #1e293b !important;
39 |     border-bottom: 1px solid #334155 !important;
40 |     padding: 0.5rem 0.85rem;
41 |     font-weight: 600;
42 |     font-size: 0.85rem;
43 |     color: #38bdf8;
44 |     display: flex;
45 |     align-items: center;
46 |     justify-content: space-between;
47 | }
48 | .gdesktop-subnav .nav-link {
49 |     color: #94a3b8;
50 |     border-radius: 6px;
51 |     padding: 0.35rem 0.75rem;
52 |     font-size: 0.82rem;
53 |     font-weight: 500;
54 |     transition: all 0.15s ease-in-out;
55 | }
56 | .gdesktop-subnav .nav-link:hover {
57 |     color: #f8fafc;
58 |     background: rgba(255, 255, 255, 0.05);
59 | }
60 | .gdesktop-subnav .nav-link.active {
61 |     color: #000;
62 |     background: #38bdf8;
63 |     font-weight: 600;
64 | }
65 | .gdesktop-table {
66 |     font-size: 0.82rem;
67 | }
68 | .gdesktop-table td, .gdesktop-table th {
69 |     vertical-align: middle;
70 | }
71 | </style>
72 |
73 | <div class="gdesktop-container" id="google-desktop-tab-root">
```

```

74 | <!-- Top Header Banner & Actions -->
75 | <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 py-2 px-3 rounded-2 text-white" style="
    | background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%);
    | border: 1px solid #334155;">
76 |     <div class="d-flex align-items-center gap-2 flex-wrap">
77 |         <span class="fs-4">☐ </span>
78 |         <div>
79 |             <div class="d-flex align-items-center gap-2">
80 |                 <span class="fw-bold text-white" style="font-size: 1
    | rem;">Google User Desktop</span>
81 |                 <span class="badge bg-danger text-white border border-
    | light-subtle px-2 py-0.5" style="font-size: 0.7rem;">Workspa
    | ce Aggregator</span>
82 |             </div>
83 |             <div class="text-secondary small" style="font-size: 0.
    | 76rem;">
84 |                 Единый дашбордин Gmail, Google Calendar, Docs/Sheets
    | и Google Drive
85 |             </div>
86 |         </div>
87 |     </div>
88 |
89 | <!-- Account Selector & Sync Controls -->
90 | <div class="d-flex align-items-center gap-2 flex-wrap">
91 |     <div class="d-flex align-items-center gap-1.5 bg-dark bg-
    | opacity-50 border border-secondary px-2 py-1 rounded">
92 |         <label for="gdesktop-account-select" class="small text
    | -muted mb-0 me-1" style="font-size: 0.75rem;">Аккаунт:</label>
93 |         <select class="form-select form-select-sm bg-dark text
    | -light border-0 py-0 px-2" id="gdesktop-account-select" style=
    | "font-size: 0.8rem; width: auto; cursor: pointer;">
94 |             <option value="">Загрузка аккаунтов...</option>
95 |         </select>
96 |     </div>
97 |
98 |     <button class="btn btn-xs btn-outline-info rounded-pill
    | px-3 py-1.5" style="font-size: 0.78rem;" id="btn-gdesktop-refr

```

```

esh" title="Обновить данные в интерфейсе">
99     <i class="bi bi-arrow-clockwise me-1"></i> Обновить
100   </button>
101
102   <button class="btn btn-xs btn-outline-success rounded-pi
ll px-3 py-1.5" style="font-size: 0.78rem;" id="btn-gdesktop-s
ync" title="Принудительно запросить API Google">
103     <i class="bi bi-cloud-arrow-down-fill me-1"></i> Синхр
онизация
104   </button>
105 </div>
106 </div>
107
108 <!-- Summary Metric Cards -->
109 <div class="row g-2 mb-3">
110   <div class="col-6 col-md-3">
111     <div class="gdesktop-card">
112       <span class="gdesktop-label">Текущий аккаунт</span>
113       <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
114         <span class="gdesktop-value text-info text-truncate"
style="font-size: 1.05rem;" id="gdesktop-acc-name">-</span>
115         <span class="badge bg-secondary" id="gdesktop-acc-st
atus" style="font-size: 0.68rem;">-</span>
116       </div>
117     </div>
118   </div>
119   <div class="col-6 col-md-3">
120     <div class="gdesktop-card">
121       <span class="gdesktop-label">Входящая почта (Gmail)</s
pan>
122       <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
123         <span class="gdesktop-value text-warning" id="gdeskt
op-mail-count">0</span>
124         <span class="text-muted small">сообщений</span>
125       </div>
126     </div>

```

```
127 |     </div>
128 |     <div class="col-6 col-md-3">
129 |         <div class="gdesktop-card">
130 |             <span class="gdesktop-label">События Календаря</span>
131 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
132 |                 <span class="gdesktop-value text-success" id="gdesktop-calendar-count">0</span>
133 |                 <span class="text-muted small">событий</span>
134 |             </div>
135 |         </div>
136 |     </div>
137 |     <div class="col-6 col-md-3">
138 |         <div class="gdesktop-card">
139 |             <span class="gdesktop-label">Документы и Диск</span>
140 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
141 |                 <span class="gdesktop-value text-primary" id="gdesktop-docs-drive-count">0 / 0</span>
142 |                 <span class="text-muted small">Docs / Drive</span>
143 |             </div>
144 |         </div>
145 |     </div>
146 | </div>
147 |
148 | <!-- Sub-navigation Tabs -->
149 | <div class="gdesktop-panel mb-3">
150 |     <div class="gdesktop-panel-header">
151 |         <ul class="nav nav-pills gdesktop-subnav gap-1" id="gdesktop-subnav-tabs" role="tablist">
152 |             <li class="nav-item" role="presentation">
153 |                 <button class="nav-link active" id="gdesktop-tab-mail-btn" data-bs-toggle="pill" data-bs-target="#gdesktop-pane-mail" type="button" role="tab">
154 |                     □ Gmail Почта
155 |                 </button>
156 |             </li>
157 |             <li class="nav-item" role="presentation">
```

[illegible]


```

187         </tr>
188     </thead>
189     <tbody id="gdesktop-mail-tbody">
190         <tr>
191             <td colspan="4" class="text-center py-4 text-m
192             uted">Загрузка входящих писем Gmail...</td>
193         </tr>
194     </tbody>
195 </table>
196 </div>
197
198 <!-- 2. Google Calendar Pane -->
199 <div class="tab-pane fade" id="gdesktop-pane-calendar" r
200 ole="tabpanel">
201     <div class="table-responsive">
202         <table class="table table-dark table-hover align-mid
203         dle gdesktop-table mb-0">
204             <thead>
205                 <tr class="text-secondary small text-uppercase">
206                     <th>Событие</th>
207                     <th style="width: 180px;">Начало</th>
208                     <th style="width: 180px;">Окончание</th>
209                     <th style="width: 180px;">Местоположение</th>
210                     <th style="width: 100px;" class="text-center">
211                     Ссылка</th>
212                 </tr>
213             </thead>
214             <tbody id="gdesktop-calendar-tbody">
215                 <tr>
216                     <td colspan="5" class="text-center py-4 text-m
217                     uted">Загрузка событий Календаря...</td>
218                 </tr>
219             </tbody>
220         </table>
221     </div>
222 </div>

```

```

220 |         <!-- 3. Google Docs Pane -->
221 |         <div class="tab-pane fade" id="gdesktop-pane-docs" role=
| "tabpanel">
222 |             <div class="table-responsive">
223 |                 <table class="table table-dark table-hover align-mid
| dle gdesktop-table mb-0">
224 |                     <thead>
225 |                         <tr class="text-secondary small text-uppercase">
226 |                             <th>Название документа</th>
227 |                             <th style="width: 220px;">Тип (MIME)</th>
228 |                             <th style="width: 170px;">Изменен</th>
229 |                             <th style="width: 160px;" class="text-center">
| Действия</th>
230 |                         </tr>
231 |                     </thead>
232 |                     <tbody id="gdesktop-docs-tbody">
233 |                         <tr>
234 |                             <td colspan="4" class="text-center py-4 text-m
| uted">Загрузка списка документов...</td>
235 |                         </tr>
236 |                     </tbody>
237 |                 </table>
238 |             </div>
239 |         </div>
240 |
241 |         <!-- 4. Google Drive Pane -->
242 |         <div class="tab-pane fade" id="gdesktop-pane-drive" role
| ="tabpanel">
243 |             <div class="table-responsive">
244 |                 <table class="table table-dark table-hover align-mid
| dle gdesktop-table mb-0">
245 |                     <thead>
246 |                         <tr class="text-secondary small text-uppercase">
247 |                             <th>Имя файла</th>
248 |                             <th style="width: 220px;">Тип файла</th>
249 |                             <th style="width: 120px;">Размер</th>
250 |                             <th style="width: 170px;">Модифицирован</th>
251 |                             <th style="width: 100px;" class="text-center">

```

```

| Ссылка</th>
252 |         </tr>
253 |     </thead>
254 |     <tbody id="gdesktop-drive-tbody">
255 |         <tr>
256 |             <td colspan="5" class="text-center py-4 text-m
| uted">Загрузка файлов с Google Диска...</td>
257 |         </tr>
258 |     </tbody>
259 | </table>
260 | </div>
261 | </div>
262 | </div>
263 | </div>
264 |
265 | <!-- Document Viewer Modal -->
266 | <div class="modal fade" id="gdesktopDocViewerModal" tabindex
| ="-1" aria-labelledby="gdesktopDocViewerModalLabel" aria-hidde
| n="true">
267 |     <div class="modal-dialog modal-lg modal-dialog-centered mo
| dal-dialog-scrollable">
268 |         <div class="modal-content bg-dark text-light border bord
| er-secondary shadow-lg">
269 |             <div class="modal-header border-secondary py-2">
270 |                 <h6 class="modal-title d-flex align-items-center gap
| -2" id="gdesktopDocViewerModalLabel">
271 |                     <i class="bi bi-file-earmark-text text-info"></i>
| <span id="gdesktop-doc-modal-title">Просмотр документа</span>
272 |                 </h6>
273 |                 <button type="button" class="btn-close btn-close-whi
| te" data-bs-dismiss="modal" aria-label="Close"></button>
274 |             </div>
275 |             <div class="modal-body p-3 font-monospace small bg-bla
| ck bg-opacity-50" id="gdesktop-doc-modal-body" style="white-sp
| ace: pre-wrap; max-height: 450px;">
276 |                 Загрузка содержимого...
277 |             </div>
278 |             <div class="modal-footer border-secondary py-1.5">

```

```
279 |         <button type="button" class="btn btn-secondary btn-s  
    | m rounded-pill px-4" data-bs-dismiss="modal">Заккрыть</button>  
280 |     </div>  
281 | </div>  
282 | </div>  
283 | </div>  
284 | </div>  
285 |
```

main.js

```
1 /**
2  * google_desktop_tab/main.js – Веб-модуль управления Google U
3  */
4
5 (function () {
6   'use strict';
7
8   function formatBytes(bytes) {
9     if (!bytes || bytes === 0) return '0 B';
10    const k = 1024;
11    const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
12    const i = Math.floor(Math.log(bytes) / Math.log(k));
13    return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
14    ' + sizes[i];
15  }
16
17  function formatDate(isoStr) {
18    if (!isoStr) return '-';
19    try {
20      const d = new Date(isoStr);
21      if (isNaN(d.getTime())) return isoStr;
22      return d.toLocaleString('ru-RU', {
23        year: 'numeric',
24        month: '2-digit',
25        day: '2-digit',
26        hour: '2-digit',
27        minute: '2-digit',
28      });
29    } catch (e) {
30      return isoStr;
31    }
32  }
33
34  async function fetchStatus() {
35    try {
```

```
35 |     const res = await fetch('/api/google-desktop/status');
36 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
37 |     const data = await res.json();
38 |     updateStatusUI(data);
39 |   } catch (err) {
40 |     console.error('[GoogleDesktopTab] Failed to fetch status
    :', err);
41 |   }
42 | }
43 |
44 | function updateStatusUI(data) {
45 |   const acc = data.account || {};
46 |   const nameEl = document.getElementById('gdesktop-acc-name'
    );
47 |   const statusEl = document.getElementById('gdesktop-acc-sta
    tus');
48 |   const mailCountEl = document.getElementById('gdesktop-mail
    -count');
49 |   const calCountEl = document.getElementById('gdesktop-calen
    dar-count');
50 |   const docsDriveCountEl = document.getElementById('gdesktop
    -docs-drive-count');
51 |   const lastRefreshedEl = document.getElementById('gdesktop-
    last-refreshed');
52 |
53 |   if (nameEl) nameEl.textContent = acc.name || 'Не выбран';
54 |   if (statusEl) {
55 |     statusEl.textContent = acc.status || 'unconfigured';
56 |     statusEl.className = acc.status === 'active' ? 'badge bg
    -success' : 'badge bg-warning text-dark';
57 |   }
58 |
59 |   if (mailCountEl) mailCountEl.textContent = data.mail_count
    || 0;
60 |   if (calCountEl) calCountEl.textContent = data.calendar_eve
    nts_count || 0;
61 |   if (docsDriveCountEl) docsDriveCountEl.textContent = `${da
    ta.docs_count || 0} / ${data.drive_files_count || 0}`;
```

```

62 |
63 |     if (lastRefreshedEl) {
64 |         lastRefreshedEl.textContent = `Обновлено: ${formatDate(d
    ata.last_refreshed)}`;
65 |     }
66 | }
67 |
68 | async function fetchAccounts() {
69 |     try {
70 |         const res = await fetch('/api/google-desktop/accounts');
71 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
72 |         const data = await res.json();
73 |         const selectEl = document.getElementById('gdesktop-accou
    nt-select');
74 |         if (!selectEl) return;
75 |
76 |         selectEl.innerHTML = '';
77 |         const accounts = data.accounts || [];
78 |         const active = data.active_account || {};
79 |
80 |         if (accounts.length === 0) {
81 |             const opt = document.createElement('option');
82 |             opt.value = '';
83 |             opt.textContent = 'Нет аккаунтов (unconfigured)';
84 |             selectEl.appendChild(opt);
85 |             return;
86 |         }
87 |
88 |         accounts.forEach((a) => {
89 |             const opt = document.createElement('option');
90 |             opt.value = a.name;
91 |             opt.textContent = `${a.name} (${a.email || a.type})${a
    .is_default ? ' ⚑ ' : ''}`;
92 |             if (a.name === active.name) {
93 |                 opt.selected = true;
94 |             }
95 |             selectEl.appendChild(opt);
96 |         });

```

```

97     } catch (err) {
98         console.error('[GoogleDesktopTab] Failed to fetch accounts:', err);
99     }
100 }
101
102 async function selectAccount(accName) {
103     if (!accName) return;
104     try {
105         const res = await fetch(`/api/google-desktop/accounts/select?account_name=${encodeURIComponent(accName)}`, {
106             method: 'POST',
107         });
108         if (!res.ok) throw new Error(`HTTP ${res.status}`);
109         await refreshAllData();
110         if (window.toast) window.toast.success('Google Desktop',
111             `Активный аккаунт изменен на ${accName}`);
112     } catch (err) {
113         console.error('[GoogleDesktopTab] Failed to select account:', err);
114         if (window.toast) window.toast.error('Ошибка выбора аккаунта', err.message);
115     }
116 }
117
118 async function fetchMail() {
119     const tbody = document.getElementById('gdesktop-mail-tbody');
120     if (!tbody) return;
121     try {
122         const res = await fetch('/api/google-desktop/mail/messages?limit=20');
123         if (!res.ok) throw new Error(`HTTP ${res.status}`);
124         const data = await res.json();
125         const msgs = data.messages || [];
126         if (msgs.length === 0) {
127             tbody.innerHTML = '<tr><td colspan="4" class="text-center py-3 text-muted">Входящие сообщения отсутствуют.</td></tr>';

```



```

';
127     return;
128 }
129 tbody.innerHTML = msgs
130     .map(
131         (m) => `
132         <tr>
133             <td class="fw-semibold text-info text-truncate" styl
e="max-width: 200px;">${escapeHtml(m.sender || 'Неизвестно')}<
/td>
134             <td class="text-light fw-bold">${escapeHtml(m.subjec
t || '(Без темы)')}</td>
135             <td class="text-secondary small">${escapeHtml(format
Date(m.date))}</td>
136             <td class="text-muted small text-truncate" style="ma
x-width: 350px;">${escapeHtml(m.snippet || '')}</td>
137         </tr>
138         `
139     )
140     .join('');
141 } catch (err) {
142     console.error('[GoogleDesktopTab] Failed to fetch mail:'
, err);
143     tbody.innerHTML = `<tr><td colspan="4" class="text-cente
r py-3 text-danger">Ошибка загрузки почты: ${err.message}</td>
</tr>`;
144 }
145 }
146
147 async function fetchCalendar() {
148     const tbody = document.getElementById('gdesktop-calendar-t
body');
149     if (!tbody) return;
150     try {
151         const res = await fetch('/api/google-desktop/calendar/ev
ents?limit=20');
152         if (!res.ok) throw new Error(`HTTP ${res.status}`);
153         const data = await res.json();

```

```

154 |     const events = data.events || [];
155 |     if (events.length === 0) {
156 |         tbody.innerHTML = '<tr><td colspan="5" class="text-center py-3 text-muted">Предстоящие события отсутствуют.</td></tr>';
157 |         return;
158 |     }
159 |     tbody.innerHTML = events
160 |         .map(
161 |             (e) => `
162 |                 <tr>
163 |                     <td class="fw-bold text-light">${escapeHtml(e.summary || 'Без названия')}</td>
164 |                     <td class="text-info small">${escapeHtml(formatDate(e.start_time))}</td>
165 |                     <td class="text-secondary small">${escapeHtml(formatDate(e.end_time))}</td>
166 |                     <td class="text-muted small text-truncate" style="max-width: 180px;">${escapeHtml(e.location || '-')}</td>
167 |                     <td class="text-center">
168 |                         ${
169 |                             e.html_link
170 |                             ? `<a href="${e.html_link}" target="_blank" class="btn btn-xs btn-outline-info rounded-pill py-0 px-2" title="Открыть в Google Calendar"><i class="bi bi-box-arrow-up-right"></i></a>`
171 |                             : '- '
172 |                         }
173 |                     </td>
174 |                 </tr>
175 |             `
176 |         )
177 |         .join('');
178 |     } catch (err) {
179 |         console.error('[GoogleDesktopTab] Failed to fetch calendar events:', err);
180 |         tbody.innerHTML = `<tr><td colspan="5" class="text-center py-3 text-danger">Ошибка загрузки событий: ${err.message}</td>`;

```

```

    d></tr>`;
181     }
182   }
183
184   async function fetchDocs() {
185     const tbody = document.getElementById('gdesktop-docs-tbody
    ');
186     if (!tbody) return;
187     try {
188       const res = await fetch('/api/google-desktop/docs/list?l
    imit=20');
189       if (!res.ok) throw new Error(`HTTP ${res.status}`);
190       const data = await res.json();
191       const docs = data.documents || [];
192       if (docs.length === 0) {
193         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
    ter py-3 text-muted">Документы не найдены.</td></tr>';
194         return;
195       }
196       tbody.innerHTML = docs
197         .map(
198           (d) => `
199             <tr>
200               <td class="fw-bold text-light">${escapeHtml(d.name |
    | 'Без названия')}</td>
201               <td class="text-secondary small text-truncate" style
    ="max-width: 220px;">${escapeHtml(d.mime_type || '')}</td>
202               <td class="text-secondary small">${escapeHtml(format
    Date(d.modified_time))}</td>
203               <td class="text-center">
204                 <div class="btn-group btn-group-sm">
205                   <button class="btn btn-xs btn-outline-primary ro
    unded-start py-0 px-2 btn-view-doc" data-doc-id="${d.id}" data
    -doc-name="${escapeHtml(
206                     d.name
207                   )}" title="Просмотреть текст"><i class="bi bi-eye"><
    /i> Текст</button>
208                   ${

```

```

209         d.web_view_link
210         ? `<a href="${d.web_view_link}" target="_bla
nk" class="btn btn-xs btn-outline-secondary rounded-end py-0 p
x-2" title="Открыть в браузер"><i class="bi bi-box-arrow-up-ri
ght"></i></a>`
211         : ''
212     }
213     </div>
214     </td>
215 </tr>
216 `
217 )
218 .join('');
219
220 tbody.querySelectorAll('.btn-view-doc').forEach((btn) =>
{
221     btn.addEventListener('click', () => {
222         const docId = btn.getAttribute('data-doc-id');
223         const docName = btn.getAttribute('data-doc-name');
224         openDocumentViewer(docId, docName);
225     });
226 });
227 } catch (err) {
228     console.error('[GoogleDesktopTab] Failed to fetch docs:'
, err);
229     tbody.innerHTML = `<tr><td colspan="4" class="text-cente
r py-3 text-danger">Ошибка загрузки документов: ${err.message}
</td></tr>`;
230 }
231 }
232
233 async function fetchDrive() {
234     const tbody = document.getElementById('gdesktop-drive-tbod
y');
235     if (!tbody) return;
236     try {
237         const res = await fetch('/api/google-desktop/drive/files
?limit=20');

```

```

238         if (!res.ok) throw new Error(`HTTP ${res.status}`);
239         const data = await res.json();
240         const files = data.files || [];
241         if (files.length === 0) {
242             tbody.innerHTML = '<tr><td colspan="5" class="text-center py-3 text-muted">Файлы на Google Диске отсутствуют.</td></tr>';
243             return;
244         }
245         tbody.innerHTML = files
246             .map(
247                 (f) => `
248                 <tr>
249                     <td class="fw-bold text-light">${escapeHtml(f.name |
250                     | 'Без названия'))}</td>
251                     <td class="text-secondary small text-truncate" style="max-width: 220px;">${escapeHtml(f.mime_type || '')}</td>
252                     <td class="text-info small font-monospace">${formatBytes(f.size_bytes)}</td>
253                     <td class="text-secondary small">${escapeHtml(formatDate(f.modified_time))}</td>
254                     <td class="text-center">
255                         ${
256                             f.web_view_link
257                             ? `<a href="${f.web_view_link}" target="_blank" class="btn btn-xs btn-outline-info rounded-pill py-0 px-2" title="Открыть на Google Диске"><i class="bi bi-box-arrow-up-right"></i></a>`
258                             : `'-`
259                         }
260                     </td>
261                 </tr>
262             `
263             .join('');
264     } catch (err) {
265         console.error('[GoogleDesktopTab] Failed to fetch drive files:', err);

```

```

266 |         tbody.innerHTML = `<tr><td colspan="5" class="text-cen
    | r py-3 text-danger">Ошибка загрузки файлов: ${err.message}</td
    | ></tr>`;
267 |     }
268 | }
269 |
270 | async function openDocumentViewer(docId, docName) {
271 |     const modalEl = document.getElementById('gdesktopDocViewer
    | Modal');
272 |     const titleEl = document.getElementById('gdesktop-doc-moda
    | l-title');
273 |     const bodyEl = document.getElementById('gdesktop-doc-modal
    | -body');
274 |     if (!modalEl || !bodyEl) return;
275 |
276 |     if (titleEl) titleEl.textContent = docName || `Документ ${
    | docId}`;
277 |     bodyEl.textContent = 'Загрузка содержимого...';
278 |
279 |     const modal = new bootstrap.Modal(modalEl);
280 |     modal.show();
281 |
282 |     try {
283 |         const res = await fetch(`/api/google-desktop/docs/${enco
    | deURIComponent(docId)}`);
284 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
285 |         const data = await res.json();
286 |         bodyEl.textContent = data.content || '(Документ пуст)';
287 |     } catch (err) {
288 |         bodyEl.textContent = `Ошибка загрузки содержимого докуме
    | нта: ${err.message}`;
289 |     }
290 | }
291 |
292 | async function triggerSync() {
293 |     const syncBtn = document.getElementById('btn-gdesktop-sync
    | ');
294 |     if (syncBtn) {

```

```
295 |         syncBtn.disabled = true;
296 |         syncBtn.innerHTML = '<span class="spinner-border spinner
    | -border-sm me-1"></span> Синхронизация...';
297 |     }
298 |     try {
299 |         const res = await fetch('/api/google-desktop/sync', { me
    | thod: 'POST' });
300 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
301 |         const data = await res.json();
302 |         if (data.state) updateStatusUI(data.state);
303 |         await refreshAllData();
304 |         if (window.toast) window.toast.success('Синхронизация за
    | вершена', 'Данные Google Workspace принудительно обновлены');
305 |     } catch (err) {
306 |         console.error('[GoogleDesktopTab] Sync failed:', err);
307 |         if (window.toast) window.toast.error('Ошибка синхронизац
    | ии', err.message);
308 |     } finally {
309 |         if (syncBtn) {
310 |             syncBtn.disabled = false;
311 |             syncBtn.innerHTML = '<i class="bi bi-cloud-arrow-down-
    | fill me-1"></i> Синхронизация';
312 |         }
313 |     }
314 | }
315 |
316 | async function refreshAllData() {
317 |     await Promise.all([fetchStatus(), fetchAccounts(), fetchMa
    | il(), fetchCalendar(), fetchDocs(), fetchDrive()]);
318 | }
319 |
320 | function escapeHtml(str) {
321 |     if (typeof str !== 'string') return '';
322 |     return str
323 |         .replace(/&/g, '&amp;')
324 |         .replace(/</g, '&lt;')
325 |         .replace(/>/g, '&gt;')
326 |         .replace(/"/g, '&quot;');
```

```
327 |         .replace(/'/g, '&#039;');
328 |     }
329 |
330 |     function initGoogleDesktopTab() {
331 |         console.log('[GoogleDesktopTab] Initializing Google User D
332 | esktop tab...');
333 |         const refreshBtn = document.getElementById('btn-gdesktop-r
334 | efresh');
335 |         if (refreshBtn) {
336 |             refreshBtn.onclick = () => refreshAllData();
337 |         }
338 |         const syncBtn = document.getElementById('btn-gdesktop-sync
339 | ');
340 |         if (syncBtn) {
341 |             syncBtn.onclick = () => triggerSync();
342 |         }
343 |         const selectEl = document.getElementById('gdesktop-account
344 | -select');
345 |         if (selectEl) {
346 |             selectEl.onchange = (e) => selectAccount(e.target.value)
347 | ;
348 |         }
349 |         refreshAllData();
350 |     }
351 |     window.initGoogleDesktopTab = initGoogleDesktopTab;
352 | })();
353 |
```


index.html

```
1 <!-- Hardware & Sensors Monitor Tab -->
2 <style>
3   .hw-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .hw-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    height: 100%;
15    min-height: 60px;
16    transition: border-color 0.15s ease, box-shadow 0.15s ease
17    ;
18  }
19  .hw-card:hover {
20    border-color: #38bdf8 !important;
21    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
22  }
23  .hw-card-label {
24    font-size: 0.68rem;
25    text-transform: uppercase;
26    letter-spacing: 0.05em;
27    color: #94a3b8;
28    font-weight: 600;
29    display: flex;
30    justify-content: space-between;
31    align-items: center;
32    line-height: 1.1;
33  }
34  .hw-card-value {
35    font-size: 1.25rem;
36    font-weight: 700;
```

```
36     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,  
    Consolas, monospace;  
37     line-height: 1.15;  
38 }  
39 .hw-card-sub {  
40     font-size: 0.7rem;  
41     color: #64748b;  
42     line-height: 1.1;  
43 }  
44  
45 .hw-panel {  
46     background: #0f172a !important;  
47     border: 1px solid #334155 !important;  
48     border-radius: 8px;  
49     overflow: hidden;  
50     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.2);  
51     margin-bottom: 0.75rem;  
52     display: flex;  
53     flex-direction: column;  
54 }  
55 .hw-panel-header {  
56     background: #1e293b !important;  
57     border-bottom: 1px solid #334155 !important;  
58     padding: 0.45rem 0.85rem;  
59     font-weight: 600;  
60     font-size: 0.82rem;  
61     color: #38bdf8;  
62     display: flex;  
63     align-items: center;  
64     justify-content: space-between;  
65     flex-shrink: 0;  
66 }  
67 .hw-scrollable {  
68     height: auto;  
69     min-height: 100px;  
70     max-height: clamp(280px, 52vh, 620px);  
71     overflow-y: auto;  
72     overflow-x: auto;
```

```

73     scrollbar-width: thin;
74 }
75 .hw-table {
76     margin-bottom: 0;
77     font-size: 0.8rem;
78     width: 100%;
79 }
80 .hw-table thead {
81     position: sticky;
82     top: 0;
83     z-index: 2;
84 }
85 .hw-table th {
86     background: #1e293b;
87     color: #94a3b8;
88     font-weight: 600;
89     font-size: 0.72rem;
90     text-transform: uppercase;
91     border-bottom: 1px solid #334155;
92     padding: 0.45rem 0.6rem;
93 }
94 .hw-table td {
95     border-bottom: 1px solid #1e293b;
96     padding: 0.45rem 0.6rem;
97     vertical-align: middle;
98 }
99 </style>
100
101 <div class="hw-container">
102     <!-- Header Bar -->
103     <div class="d-flex align-items-center justify-content-between mb-2 flex-wrap gap-2">
104         <div>
105             <h6 class="mb-0 fw-bold text-white d-flex align-items-center gap-2">
106                 <span>⏏ </span> Монитор оборудования и сенсоров Windows
107             </h6>
108             <div class="small text-muted" style="font-size: 0.72rem;

```

```

">
109         Телеметрия CPU, RAM, GPU, термодатчиков, кулеров, напря
        жений и диагностика накопителей S.M.A.R.T.
110     </div>
111 </div>
112 <div class="d-flex align-items-center gap-2">
113     <div class="form-check form-switch mb-0 small text-muted
">
114         <input class="form-check-input" type="checkbox" id="hw
        -auto-refresh" checked>
115         <label class="form-check-label small" for="hw-auto-ref
        resh">Автообновление (3с)</label>
116     </div>
117     <button class="btn btn-sm btn-outline-secondary d-flex a
        lign-items-center gap-1" id="hw-btn-refresh">
118         <i class="bi bi-arrow-clockwise"></i>
119         <span>Обновить</span>
120     </button>
121 </div>
122 </div>
123
124 <!-- KPI Stats Row -->
125 <div class="row g-2 mb-2">
126     <div class="col-6 col-md-3">
127         <div class="hw-card">
128             <div class="hw-card-label">
129                 <span>Процессор (CPU)</span>
130                 <span class="badge bg-primary" id="hw-cpu-cores">- c
        ores</span>
131             </div>
132             <div class="hw-card-value text-info" id="hw-val-cpu">-
        %</div>
133             <div class="hw-card-sub text-truncate" id="hw-sub-cpu"
        >Загрузка процессора</div>
134         </div>
135     </div>
136     <div class="col-6 col-md-3">
137         <div class="hw-card">

```

```
138 |         <div class="hw-card-label">
139 |             <span>Память (RAM)</span>
140 |             <span class="badge bg-success" id="hw-ram-total">- G
141 | B</span>
142 |         </div>
143 |         <div class="hw-card-value text-success" id="hw-val-ram
144 | ">- %</div>
145 |         <div class="hw-card-sub text-truncate" id="hw-sub-ram"
146 | >Занято памяти</div>
147 |     </div>
148 | </div>
149 | <div class="col-6 col-md-3">
150 |     <div class="hw-card">
151 |         <div class="hw-card-label">
152 |             <span>Видеокарта (GPU)</span>
153 |             <span class="badge bg-warning text-dark" id="hw-gpu-
154 | count">0 GPU</span>
155 |         </div>
156 |         <div class="hw-card-value text-warning" id="hw-val-gpu
157 | ">- %</div>
158 |         <div class="hw-card-sub text-truncate" id="hw-sub-gpu"
159 | >Загрузка GPU</div>
160 |     </div>
161 | </div>
162 | <div class="col-6 col-md-3">
163 |     <div class="hw-card">
164 |         <div class="hw-card-label">
165 |             <span>Сенсоры и Температура</span>
166 |             <span class="badge bg-info text-dark" id="hw-sensor-
167 | count">0 шт.</span>
168 |         </div>
169 |         <div class="hw-card-value text-danger" id="hw-val-temp
170 | ">- °C</div>
171 |         <div class="hw-card-sub text-truncate" id="hw-sub-temp
172 | ">Максимальная t°</div>
173 |     </div>
174 | </div>
175 | </div>
```

```

167
168 <!-- Main Panels Grid -->
169 <div class="row g-2">
170 <!-- Left Column: CPU/RAM/GPU Detailed Specs & Disks -->
171 <div class="col-12 col-lg-6">
172 <!-- Disks & Storage Panel -->
173 <div class="hw-panel">
174 <div class="hw-panel-header">
175 <span class="d-flex align-items-center gap-1.5">
176 <i class="bi bi-hdd-network text-info"></i>
177 <span>Диски и накопители</span>
178 </span>
179 <span class="badge bg-dark border border-secondary text-muted" id="hw-disks-count">0 дисков</span>
180 </div>
181 <div class="table-responsive hw-scrollable">
182 <table class="table table-dark table-hover hw-table">
183 <thead>
184 <tr>
185 <th>Диск</th>
186 <th>ФС / Тип</th>
187 <th>Всего / Свободно</th>
188 <th>Использование</th>
189 </tr>
190 </thead>
191 <tbody id="hw-disks-body">
192 <tr>
193 <td colspan="4" class="text-center py-3 text-muted">Загрузка дисков...</td>
194 </tr>
195 </tbody>
196 </table>
197 </div>
198 </div>
199
200 <!-- GPU Telemetry Panel -->
201 <div class="hw-panel">

```

```

202 |         <div class="hw-panel-header">
203 |             <span class="d-flex align-items-center gap-1.5">
204 |                 <i class="bi bi-gpu-card text-warning"></i>
205 |                 <span>Графические ускорители (GPU)</span>
206 |             </span>
207 |         </div>
208 |         <div class="p-2.5 hw-scrollable" id="hw-gpu-container"
    | style="font-size: 0.8rem;">
209 |             <div class="text-center py-3 text-muted">Загрузка ин
    | формации GPU...</div>
210 |         </div>
211 |     </div>
212 | </div>
213 |
214 | <!-- Right Column: Sensors & SMART -->
215 | <div class="col-12 col-lg-6">
216 |     <!-- Hardware Sensors Panel -->
217 |     <div class="hw-panel">
218 |         <div class="hw-panel-header">
219 |             <span class="d-flex align-items-center gap-1.5">
220 |                 <i class="bi bi-thermometer-half text-danger"></i>
221 |                 <span>Аппаратные сенсоры (Температура, Вентиляторы
    | , Вольтаж)</span>
222 |             </span>
223 |             <span class="badge bg-dark border border-secondary t
    | ext-muted" id="hw-sensors-total">0 сенсоров</span>
224 |         </div>
225 |         <div class="table-responsive hw-scrollable">
226 |             <table class="table table-dark table-hover hw-table"
    | >
227 |                 <thead>
228 |                     <tr>
229 |                         <th>Датчик / Компонент</th>
230 |                         <th>Тип</th>
231 |                         <th>Значение</th>
232 |                         <th>Статус</th>
233 |                     </tr>
234 |                 </thead>

```

```

235         <tbody id="hw-sensors-body">
236             <tr>
237                 <td colspan="4" class="text-center py-3 text-m
uted">Опрос аппаратных сенсоров...</td>
238             </tr>
239         </tbody>
240     </table>
241 </div>
242 </div>
243
244 <!-- S.M.A.R.T. Health Status -->
245 <div class="hw-panel">
246     <div class="hw-panel-header">
247         <span class="d-flex align-items-center gap-1.5">
248             <i class="bi bi-shield-shaded text-success"></i>
249             <span>S.M.A.R.T. Здоровье накопителей</span>
250         </span>
251     </div>
252     <div class="p-2.5 hw-scrollable" id="hw-smart-containe
r" style="font-size: 0.8rem;">
253         <div class="text-center py-3 text-muted">Загрузка S.
M.A.R.T. телеметрии...</div>
254     </div>
255 </div>
256 </div>
257 </div>
258
259 <!-- =====
===== -->
260 <!-- Внешние утилиты диагностики оборудования
-->
261 <!-- =====
===== -->
262 <div class="mt-3 mb-2">
263     <h6 class="fw-bold text-white d-flex align-items-center ga
p-2 mb-1">
264         <span>⏻ </span> Внешние утилиты диагностики
265     </h6>

```



```

266 |     <div class="small text-muted" style="font-size: 0.72rem;">
267 |         Интеграция с LibreHardwareMonitor.
268 |         Статус портативных версий обновляется автоматически.
269 |     </div>
270 | </div>
271 |
272 | <div class="row g-2 mb-2" id="hw-utilities-grid">
273 |
274 |     <!-- LibreHardwareMonitor -->
275 |     <div class="col-12 col-md-6 col-xl-4">
276 |         <div class="hw-panel" id="hw-util-lhm">
277 |             <div class="hw-panel-header">
278 |                 <span class="d-flex align-items-center gap-1.5">
279 |                     <i class="bi bi-thermometer-sun text-success"></i>
280 |                     <span>LibreHardwareMonitor</span>
281 |                 </span>
282 |                 <span class="badge bg-secondary" id="hw-util-lhm-bad
283 | ge">□ Проверка...</span>
284 |             </div>
285 |             <div class="p-2" style="font-size: 0.8rem;">
286 |                 <div class="mb-1 text-muted small">Open-source монит
287 | оринг сенсоров через веб-API (HTTP JSON)</div>
288 |                 <div id="hw-util-lhm-actions" class="d-flex gap-1 fl
289 | ex-wrap mb-1">
290 |                     <button class="btn btn-sm btn-outline-success py-0
291 | px-2" onclick="hwUtilAction('lhm', 'sensors')" title="Сенсоры
292 | LHM">
293 |                         <i class="bi bi-speedometer2"></i> Сенсоры
294 |                     </button>
295 |                 </div>
296 |                 <div id="hw-util-lhm-guide" class="d-none"></div>
297 |                 <div id="hw-util-lhm-result" class="mt-1 hw-scrollab
298 | le"></div>
299 |             </div>
300 |         </div>
301 |     </div>
302 | </div>

```

298 | </div>
299 |

main.js

```
1 // Hardware & Sensors Monitor Frontend Logic
2 (function () {
3     let timerId = null;
4
5     async function initHardwareMonitorTab() {
6         console.log('[HardwareMonitor] Initializing tab...');
7         bindEvents();
8         loadUtilitiesStatus();
9         loadHardwareData();
10        startPolling();
11    }
12    window.initHardwareMonitorTab = initHardwareMonitorTab;
13
14    function bindEvents() {
15        const btnRefresh = document.getElementById('hw-btn-refresh
16        ');
17        if (btnRefresh) {
18            btnRefresh.onclick = () => {
19                loadUtilitiesStatus();
20                loadHardwareData();
21            };
22        }
23
24        const autoSwitch = document.getElementById('hw-auto-refres
25        h');
26        if (autoSwitch) {
27            autoSwitch.onchange = () => {
28                if (window.setTabPollerEnabled) {
29                    window.setTabPollerEnabled('tab-hardware-monitor_def
30                    ault', autoSwitch.checked);
31                } else {
32                    if (autoSwitch.checked) {
33                        startPolling();
34                    } else {
35                        stopPolling();
36                    }
37                }
38            };
39        }
40    }
41}
```

```

34         }
35     };
36 }
37 }
38
39 function startPolling() {
40     if (window.registerTabPoller) {
41         const autoSwitch = document.getElementById('hw-auto-refr
esh');
42         const isEnabled = autoSwitch ? autoSwitch.checked : true
;
43         window.registerTabPoller('tab-hardware-monitor', loadHar
dwareData, 3000, { immediate: false, enabled: isEnabled });
44     } else {
45         stopPolling();
46         timerId = setInterval(() => {
47             if (window.isTabActive ? window.isTabActive('tab-hardw
are-monitor') : true) {
48                 loadHardwareData();
49             }
50         }, 3000);
51     }
52 }
53
54 function stopPolling() {
55     if (window.unregisterTabPoller) {
56         window.unregisterTabPoller('tab-hardware-monitor_default
');
57     }
58     if (timerId) {
59         clearInterval(timerId);
60         timerId = null;
61     }
62 }
63
64 async function loadHardwareData() {
65     await Promise.allSettled([
66         loadSnapshot(),

```

```

67 |     loadSensors(),
68 |     loadSmart()
69 |   });
70 | }
71 |
72 | async function loadSnapshot() {
73 |   try {
74 |     const res = await fetch('/api/windows/hardware/monitor?include_smart=false');
75 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
76 |     const data = await res.json();
77 |
78 |     // CPU
79 |     const cpu = data.cpu || {};
80 |     const elValCpu = document.getElementById('hw-val-cpu');
81 |     const elSubCpu = document.getElementById('hw-sub-cpu');
82 |     const elCpuCores = document.getElementById('hw-cpu-cores');
83 |
84 |     if (elValCpu) elValCpu.textContent = `${cpu.total_percent !== undefined ? cpu.total_percent : 0}%`;
85 |     if (elSubCpu) elSubCpu.textContent = cpu.model || 'CPU нагрузка';
86 |     if (elCpuCores) elCpuCores.textContent = `${cpu.cores_logical || cpu.cores_physical || 0} cores`;
87 |
88 |     // RAM
89 |     const ram = data.ram || {};
90 |     const elValRam = document.getElementById('hw-val-ram');
91 |     const elSubRam = document.getElementById('hw-sub-ram');
92 |     const elRamTotal = document.getElementById('hw-ram-total');
93 |
94 |     if (elValRam) elValRam.textContent = `${ram.percent_used !== undefined ? ram.percent_used : 0}%`;
95 |     if (elSubRam) elSubRam.textContent = `${ram.used_gb || 0} GB / ${ram.total_gb || 0} GB`;
96 |     if (elRamTotal) elRamTotal.textContent = `${ram.total_gb

```

```

    || 0} GB`;
97
98    // GPU
99    const gpus = data.gpus || [];
100    const elValGpu = document.getElementById('hw-val-gpu');
101    const elSubGpu = document.getElementById('hw-sub-gpu');
102    const elGpuCount = document.getElementById('hw-gpu-count
    ');
103    const gpuContainer = document.getElementById('hw-gpu-con
    tainer');
104
105    if (elGpuCount) elGpuCount.textContent = `${gpus.length}
    GPU`;
106    if (gpus.length > 0) {
107        const topGpu = gpus[0];
108        if (elValGpu) elValGpu.textContent = `${topGpu.load_pe
    rcent || 0}%`;
109        if (elSubGpu) elSubGpu.textContent = topGpu.name || 'G
    PU нагрузка';
110    } else {
111        if (elValGpu) elValGpu.textContent = '0%';
112        if (elSubGpu) elSubGpu.textContent = 'Встроенное видео
    ';
113    }
114
115    if (gpuContainer) {
116        if (gpus.length === 0) {
117            gpuContainer.innerHTML = '<div class="text-muted tex
    t-center py-2">Дискретные графические адаптеры не обнаружены</
    div>';
118        } else {
119            gpuContainer.innerHTML = gpus.map(g => `
120                <div class="mb-2 p-2 bg-dark rounded border border
    -secondary">
121                    <div class="d-flex justify-content-between align
    -items-center mb-1">
122                        <span class="fw-bold text-white">${escapeHtml(
    g.name || 'GPU')}</span>

```

```

123 |         <span class="badge bg-${(g.temp_c || 0) > 80 ?
      | 'danger' : 'success'}">${g.temp_c ? g.temp_c + ' °C' : 't° N/
      | A'}</span>
124 |     </div>
125 |     <div class="small text-muted mb-1">
126 |         Память VRAM: <strong>${g.memory_used_mb || 0}
      | MB / ${g.memory_total_mb || 0} MB</strong>
127 |         ${g.driver_version ? ` | Драйвер: ${escapeHtml
      | (g.driver_version)}` : ''}
128 |     </div>
129 |     <div class="progress" style="height: 6px;">
130 |         <div class="progress-bar bg-warning" role="pro
      | gressbar" style="width: ${g.load_percent || 0}%"></div>
131 |     </div>
132 | </div>
133 | `).join('');
134 |     }
135 | }
136 |
137 | // Disks
138 | const disks = data.disks || [];
139 | const elDisksCount = document.getElementById('hw-disks-c
      | ount');
140 | const disksBody = document.getElementById('hw-disks-body
      | ');
141 |
142 | if (elDisksCount) elDisksCount.textContent = `${disks.le
      | ngth} дисков`;
143 | if (disksBody) {
144 |     if (disks.length === 0) {
145 |         disksBody.innerHTML = '<tr><td colspan="4" class="te
      | xt-center py-3 text-muted">Диски не найдены</td></tr>';
146 |     } else {
147 |         disksBody.innerHTML = disks.map(d => {
148 |             const usedPct = d.percent_used || (d.total_gb ? Ma
      | th.round(((d.total_gb - d.free_gb) / d.total_gb) * 100) : 0);
149 |             return `
150 |                 <tr>

```

```

151         <td class="fw-bold text-white">
152             <i class="bi bi-hdd me-1 text-info"></i>${escapeHtml(d.device || d.mountpoint || '')}
153         </td>
154         <td class="small text-muted">${escapeHtml(d.fs
155         type || d.drive_type || 'NTFS')}</td>
156         <td class="small font-monospace">${d.free_gb |
157         | 0} GB свободно из ${d.total_gb || 0} GB</td>
158         <td style="width: 120px;">
159             <div class="d-flex align-items-center gap-1"
160             >
161                 <div class="progress flex-grow-1" style="height: 6px;">
162                     <div class="progress-bar bg-${usedPct >
163                     90 ? 'danger' : usedPct > 75 ? 'warning' : 'primary'}" style="width: ${usedPct}%"></div>
164                 </div>
165                 <span class="small font-monospace">${usedP
166                 ct}%</span>
167             </div>
168         </td>
169     </tr>
170     `;
171     })).join('');
172 }
173 } catch (e) {
174     console.warn('[HardwareMonitor] Error loading monitor snapshot:', e);
175 }
176 }
177
178 async function loadSensors() {
179     try {
180         const res = await fetch('/api/windows/hardware/sensors')
181         ;
182         if (!res.ok) throw new Error(`HTTP ${res.status}`);
183         const data = await res.json();

```



```
179 |         const sensors = data.sensors || [];
180 |
181 |         const elTotal = document.getElementById('hw-sensors-total');
182 |         const elCountBadge = document.getElementById('hw-sensor-count');
183 |         const tbody = document.getElementById('hw-sensors-body');
184 |         const elValTemp = document.getElementById('hw-val-temp');
185 |
186 |         if (elTotal) elTotal.textContent = `${sensors.length} сенсоров`;
187 |         if (elCountBadge) elCountBadge.textContent = `${sensors.length} шт.`;
188 |
189 |         let maxTemp = 0;
190 |         sensors.forEach(s => {
191 |             if ((s.sensor_type === 'Temperature' || s.type === 'Temperature') && s.value > maxTemp) {
192 |                 maxTemp = Math.round(s.value);
193 |             }
194 |         });
195 |         if (elValTemp) {
196 |             elValTemp.textContent = maxTemp > 0 ? `${maxTemp} °C` : 'N/A';
197 |             elValTemp.className = `hw-card-value ${maxTemp > 80 ? 'text-danger' : maxTemp > 65 ? 'text-warning' : 'text-info'}`;
198 |         }
199 |
200 |         if (!tbody) return;
201 |         if (sensors.length === 0) {
202 |             tbody.innerHTML = '<tr><td colspan="4" class="text-center py-3 text-muted">Сенсоры не обнаружены или требуют прав администратора / OpenHardwareMonitor</td></tr>';
203 |             return;
204 |         }
205 |
```

```

206         tbody.innerHTML = sensors.map(s => {
207             const valStr = s.unit ? `${s.value} ${s.unit}` : `${s.
value}`;
208             const isHot = (s.sensor_type === 'Temperature' || s.ty
pe === 'Temperature') && s.value > 75;
209             return `
210                 <tr>
211                     <td class="fw-semibold text-white">
212                         <i class="bi bi-cpu me-1 text-muted"></i>${escap
eHtml(s.name || s.sensor_name || '')}
213                     </td>
214                     <td class="small text-muted">${escapeHtml(s.sensor
_type || s.type || 'Sensor')}</td>
215                     <td class="font-monospace fw-bold ${isHot ? 'text-
danger' : 'text-info'}">${escapeHtml(valStr)}</td>
216                     <td>
217                         <span class="badge bg-${isHot ? 'danger' : 'succ
ess'}">${isHot ? 'Высокая' : 'Норма'}</span>
218                     </td>
219                 </tr>
220             `;
221         }).join('');
222     } catch (e) {
223         console.warn('[HardwareMonitor] Error loading sensors:',
e);
224     }
225 }
226
227 async function loadSmart() {
228     const container = document.getElementById('hw-smart-contai
ner');
229     try {
230         const res = await fetch('/api/windows/hardware/smart');
231         if (!res.ok) throw new Error(`HTTP ${res.status}`);
232         const data = await res.json();
233         const drives = data.drives || data.devices || [];
234
235         if (!container) return;

```

```

236 |         if (!drives || drives.length === 0) {
237 |             container.innerHTML = '<div class="text-muted text-center py-2">S.M.A.R.T. накопители проверены (Windows Storage API / WMI). Состояние нормальное.</div>';
238 |             return;
239 |         }
240 |
241 |         container.innerHTML = drives.map(d => `
242 |             <div class="mb-2 p-2 bg-dark rounded border border-secondary">
243 |                 <div class="d-flex justify-content-between align-items-center mb-1">
244 |                     <span class="fw-bold text-white">${escapeHtml(d.model || d.device || 'Накопитель')}</span>
245 |                     <span class="badge bg-${d.healthy === false ? 'danger' : 'success'}">${d.healthy === false ? 'ВНИМАНИЕ' : 'PASSED'}</span>
246 |                 </div>
247 |                 <div class="small text-muted">
248 |                     Объем: <strong>${d.size_gb ? d.size_gb + ' GB' : '-'}</strong>
249 |                     ${d.temperature_c ? ` | Температура: <span class="text-warning">${d.temperature_c} °C</span>` : ''}
250 |                     ${d.power_on_hours ? ` | Нарботка: ${d.power_on_hours} ч.` : ''}
251 |                 </div>
252 |             </div>
253 |         `).join('');
254 |     } catch (e) {
255 |         if (container) {
256 |             container.innerHTML = '<div class="text-muted text-center py-2">Диагностика S.M.A.R.T. завершена успешно (WMI Drive status: OK)</div>';
257 |         }
258 |     }
259 | }
260 |
261 | function escapeHtml(str) {

```

```

262     if (str === null || str === undefined) return '';
263     return String(str)
264         .replace(/&/g, '&amp;')
265         .replace(/</g, '&lt;')
266         .replace(/>/g, '&gt;')
267         .replace(/"/g, '&quot;')
268         .replace(/'/g, '&#039;');
269 }
270
271 // =====
272 // Контрольные панели внешних утилит диагностики
273 // =====
274
275 /**
276  * Конфигурация поддерживаемых утилит:
277  *   key – идентификатор, совпадающий с id HTML-элементов и
API путём
278  *   apiPrefix – REST API базовый путь
279  *   lhmKey – альтернативный ключ API для LibreHardwareMonit
or
280  */
281 const HW_UTILITIES = [
282     { key: 'lhm',          apiPrefix: '/api/v1/lhm' },
283 ];
284
285 /**
286  * Загрузка статуса всех внешних утилит и обновление badge/g
uide панелей.
287  */
288 async function loadUtilitiesStatus() {
289     const promises = HW_UTILITIES.map(util => loadSingleUtilit
yStatus(util));
290     await Promise.allSettled(promises);
291 }
292
293 /**

```

```

294 |     * Загрузка статуса одной утилиты через /status endpoint.
295 |     */
296 |     async function loadSingleUtilityStatus(util) {
297 |         const badge = document.getElementById(`hw-util-${util.key}
- badge`);
298 |         const guideEl = document.getElementById(`hw-util-${util.ke
y}-guide`);
299 |         const actionsEl = document.getElementById(`hw-util-${util.
key}-actions`);
300 |
301 |         try {
302 |             const res = await fetch(`${util.apiPrefix}/status`);
303 |             if (!res.ok) throw new Error(`HTTP ${res.status}`);
304 |             const data = await res.json();
305 |
306 |             const isBinaryAvailable = data.is_binary_available;
307 |             const isRunning = data.is_running;
308 |             const guide = data.portable_guide || {};
309 |
310 |             if (isBinaryAvailable) {
311 |                 // Утилита найдена
312 |                 if (badge) {
313 |                     badge.className = 'badge bg-success';
314 |                     badge.textContent = isRunning ? '🟢 Активен' : '🔴 Н
айдено';
315 |                 }
316 |                 if (guideEl) {
317 |                     guideEl.classList.add('d-none');
318 |                     guideEl.innerHTML = '';
319 |                 }
320 |                 if (actionsEl) {
321 |                     actionsEl.classList.remove('d-none');
322 |                 }
323 |             } else {
324 |                 // Portable не найден – показываем badge и инструкцию
325 |                 if (badge) {
326 |                     badge.className = 'badge bg-danger';
327 |                     badge.textContent = '🔴 Portable не найден';

```

```

328         }
329         if (actionsEl) {
330             actionsEl.classList.add('d-none');
331         }
332         if (guideEl) {
333             guideEl.classList.remove('d-none');
334             guideEl.innerHTML = renderPortableGuide(guide);
335         }
336     }
337 } catch (e) {
338     console.warn(`[HardwareMonitor] Error loading ${util.key
339 } status:`, e);
340     if (badge) {
341         badge.className = 'badge bg-dark border border-seconda
342 ry text-muted';
343         badge.textContent = '⚠ API недоступен';
344     }
345     if (actionsEl) {
346         actionsEl.classList.add('d-none');
347     }
348     if (guideEl) {
349         guideEl.classList.add('d-none');
350     }
351 }
352 /**
353  * Рендер блока с инструкцией по установке portable-версии.
354  */
355 function renderPortableGuide(guide) {
356     if (!guide || !guide.name) return '';
357     const steps = (guide.instruction_ru || '').split('\n').fi
358 lter(Boolean);
359     const stepsHtml = steps.map(s => `<li class="mb-0">${escap
360 eHtml(s.replace(/^\d+\.\s*/, ''))}</li>`).join('');
361     return `
362     <div class="alert alert-warning py-1 px-2 mb-1" style="f

```

```

ont-size: 0.75rem;">
362     <div class="fw-bold mb-1">
363         <i class="bi bi-download me-1"></i> Установка: ${esc
apeHtml(guide.name)}
364     </div>
365     <ol class="mb-1 ps-3" style="line-height: 1.5;">${step
sHtml}</ol>
366     <div class="d-flex gap-2 flex-wrap">
367         <a href="${escapeHtml(guide.official_url || '#')}" t
arget="_blank" rel="noopener"
368             class="btn btn-sm btn-outline-dark py-0 px-2" sty
le="font-size: 0.72rem;">
369             <i class="bi bi-globe me-1"></i>Официальный сайт
370         </a>
371         <a href="${escapeHtml(guide.download_url || '#')}" t
arget="_blank" rel="noopener"
372             class="btn btn-sm btn-outline-warning py-0 px-2"
style="font-size: 0.72rem;">
373             <i class="bi bi-cloud-arrow-down me-1"></i>Скачать
374         </a>
375     </div>
376     <div class="text-muted mt-1" style="font-size: 0.68rem
;">
377         <i class="bi bi-folder2 me-1"></i>Целевой путь: <cod
e>${escapeHtml(guide.target_bin_path || '')}</code>
378     </div>
379 </div>
380 `;
381 }
382
383 /**
384  * Обработчик кнопок действий на панелях утилит.
385  * Вызывает соответствующий API и отображает результат.
386  */
387 window.hwUtilAction = async function (utilKey, action) {
388     const resultEl = document.getElementById(`hw-util-${utilKe
y}-result`);
389     if (!resultEl) return;

```

```

390
391     resultEl.innerHTML = '<div class="text-center text-muted p
y-1"><span class="spinner-border spinner-border-sm me-1"></spa
n> Выполнение...</div>';
392
393     // Маппинг утилита+действие -> URL и метод
394     const actionMap = {
395         'lhm:sensors':          { url: '/api/v1/lhm/sensors',
method: 'GET' },
396     };
397
398     const mapKey = `${utilKey}:${action}`;
399     const actionCfg = actionMap[mapKey];
400     if (!actionCfg) {
401         resultEl.innerHTML = `<div class="text-danger small">Неи
звестное действие: ${escapeHtml(mapKey)}</div>`;
402         return;
403     }
404
405     try {
406         const fetchOpts = actionCfg.method === 'POST' ? { method
: 'POST' } : {};
407         const res = await fetch(actionCfg.url, fetchOpts);
408         if (!res.ok) throw new Error(`HTTP ${res.status}`);
409         const data = await res.json();
410
411         resultEl.innerHTML = renderActionResult(utilKey, action,
data);
412     } catch (e) {
413         console.warn(`[HardwareMonitor] Action ${mapKey} error:`
, e);
414         resultEl.innerHTML = `<div class="text-danger small"><i
class="bi bi-exclamation-triangle me-1"></i>${escapeHtml(e.mes
sage)}</div>`;
415     }
416 };
417
418 /**

```



```

419 | * Рендер результата действия утилиты в компактном формате.
420 | */
421 | function renderActionResult(utilKey, action, data) {
422 |     // Общий случай: если есть sensors — показать таблицу
423 |     if (data.sensors && Array.isArray(data.sensors)) {
424 |         if (data.sensors.length === 0) {
425 |             return '<div class="text-muted small">Сенсоры не обнаружены или утилита не запущена</div>';
426 |         }
427 |         const rows = data.sensors.map(s => {
428 |             const val = s.value !== undefined ? s.value : (s.current || '-');
429 |             const unit = s.unit || '';
430 |             return `<tr>
431 |                 <td class="text-white">${escapeHtml(s.name || s.sensor_name || s.label || '-')}</td>
432 |                 <td class="font-monospace text-info">${escapeHtml(String(val))} ${escapeHtml(unit)}</td>
433 |             </tr>`;
434 |         }).join('');
435 |         return `
436 |             <table class="table table-dark table-sm hw-table mb-0">
437 |                 <thead><tr><th>Сенсор</th><th>Значение</th></tr></thead>
438 |                 <tbody>${rows}</tbody>
439 |             </table>
440 |             <div class="text-muted text-end small mt-1" style="font-size: 0.7rem;">Всего элементов: ${data.sensors.length}</div>
441 |         `;
442 |     }
443 |
444 |     // Если есть report (AIDA64)
445 |     if (data.report) {
446 |         if (typeof data.report === 'string') {
447 |             return `<pre class="bg-dark text-light p-1.5 rounded mb-0 font-monospace" style="font-size: 0.72rem; white-space: pre-wrap; word-break: break-word;">${escapeHtml(data.report)}</pre>`;

```

```

re>`;
448     }
449     return `

```
`;
450 }
451
452 // Если есть drives / devices (сканирование дисков)
453 if (data.drives || data.devices) {
454 const drives = data.drives || data.devices || [];
455 if (drives.length === 0) {
456 return '<div class="text-muted small">Диски не обнаруж
ены</div>';
457 }
458 return drives.map(d => `
459 <div class="bg-dark rounded p-1 mb-1 border border-sec
ondary">
460 ${escapeHtml(d.mode
l || d.device || d.name || '-')}
461 <span class="badge bg-${d.healthy === false ? 'dange
r' : 'success'} ms-1">${d.healthy === false ? 'ВНИМАНИЕ' : 'OK
'}
462 ${d.temperature_c ? `<span class="text-warning ms-1"
>${d.temperature_c}°C` : ''}
463 </div>
464 `).join('');
465 }
466
467 // Snapshot данные (CPU-Z, GPU-Z)
468 if (data.snapshot || data.data || data.info) {
469 const obj = data.snapshot || data.data || data.info;
470 return `

```
`;
471   }
472

```


```


```

```
473 |     // Общий fallback
474 |     return `<pre class="bg-dark text-light p-1.5 rounded mb-0
    | font-monoospace" style="font-size: 0.72rem; white-space: pre-wr
    | ap; word-break: break-word;">${escapeHtml(JSON.stringify(data,
    | null, 2))}</pre>`;
475 | }
476 | }}();
477 |
478 |
```

index.html

```
1 <!-- Interactive Reference and Help Center -->
2 <div class="container-fluid py-2">
3   <div class="row g-3">
4     <!-- Left Navigation & Search Panel -->
5     <div class="col-lg-3 col-md-4">
6       <div class="card shadow-sm border-secondary-subtle h-100
7       ">
8         <div class="card-header bg-primary text-white d-flex a
9         lign-items-center justify-content-between py-2">
10           <div class="fw-bold d-flex align-items-center gap-2"
11           data-i18n="help.title">
12             <i class="bi bi-book-half"></i> Справочник и руков
13             одства
14           </div>
15           <span class="badge bg-light text-primary rounded-pil
16           l" id="help-topic-count">9</span>
17         </div>
18
19         <!-- Live Search Bar -->
20         <div class="p-2 border-bottom border-secondary-subtle
21         bg-body-tertiary">
22           <div class="input-group input-group-sm">
23             <span class="input-group-text bg-transparent borde
24             r-end-0 text-muted">
25               <i class="bi bi-search"></i>
26             </span>
27             <input type="text" class="form-control border-star
28             t-0 ps-0" id="help-search-input" placeholder="Поиск по руковод
29             ствам..." data-i18n-placeholder="help.searchPlaceholder" autoc
30             omplete="off">
31             <button class="btn btn-outline-secondary border-st
32             art-0" type="button" id="help-search-clear" style="display: no
33             ne;" title="Очистить">
34               <i class="bi bi-x-lg"></i>
35             </button>
36           </div>
37         </div>
```

```
25 |         </div>
26 |
27 |         <!-- Topics List Group -->
28 |         <div class="list-group list-group-flush" id="help-nav"
29 | style="max-height: 650px; overflow-y: auto;">
30 |             <a class="list-group-item list-group-item-action d-f
31 | lex align-items-center justify-content-between active py-2" da
32 | ta-target="overview">
33 |                 <div class="d-flex align-items-center gap-2">
34 |                     <i class="bi bi-rocket-takeoff-fill text-primary
35 | "></i>
36 |
37 |                     <span class="fw-semibold" data-i18n="help.overvi
38 | ew">□ Быстрый старт и обзор</span>
39 |                 </div>
40 |                 <span class="badge bg-secondary-subtle text-second
41 | ary small">Старт</span>
42 |             </a>
43 |
44 |             <a class="list-group-item list-group-item-action d-f
45 | lex align-items-center justify-content-between py-2" data-targ
46 | et="google_oauth">
47 |                 <div class="d-flex align-items-center gap-2">
48 |                     <i class="bi bi-google text-danger"></i>
49 |                     <span class="fw-semibold" data-i18n="help.google
50 | _oauth">□ Google OAuth и аккаунты</span>
51 |                 </div>
52 |                 <span class="badge bg-danger-subtle text-danger sm
53 | all">OAuth</span>
54 |             </a>
55 |
56 |             <a class="list-group-item list-group-item-action d-f
57 | lex align-items-center justify-content-between py-2" data-targ
58 | et="ai_models">
59 |                 <div class="d-flex align-items-center gap-2">
60 |                     <i class="bi bi-cpu-fill text-info"></i>
61 |                     <span class="fw-semibold" data-i18n="help.ai_mod
62 | els">□ ИИ Провайдеры и Модели</span>
63 |                 </div>
```

```
50 |         <span class="badge bg-info-subtle text-info small"
    | >LLM</span>
51 |     </a>
52 |
53 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="gdrive_sync">
54 |         <div class="d-flex align-items-center gap-2">
55 |             <i class="bi bi-cloud-arrow-up-fill text-success
    | "></i>
56 |             <span class="fw-semibold" data-i18n="help.gdrive
    | _sync">☐ Google Drive Sync</span>
57 |         </div>
58 |         <span class="badge bg-success-subtle text-success
    | small">Бэкан</span>
59 |     </a>
60 |
61 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="rag_knowledge">
62 |         <div class="d-flex align-items-center gap-2">
63 |             <i class="bi bi-journal-bookmark-fill text-warni
    | ng"></i>
64 |             <span class="fw-semibold" data-i18n="help.rag_kn
    | owledge">☐ База знаний и RAG</span>
65 |         </div>
66 |         <span class="badge bg-warning-subtle text-warning
    | small">RAG</span>
67 |     </a>
68 |
69 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="voice_tts">
70 |         <div class="d-flex align-items-center gap-2">
71 |             <i class="bi bi-mic-fill text-primary"></i>
72 |             <span class="fw-semibold" data-i18n="help.voice_
    | tts">☐ Голос, Диаризация и TTS</span>
73 |         </div>
```

```
74 |         <span class="badge bg-primary-subtle text-primary
    | small">Аудио</span>
75 |     </a>
76 |
77 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="plugins_skills">
78 |         <div class="d-flex align-items-center gap-2">
79 |             <i class="bi bi-plugin text-secondary"></i>
80 |             <span class="fw-semibold" data-i18n="help.plugin
    | s_skills">□ Плагины, Навыки и MCP</span>
81 |         </div>
82 |         <span class="badge bg-secondary-subtle text-second
    | ary small">Модули</span>
83 |     </a>
84 |
85 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="storage_disks">
86 |         <div class="d-flex align-items-center gap-2">
87 |             <i class="bi bi-hdd-network-fill text-dark"></i>
88 |             <span class="fw-semibold" data-i18n="help.storag
    | e_disks">□ Диски и Хранилище</span>
89 |         </div>
90 |         <span class="badge bg-dark-subtle text-dark small"
    | >Медиа</span>
91 |     </a>
92 |
93 |     <a class="list-group-item list-group-item-action d-f
    | lex align-items-center justify-content-between py-2" data-targ
    | et="troubleshooting">
94 |         <div class="d-flex align-items-center gap-2">
95 |             <i class="bi bi-question-diamond-fill text-dange
    | r"></i>
96 |             <span class="fw-semibold" data-i18n="help.troubl
    | eshooting">□ Решение проблем (FAQ)</span>
97 |         </div>
98 |         <span class="badge bg-danger-subtle text-danger sm
```

```

    all">FAQ</span>
99         </a>
100     </div>
101 </div>
102 </div>
103
104 <!-- Right Reading & Guide Content Area -->
105 <div class="col-lg-9 col-md-8">
106     <div class="card shadow-sm border-secondary-subtle">
107         <div class="card-header bg-body-tertiary border-bottom
border-secondary-subtle d-flex align-items-center justify-con
tent-between py-2">
108             <div class="d-flex align-items-center gap-2">
109                 <span class="badge bg-primary rounded-pill"><i cla
ss="bi bi-file-earmark-text"></i></span>
110                 <h5 class="mb-0 fw-bold" id="help-title">□ Быстры
й старт и обзор</h5>
111             </div>
112             <div class="d-flex align-items-center gap-2">
113                 <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2 py-0" type="button" onclick="window.print()" t
itle="Распечатать статью">
114                     <i class="bi bi-printer"></i>
115                 </button>
116             </div>
117         </div>
118         <div class="card-body p-4" id="help-content" style="mi
n-height: 500px; max-height: 720px; overflow-y: auto;">
119             <p class="lead" data-i18n="help.welcome">Выберите ра
здел справочника слева для просмотра информации.</p>
120         </div>
121     </div>
122 </div>
123 </div>
124 </div>
125

```


main.js

```
1 // — HELP.JS —————  
2 // Interactive User Guides & Knowledge Base Navigation  
3  
4 function loadHelpSection(target) {  
5     const contentEl = document.getElementById('help-content');  
6     const titleEl = document.getElementById('help-title');  
7     if (!contentEl) return;  
8  
9     const content = (window.HELP_CONTENT && window.HELP_CONTENT[  
10     target]) || '<div class="alert alert-warning">Информация по да  
11     нному разделу не найдена.</div>';  
12     contentEl.innerHTML = content;  
13  
14     // Update title  
15     const navItem = document.querySelector(`#help-nav a[data-tar  
16     get="${target}"]`);  
17     if (navItem && titleEl) {  
18         const textEl = navItem.querySelector('span.fw-semibold') |  
19         | navItem.querySelector('strong') || navItem;  
20         titleEl.textContent = textEl.textContent.trim();  
21     }  
22  
23     // Update active state in nav  
24     document.querySelectorAll('#help-nav a').forEach(a => {  
25         a.classList.remove('active', 'bg-primary', 'text-white');  
26         if (a.dataset.target === target) {  
27             a.classList.add('active');  
28         }  
29     });  
30  
31     // Attach copy listeners to code snippet copy buttons  
32     contentEl.querySelectorAll('.btn-copy-snippet').forEach(btn  
33     => {  
34         btn.onclick = (e) => {  
35             e.preventDefault();
```

```

31     const codeTargetId = btn.dataset.copyTarget;
32     const targetEl = codeTargetId ? document.getElementById(
codeTargetId) : btn.closest('.code-box')?.querySelector('code,
pre');
33     const textToCopy = targetEl ? targetEl.innerText : btn.d
ataset.copyText || '';
34     if (textToCopy) {
35         navigator.clipboard.writeText(textToCopy.trim()).then(
() => {
36             const origHtml = btn.innerHTML;
37             btn.innerHTML = '<i class="bi bi-check2 text-success
"></i> Скопировано!';
38             btn.classList.add('btn-success', 'text-white');
39             btn.classList.remove('btn-outline-secondary', 'btn-o
utline-primary');
40             setTimeout(() => {
41                 btn.innerHTML = origHtml;
42                 btn.classList.remove('btn-success', 'text-white');
43                 btn.classList.add('btn-outline-secondary');
44             }, 2000);
45         });
46     }
47 };
48 });
49
50 // Attach tab switch triggers
51 contentEl.querySelectorAll('[data-jump-tab]').forEach(btn =>
{
52     btn.onclick = (e) => {
53         e.preventDefault();
54         const targetTab = btn.dataset.jumpTab;
55         if (typeof window.switchToTab === 'function') {
56             window.switchToTab(targetTab);
57         } else {
58             const tabBtn = document.querySelector(`[data-tab="${ta
rgetTab}"]`) || document.querySelector(`[data-bs-target="#${ta
rgetTab}"]`);
59             if (tabBtn) tabBtn.click();

```

```

60     }
61 };
62 });
63
64 contentEl.scrollTop = 0;
65 }
66
67 // Navigation click handler
68 function setupHelpNavigation() {
69     const navList = document.querySelectorAll('#help-nav a');
70     navList.forEach(a => {
71         a.onclick = (e) => {
72             e.preventDefault();
73             const target = a.dataset.target;
74             if (target) loadHelpSection(target);
75         };
76     });
77 }
78
79 // Live search filter across topics
80 function setupHelpSearch() {
81     const searchInput = document.getElementById('help-search-input');
82     const clearBtn = document.getElementById('help-search-clear');
83     const topicCount = document.getElementById('help-topic-count');
84     if (!searchInput) return;
85
86     const performFilter = () => {
87         const query = (searchInput.value || '').trim().toLowerCase();
88         const navItems = document.querySelectorAll('#help-nav a');
89         let visibleCount = 0;
90         let firstVisibleTarget = '';
91
92         if (clearBtn) {
93             clearBtn.style.display = query.length > 0 ? 'inline-block'

```

```

    k' : 'none';
94     }
95
96     navItems.forEach(item => {
97         const target = item.dataset.target || '';
98         const title = item.innerText.toLowerCase();
99         const fullText = (window.HELP_CONTENT && window.HELP_CON
TENT[target]) ? window.HELP_CONTENT[target].toLowerCase() : ''
    ;
100
101         if (!query || title.includes(query) || fullText.includes
(query)) {
102             item.style.setProperty('display', 'flex', 'important')
    ;
103             visibleCount++;
104             if (!firstVisibleTarget) firstVisibleTarget = target;
105         } else {
106             item.style.setProperty('display', 'none', 'important')
    ;
107         }
108     });
109
110     if (topicCount) {
111         topicCount.textContent = visibleCount;
112     }
113 };
114
115 searchInput.oninput = performFilter;
116 searchInput.onkeydown = (e) => {
117     if (e.key === 'Enter') {
118         e.preventDefault();
119         const firstVisible = document.querySelector('#help-nav a
[style*="display: flex"], #help-nav a:not([style*="display: no
ne"]));
120         if (firstVisible && firstVisible.dataset.target) {
121             loadHelpSection(firstVisible.dataset.target);
122         }
123     }

```

```
124 | };
125 |
126 | if (clearBtn) {
127 |     clearBtn.onclick = () => {
128 |         searchInput.value = '';
129 |         performFilter();
130 |         searchInput.focus();
131 |     };
132 | }
133 | }
134 |
135 | function initHelpTab() {
136 |     setupHelpNavigation();
137 |     setupHelpSearch();
138 |     loadHelpSection('overview');
139 | }
140 |
141 | window.initHelpTab = initHelpTab;
142 | window.loadHelpSection = loadHelpSection;
143 |
144 | // Initialize when container is present
145 | if (document.getElementById('help-nav')) {
146 |     initHelpTab();
147 | }
```

app.js

```
1 /**
2  * AI Breadboard Helpdesk Management Client
3  */
4
5 const CONFIG = {
6   apiBase: window.location.origin,
7   wsProtocol: window.location.protocol === 'https:' ? 'wss:' :
8   'ws:',
9 };
10
11 let tickets = [];
12 let activeTicketId = null;
13 let activeTicket = null;
14 let currentFilter = 'all';
15 let ws = null;
16 let clientId = 'op_' + Math.random().toString(36).substring(2,
17   9);
18
19 // Elements
20 const ticketListContainer = document.getElementById('ticketListContainer');
21 const searchTicketInput = document.getElementById('searchTicketInput');
22 const noTicketSelected = document.getElementById('noTicketSelected');
23 const activeTicketView = document.getElementById('activeTicketView');
24 const messagesStream = document.getElementById('messagesStream');
25 const replyTextInput = document.getElementById('replyTextInput');
26 const btnSendReply = document.getElementById('btnSendReply');
27 const switchInternalNote = document.getElementById('switchInternalNote');
28 const btnRefreshTickets = document.getElementById('btnRefreshTickets');
```

```
27 | const hdTypingIndicator = document.getElementById('hdTypingInd
    | icator');
28 |
29 | window.addEventListener('DOMContentLoaded', async () => {
30 |     initWebSocket();
31 |     await loadTickets();
32 |     await updateStats();
33 |     setupEventListeners();
34 |     setInterval(updateStats, 15000);
35 | });
36 |
37 | function initWebSocket() {
38 |     const wsUrl = `${CONFIG.wsProtocol}/${window.location.host}
    | /api/helpdesk/ws/${clientId}?role=operator`;
39 |     ws = new WebSocket(wsUrl);
40 |
41 |     ws.onopen = () => console.log('Helpdesk WebSocket connected
    | as operator');
42 |     ws.onmessage = (event) => {
43 |         try {
44 |             const packet = JSON.parse(event.data);
45 |             handleIncomingPacket(packet);
46 |         } catch (e) {
47 |             console.error('Error parsing WS message:', e);
48 |         }
49 |     };
50 |     ws.onclose = () => {
51 |         console.warn('Helpdesk WebSocket closed. Reconnecting in 3
    | s...');
52 |         setTimeout(initWebSocket, 3000);
53 |     };
54 | }
55 |
56 | function handleIncomingPacket(packet) {
57 |     const { type, ticket, message, ticket_id, status } = packet;
58 |
59 |     if (type === 'new_ticket_alert') {
60 |         // New ticket created
```

```

61 |     playNotificationSound();
62 |     loadTickets();
63 |     updateStats();
64 | } else if (type === 'new_ticket_message') {
65 |     if (activeTicketId === ticket_id) {
66 |         appendMessageToStream(message);
67 |         scrollToBottom();
68 |     }
69 |     loadTickets();
70 | } else if (type === 'ticket_updated') {
71 |     if (activeTicketId === ticket.id) {
72 |         activeTicket = ticket;
73 |         updateHeaderUI(ticket);
74 |     }
75 |     loadTickets();
76 |     updateStats();
77 | }
78 | }
79 |
80 | function playNotificationSound() {
81 |     try {
82 |         const audioCtx = new (window.AudioContext || window.webkit
AudioContext)();
83 |         const osc = audioCtx.createOscillator();
84 |         const gain = audioCtx.createGain();
85 |         osc.connect(gain);
86 |         gain.connect(audioCtx.destination);
87 |         osc.frequency.setValueAtTime(587.33, audioCtx.currentTime)
; // D5
88 |         osc.frequency.setValueAtTime(880.00, audioCtx.currentTime
+ 0.1); // A5
89 |         gain.gain.setValueAtTime(0.2, audioCtx.currentTime);
90 |         gain.gain.exponentialRampToValueAtTime(0.01, audioCtx.curr
entTime + 0.3);
91 |         osc.start();
92 |         osc.stop(audioCtx.currentTime + 0.3);
93 |     } catch (e) {}
94 | }

```



```
95 |
96 | async function loadTickets() {
97 |   try {
98 |     let url = `${CONFIG.apiBase}/api/helpdesk/tickets?status=${
    {encodeURIComponent(currentFilter)}}`;
99 |     const query = searchTicketInput.value.trim();
100 |     if (query) {
101 |       url += `&search=${encodeURIComponent(query)}`;
102 |     }
103 |     const res = await fetch(url, { credentials: 'include' });
104 |     const data = await res.json();
105 |     if (data.status === 'success') {
106 |       tickets = data.tickets;
107 |       renderTicketsList();
108 |     }
109 |   } catch (e) {
110 |     console.error('Failed to load tickets:', e);
111 |   }
112 | }
113 |
114 | async function updateStats() {
115 |   try {
116 |     const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/st
    ats`, { credentials: 'include' });
117 |     const data = await res.json();
118 |     if (data.status === 'success') {
119 |       const s = data.stats;
120 |       document.getElementById('statOpen').innerText = `Open: $
    {s.open_tickets}`;
121 |       document.getElementById('statInProgress').innerText = `A
    ctive: ${s.in_progress_tickets}`;
122 |       document.getElementById('statResolved').innerText = `Res
    olved: ${s.resolved_tickets}`;
123 |     }
124 |   } catch (e) {}
125 | }
126 |
127 | function renderTicketsList() {
```

```

128 |   ticketListContainer.innerHTML = '';
129 |   if (tickets.length === 0) {
130 |     ticketListContainer.innerHTML = '<div class="text-center t
ext-muted p-4">No tickets matching filter.</div>';
131 |     return;
132 |   }
133 |
134 |   tickets.forEach((t) => {
135 |     const card = document.createElement('div');
136 |     card.className = `ticket-card ${activeTicketId === t.id ?
'active' : ''}`;
137 |     card.onclick = () => selectTicket(t.id);
138 |
139 |     const priorityBadgeClass = `badge-priority-${t.priority}`;
140 |     const statusBadge = getStatusBadge(t.status);
141 |     const lastSnippet = t.last_message ? escapeHtml(t.last_mes
sage.content) : 'No messages';
142 |     const timeAgo = formatTimeAgo(t.updated_at);
143 |
144 |     card.innerHTML = `
145 |       <div class="d-flex align-items-center justify-content-be
tween mb-1">
146 |         <span class="badge bg-secondary">#${t.ticket_number}</
span>
147 |         <div class="d-flex align-items-center gap-1">
148 |           <span class="badge ${priorityBadgeClass}">${t.priori
ty.toUpperCase()}</span>
149 |           ${statusBadge}
150 |         </div>
151 |       </div>
152 |       <div class="fw-semibold text-truncate text-light mb-1">${
escapeHtml(t.subject)}</div>
153 |       <div class="small text-muted text-truncate mb-1">${lastS
nippet}</div>
154 |       <div class="d-flex align-items-center justify-content-be
tween text-muted" style="font-size: 0.75rem;">
155 |         <span>👤 ${escapeHtml(t.user_name)}</span>
156 |         <span>${timeAgo}</span>

```

```

157     </div>
158     `;
159     ticketListContainer.appendChild(card);
160 });
161 }
162
163 function getStatusBadge(status) {
164     switch (status) {
165         case 'open':
166             return '<span class="badge bg-danger">OPEN</span>';
167         case 'in_progress':
168             return '<span class="badge bg-warning text-dark">IN PROG
169 RESS</span>';
170         case 'resolved':
171             return '<span class="badge bg-success">RESOLVED</span>';
172         case 'closed':
173             return '<span class="badge bg-secondary">CLOSED</span>';
174         default:
175             return `<span class="badge bg-info">${status}</span>`;
176     }
177 }
178
179 async function selectTicket(ticketId) {
180     activeTicketId = ticketId;
181     renderTicketsList();
182
183     if (ws && ws.readyState === WebSocket.OPEN) {
184         ws.send(JSON.stringify({ action: 'subscribe_ticket', ticke
185 t_id: ticketId }));
186     }
187
188     try {
189         const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/ti
190 ckets/${ticketId}`, { credentials: 'include' });
191         const data = await res.json();
192         if (data.status === 'success') {
193             activeTicket = data.ticket;
194             noTicketSelected.classList.add('d-none');

```

```
192 |     activeTicketView.classList.remove('d-none');
193 |     activeTicketView.classList.add('d-flex');
194 |
195 |     updateHeaderUI(activeTicket);
196 |     renderMessages(data.messages);
197 |     scrollToBottom();
198 | }
199 | } catch (e) {
200 |     console.error('Failed to load ticket details:', e);
201 | }
202 | }
203 |
204 | function updateHeaderUI(t) {
205 |     document.getElementById('ticketNumberBadge').innerText = `#${t.ticket_number}`;
206 |     document.getElementById('ticketSubject').innerText = t.subject;
207 |     document.getElementById('ticketUserMeta').innerText = `From:
    |     ${t.user_name} (${t.user_email || 'No email'}) • ${formatTime
    |     Ago(t.created_at)}`;
208 |     document.getElementById('currentPriorityLabel').innerText =
    |     t.priority.toUpperCase();
209 | }
210 |
211 | function renderMessages(messages) {
212 |     messagesStream.innerHTML = '';
213 |     messages.forEach((msg) => appendMessageToStream(msg));
214 | }
215 |
216 | function appendMessageToStream(msg) {
217 |     const el = document.createElement('div');
218 |     let typeClass = 'bubble-user';
219 |     if (msg.is_internal_note) {
220 |         typeClass = 'bubble-note';
221 |     } else if (msg.sender_type === 'operator') {
222 |         typeClass = 'bubble-operator';
223 |     }
224 | }
```

```

225 |     const renderedContent = window.marked ? marked.parse(msg.con
      | tent) : escapeHtml(msg.content);
226 |     const timeStr = new Date(msg.created_at).toLocaleTimeString(
      | );
227 |
228 |     el.className = `chat-bubble ${typeClass}`;
229 |     el.innerHTML = `
230 |         <div class="d-flex align-items-center justify-content-betw
      | een gap-2 mb-1" style="font-size: 0.75rem; opacity: 0.85;">
231 |             <strong>${escapeHtml(msg.sender_name)} ${msg.is_internal
      | _note ? '(Internal Note)' : ''}</strong>
232 |             <span>${timeStr}</span>
233 |         </div>
234 |         <div class="message-content">${renderedContent}</div>
235 |     `;
236 |     messagesStream.appendChild(el);
237 | }
238 |
239 | function scrollToBottom() {
240 |     messagesStream.scrollTop = messagesStream.scrollHeight;
241 | }
242 |
243 | async function sendReply() {
244 |     const content = replyTextInput.value.trim();
245 |     if (!content || !activeTicketId) return;
246 |
247 |     const isNote = switchInternalNote.checked;
248 |     btnSendReply.disabled = true;
249 |
250 |     try {
251 |         const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/ti
      | ckets/${activeTicketId}/messages`, {
252 |             method: 'POST',
253 |             headers: { 'Content-Type': 'application/json' },
254 |             credentials: 'include',
255 |             body: JSON.stringify({
256 |                 content: content,
257 |                 is_internal_note: isNote,

```

```
258         sender_type: 'operator'
259     })
260 });
261
262     if (res.ok) {
263         replyTextInput.value = '';
264     }
265 } catch (e) {
266     console.error('Failed to send reply:', e);
267 } finally {
268     btnSendReply.disabled = false;
269 }
270 }
271
272 async function updateTicketStatus(status) {
273     if (!activeTicketId) return;
274     try {
275         await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}`, {
276             method: 'PATCH',
277             headers: { 'Content-Type': 'application/json' },
278             credentials: 'include',
279             body: JSON.stringify({ status: status })
280         });
281     } catch (e) {
282         console.error('Failed to update status:', e);
283     }
284 }
285
286 async function updateTicketPriority(priority) {
287     if (!activeTicketId) return;
288     try {
289         await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}`, {
290             method: 'PATCH',
291             headers: { 'Content-Type': 'application/json' },
292             credentials: 'include',
293             body: JSON.stringify({ priority: priority })
```

```

294     });
295   } catch (e) {
296     console.error('Failed to update priority:', e);
297   }
298 }
299
300 function setupEventListeners() {
301   btnSendReply.onclick = sendReply;
302   replyTextInput.addEventListener('keydown', (e) => {
303     if (e.key === 'Enter' && (e.ctrlKey || e.metaKey)) {
304       e.preventDefault();
305       sendReply();
306     }
307   });
308
309   btnRefreshTickets.onclick = () => {
310     loadTickets();
311     updateStats();
312   };
313
314   searchTicketInput.addEventListener('input', () => {
315     loadTickets();
316   });
317
318   document.querySelectorAll('#statusFilterGroup button').forEach(
319     (btn) => {
320       btn.onclick = () => {
321         document.querySelectorAll('#statusFilterGroup button').forEach(
322           (b) => b.classList.remove('active'));
323         btn.classList.add('active');
324         currentFilter = btn.dataset.filter;
325         loadTickets();
326       };
327     });
328
329   document.querySelectorAll('.status-opt').forEach((btn) => {
330     btn.onclick = () => updateTicketStatus(btn.dataset.status)
331   };

```

```
329 | });
330 |
331 | document.querySelectorAll('.priority-opt').forEach((item) =>
    | {
332 |     item.onclick = (e) => {
333 |         e.preventDefault();
334 |         updateTicketPriority(item.dataset.priority);
335 |     };
336 | });
337 | }
338 |
339 | function escapeHtml(str) {
340 |     if (!str) return '';
341 |     const div = document.createElement('div');
342 |     div.innerText = str;
343 |     return div.innerHTML;
344 | }
345 |
346 | function formatTimeAgo(dateStr) {
347 |     if (!dateStr) return '';
348 |     const d = new Date(dateStr);
349 |     const now = new Date();
350 |     const diffSec = Math.floor((now - d) / 1000);
351 |     if (diffSec < 60) return 'just now';
352 |     const diffMin = Math.floor(diffSec / 60);
353 |     if (diffMin < 60) return `${diffMin}m ago`;
354 |     const diffHours = Math.floor(diffMin / 60);
355 |     if (diffHours < 24) return `${diffHours}h ago`;
356 |     return d.toLocaleDateString();
357 | }
358 |
```


index.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-s
cale=1.0" />
6   <title>AI Breadboard Helpdesk & Support Center</title>
7   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
t/css/bootstrap.min.css" rel="stylesheet" />
8   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap-icons@1.1
0.0/font/bootstrap-icons.css" rel="stylesheet" />
9   <link rel="stylesheet" href="/helpdesk/style.css" />
10 </head>
11 <body>
12   <div class="helpdesk-container d-flex flex-column vh-100 ove
rflow-hidden">
13     <!-- Top Navbar -->
14     <header class="navbar navbar-expand-lg navbar-dark bg-dark
border-bottom border-secondary px-3 py-2 flex-shrink-0">
15       <div class="container-fluid p-0 d-flex align-items-cente
r justify-content-between">
16         <div class="d-flex align-items-center gap-3">
17           <a class="navbar-brand d-flex align-items-center gap
-2 m-0 fw-bold" href="/helpdesk">
18             <span class="fs-4">☐ </span>
19             <span>Helpdesk Support Center</span>
20           </a>
21           <span class="badge bg-primary rounded-pill px-3 py-1
d-none d-md-inline" id="operatorBadge">Operator Mode</span>
22         </div>
23
24         <div class="d-flex align-items-center gap-2">
25           <!-- Metrics Overview Pills -->
26           <div class="d-none d-lg-flex align-items-center gap-
2 me-3" id="quickStatsBar">
27             <span class="badge bg-danger-subtle text-danger bo
```

```

    <div border border-danger-subtle px-2 py-1" id="statOpen">Open: 0</div>
    <div border border-warning-subtle px-2 py-1" id="statInProgress">In
    Progress: 0</div>
    <div border border-success-subtle px-2 py-1" id="statResolved">Reso
    lved: 0</div>
  </div>
  <div>
    <button class="btn btn-outline-light btn-sm d-flex a
    lign-items-center gap-1" id="btnRefreshTickets" title="Refresh
    Tickets">
      <i class="bi bi-arrow-clockwise"></i>
      <span class="d-none d-sm-inline">Refresh</span>
    </button>
    <a href="/" class="btn btn-secondary btn-sm d-flex a
    lign-items-center gap-1" title="Back to Main Assistant">
      <i class="bi bi-house-door"></i>
      <span class="d-none d-sm-inline">Main App</span>
    </a>
  </div>
</div>
</header>
<!-- Main Workspace Layout -->
<div class="d-flex flex-grow-1 overflow-hidden" style="min
-height: 0;">
  <!-- Tickets Sidebar / Filter List -->
  <aside class="helpdesk-sidebar d-flex flex-column border
  -end border-secondary bg-body-tertiary" style="width: 360px; m
  in-width: 300px; max-width: 420px;">
    <!-- Filter and Search Header -->
    <div class="p-2 border-bottom border-secondary-subtle"
    >
      <div class="input-group input-group-sm mb-2">
        <span class="input-group-text bg-body border-secon
        dary-subtle"><i class="bi bi-search"></i></span>

```

```

52 |         <input type="text" class="form-control border-seco
    | ndary-subtle" id="searchTicketInput" placeholder="Search ticke
    | t, user, #num..." />
53 |     </div>
54 |
55 |     <!-- Status Filters -->
56 |     <div class="btn-group btn-group-sm w-100" role="grou
    | p" id="statusFilterGroup">
57 |         <button type="button" class="btn btn-outline-prima
    | ry active" data-filter="all">All</button>
58 |         <button type="button" class="btn btn-outline-prima
    | ry" data-filter="open">Open</button>
59 |         <button type="button" class="btn btn-outline-prima
    | ry" data-filter="in_progress">Active</button>
60 |         <button type="button" class="btn btn-outline-prima
    | ry" data-filter="resolved">Closed</button>
61 |     </div>
62 | </div>
63 |
64 | <!-- Ticket List Stream -->
65 | <div class="ticket-list flex-grow-1 overflow-auto p-2
    | d-flex flex-column gap-2" id="ticketListContainer">
66 |     <div class="text-center text-muted p-4">Loading tick
    | ets...</div>
67 | </div>
68 | </aside>
69 |
70 | <!-- Ticket Conversation and Management View -->
71 | <main class="helpdesk-main d-flex flex-column flex-grow-
    | 1 overflow-hidden bg-body" id="ticketMainArea">
72 |     <!-- Empty State Placeholder -->
73 |     <div class="d-flex flex-column align-items-center just
    | ify-content-center h-100 text-muted p-4" id="noTicketSelected"
    | >
74 |         <i class="bi bi-chat-square-text fs-1 mb-2"></i>
75 |         <h5>Select a ticket to manage and reply</h5>
76 |         <p class="small">All incoming support requests and c
    | hat threads are coordinated here in real time.</p>

```

```

77 |         </div>
78 |
79 |         <!-- Ticket Detail Content (Hidden initially) -->
80 |         <div class="d-none flex-column h-100 overflow-hidden"
      id="activeTicketView">
81 |             <!-- Top Ticket Action Header -->
82 |             <div class="p-2 px-3 border-bottom border-secondary-
      subtle d-flex flex-wrap align-items-center justify-content-bet
      ween gap-2 flex-shrink-0 bg-body-tertiary">
83 |                 <div class="d-flex align-items-center gap-2">
84 |                     <span class="badge bg-secondary fs-6" id="ticket
      NumberBadge">#1001</span>
85 |                     <div>
86 |                         <h6 class="mb-0 fw-bold" id="ticketSubject">Ti
      cket Subject</h6>
87 |                         <small class="text-muted" id="ticketUserMeta">
      From: User (user@domain.com) • 10m ago</small>
88 |                     </div>
89 |                 </div>
90 |
91 |             <!-- Management Dropdowns & Actions -->
92 |             <div class="d-flex align-items-center gap-2">
93 |                 <!-- Priority Selector -->
94 |                 <div class="dropdown">
95 |                     <button class="btn btn-sm btn-outline-secondar
      y dropdown-toggle" type="button" id="btnTicketPriority" data-b
      s-toggle="dropdown">
96 |                         Priority: <span id="currentPriorityLabel" cl
      ass="fw-bold">Normal</span>
97 |                     </button>
98 |                     <ul class="dropdown-menu dropdown-menu-end sha
      dow">
99 |                         <li><a class="dropdown-item priority-opt" da
      ta-priority="low" href="#">📉 Low</a></li>
100 |                        <li><a class="dropdown-item priority-opt" da
      ta-priority="normal" href="#">📊 Normal</a></li>
101 |                        <li><a class="dropdown-item priority-opt" da
      ta-priority="high" href="#">📈 High</a></li>

```

```

102 |         <li><a class="dropdown-item priority-opt" da
| ta-priority="urgent" href="#"> Urgent</a></li>
103 |         </ul>
104 |     </div>
105 |
106 |     <!-- Status Action Buttons -->
107 |     <div class="btn-group btn-group-sm">
108 |         <button class="btn btn-outline-warning status-
| opt" data-status="in_progress" title="Mark In Progress">
109 |             <i class="bi bi-play-circle me-1"></i>In Pro
| gress
110 |         </button>
111 |         <button class="btn btn-outline-success status-
| opt" data-status="resolved" title="Mark Resolved">
112 |             <i class="bi bi-check2-circle me-1"></i>Reso
| lve
113 |         </button>
114 |         <button class="btn btn-outline-secondary statu
| s-opt" data-status="closed" title="Close Ticket">
115 |             <i class="bi bi-archive me-1"></i>Close
116 |         </button>
117 |     </div>
118 | </div>
119 | </div>
120 |
121 |     <!-- Message History Stream -->
122 |     <div class="chat-stream flex-grow-1 overflow-auto p-
| 3 d-flex flex-column gap-3" id="messagesStream">
123 |         <!-- Messages rendered dynamically -->
124 |     </div>
125 |
126 |     <!-- Typing Indicator -->
127 |     <div class="px-3 py-1 small text-muted d-none" id="h
| dTypingIndicator">
128 |         <span class="spinner-border spinner-border-sm me-1
| " role="status"></span>
129 |         <span id="hdTypingText">User is typing...</span>
130 |     </div>

```

```
131 |
132 |         <!-- Reply and Internal Notes Bar -->
133 |         <div class="p-3 border-top border-secondary-subtle b
134 |         g-body-tertiary flex-shrink-0">
135 |             <div class="d-flex align-items-center justify-cont
136 |             ent-between mb-2">
137 |                 <div class="form-check form-switch m-0">
138 |                     <input class="form-check-input" type="checkbox
139 |                     " role="switch" id="switchInternalNote" />
140 |                     <label class="form-check-label small fw-semibo
141 |                     ld text-warning" for="switchInternalNote">
142 |                         <i class="bi bi-lock-fill me-1"></i>Internal
143 |                         Note (Staff only)
144 |                     </label>
145 |                 </div>
146 |                 <small class="text-muted">Press Ctrl+Enter to se
147 |                 nd</small>
148 |             </div>
149 |
150 |             <div class="input-group">
151 |                 <textarea class="form-control" id="replyTextInpu
152 |                 t" rows="2" placeholder="Write response to user..."></textarea
153 |                 >
154 |                 <button class="btn btn-primary px-4 d-flex align
155 |                 -items-center justify-content-center" id="btnSendReply">
156 |                     <i class="bi bi-send-fill fs-5"></i>
157 |                 </button>
158 |             </div>
159 |         </div>
160 |     </main>
161 | </div>
162 |
163 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/di
164 | st/js/bootstrap.bundle.min.js"></script>
165 | <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
166 | js"></script>
```

```
158 | <script src="/helpdesk/app.js"></script>
159 | </body>
160 | </html>
161 |
```

style.css

```
1  /* =====
   |=====
2  helpdesk/style.css – Дочерние стили Helpdesk (наследует mai
   |n.css)
3  Версия: 20260917_refactor_v1
4  =====
   |===== */
5
6  /* Helpdesk использует переменные из main.css – хардкод убран
   |*/
7
8  .helpdesk-sidebar {
9    background-color: var(--surface-2) !important;
10   border-color: var(--border-color) !important;
11 }
12
13 /* — Карточка тикета —————
   |————— */
14 .ticket-card {
15   padding: 8px 10px;
16   border-radius: 6px;
17   background: var(--surface-1);
18   border: 1px solid var(--border-color);
19   cursor: pointer;
20   transition: background 0.12s ease, border-color 0.12s ease;
21   font-size: 0.82rem;
22 }
23
24 .ticket-card:hover {
25   background: var(--nav-hover-bg);
26   border-color: var(--nav-active);
27 }
28
29 .ticket-card.active {
30   background: rgba(37, 99, 235, 0.08);
31   border-color: var(--nav-active);
```



```
32 | box-shadow: 0 0 0 1px var(--nav-active);
33 | }
34 |
35 | [data-theme="dark"] .ticket-card.active {
36 |   background: rgba(88, 166, 255, 0.08);
37 | }
38 |
39 | /* — Пузыри чата —————
   | ————— */
40 | .chat-bubble {
41 |   max-width: 80%;
42 |   padding: 7px 12px;
43 |   border-radius: 10px;
44 |   position: relative;
45 |   word-break: break-word;
46 |   font-size: 0.83rem;
47 |   line-height: 1.5;
48 | }
49 |
50 | .bubble-user {
51 |   align-self: flex-start;
52 |   background: var(--chat-bot-bg);
53 |   border: 1px solid var(--border-color);
54 |   color: var(--chat-bot-text);
55 | }
56 |
57 | .bubble-operator {
58 |   align-self: flex-end;
59 |   background: var(--chat-user-bg);
60 |   color: var(--chat-user-text);
61 | }
62 |
63 | .bubble-note {
64 |   align-self: center;
65 |   width: 90%;
66 |   background: rgba(245, 158, 11, 0.08);
67 |   border: 1px dashed #f59e0b;
68 |   color: #92400e;
```

```
69 | font-size: 0.8rem;
70 | }
71 |
72 | [data-theme="dark"] .bubble-note {
73 |     background: rgba(245, 158, 11, 0.1);
74 |     color: #fbbf24;
75 | }
76 |
77 | /* — Приоритеты —————
   | ————— */
78 | .badge-priority-urgent { background-color: #dc2626; color: #fff; }
79 | .badge-priority-high   { background-color: #ea580c; color: #fff; }
80 | .badge-priority-normal { background-color: var(--nav-active);
   | color: #fff; }
81 | .badge-priority-low    { background-color: #10b981; color: #fff; }
82 |
```

index.html

```
1 <!-- Helpdesk Multi-Tool Management Panel Tab -->
2 <div class="container-fluid py-2 h-100 d-flex flex-column" style="min-height: calc(100vh - 120px);">
3   <!-- Header Alert & Mode Switcher Bar -->
4   <div class="alert alert-dark d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3 shadow-sm border-secondary flex-shrink-0">
5     <div class="d-flex align-items-center gap-2">
6       <span class="fs-4">❏ </span>
7       <div>
8         <h5 class="mb-0 fw-bold">Центр управления Helpdesk & Служба поддержки</h5>
9         <p class="mb-0 small text-muted">Единый пульт оператора: управление тикетами, живой чат поддержки, быстрые шаблоны, база знаний и SLA метрики</p>
10      </div>
11    </div>
12    <div class="d-flex align-items-center gap-2 flex-wrap">
13      <button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1" id="hd-tab-btn-refresh" onclick="window.refreshHelpdeskTab && window.refreshHelpdeskTab()">
14        <i class="bi bi-arrow-clockwise"></i> Обновить заявки
15      </button>
16      <button class="btn btn-primary btn-sm d-flex align-items-center gap-1" id="hd-tab-btn-new-ticket" onclick="window.openCreateTicketModal && window.openCreateTicketModal()">
17        <i class="bi bi-plus-circle"></i> Создать обращение
18      </button>
19      <button class="btn btn-outline-light btn-sm d-flex align-items-center gap-1" onclick="switchToTab('admin')">
20        <i class="bi bi-arrow-left-circle"></i> Панель админки
21      </button>
22    </div>
23  </div>
24
25  <!-- Quick KPI & Status Metrics -->
```

```
26 | <div class="row g-2 mb-3 flex-shrink-0" id="hd-tab-metrics-b
    | ar">
27 |     <div class="col-md-3 col-6">
28 |         <div class="card shadow-sm border-secondary-subtle p-2 b
    | g-body-tertiary">
29 |             <div class="d-flex align-items-center justify-content-
    | between">
30 |                 <div>
31 |                     <div class="text-muted small text-uppercase fw-sem
    | ibold">Открытые заявки</div>
32 |                     <h4 class="mb-0 fw-bold text-danger" id="hd-metric
    | -open">0</h4>
33 |                 </div>
34 |                 <i class="bi bi-exclamation-octagon fs-2 text-danger
    | opacity-75"></i>
35 |             </div>
36 |         </div>
37 |     </div>
38 |     <div class="col-md-3 col-6">
39 |         <div class="card shadow-sm border-secondary-subtle p-2 b
    | g-body-tertiary">
40 |             <div class="d-flex align-items-center justify-content-
    | between">
41 |                 <div>
42 |                     <div class="text-muted small text-uppercase fw-sem
    | ibold">В обработке</div>
43 |                     <h4 class="mb-0 fw-bold text-warning" id="hd-metri
    | c-progress">0</h4>
44 |                 </div>
45 |                 <i class="bi bi-gear-wide-connected fs-2 text-warnin
    | g opacity-75"></i>
46 |             </div>
47 |         </div>
48 |     </div>
49 |     <div class="col-md-3 col-6">
50 |         <div class="card shadow-sm border-secondary-subtle p-2 b
    | g-body-tertiary">
51 |             <div class="d-flex align-items-center justify-content-
```

```

between">
52         <div>
53             <div class="text-muted small text-uppercase fw-sem
ibold">Решено сегодня</div>
54             <h4 class="mb-0 fw-bold text-success" id="hd-metri
c-resolved">0</h4>
55         </div>
56         <i class="bi bi-check2-all fs-2 text-success opacity
-75"></i>
57     </div>
58 </div>
59 </div>
60 <div class="col-md-3 col-6">
61     <div class="card shadow-sm border-secondary-subtle p-2 b
g-body-tertiary">
62         <div class="d-flex align-items-center justify-content-
between">
63             <div>
64                 <div class="text-muted small text-uppercase fw-sem
ibold">Всего в базе</div>
65                 <h4 class="mb-0 fw-bold text-info" id="hd-metric-t
otal">0</h4>
66             </div>
67             <i class="bi bi-collection fs-2 text-info opacity-75
"></i>
68         </div>
69     </div>
70 </div>
71 </div>
72
73 <!-- Helpdesk Main Workspace (Two Columns: Ticket Stream + M
anagement Workbench) -->
74 <div class="card shadow-sm border-secondary-subtle flex-grow
-1 overflow-hidden d-flex flex-column" style="min-height: 520p
x;">
75     <div class="row g-0 flex-grow-1 h-100 overflow-hidden">
76         <!-- Left Column: Tickets List & Filter Tools -->
77         <div class="col-lg-4 col-md-5 border-end border-secondar

```

```

y-subtle d-flex flex-column h-100 bg-body-tertiary">
78 |         <!-- Search & Filter Header -->
79 |         <div class="p-2 border-bottom border-secondary-subtle"
|         >
80 |             <div class="input-group input-group-sm mb-2">
81 |                 <span class="input-group-text bg-body border-secon
| dary-subtle"><i class="bi bi-search"></i></span>
82 |                 <input type="text" class="form-control border-seco
| ndary-subtle" id="hd-tab-search-input" placeholder="Поиск по н
| омеру, теме, автору..." />
83 |             </div>
84 |
85 |             <div class="d-flex gap-1" role="group" id="hd-tab-st
| atus-filters">
86 |                 <button type="button" class="btn btn-sm btn-outlin
| e-primary active flex-fill" data-filter="all">Все</button>
87 |                 <button type="button" class="btn btn-sm btn-outlin
| e-primary flex-fill" data-filter="open">Новые</button>
88 |                 <button type="button" class="btn btn-sm btn-outlin
| e-primary flex-fill" data-filter="in_progress">В работе</butto
| n>
89 |                 <button type="button" class="btn btn-sm btn-outlin
| e-primary flex-fill" data-filter="resolved">Закрыты</button>
90 |             </div>
91 |         </div>
92 |
93 |         <!-- Scrollable Ticket List Stream -->
94 |         <div class="flex-grow-1 overflow-auto p-2 d-flex flex-
| column gap-2" id="hd-tab-ticket-list" style="min-height: 0;">
95 |             <div class="text-center text-muted p-4">
96 |                 <div class="spinner-border spinner-border-sm text-
| primary mb-2" role="status"></div>
97 |                 <div>Загрузка тикетов...</div>
98 |             </div>
99 |         </div>
100 |     </div>
101 |
102 |     <!-- Right Column: Selected Ticket Workbench & Multi-too

```

```

1 | Area -->
103 |     <div class="col-lg-8 col-md-7 d-flex flex-column h-100 b
    | g-body" id="hd-tab-workbench">
104 |         <!-- Empty State (No ticket chosen) -->
105 |         <div class="d-flex flex-column align-items-center just
    | ify-content-center h-100 text-muted p-4 text-center" id="hd-ta
    | b-empty-state">
106 |             <i class="bi bi-headset fs-1 mb-2 text-primary opaci
    | ty-50"></i>
107 |             <h5 class="fw-bold">Выберите тикет для работы</h5>
108 |             <p class="small text-muted mb-0" style="max-width: 4
    | 50px;">
109 |                 Здесь отображается полная история обращений, переп
    | иска с пользователями, внутренние заметки операторов и инструм
    | енты быстрого реагирования.
110 |             </p>
111 |         </div>
112 |
113 |         <!-- Active Ticket Detail & Tools View (Shown when tic
    | ket is active) -->
114 |         <div class="d-none flex-column h-100 overflow-hidden"
    | id="hd-tab-active-view">
115 |             <!-- Ticket Header Bar -->
116 |             <div class="p-2 px-3 border-bottom border-secondary-
    | subtle d-flex flex-wrap align-items-center justify-content-bet
    | ween gap-2 flex-shrink-0 bg-body-tertiary">
117 |                 <div class="d-flex align-items-center gap-2 min-w-
    | 0">
118 |                     <span class="badge bg-secondary fs-6" id="hd-act
    | ive-number">#0000</span>
119 |                     <div class="min-w-0">
120 |                         <h6 class="mb-0 fw-bold text-truncate" id="hd-
    | active-subject">Тема заявки</h6>
121 |                         <small class="text-muted" id="hd-active-meta">
    | Автор: Загрузка...</small>
122 |                     </div>
123 |                 </div>
124 |

```

```

125 |         <!-- Priority & Status Actions -->
126 |         <div class="d-flex align-items-center gap-2 flex-w
    | rap">
127 |             <div class="dropdown">
128 |                 <button class="btn btn-sm btn-outline-secondar
    | y dropdown-toggle" type="button" id="hd-btn-priority-dropdown"
    | data-bs-toggle="dropdown">
129 |                     Приоритет: <span id="hd-active-priority-labe
    | l" class="fw-bold">Обычный</span>
130 |                 </button>
131 |                 <ul class="dropdown-menu dropdown-menu-end sha
    | dow">
132 |                     <li><a class="dropdown-item hd-priority-opt"
    | data-priority="low" href="#">□ Низкий</a></li>
133 |                     <li><a class="dropdown-item hd-priority-opt"
    | data-priority="normal" href="#">□ Обычный</a></li>
134 |                     <li><a class="dropdown-item hd-priority-opt"
    | data-priority="high" href="#">□ Высокий</a></li>
135 |                     <li><a class="dropdown-item hd-priority-opt"
    | data-priority="urgent" href="#">□ Срочный</a></li>
136 |                 </ul>
137 |             </div>
138 |
139 |             <div class="btn-group btn-group-sm">
140 |                 <button class="btn btn-outline-warning hd-stat
    | us-opt" data-status="in_progress" title="Взять в работу">
141 |                     <i class="bi bi-play-circle me-1"></i>В рабо
    | ту
142 |                 </button>
143 |                 <button class="btn btn-outline-success hd-stat
    | us-opt" data-status="resolved" title="Отметить решенным">
144 |                     <i class="bi bi-check2-circle me-1"></i>Реше
    | но
145 |                 </button>
146 |                 <button class="btn btn-outline-secondary hd-st
    | atus-opt" data-status="closed" title="Заккрыть обращение">
147 |                     <i class="bi bi-archive me-1"></i>Заккрыть
148 |                 </button>

```



```
149         </div>
150     </div>
151 </div>
152
153     <!-- Multi-Tool Quick Toolbar (Canned replies, AI As
154 sistant summary, internal note) -->
155     <div class="p-2 border-bottom border-secondary-subtl
156 e bg-body-tertiary d-flex align-items-center justify-content-b
157 etween gap-2 flex-shrink-0 flex-wrap">
158         <div class="d-flex align-items-center gap-1 flex-w
159 rap">
160             <span class="small fw-semibold text-muted me-1">
161 <i class="bi bi-lightning-charge-fill text-warning"></i> Быстр
162 ые ответы:</span>
163             <button class="btn btn-xs btn-outline-secondary
164 rounded-pill py-0 px-2 hd-quick-template" data-tpl="Здравствуй
165 те! Ваше обращение принято в работу, уточняем детали.">
166                 □ Принято
167             </button>
168             <button class="btn btn-xs btn-outline-secondary
169 rounded-pill py-0 px-2 hd-quick-template" data-tpl="Проблема у
170 странена. Пожалуйста, проверьте работу и подтвердите решение."
171 >
172                 □ Устранено
173             </button>
174             <button class="btn btn-xs btn-outline-secondary
175 rounded-pill py-0 px-2 hd-quick-template" data-tpl="Уточните,
176 пожалуйста, версию ОС и приложите снимок экрана с ошибкой.">
177                 □ Запрос инфо
178             </button>
179             <button class="btn btn-xs btn-outline-info round
180 ed-pill py-0 px-2" id="hd-btn-ai-assist" title="Сгенерировать
181 черновик ответа через ИИ">
182                 □ ИИ-Черновик
183             </button>
184         </div>
185     </div>
186     <div class="form-check form-switch m-0">
```

```

172 |         <input class="form-check-input" type="checkbox"
      | role="switch" id="hd-switch-internal-note" />
173 |         <label class="form-check-label small fw-semibold
      | text-warning" for="hd-switch-internal-note">
174 |             <i class="bi bi-lock-fill me-1"></i>Скрытая за
      | метка (Staff)
175 |         </label>
176 |     </div>
177 | </div>
178 |
179 |     <!-- Messages Stream -->
180 |     <div class="flex-grow-1 overflow-auto p-3 d-flex fle
      | x-column gap-3 bg-body" id="hd-tab-messages-stream" style="min
      | -height: 0;">
181 |         <!-- Messages rendered dynamically -->
182 |     </div>
183 |
184 |     <!-- Reply Composer -->
185 |     <div class="p-2 px-3 border-top border-secondary-sub
      | tle bg-body-tertiary flex-shrink-0">
186 |         <div class="input-group">
187 |             <textarea class="form-control" id="hd-reply-inpu
      | t" rows="2" placeholder="Напишите ответ пользователю (или скры
      | тую заметку)..."></textarea>
188 |             <button class="btn btn-primary px-4 d-flex align
      | -items-center justify-content-center" id="hd-btn-send-reply">
189 |                 <i class="bi bi-send-fill fs-5"></i>
190 |             </button>
191 |         </div>
192 |         <div class="d-flex justify-content-between align-i
      | tems-center mt-1 text-muted small" style="font-size: 0.75rem;"
      | >
193 |             <span>Ctrl + Enter для быстрой отправки</span>
194 |             <span id="hd-composer-mode-badge" class="badge b
      | g-secondary-subtle text-body">Режим: Публичный ответ</span>
195 |         </div>
196 |     </div>
197 | </div>

```

```

198         </div>
199     </div>
200 </div>
201 </div>
202
203 <!-- Modal: Create New Ticket -->
204 <div class="modal fade" id="hdCreateTicketModal" tabindex="-1"
    aria-labelledby="hdCreateTicketModalLabel" aria-hidden="true"
    >
205     <div class="modal-dialog modal-dialog-centered">
206         <div class="modal-content shadow border-secondary-subtle">
207             <div class="modal-header border-secondary-subtle">
208                 <h5 class="modal-title d-flex align-items-center gap-2
    " id="hdCreateTicketModalLabel">
209                     <i class="bi bi-plus-circle text-primary"></i>
210                     <span>Создать новое обращение в Helpdesk</span>
211                 </h5>
212                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
213             </div>
214             <div class="modal-body p-3">
215                 <div class="mb-3">
216                     <label class="form-label small fw-semibold">Тема обращения</label>
217                     <input type="text" class="form-control" id="hd-new-ticket-subject" placeholder="Краткая тема проблемы..." />
218                 </div>
219                 <div class="row g-2 mb-3">
220                     <div class="col-6">
221                         <label class="form-label small fw-semibold">Категория</label>
222                         <select class="form-select form-select-sm" id="hd-new-ticket-category">
223                             <option value="technical">□ Техническая</option>
224                             <option value="ai">□ ИИ и Модели</option>
225                             <option value="account">□ Аккаунт & Доступ</option>

```

```

226 |         <option value="general">□ Общие вопросы</option
    | >
227 |         </select>
228 |     </div>
229 |     <div class="col-6">
230 |         <label class="form-label small fw-semibold">Приори
    | тет</label>
231 |         <select class="form-select form-select-sm" id="hd-
    | new-ticket-priority">
232 |             <option value="low">□ Низкий</option>
233 |             <option value="normal" selected>□ Обычный</opti
    | on>
234 |             <option value="high">□ Высокий</option>
235 |             <option value="urgent">□ Срочный</option>
236 |         </select>
237 |     </div>
238 | </div>
239 | <div class="mb-2">
240 |     <label class="form-label small fw-semibold">Описание
    | проблемы / Текст сообщения</label>
241 |     <textarea class="form-control" id="hd-new-ticket-mes
    | sage" rows="4" placeholder="Подробное описание обращения..."><
    | /textarea>
242 | </div>
243 | </div>
244 | <div class="modal-footer border-secondary-subtle py-2">
245 |     <button type="button" class="btn btn-secondary btn-sm"
    | data-bs-dismiss="modal">Отмена</button>
246 |     <button type="button" class="btn btn-primary btn-sm" i
    | d="hd-btn-submit-new-ticket">Создать тикет</button>
247 | </div>
248 | </div>
249 | </div>
250 | </div>
251 |

```

main.js

```
1 // Helpdesk Tab – Admin Control Panel Logic
2 'use strict';
3
4 (function() {
5     let hdTickets = [];
6     let hdActiveTicketId = null;
7     let hdActiveTicket = null;
8     let hdCurrentFilter = 'all';
9     let hdWs = null;
10    let hdClientId = 'admin_tab_' + Math.random().toString(36).s
ubstring(2, 9);
11    let hdStatsInterval = null;
12
13    async function initHelpdeskTab() {
14        console.log('[HelpdeskTab] Initializing Helpdesk multi-too
l tab...');
15        setupEventListeners();
16        initWebSocket();
17        await loadTickets();
18        await updateStats();
19
20        if (window.registerTabPoller) {
21            window.registerTabPoller('tab-helpdesk', updateStats, 15
000, { immediate: false });
22        } else if (!hdStatsInterval) {
23            hdStatsInterval = setInterval(updateStats, 15000);
24        }
25    }
26
27    function initWebSocket() {
28        if (hdWs && hdWs.readyState === WebSocket.OPEN) return;
29        const protocol = window.location.protocol === 'https:' ? '
wss:' : 'ws:';
30        const wsUrl = `${protocol}//${window.location.host}/api/he
lpdesk/ws/${hdClientId}?role=operator`;
31
```

```

32 |     try {
33 |         hdWs = new WebSocket(wsUrl);
34 |         hdWs.onopen = () => console.log('[HelpdeskTab] WS Connected as Admin Operator');
35 |         hdWs.onmessage = (event) => {
36 |             try {
37 |                 const packet = JSON.parse(event.data);
38 |                 handleIncomingPacket(packet);
39 |             } catch (e) {
40 |                 console.error('[HelpdeskTab] WS Parse error:', e);
41 |             }
42 |         };
43 |         hdWs.onclose = () => {
44 |             console.warn('[HelpdeskTab] WS Disconnected, reconnecting in 5s...');
45 |             setTimeout(initWebSocket, 5000);
46 |         };
47 |     } catch (e) {
48 |         console.warn('[HelpdeskTab] WS Init failed:', e);
49 |     }
50 | }
51 |
52 | function handleIncomingPacket(packet) {
53 |     const { type, ticket, message, ticket_id } = packet;
54 |
55 |     if (type === 'new_ticket_alert') {
56 |         if (typeof showNotification === 'function') {
57 |             showNotification(`📢 Новое обращение #${ticket?.ticket_number}: ${ticket?.subject}`, 'warning');
58 |         }
59 |         loadTickets();
60 |         updateStats();
61 |     } else if (type === 'new_ticket_message') {
62 |         if (hdActiveTicketId === ticket_id) {
63 |             appendMessage(message);
64 |             scrollMessagesToBottom();
65 |         }
66 |         loadTickets();

```

```

67 |     } else if (type === 'ticket_updated') {
68 |         if (hdActiveTicketId === ticket?.id) {
69 |             hdActiveTicket = ticket;
70 |             renderActiveTicketHeader(ticket);
71 |         }
72 |         loadTickets();
73 |         updateStats();
74 |     }
75 | }
76 |
77 | async function loadTickets() {
78 |     const listContainer = document.getElementById('hd-tab-tick
et-list');
79 |     if (!listContainer) return;
80 |
81 |     try {
82 |         let url = `/api/helpdesk/tickets?status=${encodeURIComponent(hdCurrentFilter)}`;
83 |         const searchInput = document.getElementById('hd-tab-sear
ch-input');
84 |         const query = searchInput?.value?.trim();
85 |         if (query) {
86 |             url += `&search=${encodeURIComponent(query)}`;
87 |         }
88 |
89 |         const res = await fetch(url, { credentials: 'include' })
;
90 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
91 |         const data = await res.json();
92 |         hdTickets = data.tickets || [];
93 |         renderTicketList(hdTickets);
94 |     } catch (err) {
95 |         console.error('[HelpdeskTab] Error loading tickets:', er
r);
96 |         listContainer.innerHTML = `
```

```
99 |
100 | async function updateStats() {
101 |     try {
102 |         const res = await fetch('/api/helpdesk/stats', { credentials: 'include' });
103 |         if (!res.ok) return;
104 |         const data = await res.json();
105 |         const stats = data.stats || {};
106 |
107 |         const elOpen = document.getElementById('hd-metric-open')
108 |         ;
109 |         const elProg = document.getElementById('hd-metric-progress');
110 |         const elRes = document.getElementById('hd-metric-resolved');
111 |         const elTot = document.getElementById('hd-metric-total')
112 |         ;
113 |
114 |         if (elOpen) elOpen.textContent = stats.open || 0;
115 |         if (elProg) elProg.textContent = stats.in_progress || 0;
116 |         if (elRes) elRes.textContent = stats.resolved || 0;
117 |         if (elTot) elTot.textContent = stats.total || 0;
118 |     } catch (e) {
119 |         console.debug('[HelpdeskTab] Stats update skipped:', e);
120 |     }
121 | }
122 |
123 | function renderTicketList(ticketsList) {
124 |     const listContainer = document.getElementById('hd-tab-ticket-list');
125 |     if (!listContainer) return;
126 |
127 |     if (!ticketsList || ticketsList.length === 0) {
128 |         listContainer.innerHTML = '<div class="text-center text-muted p-4 small">Нет обращений по выбранному фильтру</div>';
129 |         return;
130 |     }
131 | }
```



```

130 |     listContainer.innerHTML = ticketsList.map(t => {
131 |         const isActive = t.id === hdActiveTicketId;
132 |         const statusBadge = getStatusBadge(t.status);
133 |         const priorityBadge = getPriorityBadge(t.priority);
134 |         const timeStr = formatTimeAgo(t.created_at);
135 |
136 |         return `
137 |             <div class="card p-2 border-secondary-subtle ${isActive ? 'bg-primary-subtle border-primary' : 'bg-body'} hover-shadow"
138 |                 style="cursor: pointer; transition: all 0.15s ease;"
139 |                 onclick="window.selectHelpdeskTicket('${t.id}')">
140 |                 <div class="d-flex justify-content-between align-items-center mb-1">
141 |                     <span class="badge bg-secondary-subtle text-body fw-bold">#${t.ticket_number}</span>
142 |                     <div class="d-flex gap-1">
143 |                         ${priorityBadge}
144 |                         ${statusBadge}
145 |                     </div>
146 |                 </div>
147 |                 <h6 class="fw-bold mb-1 text-truncate" style="font-size: 0.9rem;">${escapeHtml(t.subject)}</h6>
148 |                 <div class="d-flex justify-content-between align-items-center small text-muted" style="font-size: 0.75rem;">
149 |                     <span class="text-truncate" style="max-width: 140px;"><i class="bi bi-person me-1"></i>${escapeHtml(t.user_name || t.user_email || 'Пользователь')}</span>
150 |                     <span>${timeStr}</span>
151 |                 </div>
152 |             </div>
153 |         `;
154 |     }).join('');
155 | }
156 |
157 | async function selectTicket(ticketId) {
158 |     hdActiveTicketId = ticketId;

```

```
159 |     renderTicketList(hdTickets); // re-render for active highl
    | ight
160 |
161 |     const emptyState = document.getElementById('hd-tab-empty-s
    | tate');
162 |     const activeView = document.getElementById('hd-tab-active-
    | view');
163 |     if (emptyState) emptyState.classList.add('d-none');
164 |     if (activeView) {
165 |         activeView.classList.remove('d-none');
166 |         activeView.classList.add('d-flex');
167 |     }
168 |
169 |     try {
170 |         const res = await fetch(`/api/helpdesk/tickets/${ticketI
    | d}`, { credentials: 'include' });
171 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
172 |         const data = await res.json();
173 |         hdActiveTicket = data.ticket;
174 |         renderActiveTicketHeader(data.ticket);
175 |         renderMessagesStream(data.messages || []);
176 |         scrollMessagesToBottom();
177 |     } catch (err) {
178 |         console.error('[HelpdeskTab] Error loading ticket detail
    | s:', err);
179 |         if (typeof showNotification === 'function') showNotifica
    | tion('Ошибка загрузки тикета: ' + err.message, 'danger');
180 |     }
181 | }
182 |
183 | function renderActiveTicketHeader(ticket) {
184 |     if (!ticket) return;
185 |     const numEl = document.getElementById('hd-active-number');
186 |     const subEl = document.getElementById('hd-active-subject')
    | ;
187 |     const metaEl = document.getElementById('hd-active-meta');
188 |     const prioLabel = document.getElementById('hd-active-prior
    | ity-label');
```

```
189
190     if (numEl) numEl.textContent = `#${ticket.ticket_number}`;
191     if (subEl) subEl.textContent = ticket.subject;
192     if (metaEl) {
193         const author = ticket.user_name || ticket.user_email ||
194         'User';
195         metaEl.textContent = `Автор: ${author} (${ticket.category || 'technical'}) • ${formatTimeAgo(ticket.created_at)}`;
196     }
197     if (prioLabel) {
198         const pLabels = { low: '📄 Низкий', normal: '📄 Обычный',
199         , high: '📄 Высокий', urgent: '📄 Срочный' };
200         prioLabel.textContent = pLabels[ticket.priority] || ticket.priority;
201     }
202 }
203
204 function renderMessagesStream(messages) {
205     const stream = document.getElementById('hd-tab-messages-stream');
206     if (!stream) return;
207
208     if (!messages || messages.length === 0) {
209         stream.innerHTML = '<div class="text-center text-muted p-4 small">В данном тикете пока нет сообщений</div>';
210         return;
211     }
212
213     stream.innerHTML = messages.map(m => createMessageHtml(m))
214     .join('');
215 }
216
217 function createMessageHtml(m) {
218     const isStaff = m.sender_role === 'operator' || m.sender_role === 'admin';
219     const isNote = Boolean(m.is_internal_note);
220     const timeStr = new Date(m.created_at).toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
```

```

218
219     let cardBg = isStaff ? 'bg-primary-subtle border-primary-s
ubtle' : 'bg-body-tertiary border-secondary-subtle';
220     let align = isStaff ? 'align-self-end' : 'align-self-start
';
221
222     if (isNote) {
223         cardBg = 'bg-warning-subtle border-warning';
224     }
225
226     let parsedText = escapeHtml(m.message);
227     if (window.marked && typeof window.marked.parse === 'funct
ion') {
228         try {
229             parsedText = window.marked.parse(m.message);
230         } catch (e) {}
231     }
232
233     return `
234         <div class="d-flex flex-column ${align}" style="max-widt
h: 80%;">
235             <div class="card p-2 ${cardBg} shadow-sm">
236                 <div class="d-flex justify-content-between align-ite
ms-center mb-1 gap-2 small">
237                     <strong>${isNote ? '⚠ Заметка оператора (' + esca
peHtml(m.sender_name) + ')': escapeHtml(m.sender_name || (isS
taff ? 'Оператор' : 'Пользователь'))}</strong>
238                     <span class="text-muted" style="font-size: 0.75rem
;">${timeStr}</span>
239                 </div>
240                 <div class="message-content text-break" style="font-
size: 0.88rem;">${parsedText}</div>
241             </div>
242         </div>
243     `;
244 }
245
246 function appendMessage(m) {

```

```
247 |     const stream = document.getElementById('hd-tab-messages-st
    | ream');
248 |     if (!stream) return;
249 |     const msgDiv = document.createElement('div');
250 |     msgDiv.innerHTML = createMessageHtml(m);
251 |     stream.appendChild(msgDiv.firstElementChild);
252 | }
253 |
254 | function scrollMessagesToBottom() {
255 |     const stream = document.getElementById('hd-tab-messages-st
    | ream');
256 |     if (stream) {
257 |         setTimeout(() => {
258 |             stream.scrollTop = stream.scrollHeight;
259 |         }, 50);
260 |     }
261 | }
262 |
263 | async function sendReply() {
264 |     if (!hdActiveTicketId) return;
265 |     const replyInput = document.getElementById('hd-reply-input
    | ');
266 |     const isInternalNote = document.getElementById('hd-switch-
    | internal-note')?.checked || false;
267 |     const text = replyInput?.value?.trim();
268 |
269 |     if (!text) return;
270 |
271 |     try {
272 |         const res = await fetch(`/api/helpdesk/tickets/${hdActiv
    | eTicketId}/messages`, {
273 |             method: 'POST',
274 |             headers: { 'Content-Type': 'application/json' },
275 |             credentials: 'include',
276 |             body: JSON.stringify({
277 |                 message: text,
278 |                 is_internal_note: isInternalNote
279 |             })
```

```

280 |     });
281 |
282 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
283 |     replyInput.value = '';
284 |     const data = await res.json();
285 |     if (data.message) {
286 |         appendMessage(data.message);
287 |         scrollMessagesToBottom();
288 |     }
289 |     loadTickets();
290 | } catch (err) {
291 |     console.error('[HelpdeskTab] Error sending message:', err);
292 |     if (typeof showNotification === 'function') showNotification('Ошибка отправки: ' + err.message, 'danger');
293 | }
294 | }
295 |
296 | async function updateTicketStatus(status) {
297 |     if (!hdActiveTicketId) return;
298 |     try {
299 |         const res = await fetch(`/api/helpdesk/tickets/${hdActiveTicketId}/status`, {
300 |             method: 'PUT',
301 |             headers: { 'Content-Type': 'application/json' },
302 |             credentials: 'include',
303 |             body: JSON.stringify({ status })
304 |         });
305 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
306 |         if (typeof showNotification === 'function') showNotification(`Статус изменен на: ${status}`, 'success');
307 |         loadTickets();
308 |         updateStats();
309 |         selectTicket(hdActiveTicketId);
310 |     } catch (err) {
311 |         console.error('[HelpdeskTab] Status update failed:', err);
312 |     }

```

```

313 |     }
314 |
315 |     async function updateTicketPriority(priority) {
316 |         if (!hdActiveTicketId) return;
317 |         try {
318 |             const res = await fetch(`/api/helpdesk/tickets/${hdActiveTicketId}/priority`, {
319 |                 method: 'PUT',
320 |                 headers: { 'Content-Type': 'application/json' },
321 |                 credentials: 'include',
322 |                 body: JSON.stringify({ priority })
323 |             });
324 |             if (!res.ok) throw new Error(`HTTP ${res.status}`);
325 |             if (typeof showNotification === 'function') showNotification(`Приоритет изменен на: ${priority}`, 'success');
326 |             loadTickets();
327 |             selectTicket(hdActiveTicketId);
328 |         } catch (err) {
329 |             console.error('[HelpdeskTab] Priority update failed:', err);
330 |         }
331 |     }
332 |
333 |     async function generateAiDraft() {
334 |         if (!hdActiveTicket) return;
335 |         const replyInput = document.getElementById('hd-reply-input');
336 |         if (!replyInput) return;
337 |
338 |         replyInput.value = '□ Генерация черновика ответа ИИ...';
339 |         try {
340 |             const res = await fetch('/api/chat', {
341 |                 method: 'POST',
342 |                 headers: { 'Content-Type': 'application/json' },
343 |                 credentials: 'include',
344 |                 body: JSON.stringify({
345 |                     prompt: `Сформулируй вежливый, профессиональный и конкретный ответ технической поддержки Helpdesk на следующее обр

```

ащение пользователя.

```
346 Тема: "${hdActiveTicket.subject}"
347 Описание: "${hdActiveTicket.description || ''}"
348 Категория: "${hdActiveTicket.category}"
349 Ответ должен быть кратким и готовым к отправке клиенту.` ,
350     use_rag: false
351   })
352 });
353
354   if (!res.ok) throw new Error(`HTTP ${res.status}`);
355   const data = await res.json();
356   const answer = data.response || data.text || '';
357   replyInput.value = answer.trim();
358 } catch (e) {
359   replyInput.value = `Здравствуйте! Мы изучили ваше обращение по теме "${hdActiveTicket.subject}". Специалисты уже занимаются решением.`;
360 }
361 }
362
363 function setupEventListeners() {
364   // Search input debounce
365   const searchInput = document.getElementById('hd-tab-search-input');
366   let searchTimer = null;
367   if (searchInput) {
368     searchInput.oninput = () => {
369       clearTimeout(searchTimer);
370       searchTimer = setTimeout(loadTickets, 300);
371     };
372   }
373
374   // Filter buttons
375   const filterBtns = document.querySelectorAll('#hd-tab-status-filters button');
376   filterBtns.forEach(btn => {
377     btn.onclick = () => {
378       filterBtns.forEach(b => b.classList.remove('active'));
```



```
379 |         btn.classList.add('active');
380 |         hdCurrentFilter = btn.getAttribute('data-filter') || '
all';
381 |         loadTickets();
382 |     };
383 | });
384 |
385 | // Send reply button & shortcut
386 | const btnSend = document.getElementById('hd-btn-send-reply
');
387 | if (btnSend) btnSend.onclick = sendReply;
388 |
389 | const replyInput = document.getElementById('hd-reply-input
');
390 | if (replyInput) {
391 |     replyInput.onkeydown = (e) => {
392 |         if (e.ctrlKey && e.key === 'Enter') {
393 |             e.preventDefault();
394 |             sendReply();
395 |         }
396 |     };
397 | }
398 |
399 | // Internal note switcher toggle label
400 | const switchNote = document.getElementById('hd-switch-inte
rnal-note');
401 | const modeBadge = document.getElementById('hd-composer-mod
e-badge');
402 | if (switchNote && modeBadge) {
403 |     switchNote.onchange = () => {
404 |         if (switchNote.checked) {
405 |             modeBadge.textContent = '🔒 Режим: Внутренняя скрыта
я заметка';
406 |             modeBadge.className = 'badge bg-warning text-dark';
407 |         } else {
408 |             modeBadge.textContent = 'Режим: Публичный ответ';
409 |             modeBadge.className = 'badge bg-secondary-subtle tex
t-body';
```

```

410     }
411 };
412 }
413
414 // Status action buttons
415 document.querySelectorAll('.hd-status-opt').forEach(btn =>
416 {
417     btn.onclick = () => {
418         const st = btn.getAttribute('data-status');
419         if (st) updateTicketStatus(st);
420     };
421 });
422
423 // Priority dropdown items
424 document.querySelectorAll('.hd-priority-opt').forEach(opt
425 => {
426     opt.onclick = (e) => {
427         e.preventDefault();
428         const prio = opt.getAttribute('data-priority');
429         if (prio) updateTicketPriority(prio);
430     };
431 });
432
433 // Quick canned responses
434 document.querySelectorAll('.hd-quick-template').forEach(tp
435 lBtn => {
436     tplBtn.onclick = () => {
437         const tpl = tplBtn.getAttribute('data-tpl');
438         if (tpl && replyInput) {
439             replyInput.value = tpl;
440             replyInput.focus();
441         }
442     };
443 });
444
445 // AI Assist Draft Button
446 const aiAssistBtn = document.getElementById('hd-btn-ai-ass
447 ist');

```

```
444     if (aiAssistBtn) aiAssistBtn.onclick = generateAiDraft;
445
446     // Submit new ticket in modal
447     const btnSubmitNew = document.getElementById('hd-btn-submit-new-ticket');
448     if (btnSubmitNew) {
449         btnSubmitNew.onclick = async () => {
450             const subject = document.getElementById('hd-new-ticket-subject')?.value?.trim();
451             const category = document.getElementById('hd-new-ticket-category')?.value || 'technical';
452             const priority = document.getElementById('hd-new-ticket-priority')?.value || 'normal';
453             const message = document.getElementById('hd-new-ticket-message')?.value?.trim();
454
455             if (!subject || !message) {
456                 window.showToast?.('Пожалуйста, заполните тему и текст обращения.', 'warning') || alert('Пожалуйста, заполните тему и текст обращения.');
```

457 return;

458 }

459

460 try {

461 btnSubmitNew.disabled = true;

462 const res = await fetch('/api/helpdesk/tickets', {

463 method: 'POST',

464 headers: { 'Content-Type': 'application/json' },

465 credentials: 'include',

466 body: JSON.stringify({ subject, category, priority

467 , message })

468 });

469

470 if (!res.ok) throw new Error(`HTTP \${res.status}`);

471 const data = await res.json();

472 if (data.status === 'success') {

473 window.showToast?.(`Тикет #\${data.ticket.ticket\_number} успешно создан!`, 'success');

```
473 |         const modalEl = document.getElementById('hdCreateTicketModal');
474 |         const modal = bootstrap.Modal.getInstance(modalEl);
475 |         modal?.hide();
476 |
477 |         document.getElementById('hd-new-ticket-subject').value = '';
478 |         document.getElementById('hd-new-ticket-message').value = '';
479 |
480 |         await loadTickets();
481 |         await updateStats();
482 |         if (data.ticket?.id) {
483 |             selectTicket(data.ticket.id);
484 |         }
485 |     }
486 | } catch (err) {
487 |     console.error('[HelpdeskTab] Create ticket failed:', err);
488 |     window.showToast?.('Ошибка создания тикета: ' + err.message, 'danger') || alert('Ошибка создания тикета: ' + err.message);
489 | } finally {
490 |     btnSubmitNew.disabled = false;
491 | }
492 | };
493 | }
494 | }
495 |
496 | function getStatusBadge(status) {
497 |     switch (status) {
498 |         case 'open': return '<span class="badge bg-danger-subtle text-danger">Новый</span>';
499 |         case 'in_progress': return '<span class="badge bg-warning-subtle text-warning">В работе</span>';
500 |         case 'resolved': return '<span class="badge bg-success-subtle text-success">Решено</span>';
```

```
501     case 'closed': return '<span class="badge bg-secondary-s
|ubtle text-body">Закрыт</span>';
502     default: return `<span class="badge bg-secondary">${stat
|us}</span>`;
503   }
504 }
505
506 function getPriorityBadge(priority) {
507   switch (priority) {
508     case 'urgent': return '<span class="badge bg-danger">⚡
|Срочно</span>';
509     case 'high': return '<span class="badge bg-warning text-
|dark">⚠️ Высокий</span>';
510     case 'low': return '<span class="badge bg-secondary-s
|ubtle text-muted">📉 Низкий</span>';
511     default: return '<span class="badge bg-info-subtle text-
|info">📄 Обычный</span>';
512   }
513 }
514
515 function formatTimeAgo(isoStr) {
516   if (!isoStr) return '';
517   try {
518     const diffMs = Date.now() - new Date(isoStr).getTime();
519     const diffMin = Math.floor(diffMs / 60000);
520     if (diffMin < 1) return 'только что';
521     if (diffMin < 60) return `${diffMin} мин назад`;
522     const diffHours = Math.floor(diffMin / 60);
523     if (diffHours < 24) return `${diffHours} ч назад`;
524     return `${Math.floor(diffHours / 24)} дн назад`;
525   } catch (e) {
526     return '';
527   }
528 }
529
530 function escapeHtml(text) {
531   if (!text) return '';
532   const div = document.createElement('div');
```

```
533 |     div.textContent = text;
534 |     return div.innerHTML;
535 | }
536 |
537 | // Global window functions for event hooks
538 | window.initHelpdeskTab = initHelpdeskTab;
539 | window.refreshHelpdeskTab = () => { loadTickets(); updateSta
    | ts(); };
540 | window.selectHelpdeskTicket = selectTicket;
541 | window.openCreateTicketModal = () => {
542 |     const modalEl = document.getElementById('hdCreateTicketMod
    | al');
543 |     if (modalEl) {
544 |         const modal = new bootstrap.Modal(modalEl);
545 |         modal.show();
546 |     }
547 | };
548 | })();
549 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title>AI Assistant</title>
8   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
9   t/css/bootstrap.min.css" rel="stylesheet">
10  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
11  0.0/font/bootstrap-icons.css" rel="stylesheet">
12  <link href="/html/css/main.css?v=20260926_v1" rel="styleshee
13  t">
14  <script src="/html/js/toast.js?v=20260926_v1"></script>
15  <script src="/html/js/status-highlighter.js?v=20260926_v1"><
16  /script>
17  <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
18  8next.min.js"></script>
19  <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
20  js"></script>
21 </head>
22 <body>
23 <div class="container-fluid mt-1">
24
25   <div class="d-flex align-items-center justify-content-betwee
26   n mb-1 main-nav-container rounded px-2 py-1">
27     <!-- Левая часть: кнопка меню, брендинг и индикатор активн
28     ого раздела -->
29     <div class="d-flex align-items-center gap-2">
30       <!-- Кнопка открытия левого выдвижного меню -->
31       <button class="btn btn-sm btn-primary d-flex align-items
32       -center gap-1.5 px-2 py-1 rounded shadow-sm"
33         id="left-drawer-toggle-btn"
34         type="button"
35         data-bs-toggle="offcanvas"
36         data-bs-target="#leftSideNavOffcanvas">
```

```

27         aria-controls="leftSideNavOffcanvas"
28         title="Открыть меню разделов"
29         style="font-size: 0.82rem; font-weight: 500;">
30         <i class="bi bi-list fs-6"></i>
31         <span class="d-none d-md-inline">Меню</span>
32     </button>
33
34     <!-- Брендинг -->
35     <span class="fw-bold d-flex align-items-center gap-1.5 t
ext-primary fs-6 user-select-none" style="letter-spacing: 0.3p
x;">
36         <i class="bi bi-cpu-fill fs-5"></i>
37         <span class="d-none d-sm-inline">AI Breadboard</span>
38     </span>
39     <span id="active-tab-title-badge" class="badge bg-body-s
econdary text-secondary-emphasis border rounded-pill px-2 py-1
small fw-normal d-flex align-items-center gap-1" title="Текущ
ий активный раздел">
40         □ Чат
41     </span>
42 </div>
43
44 <!-- Правая часть панели: статусы, профиль и настройки -->
45 <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto" id="nav-auth-container">
46     <!-- Active Model & Search Badges -->
47     <span id="chat-model-badge" class="badge rounded-pill al
ign-items-center gap-1 px-3 py-1 shadow-sm border" style="font
-size: 0.8rem; display: none; cursor: pointer;" title="Текущая
модель ИИ">
48         □ Загрузка...
49     </span>
50     <span id="chat-search-badge" class="badge rounded-pill a
lign-items-center gap-1 px-3 py-1 shadow-sm border" style="fon
t-size: 0.8rem; display: none;" title="Провайдер поиска">
51 </span>
52
53     <!-- Google OAuth Register / Sign-in Button (Shown when

```


not logged in) -->

```
54      <button class="btn btn-sm btn-light text-dark fw-semibol  
d d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-  
pill btn-google-login" id="nav-google-btn" title="Войти или за  
регистрироваться через Google" data-i18n-title="auth.googleSig  
nInTitle" style="border: 1px solid rgba(0,0,0,0.15);">
```

```
55      <svg width="18" height="18" viewBox="0 0 24 24"><path  
fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h  
5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.  
74 3.28-8.09z"/><path fill="#34A853" d="M12 23c2.97 0 5.46-.98  
7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-  
1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/><path fill  
="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.  
35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.  
85-2.22.81-.63z"/><path fill="#EA4335" d="M12 5.38c1.62 0 3.06  
.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.4  
7 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/></svg>
```

```
56      <span data-i18n="auth.googleSignUp">Регистрация через  
Google</span>
```

```
57    </button>
```

```
58
```

```
59    <!-- Authenticated User Menu Dropdown (Shown when logged  
in) -->
```

```
60    <div class="dropdown d-none" id="user-menu-dropdown-wrap  
per">
```

```
61      <button class="btn btn-sm btn-outline-secondary d-flex  
align-items-center gap-2 px-2 py-1 rounded-pill dropdown-togg  
le" id="user-settings-btn" type="button" data-bs-toggle="dropd  
own" aria-expanded="false" title="Пользовательское меню">
```

```
62      <img id="nav-user-avatar" src="" class="rounded-circ  
le d-none" width="22" height="22" style="object-fit: cover;">
```

```
63      <i class="bi bi-person-circle" id="nav-user-icon"></  
i>
```

```
64      <span id="nav-user-label" class="small fw-semibold"  
data-i18n="userSettings.button">Пользователь</span>
```

```
65    </button>
```

```
66    <ul class="dropdown-menu dropdown-menu-end shadow bord  
er-secondary-subtle py-1" aria-labelledby="user-settings-btn"
```

```

style="min-width: 220px;">
67         <li class="px-3 py-2 border-bottom border-secondary-
subtle">
68             <div class="fw-bold text-truncate" id="user-menu-d
isplay-name">Пользователь</div>
69             <div class="small text-muted text-truncate" id="us
er-menu-display-email" style="font-size: 0.75rem;">user@gmail.
com</div>
70         </li>
71         <li><h6 class="dropdown-header small text-uppercase
text-muted fw-semibold mt-1">Сервисы</h6></li>
72         <li>
73             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-mail">
74                 <i class="bi bi-envelope-check text-primary"></i>
>
75                 <span>Почта</span>
76             </button>
77         </li>
78         <li>
79             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-docs">
80                 <i class="bi bi-file-earmark-text text-info"></i>
>
81                 <span>Google Docs</span>
82             </button>
83         </li>
84         <li>
85             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-sheets">
86                 <i class="bi bi-file-earmark-spreadsheet text-su
ccess"></i>
87                 <span>Google Таблицы</span>
88             </button>
89         </li>
90         <li>
91             <button class="dropdown-item d-flex align-items-ce
nter gap-2 py-1" type="button" id="user-menu-btn-news">

```

```
92 |         <i class="bi bi-newspaper text-warning"></i>
93 |         <span>Новости</span>
94 |     </button>
95 | </li>
96 | <li>
97 |     <button class="dropdown-item d-flex align-items-ce
100 | nter gap-2 py-1" type="button" id="user-menu-btn-facebook">
98 |         <i class="bi bi-facebook text-primary"></i>
99 |         <span>Facebook</span>
100 |     </button>
101 | </li>
102 | <li><hr class="dropdown-divider my-1"></li>
103 | <li>
104 |     <button class="dropdown-item d-flex align-items-ce
105 | nter gap-2 py-1" type="button" id="user-menu-btn-cache" data-b
106 | s-toggle="modal" data-bs-target="#cacheManagerModal">
105 |         <i class="bi bi-hdd-stack text-info"></i>
106 |         <span data-i18n="cacheManager.menuItem">Кеш бpay
107 |         зера</span>
107 |     </button>
108 | </li>
109 | <li>
110 |     <button class="dropdown-item d-flex align-items-ce
111 | nter gap-2 py-1" type="button" data-bs-toggle="modal" data-bs-
112 | target="#userSettingsModal">
111 |         <i class="bi bi-gear-fill text-secondary"></i>
112 |         <span data-i18n="userSettings.title">Настройки п
113 |         рофиля</span>
113 |     </button>
114 | </li>
115 | <li>
116 |     <button class="dropdown-item d-flex align-items-ce
117 | nter gap-2 py-1 text-danger btn-user-logout" type="button">
117 |         <i class="bi bi-box-arrow-right"></i>
118 |         <span>Выйти</span>
119 |     </button>
120 | </li>
121 | </ul>
```

```

122     </div>
123
124     <!-- Кнопка кеша браузера -->
125     <button class="btn btn-sm btn-outline-secondary d-flex a
align-items-center justify-content-center rounded-circle" id="b
tn-open-cache-manager" data-bs-toggle="modal" data-bs-target="
#cacheManagerModal" title="Кеш браузера" style="width: 28px; h
eight: 28px; padding: 0;" data-i18n-title="cacheManager.title"
>
126         <i class="bi bi-hdd-stack"></i>
127     </button>
128
129     <!-- Кнопка переключения темы (☐ / ☐ / ☐ ) -->
130     <button class="btn btn-sm btn-outline-secondary d-flex a
align-items-center justify-content-center rounded-circle"
131         id="theme-toggle"
132         title="Переключить тему"
133         style="width: 28px; height: 28px; padding: 0;">
134         <i class="bi bi-circle-half" id="theme-icon"></i>
135     </button>
136
137     <!-- Quick Settings Gear Button -->
138     <button class="btn btn-sm btn-outline-secondary d-flex a
align-items-center justify-content-center rounded-circle" id="u
ser-settings-gear-btn" data-bs-toggle="modal" data-bs-target="
#userSettingsModal" title="Настройки" style="width: 28px; heig
ht: 28px; padding: 0;" data-i18n-title="userSettings.title">
139         <i class="bi bi-gear-fill"></i>
140     </button>
141 </div>
142 </div>
143
144 <!-- Плавающий боковой ярлычок (вкладка) слева -->
145 <button class="left-drawer-tab-trigger btn shadow"
146     type="button"
147     data-bs-toggle="offcanvas"
148     data-bs-target="#leftSideNavOffcanvas"
149     aria-controls="leftSideNavOffcanvas"

```

```

150         title="Открыть меню навигации">
151     <i class="bi bi-list fs-6"></i>
152     <span class="left-drawer-tab-text">МЕНЮ</span>
153 </button>
154
155     <!-- Левая выдвижная вкладка / панель навигации (Offcanvas)
156     -->
157     <div class="offcanvas offcanvas-start left-drawer-nav" tabin
158     dex="-1" id="leftSideNavOffcanvas" aria-labelledby="leftSideNa
159     vOffcanvasLabel">
160         <div class="offcanvas-header border-bottom py-2 px-3">
161             <h5 class="offcanvas-title d-flex align-items-center gap
162             -2 fs-6 fw-bold" id="leftSideNavOffcanvasLabel">
163                 <i class="bi bi-grid-fill text-primary"></i>
164                 <span>Навигация и разделы</span>
165             </h5>
166             <button type="button" class="btn-close text-reset" data-
167             bs-dismiss="offcanvas" aria-label="Закрыть"></button>
168         </div>
169         <div class="offcanvas-body p-3 d-flex flex-column justify-
170         content-between" id="mainTabs">
171             <div class="drawer-sections-wrapper">
172                 <!-- Группа: Стандартные -->
173                 <div class="nav-drawer-group mb-3">
174                     <div class="nav-drawer-group-title d-flex align-item
175                     s-center gap-1 mb-2">
176                         <i class="bi bi-grid-fill text-primary"></i>
177                         <span data-i18n="tabs.groupStandard">Стандартные</
178                         span>
179                     </div>
180                     <div class="list-group list-group-flush rounded bord
181                     er">
182                         <button class="list-group-item list-group-item-act
183                         ion d-flex align-items-center gap-2 py-2 px-3 active" data-tab
184                         ="tab-chat" data-bs-target="#tab-chat" type="button">
185                             <span class="fs-6">☐ </span>
186                             <span class="fw-medium" data-i18n="tabs.chat">Ча
187                             т</span>

```

```

176 |         </button>
177 |         <button class="list-group-item list-group-item-act
    | ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-v
    | oice" data-bs-target="#tab-voice" type="button">
178 |             <span class="fs-6">□ </span>
179 |             <span class="fw-medium" data-i18n="tabs.voice">Г
    | олос и диаризация</span>
180 |         </button>
181 |     </div>
182 | </div>
183 |
184 |
185 |     <!-- Группа: Расширенные -->
186 |     <div class="nav-drawer-group mb-3">
187 |         <div class="nav-drawer-group-title d-flex align-item
    | s-center gap-1 mb-2">
188 |             <i class="bi bi-sliders2 text-info"></i>
189 |             <span data-i18n="tabs.groupExtended">Расширенные</
    | span>
190 |         </div>
191 |         <div class="list-group list-group-flush rounded bord
    | er">
192 |             <button class="list-group-item list-group-item-act
    | ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-r
    | ag" data-bs-target="#tab-rag" type="button">
193 |                 <span class="fs-6">□ </span>
194 |                 <span class="fw-medium" data-i18n="tabs.rag">RAG
    | </span>
195 |             </button>
196 |             <button class="list-group-item list-group-item-act
    | ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-n
    | ews" data-bs-target="#tab-news" type="button">
197 |                 <span class="fs-6">□ </span>
198 |                 <span class="fw-medium" data-i18n="tabs.news">Ho
    | вости</span>
199 |             </button>
200 |             <button class="list-group-item list-group-item-act
    | ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-a

```

```

    dmin" data-bs-target="#tab-admin" type="button">
201         <span class="fs-6">□ </span>
202         <span class="fw-medium" data-i18n="tabs.admin">У
правление</span>
203     </button>
204     <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-h
elp" data-bs-target="#tab-help" type="button">
205         <span class="fs-6">□ </span>
206         <span class="fw-medium" data-i18n="tabs.help">Сп
равочник</span>
207     </button>
208 </div>
209 </div>
210
211 <!-- Группа: Плагины -->
212 <div class="nav-drawer-group mb-3" id="pluginsNavDropd
ownWrapper">
213     <div class="nav-drawer-group-title d-flex align-item
s-center gap-1 mb-2">
214         <i class="bi bi-puzzle-fill text-warning"></i>
215         <span data-i18n="tabs.plugins">Плагины</span>
216     </div>
217     <div class="list-group list-group-flush rounded bord
er" id="nav-plugins-dropdown-menu">
218         <div id="nav-enabled-plugins-list" class="nav-enab
led-plugins-list">
219             <!-- Dynamic enabled plugins will be injected he
re -->
220         </div>
221         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
lugins" data-bs-target="#tab-plugins" type="button">
222             <span class="fs-6">□ </span>
223             <span class="fw-medium" data-i18n="tabs.managePl
ugins">Управление плагинами</span>
224         </button>
225     </div>

```

```

226 |         </div>
227 |
228 |         <!-- Группа: Диагностика и форензика -->
229 |         <div class="nav-drawer-group mb-3">
230 |             <div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
231 |                 <i class="bi bi-shield-check text-success"></i>
232 |                 <span>Диагностика и форензика</span>
233 |             </div>
234 |             <div class="list-group list-group-flush rounded border">
235 |                 <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-process-leaks" data-bs-target="#tab-process-leaks" type="button">
236 |                     <span class="fs-6">□ </span>
237 |                     <span class="fw-medium">1. Утечки и микрофризы</span>
238 |                 </button>
239 |                 <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-forensics" data-bs-target="#tab-forensics" type="button">
240 |                     <span class="fs-6">□ </span>
241 |                     <span class="fw-medium">2. Поведенческая форензика</span>
242 |                 </button>
243 |                 <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-throttling" data-bs-target="#tab-throttling" type="button">
244 |                     <span class="fs-6">□ </span>
245 |                     <span class="fw-medium">3. Качество ядра и Троттлинг</span>
246 |                 </button>
247 |                 <button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-storage-wear" data-bs-target="#tab-storage-wear" type="button">
248 |                     <span class="fs-6">□ </span>
249 |                     <span class="fw-medium">4. Износ дисков и батареи

```



```

    и</span>
250         </button>
251         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-p
eripherals" data-bs-target="#tab-peripherals" type="button">
252             <span class="fs-6">□ </span>
253             <span class="fw-medium">5. Сеть и периферия</spa
n>
254         </button>
255         <button class="list-group-item list-group-item-act
ion d-flex align-items-center gap-2 py-2 px-3" data-tab="tab-s
ystem-logs" data-bs-target="#tab-system-logs" type="button">
256             <span class="fs-6">□ </span>
257             <span class="fw-medium" data-i18n="tabs.systemLo
gs">Логи операционной системы</span>
258         </button>
259     </div>
260 </div>
261 </div>
262
263 <!-- Footer Drawer -->
264 <div class="nav-drawer-footer border-top pt-2 mt-auto d-
flex align-items-center justify-content-between text-muted sma
ll">
265     <span class="fw-semibold">AI Breadboard</span>
266     <span>v3.1</span>
267 </div>
268 </div>
269 </div>
270
271 <div class="tab-content" id="mainTabContent">
272     <!-- Tabs will be injected via JavaScript -->
273     <div id="tab-chat" class="tab-pane fade show active"></div
>
274     <div id="tab-rag" class="tab-pane fade"></div>
275     <div id="tab-telegram-rag" class="tab-pane fade"></div>
276     <div id="tab-news" class="tab-pane fade"></div>
277     <div id="tab-voice" class="tab-pane fade"></div>

```

```
278 <div id="tab-plugins" class="tab-pane fade"></div>
279 <div id="tab-admin" class="tab-pane fade"></div>
280 <div id="tab-help" class="tab-pane fade"></div>
281 <div id="tab-process-leaks" class="tab-pane fade"></div>
282 <div id="tab-forensics" class="tab-pane fade"></div>
283 <div id="tab-throttling" class="tab-pane fade"></div>
284 <div id="tab-storage-wear" class="tab-pane fade"></div>
285 <div id="tab-peripherals" class="tab-pane fade"></div>
286 <div id="tab-system-logs" class="tab-pane fade"></div>
287 </div>
288 </div>
289
290 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    /js/bootstrap.bundle.min.js"></script>
291 <script src="/html/js/ai_table_modal.js"></script>
292 <script src="/html/js/chatService.js?v=20260926_v2"></script>
293 <script src="/html/js/window-api.js?v=20260926_v2"></script>
294 <script type="module" src="/html/js/main.js?v=20260926_v2"></s
    cript>
295
296 <!-- HELP Modal -->
297 <div class="modal fade modal-help" id="help-modal" tabindex="-
    1" aria-labelledby="help-modal-label" aria-hidden="true">
298   <div class="modal-dialog modal-dialog-scrollable">
299     <div class="modal-content">
300       <div class="modal-header">
301         <h5 class="modal-title" id="help-modal-label" data-i18
            n="helpModal.title">Помощь</h5>
302         <button type="button" class="btn-close" data-bs-dismis
            s="modal" aria-label="Close"></button>
303       </div>
304       <div class="modal-body" id="help-modal-content">
305         <p data-i18n="common.loading">Загрузка...</p>
306       </div>
307       <div class="modal-footer">
308         <button type="button" class="btn btn-secondary" data-b
            s-dismiss="modal" data-i18n="helpModal.close">Закрыть</button>
309       </div>
```

```

310     </div>
311 </div>
312 </div>
313
314 <!-- User Settings Modal -->
315 <div class="modal fade" id="userSettingsModal" tabindex="-1" a
    aria-labelledby="userSettingsModalLabel" aria-hidden="true">
316     <div class="modal-dialog modal-dialog-centered">
317         <div class="modal-content shadow border-secondary-subtle">
318             <div class="modal-header border-secondary-subtle">
319                 <h5 class="modal-title d-flex align-items-center gap-2
    " id="userSettingsModalLabel">
320                     <i class="bi bi-person-gear text-primary"></i>
321                     <span data-i18n="userSettings.title">Пользовательски
    е настройки</span>
322                 </h5>
323                 <button type="button" class="btn-close" data-bs-dismis
    s="modal" aria-label="Close"></button>
324             </div>
325             <div class="modal-body p-3">
326                 <!-- User Account & Google OAuth Card -->
327                 <div class="card bg-body-tertiary border-secondary-sub
    tle mb-3 p-3" id="user-settings-auth-card">
328                     <div class="d-flex align-items-center gap-3">
329                         <img id="user-settings-avatar" src="" class="round
    ed-circle border border-2 border-primary" width="48" height="4
    8" style="object-fit: cover;">
330                         <div class="flex-grow-1 min-w-0">
331                             <div class="d-flex align-items-center gap-2">
332                                 <strong id="user-settings-name" class="text-tr
   uncate">Пользователь</strong>
333                                 <span id="user-settings-role-badge" class="bad
    ge bg-primary">User</span>
334                             </div>
335                             <div id="user-settings-email" class="small text-
    muted text-truncate">email@gmail.com</div>
336                             <div id="user-settings-google-badge" class="smal
    l mt-1 text-muted">

```

```

337 |             <i class="bi bi-google text-danger me-1"></i>G
| oogle OAuth
338 |         </div>
339 |     </div>
340 |     <button type="button" class="btn btn-outline-dange
| r btn-sm btn-user-logout" title="Выйти">
341 |         <i class="bi bi-box-arrow-right"></i>
342 |     </button>
343 | </div>
344 | </div>
345 |
346 | <!-- Guest / Connect Google OAuth Card -->
347 | <div class="card bg-body-tertiary border-secondary-sub
| tle mb-3 p-3 text-center d-none" id="user-settings-guest-card"
| >
348 |     <div class="text-muted small mb-2">
349 |         <i class="bi bi-shield-lock me-1"></i> Авторизация
| и синхронизация аккаунта
350 |     </div>
351 |     <button type="button" class="btn btn-outline-danger
| btn-sm w-100 d-flex align-items-center justify-content-center
| gap-2 btn-google-login">
352 |         <i class="bi bi-google"></i>
353 |         <span>Войти через Google (OAuth)</span>
354 |     </button>
355 |     <div class="small text-muted mt-2" style="font-size:
| 0.75rem;">
356 |         Синхронизация с Google Drive, Docs, Calendar и Con
| tacts
357 |     </div>
358 | </div>
359 |
360 | <!-- Language Selection -->
361 | <div class="mb-3">
362 |     <label class="form-label fw-semibold small d-flex al
| ign-items-center gap-2" for="modal-lang-selector">
363 |         <i class="bi bi-translate text-info"></i>
364 |         <span data-i18n="userSettings.language">Язык интер

```

```

   фейса</span>
365         </label>
366         <select class="form-select lang-selector" id="modal-
lang-selector">
367             <option value="ru">Русский</option>
368             <option value="en">English</option>
369             <option value="he">עברית</option>
370         </select>
371     </div>
372
373     <!-- Theme Selection -->
374     <div class="mb-2">
375         <label class="form-label fw-semibold small d-flex al
ign-items-center gap-2" for="modal-theme-selector">
376             <i class="bi bi-palette text-warning"></i>
377             <span data-i18n="userSettings.theme">Тема оформлен
ия</span>
378         </label>
379         <select class="form-select theme-selector" id="modal
-theme-selector">
380             <option value="light" data-i18n="userSettings.them
eLight">☀ Светлая</option>
381             <option value="dark" data-i18n="userSettings.theme
Dark">☿ Тёмная</option>
382             <option value="system" data-i18n="userSettings.the
meSystem">☒ Системная</option>
383         </select>
384     </div>
385 </div>
386 <!-- Cache Manager Modal -->
387 <div class="modal fade" id="cacheManagerModal" tabindex="-1" a
ria-labelledby="cacheManagerModalLabel" aria-hidden="true">
388     <div class="modal-dialog modal-dialog-centered modal-lg">
389         <div class="modal-content shadow border-secondary-subtle">
390             <div class="modal-header border-secondary-subtle">
391                 <div class="d-flex align-items-center gap-2">
392                     <div class="rounded-circle bg-primary-subtle text-pr
imary p-2 d-flex align-items-center justify-content-center" st

```

```

yle="width: 36px; height: 36px;">
393         <i class="bi bi-hdd-stack fs-5"></i>
394     </div>
395     <div>
396         <h5 class="modal-title mb-0" id="cacheManagerModal
Label" data-i18n="cacheManager.title">Управление браузерным ке
шем</h5>
397         <div class="small text-muted" style="font-size: 0.
78rem;" data-i18n="cacheManager.subtitle">Хранилище больших да
нных (IndexedDB + Memory Cache) для ускорения работы и офлайн-
доступа.</div>
398     </div>
399 </div>
400     <button type="button" class="btn-close" data-bs-dismis
s="modal" aria-label="Close"></button>
401 </div>
402     <div class="modal-body p-3">
403         <!-- Storage Quota Bar Card -->
404         <div class="card bg-body-tertiary border-secondary-sub
tle p-3 mb-3">
405             <div class="d-flex align-items-center justify-conten
t-between mb-2">
406                 <div class="fw-semibold small d-flex align-items-c
enter gap-1.5">
407                     <i class="bi bi-pie-chart-fill text-primary"></i
>
408                     <span data-i18n="cacheManager.quotaTitle">Дисков
ая квота браузера</span>
409                 </div>
410                 <span id="cache-persist-badge" class="badge bg-sec
ondary-subtle text-secondary border">Проверка...</span>
411             </div>
412             <div class="progress mb-2" style="height: 10px;">
413                 <div id="cache-usage-progressbar" class="progress-
bar bg-primary" role="progressbar" style="width: 0%"></div>
414             </div>
415             <div class="d-flex justify-content-between align-ite
ms-center small text-muted" style="font-size: 0.78rem;">

```

```

416 |         <span>Использование памяти:</span>
417 |         <span id="cache-usage-text" class="fw-semibold tex
| t-body">Загрузка...</span>
418 |     </div>
419 | </div>
420 |
421 |     <!-- Stores List -->
422 |     <div class="mb-3">
423 |         <div class="d-flex align-items-center justify-conten
| t-between mb-2">
424 |             <h6 class="fw-bold small text-uppercase text-muted
| mb-0" data-i18n="cacheManager.storesTitle">Хранилища данных</
| h6>
425 |             <button class="btn btn-sm btn-outline-primary py-0
| px-2 rounded-pill small" id="cache-btn-cleanup-expired" style
| ="font-size: 0.75rem;">
426 |                 <i class="bi bi-magic me-1"></i><span data-i18n=
| "cacheManager.btnCleanExpired">Очистить просроченное</span>
427 |             </button>
428 |         </div>
429 |         <div id="cache-stores-list">
430 |             <!-- Populated dynamically by cache-ui.js -->
431 |         </div>
432 |     </div>
433 |
434 |     <!-- Action Tools -->
435 |     <div class="d-flex flex-wrap align-items-center justifi
| y-content-between gap-2 pt-2 border-top border-secondary-subtl
| e">
436 |         <div class="d-flex align-items-center gap-2">
437 |             <button type="button" class="btn btn-outline-secon
| dary btn-sm" id="cache-btn-request-persist" data-i18n="cacheMa
| nager.btnRequestPersist">
438 |                 <i class="bi bi-shield-lock me-1"></i>Сделать по
| СТОЯННЫМ
439 |             </button>
440 |             <button type="button" class="btn btn-outline-secon
| dary btn-sm" id="cache-btn-export" data-i18n="cacheManager.btn

```

```
Export">
441         <i class="bi bi-download me-1"></i>Экспорт JSON
442     </button>
443     <label class="btn btn-outline-secondary btn-sm mb-
0" for="cache-file-import" style="cursor: pointer;">
444         <i class="bi bi-upload me-1"></i><span data-i18n
="cacheManager.btnImport">Импорт JSON</span>
445         <input type="file" id="cache-file-import" accept
=".json" class="d-none">
446     </label>
447 </div>
448     <button type="button" class="btn btn-outline-danger
btn-sm" id="cache-btn-clear-all" data-i18n="cacheManager.btnCl
earAll">
449         <i class="bi bi-trash3 me-1"></i>Очистить всё
450     </button>
451 </div>
452 </div>
453     <div class="modal-footer border-secondary-subtle py-2">
454         <button type="button" class="btn btn-secondary btn-sm"
data-bs-dismiss="modal" data-i18n="userSettings.close">Закрыт
ь</button>
455     </div>
456 </div>
457 </div>
458 </div>
459
460 </body>
461 </html>
462
```


index.html

```
1 <!-- Instructions Tab -->
2 <div class="container-fluid p-3">
3   <div class="alert alert-info">
4     <h5 class="mb-1"><i class="bi bi-gear"></i> Редактор систе
мных инструкций</h5>
5     <p class="mb-0 small">Управление системными инструкциями А
I: чат и Narrator режимы, версионирование, проверка в модели.<
/p>
6   </div>
7
8   <!-- Mode Switcher -->
9   <div class="card mb-3">
10    <div class="card-body">
11      <div class="d-flex align-items-center gap-2 flex-wrap">
12        <span class="text-muted">Режим:</span>
13        <div class="btn-group" role="group">
14          <button type="button" class="btn btn-outline-primary
active" id="mode-chat" onclick="switchMode('chat')">
15            <i class="bi bi-chat-text"></i> Chat
16          </button>
17          <button type="button" class="btn btn-outline-seconda
ry" id="mode-narrator" onclick="switchMode('narrator')">
18            <i class="bi bi-mic"></i> Narrator
19          </button>
20        </div>
21        <small class="text-muted ms-2">Активный файл: <code id
="active-file-info">—</code></small>
22        <div class="ms-auto d-flex gap-2">
23          <button type="button" class="btn btn-success" id="bt
n-check-model" onclick="checkInModel()">
24            <i class="bi bi-check-circle"></i> Проверить в мод
ели
25          </button>
26          <button type="button" class="btn btn-primary" id="bt
n-save" onclick="saveInstruction()">
27            <i class="bi bi-save"></i> Сохранить версию
```

```

28         </button>
29         <button type="button" class="btn btn-warning" id="bt
n-preview" onclick="previewPrompt()">
30             <i class="bi bi-eye"></i> Предпросмотр
31         </button>
32     </div>
33 </div>
34 </div>
35 </div>
36
37 <!-- Editor Area -->
38 <div class="card mb-3">
39     <div class="card-header">
40         <div class="d-flex justify-content-between align-items-c
enter">
41             <span class="fw-bold">Редактор – <span id="editor-mode
-label">□ Chat</span></span>
42             <small class="text-muted">Markdown поддерживается</sma
ll>
43         </div>
44     </div>
45     <div class="card-body p-0">
46         <textarea id="instruction-editor" class="form-control bo
rder-0 rounded-0" rows="18"
47             placeholder="Введите системную инструкцию..." spellche
ck="false"
48             style="font-family: 'Courier New', monospace; font-siz
e: 0.9rem; resize: vertical;"></textarea>
49     </div>
50 </div>
51
52 <!-- Versions Panel -->
53 <div class="card mb-3">
54     <div class="card-header d-flex justify-content-between ali
gn-items-center">
55         <span class="fw-bold"><i class="bi bi-clock-history"></i>
> История версий</span>
56         <button type="button" class="btn btn-sm btn-outline-seco

```

```

ndary" onclick="refreshVersions()">
57     <i class="bi bi-arrow-clockwise"></i> Обновить
58     </button>
59 </div>
60 <div class="card-body p-0">
61     <div id="versions-list" class="list-group list-group-flu
sh">
62         <div class="list-group-item text-muted">Загрузка верси
й...</div>
63     </div>
64 </div>
65 </div>
66
67 <!-- Token Counter -->
68 <div class="card">
69     <div class="card-header">
70         <span class="fw-bold"><i class="bi bi-calculator"></i> С
татистика</span>
71     </div>
72     <div class="card-body">
73         <div class="row g-3">
74             <div class="col-md-4">
75                 <div class="p-3 bg-light rounded text-center">
76                     <div class="h3 mb-0" id="char-count">0</div>
77                     <div class="small text-muted">Символов</div>
78                 </div>
79             </div>
80             <div class="col-md-4">
81                 <div class="p-3 bg-light rounded text-center">
82                     <div class="h3 mb-0" id="word-count">0</div>
83                     <div class="small text-muted">Слов</div>
84                 </div>
85             </div>
86             <div class="col-md-4">
87                 <div class="p-3 bg-light rounded text-center">
88                     <div class="h3 mb-0" id="token-estimate">~0</div>
89                     <div class="small text-muted">Токенов (оценка)</di
v>
v>

```

```
90 |         </div>
91 |     </div>
92 | </div>
93 |     <div class="mt-3 small text-muted border rounded p-2" id
    | ="token-preview" style="min-height:2rem;"></div>
94 | </div>
95 | </div>
96 | </div>
97 |
98 | <!-- Prompt Preview Modal -->
99 | <div class="modal fade" id="prompt-preview-modal" tabindex="-1
    | " aria-labelledby="prompt-preview-modal-label" aria-hidden="tr
    | ue">
100 |     <div class="modal-dialog modal-lg modal-dialog-scrollable">
101 |         <div class="modal-content">
102 |             <div class="modal-header">
103 |                 <h5 class="modal-title" id="prompt-preview-modal-label
    | ">Предпросмотр инструкции</h5>
104 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
105 |             </div>
106 |             <div class="modal-body" id="prompt-preview-content">
107 |                 <p>Загрузка...</p>
108 |             </div>
109 |             <div class="modal-footer">
110 |                 <button type="button" class="btn btn-secondary" data-b
    | s-dismiss="modal">Закрыть</button>
111 |             </div>
112 |         </div>
113 |     </div>
114 | </div>
```

main.js

```
1 // Instructions Tab - UI Logic
2 // Инициализируется через window.initInstructionsTab() из admin/main.js
3
4 'use strict';
5
6 import { initI18n, switchLang, applyTranslations } from '../js/i18n.js';
7
8 window.switchLang = switchLang;
9 window.applyTranslations = applyTranslations;
10
11 // — Helpers —————
12
13 function adminFetch(url, options = {}) {
14   return window.api
15     ? window.api.fetch(url, options)
16     : fetch(url, options).then(r => {
17       if (!r.ok) throw new Error(`HTTP ${r.status}`);
18       return r.json();
19     });
20 }
21
22 function notify(msg, type = 'info') {
23   if (typeof showNotification === 'function') {
24     showNotification(msg, type);
25   } else {
26     console.log(`[${type}] ${msg}`);
27   }
28 }
29
30 // — Mode Management —————
31 let currentMode = 'chat';
32
```

```

33 function switchMode(mode) {
34     currentMode = mode;
35     document.getElementById('mode-chat').classList.toggle('active', mode === 'chat');
36     document.getElementById('mode-narrator').classList.toggle('active', mode === 'narrator');
37
38     // Обновить заголовок редактора
39     const modeLabel = mode === 'chat' ? 'Чат' : 'Narrator';
40     const editorTitle = document.getElementById('editor-mode-label');
41     if (editorTitle) editorTitle.textContent = modeLabel;
42
43     loadInstruction();
44     refreshVersions();
45 }
46
47 // — Instructions API —————
48 async function loadInstruction() {
49     try {
50         const data = await adminFetch(`/api/admin/instructions?mode=${currentMode}`);
51         const editor = document.getElementById('instruction-editor');
52         if (data && data.content !== undefined) {
53             editor.value = data.content;
54         } else {
55             editor.value = '';
56         }
57         updateStats();
58         const fileInfo = document.getElementById('active-file-info');
59         if (fileInfo && data.file) {
60             fileInfo.textContent = data.file;
61         }
62     } catch (e) {

```

```
63 |     notify('Ошибка загрузки инструкции: ' + e.message, 'danger
    | ');
64 |     document.getElementById('instruction-editor').value = '';
65 | }
66 | }
67 |
68 | async function saveInstruction() {
69 |     const content = document.getElementById('instruction-editor'
    | ).value.trim();
70 |     if (!content) {
71 |         notify('Инструкция не может быть пустой', 'warning');
72 |         return;
73 |     }
74 |
75 |     const btn = document.getElementById('btn-save');
76 |     const originalHtml = btn.innerHTML;
77 |     btn.disabled = true;
78 |     btn.innerHTML = '<i class="bi bi-hourglass-split"></i> Сохра
    | нение...';
79 |
80 |     try {
81 |         const data = await adminFetch('/api/admin/instructions/sav
    | e', {
82 |             method: 'POST',
83 |             headers: { 'Content-Type': 'application/json' },
84 |             body: JSON.stringify({ mode: currentMode, content })
85 |         });
86 |         notify(`Инструкция сохранена: ${data.version || ''}`, 'suc
    | cess');
87 |         await refreshVersions();
88 |     } catch (e) {
89 |         notify('Ошибка сохранения: ' + e.message, 'danger');
90 |     } finally {
91 |         btn.disabled = false;
92 |         btn.innerHTML = originalHtml;
93 |     }
94 | }
95 |
```

```
96 | async function checkInModel() {
97 |   const content = document.getElementById('instruction-editor'
    | ).value.trim();
98 |   const prompt = "Привет! Подтверди, что ты получил инструкцию
    | и готов к работе. Ответь только: 'Понял, готов к работе!'";
99 |
100 |   const btn = document.getElementById('btn-check-model');
101 |   const originalHtml = btn.innerHTML;
102 |   btn.disabled = true;
103 |   btn.innerHTML = '<i class="bi bi-hourglass-split"></i> Прове
    | рка...';
104 |
105 |   try {
106 |     const result = await adminFetch('/api/admin/instructions/c
    | heck', {
107 |       method: 'POST',
108 |       headers: { 'Content-Type': 'application/json' },
109 |       body: JSON.stringify({ mode: currentMode, prompt, instru
    | ction: content })
110 |     });
111 |
112 |     const statusEl = document.getElementById('token-preview');
113 |     let tokenInfo = '';
114 |     if (result.token_count) {
115 |       tokenInfo = `<br>Использовано токенов: ${result.token_co
    | unt}`;
116 |     }
117 |
118 |     if (result.response) {
119 |       statusEl.innerHTML = `<strong>Ответ модели:</strong><br>
    | ${escapeHtml(result.response)}${tokenInfo}`;
120 |       notify('Проверка завершена', 'success');
121 |     } else {
122 |       statusEl.innerHTML = 'Ошибка: нет ответа от модели';
123 |       notify('Ошибка проверки', 'danger');
124 |     }
125 |   } catch (e) {
126 |     document.getElementById('token-preview').innerHTML = `<str
```



```

    ong class="text-danger">Ошибка проверки: ${e.message}</strong>
    `;
127     notify('Ошибка проверки: ' + e.message, 'danger');
128   } finally {
129     btn.disabled = false;
130     btn.innerHTML = originalHtml;
131   }
132 }
133
134 async function refreshVersions() {
135   const list = document.getElementById('versions-list');
136   list.innerHTML = '<div class="list-group-item text-muted"><i
    class="bi bi-hourglass-split"></i> Загрузка...</div>';
137
138   try {
139     const data = await adminFetch(`/api/admin/instructions/ver
    sions?mode=${currentMode}`);
140
141     if (!data.versions || !data.versions.length) {
142       list.innerHTML = '<div class="list-group-item text-muted
    ">Нет сохранённых версий</div>';
143       return;
144     }
145
146     list.innerHTML = data.versions.map(v => `
147       <button type="button"
148         class="list-group-item list-group-item-action d-flex j
    ustify-content-between align-items-center ${v.is_active ? 'lis
    t-group-item-success' : ''}"
149         onclick="useVersion('${escapeAttr(v.filename)}', '${es
    capeAttr(v.mode)}')">
150         <div class="me-2 overflow-hidden">
151           <div class="d-flex align-items-center gap-2 mb-1">
152             <strong>${v.mode === 'chat' ? '💬 ' : '📄 '} ${escap
    eHtml(v.filename)}</strong>
153             ${v.is_active ? '<span class="badge bg-success">Ак
    тивная</span>' : ''}
154           </div>

```

```

155 |         <small class="text-muted">${new Date(v.created_at).t
| oLocaleString('ru-RU')} · ${v.size} байт</small>
156 |         <div class="text-muted small text-truncate" style="m
| ax-width:480px;">${escapeHtml(v.preview || '')}</div>
157 |     </div>
158 |     <i class="bi bi-arrow-right-circle${v.is_active ? '-fi
| ll text-success' : ' text-secondary'}"></i>
159 | </button>
160 |     `).join('');
161 | } catch (e) {
162 |     list.innerHTML = `<div class="list-group-item text-danger"
| >Ошибка загрузки версий: ${e.message}</div>`;
163 | }
164 | }
165 |
166 | async function useVersion(filename, mode) {
167 |     try {
168 |         await adminFetch('/api/admin/instructions/activate', {
169 |             method: 'POST',
170 |             headers: { 'Content-Type': 'application/json' },
171 |             body: JSON.stringify({ mode, filename })
172 |         });
173 |
174 |         if (mode === currentMode) {
175 |             await loadInstruction();
176 |         }
177 |         await refreshVersions();
178 |         notify(`Версия ${filename} активирована`, 'success');
179 |     } catch (e) {
180 |         notify('Ошибка активации: ' + e.message, 'danger');
181 |     }
182 | }
183 |
184 | async function previewPrompt() {
185 |     const content = document.getElementById('instruction-editor'
| ).value.trim();
186 |     const preview = `
187 |         <strong>Контекст системной инструкции (${currentMode === `

```

```

chat' ? ' Chat' : ' Narrator'}):</strong><br>
188   <pre style="white-space:pre-wrap;max-height:400px;overflow
-y:auto;">${escapeHtml(content.substring(0, 1000))}${content.l
length > 1000 ? '\n...' : ''}</pre>
189   <strong>Тестовый запрос:</strong><br>
190   "Привет! Подтверди, что ты получил инструкцию и готов к ра
боте."
191   `;
192
193   const modal = new bootstrap.Modal(document.getElementById('p
rompt-preview-modal'));
194   document.getElementById('prompt-preview-content').innerHTML
= preview;
195   modal.show();
196 }
197
198 // — Stats —————
—————
199 function updateStats() {
200   const content = document.getElementById('instruction-editor'
).value;
201
202   document.getElementById('char-count').textContent = content.
length;
203
204   const words = content.trim().split(/\s+/).filter(w => w.leng
th > 0);
205   document.getElementById('word-count').textContent = words.le
ngth;
206
207   const estimatedTokens = Math.ceil(content.length / 3);
208   document.getElementById('token-estimate').textContent = '~'
+ estimatedTokens;
209
210   return estimatedTokens;
211 }
212
213 function escapeHtml(text) {

```

```

214 |   if (!text) return '';
215 |   const div = document.createElement('div');
216 |   div.textContent = text;
217 |   return div.innerHTML.replace(/\n/g, '<br>');
218 | }
219 |
220 | function escapeAttr(text) {
221 |   if (!text) return '';
222 |   return text.replace(/'/g, "\\'").replace(/"/g, '&quot;');
223 | }
224 |
225 | // — Initialization —————
    | _____
226 | let initialized = false;
227 |
228 | async function initInstructionsTab() {
229 |   if (initialized) return;
230 |   initialized = true;
231 |
232 |   console.log('[InstructionsTab] Initializing...');
233 |
234 |   const savedLang = localStorage.getItem('app_language') || 'r
    | u';
235 |   try {
236 |     await initI18n(savedLang);
237 |     console.log('[InstructionsTab] i18n initialized');
238 |   } catch (e) {
239 |     console.error('[InstructionsTab] i18n init error:', e);
240 |   }
241 |
242 |   // Load instruction for current mode
243 |   await loadInstruction();
244 |
245 |   // Load versions
246 |   await refreshVersions();
247 |
248 |   // Stats on input
249 |   document.getElementById('instruction-editor').addEventListener

```

```
er('input', updateStats);
250
251   try {
252       applyTranslations();
253   } catch (e) {
254       console.error('[InstructionsTab] applyTranslations error:'
255       , e);
256   }
257   console.log('[InstructionsTab] Initialized successfully');
258 }
259
260 // — Экспорт в window (необходим для inline onclick в HTML пр
    и type="module") —
261 window.initInstructionsTab = initInstructionsTab;
262 window.switchMode         = switchMode;
263 window.saveInstruction     = saveInstruction;
264 window.checkInModel       = checkInModel;
265 window.refreshVersions    = refreshVersions;
266 window.useVersion         = useVersion;
267 window.previewPrompt      = previewPrompt;
```

activityTracker.js

```
1 /**
2  * activityTracker.js – Client-side Telemetry & User Activity
3  * Tracker
4  *
5  * Tracks tab views, dwell durations, and UI button/action cli
6  * cks strictly
7  *
8  * for authenticated registered users and flushes batches to /
9  * api/telemetry/events.
10 */
11
12 'use strict';
13
14 let currentTabId = 'tab-chat';
15 let tabEnterTimestamp = Date.now();
16 let eventQueue = [];
17 let flushTimer = null;
18 let isInitialized = false;
19
20 /**
21  * Check if the current user session is an authenticated regis
22  * tered user.
23  * @returns {boolean}
24  */
25 export function isUserAuthenticated() {
26   if (window.currentUser && window.currentUser.authenticated)
27   {
28     return true;
29   }
30   // Check auth_token cookie or session storage flag
31   return document.cookie.includes('auth_token=');
32 }
33
34 /**
35  * Push an event to the internal batch queue if user is authen
36  * ticated.
37  * @param {Object} event
```

```

31  */
32  export function trackEvent(event) {
33    if (!isUserAuthenticated()) {
34      return;
35    }
36
37    const payload = {
38      action: event.action || event.event_type || 'action',
39      event_type: event.event_type || 'action',
40      tab_name: event.tab_name || currentTabId,
41      target_element: event.target_element || '',
42      duration_ms: event.duration_ms || 0,
43      details: event.details || null,
44      timestamp: new Date().toISOString()
45    };
46
47    eventQueue.push(payload);
48
49    if (eventQueue.length >= 20) {
50      flushEvents();
51    }
52  }
53
54  /**
55   * Record a tab switch / tab view event with dwell time calculation.
56   * @param {string} newTabId
57   */
58  export function trackTabSwitch(newTabId) {
59    if (!newTabId) return;
60    const now = Date.now();
61    const cleanNewTab = newTabId.startsWith('#') ? newTabId.slice(1) : newTabId;
62
63    if (currentTabId && currentTabId !== cleanNewTab) {
64      const duration = now - tabEnterTimestamp;
65      trackEvent({
66        action: `Left tab ${currentTabId}`,

```

```

67 |         event_type: 'tab_view',
68 |         tab_name: currentTabId,
69 |         target_element: `tab:${currentTabId}`,
70 |         duration_ms: duration,
71 |         details: {
72 |             transition_to: cleanNewTab,
73 |             duration_sec: Math.round(duration / 1000)
74 |         }
75 |     });
76 | }
77 |
78 | currentTabId = cleanNewTab;
79 | tabEnterTimestamp = now;
80 |
81 | trackEvent({
82 |     action: `Entered tab ${cleanNewTab}`,
83 |     event_type: 'tab_view',
84 |     tab_name: cleanNewTab,
85 |     target_element: `tab:${cleanNewTab}`,
86 |     duration_ms: 0,
87 |     details: {
88 |         timestamp: new Date().toISOString()
89 |     }
90 | });
91 | }
92 |
93 | /**
94 |  * Send queued events to the backend telemetry ingestion endpo
95 |  int.
96 |  */
97 | export async function flushEvents() {
98 |     if (!isUserAuthenticated() || eventQueue.length === 0) {
99 |         return;
100 |     }
101 |
102 |     const batch = [...eventQueue];
103 |     eventQueue = [];

```



```
104 |   const body = JSON.stringify({
105 |     events: batch,
106 |     session_id: window.currentUser?.email || 'authenticated_session'
107 |   });
108 |
109 |   try {
110 |     if (navigator.sendBeacon) {
111 |       const blob = new Blob([body], { type: 'application/json'
112 |     });
113 |     const success = navigator.sendBeacon('/api/telemetry/events', blob);
114 |     if (success) return;
115 |   }
116 |
117 |   await fetch('/api/telemetry/events', {
118 |     method: 'POST',
119 |     headers: {
120 |       'Content-Type': 'application/json'
121 |     },
122 |     body,
123 |     keepalive: true
124 |   });
125 | } catch (err) {
126 |   console.debug('Telemetry flush failed, restoring queue:',
127 | err);
128 |   // Put events back in queue up to maximum limit
129 |   if (eventQueue.length < 50) {
130 |     eventQueue = [...batch, ...eventQueue];
131 |   }
132 | }
133 |
134 | /**
135 |  * Inspect click target and capture non-sensitive interactive
136 |  * element info.
137 |  * @param {MouseEvent} event
138 |  */
```

```
137 function handleGlobalClick(event) {
138   if (!isUserAuthenticated()) return;
139
140   const target = event.target;
141   if (!target) return;
142
143   // Find closest interactive element
144   const interactiveEl = target.closest(
145     'button, a, input[type="button"], input[type="submit"], [r
146     ole="button"], .nav-link, .dropdown-item, .btn, .badge[role="b
147     utton"], [data-action], [data-tab]'
148   );
149
150   if (!interactiveEl) return;
151
152   // Skip sensitive elements (passwords, auth inputs, tokens)
153   const isSensitive =
154     interactiveEl.type === 'password' ||
155     interactiveEl.name === 'password' ||
156     interactiveEl.id?.includes('password') ||
157     interactiveEl.id?.includes('token') ||
158     interactiveEl.classList?.contains('sensitive');
159
160   if (isSensitive) return;
161
162   // Extract meaningful descriptive label
163   let label = (
164     interactiveEl.getAttribute('data-action') ||
165     interactiveEl.getAttribute('data-tab') ||
166     interactiveEl.getAttribute('aria-label') ||
167     interactiveEl.getAttribute('title') ||
168     interactiveEl.innerText?.trim().slice(0, 60) ||
169     interactiveEl.id ||
170     interactiveEl.name ||
171     interactiveEl.tagName.toLowerCase()
172   );
173
174   // Clean label whitespace
```

```

173 |   label = label.replace(/\s+/g, ' ').trim();
174 |   if (!label) return;
175 |
176 |   const targetDescriptor = `${interactiveEl.tagName.toLowerCase()
    |   ${interactiveEl.id ? '#' + interactiveEl.id : ''}
    |   ${interactiveEl.className ? '.' + interactiveEl.className.split(' ').fi
    |   lter(c => !c.startsWith('show') && !c.startsWith('active')).sl
    |   ice(0, 2).join('.') : ''}`;
177 |
178 |   trackEvent({
179 |     action: `Click: ${label}`,
180 |     event_type: 'click',
181 |     tab_name: currentTabId,
182 |     target_element: targetDescriptor,
183 |     details: {
184 |       label,
185 |       tag: interactiveEl.tagName.toLowerCase(),
186 |       id: interactiveEl.id || null,
187 |       data_action: interactiveEl.getAttribute('data-action') |
    | | null,
188 |       data_tab: interactiveEl.getAttribute('data-tab') || null
189 |     }
190 |   });
191 | }
192 |
193 | /**
194 |  * Initialize Activity Tracker listeners and flush loop.
195 |  */
196 | export function initActivityTracker() {
197 |   if (isInitialized) return;
198 |   isInitialized = true;
199 |
200 |   // Listen for click interactions
201 |   document.addEventListener('click', handleGlobalClick, { capture: true, passive: true });
202 |
203 |   // Periodic flush every 5 seconds
204 |   flushTimer = setInterval(flushEvents, 5000);

```

```
205 |
206 | // Flush on visibility change / window unload
207 | document.addEventListener('visibilitychange', () => {
208 |     if (document.visibilityState === 'hidden') {
209 |         flushEvents();
210 |     }
211 | });
212 | window.addEventListener('pagehide', () => {
213 |     flushEvents();
214 | });
215 |
216 | console.log('□ Activity Tracker initialized (registered user
    | s only)');
217 | }
218 |
219 | // Global exposure for legacy scripts
220 | window.activityTracker = {
221 |     trackTabSwitch,
222 |     trackEvent,
223 |     flushEvents,
224 |     initActivityTracker
225 | };
226 |
```



```

28         <div class="row g-2 mb-3 small" id="uai-modal-me
| tadata-grid"></div>
29
30         <!-- Raw / Code Content Section -->
31         <div class="card bg-black border-secondary p-3 m
| b-3" id="uai-modal-row-container" style="display: none;">
32             <div class="d-flex justify-content-between ali
| gn-items-center mb-1">
33                 <h6 class="small text-uppercase text-muted f
| w-bold mb-0" id="uai-modal-row-title">Контекст / Данные</h6>
34             </div>
35             <div class="small font-monospace text-light" i
| d="uai-modal-row-content" style="white-space: pre-wrap; word-b
| reak: break-all;"></div>
36         </div>
37
38         <!-- AI Contextual Diagnostic Card -->
39         <div class="card border-info bg-dark p-3" id="ua
| i-modal-ai-section">
40             <div class="d-flex justify-content-between ali
| gn-items-center mb-2 flex-wrap gap-2">
41                 <div class="d-flex align-items-center gap-2"
| >
42                     <h6 class="fw-bold text-info mb-0"><i clas
| s="bi bi-robot me-1"></i> AI Contextual Explanation</h6>
43                     <span class="badge bg-info-subtle text-inf
| o border border-info-subtle small px-2 py-0.5" id="uai-modal-w
| eb-status" title="Сведения обогащаются поиском в интернете">
44                         <i class="bi bi-globe me-1"></i>Web Grou
| nding
45                     </span>
46                 </div>
47                 <div class="d-flex align-items-center gap-1.
| 5">
48                     <button class="btn btn-sm btn-outline-seco
| ndary py-0 px-2 rounded-pill" id="btn-uai-modal-edit-prompt" t
| itle="Настроить промпт для этого типа таблицы">
49                         <i class="bi bi-gear-fill me-1"></i>Пром

```

```

    пт
50         </button>
51         <button class="btn btn-sm btn-outline-info
    py-0 px-2 rounded-pill" id="btn-uai-modal-refresh" title="Обн
    овить AI-анализ">
52             <i class="bi bi-arrow-clockwise"></i>
53         </button>
54     </div>
55 </div>
56 <div class="small" id="uai-modal-ai-explanatio
    n">
57     <div class="d-flex align-items-center gap-2
    py-2 text-info">
58         <div class="spinner-border spinner-border-
    sm" role="status"></div>
59         <span>Выполняется экспертный AI-анализ и п
    оиск сведений в интернете...</span>
60     </div>
61 </div>
62 </div>
63 </div>
64 <div class="modal-footer border-secondary py-2 px-
    3 d-flex justify-content-between">
65     <div id="uai-modal-custom-actions" class="d-flex
    gap-1.5 flex-wrap"></div>
66     <button type="button" class="btn btn-sm btn-seco
    ndary" data-bs-dismiss="modal">Закрыть</button>
67 </div>
68 </div>
69 </div>
70 `;
71 document.body.appendChild(modalEl);
72 }
73 return modalEl;
74 }
75
76 function escapeHtml(str) {
77     if (str === null || str === undefined) return '';

```

```

78     return String(str)
79         .replace(/&/g, '&amp;')
80         .replace(/</g, '&lt;')
81         .replace(/>/g, '&gt;')
82         .replace(/"/g, '&quot;')
83         .replace(/'/g, '&#039;');
84     }
85
86     /**
87      * Universal method to show item inspection modal with AI ex
88      * @param {Object} options Configuration object
89      */
90     function show(options) {
91         const opts = Object.assign({
92             title: 'Детали записи',
93             subtitle: '',
94             icon: '📄',
95             badges: [],
96             metadata: [],
97             rawTitle: 'Данные / Путь',
98             rawContent: '',
99             tableType: 'generic',
100             requestData: {},
101             actions: []
102         }, options);
103
104         const modalEl = ensureModalElement();
105
106         // Icon & Title
107         const iconEl = document.getElementById('uai-modal-icon');
108         if (iconEl) iconEl.textContent = opts.icon;
109
110         const titleEl = document.getElementById('uai-modal-title')
111         ;
112         if (titleEl) titleEl.textContent = opts.title;
113
114         // Badges

```



```

114     const badgesContainer = document.getElementById('uai-modal
    -badges');
115     if (badgesContainer) {
116         badgesContainer.innerHTML = (opts.badges || []).map(b =>
    {
117             const cls = b.class || 'badge bg-secondary';
118             return `<span class="${cls}">${escapeHtml(b.text)}</sp
    an>`;
119         }).join('');
120     }
121
122     // Metadata Grid
123     const metaContainer = document.getElementById('uai-modal-m
    etadata-grid');
124     if (metaContainer) {
125         metaContainer.innerHTML = (opts.metadata || []).map(m =>
    {
126             const colSize = m.fullWidth ? 'col-sm-12' : 'col-sm-6'
    ;
127             let valHtml = '';
128             if (m.html) {
129                 valHtml = m.html;
130             } else if (m.isCode) {
131                 valHtml = `<span class="font-monospace text-info sma
    ll" style="word-break: break-all;">${escapeHtml(m.value || '-'
    )}</span>`;
132             } else {
133                 valHtml = `<span class="text-light">${escapeHtml(m.v
    alue || '-')}</span>`;
134             }
135             return `
136                 <div class="${colSize}">
137                     <strong class="text-secondary">${escapeHtml(m.labe
    l)}</strong> ${valHtml}
138                 </div>
139             `;
140         }).join('');
141     }

```

```
142 |
143 |     // Raw Content Block
144 |     const rawContainer = document.getElementById('uai-modal-ra
    w-container');
145 |     const rawTitleEl = document.getElementById('uai-modal-raw-
    title');
146 |     const rawContentEl = document.getElementById('uai-modal-ra
    w-content');
147 |     if (rawContainer && rawTitleEl && rawContentEl) {
148 |         if (opts.rawContent && opts.rawContent.trim()) {
149 |             rawTitleEl.textContent = opts.rawTitle;
150 |             rawContentEl.textContent = opts.rawContent;
151 |             rawContainer.style.display = 'block';
152 |         } else {
153 |             rawContainer.style.display = 'none';
154 |         }
155 |     }
156 |
157 |     // Reset AI Diagnostic section
158 |     const aiExplanation = document.getElementById('uai-modal-a
    i-explanation');
159 |     if (aiExplanation) {
160 |         aiExplanation.innerHTML = `Нажмите «Анализ контекста», ч
    тобы получить экспертное объяснение от языковой модели, оценку
    безопасности и рекомендации.`;
161 |     }
162 |
163 |     // Wire up Prompt Editor button
164 |     const editPromptBtn = document.getElementById('btn-uai-mod
    al-edit-prompt');
165 |     if (editPromptBtn) {
166 |         editPromptBtn.onclick = () => openPromptEditor(opts.tabl
    eType);
167 |     }
168 |
169 |     // Function to execute AI diagnostic request with Web Grou
    nding
170 |     async function executeDiagnostics() {
```

```

171     const aiExplanation = document.getElementById('uai-modal
    -ai-explanation');
172     if (!aiExplanation) return;
173
174     aiExplanation.innerHTML = `
175         <div class="d-flex align-items-center gap-2 py-2 text-
    info">
176             <div class="spinner-border spinner-border-sm" role="
    status"></div>
177             <span>Выполняется экспертный AI-анализ для «${escape
    Html(opts.title)}» с проверкой в интернете...</span>
178         </div>
179     `;
180
181     const metaDict = {};
182     (opts.metadata || []).forEach(m => {
183         if (m.label && (m.value !== undefined && m.value !== n
    ull)) {
184             metaDict[m.label] = m.value;
185         }
186     });
187
188     try {
189         const fetchFn = (window.api && window.api.fetch) ? win
    dow.api.fetch : fetch;
190         const res = await fetchFn('/api/v1/diagnostics/explain
    ', {
191             method: 'POST',
192             headers: { 'Content-Type': 'application/json' },
193             body: JSON.stringify({
194                 table_type: opts.tableType,
195                 title: opts.title,
196                 subtitle: opts.subtitle,
197                 metadata: Object.assign({}, metaDict, opts.request
    Data || {}),
198                 raw_data: opts.rawContent || '',
199                 web_search: true
200             })

```

```

201         });
202         const data = (res && typeof res.json === 'function') ?
        await res.json() : res;
203
204         aiExplanation.innerHTML = `
205             <div class="mb-2"><strong class="text-info"><i class
        ="bi bi-card-text me-1"></i>Назначение:</strong> ${escapeHtml(
        data.summary)}</div>
206             <div class="mb-2"><strong class="text-secondary"><i
        class="bi bi-building me-1"></i>Разработчик / Категория:</stro
        ng> ${escapeHtml(data.developer || 'Неизвестен')} (${escapeHtm
        l(data.category || 'Компонент')})</div>
207             <div class="mb-2"><strong class="text-warning"><i cl
        ass="bi bi-shield-lock me-1"></i>Оценка безопасности:</strong>
        ${escapeHtml(data.security_verdict)}</div>
208             <div class="mb-2"><strong class="text-info"><i class
        ="bi bi-speedometer2 me-1"></i>Влияние на ресурсы:</strong> ${
        escapeHtml(data.performance_impact)}</div>
209             <div class="p-2 mb-2 rounded bg-dark-subtle border b
        order-warning-subtle">
210                 <strong class="text-warning"><i class="bi bi-light
        bulb me-1"></i>Рекомендация:</strong> ${escapeHtml(data.recomm
        endation)}
211             </div>
212             ${data.action_steps && data.action_steps.length > 0
        ? `
213                 <h6 class="small text-uppercase text-muted fw-bold
        mb-1">Рекомендуемые действия:</h6>
214                 <ul class="mb-0 ps-3">
215                     ${data.action_steps.map(s => `<li>${escapeHtml(s
        )}</li>`).join('')}
216                 </ul>
217                 ` : ''}
218             `;
219         } catch (err) {
220             aiExplanation.innerHTML = `<div class="text-danger py-
        2"><i class="bi bi-exclamation-octagon me-1"></i>Ошибка получе
        ния AI-анализа: ${escapeHtml(err.message)}</div>`;

```

```

221     }
222 }
223
224 // Wire up refresh button
225 const refreshBtn = document.getElementById('btn-uai-modal-
refresh');
226 if (refreshBtn) {
227     refreshBtn.onclick = () => executeDiagnostics();
228 }
229
230 // Auto-run AI diagnostic on show
231 executeDiagnostics();
232
233 // Custom Footer Actions
234 const actionsContainer = document.getElementById('uai-moda
l-custom-actions');
235 if (actionsContainer) {
236     actionsContainer.innerHTML = '';
237     (opts.actions || []).forEach(act => {
238         const btn = document.createElement('button');
239         btn.type = 'button';
240         btn.className = `btn btn-sm ${act.class || 'btn-outlin
e-secondary'}`;
241         btn.innerHTML = `${act.icon ? `<i class="bi ${act.icon
} me-1"></i>` : ''}${escapeHtml(act.label)}`;
242         btn.onclick = () => act.onClick && act.onClick(opts);
243         actionsContainer.appendChild(btn);
244     });
245 }
246
247 // Open Modal
248 if (window.bootstrap && window.bootstrap.Modal) {
249     const bsModal = window.bootstrap.Modal.getOrCreateInstan
ce(modalEl);
250     bsModal.show();
251 }
252 }
253

```

```

254 | const PROMPT_MODAL_ID = 'universal-ai-prompt-editor-modal';
255 |
256 | function ensurePromptModalElement() {
257 |     let promptEl = document.getElementById(PROMPT_MODAL_ID);
258 |     if (!promptEl) {
259 |         promptEl = document.createElement('div');
260 |         promptEl.id = PROMPT_MODAL_ID;
261 |         promptEl.className = 'modal fade';
262 |         promptEl.tabIndex = -1;
263 |         promptEl.setAttribute('aria-hidden', 'true');
264 |         promptEl.innerHTML = `
265 |             <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
266 |                 <div class="modal-content bg-dark text-light border-secondary shadow-lg">
267 |                     <div class="modal-header border-secondary py-2 px-3">
268 |                         <div class="d-flex align-items-center gap-2">
269 |                             <span class="fs-5 text-warning">⚠ </span>
270 |                             <h5 class="modal-title fw-bold text-warning mb-0">Редактор промпта диагностики</h5>
271 |                         </div>
272 |                         <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
273 |                     </div>
274 |                     <div class="modal-body py-3 px-3">
275 |                         <div class="row g-2 mb-3">
276 |                             <div class="col-sm-6">
277 |                                 <label class="form-label small text-secondary fw-bold mb-1">Тип таблицы / сущности</label>
278 |                                 <select class="form-select form-select-sm bg-dark text-light border-secondary" id="uai-prompt-type-select">
279 |                                     <option value="software">software (Установленное ПО)</option>
280 |                                     <option value="process">process (Процессы Windows)</option>
281 |                                     <option value="service">service (Службы Wi

```

```

ndows)</option>
282         <option value="task">task (Задачи планиров
щика)</option>
283         <option value="network">network (Сетевой м
ониторинг)</option>
284         <option value="registry">registry (Реестр
Windows)</option>
285         <option value="user">user (Пользователи и
группы)</option>
286         <option value="website">website (Мониторин
г сайтов)</option>
287         <option value="rag_doc">rag_doc (RAG База
знаний)</option>
288         <option value="disk">disk (Диски и тома)</
option>
289         <option value="startup">startup (Автозагру
зка)</option>
290         <option value="generic">generic (Общий шаб
лон)</option>
291     </select>
292 </div>
293 <div class="col-sm-6">
294     <label class="form-label small text-secondar
y fw-bold mb-1">Название шаблона</label>
295     <input type="text" class="form-control form-
control-sm bg-dark text-light border-secondary" id="uai-prompt
-name-input" placeholder="Название шаблона">
296 </div>
297 </div>
298
299 <div class="mb-3">
300     <label class="form-label small text-secondary
fw-bold mb-1">Системная инструкция (System Instruction)</label
>
301     <textarea class="form-control form-control-sm
bg-dark text-light border-secondary font-monospace" id="uai-pr
ompt-system-input" rows="2" placeholder="Роль и поведение моде
ли..."></textarea>

```

```
302         </div>
303
304         <div class="mb-2">
305             <div class="d-flex justify-content-between align-items-center mb-1">
306                 <label class="form-label small text-secondary fw-bold mb-0">Шаблон текста промпта (Prompt Template)</label>
307             <div class="d-flex gap-1" id="uai-prompt-variables-bar">
308                 <button type="button" class="btn btn-xs btn-outline-info py-0 px-1 font-monospace" style="font-size: 0.75rem;" onclick="window.AITableModal.insertVariable('{title}')">{title}</button>
309                 <button type="button" class="btn btn-xs btn-outline-info py-0 px-1 font-monospace" style="font-size: 0.75rem;" onclick="window.AITableModal.insertVariable('{subtitle}')">{subtitle}</button>
310                 <button type="button" class="btn btn-xs btn-outline-info py-0 px-1 font-monospace" style="font-size: 0.75rem;" onclick="window.AITableModal.insertVariable('{metadata}')">{metadata}</button>
311                 <button type="button" class="btn btn-xs btn-outline-info py-0 px-1 font-monospace" style="font-size: 0.75rem;" onclick="window.AITableModal.insertVariable('{raw_data}')">{raw_data}</button>
312             </div>
313         </div>
314         <textarea class="form-control form-control-sm bg-dark text-light border-secondary font-monospace" id="uai-prompt-template-input" rows="8" placeholder="Текст промпта с плейсхолдерами..."></textarea>
315         <div class="form-text small text-secondary" style="font-size: 0.75rem;">
316             Используйте переменные <code>{title}</code>, <code>{subtitle}</code>, <code>{metadata}</code>, <code>{raw_data}</code> для автоматической подстановки контекста строки таблицы.
```



```

317         </div>
318     </div>
319
320     <div id="uai-prompt-status-msg" class="small mt-
321 2" style="display: none;"></div>
322 </div>
323     <div class="modal-footer border-secondary py-2 px-
324 3 d-flex justify-content-between">
325         <button type="button" class="btn btn-sm btn-outl
326 ine-danger" id="btn-uai-prompt-reset">
327             <i class="bi bi-arrow-counterclockwise me-1"><
328 /i>Сбросить по умолчанию
329         </button>
330         <div class="d-flex gap-2">
331             <button type="button" class="btn btn-sm btn-se
332 condary" data-bs-dismiss="modal">Закрыть</button>
333             <button type="button" class="btn btn-sm btn-pr
334 imary" id="btn-uai-prompt-save">
335                 <i class="bi bi-check2 me-1"></i>Сохранить п
336 ромпт
337             </button>
338         </div>
339     </div>
340 </div>
341 `;
342     document.body.appendChild(promptEl);
343 }
344     return promptEl;
345 }
346
347 async function openPromptEditor(tableType) {
348     const promptModalEl = ensurePromptModalElement();
349     const typeSelect = document.getElementById('uai-prompt-typ
350 e-select');
351     const nameInput = document.getElementById('uai-prompt-name
352 -input');
353     const systemInput = document.getElementById('uai-prompt-sy

```

```

stem-input');
346     const templateInput = document.getElementById('uai-prompt-
template-input');
347     const statusMsg = document.getElementById('uai-prompt-stat
us-msg');
348     const saveBtn = document.getElementById('btn-uai-prompt-sa
ve');
349     const resetBtn = document.getElementById('btn-uai-prompt-r
eset');
350
351     if (typeSelect && tableType) {
352         typeSelect.value = tableType;
353     }
354
355     async function loadTemplate(tType) {
356         if (statusMsg) statusMsg.style.display = 'none';
357         try {
358             const fetchFn = (window.api && window.api.fetch) ? win
dow.api.fetch : fetch;
359             const res = await fetchFn(`/api/v1/diagnostics/prompts
/${encodeURIComponent(tType)}`);
360             const data = (res && typeof res.json === 'function') ?
await res.json() : res;
361             if (data) {
362                 if (nameInput) nameInput.value = data.name || '';
363                 if (systemInput) systemInput.value = data.system_ins
truction || '';
364                 if (templateInput) templateInput.value = data.prompt
_template || '';
365             }
366         } catch (e) {
367             console.error('Ошибка загрузки шаблона промпта:', e);
368         }
369     }
370
371     if (typeSelect) {
372         typeSelect.onChange = () => loadTemplate(typeSelect.valu
e);

```

```

373 |     }
374 |
375 |     if (saveBtn) {
376 |         saveBtn.onclick = async () => {
377 |             const currentType = typeSelect ? typeSelect.value : 'g
eneric';
378 |             try {
379 |                 const fetchFn = (window.api && window.api.fetch) ? w
indow.api.fetch : fetch;
380 |                 await fetchFn(`/api/v1/diagnostics/prompts/${encodeU
RIComponent(currentType)}`, {
381 |                     method: 'POST',
382 |                     headers: { 'Content-Type': 'application/json' },
383 |                     body: JSON.stringify({
384 |                         table_type: currentType,
385 |                         name: nameInput ? nameInput.value : '',
386 |                         system_instruction: systemInput ? systemInput.va
lue : '',
387 |                         prompt_template: templateInput ? templateInput.v
alue : ''
388 |                     })
389 |                 });
390 |                 if (statusMsg) {
391 |                     statusMsg.className = 'small mt-2 text-success';
392 |                     statusMsg.innerHTML = '<i class="bi bi-check-circl
e me-1"></i>Шаблон промпта успешно сохранен!';
393 |                     statusMsg.style.display = 'block';
394 |                     setTimeout(() => { if (statusMsg) statusMsg.style.
display = 'none'; }, 3000);
395 |                 }
396 |             } catch (err) {
397 |                 if (statusMsg) {
398 |                     statusMsg.className = 'small mt-2 text-danger';
399 |                     statusMsg.innerHTML = '<i class="bi bi-exclamation
-triangle me-1"></i>Ошибка сохранения: ${escapeHtml(err.messag
e)}';
400 |                     statusMsg.style.display = 'block';
401 |                 }

```

```

402         }
403     };
404 }
405
406     if (resetBtn) {
407         resetBtn.onclick = async () => {
408             const currentType = typeSelect ? typeSelect.value : 'g
eneric';
409             if (!confirm(`Сбросить промпт для «${currentType}» к с
истемному значению по умолчанию?`)) return;
410             try {
411                 const fetchFn = (window.api && window.api.fetch) ? w
indow.api.fetch : fetch;
412                 const res = await fetchFn(`/api/v1/diagnostics/promp
ts/${encodeURIComponent(currentType)}/reset`, {
413                     method: 'POST'
414                 });
415                 const data = (res && typeof res.json === 'function')
? await res.json() : res;
416                 if (data) {
417                     if (nameInput) nameInput.value = data.name || '';
418                     if (systemInput) systemInput.value = data.system_i
nstruction || '';
419                     if (templateInput) templateInput.value = data.prom
pt_template || '';
420                 }
421                 if (statusMsg) {
422                     statusMsg.className = 'small mt-2 text-info';
423                     statusMsg.innerHTML = '<i class="bi bi-arrow-count
erclockwise me-1"></i>Шаблон промпта сброшен к системному по у
молчанию.';
424                     statusMsg.style.display = 'block';
425                     setTimeout(() => { if (statusMsg) statusMsg.style.
display = 'none'; }, 3000);
426                 }
427             } catch (err) {
428                 if (statusMsg) {
429                     statusMsg.className = 'small mt-2 text-danger';

```

```

430 |         statusMsg.innerHTML = `

```

```
461 |     escapeHtml: escapeHtml
462 | };
463 |})();
464 |
```

api-cache.js

```
1 /**
2  * api-cache.js - Универсальный кеш-слой для всех API-запросов
3  * веб-интерфейса
4  *
5  * Предоставляет единую точку входа для всех HTTP-запросов с а
6  * втоматическим кешированием,
7  * используя современные стратегии: Cache-First, Network-First
8  * , Stale-While-Revalidate.
9  *
10 * АРХИТЕКТУРА:
11 * - L1 Cache: In-Memory (быстрый доступ)
12 * - L2 Cache: IndexedDB (персистентный)
13 * - L3 Fallback: Direct network (если кеш недоступен)
14 *
15 * ПРИМЕНЕНИЕ:
16 * Используется всеми вкладками веб-интерфейса для оптимизации
17 * загрузки данных,
18 * снижения нагрузки на API и обеспечения offline-режима.
19 *
20 * @author hypo69
21 * @version 1.0.0
22 * @since 2026-09-24
23 */
24
25 import { browserCache, STORES } from './browser-cache.js';
26
27 // Глобальные стратегии кеширования для разных типов данных
28 export const CACHE_STRATEGIES = {
29   // Статические справочники (модели, конфигурации) - долгое T
30   // TL
31   STATIC: { ttl: 60 * 60 * 1000, strategy: 'cache-first' },
32   // 1 час
33
34   // Редко меняющиеся данные (hardware spec, user profiles) -
35   // средний TTL
36   SEMI_STATIC: { ttl: 30 * 60 * 1000, strategy: 'cache-first' }
```

```

    },          // 30 минут
30
31    // Часто меняющиеся данные (системные метрики, статусы) - ко
    роткий TTL
32    DYNAMIC: { ttl: 5 * 1000, strategy: 'stale-while-revalidate'
    },          // 5 секунд
33
34    // Реалтайм данные (live telemetry, сенсоры) - минимальный T
    TL
35    REALTIME: { ttl: 3 * 1000, strategy: 'network-first' },
    // 3 секунды
36
37    // Критичные данные (security status, errors) - всегда из се
    ти
38    CRITICAL: { ttl: 0, strategy: 'network-first' }
    // Нет кеша
39 };
40
41 // Автоматические правила сопоставления URL → стратегия
42 const URL_PATTERNS = [
43     // Статические данные
44     { pattern: /\api\v1\models/, strategy: CACHE_STRATEGIES.S
    TATIC, tag: 'models' },
45     { pattern: /\api\config/, strategy: CACHE_STRATEGIES.STATI
    C, tag: 'config' },
46     { pattern: /\api\plugins/, strategy: CACHE_STRATEGIES.SEMI
    _STATIC, tag: 'plugins' },
47
48     // Hardware и системные спецификации
49     { pattern: /\api\v1\system\hardware/, strategy: CACHE_ST
    RATEGIES.SEMI_STATIC, tag: 'hardware' },
50     { pattern: /\api\v1\windows-backup/, strategy: CACHE_STR
    ATEGIES.SEMI_STATIC, tag: 'backup' },
51
52     // Динамические системные данные
53     { pattern: /\api\v1\system\summary/, strategy: CACHE_STR
    ATEGIES.DYNAMIC, tag: 'system-summary' },
54     { pattern: /\api\system-control\status/, strategy: CACHE_

```



```

    STRATEGIES.DYNAMIC, tag: 'control-status' },
55
56    // Реалтайм телеметрия
57    { pattern: /\api\v1\system\sensors/, strategy: CACHE_STR
    ATEGIES.REALTIME, tag: 'sensors' },
58    { pattern: /\api\v1\system\processes/, strategy: CACHE_S
    TRATEGIES.REALTIME, tag: 'processes' },
59
60    // RAG и чат (частые обновления)
61    { pattern: /\api\chat/, strategy: CACHE_STRATEGIES.DYNAMIC
    , tag: 'chat' },
62    { pattern: /\api\rag/, strategy: CACHE_STRATEGIES.DYNAMIC,
    tag: 'rag' }
63 ];
64
65 /**
66  * Определение стратегии кеширования по URL
67  * @param {string} url - URL запроса
68  * @returns {Object} { ttl, strategy, tag }
69  */
70 function getCacheStrategy(url) {
71     for (const rule of URL_PATTERNS) {
72         if (rule.pattern.test(url)) {
73             return { ...rule.strategy, tag: rule.tag };
74         }
75     }
76     // По умолчанию: stale-while-revalidate с TTL 1 минута
77     return { ttl: 60 * 1000, strategy: 'stale-while-revalidate',
    tag: 'default' };
78 }
79
80 /**
81  * Генерация ключа кеша из URL и параметров
82  * @param {string} url - URL запроса
83  * @param {Object} options - Опции запроса (method, body)
84  * @returns {string} Уникальный ключ кеша
85  */
86 function generateCacheKey(url, options = {}) {

```

```

87 |   const method = options.method || 'GET';
88 |   const body = options.body ? `-${btoa(options.body).substring
    | (0, 32)}` : '';
89 |   return `${method}:${url}${body}`;
90 | }
91 |
92 | /**
93 |  * Универсальная функция для выполнения HTTP-запросов с автома
    | тическим кешированием
94 |  *
95 |  * @param {string} url - URL запроса
96 |  * @param {Object} options - Опции запроса
97 |  * @param {Object} cacheOptions - Опции кеширования (переопред
    | еляют автоматику)
98 |  * @returns {Promise<any>} Результат запроса
99 |  *
100 |  * @example
101 |  * // Автоматическое определение стратегии
102 |  * const data = await cachedApiFetch('/api/v1/system/hardware'
    | );
103 |  *
104 |  * @example
105 |  * // Ручное переопределение стратегии
106 |  * const data = await cachedApiFetch('/api/custom', {}, {
107 |  *   ttl: 10000,
108 |  *   strategy: 'cache-first',
109 |  *   tag: 'custom-data'
110 |  * });
111 |  */
112 | /**
113 |  * Оборачивает результат в полиморфный объект, поддерживающий
    | как прямое обращение к полям,
114 |  * так и стандартный интерфейс Response (res.ok, res.status, r
    | es.json()).
115 |  * @param {any} data
116 |  * @returns {any}
117 |  */
118 | function wrapCachedData(data) {

```

```

119 |   if (data && typeof data === 'object') {
120 |       try {
121 |           if (!('ok' in data)) {
122 |               Object.defineProperty(data, 'ok', { value: true, writa
123 |               ble: true, configurable: true, enumerable: false });
124 |           }
125 |           if (!('status' in data)) {
126 |               Object.defineProperty(data, 'status', { value: 200, wr
127 |               itable: true, configurable: true, enumerable: false });
128 |           }
129 |           if (typeof data.json !== 'function') {
130 |               Object.defineProperty(data, 'json', { value: async ()
131 |               => data, writable: true, configurable: true, enumerable: false
132 |               });
133 |           }
134 |       } catch (_) {}
135 |   }
136 |   return data;
137 | }
138 |
139 | /**
140 |  * Универсальная функция для выполнения HTTP-запросов с автома
141 |  * тическим кешированием
142 |  *
143 |  * @param {string} url - URL запроса
144 |  * @param {Object} options - Опции запроса
145 |  * @param {Object} cacheOptions - Опции кеширования (переопред
146 |  * еляют автоматику)
147 |  * @returns {Promise<any>} Результат запроса (поддерживает как
148 |  * прямое чтение, так и res.ok / await res.json())
149 |  */
150 | export async function cachedApiFetch(url, options = {}, cacheO
151 |   ptions = {}) {
152 |     // Определяем стратегию автоматически или используем передан
153 |     ную
154 |     const autoStrategy = getCacheStrategy(url);
155 |     const {
156 |       ttl = autoStrategy.ttl,

```

```
148     strategy = autoStrategy.strategy,
149     tag = autoStrategy.tag,
150     forceNetwork = false
151 } = cacheOptions;
152
153 const cacheKey = generateCacheKey(url, options);
154 const cache = browserCache;
155
156 // Дожидаемся готовности базы, но не блокируем если есть оши
    бка
157 if (cache) {
158     try {
159         await cache.ready();
160     } catch (_) {}
161 }
162
163 // CACHE-FIRST: сначала проверяем кеш, потом сеть
164 if (cache && strategy === 'cache-first' && !forceNetwork) {
165     try {
166         const cached = await cache.get(STORES.API_CACHE, cacheKe
    y);
167         if (cached !== null && cached !== undefined) {
168             return wrapCachedData(cached);
169         }
170     } catch (_) {}
171 }
172
173 // STALE-WHILE-REVALIDATE: показываем кеш, обновляем в фоне
174 if (cache && strategy === 'stale-while-revalidate' && !force
    Network) {
175     try {
176         const cached = await cache.get(STORES.API_CACHE, cacheKe
    y);
177         if (cached !== null && cached !== undefined) {
178             // Фоновое асинхронное обновление
179             directFetch(url, options)
180                 .then(fresh => cache.set(STORES.API_CACHE, cacheKey,
    fresh, { ttl, tags: [tag, 'api-cache'] })))
```

```

181 |         .catch(err => console.debug(`[API Cache] Background
    | update skipped for ${url}:`, err));
182 |         return wrapCachedData(cached);
183 |     }
184 | } catch (_) {}
185 | }
186 |
187 | // NETWORK-FIRST или первый запрос: сначала сеть, потом кеш
    | как fallback
188 | try {
189 |     const fresh = await directFetch(url, options);
190 |
191 |     // Сохраняем в кеш (если TTL > 0)
192 |     if (cache && ttl > 0) {
193 |         cache.set(STORES.API_CACHE, cacheKey, fresh, { ttl, tags
    | : [tag, 'api-cache'] }).catch(() => {});
194 |     }
195 |
196 |     return wrapCachedData(fresh);
197 | } catch (err) {
198 |     // При ошибке сети пытаемся вернуть устаревший кеш
199 |     if (cache) {
200 |         try {
201 |             const staleCache = await cache.get(STORES.API_CACHE, c
    | acheKey);
202 |             if (staleCache !== null && staleCache !== undefined) {
203 |                 console.warn(`[API Cache] Network failed, returning
    | STALE cache: ${url}`);
204 |                 return wrapCachedData(staleCache);
205 |             }
206 |         } catch (_) {}
207 |     }
208 |     throw err;
209 | }
210 | }
211 |
212 | /**
213 |  * Прямой HTTP-запрос без кеширования

```

```

214 | * @param {string} url - URL запроса
215 | * @param {Object} options - Опции fetch
216 | * @returns {Promise<any>}
217 | */
218 | async function directFetch(url, options = {}) {
219 |     const opts = { ...options };
220 |     if (opts.body && typeof opts.body === 'string') {
221 |         opts.headers = {
222 |             'Content-Type': 'application/json',
223 |             ...(opts.headers || {})
224 |         };
225 |     }
226 |
227 |     if (window.api && typeof window.api.fetch === 'function') {
228 |         const r = await window.api.fetch(url, opts);
229 |         return wrapCachedData(r);
230 |     }
231 |
232 |     const res = await fetch(url, opts);
233 |     if (!res.ok) {
234 |         let errMsg = `HTTP ${res.status}`;
235 |         try {
236 |             const errJson = await res.json();
237 |             if (errJson && errJson.detail) {
238 |                 errMsg += ` (${typeof errJson.detail === 'object' ? JS
239 |                 ON.stringify(errJson.detail) : errJson.detail})`;
240 |             } catch (_) {}
241 |             throw new Error(errMsg);
242 |         }
243 |         const json = await res.json();
244 |         return wrapCachedData(json);
245 |     }
246 |
247 | /**
248 |  * Инвалидация кеша по тегу (например, после обновления данных
249 |  * @param {string} tag - Тег для инвалидации

```

```

250 | * @returns {Promise<number>} Количество удаленных записей
251 | *
252 | * @example
253 | * // Очистить весь кеш моделей после обновления
254 | * await invalidateCacheByTag('models');
255 | */
256 | export async function invalidateCacheByTag(tag) {
257 |   if (!browserCache || !browserCache.isDbReady) return 0;
258 |   const count = await browserCache.invalidateByTag(STORES.API_
  CACHE, tag);
259 |   console.log(`[API Cache] □ Invalidated ${count} entries wit
  h tag: ${tag}`);
260 |   return count;
261 | }
262 |
263 | /**
264 | * Очистка всего API-кеша
265 | * @returns {Promise<boolean>}
266 | */
267 | export async function clearAllApiCache() {
268 |   if (!browserCache || !browserCache.isDbReady) return false;
269 |   await browserCache.clearStore(STORES.API_CACHE);
270 |   console.log('[API Cache] □ All API cache cleared');
271 |   return true;
272 | }
273 |
274 | /**
275 | * Получение статистики кеша
276 | * @returns {Promise<Object>}
277 | */
278 | export async function getApiCacheStats() {
279 |   if (!browserCache || !browserCache.isDbReady) {
280 |     return { available: false, metrics: null };
281 |   }
282 |
283 |   const stats = await browserCache.getDetailedStats();
284 |   const apiStore = stats.stores[STORES.API_CACHE] || {};
285 |

```

```
286 |   return {
287 |     available: true,
288 |     metrics: stats.metrics,
289 |     apiCache: {
290 |       count: apiStore.count || 0,
291 |       sizeMB: apiStore.sizeMB || '0.00',
292 |       hitRate: stats.metrics.hits + stats.metrics.misses > 0
293 |         ? Math.round((stats.metrics.hits / (stats.metrics.hits
+ stats.metrics.misses)) * 100)
294 |         : 0
295 |     },
296 |     storage: stats.storageEstimate
297 |   };
298 | }
299 |
300 | // Экспорт в глобальный контекст для доступности из всех вклад
ок
301 | if (typeof window !== 'undefined') {
302 |   window.cachedApiFetch = cachedApiFetch;
303 |   window.invalidateCacheByTag = invalidateCacheByTag;
304 |   window.clearAllApiCache = clearAllApiCache;
305 |   window.getApiCacheStats = getApiCacheStats;
306 |   window.CACHE_STRATEGIES = CACHE_STRATEGIES;
307 | }
308 |
309 | export default cachedApiFetch;
310 |
```


appConfigEditor.js

```
1 // =====
  =====
2 // Process Name: App Configuration Editor UI Module
3 // =====
  =====
4 // Description:
5 //   Reusable client-side controller for inspecting and modify
  ing config.json
6 //   across /apps microservices in the AI-Breadboard web admin
  istrative panel.
7 //
8 // File: appConfigEditor.js
9 // Project: ai-breadboard
10 // Package: src.api.webinterface.js
11 // Author: hypo69
12 // Copyright: © 2026 hypo69
13 // =====
  =====
14
15 (function() {
16   let activeConfig = null;
17   let activeAppName = '';
18
19   /**
20    * Fetch app config from server
21    * @param {string} appName
22    * @returns {Promise<Object>}
23    */
24   async function fetchAppConfig(appName) {
25     let res;
26     if (window.api && typeof window.api.fetch === 'function')
27     {
28       try {
29         return await window.api.fetch(`/api/admin/apps/${appName}/config`);
29       } catch (e) {
```

```

30         // fallback
31     }
32 }
33 res = await fetch(`/api/admin/apps/${appName}/config`);
34 if (!res.ok) {
35     throw new Error(`Failed to load config: HTTP ${res.status}`);
36 }
37 return await res.json();
38 }
39
40 /**
41  * Save app config to server
42  * @param {string} appName
43  * @param {Object} configObj
44  * @returns {Promise<Object>}
45  */
46 async function saveAppConfig(appName, configObj) {
47     const payload = { config: configObj };
48     const res = await fetch(`/api/admin/apps/${appName}/config`, {
49         method: 'POST',
50         headers: { 'Content-Type': 'application/json' },
51         body: JSON.stringify(payload),
52     });
53     if (!res.ok) {
54         const err = await res.json().catch(() => ({ detail: res.statusText }));
55         throw new Error(err.detail || `HTTP ${res.status}`);
56     }
57     return await res.json();
58 }
59
60 /**
61  * Render or update modal HTML
62  */
63 function ensureModalExists() {
64     let modalEl = document.getElementById('app-config-modal');

```

```

65 |     if (!modalEl) {
66 |         modalEl = document.createElement('div');
67 |         modalEl.id = 'app-config-modal';
68 |         modalEl.className = 'modal fade';
69 |         modalEl.tabIndex = -1;
70 |         modalEl.setAttribute('aria-hidden', 'true');
71 |         modalEl.innerHTML = `
72 |             <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
73 |                 <div class="modal-content border-secondary shadow-lg
74 | " style="background: #0f172a; color: #f8fafc;">
75 |                     <div class="modal-header border-secondary" style="
76 | background: #1e293b;">
77 |                         <h5 class="modal-title d-flex align-items-center
78 | gap-2 fw-bold text-info" id="app-config-modal-title">
79 |                             <i class="bi bi-sliders"></i> Настройка конфигурации (config.json)
80 |                         </h5>
81 |                         <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
82 |                     </div>
83 |                     <div class="modal-body p-3">
84 |                         <div id="app-config-alert" class="alert d-none p
85 | y-2 px-3 small"></div>
86 |
87 |                         <!-- Tabs: Visual Form vs Raw JSON -->
88 |                         <ul class="nav nav-pills nav-fill mb-3" id="app-
89 | config-tabs" role="tablist">
90 |                             <li class="nav-item">
91 |                                 <button class="nav-link active py-1 small fw

```

```

ld" id="app-config-tab-row-btn" data-bs-toggle="pill" data-bs-
target="#app-config-tab-row" type="button">
92         <i class="bi bi-code-square me-1"></i> Иск
одный JSON
93         </button>
94     </li>
95 </ul>
96
97     <div class="tab-content">
98         <!-- Visual Form Pane -->
99         <div class="tab-pane fade show active" id="app
-config-tab-visual">
100             <div class="card border-secondary mb-3" styl
e="background: #1e293b;">
101                 <div class="card-header border-secondary p
y-2 small fw-bold text-primary d-flex align-items-center gap-2
">
102                     <i class="bi bi-hdd-network"></i> Режим
сервера (Server Mode)
103                 </div>
104                 <div class="card-body p-3">
105                     <div class="row g-3">
106                         <div class="col-md-6">
107                             <label class="form-label small fw-se
mibold text-light mb-1">Режим работы сервера:</label>
108                             <select class="form-select form-sele
ct-sm bg-dark text-white border-secondary" id="app-cfg-server-
mode">
109                                 <option value="dedicated">Dedicate
d (Отдельный процесс / порт)</option>
110                                 <option value="shared">Shared (Мар
шрутизация через главный сервер :8000)</option>
111                             </select>
112                             <div class="form-text small text-mut
ed" id="app-cfg-server-mode-hint">
113                                 В режиме dedicated run.ps1 запуска
ет отдельный процесс микросервиса.
114                             </div>

```

```

115 |             </div>
116 |             <div class="col-md-6">
117 |                 <label class="form-label small fw-se
mibold text-light mb-1">Порт (Port):</label>
118 |                 <input type="number" class="form-con
trol form-control-sm bg-dark text-white border-secondary font-
monospace" id="app-cfg-server-port" placeholder="8100">
119 |             </div>
120 |             <div class="col-md-6">
121 |                 <label class="form-label small fw-se
mibold text-light mb-1">Хост (Host):</label>
122 |                 <input type="text" class="form-contr
ol form-control-sm bg-dark text-white border-secondary font-mo
nospace" id="app-cfg-server-host" placeholder="127.0.0.1">
123 |             </div>
124 |             <div class="col-md-6">
125 |                 <label class="form-label small fw-se
mibold text-light mb-1">Воркеры (Workers):</label>
126 |                 <input type="number" class="form-con
trol form-control-sm bg-dark text-white border-secondary font-
monospace" id="app-cfg-server-workers" placeholder="1" min="1"
max="16">
127 |             </div>
128 |             <div class="col-12">
129 |                 <div class="form-check form-switch">
130 |                     <input class="form-check-input" ty
pe="checkbox" id="app-cfg-server-ssl">
131 |                     <label class="form-check-label sma
ll text-light" for="app-cfg-server-ssl">Использовать SSL (HTTP
S)</label>
132 |                 </div>
133 |             </div>
134 |         </div>
135 |     </div>
136 | </div>
137 |
138 | <!-- Dynamic App Extra Params Card -->
139 | <div class="card border-secondary" style="ba

```

```

    ckground: #1e293b;" id="app-cfg-extra-card">
140         <div class="card-header border-secondary p
    y-2 small fw-bold text-info d-flex align-items-center gap-2">
141             <i class="bi bi-gear-wide-connected"></i
    > Параметры приложения
142         </div>
143         <div class="card-body p-3" id="app-cfg-ext
    ra-body">
144             <!-- Populated dynamically based on app
    properties -->
145             </div>
146         </div>
147     </div>
148
149     <!-- Raw JSON Pane -->
150     <div class="tab-pane fade" id="app-config-tab-
    raw">
151         <div class="d-flex justify-content-between a
    lign-items-center mb-2">
152             <span class="small text-muted font-monospa
    ce" id="app-cfg-file-path">config.json</span>
153             <button class="btn btn-sm btn-outline-seco
    ndary py-0 px-2" id="app-cfg-btn-format">
154                 <i class="bi bi-magic"></i> Форматироват
    ь
155             </button>
156         </div>
157         <textarea class="form-control font-monospace
    bg-dark text-white border-secondary" id="app-cfg-raw-json" ro
    ws="14" style="font-size: 0.82rem; line-height: 1.4; tab-size:
    2;"></textarea>
158         </div>
159     </div>
160 </div>
161 <div class="modal-footer border-secondary justify-
    content-between" style="background: #1e293b;">
162     <button type="button" class="btn btn-sm btn-outl
    ine-secondary rounded-pill px-3" id="app-cfg-btn-reload">

```

```

163         <i class="bi bi-arrow-clockwise me-1"></i> Сбp
        |   ОСИТЬ
164         </button>
165         <div class="d-flex gap-2">
166             <button type="button" class="btn btn-sm btn-se
        |   condary rounded-pill px-3" data-bs-dismiss="modal">Отмена</but
        |   ton>
167             <button type="button" class="btn btn-sm btn-pr
        |   imary rounded-pill px-4 fw-semibold" id="app-cfg-btn-save">
168                 <i class="bi bi-save me-1"></i> Сохранить
169             </button>
170         </div>
171     </div>
172 </div>
173 </div>
174 `;
175 document.body.appendChild(modalEl);
176
177 // Event handlers
178 document.getElementById('app-cfg-btn-save')?.addEventListener(
        |   'click', onSaveConfig);
179 document.getElementById('app-cfg-btn-reload')?.addEventL
        |   istener('click', onReloadConfig);
180 document.getElementById('app-cfg-btn-format')?.addEventL
        |   istener('click', () => {
181         const rawEl = document.getElementById('app-cfg-raw-jso
        |   n');
182         try {
183             const parsed = JSON.parse(rawEl.value);
184             rawEl.value = JSON.stringify(parsed, null, 2);
185         } catch (e) {
186             showModalAlert(`Ошибка JSON: ${e.message}`, 'danger'
        |   );
187         }
188     });
189
190 // Synchronize visual form to raw JSON on tab switch
191 document.getElementById('app-config-tab-raw-btn')?.addEv

```

```

entListener('click', () => {
192     syncFormToRawJson();
193 });
194 document.getElementById('app-config-tab-visual-btn')?.ad
dEventListener('click', () => {
195     syncRawJsonToForm();
196 });
197 }
198 }
199
200 function showModalAlert(message, type = 'info') {
201     const el = document.getElementById('app-config-alert');
202     if (!el) return;
203     el.className = `alert alert-${type} py-2 px-3 small`;
204     el.innerHTML = message;
205     el.classList.remove('d-none');
206     if (type === 'success') {
207         setTimeout(() => el.classList.add('d-none'), 3500);
208     }
209 }
210
211 function populateVisualForm(config) {
212     if (!config) return;
213
214     // Server section
215     let serverMode = 'dedicated';
216     let host = '127.0.0.1';
217     let port = 8100;
218     let workers = 1;
219     let ssl = false;
220
221     if (typeof config.server === 'string') {
222         serverMode = config.server.toLowerCase();
223     } else if (config.server && typeof config.server === 'obje
ct') {
224         if (typeof config.server.dedicated === 'boolean') {
225             serverMode = config.server.dedicated ? 'dedicated' : '
shared';

```



```

226     } else if (config.server.dedicated !== undefined) {
227         serverMode = String(config.server.dedicated).toLowerCase() === 'true' ? 'dedicated' : 'shared';
228     } else {
229         serverMode = config.server.mode || config.server.type || 'dedicated';
230     }
231     host = config.server.host || '127.0.0.1';
232     port = config.server.port || port;
233     workers = config.server.workers || 1;
234     ssl = !!config.server.use_ssl;
235 }
236
237 const modeSelect = document.getElementById('app-cfg-server-mode');
238 if (modeSelect) modeSelect.value = serverMode;
239 const portInput = document.getElementById('app-cfg-server-port');
240 if (portInput) portInput.value = port;
241 const hostInput = document.getElementById('app-cfg-server-host');
242 if (hostInput) hostInput.value = host;
243 const workersInput = document.getElementById('app-cfg-server-workers');
244 if (workersInput) workersInput.value = workers;
245 const sslCheck = document.getElementById('app-cfg-server-ssl');
246 if (sslCheck) sslCheck.checked = ssl;
247
248 // Extra properties
249 const extraBody = document.getElementById('app-cfg-extra-body');
250 if (extraBody) {
251     let fieldsHtml = '<div class="row g-2">';
252     let hasExtra = false;
253
254     for (const [key, val] of Object.entries(config)) {
255         if (key === 'server' || key === 'cors' || key === 'ser

```

```

    ver_cors_origins' || key === 'server_cors_allow_credentials')
    {
256         continue;
257     }
258     hasExtra = true;
259     if (typeof val === 'object' && val !== null && !Array.
isArray(val)) {
260         fieldsHtml += `<div class="col-12"><h6 class="text-w
arning small fw-bold mt-2 mb-1">${key}</h6></div>`;
261         for (const [subK, subV] of Object.entries(val)) {
262             fieldsHtml += `
263                 <div class="col-md-6">
264                     <label class="form-label small text-muted mb-0
">${key}.${subK}</label>
265                     <input type="text" class="form-control form-co
ntrol-sm bg-dark text-white border-secondary font-monospace ap
p-cfg-extra-input" data-path="${key}.${subK}" value="${subV !=
= undefined ? String(subV) : ''}">
266                 </div>
267             `;
268         }
269     } else {
270         fieldsHtml += `
271             <div class="col-md-6">
272                 <label class="form-label small text-muted mb-0">
${key}</label>
273                 <input type="text" class="form-control form-cont
rol-sm bg-dark text-white border-secondary font-monospace app-
cfg-extra-input" data-path="${key}" value="${val != undefined
? String(val) : ''}">
274             </div>
275             `;
276         }
277     }
278
279     fieldsHtml += '</div>';
280     extraBody.innerHTML = hasExtra ? fieldsHtml : '<div clas
s="text-muted small">Дополнительные параметры отсутствуют.</di

```

```

    v>';
281     }
282
283     // Raw JSON
284     const rawEl = document.getElementById('app-cfg-raw-json');
285     if (rawEl) {
286         rawEl.value = JSON.stringify(config, null, 2);
287     }
288 }
289
290 function syncFormToRawJson() {
291     try {
292         const rawEl = document.getElementById('app-cfg-raw-json'
293 );
294         let currentObj = {};
295         try {
296             currentObj = JSON.parse(rawEl.value || '{}');
297         } catch {}
298
299         const serverMode = document.getElementById('app-cfg-server-mode')?.value || 'dedicated';
300         const isDedicated = serverMode === 'dedicated';
301         const port = parseInt(document.getElementById('app-cfg-server-port')?.value || '8100', 10);
302         const host = document.getElementById('app-cfg-server-host')?.value || '127.0.0.1';
303         const workers = parseInt(document.getElementById('app-cfg-server-workers')?.value || '1', 10);
304         const ssl = !!document.getElementById('app-cfg-server-ssl')?.checked;
305
306         if (!currentObj.server || typeof currentObj.server !== 'object') {
307             currentObj.server = {};
308         }
309         currentObj.server.dedicated = isDedicated;
310         if (currentObj.server.mode !== undefined) {
311             delete currentObj.server.mode;

```

```

311     }
312     currentObj.server.host = host;
313     currentObj.server.port = port;
314     currentObj.server.use_ssl = ssl;
315     currentObj.server.workers = workers;
316
317     // Sync extra inputs
318     document.querySelectorAll('.app-cfg-extra-input').forEach
319     h((input) => {
320         const path = input.getAttribute('data-path');
321         const val = input.value;
322         if (!path) return;
323
324         let parsedVal = val;
325         if (val === 'true') parsedVal = true;
326         else if (val === 'false') parsedVal = false;
327         else if (/^\d+$/.test(val)) parsedVal = parseInt(val,
328         10);
329         else if (/^\d+\.\d+$/.test(val)) parsedVal = parseFloat(
330         val);
331
332         const parts = path.split('.');
333         if (parts.length === 1) {
334             currentObj[parts[0]] = parsedVal;
335         } else if (parts.length === 2) {
336             if (!currentObj[parts[0]] || typeof currentObj[parts
337             [0]] !== 'object') {
338                 currentObj[parts[0]] = {};
339             }
340             currentObj[parts[0]][parts[1]] = parsedVal;
341         }
342     });
343
344     rawEl.value = JSON.stringify(currentObj, null, 2);
345     activeConfig = currentObj;
346 } catch (e) {
347     console.error('[AppConfigEditor] Form sync error:', e);
348 }

```

```
345 | }
346 |
347 | function syncRawJsonToForm() {
348 |     try {
349 |         const rawEl = document.getElementById('app-cfg-raw-json'
350 | );
351 |         const parsed = JSON.parse(rawEl.value || '{}');
352 |         activeConfig = parsed;
353 |         populateVisualForm(parsed);
354 |     } catch (e) {
355 |         showModalAlert(`Ошибка парсинга JSON: ${e.message}`, 'da
356 | nger');
357 |     }
358 | }
359 |
360 | async function onSaveConfig() {
361 |     try {
362 |         syncFormToRawJson();
363 |         const rawEl = document.getElementById('app-cfg-raw-json'
364 | );
365 |         const parsed = JSON.parse(rawEl.value);
366 |         const res = await saveAppConfig(activeAppName, parsed);
367 |         showModalAlert(`☑ ${res.message || 'Конфигурация успешно
368 | сохранена!'}`, 'success');
369 |         activeConfig = parsed;
370 |         if (typeof window.showNotification === 'function') {
371 |             window.showNotification(`Конфигурация ${activeAppName}
372 | сохранена`, 'success');
373 |         }
374 |     } catch (e) {
375 |         showModalAlert(`Ошибка при сохранении: ${e.message}`, 'd
376 | anger');
377 |     }
378 | }
379 |
380 | async function onReloadConfig() {
```

```

377 |     try {
378 |         const data = await fetchAppConfig(activeAppName);
379 |         activeConfig = data.config;
380 |         populateVisualForm(activeConfig);
381 |         showModalAlert('Конфигурация перезагружена с диска', 'in
fo');
382 |     } catch (e) {
383 |         showModalAlert(`Ошибка загрузки: ${e.message}`, 'danger'
);
384 |     }
385 | }
386 |
387 | /**
388 |  * Open the configuration modal for an app
389 |  * @param {string} appName e.g. 'trading_terminal', 'network
_terminal', etc.
390 |  * @param {string} displayName e.g. 'Exchange Trading Termin
al'
391 |  */
392 | window.openAppConfigModal = async function(appName, displayN
ame = '') {
393 |     ensureModalExists();
394 |     activeAppName = appName;
395 |
396 |     const titleEl = document.getElementById('app-config-modal-
title');
397 |     if (titleEl) {
398 |         titleEl.innerHTML = `<i class="bi bi-sliders"></i> ${dis
playName || appName} – config.json`;
399 |     }
400 |
401 |     const pathEl = document.getElementById('app-cfg-file-path'
);
402 |     if (pathEl) {
403 |         pathEl.textContent = `apps/${appName}/config.json`;
404 |     }
405 |
406 |     const alertEl = document.getElementById('app-config-alert'

```

```
    );  
407 |     if (alertEl) alertEl.classList.add('d-none');  
408 |  
409 |     // Show modal  
410 |     const modalEl = document.getElementById('app-config-modal'  
    );  
411 |     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)  
    ;  
412 |     modal.show();  
413 |  
414 |     try {  
415 |         const data = await fetchAppConfig(appName);  
416 |         activeConfig = data.config;  
417 |         populateVisualForm(activeConfig);  
418 |     } catch (e) {  
419 |         showModalAlert(`Не удалось загрузить конфигурацию: ${e.m  
    essage}`, 'danger');  
420 |     }  
421 | };  
422 |})();  
423 |
```

browser-cache.js

```
1  /**
2  * browser-cache.js - Универсальный менеджер браузерного кеша
  и долговременного хранилища
3  *
4  * Предоставляет многоуровневое хранилище (L1 In-Memory + L2 I
  ndexedDB + L3 Fallback)
5  * для эффективного хранения больших объемов клиентских данных
  :
6  * - Ответы API и справочники
7  * - RAG-индексы, эмбединги и поисковые чанки
8  * - История чатов и сообщений ассистентов
9  * - Аудиофайлы TTS и медиа-блобы (Blob / ArrayBuffer)
10 * - Реестры моделей и системные конфигурации
11 *
12 * Включает автоматическую очистку по TTL, групповую инвалидац
  ию по тегам,
13 * мониторинг дисковой квоты браузера и персистентность.
14 *
15 * ПРИМЕЧАНИЕ: Приложение запускается в своем окне через Edge
  --app=url,
16 * что может ограничивать доступ к Storage API. Проверьте конс
  оль браузера
17 * на наличие предупреждений о недоступности IndexedDB.
18 */
19
20 const DB_NAME = 'AI_Breadboard_DB';
21 const DB_VERSION = 1;
22
23 // Стандартные хранилища объектов IndexedDB
24 export const STORES = {
25   API_CACHE: 'api_cache',
26   RAG_EMBEDDINGS: 'rag_embeddings',
27   CHAT_HISTORY: 'chat_history',
28   MEDIA_BLOBS: 'media_blobs',
29   MODELS_REGISTRY: 'models_registry',
30   KEY_VALUE: 'key_value'
```



```

31 };
32
33 export class BrowserCacheManager {
34   /**
35    * Инициализация менеджера браузерного кеша
36    * @param {Object} options - Параметры конфигурации
37    * @param {number} options.defaultTTL - Время жизни по умолчанию (в мс, 1 час = 3600000)
38    * @param {number} options.maxMemoryEntries - Максимальное число записей в L1 памяти
39    */
40   constructor(options = {}) {
41     this.defaultTTL = options.defaultTTL || 60 * 60 * 1000; // 1 час
42     this.maxMemoryEntries = options.maxMemoryEntries || 200;
43     this.memoryCache = new Map(); // L1 Cache: key -> { value, expiresAt, storeName, tags }
44     this.db = null;
45     this.isDbReady = false;
46     this.initPromise = this._initIndexedDB();
47
48     // Метрики
49     this.stats = {
50       hits: 0,
51       misses: 0,
52       writes: 0,
53       deletes: 0
54     };
55
56     // Периодическая фоновая очистка просроченных записей каждые 10 минут
57     if (typeof window !== 'undefined') {
58       window.setInterval(() => this.cleanupExpired(), 10 * 60 * 1000);
59     }
60   }
61
62   /**

```

```

63 * Инициализация базы данных IndexedDB
64 * @private
65 */
66 async _initIndexedDB() {
67     if (typeof window === 'undefined' || !window.indexedDB) {
68         console.warn('[BrowserCache] IndexedDB недоступен в текущем окружении. Используется fallback на память.');
```

Используется fallback на память.

```

69         console.warn('[BrowserCache] Возможные причины:');
70         console.warn('[BrowserCache] 1. Приложение запущено в PW A-окне (Edge --app=url)');
71         console.warn('[BrowserCache] 2. Ограничения безопасности браузера');
72         console.warn('[BrowserCache] 3. Поврежденный user-data-dir');
73         this.isDbReady = false;
74         return null;
75     }
76
77     return new Promise((resolve) => {
78         try {
79             const request = indexedDB.open(DB_NAME, DB_VERSION);
80
81             request.onupgradeneeded = (event) => {
82                 const db = event.target.result;
83
84                 Object.values(STORES).forEach((storeName) => {
85                     if (!db.objectStoreNames.contains(storeName)) {
86                         const store = db.createObjectStore(storeName, {
87                             keyPath: 'key' });
88                         store.createIndex('expiresAt', 'expiresAt', { unique: false });
89                         store.createIndex('updatedAt', 'updatedAt', { unique: false });
90                         store.createIndex('tags', 'tags', { unique: false, multiEntry: true });
91                     }
92                 });
93                 console.log('[BrowserCache] Структура IndexedDB успешно создана');
94             }
95             request.onsuccess = () => {
96                 resolve(db);
97             }
98         } catch (error) {
99             console.error('[BrowserCache] Ошибка при инициализации IndexedDB:', error);
100         }
101     });
102 }

```

```

    шно обновлена до версии', DB_VERSION);
93         };
94
95         request.onsuccess = (event) => {
96             this.db = event.target.result;
97             this.isDbReady = true;
98             console.log('[BrowserCache] IndexedDB успешно подклю
    чена:', DB_NAME);
99             console.log('[BrowserCache] Данные будут сохраняться
    в профиле пользователя:', this._getUserDataDir());
100             this.cleanupExpired().catch(() => {});
101             resolve(this.db);
102         };
103
104         request.onerror = (event) => {
105             console.error('[BrowserCache] Ошибка открытия Indexe
    dDB:', event.target.error);
106             console.error('[BrowserCache] Проверьте:');
107             console.error('[BrowserCache] 1. Есть ли права на за
    пись в user-data-dir');
108             console.error('[BrowserCache] 2. Не заблокирован ли
    IndexedDB политикой браузера');
109             this.isDbReady = false;
110             resolve(null);
111         };
112     } catch (err) {
113         console.error('[BrowserCache] Исключение при инициализ
    ации IndexedDB:', err);
114         this.isDbReady = false;
115         resolve(null);
116     }
117     });
118 }
119
120 /**
121  * Ожидание готовности базы данных
122  */
123 async ready() {

```

```

124     await this.initPromise;
125     return this.isDbReady;
126 }
127
128 /**
129  * Сохранить значение в кеш (L1 память + L2 IndexedDB)
130  * @param {string} storeName - Имя хранилища (из STORES)
131  * @param {string} key - Уникальный ключ
132  * @param {any} value - Данные (объект, массив, строка, Blob
    , ArrayBuffer)
133  * @param {Object} options - Опции записи (ttl, tags, persis
    t)
134  * @returns {Promise<boolean>}
135  */
136 async set(storeName = STORES.KEY_VALUE, key, value, options
    = {}) {
137     if (!key) return false;
138     await this.ready();
139
140     const ttl = options.ttl !== undefined ? options.ttl : this
        .defaultTTL;
141     const now = Date.now();
142     const expiresAt = ttl > 0 ? now + ttl : 0; // 0 = бессрочн
        o
143     const tags = Array.isArray(options.tags) ? options.tags :
        (options.tags ? [options.tags] : []);
144
145     const record = {
146         key,
147         value,
148         expiresAt,
149         createdAt: now,
150         updatedAt: now,
151         tags,
152         size: this._estimateSize(value)
153     };
154
155     // 1. Сохранение в L1 In-Memory Cache

```

```

156     const memKey = `${storeName}:${key}`;
157     this._setMemoryCache(memKey, record);
158     this.stats.writes++;
159
160     // Если persist = false, сохраняем только в L1 памяти
161     if (options.persist === false) {
162         return true;
163     }
164
165     // 2. Сохранение в L2 IndexedDB
166     if (this.isDbReady && this.db) {
167         try {
168             await new Promise((resolve, reject) => {
169                 const transaction = this.db.transaction([storeName],
170                 'readwrite');
171                 const store = transaction.objectStore(storeName);
172                 const request = store.put(record);
173
174                 request.onsuccess = () => resolve(true);
175                 request.onerror = (e) => reject(e.target.error);
176             });
177             return true;
178         } catch (err) {
179             console.warn(`[BrowserCache] Ошибка записи в IndexedDB
180             (${storeName}/${key}):`, err);
181         }
182     }
183
184     return true;
185 }
186
187 /**
188  * Получить значение из кеша (L1 -> L2)
189  * @param {string} storeName - Имя хранилища
190  * @param {string} key - Ключ
191  * @returns {Promise<any|null>}
192  */
193 async get(storeName = STORES.KEY_VALUE, key) {

```

```
192 |     if (!key) return null;
193 |     await this.ready();
194 |
195 |     const now = Date.now();
196 |     const memKey = `${storeName}:${key}`;
197 |
198 |     // 1. Проверяем L1 In-Memory Cache
199 |     const memRecord = this.memoryCache.get(memKey);
200 |     if (memRecord) {
201 |         if (memRecord.expiresAt > 0 && now > memRecord.expiresAt
202 | ) {
203 |             this.memoryCache.delete(memKey);
204 |             this.delete(storeName, key).catch(() => {});
205 |             this.stats.misses++;
206 |             return null;
207 |         }
208 |         this.stats.hits++;
209 |         return memRecord.value;
210 |     }
211 |
212 |     // 2. Проверяем L2 IndexedDB
213 |     if (this.isDbReady && this.db) {
214 |         try {
215 |             const record = await new Promise((resolve, reject) =>
216 | {
217 |                 const transaction = this.db.transaction([storeName],
218 | 'readonly');
219 |                 const store = transaction.objectStore(storeName);
220 |                 const request = store.get(key);
221 |
222 |                 request.onsuccess = () => resolve(request.result);
223 |                 request.onerror = (e) => reject(e.target.error);
224 |             });
225 |
226 |             if (!record) {
227 |                 this.stats.misses++;
228 |                 return null;
229 |             }
230 |         }
231 |     }
```

```
227 |
228 |     // Проверяем срок действия
229 |     if (record.expiresAt > 0 && now > record.expiresAt) {
230 |         this.delete(storeName, key).catch(() => {});
231 |         this.stats.misses++;
232 |         return null;
233 |     }
234 |
235 |     // Добавляем в L1 для ускорения повторных обращений
236 |     this._setMemoryCache(memKey, record);
237 |     this.stats.hits++;
238 |     return record.value;
239 | } catch (err) {
240 |     console.warn(`[BrowserCache] Ошибка чтения из IndexedD
241 | B (${storeName}/${key}):`, err);
242 | }
243 |
244 |     this.stats.misses++;
245 |     return null;
246 | }
247 |
248 | /**
249 |  * Проверить наличие актуального ключа в кеше
250 |  */
251 | async has(storeName = STORES.KEY_VALUE, key) {
252 |     const val = await this.get(storeName, key);
253 |     return val !== null;
254 | }
255 |
256 | /**
257 |  * Удалить запись из кеша
258 |  * @param {string} storeName - Имя хранилища
259 |  * @param {string} key - Ключ
260 |  */
261 | async delete(storeName = STORES.KEY_VALUE, key) {
262 |     if (!key) return false;
263 |     await this.ready();
```

```

264
265     const memKey = `${storeName}:${key}`;
266     this.memoryCache.delete(memKey);
267     this.stats.deletes++;
268
269     if (this.isDbReady && this.db) {
270         try {
271             await new Promise((resolve, reject) => {
272                 const transaction = this.db.transaction([storeName],
273                 'readwrite');
274                 const store = transaction.objectStore(storeName);
275                 const request = store.delete(key);
276
277                 request.onsuccess = () => resolve(true);
278                 request.onerror = (e) => reject(e.target.error);
279             });
280             return true;
281         } catch (err) {
282             console.warn(`[BrowserCache] Ошибка удаления из Indexe
283             dDB (${storeName}/${key}):`, err);
284         }
285     }
286
287     /**
288     * Инвалидировать записи по тегу (например: 'models', 'chat_
289     session_1')
290     * @param {string} storeName - Имя хранилища
291     * @param {string} tag - Тег для поиска
292     */
293     async invalidateByTag(storeName = STORES.KEY_VALUE, tag) {
294         await this.ready();
295
296         // Удаляем из L1
297         for (const [memKey, record] of this.memoryCache.entries())
298         {
299             if (memKey.startsWith(`${storeName}:`) && record.tags &&

```



```

    record.tags.includes(tag)) {
298         this.memoryCache.delete(memKey);
299     }
300 }
301
302 if (!this.isDbReady || !this.db) return 0;
303
304 let count = 0;
305 try {
306     count = await new Promise((resolve, reject) => {
307         const transaction = this.db.transaction([storeName], '
readwrite');
308         const store = transaction.objectStore(storeName);
309         const index = store.index('tags');
310         const request = index.getAllKeys(tag);
311
312         request.onsuccess = () => {
313             const keys = request.result || [];
314             keys.forEach((k) => store.delete(k));
315             resolve(keys.length);
316         };
317         request.onerror = (e) => reject(e.target.error);
318     });
319     console.log(`[BrowserCache] Инвалидировано ${count} запи
сей по тегу '${tag}' в '${storeName}'`);
320 } catch (err) {
321     console.warn(`[BrowserCache] Ошибка инвалидации по тегу
'${tag}':`, err);
322 }
323 return count;
324 }
325
326 /**
327  * Инвалидировать записи по регулярному выражению или префик
су ключа
328  * @param {string} storeName - Имя хранилища
329  * @param {string|RegExp} pattern - Регулярное выражение или
строка префикса

```

```

330     */
331     async invalidateByPattern(storeName = STORES.KEY_VALUE, pattern) {
332         await this.ready();
333
334         const isRegExp = pattern instanceof RegExp;
335         const testMatch = (key) => isRegExp ? pattern.test(key) :
336         (key === pattern || key.startsWith(pattern));
337
338         // Очистка L1
339         for (const [memKey] of this.memoryCache.entries()) {
340             if (memKey.startsWith(`${storeName}:`)) {
341                 const rawKey = memKey.replace(`${storeName}:`, '');
342                 if (testMatch(rawKey)) {
343                     this.memoryCache.delete(memKey);
344                 }
345             }
346         }
347
348         if (!this.isDbReady || !this.db) return 0;
349
350         let deletedCount = 0;
351         try {
352             deletedCount = await new Promise((resolve, reject) => {
353                 const transaction = this.db.transaction([storeName], '
354                 readwrite');
355                 const store = transaction.objectStore(storeName);
356                 const request = store.openCursor();
357
358                 request.onsuccess = (event) => {
359                     const cursor = event.target.result;
360                     if (cursor) {
361                         if (testMatch(cursor.key)) {
362                             cursor.delete();
363                             deletedCount++;
364                         }
365                         cursor.continue();
366                     } else {

```

```

365         resolve(deletedCount);
366     }
367 };
368     request.onerror = (e) => reject(e.target.error);
369 });
370     console.log(`[BrowserCache] Инвалидировано ${deletedCount}
    t} записей по паттерну в '${storeName}'`);
371 } catch (err) {
372     console.warn(`[BrowserCache] Ошибка инвалидации по патте
    рну в '${storeName}':`, err);
373 }
374     return deletedCount;
375 }
376
377 /**
378  * Полностью очистить указанное хранилище
379  * @param {string} storeName - Имя хранилища
380  */
381 async clearStore(storeName = STORES.KEY_VALUE) {
382     await this.ready();
383
384     // Очистка L1
385     for (const [memKey] of this.memoryCache.entries()) {
386         if (memKey.startsWith(`${storeName}:`)) {
387             this.memoryCache.delete(memKey);
388         }
389     }
390
391     if (this.isDbReady && this.db) {
392         try {
393             await new Promise((resolve, reject) => {
394                 const transaction = this.db.transaction([storeName],
    'readwrite');
395                 const store = transaction.objectStore(storeName);
396                 const request = store.clear();
397
398                 request.onsuccess = () => resolve(true);
399                 request.onerror = (e) => reject(e.target.error);

```

```
400         });
401         console.log(`[BrowserCache] Хранилище '${storeName}' п
| олностью очищено.`);
402         return true;
403     } catch (err) {
404         console.warn(`[BrowserCache] Ошибка очистки хранилища
| '${storeName}':`, err);
405     }
406 }
407 return true;
408 }
409
410 /**
411  * Очистить абсолютно все хранилища
412  */
413 async clearAll() {
414     this.memoryCache.clear();
415     for (const storeName of Object.values(STORES)) {
416         await this.clearStore(storeName);
417     }
418     console.log(`[BrowserCache] Все хранилища браузерного кеша
| очищены.`);
419 }
420
421 /**
422  * Очистить просроченные записи во всех хранилищах
423  */
424 async cleanupExpired() {
425     await this.ready();
426     const now = Date.now();
427
428     // L1
429     for (const [memKey, record] of this.memoryCache.entries())
| {
430         if (record.expiresAt > 0 && now > record.expiresAt) {
431             this.memoryCache.delete(memKey);
432         }
433     }
```

```
434
435     if (!this.isDbReady || !this.db) return 0;
436
437     let totalDeleted = 0;
438     for (const storeName of Object.values(STORES)) {
439         try {
440             const deletedInStore = await new Promise((resolve, reject) => {
441                 const transaction = this.db.transaction([storeName],
442                 'readwrite');
443                 const store = transaction.objectStore(storeName);
444                 const index = store.index('expiresAt');
445                 const range = IDBKeyRange.bound(1, now);
446                 const request = index.openCursor(range);
447                 let count = 0;
448
449                 request.onsuccess = (event) => {
450                     const cursor = event.target.result;
451                     if (cursor) {
452                         cursor.delete();
453                         count++;
454                         cursor.continue();
455                     } else {
456                         resolve(count);
457                     }
458                 };
459                 request.onerror = (e) => reject(e.target.error);
460             });
461             totalDeleted += deletedInStore;
462         } catch (e) {
463             // Игнорируем возможные локальные ошибки курсора
464         }
465     }
466
467     if (totalDeleted > 0) {
468         console.log(`[BrowserCache] Фоновый сборщик мусора удалил ${totalDeleted} просроченных записей.`);
469     }
```

```

469     return totalDeleted;
470 }
471
472 /**
473  * Получить детальную статистику использования кеша и дисков
    ой квоты
474  */
475 async getDetailedStats() {
476     await this.ready();
477     const stats = {
478         metrics: { ...this.stats },
479         memoryCacheSize: this.memoryCache.size,
480         stores: {},
481         storageEstimate: {
482             usage: 0,
483             quota: 0,
484             usageMB: '0.00',
485             quotaMB: '0.00',
486             percentUsed: 0,
487             isPersisted: false
488         }
489     };
490
491     // Оценка дискового пространства через navigator.storage A
    PI
492     if (typeof navigator !== 'undefined' && navigator.storage
    && navigator.storage.estimate) {
493         try {
494             const estimate = await navigator.storage.estimate();
495             const usage = estimate.usage || 0;
496             const quota = estimate.quota || 0;
497             const usageMB = (usage / (1024 * 1024)).toFixed(2);
498             const quotaMB = (quota / (1024 * 1024)).toFixed(2);
499             const percentUsed = quota > 0 ? ((usage / quota) * 100
    ).toFixed(2) : 0;
500
501             let isPersisted = false;
502             if (navigator.storage.persisted) {

```

```

503         isPersisted = await navigator.storage.persisted();
504     }
505
506     stats.storageEstimate = {
507         usage,
508         quota,
509         usageMB,
510         quotaMB,
511         percentUsed: Number(percentUsed),
512         isPersisted
513     };
514 } catch (err) {
515     console.warn('[BrowserCache] Не удалось получить navig
516 ator.storage.estimate:', err);
517 }
518
519 // Подсчет записей и объема по каждому хранилищу
520 if (this.isDbReady && this.db) {
521     for (const storeName of Object.values(STORES)) {
522         try {
523             const storeData = await new Promise((resolve) => {
524                 const transaction = this.db.transaction([storeName
525 ], 'readonly');
526                 const store = transaction.objectStore(storeName);
527                 const countReq = store.count();
528                 const allReq = store.getAll();
529
530                 let count = 0;
531                 let totalBytes = 0;
532
533                 countReq.onsuccess = () => { count = countReq.resu
534 lt || 0; };
535                 allReq.onsuccess = () => {
536                     const records = allReq.result || [];
537                     records.forEach(r => {
538                         totalBytes += r.size || this._estimateSize(r.v
539 alue);

```

```

537         });
538         resolve({ count, sizeBytes: totalBytes });
539     };
540     allReq.onerror = () => resolve({ count: 0, sizeBytes: 0 });
541     });
542
543     stats.stores[storeName] = {
544         count: storeData.count,
545         sizeBytes: storeData.sizeBytes,
546         sizeKB: (storeData.sizeBytes / 1024).toFixed(2),
547         sizeMB: (storeData.sizeBytes / (1024 * 1024)).toFixed(2)
548     };
549     } catch (e) {
550         stats.stores[storeName] = { count: 0, sizeBytes: 0,
551         sizeKB: '0', sizeMB: '0' };
552     }
553 }
554
555 return stats;
556 }
557
558 /**
559  * Запросить у браузера постоянное хранилище (защита от случайной очистки)
560  */
561 async requestPersistence() {
562     if (typeof navigator !== 'undefined' && navigator.storage
563     && navigator.storage.persist) {
564         const isPersisted = await navigator.storage.persist();
565         console.log(`[BrowserCache] Запрос персистентности хранилища: ${isPersisted ? 'Одобрено' : 'Отклонено'}`);
566         return isPersisted;
567     }
568     return false;
569 }

```



```

569
570  /**
571   * Экспортировать данные хранилищ в JSON-структуру
572   * @param {Array<string>} storeNames - Список хранилищ для э
кспорта
573   */
574   async exportData(storeNames = [STORES.API_CACHE, STORES.MODE
LS_REGISTRY, STORES.CHAT_HISTORY]) {
575       await this.ready();
576       const exportResult = {
577           version: DB_VERSION,
578           timestamp: Date.now(),
579           exportedAt: new Date().toISOString(),
580           stores: {}
581       };
582
583       if (!this.isDbReady || !this.db) {
584           console.warn('[BrowserCache] Cannot export data: Indexed
DB not ready');
585           return exportResult;
586       }
587
588       for (const storeName of storeNames) {
589           try {
590               const records = await new Promise((resolve, reject) =>
{
591                   const transaction = this.db.transaction([storeName],
'readonly');
592                   const store = transaction.objectStore(storeName);
593                   const request = store.getAll();
594                   request.onsuccess = () => resolve(request.result ||
[]);
595                   request.onerror = (e) => reject(e.target.error);
596               });
597               exportResult.stores[storeName] = records;
598           } catch (err) {
599               console.warn(`[BrowserCache] Ошибка экспорта из ${stor
eName}:`, err);

```

```

600     }
601 }
602
603     return exportResult;
604 }
605
606 /**
607  * Импортировать данные в хранилища
608  * @param {Object} data - Экспортированный объект данных
609  */
610 async importData(data) {
611     if (!data || !data.stores) return false;
612     await this.ready();
613
614     for (const [storeName, records] of Object.entries(data.stores)) {
615         if (!Object.values(STORES).includes(storeName)) continue
        ;
616         if (!Array.isArray(records)) continue;
617
618         for (const record of records) {
619             if (record && record.key) {
620                 await this.set(storeName, record.key, record.value,
621 {
622                     ttl: record.expiresAt ? Math.max(0, record.expires
623 At - Date.now()) : 0,
624                     tags: record.tags || []
625                 });
626             }
627         }
628     }
629     console.log('[BrowserCache] Импорт данных успешно завершен
630 .');
631     return true;
632 }
633
634 /**
635  * Добавить запись в L1 In-Memory Cache с контролем размера

```

```

(LRU eviction)
633     * @private
634     */
635     _setMemoryCache(key, record) {
636         if (this.memoryCache.size >= this.maxMemoryEntries) {
637             const firstKey = this.memoryCache.keys().next().value;
638             if (firstKey) this.memoryCache.delete(firstKey);
639         }
640         this.memoryCache.set(key, record);
641     }
642
643     /**
644     * Приблизительная оценка размера объекта в байтах
645     * @private
646     */
647     _estimateSize(value) {
648         if (value === null || value === undefined) return 0;
649         if (typeof value === 'string') return value.length * 2;
650         if (typeof value === 'number') return 8;
651         if (typeof value === 'boolean') return 4;
652         if (typeof Blob !== 'undefined' && value instanceof Blob)
653             return value.size;
654         if (typeof ArrayBuffer !== 'undefined' && value instanceof
655             ArrayBuffer) return value.byteLength;
656         try {
657             return JSON.stringify(value).length * 2;
658         } catch {
659             return 1024;
660         }
661     }
662
663     /**
664     * Попытка получить путь к user-data-dir (для отладки)
665     * @private
666     */
667     _getUserDataDir() {
668         // В браузере этот путь недоступен из-за ограничений безоп
669         асности

```

```
667 |     // Но мы можем попытаться определить, запущено ли приложен
    | ие в PWA-окне
668 |     const isPWA = window.matchMedia('(display-mode: standalone
    |)').matches;
669 |     return isPWA ? '[PWA-окно - путь недоступен]' : '[Обычное
    |окно]';
670 | }
671 | }
672 |
673 | // Создаем глобальный singleton экземпляр
674 | export const browserCache = new BrowserCacheManager();
675 |
676 | // Регистрация в глобальном объекте window для доступности во
    | всех вкладках и скриптах
677 | if (typeof window !== 'undefined') {
678 |     window.BrowserCacheManager = BrowserCacheManager;
679 |     window.browserCache = browserCache;
680 |     window.BROWSER_STORES = STORES;
681 | }
682 |
683 | export default browserCache;
684 |
```

cache-ui.js

```
1 /**
2  * cache-ui.js - Контроллер пользовательского интерфейса управ
3  * ления браузерным кешем
4  *
5  * Отвечает за:
6  * - Отображение модального окна управления кешем
7  * - Получение и визуализацию статистики хранилищ и квот браузер
8  * а
9  * - Обработку действий очистки, экспорта, импорта и запроса п
10 ерсистентности
11 */
12
13 import { browserCache, STORES } from './browser-cache.js';
14
15 // Человекопонятные названия хранилищ
16 const STORE_LABELS = {
17   [STORES.API_CACHE]: { title: 'Кеш ответов API', icon: 'bi-hd
18     d-network', desc: 'Кешированные ответы сервера и справочники'
19   },
20   [STORES.RAG_EMBEDDINGS]: { title: 'RAG и Векторы', icon: 'bi-
21     database-fill-gear', desc: 'Эмбединги, векторные чанки и пои
22     сковый индекс' },
23   [STORES.CHAT_HISTORY]: { title: 'История чатов', icon: 'bi-c
24     hat-dots-fill', desc: 'Диалоги, сообщения ассистентов и контек
25     ст' },
26   [STORES.MEDIA_BLOBS]: { title: 'Медиа и Аудио TTS', icon: 'b
27     i-file-earmark-music-fill', desc: 'Сгенерированные аудиодорожк
28     и, картинки и бинарные файлы' },
29   [STORES.MODELS_REGISTRY]: { title: 'Реестр моделей', icon: '
30     bi-robot', desc: 'Метаданные локальных и облачных моделей' },
31   [STORES.KEY_VALUE]: { title: 'Пользовательские данные', icon
32     : 'bi-key-fill', desc: 'Локальные параметры и временные значен
33     ия' }
34 };
35
36 /**
```

```
23 | * Инициализация интерфейса управления кешем
24 | */
25 | export function initCacheUI() {
26 |     const openButtons = document.querySelectorAll('#btn-open-cache-manager, .btn-open-cache-manager, #user-menu-btn-cache');
27 |     const modalEl = document.getElementById('cacheManagerModal')
    | ;
28 |
29 |     if (!modalEl) {
30 |         console.warn('[CacheUI] Элемент #cacheManagerModal не найден в DOM. ');
31 |         return;
32 |     }
33 |
34 |     // Привязка кнопок открытия
35 |     openButtons.forEach((btn) => {
36 |         btn.addEventListener('click', (e) => {
37 |             e.preventDefault();
38 |             openCacheModal();
39 |         });
40 |     });
41 |
42 |     // Обновление статистики при открытии модального окна
43 |     modalEl.addEventListener('show.bs.modal', () => {
44 |         refreshCacheModalStats();
45 |     });
46 |
47 |     // Привязка действий кнопок
48 |     const btnCleanupExpired = document.getElementById('cache-btn-cleanup-expired');
49 |     if (btnCleanupExpired) {
50 |         btnCleanupExpired.addEventListener('click', async () => {
51 |             btnCleanupExpired.disabled = true;
52 |             const count = await browserCache.cleanupExpired();
53 |             window.showToast?.(`Очищено просроченных записей: ${count}`, 'info') || alert(`Очищено просроченных записей: ${count}`);
54 |             btnCleanupExpired.disabled = false;
```

```
55 |     await refreshCacheModalStats();
56 | });
57 | }
58 |
59 | const btnClearAll = document.getElementById('cache-btn-clear
    -all');
60 | if (btnClearAll) {
61 |     btnClearAll.addEventListener('click', async () => {
62 |         if (confirm('Вы действительно хотите полностью очистить
    весь браузерный кеш?')) {
63 |             btnClearAll.disabled = true;
64 |             await browserCache.clearAll();
65 |             window.showToast?.('Все хранилища браузерного кеша усп
    ешно очищены.', 'success') || alert('Все хранилища браузерного
    кеша успешно очищены.');
```

```
66 |             btnClearAll.disabled = false;
67 |             await refreshCacheModalStats();
68 |         }
69 |     });
70 | }
71 |
72 | const btnRequestPersist = document.getElementById('cache-btn
    -request-persist');
73 | if (btnRequestPersist) {
74 |     btnRequestPersist.addEventListener('click', async () => {
75 |         const granted = await browserCache.requestPersistence();
76 |         const msg = granted ? 'Постоянное хранилище успешно вклю
    чено браузером.' : 'Браузер отклонил запрос персистентности.';
77 |         window.showToast?.(msg, granted ? 'success' : 'warning')
    || alert(msg);
78 |         await refreshCacheModalStats();
79 |     });
80 | }
81 |
82 | const btnExport = document.getElementById('cache-btn-export'
    );
83 | if (btnExport) {
84 |     btnExport.addEventListener('click', async () => {
```

```

85     const data = await browserCache.exportData();
86     const blob = new Blob([JSON.stringify(data, null, 2)], {
      type: 'application/json' });
87     const url = URL.createObjectURL(blob);
88     const a = document.createElement('a');
89     a.href = url;
90     a.download = `ai_breadboard_cache_${new Date().toISOString().slice(0, 10)}.json`;
91     document.body.appendChild(a);
92     a.click();
93     document.body.removeChild(a);
94     URL.revokeObjectURL(url);
95     window.showToast?.('Экспорт кеша успешно сохранён', 'suc
      cess');
96   });
97 }
98
99 const fileInput = document.getElementById('cache-file-import
      ');
100 if (fileInput) {
101   fileInput.addEventListener('change', async (e) => {
102     const file = e.target.files?.[0];
103     if (!file) return;
104
105     try {
106       const text = await file.text();
107       const data = JSON.parse(text);
108       await browserCache.importData(data);
109       window.showToast?.('Данные кеша успешно импортированы.
        ', 'success') || alert('Данные кеша успешно импортированы. ');
110       await refreshCacheModalStats();
111     } catch (err) {
112       window.showToast?.('Ошибка при чтении файла импорта: '
        + err.message, 'danger') || alert('Ошибка при чтении файла им
        порта: ' + err.message);
113     } finally {
114       fileInput.value = '';
115     }

```



```

116     });
117 }
118 }
119
120 /**
121  * Открыть модальное окно управления кешем
122  */
123 export function openCacheModal() {
124     const modalEl = document.getElementById('cacheManagerModal')
125     ;
126     if (modalEl && typeof bootstrap !== 'undefined' && bootstrap
127     .Modal) {
128         const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl
129         );
130         bsModal.show();
131     }
132 }
133
134 /**
135  * Обновить и отрендерить статистику в модальном окне
136  */
137 export async function refreshCacheModalStats() {
138     const statsContainer = document.getElementById('cache-stores
139     -list');
140     const usageText = document.getElementById('cache-usage-text'
141     );
142     const usageProgress = document.getElementById('cache-usage-p
143     rogressbar');
144     const persistBadge = document.getElementById('cache-persist-
145     badge');
146
147     if (!statsContainer) return;
148
149     statsContainer.innerHTML = `
150         <div class="text-center py-3 text-muted">
151             <div class="spinner-border spinner-border-sm text-primar
152             y me-2" role="status"></div>
153             Анализ данных браузерного хранилища...

```

```
146     </div>
147     `;
148
149     try {
150         const stats = await browserCache.getDetailedStats();
151
152         // 1. Отображение квоты
153         if (usageText && usageProgress) {
154             const usageMB = stats.storageEstimate.usageMB;
155             const quotaMB = stats.storageEstimate.quotaMB;
156             const percent = stats.storageEstimate.percentUsed;
157
158             usageText.textContent = `${usageMB} МБ из ${quotaMB} МБ
159             (${percent}%)`;
160             usageProgress.style.width = `${Math.min(100, Math.max(1,
161             percent))}%`;
162             usageProgress.className = `progress-bar ${percent > 80 ?
163             'bg-danger' : (percent > 50 ? 'bg-warning' : 'bg-primary')}`;
164         }
165
166         // 2. Статус персистентности
167         if (persistBadge) {
168             if (stats.storageEstimate.isPersisted) {
169                 persistBadge.className = 'badge bg-success-subtle text
170                 -success border border-success-subtle';
171                 persistBadge.innerHTML = '<i class="bi bi-shield-check
172                 me-1"></i>Постоянное хранилище активно';
173             } else {
174                 persistBadge.className = 'badge bg-warning-subtle text
175                 -warning-emphasis border border-warning-subtle';
176                 persistBadge.innerHTML = '<i class="bi bi-shield-excla
177                 mation me-1"></i>Временное хранилище';
178             }
179         }
180
181         // 3. Таблица хранилищ
182         let html = '<div class="list-group list-group-flush border
183         rounded overflow-hidden">';
```

```

176     for (const [storeName, storeInfo] of Object.entries(stats.
stores)) {
177         const meta = STORE_LABELS[storeName] || { title: storeNa
me, icon: 'bi-folder', desc: 'Хранилище данных' };
178         html += `
179             <div class="list-group-item d-flex align-items-center
justify-content-between p-2.5">
180                 <div class="d-flex align-items-center gap-3">
181                     <div class="rounded-circle bg-primary-subtle text-
primary p-2 d-flex align-items-center justify-content-center"
style="width: 38px; height: 38px;">
182                         <i class="bi ${meta.icon} fs-5"></i>
183                     </div>
184                     <div>
185                         <div class="fw-semibold text-body">${meta.title}
</div>
186                         <div class="small text-muted" style="font-size:
0.75rem;">${meta.desc}</div>
187                     </div>
188                 </div>
189                 <div class="d-flex align-items-center gap-3">
190                     <div class="text-end">
191                         <span class="badge bg-secondary-subtle text-seco
ndary border px-2 py-1">${storeInfo.count} зап.</span>
192                         <div class="small fw-semibold text-muted mt-0.5"
>${storeInfo.sizeKB} КБ</div>
193                     </div>
194                     <button class="btn btn-outline-danger btn-sm px-2
py-1 rounded"
195                         title="Очистить это хранилище"
196                         onclick="window.clearSingleStore('${storeN
ame}')">
197                         <i class="bi bi-trash3"></i>
198                     </button>
199                 </div>
200             </div>
201         `;
202     }

```

```
203 |     html += '</div>';
204 |
205 |     statsContainer.innerHTML = html;
206 | } catch (err) {
207 |     console.error('[CacheUI] Ошибка обновления статистики:', e
rr);
208 |     statsContainer.innerHTML = `<div class="alert alert-danger
p-2 small">Ошибка получения статистики кеша: ${err.message}</
div>`;
209 | }
210 | }
211 |
212 | // Глобальная функция для очистки одного хранилища из UI
213 | if (typeof window !== 'undefined') {
214 |     window.clearSingleStore = async function(storeName) {
215 |         if (confirm(`Очистить данные хранилища "${storeName}"?`))
{
216 |             await browserCache.clearStore(storeName);
217 |             await refreshCacheModalStats();
218 |         }
219 |     };
220 | }
221 |
```

chatService.js

```
1 // Shared Chat Service Module
2 // Exposes window.chatService to unify API calling and message
  /error formatting.
3
4 class AudioQueuePlayer {
5   constructor(chunks, userSettings) {
6     this.chunks = chunks;
7     this.userSettings = userSettings;
8     this.currentIndex = 0;
9     this.isPlaying = false;
10    this.audioElements = new Map(); // index -> Audio
11    this.activeAudio = null;
12
13    // Prefetch first few chunks
14    this._prefetch(0);
15    this._prefetch(1);
16    this._prefetch(2);
17  }
18
19  _prefetch(index) {
20    if (index >= this.chunks.length || this.audioElements.has(
index)) return;
21
22    const text = this.chunks[index];
23    const queryParams = new URLSearchParams({
24      text: text,
25      system: this.userSettings.tts_system,
26      voice: this.userSettings.tts_voice
27    });
28    const audioUrl = `/api/tts/synthesize?${queryParams.toStri
ng()}`;
29    const audio = new Audio(audioUrl);
30    audio.load();
31    this.audioElements.set(index, audio);
32  }
33
```

```

34 |   async play() {
35 |       this.isPlaying = true;
36 |       while (this.currentIndex < this.chunks.length && this.isPl
    | aying) {
37 |           // Prefetch upcoming chunks
38 |           this._prefetch(this.currentIndex);
39 |           this._prefetch(this.currentIndex + 1);
40 |           this._prefetch(this.currentIndex + 2);
41 |           this._prefetch(this.currentIndex + 3);
42 |
43 |           const audio = this.audioElements.get(this.currentIndex);
44 |           if (!audio) {
45 |               this.currentIndex++;
46 |               continue;
47 |           }
48 |
49 |           this.activeAudio = audio;
50 |           window.chatServiceAudio = audio; // for global pause/sto
    | p integration
51 |
52 |           const playPromise = new Promise((resolve) => {
53 |               audio.onended = () => resolve();
54 |               audio.onerror = (e) => {
55 |                   console.error(`Audio error for chunk ${this.currentI
    | ndex}:`, e);
56 |                   this.stop(); // Останавливаем всю очередь при ошибке
    | загрузки
57 |                   resolve();
58 |               };
59 |           });
60 |
61 |           try {
62 |               await audio.play();
63 |               await playPromise;
64 |           } catch (err) {
65 |               console.error("Audio playback interrupted or failed:",
    | err);
66 |           }

```

```
67 |
68 |     this.currentIndex++;
69 | }
70 | this.isPlaying = false;
71 | }
72 |
73 | stop() {
74 |     this.isPlaying = false;
75 |     if (this.activeAudio) {
76 |         try {
77 |             this.activeAudio.pause();
78 |             this.activeAudio.src = "";
79 |         } catch (e) {}
80 |     }
81 |     this.audioElements.clear();
82 | }
83 | }
84 |
85 | window.chatService = {
86 |     /**
87 |      * Stops any currently playing audio or speech synthesis.
88 |      */
89 |     stop() {
90 |         if (window.chatServiceQueue) {
91 |             try {
92 |                 window.chatServiceQueue.stop();
93 |                 window.chatServiceQueue = null;
94 |             } catch (e) {
95 |                 console.error("Error stopping chatServiceQueue:", e);
96 |             }
97 |         }
98 |         if (window.chatServiceAudio) {
99 |             try {
100 |                 window.chatServiceAudio.pause();
101 |                 window.chatServiceAudio.src = "";
102 |                 window.chatServiceAudio = null;
103 |             } catch (e) {
104 |                 console.error("Error stopping chatServiceAudio:", e);
```

```
105 |     }
106 | }
107 | if ('speechSynthesis' in window) {
108 |     try {
109 |         window.speechSynthesis.cancel();
110 |     } catch (e) {
111 |         console.error("Error cancelling speechSynthesis:", e);
112 |     }
113 | }
114 | },
115 | _abortController: null,
116 |
117 | /**
118 |  * Sends a chat message to the /api/chat backend.
119 |  * Resolves to the text reply or throws the parsed error object/string.
120 |  *
121 |  * @param {string} message
122 |  * @returns {Promise<any>}
123 |  */
124 | async sendChatMessage(message, onChunk, history = [], generationConfig = {}) {
125 |     // Останавливаем любую активную озвучку перед отправкой нового сообщения
126 |     this.stop();
127 |
128 |     // Прерываем предыдущий запрос, если он еще выполняется
129 |     if (this._abortController) {
130 |         try {
131 |             this._abortController.abort();
132 |         } catch (e) {}
133 |     }
134 |     this._abortController = new AbortController();
135 |
136 |     let fullText = '';
137 |     let voiceText = '';
138 |
139 |     const effectiveGenConfig = { ...generationConfig };
```



```
140     if (!effectiveGenConfig.model && window.activeModelName) {
141         effectiveGenConfig.model = window.activeModelName;
142     }
143     if (!effectiveGenConfig.search_engine && window.activeSearchEngine) {
144         effectiveGenConfig.search_engine = window.activeSearchEngine;
145     }
146
147     try {
148         const response = await fetch('/api/chat', {
149             method: 'POST',
150             headers: { 'Content-Type': 'application/json' },
151             body: JSON.stringify({ message, history, generation_config: effectiveGenConfig }),
152             signal: this._abortController.signal
153         });
154
155         if (!response.ok) {
156             let data;
157             try {
158                 data = await response.json();
159             } catch (e) {
160                 throw new Error(`Ошибка сервера (HTTP ${response.status}): ${response.statusText}`);
161             }
162             throw (data.detail || data);
163         }
164
165         const reader = response.body.getReader();
166         const decoder = new TextDecoder('utf-8');
167         let buffer = '';
168
169         while (true) {
170             const { value, done } = await reader.read();
171             if (done) break;
172
173             buffer += decoder.decode(value, { stream: true });
```

```

174 |     const lines = buffer.split('\n');
175 |     buffer = lines.pop();
176 |
177 |     for (const line of lines) {
178 |         const trimmed = line.trim();
179 |         if (!trimmed) continue;
180 |         if (trimmed.startsWith('data: ')) {
181 |             try {
182 |                 const data = JSON.parse(trimmed.substring(6));
183 |                 if (data.error) {
184 |                     throw new Error(data.error);
185 |                 }
186 |                 if (data.text || data.status || data.voice || data.prompt_dump) {
187 |                     if (data.text) fullText += data.text;
188 |                     if (data.voice) {
189 |                         voiceText += data.voice;
190 |                     }
191 |                     if (onChunk) {
192 |                         onChunk(data.text || '', data.status || '',
193 | data.voice || '', data.prompt_dump || '');
194 |                     }
195 |                 } catch (e) {
196 |                     console.error('Parsing SSE error:', e, trimmed);
197 |                     throw e;
198 |                 }
199 |             }
200 |         }
201 |     }
202 |
203 |     if (buffer && buffer.trim().startsWith('data: ')) {
204 |         const data = JSON.parse(buffer.trim().substring(6));
205 |         if (data.error) throw new Error(data.error);
206 |         if (data.text || data.status || data.voice) {
207 |             if (data.text) fullText += data.text;
208 |             if (data.voice) voiceText += data.voice;
209 |             if (onChunk) onChunk(data.text || '', data.status ||

```

```

    '', data.voice || '');
210     }
211   }
212   } catch (e) {
213     if (e.name === 'AbortError') {
214       // Возвращаем Promise, который никогда не зарезолвится
215       .
216       // Это остановит старый вызов sendChatMessage() на мес
217       те, не вызывая catch/finally в UI-скриптах,
218       // так как они уже перекрыты новым запросом.
219       return new Promise(() => {});
220     }
221     throw e;
222   }
223   let followUpQuery = null;
224   const nextQueryRegex = /\[NEXT_QUERY\](.*?)\[\/NEXT_QUERY\]
225   \]/i;
226   const match = fullText.match(nextQueryRegex);
227   if (match) {
228     followUpQuery = match[1].trim();
229     fullText = fullText.replace(match[0], '');
230   }
231   return { text: fullText, voice: voiceText, followUp: follo
232   wUpQuery };
233 },
234 /**
235  * Formats chat messages.
236  * Converts objects into stringified JSON inside pre tags fo
237  r rich rendering.
238  *
239  * @param {any} text
240  * @returns {string|HTMLElement}
241  */
242   formatMessage(text) {

```

```

242     if (typeof text === 'object' && text !== null) {
243         // Если это просто объект с ответом { text, voice }, ото
бражаем только текст
244         if (text.text !== undefined) {
245             return text.text;
246         }
247         return `

```
${JSON.stringify(text, nu
ll, 2)}
```

`;
248     }
249     return text;
250 },
251
252 /**
253  * Reads the given text aloud using the browser SpeechSynthe
sis API or backend TTS.
254  *
255  * @param {string|object} text
256  */
257 async speak(text) {
258     if (!text) return;
259
260     // Останавливаем все предыдущие аудиопотоки
261     this.stop();
262
263     let cleanText = '';
264     if (typeof text === 'object' && text !== null) {
265         if (text.voice || text.text) {
266             // Если это объект { text, voice } из sendMessage
cleanText = text.voice || text.text;
267         } else {
268             // Если это карточка фильма
269             const title = text.title_ru || text.title;
270             const rec = text.why_watch || text.plot || '';
271             cleanText = `Найдено: ${title}. ${rec}`;
272         }
273     }
274     } else {

```

```

275 |     const trimmed = text.trim();
276 |     if (trimmed.startsWith('{') && trimmed.endsWith('}')) {
277 |         try {
278 |             const card = JSON.parse(trimmed);
279 |             const title = card.title_ru || card.title;
280 |             const rec = card.why_watch || card.plot || '';
281 |             cleanText = `Найдено: ${title}. ${rec}`;
282 |         } catch (e) {
283 |             cleanText = trimmed;
284 |         }
285 |     } else {
286 |         // Strip code blocks
287 |         cleanText = trimmed.replace(/```\s\S]*?```/g, '');
288 |         // Strip HTML tags
289 |         cleanText = cleanText.replace(/<[^>]*>/g, '');
290 |         // Strip Markdown formatting
291 |         cleanText = cleanText.replace(/[*_`#]/g, '');
292 |     }
293 | }
294 |
295 | if (!cleanText.trim()) return;
296 |
297 | let userSettings = null;
298 | try {
299 |     const response = await fetch('/auth/settings');
300 |     if (response.ok) {
301 |         userSettings = await response.json();
302 |         if (userSettings.tts_enabled === 1 && userSettings.tts
    _system && userSettings.tts_system !== 'browser') {
303 |             // Разбиваем текст по абзацам и предложениям на боле
    е мелкие чанки (~150-200 символов)
304 |             const paragraphs = cleanText
305 |                 .split(/\n+/)
306 |                 .map(p => p.trim())
307 |                 .filter(p => p.length > 0);
308 |
309 |             let chunks = [];
310 |             for (const para of paragraphs) {

```

```

311 |         const sentences = para.match(/^[.!?]+[.!?]*\/g) ||
    | [para];
312 |         let currentChunk = "";
313 |         for (const sentence of sentences) {
314 |             const trimmedSentence = sentence.trim();
315 |             if (!trimmedSentence) continue;
316 |             if ((currentChunk + " " + trimmedSentence).length
    | h > 200) {
317 |                 chunks.push(currentChunk.trim());
318 |                 currentChunk = trimmedSentence;
319 |             } else {
320 |                 currentChunk = (currentChunk + " " + trimmedSe
    | ntence).trim();
321 |             }
322 |         }
323 |         if (currentChunk) {
324 |             chunks.push(currentChunk.trim());
325 |         }
326 |     }
327 |     chunks = chunks.filter(c => c.length > 0);
328 |
329 |     if (chunks.length > 0) {
330 |         window.chatServiceQueue = new AudioQueuePlayer(chu
    | nks, userSettings);
331 |         await window.chatServiceQueue.play();
332 |     }
333 |     return;
334 | }
335 | }
336 | } catch (e) {
337 |     console.error('Failed to speak using backend TTS, fallin
    | g back to browser:', e);
338 | }
339 |
340 | if ('speechSynthesis' in window) {
341 |     window.currentUtterance = new SpeechSynthesisUtterance(c
    | leanText);
342 |

```

```

343 |         // Dynamic language detection for Web Speech API fallback
      | k
344 |         const hasHebrew = /[\\u0590-\\u05FF]/.test(cleanText);
345 |         const hasCyrillic = /[\\u0400-\\u04FF]/.test(cleanText);
346 |         window.currentUtterance.lang = hasHebrew ? 'he-IL' : (has
      | sCyrillic ? 'ru-RU' : 'en-US');
347 |
348 |         if (userSettings && userSettings.tts_voice) {
349 |             const voices = window.speechSynthesis.getVoices();
350 |             const selectedVoice = voices.find(v => v.name === user
      | Settings.tts_voice);
351 |             if (selectedVoice) {
352 |                 window.currentUtterance.voice = selectedVoice;
353 |                 if (selectedVoice.lang) {
354 |                     window.currentUtterance.lang = selectedVoice.lang;
355 |                 }
356 |             }
357 |         }
358 |
359 |         window.speechSynthesis.speak(window.currentUtterance);
360 |     }
361 | }
362 | };
363 |
364 | // Форматирование отображаемого названия поискового движка
365 | function formatSearchEngine(engine) {
366 |     if (!engine) return '';
367 |     let eng = engine;
368 |     let mdl = '';
369 |     if (engine.includes(':') && !engine.startsWith('ollama:') &&
      | !engine.startsWith('foundry:')) {
370 |         const parts = engine.split(':');
371 |         eng = parts[0];
372 |         mdl = parts.slice(1).join(':');
373 |     }
374 |     const map = {
375 |         'gemini_cli': '□ gemini_cli',
376 |         'gemini': '□ gemini',

```

```

377     'agy': '🔗 agy',
378     'langchain': '🔗 langchain',
379     'playwright': '🔗 playwright'
380 };
381 const iconAndName = map[eng] || `🔗 ${eng}`;
382 if (mdl && !mdl.startsWith('chromium')) {
383     return `${iconAndName} (${mdl})`;
384 }
385 return iconAndName;
386 }
387
388 window.formatSearchEngine = formatSearchEngine;
389
390 window.updateChatBadges = function(modelName, searchEngine) {
391     if (modelName !== undefined && modelName !== null) {
392         window.activeModelName = modelName;
393     }
394     if (searchEngine !== undefined && searchEngine !== null) {
395         window.activeSearchEngine = searchEngine;
396     }
397
398     const curModel = window.activeModelName || '';
399     const curSearch = window.activeSearchEngine || '';
400
401     const modelBadges = document.querySelectorAll('#chat-model-b
402     adge, #chat-popup-model-badge');
403     modelBadges.forEach(badge => {
404         if (curModel) {
405             badge.innerHTML = `

```



```

(клик для выбора модели)`;
410     badge.style.display = 'inline-flex';
411 }
412
413 if (!badge._hasClickBound) {
414     badge._hasClickBound = true;
415     badge.style.cursor = 'pointer';
416     badge.onclick = () => {
417         // Try activating models tab if present
418         const modelsTabBtn = document.querySelector('button[da
ta-bs-target="#tab-models"]');
419         if (modelsTabBtn) {
420             const tab = new bootstrap.Tab(modelsTabBtn);
421             tab.show();
422         } else {
423             // Open settings modal
424             const modalEl = document.getElementById('userSetting
sModal');
425             if (modalEl && window.bootstrap) {
426                 const modal = bootstrap.Modal.getInstance(modalEl)
|| new bootstrap.Modal(modalEl);
427                 modal.show();
428             }
429         }
430     };
431 }
432 });
433
434 const searchBadges = document.querySelectorAll('#chat-search
-badge, #chat-popup-search-badge');
435 searchBadges.forEach(badge => {
436     if (curSearch) {
437         badge.innerHTML = `

```

```
442     }
443   });
444
445   const savedCardBadge = document.getElementById('saved-default-model-badge');
446   if (savedCardBadge) {
447     if (curModel) {
448       savedCardBadge.textContent = curModel;
449       savedCardBadge.className = 'badge bg-success font-monospace px-2 py-1';
450       savedCardBadge.title = `В вашем профиле сохранена модель: ${curModel}`;
451     } else {
452       savedCardBadge.textContent = 'Не задана (системный fallback)';
453       savedCardBadge.className = 'badge bg-secondary font-monospace px-2 py-1';
454       savedCardBadge.title = 'Модель в профиле не выбрана, сервер использует системный дефолт';
455     }
456   }
457
458   // Update chat message input state based on model selection
459   const msgInputs = document.querySelectorAll('#message-input, #chat-popup-input');
460   const sendButtons = document.querySelectorAll('#send-button, #chat-popup-send-btn');
461
462   msgInputs.forEach(input => {
463     if (!curModel) {
464       input.disabled = true;
465       input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
466       input.classList.add('is-invalid');
467     } else {
468       input.disabled = false;
469       input.placeholder = 'Введите сообщение...';
470       input.classList.remove('is-invalid');
```

```
471     }
472   });
473
474   sendButtons.forEach(btn => {
475     btn.disabled = !curModel;
476   });
477 };
478
479 // Автоматически загружаем настройки и обновляем бейджи модели
    и поиска при загрузке страницы
480 document.addEventListener('DOMContentLoaded', async () => {
481   try {
482     let modelName = '';
483     let searchEngine = '';
484     const response = await fetch('/auth/settings');
485     if (response.ok) {
486       const settings = await response.json();
487       modelName = settings.model || '';
488       searchEngine = settings.search_engine || '';
489     }
490     if (!modelName) {
491       try {
492         const activeResp = await fetch('/api/chat/active-model
493         ');
494         if (activeResp.ok) {
495           const activeData = await activeResp.json();
496           if (activeData && activeData.model) {
497             modelName = activeData.display || (activeData.prov
498             ider ? `${activeData.provider}: ${activeData.model}` : activeD
499             ata.model);
500           }
501         }
502       } catch {}
503     }
504     window.activeModelName = modelName;
505     window.activeSearchEngine = searchEngine;
506
507     window.updateChatBadges(modelName, searchEngine);
```

```
505 |     // На случай если DOM элементы добавились/отрендерились по
    |     зже
506 |     setTimeout(() => window.updateChatBadges(), 500);
507 |     setTimeout(() => window.updateChatBadges(), 1500);
508 | } catch (e) {
509 |     console.error('Failed to load active model / search badge:
    |     ', e);
510 | }
511 | });
512 |
```

debounce-integration.js

```
1 /**
2  * Debounce Integration - Интеграция дебаунсинга в популярные
  | операции
3  *
4  * Этот модуль предоставляет готовые функции дебаунса/батчинга
  | для:
5  * - Поиска пользователей
6  * - Поиска плагинов
7  * - Автосохранения инструкций
8  * - Обновления статуса
9  */
10
11 import { RequestManager } from './request-manager.js';
12
13 // Создаем глобальный экземпляр request manager
14 const globalRequestManager = new RequestManager();
15
16 window.globalRequestManager = globalRequestManager;
17
18 /**
19  * Дебаунс поиска пользователей (500ms)
20  */
21 export function createUserSearchDebounce(searchFn) {
22   return (query) => {
23     if (window.optimizationModule?.stats) {
24       window.optimizationModule.stats.requestsDebounced++;
25     }
26     return globalRequestManager.debounce(
27       'user-search',
28       () => searchFn(query),
29       500
30     );
31   };
32 }
33
34 /**
```

```
35 | * Дебаунс поиска плагинов (500ms)
36 | */
37 | export function createPluginSearchDebounce(searchFn) {
38 |   return (query) => {
39 |     if (window.optimizationModule?.stats) {
40 |       window.optimizationModule.stats.requestsDebounced++;
41 |     }
42 |     return globalRequestManager.debounce(
43 |       'plugin-search',
44 |       () => searchFn(query),
45 |       500
46 |     );
47 |   };
48 | }
49 |
50 | /**
51 | * Дебаунс автосохранения инструкций (1000ms)
52 | */
53 | export function createInstructionAutoSaveDebounce(saveFn) {
54 |   return (content) => {
55 |     if (window.optimizationModule?.stats) {
56 |       window.optimizationModule.stats.requestsDebounced++;
57 |     }
58 |     return globalRequestManager.debounce(
59 |       'instruction-autosave',
60 |       () => saveFn(content),
61 |       1000
62 |     );
63 |   };
64 | }
65 |
66 | /**
67 | * Дебаунс обновления статуса (2000ms)
68 | */
69 | export function createStatusUpdateDebounce(updateFn) {
70 |   return (status) => {
71 |     if (window.optimizationModule?.stats) {
72 |       window.optimizationModule.stats.requestsDebounced++;
```

```
73     }
74     return globalRequestManager.debounce(
75         'status-update',
76         () => updateFn(status),
77         2000
78     );
79 };
80 }
81
82 /**
83  * Создает throttle функцию для real-time обновлений (1000ms)
84  */
85 export function createRealtimeUpdateThrottle(updateFn) {
86     return globalRequestManager.throttle(
87         'realtime-update',
88         updateFn,
89         1000
90     );
91 }
92
93 /**
94  * Батчинг загрузки пользователей по ID
95  * Вместо 5 отдельных запросов - один запрос с массивом ID
96  */
97 export function createUserBatchLoader(loadFn) {
98     return (userId) => {
99         if (window.optimizationModule?.stats) {
100             window.optimizationModule.stats.requestsDebounced++;
101         }
102         return globalRequestManager.batch(
103             'user-batch-load',
104             async (userIds) => {
105                 return loadFn(userIds);
106             },
107             userId,
108             50 // Батч отправляется через 50ms
109         );
110     };
111 }
```

```
111 }
112
113 /**
114  * Дублирование запросов - если один и тот же запрос идет два-
115  *   ды,
116  *   * используем результат первого
117  */
118 export function createDedupeFetch(url, fetchFn) {
119   const dedupeKey = `fetch:${url}`;
120   return () => {
121     if (window.optimizationModule?.stats) {
122       window.optimizationModule.stats.requestsDebounced++;
123     }
124     return globalRequestManager.deduplicate(dedupeKey, fetchFn);
125   };
126 }
127
128 /**
129  * Вспомогательная функция для измерения эффективности
130  */
131 export function logDebouncingStats() {
132   const stats = globalRequestManager.getStats();
133   console.group('▯ Debouncing Statistics');
134   console.log('Pending Debounces:', stats.pendingDebounces);
135   console.log('Pending Requests:', stats.pendingRequests);
136   console.log('Cached Results:', stats.cachedResults);
137   console.table(stats.cacheKeys.slice(0, 5));
138   console.groupEnd();
139 }
140
141 // Делаем доступным из консоли
142 window.logDebouncingStats = logDebouncingStats;
143
144 export default {
145   createUserSearchDebounce,
146   createPluginSearchDebounce,
147   createInstructionAutoSaveDebounce,
```



```
147 |   createStatusUpdateDebounce,  
148 |   createRealtimeUpdateThrottle,  
149 |   createUserBatchLoader,  
150 |   createDedupeFetch,  
151 |   logDebouncingStats,  
152 |   globalRequestManager  
153 | };  
154 |
```

i18n.js

```
1 /**
2  * i18n.js – Internationalization module
3  *
4  * Uses i18next (loaded via CDN in index.html).
5  *
6  * HTML attribute conventions:
7  *   data-i18n="key" – sets element innerHTML
8  *   data-i18n-placeholder="key" – sets input/textarea placeholder
9  *   data-i18n-title="key" – sets title attribute
10 *
11 * Selected language is saved to localStorage.
12 * On load: reads language from localStorage, falls back to browser lang, then 'en'.
13 */
14
15 const SUPPORTED = ['en', 'ru', 'he'];
16
17 let i18nInitialized = false;
18
19 let eventsBound = false;
20
21 function setupLanguageEvents() {
22   if (eventsBound) return;
23   eventsBound = true;
24   document.addEventListener('change', (e) => {
25     if (e.target && e.target.classList.contains('lang-selector')) {
26       switchLang(e.target.value);
27     }
28   });
29 }
30
31 /**
32  * Initialize i18next and apply translations to the DOM.
33  * Called once from main.js after DOMContentLoaded.
```

```

34 * @param {string} lang - language code (may be empty)
35 */
36 export async function initI18n(lang) {
37     // If lang is explicitly set – use it.
38     // If empty/null – detect from localStorage, browser, fall
    back to 'en'.
39     const resolved = SUPPORTED.includes(lang) ? lang : detectL
    ang();
40
41     // Always load English as fallback bundle
42     const [mainRes, enRes] = await Promise.all([
43         loadLocale(resolved),
44         resolved !== 'en' ? loadLocale('en') : Promise.resolve
    (null),
45     ]);
46
47     await i18next.init({
48         lng: resolved,
49         fallbackLng: 'en',
50         resources: { [resolved]: { translation: mainRes } },
51         interpolation: { escapeValue: false },
52     });
53
54     if (enRes) {
55         i18next.addResourceBundle('en', 'translation', enRes,
    true, true);
56     }
57
58     document.documentElement.setAttribute('lang', resolved);
59     document.documentElement.setAttribute('dir', resolved ===
    'he' ? 'rtl' : 'ltr');
60
61     applyTranslations();
62     syncSelectors(resolved);
63     setupLanguageEvents();
64     i18nInitialized = true;
65 }
66

```

```

67 /**
68  * Switch language, re-apply all translations, save to localSt
   orage.
69  * @param {string} lang
70  */
71 export async function switchLang(lang) {
72     if (!SUPPORTED.includes(lang)) return;
73
74     // Lazy-load locale bundle if not yet loaded
75     if (!i18next.hasResourceBundle(lang, 'translation')) {
76         const res = await loadLocale(lang);
77         i18next.addResourceBundle(lang, 'translation', res, tr
   ue, true);
78     }
79
80     await i18next.changeLanguage(lang);
81
82     document.documentElement.setAttribute('lang', lang);
83     document.documentElement.setAttribute('dir', lang === 'he'
   ? 'rtl' : 'ltr');
84
85     applyTranslations();
86     syncSelectors(lang);
87
88     // Persist to localStorage
89     localStorage.setItem('app_language', lang);
90
91     // Dispatch global event for listeners across tabs/compone
   nts
92     window.dispatchEvent(new CustomEvent('languageChanged', {
   detail: { lang } }));
93 }
94
95 /**
96  * Walk the DOM and update all elements with data-i18n* attrib
   utes.
97  * Safe to call multiple times – idempotent.
98  */

```

```

99 export function applyTranslations() {
100     // Text / HTML content
101     document.querySelectorAll('[data-i18n]').forEach(el => {
102         const key = el.getAttribute('data-i18n');
103         if (!key) return;
104         const val = i18next.t(key);
105         if (val && val !== key) el.innerHTML = val;
106     });
107
108     // Input / textarea placeholder
109     document.querySelectorAll('[data-i18n-placeholder]').forEa
110 ch(el => {
111         const key = el.getAttribute('data-i18n-placeholder');
112         if (!key) return;
113         const val = i18next.t(key);
114         if (val && val !== key) el.placeholder = val;
115     });
116
117     // Tooltip / title attribute
118     document.querySelectorAll('[data-i18n-title]').forEach(el
119 => {
120         const key = el.getAttribute('data-i18n-title');
121         if (!key) return;
122         const val = i18next.t(key);
123         if (val && val !== key) el.title = val;
124     });
125 }
126
127 async function loadLocale(lang, bundle = 'main') {
128     const pathName = window.location.pathname;
129     let paths = [];
130
131     if (pathName.includes('/user_tts')) {
132         paths = [
133             `/html/locales/${lang}.json`,
134             `/html/user_tts/locales/user_${lang}.json`
135         ];
136     } else {

```

```

135 |     paths = [
136 |         `/html/locales/${lang}.json`,
137 |         `/html/locales/user_${lang}.json`
138 |     ];
139 | }
140 |
141 | for (const path of paths) {
142 |     try {
143 |         const cacheBustUrl = `${path}?v=${Date.now()}`;
144 |         const r = await fetch(cacheBustUrl);
145 |         if (r.ok) return await r.json();
146 |     } catch (e) {
147 |         console.log(`[i18n] Try next path: ${path}`);
148 |     }
149 | }
150 | console.error(`[i18n] Failed to load locale: ${lang}`);
151 | return {};
152 | }
153 |
154 | function detectLang() {
155 |     // Check localStorage first
156 |     const saved = localStorage.getItem('app_language');
157 |     if (SUPPORTED.includes(saved)) return saved;
158 |
159 |     // Fall back to browser language
160 |     const browser = (navigator.language || 'en').split('-')[0]
161 |     ;
162 |     return SUPPORTED.includes(browser) ? browser : 'en';
163 | }
164 | /**
165 |  * Sync all language selector elements to the current language
166 |  *
167 |  */
168 | function syncSelectors(lang) {
169 |     document.querySelectorAll('.lang-selector').forEach(sel =>

```

```
170 |     });
171 | }
172 |
173 | /**
174 |  * Get current language
175 |  */
176 | export function getCurrentLang() {
177 |     return i18next?.language || detectLang();
178 | }
```

main.js

```
1 /**
2  * js/main.js – оркестратор главной страницы (run.ps1, маршрут
3  * /)
4  * Архитектура: UI_ARCHITECTURE.md
5  */
6 import { initI18n, switchLang, applyTranslations } from './i18
7 n.js';
8 import { initTheme } from './theme.js';
9 import { initUserSettings } from './userSettings.js';
10 import { initActivityTracker } from './activityTracker.js';
11 import { initCacheUI } from './cache-ui.js';
12 import { switchTab, loadTab, setupTabClicks } from './tab-core
13 .js';
14 import './api-cache.js'; // Подключаем универсальный API-кеш с
15 лой
16 import './test-api-cache.js'; // Подключаем тестовый скрипт
17
18 // Глобальный экспорт для вызова из вкладок
19 window.switchTab = switchTab;
20 window.switchToTab = switchTab;
21 window.applyTranslations = applyTranslations;
22
23 // Карта вкладок: имя → [htmlUrl, jsUrl]
24 // Пути явные – не генерируются из имени
25 const TABS = {
26   'chat':          ['/html/chat/index.html',
27                     '/html/chat/main.js'],
28   'rag':           ['/html/rag_tab/index.html',
29                     '/html/rag_tab/main.js'],
30   'telegram-rag':  ['/html/telegram_rag_tab/index.html',
31                     '/html/telegram_rag_tab/main.js'],
32   'news':          ['/html/news_tab/index.html',
33                     '/html/news_tab/main.js'],
34   'voice':         ['/html/voice_tab/index.html',
35                     '/html/voice_tab/main.js'],
36 }
```



```

28 |   'plugins':      ['/html/plugins_tab/index.html',
   |   '/html/plugins_tab/main.js'],
29 |   'admin':        ['/html/admin_tab/index.html',
   |   '/html/admin_tab/main.js'],
30 |   'help':         ['/html/help/index.html',
   |   '/html/help/main.js'],
31 |   'process-leaks': ['/html/process_leaks_tab/index.html',
   |   '/html/process_leaks_tab/main.js'],
32 |   'forensics':    ['/html/forensics_tab/index.html',
   |   '/html/forensics_tab/main.js'],
33 |   'throttling':   ['/html/throttling_tab/index.html',
   |   '/html/throttling_tab/main.js'],
34 |   'storage-wear': ['/html/storage_wear_tab/index.html',
   |   '/html/storage_wear_tab/main.js'],
35 |   'peripherals':  ['/html/peripherals_tab/index.html',
   |   '/html/peripherals_tab/main.js'],
36 |   'system-logs':  ['/html/system_logs_tab/index.html',
   |   '/html/system_logs_tab/main.js'],
37 | };
38 |
39 | // Lazy-загрузка: вкладка грузится при первом открытии
40 | const loaded = new Set();
41 |
42 | async function lazyLoad(tabName) {
43 |   if (loaded.has(tabName) || !TABS[tabName]) return;
44 |   loaded.add(tabName);
45 |   const v = Date.now();
46 |   const [html, js] = TABS[tabName];
47 |   await loadTab(tabName, `${html}?v=${v}`, `${js}?v=${v}`);
48 |   applyTranslations();
49 | }
50 |
51 | // Переопределяем switchTab с приоритетной мгновенной навигац
   | ей
52 | const _switchTab = switchTab;
53 | export function switchMainTab(tabId) {
54 |   if (!tabId) return;
55 |   // 1. Мгновенное переключение UI без задержки

```

```

56 | _switchTab(tabId);
57 |
58 | // 2. Фоновая асинхронная подгрузка разметки и скрипта вклад
    | ки (если еще не загружена)
59 | const name = (tabId.startsWith('tab-') ? tabId.slice(4) : ta
    | bId);
60 | if (!loaded.has(name) && TABS[name]) {
61 |     lazyLoad(name).catch(err => console.warn(`[Main] Ошибка la
    | zyLoad для ${name}:`, err));
62 | }
63 | }
64 | window.switchTab = switchMainTab;
65 | window.switchToTab = switchMainTab;
66 |
67 | // Открыть плагин по имени (вызывается из плагинов)
68 | window.openPluginFromDropdown = function(pluginName) {
69 |     if (!pluginName) return;
70 |     const n = pluginName.toLowerCase().replace(/-/g, '_');
71 |     if (n === 'telegram_channel_rag' || n === 'telegram_rag') {
72 |         window.switchTab('tab-telegram-rag'); return;
73 |     }
74 |     if (n === 'news_feed' || n === 'news' || n === 'smart_news')
    | {
75 |         window.switchTab('tab-news'); return;
76 |     }
77 |     window.switchTab('tab-plugins');
78 |     const sel = () => window.selectPlugin?.(pluginName);
79 |     sel(); setTimeout(sel, 200);
80 | };
81 |
82 | async function init() {
83 |     // 1. Тема
84 |     initTheme();
85 |
86 |     // 2. i18n
87 |     const lang = localStorage.getItem('app_language') || 'ru';
88 |     await initI18n(lang);
89 |     document.querySelectorAll('.lang-selector').forEach(sel => {

```

```
90     sel.value = lang;
91     sel.addEventListener('change', e => switchLang(e.target.va
| lue));
92   });
93
94   // 3. Пользователь, активность, кеш
95   await initUserSettings();
96   initActivityTracker();
97   initCacheUI();
98
99   // 4. Единый обработчик кликов
100  setupTabClicks();
101
102  // 5. Загрузить первую вкладку (чат)
103  await lazyLoad('chat');
104  applyTranslations();
105
106  // 6. Синхронизация видимости плагинов
107  try {
108    const r = await fetch('/api/admin/plugins');
109    if (r.ok) {
110      const data = await r.json();
111      window.syncPluginTabsVisibility?.(data.plugins);
112    }
113  } catch {}
114
115  // 7. Активировать вкладку по hash или чат
116  const hash = location.hash.replace('#', '');
117  await switchMainTab(hash || 'tab-chat');
118
119  // 8. Фоновый WebSocket
120  initComputerStream();
121 }
122
123 // Фоновый WebSocket для remote transcript
124 function initComputerStream() {
125   const proto = location.protocol === 'https:' ? 'wss:' : 'ws:
| ';
```

```
126 |   const ws = new WebSocket(`${proto}://${location.host}/api/con
    |   trol/ws?role=computer`);
127 |   ws.onmessage = e => {
128 |     try {
129 |       const msg = JSON.parse(e.data);
130 |       if (msg.event === 'transcript') window.handleRemoteTrans
    |   cript?.(msg.text);
131 |     } catch {}
132 |   };
133 |   ws.onclose = () => setTimeout(initComputerStream, 3000);
134 | }
135 |
136 |
137 |
138 | document.readyState === 'loading'
139 |   ? document.addEventListener('DOMContentLoaded', init)
140 |   : init();
141 |
```

optimization-test-suite.js

```
1 /**
2  * Optimization Test Suite - Комплексное тестирование оптимизаций
3  *
4  * Тестирует:
5  * - Ленивую загрузку вкладок
6  * - Кеширование API
7  * - Дебаунсинг запросов
8  * - Сохранение состояния
9  * - Производительность переключения
10 *
11 * Usage:
12 *   const tester = new OptimizationTestSuite();
13 *   await tester.runAllTests();
14 *   tester.printReport();
15 */
16
17 class OptimizationTestSuite {
18   constructor() {
19     this.tests = [];
20     this.results = [];
21     this.startTime = null;
22   }
23
24   /**
25    * Добавить тест
26    */
27   addTest(name, testFn) {
28     this.tests.push({ name, testFn });
29   }
30
31   /**
32    * Запустить все тесты
33    */
34   async runAllTests() {
35     console.log('□ Starting Optimization Test Suite...\n');
```

```
36 |     this.startTime = performance.now();
37 |
38 |     for (const test of this.tests) {
39 |         await this._runTest(test);
40 |     }
41 |
42 |     const totalTime = performance.now() - this.startTime;
43 |     console.log(`\n   All tests completed in ${totalTime.toFixed(
44 | d(2)}ms\n`);
45 | }
46 | /**
47 |  * Запустить отдельный тест
48 |  */
49 | async _runTest(test) {
50 |     const startTime = performance.now();
51 |     const result = {
52 |         name: test.name,
53 |         passed: false,
54 |         time: 0,
55 |         error: null,
56 |         details: {}
57 |     };
58 |
59 |     try {
60 |         console.log(`▶ Running: ${test.name}...`);
61 |         const testResult = await test.testFn();
62 |         result.passed = true;
63 |         result.details = testResult;
64 |         result.time = performance.now() - startTime;
65 |         console.log(`   PASSED (${result.time.toFixed(2)}ms)\n
66 | `);
67 |     } catch (error) {
68 |         result.passed = false;
69 |         result.error = error.message;
70 |         result.time = performance.now() - startTime;
71 |         console.log(`   FAILED: ${error.message}\n`);
72 |     }
```

```

72 |
73 |     this.results.push(result);
74 | }
75 |
76 | /**
77 |  * Печать отчета
78 |  */
79 | printReport() {
80 |     const passed = this.results.filter(r => r.passed).length;
81 |     const failed = this.results.length - passed;
82 |     const passRate = ((passed / this.results.length) * 100).to
Fixed(1);
83 |
84 |     console.group('□ Test Report');
85 |     console.log(`Total: ${this.results.length} tests`);
86 |     console.log(`□ Passed: ${passed}`);
87 |     console.log(`□ Failed: ${failed}`);
88 |     console.log(`□ Pass Rate: ${passRate}%`);
89 |     console.table(this.results.map(r => ({
90 |         Test: r.name,
91 |         Status: r.passed ? '□ PASS' : '□ FAIL',
92 |         Time: `${r.time.toFixed(2)}ms`,
93 |         Details: r.error || 'OK'
94 |     })));
95 |     console.groupEnd();
96 | }
97 |
98 | /**
99 |  * Получить результаты в JSON
100 |  */
101 | getResults() {
102 |     return this.results;
103 | }
104 | }
105 |
106 | /**
107 |  * Предварительно определенные тесты
108 |  */

```

```

109 export async function createDefaultTestSuite() {
110     const suite = new OptimizationTestSuite();
111
112     // =====
113     // Тест 1: Проверка ленивой загрузки вкладок
114     // =====
115     suite.addTest('Lazy Tab Loading', async () => {
116         if (!window.optimizationModule?.tabLoader) {
117             throw new Error('TabLoader not initialized');
118         }
119
120         const tabLoader = window.optimizationModule.tabLoader;
121         const initialLoaded = tabLoader.loaded.size;
122
123         // Проверяем что не все вкладки загружены сразу
124         if (initialLoaded > 5) {
125             throw new Error(`Too many tabs loaded at start: ${initialLoaded}`);
126         }
127
128         return {
129             initiallyLoadedTabs: initialLoaded,
130             totalRegisteredTabs: tabLoader.tabs.size,
131             lazyLoadingActive: true
132         };
133     });
134
135     // =====
136     // Тест 2: Проверка API кеширования
137     // =====
138     suite.addTest('API Response Caching', async () => {
139         if (!window.optimizationModule?.apiCache) {
140             throw new Error('APICache not initialized');
141         }

```



```

142
143     const apiCache = window.optimizationModule.apiCache;
144
145     // Сохраняем тестовые данные
146     const testKey = 'test:caching';
147     const testData = { test: true, timestamp: Date.now() };
148
149     apiCache.set(testKey, testData, 1000);
150
151     // Проверяем что данные кешированы
152     const cached = apiCache.get(testKey);
153     if (!cached || cached.test !== true) {
154         throw new Error('Cache set/get failed');
155     }
156
157     // Проверяем инвалидацию
158     apiCache.invalidate('test:');
159     const invalidated = apiCache.get(testKey);
160     if (invalidated !== null) {
161         throw new Error('Cache invalidation failed');
162     }
163
164     const stats = apiCache.getStats();
165     return {
166         cacheSize: stats.size,
167         cacheWorking: true,
168         testsPassed: 2
169     };
170 });
171
172     // =====
173     // =====
174     // Тест 3: Проверка дебаунсинга запросов
175     // =====
176     // =====
177     suite.addTest('Request Debouncing', async () => {
178         if (!window.globalRequestManager) {
179             throw new Error('RequestManager not initialized');

```

```

178 |     }
179 |
180 |     const manager = window.globalRequestManager;
181 |     let executeCount = 0;
182 |
183 |     const testFn = () => new Promise(resolve => {
184 |         executeCount++;
185 |         resolve();
186 |     });
187 |
188 |     // Вызываем функцию 10 раз быстро
189 |     const promises = [];
190 |     for (let i = 0; i < 10; i++) {
191 |         promises.push(manager.debounce('test-debounce', testFn,
192 | 50));
193 |     }
194 |
195 |     await Promise.all(promises);
196 |
197 |     // Функция должна выполняться только 1 раз вместо 10
198 |     if (executeCount > 2) {
199 |         throw new Error(`Debounce failed: executed ${executeCount}
200 | times instead of 1`);
201 |     }
202 |
203 |     return {
204 |         callsAttempted: 10,
205 |         actualExecutions: executeCount,
206 |         debounceEffective: executeCount <= 2,
207 |         reductionRate: `${((1 - executeCount / 10) * 100).toFixed(1)}%`
208 |     };
209 | });
210 |
211 | // =====
212 |
213 | // Тест 4: Проверка сохранения состояния
214 | // =====

```

```

=====
212 suite.addTest('Tab State Persistence', async () => {
213   if (!window.tabPersistence) {
214     throw new Error('TabStatePersistence not initialized');
215   }
216
217   const persistence = window.tabPersistence;
218   const testTabId = 'tab-test';
219
220   // Сохраняем состояние
221   persistence.saveActiveTab(testTabId);
222   const restored = persistence.getLastActiveTab();
223
224   if (restored !== testTabId) {
225     throw new Error('State persistence failed');
226   }
227
228   // Проверяем историю
229   const history = persistence.getHistory();
230   if (!history.some(h => h.tabId === testTabId)) {
231     throw new Error('History tracking failed');
232   }
233
234   const storageInfo = persistence.getStorageInfo();
235   return {
236     persistenceWorking: true,
237     totalStorageEntries: storageInfo.totalEntries,
238     historySize: history.length,
239     currentActiveTab: restored
240   };
241 });
242
243 // =====
=====
244 // Тест 5: Проверка производительности переключения вкладок
245 // =====
=====
246 suite.addTest('Tab Switch Performance', async () => {

```

```

247 |     if (!window.tabSwitchOptimizer) {
248 |         throw new Error('TabSwitchOptimizer not initialized');
249 |     }
250 |
251 |     const optimizer = window.tabSwitchOptimizer;
252 |
253 |     // Проверяем что оптимизация активна
254 |     if (!optimizer.isOptimized) {
255 |         throw new Error('TabSwitchOptimizer not enabled');
256 |     }
257 |
258 |     const stats = optimizer.getStats();
259 |
260 |     // Проверяем что время переключения в норме (<500ms в сред
    нем)
261 |     if (stats.avgSwitchTime > 500 && stats.totalSwitches > 0)
    {
262 |         throw new Error(`Tab switch too slow: ${stats.avgSwitchT
    ime.toFixed(2)}ms`);
263 |     }
264 |
265 |     return {
266 |         optimizationEnabled: optimizer.isOptimized,
267 |         totalSwitches: stats.totalSwitches,
268 |         avgSwitchTime: `${stats.avgSwitchTimeFormatted}`,
269 |         cachedTabs: stats.cachedTabs,
270 |         performanceTarget: '<100ms',
271 |         targetMet: stats.avgSwitchTime < 100 || stats.totalSwitc
    hes === 0
272 |     };
273 | });
274 |
275 |     // =====
    =====
276 |     // Тест 6: Проверка глобальной конфигурации оптимизаций
277 |     // =====
    =====
278 |     suite.addTest('Global Optimization Module', async () => {

```

```

279 |     if (!window.optimizationModule) {
280 |         throw new Error('OptimizationModule not initialized');
281 |     }
282 |
283 |     const module = window.optimizationModule;
284 |
285 |     // Проверяем наличие всех компонентов
286 |     const requiredComponents = ['tabLoader', 'apiCache', 'apiF
    etcher', 'requestManager'];
287 |     const missingComponents = requiredComponents.filter(comp =
    > !module[comp]);
288 |
289 |     if (missingComponents.length > 0) {
290 |         throw new Error(`Missing components: ${missingComponents
    .join(', ')}`);
291 |     }
292 |
293 |     const stats = module.stats;
294 |     return {
295 |         componentsInitialized: requiredComponents.length,
296 |         apiCallsUsed: stats.apiCallsUsed,
297 |         cachedCallsUsed: stats.cachedCallsUsed,
298 |         cacheHitRate: stats.apiCallsUsed > 0 ?
299 |             `${((stats.cachedCallsUsed / (stats.apiCallsUsed + sta
    ts.cachedCallsUsed)) * 100).toFixed(1)}%` :
300 |             'N/A',
301 |         requestsDebounced: stats.requestsDebounced,
302 |         tabsPreloaded: stats.tabsLoaded.size
303 |     };
304 | });
305 |
306 | // =====
    =====
307 | // Тест 7: Проверка отсутствия консольных ошибок
308 | // =====
    =====
309 | suite.addTest('Console Error Check', async () => {
310 |     // Проверяем нет ли критических ошибок в логе

```

```
311 |     const errorCount = window.optimizationModule?.stats?.consoleErrors || 0;
312 |
313 |     // Это примерная проверка, реально надо отслеживать console.error
314 |     return {
315 |         consoleErrorsDetected: 0,
316 |         warningsOk: true,
317 |         noFatalErrors: true
318 |     };
319 | });
320 |
321 | return suite;
322 | }
323 |
324 | /**
325 |  * Глобальная функция для запуска тестов
326 |  */
327 | window.runOptimizationTests = async function() {
328 |     const suite = await createDefaultTestSuite();
329 |     await suite.runAllTests();
330 |     suite.printReport();
331 |     return suite.getResults();
332 | };
333 |
334 | /**
335 |  * Глобальная функция для вывода статистики оптимизаций
336 |  */
337 | window.printOptimizationStatus = function() {
338 |     console.group('□ Optimization Status');
339 |
340 |     if (window.optimizationModule) {
341 |         console.log('□ Optimization Module:', 'Active');
342 |         console.log('    - LazyTabLoader:', window.optimizationModule.tabLoader ? 'Enabled' : 'Disabled');
343 |         console.log('    - APICache:', window.optimizationModule.apiCache ? 'Enabled' : 'Disabled');
344 |         console.log('    - RequestManager:', window.optimizationMod
```

```
    ule.requestManager ? 'Enabled' : 'Disabled');
345 |   } else {
346 |     console.log('❑ Optimization Module:', 'Not initialized');
347 |   }
348 |
349 |   if (window.tabPersistence) {
350 |     console.log('❑ Tab Persistence:', 'Active');
351 |   } else {
352 |     console.log('❑ Tab Persistence:', 'Not initialized');
353 |   }
354 |
355 |   if (window.tabSwitchOptimizer) {
356 |     console.log('❑ Tab Switch Optimizer:', window.tabSwitchOpt
    imizer.isOptimized ? 'Active' : 'Inactive');
357 |   } else {
358 |     console.log('❑ Tab Switch Optimizer:', 'Not initialized');
359 |   }
360 |
361 |   console.groupEnd();
362 | };
363 |
364 | export { OptimizationTestSuite };
365 |
366 |
```

request-manager.js

```
1 /**
2  * RequestManager - Умное управление запросами
3  *
4  * Предотвращает дублирующиеся запросы через дебаунсинг и батч
   инг
5  *
6  * Usage:
7  *   const manager = new RequestManager();
8  *   manager.debounce('search-users', async () => {
9  *     const result = await api.search(term);
10  *   }, 500);
11 */
12
13 class RequestManager {
14   constructor() {
15     this.debounceTimers = new Map(); // { key: timeoutId }
16     this.pendingRequests = new Map(); // { key: Promise }
17     this.requestCache = new Map(); // { key: result }
18   }
19
20   /**
21    * Дебаунс функции - откладывает выполнение, если вызов повт
   оряется
22    *
23    * @param {string} key - Уникальный ключ для дебаунса
24    * @param {function} fn - Функция для выполнения
25    * @param {number} delay - Задержка в миллисекундах
26    * @returns {Promise} - Результат выполнения функции
27    */
28   debounce(key, fn, delay = 300) {
29     return new Promise((resolve, reject) => {
30       // Отмены предыдущий таймер если был
31       if (this.debounceTimers.has(key)) {
32         clearTimeout(this.debounceTimers.get(key));
33       }
34     });
35   }
36 }
```



```
35 |         // Устанавливаем новый таймер
36 |         const timeoutId = setTimeout(async () => {
37 |             try {
38 |                 const result = await fn();
39 |                 this.requestCache.set(key, result);
40 |                 resolve(result);
41 |             } catch (error) {
42 |                 reject(error);
43 |             } finally {
44 |                 this.debounceTimers.delete(key);
45 |             }
46 |         }, delay);
47 |
48 |         this.debounceTimers.set(key, timeoutId);
49 |     });
50 | }
51 |
52 | /**
53 |  * Отмена дебаунса
54 |  */
55 | cancelDebounce(key) {
56 |     if (this.debounceTimers.has(key)) {
57 |         clearTimeout(this.debounceTimers.get(key));
58 |         this.debounceTimers.delete(key);
59 |     }
60 | }
61 |
62 | /**
63 |  * Отмена всех дебаунсов
64 |  */
65 | cancelAll() {
66 |     this.debounceTimers.forEach(timeoutId => clearTimeout(time
67 | outId));
68 |     this.debounceTimers.clear();
69 | }
70 | /**
71 |  * Батчинг - объединяет несколько запросов в один
```

```

72 | *
73 | * @param {string} key - Уникальный ключ батча
74 | * @param {function} batchFn - Функция которая выполняет бат
    | ч операцию
75 | * @param {any} item - Элемент для добавления в батч
76 | * @param {number} flushDelay - Задержка перед отправкой бат
    | ча
77 | */
78 | batch(key, batchFn, item, flushDelay = 100) {
79 |   if (!this.pendingRequests.has(key)) {
80 |     this.pendingRequests.set(key, { items: [], promise: null
    | });
81 |   }
82 |
83 |   const batch = this.pendingRequests.get(key);
84 |   batch.items.push(item);
85 |
86 |   if (!batch.promise) {
87 |     batch.promise = new Promise((resolve, reject) => {
88 |       setTimeout(async () => {
89 |         try {
90 |           const items = batch.items;
91 |           this.pendingRequests.delete(key);
92 |
93 |           console.log(`[RequestManager] Batching ${items.len
    | gth} items for ${key}`);
94 |           const result = await batchFn(items);
95 |           resolve(result);
96 |         } catch (error) {
97 |           this.pendingRequests.delete(key);
98 |           reject(error);
99 |         }
100 |       }, flushDelay);
101 |     });
102 |   }
103 |
104 |   return batch.promise;
105 | }

```

```

106
107  /**
108   * Throttle функции - вызывает функцию максимум один раз за
    period ms
109   *
110   * @param {string} key - Уникальный ключ для throttle
111   * @param {function} fn - Функция для выполнения
112   * @param {number} period - Минимальный интервал между вызов
    ами
113   * @returns {function} - Завернутая функция
114   */
115  throttle(key, fn, period = 1000) {
116    let lastCall = 0;
117    let timeoutId = null;
118
119    return async (...args) => {
120      const now = Date.now();
121      const timeSinceLastCall = now - lastCall;
122
123      if (timeSinceLastCall >= period) {
124        lastCall = now;
125        return fn.apply(this, args);
126      } else {
127        // Запланируем вызов на конец периода
128        if (timeoutId) clearTimeout(timeoutId);
129
130        return new Promise((resolve) => {
131          timeoutId = setTimeout(() => {
132            lastCall = Date.now();
133            fn.apply(this, args).then(resolve);
134          }, period - timeSinceLastCall);
135        });
136      }
137    };
138  }
139
140  /**
141   * Дублирование запросов - если один и тот же запрос идет дв

```

```

ажды,
142 * вернуть результат первого вместо создания второго
143 *
144 * @param {string} key - Уникальный ключ запроса
145 * @param {function} fn - Функция для выполнения
146 * @returns {Promise}
147 */
148 deduplicate(key, fn) {
149     // Если такой запрос уже в процессе - вернуть его
150     if (this.pendingRequests.has(key)) {
151         console.log(`[RequestManager] Deduplicating request: ${key}`);
152         return this.pendingRequests.get(key);
153     }
154
155     // Выполнить запрос
156     const promise = fn();
157     this.pendingRequests.set(key, promise);
158
159     promise
160         .then(result => {
161             this.requestCache.set(key, result);
162             return result;
163         })
164         .finally(() => {
165             this.pendingRequests.delete(key);
166         });
167
168     return promise;
169 }
170
171 /**
172 * Получить кешированный результат
173 */
174 getCache(key) {
175     return this.requestCache.get(key) || null;
176 }
177

```

```
178 | /**
179 |  * Очистить кеш
180 |  */
181 | clearCache() {
182 |     this.requestCache.clear();
183 | }
184 |
185 | /**
186 |  * Получить статистику
187 |  */
188 | getStats() {
189 |     return {
190 |         pendingDebounces: this.debounceTimers.size,
191 |         pendingRequests: this.pendingRequests.size,
192 |         cachedResults: this.requestCache.size,
193 |         debounceKeys: Array.from(this.debounceTimers.keys()),
194 |         requestKeys: Array.from(this.pendingRequests.keys()),
195 |         cacheKeys: Array.from(this.requestCache.keys())
196 |     };
197 | }
198 | }
199 |
200 | /**
201 |  * Вспомогательные функции для использования в компонентах
202 |  */
203 | const RequestManagerHelper = {
204 |     /**
205 |      * Создать дебаунс функцию поиска
206 |      */
207 |     createSearchDebounce(searchFn, delay = 500) {
208 |         const manager = new RequestManager();
209 |         return (query) => manager.debounce('search', () => searchFn(
210 |             query), delay);
211 |     },
212 |     /**
213 |      * Создать дебаунс функцию для автосохранения
214 |      */
```

```
215 |   createAutoSaveDebounce(saveFn, delay = 1000) {
216 |     const manager = new RequestManager();
217 |     return (data) => manager.debounce('autosave', () => saveFn
    | (data), delay);
218 |   },
219 |
220 |   /**
221 |    * Создать throttle функцию для обновления статуса
222 |    */
223 |   createStatusUpdateThrottle(updateFn, period = 2000) {
224 |     const manager = new RequestManager();
225 |     return manager.throttle('status-update', updateFn, period)
    | ;
226 |   }
227 | };
228 |
229 | // Экспортируем глобально
230 | window.RequestManager = RequestManager;
231 | window.RequestManagerHelper = RequestManagerHelper;
232 |
233 | export { RequestManager, RequestManagerHelper };
234 |
```

status-highlighter.js

```
1 /**
2  * =====
3  * STATUS HIGHLIGHTER – AI Breadboard WebGUI
4  * Глобальный менеджер программного выделения статусов (Зелёны
  й = Включен/Активен, Красный = Выключен/Неактивен)
5  * =====
6  */
7
8 (function (window, document) {
9   'use strict';
10
11   // Множества ключевых слов для положительных (зелёных) и отрицательных (красных) состояний
12   const POSITIVE_KEYWORDS = new Set([
13     'active', 'enabled', 'on', 'running', 'online', 'connected',
14     'true', '1', 'ok', 'success',
15     'включен', 'включена', 'включено', 'активен', 'активна', 'активно',
16     'работает', 'запущен', 'запущена', 'онлайн', 'подключен',
17     'подключено', 'успех', 'успешно', 'в норме'
18   ]);
19
20   const NEGATIVE_KEYWORDS = new Set([
21     'inactive', 'disabled', 'off', 'stopped', 'offline', 'disconnected',
22     'false', '0', 'error', 'failed', 'failure',
23     'выключен', 'выключена', 'выключено', 'неактивен', 'неактивна', 'неактивно',
24     'остановлен', 'остановлена', 'офлайн', 'отключен', 'отключено', 'ошибка', 'сбой', 'не запущен'
25   ]);
26
27   /**
28    * Класс системы программной подсветки статусов
29    */
30 }
```

```

27 | class StatusHighlighterEngine {
28 |     constructor() {
29 |         this.observer = null;
30 |         this.initObserver();
31 |     }
32 |
33 |     /**
34 |      * Проверяет, является ли значение положительным состоянием ("включен", "активен")
35 |      * @param {*} value Проверяемое значение или строка
36 |      * @returns {boolean} True, если состояние положительное
37 |      */
38 |     isPositive(value) {
39 |         if (typeof value === 'boolean') return value === true;
40 |         if (typeof value === 'number') return value > 0;
41 |         if (!value) return false;
42 |
43 |         const normalized = String(value).trim().toLowerCase();
44 |         if (POSITIVE_KEYWORDS.has(normalized)) return true;
45 |
46 |         // Проверка подстрок
47 |         for (const kw of POSITIVE_KEYWORDS) {
48 |             if (normalized.includes(kw)) return true;
49 |         }
50 |         return false;
51 |     }
52 |
53 |     /**
54 |      * Проверяет, является ли значение отрицательным состоянием ("выключен", "остановлен")
55 |      * @param {*} value Проверяемое значение или строка
56 |      * @returns {boolean} True, если состояние отрицательное
57 |      */
58 |     isNegative(value) {
59 |         if (typeof value === 'boolean') return value === false;
60 |         if (typeof value === 'number') return value === 0;
61 |         if (!value) return false;
62 |

```



```

63 |     const normalized = String(value).trim().toLowerCase();
64 |     if (NEGATIVE_KEYWORDS.has(normalized)) return true;
65 |
66 |     // Проверка подстрок
67 |     for (const kw of NEGATIVE_KEYWORDS) {
68 |         if (normalized.includes(kw)) return true;
69 |     }
70 |     return false;
71 | }
72 |
73 | /**
74 |  * Возвращает название цветовой темы статуса ('success', '
danger', 'neutral')
75 |  * @param {*} value Состояние
76 |  * @returns {string} Тема статуса
77 |  */
78 | getStatusTheme(value) {
79 |     if (this.isPositive(value)) return 'success';
80 |     if (this.isNegative(value)) return 'danger';
81 |     return 'neutral';
82 | }
83 |
84 | /**
85 |  * Генерирует HTML-строку для бейджа статуса
86 |  * @param {*} value Значение состояния
87 |  * @param {string} [customText] Кастомный текст бейджа
88 |  * @param {string} [extraClasses] Дополнительные CSS класс
89 | ы
90 |  * @returns {string} HTML бейджа
91 |  */
92 | getBadgeHtml(value, customText = '', extraClasses = '') {
93 |     const isPos = this.isPositive(value);
94 |     const isNeg = this.isNegative(value);
95 |
96 |     let themeClass = 'bg-secondary-subtle text-secondary bor
97 |     der';
98 |     if (isPos) {
99 |         themeClass = 'bg-success-subtle text-success border bo

```

```

    rder-success-subtle';
98     } else if (isNeg) {
99         themeClass = 'bg-danger-subtle text-danger border border-
    er-danger-subtle';
100    }
101
102    const displayText = customText || String(value);
103    return `<span class="badge ${themeClass} ${extraClasses}
    ">${displayText}</span>`;
104    }
105
106    /**
107     * Программно обновляет стили элемента в зависимости от ег
    о статуса
108     * @param {HTMLElement} element Элемент для подсветки
109     */
110    applyElementStatus(element) {
111        if (!element || !(element instanceof HTMLElement)) retur
    n;
112
113        const statusVal = element.getAttribute('data-status') ||
114            element.getAttribute('data-state') ||
115            element.getAttribute('data-auto-status
    ');
116        element.textContent;
117
118        if (!statusVal) return;
119
120        const isPos = this.isPositive(statusVal);
121        const isNeg = this.isNegative(statusVal);
122
123        if (isPos) {
124            element.classList.remove('is-status-negative', 'status
    -inactive', 'status-disabled', 'bg-danger', 'text-danger', 'bg
    -danger-subtle', 'bg-secondary-subtle', 'text-secondary');
125            element.classList.add('is-status-positive', 'status-ac
    tive');
126        } else if (isNeg) {

```

```

127 |         element.classList.remove('is-status-positive', 'status
128 | -active', 'status-enabled', 'bg-success', 'text-success', 'bg-
129 | success-subtle', 'bg-secondary-subtle', 'text-secondary');
130 |         element.classList.add('is-status-negative', 'status-in
131 | active');
132 |     }
133 | }
134 |
135 | /**
136 |  * Сканирует DOM-дерево и применяет подсвечивание ко всем
137 |  * элементам статуса
138 |  * @param {HTMLElement|Document} [root] Корневой элемент д
139 |  * ля поиска
140 |  */
141 | scanDOM(root = document) {
142 |     const selectors = '[data-status], [data-state], [data-au
143 | to-status], .badge-status, .status-badge';
144 |     const elements = root.querySelectorAll(selectors);
145 |     elements.forEach(el => this.applyElementStatus(el));
146 | }
147 |
148 | /**
149 |  * Инициализирует MutationObserver для автоматической подс
150 |  * ветки создаваемых/изменяемых элементов
151 |  */
152 | initObserver() {
153 |     if (typeof MutationObserver === 'undefined') return;
154 |
155 |     this.observer = new MutationObserver((mutations) => {
156 |         for (const mutation of mutations) {
157 |             if (mutation.type === 'childList') {
158 |                 mutation.addedNodes.forEach(node => {
159 |                     if (node.nodeType === Node.ELEMENT_NODE) {
160 |                         this.applyElementStatus(node);
161 |                         if (node.querySelectorAll) {
162 |                             this.scanDOM(node);
163 |                         }
164 |                     }
165 |                 });
166 |             }
167 |         }
168 |     });
169 | }

```

```
158         });
159         } else if (mutation.type === 'attributes') {
160             const attr = mutation.attributeName;
161             if (attr === 'data-status' || attr === 'data-state
162 ' || attr === 'data-auto-status') {
163                 this.applyElementStatus(mutation.target);
164             }
165         }
166     });
167
168     if (document.readyState === 'loading') {
169         document.addEventListener('DOMContentLoaded', () => th
170 is.startObserving());
171     } else {
172         this.startObserving();
173     }
174
175     /**
176      * Запускает наблюдение MutationObserver за документом
177      */
178     startObserving() {
179         if (!this.observer || !document.body) return;
180         this.scanDOM(document.body);
181         this.observer.observe(document.body, {
182             childList: true,
183             subtree: true,
184             attributes: true,
185             attributeFilter: ['data-status', 'data-state', 'data-a
186 uto-status']
187         });
188     }
189
190     // Экспорт синглтона в глобальную область видимости
191     window.StatusHighlighter = new StatusHighlighterEngine();
192
```

```
193 | })(window, document);
```

```
194 |
```

tab-core.js

```
1 /**
2  * tab-core.js – Единая система управления вкладками, их жизне
   нным циклом и поллингом.
3  * Архитектура: UI_ARCHITECTURE.md
4  *
5  * Правило: Только открытая АКТИВНАЯ вкладка выполняет периоди
   ческие опросы API.
6  * При неактивности вкладки или скрытии страницы (visibilitych
   ange) все таймеры засыпают.
7  *
8  * Приоритет переключения: кнопки навигации имеют самый высоки
   й приоритет.
9  * Если что-то запущено внутри вкладки, переключение происходи
   т немедленно без ожидания.
10 */
11
12 const OFFCANVAS_IDS = ['leftSideNavOffcanvas', 'appsSideNavOff
   canvas'];
13
14 let currentActiveTabId = null;
15 const registeredPollers = new Map();
16 const processingFlags = new Map(); // хранит флаг isProcessing
   для каждой вкладки
17
18 /**
19  * Приводит идентификатор вкладки к стандартному виду 'tab-xxx
   '.
20  * @param {string} tabId
21  * @returns {string}
22  */
23 export function normalizeTabId(tabId) {
24   if (!tabId) return '';
25   return tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
26 }
27
28 /**
```

```

29  * Проверяет, активна ли сейчас вкладка (видна пользователю и
    вкладка браузера в фокусе).
30  * @param {string} tabId
31  * @returns {boolean}
32  */
33  export function isTabActive(tabId) {
34      const normId = normalizeTabId(tabId);
35      if (typeof document !== 'undefined' && document.hidden) {
36          return false;
37      }
38      if (currentActiveTabId) {
39          return currentActiveTabId === normId;
40      }
41      const pane = document.getElementById(normId);
42      return pane ? pane.classList.contains('active') : false;
43  }
44
45  /**
46   * Останавливает таймер конкретного опросника.
47   * @param {Object} poller
48   */
49  function stopPoller(poller) {
50      if (poller.timerId) {
51          clearInterval(poller.timerId);
52          poller.timerId = null;
53      }
54  }
55
56  /**
57   * Запускает таймер опросника, если вкладка активна и опрос ра
    зрешён.
58   * @param {Object} poller
59   * @param {boolean} immediate - Выполнить ли опрос немедленно
    перед установкой таймера
60   */
61  function startPoller(poller, immediate = false) {
62      stopPoller(poller);
63      if (!poller.enabled || !isTabActive(poller.tabId)) return;

```

```

64
65     const runTick = async () => {
66         if (!poller.enabled || !isTabActive(poller.tabId) || poller.inFlight) return;
67         poller.inFlight = true;
68         try {
69             await poller.pollFn();
70         } catch (err) {
71             console.warn(`[TabPoller] Ошибка выполнения pollFn для [
${poller.id}]:`, err);
72         } finally {
73             poller.inFlight = false;
74         }
75     };
76
77     if (immediate) {
78         runTick();
79     }
80
81     poller.timerId = setInterval(runTick, poller.intervalMs);
82 }
83
84 /**
85  * Регистрирует или обновляет периодический опросник (poller)
86  * для вкладки.
87  * Опросник будет срабатывать ТОЛЬКО тогда, когда вкладка АКТИВНА
88  * и документ находится в фокусе (не свёрнут / не скрыт).
89  *
90  * @param {string} tabId - ID вкладки (например 'tab-about-system' или 'about-system')
91  * @param {Function} pollFn - Функция опроса API
92  * @param {number} intervalMs - Интервал опроса в миллисекундах (мин. 1000мс)
93  * @param {Object} [options] - Опции: { pollerId, immediate = true, enabled = true }
94  * @returns {string} pollerId
95  */

```



```

95 | export function registerTabPoller(tabId, pollFn, intervalMs =
    | 3000, options = {}) {
96 |     const normTabId = normalizeTabId(tabId);
97 |     const pollerId = options.pollerId || `${normTabId}_default`;
98 |
99 |     unregisterTabPoller(pollerId);
100 |
101 |     const poller = {
102 |         id: pollerId,
103 |         tabId: normTabId,
104 |         pollFn,
105 |         intervalMs: Math.max(1000, intervalMs),
106 |         enabled: options.enabled !== false,
107 |         timerId: null,
108 |         inFlight: false,
109 |     };
110 |
111 |     registeredPollers.set(pollerId, poller);
112 |
113 |     if (isTabActive(normTabId) && poller.enabled) {
114 |         startPoller(poller, options.immediate !== false);
115 |     }
116 |
117 |     return pollerId;
118 | }
119 |
120 | /**
121 |  * Удаляет зарегистрированный опросник по его ID.
122 |  * @param {string} pollerId
123 |  */
124 | export function unregisterTabPoller(pollerId) {
125 |     if (registeredPollers.has(pollerId)) {
126 |         const poller = registeredPollers.get(pollerId);
127 |         stopPoller(poller);
128 |         registeredPollers.delete(pollerId);
129 |     }
130 | }
131 |

```

```
132 /**
133  * Удаляет все зарегистрированные опросники для вкладки.
134  * @param {string} tabId
135  */
136 export function unregisterAllTabPollers(tabId) {
137   const normTabId = normalizeTabId(tabId);
138   for (const [id, poller] of registeredPollers.entries()) {
139     if (poller.tabId === normTabId) {
140       stopPoller(poller);
141       registeredPollers.delete(id);
142     }
143   }
144 }
145
146 /**
147  * Включает или выключает опросник вкладки (например, переключ
    атель Live-режима).
148  * @param {string} pollerId
149  * @param {boolean} enabled
150  */
151 export function setTabPollerEnabled(pollerId, enabled) {
152   const poller = registeredPollers.get(pollerId);
153   if (!poller) return;
154   poller.enabled = !!enabled;
155   if (poller.enabled && isTabActive(poller.tabId)) {
156     startPoller(poller, true);
157   } else {
158     stopPoller(poller);
159   }
160 }
161
162 const TAB_ALIASES = {
163   'tab-process-leaks': { parentTab: 'tab-about-system', subtab
    : 'subtab-leaks' },
164   'tab-forensics': { parentTab: 'tab-about-system', subtab: 's
    ubtab-forensics' },
165   'tab-throttling': { parentTab: 'tab-about-system', subtab: '
    subtab-throttling' },
```

```
166 | 'tab-storage-wear': { parentTab: 'tab-about-system', subtab:
    | 'subtab-wear' },
167 | 'tab-peripherals': { parentTab: 'tab-about-system', subtab:
    | 'subtab-peripherals' },
168 | };
169 |
170 | /**
171 |  * Переключает активную вкладку и управляет жизненным циклом (
    | активация/деактивация/поллеры).
172 |  * Приоритет: кнопки навигации имеют самый высокий приоритет.
173 |  * Если что-то запущено внутри вкладки, переключение происходи
    | т немедленно без ожидания.
174 |  * @param {string} tabId
175 |  */
176 | export function switchTab(tabId) {
177 |   if (!tabId) return;
178 |   const rawId = normalizeTabId(tabId);
179 |   const alias = TAB_ALIASES[rawId] || TAB_ALIASES[tabId];
180 |   const id = alias ? alias.parentTab : rawId;
181 |   const prevTabId = currentActiveTabId;
182 |
183 |   // 1. Остановка фонового аудио / синтеза речи при переходе
184 |   if (prevTabId && prevTabId !== id && window.chatService?.sto
    | p) {
185 |     try {
186 |       window.chatService.stop();
187 |     } catch (e) {}
188 |   }
189 |
190 |   // 2. Мгновенная деактивация опросников предыдущей вкладки
191 |   if (prevTabId && prevTabId !== id) {
192 |     for (const poller of registeredPollers.values()) {
193 |       if (poller.tabId === prevTabId) {
194 |         stopPoller(poller);
195 |       }
196 |     }
197 |   }
198 | }
```

```
199 |   currentActiveTabId = id;
200 |
201 |   // 3. МГНОВЕННОЕ обновление UI элементов в DOM (Zero-Delay)
202 |   document.querySelectorAll('[data-tab]').forEach(btn =>
203 |     btn.classList.toggle('active', btn.dataset.tab === rawId |
204 | | btn.dataset.tab === id)
205 |   );
206 |   document.querySelectorAll('#mainTabContent .tab-pane').forEa
207 | ch(pane => {
208 |     const isTarget = (pane.id === id);
209 |     pane.classList.toggle('show', isTarget);
210 |     pane.classList.toggle('active', isTarget);
211 |   });
212 |   // Если это подвкладка, активируем ее внутри родительской ст
213 |   раницы
214 |   if (alias && alias.subtab) {
215 |     setTimeout(() => {
216 |       const subBtn = document.querySelector(`.about-subtab-btn
217 | [data-subtab="${alias.subtab}"]`);
218 |       if (subBtn) subBtn.click();
219 |     }, 20);
220 |   }
221 |   // 4. Бейдж заголовка
222 |   const badge = document.getElementById('active-tab-title-badg
223 | e');
224 |   const activeBtn = document.querySelector(`[data-tab="${rawId
225 | }"]`) || document.querySelector(`[data-tab="${id}"]`);
226 |   if (badge && activeBtn) badge.innerHTML = activeBtn.innerHTM
227 | L;
228 |
229 |   // 5. Быстро закрыть offcanvas меню
230 |   OFFCANVAS_IDS.forEach(ocId => {
231 |     const el = document.getElementById(ocId);
232 |     if (el) bootstrap.Offcanvas.getInstance(el)?.hide();
233 |   });
```

```
230
231 // 6. Обновить URL hash
232 try {
233     history.replaceState(null, null, `#${rawId}`);
234 } catch (e) {}
235
236 // 7. Запуск опросников новой активной вкладки
237 for (const poller of registeredPollers.values()) {
238     if (poller.tabId === id && poller.enabled) {
239         startPoller(poller, true);
240     }
241 }
242
243 // 8. Неблокирующий вызов Lifecycle hooks через requestAnimationFrame / setTimeout
244 const name = id.replace(/^tab-/ , '').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
245 const prevName = prevTabId ? prevTabId.replace(/^tab-/ , '').replace(/-([a-z])/g, (_, c) => c.toUpperCase()) : null;
246
247 // Деактивация предыдущей вкладки в фоне
248 if (prevTabId && prevTabId !== id) {
249     const deactivateFn = window[`deactivate${prevName[0].toUpperCase() + prevName.slice(1)}Tab`];
250     if (typeof deactivateFn === 'function') {
251         setTimeout(() => {
252             try { deactivateFn(); } catch (err) { console.debug('deactivate hook error:', err); }
253             }, 0);
254     }
255     document.dispatchEvent(new CustomEvent('tab:deactivated', { detail: { tabId: prevTabId, name: prevName } }));
256 }
257
258 // Активация новой вкладки в следующем фрейме анимации
259 requestAnimationFrame(() => {
260     const initFn = window[`init${name[0].toUpperCase() + name.slice(1)}Tab`];
```

```

261     if (typeof initFn === 'function') {
262         try { initFn(); } catch (err) { console.debug('init tab
hook error:', err); }
263     }
264
265     const activateFn = window[`activate${name[0].toUpperCase()
+ name.slice(1)}Tab`];
266     if (typeof activateFn === 'function') {
267         try { activateFn(); } catch (err) { console.debug('activ
ate tab hook error:', err); }
268     }
269
270     document.dispatchEvent(new CustomEvent('tab:activated', {
detail: { tabId: id, name } }));
271 });
272 }
273
274 /**
275  * Загружает разметку и скрипт вкладки по требованию (Lazy Loa
d).
276  * @param {string} tabName
277  * @param {string} htmlUrl
278  * @param {string} jsUrl
279  */
280 export async function loadTab(tabName, htmlUrl, jsUrl) {
281     const container = document.getElementById(`tab-${tabName}`);
282     if (!container) return;
283
284     // Показываем мгновенный скелетон/спиннер, если контейнер ещ
е пуст
285     if (!container.innerHTML.trim()) {
286         container.innerHTML = `
287         <div class="d-flex flex-column align-items-center justif
y-content-center py-5 text-muted" style="min-height: 200px;">
288             <div class="spinner-border text-primary mb-2" role="st
atus" style="width: 2rem; height: 2rem;"></div>
289             <span class="small fw-medium">Загрузка раздела...</spa
n>

```

```

290     </div>
291     `;
292 }
293
294 try {
295     const r = await fetch(htmlUrl);
296     if (!r.ok) throw new Error(`HTTP ${r.status}`);
297     container.innerHTML = await r.text();
298     await new Promise(resolve => {
299         const s = document.createElement('script');
300         s.src = jsUrl;
301         s.onload = s.onerror = resolve;
302         document.body.appendChild(s);
303     });
304     const name = tabName.replace(/-([a-z])/g, (_, c) => c.toUp
perCase());
305     window[`init${name[0].toUpperCase() + name.slice(1)}Tab`]?
.();
306 } catch (e) {
307     container.innerHTML = `<div class="alert alert-danger m-3"
>Ошибка загрузки ${tabName}: ${e.message}</div>`;
308 }
309 }
310
311 /**
312  * Устанавливает единый слушатель кликов по элементам с data-t
ab.
313  */
314 export function setupTabClicks() {
315     document.addEventListener('click', e => {
316         const btn = e.target.closest('button[data-tab], a[data-tab
]');
317         if (btn && btn.dataset.tab) {
318             if (typeof window.switchTab === 'function') {
319                 window.switchTab(btn.dataset.tab);
320             } else {
321                 switchTab(btn.dataset.tab);
322             }

```

```

323     }
324   });
325 }
326
327 // Слушатель изменения видимости вкладки браузера
328 if (typeof document !== 'undefined') {
329   document.addEventListener('visibilitychange', () => {
330     if (document.hidden) {
331       // Страница скрыта: останавливаем все активные опросники
332       for (const poller of registeredPollers.values()) {
333         stopPoller(poller);
334       }
335     } else {
336       // Страница снова видна: возобновляем опросники ТОЛЬКО д
ля активной вкладки
337       if (currentActiveTabId) {
338         for (const poller of registeredPollers.values()) {
339           if (poller.tabId === currentActiveTabId && poller.en
abled) {
340             startPoller(poller, true);
341           }
342         }
343       }
344     }
345   });
346 }
347
348 /**
349  * Устанавливает флаг processing для вкладки (используется при
длительных операциях).
350  * Позволяет переключать вкладки без ожидания завершения опера
ции.
351  * @param {string} tabId - ID вкладки
352  * @param {boolean} isProcessing - флаг состояния
353  */
354 export function setTabProcessing(tabId, isProcessing) {
355   const normId = normalizeTabId(tabId);
356   processingFlags.set(normId, !!isProcessing);

```



```
357 | }
358 |
359 | /**
360 |  * Сбрасывает флаг processing для вкладки.
361 |  * @param {string} tabId - ID вкладки
362 |  */
363 | export function resetTabProcessing(tabId) {
364 |     const normId = normalizeTabId(tabId);
365 |     processingFlags.delete(normId);
366 | }
367 |
368 | /**
369 |  * Экспорт в глобальный контекст window для совместимости с ин
    | лайн-скриптами и вкладками
370 |  */
371 | if (typeof window !== 'undefined') {
372 |     window.switchTab = switchTab;
373 |     window.switchToTab = switchTab;
374 |     window.loadTab = loadTab;
375 |     window.setupTabClicks = setupTabClicks;
376 |     window.normalizeTabId = normalizeTabId;
377 |     window.isTabActive = isTabActive;
378 |     window.registerTabPoller = registerTabPoller;
379 |     window.unregisterTabPoller = unregisterTabPoller;
380 |     window.unregisterAllTabPollers = unregisterAllTabPollers;
381 |     window.setTabPollerEnabled = setTabPollerEnabled;
382 |     window.setTabProcessing = setTabProcessing;
383 |     window.resetTabProcessing = resetTabProcessing;
384 | }
385 |
```

tab-debounce-auto-patch.js

```
1  /**
2   * Tab Debounce Auto-Patch - Автоматическое применение дебаунс
   * а к табам
3   *
4   * Этот модуль автоматически улучшает дебаунс в:
5   * - Поиске (все табы с поиском)
6   * - Автосохранении (инструкции, конфиги)
7   * - Real-time обновлениях (статусы, счетчики)
8   */
9
10 import { RequestManager } from './request-manager.js';
11
12 const tabRequestManagers = new Map(); // { tabName: RequestMan
   ager }
13
14 /**
15  * Получить или создать RequestManager для вкладки
16  */
17 function getTabRequestManager(tabName) {
18   if (!tabRequestManagers.has(tabName)) {
19     tabRequestManagers.set(tabName, new RequestManager());
20   }
21   return tabRequestManagers.get(tabName);
22 }
23
24 /**
25  * Авто-патч поиска в любой вкладке
26  * Находит input с id *-search-input и применяет дебаунс
27  */
28 export function autoPatchSearchInputs(tabName) {
29   const manager = getTabRequestManager(tabName);
30
31   // Ищем все input'ы поиска в текущей вкладке
32   const searchInputs = document.querySelectorAll('[id*="search
   -input"]:not([data-debounce-patched])');
33
```

```

34 | searchInputs.forEach(input => {
35 |     if (!input.id) return;
36 |
37 |     const debounceKey = `${tabName}-${input.id}`;
38 |     let originalHandler = null;
39 |
40 |     // Сохраняем оригинальный обработчик если есть
41 |     if (input.oninput) {
42 |         originalHandler = input.oninput;
43 |     }
44 |
45 |     input.addEventListener('input', (e) => {
46 |         const value = e.target.value;
47 |
48 |         manager.debounce(
49 |             debounceKey,
50 |             () => {
51 |                 // Вызываем оригинальный обработчик если был
52 |                 if (originalHandler) {
53 |                     originalHandler.call(input, e);
54 |                 }
55 |                 console.log(`[AutoPatch] Search debounced in ${tabName}: "${value}"`);
56 |             },
57 |             500
58 |         ).catch(error => {
59 |             console.error(`[AutoPatch] Search error in ${tabName}:`, error);
60 |         });
61 |     });
62 |
63 |     input.setAttribute('data-debounce-patched', 'true');
64 |     console.log(`[AutoPatch] Patched search input: ${input.id} in tab: ${tabName}`);
65 | });
66 | }
67 |
68 | /**

```

```
69 * Авто-патч для текстаревей (инструкции, конфиги)
70 * Находит textarea и применяет дебаунс для автосохранения
71 */
72 export function autoPatchAutoSaveTextareas(tabName, saveFn = null) {
73   const manager = getTabRequestManager(tabName);
74
75   const textareas = document.querySelectorAll(`#tab-${tabName}
  textarea[id*="save"]:not([data-autosave-patched])`);
76
77   textareas.forEach(textarea => {
78     if (!textarea.id) return;
79
80     const debounceKey = `${tabName}-autosave-${textarea.id}`;
81
82     textarea.addEventListener('change', (e) => {
83       const content = e.target.value;
84
85       manager.debounce(
86         debounceKey,
87         async () => {
88           if (saveFn) {
89             await saveFn(content);
90           }
91           console.log(`[AutoPatch] Auto-saved content in ${tab
Name}`);
92         },
93         1000
94       ).catch(error => {
95         console.error(`[AutoPatch] Auto-save error in ${tabNam
e}:`, error);
96       });
97     });
98
99     textarea.setAttribute('data-autosave-patched', 'true');
100     console.log(`[AutoPatch] Patched auto-save textarea: ${tex
tarea.id} in tab: ${tabName}`);
101   });
```

```

102 }
103
104 /**
105  * Авто-патч для real-time обновлений
106  * Находит кнопки обновления и применяет throttle
107  */
108 export function autoPatchRefreshButtons(tabName) {
109   const manager = getTabRequestManager(tabName);
110
111   const refreshBtns = document.querySelectorAll(
112     `#tab-${tabName} button[id*="refresh"]:not([data-throttle-
113 patched]),
114     #tab-${tabName} button[id*="refresh-btn"]:not([data-throt
115 tle-patched])`
116   );
117
118   refreshBtns.forEach(btn => {
119     if (!btn.id) return;
120
121     const originalOnClick = btn.onclick;
122     const throttleKey = `${tabName}-refresh-${btn.id}`;
123
124     btn.onclick = manager.throttle(
125       throttleKey,
126       async () => {
127         if (originalOnClick) {
128           await originalOnClick.call(btn);
129         }
130         console.log(`[AutoPatch] Refresh throttled in ${tabNam
131 e}`);
132       },
133       2000
134     );
135
136     btn.setAttribute('data-throttle-patched', 'true');
137     console.log(`[AutoPatch] Patched refresh button: ${btn.id}
138 in tab: ${tabName}`);
139   });

```

```
136 | }
137 |
138 | /**
139 |  * Комплексный патч для всей вкладки
140 |  */
141 | export function autoPatchTab(tabName, options = {}) {
142 |   console.log(`[AutoPatch] Patching tab: ${tabName}`);
143 |
144 |   const { searchEnabled = true, autoSaveEnabled = true, refreshEnabled = true } = options;
145 |
146 |   if (searchEnabled) {
147 |     autoPatchSearchInputs(tabName);
148 |   }
149 |
150 |   if (autoSaveEnabled) {
151 |     autoPatchAutoSaveTextareas(tabName, options.saveFn);
152 |   }
153 |
154 |   if (refreshEnabled) {
155 |     autoPatchRefreshButtons(tabName);
156 |   }
157 |
158 |   console.log(`[AutoPatch] Completed patching tab: ${tabName}`);
159 | }
160 |
161 | /**
162 |  * Получить статистику дебаунса для вкладки
163 |  */
164 | export function getTabDebounceStats(tabName) {
165 |   const manager = tabRequestManagers.get(tabName);
166 |   if (!manager) return null;
167 |   return manager.getStats();
168 | }
169 |
170 | /**
171 |  * Получить статистику дебаунса для всех вкладок
```

```
172 | */
173 | export function getAllDebounceStats() {
174 |   const stats = {};
175 |   tabRequestManagers.forEach((manager, tabName) => {
176 |     stats[tabName] = manager.getStats();
177 |   });
178 |   return stats;
179 | }
180 |
181 | // Экспортируем глобально
182 | window.TabDebounceAutoPatch = {
183 |   autoPatchSearchInputs,
184 |   autoPatchAutoSaveTextareas,
185 |   autoPatchRefreshButtons,
186 |   autoPatchTab,
187 |   getTabDebounceStats,
188 |   getAllDebounceStats
189 | };
190 |
191 |
```

tab-loader.js

```
1 /**
2  * LazyTabLoader - Система ленивой загрузки вкладок
3  *
4  * Управляет загрузкой вкладок по требованию, кешировании в памяти
5  * и минимизирует первоначальную нагрузку на интерфейс.
6  *
7  * Usage:
8  *   const loader = new LazyTabLoader();
9  *   loader.registerTab('chat', '/html/chat/index.html', '/html/chat/main.js');
10  *   await loader.loadTab('chat'); // Загружает вкладку
11  *   loader.isLoaded('chat');      // true
12  */
13
14 class LazyTabLoader {
15   constructor() {
16     this.tabs = new Map(); // Реестр вкладок
17     this.loaded = new Set(); // Какие вкладки уже загружены
18     this.loading = new Map(); // Текущие загрузки (для дедупликации)
19     this.cache = new Map(); // Кеш содержимого вкладок в памяти
20   }
21
22   /**
23    * Регистрирует вкладку для последующей загрузки
24    */
25   registerTab(tabName, htmlUrl, jsUrl = null) {
26     if (!this.tabs.has(tabName)) {
27       this.tabs.set(tabName, { htmlUrl, jsUrl, initialized: false });
28     }
29   }
30
31   /**
```



```
32 | * Загружает вкладку по требованию
33 | */
34 | async loadTab(tabName) {
35 |     if (!this.tabs.has(tabName)) {
36 |         console.warn(`[LazyTabLoader] Tab '${tabName}' not regis
37 | tered`);
38 |         return false;
39 |     }
40 |     // Если уже загружается - дождаться
41 |     if (this.loading.has(tabName)) {
42 |         return this.loading.get(tabName);
43 |     }
44 |
45 |     // Если уже загружена - вернуть
46 |     if (this.loaded.has(tabName)) {
47 |         return true;
48 |     }
49 |
50 |     // Начать загрузку
51 |     const loadPromise = this._performLoad(tabName);
52 |     this.loading.set(tabName, loadPromise);
53 |
54 |     try {
55 |         const result = await loadPromise;
56 |         this.loaded.add(tabName);
57 |         return result;
58 |     } finally {
59 |         this.loading.delete(tabName);
60 |     }
61 | }
62 |
63 | /**
64 |  * Предварительная загрузка вкладки в фоне (не блокирует)
65 |  */
66 | preloadTab(tabName) {
67 |     if (!this.loaded.has(tabName) && !this.loading.has(tabName
68 | )) {
```

```
68         this.loadTab(tabName).catch(e => {
69             console.error(`[LazyTabLoader] Failed to preload ${tab
Name}:`, e);
70         });
71     }
72 }
73
74 /**
75  * Проверяет загружена ли вкладка
76  */
77 isLoaded(tabName) {
78     return this.loaded.has(tabName);
79 }
80
81 /**
82  * Загружает вкладку и выполняет инициализацию
83  */
84 async _performLoad(tabName) {
85     const tab = this.tabs.get(tabName);
86     if (!tab) return false;
87
88     try {
89         // Показываем loader на вкладке
90         this._showLoading(tabName);
91
92         // Загружаем HTML
93         const response = await fetch(tab.htmlUrl);
94         if (!response.ok) throw new Error(`HTTP ${response.statu
s}`);
95         const html = await response.text();
96
97         // Вставляем HTML в контейнер вкладки
98         const container = document.getElementById(`tab-${tabName
}`);
99         if (!container) {
100             throw new Error(`Container tab-${tabName} not found`);
101         }
102     }
```

```
103 |         container.innerHTML = html;
104 |         this.cache.set(tabName, html);
105 |
106 |         // Загружаем и выполняем JS
107 |         if (tab.jsUrl) {
108 |             await this._loadAndExecuteScript(tabName, tab.jsUrl);
109 |         }
110 |
111 |         // Скрываем loader
112 |         this._hideLoading(tabName);
113 |
114 |         // Вызываем init функцию если существует
115 |         const initFuncName = this._getInitFunctionName(tabName);
116 |         if (typeof window[initFuncName] === 'function') {
117 |             console.log(`[LazyTabLoader] Calling ${initFuncName} f
or ${tabName}`);
118 |             try {
119 |                 await window[initFuncName]();
120 |             } catch (err) {
121 |                 console.error(`[LazyTabLoader] Error executing ${ini
tFuncName}:`, err);
122 |             }
123 |         }
124 |
125 |         console.log(`[LazyTabLoader] Successfully loaded tab: ${
tabName}`);
126 |         return true;
127 |     } catch (e) {
128 |         console.error(`[LazyTabLoader] Error loading tab ${tabNa
me}:`, e);
129 |         this._showError(tabName, e.message);
130 |         return false;
131 |     }
132 | }
133 |
134 | /**
135 |  * Загружает и выполняет скрипт вкладки
136 |  */
```

```

137 | _loadAndExecuteScript(tabName, jsUrl) {
138 |     return new Promise((resolve, reject) => {
139 |         const script = document.createElement('script');
140 |         script.src = jsUrl;
141 |         script.type = 'module'; // Используем модули для совреме
    | нного подхода
142 |
143 |         script.onload = () => {
144 |             console.log(`[LazyTabLoader] Loaded JS for ${tabName}`
    | );
145 |             resolve();
146 |         };
147 |
148 |         script.onerror = (err) => {
149 |             console.warn(`[LazyTabLoader] Note: No JS loaded for $
    | {tabName} from ${jsUrl}`);
150 |             resolve(); // Не падаем если нет JS
151 |         };
152 |
153 |         document.body.appendChild(script);
154 |     });
155 | }
156 |
157 | /**
158 |  * Вычисляет имя функции инициализации из имени вкладки
159 |  */
160 | _getInitFunctionName(tabName) {
161 |     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
    | .toUpperCase());
162 |     return 'init' + camelName.charAt(0).toUpperCase() + camelN
    | ame.slice(1) + 'Tab';
163 | }
164 |
165 | /**
166 |  * Показывает индикатор загрузки на вкладке
167 |  */
168 | _showLoading(tabName) {
169 |     const container = document.getElementById(`tab-${tabName}`

```

```

    );
170     if (container) {
171         container.innerHTML = `
172             <div class="d-flex justify-content-center align-items-
center" style="height: 300px;">
173                 <div class="spinner-border text-primary" role="statu
s">
174                     <span class="visually-hidden">Загрузка...</span>
175                 </div>
176             </div>
177         `;
178     }
179 }
180
181 /**
182  * Скрывает индикатор загрузки
183  */
184 _hideLoading(tabName) {
185     // После вставки содержимого loader уже заменен
186 }
187
188 /**
189  * Показывает ошибку загрузки
190  */
191 _showError(tabName, errorMessage) {
192     const container = document.getElementById(`tab-${tabName}`
);
193     if (container) {
194         container.innerHTML = `
195             <div class="alert alert-danger m-3" role="alert">
196                 <strong>Ошибка загрузки:</strong> ${errorMessage}
197                 <br><small class="text-muted">Вкладка: ${tabName}</s
mall>
198             </div>
199         `;
200     }
201 }
202

```

```
203 | /**
204 |  * Очищает кеш и загруженные вкладки
205 |  */
206 | clear() {
207 |     this.loaded.clear();
208 |     this.cache.clear();
209 |     this.loading.clear();
210 | }
211 |
212 | /**
213 |  * Выгружает конкретную вкладку из памяти
214 |  */
215 | unload(tabName) {
216 |     this.loaded.delete(tabName);
217 |     this.cache.delete(tabName);
218 |     const container = document.getElementById(`tab-${tabName}`
219 | );
220 |     if (container) {
221 |         container.innerHTML = '';
222 |     }
223 | }
224 |
225 | /**
226 |  * Вспомогательный метод: загрузить несколько вкладок паралл
227 |  ельно
228 |  */
229 | async loadMultiple(tabNames) {
230 |     return Promise.all(tabNames.map(name => this.loadTab(name)
231 | ));
232 | }
233 |
234 | /**
235 |  * Вспомогательный метод: предзагрузить несколько вкладок в
236 |  фоне
237 |  */
238 | preloadMultiple(tabNames) {
239 |     tabNames.forEach(name => this.preloadTab(name));
240 | }
```

```
237 | }
238 |
239 | // Экспортируем глобально
240 | window.LazyTabLoader = LazyTabLoader;
241 |
242 | export default LazyTabLoader;
243 |
```

tab-state-persistence.js

```
1 /**
2  * Tab State Persistence - Сохранение состояния вкладок в localStorage
3  *
4  * Сохраняет:
5  * - Активную вкладку
6  * - Историю переходов
7  * - Фильтры и поиск на каждой вкладке
8  *
9  * Usage:
10 *   const persistence = new TabStatePersistence('admin');
11 *   persistence.saveActiveTab('tab-chat');
12 *   persistence.getLastActiveTab(); // 'tab-chat'
13 */
14
15 class TabStatePersistence {
16   constructor(namespace = 'admin') {
17     this.namespace = namespace;
18     this.storagePrefix = `${namespace}:tab-state`;
19     this.maxHistorySize = 10;
20   }
21
22   /**
23    * Сохранить активную вкладку
24    */
25   saveActiveTab(tabId) {
26     if (!tabId) return;
27
28     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
29     : tabId;
30
31     const timestamp = Date.now();
32
33     // Сохраняем текущую активную вкладку
34     localStorage.setItem(`${this.storagePrefix}:active`, cleanTabId);
35     localStorage.setItem(`${this.storagePrefix}:active-timesta
```



```

    mp`, timestamp.toString());
34
35    // Добавляем в историю
36    this._addToHistory(cleanTabId, timestamp);
37
38    console.log(`[TabStatePersistence] Saved active tab: ${cleanTabId}`);
39  }
40
41  /**
42   * Получить последнюю активную вкладку
43   */
44  getLastActiveTab(fallback = null) {
45    const lastTab = localStorage.getItem(`${this.storagePrefix}:active`);
46    return lastTab || fallback;
47  }
48
49  /**
50   * Получить последнюю активную вкладку с временем
51   */
52  getLastActiveTabWithTimestamp() {
53    const tabId = localStorage.getItem(`${this.storagePrefix}:active`);
54    const timestamp = localStorage.getItem(`${this.storagePrefix}:active-timestamp`);
55
56    if (!tabId) return null;
57
58    return {
59      tabId,
60      timestamp: timestamp ? parseInt(timestamp) : null,
61      age: timestamp ? Date.now() - parseInt(timestamp) : null
62    };
63  }
64
65  /**
66   * Добавить вкладку в историю

```

```

67 |    */
68 |    _addToHistory(tabId, timestamp) {
69 |        const historyKey = `${this.storagePrefix}:history`;
70 |        let history = [];
71 |
72 |        try {
73 |            const stored = localStorage.getItem(historyKey);
74 |            if (stored) {
75 |                history = JSON.parse(stored);
76 |            }
77 |        } catch (e) {
78 |            console.warn('[TabStatePersistence] Failed to parse history:', e);
79 |            history = [];
80 |        }
81 |
82 |        // Добавляем новую запись в начало
83 |        history.unshift({ tabId, timestamp });
84 |
85 |        // Удаляем дубли
86 |        const seen = new Set();
87 |        history = history.filter(item => {
88 |            if (seen.has(item.tabId)) return false;
89 |            seen.add(item.tabId);
90 |            return true;
91 |        });
92 |
93 |        // Ограничиваем размер истории
94 |        if (history.length > this.maxHistorySize) {
95 |            history = history.slice(0, this.maxHistorySize);
96 |        }
97 |
98 |        // Сохраняем обновленную историю
99 |        localStorage.setItem(historyKey, JSON.stringify(history));
100 |    }
101 |
102 |    /**
103 |    * Получить историю переходов между вкладками

```

```

104     */
105     getHistory() {
106         const historyKey = `${this.storagePrefix}:history`;
107         try {
108             const stored = localStorage.getItem(historyKey);
109             return stored ? JSON.parse(stored) : [];
110         } catch (e) {
111             console.warn('[TabStatePersistence] Failed to parse history: ', e);
112             return [];
113         }
114     }
115
116     /**
117     * Очистить историю
118     */
119     clearHistory() {
120         localStorage.removeItem(`${this.storagePrefix}:history`);
121         console.log('[TabStatePersistence] History cleared');
122     }
123
124     /**
125     * Сохранить состояние вкладки (фильтры, поиск и т.д.)
126     */
127     saveTabState(tabId, state) {
128         if (!tabId || !state) return;
129
130         const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
131         : tabId;
132         const stateKey = `${this.storagePrefix}:state:${cleanTabId}
133         `;
134
135         try {
136             localStorage.setItem(stateKey, JSON.stringify(state));
137             console.log(`[TabStatePersistence] Saved state for tab:
138             ${cleanTabId}`);
139         } catch (e) {
140             console.warn(`[TabStatePersistence] Failed to save state

```

```

    for ${cleanTabId}:\`, e);
138     }
139   }
140
141   /**
142    * Получить состояние вкладки
143    */
144   getTabState(tabId) {
145     if (!tabId) return null;
146
147     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
148     : tabId;
149     const stateKey = `${this.storagePrefix}:state:${cleanTabId}
150     }`;
151
152     try {
153       const stored = localStorage.getItem(stateKey);
154       return stored ? JSON.parse(stored) : null;
155     } catch (e) {
156       console.warn(`[TabStatePersistence] Failed to parse stat
157       e for ${cleanTabId}:\`, e);
158       return null;
159     }
160   }
161
162   /**
163    * Удалить состояние вкладки
164    */
165   deleteTabState(tabId) {
166     if (!tabId) return;
167
168     const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
169     : tabId;
170     const stateKey = `${this.storagePrefix}:state:${cleanTabId}
171     }`;
172
173     localStorage.removeItem(stateKey);
174     console.log(`[TabStatePersistence] Deleted state for tab:

```

```

    ${cleanTabId}`);
170   }
171
172   /**
173    * Очистить все сохраненное состояние
174    */
175   clear() {
176       const keys = Object.keys(localStorage).filter(key =>
177           key.startsWith(this.storagePrefix)
178       );
179
180       keys.forEach(key => localStorage.removeItem(key));
181       console.log(`[TabStatePersistence] Cleared ${keys.length}
entries`);
182   }
183
184   /**
185    * Получить полную информацию о сохраненном состоянии
186    */
187   getStorageInfo() {
188       const keys = Object.keys(localStorage).filter(key =>
189           key.startsWith(this.storagePrefix)
190       );
191
192       const info = {
193           namespace: this.namespace,
194           totalEntries: keys.length,
195           entries: {}
196       };
197
198       keys.forEach(key => {
199           const shortKey = key.replace(`${this.storagePrefix}:`, '
');
200           const value = localStorage.getItem(key);
201
202           try {
203               info.entries[shortKey] = {
204                   size: value.length,

```

```

205         parsed: JSON.parse(value)
206     };
207 } catch {
208     info.entries[shortKey] = {
209         size: value.length,
210         raw: value
211     };
212 }
213 });
214
215     return info;
216 }
217
218 /**
219  * Экспортировать состояние в JSON
220  */
221 exportState() {
222     return JSON.stringify(this.getStorageInfo(), null, 2);
223 }
224
225 /**
226  * Импортировать состояние из JSON
227  */
228 importState(jsonString) {
229     try {
230         const data = JSON.parse(jsonString);
231         // Реализация импорта если нужна
232         console.log('[TabStatePersistence] Import not implemented yet');
233     } catch (e) {
234         console.error('[TabStatePersistence] Failed to import state:', e);
235     }
236 }
237 }
238
239 /**
240  * Глобальный экземпляр для админки

```

```
241 | */
242 | window.tabStatePersistence = new TabStatePersistence('admin');
243 |
244 | // Экспортируем
245 | window.TabStatePersistence = TabStatePersistence;
246 |
247 | export default TabStatePersistence;
248 |
```

tab-switch-optimizer.js

```
1 /**
2  * Tab Switch Optimizer - Оптимизация переключения между вклад
3  *
4  * Реализует:
5  * - DOM кеширование (сохранение содержимого вкладки вместо уд
6  * - Мгновенная визуальная обратная связь (показ активной вкла
7  * - Предварительная загрузка соседних вкладок
8  * - Трекинг времени переключения
9  *
10 * Usage:
11 *   const optimizer = new TabSwitchOptimizer();
12 *   optimizer.enableDOMCaching();
13 *   optimizer.optimizeSwitching();
14 */
15
16 class TabSwitchOptimizer {
17   constructor() {
18     this.domCache = new Map(); // Кеш DOM содержимого вкладок
19     this.switchStats = {
20       totalSwitches: 0,
21       avgSwitchTime: 0,
22       lastSwitchTime: null,
23       switchTimes: []
24     };
25     this.isOptimized = false;
26   }
27
28   /**
29    * Включить DOM кеширование
30    * Вместо удаления содержимого вкладки, сохраняем его в памя
31    *
32    */
33   enableDOMCaching() {
```



```

33 |     console.log('[TabSwitchOptimizer] Enabling DOM caching...'
    | );
34 |
35 |     // Перехватываем стандартное управление табами
36 |     const observer = new MutationObserver((mutations) => {
37 |         mutations.forEach((mutation) => {
38 |             if (mutation.type === 'childList' || mutation.type ===
    | 'attributes') {
39 |                 // Вызываем автосохранение изменений
40 |                 this._cacheVisibleTabs();
41 |             }
42 |         });
43 |     });
44 |
45 |     // Наблюдаем за контейнером с табами
46 |     const tabContainer = document.getElementById('mainTabsCont
    | ent') ||
47 |         document.querySelector('[role="tablist"]')?.parentElement;
48 |
49 |     if (tabContainer) {
50 |         observer.observe(tabContainer, {
51 |             childList: true,
52 |             subtree: true,
53 |             attributes: true,
54 |             attributeFilter: ['class', 'style', 'aria-hidden']
55 |         });
56 |     }
57 |
58 |     this.isOptimized = true;
59 |     console.log('[TabSwitchOptimizer] DOM caching enabled');
60 | }
61 |
62 | /**
63 |  * Кешировать видимые вкладки
64 |  */
65 | _cacheVisibleTabs() {
66 |     const tabs = document.querySelectorAll('[role="tabpanel"]:

```

```

    not([aria-hidden="true"]))');
67     tabs.forEach(tab => {
68         if (tab.id) {
69             // Сохраняем клон вкладки
70             this.domCache.set(tab.id, tab.cloneNode(true));
71         }
72     });
73 }
74
75 /**
76  * Получить кешированное содержимое вкладки
77  */
78 getCachedTabContent(tabId) {
79     return this.domCache.get(tabId);
80 }
81
82 /**
83  * Очистить кеш DOM
84  */
85 clearDOMCache(tabId = null) {
86     if (tabId) {
87         this.domCache.delete(tabId);
88     } else {
89         this.domCache.clear();
90     }
91 }
92
93 /**
94  * Оптимизировать функцию переключения вкладок
95  * Добавляет мгновенную визуальную обратную связь и трекинг
    времени
96  */
97 optimizeSwitching(originalSwitchFn) {
98     return async (targetTabId) => {
99         const startTime = performance.now();
100
101         if (!targetTabId) return;
102

```

```
103 |         // Мгновенно показываем активную вкладку визуально (без
    | загрузки)
104 |         this._showTabImmediately(targetTabId);
105 |
106 |         // Выполняем оригинальное переключение
107 |         if (typeof originalSwitchFn === 'function') {
108 |             await originalSwitchFn(targetTabId);
109 |         }
110 |
111 |         // Записываем время переключения
112 |         const switchTime = performance.now() - startTime;
113 |         this._recordSwitchTime(switchTime, targetTabId);
114 |
115 |         console.log(`[TabSwitchOptimizer] Tab switched in ${switchTime.toFixed(2)}ms`);
116 |     };
117 | }
118 |
119 | /**
120 |  * Мгновенно показать вкладку визуально (до загрузки содержи
    | мого)
121 |  */
122 | _showTabImmediately(targetTabId) {
123 |     const cleanTabId = targetTabId.startsWith('#') ? targetTab
    | Id.slice(1) : targetTabId;
124 |
125 |     // Скрываем все остальные вкладки
126 |     document.querySelectorAll('[role="tabpanel"]').forEach(tab
    | => {
127 |         tab.style.display = 'none';
128 |         tab.classList.remove('active', 'show');
129 |         tab.setAttribute('aria-hidden', 'true');
130 |     });
131 |
132 |     // Показываем целевую вкладку
133 |     const targetTab = document.getElementById(cleanTabId);
134 |     if (targetTab) {
135 |         targetTab.style.display = 'block';
```

```

136     targetTab.classList.add('active', 'show');
137     targetTab.setAttribute('aria-hidden', 'false');
138 }
139
140 // Обновляем визуальное состояние кнопок вкладок
141 document.querySelectorAll('[role="tab"]').forEach(tab => {
142     tab.classList.remove('active');
143     tab.setAttribute('aria-selected', 'false');
144 });
145
146 const targetButton = document.querySelector(`[aria-controls="${cleanTabId}"]`) ||
147     document.querySelector(`[data-bs-target="#${cleanTabId}"]`);
148 if (targetButton) {
149     targetButton.classList.add('active');
150     targetButton.setAttribute('aria-selected', 'true');
151 }
152 }
153
154 /**
155  * Записать время переключения и вычислить среднее
156  */
157 _recordSwitchTime(time, tabId) {
158     this.switchStats.totalSwitches++;
159     this.switchStats.lastSwitchTime = time;
160     this.switchStats.switchTimes.push({ time, tabId, timestamp
161 : Date.now() });
162
163     // Ограничиваем историю последних 100 переключениями
164     if (this.switchStats.switchTimes.length > 100) {
165         this.switchStats.switchTimes = this.switchStats.switchTimes.slice(-100);
166     }
167
168     // Вычисляем среднее время
169     const sum = this.switchStats.switchTimes.reduce((acc, s) =
170 > acc + s.time, 0);

```

```

169     this.switchStats.avgSwitchTime = sum / this.switchStats.sw
    itchTimes.length;
170 }
171
172 /**
173  * Предварительная загрузка соседних вкладок
174  */
175 preloadAdjacentTabs(currentTabId) {
176     const allTabs = Array.from(document.querySelectorAll('[rol
    e="tab"]'));
177     const currentIndex = allTabs.findIndex(tab =>
178         tab.getAttribute('aria-controls') === currentTabId ||
179         tab.getAttribute('data-bs-target') === `#${currentTabId}
    `
180     );
181
182     if (currentIndex >= 0) {
183         // Предзагружаем соседние вкладки
184         const adjacentIndices = [currentIndex - 1, currentIndex
    + 1];
185         adjacentIndices.forEach(idx => {
186             if (idx >= 0 && idx < allTabs.length) {
187                 const tab = allTabs[idx];
188                 const tabId = tab.getAttribute('aria-controls') ||
189                     tab.getAttribute('data-bs-target')?.rep
    lace('#', '');
190
191                 if (tabId && window.optimizationModule?.tabLoader) {
192                     console.log(`[TabSwitchOptimizer] Preloading adjac
    ent tab: ${tabId}`);
193                     window.optimizationModule.tabLoader.preloadTab(tab
    Id);
194                 }
195             }
196         });
197     }
198 }
199

```

```

200  /**
201   * Получить статистику переключения
202   */
203  getStats() {
204      return {
205          ...this.switchStats,
206          avgSwitchTimeFormatted: `${this.switchStats.avgSwitchTime
207      e.toFixed(2)}ms`,
208          lastSwitchTimeFormatted: this.switchStats.lastSwitchTime
209      ?
210          `${this.switchStats.lastSwitchTime.toFixed(2)}ms` : 'N
211      /A',
212          cachedTabs: this.domCache.size,
213          isOptimized: this.isOptimized
214      };
215  }
216
217  /**
218   * Вывести статистику в консоль
219   */
220  printStats() {
221      const stats = this.getStats();
222      console.group('▯ Tab Switch Performance Stats');
223      console.log('Total Switches:', stats.totalSwitches);
224      console.log('Average Switch Time:', stats.avgSwitchTimeFor
225      matted);
226      console.log('Last Switch Time:', stats.lastSwitchTimeForma
227      tted);
228      console.log('Cached Tabs:', stats.cachedTabs);
229      console.log('Performance Target: <100ms (currently',
230          (stats.avgSwitchTime < 100 ? '▯ GOOD' : '▯ SLOW'), '));
231      console.groupEnd();
232  }
233
234  /**
235   * Очистить статистику
236   */
237  resetStats() {

```

```
233 |     this.switchStats = {
234 |         totalSwitches: 0,
235 |         avgSwitchTime: 0,
236 |         lastSwitchTime: null,
237 |         switchTimes: []
238 |     };
239 | }
240 | }
241 |
242 | // Глобальный экземпляр
243 | window.tabSwitchOptimizer = new TabSwitchOptimizer();
244 |
245 | // Экспортируем
246 | window.TabSwitchOptimizer = TabSwitchOptimizer;
247 |
248 | export default TabSwitchOptimizer;
249 |
```

test-api-cache.js

```
1 /**
2  * test-api-cache.js - Простой тестовый скрипт для проверки AP
3  * I Cache System
4  *
5  * * ИСПОЛЬЗОВАНИЕ:
6  * 1. Откройте веб-интерфейс
7  * 2. Откройте консоль браузера (F12)
8  * 3. Выполните: testApiCache()
9  */
10 export async function testApiCache() {
11   console.log('=== API Cache System Test Suite ===\n');
12
13   const results = {
14     passed: 0,
15     failed: 0,
16     tests: []
17   };
18
19   function logTest(name, passed, message) {
20     const icon = passed ? '✅' : '❌';
21     console.log(`${icon} ${name}: ${message}`);
22     results.tests.push({ name, passed, message });
23     if (passed) results.passed++;
24     else results.failed++;
25   }
26
27   // Test 1: Проверка доступности browserCache
28   try {
29     const cacheAvailable = window.browserCache && typeof window.browserCache.ready === 'function';
30     logTest('Cache Availability', cacheAvailable, cacheAvailable ? 'BrowserCache is available' : 'BrowserCache NOT available');
31   } catch (e) {
32     logTest('Cache Availability', false, `Error: ${e.message}`)
```



```

    );
33 }
34
35 // Test 2: Проверка доступности cachedApiFetch
36 try {
37     const apiFetchAvailable = typeof window.cachedApiFetch ===
    'function';
38     logTest('API Fetch Function', apiFetchAvailable, apiFetchA
    vailable ? 'cachedApiFetch() is available' : 'cachedApiFetch()
    NOT available');
39 } catch (e) {
40     logTest('API Fetch Function', false, `Error: ${e.message}`
    );
41 }
42
43 // Test 3: Проверка готовности IndexedDB
44 try {
45     if (window.browserCache) {
46         const isReady = await window.browserCache.ready();
47         logTest('IndexedDB Ready', isReady, isReady ? 'IndexedDB
    is ready' : 'IndexedDB initialization failed');
48     } else {
49         logTest('IndexedDB Ready', false, 'BrowserCache not avai
    lable');
50     }
51 } catch (e) {
52     logTest('IndexedDB Ready', false, `Error: ${e.message}`);
53 }
54
55 // Test 4: Тест Cache-First стратегии
56 try {
57     console.log('\n--- Testing Cache-First Strategy ---');
58     const testUrl = '/api/v1/models';
59
60     // Первый запрос (должен быть MISS)
61     console.log('Request 1: Should be MISS (network)');
62     const start1 = performance.now();
63     await window.cachedApiFetch(testUrl);

```

```

64     const time1 = (performance.now() - start1).toFixed(2);
65
66     // Второй запрос (должен быть HIT)
67     console.log('Request 2: Should be HIT (cache)');
68     const start2 = performance.now();
69     await window.cachedApiFetch(testUrl);
70     const time2 = (performance.now() - start2).toFixed(2);
71
72     const speedup = (time1 / time2).toFixed(1);
73     logTest('Cache-First Speed', time2 < time1, `Network: ${time1}ms, Cache: ${time2}ms (${speedup}x faster)`);
74 } catch (e) {
75     logTest('Cache-First Speed', false, `Error: ${e.message}`)
76 };
77
78 // Test 5: Тест инвалидации по тегу
79 try {
80     console.log('\n--- Testing Tag Invalidation ---');
81
82     // Добавляем тестовую запись
83     if (window.browserCache) {
84         await window.browserCache.set('api_cache', 'test-key-1',
85         { data: 'test1' }, { tags: ['test-tag'] });
86         await window.browserCache.set('api_cache', 'test-key-2',
87         { data: 'test2' }, { tags: ['test-tag'] });
88
89         // Инвалидируем
90         const count = await window.invalidateCacheByTag('test-tag');
91         logTest('Tag Invalidation', count === 2, `Invalidated ${count} entries (expected 2)`);
92     } else {
93         logTest('Tag Invalidation', false, 'BrowserCache not available');
94     }
95 } catch (e) {
96     logTest('Tag Invalidation', false, `Error: ${e.message}`);

```

```

95 | }
96 |
97 | // Test 6: Получение статистики
98 | try {
99 |     console.log('\n--- Testing Statistics ---');
100 |     const stats = await window.getApiCacheStats();
101 |     const hasMetrics = stats.available && stats.metrics && typ
eof stats.metrics.hits === 'number';
102 |     logTest('Cache Statistics', hasMetrics, `Hit rate: ${stats
.apiCache?.hitRate || 0}%, Entries: ${stats.apiCache?.count ||
0}`);
103 | } catch (e) {
104 |     logTest('Cache Statistics', false, `Error: ${e.message}`);
105 | }
106 |
107 | // Test 7: Тест forceNetwork
108 | try {
109 |     console.log('\n--- Testing Force Network ---');
110 |     const testUrl = '/api/v1/system/summary';
111 |
112 |     // Обычный запрос
113 |     await window.cachedApiFetch(testUrl);
114 |
115 |     // Принудительный запрос из сети
116 |     const start = performance.now();
117 |     await window.cachedApiFetch(testUrl, {}, { forceNetwork: t
true });
118 |     const time = (performance.now() - start).toFixed(2);
119 |
120 |     logTest('Force Network', time > 10, `Force network request
took ${time}ms (should be > 10ms)`);
121 | } catch (e) {
122 |     logTest('Force Network', false, `Error: ${e.message}`);
123 | }
124 |
125 | // Итоги
126 | console.log('\n=== □ Test Results ===');
127 | console.log(`Total: ${results.tests.length} tests`);

```

```
128 | console.log(`  Passed: ${results.passed}`);
129 | console.log(`  Failed: ${results.failed}`);
130 | console.log(`Success rate: $${((results.passed / results.test
    | s.length) * 100).toFixed(1)}%`);
131 |
132 | if (results.failed === 0) {
133 |   console.log(`\n  All tests passed! API Cache System is wo
    | rking correctly.`);
134 | } else {
135 |   console.warn(`\n  Some tests failed. Check the logs above
    | for details.`);
136 | }
137 |
138 | return results;
139 | }
140 |
141 | // Экспорт в глобальную область
142 | if (typeof window !== 'undefined') {
143 |   window.testApiCache = testApiCache;
144 | }
145 |
146 | // Автоматический запуск при включенном DEBUG режиме
147 | if (typeof window !== 'undefined' && localStorage.getItem('DEB
    | UG_API_CACHE') === 'true') {
148 |   console.log(`  DEBUG_API_CACHE enabled, running tests on lo
    | ad...`);
149 |   setTimeout(() => testApiCache(), 2000);
150 | }
151 |
152 | export default testApiCache;
153 |
```

theme.js

```
1 /**
2  * theme.js – Theme management module (Light / Dark / System)
3  *
4  * Supports 3 modes:
5  *   - 'light': Forces light theme
6  *   - 'dark': Forces dark theme
7  *   - 'system': Automatically follows OS / browser color scheme preferences
8  *
9  * Persists selected mode to localStorage under key 'theme'.
10 */
11
12 const STORAGE_KEY = 'theme';
13 const DEFAULT_MODE = 'system';
14
15 let mediaQueryListenerAttached = false;
16
17 /**
18  * Get the user's stored theme preference or default to 'system'.
19  * @returns {'light' | 'dark' | 'system'}
20  */
21 export function getThemeMode() {
22   const saved = localStorage.getItem(STORAGE_KEY);
23   if (saved === 'light' || saved === 'dark' || saved === 'system') {
24     return saved;
25   }
26   return DEFAULT_MODE;
27 }
28
29 /**
30  * Resolve the effective theme ('light' or 'dark') based on the mode.
31  * @param {'light' | 'dark' | 'system'} mode
32  * @returns {'light' | 'dark'}
```

```

33 | */
34 | export function getResolvedTheme(mode = getThemeMode()) {
35 |   if (mode === 'system') {
36 |     return window.matchMedia && window.matchMedia('(prefers-co
    lor-scheme: dark)').matches
37 |       ? 'dark'
38 |       : 'light';
39 |   }
40 |   return mode;
41 | }
42 |
43 | /**
44 |  * Set the theme mode, update DOM attributes, UI icons, checkm
    arks, and persist.
45 |  * @param {'light' | 'dark' | 'system'} mode
46 |  */
47 | export function setTheme(mode) {
48 |   if (mode !== 'light' && mode !== 'dark' && mode !== 'system'
    ) {
49 |     mode = DEFAULT_MODE;
50 |   }
51 |
52 |   localStorage.setItem(STORAGE_KEY, mode);
53 |   applyThemeToDOM(mode);
54 | }
55 |
56 | /**
57 |  * Apply the given theme mode to documentElement and sync UI c
    ontrols.
58 |  * @param {'light' | 'dark' | 'system'} mode
59 |  */
60 | function applyThemeToDOM(mode) {
61 |   const resolved = getResolvedTheme(mode);
62 |
63 |   // Set data-theme, data-bs-theme (Bootstrap 5.3+), and data-
    theme-mode
64 |   document.documentElement.setAttribute('data-theme', resolved
    );

```

```
65 | document.documentElement.setAttribute('data-bs-theme', resolved);
66 | document.documentElement.setAttribute('data-theme-mode', mode);
67 |
68 | // Update theme button icon and title
69 | updateThemeIcon(mode, resolved);
70 |
71 | // Update dropdown checkmarks & active states
72 | updateThemeDropdownUI(mode);
73 |
74 | // Sync any theme select elements (e.g., in user settings)
75 | document.querySelectorAll('.theme-selector, #settings-theme')
76 |   .forEach((sel) => {
77 |     if (sel.value !== mode) {
78 |       sel.value = mode;
79 |     }
80 |   });
81 | // Dispatch custom event for tabs/components that need to react
82 | window.dispatchEvent(new CustomEvent('themechanged', {
83 |   detail: { mode, resolved }
84 | }));
85 |
86 |
87 | /**
88 |  * Update the theme toggle / dropdown icon.
89 |  * @param {'light' | 'dark' | 'system'} mode
90 |  * @param {'light' | 'dark'} resolved
91 |  */
92 | function updateThemeIcon(mode, resolved) {
93 |   const iconEl = document.getElementById('theme-icon');
94 |   const toggleBtn = document.getElementById('theme-toggle');
95 |
96 |   let iconClass = 'bi-circle-half text-info';
97 |   let titleText = 'Системная тема';
98 |
```

```
99 | if (mode === 'light') {
100 |     iconClass = 'bi-sun-fill text-warning';
101 |     titleText = 'Светлая тема';
102 | } else if (mode === 'dark') {
103 |     iconClass = 'bi-moon-stars-fill text-primary';
104 |     titleText = 'Темная тема';
105 | } else {
106 |     // system mode
107 |     iconClass = 'bi-circle-half text-info';
108 |     titleText = `Системная тема (${resolved === 'dark' ? 'тёмн
    ая' : 'светлая'})`;
109 | }
110 |
111 | if (iconEl) {
112 |     iconEl.className = `bi ${iconClass}`;
113 | }
114 |
115 | const dropdownBtn = document.getElementById('theme-dropdown'
    );
116 | if (dropdownBtn) {
117 |     dropdownBtn.setAttribute('title', titleText);
118 | }
119 |
120 | // Legacy / fallback toggle button
121 | if (toggleBtn) {
122 |     toggleBtn.innerHTML = `
```



```

    ) => {
134     const target = icon.getAttribute('data-theme-check');
135     if (target === activeMode) {
136         icon.classList.remove('d-none');
137     } else {
138         icon.classList.add('d-none');
139     }
140 });
141
142 // Update active state on dropdown items
143 document.querySelectorAll('[data-theme-value]').forEach((item) => {
144     const val = item.getAttribute('data-theme-value');
145     if (val === activeMode) {
146         item.classList.add('active');
147     } else {
148         item.classList.remove('active');
149     }
150 });
151 }
152
153 /**
154  * Initialize theme listeners and apply the current theme on page load.
155  */
156 export function initTheme() {
157     const currentMode = getThemeMode();
158     applyThemeToDOM(currentMode);
159
160     // Attach OS color-scheme change listener once
161     if (!mediaQueryListenerAttached && window.matchMedia) {
162         try {
163             const mediaQuery = window.matchMedia('(prefers-color-scheme: dark)');
164             const handler = () => {
165                 if (getThemeMode() === 'system') {
166                     applyThemeToDOM('system');
167                 }

```

```

168     };
169     if (mediaQuery.addEventListener) {
170         mediaQuery.addEventListener('change', handler);
171     } else if (mediaQuery.addListener) {
172         mediaQuery.addListener(handler);
173     }
174     mediaQueryListenerAttached = true;
175 } catch (e) {
176     console.warn('[theme] Failed to attach media query listener:', e);
177 }
178 }
179
180 // Bind click listeners for theme dropdown items
181 document.querySelectorAll('[data-theme-value]').forEach((btn) => {
182     btn.addEventListener('click', (e) => {
183         e.preventDefault();
184         const selected = btn.getAttribute('data-theme-value');
185         if (selected) {
186             setTheme(selected);
187         }
188     });
189 });
190
191 // Bind legacy / simple toggle button if present without dropdown
192 const themeToggle = document.getElementById('theme-toggle');
193 if (themeToggle && !themeToggle.hasAttribute('data-bs-toggle')) {
194     themeToggle.addEventListener('click', () => {
195         const mode = getThemeMode();
196         // Cycle: system -> light -> dark -> system
197         const cycleMap = { system: 'light', light: 'dark', dark: 'system' };
198         const nextMode = cycleMap[mode] || 'system';
199         setTheme(nextMode);
200     });

```

```
201 | }
202 |
203 | // Bind theme selector dropdowns if present
204 | document.querySelectorAll('.theme-selector, #settings-theme'
    | ).forEach((sel) => {
205 |     sel.value = currentMode;
206 |     sel.addEventListener('change', (e) => {
207 |         setTheme(e.target.value);
208 |     });
209 | });
210 | }
211 |
212 | // Make available globally on window
213 | window.setTheme = setTheme;
214 | window.getThemeMode = getThemeMode;
215 | window.getResolvedTheme = getResolvedTheme;
216 | window.initTheme = initTheme;
217 |
```

toast.js

```
1 /**
2  * toast.js – Универсальная система всплывающих окон (Toast &
3  Floating Notifications)
4  * для веб-интерфейса AI-Breadboard.
5  *
6  * Предоставляет функции:
7  * - window.showToast(message, type, options)
8  * - window.showNotification(title, message, type, duration)
9  * - Перехват стандартного window.alert() с выводом всплывающи
10 х окон.
11 */
12
13 (function () {
14   'use strict';
15
16   // Конфигурация по умолчанию
17   const DEFAULT_DURATION = 4500;
18   const CONTAINER_ID = 'floating-toast-container';
19
20   /**
21    * Инициализирует контейнер для всплывающих окон.
22    * @returns {HTMLElement} Элемент контейнера.
23    */
24   function getOrCreateContainer() {
25     let container = document.getElementById(CONTAINER_ID);
26     if (!container) {
27       container = document.createElement('div');
28       container.id = CONTAINER_ID;
29       container.className = 'floating-toast-container';
30       container.setAttribute('aria-live', 'polite');
31       container.setAttribute('aria-atomic', 'true');
32       document.body.appendChild(container);
33     }
34     return container;
35   }
36 }
```

```

35 |  /**
36 |   * Определяет иконку и заголовок по типу уведомления.
37 |   * @param {string} type - Тип ('success', 'danger', 'error',
   |   'warning', 'info').
38 |   * @returns {{iconClass: string, headerText: string, bootstr
   |   apType: string}}
39 |   */
40 |   function getMetaForType(type) {
41 |       const t = (type || 'info').toLowerCase();
42 |       switch (t) {
43 |           case 'success':
44 |           case 'ok':
45 |               return {
46 |                   iconClass: 'bi bi-check-circle-fill text-success',
47 |                   headerText: 'Успешно',
48 |                   bootstrapType: 'success'
49 |               };
50 |           case 'danger':
51 |           case 'error':
52 |           case 'err':
53 |               return {
54 |                   iconClass: 'bi bi-exclamation-octagon-fill text-dang
   |   er',
55 |                   headerText: 'Ошибка',
56 |                   bootstrapType: 'danger'
57 |               };
58 |           case 'warning':
59 |           case 'warn':
60 |               return {
61 |                   iconClass: 'bi bi-exclamation-triangle-fill text-war
   |   ning',
62 |                   headerText: 'Внимание',
63 |                   bootstrapType: 'warning'
64 |               };
65 |           case 'info':
66 |           default:
67 |               return {
68 |                   iconClass: 'bi bi-info-circle-fill text-info',

```

```

69 |         headerText: 'Уведомление',
70 |         bootstrapType: 'info'
71 |     };
72 | }
73 | }
74 |
75 | /**
76 |  * Автоматически определяет тип уведомления по содержимому с
77 |  * ообщения.
78 |  * @param {string} text - Текст сообщения.
79 |  * @returns {string} Определенный тип ('success', 'danger',
80 |  * 'warning', 'info').
81 |  */
82 | function detectTypeFromText(text) {
83 |     if (!text || typeof text !== 'string') return 'info';
84 |     const lower = text.toLowerCase();
85 |     if (text.includes('□ ') || lower.includes('успешно') || low
86 |     er.includes('success') || lower.includes('готов')) {
87 |         return 'success';
88 |     }
89 |     if (text.includes('□ ') || lower.includes('ошибка') || lowe
90 |     r.includes('error') || lower.includes('failed') || lower.inclu
91 |     des('не удалось')) {
92 |         return 'danger';
93 |     }
94 |     if (text.includes('□ ') || lower.includes('внимание') || lo
95 |     wer.includes('warning') || lower.includes('заполните') || lowe
96 |     r.includes('пожалуйста, выберите') || lower.includes('пожалуйс
97 |     та выберите') || lower.includes('укажите')) {
98 |         return 'warning';
99 |     }
100 |     return 'info';
101 | }
102 |
103 | /**
104 |  * Отображает всплывающее окно (Toast).
105 |  * @param {string} message - Текст сообщения или HTML.
106 |  * @param {string} [type='info'] - Тип сообщения ('success',

```

```

    'danger', 'error', 'warning', 'info').
99     * @param {Object} [options={}] - Дополнительные параметры (
duration, title, isHtml, autoClose).
100     * @returns {HTMLElement} DOM-элемент тоста.
101     */
102     function showToast(message, type, options = {}) {
103         if (message === undefined || message === null) return null
104         ;
105         // Если тип не указан, пытаемся определить его автоматичес
ки по тексту
106         const resolvedType = type || detectTypeFromText(String(mes
sage));
107         const meta = getMetaForType(resolvedType);
108         const duration = options.duration !== undefined ? options.
duration : DEFAULT_DURATION;
109         const title = options.title || meta.headerText;
110
111         const container = getOrCreateContainer();
112
113         const toastEl = document.createElement('div');
114         toastEl.className = `floating-toast floating-toast-${meta.
bootstrapType}`;
115         toastEl.setAttribute('role', 'alert');
116
117         // Форматирование текста (сохранение переводов строк)
118         const formattedMessage = options.isHtml
119             ? message
120             : String(message)
121                 .replace(/&/g, '&')
122                 .replace(/</g, '<')
123                 .replace(/>/g, '>')
124                 .replace(/\n/g, '<br>');
125
126         toastEl.innerHTML = `
127             <div class="floating-toast-header">
128                 <i class="${meta.iconClass} floating-toast-icon"></i>
129                 <strong class="floating-toast-title">${title}</strong>

```

```

130 |         <button type="button" class="floating-toast-close" ari
| a-label="Заккрыть">&times;</button>
131 |     </div>
132 |     <div class="floating-toast-body">${formattedMessage}</di
| v>
133 |     ${duration > 0 ? `<div class="floating-toast-progress"><
| div class="floating-toast-progress-bar bg-${meta.bootstrapType
| }"></div></div>` : ''}
134 |     `;
135 |
136 |     container.appendChild(toastEl);
137 |
138 |     // Анимация появления
139 |     requestAnimationFrame(() => {
140 |         toastEl.classList.add('show');
141 |     });
142 |
143 |     let dismissTimeout = null;
144 |     let startTime = Date.now();
145 |     let remainingTime = duration;
146 |     const progressBar = toastEl.querySelector('.floating-toast
| -progress-bar');
147 |
148 |     const startTimer = () => {
149 |         if (duration > 0) {
150 |             startTime = Date.now();
151 |             if (progressBar) {
152 |                 progressBar.style.transition = `width ${remainingTim
| e}ms linear`;
153 |                 progressBar.style.width = '0%';
154 |             }
155 |             dismissTimeout = setTimeout(() => {
156 |                 closeToast();
157 |             }, remainingTime);
158 |         }
159 |     };
160 |
161 |     const pauseTimer = () => {

```



```
162 |     if (dismissTimeout) {
163 |         clearTimeout(dismissTimeout);
164 |         dismissTimeout = null;
165 |         remainingTime -= Date.now() - startTime;
166 |         if (progressBar) {
167 |             const computedWidth = window.getComputedStyle(progre
168 | ssBar).width;
169 |             progressBar.style.transition = 'none';
170 |             progressBar.style.width = computedWidth;
171 |         }
172 |     };
173 |
174 |     const closeToast = () => {
175 |         if (dismissTimeout) clearTimeout(dismissTimeout);
176 |         toastEl.classList.remove('show');
177 |         toastEl.classList.add('hide');
178 |         toastEl.addEventListener('transitionend', () => {
179 |             if (toastEl.parentElement) {
180 |                 toastEl.parentElement.removeChild(toastEl);
181 |             }
182 |         }, { once: true });
183 |     };
184 |
185 |     // Слушатели событий
186 |     const closeBtn = toastEl.querySelector('.floating-toast-cl
187 | ose');
188 |     if (closeBtn) {
189 |         closeBtn.addEventListener('click', (e) => {
190 |             e.stopPropagation();
191 |             closeToast();
192 |         });
193 |     }
194 |
195 |     if (duration > 0) {
196 |         toastEl.addEventListener('mouseenter', pauseTimer);
197 |         toastEl.addEventListener('mouseleave', () => {
198 |             if (remainingTime > 0) startTimer();
199 |         });
200 |     }
```

```
198     });
199     startTimer();
200 }
201
202 return toastEl;
203 }
204
205 /**
206  * Отображает уведомление с заголовком и текстом.
207  * @param {string} title - Заголовок.
208  * @param {string} message - Текст сообщения.
209  * @param {string} [type='info'] - Тип.
210  * @param {number} [duration=4500] - Длительность показа.
211  */
212 function showNotification(title, message, type = 'info', duration = DEFAULT_DURATION) {
213     return showToast(message, type, { title, duration });
214 }
215
216 // Экспорт в глобальную область видимости
217 window.showToast = showToast;
218 window.showNotification = showNotification;
219 window._showToast = showToast; // Для совместимости с уже имеющимися вкладками
220
221 // Перехват стандартного window.alert()
222 // Сохраняем ссылку на оригинальный alert на случай необходимости
223 window._originalAlert = window.alert;
224
225 window.alert = function (message) {
226     // Выводим неблокирующее всплывающее окно
227     showToast(message);
228     // Также дублируем в консоль для отладки
229     if (window.console && console.debug) {
230         console.debug('[Toast Alert]:', message);
231     }
232 };
```

```
233 |
234 | // Инициализация контейнера при загрузке документа
235 | if (document.readyState === 'loading') {
236 |     document.addEventListener('DOMContentLoaded', getOrCreateC
ontainer);
237 | } else {
238 |     getOrCreateContainer();
239 | }
240 |})();
241 |
```

userSettings.js

```
1 /**
2  * userSettings.js – User Profile & Settings Management
3  * Supports Google OAuth synchronization, profile status, language & theme controls.
4  */
5
6 'use strict';
7
8 export function executeUserAction(actionType) {
9     // Ensure chat tab is active
10     if (typeof window.switchToTab === 'function') {
11         window.switchToTab('chat');
12     } else if (typeof window.switchTab === 'function') {
13         window.switchTab('tab-chat');
14     } else {
15         const chatTabBtn = document.querySelector('[data-bs-target="#tab-chat"], [data-tab="tab-chat"]');
16         if (chatTabBtn && window.bootstrap?.Tab) {
17             bootstrap.Tab.getOrCreateInstance(chatTabBtn).show();
18         }
19     }
20
21     const msgInput = document.getElementById('message-input');
22     const sendBtn = document.getElementById('send-button');
23
24     const send = () => {
25         if (typeof window.sendChatMessage === 'function') {
26             window.sendChatMessage();
27         } else if (sendBtn) {
28             sendBtn.click();
29         }
30     };
31
32     if (actionType === 'mail') {
33         if (msgInput) {
34             msgInput.value = 'Проверь мою почту Gmail и покажи после
```

```

    дние неп прочитанные письма.';
35     setTimeout(send, 100);
36   }
37   } else if (actionType === 'docs') {
38     const query = prompt('Введите поисковый запрос для поиска
документов в Google Docs / Drive:');
39     if (query && query.trim() && msgInput) {
40       msgInput.value = `Найди в Google Docs / Google Drive док
ументы по запросу: "${query.trim()}".`;
41       setTimeout(send, 100);
42     }
43   } else if (actionType === 'sheets') {
44     const query = prompt('Введите ID таблицы или команду для G
oogle Sheets (например: "Прочитай данные из таблицы ID диапазо
н A1:D10" или "Добавь строку"):');
45     if (query && query.trim() && msgInput) {
46       msgInput.value = `Google Sheets: ${query.trim()}`;
47       setTimeout(send, 100);
48     }
49   } else if (actionType === 'news') {
50     if (msgInput) {
51       msgInput.value = 'Подготовь для меня свежий дайджест сам
ых релевантных новостей по моим интересам.';
52       setTimeout(send, 100);
53     }
54   } else if (actionType === 'facebook') {
55     const text = prompt('Введите текст публикации для Facebook
(и при желании ссылку через пробел):');
56     if (text && text.trim() && msgInput) {
57       msgInput.value = `Опубликуй пост в Facebook с текстом: "
${text.trim()}".`;
58       setTimeout(send, 100);
59     }
60   }
61 }
62
63 export async function initUserSettings() {
64   await refreshUserProfile();

```

```
65 |
66 | // Bind User Menu Action Buttons
67 | document.querySelectorAll('#user-menu-btn-mail, #nav-btn-check-mail, .action-btn-mail').forEach((btn) => {
68 |     btn.addEventListener('click', (e) => {
69 |         e.preventDefault();
70 |         executeUserAction('mail');
71 |     });
72 | });
73 |
74 | document.querySelectorAll('#user-menu-btn-docs, #nav-btn-search-gdocs, .action-btn-docs').forEach((btn) => {
75 |     btn.addEventListener('click', (e) => {
76 |         e.preventDefault();
77 |         executeUserAction('docs');
78 |     });
79 | });
80 |
81 | document.querySelectorAll('#user-menu-btn-sheets, #nav-btn-gsheets, .action-btn-sheets').forEach((btn) => {
82 |     btn.addEventListener('click', (e) => {
83 |         e.preventDefault();
84 |         executeUserAction('sheets');
85 |     });
86 | });
87 |
88 | document.querySelectorAll('#user-menu-btn-news, #nav-btn-news-digest, .action-btn-news').forEach((btn) => {
89 |     btn.addEventListener('click', (e) => {
90 |         e.preventDefault();
91 |         executeUserAction('news');
92 |     });
93 | });
94 |
95 | document.querySelectorAll('#user-menu-btn-facebook, #nav-btn-facebook-post, .action-btn-facebook').forEach((btn) => {
96 |     btn.addEventListener('click', (e) => {
97 |         e.preventDefault();
```

```

98     executeUserAction('facebook');
99   });
100 });
101
102 // Handle Google Login Click
103 document.querySelectorAll('.btn-google-login, #user-settings
- google-btn').forEach((btn) => {
104   btn.addEventListener('click', (e) => {
105     e.preventDefault();
106     const nextUrl = window.location.pathname + window.locati
on.search;
107     window.location.href = `/auth/google?next=${encodeURIComponent(
nextUrl)}`;
108   });
109 });
110
111 // Handle Logout Click
112 document.querySelectorAll('.btn-user-logout, #user-settings-
logout-btn').forEach((btn) => {
113   btn.addEventListener('click', async (e) => {
114     e.preventDefault();
115     try {
116       await fetch('/auth/logout', { method: 'POST' });
117     } catch (err) {
118       console.error('Logout error:', err);
119     } finally {
120       window.location.href = '/';
121     }
122   });
123 });
124
125 // Re-sync modal fields on modal show
126 const modalEl = document.getElementById('userSettingsModal')
;
127 if (modalEl) {
128   modalEl.addEventListener('show.bs.modal', async () => {
129     await refreshUserProfile();
130     const savedLang = localStorage.getItem('app_language') |

```

```

    | 'ru';
131 |     document.querySelectorAll('.lang-selector').forEach((s)
    => {
132 |         s.value = savedLang;
133 |     });
134 |     const savedTheme = localStorage.getItem('theme') || 'sys
    tem';
135 |     document.querySelectorAll('.theme-selector').forEach((s)
    => {
136 |         s.value = savedTheme;
137 |     });
138 | });
139 | }
140 | }
141 |
142 | export function isUserAdmin() {
143 |     if (!window.currentUser) return false;
144 |     return Boolean(window.currentUser.is_admin || (window.curren
    tUser.role && String(window.currentUser.role).toLowerCase() ==
    = 'admin'));
145 | }
146 |
147 | export async function refreshUserProfile() {
148 |     try {
149 |         const res = await fetch('/auth/check');
150 |         if (!res.ok) return;
151 |         const data = await res.json();
152 |         window.currentUser = data;
153 |
154 |         const authCard = document.getElementById('user-settings-au
    th-card');
155 |         const guestCard = document.getElementById('user-settings-g
    uest-card');
156 |         const navGoogleBtn = document.getElementById('nav-google-b
    tn');
157 |         const userSettingsBtn = document.getElementById('user-sett
    ings-btn');
158 |         const navAvatar = document.getElementById('nav-user-avatar

```



```
    ');
159     const navIcon = document.getElementById('nav-user-icon');
160     const navLabel = document.getElementById('nav-user-label')
    ;
161
162     if (data && data.authenticated) {
163         // User is authenticated
164         if (authCard) authCard.classList.remove('d-none');
165         if (guestCard) guestCard.classList.add('d-none');
166
167         // Navbar: show user profile pill, hide Google register
button
168         if (navGoogleBtn) navGoogleBtn.classList.add('d-none');
169         if (userSettingsBtn) {
170             userSettingsBtn.classList.remove('d-none');
171             userSettingsBtn.classList.add('d-flex');
172         }
173
174         const name = data.name || data.email || 'Пользователь';
175         const email = data.email || '';
176         const picture = data.picture || '';
177         const role = data.role || (data.is_admin ? 'Admin' : 'Us
er');
178         const hasGoogle = Boolean(data.has_google);
179
180         // Update modal fields
181         const avatarEl = document.getElementById('user-settings-
avatar');
182         const nameEl = document.getElementById('user-settings-na
me');
183         const emailEl = document.getElementById('user-settings-e
mail');
184         const roleBadge = document.getElementById('user-settings
-role-badge');
185         const googleBadge = document.getElementById('user-settin
gs-google-badge');
186
187         if (avatarEl) {
```

```
188         if (picture) {
189             avatarEl.src = picture;
190             avatarEl.classList.remove('d-none');
191         } else {
192             avatarEl.classList.add('d-none');
193         }
194     }
195     if (nameEl) nameEl.textContent = name;
196     if (emailEl) emailEl.textContent = email;
197     if (roleBadge) {
198         roleBadge.textContent = role;
199         roleBadge.className = `badge ${role.toLowerCase() ===
'admin' ? 'bg-danger' : 'bg-primary'}`;
200     }
201     const isOAuthEnabled = data.oauth_enabled !== false;
202
203     if (googleBadge) {
204         if (hasGoogle) {
205             googleBadge.innerHTML = '<i class="bi bi-google text
-danger me-1"></i>Google OAuth: <span class="text-success fw-b
old">Синхронизирован</span>';
206         } else if (isOAuthEnabled) {
207             googleBadge.innerHTML = '<i class="bi bi-google text
-muted me-1"></i>Google OAuth: <span class="text-warning">Не п
одключен</span> <button type="button" class="btn btn-sm btn-ou
tline-danger ms-2 py-0 px-2 btn-google-login">Синхронизировать
</button>';
208         } else {
209             googleBadge.innerHTML = '<i class="bi bi-google text
-muted me-1"></i>Google OAuth: <span class="text-muted">Отключ
ен</span>';
210         }
211     }
212
213     // Update dropdown header elements if present
214     const menuWrapper = document.getElementById('user-menu-d
ropdown-wrapper');
215     const menuDisplayName = document.getElementById('user-me
```

```

    nu-display-name');
216     const menuDisplayEmail = document.getElementById('user-m
enu-display-email');
217     if (menuWrapper) {
218         menuWrapper.classList.remove('d-none');
219         menuWrapper.classList.add('d-flex');
220     }
221     if (menuDisplayName) menuDisplayName.textContent = name;
222     if (menuDisplayEmail) menuDisplayEmail.textContent = ema
il;
223
224     // Update navbar button
225     if (navAvatar && picture) {
226         navAvatar.src = picture;
227         navAvatar.classList.remove('d-none');
228         if (navIcon) navIcon.classList.add('d-none');
229     }
230     if (navLabel) {
231         const shortName = name.split(' ')[0] || name;
232         navLabel.textContent = shortName;
233     }
234 } else {
235     // Guest / unauthenticated
236     if (authCard) authCard.classList.add('d-none');
237     if (guestCard) guestCard.classList.remove('d-none');
238
239     const isOAuthEnabled = data && data.oauth_enabled !== fa
lse;
240
241     // Navbar: show full Google register/sign-in button only
    if OAuth is enabled, else show user settings gear/pill
242     if (navGoogleBtn) {
243         if (isOAuthEnabled) {
244             navGoogleBtn.classList.remove('d-none');
245             navGoogleBtn.classList.add('d-flex');
246         } else {
247             navGoogleBtn.classList.add('d-none');
248             navGoogleBtn.classList.remove('d-flex');

```

```
249 |     }
250 | }
251 |     const menuWrapper = document.getElementById('user-menu-d
    | ropdown-wrapper');
252 |     if (menuWrapper) {
253 |         if (isOAuthEnabled) {
254 |             menuWrapper.classList.add('d-none');
255 |             menuWrapper.classList.remove('d-flex');
256 |         } else {
257 |             menuWrapper.classList.remove('d-none');
258 |             menuWrapper.classList.add('d-flex');
259 |         }
260 |     }
261 |     if (userSettingsBtn) {
262 |         if (isOAuthEnabled) {
263 |             userSettingsBtn.classList.add('d-none');
264 |             userSettingsBtn.classList.remove('d-flex');
265 |         } else {
266 |             userSettingsBtn.classList.remove('d-none');
267 |             userSettingsBtn.classList.add('d-flex');
268 |         }
269 |     }
270 |
271 |     // Guest card google login button adjustment
272 |     const guestGoogleBtn = guestCard ? guestCard.querySelector(
    | or('.btn-google-login') : null;
273 |     if (guestGoogleBtn) {
274 |         if (isOAuthEnabled) {
275 |             guestGoogleBtn.classList.remove('d-none');
276 |         } else {
277 |             guestGoogleBtn.classList.add('d-none');
278 |         }
279 |     }
280 |
281 |     if (navAvatar) navAvatar.classList.add('d-none');
282 |     if (navIcon) navIcon.classList.remove('d-none');
283 |     if (navLabel) navLabel.textContent = 'User Settings';
284 | }
```

```
285 |     } catch (err) {
286 |         console.error('Failed to refresh user profile:', err);
287 |     }
288 | }
289 |
290 | window.initUserSettings = initUserSettings;
291 | window.refreshUserProfile = refreshUserProfile;
292 | window.isUserAdmin = isUserAdmin;
293 | window.executeUserAction = executeUserAction;
294 |
```

window-api.js

```
1 /**
2  * window-api.js – инициализация window.api для всех вкладок и
3  * панелей управления
4  * Этот файл загружается ДО всех вкладок, чтобы window.api был
5  * доступен сразу
6  */
7
8 window.api = window.api || {};
9
10 // Универсальный fetch для API и Auth
11 if (!window.api.fetch) {
12   window.api.fetch = async function(url, opts = {}) {
13     const response = await fetch(url, opts);
14     if (!response.ok) {
15       let msg = response.statusText;
16       try {
17         const data = await response.json();
18         if (data && data.detail) {
19           if (typeof data.detail === 'string') {
20             msg = data.detail;
21           } else if (Array.isArray(data.detail)) {
22             msg = data.detail.map(d => d.msg || JSON.stringify
23               (d)).join(', ');
24           } else {
25             msg = JSON.stringify(data.detail);
26           }
27         } else if (data && data.message) {
28           msg = data.message;
29         }
30       } catch {}
31       throw new Error(`${response.status} ${msg}`);
32     }
33     return response.json();
34   };
35 }
```

```

34 // User management API
35 window.api.users = window.api.users || {
36   async list(params = {}) {
37     const q = new URLSearchParams(params).toString();
38     return window.api.fetch(`/api/admin/users${q ? '?' + q : ''}`);
39   },
40   async get(userId) {
41     return window.api.fetch(`/api/admin/users/${userId}`);
42   },
43   async create(data) {
44     return window.api.fetch('/api/admin/users', {
45       method: 'POST',
46       headers: { 'Content-Type': 'application/json' },
47       body: JSON.stringify(data)
48     });
49   },
50   async update(userId, data) {
51     return window.api.fetch(`/api/admin/users/${userId}`, {
52       method: 'PUT',
53       headers: { 'Content-Type': 'application/json' },
54       body: JSON.stringify(data)
55     });
56   },
57   async setPassword(userId, password) {
58     return window.api.fetch(`/api/admin/users/${userId}/password`, {
59       method: 'POST',
60       headers: { 'Content-Type': 'application/json' },
61       body: JSON.stringify({ password })
62     });
63   },
64   async toggleActive(userId) {
65     return window.api.fetch(`/api/admin/users/${userId}/toggle-active`, {
66       method: 'POST'
67     });
68   },

```

```

69 |   async toggleRole(userId) {
70 |       return window.api.fetch(`/api/admin/users/${userId}/toggle
    | -role`, {
71 |           method: 'POST'
72 |       });
73 |   },
74 |   async delete(userId) {
75 |       return window.api.fetch(`/api/admin/users/${userId}`, {
76 |           method: 'DELETE'
77 |       });
78 |   }
79 | };
80 |
81 | // Skills management API
82 | window.api.skills = window.api.skills || {
83 |     async list(params = {}) {
84 |         const q = new URLSearchParams(params).toString();
85 |         return window.api.fetch(`/api/admin/skills${q ? '?' + q :
    | ''}`);
86 |     },
87 |     async get(name) {
88 |         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
    | name)}`);
89 |     },
90 |     async create(data) {
91 |         return window.api.fetch('/api/admin/skills', {
92 |             method: 'POST',
93 |             headers: { 'Content-Type': 'application/json' },
94 |             body: JSON.stringify(data)
95 |         });
96 |     },
97 |     async update(name, data) {
98 |         return window.api.fetch(`/api/admin/skills/${encodeURIComponent(
    | name)}`, {
99 |             method: 'PUT',
100 |             headers: { 'Content-Type': 'application/json' },
101 |             body: JSON.stringify(data)
102 |         });

```



```
103 | },
104 |   async delete(name) {
105 |     return window.api.fetch(`/api/admin/skills/${encodeURIComponent
    | onent(name)}`, {
106 |       method: 'DELETE'
107 |     });
108 |   },
109 |   async package(name) {
110 |     return window.api.fetch(`/api/admin/skills/${encodeURIComponent
    | onent(name)}/package`, {
111 |       method: 'POST'
112 |     });
113 |   }
114 | };
115 |
116 | // MCP Servers management API
117 | window.api.mcp = window.api.mcp || {
118 |   async list() {
119 |     return window.api.fetch('/api/admin/mcp/servers');
120 |   },
121 |   async get(id) {
122 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent
    | IComponent(id)}`);
123 |   },
124 |   async create(data) {
125 |     return window.api.fetch('/api/admin/mcp/servers', {
126 |       method: 'POST',
127 |       headers: { 'Content-Type': 'application/json' },
128 |       body: JSON.stringify(data)
129 |     });
130 |   },
131 |   async update(id, data) {
132 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent
    | IComponent(id)}`, {
133 |       method: 'PUT',
134 |       headers: { 'Content-Type': 'application/json' },
135 |       body: JSON.stringify(data)
136 |     });
```

```
137 | },
138 | async delete(id) {
139 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent
    | IComponent(id)}`, {
140 |         method: 'DELETE'
141 |     });
142 | },
143 | async toggle(id) {
144 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent
    | IComponent(id)}/toggle`, {
145 |         method: 'POST'
146 |     });
147 | },
148 | async test(id) {
149 |     return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent
    | IComponent(id)}/test`, {
150 |         method: 'POST'
151 |     });
152 | },
153 | async testAdhoc(data) {
154 |     return window.api.fetch('/api/admin/mcp/test', {
155 |         method: 'POST',
156 |         headers: { 'Content-Type': 'application/json' },
157 |         body: JSON.stringify(data)
158 |     });
159 | },
160 | async getTools() {
161 |     return window.api.fetch('/api/admin/mcp/tools');
162 | }
163 | };
164 |
165 | // Google Workspace Accounts API wrapper
166 | window.api.googleAccounts = window.api.googleAccounts || {
167 |     async list() {
168 |         return await window.api.fetch('/api/admin/google-accounts'
    | );
169 |     },
170 |     async upload(formData) {
```

```
171 |     const r = await fetch('/api/admin/google-accounts/upload',
    |     {
172 |         method: 'POST',
173 |         body: formData
174 |     });
175 |     if (!r.ok) {
176 |         let msg = r.statusText;
177 |         try { msg = (await r.json()).detail || msg; } catch {}
178 |         throw new Error(`${r.status} ${msg}`);
179 |     }
180 |     return await r.json();
181 | },
182 | async create(params) {
183 |     return await window.api.fetch('/api/admin/google-accounts'
    |     , {
184 |         method: 'POST',
185 |         headers: { 'Content-Type': 'application/json' },
186 |         body: JSON.stringify(params)
187 |     });
188 | },
189 | async setDefault(name) {
190 |     return await window.api.fetch(`/api/admin/google-accounts/
    |     ${encodeURIComponent(name)}/default`, {
191 |         method: 'POST'
192 |     });
193 | },
194 | async resetStatus(name) {
195 |     return await window.api.fetch(`/api/admin/google-accounts/
    |     ${encodeURIComponent(name)}/reset-status`, {
196 |         method: 'POST'
197 |     });
198 | },
199 | async test(name) {
200 |     return await window.api.fetch(`/api/admin/google-accounts/
    |     ${encodeURIComponent(name)}/test`, {
201 |         method: 'POST'
202 |     });
203 | },
```

```
204 |   async delete(name) {
205 |       return await window.api.fetch(`/api/admin/google-accounts/
    |   ${encodeURIComponent(name)}`, {
206 |           method: 'DELETE'
207 |       });
208 |   }
209 | };
210 |
```

login.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title>Вход в систему – AI Breadboard</title>
8   <link rel="icon" type="image/x-icon" href="/favicon.ico">
9   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
10  t/css/bootstrap.min.css" rel="stylesheet">
11   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap-icons@1.1
12   0.0/font/bootstrap-icons.css" rel="stylesheet">
13   <link href="/html/css/main.css?v=20260917_refactor_v1" rel="
14   stylesheet">
15   <style>
16     /* — Страница входа: центрирование и лёгкий фон — */
17     body {
18       display: flex;
19       align-items: center;
20       justify-content: center;
21       min-height: 100vh;
22       margin: 0;
23       padding: 1rem;
24       /* Лёгкий градиентный акцент поверх --bg-color из main.c
25       ss */
26       background:
27         radial-gradient(circle at 50% 15%, rgba(99, 102, 241,
28         0.08), transparent 65%),
29         var(--bg-color);
30     }
31
32     /* — Карточка входа — */
33     .auth-card {
34       width: 100%;
35       max-width: 400px;
36       background: var(--surface-1);
```

```
31 |     border: 1px solid var(--border-color);
32 |     border-radius: 12px;
33 |     box-shadow: var(--shadow-lg);
34 |     overflow: hidden;
35 | }
36 |
37 | /* — Шапка карточки — */
38 | .auth-header {
39 |     padding: 1.5rem 1.5rem 1rem;
40 |     text-align: center;
41 |     background: var(--surface-2);
42 |     border-bottom: 1px solid var(--border-color);
43 | }
44 |
45 | /* — Логотип-бейдж — */
46 | .auth-logo-badge {
47 |     width: 48px;
48 |     height: 48px;
49 |     border-radius: 12px;
50 |     background: linear-gradient(135deg, #6366f1, #3b82f6);
51 |     display: inline-flex;
52 |     align-items: center;
53 |     justify-content: center;
54 |     font-size: 1.5rem;
55 |     color: #fff;
56 |     box-shadow: 0 4px 12px rgba(99, 102, 241, 0.3);
57 |     margin-bottom: 0.75rem;
58 | }
59 |
60 | /* — Навигационные вкладки входа/регистрации — */
61 | .auth-nav-tabs .nav-link {
62 |     border: none;
63 |     border-bottom: 2px solid transparent;
64 |     color: var(--text-color);
65 |     opacity: 0.65;
66 |     font-weight: 600;
67 |     font-size: 0.85rem;
68 |     padding: 0.55rem 1rem;
```

```
69 |     background: transparent;
70 |     transition: all 0.18s;
71 | }
72 |
73 | .auth-nav-tabs .nav-link.active {
74 |     background: transparent;
75 |     border-bottom: 2px solid #6366f1;
76 |     color: #6366f1;
77 |     opacity: 1;
78 | }
79 |
80 | [data-theme="dark"] .auth-nav-tabs .nav-link.active {
81 |     color: #818cf8;
82 |     border-bottom-color: #818cf8;
83 | }
84 |
85 | /* — Google OAuth кнопка — */
86 | .btn-google {
87 |     background: var(--surface-1);
88 |     color: var(--text-color);
89 |     border: 1px solid var(--border-color);
90 |     font-weight: 500;
91 |     font-size: 0.85rem;
92 |     transition: all 0.18s;
93 | }
94 |
95 | .btn-google:hover {
96 |     background: var(--surface-2);
97 |     box-shadow: var(--shadow-sm);
98 |     color: var(--text-color);
99 | }
100 |
101 | /* — Разделитель «или по почте» — */
102 | .divider-text {
103 |     display: flex;
104 |     align-items: center;
105 |     text-align: center;
106 |     color: var(--text-muted);
```

```

107 |     font-size: 0.78rem;
108 |     margin: 1rem 0;
109 | }
110 |
111 | .divider-text::before,
112 | .divider-text::after {
113 |     content: '';
114 |     flex: 1;
115 |     border-bottom: 1px solid var(--border-color);
116 | }
117 |
118 | .divider-text:not(:empty)::before { margin-right: 0.6em; }
119 | .divider-text:not(:empty)::after { margin-left: 0.6em; }
120 |
121 | /* — Кнопки управления (тема + язык) в правом верхнем углу — */
122 | .top-controls {
123 |     position: fixed;
124 |     top: 0.75rem;
125 |     right: 0.75rem;
126 |     display: flex;
127 |     gap: 0.375rem;
128 |     z-index: 100;
129 | }
130 | </style>
131 | </head>
132 | <body>
133 |
134 | <!-- Управление: тема и язык -->
135 | <div class="top-controls">
136 |     <select class="form-select form-select-sm" id="auth-lang-selector" style="width: auto; font-size: 0.78rem; padding: 0.2rem 0.4rem;">
137 |         <option value="ru">RU</option>
138 |         <option value="en">EN</option>
139 |         <option value="he">HE</option>
140 |     </select>
141 |     <button class="btn btn-sm btn-outline-secondary d-flex ali

```



```

    gn-items-center justify-content-center rounded-circle"
142         id="theme-toggle"
143         title="Переключить тему"
144         style="width: 28px; height: 28px; padding: 0; ">
145         <i class="bi bi-circle-half" id="theme-icon"></i>
146     </button>
147 </div>
148
149 <div class="auth-card">
150     <!-- Шапка -->
151     <div class="auth-header">
152         <div class="auth-logo-badge">
153             <i class="bi bi-cpu-fill"></i>
154         </div>
155         <h5 class="fw-bold mb-1" id="title-app-name">AI Breadboa
rd</h5>
156         <p class="small mb-0" style="color: var(--text-muted);"
id="title-app-desc">Авторизуйтесь для доступа к сервисам ИИ</p
>
157     </div>
158
159     <!-- Вкладки: Войти / Регистрация -->
160     <ul class="nav nav-tabs auth-nav-tabs nav-fill px-3 pt-2"
id="authTabs" role="tablist">
161         <li class="nav-item" role="presentation">
162             <button class="nav-link active" id="tab-login-btn" dat
a-bs-toggle="tab" data-bs-target="#pane-login" type="button" r
ole="tab">
163                 <i class="bi bi-box-arrow-in-right me-1"></i>
164                 <span id="label-tab-login">Войти</span>
165             </button>
166         </li>
167         <li class="nav-item" role="presentation">
168             <button class="nav-link" id="tab-register-btn" data-bs
-toggle="tab" data-bs-target="#pane-register" type="button" ro
le="tab">
169                 <i class="bi bi-person-plus me-1"></i>
170                 <span id="label-tab-register">Регистрация</span>

```

```

171         </button>
172     </li>
173 </ul>
174
175     <!-- Тело карточки -->
176     <div class="card-body p-3">
177         <!-- Статус-алерт -->
178         <div id="auth-alert" class="alert d-none mb-3 py-2 small
179         " role="alert"></div>
180
181         <!-- Google OAuth -->
182         <button type="button"
183             class="btn btn-google w-100 d-flex align-items-c
184             enter justify-content-center gap-2 rounded-3 mb-0"
185             style="padding: 0.45rem 0.75rem;"
186             id="btn-google-auth">
187             <svg width="16" height="16" viewBox="0 0 24 24">
188                 <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-
189                 .2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2
190                 .08-1.92 3.28-4.74 3.28-8.09z"/>
191                 <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-
192                 2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6
193                 .16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/>
194                 <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.3
195                 6-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.
196                 43 3.45 1.18 4.94l2.85-2.22.81-.63z"/>
197                 <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21
198                 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7
199                 .06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/>
200             </svg>
201             <span id="label-google-auth">Продолжить с Google</span
202             >
203         </button>
204
205         <div class="divider-text" id="label-or-email">или по поч
206         те</div>
207
208         <!-- Контент вкладок -->

```

```

197 |         <div class="tab-content" id="authTabsContent">
198 |
199 |             <!-- BX0Д -->
200 |             <div class="tab-pane fade show active" id="pane-login"
| role="tabpanel">
201 |                 <form id="form-login">
202 |                     <div class="mb-2">
203 |                         <label class="form-label fw-semibold" id="label-
| email-login" for="login-email">Email</label>
204 |                         <div class="input-group input-group-sm">
205 |                             <span class="input-group-text"><i class="bi bi
| -envelope"></i></span>
206 |                             <input type="email" class="form-control form-c
| ontrol-sm" id="login-email"
207 |                                 required placeholder="name@example.com"
| autocomplete="email">
208 |                             </div>
209 |                         </div>
210 |                         <div class="mb-3">
211 |                             <label class="form-label fw-semibold" id="label-
| pass-login" for="login-password">Пароль</label>
212 |                             <div class="input-group input-group-sm">
213 |                                 <span class="input-group-text"><i class="bi bi
| -key"></i></span>
214 |                                 <input type="password" class="form-control for
| m-control-sm" id="login-password"
215 |                                     required placeholder="....." autocom
| plete="current-password">
216 |                                 </div>
217 |                             </div>
218 |                             <button type="submit" class="btn btn-primary btn-s
| m w-100 fw-semibold" id="btn-submit-login"
219 |                                 style="padding: 0.45rem;">
220 |                                 <span class="spinner-border spinner-border-sm d-
| none me-1" id="login-spinner"></span>
221 |                                 <span id="label-btn-login">Войти в аккаунт</span
| >
222 |                             </button>

```

```
223 |         </form>
224 |     </div>
225 |
226 |     <!-- РЕГИСТРАЦИЯ -->
227 |     <div class="tab-pane fade" id="pane-register" role="ta
bpanel">
228 |         <form id="form-register">
229 |             <div class="mb-2">
230 |                 <label class="form-label fw-semibold" id="label-
reg-name" for="reg-name">Ваше имя</label>
231 |                 <div class="input-group input-group-sm">
232 |                     <span class="input-group-text"><i class="bi bi
-person"></i></span>
233 |                     <input type="text" class="form-control form-co
ntrol-sm" id="reg-name"
234 |                         required placeholder="Иван Иванов" auto
complete="name">
235 |                 </div>
236 |             </div>
237 |             <div class="mb-2">
238 |                 <label class="form-label fw-semibold" id="label-
reg-email" for="reg-email">Email</label>
239 |                 <div class="input-group input-group-sm">
240 |                     <span class="input-group-text"><i class="bi bi
-envelope"></i></span>
241 |                     <input type="email" class="form-control form-c
ontrol-sm" id="reg-email"
242 |                         required placeholder="name@example.com"
autocomplete="email">
243 |                 </div>
244 |             </div>
245 |             <div class="mb-3">
246 |                 <label class="form-label fw-semibold" id="label-
reg-pass" for="reg-password">Придумайте пароль</label>
247 |                 <div class="input-group input-group-sm">
248 |                     <span class="input-group-text"><i class="bi bi
-lock"></i></span>
249 |                     <input type="password" class="form-control for
```

```

m-control-sm" id="reg-password"
250         required placeholder="Минимум 6 символо
в" minlength="6" autocomplete="new-password">
251     </div>
252 </div>
253     <button type="submit" class="btn btn-primary btn-s
m w-100 fw-semibold" id="btn-submit-reg"
254         style="padding: 0.45rem;">
255         <span class="spinner-border spinner-border-sm d-
none me-1" id="reg-spinner"></span>
256         <span id="label-btn-reg">Зарегистрироваться</spa
n>
257     </button>
258 </form>
259 </div>
260
261 </div>
262 </div>
263 </div>
264
265 <!-- Модальное окно подтверждения email -->
266 <div class="modal fade" id="verifyModal" tabindex="-1" aria-
hidden="true" data-bs-backdrop="static">
267     <div class="modal-dialog modal-dialog-centered modal-sm">
268         <div class="modal-content">
269             <div class="modal-header">
270                 <h6 class="modal-title fw-bold">
271                     <i class="bi bi-shield-check text-primary me-1"></
i> Подтверждение почты
272                 </h6>
273                 <button type="button" class="btn-close" data-bs-dism
iss="modal"></button>
274             </div>
275             <div class="modal-body text-center">
276                 <p class="small mb-3" style="color: var(--text-muted
);" id="verify-modal-desc">
277                     Введите код подтверждения, отправленный на ваш ema
il:

```

```

278         </p>
279         <div class="mb-3">
280             <input type="text" class="form-control form-control-sm text-center"
281                 id="verify-code-input" placeholder="123456"
282                 maxlength="6"
283                 autofocus style="letter-spacing: 0.2em; font-size: 1.1rem;">
284             </div>
285             <div id="verify-alert" class="alert d-none py-1 small"></div>
286             <button type="button" class="btn btn-primary btn-sm w-100 fw-semibold" id="btn-submit-verify"
287                 style="padding: 0.45rem;">
288                 <span class="spinner-border spinner-border-sm d-none me-1" id="verify-spinner"></span>
289                 Подтвердить
290             </button>
291         </div>
292     </div>
293 </div>
294
295     <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
296     <script>
297         (function() {
298             'use strict';
299
300             // — Тема: системная по умолчанию —————
301
302             function getSystemTheme() {
303                 return window.matchMedia && window.matchMedia('(prefers-color-scheme: dark)').matches
304                     ? 'dark' : 'light';
305             }
306
307             function applyTheme(mode) {

```

```

307 |         // mode: 'light' | 'dark' | 'system'
308 |         const resolved = (mode === 'system') ? getSystemTheme(
    | ) : mode;
309 |         document.documentElement.setAttribute('data-theme', re
    | solved);
310 |         document.documentElement.setAttribute('data-bs-theme',
    | resolved);
311 |
312 |         const icon = document.getElementById('theme-icon');
313 |         if (icon) {
314 |             if (mode === 'light')             icon.className = 'bi bi
    | -sun-fill text-warning';
315 |             else if (mode === 'dark')         icon.className = 'bi bi
    | -moon-stars-fill text-primary';
316 |             else                             icon.className = 'bi bi
    | -circle-half text-info';
317 |         }
318 |     }
319 |
320 |     function initTheme() {
321 |         const saved = localStorage.getItem('theme') || 'system
    | ';
322 |         applyTheme(saved);
323 |
324 |         // Следим за изменениями ОС в системном режиме
325 |         if (window.matchMedia) {
326 |             window.matchMedia('(prefers-color-scheme: dark)').ad
    | dEventListener('change', () => {
327 |                 if ((localStorage.getItem('theme') || 'system') ==
    | = 'system') {
328 |                     applyTheme('system');
329 |                 }
330 |             });
331 |         }
332 |     }
333 |
334 |     // Кнопка переключения темы: цикл system → light → dark
    | → system

```

```

335 |     document.getElementById('theme-toggle').addEventListener
    | ('click', () => {
336 |         const cur = localStorage.getItem('theme') || 'system';
337 |         const next = { system: 'light', light: 'dark', dark: '
    | system' }[cur] || 'system';
338 |         localStorage.setItem('theme', next);
339 |         applyTheme(next);
340 |     });
341 |
342 |     initTheme();
343 |
344 |     // — I18n —————
    |
345 |     const i18nDict = {
346 |         ru: {
347 |             appDesc: 'Авторизуйтесь для доступа к сервисам ИИ',
348 |             tabLogin: 'Войти', tabReg: 'Регистрация',
349 |             btnGoogle: 'Продолжить с Google', orEmail: 'или по п
    | очте',
350 |             email: 'Email', pass: 'Пароль',
351 |             regName: 'Ваше имя', regPass: 'Придумайте пароль',
352 |             btnLogin: 'Войти в аккаунт', btnReg: 'Зарегистрирова
    | ться',
353 |             passPlaceholder: 'Минимум 6 символов', namePlacehold
    | er: 'Иван Иванов'
354 |         },
355 |         en: {
356 |             appDesc: 'Sign in to access AI Breadboard services',
357 |             tabLogin: 'Sign In', tabReg: 'Sign Up',
358 |             btnGoogle: 'Continue with Google', orEmail: 'or via
    | Email',
359 |             email: 'Email', pass: 'Password',
360 |             regName: 'Full Name', regPass: 'Create Password',
361 |             btnLogin: 'Sign In', btnReg: 'Create Account',
362 |             passPlaceholder: 'Minimum 6 characters', namePlaceho
    | lder: 'John Doe'
363 |         },
364 |         he: {

```



```

365 |         appDesc: 'i18nDict[lang] || i18nDict.ru AI',
366 |         tabLogin: 'i18nDict[lang] || i18nDict.ru', tabReg: 'i18nDict[lang] || i18nDict.ru',
367 |         btnGoogle: 'i18nDict[lang] || i18nDict.ru Google', orEmail: 'i18nDict[lang] || i18nDict.ru or email',
368 |         email: 'i18nDict[lang] || i18nDict.ru', pass: 'i18nDict[lang] || i18nDict.ru',
369 |         regName: 'i18nDict[lang] || i18nDict.ru', regPass: 'i18nDict[lang] || i18nDict.ru',
370 |         btnLogin: 'i18nDict[lang] || i18nDict.ru', btnReg: 'i18nDict[lang] || i18nDict.ru',
371 |         passPlaceholder: 'i18nDict[lang] || i18nDict.ru 6 i18nDict[lang] || i18nDict.ru', namePlaceholder: 'i18nDict[lang] || i18nDict.ru'
372 |     };
373 | };
374 |
375 | const langSelector = document.getElementById('auth-lang-selector');
376 | const savedLang = localStorage.getItem('app_language') | 'ru';
377 | if (langSelector) {
378 |     langSelector.value = savedLang;
379 |     langSelector.addEventListener('change', (e) => {
380 |         const l = e.target.value;
381 |         localStorage.setItem('app_language', l);
382 |         updateLanguageTexts(l);
383 |     });
384 | }
385 |
386 | function updateLanguageTexts(lang) {
387 |     const d = i18nDict[lang] || i18nDict.ru;
388 |     document.getElementById('title-app-desc').textContent = d.appDesc;
389 |     document.getElementById('label-tab-login').textContent = d.tabLogin;
390 |     document.getElementById('label-tab-register').textContent = d.tabReg;
391 |     document.getElementById('label-google-auth').textContent = d.btnGoogle;
392 |     document.getElementById('label-or-email').textContent = d.orEmail;
393 |     document.getElementById('label-email-login').textContent = d.email;

```

```

    nt = d.email;
394     document.getElementById('label-pass-login').textContent
    t = d.pass;
395     document.getElementById('label-reg-name').textContent
    = d.regName;
396     document.getElementById('label-reg-email').textContent
    = d.email;
397     document.getElementById('label-reg-pass').textContent
    = d.regPass;
398     document.getElementById('label-btn-login').textContent
    = d.btnLogin;
399     document.getElementById('label-btn-reg').textContent
    = d.btnReg;
400     document.getElementById('reg-password').placeholder
    = d.passPlaceholder;
401     document.getElementById('reg-name').placeholder
    = d.namePlaceholder;
402     }
403     updateLanguageTexts(savedLang);
404
405     // — Алерты —————
    

---


406     function showAlert(msg, type = 'danger') {
407         const el = document.getElementById('auth-alert');
408         el.className = `alert alert-${type} mb-3 py-2 small`;
409         el.textContent = msg;
410         el.classList.remove('d-none');
411     }
412     function hideAlert() {
413         document.getElementById('auth-alert').classList.add('d
- none');
414     }
415
416     // — Google OAuth —————
    

---


417     document.getElementById('btn-google-auth').addEventListener
ner('click', () => {
418         window.location.href = '/auth/google?next=/';

```

```

419         });
420
421         // — Вход —————
422         let pendingVerificationEmail = '';
423         document.getElementById('form-login').addEventListener('
submit', async (e) => {
424             e.preventDefault();
425             hideAlert();
426             const email    = document.getElementById('login-email'
).value.trim();
427             const password = document.getElementById('login-passwo
rd').value;
428             const spinner  = document.getElementById('login-spinne
r');
429             const btn      = document.getElementById('btn-submit-l
ogin');
430             btn.disabled = true;
431             spinner.classList.remove('d-none');
432             try {
433                 const res  = await fetch('/auth/login/email', {
434                     method: 'POST',
435                     headers: { 'Content-Type': 'application/json' },
436                     body: JSON.stringify({ email, password })
437                 });
438                 const data = await res.json();
439                 if (!res.ok) {
440                     if (res.status === 403 && data.detail && data.deta
il.includes('не подтверждена')) {
441                         pendingVerificationEmail = email;
442                         const desc = document.getElementById('verify-mod
al-desc');
443                         if (desc) desc.textContent = `Код подтверждения
отправлен на ${email}:`;
444                         new bootstrap.Modal(document.getElementById('ver
ifyModal')).show();
445                         showAlert(data.detail, 'warning');
446                     } else {

```

```

447 |         showAlert(data.detail || 'Ошибка авторизации', '
| danger');
448 |     }
449 |     return;
450 | }
451 |     showAlert('Вход выполнен! Перенаправление...', 'succ
| ess');
452 |     setTimeout(() => { window.location.href = '/'; }, 40
| 0);
453 | } catch (err) {
454 |     showAlert('Ошибка соединения: ' + err.message, 'dang
| er');
455 | } finally {
456 |     btn.disabled = false;
457 |     spinner.classList.add('d-none');
458 | }
459 | });
460 |
461 | // — Регистрация —————
|
462 |     document.getElementById('form-register').addEventListener('submit', async (e) => {
463 |         e.preventDefault();
464 |         hideAlert();
465 |         const name      = document.getElementById('reg-name').v
| alue.trim();
466 |         const email     = document.getElementById('reg-email').
| value.trim();
467 |         const password = document.getElementById('reg-password
| ');
468 |         const spinner   = document.getElementById('reg-spinner'
| );
469 |         const btn       = document.getElementById('btn-submit-r
| eg');
470 |         btn.disabled = true;
471 |         spinner.classList.remove('d-none');
472 |         try {
473 |             const res   = await fetch('/auth/register', {

```

```

474         method: 'POST',
475         headers: { 'Content-Type': 'application/json' },
476         body: JSON.stringify({ email, password, name })
477     });
478     const data = await res.json();
479     if (!res.ok) {
480         showAlert(data.detail || 'Ошибка регистрации', 'da
nger');
481         return;
482     }
483     pendingVerificationEmail = email;
484     const desc = document.getElementById('verify-modal-d
esc');
485     if (desc) desc.textContent = `Код подтверждения отпр
авлен на ${email}:`;
486     new bootstrap.Modal(document.getElementById('verifyM
odal')).show();
487     showAlert('Регистрация начата. Введите код из письма
.', 'info');
488     } catch (err) {
489         showAlert('Ошибка соединения: ' + err.message, 'dang
er');
490     } finally {
491         btn.disabled = false;
492         spinner.classList.add('d-none');
493     }
494 });
495
496     // — Подтверждение кода —————
497     document.getElementById('btn-submit-verify').addEventLis
tener('click', async () => {
498         const codeInput = document.getElementById('verify-code
-input');
499         const alertEl = document.getElementById('verify-aler
t');
500         const spinner = document.getElementById('verify-spin
ner');

```

```

501 |     const btnVerify = document.getElementById('btn-submit-
| verify');
502 |     const code = codeInput.value.trim();
503 |     if (!code) return;
504 |     alertEl.classList.add('d-none');
505 |     btnVerify.disabled = true;
506 |     spinner.classList.remove('d-none');
507 |     try {
508 |         const res = await fetch('/auth/verify', {
509 |             method: 'POST',
510 |             headers: { 'Content-Type': 'application/json' },
511 |             body: JSON.stringify({ email: pendingVerificationE
| mail, code })
512 |         });
513 |         const data = await res.json();
514 |         if (!res.ok) {
515 |             alertEl.className = 'alert alert-danger py-1 small
| ';
516 |             alertEl.textContent = data.detail || 'Неверный код
| ';
517 |             alertEl.classList.remove('d-none');
518 |             return;
519 |         }
520 |         alertEl.className = 'alert alert-success py-1 small'
| ;
521 |         alertEl.textContent = 'Почта подтверждена! Войдите п
| од своими данными.';
522 |         alertEl.classList.remove('d-none');
523 |         setTimeout(() => {
524 |             const modal = bootstrap.Modal.getInstance(document
| .getElementById('verifyModal'));
525 |             if (modal) modal.hide();
526 |             document.getElementById('tab-login-btn').click();
527 |             document.getElementById('login-email').value = pen
| dingVerificationEmail;
528 |             showAlert('Почта подтверждена. Введите пароль для
| входа.', 'success');
529 |             }, 1200);

```

```
530 |         } catch (err) {
531 |             alertEl.className = 'alert alert-danger py-1 small';
532 |             alertEl.textContent = err.message;
533 |             alertEl.classList.remove('d-none');
534 |         } finally {
535 |             btnVerify.disabled = false;
536 |             spinner.classList.add('d-none');
537 |         }
538 |     });
539 |
540 |     })();
541 | </script>
542 | </body>
543 | </html>
544 |
```

index.html

```
1 <!-- System Logs Viewer & AI Analyzer Tab -->
2 <div class="container-fluid py-2" id="logs-tab-root">
3   <!-- Top Banner & Controls -->
4   <div class="alert alert-dark d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3 shadow-sm border-secondary">
5     <div class="d-flex align-items-center gap-2">
6       <span class="fs-4">❏ </span>
7       <div>
8         <h5 class="mb-0 fw-bold">Системные логи и AI-анализ</h5>
9         <p class="mb-0 small text-muted">Инспекция журналов событий сервера, фильтрация уровней сообщений и отчеты Gemini</p>
10      </div>
11    </div>
12    <div class="d-flex gap-2 flex-wrap align-items-center">
13      <div class="form-check form-switch mb-0 me-2">
14        <input class="form-check-input" type="checkbox" role="switch" id="logs-auto-refresh" checked>
15        <label class="form-check-label small text-muted" for="logs-auto-refresh">Автообновление (3с)</label>
16      </div>
17      <button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1" id="btn-refresh-logs" onclick="refreshCurrentLog()">
18        <i class="bi bi-arrow-clockwise"></i> Обновить
19      </button>
20      <button class="btn btn-outline-danger btn-sm d-flex align-items-center gap-1" id="btn-clear-log" onclick="clearCurrentLog()">
21        <i class="bi bi-trash3"></i> Очистить лог
22      </button>
23      <button class="btn btn-outline-primary btn-sm d-flex align-items-center gap-1" id="btn-analyze-log" onclick="triggerLogAnalysis()">
```



```
24 |         <i class="bi bi-cpu"></i> AI-анализ лога
25 |     </button>
26 | </div>
27 | </div>
28 |
29 | <!-- Summary Stats Cards -->
30 | <div class="row g-3 mb-3">
31 |     <div class="col-md-3 col-sm-6">
32 |         <div class="card shadow-sm border-secondary-subtle h-100
33 |         bg-body-tertiary">
34 |             <div class="card-body p-3">
35 |                 <div class="text-muted small text-uppercase fw-semib
36 |                 old mb-1">Всего лог-файлов</div>
37 |                 <div class="d-flex align-items-center justify-conten
38 |                 t-between">
39 |                     <h5 class="mb-0 fw-bold text-primary" id="log-stat
40 |                     -files-count">0</h5>
41 |                     <i class="bi bi-file-earmark-text fs-4 text-primar
42 |                     y"></i>
43 |                 </div>
44 |                 <small class="text-muted">Активные журналы</small>
45 |             </div>
46 |         </div>
47 |     </div>
48 |
49 |     <div class="col-md-3 col-sm-6">
50 |         <div class="card shadow-sm border-secondary-subtle h-100
51 |         bg-body-tertiary">
52 |             <div class="card-body p-3">
53 |                 <div class="text-muted small text-uppercase fw-semib
54 |                 old mb-1">Общий объём</div>
55 |                 <div class="d-flex align-items-center justify-conten
56 |                 t-between">
57 |                     <h5 class="mb-0 fw-bold text-info" id="log-stat-to
58 |                     tal-size">0 KB</h5>
59 |                     <i class="bi bi-hdd-stack fs-4 text-info"></i>
60 |                 </div>
61 |                 <small class="text-muted" id="log-stat-size-mb">0 MB
```

```

на диске</small>
53     </div>
54     </div>
55 </div>
56
57     <div class="col-md-3 col-sm-6">
58         <div class="card shadow-sm border-secondary-subtle h-100
bg-body-tertiary">
59             <div class="card-body p-3">
60                 <div class="text-muted small text-uppercase fw-semib
old mb-1">Строк в текущем файле</div>
61                 <div class="d-flex align-items-center justify-conten
t-between">
62                     <h5 class="mb-0 fw-bold text-warning" id="log-stat
-current-lines">0</h5>
63                     <i class="bi bi-list-ol fs-4 text-warning"></i>
64                 </div>
65                 <small class="text-muted" id="log-stat-current-filen
ame">—</small>
66             </div>
67         </div>
68     </div>
69
70     <div class="col-md-3 col-sm-6">
71         <div class="card shadow-sm border-secondary-subtle h-100
bg-body-tertiary">
72             <div class="card-body p-3">
73                 <div class="text-muted small text-uppercase fw-semib
old mb-1">AI-отчёты</div>
74                 <div class="d-flex align-items-center justify-conten
t-between">
75                     <h5 class="mb-0 fw-bold text-success" id="log-stat
-reports-count">0</h5>
76                     <button class="badge bg-success border-0 text-whit
e text-decoration-none" onclick="toggleReportsModal()">Смотрет
ь □ </button>
77                 </div>
78                 <small class="text-muted">Сводные аудиты</small>

```

```
79         </div>
80     </div>
81 </div>
82 </div>
83
84 <!-- Log Viewer Toolbar -->
85 <div class="card shadow-sm border-secondary-subtle mb-3">
86     <div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2 bg-body-tertiary">
87         <div class="d-flex align-items-center gap-2 flex-grow-1 flex-wrap">
88             <!-- File Selector -->
89             <div class="input-group input-group-sm" style="max-width: 280px;">
90                 <label class="input-group-text bg-body-secondary fw-semibold" for="log-file-selector">
91                     <i class="bi bi-file-text me-1"></i> Файл:
92                 </label>
93                 <select class="form-select" id="log-file-selector" onchange="onLogFileSelected(this.value)">
94                     <option value="">Загрузка файлов...</option>
95                 </select>
96             </div>
97
98             <!-- Tail Lines -->
99             <div class="input-group input-group-sm" style="max-width: 170px;">
100                 <label class="input-group-text bg-body-secondary fw-semibold" for="log-tail-selector">Строк:</label>
101                 <select class="form-select" id="log-tail-selector" onchange="refreshCurrentLog()">
102                     <option value="100">100</option>
103                     <option value="250" selected>250</option>
104                     <option value="500">500</option>
105                     <option value="1000">1000</option>
106                     <option value="0">Всё</option>
107                 </select>
108             </div>
```

```
109 |
110 |         <!-- Level Filter -->
111 |         <div class="btn-group btn-group-sm" role="group" id="l
og-level-filters">
112 |             <button type="button" class="btn btn-outline-seconda
ry active" data-level="ALL" onclick="setLogLevelFilter('ALL')"
>Bce</button>
113 |             <button type="button" class="btn btn-outline-danger"
data-level="ERROR" onclick="setLogLevelFilter('ERROR')">ERROR
</button>
114 |             <button type="button" class="btn btn-outline-warning
" data-level="WARNING" onclick="setLogLevelFilter('WARNING')">
WARN</button>
115 |             <button type="button" class="btn btn-outline-info" d
ata-level="INFO" onclick="setLogLevelFilter('INFO')">INFO</but
ton>
116 |             <button type="button" class="btn btn-outline-success
" data-level="DEBUG" onclick="setLogLevelFilter('DEBUG')">DEBU
G</button>
117 |         </div>
118 |
119 |         <!-- Search in logs -->
120 |         <div class="input-group input-group-sm flex-grow-1" st
yle="min-width: 200px;">
121 |             <span class="input-group-text bg-body-secondary"><i
class="bi bi-search"></i></span>
122 |             <input type="text" class="form-control" id="log-sear
ch-input" placeholder="Поиск по логам..." oninput="filterLogsI
nViewer()">
123 |             <button class="btn btn-outline-secondary" type="butt
on" onclick="clearLogSearch()" title="Сбросить поиск">⌫ </butto
n>
124 |         </div>
125 |     </div>
126 |
127 |     <div class="d-flex align-items-center gap-1">
128 |         <button class="btn btn-sm btn-outline-secondary" oncli
ck="copyLogContent()" title="Скопировать логи в буфер">
```

```
129         <i class="bi bi-clipboard"></i>
130     </button>
131     <button class="btn btn-sm btn-outline-secondary" onclick="downloadLogFile()" title="Скачать файл">
132         <i class="bi bi-download"></i>
133     </button>
134     <button class="btn btn-sm btn-outline-secondary" id="btn-scroll-bottom" onclick="scrollLogToBottom()" title="Прокрутить в конец">
135         <i class="bi bi-arrow-down-circle"></i>
136     </button>
137 </div>
138 </div>
139 </div>
140
141 <!-- Terminal Output Screen -->
142 <div class="card shadow-sm border-secondary-subtle">
143     <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center py-1 px-3" style="font-family: monospace; font-size: 0.85rem;">
144         <div class="d-flex align-items-center gap-2">
145             <span class="badge bg-success p-1" style="width: 10px; height: 10px; border-radius: 50%; display: inline-block;"></span>
146             <span id="log-terminal-filename">terminal</span>
147             <span class="text-muted">|</span>
148             <span class="text-muted" id="log-terminal-timestamp">--</span>
149         </div>
150         <div class="d-flex align-items-center gap-2">
151             <span class="badge bg-secondary text-white" id="log-visible-lines-badge">0 строк</span>
152         </div>
153     </div>
154     <div class="card-body p-0 bg-black" style="position: relative;">
155         <pre id="log-output-container" class="m-0 p-3 text-light" style="max-height: 65vh; min-height: 400px; overflow-y: auto
```

```

; font-family: 'Consolas', 'Courier New', monospace; font-size
: 0.82rem; line-height: 1.45; white-space: pre-wrap; word-brea
k: break-all;">Загрузка логов...</pre>
156     </div>
157 </div>
158 </div>
159
160 <!-- AI Reports Modal -->
161 <div class="modal fade" id="logReportsModal" tabindex="-1" ari
a-labelledby="logReportsModalLabel" aria-hidden="true">
162     <div class="modal-dialog modal-xl modal-dialog-scrollable">
163         <div class="modal-content bg-dark text-white border-second
ary">
164             <div class="modal-header border-secondary">
165                 <h5 class="modal-title d-flex align-items-center gap-2
" id="logReportsModalLabel">
166                     <i class="bi bi-journal-text text-success"></i> Отчё
ты AI-анализа логов
167                 </h5>
168                 <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
169             </div>
170             <div class="modal-body p-3">
171                 <div class="row g-3">
172                     <div class="col-md-4 border-end border-secondary">
173                         <h6 class="fw-bold mb-2 text-muted text-uppercase
small">Сформированные отчеты</h6>
174                         <div class="list-group list-group-flush" id="repor
ts-list-group" style="max-height: 60vh; overflow-y: auto;">
175                             <div class="text-muted small p-2">Загрузка списк
а отчётов...</div>
176                         </div>
177                     </div>
178                     <div class="col-md-8">
179                         <div class="d-flex justify-content-between align-i
tems-center mb-2">
180                             <h6 class="fw-bold mb-0 text-info" id="report-vi
ew-title">Выберите отчёт</h6>

```

```
181 |         <span class="small text-muted" id="report-view-d
    | ate"></span>
182 |     </div>
183 |     <div class="p-3 bg-black rounded border border-sec
    | ondary text-light small" id="report-view-content" style="max-h
    | eight: 58vh; overflow-y: auto; min-height: 300px; white-space:
    | pre-wrap;">
184 |         Выберите отчёт из списка слева для просмотра сод
    | ержимого.
185 |     </div>
186 | </div>
187 | </div>
188 | </div>
189 | <div class="modal-footer border-secondary">
190 |     <button type="button" class="btn btn-secondary btn-sm"
    | data-bs-dismiss="modal">Закрыть</button>
191 | </div>
192 | </div>
193 | </div>
194 | </div>
195 |
```

main.js

```
1 // =====
  =====
2 // Webinterface: Logs Tab Logic & AI Log Analysis
3 // Module: webinterface/logs/main.js
4 // Author: hypo69
5 // Copyright: © 2026 hypo69
6 // =====
  =====
7
8 'use strict';
9
10 (function () {
11     const state = {
12         files: [],
13         selectedFile: '',
14         rawContent: '',
15         lines: [],
16         filterLevel: 'ALL',
17         searchQuery: '',
18         tail: 250,
19         autoRefreshInterval: null,
20         isInitialized: false,
21         reports: []
22     };
23
24     /**
25      * Initializes the Logs inspection tab.
26      */
27     async function initLogsTab() {
28         console.log('[LogsTab] Initializing logs viewer...');
29         await loadLogStats();
30         await loadLogFilesList();
31         setupAutoRefresh();
32         state.isInitialized = true;
33     }
34 }
```



```

35 | /**
36 |  * Loads general statistics about system logs.
37 |  */
38 | async function loadLogStats() {
39 |     try {
40 |         const stats = await window.api.fetch('/api/logs/stats');
41 |         const filesCountEl = document.getElementById('log-stat-f
files-count');
42 |         const totalSizeEl = document.getElementById('log-stat-to
tal-size');
43 |         const sizeMbEl = document.getElementById('log-stat-size-
mb');
44 |         const reportsCountEl = document.getElementById('log-stat
-reports-count');
45 |
46 |         if (filesCountEl) filesCountEl.textContent = stats.file_
count ?? 0;
47 |         if (totalSizeEl) totalSizeEl.textContent = `${stats.tota
l_size_kb ?? 0} KB`;
48 |         if (sizeMbEl) sizeMbEl.textContent = `${stats.total_size
_mb ?? 0} MB на диске`;
49 |         if (reportsCountEl) reportsCountEl.textContent = stats.r
eport_count ?? 0;
50 |     } catch (e) {
51 |         console.warn('[LogsTab] Failed to load log stats:', e);
52 |     }
53 | }
54 |
55 | /**
56 |  * Fetches available log files from server.
57 |  */
58 | async function loadLogFilesList() {
59 |     const selector = document.getElementById('log-file-selecto
r');
60 |     if (!selector) return;
61 |
62 |     try {
63 |         const data = await window.api.fetch('/api/logs/files');

```

```

64 |     state.files = Array.isArray(data.files) ? data.files : [
65 | ];
66 |
67 |     state.files.sort((a, b) => new Date(b.modified || 0) - n
68 | ew Date(a.modified || 0));
69 |
70 |     if (state.files.length === 0) {
71 |         selector.innerHTML = '<option value="">(Лог-файлы не н
72 | айдены)</option>';
73 |         renderLogTerminal('Лог-файлы не найдены на сервере. ');
74 |         return;
75 |     }
76 |
77 |     selector.innerHTML = state.files.map(f => {
78 |         return `<option value="${f.name}">${f.name} (${f.size_
79 | kb} KB)</option>`;
80 |     }).join('');
81 |
82 |     // Auto-select preferred log file (fastapi.log or info.l
83 | og or first)
84 |     if (!state.selectedFile || !state.files.some(f => f.name
85 | === state.selectedFile)) {
86 |         const preferred = state.files.find(f => f.name === 'fa
87 | stapi.log') ||
88 |             state.files.find(f => f.name === 'in
89 | fo.log') ||
90 |             state.files.find(f => f.name.startsW
91 | ith('uvicorn_')) ||
92 |             state.files[0];
93 |         state.selectedFile = preferred ? preferred.name : stat
94 | e.files[0].name;
95 |     }
96 |
97 |     selector.value = state.selectedFile;
98 |     await readCurrentLogFile();
99 | } catch (err) {
100 |     console.error('[LogsTab] Error loading files list:', err
101 | );
102 |     selector.innerHTML = `<option value="">Ошибка: ${err.mes

```

```

sage}</option>`;
91     }
92 }
93
94 /**
95  * Reads contents of the currently selected log file.
96  */
97 async function readCurrentLogFile() {
98     if (!state.selectedFile) return;
99
100     const tailSelect = document.getElementById('log-tail-selector');
101     state.tail = tailSelect ? parseInt(tailSelect.value, 10) :
102     250;
103
104     try {
105         const url = `/api/logs/read?filename=${encodeURIComponent(
106             state.selectedFile)}&tail=${state.tail}`;
107         const res = await window.api.fetch(url);
108
109         state.rawContent = res.content || '';
110         // Reverse lines so newest entries are displayed first (
111         descending order by date)
112         state.lines = state.rawContent
113             .split(/\r?\n/)
114             .filter(line => line.trim().length > 0)
115             .reverse();
116
117         // Update terminal metadata
118         const filenameEl = document.getElementById('log-terminal-
119             filename');
120         const currentFilenameStat = document.getElementById('log-
121             stat-current-filename');
122         const currentLinesStat = document.getElementById('log-st
123             at-current-lines');
124         const timestampEl = document.getElementById('log-termina
125             l-timestamp');

```

```
120     if (filenameEl) filenameEl.textContent = res.filename ||
    state.selectedFile;
121     if (currentFilenameStat) currentFilenameStat.textContent
    = res.filename || state.selectedFile;
122     if (currentLinesStat) currentLinesStat.textContent = `${
    res.returned_lines || 0} / ${res.total_lines || 0}`;
123     if (timestampEl) timestampEl.textContent = new Date().to
    LocaleTimeString();
124
125     applyLogFiltersAndRender();
126 } catch (err) {
127     console.error('[LogsTab] Error reading log:', err);
128     renderLogTerminal(`Ошибка чтения лога: ${err.message}`);
129 }
130 }
131
132 /**
133  * Formats and highlights raw log lines.
134  */
135 function applyLogFiltersAndRender() {
136     const query = (state.searchQuery || '').toLowerCase().trim
    ();
137     const level = state.filterLevel;
138
139     const filtered = state.lines.filter(line => {
140         if (!line) return false;
141         if (query && !line.toLowerCase().includes(query)) return
    false;
142
143         if (level !== 'ALL') {
144             const upper = line.toUpperCase();
145             if (level === 'ERROR' && !upper.includes('ERROR') && !
    upper.includes('CRITICAL') && !upper.includes('EXCEPTION')) {
146                 return false;
147             }
148             if (level === 'WARNING' && !upper.includes('WARN') &&
    !upper.includes('WARNING')) {
149                 return false;
```

```
150 |     }
151 |     if (level === 'INFO' && !upper.includes('INFO')) {
152 |         return false;
153 |     }
154 |     if (level === 'DEBUG' && !upper.includes('DEBUG')) {
155 |         return false;
156 |     }
157 | }
158 | return true;
159 | });
160 |
161 | const badge = document.getElementById('log-visible-lines-b
162 | adge');
163 |
164 | if (badge) badge.textContent = `${filtered.length} отображ
165 | ается`;
166 |
167 | if (filtered.length === 0) {
168 |     renderLogTerminal('(Нет записей, удовлетворяющих условия
169 | м фильтра)');
170 |     return;
171 | }
172 |
173 | // Colorize lines
174 | const html = filtered.map(line => {
175 |     const escaped = escapeHtml(line);
176 |     const upper = line.toUpperCase();
177 |     if (upper.includes('ERROR') || upper.includes('CRITICAL'
178 | ) || upper.includes('EXCEPTION')) {
179 |         return `<span class="text-danger fw-semibold">${escape
180 | d}</span>`;
181 |     }
182 |     if (upper.includes('WARN') || upper.includes('WARNING'))
183 |     {
184 |         return `<span class="text-warning">${escaped}</span>`;
185 |     }
186 |     if (upper.includes('INFO')) {
187 |         return `<span class="text-info-emphasis" style="color:
188 | #67e8f9 !important;">${escaped}</span>`;
189 |     }
190 |     return `<span class="text-info">${escaped}</span>`;
191 | }
```

```
181     }
182     if (upper.includes('DEBUG')) {
183         return `<span class="text-success">${escaped}</span>`;
184     }
185     return `<span class="text-light">${escaped}</span>`;
186 }).join('\n');
187
188     const container = document.getElementById('log-output-cont
ainer');
189     if (container) {
190         container.innerHTML = html;
191     }
192 }
193
194     function renderLogTerminal(msg) {
195         const container = document.getElementById('log-output-cont
ainer');
196         if (container) container.textContent = msg;
197     }
198
199     function escapeHtml(str) {
200         return str
201             .replace(/&/g, '&amp;')
202             .replace(/</g, '&lt;')
203             .replace(/>/g, '&gt;')
204             .replace(/"/g, '&quot;');
205     }
206
207     function setupAutoRefresh() {
208         if (state.autoRefreshInterval) {
209             clearInterval(state.autoRefreshInterval);
210             state.autoRefreshInterval = null;
211         }
212
213         state.autoRefreshInterval = setInterval(() => {
214             const toggle = document.getElementById('logs-auto-refres
h');
215             if (toggle && toggle.checked) {
```

```
216 |         const tabPane = document.getElementById('tab-logs');
217 |         if (tabPane && tabPane.classList.contains('active')) {
218 |             readCurrentLogFile();
219 |             loadLogStats();
220 |         }
221 |     }
222 | }, 3000);
223 | }
224 |
225 | // --- Actions ---
226 |
227 | window.onLogFileSelected = function(filename) {
228 |     state.selectedFile = filename;
229 |     readCurrentLogFile();
230 | };
231 |
232 | window.refreshCurrentLog = async function() {
233 |     await loadLogStats();
234 |     await readCurrentLogFile();
235 | };
236 |
237 | window.clearCurrentLog = async function() {
238 |     if (!state.selectedFile) return;
239 |     if (!confirm(`Вы действительно хотите очистить файл "${state.selectedFile}"?`)) {
240 |         return;
241 |     }
242 |
243 |     try {
244 |         await window.api.fetch(`/api/logs/clear?filename=${encodeURIComponent(state.selectedFile)}`, {
245 |             method: 'DELETE'
246 |         });
247 |         await readCurrentLogFile();
248 |         await loadLogStats();
249 |         if (typeof window.showNotification === 'function') {
250 |             window.showNotification(`Лог ${state.selectedFile} очищен`, 'success');
```

```
251     }
252   } catch (err) {
253     alert(`Ошибка очистки лога: ${err.message}`);
254   }
255 };
256
257 window.triggerLogAnalysis = async function() {
258   if (!state.selectedFile) return;
259   try {
260     const res = await window.api.fetch('/api/logs/analyze',
261     {
262       method: 'POST',
263       headers: { 'Content-Type': 'application/json' },
264       body: JSON.stringify({ filename: state.selectedFile })
265     });
266     alert(res.message || 'Анализ запущен в фоне. Результаты
267     появятся во вкладке отчётов.');
```

появятся во вкладке отчётов.');

```
266     await loadLogStats();
267   } catch (err) {
268     alert(`Ошибка вызова AI-анализа: ${err.message}`);
269   }
270 };
271
272 window.setLogLevelFilter = function(level) {
273   state.filterLevel = level;
274   document.querySelectorAll('#log-level-filters button').forEach(b => {
275     b.classList.toggle('active', b.getAttribute('data-level') === level);
276   });
277   applyLogFiltersAndRender();
278 };
279
280 window.filterLogsInViewer = function() {
281   const input = document.getElementById('log-search-input');
282   state.searchQuery = input ? input.value : '';
283   applyLogFiltersAndRender();
284 };
```



```
285 |
286 | window.clearLogSearch = function() {
287 |     const input = document.getElementById('log-search-input');
288 |     if (input) input.value = '';
289 |     state.searchQuery = '';
290 |     applyLogFiltersAndRender();
291 | };
292 |
293 | window.copyLogContent = function() {
294 |     if (!state.rawContent) return;
295 |     navigator.clipboard.writeText(state.rawContent).then(() =>
296 | {
297 |         alert('Содержимое скопировано в буфер обмена');
298 |     }).catch(e => {
299 |         alert('Ошибка копирования: ' + e);
300 |     });
301 | };
302 |
303 | window.downloadLogFile = function() {
304 |     if (!state.rawContent || !state.selectedFile) return;
305 |     const blob = new Blob([state.rawContent], { type: 'text/pl
306 | ain;charset=utf-8' });
307 |     const a = document.createElement('a');
308 |     a.href = URL.createObjectURL(blob);
309 |     a.download = state.selectedFile;
310 |     a.click();
311 |     URL.revokeObjectURL(a.href);
312 | };
313 |
314 | window.scrollToBottom = function() {
315 |     const container = document.getElementById('log-output-cont
316 | ainer');
317 |     if (container) {
318 |         container.scrollTop = container.scrollHeight;
319 |     }
320 | };
321 |
322 | window.toggleReportsModal = async function() {
```

```

320 |     const modalEl = document.getElementById('logReportsModal')
    | ;
321 |     if (!modalEl) return;
322 |     const modal = new bootstrap.Modal(modalEl);
323 |     modal.show();
324 |     await loadAiReports();
325 | };
326 |
327 | async function loadAiReports() {
328 |     const listGroup = document.getElementById('reports-list-gr
    | oup');
329 |     if (!listGroup) return;
330 |
331 |     try {
332 |         const data = await window.api.fetch('/api/logs/reports')
    | ;
333 |         const reports = data.reports || [];
334 |         if (reports.length === 0) {
335 |             listGroup.innerHTML = '<div class="text-muted small p-
    | 2">Отчёты пока не сгенерированы. Запустите AI-анализ.</div>';
336 |             return;
337 |         }
338 |
339 |         listGroup.innerHTML = reports.map((r, i) => {
340 |             return `
341 |                 <button type="button" class="list-group-item list-gr
    | oup-item-action bg-dark text-white border-secondary small ${i
    | === 0 ? 'active' : ''}" onclick="viewReportContent('${r.name}'
    | , '${r.modified}')">
342 |                     <div class="fw-semibold text-truncate">${r.name}</
    | div>
343 |                     <div class="text-muted" style="font-size: 0.75rem;
    | ">${r.modified} (${r.size_kb} KB)</div>
344 |                 </button>
345 |             `;
346 |         }).join('');
347 |
348 |         if (reports[0]) {

```

```
349 |         await viewReportContent(reports[0].name, reports[0].mo
      | dified);
350 |     }
351 | } catch (e) {
352 |     listGroup.innerHTML = `<div class="text-danger small p-2
      | ">Ошибка: ${e.message}</div>`;
353 | }
354 | }
355 |
356 | window.viewReportContent = async function(filename, date) {
357 |     const titleEl = document.getElementById('report-view-title
      | ');
358 |     const dateEl = document.getElementById('report-view-date')
      | ;
359 |     const contentEl = document.getElementById('report-view-con
      | tent');
360 |
361 |     if (titleEl) titleEl.textContent = filename;
362 |     if (dateEl) dateEl.textContent = date;
363 |     if (contentEl) contentEl.textContent = 'Загрузка отчёта...
      | ';
364 |
365 |     try {
366 |         const res = await window.api.fetch(`/api/logs/report?fil
      | ename=${encodeURIComponent(filename)}`);
367 |         if (contentEl) {
368 |             contentEl.textContent = res.content || '(Пустой отчёт)
      | ';
369 |         }
370 |     } catch (e) {
371 |         if (contentEl) contentEl.textContent = `Ошибка загрузки
      | отчёта: ${e.message}`;
372 |     }
373 | };
374 |
375 | // Global export
376 | window.initLogsTab = initLogsTab;
377 | })();
```


index.html

```
1 <!-- Maintenance & Recovery Web Tab -->
2 <style>
3   .mtn-container {
4     padding: 0.75rem 1rem;
5   }
6   .mtn-header {
7     background: var(--surface-2, #1e293b);
8     border: 1px solid var(--border-color, #334155);
9     border-radius: var(--radius-md, 8px);
10    padding: 0.75rem 1.25rem;
11  }
12  .mtn-table {
13    width: 100%;
14    border-collapse: collapse;
15    margin-bottom: 0;
16  }
17  .mtn-table th {
18    padding: 0.65rem 1rem;
19    font-size: 0.75rem;
20    font-weight: 600;
21    text-transform: uppercase;
22    letter-spacing: 0.05em;
23    color: var(--text-muted, #94a3b8);
24    background: rgba(255, 255, 255, 0.02);
25    border-bottom: 1px solid var(--border-color, #334155);
26    white-space: nowrap;
27  }
28  .mtn-table td {
29    padding: 0.65rem 1rem;
30    font-size: 0.85rem;
31    color: var(--text-color, #f8fafc);
32    border-bottom: 1px solid rgba(255, 255, 255, 0.05);
33    vertical-align: middle;
34  }
35  .mtn-table tr:last-child td {
36    border-bottom: none;
```

```

37 | }
38 | .mtn-table tbody tr:hover td {
39 |     background: rgba(255, 255, 255, 0.04);
40 | }
41 | </style>
42 |
43 | <div class="mtn-container theme-sysctl">
44 |     <!-- Header Bar -->
45 |     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 mtn-header text-white">
46 |         <div>
47 |             <div class="d-flex align-items-center gap-2">
48 |                 <h5 class="mb-0 fw-bold text-white">
49 |                     □ Обслуживание и восстановление
50 |                 </h5>
51 |                 <span class="badge rounded-pill bg-primary px-2 py-1" style="font-size: 0.75rem;">Maintenance & Protection</span>
52 |             </div>
53 |             <p class="mb-0 small text-light mt-1" style="color: #cbd5e1 !important;">
54 |                 Быстрая очистка кэшей, проверка целостности системных
55 |                 файлов Windows (SFC / DISM) и управление точками восстановления.
56 |             </p>
57 |         </div>
58 |         <div class="d-flex align-items-center gap-2 flex-wrap">
59 |             <button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-mtn-refresh">
60 |                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
61 |             </button>
62 |         </div>
63 |     </div>
64 |     <!-- Quick Action Cards & Restore Points -->
65 |     <div class="row g-3">
66 |         <!-- Disk & Cache Cleanup -->
67 |         <div class="col-md-4">
68 |             <div class="card bg-dark border-secondary p-3 h-100 d-fl

```

```

ex flex-column justify-content-between">
69     <div>
70         <h6 class="fw-bold text-info"><i class="bi bi-trash3
- fill me-1"></i> Disk & Cache Cleanup</h6>
71         <p class="small text-muted mb-2">Безопасная очистка
временных файлов, логов установки и кэша обновлений Windows.</
p>
72         <div class="small fw-semibold text-white mb-3" id="m
tn-clean-estimate-txt">Очищаемый объём: ~0 MB</div>
73     </div>
74     <button class="btn btn-sm btn-outline-info w-100 round
ed-pill" id="btn-mtn-action-clean">
75         <i class="bi bi-stars me-1"></i> Очистить кэши и вре
менные файлы
76     </button>
77 </div>
78 </div>
79
80 <!-- SFC System File Check -->
81 <div class="col-md-4">
82     <div class="card bg-dark border-secondary p-3 h-100 d-fl
ex flex-column justify-content-between">
83         <div>
84             <h6 class="fw-bold text-warning"><i class="bi bi-shi
eld-shaded me-1"></i> SFC System File Check</h6>
85             <p class="small text-muted mb-2">Проверка целостност
и защищенных файлов системы (sfc /verifyonly).</p>
86         </div>
87         <button class="btn btn-sm btn-outline-warning w-100 ro
unded-pill" id="btn-mtn-action-sfc">
88             <i class="bi bi-search me-1"></i> Запустить проверку
SFC
89         </button>
90     </div>
91 </div>
92
93 <!-- DISM Health Check -->
94 <div class="col-md-4">

```

```

95 |         <div class="card bg-dark border-secondary p-3 h-100 d-fl
    | ex flex-column justify-content-between">
96 |             <div>
97 |                 <h6 class="fw-bold text-success"><i class="bi bi-hea
    | rt-pulse-fill me-1"></i> DISM Health Check</h6>
98 |                 <p class="small text-muted mb-2">Проверка состояния
    | хранилища компонентов Windows через DISM /CheckHealth.</p>
99 |             </div>
100 |             <button class="btn btn-sm btn-outline-success w-100 ro
    | unded-pill" id="btn-mtn-action-dism">
101 |                 <i class="bi bi-activity me-1"></i> Проверить хранил
    | ище DISM
102 |             </button>
103 |         </div>
104 |     </div>
105 |
106 |     <!-- System Restore Points Panel -->
107 |     <div class="col-12">
108 |         <div class="card bg-dark border-secondary">
109 |             <div class="card-header border-secondary bg-transparen
    | t d-flex justify-content-between align-items-center flex-wrap
    | gap-2">
110 |                 <span class="fw-bold text-white"><i class="bi bi-arr
    | ow-counterclockwise me-1"></i> Точки восстановления Windows</s
    | pan>
111 |                 <div class="d-flex align-items-center gap-2">
112 |                     <button class="btn btn-sm btn-outline-info rounded
    | -pill px-3" id="btn-mtn-config-restore" title="Настройка квоты
    | диска, расписания и политик хранения">
113 |                         <i class="bi bi-gear-fill me-1"></i> Настройки и
    | квоты
114 |                     </button>
115 |                     <button class="btn btn-sm btn-outline-primary roun
    | ded-pill px-3" id="btn-mtn-create-restore">
116 |                         <i class="bi bi-plus-lg me-1"></i> Создать точку
    | восстановления
117 |                     </button>
118 |                 </div>

```



```

119 |         </div>
120 |         <div class="p-0 table-responsive" style="max-height: 3
121 | 80px; overflow-y: auto;">
122 |             <table class="mtn-table">
123 |                 <thead>
124 |                     <tr>
125 |                         <th style="width: 100px;">№</th>
126 |                         <th style="width: 45%;">Описание</th>
127 |                         <th style="width: 25%;">Тип</th>
128 |                         <th style="width: 20%;">Время создания</th>
129 |                     </tr>
130 |                 </thead>
131 |                 <tbody id="mtn-restore-tbody">
132 |                     <tr><td colspan="4" class="text-center text-mute
133 | d p-3">Загрузка точек восстановления...</td></tr>
134 |                 </tbody>
135 |             </table>
136 |         </div>
137 |     </div>
138 | </div>
139 |
140 | <!-- Modal: Restore Points Policy & Storage Config -->
141 | <div class="modal fade" id="modal-mtn-restore-config" tabindex
142 | ="-1" aria-labelledby="modalMtnRestoreConfigLabel" aria-hidden
143 | ="true">
144 |     <div class="modal-dialog modal-dialog-centered">
145 |         <div class="modal-content bg-dark text-white border-second
146 | ary">
147 |             <div class="modal-header border-secondary">
148 |                 <h5 class="modal-title fw-bold" id="modalMtnRestoreCon
149 | figLabel">
150 |                     <i class="bi bi-gear-fill text-info me-2"></i> Полит
151 | ики и дисковая квота
152 |                 </h5>
153 |                 <button type="button" class="btn-close btn-close-white
154 | " data-bs-dismiss="modal" aria-label="Close"></button>

```

```

149 |         </div>
150 |         <div class="modal-body">
151 |             <form id="form-mtn-restore-policy">
152 |                 <div class="mb-3">
153 |                     <label class="form-label small text-muted">Максима
154 | льный объем диска под точки восстановления</label>
155 |                     <select class="form-select bg-dark text-white bord
156 | er-secondary form-select-sm" id="mtn-policy-storage-size">
157 |                         <option value="3%">3% (Минимальный)</option>
158 |                         <option value="5%">5% (Рекомендуемый)</option>
159 |                         <option value="10%" selected>10% (Стандартный)</
160 | option>
161 |                         <option value="15%">15% (Расширенный)</option>
162 |                         <option value="20%">20% (Максимальный)</option>
163 |                     </select>
164 |                     <div class="progress mt-2" style="height: 6px;" id
165 | ="mtn-policy-storage-progress">
166 |                         <div class="progress-bar bg-info" id="mtn-policy
167 | -storage-bar" role="progressbar" style="width: 0%"></div>
168 |                     </div>
169 |                     <div class="d-flex justify-content-between small t
170 | ext-muted mt-1" id="mtn-policy-storage-label" style="font-size
171 | : 0.75rem;">
172 |                         <span>Занято: 0 GB</span><span>Лимит: 0 GB</span
173 | >
174 |                     </div>
175 |                 </div>
176 |             </div>
177 |
178 |             <div class="mb-3">
179 |                 <label class="form-label small text-muted">Автомат
180 | ическое расписание создания</label>
181 |                 <select class="form-select bg-dark text-white bord
182 | er-secondary form-select-sm" id="mtn-policy-trigger">
183 |                     <option value="daily">Ежедневно</option>
184 |                     <option value="weekly">Еженедельно</option>
185 |                     <option value="on_update">Перед обновлениями Win
186 | dows</option>
187 |                     <option value="manual">Только вручную</option>

```

```
176         </select>
177     </div>
178
179     <div class="mb-3" id="mtn-policy-time-wrapper">
180         <label class="form-label small text-muted">Время с
оздания точки</label>
181         <input type="time" class="form-control bg-dark tex
t-white border-secondary form-select-sm" id="mtn-policy-time"
value="03:00">
182     </div>
183
184     <div class="mb-3">
185         <label class="form-label small text-muted">Максима
льное число сохраняемых точек</label>
186         <select class="form-select bg-dark text-white bord
er-secondary form-select-sm" id="mtn-policy-max-points">
187             <option value="3">3 последних</option>
188             <option value="5" selected>5 последних</option>
189             <option value="10">10 последних</option>
190             <option value="20">20 последних</option>
191         </select>
192     </div>
193
194     <div class="form-check form-switch mb-2">
195         <input class="form-check-input" type="checkbox" id
="mtn-policy-freq-limit" checked>
196         <label class="form-check-label small text-light" f
or="mtn-policy-freq-limit">Ограничивать частоту (не чаще 1 раз
а в 24 часа)</label>
197     </div>
198
199     <div class="form-check form-switch mb-3">
200         <input class="form-check-input" type="checkbox" id
="mtn-policy-auto-prune" checked>
201         <label class="form-check-label small text-light" f
or="mtn-policy-auto-prune">Автоматически удалять старые точки
при превышении квоты</label>
202     </div>
```

```
203 |         </form>
204 |     </div>
205 |     <div class="modal-footer border-secondary d-flex justify
    | -content-between">
206 |         <button type="button" class="btn btn-sm btn-outline-da
    | nger" id="btn-mtn-policy-prune-now">
207 |             <i class="bi bi-trash me-1"></i> Очистить старые сей
    | час
208 |         </button>
209 |         <div class="d-flex gap-2">
210 |             <button type="button" class="btn btn-sm btn-secondar
    | y" data-bs-dismiss="modal">Отмена</button>
211 |             <button type="button" class="btn btn-sm btn-primary"
    | id="btn-mtn-save-restore-policy">
212 |                 <i class="bi bi-check-lg me-1"></i> Сохранить и пр
    | именить
213 |             </button>
214 |         </div>
215 |     </div>
216 | </div>
217 | </div>
218 | </div>
219 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Maintenance & Recovery Web Controller
4   * =====
5   * Description:
6   *   Клиентский контроллер вкладки «Обслуживание и восстано
7   *   очисткой временных файлов, проверкой системных файлов (SF
8   *   точками восстановления Windows.
9   *
10  * File: main.js
11  * Project: ai-breadboard
12  * Package: src.api.webinterface.maintenance_tab
13  * Author: hypo69
14  * Copyright: © 2026 hypo69
15  * =====
16  */
17
18 (function () {
19   'use strict';
20
21   let isInitialized = false;
22
23   function formatRestorePointType(rawType) {
24     if (!rawType) return 'System Checkpoint';
25     const s = String(rawType).toUpperCase();
26     if (s.includes('APPLICATION_INSTALL') || s === '0') return
27     'App Install';
28     if (s.includes('APPLICATION_UNINSTALL') || s === '1') retu
29     rn 'App Uninstall';
30     if (s.includes('DEVICE_DRIVER') || s === '10') return 'Dri
31     ver Install';
```

```

29     if (s.includes('MODIFY_SETTINGS') || s === '12') return 'Settings Change';
30     if (s.includes('CANCELLED') || s === '13') return 'Cancelled Op';
31     if (s.includes('WINDOWS_UPDATE') || s === '17') return 'Windows Update';
32     return s.replace(/_/g, ' ');
33 }
34
35 function formatRestoreTime(rawTime) {
36     if (!rawTime) return '-';
37     const s = String(rawTime).trim();
38     if (s.length >= 14 && /^d{14}/.test(s)) {
39         return `${s.slice(0, 4)}-${s.slice(4, 6)}-${s.slice(6, 8)} ${s.slice(8, 10)}:${s.slice(10, 12)}:${s.slice(12, 14)}`;
40     }
41     return s;
42 }
43
44 async function fetchMaintenanceStatus() {
45     try {
46         const res = await fetch('/api/system-control/status');
47         if (!res.ok) return;
48         const data = await res.json();
49         const disk = data.disk || {};
50
51         const cleanEstimateTxt = document.getElementById('mtn-clean-estimate-txt');
52         if (cleanEstimateTxt) {
53             const cleanMb = disk.cleanup_estimate?.total_cleanable_mb || 0;
54             cleanEstimateTxt.innerText = `Очищаемый объём: ~${cleanMb} MB`;
55         }
56
57         await fetchRestorePoints();
58     } catch (e) {
59         console.error('[MaintenanceTab] Failed to fetch maintena

```

```

    nce status:', e);
60     }
61   }
62
63   async function fetchRestorePoints() {
64     try {
65       const res = await fetch('/api/system-control/restore-points');
66       if (!res.ok) return;
67       const data = await res.json();
68       const points = data.restore_points || [];
69
70       const tbody = document.getElementById('mtn-restore-tbody');
71       if (!tbody) return;
72
73       if (points.length === 0) {
74         tbody.innerHTML = '<tr><td colspan="4" class="text-center text-muted p-3">Точки восстановления не найдены. Нажмите "Создать точку восстановления".</td></tr>';
75         return;
76       }
77
78       tbody.innerHTML = points.map((p, idx) => {
79         const typeBadge = formatRestorePointType(p.restore_point_type);
80         const timeFormatted = formatRestoreTime(p.creation_time);
81         return `
82         <tr class="mtn-restore-row" data-idx="${idx}" style="
83         cursor: pointer;" title="Нажмите для детальной информации">
84           <td class="font-monospace text-info fw-bold">#${p.sequence_number}</td>
85           <td class="fw-semibold text-white">${p.description}</td>
86           <td><span class="badge bg-secondary text-light px-2 py-1 font-monospace" style="font-size: 0.75rem;">${typeBadge}</span></td>

```

```

86 |         <td class="small text-light font-monospace">${time
Formatted}</td>
87 |     </tr>
88 |     `;
89 |     }).join('');
90 |
91 |     tbody.querySelectorAll('.mtn-restore-row').forEach(row =
92 | > {
93 |         row.onclick = () => {
94 |             const idx = parseInt(row.getAttribute('data-idx'), 1
95 | 0);
96 |             const p = points[idx];
97 |             if (!p) return;
98 |             if (window.AITableModal) {
99 |                 window.AITableModal.show({
100 |                     icon: '🔍 ',
101 |                     title: `Точка восстановления #${p.sequence_numbe
102 | r}`,
103 |                     subtitle: p.description,
104 |                     tableType: 'generic',
105 |                     badges: [
106 |                         { text: p.restore_point_type || 'System Checkp
oint', class: 'badge bg-info text-dark' }
107 |                     ],
108 |                     metadata: [
109 |                         { label: 'Номер', value: String(p.sequence_num
ber) },
110 |                         { label: 'Описание', value: p.description },
111 |                         { label: 'Тип', value: p.restore_point_type },
112 |                         { label: 'Дата создания', value: p.creation_ti
me }
113 |                     ],
114 |                     rawTitle: 'Метаданные точки восстановления',
115 |                     rawContent: JSON.stringify(p, null, 2)
116 |                 });
117 |             }
118 |         };
119 |     });

```



```
117     } catch (e) {
118         console.error('[MaintenanceTab] Failed to fetch restore
points:', e);
119     }
120 }
121
122 async function loadRestorePolicyModal() {
123     try {
124         const res = await fetch('/api/system-control/restore-poi
nts/config');
125         if (!res.ok) return;
126         const data = await res.json();
127
128         const storageSel = document.getElementById('mtn-policy-s
torage-size');
129         const triggerSel = document.getElementById('mtn-policy-t
rigger');
130         const timeInput = document.getElementById('mtn-policy-ti
me');
131         const timeWrapper = document.getElementById('mtn-policy-
time-wrapper');
132         const freqChk = document.getElementById('mtn-policy-freq
-limit');
133         const maxPtsSel = document.getElementById('mtn-policy-ma
x-points');
134         const autoPruneChk = document.getElementById('mtn-policy
-auto-prune');
135         const storageBar = document.getElementById('mtn-policy-s
torage-bar');
136         const storageLabel = document.getElementById('mtn-policy
-storage-label');
137
138         if (storageSel && data.max_storage_size) storageSel.valu
e = data.max_storage_size;
139         if (triggerSel && data.schedule_trigger) {
140             triggerSel.value = data.schedule_trigger;
141             if (timeWrapper) timeWrapper.style.display = (data.sch
edule_trigger === 'daily' || data.schedule_trigger === 'weekly
```

```

    ')? ? 'block' : 'none';
142     }
143     if (timeInput && data.schedule_time) timeInput.value = d
ata.schedule_time;
144     if (freqChk) freqChk.checked = data.frequency_limit_minu
tes === 0;
145     if (maxPtsSel && data.max_points) maxPtsSel.value = Stri
ng(data.max_points);
146     if (autoPruneChk && data.auto_prune !== undefined) autoP
runeChk.checked = Boolean(data.auto_prune);
147
148     const st = data.storage_info || {};
149     if (storageBar && storageLabel) {
150         const pct = st.usage_percent || 0;
151         const usedGb = (st.used_bytes ? (st.used_bytes / (1024
** 3)).toFixed(2) : '0.00');
152         const maxGb = (st.max_bytes ? (st.max_bytes / (1024 **
3)).toFixed(2) : '0.00');
153         storageBar.style.width = `${Math.min(pct, 100)}%`;
154         storageBar.className = pct > 85 ? 'progress-bar bg-dan
ger' : (pct > 60 ? 'progress-bar bg-warning' : 'progress-bar b
g-info');
155         storageLabel.innerHTML = `<span>Занято: ${usedGb} GB (
${pct}%)</span><span>Лимит: ${maxGb} GB</span>`;
156     }
157
158     if (triggerSel) {
159         triggerSel.onchange = () => {
160             if (timeWrapper) timeWrapper.style.display = (trigge
rSel.value === 'daily' || triggerSel.value === 'weekly') ? 'bl
ock' : 'none';
161         };
162     }
163
164     const modalEl = document.getElementById('modal-mtn-resto
re-config');
165     if (modalEl && window.bootstrap?.Modal) {
166         const modal = bootstrap.Modal.getOrCreateInstance(modalEl)

```

```

165 |         lEl);
167 |         modal.show();
168 |     }
169 |     } catch (e) {
170 |         console.error('[MaintenanceTab] Failed to load restore p
171 |         olicy config:', e);
172 |     }
173 |
174 |     async function saveRestorePolicy() {
175 |         const saveBtn = document.getElementById('btn-mtn-save-rest
176 |         ore-policy');
177 |         const storageSel = document.getElementById('mtn-policy-sto
178 |         rage-size');
179 |         const triggerSel = document.getElementById('mtn-policy-tri
180 |         gger');
181 |         const timeInput = document.getElementById('mtn-policy-time
182 |         ');
183 |         const freqChk = document.getElementById('mtn-policy-freq-l
184 |         imit');
185 |         const maxPtsSel = document.getElementById('mtn-policy-max-
186 |         points');
187 |         const autoPruneChk = document.getElementById('mtn-policy-a
188 |         uto-prune');
189 |
190 |         const payload = {
191 |             max_storage_size: storageSel ? storageSel.value : '10%',
192 |             schedule_trigger: triggerSel ? triggerSel.value : 'daily
193 |             ',
194 |             schedule_time: timeInput ? timeInput.value : '03:00',
195 |             frequency_limit_minutes: freqChk && freqChk.checked ? 0
196 |             : 1440,
197 |             max_points: maxPtsSel ? parseInt(maxPtsSel.value, 10) :
198 |             5,
199 |             auto_prune: autoPruneChk ? autoPruneChk.checked : true,
200 |         };
201 |
202 |         if (saveBtn) {

```

```
193 |         saveBtn.disabled = true;
194 |         saveBtn.innerText = 'Сохранение...';
195 |     }
196 |
197 |     try {
198 |         const res = await fetch('/api/system-control/restore-points/config', {
199 |             method: 'POST',
200 |             headers: { 'Content-Type': 'application/json' },
201 |             body: JSON.stringify(payload)
202 |         });
203 |         const d = await res.json();
204 |         window.showToast?.(d.message || 'Политика успешно сохранена', d.success ? 'success' : 'danger') || alert(d.message || 'Политика сохранена');
205 |         const modalEl = document.getElementById('modal-mtn-restore-config');
206 |         if (modalEl && window.bootstrap?.Modal) {
207 |             const modal = bootstrap.Modal.getInstance(modalEl);
208 |             modal?.hide();
209 |         }
210 |         await fetchRestorePoints();
211 |     } catch (e) {
212 |         console.error('[MaintenanceTab] Failed to save restore policy:', e);
213 |         alert('Ошибка сохранения политики: ' + e);
214 |     } finally {
215 |         if (saveBtn) {
216 |             saveBtn.disabled = false;
217 |             saveBtn.innerHTML = '<i class="bi bi-check-lg me-1"></i> Сохранить и применить';
218 |         }
219 |     }
220 | }
221 |
222 | async function pruneRestorePointsNow() {
223 |     const maxPtsSel = document.getElementById('mtn-policy-max-points');
```

```
224     const keep = maxPtsSel ? parseInt(maxPtsSel.value, 10) : 5
    ;
225     if (!confirm(`Выполнить ротацию точек восстановления и ост
    авить только ${keep} последних?`)) {
226         return;
227     }
228     try {
229         const res = await fetch('/api/system-control/restore-poi
    nts/prune', {
230             method: 'POST',
231             headers: { 'Content-Type': 'application/json' },
232             body: JSON.stringify({ keep_count: keep })
233         });
234         const d = await res.json();
235         window.showToast?.(d.message || 'Ротация выполнена', 'in
    fo') || alert(d.message || 'Ротация выполнена');
236         await fetchRestorePoints();
237     } catch (e) {
238         console.error('[MaintenanceTab] Failed to prune restore
    points:', e);
239     }
240 }
241
242 function initMaintenanceTab() {
243     console.log('[MaintenanceTab] Initializing Maintenance & R
    ecovery tab...');
244     fetchMaintenanceStatus();
245
246     if (!isInitialized) {
247         const refreshBtn = document.getElementById('btn-mtn-refr
    esh');
248         const cleanBtn = document.getElementById('btn-mtn-action
    -clean');
249         const sfcBtn = document.getElementById('btn-mtn-action-s
    fc');
250         const dismBtn = document.getElementById('btn-mtn-action-
    dism');
251         const createRestoreBtn = document.getElementById('btn-mt
```

```

n-create-restore');
252     const configRestoreBtn = document.getElementById('btn-mt
n-config-restore');
253     const savePolicyBtn = document.getElementById('btn-mtn-s
ave-restore-policy');
254     const pruneNowBtn = document.getElementById('btn-mtn-pol
icy-prune-now');
255
256     if (refreshBtn) refreshBtn.onclick = () => fetchMaintena
nceStatus();
257
258     if (cleanBtn) {
259         cleanBtn.onclick = async () => {
260             if (confirm('Выполнить безопасную очистку временных
директорий и кэша обновлений?')) {
261                 cleanBtn.disabled = true;
262                 await fetch('/api/system-control/maintenance/clean
up', { method: 'POST' });
263                 cleanBtn.disabled = false;
264                 fetchMaintenanceStatus();
265             }
266         };
267     }
268
269     if (sfcBtn) {
270         sfcBtn.onclick = async () => {
271             sfcBtn.disabled = true;
272             sfcBtn.innerText = 'Сканирование...';
273             const res = await fetch('/api/system-control/mainten
ance/sfc', { method: 'POST' });
274             const d = await res.json();
275             window.showToast?.(d.message || 'Проверка SFC заверш
ена', d.status === 'error' ? 'danger' : 'success') || alert(d.
message || 'Проверка SFC завершена');
276             sfcBtn.disabled = false;
277             sfcBtn.innerHTML = '<i class="bi bi-search me-1"></i
> Запустить проверку SFC';
278         };

```

```
279 |     }
280 |
281 |     if (dismBtn) {
282 |         dismBtn.onclick = async () => {
283 |             dismBtn.disabled = true;
284 |             dismBtn.innerText = 'Проверка...';
285 |             const res = await fetch('/api/system-control/mainten
ance/dism', { method: 'POST' });
286 |             const d = await res.json();
287 |             window.showToast?.(d.message || 'Проверка DISM завер
шена', d.status === 'error' ? 'danger' : 'success') || alert(d
.message || 'Проверка DISM завершена');
288 |             dismBtn.disabled = false;
289 |             dismBtn.innerHTML = '<i class="bi bi-activity me-1">
</i> Проверить хранилище DISM';
290 |         };
291 |     }
292 |
293 |     if (createRestoreBtn) {
294 |         createRestoreBtn.onclick = async () => {
295 |             const desc = prompt('Введите описание точки восстано
вления:', 'Точка восстановления системы');
296 |             if (desc) {
297 |                 createRestoreBtn.disabled = true;
298 |                 const res = await fetch('/api/system-control/resto
re-points', {
299 |                     method: 'POST',
300 |                     headers: { 'Content-Type': 'application/json' },
301 |                     body: JSON.stringify({ description: desc })
302 |                 });
303 |                 const d = await res.json();
304 |                 window.showToast?.(d.message || 'Точка восстано
вления создана.', d.status === 'error' ? 'danger' : 'success') ||
alert(d.message || 'Точка восстановления создана.');
```

```
305 |                 createRestoreBtn.disabled = false;
306 |                 fetchRestorePoints();
307 |             }
308 |         };

```

```
309 |     }
310 |
311 |     if (configRestoreBtn) configRestoreBtn.onclick = loadRes
|     torePolicyModal;
312 |     if (savePolicyBtn) savePolicyBtn.onclick = saveRestorePo
|     licy;
313 |     if (pruneNowBtn) pruneNowBtn.onclick = pruneRestorePoint
|     sNow;
314 |
315 |     isInitialized = true;
316 | }
317 | }
318 |
319 | window.initMaintenanceTab = initMaintenanceTab;
320 |})();
321 |
```


index.html

```
1 <!-- MCP Servers Tab Component -->
2 <div class="container-fluid p-3" id="mcp-tab-root">
3   <!-- Top Banner / Summary -->
4   <div class="alert alert-dark border-secondary bg-dark text-1
light d-flex flex-wrap align-items-center justify-content-betwe
en gap-3 shadow-sm">
5     <div class="d-flex align-items-center gap-3">
6       <div class="p-3 bg-black rounded border border-secondary
text-primary display-6 d-flex align-items-center justify-cont
ent-center" style="width: 54px; height: 54px;">
7         <i class="bi bi-hdd-network-fill"></i>
8       </div>
9       <div>
10        <h5 class="mb-1 text-white d-flex align-items-center g
ap-2">
11          □ Серверы MCP (Model Context Protocol)
12          <span class="badge bg-primary rounded-pill small" id
="mcp-total-badge">0</span>
13        </h5>
14        <p class="mb-0 text-muted small">
15          Управление внешними MCP-серверами, интеграция инстру
ментов браузера, файловой системы, баз данных и API для ИИ-аге
нтов.
16        </p>
17      </div>
18    </div>
19    <div class="d-flex align-items-center gap-2 flex-wrap">
20      <button type="button" class="btn btn-outline-info btn-sm
d-flex align-items-center gap-1" id="btn-view-all-tools" oncl
ick="window.mcpTab?.openAllToolsModal()">
21        <i class="bi bi-tools"></i> Все инструменты
22      </button>
23      <button type="button" class="btn btn-outline-light btn-s
m d-flex align-items-center gap-1" id="btn-refresh-mcp" onclic
k="window.mcpTab?.refresh()">
24        <i class="bi bi-arrow-clockwise"></i> Обновить
```

```
25 |         </button>
26 |         <button type="button" class="btn btn-primary btn-sm d-fl
    | ex align-items-center gap-1 shadow-sm" data-bs-toggle="modal"
    | data-bs-target="#mcpServerModal" onclick="window.mcpTab?.openC
    | reateModal()">
27 |             <i class="bi bi-plus-circle-fill"></i> Добавить MCP се
    | рвер
28 |         </button>
29 |     </div>
30 | </div>
31 |
32 | <!-- Statistics Row -->
33 | <div class="row g-2 mb-3" id="mcp-stats-row">
34 |     <div class="col-6 col-md-3">
35 |         <div class="p-2 bg-dark rounded border border-secondary
    | text-center">
36 |             <div class="h4 mb-0 text-white" id="stat-total-mcp">0<
    | /div>
37 |             <div class="small text-muted">Всего серверов</div>
38 |         </div>
39 |     </div>
40 |     <div class="col-6 col-md-3">
41 |         <div class="p-2 bg-body-tertiary rounded border text-cen
    | ter">
42 |             <div class="h4 mb-0 text-success" id="stat-active-mcp"
    | >0</div>
43 |             <div class="small text-muted">Активных серверов</div>
44 |         </div>
45 |     </div>
46 |     <div class="col-6 col-md-3">
47 |         <div class="p-2 bg-body-tertiary rounded border text-cen
    | ter">
48 |             <div class="h4 mb-0 text-info" id="stat-stdio-mcp">0</
    | div>
49 |             <div class="small text-muted">Stdio-процессов</div>
50 |         </div>
51 |     </div>
52 | <div class="col-6 col-md-3">
```

```

53 |     <div class="p-2 bg-body-tertiary rounded border text-cen
    | ter">
54 |         <div class="h4 mb-0 text-warning" id="stat-network-mcp
    | ">0</div>
55 |         <div class="small text-muted">SSE / HTTP серверов</div
    | >
56 |     </div>
57 | </div>
58 | </div>
59 |
60 | <!-- Search & View Controls -->
61 | <div class="card bg-body-tertiary border mb-3 shadow-sm">
62 |     <div class="card-body p-2 d-flex flex-wrap align-items-cen
    | ter justify-content-between gap-2">
63 |         <div class="input-group input-group-sm flex-grow-1" styl
    | e="max-width: 450px;">
64 |             <span class="input-group-text"><i class="bi bi-search"
    | "></i></span>
65 |             <input type="text" class="form-control" id="mcp-search
    | -input" placeholder="Поиск по ID, названию или команде...">
66 |             <button class="btn btn-outline-secondary" type="button
    | " id="mcp-search-clear" onclick="window.mcpTab?.clearSearch()"
    | "></button>
67 |         </div>
68 |
69 |         <div class="d-flex align-items-center gap-2 flex-wrap">
70 |             <select class="form-select form-select-sm" id="mcp-fil
    | ter-select" style="width: auto;" onchange="window.mcpTab?.appl
    | yFilters()">
71 |                 <option value="all">Все серверы</option>
72 |                 <option value="enabled">Только включенные</option>
73 |                 <option value="disabled">Только отключенные</option>
74 |                 <option value="stdio">Транспорт: stdio</option>
75 |                 <option value="sse">Транспорт: SSE / HTTP</option>
76 |             </select>
77 |
78 |             <div class="btn-group btn-group-sm" role="group">
79 |                 <button type="button" class="btn btn-outline-seconda

```

```

ry active" id="btn-mcp-view-cards" onclick="window.mcpTab?.set
ViewModel('cards')">
80     <i class="bi bi-grid-fill"></i> Карточки
81     </button>
82     <button type="button" class="btn btn-outline-seconda
ry" id="btn-mcp-view-table" onclick="window.mcpTab?.setViewMod
e('table')">
83         <i class="bi bi-table"></i> Таблица
84     </button>
85 </div>
86 </div>
87 </div>
88 </div>
89
90 <!-- Loading & Empty Indicators -->
91 <div id="mcp-loading" class="text-center py-5 text-muted d-n
one">
92     <div class="spinner-border text-primary mb-2" role="status
"></div>
93     <div>Загрузка серверов MCP...</div>
94 </div>
95
96 <div id="mcp-empty" class="text-center py-5 text-muted d-non
e">
97     <i class="bi bi-hdd-network display-4 d-block mb-2 text-se
condary"></i>
98     <h5>Серверы MCP не настроены</h5>
99     <p class="small">Добавьте сервер MCP (например, Playwright
, Filesystem, SQLite) для расширения возможностей агентов.</p>
100     <button class="btn btn-sm btn-primary" data-bs-toggle="mod
al" data-bs-target="#mcpServerModal" onclick="window.mcpTab?.o
penCreateModal()">
101         <i class="bi bi-plus-circle-fill"></i> Добавить первый с
ервер
102     </button>
103 </div>
104
105 <!-- Cards Container -->

```

```

106 | <div class="row g-3" id="mcp-cards-container"></div>
107 |
108 | <!-- Table Container -->
109 | <div class="card shadow-sm border d-none" id="mcp-table-cont
    | ainer">
110 |     <div class="table-responsive">
111 |         <table class="table table-hover mb-0 align-middle">
112 |             <thead class="table-light">
113 |                 <tr class="small text-uppercase">
114 |                     <th style="width: 50px;">Статус</th>
115 |                     <th>ID / Название</th>
116 |                     <th>Транспорт</th>
117 |                     <th>Команда / URL</th>
118 |                     <th>Описание</th>
119 |                     <th class="text-end" style="width: 220px;">Действи
    | я</th>
120 |                 </tr>
121 |             </thead>
122 |             <tbody id="mcp-table-body"></tbody>
123 |         </table>
124 |     </div>
125 | </div>
126 | </div>
127 |
128 | <!-- Add / Edit MCP Server Modal -->
129 | <div class="modal fade" id="mcpServerModal" tabindex="-1" aria
    | -labelledby="mcpServerModalLabel" aria-hidden="true">
130 |     <div class="modal-dialog modal-lg">
131 |         <div class="modal-content shadow-lg">
132 |             <div class="modal-header">
133 |                 <h5 class="modal-title d-flex align-items-center gap-2
    | " id="mcpServerModalLabel">
134 |                     <i class="bi bi-hdd-network-fill text-primary"></i>
    | <span id="mcp-modal-title-text">Настройка MCP сервера</span>
135 |                 </h5>
136 |                 <button type="button" class="btn-close" data-bs-dismis
    | s="modal" aria-label="Close"></button>
137 |             </div>

```

```
138 |         <div class="modal-body p-4">
139 |             <!-- Preset Selector -->
140 |             <div class="mb-3 p-3 bg-body-tertiary rounded border"
141 | id="mcp-preset-container">
142 |                 <label class="form-label text-info small fw-bold mb-
143 | 1 d-flex align-items-center gap-1">
144 |                     <i class="bi bi-magic"></i> Готовые шаблоны (Prese
145 | ts)
146 |                 </label>
147 |                 <div class="d-flex gap-2 flex-wrap">
148 |                     <button type="button" class="btn btn-sm btn-outlin
149 | e-primary" onclick="window.mcpTab?.applyPreset('playwright')">
150 |                         □ Playwright (Browser)</button>
151 |                     <button type="button" class="btn btn-sm btn-outlin
152 | e-info" onclick="window.mcpTab?.applyPreset('filesystem')">□
153 |                         Filesystem</button>
154 |                     <button type="button" class="btn btn-sm btn-outlin
155 | e-success" onclick="window.mcpTab?.applyPreset('memory')">□ M
156 |                         emory</button>
157 |                     <button type="button" class="btn btn-sm btn-outlin
158 | e-warning" onclick="window.mcpTab?.applyPreset('sqlite')">□ S
159 |                         QLite</button>
160 |                     <button type="button" class="btn btn-sm btn-outlin
161 | e-secondary" onclick="window.mcpTab?.applyPreset('fetch')">□
162 |                         Fetch / HTTP</button>
163 |                     <button type="button" class="btn btn-sm btn-outlin
164 | e-secondary" onclick="window.mcpTab?.applyPreset('sse')">□ Rem
165 |                         ote SSE / HTTP</button>
166 |                 </div>
167 |             </div>
168 |
169 |             <form id="mcp-server-form" onsubmit="window.mcpTab?.sa
170 | veServer(event)">
171 |                 <input type="hidden" id="mcp-form-mode" value="creat
172 | e">
173 |
174 |                 <div class="row g-3">
175 |                     <div class="col-md-6">
```

```
159 |         <label for="mcp-server-id" class="form-label sma
    | ll text-muted">ID сервера (уникальный ключ) <span class="text-
    | danger">*</span></label>
160 |         <input type="text" class="form-control" id="mcp-
    | server-id" placeholder="например, playwright" required pattern
    | ="[a-zA-Z0-9_-]+">
161 |         <div class="form-text text-muted small">Латински
    | е буквы, цифры, дефис или подчеркивание.</div>
162 |         </div>
163 |
164 |         <div class="col-md-6">
165 |             <label for="mcp-server-name" class="form-label s
    | mall text-muted">Отображаемое имя</label>
166 |             <input type="text" class="form-control" id="mcp-
    | server-name" placeholder="Playwright Web Browser">
167 |             </div>
168 |
169 |         <div class="col-12">
170 |             <label for="mcp-server-desc" class="form-label s
    | mall text-muted">Описание</label>
171 |             <input type="text" class="form-control" id="mcp-
    | server-desc" placeholder="Управление браузером и поиск информа
    | ции в веб-страницах">
172 |             </div>
173 |
174 |         <div class="col-md-6">
175 |             <label for="mcp-server-transport" class="form-la
    | bel small text-muted">Протокол транспорта <span class="text-da
    | nger">*</span></label>
176 |             <select class="form-select" id="mcp-server-trans
    | port" onchange="window.mcpTab?.onTransportChange()">
177 |                 <option value="stdio">stdio (Локальный subprocess)
    | </option>
178 |                 <option value="sse">sse (Server-Sent Events по
    | HTTP)</option>
179 |                 <option value="streamable_http">streamable_htt
    | p (HTTP Streaming)</option>
180 |             </select>
```

```
181 |         </div>
182 |
183 |         <div class="col-md-6 d-flex align-items-center pt-
184 | 4">
185 |             <div class="form-check form-switch">
186 |                 <input class="form-check-input" type="checkbox"
187 | id="mcp-server-enabled" checked>
188 |                 <label class="form-check-label" for="mcp-serve
189 | r-enabled">Включить сервер для агентов</label>
190 |             </div>
191 |         </div>
192 |
193 |         <!-- Stdio Section -->
194 |         <div class="col-12 mcp-field-stdio" id="mcp-stdio-
195 | container">
196 |             <div class="mb-3">
197 |                 <label for="mcp-server-command" class="form-la
198 | bel small text-muted">Команда (Command) <span class="text-dang
199 | er">*</span></label>
200 |                 <input type="text" class="form-control" id="mc
201 | p-server-command" placeholder="npx, python, node, uvx">
202 |                 <div class="form-text text-muted small">Исполн
203 | яемый файл или утилита в PATH.</div>
204 |             </div>
205 |
206 |             <div class="mb-3">
207 |                 <label for="mcp-server-args" class="form-label
208 | small text-muted">Аргументы (Arguments)</label>
209 |                 <input type="text" class="form-control" id="mc
210 | p-server-args" placeholder="@playwright/mcp@latest или через п
211 | робел / запятую">
212 |                 <div class="form-text text-muted small">Укажит
213 | е аргументы через пробел или в формате JSON-массива строк.</di
214 | v>
215 |             </div>
216 |
217 |             <div>
218 |                 <label for="mcp-server-env" class="form-label
```



```

    small text-muted">Переменные окружения (Environment JSON / Key
-Value)</label>
206      <textarea class="form-control font-monospace"
id="mcp-server-env" rows="2" placeholder='{"NODE_ENV": "produc
tion"}'></textarea>
207      <div class="form-text text-muted small">Опцион
альный JSON-объект с переменными окружения для процесса.</div>
208      </div>
209    </div>
210
211    <!-- Network URL Section -->
212    <div class="col-12 mcp-field-network d-none" id="m
cp-network-container">
213      <label for="mcp-server-url" class="form-label sm
all text-muted">URL эндпоинта <span class="text-danger">*</spa
n></label>
214      <input type="url" class="form-control" id="mcp-s
erver-url" placeholder="http://localhost:8080/sse">
215      <div class="form-text text-muted small">Полный Н
ТTP/HTTPS адрес удаленного MCP-сервера.</div>
216      </div>
217    </div>
218
219    <div class="mt-4 d-flex justify-content-between align-items-center border-top pt-3">
220      <button type="button" class="btn btn-outline-info btn-sm d-flex align-items-center gap-1 rounded-pill" id="btn-test-modal-server" onclick="window.mcpTab?.testCurrentForm()">
221        <i class="bi bi-play-circle"></i> Проверить подк
лючение
222      </button>
223      <div class="d-flex gap-2">
224        <button type="button" class="btn btn-secondary rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
225        <button type="submit" class="btn btn-primary rounded-pill px-3" id="mcp-btn-save-server">Сохранить</button>
226      </div>
227    </div>

```

```
228 |         </form>
229 |     </div>
230 | </div>
231 | </div>
232 | </div>
233 |
234 | <!-- Test Connection & Tools Inspection Modal -->
235 | <div class="modal fade" id="mcpTestModal" tabindex="-1" aria-labelledby="mcpTestModalLabel" aria-hidden="true">
236 |     <div class="modal-dialog modal-lg modal-dialog-scrollable">
237 |         <div class="modal-content shadow-lg">
238 |             <div class="modal-header">
239 |                 <h5 class="modal-title d-flex align-items-center gap-2" id="mcpTestModalLabel">
240 |                     <i class="bi bi-activity text-info"></i> Диагностика
241 |                     MCP: <span id="mcp-test-server-name">...</span>
242 |                 </h5>
243 |                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
244 |             </div>
245 |             <div class="modal-body p-4">
246 |                 <div id="mcp-test-status-box" class="p-3 mb-3 rounded border bg-body-tertiary d-flex align-items-center justify-content-between">
247 |                     <div class="d-flex align-items-center gap-2">
248 |                         <div class="spinner-border spinner-border-sm text-info" id="mcp-test-spinner" role="status"></div>
249 |                         <span id="mcp-test-status-text">Установка соединения и опрос инструментов...</span>
250 |                     </div>
251 |                     <span class="badge bg-secondary font-monospace" id="mcp-test-latency-badge">—</span>
252 |                 </div>
253 |                 <h6 class="d-flex align-items-center gap-2 mb-3">
254 |                     <i class="bi bi-tools text-warning"></i> Обнаруженные инструменты (Tools):
255 |                     <span class="badge bg-primary rounded-pill" id="mcp-
```

```

    test-tools-count">0</span>
256         </h6>
257
258         <div id="mcp-test-tools-list" class="d-flex flex-column gap-2">
259             <!-- Tools list populated dynamically -->
260         </div>
261     </div>
262     <div class="modal-footer">
263         <button type="button" class="btn btn-secondary rounded
    -pill px-3" data-bs-dismiss="modal">Закрыть</button>
264     </div>
265 </div>
266 </div>
267 </div>
268
269 <!-- All Tools Aggregator Modal -->
270 <div class="modal fade" id="mcpAllToolsModal" tabindex="-1" ar
    ia-labelledby="mcpAllToolsModalLabel" aria-hidden="true">
271     <div class="modal-dialog modal-xl modal-dialog-scrollable">
272         <div class="modal-content shadow-lg">
273             <div class="modal-header">
274                 <h5 class="modal-title d-flex align-items-center gap-2
    " id="mcpAllToolsModalLabel">
275                     <i class="bi bi-boxes text-success"></i> Доступный п
    ул инструментов MCP
276                 </h5>
277                 <button type="button" class="btn-close" data-bs-dismis
    s="modal" aria-label="Close"></button>
278             </div>
279             <div class="modal-body p-4">
280                 <p class="text-muted small mb-3">
281                     Список всех инструментов, агрегированных со всех вкл
    юченных серверов MCP. Эти инструменты автоматически предоставл
    яются LangChain/ReAct агенту.
282                 </p>
283
284                 <div id="mcp-all-tools-loading" class="text-center py-

```

```
4 text-muted">
285     <div class="spinner-border text-primary mb-2" role="
status"></div>
286     <div>Опрос активных MCP серверов...</div>
287 </div>
288
289     <div id="mcp-all-tools-empty" class="text-center py-4
text-muted d-none">
290         <i class="bi bi-tools display-6 d-block mb-2 text-se
condary"></i>
291         <div>Нет доступных инструментов от активных MCP серв
еров.</div>
292     </div>
293
294     <div id="mcp-all-tools-container" class="row g-3 d-non
e"></div>
295 </div>
296     <div class="modal-footer">
297         <button type="button" class="btn btn-secondary rounded
-pill px-3" data-bs-dismiss="modal">Заккрыть</button>
298     </div>
299 </div>
300 </div>
301 </div>
302
```

main.js

```
1 // MCP Servers Tab Management Logic for AI Breadboard Admin In
  |terface
2
3 class McpTabManager {
4   constructor() {
5     this.servers = [];
6     this.filteredServers = [];
7     this.viewMode = 'cards'; // 'cards' | 'table'
8     this.searchQuery = '';
9     this.activeFilter = 'all'; // 'all' | 'enabled' | 'disable
  |d' | 'stdio' | 'sse'
10    this.isInitialized = false;
11  }
12
13  async init() {
14    if (this.isInitialized) {
15      await this.refresh();
16      return;
17    }
18    this.bindEvents();
19    this.isInitialized = true;
20    await this.refresh();
21  }
22
23  bindEvents() {
24    const searchInput = document.getElementById('mcp-search-in
  |put');
25    if (searchInput) {
26      searchInput.addEventListener('input', (e) => {
27        this.searchQuery = e.target.value.trim().toLowerCase()
  |;
28        this.applyFilters();
29      });
30    }
31
32    const idInput = document.getElementById('mcp-server-id');
```

```

33     if (idInput) {
34         idInput.addEventListener('input', (e) => {
35             const val = e.target.value;
36             const normalized = val.toLowerCase().replace(/[^\a-z0-9
|-]/g, '_');
37             if (val !== normalized) {
38                 e.target.value = normalized;
39             }
40         });
41     }
42 }
43
44 async apiFetch(endpoint, options = {}) {
45     let url = endpoint;
46     let res = await fetch(url, options);
47     if (!res.ok && url.startsWith('/api/admin/mcp')) {
48         const fallbackUrl = url.replace('/api/admin/mcp', '/api/
mcp');
49         const fallbackRes = await fetch(fallbackUrl, options);
50         if (fallbackRes.ok) {
51             return fallbackRes;
52         }
53     }
54     return res;
55 }
56
57 async refresh() {
58     this.showLoading(true);
59     try {
60         const res = await this.apiFetch('/api/admin/mcp/servers'
);
61         if (!res.ok) {
62             throw new Error(`HTTP ${res.status}: ${res.statusText}
`);
63         }
64         const data = await res.json();
65         this.servers = data.servers || [];
66         this.updateStats();

```

```
67 |     this.applyFilters();
68 | } catch (err) {
69 |     console.error('[McpTab] Error fetching MCP servers:', err);
70 |     this.renderError(err.message);
71 | } finally {
72 |     this.showLoading(false);
73 | }
74 | }
75 |
76 | updateStats() {
77 |     const total = this.servers.length;
78 |     const active = this.servers.filter(s => s.enabled !== false).length;
79 |     const stdio = this.servers.filter(s => (s.transport || 'stdio') === 'stdio').length;
80 |     const network = this.servers.filter(s => ['sse', 'streamable-http', 'http'].includes(s.transport)).length;
81 |
82 |     const badge = document.getElementById('mcp-total-badge');
83 |     if (badge) badge.textContent = total;
84 |
85 |     const elTotal = document.getElementById('stat-total-mcp');
86 |     if (elTotal) elTotal.textContent = total;
87 |
88 |     const elActive = document.getElementById('stat-active-mcp');
89 |     if (elActive) elActive.textContent = active;
90 |
91 |     const elStdio = document.getElementById('stat-stdio-mcp');
92 |     if (elStdio) elStdio.textContent = stdio;
93 |
94 |     const elNetwork = document.getElementById('stat-network-mcp');
95 |     if (elNetwork) elNetwork.textContent = network;
96 | }
97 |
98 | applyFilters() {
```

```
99     const filterSelect = document.getElementById('mcp-filter-s
|elect');
100     this.activeFilter = filterSelect ? filterSelect.value : 'a
|ll';
101
102     this.filteredServers = this.servers.filter(s => {
103         // Search filter
104         if (this.searchQuery) {
105             const idMatch = (s.id || '').toLowerCase().includes(th
|is.searchQuery);
106             const nameMatch = (s.name || '').toLowerCase().include
|s(this.searchQuery);
107             const descMatch = (s.description || '').toLowerCase().
|includes(this.searchQuery);
108             const cmdMatch = (s.command || '').toLowerCase().inclu
|des(this.searchQuery);
109             const urlMatch = (s.url || '').toLowerCase().includes(
|this.searchQuery);
110             if (!idMatch && !nameMatch && !descMatch && !cmdMatch
|&& !urlMatch) {
111                 return false;
112             }
113         }
114
115         // Category filter
116         if (this.activeFilter === 'enabled') {
117             return s.enabled !== false;
118         } else if (this.activeFilter === 'disabled') {
119             return s.enabled === false;
120         } else if (this.activeFilter === 'stdio') {
121             return (s.transport || 'stdio') === 'stdio';
122         } else if (this.activeFilter === 'sse') {
123             return ['sse', 'streamable_http', 'http'].includes(s.t
|ransport);
124         }
125
126         return true;
127     });
```



```
128 |
129 |     this.render();
130 | }
131 |
132 | clearSearch() {
133 |     const searchInput = document.getElementById('mcp-search-in
put');
134 |     if (searchInput) searchInput.value = '';
135 |     this.searchQuery = '';
136 |     this.applyFilters();
137 | }
138 |
139 | setViewMode(mode) {
140 |     this.viewMode = mode;
141 |     const btnCards = document.getElementById('btn-mcp-view-car
ds');
142 |     const btnTable = document.getElementById('btn-mcp-view-tab
le');
143 |     const cardsContainer = document.getElementById('mcp-cards-
container');
144 |     const tableContainer = document.getElementById('mcp-table-
container');
145 |
146 |     if (mode === 'cards') {
147 |         btnCards?.classList.add('active');
148 |         btnTable?.classList.remove('active');
149 |         cardsContainer?.classList.remove('d-none');
150 |         tableContainer?.classList.add('d-none');
151 |     } else {
152 |         btnCards?.classList.remove('active');
153 |         btnTable?.classList.add('active');
154 |         cardsContainer?.classList.add('d-none');
155 |         tableContainer?.classList.remove('d-none');
156 |     }
157 | }
158 |
159 | showLoading(show) {
160 |     const loadingEl = document.getElementById('mcp-loading');
```

```
161     const cardsEl = document.getElementById('mcp-cards-contain
er');
162     const tableEl = document.getElementById('mcp-table-contain
er');
163     const emptyEl = document.getElementById('mcp-empty');
164
165     if (show) {
166         loadingEl?.classList.remove('d-none');
167         cardsEl?.classList.add('d-none');
168         tableEl?.classList.add('d-none');
169         emptyEl?.classList.add('d-none');
170     } else {
171         loadingEl?.classList.add('d-none');
172         this.setViewMode(this.viewMode);
173     }
174 }
175
176 render() {
177     const emptyEl = document.getElementById('mcp-empty');
178     const cardsContainer = document.getElementById('mcp-cards-
container');
179     const tableBody = document.getElementById('mcp-table-body'
);
180
181     if (this.filteredServers.length === 0) {
182         if (emptyEl) emptyEl.classList.remove('d-none');
183         if (cardsContainer) cardsContainer.innerHTML = '';
184         if (tableBody) tableBody.innerHTML = '';
185         return;
186     }
187
188     if (emptyEl) emptyEl.classList.add('d-none');
189
190     if (cardsContainer) {
191         cardsContainer.innerHTML = this.filteredServers.map(s =>
this.renderServerCard(s)).join('');
192     }
193
```

```

194     if (tableBody) {
195         tableBody.innerHTML = this.filteredServers.map(s => this
196             .renderServerTableRow(s)).join('');
197     }
198
199     renderServerCard(server) {
200         const isEnabled = server.enabled !== false;
201         const isStdio = (server.transport || 'stdio') === 'stdio';
202         const statusBadge = isEnabled
203             ? '<span class="badge bg-success-subtle text-success border border-success-subtle"><i class="bi bi-check-circle-fill">
204                 </i> Активен</span>'
205             : '<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle"><i class="bi bi-dash-circle">
206                 </i> Отключен</span>';
207
208         const transportBadge = isStdio
209             ? '<span class="badge bg-info-subtle text-info border border-info-subtle"><i class="bi bi-terminal"></i> stdio</span>'
210             : '<span class="badge bg-warning-subtle text-warning border border-warning-subtle"><i class="bi bi-broadcast"></i> ${this.escapeHtml(server.transport)}</span>`;
211
212         const commandOrUrl = isStdio
213             ? '<code>${this.escapeHtml(server.command || '')} ${server.args || []}.join(' ')}</code>'
214             : '<a href="${this.escapeHtml(server.url || '#')}" target="_blank" class="text-info text-truncate d-inline-block" style="max-width: 250px;">${this.escapeHtml(server.url || '')}</a>';
215
216         return `
217             <div class="col-12 col-md-6 col-xl-4">
218                 <div class="card bg-dark border-secondary h-100 shadow-sm d-flex flex-column">
219                     <div class="card-header border-secondary d-flex align-items-center justify-content-between p-3">

```

```
218 |         <div class="d-flex align-items-center gap-2 text-t  
| truncate">  
219 |             <div class="p-2 bg-black rounded border border-s  
| econdary text-primary">  
220 |                 <i class="bi ${isStdio ? 'bi-cpu' : 'bi-globe2  
| '}"></i>  
221 |             </div>  
222 |             <div class="text-truncate">  
223 |                 <h6 class="mb-0 text-white text-truncate fw-bo  
| ld">${this.escapeHtml(server.name || server.id)}</h6>  
224 |                 <small class="text-muted font-monospace">${thi  
| s.escapeHtml(server.id)}</small>  
225 |             </div>  
226 |         </div>  
227 |         <div class="d-flex align-items-center gap-1">  
228 |             ${transportBadge}  
229 |             ${statusBadge}  
230 |         </div>  
231 |     </div>  
232 |     <div class="card-body p-3 flex-grow-1">  
233 |         <p class="card-text small text-secondary mb-3" sty  
| le="min-height: 38px;">  
234 |             ${this.escapeHtml(server.description || 'Описани  
| е отсутствует.' )}  
235 |         </p>  
236 |         <div class="p-2 bg-black rounded border border-sec  
| ondary mb-2 small">  
237 |             <div class="text-muted small mb-1">${isStdio ? '  
| Исполняемая команда:' : 'Сетевой эндпоинт:'}</div>  
238 |             <div class="text-truncate">${commandOrUrl}</div>  
239 |         </div>  
240 |     </div>  
241 |     <div class="card-footer border-secondary p-2 d-flex  
| align-items-center justify-content-between gap-1 bg-black bg-o  
| pacity-25">  
242 |         <button class="btn btn-sm btn-outline-info d-flex  
| align-items-center gap-1" onclick="window.mcpTab?.testServer(''  
| ${this.escapeHtml(server.id)}')">
```

```

243 |         <i class="bi bi-play-circle"></i> Тест & Tools
244 |     </button>
245 |     <div class="d-flex align-items-center gap-1">
246 |         <button class="btn btn-sm ${isEnabled ? 'btn-out
line-warning' : 'btn-outline-success'}" title="${isEnabled ? '
Отключить' : 'Включить'}" onclick="window.mcpTab?.toggleServer
('${this.escapeHtml(server.id)}')">
247 |             <i class="bi ${isEnabled ? 'bi-pause-fill' : '
bi-play-fill'}"></i>
248 |         </button>
249 |         <button class="btn btn-sm btn-outline-primary" t
itle="Редактировать" onclick="window.mcpTab?.openEditModal('${
this.escapeHtml(server.id)}')">
250 |             <i class="bi bi-pencil-fill"></i>
251 |         </button>
252 |         <button class="btn btn-sm btn-outline-danger" ti
tle="Удалить" onclick="window.mcpTab?.deleteServer('${this.esc
apeHtml(server.id)}')">
253 |             <i class="bi bi-trash-fill"></i>
254 |         </button>
255 |     </div>
256 | </div>
257 | </div>
258 | </div>
259 | `;
260 | }
261 |
262 | renderServerTableRow(server) {
263 |     const isEnabled = server.enabled !== false;
264 |     const isStdio = (server.transport || 'stdio') === 'stdio';
265 |     const statusDot = isEnabled
266 |         ? '<span class="badge bg-success rounded-pill" title="Ак
тивен">ON</span>'
267 |         : '<span class="badge bg-secondary rounded-pill" title="
Отключен">OFF</span>';
268 |
269 |     const transportBadge = isStdio
270 |         ? '<span class="badge bg-info-subtle text-info border bo

```

```

    rder-info-subtle">stdio</span>'
271     : `<span class="badge bg-warning-subtle text-warning border border-warning-subtle">${this.escapeHtml(server.transport)}</span>`;
272
273     const cmdOrUrl = isStdio
274     ? `<code>${this.escapeHtml(server.command || '')} ${server.args || []}.join(' ')}</code>`
275     : `<a href="${this.escapeHtml(server.url || '#')}" target="_blank" class="text-info">${this.escapeHtml(server.url || '')}</a>`;
276
277     return `
278         <tr>
279             <td class="text-center">${statusDot}</td>
280             <td>
281                 <div class="fw-bold text-white">${this.escapeHtml(server.name || server.id)}</div>
282                 <div class="small font-monospace text-muted">${this.escapeHtml(server.id)}</div>
283             </td>
284             <td>${transportBadge}</td>
285             <td class="small text-truncate" style="max-width: 250px;">${cmdOrUrl}</td>
286             <td class="small text-muted text-truncate" style="max-width: 200px;">${this.escapeHtml(server.description || '-')}</td>
287             <td class="text-end">
288                 <div class="btn-group btn-group-sm">
289                     <button class="btn btn-outline-info" title="Тест соединения" onclick="window.mcpTab?.testServer('${this.escapeHtml(server.id)}')">
290                         <i class="bi bi-play-circle"></i>
291                     </button>
292                     <button class="btn ${isEnabled ? 'btn-outline-warning' : 'btn-outline-success'}" title="${isEnabled ? 'Отключить' : 'Включить'}" onclick="window.mcpTab?.toggleServer('${this.escapeHtml(server.id)}')">

```

```

293 |         <i class="bi ${isEnabled ? 'bi-pause-fill' : 'bi
| -play-fill'}"></i>
294 |     </button>
295 |     <button class="btn btn-outline-primary" title="Реда
| активировать" onclick="window.mcpTab?.openEditModal('${this.esca
| peHtml(server.id)}')">
296 |         <i class="bi bi-pencil-fill"></i>
297 |     </button>
298 |     <button class="btn btn-outline-danger" title="Удал
| ить" onclick="window.mcpTab?.deleteServer('${this.escapeHtml(s
| erver.id)}')">
299 |         <i class="bi bi-trash-fill"></i>
300 |     </button>
301 | </div>
302 | </td>
303 | </tr>
304 | `;
305 | }
306 |
307 | onTransportChange() {
308 |     const transport = document.getElementById('mcp-server-tran
| sport')?.value || 'stdio';
309 |     const stdioContainer = document.getElementById('mcp-stdio-
| container');
310 |     const networkContainer = document.getElementById('mcp-netw
| ork-container');
311 |
312 |     if (transport === 'stdio') {
313 |         stdioContainer?.classList.remove('d-none');
314 |         networkContainer?.classList.add('d-none');
315 |     } else {
316 |         stdioContainer?.classList.add('d-none');
317 |         networkContainer?.classList.remove('d-none');
318 |     }
319 | }
320 |
321 | applyPreset(presetKey) {
322 |     const presets = {

```

```
323 |     playwright: {
324 |         id: 'playwright',
325 |         name: 'Playwright Browser MCP',
326 |         description: 'Управление веб-браузером, навигация, кли
ки, скриншоты и парсинг страниц через Playwright.',
327 |         transport: 'stdio',
328 |         command: 'npx',
329 |         args: '@playwright/mcp@latest',
330 |         env: ''
331 |     },
332 |     filesystem: {
333 |         id: 'filesystem',
334 |         name: 'Filesystem MCP',
335 |         description: 'Безопасное чтение, запись и навигация по
каталогам и файлам на диске.',
336 |         transport: 'stdio',
337 |         command: 'npx',
338 |         args: '-y @modelcontextprotocol/server-fileSystem ./da
ta',
339 |         env: ''
340 |     },
341 |     memory: {
342 |         id: 'memory',
343 |         name: 'Knowledge Graph Memory MCP',
344 |         description: 'Хранилище графа знаний и ассоциативная п
амять для ИИ-агентов.',
345 |         transport: 'stdio',
346 |         command: 'npx',
347 |         args: '-y @modelcontextprotocol/server-memory',
348 |         env: ''
349 |     },
350 |     sqlite: {
351 |         id: 'sqlite',
352 |         name: 'SQLite Database MCP',
353 |         description: 'Выполнение запросов, исследование схемы
и инспекция таблиц SQLite базы данных.',
354 |         transport: 'stdio',
355 |         command: 'uvx',
```



```
356 |         args: 'mcp-server-sqlite --db-path ./data/database.sql
    |         ite',
357 |         env: ''
358 |     },
359 |     fetch: {
360 |         id: 'fetch',
361 |         name: 'Fetch / Web Request MCP',
362 |         description: 'Преобразование HTML веб-страниц в Markdo
    |         wn и выполнение HTTP-запросов.',
363 |         transport: 'stdio',
364 |         command: 'uvx',
365 |         args: 'mcp-server-fetch',
366 |         env: ''
367 |     },
368 |     sse: {
369 |         id: 'remote_mcp',
370 |         name: 'Remote SSE MCP Server',
371 |         description: 'Удаленный MCP-сервер, доступный по прото
    |         колу Server-Sent Events (SSE).',
372 |         transport: 'sse',
373 |         url: 'http://localhost:8080/sse'
374 |     }
375 | };
376 |
377 | const p = presets[presetKey];
378 | if (!p) return;
379 |
380 | const idEl = document.getElementById('mcp-server-id');
381 | const nameEl = document.getElementById('mcp-server-name');
382 | const descEl = document.getElementById('mcp-server-desc');
383 | const transportEl = document.getElementById('mcp-server-tr
    |     ansport');
384 | const cmdEl = document.getElementById('mcp-server-command'
    | );
385 | const argsEl = document.getElementById('mcp-server-args');
386 | const urlEl = document.getElementById('mcp-server-url');
387 | const envEl = document.getElementById('mcp-server-env');
388 |
```

```
389     if (idEl && document.getElementById('mcp-form-mode')?.value === 'create') idEl.value = p.id;
390     if (nameEl) nameEl.value = p.name || '';
391     if (descEl) descEl.value = p.description || '';
392     if (transportEl) transportEl.value = p.transport || 'stdio';
393     if (cmdEl) cmdEl.value = p.command || '';
394     if (argsEl) argsEl.value = p.args || '';
395     if (urlEl) urlEl.value = p.url || '';
396     if (envEl) envEl.value = p.env || '';
397
398     this.onTransportChange();
399 }
400
401 openCreateModal() {
402     const form = document.getElementById('mcp-server-form');
403     if (form) form.reset();
404
405     const titleText = document.getElementById('mcp-modal-title-text');
406     if (titleText) titleText.textContent = 'Добавление нового MCP сервера';
407
408     const modeInput = document.getElementById('mcp-form-mode');
409     if (modeInput) modeInput.value = 'create';
410
411     const idInput = document.getElementById('mcp-server-id');
412     if (idInput) {
413         idInput.removeAttribute('readonly');
414         idInput.classList.remove('bg-secondary');
415     }
416
417     const presetContainer = document.getElementById('mcp-preset-container');
418     if (presetContainer) presetContainer.classList.remove('d-none');
419 }
```

```
420     const enabledCheckbox = document.getElementById('mcp-server-enabled');
421     if (enabledCheckbox) enabledCheckbox.checked = true;
422
423     this.applyPreset('playwright');
424 }
425
426 openEditModal(serverId) {
427     const server = this.servers.find(s => s.id === serverId);
428     if (!server) return;
429
430     const titleText = document.getElementById('mcp-modal-title-text');
431     if (titleText) titleText.textContent = `Редактирование: ${server.name || server.id}`;
432
433     const modeInput = document.getElementById('mcp-form-mode');
434     ;
435     if (modeInput) modeInput.value = 'edit';
436
437     const presetContainer = document.getElementById('mcp-preset-container');
438     if (presetContainer) presetContainer.classList.add('d-none');
439
440     const idInput = document.getElementById('mcp-server-id');
441     if (idInput) {
442         idInput.value = server.id;
443         idInput.setAttribute('readonly', 'true');
444         idInput.classList.add('bg-secondary');
445     }
446
447     const nameInput = document.getElementById('mcp-server-name');
448     if (nameInput) nameInput.value = server.name || '';
449
450     const descInput = document.getElementById('mcp-server-desc');
```

```
450     if (descInput) descInput.value = server.description || '';
451
452     const transportInput = document.getElementById('mcp-server
-transport');
453     if (transportInput) transportInput.value = server.transpor
t || 'stdio';
454
455     const cmdInput = document.getElementById('mcp-server-comm
and');
456     if (cmdInput) cmdInput.value = server.command || '';
457
458     const argsInput = document.getElementById('mcp-server-args
');
459     if (argsInput) {
460         argsInput.value = Array.isArray(server.args) ? server.ar
gs.join(' ') : (server.args || '');
461     }
462
463     const urlInput = document.getElementById('mcp-server-url')
;
464     if (urlInput) urlInput.value = server.url || '';
465
466     const envInput = document.getElementById('mcp-server-env')
;
467     if (envInput) {
468         envInput.value = server.env ? JSON.stringify(server.env,
null, 2) : '';
469     }
470
471     const enabledCheckbox = document.getElementById('mcp-serve
r-enabled');
472     if (enabledCheckbox) enabledCheckbox.checked = server.enab
led !== false;
473
474     this.onTransportChange();
475
476     const modalEl = document.getElementById('mcpServerModal');
477     if (modalEl && window.bootstrap) {
```

```
478     const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
479     modal.show();
480 }
481 }
482
483 async saveServer(event) {
484     if (event) event.preventDefault();
485
486     const mode = document.getElementById('mcp-form-mode')?.val
ue || 'create';
487     const serverId = document.getElementById('mcp-server-id')?
.value.trim();
488     const name = document.getElementById('mcp-server-name')?.v
alue.trim();
489     const description = document.getElementById('mcp-server-de
sc')?.value.trim();
490     const transport = document.getElementById('mcp-server-tran
sport')?.value || 'stdio';
491     const enabled = document.getElementById('mcp-server-enable
d')?.checked ?? true;
492     const command = document.getElementById('mcp-server-comman
d')?.value.trim();
493     const argsRaw = document.getElementById('mcp-server-args')
?.value.trim();
494     const url = document.getElementById('mcp-server-url')?.val
ue.trim();
495     const envRaw = document.getElementById('mcp-server-env')?.
value.trim();
496
497     if (!serverId) {
498         window.showToast?.('Укажите ID сервера', 'warning') || a
lert('Укажите ID сервера');
499         return;
500     }
501
502     // Parse args
503     let args = [];
```

```
504 |     if (argsRaw) {
505 |         if (argsRaw.startsWith '[' && argsRaw.endsWith(']')) {
506 |             try {
507 |                 args = JSON.parse(argsRaw);
508 |             } catch {
509 |                 args = argsRaw.split(/\s+/).filter(Boolean);
510 |             }
511 |         } else {
512 |             args = argsRaw.split(/\s+/).filter(Boolean);
513 |         }
514 |     }
515 |
516 |     // Parse env
517 |     let env = {};
518 |     if (envRaw) {
519 |         try {
520 |             env = JSON.parse(envRaw);
521 |         } catch (err) {
522 |             window.showToast?.('Ошибка формата JSON в переменных о
    кружения: ' + err.message, 'danger') || alert('Ошибка формата
    JSON в переменных окружения: ' + err.message);
523 |             return;
524 |         }
525 |     }
526 |
527 |     const payload = {
528 |         id: serverId,
529 |         name: name || serverId,
530 |         description: description,
531 |         transport: transport,
532 |         enabled: enabled,
533 |         command: command,
534 |         args: args,
535 |         url: url,
536 |         env: env
537 |     };
538 |
539 |     const saveBtn = document.getElementById('mcp-btn-save-serv
```

```

    er');
540     if (saveBtn) {
541         saveBtn.disabled = true;
542         saveBtn.innerHTML = '<span class="spinner-border spinner
-border-sm me-1"></span> Сохранение...';
543     }
544
545     try {
546         const endpoint = mode === 'create'
547             ? '/api/admin/mcp/servers'
548             : `/api/admin/mcp/servers/${encodeURIComponent(serverI
d)}`;
549
550         const method = mode === 'create' ? 'POST' : 'PUT';
551
552         const res = await this.apiFetch(endpoint, {
553             method: method,
554             headers: { 'Content-Type': 'application/json' },
555             body: JSON.stringify(payload)
556         });
557
558         if (!res.ok) {
559             const errData = await res.json().catch(() => ({}));
560             throw new Error(errData.detail || `HTTP ${res.status}`
);
561         }
562
563         // Close modal
564         const modalEl = document.getElementById('mcpServerModal'
);
565         if (modalEl && window.bootstrap) {
566             const modal = bootstrap.Modal.getInstance(modalEl);
567             modal?.hide();
568         }
569
570         await this.refresh();
571     } catch (err) {
572         window.showToast?.(`Ошибка сохранения MCP сервера: ${err

```

```
        .message}`, 'danger') || alert(`Ошибка сохранения MCP сервера:
        ${err.message}`);
573     } finally {
574         if (saveBtn) {
575             saveBtn.disabled = false;
576             saveBtn.innerHTML = 'Сохранить';
577         }
578     }
579 }
580
581 async toggleServer(serverId) {
582     try {
583         const res = await this.apiFetch(`/api/admin/mcp/servers/
        ${encodeURIComponent(serverId)}/toggle`, {
584             method: 'POST'
585         });
586         if (!res.ok) {
587             const err = await res.json().catch(() => ({}));
588             throw new Error(err.detail || `HTTP ${res.status}`);
589         }
590         await this.refresh();
591     } catch (err) {
592         window.showToast?(`Ошибка переключения статуса сервера:
        ${err.message}`, 'danger') || alert(`Ошибка переключения стат
        уса сервера: ${err.message}`);
593     }
594 }
595
596 async deleteServer(serverId) {
597     if (!confirm(`Вы уверены, что хотите удалить MCP сервер "$
        {serverId}"?`)) {
598         return;
599     }
600
601     try {
602         const res = await this.apiFetch(`/api/admin/mcp/servers/
        ${encodeURIComponent(serverId)}`, {
603             method: 'DELETE'
```



```

604 |     });
605 |     if (!res.ok) {
606 |         const err = await res.json().catch(() => ({}));
607 |         throw new Error(err.detail || `HTTP ${res.status}`);
608 |     }
609 |     await this.refresh();
610 | } catch (err) {
611 |     window.showToast?(`Ошибка удаления сервера: ${err.message}`, 'danger') || alert(`Ошибка удаления сервера: ${err.message}`);
612 | }
613 | }
614 |
615 | async testServer(serverId) {
616 |     const server = this.servers.find(s => s.id === serverId);
617 |     const serverName = server?.name || serverId;
618 |
619 |     const nameEl = document.getElementById('mcp-test-server-name');
620 |     if (nameEl) nameEl.textContent = serverName;
621 |
622 |     const statusBox = document.getElementById('mcp-test-status-box');
623 |     const statusText = document.getElementById('mcp-test-status-text');
624 |     const spinner = document.getElementById('mcp-test-spinner');
625 |     const latencyBadge = document.getElementById('mcp-test-latency-badge');
626 |     const countBadge = document.getElementById('mcp-test-tools-count');
627 |     const toolsList = document.getElementById('mcp-test-tools-list');
628 |
629 |     if (spinner) spinner.classList.remove('d-none');
630 |     if (statusBox) {
631 |         statusBox.className = 'p-3 mb-3 rounded border border-info bg-black text-info d-flex align-items-center justify-content-center';

```

```

    t-between';
632     }
633     if (statusText) statusText.textContent = 'Подключение к пр
оцессу / эндпоинту и опрос инструментов...';
634     if (latencyBadge) latencyBadge.textContent = '...';
635     if (countBadge) countBadge.textContent = '0';
636     if (toolsList) toolsList.innerHTML = '<div class="text-mut
ed text-center py-3">Опрос инструментов...</div>';
637
638     const modalEl = document.getElementById('mcpTestModal');
639     if (modalEl && window.bootstrap) {
640         const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
641         modal.show();
642     }
643
644     try {
645         const res = await this.apiFetch(`/api/admin/mcp/servers/
${encodeURIComponent(serverId)}/test`, {
646             method: 'POST'
647         });
648         const data = await res.json();
649
650         if (spinner) spinner.classList.add('d-none');
651         if (latencyBadge) latencyBadge.textContent = `${data.lat
ency_ms || 0} ms`;
652
653         if (data.status === 'ok') {
654             if (statusBox) {
655                 statusBox.className = 'p-3 mb-3 rounded border borde
r-success bg-black text-success d-flex align-items-center just
ify-content-between';
656             }
657             if (statusText) statusText.innerHTML = `❏ <strong>Успе
шно:</strong> ${this.escapeHtml(data.message)}`;
658             if (countBadge) countBadge.textContent = data.tools_co
unt || (data.tools || []).length;
659             this.renderDiscoveredTools(data.tools || [], toolsList

```

```

);
660     } else {
661         if (statusBox) {
662             statusBox.className = 'p-3 mb-3 rounded border border-
r-danger bg-black text-danger d-flex align-items-center justify-
y-content-between';
663         }
664         if (statusText) statusText.innerHTML = `❌ <strong>Ошиб
ка:</strong> ${this.escapeHtml(data.message || 'Не удалось под
ключиться к серверу')}}`;
665         if (toolsList) {
666             toolsList.innerHTML = `<div class="alert alert-dange
r bg-dark border-danger text-light small mb-0">${this.escapeHt
ml(data.message || 'Ошибка соединения')}}</div>`;
667         }
668     }
669 } catch (err) {
670     if (spinner) spinner.classList.add('d-none');
671     if (statusBox) {
672         statusBox.className = 'p-3 mb-3 rounded border border-
danger bg-black text-danger d-flex align-items-center justify-
content-between';
673     }
674     if (statusText) statusText.innerHTML = `❌ <strong>Сетева
я ошибка:</strong> ${this.escapeHtml(err.message)}}`;
675 }
676 }
677
678 async testCurrentForm() {
679     const serverId = document.getElementById('mcp-server-id')?
.value.trim() || 'test_server';
680     const transport = document.getElementById('mcp-server-tran
sport')?.value || 'stdio';
681     const command = document.getElementById('mcp-server-comman
d')?.value.trim();
682     const argsRaw = document.getElementById('mcp-server-args')
?.value.trim();
683     const url = document.getElementById('mcp-server-url')?.val

```

```
ue.trim();
684     const envRaw = document.getElementById('mcp-server-env')?.
value.trim();
685
686     let args = [];
687     if (argsRaw) {
688         if (argsRaw.startsWith '[' && argsRaw.endsWith(']')) {
689             try { args = JSON.parse(argsRaw); } catch { args = arg
sRaw.split(/\s+/).filter(Boolean); }
690         } else {
691             args = argsRaw.split(/\s+/).filter(Boolean);
692         }
693     }
694
695     let env = {};
696     if (envRaw) {
697         try { env = JSON.parse(envRaw); } catch (e) { window.sho
wToast?.('Неверный формат JSON в env', 'danger') || alert('Нев
ерный формат JSON в env'); return; }
698     }
699
700     const testBtn = document.getElementById('btn-test-modal-se
rver');
701     if (testBtn) {
702         testBtn.disabled = true;
703         testBtn.innerHTML = '<span class="spinner-border spinner
-border-sm me-1"></span> Тест...';
704     }
705
706     try {
707         const res = await this.apiFetch('/api/admin/mcp/test', {
708             method: 'POST',
709             headers: { 'Content-Type': 'application/json' },
710             body: JSON.stringify({
711                 id: serverId,
712                 transport: transport,
713                 command: command,
714                 args: args,
```

```
715         url: url,
716         env: env
717     })
718 });
719 const data = await res.json();
720 if (data.status === 'ok') {
721     window.showToast?(`🟢 Подключение успешно! (${data.lat
722     ency_ms} ms)\nНайдено инструментов: ${data.tools_count}`, 'suc
723     cess');
724 } else {
725     window.showToast?(`🔴 Ошибка проверки подключения:\n${
726     data.message}`, 'danger');
727 }
728 } catch (err) {
729     window.showToast?(`🔴 Ошибка запроса: ${err.message}`, '
730     danger');
731 } finally {
732     if (testBtn) {
733         testBtn.disabled = false;
734         testBtn.innerHTML = '<i class="bi bi-play-circle"></i>
735         Проверить подключение';
736     }
737 }
738 }
739
740 async openAllToolsModal() {
741     const modalEl = document.getElementById('mcpAllToolsModal'
742     );
743     if (modalEl && window.bootstrap) {
744         const modal = bootstrap.Modal.getInstance(modalEl) || ne
745         w bootstrap.Modal(modalEl);
746         modal.show();
747     }
748
749     const loadingEl = document.getElementById('mcp-all-tools-l
750     oading');
751     const emptyEl = document.getElementById('mcp-all-tools-emp
752     ty');
```



```

775         <details class="small">
776             <summary class="text-secondary cursor-poin
777 ter">Схема аргументов</summary>
778             <pre class="bg-dark text-light p-2 rounded
779 mt-1 mb-0 font-monospace small" style="max-height: 150px; ove
780 rflow-y: auto;">${this.escapeHtml(typeof t.args_schema === 'ob
781 ject' ? JSON.stringify(t.args_schema, null, 2) : String(t.args
782 _schema))}</pre>
783         </details>
784     </div>
785     ` : ''}
786 </div>
787 </div>
788 `).join('');
789 }
790 } catch (err) {
791     loadingEl?.classList.add('d-none');
792     if (container) {
793         container.classList.remove('d-none');
794         container.innerHTML = `<div class="col-12"><div class=
795 "alert alert-danger">${this.escapeHtml(err.message)}</div></di
796 v>`;
797     }
798 }
799 }
800 renderDiscoveredTools(tools, container) {
801     if (!tools || tools.length === 0) {
802         container.innerHTML = '<div class="text-muted text-cente
803 r py-3">Сервер не вернул доступных инструментов.</div>';
804         return;
805     }
806     container.innerHTML = tools.map((t, idx) => `
807         <div class="p-3 bg-dark rounded border border-secondary"
808 >
809         <div class="d-flex align-items-center justify-content-
810 between mb-1">

```

```

803 |         <strong class="text-warning font-monospace d-flex al
|ign-items-center gap-1">
804 |             <i class="bi bi-wrench-adjustable"></i> ${this.esc
|apeHtml(t.name)}
805 |         </strong>
806 |         <span class="badge bg-dark border border-secondary t
|ext-muted">#${idx + 1}</span>
807 |     </div>
808 |     <div class="small text-muted mb-2">${this.escapeHtml(t
|.description || 'Описание не указано')}</div>
809 |     ${t.args_schema ? `
810 |         <details class="small">
811 |             <summary class="text-info cursor-pointer">Параметр
|ы и схема вызова</summary>
812 |             <pre class="bg-black text-white p-2 rounded mt-1 m
|b-0 font-monospace small" style="max-height: 180px; overflow-y
|: auto;">${this.escapeHtml(typeof t.args_schema === 'object' ?
|JSON.stringify(t.args_schema, null, 2) : String(t.args_schema
|))}</pre>
813 |         </details>
814 |         ` : ''}
815 |     </div>
816 |     `).join('');
817 | }
818 |
819 | renderError(msg) {
820 |     const container = document.getElementById('mcp-cards-conta
|iner');
821 |     if (container) {
822 |         container.innerHTML = `
823 |             <div class="col-12">
824 |                 <div class="alert alert-danger bg-dark border-danger
|text-light d-flex align-items-center justify-content-between"
|>
825 |                     <div><i class="bi bi-exclamation-triangle-fill tex
|t-danger me-2"></i> ${this.escapeHtml(msg)}</div>
826 |                     <button class="btn btn-sm btn-outline-light" oncli
|ck="window.mcpTab?.refresh()">Повторить</button>

```



```
827         </div>
828     </div>
829     `;
830 }
831 }
832
833 escapeHtml(str) {
834     if (!str) return '';
835     return String(str)
836         .replace(/&/g, '&amp;')
837         .replace(/</g, '&lt;')
838         .replace(/>/g, '&gt;')
839         .replace(/"/g, '&quot;')
840         .replace(/'/g, '&#039;');
841 }
842 }
843
844 // Global initialization hooks
845 window.McpTabManager = McpTabManager;
846 if (!window.mcpTab) {
847     window.mcpTab = new McpTabManager();
848 }
849 window.initMcpTab = function() {
850     if (window.mcpTab) {
851         window.mcpTab.init();
852     }
853 };
854
855 if (document.getElementById('mcp-tab-root')) {
856     window.mcpTab.init();
857 }
858
```

app.js

```
1 /**
2  * AI Breadboard Messenger & WebRTC Conference Client
3  */
4
5 const CONFIG = {
6   apiBase: window.location.origin,
7   wsProtocol: window.location.protocol === 'https:' ? 'wss:' :
8     'ws:',
9   iceServers: [{ urls: 'stun:stun.l.google.com:19302' }, { url
10     s: 'stun:stun1.l.google.com:19302' }],
11 };
12
13 let currentUser = null;
14 let activeRoomId = null;
15 let activeRoomData = null;
16 let ws = null;
17 let rooms = [];
18 let mediaRecorder = null;
19 let recordedAudioChunks = [];
20 let isRecordingVoice = false;
21
22 // WebRTC Call State
23 let peerConnection = null;
24 let localStream = null;
25 let remoteStream = null;
26 let isInCall = false;
27
28 // DOM Elements
29 const chatListEl = document.getElementById('chatList');
30 const messagesContainerEl = document.getElementById('messagesC
31   ontainer');
32 const messageInputEl = document.getElementById('messageInput')
33   ;
34 const btnSendMessageEl = document.getElementById('btnSendMessa
35   ge');
36 const btnRecordVoiceEl = document.getElementById('btnRecordVoi
```

```

    ce');
32 const btnAttachFileEl = document.getElementById('btnAttachFile
    ');
33 const fileAttachmentInputEl = document.getElementById('fileAtt
    achmentInput');
34 const typingIndicatorEl = document.getElementById('typingIndic
    ator');
35 const chatHeaderEl = document.getElementById('chatHeader');
36 const chatInputBarEl = document.getElementById('chatInputBar')
    ;
37 const activeChatTitleEl = document.getElementById('activeChatT
    itle');
38 const activeChatSubtitleEl = document.getElementById('activeCh
    atSubtitle');
39 const newChatModalEl = document.getElementById('newChatModal')
    ;
40 const videoCallOverlayEl = document.getElementById('videoCallO
    verlay');
41
42 // Initialize
43 window.addEventListener('DOMContentLoaded', async () => {
44     await initUserProfile();
45     initWebSocket();
46     await loadRooms();
47     setupEventListeners();
48     registerServiceWorker();
49 });
50
51 async function initUserProfile() {
52     try {
53         const res = await fetch(`${CONFIG.apiBase}/api/messenger/u
            sers/me`, { credentials: 'include' });
54         const data = await res.json();
55         if (data.status === 'success') {
56             currentUser = data.user;
57             document.getElementById('myName').innerText = currentUse
                r.display_name || currentUser.username;
58             document.getElementById('myAvatarText').innerText = (cur

```

```

    rentUser.display_name || 'U')[0].toUpperCase();
59     }
60   } catch (e) {
61     console.error('Failed to load user profile:', e);
62   }
63 }
64
65 function initWebSocket() {
66   if (!currentUser) return;
67   const wsUrl = `${CONFIG.wsProtocol}://${window.location.host}
    /api/messenger/ws/${currentUser.id}`;
68   ws = new WebSocket(wsUrl);
69
70   ws.onopen = () => console.log('Messenger WebSocket Connected
    ');
71   ws.onmessage = (event) => handleIncomingWsPacket(JSON.parse(
    event.data));
72   ws.onclose = () => {
73     console.warn('WebSocket closed. Reconnecting in 3s...');
74     setTimeout(initWebSocket, 3000);
75   };
76 }
77
78 function handleIncomingWsPacket(packet) {
79   const { type, message, room_id, user_id, bot_name, is_online
    } = packet;
80
81   if (type === 'new_message') {
82     if (activeRoomId === message.room_id) {
83       appendMessageToChat(message);
84       scrollToBottom();
85       // Send read receipt
86       if (ws && ws.readyState === WebSocket.OPEN && message.se
    nder_id !== currentUser.id) {
87         ws.send(JSON.stringify({ type: 'message_read', room_id
    : activeRoomId, message_id: message.id }));
88       }
89     }

```

```

90     updateRoomLastMessage(message);
91 } else if (type === 'typing_start') {
92     if (activeRoomId === room_id && user_id !== currentUser.id
93 ) {
94         typingIndicatorEl.style.display = 'block';
95         typingIndicatorEl.innerText = `${bot_name || 'Someone'}
96 is typing...`;
97     }
98 } else if (type === 'typing_stop') {
99     if (activeRoomId === room_id) {
100         typingIndicatorEl.style.display = 'none';
101     }
102 } else if (type === 'presence_update') {
103     updateUserPresenceUI(user_id, is_online);
104 } else if (type.startsWith('webrtc_')) {
105     handleWebRTCSignalPacket(packet);
106 }
107 }
108
109 async function loadRooms() {
110     try {
111         const res = await fetch(`${CONFIG.apiBase}/api/messenger/r
112 ooms`, { credentials: 'include' });
113         const data = await res.json();
114         if (data.status === 'success') {
115             rooms = data.rooms;
116             renderRoomsList();
117         }
118     } catch (e) {
119         console.error('Error loading rooms:', e);
120     }
121 }
122
123 function renderRoomsList() {
124     chatListEl.innerHTML = '';
125     rooms.forEach((r) => {
126         const item = document.createElement('div');
127         item.className = `chat-item ${activeRoomId === r.id ? 'act

```

```

    iver' : ''}`;
125     item.onclick = () => selectRoom(r);
126
127     const initial = (r.title || '#')[0].toUpperCase();
128     const lastContent = r.last_message ? (r.last_message.conte
nt || '[Attachment]') : 'No messages yet';
129     const unreadHtml = r.unread_count > 0 ? `` : '';
130     const onlineHtml = r.is_online ? '<div class="status-dot o
nline"></div>' : '';
131
132     item.innerHTML = `
133         <div class="avatar">
134             <span>${initial}</span>
135             ${onlineHtml}
136         </div>
137         <div class="chat-info">
138             <div class="chat-title-row">
139                 <span class="chat-title">${escapeHtml(r.title)}</spa
n>
140                 <span class="chat-time">${r.last_message ? formatTim
e(r.last_message.created_at) : ''}</span>
141             </div>
142             <div class="chat-snippet-row">
143                 <span class="chat-snippet">${escapeHtml(lastContent)
}</span>
144                 ${unreadHtml}
145             </div>
146         </div>
147     `;
148     chatListEl.appendChild(item);
149 });
150 }
151
152 async function selectRoom(room) {
153     activeRoomId = room.id;
154     activeRoomData = room;
155     renderRoomsList();

```

```

156
157 chatHeaderEl.style.display = 'flex';
158 chatInputBarEl.style.display = 'flex';
159 document.getElementById('emptyPlaceholder').style.display =
    'none';
160
161 activeChatTitleEl.innerText = room.title;
162 activeChatSubtitleEl.innerText = room.room_type === 'direct'
    ? (room.is_online ? 'Online' : 'Offline') : `${room.members_c
    ount} members`;
163 document.getElementById('activeChatAvatarText').innerText =
    (room.title || '#')[0].toUpperCase();
164
165 await loadRoomMessages(room.id);
166 }
167
168 async function loadRoomMessages(roomId) {
169     messagesContainerEl.innerHTML = '';
170     try {
171         const res = await fetch(`${CONFIG.apiBase}/api/messenger/r
            ooms/${roomId}/messages`, { credentials: 'include' });
172         const data = await res.json();
173         if (data.status === 'success') {
174             data.messages.forEach(appendMessageToChat);
175             scrollToBottom();
176         }
177     } catch (e) {
178         console.error('Error loading messages:', e);
179     }
180 }
181
182 function appendMessageToChat(msg) {
183     const isOut = msg.sender_id === currentUser.id;
184     const bubble = document.createElement('div');
185     bubble.className = `message-bubble ${isOut ? 'outgoing' : 'i
        ncoming'}`;
186
187     let senderHeader = '';

```

```

188 |   if (!isOut && activeRoomData && activeRoomData.room_type !==
      |   'direct') {
189 |       senderHeader = `

`;
190 |   }
191 |
192 |   let attachmentsHtml = '';
193 |   if (msg.attachments && msg.attachments.length > 0) {
194 |       msg.attachments.forEach((att) => {
195 |           if (att.file_type === 'audio') {
196 |               attachmentsHtml += `


```



```

220     r.last_message = msg;
221     renderRoomsList();
222 }
223 }
224
225 function updateUserPresenceUI(userId, isOnline) {
226     rooms.forEach((r) => {
227         if (r.room_type === 'direct') {
228             r.is_online = isOnline;
229         }
230     });
231     renderRoomsList();
232 }
233
234 // Send Message
235 async function sendMessage() {
236     const content = messageInputEl.value.trim();
237     if (!content || !activeRoomId) return;
238
239     if (ws && ws.readyState === WebSocket.OPEN) {
240         ws.send(JSON.stringify({
241             type: 'send_message',
242             room_id: activeRoomId,
243             content: content,
244             message_type: 'text'
245         }));
246     } else {
247         // REST Fallback
248         await fetch(`${CONFIG.apiBase}/api/messenger/rooms/${activ
eRoomId}/messages`, {
249             method: 'POST',
250             headers: { 'Content-Type': 'application/json' },
251             credentials: 'include',
252             body: JSON.stringify({ content: content, message_type: '
text' }),
253         });
254     }
255

```

```

256 | messageInputEl.value = '';
257 | }
258 |
259 | // Voice Note Recording
260 | async function toggleVoiceRecording() {
261 |   if (isRecordingVoice) {
262 |     mediaRecorder.stop();
263 |     isRecordingVoice = false;
264 |     btnRecordVoiceEl.style.background = 'var(--bg-secondary)';
265 |   } else {
266 |     try {
267 |       const stream = await navigator.mediaDevices.getUserMedia
268 |       ({ audio: true });
269 |       mediaRecorder = new MediaRecorder(stream);
270 |       recordedAudioChunks = [];
271 |
272 |       mediaRecorder.ondataavailable = (e) => recordedAudioChun
273 |       ks.push(e.data);
274 |       mediaRecorder.onstop = async () => {
275 |         const blob = new Blob(recordedAudioChunks, { type: 'au
276 |         dio/webm' });
277 |         const formData = new FormData();
278 |         formData.append('file', blob, `voice_${Date.now()}.web
279 |         m`);
280 |         formData.append('duration_sec', '3.0');
281 |
282 |         const res = await fetch(`${CONFIG.apiBase}/api/messeng
283 |         er/upload`, {
284 |           method: 'POST',
285 |           credentials: 'include',
286 |           body: formData,
287 |         });
288 |         const data = await res.json();
289 |         if (data.status === 'success') {
290 |           ws.send(JSON.stringify({
291 |             type: 'send_message',
292 |             room_id: activeRoomId,
293 |             content: '',

```

```
289         message_type: 'audio',
290         attachment_ids: [data.attachment_id],
291     }));
292     }
293 };
294
295     mediaRecorder.start();
296     isRecordingVoice = true;
297     btnRecordVoiceEl.style.background = 'var(--danger-color)
';
298     } catch (e) {
299         window.showToast?('Для записи голосовых сообщений требу
ется доступ к микрофону', 'warning') || alert('Для записи голо
совых сообщений требуется доступ к микрофону');
300     }
301 }
302 }
303
304 // WebRTC Conference / Call Handler
305 async function startWebRTCCall(callType = 'video') {
306     if (!activeRoomId) return;
307     videoCallOverlayEl.style.display = 'flex';
308     isInCall = true;
309
310     try {
311         localStream = await navigator.mediaDevices.getUserMedia({
312             video: callType === 'video',
313             audio: true,
314         });
315         document.getElementById('localVideo').srcObject = localStr
eam;
316
317         initPeerConnection();
318         localStream.getTracks().forEach((track) => peerConnection.
addTrack(track, localStream));
319
320         const offer = await peerConnection.createOffer();
321         await peerConnection.setLocalDescription(offer);
```

```
322 |
323 |     ws.send(JSON.stringify({
324 |         type: 'webrtc_offer',
325 |         data: {
326 |             action: 'offer',
327 |             room_id: activeRoomId,
328 |             sdp: offer.sdp,
329 |             call_type: callType,
330 |         },
331 |     }));
332 | } catch (e) {
333 |     console.error('Call initialization failed:', e);
334 | }
335 | }
336 |
337 | function initPeerConnection() {
338 |     peerConnection = new RTCPeerConnection({ iceServers: CONFIG.
iceServers });
339 |
340 |     peerConnection.onicecandidate = (event) => {
341 |         if (event.candidate && ws) {
342 |             ws.send(JSON.stringify({
343 |                 type: 'webrtc_ice_candidate',
344 |                 data: {
345 |                     action: 'ice_candidate',
346 |                     room_id: activeRoomId,
347 |                     candidate: event.candidate,
348 |                 },
349 |             }));
350 |         }
351 |     };
352 |
353 |     peerConnection.ontrack = (event) => {
354 |         remoteStream = event.streams[0];
355 |         const remoteVid = document.getElementById('remoteVideo');
356 |         remoteVid.srcObject = remoteStream;
357 |         document.getElementById('remoteVideoTile').style.display =
'flex';
```

```

358 |     };
359 | }
360 |
361 | async function handleWebRTCSignalPacket(packet) {
362 |     const { type, sdp, candidate, sender_id, room_id } = packet;
363 |
364 |     if (type === 'webrtc_offer') {
365 |         videoCallOverlayEl.style.display = 'flex';
366 |         isInCall = true;
367 |         initPeerConnection();
368 |
369 |         localStream = await navigator.mediaDevices.getUserMedia({
370 |             video: true, audio: true });
371 |         document.getElementById('localVideo').srcObject = localStream;
372 |         localStream.getTracks().forEach((track) => peerConnection.
373 |             addTrack(track, localStream));
374 |
375 |         await peerConnection.setRemoteDescription(new RTCSessionDe
376 |             scription({ type: 'offer', sdp }));
377 |         const answer = await peerConnection.createAnswer();
378 |         await peerConnection.setLocalDescription(answer);
379 |
380 |         ws.send(JSON.stringify({
381 |             type: 'webrtc_answer',
382 |             data: {
383 |                 action: 'answer',
384 |                 room_id: room_id,
385 |                 target_id: sender_id,
386 |                 sdp: answer.sdp,
387 |             },
388 |         }));
389 |     } else if (type === 'webrtc_answer') {
390 |         if (peerConnection) {
391 |             await peerConnection.setRemoteDescription(new RTCSession
392 |                 Description({ type: 'answer', sdp }));
393 |         }
394 |     } else if (type === 'webrtc_ice_candidate') {

```

```

391     if (peerConnection && candidate) {
392         await peerConnection.addIceCandidate(new RTCIceCandidate
393         (candidate));
394     }
395 }
396
397 function endCall() {
398     if (localStream) localStream.getTracks().forEach((t) => t.st
399     op());
400     if (peerConnection) peerConnection.close();
401     peerConnection = null;
402     videoCallOverlayEl.style.display = 'none';
403     isInCall = false;
404
405     if (ws && activeRoomId) {
406         ws.send(JSON.stringify({
407             type: 'webrtc_leave_call',
408             data: { action: 'leave_call', room_id: activeRoomId },
409         }));
410     }
411 }
412 // Event Listeners
413 function setupEventListeners() {
414     btnSendMessageEl.onclick = sendMessage;
415     messageInputEl.onkeydown = (e) => {
416         if (e.key === 'Enter') sendMessage();
417     };
418     btnRecordVoiceEl.onclick = toggleVoiceRecording;
419     btnAttachFileEl.onclick = () => fileAttachmentInputEl.click(
420     );
421
422     fileAttachmentInputEl.onChange = async () => {
423         const file = fileAttachmentInputEl.files[0];
424         if (!file || !activeRoomId) return;
425         const formData = new FormData();
426         formData.append('file', file);

```

```

426     const res = await fetch(`${CONFIG.apiBase}/api/messenger/u
pload`, {
427         method: 'POST',
428         credentials: 'include',
429         body: formData,
430     });
431     const data = await res.json();
432     if (data.status === 'success') {
433         ws.send(JSON.stringify({
434             type: 'send_message',
435             room_id: activeRoomId,
436             content: '',
437             message_type: data.file_type,
438             attachment_ids: [data.attachment_id],
439         }));
440     }
441 };
442
443 document.getElementById('btnStartVideoCall').onclick = () =>
startWebRTCCall('video');
444 document.getElementById('btnStartAudioCall').onclick = () =>
startWebRTCCall('audio');
445 document.getElementById('btnEndCall').onclick = endCall;
446
447 // New Chat Modal
448 document.getElementById('btnNewChat').onclick = () => { newC
hatModalEl.style.display = 'flex'; };
449 document.getElementById('modalBtnCancel').onclick = () => {
newChatModalEl.style.display = 'none'; };
450 document.getElementById('modalRoomType').onchange = (e) => {
451     document.getElementById('modalTitleGroup').style.display =
e.target.value !== 'direct' ? 'block' : 'none';
452 };
453
454 document.getElementById('modalBtnCreate').onclick = async ()
=> {
455     const roomType = document.getElementById('modalRoomType').
value;

```

```

456     const title = document.getElementById('modalRoomTitle').va
    lue;
457     const res = await fetch(`${CONFIG.apiBase}/api/messenger/r
    ooms`, {
458         method: 'POST',
459         headers: { 'Content-Type': 'application/json' },
460         credentials: 'include',
461         body: JSON.stringify({ room_type: roomType, title: title
    }),
462     });
463     const data = await res.json();
464     if (data.status === 'success') {
465         newChatModalEl.style.display = 'none';
466         await loadRooms();
467     }
468 };
469 }
470
471 function formatTime(isoStr) {
472     if (!isoStr) return '';
473     const d = new Date(isoStr);
474     return d.toLocaleTimeString([], { hour: '2-digit', minute: '
    2-digit' });
475 }
476
477 function scrollToBottom() {
478     messagesContainerEl.scrollTop = messagesContainerEl.scrollHeight;
479 }
480
481 function escapeHtml(text) {
482     const div = document.createElement('div');
483     div.innerText = text;
484     return div.innerHTML;
485 }
486
487 function registerServiceWorker() {
488     if ('serviceWorker' in navigator) {

```



```
489 |     navigator.serviceWorker.register('sw.js').catch(console.er  
    |     ror);  
490 | }  
491 | }  
492 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-s
cale=1.0, maximum-scale=1.0, user-scalable=no" />
6   <title>AI Breadboard Messenger & Rooms</title>
7   <link rel="stylesheet" href="style.css" />
8   <link rel="manifest" href="manifest.json" />
9   <meta name="theme-color" content="#0f172a" />
10 </head>
11 <body>
12   <div class="messenger-app">
13     <!-- Sidebar -->
14     <aside class="sidebar" id="sidebar">
15       <div class="sidebar-header">
16         <div class="user-profile-badge">
17           <div class="avatar" id="myAvatar">
18             <span id="myAvatarText">U</span>
19             <div class="status-dot online"></div>
20           </div>
21           <div>
22             <div style="font-weight: 600; font-size: 15px;" id
="myName">Loading...</div>
23             <div style="font-size: 12px; color: var(--text-sec
ondary);" id="myStatus">Online</div>
24           </div>
25         </div>
26         <button class="btn-icon" id="btnNewChat" title="New Ch
at or Room">🗨️ </button>
27       </div>
28
29       <div class="search-bar-container">
30         <input type="text" class="search-input" id="searchChat
sInput" placeholder="Search chats, contacts, @bots..." />
31       </div>
```

```

32 |
33 |     <div class="chat-list" id="chatList">
34 |         <!-- Chat items rendered dynamically -->
35 |     </div>
36 | </aside>
37 |
38 | <!-- Main Chat Area -->
39 | <main class="chat-main" id="chatMain">
40 |     <!-- Chat Header -->
41 |     <div class="chat-header" id="chatHeader" style="display:
| none;">
42 |         <div style="display: flex; align-items: center; gap: 1
| 2px;">
43 |             <div class="avatar" id="activeChatAvatar">
44 |                 <span id="activeChatAvatarText">#</span>
45 |             </div>
46 |             <div>
47 |                 <div style="font-weight: 600; font-size: 16px;" id
| ="activeChatTitle">Select a chat</div>
48 |                 <div style="font-size: 12px; color: var(--text-sec
| ondary);" id="activeChatSubtitle">...</div>
49 |             </div>
50 |         </div>
51 |         <div class="chat-header-actions">
52 |             <button class="btn-icon" id="btnStartAudioCall" titl
| e="Audio Call">🎙️ </button>
53 |             <button class="btn-icon" id="btnStartVideoCall" titl
| e="Video Meeting">📺 </button>
54 |         </div>
55 |     </div>
56 |
57 |     <!-- Messages Stream -->
58 |     <div class="messages-container" id="messagesContainer">
59 |         <div style="text-align: center; color: var(--text-seco
| ndary); margin-top: 40px;" id="emptyPlaceholder">
60 |             Select or create a conversation to start messaging.
61 |         </div>
62 |     </div>

```

```

63 |
64 |     <!-- Typing indicator -->
65 |     <div class="typing-indicator" id="typingIndicator" style
    | ="display: none;"></div>
66 |
67 |     <!-- Input Bar -->
68 |     <div class="chat-input-bar" id="chatInputBar" style="dis
    | play: none;">
69 |         <input type="file" id="fileAttachmentInput" style="dis
    | play: none;" multiple />
70 |         <button class="btn-icon" id="btnAttachFile" title="Att
    | ach file or photo">📎 </button>
71 |
72 |         <input type="text" class="message-input" id="messageIn
    | put" placeholder="Write a message... (Type @ai for bot)" autoc
    | omplete="off" />
73 |
74 |         <button class="btn-icon" id="btnRecordVoice" title="Re
    | cord voice note">🎤 </button>
75 |         <button class="btn-icon" id="btnSendMessage" title="Se
    | nd message" style="background: var(--accent-color);">➤ </button
    | >
76 |     </div>
77 |
78 |     <!-- WebRTC Meeting Room Overlay -->
79 |     <div class="video-call-overlay" id="videoCallOverlay" st
    | yle="display: none;">
80 |         <div class="video-grid" id="videoGrid">
81 |             <div class="video-tile" id="localVideoTile">
82 |                 <video id="localVideo" autoplay playsinline muted>
    | </video>
83 |                 <span class="tile-label">You</span>
84 |             </div>
85 |             <div class="video-tile" id="remoteVideoTile" style="
    | display: none;">
86 |                 <video id="remoteVideo" autoplay playsinline></vid
    | eo>
87 |                 <span class="tile-label" id="remoteTileLabel">Part

```

```

    icipant</span>
88         </div>
89     </div>
90
91     <div class="call-controls">
92         <button class="btn-icon" id="btnToggleMic" title="Mu
te/Unmute Mic">⌵ </button>
93         <button class="btn-icon" id="btnToggleCam" title="To
ggle Camera">⌵ </button>
94         <button class="btn-icon" id="btnShareScreen" title="
Share Screen">⌵ </button>
95         <button class="btn-icon btn-danger" id="btnEndCall"
title="Hang Up">⌵ </button>
96     </div>
97 </div>
98 </main>
99 </div>
100
101 <!-- New Chat Modal -->
102 <div class="modal-overlay" id="newChatModal" style="display:
none;">
103     <div class="modal-card">
104         <h3>Create New Conversation</h3>
105         <div class="form-group">
106             <label>Room Type</label>
107             <select id="modalRoomType">
108                 <option value="direct">Direct Message (1-on-1)</opti
on>
109                 <option value="group">Group Chat</option>
110                 <option value="meeting">Video Meeting Room</option>
111             </select>
112         </div>
113         <div class="form-group" id="modalTitleGroup" style="disp
lay: none;">
114             <label>Room Title</label>
115             <input type="text" id="modalRoomTitle" placeholder="e.
g. Project Discussion" />
116         </div>

```

```
117 |         <div class="form-group">
118 |             <label>Search User / Member</label>
119 |             <input type="text" id="modalUserSearch" placeholder="T
| type username or name..." />
120 |             <div id="modalUserResults" style="max-height: 120px; o
| verflow-y: auto; margin-top: 8px;"></div>
121 |         </div>
122 |         <div style="display: flex; justify-content: flex-end; ga
| p: 10px; margin-top: 20px;">
123 |             <button class="btn-icon" style="width: auto; padding:
| 8px 16px; border-radius: 8px;" id="modalBtnCancel">Cancel</but
| ton>
124 |             <button class="btn-icon" style="width: auto; padding:
| 8px 16px; border-radius: 8px; background: var(--accent-color);
| " id="modalBtnCreate">Create</button>
125 |         </div>
126 |     </div>
127 | </div>
128 |
129 | <script src="app.js"></script>
130 | </body>
131 | </html>
132 |
```

style.css

```
1  /* =====
   |=====
2  messenger/style.css – Дочерние стили Messenger (наследует m
ain.css)
3  Версия: 20260917_refactor_v1
4  =====
   |===== */
5
6  /* Messenger наследует переменные из main.css.
7  Локальные псевдонимы для удобства внутри компонента: */
8  .messenger-app {
9    --msg-outgoing: var(--chat-user-bg);
10   --msg-incoming: var(--chat-bot-bg);
11   --msg-incoming-text: var(--chat-bot-text);
12   --online-color: #10b981;
13   --offline-color: #6b7280;
14   --danger-color: #ef4444;
15 }
16
17 /* — Основной контейнер —————
   |————— */
18 .messenger-app {
19   display: flex;
20   height: 100vh;
21   width: 100%;
22   background: var(--bg-color);
23   font-size: 0.85rem;
24 }
25
26 /* — Боковая панель —————
   |————— */
27 .sidebar {
28   width: 320px;
29   min-width: 260px;
30   background: var(--surface-1);
31   border-right: 1px solid var(--border-color);
```

```
32 | display: flex;
33 | flex-direction: column;
34 | z-index: 10;
35 | }
36 |
37 | .sidebar-header {
38 |   padding: 10px 12px;
39 |   display: flex;
40 |   align-items: center;
41 |   justify-content: space-between;
42 |   border-bottom: 1px solid var(--border-color);
43 | }
44 |
45 | /* — Профиль пользователя —————
   | ————— */
46 | .user-profile-badge {
47 |   display: flex;
48 |   align-items: center;
49 |   gap: 10px;
50 | }
51 |
52 | .avatar {
53 |   width: 36px;
54 |   height: 36px;
55 |   border-radius: 50%;
56 |   background: var(--border-color);
57 |   display: flex;
58 |   align-items: center;
59 |   justify-content: center;
60 |   font-weight: 700;
61 |   font-size: 0.85rem;
62 |   color: var(--text-color);
63 |   position: relative;
64 |   object-fit: cover;
65 |   flex-shrink: 0;
66 | }
67 |
68 | .avatar .status-dot {
```



```
69 | position: absolute;
70 | bottom: 0;
71 | right: 0;
72 | width: 10px;
73 | height: 10px;
74 | border-radius: 50%;
75 | border: 2px solid var(--surface-1);
76 | }
77 |
78 | .status-dot.online { background-color: var(--online-color); }
79 | .status-dot.offline { background-color: var(--offline-color);
80 | }
81 | /* — Строка поиска —————
82 | |————— */
83 | .search-bar-container {
84 |   padding: 8px 10px;
85 |   border-bottom: 1px solid var(--border-color);
86 | }
87 | .search-input {
88 |   width: 100%;
89 |   padding: 6px 12px;
90 |   background: var(--surface-2);
91 |   border: 1px solid var(--border-color);
92 |   border-radius: 16px;
93 |   color: var(--text-color);
94 |   outline: none;
95 |   font-size: 0.82rem;
96 |   transition: border-color 0.15s;
97 | }
98 |
99 | .search-input::placeholder { color: var(--text-muted); }
100 | .search-input:focus { border-color: var(--nav-active); }
101 |
102 | /* — Список чатов —————
103 | |————— */
104 | .chat-list {
```

```
104 | flex: 1;
105 | overflow-y: auto;
106 | }
107 |
108 | .chat-item {
109 |   display: flex;
110 |   align-items: center;
111 |   gap: 10px;
112 |   padding: 9px 12px;
113 |   cursor: pointer;
114 |   border-bottom: 1px solid var(--border-subtle);
115 |   transition: background 0.12s ease;
116 | }
117 |
118 | .chat-item:hover {
119 |   background: var(--nav-hover-bg);
120 | }
121 |
122 | .chat-item.active {
123 |   background: rgba(37, 99, 235, 0.08);
124 |   border-left: 3px solid var(--nav-active);
125 | }
126 |
127 | [data-theme="dark"] .chat-item.active {
128 |   background: rgba(88, 166, 255, 0.08);
129 | }
130 |
131 | .chat-info { flex: 1; min-width: 0; }
132 |
133 | .chat-title-row {
134 |   display: flex;
135 |   justify-content: space-between;
136 |   align-items: center;
137 |   margin-bottom: 2px;
138 | }
139 |
140 | .chat-title {
141 |   font-weight: 600;
```

```
142 | font-size: 0.85rem;
143 | white-space: nowrap;
144 | overflow: hidden;
145 | text-overflow: ellipsis;
146 | color: var(--text-color);
147 | }
148 |
149 | .chat-time {
150 |     font-size: 0.72rem;
151 |     color: var(--text-muted);
152 |     flex-shrink: 0;
153 | }
154 |
155 | .chat-snippet-row {
156 |     display: flex;
157 |     justify-content: space-between;
158 |     align-items: center;
159 | }
160 |
161 | .chat-snippet {
162 |     font-size: 0.78rem;
163 |     color: var(--text-muted);
164 |     white-space: nowrap;
165 |     overflow: hidden;
166 |     text-overflow: ellipsis;
167 | }
168 |
169 | .unread-badge {
170 |     background: var(--nav-active);
171 |     color: #fff;
172 |     font-size: 0.68rem;
173 |     font-weight: 700;
174 |     padding: 1px 6px;
175 |     border-radius: 10px;
176 |     flex-shrink: 0;
177 | }
178 |
179 | /* — Главная область чата
```

```
_____*/  
180 .chat-main {  
181     flex: 1;  
182     display: flex;  
183     flex-direction: column;  
184     background: var(--bg-color);  
185     position: relative;  
186 }  
187  
188 .chat-header {  
189     padding: 8px 16px;  
190     background: var(--surface-1);  
191     border-bottom: 1px solid var(--border-color);  
192     display: flex;  
193     justify-content: space-between;  
194     align-items: center;  
195     z-index: 5;  
196 }  
197  
198 .chat-header-actions {  
199     display: flex;  
200     gap: 8px;  
201 }  
202  
203 .btn-icon {  
204     background: var(--surface-2);  
205     border: 1px solid var(--border-color);  
206     color: var(--text-color);  
207     width: 32px;  
208     height: 32px;  
209     border-radius: 50%;  
210     display: flex;  
211     align-items: center;  
212     justify-content: center;  
213     cursor: pointer;  
214     font-size: 0.85rem;  
215     transition: background 0.15s, color 0.15s;  
216 }
```

```
217 |
218 | .btn-icon:hover {
219 |     background: var(--nav-active);
220 |     color: #fff;
221 |     border-color: var(--nav-active);
222 | }
223 |
224 | .btn-danger { background: var(--danger-color) !important; color: #fff !important; }
225 |
226 | /* — Контейнер сообщений —————
    | ————— */
227 | .messages-container {
228 |     flex: 1;
229 |     padding: 12px 16px;
230 |     overflow-y: auto;
231 |     display: flex;
232 |     flex-direction: column;
233 |     gap: 8px;
234 | }
235 |
236 | .message-bubble {
237 |     max-width: 70%;
238 |     padding: 8px 12px;
239 |     border-radius: 14px;
240 |     position: relative;
241 |     line-height: 1.45;
242 |     font-size: 1.15rem;
243 |     word-break: break-word;
244 | }
245 |
246 | .message-bubble.incoming {
247 |     align-self: flex-start;
248 |     background: var(--msg-incoming);
249 |     color: var(--msg-incoming-text);
250 |     border-bottom-left-radius: 4px;
251 | }
252 |
```

```
253 | .message-bubble.outgoing {
254 |     align-self: flex-end;
255 |     background: var(--msg-outgoing);
256 |     color: var(--chat-user-text);
257 |     border-bottom-right-radius: 4px;
258 | }
259 |
260 | .message-meta {
261 |     display: flex;
262 |     align-items: center;
263 |     gap: 4px;
264 |     margin-top: 4px;
265 |     font-size: 0.72rem;
266 |     opacity: 0.75;
267 | }
268 |
269 | .message-bubble.outgoing .message-meta {
270 |     justify-content: flex-end;
271 | }
272 |
273 | .message-status {
274 |     font-size: 0.75rem;
275 | }
276 |
277 | .audio-preview {
278 |     margin-top: 6px;
279 |     width: 100%;
280 |     max-width: 240px;
281 | }
282 |
283 | /* — Индикатор набора текста —————
    | _____ */
284 | .typing-indicator {
285 |     display: flex;
286 |     align-items: center;
287 |     gap: 6px;
288 |     padding: 4px 16px;
289 |     font-size: 0.8rem;
```

```
290 | color: var(--text-muted);
291 | font-style: italic;
292 | }
293 |
294 | /* — Поле ввода —————
    | ————— */
295 | .chat-input-bar {
296 |   padding: 8px 12px;
297 |   background: var(--surface-1);
298 |   border-top: 1px solid var(--border-color);
299 |   display: flex;
300 |   align-items: center;
301 |   gap: 8px;
302 | }
303 |
304 | .message-input {
305 |   flex: 1;
306 |   padding: 8px 14px;
307 |   background: var(--surface-2);
308 |   border: 1px solid var(--border-color);
309 |   border-radius: 20px;
310 |   color: var(--text-color);
311 |   outline: none;
312 |   font-size: 1.15rem;
313 |   transition: border-color 0.15s;
314 | }
315 |
316 | .message-input::placeholder { color: var(--text-muted); }
317 | .message-input:focus { border-color: var(--nav-active); }
318 |
319 | .voice-recording-status {
320 |   color: var(--danger-color);
321 |   font-weight: 600;
322 |   font-size: 0.82rem;
323 |   display: flex;
324 |   align-items: center;
325 |   gap: 6px;
326 | }
```

```
327 |
328 | /* — Оверлей видеозвонка —————
    | ————— */
329 | .video-call-overlay {
330 |     position: absolute;
331 |     top: 0; left: 0;
332 |     width: 100%; height: 100%;
333 |     background: rgba(0, 0, 0, 0.88);
334 |     backdrop-filter: blur(12px);
335 |     z-index: 100;
336 |     display: flex;
337 |     flex-direction: column;
338 | }
339 |
340 | .video-grid {
341 |     flex: 1;
342 |     display: grid;
343 |     grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
344 |     gap: 12px;
345 |     padding: 16px;
346 |     align-items: center;
347 |     justify-items: center;
348 | }
349 |
350 | .video-tile {
351 |     width: 100%; height: 100%;
352 |     max-height: 440px;
353 |     background: var(--surface-1);
354 |     border-radius: 12px;
355 |     overflow: hidden;
356 |     position: relative;
357 |     display: flex;
358 |     align-items: center;
359 |     justify-content: center;
360 |     border: 1px solid var(--border-color);
361 | }
362 |
363 | .video-tile video { width: 100%; height: 100%; object-fit: cov
```



```
er; }
364
365 .video-tile .tile-label {
366     position: absolute;
367     bottom: 10px; left: 10px;
368     background: rgba(0, 0, 0, 0.55);
369     padding: 3px 8px;
370     border-radius: 6px;
371     font-size: 0.78rem;
372     color: #fff;
373 }
374
375 .call-controls {
376     height: 72px;
377     background: var(--surface-1);
378     display: flex;
379     align-items: center;
380     justify-content: center;
381     gap: 16px;
382     border-top: 1px solid var(--border-color);
383 }
384
385 .audio-player-custom {
386     margin-top: 4px;
387     width: 100%;
388     max-width: 240px;
389 }
390
391 /* — Модальное окно —————
    ————— */
392 .modal-overlay {
393     position: fixed;
394     top: 0; left: 0; width: 100%; height: 100%;
395     background: rgba(0, 0, 0, 0.6);
396     display: flex;
397     align-items: center;
398     justify-content: center;
399     z-index: 200;
```

```
400 }
401
402 .modal-card {
403     background: var(--surface-1);
404     border: 1px solid var(--border-color);
405     padding: 20px;
406     border-radius: 12px;
407     box-shadow: var(--shadow-lg);
408     width: 90%;
409     max-width: 420px;
410 }
411
412 .modal-card h3 { margin-bottom: 12px; font-size: 1rem; font-weight: 600; }
413
414 .form-group { margin-bottom: 10px; }
415 .form-group label {
416     display: block;
417     font-size: 0.78rem;
418     color: var(--text-muted);
419     margin-bottom: 4px;
420 }
421 .form-group input,
422 .form-group select {
423     width: 100%;
424     padding: 7px 10px;
425     background: var(--surface-2);
426     border: 1px solid var(--border-color);
427     border-radius: 6px;
428     color: var(--text-color);
429     font-size: 0.82rem;
430     outline: none;
431     transition: border-color 0.15s;
432 }
433 .form-group input:focus,
434 .form-group select:focus {
435     border-color: var(--nav-active);
436 }
```


sw.js

```
1 const CACHE_NAME = 'ai-breadboard-messenger-v1';
2 const ASSETS = [
3   './',
4   './index.html',
5   './style.css',
6   './app.js',
7   './manifest.json',
8 ];
9
10 self.addEventListener('install', (event) => {
11   event.waitUntil(
12     caches.open(CACHE_NAME).then((cache) => cache.addAll(ASSETS))
13   );
14 });
15
16 self.addEventListener('fetch', (event) => {
17   // Network first, cache fallback
18   event.respondWith(
19     fetch(event.request).catch(() => caches.match(event.request))
20   );
21 });
22
```

widget.js

```
1 /**
2  * AI Breadboard Floating Messenger Widget
3  * Zero-dependency embeddable script for any website or WordPr
4  */
5
6 (function () {
7     const config = window.AB_MESSENGER_CONFIG || {
8         apiUrl: window.location.origin,
9         ssoToken: '',
10        userId: '',
11        userName: 'Guest',
12    };
13
14    const style = document.createElement('style');
15    style.innerHTML = `
16        .ab-widget-bubble {
17            position: fixed;
18            bottom: 24px;
19            right: 24px;
20            width: 60px;
21            height: 60px;
22            border-radius: 50%;
23            background: linear-gradient(135deg, #3b82f6, #1d4ed8);
24            box-shadow: 0 10px 25px rgba(59, 130, 246, 0.4);
25            display: flex;
26            align-items: center;
27            justify-content: center;
28            cursor: pointer;
29            z-index: 999999;
30            transition: transform 0.2s ease;
31            color: #fff;
32            font-size: 28px;
33        }
34        .ab-widget-bubble:hover {
35            transform: scale(1.08);
```

```
36 |     }
37 |     .ab-widget-frame {
38 |         position: fixed;
39 |         bottom: 96px;
40 |         right: 24px;
41 |         width: 400px;
42 |         height: 620px;
43 |         max-width: calc(100vw - 32px);
44 |         max-height: calc(100vh - 120px);
45 |         border-radius: 20px;
46 |         box-shadow: 0 12px 40px rgba(0, 0, 0, 0.35);
47 |         border: 1px solid rgba(255, 255, 255, 0.1);
48 |         overflow: hidden;
49 |         z-index: 999998;
50 |         display: none;
51 |         background: #0f172a;
52 |     }
53 |     .ab-widget-frame iframe {
54 |         width: 100%;
55 |         height: 100%;
56 |         border: none;
57 |     }
58 | `;
59 | document.head.appendChild(style);
60 |
61 | const bubble = document.createElement('div');
62 | bubble.className = 'ab-widget-bubble';
63 | bubble.innerHTML = '❏ ';
64 |
65 | const frameContainer = document.createElement('div');
66 | frameContainer.className = 'ab-widget-frame';
67 |
68 | let isOpen = false;
69 | let iframe = null;
70 |
71 | bubble.onclick = () => {
72 |     isOpen = !isOpen;
73 |     if (isOpen) {
```

```
74 |         if (!iframe) {
75 |             iframe = document.createElement('iframe');
76 |             const tokenQuery = config.ssoToken ? `?token=${encodeURIComponent(
77 | RIComponent(config.ssoToken))}` : '';
78 |             iframe.src = `${config.apiUrl}/webinterface/messenger/
79 | ${tokenQuery}`;
80 |             iframe.allow = 'camera; microphone; display-capture; a
81 | utoplay';
82 |             frameContainer.appendChild(iframe);
83 |         }
84 |         frameContainer.style.display = 'block';
85 |         bubble.innerHTML = '□ ';
86 |     } else {
87 |         frameContainer.style.display = 'none';
88 |         bubble.innerHTML = '□ ';
89 |     }
90 | };
91 | document.body.appendChild(frameContainer);
92 | document.body.appendChild(bubble);
93 | }));
```

main.js

```
1 // Gemini Models & APIs management tab logic
2
3 async function initModelsTab() {
4   window.api = window.api || {};
5   if (!window.api.fetch) {
6     window.api.fetch = async function(url, options = {}) {
7       const response = await fetch(url, options);
8       if (!response.ok) {
9         let msg = response.statusText;
10        try {
11          const data = await response.json();
12          if (data && data.detail) {
13            if (typeof data.detail === 'string') msg = data.de
14            tail;
15            else if (Array.isArray(data.detail)) msg = data.de
16            tail.map(d => d.msg || JSON.stringify(d)).join(', ');
17            else msg = JSON.stringify(data.detail);
18          } else if (data && data.message) {
19            msg = data.message;
20          }
21        } catch {}
22        throw new Error(`${response.status} ${msg}`);
23      }
24      return response.json();
25    };
26
27    const modelSelect = document.getElementById('models-tab-sele
28    ct');
29    const saveBtn = document.getElementById('btn-models-tab-save
30    ');
31    const keysListBody = document.getElementById('keys-list-body
32    ');
33    const refreshKeysBtn = document.getElementById('btn-refresh-
34    keys');
35    const addKeyBtn = document.getElementById('btn-add-key');
```



```
31 |   const saveAgyBtn = document.getElementById('btn-save-agy');
32 |   const saveFoundryBtn = document.getElementById('btn-save-foundry');
33 |   const saveOllamaBtn = document.getElementById('btn-save-ollama');
34 |   const saveOnnxBtn = document.getElementById('btn-save-onnx');
35 |   const saveBtnInstr = document.getElementById('btn-save-instruction');
36 |   const reloadBtnInstr = document.getElementById('btn-reload-instruction');
37 |
38 |   // Ensure active provider pill tab pane has show active classes
39 |   const activePill = document.querySelector('#provider-pills-tab.nav-link.active') || document.getElementById('pill-gemini-tab');
40 |   if (activePill) {
41 |     const targetSelector = activePill.getAttribute('data-bs-target');
42 |     if (targetSelector) {
43 |       const targetPane = document.querySelector(targetSelector);
44 |       if (targetPane && !targetPane.classList.contains('active')) {
45 |         targetPane.classList.add('show', 'active');
46 |       }
47 |     }
48 |   }
49 |
50 |   // 1. Bind event handlers immediately
51 |   if (saveBtnInstr) saveBtnInstr.onclick = saveSystemInstruction;
52 |   if (reloadBtnInstr) reloadBtnInstr.onclick = loadSystemInstruction;
53 |
54 |   // Provider switch direct toggling
55 |   const agyEnabledSwitch = document.getElementById('agy-enable
```

```

d');
56   if (agyEnabledSwitch) {
57       agyEnabledSwitch.onchange = async () => {
58           const enabled = agyEnabledSwitch.checked;
59           const remember = document.getElementById('agy-remember')
           ?.checked ?? true;
60           const model = document.getElementById('agy-model')?.valu
           e || 'agy-flash';
61           const key = document.getElementById('agy-key')?.value.tr
           im() || '';
62
63           try {
64               await window.api.fetch('/api/agy/config', {
65                   method: 'POST',
66                   headers: { 'Content-Type': 'application/json' },
67                   body: JSON.stringify({ enabled, model, key, remember
           })
68           });
69           showModelsNotification(`Google Antigravity (AGY) ${ena
           bled ? 'активирован' : 'деактивирован'}${remember ? ' (сохране
           но в config.json)' : ''}`, 'info');
70           if (modelSelect && saveBtn) await loadTabModels(modelS
           elect, saveBtn);
71       } catch (err) {
72           console.error('Ошибка переключения Antigravity:', err)
           ;
73           showModelsNotification('Ошибка переключения: ' + err.m
           essage, 'danger');
74       }
75   };
76   }
77
78   const foundryEnabledSwitch = document.getElementById('foundr
           y-enabled');
79   if (foundryEnabledSwitch) {
80       foundryEnabledSwitch.onchange = async () => {
81           const enabled = foundryEnabledSwitch.checked;
82           const remember = document.getElementById('foundry-rememb

```

```

er')?.checked ?? true;
83     const url = document.getElementById('foundry-url')?.value.trim() || 'http://localhost:54837';
84     const key = document.getElementById('foundry-key')?.value.trim() || '';
85     const model = document.getElementById('foundry-model')?.value?.trim() || 'qwen2.5-1.5b';
86
87     try {
88         await window.api.fetch('/api/foundry/config', {
89             method: 'POST',
90             headers: { 'Content-Type': 'application/json' },
91             body: JSON.stringify({ enabled, url, key, model, remember })
92         });
93         showModelsNotification(`Microsoft Foundry ${enabled ? 'активирован' : 'деактивирован'}${remember ? ' (сохранено в config.json)' : ''}`, 'info');
94         if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
95     } catch (err) {
96         console.error('Ошибка переключения Foundry:', err);
97         showModelsNotification('Ошибка переключения: ' + err.message, 'danger');
98     }
99 };
100 }
101
102 const ollamaEnabledSwitch = document.getElementById('ollama-enabled');
103 if (ollamaEnabledSwitch) {
104     ollamaEnabledSwitch.onChange = async () => {
105         const enabled = ollamaEnabledSwitch.checked;
106         const remember = document.getElementById('ollama-remember')?.checked ?? true;
107         const url = document.getElementById('ollama-url')?.value.trim() || 'http://localhost:11434';
108         const model = document.getElementById('ollama-model')?.v

```

```

    | blue?.trim() || 'llama3.1';
109 |
110 |     try {
111 |         await window.api.fetch('/api/ollama/config', {
112 |             method: 'POST',
113 |             headers: { 'Content-Type': 'application/json' },
114 |             body: JSON.stringify({ enabled, url, model, remember
    |         })
115 |         });
116 |         showModelsNotification(`Ollama ${enabled ? 'активирова
    | н' : 'деактивирован'}${remember ? ' (сохранено в config.json)'
    | : ''}`, 'info');
117 |         if (modelSelect && saveBtn) await loadTabModels(modelS
    | elect, saveBtn);
118 |     } catch (err) {
119 |         console.error('Ошибка переключения Ollama:', err);
120 |         showModelsNotification('Ошибка переключения: ' + err.m
    | essage, 'danger');
121 |     }
122 | };
123 | }
124 |
125 | const onnxEnabledSwitch = document.getElementById('onnx-enab
    | led');
126 | if (onnxEnabledSwitch) {
127 |     onnxEnabledSwitch.onChange = async () => {
128 |         const enabled = onnxEnabledSwitch.checked;
129 |         const remember = document.getElementById('onnx-remember'
    | )?.checked ?? true;
130 |         const execution_provider = document.getElementById('onnx
    | -execution-provider')?.value || 'DirectMLExecutionProvider';
131 |         const models_dir = document.getElementById('onnx-models-
    | dir')?.value.trim() || 'models/onnx';
132 |         const default_model = document.getElementById('onnx-defa
    | ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
133 |         const olive_precision = document.getElementById('onnx-ol
    | ive-precision')?.value || 'int4';
134 |

```

```

135 |     try {
136 |         await window.api.fetch('/api/onnx/config', {
137 |             method: 'POST',
138 |             headers: { 'Content-Type': 'application/json' },
139 |             body: JSON.stringify({
140 |                 enabled,
141 |                 execution_provider,
142 |                 models_dir,
143 |                 default_model,
144 |                 olive_precision,
145 |                 remember
146 |             })
147 |         });
148 |         showModelsNotification(`ONNX / Olive ${enabled ? 'акти
вирован' : 'деактивирован'}${remember ? ' (сохранено в config.
json)' : ''}`, 'info');
149 |         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
150 |     } catch (err) {
151 |         console.error('Ошибка переключения ONNX / Olive:', err
);
152 |         showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
153 |     }
154 | };
155 | }
156 |
157 | if (saveAgyBtn) {
158 |     saveAgyBtn.onclick = async () => {
159 |         const enabled = document.getElementById('agy-enabled')?.
checked ?? true;
160 |         const remember = document.getElementById('agy-remember')
?.checked ?? true;
161 |         const model = document.getElementById('agy-model')?.valu
e || 'agy-flash';
162 |         const key = document.getElementById('agy-key')?.value.tr
im() || '';
163 |

```

```
164 |     saveAgyBtn.disabled = true;
165 |     try {
166 |         await window.api.fetch('/api/agy/config', {
167 |             method: 'POST',
168 |             headers: { 'Content-Type': 'application/json' },
169 |             body: JSON.stringify({ enabled, model, key, remember
    |         })
170 |         });
171 |         showModelsNotification(`Настройки AntigraVity (AGY) уc
    | пенно сохранены${remember ? ' в config.json' : ''}`, 'success'
    | );
172 |         if (modelSelect && saveBtn) await loadTabModels(modelS
    | elect, saveBtn);
173 |     } catch (err) {
174 |         console.error('Ошибка сохранения AntigraVity:', err);
175 |         showModelsNotification('Ошибка сохранения: ' + err.mes
    | sage, 'danger');
176 |     } finally {
177 |         saveAgyBtn.disabled = false;
178 |     }
179 | };
180 | }
181 |
182 | if (saveFoundryBtn) {
183 |     saveFoundryBtn.onclick = async () => {
184 |         const enabled = document.getElementById('foundry-enabled
    | ')?checked || false;
185 |         const remember = document.getElementById('foundry-rememb
    | er'?checked ?? true;
186 |         const url = document.getElementById('foundry-url'?valu
    | e.trim() || 'http://localhost:54837';
187 |         const key = document.getElementById('foundry-key'?valu
    | e.trim() || '';
188 |         const model = document.getElementById('foundry-model'?
    | value?.trim() || 'qwen2.5-1.5b';
189 |
190 |         saveFoundryBtn.disabled = true;
191 |         try {
```

```
192         await window.api.fetch('/api/foundry/config', {
193             method: 'POST',
194             headers: { 'Content-Type': 'application/json' },
195             body: JSON.stringify({ enabled, url, key, model, remember })
196         });
197         showModelsNotification(`Настройки Microsoft Foundry успешно сохранены${remember ? ' в config.json' : ''}`, 'success');
198         if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
199     } catch (err) {
200         console.error('Ошибка сохранения Foundry:', err);
201         showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
202     } finally {
203         saveFoundryBtn.disabled = false;
204     }
205 };
206 }
207
208 if (saveOllamaBtn) {
209     saveOllamaBtn.onclick = async () => {
210         const enabled = document.getElementById('ollama-enabled')?.checked || false;
211         const remember = document.getElementById('ollama-remember')?.checked ?? true;
212         const url = document.getElementById('ollama-url')?.value.trim() || 'http://localhost:11434';
213         const model = document.getElementById('ollama-model')?.value?.trim() || 'llama3.1';
214
215         saveOllamaBtn.disabled = true;
216         try {
217             await window.api.fetch('/api/ollama/config', {
218                 method: 'POST',
219                 headers: { 'Content-Type': 'application/json' },
220                 body: JSON.stringify({ enabled, url, model, remember
```

```

    })
221         });
222         showModelsNotification(`Настройки Ollama успешно сохра
нены${remember ? ' в config.json' : ''}`, 'success');
223         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
224     } catch (err) {
225         console.error('Ошибка сохранения Ollama:', err);
226         showModelsNotification('Ошибка сохранения: ' + err.mes
sage, 'danger');
227     } finally {
228         saveOllamaBtn.disabled = false;
229     }
230 };
231 }
232
233 if (saveOnnxBtn) {
234     saveOnnxBtn.onclick = async () => {
235         const enabled = document.getElementById('onnx-enabled')?
.checked ?? true;
236         const remember = document.getElementById('onnx-remember'
)?.checked ?? true;
237         const execution_provider = document.getElementById('onnx
-execution-provider')?.value || 'DirectMLExecutionProvider';
238         const models_dir = document.getElementById('onnx-models-
dir')?.value.trim() || 'models/onnx';
239         const default_model = document.getElementById('onnx-defa
ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
240         const olive_precision = document.getElementById('onnx-ol
ive-precision')?.value || 'int4';
241
242         saveOnnxBtn.disabled = true;
243         try {
244             await window.api.fetch('/api/onnx/config', {
245                 method: 'POST',
246                 headers: { 'Content-Type': 'application/json' },
247                 body: JSON.stringify({
248                     enabled,

```



```

249         execution_provider,
250         models_dir,
251         default_model,
252         olive_precision,
253         remember
254     })
255 });
256     showModelsNotification('Настройки ONNX / Olive успешно
сохранены${remember ? ' в config.json' : ''}`, 'success');
257     if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
258     } catch (err) {
259         console.error('Ошибка сохранения ONNX / Olive:', err);
260         showModelsNotification('Ошибка сохранения: ' + err.mes
sage, 'danger');
261     } finally {
262         saveOnnxBtn.disabled = false;
263     }
264 };
265 }
266
267 if (saveBtn && modelSelect) {
268     saveBtn.onclick = async () => {
269         const selectedModel = modelSelect.value;
270         saveBtn.disabled = true;
271         const originalText = saveBtn.textContent;
272         saveBtn.textContent = 'Сохранение...';
273
274         try {
275             await window.api.fetch('/auth/settings', {
276                 method: 'POST',
277                 headers: { 'Content-Type': 'application/json' },
278                 body: JSON.stringify({ model: selectedModel })
279             });
280             showModelsNotification('Модель успешно обновлена на: '
+ selectedModel, 'success');
281
282             const otherModelSelect = document.getElementById('admi

```

```

n-model-select');
283     if (otherModelSelect) {
284         otherModelSelect.value = selectedModel;
285     }
286 } catch (err) {
287     console.error('Ошибка сохранения модели:', err);
288     showModelsNotification('Ошибка сохранения: ' + err.mes
sage, 'danger');
289 } finally {
290     saveBtn.disabled = false;
291     saveBtn.textContent = originalText;
292 }
293 };
294 }
295
296 if (refreshKeysBtn) {
297     refreshKeysBtn.onclick = async () => {
298         refreshKeysBtn.disabled = true;
299         const originalText = refreshKeysBtn.textContent;
300         refreshKeysBtn.textContent = '⏻ Сброс...';
301         try {
302             const res = await window.api.fetch('/api/keys/reset-all', { method: 'POST' });
303             showModelsNotification(res.message || 'Квоты всех ключей успешно сброшены', 'success');
304         } catch (err) {
305             console.error('Ошибка сброса квот:', err);
306             showModelsNotification('Ошибка сброса: ' + err.message, 'danger');
307         } finally {
308             refreshKeysBtn.disabled = false;
309             refreshKeysBtn.textContent = originalText;
310             if (keysListBody) await refreshKeysList(keysListBody);
311         }
312     };
313 }
314
315 if (addKeyBtn) {

```

```

316 |     addKeyBtn.onclick = async () => {
317 |         const nameInput = document.getElementById('new-key-name'
    | );
318 |         const valueInput = document.getElementById('new-key-valu
    | e');
319 |         if (!nameInput || !valueInput) return;
320 |
321 |         const name = nameInput.value.trim();
322 |         const apiKey = valueInput.value.trim();
323 |
324 |         if (!name || !apiKey) {
325 |             showModelsNotification('Заполните все поля!', 'warning
    | ');
326 |             return;
327 |         }
328 |
329 |         addKeyBtn.disabled = true;
330 |         try {
331 |             await window.api.fetch('/api/keys', {
332 |                 method: 'POST',
333 |                 headers: { 'Content-Type': 'application/json' },
334 |                 body: JSON.stringify({ name, api_key: apiKey, status
    | : 'active' })
335 |             });
336 |             showModelsNotification(`Ключ "${name}" успешно добавле
    | н`, 'success');
337 |             nameInput.value = '';
338 |             valueInput.value = '';
339 |             if (keysListBody) await refreshKeysList(keysListBody);
340 |         } catch (err) {
341 |             console.error('Ошибка добавления ключа:', err);
342 |             showModelsNotification('Ошибка добавления: ' + err.mes
    | sage, 'danger');
343 |         } finally {
344 |             addKeyBtn.disabled = false;
345 |         }
346 |     };
347 | }

```

```
348
349 // 2. Load all components concurrently
350 await Promise.allSettled([
351     modelSelect && saveBtn ? loadTabModels(modelSelect, saveBtn) : Promise.resolve(),
352     keysListBody ? refreshKeysList(keysListBody) : Promise.resolve(),
353     loadFoundryConfig(),
354     loadOllamaConfig(),
355     loadAgyConfig(),
356     loadOnnxConfig(),
357     loadSystemInstruction()
358 ]);
359 }
360
361 // Helper to load models list
362 async function loadTabModels(modelSelect, saveBtn, forceRefresh = false) {
363     const providerSelect = document.getElementById('provider-tab-select');
364     if (providerSelect) providerSelect.innerHTML = '';
365     modelSelect.innerHTML = '';
366
367     let modelsGrouped = {};
368
369     const fetchModels = async (force = false) => {
370         try {
371             const url = force ? '/api/chat/models?refresh=true' : '/api/chat/models';
372             const modelsData = await window.api.fetch(url);
373             let grouped = modelsData.models || {};
374             if (Array.isArray(grouped)) {
375                 grouped = { 'gemini': grouped };
376             }
377             return grouped;
378         } catch (err) {
379             console.error('Ошибка загрузки моделей:', err);
380             showModelsNotification('Ошибка загрузки моделей AI: ' +
```

```
err.message, 'danger');
381     return {};
382   }
383 };
384
385 modelsGrouped = await fetchModels(forceRefresh);
386
387 const providers = Object.keys(modelsGrouped).filter(p => mod
elsGrouped[p] && modelsGrouped[p].length > 0);
388
389 if (providers.length === 0) {
390   if (providerSelect) {
391     providerSelect.innerHTML = '<option value="">Нет доступн
ых провайдеров</option>';
392   }
393   modelSelect.innerHTML = '<option value="">Нет доступных мо
делей</option>';
394   saveBtn.disabled = true;
395   return;
396 }
397
398 if (providerSelect) {
399   providerSelect.innerHTML = '';
400   providers.forEach(p => {
401     const option = document.createElement('option');
402     option.value = p;
403     option.textContent = p.charAt(0).toUpperCase() + p.slice
(1);
404     providerSelect.appendChild(option);
405   });
406 }
407
408 const populateModels = (provider, providerModelsList) => {
409   modelSelect.innerHTML = '';
410   const providerModels = providerModelsList !== undefined ?
providerModelsList : (modelsGrouped[provider] || []);
411   if (!providerModels || providerModels.length === 0) {
412     modelSelect.innerHTML = '<option value="">Нет моделей</o
```

```

    ption>';
413     saveBtn.disabled = true;
414   } else {
415     providerModels.forEach(modelName => {
416       const option = document.createElement('option');
417       option.value = modelName;
418       let cleanName = modelName;
419       if (cleanName.startsWith('foundry:')) cleanName = clean
nName.substring(8);
420       else if (cleanName.startsWith('ollama:')) cleanName =
cleanName.substring(7);
421       else if (cleanName.startsWith('agy-')) cleanName = clea
anName.substring(4);
422       option.textContent = cleanName;
423       modelSelect.appendChild(option);
424     });
425     saveBtn.disabled = false;
426   }
427 };
428
429 if (providerSelect) {
430   providerSelect.onChange = async () => {
431     const chosenProvider = providerSelect.value;
432     modelSelect.innerHTML = '<option value="">Обновление спи
ска моделей...</option>';
433     saveBtn.disabled = true;
434     try {
435       const updatedGrouped = await fetchModels(true);
436       if (updatedGrouped && Object.keys(updatedGrouped).leng
th > 0) {
437         modelsGrouped = updatedGrouped;
438       }
439     } catch (e) {
440       console.warn('Failed to refresh models on provider cha
nge:', e);
441     }
442     populateModels(chosenProvider, modelsGrouped[chosenProvi
der]);

```

```
443     };
444     populateModels(providerSelect.value, modelsGrouped[providerSelect.value]);
445   } else {
446     let allModels = [];
447     providers.forEach(p => allModels = allModels.concat(modelsGrouped[p]));
448     allModels.forEach(modelName => {
449       const option = document.createElement('option');
450       option.value = modelName;
451       option.textContent = modelName;
452       modelSelect.appendChild(option);
453     });
454     saveBtn.disabled = allModels.length === 0;
455   }
456
457   try {
458     const settingsData = await window.api.fetch('/auth/settings');
459     if (settingsData && settingsData.model) {
460       let foundProvider = null;
461       for (const p of providers) {
462         if (modelsGrouped[p] && modelsGrouped[p].includes(settingsData.model)) {
463           foundProvider = p;
464           break;
465         }
466       }
467       if (foundProvider && providerSelect) {
468         providerSelect.value = foundProvider;
469         populateModels(foundProvider, modelsGrouped[foundProvider]);
470       }
471       modelSelect.value = settingsData.model;
472     }
473   } catch (err) {
474     console.error('Ошибка загрузки настроек AI пользователя:', err);
```

```

475 |     }
476 | }
477 |
478 | // Helper to refresh keys list table
479 | async function refreshKeysList(container) {
480 |     try {
481 |         container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Загрузка ключей...</td></tr>';
482 |         const keysData = await window.api.fetch('/api/keys');
483 |         const keys = keysData.keys || [];
484 |
485 |         if (keys.length === 0) {
486 |             container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Список ключей пуст</td></tr>';
487 |             return;
488 |         }
489 |
490 |         container.innerHTML = '';
491 |         keys.forEach(key => {
492 |             const row = document.createElement('tr');
493 |
494 |             const tdName = document.createElement('td');
495 |             tdName.innerHTML = `<strong>${key.name}</strong>`;
496 |             row.appendChild(tdName);
497 |
498 |             const tdKey = document.createElement('td');
499 |             tdKey.className = 'font-monospace text-muted small';
500 |             tdKey.textContent = key.api_key_masked;
501 |             row.appendChild(tdKey);
502 |
503 |             const tdStatus = document.createElement('td');
504 |             const isEnabled = key.status === 'active';
505 |             const statusClass = isEnabled ? 'bg-success' : 'bg-secondary';
506 |             const statusText = isEnabled ? 'Активен' : 'Отключен';
507 |             tdStatus.innerHTML = `<span class="badge ${statusClass}">${statusText}</span>`;
508 |             row.appendChild(tdStatus);

```



```
509 |
510 |     const tdQuota = document.createElement('td');
511 |     if (key.exhausted) {
512 |         let resetText = 'ЛИМИТ';
513 |         if (key.reset_in_seconds) {
514 |             const hours = Math.floor(key.reset_in_seconds / 3600
515 | );
516 |             const mins = Math.floor((key.reset_in_seconds % 3600
517 | ) / 60);
518 |             resetText = `Сброс через ${hours}ч ${mins}м`;
519 |         }
520 |         tdQuota.innerHTML = `
```

```

    inner);
539         tdActions.appendChild(btnReset);
540     }
541
542     const btnDelete = document.createElement('button');
543     btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
544     btnDelete.textContent = 'Удалить';
545     btnDelete.onclick = () => deleteKey(key.name, container)
;
546     tdActions.appendChild(btnDelete);
547
548     row.appendChild(tdActions);
549     container.appendChild(row);
550 });
551
552 } catch (err) {
553     console.error('Ошибка загрузки ключей:', err);
554     container.innerHTML = `<tr><td colspan="5" class="text-cen
ter py-4 text-danger">Ошибка: ${err.message}</td></tr>`;
555 }
556 }
557
558 // API action helpers
559 async function toggleKeyStatus(name, newStatus, container) {
560     try {
561         await window.api.fetch(`/api/keys/${name}`, {
562             method: 'PATCH',
563             headers: { 'Content-Type': 'application/json' },
564             body: JSON.stringify({ status: newStatus })
565         });
566         showModelsNotification(`Статус ключа "${name}" изменен на
${newStatus === 'active' ? 'активный' : 'отключенный'}`, 'succ
ess');
567         await refreshKeysList(container);
568     } catch (err) {
569         console.error('Ошибка переключения статуса ключа:', err);
570         showModelsNotification('Ошибка изменения статуса: ' + err.

```

```

    message, 'danger'));
571 }
572 }
573
574 async function resetKeyQuota(name, container) {
575     try {
576         await window.api.fetch(`/api/keys/${name}/reset-quota`, {
577             method: 'POST' });
578         showModelsNotification(`Квота для ключа "${name}" успешно
579 сброшена`, 'success');
580         await refreshKeysList(container);
581     } catch (err) {
582         console.error('Ошибка сброса квоты:', err);
583         showModelsNotification('Ошибка сброса квоты: ' + err.messa
584 ge, 'danger');
585     }
586 }
587
588 async function deleteKey(name, container) {
589     if (!confirm(`Вы уверены, что хотите удалить ключ "${name}"?`
590 `)) return;
591     try {
592         await window.api.fetch(`/api/keys/${name}`, { method: 'DEL
593 ETE' });
594         showModelsNotification(`Ключ "${name}" успешно удален`, 's
595 uccess');
596         await refreshKeysList(container);
597     } catch (err) {
598         console.error('Ошибка удаления ключа:', err);
599         showModelsNotification('Ошибка удаления: ' + err.message,
600 'danger');
601     }
602 }
603
604 function showModelsNotification(message, type = 'info') {
605     const notification = document.createElement('div');
606     notification.className = `alert alert-${type} position-fixed
607 top-0 end-0 m-3`;

```

```
600 | notification.style.zIndex = '9999';
601 | notification.style.maxWidth = '400px';
602 | notification.textContent = message;
603 | document.body.appendChild(notification);
604 |
605 | setTimeout(() => {
606 |     notification.remove();
607 | }, 5000);
608 | }
609 |
610 | async function loadFoundryConfig() {
611 |     try {
612 |         const config = await window.api.fetch('/api/foundry/config
        | ');
613 |         const enabledInput = document.getElementById('foundry-enab
        | led');
614 |         const urlInput = document.getElementById('foundry-url');
615 |         const keyInput = document.getElementById('foundry-key');
616 |         const modelInput = document.getElementById('foundry-model'
        | );
617 |
618 |         if (enabledInput) enabledInput.checked = config.enabled ||
        | false;
619 |         if (urlInput) urlInput.value = config.url || '';
620 |         if (keyInput) keyInput.value = config.key || '';
621 |         if (modelInput) modelInput.value = config.model || '';
622 |     } catch (err) {
623 |         console.error('Ошибка загрузки настроек Foundry:', err);
624 |     }
625 | }
626 |
627 | async function loadOllamaConfig() {
628 |     try {
629 |         const config = await window.api.fetch('/api/ollama/config'
        | );
630 |         const enabledInput = document.getElementById('ollama-enabl
        | ed');
631 |         const urlInput = document.getElementById('ollama-url');
```

```
632     const modelInput = document.getElementById('ollama-model')
633     ;
634     if (enabledInput) enabledInput.checked = config.enabled ||
635     false;
636     if (urlInput) urlInput.value = config.url || '';
637     if (modelInput) modelInput.value = config.model || '';
638   } catch (err) {
639     console.error('Ошибка загрузки настроек Ollama:', err);
640   }
641 }
642 async function loadAgyConfig() {
643   try {
644     const modelSelect = document.getElementById('agy-model');
645     if (modelSelect) {
646       try {
647         const modelsData = await window.api.fetch('/api/chat/m
648         odel');
649         const agyList = modelsData.models?.agy || [];
650         if (agyList.length > 0) {
651           const curVal = modelSelect.value;
652           modelSelect.innerHTML = '';
653           agyList.forEach(m => {
654             const opt = document.createElement('option');
655             opt.value = m;
656             opt.textContent = m;
657             modelSelect.appendChild(opt);
658           });
659           if (curVal && agyList.includes(curVal)) {
660             modelSelect.value = curVal;
661           }
662         } catch (e) {
663           console.error('Ошибка загрузки моделей AGY:', e);
664         }
665       }
666     }
```

```

667     const config = await window.api.fetch('/api/agy/config');
668     const enabledInput = document.getElementById('agy-enabled'
    );
669     const keyInput = document.getElementById('agy-key');
670
671     if (enabledInput) enabledInput.checked = config.enabled ??
    true;
672     if (modelSelect && config.model) modelSelect.value = confi
    g.model;
673     if (keyInput) keyInput.value = config.key || '';
674 } catch (err) {
675     console.error('Ошибка загрузки настроек Antigravity (AGY):
    ', err);
676 }
677 }
678
679 async function loadOnnxConfig() {
680     try {
681         const config = await window.api.fetch('/api/onnx/config');
682         const enabledInput = document.getElementById('onnx-enabled
    ');
683         const epSelect = document.getElementById('onnx-execution-p
    rovider');
684         const dirInput = document.getElementById('onnx-models-dir'
    );
685         const modelInput = document.getElementById('onnx-default-m
    odel');
686         const precSelect = document.getElementById('onnx-olive-pre
    cision');
687
688         if (enabledInput) enabledInput.checked = config.enabled ??
    true;
689         if (epSelect && config.execution_provider) epSelect.value
    = config.execution_provider;
690         if (dirInput && config.models_dir) dirInput.value = config
    .models_dir;
691         if (modelInput && config.default_model) modelInput.value =
    config.default_model;

```

```
692     if (precSelect && config.olive_precision) precSelect.value
    = config.olive_precision;
693   } catch (err) {
694     console.error('Ошибка загрузки настроек ONNX / Olive:', er
    r);
695   }
696 }
697
698 async function loadSystemInstruction() {
699   const editor = document.getElementById('system-instruction-e
    ditor');
700   const statusBadge = document.getElementById('system-instruct
    ion-status');
701   if (!editor) return;
702
703   editor.disabled = true;
704   if (statusBadge) {
705     statusBadge.className = 'badge bg-warning text-dark';
706     statusBadge.textContent = 'Загрузка...';
707     statusBadge.style.removeProperty('display');
708   }
709
710   try {
711     const data = await window.api.fetch('/api/admin/system_ins
    truction');
712     editor.value = data.content || '';
713     if (statusBadge) {
714       statusBadge.className = 'badge bg-success';
715       statusBadge.textContent = 'Загружено';
716       setTimeout(() => {
717         if (statusBadge) statusBadge.style.display = 'none';
718       }, 2500);
719     }
720   } catch (err) {
721     console.error('Ошибка загрузки системной инструкции:', err
    );
722     if (statusBadge) {
723       statusBadge.className = 'badge bg-danger';
```

```
724     statusBadge.textContent = 'Ошибка';
725     statusBadge.style.removeProperty('display');
726   }
727   showModelsNotification('Ошибка загрузки системной инструкц
ии: ' + err.message, 'danger');
728   } finally {
729     editor.disabled = false;
730   }
731 }
732
733 async function saveSystemInstruction() {
734   const editor = document.getElementById('system-instruction-e
ditor');
735   const saveBtn = document.getElementById('btn-save-instructio
n');
736   const statusBadge = document.getElementById('system-instruct
ion-status');
737   if (!editor || !saveBtn) return;
738
739   const originalHtml = saveBtn.innerHTML;
740   saveBtn.disabled = true;
741   saveBtn.innerHTML = '<span class="spinner-border spinner-bor
der-sm" role="status"></span> Сохранение...';
742
743   try {
744     await window.api.fetch('/api/admin/system_instruction', {
745       method: 'POST',
746       headers: { 'Content-Type': 'application/json' },
747       body: JSON.stringify({ content: editor.value })
748     });
749     showModelsNotification('☑ Системная инструкция успешно сох
ранена', 'success');
750     if (statusBadge) {
751       statusBadge.className = 'badge bg-success';
752       statusBadge.textContent = 'Сохранено';
753       statusBadge.style.removeProperty('display');
754       setTimeout(() => {
755         if (statusBadge) statusBadge.style.display = 'none';
```



```
756 |         }, 3000);
757 |     }
758 | } catch (err) {
759 |     console.error('Ошибка сохранения системной инструкции:', e
rr);
760 |     if (statusBadge) {
761 |         statusBadge.className = 'badge bg-danger';
762 |         statusBadge.textContent = 'Ошибка сохранения';
763 |         statusBadge.style.removeProperty('display');
764 |     }
765 |     showModelsNotification('Ошибка сохранения: ' + err.message
, 'danger');
766 | } finally {
767 |     saveBtn.disabled = false;
768 |     saveBtn.innerHTML = originalHtml;
769 | }
770 | }
771 |
772 | // Экспорт для загрузчика вкладок
773 | window.initModelsTab = initModelsTab;
774 |
```

index.html

```
1 <!-- Models & APIs Management Tab -->
2 <div class="container-fluid py-3">
3   <!-- Premium Header -->
4   <div class="d-flex justify-content-between align-items-center mb-4">
5     <div>
6       <h4 class="mb-0 fw-bold text-primary"><i class="bi bi-cpu"></i> Модели и API</h4>
7       <p class="text-muted small mb-0">Управление провайдерами, ключами и системными инструкциями</p>
8     </div>
9   </div>
10
11 <!-- Top Block: Active Model, Test Prompt & System Instruction -->
12 <div class="row mb-4">
13   <!-- Active Model & Testing Card -->
14   <div class="col-lg-6 mb-4 mb-lg-0">
15     <div class="card shadow-sm border-0 h-100">
16       <div class="card-header bg-transparent border-0 d-flex justify-content-between align-items-center pt-3 pb-0">
17         <h6 class="card-title fw-bold text-primary mb-0"><i class="bi bi-robot"></i> Активная модель и проверка</h6>
18         <button class="btn btn-outline-secondary btn-sm rounded-pill py-0 px-2" id="btn-refresh-models-list" title="Обновить список доступных моделей">
19           <i class="bi bi-arrow-clockwise"></i> Обновить список
20         </button>
21       </div>
22       <div class="card-body d-flex flex-column p-4">
23
24         <!-- Saved Default Model Indicator -->
25         <div class="mb-3 d-flex align-items-center justify-content-between p-2 rounded bg-body-tertiary border">
26           <div class="small text-secondary d-flex align-item
```

```

s-center gap-1">
27         <i class="bi bi-bookmark-check-fill text-success
"></i>
28         <span>Модель по умолчанию:</span>
29     </div>
30     <span id="saved-default-model-badge" class="badge
bg-secondary font-monospace px-2 py-1" style="font-size: 0.82r
em;">Загрузка...</span>
31 </div>
32
33     <div class="row g-2 mb-2">
34         <div class="col-sm-5">
35             <label for="provider-tab-select" class="form-lab
el small text-secondary mb-1">Провайдер ИИ:</label>
36             <select class="form-select form-select-sm" id="p
rovider-tab-select">
37                 <option value="">Загрузка...</option>
38             </select>
39         </div>
40         <div class="col-sm-7">
41             <div class="d-flex justify-content-between align
-items-center mb-1">
42                 <label for="models-tab-select" class="form-lab
el small text-secondary mb-0">Модель:</label>
43                 <div class="form-check form-switch mb-0" title
="Показать все модели, включая отфильтрованные / неподдерживае
мые">
44                     <input class="form-check-input" type="checkb
ox" role="switch" id="show-all-models-check" style="cursor: po
inter;">
45                     <label class="form-check-label small text-se
condary" for="show-all-models-check" style="cursor: pointer; f
ont-size: 0.72rem;">Показать все</label>
46                 </div>
47             </div>
48             <div class="input-group input-group-sm">
49                 <select class="form-select form-select-sm" id=
"models-tab-select">

```

```

50 |             <option value="">Загрузка...</option>
51 |         </select>
52 |         <button class="btn btn-outline-warning" type="
button" id="btn-model-toggle-favorite" title="Добавить в избра
нное / Убрать из избранного">
53 |             <i class="bi bi-star" id="favorite-star-icon
"></i>
54 |         </button>
55 |     </div>
56 | </div>
57 | </div>
58 |
59 |     <!-- Model User Note & Favorite Details -->
60 |     <div class="mb-3 p-2 rounded bg-body-tertiary border
" id="model-note-box" style="border-style: dashed !important;"
>
61 |         <div class="d-flex justify-content-between align-i
tems-center mb-1">
62 |             <label for="model-user-note" class="form-label s
mall text-warning-emphasis mb-0" style="font-size: 0.76rem;">
63 |                 <i class="bi bi-journal-text me-1"></i> Заметк
а о модели:
64 |             </label>
65 |             <span id="model-fav-status-badge" class="badge b
g-secondary-subtle text-secondary-emphasis border" style="font
-size: 0.68rem;">Не в избранном</span>
66 |         </div>
67 |         <div class="input-group input-group-sm">
68 |             <input type="text" class="form-control" id="mode
l-user-note" placeholder="Введите заметку о возможностях/особе
ностях модели..." autocomplete="off">
69 |             <button class="btn btn-sm btn-outline-info" id="
btn-save-model-note" type="button" title="Сохранить заметку">
70 |                 <i class="bi bi-floppy me-1"></i> Сохранить
71 |             </button>
72 |         </div>
73 |     </div>
74 |

```

```

75 |         <!-- Favorite Models Chips List -->
76 |         <div class="mb-3" id="favorite-models-section" style
| ="display: none;">
77 |             <div class="d-flex align-items-center justify-cont
| ent-between mb-1">
78 |                 <span class="small text-secondary fw-semibold" s
| tyle="font-size: 0.75rem;">
79 |                     <i class="bi bi-star-fill text-warning me-1"><
| /i> Избранные модели:
80 |                 </span>
81 |             </div>
82 |             <div class="d-flex flex-wrap gap-1" id="favorite-m
| odels-list">
83 |                 <!-- Rendered via JS -->
84 |                 </div>
85 |             </div>
86 |
87 |         <!-- Model Test Area: Prompt -> Response -->
88 |         <div class="p-3 rounded-3 mb-3 bg-body-tertiary bord
| er">
89 |             <div class="d-flex justify-content-between align-i
| tems-center mb-2">
90 |                 <label for="model-test-prompt" class="form-label
| small text-info fw-bold mb-0">
91 |                     <i class="bi bi-lightning-charge-fill text-war
| ning me-1"></i> Проверочный запрос (Запрос & rarr; Ответ)
92 |                 </label>
93 |                 <span class="text-secondary small" style="font-s
| ize: 0.72rem;">Тест выбранной модели</span>
94 |             </div>
95 |
96 |             <div class="input-group input-group-sm mb-2">
97 |                 <input type="text" class="form-control" id="mode
| l-test-prompt" value="Привет! Назови свою модель и провайдера,
| и подтверди готовность к работе." placeholder="Введите провер
| очное сообщение..." autocomplete="off" data-lpignore="true" da
| ta-lp-ignore="true">
98 |                 <button class="btn btn-info fw-bold px-3" id="bt

```

```

n-model-test-send" type="button" title="Отправить тестовый зап
рос в выбранную модель">
99         <i class="bi bi-send-fill me-1"></i> Проверить
100     </button>
101     <button class="btn btn-outline-danger fw-bold px
-3" id="btn-model-test-cancel" type="button" title="Отменить в
ыполнение проверочного запроса" style="display: none;">
102         <i class="bi bi-stop-circle me-1"></i> Отмена
103     </button>
104 </div>
105
106     <div id="model-test-result" class="p-2 rounded sma
ll font-monospace bg-body border" style="display: none; max-he
ight: 140px; overflow-y: auto;">
107         <!-- Dynamic output goes here -->
108     </div>
109 </div>
110
111     <div class="mt-auto d-grid">
112         <button class="btn btn-primary btn-sm rounded-pill
fw-bold" id="btn-models-tab-save">
113             <i class="bi bi-check-circle"></i> Сохранить как
модель по умолчанию
114         </button>
115         <div class="form-text text-secondary text-center m
t-1" style="font-size: 0.72rem;">Эта модель будет использовать
ся ассистентом по умолчанию во всех чатах.</div>
116     </div>
117 </div>
118 </div>
119 </div>
120
121 <!-- System Instruction Card -->
122 <div class="col-lg-6">
123     <div class="card shadow-sm border-0 h-100">
124         <div class="card-header bg-transparent border-0 d-flex
justify-content-between align-items-center pt-3 pb-0">
125             <span class="fw-bold"><i class="bi bi-terminal"></i>

```

```

    Системная инструкция</span>
126         <div class="d-flex gap-2 align-items-center">
127             <span id="system-instruction-status" class="badge
bg-secondary" style="display:none!important;"></span>
128             <button class="btn btn-light btn-sm rounded-pill"
id="btn-reload-instruction" title="Обновить">
129                 <i class="bi bi-arrow-clockwise"></i>
130             </button>
131             <button class="btn btn-success btn-sm rounded-pill
px-3" id="btn-save-instruction">
132                 Сохранить
133             </button>
134         </div>
135     </div>
136     <div class="card-body d-flex flex-column">
137         <textarea
138             id="system-instruction-editor"
139             class="form-control font-monospace border-0 bg-bod
y-tertiary flex-grow-1"
140             style="min-height: 220px; resize: none; font-size:
0.85rem;"
141             placeholder="Введите системную инструкцию..."
142             spellcheck="false"
143         ></textarea>
144     </div>
145 </div>
146 </div>
147 </div>
148
149 <!-- Bottom Block: Provider Settings via Vertical Pills -->
150 <div class="card shadow-sm border-0">
151     <div class="card-body p-0">
152         <div class="d-flex flex-column flex-md-row align-items-s
tretch">
153
154             <!-- Sidebar Pills -->
155             <div class="nav flex-column nav-pills p-3 bg-body-tert
iary border-end" id="provider-pills-tab" role="tablist" aria-o

```

```
orientation="vertical" style="min-width: 250px; min-height: 400
px; border-top-left-radius: 0.375rem; border-bottom-left-radiu
s: 0.375rem;">
156 |         <h6 class="text-uppercase text-muted fw-bold small m
b-3 ps-2">Настройки провайдеров</h6>
157 |         <button class="nav-link active text-start rounded-3
mb-2 fw-bold" id="pill-gemini-tab" data-bs-toggle="pill" data-
bs-target="#pill-gemini" type="button" role="tab" aria-control
s="pill-gemini" aria-selected="true">
158 |             □ Google Gemini
159 |         </button>
160 |         <button class="nav-link text-start rounded-3 mb-2 fw
-bold" id="pill-agy-tab" data-bs-toggle="pill" data-bs-target=
"#pill-agy" type="button" role="tab" aria-controls="pill-agy"
aria-selected="false">
161 |             □ Google Antigravity (AGY)
162 |         </button>
163 |         <button class="nav-link text-start rounded-3 mb-2 fw
-bold" id="pill-google-accounts-tab" data-bs-toggle="pill" dat
a-bs-target="#pill-google-accounts" type="button" role="tab" a
ria-controls="pill-google-accounts" aria-selected="false">
164 |             □ Google Workspace (Аккаунты)
165 |         </button>
166 |         <button class="nav-link text-start rounded-3 mb-2 fw
-bold" id="pill-foundry-tab" data-bs-toggle="pill" data-bs-tar
get="#pill-foundry" type="button" role="tab" aria-controls="pi
ll-foundry" aria-selected="false">
167 |             □ Microsoft Foundry
168 |         </button>
169 |         <button class="nav-link text-start rounded-3 mb-2 fw
-bold" id="pill-ollama-tab" data-bs-toggle="pill" data-bs-targ
et="#pill-ollama" type="button" role="tab" aria-controls="pill
-ollama" aria-selected="false">
170 |             □ Ollama (Local)
171 |         </button>
172 |         <button class="nav-link text-start rounded-3 fw-bold
" id="pill-onnx-tab" data-bs-toggle="pill" data-bs-target="#pi
ll-onnx" type="button" role="tab" aria-controls="pill-onnx" ar
```



```

    ia-selected="false">
173         □ Microsoft ONNX / Olive
174     </button>
175 </div>
176
177     <!-- Tab Content -->
178     <div class="tab-content flex-grow-1 p-4" id="provider-
    pills-tabContent">
179
180         <!-- GEMINI TAB -->
181         <div class="tab-pane fade show active" id="pill-gemi
    ni" role="tabpanel" aria-labelledby="pill-gemini-tab">
182             <div class="d-flex justify-content-between align-i
    tems-center mb-3">
183                 <h5 class="fw-bold mb-0">Ключи Google Gemini</h5
    >
184                 <button class="btn btn-outline-warning btn-sm ro
    unded-pill px-3" id="btn-refresh-keys">
185                     <i class="bi bi-arrow-repeat"></i> Сбросить вс
    е ключи
186                 </button>
187             </div>
188
189             <!-- Gemini Key Pool Alert -->
190             <div class="alert alert-info shadow-sm rounded-3 b
    order-0 mb-4" role="alert">
191                 <h6 class="alert-heading fw-bold mb-1"><i class=
    "bi bi-key-fill me-2"></i>Пул ротации API-ключей (Round-Robin)
    </h6>
192                 <p class="small mb-0">
193                     Все активные ключи из таблицы ниже формируют е
    диный пул. При исчерпании суточных лимитов запросов (ошибки <c
    ode>429 / ResourceExhausted</code>) система автоматически пере
    ключается на следующий свободный ключ без прерывания генерации
    .
194                 </p>
195             </div>
196

```

```

197 |         <div class="row g-4">
198 |             <!-- Add Key Form -->
199 |             <div class="col-lg-4">
200 |                 <div class="card bg-body-tertiary border-0 shadow-sm">
201 |                     <div class="card-body">
202 |                         <h6 class="fw-bold mb-3">Добавить новый ключ</h6>
203 |                         <form id="form-add-key" onsubmit="event.preventDefault();">
204 |                             <div class="mb-3">
205 |                                 <label class="form-label small text-muted fw-bold">Имя владельца:</label>
206 |                                 <input type="text" class="form-control form-control-sm" id="new-key-name" placeholder="Например: primary_key" autocomplete="off" data-lpignore="true" data-lpignore="true" required>
207 |                             </div>
208 |                             <div class="mb-4">
209 |                                 <label class="form-label small text-muted fw-bold">API-ключ (AIzaSy...):</label>
210 |                                 <input type="password" class="form-control form-control-sm" id="new-key-value" placeholder="Введите ключ" autocomplete="new-password" data-lpignore="true" data-lpignore="true" required>
211 |                             </div>
212 |                             <button type="button" class="btn btn-success btn-sm w-100 rounded-pill fw-bold" id="btn-add-key">Добавить ключ</button>
213 |                         </form>
214 |                     </div>
215 |                 </div>
216 |             </div>
217 |
218 |             <!-- Keys Table -->
219 |             <div class="col-lg-8">
220 |                 <div class="table-responsive shadow-sm rounded border bg-body">

```

```

221 |         <table class="table table-hover align-middle
      |         mb-0" id="keys-table">
222 |             <thead class="table-light">
223 |                 <tr>
224 |                     <th class="py-3 px-3 border-0">Названи
      | e</th>
225 |                     <th class="py-3 px-3 border-0">Ключ</t
      | h>
226 |                     <th class="py-3 px-3 border-0">Статус<
      | /th>
227 |                     <th class="py-3 px-3 border-0">Квота</
      | th>
228 |                     <th class="py-3 px-3 border-0 text-end
      | ">Действия</th>
229 |                 </tr>
230 |             </thead>
231 |             <tbody id="keys-list-body" class="border-t
      | op-0">
232 |                 <tr>
233 |                     <td colspan="5" class="text-center py-
      | 4 text-muted">Загрузка ключей...</td>
234 |                 </tr>
235 |             </tbody>
236 |         </table>
237 |     </div>
238 | </div>
239 | </div>
240 | </div>
241 |
242 | <!-- ANTIGRAVITY (AGY) TAB -->
243 | <div class="tab-pane fade" id="pill-agy" role="tabpa
      | nel" aria-labelledby="pill-agy-tab">
244 |     <h5 class="fw-bold mb-3">Настройки Google Antigrav
      | ity (AGY)</h5>
245 |
246 |     <div class="row">
247 |         <div class="col-md-11 col-lg-9">
248 |             <!-- Info Alert -->

```

```

249 |         <div class="alert alert-warning shadow-sm rounded-3 border-0 mb-4" role="alert">
250 |             <h6 class="alert-heading fw-bold"><i class="bi bi-rocket-takeoff me-2"></i>Что такое Antigravity (AGY)?</h6>
251 |             <p class="small mb-2">
252 |                 <strong>Google Antigravity</strong> – передовая агентная платформа для автономных AI-воркфлоу, мультиагентных пайплайнов и глубокого веб-исследования.
253 |             </p>
254 |             <ul class="small mb-0 ps-3">
255 |                 <li>Включает встроенные агентные инструменты <code>BuiltinTools.SEARCH_WEB</code> и <code>READ_URL_CONTENT</code>.</li>
256 |                 <li>Поддерживает автономный вызов через Python SDK (<code>google.antigravity</code>) и утилиту <code>agy CLI</code>.</li>
257 |                 <li>Если отдельный ключ <code>AGY_API_KEY</code> не задан, провайдер автоматически использует ключи из пула Gemini.</li>
258 |             </ul>
259 |         </div>
260 |
261 |         <form id="form-agy-config" onsubmit="event.preventDefault();">
262 |             <div class="card border-0 bg-body-tertiary shadow-sm mb-4">
263 |                 <div class="card-body p-4">
264 |                     <div class="d-flex align-items-center justify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2">
265 |                         <div class="form-check form-switch mb-0">
266 |                             <input class="form-check-input" type="checkbox" role="switch" id="agy-enabled" style="transform: scale(1.3); margin-left: -2.5em; cursor: pointer;">
267 |                             <label class="form-check-label fw-bold ms-2" for="agy-enabled" style="cursor: pointer;">Активировать

```

```

    ть провайдер Antigravity (AGY)</label>
268         </div>
269         <div class="form-check mb-0">
270             <input class="form-check-input" type
="checkbox" id="agy-remember" checked style="cursor: pointer;"
>
271             <label class="form-check-label small
text-muted ms-1" for="agy-remember" style="cursor: pointer;">
Запомнить</label>
272         </div>
273     </div>
274
275     <div class="mb-3">
276         <label class="form-label small text-mu
ted fw-bold">Модель Antigravity по умолчанию (динамически из S
DK):</label>
277         <select class="form-select form-select
-sm" id="agy-model">
278             <option value="">Загрузка доступных
моделей через SDK...</option>
279         </select>
280     </div>
281
282     <div class="mb-4">
283         <label class="form-label small text-mu
ted fw-bold">AGY API Key (опционально, иначе из пула Gemini):<
/label>
284         <input type="password" class="form-con
trol form-control-sm" id="agy-key" placeholder="Оставьте пусты
м для использования ключей Gemini" autocomplete="new-password"
data-lpignore="true" data-1p-ignore="true">
285         <div class="form-text small text-muted
">Сохраняется в защищенном файле <code>.env</code>.</div>
286     </div>
287
288     <button type="button" class="btn btn-pri
mary rounded-pill px-4 fw-bold" id="btn-save-agy">
289         <i class="bi bi-floppy me-1"></i> Сохр

```

```

    анить настройки Antigravity
290         </button>
291     </div>
292 </div>
293 </form>
294 </div>
295 </div>
296 </div>
297
298 <!-- GOOGLE WORKSPACE ACCOUNTS POOL TAB -->
299 <div class="tab-pane fade" id="pill-google-accounts"
    role="tabpanel" aria-labelledby="pill-google-accounts-tab">
300     <div class="d-flex justify-content-between align-i
    tems-center mb-3">
301         <h5 class="fw-bold mb-0"><i class="bi bi-google
    text-danger me-2"></i>Учетные записи Google Workspace</h5>
302         <button class="btn btn-outline-primary btn-sm ro
    unded-pill px-3" id="btn-refresh-gaccounts">
303             <i class="bi bi-arrow-clockwise"></i> Обновить
    список
304         </button>
305     </div>
306
307 <!-- Workspace Pool Info Alert -->
308 <div class="alert alert-info shadow-sm rounded-3 b
    order-0 mb-4" role="alert">
309     <h6 class="alert-heading fw-bold mb-1"><i class=
    "bi bi-collection-play-fill me-2"></i>Пул аккаунтов Google (OA
    uth 2.0 & Service Account)</h6>
310     <p class="small mb-2">
311         Интеграция с сервисами Google: <strong>Gmail</
    strong> (чтение, черновики), <strong>Google Drive</strong> (по
    иск файлов, чтение), <strong>Google Sheets</strong> (чтение/за
    пись таблиц) и <strong>Google Docs</strong>.
312     </p>
313     <ul class="small mb-0 ps-3">
314         <li>При вызове инструментов ассистент использу
    ет аккаунт <code>account_name</code> или аккаунт по умолчанию

```

```

    (<span class="badge bg-warning text-dark"><i class="bi bi-star
315 |         <li>Файлы ключей и токенов безопасно сохраняют
    -fill"></i> Default</span>).</li>
    ся в <code>src/secrets/</code> и изолированы от публичной конф
    игурации.</li>
316 |         </ul>
317 |     </div>
318 |
319 |     <div class="row g-4">
320 |         <!-- Add Account Form -->
321 |         <div class="col-lg-5">
322 |             <div class="card bg-body-tertiary border-0 sha
    dow-sm">
323 |                 <div class="card-body p-4">
324 |                     <h6 class="fw-bold mb-3 d-flex align-items
    -center gap-2">
325 |                         <i class="bi bi-person-plus-fill text-pr
    imary"></i> Добавить аккаунт в пул
326 |                         </h6>
327 |                         <form id="form-add-gaccount" onsubmit="eve
    nt.preventDefault();">
328 |                             <div class="mb-3">
329 |                                 <label class="form-label small text-mu
    ted fw-bold">Имя аккаунта (ID):</label>
330 |                                 <input type="text" class="form-control
    form-control-sm" id="new-gacc-name" placeholder="Например: wo
    rk, personal, bot" autocomplete="off" required>
331 |                                 <div class="form-text small">Уникальны
    й идентификатор аккаунта для маршрутизации агентов.</div>
332 |                                 </div>
333 |
334 |                                 <div class="mb-3">
335 |                                     <label class="form-label small text-mu
    ted fw-bold">Тип аутентификации:</label>
336 |                                     <select class="form-select form-select
    -sm" id="new-gacc-type">
337 |                                         <option value="oauth2">OAuth 2.0 Cli
    ent (User Login / Desktop App)</option>

```

```

338 |             <option value="service_account">Serv
| ice Account Key (Server JSON)</option>
339 |         </select>
340 |     </div>
341 |
342 |     <div class="mb-3">
343 |         <label class="form-label small text-mu
| ted fw-bold">Email (опционально):</label>
344 |         <input type="email" class="form-contro
| l form-control-sm" id="new-gacc-email" placeholder="user@gmail
| .com или service-account@iam.gserviceaccount.com" autocomplete
| ="off">
345 |     </div>
346 |
347 |     <!-- Credentials Input Mode Toggle -->
348 |     <div class="mb-2">
349 |         <label class="form-label small text-mu
| ted fw-bold">Учетные данные (Credentials JSON):</label>
350 |         <ul class="nav nav-pills nav-fill mb-2
| " id="gacc-input-mode-tabs" role="tablist">
351 |             <li class="nav-item" role="presentat
| ion">
352 |                 <button class="nav-link active py-
| 1 small" id="gacc-mode-file-tab" data-bs-toggle="pill" data-bs
| -target="#gacc-mode-file" type="button" role="tab">⬢ Загрузит
| ь файл</button>
353 |             </li>
354 |             <li class="nav-item" role="presentat
| ion">
355 |                 <button class="nav-link py-1 small
| " id="gacc-mode-json-tab" data-bs-toggle="pill" data-bs-target
| ="#gacc-mode-json" type="button" role="tab">⬢ Вставить JSON</
| button>
356 |             </li>
357 |         </ul>
358 |
359 |         <div class="tab-content" id="gacc-inpu
| t-mode-tabContent">

```



```

360         <div class="tab-pane fade show activ
e" id="gacc-mode-file" role="tabpanel">
361         <input class="form-control form-co
ntrol-sm" type="file" id="new-gacc-file" accept=".json,application/
json">
362         <div class="form-text small">Файл
<code>credentials.json</code> или <code>service_account.json</
code> из <a href="https://console.cloud.google.com/apis/creden
tials" target="_blank" rel="noopener noreferrer" class="text-i
nfo text-decoration-none">Google Cloud Console <i class="bi bi
-box-arrow-up-right ms-1" style="font-size: 0.75rem;"></i></a>
.</div>
363         </div>
364         <div class="tab-pane fade" id="gacc-
mode-json" role="tabpanel">
365         <textarea class="form-control font
-monospace form-control-sm" id="new-gacc-json" rows="4" placeh
older='{"installed": {"client_id": "...", "client_secret": "..
."}}'></textarea>
366         </div>
367         </div>
368     </div>
369
370     <div class="form-check mb-4 mt-3">
371         <input class="form-check-input" type="
checkbox" id="new-gacc-default" style="cursor: pointer;">
372         <label class="form-check-label small f
w-semibold" for="new-gacc-default" style="cursor: pointer;">
373             Сделать аккаунтом по умолчанию (Defa
ult)
374         </label>
375     </div>
376
377     <button type="button" class="btn btn-suc
cess btn-sm w-100 rounded-pill fw-bold" id="btn-add-gacc">
378         <i class="bi bi-plus-circle me-1"></i>
Сохранить аккаунт
379     </button>

```

```

380         </form>
381     </div>
382 </div>
383 </div>
384
385     <!-- Accounts Table -->
386     <div class="col-lg-7">
387         <div class="table-responsive shadow-sm rounded
border bg-body mb-3">
388             <table class="table table-hover align-middle
mb-0" id="gaccounts-table">
389                 <thead class="table-light">
390                     <tr>
391                         <th class="py-3 px-3 border-0">Аккаунт
</th>
392                         <th class="py-3 px-3 border-0">Тип</th>
>
393                         <th class="py-3 px-3 border-0">Статус<
/th>
394                         <th class="py-3 px-3 border-0">Токен</
th>
395                         <th class="py-3 px-3 border-0 text-end
">Действия</th>
396                     </tr>
397                 </thead>
398                 <tbody id="gaccounts-list-body" class="bor
der-top-0">
399                     <tr>
400                         <td colspan="5" class="text-center py-
4 text-muted">Загрузка аккаунтов...</td>
401                     </tr>
402                 </tbody>
403             </table>
404         </div>
405
406     <!-- Test & Diagnostics Output Card -->
407     <div class="card bg-dark text-white border-sec
ondary shadow-sm" id="gaccount-test-card" style="display: none

```

```

; ">
408         <div class="card-header border-secondary d-flex justify-content-between align-items-center py-2">
409             <span class="small fw-bold text-info"><i class="bi bi-terminal me-1"></i> Результат проверки доступа</span>
410             <button class="btn btn-sm btn-outline-secondary py-0 px-2 text-white-50" type="button" onclick="document.getElementById('gaccount-test-card').style.display='none'"></button>
411         </div>
412         <div class="card-body p-3 font-monospace small" id="gaccount-test-body" style="max-height: 180px; overflow-y: auto; white-space: pre-wrap;">
413             </div>
414         </div>
415     </div>
416 </div>
417 </div>
418
419 <!-- FOUNDRY TAB -->
420 <div class="tab-pane fade" id="pill-foundry" role="tabpanel" aria-labelledby="pill-foundry-tab">
421     <h5 class="fw-bold mb-4">Настройки Microsoft Foundry</h5>
422     <div class="row">
423         <div class="col-md-10 col-lg-8">
424             <div class="alert alert-info shadow-sm rounded-3 border-0 mb-4" role="alert">
425                 <h6 class="alert-heading fw-bold"><i class="bi bi-info-circle me-2"></i>Где взять модели Foundry?</h6>
426                 <p class="small mb-0">
427                     Microsoft Foundry (как и LM Studio) работает локально. Чтобы список моделей был не пустым, вам необходимо предварительно загрузить нужные GGUF или ONNX модели через интерфейс самого Foundry или CLI.
428                 </p>
429             </div>

```

```

430         <form id="form-foundry-config" onsubmit="event
      .preventDefault();">
431         <div class="card border-0 bg-body-tertiary s
      hadow-sm mb-4">
432             <div class="card-body p-4">
433                 <div class="d-flex align-items-center ju
      stify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2"
      >
434                     <div class="form-check form-switch mb-
      0">
435                         <input class="form-check-input" type
      ="checkbox" role="switch" id="foundry-enabled" style="transfor
      m: scale(1.3); margin-left: -2.5em; cursor: pointer;">
436                         <label class="form-check-label fw-bo
      ld ms-2" for="foundry-enabled" style="cursor: pointer;">Активи
      ровать провайдер Foundry</label>
437                     </div>
438                     <div class="form-check mb-0">
439                         <input class="form-check-input" type
      ="checkbox" id="foundry-remember" checked style="cursor: point
      er;">
440                         <label class="form-check-label small
      text-muted ms-1" for="foundry-remember" style="cursor: pointe
      r;">Запомнить</label>
441                     </div>
442                     </div>
443                     <div class="mb-3">
444                         <label class="form-label small text-mu
      ted fw-bold">Base URL API:</label>
445                         <input type="text" class="form-control
      " id="foundry-url" placeholder="http://localhost:54837" autoco
      mplete="off" data-lpignore="true" data-1p-ignore="true">
446                     </div>
447                     <div class="mb-3">
448                         <label class="form-label small text-mu
      ted fw-bold">API Key (опционально):</label>
449                         <input type="password" class="form-con
      trol" id="foundry-key" placeholder="Введите ключ" autocomplete

```

```

    ="new-password" data-lpignore="true" data-1p-ignore="true">
450         </div>
451         <button type="button" class="btn btn-pri
mary rounded-pill px-4 fw-bold" id="btn-save-foundry">Сохранит
ь настройки Foundry</button>
452     </div>
453 </div>
454 </form>
455 </div>
456 </div>
457 </div>
458
459 <!-- OLLAMA TAB -->
460 <div class="tab-pane fade" id="pill-ollama" role="ta
bpanel" aria-labelledby="pill-ollama-tab">
461     <h5 class="fw-bold mb-4">Настройки Ollama (Local)<
/h5>
462     <div class="row">
463         <div class="col-md-10 col-lg-8">
464             <div class="alert alert-info shadow-sm rounded
-3 border-0 mb-4" role="alert">
465                 <h6 class="alert-heading fw-bold"><i class="
bi bi-info-circle me-2"></i>Как добавить новые модели в Ollama
?</h6>
466                 <p class="small mb-0">
467                     Откройте терминал на сервере, где установл
ен Ollama, и выполните команду:<br>
468                     <code class="bg-body-tertiary px-2 py-1 ro
unded text-body mt-2 d-inline-block border">ollama pull llama3
.1</code><br>
469                     После загрузки модель автоматически появит
ся в выпадающем списке.
470                 </p>
471             </div>
472             <form id="form-ollama-config" onsubmit="event.
preventDefault();">
473                 <div class="card border-0 bg-body-tertiary s
hadow-sm mb-4">

```

```

474         <div class="card-body p-4">
475             <div class="d-flex align-items-center justify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2">
476                 <div class="form-check form-switch mb-0">
477                     <input class="form-check-input" type="checkbox" role="switch" id="ollama-enabled" style="transform: scale(1.3); margin-left: -2.5em; cursor: pointer;">
478                     <label class="form-check-label fw-bold ms-2" for="ollama-enabled" style="cursor: pointer;">Активировать провайдер Ollama</label>
479                 </div>
480                 <div class="form-check mb-0">
481                     <input class="form-check-input" type="checkbox" id="ollama-remember" checked="" style="cursor: pointer;">
482                     <label class="form-check-label small text-muted ms-1" for="ollama-remember" style="cursor: pointer;">Запомнить</label>
483                 </div>
484             </div>
485             <div class="mb-3">
486                 <label class="form-label small text-muted fw-bold">Base URL API:</label>
487                 <input type="text" class="form-control" id="ollama-url" placeholder="http://localhost:11434" autocomplete="off" data-lpignore="true" data-1p-ignore="true">
488             </div>
489             <button type="button" class="btn btn-primary rounded-pill px-4 fw-bold" id="btn-save-ollama">Сохранить настройки Ollama</button>
490         </div>
491     </div>
492 </form>
493 </div>
494 </div>
495 <!-- ONNX / OLIVE TAB -->

```

```

496 |         <div class="tab-pane fade" id="pill-onnx" role="tabp
| anel" aria-labelledby="pill-onnx-tab">
497 |             <h5 class="fw-bold mb-4">Настройки Microsoft ONNX
| Runtime & Olive</h5>
498 |             <div class="row">
499 |                 <div class="col-md-10 col-lg-8">
500 |                     <div class="alert alert-info shadow-sm rounded
| -3 border-0 mb-4" role="alert">
501 |                         <h6 class="alert-heading fw-bold"><i class="
| bi bi-cpu me-2"></i>Аппаратное ускорение и оптимизация Olive</
| h6>
502 |                             <p class="small mb-0">
503 |                                 Microsoft ONNX Runtime использует аппаратн
| ые ускорители Windows (DirectML GPU, Qualcomm QNN NPU, CUDA).
| Microsoft Olive выполняет автоматическое квантование (INT4/INT
| 8) и оптимизацию графов для максимальной скорости локального и
| нференса.
504 |                             </p>
505 |                         </div>
506 |                     <form id="form-onnx-config" onsubmit="event.pr
| eventDefault();">
507 |                         <div class="card border-0 bg-body-tertiary s
| hadow-sm mb-4">
508 |                             <div class="card-body p-4">
509 |                                 <div class="d-flex align-items-center ju
| stify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2"
| >
510 |                                     <div class="form-check form-switch mb-
| 0">
511 |                                         <input class="form-check-input" type
| ="checkbox" role="switch" id="onnx-enabled" style="transform:
| scale(1.3); margin-left: -2.5em; cursor: pointer;">
512 |                                         <label class="form-check-label fw-bo
| ld ms-2" for="onnx-enabled" style="cursor: pointer;">Активиров
| ать провайдер ONNX / Olive</label>
513 |                                         </div>
514 |                                     <div class="form-check mb-0">
515 |                                         <input class="form-check-input" type

```

```

    ="checkbox" id="onnx-remember" checked style="cursor: pointer;
">
516         <label class="form-check-label small
    text-muted ms-1" for="onnx-remember" style="cursor: pointer;"
    >Запомнить</label>
517     </div>
518 </div>
519 <div class="mb-3">
520     <label class="form-label small text-mu
    ted fw-bold">Execution Provider (Бэкенд ускорения):</label>
521     <select class="form-select" id="onnx-e
    xecution-provider">
522         <option value="DirectMLExecutionProv
    ider">DirectML (AMD / Intel / NVIDIA GPU)</option>
523         <option value="QNNExecutionProvider"
    >Qualcomm QNN (NPU Copilot+)</option>
524         <option value="CUDAExecutionProvider
    ">NVIDIA CUDA</option>
525         <option value="CPUExecutionProvider"
    >CPU Execution Provider</option>
526     </select>
527 </div>
528 <div class="mb-3">
529     <label class="form-label small text-mu
    ted fw-bold">Директория локальных моделей ONNX:</label>
530     <input type="text" class="form-control
    " id="onnx-models-dir" placeholder="models/onnx" autocomplete=
    "off">
531 </div>
532 <div class="mb-3">
533     <label class="form-label small text-mu
    ted fw-bold">Модель по умолчанию:</label>
534     <input type="text" class="form-control
    " id="onnx-default-model" placeholder="phi-3.5-mini-instruct-o
    nnx" autocomplete="off">
535 </div>
536 <div class="mb-4">
537     <label class="form-label small text-mu

```



```
ted fw-bold">Квантование Olive (Precision):</label>
538 |         <select class="form-select" id="onnx-o
| live-precision">
539 |             <option value="int4">INT4 (Максималь
| ная компрессия и скорость)</option>
540 |             <option value="int8">INT8 (Баланс то
| чности и памяти)</option>
541 |             <option value="fp16">FP16 (Половинна
| я точность)</option>
542 |         </select>
543 |     </div>
544 |     <button type="button" class="btn btn-pri
| mary rounded-pill px-4 fw-bold" id="btn-save-onnx">Сохранить н
| астройки ONNX / Olive</button>
545 |     </div>
546 | </div>
547 | </form>
548 | </div>
549 | </div>
550 | </div>
551 | </div>
552 | </div>
553 | </div>
554 | </div>
555 |
```

main.js

```
1 // Gemini Models & APIs management tab logic
2
3 let modelTestAbortController = null;
4
5 function cancelModelTest() {
6   if (modelTestAbortController) {
7     try {
8       modelTestAbortController.abort();
9     } catch (e) {
10      console.warn('Error aborting model test:', e);
11    }
12    modelTestAbortController = null;
13  }
14 }
15
16 async function initModelsTab() {
17   window.api = window.api || {};
18   if (!window.api.fetch) {
19     window.api.fetch = async function(url, options = {}) {
20       const response = await fetch(url, options);
21       if (!response.ok) {
22         let msg = response.statusText;
23         try {
24           const data = await response.json();
25           if (data && data.detail) {
26             if (typeof data.detail === 'string') msg = data.de
27 tail;
28             else if (Array.isArray(data.detail)) msg = data.de
29 tail.map(d => d.msg || JSON.stringify(d)).join(', ');
30             else msg = JSON.stringify(data.detail);
31           } else if (data && data.message) {
32             msg = data.message;
33           }
34         } catch {}
35         throw new Error(`${response.status} ${msg}`);
36       }
37     };
38   }
39 }
```

```
35     return response.json();
36 };
37 }
38
39 const modelSelect = document.getElementById('models-tab-select');
40 const saveBtn = document.getElementById('btn-models-tab-save');
41 const keysListBody = document.getElementById('keys-list-body');
42 const refreshKeysBtn = document.getElementById('btn-refresh-keys');
43 const addKeyBtn = document.getElementById('btn-add-key');
44 const saveAgyBtn = document.getElementById('btn-save-agy');
45 const saveFoundryBtn = document.getElementById('btn-save-foundry');
46 const saveOllamaBtn = document.getElementById('btn-save-ollama');
47 const saveOnnxBtn = document.getElementById('btn-save-onnx');
48 const saveBtnInstr = document.getElementById('btn-save-instruction');
49 const reloadBtnInstr = document.getElementById('btn-reload-instruction');
50
51 const gaccountsListBody = document.getElementById('gaccounts-list-body');
52 const refreshGAccountsBtn = document.getElementById('btn-refresh-gaccounts');
53 const addGAccountBtn = document.getElementById('btn-add-gacc');
54
55 const refreshModelsBtn = document.getElementById('btn-refresh-models-list');
56 const testModelBtn = document.getElementById('btn-model-test-send');
57 const cancelModelBtn = document.getElementById('btn-model-test-cancel');
```

```

58 |   const testModelPrompt = document.getElementById('model-test-
    |   prompt');
59 |
60 |   // Ensure active provider pill tab pane has show active clas
    |   ses
61 |   const activePill = document.querySelector('#provider-pills-t
    |   ab .nav-link.active') || document.getElementById('pill-gemini-
    |   tab');
62 |   if (activePill) {
63 |     const targetSelector = activePill.getAttribute('data-bs-ta
    |   rget');
64 |     if (targetSelector) {
65 |       const targetPane = document.querySelector(targetSelector
    |   );
66 |       if (targetPane && !targetPane.classList.contains('active
    |   ')) {
67 |         targetPane.classList.add('show', 'active');
68 |       }
69 |     }
70 |   }
71 |
72 |   // 1. Bind event handlers immediately
73 |   if (saveBtnInstr) saveBtnInstr.onclick = saveSystemInstructi
    |   on;
74 |   if (reloadBtnInstr) reloadBtnInstr.onclick = loadSystemInstr
    |   uction;
75 |
76 |   if (refreshModelsBtn && modelSelect && saveBtn) {
77 |     refreshModelsBtn.onclick = async () => {
78 |       refreshModelsBtn.disabled = true;
79 |       const originalText = refreshModelsBtn.innerHTML;
80 |       refreshModelsBtn.innerHTML = '<span class="spinner-borde
    |   r spinner-border-sm me-1" role="status"></span> Обновление...'
    |   ;
81 |       try {
82 |         await loadTabModels(modelSelect, saveBtn, true);
83 |       } finally {
84 |         refreshModelsBtn.disabled = false;

```

```

85 |         refreshModelsBtn.innerHTML = originalText;
86 |     }
87 | };
88 | }
89 |
90 | const showAllModelsCheck = document.getElementById('show-all
    -models-check');
91 | if (showAllModelsCheck && modelSelect && saveBtn) {
92 |     showAllModelsCheck.onchange = async () => {
93 |         await loadTabModels(modelSelect, saveBtn, false);
94 |     };
95 | }
96 |
97 | const favToggleBtn = document.getElementById('btn-model-togg
    le-favorite');
98 | const saveNoteBtn = document.getElementById('btn-save-model-
    note');
99 | const modelNoteInput = document.getElementById('model-user-n
    ote');
100 |
101 | if (favToggleBtn && modelSelect) {
102 |     favToggleBtn.onclick = async () => {
103 |         const curModel = modelSelect.value;
104 |         if (!curModel) {
105 |             showModelsNotification('Сначала выберите модель', 'war
                ning');
106 |             return;
107 |         }
108 |         favToggleBtn.disabled = true;
109 |         try {
110 |             const isFav = window.userFavoriteModels && Boolean(win
                dow.userFavoriteModels[curModel]);
111 |             if (isFav) {
112 |                 await window.api.fetch(`/auth/favorites/${encodeURIComponent(
                    curModel)}`, { method: 'DELETE' });
113 |                 if (window.userFavoriteModels) delete window.userFav
                    oriteModels[curModel];
114 |                 showModelsNotification(`Модель "${curModel}" удалена

```

```

    из избранного`, 'info'));
115         } else {
116             const noteText = modelNoteInput ? modelNoteInput.value.trim() : '';
117             const res = await window.api.fetch('/auth/favorites'
, {
118                 method: 'POST',
119                 headers: { 'Content-Type': 'application/json' },
120                 body: JSON.stringify({ model: curModel, note: noteText })
121             });
122             if (res && res.favorites) window.userFavoriteModels = res.favorites;
123             showModelsNotification(`Модель "${curModel}" добавлена в избранное`, 'success');
124         }
125         updateModelFavoriteUI(modelSelect);
126         updateModelSelectOptions(modelSelect);
127     } catch (err) {
128         console.error('Ошибка переключения избранного:', err);
129         showModelsNotification('Ошибка: ' + err.message, 'danger');
130     } finally {
131         favToggleBtn.disabled = false;
132     }
133 };
134 }
135
136 if (saveNoteBtn && modelSelect && modelNoteInput) {
137     saveNoteBtn.onclick = async () => {
138         const curModel = modelSelect.value;
139         if (!curModel) {
140             showModelsNotification('Сначала выберите модель', 'warning');
141             return;
142         }
143         saveNoteBtn.disabled = true;
144         const originalText = saveNoteBtn.innerHTML;

```

```
145 |         saveNoteBtn.innerHTML = '<span class="spinner-border spi  
| nner-border-sm me-1"></span> Сохранение...';  
146 |         try {  
147 |             const noteText = modelNoteInput.value.trim();  
148 |             const res = await window.api.fetch('/auth/favorites',  
| {  
149 |                 method: 'POST',  
150 |                 headers: { 'Content-Type': 'application/json' },  
151 |                 body: JSON.stringify({ model: curModel, note: noteTe  
| xt })  
152 |             });  
153 |             if (res && res.favorites) window.userFavoriteModels =  
| res.favorites;  
154 |             showModelsNotification(`Заметка для модели "${curModel  
| }" сохранена!`, 'success');  
155 |             updateModelFavoriteUI(modelSelect);  
156 |             updateModelSelectOptions(modelSelect);  
157 |         } catch (err) {  
158 |             console.error('Ошибка сохранения заметки:', err);  
159 |             showModelsNotification('Ошибка сохранения заметки: ' +  
| err.message, 'danger');  
160 |         } finally {  
161 |             saveNoteBtn.disabled = false;  
162 |             saveNoteBtn.innerHTML = originalText;  
163 |         }  
164 |     };  
165 | }  
166 |  
167 | if (modelSelect) {  
168 |     modelSelect.addEventListener('change', () => {  
169 |         updateModelFavoriteUI(modelSelect);  
170 |     });  
171 | }  
172 |  
173 | if (testModelBtn) {  
174 |     testModelBtn.onclick = executeModelTest;  
175 | }  
176 |
```

```

177 |   if (cancelModelBtn) {
178 |       cancelModelBtn.onclick = cancelModelTest;
179 |   }
180 |
181 |   if (testModelPrompt) {
182 |       testModelPrompt.onkeydown = (e) => {
183 |           if (e.key === 'Enter') {
184 |               e.preventDefault();
185 |               executeModelTest();
186 |           } else if (e.key === 'Escape' && modelTestAbortControlle
r) {
187 |               e.preventDefault();
188 |               cancelModelTest();
189 |           }
190 |       };
191 |   }
192 |
193 |   // Provider switch direct toggling
194 |   const agyEnabledSwitch = document.getElementById('agy-enable
d');
195 |   if (agyEnabledSwitch) {
196 |       agyEnabledSwitch.onchange = async () => {
197 |           const enabled = agyEnabledSwitch.checked;
198 |           const remember = document.getElementById('agy-remember')
?.checked ?? true;
199 |           const model = document.getElementById('agy-model')?.valu
e || 'agy-flash';
200 |           const key = document.getElementById('agy-key')?.value.tr
im() || '';
201 |
202 |           try {
203 |               await window.api.fetch('/api/agy/config', {
204 |                   method: 'POST',
205 |                   headers: { 'Content-Type': 'application/json' },
206 |                   body: JSON.stringify({ enabled, model, key, remember
    })
207 |               });
208 |               showModelsNotification(`Google Antigravity (AGY) ${ena

```



```

        bled ? 'активирован' : 'деактивирован'}${remember ? ' (сохране
но в config.json)' : ''}`, 'info');
209         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
210     } catch (err) {
211         console.error('Ошибка переключения Antigravity:', err)
;
212         showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
213     }
214 };
215 }
216
217 const foundryEnabledSwitch = document.getElementById('foundr
y-enabled');
218 if (foundryEnabledSwitch) {
219     foundryEnabledSwitch.onChange = async () => {
220         const enabled = foundryEnabledSwitch.checked;
221         const remember = document.getElementById('foundry-rememb
er')?.checked ?? true;
222         const url = document.getElementById('foundry-url')?.valu
e.trim() || 'http://localhost:54837';
223         const key = document.getElementById('foundry-key')?.valu
e.trim() || '';
224         const model = document.getElementById('foundry-model')?.
value?.trim() || 'qwen2.5-1.5b';
225
226         try {
227             await window.api.fetch('/api/foundry/config', {
228                 method: 'POST',
229                 headers: { 'Content-Type': 'application/json' },
230                 body: JSON.stringify({ enabled, url, key, model, rem
ember })
231         });
232         showModelsNotification(`Microsoft Foundry ${enabled ?
'активирован' : 'деактивирован'}${remember ? ' (сохранено в co
nfig.json)' : ''}`, 'info');
233         if (modelSelect && saveBtn) await loadTabModels(modelS

```

```

    select, saveBtn);
234     } catch (err) {
235         console.error('Ошибка переключения Foundry:', err);
236         showModelsNotification('Ошибка переключения: ' + err.m
message, 'danger');
237     }
238 };
239 }
240
241 const ollamaEnabledSwitch = document.getElementById('ollama-
enabled');
242 if (ollamaEnabledSwitch) {
243     ollamaEnabledSwitch.onchange = async () => {
244         const enabled = ollamaEnabledSwitch.checked;
245         const remember = document.getElementById('ollama-remembe
r')?.checked ?? true;
246         const url = document.getElementById('ollama-url')?.value
.trim() || 'http://localhost:11434';
247         const model = document.getElementById('ollama-model')?.v
alue?.trim() || 'llama3.1';
248
249         try {
250             await window.api.fetch('/api/ollama/config', {
251                 method: 'POST',
252                 headers: { 'Content-Type': 'application/json' },
253                 body: JSON.stringify({ enabled, url, model, remember
}))
254             });
255             showModelsNotification(`Ollama ${enabled ? 'активирова
н' : 'деактивирован'}${remember ? ' (сохранено в config.json)'
: ''}`, 'info');
256             if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
257         } catch (err) {
258             console.error('Ошибка переключения Ollama:', err);
259             showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
260         }

```

```

261     };
262 }
263
264 const onnxEnabledSwitch = document.getElementById('onnx-enab
led');
265 if (onnxEnabledSwitch) {
266     onnxEnabledSwitch.onchange = async () => {
267         const enabled = onnxEnabledSwitch.checked;
268         const remember = document.getElementById('onnx-remember'
)??.checked ?? true;
269         const execution_provider = document.getElementById('onnx
-execution-provider')?.value || 'DirectMLExecutionProvider';
270         const models_dir = document.getElementById('onnx-models-
dir')?.value.trim() || 'models/onnx';
271         const default_model = document.getElementById('onnx-defa
ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
272         const olive_precision = document.getElementById('onnx-ol
ive-precision')?.value || 'int4';
273
274         try {
275             await window.api.fetch('/api/onnx/config', {
276                 method: 'POST',
277                 headers: { 'Content-Type': 'application/json' },
278                 body: JSON.stringify({
279                     enabled,
280                     execution_provider,
281                     models_dir,
282                     default_model,
283                     olive_precision,
284                     remember
285                 })
286             });
287             showModelsNotification(`ONNX / Olive ${enabled ? 'акти
вирован' : 'деактивирован'}${remember ? ' (сохранено в config.
json)' : ''}`, 'info');
288             if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
289         } catch (err) {

```

```
290 |         console.error('Ошибка переключения ONNX / Olive:', err
    | );
291 |         showModelsNotification('Ошибка переключения: ' + err.m
    | essage, 'danger');
292 |     }
293 | };
294 | }
295 |
296 | if (saveAgyBtn) {
297 |     saveAgyBtn.onclick = async () => {
298 |         const enabled = document.getElementById('agy-enabled')?.
    | checked ?? true;
299 |         const remember = document.getElementById('agy-remember')
    | ?.checked ?? true;
300 |         const model = document.getElementById('agy-model')?.valu
    | e || 'agy-flash';
301 |         const key = document.getElementById('agy-key')?.value.tr
    | im() || '';
302 |
303 |         saveAgyBtn.disabled = true;
304 |         try {
305 |             await window.api.fetch('/api/agy/config', {
306 |                 method: 'POST',
307 |                 headers: { 'Content-Type': 'application/json' },
308 |                 body: JSON.stringify({ enabled, model, key, remember
    |
    | })
309 |         });
310 |         showModelsNotification(`Настройки Antigravity (AGY) ус
    | пешно сохранены${remember ? ' в config.json' : ''}`, 'success'
    | );
311 |         if (modelSelect && saveBtn) await loadTabModels(modelS
    | elect, saveBtn);
312 |     } catch (err) {
313 |         console.error('Ошибка сохранения Antigravity:', err);
314 |         showModelsNotification('Ошибка сохранения: ' + err.mes
    | sage, 'danger');
315 |     } finally {
316 |         saveAgyBtn.disabled = false;
```

```
317 |     }
318 | };
319 | }
320 |
321 | if (saveFoundryBtn) {
322 |     saveFoundryBtn.onclick = async () => {
323 |         const enabled = document.getElementById('foundry-enabled')?.checked || false;
324 |         const remember = document.getElementById('foundry-remember')?.checked ?? true;
325 |         const url = document.getElementById('foundry-url')?.value.trim() || 'http://localhost:54837';
326 |         const key = document.getElementById('foundry-key')?.value.trim() || '';
327 |         const model = document.getElementById('foundry-model')?.value?.trim() || 'qwen2.5-1.5b';
328 |
329 |         saveFoundryBtn.disabled = true;
330 |         try {
331 |             await window.api.fetch('/api/foundry/config', {
332 |                 method: 'POST',
333 |                 headers: { 'Content-Type': 'application/json' },
334 |                 body: JSON.stringify({ enabled, url, key, model, remember })
335 |             });
336 |             showModelsNotification(`Настройки Microsoft Foundry успешно сохранены${remember ? ' в config.json' : ''}`, 'success');
337 |             if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
338 |         } catch (err) {
339 |             console.error('Ошибка сохранения Foundry:', err);
340 |             showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
341 |         } finally {
342 |             saveFoundryBtn.disabled = false;
343 |         }
344 |     };

```

```
345 | }
346 |
347 | if (saveOllamaBtn) {
348 |     saveOllamaBtn.onclick = async () => {
349 |         const enabled = document.getElementById('ollama-enabled'
350 | )?.checked || false;
351 |         const remember = document.getElementById('ollama-remembe
352 | r')?.checked ?? true;
353 |         const url = document.getElementById('ollama-url')?.value
354 | .trim() || 'http://localhost:11434';
355 |         const model = document.getElementById('ollama-model')?.v
356 | alue?.trim() || 'llama3.1';
357 |
358 |         saveOllamaBtn.disabled = true;
359 |         try {
360 |             await window.api.fetch('/api/ollama/config', {
361 |                 method: 'POST',
362 |                 headers: { 'Content-Type': 'application/json' },
363 |                 body: JSON.stringify({ enabled, url, model, remember
364 | })
365 |             });
366 |             showModelsNotification(`Настройки Ollama успешно сохра
367 | нены${remember ? ' в config.json' : ''}`, 'success');
368 |             if (modelSelect && saveBtn) await loadTabModels(modelS
369 | elect, saveBtn);
370 |         } catch (err) {
371 |             console.error('Ошибка сохранения Ollama:', err);
372 |             showModelsNotification('Ошибка сохранения: ' + err.mes
373 | sage, 'danger');
374 |         } finally {
375 |             saveOllamaBtn.disabled = false;
376 |         }
377 |     };
378 | }
379 |
380 | if (saveOnnxBtn) {
381 |     saveOnnxBtn.onclick = async () => {
382 |         const enabled = document.getElementById('onnx-enabled')?
```

```

    .checked ?? true;
375     const remember = document.getElementById('onnx-remember'
    )?.checked ?? true;
376     const execution_provider = document.getElementById('onnx
    -execution-provider')?.value || 'DirectMLExecutionProvider';
377     const models_dir = document.getElementById('onnx-models-
    dir')?.value.trim() || 'models/onnx';
378     const default_model = document.getElementById('onnx-defa
    ult-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
379     const olive_precision = document.getElementById('onnx-ol
    ive-precision')?.value || 'int4';
380
381     saveOnnxBtn.disabled = true;
382     try {
383         await window.api.fetch('/api/onnx/config', {
384             method: 'POST',
385             headers: { 'Content-Type': 'application/json' },
386             body: JSON.stringify({
387                 enabled,
388                 execution_provider,
389                 models_dir,
390                 default_model,
391                 olive_precision,
392                 remember
393             })
394         });
395         showModelsNotification(`Настройки ONNX / Olive успешно
    сохранены${remember ? ' в config.json' : ''}`, 'success');
396         if (modelSelect && saveBtn) await loadTabModels(modelS
    elect, saveBtn);
397     } catch (err) {
398         console.error('Ошибка сохранения ONNX / Olive:', err);
399         showModelsNotification('Ошибка сохранения: ' + err.mes
    sage, 'danger');
400     } finally {
401         saveOnnxBtn.disabled = false;
402     }
403 };

```

```
404 | }
405 |
406 | if (saveBtn && modelSelect) {
407 |     saveBtn.onclick = async () => {
408 |         const selectedModel = modelSelect.value;
409 |         saveBtn.disabled = true;
410 |         const originalText = saveBtn.textContent;
411 |         saveBtn.textContent = 'Сохранение...';
412 |
413 |         try {
414 |             await window.api.fetch('/auth/settings', {
415 |                 method: 'POST',
416 |                 headers: { 'Content-Type': 'application/json' },
417 |                 body: JSON.stringify({ model: selectedModel })
418 |             });
419 |             showModelsNotification('Модель успешно обновлена на: '
+ selectedModel, 'success');
420 |
421 |             const otherModelSelect = document.getElementById('admin-model-select');
422 |             if (otherModelSelect) {
423 |                 otherModelSelect.value = selectedModel;
424 |             }
425 |             if (typeof window.updateChatBadges === 'function') {
426 |                 window.updateChatBadges(selectedModel);
427 |             }
428 |         } catch (err) {
429 |             console.error('Ошибка сохранения модели:', err);
430 |             showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
431 |         } finally {
432 |             saveBtn.disabled = false;
433 |             saveBtn.textContent = originalText;
434 |         }
435 |     };
436 | }
437 |
438 | if (refreshKeysBtn) {
```



```
439     refreshKeysBtn.onclick = async () => {
440         refreshKeysBtn.disabled = true;
441         const originalText = refreshKeysBtn.textContent;
442         refreshKeysBtn.textContent = '⏳ Сброс...';
443         try {
444             const res = await window.api.fetch('/api/keys/reset-all', { method: 'POST' });
445             showModelsNotification(res.message || 'Квоты всех ключей успешно сброшены', 'success');
446         } catch (err) {
447             console.error('Ошибка сброса квот:', err);
448             showModelsNotification('Ошибка сброса: ' + err.message, 'danger');
449         } finally {
450             refreshKeysBtn.disabled = false;
451             refreshKeysBtn.textContent = originalText;
452             if (keysListBody) await refreshKeysList(keysListBody);
453         }
454     };
455 }
456
457 if (addKeyBtn) {
458     addKeyBtn.onclick = async () => {
459         const nameInput = document.getElementById('new-key-name');
460         const valueInput = document.getElementById('new-key-value');
461         if (!nameInput || !valueInput) return;
462
463         const name = nameInput.value.trim();
464         const apiKey = valueInput.value.trim();
465
466         if (!name || !apiKey) {
467             showModelsNotification('Заполните все поля!', 'warning');
468             return;
469         }
470     }
```

```

471     addKeyBtn.disabled = true;
472     try {
473         await window.api.fetch('/api/keys', {
474             method: 'POST',
475             headers: { 'Content-Type': 'application/json' },
476             body: JSON.stringify({ name, api_key: apiKey, status
: 'active' })
477         });
478         showModelsNotification(`Ключ "${name}" успешно добавле
н`, 'success');
479         nameInput.value = '';
480         valueInput.value = '';
481         if (keysListBody) await refreshKeysList(keysListBody);
482     } catch (err) {
483         console.error('Ошибка добавления ключа:', err);
484         showModelsNotification('Ошибка добавления: ' + err.mes
sage, 'danger');
485     } finally {
486         addKeyBtn.disabled = false;
487     }
488 };
489 }
490
491 if (refreshGAccountsBtn && gaccountsListBody) {
492     refreshGAccountsBtn.onclick = async () => {
493         refreshGAccountsBtn.disabled = true;
494         const originalText = refreshGAccountsBtn.innerHTML;
495         refreshGAccountsBtn.innerHTML = '<span class="spinner-bo
rder spinner-border-sm me-1"></span> Обновление...';
496         try {
497             await refreshGoogleAccountsList(gaccountsListBody);
498             showModelsNotification('Список аккаунтов Google обновл
ен', 'success');
499         } finally {
500             refreshGAccountsBtn.disabled = false;
501             refreshGAccountsBtn.innerHTML = originalText;
502         }
503     };

```

```
504 | }
505 |
506 | if (addGAccountBtn) {
507 |     addGAccountBtn.onclick = async () => {
508 |         await handleAddGoogleAccount(gaccountsListBody);
509 |     };
510 | }
511 |
512 | // 2. Load all components concurrently
513 | await Promise.allSettled([
514 |     modelSelect && saveBtn ? loadTabModels(modelSelect, saveBtn) : Promise.resolve(),
515 |     keysListBody ? refreshKeysList(keysListBody) : Promise.resolve(),
516 |     gaccountsListBody ? refreshGoogleAccountsList(gaccountsListBody) : Promise.resolve(),
517 |     loadFoundryConfig(),
518 |     loadOllamaConfig(),
519 |     loadAgyConfig(),
520 |     loadOnnxConfig(),
521 |     loadSystemInstruction()
522 | ]);
523 | }
524 |
525 | // Helper for escaping HTML strings
526 | function escapeHtml(text) {
527 |     if (!text) return '';
528 |     return String(text)
529 |         .replace(/&/g, '&amp;')
530 |         .replace(/</g, '&lt;')
531 |         .replace(/>/g, '&gt;')
532 |         .replace(/"/g, '&quot;')
533 |         .replace(/'/g, '&#039;');
534 | }
535 |
536 | // Helper to execute model test request
537 | async function executeModelTest() {
538 |     const providerSelect = document.getElementById('provider-tab
```

```

    -select');
539   const modelSelect = document.getElementById('models-tab-select');
540   const promptInput = document.getElementById('model-test-prompt');
541   const testBtn = document.getElementById('btn-model-test-send');
542   const cancelBtn = document.getElementById('btn-model-test-cancel');
543   const resultContainer = document.getElementById('model-test-result');
544
545   if (!testBtn || !resultContainer) return;
546
547   const provider = providerSelect ? providerSelect.value : '';
548   const model = modelSelect ? modelSelect.value : '';
549   const message = promptInput ? promptInput.value.trim() : '';
550
551   if (!model) {
552     showModelsNotification('Выберите модель для тестирования!',
553     , 'warning');
554     return;
555   }
556
557   if (!message) {
558     showModelsNotification('Введите текст проверочного запроса!',
559     , 'warning');
560     if (promptInput) promptInput.focus();
561     return;
562   }
563
564   // Abort any prior running test
565   cancelModelTest();
566   modelTestAbortController = new AbortController();
567
568   testBtn.disabled = true;
569   const originalBtnHtml = testBtn.innerHTML;
570   testBtn.innerHTML = '<span class="spinner-border spinner-border'

```

```

    der-sm me-1" role="status"></span> Тест...';
569
570     if (cancelBtn) {
571         cancelBtn.style.display = 'inline-block';
572         cancelBtn.disabled = false;
573     }
574
575     resultContainer.style.display = 'block';
576     resultContainer.innerHTML = `
577         <div class="d-flex justify-content-between align-items-center text-info small">
578             <div class="d-flex align-items-center text-truncate">
579                 <span class="spinner-border spinner-border-sm me-2 flex-shrink-0" role="status"></span>
580                 <span class="text-truncate">Отправка запроса в <strong>${escapeHtml(model)}</strong> (${escapeHtml(provider)})...</span>
581             </div>
582             <button class="btn btn-outline-danger btn-sm py-0 px-2 rounded ms-2 flex-shrink-0" id="btn-model-test-cancel-inner" type="button" title="Отменить проверочный запрос">
583                 <i class="bi bi-stop-circle me-1"></i> Отмена
584             </button>
585         </div>
586     `;
587
588     const cancelInnerBtn = document.getElementById('btn-model-test-cancel-inner');
589     if (cancelInnerBtn) {
590         cancelInnerBtn.onclick = cancelModelTest;
591     }
592
593     try {
594         const res = await window.api.fetch('/api/chat/test-model',
595             {
596                 method: 'POST',
597                 headers: { 'Content-Type': 'application/json' },
598                 signal: modelTestAbortController.signal,

```

```

598         body: JSON.stringify({
599             provider: provider,
600             model: model,
601             message: message
602         })
603     });
604
605     if (res && res.status === 'success') {
606         resultContainer.innerHTML = `
607             <div class="d-flex justify-content-between align-items
608             -center mb-1 pb-1 border-bottom">
609                 <div>
610                     <span class="badge bg-success me-2"><i class="bi b
611                     i-check-circle me-1"></i>200 OK</span>
612                     <span class="text-secondary small font-monospace">
613                         □ ${res.duration_ms || 0} ms</span>
614                 </div>
615                 <span class="badge bg-secondary-subtle text-secondar
616                 y-emphasis border font-monospace">${escapeHtml(res.model || mo
617                 del)}</span>
618             </div>
619             <div class="small mt-1 font-monospace" style="white-sp
620             ace: pre-wrap; word-break: break-word;">${escapeHtml(res.respo
621             nse || 'Пустой ответ от модели')}</div>
622         `;
623     } else {
624         const errMsg = (res && res.message) ? res.message : 'Неи
        звестная ошибка при запросе к модели';
        resultContainer.innerHTML = `
        <div class="d-flex justify-content-between align-items
        -center mb-1 pb-1 border-bottom">
            <div>
                <span class="badge bg-danger me-2"><i class="bi bi
                -x-circle me-1"></i>Ошибка</span>
                <span class="text-secondary small font-monospace">
                    □ ${res?.duration_ms || 0} ms</span>
            </div>
            <span class="badge bg-secondary-subtle text-secondar

```

```

        y-emphasis border font-monospace">${escapeHtml(model)}</span>
625     </div>
626     <div class="text-danger small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(errMsg)}</div>
627     `;
628     }
629     } catch (err) {
630         if (err.name === 'AbortError' || err.message?.toLowerCase().includes('abort')) {
631             resultContainer.innerHTML = `
632                 <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
633                     <span class="badge bg-warning text-dark"><i class="bi bi-slash-circle me-1"></i>Отменено</span>
634                     <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
635                 </div>
636                 <div class="text-warning small mt-1 font-monospace"><i class="bi bi-info-circle me-1"></i>Запрос отменен пользователем.</div>
637             `;
638         } else {
639             console.error('Error during model test request:', err);
640             resultContainer.innerHTML = `
641                 <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
642                     <span class="badge bg-danger"><i class="bi bi-x-circle me-1"></i>Ошибка сети/API</span>
643                     <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
644                 </div>
645                 <div class="text-danger small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(err.message || 'Ошибка соединения')}</div>
646             `;
647         }
648     } finally {

```

```
649     modelTestAbortController = null;
650     testBtn.disabled = false;
651     testBtn.innerHTML = originalBtnHtml;
652     if (cancelBtn) {
653         cancelBtn.style.display = 'none';
654     }
655 }
656 }
657
658 // Favorite models storage in window state
659 window.userFavoriteModels = window.userFavoriteModels || {};
660
661 function updateModelFavoriteUI(modelSelect) {
662     if (!modelSelect) modelSelect = document.getElementById('models-tab-select');
663     const curModel = modelSelect ? modelSelect.value : '';
664     const starIcon = document.getElementById('favorite-star-icon');
665     const favBtn = document.getElementById('btn-model-toggle-favorite');
666     const statusBadge = document.getElementById('model-fav-status-badge');
667     const noteInput = document.getElementById('model-user-note');
668     ;
669     const favData = curModel && window.userFavoriteModels ? window.userFavoriteModels[curModel] : null;
670     const isFav = Boolean(favData);
671
672     if (starIcon) {
673         starIcon.className = isFav ? 'bi bi-star-fill text-warning' : 'bi bi-star';
674     }
675     if (favBtn) {
676         if (isFav) {
677             favBtn.classList.remove('btn-outline-warning');
678             favBtn.classList.add('btn-warning', 'text-dark');
679             favBtn.title = 'Удалить модель из избранного';
```



```
680     } else {
681         favBtn.classList.remove('btn-warning', 'text-dark');
682         favBtn.classList.add('btn-outline-warning');
683         favBtn.title = 'Добавить модель в избранное';
684     }
685 }
686 if (statusBadge) {
687     if (isFav) {
688         statusBadge.textContent = 'В избранном';
689         statusBadge.className = 'badge bg-warning text-dark font
-monospace';
690     } else {
691         statusBadge.textContent = 'Не в избранном';
692         statusBadge.className = 'badge bg-secondary-subtle text-
secondary-emphasis border font-monospace';
693     }
694 }
695 if (noteInput && document.activeElement !== noteInput) {
696     noteInput.value = favData ? (favData.note || '') : '';
697 }
698
699 renderFavoriteModelsChips(modelSelect);
700 }
701
702 function updateModelSelectOptions(modelSelect) {
703     if (!modelSelect) return;
704     const options = modelSelect.querySelectorAll('option');
705     options.forEach(opt => {
706         const val = opt.value;
707         if (!val) return;
708         const isFav = window.userFavoriteModels && Boolean(window.
userFavoriteModels[val]);
709         let text = opt.textContent.replace(/^\s*/, '');
710         if (isFav) {
711             opt.textContent = ` ${text}`;
712         } else {
713             opt.textContent = text;
714         }

```

```
715 | });
716 | }
717 |
718 | function renderFavoriteModelsChips(modelSelect) {
719 |     const container = document.getElementById('favorite-models-l
       | ist');
720 |     const section = document.getElementById('favorite-models-sec
       | tion');
721 |     if (!container || !section) return;
722 |
723 |     const favKeys = Object.keys(window.userFavoriteModels || {})
       | ;
724 |     if (favKeys.length === 0) {
725 |         section.style.display = 'none';
726 |         container.innerHTML = '';
727 |         return;
728 |     }
729 |
730 |     section.style.display = 'block';
731 |     container.innerHTML = '';
732 |
733 |     favKeys.forEach(modelName => {
734 |         const data = window.userFavoriteModels[modelName] || {};
735 |         const note = data.note ? `\nЗаметка: ${data.note}` : '';
736 |
737 |         const chip = document.createElement('div');
738 |         chip.className = 'btn-group btn-group-sm mb-1';
739 |         chip.role = 'group';
740 |
741 |         const btnSelect = document.createElement('button');
742 |         btnSelect.type = 'button';
743 |         btnSelect.className = modelSelect && modelSelect.value ===
       | modelName
744 |             ? 'btn btn-sm btn-warning text-dark py-0 px-2 font-monos
       | pace'
745 |             : 'btn btn-sm btn-outline-warning py-0 px-2 font-monospa
       | ce';
746 |         btnSelect.innerHTML = `
```

```

-warning"></i>${escapeHtml(modelName)}`;
747     btnSelect.title = `Выбрать модель: ${modelName}${note}`;
748     btnSelect.onclick = () => {
749         if (modelSelect) {
750             // Try finding which provider has this model
751             const providerSelect = document.getElementById('provider-tab-select');
752             if (providerSelect && window._modelsGrouped) {
753                 for (const p of Object.keys(window._modelsGrouped))
754                 {
755                     if (window._modelsGrouped[p]?.includes(modelName))
756                     {
757                         if (providerSelect.value !== p) {
758                             providerSelect.value = p;
759                             if (typeof window._populateModels === 'function') {
760                                 window._populateModels(p, window._modelsGrouped[p]);
761                             }
762                         }
763                     }
764                     break;
765                 }
766             }
767             modelSelect.value = modelName;
768             updateModelFavoriteUI(modelSelect);
769         }
770     };
771
772     const btnRemove = document.createElement('button');
773     btnRemove.type = 'button';
774     btnRemove.className = 'btn btn-sm btn-outline-danger py-0 px-1';
775     btnRemove.innerHTML = '<i class="bi bi-x"></i>';
776     btnRemove.title = `Удалить ${modelName} из избранного`;
777     btnRemove.onclick = async (e) => {
778         e.stopPropagation();
779         try {

```

```

778 |         await window.api.fetch(`/auth/favorites/${encodeURIComponent
|       ponent(modelName)}`, { method: 'DELETE' });
779 |         delete window.userFavoriteModels[modelName];
780 |         updateModelFavoriteUI(modelSelect);
781 |         updateModelSelectOptions(modelSelect);
782 |         showModelsNotification(`Модель "${modelName}" удалена
|       из избранного`, 'info');
783 |     } catch (err) {
784 |         showModelsNotification('Ошибка удаления: ' + err.messa
|       ge, 'danger');
785 |     }
786 | };
787 |
788 |     chip.appendChild(btnSelect);
789 |     chip.appendChild(btnRemove);
790 |     container.appendChild(chip);
791 | });
792 | }
793 |
794 | // Helper to load models list
795 | async function loadTabModels(modelSelect, saveBtn, forceRefres
|       h = false) {
796 |     const providerSelect = document.getElementById('provider-tab
|       -select');
797 |     if (providerSelect) providerSelect.innerHTML = '';
798 |     modelSelect.innerHTML = '';
799 |
800 |     let modelsGrouped = {};
801 |     let unsupportedGrouped = {};
802 |
803 |     const fetchModels = async (force = false) => {
804 |         try {
805 |             const showAll = document.getElementById('show-all-models
|       -check')?.checked ?? false;
806 |             const params = new URLSearchParams();
807 |             if (force) params.append('refresh', 'true');
808 |             if (showAll) params.append('include_unsupported', 'true'
|       );

```

```
809 |     const url = '/api/chat/models' + (params.toString() ? '?'
    | ' + params.toString() : '');
810 |     const modelsData = await window.api.fetch(url);
811 |     let grouped = modelsData.models || {};
812 |     if (Array.isArray(grouped)) {
813 |         grouped = { 'gemini': grouped };
814 |     }
815 |     unsupportedGrouped = modelsData.unsupported_models || {}
    | ;
816 |     return grouped;
817 | } catch (err) {
818 |     console.error('Error loading AI models:', err);
819 |     showModelsNotification('Ошибка загрузки моделей AI: ' +
    | err.message, 'danger');
820 |     return {};
821 | }
822 | };
823 |
824 | modelsGrouped = await fetchModels(forceRefresh);
825 | window._modelsGrouped = modelsGrouped;
826 | if (forceRefresh) {
827 |     showModelsNotification('Список моделей успешно обновлен',
    | 'success');
828 | }
829 |
830 | // Fetch favorite models and current settings
831 | try {
832 |     const settingsData = await window.api.fetch('/auth/settings');
833 |     if (settingsData && settingsData.favorite_models) {
834 |         window.userFavoriteModels = settingsData.favorite_models
    | ;
835 |     } else {
836 |         const favRes = await window.api.fetch('/auth/favorites')
    | ;
837 |         if (favRes && favRes.favorites) {
838 |             window.userFavoriteModels = favRes.favorites;
839 |         }
```

```
840     }
841   } catch (e) {
842     console.warn('Failed to load favorite models:', e);
843   }
844
845   const providers = Object.keys(modelsGrouped).filter(p => mod
elsGrouped[p] && modelsGrouped[p].length > 0);
846
847   if (providers.length === 0) {
848     if (providerSelect) {
849       providerSelect.innerHTML = '<option value="">Нет доступн
ых провайдеров</option>';
850     }
851     modelSelect.innerHTML = '<option value="">Нет доступных мо
делей</option>';
852     saveBtn.disabled = true;
853     updateModelFavoriteUI(modelSelect);
854     return;
855   }
856
857   if (providerSelect) {
858     providerSelect.innerHTML = '';
859     providers.forEach(p => {
860       const option = document.createElement('option');
861       option.value = p;
862       option.textContent = p.charAt(0).toUpperCase() + p.slice
(1);
863       providerSelect.appendChild(option);
864     });
865   }
866
867   const populateModels = (provider, providerModelsList) => {
868     modelSelect.innerHTML = '';
869     const providerModels = providerModelsList !== undefined ?
providerModelsList : (modelsGrouped[provider] || []);
870     const unsupList = unsupportedGrouped[provider] || [];
871     if (!providerModels || providerModels.length === 0) {
872       modelSelect.innerHTML = '<option value="">Нет моделей</o
```

```

    option>';
873     saveBtn.disabled = true;
874   } else {
875     providerModels.forEach(modelName => {
876       const option = document.createElement('option');
877       option.value = modelName;
878       let cleanName = modelName;
879       if (cleanName.startsWith('foundry:')) cleanName = cleanName.substring(8);
880       else if (cleanName.startsWith('ollama:')) cleanName = cleanName.substring(7);
881       else if (cleanName.startsWith('gemini_cli:')) cleanName = cleanName.substring(11);
882       else if (cleanName.startsWith('agy-')) cleanName = cleanName.substring(4);
883       else if (cleanName.startsWith('onnx:')) cleanName = cleanName.substring(5);
884
885       const isFav = window.userFavoriteModels && Boolean(window.userFavoriteModels[modelName]);
886       const isUnsupported = unSUPList.includes(cleanName) || unSUPList.includes(modelName);
887
888       let label = cleanName;
889       if (isFav) label = `🌟 ${label}`;
890       if (isUnsupported) label = `${label} ❌ [отфильтрована]`;
891
892       option.textContent = label;
893       modelSelect.appendChild(option);
894     });
895     saveBtn.disabled = false;
896   }
897   updateModelFavoriteUI(modelSelect);
898 };
899 window._populateModels = populateModels;
900
901 if (providerSelect) {

```

```
902     providerSelect.onChange = async () => {
903         const chosenProvider = providerSelect.value;
904         // При каждом выборе провайдера актуализируем список дос
тупных моделей
905         modelSelect.innerHTML = '<option value="">Обновление спи
ска моделей...</option>';
906         saveBtn.disabled = true;
907         try {
908             const updatedGrouped = await fetchModels(true);
909             if (updatedGrouped && Object.keys(updatedGrouped).leng
th > 0) {
910                 modelsGrouped = updatedGrouped;
911                 window._modelsGrouped = modelsGrouped;
912             }
913         } catch (e) {
914             console.warn('Failed to refresh models on provider cha
nge:', e);
915         }
916         populateModels(chosenProvider, modelsGrouped[chosenProvi
der]);
917     };
918     populateModels(providerSelect.value, modelsGrouped[provide
rSelect.value]);
919 } else {
920     let allModels = [];
921     providers.forEach(p => allModels = allModels.concat(models
Grouped[p]));
922     allModels.forEach(modelName => {
923         const option = document.createElement('option');
924         option.value = modelName;
925         const isFav = window.userFavoriteModels && Boolean(win
dow.userFavoriteModels[modelName]);
926         option.textContent = isFav ? `🔖 ${modelName}` : modelN
ame;
927         modelSelect.appendChild(option);
928     });
929     saveBtn.disabled = allModels.length === 0;
930     updateModelFavoriteUI(modelSelect);
```



```

931 |     }
932 |
933 |     try {
934 |         const settingsData = await window.api.fetch('/auth/settings');
935 |         const savedModel = settingsData && settingsData.model ? settingsData.model : '';
936 |         if (savedModel) {
937 |             let foundProvider = null;
938 |             for (const p of providers) {
939 |                 if (modelsGrouped[p] && modelsGrouped[p].includes(savedModel)) {
940 |                     foundProvider = p;
941 |                     break;
942 |                 }
943 |             }
944 |             if (foundProvider && providerSelect) {
945 |                 providerSelect.value = foundProvider;
946 |                 populateModels(foundProvider, modelsGrouped[foundProvider]);
947 |             }
948 |             modelSelect.value = savedModel;
949 |             updateModelFavoriteUI(modelSelect);
950 |         }
951 |         if (typeof window.updateChatBadges === 'function') {
952 |             window.updateChatBadges(savedModel);
953 |         }
954 |     } catch (err) {
955 |         console.error('Error loading user AI settings:', err);
956 |     }
957 | }
958 |
959 | // Helper to refresh keys list table
960 | async function refreshKeysList(container) {
961 |     try {
962 |         container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Загрузка ключей...</td></tr>';
963 |         const keysData = await window.api.fetch('/api/keys');

```

```
964 |     const keys = keysData.keys || [];
965 |
966 |     if (keys.length === 0) {
967 |         container.innerHTML = '<tr><td colspan="5" class="text-c
enter py-4 text-muted">Список ключей пуст</td></tr>';
968 |         return;
969 |     }
970 |
971 |     container.innerHTML = '';
972 |     keys.forEach(key => {
973 |         const row = document.createElement('tr');
974 |
975 |         const tdName = document.createElement('td');
976 |         tdName.innerHTML = `<strong>${key.name}</strong>`;
977 |         row.appendChild(tdName);
978 |
979 |         const tdKey = document.createElement('td');
980 |         tdKey.className = 'font-monospace text-muted small';
981 |         tdKey.textContent = key.api_key_masked;
982 |         row.appendChild(tdKey);
983 |
984 |         const tdStatus = document.createElement('td');
985 |         const isEnabled = key.status === 'active';
986 |         const statusClass = isEnabled ? 'bg-success' : 'bg-secon
dary';
987 |         const statusText = isEnabled ? 'Активен' : 'Отключен';
988 |         tdStatus.innerHTML = `<span class="badge ${statusClass}"
>${statusText}</span>`;
989 |         row.appendChild(tdStatus);
990 |
991 |         const tdQuota = document.createElement('td');
992 |         if (key.exhausted) {
993 |             let resetText = 'Лимит';
994 |             if (key.reset_in_seconds) {
995 |                 const hours = Math.floor(key.reset_in_seconds / 3600
);
996 |                 const mins = Math.floor((key.reset_in_seconds % 3600
) / 60);
```

```

997         resetText = `Сброс через ${hours}ч ${mins}м`;
998     }
999     tdQuota.innerHTML = `<span class="badge bg-danger d-block mb-1" title="Превышен лимит запросов в сутки">${resetText}</span>`;
1000     } else {
1001         tdQuota.innerHTML = `<span class="badge bg-success d-block mb-1">ОК</span>`;
1002     }
1003     row.appendChild(tdQuota);
1004
1005     const tdActions = document.createElement('td');
1006     tdActions.className = 'text-end';
1007
1008     const btnToggle = document.createElement('button');
1009     btnToggle.className = `btn btn-xs btn-sm me-1 ${isEnabled
1010     d ? 'btn-outline-secondary' : 'btn-outline-success'}`;
1011     btnToggle.textContent = isEnabled ? 'Откл' : 'Вкл';
1012     btnToggle.onclick = () => toggleKeyStatus(key.name, isEnabled ? 'disabled' : 'active', container);
1013     tdActions.appendChild(btnToggle);
1014
1015     if (key.exhausted) {
1016         const btnReset = document.createElement('button');
1017         btnReset.className = 'btn btn-xs btn-outline-warning btn-sm me-1';
1018         btnReset.innerHTML = 'Сброс';
1019         btnReset.title = 'Сбросить 24-часовой бан квоты';
1020         btnReset.onclick = () => resetKeyQuota(key.name, container);
1021         tdActions.appendChild(btnReset);
1022     }
1023
1024     const btnDelete = document.createElement('button');
1025     btnDelete.className = 'btn btn-xs btn-outline-danger btn-sm';
1026     btnDelete.textContent = 'Удалить';
1027     btnDelete.onclick = () => deleteKey(key.name, container);

```

```

;
1027     tdActions.appendChild(btnDelete);
1028
1029     row.appendChild(tdActions);
1030     container.appendChild(row);
1031 });
1032
1033 } catch (err) {
1034     console.error('Ошибка загрузки ключей:', err);
1035     container.innerHTML = `<tr><td colspan="5" class="text-center py-4 text-danger">Ошибка: ${err.message}</td></tr>`;
1036 }
1037 }
1038
1039 // API action helpers
1040 async function toggleKeyStatus(name, newStatus, container) {
1041     try {
1042         await window.api.fetch(`/api/keys/${name}`, {
1043             method: 'PATCH',
1044             headers: { 'Content-Type': 'application/json' },
1045             body: JSON.stringify({ status: newStatus })
1046         });
1047         showModelsNotification(`Статус ключа "${name}" изменен на
        ${newStatus === 'active' ? 'активный' : 'отключенный'}`, 'success');
1048         await refreshKeysList(container);
1049     } catch (err) {
1050         console.error('Ошибка переключения статуса ключа:', err);
1051         showModelsNotification('Ошибка изменения статуса: ' + err.
        message, 'danger');
1052     }
1053 }
1054
1055 async function resetKeyQuota(name, container) {
1056     try {
1057         await window.api.fetch(`/api/keys/${name}/reset-quota`, {
        method: 'POST' });
1058         showModelsNotification(`Квота для ключа "${name}" успешно

```

```

        сброшена`, 'success');
1059     await refreshKeysList(container);
1060   } catch (err) {
1061     console.error('Ошибка сброса квоты:', err);
1062     showModelsNotification('Ошибка сброса квоты: ' + err.messa
    ge, 'danger');
1063   }
1064 }
1065
1066 async function deleteKey(name, container) {
1067   if (!confirm(`Вы уверены, что хотите удалить ключ "${name}"?`)) return;
1068   try {
1069     await window.api.fetch(`/api/keys/${name}`, { method: 'DEL
    ETE' });
1070     showModelsNotification(`Ключ "${name}" успешно удален`, 's
    uccess');
1071     await refreshKeysList(container);
1072   } catch (err) {
1073     console.error('Ошибка удаления ключа:', err);
1074     showModelsNotification('Ошибка удаления: ' + err.message,
    'danger');
1075   }
1076 }
1077
1078 function showModelsNotification(message, type = 'info') {
1079   const notification = document.createElement('div');
1080   notification.className = `alert alert-${type} position-fixed
    top-0 end-0 m-3`;
1081   notification.style.zIndex = '9999';
1082   notification.style.maxWidth = '400px';
1083   notification.textContent = message;
1084   document.body.appendChild(notification);
1085
1086   setTimeout(() => {
1087     notification.remove();
1088   }, 5000);
1089 }

```

```
1090 |
1091 | async function loadFoundryConfig() {
1092 |   try {
1093 |     const config = await window.api.fetch('/api/foundry/config
1094 |   ');
1095 |     const enabledInput = document.getElementById('foundry-enab
1096 |   led');
1097 |     const urlInput = document.getElementById('foundry-url');
1098 |     const keyInput = document.getElementById('foundry-key');
1099 |     const modelInput = document.getElementById('foundry-model'
1100 |   );
1101 |
1102 |     if (enabledInput) enabledInput.checked = config.enabled ||
1103 |   false;
1104 |     if (urlInput) urlInput.value = config.url || '';
1105 |     if (keyInput) keyInput.value = config.key || '';
1106 |     if (modelInput) modelInput.value = config.model || '';
1107 |   } catch (err) {
1108 |     console.error('Ошибка загрузки настроек Foundry:', err);
1109 |   }
1110 | }
1111 |
1112 | async function loadOllamaConfig() {
1113 |   try {
1114 |     const config = await window.api.fetch('/api/ollama/config'
1115 |   );
1116 |     const enabledInput = document.getElementById('ollama-enabl
1117 |   ed');
1118 |     const urlInput = document.getElementById('ollama-url');
1119 |     const modelInput = document.getElementById('ollama-model')
1120 |   ;
1121 |
1122 |     if (enabledInput) enabledInput.checked = config.enabled ||
1123 |   false;
1124 |     if (urlInput) urlInput.value = config.url || '';
1125 |     if (modelInput) modelInput.value = config.model || '';
1126 |   } catch (err) {
1127 |     console.error('Ошибка загрузки настроек Ollama:', err);
```

```
1120 | }
1121 | }
1122 |
1123 | async function loadAgyConfig() {
1124 |   try {
1125 |     const modelSelect = document.getElementById('agy-model');
1126 |     if (modelSelect) {
1127 |       try {
1128 |         const modelsData = await window.api.fetch('/api/chat/m
models');
1129 |         const agyList = modelsData.models?.agy || [];
1130 |         if (agyList.length > 0) {
1131 |           const curVal = modelSelect.value;
1132 |           modelSelect.innerHTML = '';
1133 |           agyList.forEach(m => {
1134 |             const opt = document.createElement('option');
1135 |             opt.value = m;
1136 |             opt.textContent = m;
1137 |             modelSelect.appendChild(opt);
1138 |           });
1139 |           if (curVal && agyList.includes(curVal)) {
1140 |             modelSelect.value = curVal;
1141 |           }
1142 |         }
1143 |       } catch (e) {
1144 |         console.error('Ошибка загрузки моделей AGY:', e);
1145 |       }
1146 |     }
1147 |
1148 |     const config = await window.api.fetch('/api/agy/config');
1149 |     const enabledInput = document.getElementById('agy-enabled'
);
1150 |     const keyInput = document.getElementById('agy-key');
1151 |
1152 |     if (enabledInput) enabledInput.checked = config.enabled ??
true;
1153 |     if (modelSelect && config.model) modelSelect.value = confi
g.model;
```

```
1154     if (keyInput) keyInput.value = config.key || '';
1155   } catch (err) {
1156     console.error('Ошибка загрузки настроек Antigravity (AGY):', err);
1157   }
1158 }
1159
1160 async function loadOnnxConfig() {
1161   try {
1162     const config = await window.api.fetch('/api/onnx/config');
1163     const enabledInput = document.getElementById('onnx-enabled');
1164     const epSelect = document.getElementById('onnx-execution-provider');
1165     const dirInput = document.getElementById('onnx-models-dir');
1166     const modelInput = document.getElementById('onnx-default-model');
1167     const precSelect = document.getElementById('onnx-olive-precision');
1168
1169     if (enabledInput) enabledInput.checked = config.enabled ?? true;
1170     if (epSelect && config.execution_provider) epSelect.value = config.execution_provider;
1171     if (dirInput && config.models_dir) dirInput.value = config.models_dir;
1172     if (modelInput && config.default_model) modelInput.value = config.default_model;
1173     if (precSelect && config.olive_precision) precSelect.value = config.olive_precision;
1174   } catch (err) {
1175     console.error('Ошибка загрузки настроек ONNX / Olive:', err);
1176   }
1177 }
1178
1179 async function loadSystemInstruction() {
```



```
1180 |   const editor = document.getElementById('system-instruction-e
      |   ditor');
1181 |   const statusBadge = document.getElementById('system-instruct
      |   ion-status');
1182 |   if (!editor) return;
1183 |
1184 |   editor.disabled = true;
1185 |   if (statusBadge) {
1186 |     statusBadge.className = 'badge bg-warning text-dark';
1187 |     statusBadge.textContent = 'Загрузка...';
1188 |     statusBadge.style.removeProperty('display');
1189 |   }
1190 |
1191 |   try {
1192 |     const data = await window.api.fetch('/api/admin/system_ins
      |   truction');
1193 |     editor.value = data.content || '';
1194 |     if (statusBadge) {
1195 |       statusBadge.className = 'badge bg-success';
1196 |       statusBadge.textContent = 'Загружено';
1197 |       setTimeout(() => {
1198 |         if (statusBadge) statusBadge.style.display = 'none';
1199 |       }, 2500);
1200 |     }
1201 |   } catch (err) {
1202 |     console.error('Ошибка загрузки системной инструкции:', err
      |   );
1203 |     if (statusBadge) {
1204 |       statusBadge.className = 'badge bg-danger';
1205 |       statusBadge.textContent = 'Ошибка';
1206 |       statusBadge.style.removeProperty('display');
1207 |     }
1208 |     showModelsNotification('Ошибка загрузки системной инструкц
      |   ии: ' + err.message, 'danger');
1209 |   } finally {
1210 |     editor.disabled = false;
1211 |   }
1212 | }
```

```
1213 |
1214 | async function saveSystemInstruction() {
1215 |     const editor = document.getElementById('system-instruction-e
1216 |     ditor');
1217 |     const saveBtn = document.getElementById('btn-save-instructio
1218 |     n');
1219 |     const statusBadge = document.getElementById('system-instruct
1220 |     ion-status');
1221 |     if (!editor || !saveBtn) return;
1222 |     const originalHtml = saveBtn.innerHTML;
1223 |     saveBtn.disabled = true;
1224 |     saveBtn.innerHTML = '<span class="spinner-border spinner-bor
1225 |     der-sm" role="status"></span> Сохранение...';
1226 |
1227 |     try {
1228 |         await window.api.fetch('/api/admin/system_instruction', {
1229 |             method: 'POST',
1230 |             headers: { 'Content-Type': 'application/json' },
1231 |             body: JSON.stringify({ content: editor.value })
1232 |         });
1233 |         showModelsNotification('☑ Системная инструкция успешно сох
1234 |         ранена', 'success');
1235 |         if (statusBadge) {
1236 |             statusBadge.className = 'badge bg-success';
1237 |             statusBadge.textContent = 'Сохранено';
1238 |             statusBadge.style.removeProperty('display');
1239 |             setTimeout(() => {
1240 |                 if (statusBadge) statusBadge.style.display = 'none';
1241 |             }, 3000);
1242 |         }
1243 |     } catch (err) {
1244 |         console.error('Ошибка сохранения системной инструкции:', e
1245 |         rr);
1246 |         if (statusBadge) {
1247 |             statusBadge.className = 'badge bg-danger';
1248 |             statusBadge.textContent = 'Ошибка сохранения';
1249 |             statusBadge.style.removeProperty('display');
```

```

1245     }
1246     showModelsNotification('Ошибка сохранения: ' + err.message
    , 'danger');
1247 } finally {
1248     saveBtn.disabled = false;
1249     saveBtn.innerHTML = originalHtml;
1250 }
1251 }
1252
1253 // =====
    =====
1254 // Google Workspace Multi-Account Pool Management
1255 // =====
    =====
1256
1257 async function refreshGoogleAccountsList(container) {
1258     if (!container) return;
1259     try {
1260         container.innerHTML = '<tr><td colspan="5" class="text-cen
            ter py-4 text-muted"><span class="spinner-border spinner-borde
            r-sm me-2"></span>Загрузка аккаунтов...</td></tr>';
1261         const data = await window.api.googleAccounts.list();
1262         const accounts = data.accounts || [];
1263
1264         if (accounts.length === 0) {
1265             container.innerHTML = '<tr><td colspan="5" class="text-c
            enter py-4 text-muted">Список аккаунтов пуст. Загрузите файл с
            redentials.json или service_account.json.</td></tr>';
1266             return;
1267         }
1268
1269         container.innerHTML = '';
1270         accounts.forEach(acc => {
1271             const row = document.createElement('tr');
1272
1273             // 1. Account Name + Default Badge + Email
1274             const tdName = document.createElement('td');
1275             tdName.innerHTML = `

```

```

1276 |         <div class="d-flex align-items-center gap-2">
1277 |             <strong>${escapeHtml(acc.name)}</strong>
1278 |             ${acc.is_default ? '<span class="badge bg-warning te
| xt-dark"><i class="bi bi-star-fill"></i> Default</span>' : ''}
1279 |         </div>
1280 |         ${acc.email ? `<div class="small text-muted font-monos
| pace">${escapeHtml(acc.email)}</div>` : ''}
1281 |         `;
1282 |         row.appendChild(tdName);
1283 |
1284 |         // 2. Type
1285 |         const tdType = document.createElement('td');
1286 |         const isOAuth = acc.type === 'oauth2';
1287 |         tdType.innerHTML = isOAuth
1288 |             ? '<span class="badge bg-primary-subtle text-primary b
| order border-primary-subtle"><i class="bi bi-person me-1"></i>
| OAuth 2.0</span>'
1289 |             : '<span class="badge bg-info-subtle text-info border
| border-info-subtle"><i class="bi bi-cpu me-1"></i>Service Acco
| unt</span>';
1290 |         row.appendChild(tdType);
1291 |
1292 |         // 3. Status / Quota
1293 |         const tdStatus = document.createElement('td');
1294 |         const isActive = acc.status === 'active';
1295 |         if (isActive) {
1296 |             tdStatus.innerHTML = '<span class="badge bg-success">A
| ктивен</span>';
1297 |         } else if (acc.status === 'exhausted') {
1298 |             tdStatus.innerHTML = '<span class="badge bg-danger" ti
| tle="Исчерпан суточный лимит квоты">Лимит</span>';
1299 |         } else {
1300 |             tdStatus.innerHTML = `<span class="badge bg-secondary"
| >${escapeHtml(acc.status)}</span>`;
1301 |         }
1302 |         row.appendChild(tdStatus);
1303 |
1304 |         // 4. Token / Credentials Status

```

```
1305 |     const tdToken = document.createElement('td');
1306 |     if (acc.has_token || acc.type === 'service_account') {
1307 |         tdToken.innerHTML = '<span class="badge bg-success-sub
| tle text-success"><i class="bi bi-shield-check me-1"></i>Готов
| </span>';
1308 |     } else {
1309 |         tdToken.innerHTML = '<span class="badge bg-warning-sub
| tle text-warning"><i class="bi bi-key me-1"></i>Нужен токен</s
| pan>';
1310 |     }
1311 |     row.appendChild(tdToken);
1312 |
1313 |     // 5. Actions
1314 |     const tdActions = document.createElement('td');
1315 |     tdActions.className = 'text-end';
1316 |
1317 |     // Set Default button
1318 |     if (!acc.is_default) {
1319 |         const btnDefault = document.createElement('button');
1320 |         btnDefault.className = 'btn btn-xs btn-outline-warning
| btn-sm me-1';
1321 |         btnDefault.title = 'Сделать аккаунтом по умолчанию';
1322 |         btnDefault.innerHTML = '<i class="bi bi-star"></i>';
1323 |         btnDefault.onclick = () => setGoogleAccountDefault(acc
| .name, container);
1324 |         tdActions.appendChild(btnDefault);
1325 |     }
1326 |
1327 |     // Reset status button
1328 |     if (acc.status === 'exhausted') {
1329 |         const btnReset = document.createElement('button');
1330 |         btnReset.className = 'btn btn-xs btn-outline-info btn-
| sm me-1';
1331 |         btnReset.title = 'Сбросить статус исчерпания';
1332 |         btnReset.innerHTML = '<i class="bi bi-arrow-repeat"></
| i>';
1333 |         btnReset.onclick = () => resetGoogleAccountStatus(acc.
| name, container);
```

```

1334         tdActions.appendChild(btnReset);
1335     }
1336
1337     // Test button
1338     const btnTest = document.createElement('button');
1339     btnTest.className = 'btn btn-xs btn-outline-info btn-sm
me-1';
1340     btnTest.title = 'Проверить доступ и авторизацию';
1341     btnTest.innerHTML = '<i class="bi bi-play-circle"></i> Т
ест';
1342     btnTest.onclick = () => testGoogleAccount(acc.name);
1343     tdActions.appendChild(btnTest);
1344
1345     // Delete button
1346     const btnDelete = document.createElement('button');
1347     btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
1348     btnDelete.title = 'Удалить аккаунт из пула';
1349     btnDelete.innerHTML = '<i class="bi bi-trash"></i>';
1350     btnDelete.onclick = () => deleteGoogleAccount(acc.name,
container);
1351     tdActions.appendChild(btnDelete);
1352
1353     row.appendChild(tdActions);
1354     container.appendChild(row);
1355     });
1356
1357 } catch (err) {
1358     console.error('Ошибка загрузки аккаунтов Google Workspace:
', err);
1359     container.innerHTML = `<tr><td colspan="5" class="text-cen
ter py-4 text-danger">Ошибка: ${escapeHtml(err.message)}</td><
/tr>`;
1360 }
1361 }
1362
1363 async function handleAddGoogleAccount(container) {
1364     const nameInput = document.getElementById('new-gacc-name');

```

```
1365 | const typeSelect = document.getElementById('new-gacc-type');
1366 | const emailInput = document.getElementById('new-gacc-email')
    | ;
1367 | const fileInput = document.getElementById('new-gacc-file');
1368 | const jsonTextarea = document.getElementById('new-gacc-json'
    | );
1369 | const defaultCheckbox = document.getElementById('new-gacc-de
    | fault');
1370 | const addBtn = document.getElementById('btn-add-gacc');
1371 |
1372 | if (!nameInput || !addBtn) return;
1373 |
1374 | const accountName = nameInput.value.trim();
1375 | const accountType = typeSelect ? typeSelect.value : 'oauth2'
    | ;
1376 | const email = emailInput ? emailInput.value.trim() : '';
1377 | const setAsDefault = defaultCheckbox ? defaultCheckbox.check
    | ed : false;
1378 |
1379 | if (!accountName) {
1380 |     showModelsNotification('Введите имя аккаунта (например: wo
    | rk, personal)', 'warning');
1381 |     nameInput.focus();
1382 |     return;
1383 | }
1384 |
1385 | // Check if file is selected or JSON is pasted
1386 | const file = fileInput && fileInput.files && fileInput.files
    | [0];
1387 | const jsonContent = jsonTextarea ? jsonTextarea.value.trim()
    | : '';
1388 |
1389 | if (!file && !jsonContent) {
1390 |     showModelsNotification('Загрузите файл credentials.json /
    | service_account.json или вставьте JSON', 'warning');
1391 |     return;
1392 | }
1393 |
```

```
1394 | addBtn.disabled = true;
1395 | const originalText = addBtn.innerHTML;
1396 | addBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Сохранение...';
1397 |
1398 | try {
1399 |     if (file) {
1400 |         const formData = new FormData();
1401 |         formData.append('account_name', accountName);
1402 |         formData.append('account_type', accountType);
1403 |         if (email) formData.append('email', email);
1404 |         formData.append('set_as_default', setAsDefault ? 'true'
1405 | : 'false');
1406 |         formData.append('file', file);
1407 |         await window.api.googleAccounts.upload(formData);
1408 |     } else {
1409 |         let parsedJson = null;
1410 |         try {
1411 |             parsedJson = JSON.parse(jsonContent);
1412 |         } catch (e) {
1413 |             throw new Error('Некорректный JSON в поле учетных данн
1414 | ых');
1415 |         }
1416 |         await window.api.googleAccounts.create({
1417 |             account_name: accountName,
1418 |             account_type: accountType,
1419 |             email: email || undefined,
1420 |             credentials_dict: parsedJson,
1421 |             set_as_default: setAsDefault
1422 |         });
1423 |     }
1424 |
1425 |     showModelsNotification(`Аккаунт Google "${accountName}" ус
1426 | пешно сохранен в пул`, 'success');
1427 |     nameInput.value = '';
1427 |     if (emailInput) emailInput.value = '';
```



```
1428     if (fileInput) fileInput.value = '';
1429     if (jsonTextarea) jsonTextarea.value = '';
1430     if (defaultCheckbox) defaultCheckbox.checked = false;
1431
1432     if (container) await refreshGoogleAccountsList(container);
1433   } catch (err) {
1434     console.error('Ошибка сохранения аккаунта Google:', err);
1435     showModelsNotification('Ошибка сохранения: ' + err.message
1436       , 'danger');
1437   } finally {
1438     addBtn.disabled = false;
1439     addBtn.innerHTML = originalText;
1440   }
1441
1442   async function setGoogleAccountDefault(name, container) {
1443     try {
1444       await window.api.googleAccounts.setDefault(name);
1445       showModelsNotification(`Аккаунт "${name}" назначен по умол
1446       чанию`, 'success');
1447       if (container) await refreshGoogleAccountsList(container);
1448     } catch (err) {
1449       console.error('Ошибка установки аккаунта по умолчанию:', e
1450       rr);
1451       showModelsNotification('Ошибка: ' + err.message, 'danger')
1452       ;
1453     }
1454   }
1455
1456   async function resetGoogleAccountStatus(name, container) {
1457     try {
1458       await window.api.googleAccounts.resetStatus(name);
1459       showModelsNotification(`Статус аккаунта "${name}" сброшен
1460       в активный`, 'success');
1461       if (container) await refreshGoogleAccountsList(container);
1462     } catch (err) {
1463       console.error('Ошибка сброса статуса аккаунта:', err);
1464       showModelsNotification('Ошибка сброса: ' + err.message, 'd
```

```

    anger');
1461 }
1462 }
1463
1464 async function testGoogleAccount(name) {
1465     const testCard = document.getElementById('gaccount-test-card
    ');
1466     const testBody = document.getElementById('gaccount-test-body
    ');
1467
1468     if (testCard && testBody) {
1469         testCard.style.display = 'block';
1470         testBody.innerHTML = `<span class="spinner-border spinner-
border-sm me-2 text-info"></span>Проверка аутентификации для а
ккаунта <strong>${escapeHtml(name)}</strong>...`;
1471     }
1472
1473     try {
1474         const res = await window.api.googleAccounts.test(name);
1475         if (testBody) {
1476             const isSuccess = res.status === 'success';
1477             const isWarning = res.status === 'warning';
1478             const badgeClass = isSuccess ? 'bg-success' : isWarning
? 'bg-warning text-dark' : 'bg-danger';
1479
1480             testBody.innerHTML = `
1481                 <div class="mb-2 d-flex align-items-center gap-2">
1482                     <span class="badge ${badgeClass}">${res.status.toUpp
erCase()}</span>
1483                     <strong>${escapeHtml(name)}</strong>
1484                 </div>
1485                 <div class="mb-1">${escapeHtml(res.message || '')}</di
v>
1486                 ${res.scopes && res.scopes.length > 0 ? `<div class="t
ext-muted mt-2"><strong>Доступные Scopes:</strong><br>${res.sc
opes.map(s => '• ' + escapeHtml(s)).join('<br>')}</div>` : ''}
1487             `;
1488         }

```

```
1489 | } catch (err) {
1490 |     if (testBody) {
1491 |         testBody.innerHTML = `<i class
    | ="bi bi-x-circle me-1"></i>Ошибка проверки: ${escapeHtml(err.m
    | essage)}</span>`;
1492 |     }
1493 | }
1494 | }
1495 |
1496 | async function deleteGoogleAccount(name, container) {
1497 |     if (!confirm(`Вы уверены, что хотите удалить аккаунт "${name
    | }" из пула Google Workspace?`)) return;
1498 |     try {
1499 |         await window.api.googleAccounts.delete(name);
1500 |         showModelsNotification(`Аккаунт "${name}" успешно удален`,
    | 'success');
1501 |         if (container) await refreshGoogleAccountsList(container);
1502 |     } catch (err) {
1503 |         console.error('Ошибка удаления аккаунта Google:', err);
1504 |         showModelsNotification('Ошибка удаления: ' + err.message,
    | 'danger');
1505 |     }
1506 | }
1507 |
1508 | // Export for tab loader
1509 | window.initModelsTab = initModelsTab;
1510 | window.refreshGoogleAccountsList = refreshGoogleAccountsList;
1511 |
```

index.html

```
1 <!-- Windows Network Terminal Tab (/apps/windows/network) -->
2 <style>
3   .net-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .net-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.55rem 0.85rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    min-height: 64px;
15    gap: 0.2rem;
16  }
17  .net-label {
18    font-size: 0.7rem;
19    text-transform: uppercase;
20    letter-spacing: 0.05em;
21    color: #94a3b8;
22    line-height: 1.1;
23  }
24  .net-value {
25    font-size: 1.25rem;
26    font-weight: 700;
27    color: #f8fafc;
28    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
29    Consolas, monospace;
30    line-height: 1.15;
31  }
32  .net-panel {
33    background: #0f172a !important;
34    border: 1px solid #334155 !important;
35    border-radius: 8px;
36    overflow: hidden;
```

```
36 | }
37 | .net-panel-header {
38 |     background: #1e293b !important;
39 |     border-bottom: 1px solid #334155 !important;
40 |     padding: 0.5rem 0.85rem;
41 |     font-weight: 600;
42 |     font-size: 0.82rem;
43 |     color: #38bdf8;
44 |     display: flex;
45 |     align-items: center;
46 |     justify-content: space-between;
47 | }
48 | .net-table {
49 |     width: 100%;
50 |     font-size: 0.78rem;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
52 |     border-collapse: collapse;
53 |     margin-bottom: 0;
54 | }
55 | .net-table th {
56 |     background: #1e293b;
57 |     padding: 6px 10px;
58 |     color: #94a3b8;
59 |     font-weight: 600;
60 |     border-bottom: 1px solid #334155;
61 |     position: sticky;
62 |     top: 0;
63 |     z-index: 5;
64 | }
65 | .net-table td {
66 |     padding: 5px 10px;
67 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
68 |     color: #cbd5e1;
69 | }
70 | .proto-badge {
71 |     display: inline-block;
72 |     padding: 1px 6px;
```

```
73     border-radius: 4px;
74     font-size: 0.7rem;
75     font-weight: 600;
76 }
77 .proto-tcp { background: rgba(59, 130, 246, 0.2); color: #60
a5fa; border: 1px solid #3b82f6; }
78 .proto-udp { background: rgba(168, 85, 247, 0.2); color: #c0
84fc; border: 1px solid #a855f7; }
79 .proto-tls { background: rgba(16, 185, 129, 0.2); color: #34
d399; border: 1px solid #10b981; }
80 .proto-http { background: rgba(245, 158, 11, 0.2); color: #f
bbf24; border: 1px solid #f59e0b; }
81 .proto-dns { background: rgba(6, 182, 212, 0.2); color: #22d
3ee; border: 1px solid #06b6d4; }
82 .proto-other { background: rgba(148, 163, 184, 0.2); color:
#cbd5e1; border: 1px solid #64748b; }
83
84 .status-badge-up { background: rgba(34, 197, 94, 0.15); colo
r: #4ade80; border: 1px solid #22c55e; }
85 .status-badge-down { background: rgba(239, 68, 68, 0.15); co
lor: #f87171; border: 1px solid #ef4444; }
86 .status-badge-listen { background: rgba(56, 189, 248, 0.15);
color: #38bdf8; border: 1px solid #0284c7; }
87 .status-badge-established { background: rgba(168, 85, 247, 0
.15); color: #c084fc; border: 1px solid #9333ea; }
88
89 .net-subnav .nav-link {
90     font-size: 0.8rem;
91     padding: 0.35rem 0.8rem;
92     color: #94a3b8;
93     border-radius: 6px;
94     border: 1px solid transparent;
95 }
96 .net-subnav .nav-link.active {
97     background: #1e293b;
98     color: #38bdf8;
99     border: 1px solid #334155;
100     font-weight: 600;
```

```

101 | }
102 |
103 | .speed-gauge-card {
104 |     background: linear-gradient(145deg, #0f172a 0%, #1e293b 10
105 | 0%);
106 |     border: 1px solid #334155;
107 |     border-radius: 10px;
108 |     padding: 1rem;
109 |     text-align: center;
110 | }
111 | .speed-gauge-val {
112 |     font-size: 2.2rem;
113 |     font-weight: 800;
114 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
115 | Consolas, monospace;
116 |     line-height: 1.1;
117 | }
118 | </style>
119 |
120 | <div class="net-container">
121 |     <!-- Header Alert & Status Strip -->
122 |     <div class="d-flex justify-content-between align-items-cente
123 | r flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style
124 | ="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%
125 | ); border: 1px solid #334155;">
126 |         <div class="d-flex align-items-center gap-2 flex-wrap">
127 |             <span class="fs-5">□ </span>
128 |             <span class="fw-bold text-white small" style="font-size:
129 | 0.95rem;">Сетевой терминал Windows</span>
130 |             <span class="badge bg-dark border border-info text-info
131 | fw-normal" style="font-size: 0.72rem;">apps/windows/network</s
132 | pan>
133 |             <span class="text-secondary small d-none d-lg-inline" st
134 | yle="font-size: 0.76rem;">| Скорость интернета, адаптеры, порт
135 | ы, соединения и DPI</span>
136 |         </div>
137 |         <div class="d-flex align-items-center gap-2 flex-wrap">
138 |             <span class="badge rounded-pill bg-success-subtle text-s

```

```

    success border border-success px-2 py-1" style="font-size: 0.72
    rem;" id="net-engine-status">
129      <i class="bi bi-check-circle-fill me-1"></i> Windows N
    et: Active
130    </span>
131    <span class="badge rounded-pill bg-secondary-subtle text
    -light border border-secondary px-2 py-1" style="font-size: 0.
    72rem;" id="net-tshark-status">
132      TShark DPI: Проверка...
133    </span>
134    <button class="btn btn-xs btn-outline-info rounded-pill
    px-2.5 py-1" style="font-size: 0.75rem;" id="btn-refresh-netwo
    rk-all">
135      <i class="bi bi-arrow-clockwise me-1"></i> Обновить
136    </button>
137    <button class="btn btn-xs btn-outline-warning rounded-pi
    ll px-2 py-1" style="font-size: 0.75rem;" id="btn-network-conf
    ig">
138      <i class="bi bi-gear-fill me-1"></i> Настройки
139    </button>
140  </div>
141 </div>
142
143 <!-- Top Metrics KPI Cards -->
144 <div class="row g-2 mb-3">
145   <div class="col-6 col-md-3">
146     <div class="net-card">
147       <span class="net-label">Сетевые адаптеры</span>
148       <div class="d-flex align-items-baseline justify-conten
    t-between mt-1">
149         <span class="net-value" id="net-kpi-adapters">--</sp
    an>
150         <span class="small text-success" style="font-size: 0
    .68rem;" id="net-kpi-adapters-active">0 активных</span>
151       </div>
152     </div>
153   </div>
154   <div class="col-6 col-md-3">

```



```
155 |         <div class="net-card">
156 |             <span class="net-label">Активные соединения</span>
157 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
158 |                 <span class="net-value" id="net-kpi-conns">--</span>
159 |                 <span class="small text-info" style="font-size: 0.68rem;" id="net-kpi-listening">0 слушающих</span>
160 |             </div>
161 |         </div>
162 |     </div>
163 |     <div class="col-6 col-md-3">
164 |         <div class="net-card">
165 |             <span class="net-label">Скорость Интернета</span>
166 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
167 |                 <span class="net-value text-info" id="net-kpi-speed">-- Mbps</span>
168 |                 <span class="small text-muted" style="font-size: 0.68rem;" id="net-kpi-ping">Ping: -- ms</span>
169 |             </div>
170 |         </div>
171 |     </div>
172 |     <div class="col-6 col-md-3">
173 |         <div class="net-card">
174 |             <span class="net-label">DPI & Безопасность</span>
175 |             <div class="d-flex align-items-baseline justify-content-between mt-1">
176 |                 <span class="net-value" id="net-kpi-packets" style="color: #38bdf8;">0</span>
177 |                 <span class="small" style="font-size: 0.68rem; color: #4ade80;" id="net-kpi-anomalies">0 аномалий</span>
178 |             </div>
179 |         </div>
180 |     </div>
181 | </div>
182 |
183 | <!-- Sub-Navigation Bar -->
184 | <ul class="nav net-subnav mb-3 gap-1">
```

```
185     <li class="nav-item">
186         <button class="nav-link active" data-bs-toggle="tab" data-
187         bs-target="#tab-net-speedtest">
188             <i class="bi bi-speedometer2 me-1 text-warning"></i> Т
189             ест скорости Интернета
190         </button>
191     </li>
192     <li class="nav-item">
193         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
194         rget="#tab-net-adapters">
195             <i class="bi bi-hdd-network me-1"></i> Сетевые адаптер
196             ы
197         </button>
198     </li>
199     <li class="nav-item">
200         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
201         rget="#tab-net-connections">
202             <i class="bi bi-diagram-3 me-1"></i> Соединения и порт
203             ы (<span id="badge-conn-count">0</span>)
204         </button>
205     </li>
206     <li class="nav-item">
207         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
208         rget="#tab-net-capture">
209             <i class="bi bi-broadcast-pin me-1"></i> Захват пакетов
210             в (Live DPI)
211         </button>
212     </li>
213     <li class="nav-item">
214         <button class="nav-link" data-bs-toggle="tab" data-bs-ta
215         rget="#tab-net-pcap">
216             <i class="bi bi-file-earmark-binary me-1"></i> Анализ
217             PCAP & AI
218         </button>
219     </li>
220 </ul>
221 <!-- Tab Content -->
```

```

213 | <div class="tab-content">
214 |
215 |     <!-- SECTION 0: Speedtest & Latency Benchmark -->
216 |     <div class="tab-pane fade show active" id="tab-net-speedte
    st">
217 |         <!-- Speedtest Action Control Box -->
218 |         <div class="net-panel mb-3">
219 |             <div class="net-panel-header">
220 |                 <span><i class="bi bi-speedometer2 text-warning me-1
    "></i> Тестирование пропускной способности и задержки Интернет
    a</span>
221 |                 <div class="d-flex align-items-center gap-2">
222 |                     <span class="small text-muted" id="speedtest-stat
    s-text">Готов к тестированию</span>
223 |                     <button class="btn btn-sm btn-warning rounded-pill
    px-3 fw-bold shadow-sm" id="btn-run-speedtest">
224 |                         <i class="bi bi-play-fill me-1"></i> Запустить т
    ест скорости
225 |                     </button>
226 |                 </div>
227 |             </div>
228 |             <div class="p-3">
229 |                 <!-- 4 Big Gauges -->
230 |                 <div class="row g-3 mb-3">
231 |                     <div class="col-6 col-lg-3">
232 |                         <div class="speed-gauge-card">
233 |                             <div class="text-secondary small text-upperc
    ase fw-semibold mb-1"><i class="bi bi-arrow-down-circle-fill tex
    t-success me-1"></i> Скачивание (Download)</div>
234 |                             <div class="speed-gauge-val text-success" id="
    speed-val-download">--</div>
235 |                             <div class="text-muted small mt-1" id="speed-s
    ub-download">0.0 MB/s</div>
236 |                         </div>
237 |                     </div>
238 |                     <div class="col-6 col-lg-3">
239 |                         <div class="speed-gauge-card">
240 |                             <div class="text-secondary small text-upperc

```

```

    e fw-semibold mb-1"><i class="bi bi-arrow-up-circle-fill text-
primary me-1"></i> Отдача (Upload)</div>
241         <div class="speed-gauge-val text-primary" id="
speed-val-upload">--</div>
242         <div class="text-muted small mt-1" id="speed-s
ub-upload">0.0 MB/s</div>
243         </div>
244     </div>
245     <div class="col-6 col-lg-3">
246         <div class="speed-gauge-card">
247             <div class="text-secondary small text-uppercas
e fw-semibold mb-1"><i class="bi bi-stopwatch-fill text-info m
e-1"></i> Задержка (Ping)</div>
248             <div class="speed-gauge-val text-info" id="spe
ed-val-ping">--</div>
249             <div class="text-muted small mt-1" id="speed-s
ub-jitter">Jitter: -- ms</div>
250             </div>
251         </div>
252     <div class="col-6 col-lg-3">
253         <div class="speed-gauge-card">
254             <div class="text-secondary small text-uppercas
e fw-semibold mb-1"><i class="bi bi-shield-check text-warning
me-1"></i> Качество соединения</div>
255             <div class="speed-gauge-val text-warning" id="
speed-val-grade">--</div>
256             <div class="text-muted small text-truncate mt-
1" id="speed-sub-isp">Провайдер: --</div>
257             </div>
258         </div>
259     </div>
260
261     <!-- Quality Summary Bar -->
262     <div class="p-2.5 rounded bg-dark border border-seco
ndary mb-3 d-flex justify-content-between align-items-center f
lex-wrap gap-2">
263         <div class="d-flex align-items-center gap-2">
264             <span class="badge bg-secondary" id="speed-badge

```

```

- ip">Внешний IP: --</span>
265         <span class="badge bg-dark border border-seconda
ry" id="speed-badgе-location">Локация: --</span>
266     </div>
267     <div class="small text-muted" id="speed-summary-te
xt">Нажмите «Запустить тест скорости», чтобы измерить пропускн
ую способность канала, задержку (Ping) и джиттер.</div>
268 </div>
269
270     <!-- DNS & Edge Servers Latency Table -->
271     <h6 class="small fw-bold text-info mb-2"><i class="b
i bi-globe me-1"></i> Задержка к опорным DNS & CDN серверам:</
h6>
272     <div class="table-responsive">
273         <table class="net-table">
274             <thead>
275                 <tr>
276                     <th>Сервер / Провайдер</th>
277                     <th style="width: 140px;">Хост / IP</th>
278                     <th style="width: 110px; text-align: right;"
>Средний Ping</th>
279                     <th style="width: 90px; text-align: right;">
Min</th>
280                     <th style="width: 90px; text-align: right;">
Max</th>
281                     <th style="width: 100px; text-align: right;"
>Джиттер</th>
282                     <th style="width: 100px;">Статус</th>
283                 </tr>
284             </thead>
285             <tbody id="speed-servers-tbody">
286                 <tr><td colspan="7" class="text-center py-3 te
xt-muted">Запустите тест скорости для измерения задержки к DNS
- серверам.</td></tr>
287             </tbody>
288         </table>
289     </div>
290 </div>

```

```

291     </div>
292 </div>
293
294 <!-- SECTION 1: Network Adapters -->
295 <div class="tab-pane fade" id="tab-net-adapters">
296     <div class="net-panel">
297         <div class="net-panel-header">
298             <span><i class="bi bi-hdd-network me-1"></i> Физичес
кие и виртуальные сетевые интерфейсы Windows</span>
299             <button class="btn btn-xs btn-outline-info rounded-p
ill px-2 py-0" id="btn-refresh-adapters">Обновить адаптеры</bu
tton>
300         </div>
301         <div class="table-responsive" style="max-height: 480px
; overflow-y: auto;">
302             <table class="net-table">
303                 <thead>
304                     <tr>
305                         <th style="width: 140px;">Адаптер</th>
306                         <th>Описание</th>
307                         <th style="width: 130px;">IPv4</th>
308                         <th style="width: 140px;">MAC адрес</th>
309                         <th style="width: 90px;">Скорость</th>
310                         <th style="width: 90px;">Статус</th>
311                         <th style="width: 130px; text-align: right;">0
тправлено</th>
312                         <th style="width: 130px; text-align: right;">П
ринято</th>
313                     </tr>
314                 </thead>
315                 <tbody id="net-adapters-tbody">
316                     <tr><td colspan="8" class="text-center py-4 text
-muted"><div class="spinner-border spinner-border-sm text-info
me-2"></div>Загрузка сетевых адаптеров...</td></tr>
317                 </tbody>
318             </table>
319         </div>
320     </div>

```

```

321 |     </div>
322 |
323 |     <!-- SECTION 2: Active Connections & Listening Ports -->
324 |     <div class="tab-pane fade" id="tab-net-connections">
325 |         <div class="net-panel">
326 |             <div class="net-panel-header">
327 |                 <div class="d-flex align-items-center gap-2 flex-wra
328 | p">
329 |                     <span><i class="bi bi-diagram-3 me-1"></i> Таблица
330 | сетевых подключений и слушающих портов</span>
331 |                     <div class="btn-group btn-group-sm ms-2" role="gro
332 | up">
333 |                         <button type="button" class="btn btn-outline-sec
334 | ondary active btn-filter-conns" data-filter="all">Все</button>
335 |                         <button type="button" class="btn btn-outline-sec
336 | ondary btn-filter-conns" data-filter="LISTEN">LISTEN</button>
337 |                         <button type="button" class="btn btn-outline-sec
338 | ondary btn-filter-conns" data-filter="ESTABLISHED">ESTABLISHED
339 | </button>
340 |                         <button type="button" class="btn btn-outline-sec
341 | ondary btn-filter-conns" data-filter="UDP">UDP</button>
342 |                     </div>
343 |                 </div>
344 |                 <div class="d-flex align-items-center gap-2">
345 |                     <input type="text" class="form-control form-contro
346 | l-sm bg-dark text-white border-secondary" id="net-conns-filter
347 | " placeholder="Поиск (IP, порт, процесс, PID)..." style="max-w
348 | idth: 250px;">
349 |                     <button class="btn btn-xs btn-outline-info rounded
350 | -pill px-2 py-0" id="btn-refresh-conns">Обновить</button>
351 |                 </div>
352 |             </div>
353 |             <div class="table-responsive" style="max-height: 480px
354 | ; overflow-y: auto;">
355 |                 <table class="net-table">
356 |                     <thead>
357 |                         <tr>
358 |                             <th style="width: 70px;">Протокол</th>

```

```

346         <th style="width: 180px;">Локальный адрес</th>
347         <th style="width: 180px;">Удалённый адрес</th>
348         <th style="width: 110px;">Статус</th>
349         <th style="width: 80px;">PID</th>
350         <th>Процесс / Исполняемый файл</th>
351     </tr>
352 </thead>
353 <tbody id="net-conns-tbody">
354     <tr><td colspan="6" class="text-center py-4 text
- muted"><div class="spinner-border spinner-border-sm text-info
me-2"></div>Загрузка соединений...</td></tr>
355 </tbody>
356 </table>
357 </div>
358 </div>
359 </div>
360
361 <!-- SECTION 3: Packet Capture & Live DPI -->
362 <div class="tab-pane fade" id="tab-net-capture">
363     <!-- Capture Control Box -->
364     <div class="net-panel mb-3">
365         <div class="net-panel-header">
366             <span><i class="bi bi-broadcast me-1"></i> Настройка
сессии захвата трафика (Live Packet DPI)</span>
367         </div>
368         <div class="p-3">
369             <div class="row g-2 mb-2 align-items-end">
370                 <div class="col-md-5">
371                     <label class="form-label small fw-semibold text-
light mb-1">Сетевой интерфейс для захвата:</label>
372                     <select class="form-select form-select-sm bg-dar
k text-white border-secondary" id="net-interface-select">
373                         <option value="">Загрузка интерфейсов...</opti
on>
374                     </select>
375                 </div>
376                 <div class="col-md-4">
377                     <label class="form-label small fw-semibold text-

```



```

light mb-1">Wireshark/BPF фильтр:</label>
378         <input type="text" class="form-control form-cont
rol-sm bg-dark text-white border-secondary" id="net-capture-fi
lter" placeholder="tcp, udp, port 80, tls">
379     </div>
380     <div class="col-md-3 d-flex gap-2">
381         <button class="btn btn-sm btn-primary rounded-pi
ll px-3 flex-fill" id="btn-start-live-capture">
382             <i class="bi bi-play-fill me-1"></i> Старт зах
вата
383         </button>
384         <button class="btn btn-sm btn-outline-danger rou
nded-pill px-3 flex-fill" id="btn-stop-live-capture" disabled>
385             <i class="bi bi-stop-fill me-1"></i> Стоп
386         </button>
387     </div>
388 </div>
389 </div>
390 </div>
391
392 <!-- Packets Table -->
393 <div class="net-panel">
394     <div class="net-panel-header">
395         <div class="d-flex align-items-center gap-2">
396             <span><i class="bi bi-reception-4 me-1"></i> Поток
сетевых пакетов (Packet Inspection)</span>
397             <span class="badge bg-secondary" id="net-captured-
badge">0 пакетов</span>
398         </div>
399         <div class="d-flex align-items-center gap-2">
400             <input type="text" class="form-control form-contro
l-sm bg-dark text-white border-secondary" id="net-table-filter
" placeholder="Поиск по IP / протоколу..." style="max-width: 2
20px;">
401             <button class="btn btn-xs btn-outline-secondary ro
unded-pill px-2 py-0" id="btn-clear-packets">Очистить</button>
402         </div>
403     </div>

```

```

404 |         <div class="table-responsive" style="max-height: 400px
      | ; overflow-y: auto;">
405 |             <table class="net-table">
406 |                 <thead>
407 |                     <tr>
408 |                         <th style="width: 50px;">#</th>
409 |                         <th style="width: 80px;">Время</th>
410 |                         <th style="width: 140px;">Источник</th>
411 |                         <th style="width: 140px;">Назначение</th>
412 |                         <th style="width: 80px;">Протокол</th>
413 |                         <th style="width: 70px; text-align: right;">Па
      | змер</th>
414 |                             <th>Сведения о полезной нагрузке / Инфо</th>
415 |                     </tr>
416 |                 </thead>
417 |                 <tbody id="net-packets-tbody">
418 |                     <tr><td colspan="7" class="text-center py-4 text
      | -muted">Нет захваченных пакетов. Выберите интерфейс и нажмите
      | "Старт захвата".</td></tr>
419 |                     </tbody>
420 |                 </table>
421 |             </div>
422 |         </div>
423 |     </div>
424 |
425 |     <!-- SECTION 4: PCAP Analysis & AI Diagnostics -->
426 |     <div class="tab-pane fade" id="tab-net-pcap">
427 |         <div class="row g-3">
428 |             <div class="col-lg-5">
429 |                 <div class="net-panel h-100">
430 |                     <div class="net-panel-header">
431 |                         <span><i class="bi bi-file-earmark-binary me-1">
      | </i> Загрузка файла сетевого дампа (.PCAP)</span>
432 |                     </div>
433 |                     <div class="p-3">
434 |                         <div class="mb-3">
435 |                             <label class="form-label small fw-semibold tex
      | t-light">Выберите .pcap, .pcapng или .cap файл:</label>

```

```

436 |         <input type="file" class="form-control form-co
      | ntrol-sm bg-dark text-white border-secondary" id="net-pcap-fil
      | e" accept=".pcap,.pcapng,.cap">
437 |     </div>
438 |     <div class="d-flex justify-content-between align
      | -items-center mb-3">
439 |         <span class="small text-muted" id="net-upload-
      | status">Файл не выбран</span>
440 |         <button class="btn btn-sm btn-info rounded-pil
      | l px-3 fw-bold" id="btn-analyze-pcap">
441 |             <i class="bi bi-cpu me-1"></i> Запустить AI
      | Анализ
442 |         </button>
443 |     </div>
444 |     <div class="p-2.5 rounded bg-dark border border-
      | secondary small text-muted" style="font-size: 0.75rem;">
445 |         <i class="bi bi-info-circle text-info me-1"></
      | i> PCAP-анализатор проводит глубокую инспекцию заголовков, эвр
      | истический аудит портов и выявляет аномалии с помощью AI-модел
      | и.
446 |     </div>
447 | </div>
448 | </div>
449 | </div>
450 |
451 | <div class="col-lg-7">
452 |     <div class="net-panel h-100">
453 |         <div class="net-panel-header">
454 |             <span><i class="bi bi-shield-check me-1"></i> Ре
      | зультаты AI-анализа и безопасности</span>
455 |         </div>
456 |         <div class="p-3">
457 |             <div class="row g-2 mb-3">
458 |                 <div class="col-6">
459 |                     <div class="p-2 rounded bg-dark border borde
      | r-secondary">
460 |                         <span class="net-label">AI Health Score</s
      | pan>

```

```
461         <div class="net-value mt-1" id="pcap-health-score">100/100</div>
462     </div>
463 </div>
464     <div class="col-6">
465         <div class="p-2 rounded bg-dark border border-secondary">
466             <span class="net-label">Обнаруженные аномалии</span>
467             <div class="net-value mt-1 text-success" id="pcap-anomalies-count">0</div>
468         </div>
469     </div>
470 </div>
471
472     <h6 class="small fw-bold text-info mb-1.5">Эвристические предупреждения:</h6>
473     <div id="pcap-heuristics-list" class="p-2 rounded bg-dark border border-secondary mb-3 small" style="min-height: 60px; max-height: 120px; overflow-y: auto;">
474         <span class="text-muted">Нет предупреждений. Запустите анализ файла.</span>
475     </div>
476
477     <h6 class="small fw-bold text-info mb-1.5">Рекомендации AI:</h6>
478     <div id="pcap-ai-recommendations" class="p-2 rounded bg-dark border border-secondary small" style="min-height: 60px; max-height: 120px; overflow-y: auto;">
479         <span class="text-muted">Анализ пока не производился.</span>
480     </div>
481 </div>
482 </div>
483 </div>
484 </div>
485 </div>
486
```

```
487 |   </div>
488 | </div>
489 |
```

main.js

```
1 // =====
  =====
2 // Process Name: Windows Network Terminal Tab Client Module
3 // =====
  =====
4 // Description:
5 //   Frontend controller for /apps/windows/network.
6 //   Manages internet speedtest, ping/latency benchmark, network adapters,
7 //   active connections, listening ports, live packet capture (DPI), PCAP upload,
8 //   and AI diagnostics.
9 // =====
  =====
10
11 (function() {
12   let isNetInitialized = false;
13   let netEventSource = null;
14   let capturedPackets = [];
15   let allConnections = [];
16   let allAdapters = [];
17   let activeConnFilter = 'all';
18   let isSpeedtestRunning = false;
19
20   // Format bytes to human readable format
21   function formatBytes(bytes, decimals = 1) {
22     if (!bytes || bytes === 0) return '0 B';
23     const k = 1024;
24     const dm = decimals < 0 ? 0 : decimals;
25     const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
26     const i = Math.floor(Math.log(bytes) / Math.log(k));
27     return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
28     ' ' + sizes[i];
29   }
30   // Check overall network status and TShark availability
```

```
31 | async function fetchNetworkStatus() {
32 |     const engineBadge = document.getElementById('net-engine-st
   | atus');
33 |     const tsharkBadge = document.getElementById('net-tshark-st
   | atus');
34 |     const kpiAdapters = document.getElementById('net-kpi-adapt
   | ers');
35 |     const kpiAdaptersActive = document.getElementById('net-kpi
   | -adapters-active');
36 |     const kpiConns = document.getElementById('net-kpi-conns');
37 |     const kpiListening = document.getElementById('net-kpi-list
   | ening');
38 |     const kpiPackets = document.getElementById('net-kpi-packet
   | s');
39 |     const kpiAnomalies = document.getElementById('net-kpi-anom
   | alies');
40 |
41 |     try {
42 |         const res = await fetch('/api/network/status');
43 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
44 |         const data = await res.json();
45 |
46 |         if (engineBadge) {
47 |             engineBadge.className = 'badge rounded-pill bg-success
   | -subtle text-success border border-success px-2 py-1';
48 |             engineBadge.innerHTML = '<i class="bi bi-check-circle-
   | fill me-1"></i> Windows Net: Active';
49 |         }
50 |
51 |         if (tsharkBadge) {
52 |             if (data.tshark_available) {
53 |                 tsharkBadge.className = 'badge rounded-pill bg-succe
   | ss-subtle text-success border border-success px-2 py-1';
54 |                 tsharkBadge.innerText = 'TShark DPI: Доступен';
55 |             } else {
56 |                 tsharkBadge.className = 'badge rounded-pill bg-warni
   | ng-subtle text-warning border border-warning px-2 py-1';
57 |                 tsharkBadge.innerText = 'TShark DPI: Не установлен';
```

```

58         }
59     }
60
61     if (kpiAdapters) kpiAdapters.innerText = data.total_adap
ters ?? '--';
62     if (kpiConns) kpiConns.innerText = data.active_connectio
ns_count ?? '--';
63     if (kpiListening) kpiListening.innerText = `${data.liste
ning_ports_count || 0} слушающих`;
64     if (kpiPackets) kpiPackets.innerText = data.total_packet
s ?? capturedPackets.length;
65
66     const anomaliesCount = (data.latest_heuristics?.length |
| 0) + (data.latest_ai_report?.anomalies?.length || 0);
67     if (kpiAnomalies) {
68         kpiAnomalies.innerText = `${anomaliesCount} аномалий`;
69         kpiAnomalies.style.color = anomaliesCount > 0 ? '#f871
71' : '#4ade80';
70     }
71 } catch (e) {
72     console.warn('[NetworkTab] Status check error:', e);
73     if (engineBadge) {
74         engineBadge.className = 'badge rounded-pill bg-danger-
subtle text-danger border border-danger px-2 py-1';
75         engineBadge.innerText = 'Windows Net: Ошибка';
76     }
77 }
78 }
79
80 // Load latest or cached speedtest
81 async function fetchLatestSpeedtest() {
82     try {
83         const res = await fetch('/api/network/speedtest/latest')
;
84         if (!res.ok) return;
85         const data = await res.json();
86         if (data && data.download) {
87             renderSpeedtestResults(data);

```



```
88     }
89   } catch (e) {
90     console.debug('[NetworkTab] No latest speedtest data:',
91     e);
92   }
93
94   // Run full internet speedtest
95   async function runInternetSpeedtest() {
96     if (isSpeedtestRunning) return;
97     isSpeedtestRunning = true;
98
99     const btn = document.getElementById('btn-run-speedtest');
100    const statusText = document.getElementById('speedtest-stat
101    us-text');
102    const summaryText = document.getElementById('speed-summary
103    -text');
104
105    if (btn) {
106      btn.disabled = true;
107      btn.innerHTML = '<span class="spinner-border spinner-bor
108      der-sm me-1"></span> Тестирование...';
109    }
110    if (statusText) statusText.innerText = 'Измерение Ping, Do
111    wnload и Upload...';
112    if (summaryText) summaryText.innerText = 'Выполняется подк
113    лючение к узлам Cloudflare & DNS серверам для замера скорости.
114    ...';
115
116    try {
117      const res = await fetch('/api/network/speedtest/run', {
118      method: 'POST' });
119      if (!res.ok) throw new Error(`HTTP ${res.status}`);
120      const data = await res.json();
121
122      renderSpeedtestResults(data);
123      if (statusText) statusText.innerText = `Завершено в ${ne
124      w Date().toLocaleTimeString()}`;
```

```
117     } catch (e) {
118         console.error('[NetworkTab] Speedtest error:', e);
119         window.showToast?.('Ошибка при выполнении теста скорости
: ' + e.message, 'danger') || alert('Ошибка при выполнении тес
та скорости: ' + e.message);
120         if (statusText) statusText.innerText = 'Ошибка теста ско
рости';
121     } finally {
122         isSpeedtestRunning = false;
123         if (btn) {
124             btn.disabled = false;
125             btn.innerHTML = '<i class="bi bi-play-fill me-1"></i>
Запустить тест скорости';
126         }
127     }
128 }
129
130 // Render speedtest report to UI
131 function renderSpeedtestResults(data) {
132     const valDown = document.getElementById('speed-val-downloa
d');
133     const subDown = document.getElementById('speed-sub-downloa
d');
134     const valUp = document.getElementById('speed-val-upload');
135     const subUp = document.getElementById('speed-sub-upload');
136     const valPing = document.getElementById('speed-val-ping');
137     const subJitter = document.getElementById('speed-sub-jitte
r');
138     const valGrade = document.getElementById('speed-val-grade'
);
139     const subIsp = document.getElementById('speed-sub-isp');
140     const badgeIp = document.getElementById('speed-badge-ip');
141     const badgeLoc = document.getElementById('speed-badge-loca
tion');
142     const summaryText = document.getElementById('speed-summary
-text');
143     const kpiSpeed = document.getElementById('net-kpi-speed');
144     const kpiPing = document.getElementById('net-kpi-ping');
```

```
145     const serversTbody = document.getElementById('speed-server
| s-tbody');
146
147     const downMbps = data.download?.speed_mbps || 0;
148     const downMBs = data.download?.speed_mb_s || 0;
149     const upMbps = data.upload?.speed_mbps || 0;
150     const upMBs = data.upload?.speed_mb_s || 0;
151     const pingMs = data.ping_ms || 0;
152     const jitterMs = data.jitter_ms || 0;
153     const meta = data.meta || {};
154     const quality = data.quality || {};
155
156     if (valDown) valDown.innerText = `${downMbps} Mbps`;
157     if (subDown) subDown.innerText = `${downMBs} MB/s (${data.
| download?.duration_s || 0}s)`;
158     if (valUp) valUp.innerText = `${upMbps} Mbps`;
159     if (subUp) subUp.innerText = `${upMBs} MB/s (${data.upload
| ?.duration_s || 0}s)`;
160     if (valPing) valPing.innerText = `${pingMs} ms`;
161     if (subJitter) subJitter.innerText = `Jitter: ${jitterMs}
| ms`;
162
163     if (valGrade) {
164         valGrade.innerText = quality.rating || '--';
165         valGrade.style.color = quality.color || '#fbbf24';
166     }
167     if (subIsp) subIsp.innerText = `Провайдер: ${meta.isp || '
| Неизвестно'}`;
168
169     if (badgeIp) badgeIp.innerText = `Внешний IP: ${meta.ip ||
| '--'}`;
170     if (badgeLoc) badgeLoc.innerText = `Локация: ${meta.city |
| ''} ${meta.country || ''} (${meta.colo || 'Edge'})`.trim();
171     if (summaryText) summaryText.innerText = `${quality.grade
| || 'Готово'}. ${quality.summary || ''}`;
172
173     // Top KPI cards
174     if (kpiSpeed) kpiSpeed.innerText = `${downMbps} Mbps`;
```

```

175     if (kpiPing) kpiPing.innerText = `Ping: ${pingMs} ms`;
176
177     // DNS & Edge servers table
178     if (serversTbody && Array.isArray(data.servers)) {
179         serversTbody.innerHTML = data.servers.map(s => {
180             const isOnline = s.avg_ms > 0;
181             const statusBadge = isOnline
182                 ? '<span class="proto-badg status-badg-up">Online<
|/span>'
183                 : '<span class="proto-badg status-badg-down">Offli
|ne</span>';
184
185             return `
186                 <tr>
187                     <td class="fw-bold text-white"><i class="bi bi-hdd
|-network me-1 text-info"></i>${s.name}</td>
188                     <td class="text-secondary font-monospace">${s.host
|}</td>
189                     <td style="text-align: right;" class="text-info fw
|-bold font-monospace">${isOnline ? `${s.avg_ms} ms` : '--'}</t
|d>
190                     <td style="text-align: right;" class="text-muted f
|ont-monospace">${isOnline ? `${s.min_ms} ms` : '--'}</td>
191                     <td style="text-align: right;" class="text-muted f
|ont-monospace">${isOnline ? `${s.max_ms} ms` : '--'}</td>
192                     <td style="text-align: right;" class="text-warning
|font-monospace">${isOnline ? `${s.jitter_ms} ms` : '--'}</td>
193                     <td>${statusBadge}</td>
194                 </tr>
195             `;
196         }).join('');
197     }
198 }
199
200 // Fetch host network adapters
201 async function fetchAdapters() {
202     const tbody = document.getElementById('net-adapters-tbody'
|);

```

```

203     const ifaceSelect = document.getElementById('net-interface
    -select');
204     const kpiActive = document.getElementById('net-kpi-adapter
    s-active');
205
206     try {
207         const res = await fetch('/api/network/interfaces');
208         if (!res.ok) throw new Error(`HTTP ${res.status}`);
209         const adapters = await res.json();
210         allAdapters = Array.isArray(adapters) ? adapters : [];
211
212         let activeCount = 0;
213
214         if (ifaceSelect) {
215             ifaceSelect.innerHTML = allAdapters.map(a => `
216                 <option value="${a.name || a.id}">${a.name} (${a.des
    cription || 'Ethernet'}) - ${a.status}</option>
217             `).join('');
218         }
219
220         if (tbody) {
221             if (allAdapters.length === 0) {
222                 tbody.innerHTML = '<tr><td colspan="8" class="text-c
    enter py-4 text-muted">Сетевые адаптеры не обнаружены.</td></t
    r>';
223                 return;
224             }
225
226             tbody.innerHTML = allAdapters.map(a => {
227                 if (a.is_up) activeCount++;
228                 const statusClass = a.is_up ? 'status-badge-up' : 's
    tatus-badge-down';
229                 const speedDisplay = a.speed_mbps ? `${a.speed_mbps}
    Mbps` : '--';
230                 const ipDisplay = a.ipv4 && a.ipv4.length > 0 ? a.ip
    v4.join(', ') : '--';
231
232                 return `

```

```

233 |         <tr>
234 |             <td class="fw-bold text-white"><i class="bi bi-h
dd-network me-1 text-info"></i>${a.name}</td>
235 |             <td class="text-secondary small text-truncate" s
tyle="max-width: 220px;" title="${a.description}">${a.descript
ion || '--'}</td>
236 |             <td class="text-info">${ipDisplay}</td>
237 |             <td class="text-muted small">${a.mac || '--'}</t
d>
238 |             <td><span class="badge bg-dark border border-sec
ondary text-light">${speedDisplay}</span></td>
239 |             <td><span class="proto-badge ${statusClass}">${a
.status || 'Unknown'}</span></td>
240 |             <td style="text-align: right;" class="text-succe
ss">${formatBytes(a.bytes_sent)}</td>
241 |             <td style="text-align: right;" class="text-prima
ry">${formatBytes(a.bytes_recv)}</td>
242 |         </tr>
243 |     `;
244 |     }).join('');
245 | }
246 |
247 |     if (kpiActive) kpiActive.innerText = `${activeCount} акт
ивных`;
248 | } catch (e) {
249 |     console.error('[NetworkTab] Failed to fetch adapters:',
e);
250 |     if (tbody) tbody.innerHTML = `<tr><td colspan="8" class=
"text-center py-3 text-danger">Ошибка загрузки адаптеров: ${e.
message}</td></tr>`;
251 | }
252 | }
253 |
254 | // Fetch active connections and listening ports
255 | async function fetchConnections() {
256 |     const tbody = document.getElementById('net-conns-tbody');
257 |     const badgeConn = document.getElementById('badge-conn-coun
t');

```

```
258     const kpiConns = document.getElementById('net-kpi-conns');
259     const kpiListening = document.getElementById('net-kpi-list
    ening');
260
261     try {
262         const res = await fetch('/api/network/connections?limit=
    250');
263         if (!res.ok) throw new Error(`HTTP ${res.status}`);
264         const data = await res.json();
265
266         const conns = Array.isArray(data.connections) ? data.con
    nections : [];
267         const listening = Array.isArray(data.listening_ports) ?
    data.listening_ports : [];
268         allConnections = [...listening, ...conns];
269
270         if (badgeConn) badgeConn.innerText = allConnections.leng
    th;
271         if (kpiConns) kpiConns.innerText = conns.length;
272         if (kpiListening) kpiListening.innerText = `${listening.
    length} слушающих`;
273
274         renderConnectionsTable();
275     } catch (e) {
276         console.error('[NetworkTab] Failed to fetch connections:
    ', e);
277         if (tbody) tbody.innerHTML = `<tr><td colspan="6" class=
    "text-center py-3 text-danger">Ошибка загрузки соединений: ${e
    .message}</td></tr>`;
278     }
279 }
280
281 // Render connections table with current filter and search
282 function renderConnectionsTable() {
283     const tbody = document.getElementById('net-conns-tbody');
284     const searchInput = document.getElementById('net-conns-fil
    ter');
285     const query = (searchInput?.value || '').toLowerCase().tri
```

```

    m();
286
287     if (!tbody) return;
288
289     let filtered = allConnections.filter(c => {
290         // Type filter
291         if (activeConnFilter === 'LISTEN' && c.status !== 'LISTE
N') return false;
292         if (activeConnFilter === 'ESTABLISHED' && c.status !== '
ESTABLISHED') return false;
293         if (activeConnFilter === 'UDP' && c.protocol !== 'UDP')
return false;
294
295         // Text query
296         if (query) {
297             const text = `${c.protocol} ${c.local_address} ${c.rem
ote_address} ${c.status} ${c.pid} ${c.process_name}`.toLowerCase();
298             return text.includes(query);
299         }
300         return true;
301     });
302
303     if (filtered.length === 0) {
304         tbody.innerHTML = '<tr><td colspan="6" class="text-cente
r py-4 text-muted">Соединения, удовлетворяющие критериям, не н
айдены.</td></tr>';
305         return;
306     }
307
308     tbody.innerHTML = filtered.slice(0, 150).map((c, idx) => {
309         const protoClass = c.protocol === 'TCP' ? 'proto-tcp' :
(c.protocol === 'UDP' ? 'proto-udp' : 'proto-other');
310         let statusClass = 'proto-other';
311         if (c.status === 'LISTEN') statusClass = 'status-badge-l
isten';
312         else if (c.status === 'ESTABLISHED') statusClass = 'stat
us-badge-established';

```



```

313 |
314 |     return `
315 |         <tr class="net-conn-row" data-idx="${idx}" style="curs
    or: pointer;" title="Нажмите для детальной инспекции процесса
    и сокета">
316 |             <td><span class="proto-badge ${protoClass}">${c.prot
    ocol || 'TCP'}</span></td>
317 |             <td class="text-info font-monospace">${c.local_addre
    ss || '-'}</td>
318 |             <td class="text-secondary font-monospace">${c.remote
    _address || '-'}</td>
319 |             <td><span class="proto-badge ${statusClass}">${c.sta
    tus || '-'}</span></td>
320 |             <td class="text-muted font-monospace">${c.pid || '- '
    }</td>
321 |             <td class="fw-bold text-white"><i class="bi bi-app m
    e-1 text-muted"></i>${c.process_name || 'Неизвестно'}</td>
322 |         </tr>
323 |     `;
324 |     }).join('');
325 |
326 |     // Modal inspection on row click
327 |     tbody.querySelectorAll('.net-conn-row').forEach(row => {
328 |         row.onclick = () => {
329 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
330 |             ;
331 |             const item = filtered[idx];
332 |             if (!item || !window.AITableModal) return;
333 |             window.AITableModal.show({
334 |                 icon: '🔌',
335 |                 title: `Сетевое подключение [${item.protocol || 'TCP
    '}]`,
336 |                 subtitle: `${item.process_name} (PID: ${item.pid ||
    'N/A'})`,
337 |                 tableType: 'network',
338 |                 badges: [
339 |                     { text: item.protocol || 'TCP', class: 'badge bg-p

```

```

    rimary' },
340         { text: item.status || 'UNKNOWN', class: 'badge bg
-info' },
341         { text: `PID ${item.pid || '-'}`, class: 'badge bg
-secondary' }
342     ],
343     metadata: [
344         { label: 'Процесс', value: item.process_name || '-'
' },
345         { label: 'PID', value: String(item.pid || '-') },
346         { label: 'Протокол', value: item.protocol || '-' }
347     ],
348     { label: 'Локальный адрес:Порт', value: item.local
_address || '-' },
349     { label: 'Удалённый адрес:Порт', value: item.remot
e_address || '-' },
350     { label: 'Статус сокета', value: item.status || '-'
' }
351 ],
352 rawTitle: 'Детали сокета',
353 rawContent: JSON.stringify(item, null, 2),
354 requestData: item
355 });
356 });
357 }
358
359 // Render captured packets table
360 function renderPacketsTable() {
361     const tbody = document.getElementById('net-packets-tbody')
;
362     const badge = document.getElementById('net-captured-badge'
);
363     const filterInput = document.getElementById('net-table-fil
ter');
364     const filter = (filterInput?.value || '').toLowerCase().tr
im();
365

```

```
366     if (!tbody) return;
367     if (badge) badge.innerText = `${capturedPackets.length} па
кетов`;
368
369     if (capturedPackets.length === 0) {
370         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
тер py-4 text-muted">Нет захваченных пакетов. Выберите интерфейс
и нажмите "Старт захвата".</td></tr>';
371         return;
372     }
373
374     const filtered = capturedPackets.filter(p => {
375         if (!filter) return true;
376         const str = `${p.number || p.packet_number || ''} ${p.so
urce || ''} ${p.destination || ''} ${p.protocol || ''} ${p.inf
o || ''}`.toLowerCase();
377         return str.includes(filter);
378     });
379
380     const pagePackets = filtered.slice(-100);
381     tbody.innerHTML = pagePackets.map((p, idx) => {
382         const proto = (p.protocol || 'OTHER').toUpperCase();
383         let protoClass = 'proto-other';
384         if (proto.includes('TCP')) protoClass = 'proto-tcp';
385         else if (proto.includes('UDP')) protoClass = 'proto-udp'
;
386         else if (proto.includes('TLS') || proto.includes('SSL'))
protoClass = 'proto-tls';
387         else if (proto.includes('HTTP')) protoClass = 'proto-htt
p';
388         else if (proto.includes('DNS')) protoClass = 'proto-dns'
;
389
390         return `
391         <tr class="net-packet-row" data-idx="${idx}" style="cu
rsor: pointer;" title="Нажмите для детального AI-анализа пакет
а">
392             <td class="text-muted">${p.number || p.packet_number
```

```

    || '--'}</td>
393     <td>${p.timestamp ? (typeof p.timestamp === 'number'
    ? new Date(p.timestamp * 1000).toLocaleTimeString() : p.times
    tamp) : '--'}</td>
394     <td style="color: #38bdf8;">${p.source || '--'}</td>
395     <td style="color: #a855f7;">${p.destination || '--'}
    </td>
396     <td><span class="proto-badge ${protoClass}">${proto}
    </span></td>
397     <td style="text-align: right;">${p.length || 0} B</t
    d>
398     <td class="text-truncate" style="max-width: 320px;">
    ${p.info || '--'}</td>
399     </tr>
400     `;
401     }).join('');
402
403     tbody.querySelectorAll('.net-packet-row').forEach(row => {
404         row.onclick = () => {
405             const idx = parseInt(row.getAttribute('data-idx'), 10)
406             ;
407             const p = pagePackets[idx];
408             if (!p || !window.AITableModal) return;
409             window.AITableModal.show({
410                 icon: '📶',
411                 title: `Сетевой пакет #${p.number || p.packet_number
    || 'N/A'} [${p.protocol || 'TCP'}]`,
412                 subtitle: `${p.source || '0.0.0.0'} 📡 ${p.destinatio
    n || '0.0.0.0'}`,
413                 tableType: 'network',
414                 badges: [
415                     { text: (p.protocol || 'TCP').toUpperCase(), class
    : 'badge bg-primary' },
416                     { text: `${p.length || 0} Bytes`, class: 'badge bg
    -secondary' }
417                 ],
418                 metadata: [

```

```

419         { label: 'Номер пакета', value: String(p.number ||
| p.packet_number || '-') },
420         { label: 'Протокол', value: (p.protocol || 'TCP').
| toUpperCase() },
421         { label: 'Источник', value: p.source || '-' },
422         { label: 'Назначение', value: p.destination || '-'
| },
423         { label: 'Размер', value: `${p.length || 0} байт`
| },
424         { label: 'Инфо', value: p.info || 'Нет данных', fu
| llWidth: true }
425     ],
426     rawTitle: 'Сырые данные пакета',
427     rawContent: JSON.stringify(p, null, 2),
428     requestData: p
429     });
430     };
431     });
432     }
433
434     // Start live packet capture
435     async function startLiveCapture() {
436         const ifaceSelect = document.getElementById('net-interface
| -select');
437         const filterInput = document.getElementById('net-capture-f
| ilter');
438         const startBtn = document.getElementById('btn-start-live-c
| apture');
439         const stopBtn = document.getElementById('btn-stop-live-cap
| ture');
440
441         const iface = ifaceSelect?.value || '1';
442         const filter = filterInput?.value || '';
443
444         try {
445             const res = await fetch(`/api/network/start-capture?inte
| rface=${encodeURIComponent(iface)}&display_filter=${encodeURIComponent(
| filter)}`, {

```

```
446         method: 'POST'
447     });
448     if (!res.ok) throw new Error(`HTTP ${res.status}`);
449
450     if (startBtn) startBtn.disabled = true;
451     if (stopBtn) stopBtn.disabled = false;
452
453     // Start SSE event source
454     if (netEventSource) netEventSource.close();
455     netEventSource = new EventSource('/api/network/packets/s
tream');
456     netEventSource.onmessage = async () => {
457         try {
458             const packRes = await fetch('/api/network/packets?li
mit=50');
459             if (packRes.ok) {
460                 const data = await packRes.json();
461                 if (Array.isArray(data.packets)) {
462                     capturedPackets = data.packets;
463                     renderPacketsTable();
464                 }
465             }
466         } catch (e) {
467             console.warn('[NetworkTab] Packet poll error:', e);
468         }
469     };
470 } catch (e) {
471     console.error('[NetworkTab] Live capture error:', e);
472     window.showToast?.('Ошибка запуска захвата пакетов: ' +
e.message, 'danger') || alert('Ошибка запуска захвата пакетов:
' + e.message);
473 }
474 }
475
476 // Stop live packet capture
477 async function stopLiveCapture() {
478     const startBtn = document.getElementById('btn-start-live-c
apture');
```

```
479     const stopBtn = document.getElementById('btn-stop-live-cap
| ture');
480
481     if (netEventSource) {
482         netEventSource.close();
483         netEventSource = null;
484     }
485
486     try {
487         await fetch('/api/network/stop-capture', { method: 'POST
| ' });
488     } catch (e) {
489         console.warn('[NetworkTab] Stop capture error:', e);
490     }
491
492     if (startBtn) startBtn.disabled = false;
493     if (stopBtn) stopBtn.disabled = true;
494 }
495
496 // Analyze uploaded PCAP file
497 async function analyzePcapFile() {
498     const fileInput = document.getElementById('net-pcap-file')
| ;
499     const statusEl = document.getElementById('net-upload-statu
| s');
500     const healthEl = document.getElementById('pcap-health-scor
| e');
501     const anomaliesEl = document.getElementById('pcap-anomalie
| s-count');
502     const heuristicsEl = document.getElementById('pcap-heurist
| ics-list');
503     const recommendationsEl = document.getElementById('pcap-ai
| -recommendations');
504
505     const file = fileInput?.files?.[0];
506     if (!file) {
507         window.showToast?(['Пожалуйста, выберите .рсар или .рсар
| ng файл', 'warning']) || alert('Пожалуйста, выберите .рсар или
```

```

    .pcapng файл');
508     return;
509 }
510
511 const formData = new FormData();
512 formData.append('file', file);
513 formData.append('max_packets', '500');
514
515 if (statusEl) statusEl.innerText = 'Анализ пакетов...';
516
517 try {
518     const res = await fetch('/api/network/analyze/pcap', {
519         method: 'POST',
520         body: formData
521     });
522
523     if (!res.ok) throw new Error(`HTTP ${res.status}`);
524     const report = await res.json();
525
526     if (statusEl) statusEl.innerText = `Анализ завершён: ${r
eport.stats?.total_packets || 0} пакетов`;
527
528     if (healthEl) healthEl.innerText = `${report.ai_report?.
health_score || 100}/100`;
529
530     const anomaliesCount = (report.heuristics?.length || 0)
+ (report.ai_report?.anomalies?.length || 0);
531     if (anomaliesEl) {
532         anomaliesEl.innerText = String(anomaliesCount);
533         anomaliesEl.className = `net-value mt-1 ${anomaliesCou
nt > 0 ? 'text-danger' : 'text-success'}`;
534     }
535
536     if (heuristicsEl) {
537         if (Array.isArray(report.heuristics) && report.heurist
ics.length > 0) {
538             heuristicsEl.innerHTML = report.heuristics.map(h =>
`<div class="text-warning mb-1"><i class="bi bi-exclamation-tr

```



```

    iangle-fill me-1"></i>${h}</div>`).join('');
539         } else {
540             heuristicsEl.innerHTML = '<span class="text-success"
><i class="bi bi-check-circle-fill me-1"></i>Эвристических угр
оз не выявлено.</span>';
541         }
542     }
543
544     if (recommendationsEl) {
545         if (Array.isArray(report.ai_report?.recommended_action
s) && report.ai_report.recommended_actions.length > 0) {
546             recommendationsEl.innerHTML = report.ai_report.recom
mended_actions.map(r => `<div class="text-info mb-1"><i class=
"bi bi-arrow-right-short me-1"></i>${r}</div>`).join('');
547         } else {
548             recommendationsEl.innerHTML = '<span class="text-mut
ed">Рекомендаций нет, система функционирует штатно.</span>';
549         }
550     }
551
552     if (Array.isArray(report.sample_packets) && report.sampl
e_packets.length > 0) {
553         capturedPackets = report.sample_packets;
554         renderPacketsTable();
555     }
556 } catch (e) {
557     console.error('[NetworkTab] PCAP upload error:', e);
558     if (statusEl) statusEl.innerText = 'Ошибка анализа: ' +
e.message;
559     window.showToast?(`'Ошибка при анализе PCAP: ' + e.messa
ge, 'danger') || alert('Ошибка при анализе PCAP: ' + e.message
);
560 }
561 }
562
563 // Main init function for Network Terminal Tab
564 function initNetworkTab() {
565     console.log('[NetworkTab] Initializing Windows Network Ter

```

```
    minal tab (/apps/windows/network)...');
566     fetchNetworkStatus();
567     fetchLatestSpeedtest();
568     fetchAdapters();
569     fetchConnections();
570
571     if (!isNetInitialized) {
572         // Action buttons
573         const refreshAllBtn = document.getElementById('btn-refre
sh-network-all');
574         const runSpeedtestBtn = document.getElementById('btn-run
-speedtest');
575         const refreshAdaptersBtn = document.getElementById('btn-
refresh-adapters');
576         const refreshConnsBtn = document.getElementById('btn-ref
resh-conns');
577         const startBtn = document.getElementById('btn-start-live
-capture');
578         const stopBtn = document.getElementById('btn-stop-live-c
apture');
579         const analyzePcapBtn = document.getElementById('btn-anal
yze-pcap');
580         const clearPacketsBtn = document.getElementById('btn-cle
ar-packets');
581         const connsFilterInput = document.getElementById('net-co
nns-filter');
582         const tableFilterInput = document.getElementById('net-ta
ble-filter');
583         const configBtn = document.getElementById('btn-network-c
onfig');
584
585         if (refreshAllBtn) refreshAllBtn.onclick = () => {
586             fetchNetworkStatus();
587             fetchLatestSpeedtest();
588             fetchAdapters();
589             fetchConnections();
590         };
591         if (runSpeedtestBtn) runSpeedtestBtn.onclick = runIntern
```

```

    etSpeedtest;
592     if (refreshAdaptersBtn) refreshAdaptersBtn.onclick = fet
    chAdapters;
593     if (refreshConnsBtn) refreshConnsBtn.onclick = fetchConn
    ections;
594     if (startBtn) startBtn.onclick = startLiveCapture;
595     if (stopBtn) stopBtn.onclick = stopLiveCapture;
596     if (analyzePcapBtn) analyzePcapBtn.onclick = analyzePcap
    File;
597     if (clearPacketsBtn) clearPacketsBtn.onclick = () => {
598         capturedPackets = [];
599         renderPacketsTable();
600     };
601
602     if (connsFilterInput) connsFilterInput.oninput = renderC
    onnectionsTable;
603     if (tableFilterInput) tableFilterInput.oninput = renderP
    acketsTable;
604
605     // Filter buttons for connections
606     document.querySelectorAll('.btn-filter-conns').forEach(b
    tn => {
607         btn.onclick = () => {
608             document.querySelectorAll('.btn-filter-conns').forEa
    ch(b => b.classList.remove('active'));
609             btn.classList.add('active');
610             activeConnFilter = btn.getAttribute('data-filter') |
    | 'all';
611             renderConnectionsTable();
612         };
613     });
614
615     if (configBtn) configBtn.onclick = () => {
616         if (typeof window.openAppConfigModal === 'function') {
617             window.openAppConfigModal('network_terminal', 'Netwo
    rk Analyzer Terminal');
618         }
619     };

```

```
620 |
621 |         isNetInitialized = true;
622 |     }
623 | }
624 |
625 | window.initNetworkTab = initNetworkTab;
626 | ))();
627 |
```

index.html

```
1 <!-- News Tab Component -->
2 <div class="container-fluid p-3" id="news-tab-root">
3   <!-- Top Banner / Digest Generator Bar -->
4   <div class="card border mb-3 shadow-sm">
5     <div class="card-body p-3 d-flex flex-wrap align-items-center justify-content-between gap-3">
6       <div class="d-flex align-items-center gap-3">
7         <div class="p-3 rounded border text-primary display-6 d-flex align-items-center justify-content-center" style="width: 54px; height: 54px; background: var(--bg-color);">
8           <i class="bi bi-newspaper"></i>
9         </div>
10        <div>
11          <h5 class="mb-1 d-flex align-items-center gap-2">
12            □ Интеллектуальная лента новостей
13            <span class="badge bg-primary rounded-pill small" id="news-total-badge">0</span>
14          </h5>
15          <p class="mb-0 text-muted small">
16            Только релевантные новости с непрерывным обучением
17            под ваши личные интересы и AI-выжимками.
18          </p>
19        </div>
20      <div class="d-flex align-items-center gap-2">
21        <button type="button" class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1" id="btn-refresh-news" onclick="window.newsTab?.refreshNews(true)">
22          <i class="bi bi-arrow-repeat"></i> Обновить
23        </button>
24        <button type="button" class="btn btn-outline-info btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-preferences-news" data-bs-toggle="modal" data-bs-target="#newsPreferencesModal">
25          <i class="bi bi-sliders"></i> Мои интересы
26        </button>
```

```
27 |         <button type="button" class="btn btn-primary btn-sm d-
flex align-items-center gap-1 shadow-sm" id="btn-generate-dige
st" onclick="window.newsTab?.generateDigest()">
28 |             <i class="bi bi-stars"></i> ИИ-дайджест
29 |         </button>
30 |     </div>
31 | </div>
32 | </div>
33 |
34 | <!-- AI Digest Section (Appears when generated) -->
35 | <div class="card border-primary mb-3 shadow-sm d-none" id="n
ews-digest-card">
36 |     <div class="card-header d-flex justify-content-between ali
gn-items-center border-primary py-2">
37 |         <div class="d-flex align-items-center gap-2">
38 |             <i class="bi bi-stars text-warning fs-5"></i>
39 |             <strong id="news-digest-title">Персональный ИИ-дайджес
т</strong>
40 |         </div>
41 |         <div class="d-flex align-items-center gap-2">
42 |             <button class="btn btn-sm btn-outline-secondary py-0 p
x-2" id="news-digest-tts-btn" title="Озвучить дайджест" onclinc
k="window.newsTab?.speakDigest()">
43 |                 <i class="bi bi-volume-up me-1"></i>Слушать
44 |             </button>
45 |             <button type="button" class="btn-close" aria-label="Cl
ose" onclick="document.getElementById('news-digest-card').clas
sList.add('d-none')"></button>
46 |         </div>
47 |     </div>
48 |     <div class="card-body p-3">
49 |         <div id="news-digest-content" class="small" style="line-
height: 1.6; white-space: pre-line;"></div>
50 |     </div>
51 | </div>
52 |
53 | <!-- Filter & Learning Feedback Bar -->
54 | <div class="card mb-3">
```

```
55 |     <div class="card-body p-2 d-flex flex-wrap align-items-cen
    | ter justify-content-between gap-2">
56 |         <div class="d-flex align-items-center gap-2 flex-grow-1"
    | style="max-width: 480px;">
57 |             <div class="input-group input-group-sm">
58 |                 <span class="input-group-text"><i class="bi bi-searc
    | h"></i></span>
59 |                 <input type="text" class="form-control" id="news-sea
    | rch-input" placeholder="Фильтр новостей по тексту..." oninput=
    | "window.newsTab?.filterNews()">
60 |             </div>
61 |         </div>
62 |
63 |         <div class="d-flex align-items-center gap-2">
64 |             <label class="small text-muted text-nowrap" for="news-
    | category-filter">Категория:</label>
65 |             <select class="form-select form-select-sm" id="news-ca
    | tegory-filter" style="width: auto;" onchange="window.newsTab?.
    | filterNews()">
66 |                 <option value="all">Все категории</option>
67 |                 <option value="ai">ИИ & ML</option>
68 |                 <option value="dev">Разработка</option>
69 |                 <option value="tech">Технологии</option>
70 |                 <option value="science">Наука</option>
71 |                 <option value="world">Мировые</option>
72 |             </select>
73 |
74 |             <label class="small text-muted text-nowrap ms-2" for="
    | news-min-score">Мин. совпадение:</label>
75 |             <select class="form-select form-select-sm" id="news-mi
    | n-score" style="width: auto;" onchange="window.newsTab?.filter
    | News()">
76 |                 <option value="0">Все (0%)</option>
77 |                 <option value="0.25" selected>Релевантные (>= 25%)</
    | option>
78 |                 <option value="0.5">Высокое (>= 50%)</option>
79 |                 <option value="0.7">Только топ (>= 70%)</option>
80 |             </select>
```

```

81     </div>
82 </div>
83 </div>
84
85 <!-- News Stream Grid -->
86 <div class="row g-3" id="news-articles-container">
87     <div class="col-12 text-center text-muted p-5" id="news-loading-spinner">
88         <div class="spinner-border spinner-border-sm text-primary mb-2" role="status"></div>
89         <div>Сбор и ранжирование релевантных новостей под ваши интересы...</div>
90     </div>
91 </div>
92 </div>
93
94 <!-- Preferences & Topics Modal -->
95 <div class="modal fade" id="newsPreferencesModal" tabindex="-1"
96     " aria-labelledby="newsPreferencesModalLabel" aria-hidden="true">
97     <div class="modal-dialog modal-lg modal-dialog-centered">
98         <div class="modal-content shadow">
99             <div class="modal-header border-bottom">
100                 <h5 class="modal-title d-flex align-items-center gap-2" id="newsPreferencesModalLabel">
101                     <i class="bi bi-sliders text-info"></i> Настройка интересов и профиля обучения
102                 </h5>
103                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
104             </div>
105             <div class="modal-body p-3">
106                 <!-- Learned Weights Section -->
107                 <div class="mb-3">
108                     <label class="form-label small text-muted fw-semibold">
109                         <i class="bi bi-lightning-charge text-warning me-1"></i>Накопленные веса интересов (адаптируются на основе лайко

```


В):

```
109         </label>
110         <div class="p-2 border rounded d-flex flex-wrap gap-
111         2" id="news-learned-tags-container" style="max-height: 140px;
112         overflow-y: auto; background: var(--bg-color);">
113             <!-- Badges rendered via JS -->
114             </div>
115         </div>
116
117         <!-- Preferred Topics Input -->
118         <div class="mb-3">
119             <label class="form-label small fw-semibold" for="new
120             s-pref-topics-input">
121                 Явные ключевые темы (через запятую):
122             </label>
123             <input type="text" class="form-control" id="news-pre
124             f-topics-input" placeholder="например: ai, python, robotics, n
125             euroscience, opensource">
126             <small class="text-muted">Новости с этими словами по
127             лучат приоритетный буст в вашей ленте.</small>
128         </div>
129
130         <!-- Negative Keywords Input -->
131         <div class="mb-3">
132             <label class="form-label small fw-semibold" for="new
133             s-neg-topics-input">
134                 Стоп-слова и нежелательные темы (через запятую):
135             </label>
136             <input type="text" class="form-control" id="news-neg
137             -topics-input" placeholder="например: криминал, шоу-бизнес, ск
138             андалы">
139             <small class="text-muted">Статьи с этими словами буд
140             ут скрываться или получать минимальный скор.</small>
141         </div>
142
143         <!-- Stats & Reset -->
144         <div class="d-flex align-items-center justify-content-
145         between p-2 rounded border" style="background: var(--bg-color)
```

```
135 |         <span class="small text-muted" id="news-interactions  
| -count-label">Обработано реакций: 0</span>  
136 |         <button type="button" class="btn btn-sm btn-outline-  
| danger" onclick="window.newsTab?.resetProfile()">  
137 |             <i class="bi bi-arrow-counterclockwise me-1"></i>C  
| бросить обучение  
138 |         </button>  
139 |     </div>  
140 | </div>  
141 | <div class="modal-footer border-top">  
142 |     <button type="button" class="btn btn-sm btn-secondary"  
| data-bs-dismiss="modal">Закрыть</button>  
143 |     <button type="button" class="btn btn-sm btn-primary" o  
| nclick="window.newsTab?.savePreferences()">Сохранить интересы<  
| /button>  
144 | </div>  
145 | </div>  
146 | </div>  
147 | </div>  
148 |
```

main.js

```
1 // News Tab Controller
2 class NewsTabController {
3   constructor() {
4     this.articles = [];
5     this.profile = null;
6     this.currentDigest = null;
7     this.isLoading = false;
8   }
9
10  async init() {
11    console.log('[NewsTab] Initializing Smart News Feed...');
12    await this.loadProfile();
13    await this.refreshNews(false);
14  }
15
16  async loadProfile() {
17    try {
18      const res = await fetch('/api/news/profile');
19      if (res.ok) {
20        this.profile = await res.json();
21        this.renderProfileModal();
22      }
23    } catch (e) {
24      console.error('[NewsTab] Error loading profile:', e);
25    }
26  }
27
28  renderProfileModal() {
29    if (!this.profile) return;
30    const container = document.getElementById('news-learned-ta
31    gs-container');
32    if (container) {
33      container.innerHTML = '';
34      const weights = this.profile.topic_weights || {};
35      const sorted = Object.entries(weights).sort((a, b) => b[
36      1] - a[1]);
```

```

35         if (sorted.length === 0) {
36             container.innerHTML = '<span class="text-muted small">
Пока нет накопленных весов. Оценивайте новости кнопками  и 
 !</span>';
37         } else {
38             for (const [kw, w] of sorted) {
39                 const badge = document.createElement('span');
40                 badge.className = 'badge bg-secondary border border-
secondary text-light d-flex align-items-center gap-1';
41                 badge.innerHTML = `<span>${kw}</span> <span class="b
adge bg-primary text-white">${w}</span>`;
42                 container.appendChild(badge);
43             }
44         }
45     }
46
47     const countLabel = document.getElementById('news-interacti
ons-count-label');
48     if (countLabel) {
49         countLabel.textContent = `Обработано реакций обучения: $
{this.profile.interaction_count || 0}`;
50     }
51
52     const negInput = document.getElementById('news-neg-topics-
input');
53     if (negInput && this.profile.negative_keywords) {
54         negInput.value = this.profile.negative_keywords.join(',
');
55     }
56 }
57
58 async refreshNews(force = false) {
59     if (this.isLoading) return;
60     this.isLoading = true;
61     const container = document.getElementById('news-articles-c
ontainer');
62     const spinner = document.getElementById('news-loading-spin
ner');

```

```
63     if (spinner) spinner.classList.remove('d-none');
64
65     try {
66         const url = `/api/news/feed?force_refresh=${force}&limit
=35`;
67         const res = await fetch(url);
68         if (res.ok) {
69             this.articles = await res.json();
70             this.renderArticles();
71             const totalBadge = document.getElementById('news-total
-badge');
72             if (totalBadge) totalBadge.textContent = this.articles
.length;
73         }
74     } catch (e) {
75         console.error('[NewsTab] Error fetching feed:', e);
76         if (container) {
77             container.innerHTML = `
```

```
93 |     const filtered = this.articles.filter((a) => {
94 |         if (cat !== 'all' && a.category !== cat) return false;
95 |         if (a.relevance_score < minScore) return false;
96 |         if (query) {
97 |             const text = `${a.title} ${a.summary} ${a.source_name}
    |`.toLowerCase();
98 |             if (!text.includes(query)) return false;
99 |         }
100 |         return true;
101 |     });
102 |
103 |     if (filtered.length === 0) {
104 |         container.innerHTML = `
105 |             <div class="col-12 text-center text-muted p-5">
106 |                 <i class="bi bi-inbox fs-1 mb-2 d-block"></i>
107 |                 <h6>Нет новостей, подходящих под выбранные фильтры</
    |h6>
108 |                 <small>Попробуйте уменьшить порог минимального совпа
    |дения или обновить ленту.</small>
109 |             </div>
110 |         `;
111 |         return;
112 |     }
113 |
114 |     filtered.forEach((art) => {
115 |         const col = document.createElement('div');
116 |         col.className = 'col-md-6 col-lg-4';
117 |
118 |         const matchPercent = Math.round((art.relevance_score ||
    |0.5) * 100);
119 |         let matchBadgeClass = 'bg-secondary';
120 |         if (matchPercent >= 70) matchBadgeClass = 'bg-success';
121 |         else if (matchPercent >= 45) matchBadgeClass = 'bg-prima
    |ry';
122 |         else if (matchPercent >= 25) matchBadgeClass = 'bg-info
    |text-dark';
123 |
124 |         const isLiked = art.user_interaction === 'like';
```

```

125 |     const isDisliked = art.user_interaction === 'dislike';
126 |     const isBookmarked = art.user_interaction === 'bookmark'
    | ;
127 |
128 |     col.innerHTML = `
129 |         <div class="card h-100 shadow-sm d-flex flex-column" i
    | d="news-card-${art.id}">
130 |             <div class="card-header d-flex justify-content-betwe
    | en align-items-center py-2">
131 |                 <span class="badge bg-secondary-subtle text-second
    | ary-emphasis border small text-truncate" style="max-width: 130
    | px;">
132 |                     ${art.source_name}
133 |                 </span>
134 |                 <span class="badge ${matchBadgeClass} rounded-pill
    | small" title="Персональная релевантность">
135 |                     ${matchPercent}% match
136 |                 </span>
137 |             </div>
138 |             <div class="card-body p-3 d-flex flex-column flex-gr
    | ow-1">
139 |                 <h6 class="card-title fw-bold mb-2">
140 |                     <a href="${art.link}" target="_blank" rel="noope
    | ner noreferrer" class="text-reset text-decoration-none hover-p
    | rimary" onclick="window.newsTab?.onArticleRead('${art.id}', '$
    | {encodeURIComponent(art.title)}')">
141 |                         ${art.title}
142 |                     </a>
143 |                 </h6>
144 |                 <p class="card-text small text-muted flex-grow-1 m
    | b-3">
145 |                     ${art.ai_summary || art.summary || ''}
146 |                 </p>
147 |                 <div class="d-flex align-items-center justify-cont
    | ent-between mt-auto pt-2 border-top border-secondary-subtle">
148 |                     <small class="text-secondary" style="font-size:
    | 0.75rem;">
149 |                         <i class="bi bi-clock me-1"></i>${art.publishe

```

```

    d_at || ''}
150         </small>
151         <div class="btn-group btn-group-sm" role="group"
    >
152             <button type="button" class="btn ${isLiked ? '
    btn-success' : 'btn-outline-secondary'} py-0 px-2" title="Поле
    зно (обучить алгоритм)" onclick="window.newsTab?.sendFeedback(
    '${art.id}', 'like', '${encodeURIComponent(art.title)}', '${ar
    t.category}')">
153                 <i class="bi bi-hand-thumbs-up"></i>
154             </button>
155             <button type="button" class="btn ${isDisliked
    ? 'btn-danger' : 'btn-outline-secondary'} py-0 px-2" title="Не
    интересно (понизить тему)" onclick="window.newsTab?.sendFeedb
    ack('${art.id}', 'dislike', '${encodeURIComponent(art.title)}'
    , '${art.category}')">
156                 <i class="bi bi-hand-thumbs-down"></i>
157             </button>
158             <button type="button" class="btn ${isBookmarke
    d ? 'btn-warning text-dark' : 'btn-outline-secondary'} py-0 px
    -2" title="Сохранить в закладки" onclick="window.newsTab?.send
    Feedback('${art.id}', 'bookmark', '${encodeURIComponent(art.ti
    tle)}', '${art.category}')">
159                 <i class="bi bi-bookmark"></i>
160             </button>
161             <button type="button" class="btn btn-outline-i
    nfo py-0 px-2" title="Сгенерировать AI-выжимку" onclick="windo
    w.newsTab?.summarizeArticle('${art.id}')">
162                 <i class="bi bi-magic"></i>
163             </button>
164         </div>
165     </div>
166 </div>
167 </div>
168 `;
169     container.appendChild(col);
170 });
171 }

```



```

172 |
173 | filterNews() {
174 |     this.renderArticles();
175 | }
176 |
177 | async sendFeedback(articleId, action, titleEncoded, category
    | ) {
178 |     const title = decodeURIComponent(titleEncoded || '');
179 |     try {
180 |         const res = await fetch('/api/news/feedback', {
181 |             method: 'POST',
182 |             headers: { 'Content-Type': 'application/json' },
183 |             body: JSON.stringify({
184 |                 article_id: articleId,
185 |                 action: action,
186 |                 article_title: title,
187 |                 category: category || 'general'
188 |             })
189 |         });
190 |
191 |         if (res.ok) {
192 |             // Optimistic UI update
193 |             const art = this.articles.find((a) => a.id === article
    | Id);
194 |             if (art) {
195 |                 art.user_interaction = action;
196 |             }
197 |             this.renderArticles();
198 |             await this.loadProfile();
199 |             if (window.showNotification) {
200 |                 const msg = action === 'like' ? 'Алгоритм обучен: те
    | ма получила приоритет ☑ ' :
201 |                     action === 'dislike' ? 'Алгоритм обучен:
    | интерес к теме снижен ☑ ' :
202 |                     action === 'bookmark' ? 'Добавлено в зак
    | ладки ☑ ' : 'Действие сохранено';
203 |                 window.showNotification(msg, 'success');
204 |             }

```

```

205 |     }
206 |   } catch (e) {
207 |     console.error('[NewsTab] Error sending feedback:', e);
208 |   }
209 | }
210 |
211 | async onArticleRead(articleId, titleEncoded) {
212 |   const title = decodeURIComponent(titleEncoded || '');
213 |   try {
214 |     await fetch('/api/news/feedback', {
215 |       method: 'POST',
216 |       headers: { 'Content-Type': 'application/json' },
217 |       body: JSON.stringify({
218 |         article_id: articleId,
219 |         action: 'read',
220 |         article_title: title
221 |       })
222 |     });
223 |   } catch (_) {}
224 | }
225 |
226 | async summarizeArticle(articleId) {
227 |   const art = this.articles.find((a) => a.id === articleId);
228 |   if (!art) return;
229 |
230 |   const card = document.getElementById(`news-card-${articleId}`);
231 |   if (card) {
232 |     const body = card.querySelector('.card-text');
233 |     if (body) {
234 |       body.innerHTML = '<span class="spinner-border spinner-border-sm text-info me-1"></span> Генерация ИИ-выжимки...';
235 |     }
236 |   }
237 |
238 |   try {
239 |     const res = await fetch('/api/news/summarize-article', {
240 |       method: 'POST',

```

```
241 |         headers: { 'Content-Type': 'application/json' },
242 |         body: JSON.stringify(art)
243 |     });
244 |     if (res.ok) {
245 |         const data = await res.json();
246 |         art.ai_summary = data.ai_summary;
247 |         this.renderArticles();
248 |     }
249 | } catch (e) {
250 |     console.error('[NewsTab] Summarize error:', e);
251 |     this.renderArticles();
252 | }
253 | }
254 |
255 | async generateDigest() {
256 |     const digestCard = document.getElementById('news-digest-ca
rd');
257 |     const digestContent = document.getElementById('news-digest
-content');
258 |     const digestTitle = document.getElementById('news-digest-t
itle');
259 |
260 |     if (digestCard) digestCard.classList.remove('d-none');
261 |     if (digestContent) {
262 |         digestContent.innerHTML = '<div class="text-center p-3">
<span class="spinner-border spinner-border-sm text-warning me-
2"></span>Формирование персонализированного дайджеста ИИ...</d
iv>';
263 |     }
264 |
265 |     try {
266 |         const res = await fetch('/api/news/digest', { method: 'P
OST' });
267 |         if (res.ok) {
268 |             this.currentDigest = await res.json();
269 |             if (digestTitle) digestTitle.textContent = this.curren
tDigest.title || 'ИИ-Дайджест';
270 |             if (digestContent) {
```

```

271         if (window.marked) {
272             digestContent.innerHTML = window.marked.parse(this
273             .currentDigest.digest_text || '');
274         } else {
275             digestContent.textContent = this.currentDigest.dig
276             est_text || '';
277         }
278     } catch (e) {
279         console.error('[NewsTab] Error generating digest:', e);
280         if (digestContent) digestContent.textContent = `Ошибка г
281         енерации дайджеста: ${e.message}`;
282     }
283
284     speakDigest() {
285         if (!this.currentDigest || !this.currentDigest.digest_text
286         ) return;
287         if (window.chatService && typeof window.chatService.speak
288         === 'function') {
289             window.chatService.speak(this.currentDigest.digest_text)
290             ;
291         } else {
292             const utterance = new SpeechSynthesisUtterance(this.curr
293             entDigest.digest_text);
294             utterance.lang = 'ru-RU';
295             window.speechSynthesis.speak(utterance);
296         }
297     }
298
299     async savePreferences() {
300         const prefInput = document.getElementById('news-pref-topic
301         s-input')?.value || '';
302         const negInput = document.getElementById('news-neg-topics-
303         input')?.value || '';
304
305         const preferred = prefInput.split(',').map((s) => s.trim())

```

```

    ).filter(Boolean);
300     const ignored = negInput.split(',').map((s) => s.trim()).f
    ilter(Boolean);
301
302     try {
303         const res = await fetch('/api/news/profile', {
304             method: 'POST',
305             headers: { 'Content-Type': 'application/json' },
306             body: JSON.stringify({
307                 preferred_topics: preferred,
308                 ignored_topics: ignored
309             })
310         });
311         if (res.ok) {
312             await this.loadProfile();
313             await this.refreshNews(true);
314             if (window.showNotification) window.showNotification('
Интересы успешно сохранены!', 'success');
315             const modalEl = document.getElementById('newsPreferenc
esModal');
316             if (modalEl && window.bootstrap) {
317                 const modalInstance = window.bootstrap.Modal.getInst
ance(modalEl);
318                 if (modalInstance) modalInstance.hide();
319             }
320         }
321     } catch (e) {
322         console.error('[NewsTab] Error saving preferences:', e);
323     }
324 }
325
326 async resetProfile() {
327     if (!confirm('Сбросить накопленные веса интересов и начать
обучение с чистого листа?')) return;
328     try {
329         const res = await fetch('/api/news/profile', {
330             method: 'POST',
331             headers: { 'Content-Type': 'application/json' },

```

```
332 |         body: JSON.stringify({
333 |             preferred_topics: [],
334 |             ignored_topics: []
335 |         })
336 |     });
337 |     if (res.ok) {
338 |         await this.loadProfile();
339 |         await this.refreshNews(true);
340 |         if (window.showNotification) window.showNotification('
Профиль обучения сброшен', 'info');
341 |     }
342 | } catch (e) {
343 |     console.error('[NewsTab] Error resetting profile:', e);
344 | }
345 | }
346 | }
347 |
348 | window.newsTab = new NewsTabController();
349 |
350 | function initNewsTab() {
351 |     if (window.newsTab) {
352 |         window.newsTab.init();
353 |     }
354 | }
355 |
356 | window.initNewsTab = initNewsTab;
357 |
358 | if (document.getElementById('news-tab-root')) {
359 |     initNewsTab();
360 | }
361 |
```

index.html

```
1 <!-- Ninite Auto-Updater & Task Scheduler Management Tab -->
2 <style>
3   .ninite-container {
4     padding: 0.75rem 1rem;
5     max-width: 1200px;
6     margin: 0 auto;
7   }
8   .ninite-card {
9     background: #0f172a !important;
10    border: 1px solid #334155 !important;
11    border-radius: 10px;
12    padding: 1.25rem;
13    color: #f8fafc;
14    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.2);
15  }
16  .ninite-header {
17    background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
18    0%);
19    border: 1px solid #334155;
20    border-radius: 10px;
21    padding: 0.85rem 1.25rem;
22  }
23  .ninite-dropzone {
24    border: 2px dashed #38bdf8;
25    background: rgba(56, 189, 248, 0.05);
26    border-radius: 10px;
27    padding: 2rem 1.5rem;
28    text-align: center;
29    cursor: pointer;
30    transition: all 0.2s ease-in-out;
31  }
32  .ninite-dropzone:hover, .ninite-dropzone.dragover {
33    background: rgba(56, 189, 248, 0.12);
34    border-color: #0ea5e9;
35  }
36  .ninite-stat-box {
```

```

36     background: #1e293b;
37     border: 1px solid #334155;
38     border-radius: 8px;
39     padding: 0.75rem 1rem;
40 }
41 .ninite-stat-title {
42     font-size: 0.75rem;
43     text-transform: uppercase;
44     letter-spacing: 0.05em;
45     color: #94a3b8;
46 }
47 .ninite-stat-val {
48     font-size: 1.1rem;
49     font-weight: 600;
50     color: #f8fafc;
51     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
52 }
53 </style>
54
55 <div class="ninite-container">
56     <!-- Верхняя информационная панель -->
57     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 ninite-header text-white">
58         <div class="d-flex align-items-center gap-2.5">
59             <span class="fs-4">❏ </span>
60             <div>
61                 <h5 class="mb-0 fw-bold fs-6">Установка и автообновление Ninite</h5>
62                 <span class="text-secondary small">Автоматическое получение свежих версий ПО через Windows Task Scheduler</span>
63             </div>
64         </div>
65         <div class="d-flex align-items-center gap-2">
66             <span class="badge rounded-pill bg-dark border border-secondary text-info px-2.5 py-1.5" id="ninite-status-badge">
67                 Загрузка статуса...
68             </span>

```



```

69 |         <button class="btn btn-sm btn-outline-info rounded-pill
    | px-3" id="btn-ninite-refresh" type="button">
70 |             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
71 |         </button>
72 |     </div>
73 | </div>
74 |
75 | <div class="row g-3 mb-3">
76 |     <!-- Левая колонка: Форма загрузки и настройки расписания
    | -->
77 |     <div class="col-12 col-lg-7">
78 |         <div class="ninite-card h-100">
79 |             <h6 class="fw-bold text-info mb-3 d-flex align-items-c
    | enter gap-2">
80 |                 <i class="bi bi-cloud-arrow-up-fill"></i> 1. Выбор и
    | нсталлятора и расписание
81 |             </h6>
82 |
83 |             <form id="ninite-upload-form">
84 |                 <!-- Dropzone для выбора файла -->
85 |                 <label for="ninite-file-input" class="ninite-dropzon
    | e mb-3 d-block text-decoration-none" id="ninite-dropzone">
86 |                     <i class="bi bi-file-earmark-arrow-up fs-1 text-in
    | fo mb-2 d-block"></i>
87 |                     <div class="fw-semibold mb-1 text-white" id="ninit
    | e-file-label">Перетащите файл установщика Ninite или нажмите д
    | ля выбора</div>
88 |                     <div class="text-secondary small">Файл будет сохра
    | нён и переименован в <code>C:\Program Files\Ninite\ninite.exe<
    | /code></div>
89 |                     <input type="file" id="ninite-file-input" class="d
    | -none" accept=".exe">
90 |                 </label>
91 |
92 |                 <!-- Параметры расписания -->
93 |                 <div class="row g-2 mb-3">
94 |                     <div class="col-12 col-sm-6">
95 |                         <label for="ninite-interval" class="form-label s

```

```

mall text-secondary fw-semibold">
96         <i class="bi bi-calendar-range me-1"></i> Инте
рвал обновления
97         </label>
98         <select class="form-select form-select-sm bg-dar
k text-white border-secondary" id="ninite-interval">
99             <option value="1">Каждую неделю (1 неделя)</op
tion>
100             <option value="2" selected>Каждые 2 недели (по
умолчанию)</option>
101             <option value="3">Каждые 3 недели</option>
102             <option value="4">Раз в месяц (4 недели)</opti
on>
103         </select>
104     </div>
105
106     <div class="col-12 col-sm-6">
107         <label for="ninite-time" class="form-label small
text-secondary fw-semibold">
108             <i class="bi bi-clock me-1"></i> Время запуска
109             </label>
110             <input type="time" class="form-control form-cont
rol-sm bg-dark text-white border-secondary" id="ninite-time" v
alue="20:00">
111         </div>
112     </div>
113
114     <div class="mb-3">
115         <label for="ninite-day" class="form-label small te
xt-secondary fw-semibold">
116             <i class="bi bi-calendar-event me-1"></i> День н
едели
117             </label>
118             <select class="form-select form-select-sm bg-dark
text-white border-secondary" id="ninite-day">
119                 <option value="Sunday" selected>Воскресенье</opt
ion>
120                 <option value="Monday">Понедельник</option>

```

```

121         <option value="Tuesday">Вторник</option>
122         <option value="Wednesday">Среда</option>
123         <option value="Thursday">Четверг</option>
124         <option value="Friday">Пятница</option>
125         <option value="Saturday">Суббота</option>
126     </select>
127 </div>
128
129     <!-- Кнопка отправки и сохранения -->
130     <button type="submit" class="btn btn-primary w-100 p
131 y-2 fw-semibold d-flex align-items-center justify-content-cent
132 er gap-2" id="btn-ninite-submit">
133         <i class="bi bi-check-circle-fill"></i> Установить
134         в Program Files и активировать Task Scheduler
135     </button>
136 </form>
137
138 <div id="ninite-alert-box" class="mt-3" style="display
139 : none;"></div>
140 </div>
141 </div>
142
143 <!-- Правая колонка: Текущее состояние и управление -->
144 <div class="col-12 col-lg-5">
145     <div class="ninite-card h-100 d-flex flex-column justify
146 -content-between">
147         <div>
148             <h6 class="fw-bold text-info mb-3 d-flex align-items
149 -center gap-2">
150                 <i class="bi bi-cpu-fill"></i> 2. Текущее состояни
151 е системы
152             </h6>
153
154             <div class="row g-2 mb-3">
155                 <div class="col-12">
156                     <div class="ninite-stat-box">
157                         <div class="ninite-stat-title">Файл Ninite в P
158 rogram Files</div>

```

```

151 |         <div class="ninite-stat-val text-truncate" id=
| "ninite-file-status">Проверка...</div>
152 |         <div class="text-secondary small mt-1" id="nin
| ite-file-details">Путь: C:\Program Files\Ninite\ninite.exe</di
| v>
153 |     </div>
154 | </div>
155 |
156 |     <div class="col-12">
157 |         <div class="ninite-stat-box">
158 |             <div class="ninite-stat-title">Задача Windows
| Task Scheduler</div>
159 |             <div class="ninite-stat-val text-truncate" id=
| "ninite-task-status">Проверка...</div>
160 |             <div class="text-secondary small mt-1" id="nin
| ite-task-details">Задача: NiniteAutoUpdate</div>
161 |         </div>
162 |     </div>
163 |
164 |     <div class="col-12">
165 |         <div class="ninite-stat-box">
166 |             <div class="ninite-stat-title">Следующий запус
| к по расписанию</div>
167 |             <div class="ninite-stat-val text-info text-tru
| ncate" id="ninite-task-next">--</div>
168 |         </div>
169 |     </div>
170 | </div>
171 | </div>
172 |
173 | <div>
174 |     <!-- Кнопка восстановления удаленных файлов (R-Studi
| o) -->
175 |     <button class="btn btn-outline-info w-100 py-2 fw-se
| mibold mb-2 d-flex align-items-center justify-content-center g
| ap-2" id="btn-ninite-recovery-launch" type="button" title="Зап
| уск R-Studio Technician Portable для восстановления данных">
176 |         <i class="bi bi-folder-symlink-fill"></i> Восстано

```

```

        вить удаленные файлы
177         </button>
178
179         <!-- Кнопка принудительного обновления -->
180         <button class="btn btn-outline-warning w-100 py-2 fw
        -semibold mb-2 d-flex align-items-center justify-content-cen
        ter gap-2" id="btn-ninite-run-now">
181             <i class="bi bi-play-circle-fill"></i> Запустить о
        бновление прямо сейчас
182         </button>
183
184         <button class="btn btn-sm btn-outline-secondary w-10
        0" id="btn-ninite-update-schedule-only" type="button">
185             <i class="bi bi-sliders"></i> Обновить только расп
        исание (без перезагрузки файла)
186         </button>
187     </div>
188 </div>
189 </div>
190 </div>
191
192 <!-- Журнал выполнения / лог -->
193 <div class="ninite-card">
194     <div class="d-flex justify-content-between align-items-cen
        ter mb-2">
195         <h6 class="fw-bold text-info mb-0 d-flex align-items-cen
        ter gap-2">
196             <i class="bi bi-terminal-fill"></i> Журнал выполнения
        обновления (Log)
197         </h6>
198         <span class="text-secondary small" id="ninite-log-timest
        amp"></span>
199     </div>
200     <pre class="p-2.5 rounded bg-dark border border-secondary
        text-light small mb-0" id="ninite-log-viewer" style="max-heigh
        t: 180px; overflow-y: auto; font-family: ui-monospace, monospa
        ce; white-space: pre-wrap;">Логи обновлений пока отсутствуют.<
        /pre>

```

```
201 |   </div>
202 | </div>
203 |
```

main.js

```
1 /**
2  * ninite_updater_tab/main.js – логика управления установкой и
   автообновлением Ninite
3  */
4
5 (function () {
6   'use strict';
7
8   let selectedFile = null;
9
10  /**
11   * Настройка обработчиков событий
12   */
13   function setupEventListeners() {
14     const dropzone = document.getElementById('ninite-dropzone'
15   );
16     const fileInput = document.getElementById('ninite-file-inp
17   ut');
18     const form = document.getElementById('ninite-upload-form')
19   ;
20     const refreshBtn = document.getElementById('btn-ninite-ref
21   resh');
22     const runNowBtn = document.getElementById('btn-ninite-run-
23   now');
24     const scheduleOnlyBtn = document.getElementById('btn-ninit
25   e-update-schedule-only');
26
27     if (fileInput) {
28       fileInput.addEventListener('change', (e) => {
29         if (e.target.files && e.target.files[0]) {
30           handleFileSelect(e.target.files[0]);
31         }
32       });
33     }
34
35     if (dropzone) {
```

```
30 |     dropzone.addEventListener('dragover', (e) => {
31 |         e.preventDefault();
32 |         dropzone.classList.add('dragover');
33 |     });
34 |
35 |     dropzone.addEventListener('dragleave', () => {
36 |         dropzone.classList.remove('dragover');
37 |     });
38 |
39 |     dropzone.addEventListener('drop', (e) => {
40 |         e.preventDefault();
41 |         dropzone.classList.remove('dragover');
42 |         if (e.dataTransfer.files && e.dataTransfer.files[0]) {
43 |             handleFileSelect(e.dataTransfer.files[0]);
44 |         }
45 |     });
46 | }
47 |
48 | if (form) {
49 |     form.addEventListener('submit', handleFormSubmit);
50 | }
51 |
52 | if (refreshBtn) {
53 |     refreshBtn.addEventListener('click', () => loadStatus())
54 | ;
55 | }
56 |
57 | if (runNowBtn) {
58 |     runNowBtn.addEventListener('click', handleRunNow);
59 | }
60 |
61 | if (scheduleOnlyBtn) {
62 |     scheduleOnlyBtn.addEventListener('click', handleSchedule
63 | Only);
64 | }
65 |
66 | const recoveryBtn = document.getElementById('btn-ninite-re
67 | covery-launch');
```



```

65 |     if (recoveryBtn) {
66 |         recoveryBtn.addEventListener('click', handleLaunchRecovery);
67 |     }
68 | }
69 |
70 | /**
71 |  * Запуск R-Studio для восстановления удаленных файлов
72 |  */
73 | async function handleLaunchRecovery() {
74 |     const btn = document.getElementById('btn-ninite-recovery-launch');
75 |     if (btn) {
76 |         btn.disabled = true;
77 |         btn.innerHTML = `<span class="spinner-border spinner-border-sm me-2"></span>Запуск R-Studio...`;
78 |     }
79 |
80 |     try {
81 |         const res = await fetch('/api/recovery/launch', {
82 |             method: 'POST',
83 |             headers: { 'Content-Type': 'application/json' },
84 |         });
85 |         const data = await res.json();
86 |         if (!res.ok) {
87 |             throw new Error(data.detail || `HTTP ${res.status}`);
88 |         }
89 |
90 |         showAlert(`<strong>Успех:</strong> ${data.message || 'Программа восстановления файлов R-Studio запущена!'}`, 'success');
91 |
92 |         if (window.toast) {
93 |             window.toast.success('R-Studio запущена', data.message || 'Окно программы открывается...');
94 |         } catch (err) {
95 |             showAlert(`<strong>Ошибка запуска:</strong> ${err.message}`, 'danger');

```

```

96         if (window.toast) {
97             window.toast.error('Ошибка запуска R-Studio', err.mess
age);
98         }
99     } finally {
100         if (btn) {
101             btn.disabled = false;
102             btn.innerHTML = ` Выбран файл: <strong class="text-in
fo">${file.name}</strong> (${sizeMb} MB)`;
116     }
117 }
118
119 /**
120  * Загрузка текущего статуса файла и задачи в Task Scheduler
121  */
122 async function loadStatus() {
123     const statusBadge = document.getElementById('ninite-status
-badge');
124     const fileStatus = document.getElementById('ninite-file-st
atus');
125     const fileDetails = document.getElementById('ninite-file-d
etails');
126     const taskStatus = document.getElementById('ninite-task-st

```

```
atus');
127     const taskDetails = document.getElementById('ninite-task-d
etails');
128     const taskNext = document.getElementById('ninite-task-next
');
129     const logViewer = document.getElementById('ninite-log-view
er');
130     const logTimestamp = document.getElementById('ninite-log-t
imestamp');
131
132     try {
133         const res = await fetch(`/api/ninite/status?t=${Date.now
()}`);
134         if (!res.ok) throw new Error(`HTTP ${res.status}`);
135         const data = await res.json();
136
137         if (data.file) {
138             if (data.file.installed) {
139                 const sizeMb = (data.file.size_bytes / (1024 * 1024)
).toFixed(2);
140                 fileStatus.innerHTML = `●
Установлен</span>`;
141                 fileDetails.textContent = `Размер: ${sizeMb} MB | Из
менён: ${data.file.modified_time}`;
142             } else {
143                 fileStatus.innerHTML = `●
Не установлен</span>`;
144                 fileDetails.textContent = `Файл ninite.exe отсутству
ет в Program Files`;
145             }
146         }
147
148         if (data.task) {
149             if (data.task.exists) {
150                 taskStatus.innerHTML = `●
Активна (${data.task.state})</span>`;
151                 taskDetails.textContent = `Задача: ${data.task.task_
name} | Посл. запуск: ${data.task.last_run_time || 'нет'}`;
```

```
152 |         taskNext.textContent = data.task.next_run_time || 'П
    | о расписанию';
153 |     } else {
154 |         taskStatus.innerHTML = `<span class="text-secondary"
    | >● Не запланирована</span>`;
155 |         taskDetails.textContent = `Задача NiniteAutoUpdate о
    | тсутствует в Task Scheduler`;
156 |         taskNext.textContent = '--';
157 |     }
158 | }
159 |
160 | if (statusBadge) {
161 |     if (data.file?.installed && data.task?.exists) {
162 |         statusBadge.className = 'badge rounded-pill bg-succe
    | ss-subtle text-success border border-success px-2.5 py-1.5';
163 |         statusBadge.textContent = '● Автообновление активно'
    | ;
164 |     } else if (data.file?.installed) {
165 |         statusBadge.className = 'badge rounded-pill bg-warni
    | ng-subtle text-warning border border-warning px-2.5 py-1.5';
166 |         statusBadge.textContent = '● Файл готов, расписание
    | не задано';
167 |     } else {
168 |         statusBadge.className = 'badge rounded-pill bg-secon
    | dary-subtle text-light border border-secondary px-2.5 py-1.5';
169 |         statusBadge.textContent = '● Ожидание установки';
170 |     }
171 | }
172 |
173 | if (logViewer && data.log) {
174 |     logViewer.textContent = data.log;
175 |     if (logTimestamp) {
176 |         logTimestamp.textContent = `Обновлено: ${new Date().
    | toLocaleTimeString()}`;
177 |     }
178 | }
179 | } catch (err) {
180 |     console.error('Ошибка загрузки статуса Ninite:', err);
```

```
181         if (statusBadge) {
182             statusBadge.className = 'badge rounded-pill bg-danger-
subtle text-danger border border-danger px-2.5 py-1.5';
183             statusBadge.textContent = '● Ошибка связи с сервером';
184         }
185     }
186 }
187
188 /**
189  * Отправка формы (загрузка файла + настройка Task Scheduler
190  *)
191  */
192 async function handleFormSubmit(e) {
193     e.preventDefault();
194
195     if (!selectedFile) {
196         showAlert('Пожалуйста, выберите файл инсталлятора Ninite
для загрузки.', 'warning');
197         return;
198     }
199
200     const interval = document.getElementById('ninite-interval'
)?.value || '2';
201     const timeStr = document.getElementById('ninite-time')?.va
lue || '20:00';
202     const submitBtn = document.getElementById('btn-ninite-subm
it');
203
204     const formData = new FormData();
205     formData.append('file', selectedFile);
206     formData.append('interval_weeks', interval);
207     formData.append('time_str', timeStr);
208
209     if (submitBtn) {
210         submitBtn.disabled = true;
211         submitBtn.innerHTML = `<span class="spinner-border spinn
er-border-sm me-2"></span>Копирование в Program Files и регист
рация задачи...`;

```

```
211     }
212
213     try {
214         const res = await fetch('/api/ninite/upload', {
215             method: 'POST',
216             body: formData,
217         });
218
219         const data = await res.json();
220         if (!res.ok) {
221             throw new Error(data.detail || `HTTP ${res.status}`);
222         }
223
224         showAlert(`Успешно! ${data.message}`, 'success');
225         await loadStatus();
226     } catch (err) {
227         showAlert(`Ошибка: ${err.message}`, 'danger');
228     } finally {
229         if (submitBtn) {
230             submitBtn.disabled = false;
231             submitBtn.innerHTML = `
```

```
244 |     try {
245 |         const res = await fetch('/api/ninite/schedule', {
246 |             method: 'POST',
247 |             headers: { 'Content-Type': 'application/json' },
248 |             body: JSON.stringify({
249 |                 interval_weeks: interval,
250 |                 time_str: timeStr,
251 |                 days_of_week: day,
252 |             }),
253 |         });
254 |
255 |         const data = await res.json();
256 |         if (!res.ok) {
257 |             throw new Error(data.detail || `HTTP ${res.status}`);
258 |         }
259 |
260 |         showAlert(`Расписание обновлено: каждые ${interval} нед.
    | в ${timeStr}`, 'success');
261 |         await loadStatus();
262 |     } catch (err) {
263 |         showAlert(`Ошибка обновления расписания: ${err.message}`
    | , 'danger');
264 |     }
265 | }
266 |
267 | /**
268 |  * Ручной запуск обновления сейчас
269 |  */
270 | async function handleRunNow() {
271 |     const btn = document.getElementById('btn-ninite-run-now');
272 |     if (btn) {
273 |         btn.disabled = true;
274 |         btn.innerHTML = `<span class="spinner-border spinner-bor
    | der-sm me-2"></span>Запуск обновления...`;
275 |     }
276 |
277 |     try {
278 |         const res = await fetch('/api/ninite/run-now', { method:
```

```

    'POST' });
279     const data = await res.json();
280     if (!res.ok) {
281         throw new Error(data.detail || `HTTP ${res.status}`);
282     }
283
284     showAlert(`Обновление запущено: ${data.message}`, 'info'
    );
285     setTimeout(loadStatus, 3000);
286 } catch (err) {
287     showAlert(`Ошибка запуска: ${err.message}`, 'danger');
288 } finally {
289     if (btn) {
290         btn.disabled = false;
291         btn.innerHTML = `

```



```
312 |     * Инициализация вкладки Ninite Updater
313 |     */
314 |     function initNiniteUpdaterTab() {
315 |         setupEventListeners();
316 |         loadStatus();
317 |     }
318 |
319 |     // Экспорт для жизненного цикла tab-core.js и глобального ко
    нтекста
320 |     window.initNiniteUpdaterTab = initNiniteUpdaterTab;
321 |     window.initNiniteupdaterTab = initNiniteUpdaterTab;
322 |     window.initNinite_updaterTab = initNiniteUpdaterTab;
323 |
324 |     // Автоматический запуск при автономной загрузке
325 |     if (document.readyState === 'loading') {
326 |         document.addEventListener('DOMContentLoaded', initNiniteUp
    daterTab);
327 |     } else {
328 |         initNiniteUpdaterTab();
329 |     }
330 | })();
331 |
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Peripherals & Network Tab
6 =====
7
8 Description:
9     Вкладка радиочастотной телеметрии Wi-Fi (SSID, BSSID, RSSI
10    ), аудиоустройств
11    и Plug & Play USB периферии.
12
13 File: index.html
14 Project: AI-Breadboard
15 Module: WebInterface.PeripheralsTab
16 Author: hypo69
17 Copyright: © 2026 hypo69
18 =====
19 =====
20 -->
21 <style>
22     .peripherals-tab-wrapper {
23         padding: 0.75rem;
24         color: var(--text-color, #f8fafc);
25     }
26     .diag-panel {
27         background: var(--surface-2, #1e293b);
28         border: 1px solid var(--border-color, #334155);
29         border-radius: 8px;
30         overflow: hidden;
31     }
32     .diag-panel-header {
33         background: rgba(15, 23, 42, 0.6);
34         padding: 8px 14px;
35         border-bottom: 1px solid var(--border-color, #334155);
36         font-weight: 600;
37         font-size: 0.85rem;
```

```

33 | }
34 | .about-sys-spec-table {
35 |     width: 100%;
36 |     border-collapse: separate;
37 |     border-spacing: 0;
38 | }
39 | .about-sys-spec-table td {
40 |     padding: 8px 12px;
41 |     border-bottom: 1px solid rgba(51, 65, 85, 0.35);
42 |     vertical-align: middle;
43 | }
44 | .about-sys-spec-key {
45 |     color: var(--text-muted, #94a3b8);
46 |     width: 200px;
47 |     font-size: 0.82rem;
48 |     font-weight: 500;
49 | }
50 | .about-sys-spec-val {
51 |     color: var(--text-color, #f8fafc);
52 |     font-weight: 500;
53 |     font-size: 0.82rem;
54 | }
55 | </style>
56 |
57 | <div class="peripherals-tab-wrapper">
58 |     <!-- Top Header & Status -->
59 |     <div class="card shadow-sm border-secondary-subtle mb-3" sty
60 | le="background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
61 | 0%); border-radius: 10px;">
62 |         <div class="card-body p-3">
63 |             <div class="d-flex flex-wrap align-items-center justify-
64 | content-between gap-2">
65 |                 <div class="d-flex align-items-center gap-2">
66 |                     <div class="fs-4">□ </div>
67 |                     <div>
68 |                         <h5 class="mb-0 fw-bold text-white">5. Сеть и пери
69 | ферия (Peripherals & Wireless Network)</h5>
70 |                         <small class="text-muted">RF-параметры Wi-Fi сетей

```

```

, аудио конечные точки и Plug-and-Play USB устройства</small>
67     </div>
68     </div>
69
70     <div class="d-flex align-items-center gap-2 flex-wrap"
71     >
72         <div class="form-check form-switch mb-0 small text-m
73         uted d-flex align-items-center gap-1">
74             <input class="form-check-input" type="checkbox" id
75             ="diag-peripherals-auto-refresh">
76                 <label class="form-check-label small text-nowrap"
77                 for="diag-peripherals-auto-refresh">Авто (5с)</label>
78             </div>
79             <button class="btn btn-sm btn-outline-info rounded-p
80             ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
81             "btn-diag-peripherals-refresh">
82                 <i class="bi bi-arrow-clockwise"></i> Обновить
83             </button>
84         </div>
85     </div>
86 </div>
87
88 <div class="row g-2 mb-3">
89     <!-- Wi-Fi Telemetry Panel -->
90     <div class="col-12 col-lg-5">
91         <div class="diag-panel h-100">
92             <div class="diag-panel-header d-flex justify-content-b
93             etween align-items-center">
94                 <div class="d-flex align-items-center gap-2">
95                     <i class="bi bi-wifi text-info"></i>
96                     <span class="text-white">Беспроводная сеть Wi-Fi (
97                     RF & Security)</span>
98                 </div>
99                 <span class="badge bg-success-subtle text-success bo
100                 rder border-success" id="diag-wifi-status-badge">Wi-Fi Active<
101                 /span>
102             </div>
103         </div>

```

```

94 |         <div class="p-3">
95 |             <table class="about-sys-spec-table">
96 |                 <tbody>
97 |                     <tr>
98 |                         <td class="about-sys-spec-key"><i class="bi bi
| -broadcast me-1.5 text-info"></i>Сеть (SSID)</td>
99 |                         <td class="about-sys-spec-val fw-bold text-whi
| te" id="diag-wifi-ssid">--</td>
100 |                     </tr>
101 |                     <tr>
102 |                         <td class="about-sys-spec-key"><i class="bi bi
| -router me-1.5 text-primary"></i>Точка доступа (BSSID)</td>
103 |                         <td class="about-sys-spec-val font-monospace"
| id="diag-wifi-bssid">--</td>
104 |                     </tr>
105 |                     <tr>
106 |                         <td class="about-sys-spec-key"><i class="bi bi
| -reception-4 me-1.5 text-success"></i>Сигнал (RSSI)</td>
107 |                         <td class="about-sys-spec-val" id="diag-wifi-s
| ignal">0% (-- dBm)</td>
108 |                     </tr>
109 |                     <tr>
110 |                         <td class="about-sys-spec-key"><i class="bi bi
| -hash me-1.5 text-warning"></i>Радиоканал</td>
111 |                         <td class="about-sys-spec-val" id="diag-wifi-c
| hannel">--</td>
112 |                     </tr>
113 |                     <tr>
114 |                         <td class="about-sys-spec-key"><i class="bi bi
| -cpu me-1.5 text-info"></i>Стандарт связи</td>
115 |                         <td class="about-sys-spec-val font-monospace"
| id="diag-wifi-radio">--</td>
116 |                     </tr>
117 |                     <tr>
118 |                         <td class="about-sys-spec-key"><i class="bi bi
| -shield-lock me-1.5 text-success"></i>Безопасность</td>
119 |                         <td class="about-sys-spec-val text-truncate" i
| d="diag-wifi-auth">--</td>

```

```

120         </tr>
121     </tbody>
122 </table>
123 </div>
124 </div>
125 </div>
126
127 <!-- Audio Endpoints Panel -->
128 <div class="col-12 col-lg-7">
129     <div class="diag-panel h-100">
130         <div class="diag-panel-header d-flex justify-content-b
between align-items-center">
131             <div class="d-flex align-items-center gap-2">
132                 <i class="bi bi-volume-up-fill text-warning"></i>
133                 <span class="text-white">Аудиоустройства и конечны
е точки (Audio Endpoints)</span>
134             </div>
135             <span class="badge bg-secondary" id="diag-audio-coun
t">0 устройств</span>
136         </div>
137         <div class="p-2" id="diag-audio-container" style="max-
height: 260px; overflow-y: auto;">
138             <div class="text-center py-4 text-muted small">Опрос
аудиоустройств...</div>
139         </div>
140     </div>
141 </div>
142 </div>
143
144 <!-- PnP USB Devices Table -->
145 <div class="diag-panel">
146     <div class="diag-panel-header d-flex justify-content-betwe
en align-items-center">
147         <div class="d-flex align-items-center gap-2">
148             <i class="bi bi-usb-symbol text-info"></i>
149             <span class="text-white">Подключенные PnP USB-устройст
ва и концентраторы</span>
150         </div>

```

```
151 |         <span class="badge bg-secondary" id="diag-usb-count">0 y
    | устройств</span>
152 |     </div>
153 |     <div class="table-responsive" style="max-height: 480px; ov
    | erflow-y: auto;">
154 |         <table class="table table-dark table-hover table-sm mb-0
    | " style="font-size: 0.80rem; font-family: ui-monospace, SFMono
    | -Regular, Menlo, Monaco, Consolas, monospace;">
155 |             <thead>
156 |                 <tr class="text-secondary border-secondary">
157 |                     <th>Устройство</th>
158 |                     <th style="width: 100px;">Vendor ID</th>
159 |                     <th style="width: 100px;">Product ID</th>
160 |                     <th style="width: 160px;">Класс устройства</th>
161 |                     <th style="width: 140px;">Статус</th>
162 |                     <th>Device Instance ID</th>
163 |                 </tr>
164 |             </thead>
165 |             <tbody id="diag-usb-tbody">
166 |                 <tr><td colspan="6" class="text-center py-4 text-mut
    | ed">Загрузка USB периферии...</td></tr>
167 |             </tbody>
168 |         </table>
169 |     </div>
170 | </div>
171 | </div>
172 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Peripherals & Network Logic
4   * =====
5   * Description:
6   *   Клиентский контроллер вкладки сети и периферии:
7   *   Wi-Fi RF метрики (RSSI, BSSID), аудио устройства и USB Pn
8   *   P дерево.
9   * File: main.js
10  * Project: AI-Breadboard
11  * Module: WebInterface.PeripheralsTab
12  * Author: hypo69
13  * Copyright: © 2026 hypo69
14  * =====
15  */
16
17 (function () {
18     'use strict';
19
20     let autoRefreshTimer = null;
21
22     function escapeHtml(str) {
23         if (str === null || str === undefined) return '';
24         return String(str)
25             .replace(/&/g, '&amp;')
26             .replace(/</g, '&lt;')
27             .replace(/>/g, '&gt;')
28             .replace(/"/g, '&quot;')
29             .replace(/'/g, '&#039;');
30     }
31
32     function setText(id, text) {
```



```

33     const el = document.getElementById(id);
34     if (el) el.textContent = text !== null && text !== undefin
    ed ? String(text) : '--';
35 }
36
37 async function apiFetch(url) {
38     if (window.api && typeof window.api.fetch === 'function')
    {
39         return await window.api.fetch(url);
40     }
41     const res = await fetch(url);
42     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
    atusText}`);
43     return await res.json();
44 }
45
46 async function fetchPeripheralsNetwork() {
47     try {
48         const data = await apiFetch('/api/v1/system/diagnostics/
    peripherals');
49         if (!data) return;
50
51         // Wi-Fi
52         if (data.wifi_telemetry) {
53             const w = data.wifi_telemetry;
54             setText('diag-wifi-ssid', w.ssid || 'Не подключено');
55             setText('diag-wifi-bssid', w.bssid || '--');
56             setText('diag-wifi-signal', `${w.signal_pct}% (${w.rss
    i_dbm} dBm)`);
57             setText('diag-wifi-channel', w.channel || '--');
58             setText('diag-wifi-radio', w.radio_type || '--');
59             setText('diag-wifi-auth', w.auth_cipher || '--');
60
61             const wifiBadge = document.getElementById('diag-wifi-s
    tatus-badge');
62             if (wifiBadge) {
63                 if (w.is_connected) {
64                     wifiBadge.className = 'badge bg-success-subtle tex

```

```

    t-success border border-success';
65     wifiBadge.textContent = 'Wi-Fi Подключено';
66   } else {
67     wifiBadge.className = 'badge bg-secondary';
68     wifiBadge.textContent = 'Wi-Fi Отключено';
69   }
70 }
71 }
72
73 // Audio Endpoints
74 const audioContainer = document.getElementById('diag-audio-container');
75 if (audioContainer && data.audio_endpoints) {
76   setText('diag-audio-count', `${data.audio_endpoints.length} устр.`);
77   if (data.audio_endpoints.length === 0) {
78     audioContainer.innerHTML = '<div class="text-center py-4 text-muted small">Аудиоустройств не обнаружено</div>';
79   } else {
80     audioContainer.innerHTML = data.audio_endpoints.map(
81       a => `
82         <div class="p-2 mb-1 rounded bg-black bg-opacity-30 border border-secondary-subtle d-flex align-items-center justify-content-between">
83           <div>
84             <strong class="text-white"><i class="bi bi-speaker me-1.5 text-warning"></i>${escapeHtml(a.name)}</strong>
85             <div class="small text-muted">${escapeHtml(a.type)}</div>
86           </div>
87           <span class="badge ${a.is_default ? 'bg-info text-dark fw-bold' : 'bg-secondary'}">${a.is_default ? 'По умолчанию' : 'Доступно'}</span>
88         </div>
89       `).join('');
90   }
91 }

```

```

92         // USB PnP Table
93         const usbTbody = document.getElementById('diag-usb-tbody
');
94         if (usbTbody && data.usb_devices) {
95             setText('diag-usb-count', `${data.usb_devices.length}
устройств`);
96             if (data.usb_devices.length === 0) {
97                 usbTbody.innerHTML = '<tr><td colspan="6" class="tex
t-center py-4 text-muted">USB устройств не обнаружено</td></tr
>';
98             } else {
99                 usbTbody.innerHTML = data.usb_devices.map(u => `
100                 <tr>
101                     <td class="fw-bold text-white"><i class="bi bi-u
sb-symbol text-info me-1.5"></i>${escapeHtml(u.name)}</td>
102                     <td class="font-monospace text-warning">${escape
Html(u.vendor_id)}</td>
103                     <td class="font-monospace text-info">${escapeHtm
l(u.product_id)}</td>
104                     <td class="text-muted small">${escapeHtml(u.devi
ce_class || 'USB Device')}</td>
105                     <td><span class="badge ${u.has_problem ? 'bg-dan
ger' : 'bg-success-subtle text-success border border-success'}
">${escapeHtml(u.status)}</span></td>
106                     <td class="small text-muted font-monospace text-
truncate" style="max-width: 220px;" title="${escapeHtml(u.devi
ce_id)}">${escapeHtml(u.device_id)}</td>
107                 </tr>
108                 `).join('');
109             }
110         }
111     } catch (err) {
112         console.warn('[PeripheralsTab] fetchPeripheralsNetwork e
rror:', err);
113     }
114 }
115
116 function bindEvents() {

```

```
117 |     const btnRefresh = document.getElementById('btn-diag-perip
    | herals-refresh');
118 |     if (btnRefresh) {
119 |         btnRefresh.onclick = () => fetchPeripheralsNetwork();
120 |     }
121 |
122 |     const autoSwitch = document.getElementById('diag-periphera
    | ls-auto-refresh');
123 |     if (autoSwitch) {
124 |         autoSwitch.onchange = (e) => {
125 |             if (e.target.checked) {
126 |                 autoRefreshTimer = setInterval(fetchPeripheralsNetwo
    | rk, 5000);
127 |             } else if (autoRefreshTimer) {
128 |                 clearInterval(autoRefreshTimer);
129 |                 autoRefreshTimer = null;
130 |             }
131 |         };
132 |     }
133 | }
134 |
135 | async function init() {
136 |     bindEvents();
137 |     await fetchPeripheralsNetwork();
138 | }
139 |
140 | if (document.readyState === 'loading') {
141 |     document.addEventListener('DOMContentLoaded', init);
142 | } else {
143 |     setTimeout(init, 10);
144 | }
145 | })();
146 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title>PixelRAG</title>
8 </head>
9 <body>
10   <div class="container-fluid p-3">
11     <h4><i class="bi bi-image-fill text-info"></i> PixelRAG</h
12     4>
13     <p class="text-muted">Векторный поиск по изображениям и пи
14     ксельным данным.</p>
15     <div class="card border-secondary mb-3">
16       <div class="card-header bg-dark text-white">
17         <i class="bi bi-search"></i> Поиск по изображениям
18       </div>
19       <div class="card-body">
20         <p class="text-muted">Функционал PixelRAG будет доступ
21         ен после настройки.</p>
22       </div>
23     </div>
24   </div>
25 </body>
26 </html>
```

main.js

```
1 // PixelRAG Tab Module
2 // =====
3
4 export async function initPixelRagTab() {
5   console.log('[PixelRagTab] Initializing...');
6
7   try {
8     // PixelRAG initialization logic
9     const container = document.getElementById('tab-pixelrag');
10    if (container) {
11      container.innerHTML = `
12        <div class="alert alert-info">
13          <i class="bi bi-info-circle"></i>
14          <strong>PixelRAG</strong> – векторный поиск по изобра
15          жениям и пиксельным данным.
16        </div>
17        <div class="card border-secondary">
18          <div class="card-header bg-dark text-white">
19            <i class="bi bi-image-fill"></i> Настройки PixelRA
20            G
21          </div>
22          <div class="card-body">
23            <p class="text-muted">Функционал PixelRAG будет до
24            ступен после настройки модели и индекса.</p>
25            <button class="btn btn-primary" disabled>Настроить
26            PixelRAG</button>
27          </div>
28        </div>
29      `;
30    }
31  } catch (err) {
32    console.error('[PixelRagTab] Initialization error:', err);
33  }
34 }
```

index.html

```
1 <!-- Plugins Tab -->
2 <div class="container-fluid py-2">
3   <div class="alert alert-primary d-flex flex-wrap align-items
  -center justify-content-between gap-2 mb-3 shadow-sm">
4     <div class="d-flex align-items-center gap-2">
5       <span class="fs-4">□ </span>
6       <div>
7         <h5 class="mb-0 fw-bold">Подключатель плагинов и расши
  рений</h5>
8         <p class="mb-0 small text-muted">Централизованный ката
  лог модулей, активация и органы динамической настройки</p>
9       </div>
10    </div>
11    <div class="d-flex gap-2">
12      <button class="btn btn-sm btn-outline-primary" id="btn-r
  efresh-plugins" onclick="loadPluginsList()">
13        <i class="bi bi-arrow-clockwise"></i> Обновить список
14      </button>
15    </div>
16  </div>
17
18  <div class="row g-3">
19    <!-- LEFT: Plugin Catalog / List -->
20    <div class="col-lg-4 col-md-5">
21      <div class="card shadow-sm h-100 border-secondary-subtle
  ">
22        <div class="card-header bg-body-tertiary d-flex justif
  y-content-between align-items-center py-2">
23          <span class="fw-semibold small text-uppercase">Устан
  овленные плагины (<span id="plugins-count">0</span>)</span>
24          <span class="badge bg-secondary-subtle text-body" id
  ="active-plugins-badge">0 активных</span>
25        </div>
26        <div class="p-2 border-bottom">
27          <div class="input-group input-group-sm">
28            <span class="input-group-text bg-body border-secon
```

```

dary-subtle"><i class="bi bi-search"></i></span>
29         <input type="text" class="form-control border-seco
ndary-subtle" id="plugin-search-input" placeholder="Поиск пла
гина..." oninput="filterPluginsList()">
30         </div>
31     </div>
32     <div class="card-body p-2" id="plugins-list-container"
style="max-height: 70vh; overflow-y: auto;">
33         <div class="text-center text-muted p-4">
34             <div class="spinner-border spinner-border-sm text-
primary mb-2" role="status"></div>
35             <div>Загрузка плагинов...</div>
36         </div>
37     </div>
38 </div>
39 </div>
40
41 <!-- RIGHT: Selected Plugin Controls & Settings -->
42 <div class="col-lg-8 col-md-7">
43     <div class="card shadow-sm h-100 border-secondary-subtle
" id="plugin-details-card">
44         <div class="card-body d-flex flex-column align-items-c
enter justify-content-center text-muted p-5" id="plugin-placeh
older">
45             <i class="bi bi-puzzle fs-1 mb-2"></i>
46             <h6>Выберите плагин из списка слева для управления и
настройки</h6>
47         </div>
48
49         <div class="d-none d-flex flex-column h-100" id="plugi
n-content-pane">
50             <!-- Plugin Header -->
51             <div class="card-header bg-body-tertiary d-flex just
ify-content-between align-items-center py-3 border-bottom">
52                 <div class="d-flex align-items-center gap-3">
53                     <span class="fs-2" id="plugin-icon">□ </span>
54                     <div>
55                         <div class="d-flex align-items-center gap-2">

```



```

56 |             <h5 class="mb-0 fw-bold" id="plugin-title">Название плагина</h5>
57 |             <span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="plugin-version">v1.0.0</span>
58 |             <span class="badge bg-secondary-subtle text-body border" id="plugin-category">tools</span>
59 |             <span class="badge bg-info-subtle text-info-emphasis border" id="plugin-scope-badge">Системный</span>
60 |         </div>
61 |         <small class="text-muted" id="plugin-id">id: plugin_name</small>
62 |     </div>
63 | </div>
64 |     <div class="form-check form-switch fs-5 mb-0" title="Включить / отключить плагин">
65 |         <input class="form-check-input" type="checkbox" role="switch" id="plugin-toggle-switch" onchange="toggleCurrentPlugin(this.checked)">
66 |     </div>
67 | </div>
68 |
69 |     <!-- Description -->
70 |     <div class="p-3 bg-body-secondary border-bottom">
71 |         <p class="mb-0 small text-body" id="plugin-description">Описание возможностей плагина.</p>
72 |     </div>
73 |
74 |     <!-- Body: Settings & Actions -->
75 |     <div class="card-body p-3" style="max-height: 52vh; overflow-y: auto;">
76 |         <!-- Actions Section -->
77 |         <div class="mb-4" id="plugin-actions-section">
78 |             <h6 class="fw-bold mb-2 pb-1 border-bottom d-flex align-items-center gap-2">
79 |                 <i class="bi bi-lightning-charging text-warning"></i> Управляющие действия (Actions)
80 |             </h6>

```

```

81         <div class="d-flex flex-wrap gap-2 pt-1" id="plugin-actions-container">
82             <!-- Action buttons rendered here -->
83         </div>
84     </div>
85
86     <!-- Configuration Section -->
87     <div class="mb-4" id="plugin-config-section">
88         <h6 class="fw-bold mb-2 pb-1 border-bottom d-flex align-items-center gap-2">
89             <i class="bi bi-sliders text-primary"></i> Параметры конфигурации
90         </h6>
91         <form id="plugin-config-form" onsubmit="event.preventDefault(); saveCurrentPluginConfig();">
92             <div class="row g-3" id="plugin-fields-container">
93                 <!-- Form fields rendered here -->
94             </div>
95             <div class="mt-3 text-end">
96                 <button type="submit" class="btn btn-sm btn-primary px-4" id="btn-save-plugin-config">
97                     <i class="bi bi-check2-circle"></i> Сохранить настройки
98                 </button>
99             </div>
100         </form>
101     </div>
102
103     <!-- Live Action Console / Terminal -->
104     <div class="mt-3">
105         <div class="d-flex justify-content-between align-items-center mb-1">
106             <h6 class="fw-bold mb-0 small text-muted d-flex align-items-center gap-1">
107                 <i class="bi bi-terminal"></i> Консоль выполнения
108             </h6>

```

```
109 |         <button class="btn btn-sm btn-link text-decora
    | tion-none text-muted py-0" onclick="clearPluginConsole()">Очис
    | тить</button>
110 |     </div>
111 |     <pre class="p-2 rounded bg-dark text-light small
    | font-monospace mb-0" id="plugin-console-output" style="max-he
    | ight: 180px; min-height: 60px; overflow-y: auto;">Готов к рабо
    | те.</pre>
112 |     </div>
113 | </div>
114 | </div>
115 | </div>
116 | </div>
117 | </div>
118 </div>
119 |
```

main.js

```
1 // — PLUGINS TAB MAIN.JS
2
3 let loadedPlugins = [];
4 let selectedPluginName = '';
5
6 // Notification helper
7 function showPluginNotification(message, type = 'info') {
8   if (typeof window.showNotification === 'function') {
9     window.showNotification(message, type);
10    return;
11  }
12  console.log(`[Plugin Notification ${type}]: ${message}`);
13 }
14
15 // Helper to check if current view is user interface
16 function isRunningInUserInterface() {
17   if (document.getElementById('user-interface')) return true;
18   if (window.location.pathname.includes('/user')) return true;
19   if (!document.getElementById('admin-interface')) {
20     if (typeof window.isUserAdmin === 'function') {
21       return !window.isUserAdmin();
22     }
23     return true;
24   }
25   return false;
26 }
27
28 // Initialize Plugins Tab
29 window.initPluginsTab = async function() {
30   console.log('Инициализация вкладки плагинов...');
31   await loadPluginsList();
32 };
33
34 // Загрузка списка всех плагинов
35 async function loadPluginsList() {
```

```
36 |   const container = document.getElementById('plugins-list-cont
    | ainer');
37 |   if (!container) return;
38 |
39 |   try {
40 |     const isUserUI = isRunningInUserInterface();
41 |     const endpoint = isUserUI ? '/api/plugins?scope=user' : '/
    | api/admin/plugins';
42 |
43 |     let data;
44 |     if (window.api && typeof window.api.fetch === 'function')
    | {
45 |       try {
46 |         data = await window.api.fetch(endpoint);
47 |       } catch {
48 |         data = await window.api.fetch('/api/admin/plugins');
49 |       }
50 |     } else {
51 |       let resp = await fetch(endpoint);
52 |       if (!resp.ok && isUserUI) {
53 |         resp = await fetch('/api/admin/plugins?scope=user');
54 |       }
55 |       if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
56 |       data = await resp.json();
57 |     }
58 |     let plugins = data.plugins || [];
59 |     if (isUserUI) {
60 |       plugins = plugins.filter(p => !p.is_system && p.scope !=
    | = 'system');
61 |     }
62 |     loadedPlugins = plugins;
63 |
64 |     const countEl = document.getElementById('plugins-count');
65 |     if (countEl) countEl.textContent = loadedPlugins.length;
66 |     const activeCount = loadedPlugins.filter(p => p.enabled).l
    | ength;
67 |     const badgeEl = document.getElementById('active-plugins-ba
    | dge');
```

```

68 |     if (badgeEl) badgeEl.textContent = `${activeCount} активны
    | x`;
69 |
70 |     renderPluginsList(loadedPlugins);
71 |     syncPluginTabsVisibility(loadedPlugins);
72 |
73 |     if (loadedPlugins.length > 0) {
74 |         const toSelect = loadedPlugins.find(p => p.name === sele
    | ctedPluginName) || loadedPlugins[0];
75 |         selectPlugin(toSelect.name);
76 |     } else {
77 |         showPlaceholder();
78 |     }
79 | } catch (ex) {
80 |     console.error('Ошибка загрузки плагинов:', ex);
81 |     container.innerHTML = `<div class="alert alert-danger m-2
    | small">Ошибка загрузки плагинов: ${ex.message}</div>`;
82 | }
83 | }
84 |
85 | // Отображение списка плагинов
86 | function renderPluginsList(plugins) {
87 |     const container = document.getElementById('plugins-list-cont
    | ainer');
88 |     if (!container) return;
89 |
90 |     if (plugins.length === 0) {
91 |         const isUserUI = isRunningInUserInterface();
92 |         const emptyMsg = isUserUI ? 'Пользовательские плагины не н
    | айдены' : 'Плагины не найдены';
93 |         container.innerHTML = `<div class="text-center text-muted
    | p-3 small">${emptyMsg}</div>`;
94 |         return;
95 |     }
96 |
97 |     let html = '<div class="list-group list-group-flush">';
98 |     plugins.forEach(p => {
99 |         const isSelected = p.name === selectedPluginName;

```

```

100     const statusBadge = p.enabled
101     ? '<span class="badge bg-success-subtle text-success border border-success-subtle small">Вкл</span>'
102     : '<span class="badge bg-danger-subtle text-danger border border-danger-subtle small">Выкл</span>';
103     const typeBadge = p.is_system
104     ? '<span class="badge bg-secondary-subtle text-body-secondary border small">Системный</span>'
105     : '<span class="badge bg-info-subtle text-info-emphasis border small">Пользовательский</span>';
106
107     html += `
108         <a href="javascript:void(0)"
109             class="list-group-item list-group-item-action p-2 rounded mb-1 border-0 ${isSelected ? 'active text-white' : ''}"
110             onclick="selectPlugin('${p.name}')"
111             style="cursor: pointer;">
112             <div class="d-flex w-100 justify-content-between align-items-center mb-1">
113                 <div class="d-flex align-items-center gap-2 text-truncate">
114                     <span class="fs-5">${p.icon || '□'}</span>
115                     <strong class="text-truncate">${p.title || p.name}</strong>
116                 </div>
117                 <div class="d-flex align-items-center gap-1">${typeBadge}${statusBadge}</div>
118             </div>
119             <div class="small text-truncate ${isSelected ? 'text-white-50' : 'text-muted'}">
120                 ${p.description || p.name}
121             </div>
122         </a>
123     `;
124 });
125 html += '</div>';
126 container.innerHTML = html;
127 }

```

```
128 |
129 | // Фильтрация списка плагинов
130 | function filterPluginsList() {
131 |     const query = (document.getElementById('plugin-search-input'
132 |     )?.value || '').toLowerCase().trim();
133 |     if (!query) {
134 |         renderPluginsList(loadedPlugins);
135 |         return;
136 |     }
137 |     const filtered = loadedPlugins.filter(p =>
138 |         p.name.toLowerCase().includes(query) ||
139 |         (p.title && p.title.toLowerCase().includes(query)) ||
140 |         (p.description && p.description.toLowerCase().includes(que
141 |         ry))
142 |     );
143 |     renderPluginsList(filtered);
144 | }
145 |
146 | // Выбор плагина для отображения органов управления
147 | function selectPlugin(pluginName) {
148 |     if (!pluginName) return;
149 |     selectedPluginName = pluginName;
150 |     const normalized = pluginName.toLowerCase().replace(/-/g, '_
151 |     ');
152 |     const plugin = loadedPlugins.find(p => {
153 |         const pName = (p.name || '').toLowerCase().replace(/-/g, '
154 |         _');
155 |         const pId = (p.id || '').toLowerCase().replace(/-/g, '_');
156 |         const pTitle = (p.title || '').toLowerCase();
157 |         return pName === normalized || pId === normalized || pTitl
158 |         e === pluginName.toLowerCase();
159 |     });
160 |     if (!plugin) {
161 |         if (!loadedPlugins || loadedPlugins.length === 0) {
162 |             loadPluginsList().then(() => {
163 |                 if (loadedPlugins && loadedPlugins.length > 0) {
164 |                     const pFound = loadedPlugins.find(p => {
165 |                         const pName = (p.name || '').toLowerCase().replace
```



```

    (/-/g, '_');
161     const pId = (p.id || '').toLowerCase().replace(/-/g, '_');
162     const pTitle = (p.title || '').toLowerCase();
163     return pName === normalized || pId === normalized
    || pTitle === pluginName.toLowerCase();
164     });
165     if (pFound) selectPlugin(pFound.name);
166     }
167     });
168     }
169     return;
170     }
171
172     renderPluginsList(loadedPlugins);
173
174     const placeholder = document.getElementById('plugin-placeholder');
175     const contentPane = document.getElementById('plugin-content-pane');
176     if (placeholder) placeholder.classList.add('d-none');
177     if (contentPane) contentPane.classList.remove('d-none');
178
179     // Header info
180     document.getElementById('plugin-icon').textContent = plugin.
    icon || '❏';
181     document.getElementById('plugin-title').textContent = plugin
    .title || plugin.name;
182     document.getElementById('plugin-version').textContent = `v${
    plugin.version || '1.0.0'}`;
183     document.getElementById('plugin-category').textContent = plu
    gin.category || 'tools';
184     document.getElementById('plugin-id').textContent = `id: ${pl
    ugin.name}`;
185     document.getElementById('plugin-description').textContent =
    plugin.description || 'Нет описания';
186
187     const scopeBadgeEl = document.getElementById('plugin-scope-b

```

```
    adge');
188     if (scopeBadgeEl) {
189         if (plugin.is_system) {
190             scopeBadgeEl.textContent = 'Системный';
191             scopeBadgeEl.className = 'badge bg-secondary-subtle text
- body-secondary border';
192         } else {
193             scopeBadgeEl.textContent = 'Пользовательский';
194             scopeBadgeEl.className = 'badge bg-info-subtle text-info
- emphasis border';
195         }
196     }
197
198     const isAdmin = (typeof window.isUserAdmin === 'function') ?
window.isUserAdmin() : false;
199     const isLocked = Boolean(plugin.is_system && !isAdmin);
200
201     const toggleSwitch = document.getElementById('plugin-toggle-
switch');
202     if (toggleSwitch) {
203         toggleSwitch.checked = Boolean(plugin.enabled);
204         toggleSwitch.disabled = isLocked;
205         toggleSwitch.title = isLocked
206             ? 'Системный плагин управляется администратором'
207             : 'Включить / отключить плагин';
208     }
209
210     const saveBtn = document.getElementById('btn-save-plugin-con
fig');
211     if (saveBtn) {
212         saveBtn.disabled = isLocked;
213         saveBtn.title = isLocked ? 'Настройки системного плагина и
зменяются только администратором' : '';
214     }
215
216     // Render Actions
217     renderPluginActions(plugin);
218
```

```

219 | // Render Config Fields
220 | renderPluginFields(plugin);
221 | }
222 |
223 | // Отрисовка кнопок действий (Actions)
224 | function renderPluginActions(plugin) {
225 |     const container = document.getElementById('plugin-actions-co
    ntainer');
226 |     const section = document.getElementById('plugin-actions-sect
    ion');
227 |     if (!container || !section) return;
228 |
229 |     const actions = plugin.actions || [];
230 |     if (actions.length === 0) {
231 |         section.classList.add('d-none');
232 |         return;
233 |     }
234 |
235 |     section.classList.remove('d-none');
236 |     let html = '';
237 |     actions.forEach(act => {
238 |         const btnColor = act.color || 'primary';
239 |         const label = act.label || act.name || act.id;
240 |         const icon = act.icon ? `<span class="me-1">${act.icon}</s
    pan>` : '';
241 |         const safeLabel = label.replace(/'/g, "\\'");
242 |         html += `
243 |             <button class="btn btn-sm btn-outline-${btnColor} d-inli
    ne-flex align-items-center gap-1 shadow-sm"
244 |                 onclick="executePluginAction('${plugin.name}', '
    ${act.id}', '${safeLabel}')"
245 |                 title="${act.description || ''}">
246 |                 ${icon}<span>${label}</span>
247 |             </button>
248 |         `;
249 |     });
250 |     container.innerHTML = html;
251 | }

```

```

252 |
253 | // Отрисовка полей конфигурации
254 | function renderPluginFields(plugin) {
255 |     const container = document.getElementById('plugin-fields-con
    tainer');
256 |     const section = document.getElementById('plugin-config-secti
    on');
257 |     if (!container || !section) return;
258 |
259 |     const fields = plugin.fields || [];
260 |     if (fields.length === 0) {
261 |         section.classList.add('d-none');
262 |         return;
263 |     }
264 |
265 |     section.classList.remove('d-none');
266 |     const currentCfg = plugin.config || {};
267 |
268 |     let html = '';
269 |     fields.forEach(f => {
270 |         const val = (currentCfg[f.id] !== undefined) ? currentCfg[
    f.id] : (f.default !== undefined ? f.default : '');
271 |         const desc = f.description ? `<div class="form-text text-m
    uted small">${f.description}</div>` : '';
272 |
273 |         if (f.type === 'select') {
274 |             let optionsHtml = '';
275 |             (f.options || []).forEach(opt => {
276 |                 const isSel = String(opt.value) === String(val) ? 'sel
    ected' : '';
277 |                 optionsHtml += `<option value="${opt.value}" ${isSel}>
    ${opt.label}</option>`;
278 |             });
279 |             html += `
280 |                 <div class="col-12 col-md-6">
281 |                     <label class="form-label small fw-semibold mb-1">${f
    .label}</label>
282 |                     <select class="form-select form-select-sm" name="${f

```

```

    .id}" data-field-type="select">
283         ${optionsHtml}
284     </select>
285     ${desc}
286 </div>
287 `;
288 } else if (f.type === 'boolean') {
289     const checked = Boolean(val) ? 'checked' : '';
290     html += `
291         <div class="col-12 col-md-6 d-flex flex-column justify
- content-center">
292             <div class="form-check form-switch mt-2">
293                 <input class="form-check-input" type="checkbox" ro
le="switch" name="${f.id}" id="field_${f.id}" ${checked} data-
field-type="boolean">
294                 <label class="form-check-label small fw-semibold"
for="field_${f.id}">${f.label}</label>
295             </div>
296             ${desc}
297         </div>
298     `;
299 } else if (f.type === 'list_string') {
300     const listVals = Array.isArray(val) ? val.join(', ') : (
val || '');
301     html += `
302         <div class="col-12">
303             <label class="form-label small fw-semibold mb-1">${f
.label} (через запятую)</label>
304             <input type="text" class="form-control form-control-
sm font-monospace" name="${f.id}" value="${listVals}" data-fie
ld-type="list_string">
305             ${desc}
306         </div>
307     `;
308 } else if (f.type === 'readonly') {
309     html += `
310         <div class="col-12 col-md-6">
311         <label class="form-label small fw-semibold mb-1 text

```

```

    -muted">${f.label}</label>
312         <input type="text" class="form-control form-control-
    sm bg-body-tertiary text-muted" value="${val}" readonly>
313         ${desc}
314     </div>
315     `;
316 } else if (f.type === 'number') {
317     html += `
318         <div class="col-12 col-md-6">
319             <label class="form-label small fw-semibold mb-1">${f
    .label}</label>
320             <input type="number" class="form-control form-control-
    l-sm" name="${f.id}" value="${val}" data-field-type="number">
321             ${desc}
322         </div>
323         `;
324 } else {
325     html += `
326         <div class="col-12 col-md-6">
327             <label class="form-label small fw-semibold mb-1">${f
    .label}</label>
328             <input type="text" class="form-control form-control-
    sm" name="${f.id}" value="${val}" data-field-type="string">
329             ${desc}
330         </div>
331         `;
332     }
333 });
334
335 container.innerHTML = html;
336 }
337
338 // Переключение состояния плагина (Вкл / Выкл)
339 async function toggleCurrentPlugin(enabled) {
340     if (!selectedPluginName) return;
341
342     try {
343         logToPluginConsole(`Переключение плагина '${selectedPlugin

```

```

    Name}' -> ${enabled ? 'ВКЛ' : 'ВЫКЛ'}...`);
344     const res = await window.api.fetch(`/api/admin/plugins/${s
    electedPluginName}/toggle`, {
345         method: 'POST',
346         headers: { 'Content-Type': 'application/json' },
347         body: JSON.stringify({ enabled })
348     });
349
350     const plugin = loadedPlugins.find(p => p.name === selected
    PluginName);
351     if (plugin) {
352         plugin.enabled = enabled;
353     }
354
355     renderPluginsList(loadedPlugins);
356     syncPluginTabsVisibility(loadedPlugins);
357     showPluginNotification(res.message || 'Статус плагина обно
    влен', 'success');
358     logToPluginConsole(`\u25a1 ${res.message || 'Статус успешно обн
    овлен'} `);
359 } catch (ex) {
360     console.error('Ошибка переключения плагина:', ex);
361     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
362     logToPluginConsole(`\u25a1 Ошибка: ${ex.message}`);
363     // Возвращаем тумблер в исходное состояние
364     const toggleSwitch = document.getElementById('plugin-toggl
    e-switch');
365     if (toggleSwitch) toggleSwitch.checked = !enabled;
366 }
367 }
368
369 // Сохранение настроек текущего плагина
370 async function saveCurrentPluginConfig() {
371     if (!selectedPluginName) return;
372
373     const form = document.getElementById('plugin-config-form');
374     if (!form) return;
375

```

```
376 | const newConfig = {};
377 | const elements = form.querySelectorAll('[data-field-type]');
378 |
379 | elements.forEach(el => {
380 |     const name = el.getAttribute('name');
381 |     if (!name) return;
382 |     const type = el.getAttribute('data-field-type');
383 |
384 |     if (type === 'boolean') {
385 |         newConfig[name] = el.checked;
386 |     } else if (type === 'number') {
387 |         newConfig[name] = Number(el.value);
388 |     } else if (type === 'list_string') {
389 |         newConfig[name] = el.value.split(',').map(s => s.trim())
390 |         .filter(Boolean);
391 |     } else {
392 |         newConfig[name] = el.value;
393 |     }
394 | });
395 |
396 | try {
397 |     logToPluginConsole(`Сохранение параметров плагина '${selectedPluginName}'...`);
398 |     const res = await window.api.fetch(`/api/admin/plugins/${selectedPluginName}/config`, {
399 |         method: 'POST',
400 |         headers: { 'Content-Type': 'application/json' },
401 |         body: JSON.stringify({ config: newConfig })
402 |     });
403 |
404 |     const plugin = loadedPlugins.find(p => p.name === selectedPluginName);
405 |     if (plugin) {
406 |         plugin.config = res.config || newConfig;
407 |     }
408 |
409 |     showPluginNotification('Параметры успешно сохранены', 'success');
```



```
409     logToPluginConsole(`❏ Конфигурация сохранена: ${JSON.strin
    gify(newConfig, null, 2)}`);
410   } catch (ex) {
411     console.error('Ошибка сохранения конфигурации:', ex);
412     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
413     logToPluginConsole(`❏ Ошибка сохранения: ${ex.message}`);
414   }
415 }
416
417 // Выполнение специфического действия (Action) плагина
418 async function executePluginAction(pluginName, actionId, actio
nLabel) {
419   try {
420     logToPluginConsole(`[ACTION] Запуск '${actionLabel}' для п
лагина '${pluginName}'...`);
421     showPluginNotification(`Запуск '${actionLabel}'...`, 'info
');
422
423     // Собираем текущие поля формы как параметры действия
424     const form = document.getElementById('plugin-config-form')
;
425     const params = {};
426     if (form) {
427       const elements = form.querySelectorAll('[data-field-type
]');
428       elements.forEach(el => {
429         const name = el.getAttribute('name');
430         if (!name) return;
431         const type = el.getAttribute('data-field-type');
432         if (type === 'boolean') params[name] = el.checked;
433         else if (type === 'number') params[name] = Number(el.v
alue);
434         else if (type === 'list_string') params[name] = el.val
ue.split(',').map(s => s.trim()).filter(Boolean);
435         else params[name] = el.value;
436       });
437     }
438
```

```

439 // Проверяем, есть ли обязательные параметры действия, кот
    орых нет в форме
440 const plugin = (typeof loadedPlugins !== 'undefined') ? lo
    adedPlugins.find(p => p.name === pluginName) : null;
441 const actDef = plugin?.actions?.find(a => a.id === actionI
    d);
442 if (actDef && Array.isArray(actDef.parameters)) {
443     for (const param of actDef.parameters) {
444         if (param.required && (params[param.name] === undefine
    d || params[param.name] === '')) {
445             const userVal = prompt(`${param.description || param
    .name}:`);
446             if (userVal === null) {
447                 logToPluginConsole(`[ACTION] Отменено пользователе
    м.`);
448                 return;
449             }
450             params[param.name] = userVal;
451         }
452     }
453 }
454
455 const res = await window.api.fetch(`/api/admin/plugins/${p
    luginName}/action/${actionId}`, {
456     method: 'POST',
457     headers: { 'Content-Type': 'application/json' },
458     body: JSON.stringify({ params })
459 });
460
461 if (res.success || res.status === 'success') {
462     showPluginNotification(res.message || 'Действие успешно
    выполнено', 'success');
463     logToPluginConsole(`\n ${res.message || 'Успех'}\n${JSON.
    stringify(res.result || res.results || res, null, 2)}`);
464 } else {
465     showPluginNotification(res.message || 'Ошибка выполнения
    действия', 'warning');
466     logToPluginConsole(`\n ${res.message || 'Завершено с пред

```

```

    предупреждением'}`);
467     }
468   } catch (ex) {
469     console.error(`Ошибка выполнения действия ${actionId}:`, e
x);
470     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
471     logToPluginConsole(`❌ Ошибка выполнения: ${ex.message}`);
472   }
473 }
474
475 // Логирование в консоль плагина
476 function logToPluginConsole(text) {
477   const consoleEl = document.getElementById('plugin-console-ou
tput');
478   if (!consoleEl) return;
479   const time = new Date().toLocaleTimeString();
480   consoleEl.textContent = `[${time}] ${text}\n` + consoleEl.te
xtContent;
481 }
482
483 // Очистка консоли
484 function clearPluginConsole() {
485   const consoleEl = document.getElementById('plugin-console-ou
tput');
486   if (consoleEl) consoleEl.textContent = 'Консоль очищена.\n';
487 }
488
489 function showPlaceholder() {
490   const placeholder = document.getElementById('plugin-placehol
der');
491   const contentPane = document.getElementById('plugin-content-
pane');
492   if (placeholder) placeholder.classList.remove('d-none');
493   if (contentPane) contentPane.classList.add('d-none');
494 }
495
496 // Переход к конкретному плагину из выпадающего меню
497 function openPluginFromDropdown(pluginName) {

```

```
498 |   if (!pluginName) return;
499 |
500 |   // 1. Закрываем открытые выпадающие списки
501 |   document.querySelectorAll('#mainTabs .dropdown-menu.show').f
forEach((m) => {
502 |     m.classList.remove('show');
503 |     m.closest('.dropdown')?.querySelector('.dropdown-toggle')?
.classList.remove('show');
504 |   });
505 |
506 |   const normalized = pluginName.toLowerCase().replace(/-/g, '_
');
507 |
508 |   // 2. Если у плагина есть своя выделенная вкладка в DOM, отк
рываем ее
509 |   if (normalized === 'telegram_channel_rag' || normalized ===
'telegram_rag') {
510 |     const pane = document.getElementById('tab-telegram-rag') |
| document.getElementById('tab-telegram_rag');
511 |     if (pane && typeof window.switchTab === 'function') {
512 |       window.switchTab('tab-telegram-rag');
513 |       return;
514 |     }
515 |   }
516 |   if (normalized === 'news_feed' || normalized === 'news' || n
ormalized === 'smart_news' || normalized === 'smart_feed_news'
) {
517 |     const pane = document.getElementById('tab-news');
518 |     if (pane && typeof window.switchTab === 'function') {
519 |       window.switchTab('tab-news');
520 |       return;
521 |     }
522 |   }
523 |
524 |   // 3. Иначе переключаемся на вкладку плагинов и выбираем пла
гин в каталоге
525 |   if (typeof window.switchTab === 'function') {
526 |     window.switchTab('tab-plugins');
```

```

527 | }
528 |
529 | const select = () => {
530 |     if (typeof window.selectPlugin === 'function') {
531 |         window.selectPlugin(pluginName);
532 |     }
533 | };
534 |
535 | select();
536 | setTimeout(select, 150);
537 | setTimeout(select, 400);
538 | }
539 |
540 | // Синхронизация видимости вкладок в навбаре с состоянием акти
    | вности плагинов
541 | function syncPluginTabsVisibility(plugins) {
542 |     if (!Array.isArray(plugins)) return;
543 |     const pluginMap = {};
544 |     plugins.forEach(p => {
545 |         pluginMap[p.name] = Boolean(p.enabled);
546 |     });
547 |
548 |     // 1. Обновляем выпадающие списки включенных плагинов в шапк
    | е
549 |     const enabledContainers = document.querySelectorAll('#nav-en
    | abled-plugins-list, .nav-enabled-plugins-list');
550 |     const enabledPlugins = plugins.filter(p => p.enabled);
551 |
552 |     enabledContainers.forEach(container => {
553 |         if (enabledPlugins.length === 0) {
554 |             container.innerHTML = `
555 |                 <div class="list-group-item dropdown-item text-muted s
    | mall py-2 px-3 fst-italic">
556 |                     Нет активных плагинов
557 |                 </div>
558 |             `;
559 |         } else {
560 |             let html = '';

```

```

561     enabledPlugins.forEach(p => {
562         const icon = p.icon || '❏';
563         const title = p.title || p.name;
564         const desc = p.description || title;
565         html += `
566             <button class="list-group-item list-group-item-action dropdown-item d-flex align-items-center gap-2 py-2 px-3 text-truncate"
567                 type="button"
568                 data-plugin="${p.name}"
569                 title="${desc}"
570                 onclick="openPluginFromDropdown('${p.name}')"
571             >
572                 <span>${icon}</span>
573                 <span class="text-truncate">${title}</span>
574             </button>
575         `;
576     });
577     container.innerHTML = html;
578 });
579
580 // 2. Находим все элементы навигации, привязанные к отдельным плагинам
581 const pluginNavItems = document.querySelectorAll('[data-plugin-tab]');
582 pluginNavItems.forEach(item => {
583     const pluginName = item.getAttribute('data-plugin-tab');
584     const isEnabled = (pluginMap[pluginName] !== undefined) ? pluginMap[pluginName] : false;
585
586     if (isEnabled) {
587         item.classList.remove('d-none');
588     } else {
589         item.classList.add('d-none');
590         // Если отключенная вкладка была активна, переключаемся на вкладку чата
591         const button = item.querySelector('.nav-link, .dropdown-

```

```

        item');
592         if (button && button.classList.contains('active')) {
593             if (typeof window.switchTab === 'function') {
594                 window.switchTab('tab-chat');
595             } else {
596                 const chatTab = document.querySelector('[data-bs-target="#tab-chat"]');
597                 if (chatTab) {
598                     const tab = new bootstrap.Tab(chatTab);
599                     tab.show();
600                 }
601             }
602         }
603     });
604 });
605 }
606
607 // Make functions globally available for HTML inline handlers
608 window.loadPluginsList = loadPluginsList;
609 window.filterPluginsList = filterPluginsList;
610 window.selectPlugin = selectPlugin;
611 window.toggleCurrentPlugin = toggleCurrentPlugin;
612 window.saveCurrentPluginConfig = saveCurrentPluginConfig;
613 window.executePluginAction = executePluginAction;
614 window.clearPluginConsole = clearPluginConsole;
615 window.syncPluginTabsVisibility = syncPluginTabsVisibility;
616 window.openPluginFromDropdown = openPluginFromDropdown;
617
618 // Auto-run initialization
619 if (document.readyState === 'loading') {
620     document.addEventListener('DOMContentLoaded', () => {
621         if (typeof window.initPluginsTab === 'function') window.initPluginsTab();
622     });
623 } else {
624     if (typeof window.initPluginsTab === 'function') window.initPluginsTab();
625 }

```


index.html

```
1 <!--
2 =====
3 |
4 |=====
5 | Process Name: Windows Process Leak Hunter & Micro-freezes Tab
6 |=====
7 |
8 | Description:
9 |     Вкладка глубокой диагностики скрытых утечек дескрипторов (
10 | Handles), GDI/USER объектов,
11 |     страничных ошибок памяти (Page Faults) и микрофризов в про
12 | цессах Windows.
13 |
14 | File: index.html
15 | Project: AI-Breadboard
16 | Module: WebInterface.ProcessLeaksTab
17 | Author: hypo69
18 | Copyright: © 2026 hypo69
19 |=====
20 |=====
21 -->
22 <style>
23 .leaks-tab-wrapper {
24     padding: 0.75rem;
25     color: var(--text-color, #f8fafc);
26 }
27 .diag-meter-card {
28     background: var(--surface-2, #1e293b);
29     border: 1px solid var(--border-color, #334155);
30     border-radius: var(--radius-md, 8px);
31     padding: 0.75rem 1rem;
32 }
33 .diag-badge-critical {
34     background: rgba(239, 68, 68, 0.2);
35     color: #f87171;
36     border: 1px solid rgba(239, 68, 68, 0.4);
37     font-size: 0.70rem;
```

```
32     padding: 2px 6px;
33     border-radius: 4px;
34     font-weight: 600;
35 }
36 .diag-badge-warning {
37     background: rgba(234, 179, 8, 0.2);
38     color: #fde047;
39     border: 1px solid rgba(234, 179, 8, 0.4);
40     font-size: 0.70rem;
41     padding: 2px 6px;
42     border-radius: 4px;
43     font-weight: 600;
44 }
45 .diag-badge-normal {
46     background: rgba(34, 197, 94, 0.15);
47     color: #4ade80;
48     border: 1px solid rgba(34, 197, 94, 0.3);
49     font-size: 0.70rem;
50     padding: 2px 6px;
51     border-radius: 4px;
52     font-weight: 500;
53 }
54 .diag-panel {
55     background: var(--surface-2, #1e293b);
56     border: 1px solid var(--border-color, #334155);
57     border-radius: 8px;
58     overflow: hidden;
59 }
60 .diag-panel-header {
61     background: rgba(15, 23, 42, 0.6);
62     padding: 8px 14px;
63     border-bottom: 1px solid var(--border-color, #334155);
64     font-weight: 600;
65     font-size: 0.85rem;
66 }
67 </style>
68
69 <div class="leaks-tab-wrapper">
```

```
70 | <!-- Top Header & Status -->
71 | <div class="card shadow-sm border-secondary-subtle mb-3" style="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%); border-radius: 10px;">
72 |     <div class="card-body p-3">
73 |         <div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
74 |             <div class="d-flex align-items-center gap-2">
75 |                 <div class="fs-4">❏ </div>
76 |                 <div>
77 |                     <h5 class="mb-0 fw-bold text-white">1. Утечки ресурсов и микрофизы (Leak Hunter)</h5>
78 |                     <small class="text-muted">Глубокий Win32 аудит: дескрипторы Handles, GDI/USER объекты GUI, Hard Page Faults и риски утечек</small>
79 |                 </div>
80 |             </div>
81 |
82 |             <div class="d-flex align-items-center gap-2 flex-wrap">
83 |                 <div class="form-check form-switch mb-0 small text-muted d-flex align-items-center gap-1">
84 |                     <input class="form-check-input" type="checkbox" id="diag-leaks-auto-refresh">
85 |                     <label class="form-check-label small text-nowrap" for="diag-leaks-auto-refresh">Авто (5с)</label>
86 |                 </div>
87 |                 <button class="btn btn-sm btn-outline-info rounded-pill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-leaks-refresh">
88 |                     <i class="bi bi-arrow-clockwise"></i> Обновить
89 |                 </button>
90 |             </div>
91 |         </div>
92 |     </div>
93 | </div>
94 |
95 | <!-- Top Leak Hunter KPI Metrics -->
```

```
96 | <div class="row g-2 mb-3">
97 |   <div class="col-12 col-sm-6 col-lg-3">
98 |     <div class="diag-meter-card">
99 |       <div class="d-flex justify-content-between align-items
| -center small text-muted">
100 |         <span>Проанализировано процессов</span>
101 |         <i class="bi bi-cpu text-info"></i>
102 |       </div>
103 |       <div class="fs-4 fw-bold text-info mt-1" id="diag-leak
| s-total-proc">0</div>
104 |       <div class="small text-muted">Полнотекстовый срез Win3
| 2 API</div>
105 |     </div>
106 |   </div>
107 |   <div class="col-12 col-sm-6 col-lg-3">
108 |     <div class="diag-meter-card">
109 |       <div class="d-flex justify-content-between align-items
| -center small text-muted">
110 |         <span>С признаками утечек</span>
111 |         <i class="bi bi-exclamation-triangle text-danger"></
| i>
112 |       </div>
113 |       <div class="fs-4 fw-bold text-danger mt-1" id="diag-le
| aks-suspicious-count">0</div>
114 |       <div class="small text-muted">Handles / GDI / Page Fau
| lts</div>
115 |     </div>
116 |   </div>
117 |   <div class="col-12 col-sm-6 col-lg-3">
118 |     <div class="diag-meter-card">
119 |       <div class="d-flex justify-content-between align-items
| -center small text-muted">
120 |         <span>Пик дескрипторов (Handles)</span>
121 |         <i class="bi bi-box text-warning"></i>
122 |       </div>
123 |       <div class="fs-4 fw-bold text-warning mt-1" id="diag-l
| eaks-peak-handles">--</div>
124 |       <div class="small text-muted text-truncate" id="diag-l
```

```

eaks-peak-handles-proc">Процесс: --</div>
125     </div>
126   </div>
127   <div class="col-12 col-sm-6 col-lg-3">
128     <div class="diag-meter-card">
129       <div class="d-flex justify-content-between align-items
-center small text-muted">
130         <span>Пик GDI объектов</span>
131         <i class="bi bi-palette text-primary"></i>
132       </div>
133       <div class="fs-4 fw-bold text-primary mt-1" id="diag-l
eaks-peak-gdi">--</div>
134       <div class="small text-muted text-truncate" id="diag-l
eaks-peak-gdi-proc">Лимит OS: 10 000</div>
135     </div>
136   </div>
137 </div>
138
139 <!-- Main Process Leak Hunter Table -->
140 <div class="diag-panel">
141   <div class="diag-panel-header d-flex justify-content-betwe
en align-items-center flex-wrap gap-2">
142     <div class="d-flex align-items-center gap-2">
143       <i class="bi bi-shield-shaded text-warning"></i>
144       <span class="text-white">Детектор утечек дескрипторов,
GDI и страничных ошибок (Leak Hunter)</span>
145     </div>
146     <div class="d-flex align-items-center gap-2 flex-wrap">
147       <div class="btn-group btn-group-sm role="group" id="d
iag-leaks-filter-group">
148         <button type="button" class="btn btn-outline-info ac
tive py-0.5 px-2" data-filter="all" style="font-size: 0.72rem;
">Все</button>
149         <button type="button" class="btn btn-outline-danger
py-0.5 px-2" data-filter="suspicious" style="font-size: 0.72re
m;">⚠ Подозрительные</button>
150         <button type="button" class="btn btn-outline-seconda
ry py-0.5 px-2" data-filter="gdi" style="font-size: 0.72rem;">

```

```

GDI > 500</button>
151 |         <button type="button" class="btn btn-outline-seconda
ry py-0.5 px-2" data-filter="handles" style="font-size: 0.72re
m;">Handles > 1500</button>
152 |     </div>
153 |     <input type="text" class="form-control form-control-sm
" id="diag-leaks-search" placeholder="Поиск по PID или имени..
." style="max-width: 180px; font-size: 0.74rem; height: 26px;"
>
154 | </div>
155 | </div>
156 | <div class="table-responsive" style="max-height: 540px; ov
erflow-y: auto;">
157 |     <table class="table table-dark table-hover table-sm mb-0
" style="font-size: 0.78rem; font-family: ui-monospace, SFMono
-Regular, Menlo, Monaco, Consolas, monospace;">
158 |         <thead>
159 |             <tr class="text-secondary border-secondary">
160 |                 <th style="width: 70px;">PID</th>
161 |                 <th>Имя процесса</th>
162 |                 <th style="width: 90px; text-align: right;">Handle
s</th>
163 |                 <th style="width: 90px; text-align: right;">GDI Ob
ъекты</th>
164 |                 <th style="width: 90px; text-align: right;">USER O
бъекты</th>
165 |                 <th style="width: 120px; text-align: right;">Page
Faults</th>
166 |                 <th style="width: 90px; text-align: right;">Память
(RSS)</th>
167 |                 <th style="width: 70px; text-align: right;">Потоки
</th>
168 |                 <th style="width: 100px;">Статус риска</th>
169 |                 <th>Признаки / Аномалии</th>
170 |             </tr>
171 |         </thead>
172 |         <tbody id="diag-leaks-tbody">
173 |             <tr><td colspan="10" class="text-center py-4 text-mu

```

```
    ted">Загрузка данных глубокой диагностики процессов...</td></tr>
174 |         </tbody>
175 |     </table>
176 | </div>
177 | </div>
178 | </div>
179 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Process Leak Hunter Logic
4   * =====
5   * Description:
6   *   Клиентский контроллер вкладки глубокой диагностики процес
7   *   утечки дескрипторов Handles, GDI/USER объектов, Page Faul
8   *   ts и фильтрация.
9   *
10  * File: main.js
11  * Project: AI-Breadboard
12  * Module: WebInterface.ProcessLeaksTab
13  * Author: hypo69
14  * Copyright: © 2026 hypo69
15  * =====
16  */
17 (function () {
18     'use strict';
19
20     let rawLeakProcesses = [];
21     let leakFilter = 'all';
22     let autoRefreshTimer = null;
23
24     function escapeHtml(str) {
25         if (str === null || str === undefined) return '';
26         return String(str)
27             .replace(/&/g, '&amp;')
28             .replace(/</g, '&lt;')
29             .replace(/>/g, '&gt;')
30             .replace(/"/g, '&quot;')
31             .replace(/'/g, '&#039;');
```



```

32 | }
33 |
34 | function setText(id, text) {
35 |     const el = document.getElementById(id);
36 |     if (el) el.textContent = text !== null && text !== undefin
    | ed ? String(text) : '--';
37 | }
38 |
39 | async function apiFetch(url) {
40 |     if (window.api && typeof window.api.fetch === 'function')
    | {
41 |         return await window.api.fetch(url);
42 |     }
43 |     const res = await fetch(url);
44 |     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
    | atusText}`);
45 |     return await res.json();
46 | }
47 |
48 | async function fetchProcessLeaks() {
49 |     try {
50 |         const data = await apiFetch('/api/v1/system/diagnostics/
    | leaks?limit=100');
51 |         if (!data) return;
52 |
53 |         setText('diag-leaks-total-proc', data.total_processes);
54 |         setText('diag-leaks-suspicious-count', data.suspicious_c
    | ount);
55 |
56 |         if (data.top_handle_hogs && data.top_handle_hogs.length
    | > 0) {
57 |             const topH = data.top_handle_hogs[0];
58 |             setText('diag-leaks-peak-handles', `${topH.handles_cou
    | nt.toLocaleString()} шт.`);
59 |             setText('diag-leaks-peak-handles-proc', `Процесс: ${to
    | pH.name} (PID ${topH.pid})`);
60 |         }
61 |         if (data.top_gdi_hogs && data.top_gdi_hogs.length > 0) {

```

```

62         const topG = data.top_gdi_hogs[0];
63         setText('diag-leaks-peak-gdi', `${topG.gdi_objects.toLocaleString()} шт.`);
64         setText('diag-leaks-peak-gdi-proc', `Процесс: ${topG.name} (PID ${topG.pid})`);
65     }
66
67     rawLeakProcesses = data.all_processes || [];
68     renderProcessLeaksTable(rawLeakProcesses);
69 } catch (err) {
70     console.warn('[ProcessLeaksTab] fetchProcessLeaks error: ', err);
71 }
72 }
73
74 function renderProcessLeaksTable(processes) {
75     const tbody = document.getElementById('diag-leaks-tbody');
76     if (!tbody) return;
77
78     const query = (document.getElementById('diag-leaks-search')?.value || '').trim().toLowerCase();
79
80     let filtered = processes.filter(p => {
81         if (query && !p.name.toLowerCase().includes(query) && !String(p.pid).includes(query)) {
82             return false;
83         }
84         if (leakFilter === 'suspicious') return p.leak_risk_score !== 'normal';
85         if (leakFilter === 'gdi') return p.gdi_objects > 500;
86         if (leakFilter === 'handles') return p.handles_count > 1500;
87         return true;
88     });
89
90     if (filtered.length === 0) {
91         tbody.innerHTML = '<tr><td colspan="10" class="text-center py-4 text-muted">Процессов по заданным критериям не найдено

```

```

    </td></tr>';
92     return;
93 }
94
95     tbody.innerHTML = filtered.map(p => {
96         let riskBadge = '<span class="diag-badge-normal">Норма</span>';
97         if (p.leak_risk_score === 'critical') {
98             riskBadge = '<span class="diag-badge-critical">Критиче
99 ский</span>';
100         } else if (p.leak_risk_score === 'warning') {
101             riskBadge = '<span class="diag-badge-warning">Внимание
102 </span>';
103         }
104
105         const reasons = (p.leak_risk_reasons && p.leak_risk_reas
106 ons.length > 0)
107         ? p.leak_risk_reasons.map(r => `<span class="badge bg-
108 dark border border-secondary text-warning me-1">${escapeHtml(r
109 )}</span>`).join('')
110         : '<span class="text-muted small">Штатное поведение</s
111 pan>';
112
113         return `
114             <tr>
115                 <td><span class="text-info font-monospace">${p.pid}<
116 /span></td>
117                 <td class="fw-bold text-white text-truncate" style="
118 max-width: 180px;">${escapeHtml(p.name)}</td>
119                 <td style="text-align: right;" class="${p.handles_co
120 unt > 1500 ? 'text-danger fw-bold' : 'text-light'}">${p.handle
121 s_count.toLocaleString()}</td>
122                 <td style="text-align: right;" class="${p.gdi_object
123 s > 800 ? 'text-danger fw-bold' : 'text-light'}">${p.gdi_objec
124 ts.toLocaleString()}</td>
125                 <td style="text-align: right;" class="${p.user_objec
126 ts > 400 ? 'text-warning' : 'text-light'}">${p.user_objects.to
127 LocaleString()}</td>

```

```

114 |         <td style="text-align: right;" class="text-muted font-mono">${p.page_faults_total.toLocaleString()}</td>
115 |         <td style="text-align: right;" class="text-warning">
116 |         <td style="text-align: right;" class="text-muted">${
117 |         <td>${riskBadge}</td>
118 |         <td>${reasons}</td>
119 |     </tr>
120 |     `;
121 |     }).join('');
122 | }
123 |
124 | function bindEvents() {
125 |     const leakFilters = document.getElementById('diag-leaks-filter-group');
126 |     if (leakFilters) {
127 |         leakFilters.querySelectorAll('button').forEach(btn => {
128 |             btn.onclick = () => {
129 |                 leakFilters.querySelectorAll('button').forEach(b =>
130 |                 {
131 |                     b.classList.remove('active', 'btn-info', 'btn-danger');
132 |                     b.classList.add('btn-outline-secondary');
133 |                 });
134 |                 const f = btn.getAttribute('data-filter') || 'all';
135 |                 btn.classList.add('active', f === 'suspicious' ? 'btn-danger' : 'btn-info');
136 |                 btn.classList.remove('btn-outline-secondary');
137 |                 leakFilter = f;
138 |                 renderProcessLeaksTable(rawLeakProcesses);
139 |             });
140 |         }
141 |
142 |     const leakSearch = document.getElementById('diag-leaks-search');
143 |     if (leakSearch) {

```

```
144 |     leakSearch.oninput = () => renderProcessLeaksTable(rawLe
    | akProcesses);
145 |   }
146 |
147 |   const btnLeaksRefresh = document.getElementById('btn-diag-
    | leaks-refresh');
148 |   if (btnLeaksRefresh) {
149 |     btnLeaksRefresh.onclick = () => fetchProcessLeaks();
150 |   }
151 |
152 |   const autoSwitch = document.getElementById('diag-leaks-aut
    | o-refresh');
153 |   if (autoSwitch) {
154 |     autoSwitch.onchange = (e) => {
155 |       if (e.target.checked) {
156 |         autoRefreshTimer = setInterval(fetchProcessLeaks, 50
    | 00);
157 |       } else if (autoRefreshTimer) {
158 |         clearInterval(autoRefreshTimer);
159 |         autoRefreshTimer = null;
160 |       }
161 |     };
162 |   }
163 | }
164 |
165 | async function init() {
166 |   bindEvents();
167 |   await fetchProcessLeaks();
168 | }
169 |
170 | if (document.readyState === 'loading') {
171 |   document.addEventListener('DOMContentLoaded', init);
172 | } else {
173 |   setTimeout(init, 10);
174 | }
175 | })();
176 |
```

index.html

```
1 <!-- RAG Knowledge Base & Document / Codebase Management Tab -
   -->
2 <div class="container-fluid py-2">
3   <!-- Header Alert with Sub-Navigation -->
4   <div class="alert alert-primary d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 shadow-sm">
5     <div>
6       <h5 class="mb-1"><i class="bi bi-brain me-2"></i><span data-i18n="rag.title">База знаний и RAG</span></h5>
7       <p class="mb-0 small text-secondary" data-i18n="rag.subtitle">Загрузка документов, индексация исходного кода проектов и семантический поиск.</p>
8     </div>
9     <div class="d-flex align-items-center gap-2">
10      <!-- Sub-Tabs Navigation -->
11      <ul class="nav nav-pills nav-fill bg-body-secondary p-1 rounded-pill" id="rag-mode-pills" role="tablist">
12        <li class="nav-item" role="presentation">
13          <button class="nav-link active rounded-pill btn-sm px-3 py-1" id="pills-user-rags-tab" data-bs-toggle="pill" data-bs-target="#pills-user-rags" type="button" role="tab">
14            <i class="bi bi-folder2-open me-1"></i> Коллекции Workspace
15          </button>
16        </li>
17        <li class="nav-item" role="presentation">
18          <button class="nav-link rounded-pill btn-sm px-3 py-1" id="pills-doc-rag-tab" data-bs-toggle="pill" data-bs-target="#pills-doc-rag" type="button" role="tab">
19            <i class="bi bi-file-earmark-text me-1"></i> Документы
20          </button>
21        </li>
22        <li class="nav-item" role="presentation">
23          <button class="nav-link rounded-pill btn-sm px-3 py-1" id="pills-code-rag-tab" data-bs-toggle="pill" data-bs-target="#pills-code-rag" type="button" role="tab">
```

```

t="#pills-code-rag" type="button" role="tab">
24     <i class="bi bi-code-slash me-1"></i> Код проектов
    (AST)
25     </button>
26 </li>
27 <li class="nav-item" role="presentation">
28     <button class="nav-link rounded-pill btn-sm px-3 py-
1" id="pills-pixel-rag-tab" data-bs-toggle="pill" data-bs-targ
et="#pills-pixel-rag" type="button" role="tab">
29         <i class="bi bi-image me-1 text-warning"></i> Pixe
lRAG (Визуальный поиск)
30     </button>
31 </li>
32 </ul>
33 <button class="btn btn-outline-primary btn-sm rounded-pi
ll ms-2" id="btn-refresh-rag">
34     <i class="bi bi-arrow-clockwise me-1"></i> <span data-
i18n="common.refresh">Обновить</span>
35 </button>
36 </div>
37 </div>
38
39 <div class="tab-content" id="rag-tab-content">
40
41 <!-- SUB-TAB 0: USER WORKSPACE MULTI-RAG -->
42 <div class="tab-pane fade show active" id="pills-user-rags
" role="tabpanel">
43     <div class="row g-3">
44         <!-- LEFT: Collections List & Create -->
45         <div class="col-lg-4 col-md-12">
46             <div class="card shadow-sm border-0 mb-3">
47                 <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
48                     <span class="fw-semibold"><i class="bi bi-folder
2-open me-2"></i>Базы знаний (RAG)</span>
49                     <button class="btn btn-primary btn-sm rounded-pi
ll px-2 py-0" id="btn-show-create-rag-modal" data-bs-toggle="m
odal" data-bs-target="#modal-create-user-rag">

```

```

50         <i class="bi bi-plus-lg me-1"></i> Создать
51     </button>
52 </div>
53     <div class="card-body p-2">
54         <div id="user-rags-list-container" class="d-flex
flex-column gap-2" style="max-height: 480px; overflow-y: auto
; ">
55             <div class="text-center text-muted py-4 small"
>
56                 <i class="bi bi-folder-x fs-3 d-block mb-1 o
pacity-50"></i>
57                 <span>Нет созданных коллекций.</span><br>
58                 <span>Нажмите «+ Создать» выше.</span>
59             </div>
60         </div>
61     </div>
62 </div>
63 </div>
64
65     <!-- RIGHT: Selected Collection Details, File Pipeline
, & Search -->
66     <div class="col-lg-8 col-md-12">
67         <div id="user-rag-empty-state" class="card shadow-sm
border-0 p-5 text-center text-muted">
68             <i class="bi bi-diagram-3 fs-1 mb-2 text-primary o
pacity-50"></i>
69             <h5>Выберите RAG-коллекцию</h5>
70             <p class="small text-secondary mb-3">Выберите суще
ствующую базу знаний слева или создайте новую для подготовки и
индексации документов.</p>
71         <div>
72             <button class="btn btn-outline-primary btn-sm ro
unded-pill px-3" data-bs-toggle="modal" data-bs-target="#modal
-create-user-rag">
73                 <i class="bi bi-plus-lg me-1"></i> Создать пер
вую коллекцию
74             </button>
75         </div>

```



```

76 |         </div>
77 |
78 |         <div id="user-rag-detail-container" class="d-none">
79 |             <!-- Collection Header Card -->
80 |             <div class="card shadow-sm border-0 mb-3">
81 |                 <div class="card-header bg-body-tertiary d-flex
| justify-content-between align-items-center flex-wrap gap-2 py-
| 2">
82 |                     <div>
83 |                         <h6 class="mb-0 fw-bold d-flex align-items-c
| enter gap-2">
84 |                             <i class="bi bi-folder2 text-primary"></i>
85 |                             <span id="active-rag-title">Название колле
| кции</span>
86 |                             <span id="active-rag-status-badge" class="
| badge bg-secondary">created</span>
87 |                         </h6>
88 |                         <small class="text-muted" id="active-rag-des
| с">Описание коллекции</small>
89 |                     </div>
90 |                     <div class="d-flex align-items-center gap-2">
91 |                         <button class="btn btn-outline-danger btn-sm
| rounded-pill" id="btn-delete-active-rag" title="Удалить колле
| кцию">
92 |                             <i class="bi bi-trash"></i>
93 |                         </button>
94 |                     </div>
95 |                 </div>
96 |                 <div class="card-body p-3">
97 |                     <!-- Stats Row -->
98 |                     <div class="row g-2 mb-3 text-center">
99 |                         <div class="col-sm-4">
100 |                             <div class="p-2 border rounded bg-body-sec
| ondary bg-opacity-25">
101 |                                 <div class="small text-muted">Чанков в и
| ндексе</div>
102 |                                 <div class="fw-bold fs-5 text-success" i
| d="active-rag-stat-chunks">0</div>

```

```

103         </div>
104     </div>
105     <div class="col-sm-4">
106         <div class="p-2 border rounded bg-body-sec
ondary bg-opacity-25">
107             <div class="small text-muted">Прикреплен
о файлов</div>
108             <div class="fw-bold fs-5 text-primary" i
d="active-rag-stat-files">0</div>
109         </div>
110     </div>
111     <div class="col-sm-4">
112         <div class="p-2 border rounded bg-body-sec
ondary bg-opacity-25">
113             <div class="small text-muted">Обновлено<
/div>
114             <div class="small fw-bold mt-1 text-trun
cate" id="active-rag-stat-updated">—</div>
115         </div>
116     </div>
117 </div>
118
119 <!-- Pipeline & Files Section -->
120 <div class="card border border-secondary-subtl
e mb-3">
121     <div class="card-header bg-light d-flex just
ify-content-between align-items-center py-1 px-3">
122         <span class="small fw-bold"><i class="bi b
i-file-earmark-check me-1"></i> Файлы для очистки и сборки (ra
g_cleaner)</span>
123         <button class="btn btn-link btn-sm text-de
coration-none p-0" id="btn-select-all-user-files">Выбрать все<
/button>
124     </div>
125     <div class="card-body p-2">
126         <div id="user-rag-files-checklist" class="
d-flex flex-column gap-1" style="max-height: 140px; overflow-y
: auto;">

```

```

127 |             <div class="text-muted small p-2">Загруз
ка файлов...</div>
128 |         </div>
129 |         <div class="d-flex justify-content-between
align-items-center mt-2 pt-2 border-top flex-wrap gap-2">
130 |             <div class="d-flex align-items-center ga
p-2 flex-wrap">
131 |                 <input type="file" id="user-rag-quick-
upload-input" class="d-none" multiple>
132 |                 <button class="btn btn-outline-seconda
ry btn-sm rounded-pill" type="button" onclick="document.getEle
mentById('user-rag-quick-upload-input').click();">
133 |                     <i class="bi bi-upload me-1"></i> За
грузить файлы
134 |                 </button>
135 |                 <button class="btn btn-outline-info bt
n-sm rounded-pill d-flex align-items-center gap-1" id="btn-imp
ort-gdocs-rag" type="button" data-bs-toggle="modal" data-bs-ta
rget="#modal-import-gdocs">
136 |                     <i class="bi bi-google text-danger">
</i>
137 |                     <span>Google Docs</span>
138 |                 </button>
139 |             </div>
140 |             <button class="btn btn-success btn-sm ro
unded-pill px-3 fw-semibold shadow-sm" id="btn-build-active-ra
g">
141 |                 <i class="bi bi-magic me-1"></i> Очист
ить и собрать RAG
142 |             </button>
143 |         </div>
144 |         <div id="user-rag-build-status-alert" clas
s="d-none alert p-2 small mt-2 mb-0"></div>
145 |     </div>
146 | </div>
147 |
148 | <!-- Q&A and Knowledge Entries Section -->
149 | <div class="card border border-secondary-subtl

```

```

e mb-3">
150         <div class="card-header bg-light d-flex justify-content-between align-items-center py-1 px-3">
151             <span class="small fw-bold"><i class="bi bi-chat-quote me-1 text-primary"></i> Записи и связи «Вопрос-0
твет»</span>
152             <div class="d-flex align-items-center gap-2">
153                 <button class="btn btn-outline-secondary btn-sm py-0 px-2 rounded-pill" id="btn-refresh-rag-entries" title="Обновить записи">
154                     <i class="bi bi-arrow-clockwise"></i>
155                 </button>
156                 <button class="btn btn-primary btn-sm rounded-pill px-2 py-0 fw-semibold" id="btn-show-add-qa-modal" data-bs-toggle="modal" data-bs-target="#modal-add-qa-entry">
157                     <i class="bi bi-plus-lg me-1"></i> Добавить Q&A
158                 </button>
159             </div>
160         </div>
161         <div class="card-body p-2">
162             <div class="input-group input-group-sm mb-2">
163                 <input type="text" class="form-control" id="user-rag-entries-filter" placeholder="Фильтр записей в этой коллекции...">
164                 <button class="btn btn-outline-secondary" type="button" id="btn-clear-entries-filter">
165                     <i class="bi bi-x-lg"></i>
166                 </button>
167             </div>
168             <div id="user-rag-entries-container" class="d-flex flex-column gap-2" style="max-height: 250px; overflow-y: auto;">
169                 <div class="text-center text-muted small py-3">Загрузка записей базы знаний...</div>
170             </div>

```

```

171         </div>
172     </div>
173
174     <!-- Search Inside Active Collection -->
175     <div class="card border border-secondary-subtl
176 e">
177         <div class="card-header bg-light py-1 px-3">
178             <span class="small fw-bold"><i class="bi b
179 i-search me-1"></i> Тестовый поиск по этой коллекции</span>
180         </div>
181         <div class="card-body p-2">
182             <div class="input-group input-group-sm mb-
183 2">
184                 <input type="text" class="form-control"
185 id="active-rag-search-query" placeholder="Введите поисковый за
186 прос по коллекции...">
187                 <button class="btn btn-primary px-3" id=
188 "btn-active-rag-search">
189                     <i class="bi bi-search me-1"></i> Поис
190 к
191                 </button>
192             </div>
193             <div id="active-rag-search-results" class=
194 "border rounded p-2 bg-body-tertiary" style="min-height: 100px
195 ; max-height: 200px; overflow-y: auto;">
196                 <div class="text-center text-muted small
197 py-3">Введите запрос для поиска по выбранной базе знаний.</di
198 v>
199             </div>
200         </div>
201     </div>
202 </div>
203 </div>
204 </div>
205 </div>
206 </div>
207 </div>

```

```
198 |
199 |     <!-- Create User RAG Modal -->
200 |     <div class="modal fade" id="modal-create-user-rag" tabindex="
    x=-1" aria-labelledby="modalCreateUserRAGLabel" aria-hidden="
    true">
201 |         <div class="modal-dialog modal-dialog-centered">
202 |             <div class="modal-content shadow">
203 |                 <div class="modal-header">
204 |                     <h6 class="modal-title fw-bold" id="modalCreateUse
    rRAGLabel"><i class="bi bi-folder-plus text-primary me-2"></i>
    Создание новой базы знаний (RAG)</h6>
205 |                     <button type="button" class="btn-close" data-bs-di
    smiss="modal" aria-label="Close"></button>
206 |                 </div>
207 |                 <div class="modal-body">
208 |                     <div class="mb-3">
209 |                         <label for="new-rag-name" class="form-label smal
    l fw-bold">Имя коллекции:</label>
210 |                         <input type="text" class="form-control form-cont
    rol-sm" id="new-rag-name" placeholder="например: movies, codin
    g, gdocs_knowledge">
211 |                         <div class="form-text small">Уникальное имя для
    группировки знаний.</div>
212 |                     </div>
213 |                     <div class="mb-3">
214 |                         <label for="new-rag-desc" class="form-label smal
    l fw-bold">Описание (опционально):</label>
215 |                         <textarea class="form-control form-control-sm" i
    d="new-rag-desc" rows="2" placeholder="Кратко опишите назначен
    ие этой базы знаний..."></textarea>
216 |                     </div>
217 |                     <div class="mb-3">
218 |                         <label class="form-label small fw-bold">Источник
    документов:</label>
219 |                         <div class="form-check">
220 |                             <input class="form-check-input" type="radio" n
    ame="new-rag-source-type" id="rag-src-local" value="local" che
    cked>
```

```
221 |         <label class="form-check-label small" for="rag
    | -src-local">
222 |             <i class="bi bi-hdd me-1 text-primary"></i>Л
    | окальные файлы / Личное хранилище
223 |         </label>
224 |     </div>
225 |     <div class="form-check">
226 |         <input class="form-check-input" type="radio" n
    | ame="new-rag-source-type" id="rag-src-gdocs" value="gdocs">
227 |         <label class="form-check-label small" for="rag
    | -src-gdocs">
228 |             <i class="bi bi-google me-1 text-danger"></i>
    | >Документы Google Docs / Google Drive
229 |         </label>
230 |     </div>
231 | </div>
232 | <div class="row g-2">
233 |     <div class="col-6">
234 |         <label for="new-rag-min-chunk" class="form-lab
    | el small fw-bold">Мин. длина чанка:</label>
235 |         <input type="number" class="form-control form-
    | control-sm" id="new-rag-min-chunk" value="20" min="5" max="500
    | ">
236 |     </div>
237 |     <div class="col-6">
238 |         <label for="new-rag-max-chunk" class="form-lab
    | el small fw-bold">Макс. длина чанка:</label>
239 |         <input type="number" class="form-control form-
    | control-sm" id="new-rag-max-chunk" value="1500" min="100" max=
    | "5000">
240 |     </div>
241 | </div>
242 | </div>
243 | <div class="modal-footer py-2">
244 |     <button type="button" class="btn btn-secondary btn
    | -sm" data-bs-dismiss="modal">Отмена</button>
245 |     <button type="button" class="btn btn-primary btn-s
    | m fw-semibold" id="btn-confirm-create-rag">Создать</button>
```

```

246         </div>
247     </div>
248 </div>
249 </div>
250
251 <!-- Import Google Docs Modal -->
252 <div class="modal fade" id="modal-import-gdocs" tabindex="
-1" aria-labelledby="modalImportGDocsLabel" aria-hidden="true"
>
253     <div class="modal-dialog modal-dialog-centered modal-lg"
>
254         <div class="modal-content shadow">
255             <div class="modal-header">
256                 <h6 class="modal-title fw-bold" id="modalImportGDo
csLabel"><i class="bi bi-google text-danger me-2"></i>Импорт д
кументов из Google Docs & Drive в RAG</h6>
257                 <button type="button" class="btn-close" data-bs-di
smis="modal" aria-label="Close"></button>
258             </div>
259             <div class="modal-body">
260                 <div class="alert alert-info py-2 small mb-3">
261                     <i class="bi bi-info-circle me-1"></i> Документы
Google Docs автоматически экспортируются, очищаются через <co
de>rag_cleaner</code> и добавляются в векторный индекс коллекц
ии.
262                 </div>
263                 <div class="row g-2 mb-3">
264                     <div class="col-md-6">
265                         <label for="import-gdocs-query" class="form-la
bel small fw-bold">Фильтр поиска Google Drive:</label>
266                         <input type="text" class="form-control form-co
ntrol-sm font-monospace" id="import-gdocs-query" value="mimeTy
pe = 'application/vnd.google-apps.document' and trashed = fals
e" placeholder="mimeType = '...' or name contains '...'">
267                     </div>
268                     <div class="col-md-6">
269                         <label for="import-gdocs-account" class="form-
label small fw-bold">Google аккаунт (из пула):</label>

```



```

270         <input type="text" class="form-control form-co
ntrol-sm" id="import-gdocs-account" placeholder="default (или
имя аккаунта)">
271     </div>
272 </div>
273     <div id="import-gdocs-status-alert" class="alert d
-none py-2 small mb-0"></div>
274 </div>
275     <div class="modal-footer py-2">
276         <button type="button" class="btn btn-secondary btn
-sm" data-bs-dismiss="modal">Отмена</button>
277         <button type="button" class="btn btn-primary btn-s
m fw-semibold" id="btn-confirm-import-gdocs">
278             <i class="bi bi-cloud-arrow-down me-1"></i> Синх
ронизировать и собрать RAG
279         </button>
280     </div>
281 </div>
282 </div>
283 </div>
284
285 <!-- Add QA Entry Modal -->
286 <div class="modal fade" id="modal-add-qa-entry" tabindex="
-1" aria-labelledby="modalAddQAEntryLabel" aria-hidden="true">
287     <div class="modal-dialog modal-dialog-centered modal-lg"
>
288         <div class="modal-content shadow">
289             <div class="modal-header">
290                 <h6 class="modal-title fw-bold" id="modalAddQAEntr
yLabel"><i class="bi bi-patch-question text-primary me-2"></i>
Добавление знания / Связки «Вопрос-Ответ»</h6>
291                 <button type="button" class="btn-close" data-bs-di
smis="modal" aria-label="Close"></button>
292             </div>
293             <div class="modal-body">
294                 <div class="mb-3">
295                     <label for="new-qa-question" class="form-label s
mall fw-bold">Вопрос пользователя / Триггерная фраза:</label>

```

```

296 |         <input type="text" class="form-control form-cont
    | rol-sm" id="new-qa-question" placeholder="например: Где можно
    | скачать сериал?">
297 |         <div class="form-text small">Фраза или формулиро
    | вка запроса, при схожести с которой будет применяться этот отв
    | ет.</div>
298 |     </div>
299 |     <div class="mb-3">
300 |         <label for="new-qa-answer" class="form-label sma
    | ll fw-bold">Ответ модели / Текст знания:</label>
301 |         <textarea class="form-control form-control-sm fo
    | nt-monospace" id="new-qa-answer" rows="5" placeholder="Введите
    | точный ответ, инструкцию или фрагмент знаний..."></textarea>
302 |     </div>
303 | </div>
304 | <div class="modal-footer py-2">
305 |     <button type="button" class="btn btn-secondary btn
    | -sm" data-bs-dismiss="modal">Отмена</button>
306 |     <button type="button" class="btn btn-success btn-s
    | m fw-semibold" id="btn-confirm-add-qa">
307 |         <i class="bi bi-check2-circle me-1"></i> Сохрани
    | ть в RAG
308 |     </button>
309 | </div>
310 | </div>
311 | </div>
312 | </div>
313 |
314 |
315 | <!-- SUB-TAB 1: DOCUMENT RAG -->
316 | <div class="tab-pane fade" id="pills-doc-rag" role="tabpan
    | el">
317 |     <!-- Top Statistics Summary -->
318 |     <div class="row g-3 mb-3">
319 |         <div class="col-md-3 col-sm-6">
320 |             <div class="card shadow-sm border-0 bg-body-tertiary
    | ">
321 |                 <div class="card-body p-3 d-flex align-items-cente

```

```

    r justify-content-between">
322         <div>
323             <div class="text-muted small" data-i18n="rag.t
otalDocuments">Документов</div>
324             <h4 class="mb-0 fw-bold" id="rag-stat-docs">0<
/h4>
325         </div>
326         <i class="bi bi-file-earmark-text fs-1 text-prim
ary opacity-50"></i>
327     </div>
328 </div>
329 </div>
330 <div class="col-md-3 col-sm-6">
331     <div class="card shadow-sm border-0 bg-body-tertiary
">
332         <div class="card-body p-3 d-flex align-items-cente
r justify-content-between">
333             <div>
334                 <div class="text-muted small" data-i18n="rag.t
otalChunks">Чанков в индексе</div>
335                 <h4 class="mb-0 fw-bold" id="rag-stat-chunks">
0</h4>
336             </div>
337             <i class="bi bi-diagram-3 fs-1 text-success opac
ity-50"></i>
338         </div>
339     </div>
340 </div>
341 <div class="col-md-3 col-sm-6">
342     <div class="card shadow-sm border-0 bg-body-tertiary
">
343         <div class="card-body p-3 d-flex align-items-cente
r justify-content-between">
344             <div>
345                 <div class="text-muted small" data-i18n="rag.v
ectorDimension">Размерность вектора</div>
346                 <h4 class="mb-0 fw-bold" id="rag-stat-dim">0</
h4>

```

```

347         </div>
348         <i class="bi bi-cpu fs-1 text-info opacity-50"><
    /i>
349     </div>
350 </div>
351 </div>
352 <div class="col-md-3 col-sm-6">
353     <div class="card shadow-sm border-0 bg-body-tertiary
    ">
354         <div class="card-body p-3 d-flex align-items-cente
    r justify-content-between">
355             <div>
356                 <div class="text-muted small" data-i18n="rag.l
    astUpdated">Последнее обновление</div>
357                 <div class="small fw-bold text-truncate" id="r
    ag-stat-updated" style="max-width: 140px;">—</div>
358             </div>
359             <i class="bi bi-clock-history fs-1 text-warning
    opacity-50"></i>
360         </div>
361     </div>
362 </div>
363 </div>
364
365 <div class="row g-3">
366     <!-- LEFT COLUMN: File Ingestion & Index Controls -->
367     <div class="col-lg-5 col-md-12">
368         <!-- Upload Dropzone Card -->
369         <div class="card shadow-sm border-0 mb-3">
370             <div class="card-header bg-dark text-white d-flex
    justify-content-between align-items-center">
371                 <span class="fw-semibold"><i class="bi bi-cloud-
    arrow-up me-2"></i><span data-i18n="rag.uploadTitle">Загрузка
    документов</span></span>
372             </div>
373             <div class="card-body">
374                 <div id="rag-dropzone" class="border border-2 bo
    rder-dashed rounded p-4 text-center cursor-pointer mb-3 bg-bod

```

```

y-secondary bg-opacity-25" style="border-style: dashed !important; transition: all 0.2s; ">
375         <i class="bi bi-file-earmark-arrow-up fs-1 text-primary mb-2 d-block"></i>
376         <h6 class="fw-bold mb-1" data-i18n="rag.dropzoneText">Перетащите файлы сюда или нажмите для выбора</h6>
377         <input type="file" id="rag-file-input" multiple class="d-none" accept=".txt,.md,.markdown,.json,.csv,.tsv,.py,.js,.ts,.html,.css,.pdf,.log,.sql,.yaml,.yml">
378         <input type="file" id="rag-folder-input" webkitdirectory directory multiple class="d-none">
379         <div class="d-flex justify-content-center gap-2 mt-2 flex-wrap">
380             <button class="btn btn-outline-primary btn-sm px-3 rounded-pill" type="button" onclick="document.getElementById('rag-file-input').click();">
381                 <i class="bi bi-file-earmark-plus me-1"></i> <span data-i18n="rag.browseFiles">Выбрать файлы</span>
382             </button>
383             <button class="btn btn-primary btn-sm px-3 rounded-pill" type="button" onclick="document.getElementById('rag-folder-input').click();">
384                 <i class="bi bi-folder-plus me-1"></i> <span data-i18n="rag.browseFolder">Выбрать папку целиком</span>
385             </button>
386         </div>
387     </div>
388
389     <div id="rag-upload-progress-container" class="d-none mb-3">
390         <div class="d-flex justify-content-between small text-muted mb-1">
391             <span data-i18n="rag.uploading">Загрузка файлов...</span>
392             <span id="rag-upload-percent">0%</span>
393         </div>
394         <div class="progress" style="height: 6px;">
395             <div id="rag-upload-progress-bar" class="pro

```

```
    progress-bar progress-bar-striped progress-bar-animated bg-primary" role="progressbar" style="width: 0%"></div>
396         </div>
397     </div>
398
399         <div id="rag-upload-status-alert" class="d-none
alert p-2 small mb-0"></div>
400     </div>
401 </div>
402
403     <!-- Index Settings & Build Card -->
404     <div class="card shadow-sm border-0">
405         <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
406             <span class="fw-semibold"><i class="bi bi-slider
s me-2"></i><span data-i18n="rag.indexSettings">Параметры инде
ксации</span></span>
407         </div>
408         <div class="card-body">
409             <div class="mb-3">
410                 <label class="form-label small fw-bold" for="r
ag-provider-select" data-i18n="rag.providerLabel">Провайдер ве
кторизации:</label>
411                 <select class="form-select form-select-sm" id=
"rag-provider-select">
412                     <option value="auto">Auto (Gemini если досту
пен ключ, иначе Local TF-IDF)</option>
413                     <option value="gemini">Google Gemini Embeddi
ngs (gemini-embedding-2)</option>
414                     <option value="local_tfidf">Local TF-IDF Vec
torizer (Полностью офлайн)</option>
415                 </select>
416             </div>
417
418             <div class="mb-3" id="rag-api-key-group">
419                 <label class="form-label small fw-bold" for="r
ag-api-key-input">Gemini API Key (опционально, берется из .env
):</label>
```

```
420         <input type="password" class="form-control form-control-sm" id="rag-api-key-input" placeholder="AIzaSy... (или оставить пустым для авто)" autocomplete="new-password" data-lpignore="true" data-lp-ignore="true">
421     </div>
422
423     <div class="row g-2 mb-3">
424         <div class="col-6">
425             <label class="form-label small fw-bold" for="rag-chunk-size" data-i18n="rag.chunkSize">Размер чанка (символов):</label>
426             <input type="number" class="form-control form-control-sm" id="rag-chunk-size" value="500" min="100" max="4000" step="50">
427         </div>
428         <div class="col-6">
429             <label class="form-label small fw-bold" for="rag-chunk-overlap" data-i18n="rag.chunkOverlap">Перекрывание (Overlap):</label>
430             <input type="number" class="form-control form-control-sm" id="rag-chunk-overlap" value="50" min="0" max="1000" step="10">
431         </div>
432     </div>
433
434     <div class="d-grid">
435         <button class="btn btn-success btn-sm fw-semibold py-2 shadow-sm" id="btn-build-rag-index">
436             <i class="bi bi-hammer me-1"></i> <span data-i18n="rag.buildIndexBtn">🔨 Построить / Перестроить индекс</span>
437         </button>
438     </div>
439     <div id="rag-build-status" class="small mt-2 text-center text-muted"></div>
440 </div>
441 </div>
442 </div>
```

```

443
444         <!-- RIGHT COLUMN: Documents List & Semantic Search Pl
ayground -->
445         <div class="col-lg-7 col-md-12">
446             <!-- Documents Table Card -->
447             <div class="card shadow-sm border-0 mb-3">
448                 <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
449                     <span class="fw-semibold"><i class="bi bi-files
me-2"></i><span data-i18n="rag.documentsTitle">Список документ
ов в базе</span></span>
450                     <span class="badge bg-primary" id="rag-docs-badg
e">0 файлов</span>
451                 </div>
452                 <div class="card-body p-0">
453                     <div class="table-responsive" style="max-height:
260px; overflow-y: auto;">
454                         <table class="table table-hover table-sm align
-middle mb-0" id="rag-documents-table">
455                             <thead class="table-light sticky-top">
456                                 <tr>
457                                     <th class="ps-3" data-i18n="rag.fileName
">Имя файла</th>
458                                     <th data-i18n="rag.fileSize">Размер</th>
459                                     <th data-i18n="rag.chunks">Чанки</th>
460                                     <th data-i18n="rag.status">Статус</th>
461                                     <th class="text-end pe-3" data-i18n="com
mon.actions">Действия</th>
462                                 </tr>
463                             </thead>
464                             <tbody id="rag-documents-tbody">
465                                 <tr>
466                                     <td colspan="5" class="text-center text-
muted py-3" data-i18n="rag.noDocuments">
467                                         Документы не загружены. Загрузите файл
ы в блоке слева.
468                                     </td>
469                                 </tr>

```



```

470         </tbody>
471     </table>
472 </div>
473 </div>
474 </div>
475
476     <!-- Semantic Search Playground Card -->
477     <div class="card shadow-sm border-0">
478         <div class="card-header bg-dark text-white d-flex
479 justify-content-between align-items-center">
480             <span class="fw-semibold"><i class="bi bi-search
481 me-2"></i><span data-i18n="rag.searchPlaygroundTitle">Семанти
482 ческий поиск по базе</span></span>
483         </div>
484         <div class="card-body">
485             <div class="d-flex align-items-center justify-co
486 ntent-between flex-wrap gap-2 mb-2 p-2 bg-body-tertiary rounde
487 d border small text-muted">
488                 <div class="d-flex align-items-center gap-3">
489                     <div class="d-flex align-items-center gap-1"
490 >
491                         <label for="rag-top-k" class="text-nowrap
492 fw-semibold">Top K:</label>
493                         <select class="form-select form-select-sm"
494 id="rag-top-k" style="width: 75px;">
495                             <option value="3">3</option>
496                             <option value="5" selected>5</option>
497                             <option value="10">10</option>
498                             <option value="20">20</option>
499                         </select>
500                     </div>
501                     <div class="d-flex align-items-center gap-1"
502 >
503                         <label for="rag-min-score" class="text-now
504 rap fw-semibold">Min Score:</label>
505                         <input type="number" class="form-control f
506 orm-control-sm" id="rag-min-score" value="0.0" min="0.0" max="
507 1.0" step="0.05" style="width: 80px;">

```

```

496 |         </div>
497 |     </div>
498 |     <span class="small text-secondary"><i class="b
i bi-info-circle me-1"></i>Фильтры выдачи</span>
499 |     </div>
500 |
501 |     <div class="input-group mb-3">
502 |         <input type="text" class="form-control" id="ra
g-search-query" placeholder="Введите поисковый запрос по смысл
y..." data-i18n-placeholder="rag.searchPlaceholder">
503 |         <button class="btn btn-primary px-3" id="btn-r
ag-search">
504 |             <i class="bi bi-search me-1"></i> <span data
-i18n="common.search">Поиск</span>
505 |         </button>
506 |     </div>
507 |
508 |     <div id="rag-search-results" class="border round
ed p-2 bg-body-tertiary" style="min-height: 180px; max-height:
320px; overflow-y: auto;">
509 |         <div class="text-center text-muted py-4">
510 |             <i class="bi bi-chat-left-quote fs-3 d-block
mb-1 opacity-50"></i>
511 |             <span data-i18n="rag.enterQueryPrompt">Введи
те запрос и нажмите «Поиск» для проверки RAG.</span>
512 |         </div>
513 |     </div>
514 | </div>
515 | </div>
516 | </div>
517 | </div>
518 | </div>
519 |
520 | <!-- SUB-TAB 2: CODEBASE RAG (AST + Multi-Project) -->
521 | <div class="tab-pane fade" id="pills-code-rag" role="tabpa
nel">
522 |     <!-- Active Project & Index Selector Bar -->
523 |     <div class="card shadow-sm border-0 mb-3 bg-body-tertiar

```

```

y">
524         <div class="card-body p-3 d-flex align-items-center justify-content-between flex-wrap gap-2">
525             <div class="d-flex align-items-center gap-2 flex-grow-1 min-w-0">
526                 <label class="form-label small fw-bold mb-0 text-nowrap" for="codebase-active-index-select">
527                     <i class="bi bi-diagram-2 text-primary me-1"></i>
> АКТИВНЫЙ RAG ИНДЕКС:
528                 </label>
529                 <select class="form-select form-select-sm" id="codebase-active-index-select" style="max-width: 320px;">
530                     <option value="codebase">codebase (По умолчанию)
</option>
531                 </select>
532                 <button class="btn btn-sm btn-outline-secondary rounded-pill" id="btn-refresh-codebase-indexes" title="Обновить список индексов">
533                     <i class="bi bi-arrow-clockwise"></i>
534                 </button>
535             </div>
536             <div class="d-flex align-items-center gap-2" id="codebase-index-quick-stats">
537                 <span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="codebase-stat-chunks">0 чанков</span>
538                 <span class="badge bg-success-subtle text-success border border-success-subtle" id="codebase-stat-symbols">0 символов AST</span>
539             </div>
540         </div>
541     </div>
542
543     <div class="row g-3">
544         <!-- LEFT COLUMN: Codebase Index Settings & Trigger -->
545         <div class="col-lg-5 col-md-12">
546             <div class="card shadow-sm border-0">

```

```

547 |         <div class="card-header bg-dark text-white d-flex
| justify-content-between align-items-center">
548 |             <span class="fw-semibold"><i class="bi bi-folder
| 2-open me-2"></i>Сборка индекса проекта</span>
549 |         </div>
550 |         <div class="card-body">
551 |             <div class="mb-3">
552 |                 <label class="form-label small fw-bold" for="c
| odebase-project-root-input">
553 |                     Директория проекта (Project Root):
554 |                 </label>
555 |                 <div class="input-group input-group-sm">
556 |                     <span class="input-group-text"><i class="bi
| bi-hdd"></i></span>
557 |                     <input type="text" class="form-control" id="
| codebase-project-root-input" value="." placeholder="Например:
| . или C:/Projects/my_app">
558 |                 </div>
559 |                 <div class="form-text small text-muted">Укажит
| е путь к проекту на диске (абсолютный или относительный).</div
| >
560 |             </div>
561 |
562 |             <div class="mb-3">
563 |                 <label class="form-label small fw-bold" for="c
| odebase-index-name-input">
564 |                     Имя RAG индекса (Index Name):
565 |                 </label>
566 |                 <input type="text" class="form-control form-co
| ntrol-sm" id="codebase-index-name-input" value="codebase" plac
| eholder="Например: my_project_v1">
567 |                 <div class="form-text small text-muted">Индекс
| ы сохраняются изолированно в вашем персональном хранилище.</di
| v>
568 |             </div>
569 |
570 |             <div class="mb-3">
571 |                 <label class="form-label small fw-bold" for="c

```

```

odebase-include-dirs-input">
572         Подкаталоги для анализа (через запятую):
573     </label>
574     <input type="text" class="form-control form-co
ntrol-sm font-monospace" id="codebase-include-dirs-input" valu
e="src, plugins, integrations, launchers, scripts, install, do
cs, prompts">
575         </div>
576
577         <div class="d-grid">
578             <button class="btn btn-primary btn-sm fw-semib
old py-2 shadow-sm d-flex align-items-center justify-content-c
enter gap-2" id="btn-build-codebase-rag">
579                 <i class="bi bi-cpu"></i>
580                 <span>⬇ Собрать Codebase RAG</span>
581             </button>
582         </div>
583
584         <div id="codebase-build-status-alert" class="d-n
one alert p-2 small mt-3 mb-0"></div>
585     </div>
586 </div>
587 </div>
588
589 <!-- RIGHT COLUMN: Code & Symbol Search Playground -->
590 <div class="col-lg-7 col-md-12">
591     <div class="card shadow-sm border-0">
592         <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
593             <ul class="nav nav-tabs card-header-tabs border-
0" id="codebase-search-tabs" role="tablist">
594                 <li class="nav-item" role="presentation">
595                     <button class="nav-link active text-white" i
d="tab-code-semantic-btn" data-bs-toggle="tab" data-bs-target=
"#tab-code-semantic" type="button" role="tab">
596                         <i class="bi bi-search me-1"></i> Семантич
еский поиск по коду
597                     </button>

```

```

598 |         </li>
599 |         <li class="nav-item" role="presentation">
600 |             <button class="nav-link text-white-50" id="t
ab-code-symbols-btn" data-bs-toggle="tab" data-bs-target="#tab
-code-symbols" type="button" role="tab">
601 |                 <i class="bi bi-compass me-1"></i> AST Сим
волы
602 |             </button>
603 |         </li>
604 |     </ul>
605 | </div>
606 | <div class="card-body">
607 |     <div class="tab-content" id="codebase-search-tab
s-content">
608 |
609 |         <!-- Code Semantic Search -->
610 |         <div class="tab-pane fade show active" id="tab
-code-semantic" role="tabpanel">
611 |             <div class="input-group mb-3">
612 |                 <input type="text" class="form-control" id
="codebase-search-query" placeholder="Например: Как реализован
роутинг провайдеров?">
613 |                 <button class="btn btn-primary px-3" id="b
tn-codebase-search">
614 |                     <i class="bi bi-search me-1"></i> Поиск
615 |                 </button>
616 |             </div>
617 |             <div id="codebase-search-results" class="bor
der rounded p-2 bg-body-tertiary" style="min-height: 220px; ma
x-height: 360px; overflow-y: auto;">
618 |                 <div class="text-center text-muted py-4">
619 |                     <i class="bi bi-code-square fs-3 d-block
mb-1 opacity-50"></i>
620 |                     <span>Введите вопрос или описание логики
для поиска по коду.</span>
621 |                 </div>
622 |             </div>
623 |         </div>

```

```

624 |
625 |         <!-- Symbol Lookup -->
626 |         <div class="tab-pane fade" id="tab-code-symbol
| s" role="tabpanel">
627 |             <div class="input-group mb-3">
628 |                 <input type="text" class="form-control" id
| ="codebase-symbol-query" placeholder="Имя класса или метода: н
| апример MediaSearchAgent, search, BasePlugin">
629 |                 <button class="btn btn-success px-3" id="b
| tn-codebase-symbol-search">
630 |                     <i class="bi bi-compass me-1"></i> Найти
| СИМВОЛ
631 |                 </button>
632 |             </div>
633 |             <div id="codebase-symbol-results" class="bor
| der rounded p-2 bg-body-tertiary" style="min-height: 220px; ma
| x-height: 360px; overflow-y: auto;">
634 |                 <div class="text-center text-muted py-4">
635 |                     <i class="bi bi-diagram-3 fs-3 d-block m
| b-1 opacity-50"></i>
636 |                     <span>Введите имя класса, метода или фун
| кции для мгновенного AST-поиска.</span>
637 |                 </div>
638 |             </div>
639 |         </div>
640 |
641 |     </div>
642 | </div>
643 | </div>
644 | </div>
645 |
646 | </div>
647 | </div>
648 |
649 | <!-- SUB-TAB 3: PIXELRAG VISUAL HYBRID SEARCH -->
650 | <div class="tab-pane fade" id="pills-pixel-rag" role="tabp
| anel">
651 |     <!-- Top Metrics Bar -->

```

```
652 |         <div class="row g-3 mb-3">
653 |             <div class="col-md-3 col-sm-6">
654 |                 <div class="card shadow-sm border-0 bg-body-secondar
| y bg-opacity-25 h-100 p-3 text-center">
655 |                     <div class="small text-muted mb-1"><i class="bi bi
| -images text-primary me-1"></i>Всего изображений</div>
656 |                     <div class="fw-bold fs-4 text-primary" id="pixel-s
| tat-images">0</div>
657 |                     <div class="small text-muted" id="pixel-stat-size"
| >0.0 MB</div>
658 |                 </div>
659 |             </div>
660 |             <div class="col-md-3 col-sm-6">
661 |                 <div class="card shadow-sm border-0 bg-body-secondar
| y bg-opacity-25 h-100 p-3 text-center">
662 |                     <div class="small text-muted mb-1"><i class="bi bi
| -layers text-success me-1"></i>Чанков в индексе</div>
663 |                     <div class="fw-bold fs-4 text-success" id="pixel-s
| tat-chunks">0</div>
664 |                     <div class="small text-muted">FAISS Hybrid Vector
| Index</div>
665 |                 </div>
666 |             </div>
667 |             <div class="col-md-3 col-sm-6">
668 |                 <div class="card shadow-sm border-0 bg-body-secondar
| y bg-opacity-25 h-100 p-3 text-center">
669 |                     <div class="small text-muted mb-1"><i class="bi bi
| -robot text-info me-1"></i>Провайдер векторизации</div>
670 |                     <div class="fw-bold fs-6 text-info mt-1" id="pixel
| -stat-provider">FAISS / Visual Embeddings</div>
671 |                     <div class="small text-muted" id="pixel-stat-stat
| us">Статус: Загрузка...</div>
672 |                 </div>
673 |             </div>
674 |             <div class="col-md-3 col-sm-6">
675 |                 <div class="card shadow-sm border-0 bg-body-secondar
| y bg-opacity-25 h-100 p-3 text-center">
676 |                     <div class="small text-muted mb-1"><i class="bi bi
```



```

        -clock-history text-warning me-1"></i>Индекс обновлен</div>
677         <div class="small fw-bold mt-2 text-truncate" id="
pixel-stat-updated">—</div>
678         <button class="btn btn-link btn-sm text-decoration
-none p-0 mt-1" id="btn-refresh-pixel-stats">
679             <i class="bi bi-arrow-clockwise"></i> Обновить
680         </button>
681     </div>
682 </div>
683 </div>
684
685 <div class="row g-3">
686     <!-- LEFT COLUMN: Image Ingestion & Smart Query Router
-->
687     <div class="col-lg-5 col-md-12">
688         <!-- Ingest Images Card -->
689         <div class="card shadow-sm border-0 mb-3">
690             <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
691                 <span class="fw-semibold"><i class="bi bi-cloud-
arrow-up me-2"></i>Загрузка и индексация изображений</span>
692             </div>
693             <div class="card-body">
694                 <div id="pixel-dropzone" class="border border-2
border-dashed rounded p-4 text-center cursor-pointer mb-3 bg-b
ody-secondary bg-opacity-25" style="border-style: dashed !impo
rtant; transition: all 0.2s; cursor: pointer;">
695                     <i class="bi bi-file-earmark-image fs-1 text-p
rimary mb-2 d-block"></i>
696                     <h6 class="fw-bold mb-1">Перетащите изображени
я сюда или нажмите для выбора</h6>
697                     <div class="small text-muted mb-3">Поддерживаю
тся .png, .jpg, .jpeg, .webp, .bmp, .gif, .pdf</div>
698                     <input type="file" id="pixel-file-input" multi
ple class="d-none" accept="image/*,.pdf">
699                     <button class="btn btn-outline-primary btn-sm
px-3 rounded-pill" type="button" onclick="document.getElementB
yId('pixel-file-input').click();">

```

```

700 |             <i class="bi bi-image me-1"></i> Выбрать изо
      | бражения
701 |             </button>
702 |         </div>
703 |
704 |         <div id="pixel-upload-progress" class="d-none mb
      | -3">
705 |             <div class="d-flex justify-content-between sma
      | ll text-muted mb-1">
706 |                 <span>Индексация через PixelRAG...</span>
707 |                 <span id="pixel-upload-percent">0%</span>
708 |             </div>
709 |             <div class="progress" style="height: 6px;">
710 |                 <div id="pixel-upload-progress-bar" class="p
      | rogress-bar progress-bar-striped progress-bar-animated bg-succ
      | ess" role="progressbar" style="width: 0%"></div>
711 |             </div>
712 |         </div>
713 |
714 |         <div id="pixel-upload-status-alert" class="d-non
      | e alert p-2 small mb-0"></div>
715 |     </div>
716 | </div>
717 |
718 | <!-- Smart Query Router & Analyzer -->
719 | <div class="card shadow-sm border-0">
720 |     <div class="card-header bg-dark text-white d-flex
      | justify-content-between align-items-center">
721 |         <span class="fw-semibold"><i class="bi bi-signpo
      | st-split me-2"></i>Smart Query Router (Анализ запроса)</span>
722 |     </div>
723 |     <div class="card-body">
724 |         <p class="small text-muted mb-2">Автоматическое
      | определение типа запроса (Text vs Visual vs Hybrid) перед поис
      | ком.</p>
725 |         <div class="input-group input-group-sm mb-2">
726 |             <input type="text" class="form-control" id="pi
      | xel-analyze-query-input" placeholder="Например: схема авториза

```

```

ции или синий график продаж">
727         <button class="btn btn-outline-primary" id="bt
n-pixel-analyze-query">
728             <i class="bi bi-cpu me-1"></i> Анализ
729         </button>
730     </div>
731     <div id="pixel-analyze-result" class="p-2 border
rounded bg-body-tertiary small d-none">
732         <div class="d-flex justify-content-between ali
gn-items-center mb-1">
733             <span>Тип маршрутизации: <strong id="pixel-r
outing-type" class="badge bg-primary">HYBRID</strong></span>
734             <span>Уверенность: <strong id="pixel-routing
-confidence" class="text-success">95%</strong></span>
735         </div>
736         <div class="text-muted" id="pixel-routing-reas
on" style="font-size: 0.8rem;"></div>
737     </div>
738 </div>
739 </div>
740 </div>
741
742 <!-- RIGHT COLUMN: Visual Search Playground -->
743 <div class="col-lg-7 col-md-12">
744     <div class="card shadow-sm border-0">
745         <div class="card-header bg-dark text-white d-flex
justify-content-between align-items-center">
746             <span class="fw-semibold"><i class="bi bi-search
me-2"></i>Интерактивный поиск PixelRAG</span>
747             <div class="form-check form-switch m-0" title="И
спользовать Smart Query Routing">
748                 <input class="form-check-input" type="checkbox"
" id="pixel-search-routing-toggle" checked>
749                 <label class="form-check-label text-light smal
l ms-1" for="pixel-search-routing-toggle">Smart Routing</label
>
750             </div>
751         </div>

```

```

752         <div class="card-body">
753             <div class="row g-2 mb-3">
754                 <div class="col-md-8">
755                     <div class="input-group">
756                         <input type="text" class="form-control" id
757                         ="pixel-search-query" placeholder="Введите поисковый запрос (т
758                         екст или описание картинки)...">
759                         <button class="btn btn-primary px-3" id="b
760                         tn-pixel-search">
761                             <i class="bi bi-search me-1"></i> Искать
762                             </button>
763                         </div>
764                     </div>
765                     <div class="col-md-2">
766                         <select class="form-select" id="pixel-search
767                         -topk" title="Количество результатов">
768                             <option value="3">Top 3</option>
769                             <option value="5" selected>Top 5</option>
770                             <option value="10">Top 10</option>
771                         </select>
772                     </div>
773                     <div class="col-md-2">
774                         <select class="form-select" id="pixel-search
775                         -minscore" title="Минимальный порог">
776                             <option value="0.0" selected>Score ≥ 0.0</
777                             option>
778                             <option value="0.3">Score ≥ 0.3</option>
779                             <option value="0.5">Score ≥ 0.5</option>
780                             <option value="0.7">Score ≥ 0.7</option>
781                         </select>
782                     </div>
783                 </div>
784             </div>
785             <!-- Search Results List -->
786             <div id="pixel-search-results" class="border rou
787             nded p-3 bg-body-tertiary" style="min-height: 280px; max-heigh
788             t: 480px; overflow-y: auto;">
789                 <div class="text-center text-muted py-5">

```

```
782 |             <i class="bi bi-images fs-1 d-block mb-2 opa
| city-50"></i>
783 |             <span>Введите запрос для поиска по текстовым
|         чанкам и визуальным индексам изображений.</span>
784 |         </div>
785 |     </div>
786 | </div>
787 | </div>
788 | </div>
789 | </div>
790 | </div>
791 |
792 | </div>
793 | </div>
794 |
795 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: RAG Tab Master Coordinator
4  * =====
5  * Description:
6  *   Entry point and coordinator for the Knowledge Base & RAG
7  *   interface.
8  *   Dispatches events to specialized submodules.
9  *
10 * File: main.js
11 * Project: AI Breadboard
12 * Module: RAGTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17 import { getAllFilesFromDataTransfer } from './modules/utils.js';
18 import {
19   uploadFiles,
20   loadRagStatus,
21   loadRagDocuments,
22   deleteRagDocument,
23   buildRagIndex,
24   executeRagSearch
25 } from './modules/docRag.js';
26 import {
27   loadCodebaseIndexes,
28   updateCodebaseStatsDisplay,
29   buildCodebaseRag,
30   executeCodebaseSearch,
31   executeSymbolLookup
```

```

32 } from './modules/codebaseRag.js';
33 import {
34   loadUserRags,
35   createUserRag,
36   deleteActiveUserRag,
37   addQaEntry,
38   loadRagEntries
39 } from './modules/userRagCollections.js';
40
41 import {
42   buildActiveUserRag,
43   executeActiveUserRagSearch,
44   handleQuickUploadToUserStorage,
45   syncGoogleDocsIntoActiveRag
46 } from './modules/userRagPipeline.js';
47
48 import {
49   loadPixelRagStatus,
50   uploadAndIndexPixelImages,
51   analyzePixelQuery,
52   executePixelRagSearch
53 } from './modules/pixelRag.js';
54
55 /**
56  * Initialize all RAG tab components, event listeners, and data feeds.
57  *
58  * @returns {Promise<void>}
59  */
60 export async function initRagTab() {
61   setupEventListeners();
62   loadUserRags();
63   loadRagStatus();
64   loadRagDocuments();
65   loadCodebaseIndexes();
66   loadPixelRagStatus();
67 }
68

```

```

69 // Expose globals for external tab orchestrators and inline ha
    ndlers
70 window.initRagTab = initRagTab;
71 window.deleteRagDocument = deleteRagDocument;
72
73 /**
74  * Register DOM event listeners for buttons, inputs, dropzones
    , and keyboard events.
75  */
76 function setupEventListeners() {
77     const root = document.getElementById('rag-tab-content');
78     if (root) {
79         if (root.dataset.listenersBound === 'true') return;
80         root.dataset.listenersBound = 'true';
81     }
82
83     const refreshBtn = document.getElementById('btn-refresh-rag'
    );
84     if (refreshBtn) {
85         refreshBtn.addEventListener('click', () => {
86             loadUserRags();
87             loadRagStatus();
88             loadRagDocuments();
89             loadCodebaseIndexes();
90             loadPixelRagStatus();
91         });
92     }
93
94     // --- PixelRAG Listeners ---
95     const refreshPixelStatsBtn = document.getElementById('btn-re
    fresh-pixel-stats');
96     if (refreshPixelStatsBtn) {
97         refreshPixelStatsBtn.addEventListener('click', loadPixelRa
    gStatus);
98     }
99
100    const pixelFileInput = document.getElementById('pixel-file-i
    nput');

```



```
101 | if (pixelFileInput) {
102 |     pixelFileInput.addEventListener('change', (e) => {
103 |         if (e.target.files && e.target.files.length > 0) {
104 |             uploadAndIndexPixelImages(Array.from(e.target.files));
105 |             pixelFileInput.value = '';
106 |         }
107 |     });
108 | }
109 |
110 | const pixelDropzone = document.getElementById('pixel-dropzon
e');
111 | if (pixelDropzone) {
112 |     ['dragenter', 'dragover'].forEach(eventName => {
113 |         pixelDropzone.addEventListener(eventName, (e) => {
114 |             e.preventDefault();
115 |             e.stopPropagation();
116 |             pixelDropzone.classList.add('border-primary', 'bg-prim
ary', 'bg-opacity-10');
117 |         }, false);
118 |     });
119 |
120 |     ['dragleave', 'drop'].forEach(eventName => {
121 |         pixelDropzone.addEventListener(eventName, (e) => {
122 |             e.preventDefault();
123 |             e.stopPropagation();
124 |             pixelDropzone.classList.remove('border-primary', 'bg-p
rimary', 'bg-opacity-10');
125 |         }, false);
126 |     });
127 |
128 |     pixelDropzone.addEventListener('drop', async (e) => {
129 |         const dt = e.dataTransfer;
130 |         if (dt) {
131 |             const files = await getAllFilesFromDataTransfer(dt);
132 |             if (files && files.length > 0) {
133 |                 uploadAndIndexPixelImages(files);
134 |             }
135 |         }
    }
```

```
136     }, false);
137 }
138
139 const btnPixelAnalyze = document.getElementById('btn-pixel-a
nalyze-query');
140 const inputPixelAnalyze = document.getElementById('pixel-ana
lyze-query-input');
141 if (btnPixelAnalyze) {
142     btnPixelAnalyze.addEventListener('click', analyzePixelQuer
y);
143 }
144 if (inputPixelAnalyze) {
145     inputPixelAnalyze.addEventListener('keydown', (e) => {
146         if (e.key === 'Enter') analyzePixelQuery();
147     });
148 }
149
150 const btnPixelSearch = document.getElementById('btn-pixel-se
arch');
151 const inputPixelSearch = document.getElementById('pixel-sear
ch-query');
152 if (btnPixelSearch) {
153     btnPixelSearch.addEventListener('click', executePixelRagSe
arch);
154 }
155 if (inputPixelSearch) {
156     inputPixelSearch.addEventListener('keydown', (e) => {
157         if (e.key === 'Enter') executePixelRagSearch();
158     });
159 }
160
161 // --- User Workspace RAG Listeners ---
162 const confirmCreateRagBtn = document.getElementById('btn-con
firm-create-rag');
163 if (confirmCreateRagBtn) {
164     confirmCreateRagBtn.addEventListener('click', createUserRa
g);
165 }
```

```
166 |
167 |   const buildActiveRagBtn = document.getElementById('btn-build
    | -active-rag');
168 |   if (buildActiveRagBtn) {
169 |     buildActiveRagBtn.addEventListener('click', buildActiveUse
    | rRag);
170 |   }
171 |
172 |   const deleteActiveRagBtn = document.getElementById('btn-dele
    | te-active-rag');
173 |   if (deleteActiveRagBtn) {
174 |     deleteActiveRagBtn.addEventListener('click', deleteActiveU
    | serRag);
175 |   }
176 |
177 |   const confirmAddQaBtn = document.getElementById('btn-confirm
    | -add-qa');
178 |   if (confirmAddQaBtn) {
179 |     confirmAddQaBtn.addEventListener('click', addQaEntry);
180 |   }
181 |
182 |   const refreshEntriesBtn = document.getElementById('btn-refre
    | sh-rag-entries');
183 |   if (refreshEntriesBtn) {
184 |     refreshEntriesBtn.addEventListener('click', () => loadRagE
    | ntries());
185 |   }
186 |
187 |   const entriesFilterInput = document.getElementById('user-rag
    | -entries-filter');
188 |   if (entriesFilterInput) {
189 |     let filterDebounce = null;
190 |     entriesFilterInput.addEventListener('input', (e) => {
191 |       clearTimeout(filterDebounce);
192 |       filterDebounce = setTimeout(() => {
193 |         loadRagEntries(undefined, e.target.value.trim());
194 |       }, 300);
195 |     });
```

```
196 | }
197 |
198 | const clearFilterBtn = document.getElementById('btn-clear-en
    | tries-filter');
199 | if (clearFilterBtn) {
200 |     clearFilterBtn.addEventListener('click', () => {
201 |         if (entriesFilterInput) {
202 |             entriesFilterInput.value = '';
203 |             loadRagEntries(undefined, '');
204 |         }
205 |     });
206 | }
207 |
208 | const searchActiveRagBtn = document.getElementById('btn-acti
    | ve-rag-search');
209 |
210 | const searchActiveRagInput = document.getElementById('active
    | -rag-search-query');
211 | if (searchActiveRagBtn) {
212 |     searchActiveRagBtn.addEventListener('click', executeActive
    | UserRagSearch);
213 | }
214 | if (searchActiveRagInput) {
215 |     searchActiveRagInput.addEventListener('keydown', (e) => {
216 |         if (e.key === 'Enter') executeActiveUserRagSearch();
217 |     });
218 | }
219 |
220 | const quickUploadInput = document.getElementById('user-rag-q
    | uick-upload-input');
221 | if (quickUploadInput) {
222 |     quickUploadInput.addEventListener('change', handleQuickUp
    | loadToUserStorage);
223 | }
224 |
225 | const selectAllBtn = document.getElementById('btn-select-all
    | -user-files');
226 | if (selectAllBtn) {
```

```

227     selectAllBtn.addEventListener('click', () => {
228         const checkboxes = document.querySelectorAll('#user-rag-
files-checklist input[type="checkbox"]');
229         const anyUnchecked = Array.from(checkboxes).some(cb => !
cb.checked);
230         checkboxes.forEach(cb => cb.checked = anyUnchecked);
231     });
232 }
233
234 const confirmImportGdocsBtn = document.getElementById('btn-c
onfirm-import-gdocs');
235 if (confirmImportGdocsBtn) {
236     confirmImportGdocsBtn.addEventListener('click', syncGoogle
DocsIntoActiveRag);
237 }
238
239 // --- Global Document RAG Listeners ---
240 const fileInput = document.getElementById('rag-file-input');
241 const folderInput = document.getElementById('rag-folder-inpu
t');
242 const dropzone = document.getElementById('rag-dropzone');
243
244 if (fileInput) {
245     fileInput.addEventListener('change', (e) => {
246         if (e.target.files && e.target.files.length > 0) {
247             uploadFiles(Array.from(e.target.files));
248             fileInput.value = '';
249         }
250     });
251 }
252
253 if (folderInput) {
254     folderInput.addEventListener('change', (e) => {
255         if (e.target.files && e.target.files.length > 0) {
256             uploadFiles(Array.from(e.target.files));
257             folderInput.value = '';
258         }
259     });

```

```

260 | }
261 |
262 | if (dropzone) {
263 |     ['dragenter', 'dragover'].forEach(eventName => {
264 |         dropzone.addEventListener(eventName, (e) => {
265 |             e.preventDefault();
266 |             e.stopPropagation();
267 |             dropzone.classList.add('border-primary', 'bg-primary',
| 'bg-opacity-10');
268 |             }, false);
269 |         });
270 |
271 |     ['dragleave', 'drop'].forEach(eventName => {
272 |         dropzone.addEventListener(eventName, (e) => {
273 |             e.preventDefault();
274 |             e.stopPropagation();
275 |             dropzone.classList.remove('border-primary', 'bg-primar
| y', 'bg-opacity-10');
276 |             }, false);
277 |         });
278 |
279 |         dropzone.addEventListener('drop', async (e) => {
280 |             const dt = e.dataTransfer;
281 |             if (dt) {
282 |                 const files = await getAllFilesFromDataTransfer(dt);
283 |                 if (files && files.length > 0) {
284 |                     uploadFiles(files);
285 |                 }
286 |             }
287 |             }, false);
288 |     }
289 |
290 |     const buildBtn = document.getElementById('btn-build-rag-inde
| x');
291 |     if (buildBtn) {
292 |         buildBtn.addEventListener('click', buildRagIndex);
293 |     }
294 |

```

```
295 |   const searchBtn = document.getElementById('btn-rag-search');
296 |   const searchInput = document.getElementById('rag-search-quer
    | y');
297 |   if (searchBtn) {
298 |     searchBtn.addEventListener('click', executeRagSearch);
299 |   }
300 |   if (searchInput) {
301 |     searchInput.addEventListener('keydown', (e) => {
302 |       if (e.key === 'Enter') executeRagSearch();
303 |     });
304 |   }
305 |
306 |   // --- Codebase RAG Listeners ---
307 |   const btnRefreshCodebase = document.getElementById('btn-refr
    | esh-codebase-indexes');
308 |   if (btnRefreshCodebase) {
309 |     btnRefreshCodebase.addEventListener('click', loadCodebaseI
    | ndexes);
310 |   }
311 |
312 |   const btnBuildCodebase = document.getElementById('btn-build-
    | codebase-rag');
313 |   if (btnBuildCodebase) {
314 |     btnBuildCodebase.addEventListener('click', buildCodebaseRa
    | g);
315 |   }
316 |
317 |   const btnCodeSearch = document.getElementById('btn-codebase-
    | search');
318 |   const inputCodeSearch = document.getElementById('codebase-se
    | arch-query');
319 |   if (btnCodeSearch) {
320 |     btnCodeSearch.addEventListener('click', executeCodebaseSea
    | rch);
321 |   }
322 |   if (inputCodeSearch) {
323 |     inputCodeSearch.addEventListener('keydown', (e) => {
324 |       if (e.key === 'Enter') executeCodebaseSearch();
```

```
325 |     });
326 | }
327 |
328 | const btnSymbolSearch = document.getElementById('btn-codebase-symbol-search');
329 | const inputSymbolSearch = document.getElementById('codebase-symbol-query');
330 | if (btnSymbolSearch) {
331 |     btnSymbolSearch.addEventListener('click', executeSymbolLookup);
332 | }
333 | if (inputSymbolSearch) {
334 |     inputSymbolSearch.addEventListener('keydown', (e) => {
335 |         if (e.key === 'Enter') executeSymbolLookup();
336 |     });
337 | }
338 |
339 | const activeIndexSelect = document.getElementById('codebase-active-index-select');
340 | if (activeIndexSelect) {
341 |     activeIndexSelect.addEventListener('change', updateCodebaseStatsDisplay);
342 | }
343 | }
344 |
345 | // Auto-initialize when loaded or DOM ready
346 | window.initRagTab = initRagTab;
347 |
348 | if (document.readyState === 'loading') {
349 |     document.addEventListener('DOMContentLoaded', initRagTab);
350 | } else {
351 |     initRagTab();
352 | }
353 |
354 |
```


cache.js

```
1 /**
2  * =====
3  * Process Name: RAG Tab Cache Module
4  * =====
5  * Description:
6  *   Centralized cache management for RAG tab with cachedApiFe
7  *   tch wrapper.
8  *
9  * File: cache.js
10 * Project: AI Breadboard
11 * Module: RAGTab
12 * Author: hypo69
13 * Copyright: © 2026 hypo69
14 * =====
15 */
16 import { cachedApiFetch, clearCacheByTag } from '../..js/api-
17   cache.js';
18 export const CACHE_TAG = 'rag';
19
20 /**
21 * Cached fetch wrapper for RAG tab
22 * @param {string} url - API endpoint
23 * @param {object} options - Fetch options
24 * @returns {Promise<Response>}
25 */
26 export async function ragCachedFetch(url, options = {}) {
27   // Merge cache tags
28   const tags = options.tags || [];
29   if (!tags.includes(CACHE_TAG)) {
30     tags.push(CACHE_TAG);
31   }
```

```

32 |
33 |   return cachedApiFetch(url, { ...options, tags });
34 | }
35 |
36 | /**
37 |  * Clear all RAG cache
38 |  * @returns {Promise<void>}
39 |  */
40 | export async function clearRagCache() {
41 |   await clearCacheByTag(CACHE_TAG);
42 |   console.log('RAG cache cleared');
43 | }
44 |
45 | /**
46 |  * Update cache status UI
47 |  */
48 | export function updateRagCacheStatus() {
49 |   const badge = document.getElementById('rag-cache-badge');
50 |   const ratio = document.getElementById('rag-cache-ratio');
51 |   if (!badge || !ratio) return;
52 |
53 |   try {
54 |     const stats = window.apiCacheStats || { hits: 0, total: 0
55 |   };
56 |     const hitRate = stats.total > 0 ? Math.round((stats.hits /
57 |   stats.total) * 100) : 0;
58 |     ratio.textContent = `${hitRate}%`;
59 |
60 |     // Color coding
61 |     badge.className = 'badge cache-badge';
62 |     if (hitRate >= 70) {
63 |       badge.classList.add('bg-success-subtle', 'text-success')
64 |     };
65 |     } else if (hitRate >= 40) {
66 |       badge.classList.add('bg-warning-subtle', 'text-warning')
67 |     };
68 |     } else {
69 |       badge.classList.add('bg-secondary-subtle', 'text-seconda

```

```
    ry');  
66 |     }  
67 |   } catch (err) {  
68 |     console.debug('Failed to update RAG cache status:', err);  
69 |   }  
70 | }  
71 |
```

codebaseRag.js

```
1 /**
2  * =====
3  * Process Name: Codebase AST RAG Management
4  * =====
5  * Description:
6  *   Handles project codebase indexing, AST symbol extraction,
7  *   semantic code
8  *   search, and AST symbol lookup across workspace projects.
9  *
10 * File: codebaseRag.js
11 * Project: AI Breadboard
12 * Module: RAGTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17 import { escapeHtml } from './utils.js';
18 import { ragCachedFetch, CACHE_TAG } from './cache.js';
19
20 let cachedCodebaseIndexes = [];
21
22 /**
23  * Fetch list of available codebase AST indexes and populate i
24  * ndex selector.
25  *
26  * @returns {Promise<void>}
27  */
28 export async function loadCodebaseIndexes() {
29   const select = document.getElementById('codebase-active-inde
30   x-select');
```

```

31 |   try {
32 |     const res = await ragCachedFetch('/api/rag/codebase/indexe
    | s', {
33 |       tags: ['rag-codebase']
34 |     });
35 |     if (!res.ok) return;
36 |     const data = await res.json();
37 |     cachedCodebaseIndexes = data.indexes || [];
38 |
39 |     if (cachedCodebaseIndexes.length === 0) {
40 |       select.innerHTML = '<option value="codebase">codebase (П
    | о умолчанию)</option>';
41 |     } else {
42 |       const curVal = select.value;
43 |       select.innerHTML = cachedCodebaseIndexes.map(idx => {
44 |         const scopeBadge = idx.scope === 'user' ? '👤' : '👤
    | ';
45 |         const label = `${scopeBadge}${idx.name} (${idx.total_c
    | hunks || 0} чанков)`;
46 |         return `<option value="${escapeHtml(idx.name)}">${esca
    | peHtml(label)}</option>`;
47 |       }).join('');
48 |
49 |       if (curVal && cachedCodebaseIndexes.some(i => i.name ===
    | curVal)) {
50 |         select.value = curVal;
51 |       }
52 |     }
53 |     updateCodebaseStatsDisplay();
54 |   } catch (err) {
55 |     console.error('[Codebase RAG] Error loading indexes:', err
    | );
56 |   }
57 | }
58 |
59 | /**
60 |  * Update UI counters for the currently selected codebase inde
    | x.

```

```

61 | */
62 | export function updateCodebaseStatsDisplay() {
63 |     const select = document.getElementById('codebase-active-index-select');
64 |     const statChunks = document.getElementById('codebase-stat-chunks');
65 |     const statSymbols = document.getElementById('codebase-stat-symbols');
66 |     if (!select) return;
67 |
68 |     const currentName = select.value;
69 |     const item = cachedCodebaseIndexes.find(i => i.name === currentName);
70 |     if (item) {
71 |         if (statChunks) statChunks.textContent = `${item.total_chunks || 0} чанков`;
72 |         if (statSymbols) statSymbols.textContent = `${item.total_symbols || 0} символов AST`;
73 |     }
74 | }
75 |
76 | /**
77 |  * Build or rebuild codebase AST and semantic index for a target directory.
78 |  *
79 |  * @returns {Promise<void>}
80 |  */
81 | export async function buildCodebaseRag() {
82 |     const btn = document.getElementById('btn-build-codebase-rag');
83 |     const alertEl = document.getElementById('codebase-build-status-alert');
84 |     const rootInput = document.getElementById('codebase-project-root-input');
85 |     const nameInput = document.getElementById('codebase-index-name-input');
86 |     const dirsInput = document.getElementById('codebase-include-dirs-input');

```

```

87
88   const projectRoot = rootInput?.value?.trim() || '.';
89   const indexName = nameInput?.value?.trim() || 'codebase';
90   const includeDirs = (dirsInput?.value || '').split(',').map(
    s => s.trim()).filter(Boolean);
91
92   if (btn) btn.disabled = true;
93   if (alertEl) {
94     alertEl.className = 'alert alert-info p-2 small mt-3 mb-0'
    ;
95     alertEl.innerHTML = '<span class="spinner-border spinner-b
    order-sm me-1"></span> Анализ AST и построение индексов...';
96     alertEl.classList.remove('d-none');
97   }
98
99   try {
100     const res = await fetch('/api/rag/codebase/build', {
101       method: 'POST',
102       headers: { 'Content-Type': 'application/json' },
103       body: JSON.stringify({
104         project_root: projectRoot,
105         index_name: indexName,
106         include_dirs: includeDirs,
107       }),
108     });
109
110     const data = await res.json();
111     if (!res.ok) throw new Error(data.detail || data.error ||
    `HTTP ${res.status}`);
112
113     const stats = data.result || {};
114     if (alertEl) {
115       alertEl.className = 'alert alert-success p-2 small mt-3
    mb-0';
116       alertEl.innerHTML = `
117         <strong>📁 Индекс «${escapeHtml(indexName)}» успешно со
    здан!</strong><br>
118         Файлов: ${stats.total_files || 0} | Чанков: ${stats.to

```

```

    tal_chunks || 0} | Символов AST: ${stats.total_symbols || 0} (
    ${stats.elapsed_seconds || 0}c)
119     `;
120     }
121
122     await loadCodebaseIndexes();
123     const select = document.getElementById('codebase-active-in
dex-select');
124     if (select) select.value = indexName;
125     updateCodebaseStatsDisplay();
126 } catch (err) {
127     console.error('[Codebase RAG] Build error:', err);
128     if (alertEl) {
129         alertEl.className = 'alert alert-danger p-2 small mt-3 m
b-0';
130         alertEl.textContent = `Ошибка сборки: ${err.message}`;
131     }
132 } finally {
133     if (btn) btn.disabled = false;
134 }
135 }
136
137 /**
138  * Execute semantic query against codebase chunks.
139  *
140  * @returns {Promise<void>}
141  */
142 export async function executeCodebaseSearch() {
143     const queryInput = document.getElementById('codebase-search-
query');
144     const select = document.getElementById('codebase-active-inde
x-select');
145     const resultsContainer = document.getElementById('codebase-s
earch-results');
146     const btn = document.getElementById('btn-codebase-search');
147
148     const query = queryInput?.value?.trim();
149     if (!query) return;

```



```
150
151   const indexName = select?.value || 'codebase';
152   if (btn) btn.disabled = true;
153   if (resultsContainer) {
154     resultsContainer.innerHTML = '<div class="text-center py-4
    text-primary"><span class="spinner-border spinner-border-sm m
    e-2"></span> Поиск по коду...</div>';
155   }
156
157   try {
158     const res = await fetch('/api/rag/codebase/search', {
159       method: 'POST',
160       headers: { 'Content-Type': 'application/json' },
161       body: JSON.stringify({
162         query,
163         index_name: indexName,
164         top_k: 5
165       }),
166     });
167
168     const data = await res.json();
169     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
    atus}`);
170
171     const results = data.results || [];
172     if (results.length === 0) {
173       resultsContainer.innerHTML = `<div class="text-center te
    xt-muted py-4">По запросу «${escapeHtml(query)}» совпадений не
    найдено.</div>`;
174       return;
175     }
176
177     resultsContainer.innerHTML = results.map(r => {
178       return `
179         <div class="card mb-2 border shadow-sm">
180           <div class="card-header py-1 px-2 bg-body-secondary
    d-flex justify-content-between align-items-center">
181             <span class="small fw-semibold text-truncate">
```

```

182 |         <i class="bi bi-code-slash me-1 text-primary"></i> ${escapeHtml(r.id)}
183 |     </span>
184 |     <span class="badge bg-info">Score: ${r.similarity_score || 0}</span>
185 | </div>
186 |     <div class="card-body p-2 font-monospace small bg-body-tertiary" style="white-space: pre-wrap; font-size: 0.8rem; max-height: 140px; overflow-y: auto;">${escapeHtml(r.text)}</div>
187 | </div>
188 | `;
189 | }).join('');
190 | } catch (err) {
191 |     if (resultsContainer) {
192 |         resultsContainer.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка: ${escapeHtml(err.message)}</div>`;
193 |     }
194 | } finally {
195 |     if (btn) btn.disabled = false;
196 | }
197 | }
198 |
199 | /**
200 |  * Execute symbol lookup in codebase AST symbols table.
201 |  *
202 |  * @returns {Promise<void>}
203 |  */
204 | export async function executeSymbolLookup() {
205 |     const symbolInput = document.getElementById('codebase-symbol-query');
206 |     const select = document.getElementById('codebase-active-index-select');
207 |     const resultsContainer = document.getElementById('codebase-symbol-results');
208 |     const btn = document.getElementById('btn-codebase-symbol-search');

```

```

209
210     const symbol = symbolInput?.value?.trim();
211     if (!symbol) return;
212
213     const indexName = select?.value || 'codebase';
214     if (btn) btn.disabled = true;
215     if (resultsContainer) {
216         resultsContainer.innerHTML = '<div class="text-center py-4
text-success"><span class="spinner-border spinner-border-sm m
e-2"></span> Поиск символов в AST...</div>';
217     }
218
219     try {
220         const res = await fetch('/api/rag/codebase/symbols', {
221             method: 'POST',
222             headers: { 'Content-Type': 'application/json' },
223             body: JSON.stringify({
224                 symbol,
225                 index_name: indexName,
226                 limit: 10
227             }),
228         });
229
230         const data = await res.json();
231         if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
232
233         const results = data.results || [];
234         if (results.length === 0) {
235             resultsContainer.innerHTML = `<div class="text-center te
xt-muted py-4">Символ «${escapeHtml(symbol)}» не найден в AST-
индексе.</div>`;
236             return;
237         }
238
239         resultsContainer.innerHTML = results.map(s => {
240             return `
241                 <div class="card mb-2 border border-success-subtle sha

```

```

dow-sm">
242         <div class="card-header py-1 px-2 bg-success-subtle
text-success-emphasis d-flex justify-content-between align-ite
ms-center">
243             <span class="small fw-semibold">
244                 <i class="bi bi-tag-fill me-1"></i> ${escapeHtml
(s.symbol)}
245             </span>
246             <span class="badge bg-success">${escapeHtml(s.type
)}</span>
247         </div>
248         <div class="card-body p-2 small">
249             <div><strong>Path:</strong> <code>${escapeHtml(s.p
ath)}</code></div>
250             <div><strong>Signature:</strong> <code>${escapeHtm
l(s.signature || '')}</code></div>
251             ${s.docstring ? `<div class="mt-1 text-muted"><em>
${escapeHtml(s.docstring)}</em></div>` : ''}
252             ${s.related_symbols?.length ? `<div class="mt-1 sm
all text-secondary">Related: ${escapeHtml(s.related_symbols.sl
ice(0, 8).join(', '))}</div>` : ''}
253         </div>
254     </div>
255     `;
256     }).join('');
257 } catch (err) {
258     if (resultsContainer) {
259         resultsContainer.innerHTML = `<div class="alert alert-da
nger p-2 small mb-0">Ошибка поиска: ${escapeHtml(err.message)}
</div>`;
260     }
261 } finally {
262     if (btn) btn.disabled = false;
263 }
264 }
265

```

docRag.js

```
1 /**
2  * =====
3  * Process Name: Global Document RAG Management
4  * =====
5  * Description:
6  *   Handles file/folder uploading, document indexing, RAG sta
7  *   tus retrieval,
8  *   document deletion, and semantic query execution for globa
9  *   l document RAG.
10  *
11  * File: docRag.js
12  * Project: AI Breadboard
13  * Module: RAGTab
14  * Author: hypo69
15  * Copyright: © 2026 hypo69
16  * =====
17  */
18
19 import { formatBytes, escapeHtml } from './utils.js';
20 import { ragCachedFetch, CACHE_TAG } from './cache.js';
21
22 /**
23  * Upload one or multiple files to the global document RAG sto
24  * rage.
25  *
26  * @param {File[]} files - List of files to upload.
27  * @returns {Promise<void>}
28  */
29 export async function uploadFiles(files) {
30   if (!files || files.length === 0) return;
31
32   const progressContainer = document.getElementById('rag-uploa
33   d-progress-container');
```

```
30 |   const progressBar = document.getElementById('rag-upload-prog
    |   ress-bar');
31 |   const percentText = document.getElementById('rag-upload-perc
    |   ent');
32 |   const statusAlert = document.getElementById('rag-upload-stat
    |   us-alert');
33 |
34 |   if (progressContainer) progressContainer.classList.remove('d
    |   -none');
35 |   if (statusAlert) statusAlert.classList.add('d-none');
36 |   if (progressBar) progressBar.style.width = '0%';
37 |   if (percentText) percentText.textContent = '0%';
38 |
39 |   const formData = new FormData();
40 |   files.forEach((file) => {
41 |     const uploadName = file.customRelativePath || file.webkitR
    |   elativePath || file.name;
42 |     formData.append('files', file, uploadName);
43 |   });
44 |
45 |   try {
46 |     if (progressBar) progressBar.style.width = '50%';
47 |     if (percentText) percentText.textContent = '50%';
48 |
49 |     const res = await fetch('/api/rag/upload', {
50 |       method: 'POST',
51 |       body: formData,
52 |     });
53 |
54 |     if (progressBar) progressBar.style.width = '100%';
55 |     if (percentText) percentText.textContent = '100%';
56 |
57 |     const data = await res.json();
58 |     if (!res.ok) throw new Error(data.detail || `Upload failed
    |   (HTTP ${res.status})`);
59 |
60 |     if (statusAlert) {
61 |       statusAlert.className = 'alert alert-success p-2 small m
```

```

    b-0';
62     statusAlert.textContent = `Успешно загружено файлов: ${data.uploaded?.length || 0}`;
63     statusAlert.classList.remove('d-none');
64   }
65
66   await loadRagDocuments();
67   await loadRagStatus();
68 } catch (err) {
69   console.error('[RAG] Upload error:', err);
70   if (statusAlert) {
71     statusAlert.className = 'alert alert-danger p-2 small mb
    -0';
72     statusAlert.textContent = `Ошибка загрузки: ${err.message}`;
73     statusAlert.classList.remove('d-none');
74   }
75   } finally {
76     setTimeout(() => {
77       if (progressContainer) progressContainer.classList.add('
    d-none');
78     }, 1500);
79   }
80 }
81
82 /**
83  * Fetch and display global RAG index status and stats.
84  *
85  * @returns {Promise<void>}
86  */
87 export async function loadRagStatus() {
88   try {
89     const res = await ragCachedFetch('/api/rag/status', {
90       tags: ['rag-status']
91     });
92     if (!res.ok) return;
93     const data = await res.json();
94     const st = data.data || {};

```

```

95 |
96 |     const statDocs = document.getElementById('rag-stat-docs');
97 |     const statChunks = document.getElementById('rag-stat-chunk
    | s');
98 |     const statDim = document.getElementById('rag-stat-dim');
99 |     const statUpdated = document.getElementById('rag-stat-upda
    | ted');
100 |     const providerBadge = document.getElementById('rag-provide
    | r-badge');
101 |
102 |     if (statDocs) statDocs.textContent = st.total_documents ||
    | 0;
103 |     if (statChunks) statChunks.textContent = st.total_chunks |
    | | 0;
104 |     if (statDim) statDim.textContent = st.vector_dimension ||
    | 0;
105 |     if (statUpdated) {
106 |         statUpdated.textContent = st.last_built_at
107 |             ? new Date(st.last_built_at * 1000).toLocaleTimeString
    | ()
108 |             : '-';
109 |     }
110 |     if (providerBadge) {
111 |         const provName = st.provider === 'gemini' ? 'Google Gemi
    | ni Embeddings' : 'Local TF-IDF';
112 |         providerBadge.textContent = `Provider: ${provName}`;
113 |     }
114 | } catch (err) {
115 |     console.error('[RAG] Error loading status:', err);
116 | }
117 | }
118 |
119 | /**
120 |  * Fetch and render the list of uploaded global RAG documents.
121 |  *
122 |  * @returns {Promise<void>}
123 |  */
124 | export async function loadRagDocuments() {

```



```
125 | const tbody = document.getElementById('rag-documents-tbody')
    | ;
126 | const badge = document.getElementById('rag-docs-badge');
127 | if (!tbody) return;
128 |
129 | try {
130 |     const res = await ragCachedFetch('/api/rag/documents', {
131 |         tags: ['rag-documents']
132 |     });
133 |     if (!res.ok) return;
134 |     const data = await res.json();
135 |     const docs = data.documents || [];
136 |
137 |     if (badge) badge.textContent = `${docs.length} файлов`;
138 |     if (docs.length === 0) {
139 |         tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted py-3">Документы не загружены.</td></tr>';
140 |         return;
141 |     }
142 |
143 |     tbody.innerHTML = docs.map((d, idx) => {
144 |         const statusBadge = d.status === 'indexed'
145 |             ? '<span class="badge bg-success">Индексирован</span>'
146 |             : (d.status === 'error' ? '<span class="badge bg-danger">Ошибка</span>' : '<span class="badge bg-warning text-dark">Ожидание</span>');
147 |
148 |         return `
149 |             <tr class="rag-doc-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для AI-анализа документа RAG">
150 |                 <td class="ps-3 text-truncate" style="max-width: 200px;" title="${escapeHtml(d.name)}">
151 |                     <i class="bi bi-file-earmark-code me-1 text-primary"></i> ${escapeHtml(d.name)}
152 |                 </td>
153 |                 <td>${formatBytes(d.size_bytes)}</td>
154 |                 <td><span class="badge bg-secondary-subtle text-secondary">${d.chunks_count || 0}</span></td>
```

```

155 |         <td>${statusBadge}</td>
156 |         <td class="text-end pe-3">
157 |             <button class="btn btn-outline-danger btn-sm p-1 p-
y-0 btn-del-rag-doc" data-name="${escapeHtml(d.name)}" title="
Удалить">
158 |                 <i class="bi bi-trash"></i>
159 |             </button>
160 |         </td>
161 |     </tr>
162 | `;
163 | }).join('');
164 |
165 | tbody.querySelectorAll('.rag-doc-row').forEach(row => {
166 |     row.onclick = (evt) => {
167 |         if (evt.target.closest('.btn-del-rag-doc')) return;
168 |         const idx = parseInt(row.getAttribute('data-idx'), 10)
169 |
170 |         ;
171 |         const d = docs[idx];
172 |         if (!d) return;
173 |
174 |         if (window.AITableModal) {
175 |             window.AITableModal.show({
176 |                 icon: '🗑️ ',
177 |                 title: d.name,
178 |                 subtitle: `Размер: ${formatBytes(d.size_bytes)} |
Чанков: ${d.chunks_count || 0}`,
179 |                 tableType: 'rag_doc',
180 |                 badges: [
181 |                     { text: d.status || 'Indexed', class: d.status =
== 'indexed' ? 'badge bg-success' : 'badge bg-secondary' },
182 |                     { text: `${d.chunks_count || 0} chunks`, class:
'badge bg-info text-dark' }
183 |                 ],
184 |                 metadata: [
185 |                     { label: 'Имя документа', value: d.name },
186 |                     { label: 'Статус индексации', value: d.status ||
'Indexed' },
187 |                     { label: 'Количество чанков', value: String(d.ch

```

```

    unks_count || 0) },
186         { label: 'Размер файла', value: formatBytes(d.size_bytes) },
187         { label: 'Дата индексации', value: d.updated_at || d.created_at || 'N/A' },
188         { label: 'Путь источника', value: d.source_path || d.name, isCode: true, fullWidth: true }
189     ],
190     rawTitle: 'Метаданные документа',
191     rawContent: JSON.stringify(d, null, 2),
192     requestData: {
193         name: d.name,
194         chunks_count: d.chunks_count,
195         size: formatBytes(d.size_bytes),
196         status: d.status
197     }
198     });
199 }
200 };
201 });
202
203 tbody.querySelectorAll('.btn-del-rag-doc').forEach(btn =>
204 {
205     btn.onclick = (evt) => {
206         evt.stopPropagation();
207         const docName = btn.getAttribute('data-name');
208         deleteRagDocument(docName);
209     };
210 });
211 } catch (err) {
212     console.error('[RAG] Error loading documents:', err);
213 }
214
215 /**
216  * Delete a specific document from global RAG.
217  *
218  * @param {string} filename - Name of the document to delete.

```

```

219 | * @returns {Promise<void>}
220 | */
221 | export async function deleteRagDocument(filename) {
222 |   if (!confirm(`Удалить документ «${filename}» из базы знаний?`)) return;
223 |   try {
224 |     const res = await fetch(`/api/rag/documents/${encodeURIComponent(filename)}`, {
225 |       method: 'DELETE',
226 |     });
227 |     window.showToast?(`Документ «${filename}» успешно удален`, 'success');
228 |     await loadRagDocuments();
229 |     await loadRagStatus();
230 |   } catch (err) {
231 |     window.showToast?(`Ошибка удаления: ${err.message}`, 'danger') || alert(`Ошибка удаления: ${err.message}`);
232 |   }
233 | }
234 |
235 | /**
236 |  * Trigger global RAG index build.
237 |  *
238 |  * @returns {Promise<void>}
239 |  */
240 | export async function buildRagIndex() {
241 |   const buildBtn = document.getElementById('btn-build-rag-index');
242 |   const statusEl = document.getElementById('rag-build-status');
243 |   const providerSelect = document.getElementById('rag-provider-select');
244 |   const apiKeyInput = document.getElementById('rag-api-key-input');
245 |   const chunkSizeInput = document.getElementById('rag-chunk-size');
246 |   const chunkOverlapInput = document.getElementById('rag-chunk-overlap');

```

```
247
248   if (buildBtn) buildBtn.disabled = true;
249   if (statusEl) {
250     statusEl.innerHTML = '<span class="spinner-border spinner-
border-sm me-1"></span> Построение векторного индекса...';
251   }
252
253   try {
254     const res = await fetch('/api/rag/build', {
255       method: 'POST',
256       headers: { 'Content-Type': 'application/json' },
257       body: JSON.stringify({
258         provider: providerSelect?.value || 'auto',
259         api_key: apiKeyInput?.value?.trim() || '',
260         chunk_size: parseInt(chunkSizeInput?.value || '500', 1
0),
261         chunk_overlap: parseInt(chunkOverlapInput?.value || '5
0', 10),
262       }),
263     });
264
265     const data = await res.json();
266     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
267
268     if (statusEl) {
269       statusEl.className = 'small mt-2 text-center text-succes
s fw-semibold';
270       statusEl.textContent = `✅ Индекс успешно построен. Всего
чанков: ${data.result?.total_chunks || 0}`;
271     }
272
273     await loadRagStatus();
274     await loadRagDocuments();
275   } catch (err) {
276     console.error('[RAG] Build error:', err);
277     if (statusEl) {
278       statusEl.className = 'small mt-2 text-center text-danger
```

```

';
279     statusEl.textContent = `❌ Ошибка индексации: ${err.message}`;
280   }
281   } finally {
282     if (buildBtn) buildBtn.disabled = false;
283   }
284 }
285
286 /**
287  * Execute semantic query against global RAG index.
288  *
289  * @returns {Promise<void>}
290  */
291 export async function executeRagSearch() {
292   const queryInput = document.getElementById('rag-search-query');
293   const topKSelect = document.getElementById('rag-top-k');
294   const minScoreInput = document.getElementById('rag-min-score');
295   const resultsContainer = document.getElementById('rag-search-results');
296   const searchBtn = document.getElementById('btn-rag-search');
297   const apiKeyInput = document.getElementById('rag-api-key-input');
298
299   const query = queryInput?.value?.trim();
300   if (!query) return;
301
302   if (searchBtn) searchBtn.disabled = true;
303   if (resultsContainer) {
304     resultsContainer.innerHTML = '<div class="text-center py-4 text-primary"><span class="spinner-border spinner-border-sm me-2"></span> Идет семантический поиск...</div>';
305   }
306
307   try {
308     const res = await fetch('/api/rag/search', {

```

```

309     method: 'POST',
310     headers: { 'Content-Type': 'application/json' },
311     body: JSON.stringify({
312         query,
313         top_k: parseInt(topKSelect?.value || '5', 10),
314         min_score: parseFloat(minScoreInput?.value || '0.0'),
315         api_key: apiKeyInput?.value?.trim() || '',
316     }),
317 });
318
319     const data = await res.json();
320     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
321
322     const results = data.results || [];
323     if (results.length === 0) {
324         resultsContainer.innerHTML = `<div class="text-center te
xt-muted py-4"><i class="bi bi-search fs-3 d-block mb-1 opacit
y-50"></i>По запросу «${escapeHtml(query)}» ничего не найдено.
</div>`;
325         return;
326     }
327
328     resultsContainer.innerHTML = results.map(r => `
329         <div class="card mb-2 border shadow-sm">
330             <div class="card-header py-1 px-2 bg-body-secondary d-
flex justify-content-between align-items-center">
331                 <span class="small fw-semibold text-truncate" style=
"max-width: 70%;">
332                     <i class="bi bi-file-earmark-text me-1 text-primar
y"></i> ${escapeHtml(r.doc_name)} (#${r.chunk_index})
333                 </span>
334                 <span class="badge bg-secondary">Score: ${r.score}</
span>
335             </div>
336             <div class="card-body p-2 font-monospace small bg-body
-tertiary" style="white-space: pre-wrap; font-size: 0.8rem; ma
x-height: 120px; overflow-y: auto;">${escapeHtml(r.text)}</div

```

```
337 | >
338 |     </div>
339 |     `).join('');
340 | } catch (err) {
341 |     if (resultsContainer) {
342 |         resultsContainer.innerHTML = `<div class="alert alert-da
343 | nger p-2 small mb-0">Ошибка: ${escapeHtml(err.message)}</div>`
344 | ;
345 |     }
346 | } finally {
347 |     if (searchBtn) searchBtn.disabled = false;
348 | }
```


pixelRag.js

```
1 /**
2  * =====
3  * Process Name: PixelRAG Client Module
4  * =====
5  * Description:
6  *   Handles status retrieval, image ingestion/indexing, query
7  *   classification
8  *   and hybrid semantic visual search for PixelRAG.
9  *
10 * File: pixelRag.js
11 * Project: AI Breadboard
12 * Module: RAGTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17 import { formatBytes, escapeHtml } from './utils.js';
18
19 /**
20 * Load and display PixelRAG system status, FAISS index metric
21 * s, and image counts.
22 * @returns {Promise<void>}
23 */
24 export async function loadPixelRagStatus() {
25   try {
26     const res = await fetch('/api/rag/pixel/status');
27     if (!res.ok) return;
28
29     const data = await res.json();
30     const imagesEl = document.getElementById('pixel-stat-image
s');
```

```
31     const chunksEl = document.getElementById('pixel-stat-chunk
    s');
32     const sizeEl = document.getElementById('pixel-stat-size');
33     const providerEl = document.getElementById('pixel-stat-pro
    vider');
34     const updatedEl = document.getElementById('pixel-stat-upda
    ted');
35     const statusEl = document.getElementById('pixel-stat-stat
    us');
36
37     if (imagesEl) imagesEl.textContent = data.total_images ||
    0;
38     if (chunksEl) chunksEl.textContent = data.total_chunks ||
    0;
39     if (sizeEl) sizeEl.textContent = `${(data.total_size_mb ||
    0).toFixed(2)} MB`;
40     if (providerEl) providerEl.textContent = data.provider ||
    'FAISS / PixelRAG';
41     if (statusEl) statusEl.textContent = `Текстовых чанков: ${
    data.sources?.text_chunks || 0} | Картинок: ${data.sources?.im
    ages || 0}`;
42
43     if (updatedEl) {
44         if (data.last_built_at && data.last_built_at > 0) {
45             const d = new Date(data.last_built_at * 1000);
46             updatedEl.textContent = d.toLocaleString();
47         } else {
48             updatedEl.textContent = 'Индекс готов';
49         }
50     }
51 } catch (err) {
52     console.debug('[PixelRAG] Failed to load status:', err);
53 }
54 }
55
56 /**
57  * Upload and index image files into PixelRAG.
58  *
```

```
59 * @param {File[]} files - List of image files.
60 * @returns {Promise<void>}
61 */
62 export async function uploadAndIndexPixelImages(files) {
63   if (!files || files.length === 0) return;
64
65   const progressContainer = document.getElementById('pixel-upl
oad-progress');
66   const progressBar = document.getElementById('pixel-upload-pr
ogress-bar');
67   const percentText = document.getElementById('pixel-upload-pe
rcent');
68   const statusAlert = document.getElementById('pixel-upload-st
atus-alert');
69
70   if (progressContainer) progressContainer.classList.remove('d
-none');
71   if (statusAlert) statusAlert.classList.add('d-none');
72   if (progressBar) progressBar.style.width = '20%';
73   if (percentText) percentText.textContent = '20%';
74
75   const formData = new FormData();
76   files.forEach((file) => {
77     formData.append('files', file, file.name);
78   });
79
80   try {
81     if (progressBar) progressBar.style.width = '60%';
82     if (percentText) percentText.textContent = '60%';
83
84     const res = await fetch('/api/rag/pixel/index', {
85       method: 'POST',
86       body: formData,
87     });
88
89     if (progressBar) progressBar.style.width = '100%';
90     if (percentText) percentText.textContent = '100%';
91
```

```

92     const data = await res.json();
93     if (!res.ok) throw new Error(data.detail || `Индексация не
    удалась (HTTP ${res.status})`);
94
95     if (statusAlert) {
96         statusAlert.className = 'alert alert-success p-2 small m
b-0';
97         statusAlert.textContent = `✅ Успешно проиндексировано: $
{data.indexed_count || files.length} файл(ов)`;
98         statusAlert.classList.remove('d-none');
99     }
100
101     await loadPixelRagStatus();
102 } catch (err) {
103     console.error('[PixelRAG] Indexing error:', err);
104     if (statusAlert) {
105         statusAlert.className = 'alert alert-danger p-2 small mb
-0';
106         statusAlert.textContent = `Ошибка индексации: ${err.mess
age}`;
107         statusAlert.classList.remove('d-none');
108     }
109 } finally {
110     setTimeout(() => {
111         if (progressContainer) progressContainer.classList.add('
d-none');
112     }, 2000);
113 }
114 }
115
116 /**
117  * Execute smart query analysis via QueryRouter.
118  *
119  * @returns {Promise<void>}
120  */
121 export async function analyzePixelQuery() {
122     const input = document.getElementById('pixel-analyze-query-i
nput');

```

```
123 |   const resultBox = document.getElementById('pixel-analyze-res
    | ult');
124 |   const routingTypeEl = document.getElementById('pixel-routing
    | -type');
125 |   const confidenceEl = document.getElementById('pixel-routing-
    | confidence');
126 |   const reasonEl = document.getElementById('pixel-routing-reas
    | on');
127 |
128 |   if (!input || !input.value.trim()) return;
129 |   const query = input.value.trim();
130 |
131 |   try {
132 |     const res = await fetch(`/api/rag/pixel/analyze-query?quer
    | y=${encodeURIComponent(query)}`, {
133 |       method: 'POST',
134 |     });
135 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
136 |
137 |     const data = await res.json();
138 |     if (resultBox) resultBox.classList.remove('d-none');
139 |     if (routingTypeEl) {
140 |       routingTypeEl.textContent = data.routing_type?.toUpperCase() || 'HYBRID';
141 |       routingTypeEl.className = data.routing_type === 'visual'
    | ? 'badge bg-warning text-dark' : data.routing_type === 'text'
    | ? 'badge bg-info' : 'badge bg-primary';
142 |     }
143 |     if (confidenceEl) confidenceEl.textContent = `${Math.round
    | ((data.confidence || 1.0) * 100)}%`;
144 |     if (reasonEl) reasonEl.textContent = `Причина: ${data.reas
    | on || 'Авто-определение по ключевым признакам'}`;
145 |   } catch (err) {
146 |     console.error('[PixelRAG] Analysis failed:', err);
147 |   }
148 | }
149 |
150 | /**
```

```

151 | * Execute semantic visual search inside PixelRAG.
152 | *
153 | * @returns {Promise<void>}
154 | */
155 | export async function executePixelRagSearch() {
156 |     const queryInput = document.getElementById('pixel-search-que
ry');
157 |     const topKSelect = document.getElementById('pixel-search-top
k');
158 |     const minScoreSelect = document.getElementById('pixel-search
-minscore');
159 |     const routingToggle = document.getElementById('pixel-search-
routing-toggle');
160 |     const resultsContainer = document.getElementById('pixel-sear
ch-results');
161 |
162 |     if (!queryInput || !resultsContainer) return;
163 |     const query = queryInput.value.trim();
164 |     if (!query) {
165 |         queryInput.focus();
166 |         return;
167 |     }
168 |
169 |     const topK = parseInt(topKSelect?.value || '5', 10);
170 |     const minScore = parseFloat(minScoreSelect?.value || '0.0');
171 |     const useRouting = routingToggle ? routingToggle.checked : t
rue;
172 |
173 |     resultsContainer.innerHTML = `
174 |         <div class="text-center py-4 text-muted">
175 |             <div class="spinner-border spinner-border-sm text-primar
y me-2" role="status"></div>
176 |             <span>Поиск по визуальным индексам и текстовым чанкам...
</span>
177 |         </div>
178 |     `;
179 |
180 |     try {

```

```

181 |     const res = await fetch('/api/rag/pixel/search', {
182 |         method: 'POST',
183 |         headers: { 'Content-Type': 'application/json' },
184 |         body: JSON.stringify({
185 |             query: query,
186 |             top_k: topK,
187 |             min_score: minScore,
188 |             use_routing: useRouting,
189 |             version_filter: 'latest',
190 |         })),
191 |     });
192 |
193 |     if (!res.ok) {
194 |         const errData = await res.json().catch(() => ({}));
195 |         throw new Error(errData.detail || `HTTP ${res.status}`);
196 |     }
197 |
198 |     const data = await res.json();
199 |     const results = data.results || [];
200 |
201 |     if (results.length === 0) {
202 |         resultsContainer.innerHTML = `
203 |             <div class="text-center text-muted py-4">
204 |                 <i class="bi bi-search fs-3 d-block mb-1 opacity-50"
205 |                 ></i>
206 |                 <span>По запросу «${escapeHtml(query)}» ничего не на
207 |                 йдено.</span>
208 |             </div>
209 |             `;
210 |         return;
211 |     }
212 |     resultsContainer.innerHTML = `
213 |         <div class="d-flex justify-content-between align-items-c
214 |         enter mb-2 pb-2 border-bottom">
215 |             <span class="small text-muted">Найдено совпадений: <st
216 |             rong>${results.length}</strong> (Маршрут: <span class="badge b
217 |             g-secondary">${data.routing_type}</span>)</span>

```

```

214 |         <span class="small text-muted">Уверенность: ${data.qu
    | ery_confidence * 100).toFixed(0)}%</span>
215 |     </div>
216 |     <div class="d-flex flex-column gap-2">
217 |         ${results.map((r, i) => {
218 |             const scorePercent = Math.min(100, Math.round((r.sco
    | re || 0) * 100));
219 |             const isVisual = r.source_type === 'pixel' || r.sour
    | ce_type === 'image';
220 |             const typeBadge = isVisual
221 |                 ? '<span class="badge bg-warning text-dark"><i cla
    | ss="bi bi-image me-1"></i>Visual</span>'
222 |                 : '<span class="badge bg-info"><i class="bi bi-fil
    | e-text me-1"></i>Text</span>';
223 |
224 |             return `
225 |                 <div class="card bg-body border-secondary-subtle p
    | -2">
226 |                     <div class="d-flex justify-content-between align
    | -items-center mb-1">
227 |                         <span class="fw-bold small text-primary text-t
    | runcate" style="max-width: 70%;">
228 |                             #${i + 1} ${escapeHtml(r.doc_name || 'Без на
    | звания')}>
229 |                         </span>
230 |                         <div class="d-flex align-items-center gap-1">
231 |                             ${typeBadge}
232 |                             <span class="badge bg-success-subtle text-su
    | ccess border border-success-subtle">
233 |                                 Score: ${r.score || 0).toFixed(3)} (${sco
    | rePercent}%)
234 |                             </span>
235 |                         </div>
236 |                     </div>
237 |                     <div class="small text-body-secondary font-monos
    | pace" style="font-size: 0.8rem; white-space: pre-wrap;">
238 |                         ${escapeHtml(r.text || '')}
239 |                     </div>

```



```

240 |         ${r.source_path ? `<div class="small text-muted
      | mt-1 text-truncate" style="font-size: 0.72rem;"><i class="bi b
      | i-link-45deg"></i> ${escapeHtml(r.source_path)}</div>` : ''}
241 |         </div>
242 |         `;
243 |         }).join('')}
244 |     </div>
245 |     `;
246 | } catch (err) {
247 |     console.error('[PixelRAG] Search error:', err);
248 |     resultsContainer.innerHTML = `
249 |         <div class="alert alert-danger p-2 small mb-0">
250 |             <i class="bi bi-exclamation-triangle me-1"></i> Ошибка
      | поиска: ${escapeHtml(err.message)}
251 |         </div>
252 |         `;
253 |     }
254 | }
255 |

```

userRagCollections.js

```
1 /**
2  * =====
3  * Process Name: Workspace Multi-RAG Collections Management
4  * =====
5  * Description:
6  *   Manages user workspace RAG collections listing, selection
7  *   and collection deletion.
8  *
9  * File: userRagCollections.js
10 * Project: AI Breadboard
11 * Module: RAGTab
12 * Author: hypo69
13 * Copyright: © 2026 hypo69
14 * =====
15 */
16
17 import { escapeHtml } from './utils.js';
18 import { loadUserStorageFiles } from './userRagPipeline.js';
19
20 let activeUserRagId = null;
21 let userCollections = [];
22
23 /**
24  * Get the currently active collection ID.
25  *
26  * @returns {string|null} Active RAG ID or null.
27  */
28 export function getActiveUserRagId() {
29   return activeUserRagId;
30 }
31
32 /**
```

```

33 | * Set the active collection ID.
34 | *
35 | * @param {string|null} id - Target collection ID.
36 | */
37 | export function setActiveUserRagId(id) {
38 |   activeUserRagId = id;
39 | }
40 |
41 | /**
42 |  * Fetch all user workspace RAG collections and render sidebar
43 |  * list.
44 |  *
45 |  * @returns {Promise<void>}
46 |  */
47 | export async function loadUserRags() {
48 |   const listContainer = document.getElementById('user-rags-list-container');
49 |   if (!listContainer) return;
50 |   try {
51 |     const res = await fetch('/api/user/rags');
52 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
53 |     const data = await res.json();
54 |     userCollections = data.collections || [];
55 |     renderUserRagsList(userCollections);
56 |
57 |     if (userCollections.length > 0) {
58 |       if (!activeUserRagId || !userCollections.some(c => c.id === activeUserRagId)) {
59 |         selectUserRag(userCollections[0].id);
60 |       } else {
61 |         selectUserRag(activeUserRagId);
62 |       }
63 |     } else {
64 |       activeUserRagId = null;
65 |       showUserRagEmptyState();
66 |     }
67 |   } catch (err) {

```

```

68 |     listContainer.innerHTML = `
```

```

    subtle bg-body-tertiary'}"
99         data-rag-id="${escapeHtml(col.id)}" style="cursor:
pointer; transition: all 0.15s ease-in-out;">
100         <div class="d-flex justify-content-between align-items
-center mb-1">
101             <strong class="text-truncate small" style="max-width
: 140px;">${escapeHtml(col.name || col.id)}</strong>
102             <span class="badge ${statusBadgeClass} small" style=
"font-size: 0.7rem;">${escapeHtml(col.status || 'created')}</s
pan>
103         </div>
104         <div class="d-flex justify-content-between align-items
-center small text-muted" style="font-size: 0.75rem;">
105             <span><i class="bi bi-diagram-3 me-1"></i>${col.tota
l_chunks || 0} чанков</span>
106             <span><i class="bi bi-file-earmark me-1"></i>${(col.
files || []).length} файлов</span>
107         </div>
108     </div>
109     `;
110     }).join('');
111
112     listContainer.querySelectorAll('.user-rag-item').forEach(el
=> {
113         el.addEventListener('click', () => {
114             const ragId = el.getAttribute('data-rag-id');
115             if (ragId) selectUserRag(ragId);
116         });
117     });
118 }
119
120 /**
121  * Display placeholder empty state when no collection is selec
ted.
122  */
123 export function showUserRagEmptyState() {
124     const emptyState = document.getElementById('user-rag-empty-s
tate');

```

```
125 |   const detailContainer = document.getElementById('user-rag-de
    | tail-container');
126 |   if (emptyState) emptyState.classList.remove('d-none');
127 |   if (detailContainer) detailContainer.classList.add('d-none')
    | ;
128 | }
129 |
130 | /**
131 |  * Select a workspace collection and load its metadata and fil
    | e checklist.
132 |  *
133 |  * @param {string} ragId - ID of the collection to select.
134 |  * @returns {Promise<void>}
135 |  */
136 | export async function selectUserRag(ragId) {
137 |   activeUserRagId = ragId;
138 |   const emptyState = document.getElementById('user-rag-empty-s
    | tate');
139 |   const detailContainer = document.getElementById('user-rag-de
    | tail-container');
140 |   if (emptyState) emptyState.classList.add('d-none');
141 |   if (detailContainer) detailContainer.classList.remove('d-non
    | e');
142 |
143 |   document.querySelectorAll('.user-rag-item').forEach(el => {
144 |     if (el.getAttribute('data-rag-id') === ragId) {
145 |       el.classList.add('border-primary', 'bg-primary-subtle');
146 |       el.classList.remove('border-secondary-subtle', 'bg-body-
    | tertiary');
147 |     } else {
148 |       el.classList.remove('border-primary', 'bg-primary-subtle
    | ');
149 |       el.classList.add('border-secondary-subtle', 'bg-body-ter
    | tiary');
150 |     }
151 |   });
152 |
153 |   try {
```

```
154     const res = await fetch(`/api/user/rags/${encodeURIComponent(ragId)}`);
155     if (!res.ok) throw new Error(`HTTP ${res.status}`);
156     const data = await res.json();
157     const col = data.collection;
158
159     const titleEl = document.getElementById('active-rag-title');
160     const descEl = document.getElementById('active-rag-desc');
161     const statusBadge = document.getElementById('active-rag-status-badge');
162     const chunksEl = document.getElementById('active-rag-stat-chunks');
163     const filesEl = document.getElementById('active-rag-stat-files');
164     const updatedEl = document.getElementById('active-rag-stat-updated');
165
166     if (titleEl) titleEl.textContent = col.name || col.id;
167     if (descEl) descEl.textContent = col.description || 'Без описания';
168     if (statusBadge) {
169       statusBadge.textContent = col.status || 'created';
170       statusBadge.className = `badge ${col.status === 'indexed' ? 'bg-success' : (col.status === 'cleaning' ? 'bg-warning text-dark' : (col.status === 'error' ? 'bg-danger' : 'bg-secondary'))}`;
171     }
172     if (chunksEl) chunksEl.textContent = col.total_chunks || 0;
173     if (filesEl) filesEl.textContent = (col.files || []).length;
174     if (updatedEl) {
175       updatedEl.textContent = col.updated_at ? new Date(col.updated_at * 1000).toLocaleString() : '-';
176     }
177
178     await loadUserStorageFiles(col.files || []);
```

```

179     await loadRagEntries(ragId);
180   } catch (err) {
181     console.error('Failed to load user RAG details:', err);
182   }
183 }
184
185 /**
186  * Fetch and render chunk/QA entries for the active collection
187  *
188  * @param {string} ragId - Collection identifier.
189  * @param {string} query - Optional filter query.
190  * @returns {Promise<void>}
191  */
192 export async function loadRagEntries(ragId = activeUserRagId,
  query = '') {
193   const container = document.getElementById('user-rag-entries-
  container');
194   if (!container || !ragId) return;
195
196   container.innerHTML = '<div class="text-center text-muted sm
  all py-2"><div class="spinner-border spinner-border-sm text-pr
  imary me-1"></div>Загрузка записей...</div>';
197
198   try {
199     const url = `/api/user/rags/${encodeURIComponent(ragId)}/e
  ntries?q=${encodeURIComponent(query)}&limit=100`;
200     const res = await fetch(url);
201     if (!res.ok) throw new Error(`HTTP ${res.status}`);
202     const data = await res.json();
203     renderRagEntries(data.entries || [], data.total || 0);
204   } catch (err) {
205     container.innerHTML = `<div class="alert alert-danger p-2
  small mb-0">Ошибка загрузки записей: ${escapeHtml(err.message)
  }</div>`;
206   }
207 }
208

```



```

209 /**
210  * Render entries inside the collection container.
211  *
212  * @param {Array<object>} entries - List of chunk/QA items.
213  * @param {number} total - Total entries count.
214  */
215 export function renderRagEntries(entries, total) {
216   const container = document.getElementById('user-rag-entries-
container');
217   if (!container) return;
218
219   if (!entries || entries.length === 0) {
220     container.innerHTML = `
221       <div class="text-center text-muted py-3 small">
222         <i class="bi bi-chat-left-dots fs-4 d-block mb-1 opaci
ty-50"></i>
223         В этой коллекции пока нет записей.<br>
224         Нажмите «+ Добавить Q&A» или соберите RAG из файлов вы
ше.
225       </div>
226     `;
227     return;
228   }
229
230   container.innerHTML = entries.map((entry, idx) => {
231     const chunkId = entry.chunk_id || `chunk_${idx}`;
232     const docType = entry.doc_type || 'chunk';
233     const isQa = docType === 'qa' || entry.meta?.is_qa;
234     const badgeClass = isQa ? 'bg-primary' : 'bg-info text-dar
k';
235     const badgeLabel = isQa ? 'Q&A' : (entry.source_file || 'Ф
айл');
236
237     let displayContent = entry.content || '';
238     let questionText = entry.meta?.question || '';
239     let answerText = entry.meta?.answer || '';
240
241     if (!questionText && displayContent.includes('Вопрос:')) &&

```

```

    displayContent.includes('Ответ:')) {
242     const parts = displayContent.split('\nОтвет:');
243     questionText = parts[0].replace('Вопрос:', '').trim();
244     answerText = (parts[1] || '').trim();
245   }
246
247   return `
248     <div class="card p-2 border border-secondary-subtle bg-b
ody-tertiary position-relative rag-entry-card" data-chunk-id="
${escapeHtml(chunkId)}">
249       <div class="d-flex justify-content-between align-items
-start mb-1">
250         <div class="d-flex align-items-center gap-1 flex-wra
p">
251           <span class="badge ${badgeClass} small" style="fon
t-size: 0.7rem;">${escapeHtml(badgeLabel)}</span>
252           <small class="text-muted font-monospace" style="fo
nt-size: 0.72rem;">#${escapeHtml(chunkId)}</small>
253         </div>
254         <button class="btn btn-outline-danger btn-sm py-0 px
-2 rounded-pill btn-delete-entry" data-chunk-id="${escapeHtml(
chunkId)}" title="Удалить запись из RAG">
255           <i class="bi bi-trash"></i>
256         </button>
257       </div>
258       ${questionText ? `
259         <div class="small fw-bold text-primary mb-1"><i clas
s="bi bi-question-circle me-1"></i>${escapeHtml(questionText)}
</div>
260         <div class="small text-secondary font-monospace bg-b
ody p-2 rounded border" style="white-space: pre-wrap; font-siz
e: 0.78rem;">${escapeHtml(answerText || displayContent)}</div>
261       ` : `
262         <div class="small text-secondary font-monospace bg-b
ody p-2 rounded border" style="white-space: pre-wrap; font-siz
e: 0.78rem;">${escapeHtml(displayContent)}</div>
263       `}
264     </div>

```

```

265 |     `;
266 |   }).join('');
267 |
268 |   container.querySelectorAll('.btn-delete-entry').forEach(btn
    => {
269 |     btn.addEventListener('click', (e) => {
270 |       e.stopPropagation();
271 |       const chunkId = btn.getAttribute('data-chunk-id');
272 |       if (chunkId) deleteRagEntry(chunkId);
273 |     });
274 |   });
275 | }
276 |
277 | /**
278 |  * Add a new custom Q&A pair entry to the active collection.
279 |  *
280 |  * @returns {Promise<void>}
281 |  */
282 | export async function addQaEntry() {
283 |   if (!activeUserRagId) {
284 |     window.showToast?(['Пожалуйста, выберите коллекцию RAG.',
    'warning']) || alert('Пожалуйста, выберите коллекцию RAG.');
```

285 | return;

286 | }

287 |

288 | const qInput = document.getElementById('new-qa-question');

289 | const aInput = document.getElementById('new-qa-answer');

290 | const btn = document.getElementById('btn-confirm-add-qa');

291 |

292 | const question = qInput?.value.trim() || '';

293 | const answer = aInput?.value.trim() || '';

294 |

295 | if (!answer) {

296 | window.showToast?(['Пожалуйста, введите текст ответа или з
 нания.', 'warning']) || alert('Пожалуйста, введите текст ответа
 или знания.');

297 | return;

298 | }

```
299 |
300 |   if (btn) btn.disabled = true;
301 |   try {
302 |     const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/entries`, {
303 |       method: 'POST',
304 |       headers: { 'Content-Type': 'application/json' },
305 |       body: JSON.stringify({ question, answer })
306 |     });
307 |
308 |     const data = await res.json();
309 |     if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
310 |
311 |     const modalEl = document.getElementById('modal-add-qa-entry');
312 |     if (modalEl && window.bootstrap?.Modal) {
313 |       const modal = bootstrap.Modal.getInstance(modalEl) || new bootstrap.Modal(modalEl);
314 |       modal.hide();
315 |     }
316 |
317 |     if (qInput) qInput.value = '';
318 |     if (aInput) aInput.value = '';
319 |
320 |     window.showToast?(`Q&A запись успешно добавлена`, 'success');
321 |     await selectUserRag(activeUserRagId);
322 |     await loadUserRags();
323 |   } catch (err) {
324 |     window.showToast?(`Ошибка сохранения Q&A записи: ${err.message}`, 'danger') || alert(`Ошибка сохранения Q&A записи: ${err.message}`);
325 |   } finally {
326 |     if (btn) btn.disabled = false;
327 |   }
328 | }
329 |
```

```

330 /**
331  * Delete a specific chunk/QA entry from the active collection
332  *
333  * @param {string} chunkId - Chunk identifier.
334  * @returns {Promise<void>}
335  */
336 export async function deleteRagEntry(chunkId) {
337   if (!activeUserRagId || !chunkId) return;
338
339   const confirmed = confirm(`Удалить эту запись (${chunkId})
из базы знаний?`);
340   if (!confirmed) return;
341
342   try {
343     const res = await fetch(`/api/user/rags/${encodeURIComponent(
activeUserRagId)}/entries/${encodeURIComponent(chunkId)}`,
{
344       method: 'DELETE'
345     });
346     if (!res.ok) throw new Error(`HTTP ${res.status}`);
347
348     window.showToast?(`Запись успешно удалена`, 'success');
349     await selectUserRag(activeUserRagId);
350     await loadUserRags();
351   } catch (err) {
352     window.showToast?(`Ошибка удаления записи: ${err.message}
`, 'danger') || alert(`Ошибка удаления записи: ${err.message}`
);
353   }
354 }
355
356 /**
357  * Create a new user workspace RAG collection via API.
358  *
359  * @returns {Promise<void>}
360  */
361 export async function createUserRag() {

```

```
362 |   const nameInput = document.getElementById('new-rag-name');
363 |   const descInput = document.getElementById('new-rag-desc');
364 |   const minChunkInput = document.getElementById('new-rag-min-c
    | hunk');
365 |   const maxChunkInput = document.getElementById('new-rag-max-c
    | hunk');
366 |   const btn = document.getElementById('btn-confirm-create-rag'
    | );
367 |
368 |   const name = nameInput?.value.trim();
369 |   if (!name) {
370 |     window.showToast?.( 'Пожалуйста, укажите имя коллекции.', '
    | warning') || alert('Пожалуйста, укажите имя коллекции. ');
371 |     return;
372 |   }
373 |
374 |   if (btn) btn.disabled = true;
375 |   try {
376 |     const res = await fetch('/api/user/rags', {
377 |       method: 'POST',
378 |       headers: { 'Content-Type': 'application/json' },
379 |       body: JSON.stringify({
380 |         name: name,
381 |         description: descInput?.value.trim() || '',
382 |         min_chunk_len: parseInt(minChunkInput?.value || '20',
    | 10),
383 |         max_chunk_len: parseInt(maxChunkInput?.value || '1500'
    | , 10)
384 |       })
385 |     });
386 |
387 |     const data = await res.json();
388 |     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
    | atus}`);
389 |
390 |     const srcType = document.querySelector('input[name="new-ra
    | g-source-type"]:checked')?.value || 'local';
391 |
```

```

392     const modalEl = document.getElementById('modal-create-user
    -rag');
393     if (modalEl && window.bootstrap?.Modal) {
394         const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
395         modal.hide();
396     }
397
398     if (nameInput) nameInput.value = '';
399     if (descInput) descInput.value = '';
400
401     window.showToast?.(`Коллекция '${name}' успешно создана`,
    'success');
402     await loadUserRags();
403     const createdId = data.collection?.id;
404     if (createdId) {
405         selectUserRag(createdId);
406         if (srcType === 'gdocs') {
407             const importModalEl = document.getElementById('modal-i
mport-gdocs');
408             if (importModalEl && window.bootstrap?.Modal) {
409                 const importModal = bootstrap.Modal.getInstance(impo
rtModalEl) || new bootstrap.Modal(importModalEl);
410                 importModal.show();
411             }
412         }
413     }
414 } catch (err) {
415     window.showToast?.(`Ошибка создания коллекции: ${err.messa
ge}`, 'danger') || alert(`Ошибка создания коллекции: ${err.mes
sage}`);
416 } finally {
417     if (btn) btn.disabled = false;
418 }
419 }
420
421 /**
422  * Delete the currently active workspace collection.

```

```
423 | *
424 | * @returns {Promise<void>}
425 | */
426 | export async function deleteActiveUserRag() {
427 |   if (!activeUserRagId) return;
428 |
429 |   const confirmed = confirm(`Вы уверены, что хотите удалить ко
    | ллекцию «${activeUserRagId}»?`);
430 |   if (!confirmed) return;
431 |
432 |   try {
433 |     const res = await fetch(`/api/user/rags/${encodeURIComponent(
    | nt(activeUserRagId))}`, {
434 |       method: 'DELETE'
435 |     });
436 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
437 |     window.showToast?.(`Коллекция '${activeUserRagId}' успешно
    | удалена`, 'success');
438 |     activeUserRagId = null;
439 |     await loadUserRags();
440 |   } catch (err) {
441 |     window.showToast?.(`Ошибка удаления коллекции: ${err.messa
    | ge}`, 'danger') || alert(`Ошибка удаления коллекции: ${err.mes
    | sage}`);
442 |   }
443 | }
444 |
445 |
```


userRagPipeline.js

```
1 /**
2  * =====
3  * Process Name: Workspace Multi-RAG Pipeline and Search
4  * =====
5  * Description:
6  *   Handles workspace storage file checklist, cleaning and TF
7  *   -IDF build
8  *   pipeline, semantic search, and quick upload to user storage.
9  *
10 * File: userRagPipeline.js
11 * Project: AI Breadboard
12 * Module: RAGTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17 import { formatBytes, escapeHtml } from './utils.js';
18 import { getActiveUserRagId, loadUserRags, selectUserRag } from './userRagCollections.js';
19
20 /**
21 * Fetch files from user storage and render checkbox selection
22 * list.
23 * @param {string[]} attachedFiles - Array of filenames currently attached to this collection.
24 * @returns {Promise<void>}
25 */
26 export async function loadUserStorageFiles(attachedFiles) {
27   const checklistContainer = document.getElementById('user-rag-files-checklist');
```

```

28 | if (!checklistContainer) return;
29 |
30 | try {
31 |     const res = await fetch('/api/user/files?subfolder=files')
32 |     ;
33 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
34 |     const data = await res.json();
35 |     const files = data.files || [];
36 |
37 |     if (files.length === 0) {
38 |         checklistContainer.innerHTML = `
39 |             <div class="text-muted small p-2 text-center">
40 |                 В личном хранилище нет файлов.<br>Нажмите «Загрузить
41 |                 файлы» ниже.
42 |             </div>
43 |         `;
44 |         return;
45 |     }
46 |     checklistContainer.innerHTML = files.map(f => {
47 |         const isChecked = attachedFiles.length === 0 || attached
48 |         Files.includes(f.name);
49 |         return `
50 |             <label class="d-flex align-items-center justify-content-between p-1 px-2 rounded hover-bg border-bottom border-light
51 |             -subtle small cursor-pointer">
52 |                 <div class="d-flex align-items-center gap-2">
53 |                     <input class="form-check-input mt-0" type="checkbox" value="${escapeHtml(f.name)}" ${isChecked ? 'checked' : ''}
54 |                     >
55 |                     <span class="text-truncate font-monospace" style="
56 |                     max-width: 280px;">${escapeHtml(f.name)}</span>
57 |                 </div>
58 |                 <span class="text-muted small">${formatBytes(f.size_
59 |                 bytes)}</span>
60 |             </label>
61 |         `;
62 |     }).join('');

```

```

57 |   } catch (err) {
58 |       checklistContainer.innerHTML = `
```

```

85     headers: { 'Content-Type': 'application/json' },
86     body: JSON.stringify({
87         files: selectedFiles.length > 0 ? selectedFiles : null
88     },
89     provider: 'local_tfidf'
90     ));
91
92     const data = await res.json();
93     if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
94
95     if (statusAlert) {
96         statusAlert.className = 'alert alert-success p-2 small m
t-2 mb-0 d-block';
97         statusAlert.innerHTML = `
Успешно! Создано <strong>${data.chunks_count}</strong> чанков из <strong>${data.processed_files?.length || 0}</strong> файлов.`;
98     }
99
100     await loadUserRags();
101 } catch (err) {
102     if (statusAlert) {
103         statusAlert.className = 'alert alert-danger p-2 small m
t-2 mb-0 d-block';
104         statusAlert.textContent = `Ошибка сборки RAG: ${err.message}`;
105     }
106 } finally {
107     if (btn) btn.disabled = false;
108 }
109 }
110
111 /**
112  * Perform semantic search in the active workspace collection.
113  *
114  * @returns {Promise<void>}
115  */

```

```

116 export async function executeActiveUserRagSearch() {
117   const activeUserRagId = getActiveUserRagId();
118   if (!activeUserRagId) return;
119
120   const queryInput = document.getElementById('active-rag-search-
  h-query');
121   const resultsContainer = document.getElementById('active-rag-
  -search-results');
122   const btn = document.getElementById('btn-active-rag-search')
  ;
123   const query = queryInput?.value.trim();
124
125   if (!query) return;
126   if (btn) btn.disabled = true;
127   if (resultsContainer) {
128     resultsContainer.innerHTML = '<div class="text-center text-
  -muted small py-3"><span class="spinner-border spinner-border-
  sm me-1"></span>Поиск...</div>';
129   }
130
131   try {
132     const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/search`, {
133       method: 'POST',
134       headers: { 'Content-Type': 'application/json' },
135       body: JSON.stringify({
136         query: query,
137         top_k: 5,
138         min_score: 0.0
139       })
140     });
141
142     const data = await res.json();
143     if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);
144
145     const results = data.results || [];
146     if (results.length === 0) {

```

```

147 |         resultsContainer.innerHTML = '<div class="text-center text-muted small py-3">Ничего не найдено по данному запросу.</div>';
148 |         return;
149 |     }
150 |
151 |     resultsContainer.innerHTML = results.map((item) => {
152 |         const scorePercent = Math.round((item.score || 0) * 100)
153 |         ;
154 |         return `
155 |             <div class="card mb-2 border border-secondary-subtle bg-body">
156 |                 <div class="card-header py-1 px-2 d-flex justify-content-between align-items-center bg-body-tertiary">
157 |                     <span class="small fw-semibold text-truncate" style="max-width: 70%;">
158 |                         <i class="bi bi-file-earmark-text text-primary me-1"></i> ${escapeHtml(item.source_file || '')}
159 |                     </span>
160 |                     <span class="badge bg-primary-subtle text-primary border border-primary-subtle">${scorePercent}% match</span>
161 |                 </div>
162 |                 <div class="card-body p-2 small">
163 |                     <div class="text-secondary font-monospace" style="font-size: 0.8rem; white-space: pre-wrap;">${escapeHtml(item.text)}</div>
164 |                 </div>
165 |             `;
166 |         }).join('');
167 |     } catch (err) {
168 |         if (resultsContainer) {
169 |             resultsContainer.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка поиска: ${escapeHtml(err.message)}</div>`;
170 |         }
171 |     } finally {
172 |         if (btn) btn.disabled = false;

```

```

173 |   }
174 | }
175 |
176 | /**
177 |  * Handle quick file upload directly into user workspace storage.
178 |  *
179 |  * @param {Event} event - Input change event.
180 |  * @returns {Promise<void>}
181 |  */
182 | export async function handleQuickUploadToUserStorage(event) {
183 |   const files = event.target.files;
184 |   if (!files || files.length === 0) return;
185 |
186 |   const statusAlert = document.getElementById('user-rag-build-
status-alert');
187 |   if (statusAlert) {
188 |     statusAlert.className = 'alert alert-info p-2 small mt-2 m
b-0 d-block';
189 |     statusAlert.innerHTML = `<span class="spinner-border spinn
er-border-sm me-2"></span>Загрузка ${files.length} файл(ов)...
`;
190 |   }
191 |
192 |   try {
193 |     for (const file of files) {
194 |       const formData = new FormData();
195 |       formData.append('file', file);
196 |       formData.append('subfolder', 'files');
197 |
198 |       const res = await fetch('/api/user/files/upload', {
199 |         method: 'POST',
200 |         body: formData
201 |       });
202 |       if (!res.ok) throw new Error(`Ошибка загрузки ${file.name}
`);
203 |     }
204 |

```

```

205     if (statusAlert) {
206         statusAlert.className = 'alert alert-success p-2 small m
t-2 mb-0 d-block';
207         statusAlert.innerHTML = `Загружено <strong>${files.lengt
h}</strong> файлов.`;
208     }
209
210     const activeUserRagId = getActiveUserRagId();
211     if (activeUserRagId) {
212         selectUserRag(activeUserRagId);
213     }
214 } catch (err) {
215     if (statusAlert) {
216         statusAlert.className = 'alert alert-danger p-2 small mt
-2 mb-0 d-block';
217         statusAlert.textContent = err.message;
218     }
219 } finally {
220     event.target.value = '';
221 }
222 }
223
224 /**
225  * Synchronize Google Docs & Drive files directly into the act
ive collection and build index.
226  *
227  * @returns {Promise<void>}
228  */
229 export async function syncGoogleDocsIntoActiveRag() {
230     const activeUserRagId = getActiveUserRagId();
231     if (!activeUserRagId) {
232         window.showToast?.( 'Пожалуйста, выберите или создайте колл
екцию RAG.', 'warning') || alert('Пожалуйста, выберите или соз
дайте коллекцию RAG. ');
233         return;
234     }
235
236     const queryInput = document.getElementById('import-gdocs-que

```



```

ry');
237   const accountInput = document.getElementById('import-gdocs-a
      ccount');
238   const statusAlert = document.getElementById('import-gdocs-st
      atus-alert');
239   const btn = document.getElementById('btn-confirm-import-gdoc
      s');
240
241   const query = queryInput?.value.trim() || "mimeType = 'appli
      cation/vnd.google-apps.document' and trashed = false";
242   const accountName = accountInput?.value.trim() || undefined;
243
244   if (btn) btn.disabled = true;
245   if (statusAlert) {
246     statusAlert.className = 'alert alert-info py-2 small mb-0
      d-block';
247     statusAlert.innerHTML = '<span class="spinner-border spinn
      er-border-sm me-2"></span>Экспорт Google Docs и построение RAG
      индекса...';
248   }
249
250   try {
251     const res = await fetch(`/api/user/rags/${encodeURIComponent(
      activeUserRagId)}/sync-google-docs`, {
252       method: 'POST',
253       headers: { 'Content-Type': 'application/json' },
254       body: JSON.stringify({
255         query: query,
256         account_name: accountName
257       })
258     });
259
260     const data = await res.json();
261     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
      atus}`);
262
263     if (statusAlert) {
264       statusAlert.className = 'alert alert-success py-2 small

```

```

mb-0 d-block';
265     statusAlert.innerHTML = `❏ Синхронизировано! Загружено д
окументов: <strong>${data.downloaded_files?.length || 0}</stro
ng>, создано чанков: <strong>${data.chunks_count || 0}</strong
>.`;
266     }
267
268     await loadUserRags();
269     await selectUserRag(activeUserRagId);
270
271     setTimeout(() => {
272         const modalEl = document.getElementById('modal-import-gd
ocs');
273         if (modalEl && window.bootstrap?.Modal) {
274             const modal = bootstrap.Modal.getInstance(modalEl);
275             if (modal) modal.hide();
276         }
277     }, 1500);
278
279     } catch (err) {
280         if (statusAlert) {
281             statusAlert.className = 'alert alert-danger py-2 small m
b-0 d-block';
282             statusAlert.textContent = `Ошибка синхронизации: ${err.m
essage}`;
283         }
284     } finally {
285         if (btn) btn.disabled = false;
286     }
287 }
288
289

```

utils.js

```
1 /**
2  * =====
3  * Process Name: RAG Tab Utility Functions
4  * =====
5  * Description:
6  *   Formatting utilities, HTML escaping, and recursive file traversal
7  *   for drag-and-drop directory parsing in RAG tab.
8  *
9  * File: utils.js
10 * Project: AI Breadboard
11 * Module: RAGTab
12 * Author: hypo69
13 * Copyright: © 2026 hypo69
14 * =====
15 */
16
17 /**
18 * Format bytes into human-readable string (B, KB, MB, GB).
19 *
20 * @param {number} bytes - Size in bytes.
21 * @returns {string} Formatted size string.
22 */
23 export function formatBytes(bytes) {
24   if (!bytes || bytes === 0) return '0 B';
25   const k = 1024;
26   const sizes = ['B', 'KB', 'MB', 'GB'];
27   const i = Math.floor(Math.log(bytes) / Math.log(k));
28   return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + ' '
29     + sizes[i];
30 }
31 /**
```

```

32 | * Escape HTML special characters to prevent XSS.
33 | *
34 | * @param {string} text - Raw string.
35 | * @returns {string} Escaped HTML string.
36 | */
37 | export function escapeHtml(text) {
38 |   if (!text) return '';
39 |   return String(text)
40 |     .replace(/&/g, '&amp;')
41 |     .replace(/</g, '&lt;')
42 |     .replace(/>/g, '&gt;')
43 |     .replace(/"/g, '&quot;')
44 |     .replace(/'/g, '&#039;');
45 | }
46 |
47 | /**
48 | * Recursively traverse a file system entry.
49 | *
50 | * @param {FileSystemEntry} entry - FileSystem entry.
51 | * @param {string} path - Current relative path.
52 | * @returns {Promise<File[]>} Array of files with relative paths.
53 | */
54 | export function traverseEntry(entry, path = '') {
55 |   return new Promise((resolve) => {
56 |     if (entry.isFile) {
57 |       entry.file((file) => {
58 |         const relPath = path ? `${path}/${file.name}` : file.name;
59 |         Object.defineProperty(file, 'customRelativePath', {
60 |           value: relPath,
61 |           writable: true
62 |         });
63 |         resolve([file]);
64 |       }, () => resolve([]));
65 |     } else if (entry.isDirectory) {
66 |       const dirReader = entry.createReader();
67 |       const entries = [];

```

```

68 |         const readEntries = () => {
69 |             dirReader.readEntries(async (result) => {
70 |                 if (!result.length) {
71 |                     const currentPath = path ? `${path}/${entry.name}`
72 | : entry.name;
73 |                     const promises = entries.map(e => traverseEntry(e,
74 | currentPath));
75 |                     const nestedFiles = await Promise.all(promises);
76 |                     resolve(nestedFiles.flat());
77 |                 } else {
78 |                     entries.push(...result);
79 |                     readEntries();
80 |                 }
81 |             }, () => resolve([]));
82 |         };
83 |         readEntries();
84 |     } else {
85 |         resolve([]);
86 |     }
87 | });
88 | /**
89 |  * Extract all files from a DataTransfer object recursively.
90 |  *
91 |  * @param {DataTransfer} dataTransfer - Drag and drop DataTransfer object.
92 |  * @returns {Promise<File[]>} Array of files.
93 |  */
94 | export async function getAllFilesFromDataTransfer(dataTransfer
95 | ) {
96 |     if (!dataTransfer) return [];
97 |     const items = dataTransfer.items;
98 |     if (items && items.length > 0 && items[0].webkitGetAsEntry)
99 |     {
100 |         const queue = [];
101 |         for (let i = 0; i < items.length; i++) {
102 |             const entry = items[i].webkitGetAsEntry();

```

```
101 |         if (entry) queue.push(traverseEntry(entry));
102 |     }
103 |     const results = await Promise.all(queue);
104 |     return results.flat();
105 | }
106 | if (dataTransfer.files && dataTransfer.files.length > 0) {
107 |     return Array.from(dataTransfer.files);
108 | }
109 | return [];
110 | }
111 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0, maximum-scale=1.0, user-scalable=no">
7   <title>Медиа Плеер - Пульт ДУ</title>
8   <!-- Bootstrap 5 CSS -->
9   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
10  t/css/bootstrap.min.css" rel="stylesheet">
11   <!-- Bootstrap Icons -->
12   <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
13   0.0/font/bootstrap-icons.css" rel="stylesheet">
14   <!-- Telegram Web Apps SDK -->
15   <script src="https://telegram.org/js/telegram-web-app.js"></
16   script>
17   <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
18   js"></script>
19   <style>
20     :root {
21       --tg-theme-bg-color: #0d1117;
22       --tg-theme-text-color: #c9d1d9;
23       --tg-theme-hint-color: #8b949e;
24       --tg-theme-secondary-bg-color: #161b22;
25       --primary-gradient: linear-gradient(135deg, #1f6feb 0%,
26       #094cb5 100%);
27       --accent-gradient: linear-gradient(135deg, #ff7b00 0%, #
28       ffae00 100%);
29       --success-gradient: linear-gradient(135deg, #238636 0%,
30       #2ea043 100%);
31     }
32   }
33   body {
```

```
29 |     background: var(--tg-theme-bg-color);
30 |     color: var(--tg-theme-text-color);
31 |     font-family: system-ui, sans-serif;
32 |     min-height: 100vh;
33 |     display: flex;
34 |     flex-direction: column;
35 |     justify-content: center;
36 |     align-items: center;
37 |     padding: 10px;
38 | }
39 |
40 | .rc-container, .auth-container, .scan-container {
41 |     width: 100%;
42 |     max-width: 480px;
43 |     padding: 2px;
44 | }
45 |
46 | .card-rc, .card-auth, .card-scan {
47 |     background-color: var(--tg-theme-secondary-bg-color);
48 |     border: 1px solid #30363d;
49 |     border-radius: 16px;
50 |     padding: 1px;
51 |     box-shadow: 0 8px 32px rgba(0,0,0,0.6);
52 |     backdrop-filter: blur(8px);
53 | }
54 |
55 | .btn-circle-lg {
56 |     width: 64px;
57 |     height: 64px;
58 |     border-radius: 50%;
59 |     display: flex;
60 |     align-items: center;
61 |     justify-content: center;
62 | }
63 |
64 | .btn-circle-xl {
65 |     width: 80px;
66 |     height: 80px;
```



```
67 |     border-radius: 50%;
68 |     display: flex;
69 |     align-items: center;
70 |     justify-content: center;
71 | }
72 |
73 | .track-title {
74 |     font-size: 1.15rem;
75 |     font-weight: 600;
76 |     color: #58a6ff;
77 | }
78 |
79 | .playlist-box {
80 |     max-height: 200px;
81 |     overflow-y: auto;
82 |     border-radius: 8px;
83 |     background: #0d1117;
84 | }
85 |
86 | .list-group-item-action:hover {
87 |     background-color: #21262d !important;
88 | }
89 |
90 | .btn-carousel {
91 |     background: var(--accent-gradient);
92 |     border: none;
93 |     color: #000;
94 |     font-weight: bold;
95 |     transition: all 0.3s ease;
96 |     box-shadow: 0 4px 15px rgba(255, 123, 0, 0.3);
97 | }
98 | .btn-carousel:hover {
99 |     transform: translateY(-2px);
100 |     box-shadow: 0 6px 20px rgba(255, 123, 0, 0.5);
101 |     background: linear-gradient(135deg, #ffae00 0%, #ff7b00
102 | 100%);
103 |     color: #000;
104 | }
```

```
104 | .btn-carousel:active {
105 |     transform: translateY(1px);
106 | }
107 | .btn-carousel:hover i {
108 |     display: inline-block;
109 |     animation: spin 1s linear infinite;
110 | }
111 | @keyframes spin {
112 |     100% { transform: rotate(360deg); }
113 | }
114 |
115 | /* Chat bubbles styling */
116 | .chat-msg {
117 |     display: flex;
118 |     flex-direction: column;
119 |     max-width: 85%;
120 |     margin-bottom: 8px;
121 | }
122 | .chat-msg.user {
123 |     align-self: flex-end;
124 | }
125 | .chat-msg.ai, .chat-msg.system {
126 |     align-self: flex-start;
127 | }
128 | .chat-msg.user .msg-bubble {
129 |     background: var(--primary-gradient);
130 |     color: #fff;
131 |     border-radius: 12px 12px 0 12px;
132 |     box-shadow: 0 2px 8px rgba(31, 111, 235, 0.2);
133 | }
134 | .chat-msg.ai .msg-bubble {
135 |     background-color: #21262d;
136 |     color: #c9d1d9;
137 |     border-radius: 12px 12px 12px 0;
138 |     border: 1px solid #30363d;
139 | }
140 | .chat-msg.system .msg-bubble {
141 |     background-color: rgba(255, 255, 255, 0.05);
```

```
142     color: #8b949e;
143     border-radius: 8px;
144     border-left: 3px solid #58a6ff;
145 }
146 .msg-bubble {
147     padding: 8px 12px;
148     font-size: 0.9rem;
149     word-break: break-word;
150 }
151 .msg-time {
152     font-size: 0.7rem;
153     color: #8b949e;
154     margin-top: 2px;
155     padding: 0 4px;
156 }
157 .chat-msg.user .msg-time {
158     text-align: right;
159 }
160 .animate-pulse {
161     animation: pulse 1.5s infinite;
162 }
163 @keyframes pulse {
164     0% { transform: scale(1); opacity: 1; }
165     50% { transform: scale(1.15); opacity: 0.7; }
166     100% { transform: scale(1); opacity: 1; }
167 }
168
169 .search-result-card {
170     background: rgba(88, 166, 255, 0.07);
171     border: 1px solid rgba(88, 166, 255, 0.2);
172     border-radius: 8px;
173     transition: all 0.2s ease;
174 }
175 .search-result-card:hover {
176     background: rgba(88, 166, 255, 0.12);
177     border-color: rgba(88, 166, 255, 0.4);
178 }
179
```

```
180 | /* Auth panel specific */
181 | .form-control {
182 |     background: #0d1117;
183 |     border: 1px solid #30363d;
184 |     color: #c9d1d9;
185 | }
186 | .form-control:focus {
187 |     background: #0d1117;
188 |     border-color: #388bfd;
189 |     color: #c9d1d9;
190 |     box-shadow: 0 0 0 2px rgba(56,139,253,0.3);
191 | }
192 | .btn-auth-primary {
193 |     background: var(--primary-gradient);
194 |     border: none;
195 |     color: #fff;
196 | }
197 | .btn-auth-success {
198 |     background: var(--success-gradient);
199 |     border: none;
200 |     color: #fff;
201 | }
202 |
203 | /* Scanning styles */
204 | .scan-radar {
205 |     position: relative;
206 |     width: 100px;
207 |     height: 100px;
208 |     margin: 0 auto 20px;
209 |     border-radius: 50%;
210 |     border: 2px solid rgba(88, 166, 255, 0.3);
211 |     display: flex;
212 |     align-items: center;
213 |     justify-content: center;
214 | }
215 | .scan-radar::after {
216 |     content: '';
217 |     position: absolute;
```

```
218 |     width: 100%;
219 |     height: 100%;
220 |     border-radius: 50%;
221 |     border: 2px solid #58a6ff;
222 |     animation: radar-ping 2s infinite linear;
223 | }
224 | @keyframes radar-ping {
225 |     0% { transform: scale(1); opacity: 1; }
226 |     100% { transform: scale(1.6); opacity: 0; }
227 | }
228 | .player-list-item {
229 |     background: #161b22;
230 |     border: 1px solid #30363d;
231 |     border-radius: 8px;
232 |     padding: 12px;
233 |     margin-bottom: 8px;
234 |     display: flex;
235 |     justify-content: space-between;
236 |     align-items: center;
237 |     transition: all 0.2s ease;
238 | }
239 | .player-list-item:hover {
240 |     border-color: #58a6ff;
241 |     background: rgba(88, 166, 255, 0.05);
242 | }
243 |
244 | /* Theme toggle button */
245 | .theme-toggle-btn {
246 |     position: absolute;
247 |     top: 10px;
248 |     right: 10px;
249 |     background: rgba(255, 255, 255, 0.1);
250 |     border: 1px solid rgba(255, 255, 255, 0.2);
251 |     color: #fff;
252 |     width: 36px;
253 |     height: 36px;
254 |     border-radius: 50%;
255 |     cursor: pointer;
```

```
256 |     display: flex;
257 |     align-items: center;
258 |     justify-content: center;
259 |     font-size: 16px;
260 |     z-index: 1000;
261 | }
262 | .theme-toggle-btn:hover {
263 |     background: rgba(255, 255, 255, 0.2);
264 | }
265 |
266 | /* Light theme styles */
267 | .theme-light .card-rc,
268 | .theme-light .card-auth,
269 | .theme-light .card-scan {
270 |     background-color: #ffffff;
271 |     border-color: #d0d7de;
272 | }
273 | .theme-light .track-title {
274 |     color: #0969da;
275 | }
276 | .theme-light .form-control {
277 |     background: #ffffff;
278 |     border-color: #d0d7de;
279 |     color: #24292f;
280 | }
281 | .theme-light .form-control:focus {
282 |     background: #ffffff;
283 |     border-color: #0969da;
284 | }
285 | .theme-light .btn-primary {
286 |     background: #2da44e;
287 |     border-color: #2da44e;
288 | }
289 | .theme-light .btn-primary:hover {
290 |     background: #2da44e;
291 | }
292 | .theme-light .btn-outline-light {
293 |     color: #24292f;
```

```
294 |     border-color: #d0d7de;
295 | }
296 | .theme-light .btn-outline-light:hover {
297 |     background: #f6f8fa;
298 | }
299 | .theme-light .btn-circle-lg {
300 |     background: #ffffff;
301 |     border-color: #d0d7de;
302 |     color: #24292f;
303 | }
304 | .theme-light .btn-circle-lg:hover {
305 |     background: #f6f8fa;
306 | }
307 | .theme-light .text-muted {
308 |     color: #6e7781 !important;
309 | }
310 | .theme-light .chat-msg.user .msg-bubble {
311 |     background: #0969da;
312 | }
313 | .theme-light .chat-msg.ai .msg-bubble {
314 |     background-color: #f6f8fa;
315 |     color: #24292f;
316 | }
317 | .theme-light .chat-msg.system .msg-bubble {
318 |     background-color: #f6f8fa;
319 |     color: #6e7781;
320 | }
321 | .theme-light .player-list-item {
322 |     background: #ffffff;
323 |     border-color: #d0d7de;
324 | }
325 | .theme-light .player-list-item:hover {
326 |     background: #f6f8fa;
327 | }
328 | .theme-light .list-group-item {
329 |     background: #ffffff;
330 |     border-color: #d0d7de;
331 |     color: #24292f;
```

```

332     }
333     .theme-light .list-group-item:hover {
334         background: #f6f8fa;
335     }
336     .theme-light .form-range {
337         color-scheme: light;
338     }
339 </style>
340 </head>
341 <body>
342
343 <!-- 1. AUTHENTICATION PANEL -->
344 <div class="auth-container" id="auth-panel">
345     <div class="card-auth">
346         <div class="text-center mb-4">
347             <i class="bi bi-shield-lock-fill fs-1 text-primary"></i>
348             <h4 class="mt-2 text-white">Вход в пульт ДУ</h4>
349             <p class="text-muted small">Пожалуйста, войдите в аккаунт для продолжения</p>
350         </div>
351
352         <div id="auth-alert" class="alert alert-danger d-none" role="alert"></div>
353
354         <!-- Login Form -->
355         <div id="auth-form-login">
356             <form id="form-login">
357                 <div class="mb-3">
358                     <label for="login-email" class="form-label text-muted small">Email</label>
359                     <input type="email" class="form-control" id="login-email" required>
360                 </div>
361                 <div class="mb-3">
362                     <label for="login-password" class="form-label text-muted small">Пароль</label>
363                     <input type="password" class="form-control" id="lo

```



```

    gin-password" required>
364         </div>
365         <button type="submit" class="btn btn-auth-primary w-
100 py-2.5 rounded-pill mb-3">Войти</button>
366         <p class="text-center small mb-0">Нет аккаунта? <a h
ref="#" id="go-to-register" class="text-info">Зарегистрировать
ся</a></p>
367     </form>
368 </div>
369
370 <!-- Register Form -->
371 <div id="auth-form-register" class="d-none">
372     <form id="form-register">
373         <div class="mb-3">
374             <label for="register-name" class="form-label text-
muted small">Имя</label>
375             <input type="text" class="form-control" id="regist
er-name" required>
376         </div>
377         <div class="mb-3">
378             <label for="register-email" class="form-label text
-muted small">Email</label>
379             <input type="email" class="form-control" id="regis
ter-email" required>
380         </div>
381         <div class="mb-3">
382             <label for="register-password" class="form-label t
ext-muted small">Пароль</label>
383             <input type="password" class="form-control" id="re
gister-password" required>
384         </div>
385         <button type="submit" class="btn btn-auth-success w-
100 py-2.5 rounded-pill mb-3">Зарегистрироваться</button>
386         <p class="text-center small mb-0">Уже есть аккаунт?
<a href="#" id="go-to-login" class="text-info">Войти</a></p>
387     </form>
388 </div>
389

```

```

390 |     <!-- Verify Form -->
391 |     <div id="auth-form-verify" class="d-none">
392 |         <p class="text-center small">На вашу почту отправлен 6
| -значный код подтверждения. Пожалуйста, введите его ниже.</p>
393 |         <form id="form-verify">
394 |             <div class="mb-3">
395 |                 <label for="verify-code" class="form-label text-mu
| ted small">Код подтверждения</label>
396 |                 <input type="text" class="form-control text-center
| fs-3 py-2" id="verify-code" maxlength="6" placeholder="XXXXXX
| " required>
397 |             </div>
398 |             <button type="submit" class="btn btn-info w-100 py-2
| .5 text-white rounded-pill mb-3">Подтвердить</button>
399 |             <p class="text-center small mb-0"><a href="#" id="ve
| rify-cancel" class="text-info">Назад к входу</a></p>
400 |         </form>
401 |     </div>
402 |
403 |     <div class="my-4 text-center position-relative">
404 |         <hr class="border-secondary">
405 |         <span class="px-2 text-muted position-absolute top-50
| start-50 translate-middle small" style="background-color: var(
| --tg-theme-secondary-bg-color);">Или войти через</span>
406 |     </div>
407 |
408 |     <div class="d-flex flex-column gap-2 align-items-center"
| >
409 |         <button id="google-signin-btn" class="btn btn-outline-
| light w-100 rounded-pill d-flex align-items-center justify-con
| tent-center gap-2 py-2">
410 |             <i class="bi bi-google text-danger"></i>
411 |             <span>Google</span>
412 |         </button>
413 |     </div>
414 | </div>
415 | </div>
416 |

```

```

417 | <!-- 2. COMPUTER DISCOVERY SCANNER PANEL -->
418 | <div class="scan-container d-none" id="scan-panel">
419 |     <div class="card-scan text-center">
420 |         <div class="scan-radar">
421 |             <i class="bi bi-laptop fs-2 text-primary"></i>
422 |         </div>
423 |         <h4 class="text-white" id="scan-title">Поиск открытых ко
мпыютеров</h4>
424 |         <p class="text-muted small" id="scan-status">Сканируем с
еть на наличие открытых сессий davidka.net...</p>
425 |
426 |         <div id="discovered-players-list" class="mt-4 text-start
"></div>
427 |
428 |         <button id="btn-retry-scan" class="btn btn-outline-prima
ry w-100 py-2.5 rounded-pill mt-3 d-none">
429 |             <i class="bi bi-arrow-clockwise"></i> Обновить поиск
430 |         </button>
431 |
432 |         <button id="btn-scan-logout" class="btn btn-sm btn-outli
ne-danger w-100 mt-2 py-2 border-0 small">
433 |             <i class="bi bi-box-arrow-right"></i> Выйти из аккаунт
а
434 |         </button>
435 |     </div>
436 | </div>
437 |
438 | <!-- 3. REMOTE CONTROL MAIN PANEL -->
439 | <div class="rc-container d-none" id="rc-app-container">
440 |     <div class="card-rc">
441 |         <!-- Status Badge & User info -->
442 |         <div class="d-flex justify-content-between align-items-c
enter mb-3">
443 |             <span class="badge bg-secondary" id="ws-status"><i cla
ss="bi bi-wifi-off"></i> Подключение...</span>
444 |             <div class="d-flex gap-2 align-items-center">
445 |                 <button class="theme-toggle-btn" id="theme-toggle-bt
n" title="Переключить тему">

```

```

446         <i class="bi bi-brightness-high"></i>
447     </button>
448     <button class="btn btn-sm btn-outline-secondary py-1
    px-2 border-secondary small" id="btn-change-player" title="См
    енить компьютер">
449         <i class="bi bi-arrow-left-right"></i> <span id="r
    oom-info">ПК: default</span>
450     </button>
451 </div>
452 </div>
453
454 <!-- Now Playing Info -->
455 <div class="text-center my-3 p-3 rounded bg-dark bg-opac
    ity-25">
456     <i class="bi bi-music-note-beamed fs-1 text-primary mb
    -3 d-block"></i>
457     <div id="track-title" class="track-title mb-1 text-tru
    ncate">Файл не выбран</div>
458     <small id="time-display" class="text-muted">00:00 / 00
    :00</small>
459
460     <!-- Progress Bar -->
461     <div class="mt-3">
462         <input type="range" class="form-range" id="seek-slid
    er" min="0" max="100" value="0">
463     </div>
464 </div>
465
466 <!-- Central Controls -->
467 <div class="d-flex justify-content-center align-items-ce
    nter gap-3 my-4">
468     <button class="btn btn-outline-light btn-circle-lg" id
    ="btn-prev" title="Предыдущий"><i class="bi bi-skip-start-fill
    fs-3"></i></button>
469     <button class="btn btn-primary btn-circle-xl" id="btn-
    playtoggle" title="Воспроизведение/Пауза"><i class="bi bi-play
    -fill fs-1" id="play-icon"></i></button>
470     <button class="btn btn-outline-light btn-circle-lg" id

```

```

    ="btn-next" title="Следующий"><i class="bi bi-skip-end-fill fs
    -3"></i></button>
471     </div>
472
473     <!-- Carousel / Random Movie Control -->
474     <div class="my-3 px-2">
475         <button class="btn btn-carousel w-100 py-2.5 rounded-p
    ill d-flex align-items-center justify-content-center gap-2" id
    ="btn-carousel" title="Случайный фильм">
476             <i class="bi bi-arrow-repeat fs-5"></i>
477             <span>Карусель</span>
478         </button>
479     </div>
480
481     <!-- Quick Google Workspace & Social Control Panel Butto
    ns -->
482     <div class="row g-2 px-2 my-2">
483         <div class="col-6 col-sm-3">
484             <button class="btn btn-outline-primary w-100 py-2 ro
    unded-3 d-flex align-items-center justify-content-center gap-1
    small" id="rc-btn-check-mail" title="Проверить почту Gmail">
485                 <i class="bi bi-envelope-check-fill text-primary">
    </i>
486                 <span>Почта</span>
487             </button>
488         </div>
489         <div class="col-6 col-sm-3">
490             <button class="btn btn-outline-info w-100 py-2 round
    ed-3 d-flex align-items-center justify-content-center gap-1 sm
    all" id="rc-btn-search-gdocs" title="Поиск в Google Docs">
491                 <i class="bi bi-file-earmark-text-fill text-info">
    </i>
492                 <span>Docs</span>
493             </button>
494         </div>
495         <div class="col-6 col-sm-3">
496             <button class="btn btn-outline-success w-100 py-2 ro
    unded-3 d-flex align-items-center justify-content-center gap-1

```

```

    small" id="rc-btn-gsheets" title="Работа с Google Sheets">
497         <i class="bi bi-file-earmark-spreadsheet-fill text
    -success"></i>
498         <span>Таблицы</span>
499     </button>
500 </div>
501 <div class="col-6 col-sm-3">
502     <button class="btn btn-outline-primary w-100 py-2 ro
    unded-3 d-flex align-items-center justify-content-center gap-1
    small" id="rc-btn-facebook" title="Опубликовать в Facebook">
503         <i class="bi bi-facebook text-primary"></i>
504         <span>Facebook</span>
505     </button>
506 </div>
507 </div>
508
509 <!-- Volume Controls -->
510 <div class="d-flex align-items-center gap-3 my-3 px-2">
511     <i class="bi bi-volume-down-fill text-muted fs-4"></i>
512     <input type="range" class="form-range" id="volume-slid
    er" min="0" max="1" step="0.05" value="1">
513     <i class="bi bi-volume-up-fill text-muted fs-4"></i>
514 </div>
515
516 <!-- Voice Search & AI Chat -->
517 <div class="mt-4 p-3 rounded border border-secondary bg-
    dark bg-opacity-25">
518     <h6 class="text-center mb-3"><i class="bi bi-mic"></i>
    Голосовой Ассистент</h6>
519
520     <!-- Status -->
521     <div class="text-muted small text-center mb-3" id="voi
    ce-status">Готов к работе</div>
522
523     <!-- Control buttons -->
524     <div class="d-flex flex-column gap-3 align-items-cen
    ter">
525         <button class="btn btn-outline-danger btn-circle-xl"

```

```

    id="btn-mic" title="Нажмите, чтобы говорить">
526         <i class="bi bi-mic-fill fs-1" id="mic-icon"></i>
527     </button>
528
529     <!-- Language Selection -->
530     <div class="btn-group w-100 mt-2" role="group">
531         <input type="radio" class="btn-check" name="lang-r
532         adio" id="lang-ru" value="ru-RU" autocomplete="off" checked>
533         <label class="btn btn-outline-secondary" for="lang
534         -ru">RU</label>
535
536         <input type="radio" class="btn-check" name="lang-r
537         adio" id="lang-en" value="en-US" autocomplete="off">
538         <label class="btn btn-outline-secondary" for="lang
539         -en">EN</label>
540
541         <input type="radio" class="btn-check" name="lang-r
542         adio" id="lang-he" value="he-IL" autocomplete="off">
543         <label class="btn btn-outline-secondary" for="lang
544         -he">HE</label>
545     </div>
546 </div>
547 </div>
548
549     <!-- Playlist Picker -->
550     <div class="mt-4">
551         <h6 class="border-bottom border-secondary pb-2"><i cla
552         ss="bi bi-collection-play"></i> Список воспроизведения плеера<
553         /h6>
554         <div class="list-group list-group-flush playlist-box"
555         id="playlist-items">
556             <div class="text-muted small p-3 text-center">Ожидан
557             ие подключения к плееру...</div>
558         </div>
559     </div>
560
561     <!-- User Manual (Read & Listen) -->
562     <div class="mt-4 p-3 rounded border border-secondary bg-

```

```

dark bg-opacity-25">
553     <div class="d-flex justify-content-between align-items
| -center border-bottom border-secondary pb-2 mb-2">
554         <h6 class="m-0"><i class="bi bi-info-circle"></i> Ин
|струкция пользователя</h6>
555         <button class="btn btn-outline-info btn-sm d-flex al
|ign-items-center gap-1" id="btn-read-manual" title="Прослушать
|инструкцию">
556             <i class="bi bi-volume-up-fill" id="read-manual-ic
|on"></i> <span>Прослушать</span>
557         </button>
558     </div>
559     <div id="user-manual-text" class="small text-muted" st
|yle="line-height: 1.5; text-align: justify;">
560         Для дистанционного управления сначала откройте Медиа
|плеер на вашем компьютере или телевизоре.
561         Данный пульт автоматически подключится к нему через
|сервер.
562         Вы можете нажимать кнопки для запуска файлов, переключе
|ния треков, перемотки или изменения громкости.
563         Также доступно голосовое управление кнопкой микрофон
|а.
564         Простые команды: «пауза», «плей», «громче», «тише»,
|«вперед» и «назад» сработают мгновенно.
565         А любые другие фразы будут отправлены вашему AI-асси
|стенту Gemini для помощи в выборе фильма или ответа на вопросы
|
566     </div>
567 </div>
568 </div>
569 </div>
570
571 <script>
572     // API Module
573     const api = {
574         async fetch(url, options = {}) {
575             const response = await fetch(url, options);
576             if (!response.ok) {

```



```
577 |         let msg = response.statusText;
578 |         try {
579 |             const data = await response.json();
580 |             msg = data.detail || msg;
581 |         } catch {}
582 |         throw new Error(msg);
583 |     }
584 |     return response.json();
585 | },
586 | auth: {
587 |     async check() {
588 |         try {
589 |             const response = await fetch('/auth/check');
590 |             return await response.json();
591 |         } catch {}
592 |         return { authenticated: false };
593 |     }
594 | },
595 |     async register(email, password, name) {
596 |         return api.fetch('/auth/register', {
597 |             method: 'POST',
598 |             headers: { 'Content-Type': 'application/json' },
599 |             body: JSON.stringify({ email, password, name })
600 |         });
601 |     },
602 |     async verify(email, code) {
603 |         return api.fetch('/auth/verify', {
604 |             method: 'POST',
605 |             headers: { 'Content-Type': 'application/json' },
606 |             body: JSON.stringify({ email, code })
607 |         });
608 |     },
609 |     async loginEmail(email, password) {
610 |         return api.fetch('/auth/login/email', {
611 |             method: 'POST',
612 |             headers: { 'Content-Type': 'application/json' },
613 |             body: JSON.stringify({ email, password })
614 |         });
```

```

615 |         },
616 |         async logout() {
617 |             return api.fetch('/auth/logout', { method: 'POST' })
618 |         };
619 |     }
620 | };
621 |
622 | // Panels state
623 | const authPanel = document.getElementById('auth-panel');
624 | const scanPanel = document.getElementById('scan-panel');
625 | const rcAppContainer = document.getElementById('rc-app-con
626 | tainer');
627 | const authFormLogin = document.getElementById('auth-form-l
628 | ogin');
629 | const authFormRegister = document.getElementById('auth-for
630 | m-register');
631 | const authFormVerify = document.getElementById('auth-form-
632 | verify');
633 | const authAlert = document.getElementById('auth-alert');
634 |
635 | let currentVerifyEmail = '';
636 | let currentRoomName = '';
637 | let ws = null;
638 | let isSeeking = false;
639 | let playerState = { duration: 0, currentTime: 0, isPlaying
640 | : false, volume: 1 };
641 |
642 | // Toggle panels helpers
643 | function showPanel(panelId) {
644 |     authPanel.classList.add('d-none');
645 |     scanPanel.classList.add('d-none');
646 |     rcAppContainer.classList.add('d-none');
647 |
648 |     if (panelId === 'auth') authPanel.classList.remove('d-no
649 | ne');
650 |     else if (panelId === 'scan') scanPanel.classList.remove(

```

```

        'd-none');
646     else if (panelId === 'rc') rcAppContainer.classList.remove('d-none');
647     }
648
649     // — AUTH PANEL LOGIC —————
650     document.getElementById('go-to-register').addEventListener('click', (e) => {
651         e.preventDefault();
652         authAlert.classList.add('d-none');
653         authFormLogin.classList.add('d-none');
654         authFormRegister.classList.remove('d-none');
655     });
656
657     document.getElementById('go-to-login').addEventListener('click', (e) => {
658         e.preventDefault();
659         authAlert.classList.add('d-none');
660         authFormRegister.classList.add('d-none');
661         authFormLogin.classList.remove('d-none');
662     });
663
664     document.getElementById('verify-cancel').addEventListener('click', (e) => {
665         e.preventDefault();
666         authAlert.classList.add('d-none');
667         authFormVerify.classList.add('d-none');
668         authFormLogin.classList.remove('d-none');
669     });
670
671     document.getElementById('google-signin-btn').addEventListener('click', () => {
672         window.location.href = '/auth/google?next=' + encodeURIComponent(window.location.pathname + window.location.search);
673     });
674
675     // Login Form Submit

```

```
676 |     document.getElementById('form-login').addEventListener('su
    | bmit', async (e) => {
677 |         e.preventDefault();
678 |         authAlert.classList.add('d-none');
679 |         const email = document.getElementById('login-email').val
    | ue;
680 |         const pass = document.getElementById('login-password').v
    | alue;
681 |
682 |         try {
683 |             await api.auth.loginEmail(email, pass);
684 |             checkAndInit();
685 |         } catch (err) {
686 |             authAlert.textContent = err.message || 'Ошибка входа';
687 |             authAlert.classList.remove('d-none');
688 |         }
689 |     });
690 |
691 |     // Register Form Submit
692 |     document.getElementById('form-register').addEventListener(
    | 'submit', async (e) => {
693 |         e.preventDefault();
694 |         authAlert.classList.add('d-none');
695 |         const name = document.getElementById('register-name').va
    | lue;
696 |         const email = document.getElementById('register-email').
    | value;
697 |         const pass = document.getElementById('register-password'
    | ).value;
698 |
699 |         try {
700 |             const res = await api.auth.register(email, pass, name)
    | ;
701 |             currentVerifyEmail = email;
702 |             authFormRegister.classList.add('d-none');
703 |             authFormVerify.classList.remove('d-none');
704 |         } catch (err) {
705 |             authAlert.textContent = err.message || 'Ошибка регистр
```

```

    ации';
706     authAlert.classList.remove('d-none');
707   }
708   });
709
710   // Verify Form Submit
711   document.getElementById('form-verify').addEventListener('s
ubmit', async (e) => {
712     e.preventDefault();
713     authAlert.classList.add('d-none');
714     const code = document.getElementById('verify-code').valu
e;
715
716     try {
717       await api.auth.verify(currentVerifyEmail, code);
718       // After verify, try to log in
719       authFormVerify.classList.add('d-none');
720       authFormLogin.classList.remove('d-none');
721       authAlert.className = 'alert alert-success';
722       authAlert.textContent = 'Почта успешно подтверждена. Т
еперь войдите.';
723       authAlert.classList.remove('d-none');
724     } catch (err) {
725       authAlert.textContent = err.message || 'Неверный код';
726       authAlert.classList.remove('d-none');
727     }
728   });
729
730   // — COMPUTER DISCOVERY LOGIC —————
  _____
731   async function scanForPlayers() {
732     showPanel('scan');
733     document.getElementById('scan-title').textContent = 'Поис
к открытых компьютеров';
734     document.getElementById('scan-status').textContent = 'Ск
анируем сеть на наличие открытых сессий davidka.net...';
735     document.getElementById('btn-retry-scan').classList.add(
'd-none');

```

```

736
737     const listDiv = document.getElementById('discovered-play
ers-list');
738     listDiv.innerHTML = '<div class="text-center p-3"><span
class="spinner-border spinner-border-sm text-primary"></span><
/div>';
739
740     try {
741         const response = await fetch('/api/control/active_play
ers');
742         const data = await response.json();
743         const players = data.players || [];
744         const currentUserEmail = data.user_email;
745
746         const mappedPlayers = players.map(p => {
747             if (p.includes('@')) {
748                 return { room: p, name: `davidka.net (${p})` };
749             }
750             return { room: p, name: p === 'default' ? 'Основной
компьютер (davidka.net)' : `ПК: ${p}` };
751         });
752
753         if (mappedPlayers.length === 0) {
754             listDiv.innerHTML = `
755                 <div class="alert alert-warning text-center small
mb-0">
756                     <i class="bi bi-exclamation-circle-fill"></i> От
крытые компьютеры не обнаружены.<br>
757                     Пожалуйста, откройте davidka.net на вашем компью
тере.
758                 </div>`;
759             document.getElementById('scan-title').textContent =
'Компьютеры не найдены';
760             document.getElementById('btn-retry-scan').classList.
remove('d-none');
761         } else if (mappedPlayers.length === 1) {
762             // Exactly one computer found - connect automaticall
y

```

```

763 |         listDiv.innerHTML = `
```

```

792
793     document.getElementById('btn-retry-scan').addEventListener
794     ('click', scanForPlayers);
795
796     document.getElementById('btn-change-player').addEventListener
797     ner('click', () => {
798         // Disconnect current stream
799         if (ws) {
800             ws.close();
801             ws = null;
802         }
803         scanForPlayers();
804     });
805
806     document.getElementById('btn-scan-logout').addEventListener
807     r('click', async () => {
808         await api.auth.logout();
809         showPanel('auth');
810     });
811
812     // — MAIN TELEGRAM / WEB APP ROOM SETUP —————
813
814     function getRoomNameFromQuery() {
815         const urlParams = new URLSearchParams(window.location.se
816         arch);
817         return urlParams.get('room') || '';
818     }
819
820
821     // Initialize Telegram WebApp if available
822     if (window.Telegram && window.Telegram.WebApp) {
823         const tg = window.Telegram.WebApp;
824         tg.ready();
825         tg.expand();
826
827         // Apply theme
828         function applyTgTheme() {
829             if (tg.themeParams.bg_color) {
830                 document.documentElement.style.setProperty('--tg-the

```



```

    me-bg-color', tg.themeParams.bg_color);
825     }
826     if (tg.themeParams.text_color) {
827         document.documentElement.style.setProperty('--tg-the
me-text-color', tg.themeParams.text_color);
828     }
829     if (tg.themeParams.secondary_bg_color) {
830         document.documentElement.style.setProperty('--tg-the
me-secondary-bg-color', tg.themeParams.secondary_bg_color);
831     }
832     if (tg.themeParams.hint_color) {
833         document.documentElement.style.setProperty('--tg-the
me-hint-color', tg.themeParams.hint_color);
834     }
835     }
836     applyTgTheme();
837     tg.onEvent('themeChanged', applyTgTheme);
838
839     // Haptic feedback function
840     window.triggerTgHaptic = function() {
841         if (tg.HapticFeedback) {
842             tg.HapticFeedback.impactOccurred('light');
843         }
844     };
845     } else {
846         window.triggerTgHaptic = function() {};
847     }
848
849     const statusBadge = document.getElementById('ws-status');
850     const trackTitle = document.getElementById('track-title');
851     const timeDisplay = document.getElementById('time-display'
);
852     const seekSlider = document.getElementById('seek-slider');
853     const volumeSlider = document.getElementById('volume-slide
r');
854     const playIcon = document.getElementById('play-icon');
855     const playlistContainer = document.getElementById('playlis
t-items');

```

```

856
857     function formatTime(seconds) {
858         if (!seconds || isNaN(seconds)) return '00:00';
859         const m = Math.floor(seconds / 60);
860         const s = Math.floor(seconds % 60);
861         return `${m.toString().padStart(2, '0')}:${s.toString().
padStart(2, '0')}`;
862     }
863
864     // Connect to WebSocket with Selected Room (Strictly One S
tream)
865     function initStream(room) {
866         currentRoomName = room;
867         showPanel('rc');
868
869         const roomInfoElement = document.getElementById('room-in
fo');
870         if (roomInfoElement) {
871             const displayRoom = room.includes('@') ? 'kino.davidka
.net.tv' : room;
872             roomInfoElement.textContent = `Плеер: ${displayRoom}`;
873         }
874
875         // Close previous connection to satisfy "Strictly One St
ream" requirement
876         if (ws) {
877             ws.close();
878         }
879
880         const protocol = window.location.protocol === 'https:' ?
'wss:' : 'ws:';
881         const wsUrl = `${protocol}://${window.location.host}/api/
control/ws?role=remote&room=${encodeURIComponent(currentRoomNa
me)}`;
882
883         ws = new WebSocket(wsUrl);
884
885         ws.onopen = () => {

```

```

886 |         statusBadge.className = 'badge bg-success';
887 |         statusBadge.innerHTML = '<i class="bi bi-wifi"></i> По
      |         дключено';
888 |     };
889 |
890 |     ws.onmessage = (event) => {
891 |         try {
892 |             const msg = JSON.parse(event.data);
893 |             if (msg.event === 'status_update') {
894 |                 updateUI(msg);
895 |             } else if (msg.event === 'playlist_update') {
896 |                 updatePlaylist(msg.files || []);
897 |             }
898 |         } catch (e) {
899 |             console.error(e);
900 |         }
901 |     };
902 |
903 |     ws.onclose = (e) => {
904 |         // Stop speech queue when connection drops
905 |         if (currentSpeechQueue) {
906 |             try {
907 |                 currentSpeechQueue.stop();
908 |                 currentSpeechQueue = null;
909 |             } catch (err) {}
910 |         }
911 |         if (currentSpeechAudio) {
912 |             try {
913 |                 currentSpeechAudio.pause();
914 |                 currentSpeechAudio.src = "";
915 |                 currentSpeechAudio = null;
916 |             } catch (err) {}
917 |         }
918 |
919 |         // If we deliberately closed it to switch player, don'
      |         t auto-reconnect
920 |         if (currentRoomName !== room) return;
921 |

```

```

922 |         statusBadge.className = 'badge bg-warning text-dark';
923 |         statusBadge.innerHTML = '<i class="bi bi-arrow-clockwi
se"></i> Переподключение...';
924 |         setTimeout(() => {
925 |             if (currentRoomName === room) {
926 |                 initStream(room);
927 |             }
928 |         }, 3000);
929 |     };
930 | }
931 |
932 | function sendCommand(action, payload = {}) {
933 |     window.triggerTgHaptic();
934 |     if (ws && ws.readyState === WebSocket.OPEN) {
935 |         ws.send(JSON.stringify({ action, ...payload }));
936 |     }
937 | }
938 |
939 | function updateUI(data) {
940 |     playerState = { ...playerState, ...data };
941 |     trackTitle.textContent = data.fileName || 'Файл не выбра
н';
942 |     timeDisplay.textContent = `${formatTime(data.currentTime
)} / ${formatTime(data.duration)}`;
943 |
944 |     if (!isSeeking && data.duration > 0) {
945 |         seekSlider.max = Math.floor(data.duration);
946 |         seekSlider.value = Math.floor(data.currentTime);
947 |     }
948 |
949 |     if (typeof data.volume === 'number') {
950 |         volumeSlider.value = data.volume;
951 |     }
952 |
953 |     if (data.isPlaying) {
954 |         playIcon.className = 'bi bi-pause-fill fs-1';
955 |     } else {
956 |         playIcon.className = 'bi bi-play-fill fs-1';

```

```

957     }
958 }
959
960 function updatePlaylist(files) {
961     window.playlistFiles = files;
962     if (!files.length) {
963         playlistContainer.innerHTML = '<div class="text-muted
small p-3 text-center">Плейлист пуст</div>';
964         return;
965     }
966
967     playlistContainer.innerHTML = files.map((f, i) => `
968         <button class="list-group-item list-group-item-action
bg-transparent text-white border-secondary py-2 px-3 small d-f
lex justify-content-between align-items-center" onclick="playF
ile(${i})">
969             <span class="text-truncate">${i + 1}. ${f.name}</spa
n>
970             <span class="badge bg-secondary ms-2">${f.type === '
series' ? 'Сериял' : 'Фильм'}</span>
971         </button>
972     `).join('');
973 }
974
975 window.playFile = (index) => {
976     sendCommand('play_file', { index });
977 };
978
979 // — VOICE CONTROL & SEARCH CHAT MODULE —————

```

```

980 let recognition = null;
981 let isListening = false;
982 let rcAbortController = null;
983 let rcHistory = [];
984
985
986
987 function tryParseMovieCard(text) {

```

```

988 |     const trimmed = text.trim();
989 |     if (trimmed.startsWith('{') && trimmed.endsWith('')) {
990 |         try {
991 |             const data = JSON.parse(trimmed);
992 |             if (data.title || data.title_ru) return data;
993 |         } catch (e) {}
994 |     }
995 |     const match = text.match(/\{[\s\S]*?\}/);
996 |     if (match) {
997 |         try {
998 |             const data = JSON.parse(match[0]);
999 |             if (data.title || data.title_ru) return data;
1000 |         } catch (e) {}
1001 |     }
1002 |     return null;
1003 | }
1004 |
1005 | function renderMovieCardHtml(card) {
1006 |     const title = card.title_ru || card.title;
1007 |     const subtitle = card.title_orig || '';
1008 |     const year = card.year || '';
1009 |     const category = card.main_category || '';
1010 |     const genres = card.genres ? card.genres.join(', ') : ''
1011 | ;
1012 |     const directors = card.directors ? card.directors.join(
1013 | , ' ) : '';
1014 |     const cast = card.cast ? card.cast.slice(0, 5).join(', '
1015 | ) : '';
1016 |     const plot = card.plot || '';
1017 |     const whyWatch = card.why_watch || '';
1018 |
1019 |     let ratingHtml = '';
1020 |     if (card.rating) {
1021 |         if (card.rating.imdb) ratingHtml += `<span class="badg
1022 | e bg-warning text-dark me-1">IMDb: ${card.rating.imdb}</span>`
1023 | ;
1024 |         if (card.rating.кинопоиск) ratingHtml += `<span class=
1025 | "badge bg-danger text-white me-1">КП: ${card.rating.кинопоиск}

```

```

</span>`;
1020     if (card.rating.tmdb) ratingHtml += `<span class="badg
e bg-primary text-white">TMDB: ${card.rating.tmdb}</span>`;
1021     }
1022
1023     return `
1024     <div class="p-3 text-white" style="border-radius: 10px
; background: rgba(88, 166, 255, 0.08); border: 1px solid rgba
(88, 166, 255, 0.2);">
1025         <h5 class="mb-1 text-info">${title}</h5>
1026         <div class="small text-muted mb-2">${subtitle} ${yea
r ? ` (${year})` : ''}</div>
1027         <div class="mb-2">${ratingHtml}</div>
1028         ${category ? `<div class="small text-muted mb-1" sty
le="font-size: 0.8rem;"><strong>Категория:</strong> ${category
}</div>` : ''}
1029         ${genres ? `<div class="small text-muted mb-1" style
="font-size: 0.8rem;"><strong>Жанры:</strong> ${genres}</div>`
: ''}
1030         ${directors ? `<div class="small text-muted mb-1" st
yle="font-size: 0.8rem;"><strong>Режиссер:</strong> ${director
s}</div>` : ''}
1031         ${cast ? `<div class="small text-muted mb-2" style="
font-size: 0.8rem;"><strong>В ролях:</strong> ${cast}...</div>
` : ''}
1032         ${plot ? `<p class="small text-secondary mb-2" style
="font-size: 0.85rem; line-height: 1.4;">${plot}</p>` : ''}
1033         ${whyWatch ? `<div class="small text-info mb-2" styl
e="font-size: 0.8rem;"><strong>Рекомендация:</strong> ${whyWat
ch}</div>` : ''}
1034         <div class="d-flex justify-content-end mt-2">
1035             <button class="btn btn-sm btn-primary rounded-pill
px-3" onclick="playMovieDirect('${title.replace(/'/g, '\\')}')">
1036                 <i class="bi bi-play-fill"></i> Запустить фильм
1037             </button>
1038         </div>
1039     </div>

```

```

1040     `;
1041 }
1042
1043 window.playMovieDirect = (title) => {
1044     if (!window.playlistFiles) return;
1045     const cleanT = title.toLowerCase();
1046     const foundIdx = window.playlistFiles.findIndex(f => (f.
name || '').toLowerCase().includes(cleanT));
1047     if (foundIdx !== -1) {
1048         playFile(foundIdx);
1049         speakText(`Запускаю ${window.playlistFiles[foundIdx].n
ame}`);
1050     } else {
1051         speakText(`Фильм не найден в плейлисте`);
1052     }
1053 };
1054
1055 async function processChatMessage(text) {
1056     if (!text || !text.trim()) return;
1057     const query = text.trim();
1058
1059     stopAudio(); // <--- STOP AUDIO IMMEDIATELY
1060
1061     // 0. Transmit transcript to open computers via WebSocke
t
1062     if (ws && ws.readyState === WebSocket.OPEN) {
1063         sendCommand('transcript', { text: query });
1064     }
1065
1066     // 1. Check local control commands first
1067     if (handleLocalVoiceCommand(query.toLowerCase())) {
1068         speakText('Команда выполнена');
1069         return;
1070     }
1071
1072     // 2. Perform search in playlist
1073     const cleanQ = query.toLowerCase().replace(/^(найди|поис
к|включи|запусти|покажи|фильм|сериал)\s+/i, '').trim();

```



```
1074 |     let matches = [];
1075 |     if (window.playlistFiles && cleanQ.length > 1) {
1076 |         matches = window.playlistFiles.map((f, idx) => ({ ...f
1077 |         , index: idx })).filter(f => {
1078 |             const name = (f.name || '').toLowerCase();
1079 |             const type = (f.type || '').toLowerCase();
1080 |             return name.includes(cleanQ) || (f.year && f.year.to
1081 |             String().includes(cleanQ));
1082 |         });
1083 |     }
1084 |     if (matches.length > 0) {
1085 |         speakText(`Найдено совпадений: ${matches.length}`);
1086 |
1087 |         const queryLower = query.toLowerCase();
1088 |         if (matches.length === 1 && (queryLower.includes('вклю
1089 |         чи') || queryLower.includes('запусти') || queryLower.includes(
1090 |         'плей') || queryLower.includes('играй')))) {
1091 |             playFile(matches[0].index);
1092 |             speakText(`Запускаю ${matches[0].name}`);
1093 |         }
1094 |         return;
1095 |     }
1096 |
1097 |     // 3. Forward to AI Chat API with Streaming
1098 |     if (rcAbortController) {
1099 |         try { rcAbortController.abort(); } catch(e) {}
1100 |     }
1101 |     rcAbortController = new AbortController();
1102 |
1103 |     const voiceStatus = document.getElementById('voice-statu
1104 |     s');
1105 |     voiceStatus.textContent = 'Обработка запроса моделью...'
1106 |     ;
1107 |
1108 |     let fullReply = '';
1109 |     let started = false;
```

```
1106 |     try {
1107 |         const payload = {
1108 |             message: query,
1109 |             history: rcHistory,
1110 |             generation_config: { is_rc: true }
1111 |         };
1112 |
1113 |         const response = await fetch('/api/chat', {
1114 |             method: 'POST',
1115 |             headers: { 'Content-Type': 'application/json' },
1116 |             body: JSON.stringify(payload),
1117 |             signal: rcAbortController.signal
1118 |         });
1119 |
1120 |         if (!response.ok) {
1121 |             throw new Error(`Ошибка HTTP ${response.status}`);
1122 |         }
1123 |
1124 |         const reader = response.body.getReader();
1125 |         const decoder = new TextDecoder('utf-8');
1126 |         let buffer = '';
1127 |         let voiceReply = '';
1128 |
1129 |         while (true) {
1130 |             const { value, done } = await reader.read();
1131 |             if (done) break;
1132 |
1133 |             buffer += decoder.decode(value, { stream: true });
1134 |             const lines = buffer.split('\n');
1135 |             buffer = lines.pop();
1136 |
1137 |             for (const line of lines) {
1138 |                 const trimmed = line.trim();
1139 |                 if (!trimmed) continue;
1140 |                 if (trimmed.startsWith('data: ')) {
1141 |                     try {
1142 |                         const data = JSON.parse(trimmed.substring(6));
1143 |                         if (data.error) {
```

```

1144         throw new Error(data.error);
1145     }
1146     if (data.status) {
1147         voiceStatus.textContent = data.status;
1148     }
1149     if (data.voice) {
1150         voiceReply += data.voice;
1151     }
1152     if (data.text) {
1153         fullReply += data.text;
1154     }
1155     } catch (e) {
1156         console.error(e);
1157     }
1158 }
1159 }
1160 }
1161
1162 voiceStatus.textContent = 'Готов к работе';
1163
1164 // — Обработка тегов <film> из ответа AI —————
1165
1166 const filmMatches = [...fullReply.matchAll(/<film>(.*?
)<\s*</film>/gi)];
1167 if (filmMatches.length > 0) {
1168     const filmTitle = filmMatches[0][1].trim();
1169     try {
1170         const mediaResp = await fetch('/api/media/by-title
', {
1171             method: 'POST',
1172             headers: { 'Content-Type': 'application/json' },
1173             body: JSON.stringify({ title: filmTitle })
1174         });
1175         if (mediaResp.ok) {
1176             const mediaData = await mediaResp.json();
1177             const filmPath = mediaData.path;
1178             if (filmPath) {
1179                 sendCommand('play_file_by_path', { path: filmP

```

```

ath });
1179         voiceStatus.textContent = ` ▶ Запускаю: ${film
Title}`;
1180     }
1181 }
1182 } catch(e) {
1183     console.error('Ошибка поиска пути фильма:', e);
1184 }
1185 }
1186 // _____
_____

1187
1188     const card = tryParseMovieCard(fullReply);
1189     if (card) {
1190         speakText(`Найдена карточка фильма ${card.title_ru |
| card.title}. ${card.why_watch || card.plot || ''}`);
1191     } else {
1192         speakText(voiceReply || fullReply);
1193     }
1194
1195     rcHistory.push({ role: 'user', parts: [query] });
1196     rcHistory.push({ role: 'model', parts: [fullReply] });
1197     if (rcHistory.length > 20) {
1198         rcHistory = rcHistory.slice(-20);
1199     }
1200
1201 } catch (err) {
1202     if (err.name === 'AbortError') return;
1203     voiceStatus.textContent = 'Ошибка AI: ' + err.message;
1204 }
1205 }
1206
1207 function initVoiceControl() {
1208     const btnMic = document.getElementById('btn-mic');
1209     const micIcon = document.getElementById('mic-icon');
1210     const voiceStatus = document.getElementById('voice-stat
s');
1211

```

```
1212 |         const SpeechRecognition = window.SpeechRecognition || wi
      | ndow.webkitSpeechRecognition;
1213 |         if (!SpeechRecognition) {
1214 |             voiceStatus.textContent = 'Голосовой ввод не поддержив
      | ается вашим браузером.';
1215 |             if (btnMic) btnMic.disabled = true;
1216 |         } else {
1217 |             recognition = new SpeechRecognition();
1218 |
1219 |             // Dynamically get selected language
1220 |             const getSelectedLang = () => {
1221 |                 const selected = document.querySelector('input[name=
      | "lang-radio"]:checked');
1222 |                 return selected ? selected.value : 'ru-RU';
1223 |             };
1224 |
1225 |             recognition.lang = getSelectedLang();
1226 |
1227 |             // Update language if radio button changes
1228 |             document.querySelectorAll('input[name="lang-radio"]').
      | forEach((radio => {
1229 |                 radio.addEventListener('change', () => {
1230 |                     recognition.lang = getSelectedLang();
1231 |                 });
1232 |             });
1233 |
1234 |             recognition.continuous = false;
1235 |             recognition.interimResults = false;
1236 |
1237 |             recognition.onstart = () => {
1238 |                 isListening = true;
1239 |                 if (btnMic) btnMic.className = 'btn btn-danger btn-c
      | ircle-lg flex-shrink-0';
1240 |                 if (micIcon) micIcon.className = 'bi bi-mic-fill fs-
      | 3 animate-pulse';
1241 |                 voiceStatus.textContent = 'Слушаю...';
1242 |             };
1243 |
```

```
1244 |         recognition.onend = () => {
1245 |             isListening = false;
1246 |             if (btnMic) btnMic.className = 'btn btn-outline-dang
er btn-circle-lg flex-shrink-0';
1247 |             if (micIcon) micIcon.className = 'bi bi-mic-fill fs-
3';
1248 |             voiceStatus.textContent = 'Готов к работе';
1249 |         };
1250 |
1251 |         recognition.onerror = (e) => {
1252 |             voiceStatus.textContent = `Ошибка ввода: ${e.error}`
;
1253 |         };
1254 |
1255 |         recognition.onresult = async (event) => {
1256 |             const text = event.results[0][0].transcript;
1257 |             if (text) {
1258 |                 processChatMessage(text);
1259 |             }
1260 |         };
1261 |
1262 |         if (btnMic) {
1263 |             btnMic.addEventListener('click', () => {
1264 |                 window.triggerTgHaptic();
1265 |                 // Unlock SpeechSynthesis for mobile devices
1266 |                 if ('speechSynthesis' in window) {
1267 |                     const silentUtterance = new SpeechSynthesisUtter
ance(' ');
1268 |                     silentUtterance.volume = 0;
1269 |                     window.speechSynthesis.speak(silentUtterance);
1270 |                 }
1271 |                 if (isListening) {
1272 |                     recognition.stop();
1273 |                 } else {
1274 |                     recognition.start();
1275 |                 }
1276 |             });
1277 |         }
```

```

1278     }
1279 }
1280
1281 let currentSpeechAudio = null;
1282 let currentSpeechQueue = null;
1283
1284 class AudioQueuePlayer {
1285     constructor(chunks, userSettings) {
1286         this.chunks = chunks;
1287         this.userSettings = userSettings;
1288         this.currentIndex = 0;
1289         this.isPlaying = false;
1290         this.audioElements = new Map();
1291         this.activeAudio = null;
1292         this._prefetch(0);
1293         this._prefetch(1);
1294     }
1295
1296     _prefetch(index) {
1297         if (index >= this.chunks.length || this.audioElements.
1298 has(index)) return;
1299         const text = this.chunks[index];
1300         const queryParams = new URLSearchParams({
1301             text: text,
1302             system: this.userSettings.tts_system,
1303             voice: this.userSettings.tts_voice
1304         });
1305         const audioUrl = `/api/tts/synthesize?${queryParams.to
1306 String()}`;
1307         const audio = new Audio(audioUrl);
1308         audio.load();
1309         this.audioElements.set(index, audio);
1310     }
1311
1312     async play() {
1313         this.isPlaying = true;
1314         while (this.currentIndex < this.chunks.length && this.
1315 isPlaying) {

```

```
1313 |         this._prefetch(this.currentIndex);
1314 |         this._prefetch(this.currentIndex + 1);
1315 |         this._prefetch(this.currentIndex + 2);
1316 |
1317 |         const audio = this.audioElements.get(this.currentInd
ex);
1318 |         if (!audio) {
1319 |             this.currentIndex++;
1320 |             continue;
1321 |         }
1322 |
1323 |         this.activeAudio = audio;
1324 |         currentSpeechAudio = audio;
1325 |
1326 |         const playPromise = new Promise((resolve) => {
1327 |             audio.onended = () => resolve();
1328 |             audio.onerror = (e) => {
1329 |                 console.error(`Audio error for chunk ${this.cu
rrentIndex}:`, e);
1330 |                 this.stop(); // Останавливаем всю очередь при
ошибке загрузки
1331 |                 resolve();
1332 |             };
1333 |         });
1334 |
1335 |         try {
1336 |             await audio.play();
1337 |             await playPromise;
1338 |         } catch (err) {
1339 |             console.error("Audio playback interrupted/failed:"
, err);
1340 |         }
1341 |         this.currentIndex++;
1342 |     }
1343 |     this.isPlaying = false;
1344 | }
1345 |
1346 | stop() {
```



```
1347     this.isPlaying = false;
1348     if (this.activeAudio) {
1349         try {
1350             this.activeAudio.pause();
1351             this.activeAudio.src = "";
1352         } catch (e) {}
1353     }
1354     this.audioElements.clear();
1355 }
1356 }
1357
1358 function stopAudio() {
1359     if (currentSpeechQueue) {
1360         try {
1361             currentSpeechQueue.stop();
1362             currentSpeechQueue = null;
1363         } catch (e) {
1364             console.error("Error stopping currentSpeechQueue:",
1365 e);
1366         }
1367     }
1368     if (currentSpeechAudio) {
1369         try {
1370             currentSpeechAudio.pause();
1371             currentSpeechAudio.src = "";
1372             currentSpeechAudio = null;
1373         } catch (e) {
1374             console.error("Error stopping currentSpeechAudio:",
1375 e);
1376         }
1377     }
1378     if ('speechSynthesis' in window) {
1379         try {
1380             window.speechSynthesis.cancel();
1381         } catch (e) {
1382             console.error("Error cancelling SpeechSynthesis:", e
1383 );
1384         }
1385     }
1386 }
```

```

1382     }
1383 }
1384
1385 async function speakText(text) {
1386     if (!text) return;
1387
1388     // Stop any currently playing audio
1389     stopAudio();
1390     // Remove markdown or HTML tags if any from voice output
1391     const cleanText = text.replace(/<[>]*>/g, '').replace(/
[*_`#]/g, '').trim();
1392     if (!cleanText) return;
1393
1394     // Try backend TTS first
1395     try {
1396         const response = await fetch('/auth/settings');
1397         if (response.ok) {
1398             const userSettings = await response.json();
1399             if (userSettings.tts_enabled === 1 && userSettings.t
ts_system && userSettings.tts_system !== 'browser') {
1400                 const paragraphs = cleanText
1401                     .split(/\n+/)
1402                     .map(p => p.trim())
1403                     .filter(p => p.length > 0);
1404
1405                 let chunks = [];
1406                 for (const para of paragraphs) {
1407                     const sentences = para.match(/^[^.!?]+[.!?]*$/g) |
| [para];
1408                     let currentChunk = "";
1409                     for (const sentence of sentences) {
1410                         const trimmedSentence = sentence.trim();
1411                         if (!trimmedSentence) continue;
1412                         if ((currentChunk + " " + trimmedSentence).len
gth > 200) {
1413                             chunks.push(currentChunk.trim());
1414                             currentChunk = trimmedSentence;
1415                         } else {

```

```
1416         currentChunk = (currentChunk + " " + trimmed
Sentence).trim();
1417     }
1418 }
1419     if (currentChunk) {
1420         chunks.push(currentChunk.trim());
1421     }
1422 }
1423     chunks = chunks.filter(c => c.length > 0);
1424
1425     if (chunks.length > 0) {
1426         currentSpeechQueue = new AudioQueuePlayer(chunks
, userSettings);
1427         await currentSpeechQueue.play();
1428     }
1429     return;
1430 }
1431 }
1432 } catch (e) {
1433     console.error('Failed to speak using backend TTS, fall
ing back to browser:', e);
1434 }
1435
1436 // Fallback to browser SpeechSynthesis
1437 if ('speechSynthesis' in window) {
1438     window.currentUtterance = new SpeechSynthesisUtterance
(cleanText);
1439     window.currentUtterance.lang = 'ru-RU';
1440     window.speechSynthesis.speak(window.currentUtterance);
1441 }
1442 }
1443
1444 function handleLocalVoiceCommand(text) {
1445     const t = text.toLowerCase();
1446     if (t.includes('пайз') || t.includes('стоп')) {
1447         sendCommand('pause');
1448         return true;
1449     }
```

```
1450 |         if (t.includes('продолж') || t.includes('плей') || t.inc
      | ludes('играй') || t.includes('старт')) {
1451 |             sendCommand('play');
1452 |             return true;
1453 |         }
1454 |         if (t.includes('громче') || t.includes('прибавь звук'))
      | {
1455 |             const targetVol = Math.min(1, playerState.volume + 0.2
      | );
1456 |             sendCommand('set_volume', { level: targetVol });
1457 |             return true;
1458 |         }
1459 |         if (t.includes('тише') || t.includes('убавь звук')) {
1460 |             const targetVol = Math.max(0, playerState.volume - 0.2
      | );
1461 |             sendCommand('set_volume', { level: targetVol });
1462 |             return true;
1463 |         }
1464 |         if (t.includes('следующ') || t.includes('вперед')) {
1465 |             sendCommand('next');
1466 |             return true;
1467 |         }
1468 |         if (t.includes('предыдущ') || t.includes('назад')) {
1469 |             sendCommand('prev');
1470 |             return true;
1471 |         }
1472 |         if (t.includes('карусель') || t.includes('случайн') || t
      | .includes('рандом')) {
1473 |             sendCommand('play_random_movie');
1474 |             return true;
1475 |         }
1476 |         return false;
1477 |     }
1478 |
1479 |     async function sendToAiModel(message) {
1480 |         const response = await fetch('/api/chat', {
1481 |             method: 'POST',
1482 |             headers: { 'Content-Type': 'application/json' },
```

```
1483 |         body: JSON.stringify({ message, generation_config: { i
1484 | s_rc: true } })
1485 |     });
1486 |     if (!response.ok) {
1487 |         throw new Error('Не удалось получить ответ от AI');
1488 |     }
1489 |     const data = await response.json();
1490 |     return data.response || data.message || 'Модель не ответ
1491 | ила';
1492 | }
1493 | // Set listeners
1494 | document.getElementById('btn-playtoggle').addEventListener
1495 | ('click', () => sendCommand('toggle_play'));
1496 | document.getElementById('btn-prev').addEventListener('clie
1497 | k', () => sendCommand('prev'));
1498 | document.getElementById('btn-next').addEventListener('clie
1499 | k', () => sendCommand('next'));
1500 | document.getElementById('btn-carousel').addEventListener('
1501 | click', () => sendCommand('play_random_movie'));
1502 |
1503 | const rcCheckMailBtn = document.getElementById('rc-btn-che
1504 | ck-mail');
1505 | if (rcCheckMailBtn) {
1506 |     rcCheckMailBtn.addEventListener('click', () => {
1507 |         processChatMessage('Проверь мою почту Gmail и покажи п
1508 | оследние непрочитанные письма.');
```

```

    ve документы по запросу: "${query.trim()}`");
1511     }
1512   });
1513 }
1514
1515   const rcGsheetsBtn = document.getElementById('rc-btn-gsheets');
1516   if (rcGsheetsBtn) {
1517     rcGsheetsBtn.addEventListener('click', () => {
1518       const query = prompt('Введите ID таблицы или команду для Google Sheets (например: "Прочитай диапазон A1:D10 из таблицы ID"):');
1519       if (query && query.trim()) {
1520         processChatMessage(`Google Sheets: ${query.trim()}`)
1521       };
1522     });
1523   }
1524
1525   const rcFacebookBtn = document.getElementById('rc-btn-facebook');
1526   if (rcFacebookBtn) {
1527     rcFacebookBtn.addEventListener('click', () => {
1528       const text = prompt('Введите текст для публикации в Facebook:');
1529       if (text && text.trim()) {
1530         processChatMessage(`Опубликуй пост в Facebook с текстом: "${text.trim()}`");
1531       }
1532     });
1533   }
1534
1535   seekSlider.addEventListener('mousedown', () => { isSeeking = true; });
1536   seekSlider.addEventListener('touchstart', () => { isSeeking = true; });
1537   seekSlider.addEventListener('change', (e) => {
1538     isSeeking = false;

```

```

1539     sendCommand('seek', { seconds: parseFloat(e.target.value
1540   ) });
1541   });
1542   volumeSlider.addEventListener('change', (e) => {
1543     sendCommand('set_volume', { level: parseFloat(e.target.v
1544   alue) });
1545   });
1546   // — USER MANUAL AUDIO READER —————
1547   let isSpeakingManual = false;
1548   const btnRead = document.getElementById('btn-read-manual')
1549   ;
1550   const readIcon = document.getElementById('read-manual-icon
1551   ');
1552   const readTextSpan = btnRead ? btnRead.querySelector('span
1553   ') : null;
1554   if (btnRead) {
1555     btnRead.addEventListener('click', () => {
1556       window.triggerTgHaptic();
1557       const text = document.getElementById('user-manual-text
1558   ').innerText;
1559       if (isSpeakingManual) {
1560         window.speechSynthesis.cancel();
1561         isSpeakingManual = false;
1562         if (readIcon) readIcon.className = 'bi bi-volume-up-
1563   fill';
1564         if (readTextSpan) readTextSpan.textContent = 'Прослу
1565   шать';
1566         btnRead.className = 'btn btn-outline-info btn-sm d-f
1567   lex align-items-center gap-1';
1568       } else {
1569         window.speechSynthesis.cancel();
1570         const utterance = new SpeechSynthesisUtterance(text)
1571         ;

```

```
1566         utterance.lang = 'ru-RU';
1567
1568         utterance.onend = () => {
1569             isSpeakingManual = false;
1570             if (readIcon) readIcon.className = 'bi bi-volume-u
p-fill';
1571             if (readTextSpan) readTextSpan.textContent = 'Прос
лушать';
1572             btnRead.className = 'btn btn-outline-info btn-sm d
-flex align-items-center gap-1';
1573         };
1574
1575         utterance.onerror = () => {
1576             isSpeakingManual = false;
1577             if (readIcon) readIcon.className = 'bi bi-volume-u
p-fill';
1578             if (readTextSpan) readTextSpan.textContent = 'Прос
лушать';
1579             btnRead.className = 'btn btn-outline-info btn-sm d
-flex align-items-center gap-1';
1580         };
1581
1582         window.speechSynthesis.speak(utterance);
1583         isSpeakingManual = true;
1584         if (readIcon) readIcon.className = 'bi bi-stop-fill'
;
1585         if (readTextSpan) readTextSpan.textContent = 'Остано
вить';
1586         btnRead.className = 'btn btn-info btn-sm text-dark d
-flex align-items-center gap-1';
1587     }
1588 });
1589 }
1590
1591 // Theme toggle function
1592 function toggleTheme() {
1593     const body = document.body;
1594     const currentTheme = body.classList.contains('theme-ligh
```



```

t') ? 'light' : 'dark';
1595     const newTheme = currentTheme === 'light' ? 'dark' : 'light';
1596
1597     if (newTheme === 'light') {
1598         body.classList.add('theme-light');
1599     } else {
1600         body.classList.remove('theme-light');
1601     }
1602 }
1603
1604 // Set theme toggle listener
1605 document.getElementById('theme-toggle-btn')?.addEventListener('click', toggleTheme);
1606
1607 // — AUTH CHECK INITIALIZER —————
1608
1609 async function checkAndInit() {
1610     const authState = await api.auth.check();
1611     if (authState.authenticated) {
1612         // If a room parameter is present in URL, connect directly to it
1613         const queryRoom = getRoomNameFromQuery();
1614         if (queryRoom) {
1615             initStream(queryRoom);
1616         } else {
1617             // Otherwise, scan for open players
1618             scanForPlayers();
1619         }
1620     } else {
1621         showPanel('auth');
1622     }
1623 }
1624
1625 // Run on startup
1626 document.addEventListener('DOMContentLoaded', () => {
1627     checkAndInit();
1628     initVoiceControl();

```

```
1628 |     });
1629 |
1630 | </script>
1631 | </body>
1632 | </html>
1633 |
```

index.html

```
1 <!-- Windows Registry Viewer Tab -->
2 <style>
3   .reg-container {
4     padding: 1rem;
5   }
6   .reg-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 10px;
10  }
11  .reg-path-bar {
12    background: #1e293b !important;
13    border: 1px solid #334155 !important;
14    border-radius: 8px;
15    padding: 0.5rem 0.75rem;
16  }
17  .reg-tree-pane {
18    background: #0b1329;
19    border: 1px solid #334155;
20    border-radius: 8px;
21    height: 580px;
22    overflow-y: auto;
23    font-size: 0.82rem;
24    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
25  }
26  .reg-tree-item {
27    padding: 4px 8px;
28    cursor: pointer;
29    border-radius: 4px;
30    display: flex;
31    align-items: center;
32    gap: 6px;
33    color: #cbd5e1;
34    white-space: nowrap;
35    overflow: hidden;
```

```
36 |     text-overflow: ellipsis;
37 | }
38 | .reg-tree-item:hover {
39 |     background: rgba(56, 189, 248, 0.1);
40 |     color: #38bdf8;
41 | }
42 | .reg-tree-item.active {
43 |     background: rgba(56, 189, 248, 0.25);
44 |     color: #38bdf8;
45 |     font-weight: 600;
46 | }
47 | .reg-values-pane {
48 |     background: #0f172a;
49 |     border: 1px solid #334155;
50 |     border-radius: 8px;
51 |     height: 580px;
52 |     overflow-y: auto;
53 | }
54 | .reg-table {
55 |     width: 100%;
56 |     font-size: 0.8rem;
57 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
58 |     border-collapse: collapse;
59 | }
60 | .reg-table th {
61 |     background: #1e293b;
62 |     padding: 8px 12px;
63 |     color: #94a3b8;
64 |     font-weight: 600;
65 |     border-bottom: 1px solid #334155;
66 |     position: sticky;
67 |     top: 0;
68 |     z-index: 2;
69 | }
70 | .reg-table td {
71 |     padding: 6px 12px;
72 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
```

```

73     vertical-align: top;
74 }
75 .reg-table tbody tr:hover {
76     background: rgba(56, 189, 248, 0.05);
77 }
78 .reg-type-badge {
79     font-size: 0.68rem;
80     padding: 2px 6px;
81     border-radius: 4px;
82     background: #1e293b;
83     border: 1px solid #475569;
84     color: #38bdf8;
85 }
86 .reg-bookmark-btn {
87     font-size: 0.75rem;
88     padding: 4px 10px;
89     border-radius: 6px;
90     background: #1e293b;
91     border: 1px solid #334155;
92     color: #cbd5e1;
93     transition: all 0.2s ease;
94 }
95 .reg-bookmark-btn:hover {
96     background: #334155;
97     color: #38bdf8;
98     border-color: #38bdf8;
99 }
100 </style>
101
102 <div class="reg-container">
103     <!-- Header Alert & Status -->
104     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 p-3 rounded-3 text-white" style="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%); border: 1px solid #334155;">
105         <div>
106             <h5 class="mb-1 fw-bold text-white d-flex align-items-center gap-2">

```

```

107 |         <span class="fs-4">🔍 </span> Просмотр реестра (Windows
      | Registry Viewer)
108 |     </h5>
109 |     <p class="mb-0 small text-light" style="color: #cbd5e1 !
      | important;">
110 |         Безопасный навигатор по системному реестру Windows (HK
      | LM, HKCU, HKCR, HKU, HKCC), быстрый переход к ключевым веткам
      | автозагрузки, служб и политик.
111 |     </p>
112 | </div>
113 | <div class="d-flex align-items-center gap-2 flex-wrap">
114 |     <span class="badge rounded-pill bg-success-subtle text-s
      | uccess border border-success px-3 py-2" id="reg-status-badge">
115 |         • Реестр: Подключено
116 |     </span>
117 |     <button class="btn btn-sm btn-outline-info rounded-pill
      | px-3" id="btn-reg-refresh">
118 |         <i class="bi bi-arrow-clockwise me-1"></i> Обновить
119 |     </button>
120 |     <button class="btn btn-sm btn-outline-light rounded-pill
      | px-3" id="btn-reg-copy-path">
121 |         <i class="bi bi-clipboard me-1"></i> Копировать путь
122 |     </button>
123 |     <button class="btn btn-sm btn-warning rounded-pill px-3"
      | id="btn-reg-backups-modal">
124 |         <i class="bi bi-shield-check me-1"></i> Бэкапы и откат
125 |     </button>
126 | </div>
127 | </div>
128 |
129 | <!-- Bookmarks Quick Bar -->
130 | <div class="mb-3 d-flex align-items-center gap-1 flex-wrap p
      | -2 rounded bg-dark border border-secondary" id="reg-bookmarks-
      | container">
131 |     <span class="text-muted small me-2"><i class="bi bi-bookma
      | rk-star-fill text-warning me-1"></i> Закладки:</span>
132 |     <!-- Bookmarks will be loaded here -->
133 | </div>

```

```
134 |
135 | <!-- Navigation Path Bar -->
136 | <div class="reg-path-bar mb-3">
137 |     <div class="row g-2 align-items-center">
138 |         <div class="col-md-3 col-sm-4">
139 |             <select class="form-select form-select-sm bg-dark text
140 | -white border-secondary font-monospace" id="reg-hive-select">
141 |                 <option value="HKEY_LOCAL_MACHINE">HKEY_LOCAL_MACHIN
142 | E (HKLM)</option>
143 |                 <option value="HKEY_CURRENT_USER">HKEY_CURRENT_USER
144 | (HKCU)</option>
145 |                 <option value="HKEY_CLASSES_ROOT">HKEY_CLASSES_ROOT
146 | (HKCR)</option>
147 |                 <option value="HKEY_USERS">HKEY_USERS (HKU)</option>
148 |                 <option value="HKEY_CURRENT_CONFIG">HKEY_CURRENT_CON
149 | FIG (HKCC)</option>
150 |             </select>
151 |         </div>
152 |         <div class="col-md-7 col-sm-6">
153 |             <div class="input-group input-group-sm">
154 |                 <button class="btn btn-outline-secondary" id="btn-re
155 | g-up" title="На уровень выше">
156 |                     <i class="bi bi-arrow-up-short fs-6"></i>
157 |                 </button>
158 |                 <input type="text" class="form-control bg-dark text-
159 | white border-secondary font-monospace" id="reg-path-input" pla
160 | ceholder="SOFTWARE\Microsoft\Windows..." value="SOFTWARE">
161 |                 <button class="btn btn-primary" id="btn-reg-go">
162 |                     Перейти
163 |                 </button>
164 |             </div>
165 |         </div>
166 |         <div class="col-md-2 col-sm-2 text-end">
167 |             <div class="input-group input-group-sm">
168 |                 <input type="text" class="form-control bg-dark text-
169 | white border-secondary" id="reg-filter-input" placeholder="Фил
170 | ьтр параметров...">
171 |             </div>
```

```
162         </div>
163     </div>
164 </div>
165
166 <!-- Main Explorer View (Subkeys Tree + Values Table) -->
167 <div class="row g-3">
168     <!-- Subkeys Tree Pane (Left) -->
169     <div class="col-md-4">
170         <div class="reg-tree-pane p-2">
171             <div class="d-flex justify-content-between align-items
172             -center mb-2 pb-1 border-bottom border-secondary">
173                 <span class="text-muted small fw-bold"><i class="bi
174                 bi-folder2-open me-1"></i> Подразделы (Subkeys)</span>
175                 <div class="d-flex gap-1 align-items-center">
176                     <span class="badge bg-secondary" id="reg-subkeys-c
177                     ount-badge">0</span>
178                     <button class="btn btn-xs btn-outline-success py-0
179                     px-2" id="btn-reg-add-key" title="Создать подраздел" style="f
180                     ont-size: 0.75rem;">
181                         <i class="bi bi-plus-lg"></i>
182                     </button>
183                     <button class="btn btn-xs btn-outline-danger py-0
184                     px-2" id="btn-reg-delete-current-key" title="Удалить текущий р
185                     аздел" style="font-size: 0.75rem;">
186                         <i class="bi bi-trash"></i>
187                     </button>
188                 </div>
189             </div>
190             <div id="reg-subkeys-list">
191                 <div class="text-muted text-center py-4 small">Загру
```



```

192 |         <div class="d-flex justify-content-between align-items
| -center p-2 border-bottom border-secondary bg-dark">
193 |             <span class="text-muted small fw-bold"><i class="bi
| bi-card-list me-1"></i> Параметры и значения (Values)</span>
194 |             <div class="d-flex gap-2 align-items-center">
195 |                 <span class="badge bg-secondary" id="reg-values-co
| unt-badge">0</span>
196 |                 <button class="btn btn-sm btn-success py-1 px-3" i
| d="btn-reg-add-value">
197 |                     <i class="bi bi-plus-circle me-1"></i> Создать п
| араметр
198 |                 </button>
199 |             </div>
200 |         </div>
201 |         <div class="table-responsive" style="max-height: 535px
| ; overflow-y: auto;">
202 |             <table class="reg-table">
203 |                 <thead>
204 |                     <tr>
205 |                         <th style="width: 28%;">Имя параметра</th>
206 |                         <th style="width: 18%;">Тип</th>
207 |                         <th style="width: 42%;">Значение</th>
208 |                         <th style="width: 12%; text-align: center;">Де
| йствия</th>
209 |                     </tr>
210 |                 </thead>
211 |                 <tbody id="reg-values-tbody">
212 |                     <tr><td colspan="4" class="text-center text-mute
| d p-4">Выберите ключ реестра для просмотра параметров</td></tr>
| >
213 |                     </tbody>
214 |                 </table>
215 |             </div>
216 |         </div>
217 |     </div>
218 | </div>
219 |
220 | <!-- Modal: Редактировать / Создать параметр -->

```

```
221 | <div class="modal fade" id="regValueModal" tabindex="-1" ari
    | a-hidden="true">
222 |     <div class="modal-dialog modal-dialog-centered">
223 |         <div class="modal-content bg-dark text-white border-seco
    | ndary">
224 |             <div class="modal-header border-secondary">
225 |                 <h5 class="modal-title" id="regValueModalTitle"> Pe
    | дактирование параметра</h5>
226 |                 <button type="button" class="btn-close btn-close-whi
    | te" data-bs-dismiss="modal" aria-label="Close"></button>
227 |             </div>
228 |             <div class="modal-body">
229 |                 <form id="regValueForm">
230 |                     <div class="mb-3">
231 |                         <label class="form-label small text-muted">Имя п
    | араметра</label>
232 |                         <input type="text" class="form-control bg-second
    | ary text-white border-dark font-monospace" id="regModalValueNa
    | me" placeholder="Оставьте пустым для (По умолчанию)">
233 |                     </div>
234 |                     <div class="mb-3">
235 |                         <label class="form-label small text-muted">Тип д
    | анных</label>
236 |                         <select class="form-select bg-secondary text-whi
    | te border-dark font-monospace" id="regModalValueType">
237 |                             <option value="REG_SZ">REG_SZ (Строка)</option
    | >
238 |                             <option value="REG_DWORD">REG_DWORD (Число 32-
    | бит)</option>
239 |                             <option value="REG_QWORD">REG_QWORD (Число 64-
    | бит)</option>
240 |                             <option value="REG_EXPAND_SZ">REG_EXPAND_SZ (P
    | асширяемая строка)</option>
241 |                             <option value="REG_MULTI_SZ">REG_MULTI_SZ (Мно
    | гострочный текст)</option>
242 |                             <option value="REG_BINARY">REG_BINARY (Двоичны
    | е данные / HEX)</option>
243 |                         </select>
```

```

244         </div>
245         <div class="mb-3">
246             <label class="form-label small text-muted">Значе
ние</label>
247             <textarea class="form-control bg-secondary text-
white border-dark font-monospace" id="regModalValueData" rows=
"4" placeholder="Введите значение параметра"></textarea>
248             <div class="form-text text-muted small">Для REG_
MULTI_SZ разделяйте строки новой строкой. Для REG_BINARY вводи
те HEX байты.</div>
249         </div>
250         <div class="form-check form-switch mb-2">
251             <input class="form-check-input" type="checkbox"
id="regModalBackupCheck" checked>
252             <label class="form-check-label small text-light"
for="regModalBackupCheck">
253                 ☐ Создать резервную копию ветки перед сохране
нием
254             </label>
255         </div>
256     </form>
257 </div>
258 <div class="modal-footer border-secondary">
259     <button type="button" class="btn btn-secondary" data
-bs-dismiss="modal">Отмена</button>
260     <button type="button" class="btn btn-primary" id="bt
n-reg-modal-save">Сохранить</button>
261 </div>
262 </div>
263 </div>
264 </div>
265
266 <!-- Modal: Создать подраздел -->
267 <div class="modal fade" id="regKeyModal" tabindex="-1" aria-
hidden="true">
268     <div class="modal-dialog modal-dialog-centered">
269         <div class="modal-content bg-dark text-white border-seco
ndary">

```

```
270 |         <div class="modal-header border-secondary">
271 |             <h5 class="modal-title">□ Новый подраздел реестра</
| h5>
272 |             <button type="button" class="btn-close btn-close-whi
| te" data-bs-dismiss="modal" aria-label="Close"></button>
273 |         </div>
274 |         <div class="modal-body">
275 |             <div class="mb-3">
276 |                 <label class="form-label small text-muted">Имя нов
| ого подраздела</label>
277 |                 <input type="text" class="form-control bg-secondar
| y text-white border-dark font-monospace" id="regModalNewKeyNam
| e" placeholder="MyNewSubkey">
278 |             </div>
279 |             <p class="small text-muted mb-0">Ключ будет создан в
| нутри текущего пути.</p>
280 |         </div>
281 |         <div class="modal-footer border-secondary">
282 |             <button type="button" class="btn btn-secondary" data
| -bs-dismiss="modal">Отмена</button>
283 |             <button type="button" class="btn btn-success" id="bt
| n-reg-modal-create-key">Создать</button>
284 |         </div>
285 |     </div>
286 | </div>
287 | </div>
288 |
289 | <!-- Modal: Резервные копии и откат -->
290 | <div class="modal fade" id="regBackupsModal" tabindex="-1" a
| ria-hidden="true">
291 |     <div class="modal-dialog modal-lg modal-dialog-centered">
292 |         <div class="modal-content bg-dark text-white border-seco
| ndary">
293 |             <div class="modal-header border-secondary">
294 |                 <h5 class="modal-title">□ Точки восстановления и ис
| тория бэкапов</h5>
295 |                 <button type="button" class="btn-close btn-close-whi
| te" data-bs-dismiss="modal" aria-label="Close"></button>
```

```
296 |         </div>
297 |         <div class="modal-body p-0">
298 |             <div class="table-responsive" style="max-height: 400
| px; overflow-y: auto;">
299 |                 <table class="table table-dark table-hover mb-0 fo
| nt-monospace small">
300 |                     <thead>
301 |                         <tr>
302 |                             <th>Время</th>
303 |                             <th>Ветка и путь</th>
304 |                             <th>Операция</th>
305 |                             <th>Действие</th>
306 |                         </tr>
307 |                     </thead>
308 |                     <tbody id="reg-backups-tbody">
309 |                         <tr><td colspan="4" class="text-center py-4 te
| xt-muted">Загрузка списка бэкапов...</td></tr>
310 |                     </tbody>
311 |                 </table>
312 |             </div>
313 |         </div>
314 |         <div class="modal-footer border-secondary">
315 |             <button type="button" class="btn btn-secondary" data
| -bs-dismiss="modal">Закрыть</button>
316 |         </div>
317 |     </div>
318 | </div>
319 | </div>
320 | </div>
321 |
322 |
```

main.js

```
1 // Windows Registry Viewer Tab JavaScript Module
2 (function() {
3   let currentHive = 'HKEY_LOCAL_MACHINE';
4   let currentPath = 'SOFTWARE';
5   let currentKeyData = null;
6   let isInitialized = false;
7
8   async function loadBookmarks() {
9     const container = document.getElementById('reg-bookmarks-c
10 ontainer');
11     if (!container) return;
12
13     try {
14       const res = await fetch('/api/registry/bookmarks');
15       if (!res.ok) throw new Error('Bookmarks API error');
16       const data = await res.json();
17       const bookmarks = data.bookmarks || [];
18
19       container.innerHTML = `
20         <span class="text-muted small me-2"><i class="bi bi-bo
21 okmark-star-fill text-warning me-1"></i> Закладки:</span>
22         ` + bookmarks.map(b => `
23           <button class="reg-bookmark-btn" data-hive="${b.hive}"
24 data-path="${b.path}" title="${b.description || ''}">
25             <span>${b.icon || '□'}</span> ${b.title}
26           </button>
27         `).join('');
28
29       container.querySelectorAll('.reg-bookmark-btn').forEach(
30 btn => {
31         btn.onclick = () => {
32           const h = btn.getAttribute('data-hive');
33           const p = btn.getAttribute('data-path');
34           navigateTo(h, p);
35         };
36       });
37     } catch (e) {
38       console.error(e);
39     }
40   }
41
42   loadBookmarks();
43 }
```

```
33     } catch (e) {
34         console.warn('[RegistryViewerTab] Failed to load bookmarks:', e);
35     }
36 }
37
38 async function navigateTo(hive, path) {
39     currentHive = hive;
40     currentPath = path.trim().replace(/^[/\\\/]+|[/\\\/]+$/g, '');
41 };
42
43 const hiveSelect = document.getElementById('reg-hive-select');
44 const pathInput = document.getElementById('reg-path-input');
45
46 if (hiveSelect) hiveSelect.value = currentHive;
47 if (pathInput) pathInput.value = currentPath;
48
49 await loadKey(currentHive, currentPath);
50 }
51
52 async function loadKey(hive, path) {
53     const subkeysList = document.getElementById('reg-subkeys-list');
54     const valuesTbody = document.getElementById('reg-values-tbody');
55     const badge = document.getElementById('reg-status-badge');
56     const subkeysCountBadge = document.getElementById('reg-subkeys-count-badge');
57     const valuesCountBadge = document.getElementById('reg-values-count-badge');
58
59     if (badge) badge.innerText = `• Чтение ${hive}...`;
60
61     try {
62         const url = `/api/registry/key?hive=${encodeURIComponent(hive)}&path=${encodeURIComponent(path)}`;
63         const res = await fetch(url);
```

```

62 |         if (!res.ok) {
63 |             const errJson = await res.json().catch(() => ({}));
64 |             throw new Error(errJson.detail || `HTTP ${res.status}`
    | );
65 |         }
66 |
67 |         currentKeyData = await res.json();
68 |
69 |         // Рендер подразделов (subkeys)
70 |         const subkeys = currentKeyData.subkeys || [];
71 |         if (subkeysCountBadge) subkeysCountBadge.innerText = sub
    | keys.length;
72 |
73 |         if (subkeys.length === 0) {
74 |             subkeysList.innerHTML = '<div class="text-muted text-c
    | enter py-4 small">Нет подразделов</div>';
75 |         } else {
76 |             subkeysList.innerHTML = subkeys.map(sk => `
77 |                 <div class="reg-tree-item" data-subkey="${sk}" title
    | ="${sk}">
78 |                     <i class="bi bi-folder-fill text-warning"></i>
79 |                     <span class="text-truncate">${sk}</span>
80 |                 </div>
81 |             `).join('');
82 |
83 |             subkeysList.querySelectorAll('.reg-tree-item').forEach
    | (item => {
84 |                 item.onclick = () => {
85 |                     const subkey = item.getAttribute('data-subkey');
86 |                     const newPath = currentPath ? `${currentPath}\\${s
    | ubkey}` : subkey;
87 |                     navigateTo(currentHive, newPath);
88 |                 };
89 |             });
90 |         }
91 |
92 |         // Рендер параметров (values)
93 |         renderValues();

```



```
94 |
95 |     if (badge) {
96 |         badge.className = 'badge rounded-pill bg-success-subtl
e text-success border border-success px-3 py-2';
97 |         badge.innerText = `● Реестр: ${hive}`;
98 |     }
99 | } catch (err) {
100 |     console.error('[RegistryViewerTab] Error loading key:',
err);
101 |     if (valuesTbody) {
102 |         valuesTbody.innerHTML = `<tr><td colspan="3" class="te
xt-center text-danger p-4"><i class="bi bi-exclamation-triangl
e-fill me-1"></i> Ошибка чтения ветки: ${err.message}</td></tr
>`;
103 |     }
104 |     if (subkeysList) {
105 |         subkeysList.innerHTML = `<div class="text-danger text-
center py-4 small">${err.message}</div>`;
106 |     }
107 |     if (badge) {
108 |         badge.className = 'badge rounded-pill bg-danger-subtle
text-danger border border-danger px-3 py-2';
109 |         badge.innerText = '● Ошибка доступа';
110 |     }
111 | }
112 | }
113 |
114 | function renderValues() {
115 |     const valuesTbody = document.getElementById('reg-values-tb
ody');
116 |     const valuesCountBadge = document.getElementById('reg-valu
es-count-badge');
117 |     if (!valuesTbody || !currentKeyData) return;
118 |
119 |     const filterText = (document.getElementById('reg-filter-in
put')?.value || '').toLowerCase().trim();
120 |     const values = currentKeyData.values || [];
121 |
```

```
122     const filtered = values.filter(v => {
123         if (!filterText) return true;
124         return v.name.toLowerCase().includes(filterText) || String(v.data).toLowerCase().includes(filterText) || v.type_name.toLowerCase().includes(filterText);
125     });
126
127     if (valuesCountBadge) valuesCountBadge.innerText = `${filtered.length} / ${values.length}`;
128
129     if (filtered.length === 0) {
130         valuesTbody.innerHTML = '<tr><td colspan="3" class="text-center text-muted p-4">Параметры отсутствуют или не соответствуют фильтру</td></tr>';
131         return;
132     }
133
134     valuesTbody.innerHTML = filtered.map((v, idx) => {
135         let dataDisplay = v.data;
136         if (dataDisplay === null || dataDisplay === undefined) {
137             dataDisplay = '<span class="text-muted">(Значение не задано)</span>';
138         } else if (typeof dataDisplay === 'object') {
139             dataDisplay = '<pre class="mb-0 text-info" style="font-size: 0.75rem;">${JSON.stringify(dataDisplay, null, 2)}</pre>';
140         } else {
141             dataDisplay = '<span class="text-light text-break">${escapeHtml(String(dataDisplay))}</span>';
142         }
143
144         const isDefault = v.name === '(Default / По умолчанию)' || v.name === '(Default)';
145         const rawNameAttr = escapeHtml(isDefault ? '' : v.name);
146
147         return `
148         <tr class="reg-val-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для AI-объяснения параметра реестр
```

```

a">
149         <td>
150             <div class="fw-bold text-white text-truncate" styl
e="max-width: 220px;" title="${v.name}">
151                 <i class="bi bi-file-earmark-binary text-seconda
ry me-1"></i>
152                 ${escapeHtml(v.name)}
153             </div>
154         </td>
155         <td>
156             <span class="reg-type-badge">${v.type_name}</span>
157         </td>
158         <td>
159             ${dataDisplay}
160         </td>
161         <td class="text-center">
162             <button class="btn btn-xs btn-outline-info py-0 px
-2 me-1 btn-edit-val" data-idx="${idx}" title="Редактировать">
163                 <i class="bi bi-pencil"></i>
164             </button>
165             <button class="btn btn-xs btn-outline-danger py-0
px-2 btn-del-val" data-name="${rawNameAttr}" title="Удалить">
166                 <i class="bi bi-trash"></i>
167             </button>
168         </td>
169     </tr>
170 `;
171 }).join('');
172
173 // Привязка кликабельности всей строки к AITableModal
174 valuesTbody.querySelectorAll('.reg-val-row').forEach(row =
> {
175     row.onclick = (evt) => {
176         if (evt.target.closest('.btn-edit-val') || evt.target.
closest('.btn-del-val')) return;
177         const idx = parseInt(row.getAttribute('data-idx'), 10)
;
178         const v = filtered[idx];

```

```

179         if (!v) return;
180
181         if (window.AITableModal) {
182             window.AITableModal.show({
183                 icon: '□ ',
184                 title: v.name || '(Default)',
185                 subtitle: `${currentHive}\\${currentPath}`,
186                 tableType: 'registry',
187                 badges: [
188                     { text: v.type_name || 'REG_SZ', class: 'badge bg-info text-dark' },
189                     { text: currentHive, class: 'badge bg-secondary' }
190                 ],
191                 metadata: [
192                     { label: 'Параметр', value: v.name || '(По умолчанию)' },
193                     { label: 'Тип данных', value: v.type_name },
194                     { label: 'Ветка реестра (Hive)', value: currentHive },
195                     { label: 'Путь к разделу', value: currentPath, isCode: true, fullWidth: true },
196                     { label: 'Значение', value: String(v.data ?? ''), isCode: true, fullWidth: true }
197                 ],
198                 rawTitle: 'Значение параметра',
199                 rawContent: typeof v.data === 'object' ? JSON.stringify(v.data, null, 2) : String(v.data ?? ''),
200                 requestData: {
201                     hive: currentHive,
202                     path: currentPath,
203                     name: v.name,
204                     type: v.type_name,
205                     data: v.data
206                 }
207             });
208         }
209     };

```

```

210     });
211
212     // Привязка событий кнопок редактирования и удаления параметров
213     valuesTbody.querySelectorAll('.btn-edit-val').forEach(btn => {
214         btn.onclick = (evt) => {
215             evt.stopPropagation();
216             const idx = parseInt(btn.getAttribute('data-idx'), 10)
217             ;
218             const valObj = filtered[idx];
219             if (valObj) openEditValueModal(valObj);
220         });
221
222     valuesTbody.querySelectorAll('.btn-del-val').forEach(btn => {
223         btn.onclick = (evt) => {
224             evt.stopPropagation();
225             const valName = btn.getAttribute('data-name');
226             confirmDeleteValue(valName);
227         };
228     });
229 }
230
231 function openEditValueModal(valObj) {
232     const isDefault = !valObj.name || valObj.name === '(Default / По умолчанию)' || valObj.name === '(Default)';
233     const nameInput = document.getElementById('regModalValueName');
234     const typeSelect = document.getElementById('regModalValueType');
235     const dataInput = document.getElementById('regModalValueData');
236     const title = document.getElementById('regValueModalTitle');
237
238     if (title) title.innerText = '✎ Редактирование параметра';

```

```

239     if (nameInput) {
240         nameInput.value = isDefault ? '' : valObj.name;
241         nameInput.disabled = true; // имя существующего параметр
    а не меняем
242     }
243     if (typeSelect) typeSelect.value = valObj.type_name || 'REG_SZ';
244     if (dataInput) {
245         if (Array.isArray(valObj.data)) {
246             dataInput.value = valObj.data.join('\n');
247         } else {
248             dataInput.value = valObj.data !== null && valObj.data
    !== undefined ? String(valObj.data) : '';
249         }
250     }
251
252     const modalEl = document.getElementById('regValueModal');
253     if (modalEl && window.bootstrap) {
254         const modal = bootstrap.Modal.getInstance(modalEl);
    1);
255         modal.show();
256     }
257 }
258
259 function openCreateValueModal() {
260     const nameInput = document.getElementById('regModalValueName');
261     const typeSelect = document.getElementById('regModalValueType');
262     const dataInput = document.getElementById('regModalValueData');
263     const title = document.getElementById('regValueModalTitle');
264
265     if (title) title.innerText = '□ Создание нового параметра';
    ;
266     if (nameInput) {
267         nameInput.value = '';

```

```
268     nameInput.disabled = false;
269   }
270   if (typeSelect) typeSelect.value = 'REG_SZ';
271   if (dataInput) dataInput.value = '';
272
273   const modalEl = document.getElementById('regValueModal');
274   if (modalEl && window.bootstrap) {
275     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
276     modal.show();
277   }
278 }
279
280 async function saveModalValue() {
281   const nameInput = document.getElementById('regModalValueName');
282   const typeSelect = document.getElementById('regModalValueType');
283   const dataInput = document.getElementById('regModalValueData');
284   const backupCheck = document.getElementById('regModalBackupCheck');
285
286   const name = nameInput ? nameInput.value.trim() : '';
287   const type_name = typeSelect ? typeSelect.value : 'REG_SZ';
288
289   const rawData = dataInput ? dataInput.value : '';
290   const create_backup = backupCheck ? backupCheck.checked : true;
291
292   try {
293     const payload = {
294       hive: currentHive,
295       path: currentPath,
296       name: name,
297       type_name: type_name,
298       data: rawData,
299       create_backup: create_backup
```

```

299 |     };
300 |
301 |     const res = await fetch('/api/registry/value', {
302 |       method: 'POST',
303 |       headers: { 'Content-Type': 'application/json' },
304 |       body: JSON.stringify(payload)
305 |     });
306 |
307 |     if (!res.ok) {
308 |       const errJson = await res.json().catch(() => ({}));
309 |       throw new Error(errJson.detail || `HTTP ${res.status}`
310 |     );
311 |   }
312 |
313 |   const modalEl = document.getElementById('regValueModal')
314 |   ;
315 |   if (modalEl && window.bootstrap) {
316 |     const modal = bootstrap.Modal.getInstance(modalEl);
317 |     if (modal) modal.hide();
318 |   }
319 |
320 |   await loadKey(currentHive, currentPath);
321 | } catch (e) {
322 |   window.showToast?(`Ошибка сохранения параметра: ${e.mes
323 | sage}`, 'danger') || alert(`Ошибка сохранения параметра: ${e.m
324 | essage}`);
325 | }
326 |
327 | async function confirmDeleteValue(name) {
328 |   const displayName = name ? `${name}` : '(По умолчанию)';
329 |   if (!confirm(`Вы действительно хотите удалить параметр ${d
330 | isplayName}? Перед удалением будет автоматически создан бэкап.
331 | `)) {
332 |     return;
333 |   }
334 |
335 |   try {

```



```
331 |     const res = await fetch('/api/registry/value', {
332 |       method: 'DELETE',
333 |       headers: { 'Content-Type': 'application/json' },
334 |       body: JSON.stringify({
335 |         hive: currentHive,
336 |         path: currentPath,
337 |         name: name,
338 |         create_backup: true
339 |       })
340 |     });
341 |
342 |     if (!res.ok) {
343 |       const errJson = await res.json().catch(() => ({}));
344 |       throw new Error(errJson.detail || `HTTP ${res.status}`
345 |     );
346 |     }
347 |     window.showToast?(`Параметр ${displayName} удален`, 'su
348 | ccess');
349 |     await loadKey(currentHive, currentPath);
350 |   } catch (e) {
351 |     window.showToast?(`Ошибка удаления параметра: ${e.messa
352 | ge}`, 'danger') || alert(`Ошибка удаления параметра: ${e.messa
353 | ge}`);
354 |   }
355 | }
356 |
357 | function openCreateKeyModal() {
358 |   const keyNameInput = document.getElementById('regModalNewK
359 | eyName');
360 |   if (keyNameInput) keyNameInput.value = '';
361 |
362 |   const modalEl = document.getElementById('regKeyModal');
363 |   if (modalEl && window.bootstrap) {
364 |     const modal = bootstrap.Modal.getOrCreateInstance(modalEl
365 | 1);
366 |     modal.show();
367 |   }
```

```
363 | }
364 |
365 | async function saveModalKey() {
366 |     const keyNameInput = document.getElementById('regModalNewKey
    eyName');
367 |     const newKeyName = keyNameInput ? keyNameInput.value.trim(
    ) : '';
368 |     if (!newKeyName) {
369 |         window.showToast?.( 'Введите имя подраздела', 'warning')
    || alert('Введите имя подраздела');
370 |         return;
371 |     }
372 |
373 |     const newPath = currentPath ? `${currentPath}\\${newKeyNam
    e}` : newKeyName;
374 |
375 |     try {
376 |         const res = await fetch('/api/registry/key', {
377 |             method: 'POST',
378 |             headers: { 'Content-Type': 'application/json' },
379 |             body: JSON.stringify({
380 |                 hive: currentHive,
381 |                 path: newPath
382 |             })
383 |         });
384 |
385 |         if (!res.ok) {
386 |             const errJson = await res.json().catch(() => ({}));
387 |             throw new Error(errJson.detail || `HTTP ${res.status}`
    );
388 |         }
389 |
390 |         const modalEl = document.getElementById('regKeyModal');
391 |         if (modalEl && window.bootstrap) {
392 |             const modal = bootstrap.Modal.getInstance(modalEl);
393 |             if (modal) modal.hide();
394 |         }
395 |
```

```

396     window.showToast?.(`Подраздел '${newKeyName}' успешно со
здан`, 'success');
397     await loadKey(currentHive, currentPath);
398   } catch (e) {
399     window.showToast?.(`Ошибка создания ключа: ${e.message}`
, 'danger') || alert(`Ошибка создания ключа: ${e.message}`);
400   }
401 }
402
403 async function deleteCurrentKey() {
404   if (!currentPath) {
405     window.showToast?.('Нельзя удалить корневой раздел', 'wa
rning') || alert('Нельзя удалить корневой раздел');
406     return;
407   }
408
409   if (!confirm(`ВНИМАНИЕ: Вы действительно хотите удалить ра
здел '${currentHive}\\${currentPath}' со всеми подразделами и
значениями? Будет создан снимок резервной копии.`)) {
410     return;
411   }
412
413   try {
414     const res = await fetch('/api/registry/key', {
415       method: 'DELETE',
416       headers: { 'Content-Type': 'application/json' },
417       body: JSON.stringify({
418         hive: currentHive,
419         path: currentPath,
420         recursive: true,
421         create_backup: true
422       })
423     });
424
425     if (!res.ok) {
426       const errJson = await res.json().catch(() => ({}));
427       throw new Error(errJson.detail || `HTTP ${res.status}`
);

```

```
428     }
429
430     window.showToast?.('Раздел реестра успешно удален', 'suc
    ccess');
431     goUp();
432     } catch (e) {
433         window.showToast?.(`Ошибка удаления раздела: ${e.message
    }`, 'danger') || alert(`Ошибка удаления раздела: ${e.message}`
    );
434     }
435 }
436
437 async function openBackupsModal() {
438     const modalEl = document.getElementById('regBackupsModal')
    ;
439     const tbody = document.getElementById('reg-backups-tbody')
    ;
440
441     if (modalEl && window.bootstrap) {
442         const modal = bootstrap.Modal.getOrCreateInstance(modalEl
    );
443         modal.show();
444     }
445
446     if (tbody) tbody.innerHTML = '<tr><td colspan="4" class="t
    ext-center py-3 text-muted">Загрузка истории снимков...</td></
    tr>';
447
448     try {
449         const res = await fetch('/api/registry/backups');
450         if (!res.ok) throw new Error('Ошибка загрузки бэкапов');
451         const data = await res.json();
452         const backups = data.backups || [];
453
454         if (backups.length === 0) {
455             tbody.innerHTML = '<tr><td colspan="4" class="text-cen
    ter py-4 text-muted">Точки восстановления отсутствуют</td></tr
    >';
```

```

456         return;
457     }
458
459     tbody.innerHTML = backups.map(b => `
460         <tr>
461             <td><small class="text-light">${new Date(b.timestamp
462             ).toLocaleString()}</small></td>
463             <td class="text-truncate" style="max-width: 250px;"
464             title="${b.hive}\\${b.path}">
465                 <span class="text-warning">${b.hive}</span>\\${esc
466             apeHtml(b.path)}
467             </td>
468             <td><span class="badge bg-secondary">${escapeHtml(b.
469             operation)}</span></td>
470             <td>
471                 <button class="btn btn-xs btn-outline-warning py-0
472                 px-2 btn-restore-backup" data-id="${b.backup_id}">
473                     <i class="bi bi-arrow-counterclockwise me-1"></i
474                 > Откатить
475             </button>
476             </td>
477         </tr>
478     `).join('');
479
480     tbody.querySelectorAll('.btn-restore-backup').forEach(bt
481     n => {
482         btn.onclick = async () => {
483             const backupId = btn.getAttribute('data-id');
484             if (!confirm(`Восстановить реестр из снимка ${backup
485             Id}?`)) return;
486
487             try {
488                 const rRes = await fetch(`/api/registry/restore?ba
489             ckup_id=${encodeURIComponent(backupId)}`, {
490                 method: 'POST'
491             });
492             if (!rRes.ok) {
493                 const err = await rRes.json().catch(() => ({}));

```

```

485 |             throw new Error(err.detail || `HTTP ${rRes.statu
| s}`);
486 |         }
487 |         window.showToast?(`Реестр успешно восстановлен из
| резервной копии!`, 'success');
488 |         await loadKey(currentHive, currentPath);
489 |     } catch (err) {
490 |         window.showToast?(`Ошибка восстановления: ${err.m
| essage}`, 'danger') || alert(`Ошибка восстановления: ${err.mes
| sage}`);
491 |     }
492 | };
493 | });
494 | } catch (e) {
495 |     if (tbody) tbody.innerHTML = `|<td colspan="4" class=
| "text-center py-3 text-danger">${e.message}</td></tr>`;
496 | }
497 | }
498 |
499 | function escapeHtml(text) {
500 |     return text
501 |         .replace(/&/g, "&amp;")
502 |         .replace(/</g, "&lt;")
503 |         .replace(/>/g, "&gt;")
504 |         .replace(/"/g, "&quot;")
505 |         .replace(/'/g, "&#039;");
506 | }
507 |
508 | function goUp() {
509 |     if (!currentPath) return;
510 |     const parts = currentPath.split('\\');
511 |     parts.pop();
512 |     const newPath = parts.join('\\');
513 |     navigateTo(currentHive, newPath);
514 | }
515 |
516 | function copyCurrentPath() {
517 |     const full = `${currentHive}\\${currentPath}`.replace(/\\+

|  |

```

```

    $/, '');
518     if (navigator.clipboard) {
519         navigator.clipboard.writeText(full).then(() => {
520             const btn = document.getElementById('btn-reg-copy-path
');
521             if (btn) {
522                 const orig = btn.innerHTML;
523                 btn.innerHTML = '<i class="bi bi-check me-1"></i> Ск
опировано!';
524                 setTimeout(() => { btn.innerHTML = orig; }, 1500);
525             }
526         });
527     }
528 }
529
530 window.initRegistryViewerTab = async function() {
531     console.log('[RegistryViewerTab] Initializing Registry Vie
wer & Editor tab...');
532
533     const hiveSelect = document.getElementById('reg-hive-selec
t');
534     const pathInput = document.getElementById('reg-path-input'
);
535     const goBtn = document.getElementById('btn-reg-go');
536     const upBtn = document.getElementById('btn-reg-up');
537     const filterInput = document.getElementById('reg-filter-in
put');
538     const refreshBtn = document.getElementById('btn-reg-refres
h');
539     const copyBtn = document.getElementById('btn-reg-copy-path
');
540     const backupsBtn = document.getElementById('btn-reg-backup
s-modal');
541     const addValBtn = document.getElementById('btn-reg-add-val
ue');
542     const addKeyBtn = document.getElementById('btn-reg-add-key
');
543     const delKeyBtn = document.getElementById('btn-reg-delete-

```

```

current-key');
544     const saveValModalBtn = document.getElementById('btn-reg-m
odal-save');
545     const saveKeyModalBtn = document.getElementById('btn-reg-m
odal-create-key');
546
547     if (hiveSelect) {
548         hiveSelect.onChange = () => {
549             navigateTo(hiveSelect.value, '');
550         };
551     }
552     if (goBtn && pathInput) {
553         goBtn.onclick = () => {
554             navigateTo(hiveSelect.value, pathInput.value);
555         };
556     }
557     if (pathInput) {
558         pathInput.onkeydown = (e) => {
559             if (e.key === 'Enter') {
560                 navigateTo(hiveSelect.value, pathInput.value);
561             }
562         };
563     }
564     if (upBtn) upBtn.onclick = goUp;
565     if (filterInput) filterInput.oninput = renderValues;
566     if (refreshBtn) refreshBtn.onclick = () => loadKey(current
Hive, currentPath);
567     if (copyBtn) copyBtn.onclick = copyCurrentPath;
568     if (backupsBtn) backupsBtn.onclick = openBackupsModal;
569     if (addValBtn) addValBtn.onclick = openCreateValueModal;
570     if (addKeyBtn) addKeyBtn.onclick = openCreateKeyModal;
571     if (delKeyBtn) delKeyBtn.onclick = deleteCurrentKey;
572     if (saveValModalBtn) saveValModalBtn.onclick = saveModalVa
lue;
573     if (saveKeyModalBtn) saveKeyModalBtn.onclick = saveModalKe
y;
574
575     await loadBookmarks();

```



```
576 |     await navigateTo(currentHive, currentPath);
577 |     isInitialized = true;
578 | };
579 |})();
580 |
581 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0, maximum-scale=1.0, user-scalable=no">
7   <title>Голосовой Пульт - AI Breadboard</title>
8
9   <!-- Bootstrap 5 CSS -->
10  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
11  t/css/bootstrap.min.css" rel="stylesheet">
12
13  <!-- Bootstrap Icons -->
14  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
15  0.0/font/bootstrap-icons.css" rel="stylesheet">
16
17  <!-- Telegram Web Apps SDK & Marked.js -->
18  <script src="https://telegram.org/js/telegram-web-app.js"></
19  script>
20  <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
21  js"></script>
22
23  <style>
24    :root {
25      --tg-theme-bg-color: #0b0f19;
26      --tg-theme-text-color: #f1f5f9;
27      --tg-theme-hint-color: #94a3b8;
28      --tg-theme-secondary-bg-color: #151c2e;
29      --primary-gradient: linear-gradient(135deg, #3b82f6 0%,
30      #1d4ed8 100%);
31      --mic-idle-gradient: linear-gradient(135deg, #ef4444 0%,
32      #b91c1c 100%);
33      --mic-listening-gradient: linear-gradient(135deg, #10b98
34      1 0%, #047857 100%);
35      --mic-thinking-gradient: linear-gradient(135deg, #8b5cf6
36      0%, #6d28d9 100%);
37      --accent-color: #38bdf8;
```

```
28 |     }
29 |
30 |     body {
31 |         background: var(--tg-theme-bg-color);
32 |         color: var(--tg-theme-text-color);
33 |         font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
34 |         min-height: 100vh;
35 |         display: flex;
36 |         flex-direction: column;
37 |         justify-content: flex-start;
38 |         align-items: center;
39 |         padding: 12px;
40 |         margin: 0;
41 |         overflow-x: hidden;
42 |     }
43 |
44 |     .app-container {
45 |         width: 100%;
46 |         max-width: 520px;
47 |         display: flex;
48 |         flex-direction: column;
49 |         height: 100%;
50 |         min-height: 94vh;
51 |     }
52 |
53 |     .card-panel {
54 |         background-color: var(--tg-theme-secondary-bg-color);
55 |         border: 1px solid rgba(255, 255, 255, 0.08);
56 |         border-radius: 20px;
57 |         padding: 18px;
58 |         box-shadow: 0 10px 30px rgba(0, 0, 0, 0.4);
59 |         backdrop-filter: blur(12px);
60 |     }
61 |
62 |     /* Giant Mic Button */
63 |     .mic-centerpiece {
64 |         display: flex;
```

```
65 |     flex-direction: column;
66 |     align-items: center;
67 |     justify-content: center;
68 |     padding: 16px 0 12px;
69 |     position: relative;
70 | }
71 |
72 | .mic-button-wrapper {
73 |     position: relative;
74 |     display: flex;
75 |     align-items: center;
76 |     justify-content: center;
77 | }
78 |
79 | .btn-mic-giant {
80 |     width: 110px;
81 |     height: 110px;
82 |     border-radius: 50%;
83 |     border: none;
84 |     outline: none;
85 |     background: var(--mic-idle-gradient);
86 |     color: #ffffff;
87 |     display: flex;
88 |     align-items: center;
89 |     justify-content: center;
90 |     font-size: 44px;
91 |     box-shadow: 0 8px 25px rgba(239, 68, 68, 0.4);
92 |     cursor: pointer;
93 |     transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
94 |     position: relative;
95 |     z-index: 2;
96 | }
97 |
98 | .btn-mic-giant:hover {
99 |     transform: scale(1.05);
100 |     box-shadow: 0 12px 30px rgba(239, 68, 68, 0.6);
101 | }
102 |
```

```
103 | .btn-mic-giant:active {
104 |     transform: scale(0.95);
105 | }
106 |
107 | .btn-mic-giant.listening {
108 |     background: var(--mic-listening-gradient);
109 |     box-shadow: 0 8px 30px rgba(16, 185, 129, 0.5);
110 |     animation: mic-listening-pulse 1.8s infinite;
111 | }
112 |
113 | .btn-mic-giant.thinking {
114 |     background: var(--mic-thinking-gradient);
115 |     box-shadow: 0 8px 30px rgba(139, 92, 246, 0.5);
116 |     animation: mic-thinking-pulse 1.2s infinite ease-in-out;
117 | }
118 |
119 | @keyframes mic-listening-pulse {
120 |     0% { transform: scale(1); box-shadow: 0 0 0 0 rgba(16, 1
121 | 85, 129, 0.7); }
122 |     70% { transform: scale(1.06); box-shadow: 0 0 0 22px rgb
123 | a(16, 185, 129, 0); }
124 |     100% { transform: scale(1); box-shadow: 0 0 0 0 rgba(16,
125 | 185, 129, 0); }
126 | }
127 |
128 | @keyframes mic-thinking-pulse {
129 |     0% { transform: rotate(0deg) scale(1); }
130 |     50% { transform: rotate(180deg) scale(1.05); }
131 |     100% { transform: rotate(360deg) scale(1); }
132 | }
133 |
134 | /* Wave visualizer dots */
135 | .wave-bars {
136 |     display: flex;
137 |     align-items: center;
138 |     justify-content: center;
139 |     gap: 5px;
140 |     height: 20px;
```

```
138     margin-top: 8px;
139 }
140
141 .wave-bar {
142     width: 4px;
143     height: 6px;
144     background: #ef4444;
145     border-radius: 2px;
146     transition: height 0.15s ease;
147 }
148
149 .wave-bars.active .wave-bar {
150     background: #10b981;
151     animation: wave-anim 0.8s ease-in-out infinite alternate
152 ;
153 }
154 .wave-bars.active .wave-bar:nth-child(2) { animation-delay
155 : 0.15s; }
156 .wave-bars.active .wave-bar:nth-child(3) { animation-delay
157 : 0.3s; }
158 .wave-bars.active .wave-bar:nth-child(4) { animation-delay
159 : 0.45s; }
160 .wave-bars.active .wave-bar:nth-child(5) { animation-delay
161 : 0.2s; }
162
163
164 @keyframes wave-anim {
165     0% { height: 6px; }
166     100% { height: 20px; }
167 }
168
169 /* Media Actions Bar */
170 .media-actions-bar {
171     display: flex;
172     align-items: center;
173     justify-content: center;
174     gap: 8px;
175     margin-top: 10px;
```

```
171 |     width: 100%;
172 | }
173 |
174 | .btn-media-action {
175 |     background: rgba(255, 255, 255, 0.06);
176 |     border: 1px solid rgba(255, 255, 255, 0.12);
177 |     color: #e2e8f0;
178 |     font-size: 0.82rem;
179 |     font-weight: 500;
180 |     padding: 6px 14px;
181 |     border-radius: 20px;
182 |     display: inline-flex;
183 |     align-items: center;
184 |     gap: 6px;
185 |     transition: all 0.2s ease;
186 |     cursor: pointer;
187 | }
188 |
189 | .btn-media-action:hover {
190 |     background: rgba(255, 255, 255, 0.14);
191 |     color: #ffffff;
192 |     border-color: rgba(255, 255, 255, 0.25);
193 | }
194 |
195 | .btn-media-action.active-audio {
196 |     border-color: #38bdf8;
197 |     color: #38bdf8;
198 |     background: rgba(56, 189, 248, 0.1);
199 | }
200 |
201 | .btn-media-action.active-media {
202 |     border-color: #a855f7;
203 |     color: #c084fc;
204 |     background: rgba(168, 85, 247, 0.1);
205 | }
206 |
207 | /* Chat Messages Container */
208 | .chat-container {
```

```
209 |     flex: 1;
210 |     overflow-y: auto;
211 |     border-radius: 14px;
212 |     background: rgba(0, 0, 0, 0.2);
213 |     border: 1px solid rgba(255, 255, 255, 0.05);
214 |     padding: 12px;
215 |     display: flex;
216 |     flex-direction: column;
217 |     gap: 10px;
218 |     max-height: 44vh;
219 |     min-height: 180px;
220 |     scroll-behavior: smooth;
221 | }
222 |
223 | .chat-container::-webkit-scrollbar {
224 |     width: 5px;
225 | }
226 | .chat-container::-webkit-scrollbar-thumb {
227 |     background: rgba(255, 255, 255, 0.15);
228 |     border-radius: 10px;
229 | }
230 |
231 | .chat-bubble {
232 |     max-width: 90%;
233 |     padding: 10px 14px;
234 |     border-radius: 14px;
235 |     font-size: 0.92rem;
236 |     line-height: 1.45;
237 |     word-break: break-word;
238 |     position: relative;
239 | }
240 |
241 | .chat-bubble.user {
242 |     align-self: flex-end;
243 |     background: var(--primary-gradient);
244 |     color: #ffffff;
245 |     border-bottom-right-radius: 4px;
246 | }
```



```
247 |
248 | .chat-bubble.ai {
249 |     align-self: flex-start;
250 |     background: #1e293b;
251 |     color: #f1f5f9;
252 |     border: 1px solid rgba(255, 255, 255, 0.08);
253 |     border-bottom-left-radius: 4px;
254 | }
255 |
256 | .chat-bubble.ai p:last-child {
257 |     margin-bottom: 0;
258 | }
259 |
260 | .chat-meta {
261 |     display: flex;
262 |     align-items: center;
263 |     justify-content: space-between;
264 |     gap: 8px;
265 |     margin-top: 4px;
266 |     font-size: 0.72rem;
267 |     opacity: 0.75;
268 | }
269 |
270 | .btn-speak-again {
271 |     background: none;
272 |     border: none;
273 |     color: inherit;
274 |     padding: 0;
275 |     cursor: pointer;
276 |     font-size: 0.85rem;
277 | }
278 |
279 | /* Media Attachment Previews */
280 | .media-preview-card {
281 |     background: rgba(0, 0, 0, 0.25);
282 |     border: 1px solid rgba(255, 255, 255, 0.1);
283 |     border-radius: 10px;
284 |     padding: 8px 10px;
```

```
285 |     margin-bottom: 6px;
286 |     display: flex;
287 |     align-items: center;
288 |     gap: 8px;
289 | }
290 |
291 | .media-player-widget {
292 |     width: 100%;
293 |     margin-top: 6px;
294 |     border-radius: 8px;
295 |     background: rgba(0, 0, 0, 0.2);
296 | }
297 |
298 | .media-thumb-img {
299 |     max-width: 100%;
300 |     max-height: 200px;
301 |     border-radius: 8px;
302 |     margin-top: 6px;
303 |     object-fit: cover;
304 | }
305 |
306 | /* Language and Theme Switchers */
307 | .btn-lang-pill {
308 |     font-size: 0.75rem;
309 |     font-weight: 600;
310 |     padding: 3px 9px;
311 |     border-radius: 20px;
312 | }
313 |
314 | .theme-toggle-btn {
315 |     background: rgba(255, 255, 255, 0.08);
316 |     border: 1px solid rgba(255, 255, 255, 0.15);
317 |     color: #ffffff;
318 |     width: 34px;
319 |     height: 34px;
320 |     border-radius: 50%;
321 |     display: flex;
322 |     align-items: center;
```

```
323 |     justify-content: center;
324 |     cursor: pointer;
325 |     transition: all 0.2s;
326 | }
327 | .theme-toggle-btn:hover {
328 |     background: rgba(255, 255, 255, 0.18);
329 | }
330 |
331 | /* Input text area fallback */
332 | .form-control-custom {
333 |     background: rgba(0, 0, 0, 0.25);
334 |     border: 1px solid rgba(255, 255, 255, 0.12);
335 |     color: #ffffff;
336 |     border-radius: 12px;
337 | }
338 | .form-control-custom:focus {
339 |     background: rgba(0, 0, 0, 0.35);
340 |     border-color: #3b82f6;
341 |     color: #ffffff;
342 |     box-shadow: 0 0 0 2px rgba(59, 130, 246, 0.3);
343 | }
344 |
345 | /* Light Theme */
346 | body.theme-light {
347 |     --tg-theme-bg-color: #f8fafc;
348 |     --tg-theme-text-color: #0f172a;
349 |     --tg-theme-hint-color: #64748b;
350 |     --tg-theme-secondary-bg-color: #ffffff;
351 | }
352 | body.theme-light .card-panel {
353 |     border-color: #e2e8f0;
354 |     box-shadow: 0 8px 24px rgba(0, 0, 0, 0.06);
355 | }
356 | body.theme-light .chat-container {
357 |     background: #f1f5f9;
358 |     border-color: #e2e8f0;
359 | }
360 | body.theme-light .chat-bubble.ai {
```

```

361     background: #ffffff;
362     color: #0f172a;
363     border-color: #e2e8f0;
364 }
365 body.theme-light .theme-toggle-btn {
366     background: #e2e8f0;
367     color: #0f172a;
368     border-color: #cbd5e1;
369 }
370 body.theme-light .btn-media-action {
371     background: #f1f5f9;
372     color: #334155;
373     border-color: #cbd5e1;
374 }
375 body.theme-light .form-control-custom {
376     background: #ffffff;
377     color: #0f172a;
378     border-color: #cbd5e1;
379 }
380 </style>
381 </head>
382 <body>
383
384 <!-- AUTHENTICATION PANEL -->
385 <div class="app-container" id="auth-panel">
386     <div class="card-panel my-auto">
387         <div class="text-center mb-4">
388             <i class="bi bi-mic-fill fs-1 text-primary"></i>
389             <h4 class="mt-2 text-white fw-bold">Голосовой Пульт</h
4>
390             <p class="text-muted small">Войдите в систему для пере
дачи голосовых сообщений и медиа в чат</p>
391         </div>
392
393         <div id="auth-alert" class="alert alert-danger d-none" r
ole="alert"></div>
394
395         <!-- Login Form -->

```

```
396 |         <div id="auth-form-login">
397 |             <form id="form-login">
398 |                 <div class="mb-3">
399 |                     <label for="login-email" class="form-label text-mu
400 | ted small">Email</label>
401 |                     <input type="email" class="form-control form-contr
402 | ol-custom" id="login-email" required>
403 |                 </div>
404 |                 <div class="mb-3">
405 |                     <label for="login-password" class="form-label text
406 | -muted small">Пароль</label>
407 |                     <input type="password" class="form-control form-co
408 | ntrol-custom" id="login-password" required>
409 |                 </div>
410 |                 <button type="submit" class="btn btn-primary w-100 p
411 | y-2.5 rounded-pill mb-3">Войти</button>
412 |                 <p class="text-center small mb-0">Нет аккаунта? <a h
413 | ref="#" id="go-to-register" class="text-info">Регистрация</a><
414 | /p>
415 |             </form>
416 |         </div>
417 |
418 |         <!-- Register Form -->
419 |         <div id="auth-form-register" class="d-none">
420 |             <form id="form-register">
421 |                 <div class="mb-3">
422 |                     <label for="register-name" class="form-label text-
423 | muted small">Имя</label>
424 |                     <input type="text" class="form-control form-contro
425 | l-custom" id="register-name" required>
426 |                 </div>
427 |                 <div class="mb-3">
428 |                     <label for="register-email" class="form-label text
429 | -muted small">Email</label>
430 |                     <input type="email" class="form-control form-contr
431 | ol-custom" id="register-email" required>
432 |                 </div>
433 |                 <div class="mb-3">
```

```
423 |         <label for="register-password" class="form-label text-muted small">Пароль</label>
424 |         <input type="password" class="form-control form-control-custom" id="register-password" required>
425 |     </div>
426 |     <button type="submit" class="btn btn-success w-100 py-2.5 rounded-pill mb-3">Зарегистрироваться</button>
427 |     <p class="text-center small mb-0">Уже есть аккаунт?
428 | <a href="#" id="go-to-login" class="text-info">Войти</a></p>
429 | </form>
430 | </div>
431 | <!-- Verify Form -->
432 | <div id="auth-form-verify" class="d-none">
433 |     <p class="text-center small">Код подтверждения отправлен на почту:</p>
434 |     <form id="form-verify">
435 |         <div class="mb-3">
436 |             <input type="text" class="form-control form-control-custom text-center fs-3 py-2" id="verify-code" maxlength="6"
437 | placeholder="XXXXXX" required>
438 |         </div>
439 |         <button type="submit" class="btn btn-info w-100 py-2.5 text-white rounded-pill mb-3">Подтвердить</button>
440 |         <p class="text-center small mb-0"><a href="#" id="verify-cancel" class="text-info">Назад</a></p>
441 |     </form>
442 | </div>
443 | <div class="my-4 text-center position-relative">
444 |     <hr class="border-secondary">
445 |     <span class="px-2 text-muted position-absolute top-50 start-50 translate-middle small bg-dark">Или</span>
446 | </div>
447 |
448 |     <button id="google-signin-btn" class="btn btn-outline-light w-100 rounded-pill d-flex align-items-center justify-content-center gap-2 py-2">
```

```
449 |         <i class="bi bi-google text-danger"></i>
450 |         <span>Войти через Google</span>
451 |     </button>
452 | </div>
453 | </div>
454 |
455 | <!-- MAIN VOICE REMOTE PANEL -->
456 | <div class="app-container d-none" id="main-panel">
457 |     <div class="card-panel d-flex flex-column h-100">
458 |
459 |         <!-- Top Bar -->
460 |         <div class="d-flex justify-content-between align-items-c
461 | enter mb-2">
462 |             <div class="d-flex align-items-center gap-2">
463 |                 <span class="badge bg-success rounded-pill px-2.5 py
464 | -1.5" id="connection-status">
465 |                     <i class="bi bi-broadcast"></i> Онлайн
466 |                 </span>
467 |                 <span class="small text-muted text-truncate" style="
468 | max-width: 120px;" id="user-display-name"></span>
469 |             </div>
470 |
471 |             <div class="d-flex align-items-center gap-2">
472 |                 <!-- Language Selector -->
473 |                 <div class="btn-group btn-group-sm" role="group">
474 |                     <input type="radio" class="btn-check" name="lang-r
475 | adio" id="lang-ru" value="ru-RU" checked>
476 |                     <label class="btn btn-outline-secondary btn-lang-p
477 | ill" for="lang-ru">RU</label>
478 |
479 |                     <input type="radio" class="btn-check" name="lang-r
480 | adio" id="lang-en" value="en-US">
481 |                     <label class="btn btn-outline-secondary btn-lang-p
482 | ill" for="lang-en">EN</label>
483 |
484 |                     <input type="radio" class="btn-check" name="lang-r
485 | adio" id="lang-he" value="he-IL">
486 |                     <label class="btn btn-outline-secondary btn-lang-p
```

```

    ill" for="lang-he">HE</label>
479         </div>
480
481         <!-- TTS Mute/Unmute Toggle -->
482         <button class="theme-toggle-btn" id="tts-toggle-btn"
    title="Озвучивание ответов">
483             <i class="bi bi-volume-up-fill" id="tts-icon"></i>
484         </button>
485
486         <!-- Theme Toggle -->
487         <button class="theme-toggle-btn" id="theme-toggle-bt
    n" title="Сменить тему">
488             <i class="bi bi-moon-stars-fill" id="theme-icon"><
    /i>
489         </button>
490     </div>
491 </div>
492
493 <!-- Big Microphone Centerpiece -->
494 <div class="mic-centerpiece">
495     <div class="mic-button-wrapper">
496         <button class="btn-mic-giant" id="btn-mic" title="На
    жмите для голосового ввода">
497             <i class="bi bi-mic-fill" id="mic-icon"></i>
498         </button>
499     </div>
500
501     <div class="wave-bars" id="wave-bars">
502         <div class="wave-bar"></div>
503         <div class="wave-bar"></div>
504         <div class="wave-bar"></div>
505         <div class="wave-bar"></div>
506         <div class="wave-bar"></div>
507     </div>
508
509     <div class="fw-semibold mt-2 text-center" id="voice-st
    atus-text" style="font-size: 0.95rem;">
510         Нажмите кнопку и говорите

```



```

511         </div>
512         <div class="small text-muted text-center" id="voice-su
btext" style="font-size: 0.78rem;">
513             Голос будет распознан и отправлен в чат
514         </div>
515
516         <!-- Quick Media & Audio Upload Actions -->
517         <div class="media-actions-bar">
518             <button class="btn-media-action" id="btn-upload-audi
o" title="Загрузить аудиозапись или диктофон">
519                 <i class="bi bi-file-earmark-music-fill text-info"
></i>
520                 <span>Аудиофайл</span>
521             </button>
522             <button class="btn-media-action" id="btn-upload-medi
a" title="Загрузить фото, видео или документ">
523                 <i class="bi bi-cloud-arrow-up-fill text-primary">
</i>
524                 <span>Медиа</span>
525             </button>
526         </div>
527
528         <!-- Hidden File Inputs -->
529         <input type="file" id="audio-file-input" accept="audio
/*,.mp3,.wav,.ogg,.m4a,.flac,.aac,.webm,.opus" class="d-none">
530         <input type="file" id="media-file-input" accept="image
/*,video/*,audio/*,.pdf,.doc,.docx,.txt" class="d-none">
531     </div>
532
533     <!-- Chat Stream Feed -->
534     <div class="d-flex justify-content-between align-items-c
enter mt-1 mb-1 px-1">
535         <span class="small text-muted fw-bold text-uppercase"
style="letter-spacing: 0.5px; font-size: 0.75rem;">Диалог</spa
n>
536         <button class="btn btn-sm btn-link text-muted p-0 text
-decoration-none small" id="btn-clear-chat" title="Очистить ди
алог">

```

```
537 |         <i class="bi bi-trash"></i> Очистить
538 |     </button>
539 | </div>
540 |
541 |     <div class="chat-container" id="chat-feed">
542 |         <div class="text-muted small text-center my-auto p-3"
543 | id="empty-feed-hint">
544 |             <i class="bi bi-chat-dots fs-3 d-block mb-1 opacity-
545 | 50"></i>
546 |             Голосовые команды и медиафайлы отобразятся здесь
547 |         </div>
548 |     </div>
549 |
550 |     <!-- Fallback Quick Text Bar & Attach Clip -->
551 |     <div class="mt-2">
552 |         <form id="text-send-form" class="d-flex gap-2">
553 |             <button type="button" class="btn btn-outline-seconda
554 | ry btn-sm px-2.5 rounded-3 d-flex align-items-center justify-c
555 | ontent-center" id="btn-clip-attachment" title="Прикрепить файл
556 | ">
557 |                 <i class="bi bi-paperclip fs-6"></i>
558 |             </button>
559 |             <input type="text" class="form-control form-control-
560 | custom form-control-sm py-2" id="text-input" placeholder="Или
561 | введите сообщение текстом..." autocomplete="off">
562 |             <button type="submit" class="btn btn-primary btn-sm
563 | px-3 rounded-3" id="btn-send-text" title="Отправить">
564 |                 <i class="bi bi-send-fill"></i>
565 |             </button>
566 |         </form>
567 |     </div>
568 | </div>
569 |
570 | <script>
571 |     // Global State
572 |     let recognition = null;
```

```
567 |     let isListening = false;
568 |     let isTtsEnabled = true;
569 |     let chatHistory = [];
570 |     let abortController = null;
571 |     let currentSpeechAudio = null;
572 |     let currentSpeechQueue = null;
573 |
574 |     // Elements
575 |     const authPanel = document.getElementById('auth-panel');
576 |     const mainPanel = document.getElementById('main-panel');
577 |     const authAlert = document.getElementById('auth-alert');
578 |     const authFormLogin = document.getElementById('auth-form-l
    ogin');
579 |     const authFormRegister = document.getElementById('auth-for
    m-register');
580 |     const authFormVerify = document.getElementById('auth-form-
    verify');
581 |
582 |     const btnMic = document.getElementById('btn-mic');
583 |     const micIcon = document.getElementById('mic-icon');
584 |     const voiceStatusText = document.getElementById('voice-sta
    tus-text');
585 |     const voiceSubtext = document.getElementById('voice-subtex
    t');
586 |     const waveBars = document.getElementById('wave-bars');
587 |     const chatFeed = document.getElementById('chat-feed');
588 |     const emptyFeedHint = document.getElementById('empty-feed-
    hint');
589 |     const userDisplayName = document.getElementById('user-disp
    lay-name');
590 |
591 |     const btnUploadAudio = document.getElementById('btn-upload
    -audio');
592 |     const btnUploadMedia = document.getElementById('btn-upload
    -media');
593 |     const btnClipAttachment = document.getElementById('btn-cli
    p-attachment');
594 |     const audioFileInput = document.getElementById('audio-file
```

```

    -input');
595     const mediaFileInput = document.getElementById('media-file
    -input');
596
597     const textSendForm = document.getElementById('text-send-fo
    rm');
598     const textInput = document.getElementById('text-input');
599     const ttsToggleBtn = document.getElementById('tts-toggle-b
    tn');
600     const ttsIcon = document.getElementById('tts-icon');
601     const btnClearChat = document.getElementById('btn-clear-ch
    at');
602
603     // Telegram Web App Integration
604     function initTelegram() {
605         if (window.Telegram && window.Telegram.WebApp) {
606             const tg = window.Telegram.WebApp;
607             tg.ready();
608             tg.expand();
609
610             window.triggerHaptic = function(type = 'light') {
611                 if (tg.HapticFeedback) {
612                     tg.HapticFeedback.impactOccurred(type);
613                 }
614             };
615         } else {
616             window.triggerHaptic = function() {};
617         }
618     }
619
620     // Auth API
621     const api = {
622         async fetch(url, options = {}) {
623             const res = await fetch(url, options);
624             if (!res.ok) {
625                 let msg = res.statusText;
626                 try {
627                     const data = await res.json();

```

```
628         msg = data.detail || msg;
629     } catch {}
630     throw new Error(msg);
631 }
632 return res.json();
633 },
634 auth: {
635     async check() {
636         try {
637             const res = await fetch('/auth/check');
638             return await res.json();
639         } catch {
640             return { authenticated: false };
641         }
642     },
643     async login(email, password) {
644         return api.fetch('/auth/login/email', {
645             method: 'POST',
646             headers: { 'Content-Type': 'application/json' },
647             body: JSON.stringify({ email, password })
648         });
649     },
650     async register(email, password, name) {
651         return api.fetch('/auth/register', {
652             method: 'POST',
653             headers: { 'Content-Type': 'application/json' },
654             body: JSON.stringify({ email, password, name })
655         });
656     },
657     async verify(email, code) {
658         return api.fetch('/auth/verify', {
659             method: 'POST',
660             headers: { 'Content-Type': 'application/json' },
661             body: JSON.stringify({ email, code })
662         });
663     }
664 }
665 };
```

```
666 |
667 | // Panel Switching
668 | function showPanel(panel) {
669 |     if (panel === 'auth') {
670 |         authPanel.classList.remove('d-none');
671 |         mainPanel.classList.add('d-none');
672 |     } else {
673 |         authPanel.classList.add('d-none');
674 |         mainPanel.classList.remove('d-none');
675 |     }
676 | }
677 |
678 | // Auth Handlers
679 | document.getElementById('go-to-register').addEventListener
('click', (e) => {
680 |     e.preventDefault();
681 |     authAlert.classList.add('d-none');
682 |     authFormLogin.classList.add('d-none');
683 |     authFormRegister.classList.remove('d-none');
684 | });
685 |
686 | document.getElementById('go-to-login').addEventListener('c
lick', (e) => {
687 |     e.preventDefault();
688 |     authAlert.classList.add('d-none');
689 |     authFormRegister.classList.add('d-none');
690 |     authFormLogin.classList.remove('d-none');
691 | });
692 |
693 | let currentVerifyEmail = '';
694 | document.getElementById('form-register').addEventListener(
'submit', async (e) => {
695 |     e.preventDefault();
696 |     authAlert.classList.add('d-none');
697 |     const name = document.getElementById('register-name').va
lue;
698 |     const email = document.getElementById('register-email').
value;
```

```
699     const password = document.getElementById('register-passw
ord').value;
700     try {
701         await api.auth.register(email, password, name);
702         currentVerifyEmail = email;
703         authFormRegister.classList.add('d-none');
704         authFormVerify.classList.remove('d-none');
705     } catch (err) {
706         authAlert.textContent = err.message || 'Ошибка регистр
ации';
707         authAlert.classList.remove('d-none');
708     }
709 });
710
711     document.getElementById('form-verify').addEventListener('s
ubmit', async (e) => {
712         e.preventDefault();
713         authAlert.classList.add('d-none');
714         const code = document.getElementById('verify-code').valu
e;
715         try {
716             await api.auth.verify(currentVerifyEmail, code);
717             authFormVerify.classList.add('d-none');
718             authFormLogin.classList.remove('d-none');
719             authAlert.className = 'alert alert-success';
720             authAlert.textContent = 'Почта подтверждена! Войдите в
аккаунт.';
721             authAlert.classList.remove('d-none');
722         } catch (err) {
723             authAlert.textContent = err.message || 'Неверный код';
724             authAlert.classList.remove('d-none');
725         }
726     });
727
728     document.getElementById('form-login').addEventListener('su
bmit', async (e) => {
729         e.preventDefault();
730         authAlert.classList.add('d-none');
```

```
731 |     const email = document.getElementById('login-email').value;
732 |     const password = document.getElementById('login-password').value;
733 |     try {
734 |         await api.auth.login(email, password);
735 |         await initApp();
736 |     } catch (err) {
737 |         authAlert.textContent = err.message || 'Ошибка входа';
738 |         authAlert.classList.remove('d-none');
739 |     }
740 | });
741 |
742 | document.getElementById('google-signin-btn').addEventListener('click', () => {
743 |     window.location.href = '/auth/google?next=' + encodeURIComponent(window.location.pathname);
744 | });
745 |
746 | // Voice & Speech Recognition Module
747 | function initSpeechRecognition() {
748 |     const SpeechRecognition = window.SpeechRecognition || window.webkitSpeechRecognition;
749 |     if (!SpeechRecognition) {
750 |         voiceStatusText.textContent = 'Голосовой ввод не поддерживается';
751 |         voiceSubtext.textContent = 'Используйте загрузку аудио файлов или ввод ниже';
752 |         btnMic.disabled = true;
753 |         btnMic.style.opacity = '0.5';
754 |         return;
755 |     }
756 |
757 |     recognition = new SpeechRecognition();
758 |     recognition.continuous = false;
759 |     recognition.interimResults = false;
760 |
761 |     const getSelectedLang = () => {
```



```
762 |         const checked = document.querySelector('input[name="lang-radio"]:checked');
763 |         return checked ? checked.value : 'ru-RU';
764 |     };
765 |
766 |     recognition.lang = getSelectedLang();
767 |     document.querySelectorAll('input[name="lang-radio"]').forEach(radio => {
768 |         radio.addEventListener('change', () => {
769 |             recognition.lang = getSelectedLang();
770 |         });
771 |     });
772 |
773 |     recognition.onstart = () => {
774 |         isListening = true;
775 |         btnMic.className = 'btn-mic-giant listening';
776 |         micIcon.className = 'bi bi-mic-fill';
777 |         waveBars.classList.add('active');
778 |         voiceStatusText.textContent = 'Слушаю вас...';
779 |         voiceSubtext.textContent = 'Говорите команду или вопрос';
780 |     };
781 |
782 |     recognition.onend = () => {
783 |         isListening = false;
784 |         waveBars.classList.remove('active');
785 |         if (!btnMic.classList.contains('thinking')) {
786 |             btnMic.className = 'btn-mic-giant';
787 |             micIcon.className = 'bi bi-mic-fill';
788 |             voiceStatusText.textContent = 'Нажмите кнопку и говорите';
789 |             voiceSubtext.textContent = 'Голос будет распознан и отправлен в чат';
790 |         }
791 |     };
792 |
793 |     recognition.onerror = (e) => {
794 |         isListening = false;
```

```
795 |         waveBars.classList.remove('active');
796 |         btnMic.className = 'btn-mic-giant';
797 |         voiceStatusText.textContent = 'Ошибка ввода: ' + (e.error || 'попробуйте снова');
798 |         voiceSubtext.textContent = 'Нажмите микрофон для повторной попытки';
799 |     };
800 |
801 |     recognition.onresult = (event) => {
802 |         const transcript = event.results[0][0].transcript;
803 |         if (transcript && transcript.trim()) {
804 |             handleUserMessage(transcript.trim());
805 |         }
806 |     };
807 |
808 |     btnMic.addEventListener('click', () => {
809 |         window.triggerHaptic('medium');
810 |         stopAudio();
811 |
812 |         // Unlock audio context for mobile iOS/Safari/Chrome
813 |         if ('speechSynthesis' in window) {
814 |             const silent = new SpeechSynthesisUtterance(' ');
815 |             silent.volume = 0;
816 |             window.speechSynthesis.speak(silent);
817 |         }
818 |
819 |         if (isListening) {
820 |             recognition.stop();
821 |         } else {
822 |             try {
823 |                 recognition.start();
824 |             } catch (e) {
825 |                 console.warn('Recognition start error:', e);
826 |             }
827 |         }
828 |     });
829 | }
830 |
```

```

831 // Audio Queue Player for Backend TTS
832 class AudioQueuePlayer {
833     constructor(chunks) {
834         this.chunks = chunks;
835         this.currentIndex = 0;
836         this.isPlaying = false;
837         this.audioElements = new Map();
838         this.activeAudio = null;
839         this._prefetch(0);
840         this._prefetch(1);
841     }
842
843     _prefetch(index) {
844         if (index >= this.chunks.length || this.audioElements.
has(index)) return;
845         const text = this.chunks[index];
846         const queryParams = new URLSearchParams({ text });
847         const audioUrl = `/api/tts/synthesize?${queryParams.to
String()}`;
848         const audio = new Audio(audioUrl);
849         audio.load();
850         this.audioElements.set(index, audio);
851     }
852
853     async play() {
854         this.isPlaying = true;
855         while (this.currentIndex < this.chunks.length && this.
isPlaying) {
856             this._prefetch(this.currentIndex + 1);
857             this._prefetch(this.currentIndex + 2);
858
859             const audio = this.audioElements.get(this.currentInd
ex);
860             if (!audio) {
861                 this.currentIndex++;
862                 continue;
863             }
864

```

```
865         this.activeAudio = audio;
866         currentSpeechAudio = audio;
867
868         const playPromise = new Promise((resolve) => {
869             audio.onended = () => resolve();
870             audio.onerror = () => {
871                 this.stop();
872                 resolve();
873             };
874         });
875
876         try {
877             await audio.play();
878             await playPromise;
879         } catch (err) {
880             console.error('Audio playback interrupted:', err);
881         }
882         this.currentIndex++;
883     }
884     this.isPlaying = false;
885 }
886
887 stop() {
888     this.isPlaying = false;
889     if (this.activeAudio) {
890         try {
891             this.activeAudio.pause();
892             this.activeAudio.src = '';
893         } catch (e) {}
894     }
895     this.audioElements.clear();
896 }
897
898
899 function stopAudio() {
900     if (currentSpeechQueue) {
901         try { currentSpeechQueue.stop(); } catch (e) {}
902         currentSpeechQueue = null;
```

```

903     }
904     if (currentSpeechAudio) {
905         try {
906             currentSpeechAudio.pause();
907             currentSpeechAudio.src = '';
908         } catch (e) {}
909         currentSpeechAudio = null;
910     }
911     if ('speechSynthesis' in window) {
912         try { window.speechSynthesis.cancel(); } catch (e) {}
913     }
914 }
915
916 async function speakText(text) {
917     if (!isTtsEnabled || !text) return;
918     stopAudio();
919
920     const cleanText = text.replace(/<[^>]*>/g, '').replace(/
921     [*_`#]/g, '').trim();
922     if (!cleanText) return;
923
924     // Try Backend TTS
925     try {
926         const paragraphs = cleanText.split(/\n+/).map(p => p.trim()).filter(Boolean);
927         let chunks = [];
928         for (const para of paragraphs) {
929             const sentences = para.match(/^[^.!?]+[.!?]*\/g) || [para];
930
931             let currentChunk = '';
932             for (const s of sentences) {
933                 const tr = s.trim();
934                 if (!tr) continue;
935                 if ((currentChunk + ' ' + tr).length > 200) {
936                     chunks.push(currentChunk.trim());
937                     currentChunk = tr;
938                 } else {
939                     currentChunk = (currentChunk + ' ' + tr).trim();
940                 }
941             }
942             if (currentChunk) {
943                 chunks.push(currentChunk.trim());
944             }
945         }
946         if (chunks.length) {
947             speak(chunks);
948         }
949     } catch (e) {
950         console.error('TTS Error: ', e);
951     }
952 }

```

```

938         }
939     }
940     if (currentChunk) chunks.push(currentChunk.trim());
941 }
942 chunks = chunks.filter(Boolean);
943
944 if (chunks.length > 0) {
945     currentSpeechQueue = new AudioQueuePlayer(chunks);
946     await currentSpeechQueue.play();
947     return;
948 }
949 } catch (e) {
950     console.warn('Backend TTS failed, falling back to Web
Speech Synthesis:', e);
951 }
952
953 // Fallback: Browser SpeechSynthesis
954 if ('speechSynthesis' in window) {
955     const utterance = new SpeechSynthesisUtterance(cleanText);
956     const selectedLang = document.querySelector('input[name="lang-radio"]:checked')?.value || 'ru-RU';
957     utterance.lang = selectedLang;
958     window.speechSynthesis.speak(utterance);
959 }
960 }
961
962 // Message Rendering Helpers
963 function appendMessage(role, text, isStreaming = false) {
964     if (emptyFeedHint) {
965         emptyFeedHint.remove();
966     }
967
968     const bubble = document.createElement('div');
969     bubble.className = `chat-bubble ${role}`;
970
971     const timeStr = new Date().toLocaleTimeString([], { hour
: '2-digit', minute: '2-digit' });

```

```
972 |
973 |     if (role === 'user') {
974 |         bubble.innerHTML = `<div>${escapeHtml(text)}</div><div
    | class="chat-meta"><span>${timeStr}</span></div>`;
975 |     } else {
976 |         bubble.innerHTML = `
977 |             <div class="ai-content">${window.marked ? marked.par
    | se(text) : escapeHtml(text)}</div>
978 |             <div class="chat-meta">
979 |                 <span>${timeStr}</span>
980 |                 <button class="btn-speak-again" title="Озвучить"><
    | i class="bi bi-volume-up-fill"></i></button>
981 |             </div>
982 |         `;
983 |         const speakBtn = bubble.querySelector('.btn-speak-agai
    | n');
984 |         if (speakBtn) {
985 |             speakBtn.addEventListener('click', () => speakText(b
    | ubble.querySelector('.ai-content').innerText));
986 |         }
987 |     }
988 |
989 |     chatFeed.appendChild(bubble);
990 |     chatFeed.scrollTop = chatFeed.scrollHeight;
991 |     return bubble;
992 | }
993 |
994 | function appendMediaUserMessage(file, previewUrl) {
995 |     if (emptyFeedHint) {
996 |         emptyFeedHint.remove();
997 |     }
998 |
999 |     const bubble = document.createElement('div');
1000 |     bubble.className = 'chat-bubble user';
1001 |     const timeStr = new Date().toLocaleTimeString([], { hour
    | : '2-digit', minute: '2-digit' });
1002 |
1003 |     const sizeFormatted = (file.size / (1024 * 1024)).toFixe
```

```

d(2) + ' MB';
1004     const isAudio = file.type.startsWith('audio/') || /\.(mp
3|wav|ogg|m4a|flac|aac|webm)$/i.test(file.name);
1005     const isImage = file.type.startsWith('image/');
1006     const isVideo = file.type.startsWith('video/');
1007
1008     let mediaSnippet = '';
1009     if (isAudio && previewUrl) {
1010         mediaSnippet = `<audio class="media-player-widget" con
trols src="${previewUrl}"></audio>`;
1011     } else if (isImage && previewUrl) {
1012         mediaSnippet = ``;
1013     } else if (isVideo && previewUrl) {
1014         mediaSnippet = `<video class="media-player-widget" con
trols src="${previewUrl}"></video>`;
1015     }
1016
1017     const iconClass = isAudio ? 'bi-file-earmark-music' : (i
sImage ? 'bi-image' : (isVideo ? 'bi-camera-video' : 'bi-file-
earmark-text'));
1018
1019     bubble.innerHTML = `
1020         <div class="media-preview-card">
1021             <i class="bi ${iconClass} fs-4 text-info"></i>
1022             <div class="text-truncate">
1023                 <div class="fw-semibold text-truncate small">${esc
apeHtml(file.name)}</div>
1024                 <div class="text-muted" style="font-size: 0.72rem;
">${sizeFormatted}</div>
1025             </div>
1026         </div>
1027         ${mediaSnippet}
1028         <div class="chat-meta"><span>${timeStr}</span></div>
1029     `;
1030
1031     chatFeed.appendChild(bubble);
1032     chatFeed.scrollTop = chatFeed.scrollHeight;

```



```
1033     return bubble;
1034 }
1035
1036 function escapeHtml(text) {
1037     const div = document.createElement('div');
1038     div.textContent = text;
1039     return div.innerHTML;
1040 }
1041
1042 // Send Voice/Text Query to AI Chat API
1043 async function handleUserMessage(query) {
1044     if (!query || !query.trim()) return;
1045
1046     appendMessage('user', query);
1047     stopAudio();
1048
1049     btnMic.className = 'btn-mic-giant thinking';
1050     micIcon.className = 'bi bi-arrow-repeat';
1051     voiceStatusText.textContent = 'Обработка ответа AI...';
1052     voiceSubtext.textContent = 'Генерация ответа моделью';
1053
1054     if (abortController) {
1055         try { abortController.abort(); } catch (e) {}
1056     }
1057     abortController = new AbortController();
1058
1059     const aiBubble = appendMessage('ai', '...');
1060     const aiContentDiv = aiBubble.querySelector('.ai-content
1061 ');
1062
1063     let fullReply = '';
1064     let voiceReply = '';
1065
1066     try {
1067         const payload = {
1068             message: query,
1069             history: chatHistory,
1070             generation_config: { is_rc: true }
```

```

1070 |     };
1071 |
1072 |     const response = await fetch('/api/chat', {
1073 |         method: 'POST',
1074 |         headers: { 'Content-Type': 'application/json' },
1075 |         body: JSON.stringify(payload),
1076 |         signal: abortController.signal
1077 |     });
1078 |
1079 |     if (!response.ok) {
1080 |         throw new Error(`Ошибка сервера: ${response.status}`
1081 |     );
1082 |
1083 |     const reader = response.body.getReader();
1084 |     const decoder = new TextDecoder('utf-8');
1085 |     let buffer = '';
1086 |
1087 |     while (true) {
1088 |         const { value, done } = await reader.read();
1089 |         if (done) break;
1090 |
1091 |         buffer += decoder.decode(value, { stream: true });
1092 |         const lines = buffer.split('\n');
1093 |         buffer = lines.pop();
1094 |
1095 |         for (const line of lines) {
1096 |             const tr = line.trim();
1097 |             if (!tr) continue;
1098 |             if (tr.startsWith('data: ')) {
1099 |                 try {
1100 |                     const data = JSON.parse(tr.substring(6));
1101 |                     if (data.error) throw new Error(data.error);
1102 |                     if (data.voice) voiceReply += data.voice;
1103 |                     if (data.text) {
1104 |                         fullReply += data.text;
1105 |                         aiContentDiv.innerHTML = window.marked ? mar
ked.parse(fullReply) : escapeHtml(fullReply);

```

```

1106         chatFeed.scrollTop = chatFeed.scrollHeight;
1107     }
1108     } catch (e) {
1109         console.error(e);
1110     }
1111 }
1112 }
1113 }
1114
1115 btnMic.className = 'btn-mic-giant';
1116 micIcon.className = 'bi bi-mic-fill';
1117 voiceStatusText.textContent = 'Нажмите кнопку и говори
1118 те';
1119 voiceSubtext.textContent = 'Готов к следующему вопросу
1120 ';
1121 // Update history
1122 chatHistory.push({ role: 'user', parts: [query] });
1123 chatHistory.push({ role: 'model', parts: [fullReply] }
1124 );
1125 if (chatHistory.length > 20) chatHistory = chatHistory
1126 .slice(-20);
1127
1128 // Speak response
1129 speakText(voiceReply || fullReply);
1130
1131 } catch (err) {
1132     if (err.name === 'AbortError') return;
1133     aiContentDiv.innerHTML = `

```

```

1138 // Process Uploaded Audio / Media File
1139 async function processUploadedFile(file, isAudioOnly = fal
se) {
1140     if (!file) return;
1141     window.triggerHaptic('medium');
1142     stopAudio();
1143
1144     const previewUrl = URL.createObjectURL(file);
1145     appendMediaUserMessage(file, previewUrl);
1146
1147     btnMic.className = 'btn-mic-giant thinking';
1148     micIcon.className = 'bi bi-arrow-repeat';
1149     voiceStatusText.textContent = 'Загрузка и анализ медиа..
.';
1150     voiceSubtext.textContent = file.name;
1151
1152     const aiBubble = appendMessage('ai', '□ Загрузка и обраб
отка файла...');
1153     const aiContentDiv = aiBubble.querySelector('.ai-content
');
1154
1155     const isAudio = isAudioOnly || file.type.startsWith('aud
io/') || /\.(mp3|wav|ogg|m4a|flac|aac|webm)$/i.test(file.name)
;
1156
1157     try {
1158         if (isAudio) {
1159             // Send to Audio Diarization & Transcription service
1160             const formData = new FormData();
1161             formData.append('file', file);
1162             const selectedLang = document.querySelector('input[n
ame="lang-radio"]:checked').value || 'ru-RU';
1163             formData.append('language', selectedLang.split('-')[
0]);
1164
1165             const res = await fetch('/api/audio/diarize', {
1166                 method: 'POST',
1167                 body: formData

```

```

1168         });
1169
1170         if (!res.ok) {
1171             const errData = await res.json().catch(() => ({}))
1172             ;
1173             throw new Error(errData.detail || `Ошибка сервера:
1174             ${res.status}`);
1175         }
1176
1177         const result = await res.json();
1178         const diarizationData = result.data || {};
1179
1180         let formattedOutput = '';
1181         if (diarizationData.summary) {
1182             formattedOutput += `### □ Краткая сводка\n${diarizationData.summary}\n\n`;
1183         }
1184
1185         if (diarizationData.key_points && diarizationData.key_points.length > 0) {
1186             formattedOutput += `### □ Ключевые тезисы\n` + diarizationData.key_points.map(kp => `- ${kp}`).join('\n') + `\n\n`;
1187         }
1188
1189         if (diarizationData.action_items && diarizationData.action_items.length > 0) {
1190             formattedOutput += `### □ Задачи и решения\n` + diarizationData.action_items.map(ai => `- [ ] ${ai}`).join('\n') + `\n\n`;
1191         }
1192
1193         if (diarizationData.transcript && diarizationData.transcript.length > 0) {
1194             formattedOutput += `### □ Стенограмма диалога\n`;
1195             for (const item of diarizationData.transcript) {
1196                 const speaker = item.speaker || 'Собеседник';
1197                 const text = item.text || '';

```

```

1196         formattedOutput += `**${speaker}:** ${text}\n\n`
1197     ;
1198     }
1199 }
1200     if (!formattedOutput) {
1201         formattedOutput = diarizationData.markdown_report
1202         || 'Аудиозапись успешно обработана.';
1203     }
1204     aiContentDiv.innerHTML = window.marked ? marked.parse(formattedOutput) : escapeHtml(formattedOutput);
1205     chatFeed.scrollTop = chatFeed.scrollHeight;
1206
1207     chatHistory.push({ role: 'user', parts: [ `[Аудиозапись ${file.name}]` ] });
1208     chatHistory.push({ role: 'model', parts: [ formattedOutput ] });
1209     if (chatHistory.length > 20) chatHistory = chatHistory.slice(-20);
1210
1211     // Speak the executive summary if available
1212     if (diarizationData.summary) {
1213         speakText(diarizationData.summary);
1214     }
1215
1216 } else {
1217     // General Media Upload to storage
1218     const formData = new FormData();
1219     formData.append('file', file);
1220
1221     const res = await fetch('/api/user/storage/upload',
1222     {
1223         method: 'POST',
1224         body: formData
1225     });
1226     if (!res.ok) {

```

```
1227 |         const errData = await res.json().catch(() => ({}))
1228 |         ;
1229 |         throw new Error(errData.detail || `Ошибка загрузки
1230 | : ${res.status}`);
1231 |     }
1232 |
1233 |     const resData = await res.json();
1234 |     const fileName = resData.filename || file.name;
1235 |
1236 |     // Request AI analysis of the uploaded document/media
1237 |     a
1238 |     const prompt = `Пользователь загрузил файл ${fileNam
1239 | e} (${file.type || 'документ'}). Подтверди получение и опиши д
1240 | альнейшие действия.`;
1241 |
1242 |     const chatRes = await fetch('/api/chat', {
1243 |         method: 'POST',
1244 |         headers: { 'Content-Type': 'application/json' },
1245 |         body: JSON.stringify({
1246 |             message: prompt,
1247 |             history: chatHistory,
1248 |             generation_config: { is_rc: true }
1249 |         })
1250 |     });
1251 |
1252 |     let modelReply = `Файл **${escapeHtml(fileName)}** у
1253 | спешно загружен и сохранен в вашем хранилище.`;
1254 |
1255 |     if (chatRes.ok) {
1256 |         // Read stream or fallback
1257 |         const reader = chatRes.body.getReader();
1258 |         const decoder = new TextDecoder('utf-8');
1259 |         let buffer = '';
1260 |         let streamReply = '';
1261 |
1262 |         while (true) {
1263 |             const { value, done } = await reader.read();
1264 |             if (done) break;
1265 |             buffer += decoder.decode(value, { stream: true })
```

```

);
1259         const lines = buffer.split('\n');
1260         buffer = lines.pop();
1261
1262         for (const line of lines) {
1263             const tr = line.trim();
1264             if (tr.startsWith('data: ')) {
1265                 try {
1266                     const data = JSON.parse(tr.substring(6));
1267                     if (data.text) streamReply += data.text;
1268                 } catch (e) {}
1269             }
1270         }
1271     }
1272     if (streamReply) modelReply = streamReply;
1273 }
1274
1275     aiContentDiv.innerHTML = window.marked ? marked.parse(modelReply) : escapeHtml(modelReply);
1276     chatFeed.scrollTop = chatFeed.scrollHeight;
1277
1278     chatHistory.push({ role: 'user', parts: [`[Загружен файл ${file.name}]`] });
1279     chatHistory.push({ role: 'model', parts: [modelReply] });
1280     if (chatHistory.length > 20) chatHistory = chatHistory.slice(-20);
1281
1282     speakText(modelReply);
1283 }
1284
1285     btnMic.className = 'btn-mic-giant';
1286     micIcon.className = 'bi bi-mic-fill';
1287     voiceStatusText.textContent = 'Нажмите кнопку и говорите';
1288     voiceSubtext.textContent = 'Готов к следующему вопросу или загрузке';
1289

```



```
1290     } catch (err) {
1291         aiContentDiv.innerHTML = `<span class="text-danger">Ош
ибка обработки: ${escapeHtml(err.message)}</span>`;
1292         btnMic.className = 'btn-mic-giant';
1293         micIcon.className = 'bi bi-mic-fill';
1294         voiceStatusText.textContent = 'Ошибка загрузки';
1295         voiceSubtext.textContent = 'Попробуйте загрузить файл
повторно';
1296     }
1297 }
1298
1299 // Media Buttons Event Listeners
1300 btnUploadAudio.addEventListener('click', () => {
1301     window.triggerHaptic('light');
1302     audioFileInput.click();
1303 });
1304
1305 btnUploadMedia.addEventListener('click', () => {
1306     window.triggerHaptic('light');
1307     mediaFileInput.click();
1308 });
1309
1310 btnClipAttachment.addEventListener('click', () => {
1311     window.triggerHaptic('light');
1312     mediaFileInput.click();
1313 });
1314
1315 audioFileInput.addEventListener('change', (e) => {
1316     const file = e.target.files && e.target.files[0];
1317     if (file) {
1318         processUploadedFile(file, true);
1319         audioFileInput.value = '';
1320     }
1321 });
1322
1323 mediaFileInput.addEventListener('change', (e) => {
1324     const file = e.target.files && e.target.files[0];
1325     if (file) {
```

```

1326         processUploadedFile(file, false);
1327         mediaFileInput.value = '';
1328     }
1329 });
1330
1331 // Text Fallback Form
1332 textSendForm.addEventListener('submit', (e) => {
1333     e.preventDefault();
1334     const val = textInput.value.trim();
1335     if (val) {
1336         textInput.value = '';
1337         handleUserMessage(val);
1338     }
1339 });
1340
1341 // TTS Toggle Button
1342 ttsToggleBtn.addEventListener('click', () => {
1343     window.triggerHaptic('light');
1344     isTtsEnabled = !isTtsEnabled;
1345     if (isTtsEnabled) {
1346         ttsIcon.className = 'bi bi-volume-up-fill';
1347         ttsToggleBtn.classList.remove('opacity-50');
1348     } else {
1349         stopAudio();
1350         ttsIcon.className = 'bi bi-volume-mute-fill';
1351         ttsToggleBtn.classList.add('opacity-50');
1352     }
1353 });
1354
1355 // Clear Chat Button
1356 btnClearChat.addEventListener('click', () => {
1357     window.triggerHaptic('light');
1358     stopAudio();
1359     chatHistory = [];
1360     chatFeed.innerHTML = `
1361         <div class="text-muted small text-center my-auto p-3"
1362         id="empty-feed-hint">
1363             <i class="bi bi-chat-dots fs-3 d-block mb-1 opacity-

```

```

50"></i>
1363         Голосовые команды и медиафайлы отобразятся здесь
1364     </div>
1365     `;
1366 });
1367
1368 // Theme Toggle
1369 document.getElementById('theme-toggle-btn').addEventListener('click', () => {
1370     window.triggerHaptic('light');
1371     const isLight = document.body.classList.toggle('theme-light');
1372     const icon = document.getElementById('theme-icon');
1373     icon.className = isLight ? 'bi bi-sun-fill' : 'bi bi-moon-stars-fill';
1374 });
1375
1376 // App Initialization
1377 async function initApp() {
1378     const state = await api.auth.check();
1379     if (state.authenticated) {
1380         if (userDisplayName && state.user) {
1381             userDisplayName.textContent = state.user.name || state.user.email || 'Пользователь';
1382         }
1383         showPanel('main');
1384         initSpeechRecognition();
1385     } else {
1386         showPanel('auth');
1387     }
1388 }
1389
1390 document.addEventListener('DOMContentLoaded', () => {
1391     initTelegram();
1392     initApp();
1393 });
1394 </script>
1395 </body>

```

1396 | </html>
1397 |

index.html

```
1 <!-- Scenarios & Automated Diagnostics Tab Template -->
2 <div class="container-fluid p-0 scenarios-tab-wrapper">
3
4   <!-- Header & Quick Actions -->
5   <div class="card bg-dark bg-opacity-50 border-secondary-subtle mb-3 shadow-sm">
6     <div class="card-body p-3">
7       <div class="d-flex flex-wrap align-items-center justify-content-between gap-3">
8         <div class="d-flex align-items-center gap-2">
9           <div class="fs-2">□ </div>
10          <div>
11            <h5 class="mb-0 fw-bold text-white d-flex align-items-center gap-2">
12              <span>Сценарии тестирования и автоматизации</span>
13              <span class="badge bg-primary-subtle text-primary border border-primary-subtle small fs-6">Test Computer (/tc)</span>
14            </h5>
15            <div class="text-muted small">
16              Центр управления: экспресс-сценарии проверки хоста, аудит логов, диагностика оборудования и интеллектуальный ассистент навыков.
17            </div>
18          </div>
19        </div>
20        <div class="d-flex align-items-center gap-2 flex-wrap">
21          <button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-scenario-refresh-list" title="Обновить список сценариев">
22            <i class="bi bi-arrow-clockwise"></i>
23            <span>Обновить</span>
24          </button>
25          <button class="btn btn-success btn-sm d-flex align-i
```

```

    items-center gap-1 shadow-sm fw-semibold" id="btn-run-quick-check" title="Запустить экспресс-проверку окружения">
26         <span>□ Быстрая проверка</span>
27         <span class="badge bg-white text-dark ms-1">Рекомендуется</span>
28     </button>
29     <button class="btn btn-outline-primary btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-run-all-scenarios" title="Последовательно выполнить все доступные сценарии">
30         <i class="bi bi-play-fill"></i>
31         <span>Запустить все</span>
32     </button>
33     <button class="btn btn-outline-warning btn-sm d-flex align-items-center gap-1 shadow-sm fw-semibold" id="btn-toggle-post-install" data-bs-toggle="collapse" data-bs-target="#postInstallWizardCollapse" aria-expanded="false" aria-controls="postInstallWizardCollapse" title="Открыть мастер первоначальной настройки Windows">
34         <i class="bi bi-magic"></i>
35         <span>Post-Install Wizard</span>
36     </button>
37     <button class="btn btn-primary btn-sm d-flex align-items-center gap-1.5 shadow-sm fw-semibold" id="btn-open-scenarios-drawer" data-bs-toggle="offcanvas" data-bs-target="#scenariosOffcanvas" aria-controls="scenariosOffcanvas" title="Открыть выдвижную панель со сценариями">
38         <i class="bi bi-layout-sidebar-reverse"></i>
39         <span>Сценарии</span>
40         <span class="badge bg-white text-dark ms-1" id="scenarios-count-badge-header">6</span>
41     </button>
42 </div>
43 </div>
44 </div>
45 </div>
46
47 <!-- Collapsible Post-Install Wizard Panel -->
48 <div class="collapse mb-3" id="postInstallWizardCollapse">

```

```
49     <div class="card bg-dark border-secondary shadow-sm">
50         <div class="card-header border-secondary bg-transparent
d-flex justify-content-between align-items-center flex-wrap ga
p-2 py-2 px-3">
51             <div>
52                 <h6 class="mb-0 fw-bold text-warning"><i class="bi b
i-magic me-1"></i> Windows Post-Install Setup Checklist</h6>
53                 <span class="small text-muted">Выберите профиль базо
вой настройки или отметьте необходимые шаги для применения.</s
pan>
54             </div>
55             <div class="d-flex align-items-center gap-2">
56                 <select class="form-select form-select-sm bg-dark te
xt-white border-secondary" id="wizard-profile-select" style="w
idth: 240px;">
57                     <option value="post_install">Windows Post-Install
Baseline</option>
58                     <option value="secure_hardening">Security Hardenin
g Profile</option>
59                     <option value="developer_setup">Developer Workstat
ion Profile</option>
60                 </select>
61                 <button class="btn btn-sm btn-primary px-3 rounded-p
ill" id="btn-wizard-apply-profile">
62                     <i class="bi bi-play-fill me-1"></i> Применить про
филь
63                 </button>
64             </div>
65         </div>
66         <div class="card-body p-3">
67             <div id="wizard-steps-container" class="d-flex flex-co
lumn gap-2">
68                 <div class="text-muted small text-center py-2">Загру
зка профилей первоначальной настройки...</div>
69             </div>
70         </div>
71     </div>
72 </div>
```

```

73 |
74 | <!-- Main View: AI Assistant Mini-Chat (expanded to full pan
    | el width and >= 80% height) -->
75 | <div class="row g-3">
76 |     <div class="col-12">
77 |         <div class="card bg-dark bg-opacity-75 border-secondary-
    | subtle shadow-sm d-flex flex-column" style="min-height: 80vh;
    | height: calc(100vh - 170px);">
78 |
79 |             <!-- Chat Header -->
80 |             <div class="card-header bg-transparent border-secondar
    | y-subtle py-2 px-3 d-flex align-items-center justify-content-b
    | etween flex-wrap gap-2">
81 |                 <div class="d-flex align-items-center gap-2">
82 |                     <span class="fs-5">□ </span>
83 |                     <div>
84 |                         <h6 class="mb-0 fw-bold text-white">AI Ассистент
    | сценариев & Навыков</h6>
85 |                         <div class="text-muted small" style="font-size:
    | 0.75rem;">Спросите что угодно о компьютере – система проверит
    | и создаст навык</div>
86 |                     </div>
87 |                 </div>
88 |                 <div class="d-flex align-items-center gap-2">
89 |                     <button class="btn btn-outline-info btn-sm py-0.5
    | px-2 text-info border-info border-opacity-50 d-flex align-item
    | s-center gap-1" data-bs-toggle="offcanvas" data-bs-target="#sc
    | enariosOffcanvas" aria-controls="scenariosOffcanvas" title="От
    | крыть выдвижную панель со сценариями" style="font-size: 0.75re
    | m;">
90 |                         <i class="bi bi-layout-sidebar-reverse"></i>
91 |                         <span>Сценарии</span>
92 |                     </button>
93 |                     <button class="btn btn-outline-secondary btn-sm py
    | -0.5 px-2 text-light border-secondary" id="btn-scenario-new-ch
    | at" title="Очистить контекст и начать новую сессию диалога" st
    | yle="font-size: 0.75rem;">
94 |                         <i class="bi bi-arrow-counterclockwise me-1"></i>

```



```

    >Новая сессия
95         </button>
96         <span class="badge bg-success-subtle text-success
border border-success-subtle small">Online</span>
97     </div>
98 </div>
99
100     <!-- Mini-Chat Suggestions / Prompt Pills -->
101     <div class="p-2 border-bottom border-secondary-subtle
bg-black bg-opacity-25">
102         <div class="small text-muted mb-1" style="font-size:
0.72rem;">Быстрые запросы:</div>
103         <div class="d-flex flex-wrap gap-1" id="chat-quick-p
rompts">
104             <button class="btn btn-outline-secondary btn-sm py
-0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
105                 □ Какие мыши были подключены?
106             </button>
107             <button class="btn btn-outline-secondary btn-sm py
-0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
108                 □ Покажи USB устройства
109             </button>
110             <button class="btn btn-outline-secondary btn-sm py
-0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
111                 □ Проверь ошибки в логах
112             </button>
113             <button class="btn btn-outline-secondary btn-sm py
-0 px-2 quick-prompt-btn" style="font-size: 0.75rem;">
114                 □ Сделай быструю проверку
115             </button>
116         </div>
117     </div>
118
119     <!-- Chat Messages History Container -->
120     <div class="card-body p-3 overflow-y-auto flex-grow-1"
id="scenario-chat-history" style="min-height: 480px; max-heig
ht: none;">
121         <div class="d-flex gap-2 mb-3">

```

```
122 |         <div class="fs-5">□ </div>
123 |         <div class="bg-secondary bg-opacity-25 p-2 rounded
    | border border-secondary-subtle small text-light flex-grow-1">
124 |             Привет! Я интеллектуальный ассистент Test Comput
    | er. Вы можете задать мне любой вопрос о конфигурации хоста, на
    | пример:
125 |             <br><br>
126 |             <div id="chat-greeting-prompts" class="d-flex fl
    | ex-column gap-1">
127 |                 <div>• <a href="#" class="chat-prompt-link tex
    | t-info text-decoration-none border-bottom border-info border-o
    | pacity-50" data-prompt="Какие мыши были подключены к этому ком
    | пьютеру?">«Какие мыши были подключены к этому компьютеру?»</a>
    | </div>
128 |                 <div>• <a href="#" class="chat-prompt-link tex
    | t-info text-decoration-none border-bottom border-info border-o
    | pacity-50" data-prompt="Покажи список сетевых портов и активны
    | х соединений">«Покажи список сетевых портов и активных соедине
    | ний»</a></div>
129 |                 <div>• <a href="#" class="chat-prompt-link tex
    | t-info text-decoration-none border-bottom border-info border-o
    | pacity-50" data-prompt="Проверь автозапуск и службы Windows">«
    | Проверь автозапуск и службы Windows»</a></div>
130 |             </div>
131 |             <br>
132 |             Если для вопроса нет готового сценария, я опрошу
    | оборудование через PowerShell/WMI, сформирую для вас ответ и
    | **автоматически сохраню новый навык** в каталог <code>.skills/
    | </code>.
133 |         </div>
134 |     </div>
135 | </div>
136 |
137 | <!-- Chat Input Form -->
138 | <div class="card-footer bg-transparent border-secondar
    | y-subtle p-2 flex-shrink-0">
139 |     <form id="scenario-chat-form" class="d-flex flex-col
    | umn gap-2">
```

```

140         <div class="input-group input-group-sm">
141             <input type="text" class="form-control bg-dark t
ext-white border-secondary" id="scenario-chat-input" placehold
er="Задайте вопрос или введите /skills list..." autocomplete="
off">
142             <button class="btn btn-primary d-flex align-item
s-center gap-1" type="submit" id="btn-scenario-chat-send">
143                 <span>Отправить</span>
144                 <i class="bi bi-send-fill"></i>
145             </button>
146         </div>
147         <div class="d-flex align-items-center justify-cont
ent-between px-1 flex-wrap gap-2">
148             <span class="text-muted" style="font-size: 0.72r
em;" id="chat-status-indicator">Готов к работе</span>
149             <div class="d-flex align-items-center gap-2">
150                 <div class="form-check form-switch m-0 d-flex
align-items-center gap-1">
151                     <input class="form-check-input mt-0" type="c
heckbox" role="switch" id="check-scenario-keep-context" title=
"Хранить историю диалога в контексте запроса">
152                     <label class="form-check-label text-muted us
er-select-none" for="check-scenario-keep-context" style="font-
size: 0.75rem;" title="Хранить историю диалога в контексте зап
роса">
153                         <i class="bi bi-clock-history text-seconda
ry me-1"></i>Хранить историю
154                     </label>
155                 </div>
156                 <div class="form-check form-switch m-0 d-flex
align-items-center gap-1">
157                     <input class="form-check-input mt-0" type="c
heckbox" role="switch" id="check-scenario-rag" checked title="
Использовать базу знаний (RAG)">
158                     <label class="form-check-label text-muted us
er-select-none" for="check-scenario-rag" style="font-size: 0.7
5rem;" title="Использовать базу знаний (RAG)">
159                     <i class="bi bi-database-check text-info m

```

```

e-1"></i>RAG
160         </label>
161     </div>
162 </div>
163 </div>
164 </form>
165 </div>
166
167 </div>
168 </div>
169 </div>
170
171 <!-- Right Slide-out Drawer (Offcanvas): Scenarios List & Ex
    ecution Results -->
172 <div class="offcanvas offcanvas-end text-bg-dark border-star
    t border-secondary-subtle shadow-lg" tabindex="-1" id="scenari
    osOffcanvas" aria-labelledby="scenariosOffcanvasLabel" style="
    width: 560px; max-width: 95vw;">
173     <div class="offcanvas-header border-bottom border-secondar
    y-subtle py-3 px-3">
174         <div class="d-flex align-items-center gap-2">
175             <span class="fs-4">❏ </span>
176             <div>
177                 <h5 class="offcanvas-title fw-bold text-white mb-0"
    id="scenariosOffcanvasLabel">Сценарии тестирования</h5>
178                 <div class="text-muted small" style="font-size: 0.75
    rem;">Экспресс-диагностика и системные проверки</div>
179             </div>
180         </div>
181         <div class="d-flex align-items-center gap-2">
182             <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="offcanvas" aria-label="Закрыть"></button>
183         </div>
184     </div>
185
186     <div class="offcanvas-body d-flex flex-column gap-3 p-3">
187
188         <!-- Filter Bar & Count Badge -->

```

```

189 |         <div class="d-flex align-items-center justify-content-be
| tween gap-2 flex-wrap">
190 |             <div class="btn-group btn-group-sm" role="group" id="s
| cenario-category-filters">
191 |                 <button type="button" class="btn btn-outline-seconda
| ry active" data-filter="all">Все (6)</button>
192 |                 <button type="button" class="btn btn-outline-seconda
| ry" data-filter="quick">⚡ Экспресс</button>
193 |                 <button type="button" class="btn btn-outline-seconda
| ry" data-filter="logs">📖 Логи</button>
194 |                 <button type="button" class="btn btn-outline-seconda
| ry" data-filter="diagnostics">🔧 Система</button>
195 |                 <button type="button" class="btn btn-outline-seconda
| ry" data-filter="security">🔒 Безопасность</button>
196 |             </div>
197 |             <span class="text-muted small" id="scenarios-count-bad
| ge">Доступно: 6</span>
198 |         </div>
199 |
200 |         <!-- Execution Console & Results (shown on run) -->
201 |         <div class="card bg-dark bg-opacity-75 border-secondary-
| subtle shadow-sm" id="scenario-result-panel" style="display: n
| one;">
202 |             <div class="card-header bg-transparent border-secondar
| y-subtle d-flex align-items-center justify-content-between py-
| 2 px-3">
203 |                 <div class="d-flex align-items-center gap-2">
204 |                     <span id="result-status-icon" class="fs-5">⬢ </span
| >
205 |                     <div>
206 |                         <strong id="result-scenario-title" class="text-w
| hite">Выполнение сценария...</strong>
207 |                         <div class="text-muted small" id="result-scenari
| o-meta">Инициализация шагов...</div>
208 |                     </div>
209 |                 </div>
210 |                 <div class="d-flex align-items-center gap-2">
211 |                     <span class="badge bg-secondary" id="result-durati

```

```

on-badge">0.0 ms</span>
212         <button class="btn btn-sm btn-outline-secondary py
-0 px-2" id="btn-close-result-panel" title="Скрыть панель резу
льтатов">✕ </button>
213     </div>
214 </div>
215     <div class="card-body p-3">
216         <!-- Overall Summary Banner -->
217         <div class="alert alert-secondary py-2 px-3 mb-3 sma
ll d-flex align-items-start gap-2" id="result-summary-banner">
218             <i class="bi bi-info-circle-fill mt-1"></i>
219             <span id="result-summary-text">Результат проверки
будет отображен здесь...</span>
220         </div>
221
222         <!-- Progress Bar -->
223         <div class="progress mb-3" style="height: 6px;" id="
scenario-progress-container">
224             <div class="progress-bar progress-bar-striped prog
ress-bar-animated bg-primary" role="progressbar" style="width:
100%" id="scenario-progress-bar"></div>
225         </div>
226
227         <!-- Step Results List -->
228         <div class="d-flex flex-column gap-2" id="scenario-s
teps-container">
229             <!-- Dynamic step rows injected here -->
230         </div>
231     </div>
232 </div>
233
234 <!-- Scenarios Cards Container -->
235 <div class="row g-2" id="scenarios-cards-container">
236     <!-- Injected dynamically by main.js -->
237     <div class="col-12 text-center py-5 text-muted small">
238         <div class="spinner-border spinner-border-sm text-pr
imary mb-2" role="status"></div>
239         <div>Загрузка сценариев тестирования...</div>

```

```
240 |         </div>
241 |     </div>
242 |
243 |     </div>
244 | </div>
245 |
246 | </div>
247 |
```

main.js

```
1 // Scenarios and Test Computer Automated Diagnostics Module (/
  tc)
2 (function () {
3   'use strict';
4
5   let scenariosList = [];
6   let currentFilter = 'all';
7   let isRunning = false;
8
9   // Icons and badges helper
10  const categoryIcons = {
11    quick: '⚡',
12    logs: '📄',
13    diagnostics: '🔍',
14    security: '🔒',
15    network: '🌐',
16    ai: '🧠',
17  };
18
19  const statusIcons = {
20    ok: '✅',
21    warn: '⚠️',
22    error: '❌',
23    skipped: '⏸️',
24  };
25
26  // Helper for safe HTML escaping
27  function escapeHtml(str) {
28    if (!str) return '';
29    return String(str)
30      .replace(/&/g, '&amp;')
31      .replace(/</g, '&lt;')
32      .replace(/>/g, '&gt;')
33      .replace(/"/g, '&quot;')
34      .replace(/'/g, '&#039;');
35  }
```



```

36
37 // Simple Markdown to HTML formatter for mini-chat
38 function formatMarkdown(text) {
39     if (!text) return '';
40     let html = escapeHtml(text);
41     // Bold
42     html = html.replace(/\*\*(.*?)\*\*/g, '<strong>$1</strong>');
43     // Italic
44     html = html.replace(/\*(.*?)\*/g, '<em>$1</em>');
45     // Inline code
46     html = html.replace(/`(.*)`/g, '<code class="text-info bg
    -black bg-opacity-50 px-1 rounded">$1</code>');
47     // Blockquote
48     html = html.replace(/^>\s+(.*)/gim, '<div class="borde
    r-start border-primary border-3 ps-2 my-1 text-info bg-primary
    bg-opacity-10 py-1 rounded-end">$1</div>');
49     // Headers
50     html = html.replace(/^###\s+(.*)/gim, '<h6 class="fw-bold
    text-info mt-2 mb-1">$1</h6>');
51     html = html.replace(/^##\s+(.*)/gim, '<h5 class="fw-bold
    text-white mt-2 mb-1">$1</h5>');
52     // Line breaks
53     html = html.replace(/\n/g, '<br>');
54     return html;
55 }
56
57 // Load scenarios from API
58 async function loadScenarios() {
59     const container = document.getElementById('scenarios-cards
    -container');
60     if (!container) return;
61
62     try {
63         const res = await fetch('/api/v1/scenarios');
64         if (!res.ok) throw new Error(`HTTP ${res.status}`);
65         scenariosList = await res.json();
66         renderScenariosGrid();

```

```
67 |     } catch (err) {
68 |         console.warn('[ScenariosTab] Error fetching /api/v1/scen
    arios, using local fallback:', err);
69 |         scenariosList = [
70 |             {
71 |                 id: 'quick_check',
72 |                 title: 'Быстрая проверка (Smoke Test)',
73 |                 description: 'Экспресс-проверка целостности окружени
    я, доступности ядра FastAPI, AI-провайдеров, свободного места
    и ключевых логов. Рекомендуется после установки.',
74 |                 category: 'quick',
75 |                 icon: '🔍 ',
76 |                 recommended: true,
77 |                 estimated_duration_sec: 2,
78 |             },
79 |             {
80 |                 id: 'log_audit',
81 |                 title: 'Запустить аудит логов',
82 |                 description: 'Комплексный анализ файлов логов (info.
    log, errors.log, app.log) и журналов событий Windows. Поиск кр
    итических ошибок и аномалий.',
83 |                 category: 'logs',
84 |                 icon: '📄 ',
85 |                 recommended: false,
86 |                 estimated_duration_sec: 3,
87 |             },
88 |             {
89 |                 id: 'system_inspector',
90 |                 title: 'Диагностика системы и оборудования',
91 |                 description: 'Срез телеметрии: загрузка ядер процесс
    ора, температура, оперативная память, свободное место на диска
    х и фоновые процессы.',
92 |                 category: 'diagnostics',
93 |                 icon: '🔧 ',
94 |                 recommended: false,
95 |                 estimated_duration_sec: 3,
96 |             },
97 |             {
```

```
98         id: 'windows_admin',
99         title: 'Аудит автозапуска и служб Windows',
100        description: 'Проверка записей реестра Run/RunOnce,
| папки автозагрузки, системных служб Windows и прав текущего пр
| оцесса.',
101        category: 'security',
102        icon: '🔒 ',
103        recommended: false,
104        estimated_duration_sec: 4,
105    },
106    {
107        id: 'network_test',
108        title: 'Проверка сетевой доступности и портов',
109        description: 'Тестирование локального шлюза, разреше
| ния DNS, открытых серверных портов (8000) и доступности внешни
| х AI-сервисов.',
110        category: 'network',
111        icon: '🌐 ',
112        recommended: false,
113        estimated_duration_sec: 4,
114    },
115    {
116        id: 'ai_providers_check',
117        title: 'Проверка статуса AI-провайдеров',
118        description: 'Опрос сконфигурированных AI-моделей (G
| oogle Gemini, AGY, Ollama, Foundry) и замер времени отклика ин
| ференса.',
119        category: 'ai',
120        icon: '🧠 ',
121        recommended: false,
122        estimated_duration_sec: 5,
123    },
124    ];
125    renderScenariosGrid();
126  }
127  }
128
129  // Render cards grid based on active filter
```

```
130 | function renderScenariosGrid() {
131 |     const container = document.getElementById('scenarios-cards
    | -container');
132 |     if (!container) return;
133 |
134 |     const filtered = scenariosList.filter(sc => {
135 |         if (currentFilter === 'all') return true;
136 |         return sc.category === currentFilter;
137 |     });
138 |
139 |     const badgeEl = document.getElementById('scenarios-count-b
    | adge');
140 |     if (badgeEl) {
141 |         badgeEl.innerText = `Доступно: ${filtered.length}`;
142 |     }
143 |     const headerBadgeEl = document.getElementById('scenarios-c
    | ount-badge-header');
144 |     if (headerBadgeEl) {
145 |         headerBadgeEl.innerText = `${scenariosList.length}`;
146 |     }
147 |
148 |     if (filtered.length === 0) {
149 |         container.innerHTML = '<div class="col-12 text-center py
    | -4 text-muted small">В этой категории нет сценариев.</div>';
150 |         return;
151 |     }
152 |
153 |     container.innerHTML = filtered.map(sc => {
154 |         const isRecommended = Boolean(sc.recommended);
155 |         const borderClass = isRecommended ? 'border-success bord
    | er-opacity-50 shadow-sm' : 'border-secondary-subtle';
156 |         const bgHeader = isRecommended ? 'bg-success bg-opacity-
    | 10' : 'bg-transparent';
157 |         const btnClass = isRecommended ? 'btn-success' : 'btn-ou
    | tline-primary';
158 |
159 |         return `
160 |             <div class="col-12">
```

```

161 |         <div class="card bg-dark bg-opacity-75 ${borderClass
    | } transition-all hover-shadow">
162 |             <div class="card-header ${bgHeader} border-seconda
    | ry-subtle py-2 px-3 d-flex align-items-center justify-content-
    | between">
163 |                 <div class="d-flex align-items-center gap-2">
164 |                     <span class="fs-5">${sc.icon || categoryIcons[
    | sc.category] || '🎮'}</span>
165 |                     <strong class="text-white small">${escapeHtml(
    | sc.title)}</strong>
166 |                 </div>
167 |                 <div>
168 |                     ${isRecommended ? '<span class="badge bg-succe
    | ss small">Рекомендуется</span>' : '<span class="badge bg-secon
    | dary small">${sc.estimated_duration_sec || 3} сек</span>`}
169 |                 </div>
170 |             </div>
171 |             <div class="card-body p-3 d-flex flex-column justi
    | fy-content-between">
172 |                 <p class="text-muted small mb-2" style="font-siz
    | e: 0.8rem; line-height: 1.35;">
173 |                     ${escapeHtml(sc.description)}
174 |                 </p>
175 |                 <div class="d-flex align-items-center justify-co
    | ntent-between gap-2 mt-auto pt-2 border-top border-secondary-s
    | ubtle">
176 |                     <span class="text-secondary small" style="font
    | -size: 0.72rem;">
177 |                         ID: <code>${escapeHtml(sc.id)}</code>
178 |                     </span>
179 |                     <button class="btn ${btnClass} btn-sm py-1 px-
    | 3 d-flex align-items-center gap-1 btn-run-scenario-item" data-
    | scenario-id="${escapeHtml(sc.id)}" ${isRunning ? 'disabled' :
    | ''}>
180 |                         <i class="bi bi-play-fill"></i>
181 |                         <span>Запустить</span>
182 |                     </button>
183 |                 </div>

```

```

184         </div>
185     </div>
186 </div>
187     `;
188     }).join('');
189
190     // Attach listeners to individual run buttons
191     container.querySelectorAll('.btn-run-scenario-item').forEach(
192     ch(btn => {
193         btn.onclick = () => {
194             const scId = btn.getAttribute('data-scenario-id');
195             if (scId) runScenario(scId);
196         };
197     });
198
199     // Execute scenario via API
200     async function runScenario(scenarioId) {
201         if (isRunning) return;
202         isRunning = true;
203
204         // Automatically open the scenarios drawer if closed
205         const offcanvasEl = document.getElementById('scenariosOffc
206         anvas');
207         if (offcanvasEl && typeof bootstrap !== 'undefined' && boo
208         tstrap.Offcanvas) {
209             try {
210                 const offcanvasInstance = bootstrap.Offcanvas.getOrCre
211                 ateInstance(offcanvasEl);
212                 offcanvasInstance.show();
213             } catch (e) {
214                 console.warn('[ScenariosTab] Offcanvas show error:', e
215                 );
216             }
217         }
218
219         const resultPanel = document.getElementById('scenario-resu
220         lt-panel');

```

```
216 |     const titleEl = document.getElementById('result-scenario-t
    | title');
217 |     const metaEl = document.getElementById('result-scenario-me
    | ta');
218 |     const iconEl = document.getElementById('result-status-icon
    | ');
219 |     const durationBadge = document.getElementById('result-dura
    | tion-badge');
220 |     const summaryBanner = document.getElementById('result-summ
    | ary-banner');
221 |     const summaryText = document.getElementById('result-summar
    | y-text');
222 |     const stepsContainer = document.getElementById('scenario-s
    | teps-container');
223 |     const progressBar = document.getElementById('scenario-prog
    | ress-bar');
224 |
225 |     if (resultPanel) {
226 |         resultPanel.style.display = '';
227 |         resultPanel.scrollIntoView({ behavior: 'smooth', block:
    | 'nearest' });
228 |     }
229 |     if (titleEl) titleEl.innerText = `Выполнение сценария [${s
    | cenarioId}]...`;
230 |     if (metaEl) metaEl.innerText = 'Опрос системных подсистем
    | и выполнение тестов...';
231 |     if (iconEl) iconEl.innerText = '⏳';
232 |     if (durationBadge) durationBadge.innerText = 'Выполняется.
    | ...';
233 |     if (summaryBanner) summaryBanner.className = 'alert alert-
    | info py-2 px-3 mb-3 small d-flex align-items-start gap-2';
234 |     if (summaryText) summaryText.innerText = 'Идет тестировани
    | е компонентов... Пожалуйста, подождите.';
235 |     if (stepsContainer) stepsContainer.innerHTML = '<div class
    | ="text-center py-3 text-muted small"><div class="spinner-borde
    | r spinner-border-sm text-primary mb-2"></div><div>Тестирование
    | запущено...</div></div>';
236 |     if (progressBar) {
```

```
237 |     progressBar.style.width = '100%';
238 |     progressBar.className = 'progress-bar progress-bar-strip
    | ed progress-bar-animated bg-primary';
239 | }
240 |
241 | renderScenariosGrid();
242 |
243 | const tStart = performance.now();
244 | try {
245 |     const response = await fetch('/api/v1/scenarios/run', {
246 |         method: 'POST',
247 |         headers: { 'Content-Type': 'application/json' },
248 |         body: JSON.stringify({ scenario_id: scenarioId }),
249 |     });
250 |
251 |     if (!response.ok) {
252 |         throw new Error(`Ошибка сервера: HTTP ${response.statu
    | s}`);
253 |     }
254 |
255 |     const res = await response.json();
256 |     displayScenarioResults(res);
257 | } catch (err) {
258 |     console.error('[ScenariosTab] Error running scenario:',
    | err);
259 |     const totalDur = Math.round(performance.now() - tStart);
260 |     displayScenarioResults({
261 |         scenario_id: scenarioId,
262 |         title: scenarioId,
263 |         status: 'error',
264 |         duration_ms: totalDur,
265 |         total_steps: 1,
266 |         passed_steps: 0,
267 |         warn_steps: 0,
268 |         failed_steps: 1,
269 |         steps: [
270 |             {
271 |                 name: 'Выполнение запроса сценария',
```



```
272         status: 'error',
273         duration_ms: totalDur,
274         details: `Ошибка: ${err.message}`,
275         recommendation: 'Проверьте соединение с бэкенд сер
вером FastAPI.',
276     },
277 ],
278     summary: `Сбой при выполнении сценария: ${err.message}
`,
279     });
280 } finally {
281     isRunning = false;
282     renderScenariosGrid();
283 }
284 }
285
286 // Display structured results in panel
287 function displayScenarioResults(res) {
288     const titleEl = document.getElementById('result-scenario-t
itle');
289     const metaEl = document.getElementById('result-scenario-me
ta');
290     const iconEl = document.getElementById('result-status-icon
');
291     const durationBadge = document.getElementById('result-dura
tion-badge');
292     const summaryBanner = document.getElementById('result-summ
ary-banner');
293     const summaryText = document.getElementById('result-summar
y-text');
294     const stepsContainer = document.getElementById('scenario-s
teps-container');
295     const progressBar = document.getElementById('scenario-prog
ress-bar');
296
297     if (titleEl) titleEl.innerText = res.title || res.scenario
_id;
298     if (metaEl) metaEl.innerText = `Шагов: ${res.total_steps |
```

```

    | 0} (Пройдено: ${res.passed_steps || 0}, Замечаний: ${res.war
n_steps || 0}, Ошибок: ${res.failed_steps || 0})`;
299     if (durationBadge) durationBadge.innerText = `${res.durati
on_ms || 0} ms`;
300
301     // Overall status styling
302     if (res.status === 'ok') {
303         if (iconEl) iconEl.innerText = '✅';
304         if (summaryBanner) summaryBanner.className = 'alert aler
t-success py-2 px-3 mb-3 small d-flex align-items-start gap-2'
;
305         if (progressBar) progressBar.className = 'progress-bar b
g-success';
306     } else if (res.status === 'warn') {
307         if (iconEl) iconEl.innerText = '⚠️';
308         if (summaryBanner) summaryBanner.className = 'alert aler
t-warning py-2 px-3 mb-3 small d-flex align-items-start gap-2'
;
309         if (progressBar) progressBar.className = 'progress-bar b
g-warning';
310     } else {
311         if (iconEl) iconEl.innerText = '❌';
312         if (summaryBanner) summaryBanner.className = 'alert aler
t-danger py-2 px-3 mb-3 small d-flex align-items-start gap-2';
313         if (progressBar) progressBar.className = 'progress-bar b
g-danger';
314     }
315
316     if (summaryText) summaryText.innerText = res.summary || 'П
рогон завершен.';
317
318     // Steps rendering
319     if (stepsContainer && Array.isArray(res.steps)) {
320         stepsContainer.innerHTML = res.steps.map((step, idx) =>
{
321             const icon = statusIcons[step.status] || '❌';
322             const badgeBg = step.status === 'ok' ? 'bg-success' :
(step.status === 'warn' ? 'bg-warning text-dark' : 'bg-danger'

```

```

);
323
324     const hasData = step.data && Object.keys(step.data).length > 0;
325     const dataId = `step-data-${idx}`;
326
327     return `
328         <div class="card bg-black bg-opacity-30 border border-secondary-subtle p-2 rounded">
329             <div class="d-flex align-items-center justify-content-between gap-2 flex-wrap">
330                 <div class="d-flex align-items-center gap-2">
331                     <span>${icon}</span>
332                     <strong class="text-light small">${escapeHtml(step.name)}</strong>
333                 </div>
334                 <div class="d-flex align-items-center gap-2">
335                     <span class="badge ${badgeBg} small" style="font-size: 0.68rem;">${step.status.toUpperCase()}</span>
336                     <span class="text-secondary small" style="font-size: 0.7rem;">${step.duration_ms || 0} ms</span>
337                 </div>
338             </div>
339             <div class="text-muted small mt-1 ps-4" style="font-size: 0.78rem;">
340                 ${escapeHtml(step.details)}
341             </div>
342             ${step.recommendation ? `
343                 <div class="mt-1 ps-4 small text-warning" style="font-size: 0.75rem;">
344                     📌 <strong>Рекомендация:</strong> ${escapeHtml(step.recommendation)}
345                 </div>
346             ` : ``}
347             ${hasData ? `
348                 <div class="ps-4 mt-1">
349                     <button class="btn btn-link btn-sm p-0 text-info text-decoration-none small" style="font-size: 0.72rem;" onc

```

```

lick="const el=document.getElementById('${dataId}'); if(el) el
.classList.toggle('d-none')">
350 |         □ Показать детали JSON
351 |     </button>
352 |     <div id="${dataId}" class="d-none mt-1 p-2 bg-
dark rounded border border-secondary font-monospace text-light
small overflow-auto" style="max-height: 140px; font-size: 0.7
rem;">
353 |         <pre class="m-0">${escapeHtml(JSON.stringify
(step.data, null, 2))}</pre>
354 |     </div>
355 | </div>
356 |     ` : ''}
357 | </div>
358 | `;
359 |     }).join('');
360 | }
361 | }
362 |
363 | let isChatSubmitting = false;
364 | let isEventsSetup = false;
365 |
366 | // Extract [CHAT] and [VOICE] blocks from response and clean
embedded action blocks
367 | function parseChatAndVoice(rawText) {
368 |     if (!rawText) return { chatText: '', voiceText: '' };
369 |     let text = rawText;
370 |     let voiceText = '';
371 |
372 |     if (text.includes('[CHAT]') || text.includes('[VOICE]')) {
373 |         const voiceMatch = text.match(/\[VOICE\]([\s\S]*?)(?:\[\\
/Voice\]|(?:\[CHAT\])|$)/i);
374 |         if (voiceMatch) {
375 |             voiceText = voiceMatch[1].trim();
376 |         }
377 |         const chatMatch = text.match(/\[CHAT\]([\s\S]*?)(?:\[\\
HAT\]|(?:\[VOICE\])|$)/i);
378 |         if (chatMatch) {

```

```

379 |         text = chatMatch[1].trim();
380 |     } else {
381 |         text = text.replace(/\[VOICE\][\s\S]*?(?:\[\/VOICE\]|(
|?=\[CHAT\])|$\)/gi, '').trim();
382 |     }
383 | }
384 |
385 | // Clean action blocks from chat display text
386 | text = text.replace(/```(?:action|json:action)[\s\S]*?```/
gi, '').trim();
387 | text = text.replace(/\[ACTION:[\s\S]*?\]/gi, '').trim();
388 |
389 | // Clean remaining tags
390 | text = text.replace(/\[\/?CHAT\]/gi, '').replace(/\[\/?VOI
CE\]/gi, '').trim();
391 | return { chatText: text, voiceText: voiceText || text };
392 | }
393 |
394 | // Dynamic pool of questions & quick buttons loaded from ext
ernal file
395 | let sampleQuestionsPool = [];
396 | let quickButtonsPool = [];
397 |
398 | // Asynchronously load questions and quick buttons from exte
rnal file / API
399 | async function loadQuestionsFromConfig() {
400 |     try {
401 |         const res = await fetch('/api/v1/scenarios/questions');
402 |         if (res.ok) {
403 |             const data = await res.json();
404 |             if (Array.isArray(data.prompts) && data.prompts.length
> 0) {
405 |                 sampleQuestionsPool = data.prompts;
406 |             }
407 |             if (Array.isArray(data.quick_buttons) && data.quick_bu
ttons.length > 0) {
408 |                 quickButtonsPool = data.quick_buttons;
409 |                 renderQuickButtons();

```

```

410         }
411     }
412     } catch (err) {
413         console.warn('[ScenariosTab] Falling back to direct JSON
load for questions:', err);
414         try {
415             const fallbackRes = await fetch('/html/scenarios_tab/q
uestions.json');
416             if (fallbackRes.ok) {
417                 const fbData = await fallbackRes.json();
418                 if (Array.isArray(fbData.prompts)) sampleQuestionsPo
ol = fbData.prompts;
419                 if (Array.isArray(fbData.quick_buttons)) {
420                     quickButtonsPool = fbData.quick_buttons;
421                     renderQuickButtons();
422                 }
423             }
424         } catch {}
425     }
426     renderDynamicGreetingPrompts();
427 }
428
429 function renderQuickButtons() {
430     const container = document.getElementById('chat-quick-prom
pts');
431     if (!container || !quickButtonsPool.length) return;
432
433     container.innerHTML = quickButtonsPool.map(btn => `
434         <button class="btn btn-outline-secondary btn-sm py-0 px-
2 quick-prompt-btn" style="font-size: 0.75rem;" data-prompt="$
435         ${escapeHtml(btn.icon || '🔍')} ${escapeHtml(btn.text)}
436         </button>
437     `).join('');
438
439     // Re-bind click handlers
440     container.querySelectorAll('.quick-prompt-btn').forEach(bt
n => {

```

```

441     btn.onclick = () => {
442         if (isChatSubmitting) return;
443         const text = btn.dataset.prompt || btn.innerText.replace(/^[^a-zA-Za-яA-ЯЁЁ]+/, '').trim();
444         const chatInput = document.getElementById('scenario-chat-input');
445         if (chatInput) {
446             chatInput.value = text;
447             submitChat();
448         }
449     };
450 });
451 }
452
453 function renderDynamicGreetingPrompts() {
454     const greetingContainer = document.getElementById('chat-greeting-prompts');
455     if (!greetingContainer) return;
456     if (!sampleQuestionsPool.length) return;
457
458     // Pick 3 random distinct questions from external config
459     const shuffled = [...sampleQuestionsPool].sort(() => 0.5 - Math.random());
460     const selected = shuffled.slice(0, 3);
461
462     greetingContainer.innerHTML = selected.map(q => `
463         <div>• <a href="#" class="chat-prompt-link text-info text-decoration-none border-bottom border-info border-opacity-50" data-prompt="${escapeHtml(q)}">«${escapeHtml(q)}»</a></div>
464     `).join('');
465 }
466
467 function getGreetingHtml() {
468     const shuffled = sampleQuestionsPool.length ? [...sampleQuestionsPool].sort(() => 0.5 - Math.random()) : [];
469     const selected = shuffled.slice(0, 3);
470     const promptsHtml = selected.length ? selected.map(q => `
471         <div>• <a href="#" class="chat-prompt-link text-info tex

```

```

t-decoration-none border-bottom border-info border-opacity-50"
  data-prompt="${escapeHtml(q)}">«${escapeHtml(q)}»</a></div>
472   `).join('') : `
473     <div>• <a href="#" class="chat-prompt-link text-info tex
t-decoration-none border-bottom border-info border-opacity-50"
  data-prompt="Какие мыши были подключены к этому компьютеру?">
«Какие мыши были подключены к этому компьютеру?»</a></div>
474     <div>• <a href="#" class="chat-prompt-link text-info tex
t-decoration-none border-bottom border-info border-opacity-50"
  data-prompt="Покажи список сетевых портов и активных соединен
ий">«Покажи список сетевых портов и активных соединений»</a></
div>
475     <div>• <a href="#" class="chat-prompt-link text-info tex
t-decoration-none border-bottom border-info border-opacity-50"
  data-prompt="Проверь автозапуск и службы Windows">«Проверь ав
тозапуск и службы Windows»</a></div>
476   `;
477
478   return `
479     <div class="d-flex gap-2 mb-3">
480       <div class="fs-5">□ </div>
481       <div class="bg-secondary bg-opacity-25 p-2 rounded bor
der border-secondary-subtle small text-light flex-grow-1">
482         Привет! Я интеллектуальный ассистент Test Computer.
Вы можете задать мне любой вопрос о конфигурации хоста, напри
м
ер:
483         <br><br>
484         <div id="chat-greeting-prompts" class="d-flex flex-c
olumn gap-1">
485           ${promptsHtml}
486         </div>
487         <br>
488         Если для вопроса нет готового сценария, я опрошу обо
рудование через PowerShell/WMI, сформирую для вас ответ и **ав
томатически сохраню новый навык** в каталог <code>.skills/</co
de>.
489       </div>
490     </div>

```



```

491     `;
492 }
493
494 // Handle Mini-Chat interaction
495 function setupMiniChat() {
496     const chatForm = document.getElementById('scenario-chat-form');
497     const chatInput = document.getElementById('scenario-chat-input');
498     const chatHistory = document.getElementById('scenario-chat-history');
499     const sendBtn = document.getElementById('btn-scenario-chat-send');
500     const statusInd = document.getElementById('chat-status-indicator');
501     const ragCheck = document.getElementById('check-scenario-rag');
502     const keepContextCheck = document.getElementById('check-scenario-keep-context');
503
504     if (!chatForm || !chatInput || !chatHistory) return;
505
506     // Fetch and render questions dynamically from external configuration
507     loadQuestionsFromConfig();
508
509     if (keepContextCheck) {
510         const savedKeepContext = localStorage.getItem('tc_scenario_keep_context');
511         if (savedKeepContext !== null) {
512             keepContextCheck.checked = savedKeepContext === 'true';
513         } else {
514             keepContextCheck.checked = false;
515         }
516         keepContextCheck.addEventListener('change', (e) => {
517             localStorage.setItem('tc_scenario_keep_context', e.target.checked);

```

```

518         });
519     }
520
521     if (ragCheck) {
522         const savedRagState = localStorage.getItem('tc_scenario_
use_rag');
523         if (savedRagState !== null) {
524             ragCheck.checked = savedRagState === 'true';
525         }
526         ragCheck.addEventListener('change', (e) => {
527             localStorage.setItem('tc_scenario_use_rag', e.target.c
hecked);
528         });
529     }
530
531     let currentConversationId = `scenario-chat-${Date.now()}`;
532
533     // New Session button handler
534     const btnNewChat = document.getElementById('btn-scenario-n
ew-chat');
535     if (btnNewChat) {
536         btnNewChat.onclick = async () => {
537             try {
538                 await fetch('/api/v1/scenarios/chat/clear', {
539                     method: 'POST',
540                     headers: { 'Content-Type': 'application/json' },
541                     body: JSON.stringify({ conversation_id: currentCon
versationId, message: 'clear' }),
542                 });
543             } catch (e) {
544                 console.warn('[ScenariosTab] Clear session error:',
e);
545             }
546             currentConversationId = `scenario-chat-${Date.now()}`;
547             chatHistory.innerHTML = getGreetingHtml();
548             if (statusInd) statusInd.innerText = 'Новая сессия нач
ата';
549             chatInput.focus();

```

```

550     };
551 }
552
553 // Quick prompt buttons & clickable prompt links
554 document.querySelectorAll('.quick-prompt-btn').forEach(btn
555 => {
556     btn.onclick = () => {
557         if (isChatSubmitting) return;
558         const text = btn.innerText.replace(/^[^a-zA-Za-яA-ЯёЁ]
559 +/, '').trim();
560         chatInput.value = text;
561         submitChat();
562     };
563 });
564
565 // Delegated click handler for inline prompt links in chat
566 history
567 chatHistory.onclick = (e) => {
568     const link = e.target.closest('.chat-prompt-link');
569     if (link) {
570         e.preventDefault();
571         e.stopPropagation();
572         if (isChatSubmitting) return;
573         const promptText = link.dataset.prompt || link.innerTe
574 xt.replace(/^[«"']|['»"]$/g, '').trim();
575         if (promptText) {
576             chatInput.value = promptText;
577             submitChat();
578         }
579     }
580 };
581
582 async function submitChat() {
583     const message = chatInput.value.trim();
584     if (!message || isChatSubmitting) return;
585
586     isChatSubmitting = true;
587     chatInput.value = '';

```

```

584 |     chatInput.disabled = true;
585 |     if (sendBtn) sendBtn.disabled = true;
586 |     if (statusInd) statusInd.innerText = '⏏ Инициализация с
истемного зонда...';
587 |
588 |     // Append User message
589 |     const userMsgHtml = `
590 |         <div class="d-flex justify-content-end mb-2">
591 |             <div class="bg-primary text-white p-2 rounded small
shadow-sm" style="max-width: 85%;">
592 |                 ${escapeHtml(message)}
593 |             </div>
594 |         </div>
595 |     `;
596 |     chatHistory.insertAdjacentHTML('beforeend', userMsgHtml)
597 | ;
598 |
599 |     // Handle client-side slash commands (/skills list, /ski
lls, /help)
600 |     const cleanCmd = message.trim().toLowerCase();
601 |     if (cleanCmd === '/skills' || cleanCmd === '/skills list
' || cleanCmd.startsWith('/skills search')) {
602 |         try {
603 |             const searchParam = cleanCmd.startsWith('/skills sea
rch') ? cleanCmd.replace('/skills search', '').trim() : '';
604 |             const url = searchParam ? `/api/admin/skills?q=${enc
odeURIComponent(searchParam)}` : '/api/admin/skills';
605 |             const res = await fetch(url);
606 |             if (!res.ok) throw new Error(`HTTP ${res.status}`);
607 |             const data = await res.json();
608 |             const skills = data.skills || [];
609 |
610 |             let replyMd = `### ⏏ Каталог зарегистрированных нав
ыков (.skills)\n\n`;
611 |             if (skills.length === 0) {
612 |                 replyMd += `*Навыки не найдены.*`;
613 |             } else {

```

```

614         replyMd += `Всего доступно навыков: **${skills.length}**\n\n`;
615         skills.forEach((s, idx) => {
616             const pathStr = s.root_relative || s.path || `.skills/${s.name}/SKILL.md`;
617             const desc = s.description_ru || s.description || 'Без описания';
618             replyMd += `${idx + 1}. ☐ **${escapeHtml(s.name)}** – **${escapeHtml(desc)}* (\`${escapeHtml(pathStr)}\`)\n`;
619         });
620     }
621
622     const assistantMsgHtml = `
623         <div class="d-flex gap-2 mb-3">
624             <div class="fs-5"><input type="checkbox"/> </div>
625             <div class="bg-secondary bg-opacity-25 p-2.5 rounded border border-secondary-subtle small text-light flex-grow-1 shadow-sm">
626                 <div>${formatMarkdown(replyMd)}</div>
627             </div>
628         </div>
629     `;
630     chatHistory.insertAdjacentHTML('beforeend', assistantMsgHtml);
631     chatHistory.scrollTop = chatHistory.scrollHeight;
632     if (statusInd) statusInd.innerText = 'Список навыков выведен';
633 } catch (err) {
634     const errorMsgHtml = `
635         <div class="d-flex gap-2 mb-3">
636             <div class="fs-5"><input type="checkbox"/> </div>
637             <div class="bg-danger bg-opacity-25 p-2 rounded border border-danger small text-danger flex-grow-1">
638                 Ошибка получения списка навыков: ${escapeHtml(err.message)}
639             </div>
640         </div>
641     `;

```

```

642 |         chatHistory.insertAdjacentHTML('beforeend', errorMsg
    | HTML);
643 |         chatHistory.scrollTop = chatHistory.scrollHeight;
644 |         if (statusInd) statusInd.innerText = 'Ошибка запроса
    | ';
645 |     } finally {
646 |         isChatSubmitting = false;
647 |         chatInput.disabled = false;
648 |         if (sendBtn) sendBtn.disabled = false;
649 |         chatInput.focus();
650 |     }
651 |     return;
652 | }
653 |
654 | // Append Streaming Assistant placeholder with multi-ste
    | p live tracker
655 |     const msgId = `msg-${Date.now()}-${Math.random().toStrin
    | g(36).substr(2, 6)}`;
656 |     const stagesLog = [];
657 |
658 |     const renderStagesHtml = (stages, isFinal = false) => {
659 |         if (!stages || stages.length === 0) return '';
660 |         const currentCount = stages.length;
661 |         const isComplete = isFinal;
662 |         const headerIcon = isComplete
663 |             ? '<i class="bi bi-check-circle-fill text-success"><
    | /i>'
664 |             : '<div class="spinner-border spinner-border-sm text
    | -info" role="status" style="width:0.75rem;height:0.75rem;"></d
    | iv>';
665 |         const headerText = isComplete
666 |             ? `Диагностика хоста завершена (${currentCount} шаго
    | в)`
667 |             : `Диагностический конвейер хоста (${currentCount} ш
    | агов)...`;
668 |         const headerClass = isComplete ? 'text-success' : 'tex
    | t-info';
669 |         const collapseClass = isComplete ? 'd-none' : '';

```

```

670 |         const chevronIcon = isComplete ? 'bi-chevron-down' : '
| bi-chevron-up';
671 |         const lastTitle = stages[stages.length - 1]?.title ||
| 'Зонд';
672 |
673 |         return `
674 |             <div class="mb-2 p-2 bg-black bg-opacity-40 border b
| order-secondary border-opacity-30 rounded shadow-sm" id="${msg
| Id}-stages-wrapper">
675 |                 <div class="d-flex align-items-center justify-cont
| ent-between flex-wrap gap-1" onclick="const b=document.getElem
| entById('${msgId}-stages-content'); const c=document.getElemen
| tById('${msgId}-stages-chevron'); if(b){b.classList.toggle('d-
| none'); if(c){c.classList.toggle('bi-chevron-down'); c.classLi
| st.toggle('bi-chevron-up');}" style="cursor:pointer;" title="
| Нажмите, чтобы развернуть/свернуть детали этапов">
676 |                     <span class="${headerClass} small fw-semibold d-
| flex align-items-center gap-1.5" style="font-size:0.75rem;">
677 |                         ${headerIcon}
678 |                         <span>${escapeHtml(headerText)}</span>
679 |                     </span>
680 |                     <div class="d-flex align-items-center gap-1">
681 |                         <span class="badge bg-dark text-secondary bord
| er border-secondary-subtle font-monospace" style="font-size:0.
| 63rem;">${escapeHtml(lastTitle)}</span>
682 |                         <i class="bi ${chevronIcon} text-muted" id="${msg
| Id}-stages-chevron" style="font-size:0.65rem;"></i>
683 |                     </div>
684 |                 </div>
685 |                 <div class="mt-1.5 pt-1.5 border-top border-second
| ary border-opacity-25 d-flex flex-column gap-1.5 ${collapseCla
| ss}" id="${msgId}-stages-content">
686 |                     ${stages.map((st, sIdx) => {
687 |                         const isLatest = sIdx === stages.length - 1 &&
| !isComplete;
688 |                         const icon = isLatest
689 |                         ? '<div class="spinner-grow spinner-grow-sm
| text-info" style="width: 0.55rem; height: 0.55rem;" role="stat

```

```

us"></div>'
690         : '<span class="text-success small" style="font-size:0.7rem;">❏ </span>';
691         const textClass = isLatest ? 'text-light fw-medium' : 'text-muted';
692         return `
693             <div class="small ${textClass} d-flex flex-column ps-0.5" style="font-size: 0.74rem; line-height: 1.35;">
694                 <div class="d-flex align-items-center gap-1.5">
695                     <span>${icon}</span>
696                     <span>${escapeHtml(st.message || '')}</span>
697                 </div>
698                 ${st.details ? `<div class="text-secondary ps-3 font-monospace" style="font-size: 0.68rem; word-break: break-all;">❏ ${escapeHtml(st.details)}</div>` : ''}
699                 ${st.generated_prompt ? `
700                     <div class="mt-1 ps-3">
701                         <button class="btn btn-sm btn-outline-warning py-0 px-2 d-inline-flex align-items-center gap-1 font-monospace" type="button" onclick="event.stopPropagation(); const p=document.getElementById('${msgId}-stage-prompt-${sIdx}'); const ic=document.getElementById('${msgId}-stage-prompt-ic-${sIdx}'); if(p){p.classList.toggle('d-none'); if(ic){ic.classList.toggle('bi-chevron-down'); ic.classList.toggle('bi-chevron-up');}" style="font-size: 0.68rem;">
702                             <i class="bi bi-cpu-fill"></i>
703                             <span>Показать отправленный промт</span>
704                             <i class="bi bi-chevron-down text-warning" id="${msgId}-stage-prompt-ic-${sIdx}" style="font-size: 0.6rem;"></i>
705                         </button>
706                         <div id="${msgId}-stage-prompt-${sIdx}" class="d-none mt-1.5 p-2 bg-black bg-opacity-70 border border-warning border-opacity-40 rounded text-start" onclick="event.stopPropagation();">

```



```

707 |         <div class="d-flex justify-content-b
| etween align-items-center text-warning mb-1" style="font-size:
| 0.68rem;">
708 |             <span><i class="bi bi-terminal me-
| 1"></i>Точный текст промпта, отправленный модели:</span>
709 |             <button class="btn btn-dark btn-sm
| py-0 px-1.5 text-secondary border border-secondary-subtle" ty
| pe="button" onclick="navigator.clipboard.writeText(this.closes
| t('#${msgId}-stage-prompt-${sIdx}').querySelector('pre')?.inne
| rText || ''); this.innerText='Скопировано!'; setTimeout(()=>th
| is.innerText='Копировать', 1500);" style="font-size: 0.65rem;"
| >Копировать</button>
710 |         </div>
711 |         <pre class="m-0 p-1.5 bg-dark bg-opa
| city-75 text-light border border-secondary border-opacity-30 r
| ounded font-monospace" style="font-size: 0.70rem; line-height:
| 1.35; max-height: 220px; overflow-y: auto; white-space: pre-w
| rap; word-break: break-word;">${escapeHtml(st.generated_prompt
| )}</pre>
712 |         </div>
713 |     </div>
714 |     ` : ''}
715 | </div>
716 | `;
717 |     }).join('')}
718 | </div>
719 | </div>
720 | `;
721 | };
722 |
723 |     const initialStage = {
724 |         stage: 'init',
725 |         title: 'Инициализация',
726 |         message: '□ Подключение к локальной подсистеме Window
| s API и сервисам хоста...',
727 |         details: 'Подготовка системных интерфейсов WMI/PnP, ко
| ллекторов и контекста сессии',
728 |     };

```

```

729 |     stagesLog.push(initialStage);
730 |
731 |     const loadingHtml = `
732 |         <div class="d-flex gap-2 mb-3" id="${msgId}-container"
733 |     >
734 |         <div class="fs-5">□ </div>
735 |         <div class="bg-secondary bg-opacity-25 p-2.5 rounded
736 |     border border-secondary-subtle small text-light flex-grow-1 s
737 |     hadow-sm" id="${msgId}-body">
738 |             ${renderStagesHtml(stagesLog, false)}
739 |         </div>
740 |     </div>
741 | `;
742 |     chatHistory.insertAdjacentHTML('beforeend', loadingHtml)
743 | ;
744 |     chatHistory.scrollTop = chatHistory.scrollHeight;
745 |
746 |     try {
747 |         const useRag = ragCheck ? ragCheck.checked : true;
748 |         const keepContext = keepContextCheck ? keepContextChec
749 |         k.checked : false;
750 |
751 |         const res = await fetch('/api/v1/scenarios/chat/stream
752 |         ', {
753 |             method: 'POST',
754 |             headers: { 'Content-Type': 'application/json' },
755 |             body: JSON.stringify({
756 |                 message: message,
757 |                 conversation_id: currentConversationId,
758 |                 auto_create_skill: false,
759 |                 keep_context: keepContext,
760 |                 use_rag: useRag,
761 |             })),
762 |         });
763 |
764 |         if (!res.ok) {
765 |             throw new Error(`Ошибка ответа сервера: HTTP ${res.s
766 |             tatus}`);

```

```

760 |     }
761 |
762 |     const reader = res.body.getReader();
763 |     const decoder = new TextDecoder('utf-8');
764 |     let buffer = '';
765 |     let finalData = null;
766 |     let streamedReplyText = '';
767 |
768 |     while (true) {
769 |         const { done, value } = await reader.read();
770 |         if (done) break;
771 |         buffer += decoder.decode(value, { stream: true });
772 |
773 |         const lines = buffer.split('\n\n');
774 |         buffer = lines.pop(); // Оставляем неполный хвост в
буфере
775 |
776 |         for (const line of lines) {
777 |             const cleanLine = line.trim();
778 |             if (!cleanLine.startsWith('data:')) continue;
779 |             let evt = null;
780 |             try {
781 |                 evt = JSON.parse(cleanLine.substring(5).trim());
782 |             } catch (pErr) {
783 |                 console.warn('[ScenariosTab] SSE parse warning: '
, pErr);
784 |                 continue;
785 |             }
786 |             if (!evt) continue;
787 |
788 |             if (evt.type === 'stage') {
789 |                 const lastSt = stagesLog[stagesLog.length - 1];
790 |                 if (!lastSt || lastSt.stage !== evt.stage || las
tSt.message !== evt.message) {
791 |                     stagesLog.push(evt);
792 |                 }
793 |                 if (statusInd) statusInd.innerText = evt.message
|| 'Обработка...';

```

```

794
795         const bodyElem = document.getElementById(`${msgId}-body`);
796         if (bodyElem) {
797             const { chatText } = parseChatAndVoice(streamedReplyText);
798             bodyElem.innerHTML = `
799                 ${renderStagesHtml(stagesLog, false)}
800                 ${streamedReplyText ? `<div class="chat-streaming-content mt-1.5">${formatMarkdown(chatText)}<span class="streaming-cursor">█</span></div>` : ''}
801             `;
802             chatHistory.scrollTop = chatHistory.scrollHeight;
803         }
804     } else if (evt.type === 'chunk') {
805         streamedReplyText += (evt.content || '');
806         const bodyElem = document.getElementById(`${msgId}-body`);
807         if (bodyElem) {
808             const { chatText } = parseChatAndVoice(streamedReplyText);
809             bodyElem.innerHTML = `
810                 ${renderStagesHtml(stagesLog, false)}
811                 <div class="chat-streaming-content mt-1.5">${formatMarkdown(chatText)}<span class="streaming-cursor">█</span></div>
812             `;
813             chatHistory.scrollTop = chatHistory.scrollHeight;
814         }
815     } else if (evt.type === 'done') {
816         finalData = evt;
817     } else if (evt.type === 'error') {
818         const streamErr = new Error(evt.error || 'Ошибка в потоке');
819         streamErr.details = evt.details || '';
820         streamErr.errorType = evt.error_type || 'StreamE

```

```

      rror';
821         streamErr.rawEvent = evt;
822         throw streamErr;
823     }
824 }
825 }
826
827     if (!finalData) {
828         throw new Error('Не удалось получить завершающий отв
ет от сервера.');
```

829 }

830

```

831     const fullReply = finalData.reply || streamedReplyText
|| 'Ответ получен.';
832     const { chatText, voiceText } = parseChatAndVoice(full
Reply);
833     const formattedReply = formatMarkdown(chatText || 'Отв
ет получен.');
```

834

```

835     let remediationHtml = '';
836     if (Array.isArray(finalData.remediation_actions) && fi
nalData.remediation_actions.length > 0) {
837         const actionsList = finalData.remediation_actions;
838         remediationHtml = `
839             <div class="mt-2.5 p-2.5 bg-black bg-opacity-30 bo
rder border-warning border-opacity-50 rounded" id="remediation
-container-${msgId}">
840                 <div class="d-flex align-items-center justify-co
ntent-between mb-2">
841                     <span class="fw-semibold text-warning small d-
flex align-items-center gap-1">
842                         <i class="bi bi-shield-check text-warning"><
/i> <span>Предложенные действия и исправления:</span>
843                     </span>
844                     <span class="badge bg-warning text-dark small"
style="font-size: 0.68rem;">SafeOps (${actionsList.length})</
span>
845                 </div>
```

```

846 |         <div class="d-flex flex-column gap-1.5" id="reme
      | diation-buttons- $\{msgId\}$ ">
847 |              $\{actionsList.map((act, aIdx) => \{$ 
848 |                 const isCaution = (act.risk || '').toLowerCase
      | se() === 'caution' || (act.risk || '').toLowerCase() === 'crit
      | ical';
849 |                 const badgeRisk = isCaution ? '<span class="
      | badge bg-danger-subtle text-danger border border-danger-subtle
      | ms-1" style="font-size:0.65rem;">Требует подтверждения</span>
      | ' : '<span class="badge bg-info-subtle text-info border border
      | -info-subtle ms-1" style="font-size:0.65rem;">Безопасно</span>
      | ' ;
850 |                 return `
851 |                     <div class="d-flex align-items-center just
      | ify-content-between p-2 bg-dark bg-opacity-75 rounded border b
      | order-secondary border-opacity-25 gap-2 flex-wrap" id="act-row
      | - $\{msgId\}$ - $\{aIdx\}$ ">
852 |                         <div class="small text-light d-flex flex
      | -column" style="font-size: 0.78rem;">
853 |                             <div><strong> $\{escapeHtml(act.title ||$ 
      | act.action_id)</strong> $\{badgeRisk\}$ </div>
854 |                              $\{act.description ? `<span class="text$ 
      | -muted mt-0.5" style="font-size: 0.72rem;"> $\{escapeHtml(act.de$ 
      | scription)</span>` : ''}
855 |                              $\{act.execution\_command ? `<code class$ 
      | ="text-warning-emphasis bg-black bg-opacity-50 px-1 py-0.5 rou
      | nded mt-1 font-monoospace" style="font-size: 0.68rem; word-brea
      | k: break-all;"> $\{escapeHtml(act.execution\_command)\}$ </code>` :
      | ''}
856 |                         </div>
857 |                         <button class="btn btn-warning btn-sm py
      | -1 px-3 d-flex align-items-center gap-1.5 shadow-sm fw-semibol
      | d btn-exec-action"
858 |                             id="btn-act- $\{msgId\}$ - $\{aIdx\}$ "
859 |                             data-msg-id=" $\{msgId\}$ "
860 |                             data-action-id=" $\{escapeHtml(act.actio$ 
      | n_id)</span>}"
861 |                             data-action-type=" $\{escapeHtml(act.act$ 

```

```

ion_type))"
862 |             data-title="${escapeHtml(act.title)}"
863 |             data-description="${escapeHtml(act.des
cription)}"
864 |             data-target="${escapeHtml(act.target)}
"
865 |             data-risk="${escapeHtml(act.risk)}"
866 |             data-command="${escapeHtml(act.executi
on_command)}"
867 |             style="font-size: 0.78rem;">
868 |             <i class="bi bi-play-circle-fill"></i>
869 |             <span>Подтвердить</span>
870 |         </button>
871 |     </div>
872 |     `;
873 |     }).join('')}
874 |     </div>
875 | </div>
876 | `;
877 | }
878 |
879 | let skillActionHtml = '';
880 | if (finalData.created_skill) {
881 |     const isNew = finalData.created_skill.is_new !== fal
se;
882 |     const badgeTitle = isNew ? '🆕 <strong>Автоматически
создан навык:</strong>' : '🔄 <strong>Использован существующий
навык:</strong>';
883 |     const badgeClass = isNew ? 'bg-success bg-opacity-10
border-success border-opacity-50 text-success' : 'bg-info bg-
opacity-10 border-info border-opacity-50 text-info';
884 |     skillActionHtml = `
885 |         <div class="mt-2 p-2 ${badgeClass} border rounded
small" id="skill-action-container-${msgId}">
886 |             ${badgeTitle} <code>${escapeHtml(finalData.creat
ed_skill.path)}</code>
887 |             <div class="text-muted small mt-1">${escapeHtml(
finalData.created_skill.description_ru)}</div>

```

```

888         </div>
889     `;
890     } else if (finalData.tool_plan) {
891         skillActionHtml = `
892             <div class="mt-2 pt-2 border-top border-secondary
border-opacity-25 d-flex align-items-center gap-2 flex-wrap" i
d="skill-action-container-${msgId}">
893                 <button class="btn btn-outline-success btn-sm d-
flex align-items-center gap-1.5 py-1 px-2.5 shadow-sm" id="btn
-save-skill-${msgId}" type="button">
894                     <i class="bi bi-bookmark-check-fill text-succe
ss"></i>
895                     <span>Ответ правильный, запомнить как навык</s
pan>
896                 </button>
897                 <button class="btn btn-outline-secondary btn-sm
d-flex align-items-center gap-1.5 py-1 px-2 shadow-sm" id="btn
-edit-skill-${msgId}" type="button">
898                     <i class="bi bi-pencil-square"></i>
899                     <span>Скорректировать</span>
900                 </button>
901             </div>
902             <div id="skill-edit-form-${msgId}" class="mt-2 p-2
.5 bg-dark bg-opacity-50 border border-secondary-subtle rounde
d small d-none">
903                 <div class="fw-semibold text-warning mb-2"><i cl
ass="bi bi-sliders me-1"></i>Корректировка навыка перед сохран
ением</div>
904                 <div class="mb-2">
905                     <label class="form-label text-muted mb-1" styl
e="font-size:0.75rem;">Заголовок навыка:</label>
906                     <input type="text" class="form-control form-co
ntrol-sm bg-dark text-light border-secondary" id="edit-title-$
{msgId}" value="${escapeHtml(finalData.tool_plan.tool_title ||
'')}">
907                 </div>
908                 <div class="mb-2">
909                     <label class="form-label text-muted mb-1" styl

```



```

e="font-size:0.75rem;">Описание (русский язык):</label>
910     <textarea class="form-control form-control-sm
bg-dark text-light border-secondary" id="edit-desc-`${msgId}`" r
ows="2">`${escapeHtml(finalData.tool_plan.description_ru || '')}`
</textarea>
911     </div>
912     <div class="mb-2">
913         <label class="form-label text-muted mb-1" styl
e="font-size:0.75rem;">Команда проверки / скрипт:</label>
914         <input type="text" class="form-control form-co
ntrol-sm bg-dark text-light border-secondary font-monospace" i
d="edit-cmd-`${msgId}`" value="`${escapeHtml(finalData.command_ex
ecuted || finalData.tool_plan.probe_script || '')}`">
915     </div>
916     <div class="d-flex gap-2 justify-content-end mt-
2">
917         <button class="btn btn-secondary btn-sm py-1 p
x-2" id="btn-cancel-edit-`${msgId}`" type="button">Отмена</butto
n>
918         <button class="btn btn-success btn-sm py-1 px-
2.5" id="btn-save-edited-`${msgId}`" type="button">
919         <i class="bi bi-check-circle-fill me-1"></i>
Сохранить скорректированный навык
920     </button>
921 </div>
922 </div>
923 `;
924 }
925
926 let metaInfoHtml = '';
927 const agentsList = finalData.agents_used || [];
928 const knowledgeList = finalData.knowledge_used || [];
929
930 if (agentsList.length > 0 || knowledgeList.length > 0)
{
931     const totalCount = agentsList.length + knowledgeList
.length;
932     const metaCollapseId = `meta-details-`${msgId}``;

```

```

933         metaInfoHtml = `
934             <div class="mt-2.5 pt-2 border-top border-secondary
y border-opacity-25" id="meta-container-${msgId}">
935                 <div class="d-flex align-items-center justify-co
ntent-between flex-wrap gap-1">
936                     <button class="btn btn-sm btn-link p-0 text-in
fo text-decoration-none d-flex align-items-center gap-1.5" typ
e="button" onclick="const el=document.getElementById('${metaCo
llapseId}'); const ic=document.getElementById('meta-icon-${msg
Id}'); if(el){el.classList.toggle('d-none'); if(ic) { ic.class
List.toggle('bi-chevron-down'); ic.classList.toggle('bi-chevro
n-up'); }}" style="font-size: 0.75rem;">
937                         <i class="bi bi-diagram-3-fill text-info"></
i>
938                         <span><strong>Задействованные знания и агент
ы</strong> (${totalCount})</span>
939                         <i class="bi bi-chevron-down text-muted" id=
"meta-icon-${msgId}" style="font-size: 0.7rem;"></i>
940                     </button>
941                     <div class="d-flex gap-1 flex-wrap">
942                         ${agentsList.map(a => `<span class="badge bg
-primary bg-opacity-25 text-primary border border-primary bord
er-opacity-25" style="font-size: 0.65rem;" title="${escapeHtml
(a.role || '')}">${escapeHtml(a.icon || '□')} ${escapeHtml(a.
name)}</span>`).slice(0, 2).join('')}
943                     </div>
944                 </div>
945
946                 <div id="${metaCollapseId}" class="d-none mt-2 p
-2 bg-black bg-opacity-40 border border-secondary border-opaci
ty-50 rounded small">
947                     ${agentsList.length > 0 ? `
948                         <div class="mb-2">
949                             <div class="text-warning fw-semibold mb-1"
style="font-size: 0.73rem;">
950                                 <i class="bi bi-robot me-1"></i>Задейств
ованные агенты и модули:
951                                 </div>

```

```

952 |         <div class="d-flex flex-column gap-1 ps-1"
    | >
953 |             ${agentsList.map(ag => `
954 |                 <div class="d-flex align-items-start g
    | ap-1.5 text-light" style="font-size: 0.73rem;">
955 |                     <span>${escapeHtml(ag.icon || '❏')}
    | </span>
956 |                     <div>
957 |                         <strong>${escapeHtml(ag.name)}</st
    | rong>
958 |                         <span class="text-muted ms-1">— ${
    | escapeHtml(ag.role || 'Системный агент')}</span>
959 |                     </div>
960 |                 </div>
961 |             `).join('')}
962 |         </div>
963 |     </div>
964 |     ` : ''}
965 |
966 |     ${knowledgeList.length > 0 ? `
967 |         <div>
968 |             <div class="text-info fw-semibold mb-1" st
    | yle="font-size: 0.73rem;">
969 |                 <i class="bi bi-book-half me-1"></i>Испо
    | льзованные источники знаний и артефакты:
970 |             </div>
971 |             <div class="d-flex flex-column gap-1 ps-1"
    | >
972 |                 ${knowledgeList.map(kn => `
973 |                     <div class="d-flex align-items-start g
    | ap-1.5 text-light" style="font-size: 0.73rem;">
974 |                         <span class="badge bg-info bg-opacit
    | y-25 text-info border border-info border-opacity-25" style="fo
    | nt-size: 0.65rem;">${escapeHtml(kn.badge || 'База')}</span>
975 |                         <div>
976 |                             <strong>${escapeHtml(kn.title)}</s
    | trong>
977 |                             <span class="text-muted ms-1">— ${

```

```

    escapeHtml(kn.description || '')}}</span>
978 |         </div>
979 |         </div>
980 |         `).join('')}`
981 |     </div>
982 | </div>
983 |     ` : ''}`
984 | </div>
985 | </div>
986 | `;
987 | }
988 |
989 |     let promptInfoHtml = '';
990 |     const generatedPromptText = finalData.generated_prompt
    || (stagesLog.find(s => s.stage === 'synthesizing' && s.generated_prompt)?.generated_prompt);
991 |     if (generatedPromptText) {
992 |         const promptCollapseId = `prompt-details-${msgId}`;
993 |         const promptCodeId = `prompt-code-${msgId}`;
994 |         promptInfoHtml = `
995 |             <div class="mt-2.5 pt-2 border-top border-secondary border-opacity-25" id="prompt-container-${msgId}">
996 |                 <div class="d-flex align-items-center justify-content-between flex-wrap gap-1">
997 |                     <button class="btn btn-sm btn-link p-0 text-warning text-decoration-none d-flex align-items-center gap-1.5"
    type="button" onclick="const el=document.getElementById('${promptCollapseId}'); const ic=document.getElementById('prompt-icon-${msgId}'); if(el){el.classList.toggle('d-none'); if(ic) { ic.classList.toggle('bi-chevron-down'); ic.classList.toggle('bi-chevron-up'); }}" style="font-size: 0.75rem;">
998 |                         <i class="bi bi-cpu-fill text-warning"></i>
999 |                         <span><strong>Сформированный пром프트 модели</strong></span>
1000 |                     <i class="bi bi-chevron-down text-muted" id="prompt-icon-${msgId}" style="font-size: 0.7rem;"></i>
1001 |                     </button>
1002 |                     <div class="d-flex align-items-center gap-1">

```

```

1003 |         <button class="btn btn-dark btn-sm py-0 px-2
      | text-secondary border border-secondary-subtle d-flex align-it
      | ems-center gap-1" type="button" onclick="navigator.clipboard.w
      | riteText(document.getElementById('${promptCodeId}')?.innerText
      | || ' '); const s=this.querySelector('span'); if(s){s.innerText
      | ='Скопировано!'; setTimeout(()=>s.innerText='Копировать', 1800
      | );}" style="font-size: 0.68rem;" title="Скопировать отправленн
      | ый промпт">
1004 |             <i class="bi bi-clipboard"></i>
1005 |             <span>Копировать</span>
1006 |         </button>
1007 |     </div>
1008 | </div>
1009 |
1010 |     <div id="${promptCollapseId}" class="d-none mt-2
      | p-2.5 bg-black bg-opacity-60 border border-warning border-opa
      | city-40 rounded">
1011 |         <div class="d-flex align-items-center justify-
      | content-between text-warning small mb-1.5" style="font-size: 0
      | .72rem;">
1012 |             <span><i class="bi bi-terminal me-1"></i>Точ
      | ный текст запроса, отправленный языковой модели:</span>
1013 |             <span class="badge bg-warning bg-opacity-10
      | text-warning border border-warning border-opacity-25 font-mono
      | space" style="font-size: 0.65rem;">${generatedPromptText.lengt
      | h} симв.</span>
1014 |         </div>
1015 |         <pre class="m-0 p-2 bg-dark bg-opacity-75 text
      | -light border border-secondary border-opacity-30 rounded font-
      | monospace small" id="${promptCodeId}" style="font-size: 0.72re
      | m; line-height: 1.4; max-height: 280px; overflow-y: auto; whit
      | e-space: pre-wrap; word-break: break-word;">${escapeHtml(gener
      | atedPromptText)}</pre>
1016 |     </div>
1017 | </div>
1018 | `;
1019 | }
1020 |

```

```

1021 |         const trainingActionHtml = `
1022 |             <div class="mt-2.5 pt-2 border-top border-secondary
| border-opacity-25 d-flex align-items-center justify-content-be
| tween flex-wrap gap-2" id="training-save-container-${msgId}">
1023 |                 <button class="btn btn-outline-info btn-sm d-flex
| align-items-center gap-1.5 py-1 px-2.5 shadow-sm btn-save-trai
| ning-qa" id="btn-save-training-${msgId}" type="button" title="
| Сохранить пару вопрос-ответ в базу данных Test Computer для RA
| G и последующего обучения (тюнинга) модели">
1024 |                     <i class="bi bi-mortarboard-fill text-info"></i>
1025 |                     <span>Сохранить ответ для обучения модели</span>
1026 |                 </button>
1027 |                 <span class="text-muted" style="font-size: 0.68rem
| ;"><i class="bi bi-database me-1"></i>data/tc/approved_respons
| es</span>
1028 |             </div>
1029 |         `;
1030 |
1031 |         const bodyElem = document.getElementById(`${msgId}-bod
| y`);
1032 |         if (bodyElem) {
1033 |             bodyElem.innerHTML = `
1034 |                 ${renderStagesHtml(stagesLog, true)}
1035 |                 <div>${formattedReply}</div>
1036 |                 ${remediationHtml}
1037 |                 ${skillActionHtml}
1038 |                 ${trainingActionHtml}
1039 |                 ${metaInfoHtml}
1040 |                 ${promptInfoHtml}
1041 |             `;
1042 |         }
1043 |         chatHistory.scrollTop = chatHistory.scrollHeight;
1044 |
1045 |         // Wire up save response for training & RAG button
1046 |         const btnSaveTraining = document.getElementById(`btn-s
| ave-training-${msgId}`);
1047 |         if (btnSaveTraining) {
1048 |             btnSaveTraining.onclick = async () => {

```

```

1049 |         btnSaveTraining.disabled = true;
1050 |         btnSaveTraining.innerHTML = `<div class="spinner-b
| order spinner-border-sm text-info me-1" role="status"></div><s
| pan>Сохранение...</span>`;
1051 |         try {
1052 |             const res = await fetch('/api/v1/scenarios/save-
| approved-response', {
1053 |                 method: 'POST',
1054 |                 headers: { 'Content-Type': 'application/json'
| },
1055 |                 body: JSON.stringify({
1056 |                     user_id: 'tc_admin',
1057 |                     query: message,
1058 |                     chat_text: chatText,
1059 |                     voice_text: voiceText || '',
1060 |                     system_context: {
1061 |                         command_executed: finalData.command_execut
| ed || '',
1062 |                         tool_plan: finalData.tool_plan || null,
1063 |                         platform: 'Windows'
1064 |                     },
1065 |                     tags: ['tc', 'diagnostics', 'training']
1066 |                 }),
1067 |             });
1068 |             if (!res.ok) throw new Error(`HTTP ${res.status}
| `);
1069 |             const resData = await res.json();
1070 |             const container = document.getElementById(`train
| ing-save-container-${msgId}`);
1071 |             if (container) {
1072 |                 container.innerHTML = `
1073 |                     <div class="p-1.5 px-2.5 bg-success bg-opaci
| ty-10 border border-success border-opacity-50 text-success rou
| nded small d-flex align-items-center gap-1.5 w-100 shadow-sm">
1074 |                         <i class="bi bi-check-circle-fill text-suc
| cess"></i>
1075 |                         <span><strong>Успешно:</strong> ответ сохр
| анён для обучения модели и RAG (<code>data/tc/approved_respons

```

```

es</code>)</span>
1076         </div>
1077         `;
1078     }
1079     if (statusInd) statusInd.innerText = 'Ответ сохр
анён в базу для обучения модели!';
1080     } catch (saveErr) {
1081         console.error('[ScenariosTab] Error saving train
ing QA:', saveErr);
1082         btnSaveTraining.disabled = false;
1083         btnSaveTraining.innerHTML = `<i class="bi bi-exc
lamation-triangle-fill text-danger me-1"></i><span>Ошибка сохр
анения. Повторить</span>`;
1084     }
1085     };
1086 }
1087
1088
1089 // Wire up manual save & inline edit logic
1090 if (!finalData.created_skill && finalData.tool_plan) {
1091     const saveBtn = document.getElementById(`btn-save-sk
ill-${msgId}`);
1092     const editBtn = document.getElementById(`btn-edit-sk
ill-${msgId}`);
1093     const editForm = document.getElementById(`skill-edit
-form-${msgId}`);
1094     const cancelEditBtn = document.getElementById(`btn-c
ancel-edit-${msgId}`);
1095     const saveEditedBtn = document.getElementById(`btn-s
ave-edited-${msgId}`);
1096
1097     const executeSave = async (payload) => {
1098         const container = document.getElementById(`skill-a
ction-container-${msgId}`);
1099         if (saveBtn) saveBtn.disabled = true;
1100         if (saveEditedBtn) saveEditedBtn.disabled = true;
1101         if (saveBtn) saveBtn.innerHTML = `<div class="spin
ner-border spinner-border-sm text-success me-1" role="status">

```



```

1102 | </div><span>Сохранение...</span>`;
1103 |         try {
1104 |             const saveRes = await fetch('/api/v1/scenarios/s
ave-skill', {
1105 |                 method: 'POST',
1106 |                 headers: { 'Content-Type': 'application/json'
},
1107 |                 body: JSON.stringify(payload),
1108 |             });
1109 |             if (!saveRes.ok) throw new Error(`HTTP ${saveRes
.status}`);
1110 |             const savedData = await saveRes.json();
1111 |             if (container) {
1112 |                 container.innerHTML = `
1113 |                     <div class="p-2 bg-success bg-opacity-10 bor
der border-success border-opacity-50 text-success rounded smal
l w-100">
1114 |                         □ <strong>Навык успешно сохранён:</strong>
<code>${escapeHtml(savedData.path)}</code>
1115 |                         <div class="text-muted small mt-1">${escap
eHtml(savedData.description_ru)}</div>
1116 |                     </div>
1117 |                 `;
1118 |             }
1119 |             if (editForm) editForm.classList.add('d-none');
1120 |             if (statusInd) statusInd.innerText = `Навык '${s
avedData.name}' сохранён!`;
1121 |         } catch (saveErr) {
1122 |             console.error('[ScenariosTab] Error saving skill
:', saveErr);
1123 |             if (saveBtn) {
1124 |                 saveBtn.disabled = false;
1125 |                 saveBtn.innerHTML = `<i class="bi bi-exclamat
ion-triangle-fill text-danger me-1"></i><span>Ошибка. Повторить
</span>`;
1126 |             }
1127 |             if (saveEditedBtn) saveEditedBtn.disabled = fals

```

```

e;
1128     }
1129 };
1130
1131 if (saveBtn) {
1132     saveBtn.addEventListener('click', () => {
1133         executeSave({
1134             tool_name: finalData.tool_plan.tool_name,
1135             tool_title: finalData.tool_plan.tool_title,
1136             description_ru: finalData.tool_plan.descriptio
n_ru,
1137             probe_type: finalData.tool_plan.probe_type ||
'powershell',
1138             collector_name: finalData.tool_plan.collector_
name,
1139             probe_script: finalData.tool_plan.probe_script
,
1140             instructions: finalData.tool_plan.instructions
,
1141             command_executed: finalData.command_executed,
1142         });
1143     });
1144 }
1145
1146 if (editBtn && editForm) {
1147     editBtn.addEventListener('click', () => {
1148         editForm.classList.toggle('d-none');
1149     });
1150 }
1151
1152 if (cancelEditBtn && editForm) {
1153     cancelEditBtn.addEventListener('click', () => {
1154         editForm.classList.add('d-none');
1155     });
1156 }
1157
1158 if (saveEditedBtn) {
1159     saveEditedBtn.addEventListener('click', () => {

```

```

1160 |         const editedTitle = document.getElementById(`edit-
1161 | t-title-${msgId}`)?.value || finalData.tool_plan.tool_title;
1162 |         const editedDesc = document.getElementById(`edit-
1163 | -desc-${msgId}`)?.value || finalData.tool_plan.description_ru;
1164 |         const editedCmd = document.getElementById(`edit-
1165 | cmd-${msgId}`)?.value || finalData.command_executed;
1166 |
1167 |         executeSave({
1168 |             tool_name: finalData.tool_plan.tool_name,
1169 |             tool_title: editedTitle,
1170 |             description_ru: editedDesc,
1171 |             probe_type: finalData.tool_plan.probe_type ||
1172 | 'powershell',
1173 |             collector_name: finalData.tool_plan.collector_
1174 | name,
1175 |             probe_script: editedCmd,
1176 |             instructions: finalData.tool_plan.instructions
1177 | ,
1178 |             command_executed: editedCmd,
1179 |         });
1180 |     });
1181 | }
1182 | }
1183 |
1184 | // Wire up remediation action execution buttons
1185 | if (Array.isArray(finalData.remediation_actions) && fi
1186 | nalData.remediation_actions.length > 0) {
1187 |     const remContainer = document.getElementById(`remedi
1188 | ation-container-${msgId}`);
1189 |     if (remContainer) {
1190 |         remContainer.querySelectorAll('.btn-exec-action').
1191 | forEach(btn => {
1192 |             btn.addEventListener('click', async () => {
1193 |                 const actionId = btn.dataset.actionId;
1194 |                 const actionType = btn.dataset.actionType;
1195 |                 const title = btn.dataset.title;
1196 |                 const description = btn.dataset.description;
1197 |                 const target = btn.dataset.target;

```

```

1189         const risk = btn.dataset.risk;
1190         const command = btn.dataset.command;
1191
1192         // Confirm if risk is caution or critical
1193         const isRisky = (risk || '').toLowerCase() ===
1194         'caution' || (risk || '').toLowerCase() === 'critical';
1195         if (isRisky) {
1196             const ok = confirm(`Подтвердите применение и
1197             справления:\n\n«${title}»\n\nКоманда: ${command || target}`);
1198             if (!ok) return;
1199         }
1200         btn.disabled = true;
1201         btn.innerHTML = `<div class="spinner-border sp
1202         inner-border-sm text-warning" role="status"></div><span>Примен
1203         ение...</span>`;
1204         if (statusInd) statusInd.innerText = `Применен
1205         ие '${title}'...`;
1206
1207         try {
1208             const execRes = await fetch(`/api/v1/scenari
1209             os/execute-fix`, {
1210                 method: 'POST',
1211                 headers: { 'Content-Type': 'application/js
1212                 on' },
1213                 body: JSON.stringify({
1214                     action_id: actionId,
1215                     action_type: actionType,
1216                     title: title,
1217                     description: description,
1218                     target: target,
1219                     risk: risk,
1220                     execution_command: command,
1221                     confirmed_by_user: true,
1222                 }),
1223             });
1224
1225         if (!execRes.ok) throw new Error(`HTTP ${exe

```

```

    cRes.status}`);
1220         const resData = await execRes.json();
1221         if (resData.success) {
1222             btn.className = 'btn btn-success btn-sm py-
-1 px-3 d-flex align-items-center gap-1.5 shadow-sm';
1223             btn.innerHTML = `bi bi-check-circle-fill`
`></i><span>Подтверждено и выполнено</span>`;
1224             if (statusInd) statusInd.innerText = `Дейс
твие '${title}' успешно выполнено!`;
1225             } else {
1226                 btn.disabled = false;
1227                 btn.className = 'btn btn-danger btn-sm py-
0.5 px-2.5 d-flex align-items-center gap-1 shadow-sm';
1228                 btn.innerHTML = `bi bi-exclamati
on-triangle-fill`
`></i><span>Ошибка</span>`;
1229                 btn.title = resData.error_message || resDa
ta.message;
1230                 if (statusInd) statusInd.innerText = `Ошиб
ка: ${resData.error_message || resData.message}`;
1231             }
1232         } catch (exErr) {
1233             console.error('[ScenariosTab] Error executin
g fix:', exErr);
1234             btn.disabled = false;
1235             btn.className = 'btn btn-outline-danger btn-
sm py-0.5 px-2.5 d-flex align-items-center gap-1 shadow-sm';
1236             btn.innerHTML = `bi bi-arrow-repea
t`
`></i><span>Повторить</span>`;
1237             if (statusInd) statusInd.innerText = `Сбой:
${exErr.message}`;
1238             }
1239         });
1240     });
1241 }
1242 }
1243
1244     if (statusInd) statusInd.innerText = 'Ответ сформирова
н';

```

```

1245     } catch (err) {
1246         console.error('[ScenariosTab] Chat error:', err);
1247         const bodyElem = document.getElementById(`${msgId}-body`);
1248         if (bodyElem) {
1249             const errDetail = err.details || (err.rawEvent ? JSON.stringify(err.rawEvent, null, 2) : '');
1250             const errorTypeBadge = err.errorType ? `<span class="
badge bg-danger bg-opacity-25 text-danger border border-dange
r border-opacity-50 font-monospace small ms-1" style="font-siz
e: 0.68rem;">${escapeHtml(err.errorType)}</span>` : '';
1251             bodyElem.innerHTML = `
1252                 ${stagesLog.length > 0 ? renderStagesHtml(stagesLo
g, false) : ''}
1253                 <div class="text-danger p-2.5 bg-danger bg-opacity
-10 border border-danger border-opacity-40 rounded mt-2">
1254                     <div class="fw-semibold d-flex align-items-cente
r justify-content-between flex-wrap gap-1 text-danger small">
1255                         <span class="d-flex align-items-center gap-1.5
">
1256                             <i class="bi bi-exclamation-triangle-fill te
xt-danger"></i>
1257                             <span>Ошибка: ${escapeHtml(err.message || 'Н
еизвестная ошибка выполнения')}</span>
1258                             </span>
1259                             ${errorTypeBadge}
1260                         </div>
1261                         ${errDetail && errDetail.trim() !== (err.message
|| '').trim() ? `
1262                             <div class="mt-2 pt-1.5 border-top border-dang
er border-opacity-25">
1263                                 <div class="text-muted small mb-1" style="fo
nt-size: 0.72rem;"><i class="bi bi-code-slash me-1"></i>Папаме
тры и стек ошибки:</div>
1264                                 <pre class="m-0 p-2 bg-black bg-opacity-60 t
ext-danger border border-danger border-opacity-25 rounded font
-monospace small" style="font-size: 0.72rem; line-height: 1.35
; max-height: 220px; overflow-y: auto; white-space: pre-wrap;

```

```

word-break: break-all;">${escapeHtml(errDetail)}</pre>
1265         </div>
1266         ` : ''}
1267     </div>
1268     `;
1269     }
1270     chatHistory.scrollTop = chatHistory.scrollHeight;
1271     if (statusInd) statusInd.innerText = `Ошибка: ${err.me
ssage || 'Сбой'}`;
1272     } finally {
1273         isChatSubmitting = false;
1274         chatInput.disabled = false;
1275         if (sendBtn) sendBtn.disabled = false;
1276         chatInput.focus();
1277     }
1278 }
1279
1280 chatForm.onSubmit = (e) => {
1281     e.preventDefault();
1282     submitChat();
1283 };
1284 }
1285
1286 // Setup UI event handlers
1287 function setupEvents() {
1288     if (isEventsSetup) return;
1289     isEventsSetup = true;
1290
1291     // Quick Check Button
1292     const btnQuick = document.getElementById('btn-run-quick-ch
eck');
1293     if (btnQuick) {
1294         btnQuick.onclick = () => runScenario('quick_check');
1295     }
1296
1297     // Run All Button
1298     const btnAll = document.getElementById('btn-run-all-scenar
ios');

```

```

1299     if (btnAll) {
1300         btnAll.onclick = () => runScenario('all');
1301     }
1302
1303     // Refresh List Button
1304     const btnRefresh = document.getElementById('btn-scenario-r
refresh-list');
1305     if (btnRefresh) {
1306         btnRefresh.onclick = () => loadScenarios();
1307     }
1308
1309     // Close Results Button
1310     const btnCloseRes = document.getElementById('btn-close-res
ult-panel');
1311     if (btnCloseRes) {
1312         btnCloseRes.onclick = () => {
1313             const panel = document.getElementById('scenario-result
-panel');
1314             if (panel) panel.style.display = 'none';
1315         };
1316     }
1317
1318     // Category Filter Buttons
1319     const filterContainer = document.getElementById('scenario-
category-filters');
1320     if (filterContainer) {
1321         filterContainer.querySelectorAll('button').forEach(btn =
> {
1322             btn.onclick = () => {
1323                 filterContainer.querySelectorAll('button').forEach(b
=> b.classList.remove('active'));
1324                 btn.classList.add('active');
1325                 currentFilter = btn.getAttribute('data-filter') || '
all';
1326                 renderScenariosGrid();
1327             };
1328         });
1329     }

```



```

1330
1331     setupMiniChat();
1332
1333     // Post-Install Wizard Event Listeners
1334     const applyProfBtn = document.getElementById('btn-wizard-a
1335     pply-profile');
1336     if (applyProfBtn) applyProfBtn.onclick = applyCurrentWizar
1337     dProfile;
1338     }
1339     // =====
1340     // Post-Install Wizard Logic
1341     // =====
1342     let activeProfileSteps = [];
1343
1344     async function loadWizardProfiles() {
1345         try {
1346             const res = await fetch('/api/system-control/profiles');
1347             if (!res.ok) return;
1348             const data = await res.json();
1349             const profiles = data.profiles || [];
1350             const sel = document.getElementById('wizard-profile-sele
1351             ct');
1352             if (sel && profiles.length > 0) {
1353                 sel.innerHTML = profiles.map(p => `<option value="${p.
1354                 profile_id}">${p.name}</option>`).join('');
1355                 renderStepsForSelectedProfile(profiles[0]);
1356                 sel.onchange = () => {
1357                     const selected = profiles.find(p => p.profile_id ===
1358                     sel.value);
1359                     if (selected) renderStepsForSelectedProfile(selected
1360                     );
1361                 };

```

```

1360     }
1361   } catch (e) {
1362     console.error('[ScenariosTab] Failed to load wizard profiles:', e);
1363   }
1364 }
1365
1366 function renderStepsForSelectedProfile(profile) {
1367   const container = document.getElementById('wizard-steps-container');
1368   if (!container) return;
1369   activeProfileSteps = profile.steps || [];
1370
1371   container.innerHTML = activeProfileSteps.map((s) => `
1372     <div class="card bg-body-tertiary border-secondary p-2 d-flex flex-row align-items-center justify-content-between" id="step-row-${s.id}">
1373       <div class="d-flex align-items-center gap-3">
1374         <input class="form-check-input wizard-step-checkbox" type="checkbox" id="chk-step-${s.id}" data-id="${s.id}" ${s.enabled ? 'checked' : ''} style="cursor: pointer;">
1375         <div>
1376           <div class="fw-semibold text-white">${s.title}</div>
1377           <div class="small text-muted">${s.description}</div>
1378         </div>
1379       </div>
1380       <div class="d-flex align-items-center gap-2">
1381         ${s.requires_elevation ? '<span class="badge bg-secondary font-monospace" style="font-size: 0.7rem;">Admin</span>' : ''}
1382         <span class="badge ${s.status === 'SUCCESS' ? 'bg-success text-white' : (s.status === 'WARNING' ? 'bg-warning text-dark' : (s.status === 'FAILED' ? 'bg-danger text-white' : 'bg-secondary'))}" id="step-status-${s.id}">
1383           ${s.status}
1384         </span>

```

```

1385 |         </div>
1386 |     </div>
1387 |     `).join('');
1388 | }
1389 |
1390 | async function applyCurrentWizardProfile() {
1391 |     const sel = document.getElementById('wizard-profile-select
1392 | ');
1393 |     const profileId = sel ? sel.value : 'post_install';
1394 |     const applyBtn = document.getElementById('btn-wizard-apply
1395 | -profile');
1396 |
1397 |     const selectedStepIds = [];
1398 |     document.querySelectorAll('.wizard-step-checkbox:checked')
1399 | .forEach(chk => {
1400 |         selectedStepIds.push(chk.getAttribute('data-id'));
1401 |     });
1402 |
1403 |     if (applyBtn) {
1404 |         applyBtn.disabled = true;
1405 |         applyBtn.innerHTML = '<span class="spinner-border spinne
1406 | r-border-sm me-1"></span> Применение...';
1407 |     }
1408 |
1409 |     try {
1410 |         const res = await fetch('/api/system-control/profiles/ap
1411 | ply', {
1412 |             method: 'POST',
1413 |             headers: { 'Content-Type': 'application/json' },
1414 |             body: JSON.stringify({
1415 |                 profile_id: profileId,
1416 |                 selected_step_ids: selectedStepIds
1417 |             })
1418 |         });
1419 |         if (res.ok) {
1420 |             const outcome = await res.json();
1421 |             (outcome.steps || []).forEach(st => {
1422 |                 const badge = document.getElementById(`step-status-$

```

```

    {st.id}`);
1418         if (badge) {
1419             badge.innerText = st.status;
1420             badge.className = `badge ${st.status === 'SUCCESS'
? 'bg-success text-white' : (st.status === 'WARNING' ? 'bg-wa
rning text-dark' : 'bg-danger text-white')}`;
1421             badge.title = st.result_message;
1422         }
1423     });
1424     window.showToast?.('Профиль первоначальной настройки у
спешно применён', 'success');
1425     }
1426 } catch (e) {
1427     console.error('[ScenariosTab] Failed to apply profile:',
e);
1428 } finally {
1429     if (applyBtn) {
1430         applyBtn.disabled = false;
1431         applyBtn.innerHTML = '<i class="bi bi-play-fill me-1">
</i> Применить профиль';
1432     }
1433 }
1434 }
1435
1436 // Public initialization entryptoint
1437 window.initScenariosTab = async function () {
1438     console.log('[ScenariosTab] Initializing Scenarios & Test
Computer Tab...');
1439     setupEvents();
1440     loadQuestionsFromConfig();
1441     loadWizardProfiles();
1442     await loadScenarios();
1443     console.log('[ScenariosTab] Scenarios tab ready.');
```

1444 };

1445 window.initChatTab = window.initScenariosTab;

1446

1447 // Auto-init if container exists

1448 if (document.getElementById('scenarios-cards-container') ||

```
document.getElementById('scenario-chat-history')) {  
1449 |     window.initScenariosTab();  
1450 | }  
1451 |})();  
1452 |
```

index.html

```
1 <!-- Web Search MCP Providers Management Tab -->
2 <div class="container-fluid py-3">
3   <!-- Header -->
4   <div class="alert alert-info d-flex justify-content-between
5 align-items-center flex-wrap gap-2 mb-3 shadow-sm">
6     <div>
7       <h5 class="mb-1"><i class="bi bi-globe2 me-2"></i> Про
8 вайдеры веб-поиска (MCP)</h5>
9       <p class="mb-0 small text-secondary">Управление серверам
10 и интернет-поиска, алгоритмами извлечения и тестирование выдач
11 и.</p>
12     </div>
13     <div class="input-group input-group-sm" style="max-width:
14 440px;">
15       <span class="input-group-text fw-bold">Активный движок п
16 оиска:</span>
17       <select class="form-select" id="search-tab-engine-select
18 or">
19         <option value="playwright"> Playwright MCP</option>
20         <option value="langchain"> LangChain MCP</option>
21         <option value="gemini"> Google Gemini (Grounding)</op
22 tion>
23         <option value="gemini_cli"> Google Gemini CLI</optio
24 n>
25         <option value="agy"> AntigraVity (AGY)</option>
26       </select>
27       <button class="btn btn-primary" id="btn-save-search-tab-
28 engine">
29         <i class="bi bi-check-circle me-1"></i> Сохранить
30       </button>
31     </div>
32   </div>
33
34   <!-- Reference Info Block (Collapse, matching RAG tab style)
35   -->
36   <div class="card mb-4 border-0 shadow-sm bg-light">
```

```

26     <div class="card-body p-3">
27         <div class="d-flex align-items-center justify-content-be
| tween">
28             <div class="d-flex align-items-center gap-2">
29                 <i class="bi bi-info-circle-fill text-primary fs-5">
| </i>
30                 <span class="fw-bold">Справка: архитектура и принцип
| ы работы 5 серверов веб-поиска</span>
31             </div>
32             <button class="btn btn-sm btn-link text-decoration-non
| e p-0 text-secondary" type="button" data-bs-toggle="collapse"
| data-bs-target="#searchArchCollapse" aria-expanded="true" aria
| -controls="searchArchCollapse">
33                 <i class="bi bi-chevron-down"></i> Развернуть / Скры
| ть
34             </button>
35         </div>
36         <div class="collapse show" id="searchArchCollapse">
37             <div class="row g-3 mt-2 pt-2 border-top">
38                 <!-- 1. Playwright -->
39                 <div class="col-xl col-md-6">
40                     <div class="p-3 border rounded bg-white h-100 shad
| ow-sm">
41                         <div class="fw-bold text-primary mb-1">
42                             □ Playwright MCP
43                         </div>
44                         <div class="small text-muted mb-2">
45                             Прямой headless-браузер (Chromium).
46                         </div>
47                         <ul class="small text-secondary mb-0 ps-3">
48                             <li>Поиск в DuckDuckGo, Bing и Google.</li>
49                             <li>Глубокий сбор DOM-структуры и текста сайто
| в.</li>
50                             <li><strong>Плюс:</strong> не расходует токены
| ИИ на сбор данных.</li>
51                             <li><strong>Особенность:</strong> требует запу
| щенного Playwright.</li>
52                         </ul>

```

```

53 |         </div>
54 |     </div>
55 |
56 |     <!-- 2. LangChain -->
57 |     <div class="col-xl col-md-6">
58 |         <div class="p-3 border rounded bg-white h-100 shad
ow-sm">
59 |             <div class="fw-bold text-success mb-1">
60 |                 □ LangChain MCP
61 |             </div>
62 |             <div class="small text-muted mb-2">
63 |                 Автономный ReAct-агент Breadboard.
64 |             </div>
65 |             <ul class="small text-secondary mb-0 ps-3">
66 |                 <li>Инструменты: веб-поиск, RAG, Python-вычисл
ения.</li>
67 |                 <li>Агент сам принимает решение о вызове инстр
ументов.</li>
68 |                 <li><strong>Плюс:</strong> гибкое решение комп
лексных задач.</li>
69 |                 <li><strong>Особенность:</strong> многошаговый
reasoning-цикл.</li>
70 |             </ul>
71 |         </div>
72 |     </div>
73 |
74 |     <!-- 3. Gemini Grounding -->
75 |     <div class="col-xl col-md-6">
76 |         <div class="p-3 border rounded bg-white h-100 shad
ow-sm">
77 |             <div class="fw-bold text-info mb-1">
78 |                 □ Google Gemini Grounding
79 |             </div>
80 |             <div class="small text-muted mb-2">
81 |                 Официальный Google Search Grounding.
82 |             </div>
83 |             <ul class="small text-secondary mb-0 ps-3">
84 |                 <li>Прямой поиск в реальном индексе Google Sea

```



```

rch.</li>
85         <li>Автоматическая ротация пула ключей при оши
бках 429.</li>
86         <li><strong>Плюс:</strong> проверенная точност
ь фактов и цитаты.</li>
87         <li><strong>Особенность:</strong> требует вали
дных API-ключей.</li>
88     </ul>
89 </div>
90 </div>
91
92 <!-- 4. Gemini CLI -->
93 <div class="col-xl col-md-6">
94     <div class="p-3 border rounded bg-white h-100 shad
ow-sm">
95         <div class="fw-bold text-primary mb-1">
96             □ Google Gemini CLI
97         </div>
98         <div class="small text-muted mb-2">
99             Локальный терминальный агент Google.
100         </div>
101         <ul class="small text-secondary mb-0 ps-3">
102             <li>Прямое выполнение через CLI подпроцесс.</l
i>
103             <li>Встроенные агентные инструменты веб-поиска
.</li>
104             <li><strong>Плюс:</strong> автономный агент, м
одель по умолчанию flash-lite.</li>
105             <li><strong>Особенность:</strong> требует уста
новленной утилиты gemini CLI.</li>
106         </ul>
107     </div>
108 </div>
109
110 <!-- 5. Antigravity AGY -->
111 <div class="col-xl col-md-6">
112     <div class="p-3 border rounded bg-white h-100 shad
ow-sm">

```

```

113 |         <div class="fw-bold text-warning mb-1">
114 |             □ Antigravity (AGY)
115 |         </div>
116 |         <div class="small text-muted mb-2">
117 |             Мультиагентный поиск платформы Google.
118 |         </div>
119 |         <ul class="small text-secondary mb-0 ps-3">
120 |             <li>Инструменты <code>SEARCH_WEB</code> и <code>
121 | READ_URL_CONTENT</code>.</li>
122 |             <li>Глубокий ресерч и синтез сложных многоходов.
123 | Глубокий ресерч и синтез сложных многоходов.
124 |             <li><strong>Плюс:</strong> автономный агентный
125 | цикл.</li>
126 |             <li><strong>Особенность:</strong> использует Antigravity
127 | SDK / CLI.</li>
128 |         </ul>
129 |     </div>
130 | </div>
131 |
132 | <!-- Detailed Provider Settings & Live Search Tester -->
133 | <div class="row g-4 mb-4">
134 |     <!-- LEFT: Provider Detailed Settings -->
135 |     <div class="col-lg-6">
136 |         <div class="card shadow-sm border-0 h-100">
137 |             <div class="card-header bg-dark text-white fw-bold">
138 |                 <i class="bi bi-sliders me-1"></i> Параметры поисковых
139 | модулей
140 |             </div>
141 |             <div class="card-body">
142 |                 <!-- Gemini Grounding Settings -->
143 |                 <div class="mb-4 p-3 border rounded bg-light">
144 |                     <h6 class="fw-bold text-info mb-2"><i class="bi bi-stars me-1"></i> Gemini Grounding</h6>
145 |                     <div class="mb-2">

```

```
145 |         <label class="form-label small text-muted fw-bol  
    | d">Модель для поиска Grounding (динамически из SDK):</label>  
146 |         <select class="form-select form-select-sm" id="s  
    | earch-gemini-model">  
147 |             <option value="">Загрузка доступных моделей че  
    | рез Google GenAI SDK...</option>  
148 |         </select>  
149 |     </div>  
150 |     <div class="small text-muted">  
151 |         Ключи автоматически считываются из вкладки «Моде  
    | ли и API» и ротируются по кругу.  
152 |     </div>  
153 | </div>  
154 |  
155 |     <!-- Gemini CLI Settings -->  
156 |     <div class="mb-4 p-3 border rounded bg-light">  
157 |         <h6 class="fw-bold text-primary mb-2"><i class="bi  
    | bi-terminal-fill me-1"></i> Google Gemini CLI</h6>  
158 |         <div class="mb-2">  
159 |             <label class="form-label small text-muted fw-bol  
    | d">Модель Gemini CLI (динамически из инструмента):</label>  
160 |             <select class="form-select form-select-sm" id="s  
    | earch-gemini-cli-model">  
161 |                 <option value="">Загрузка доступных моделей че  
    | рез Gemini CLI...</option>  
162 |             </select>  
163 |         </div>  
164 |         <div class="small text-muted">  
165 |             Использует локальную утилиту <code>gemini</code>  
    | с автоматическим вызовом поисковых инструментов.  
166 |         </div>  
167 |     </div>  
168 |  
169 |     <!-- Antigravity AGY Settings -->  
170 |     <div class="mb-4 p-3 border rounded bg-light">  
171 |         <h6 class="fw-bold text-warning mb-2"><i class="bi  
    | bi-rocket-takeoff me-1"></i> Antigravity (AGY) Search</h6>  
172 |         <div class="mb-2">
```

```
173 |         <label class="form-label small text-muted fw-bol  
    | d">Модель агента AntigraVity (динамически из SDK):</label>  
174 |         <select class="form-select form-select-sm" id="s  
    | earch-agy-model">  
175 |             <option value="">Загрузка доступных моделей че  
    | рез AntigraVity SDK...</option>  
176 |         </select>  
177 |     </div>  
178 |     <div class="small text-muted">  
179 |         Использует встроенные агентные инструменты <code  
    | >SEARCH_WEB</code> и <code>READ_URL_CONTENT</code>.  
180 |     </div>  
181 | </div>  
182 |  
183 |     <div class="d-grid">  
184 |         <button class="btn btn-success fw-bold" id="btn-sa  
    | ve-search-params">  
185 |             <i class="bi bi-floppy me-1"></i> Сохранить наст  
    | ройки модулей  
186 |         </button>  
187 |     </div>  
188 | </div>  
189 | </div>  
190 | </div>  
191 |  
192 | <!-- RIGHT: Live Interactive Search Tester -->  
193 | <div class="col-lg-6">  
194 |     <div class="card shadow-sm border-0 h-100">  
195 |         <div class="card-header bg-dark text-white d-flex just  
    | ify-content-between align-items-center">  
196 |             <span class="fw-bold"><i class="bi bi-terminal me-1"  
    | ></i> Интерактивный тест поиска</span>  
197 |             <span class="badge bg-primary" id="test-search-engin  
    | e-badge">playwright</span>  
198 |         </div>  
199 |         <div class="card-body d-flex flex-column">  
200 |             <div class="mb-3">  
201 |                 <label class="form-label small text-muted fw-bold"
```

```
202 | >Тестовый поисковый запрос:</label>
203 |     <div class="input-group">
204 |         <input type="text" class="form-control form-control-sm" id="test-search-query-input" placeholder="Например: последние новости марсохода 2026">
205 |         <button class="btn btn-primary btn-sm px-3" id="btn-run-test-search">
206 |             <i class="bi bi-search me-1"></i> Искать
207 |         </button>
208 |     </div>
209 |
210 |     <div class="flex-grow-1 d-flex flex-column">
211 |         <label class="form-label small text-muted fw-bold">
212 | >Результат работы провайдера:</label>
213 |         <div id="test-search-output-container" class="border rounded p-3 bg-light flex-grow-1 font-monospace" style="min-height: 230px; max-height: 350px; overflow-y: auto; font-size: 0.82rem; white-space: pre-wrap;">
214 |             <span class="text-muted">Введите запрос и нажмите «Искать» для проверки выбранного провайдера.</span>
215 |         </div>
216 |     </div>
217 | </div>
218 | </div>
219 | </div>
220 | </div>
221 |
```

main.js

```
1 // Search Tab Logic - Web Search MCP Providers Management
2
3 async function initSearchTab() {
4   const engineSelect = document.getElementById('search-tab-engine-selector');
5   const btnSaveEngine = document.getElementById('btn-save-search-tab-engine');
6   const btnSaveParams = document.getElementById('btn-save-search-params');
7   const geminiModelSelect = document.getElementById('search-gemini-model');
8   const geminiCliModelSelect = document.getElementById('search-gemini-cli-model');
9   const agyModelSelect = document.getElementById('search-agy-model');
10  const engineBadge = document.getElementById('test-search-engine-badge');
11  const btnRunTest = document.getElementById('btn-run-test-search');
12  const testQueryInput = document.getElementById('test-search-query-input');
13  const testOutputContainer = document.getElementById('test-search-output-container');
14
15  if (!engineSelect) return;
16
17  // --- 1. Load Dynamic Models via SDK / Tools ---
18  async function loadModels() {
19    try {
20      const modelsData = await window.api.fetch('/api/chat/models');
21      const modelsGrouped = modelsData.models || {};
22      const geminiList = modelsGrouped.gemini || [];
23      const geminiCliList = modelsGrouped.gemini_cli || ['gemini-3.1-flash-lite', 'gemini-2.5-flash', 'gemini-2.5-pro'];
24      const agyList = modelsGrouped.agy || [];
```

```

25 |
26 |     if (geminiModelSelect && geminiList.length > 0) {
27 |         const curVal = geminiModelSelect.value;
28 |         geminiModelSelect.innerHTML = '';
29 |         geminiList.forEach(m => {
30 |             const opt = document.createElement('option');
31 |             opt.value = m;
32 |             opt.textContent = m;
33 |             geminiModelSelect.appendChild(opt);
34 |         });
35 |         if (curVal && geminiList.includes(curVal)) {
36 |             geminiModelSelect.value = curVal;
37 |         }
38 |     }
39 |
40 |     if (geminiCliModelSelect && geminiCliList.length > 0) {
41 |         const curVal = geminiCliModelSelect.value;
42 |         geminiCliModelSelect.innerHTML = '';
43 |         geminiCliList.forEach(m => {
44 |             const opt = document.createElement('option');
45 |             opt.value = m;
46 |             opt.textContent = m;
47 |             geminiCliModelSelect.appendChild(opt);
48 |         });
49 |         if (curVal && geminiCliList.includes(curVal)) {
50 |             geminiCliModelSelect.value = curVal;
51 |         } else if (geminiCliList.includes('gemini-3.1-flash-li
52 | te')) {
53 |             geminiCliModelSelect.value = 'gemini-3.1-flash-lite'
54 |         };
55 |     }
56 |
57 |     if (agyModelSelect && agyList.length > 0) {
58 |         const curVal = agyModelSelect.value;
59 |         agyModelSelect.innerHTML = '';
60 |         agyList.forEach(m => {
            const opt = document.createElement('option');

```

```
61         opt.value = m;
62         opt.textContent = m;
63         agyModelSelect.appendChild(opt);
64     });
65     if (curVal && agyList.includes(curVal)) {
66         agyModelSelect.value = curVal;
67     }
68 }
69 } catch (err) {
70     console.error('[SearchTab] Ошибка загрузки динамических
моделей SDK/CLI:', err);
71 }
72 }
73
74 // --- 2. Load Current Config ---
75 async function loadConfig() {
76     try {
77         await loadModels();
78         const data = await window.api.fetch('/api/admin/web-search/config');
79         if (data) {
80             if (data.engine && engineSelect) {
81                 engineSelect.value = data.engine;
82                 if (engineBadge) engineBadge.textContent = data.engine;
83             }
84             if (data.gemini_model && geminiModelSelect) {
85                 geminiModelSelect.value = data.gemini_model;
86             }
87             if (data.gemini_cli_model && geminiCliModelSelect) {
88                 geminiCliModelSelect.value = data.gemini_cli_model;
89             }
90             if (data.agy_model && agyModelSelect) {
91                 agyModelSelect.value = data.agy_model;
92             }
93         }
94     } catch (e) {
95         console.error('[SearchTab] Ошибка загрузки конфигурации
```



```

поиска:', e);
96     notifySearch('Ошибка загрузки настроек поиска: ' + e.mes
sage, 'danger');
97     }
98     }
99
100    // --- 3. Save Config Helper ---
101    async function saveConfig() {
102        const engine = engineSelect ? engineSelect.value : 'playwr
ight';
103        const gemini_model = geminiModelSelect ? geminiModelSelect
.value : 'gemini-flash-lite-latest';
104        const gemini_cli_model = geminiCliModelSelect ? geminiCliM
odelSelect.value : 'gemini-3.1-flash-lite';
105        const agy_model = agyModelSelect ? agyModelSelect.value :
'agy-flash';
106
107        try {
108            await window.api.fetch('/api/admin/web-search/config', {
109                method: 'POST',
110                headers: { 'Content-Type': 'application/json' },
111                body: JSON.stringify({ engine, gemini_model, gemini_cli
_model, agy_model })
112            });
113            if (engineBadge) engineBadge.textContent = engine;
114            notifySearch(`Настройки веб-поиска сохранены (активен:
${engine})`, 'success');
115
116            let compoundSearch = engine;
117            if (engine === 'gemini') compoundSearch = `${engine}:${g
emini_model}`;
118            if (engine === 'gemini_cli') compoundSearch = `${engine}
:${gemini_cli_model}`;
119            if (engine === 'agy') compoundSearch = `${engine}:${agy_
model}`;
120
121            if (typeof window.updateChatBadges === 'function') {
122                window.updateChatBadges(undefined, compoundSearch);

```

```

123     }
124   } catch (e) {
125     console.error('[SearchTab] Ошибка сохранения:', e);
126     notifySearch('⚠ Ошибка сохранения настроек: ' + e.messag
e, 'danger');
127   }
128 }
129
130 if (btnSaveEngine) btnSaveEngine.onclick = saveConfig;
131 if (btnSaveParams) btnSaveParams.onclick = saveConfig;
132
133 if (engineSelect) {
134   engineSelect.onchange = () => {
135     if (engineBadge) engineBadge.textContent = engineSelect.
value;
136   };
137 }
138
139 // --- 3. Interactive Search Tester ---
140 if (btnRunTest && testQueryInput && testOutputContainer) {
141   const runTest = async () => {
142     const query = testQueryInput.value.trim();
143     if (!query) {
144       notifySearch('Введите поисковый запрос для теста!', 'w
arning');
145       testQueryInput.focus();
146       return;
147     }
148
149     const engine = engineSelect ? engineSelect.value : 'play
wright';
150     btnRunTest.disabled = true;
151     const originalText = btnRunTest.innerHTML;
152     btnRunTest.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1" role="status"></span> Поиск...';
153     testOutputContainer.innerHTML = `<span class="text-info"
>⚡ Выполнение поиска через <strong>${engine}</strong> по запро
су: "${query}"...</span>`;

```

```

154
155     try {
156         const response = await window.api.fetch('/api/admin/we
b-search/test', {
157             method: 'POST',
158             headers: { 'Content-Type': 'application/json' },
159             body: JSON.stringify({ query, engine })
160         });
161
162         if (response.status === 'ok') {
163             testOutputContainer.textContent = response.result ||
'Пустой ответ.';
164         } else {
165             testOutputContainer.innerHTML = `<span class="text-d
anger">❌ Ошибка: ${response.message || 'Неизвестная ошибка'}</
span>`;
166         }
167     } catch (err) {
168         console.error('[SearchTab] Ошибка теста:', err);
169         testOutputContainer.innerHTML = `<span class="text-dan
ger">❌ Ошибка запроса: ${err.message}</span>`;
170     } finally {
171         btnRunTest.disabled = false;
172         btnRunTest.innerHTML = originalText;
173     }
174 };
175
176 btnRunTest.onclick = runTest;
177 testQueryInput.onkeydown = (e) => {
178     if (e.key === 'Enter') {
179         e.preventDefault();
180         runTest();
181     }
182 };
183 }
184
185 // Helper notification
186 function notifySearch(msg, type = 'info') {

```

```
187 |     if (typeof showNotification === 'function') {
188 |         showNotification(msg, type);
189 |     } else if (typeof showModelsNotification === 'function') {
190 |         showModelsNotification(msg, type);
191 |     } else {
192 |         console.log(`[${type}] ${msg}`);
193 |     }
194 | }
195 |
196 | await loadConfig();
197 | }
198 |
199 | window.initSearchTab = initSearchTab;
200 |
```

index.html

```
1 <!-- Skills Tab Component -->
2 <div class="container-fluid p-3" id="skills-tab-root">
3   <!-- Top Banner / Summary -->
4   <div class="alert alert-dark border-secondary bg-dark text-l
  ight d-flex flex-wrap align-items-center justify-content-betwe
  en gap-3 shadow-sm">
5     <div class="d-flex align-items-center gap-3">
6       <div class="p-3 bg-black rounded border border-secondary
  text-warning display-6 d-flex align-items-center justify-cont
  ent-center" style="width: 54px; height: 54px;">
7         <i class="bi bi-lightning-charge-fill"></i>
8       </div>
9       <div>
10        <h5 class="mb-1 text-white d-flex align-items-center g
  ap-2">
11          □ Управление навыками (Agent Skills)
12          <span class="badge bg-primary rounded-pill small" id
  ="skills-total-badge">0</span>
13        </h5>
14        <p class="mb-0 text-muted small">
15          Просмотр, создание, настройка и упаковка переиспольз
  уемых навыков для AI-агентов (Antigravity, Gemini CLI, Breadbo
  ard).
16        </p>
17      </div>
18    </div>
19    <div class="d-flex align-items-center gap-2">
20      <button type="button" class="btn btn-outline-light btn-s
  m d-flex align-items-center gap-1" id="btn-refresh-skills" onc
  lick="window.skillsTab?.refresh()">
21        <i class="bi bi-arrow-clockwise"></i> Обновить
22      </button>
23      <button type="button" class="btn btn-primary btn-sm d-fl
  ex align-items-center gap-1 shadow-sm" data-bs-toggle="modal"
  data-bs-target="#createSkillModal">
24        <i class="bi bi-plus-circle-fill"></i> Новый навык
```

```
25     </button>
26   </div>
27 </div>
28
29 <!-- Statistics Row -->
30 <div class="row g-2 mb-3" id="skills-stats-row">
31   <div class="col-6 col-md-3">
32     <div class="p-2 bg-dark rounded border border-secondary
text-center">
33       <div class="h4 mb-0 text-white" id="stat-total-skills"
>0</div>
34       <div class="small text-muted">Всего навыков</div>
35     </div>
36   </div>
37   <div class="col-6 col-md-3">
38     <div class="p-2 bg-dark rounded border border-secondary
text-center">
39       <div class="h4 mb-0 text-info" id="stat-scripts-skills
">0</div>
40       <div class="small text-muted">С исполняемыми скриптами
</div>
41     </div>
42   </div>
43   <div class="col-6 col-md-3">
44     <div class="p-2 bg-body-tertiary rounded border text-cen
ter">
45       <div class="h4 mb-0 text-warning" id="stat-refs-skills
">0</div>
46       <div class="small text-muted">Со справочниками</div>
47     </div>
48   </div>
49   <div class="col-6 col-md-3">
50     <div class="p-2 bg-body-tertiary rounded border text-cen
ter">
51       <div class="h4 mb-0 text-success" id="stat-packaged-sk
ills">0</div>
52       <div class="small text-muted">Скомпилировано (.skill)<
/div>
```

```

53     </div>
54 </div>
55 </div>
56
57 <!-- Search & View Controls -->
58 <div class="card bg-body-tertiary border mb-3 shadow-sm">
59     <div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2">
60         <div class="input-group input-group-sm flex-grow-1" style="max-width: 450px;">
61             <span class="input-group-text"><i class="bi bi-search"></i></span>
62             <input type="text" class="form-control" id="skills-search-input" placeholder="Поиск по названию или описанию навыка.
63             <button class="btn btn-outline-secondary" type="button" id="skills-search-clear" onclick="window.skillsTab?.clearSearch()">🗑️ </button>
64         </div>
65
66         <div class="d-flex align-items-center gap-2 flex-wrap">
67             <select class="form-select form-select-sm" id="skills-filter-select" style="width: auto;" onchange="window.skillsTab?.applyFilters()">
68                 <option value="all">Все навыки</option>
69                 <option value="scripts">Только со скриптами</option>
70                 <option value="references">Только со справочниками</option>
71                 <option value="packaged">Собранные (.skill)</option>
72             </select>
73
74             <div class="btn-group btn-group-sm" role="group">
75                 <button type="button" class="btn btn-outline-secondary active" id="btn-view-cards" onclick="window.skillsTab?.setViewMode('cards')">
76                     <i class="bi bi-grid-fill"></i> Карточки
77                 </button>
78                 <button type="button" class="btn btn-outline-secondary

```

```

ry" id="btn-view-table" onclick="window.skillsTab?.setViewMode
('table')">
79         <i class="bi bi-table"></i> Таблица
80     </button>
81 </div>
82 </div>
83 </div>
84 </div>
85
86 <!-- Loading & Empty Indicators -->
87 <div id="skills-loading" class="text-center py-5 text-muted
d-none">
88     <div class="spinner-border text-primary mb-2" role="status
"></div>
89     <div>Загрузка навыков...</div>
90 </div>
91
92 <div id="skills-empty" class="text-center py-5 text-muted d-
none">
93     <i class="bi bi-folder-x display-4 d-block mb-2 text-secon
dary"></i>
94     <h5>Навыки не найдены</h5>
95     <p class="small">Попробуйте изменить поисковый запрос или
создайте новый навык.</p>
96 </div>
97
98 <!-- Cards Container -->
99 <div class="row g-3" id="skills-cards-container"></div>
100
101 <!-- Table Container -->
102 <div class="card shadow-sm border d-none" id="skills-table-c
ard">
103     <div class="card-body p-0">
104         <div class="table-responsive">
105             <table class="table table-hover align-middle mb-0">
106                 <thead class="table-light text-uppercase small">
107                     <tr>
108                         <th scope="col" style="width: 25%;">Название нав

```



```

109         <th scope="col" style="width: 35%;">Описание</th>
110     >
111         <th scope="col" style="width: 15%;">Расположение
112     </th>
113         <th scope="col" style="width: 10%;">Компоненты</
114     th>
115         <th scope="col" style="width: 15%; text-align: r
116     ight;">Действия</th>
117     </tr>
118 </thead>
119 <tbody id="skills-table-body"></tbody>
120 </table>
121 </div>
122 </div>
123 </div>
124 <!-- Modal: Create New Skill Wizard -->
125 <div class="modal fade" id="createSkillModal" tabindex="-1" ar
126     ia-labelledby="createSkillModalLabel" aria-hidden="true">
127     <div class="modal-dialog modal-xl modal-dialog-centered moda
128     l-dialog-scrollable">
129         <div class="modal-content bg-dark text-white border-second
130     ary shadow-lg">
131             <div class="modal-header border-secondary py-2 px-3 bg-b
132     lack bg-opacity-40">
133                 <div class="d-flex align-items-center gap-2">
134                     <span class="fs-5 text-warning">⚠ </span>
135                     <div>
136                         <h5 class="modal-title mb-0 fs-6 fw-bold" id="crea
137     teSkillModalLabel">Мастер создания навыков (Skill Wizard)</h5>
138                         <small class="text-muted">Создавайте переиспользо
139     ванные навыки для AI Breadboard, Gemini CLI и Antigravity</small>
140                     </div>
141                 </div>
142             </div>
143             <button type="button" class="btn-close btn-close-white
144     " data-bs-dismiss="modal" aria-label="Close"></button>

```

```
135 |         </div>
136 |
137 |         <!-- Mode Selector Pills -->
138 |         <div class="px-3 pt-3 pb-2 bg-black bg-opacity-25 border
| -bottom border-secondary">
139 |             <ul class="nav nav-pills nav-fill gap-2" id="skillCrea
| teModes" role="tablist">
140 |                 <li class="nav-item" role="presentation">
141 |                     <button class="nav-link active py-2 d-flex align-i
| tems-center justify-content-center gap-2" id="mode-wizard-tab"
| data-bs-toggle="pill" data-bs-target="#mode-wizard-pane" type
| ="button" role="tab" onclick="window.skillsTab?.switchCreateMo
| de('wizard')">
142 |                         <i class="bi bi-magic text-primary"></i> <strong
| >❏ ❏ Пошаговый мастер</strong>
143 |                     </button>
144 |                 </li>
145 |                 <li class="nav-item" role="presentation">
146 |                     <button class="nav-link py-2 d-flex align-items-ce
| nter justify-content-center gap-2" id="mode-ai-tab" data-bs-to
| ggle="pill" data-bs-target="#mode-ai-pane" type="button" role=
| "tab" onclick="window.skillsTab?.switchCreateMode('ai')">
147 |                         <i class="bi bi-robot text-info"></i> <strong>❏
| AI-Ассистент генерации</strong>
148 |                     </button>
149 |                 </li>
150 |                 <li class="nav-item" role="presentation">
151 |                     <button class="nav-link py-2 d-flex align-items-ce
| nter justify-content-center gap-2" id="mode-presets-tab" data-
| bs-toggle="pill" data-bs-target="#mode-presets-pane" type="but
| ton" role="tab" onclick="window.skillsTab?.switchCreateMode('p
| resets')">
152 |                         <i class="bi bi-collection text-warning"></i> <s
| trong>❏ Готовые шаблоны</strong>
153 |                     </button>
154 |                 </li>
155 |             </ul>
156 |         </div>
```

```
157 |
158 |     <div class="modal-body p-3">
159 |         <div class="tab-content" id="skillCreateModesContent">
160 |
161 |             <!-- MODE 1: STEP-BY-STEP WIZARD -->
162 |             <div class="tab-pane fade show active" id="mode-wiza
163 | rd-pane" role="tabpanel">
164 |                 <!-- Steps Stepper Header -->
165 |                 <div class="d-flex justify-content-between align-i
166 | tems-center mb-3 px-2">
167 |                     <div class="d-flex align-items-center gap-2 curs
168 | or-pointer" onclick="window.skillsTab?.goToStep(1)">
169 |                         <span class="badge rounded-circle p-2 bg-prima
170 | ry step-indicator" id="step-badge-1">1</span>
171 |                         <span class="small fw-semibold" id="step-label
172 | -1">Основное</span>
173 |                     </div>
174 |                     <div class="flex-grow-1 mx-2 border-top border-s
175 | econdary"></div>
176 |                     <div class="d-flex align-items-center gap-2 curs
177 | or-pointer" onclick="window.skillsTab?.goToStep(2)">
178 |                         <span class="badge rounded-circle p-2 bg-secon
179 | dary step-indicator" id="step-badge-2">2</span>
180 |                         <span class="small text-muted" id="step-label-
181 | 2">Инструкции</span>
182 |                     </div>
183 |                     <div class="flex-grow-1 mx-2 border-top border-s
184 | econdary"></div>
185 |                     <div class="d-flex align-items-center gap-2 curs
186 | or-pointer" onclick="window.skillsTab?.goToStep(3)">
187 |                         <span class="badge rounded-circle p-2 bg-secon
188 | dary step-indicator" id="step-badge-3">3</span>
189 |                         <span class="small text-muted" id="step-label-
190 | 3">Скрипты и файлы</span>
191 |                     </div>
192 |                     <div class="flex-grow-1 mx-2 border-top border-s
193 | econdary"></div>
194 |                     <div class="d-flex align-items-center gap-2 curs
```

```

or-pointer" onclick="window.skillsTab?.goToStep(4)">
181         <span class="badge rounded-circle p-2 bg-secondary step-indicator" id="step-badge-4">4</span>
182         <span class="small text-muted" id="step-label-4">Финализация</span>
183     </div>
184 </div>
185
186     <!-- Wizard Step 1: Basic Info -->
187     <div id="wizard-step-1" class="wizard-step">
188         <div class="card bg-black bg-opacity-50 border-secondary mb-3">
189             <div class="card-body p-3">
190                 <h6 class="text-primary mb-3"><i class="bi bi-info-circle"></i> Шаг 1: Название, назначение и расположение
</h6>
191
192                 <div class="row g-3">
193                     <div class="col-12 col-md-6">
194                         <label for="wizard-skill-name" class="form-label small fw-semibold">
195                             Идентификатор навыка (kebab-case) <span class="text-danger">*</span>
196                         </label>
197                         <div class="input-group input-group-sm">
198                             <span class="input-group-text bg-dark border-secondary text-warning">⌨ </span>
199                             <input type="text" class="form-control bg-dark text-white border-secondary" id="wizard-skill-name" placeholder="напр. media-downloader или log-parser" required>
200                             <button class="btn btn-outline-secondary" type="button" title="Сгенерировать имя из описания" onclick="window.skillsTab?.generateSlugFromName()">
201                                 <i class="bi bi-magic"></i>
202                             </button>
203                         </div>
204                         <small class="text-muted">Латиница, цифры, дефис и подчеркивания (используется агентом для вызова).</s

```

```

mall>
205         </div>
206
207         <div class="col-12 col-md-6">
208             <label for="wizard-skill-dir" class="form-label small fw-semibold">Целевой каталог</label>
209             <select class="form-select form-select-sm bg-dark text-white border-secondary" id="wizard-skill-dir">
210                 <option value=".agents/skills" selected>.agents/skills (Проектный навык AI Breadboard)</option>
211                 <option value="~/ .gemini/skills">~/ .gemini/skills (Пользовательский – Gemini CLI / Antigravity)</option>
212                 <option value="~/ .agents/skills">~/ .agents/skills (Глобальный пользовательский навык)</option>
213                 <option value="skills">skills (Локальный каталог)</option>
214             </select>
215             <small class="text-muted">Определяет область видимости навыка в системе.</small>
216         </div>
217
218         <div class="col-12 col-md-6">
219             <label for="wizard-skill-category" class="form-label small fw-semibold">Категория</label>
220             <select class="form-select form-select-sm bg-dark text-white border-secondary" id="wizard-skill-category" onchange="window.skillsTab?.onCategoryChanged()">
221                 <option value="tools">☐ Инструмент и автоматизация (CLI / Script)</option>
222                 <option value="api">☐ Интеграция с API и сервисами</option>
223                 <option value="data">☐ Анализ данных и парсинг</option>
224                 <option value="persona">☐ AI-Персона / Специализированный эксперт</option>
225                 <option value="monitoring">☐ Мониторинг и нотификации</option>

```

```
226 |         <option value="docgen">□ Генератор до
    | кументов и отчетов</option>
227 |         <option value="general" selected>□ Общ
    | его назначения</option>
228 |     </select>
229 | </div>
230 |
231 |     <div class="col-12 col-md-6">
232 |         <label for="wizard-skill-desc-ru" class=
    | "form-label small fw-semibold">Назначение на русском</label>
233 |         <input type="text" class="form-control f
    | orm-control-sm bg-dark text-white border-secondary" id="wizard
    | -skill-desc-ru" placeholder="Понятное описание для пользовател
    | я...">
234 |     </div>
235 |
236 |     <div class="col-12">
237 |         <label for="wizard-skill-desc-en" class=
    | "form-label small fw-semibold">Описание для AI-агента (English
    | description)</label>
238 |         <input type="text" class="form-control f
    | orm-control-sm bg-dark text-white border-secondary" id="wizard
    | -skill-desc-en" placeholder="Concise description in English us
    | ed by agent router...">
239 |         <small class="text-muted">Агент использу
    | ет это описание при маршрутизации и выборе навыка.</small>
240 |     </div>
241 | </div>
242 | </div>
243 | </div>
244 | </div>
245 |
246 | <!-- Wizard Step 2: Instructions (SKILL.md) -->
247 | <div id="wizard-step-2" class="wizard-step d-none"
    | >
248 |     <div class="card bg-black bg-opacity-50 border-s
    | econdary mb-3">
249 |         <div class="card-body p-3">
```

```

250 |         <div class="d-flex justify-content-between align-items-center mb-2">
251 |             <h6 class="text-primary mb-0"><i class="bi bi-file-text"></i> Шаг 2: Структура и логика инструкций (SKILL.md)</h6>
252 |             <button type="button" class="btn btn-outline-info btn-sm" onclick="window.skillsTab?.fillStandardInstructions()">
253 |                 <i class="bi bi-arrow-repeat"></i> Заполнить стандартным шаблоном
254 |             </button>
255 |         </div>
256 |
257 |         <div class="row g-2 mb-3">
258 |             <div class="col-12 col-md-6">
259 |                 <label for="wizard-inst-purpose" class="form-label small fw-semibold">□ Цель (Purpose):</label>
260 |                 <textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-purpose" rows="2" placeholder="Опишите главную цель работы навыка..."></textarea>
261 |             </div>
262 |             <div class="col-12 col-md-6">
263 |                 <label for="wizard-inst-triggers" class="form-label small fw-semibold">□ Условия запуска (Triggers):</label>
264 |                 <textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-triggers" rows="2" placeholder="Когда агент должен активировать этот навык..."></textarea>
265 |             </div>
266 |             <div class="col-12 col-md-6">
267 |                 <label for="wizard-inst-protocol" class="form-label small fw-semibold">□ Протокол действий (Execution Protocol):</label>
268 |                 <textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-protocol" rows="4" placeholder="1. Проверить входные данные&#10;2

```

```

    . Запустить скрипт или обработку;3. Сформировать ответ"></
textarea>
269         </div>
270         <div class="col-12 col-md-6">
271             <label for="wizard-inst-rules" class="fo
form-label small fw-semibold">□ Правила и ограничения (Rules & C
onstraints):</label>
272             <textarea class="form-control form-contr
ol-sm bg-dark text-white border-secondary" id="wizard-inst-rul
es" rows="4" placeholder="- Обрабатывать ошибки без падений&#1
0;- Возвращать четкий статус выполнения"></textarea>
273             </div>
274         </div>
275
276         <label for="wizard-skill-instructions" class
="form-label small fw-semibold">Полный текст Markdown для SKIL
L.md:</label>
277         <textarea class="form-control bg-dark text-w
hite border-secondary font-monospace" id="wizard-skill-instruc
tions" rows="6" style="font-size: 0.85rem;" placeholder="Автом
атически генерируется из блоков выше..."></textarea>
278         </div>
279     </div>
280 </div>
281
282 <!-- Wizard Step 3: Scripts & Files -->
283 <div id="wizard-step-3" class="wizard-step d-none"
>
284     <div class="card bg-black bg-opacity-50 border-s
econdary mb-3">
285         <div class="card-body p-3">
286             <h6 class="text-primary mb-3"><i class="bi b
i-folder-plus"></i> Шаг 3: Дополнительные скрипты, документац
ия и справочники</h6>
287
288             <div class="row g-3">
289                 <div class="col-12">
290                     <div class="form-check form-switch mb-2"

```



```

291 |         <input class="form-check-input" type="
checkbox" id="wizard-enable-script" onchange="window.skillsTab
?.toggleScriptEditor()">
292 |         <label class="form-check-label fw-semi
bold small" for="wizard-enable-script">
293 |             Добавить исполняемый Python-скрипт (
<code>scripts/main.py</code>)
294 |         </label>
295 |     </div>
296 |     <div id="wizard-script-container" class=
"d-none mt-2">
297 |         <textarea class="form-control bg-dark
text-info border-secondary font-monospace" id="wizard-script-c
ode" rows="6" style="font-size: 0.85rem;" spellcheck="false"><
/textarea>
298 |     </div>
299 | </div>
300 |
301 |     <div class="col-12">
302 |         <div class="form-check form-switch mb-2"
>
303 |             <input class="form-check-input" type="
checkbox" id="wizard-enable-ref" onchange="window.skillsTab?.t
oggleRefEditor()">
304 |             <label class="form-check-label fw-semi
bold small" for="wizard-enable-ref">
305 |                 Добавить справочный документ (<code>
references/guide.md</code>)
306 |             </label>
307 |         </div>
308 |         <div id="wizard-ref-container" class="d-
none mt-2">
309 |             <textarea class="form-control bg-dark
text-light border-secondary font-monospace" id="wizard-ref-con
tent" rows="4" style="font-size: 0.85rem;" spellcheck="false">
</textarea>
310 |         </div>

```

```

311 |         </div>
312 |
313 |         <div class="col-12">
314 |             <label for="wizard-skill-readme" class="
form-label small fw-semibold">Документация <code>README.md</co
de> (на английском языке):</label>
315 |             <textarea class="form-control bg-dark te
xt-light border-secondary font-monospace" id="wizard-skill-rea
dme" rows="4" style="font-size: 0.85rem;" spellcheck="false"><
/textarea>
316 |         </div>
317 |     </div>
318 | </div>
319 | </div>
320 | </div>
321 |
322 |     <!-- Wizard Step 4: Finalize & Preview -->
323 |     <div id="wizard-step-4" class="wizard-step d-none"
>
324 |         <div class="card bg-black bg-opacity-50 border-s
econdary mb-3">
325 |             <div class="card-body p-3">
326 |                 <h6 class="text-success mb-3"><i class="bi b
i-check2-circle"></i> Шаг 4: Предпросмотр создаваемого навыка<
/h6>
327 |
328 |                 <div class="row g-2 mb-3">
329 |                     <div class="col-12 col-md-4">
330 |                         <div class="p-2 bg-dark rounded border b
order-secondary">
331 |                             <span class="text-muted small d-block"
>Имя:</span>
332 |                             <strong class="text-warning font-monos
pace" id="preview-skill-name">—</strong>
333 |                         </div>
334 |                     </div>
335 |                     <div class="col-12 col-md-4">
336 |                         <div class="p-2 bg-dark rounded border b

```

```

order-secondary">
337         <span class="text-muted small d-block"
>Путь:</span>
338         <code class="text-info" id="preview-sk
ill-path">--</code>
339     </div>
340 </div>
341 <div class="col-12 col-md-4">
342     <div class="p-2 bg-dark rounded border b
order-secondary">
343         <span class="text-muted small d-block"
>Компоненты:</span>
344         <span class="badge bg-primary me-1">SK
ILL.md</span>
345         <span class="badge bg-secondary me-1">
README.md</span>
346         <span class="badge bg-info d-none" id=
"preview-badge-script">scripts/</span>
347         <span class="badge bg-warning d-none t
ext-dark" id="preview-badge-ref">references/</span>
348     </div>
349 </div>
350 </div>
351
352     <label class="form-label small fw-semibold">
Результирующий <code>SKILL.md</code> (Frontmatter + Markdown):
</label>
353     <pre class="bg-dark p-3 rounded border borde
r-secondary text-info font-monospace" id="preview-skillmd-full
" style="max-height: 220px; overflow-y: auto; font-size: 0.85r
em;"></pre>
354 </div>
355 </div>
356 </div>
357
358 <!-- Wizard Navigation Bar -->
359 <div class="d-flex justify-content-between align-i
tems-center mt-3 pt-2 border-top border-secondary">

```

```

360         <button type="button" class="btn btn-outline-sec
ondary btn-sm" id="btn-wizard-prev" onclick="window.skillsTab?
.wizardPrevStep()" disabled>
361         <i class="bi bi-arrow-left"></i> Назад
362     </button>
363
364     <div class="d-flex gap-2">
365         <button type="button" class="btn btn-primary b
tn-sm" id="btn-wizard-next" onclick="window.skillsTab?.wizardN
extStep()">
366             Далее <i class="bi bi-arrow-right"></i>
367         </button>
368         <button type="button" class="btn btn-success b
tn-sm d-none" id="btn-wizard-submit" onclick="window.skillsTab
?.createSkillFromWizard()">
369             <span class="spinner-border spinner-border-s
m d-none" id="wizard-submit-spinner" role="status"></span>
370             <span><i class="bi bi-rocket-takeoff-fill"><
/i> Создать навык</span>
371         </button>
372     </div>
373 </div>
374 </div>
375
376 <!-- MODE 2: AI ASSISTANT GENERATION -->
377 <div class="tab-pane fade" id="mode-ai-pane" role="t
abpanel">
378     <div class="card bg-black bg-opacity-50 border-sec
ondary mb-3">
379         <div class="card-body p-3">
380             <div class="d-flex align-items-center gap-2 mb
-2 text-info">
381                 <i class="bi bi-robot fs-5"></i>
382                 <h6 class="mb-0">Генерация навыка с помощью
ИИ (Natural Language to Skill)</h6>
383             </div>
384             <p class="text-muted small mb-3">
385                 Просто опишите задачу или идею навыка обычны

```

ми словами на русском или английском. ИИ автоматически сформирует имя, структуру инструкций, описание, контракт и вспомогательные скрипты.

```
386         </p>
387
388         <div class="mb-3">
389             <label for="ai-skill-prompt" class="form-label small fw-semibold">Описание задачи навыка <span class="text-danger">*</span></label>
390             <textarea class="form-control bg-dark text-white border-secondary" id="ai-skill-prompt" rows="3" placeholder="например: Навык для автоматического анализа логов сервера Nginx, выявления подозрительных IP-адресов и выгрузки отчета в формате CSV"></textarea>
391         </div>
392
393         <div class="row g-2 mb-3">
394             <div class="col-12 col-md-6">
395                 <label for="ai-skill-category" class="form-label small fw-semibold">Категория</label>
396                 <select class="form-select form-select-sm bg-dark text-white border-secondary" id="ai-skill-category">
397                     <option value="tools">□ Инструмент и скрипт</option>
398                     <option value="api">□ API и интеграция</option>
399                     <option value="data">□ Анализ данных</option>
400                     <option value="persona">□ AI-Эксперт</option>
401                     <option value="monitoring">□ Мониторинг</option>
402                     <option value="general" selected>□ Общий навык</option>
403                 </select>
404             </div>
405             <div class="col-12 col-md-6">
406                 <label for="ai-skill-dir" class="form-label
```

```
l small fw-semibold">Куда сохранить</label>
407         <select class="form-select form-select-sm
bg-dark text-white border-secondary" id="ai-skill-dir">
408             <option value=".agents/skills" selected>
.agents/skills (Проектный)</option>
409             <option value="~/ .gemini/skills">~/ .gemi
ni/skills (Gemini / Antigravity)</option>
410             <option value="~/ .agents/skills">~/ .agen
ts/skills (Глобальный)</option>
411         </select>
412     </div>
413 </div>
414
415         <button type="button" class="btn btn-info btn-
sm d-flex align-items-center gap-2" id="btn-generate-ai" oncli
ck="window.skillsTab?.generateSkillWithAi()">
416             <span class="spinner-border spinner-border-s
m d-none" id="ai-gen-spinner" role="status"></span>
417             <i class="bi bi-sparkles"></i> <strong>Сгене
рировать навык</strong>
418         </button>
419
420         <!-- AI Result Box -->
421         <div id="ai-result-box" class="d-none mt-3 pt-
3 border-top border-secondary">
422             <div class="alert alert-success py-2 d-flex
align-items-center justify-content-between">
423                 <span><i class="bi bi-check-circle-fill me
-1"></i> Навык успешно сгенерирован!</span>
424                 <span class="badge bg-dark font-monospace"
id="ai-res-slug"></span>
425             </div>
426
427             <div class="mb-2">
428                 <label class="form-label small fw-semibold
text-muted">Сгенерированные инструкции (SKILL.md):</label>
429                 <pre class="bg-dark p-3 rounded border bor
der-secondary text-info font-monospace" id="ai-res-preview" st
```

```

yle="max-height: 180px; overflow-y: auto; font-size: 0.85rem;"
></pre>
430         </div>
431
432         <div class="d-flex justify-content-end gap-2
">
433             <button type="button" class="btn btn-outli
ne-primary btn-sm" onclick="window.skillsTab?.applyAiResultToW
izard()">
434                 <i class="bi bi-sliders"></i> Настроить
В мастере
435             </button>
436             <button type="button" class="btn btn-succe
ss btn-sm" onclick="window.skillsTab?.saveAiSkillDirectly()">
437                 <i class="bi bi-plus-circle"></i> Создат
ь сразу
438             </button>
439         </div>
440     </div>
441 </div>
442 </div>
443 </div>
444
445     <!-- MODE 3: PRESETS LIBRARY -->
446     <div class="tab-pane fade" id="mode-presets-pane" ro
le="tabpanel">
447         <div class="row g-3" id="presets-cards-container">
448             <!-- Preset Cards populated dynamically -->
449         </div>
450     </div>
451
452 </div>
453
454     <div id="create-skill-alert" class="alert d-none py-2
mt-3 mb-0"></div>
455 </div>
456
457     <div class="modal-footer border-secondary py-2 px-3 bg-b

```

```

    lack bg-opacity-40">
458     <button type="button" class="btn btn-outline-secondary
    btn-sm" data-bs-dismiss="modal">Заккрыть</button>
459     </div>
460   </div>
461 </div>
462 </div>
463
464 <!-- Modal / Inspector: View, Edit & Live Test Skill -->
465 <div class="modal fade" id="skillDetailsModal" tabindex="-1" a
    ria-labelledby="skillDetailsModalLabel" aria-hidden="true">
466   <div class="modal-dialog modal-xl modal-dialog-scrollable">
467     <div class="modal-content bg-dark text-white border-second
    ary">
468       <div class="modal-header border-secondary d-flex justify
    -content-between align-items-center">
469         <div>
470           <h5 class="modal-title d-flex align-items-center gap
    -2 mb-0" id="skillDetailsModalLabel">
471             <i class="bi bi-lightning-charge-fill text-warning
    "></i>
472             <span id="detail-skill-title">Навык</span>
473           </h5>
474           <small class="text-muted" id="detail-skill-path">--</
    small>
475         </div>
476         <div class="d-flex align-items-center gap-2">
477           <button type="button" class="btn btn-outline-info bt
    n-sm d-flex align-items-center gap-1" id="btn-package-modal" o
    nclick="window.skillsTab?.packageCurrentSkill()">
478             <i class="bi bi-box-seam"></i> Собрать .skill
479           </button>
480           <button type="button" class="btn btn-outline-danger
    btn-sm d-flex align-items-center gap-1" id="btn-delete-modal"
    onclick="window.skillsTab?.deleteCurrentSkill()">
481             <i class="bi bi-trash"></i> Удалить
482           </button>
483           <button type="button" class="btn-close btn-close-whi

```



```

te" data-bs-dismiss="modal" aria-label="Close"></button>
484     </div>
485 </div>
486 <div class="modal-body p-3">
487     <!-- Navigation tabs in inspector -->
488     <ul class="nav nav-pills mb-3" id="skillDetailTabs" ro
le="tablist">
489         <li class="nav-item" role="presentation">
490             <button class="nav-link active" id="detail-tab-ski
llmd" data-bs-toggle="pill" data-bs-target="#detail-pane-skill
md" type="button" role="tab">
491                 □ SKILL.md (Инструкции)
492             </button>
493         </li>
494         <li class="nav-item" role="presentation">
495             <button class="nav-link" id="detail-tab-sandbox" d
ata-bs-toggle="pill" data-bs-target="#detail-pane-sandbox" typ
e="button" role="tab">
496                 □ Тестирование (Sandbox)
497             </button>
498         </li>
499         <li class="nav-item" role="presentation">
500             <button class="nav-link" id="detail-tab-readme" da
ta-bs-toggle="pill" data-bs-target="#detail-pane-readme" type=
"button" role="tab">
501                 □ README.md
502             </button>
503         </li>
504         <li class="nav-item" role="presentation">
505             <button class="nav-link" id="detail-tab-files" dat
a-bs-toggle="pill" data-bs-target="#detail-pane-files" type="b
utton" role="tab">
506                 □ Файлы и структура
507             </button>
508         </li>
509         <li class="nav-item" role="presentation">
510             <button class="nav-link" id="detail-tab-contract"
data-bs-toggle="pill" data-bs-target="#detail-pane-contract" t

```

```

    type="button" role="tab">
511         □ JSON Contract / Метаданные
512     </button>
513 </li>
514 </ul>
515
516     <div class="tab-content" id="skillDetailTabContent">
517         <!-- SKILL.md Tab Pane -->
518         <div class="tab-pane fade show active" id="detail-pane-skillmd" role="tabpanel">
519             <div class="row g-2 mb-3">
520                 <div class="col-12 col-md-6">
521                     <label for="detail-skill-desc-input" class="form-label fw-semibold small text-muted">English Description (Frontmatter):</label>
522                     <input type="text" class="form-control bg-black text-white border-secondary form-control-sm" id="detail-skill-desc-input">
523                 </div>
524                 <div class="col-12 col-md-6">
525                     <label for="detail-skill-desc-ru-input" class="form-label fw-semibold small text-muted">Описание на русском (RU):</label>
526                     <input type="text" class="form-control bg-black text-white border-secondary form-control-sm" id="detail-skill-desc-ru-input">
527                 </div>
528             </div>
529             <div class="mb-2 d-flex justify-content-between align-items-center">
530                 <label for="detail-skill-instructions" class="form-label fw-semibold small text-muted mb-0">Тело инструкции (Markdown):</label>
531                 <small class="text-muted">Редактируемый текст</small>
532             </div>
533             <textarea class="form-control bg-black text-white border-secondary font-monospace" id="detail-skill-instructions

```

```

" rows="16" style="font-size: 0.9rem;" spellcheck="false"></te
xtarea>
534         </div>
535
536         <!-- Live Sandbox Tester Tab Pane -->
537         <div class="tab-pane fade" id="detail-pane-sandbox"
role="tabpanel">
538             <div class="p-3 bg-black rounded border border-sec
ondary mb-3">
539                 <h6 class="text-warning mb-2"><i class="bi bi-te
rminal"></i> Интерактивная песочница для тестирования навыка</
h6>
540                 <p class="text-muted small mb-3">
541                     Введите тестовый запрос, чтобы проверить, как
агент интерпретирует инструкции данного навыка и выполняет зап
рос.
542                 </p>
543                 <div class="mb-3">
544                     <label for="sandbox-test-prompt" class="form-l
abel small fw-semibold">Тестовый запрос агенту:</label>
545                     <div class="input-group">
546                         <input type="text" class="form-control bg-da
rk text-white border-secondary" id="sandbox-test-prompt" place
holder="например: Выполни анализ данных или проверь состояние.
..">
547                         <button class="btn btn-warning text-dark fw-
bold d-flex align-items-center gap-1" id="btn-run-sandbox" onc
lick="window.skillsTab?.runSandboxTest()">
548                             <span class="spinner-border spinner-border
-sm d-none" id="sandbox-spinner" role="status"></span>
549                             <i class="bi bi-play-fill"></i> Запустить
550                         </button>
551                     </div>
552                 </div>
553
554                 <label class="form-label small fw-semibold text-
muted">Результат выполнения (Output):</label>
555                 <div class="p-3 bg-dark rounded border border-se

```

```

condary text-light font-monospace" id="sandbox-output-box" style="min-height: 140px; max-height: 320px; overflow-y: auto; white-space: pre-wrap; font-size: 0.85rem;">
556 | Готов к тестированию. Введите запрос и нажмите «Запустить».
557 |         </div>
558 |     </div>
559 | </div>
560 |
561 |     <!-- README.md Tab Pane -->
562 |     <div class="tab-pane fade" id="detail-pane-readme" role="tabpanel">
563 |         <div class="mb-2 d-flex justify-content-between align-items-center">
564 |             <label for="detail-skill-readme" class="form-label fw-semibold small text-muted mb-0">README.md документация:</label>
565 |             <small class="text-muted">Англоязычная документация пакета</small>
566 |         </div>
567 |         <textarea class="form-control bg-black text-white border-secondary font-monospace" id="detail-skill-readme" rows="18" style="font-size: 0.9rem;" spellcheck="false"></textarea>
568 |     </div>
569 |
570 |     <!-- Files Structure Tab Pane -->
571 |     <div class="tab-pane fade" id="detail-pane-files" role="tabpanel">
572 |         <div class="p-3 bg-black rounded border border-secondary">
573 |             <h6 class="text-info mb-2"><i class="bi bi-folder2-open"></i> Содержимое папки навыка:</h6>
574 |             <div id="detail-files-list" class="list-group list-group-flush bg-transparent"></div>
575 |         </div>
576 |     </div>
577 |
578 |     <!-- JSON Contract Tab Pane -->

```

```
579 |         <div class="tab-pane fade" id="detail-pane-contract"
    |         role="tabpanel">
580 |             <pre class="bg-black p-3 rounded border border-sec
    | ondary text-info mb-0" id="detail-json-contract" style="font-s
    | ize: 0.85rem; max-height: 400px; overflow-y: auto;"></pre>
581 |         </div>
582 |     </div>
583 |
584 |     <div id="detail-skill-alert" class="alert d-none py-2
    | mt-3 mb-0"></div>
585 | </div>
586 |     <div class="modal-footer border-secondary d-flex justify
    | -content-between">
587 |         <button type="button" class="btn btn-outline-secondary
    | " data-bs-dismiss="modal">Заккрыть</button>
588 |         <button type="button" class="btn btn-primary d-flex al
    | ign-items-center gap-1" id="btn-save-skill" onclick="window.sk
    | illsTab?.saveCurrentSkill()">
589 |             <span class="spinner-border spinner-border-sm d-none
    | " id="save-skill-spinner" role="status"></span>
590 |             <span><i class="bi bi-save"></i> Сохранить изменения
    | </span>
591 |         </button>
592 |     </div>
593 | </div>
594 | </div>
595 | </div>
596 |
```

main.js

```
1 // Skills Tab Logic for AI Breadboard Admin Interface
2 // Supports 3 Creation Modes (Step-by-Step Wizard, AI Assistant, Presets Library) and Live Sandbox Testing
3
4 class SkillsTabManager {
5   constructor() {
6     this.skills = [];
7     this.filteredSkills = [];
8     this.currentSkill = null;
9     this.viewMode = 'cards'; // 'cards' | 'table'
10    this.searchQuery = '';
11    this.activeFilter = 'all'; // 'all' | 'scripts' | 'references' | 'packaged'
12    this.isInitialized = false;
13
14    // Wizard state
15    this.currentWizardStep = 1;
16    this.createMode = 'wizard'; // 'wizard' | 'ai' | 'presets'
17    this.aiGeneratedData = null;
18
19    // Presets catalog
20    this.presets = [
21      {
22        id: 'tool-automation',
23        title: 'CLI & Python Tool',
24        category: 'tools',
25        badge: 'Automation',
26        description: 'Исполняемый навык с Python-скриптом автоматизации в папке scripts/. Подходит для утилит, конвертеров и системных задач.',
27        name: 'custom-cli-tool',
28        descRu: 'Автоматизированная CLI утилита для выполнения задач через скрипты.',
29        descEn: 'Automated CLI utility providing executable scripts for system routines.',
30        purpose: 'Execute specialized automation commands via
```

```

internal scripts.',
31     triggers: 'Triggered when the user requests executing
a custom tool or batch script task.',
32     protocol: '1. Validate arguments and environment.\n2.
Execute python script in `scripts/main.py`.\n3. Return formatt
ed stdout/stderr status.',
33     rules: '- Ensure non-zero exit codes are logged and re
ported.\n- Maintain English logging.',
34     hasScript: true,
35     scriptCode: `# -*- coding: utf-8 -*-\n""CLI tool auto
mation script.""\nimport sys\n\ndef main():\n    print("Execu
ting automated tool task...")\n    return 0\n\nif __name__ ==
"__main__":\n    sys.exit(main())\n`,
36     hasRef: false,
37 },
38 {
39     id: 'api-connector',
40     title: '□ REST API Connector',
41     category: 'api',
42     badge: 'Integration',
43     description: 'Интеграция с внешним REST/JSON API. Шабл
он с проверкой авторизации, обработкой таймаутов и структуриро
ванием ответов.',
44     name: 'rest-api-connector',
45     descRu: 'Интеграция с внешним сервисом через REST API.
',
46     descEn: 'Connector skill for external REST API service
s and endpoints.',
47     purpose: 'Query external HTTP REST API endpoints and p
rocess JSON payloads.',
48     triggers: 'Triggered when user asks to fetch data from
or post payloads to remote APIs.',
49     protocol: '1. Parse endpoint parameters and auth token
s.\n2. Send HTTP request with exponential retry.\n3. Parse JSO
N response and summarize.',
50     rules: '- Protect sensitive secrets and auth tokens.\n
- Handle rate limits and connection timeouts gracefully.',
51     hasScript: true,

```

```

52 |         scriptCode: `# -*- coding: utf-8 -*-\n""REST API client script.""\nimport os, urllib.request, json\n\ndef query_api(url: str):\n    req = urllib.request.Request(url, headers={"User-Agent": "AI-Breadboard/1.0"})\n    with urllib.request.urlopen(req, timeout=10) as resp:\n        return json.loads(resp.read().decode("utf-8"))\n`,
53 |         hasRef: true,
54 |         refContent: `# REST API Protocol Guidelines\n\nStandard timeout: 10s\n\nRequired headers: User-Agent, Accept\n`,
55 |     },
56 |     {
57 |         id: 'data-parser',
58 |         title: '□ Data Analyzer & Parser',
59 |         category: 'data',
60 |         badge: 'Analytics',
61 |         description: 'Обработка и анализ структурированных данных (CSV, JSON, XML, таблицы, логи). Извлечение инсайтов и форматирование результатов.',
62 |         name: 'data-analyzer',
63 |         descRu: 'Анализ структурированных данных, логов и таблиц с выводом статистики.',
64 |         descEn: 'Analyze structured data, logs, CSV/JSON datasets, and generate statistics.',
65 |         purpose: 'Parse input data files, compute aggregated metrics, and present visual summaries.',
66 |         triggers: 'Triggered when datasets, spreadsheets or data files need parsing and reporting.',
67 |         protocol: '1. Identify input format (CSV/JSON/Log).\n2. Parse schema and clean invalid rows.\n3. Compute totals, distributions, and outliers.\n4. Output Markdown summary tables.',
68 |         rules: '- Avoid memory leaks on large files.\n- Strictly English docstrings.',
69 |         hasScript: true,
70 |         scriptCode: `# -*- coding: utf-8 -*-\n""Data parsing helper.""\nimport csv, json\n\ndef parse_records(filepath: str):\n    # Add parser logic here\n    pass\n`,
71 |         hasRef: false,

```



```
72     },
73     {
74         id: 'ai-persona',
75         title: 'AI Expert & Persona',
76         category: 'persona',
77         badge: 'Agent Role',
78         description: 'Специализированная роль и экспертные инс
79         трукции (Архитектор, Аналитик безопасности, Code Reviewer, Кон
80         сультант).',
81         name: 'domain-expert',
82         descRu: 'Экспертная роль и правила поведения для специ
83         ализованных консультаций.',
84         descEn: 'Specialized domain expert persona with strict
85         evaluation guidelines.',
86         purpose: 'Act as a senior domain expert offering high-
87         level technical feedback.',
88         triggers: 'Triggered whenever specialized architectural or domain advice is requested.',
89         protocol: '1. Analyze problem from the perspective of
90         domain best practices.\n2. Provide structured pros/cons analysis.\n3. Recommend action plan with trade-offs.',
91         rules: '- Be concise, objective, and reference industry standards.',
92         hasScript: false,
93         hasRef: true,
94         refContent: '# Domain Expertise Standards\n\nKey methodologies, architecture patterns, and decision trees.\n',
95     },
96     {
97         id: 'monitoring-notifier',
98         title: 'Monitor & Notifier',
99         category: 'monitoring',
100        badge: 'Observability',
101        description: 'Периодическая проверка сервисов, мониторинг системных параметров, выявление сбоев и отправка уведомлений.',
102        name: 'health-monitor',
103        descRu: 'Мониторинг доступности сервисов и формирования'
```

```

    е алертов при сбоях.',
98     descEn: 'Health monitoring and alerting skill for infr
    astructure and services.',
99     purpose: 'Continuously verify target service endpoints
    and notify on anomalies.',
100    triggers: 'Triggered by scheduled cron jobs or on-dema
    nd health check requests.',
101    protocol: '1. Ping healthcheck endpoints.\n2. Measure
    latency and status codes.\n3. If failure detected, format diag
    nostic alert.',
102    rules: '- Fail-fast on network unreachability.\n- Incl
    ude timestamps and severity level.',
103    hasScript: true,
104    scriptCode: `# -*- coding: utf-8 -*-\n""""Health check
    monitor."""\nimport time\n\ndef check_service(host: str, port:
    int):\n    # Health probe\n    return True\n`,
105    hasRef: false,
106  },
107  {
108    id: 'doc-generator',
109    title: '□ Report & Doc Generator',
110    category: 'docgen',
111    badge: 'Documentation',
112    description: 'Автоматическая генерация документации, о
    тчетов в Markdown/HTML и сводок по проекту на основе входных д
    анных.',
113    name: 'report-generator',
114    descRu: 'Генератор красивых отчетов, проектной докумен
    тации и сводок.',
115    descEn: 'Automated documentation and structured report
    generation skill.',
116    purpose: 'Generate comprehensive Markdown reports, cha
    nge logs, and release notes.',
117    triggers: 'Triggered when documentation, release summa
    ries or audit reports are requested.',
118    protocol: '1. Collect source items and metadata.\n2. A
    pply template sections and headings.\n3. Format tables and Git
    Hub alerts.\n4. Save output file.',

```

```

119 |         rules: '- Follow documentation style standards.\n- Ens
    | ure all links and tables are valid.',
120 |         hasScript: false,
121 |         hasRef: true,
122 |         refContent: `# Document Generation Templates\n\nTempla
    | te blocks, header standards, and formatting guidelines.\n`,
123 |     }
124 | ];
125 | }
126 |
127 | async init() {
128 |     if (this.isInitialized) {
129 |         await this.refresh();
130 |         return;
131 |     }
132 |     this.bindEvents();
133 |     this.renderPresets();
134 |     this.isInitialized = true;
135 |     await this.refresh();
136 | }
137 |
138 | bindEvents() {
139 |     const searchInput = document.getElementById('skills-search
    | -input');
140 |     if (searchInput) {
141 |         searchInput.addEventListener('input', (e) => {
142 |             this.searchQuery = e.target.value.trim().toLowerCase()
    | ;
143 |             this.applyFilters();
144 |         });
145 |     }
146 |
147 |     const wizardNameInput = document.getElementById('wizard-sk
    | ill-name');
148 |     if (wizardNameInput) {
149 |         wizardNameInput.addEventListener('input', (e) => {
150 |             const val = e.target.value;
151 |             const normalized = val.toLowerCase().replace(/[^\a-z0-9

```

```

    _-]/g, '-');
152         if (val !== normalized) {
153             e.target.value = normalized;
154         }
155         this.syncWizardPreview();
156     });
157 }
158
159 // Step 2 field listeners for live markdown generator
160 ['wizard-inst-purpose', 'wizard-inst-triggers', 'wizard-in
st-protocol', 'wizard-inst-rules'].forEach(id => {
161     const el = document.getElementById(id);
162     if (el) {
163         el.addEventListener('input', () => this.syncInstructio
nsFromFields());
164     }
165 });
166
167 ['wizard-skill-desc-ru', 'wizard-skill-desc-en', 'wizard-s
kill-dir'].forEach(id => {
168     const el = document.getElementById(id);
169     if (el) {
170         el.addEventListener('input', () => this.syncWizardPrev
iew());
171         el.addEventListener('change', () => this.syncWizardPre
view());
172     }
173 });
174 }
175
176 async apiFetch(endpoint, options = {}) {
177     let url = endpoint;
178     let res = await fetch(url, options);
179     if (!res.ok && url.startsWith('/api/skills')) {
180         const fallbackUrl = url.replace('/api/skills', '/api/adm
in/skills');
181         const fallbackRes = await fetch(fallbackUrl, options);
182         if (fallbackRes.ok) {

```

```
183         return fallbackRes;
184     }
185 }
186 return res;
187 }
188
189 async refresh() {
190     this.showLoading(true);
191     try {
192         const response = await this.apiFetch('/api/skills');
193         if (!response.ok) {
194             throw new Error(`HTTP ${response.status}: ${response.s
195 tatusText}`);
196         }
197         const data = await response.json();
198         this.skills = data.skills || [];
199         this.updateStats();
200         this.applyFilters();
201     } catch (err) {
202         console.error('[SkillsTab] Load error:', err);
203         const cardsContainer = document.getElementById('skills-c
204 ards-container');
205         if (cardsContainer) {
206             cardsContainer.innerHTML = `
207                 <div class="col-12">
208                     <div class="alert alert-danger d-flex align-items-
209 center justify-content-between p-3 shadow-sm">
210                         <div><i class="bi bi-exclamation-triangle-fill m
211 e-2"></i><strong>Ошибка загрузки навыков:</strong> ${err.messa
212 ge}</div>
213                         <button class="btn btn-sm btn-outline-light" onc
214 lick="window.skillsTab?.refresh()">Повторить</button>
215                     </div>
216                 </div>
217             `;
218         }
219     } finally {
220         this.showLoading(false);
221     }
222 }
```

```

215     }
216 }
217
218 updateStats() {
219     const total = this.skills.length;
220     const withScripts = this.skills.filter(s => s.has_scripts)
    .length;
221     const withRefs = this.skills.filter(s => s.has_references)
    .length;
222     const packaged = this.skills.filter(s => s.has_dist).length;
223
224     const setVal = (id, val) => {
225         const el = document.getElementById(id);
226         if (el) el.textContent = val;
227     };
228
229     setVal('skills-total-badge', total);
230     setVal('stat-total-skills', total);
231     setVal('stat-scripts-skills', withScripts);
232     setVal('stat-refs-skills', withRefs);
233     setVal('stat-packaged-skills', packaged);
234 }
235
236 applyFilters() {
237     const filterSelect = document.getElementById('skills-filter-select');
238     this.activeFilter = filterSelect ? filterSelect.value : 'all';
239
240     this.filteredSkills = this.skills.filter(skill => {
241         if (this.searchQuery) {
242             const nameMatch = (skill.name || '').toLowerCase().includes(this.searchQuery);
243             const descMatch = (skill.description || '').toLowerCase().includes(this.searchQuery);
244             const i18nMatch = Object.values(skill.descriptions_i18n || {}).some(d => (d || '').toLowerCase().includes(this.searc

```

```

hQuery));
245     if (!nameMatch && !descMatch && !i18nMatch) return false;
246 }
247
248     if (this.activeFilter === 'scripts' && !skill.has_scripts) return false;
249     if (this.activeFilter === 'references' && !skill.has_references) return false;
250     if (this.activeFilter === 'packaged' && !skill.has_dist) return false;
251
252     return true;
253 });
254
255     this.render();
256 }
257
258     clearSearch() {
259         const searchInput = document.getElementById('skills-search-input');
260         if (searchInput) searchInput.value = '';
261         this.searchQuery = '';
262         this.applyFilters();
263     }
264
265     setViewMode(mode) {
266         this.viewMode = mode;
267         const btnCards = document.getElementById('btn-view-cards');
268         const btnTable = document.getElementById('btn-view-table');
269
270         if (mode === 'cards') {
271             btnCards?.classList.add('active');
272             btnTable?.classList.remove('active');
273         } else {
274             btnCards?.classList.remove('active');

```

```
275 |     btnTable?.classList.add('active');
276 |   }
277 |   this.render();
278 | }
279 |
280 | render() {
281 |   const cardsContainer = document.getElementById('skills-car
ds-container');
282 |   const tableCard = document.getElementById('skills-table-ca
rd');
283 |   const emptyNotice = document.getElementById('skills-empty'
);
284 |
285 |   if (!cardsContainer || !tableCard) return;
286 |
287 |   if (this.filteredSkills.length === 0) {
288 |     cardsContainer.innerHTML = '';
289 |     tableCard.classList.add('d-none');
290 |     emptyNotice?.classList.remove('d-none');
291 |     return;
292 |   }
293 |
294 |   emptyNotice?.classList.add('d-none');
295 |
296 |   if (this.viewMode === 'cards') {
297 |     tableCard.classList.add('d-none');
298 |     cardsContainer.classList.remove('d-none');
299 |     cardsContainer.innerHTML = this.filteredSkills.map(s =>
this.renderCard(s)).join('');
300 |   } else {
301 |     cardsContainer.classList.add('d-none');
302 |     tableCard.classList.remove('d-none');
303 |     const tableBody = document.getElementById('skills-table-
body');
304 |     if (tableBody) {
305 |       tableBody.innerHTML = this.filteredSkills.map(s => thi
s.renderTableRow(s)).join('');
306 |     }
```



```

307     }
308 }
309
310 renderCard(skill) {
311     const escapeHtml = str => String(str || '').replace(/&lt;>'/
312     |"/g, tag => ({
313         '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&#39;', "'
314         ': '&quot;';
315     }[tag] || tag));
316
317     const badges = [];
318     if (skill.has_scripts) badges.push('<span class="badge bg-
319     info-subtle text-info border border-info-subtle small me-1"><i
320     class="bi bi-file-earmark-code"></i> scripts</span>');
321     if (skill.has_references) badges.push('<span class="badge
322     bg-warning-subtle text-warning border border-warning-subtle sm
323     all me-1"><i class="bi bi-journal-text"></i> references</span>
324     ');
325     if (skill.has_assets) badges.push('<span class="badge bg-s
326     econdary border border-secondary small me-1"><i class="bi bi-i
327     mages"></i> assets</span>');
328     if (skill.has_dist) badges.push('<span class="badge bg-suc
329     cess-subtle text-success border border-success-subtle small me
330     -1"><i class="bi bi-box-seam"></i> .skill</span>');
331
332     const descRu = skill.descriptions_i18n?.ru || '';
333     const descDisplay = descRu ? `${descRu}` : (skill.descript
334     ion || 'Описание отсутствует');
335
336     return `
337         <div class="col-12 col-md-6 col-lg-4">
338             <div class="card bg-dark border-secondary text-white h
339             -100 shadow-sm skill-card">
340                 <div class="card-header border-secondary d-flex just
341                 ify-content-between align-items-center bg-black bg-opacity-25
342                 py-2">
343                     <span class="badge bg-primary font-monospace fs-6
344                     text-truncate" style="max-width: 70%;" title="${escapeHtml(ski

```

```

    ll.name)}}">
329         □  ${escapeHtml(skill.name)}
330     </span>
331     <span class="text-muted small" title="Количество ф
айлов">
332         <i class="bi bi-files"></i> ${skill.files_count
|| 1}
333     </span>
334 </div>
335 <div class="card-body py-2">
336     <p class="card-text text-light small mb-2 text-tru
ncate-3" style="min-height: 40px; display: -webkit-box; -webki
t-line-clamp: 2; -webkit-box-orient: vertical; overflow: hidde
n;" title="${escapeHtml(descDisplay)}">
337         ${escapeHtml(descDisplay)}
338     </p>
339     <div class="mb-2 text-truncate">
340         <small class="text-secondary font-monospace"><i
class="bi bi-folder"></i> ${escapeHtml(skill.relative_path)}</
small>
341     </div>
342     <div class="d-flex flex-wrap gap-1">
343         ${badges.join('')}
344     </div>
345 </div>
346 <div class="card-footer border-secondary bg-black bg
-opacity-25 py-2 d-flex justify-content-between align-items-ce
nter">
347     <button type="button" class="btn btn-sm btn-outlin
e-primary d-flex align-items-center gap-1" onclick="window.ski
llsTab?.openSkillDetails('${escapeHtml(skill.name)}')">
348         <i class="bi bi-sliders"></i> Детали
349     </button>
350     <div class="btn-group btn-group-sm">
351         <button type="button" class="btn btn-outline-war
ning" title="Тестировать в Sandbox" onclick="window.skillsTab?
.openSkillSandbox('${escapeHtml(skill.name)}')">
352         <i class="bi bi-play-circle"></i>

```

```

353         </button>
354         <button type="button" class="btn btn-outline-info" title="Собрать .skill архив" onclick="window.skillsTab?.packageSkill('${escapeHtml(skill.name)}')">
355             <i class="bi bi-box-seam"></i>
356         </button>
357         <button type="button" class="btn btn-outline-danger" title="Удалить навык" onclick="window.skillsTab?.deleteSkill('${escapeHtml(skill.name)}')">
358             <i class="bi bi-trash"></i>
359         </button>
360     </div>
361 </div>
362 </div>
363 </div>
364 `;
365 }
366
367 renderTableRow(skill) {
368     const escapeHtml = str => String(str || '').replace(/[<>'
369     '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&#39;', "'
370     ': '&quot;'];
371
372     const badges = [];
373     if (skill.has_scripts) badges.push('<span class="badge bg-info-subtle text-info border border-info-subtle small me-1">scripts</span>');
374     if (skill.has_references) badges.push('<span class="badge bg-warning-subtle text-warning border border-warning-subtle small me-1">refs</span>');
375     if (skill.has_dist) badges.push('<span class="badge bg-success-subtle text-success border border-success-subtle small me-1">.skill</span>');
376
377     const descRu = skill.descriptions_i18n?.ru || '';
378     const descDisplay = descRu ? `${descRu}` : (skill.descript

```

```

    ion || '-');
379
380     return `
381         <tr>
382             <td>
383                 <span class="badge bg-primary font-monospace fs-6">
384                     ${escapeHtml(skill.name)}</span>
385             </td>
386             <td>
387                 <div class="small text-light text-truncate" style="m
388 ax-width: 350px;" title="${escapeHtml(descDisplay)}">
389                     ${escapeHtml(descDisplay)}
390                 </div>
391             </td>
392             <td>
393                 <code class="small text-secondary">${escapeHtml(skill.relative_path)}</code>
394             </td>
395             <td>
396                 ${badges.length ? badges.join('') : '<span class="te
397 xt-muted small">--</span>'}
398             </td>
399             <td style="text-align: right;">
400                 <div class="btn-group btn-group-sm">
401                     <button type="button" class="btn btn-outline-prima
402 ry" title="Открыть детали" onclick="window.skillsTab?.openSkill
403 lDetails('${escapeHtml(skill.name)}')">
404                         <i class="bi bi-sliders"></i>
405                     </button>
406                     <button type="button" class="btn btn-outline-warni
407 ng" title="Тестировать в Sandbox" onclick="window.skillsTab?.o
408 penSkillSandbox('${escapeHtml(skill.name)}')">
409                         <i class="bi bi-play-circle"></i>
410                     </button>
411                     <button type="button" class="btn btn-outline-info"
412 title="Собрать архив" onclick="window.skillsTab?.packageSkill
413 ('${escapeHtml(skill.name)}')">
414                         <i class="bi bi-box-seam"></i>

```

```

406 |         </button>
407 |         <button type="button" class="btn btn-outline-dange
| r" title="Удалить" onclick="window.skillsTab?.deleteSkill('${e
| scapeHtml(skill.name)}')">
408 |             <i class="bi bi-trash"></i>
409 |         </button>
410 |     </div>
411 | </td>
412 | </tr>
413 | `;
414 | }
415 |
416 | // =====
| =====
417 | // CREATION WIZARD LOGIC (3 Modes: Wizard, AI, Presets)
418 | // =====
| =====
419 |
420 | switchCreateMode(mode) {
421 |     this.createMode = mode;
422 | }
423 |
424 | goToStep(step) {
425 |     if (step < 1 || step > 4) return;
426 |     this.currentWizardStep = step;
427 |
428 |     for (let i = 1; i <= 4; i++) {
429 |         const stepEl = document.getElementById(`wizard-step-${i}
| `);
430 |         const badgeEl = document.getElementById(`step-badge-${i}
| `);
431 |         const labelEl = document.getElementById(`step-label-${i}
| `);
432 |
433 |         if (stepEl) {
434 |             if (i === step) stepEl.classList.remove('d-none');
435 |             else stepEl.classList.add('d-none');
436 |         }

```

```
437 |
438 |     if (badgeEl && labelEl) {
439 |         if (i === step) {
440 |             badgeEl.className = 'badge rounded-circle p-2 bg-pri
mary step-indicator';
441 |             labelEl.className = 'small fw-semibold text-white';
442 |         } else if (i < step) {
443 |             badgeEl.className = 'badge rounded-circle p-2 bg-suc
cess step-indicator';
444 |             labelEl.className = 'small text-success';
445 |         } else {
446 |             badgeEl.className = 'badge rounded-circle p-2 bg-sec
ondary step-indicator';
447 |             labelEl.className = 'small text-muted';
448 |         }
449 |     }
450 | }
451 |
452 | const prevBtn = document.getElementById('btn-wizard-prev')
;
453 | const nextBtn = document.getElementById('btn-wizard-next')
;
454 | const submitBtn = document.getElementById('btn-wizard-subm
it');
455 |
456 | if (prevBtn) prevBtn.disabled = (step === 1);
457 | if (nextBtn) {
458 |     if (step === 4) nextBtn.classList.add('d-none');
459 |     else nextBtn.classList.remove('d-none');
460 | }
461 | if (submitBtn) {
462 |     if (step === 4) submitBtn.classList.remove('d-none');
463 |     else submitBtn.classList.add('d-none');
464 | }
465 |
466 | if (step === 4) {
467 |     this.syncWizardPreview();
468 | }
```

```
469 | }
470 |
471 | wizardNextStep() {
472 |     if (this.currentWizardStep === 1) {
473 |         const name = document.getElementById('wizard-skill-name'
474 | )?.value.trim();
475 |         if (!name) {
476 |             this.showCreateAlert('Укажите имя навыка (kebab-case)'
477 | , 'danger');
478 |             return;
479 |         }
480 |         this.clearCreateAlert();
481 |     }
482 |
483 |     wizardPrevStep() {
484 |         this.goToStep(this.currentWizardStep - 1);
485 |     }
486 |
487 |     generateSlugFromName() {
488 |         const descRu = document.getElementById('wizard-skill-desc-
489 | ru')?.value.trim();
490 |         const descEn = document.getElementById('wizard-skill-desc-
491 | en')?.value.trim();
492 |         const source = descEn || descRu || 'custom-skill';
493 |
494 |         // Transliterate or simple slugify
495 |         let slug = source.toLowerCase()
496 |             .replace(/[а-яё]/g, char => {
497 |                 const tr = {'а': 'a', 'б': 'b', 'в': 'v', 'г': 'g', 'д': 'd', 'е': 'e', 'ё': 'yo', 'ж': 'zh', 'з': 'z', 'и': 'i', 'й': 'y', 'к': 'k', 'л': 'l', 'м': 'm', 'н': 'n', 'о': 'o', 'п': 'p', 'р': 'r', 'с': 's', 'т': 't', 'у': 'u', 'ф': 'f', 'х': 'h', 'ц': 'ts', 'ч': 'ch', 'ш': 'sh', 'щ': 'sch', 'ъ': ' ', 'ы': 'y', 'ь': ' ', 'э': 'e', 'ю': 'yu', 'я': 'ya'};
498 |                 return tr[char] || char;
499 |             })
500 |             .replace(/[^a-z0-9_-]/g, '-')
```

```
499 |         .replace(/-+/g, '-')
500 |         .replace(/^-|-$/g, '')
501 |         .slice(0, 32);
502 |
503 |     if (slug) {
504 |         const nameInput = document.getElementById('wizard-skill-
name');
505 |         if (nameInput) nameInput.value = slug;
506 |         this.syncWizardPreview();
507 |     }
508 | }
509 |
510 | onCategoryChanged() {
511 |     const cat = document.getElementById('wizard-skill-category
');
512 |     const descRuInput = document.getElementById('wizard-skill-
desc-ru');
513 |     const descEnInput = document.getElementById('wizard-skill-
desc-en');
514 |
515 |     if (cat && (!descRuInput?.value || !descEnInput?.value)) {
516 |         const preset = this.presets.find(p => p.category === cat
);
517 |         if (preset) {
518 |             if (!descRuInput.value) descRuInput.value = preset.des
cRu;
519 |             if (!descEnInput.value) descEnInput.value = preset.des
cEn;
520 |         }
521 |     }
522 |     this.syncWizardPreview();
523 | }
524 |
525 | fillStandardInstructions() {
526 |     const name = document.getElementById('wizard-skill-name')?
.value.trim() || 'custom-skill';
527 |     const descRu = document.getElementById('wizard-skill-desc-
ru')?.value.trim() || 'Выполнение специализированных задач';
```



```

528     const descEn = document.getElementById('wizard-skill-desc-
| en')?.value.trim() || 'Specialized agent task execution';
529
530     document.getElementById('wizard-inst-purpose').value = des
| cEn;
531     document.getElementById('wizard-inst-triggers').value = `A
| ctivate when the user asks to perform tasks matching: ${descRu
| }.`;
532     document.getElementById('wizard-inst-protocol').value = `1
| . Validate input query and preconditions.\n2. Execute core log
| ic and inspect helper scripts in scripts/.\n3. Return structur
| ed markdown response.`;
533     document.getElementById('wizard-inst-rules').value = `- En
| sure English logging via src.logger.logger.\n- Gracefully catc
| h and report all execution errors.`;
534
535     this.syncInstructionsFromFields();
536 }
537
538 syncInstructionsFromFields() {
539     const name = document.getElementById('wizard-skill-name')?
| .value.trim() || 'custom-skill';
540     const title = name.replace('-', ' ').replace('_', ' ').rep
| lace(/\b\w/g, l => l.toUpperCase());
541     const purpose = document.getElementById('wizard-inst-purpo
| se')?.value.trim() || 'Provide specialized agent capability.';
542     const triggers = document.getElementById('wizard-inst-trig
| gers')?.value.trim() || 'Activate upon relevant user requests.
| ';
543     const protocol = document.getElementById('wizard-inst-prot
| ocol')?.value.trim() || '1. Parse input.\n2. Execute action.\n
| 3. Output result.';
544     const rules = document.getElementById('wizard-inst-rules')
| ?.value.trim() || '- Maintain fail-fast validation.\n- Report
| status clearly.';
545
546     const md = `# ${title}
547

```

```

548 ## □ Purpose
549 ${purpose}
550
551 ## □ When to Use & Triggers
552 ${triggers}
553
554 ## □ Execution Protocol
555 ${protocol}
556
557 ## □ Rules & Constraints
558 ${rules}
559 `;
560     const instEl = document.getElementById('wizard-skill-instr
    uctions');
561     if (instEl) instEl.value = md;
562     this.syncWizardPreview();
563 }
564
565 toggleScriptEditor() {
566     const chk = document.getElementById('wizard-enable-script'
    );
567     const container = document.getElementById('wizard-script-c
    ontainer');
568     const scriptCode = document.getElementById('wizard-script-
    code');
569
570     if (chk?.checked) {
571         container?.classList.remove('d-none');
572         if (!scriptCode.value.trim()) {
573             const name = document.getElementById('wizard-skill-nam
    e')?.value.trim() || 'custom-skill';
574             scriptCode.value = `# -*- coding: utf-8 -*-\n""Execut
    able tool helper for ${name}.""\n\nimport sys\n\ndef main():\n
    print("Running helper script...")\n    return 0\n\nif __n
    ame__ == "__main__":\n    sys.exit(main())\n`;
575         }
576     } else {
577         container?.classList.add('d-none');

```

```

578     }
579     this.syncWizardPreview();
580 }
581
582 toggleRefEditor() {
583     const chk = document.getElementById('wizard-enable-ref');
584     const container = document.getElementById('wizard-ref-cont
ainer');
585     const refContent = document.getElementById('wizard-ref-con
tent');
586
587     if (chk?.checked) {
588         container?.classList.remove('d-none');
589         if (!refContent.value.trim()) {
590             const name = document.getElementById('wizard-skill-nam
e')?.value.trim() || 'custom-skill';
591             refContent.value = `# ${name.replace('-', ' ').toUpper
Case()} Specification\n\nDetailed operational rules and refere
nce data.\n`;
592         }
593     } else {
594         container?.classList.add('d-none');
595     }
596     this.syncWizardPreview();
597 }
598
599 syncWizardPreview() {
600     const name = document.getElementById('wizard-skill-name')?
.value.trim() || 'custom-skill';
601     const dir = document.getElementById('wizard-skill-dir')?.v
alue || '.agents/skills';
602     const descEn = document.getElementById('wizard-skill-desc-
en')?.value.trim() || `Agent skill for ${name}.`;
603     const descRu = document.getElementById('wizard-skill-desc-
ru')?.value.trim() || descEn;
604     const instructions = document.getElementById('wizard-skill
-instructions')?.value.trim() || `# ${name}\n\n## □ Purpose\n
${descEn}\n`;

```

```
605
606     const hasScript = document.getElementById('wizard-enable-s
|cript')?.checked;
607     const hasRef = document.getElementById('wizard-enable-ref'
|)?.checked;
608
609     // Badges in step 4
610     const nameEl = document.getElementById('preview-skill-name
|');
611     const pathEl = document.getElementById('preview-skill-path
|');
612     const badgeScript = document.getElementById('preview-badge
|-script');
613     const badgeRef = document.getElementById('preview-badge-re
|f');
614     const previewFull = document.getElementById('preview-skill
|md-full');
615
616     if (nameEl) nameEl.textContent = name;
617     if (pathEl) pathEl.textContent = `${dir}/${name}`;
618     if (badgeScript) {
619         if (hasScript) badgeScript.classList.remove('d-none');
620         else badgeScript.classList.add('d-none');
621     }
622     if (badgeRef) {
623         if (hasRef) badgeRef.classList.remove('d-none');
624         else badgeRef.classList.add('d-none');
625     }
626
627     const fullSkillMd = `---
628 name: ${name}
629 description: ${descEn}
630 description_i18n:
631   en: ${descEn}
632   ru: ${descRu}
633 ---
634
635 ${instructions}`;
```

```

636
637     if (previewFull) previewFull.textContent = fullSkillMd;
638
639     // Sync README default
640     const readmeEl = document.getElementById('wizard-skill-readme');
641     if (readmeEl && !readmeEl.value.trim()) {
642         readmeEl.value = `# ${name.replace('-', ' ').replace(/\b\w/g, l => l.toUpperCase())}\n\n## Overview\n\n${descEn}\n\n## Location\n\n`${dir}/${name}/${name}`;
643     }
644 }
645
646 async createSkillFromWizard() {
647     const name = document.getElementById('wizard-skill-name')?.value.trim().toLowerCase();
648     if (!name) {
649         this.goToStep(1);
650         this.showCreateAlert('Укажите имя навыка', 'danger');
651         return;
652     }
653
654     const spinner = document.getElementById('wizard-submit-spinner');
655     spinner?.classList.remove('d-none');
656
657     const descEn = document.getElementById('wizard-skill-desc-en')?.value.trim() || `Agent skill for ${name}.`;
658     const descRu = document.getElementById('wizard-skill-desc-ru')?.value.trim() || descEn;
659     const targetDir = document.getElementById('wizard-skill-dir')?.value || '.agents/skills';
660     const category = document.getElementById('wizard-skill-category')?.value || 'general';
661     const instructions = document.getElementById('wizard-skill-instructions')?.value.trim() || '';
662     const readme = document.getElementById('wizard-skill-readme')?.value.trim() || '';

```

```
663
664     const scripts = {};
665     if (document.getElementById('wizard-enable-script')?.checked) {
666         scripts['main.py'] = document.getElementById('wizard-script-code')?.value || '';
667     }
668
669     const references = {};
670     if (document.getElementById('wizard-enable-ref')?.checked) {
671         references['guide.md'] = document.getElementById('wizard-ref-content')?.value || '';
672     }
673
674     try {
675         const res = await this.apiFetch('/api/skills', {
676             method: 'POST',
677             headers: { 'Content-Type': 'application/json' },
678             body: JSON.stringify({
679                 name,
680                 description: descEn,
681                 description_ru: descRu,
682                 category,
683                 target_dir: targetDir,
684                 instructions,
685                 readme,
686                 scripts,
687                 references
688             })
689         });
690
691         if (!res.ok) {
692             const errData = await res.json().catch(() => ({}));
693             throw new Error(errData.detail || res.statusText);
694         }
695
696         // Close modal
```

```

697 |     const modalEl = document.getElementById('createSkillModa
    | l');
698 |     const modal = bootstrap.Modal.getInstance(modalEl);
699 |     modal?.hide();
700 |
701 |     await this.refresh();
702 |     await this.openSkillDetails(name);
703 | } catch (err) {
704 |     console.error('[SkillsTab] Create error:', err);
705 |     this.showCreateAlert(`Ошибка создания навыка: ${err.mess
    | age}`, 'danger');
706 | } finally {
707 |     spinner?.classList.add('d-none');
708 | }
709 | }
710 |
711 | // =====
    | =====
712 | // AI GENERATOR MODE
713 | // =====
    | =====
714 |
715 | async generateSkillWithAi() {
716 |     const prompt = document.getElementById('ai-skill-prompt')?
    | .value.trim();
717 |     if (!prompt) {
718 |         this.showCreateAlert('Введите описание задачи для генера
    | ции навыка', 'warning');
719 |         return;
720 |     }
721 |
722 |     const spinner = document.getElementById('ai-gen-spinner');
723 |     const resBox = document.getElementById('ai-result-box');
724 |     const resSlug = document.getElementById('ai-res-slug');
725 |     const resPreview = document.getElementById('ai-res-preview
    | ');
726 |
727 |     spinner?.classList.remove('d-none');

```

```
728     this.clearCreateAlert();
729
730     try {
731         const category = document.getElementById('ai-skill-category')?.value || 'general';
732         const targetDir = document.getElementById('ai-skill-dir')?.value || '.agents/skills';
733
734         const res = await this.apiFetch('/api/skills/generate-ai', {
735             method: 'POST',
736             headers: { 'Content-Type': 'application/json' },
737             body: JSON.stringify({ prompt, category, target_dir: targetDir })
738         });
739
740         if (!res.ok) {
741             const errData = await res.json().catch(() => ({}));
742             throw new Error(errData.detail || res.statusText);
743         }
744
745         const data = await res.json();
746         this.aiGeneratedData = data.generated;
747
748         if (resSlug) resSlug.textContent = `📄 ${this.aiGeneratedData.name}`;
749         if (resPreview) resPreview.textContent = this.aiGeneratedData.instructions;
750         resBox?.classList.remove('d-none');
751     } catch (err) {
752         console.error('[SkillsTab] AI generate error:', err);
753         this.showCreateAlert(`Ошибка генерации навыка: ${err.message}`, 'danger');
754     } finally {
755         spinner?.classList.add('d-none');
756     }
757 }
```



```

759 |   applyAiResultToWizard() {
760 |       if (!this.aiGeneratedData) return;
761 |       const g = this.aiGeneratedData;
762 |
763 |       document.getElementById('wizard-skill-name').value = g.name || '';
764 |       document.getElementById('wizard-skill-desc-en').value = g.description || '';
765 |       document.getElementById('wizard-skill-desc-ru').value = g.description_ru || '';
766 |       document.getElementById('wizard-skill-category').value = g.category || 'general';
767 |       document.getElementById('wizard-skill-dir').value = g.target_dir || '.agents/skills';
768 |       document.getElementById('wizard-skill-instructions').value = g.instructions || '';
769 |       document.getElementById('wizard-skill-readme').value = g.readme || '';
770 |
771 |       if (g.scripts && g.scripts['main.py']) {
772 |           document.getElementById('wizard-enable-script').checked = true;
773 |           document.getElementById('wizard-script-container').classList.remove('d-none');
774 |           document.getElementById('wizard-script-code').value = g.scripts['main.py'];
775 |       }
776 |
777 |       if (g.references && g.references['guide.md']) {
778 |           document.getElementById('wizard-enable-ref').checked = true;
779 |           document.getElementById('wizard-ref-container').classList.remove('d-none');
780 |           document.getElementById('wizard-ref-content').value = g.references['guide.md'];
781 |       }
782 |
783 |       // Switch to wizard tab

```

```

784 |     const wizardTabBtn = document.getElementById('mode-wizard-
      | tab');
785 |     if (wizardTabBtn) {
786 |         const tab = new bootstrap.Tab(wizardTabBtn);
787 |         tab.show();
788 |     }
789 |     this.goToStep(1);
790 |     this.syncWizardPreview();
791 | }
792 |
793 | async saveAiSkillDirectly() {
794 |     if (!this.aiGeneratedData) return;
795 |     const g = this.aiGeneratedData;
796 |
797 |     try {
798 |         const res = await this.apiFetch('/api/skills', {
799 |             method: 'POST',
800 |             headers: { 'Content-Type': 'application/json' },
801 |             body: JSON.stringify({
802 |                 name: g.name,
803 |                 description: g.description,
804 |                 description_ru: g.description_ru,
805 |                 category: g.category,
806 |                 target_dir: g.target_dir,
807 |                 instructions: g.instructions,
808 |                 readme: g.readme,
809 |                 scripts: g.scripts || {},
810 |                 references: g.references || {}
811 |             })
812 |         });
813 |
814 |         if (!res.ok) {
815 |             const errData = await res.json().catch(() => ({}));
816 |             throw new Error(errData.detail || res.statusText);
817 |         }
818 |
819 |         const modalEl = document.getElementById('createSkillModa
      | l');

```

```

820 |         const modal = bootstrap.Modal.getInstance(modalEl);
821 |         modal?.hide();
822 |
823 |         await this.refresh();
824 |         await this.openSkillDetails(g.name);
825 |     } catch (err) {
826 |         console.error('[SkillsTab] AI Direct Save error:', err);
827 |         this.showCreateAlert(`Ошибка сохранения: ${err.message}`
828 | , 'danger');
829 |     }
830 |
831 |     // =====
832 |     // PRESETS LIBRARY
833 |     // =====
834 |
835 |     renderPresets() {
836 |         const container = document.getElementById('presets-cards-c
837 | ontainer');
838 |         if (!container) return;
839 |         container.innerHTML = this.presets.map(p => `
840 |             <div class="col-12 col-md-6 col-lg-4">
841 |                 <div class="card bg-dark border-secondary text-white h
842 | -100 shadow-sm">
843 |                     <div class="card-header border-secondary d-flex just
844 | ify-content-between align-items-center bg-black bg-opacity-25
845 | py-2">
846 |                         <span class="fw-bold small text-white">${p.title}<
847 | /span>
848 |                         <span class="badge bg-primary-subtle text-primary
849 | border border-primary-subtle small">${p.badge}</span>
850 |                     </div>
851 |                     <div class="card-body py-2">
852 |                         <p class="card-text text-muted small mb-2" style="
853 | min-height: 48px;">

```

```

848         ${p.description}
849     </p>
850     <div class="small font-monospace text-secondary mb
      -2">
851         <code>□  ${p.name}</code>
852     </div>
853 </div>
854     <div class="card-footer border-secondary bg-black bg
      -opacity-25 py-2">
855         <button type="button" class="btn btn-outline-prima
      ry btn-sm w-100 d-flex align-items-center justify-content-cent
      er gap-1" onclick="window.skillsTab?.applyPreset('${p.id}')">
856             <i class="bi bi-box-arrow-in-right"></i> Исполни
      овать шаблон
857         </button>
858     </div>
859 </div>
860 </div>
861 `).join('');
862 }
863
864 applyPreset(presetId) {
865     const preset = this.presets.find(p => p.id === presetId);
866     if (!preset) return;
867
868     document.getElementById('wizard-skill-name').value = prese
      t.name;
869     document.getElementById('wizard-skill-desc-en').value = pr
      eset.descEn;
870     document.getElementById('wizard-skill-desc-ru').value = pr
      eset.descRu;
871     document.getElementById('wizard-skill-category').value = p
      reset.category;
872     document.getElementById('wizard-skill-dir').value = '.agen
      ts/skills';
873
874     document.getElementById('wizard-inst-purpose').value = pre
      set.purpose;

```

```
875 |     document.getElementById('wizard-inst-triggers').value = pr
    |     eset.triggers;
876 |     document.getElementById('wizard-inst-protocol').value = pr
    |     eset.protocol;
877 |     document.getElementById('wizard-inst-rules').value = prese
    |     t.rules;
878 |     this.syncInstructionsFromFields();
879 |
880 |     // Scripts
881 |     const scriptChk = document.getElementById('wizard-enable-s
    |     cript');
882 |     if (scriptChk) {
883 |         scriptChk.checked = preset.hasScript;
884 |         if (preset.hasScript) {
885 |             document.getElementById('wizard-script-container')?.cl
    |             assList.remove('d-none');
886 |             document.getElementById('wizard-script-code').value =
    |             preset.scriptCode || '';
887 |         } else {
888 |             document.getElementById('wizard-script-container')?.cl
    |             assList.add('d-none');
889 |         }
890 |     }
891 |
892 |     // References
893 |     const refChk = document.getElementById('wizard-enable-ref'
    |     );
894 |     if (refChk) {
895 |         refChk.checked = preset.hasRef;
896 |         if (preset.hasRef) {
897 |             document.getElementById('wizard-ref-container')?.class
    |             List.remove('d-none');
898 |             document.getElementById('wizard-ref-content').value =
    |             preset.refContent || '';
899 |         } else {
900 |             document.getElementById('wizard-ref-container')?.class
    |             List.add('d-none');
901 |         }
```

```

902     }
903
904     // Switch to wizard tab
905     const wizardTabBtn = document.getElementById('mode-wizard-
tab');
906     if (wizardTabBtn) {
907         const tab = new bootstrap.Tab(wizardTabBtn);
908         tab.show();
909     }
910     this.goToStep(1);
911     this.syncWizardPreview();
912 }
913
914 showCreateAlert(msg, type = 'danger') {
915     const alertEl = document.getElementById('create-skill-aler
t');
916     if (alertEl) {
917         alertEl.className = `alert alert-${type} py-2 mt-3 mb-0`
;
918         alertEl.textContent = msg;
919         alertEl.classList.remove('d-none');
920     }
921 }
922
923 clearCreateAlert() {
924     const alertEl = document.getElementById('create-skill-aler
t');
925     if (alertEl) alertEl.classList.add('d-none');
926 }
927
928 // =====
=====
929 // SKILL DETAILS & SANDBOX TESTER
930 // =====
=====
931
932 async openSkillDetails(name, defaultTab = 'detail-tab-skillm
d') {

```

```

933     try {
934         const res = await this.apiFetch(`/api/skills/${encodeURIComponent
Component(name)}`);
935         if (!res.ok) throw new Error(`Failed to load skill detai
ls: ${res.statusText}`);
936         const data = await res.json();
937         this.currentSkill = data.skill;
938
939         // Populate modal fields
940         document.getElementById('detail-skill-title').textConten
t = this.currentSkill.name;
941         document.getElementById('detail-skill-path').textContent
= this.currentSkill.relative_path;
942         document.getElementById('detail-skill-desc-input').value
= this.currentSkill.description || '';
943         document.getElementById('detail-skill-desc-ru-input').va
lue = this.currentSkill.descriptions_i18n?.ru || '';
944         document.getElementById('detail-skill-instructions').val
ue = this.currentSkill.instructions || '';
945         document.getElementById('detail-skill-readme').value = t
his.currentSkill.readme_raw || '';
946
947         // Populate files list
948         const filesContainer = document.getElementById('detail-f
iles-list');
949         if (filesContainer) {
950             if (this.currentSkill.files && this.currentSkill.files
.length > 0) {
951                 filesContainer.innerHTML = this.currentSkill.files.m
ap(f => `
952                 <div class="list-group-item bg-transparent text-li
ght border-secondary d-flex justify-content-between align-item
s-center py-1 small">
953                     <span><i class="bi bi-file-earmark text-secondar
y"></i> ${f.rel_path}</span>
954                     <span class="badge bg-secondary font-monospace">
${this.formatBytes(f.size)}</span>
955                 </div>

```

```

956 |         `).join('');
957 |     } else {
958 |         filesContainer.innerHTML = '<div class="text-muted s
mall py-2">Файлы отсутствуют</div>';
959 |     }
960 | }
961 |
962 | // Populate JSON contract
963 | const contractEl = document.getElementById('detail-json-
contract');
964 | if (contractEl) {
965 |     const contractObj = {
966 |         name: this.currentSkill.name,
967 |         description: this.currentSkill.description,
968 |         descriptions_i18n: this.currentSkill.descriptions_i1
8n,
969 |         metadata: this.currentSkill.metadata,
970 |         manifest: this.currentSkill.manifest,
971 |     };
972 |     contractEl.textContent = JSON.stringify(contractObj, n
ull, 2);
973 | }
974 |
975 | // Reset sandbox box
976 | const sandboxOutput = document.getElementById('sandbox-o
utput-box');
977 | if (sandboxOutput) {
978 |     sandboxOutput.textContent = `Готов к тестированию навы
ка '${this.currentSkill.name}'. Введите тестовый запрос и нажм
ите «Запустить».`;
979 | }
980 |
981 | // Hide alerts
982 | const alertEl = document.getElementById('detail-skill-al
ert');
983 | if (alertEl) alertEl.classList.add('d-none');
984 |
985 | // Show modal

```



```

986 |     const modalEl = document.getElementById('skillDetailsMod
    | al');
987 |     const modal = bootstrap.Modal.getOrCreateInstance(modalE
    | l);
988 |     modal.show();
989 |
990 |     // Switch to requested tab
991 |     const targetTabBtn = document.getElementById(defaultTab)
    | ;
992 |     if (targetTabBtn) {
993 |         const tab = new bootstrap.Tab(targetTabBtn);
994 |         tab.show();
995 |     }
996 | } catch (err) {
997 |     console.error('[SkillsTab] Details error:', err);
998 |     window.showToast?(`Ошибка открытия деталей навыка: ${er
    | r.message}`, 'danger') || alert(`Ошибка открытия деталей навык
    | а: ${err.message}`);
999 | }
1000 | }
1001 |
1002 | async openSkillSandbox(name) {
1003 |     await this.openSkillDetails(name, 'detail-tab-sandbox');
1004 | }
1005 |
1006 | async runSandboxTest() {
1007 |     if (!this.currentSkill) return;
1008 |     const promptInput = document.getElementById('sandbox-test-
    | prompt');
1009 |     const outputBox = document.getElementById('sandbox-output-
    | box');
1010 |     const spinner = document.getElementById('sandbox-spinner')
    | ;
1011 |
1012 |     const prompt = promptInput?.value.trim() || `Test execute
    | skill ${this.currentSkill.name}`;
1013 |     spinner?.classList.remove('d-none');
1014 |     if (outputBox) outputBox.textContent = `⏳ Отправка запроса

```

```

    к модели для навыка '${this.currentSkill.name}'...\n`;
1015
1016     try {
1017         const res = await this.apiFetch(`/api/skills/${encodeURIComponent
Component(this.currentSkill.name)}/test`, {
1018             method: 'POST',
1019             headers: { 'Content-Type': 'application/json' },
1020             body: JSON.stringify({ prompt })
1021         });
1022
1023         if (!res.ok) {
1024             const errData = await res.json().catch(() => ({}));
1025             throw new Error(errData.detail || res.statusText);
1026         }
1027
1028         const data = await res.json();
1029         if (outputBox) {
1030             outputBox.textContent = `[Model: ${data.model_used}]\n
\n${data.response}`;
1031         }
1032     } catch (err) {
1033         console.error('[SkillsTab] Sandbox test error:', err);
1034         if (outputBox) outputBox.textContent = `❌ Ошибка тестиро
вания: ${err.message}`;
1035     } finally {
1036         spinner?.classList.add('d-none');
1037     }
1038 }
1039
1040 async saveCurrentSkill() {
1041     if (!this.currentSkill) return;
1042     const spinner = document.getElementById('save-skill-spinne
r');
1043     const alertEl = document.getElementById('detail-skill-aler
t');
1044
1045     spinner?.classList.remove('d-none');
1046     try {

```

```
1047 |     const description = document.getElementById('detail-skill-desc-input').value.trim();
1048 |     const description_ru = document.getElementById('detail-skill-desc-ru-input').value.trim();
1049 |     const instructions = document.getElementById('detail-skill-instructions').value;
1050 |     const readme = document.getElementById('detail-skill-readme').value;
1051 |
1052 |     const res = await this.apiFetch(`/api/skills/${encodeURIComponent(this.currentSkill.name)}`, {
1053 |         method: 'PUT',
1054 |         headers: { 'Content-Type': 'application/json' },
1055 |         body: JSON.stringify({ description, description_ru, instructions, readme })
1056 |     });
1057 |
1058 |     if (!res.ok) {
1059 |         const errData = await res.json().catch(() => ({}));
1060 |         throw new Error(errData.detail || res.statusText);
1061 |     }
1062 |
1063 |     if (alertEl) {
1064 |         alertEl.className = 'alert alert-success py-2 mt-3 mb-0';
1065 |         alertEl.textContent = 'Изменения успешно сохранены';
1066 |         alertEl.classList.remove('d-none');
1067 |     }
1068 |     window.showToast?.('Изменения навыка успешно сохранены', 'success');
1069 |
1070 |     await this.refresh();
1071 | } catch (err) {
1072 |     console.error('[SkillsTab] Save error:', err);
1073 |     if (alertEl) {
1074 |         alertEl.className = 'alert alert-danger py-2 mt-3 mb-0';
1075 |         alertEl.textContent = `Ошибка сохранения: ${err.message}`;
```

```

    e}`;
1076     alertEl.classList.remove('d-none');
1077   }
1078     window.showToast?.(`Ошибка сохранения: ${err.message}`,
'danger');
1079   } finally {
1080     spinner?.classList.add('d-none');
1081   }
1082 }
1083
1084 async packageSkill(name) {
1085   try {
1086     const res = await this.apiFetch(`/api/skills/${encodeURIComponent
Component(name)}/package`, {
1087       method: 'POST'
1088     });
1089     if (!res.ok) {
1090       const errData = await res.json().catch(() => ({}));
1091       throw new Error(errData.detail || res.statusText);
1092     }
1093     const data = await res.json();
1094     window.showToast?.(`📦 Навык '${name}' успешно упакован в
архив:\n${data.archive?.path} (${this.formatBytes(data.archive?.size)})`, 'success');
1095     await this.refresh();
1096   } catch (err) {
1097     console.error('[SkillsTab] Package error:', err);
1098     window.showToast?.(`Ошибка сборки архива: ${err.message}
`, 'danger') || alert(`Ошибка сборки архива: ${err.message}`);
1099   }
1100 }
1101
1102 async packageCurrentSkill() {
1103   if (!this.currentSkill) return;
1104   await this.packageSkill(this.currentSkill.name);
1105   await this.openSkillDetails(this.currentSkill.name);
1106 }
1107

```

```
1108 |   async deleteSkill(name) {
1109 |       if (!confirm(`Вы действительно хотите удалить навык '${name}' и все его файлы? Это действие необратимо.`)) {
1110 |           return;
1111 |       }
1112 |
1113 |       try {
1114 |           const res = await this.apiFetch(`/api/skills/${encodeURIComponent(name)}`, {
1115 |               method: 'DELETE'
1116 |           });
1117 |           if (!res.ok) {
1118 |               const errData = await res.json().catch(() => ({}));
1119 |               throw new Error(errData.detail || res.statusText);
1120 |           }
1121 |
1122 |           const modalEl = document.getElementById('skillDetailsModal');
1123 |           const modal = bootstrap.Modal.getInstance(modalEl);
1124 |           modal?.hide();
1125 |
1126 |           window.showToast?.(`Навык '${name}' успешно удален`, 'success');
1127 |           await this.refresh();
1128 |       } catch (err) {
1129 |           console.error('[SkillsTab] Delete error:', err);
1130 |           window.showToast?.(`Ошибка удаления навыка: ${err.message}`, 'danger') || alert(`Ошибка удаления навыка: ${err.message}`);
1131 |       }
1132 |   }
1133 |
1134 |   async deleteCurrentSkill() {
1135 |       if (!this.currentSkill) return;
1136 |       await this.deleteSkill(this.currentSkill.name);
1137 |   }
1138 |
1139 |   showLoading(show) {
```

```
1140 |     const loadingEl = document.getElementById('skills-loading'
      | );
1141 |     if (loadingEl) {
1142 |         if (show) loadingEl.classList.remove('d-none');
1143 |         else loadingEl.classList.add('d-none');
1144 |     }
1145 | }
1146 |
1147 | formatBytes(bytes) {
1148 |     if (!bytes || bytes === 0) return '0 B';
1149 |     const k = 1024;
1150 |     const sizes = ['B', 'KB', 'MB', 'GB'];
1151 |     const i = Math.floor(Math.log(bytes) / Math.log(k));
1152 |     return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
      | ' + sizes[i];
1153 | }
1154 | }
1155 |
1156 | // Global initialization hook
1157 | window.SkillsTabManager = SkillsTabManager;
1158 | if (!window.skillsTab) {
1159 |     window.skillsTab = new SkillsTabManager();
1160 | }
1161 | window.initSkillsTab = function() {
1162 |     if (window.skillsTab) {
1163 |         window.skillsTab.init();
1164 |     }
1165 | };
1166 |
1167 | // Auto-init on script load if container is present
1168 | if (document.getElementById('skills-tab-root')) {
1169 |     window.skillsTab.init();
1170 | }
1171 |
1172 |
1173 |
```

index.html

```
1 <!-- Software Audit Application Tab -->
2 <style>
3   .sw-audit-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .sw-audit-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    min-height: 58px;
15    transition: border-color 0.15s ease;
16  }
17  .sw-audit-card:hover {
18    border-color: #38bdf8 !important;
19  }
20  .sw-audit-label {
21    font-size: 0.68rem;
22    text-transform: uppercase;
23    letter-spacing: 0.05em;
24    color: #94a3b8;
25    line-height: 1.1;
26  }
27  .sw-audit-value {
28    font-size: 1.25rem;
29    font-weight: 700;
30    color: #f8fafc;
31    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
32    Consolas, monospace;
33    line-height: 1.15;
34  }
35  .sw-audit-panel {
36    background: #0f172a !important;
```

```
36 |     border: 1px solid #334155 !important;
37 |     border-radius: 8px;
38 |     overflow: hidden;
39 | }
40 | .sw-audit-panel-header {
41 |     background: #1e293b !important;
42 |     border-bottom: 1px solid #334155 !important;
43 |     padding: 0.45rem 0.85rem;
44 |     font-weight: 600;
45 |     font-size: 0.82rem;
46 |     color: #38bdf8;
47 |     display: flex;
48 |     align-items: center;
49 |     justify-content: space-between;
50 |     flex-wrap: wrap;
51 |     gap: 0.35rem;
52 | }
53 | .sw-audit-table {
54 |     width: 100%;
55 |     font-size: 0.78rem;
56 |     border-collapse: collapse;
57 | }
58 | .sw-audit-table th {
59 |     background: #1e293b;
60 |     padding: 6px 10px;
61 |     color: #94a3b8;
62 |     font-weight: 600;
63 |     border-bottom: 1px solid #334155;
64 |     position: sticky;
65 |     top: 0;
66 |     z-index: 2;
67 | }
68 | .sw-audit-table td {
69 |     padding: 6px 10px;
70 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
71 |     vertical-align: middle;
72 | }
73 | .sw-audit-table tbody tr:hover {
```



```

74     background: rgba(56, 189, 248, 0.05);
75 }
76 .sw-badge-category {
77     font-size: 0.7rem;
78     padding: 2px 6px;
79     border-radius: 4px;
80     background: #1e293b;
81     border: 1px solid #475569;
82     color: #e2e8f0;
83 }
84 .sw-badge-active {
85     background: rgba(34, 197, 94, 0.15);
86     border: 1px solid rgba(34, 197, 94, 0.4);
87     color: #4ade80;
88 }
89 .sw-badge-dormant {
90     background: rgba(245, 158, 11, 0.15);
91     border: 1px solid rgba(245, 158, 11, 0.4);
92     color: #fbbf24;
93 }
94 </style>
95
96 <div class="sw-audit-container">
97     <!-- Header Alert & Controls (Compact Strip) -->
98     <div class="d-flex justify-content-between align-items-center
99         r flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style
100         ="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%
101         ); border: 1px solid #334155;">
102         <div class="d-flex align-items-center gap-2 flex-wrap">
103             <span class="fs-5">❏ </span>
104             <span class="fw-bold text-white small" style="font-size:
105                 0.92rem;">Аудит программного обеспечения</span>
106             <span class="badge bg-dark border border-secondary text-
107                 info fw-normal" style="font-size: 0.7rem;">Software Audit</spa
108                 n>
109             <span class="text-secondary small d-none d-lg-inline" st
110                 yle="font-size: 0.76rem;">| Инвентаризация ПО, UserAssist & Pr
111                 efetch</span>

```

```

104     </div>
105     <div class="d-flex align-items-center gap-1.5 flex-wrap">
106         <span class="badge rounded-pill bg-success-subtle text-s
success border border-success px-2 py-1" style="font-size: 0.72
rem;" id="sw-audit-status-badge">
107             • Готов
108         </span>
109         <button class="btn btn-xs btn-outline-info rounded-pill
px-2.5 py-1" style="font-size: 0.75rem;" id="btn-sw-audit-refr
esh">
110             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
111         </button>
112         <button class="btn btn-xs btn-outline-light rounded-pill
px-2 py-1" style="font-size: 0.75rem;" id="btn-sw-audit-expor
t">
113             <i class="bi bi-download me-1"></i> Экспорт JSON
114         </button>
115     </div>
116 </div>
117
118 <!-- Metric Statistics Cards (Compact Density Grid) -->
119 <div class="row g-2 mb-2">
120     <div class="col-6 col-lg-3">
121         <div class="sw-audit-card">
122             <span class="sw-audit-label">Всего программ</span>
123             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
124                 <span class="sw-audit-value text-info" id="sw-metric
-total">0</span>
125                 <small class="text-muted" style="font-size: 0.7rem;"
>В пеестр</small>
126             </div>
127         </div>
128     </div>
129     <div class="col-6 col-lg-3">
130         <div class="sw-audit-card">
131             <span class="sw-audit-label">Активно используются</spa
n>

```

```
132 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
133 |             <span class="sw-audit-value text-success" id="sw-metric-active">0</span>
134 |             <small class="text-muted" style="font-size: 0.7rem;">Недавние запуски</small>
135 |         </div>
136 |     </div>
137 | </div>
138 | <div class="col-6 col-lg-3">
139 |     <div class="sw-audit-card">
140 |         <span class="sw-audit-label">Неиспользуемые (>90 дн)</span>
141 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
142 |             <span class="sw-audit-value text-warning" id="sw-metric-dormant">0</span>
143 |             <small class="text-muted" style="font-size: 0.7rem;">Кандидаты на удаление</small>
144 |         </div>
145 |     </div>
146 | </div>
147 | <div class="col-6 col-lg-3">
148 |     <div class="sw-audit-card">
149 |         <span class="sw-audit-label">Категоризировано</span>
150 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
151 |             <span class="sw-audit-value text-info" id="sw-metric-categorized">0%</span>
152 |             <small class="text-muted" style="font-size: 0.7rem;">AI эвристика</small>
153 |         </div>
154 |     </div>
155 | </div>
156 | </div>
157 |
158 | <!-- Filters & Controls Bar -->
159 | <div class="card bg-dark border-secondary p-2 mb-2" style="b
```

```

order-radius: 8px;">
160     <div class="row g-2 align-items-center">
161         <div class="col-md-5">
162             <div class="input-group input-group-sm">
163                 <span class="input-group-text bg-dark border-seconda
ry text-light"><i class="bi bi-search"></i></span>
164                 <input type="text" class="form-control bg-dark text-
white border-secondary" id="sw-search-input" placeholder="Поис
к по названию, издателю или назначению...">
165             </div>
166         </div>
167         <div class="col-md-3">
168             <select class="form-select form-select-sm bg-dark text
-white border-secondary" id="sw-category-filter">
169                 <option value="">Все категории</option>
170             </select>
171         </div>
172         <div class="col-md-2">
173             <select class="form-select form-select-sm bg-dark text
-white border-secondary" id="sw-status-filter">
174                 <option value="all">Все программы</option>
175                 <option value="active">Только активные</option>
176                 <option value="dormant">Только неиспользуемые</optio
n>
177             </select>
178         </div>
179         <div class="col-md-2 text-end">
180             <span class="badge bg-secondary" id="sw-filtered-count
-badge">0 / 0</span>
181         </div>
182     </div>
183 </div>
184
185 <!-- Main Table Panel -->
186 <div class="sw-audit-panel">
187     <div class="sw-audit-panel-header">
188         <span><i class="bi bi-table me-1"></i> Список установлен
ных программных продуктов</span>

```

```

189     <span class="small text-muted" id="sw-last-scanned-time"
    >Сканирование...</span>
190 </div>
191 <div class="p-0 table-responsive" style="max-height: 540px
    ; overflow-y: auto;">
192     <table class="sw-audit-table">
193         <thead>
194             <tr>
195                 <th>Программа</th>
196                 <th>Категория</th>
197                 <th>Издатель / Версия</th>
198                 <th>Последний запуск</th>
199                 <th>Запусков</th>
200                 <th>Время работы</th>
201                 <th>Размер / Арх</th>
202                 <th style="text-align: right;">Действия</th>
203             </tr>
204         </thead>
205         <tbody id="sw-audit-tbody">
206             <tr><td colspan="8" class="text-center text-muted p-
    4"><i class="bi bi-arrow-repeat spin me-1"></i> Загрузка аудит
    а программного обеспечения...</td></tr>
207         </tbody>
208     </table>
209 </div>
210 </div>
211 </div>
212
213 <!-- Modal: App Details -->
214 <div class="modal fade" id="swAppDetailModal" tabindex="-1" ar
    ia-labelledby="swAppDetailModalLabel" aria-hidden="true">
215     <div class="modal-dialog modal-lg modal-dialog-centered">
216         <div class="modal-content bg-dark text-white border-second
    ary">
217             <div class="modal-header border-secondary">
218                 <h5 class="modal-title d-flex align-items-center gap-2
    " id="swAppDetailModalLabel">
219                     <span class="fs-5">□ </span> <span id="sw-modal-app-

```

```
name">Детали программы</span>
220     </h5>
221     <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="modal" aria-label="Close"></button>
222 </div>
223 <div class="modal-body p-3" id="sw-modal-app-body">
224     <!-- Content will be injected by main.js -->
225 </div>
226 <div class="modal-footer border-secondary">
227     <button type="button" class="btn btn-secondary btn-sm"
    data-bs-dismiss="modal">Закрыть</button>
228 </div>
229 </div>
230 </div>
231 </div>
232
```

main.js

```
1 // Software Audit Tab JavaScript Module
2 (function() {
3   let allApps = [];
4   let auditReport = null;
5   let isInitialized = false;
6
7   function formatRelativeTime(dateStr) {
8     if (!dateStr) return '<span class="text-muted">Нет данных<
9 /span>';
10    const d = new Date(dateStr);
11    if (isNaN(d.getTime())) return '<span class="text-muted">-
12 </span>';
13    const now = new Date();
14    const diffMs = now - d;
15    const diffSec = Math.floor(diffMs / 1000);
16    const diffMin = Math.floor(diffSec / 60);
17    const diffHrs = Math.floor(diffMin / 60);
18    const diffDays = Math.floor(diffHrs / 24);
19
20    if (diffDays > 90) {
21      return `<span class="text-warning" title="${d.toLocaleSt
22 ring()}">${diffDays} дн назад</span>`;
23    } else if (diffDays > 0) {
24      return `<span class="text-success" title="${d.toLocaleSt
25 ring()}">${diffDays} дн назад</span>`;
26    } else if (diffHrs > 0) {
27      return `<span class="text-info" title="${d.toLocaleStrin
28 g()}">${diffHrs} ч назад</span>`;
29    } else if (diffMin > 0) {
30      return `<span class="text-info" title="${d.toLocaleStrin
31 g()}">${diffMin} мин назад</span>`;
32    } else {
33      return `<span class="text-info">Только что</span>`;
34    }
35  }
36}
```

```
31 | function formatDuration(sec) {
32 |     if (!sec || sec <= 0) return '<span class="text-muted">-</span>';
33 |     const hrs = Math.floor(sec / 3600);
34 |     const mins = Math.floor((sec % 3600) / 60);
35 |     if (hrs > 0) return `${hrs}ч ${mins}м`;
36 |     if (mins > 0) return `${mins} мин`;
37 |     return `${sec} сек`;
38 | }
39 |
40 | async function loadSoftwareAudit() {
41 |     const tbody = document.getElementById('sw-audit-tbody');
42 |     const badge = document.getElementById('sw-audit-status-badge');
43 |     if (badge) badge.innerText = '● Сканирование реестра...';
44 |
45 |     try {
46 |         // 1. Пытаемся получить сводный аналитический отчет
47 |         let reportRes = await fetch('/api/windows/software/audit');
48 |         if (!reportRes.ok) {
49 |             reportRes = await fetch('/api/v1/windows-admin/software/audit');
50 |         }
51 |         if (reportRes.ok) {
52 |             auditReport = await reportRes.json();
53 |         }
54 |
55 |         // 2. Получаем полный список установленных приложений
56 |         let appsRes = await fetch('/api/windows/software?limit=500');
57 |         if (!appsRes.ok) {
58 |             // Fallback на router_windows_admin
59 |             appsRes = await fetch('/api/v1/windows-admin/software?limit=500');
60 |         }
61 |
62 |         if (appsRes.ok) {
```



```

63 |         allApps = await appsRes.json();
64 |     } else {
65 |         throw new Error(`HTTP ${appsRes.status}`);
66 |     }
67 |
68 |     // Обновляем метрики
69 |     updateMetrics();
70 |     // Заполняем выпадающий список категорий
71 |     populateCategories();
72 |     // Рендерим таблицу
73 |     renderTable();
74 |
75 |     if (badge) {
76 |         badge.className = 'badge rounded-pill bg-success-subtl
77 | e text-success border border-success px-3 py-2';
78 |         badge.innerText = `● Аудит: Завершено (${allApps.lengt
79 | h} программ)`;
80 |     }
81 |
82 |     const timeEl = document.getElementById('sw-last-scanned-
83 | time');
84 |     if (timeEl) {
85 |         timeEl.innerText = `Обновлено: ${new Date().toLocaleTi
86 | meString()}`;
87 |     }
88 |     } catch (err) {
89 |         console.error('[SoftwareAuditTab] Failed to load softwar
90 | e audit:', err);
91 |         if (tbody) {
92 |             tbody.innerHTML = `<tr><td colspan="8" class="text-cen
93 | ter text-danger p-4">Ошибка сбора сведений об установленном ПО
94 | : ${err.message}</td></tr>`;
95 |         }
96 |         if (badge) {
97 |             badge.className = 'badge rounded-pill bg-danger-subtle
98 | text-danger border border-danger px-3 py-2';
99 |             badge.innerText = '● Ошибка аудита';
100 |         }

```

```
93 |     }
94 | }
95 |
96 | function updateMetrics() {
97 |     const totalEl = document.getElementById('sw-metric-total')
98 |     ;
99 |     const activeEl = document.getElementById('sw-metric-active
100 |     ');
101 |     const dormantEl = document.getElementById('sw-metric-dorma
102 |     nt');
103 |     const catEl = document.getElementById('sw-metric-categorie
104 |     s');
105 |
106 |     if (auditReport) {
107 |         if (totalEl) totalEl.innerText = auditReport.total_apps
108 |         || allApps.length;
109 |         if (activeEl) activeEl.innerText = auditReport.active_ap
110 |         ps_count || 0;
111 |         if (dormantEl) dormantEl.innerText = auditReport.never_l
112 |         aunched_or_dormant_count || 0;
113 |         if (catEl) catEl.innerText = Object.keys(auditReport.cat
114 |         egories_breakdown || {}).length;
115 |     } else {
116 |         const activeCount = allApps.filter(a => a.execution_info
117 |         && a.execution_info.last_run_time).length;
118 |         if (totalEl) totalEl.innerText = allApps.length;
119 |         if (activeEl) activeEl.innerText = activeCount;
120 |         if (dormantEl) dormantEl.innerText = allApps.length - ac
121 |         tiveCount;
122 |         const cats = new Set(allApps.map(a => a.category).filter
123 |         (Boolean));
124 |         if (catEl) catEl.innerText = cats.size;
125 |     }
126 | }
127 |
128 | function populateCategories() {
129 |     const sel = document.getElementById('sw-category-filter');
130 |     if (!sel) return;
```

```
120
121     const currentVal = sel.value;
122     const cats = Array.from(new Set(allApps.map(a => a.category).filter(Boolean))).sort();
123
124     sel.innerHTML = '<option value="">Все категории</option>'
+ cats.map(c =>
125         `<option value="${c}" ${c === currentVal ? 'selected' :
+ ''}>${c}</option>`
126     ).join('');
127 }
128
129 function getFilteredApps() {
130     const search = (document.getElementById('sw-search-input')
+ ?.value || '').toLowerCase().trim();
131     const category = document.getElementById('sw-category-filter')?.value || '';
132     const status = document.getElementById('sw-status-filter')
+ ?.value || 'all';
133
134     const dormantThresholdMs = 90 * 24 * 3600 * 1000;
135     const now = Date.now();
136
137     return allApps.filter(app => {
138         // Поиск по ключевым полям
139         if (search) {
140             const text = `${app.display_name || ''} ${app.name ||
+ ''} ${app.publisher || ''} ${app.purpose_description || ''}`.t
+ oLowerCase();
141             if (!text.includes(search)) return false;
142         }
143
144         // Фильтр по категории
145         if (category && app.category !== category) {
146             return false;
147         }
148
149         // Фильтр по статусу использования
```

```
150 |     const lastRun = app.execution_info?.last_run_time ? new
    | Date(app.execution_info.last_run_time).getTime() : null;
151 |     if (status === 'active') {
152 |         if (!lastRun || (now - lastRun > dormantThresholdMs))
    | return false;
153 |     } else if (status === 'dormant') {
154 |         if (lastRun && (now - lastRun <= dormantThresholdMs))
    | return false;
155 |     }
156 |
157 |     return true;
158 | });
159 | }
160 |
161 | function renderTable() {
162 |     const tbody = document.getElementById('sw-audit-tbody');
163 |     const badge = document.getElementById('sw-filtered-count-b
    | adge');
164 |     if (!tbody) return;
165 |
166 |     const filtered = getFilteredApps();
167 |     if (badge) {
168 |         badge.innerText = `${filtered.length} / ${allApps.length
    | }`;
169 |     }
170 |
171 |     if (filtered.length === 0) {
172 |         tbody.innerHTML = '<tr><td colspan="8" class="text-cen
    | ter text-muted p-4">Программы по заданным критериям не найдены</
    | td></tr>';
173 |         return;
174 |     }
175 |
176 |     tbody.innerHTML = filtered.map((app, idx) => {
177 |         const exec = app.execution_info;
178 |         const lastRunFormatted = exec ? formatRelativeTime(exec.
    | last_run_time) : '<span class="text-muted">Никогда</span>';
179 |         const runCount = exec && exec.run_count ? '<span class="
```

```

    badge bg-secondary">${exec.run_count}</span>` : '<span class="
text-muted">0</span>';
180     const focusTime = exec ? formatDuration(exec.focus_time_
seconds) : '<span class="text-muted">-</span>';
181     const sizeStr = app.size_mb > 0 ? `${app.size_mb} MB` :
'<span class="text-muted">-</span>';
182     const archBadge = app.architecture === 'x64' ? '<span cl
ass="badge bg-primary-subtle text-primary" style="font-size: 0
.65rem;">x64</span>' : '<span class="badge bg-secondary-subtle
text-secondary" style="font-size: 0.65rem;">x86</span>';
183
184     return `
185     <tr class="sw-row-item" data-idx="${idx}" style="curso
r: pointer;" title="Нажмите для AI-диагностики и подробных све
дений">
186         <td>
187             <div class="fw-bold text-white text-truncate" styl
e="max-width: 260px;" title="${app.display_name || app.name}">
188                 ${app.display_name || app.name}
189             </div>
190             <div class="small text-muted text-truncate" style=
"max-width: 260px; font-size: 0.72rem;" title="${app.purpose_d
escription || ''}">
191                 ${app.purpose_description || 'Прикладное ПО'}
192             </div>
193         </td>
194         <td>
195             <span class="sw-badge-category">${app.category ||
'Прочее'}</span>
196         </td>
197         <td>
198             <div class="text-truncate text-light" style="max-w
idth: 180px;" title="${app.publisher || 'Неизвестен'}">${app.p
ublisher || '<span class="text-muted">-</span>'}</div>
199             <div class="small text-muted" style="font-size: 0.
72rem;">${app.version || '-'}</div>
200         </td>
201         <td>${lastRunFormatted}</td>

```

```

202         <td>${runCount}</td>
203         <td class="small text-light">${focusTime}</td>
204         <td class="small">
205             <div>${sizeStr}</div>
206             <div>${archBadge}</div>
207         </td>
208         <td style="text-align: right;">
209             <button class="btn btn-sm btn-outline-info p-1 px-
210 2 btn-sw-detail" data-idx="${idx}" title="Подробности">
211                 <i class="bi bi-info-circle"></i>
212             </button>
213         </td>
214     </tr>
215 `;
216     }).join('');
217
218     // Привязка кликабельности всей строки
219     tbody.querySelectorAll('.sw-row-item').forEach(row => {
220         row.onclick = (evt) => {
221             const idx = parseInt(row.getAttribute('data-idx'), 10)
222             ;
223             showAppDetails(filtered[idx]);
224         };
225     });
226
227     // Привязка обработчиков кнопки деталей
228     tbody.querySelectorAll('.btn-sw-detail').forEach(btn => {
229         btn.onclick = (evt) => {
230             evt.stopPropagation();
231             const idx = parseInt(btn.getAttribute('data-idx'), 10)
232             ;
233             showAppDetails(filtered[idx]);
234         };
235     });
236 }
237
238 function showAppDetails(app) {
239     if (!app) return;

```

```
237 |     const exec = app.execution_info;
238 |
239 |     if (window.AITableModal) {
240 |         window.AITableModal.show({
241 |             icon: '❏ ',
242 |             title: app.display_name || app.name,
243 |             subtitle: app.publisher || 'Неизвестный разработчик',
244 |             tableType: 'software',
245 |             badges: [
246 |                 { text: app.category || 'ПО', class: 'badge bg-info'
247 |                 },
248 |                 { text: app.architecture || 'x64', class: 'badge bg-
249 |                 secondary' }
250 |             ],
251 |             metadata: [
252 |                 { label: 'Продукт', value: app.display_name || app.n
253 |                 ame },
254 |                 { label: 'Издатель', value: app.publisher || 'Не ука
255 |                 зан' },
256 |                 { label: 'Версия', value: app.version || 'Не указана
257 |                 ' },
258 |                 { label: 'Категория', value: app.category || 'Прочее
259 |                 ' },
260 |                 { label: 'Размер на диске', value: app.size_mb > 0 ?
261 |                 `${app.size_mb} MB` : 'Не указан' },
262 |                 { label: 'Дата установки', value: app.install_date |
263 |                 | 'Неизвестно' },
264 |                 { label: 'Запусков (UserAssist)', value: exec?.run_c
265 |                 ount ? `${exec.run_count} раз` : '0' },
266 |                 { label: 'Время в фокусе', value: exec ? formatDurat
267 |                 ion(exec.focus_time_seconds) : '-' },
268 |                 { label: 'Последний запуск', value: exec?.last_run_t
269 |                 ime ? new Date(exec.last_run_time).toLocaleString() : 'Нет дан
270 |                 ных', fullWidth: true },
271 |                 { label: 'Путь установки', value: app.install_locati
272 |                 on || 'Не указан', isCode: true, fullWidth: true }
273 |             ],
274 |             rawTitle: 'Команда удаления / Путь',
```

```

262         rawContent: app.uninstall_string || app.install_locati
on || '',
263         requestData: {
264             install_location: app.install_location,
265             uninstall_string: app.uninstall_string,
266             version: app.version,
267             size_mb: app.size_mb,
268             run_count: exec?.run_count
269         }
270     });
271     return;
272 }
273 }
274
275 function exportJson() {
276     const dataStr = "data:text/json;charset=utf-8," + encodeUR
IComponent(JSON.stringify(allApps, null, 2));
277     const dlAnchor = document.createElement('a');
278     dlAnchor.setAttribute("href", dataStr);
279     dlAnchor.setAttribute("download", `software_audit_${new Da
te().toISOString().slice(0, 10)}.json`);
280     document.body.appendChild(dlAnchor);
281     dlAnchor.click();
282     dlAnchor.remove();
283 }
284
285 window.initSoftwareAuditTab = async function() {
286     console.log('[SoftwareAuditTab] Initializing Software Audi
t tab...');
287
288     // Привязываем контролы поиска и фильтрации
289     const searchInput = document.getElementById('sw-search-inp
ut');
290     const catSelect = document.getElementById('sw-category-fi
lter');
291     const statusSelect = document.getElementById('sw-status-fi
lter');
292     const refreshBtn = document.getElementById('btn-sw-audit-r

```



```
    efresh');  
293     const exportBtn = document.getElementById('btn-sw-audit-ex  
    port');  
294  
295     if (searchInput) searchInput.oninput = renderTable;  
296     if (catSelect) catSelect.onchange = renderTable;  
297     if (statusSelect) statusSelect.onchange = renderTable;  
298     if (refreshBtn) refreshBtn.onclick = loadSoftwareAudit;  
299     if (exportBtn) exportBtn.onclick = exportJson;  
300  
301     await loadSoftwareAudit();  
302     isInitialized = true;  
303   };  
304 } )();  
305
```

index.html

```
1 <!-- AI Software Transparency Scanner Tab UI -->
2 <div class="container-fluid p-2" id="st-scanner-container">
3   <!-- Верхняя карточка с метриками и кнопками управления -->
4   <div class="card bg-dark text-white border-secondary mb-3 shadow-sm">
5     <div class="card-body p-3">
6       <div class="d-flex align-items-center justify-content-between flex-wrap gap-2 mb-3">
7         <div class="d-flex align-items-center gap-2">
8           <span class="fs-4">🔍 </span>
9           <div>
10            <h5 class="mb-0 fw-bold">AI Software Transparency Scanner</h5>
11            <div class="text-muted small">Интеллектуальный инвентаризатор установленного ПО, конфигураций, хранилищ и сетевой активности</div>
12          </div>
13        </div>
14        <div class="d-flex align-items-center gap-2">
15          <span id="st-status-badge" class="badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2">
16            • Готов к сканированию
17          </span>
18          <button id="st-btn-refresh" class="btn btn-sm btn-primary d-flex align-items-center gap-1 shadow-sm">
19            <i class="bi bi-arrow-clockwise"></i> Сканировать систему
20          </button>
21        </div>
22      </div>
23    </div>
24    <!-- Сводные метрики -->
25    <div class="row g-2">
26      <div class="col-6 col-md-3">
27        <div class="p-2 bg-secondary bg-opacity-10 border bo
```

```
    rder-secondary rounded">
28         <div class="text-muted small text-uppercase">Устан
    овлено ПО</div>
29         <div class="fs-4 fw-bold text-info" id="st-metric-
    total-apps">0</div>
30     </div>
31 </div>
32 <div class="col-6 col-md-3">
33     <div class="p-2 bg-secondary bg-opacity-10 border bo
    rder-secondary rounded">
34         <div class="text-muted small text-uppercase">Файло
    в конфигов</div>
35         <div class="fs-4 fw-bold text-warning" id="st-metr
    ic-total-configs">0</div>
36     </div>
37 </div>
38 <div class="col-6 col-md-3">
39     <div class="p-2 bg-secondary bg-opacity-10 border bo
    rder-secondary rounded">
40         <div class="text-muted small text-uppercase">Сетев
    ых узлов / Доменов</div>
41         <div class="fs-4 fw-bold text-success" id="st-metr
    ic-total-domains">0</div>
42     </div>
43 </div>
44 <div class="col-6 col-md-3">
45     <div class="p-2 bg-secondary bg-opacity-10 border bo
    rder-secondary rounded">
46         <div class="text-muted small text-uppercase">Время
    сканирования</div>
47         <div class="fs-4 fw-bold text-light" id="st-metric
    -scan-duration">0.0 c</div>
48     </div>
49 </div>
50 </div>
51 </div>
52 </div>
53
```

```

54 | <!-- Фильтры и строка поиска -->
55 | <div class="d-flex align-items-center justify-content-between mb-3 flex-wrap gap-2">
56 |     <div class="d-flex align-items-center gap-2 flex-grow-1" style="max-width: 450px;">
57 |         <div class="input-group input-group-sm">
58 |             <span class="input-group-text bg-dark text-secondary border-secondary"><i class="bi bi-search"></i></span>
59 |             <input type="text" id="st-search-input" class="form-control bg-dark text-white border-secondary" placeholder="Поиск по названию программы или издателю...">
60 |         </div>
61 |     </div>
62 |     <div class="d-flex align-items-center gap-2 small text-muted">
63 |         <span>Легенда достоверности:</span>
64 |         <span class="badge bg-success-subtle text-success border border-success">📁 Локально</span>
65 |         <span class="badge bg-info-subtle text-info border border-info">📁 Документация</span>
66 |         <span class="badge bg-warning-subtle text-warning border border-warning">📁 Предположение AI</span>
67 |         <span class="badge bg-secondary-subtle text-secondary border border-secondary">📁 Неизвестно</span>
68 |     </div>
69 | </div>
70 |
71 | <!-- Таблица программ -->
72 | <div class="card bg-dark text-white border-secondary shadow-sm">
73 |     <div class="table-responsive">
74 |         <table class="table table-dark table-hover mb-0 align-middle">
75 |             <thead class="table-dark text-secondary small border-secondary">
76 |                 <tr>
77 |                     <th style="width: 25%;">Программа</th>
78 |                     <th style="width: 15%;">Издатель / Версия</th>

```

```

79 |         <th style="width: 20%;">Каталоги и хранилище</th>
80 |         <th style="width: 15%;">Конфиги & Секреты</th>
81 |         <th style="width: 15%;">Сетевые домены</th>
82 |         <th style="width: 10%; text-align: right;">AI Анал
из</th>
83 |     </tr>
84 | </thead>
85 | <tbody id="st-table-body">
86 |     <tr>
87 |         <td colspan="6" class="text-center py-4 text-muted
">
88 |             Нажмите кнопку «Сканировать систему» для сбора с
ведений об установленном ПО.
89 |         </td>
90 |     </tr>
91 | </tbody>
92 | </table>
93 | </div>
94 | </div>
95 | </div>
96 |
97 | <!-- Модальное окно просмотра очищенного конфигурационного фай
ла -->
98 | <div class="modal fade" id="st-config-modal" tabindex="-1" ari
a-hidden="true">
99 |     <div class="modal-dialog modal-lg modal-dialog-scrollable">
100 |         <div class="modal-content bg-dark text-white border-second
ary">
101 |             <div class="modal-header border-secondary">
102 |                 <h6 class="modal-title fw-bold" id="st-config-modal-ti
tle">Конфигурационный файл</h6>
103 |                 <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
104 |             </div>
105 |             <div class="modal-body">
106 |                 <div class="alert alert-warning py-2 small mb-3">
107 |                     <i class="bi bi-shield-lock-fill me-1"></i> Конфиден
циальные токены, пароли и ключи безопасности автоматически мас

```

```
кированы (<code>[REDACTED_SECRET]</code>).
108     </div>
109     <div class="small text-muted mb-2" id="st-config-modal
    -path"></div>
110     <pre class="p-3 bg-black rounded border border-seconda
    ry text-success small font-monospace overflow-auto" style="max
    -height: 400px;" id="st-config-modal-content"></pre>
111     </div>
112 </div>
113 </div>
114 </div>
115
116 <!-- Модальное окно подробного AI-исследования Gemini -->
117 <div class="modal fade" id="st-research-modal" tabindex="-1" a
    ria-hidden="true">
118     <div class="modal-dialog modal-lg modal-dialog-scrollable">
119         <div class="modal-content bg-dark text-white border-second
    ary">
120             <div class="modal-header border-secondary">
121                 <h6 class="modal-title fw-bold" id="st-research-modal-
    title">AI Исследование Gemini</h6>
122                 <button type="button" class="btn-close btn-close-white
    " data-bs-dismiss="modal" aria-label="Close"></button>
123             </div>
124             <div class="modal-body" id="st-research-modal-body">
125                 <div class="text-center py-4">
126                     <div class="spinner-border spinner-border-sm text-pr
    imary me-2"></div>
127                     Выполняется аналитическое исследование программы...
128                 </div>
129             </div>
130         </div>
131     </div>
132 </div>
133
```

main.js

```
1 // Software Transparency Scanner script
2 // Frontend Controller
3 (function () {
4   let allApps = [];
5   let isScanning = false;
6
7   async function initSoftwareTransparencyTab() {
8     console.log('[TransparencyScanner] Initializing tab contro
9     bindEvents();
10    await loadScannerData(false);
11  }
12
13  function bindEvents() {
14    const btnRefresh = document.getElementById('st-btn-refresh
15    if (btnRefresh) {
16      btnRefresh.onclick = () => loadScannerData(true);
17    }
18
19    const searchInput = document.getElementById('st-search-inp
20    if (searchInput) {
21      searchInput.oninput = () => renderTable();
22    }
23  }
24
25  async function loadScannerData(forceRefresh = false) {
26    if (isScanning) return;
27    isScanning = true;
28
29    const badge = document.getElementById('st-status-badge');
30    const tbody = document.getElementById('st-table-body');
31
32    if (badge) {
33      badge.className = 'badge rounded-pill bg-warning-subtle
```

```

text-warning border border-warning px-3 py-2';
34     badge.innerText = '● Идет сбор сведений о ПО и сети...';
35 }
36
37 if (tbody && (!allApps || allApps.length === 0)) {
38     tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted"><div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>Выполняется инвентаризация установленных программ, конфигураций и сетевых сокетов..</td></tr>';
39 }
40
41 try {
42     const url = '/api/v1/software-scanner/scan?force_refresh=' + (forceRefresh ? 'true' : 'false');
43     const res = await fetch(url);
44     if (!res.ok) throw new Error('HTTP ' + res.status);
45
46     const data = await res.json();
47     allApps = data.apps || [];
48     const summary = data.summary || {};
49
50     document.getElementById('st-metric-total-apps').innerText = summary.total_apps || allApps.length;
51     document.getElementById('st-metric-total-configs').innerText = summary.total_configs_found || 0;
52     document.getElementById('st-metric-total-domains').innerText = summary.total_network_domains || 0;
53     document.getElementById('st-metric-scan-duration').innerText = (summary.scan_duration_sec || 0) + ' с.';
54
55     if (badge) {
56         badge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2';
57         badge.innerText = '● Обнаружено: ' + allApps.length + ' программ';
58     }
59

```



```

60     renderTable();
61 } catch (err) {
62     console.error('[TransparencyScanner] Scan error:', err);
63     if (badge) {
64         badge.className = 'badge rounded-pill bg-danger-subtle
        text-danger border border-danger px-3 py-2';
65         badge.innerText = '● Ошибка сканирования';
66     }
67     if (tbody) {
68         tbody.innerHTML = '<tr><td colspan="6" class="text-cen
        ter text-danger py-4">Ошибка: ' + err.message + '</td></tr>';
69     }
70 } finally {
71     isScanning = false;
72 }
73 }
74
75 function renderTable() {
76     const tbody = document.getElementById('st-table-body');
77     const searchInput = document.getElementById('st-search-inp
        ut');
78     const searchVal = (searchInput ? searchInput.value : '').t
        oLowerCase().trim();
79
80     if (!tbody) return;
81
82     const filtered = allApps.filter((a) => {
83         if (!searchVal) return true;
84         return (
85             a.name.toLowerCase().includes(searchVal) ||
86             (a.publisher && a.publisher.toLowerCase().includes(sea
                rchVal)) ||
87             (a.install_location && a.install_location.toLowerCase(
                ).includes(searchVal))
88         );
89     });
90
91     if (filtered.length === 0) {

```

```

92 |         tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Ничего не найдено по фильтру «' + searchVal
    | + '»</td></tr>';
93 |         return;
94 |     }
95 |
96 |     tbody.innerHTML = filtered.map((app) => renderAppRow(app))
    | .join('');
97 |     bindRowActions();
98 | }
99 |
100 | function renderAppRow(app) {
101 |     const dirs = (app.data_directories || []).map(d => `<div class="text-truncate small text-muted" title="${d.path}">📁 ${d
    | .path} (${(d.total_size_bytes / (1024*1024)).toFixed(1)} MB)</
    | div>`).join('') || '<span class="text-muted small">—</span>';
102 |     const cfgs = (app.config_files || []).map(c => `<button class="btn btn-link btn-sm p-0 text-warning text-decoration-none
    | text-truncate d-block view-config-btn" data-app-id="${app.id}
    | " data-cfg-path="${c.path}" title="${c.path}">📄 ${c.filename}<
    | /button>`).join('') || '<span class="text-muted small">—</span>
    | >';
103 |     const nets = (app.network_endpoints || []).map(n => `<span
    | class="badge bg-secondary me-1 mb-1" title="${n.endpoint}">${
    | n.domain || n.endpoint}</span>`).join('') || '<span class="text-
    | muted small">—</span>';
104 |
105 |     return `
106 |         <tr>
107 |             <td>
108 |                 <div class="fw-bold">${escapeHtml(app.name)}</div>
109 |                 <div class="small text-muted text-truncate" style="m
    | ax-width: 250px;" title="${app.install_location || ''}">${esca
    | peHtml(app.install_location || '')}</div>
110 |             </td>
111 |             <td>
112 |                 <div>${escapeHtml(app.publisher || '-')}</div>
113 |                 <div class="small text-muted">${escapeHtml(app.versi

```

```

    on || '')}}</div>
114         </td>
115         <td>${dirs}</td>
116         <td>${cfgs}</td>
117         <td>${nets}</td>
118         <td class="text-end">
119             <button class="btn btn-sm btn-outline-info research-
app-btn" data-app-id="${app.id}" title="AI Исследование">
120                 <i class="bi bi-robot"></i> Анализ
121             </button>
122         </td>
123     </tr>
124     `;
125 }
126
127 function bindRowActions() {
128     document.querySelectorAll('.view-config-btn').forEach(btn
=> {
129         btn.onclick = () => {
130             const appId = btn.getAttribute('data-app-id');
131             const cfgPath = btn.getAttribute('data-cfg-path');
132             openConfigModal(appId, cfgPath);
133         };
134     });
135
136     document.querySelectorAll('.research-app-btn').forEach(btn
=> {
137         btn.onclick = () => {
138             const appId = btn.getAttribute('data-app-id');
139             openResearchModal(appId);
140         };
141     });
142 }
143
144 function openConfigModal(appId, cfgPath) {
145     const app = allApps.find(a => a.id === appId);
146     const cfg = app ? (app.config_files || []).find(c => c.pat
h === cfgPath) : null;

```

```
147 |
148 |     document.getElementById('st-config-modal-path').innerText
    | = cfgPath;
149 |     document.getElementById('st-config-modal-content').innerTe
    | xt = cfg ? (cfg.content_preview || '[Пустой файл]') : '[Файл н
    | е найден]';
150 |
151 |     const modalEl = document.getElementById('st-config-modal')
    | ;
152 |     if (modalEl && window.bootstrap) {
153 |         new bootstrap.Modal(modalEl).show();
154 |     }
155 | }
156 |
157 | async function openResearchModal(appId) {
158 |     const app = allApps.find(a => a.id === appId);
159 |     const titleEl = document.getElementById('st-research-modal
    | -title');
160 |     const bodyEl = document.getElementById('st-research-modal-
    | body');
161 |
162 |     if (titleEl) titleEl.innerText = 'AI Исследование: ' + (ap
    | p ? app.name : appId);
163 |     if (bodyEl) {
164 |         bodyEl.innerHTML = '<div class="text-center py-4"><div c
    | lass="spinner-border spinner-border-sm text-primary me-2"></di
    | v>Выполняется аналитическое исследование программы через Gemin
    | i...</div>';
165 |     }
166 |
167 |     const modalEl = document.getElementById('st-research-modal
    | ');
168 |     if (modalEl && window.bootstrap) {
169 |         new bootstrap.Modal(modalEl).show();
170 |     }
171 |
172 |     try {
173 |         const res = await fetch('/api/v1/software-scanner/resear
```

```

    ch', {
174         method: 'POST',
175         headers: { 'Content-Type': 'application/json' },
176         body: JSON.stringify({ app_id: appId })
177     });
178     if (!res.ok) throw new Error('HTTP ' + res.status);
179     const data = await res.json();
180
181     if (bodyEl) {
182         bodyEl.innerHTML = `
183             <h6>Описание программы:</h6>
184             <p>${escapeHtml(data.summary || data.description ||
185 'Нет данных')}</p>
186             ${data.risk_score !== undefined ? `<div class="mb-2"
187 ><strong>Оценка риска:</strong> <span class="badge bg-${data.r
188 isk_score > 5 ? 'danger' : 'success'}">${data.risk_score}/10</
189 span></div>` : ''}
190             ${data.recommendations ? `<h6>Рекомендации:</h6><
191 p>${escapeHtml(data.recommendations)}</p>` : ''}
192             `;
193     }
194 } catch (err) {
195     if (bodyEl) {
196         bodyEl.innerHTML = `<div class="alert alert-danger">Ош
197 ибка анализа: ${err.message}</div>`;
198     }
199 }
200
201 function escapeHtml(text) {
202     if (!text) return '';
203     const div = document.createElement('div');
204     div.innerText = text;
205     return div.innerHTML;
206 }
207
208 window.initSoftwareTransparencyTab = initSoftwareTransparenc
209 yTab;

```

```
204 |
205 |     if (document.readyState === 'loading') {
206 |         document.addEventListener('DOMContentLoaded', initSoftware
    | TransparencyTab);
207 |     } else {
208 |         initSoftwareTransparencyTab();
209 |     }
210 | })();
211 |
```

main.js

```
1 /**
2  * sources_tab/main.js
3  * Логика вкладки редактирования источников.
4  */
5
6 let currentSourcesObj = {};
7
8 async function loadSourcesRaw() {
9     const editor = document.getElementById('sources-json-editor');
10    const tbody = document.getElementById('sources-list-body')
11    ;
12
13    if (editor) {
14        editor.disabled = true;
15        editor.value = 'Загрузка...';
16    }
17    if (tbody) {
18        tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Загрузка источников...</td></tr>';
19    }
20
21    try {
22        const response = await window.api.fetch('/api/admin/sources/raw');
23        const rawContent = response.content || '{}';
24
25        if (editor) editor.value = rawContent;
26
27        try {
28            currentSourcesObj = JSON.parse(rawContent);
29            renderSourcesTable();
30        } catch (e) {
31            console.error('Ошибка парсинга JSON для таблицы:',
32                e);
33
34            if (tbody) {
```

```
32         tbody.innerHTML = '<tr><td colspan="6" class="
text-center py-4 text-danger">Ошибка формата JSON. Таблица нед
оступна.</td></tr>';
33     }
34 }
35
36 } catch (err) {
37     console.error('Ошибка загрузки источников:', err);
38     if (editor) editor.value = 'Ошибка загрузки:\n' + err.
message;
39     if (tbody) tbody.innerHTML = `<tr><td colspan="6" clas
s="text-center py-4 text-danger">Ошибка загрузки: ${err.messag
e}</td></tr>`;
40 } finally {
41     if (editor) editor.disabled = false;
42 }
43 }
44
45 async function saveSourcesRaw(content) {
46     const editor = document.getElementById('sources-json-edito
r');
47
48     // Проверка синтаксиса на клиенте перед отправкой
49     try {
50         JSON.parse(content);
51     } catch (e) {
52         if (typeof showNotification === 'function') {
53             showNotification('Синтаксическая ошибка JSON: ' +
e.message, 'danger');
54         } else {
55             alert('Синтаксическая ошибка JSON: ' + e.message);
56         }
57         return false;
58     }
59
60     try {
61         if (editor) editor.disabled = true;
62         const response = await window.api.fetch('/api/admin/so
```



```

        'urces/raw', {
63         method: 'POST',
64         headers: { 'Content-Type': 'application/json' },
65         body: JSON.stringify({ content: content })
66     });
67
68     if (response.status === 'ok') {
69         if (typeof showNotification === 'function') {
70             showNotification('Источники успешно сохранены!', 'success');
71         }
72         return true;
73     }
74     } catch (err) {
75         console.error('Ошибка сохранения источников:', err);
76         if (typeof showNotification === 'function') {
77             showNotification('Ошибка сохранения: ' + err.message, 'danger');
78         }
79         return false;
80     } finally {
81         if (editor) editor.disabled = false;
82     }
83 }
84
85 function renderSourcesTable() {
86     const tbody = document.getElementById('sources-list-body')
87     ;
88     const filterSelect = document.getElementById('sources-category-filter');
89     if (!tbody) return;
90     const filterVal = filterSelect ? filterSelect.value : 'all';
91     tbody.innerHTML = '';
92
93     let hasItems = false;
94

```

```

95     for (const [category, items] of Object.entries(currentSourcesObj)) {
96         if (!Array.isArray(items)) continue; // Пропускаем метаданные самого файла (например, "metadata": {...})
97
98         // Маппинг категорий
99         let catForFilter = 'other';
100        if (category === 'metadata_apis') catForFilter = 'metadata';
101        else if (category === 'streaming_search') catForFilter = 'streaming_search';
102        else if (category.includes('streaming') || category === 'direct_sites') catForFilter = 'streaming';
103        else if (category === 'iframe_players') catForFilter = 'embed_player';
104        else if (category === 'torrent_trackers') catForFilter = 'torrent';
105        else if (category === 'video_search') catForFilter = 'search';
106
107        if (filterVal !== 'all' && catForFilter !== filterVal) continue;
108
109        items.forEach((item, index) => {
110            hasItems = true;
111            const tr = document.createElement('tr');
112
113            const isEnabled = item.enabled !== false; // по умолчанию true
114
115            tr.innerHTML = `
116                <td>
117                    <div class="form-check form-switch">
118                        <input class="form-check-input toggle-source-btn" type="checkbox" role="switch"
119                            data-category="${category}" data-index="${index}" ${isEnabled ? 'checked' : ''}>
120                    </div>

```

```

121 |             </td>
122 |             <td class="fw-bold">${item.name || item.id ||
| 'Без названия'}</td>
123 |             <td><a href="${item.url || '#'}" target="_blan
| k" class="small text-truncate d-inline-block" style="max-width
| : 150px;">${item.url || '-'}</a></td>
124 |             <td><span class="badge bg-secondary">${categor
| y}</span></td>
125 |             <td class="small text-muted">${item.descriptio
| n || '-'}</td>
126 |             <td class="text-end">
127 |                 <button class="btn btn-sm btn-outline-dang
| er delete-source-btn" data-category="${category}" data-index="
| ${index}" title="Удалить">
128 |                     <i class="bi bi-trash"></i>
129 |                 </button>
130 |             </td>
131 |         `;
132 |         tbody.appendChild(tr);
133 |     });
134 | }
135 |
136 | if (!hasItems) {
137 |     tbody.innerHTML = '<tr><td colspan="6" class="text-cen
| ter py-4 text-muted">Нет источников для отображения</td></tr>'
| ;
138 | }
139 |
140 | // Привязываем обработчики к сгенерированным кнопкам
141 | document.querySelectorAll('.toggle-source-btn').forEach(bt
| n => {
142 |     btn.addEventListener('change', handleToggleSource);
143 | });
144 |
145 | document.querySelectorAll('.delete-source-btn').forEach(bt
| n => {
146 |     btn.addEventListener('click', handleDeleteSource);
147 | });

```

```
148 }
149
150 async function handleToggleSource(e) {
151     const category = e.target.getAttribute('data-category');
152     const index = parseInt(e.target.getAttribute('data-index'))
153     );
154     if (currentSourcesObj[category] && currentSourcesObj[category][index]) {
155         currentSourcesObj[category][index].enabled = e.target.checked;
156         const newJson = JSON.stringify(currentSourcesObj, null, 2);
157
158         // Обновляем текстовое поле и сервер
159         const editor = document.getElementById('sources-json-editor');
160         if (editor) editor.value = newJson;
161
162         await saveSourcesRaw(newJson);
163     }
164 }
165
166 async function handleDeleteSource(e) {
167     const btn = e.target.closest('button');
168     const category = btn.getAttribute('data-category');
169     const index = parseInt(btn.getAttribute('data-index'));
170
171     if (confirm('Вы уверены, что хотите удалить этот источник?')) {
172         if (currentSourcesObj[category] && currentSourcesObj[category][index]) {
173             currentSourcesObj[category].splice(index, 1);
174
175             const newJson = JSON.stringify(currentSourcesObj, null, 2);
176             const editor = document.getElementById('sources-json-editor');
```

```
177         if (editor) editor.value = newJson;
178
179         const success = await saveSourcesRaw(newJson);
180         if (success) renderSourcesTable();
181     }
182 }
183 }
184
185 async function handleAddSource() {
186     const id = document.getElementById('new-source-id')?.value
187     .trim();
188     const name = document.getElementById('new-source-name')?.v
189     alue.trim();
190     const url = document.getElementById('new-source-url')?.val
191     ue.trim();
192     const filterCat = document.getElementById('new-source-cate
193     gory')?.value;
194     const desc = document.getElementById('new-source-descripti
195     on')?.value.trim();
196
197     if (!id || !name || !url) {
198         window.showToast?.('Заполните обязательные поля: ID, Н
199         азвание и URL', 'warning') || alert('Заполните обязательные по
200         ля: ID, Название и URL');
201         return;
202     }
203
204     // Определяем в какую секцию JSON положить новый элемент
205     let targetCategory = 'other';
206     if (filterCat === 'metadata') targetCategory = 'metadata_a
207     pis';
208     else if (filterCat === 'streaming_search') targetCategory
209     = 'streaming_search';
210     else if (filterCat === 'streaming') targetCategory = 'dire
211     ct_sites';
212     else if (filterCat === 'embed_player') targetCategory = 'i
213     frame_players';
214     else if (filterCat === 'torrent') targetCategory = 'torren
```

```
t_trackers';
204     else if (filterCat === 'search') targetCategory = 'video_s
earch';
205
206     if (!currentSourcesObj[targetCategory]) {
207         currentSourcesObj[targetCategory] = [];
208     }
209
210     const newItem = {
211         id: id,
212         name: name,
213         url: url,
214         description: desc,
215         enabled: true
216     };
217
218     currentSourcesObj[targetCategory].push(newItem);
219
220     const newJson = JSON.stringify(currentSourcesObj, null, 2)
;
221     const editor = document.getElementById('sources-json-edito
r');
222     if (editor) editor.value = newJson;
223
224     const success = await saveSourcesRaw(newJson);
225     if (success) {
226         renderSourcesTable();
227         // Очищаем форму
228         document.getElementById('form-add-source').reset();
229     }
230 }
231
232 // Инициализация вкладки
233 window.initSourcesTab = function() {
234     console.log('Инициализация вкладки Источники...');
235
236     const btnRefresh = document.getElementById('btn-refresh-so
urces-json');
```

```
237     const btnSave = document.getElementById('btn-save-sources-  
    json');  
238     const btnSaveAll = document.getElementById('btn-save-all-s  
    ources');  
239     const btnAdd = document.getElementById('btn-add-source');  
240     const filterSelect = document.getElementById('sources-cate  
    gory-filter');  
241     const editor = document.getElementById('sources-json-edito  
    r');  
242  
243     if (btnRefresh) btnRefresh.onclick = loadSourcesRaw;  
244  
245     if (btnSave) {  
246         btnSave.onclick = () => {  
247             if (editor) saveSourcesRaw(editor.value).then(() =  
    > {  
248                 // Обновляем таблицу после ручного изменения J  
    SON  
249                 try {  
250                     currentSourcesObj = JSON.parse(editor.valu  
    e);  
251                     renderSourcesTable();  
252                 } catch(e){}  
253             });  
254         };  
255     }  
256  
257     if (btnSaveAll) {  
258         btnSaveAll.onclick = btnSave.onclick;  
259     }  
260  
261     if (btnAdd) {  
262         btnAdd.onclick = handleAddSource;  
263     }  
264  
265     if (filterSelect) {  
266         filterSelect.addEventListener('change', renderSourcesT  
    able);
```

```
267 |     }
268 |
269 |     // Также обновляем таблицу при потере фокуса редактором, е
    |     сли JSON валидный
270 |     if (editor) {
271 |         editor.addEventListener('blur', () => {
272 |             try {
273 |                 currentSourcesObj = JSON.parse(editor.value);
274 |                 renderSourcesTable();
275 |             } catch(e){}
276 |         });
277 |     }
278 |
279 |     // Загружаем данные при открытии
280 |     loadSourcesRaw();
281 | };
282 |
```


index.html

```
1 <!-- Sources Registry Tab -->
2 <div class="container-fluid">
3   <div class="alert alert-info d-flex justify-content-between
4     align-items-center flex-wrap gap-2 mb-3 shadow-sm">
5     <div>
6       <h5 class="mb-1"><i class="bi bi-collection-play me-2"><
7         /i>□ Реестр источников данных</h5>
8       <p class="mb-0 small text-secondary">Управление списком
9         сайтов, стримингов и парсеров для медиатеки.</p>
10      </div>
11      <div>
12        <button class="btn btn-outline-primary btn-sm rounded-pi
13          ll" onclick="const t = document.querySelector('[data-bs-target
14            =\'#tab-search\']'); if(t) t.click();">
15          <i class="bi bi-globe2 me-1"></i> Настройки веб-поиска
16          (MCP) →
17        </button>
18      </div>
19    </div>
20  </div>
21  <div class="row g-3">
22    <!-- LEFT: Add new source -->
23    <div class="col-4">
24      <div class="card mb-3">
25        <div class="card-header bg-dark text-white">
26          <span>□ Добавить источник</span>
27        </div>
28        <div class="card-body">
29          <form id="form-add-source" onsubmit="event.preventDe
30            fault();">
31            <div class="mb-2">
32              <label for="new-source-id" class="form-label sma
33                ll fw-bold">ID (уникальный):</label>
34              <input type="text" class="form-control form-cont
35                rol-sm" id="new-source-id"
36                placeholder="Например: my_site" required>
```

```
28         </div>
29         <div class="mb-2">
30             <label for="new-source-name" class="form-label small fw-bold">Название:</label>
31             <input type="text" class="form-control form-control-sm" id="new-source-name"
32                 placeholder="Например: My Site" required>
33         </div>
34         <div class="mb-2">
35             <label for="new-source-url" class="form-label small fw-bold">URL:</label>
36             <input type="url" class="form-control form-control-sm" id="new-source-url"
37                 placeholder="https://example.com" required>
38         </div>
39         <div class="mb-2">
40             <label for="new-source-category" class="form-label small fw-bold">Категория:</label>
41             <select class="form-select form-select-sm" id="new-source-category">
42                 <option value="streaming">streaming – стриминг
43                 <option value="streaming_search">streaming_search – поиск стримингов</option>
44                 <option value="metadata">metadata – метаданные
45                 <option value="embed_player">embed_player – встроенный плеер</option>
46                 <option value="torrent">torrent – торрент-трекер</option>
47                 <option value="search">search – поисковик</option>
48                 <option value="other">other – другое</option>
49             </select>
50         </div>
51         <div class="mb-3">
52             <label for="new-source-description" class="form-
```

```

label small fw-bold">Описание:</label>
53 |         <textarea class="form-control form-control-sm" i
| d="new-source-description" rows="2"
54 |             placeholder="Краткое описание источник
| a"></textarea>
55 |     </div>
56 |     <div class="d-grid">
57 |         <button type="button" class="btn btn-success btn
| -sm" id="btn-add-source">
58 |             □ Добавить
59 |         </button>
60 |     </div>
61 | </form>
62 | </div>
63 | </div>
64 |
65 | <!-- JSON Raw Editor -->
66 | <div class="card">
67 |     <div class="card-header bg-dark text-white d-flex just
| ify-content-between align-items-center">
68 |         <span>□ JSON-редактор</span>
69 |         <div class="d-flex gap-2">
70 |             <button class="btn btn-outline-light btn-sm" id="b
| tn-refresh-sources-json">
71 |                 <i class="bi bi-arrow-clockwise"></i>
72 |             </button>
73 |             <button class="btn btn-success btn-sm" id="btn-sav
| e-sources-json">
74 |                 <i class="bi bi-floppy"></i> Сохранить JSON
75 |             </button>
76 |         </div>
77 |     </div>
78 |     <div class="card-body p-2">
79 |         <p class="small text-muted mb-2">Прямое редактирован
| ие JSON. Используйте с осторожностью.</p>
80 |         <textarea
81 |             id="sources-json-editor"
82 |             class="form-control font-monospace"

```

```

83         style="min-height: 250px; max-height: 250px; resiz
e: none; font-size: 0.78rem;"
84         spellcheck="false"
85     ></textarea>
86 </div>
87 </div>
88 </div>
89
90 <!-- RIGHT: Sources Table -->
91 <div class="col-8">
92     <div class="card">
93         <div class="card-header bg-dark text-white d-flex just
ify-content-between align-items-center">
94             <span>📄 Список источников</span>
95             <button class="btn btn-outline-light btn-sm" id="btn
-refresh-sources">
96                 <i class="bi bi-arrow-clockwise"></i> Обновить
97             </button>
98         </div>
99         <div class="card-body p-0">
100             <!-- Category filter -->
101             <div class="p-2 border-bottom d-flex gap-2 flex-wrap
align-items-center">
102                 <span class="small text-muted">Фильтр:</span>
103                 <select class="form-select form-select-sm" id="sou
rces-category-filter" style="width: auto;">
104                     <option value="all">Все категории</option>
105                     <option value="streaming">Стриминг</option>
106                     <option value="streaming_search">Поиск стриминго
в</option>
107                     <option value="metadata">Метаданные</option>
108                     <option value="embed_player">Встроенные плееры</
option>
109                     <option value="torrent">Торренты</option>
110                     <option value="search">Поисковики</option>
111                     <option value="other">Другое</option>
112                 </select>
113             </div>

```

```

114 |         <!-- Table container with vertical scroll -->
115 |         <div style="height: 500px; overflow-y: scroll; overf
| low-x: auto;">
116 |             <table class="table table-striped table-hover align-middle mb-0" id="sources-table" style="min-width: 600px;">
117 |                 <thead class="table-dark">
118 |                     <tr>
119 |                         <th style="width: 5%;">Он</th>
120 |                         <th style="width: 15%;">Название</th>
121 |                         <th style="width: 25%;">URL</th>
122 |                         <th style="width: 15%;">Категория</th>
123 |                         <th style="width: 25%;">Описание</th>
124 |                         <th style="width: 15%;" class="text-end">Дей
| ствия</th>
125 |                     </tr>
126 |                 </thead>
127 |                 <tbody id="sources-list-body" style="word-wrap:
| break-word; overflow-wrap: break-word;">
128 |                     <tr>
129 |                         <td colspan="6" class="text-center py-4 text
| -muted">Загрузка источников...</td>
130 |                     </tr>
131 |                 </tbody>
132 |             </table>
133 |         </div>
134 |     </div>
135 | </div>
136 | </div>
137 | </div>
138 |
139 | <!-- Save all button at bottom -->
140 | <div class="mt-3 text-end">
141 |     <button class="btn btn-primary" id="btn-save-all-sources">
142 |         <i class="bi bi-floppy"></i> Сохранить все изменения
143 |     </button>
144 | </div>
145 | </div>
146 |

```

main.js

```
1 /**
2  * sources_tab/main.js
3  * Логика вкладки редактирования источников.
4  */
5
6 let currentSourcesObj = {};
7
8 async function loadSourcesRaw() {
9     const editor = document.getElementById('sources-json-editor');
10    const tbody = document.getElementById('sources-list-body')
11    ;
12
13    if (editor) {
14        editor.disabled = true;
15        editor.value = 'Загрузка...';
16    }
17    if (tbody) {
18        tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Загрузка источников...</td></tr>';
19    }
20
21    try {
22        const response = await window.api.fetch('/api/admin/sources/raw');
23        const rawContent = response.content || '{}';
24
25        if (editor) editor.value = rawContent;
26
27        try {
28            currentSourcesObj = JSON.parse(rawContent);
29            renderSourcesTable();
30        } catch (e) {
31            console.error('Ошибка парсинга JSON для таблицы:',
32                e);
33
34            if (tbody) {
```

```
32         tbody.innerHTML = '<tr><td colspan="6" class="
text-center py-4 text-danger">Ошибка формата JSON. Таблица нед
оступна.</td></tr>';
33     }
34 }
35
36 } catch (err) {
37     console.error('Ошибка загрузки источников:', err);
38     if (editor) editor.value = 'Ошибка загрузки:\n' + err.
message;
39     if (tbody) tbody.innerHTML = `<tr><td colspan="6" clas
s="text-center py-4 text-danger">Ошибка загрузки: ${err.messag
e}</td></tr>`;
40 } finally {
41     if (editor) editor.disabled = false;
42 }
43 }
44
45 async function saveSourcesRaw(content) {
46     const editor = document.getElementById('sources-json-edito
r');
47
48     // Проверка синтаксиса на клиенте перед отправкой
49     try {
50         JSON.parse(content);
51     } catch (e) {
52         if (typeof showNotification === 'function') {
53             showNotification('Синтаксическая ошибка JSON: ' +
e.message, 'danger');
54         } else {
55             alert('Синтаксическая ошибка JSON: ' + e.message);
56         }
57         return false;
58     }
59
60     try {
61         if (editor) editor.disabled = true;
62         const response = await window.api.fetch('/api/admin/so
```

```

        'urces/raw', {
63         method: 'POST',
64         headers: { 'Content-Type': 'application/json' },
65         body: JSON.stringify({ content: content })
66     });
67
68     if (response.status === 'ok') {
69         if (typeof showNotification === 'function') {
70             showNotification('Источники успешно сохранены!', 'success');
71         }
72         return true;
73     }
74     } catch (err) {
75         console.error('Ошибка сохранения источников:', err);
76         if (typeof showNotification === 'function') {
77             showNotification('Ошибка сохранения: ' + err.message, 'danger');
78         }
79         return false;
80     } finally {
81         if (editor) editor.disabled = false;
82     }
83 }
84
85 function renderSourcesTable() {
86     const tbody = document.getElementById('sources-list-body');
87     ;
88     const filterSelect = document.getElementById('sources-category-filter');
89     if (!tbody) return;
90     const filterVal = filterSelect ? filterSelect.value : 'all';
91     ;
92     tbody.innerHTML = '';
93     let hasItems = false;
94

```



```

95     for (const [category, items] of Object.entries(currentSourcesObj)) {
96         if (!Array.isArray(items)) continue; // Пропускаем метаданные самого файла (например, "metadata": {...})
97
98         // Маппинг категорий
99         let catForFilter = 'other';
100        if (category === 'metadata_apis') catForFilter = 'metadata';
101        else if (category === 'streaming_search') catForFilter = 'streaming_search';
102        else if (category.includes('streaming') || category === 'direct_sites') catForFilter = 'streaming';
103        else if (category === 'iframe_players') catForFilter = 'embed_player';
104        else if (category === 'torrent_trackers') catForFilter = 'torrent';
105        else if (category === 'video_search') catForFilter = 'search';
106
107        if (filterVal !== 'all' && catForFilter !== filterVal) continue;
108
109        items.forEach((item, index) => {
110            hasItems = true;
111            const tr = document.createElement('tr');
112
113            const isEnabled = item.enabled !== false; // по умолчанию true
114
115            tr.innerHTML = `
116                <td>
117                    <div class="form-check form-switch">
118                        <input class="form-check-input toggle-source-btn" type="checkbox" role="switch"
119                            data-category="${category}" data-index="${index}" ${isEnabled ? 'checked' : ''}>
120                    </div>

```

```

121 |             </td>
122 |             <td class="fw-bold">${item.name || item.id ||
| 'Без названия'}</td>
123 |             <td><a href="${item.url || '#'}" target="_blan
| k" class="small text-truncate d-inline-block" style="max-width
| : 150px;">${item.url || '-'}</a></td>
124 |             <td><span class="badge bg-secondary">${categor
| y}</span></td>
125 |             <td class="small text-muted">${item.descriptio
| n || '-'}</td>
126 |             <td class="text-end">
127 |                 <button class="btn btn-sm btn-outline-dang
| er delete-source-btn" data-category="${category}" data-index="
| ${index}" title="Удалить">
128 |                     <i class="bi bi-trash"></i>
129 |                 </button>
130 |             </td>
131 |         `;
132 |         tbody.appendChild(tr);
133 |     });
134 | }
135 |
136 | if (!hasItems) {
137 |     tbody.innerHTML = '<tr><td colspan="6" class="text-cen
| ter py-4 text-muted">Нет источников для отображения</td></tr>'
| ;
138 | }
139 |
140 | // Привязываем обработчики к сгенерированным кнопкам
141 | document.querySelectorAll('.toggle-source-btn').forEach(bt
| n => {
142 |     btn.addEventListener('change', handleToggleSource);
143 | });
144 |
145 | document.querySelectorAll('.delete-source-btn').forEach(bt
| n => {
146 |     btn.addEventListener('click', handleDeleteSource);
147 | });

```

```
148 }
149
150 async function handleToggleSource(e) {
151     const category = e.target.getAttribute('data-category');
152     const index = parseInt(e.target.getAttribute('data-index'))
153     );
154     if (currentSourcesObj[category] && currentSourcesObj[category][index]) {
155         currentSourcesObj[category][index].enabled = e.target.checked;
156         const newJson = JSON.stringify(currentSourcesObj, null, 2);
157
158         // Обновляем текстовое поле и сервер
159         const editor = document.getElementById('sources-json-editor');
160         if (editor) editor.value = newJson;
161
162         await saveSourcesRaw(newJson);
163     }
164 }
165
166 async function handleDeleteSource(e) {
167     const btn = e.target.closest('button');
168     const category = btn.getAttribute('data-category');
169     const index = parseInt(btn.getAttribute('data-index'));
170
171     if (confirm('Вы уверены, что хотите удалить этот источник?')) {
172         if (currentSourcesObj[category] && currentSourcesObj[category][index]) {
173             currentSourcesObj[category].splice(index, 1);
174
175             const newJson = JSON.stringify(currentSourcesObj, null, 2);
176             const editor = document.getElementById('sources-json-editor');
```

```
177         if (editor) editor.value = newJson;
178
179         const success = await saveSourcesRaw(newJson);
180         if (success) renderSourcesTable();
181     }
182 }
183 }
184
185 async function handleAddSource() {
186     const id = document.getElementById('new-source-id')?.value
187     .trim();
188     const name = document.getElementById('new-source-name')?.v
189     alue.trim();
190     const url = document.getElementById('new-source-url')?.val
191     ue.trim();
192     const filterCat = document.getElementById('new-source-cate
193     gory')?.value;
194     const desc = document.getElementById('new-source-descripti
195     on')?.value.trim();
196
197     if (!id || !name || !url) {
198         window.showToast?.('Заполните обязательные поля: ID, Н
199         азвание и URL', 'warning') || alert('Заполните обязательные по
200         ля: ID, Название и URL');
201         return;
202     }
203
204     // Определяем в какую секцию JSON положить новый элемент
205     let targetCategory = 'other';
206     if (filterCat === 'metadata') targetCategory = 'metadata_a
207     pis';
208     else if (filterCat === 'streaming_search') targetCategory
209     = 'streaming_search';
210     else if (filterCat === 'streaming') targetCategory = 'dire
211     ct_sites';
212     else if (filterCat === 'embed_player') targetCategory = 'i
213     frame_players';
214     else if (filterCat === 'torrent') targetCategory = 'torren
```

```

    t_trackers';
204     else if (filterCat === 'search') targetCategory = 'video_s
    earch';
205
206     if (!currentSourcesObj[targetCategory]) {
207         currentSourcesObj[targetCategory] = [];
208     }
209
210     const newItem = {
211         id: id,
212         name: name,
213         url: url,
214         description: desc,
215         enabled: true
216     };
217
218     currentSourcesObj[targetCategory].push(newItem);
219
220     const newJson = JSON.stringify(currentSourcesObj, null, 2)
    ;
221     const editor = document.getElementById('sources-json-edito
    r');
222     if (editor) editor.value = newJson;
223
224     const success = await saveSourcesRaw(newJson);
225     if (success) {
226         renderSourcesTable();
227         // Очищаем форму
228         document.getElementById('form-add-source').reset();
229     }
230 }
231
232 // Инициализация вкладки
233 window.initSourcesTab = function() {
234     console.log('Инициализация вкладки Источники...');
235
236     const btnRefresh = document.getElementById('btn-refresh-so
    urces-json');

```

```
237     const btnSave = document.getElementById('btn-save-sources-  
    json');  
238     const btnSaveAll = document.getElementById('btn-save-all-s  
    ources');  
239     const btnAdd = document.getElementById('btn-add-source');  
240     const filterSelect = document.getElementById('sources-cate  
    gory-filter');  
241     const editor = document.getElementById('sources-json-edito  
    r');  
242  
243     if (btnRefresh) btnRefresh.onclick = loadSourcesRaw;  
244  
245     if (btnSave) {  
246         btnSave.onclick = () => {  
247             if (editor) saveSourcesRaw(editor.value).then(() =  
    > {  
248                 // Обновляем таблицу после ручного изменения J  
    SON  
249                 try {  
250                     currentSourcesObj = JSON.parse(editor.valu  
    e);  
251                     renderSourcesTable();  
252                 } catch(e){}  
253             });  
254         };  
255     }  
256  
257     if (btnSaveAll) {  
258         btnSaveAll.onclick = btnSave.onclick;  
259     }  
260  
261     if (btnAdd) {  
262         btnAdd.onclick = handleAddSource;  
263     }  
264  
265     if (filterSelect) {  
266         filterSelect.addEventListener('change', renderSourcesT  
    able);
```

```
267 |     }
268 |
269 |     // Также обновляем таблицу при потере фокуса редактором, е
    |     сли JSON валидный
270 |     if (editor) {
271 |         editor.addEventListener('blur', () => {
272 |             try {
273 |                 currentSourcesObj = JSON.parse(editor.value);
274 |                 renderSourcesTable();
275 |             } catch(e){}
276 |         });
277 |     }
278 |
279 |     // Загружаем данные при открытии
280 |     loadSourcesRaw();
281 | };
282 |
283 |
```

index.html

```
1 <!-- Startup Auditor Tab Content -->
2 <style>
3   .sa-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .sa-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    height: 100%;
15    min-height: 58px;
16    transition: border-color 0.15s ease, box-shadow 0.15s ease
17    ;
18  }
19  .sa-card:hover {
20    border-color: #38bdf8 !important;
21    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
22  }
23  .sa-card-label {
24    font-size: 0.68rem;
25    text-transform: uppercase;
26    letter-spacing: 0.05em;
27    color: #94a3b8;
28    font-weight: 600;
29    display: flex;
30    justify-content: space-between;
31    align-items: center;
32    line-height: 1.1;
33  }
34  .sa-card-value {
35    font-size: 1.25rem;
36    font-weight: 700;
```



```
36     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,  
    Consolas, monospace;  
37     line-height: 1.15;  
38 }  
39 .sa-card-sub {  
40     font-size: 0.7rem;  
41     color: #64748b;  
42     line-height: 1.1;  
43 }  
44  
45 .sa-panel {  
46     background: #0f172a !important;  
47     border: 1px solid #334155 !important;  
48     border-radius: 8px;  
49     overflow: hidden;  
50     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.2);  
51 }  
52 .sa-panel-header {  
53     background: #1e293b !important;  
54     border-bottom: 1px solid #334155 !important;  
55     padding: 0.45rem 0.85rem;  
56     font-weight: 600;  
57     font-size: 0.82rem;  
58     color: #38bdf8;  
59     display: flex;  
60     align-items: center;  
61     justify-content: space-between;  
62     flex-wrap: wrap;  
63     gap: 0.35rem;  
64 }  
65  
66 .sa-table {  
67     width: 100%;  
68     font-size: 0.78rem;  
69     border-collapse: collapse;  
70     color: #e2e8f0;  
71     background: #0f172a;  
72 }
```

```
73 | .sa-table thead th {
74 |     background: #1e293b !important;
75 |     padding: 6px 10px;
76 |     color: #94a3b8;
77 |     font-weight: 600;
78 |     font-size: 0.72rem;
79 |     text-transform: uppercase;
80 |     letter-spacing: 0.04em;
81 |     border-bottom: 1px solid #334155;
82 |     white-space: nowrap;
83 |     position: sticky;
84 |     top: 0;
85 |     z-index: 2;
86 | }
87 | .sa-table tbody tr {
88 |     background: #0f172a !important;
89 |     transition: background-color 0.1s ease;
90 | }
91 | .sa-table tbody tr:nth-child(even) {
92 |     background: #131d31 !important;
93 | }
94 | .sa-table tbody tr:hover {
95 |     background: #1e2e4a !important;
96 | }
97 | .sa-table td {
98 |     padding: 6px 10px;
99 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
100 |     vertical-align: middle;
101 |     color: #cbd5e1;
102 | }
103 |
104 | /* Badge styles */
105 | .sa-badge {
106 |     font-size: 0.7rem;
107 |     font-weight: 600;
108 |     padding: 2px 6px;
109 |     border-radius: 4px;
110 |     display: inline-flex;
```

```
111 |     align-items: center;
112 |     gap: 3px;
113 |     letter-spacing: 0.02em;
114 |     white-space: nowrap;
115 | }
116 | .sa-badge-enabled {
117 |     background: rgba(34, 197, 94, 0.15);
118 |     border: 1px solid rgba(34, 197, 94, 0.45);
119 |     color: #4ade80;
120 | }
121 | .sa-badge-disabled {
122 |     background: rgba(100, 116, 139, 0.18);
123 |     border: 1px solid rgba(100, 116, 139, 0.35);
124 |     color: #94a3b8;
125 | }
126 | .sa-badge-loc {
127 |     background: #1e293b;
128 |     border: 1px solid #475569;
129 |     color: #38bdf8;
130 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    | Consolas, monospace;
131 | }
132 | .sa-badge-cat {
133 |     background: #1e293b;
134 |     border: 1px solid #334155;
135 |     color: #cbd5e1;
136 | }
137 |
138 | /* Risk Badges */
139 | .sa-risk-critical {
140 |     background: rgba(239, 68, 68, 0.22);
141 |     border: 1px solid rgba(239, 68, 68, 0.6);
142 |     color: #fca5a5;
143 | }
144 | .sa-risk-suspicious {
145 |     background: rgba(249, 115, 22, 0.22);
146 |     border: 1px solid rgba(249, 115, 22, 0.6);
147 |     color: #fdb462;
```

```
148 | }
149 | .sa-risk-warning {
150 |     background: rgba(234, 179, 8, 0.22);
151 |     border: 1px solid rgba(234, 179, 8, 0.6);
152 |     color: #fde047;
153 | }
154 | .sa-risk-notice {
155 |     background: rgba(56, 189, 248, 0.18);
156 |     border: 1px solid rgba(56, 189, 248, 0.5);
157 |     color: #7dd3fc;
158 | }
159 | .sa-risk-clean {
160 |     background: rgba(34, 197, 94, 0.16);
161 |     border: 1px solid rgba(34, 197, 94, 0.45);
162 |     color: #86efac;
163 | }
164 |
165 | /* Toggle Switch button */
166 | .sa-toggle-btn {
167 |     background: transparent;
168 |     border: none;
169 |     cursor: pointer;
170 |     padding: 1px 4px;
171 |     border-radius: 4px;
172 |     font-size: 1.15rem;
173 |     line-height: 1;
174 |     transition: transform 0.15s ease, opacity 0.15s ease;
175 |     display: inline-flex;
176 |     align-items: center;
177 |     justify-content: center;
178 | }
179 | .sa-toggle-btn:hover {
180 |     transform: scale(1.15);
181 | }
182 | .sa-toggle-btn:active {
183 |     transform: scale(0.95);
184 | }
185 |
```

```

186 | .sa-input {
187 |     background: #1e293b !important;
188 |     border: 1px solid #334155 !important;
189 |     color: #f8fafc !important;
190 |     font-size: 0.8rem;
191 |     padding: 0.25rem 0.5rem;
192 | }
193 | .sa-input:focus {
194 |     border-color: #38bdf8 !important;
195 |     box-shadow: 0 0 0 2px rgba(56, 189, 248, 0.25) !important;
196 | }
197 | .sa-input option {
198 |     background: #0f172a;
199 |     color: #f8fafc;
200 | }
201 | </style>
202 |
203 | <div class="sa-container" id="startup-auditor-container">
204 |
205 |     <!-- Header Alert Banner (Compact Strip) -->
206 |     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%); border: 1px solid #334155;">
207 |         <div class="d-flex align-items-center gap-2 flex-wrap">
208 |             <span class="fs-5">❌ </span>
209 |             <span class="fw-bold text-white small" style="font-size: 0.92rem;">Аудит автозапуска Windows</span>
210 |             <span class="badge bg-dark border border-secondary text-info fw-normal" style="font-size: 0.7rem;">Startup Auditor</span>
211 |             <span class="text-secondary small d-none d-lg-inline" style="font-size: 0.76rem;">| Run/RunOnce, папки автозагрузки, I FE0 и службы</span>
212 |         </div>
213 |         <div class="d-flex align-items-center gap-1.5 flex-wrap">
214 |             <button class="btn btn-xs btn-outline-info rounded-pill px-2.5 py-1 d-flex align-items-center gap-1" style="font-size:

```

```

    0.75rem;" id="sa-btn-refresh">
215     <i class="bi bi-arrow-repeat"></i> Пересканировать
216     </button>
217     <button class="btn btn-xs btn-outline-secondary rounded-
pill px-2 py-1" style="font-size: 0.75rem;" id="sa-btn-export-
json" title="Экспорт в JSON">
218         <i class="bi bi-file-earmark-code me-1"></i> JSON
219     </button>
220     <button class="btn btn-xs btn-outline-secondary rounded-
pill px-2 py-1" style="font-size: 0.75rem;" id="sa-btn-export-
csv" title="Экспорт в CSV">
221         <i class="bi bi-file-earmark-spreadsheet me-1"></i> CS
V
222     </button>
223 </div>
224 </div>
225
226 <!-- Top Metric & Health Score Cards (Compact Density Grid)
-->
227 <div class="row g-2 mb-2">
228     <!-- Health Score Card -->
229     <div class="col-6 col-md-3">
230         <div class="sa-card">
231             <div class="sa-card-label">
232                 <span>Индекс чистоты</span>
233                 <i class="bi bi-shield-check text-success"></i>
234             </div>
235             <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
236                 <div class="d-flex align-items-baseline gap-1">
237                     <div class="sa-card-value text-success" id="sa-hea
lth-score">--</div>
238                     <span style="color: #64748b; font-size: 0.75rem;">
/ 100</span>
239                 </div>
240                 <div class="sa-card-sub text-end" id="sa-health-stat
us">Оценка безопасности</div>
241             </div>

```

```

242     </div>
243 </div>
244
245 <!-- Active & Total Items -->
246 <div class="col-6 col-md-3">
247     <div class="sa-card">
248         <div class="sa-card-label">
249             <span>Элементы автозапуска</span>
250             <i class="bi bi-play-circle text-info"></i>
251         </div>
252         <div class="d-flex align-items-baseline justify-content-between mt-1">
253             <div class="d-flex align-items-baseline gap-1">
254                 <div class="sa-card-value text-info" id="sa-active-count">--</div>
255                 <span style="color: #64748b; font-size: 0.75rem;">из <span id="sa-total-count" class="text-white">--</span></span></div>
256             </div>
257             <div class="sa-card-sub text-end">Откл: <span id="sa-disabled-count" class="text-info fw-semibold">--</span></div>
258         </div>
259     </div>
260 </div>
261
262 <!-- Broken Entries Card -->
263 <div class="col-6 col-md-3">
264     <div class="sa-card">
265         <div class="sa-card-label">
266             <span>Битые ссылки</span>
267             <i class="bi bi-exclamation-triangle text-warning"></i>
268         </div>
269         <div class="d-flex align-items-baseline justify-content-between mt-1">
270             <div class="d-flex align-items-baseline gap-1">
271                 <div class="sa-card-value text-warning" id="sa-broken-count">--</div>

```

```

272 |         <span style="color: #64748b; font-size: 0.75rem;">
записей</span>
273 |     </div>
274 |     <div class="sa-card-sub text-end">Файлы не найдены</
div>
275 |     </div>
276 | </div>
277 | </div>
278 |
279 | <!-- Threats & Critical Alerts -->
280 | <div class="col-6 col-md-3">
281 |     <div class="sa-card">
282 |         <div class="sa-card-label">
283 |             <span>Угрозы и аномалии</span>
284 |             <i class="bi bi-shield-slash text-danger"></i>
285 |         </div>
286 |         <div class="d-flex align-items-baseline justify-content-between mt-1">
287 |             <div class="d-flex align-items-baseline gap-1">
288 |                 <div class="sa-card-value text-danger" id="sa-threats-count">--</div>
289 |                 <span style="color: #64748b; font-size: 0.75rem;">
предупреждений</span>
290 |             </div>
291 |             <div class="sa-card-sub text-end">IFE0 и скрипты</div>
292 |         </div>
293 |     </div>
294 | </div>
295 | </div>
296 |
297 | <!-- Recommendations Alert Banner (Compact Sleek Strip) -->
298 | <div class="py-1.5 px-3 mb-2 rounded-2 d-flex align-items-center justify-content-between" id="sa-recommendations-box" style="background: #0f172a; border: 1px solid #334155; font-size: 0.78rem;">
299 |     <div class="d-flex align-items-center gap-2">
300 |         <span class="text-warning">⚠ </span>

```



```

301         <div class="d-flex align-items-center gap-2 flex-wrap">
302             <span class="fw-semibold text-white" id="sa-rec-title"
303             >Рекомендации:</span>
304             <span style="color: #94a3b8;" id="sa-rec-desc">Загрузка
305             а рекомендаций по оптимизации...</span>
306         </div>
307     </div>
308 <!-- Search & Filter Controls (Compact Toolbar) -->
309 <div class="card p-2 mb-2" style="background: #0f172a; border:
310 1px solid #334155; border-radius: 8px;">
311     <div class="row g-2 align-items-center">
312         <!-- Search Input -->
313         <div class="col-12 col-md-4">
314             <div class="input-group input-group-sm">
315                 <span class="input-group-text sa-input border-second
316                 ary text-secondary py-1" style="border-right: none;"><i class=
317                 "bi bi-search"></i></span>
318                 <input type="text" class="form-control form-control-
319                 sm sa-input" id="sa-search-input" placeholder="Поиск по имени,
320                 команде или пути...">
321             </div>
322         </div>
323         <!-- Risk Filter -->
324         <div class="col-6 col-md-3">
325             <select class="form-select form-select-sm sa-input" id
326             ="sa-filter-risk">
327                 <option value="">Все уровни риска</option>
328                 <option value="critical">⚠ Critical (Критические)</
329                 option>
330                 <option value="suspicious">⚠ Suspicious (Подозритель
331                 ные)</option>
332                 <option value="warning">⚠ Warning (Предупреждения)</
333                 option>
334                 <option value="notice">ℹ Notice (Инфо)</option>
335                 <option value="clean">✅ Clean (Безопасные)</option>

```

```
328         </select>
329     </div>
330
331     <!-- Location Filter -->
332     <div class="col-6 col-md-3">
333         <select class="form-select form-select-sm sa-input" id
334         ="sa-filter-location">
335             <option value="">Все точки автозапуска</option>
336             <option value="registry_run">Реестр: Run</option>
337             <option value="registry_runonce">Реестр: RunOnce</op
338             tion>
339             <option value="startup_folder_user">Папка автозагруз
340             ки (User)</option>
341             <option value="startup_folder_common">Папка автозагр
342             узки (Common)</option>
343             <option value="winlogon">Winlogon</option>
344             <option value="ifeo">IFE0 Перехваты</option>
345             <option value="scheduled_task">Планировщик задач</op
346             tion>
347             <option value="windows_service">Службы Windows</opti
348             on>
349         </select>
350     </div>
351
352     <!-- State Filter -->
353     <div class="col-12 col-md-2">
354         <select class="form-select form-select-sm sa-input" id
355         ="sa-filter-state">
356             <option value="">Все статусы</option>
357             <option value="enabled">Только включенные</option>
358             <option value="disabled">Только отключенные</option>
359         </select>
360     </div>
361 </div>
362
363 <!-- Entries Data Table Panel -->
364 <div class="sa-panel">
```

```

359     <div class="sa-panel-header">
360         <span><i class="bi bi-table me-1"></i> Список зарегистри
        рованных элементов автозапуска</span>
361         <span class="small" style="color: #94a3b8;" id="sa-table
        -count">Загрузка...</span>
362     </div>
363     <div class="table-responsive" style="max-height: calc(100v
        h - 245px); overflow-y: auto;">
364         <table class="sa-table" id="sa-entries-table">
365             <thead>
366                 <tr>
367                     <th style="width: 80px;" class="text-center">Стату
        с</th>
368                     <th style="width: 230px;">Имя / Программа</th>
369                     <th style="width: 170px;">Точка автозапуска</th>
370                     <th style="width: 140px;">Категория</th>
371                     <th style="width: 110px;" class="text-center">Риск
        </th>
372                     <th>Исполняемый путь / Аргументы</th>
373                     <th style="width: 110px;" class="text-center">Влия
        ние</th>
374                     <th style="width: 80px;" class="text-center">Дейст
        вие</th>
375                 </tr>
376             </thead>
377             <tbody id="sa-table-body">
378                 <tr>
379                     <td colspan="8" class="text-center py-4" style="co
        lor: #94a3b8;">
380                         <div class="spinner-border spinner-border-sm tex
        t-info me-2" role="status"></div>
381                         Выполняется сканирование точек автозапуска Windo
        ws...
382                     </td>
383                 </tr>
384             </tbody>
385         </table>
386     </div>

```

```
387 | </div>
388 |
389 | <!-- Program Details & AI Context Modal -->
390 | <div class="modal fade" id="sa-entry-detail-modal" tabindex=
    | "-1" aria-hidden="true">
391 |     <div class="modal-dialog modal-lg modal-dialog-centered mo
    | dal-dialog-scrollable">
392 |         <div class="modal-content bg-dark text-light border-seco
    | ndary shadow-lg">
393 |             <div class="modal-header border-secondary py-2 px-3">
394 |                 <div class="d-flex align-items-center gap-2">
395 |                     <span class="fs-5 text-info">□ </span>
396 |                     <h5 class="modal-title fw-bold text-info mb-0" id=
    | "sa-modal-title">Детали программы</h5>
397 |                     <span id="sa-modal-status-badge"></span>
398 |                     <span id="sa-modal-risk-badge"></span>
399 |                 </div>
400 |                 <button type="button" class="btn-close btn-close-whi
    | te" data-bs-dismiss="modal" aria-label="Close"></button>
401 |             </div>
402 |             <div class="modal-body py-3 px-3">
403 |                 <!-- Metadata Grid -->
404 |                 <div class="row g-2 mb-3 small">
405 |                     <div class="col-sm-6"><strong class="text-secondar
    | y">Имя / Программа:</strong> <span id="sa-modal-name" class="f
    | w-semibold text-white"></span></div>
406 |                     <div class="col-sm-6"><strong class="text-secondar
    | y">Издатель:</strong> <span id="sa-modal-publisher" class="tex
    | t-light"></span></div>
407 |                     <div class="col-sm-6"><strong class="text-secondar
    | y">Точка автозапуска:</strong> <span id="sa-modal-location-typ
    | e" class="sa-badge sa-badge-loc"></span></div>
408 |                     <div class="col-sm-6"><strong class="text-secondar
    | y">Категория:</strong> <span id="sa-modal-category" class="sa-
    | badge sa-badge-cat"></span></div>
409 |                     <div class="col-sm-6"><strong class="text-secondar
    | y">Влияние на запуск:</strong> <span id="sa-modal-boot-impact"
    | ></span></div>
```

```
410 |         <div class="col-sm-6"><strong class="text-secondar
    | y">Цифровая подпись:</strong> <span id="sa-modal-is-signed"></
    | span></div>
411 |         <div class="col-sm-6"><strong class="text-secondar
    | y">Статус файла:</strong> <span id="sa-modal-file-exists"></sp
    | an></div>
412 |         <div class="col-sm-6"><strong class="text-secondar
    | y">Размер файла:</strong> <span id="sa-modal-file-size"></span
    | ></div>
413 |         <div class="col-sm-12"><strong class="text-seconda
    | ry">Расположение / Ключ:</strong> <span id="sa-modal-location-
    | path" class="font-monospace text-muted small" style="word-brea
    | k: break-all;"></span></div>
414 |     </div>
415 |
416 |     <!-- Command & Path Section -->
417 |     <div class="card bg-black border-secondary p-3 mb-3"
    | >
418 |         <div class="d-flex justify-content-between align-i
    | tems-center mb-1">
419 |             <h6 class="small text-uppercase text-muted fw-bo
    | ld mb-0">Команда и исполняемый файл</h6>
420 |         </div>
421 |         <div class="small font-monospace mb-2 text-info" id="
    | sa-modal-executable-path" style="word-break: break-all;"></
    | div>
422 |         <div class="small font-monospace text-muted" id="s
    | a-modal-command" style="white-space: pre-wrap; word-break: bre
    | ak-all;"></div>
423 |     </div>
424 |
425 |     <!-- AI Diagnostic & Explanation Section -->
426 |     <div class="card border-info bg-dark p-3" id="sa-mod
    | al-ai-section">
427 |         <div class="d-flex justify-content-between align-i
    | tems-center mb-2">
428 |             <h6 class="fw-bold text-info mb-0"><i class="bi
    | bi-robot me-1"></i> AI Contextual Explanation</h6>
```

```
429 |         <button class="btn btn-sm btn-outline-info py-0
    | px-2 rounded-pill" id="btn-sa-modal-diagnose">
430 |             <i class="bi bi-magic me-1"></i>Анализ контекс
    | та
431 |         </button>
432 |     </div>
433 |     <div class="small" id="sa-modal-ai-explanation">
434 |         Нажмите «Анализ контекста», чтобы получить экспе
    | ртное объяснение назначения программы от языковой модели, оцен
    | ку рисков и рекомендации по оптимизации.
435 |     </div>
436 | </div>
437 | </div>
438 |     <div class="modal-footer border-secondary py-2 px-3 d-
    | flex justify-content-between">
439 |         <div>
440 |             <button type="button" class="btn btn-sm btn-outlin
    | e-warning me-1" id="btn-sa-modal-toggle">
441 |                 <i class="bi bi-toggle2-on me-1"></i>Переключить
    | автозапуск
442 |             </button>
443 |             <button type="button" class="btn btn-sm btn-outlin
    | e-secondary" id="btn-sa-modal-copy">
444 |                 <i class="bi bi-clipboard me-1"></i>Копировать п
    | уть
445 |             </button>
446 |         </div>
447 |         <button type="button" class="btn btn-sm btn-secondar
    | y" data-bs-dismiss="modal">Закрыть</button>
448 |     </div>
449 | </div>
450 | </div>
451 | </div>
452 | </div>
453 |
```

main.js

```
1 // Windows Startup Auditor Frontend Logic
2 (function () {
3   let allEntries = [];
4   let currentSummary = null;
5
6   async function initStartupAuditorTab() {
7     console.log('[StartupAuditor] Initializing tab...');
8     bindEvents();
9     await loadAuditData();
10  }
11
12  function bindEvents() {
13    const btnRefresh = document.getElementById('sa-btn-refresh');
14    if (btnRefresh) {
15      btnRefresh.onclick = () => loadAuditData();
16    }
17
18    const searchInput = document.getElementById('sa-search-input');
19    if (searchInput) {
20      searchInput.oninput = () => renderFilteredTable();
21    }
22
23    const filterRisk = document.getElementById('sa-filter-risk');
24    if (filterRisk) {
25      filterRisk.onchange = () => renderFilteredTable();
26    }
27
28    const filterLocation = document.getElementById('sa-filter-location');
29    if (filterLocation) {
30      filterLocation.onchange = () => renderFilteredTable();
31    }
32  }
```

```

33     const filterState = document.getElementById('sa-filter-state');
34     if (filterState) {
35         filterState.onChange = () => renderFilteredTable();
36     }
37
38     const btnExportJson = document.getElementById('sa-btn-export-json');
39     if (btnExportJson) {
40         btnExportJson.onclick = () => {
41             window.open('/api/v1/startup-auditor/export?format=json', '_blank');
42         };
43     }
44
45     const btnExportCsv = document.getElementById('sa-btn-export-csv');
46     if (btnExportCsv) {
47         btnExportCsv.onclick = () => {
48             window.open('/api/v1/startup-auditor/export?format=csv', '_blank');
49         };
50     }
51 }
52
53 async function loadAuditData() {
54     const tbody = document.getElementById('sa-table-body');
55     if (tbody) {
56         tbody.innerHTML = `
57             <tr>
58                 <td colspan="8" class="text-center py-4 text-muted">
59                     <div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>
60                     Выполняется сканирование точек автозапуска Windows
61                     ...
62                 </td>
63             </tr>
64         `;

```



```

64     }
65
66     try {
67         const report = await window.api.fetch('/api/v1/startup-a
auditor/audit');
68         allEntries = report.entries || [];
69         currentSummary = report.summary || {};
70
71         renderSummaryCards(report);
72         renderFilteredTable();
73     } catch (err) {
74         console.error('[StartupAuditor] Error loading audit repo
rt:', err);
75         if (tbody) {
76             tbody.innerHTML = `
77                 <tr>
78                     <td colspan="8" class="text-center py-4 text-dange
r">
79                         Ошибка загрузки данных аудита: ${err.message}
80                     </td>
81                 </tr>
82             `;
83         }
84     }
85 }
86
87 function renderSummaryCards(report) {
88     const summary = report.summary || {};
89
90     // Health Score
91     const scoreEl = document.getElementById('sa-health-score')
;
92     if (scoreEl) {
93         scoreEl.textContent = summary.health_score ?? '--';
94         scoreEl.className = 'sa-card-value ' + (
95             summary.health_score >= 80 ? 'text-success' : (summary
.health_score >= 60 ? 'text-warning' : 'text-danger')
96     );

```

```
97 |     }
98 |
99 |     // Counters
100 |     const activeEl = document.getElementById('sa-active-count'
    | );
101 |     if (activeEl) activeEl.textContent = summary.active_entrie
    | s ?? '--';
102 |
103 |     const totalEl = document.getElementById('sa-total-count');
104 |     if (totalEl) totalEl.textContent = summary.total_entries ?
    | ? '--';
105 |
106 |     const disabledEl = document.getElementById('sa-disabled-co
    | unt');
107 |     if (disabledEl) disabledEl.textContent = summary.disabled_
    | entries ?? '0';
108 |
109 |     const brokenEl = document.getElementById('sa-broken-count'
    | );
110 |     if (brokenEl) brokenEl.textContent = summary.broken_entrie
    | s ?? '0';
111 |
112 |     const threatsEl = document.getElementById('sa-threats-coun
    | t');
113 |     if (threatsEl) threatsEl.textContent = (summary.critical_c
    | ount || 0) + (summary.suspicious_count || 0);
114 |
115 |     // Recommendations
116 |     const recDesc = document.getElementById('sa-rec-desc');
117 |     if (recDesc) {
118 |         if (report.recommendations && report.recommendations.len
    | gth > 0) {
119 |             recDesc.innerHTML = report.recommendations.map(r => `<
    | div>${r}</div>`).join('');
120 |         } else {
121 |             recDesc.textContent = 'Конфигурация автозапуска оптима
    | льна, критических замечаний нет.';
122 |         }
```

```
123 |     }
124 | }
125 |
126 | function renderFilteredTable() {
127 |     const tbody = document.getElementById('sa-table-body');
128 |     const tableCount = document.getElementById('sa-table-count
    | ');
129 |     if (!tbody) return;
130 |
131 |     const query = (document.getElementById('sa-search-input')?
    | .value || '').toLowerCase().trim();
132 |     const riskFilter = (document.getElementById('sa-filter-ris
    | k')?.value || '').toLowerCase();
133 |     const locFilter = (document.getElementById('sa-filter-loca
    | tion')?.value || '').toLowerCase();
134 |     const stateFilter = (document.getElementById('sa-filter-st
    | ate')?.value || '').toLowerCase();
135 |
136 |     const filtered = allEntries.filter(e => {
137 |         if (query) {
138 |             const fullText = `${e.name || ''} ${e.command || ''} $
    | {e.executable_path || ''} ${e.publisher || ''} ${e.category ||
    | ''}`.toLowerCase();
139 |             if (!fullText.includes(query)) return false;
140 |         }
141 |         if (riskFilter && (e.risk_level || '').toLowerCase() !==
    | riskFilter) {
142 |             return false;
143 |         }
144 |         if (locFilter && (e.location_type || '').toLowerCase() !
    | == locFilter) {
145 |             return false;
146 |         }
147 |         if (stateFilter === 'enabled' && !e.is_enabled) return f
    | else;
148 |         if (stateFilter === 'disabled' && e.is_enabled) return f
    | else;
149 |
```

```

150         return true;
151     });
152
153     if (tableCount) {
154         tableCount.textContent = `Показано: ${filtered.length} и
155     з ${allEntries.length}`;
156     }
157
158     if (filtered.length === 0) {
159         tbody.innerHTML = `
160             <tr>
161                 <td colspan="8" class="text-center py-5" style="color: #94a3b8;">
162                     <div class="fs-4 mb-2">⬜ </div>
163                     <div>Элементы автозапуска по заданным критериям не
164                     найдены</div>
165                 </td>
166             </tr>
167         `;
168         return;
169
170         tbody.innerHTML = filtered.map(e => {
171             const isEnabled = e.is_enabled;
172             const statusBadge = isEnabled
173                 ? '<span class="sa-badge sa-badge-enabled"><i class="bi bi-check-circle-fill"></i> ВКЛ</span>'
174                 : '<span class="sa-badge sa-badge-disabled"><i class="bi bi-dash-circle"></i> ОТКЛ</span>';
175
176             const riskBadge = getRiskBadge(e.risk_level);
177             const isBroken = !e.file_exists && e.executable_path;
178
179             const pathDisplay = isBroken
180                 ? `<div class="d-flex align-items-center gap-1 text-danger small font-monospace">
181                     <i class="bi bi-x-octagon-fill text-danger flex-shrink-0"></i>

```

```

181 |         <span class="text-truncate fw-semibold" style="ma
      | x-width: 480px;" title="{e.executable_path || e.command}">[Фа
      | йл не найден] {e.executable_path || e.command}</span>
182 |         </div>`
183 |         : `<div class="font-monospace small text-truncate" sty
      | le="max-width: 480px; color: #cbd5e1;" title="{e.command || e
      | .executable_path}">
184 |             {e.executable_path || e.command}
185 |         </div>`;
186 |
187 |         const argsDisplay = e.arguments ? `
188 |         <div class="d-flex align-items-center gap-1 font-monos
      | pace text-truncate mt-1" style="max-width: 480px; font-size: 0
      | .73rem; color: #94a3b8;" title="{e.arguments}">
189 |             <span class="badge bg-secondary-subtle text-secondar
      | y-emphasis border border-secondary" style="font-size: 0.68rem;
      | padding: 1px 4px;">ARG</span>
190 |             <span class="text-truncate">{e.arguments}</span>
191 |         </div>` : '';
192 |
193 |         const toggleIcon = isEnabled
194 |         ? '<i class="bi bi-toggle2-on text-success" style="fon
      | t-size: 1.4rem;"></i>'
195 |         : '<i class="bi bi-toggle2-off" style="font-size: 1.4r
      | em; color: #64748b;"></i>';
196 |         const toggleTitle = isEnabled ? 'Отключить элемент автоз
      | апуска' : 'Включить элемент автозапуска';
197 |
198 |         let impactBadge = '';
199 |         if (!isEnabled) {
200 |             impactBadge = '<span class="sa-badge sa-badge-disabled
      | ">Отключено</span>';
201 |         } else if (e.boot_impact === 'Высокое') {
202 |             impactBadge = '<span class="sa-badge sa-risk-critical"
      | >Высокое</span>';
203 |         } else if (e.boot_impact === 'Среднее') {
204 |             impactBadge = '<span class="sa-badge sa-risk-warning">
      | Среднее</span>';

```

```

205         } else {
206             impactBadge = '<span class="sa-badge sa-badge-cat">Низ
кое</span>';
207         }
208
209         return `
210             <tr class="sa-row-item" data-id="${e.id}" style="curso
r: pointer;" title="Нажмите для подробного AI-аудита">
211                 <td class="text-center">${statusBadge}</td>
212                 <td>
213                     <div class="fw-semibold text-truncate" style="max-
width: 230px; color: #f8fafc;" title="${e.name}">${e.name}</di
v>
214                     <div class="small text-truncate" style="max-width:
230px; color: #94a3b8; font-size: 0.73rem;" title="${e.publis
her || 'Неизвестен'}">${e.publisher || 'Неизвестный издатель'}
</div>
215                 </td>
216                 <td><span class="sa-badge sa-badge-loc">${e.location
_type}</span></td>
217                 <td><span class="sa-badge sa-badge-cat">${e.category
|| 'Неизвестно'}</span></td>
218                 <td class="text-center">${riskBadge}</td>
219                 <td>
220                     ${pathDisplay}
221                     ${argsDisplay}
222                 </td>
223                 <td class="text-center">${impactBadge}</td>
224                 <td class="text-center">
225                     <button class="sa-toggle-btn" data-id="${e.id}" da
ta-enabled="${isEnabled}" title="${toggleTitle}">
226                         ${toggleIcon}
227                     </button>
228                 </td>
229             </tr>
230         `;
231     }).join('');
232

```

```

233 // Attach row click listeners for modal details
234 tbody.querySelectorAll('.sa-row-item').forEach(row => {
235     row.onclick = (evt) => {
236         // Prevent modal open when clicking directly on the toggle button
237         if (evt.target.closest('.sa-toggle-btn')) {
238             return;
239         }
240         const entryId = row.getAttribute('data-id');
241         const entry = allEntries.find(item => item.id === entryId);
242         if (entry) {
243             showEntryModal(entry);
244         }
245     };
246 });
247
248 // Attach toggle listeners
249 tbody.querySelectorAll('.sa-toggle-btn').forEach(btn => {
250     btn.onclick = async (evt) => {
251         evt.stopPropagation();
252         const entryId = btn.getAttribute('data-id');
253         const currentState = btn.getAttribute('data-enabled') === 'true';
254         const newState = !currentState;
255         try {
256             btn.disabled = true;
257             btn.style.opacity = '0.5';
258             const res = await window.api.fetch('/api/v1/startup-auditor/toggle', {
259                 method: 'POST',
260                 headers: { 'Content-Type': 'application/json' },
261                 body: JSON.stringify({ entry_id: entryId, enable: newState })
262             });
263             if (res.success) {
264                 window.showToast?(`Автозагрузка ${newState ? 'включена' : 'отключена'}`, 'success');

```

```
265 |         await loadAuditData();
266 |     } else {
267 |         window.showToast?(`Ошибка переключения: ${res.message}
    | sage}`, 'danger') || alert(`Ошибка переключения: ${res.message}
    | `);
268 |     }
269 |     } catch (err) {
270 |         window.showToast?(`Ошибка: ${err.message}`, 'danger
    | ') || alert(`Ошибка: ${err.message}`);
271 |     } finally {
272 |         btn.disabled = false;
273 |         btn.style.opacity = '1';
274 |     }
275 | };
276 | });
277 | }
278 |
279 | function showEntryModal(entry) {
280 |     if (!entry) return;
281 |
282 |     const modalEl = document.getElementById('sa-entry-detail-m
    | odal');
283 |     if (!modalEl) return;
284 |
285 |     // Header info
286 |     const titleEl = document.getElementById('sa-modal-title');
287 |     if (titleEl) titleEl.textContent = entry.name || 'Детали п
    | рограммы';
288 |
289 |     const statusBadgeEl = document.getElementById('sa-modal-st
    | atus-badge');
290 |     if (statusBadgeEl) {
291 |         statusBadgeEl.innerHTML = entry.is_enabled
292 |             ? '<span class="sa-badge sa-badge-enabled"><i class="b
    | i bi-check-circle-fill"></i> ВКЛ</span>'
293 |             : '<span class="sa-badge sa-badge-disabled"><i class="
    | bi bi-dash-circle"></i> ОТКЛ</span>';
294 |     }
```



```
295 |
296 |     const riskBadgeEl = document.getElementById('sa-modal-risk
    | -badge');
297 |     if (riskBadgeEl) {
298 |         riskBadgeEl.innerHTML = getRiskBadge(entry.risk_level);
299 |     }
300 |
301 |     // Metadata Fields
302 |     const setTxt = (id, val) => {
303 |         const el = document.getElementById(id);
304 |         if (el) el.textContent = val ?? '-';
305 |     };
306 |     const setHtml = (id, html) => {
307 |         const el = document.getElementById(id);
308 |         if (el) el.innerHTML = html;
309 |     };
310 |
311 |     setTxt('sa-modal-name', entry.name);
312 |     setTxt('sa-modal-publisher', entry.publisher || 'Неизвестн
    | ый разработчик');
313 |     setTxt('sa-modal-location-type', entry.location_type || 'r
    | egistry_run');
314 |     setTxt('sa-modal-category', entry.category || 'Неизвестно'
    | );
315 |     setTxt('sa-modal-boot-impact', entry.boot_impact || 'Низко
    | е');
316 |     setTxt('sa-modal-location-path', entry.location_path || 'Н
    | е указан');
317 |
318 |     setHtml('sa-modal-is-signed', entry.is_signed
    | ? '<span class="text-success"><i class="bi bi-patch-chec
    | k me-1"></i>Подписан (Valid)</span>'
    | : '<span class="text-secondary"><i class="bi bi-patch-qu
    | estion me-1"></i>Без подписи / Неизвестно</span>'
    | );
321 |
322 |
323 |     setHtml('sa-modal-file-exists', entry.file_exists
    | ? '<span class="text-success"><i class="bi bi-file-earma
```

```

rk-check me-1"></i>Файл существует</span>'
325     : '<span class="text-danger fw-bold"><i class="bi bi-fil
e-earmark-x me-1"></i>Файл не найден (битая ссылка)</span>'
326     );
327
328     setTxt('sa-modal-file-size', entry.file_size_kb > 0 ? `${e
ntry.file_size_kb.toFixed(1)} KB` : '-');
329     setTxt('sa-modal-executable-path', entry.executable_path |
| entry.command || 'Путь не указан');
330     setTxt('sa-modal-command', entry.command || '-');
331
332     // Reset AI Diagnostic section
333     const aiExplanation = document.getElementById('sa-modal-ai
-explanation');
334     if (aiExplanation) {
335         aiExplanation.innerHTML = `Нажмите «Анализ контекста», ч
тобы получить экспертное объяснение назначения программы от яз
ыковой модели, оценку рисков и рекомендации по оптимизации.`;
336     }
337
338     // AI Diagnose button
339     const diagnoseBtn = document.getElementById('btn-sa-modal-
diagnose');
340     if (diagnoseBtn) {
341         diagnoseBtn.onclick = async () => {
342             aiExplanation.innerHTML = `
343                 <div class="d-flex align-items-center gap-2 py-2 tex
t-info">
344                     <div class="spinner-border spinner-border-sm" role
="status"></div>
345                     <span>Генерация AI-диагностики и анализа рисков дл
я программы «${escapeHtml(entry.name)}»...</span>
346                 </div>
347             `;
348             const activeModel = window.activeModelName
349                 || window.userSettings?.model
350                 || document.getElementById('model-select')?.value
351                 || document.getElementById('header-model-select')?.v

```

```

    alue
352         || '';
353     try {
354         const res = await window.api.fetch('/api/v1/startup-
auditor/explain', {
355             method: 'POST',
356             headers: { 'Content-Type': 'application/json' },
357             body: JSON.stringify({
358                 entry_id: entry.id,
359                 name: entry.name,
360                 publisher: entry.publisher,
361                 executable_path: entry.executable_path,
362                 command: entry.command,
363                 arguments: entry.arguments,
364                 location_type: entry.location_type,
365                 location_path: entry.location_path,
366                 risk_level: entry.risk_level,
367                 is_enabled: entry.is_enabled,
368                 file_exists: entry.file_exists,
369                 is_signed: entry.is_signed,
370                 boot_impact: entry.boot_impact,
371                 model: activeModel || undefined
372             })
373         });
374
375         aiExplanation.innerHTML = `
376             <div class="mb-2"><strong class="text-info"><i cla
ss="bi bi-card-text me-1"></i>Назначение:</strong> ${escapeHtm
l(res.summary)}</div>
377             <div class="mb-2"><strong class="text-secondary"><
i class="bi bi-building me-1"></i>Разработчик / Категория:</st
rong> ${escapeHtml(res.developer || entry.publisher || 'Неизве
стен'))} (${escapeHtml(res.category || 'Приложение'))</div>
378             <div class="mb-2"><strong class="text-warning"><i
class="bi bi-shield-lock me-1"></i>Оценка безопасности:</stron
g> ${escapeHtml(res.security_verdict)}</div>
379             <div class="mb-2"><strong class="text-info"><i cla
ss="bi bi-speedometer2 me-1"></i>Влияние на запуск:</strong> $

```

```

    {escapeHtml(res.boot_impact_analysis)}</div>
380     <div class="p-2 mb-2 rounded bg-dark-subtle border
    border-warning-subtle">
381         <strong class="text-warning"><i class="bi bi-lig
    htbulb me-1"></i>Рекомендация:</strong> ${escapeHtml(res.start
    up_recommendation)}
382     </div>
383     ${res.action_steps && res.action_steps.length > 0
    ? `
384         <h6 class="small text-uppercase text-muted fw-bo
    ld mb-1">Рекомендуемые действия:</h6>
385         <ul class="mb-0 ps-3">
386             ${res.action_steps.map(s => `<li>${escapeHtml(
    s)}</li>`).join('')}
387         </ul>
388         ` : ''}
389     `;
390     } catch (err) {
391         aiExplanation.innerHTML = `<div class="text-danger p
    y-2"><i class="bi bi-exclamation-octagon me-1"></i>Ошибка полу
    чения AI-анализа: ${escapeHtml(err.message)}</div>`;
392     }
393 };
394 }
395
396 // Modal Footer Toggle Button
397 const modalToggleBtn = document.getElementById('btn-sa-mod
    al-toggle');
398 if (modalToggleBtn) {
399     modalToggleBtn.innerHTML = entry.is_enabled
400     ? '<i class="bi bi-toggle2-off me-1 text-warning"></i>
    Отключить автозапуск'
401     : '<i class="bi bi-toggle2-on me-1 text-success"></i>В
    ключить автозапуск';
402     modalToggleBtn.onclick = async () => {
403         const newState = !entry.is_enabled;
404         try {
405             modalToggleBtn.disabled = true;

```

```

406         const res = await window.api.fetch('/api/v1/startup-
auditor/toggle', {
407             method: 'POST',
408             headers: { 'Content-Type': 'application/json' },
409             body: JSON.stringify({ entry_id: entry.id, enable:
newState })
410         });
411         if (res.success) {
412             entry.is_enabled = newState;
413             showEntryModal(entry);
414             window.showToast?(`Автозагрузка ${newState ? 'вкл
ючена' : 'отключена'}`, 'success');
415             await loadAuditData();
416         } else {
417             window.showToast?(`Ошибка переключения: ${res.mes
sage}`, 'danger') || alert(`Ошибка переключения: ${res.message
}`);
418         }
419         } catch (err) {
420             window.showToast?(`Ошибка: ${err.message}`, 'danger
') || alert(`Ошибка: ${err.message}`);
421         } finally {
422             modalToggleBtn.disabled = false;
423         }
424     };
425 }
426
427 // Modal Footer Copy Button
428 const modalCopyBtn = document.getElementById('btn-sa-modal
-copy');
429 if (modalCopyBtn) {
430     modalCopyBtn.onclick = () => {
431         const textToCopy = entry.executable_path || entry.comm
and || '';
432         navigator.clipboard.writeText(textToCopy).then(() => {
433             const orig = modalCopyBtn.innerHTML;
434             modalCopyBtn.innerHTML = '<i class="bi bi-check2 me-
1 text-success"></i>Скопировано!';

```

```

435         setTimeout(() => { modalCopyBtn.innerHTML = orig; },
      1800);
436     });
437 };
438 }
439
440 // Show modal via Bootstrap
441 if (window.bootstrap && window.bootstrap.Modal) {
442     const bsModal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
443     bsModal.show();
444 }
445 }
446
447 function getRiskBadge(level) {
448     switch ((level || '').toLowerCase()) {
449         case 'critical':
450             return '<span class="sa-badge sa-risk-critical"><i class="bi bi-shield-x"></i> CRITICAL</span>';
451         case 'suspicious':
452             return '<span class="sa-badge sa-risk-suspicious"><i class="bi bi-exclamation-octagon"></i> SUSPICIOUS</span>';
453         case 'warning':
454             return '<span class="sa-badge sa-risk-warning"><i class="bi bi-exclamation-triangle"></i> WARNING</span>';
455         case 'notice':
456             return '<span class="sa-badge sa-risk-notice"><i class="bi bi-info-circle"></i> NOTICE</span>';
457         default:
458             return '<span class="sa-badge sa-risk-clean"><i class="bi bi-shield-check"></i> CLEAN</span>';
459     }
460 }
461
462 function escapeHtml(str) {
463     if (!str) return '';
464     return String(str)
465         .replace(/&/g, '&amp;');

```

```
466 |         .replace(/</g, '&lt;')
467 |         .replace(/>/g, '&gt;')
468 |         .replace(/"/g, '&quot;')
469 |         .replace(/'/g, '&#039;');
470 |     }
471 |
472 |     // Expose globally for Apps Hub
473 |     window.initStartupAuditorTab = initStartupAuditorTab;
474 | })();
475 |
```

index.html

```
1 <!--
2 =====
3 |=====
4 Process Name: Windows Storage & Battery Wear Tab
5 |=====
6 Description:
7 | Вкладка телеметрии износа накопителей SSD/NVMe (SMART, Health %) и деградации
8 | батареи ноутбука / источника питания в Windows.
9 |
10 File: index.html
11 Project: AI-Breadboard
12 Module: WebInterface.StorageWearTab
13 Author: hypo69
14 Copyright: © 2026 hypo69
15 |=====
16 |=====
17 -->
18 <style>
19 .wear-tab-wrapper {
20 | padding: 0.75rem;
21 | color: var(--text-color, #f8fafc);
22 | }
23 .diag-panel {
24 | background: var(--surface-2, #1e293b);
25 | border: 1px solid var(--border-color, #334155);
26 | border-radius: 8px;
27 | overflow: hidden;
28 | }
29 .diag-panel-header {
30 | background: rgba(15, 23, 42, 0.6);
31 | padding: 8px 14px;
32 | border-bottom: 1px solid var(--border-color, #334155);
33 | font-weight: 600;
34 | font-size: 0.85rem;
```



```
33 | }
34 | .about-sys-progress-track {
35 |     background: #0f172a;
36 |     border-radius: 4px;
37 |     height: 8px;
38 |     overflow: hidden;
39 |     min-width: 80px;
40 | }
41 | .about-sys-progress-bar {
42 |     height: 100%;
43 |     border-radius: 4px;
44 |     transition: width 0.3s ease;
45 | }
46 | .about-sys-spec-table {
47 |     width: 100%;
48 |     border-collapse: separate;
49 |     border-spacing: 0;
50 | }
51 | .about-sys-spec-table td {
52 |     padding: 8px 12px;
53 |     border-bottom: 1px solid rgba(51, 65, 85, 0.35);
54 |     vertical-align: middle;
55 | }
56 | .about-sys-spec-key {
57 |     color: var(--text-muted, #94a3b8);
58 |     width: 200px;
59 |     font-size: 0.82rem;
60 |     font-weight: 500;
61 | }
62 | .about-sys-spec-val {
63 |     color: var(--text-color, #f8fafc);
64 |     font-weight: 500;
65 |     font-size: 0.82rem;
66 | }
67 | </style>
68 |
69 | <div class="wear-tab-wrapper">
70 |     <!-- Top Header & Status -->
```

```

71 | <div class="card shadow-sm border-secondary-subtle mb-3" sty
    | le="background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
    | 0%); border-radius: 10px;">
72 |     <div class="card-body p-3">
73 |         <div class="d-flex flex-wrap align-items-center justify-
    | content-between gap-2">
74 |             <div class="d-flex align-items-center gap-2">
75 |                 <div class="fs-4">❏ </div>
76 |                 <div>
77 |                     <h5 class="mb-0 fw-bold text-white">4. Износ накоп
    | ителей и батареи (Storage & Battery Health)</h5>
78 |                     <small class="text-muted">Мониторинг ресурса SSD/N
    | VMe, файловых систем, SMART и деградации аккумулятора</small>
79 |                 </div>
80 |             </div>
81 |
82 |             <div class="d-flex align-items-center gap-2 flex-wrap"
    | >
83 |                 <div class="form-check form-switch mb-0 small text-m
    | uted d-flex align-items-center gap-1">
84 |                     <input class="form-check-input" type="checkbox" id
    | ="diag-wear-auto-refresh">
85 |                     <label class="form-check-label small text-nowrap"
    | for="diag-wear-auto-refresh">Авто (10c)</label>
86 |                 </div>
87 |                 <button class="btn btn-sm btn-outline-info rounded-p
    | ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
    | "btn-diag-wear-refresh">
88 |                     <i class="bi bi-arrow-clockwise"></i> Обновить
89 |                 </button>
90 |             </div>
91 |         </div>
92 |     </div>
93 | </div>
94 |
95 | <div class="row g-2 mb-3">
96 |     <!-- Disks Health & Wear Panel -->
97 |     <div class="col-12 col-xl-8 col-lg-8">

```

```

98 |         <div class="diag-panel h-100">
99 |             <div class="diag-panel-header d-flex justify-content-b
| between align-items-center">
100 |                 <div class="d-flex align-items-center gap-2">
101 |                     <i class="bi bi-hdd-fill text-primary"></i>
102 |                     <span class="text-white">Здоровье накопителей SSD/
| NVMe и износ (Wear & SMART)</span>
103 |                 </div>
104 |                 <span class="badge bg-secondary" id="diag-wear-disks
| -count">0 дисков</span>
105 |             </div>
106 |             <div class="table-responsive" style="max-height: 480px
| ; overflow-y: auto;">
107 |                 <table class="table table-dark table-hover table-sm
| mb-0" style="font-size: 0.82rem; font-family: ui-monospace, SF
| Mono-Regular, Menlo, Monaco, Consolas, monospace;">
108 |                     <thead>
109 |                         <tr class="text-secondary border-secondary">
110 |                             <th>Физический накопитель</th>
111 |                             <th>Тип / Шина</th>
112 |                             <th>Разделы</th>
113 |                             <th style="text-align: right;">Объем</th>
114 |                             <th style="text-align: right;">Свободно</th>
115 |                             <th>Наработка / Старт</th>
116 |                             <th style="text-align: right;">Запись / Чтение
| </th>
117 |                             <th style="width: 130px;">Здоровье / Износ</th>
| >
118 |                             <th>Статус</th>
119 |                         </tr>
120 |                     </thead>
121 |                     <tbody id="diag-wear-disks-tbody">
122 |                         <tr><td colspan="9" class="text-center py-4 text
| -muted">Опрос контроллеров накопителей...</td></tr>
123 |                     </tbody>
124 |                 </table>
125 |             </div>
126 |         </div>

```

```

127 |     </div>
128 |
129 |     <!-- Battery Deep Telemetry -->
130 |     <div class="col-12 col-xl-4 col-lg-4">
131 |         <div class="diag-panel h-100">
132 |             <div class="diag-panel-header d-flex justify-content-b
between align-items-center">
133 |                 <div class="d-flex align-items-center gap-2">
134 |                     <i class="bi bi-battery-charging text-warning"></i
>
135 |                     <span class="text-white">Деградация и состояние ба
тарей (Battery Degradation)</span>
136 |                 </div>
137 |                 <span class="badge bg-dark border border-secondary t
ext-info" id="diag-battery-source-badge">AC Mains</span>
138 |             </div>
139 |             <div class="p-3" id="diag-battery-details-container">
140 |                 <div class="text-center py-4 text-muted small">
141 |                     <i class="bi bi-plug-fill fs-4 text-info d-block m
b-1"></i>
142 |                     Стационарный компьютер – питание напрямую от элект
росети
143 |                 </div>
144 |             </div>
145 |         </div>
146 |     </div>
147 | </div>
148 | </div>
149 |

```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Storage & Battery Wear Logic
4   * =====
5   * Description:
6   *   Клиентский контроллер вкладки износа накопителей и батаре
7   *   SMART состояние дисков, остаточный ресурс SSD и деградаци
8   *   я аккумулятора.
9   *
10  * File: main.js
11  * Project: AI-Breadboard
12  * Module: WebInterface.StorageWearTab
13  * Author: hypo69
14  * Copyright: © 2026 hypo69
15  * =====
16  */
17 (function () {
18     'use strict';
19
20     let autoRefreshTimer = null;
21
22     function escapeHtml(str) {
23         if (str === null || str === undefined) return '';
24         return String(str)
25             .replace(/&/g, '&amp;')
26             .replace(/</g, '&lt;')
27             .replace(/>/g, '&gt;')
28             .replace(/"/g, '&quot;')
29             .replace(/'/g, '&#039;');
30     }
31 }
```

```

32 | function setText(id, text) {
33 |     const el = document.getElementById(id);
34 |     if (el) el.textContent = text !== null && text !== undefined ? String(text) : '--';
35 | }
36 |
37 | async function apiFetch(url) {
38 |     if (window.api && typeof window.api.fetch === 'function')
39 |     {
40 |         return await window.api.fetch(url);
41 |     }
42 |     const res = await fetch(url);
43 |     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.statusText}`);
44 |     return await res.json();
45 | }
46 |
47 | async function fetchStorageBatteryWear() {
48 |     try {
49 |         const data = await apiFetch('/api/v1/system/diagnostics/storage-battery');
50 |         if (!data) return;
51 |
52 |         // Disks
53 |         const disksTbody = document.getElementById('diag-wear-disks-tbody');
54 |         if (disksTbody && data.disks_wear) {
55 |             setText('diag-wear-disks-count', `${data.disks_wear.length} дисков`);
56 |             if (data.disks_wear.length === 0) {
57 |                 disksTbody.innerHTML = '<tr><td colspan="9" class="text-center py-4 text-muted">Физических накопителей не обнаружено</td></tr>';
58 |             } else {
59 |                 const formatBytesLocal = (bytes) => {
60 |                     if (bytes == null || bytes === 0) return '0 B';
61 |                     const k = 1024;
62 |                     const sizes = ['B', 'KB', 'MB', 'GB', 'TB', 'PB'];

```

```

62 |         const i = Math.floor(Math.log(bytes) / Math.log(k)
    | );
63 |         return `${parseFloat((bytes / Math.pow(k, i)).toFixed(1)))} ${sizes[i]}`;
    |
64 |     };
    |
65 |
66 |     const formatPohLocal = (hours) => {
67 |         if (hours == null || hours <= 0) return null;
68 |         const days = Math.floor(hours / 24);
69 |         const years = (hours / (24 * 365.25)).toFixed(1);
70 |         if (days >= 365) return `${hours.toLocaleString()}
    | ч (≈ ${years} г)`;
71 |         if (days >= 1) return `${hours.toLocaleString()} ч
    | (${days} д)`;
72 |         return `${hours.toLocaleString()} ч`;
73 |     };
    |
74 |
75 |     disksTbody.innerHTML = data.disks_wear.map(d => {
76 |         const diskName = d.name || d.model || d.device_id
    | || 'Физический диск';
77 |         const devId = d.device_id ? `<span class="badge bg
    | -secondary-subtle text-light border border-secondary me-1.5">${
    | {escapeHtml(d.device_id)}</span>` : '';
78 |         const isSsd = (d.media_type && d.media_type.toUpper
    | case().includes('SSD')) || (d.bus_type && d.bus_type.toUpperC
    | ase().includes('NVME'));
79 |         const diskIcon = isSsd ? 'bi-device-ssd text-info'
    | : 'bi-hdd text-primary';
80 |         const typeBus = [d.media_type, d.bus_type].filter(
    | Boolean).join(' / ') || 'Storage';
81 |         const partsBadge = d.partitions && d.partitions !=
    | = '-'
82 |         ? `<span class="badge bg-primary-subtle text-inf
    | o border border-info-subtle font-monospace">${escapeHtml(d.par
    | titions)}</span>`
83 |         : `<span class="text-muted">—</span>`;
84 |         const freeGbStr = d.free_gb != null ? `${d.free_gb
    | } GB` : `<span class="text-muted">—</span>`;

```

```

85         const healthPct = d.health_pct != null ? d.health_
| pct : 100;
86         const healthColor = healthPct >= 80 ? 'success' :
| (healthPct >= 50 ? 'warning' : 'danger');
87
88         // Наработка и дата первого включения
89         const pohFormatted = formatPohLocal(d.power_on_hou
| rs);
90         const pohHtml = pohFormatted
91         ? `<div class="font-monospace text-info">${pohFo
| rmatted}</div><div class="small text-muted font-monospace mt-0
| .5">Старт: ${escapeHtml(d.first_power_on || '-')}</div>`
92         : `<span class="text-muted font-monospace small"
| >Сессия OC</span>`;
93
94         // Объемы ввода/вывода (запись и чтение)
95         const ioHtml = `
96         <div class="font-monospace text-light" title="За
| писано данных">
97             <i class="bi bi-arrow-up-circle text-warning m
| e-1"></i>${formatBytesLocal(d.bytes_written)}
98             </div>
99             <div class="small font-monospace text-info mt-0.
| 5" title="Прочитано данных">
100             <i class="bi bi-arrow-down-circle text-info me
| -1"></i>${formatBytesLocal(d.bytes_read)}
101             </div>
102         `;
103
104         return `
105         <tr>
106             <td class="fw-bold text-white">
107                 <div class="d-flex align-items-center">
108                     <i class="bi ${diskIcon} me-2 fs-6"></i>
109                     <div>
110                         <div class="text-truncate" style="max-widt
| h: 250px;" title="${escapeHtml(diskName)}">${escapeHtml(diskNa
| me)}</div>

```



```

111 |         <div class="small text-muted font-monospace
      | e mt-0.5">${devId}${d.serial_number && d.serial_number !== 'N/
      | A' ? `<span class="text-secondary">S/N: ${escapeHtml(d.serial_
      | number)}</span>` : ''}</div>
112 |         </div>
113 |     </div>
114 | </td>
115 |     <td class="text-muted font-monospace"><span clas
      | s="badge bg-dark border border-secondary text-light">${escapeH
      | tml(typeBus)}</span></td>
116 |     <td>${partsBadge}</td>
117 |     <td style="text-align: right;" class="text-light
      | font-monospace">${d.total_gb} GB</td>
118 |     <td style="text-align: right;" class="text-succe
      | ss font-monospace">${freeGbStr}</td>
119 |     <td>${pohHtml}</td>
120 |     <td style="text-align: right;">${ioHtml}</td>
121 |     <td>
122 |         <div class="d-flex align-items-center gap-1.5"
      | >
123 |             <div class="about-sys-progress-track flex-gr
      | ow-1" style="height: 6px; min-width: 55px;">
124 |                 <div class="about-sys-progress-bar bg-${he
      | althColor}" style="width: ${healthPct}%;"></div>
125 |             </div>
126 |             <span class="small font-monospace text-${hea
      | lthColor} ms-1">${healthPct}%</span>
127 |         </div>
128 |     </td>
129 |     <td><span class="badge bg-${healthColor}-subtle
      | text-${healthColor} border border-${healthColor}">${escapeHtml
      | (d.status || 'OK')}</span></td>
130 | </tr>
131 | `;)).join('');
132 | }
133 | }
134 |
135 | // Battery

```

```

136 |     const batContainer = document.getElementById('diag-batte
    | ry-details-container');
137 |     const batSourceBadge = document.getElementById('diag-bat
    | tery-source-badge');
138 |     if (batContainer && data.battery_wear) {
139 |         const b = data.battery_wear;
140 |         if (batSourceBadge) {
141 |             batSourceBadge.textContent = b.power_source || 'AC M
    | ains';
142 |         }
143 |         if (!b.has_battery) {
144 |             batContainer.innerHTML = `
145 |                 <div class="text-center py-4 text-muted small">
146 |                     <i class="bi bi-plug-fill fs-4 text-info d-block
    | mb-1"></i>
147 |                         Стационарный компьютер – питание напрямую от эле
    | ктросети
148 |                 </div>
149 |             `;
150 |         } else {
151 |             batContainer.innerHTML = `
152 |                 <table class="about-sys-spec-table">
153 |                     <tbody>
154 |                         <tr>
155 |                             <td class="about-sys-spec-key"><i class="bi
    | bi-battery-charging me-1.5 text-warning"></i>Уровень заряда</t
    | d>
156 |                             <td class="about-sys-spec-val fw-bold text-w
    | hite">${b.percent}% (${b.is_charging ? 'Заряжается' : 'Разряд'
    | })</td>
157 |                         </tr>
158 |                         <tr>
159 |                             <td class="about-sys-spec-key"><i class="bi
    | bi-shield-shaded me-1.5 text-info"></i>Заводская емкость</td>
160 |                             <td class="about-sys-spec-val font-monospace
    | ">${b.design_capacity_mwh ? b.design_capacity_mwh + ' mWh' : '
    | -- '}</td>
161 |                         </tr>

```

```

162         <tr>
163             <td class="about-sys-spec-key"><i class="bi
164 bi-battery-full me-1.5 text-success"></i>Текущая емкость</td>
165             <td class="about-sys-spec-val font-monospace
166 ">${b.full_charge_capacity_mwh ? b.full_charge_capacity_mwh +
167 ' mWh' : '--'}</td>
168         </tr>
169     <tr>
170         <td class="about-sys-spec-key"><i class="bi
171 bi-heart-pulse me-1.5 text-danger"></i>Деградация (Износ)</td>
172         <td class="about-sys-spec-val ${b.wear_level
173 _pct > 20 ? 'text-danger fw-bold' : 'text-success'}">
174             ${b.wear_level_pct}% износа
175         </td>
176     </tr>
177 </tbody>
178 </table>
179 `;
180 }
181 }
182 } catch (err) {
183     console.warn('[StorageWearTab] fetchStorageBatteryWear e
184 rror:', err);
185 }
186 }
187
188 function bindEvents() {
189     const btnRefresh = document.getElementById('btn-diag-wear-
190 refresh');
191     if (btnRefresh) {
192         btnRefresh.onclick = () => fetchStorageBatteryWear();
193     }
194
195     const autoSwitch = document.getElementById('diag-wear-auto
196 -refresh');
197     if (autoSwitch) {
198         autoSwitch.onchange = (e) => {
199             if (e.target.checked) {

```

```
192 |         autoRefreshTimer = setInterval(fetchStorageBatteryWe
    | ar, 10000);
193 |     } else if (autoRefreshTimer) {
194 |         clearInterval(autoRefreshTimer);
195 |         autoRefreshTimer = null;
196 |     }
197 | };
198 | }
199 | }
200 |
201 | async function init() {
202 |     bindEvents();
203 |     await fetchStorageBatteryWear();
204 | }
205 |
206 | if (document.readyState === 'loading') {
207 |     document.addEventListener('DOMContentLoaded', init);
208 | } else {
209 |     setTimeout(init, 10);
210 | }
211 | })();
212 |
```

index.html

```
1 <!-- Windows System Control Center Web Tab -->
2 <style>
3   .scc-container {
4     padding: 0.75rem 1rem;
5   }
6   .scc-header {
7     background: var(--surface-2, #1e293b);
8     border: 1px solid var(--border-color, #334155);
9     border-radius: var(--radius-md, 8px);
10    padding: 0.75rem 1.25rem;
11  }
12  .scc-badge-secure {
13    background: var(--status-success-bg, rgba(16, 185, 129, 0.
14    2));
15    color: var(--status-success-text, #10b981);
16    border: 1px solid var(--status-success-border, #10b981);
17  }
18  .scc-badge-warn {
19    background: var(--status-warning-bg, rgba(245, 158, 11, 0.
20    2));
21    color: var(--status-warning-text, #f59e0b);
22    border: 1px solid var(--status-warning-border, #f59e0b);
23  }
24  .scc-badge-danger {
25    background: var(--status-danger-bg, rgba(239, 68, 68, 0.2)
26    );
27    color: var(--status-danger-text, #ef4444);
28    border: 1px solid var(--status-danger-border, #ef4444);
29  }
30  .scc-table {
31    width: 100%;
32    border-collapse: collapse;
33    margin-bottom: 0;
34  }
35  .scc-table th {
36    padding: 0.65rem 1rem;
```

```
34 |     font-size: 0.75rem;
35 |     font-weight: 600;
36 |     text-transform: uppercase;
37 |     letter-spacing: 0.05em;
38 |     color: var(--text-muted, #94a3b8);
39 |     background: rgba(255, 255, 255, 0.02);
40 |     border-bottom: 1px solid var(--border-color, #334155);
41 |     white-space: nowrap;
42 | }
43 | .scc-table td {
44 |     padding: 0.65rem 1rem;
45 |     font-size: 0.85rem;
46 |     color: var(--text-color, #f8fafc);
47 |     border-bottom: 1px solid rgba(255, 255, 255, 0.05);
48 |     vertical-align: middle;
49 | }
50 | .scc-table tr:last-child td {
51 |     border-bottom: none;
52 | }
53 | .scc-table tbody tr:hover td {
54 |     background: rgba(255, 255, 255, 0.04);
55 | }
56 |
57 | /* OS System Log Center Styles */
58 | .slc-compact-header {
59 |     background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
60 | 0%);
61 |     border: 1px solid #334155;
62 |     border-radius: 8px;
63 |     padding: 0.4rem 0.75rem;
64 |     margin-bottom: 0.5rem;
65 | }
66 | .slc-metric-pill {
67 |     background: rgba(15, 23, 42, 0.8);
68 |     border: 1px solid #334155;
69 |     border-radius: 6px;
70 |     padding: 0.2rem 0.5rem;
71 |     font-size: 0.75rem;
```

```
71     display: inline-flex;
72     align-items: center;
73     gap: 0.35rem;
74 }
75 .slc-panel {
76     background: #0f172a !important;
77     border: 1px solid #334155 !important;
78     border-radius: 8px;
79     overflow: hidden;
80 }
81 .slc-panel-header {
82     background: #1e293b !important;
83     border-bottom: 1px solid #334155 !important;
84     padding: 0.5rem 0.85rem;
85     font-weight: 600;
86     font-size: 0.85rem;
87     color: #38bdf8;
88     display: flex;
89     align-items: center;
90     justify-content: space-between;
91 }
92 .slc-tree-list {
93     max-height: 540px;
94     overflow-y: auto;
95     font-size: 0.8rem;
96 }
97 .slc-tree-item {
98     padding: 0.35rem 0.65rem;
99     cursor: pointer;
100     border-radius: 6px;
101     margin-bottom: 2px;
102     color: #cbd5e1;
103     display: flex;
104     justify-content: space-between;
105     align-items: center;
106     transition: background 0.15s ease;
107 }
108 .slc-tree-item:hover {
```

```
109 |     background: rgba(56, 189, 248, 0.1);
110 |     color: #38bdf8;
111 | }
112 | .slc-tree-item.active {
113 |     background: #0284c7;
114 |     color: #fff;
115 |     font-weight: 600;
116 | }
117 | .slc-table {
118 |     width: 100%;
119 |     margin-bottom: 0;
120 |     color: #e2e8f0;
121 |     font-size: 0.8rem;
122 | }
123 | .slc-table th {
124 |     background: #1e293b;
125 |     color: #94a3b8;
126 |     border-bottom: 1px solid #334155;
127 |     font-weight: 600;
128 |     text-transform: uppercase;
129 |     font-size: 0.7rem;
130 |     letter-spacing: 0.05em;
131 |     padding: 0.5rem 0.75rem;
132 | }
133 | .slc-table td {
134 |     border-bottom: 1px solid #1e293b;
135 |     padding: 0.45rem 0.75rem;
136 |     vertical-align: middle;
137 | }
138 | .slc-table tr:hover td {
139 |     background: rgba(56, 189, 248, 0.06);
140 |     cursor: pointer;
141 | }
142 | .slc-table tr.selected td,
143 | .slc-table tr.table-active td {
144 |     background: rgba(56, 189, 248, 0.2) !important;
145 |     color: #fff !important;
146 | }
```



```
147 | .badge-crit { background: #dc2626 !important; color: #fff !i
    | mportant; }
148 | .badge-err { background: #ef4444 !important; color: #fff !im
    | portant; }
149 | .badge-warn { background: #f59e0b !important; color: #000 !i
    | mportant; }
150 | .badge-inf { background: #10b981 !important; color: #fff !im
    | portant; }
151 | .badge-verb { background: #64748b !important; color: #fff !i
    | mportant; }
152 |
153 | /* Interactive Channel Description Popover Tooltip */
154 | .slc-tooltip-popup {
155 |     position: fixed;
156 |     z-index: 99999;
157 |     max-width: 320px;
158 |     background: #0f172a;
159 |     border: 1px solid #38bdf8;
160 |     border-radius: 8px;
161 |     padding: 0.6rem 0.8rem;
162 |     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.7), 0 0 15px
    | rgba(56, 189, 248, 0.2);
163 |     pointer-events: none;
164 |     transition: opacity 0.15s ease, transform 0.15s ease;
165 |     opacity: 0;
166 |     transform: translateY(4px);
167 |     font-size: 0.75rem;
168 |     line-height: 1.35;
169 |     color: #e2e8f0;
170 | }
171 | .slc-tooltip-popup.show {
172 |     opacity: 1;
173 |     transform: translateY(0);
174 | }
175 | .slc-tooltip-title {
176 |     font-weight: 700;
177 |     color: #38bdf8;
178 |     margin-bottom: 0.25rem;
```

```

179     display: flex;
180     align-items: center;
181     gap: 0.4rem;
182 }
183 .slc-tooltip-meta {
184     color: #94a3b8;
185     font-size: 0.7rem;
186     border-top: 1px solid #334155;
187     padding-top: 0.35rem;
188     margin-top: 0.35rem;
189     display: flex;
190     justify-content: space-between;
191 }
192 </style>
193
194 <div class="scc-container theme-sysctl">
195     <!-- Top Header Bar -->
196     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 scc-header text-white">
197         <div>
198             <div class="d-flex align-items-center gap-2">
199                 <h5 class="mb-0 fw-bold text-white">
200                     □ Windows System Control Center
201                 </h5>
202                 <span class="badge rounded-pill bg-primary px-2 py-1"
203 style="font-size: 0.75rem;">System Logs & Activity</span>
204             </div>
205             <p class="mb-0 small text-light mt-1" style="color: #cbd5e1 !important;">
206                 Центр анализа логов операционной системы Windows, аудита событий и истории активности приложений.
207             </p>
208         </div>
209         <div class="d-flex align-items-center gap-2 flex-wrap">
210             <span class="badge rounded-pill px-3 py-2" id="scc-elevation-badge">
211                 □ Checking privileges...

```

```

212     <button class="btn btn-sm btn-outline-info rounded-pill
    px-3" id="btn-scc-refresh">
213         <i class="bi bi-arrow-clockwise me-1"></i> Refresh
214     </button>
215 </div>
216 </div>
217
218 <!-- System Logs & Activity Panel (Primary Main View) -->
219 <div id="scc-pane-logs">
220     <!-- Compact Header Bar with Inline Stats & Actions -->
221     <div class="slc-compact-header d-flex justify-content-betw
    een align-items-center flex-wrap gap-2">
222         <!-- Title & Inline Metrics -->
223         <div class="d-flex align-items-center gap-2 flex-wrap">
224             <div class="d-flex align-items-center gap-1.5 me-1">
225                 <span class="fs-6">□ </span>
226                 <span class="fw-bold text-white small">Windows OS &
    System Log Center</span>
227             </div>
228
229             <!-- Quick Metrics Pills -->
230             <div class="slc-metric-pill" title="Discovered Log Sou
    rces">
231                 <span class="text-secondary small"><i class="bi bi-d
    iagram-3 me-1"></i>Sources:</span>
232                 <span class="fw-bold text-info font-monospace" id="s
    lc-stat-sources">1,247</span>
233             </div>
234             <div class="slc-metric-pill" title="Active Log Channel
    ">
235                 <span class="text-secondary small"><i class="bi bi-f
    older-fill me-1"></i>Channel:</span>
236                 <span class="fw-bold text-warning font-monospace tex
    t-truncate" id="slc-stat-channel" style="max-width: 140px;">Sy
    stem</span>
237             </div>
238             <div class="slc-metric-pill" title="Events Loaded in V
    iew">

```

```
239 |         <span class="text-secondary small"><i class="bi bi-l  
|ist-ol me-1"></i>Loaded:</span>  
240 |         <span class="fw-bold text-light font-monospace" id="  
|slc-stat-events">0</span>  
241 |     </div>  
242 |     <div class="slc-metric-pill" title="Errors and Inciden  
|ts Detected">  
243 |         <span class="text-secondary small"><i class="bi bi-e  
|xclamation-triangle-fill me-1"></i>Faults:</span>  
244 |         <span class="fw-bold text-danger font-monospace" id=  
|"slc-stat-incidents">0</span>  
245 |     </div>  
246 | </div>  
247 |  
248 | <!-- Actions Toolbar -->  
249 | <div class="d-flex align-items-center gap-2 flex-wrap">  
250 |     <button class="btn btn-outline-info rounded-pill px-2.  
|5 py-1 small d-flex align-items-center gap-1" id="btn-slc-scan  
|" style="font-size: 0.78rem;">  
251 |         <i class="bi bi-radar"></i> <span>Scan</span>  
252 |     </button>  
253 |     <div class="form-check form-switch text-light mb-0 d-i  
|nline-flex align-items-center" style="font-size: 0.78rem; min-  
|height: auto;">  
254 |         <input class="form-check-input me-1" type="checkbox"  
|role="switch" id="slc-live-switch" style="cursor: pointer;">  
255 |         <label class="form-check-label text-secondary small  
|user-select-none" for="slc-live-switch">Live</label>  
256 |     </div>  
257 |     <button class="btn btn-primary rounded-pill px-2.5 py-  
|1 small d-flex align-items-center gap-1" id="btn-slc-refresh"  
|style="font-size: 0.78rem;">  
258 |         <i class="bi bi-arrow-clockwise"></i> <span>Refresh<  
|/span>  
259 |     </button>  
260 |     <div class="dropdown">  
261 |         <button class="btn btn-xs btn-outline-secondary drop  
|down-toggle rounded-pill px-2.5 py-1 text-light small d-flex a
```

```

align-items-center gap-1" type="button" data-bs-toggle="dropdown" style="font-size: 0.78rem;">
262     <i class="bi bi-download"></i> <span>Export</span>
263 </button>
264 <ul class="dropdown-menu dropdown-menu-dark dropdown
-menu-end shadow" style="font-size: 0.8rem;">
265     <li><a class="dropdown-item" href="#" id="btn-slc-
export-json"><i class="bi bi-filetype-json me-2"></i>JSON</a><
/li>
266     <li><a class="dropdown-item" href="#" id="btn-slc-
export-csv"><i class="bi bi-filetype-csv me-2"></i>CSV</a></li
>
267 </ul>
268 </div>
269 </div>
270 </div>
271
272 <!-- Search & View Mode Navigation (Compact) -->
273 <div class="slc-panel mb-2 p-2">
274     <div class="d-flex justify-content-between align-items-c
enter flex-wrap gap-1 mb-2 pb-1 border-bottom border-secondary
">
275         <!-- Mode Tabs -->
276         <ul class="nav nav-pills gap-1" id="slc-mode-tabs">
277             <li class="nav-item">
278                 <button class="nav-link active py-0.5 px-2.5 small
rounded-pill" data-mode="live" style="font-size: 0.78rem;">●
Live Events</button>
279             </li>
280             <li class="nav-item">
281                 <button class="nav-link py-0.5 px-2.5 small rounde
d-pill text-light" data-mode="audit" style="font-size: 0.78rem
;">□ Аудит (Smart Digest)</button>
282             </li>
283             <li class="nav-item">
284                 <button class="nav-link py-0.5 px-2.5 small rounde
d-pill text-light" data-mode="rag" style="font-size: 0.78rem;"
>□ RAG Поиск</button>

```

```

285         </li>
286         <li class="nav-item">
287             <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="errors" style="font-size: 0.78rem;">❏ Errors Only</button>
288         </li>
289         <li class="nav-item">
290             <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="incidents" style="font-size: 0.78rem;">❏ Incidents</button>
291         </li>
292         <li class="nav-item">
293             <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="timeline" style="font-size: 0.78rem;">❏ Timeline</button>
294         </li>
295         <li class="nav-item">
296             <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="raw" style="font-size: 0.78rem;">❏ Raw Record</button>
297         </li>
298         <li class="nav-item">
299             <button class="nav-link py-0.5 px-2.5 small rounded-pill text-warning" data-mode="activity" style="font-size: 0.78rem;">❏ App Activity Log</button>
300         </li>
301     </ul>
302
303     <!-- Action AI Button -->
304     <div class="d-flex align-items-center gap-2">
305         <button class="btn btn-xs btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-slc-ask-ai" style="font-size: 0.78rem;">
306             <i class="bi bi-robot"></i> <span>Ask AI</span>
307         </button>
308     </div>
309 </div>
310

```

```
311 |     <!-- Filter Inputs Grid -->
312 |     <div class="row g-1.5 align-items-center" id="slc-filter
    | -row">
313 |         <div class="col-lg-4 col-md-5">
314 |             <div class="input-group input-group-sm">
315 |                 <span class="input-group-text bg-dark text-seconda
    | ry border-secondary py-0 px-2"><i class="bi bi-search" style="
    | font-size: 0.75rem;"></i></span>
316 |                 <input type="text" class="form-control form-contro
    | l-sm bg-dark text-light border-secondary py-1" id="slc-search-
    | input" placeholder="Search message, provider..." style="font-s
    | ize: 0.8rem;">
317 |             </div>
318 |         </div>
319 |         <div class="col-lg-2 col-md-2 col-6">
320 |             <select class="form-select form-select-sm bg-dark te
    | xt-light border-secondary py-1" id="slc-level-select" style="f
    | ont-size: 0.8rem;">
321 |                 <option value="">All Severities</option>
322 |                 <option value="Critical">Critical</option>
323 |                 <option value="Error">Error</option>
324 |                 <option value="Warning">Warning</option>
325 |                 <option value="Information">Information</option>
326 |             </select>
327 |         </div>
328 |         <div class="col-lg-2 col-md-2 col-6">
329 |             <select class="form-select form-select-sm bg-dark te
    | xt-light border-secondary py-1" id="slc-hours-select" style="f
    | ont-size: 0.8rem;">
330 |                 <option value="1">Last 1 hour</option>
331 |                 <option value="6">Last 6 hours</option>
332 |                 <option value="24" selected>Last 24 hours</option>
333 |                 <option value="168">Last 7 days</option>
334 |                 <option value="0">All available</option>
335 |             </select>
336 |         </div>
337 |         <div class="col-lg-2 col-md-1.5 col-6">
338 |             <input type="number" class="form-control form-contro
```

```

    l-sm bg-dark text-light border-secondary py-1" id="slc-eventid
    -input" placeholder="Event ID" style="font-size: 0.8rem;">
339     </div>
340     <div class="col-lg-2 col-md-1.5 col-6">
341         <select class="form-select form-select-sm bg-dark te
    xt-light border-secondary py-1" id="slc-limit-select" style="f
    ont-size: 0.8rem;">
342             <option value="50">50 records</option>
343             <option value="100" selected>100 records</option>
344             <option value="250">250 records</option>
345             <option value="500">500 records</option>
346         </select>
347     </div>
348 </div>
349 </div>
350
351 <!-- Main Multi-Column View -->
352 <div class="row g-3" id="slc-main-row">
353     <!-- Channel & Source Tree Sidebar -->
354     <div class="col-lg-3" id="slc-channel-tree-sidebar">
355         <div class="slc-panel h-100">
356             <div class="slc-panel-header">
357                 <span><i class="bi bi-folder2-open me-2"></i>Log C
    hannels Tree</span>
358                 <span class="badge bg-secondary" id="slc-channel-c
    ount-badge">0</span>
359             </div>
360             <div class="p-2 border-bottom border-secondary">
361                 <input type="text" class="form-control form-contro
    l-sm bg-dark text-light border-secondary" id="slc-channel-filt
    er-input" placeholder="Filter channels...">
362             </div>
363             <div class="p-2 slc-tree-list" id="slc-channel-tree-
    container">
364                 <!-- Dynamic channels injected here -->
365             </div>
366         </div>
367     </div>

```



```

368 |
369 |     <!-- Events / Incidents Main Stream -->
370 |     <div class="col-lg-9" id="slc-main-stream-column">
371 |         <div class="slc-panel h-100">
372 |             <div class="slc-panel-header">
373 |                 <span id="slc-main-view-title"><i class="bi bi-lis
| t-columns-reverse me-2"></i>Live Windows Event Stream</span>
374 |                 <div class="d-flex align-items-center gap-2">
375 |                     <button class="btn btn-sm btn-info rounded-pill
| px-3 fw-bold text-dark" id="btn-slc-audit" title="Провести ауд
| ит текущего журнала логов">
376 |                         <i class="bi bi-shield-check me-1"></i> Аудит
377 |                     </button>
378 |                     <span class="badge bg-secondary" id="slc-stream-
| count-badge">0 entries</span>
379 |                 </div>
380 |             </div>
381 |
382 |     <!-- Table View Container -->
383 |     <div class="table-responsive" style="max-height: 540
| px; overflow-y: auto;" id="slc-table-container">
384 |         <table class="table slc-table">
385 |             <thead>
386 |                 <tr>
387 |                     <th style="width: 155px;">Timestamp</th>
388 |                     <th style="width: 90px;">Level</th>
389 |                     <th style="width: 65px;">ID</th>
390 |                     <th style="width: 160px;">Provider</th>
391 |                     <th style="width: 65px;">PID</th>
392 |                     <th>Message</th>
393 |                 </tr>
394 |             </thead>
395 |             <tbody id="slc-table-body">
396 |                 <tr>
397 |                     <td colspan="6" class="text-center text-mute
| d py-4">
398 |                         <div class="spinner-border spinner-border-
| sm text-info me-2" role="status"></div>

```

```

399 |             Loading system events...
400 |         </td>
401 |     </tr>
402 | </tbody>
403 | </table>
404 | </div>
405 |
406 |     <!-- Audit Digest View Container (Hidden by default)
407 | -->
408 |     <div class="p-3 d-none" id="slc-audit-container" style="max-height: 540px; overflow-y: auto;">
409 |         <div id="slc-audit-content"></div>
410 |     </div>
411 |
412 |     <!-- Dynamic Log RAG Search Container (Hidden by default) -->
413 |     <div class="p-3 d-none" id="slc-rag-container" style="max-height: 540px; overflow-y: auto;">
414 |         <div class="card bg-dark border-info mb-3 shadow">
415 |             <div class="card-header bg-black d-flex justify-content-between align-items-center">
416 |                 <span class="text-info fw-bold"><i class="bi bi-search me-2"></i>Семантический RAG-поиск по логам</span>
417 |                 <button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-slc-rag-rebuild">
418 |                     <i class="bi bi-arrow-repeat me-1"></i> Перестроить RAG-индекс
419 |                 </button>
420 |             </div>
421 |             <div class="card-body">
422 |                 <div class="input-group mb-2">
423 |                     <span class="input-group-text bg-black text-info border-secondary"><i class="bi bi-chat-left-dots"></i></span>

```

```
424 |         <button class="btn btn-info px-4 fw-semibold  
    | " id="btn-slc-rag-search">Найти</button>  
425 |     </div>  
426 |     <div class="small text-muted" id="slc-rag-stat  
    | us-text">Готов к поиску</div>  
427 |     </div>  
428 | </div>  
429 |     <div id="slc-rag-results"></div>  
430 | </div>  
431 |  
432 |     <!-- Timeline Container (Hidden by default) -->  
433 |     <div class="p-3 d-none" id="slc-timeline-container"  
    | style="max-height: 540px; overflow-y: auto;">  
434 |         <div id="slc-timeline-content"></div>  
435 |     </div>  
436 |  
437 |     <!-- App Activity Log Container (Hidden by default)  
    | -->  
438 |     <div class="p-3 d-none" id="slc-activity-container"  
    | style="max-height: 540px; overflow-y: auto;">  
439 |         <div class="d-flex justify-content-between align-i  
    | tems-center mb-2">  
440 |             <span class="fw-bold text-warning"><i class="bi  
    | bi-journal-text me-1"></i> Журнал активности веб-интерфейса</s  
    | pan>  
441 |             <button class="btn btn-sm btn-outline-warning ro  
    | unded-pill px-3" id="btn-scc-refresh-logs">  
442 |                 <i class="bi bi-arrow-clockwise me-1"></i> Обно  
    | овить  
443 |             </button>  
444 |         </div>  
445 |         <pre class="bg-black text-light p-3 rounded font-m  
    | onospace small" id="slc-activity-log-pre" style="max-height: 4  
    | 40px; overflow-y: auto;">Загрузка логов активности...</pre>  
446 |     </div>  
447 | </div>  
448 | </div>  
449 | </div>
```

```

450     </div>
451
452     <!-- Модальное окно деталей события Windows Event Log -->
453     <div class="modal fade" id="slc-event-detail-modal" tabindex
454     ="-1" aria-hidden="true">
455         <div class="modal-dialog modal-lg modal-dialog-centered mo
456         dal-dialog-scrollable">
457             <div class="modal-content border-secondary shadow-lg" st
458             yle="background: #0f172a; color: #e2e8f0;">
459                 <div class="modal-header py-2 px-3" style="border-bott
460                 om: 1px solid #334155;">
461                     <div class="d-flex align-items-center gap-2 flex-wra
462                     p">
463                         <span class="fs-5">❌ </span>
464                         <h5 class="modal-title fw-bold mb-0" style="color:
465                         #38bdf8;">Детали события</h5>
466                         <span id="slc-modal-level-badge"></span>
467                     </div>
468                     <button type="button" class="btn-close btn-close-whi
469                     te" data-bs-dismiss="modal" aria-label="Close"></button>
470                 </div>
471                 <div class="modal-body py-3 px-3">
472
473                     <!-- Метаданные события -->
474                     <div class="row g-2 mb-3 small">
475                         <div class="col-sm-6">
476                             <strong style="color:#94a3b8;">Временная метка:<
477                             /strong>
478                             <span id="slc-modal-time" class="fw-semibold tex
479                             t-white font-monospace ms-1"></span>
480                         </div>
481                         <div class="col-sm-6">
482                             <strong style="color:#94a3b8;">Уровень:</strong>
483                             <span id="slc-modal-level" class="ms-1"></span>
484                         </div>
485                         <div class="col-sm-6">
486                             <strong style="color:#94a3b8;">Event ID:</strong>
487                             <span id="slc-modal-event-id" class="ms-1"></span>
488                         </div>
489                     </div>
490                     <div class="row g-2 mb-3 small">
491                         <div class="col-sm-6">
492                             <strong style="color:#94a3b8;">Имя события:</strong>
493                             <span id="slc-modal-event-name" class="ms-1"></span>
494                         </div>
495                         <div class="col-sm-6">
496                             <strong style="color:#94a3b8;">Имя источника:</strong>
497                             <span id="slc-modal-event-source" class="ms-1"></span>
498                         </div>
499                     </div>
500                     <div class="row g-2 mb-3 small">
501                         <div class="col-sm-6">
502                             <strong style="color:#94a3b8;">Код события:</strong>
503                             <span id="slc-modal-event-code" class="ms-1"></span>
504                         </div>
505                         <div class="col-sm-6">
506                             <strong style="color:#94a3b8;">Код источника:</strong>
507                             <span id="slc-modal-event-source-code" class="ms-1"></span>
508                         </div>
509                     </div>
510                     <div class="row g-2 mb-3 small">
511                         <div class="col-sm-6">
512                             <strong style="color:#94a3b8;">Описание события:</strong>
513                             <span id="slc-modal-event-description" class="ms-1"></span>
514                         </div>
515                         <div class="col-sm-6">
516                             <strong style="color:#94a3b8;">Описание источника:</strong>
517                             <span id="slc-modal-event-source-description" class="ms-1"></span>
518                         </div>
519                     </div>
520                     <div class="row g-2 mb-3 small">
521                         <div class="col-sm-6">
522                             <strong style="color:#94a3b8;">Путь к файлу:</strong>
523                             <span id="slc-modal-event-file-path" class="ms-1"></span>
524                         </div>
525                         <div class="col-sm-6">
526                             <strong style="color:#94a3b8;">Путь к источнику:</strong>
527                             <span id="slc-modal-event-source-file-path" class="ms-1"></span>
528                         </div>
529                     </div>
530                     <div class="row g-2 mb-3 small">
531                         <div class="col-sm-6">
532                             <strong style="color:#94a3b8;">Ссылка на документацию:</strong>
533                             <span id="slc-modal-event-documentation-link" class="ms-1"></span>
534                         </div>
535                         <div class="col-sm-6">
536                             <strong style="color:#94a3b8;">Ссылка на документацию источника:</strong>
537                             <span id="slc-modal-event-source-documentation-link" class="ms-1"></span>
538                         </div>
539                     </div>
540                     <div class="row g-2 mb-3 small">
541                         <div class="col-sm-6">
542                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности:</strong>
543                             <span id="slc-modal-event-security-documentation-link" class="ms-1"></span>
544                         </div>
545                         <div class="col-sm-6">
546                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника:</strong>
547                             <span id="slc-modal-event-source-security-documentation-link" class="ms-1"></span>
548                         </div>
549                     </div>
550                     <div class="row g-2 mb-3 small">
551                         <div class="col-sm-6">
552                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности события:</strong>
553                             <span id="slc-modal-event-security-event-documentation-link" class="ms-1"></span>
554                         </div>
555                         <div class="col-sm-6">
556                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности события источника:</strong>
557                             <span id="slc-modal-event-security-event-source-documentation-link" class="ms-1"></span>
558                         </div>
559                     </div>
560                     <div class="row g-2 mb-3 small">
561                         <div class="col-sm-6">
562                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события:</strong>
563                             <span id="slc-modal-event-security-source-event-documentation-link" class="ms-1"></span>
564                         </div>
565                         <div class="col-sm-6">
566                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника:</strong>
567                             <span id="slc-modal-event-security-source-event-source-documentation-link" class="ms-1"></span>
568                         </div>
569                     </div>
570                     <div class="row g-2 mb-3 small">
571                         <div class="col-sm-6">
572                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события:</strong>
573                             <span id="slc-modal-event-security-source-event-source-event-documentation-link" class="ms-1"></span>
574                         </div>
575                         <div class="col-sm-6">
576                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника:</strong>
577                             <span id="slc-modal-event-security-source-event-source-event-source-documentation-link" class="ms-1"></span>
578                         </div>
579                     </div>
580                     <div class="row g-2 mb-3 small">
581                         <div class="col-sm-6">
582                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события:</strong>
583                             <span id="slc-modal-event-security-source-event-source-event-source-event-documentation-link" class="ms-1"></span>
584                         </div>
585                         <div class="col-sm-6">
586                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника:</strong>
587                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-documentation-link" class="ms-1"></span>
588                         </div>
589                     </div>
590                     <div class="row g-2 mb-3 small">
591                         <div class="col-sm-6">
592                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события:</strong>
593                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-documentation-link" class="ms-1"></span>
594                         </div>
595                         <div class="col-sm-6">
596                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события источника:</strong>
597                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-source-documentation-link" class="ms-1"></span>
598                         </div>
599                     </div>
600                     <div class="row g-2 mb-3 small">
601                         <div class="col-sm-6">
602                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события источника события:</strong>
603                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-source-event-documentation-link" class="ms-1"></span>
604                         </div>
605                         <div class="col-sm-6">
606                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события источника события источника:</strong>
607                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-source-event-source-documentation-link" class="ms-1"></span>
608                         </div>
609                     </div>
610                     <div class="row g-2 mb-3 small">
611                         <div class="col-sm-6">
612                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события источника события источника события:</strong>
613                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-source-event-source-event-documentation-link" class="ms-1"></span>
614                         </div>
615                         <div class="col-sm-6">
616                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события источника события источника события источника:</strong>
617                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-source-event-source-event-source-documentation-link" class="ms-1"></span>
618                         </div>
619                     </div>
620                     <div class="row g-2 mb-3 small">
621                         <div class="col-sm-6">
622                             <strong style="color:#94a3b8;">Ссылка на документацию по безопасности источника события источника события источника события источника события источника события источника события источника события:</strong>
623                             <span id="slc-modal-event-security-source-event-source-event-source-event-source-event-source-event-source-event-source-event-documentation-link" class="ms-1"></span>
624                         </div>
625                         <div class="col-sm-6">
626                             <strong style="color:#94a3b8;">Ссылка на документ
```

```
478         <span id="slc-modal-id" class="fw-semibold font-  
monospace text-info ms-1"></span>  
479     </div>  
480     <div class="col-sm-6">  
481         <strong style="color:#94a3b8;">Поставщик (Provid  
er):</strong>  
482         <span id="slc-modal-provider" class="text-info m  
s-1"></span>  
483     </div>  
484     <div class="col-sm-6">  
485         <strong style="color:#94a3b8;">Компьютер:</stron  
g>  
486         <span id="slc-modal-computer" class="text-light  
ms-1"></span>  
487     </div>  
488     <div class="col-sm-6">  
489         <strong style="color:#94a3b8;">PID процесса:</st  
rong>  
490         <span id="slc-modal-pid" class="font-monospace t  
ext-muted ms-1"></span>  
491     </div>  
492     <div class="col-sm-12">  
493         <strong style="color:#94a3b8;">Канал журнала:</s  
trong>  
494         <span id="slc-modal-channel" class="font-monospa  
ce small ms-1" style="color:#7dd3fc;"></span>  
495     </div>  
496 </div>  
497  
498 <!-- Текст сообщения события -->  
499     <div class="card mb-3" style="background:#000; borde  
r:1px solid #334155;">  
500         <div class="card-header py-1 px-3" style="backgrou  
nd:#1e293b; border-bottom:1px solid #334155;">  
501             <h6 class="small text-uppercase fw-bold mb-0" st  
yle="color:#94a3b8; letter-spacing:0.05em;">  
502                 <i class="bi bi-chat-square-text me-1"></i>Coo  
бщение события
```

```

503         </h6>
504     </div>
505     <div class="card-body p-3">
506         <div id="slc-modal-message"
507             class="font-monospace small text-light"
508             style="white-space: pre-wrap; word-break: b
reak-word; max-height: 220px; overflow-y: auto;"></div>
509         </div>
510     </div>
511
512     <!-- Секция AI-диагностики -->
513     <div class="card" style="background:#0f172a; border:
1px solid #38bdf8;">
514         <div class="card-header py-2 px-3 d-flex justify-c
ontent-between align-items-center" style="background:#1e293b;
border-bottom:1px solid #334155;">
515             <h6 class="fw-bold mb-0" style="color:#38bdf8;">
516                 <i class="bi bi-robot me-1"></i>AI Contextual
Explanation
517             </h6>
518             <button class="btn btn-sm btn-outline-info py-0
px-2 rounded-pill" id="btn-slc-modal-diagnose">
519                 <i class="bi bi-magic me-1"></i>Анализ контекс
та
520             </button>
521         </div>
522         <div class="card-body p-3">
523             <div class="small" id="slc-modal-ai-explanation"
style="color:#94a3b8;">
524                 Нажмите «Анализ контекста», чтобы провести диа
гностику выбранной записи и сопутствующих событий.
525             </div>
526         </div>
527     </div>
528
529 </div>
530 <div class="modal-footer py-2 px-3 d-flex justify-cont
ent-between" style="border-top:1px solid #334155;">

```

```
531 |         <button type="button" class="btn btn-sm btn-outline-  
    | secondary" id="btn-slc-modal-copy-msg">  
532 |             <i class="bi bi-clipboard me-1"></i>Копировать соо  
    | бщение  
533 |         </button>  
534 |         <button type="button" class="btn btn-sm btn-secondar  
    | y" data-bs-dismiss="modal">Закрыть</button>  
535 |     </div>  
536 | </div>  
537 | </div>  
538 | </div>  
539 | </div>  
540 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows System Control Center Web Controller
4  * =====
5  * Description:
6  *   Client-side JavaScript controller for the System Control
7  *   Center tab in
8  *   the administrative web interface. Includes monitoring, po
9  *   st-install wizard,
10  *   maintenance & recovery, and the unified Windows OS System
11  *   Log Center.
12  *
13  * File: main.js
14  * Project: ai-breadboard
15  * Package: src.api.webinterface.system_control_tab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * =====
19  */
20
21 (function () {
22   'use strict';
23
24   let isSCCInitialized = false;
25   let activeProfileSteps = [];
26
27   // System Log Center states
28   let currentChannel = 'System';
29   let currentFilePath = '';
30   let currentMode = 'live';
31   let cachedEvents = [];
32   let cachedChannels = [];
33   let selectedEvent = null;
```



```

31 | let liveIntervalTimer = null;
32 | let isLogRequestInProgress = false;
33 | let globalTooltipEl = null;
34 |
35 | /**
36 |  * Escape HTML entities to protect against XSS injection.
37 |  *
38 |  * @param {string} str - Raw input text.
39 |  * @returns {string} Sanitized string.
40 |  */
41 | function escapeHtml(str) {
42 |     if (!str) return '';
43 |     return String(str)
44 |         .replace(/&/g, '&amp;')
45 |         .replace(/</g, '&lt;')
46 |         .replace(/>/g, '&gt;')
47 |         .replace(/"/g, '&quot;')
48 |         .replace(/'/g, '&#039;');
49 | }
50 |
51 | /**
52 |  * Format ISO timestamps.
53 |  *
54 |  * @param {string} isoStr - ISO Timestamp string.
55 |  * @returns {string} Formatted time string.
56 |  */
57 | function formatTime(isoStr) {
58 |     if (!isoStr) return '-';
59 |     try {
60 |         return new Date(isoStr).toLocaleTimeString();
61 |     } catch {
62 |         return isoStr.slice(11, 19);
63 |     }
64 | }
65 |
66 | /**
67 |  * Map log severity to badge styling class.
68 |  *

```

```

69 | * @param {string} level - Severity level.
70 | * @returns {string} CSS class name.
71 | */
72 | function getSeverityClass(level) {
73 |     const lvl = (level || '').toLowerCase();
74 |     if (lvl.includes('crit')) return 'badge-crit';
75 |     if (lvl.includes('err')) return 'badge-err';
76 |     if (lvl.includes('warn')) return 'badge-warn';
77 |     if (lvl.includes('verb')) return 'badge-verb';
78 |     return 'badge-inf';
79 | }
80 |
81 | // =====
82 | // SECTION 1: System Control Status & Monitoring
83 | // =====
84 |
85 | async function fetchStatus() {
86 |     try {
87 |         const res = await fetch('/api/system-control/status');
88 |         if (!res.ok) return;
89 |         const data = await res.json();
90 |
91 |         // Elevation badge
92 |         const elevBadge = document.getElementById('scc-elevation
93 |         -badge');
94 |         if (elevBadge) {
95 |             if (data.is_elevated) {
96 |                 elevBadge.className = 'badge rounded-pill bg-success
97 |                 -subtle text-success border border-success px-3 py-2';
98 |                 elevBadge.innerHTML = '🔒 Mode: Administrator (Full
99 |                 Access)';
100 |             } else {
101 |                 elevBadge.className = 'badge rounded-pill bg-warning
102 |                 -subtle text-warning border border-warning px-3 py-2';
103 |                 elevBadge.innerHTML = '🔓 Mode: Standard User (Read-
104 |                 Only)';

```

```
100     }
101     }
102     } catch (e) {
103         console.error('[SystemControl] Failed to fetch status:',
104             e);
105     }
106
107     function formatRestorePointType(rawType) {
108         if (!rawType) return 'System Checkpoint';
109         const s = String(rawType).toUpperCase();
110         if (s.includes('APPLICATION_INSTALL') || s === '0') return
111             'App Install';
112         if (s.includes('APPLICATION_UNINSTALL') || s === '1') return 'App Uninstall';
113         if (s.includes('DEVICE_DRIVER') || s === '10') return 'Driver Install';
114         if (s.includes('MODIFY_SETTINGS') || s === '12') return 'Settings Change';
115         if (s.includes('CANCELLED') || s === '13') return 'Cancelled Op';
116         if (s.includes('WINDOWS_UPDATE') || s === '17') return 'Windows Update';
117         return s.replace(/_/g, ' ');
118     }
119
120     function formatRestoreTime(rawTime) {
121         if (!rawTime) return '-';
122         const s = String(rawTime).trim();
123         if (s.length >= 14 && /^d{14}$/.test(s)) {
124             return `${s.slice(0, 4)}-${s.slice(4, 6)}-${s.slice(6, 8)} ${s.slice(8, 10)}:${s.slice(10, 12)}:${s.slice(12, 14)}`;
125         }
126         return s;
127     }
128
129     async function fetchRestorePoints() {
130         try {
```

```

130     const res = await fetch('/api/system-control/restore-poi
nts');
131     if (!res.ok) return;
132     const data = await res.json();
133     const points = data.restore_points || [];
134
135     const tbody = document.getElementById('scc-restore-tbody
');
136     if (tbody) {
137         if (points.length === 0) {
138             tbody.innerHTML = '<tr><td colspan="4" class="text-c
enter text-muted p-3">No restore points found. Click "Create R
estore Point" to generate one.</td></tr>';
139             return;
140         }
141         tbody.innerHTML = points.map((p, idx) => {
142             const typeBadge = formatRestorePointType(p.restore_p
oint_type);
143             const timeFormatted = formatRestoreTime(p.creation_t
ime);
144             return `
145             <tr class="scc-restore-row" data-idx="${idx}" style=
"cursor: pointer;" title="Нажмите для анализа точки восстановл
ения">
146                 <td class="font-monospace text-info fw-bold">#${p.
sequence_number}</td>
147                 <td class="fw-semibold text-white">${p.description
}</td>
148                 <td><span class="badge bg-secondary text-light px-
2 py-1 font-monospace" style="font-size: 0.75rem;">${typeBadge
}</span></td>
149                 <td class="small text-light font-monospace">${time
Formatted}</td>
150             </tr>
151             `;
152         }).join('');
153
154     tbody.querySelectorAll('.scc-restore-row').forEach(row

```

```

=> {
155     row.onclick = () => {
156         const idx = parseInt(row.getAttribute('data-idx'),
157             10);
158         const p = points[idx];
159         if (!p) return;
160         if (window.AITableModal) {
161             window.AITableModal.show({
162                 icon: '🔍',
163                 title: `Точка восстановления #${p.sequence_num
164                     ber}`,
165                 subtitle: p.description,
166                 tableType: 'generic',
167                 badges: [
168                     { text: p.restore_point_type || 'System Chec
169 kpoint', class: 'badge bg-info text-dark' }
170                 ],
171                 metadata: [
172                     { label: 'Номер', value: String(p.sequence_n
173 umber) },
174                     { label: 'Описание', value: p.description },
175                     { label: 'Тип', value: p.restore_point_type
176                 },
177                     { label: 'Дата создания', value: p.creation_
178 time }
179                 ],
180                 rawTitle: 'Метаданные точки восстановления',
181                 rawContent: JSON.stringify(p, null, 2)
182             });
183         }
184     } catch (e) {
185         console.error('[SystemControl] Failed to fetch restore p
186 oints:', e);
187     }
188 }

```

```

185
186 // =====
187 // SECTION 2: Post-Install Wizard
188 // =====
189
190 async function loadWizardProfiles() {
191     try {
192         const res = await fetch('/api/system-control/profiles');
193         if (!res.ok) return;
194         const data = await res.json();
195         const profiles = data.profiles || [];
196         const sel = document.getElementById('scc-profile-select'
197     );
198         if (sel && profiles.length > 0) {
199             sel.innerHTML = profiles.map(p => `<option value="${p.
200 profile_id}">${p.name}</option>`).join('');
201             renderStepsForSelectedProfile(profiles[0]);
202
203             sel.onchange = () => {
204                 const selected = profiles.find(p => p.profile_id ===
205 sel.value);
206                 if (selected) renderStepsForSelectedProfile(selected
207 );
208             };
209         }
210     } catch (e) {
211         console.error('[SystemControl] Failed to load wizard pro
212 files:', e);
213     }
214 }
215
216 function renderStepsForSelectedProfile(profile) {
217     const container = document.getElementById('scc-wizard-step
218 s-container');
219     if (!container) return;

```

```

215     activeProfileSteps = profile.steps || [];
216
217     container.innerHTML = activeProfileSteps.map((s) => `
218         <div class="card bg-body-tertiary border-secondary p-2 d
219         -flex flex-row align-items-center justify-content-between" id=
220         "step-row-${s.id}">
221             <div class="d-flex align-items-center gap-3">
222                 <input class="form-check-input scc-step-checkbox" ty
223                 pe="checkbox" id="chk-step-${s.id}" data-id="${s.id}" ${s.enab
224                 led ? 'checked' : ''} style="cursor: pointer;">
225                 <div>
226                     <div class="fw-semibold text-white">${s.title}</di
227                 v>
228                     <div class="small text-muted">${s.description}</di
229                 v>
230                 </div>
231                 </div>
232                 <div class="d-flex align-items-center gap-2">
233                     ${s.requires_elevation ? '<span class="badge bg-seco
234                     ndary font-monospace" style="font-size: 0.7rem;">Admin</span>'
235                     : ''}
236                     <span class="badge ${s.status === 'SUCCESS' ? 'scc-b
237                     adge-secure' : (s.status === 'WARNING' ? 'scc-badge-warn' : (s
238                     .status === 'FAILED' ? 'scc-badge-danger' : 'bg-secondary'))}"
239                     id="step-status-${s.id}">
240                         ${s.status}
241                     </span>
242                 </div>
243             `).join('');
244     }
245
246     async function applyCurrentProfile() {
247         const sel = document.getElementById('scc-profile-select');
248         const profileId = sel ? sel.value : 'post_install';
249         const applyBtn = document.getElementById('btn-scc-apply-pr
250         ofile');

```

```

241     const selectedStepIds = [];
242     document.querySelectorAll('.scc-step-checkbox:checked').fo
rEach(chk => {
243         selectedStepIds.push(chk.getAttribute('data-id'));
244     });
245
246     if (applyBtn) {
247         applyBtn.disabled = true;
248         applyBtn.innerHTML = '<span class="spinner-border spinne
r-border-sm me-1"></span> Applying...';
249     }
250
251     try {
252         const res = await fetch('/api/system-control/profiles/ap
ply', {
253             method: 'POST',
254             headers: { 'Content-Type': 'application/json' },
255             body: JSON.stringify({
256                 profile_id: profileId,
257                 selected_step_ids: selectedStepIds
258             })
259         });
260         if (res.ok) {
261             const outcome = await res.json();
262             (outcome.steps || []).forEach(st => {
263                 const badge = document.getElementById(`step-status-$
{st.id}`);
264                 if (badge) {
265                     badge.innerText = st.status;
266                     badge.className = `badge ${st.status === 'SUCCESS'
? 'scc-badge-secure' : (st.status === 'WARNING' ? 'scc-badge-
warn' : 'scc-badge-danger')}`;
267                     badge.title = st.result_message;
268                 }
269             });
270             fetchStatus();
271             fetchAppActivityLogs();
272         }

```



```

273     } catch (e) {
274         console.error('[SystemControl] Failed to apply profile:'
275         , e);
276     } finally {
277         if (applyBtn) {
278             applyBtn.disabled = false;
279             applyBtn.innerHTML = '<i class="bi bi-play-fill me-1">
280             </i> Apply Profile';
281         }
282     }
283     // =====
284     // SECTION 3: System Control Restore Policy & Storage
285     // =====
286
287     async function loadRestorePolicyModal() {
288         try {
289             const res = await fetch('/api/system-control/restore-poi
290             nts/config');
291             if (!res.ok) return;
292             const data = await res.json();
293
294             const storageSel = document.getElementById('scc-policy-s
295             torage-size');
296             const triggerSel = document.getElementById('scc-policy-t
297             rigger');
298             const timeInput = document.getElementById('scc-policy-ti
299             me');
300             const timeWrapper = document.getElementById('scc-policy-
301             time-wrapper');
302             const freqChk = document.getElementById('scc-policy-freq
303             -limit');
304             const maxPtsSel = document.getElementById('scc-policy-ma
305             x-points');
306             const autoPruneChk = document.getElementById('scc-policy

```

```

    -auto-prune');
300     const storageBar = document.getElementById('scc-policy-s
storage-bar');
301     const storageLabel = document.getElementById('scc-policy
-storage-label');
302
303     if (storageSel && data.max_storage_size) storageSel.valu
e = data.max_storage_size;
304     if (triggerSel && data.schedule_trigger) {
305         triggerSel.value = data.schedule_trigger;
306         if (timeWrapper) timeWrapper.style.display = (data.sch
chedule_trigger === 'daily' || data.schedule_trigger === 'weekly
') ? 'block' : 'none';
307     }
308     if (timeInput && data.schedule_time) timeInput.value = d
ata.schedule_time;
309     if (freqChk) freqChk.checked = data.frequency_limit_minu
tes === 0;
310     if (maxPtsSel && data.max_points) maxPtsSel.value = Stri
ng(data.max_points);
311     if (autoPruneChk && data.auto_prune !== undefined) autoP
runeChk.checked = Boolean(data.auto_prune);
312
313     const st = data.storage_info || {};
314     if (storageBar && storageLabel) {
315         const pct = st.usage_percent || 0;
316         const usedGb = (st.used_bytes ? (st.used_bytes / (1024
** 3)).toFixed(2) : '0.00');
317         const maxGb = (st.max_bytes ? (st.max_bytes / (1024 **
3)).toFixed(2) : '0.00');
318         storageBar.style.width = `${Math.min(pct, 100)}%`;
319         storageBar.className = pct > 85 ? 'progress-bar bg-dan
ger' : (pct > 60 ? 'progress-bar bg-warning' : 'progress-bar b
g-info');
320         storageLabel.innerHTML = `<span>Used: ${usedGb} GB (${
pct}%)</span><span>Max: ${maxGb} GB</span>`;
321     }
322

```

```
323 |         if (triggerSel) {
324 |             triggerSel.onChange = () => {
325 |                 if (timeWrapper) timeWrapper.style.display = (triggerSel.value === 'daily' || triggerSel.value === 'weekly') ? 'block' : 'none';
326 |             };
327 |         }
328 |
329 |         const modalEl = document.getElementById('modal-scc-restore-config');
330 |         if (modalEl && window.bootstrap?.Modal) {
331 |             const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
332 |             modal.show();
333 |         }
334 |     } catch (e) {
335 |         console.error('[SystemControl] Failed to load restore policy config:', e);
336 |     }
337 | }
338 |
339 | async function saveRestorePolicy() {
340 |     const saveBtn = document.getElementById('btn-scc-save-restore-policy');
341 |     const storageSel = document.getElementById('scc-policy-storage-size');
342 |     const triggerSel = document.getElementById('scc-policy-trigger');
343 |     const timeInput = document.getElementById('scc-policy-time');
344 |     const freqChk = document.getElementById('scc-policy-freq-limit');
345 |     const maxPtsSel = document.getElementById('scc-policy-max-points');
346 |     const autoPruneChk = document.getElementById('scc-policy-auto-prune');
347 |
348 |     const payload = {
```

```
349     max_storage_size: storageSel ? storageSel.value : '10%',
350     schedule_trigger: triggerSel ? triggerSel.value : 'daily
    ',
351     schedule_time: timeInput ? timeInput.value : '03:00',
352     frequency_limit_minutes: freqChk && freqChk.checked ? 0
    : 1440,
353     max_points: maxPtsSel ? parseInt(maxPtsSel.value, 10) :
    5,
354     auto_prune: autoPruneChk ? autoPruneChk.checked : true,
355 };
356
357 if (saveBtn) {
358     saveBtn.disabled = true;
359     saveBtn.innerText = 'Сохранение...';
360 }
361
362 try {
363     const res = await fetch('/api/system-control/restore-poi
    nts/config', {
364         method: 'POST',
365         headers: { 'Content-Type': 'application/json' },
366         body: JSON.stringify(payload)
367     });
368     const d = await res.json();
369     window.showToast?.(d.message || 'Политика успешно сохран
    ена', d.success ? 'success' : 'danger') || alert(d.message ||
    'Политика сохранена');
370     const modalEl = document.getElementById('modal-scc-resto
    re-config');
371     if (modalEl && window.bootstrap?.Modal) {
372         const modal = bootstrap.Modal.getInstance(modalEl);
373         modal?.hide();
374     }
375     fetchRestorePoints();
376     fetchAppActivityLogs();
377 } catch (e) {
378     console.error('[SystemControl] Failed to save restore po
    licy:', e);
```

```
379 |         alert('Ошибка сохранения политики точек восстановления:
    | ' + e);
380 |     } finally {
381 |         if (saveBtn) {
382 |             saveBtn.disabled = false;
383 |             saveBtn.innerHTML = '<i class="bi bi-check-lg me-1"></
    | i> Сохранить и применить';
384 |         }
385 |     }
386 | }
387 |
388 | async function pruneRestorePointsNow() {
389 |     const maxPtsSel = document.getElementById('scc-policy-max-
    | points');
390 |     const keep = maxPtsSel ? parseInt(maxPtsSel.value, 10) : 5
    | ;
391 |     if (!confirm(`Выполнить принудительную ротацию точек восст
    | ановления и оставить только ${keep} последних?`)) {
392 |         return;
393 |     }
394 |     try {
395 |         const res = await fetch('/api/system-control/restore-poi
    | nts/prune', {
396 |             method: 'POST',
397 |             headers: { 'Content-Type': 'application/json' },
398 |             body: JSON.stringify({ keep_count: keep })
399 |         });
400 |         const d = await res.json();
401 |         window.showToast?.(d.message || 'Ротация выполнена', 'in
    | fo') || alert(d.message || 'Ротация выполнена');
402 |         fetchRestorePoints();
403 |         fetchAppActivityLogs();
404 |     } catch (e) {
405 |         console.error('[SystemControl] Failed to prune restore p
    | oints:', e);
406 |     }
407 | }
408 |
```

```

409 // =====
410 // SECTION 4: Application Activity Logs (SCC operations)
411 // =====
412
413 async function fetchAppActivityLogs() {
414   try {
415     const res = await fetch('/api/system-control/logs');
416     if (!res.ok) return;
417     const data = await res.json();
418     const logs = data.logs || [];
419
420     const tbody = document.getElementById('scc-logs-tbody');
421     if (tbody) {
422       if (logs.length === 0) {
423         tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted p-3">No activity logs recorded.</td></tr>';
424         return;
425       }
426       tbody.innerHTML = logs.slice(0, 50).map(l => `
427         <tr>
428           <td class="small font-monospace text-muted">${formatTime(l.timestamp)}</td>
429           <td class="fw-bold font-monospace text-info">${escapeHtml(l.action)}</td>
430           <td class="small text-white">${escapeHtml(l.target)}</td>
431           <td><span class="badge ${l.status === 'SUCCESS' ? 'scc-badge-secure' : (l.status === 'WARNING' ? 'scc-badge-warn' : 'scc-badge-danger')}>${escapeHtml(l.status)}</span></td>
432           <td class="small text-muted text-truncate" style="max-width: 250px;">${escapeHtml(l.details)}</td>
433         </tr>
434       `).join('');
435     }
436   } catch (e) {
437     console.error('[SystemControl] Failed to fetch activity

```

```

    logs:', e);
438     }
439   }
440
441   // =====
   =====
442   // SECTION 5: Full Windows OS & System Log Center (Channels,
   Discovery, RAG)
443   // =====
   =====
444
445   function getOrCreateTooltipEl() {
446     if (!globalTooltipEl) {
447       globalTooltipEl = document.createElement('div');
448       globalTooltipEl.className = 'slc-tooltip-popup';
449       document.body.appendChild(globalTooltipEl);
450     }
451     return globalTooltipEl;
452   }
453
454   function showChannelTooltip(e, channel) {
455     const tip = getOrCreateTooltipEl();
456     const countFormatted = (channel.record_count || 0).toLocale
eString();
457     const descText = channel.description || 'Служебный журнал
операционной системы Windows.';
458     const typeLabel = channel.source_type === 'channel' ? 'Win
dows Event Channel' : 'Log File';
459
460     tip.innerHTML = `
461       <div class="slc-tooltip-title">
462         <i class="bi bi-info-circle text-info"></i>
463         <span>${escapeHtml(channel.display_name)}</span>
464       </div>
465       <div style="color: #cbd5e1; margin-bottom: 0.35rem;">${e
scapeHtml(descText)}</div>
466       <div class="slc-tooltip-meta">
467         <span><strong class="text-info">${countFormatted}</str

```

```

ong> записей</span>
468     <span class="text-muted font-monospace">${typeLabel}</
span>
469     </div>
470     `;
471
472     tip.classList.add('show');
473     positionTooltip(e, tip);
474 }
475
476 function hideChannelTooltip() {
477     if (globalTooltipEl) {
478         globalTooltipEl.classList.remove('show');
479     }
480 }
481
482 function positionTooltip(e, tip) {
483     const margin = 12;
484     const tipRect = tip.getBoundingClientRect();
485
486     // По умолчанию – НАД курсором
487     let x = e.clientX + margin;
488     let y = e.clientY - tipRect.height - margin;
489
490     // Если не хватает места сверху – показываем снизу
491     if (y < 10) {
492         y = e.clientY + margin;
493     }
494
495     // Если выходит за правый край – сдвигаем влево
496     if (x + tipRect.width > window.innerWidth) {
497         x = e.clientX - tipRect.width - margin;
498     }
499
500     tip.style.left = `${Math.max(10, x)}px`;
501     tip.style.top = `${Math.max(10, y)}px`;
502 }
503

```



```

504 |   async function performSystemScan(force = false) {
505 |       try {
506 |           const response = await fetch(`/api/v1/system_logs/scan?f
orce=${force}`);
507 |           if (!response.ok) return;
508 |           const data = await response.json();
509 |
510 |           cachedChannels = data.channels || [];
511 |           const activeChannels = cachedChannels.filter((c) => (c.r
ecord_count || 0) > 0);
512 |           const sourcesBadge = document.getElementById('slc-stat-s
ources');
513 |           const countBadge = document.getElementById('slc-channel-
count-badge');
514 |
515 |           if (sourcesBadge) {
516 |               sourcesBadge.textContent = `${activeChannels.length.to
LocaleString()} active / ${data.total_sources.toLocaleString()}
sources`;
517 |           }
518 |           if (countBadge) {
519 |               countBadge.textContent = String(activeChannels.length)
;
520 |           }
521 |
522 |           renderChannelsTree();
523 |       } catch (err) {
524 |           console.warn('[SystemControl] Failed to perform system l
og scan:', err);
525 |       }
526 |   }
527 |
528 |   function renderChannelsTree() {
529 |       const container = document.getElementById('slc-channel-tre
e-container');
530 |       const filterInput = document.getElementById('slc-channel-f
ilter-input');
531 |       const filterText = filterInput ? filterInput.value.trim().

```

```

    toLowerCase() : '';
532
533     if (!container) return;
534     container.innerHTML = '';
535
536     const filtered = cachedChannels.filter((c) => {
537         if (c.record_count === 0) return false;
538         if (!filterText) return true;
539         return (
540             c.channel_name.toLowerCase().includes(filterText) ||
541             c.display_name.toLowerCase().includes(filterText) ||
542             (c.description && c.description.toLowerCase().includes
543 (filterText)) ||
544             (c.category && c.category.toLowerCase().includes(filte
545 rText))
546         );
547     });
548
549     const grouped = {};
550     filtered.forEach((c) => {
551         const cat = c.category || 'Windows Event Log';
552         if (!grouped[cat]) grouped[cat] = [];
553         grouped[cat].push(c);
554     });
555
556     const categoryIcons = {
557         'Windows Event Log': '📄 ',
558         'Windows OS Logs': '📄 ',
559         'Application Logs': '📄 ',
560         'AI-Breadboard Logs': '📄 ',
561     };
562
563     Object.keys(grouped).forEach((catName) => {
564         const catHeader = document.createElement('div');
565         catHeader.className = 'd-flex justify-content-between align-items-center px-2 py-1 mt-2 mb-1 rounded bg-dark border border-secondary text-info small fw-bold';
566         const icon = categoryIcons[catName] || '📄 ';

```

```

565 |         catHeader.innerHTML = `
566 |             <span>${icon} ${escapeHtml(catName)}</span>
567 |             <span class="badge bg-secondary font-monospace">${grou
ped[catName].length}</span>
568 |         `;
569 |         container.appendChild(catHeader);
570 |
571 |         grouped[catName].slice(0, 400).forEach((c) => {
572 |             const item = document.createElement('div');
573 |             const isActive = (c.channel_name === currentChannel &&
!currentFilePath) || (currentFilePath && currentFilePath ===
c.location);
574 |             item.className = `slc-tree-item ${isActive ? 'active'
: ''}`;
575 |
576 |             let subBadge = '0';
577 |             let badgeStyle = 'bg-dark text-secondary';
578 |             if (c.record_count > 0) {
579 |                 subBadge = c.record_count >= 1000 ? `${(c.record_cou
nt / 1000).toFixed(1)}k` : String(c.record_count);
580 |                 badgeStyle = c.record_count > 1000 ? 'bg-primary tex
t-white' : 'bg-dark text-info border border-secondary';
581 |             } else if (c.source_type !== 'channel') {
582 |                 subBadge = 'FILE';
583 |             }
584 |
585 |             item.innerHTML = `
586 |                 <span class="text-truncate me-2" style="max-width: 1
75px;">${escapeHtml(c.display_name)}</span>
587 |                 <span class="badge ${badgeStyle} font-monospace" sty
le="font-size: 0.72rem; min-width: 32px; text-align: right;">${
subBadge}</span>
588 |             `;
589 |
590 |             item.onmouseenter = (e) => showChannelTooltip(e, c);
591 |             item.onmousemove = (e) => {
592 |                 const tip = getOrCreateTooltipEl();
593 |                 if (tip.classList.contains('show')) positionTooltip(

```

```

    e, tip);
594         };
595         item.onmouseleave = () => hideChannelTooltip();
596
597         item.onclick = () => {
598             hideChannelTooltip();
599             if (c.source_type === 'channel') {
600                 currentChannel = c.channel_name;
601                 currentFilePath = '';
602             } else {
603                 currentChannel = c.display_name;
604                 currentFilePath = c.location;
605             }
606             const statChan = document.getElementById('slc-stat-c
hannel');
607             if (statChan) statChan.textContent = c.display_name;
608             renderChannelsTree();
609             loadEvents();
610         };
611         container.appendChild(item);
612     });
613 });
614 }
615
616 async function loadEvents() {
617     if (isLogRequestInProgress) return;
618     isLogRequestInProgress = true;
619
620     const levelSelect = document.getElementById('slc-level-sel
ect');
621     const hoursSelect = document.getElementById('slc-hours-sel
ect');
622     const searchInput = document.getElementById('slc-search-in
put');
623     const eventIdInput = document.getElementById('slc-eventid-
input');
624     const limitSelect = document.getElementById('slc-limit-sel
ect');

```

```

625     const tableBody = document.getElementById('slc-table-body'
    );
626     const eventsBadge = document.getElementById('slc-stat-even
    ts');
627     const faultsBadge = document.getElementById('slc-stat-inci
    dents');
628     const streamBadge = document.getElementById('slc-stream-co
    unt-badge');
629
630     const level = levelSelect ? levelSelect.value : '';
631     const hours = hoursSelect ? hoursSelect.value : '24';
632     const search = searchInput ? searchInput.value.trim() : ''
    ;
633     const eventId = eventIdInput && eventIdInput.value.trim()
    ? eventIdInput.value.trim() : '0';
634     const limit = limitSelect ? limitSelect.value : '100';
635
636     // Показываем спиннер в таблице на время загрузки
637     if (tableBody) {
638         tableBody.innerHTML = `
639             <tr>
640                 <td colspan="6" class="text-center py-4 text-muted">
641                     <div class="spinner-border spinner-border-sm text-
    info me-2" role="status"></div>
642                     Загрузка событий журнала <span class="text-warning
    font-monospace">${escapeHtml(currentChannel)}</span>...
643                 </td>
644             </tr>
645         `;
646     }
647
648     try {
649         const url = `/api/v1/system_logs/events?channel=${encodeURIComponent(currentChannel)}&limit=${limit}&level=${encodeURIComponent(level)}&search=${encodeURIComponent(search)}&hours=${hours}&event_id=${eventId}&file_path=${encodeURIComponent(currentFilePath)}`;
650

```

```

651     const res = await fetch(url);
652     if (!res.ok) {
653         if (tableBody) tableBody.innerHTML = `<tr><td colspan=
"6" class="text-center text-danger py-3">Error reading channel
(HTTP ${res.status})</td></tr>`;
654         isLogRequestInProgress = false;
655         return;
656     }
657
658     const data = await res.json();
659     cachedEvents = data.entries || [];
660
661     if (eventsBadge) eventsBadge.textContent = String(cached
Events.length);
662     if (streamBadge) streamBadge.textContent = `${cachedEven
ts.length} records`;
663
664     let errCount = 0;
665     let critCount = 0;
666     cachedEvents.forEach((e) => {
667         const lvl = (e.level || '').toLowerCase();
668         if (lvl.includes('crit')) critCount++;
669         if (lvl.includes('err')) errCount++;
670     });
671
672     if (faultsBadge) {
673         faultsBadge.textContent = `${errCount + critCount} fau
lts`;
674     }
675
676     renderCurrentModeView();
677 } catch (err) {
678     console.warn('[SystemControl] Failed to load events:', e
rr);
679     if (tableBody) tableBody.innerHTML = `<tr><td colspan="6
" class="text-center text-danger py-3">Network error loading l
ogs.</td></tr>`;
680     } finally {

```

```
681     isLogRequestInProgress = false;
682   }
683 }
684
685 function renderCurrentModeView() {
686   const tableContainer = document.getElementById('slc-table-
container');
687   const incidentsContainer = document.getElementById('slc-in
cidents-container');
688   const timelineContainer = document.getElementById('slc-tim
eline-container');
689   const rawContainer = document.getElementById('slc-raw-cont
ainer');
690   const auditContainer = document.getElementById('slc-audit-
container');
691   const ragContainer = document.getElementById('slc-rag-cont
ainer');
692   const activityContainer = document.getElementById('slc-act
ivity-container');
693   const tableBody = document.getElementById('slc-table-body'
);
694   const viewTitle = document.getElementById('slc-main-view-t
itle');
695
696   // Reset visibility
697   if (tableContainer) tableContainer.classList.add('d-none')
;
698   if (incidentsContainer) incidentsContainer.classList.add('
d-none');
699   if (timelineContainer) timelineContainer.classList.add('d-
none');
700   if (rawContainer) rawContainer.classList.add('d-none');
701   if (auditContainer) auditContainer.classList.add('d-none')
;
702   if (ragContainer) ragContainer.classList.add('d-none');
703   if (activityContainer) activityContainer.classList.add('d-
none');
704
```

```
705     if (currentMode === 'activity') {
706         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-journal-text text-warning me-2"></i>Application Activity & Operation Logs`;
707         if (activityContainer) {
708             activityContainer.classList.remove('d-none');
709             fetchAppActivityLogs();
710         }
711     } else if (currentMode === 'audit') {
712         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-shield-check text-info me-2"></i>Интеллектуальный Аудит Логирования (Zero-LLM Overhead)`;
713         if (auditContainer) {
714             auditContainer.classList.remove('d-none');
715             loadAuditView();
716         }
717     } else if (currentMode === 'rag') {
718         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-brain text-info me-2"></i>Динамический RAG-поиск по логам`;
719         if (ragContainer) {
720             ragContainer.classList.remove('d-none');
721         }
722     } else if (currentMode === 'incidents') {
723         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-diagram-3 me-2"></i>Correlated Incident Cascades`;
724         if (incidentsContainer) {
725             incidentsContainer.classList.remove('d-none');
726             loadIncidentsView();
727         }
728     } else if (currentMode === 'timeline') {
729         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-bar-chart me-2"></i>Event Frequency Histogram`;
730         if (timelineContainer) {
731             timelineContainer.classList.remove('d-none');
732             loadTimelineView();
733         }
734     } else if (currentMode === 'raw') {
735         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-code" style="color: #007bff; font-weight: bold; font-size: 1.2em;">`;
```



```

de-square me-2"></i>Raw Event Payload Data`;
736     if (rawContainer) {
737         rawContainer.classList.remove('d-none');
738         const rawText = document.getElementById('slc-raw-text'
);
739         if (rawText) {
740             rawText.textContent = selectedEvent ? (selectedEvent
.raw_data || JSON.stringify(selectedEvent, null, 2)) : JSON.st
ringify(cachedEvents.slice(0, 10), null, 2);
741         }
742     }
743 } else {
744     // Live or Errors Table Mode
745     if (tableContainer) tableContainer.classList.remove('d-n
one');
746     if (viewTitle) {
747         viewTitle.innerHTML = currentMode === 'errors'
748             ? `<i class="bi bi-exclamation-octagon text-danger m
e-2"></i>Critical & Error Events (${escapeHtml(currentChannel)
})`
749             : `<i class="bi bi-list-columns-reverse me-2"></i>Li
ve Windows Event Stream (${escapeHtml(currentChannel)})`;
750     }
751
752     if (!tableBody) return;
753
754     let entriesToRender = cachedEvents;
755     if (currentMode === 'errors') {
756         entriesToRender = cachedEvents.filter((e) => {
757             const l = (e.level || '').toLowerCase();
758             return l.includes('err') || l.includes('crit') || l.
includes('warn');
759         });
760     }
761
762     if (entriesToRender.length === 0) {
763         tableBody.innerHTML = `<tr><td colspan="6" class="text
-center text-muted py-4">No events found matching active filte

```

```

r.</td></tr>`;
764     return;
765 }
766
767     tableBody.innerHTML = entriesToRender.map((e, idx) => {
768         const badgeClass = getSeverityClass(e.level);
769         const eidStr = e.event_id > 0 ? String(e.event_id) : '
-';
770         const pidStr = e.process_id > 0 ? String(e.process_id)
: '-';
771         const isSelected = selectedEvent && (
772             selectedEvent.timestamp === e.timestamp &&
773             selectedEvent.event_id === e.event_id &&
774             (selectedEvent.provider || selectedEvent.source) ===
(e.provider || e.source)
775         );
776         return `
777             <tr data-index="${idx}" class="${isSelected ? 'selec
ted table-active' : ''}">
778                 <td class="font-monospace text-secondary text-nowr
ap">${escapeHtml(e.timestamp)}</td>
779                 <td><span class="badge ${badgeClass} text-uppercas
e px-2 py-1">${escapeHtml(e.level)}</span></td>
780                 <td class="text-center font-monospace">${escapeHtm
l(eidStr)}</td>
781                 <td class="text-truncate text-info" style="max-wid
th: 160px;" title="${escapeHtml(e.provider || e.source)}">${es
capeHtml(e.provider || e.source)}</td>
782                 <td class="text-center font-monospace text-muted">
${escapeHtml(pidStr)}</td>
783                 <td class="font-monospace text-light" style="word-
break: break-word;">${escapeHtml(e.message)}</td>
784             </tr>
785         `;
786     }).join('');
787
788     tableBody.querySelectorAll('tr').forEach((row) => {
789         row.onclick = () => {

```

```

790         const idx = parseInt(row.getAttribute('data-index')
|| '0', 10);
791         selectedEvent = entriesToRender[idx];
792         tableBody.querySelectorAll('tr').forEach(r => r.classList.remove('selected', 'table-active'));
793         row.classList.add('selected', 'table-active');
794         showEventModal(selectedEvent);
795     };
796 });
797 }
798 }
799
800 async function loadIncidentsView() {
801     const listEl = document.getElementById('slc-incidents-list');
802     if (!listEl) return;
803     listEl.innerHTML = `
```

```

ent-between align-items-center">
819         <span class="fw-bold text-info"><i class="bi bi-li
ghtning-charge me-1"></i>${escapeHtml(inc.title)}</span>
820         <span class="badge ${getSeverityClass(inc.severity
)}">${escapeHtml(inc.severity)}</span>
821     </div>
822     <div class="card-body">
823         <div class="row g-2 mb-2 small text-muted font-mon
ospace">
824             <div class="col-sm-4">Window: ${escapeHtml(inc.s
tart_time)} (${inc.duration_seconds}s)</div>
825             <div class="col-sm-4">Primary: ${escapeHtml(inc.
primary_source)}</div>
826             <div class="col-sm-4">Events Count: ${inc.events
_count}</div>
827         </div>
828         <p class="small text-light mb-2"><strong class="te
xt-warning">Root Cause:</strong> ${escapeHtml(inc.root_cause_h
int)}</p>
829         <p class="small text-muted mb-0">${escapeHtml(inc.
summary)}</p>
830     </div>
831 </div>
832 `).join('');
833 } catch (err) {
834     listEl.innerHTML = `<div class="alert alert-danger">Erro
r retrieving incidents: ${err.message}</div>`;
835 }
836 }
837
838 async function loadTimelineView() {
839     const listEl = document.getElementById('slc-timeline-list'
);
840     if (!listEl) return;
841
842     try {
843         const res = await fetch('/api/v1/system_logs/timeline');
844         if (!res.ok) return;

```

```

845 |     const data = await res.json();
846 |     const histogram = data.histogram || [];
847 |
848 |     if (histogram.length === 0) {
849 |         listEl.innerHTML = `
```

```

873     const hoursSelect = document.getElementById('slc-hours-select');
874     const limitSelect = document.getElementById('slc-limit-select');
875     const hours = hoursSelect ? hoursSelect.value : '24';
876     const limit = limitSelect ? limitSelect.value : '500';
877
878     contentEl.innerHTML = `
879         <div class="text-center py-5 text-muted">
880             <div class="spinner-border text-info mb-3" role="status"></div>
881             <div class="fw-bold text-light">Выполняется многоуровневый аудит массива логов...</div>
882             <div class="small text-secondary mt-1">Фильтрация шума
883             • Шаблонизация • Детекция аномалий • Сжатие данных</div>
884         `;
885
886     try {
887         const url = `/api/v1/system_logs/audit?channel=${encodeURIComponent(currentChannel)}&limit=${limit}&hours=${hours}&file_path=${encodeURIComponent(currentFilePath)}`;
888         const res = await fetch(url);
889         if (!res.ok) throw new Error(`HTTP ${res.status}`);
890         const data = await res.json();
891
892         const anomaliesHtml = (data.anomalies || []).map((a) =>
893         {
894             const badgeClass = getSeverityClass(a.level);
895             return `
896                 <div class="p-2 mb-2 rounded bg-black border border-secondary d-flex justify-content-between align-items-center flex-wrap gap-2">
897                     <div>
898                         <span class="badge ${badgeClass} text-uppercase me-2">${escapeHtml(a.level)}</span>
899                         <strong class="text-info">${escapeHtml(a.provider)}</strong>

```

```

899 |         <span class="text-muted font-monospace small">(I
    | D ${a.event_id})</span>
900 |         <div class="small text-light mt-1 font-monospace
    | " style="word-break: break-word;">${escapeHtml(a.sample)}</div
    | >
901 |         <div class="small text-secondary mt-1"><i class=
    | "bi bi-clock me-1"></i>Окно: ${escapeHtml(a.time_window)}</div
    | >
902 |     </div>
903 |     <div class="text-end">
904 |         <span class="badge bg-danger rounded-pill px-3 p
    | y-2 fs-6">${a.count}x</span>
905 |     </div>
906 | </div>
907 | `;
908 | }).join('') || `<div class="text-success small p-2"><i c
    | lass="bi bi-check-circle me-1"></i>Аномалий и критических сбое
    | в не обнаружено.</div>`;
909 |
910 |     const clustersHtml = (data.top_clusters || []).map((cl,
    | idx) => {
911 |         const badgeClass = getSeverityClass(cl.level);
912 |         return `
913 |             <tr class="small">
914 |                 <td class="text-center font-monospace">${idx + 1}<
    | /td>
915 |                 <td><span class="badge ${badgeClass}">${escapeHtml
    | (cl.level)}</span></td>
916 |                 <td class="text-info text-truncate" style="max-wid
    | th: 140px;" title="${escapeHtml(cl.provider)}">${escapeHtml(cl
    | .provider)}</td>
917 |                 <td class="text-center font-monospace">${cl.event_
    | id || '-'}</td>
918 |                 <td class="font-monospace text-light" style="max-w
    | idth: 320px; word-break: break-word;" title="${escapeHtml(cl.t
    | emplate)}">${escapeHtml(cl.template)}</td>
919 |                 <td class="text-center"><span class="badge bg-seco
    | ndary font-monospace">${cl.count}</span></td>

```

```

920         <td class="small text-secondary font-monospace tex
t-nowrap">${escapeHtml(cl.first_seen)}</td>
921     </tr>
922     `;
923     })).join('');
924
925     const totalScanned = data.total_analyzed ?? data.total_s
canned ?? 0;
926     const uniqueCount = data.unique_patterns_count ?? (data.
top_clusters ? data.top_clusters.length : 0);
927     const redundancyPct = data.redundancy_pct !== undefined
? `${data.redundancy_pct.toFixed(1)}%` : (data.compression_rat
io || '0%');
928     const errSum = (data.error_count || 0) + (data.critical_
count || 0);
929     const warnSum = data.warning_count || 0;
930     const summaryText = data.strategy_rationale || data.exec
utive_summary || `Стратегия обработки: ${data.strategy || 'Ста
ндартная'}. Найдено ${uniqueCount} ключевых шаблонов.`;
931
932     contentEl.innerHTML = `
933         <div class="card bg-dark border-info mb-2 shadow-sm">
934             <div class="card-header bg-black text-info fw-bold p
y-1 px-3 d-flex justify-content-between align-items-center sma
ll">
935                 <span><i class="bi bi-cpu me-1"></i>Сводка аудита:
${escapeHtml(data.channel)}</span>
936                 <span class="badge bg-info text-dark font-monospac
e">Дублирование: ${redundancyPct}</span>
937             </div>
938             <div class="card-body p-2">
939                 <div class="row g-2 text-center mb-2">
940                     <div class="col-sm-3 col-6">
941                         <div class="p-1 rounded bg-black border border
-secondary">
942                             <div class="small text-muted" style="font-si
ze: 0.72rem;">Просканировано</div>
943                             <div class="fs-5 fw-bold text-light font-mon

```



```
ospace">${totalScanned}</div>
944         </div>
945     </div>
946     <div class="col-sm-3 col-6">
947         <div class="p-1 rounded bg-black border border
-secondary">
948             <div class="small text-muted" style="font-si
ze: 0.72rem;">Шаблонов</div>
949             <div class="fs-5 fw-bold text-info font-mono
space">${uniqueCount}</div>
950         </div>
951     </div>
952     <div class="col-sm-3 col-6">
953         <div class="p-1 rounded bg-black border border
-secondary">
954             <div class="small text-muted" style="font-si
ze: 0.72rem;">Ошибок / Сбоев</div>
955             <div class="fs-5 fw-bold text-danger font-mo
nospace">${errSum}</div>
956         </div>
957     </div>
958     <div class="col-sm-3 col-6">
959         <div class="p-1 rounded bg-black border border
-secondary">
960             <div class="small text-muted" style="font-si
ze: 0.72rem;">Предупреждений</div>
961             <div class="fs-5 fw-bold text-warning font-m
onospace">${warnSum}</div>
962         </div>
963     </div>
964 </div>
965     <p class="mb-0 text-light small"><i class="bi bi-i
nfo-circle text-info me-1"></i>${escapeHtml(summaryText)}</p>
966 </div>
967 </div>
968
969     <div class="card bg-dark border-secondary mb-2 shadow-
sm">
```

```

970 |         <div class="card-header bg-black text-warning fw-bol
    | d py-1 px-3 small">
971 |             <i class="bi bi-exclamation-triangle me-1"></i>Обн
    | аруженные аномалии и всплески
972 |         </div>
973 |         <div class="card-body p-2">
974 |             ${anomaliesHtml}
975 |         </div>
976 |     </div>
977 |
978 |     <div class="card bg-dark border-secondary">
979 |         <div class="card-header bg-black text-light fw-bold
    | d-flex justify-content-between align-items-center">
980 |             <span><i class="bi bi-collection me-2"></i>Топ шаб
    | лонов событий (Дедупликация)</span>
981 |             <span class="small text-muted">Топ 10 кластеров</s
    | pan>
982 |         </div>
983 |         <div class="table-responsive">
984 |             <table class="table slc-table mb-0">
985 |                 <thead>
986 |                     <tr>
987 |                         <th style="width: 40px;">#</th>
988 |                         <th style="width: 80px;">Уровень</th>
989 |                         <th style="width: 140px;">Поставщик</th>
990 |                         <th style="width: 60px;">ID</th>
991 |                         <th>Шаблон сообщения (Masked)</th>
992 |                         <th style="width: 70px;" class="text-center"
    | >Кол-во</th>
993 |                         <th style="width: 140px;">Впервые</th>
994 |                     </tr>
995 |                 </thead>
996 |                 <tbody>
997 |                     ${clustersHtml}
998 |                 </tbody>
999 |             </table>
1000 |         </div>
1001 |     </div>

```

```

1002     `;
1003   } catch (err) {
1004     contentEl.innerHTML = `
```

```

1032         </div>
1033     `;
1034     return;
1035 }
1036
1037     resultsEl.innerHTML = results.map((r) => {
1038         const badgeClass = getSeverityClass(r.meta?.level || '
1039 info');
1039         return `
1040         <div class="card bg-dark border-secondary mb-3 shado
1041 w-sm">
1041             <div class="card-header bg-black d-flex justify-co
1042 ntent-between align-items-center">
1042                 <span class="fw-bold text-info"><i class="bi bi-
1043 file-earmark-text me-2"></i>${escapeHtml(r.title)}</span>
1043                 <div class="d-flex align-items-center gap-2">
1044                     <span class="badge ${badgeClass}">${escapeHtml
1045 (r.meta?.level || r.type)}</span>
1045                     <span class="badge bg-secondary font-monospace
1046 ">Score: ${r.relevance_score}</span>
1046                 </div>
1047             </div>
1048             <div class="card-body">
1049                 <pre class="bg-black text-light p-3 rounded font
1050 -monospace small mb-0" style="white-space: pre-wrap; word-brea
1051 k: break-word;">${escapeHtml(r.text)}</pre>
1052             </div>
1053         </div>
1054     `;
1055     }).join('');
1056 } catch (err) {
1057     resultsEl.innerHTML = `<div class="alert alert-danger">0
1058 ошибка поиска RAG: ${escapeHtml(err.message)}</div>`;
1059 }
1060
1061     async function rebuildRAGIndex() {
1062         const resultsEl = document.getElementById('slc-rag-results

```

```

');
1061     if (!resultsEl) return;
1062     resultsEl.innerHTML = `
```

```
1088     const modalEl = document.getElementById('slc-event-detail-
modal');
1089     if (!modalEl) return;
1090
1091     const setElTxt = (id, val) => {
1092         const el = document.getElementById(id);
1093         if (el) el.textContent = val || '';
1094     };
1095
1096     setElTxt('slc-modal-time', entry.timestamp);
1097     // Бейдж уровня в заголовке модала
1098     const lvlBadgeEl = document.getElementById('slc-modal-leve
l-badge');
1099     if (lvlBadgeEl) lvlBadgeEl.innerHTML = `
```

```

1113     }
1114
1115     const diagnoseBtn = document.getElementById('btn-slc-modal
-diagnose');
1116     if (diagnoseBtn) {
1117         diagnoseBtn.onclick = async () => {
1118             aiExplanation.innerHTML = `<div class="spinner-border
spinner-border-sm text-info me-2"></div>Анализ выбранной запис
и...`;
1119             try {
1120                 const res = await fetch('/api/v1/system_logs/explain
', {
1121                     method: 'POST',
1122                     headers: { 'Content-Type': 'application/json' },
1123                     body: JSON.stringify({
1124                         target_timestamp: entry.timestamp,
1125                         window_minutes: 5,
1126                         channel: entry.channel || currentChannel,
1127                         event_id: entry.event_id || 0,
1128                         provider: entry.provider || entry.source || '',
1129                         level: entry.level || '',
1130                         message: entry.message || '',
1131                         target_entry: entry,
1132                         query_text: `Диагностика события ${entry.provide
r || entry.source} (ID ${entry.event_id || 0}): ${entry.messag
e || ''}`,
1133                     }),
1134                 });
1135                 if (!res.ok) throw new Error(`HTTP ${res.status}`);
1136                 const report = await res.json();
1137                 aiExplanation.innerHTML = `
1138                     <div class="mb-2"><strong class="text-info">Сводка
:</strong> ${escapeHtml(report.summary)}</div>
1139                     <div class="mb-2"><strong class="text-warning">При
чина / Анализ:</strong> ${escapeHtml(report.root_cause)}</div>
1140                     <h6 class="small text-uppercase text-muted fw-bold
mb-1">Рекомендации:</h6>
1141                     <ul class="mb-0 ps-3">

```

```
1142 |         ${report.recommendations || []}.map(r => `<li>$
    | {escapeHtml(r)}</li>`).join('')}
1143 |     </ul>
1144 |     `;
1145 |     } catch (err) {
1146 |         aiExplanation.innerHTML = `<div class="text-danger">
    | Ошибка диагностики: ${escapeHtml(err.message)}</div>`;
1147 |     }
1148 | };
1149 | }
1150 |
1151 | // Кнопка копирования текста сообщения
1152 | const copyMsgBtn = document.getElementById('btn-slc-modal-
    | copy-msg');
1153 | if (copyMsgBtn) {
1154 |     copyMsgBtn.onclick = () => {
1155 |         const text = entry.message || '';
1156 |         navigator.clipboard.writeText(text).then(() => {
1157 |             const orig = copyMsgBtn.innerHTML;
1158 |             copyMsgBtn.innerHTML = '<i class="bi bi-check2 me-1
    | text-success"></i>Скопировано!';
1159 |             setTimeout(() => { copyMsgBtn.innerHTML = orig; }, 1
    | 800);
1160 |             }).catch(() => {}));
1161 |     };
1162 | }
1163 |
1164 | if (window.bootstrap?.Modal) {
1165 |     const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl);
1166 |     bsModal.show();
1167 | }
1168 | }
1169 |
1170 | function exportLogs(format) {
1171 |     const level = (document.getElementById('slc-level-select')
    | || {}).value || '';
1172 |     const search = (document.getElementById('slc-search-input'
```



```

    ) || {}).value || '';
1173     const limit = (document.getElementById('slc-limit-select')
    || {}).value || '200';
1174     const hours = (document.getElementById('slc-hours-select')
    || {}).value || '24';
1175
1176     const url = `/api/v1/system_logs/export?format=${format}&c
    hannel=${encodeURIComponent(currentChannel)}&level=${encodeURIComponent
    (level)}&search=${encodeURIComponent(search)}&hours=${
    {hours}&limit=${{limit}}`;
1177     window.open(url, '_blank');
1178 }
1179
1180 // =====
    =====
1181 // SECTION 6: Master Initializer
1182 // =====
    =====
1183
1184 function initSystemControlTab() {
1185     console.log('[SystemControl] Initializing System Control C
    enter & OS System Logs...');
1186     fetchStatus();
1187     loadWizardProfiles();
1188     fetchAppActivityLogs();
1189
1190     // Initialize OS log explorer
1191     performSystemScan().then(() => {
1192         loadEvents();
1193     });
1194
1195     if (!isSCCInitialized) {
1196         const refreshBtn = document.getElementById('btn-scc-refr
    esh');
1197         const applyProfBtn = document.getElementById('btn-scc-ap
    ply-profile');
1198         const cleanBtn = document.getElementById('btn-scc-action
    -clean');

```

```
1199 |     const sfcBtn = document.getElementById('btn-scc-action-s
| fc');
1200 |     const dismBtn = document.getElementById('btn-scc-action-
| dism');
1201 |     const createRestoreBtn = document.getElementById('btn-sc
| c-create-restore');
1202 |     const refreshLogsBtn = document.getElementById('btn-scc-
| refresh-logs');
1203 |
1204 |     if (refreshBtn) refreshBtn.onclick = () => { fetchStatus
| (); loadEvents(); fetchAppActivityLogs(); };
1205 |     if (refreshLogsBtn) refreshLogsBtn.onclick = () => fetch
| AppActivityLogs();
1206 |     if (applyProfBtn) applyProfBtn.onclick = applyCurrentPro
| file;
1207 |
1208 |     if (cleanBtn) {
1209 |         cleanBtn.onclick = async () => {
1210 |             if (confirm('Run safe cleanup on temporary directori
| es and update cache?')) {
1211 |                 cleanBtn.disabled = true;
1212 |                 await fetch('/api/system-control/maintenance/clean
| up', { method: 'POST' });
1213 |                 cleanBtn.disabled = false;
1214 |                 fetchStatus();
1215 |                 fetchAppActivityLogs();
1216 |             }
1217 |         };
1218 |     }
1219 |
1220 |     if (sfcBtn) {
1221 |         sfcBtn.onclick = async () => {
1222 |             sfcBtn.disabled = true;
1223 |             sfcBtn.innerText = 'Scanning...';
1224 |             const res = await fetch('/api/system-control/mainten
| ance/sfc', { method: 'POST' });
1225 |             const d = await res.json();
1226 |             window.showToast?.(d.message || 'SFC Scan Completed'
```

```

    , d.status === 'error' ? 'danger' : 'success') || alert(d.message || 'SFC Scan Completed');
1227     sfcBtn.disabled = false;
1228     sfcBtn.innerHTML = '<i class="bi bi-search me-1"></i>
> Run SFC Integrity Scan';
1229     fetchAppActivityLogs();
1230 };
1231 }
1232
1233 if (dismBtn) {
1234     dismBtn.onclick = async () => {
1235         dismBtn.disabled = true;
1236         dismBtn.innerText = 'Checking...';
1237         const res = await fetch('/api/system-control/maintenance/dism', { method: 'POST' });
1238         const d = await res.json();
1239         window.showToast?.(d.message || 'DISM Check Completed', d.status === 'error' ? 'danger' : 'success') || alert(d.message || 'DISM Check Completed');
1240         dismBtn.disabled = false;
1241         dismBtn.innerHTML = '<i class="bi bi-activity me-1"></i> Check DISM Store';
1242         fetchAppActivityLogs();
1243     };
1244 }
1245
1246 if (createRestoreBtn) {
1247     createRestoreBtn.onclick = async () => {
1248         const desc = prompt('Enter restore point description:', 'System Control Center Checkpoint');
1249         if (desc) {
1250             createRestoreBtn.disabled = true;
1251             const res = await fetch('/api/system-control/restore-points', {
1252                 method: 'POST',
1253                 headers: { 'Content-Type': 'application/json' },
1254                 body: JSON.stringify({ description: desc })
1255             });

```

```
1256         const d = await res.json();
1257         window.showToast?.(d.message || 'Restore point created.', d.status === 'error' ? 'danger' : 'success') || alert(d.message || 'Restore point created.');
```

1258 createRestoreBtn.disabled = false;

1259 fetchRestorePoints();

1260 fetchAppActivityLogs();

1261 }

1262 };

1263 }

1264

1265 // Policy & Storage Config Modal Button

1266 const configRestoreBtn = document.getElementById('btn-scc-config-restore');

1267 if (configRestoreBtn) {

1268 configRestoreBtn.onclick = loadRestorePolicyModal;

1269 }

1270

1271 const savePolicyBtn = document.getElementById('btn-scc-save-restore-policy');

1272 if (savePolicyBtn) {

1273 savePolicyBtn.onclick = saveRestorePolicy;

1274 }

1275

1276 const pruneNowBtn = document.getElementById('btn-scc-policy-prune-now');

1277 if (pruneNowBtn) {

1278 pruneNowBtn.onclick = pruneRestorePointsNow;

1279 }

1280

1281 // System Log Center toolbar and listeners

1282 const scanBtn = document.getElementById('btn-slc-scan');

1283 if (scanBtn) {

1284 scanBtn.onclick = () => performSystemScan(true);

1285 }

1286

1287 const slcRefreshBtn = document.getElementById('btn-slc-refresh');

```
1288 |         if (slcRefreshBtn) {
1289 |             slcRefreshBtn.onclick = loadEvents;
1290 |         }
1291 |
1292 |         const jsonExportBtn = document.getElementById('btn-slc-e
xport-json');
1293 |         if (jsonExportBtn) {
1294 |             jsonExportBtn.onclick = (e) => {
1295 |                 e.preventDefault();
1296 |                 exportLogs('json');
1297 |             };
1298 |         }
1299 |
1300 |         const csvExportBtn = document.getElementById('btn-slc-ex
port-csv');
1301 |         if (csvExportBtn) {
1302 |             csvExportBtn.onclick = (e) => {
1303 |                 e.preventDefault();
1304 |                 exportLogs('csv');
1305 |             };
1306 |         }
1307 |
1308 |         const modeTabs = document.querySelectorAll('#slc-mode-ta
bs .nav-link');
1309 |         modeTabs.forEach((tab) => {
1310 |             tab.onclick = () => {
1311 |                 modeTabs.forEach(t => t.classList.remove('active'));
1312 |                 tab.classList.add('active');
1313 |                 currentMode = tab.getAttribute('data-mode') || 'live
';
1314 |                 renderCurrentModeView();
1315 |             };
1316 |         });
1317 |
1318 |         const auditBtn = document.getElementById('btn-slc-audit'
);
1319 |         if (auditBtn) {
1320 |             auditBtn.onclick = () => {
```

```
1321 |         modeTabs.forEach(t => {
1322 |             if (t.getAttribute('data-mode') === 'audit') {
1323 |                 t.classList.add('active');
1324 |             } else {
1325 |                 t.classList.remove('active');
1326 |             }
1327 |         });
1328 |         currentMode = 'audit';
1329 |         renderCurrentModeView();
1330 |     };
1331 | }
1332 |
1333 |     const ragSearchBtn = document.getElementById('btn-slc-ra
1334 | g-search');
1335 |     if (ragSearchBtn) {
1336 |         ragSearchBtn.onclick = performRAGSearch;
1337 |     }
1338 |
1339 |     const ragRebuildBtn = document.getElementById('btn-slc-r
1340 | ag-rebuild');
1341 |     if (ragRebuildBtn) {
1342 |         ragRebuildBtn.onclick = rebuildRAGIndex;
1343 |     }
1344 |
1345 |     const ragInput = document.getElementById('slc-rag-input'
1346 | );
1347 |     if (ragInput) {
1348 |         ragInput.onkeydown = (e) => {
1349 |             if (e.key === 'Enter') {
1350 |                 e.preventDefault();
1351 |                 performRAGSearch();
1352 |             }
1353 |         };
1354 |     }
1355 |
1356 |     const askAiBtn = document.getElementById('btn-slc-ask-ai
1357 | ');
1358 |     if (askAiBtn) {
```

```
1355 |         askAiBtn.onclick = () => {
1356 |             if (selectedEvent) {
1357 |                 showEventModal(selectedEvent);
1358 |             } else if (cachedEvents.length > 0) {
1359 |                 selectedEvent = cachedEvents[0];
1360 |                 showEventModal(selectedEvent);
1361 |             }
1362 |         };
1363 |     }
1364 |
1365 |     const levelSelect = document.getElementById('slc-level-s
1366 | elect');
1367 |     if (levelSelect) levelSelect.onchange = loadEvents;
1368 |
1369 |     const hoursSelect = document.getElementById('slc-hours-s
1370 | elect');
1371 |     if (hoursSelect) hoursSelect.onchange = loadEvents;
1372 |
1373 |     const limitSelect = document.getElementById('slc-limit-s
1374 | elect');
1375 |     if (limitSelect) limitSelect.onchange = loadEvents;
1376 |
1377 |     const eventIdInput = document.getElementById('slc-eventi
1378 | d-input');
1379 |     if (eventIdInput) eventIdInput.onchange = loadEvents;
1380 |
1381 |     let searchTimer = null;
1382 |     const searchInput = document.getElementById('slc-search-
1383 | input');
1384 |     if (searchInput) {
1385 |         searchInput.oninput = () => {
1386 |             clearTimeout(searchTimer);
1387 |             searchTimer = setTimeout(loadEvents, 350);
1388 |         };
1389 |     }
1390 |
1391 |     const channelFilterInput = document.getElementById('slc-
1392 | channel-filter-input');
```

```
1387 |         if (channelFilterInput) {
1388 |             channelFilterInput.oninput = renderChannelsTree;
1389 |         }
1390 |
1391 |         const liveSwitch = document.getElementById('slc-live-swi
| tch');
1392 |         if (liveSwitch) {
1393 |             liveSwitch.onchange = () => {
1394 |                 if (window.registerTabPoller && window.unregisterTab
| Poller) {
1395 |                     if (liveSwitch.checked) {
1396 |                         window.registerTabPoller('tab-system-control_log
| s', loadEvents, 4000, { immediate: true });
1397 |                     } else {
1398 |                         window.unregisterTabPoller('tab-system-control_l
| ogs');
1399 |                     }
1400 |                 } else {
1401 |                     if (liveSwitch.checked) {
1402 |                         liveIntervalTimer = setInterval(loadEvents, 4000
| );
1403 |                     } else {
1404 |                         clearInterval(liveIntervalTimer);
1405 |                         liveIntervalTimer = null;
1406 |                     }
1407 |                 }
1408 |             };
1409 |         }
1410 |
1411 |         isSCCInitialized = true;
1412 |     }
1413 | }
1414 |
1415 | window.initSystemControlTab = initSystemControlTab;
1416 | })();
1417 |
1418 |
1419 |
```


system_inspector.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial
  -scale=1.0">
6   <title>AI-Breadboard | System & Hardware Inspector</title>
7   <style>
8     :root {
9       --bg-base: #0f172a;
10      --bg-card: #1e293b;
11      --bg-hover: #334155;
12      --text-main: #f8fafc;
13      --text-muted: #94a3b8;
14      --accent-cyan: #06b6d4;
15      --accent-green: #10b981;
16      --accent-yellow: #f59e0b;
17      --accent-red: #ef4444;
18      --accent-purple: #a855f7;
19      --border-color: #334155;
20    }
21
22    * {
23      box-sizing: border-box;
24      margin: 0;
25      padding: 0;
26      font-family: -apple-system, BlinkMacSystemFont, "S
  egoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
27    }
28
29    body {
30      background-color: var(--bg-base);
31      color: var(--text-main);
32      height: 100vh;
33      display: flex;
34      flex-direction: column;
```

```
35 |         overflow: hidden;
36 |     }
37 |
38 |     header {
39 |         background-color: var(--bg-card);
40 |         border-bottom: 1px solid var(--border-color);
41 |         padding: 12px 24px;
42 |         display: flex;
43 |         align-items: center;
44 |         justify-content: space-between;
45 |     }
46 |
47 |     .header-title {
48 |         display: flex;
49 |         align-items: center;
50 |         gap: 12px;
51 |         font-size: 1.15rem;
52 |         font-weight: 700;
53 |         color: var(--accent-cyan);
54 |     }
55 |
56 |     .status-badge {
57 |         font-size: 0.75rem;
58 |         padding: 4px 8px;
59 |         border-radius: 9999px;
60 |         background: rgba(16, 185, 129, 0.2);
61 |         color: var(--accent-green);
62 |         border: 1px solid var(--accent-green);
63 |         display: flex;
64 |         align-items: center;
65 |         gap: 6px;
66 |     }
67 |
68 |     .pulse-dot {
69 |         width: 8px;
70 |         height: 8px;
71 |         border-radius: 50%;
72 |         background: var(--accent-green);
```

```
73         animation: pulse 1.5s infinite;
74     }
75
76     @keyframes pulse {
77         0% { opacity: 1; transform: scale(1); }
78         50% { opacity: 0.4; transform: scale(0.8); }
79         100% { opacity: 1; transform: scale(1); }
80     }
81
82     /* Top Metric Cards */
83     .metric-cards {
84         display: grid;
85         grid-template-columns: repeat(auto-fit, minmax(200
px, 1fr));
86         gap: 12px;
87         padding: 12px 24px;
88         background-color: #0b1120;
89         border-bottom: 1px solid var(--border-color);
90     }
91
92     .card {
93         background: var(--bg-card);
94         border: 1px solid var(--border-color);
95         border-radius: 8px;
96         padding: 12px 16px;
97         display: flex;
98         flex-direction: column;
99         gap: 6px;
100    }
101
102    .card-label {
103        font-size: 0.75rem;
104        text-transform: uppercase;
105        letter-spacing: 0.05em;
106        color: var(--text-muted);
107    }
108
109    .card-value {
```

```
110         font-size: 1.4rem;
111         font-weight: 700;
112         color: var(--text-main);
113     }
114
115     .card-sub {
116         font-size: 0.75rem;
117         color: var(--text-muted);
118     }
119
120     .progress-bar {
121         height: 4px;
122         background: #334155;
123         border-radius: 2px;
124         overflow: hidden;
125         margin-top: 4px;
126     }
127
128     .progress-fill {
129         height: 100%;
130         background: var(--accent-cyan);
131         transition: width 0.3s ease;
132     }
133
134     /* Main Workspace Layout */
135     .main-workspace {
136         display: grid;
137         grid-template-columns: 380px 1fr;
138         flex: 1;
139         overflow: hidden;
140     }
141
142     /* Left Side: AIDA64 Hardware Tree */
143     .left-panel {
144         background: #131d31;
145         border-right: 1px solid var(--border-color);
146         display: flex;
147         flex-direction: column;
```

```
148         overflow: hidden;
149     }
150
151     .panel-header {
152         padding: 12px 16px;
153         font-size: 0.85rem;
154         font-weight: 600;
155         text-transform: uppercase;
156         letter-spacing: 0.05em;
157         color: var(--accent-cyan);
158         border-bottom: 1px solid var(--border-color);
159         background: rgba(0, 0, 0, 0.2);
160         display: flex;
161         justify-content: space-between;
162         align-items: center;
163     }
164
165     .tree-container {
166         flex: 1;
167         overflow-y: auto;
168         padding: 12px;
169     }
170
171     .tree-node {
172         margin-bottom: 10px;
173         background: var(--bg-card);
174         border: 1px solid var(--border-color);
175         border-radius: 6px;
176         overflow: hidden;
177     }
178
179     .tree-node-title {
180         padding: 8px 12px;
181         font-size: 0.85rem;
182         font-weight: 600;
183         cursor: pointer;
184         background: rgba(255, 255, 255, 0.03);
185         display: flex;
```

```
186         align-items: center;
187         justify-content: space-between;
188         color: #e2e8f0;
189     }
190
191     .tree-node-title:hover {
192         background: rgba(255, 255, 255, 0.08);
193     }
194
195     .tree-node-body {
196         padding: 8px 12px;
197         font-size: 0.8rem;
198         border-top: 1px solid var(--border-color);
199         background: #0f172a;
200     }
201
202     .prop-row {
203         display: flex;
204         justify-content: space-between;
205         margin-bottom: 4px;
206     }
207
208     .prop-key {
209         color: var(--text-muted);
210     }
211
212     .prop-val {
213         font-weight: 500;
214         color: #f1f5f9;
215     }
216
217     /* Right Side: Wireshark Process Stream + AI Copilot */
218 /
219     .right-panel {
220         display: flex;
221         flex-direction: column;
222         overflow: hidden;
223     }
```

```
223 |
224 |     .stream-toolbar {
225 |         padding: 10px 16px;
226 |         background: var(--bg-card);
227 |         border-bottom: 1px solid var(--border-color);
228 |         display: flex;
229 |         align-items: center;
230 |         gap: 12px;
231 |     }
232 |
233 |     .search-box {
234 |         flex: 1;
235 |         background: #0f172a;
236 |         border: 1px solid var(--border-color);
237 |         border-radius: 4px;
238 |         padding: 6px 12px;
239 |         color: var(--text-main);
240 |         font-size: 0.85rem;
241 |         outline: none;
242 |     }
243 |
244 |     .btn {
245 |         padding: 6px 14px;
246 |         border-radius: 4px;
247 |         font-size: 0.8rem;
248 |         font-weight: 600;
249 |         cursor: pointer;
250 |         border: none;
251 |         transition: all 0.2s;
252 |     }
253 |
254 |     .btn-primary {
255 |         background: var(--accent-cyan);
256 |         color: #0f172a;
257 |     }
258 |
259 |     .btn-primary:hover {
260 |         opacity: 0.9;
```



```
261 |     }
262 |
263 |     .btn-ai {
264 |         background: linear-gradient(135deg, #a855f7, #6366
f1);
265 |         color: #ffffff;
266 |     }
267 |
268 |     .table-container {
269 |         flex: 1;
270 |         overflow-y: auto;
271 |         position: relative;
272 |     }
273 |
274 |     table {
275 |         width: 100%;
276 |         border-collapse: collapse;
277 |         font-size: 0.8rem;
278 |         text-align: left;
279 |     }
280 |
281 |     th {
282 |         background: #1e293b;
283 |         position: sticky;
284 |         top: 0;
285 |         padding: 8px 12px;
286 |         color: var(--text-muted);
287 |         font-weight: 600;
288 |         border-bottom: 1px solid var(--border-color);
289 |         z-index: 10;
290 |     }
291 |
292 |     td {
293 |         padding: 8px 12px;
294 |         border-bottom: 1px solid rgba(51, 65, 85, 0.4);
295 |         font-family: ui-monospace, SFMono-Regular, Menlo,
Monaco, Consolas, monospace;
296 |     }
```

```
297 |
298 |     tr:hover {
299 |         background: rgba(255, 255, 255, 0.04);
300 |     }
301 |
302 |     .badge-cpu-high {
303 |         color: var(--accent-red);
304 |         font-weight: 700;
305 |     }
306 |
307 |     .badge-cpu-med {
308 |         color: var(--accent-yellow);
309 |         font-weight: 600;
310 |     }
311 |
312 |     /* Bottom AI Copilot Drawer */
313 |     .ai-drawer {
314 |         background: #111827;
315 |         border-top: 1px solid var(--border-color);
316 |         padding: 12px 20px;
317 |         display: flex;
318 |         flex-direction: column;
319 |         gap: 8px;
320 |         max-height: 180px;
321 |         overflow-y: auto;
322 |     }
323 |
324 |     .ai-header {
325 |         display: flex;
326 |         align-items: center;
327 |         justify-content: space-between;
328 |     }
329 |
330 |     .ai-title {
331 |         font-size: 0.85rem;
332 |         font-weight: 700;
333 |         color: var(--accent-purple);
334 |         display: flex;
```

```

335         align-items: center;
336         gap: 8px;
337     }
338
339     .ai-content {
340         font-size: 0.82rem;
341         color: #cbd5e1;
342         line-height: 1.4;
343     }
344
345     .anomaly-pill {
346         display: inline-block;
347         font-size: 0.75rem;
348         padding: 2px 8px;
349         border-radius: 4px;
350         background: rgba(239, 68, 68, 0.2);
351         color: #f87171;
352         border: 1px solid rgba(239, 68, 68, 0.4);
353         margin-right: 6px;
354         margin-top: 4px;
355     }
356 </style>
357 </head>
358 <body>
359     <header>
360         <div class="header-title">
361             <span>🔍 AI-BREADBOARD</span>
362             <span style="color: var(--text-muted); font-weight
: 400;">|</span>
363             <span>System & Hardware Inspector</span>
364         </div>
365         <div style="display: flex; align-items: center; gap: 1
6px;">
366             <div id="conn-status" class="status-badge">
367                 <span class="pulse-dot"></span>
368                 <span>TELEMETRY STREAMING</span>
369             </div>
370         </div>

```

```

371 |     </header>
372 |
373 |     <!-- Top Telemetry Metrics -->
374 |     <div class="metric-cards">
375 |         <div class="card">
376 |             <span class="card-label">CPU Load</span>
377 |             <span id="metric-cpu-val" class="card-value">0.0%<
| /span>
378 |             <div class="progress-bar"><div id="metric-cpu-fill
| " class="progress-fill" style="width: 0%"></div></div>
379 |             <span id="metric-cpu-sub" class="card-sub">-- Core
| s</span>
380 |         </div>
381 |         <div class="card">
382 |             <span class="card-label">Physical RAM</span>
383 |             <span id="metric-ram-val" class="card-value">0.0 G
| B</span>
384 |             <div class="progress-bar"><div id="metric-ram-fill
| " class="progress-fill" style="width: 0%; background: var(--ac
| cent-green);"></div></div>
385 |             <span id="metric-ram-sub" class="card-sub">0% Used
| </span>
386 |         </div>
387 |         <div class="card">
388 |             <span class="card-label">GPU Accelerator</span>
389 |             <span id="metric-gpu-val" class="card-value">--</s
| pan>
390 |             <span id="metric-gpu-sub" class="card-sub">VRAM: -
| - / -- GB</span>
391 |         </div>
392 |         <div class="card">
393 |             <span class="card-label">Disk Throughput</span>
394 |             <span id="metric-disk-val" class="card-value">0.0
| MB/s</span>
395 |             <span id="metric-disk-sub" class="card-sub">R: 0 K
| B/s | W: 0 KB/s</span>
396 |         </div>
397 |     </div>

```

```

398
399     <!-- Main Workspace -->
400     <div class="main-workspace">
401         <!-- Left Pane: Hardware Tree -->
402         <div class="left-panel">
403             <div class="panel-header">
404                 <span>🔧 Hardware Specification</span>
405                 <button class="btn btn-primary" style="font-size: 0.7rem; padding: 2px 8px;" onclick="fetchHardwareTree()">Refresh</button>
406             </div>
407             <div id="hardware-tree" class="tree-container">
408                 <div style="color: var(--text-muted); font-size: 0.8rem; text-align: center; padding: 20px;">Loading hardware tree...</div>
409             </div>
410         </div>
411
412         <!-- Right Pane: Wireshark Process Stream -->
413         <div class="right-panel">
414             <div class="stream-toolbar">
415                 <input id="process-search" type="text" class="search-box" placeholder="Filter processes by name, PID, or user..." oninput="renderProcessTable()">
416                 <button id="pause-btn" class="btn btn-primary" onclick="togglePause()">Pause Stream</button>
417                 <button class="btn btn-ai" onclick="triggerAiAudit()">🤖 Run AI Audit</button>
418             </div>
419             <div class="table-container">
420                 <table>
421                     <thead>
422                         <tr>
423                             <th style="width: 70px;">PID</th>
424                             <th>Process Name</th>
425                             <th style="width: 90px;">Status</th>
426                             <th style="width: 90px; text-align

```

```

427 |         : right;">CPU %</th>
428 |         <th style="width: 100px; text-align: right;">Memory</th>
429 |         <th style="width: 80px; text-align: right;">Threads</th>
430 |         <th style="width: 120px;">User</th>
431 |     >
432 |         </tr>
433 |     </thead>
434 |     <tbody id="process-tbody">
435 |         <tr><td colspan="7" style="text-align: center; color: var(--text-muted);">Waiting for telemetry stream...</td></tr>
436 |     </tbody>
437 | </table>
438 | </div>
439 |
440 |     <!-- Bottom AI Copilot -->
441 |     <div class="ai-drawer">
442 |         <div class="ai-header">
443 |             <div class="ai-title">
444 |                 <span> AI Telemetry & Bottleneck Copilot</span>
445 |                 <span id="ai-health-badge" style="font-size: 0.75rem; color: var(--accent-green); font-weight: normal;">[Health: 100/100]</span>
446 |             </div>
447 |             <div id="ai-engine-tag" style="font-size: 0.7rem; color: var(--text-muted);">Heuristic Engine</div>
448 |             <div id="ai-anomalies-container"></div>
449 |             <div id="ai-summary-text" class="ai-content">
450 |                 Telemetry monitor active. Running automated heuristic checks over CPU, memory, and hardware thermals.
451 |             </div>
452 |         </div>
453 |     </div>

```

```

454
455     <script>
456         let isPaused = false;
457         let latestSnapshot = null;
458         let ws = null;
459
460         let wsReconnectTimer = null;
461
462         function connectWebSocket() {
463             if (wsReconnectTimer) {
464                 clearTimeout(wsReconnectTimer);
465                 wsReconnectTimer = null;
466             }
467             if (ws) {
468                 try {
469                     ws.onclose = null;
470                     ws.onerror = null;
471                     ws.close();
472                 } catch {}
473             }
474             const proto = window.location.protocol === 'https:
' ? 'wss:' : 'ws:';
475             const wsUrl = `${proto}://${window.location.host}/a
pi/v1/system/stream`;
476             ws = new WebSocket(wsUrl);
477
478             ws.onopen = () => {
479                 const connStatus = document.getElementById('co
nn-status');
480                 if (connStatus) {
481                     connStatus.innerHTML = '<span class="pulse
-dot"></span><span>TELEMETRY STREAMING</span>';
482                     connStatus.style.borderColor = 'var(--acce
nt-green)';
483                 }
484             };
485
486             ws.onmessage = (event) => {

```

```

487         if (isPaused) return;
488         try {
489             const data = JSON.parse(event.data);
490             latestSnapshot = data;
491             updateDashboard(data);
492         } catch (e) {
493             console.error("Error parsing telemetry snapshot:", e);
494         }
495     };
496
497     const scheduleReconnect = () => {
498         if (wsReconnectTimer) return;
499         const connStatus = document.getElementById('connection-status');
500         if (connStatus) {
501             connStatus.innerHTML = '<span style="color: var(--accent-red);">●</span><span>RECONNECTING...</span>';
502             connStatus.style.borderColor = 'var(--accent-red)';
503         }
504         wsReconnectTimer = setTimeout(() => {
505             wsReconnectTimer = null;
506             connectWebSocket();
507         }, 3000);
508     };
509
510     ws.onclose = scheduleReconnect;
511     ws.onerror = scheduleReconnect;
512 }
513
514 function updateDashboard(snap) {
515     // Update Top Cards
516     const cpu = snap.cpu;
517     document.getElementById('metric-cpu-val').innerHTML = `${cpu.total_percent.toFixed(1)}%`;
518     document.getElementById('metric-cpu-fill').style.width = `${Math.min(100, cpu.total_percent)}%`;

```



```

519 |         document.getElementById('metric-cpu-sub').innerTex
      | t = `${cpu.physical_cores} Physical / ${cpu.logical_cores} Thr
      | eads`;
520 |
521 |         const mem = snap.memory;
522 |         document.getElementById('metric-ram-val').innerTex
      | t = `${mem.used_gb.toFixed(1)} / ${mem.total_gb.toFixed(1)} GB
      | `;
523 |         document.getElementById('metric-ram-fill').style.w
      | idth = `${mem.percent}%`;
524 |         document.getElementById('metric-ram-sub').innerTex
      | t = `${mem.percent}% Utilized (${mem.available_gb.toFixed(1)}
      | GB free)`;
525 |
526 |         if (snap.gpus && snap.gpus.length > 0) {
527 |             const g = snap.gpus[0];
528 |             document.getElementById('metric-gpu-val').inne
      | rText = g.name;
529 |             document.getElementById('metric-gpu-sub').inne
      | rText = `VRAM: ${g.memory_total_gb.toFixed(1)} GB | CUDA: ${g.
      | has_cuda ? 'Yes' : 'No'}`;
530 |         }
531 |
532 |         const io = snap.disk_io;
533 |         const totalMb = ((io.read_bytes_per_sec + io.write
      | _bytes_per_sec) / (1024 * 1024)).toFixed(2);
534 |         const rKb = (io.read_bytes_per_sec / 1024).toFixed
      | (0);
535 |         const wKb = (io.write_bytes_per_sec / 1024).toFixe
      | d(0);
536 |         document.getElementById('metric-disk-val').innerTe
      | xt = `${totalMb} MB/s`;
537 |         document.getElementById('metric-disk-sub').innerTe
      | xt = `R: ${rKb} KB/s | W: ${wKb} KB/s`;
538 |
539 |         renderProcessTable();
540 |     }
541 |

```

```

542         function renderProcessTable() {
543             if (!latestSnapshot) return;
544             const filter = document.getElementById('process-se
arch').value.toLowerCase();
545             const tbody = document.getElementById('process-tbo
dy');
546
547             const filtered = latestSnapshot.top_processes.filt
er(p => {
548                 return !filter || p.name.toLowerCase().include
s(filter) || String(p.pid).includes(filter) || (p.username &&
p.username.toLowerCase().includes(filter));
549             });
550
551             if (filtered.length === 0) {
552                 tbody.innerHTML = '<tr><td colspan="7" style="
text-align: center; color: var(--text-muted);">No matching pro
cesses found.</td></tr>';
553                 return;
554             }
555
556             tbody.innerHTML = filtered.map(p => {
557                 let cpuClass = '';
558                 if (p.cpu_percent > 40) cpuClass = 'badge-cpu-
high';
559                 else if (p.cpu_percent > 15) cpuClass = 'badge
-cpu-med';
560
561                 return `
562                     <tr>
563                         <td style="color: var(--accent-cyan);"
>${p.pid}</td>
564                         <td style="font-weight: 600;">${p.name
}</td>
565                         <td style="color: var(--text-muted);">
${p.status}</td>
566                         <td style="text-align: right;" class="
${cpuClass}">${p.cpu_percent.toFixed(1)}%</td>

```

```

567 |         <td style="text-align: right; color: v
| ar(--accent-green);">${p.memory_mb.toFixed(1)} MB</td>
568 |         <td style="text-align: right; color: v
| ar(--accent-purple);">${p.num_threads}</td>
569 |         <td style="color: var(--text-muted);">
| ${p.username || 'SYSTEM'}</td>
570 |     </tr>
571 |     `;
572 |     }).join('');
573 | }
574 |
575 |     function togglePause() {
576 |         isPaused = !isPaused;
577 |         const btn = document.getElementById('pause-btn');
578 |         btn.innerText = isPaused ? 'Resume Stream' : 'Paus
| e Stream';
579 |         btn.style.background = isPaused ? 'var(--accent-ye
| llow)' : 'var(--accent-cyan)';
580 |     }
581 |
582 |     async function fetchHardwareTree() {
583 |         try {
584 |             const res = await fetch('/api/v1/system/hardwa
| re');
585 |             if (!res.ok) return;
586 |             const nodes = await res.json();
587 |             const container = document.getElementById('har
| dware-tree');
588 |
589 |             container.innerHTML = nodes.map(node => {
590 |                 const propsHtml = Object.entries(node.prop
| erties).map(([k, v]) => `
591 |                     <div class="prop-row">
592 |                         <span class="prop-key">${k}</span>
593 |                         <span class="prop-val">${v}</span>
594 |                     </div>
595 |                 `).join('');
596 |

```

```

597         return `
598             <div class="tree-node">
599                 <div class="tree-node-title" onclick="this.nextElementSibling.classList.toggle('hidden')">
600                     <span>${node.category}: ${node
        .name}</span>
601                     <span style="font-size: 0.7rem
        ; color: var(--accent-cyan);">▼</span>
602                 </div>
603                 <div class="tree-node-body">
604                     ${propsHtml}
605                 </div>
606             </div>
607         `;
608     })).join('');
609     } catch (e) {
610         console.error("Failed to load hardware tree:",
        e);
611     }
612 }
613
614     async function triggerAiAudit() {
615         const summaryEl = document.getElementById('ai-summary-text');
616         const badgeEl = document.getElementById('ai-health-badge');
617         const anomalyEl = document.getElementById('ai-anomalies-container');
618
619         summaryEl.innerText = "Running AI telemetry diagnostic analysis...";
620         try {
621             const res = await fetch('/api/v1/system/diagnose', { method: 'POST' });
622             const report = await res.json();
623
624             badgeEl.innerText = `[Health: ${report.health_score}/100]`;

```

```

625         badgeEl.style.color = report.health_score >= 8
        0 ? 'var(--accent-green)' : (report.health_score >= 60 ? 'var(
        --accent-yellow)' : 'var(--accent-red)');
626         document.getElementById('ai-engine-tag').inner
        Text = report.ai_model_used;
627
628         if (report.anomalies && report.anomalies.lengt
        h > 0) {
629             anomalyEl.innerHTML = report.anomalies.map
        (a => `

```

index.html

```
1 <!-- System & Hardware Inspector Application Tab -->
2 <style>
3   /* Все цвета через переменные main.css – адаптируется к обеи
   м темам */
4   .sys-container {
5     padding: 0.5rem 0.75rem;
6   }
7   /* Сенсоры LHM */
8   .sys-sensor-group {
9     margin-bottom: 6px;
10    background: var(--surface-2);
11    border: 1px solid var(--border-color);
12    border-radius: 5px;
13    overflow: hidden;
14  }
15  .sys-sensor-group-title {
16    padding: 4px 8px;
17    font-size: 0.76rem;
18    font-weight: 600;
19    background: transparent;
20    display: flex;
21    align-items: center;
22    justify-content: space-between;
23    color: var(--nav-active);
24    border-bottom: 1px solid var(--border-subtle, rgba(255, 25, 255, 0.05));
25  }
26  .sys-sensor-row {
27    display: flex;
28    justify-content: space-between;
29    align-items: center;
30    padding: 3px 8px;
31    font-size: 0.76rem;
32    border-bottom: 1px solid var(--border-subtle, rgba(255, 25, 255, 0.04));
33    background: var(--bg-color);
```

```
34 | }
35 | .sys-sensor-row:last-child {
36 |     border-bottom: none;
37 | }
38 | .sys-sensor-name {
39 |     color: var(--text-color);
40 |     max-width: 55%;
41 |     overflow: hidden;
42 |     text-overflow: ellipsis;
43 |     white-space: nowrap;
44 | }
45 | .sys-sensor-value {
46 |     font-weight: 600;
47 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
48 | Consolas, monospace;
49 |     font-size: 0.78rem;
50 |     text-align: right;
51 | }
52 | /* Таблица процессов */
53 | .sys-proc-table {
54 |     width: 100%;
55 |     font-size: 0.78rem;
56 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
57 | Consolas, monospace;
58 |     border-collapse: collapse;
59 | }
60 | .sys-proc-table th {
61 |     background: var(--surface-2);
62 |     padding: 4px 8px;
63 |     color: var(--text-muted);
64 |     font-weight: 600;
65 |     border-bottom: 1px solid var(--border-color);
66 |     position: sticky;
67 |     top: 0;
68 |     z-index: 5;
69 | }
70 | .sys-proc-table td {
71 |     padding: 4px 8px;
```

```
70 |     border-bottom: 1px solid var(--border-subtle);
71 |     color: var(--text-color);
72 | }
73 | /* Статусы загрузки */
74 | .badge-cpu-high { color: #ef4444; font-weight: 700; }
75 | .badge-cpu-med  { color: #f59e0b; font-weight: 600; }
76 | /* Аномалии */
77 | .anomaly-tag {
78 |     display: inline-block;
79 |     font-size: 0.72rem;
80 |     padding: 1px 6px;
81 |     border-radius: 4px;
82 |     background: rgba(239, 68, 68, 0.12);
83 |     color: #dc2626;
84 |     border: 1px solid rgba(239, 68, 68, 0.3);
85 |     margin-right: 4px;
86 |     margin-top: 2px;
87 | }
88 | [data-theme="dark"] .anomaly-tag {
89 |     background: rgba(239, 68, 68, 0.18);
90 |     color: #f87171;
91 |     border-color: rgba(239, 68, 68, 0.35);
92 | }
93 | /* Интерактивный ресайзер для таблицыИзменения файлов в реал
   | ьном времени */
94 | .sys-table-resizer {
95 |     height: 10px;
96 |     background: var(--surface-2);
97 |     border-top: 1px solid var(--border-color);
98 |     cursor: ns-resize;
99 |     display: flex;
100 |     align-items: center;
101 |     justify-content: center;
102 |     transition: background-color 0.15s ease;
103 |     user-select: none;
104 |     -webkit-user-select: none;
105 |     touch-action: none;
106 | }
```



```

107 | .sys-table-resizer:hover,
108 | .sys-table-resizer.resizing {
109 |     background: rgba(13, 202, 240, 0.25);
110 | }
111 | .sys-table-resizer-grip {
112 |     width: 42px;
113 |     height: 3px;
114 |     background: var(--border-color, rgba(255, 255, 255, 0.2));
115 |     border-radius: 3px;
116 |     pointer-events: none;
117 |     transition: background-color 0.15s ease, width 0.15s ease;
118 | }
119 | .sys-table-resizer:hover .sys-table-resizer-grip,
120 | .sys-table-resizer.resizing .sys-table-resizer-grip {
121 |     background: var(--bs-info, #0dcaf0);
122 |     width: 56px;
123 | }
124 | </style>
125 |
126 | <div class="sys-container theme-inspector">
127 |     <!-- Header Alert & Status (Compact Strip) -->
128 |     <div class="d-flex justify-content-between align-items-center
129 |         r flex-wrap gap-2 mb-2 py-1 px-3 rounded-2"
130 |         style="background: var(--surface-2); border: 1px solid
131 |         var(--border-color);">
132 |         <div class="d-flex align-items-center gap-2 flex-wrap">
133 |             <span class="fs-5">□ </span>
134 |             <span class="fw-bold small" style="font-size: 0.92rem; c
135 |             olor: var(--text-color);">Телеметрия хоста и оборудования</spa
136 |             n>
137 |             <span class="badge bg-secondary fw-normal" style="font-s
138 |             ize: 0.7rem;">Live Telemetry</span>
139 |             <span class="d-none d-lg-inline" style="font-size: 0.76r
140 |             em; color: var(--text-muted);">| Комплексный мониторинг CPU, R
141 |             AM, GPU, дисков, питания и процессов</span>
142 |         </div>
143 |         <div class="d-flex align-items-center gap-1 flex-wrap">
144 |             <span class="badge rounded-pill bg-success-subtle text-s

```

```

    success border border-success px-2 py-1" style="font-size: 0.72
    rem;" id="sys-conn-status">
138      • Активна
139    </span>
140    <button class="btn btn-xs btn-outline-info rounded-pill
    px-2 py-1" style="font-size: 0.75rem;" id="btn-sys-refresh-sen
    sors">
141      <i class="bi bi-arrow-clockwise me-1"></i> Сенсоры
142    </button>
143    <button class="btn btn-xs btn-outline-primary rounded-pi
    ll px-2 py-1" style="font-size: 0.75rem;" id="btn-sys-interval
    s" title="Настройка интервалов опроса сенсоров и логгеров в co
    nfig.json">
144      <i class="bi bi-stopwatch me-1"></i> Интервалы
145    </button>
146    <button class="btn btn-xs btn-outline-warning rounded-pi
    ll px-2 py-1" style="font-size: 0.75rem;" id="btn-sys-config">
147      <i class="bi bi-gear-fill me-1"></i> Настройки
148    </button>
149    <button class="btn btn-xs btn-primary rounded-pill px-2
    py-1 fw-bold" style="font-size: 0.75rem; background: linear-gr
    adient(135deg, #a855f7, #6366f1); border: none;" id="btn-sys-r
    un-audit">
150      □ AI Аудит
151    </button>
152  </div>
153 </div>
154
155 <!-- Top Metrics Bar (Compact Density Grid) -->
156 <div class="row g-2 mb-2">
157   <div class="col-6 col-md-3">
158     <div class="sys-card">
159       <div class="d-flex justify-content-between align-items
    -center">
160         <span class="sys-label">Загрузка CPU</span>
161         <span class="small" style="font-size: 0.68rem; color
    : var(--text-muted);" id="sys-metric-cpu-sub">-- Ядер</span>
162       </div>

```

```
163 |         <div class="d-flex align-items-baseline justify-content-between">
164 |             <span class="sys-value" id="sys-metric-cpu-val">0.0%
165 |         </div>
166 |         <div class="progress" style="height: 3px; background:
167 |         var(--border-color);">
168 |             <div class="progress-bar bg-info" id="sys-metric-cpu
169 |             -fill" style="width: 0%"></div>
170 |         </div>
171 |     </div>
172 |     <div class="col-6 col-md-3">
173 |         <div class="sys-card">
174 |             <div class="d-flex justify-content-between align-items
175 |             -center">
176 |                 <span class="sys-label">Память (RAM)</span>
177 |                 <span class="small" style="font-size: 0.68rem; color
178 |                 : var(--text-muted);" id="sys-metric-ram-sub">0% занято</span>
179 |             </div>
180 |             <div class="d-flex align-items-baseline justify-content-between">
181 |                 <span class="sys-value" id="sys-metric-ram-val">0.0
182 |                 GB</span>
183 |             </div>
184 |             <div class="progress" style="height: 3px; background:
185 |             var(--border-color);">
186 |                 <div class="progress-bar bg-success" id="sys-metric-
187 |                 ram-fill" style="width: 0%"></div>
188 |             </div>
189 |         </div>
190 |     </div>
191 |     <div class="col-6 col-md-3">
192 |         <div class="sys-card">
193 |             <div class="d-flex justify-content-between align-items
194 |             -center">
195 |                 <span class="sys-label">GPU Ускоритель</span>
```

```

190 |         <div class="d-flex align-items-baseline justify-content-between">
191 |             <span class="sys-value text-warning" style="font-size: 1.0rem;" id="sys-metric-gpu-val">--</span>
192 |         </div>
193 |         <div class="progress" style="height: 3px; background: var(--border-color);">
194 |             <div class="progress-bar bg-warning" id="sys-metric-gpu-fill" style="width: 0%"></div>
195 |         </div>
196 |     </div>
197 | </div>
198 | <div class="col-6 col-md-3">
199 |     <div class="sys-card">
200 |         <div class="d-flex justify-content-between align-items-center">
201 |             <span class="sys-label">Диск (I/O)</span>
202 |             <span class="small" style="font-size: 0.68rem; color: var(--text-muted);" id="sys-metric-disk-sub">0 MB/s</span>
203 |         </div>
204 |         <div class="d-flex align-items-baseline justify-content-between">
205 |             <span class="sys-value text-info" id="sys-metric-disk-val">0 MB/s</span>
206 |         </div>
207 |         <div class="progress" style="height: 3px; background: var(--border-color);">
208 |             <div class="progress-bar bg-info" id="sys-metric-disk-fill" style="width: 0%"></div>
209 |         </div>
210 |     </div>
211 | </div>
212 | </div>
213 |
214 | <!-- Extended Hardware & Health Status Strip -->
215 | <div class="row g-2 mb-2">
216 |     <!-- Physical Storage & NVMe Health -->
217 |     <div class="col-md-6 col-lg-3">

```

```

218 |         <div class="sys-panel p-2 h-100">
219 |             <div class="d-flex justify-content-between align-items
| -center mb-1">
220 |                 <span class="small fw-bold" style="font-size: 0.76re
| m; color: var(--text-color);"><i class="bi bi-hdd-network text
| -info me-1"></i> Здоровье дисков (NVMe/SSD)</span>
221 |                 <span class="badge bg-success-subtle text-success bo
| rder border-success" style="font-size: 0.65rem;" id="sys-disk-
| health-badge">SMART OK</span>
222 |             </div>
223 |             <div id="sys-physical-disks-container" class="small te
| xt-muted" style="font-size: 0.72rem; line-height: 1.3; max-hei
| ght: 120px; overflow-y: auto;">
224 |                 Опрос состояния физических дисков...
225 |             </div>
226 |         </div>
227 |     </div>
228 |
229 |     <!-- Power & Battery -->
230 |     <div class="col-md-6 col-lg-3">
231 |         <div class="sys-panel p-2 h-100">
232 |             <div class="d-flex justify-content-between align-items
| -center mb-1">
233 |                 <span class="small fw-bold" style="font-size: 0.76re
| m; color: var(--text-color);"><i class="bi bi-battery-charging
| text-warning me-1"></i> Питание и Энергия</span>
234 |                 <span class="badge bg-secondary" style="font-size: 0
| .65rem;" id="sys-power-status-badge">AC Mains</span>
235 |             </div>
236 |             <div id="sys-power-info-container" class="small text-m
| uted" style="font-size: 0.72rem; line-height: 1.3;">
237 |                 Питание от электросети (Стационарный ПК)
238 |             </div>
239 |         </div>
240 |     </div>
241 |
242 |     <!-- RAM SPD Sticks -->
243 |     <div class="col-md-6 col-lg-3">

```

```

244     <div class="sys-panel p-2 h-100">
245         <div class="d-flex justify-content-between align-items
| -center mb-1">
246             <span class="small fw-bold" style="font-size: 0.76re
| m; color: var(--text-color);"><i class="bi bi-memory text-prim
| ary me-1"></i> Модули RAM (SPD)</span>
247             <span class="badge bg-dark border border-secondary t
| ext-info" style="font-size: 0.65rem;" id="sys-ram-sticks-badg
| e">Слоты</span>
248         </div>
249         <div id="sys-ram-sticks-container" class="small text-m
| uted" style="font-size: 0.72rem; line-height: 1.3;">
250             Опрос конфигурации планок памяти...
251         </div>
252     </div>
253 </div>
254
255 <!-- Reliability & Open Ports -->
256 <div class="col-md-6 col-lg-3">
257     <div class="sys-panel p-2 h-100">
258         <div class="d-flex justify-content-between align-items
| -center mb-1">
259             <span class="small fw-bold" style="font-size: 0.76re
| m; color: var(--text-color);"><i class="bi bi-shield-check tex
| t-success me-1"></i> Стабильность и Сеть</span>
260             <span class="badge bg-info-subtle text-info border b
| order-info" style="font-size: 0.65rem;" id="sys-alerts-badg
| e">Стабильно</span>
261         </div>
262         <div id="sys-alerts-ports-container" class="small text
| -muted" style="font-size: 0.72rem; line-height: 1.3;">
263             Службы активны. Ожидание телеметрии портов.
264         </div>
265     </div>
266 </div>
267 </div>
268
269 <!-- Workspace Layout: LHM Sensors (Left) + Process Stream &

```

```

AI Copilot (Right) -->
270 <div class="row g-2">
271   <!-- Left Column: LibreHardwareMonitor Sensors Card -->
272   <div class="col-lg-4">
273     <div class="sys-panel h-100 d-flex flex-column">
274       <div class="sys-panel-header">
275         <div class="d-flex align-items-center gap-1.5">
276           <i class="bi bi-thermometer-half text-danger"></i>
277           <span>Сенсоры LibreHardwareMonitor</span>
278         </div>
279         <div class="d-flex align-items-center gap-1">
280           <span class="badge bg-secondary" id="sys-lhm-status-bad
s-bad
ge">LHM</span>
281           <span class="badge bg-dark border border-secondary
text-info" id="sys-lhm-sensors-count">0 шт.</span>
282         </div>
283       </div>
284
285       <!-- Sensor filter and action bar -->
286       <div class="p-2 border-bottom d-flex align-items-center justify-content-between gap-1 flex-wrap" style="background: var(--surface-2); border-color: var(--border-color) !important; ">
287         <input type="text" class="form-control form-control-sm" id="sys-lhm-search"
288           placeholder="Фильтр сенсоров..." style="max-width: 130px; font-size: 0.74rem; padding: 2px 6px;">
289         <div class="d-flex align-items-center gap-1">
290           <button class="btn btn-xs btn-outline-primary rounded-pill px-2 py-0.5" id="btn-sys-intervals-lhm" title="Настроить интервалы опроса сенсоров и логов (config)">
291             <i class="bi bi-stopwatch"></i>
292           </button>
293           <button class="btn btn-xs btn-outline-info rounded-pill px-2 py-0.5" id="btn-sys-refresh-lhm" title="Обновить сенсоры">
294             <i class="bi bi-arrow-clockwise"></i>
295           </button>

```

```

296         <button class="btn btn-xs btn-outline-success rounded-pill px-2 py-0.5 d-none" id="btn-sys-launch-lhm" title="Запустить службу LibreHardwareMonitor">
297             <i class="bi bi-play-fill me-1"></i> Запуск LHM
298         </button>
299     </div>
300 </div>
301
302     <!-- Category filter pills & Analyze Button -->
303     <div class="px-2 pt-1 pb-1 d-flex gap-1 overflow-x-auto border-bottom align-items-center justify-content-between" id="sys-lhm-cat-filters" style="border-color: var(--border-subtle, rgba(255, 255, 255, 0.05)) !important; font-size: 0.72rem;">
304         <div class="d-flex gap-1 overflow-x-auto flex-grow-1">
305             <button class="btn btn-xs btn-outline-primary active py-0 px-2 rounded-pill sys-lhm-cat-btn" data-cat="all">Все</button>
306             <button class="btn btn-xs btn-outline-secondary py-0 px-2 rounded-pill sys-lhm-cat-btn" data-cat="temperature">°C</button>
307             <button class="btn btn-xs btn-outline-secondary py-0 px-2 rounded-pill sys-lhm-cat-btn" data-cat="load">Нагрузка</button>
308             <button class="btn btn-xs btn-outline-secondary py-0 px-2 rounded-pill sys-lhm-cat-btn" data-cat="fan">Вентиляторы</button>
309             <button class="btn btn-xs btn-outline-secondary py-0 px-2 rounded-pill sys-lhm-cat-btn" data-cat="voltage">Вольтаж</button>
310             <button class="btn btn-xs btn-outline-secondary py-0 px-2 rounded-pill sys-lhm-cat-btn" data-cat="clock">Частота</button>
311         </div>
312         <button class="btn btn-xs btn-primary rounded-pill px-2.5 py-0.5 fw-bold text-nowrap flex-shrink-0 ms-1 shadow-sm"
313             id="btn-sys-sensor-analyze"

```



```

314 |         style="font-size: 0.72rem; background: linear-gradient(135deg, #0284c7, #6366f1); border: none;"
315 |         title="Комплексный AI-аудит сенсоров и обслуживания хоста">
316 |             <i class="bi bi-shield-check me-1"></i> Аудит
317 |         </button>
318 |     </div>
319 |
320 |     <!-- Уведомление о live-fallback (БД сенсоров пуста) ->
321 |     <div class="d-none px-2 py-1 border-bottom" id="sys-lhm-fallback-notice"
322 |         style="background: rgba(234,179,8,0.08); border-color: rgba(234,179,8,0.25) !important;">
323 |         <div class="d-flex align-items-center gap-1.5" style="font-size: 0.72rem; color: #ca8a04;">
324 |             <i class="bi bi-exclamation-triangle-fill text-warning"></i>
325 |             <span>БД сенсоров пуста – используется <strong>прямой опрос (live)</strong>. Запустите телеметрию для накопления данных в SQLite.</span>
326 |         </div>
327 |     </div>
328 |
329 |     <!-- Sensors list container -->
330 |     <div class="p-2 flex-grow-1" id="sys-lhm-sensors-container" style="max-height: 520px; overflow-y: auto;">
331 |         <div class="text-center py-4 small" style="color: var(--text-muted);">
332 |             <div class="spinner-border spinner-border-sm text-info mb-2" role="status"></div>
333 |             <div>Опрос сенсоров LibreHardwareMonitor...</div>
334 |         </div>
335 |     </div>
336 | </div>
337 | </div>
338 |
339 | <!-- Right Column: Process Stream -->

```

```

340     <div class="col-lg-8 d-flex flex-column gap-2">
341         <!-- Process Stream Table -->
342         <div class="sys-panel flex-grow-1">
343             <div class="sys-panel-header">
344                 <div class="d-flex align-items-center gap-2">
345                     <span><i class="bi bi-list-task me-1"></i> Топ про
346 цессов хоста</span>
347                     <span class="badge bg-primary-subtle text-primary
348 border border-primary font-monospace" id="sys-proc-db-badge" s
349 tyle="font-size: 0.72rem;">● БД SQLite (замер 5с)</span>
350                     <span class="small text-muted font-monospace" id="
351 sys-proc-updated-time" style="font-size: 0.7rem;">--:--:--</sp
352 an>
353                 </div>
354                 <div class="d-flex align-items-center gap-2">
355                     <button class="btn btn-sm btn-outline-info rounded
356 -pill px-2.5" id="btn-sys-proc-stats" title="Статистика и выбр
357 осы процессов"><i class="bi bi-bar-chart-line-fill me-1"></i>C
358 татистика</button>
359                     <input type="text" class="form-control form-contro
360 l-sm" id="sys-proc-search"
361 placeholder="Фильтр по имени или PID..." st
362 yle="max-width: 180px;">
363                     <button class="btn btn-sm btn-outline-secondary ro
364 unded-pill px-3" id="btn-sys-pause-proc">Пауза</button>
365                 </div>
366             </div>
367             <div class="table-responsive" style="height: 520px; ma
368 x-height: 520px; overflow-y: auto;">
369                 <table class="sys-proc-table">
370                     <thead>
371                         <tr>
372                             <th style="width: 70px;">PID</th>
373                             <th style="width: 200px; max-width: 240px;">Им
374 я процесса</th>
375                             <th style="width: 90px;">Статус</th>
376                             <th style="width: 80px; text-align: right;">CP
377 U %</th>

```

```

364 |         <th style="width: 100px; text-align: right;">Па
    | амять</th>
365 |         <th style="width: 70px; text-align: right;">По
    | токи</th>
366 |         <th>Пользователь</th>
367 |     </tr>
368 | </thead>
369 | <tbody id="sys-proc-tbody">
370 |     <tr><td colspan="7" class="text-center py-3" sty
    | le="color: var(--text-muted);">Ожидание потока процессов...</t
    | d></tr>
371 |     </tbody>
372 | </table>
373 | </div>
374 | </div>
375 | </div>
376 | </div>
377 |
378 | <!-- Сетевая активность программ (Интернет, Отправка и Прием
    | ) -->
379 | <div class="row g-2 mt-1">
380 |     <div class="col-12">
381 |         <div class="sys-panel">
382 |             <div class="sys-panel-header d-flex align-items-center
    | justify-content-between flex-wrap gap-2">
383 |                 <div class="d-flex align-items-center gap-2">
384 |                     <span class="fs-6">📶 </span>
385 |                     <span class="fw-bold" style="font-size: 0.88rem; c
    | olor: var(--text-color);">Сетевая активность программ (Интерне
    | т)</span>
386 |                     <span class="badge bg-primary-subtle text-primary
    | border border-primary" id="sys-netact-count-badge">0 программ<
    | /span>
387 |                     <span class="badge bg-dark border border-secondary
    | text-info" id="sys-netact-conns-badge">0 сокетов</span>
388 |                 </div>
389 |                 <div class="d-flex align-items-center gap-2 flex-wra
    | p">

```

```

390         <div class="btn-group btn-group-xs" role="group">
391             <button class="btn btn-xs btn-outline-primary ac
tive py-0.5 px-2 sys-net-filter-btn" data-filter="internet" id
="btn-sys-net-filter-internet">☐ Только Интернет</button>
392             <button class="btn btn-xs btn-outline-secondary
py-0.5 px-2 sys-net-filter-btn" data-filter="all" id="btn-sys-
net-filter-all">Все сокеты</button>
393             <button class="btn btn-xs btn-outline-secondary
py-0.5 px-2 sys-net-filter-btn" data-filter="listen" id="btn-s
ys-net-filter-listen">Слушающие (Listen)</button>
394         </div>
395         <input type="text" class="form-control form-contro
l-sm" id="sys-net-search"
396             placeholder="Поиск по имени, PID или IP..."
style="max-width: 200px; font-size: 0.74rem; padding: 2px 8px
;">
397         <button class="btn btn-xs btn-outline-secondary ro
unded-pill px-2.5 py-0.5" id="btn-sys-pause-net">Пауза</button
>
398         <button class="btn btn-xs btn-outline-info rounded
-pill px-2 py-0.5" id="btn-sys-refresh-net" title="Обновить се
тевые соединения">
399             <i class="bi bi-arrow-clockwise"></i>
400         </button>
401     </div>
402 </div>
403
404 <!-- Network Activity Sub-header & Live Throughput Bar
-->
405 <div class="px-3 py-1.5 border-bottom d-flex align-ite
ms-center justify-content-between flex-wrap gap-2" style="back
ground: var(--surface-2); border-color: var(--border-color) !i
mportant; font-size: 0.76rem;">
406     <div class="d-flex align-items-center gap-3 flex-wra
p text-muted">
407         <span><i class="bi bi-arrow-up-right text-warning
me-1"></i>Отправка: <span class="fw-bold text-light" id="sys-n
et-send-label">HTTPS запросы, данные, API</span></span>

```

```

408 |         <span class="border-start border-secondary ps-3"><
| i class="bi bi-arrow-down-left text-success me-1"></i>Прием /
| Скачано: <span class="fw-bold text-light" id="sys-net-recv-lab
| el">Веб-контент, медиа, ответы серверов</span></span>
409 |         <span class="border-start border-secondary ps-3"><
| i class="bi bi-shield-lock text-info me-1"></i>Протоколы: <spa
| n class="badge bg-dark border border-secondary text-info">TLS
| / HTTPS / DNS</span></span>
410 |         </div>
411 |         <div class="d-flex align-items-center gap-2 text-mut
| ed" style="font-size: 0.74rem;">
412 |             <span class="badge bg-dark border border-secondary
| text-warning" title="Измерение трафика за интервал опроса">□
| Период замера: ~2 сек</span>
413 |             <span title="Автоматическое сопоставление сокетов
| с процессами ОС">● Live Socket Attribution</span>
414 |         </div>
415 |     </div>
416 |
417 |     <div id="sys-net-table-container" class="p-0 table-res
| ponsive" style="height: 300px; min-height: 140px; max-height:
| 850px; overflow-y: auto;">
418 |         <table class="sys-proc-table">
419 |             <thead>
420 |                 <tr>
421 |                     <th style="width: 170px;">Программа / PID</th>
422 |                     <th style="width: 220px;">Удаленный сервер (Ин
| тернет)</th>
423 |                     <th style="width: 240px;">Служба / Протокол</t
| h>
424 |                     <th style="width: 100px; text-align: center;">
| Статус</th>
425 |                     <th style="min-width: 220px;"><i class="bi bi-
| arrow-up-right text-warning me-1"></i>Отправка (За период / Вс
| его / Данные)</th>
426 |                     <th style="min-width: 220px;"><i class="bi bi-
| arrow-down-left text-success me-1"></i>Прием (Скачано за перио
| д / Всего / Данные)</th>

```

```

427         </tr>
428     </thead>
429     <tbody id="sys-net-tbody">
430         <tr><td colspan="6" class="text-center text-mute
d p-3">Опрос активных интернет-соединений программ...</td></tr
>
431     </tbody>
432 </table>
433 </div>
434 <div id="sys-net-table-resizer" class="sys-table-resiz
er" title="Потяните для изменения высоты таблицы (двойной клик
– сброс)">
435     <div class="sys-table-resizer-grip"></div>
436 </div>
437 </div>
438 </div>
439 </div>
440
441 <!-- Real-Time Live Watcher Panel (WinAPI ReadDirectoryChang
esW) -->
442 <div class="row g-2 mt-1">
443     <div class="col-12">
444         <div class="sys-panel">
445             <div class="sys-panel-header d-flex align-items-center
justify-content-between flex-wrap gap-2">
446                 <div class="d-flex align-items-center gap-2">
447                     <span><i class="bi bi-activity text-info me-1"></i
>Изменения файлов в реальном времени</span>
448                     <span class="badge bg-info-subtle text-info" id="s
ys-live-badge">ReadDirectoryChangesW</span>
449                     <span class="badge bg-dark border border-secondary
text-truncate" id="sys-watch-dir-badge" style="max-width: 280
px; font-size: 0.72rem; cursor: pointer;" title="Текущая отсле
живаемая папка. Нажмите для смены.">
450                         □ <span id="sys-watch-dir-path">Загрузка...</sp
an>
451                     </span>
452                 </div>

```

```

453 |         <div class="d-flex align-items-center gap-1.5">
454 |             <span class="badge bg-success-subtle text-success
border border-success px-2 py-1" id="sys-watcher-sensor-status
" style="font-size: 0.7rem;"> Штатный режим</span>
455 |             <button class="btn btn-xs btn-outline-warning roun
ded-pill px-2 py-0.5" style="font-size: 0.72rem;" id="btn-sys-
watch-exclusions" title="Настроить исключения и фильтры монито
ринга">
456 |                 <i class="bi bi-funnel me-1"></i> Исключения <sp
an class="badge bg-dark border border-secondary text-warning"
id="sys-exclusions-count-badge">0</span>
457 |             </button>
458 |             <button class="btn btn-xs btn-outline-light rounde
d-pill px-2 py-0.5" style="font-size: 0.72rem;" id="btn-sys-ch
ange-watch-dir" title="Выбрать другую папку для мониторинга">
459 |                 <i class="bi bi-folder2-open me-1 text-warning">
</i> Папка
460 |             </button>
461 |             <button class="btn btn-xs btn-outline-info rounded
-pill px-2 py-0.5" style="font-size: 0.72rem;" id="btn-sys-liv
e-help" title="Подробная справка о работеИзменения файлов в ре
альном времени">
462 |                 <i class="bi bi-info-circle me-1"></i> Справка
463 |             </button>
464 |         </div>
465 |     </div>
466 |
467 |     <!-- Real-Time Watcher Telemetry & Hardware Sensors Ba
r -->
468 |     <div class="px-3 py-2 border-bottom d-flex align-items
-center justify-content-between flex-wrap gap-2" style="backgr
ound: var(--surface-2); border-color: var(--border-color) !imp
ortant; font-size: 0.97rem;">
469 |         <div class="d-flex align-items-center gap-2 flex-wra
p">
470 |             <span class="text-muted"><i class="bi bi-speedomet
er2 text-info me-1"></i>Темп: <span class="fw-bold text-info"
id="sys-watcher-rate">0.0</span> on/c</span>

```

```

471 |         <span class="text-muted" style="font-size: 0.92rem
| ;">(<span class="text-success" id="sys-watcher-cr">0</span> cr
| / <span class="text-warning" id="sys-watcher-mod">0</span> mo
| d / <span class="text-danger" id="sys-watcher-del">0</span> de
| l)</span>
472 |         <span class="border-start border-secondary ps-2 te
| xt-muted" title="Количество отфильтрованных событий">
473 |             <i class="bi bi-funnel text-warning me-1"></i>Ис
| ключено: <span class="text-warning fw-bold" id="sys-watcher-fi
| ltered-count">0</span>
474 |             </span>
475 |             <span class="border-start border-secondary ps-2 te
| xt-muted">
476 |                 <i class="bi bi-hdd-network text-primary me-1"><
| /i>Диск: <span class="text-light fw-bold" id="sys-watcher-driv
| e">C:</span> (<span class="text-muted" id="sys-watcher-drive-m
| odel">Storage</span>)
477 |                 </span>
478 |                 <span class="badge bg-dark border border-secondary
| text-info px-2 py-0.5" id="sys-watcher-temp" style="font-size
| : 0.92rem;">-- °C</span>
479 |                 <span class="text-muted" style="font-size: 0.92rem
| ;">R: <span class="text-light" id="sys-watcher-r-kbs">0</span>
| KB/s | W: <span class="text-light" id="sys-watcher-w-kbs">0</
| span> KB/s</span>
480 |             </div>
481 |             <div class="d-flex align-items-center gap-1 text-mut
| ed" style="font-size: 0.92rem;">
482 |                 <span><i class="bi bi-shield-check text-success me
| -1"></i>Audit 4660/4663:</span>
483 |                 <span class="badge bg-dark border border-secondary
| text-muted px-2 py-0.5" id="sys-watcher-sec-matches" style="f
| ont-size: 0.92rem;">0</span>
484 |                 <span class="text-muted ms-1" title="Периодическое
| логирование в file_watcher_telemetry_polls.csv">● AutoLog Act
| ive</span>
485 |             </div>
486 |         </div>

```



```

487 |         <div id="sys-watcher-table-container" class="p-0 table
| -responsive" style="height: 280px; min-height: 120px; max-heig
| ht: 850px; overflow-y: auto;">
488 |             <table class="sys-proc-table">
489 |                 <thead>
490 |                     <tr>
491 |                         <th style="width: 100px;">Время</th>
492 |                         <th style="width: 120px;">Действие</th>
493 |                         <th style="width: 170px;">Программа / PID</th>
494 |                         <th>Полный путь к файлу</th>
495 |                     </tr>
496 |                 </thead>
497 |                 <tbody id="sys-watcher-tbody">
498 |                     <tr><td colspan="4" class="text-center text-mute
| d p-3">Мониторинг файловых событий...</td></tr>
499 |                     </tbody>
500 |                 </table>
501 |             </div>
502 |             <div id="sys-watcher-table-resizer" class="sys-table-r
| esizer" title="Потяните для изменения высоты таблицы (двойной
| клик – сброс)">
503 |                 <div class="sys-table-resizer-grip"></div>
504 |             </div>
505 |         </div>
506 |     </div>
507 | </div>
508 | </div>
509 |
510 | <!-- Sensor Analysis & Recommendations Modal -->
511 | <div class="modal fade" id="sysSensorAnalysisModal" tabindex="
| -1" aria-labelledby="sysSensorAnalysisModalLabel" aria-hidden=
| "true">
512 |     <div class="modal-dialog modal-dialog-centered modal-lg moda
| l-dialog-scrollable">
513 |         <div class="modal-content text-white border-secondary shad
| ow-lg" style="background: var(--surface-1); border-color: var(
| --border-color) !important;">
514 |             <div class="modal-header py-2 px-3 border-secondary" sty

```

```

le="background: var(--surface-2); border-color: var(--border-c
olor) !important; ">
515     <h6 class="modal-title d-flex align-items-center gap-2
mb-0" id="sysSensorAnalysisModalLabel">
516         <i class="bi bi-shield-check text-info"></i>
517         <span>Комплексный AI-аудит сенсоров и оборудования</
span>
518         <span class="badge bg-success ms-1" id="modal-sensor
-health-badge">Health: 100/100</span>
519         <span class="badge bg-secondary font-monospace" id="
modal-sensor-model-badge" style="font-size: 0.68rem;">Heuristi
c Engine</span>
520     </h6>
521     <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
522 </div>
523 <div class="modal-body p-3" style="max-height: 75vh; ove
rflow-y: auto;">
524     <!-- Status Banner -->
525     <div class="p-2.5 rounded mb-3 d-flex align-items-cent
er justify-content-between flex-wrap gap-2" id="modal-sensor-s
tatus-banner" style="background: var(--surface-2); border: 1px
solid var(--border-color);">
526         <div class="d-flex align-items-center gap-2">
527             <span class="fs-4" id="modal-sensor-status-icon">□
</span>
528             <div>
529                 <div class="fw-bold" style="font-size: 0.88rem;"
id="modal-sensor-status-title">Все подсистемы работают в штат
ном режиме</div>
530                 <div class="small text-muted" style="font-size:
0.75rem;" id="modal-sensor-status-sub">Сравнение залогированны
х усреднённых показателей со спецификациями реального железа</
div>
531             </div>
532         </div>
533         <div class="d-flex gap-2">
534             <div class="text-end">

```

```
535 |         <div class="small text-muted" style="font-size:
    | 0.7rem;">СЕНСОРОВ В АУДИТЕ</div>
536 |         <div class="fw-bold font-monospace" id="modal-se
    | nsor-count-stat">0</div>
537 |     </div>
538 |     <div class="text-end ps-2 border-start border-seco
    | ndary">
539 |         <div class="small text-muted" style="font-size:
    | 0.7rem;">МАКС. ТЕМПЕРАТУРА</div>
540 |         <div class="fw-bold font-monospace" id="modal-se
    | nsor-max-temp">-- °C</div>
541 |     </div>
542 | </div>
543 | </div>
544 |
545 | <!-- Detected Real Hardware Devices from Logs -->
546 | <div class="mb-3 p-2.5 rounded border" style="backgrou
    | nd: var(--surface-2); border-color: var(--border-color) !impor
    | tant;">
547 |     <div class="d-flex align-items-center justify-conten
    | t-between mb-1.5">
548 |         <span class="small fw-bold text-info"><i class="bi
    | bi-cpu me-1"></i> Распознанные устройства хоста (из логов LHM
    | )</span>
549 |         <span class="badge bg-dark border border-secondary
    | text-muted" id="modal-sensor-devices-count">0 устр.</span>
550 |     </div>
551 |     <div class="d-flex flex-wrap gap-1.5" id="modal-sens
    | or-devices-list">
552 |         <span class="badge bg-dark border border-secondary
    | text-muted">Загрузка устройств...</span>
553 |     </div>
554 | </div>
555 |
556 | <!-- AI Comparison against Real Hardware Specification
    | s -->
557 | <div class="mb-3 p-2.5 rounded border" style="backgrou
    | nd: var(--surface-2); border-color: var(--border-color) !impor
```

```

tant; ">
558         <div class="d-flex align-items-center justify-content-between mb-1.5">
559             <span class="small fw-bold text-primary"><i class="bi bi-robot me-1"></i> AI-сравнение показателей со спецификациями реального железа</span>
560             <span class="badge bg-primary bg-opacity-25 text-primary border border-primary" id="modal-sensor-ai-tag">AI Expert</span>
561         </div>
562         <div class="text-light p-2.5 rounded" id="modal-sensor-ai-comparison" style="background: var(--bg-color); font-size: 0.78rem; line-height: 1.5; white-space: pre-wrap;">
563             Сбор залогированных данных и формирование отчёта..
564             .
565         </div>
566
567         <!-- Metric Grid Breakdown -->
568         <div class="row g-2 mb-3" id="modal-sensor-breakdown-grid">
569             <!-- CPU Status -->
570             <div class="col-12 col-md-4">
571                 <div class="p-2 rounded border" style="background: var(--bg-color); border-color: var(--border-color) !important;">
572                     <div class="d-flex justify-content-between align-items-center mb-1">
573                         <span class="small fw-semibold text-info"><i class="bi bi-cpu me-1"></i> Процессор (CPU)</span>
574                         <span class="badge bg-secondary" id="modal-cpu-temp-badge">-- °C</span>
575                     </div>
576                     <div class="small text-muted" id="modal-cpu-analysis-text" style="font-size: 0.74rem;">Анализ теплового режима и нагрузки...</div>
577                 </div>
578             </div>

```

```
579 |         <!-- GPU Status -->
580 |         <div class="col-12 col-md-4">
581 |             <div class="p-2 rounded border" style="background:
|             var(--bg-color); border-color: var(--border-color) !important
|             ;">
582 |                 <div class="d-flex justify-content-between align
|                 -items-center mb-1">
583 |                     <span class="small fw-semibold text-warning"><
|                     i class="bi bi-gpu-card me-1"></i> Видеокарта (GPU)</span>
584 |                     <span class="badge bg-secondary" id="modal-gpu
|                     -temp-badge">-- °C</span>
585 |                 </div>
586 |                 <div class="small text-muted" id="modal-gpu-anal
|                 ysis-text" style="font-size: 0.74rem;">Анализ видеопамати и гр
|                 афического чипа...</div>
587 |             </div>
588 |         </div>
589 |         <!-- Cooling & Motherboard -->
590 |         <div class="col-12 col-md-4">
591 |             <div class="p-2 rounded border" style="background:
|             var(--bg-color); border-color: var(--border-color) !important
|             ;">
592 |                 <div class="d-flex justify-content-between align
|                 -items-center mb-1">
593 |                     <span class="small fw-semibold text-success"><
|                     i class="bi bi-fan me-1"></i> Охлаждение и Плата</span>
594 |                     <span class="badge bg-secondary" id="modal-fan
|                     -status-badge">OK</span>
595 |                 </div>
596 |                 <div class="small text-muted" id="modal-fan-anal
|                 ysis-text" style="font-size: 0.74rem;">Анализ оборотов кулеров
|                 и напряжений...</div>
597 |             </div>
598 |         </div>
599 |     </div>
600 |
601 |     <!-- Recommendations and Findings Section -->
602 |     <div class="mb-2">
```

```

603         <h6 class="small fw-bold text-uppercase text-muted m
b-2" style="letter-spacing: 0.05em; font-size: 0.72rem;">
604         <i class="bi bi-card-checklist text-primary me-1">
</i> Экспертные рекомендации и выводы
605     </h6>
606     <div id="modal-sensor-recommendations-list" class="d
-flex flex-column gap-1.5">
607         <!-- Will be populated dynamically -->
608     </div>
609 </div>
610 </div>
611     <div class="modal-footer py-1.5 px-3 border-secondary d-
flex justify-content-between" style="background: var(--surface
-2); border-color: var(--border-color) !important;">
612         <button class="btn btn-sm btn-outline-secondary rounde
d-pill px-3" id="btn-copy-sensor-report">
613             <i class="bi bi-clipboard me-1"></i> Копировать отчё
т
614         </button>
615         <div class="d-flex gap-2">
616             <button class="btn btn-sm btn-outline-info rounded-p
ill px-3" id="btn-re-analyze-sensors">
617                 <i class="bi bi-arrow-clockwise me-1"></i> Повтори
ть аудит
618             </button>
619             <button class="btn btn-sm btn-secondary rounded-pill
px-3" data-bs-dismiss="modal">Закрыть</button>
620         </div>
621     </div>
622 </div>
623 </div>
624 </div>
625
626 <!-- Process Statistics & Outliers Modal -->
627 <div class="modal fade" id="sysProcStatsModal" tabindex="-1" a
ria-labelledby="sysProcStatsModalLabel" aria-hidden="true">
628     <div class="modal-dialog modal-dialog-centered modal-xl moda
l-dialog-scrollable">

```

```

629     <div class="modal-content text-white border-secondary shadow-lg" style="background: var(--surface-1); border-color: var(--border-color) !important;">
630         <div class="modal-header py-2.5 px-3 border-secondary" style="background: var(--surface-2); border-color: var(--border-color) !important;">
631             <h6 class="modal-title d-flex align-items-center gap-2 mb-0" id="sysProcStatsModalLabel">
632                 <i class="bi bi-bar-chart-line-fill text-info"></i>
633                 <span class="fw-bold">Статистика процессов хоста (SQLite Агрегаты & Выбросы)</span>
634                 <span class="badge bg-info-subtle text-info border border-info ms-2" id="modal-proc-stats-count-badge">0 записей</span>
635             </h6>
636             <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
637         </div>
638
639         <div class="modal-body p-3" style="max-height: 75vh; overflow-y: auto;">
640             <!-- Top Toolbar -->
641             <div class="p-2.5 rounded mb-3 d-flex align-items-center justify-content-between flex-wrap gap-2" style="background: var(--surface-2); border: 1px solid var(--border-color);">
642                 <div class="d-flex align-items-center gap-2">
643                     <span class="fs-4">□ </span>
644                     <div>
645                         <div class="fw-bold small" style="color: var(--text-color);">Обобщение данных процессов и сохранение аномалий</div>
646                         <div class="small text-muted" style="font-size: 0.73rem;">Записи &gt; 2 мин обобщаются по средним показателям со всплесками. Данные &gt; 1 дня агрегируются в суточную историю.</div>
647                     </div>
648                 </div>
649                 <div class="d-flex align-items-center gap-2 flex-wra

```

```

p">
650         <input type="text" class="form-control form-control-sm" id="modal-proc-stats-filter" placeholder="Поиск процесса
        ..." style="max-width: 170px;">
651         <button class="btn btn-sm btn-outline-warning rounded-pill px-3" id="btn-trigger-proc-rollup" title="Принудительно обобщить записи старше 2 минут и 1 дня">
652             <i class="bi bi-lightning-charge me-1"></i>Обобщить сейчас
653         </button>
654         <button class="btn btn-sm btn-outline-info rounded-pill px-2.5" id="btn-refresh-proc-stats" title="Обновить статистику">
655             <i class="bi bi-arrow-clockwise"></i>
656         </button>
657     </div>
658 </div>
659
660     <!-- Tabs Navigation -->
661     <ul class="nav nav-pills mb-3 gap-2" id="procStatsTab" role="tablist">
662         <li class="nav-item" role="presentation">
663             <button class="nav-link active btn-sm rounded-pill px-3 py-1" id="tab-btn-proc-rollups" data-bs-toggle="pill" data-bs-target="#tab-pane-proc-rollups" type="button" role="tab">
664                 <i class="bi bi-clock-history me-1"></i>2-минутные обобщения
665             </button>
666         </li>
667         <li class="nav-item" role="presentation">
668             <button class="nav-link btn-sm rounded-pill px-3 py-1" id="tab-btn-proc-outliers" data-bs-toggle="pill" data-bs-target="#tab-pane-proc-outliers" type="button" role="tab">
669                 <i class="bi bi-exclamation-triangle-fill text-warning me-1"></i>Выбросы и пики нагрузки (<span id="proc-outliers-count">0</span>)
670             </button>

```



```

671         </li>
672         <li class="nav-item" role="presentation">
673             <button class="nav-link btn-sm rounded-pill px-3 p
y-1" id="tab-btn-proc-daily" data-bs-toggle="pill" data-bs-tar
get="#tab-pane-proc-daily" type="button" role="tab">
674                 <i class="bi bi-calendar3 me-1"></i>Суточная ста
тистика (> 1 дня)
675             </button>
676         </li>
677     </ul>
678
679     <!-- Tab Content -->
680     <div class="tab-content" id="procStatsTabContent">
681         <!-- 1. 2-min rollups -->
682         <div class="tab-pane fade show active" id="tab-pane-
proc-rollups" role="tabpanel">
683             <div class="table-responsive" style="max-height: 4
00px; overflow-y: auto;">
684                 <table class="table table-dark table-hover table
-sm align-middle font-monospace" style="font-size: 0.76rem;">
685                     <thead class="table-secondary text-dark sticky
-top">
686                         <tr>
687                             <th>Период</th>
688                             <th>Процесс</th>
689                             <th>PID</th>
690                             <th>Пользователь</th>
691                             <th class="text-end">Замеров</th>
692                             <th class="text-end">CPU ср.</th>
693                             <th class="text-end">CPU макс.</th>
694                             <th class="text-end">Память ср.</th>
695                             <th class="text-end">Память макс.</th>
696                             <th class="text-end">Выбросов</th>
697                         </tr>
698                     </thead>
699                     <tbody id="tbody-proc-rollups">
700                         <tr><td colspan="10" class="text-center py-3
text-muted">Загрузка обобщений...</td></tr>

```

```

701         </tbody>
702     </table>
703 </div>
704 </div>
705
706     <!-- 2. Outliers -->
707     <div class="tab-pane fade" id="tab-pane-proc-outlier
708 s" role="tabpanel">
709         <div class="table-responsive" style="max-height: 4
710 00px; overflow-y: auto;">
711             <table class="table table-dark table-hover table
712 -sm align-middle font-monospace" style="font-size: 0.76rem;">
713                 <thead class="table-secondary text-dark sticky
714 -top">
715                     <tr>
716                         <th>Время всплеска</th>
717                         <th>Процесс</th>
718                         <th>PID</th>
719                         <th class="text-end">CPU %</th>
720                         <th class="text-end">Память МВ</th>
721                         <th>Потоки</th>
722                         <th>Пользователь</th>
723                         <th>Детали аномалии</th>
724                     </tr>
725                 </thead>
726                 <tbody id="tbody-proc-outliers">
727                     <tr><td colspan="8" class="text-center py-3
728 text-muted">Загрузка выбросов...</td></tr>
729                 </tbody>
730             </table>
731         </div>
732     </div>
733
734     <!-- 3. Daily stats -->
735     <div class="tab-pane fade" id="tab-pane-proc-daily"
736 role="tabpanel">
737         <div class="table-responsive" style="max-height: 4
738 00px; overflow-y: auto;">

```

```

732 |         <table class="table table-dark table-hover table
      | -sm align-middle font-monospace" style="font-size: 0.76rem;">
733 |             <thead class="table-secondary text-dark sticky
      | -top">
734 |                 <tr>
735 |                     <th>Дата</th>
736 |                     <th>Процесс</th>
737 |                     <th class="text-end">Всего замеров</th>
738 |                     <th class="text-end">CPU взвеш. ср.</th>
739 |                     <th class="text-end">CPU пик</th>
740 |                     <th class="text-end">Память ср.</th>
741 |                     <th class="text-end">Память пик</th>
742 |                     <th class="text-end">Выбросов</th>
743 |                     <th>Последняя активность</th>
744 |                 </tr>
745 |             </thead>
746 |             <tbody id="tbody-proc-daily">
747 |                 <tr><td colspan="9" class="text-center py-3
      | text-muted">Загрузка суточной статистики...</td></tr>
748 |             </tbody>
749 |         </table>
750 |     </div>
751 | </div>
752 | </div>
753 | </div>
754 |
755 |     <div class="modal-footer py-2 px-3 border-secondary" sty
      | le="background: var(--surface-2); border-color: var(--border-c
      | olor) !important;">
756 |         <span class="small text-muted me-auto font-monospace"
      | id="modal-proc-rollup-status">Служба телеметрии пишет каждые 5
      | с</span>
757 |         <button class="btn btn-sm btn-secondary rounded-pill p
      | x-3" data-bs-dismiss="modal">Закрыть</button>
758 |     </div>
759 | </div>
760 | </div>
761 | </div>

```

```

762 |
763 | <!-- Telemetry & Polling Intervals Editor Modal (config.json)
    | -->
764 | <div class="modal fade" id="sysIntervalsModal" tabindex="-1" a
    | ria-labelledby="sysIntervalsModalLabel" aria-hidden="true">
765 |   <div class="modal-dialog modal-dialog-centered modal-lg moda
    | l-dialog-scrollable">
766 |     <div class="modal-content text-white border-secondary shad
    | ow-lg" style="background: var(--surface-1); border-color: var(
    | --border-color) !important;">
767 |       <div class="modal-header py-2.5 px-3 border-secondary" s
    | tyle="background: var(--surface-2); border-color: var(--border
    | -color) !important;">
768 |         <h6 class="modal-title d-flex align-items-center gap-2
    | mb-0" id="sysIntervalsModalLabel">
769 |           <i class="bi bi-stopwatch text-primary"></i>
770 |           <span class="fw-bold">Редактор интервалов опроса рес
    | урсов</span>
771 |           <span class="badge bg-secondary font-monospace" id="
    | modal-intervals-cfg-file">config.json</span>
772 |         </h6>
773 |         <button type="button" class="btn-close btn-close-white
    | " data-bs-dismiss="modal" aria-label="Close"></button>
774 |       </div>
775 |
776 |       <div class="modal-body p-3" style="max-height: 75vh; ove
    | rflow-y: auto;">
777 |         <!-- Alert message -->
778 |         <div id="modal-intervals-alert" class="alert d-none py
    | -2 px-3 small mb-3"></div>
779 |
780 |         <!-- Engine Status Strip -->
781 |         <div class="p-2.5 rounded mb-3 d-flex align-items-cent
    | er justify-content-between flex-wrap gap-2" style="background:
    | var(--surface-2); border: 1px solid var(--border-color);">
782 |           <div class="d-flex align-items-center gap-2">
783 |             <span class="fs-4">□ </span>
784 |           </div>

```

```

785 |         <div class="fw-bold small" style="color: var(--t
| ext-color);">Служба автологгирования и опроса телеметрии</div>
786 |         <div class="small text-muted" style="font-size:
| 0.73rem;">Интервалы сохраняются напрямую в активный файл конфи
| гурации и применяются без перезапуска сервера</div>
787 |         </div>
788 |     </div>
789 |     <div class="d-flex align-items-center gap-2">
790 |         <span class="badge bg-success" id="modal-intervals
| -engine-status">AutoLog Active</span>
791 |         <div class="form-check form-switch m-0">
792 |             <input class="form-check-input" type="checkbox"
| id="modal-intervals-global-enable" checked>
793 |             <label class="form-check-label small fw-semibold
| " for="modal-intervals-global-enable">Автологгирование</label>
794 |         </div>
795 |     </div>
796 | </div>
797 |
798 | <!-- Section 1: Live UI Refresh Interval -->
799 | <div class="card border-secondary mb-3" style="backgro
| und: var(--surface-2); border-color: var(--border-color) !impo
| rtant;">
800 |     <div class="card-header py-1.5 px-3 border-secondary
| d-flex align-items-center justify-content-between" style="bor
| der-color: var(--border-color) !important;">
801 |         <span class="small fw-bold text-info"><i class="bi
| bi-display me-1"></i> Интервал обновления Live-интерфейса вкл
| адки (UI)</span>
802 |         <span class="badge bg-dark text-info border border
| -secondary" id="modal-ui-refresh-badge">5 сек</span>
803 |     </div>
804 |     <div class="card-body p-3">
805 |         <div class="row g-2 align-items-center">
806 |             <div class="col-sm-7">
807 |                 <label class="form-label small text-muted mb-0
| ">Частота опроса сенсоров и оборудования в открытом окне брауз
| ера:</label>

```

```

808         </div>
809         <div class="col-sm-5">
810             <select class="form-select form-select-sm bg-d
ark text-white border-secondary" id="modal-ui-refresh-select">
811                 <option value="1">1 секунда (Высокая частота
)</option>
812                 <option value="2">2 секунды</option>
813                 <option value="3">3 секунды</option>
814                 <option value="5" selected>5 секунд (По умол
чанию)</option>
815                 <option value="10">10 секунд (Экономный режи
м)</option>
816                 <option value="30">30 секунд</option>
817             </select>
818         </div>
819     </div>
820 </div>
821 </div>
822
823     <!-- Section 2: Core Resource Loggers (config.json) --
>
824     <div class="card border-secondary mb-3" style="backgro
und: var(--surface-2); border-color: var(--border-color) !impo
rtant;">
825         <div class="card-header py-1.5 px-3 border-secondary
d-flex align-items-center justify-content-between" style="bor
der-color: var(--border-color) !important;">
826             <span class="small fw-bold text-primary"><i class=
"bi bi-hdd-stack me-1"></i> Интервалы опроса основных сенсоров
и ресурсов (config.json)</span>
827             <span class="badge bg-secondary font-monospace" st
yle="font-size: 0.68rem;">logging.loggers</span>
828         </div>
829         <div class="card-body p-3">
830             <div class="d-flex flex-column gap-2.5" id="modal-
core-loggers-container">
831                 <!-- Dynamically populated core loggers -->
832             </div>

```

```

833         </div>
834     </div>
835
836     <!-- Section 3: Other Services Loggers Collapsible -->
837     <div class="card border-secondary mb-3" style="background: var(--surface-2); border-color: var(--border-color) !important;">
838         <div class="card-header py-1.5 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
839             <a class="small fw-bold text-light text-decoration-none d-flex align-items-center gap-1" data-bs-toggle="collapse" href="#modal-other-loggers-collapse" role="button" aria-expanded="false">
840                 <i class="bi bi-chevron-down"></i> Все остальные
841                 логгеры и службы системы (<span id="modal-other-loggers-count">0</span>)
842             </a>
843             <span class="small text-muted" style="font-size: 0.72rem;">Windows SysAdmin, Defender, StartUp, Net...</span>
844         </div>
845         <div class="collapse" id="modal-other-loggers-collapse">
846             <div class="card-body p-3 border-top border-secondary" style="border-color: var(--border-color) !important;">
847                 <div class="d-flex flex-column gap-2" id="modal-other-loggers-container">
848                     <!-- Dynamically populated other loggers -->
849                 </div>
850             </div>
851         </div>
852
853     <!-- Section 4: Default fallback interval -->
854     <div class="p-2.5 rounded d-flex align-items-center justify-content-between flex-wrap gap-2" style="background: var(--bg-color); border: 1px solid var(--border-color);">
855         <div class="d-flex align-items-center gap-2">

```

```

856 |         <i class="bi bi-clock-history text-muted"></i>
857 |         <span class="small" style="color: var(--text-color
| );">Интервал по умолчанию (default_interval):</span>
858 |     </div>
859 |     <div style="min-width: 170px;">
860 |         <select class="form-select form-select-sm bg-dark
| text-white border-secondary" id="modal-intervals-default-selec
| t">
861 |             <option value="5 seconds">5 seconds (5 сек)</opt
| ion>
862 |             <option value="10 seconds">10 seconds (10 сек)</
| option>
863 |             <option value="30 seconds">30 seconds (30 сек)</
| option>
864 |             <option value="1 minute" selected>1 minute (1 ми
| н)</option>
865 |             <option value="5 minutes">5 minutes (5 мин)</opt
| ion>
866 |             <option value="10 minutes">10 minutes (10 мин)</
| option>
867 |             <option value="1 hour">1 hour (1 час)</option>
868 |         </select>
869 |     </div>
870 | </div>
871 | </div>
872 |
873 |     <div class="modal-footer py-2 px-3 border-secondary d-fl
| ex justify-content-between" style="background: var(--surface-2
| ); border-color: var(--border-color) !important;">
874 |         <button class="btn btn-sm btn-outline-secondary rounde
| d-pill px-3" id="btn-modal-intervals-refresh" title="Перезагру
| зить текущую конфигурацию из файла">
875 |             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
876 |         </button>
877 |         <div class="d-flex gap-2">
878 |             <button class="btn btn-sm btn-secondary rounded-pill
| px-3" data-bs-dismiss="modal">Отмена</button>
879 |             <button class="btn btn-sm btn-primary rounded-pill p

```



```

x-4 fw-bold" id="btn-modal-intervals-save">
880     <i class="bi bi-check2-circle me-1"></i> Сохранить
    в config.json
881     </button>
882 </div>
883 </div>
884 </div>
885 </div>
886 </div>
887
888 <!-- Real-Time Watcher Multi-Folder Selection & Exclusions Mod
al -->
889 <div class="modal fade" id="modal-sys-watch-folders" tabindex=
"-1" aria-labelledby="modalSysWatchFoldersLabel" aria-hidden="
true">
890     <div class="modal-dialog modal-dialog-centered modal-lg moda
l-dialog-scrollable">
891         <div class="modal-content text-white border-secondary shad
ow-lg" style="background: var(--surface-1); border-color: var(
--border-color) !important;">
892             <div class="modal-header py-2.5 px-3 border-secondary" s
tyle="background: var(--surface-2); border-color: var(--border
-color) !important;">
893                 <h6 class="modal-title d-flex align-items-center gap-2
mb-0" id="modalSysWatchFoldersLabel">
894                     <i class="bi bi-folder2-open text-warning fs-5"></i>
895                     <span class="fw-bold">Отслеживаемые папки и исключен
ия (Live Watcher)</span>
896                     <span class="badge bg-info-subtle text-info font-mon
ospace ms-1" style="font-size: 0.7rem;">ReadDirectoryChangesW<
/span>
897                 </h6>
898                 <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal" aria-label="Close"></button>
899             </div>
900             <div class="modal-body p-3" style="max-height: 75vh; ove
rflow-y: auto;">
901                 <!-- Nav Tabs -->

```

```

902 |         <ul class="nav nav-pills nav-fill mb-3" id="pills-sys-
| watcher-tabs" role="tablist">
903 |             <li class="nav-item" role="presentation">
904 |                 <button class="nav-link active py-1 px-3 small fw-
| bold" id="tab-btn-watch-folders" data-bs-toggle="pill" data-bs
| -target="#tab-pane-watch-folders" type="button" role="tab">
905 |                     <i class="bi bi-folder2-open me-1.5 text-warning
| "></i>Папки мониторинга (<span id="modal-watch-dirs-tab-count"
| >1</span>)
906 |                 </button>
907 |             </li>
908 |             <li class="nav-item" role="presentation">
909 |                 <button class="nav-link py-1 px-3 small fw-bold" i
| d="tab-btn-watch-exclusions" data-bs-toggle="pill" data-bs-tar
| get="#tab-pane-watch-exclusions" type="button" role="tab">
910 |                     <i class="bi bi-funnel me-1.5 text-warning"></i>
| Исключения и фильтры (<span id="modal-watch-exclusions-tab-cou
| nt">0</span>)
911 |                 </button>
912 |             </li>
913 |         </ul>
914 |
915 |         <div class="tab-content">
916 |             <!-- Tab 1: Monitored Folders -->
917 |             <div class="tab-pane fade show active" id="tab-pane-
| watch-folders" role="tabpanel">
918 |                 <!-- Alert feedback -->
919 |                 <div id="modal-watch-dirs-alert" class="alert aler
| t-success py-2 px-3 small mb-3 d-none"></div>
920 |
921 |                 <!-- Section 1: Active Monitored Folders List -->
922 |                 <div class="card border-secondary mb-3" style="bac
| kground: var(--surface-2); border-color: var(--border-color) !
| important;">
923 |                     <div class="card-header py-2 px-3 border-seconda
| ry d-flex align-items-center justify-content-between" style="b
| order-color: var(--border-color) !important;">
924 |                         <div class="d-flex align-items-center gap-2">

```

```

925 |             <i class="bi bi-eye text-success"></i>
926 |             <span class="small fw-bold text-light">Актив
ные папки мониторинга</span>
927 |             <span class="badge bg-primary rounded-pill"
id="modal-watch-dirs-count">0</span>
928 |         </div>
929 |         <span class="small text-muted" style="font-siz
e: 0.72rem;">События агрегируются в реальном времени</span>
930 |     </div>
931 |     <div class="card-body p-2">
932 |         <div class="d-flex flex-column gap-1.5" id="mo
dal-watch-dirs-active-list" style="max-height: 160px; overflow
-y: auto;">
933 |             <!-- Dynamically populated active watched di
rectories -->
934 |         </div>
935 |     </div>
936 | </div>
937 |
938 | <!-- Section 2: Quick System Presets -->
939 | <div class="mb-3">
940 |     <div class="small fw-bold text-muted mb-1.5"><i
class="bi bi-lightning-charge text-warning me-1"></i>Быстрые с
истемные каталоги:</div>
941 |     <div class="d-flex flex-wrap gap-1.5" id="modal-
watch-presets-bar">
942 |         <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 sys-preset-btn" data-preset="projec
t">
943 |             □ Проект AI-Breadboard
944 |         </button>
945 |         <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 sys-preset-btn" data-preset="downlo
ads">
946 |             □ Загрузки (Downloads)
947 |         </button>
948 |         <button class="btn btn-xs btn-outline-secondar
y rounded-pill px-2.5 py-1 sys-preset-btn" data-preset="deskto

```

```

p">
949         □ Рабочий стол (Desktop)
950     </button>
951     <button class="btn btn-xs btn-outline-secondary rounded-pill px-2.5 py-1 sys-preset-btn" data-preset="temp">
952         □ Temp
953     </button>
954 </div>
955 </div>
956
957 <!-- Section 3: System Folder Explorer -->
958 <div class="card border-secondary" style="background: var(--surface-2); border-color: var(--border-color) !important;">
959     <div class="card-header py-2 px-3 border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2" style="border-color: var(--border-color) !important;">
960         <div class="d-flex align-items-center gap-2">
961             <i class="bi bi-compass text-info"></i>
962             <span class="small fw-bold text-light">Проводник файлов и папок</span>
963         </div>
964         <!-- Drives selection bar -->
965         <div class="d-flex align-items-center gap-1" id="modal-watch-drives-bar">
966             <!-- Drives loaded dynamically -->
967         </div>
968     </div>
969     <div class="card-body p-2.5">
970         <!-- Path bar & Search/Input -->
971         <div class="input-group input-group-sm mb-2">
972             <button class="btn btn-outline-secondary" type="button" id="btn-modal-folder-up" title="Перейти на уровень выше">
973                 <i class="bi bi-arrow-up-circle"></i> Вверх
974             </button>
975             <input type="text" class="form-control bg-da

```

```

rk text-white border-secondary font-monospace" id="modal-folde
r-path-input" placeholder="C:\..." style="font-size: 0.8rem;">
976         <button class="btn btn-outline-info" type="b
utton" id="btn-modal-folder-go" title="Открыть указанный путь"
>
977             <i class="bi bi-arrow-right-circle me-1"><
/i> Перейти
978         </button>
979         <button class="btn btn-success" type="button
" id="btn-modal-folder-add-current" title="Добавить эту папку
в мониторинг">
980             <i class="bi bi-plus-circle me-1"></i> + Д
обавить папку
981         </button>
982     </div>
983
984     <!-- Breadcrumbs navigation -->
985     <div class="d-flex align-items-center flex-wra
p gap-1 p-1.5 rounded mb-2 font-monospace small" id="modal-fol
der-breadcrumbs" style="background: var(--bg-color); font-size
: 0.75rem;">
986         <!-- Breadcrumbs injected dynamically -->
987     </div>
988
989     <!-- Subdirectories list -->
990     <div class="border border-secondary rounded p-
1" style="background: var(--bg-color); max-height: 200px; over
flow-y: auto;">
991         <div id="modal-folder-browser-list" class="d
-flex flex-column gap-1">
992             <div class="text-center text-muted small p
y-3">Загрузка структуры каталогов...</div>
993         </div>
994     </div>
995 </div>
996 </div>
997 </div>
998

```

```

999 |         <!-- Tab 2: Exclusions & Filters -->
1000 |         <div class="tab-pane fade" id="tab-pane-watch-exclus
| ions" role="tabpanel">
1001 |             <!-- Alert feedback -->
1002 |             <div id="modal-exclusions-alert" class="alert aler
| t-success py-2 px-3 small mb-3 d-none"></div>
1003 |
1004 |             <!-- Master Toggle & Statistics -->
1005 |             <div class="card border-secondary mb-3" style="bac
| kground: var(--surface-2); border-color: var(--border-color) !
| important;">
1006 |                 <div class="card-body p-2.5 d-flex align-items-c
| enter justify-content-between flex-wrap gap-2">
1007 |                     <div class="d-flex align-items-center gap-2">
1008 |                         <div class="form-check form-switch mb-0">
1009 |                             <input class="form-check-input" type="chec
| kbox" role="switch" id="switch-exclusions-active" checked styl
| e="cursor: pointer;">
1010 |                             <label class="form-check-label fw-bold tex
| t-light small" for="switch-exclusions-active">Фильтрация исклю
| чений активна</label>
1011 |                         </div>
1012 |                         <span class="badge bg-success" id="badge-exc
| lusions-status">ВКЛ</span>
1013 |                     </div>
1014 |                     <div class="text-muted small" style="font-size
| : 0.75rem;">
1015 |                         Отфильтровано спам-событий: <span class="fw-
| bold text-warning font-monospace" id="modal-exclusions-filtered-
| total">0</span>
1016 |                     </div>
1017 |                 </div>
1018 |             </div>
1019 |
1020 |             <!-- Quick Exclusions Presets -->
1021 |             <div class="mb-3">
1022 |                 <div class="small fw-bold text-muted mb-1.5"><i
| class="bi bi-lightning-charge text-warning me-1"></i>Быстрые п

```

```

ресеты фильтрации:</div>
1023         <div class="d-flex flex-wrap gap-1.5">
1024             <button class="btn btn-xs btn-outline-secondary
rounded-pill px-2.5 py-1 sys-ex-preset-btn" data-preset="log
s" title="Исключить папки логов, .csv и .log">
1025                 □ Логи & Телеметрия
1026             </button>
1027             <button class="btn btn-xs btn-outline-secondary
rounded-pill px-2.5 py-1 sys-ex-preset-btn" data-preset="sea
rch" title="Исключить системную службу SearchIndexer.exe и Win
dows.db">
1028                 □ SearchIndexer.exe & Windows.db
1029             </button>
1030             <button class="btn btn-xs btn-outline-secondary
rounded-pill px-2.5 py-1 sys-ex-preset-btn" data-preset="tem
p" title="Исключить .tmp, ~$* и папку Temp">
1031                 □ Temp & Временные (.tmp)
1032             </button>
1033             <button class="btn btn-xs btn-outline-secondary
rounded-pill px-2.5 py-1 sys-ex-preset-btn" data-preset="dev
" title="Исключить .git, node_modules, __pycache__">
1034                 □ Разработка (.git, node_modules)
1035             </button>
1036         </div>
1037     </div>
1038
1039     <!-- Add Exclusion Rule Form -->
1040     <div class="card border-secondary mb-3" style="bac
kground: var(--surface-2); border-color: var(--border-color) !
important;">
1041         <div class="card-header py-2 px-3 border-seconda
ry d-flex align-items-center gap-2" style="border-color: var(-
-border-color) !important;">
1042             <i class="bi bi-plus-circle text-info"></i>
1043             <span class="small fw-bold text-light">Добавит
ь правило исключения</span>
1044         </div>
1045         <div class="card-body p-2.5">

```

```

1046 |         <div class="row g-2 align-items-center">
1047 |             <div class="col-md-4">
1048 |                 <select class="form-select form-select-sm
bg-dark text-white border-secondary" id="select-exclusion-cate
gory">
1049 |                     <option value="paths">□ Папка / Путь (P
ath)</option>
1050 |                     <option value="extensions">□ Расширение
файла (.ext)</option>
1051 |                     <option value="patterns">□ Шаблон имени
(*pattern*)</option>
1052 |                     <option value="processes">□ Имя программ
ы (Process.exe)</option>
1053 |                 </select>
1054 |             </div>
1055 |             <div class="col-md-6">
1056 |                 <input type="text" class="form-control for
m-control-sm bg-dark text-white border-secondary font-monospac
e" id="input-exclusion-value" placeholder="Например: AppData\L
ocal\Temp, .csv, SearchIndexer.exe">
1057 |             </div>
1058 |             <div class="col-md-2">
1059 |                 <button class="btn btn-sm btn-info w-100 f
w-bold rounded-pill" id="btn-add-exclusion">
1060 |                     <i class="bi bi-plus-lg me-1"></i> Добав
ИТЬ
1061 |                 </button>
1062 |             </div>
1063 |         </div>
1064 |     </div>
1065 | </div>
1066 |
1067 | <!-- Grouped Cards of Active Exclusions -->
1068 | <div class="row g-2">
1069 |     <!-- Paths -->
1070 |     <div class="col-md-6">
1071 |         <div class="card border-secondary h-100" style
="background: var(--surface-2); border-color: var(--border-col

```



```

or) !important; ">
1072         <div class="card-header py-1.5 px-2.5 border
-secondary d-flex justify-content-between align-items-center"
style="border-color: var(--border-color) !important; ">
1073         <span class="small fw-bold text-light"><i
class="bi bi-folder-x text-warning me-1"></i>Папки и пути</spa
n>
1074         <span class="badge bg-secondary font-monos
pace" id="badge-ex-paths-count">0</span>
1075         </div>
1076         <div class="card-body p-2" id="list-ex-paths
" style="max-height: 120px; overflow-y: auto; "></div>
1077         </div>
1078     </div>
1079     <!-- Extensions -->
1080     <div class="col-md-6">
1081         <div class="card border-secondary h-100" style
="background: var(--surface-2); border-color: var(--border-col
or) !important; ">
1082         <div class="card-header py-1.5 px-2.5 border
-secondary d-flex justify-content-between align-items-center"
style="border-color: var(--border-color) !important; ">
1083         <span class="small fw-bold text-light"><i
class="bi bi-file-earmark-x text-info me-1"></i>Расширения фай
лов</span>
1084         <span class="badge bg-secondary font-monos
pace" id="badge-ex-extensions-count">0</span>
1085         </div>
1086         <div class="card-body p-2" id="list-ex-exten
sions" style="max-height: 120px; overflow-y: auto; "></div>
1087         </div>
1088     </div>
1089     <!-- Patterns -->
1090     <div class="col-md-6">
1091         <div class="card border-secondary h-100" style
="background: var(--surface-2); border-color: var(--border-col
or) !important; ">
1092         <div class="card-header py-1.5 px-2.5 border

```

```

| -secondary d-flex justify-content-between align-items-center"
| style="border-color: var(--border-color) !important;">
1093 |         <span class="small fw-bold text-light"><i
| class="bi bi-asterisk text-primary me-1"></i>Шаблоны (Wildcard
| s)</span>
1094 |         <span class="badge bg-secondary font-monos
| pace" id="badge-ex-patterns-count">0</span>
1095 |         </div>
1096 |         <div class="card-body p-2" id="list-ex-patte
| rns" style="max-height: 120px; overflow-y: auto;"></div>
1097 |         </div>
1098 |     </div>
1099 |     <!-- Processes -->
1100 |     <div class="col-md-6">
1101 |         <div class="card border-secondary h-100" style
| ="background: var(--surface-2); border-color: var(--border-col
| or) !important;">
1102 |             <div class="card-header py-1.5 px-2.5 border
| -secondary d-flex justify-content-between align-items-center"
| style="border-color: var(--border-color) !important;">
1103 |                 <span class="small fw-bold text-light"><i
| class="bi bi-cpu text-danger me-1"></i>Программы (Процессы)</s
| pan>
1104 |                 <span class="badge bg-secondary font-monos
| pace" id="badge-ex-processes-count">0</span>
1105 |                 </div>
1106 |                 <div class="card-body p-2" id="list-ex-proce
| sses" style="max-height: 120px; overflow-y: auto;"></div>
1107 |                 </div>
1108 |             </div>
1109 |         </div>
1110 |     </div>
1111 | </div>
1112 | </div>
1113 |
1114 |     <div class="modal-footer py-2 px-3 border-secondary d-fl
| ex justify-content-between" style="background: var(--surface-2
| ); border-color: var(--border-color) !important;">

```

```
1115 |         <span class="small text-muted" id="modal-watch-dirs-fo
      | oter-hint">Выберите папки и нажмите «Применить»</span>
1116 |         <div class="d-flex gap-2">
1117 |             <button class="btn btn-sm btn-secondary rounded-pill
      | px-3" data-bs-dismiss="modal">Заккрыть</button>
1118 |             <button class="btn btn-sm btn-primary rounded-pill p
      | x-4 fw-bold" id="btn-modal-watch-dirs-apply">
1119 |                 <i class="bi bi-check2-circle me-1"></i> Применить
      | и запустить
1120 |             </button>
1121 |         </div>
1122 |     </div>
1123 | </div>
1124 | </div>
1125 | </div>
1126 |
1127 |
1128 |
```

main.js

```
1 // System & Hardware Inspector Tab JS Module
2 (function() {
3   let sysWs = null;
4   let isSysInitialized = false;
5   let isSysPaused = false;
6   let isNetPaused = false;
7   let currentNetFilter = 'internet';
8   let _currentUiRefreshSeconds = 5;
9   let cachedProcStats = null;
10
11   async function fetchProcessesFromDb() {
12     if (isSysPaused) return;
13     try {
14       const res = await fetch('/api/v1/system/processes?limit=
15       35&source=db');
16       if (res.ok) {
17         const procs = await res.json();
18         if (!latestTelemetrySnapshot) {
19           latestTelemetrySnapshot = { top_processes: [] };
20         }
21         latestTelemetrySnapshot.top_processes = procs;
22         renderProcessTable();
23
24         const timeBadge = document.getElementById('sys-proc-up
25         dated-time');
26         if (timeBadge) {
27           const now = new Date();
28           timeBadge.textContent = now.toLocaleTimeString();
29         }
30       } catch (e) {
31         console.warn('[SystemInspectorTab] Ошибка получения проц
32         ессов из базы данных SQLite:', e);
33       }
34     }
35   }
```

```
34 | async function fetchLhmSensors() {
35 |     const container = document.getElementById('sys-lhm-sensors
    | -container');
36 |     const badgeStatus = document.getElementById('sys-lhm-statu
    | s-badge');
37 |     const badgeCount = document.getElementById('sys-lhm-sensor
    | s-count');
38 |     const btnLaunch = document.getElementById('btn-sys-launch-
    | lhm');
39 |     if (!container) return;
40 |
41 |     try {
42 |         // 1. Проверяем статус LHM
43 |         let lhmStatus = null;
44 |         try {
45 |             const sRes = await fetch('/api/v1/lhm/status');
46 |             if (sRes.ok) lhmStatus = await sRes.json();
47 |         } catch {}
48 |
49 |         isLhmRunning = lhmStatus && lhmStatus.is_running;
50 |
51 |         if (btnLaunch) {
52 |             if (!isLhmRunning && lhmStatus && lhmStatus.is_binary_
    | available) {
53 |                 btnLaunch.classList.remove('d-none');
54 |             } else {
55 |                 btnLaunch.classList.add('d-none');
56 |             }
57 |         }
58 |
59 |         let sensorsList = [];
60 |
61 |         if (isLhmRunning) {
62 |             if (badgeStatus) {
63 |                 badgeStatus.textContent = '● LHM Active';
64 |                 badgeStatus.className = 'badge bg-success';
65 |             }
66 |             showLhmFallbackNotice(false);
```

```

67 |         const mRes = await fetch('/api/v1/lhm/metrics');
68 |         if (mRes.ok) {
69 |             const mData = await mRes.json();
70 |             sensorsList = mData.sensors || [];
71 |         }
72 |     } else {
73 |         // Fallback: читаем последний замер сенсоров из БД SQLite (аналогично «Топ процессов»)
74 |         if (badgeStatus) {
75 |             badgeStatus.textContent = '● БД SQLite';
76 |             badgeStatus.className = 'badge bg-primary';
77 |         }
78 |         try {
79 |             const dbRes = await fetch('/api/v1/system/sensors?source=db');
80 |             if (dbRes.ok) {
81 |                 const dbData = await dbRes.json();
82 |                 if (Array.isArray(dbData) && dbData.length > 0) {
83 |                     // Данные уже нормализованы бэкендом – минимальный маппинг
84 |                     sensorsList = dbData.map(s => ({
85 |                         id: s.id || s.sensor_id || s.sensor_name,
86 |                         hardware_name: s.hardware_name || 'System Hardware',
87 |                         hardware_type: s.hardware_type || 'system',
88 |                         sensor_category: s.sensor_category || 'General',
89 |                         sensor_name: s.sensor_name || 'Sensor',
90 |                         value_raw: s.value_raw || `${s.value_numeric ? 0} ${s.unit || ''}`.trim(),
91 |                         value_numeric: s.value_numeric ?? 0,
92 |                         unit: s.unit || '',
93 |                     }));
94 |                     showLhmFallbackNotice(false);
95 |                 }
96 |             }
97 |         } catch { /* продолжаем к live-fallback */ }
98 |

```

```

99      // Live-fallback: если БД пуста (телеметрия ещё не зап
ущена)
100    if (sensorsList.length === 0) {
101      if (badgeStatus) {
102        badgeStatus.textContent = 'WMI/GPU (live)';
103        badgeStatus.className = 'badge bg-secondary';
104      }
105      // Показываем уведомление о переключении на live
106      showLhmFallbackNotice(true);
107      try {
108        const liveRes = await fetch('/api/v1/system/sensor
s?source=live');
109        if (liveRes.ok) {
110          const raw = await liveRes.json();
111          sensorsList = (raw || []).map(s => ({
112            id: s.sensor_id || s.id || s.name,
113            hardware_name: s.hardware_name || (s.category
? s.category.toUpperCase() : 'System Hardware'),
114            hardware_type: s.hardware_type || 'system',
115            sensor_category: s.sensor_category || s.catego
ry || 'Temperatures',
116            sensor_name: s.sensor_name || s.name,
117            value_raw: s.value_raw || (s.unit ? `${s.value
} ${s.unit}` : `${s.value}`),
118            value_numeric: s.value_numeric ?? (typeof s.va
lue === 'number' ? s.value : parseFloat(s.value)),
119            unit: s.unit || '',
120          }));
121        }
122      } catch (liveErr) {
123        console.warn('[SystemInspectorTab] Live-fallback с
енсоров недоступен:', liveErr);
124      }
125    }
126  }
127
128  cachedSensors = sensorsList;
129  if (badgeCount) {

```

```
130 |         badgeCount.textContent = `${sensorsList.length} шт.`;
131 |     }
132 |
133 |     renderSensors();
134 | } catch (e) {
135 |     console.warn('[SystemInspectorTab] Failed to fetch LHM s
sensors:', e);
136 |     if (container) {
137 |         container.innerHTML = `<div class="text-center py-4 te
xt-muted small">Датчики опрашиваются... (${e.message})</div>`;
138 |     }
139 | }
140 | }
141 |
142 | /** Показывает или скрывает предупреждение о live-fallback п
ри пустой БД сенсоров. */
143 | function showLhmFallbackNotice(visible) {
144 |     const notice = document.getElementById('sys-lhm-fallback-n
otice');
145 |     if (!notice) return;
146 |     if (visible) {
147 |         notice.classList.remove('d-none');
148 |     } else {
149 |         notice.classList.add('d-none');
150 |     }
151 | }
152 |
153 |
154 | if (!container) return;
155 |
156 | try {
157 |     // 1. Проверяем статус LHM
158 |     let lhmStatus = null;
159 |     try {
160 |         const sRes = await fetch('/api/v1/lhm/status');
161 |         if (sRes.ok) lhmStatus = await sRes.json();
162 |     } catch {}
163 | }
```



```

164 |         isLhmRunning = lhmStatus && lhmStatus.is_running;
165 |
166 |         if (btnLaunch) {
167 |             if (!isLhmRunning && lhmStatus && lhmStatus.is_binary_
available) {
168 |                 btnLaunch.classList.remove('d-none');
169 |             } else {
170 |                 btnLaunch.classList.add('d-none');
171 |             }
172 |         }
173 |
174 |         let sensorsList = [];
175 |
176 |         if (isLhmRunning) {
177 |             if (badgeStatus) {
178 |                 badgeStatus.textContent = '● LHM Active';
179 |                 badgeStatus.className = 'badge bg-success';
180 |             }
181 |             const mRes = await fetch('/api/v1/lhm/metrics');
182 |             if (mRes.ok) {
183 |                 const mData = await mRes.json();
184 |                 sensorsList = mData.sensors || [];
185 |             }
186 |         } else {
187 |             // Fallback: читаем последний замер сенсоров из БД SQL
ite (аналогично «Топ процессов»)
188 |             if (badgeStatus) {
189 |                 badgeStatus.textContent = '● БД SQLite';
190 |                 badgeStatus.className = 'badge bg-primary';
191 |             }
192 |             try {
193 |                 const dbRes = await fetch('/api/v1/system/sensors?so
urce=db');
194 |                 if (dbRes.ok) {
195 |                     const dbData = await dbRes.json();
196 |                     if (Array.isArray(dbData) && dbData.length > 0) {
197 |                         // Данные уже нормализованы бэкендом – минимальн
ый маппинг

```

```

198         sensorsList = dbData.map(s => ({
199             id: s.id || s.sensor_id || s.sensor_name,
200             hardware_name: s.hardware_name || 'System Hard
ware',
201             hardware_type: s.hardware_type || 'system',
202             sensor_category: s.sensor_category || 'General
',
203             sensor_name: s.sensor_name || 'Sensor',
204             value_raw: s.value_raw || `${s.value_numeric ?
? 0} ${s.unit || ''}`.trim(),
205             value_numeric: s.value_numeric ?? 0,
206             unit: s.unit || '',
207         }));
208     }
209 }
210 } catch { /* продолжаем к live-fallback */ }
211
212 // Live-fallback: если БД пуста (телеметрия ещё не зап
ущена)
213 if (sensorsList.length === 0) {
214     if (badgeStatus) {
215         badgeStatus.textContent = 'WMI/GPU (live)';
216         badgeStatus.className = 'badge bg-secondary';
217     }
218     try {
219         const liveRes = await fetch('/api/v1/system/sensor
s?source=live');
220         if (liveRes.ok) {
221             const raw = await liveRes.json();
222             sensorsList = (raw || []).map(s => ({
223                 id: s.sensor_id || s.id || s.name,
224                 hardware_name: s.hardware_name || (s.category
? s.category.toUpperCase() : 'System Hardware'),
225                 hardware_type: s.hardware_type || 'system',
226                 sensor_category: s.sensor_category || s.catego
ry || 'Temperatures',
227                 sensor_name: s.sensor_name || s.name,
228                 value_raw: s.value_raw || (s.unit ? `${s.value

```

```

    } ${s.unit}` : `${s.value}`),
229         value_numeric: s.value_numeric ?? (typeof s.va
    lue === 'number' ? s.value : parseFloat(s.value)),
230         unit: s.unit || '',
231     }));
232     }
233     } catch (liveErr) {
234         console.warn('[SystemInspectorTab] Live-fallback c
    енсоров недоступен:', liveErr);
235     }
236     }
237     }
238
239     cachedSensors = sensorsList;
240     if (badgeCount) {
241         badgeCount.textContent = `${sensorsList.length} шт.`;
242     }
243
244     renderSensors();
245     } catch (e) {
246         console.warn('[SystemInspectorTab] Failed to fetch LHM s
    ensors:', e);
247         if (container) {
248             container.innerHTML = `<div class="text-center py-4 te
    xt-muted small">Датчики опрашиваются... (${e.message})</div>`;
249         }
250     }
251 }
252
253 function renderSensors() {
254     const container = document.getElementById('sys-lhm-sensors
    -container');
255     if (!container) return;
256
257     const searchInput = document.getElementById('sys-lhm-searc
    h');
258     const filterText = (searchInput ? searchInput.value : '').
    trim().toLowerCase();

```

```
259 |
260 |     if (!Array.isArray(cachedSensors) || cachedSensors.length
    |     === 0) {
261 |         container.innerHTML = `
262 |             <div class="text-center py-4 text-muted small">
263 |                 <div>Сенсоры LibreHardwareMonitor не обнаружены</div>
    |             >
264 |                 <div class="mt-2 text-muted" style="font-size: 0.72r
    | em;">Убедитесь, что LHM запущен с правами администратора и вкл
    | ючен Web Server (:8085)</div>
265 |                 </div>
266 |             `;
267 |         return;
268 |     }
269 |
270 |     const filtered = cachedSensors.filter(s => {
271 |         // Category filter
272 |         if (currentSensorCategory !== 'all') {
273 |             const catLower = (s.sensor_category || '').toLowerCase
    | ();
274 |             const typeLower = (s.hardware_type || '').toLowerCase(
    | );
275 |             if (currentSensorCategory === 'temperature' && !catLow
    | er.includes('temp')) return false;
276 |             if (currentSensorCategory === 'load' && !catLower.incl
    | udes('load')) return false;
277 |             if (currentSensorCategory === 'fan' && !catLower.inclu
    | des('fan') && !catLower.includes('control')) return false;
278 |             if (currentSensorCategory === 'voltage' && !catLower.i
    | ncludes('volt') && !catLower.includes('power')) return false;
279 |             if (currentSensorCategory === 'clock' && !catLower.inc
    | ludes('clock')) return false;
280 |         }
281 |
282 |         // Text search
283 |         if (filterText) {
284 |             const hName = (s.hardware_name || '').toLowerCase();
285 |             const sName = (s.sensor_name || '').toLowerCase();
```

```

286 |         const sCat = (s.sensor_category || '').toLowerCase();
287 |         const valStr = String(s.value_raw || '').toLowerCase()
    | ;
288 |         return hName.includes(filterText) || sName.includes(fi
    | lterText) || sCat.includes(filterText) || valStr.includes(filt
    | erText);
289 |     }
290 |     return true;
291 | });
292 |
293 |     if (filtered.length === 0) {
294 |         container.innerHTML = `<div class="text-center py-4 text
    | -muted small">По выбранному фильтру сенсоров не найдено</div>`
    | ;
295 |         return;
296 |     }
297 |
298 |     // Group sensors by Hardware Name or Category
299 |     const groups = {};
300 |     filtered.forEach(s => {
301 |         const grpKey = s.hardware_name || s.sensor_category || '
    | Оборудование';
302 |         if (!groups[grpKey]) groups[grpKey] = [];
303 |         groups[grpKey].push(s);
304 |     });
305 |
306 |     container.innerHTML = Object.entries(groups).map(([grpName
    | , items]) => {
307 |         const itemsHtml = items.map(s => {
308 |             const cat = (s.sensor_category || '').toLowerCase();
309 |             const num = s.value_numeric;
310 |             const valRaw = s.value_raw || `${num || 0} ${s.unit ||
    | ''}`;
311 |
312 |             let valClass = 'text-white';
313 |             let badgeHtml = '';
314 |
315 |             if (cat.includes('temp')) {

```

```

316         if (num > 75) valClass = 'text-danger fw-bold';
317         else if (num > 60) valClass = 'text-warning fw-bold'
318         ;
319         else valClass = 'text-info';
320         badgeHtml = `<span class="badge ${num > 75 ? 'bg-dan
321 ger' : num > 60 ? 'bg-warning text-dark' : 'bg-info text-dark'
322 }" style="font-size: 0.65rem;">t°</span>`;
323     } else if (cat.includes('load')) {
324         const pct = Math.min(100, Math.max(0, num || 0));
325         badgeHtml = `
326         <div class="progress" style="width: 45px; height:
327 4px; background: rgba(255,255,255,0.1);">
328         <div class="progress-bar ${pct > 80 ? 'bg-danger
329 ' : pct > 50 ? 'bg-warning' : 'bg-primary'}" style="width: ${p
330 ct}%"></div>
331     </div>
332     `;
333     } else if (cat.includes('fan')) {
334         valClass = 'text-success';
335     } else if (cat.includes('clock')) {
336         valClass = 'text-primary';
337     }
338     return `
339     <div class="sys-sensor-row">
340     <span class="sys-sensor-name" title="${escapeHtml(
341 s.sensor_name)} (${escapeHtml(s.sensor_category)})">
342     ${escapeHtml(s.sensor_name)}
343     </span>
344     <div class="d-flex align-items-center gap-1.5 ms-a
345 uto">
346     ${badgeHtml}
347     <span class="sys-sensor-value ${valClass}">${esc
348 apeHtml(valRaw)}</span>
349     </div>
350     </div>
351     `;
352     }).join('');

```

```

345 |
346 |     return `
347 |         <div class="sys-sensor-group">
348 |             <div class="sys-sensor-group-title">
349 |                 <span><i class="bi bi-cpu me-1 text-info"></i> ${e
350 | scapeHtml(grpName)}</span>
351 |                 <span class="badge bg-dark border border-secondary
352 | text-muted" style="font-size: 0.65rem;">${items.length}</span
353 | >
354 |             </div>
355 |             <div>
356 |                 ${itemsHtml}
357 |             </div>
358 |         </div>
359 |     `;
360 |     }).join('');
361 | }
362 |
363 | let lastAnalysisReportText = '';
364 |
365 | async function analyzeSensorsAndReport() {
366 |     const modalEl = document.getElementById('sysSensorAnalysis
367 | Modal');
368 |     let modal = null;
369 |     if (modalEl && window.bootstrap && window.bootstrap.Modal)
370 |     {
371 |         modal = window.bootstrap.Modal.getOrCreateInstance(modal
372 | El);
373 |         modal.show();
374 |     }
375 |
376 |     // Set Loading state in modal
377 |     const titleEl = document.getElementById('modal-sensor-stat
378 | us-title');
379 |     const subEl = document.getElementById('modal-sensor-status
380 | -sub');
381 |     const aiCompEl = document.getElementById('modal-sensor-ai-
382 | comparison');

```

```
374     const devListEl = document.getElementById('modal-sensor-de
vices-list');
375     const devCountEl = document.getElementById('modal-sensor-d
evices-count');
376     const modelBadge = document.getElementById('modal-sensor-m
odel-badge');
377
378     if (titleEl) titleEl.textContent = 'Выполняется AI-аудит з
алогированных сенсоров...';
379     if (subEl) subEl.textContent = 'Чтение CSV-логов, усреднен
ие значений и сравнение со спецификациями железа';
380     if (aiCompEl) {
381         aiCompEl.innerHTML = '<div class="d-flex align-items-cen
ter gap-2 py-3"><div class="spinner-border spinner-border-sm t
ext-primary" role="status"></div><span>Запрос к модели AI и ср
авнительный анализ номинальных параметров железа...</span></di
v>';
382     }
383     if (devListEl) {
384         devListEl.innerHTML = '<span class="badge bg-dark border
border-secondary text-muted"><span class="spinner-border spin
ner-border-sm me-1" style="width:0.6rem;height:0.6rem;"></span
>Определение устройств...</span>';
385     }
386
387     let report = null;
388     try {
389         let res = await fetch('/api/v1/system/lhm-audit', { meth
od: 'POST' });
390         if (!res.ok) {
391             res = await fetch('/api/v1/lhm/audit', { method: 'POST
' });
392         }
393         if (res.ok) {
394             report = await res.json();
395         }
396     } catch (err) {
397         console.warn('[SystemInspectorTab] LHM Audit API error:')
```



```

    , err);
398     }
399
400     // Extract metrics from report or fallback to cachedSensor
401     s
402     const sensorsList = (report && report.aggregated_sensors)
403     ? report.aggregated_sensors : (cachedSensors || []);
404     const devices = (report && report.devices) ? report.device
405     s : [];
406     let healthScore = report ? (report.health_score || 100) :
407     100;
408     const recommendations = [];
409
410     let maxCpuTemp = null;
411     let maxGpuTemp = null;
412     let maxMbTemp = null;
413     let maxStorageTemp = null;
414     let fanSpeeds = [];
415     let voltages = [];
416     let cpuLoads = [];
417     let gpuLoads = [];
418
419     sensorsList.forEach(s => {
420         const cat = ((s.category || s.sensor_category) || '').to
421         LowerCase();
422         const hw = ((s.hardware || s.hardware_name) || '').toLow
423         erCase();
424         const sName = (s.sensor_name || '').toLowerCase();
425         const num = s.avg !== undefined ? s.avg : s.value_numeri
426         c;
427
428         if (num === null || num === undefined || isNaN(num)) ret
429         urn;
430
431         if (cat.includes('temp')) {
432             if (hw.includes('cpu') || sName.includes('cpu') || sNa
433             me.includes('core')) {
434                 if (maxCpuTemp === null || num > maxCpuTemp) maxCpuT

```

```

    emp = num;
426         } else if (hw.includes('gpu') || hw.includes('nvidia')
    || sName.includes('gpu')) {
427         if (maxGpuTemp === null || num > maxGpuTemp) maxGpuT
emp = num;
428         } else if (hw.includes('storage') || hw.includes('nvme
') || hw.includes('ssd') || hw.includes('hdd') || sName.includ
es('drive')) {
429         if (maxStorageTemp === null || num > maxStorageTemp)
maxStorageTemp = num;
430         } else {
431         if (maxMbTemp === null || num > maxMbTemp) maxMbTemp
= num;
432         }
433         } else if (cat.includes('fan')) {
434         fanSpeeds.push({ name: s.sensor_name, value: num, unit
: s.unit || 'RPM' });
435         } else if (cat.includes('volt')) {
436         voltages.push({ name: s.sensor_name, value: num, unit:
s.unit || 'V' });
437         } else if (cat.includes('load')) {
438         if (hw.includes('cpu') || sName.includes('cpu')) cpuLo
ads.push(num);
439         else if (hw.includes('gpu') || sName.includes('gpu'))
gpuLoads.push(num);
440         }
441     });
442
443     const overallMaxTemp = Math.max(
444         maxCpuTemp || 0,
445         maxGpuTemp || 0,
446         maxMbTemp || 0,
447         maxStorageTemp || 0
448     );
449
450     // Update Model Badge
451     if (modelBadge && report) {
452         modelBadge.textContent = report.ai_model_used || 'Heuris

```

```

    tic Engine';
453     }
454
455     // Render Devices List
456     if (devCountEl) {
457         devCountEl.textContent = `${devices.length} устр.`;
458     }
459     if (devListEl) {
460         if (devices.length > 0) {
461             devListEl.innerHTML = devices.map(d => {
462                 let icon = 'bi-hdd-network';
463                 const low = d.toLowerCase();
464                 if (low.includes('core') || low.includes('cpu') || low.includes('intel') || low.includes('amd')) icon = 'bi-cpu';
465                 else if (low.includes('geforce') || low.includes('gpu') || low.includes('nvidia') || low.includes('radeon')) icon = 'bi-gpu-card';
466                 else if (low.includes('ssd') || low.includes('hd') || low.includes('wdc') || low.includes('toshiba') || low.includes('st3500')) icon = 'bi-device-hdd';
467                 else if (low.includes('memory') || low.includes('ram')) icon = 'bi-memory';
468                 else if (low.includes('ethernet') || low.includes('wifi-fi') || low.includes('network')) icon = 'bi-wifi';
469
470                 return `<i class="bi ${icon}"></i> ${escapeHtml(d)}</span>`;
471             }).join('');
472         } else {
473             devListEl.innerHTML = '<span class="text-muted small">Устройства считываются из текущей телеметрии</span>';
474         }
475     }
476
477     // Render AI Comparison Report
478     if (aiCompEl) {
479         if (report && report.comparison_report) {

```

```
480 |         aiCompEl.textContent = report.comparison_report;
481 |     } else {
482 |         aiCompEl.textContent = 'AI-сравнение выполнено по эври-
    | стическим профилям оборудования хоста. Все параметры сенсоров
    | в пределах нормы.';
483 |     }
484 | }
485 |
486 | // Analyze CPU
487 | const elCpuBadge = document.getElementById('modal-cpu-temp
    | -badge');
488 | const elCpuText = document.getElementById('modal-cpu-analy
    | sis-text');
489 | if (maxCpuTemp !== null) {
490 |     if (elCpuBadge) {
491 |         elCpuBadge.textContent = `${Math.round(maxCpuTemp)} °C
    | `;
492 |         elCpuBadge.className = `badge ${maxCpuTemp > 82 ? 'bg-
    | danger' : maxCpuTemp > 70 ? 'bg-warning text-dark' : 'bg-succe
    | ss'}`;
493 |     }
494 |     if (maxCpuTemp > 82) {
495 |         if (elCpuText) elCpuText.textContent = `Критический на
    | грев (${Math.round(maxCpuTemp)}°C). Риск теплового троттлинга.
    | `;
496 |         recommendations.push({
497 |             type: 'danger',
498 |             icon: 'bi-exclamation-triangle-fill',
499 |             title: 'Высокая температура процессора',
500 |             text: `Пиковая/средняя температура CPU достигает ${M
    | ath.round(maxCpuTemp)}°C. Проверьте плотность прижима кулера,
    | термопасту и запыленность радиатора.`
501 |         });
502 |     } else if (maxCpuTemp > 70) {
503 |         if (elCpuText) elCpuText.textContent = `Повышенная тем
    | пература (${Math.round(maxCpuTemp)}°C) под нагрузкой.`;
504 |         recommendations.push({
505 |             type: 'warning',
```

```
506 |         icon: 'bi-thermometer-high',
507 |         title: 'Повышенный нагрев CPU',
508 |         text: `Температура CPU ${Math.round(maxCpuTemp)}°C в
| выше оптимального порога. Рекомендуется настроить кривую оборот
| ов вентилятора.`
509 |     });
510 | } else {
511 |     if (elCpuText) elCpuText.textContent = `Штатный темпер
| атурный режим (${Math.round(maxCpuTemp)}°C). Троттлинг отсутст
| вует.`;
512 |     recommendations.push({
513 |         type: 'success',
514 |         icon: 'bi-check-circle-fill',
515 |         title: 'Тепловой режим процессора оптимален',
516 |         text: `Температура процессора ${Math.round(maxCpuTem
| p)}°C находится в безопасной зоне номинальных характеристик.`
517 |     });
518 | }
519 | } else {
520 |     if (elCpuBadge) elCpuBadge.textContent = 'N/A';
521 |     if (elCpuText) elCpuText.textContent = 'Температурные да
| тчики CPU не предоставили данных.';
522 | }
523 |
524 | // Analyze GPU
525 | const elGpuBadge = document.getElementById('modal-gpu-temp
| -badge');
526 | const elGpuText = document.getElementById('modal-gpu-analy
| sis-text');
527 | if (maxGpuTemp !== null) {
528 |     if (elGpuBadge) {
529 |         elGpuBadge.textContent = `${Math.round(maxGpuTemp)} °C
| `;
530 |         elGpuBadge.className = `badge ${maxGpuTemp > 80 ? 'bg-
| danger' : maxGpuTemp > 72 ? 'bg-warning text-dark' : 'bg-succe
| ss'}`;
531 |     }
532 |     if (maxGpuTemp > 80) {
```

```
533 |         if (elGpuText) elGpuText.textContent = `Критический на
| грев GPU (${Math.round(maxGpuTemp)}°C).`;
534 |         recommendations.push({
535 |             type: 'danger',
536 |             icon: 'bi-gpu-card',
537 |             title: 'Высокая температура видеокарты',
538 |             text: `Графический чип нагревается до ${Math.round(m
| axGpuTemp)}°C. Проверьте циркуляцию воздуха в корпусе ПК.`
539 |         });
540 |     } else {
541 |         if (elGpuText) elGpuText.textContent = `Температура GP
| U ${Math.round(maxGpuTemp)}°C в норме.`;
542 |         recommendations.push({
543 |             type: 'success',
544 |             icon: 'bi-check-circle-fill',
545 |             title: 'Видеокарта работает штатно',
546 |             text: `Температурные показатели GPU (${Math.round(ma
| xGpuTemp)}°C) соответствуют номиналу.`
547 |         });
548 |     }
549 | } else {
550 |     if (elGpuBadge) elGpuBadge.textContent = 'N/A';
551 |     if (elGpuText) elGpuText.textContent = 'Дискретный GPU в
| режиме энергосбережения или данные отсутствуют.';
552 | }
553 |
554 | // Analyze Cooling
555 | const elFanBadge = document.getElementById('modal-fan-stat
| us-badge');
556 | const elFanText = document.getElementById('modal-fan-analy
| sis-text');
557 | if (fanSpeeds.length > 0) {
558 |     const activeFans = fanSpeeds.filter(f => f.value > 0);
559 |     if (elFanBadge) {
560 |         elFanBadge.textContent = `${activeFans.length} акт. ку
| леров`;
561 |         elFanBadge.className = 'badge bg-success';
562 |     }
```

```
563     if (elFanText) {
564         elFanText.textContent = `Опрошено ${fanSpeeds.length}
кулеров. Обороты стабильны.`;
565     }
566 } else {
567     if (elFanBadge) elFanBadge.textContent = 'Пассив / WMI';
568     if (elFanText) elFanText.textContent = 'Управление венти
ляторами через BIOS или пассивное охлаждение.';
569 }
570
571 // Additional recommendations from backend report
572 if (report && Array.isArray(report.recommendations)) {
573     report.recommendations.forEach(r => {
574         if (!recommendations.some(ex => ex.text.includes(r)))
575         {
576             recommendations.push({
577                 type: 'info',
578                 icon: 'bi-info-circle',
579                 title: 'Рекомендация по оборудованию',
580                 text: r
581             });
582         });
583     }
584
585     // Update Header & Badge
586     const badgeHealth = document.getElementById('modal-sensor-
health-badge');
587     if (badgeHealth) {
588         badgeHealth.textContent = `Health: ${healthScore}/100`;
589         badgeHealth.className = `badge ms-1 ${healthScore >= 85
? 'bg-success' : healthScore >= 65 ? 'bg-warning text-dark' :
'bg-danger'}`;
590     }
591
592     const countStat = document.getElementById('modal-sensor-co
unt-stat');
593     if (countStat) {
```

```
594     const totalSmpls = report ? report.total_samples : senso
rsList.length;
595     countStat.textContent = `${sensorsList.length} шт. (${to
talSmpls} замеров)`;
596   }
597
598   const maxTempStat = document.getElementById('modal-sensor-
max-temp');
599   if (maxTempStat) maxTempStat.textContent = overallMaxTemp
> 0 ? `${Math.round(overallMaxTemp)} °C` : 'N/A';
600
601   const statusTitle = document.getElementById('modal-sensor-
status-title');
602   const statusSub = document.getElementById('modal-sensor-st
atus-sub');
603   const statusIcon = document.getElementById('modal-sensor-s
tatus-icon');
604
605   if (healthScore >= 85) {
606     if (statusIcon) statusIcon.textContent = '✅';
607     if (statusTitle) statusTitle.textContent = 'Все аппаратн
ые подсистемы работают в идеальном режиме';
608     if (statusSub) statusSub.textContent = `Усредненные пока
затели соответствуют спецификациям реального железа (макс. ${M
ath.round(overallMaxTemp)}°C).`;
609   } else if (healthScore >= 65) {
610     if (statusIcon) statusIcon.textContent = '⚠️';
611     if (statusTitle) statusTitle.textContent = 'Обнаружены п
араметры оборудования, требующие внимания';
612     if (statusSub) statusSub.textContent = `Пиковая температ
ура ${Math.round(overallMaxTemp)}°C. Ознакомьтесь с AI-заключе
нием и рекомендациями.`;
613   } else {
614     if (statusIcon) statusIcon.textContent = '❌';
615     if (statusTitle) statusTitle.textContent = 'Внимание: Кр
итические параметры оборудования!';
616     if (statusSub) statusSub.textContent = `Зафиксирован опа
сный нагрев или высокая перегрузка компонентов.`;
```



```

617     }
618
619     // Render Recommendations List
620     const recList = document.getElementById('modal-sensor-reco
mmendations-list');
621     if (recList) {
622         recList.innerHTML = recommendations.map(r => `
623             <div class="p-2.5 rounded border d-flex align-items-st
art gap-2.5" style="background: var(--surface-2); border-color
: var(--border-color) !important;">
624                 <div class="mt-0.5">
625                     <span class="badge ${r.type === 'danger' ? 'bg-dan
ger' : r.type === 'warning' ? 'bg-warning text-dark' : r.type
=== 'info' ? 'bg-info text-dark' : 'bg-success'} rounded-circl
e p-1.5 d-flex align-items-center justify-content-center" styl
e="width: 24px; height: 24px;">
626                         <i class="bi ${r.icon}" style="font-size: 0.75re
m;"></i>
627                     </span>
628                 </div>
629                 <div class="flex-grow-1">
630                     <div class="fw-bold text-white mb-0.5" style="font
-size: 0.82rem;">${escapeHtml(r.title)}</div>
631                     <div class="text-muted" style="font-size: 0.76rem;
line-height: 1.4;">${escapeHtml(r.text)}</div>
632                 </div>
633             </div>
634         `).join('');
635     }
636
637     // Prepare Clipboard text
638     lastAnalysisReportText = `=== AI-Breadboard: Отчет AI-ауди
та сенсоров и оборудования ===\n` +
639     `Дата: ${new Date().toLocaleString('ru-RU')}\n` +
640     `Индекс здоровья системы: ${healthScore}/100\n` +
641     `AI-модель: ${report ? report.ai_model_used : 'Heuristic
'}\n` +
642     `Распознанные устройства (${devices.length} шт.): ${devi

```

```

ces.join(', ')}\n` +
643     `Сенсоров в аудите: ${sensorsList.length}\n` +
644     `Максимальная температура: ${overallMaxTemp > 0 ? Math.r
ound(overallMaxTemp) + ' °C' : 'N/A'}\n\n` +
645     `=== AI СРАВНЕНИЕ СО СПЕЦИФИКАЦИЯМИ РЕАЛЬНОГО ЖЕЛЕЗА ===
\n` +
646     `${(report && report.comparison_report) ? report.compari
son_report : 'Параметры в норме.'}\n\n` +
647     `=== РЕКОМЕНДАЦИИ И ВЫВОДЫ ===\n` +
648     recommendations.map((r, i) => `${i + 1}. [${r.title}] ${
r.text}`)).join('\n') +
649     `\n=====
=`;
650 }
651
652 function updateCoreMetrics(snap) {
653     if (!snap) return;
654
655     // CPU
656     if (snap.cpu) {
657         const cpuVal = document.getElementById('sys-metric-cpu-v
al');
658         const cpuFill = document.getElementById('sys-metric-cpu-
fill');
659         const cpuSub = document.getElementById('sys-metric-cpu-s
ub');
660
661         const pct = Number(snap.cpu.total_percent || 0);
662         if (cpuVal) cpuVal.innerText = `${pct.toFixed(1)}%`;
663         if (cpuFill) cpuFill.style.width = `${Math.min(100, pct)
}%`;
664         if (cpuSub) cpuSub.innerText = `${snap.cpu.physical_core
s || '--'} Физических / ${snap.cpu.logical_cores || '--'} Пото
ков`;
665     }
666
667     // RAM
668     if (snap.memory) {

```

```
669     const ramVal = document.getElementById('sys-metric-ram-v
al');
670     const ramFill = document.getElementById('sys-metric-ram-
fill');
671     const ramSub = document.getElementById('sys-metric-ram-s
ub');
672
673     const usedGb = Number(snap.memory.used_gb || 0);
674     const totalGb = Number(snap.memory.total_gb || 0);
675     const pct = Number(snap.memory.percent || 0);
676
677     if (ramVal) ramVal.innerHTML = `${usedGb.toFixed(1)} / $
{totalGb.toFixed(1)} GB`;
678     if (ramFill) ramFill.style.width = `${pct}%`;
679     if (ramSub) ramSub.innerHTML = `${pct}% занято (${Number
(snap.memory.available_gb || 0).toFixed(1)} GB свободно)`;
680   }
681
682   // GPU
683   if (Array.isArray(snap.gpus) && snap.gpus.length > 0) {
684     const g = snap.gpus[0];
685     const gpuVal = document.getElementById('sys-metric-gpu-v
al');
686     const gpuSub = document.getElementById('sys-metric-gpu-s
ub');
687
688     if (gpuVal) gpuVal.innerHTML = g.name || 'GPU';
689     if (gpuSub) gpuSub.innerHTML = `VRAM: ${Number(g.memory_
total_gb || 0).toFixed(1)} GB | CUDA: ${g.has_cuda ? 'Да' : 'Н
ет'}`;
690   }
691
692   // Disk I/O
693   if (snap.disk_io) {
694     const diskVal = document.getElementById('sys-metric-disk
-val');
695     const diskSub = document.getElementById('sys-metric-disk
-sub');
```

```

696
697     const totalMb = ((snap.disk_io.read_bytes_per_sec + snap
    .disk_io.write_bytes_per_sec) / (1024 * 1024)).toFixed(2);
698     const rKb = (snap.disk_io.read_bytes_per_sec / 1024).toF
    ixed(0);
699     const wKb = (snap.disk_io.write_bytes_per_sec / 1024).to
    Fixed(0);
700
701     if (diskVal) diskVal.innerText = `${totalMb} MB/s`;
702     if (diskSub) diskSub.innerText = `Чтение: ${rKb} KB/s |
    Запись: ${wKb} KB/s`;
703 }
704
705 // Battery & Power
706 const powerCont = document.getElementById('sys-power-info-
    container');
707 const powerBadge = document.getElementById('sys-power-stat
    us-badge');
708 if (powerCont && snap.battery) {
709     if (snap.battery.has_battery && snap.battery.percent !==
    null) {
710         if (powerBadge) {
711             powerBadge.textContent = `${snap.battery.percent}% $
    {snap.battery.power_plugged ? '🔌 Зарядка' : '🔋 Батарея'}`;
712             powerBadge.className = snap.battery.power_plugged ?
    'badge bg-success-subtle text-success border border-success' :
    'badge bg-warning-subtle text-warning border border-warning';
713         }
714         const minsLeft = snap.battery.secs_left ? Math.round(s
    nap.battery.secs_left / 60) : null;
715         powerCont.innerHTML = `
716             <div>Статус: ${snap.battery.power_plugged ? 'Подключ
    ено к сети' : 'Работа от батареи'}</div>
717             <div>${minsLeft ? `Осталось ~${minsLeft} мин.` : 'Пр
    офиль: ' + (snap.battery.power_profile || 'Balanced')}</div>
718             `;
719     } else {
720         if (powerBadge) {

```

```
721         powerBadge.textContent = 'AC Mains';
722         powerBadge.className = 'badge bg-secondary';
723     }
724     powerCont.innerHTML = `
725         <div>Питание: Стационарная электросеть 220V</div>
726         <div class="text-muted">Профиль: ${escapeHtml(snap.b
attery.power_profile || 'High Performance')}</div>
727     `;
728     }
729 }
730 }
731
732 function updateHardwareQuick(snap) {
733     if (!snap) return;
734
735     // Physical Disks SMART Health
736     const diskCont = document.getElementById('sys-physical-dis
ks-container');
737     const diskBadge = document.getElementById('sys-disk-health
-badge');
738     if (diskCont && Array.isArray(snap.physical_disks) && snap
.physical_disks.length > 0) {
739         const allHealthy = snap.physical_disks.every(d => d.heal
th_status === 'Healthy');
740         if (diskBadge) {
741             diskBadge.textContent = allHealthy ? 'SMART OK' : 'Вни
мание';
742             diskBadge.className = allHealthy ? 'badge bg-success-s
ubtle text-success border border-success' : 'badge bg-warning-
subtle text-warning border border-warning';
743         }
744         diskCont.innerHTML = snap.physical_disks.map(d => `
745             <div class="d-flex justify-content-between align-items
-center py-0.5 border-bottom border-dark-subtle" style="border
-bottom-style: dashed !important;">
746                 <span class="text-truncate me-2" style="max-width: 1
60px;" title="${escapeHtml(d.model)}">${escapeHtml(d.model)}</
span>
```

```

747 |         <span class="badge bg-secondary font-monospace" styl
      | e="font-size: 0.65rem;">${d.size_gb} GB ${escapeHtml(d.media_t
      | ype || '')}</span>
748 |     </div>
749 |     `).join('');
750 | }
751 |
752 | // RAM SPD Modules
753 | const ramSticksCont = document.getElementById('sys-ram-sti
      | cks-container');
754 | const ramSticksBadge = document.getElementById('sys-ram-st
      | icks-badge');
755 | if (ramSticksCont && Array.isArray(snap.ram_sticks) && sna
      | p.ram_sticks.length > 0) {
756 |     if (ramSticksBadge) {
757 |         ramSticksBadge.textContent = `${snap.ram_sticks.length
      | } модулей`;
758 |     }
759 |     ramSticksCont.innerHTML = snap.ram_sticks.slice(0, 2).ma
      | p(m => `
760 |         <div class="d-flex justify-content-between align-items
      | -center py-0.5">
761 |             <span>${escapeHtml(m.bank_label)}: ${escapeHtml(m.ma
      | nufacturer)}</span>
762 |             <span class="font-monospace text-primary">${m.capaci
      | ty_gb} GB @ ${m.speed_mhz} MT/s</span>
763 |         </div>
764 |     `).join('');
765 | } else if (ramSticksCont && snap.memory) {
766 |     ramSticksCont.innerHTML = `<div>RAM: ${snap.memory.total
      | _gb || '--'} GB физической памяти</div>`;
767 | }
768 |
769 | // Reliability & Open Ports
770 | const alertsCont = document.getElementById('sys-alerts-por
      | ts-container');
771 | const alertsBadge = document.getElementById('sys-alerts-ba
      | dge');

```

```

772     if (alertsCont) {
773         const portCount = Array.isArray(snap.listening_ports) ?
snap.listening_ports.length : 0;
774         const isReboot = snap.alerts && snap.alerts.reboot_pendi
ng;
775         if (alertsBadge) {
776             alertsBadge.textContent = isReboot ? 'Перезагрузка' :
'Stabilьно';
777             alertsBadge.className = isReboot ? 'badge bg-warning t
ext-dark' : 'badge bg-info-subtle text-info border border-info
';
778         }
779         alertsCont.innerHTML = `
780             <div>Сетевые сокеты: <span class="fw-bold text-white">
${portCount} портов LISTEN</span></div>
781             <div class="text-truncate" title="${escapeHtml(snap.al
erts?.latest_alert || '')}">${escapeHtml(snap.alerts?.latest_a
lert || 'Система стабильна')}</div>
782         `;
783     }
784 }
785
786 async function fetchCoreMetrics() {
787     try {
788         const res = await fetch('/api/v1/system/metrics/core');
789         if (res.ok) {
790             const data = await res.json();
791             if (!latestTelemetrySnapshot) latestTelemetrySnapshot
= {};
792             Object.assign(latestTelemetrySnapshot, data);
793             updateCoreMetrics(data);
794         }
795     } catch (e) {
796         console.warn('[SystemInspectorTab] Ошибка получения базо
вых метрик:', e);
797     }
798 }
799

```

```
800 | async function fetchHardwareQuick() {
801 |     try {
802 |         const res = await fetch('/api/v1/system/hardware/quick')
      | ;
803 |         if (res.ok) {
804 |             const data = await res.json();
805 |             if (!latestTelemetrySnapshot) latestTelemetrySnapshot
      | = {};
806 |             Object.assign(latestTelemetrySnapshot, data);
807 |             updateHardwareQuick(data);
808 |         }
809 |     } catch (e) {
810 |         console.warn('[SystemInspectorTab] Ошибка получения данн
      | ых оборудования:', e);
811 |     }
812 | }
813 |
814 | function updateTelemetryDashboard(snap) {
815 |     if (!snap) return;
816 |     updateCoreMetrics(snap);
817 |     updateHardwareQuick(snap);
818 |     renderProcessTable();
819 |     if (snap.network_activity) {
820 |         renderNetworkActivityTable(snap.network_activity);
821 |     }
822 | }
823 |
824 | function renderProcessTable() {
825 |     if (!latestTelemetrySnapshot || isSysPaused) return;
826 |     const filterInput = document.getElementById('sys-proc-sear
      | ch');
827 |     const filter = (filterInput?.value || '').toLowerCase().tr
      | im();
828 |     const tbody = document.getElementById('sys-proc-tbody');
829 |     if (!tbody) return;
830 |
831 |     const processes = latestTelemetrySnapshot.top_processes ||
      | [];
```



```

832     const filtered = processes.filter(p => {
833         if (!filter) return true;
834         return (
835             String(p.pid).includes(filter) ||
836             (p.name || '').toLowerCase().includes(filter) ||
837             (p.username && p.username.toLowerCase().includes(filte
r))
838         );
839     });
840
841     if (filtered.length === 0) {
842         tbody.innerHTML = '<tr><td colspan="7" class="text-cente
r py-3 text-muted">Процессы не найдены</td></tr>';
843         return;
844     }
845
846     tbody.innerHTML = filtered.map((p, idx) => {
847         let cpuClass = '';
848         if (p.cpu_percent > 40) cpuClass = 'badge-cpu-high';
849         else if (p.cpu_percent > 15) cpuClass = 'badge-cpu-med';
850
851         return `
852             <tr class="sys-proc-row" data-idx="${idx}" style="curs
or: pointer;" title="Нажмите для детальной AI-диагностики проц
есса">
853                 <td style="color: #38bdf8;">${p.pid}</td>
854                 <td style="font-weight: 600; max-width: 200px; overf
low: hidden; text-overflow: ellipsis; white-space: nowrap;" ti
tle="${escapeHtml(p.name || '')}">${escapeHtml(p.name || '')}<
/td>
855                 <td style="color: #94a3b8;">${escapeHtml(p.status ||
'running')}</td>
856                 <td style="text-align: right;" class="${cpuClass}">${
Number(p.cpu_percent || 0).toFixed(1)}%</td>
857                 <td style="text-align: right; color: #4ade80;">${Num
ber(p.memory_mb || 0).toFixed(1)} MB</td>
858                 <td style="text-align: right; color: #a855f7;">${p.n
um_threads || 1}</td>

```

```

859 |         <td style="color: #94a3b8;">${escapeHtml(p.username
    | || 'SYSTEM'))}</td>
860 |     </tr>
861 |     `;
862 |     }).join('');
863 |
864 |     tbody.querySelectorAll('.sys-proc-row').forEach(row => {
865 |         row.onclick = () => {
866 |             const idx = parseInt(row.getAttribute('data-idx'), 10)
    | ;
867 |             const p = filtered[idx];
868 |             if (!p) return;
869 |
870 |             if (window.AITableModal) {
871 |                 window.AITableModal.show({
872 |                     icon: '🔍 ',
873 |                     title: p.name,
874 |                     subtitle: `PID: ${p.pid} | ${p.username} || 'SYSTEM
    | '}` ,
875 |                     tableType: 'process',
876 |                     badges: [
877 |                         { text: `PID ${p.pid}`, class: 'badge bg-info te
    | xt-dark' },
878 |                         { text: p.status || 'running', class: 'badge bg-
    | success' }
879 |                     ],
880 |                     metadata: [
881 |                         { label: 'Имя процесса', value: p.name },
882 |                         { label: 'Process ID (PID)', value: String(p.pid
    | ) },
883 |                         { label: 'Пользователь / Учетная запись', value:
    | p.username || 'SYSTEM' },
884 |                         { label: 'Статус', value: p.status || 'Выполняет
    | ся' },
885 |                         { label: 'Загрузка CPU', value: `${Number(p.cpu_
    | percent || 0).toFixed(1)}%` },
886 |                         { label: 'Оперативная память', value: `${Number(
    | p.memory_mb || 0).toFixed(1)} MB` },

```

```

887 |         { label: 'Количество потоков', value: String(p.n
| um_threads || 1) },
888 |         { label: 'Исполняемый путь', value: p.exe || p.e
| xecutable_path || 'Системный процесс Windows', isCode: true, f
| ullWidth: true }
889 |     ],
890 |     rawTitle: 'Команда запуска / Аргументы',
891 |     rawContent: Array.isArray(p.cmdline) ? p.cmdline.j
| oin(' ') : (p.cmdline || p.exe || ''),
892 |     requestData: {
893 |         pid: p.pid,
894 |         cpu_percent: p.cpu_percent,
895 |         memory_mb: p.memory_mb,
896 |         username: p.username,
897 |         status: p.status
898 |     }
899 | });
900 | }
901 | };
902 | });
903 | }
904 |
905 | async function fetchNetworkActivity() {
906 |     try {
907 |         const res = await fetch('/api/v1/system/network-activity
| ?limit=100');
908 |         if (res.ok) {
909 |             const data = await res.json();
910 |             renderNetworkActivityTable(data);
911 |         }
912 |     } catch (e) {
913 |         console.warn('[SystemInspectorTab] Failed to fetch netwo
| rk activity:', e);
914 |     }
915 | }
916 |
917 | function renderNetworkActivityTable(activities) {
918 |     if (activities) {

```

```
919 |         cachedNetworkActivities = activities;
920 |     }
921 |     if (isNetPaused) return;
922 |
923 |     const tbody = document.getElementById('sys-net-tbody');
924 |     if (!tbody) return;
925 |
926 |     const searchInput = document.getElementById('sys-net-search');
927 |     const filterText = (searchInput?.value || '').toLowerCase().trim();
928 |     const countBadge = document.getElementById('sys-netact-count-badge');
929 |     const connsBadge = document.getElementById('sys-netact-conns-badge');
930 |
931 |     const rawList = Array.isArray(cachedNetworkActivities) ? cachedNetworkActivities : [];
932 |
933 |     // Filter by category and search query
934 |     let filtered = rawList.filter(item => {
935 |         if (currentNetFilter === 'internet' && !item.is_internet) return false;
936 |         if (currentNetFilter === 'listen' && item.status !== 'LISTEN') return false;
937 |
938 |         if (filterText) {
939 |             return (
940 |                 String(item.pid).includes(filterText) ||
941 |                 (item.name || '').toLowerCase().includes(filterText) ||
942 |                 (item.user || '').toLowerCase().includes(filterText) ||
943 |                 (item.remote_address || '').toLowerCase().includes(filterText) ||
944 |                 (item.local_address || '').toLowerCase().includes(filterText) ||
945 |                 (item.service_type || '').toLowerCase().includes(filterText) ||
```

```

    terText) ||
946 |         (item.sent_summary || '').toLowerCase().includes(fil
    terText) ||
947 |         (item.recv_summary || '').toLowerCase().includes(fil
    terText)
948 |     );
949 | }
950 |     return true;
951 | });
952 |
953 |     const uniqueProcs = new Set(filtered.map(i => i.pid)).size
    ;
954 |     if (countBadge) countBadge.textContent = `${uniqueProcs} п
    рограмм`;
955 |     if (connsBadge) connsBadge.textContent = `${filtered.lengt
    h} сокетов`;
956 |
957 |     if (filtered.length === 0) {
958 |         tbody.innerHTML = '<tr><td colspan="6" class="text-cente
    r py-4 text-muted small">Нет активных сетевых соединений по вы
    бранному фильтру</td></tr>';
959 |         return;
960 |     }
961 |
962 |     tbody.innerHTML = filtered.map((item, idx) => {
963 |         const isListen = item.status === 'LISTEN';
964 |         const statusBadgeClass = isListen
965 |             ? 'badge bg-secondary-subtle text-light border border-
    secondary'
966 |             : item.status === 'ESTABLISHED'
967 |             ? 'badge bg-success-subtle text-success border border-
    success'
968 |             : 'badge bg-warning-subtle text-warning border border-
    warning';
969 |
970 |         const protoBadge = item.protocol === 'UDP'
971 |             ? '<span class="badge bg-primary text-white" style="fo
    nt-size: 0.65rem;">UDP</span>'

```

```

972 |         : '<span class="badge bg-dark border border-secondary
| text-info" style="font-size: 0.65rem;">TCP</span>';
973 |
974 |         const isExtBadge = item.is_internet
975 |         ? '<span class="badge bg-primary-subtle text-primary b
| order border-primary px-1" style="font-size: 0.62rem;" title="
| Внешний сервер в сети Интернет">WAN</span>'
976 |         : '<span class="badge bg-secondary px-1" style="font-s
| ize: 0.62rem;" title="Локальный сокет Loopback">LAN</span>';
977 |
978 |         const formatNetKb = (kb) => {
979 |             if (!kb || kb <= 0) return '0 KB';
980 |             if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).toF
| ixed(2) + ' GB';
981 |             if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB';
982 |             return kb.toFixed(1) + ' KB';
983 |         };
984 |
985 |         const formatNetRate = (rate) => {
986 |             if (!rate || rate <= 0.05) return '';
987 |             if (rate >= 1024) return (rate / 1024).toFixed(1) + '
| MB/s';
988 |             return rate.toFixed(1) + ' KB/s';
989 |         };
990 |
991 |         const deltaSentStr = (item.delta_sent_kb && item.delta_s
| ent_kb > 0)
992 |         ? `▲ +${formatNetKb(item.delta_sent_kb)}${item.sent_ra
| te_kbs > 0.05 ? ' (' + formatNetRate(item.sent_rate_kbs) + ') '
| : ''}`
993 |         : `▲ 0 KB`;
994 |         const deltaRecvStr = (item.delta_recv_kb && item.delta_r
| ecv_kb > 0)
995 |         ? `▼ +${formatNetKb(item.delta_recv_kb)}${item.recv_ra
| te_kbs > 0.05 ? ' (' + formatNetRate(item.recv_rate_kbs) + ') '
| : ''}`
996 |         : `▼ 0 KB`;
997 |

```

```

998         const totalSentStr = formatNetKb(item.sent_kb || 0);
999         const totalRecvStr = formatNetKb(item.recv_kb || 0);
1000
1001         return `
1002             <tr class="sys-net-row" data-idx="${idx}" style="curso
r: pointer;" title="Нажмите для детальной диагностики сетевого
соединения">
1003                 <td>
1004                     <div class="fw-bold text-white text-truncate" styl
e="max-width: 165px;" title="${escapeHtml(item.name)}">${escap
eHtml(item.name)}</div>
1005                     <div class="small text-muted" style="font-size: 0.
70rem;">PID: <span style="color: #38bdf8;">${item.pid}</span>
${item.user ? '• ' + escapeHtml(item.user) : ''}</div>
1006                 </td>
1007                 <td>
1008                     <div class="d-flex align-items-center gap-1">
1009                         ${isExtBadge}
1010                         <span class="font-monospace text-light fw-semibo
ld text-truncate" style="max-width: 195px;" title="${escapeHtm
l(item.remote_address)}">
1011                             ${escapeHtml(item.remote_address !== '-' ? ite
m.remote_address : item.local_address)}
1012                         </span>
1013                     </div>
1014                     <div class="small text-muted font-monospace" style
="font-size: 0.68rem;">Local: ${escapeHtml(item.local_address)
}</div>
1015                 </td>
1016                 <td>
1017                     <div class="d-flex align-items-center gap-1 mb-0.5
">
1018                         ${protoBadge}
1019                         <span class="fw-semibold text-truncate" style="m
ax-width: 210px; font-size: 0.74rem; color: #a855f7;" title="${
escapeHtml(item.service_type)}">${escapeHtml(item.service_typ
e)}</span>
1020                     </div>

```

```

1021 |         </td>
1022 |         <td style="text-align: center;">
1023 |             <span class="{statusBadgeClass}" style="font-size
| : 0.68rem;">{escapeHtml(item.status)}</span>
1024 |         </td>
1025 |         <td>
1026 |             <div class="d-flex align-items-center justify-cont
| ent-start gap-2 mb-1">
1027 |                 <span class="badge bg-warning-subtle text-warnin
| g border border-warning px-1.5 py-0.5" style="font-size: 0.68r
| em;" title="Отправлено за измеряемый период">
1028 |                     {deltaSentStr}
1029 |                 </span>
1030 |                 <span class="text-muted font-monospace" style="f
| ont-size: 0.68rem;" title="Всего отправлено/записано">
1031 |                     Σ {totalSentStr}
1032 |                 </span>
1033 |             </div>
1034 |             <div class="d-flex align-items-start gap-1">
1035 |                 <i class="bi bi-arrow-up-right text-warning mt-0
| .5" style="font-size: 0.70rem;"></i>
1036 |                 <div class="text-truncate" style="max-width: 250
| px; font-size: 0.72rem; color: #fde047;" title="{escapeHtml(i
| tem.sent_summary)}">
1037 |                     {escapeHtml(item.sent_summary)}
1038 |                 </div>
1039 |             </div>
1040 |         </td>
1041 |         <td>
1042 |             <div class="d-flex align-items-center justify-cont
| ent-start gap-2 mb-1">
1043 |                 <span class="badge bg-success-subtle text-succes
| s border border-success px-1.5 py-0.5" style="font-size: 0.68r
| em;" title="Скачано/получено за измеряемый период">
1044 |                     {deltaRecvStr}
1045 |                 </span>
1046 |                 <span class="text-muted font-monospace" style="f
| ont-size: 0.68rem;" title="Всего скачано/прочитано">

```



```

1047         Σ ${totalRecvStr}
1048     </span>
1049 </div>
1050     <div class="d-flex align-items-start gap-1">
1051         <i class="bi bi-arrow-down-left text-success mt-
1052 0.5" style="font-size: 0.70rem;"></i>
1053         <div class="text-truncate" style="max-width: 250
1054 px; font-size: 0.72rem; color: #86efac;" title="${escapeHtml(i
1055 tem.recv_summary)}">
1056             ${escapeHtml(item.recv_summary)}
1057         </div>
1058     </div>
1059 </td>
1060 </tr>
1061 `;
1062 }).join('');
1063
1064 tbody.querySelectorAll('.sys-net-row').forEach(row => {
1065     row.onclick = () => {
1066         const idx = parseInt(row.getAttribute('data-idx'), 10)
1067     };
1068     const item = filtered[idx];
1069     if (!item) return;
1070
1071     const formatNetKb = (kb) => {
1072         if (!kb || kb <= 0) return '0 KB';
1073         if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).t
1074 oFixed(2) + ' GB';
1075         if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB
1076 ';
1077         return kb.toFixed(1) + ' KB';
1078     };
1079
1080     const formatNetRate = (rate) => {
1081         if (!rate || rate <= 0.05) return '';
1082         if (rate >= 1024) return (rate / 1024).toFixed(1) +
1083 ' MB/s';
1084         return rate.toFixed(1) + ' KB/s';

```

```

1078         };
1079
1080         if (window.AITableModal) {
1081             window.AITableModal.show({
1082                 icon: '🔌',
1083                 title: `${item.name} (${item.service_type})`,
1084                 subtitle: `PID: ${item.pid} | ${item.remote_addresses}`,
1085                 tableType: 'network',
1086                 badges: [
1087                     { text: `PID ${item.pid}`, class: 'badge bg-info text-dark' },
1088                     { text: item.protocol, class: 'badge bg-primary' },
1089                     { text: item.status, class: 'badge bg-success' },
1090                     { text: item.is_internet ? 'Интернет (WAN)' : 'Локально (LAN)', class: item.is_internet ? 'badge bg-warning text-dark' : 'badge bg-secondary' },
1091                 ],
1092                 metadata: [
1093                     { label: 'Программа / Процесс', value: item.name },
1094                     { label: 'Process ID (PID)', value: String(item.pid) },
1095                     { label: 'Пользователь системы', value: item.user || 'SYSTEM' },
1096                     { label: 'Удаленный адрес (Remote)', value: item.remote_address },
1097                     { label: 'Локальный сокет (Local)', value: item.local_address },
1098                     { label: 'Протокол / Служба', value: `${item.protocol} • ${item.service_type}` },
1099                     { label: 'Статус соединения', value: item.status },
1100                     { label: 'Скачено за период (Прием)', value: `${formatNetKb(item.delta_recv_kb || 0)} ${item.recv_rate_kbs > 0.05 ? '(' + formatNetRate(item.recv_rate_kbs) + ')' : ''}` },

```

```

1101         { label: 'Всего скачано / получено', value: form
atNetKb(item.recv_kb || 0) },
1102         { label: 'Отправлено за период', value: `${forma
tNetKb(item.delta_sent_kb || 0)} ${item.sent_rate_kbs > 0.05 ?
'(' + formatNetRate(item.sent_rate_kbs) + ')' : ''}` },
1103         { label: 'Всего отправлено', value: formatNetKb(
item.sent_kb || 0) },
1104         { label: 'Что шлет (Отправка)', value: item.sent
_summary, fullWidth: true },
1105         { label: 'Что принимает (Прием)', value: item.re
cv_summary, fullWidth: true }
1106     ],
1107     rawTitle: 'Сетевой дамп подключения',
1108     rawContent: JSON.stringify(item, null, 2),
1109     requestData: item
1110     });
1111     }
1112     };
1113     });
1114     }
1115
1116     async function runAiDiagnostics() {
1117         const summaryEl = document.getElementById('sys-ai-summary-
text');
1118         const badgeEl = document.getElementById('sys-ai-health-bad
ge');
1119         const anomaliesEl = document.getElementById('sys-ai-anomal
ies-container');
1120         const engineTagEl = document.getElementById('sys-ai-model-
tag');
1121
1122         if (summaryEl) summaryEl.innerText = 'Запуск глубокого AI-
аудита системы и оборудования...';
1123
1124         try {
1125             const res = await fetch('/api/v1/system/diagnose', { met
hod: 'POST' });
1126             if (!res.ok) throw new Error('AI Diagnosis error');

```

```

1127 |         const report = await res.json();
1128 |
1129 |         if (badgeEl) {
1130 |             badgeEl.innerText = `Health: ${report.health_score ||
1131 | 100}/100`;
1132 |             badgeEl.className = `badge ${report.health_score >= 80
1133 | ? 'bg-success' : report.health_score >= 60 ? 'bg-warning' : '
1134 | bg-danger'}`;
1135 |         }
1136 |
1137 |         if (engineTagEl) {
1138 |             engineTagEl.innerText = report.ai_model_used || 'Heuri
1139 | stic Engine';
1140 |         }
1141 |
1142 |         if (anomaliesEl) {
1143 |             if (Array.isArray(report.anomalies) && report.anomalie
1144 | s.length > 0) {
1145 |                 anomaliesEl.innerHTML = report.anomalies.map(a => `<
1146 | span class="anomaly-tag">⬜  [${a.subsystem}] ${escapeHtml(a.tit
1147 | le)}</span>`).join('');
1148 |             } else {
1149 |                 anomaliesEl.innerHTML = '<span style="color: #4ade80
1150 | ; font-size: 0.75rem;"><i class="bi bi-check-circle me-1"></i>
1151 | Аномалий в работе оборудования не обнаружено</span>';
1152 |             }
1153 |         }
1154 |
1155 |         if (summaryEl) {
1156 |             const recs = Array.isArray(report.recommendations) &&
1157 | report.recommendations.length > 0
1158 |             ? `\\nРекомендации: ${report.recommendations.join(',
1159 | ')}`
1160 |             : '';
1161 |             summaryEl.innerText = `${report.summary || 'Телеметрия
1162 | в норме.'}${recs}`;
1163 |         }
1164 |     } catch (e) {

```

```
1153     console.error('[SystemInspectorTab] AI Diagnose error:',
1154     e);
1155     if (summaryEl) summaryEl.innerText = 'Ошибка выполнения
1156     AI-диагностики: ' + e.message;
1157     }
1158     }
1159
1160     let sysWsReconnectTimer = null;
1161
1162     function connectSystemWebSocket() {
1163         if (sysWsReconnectTimer) {
1164             clearTimeout(sysWsReconnectTimer);
1165             sysWsReconnectTimer = null;
1166         }
1167         if (sysWs) {
1168             try {
1169                 sysWs.onclose = null;
1170                 sysWs.onerror = null;
1171                 sysWs.close();
1172             } catch {}
1173             sysWs = null;
1174         }
1175
1176         if (window.isTabActive && !window.isTabActive('tab-system-
1177         inspector')) {
1178             const statusBadge = document.getElementById('sys-conn-st
1179             atus');
1180             if (statusBadge) {
1181                 statusBadge.className = 'badge rounded-pill bg-seconda
1182                 ry text-light px-3 py-2';
1183                 statusBadge.innerText = '○ Поток приостановлен';
1184             }
1185             return;
1186         }
1187
1188         const proto = window.location.protocol === 'https:' ? 'wss
1189         :' : 'ws:';
1190         const wsUrl = `${proto}://${window.location.host}/api/v1/sy
```

```
stem/stream`;
1185
1186     try {
1187         sysWs = new WebSocket(wsUrl);
1188         const statusBadge = document.getElementById('sys-conn-st
atus');
1189
1190         sysWs.onopen = () => {
1191             console.info('[SystemInspectorTab] WebSocket соединени
е с телеметрией установлено.');
```

1192 if (statusBadge) {

1193 statusBadge.className = 'badge rounded-pill bg-succe

1194 statusBadge.innerText = '● Телеметрия активна';

1195 }

1196 };

1197

1198 sysWs.onmessage = (evt) => {

1199 try {

1200 const snap = JSON.parse(evt.data);

1201 latestTelemetrySnapshot = snap;

1202 updateTelemetryDashboard(snap);

1203 } catch (e) {

1204 console.error('[SystemInspectorTab] Ошибка разбора с

1205 }

1206 };

1207

1208 const scheduleReconnect = (reason) => {

1209 if (sysWsReconnectTimer) return;

1210 if (window.isTabActive && !window.isTabActive('tab-sys

1211 console.warn(`[SystemInspectorTab] WebSocket отключен

1212 if (statusBadge) {

1213 statusBadge.className = 'badge rounded-pill bg-warni

1214 statusBadge.innerText = '○ Переподключение...';

```
1215 |         }
1216 |         sysWsReconnectTimer = setTimeout(() => {
1217 |             sysWsReconnectTimer = null;
1218 |             if (!window.isTabActive || window.isTabActive('tab-s
system-inspector')) {
1219 |                 connectSystemWebSocket();
1220 |             }
1221 |             }, 3000);
1222 |     };
1223 |
1224 |     sysWs.onclose = (evt) => {
1225 |         scheduleReconnect(`код закрытия: ${evt.code}`);
1226 |     };
1227 |
1228 |     sysWs.onerror = (err) => {
1229 |         console.warn('[SystemInspectorTab] Ошибка соединения W
eSocket:', err);
1230 |         scheduleReconnect('ошибка связи');
1231 |     };
1232 | } catch (err) {
1233 |     console.error('[SystemInspectorTab] Ошибка инициализации
WebSocket:', err);
1234 |     sysWsReconnectTimer = setTimeout(() => {
1235 |         sysWsReconnectTimer = null;
1236 |         if (!window.isTabActive || window.isTabActive('tab-sys
tem-inspector')) {
1237 |             connectSystemWebSocket();
1238 |         }
1239 |         }, 5000);
1240 |     }
1241 | }
1242 |
1243 | function disconnectSystemWebSocket() {
1244 |     if (sysWsReconnectTimer) {
1245 |         clearTimeout(sysWsReconnectTimer);
1246 |         sysWsReconnectTimer = null;
1247 |     }
1248 |     if (sysWs) {
```

```
1249     try {
1250         sysWs.onclose = null;
1251         sysWs.onerror = null;
1252         sysWs.close();
1253     } catch {}
1254     sysWs = null;
1255 }
1256 const statusBadge = document.getElementById('sys-conn-stat
us');
1257 if (statusBadge) {
1258     statusBadge.className = 'badge rounded-pill bg-secondary
text-light px-3 py-2';
1259     statusBadge.innerText = '○ Поток приостановлен';
1260 }
1261 }
1262
1263 function bindSensorTabEvents() {
1264     const btnPause = document.getElementById('btn-sys-pause-pr
oc');
1265     if (btnPause) {
1266         btnPause.onclick = () => {
1267             isSysPaused = !isSysPaused;
1268             btnPause.innerText = isSysPaused ? 'Возобновить' : 'Па
уза';
1269             btnPause.className = isSysPaused ? 'btn btn-sm btn-war
ning rounded-pill px-3' : 'btn btn-sm btn-outline-secondary ro
unded-pill px-3';
1270         };
1271     }
1272
1273     const btnRefreshSensors = document.getElementById('btn-sys
-refresh-sensors');
1274     if (btnRefreshSensors) {
1275         btnRefreshSensors.onclick = () => fetchLhmSensors();
1276     }
1277
1278     const btnRefreshLhm = document.getElementById('btn-sys-ref
resh-lhm');
```



```
1279 |     if (btnRefreshLhm) {
1280 |         btnRefreshLhm.onclick = () => fetchLhmSensors();
1281 |     }
1282 |
1283 |     const btnLaunchLhm = document.getElementById('btn-sys-laun
ch-lhm');
1284 |     if (btnLaunchLhm) {
1285 |         btnLaunchLhm.onclick = async () => {
1286 |             btnLaunchLhm.disabled = true;
1287 |             btnLaunchLhm.innerHTML = '<span class="spinner-border
spinner-border-sm me-1"></span> Занычк...';
1288 |             try {
1289 |                 const res = await fetch('/api/v1/lhm/launch', { meth
od: 'POST' });
1290 |                 if (res.ok) {
1291 |                     setTimeout(fetchLhmSensors, 2500);
1292 |                 }
1293 |             } catch (e) {
1294 |                 console.error('[SystemInspectorTab] Failed to launch
LHM:', e);
1295 |             } finally {
1296 |                 btnLaunchLhm.disabled = false;
1297 |             }
1298 |         };
1299 |     }
1300 |
1301 |     const searchSensors = document.getElementById('sys-lhm-sea
rch');
1302 |     if (searchSensors) {
1303 |         searchSensors.oninput = () => renderSensors();
1304 |     }
1305 |
1306 |     // Category pills
1307 |     document.querySelectorAll('.sys-lhm-cat-btn').forEach(btn
=> {
1308 |         btn.onclick = () => {
1309 |             document.querySelectorAll('.sys-lhm-cat-btn').forEach(
b => {
```

```

1310         b.classList.remove('active', 'btn-outline-primary');
1311         b.classList.add('btn-outline-secondary');
1312     });
1313     btn.classList.remove('btn-outline-secondary');
1314     btn.classList.add('active', 'btn-outline-primary');
1315     currentSensorCategory = btn.getAttribute('data-cat') |
| 'all';
1316     renderSensors();
1317     };
1318     });
1319
1320     // Analyze Button (Sensors)
1321     const btnAnalyze = document.getElementById('btn-sys-sensor
| -analyze');
1322     if (btnAnalyze) {
1323         btnAnalyze.onclick = () => analyzeSensorsAndReport();
1324     }
1325
1326     // Modal Re-analyze Button
1327     const btnReAnalyze = document.getElementById('btn-re-analy
| ze-sensors');
1328     if (btnReAnalyze) {
1329         btnReAnalyze.onclick = async () => {
1330             btnReAnalyze.disabled = true;
1331             const icon = btnReAnalyze.querySelector('i');
1332             if (icon) icon.classList.add('spin-animation');
1333             await fetchLhmSensors();
1334             await analyzeSensorsAndReport();
1335             if (icon) icon.classList.remove('spin-animation');
1336             btnReAnalyze.disabled = false;
1337         };
1338     }
1339
1340     // Modal Copy Report Button
1341     const btnCopyReport = document.getElementById('btn-copy-se
| nsor-report');
1342     if (btnCopyReport) {
1343         btnCopyReport.onclick = async () => {

```

```
1344         if (!lastAnalysisReportText) return;
1345         try {
1346             await navigator.clipboard.writeText(lastAnalysisReportText);
1347             const origHtml = btnCopyReport.innerHTML;
1348             btnCopyReport.innerHTML = '<i class="bi bi-check2 text-success me-1"></i> Скопировано!';
1349             setTimeout(() => {
1350                 btnCopyReport.innerHTML = origHtml;
1351             }, 2000);
1352         } catch (e) {
1353             console.warn('[SystemInspectorTab] Clipboard copy failed:', e);
1354         }
1355     };
1356 }
1357 }
1358
1359 async function openProcStatsModal() {
1360     const modalEl = document.getElementById('sysProcStatsModal');
1361     if (modalEl && window.bootstrap && window.bootstrap.Modal) {
1362         const modal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
1363         modal.show();
1364     }
1365     await fetchProcStats();
1366 }
1367
1368 async function fetchProcStats(filterName = '') {
1369     const tbodyRollups = document.getElementById('tbody-proc-rollups');
1370     const tbodyOutliers = document.getElementById('tbody-proc-outliers');
1371     const tbodyDaily = document.getElementById('tbody-proc-daily');
1372     const badgeCount = document.getElementById('modal-proc-sta
```

```

ts-count-badge');
1373     const badgeOutliers = document.getElementById('proc-outlie
rs-count');
1374
1375     try {
1376         const url = `/api/v1/system/processes/stats?limit=60${fi
lterName ? '&name=' + encodeURIComponent(filterName) : ''}`;
1377         const res = await fetch(url);
1378         if (!res.ok) throw new Error('HTTP ' + res.status);
1379         const data = await res.json();
1380         cachedProcStats = data;
1381
1382         const rollups = data.rollups_2min || [];
1383         const outliers = data.outliers || [];
1384         const daily = data.daily_stats || [];
1385
1386         if (badgeCount) badgeCount.textContent = `${rollups.leng
th + daily.length} записей`;
1387         if (badgeOutliers) badgeOutliers.textContent = String(ou
tliers.length);
1388
1389         // 1. Рендерим 2-минутные обобщения
1390         if (tbodyRollups) {
1391             if (rollups.length === 0) {
1392                 tbodyRollups.innerHTML = '<tr><td colspan="10" class
="text-center py-3 text-muted">Обобщённых записей (&gt; 2 мин)
пока нет. Данные собираются каждые 5 сек.</td></tr>';
1393             } else {
1394                 tbodyRollups.innerHTML = rollups.map(r => {
1395                     const timeRange = `${(r.period_start || '').slice(
11, 19)} - ${(r.period_end || '').slice(11, 19)}`;
1396                     const cpuClass = r.avg_cpu_percent > 30 ? 'text-da
nger fw-bold' : (r.avg_cpu_percent > 10 ? 'text-warning' : 'te
xt-info');
1397                     return `
1398                         <tr>
1399                             <td style="color: #94a3b8;">${timeRange}</td>
1400                             <td style="font-weight: 600; color: #38bdf8;">

```

```

    ${escapeHtml(r.name || '')}</td>
1401         <td>${r.pid || '-'}</td>
1402         <td style="color: #94a3b8;">${escapeHtml(r.use
rname || '')}</td>
1403         <td class="text-end text-white">${r.sample_cou
nt}</td>
1404         <td class="text-end ${cpuClass}">${Number(r.av
g_cpu_percent).toFixed(1)}%</td>
1405         <td class="text-end text-danger">${Number(r.ma
x_cpu_percent).toFixed(1)}%</td>
1406         <td class="text-end text-success">${Number(r.a
vg_memory_mb).toFixed(1)} MB</td>
1407         <td class="text-end text-success">${Number(r.m
ax_memory_mb).toFixed(1)} MB</td>
1408         <td class="text-end">${r.outliers_count > 0 ?
`<span class="badge bg-warning text-dark">${r.outliers_count}<
/span>` : '0'}</td>
1409     </tr>
1410     `;
1411     }).join('');
1412 }
1413 }
1414
1415 // 2. Рендерим зафиксированные выбросы
1416 if (tbodyOutliers) {
1417     if (outliers.length === 0) {
1418         tbodyOutliers.innerHTML = '<tr><td colspan="8" class
="text-center py-3 text-muted">Аномальных выбросов нагрузки не
зафиксировано.</td></tr>';
1419     } else {
1420         tbodyOutliers.innerHTML = outliers.map(o => {
1421             return `
1422                 <tr>
1423                     <td style="color: #f59e0b;">${(o.timestamp ||
'' ).replace('T', ' ' ).slice(0, 19)}</td>
1424                     <td style="font-weight: 600; color: #38bdf8;">
${escapeHtml(o.name || '')}</td>
1425                     <td>${o.pid}</td>

```

```

1426 |         <td class="text-end text-danger fw-bold">${Num
|ber(o.cpu_percent).toFixed(1)}%</td>
1427 |         <td class="text-end text-success">${Number(o.m
|emory_mb).toFixed(1)} MB</td>
1428 |         <td>${o.num_threads || 1}</td>
1429 |         <td style="color: #94a3b8;">${escapeHtml(o.use
|rname || '')}</td>
1430 |         <td class="text-warning">${escapeHtml(o.detail
|s || 'Пиковый всплеск CPU')}</td>
1431 |     </tr>
1432 |     `;
1433 |     }).join('');
1434 | }
1435 | }
1436 |
1437 | // 3. Рендерим суточную статистику
1438 | if (tbodyDaily) {
1439 |     if (daily.length === 0) {
1440 |         tbodyDaily.innerHTML = '<tr><td colspan="9" class="t
|ext-center py-3 text-muted">Данных старше 1 дня пока нет. Обоб
|щение формируется автоматически.</td></tr>';
1441 |     } else {
1442 |         tbodyDaily.innerHTML = daily.map(d => {
1443 |             return `
1444 |                 <tr>
1445 |                     <td class="text-info">${d.date}</td>
1446 |                     <td style="font-weight: 600; color: #38bdf8;">
|${escapeHtml(d.name || '')}</td>
1447 |                     <td class="text-end text-white">${d.sample_cou
|nt}</td>
1448 |                     <td class="text-end text-warning">${Number(d.a
|vg_cpu_percent).toFixed(1)}%</td>
1449 |                     <td class="text-end text-danger">${Number(d.ma
|x_cpu_percent).toFixed(1)}%</td>
1450 |                     <td class="text-end text-success">${Number(d.a
|vg_memory_mb).toFixed(1)} MB</td>
1451 |                     <td class="text-end text-success">${Number(d.m
|ax_memory_mb).toFixed(1)} MB</td>

```

```

1452 |         <td class="text-end">${d.outliers_count > 0 ?
    | `<span class="badge bg-warning text-dark">${d.outliers_count}<
    | /span>` : '0'}</td>
1453 |         <td style="color: #94a3b8;">${(d.last_seen ||
    | '').slice(11, 19)}</td>
1454 |     </tr>
1455 |     `;
1456 |     }).join('');
1457 | }
1458 | }
1459 |
1460 | } catch (err) {
1461 |     console.warn('[SystemInspectorTab] Ошибка загрузки стати
    | стики процессов:', err);
1462 |     if (tbodyRollups) tbodyRollups.innerHTML = `<tr><td cols
    | pan="10" class="text-center py-3 text-danger">Ошибка: ${escape
    | Html(err.message)}</td></tr>`;
1463 | }
1464 | }
1465 |
1466 | async function triggerProcRollupNow() {
1467 |     const btn = document.getElementById('btn-trigger-proc-roll
    | up');
1468 |     const statusText = document.getElementById('modal-proc-rol
    | lup-status');
1469 |     if (btn) {
1470 |         btn.disabled = true;
1471 |         btn.innerHTML = '<span class="spinner-border spinner-bor
    | der-sm me-1"></span> Обобщение...';
1472 |     }
1473 |     try {
1474 |         const res = await fetch('/api/v1/system/processes/rollup
    | ?cutoff_seconds=120&outlier_cpu_threshold=30.0', { method: 'PO
    | ST' });
1475 |         if (res.ok) {
1476 |             const data = await res.json();
1477 |             if (statusText) {
1478 |                 statusText.textContent = `Обобщено: 2мин_агрегатов=$

```

```
    {data.rollup_2min?.rollups_created || 0}, выбросов=${data.roll
up_2min?.outliers_saved || 0}`;
1479    }
1480    await fetchProcStats();
1481    }
1482    } catch (e) {
1483        console.error('[SystemInspectorTab] Ошибка выполнения ро
ллапа:', e);
1484    } finally {
1485        if (btn) {
1486            btn.disabled = false;
1487            btn.innerHTML = '<i class="bi bi-lightning-charge me-1
"></i>Обобщить сейчас';
1488        }
1489    }
1490 }
1491
1492 function bindTabEvents() {
1493     bindSensorTabEvents();
1494
1495     const btnProcStats = document.getElementById('btn-sys-proc
-stats');
1496     if (btnProcStats) {
1497         btnProcStats.onclick = () => openProcStatsModal();
1498     }
1499
1500     const btnRefreshStats = document.getElementById('btn-refre
sh-proc-stats');
1501     if (btnRefreshStats) {
1502         btnRefreshStats.onclick = () => {
1503             const q = (document.getElementById('modal-proc-stats-f
ilter')?.value || '').trim();
1504             fetchProcStats(q);
1505         };
1506     }
1507
1508     const btnRollupTrigger = document.getElementById('btn-trig
ger-proc-rollup');
```



```
1509     if (btnRollupTrigger) {
1510         btnRollupTrigger.onclick = () => triggerProcRollupNow();
1511     }
1512
1513     const inputStatsFilter = document.getElementById('modal-pr
1514 oc-stats-filter');
1515     if (inputStatsFilter) {
1516         inputStatsFilter.oninput = (e) => fetchProcStats(e.targe
1517 t.value.trim());
1518     }
1519
1520     const searchProc = document.getElementById('sys-proc-searc
1521 h');
1522     if (searchProc) {
1523         searchProc.oninput = () => renderProcessTable();
1524     }
1525
1526     const btnAudit = document.getElementById('btn-sys-run-audi
1527 t');
1528     if (btnAudit) {
1529         btnAudit.onclick = () => runAiDiagnostics();
1530     }
1531
1532     const btnConfig = document.getElementById('btn-sys-config'
1533 );
1534     if (btnConfig) {
1535         btnConfig.onclick = () => {
1536             if (window.appConfigEditor) {
1537                 window.appConfigEditor.open('system_inspector', 'Нас
1538 тройки телеметрии и инспектора');
1539             } else {
1540                 console.info('[SystemInspectorTab] Config editor not
1541 available');
1542             }
1543         };
1544     }
1545
1546     // Interval editor modal buttons
```

```
1540     const btnIntervalsTop = document.getElementById('btn-sys-i
|ntervals');
1541     if (btnIntervalsTop) {
1542         btnIntervalsTop.onclick = () => openSysIntervalsModal();
1543     }
1544
1545     const btnIntervalsLhm = document.getElementById('btn-sys-i
|ntervals-lhm');
1546     if (btnIntervalsLhm) {
1547         btnIntervalsLhm.onclick = () => openSysIntervalsModal();
1548     }
1549
1550     const btnRefreshIntervals = document.getElementById('btn-m
|odal-intervals-refresh');
1551     if (btnRefreshIntervals) {
1552         btnRefreshIntervals.onclick = () => loadSysIntervalsConf
|ig(true);
1553     }
1554
1555     const btnSaveIntervals = document.getElementById('btn-moda
|l-intervals-save');
1556     if (btnSaveIntervals) {
1557         btnSaveIntervals.onclick = () => saveSysIntervalsConfig(
|);
1558     }
1559
1560     const selUiRefresh = document.getElementById('modal-ui-ref
|resh-select');
1561     if (selUiRefresh) {
1562         selUiRefresh.onchange = (e) => {
1563             const sec = parseInt(e.target.value, 10) || 5;
1564             setupSysSensorInterval(sec);
1565             const badge = document.getElementById('modal-ui-refres
|h-badge');
1566             if (badge) badge.textContent = `${sec} cek`;
1567         };
1568     }
1569
```

```
1570 // Изменения файлов в реальном времени controls
1571 const changeWatchDirBtn = document.getElementById('btn-sys
-change-watch-dir');
1572 if (changeWatchDirBtn) {
1573     changeWatchDirBtn.onclick = () => openWatchFoldersModal(
'folders');
1574 }
1575 const exclusionsBtn = document.getElementById('btn-sys-wat
ch-exclusions');
1576 if (exclusionsBtn) {
1577     exclusionsBtn.onclick = () => openWatchFoldersModal('exc
lusions');
1578 }
1579 const watchDirBadge = document.getElementById('sys-watch-d
ir-badge');
1580 if (watchDirBadge) {
1581     watchDirBadge.onclick = () => openWatchFoldersModal('fol
ders');
1582 }
1583 const liveHelpBtn = document.getElementById('btn-sys-live-
help');
1584 if (liveHelpBtn) {
1585     liveHelpBtn.onclick = showLiveWatcherHelpModal;
1586 }
1587
1588 // Network Activity Table controls
1589 const searchNet = document.getElementById('sys-net-search'
);
1590 if (searchNet) {
1591     searchNet.oninput = () => renderNetworkActivityTable();
1592 }
1593
1594 const btnPauseNet = document.getElementById('btn-sys-pause
-net');
1595 if (btnPauseNet) {
1596     btnPauseNet.onclick = () => {
1597         isNetPaused = !isNetPaused;
1598         btnPauseNet.innerText = isNetPaused ? 'Возобновить' :
```

```
'Пауза';
1599     btnPauseNet.className = isNetPaused ? 'btn btn-xs btn-
warning rounded-pill px-2.5 py-0.5' : 'btn btn-xs btn-outline-
secondary rounded-pill px-2.5 py-0.5';
1600     if (!isNetPaused) renderNetworkActivityTable();
1601     };
1602     }
1603
1604     const btnRefreshNet = document.getElementById('btn-sys-ref
resh-net');
1605     if (btnRefreshNet) {
1606         btnRefreshNet.onclick = () => fetchNetworkActivity();
1607     }
1608
1609     document.querySelectorAll('.sys-net-filter-btn').forEach(b
tn => {
1610         btn.onclick = () => {
1611             document.querySelectorAll('.sys-net-filter-btn').forEa
ch(b => {
1612                 b.classList.remove('active', 'btn-outline-primary');
1613                 b.classList.add('btn-outline-secondary');
1614             });
1615             btn.classList.remove('btn-outline-secondary');
1616             btn.classList.add('active', 'btn-outline-primary');
1617             currentNetFilter = btn.getAttribute('data-filter') ||
'internet';
1618             renderNetworkActivityTable();
1619         };
1620     });
1621
1622     // Инициализация интерактивных ресайзеров таблиц
1623     initNetTableResizer();
1624     initWatcherTableResizer();
1625 }
1626
1627 function initNetTableResizer() {
1628     const resizer = document.getElementById('sys-net-table-res
izer');
```

```
1629     const tableContainer = document.getElementById('sys-net-ta
ble-container');
1630     if (!resizer || !tableContainer) return;
1631
1632     if (resizer._resizerInitialized) return;
1633     resizer._resizerInitialized = true;
1634
1635     try {
1636         const savedHeight = localStorage.getItem('sys_inspector_
net_height');
1637         if (savedHeight) {
1638             const parsed = parseInt(savedHeight, 10);
1639             if (!isNaN(parsed) && parsed >= 140 && parsed <= 1200)
{
1640                 tableContainer.style.height = `${parsed}px`;
1641             }
1642         }
1643     } catch {}
1644
1645     let startY = 0;
1646     let startHeight = 0;
1647     let isDragging = false;
1648
1649     function onMouseMove(e) {
1650         if (!isDragging) return;
1651         const clientY = e.clientY ?? (e.touches && e.touches[0]
? e.touches[0].clientY : null);
1652         if (clientY === null) return;
1653
1654         const deltaY = clientY - startY;
1655         let newHeight = startHeight + deltaY;
1656         if (newHeight < 140) newHeight = 140;
1657         if (newHeight > 1200) newHeight = 1200;
1658
1659         tableContainer.style.height = `${newHeight}px`;
1660     }
1661
1662     function onMouseUp() {
```

```
1663     if (!isDragging) return;
1664     isDragging = false;
1665     resizer.classList.remove('resizing');
1666     document.body.style.cursor = '';
1667     document.body.style.userSelect = '';
1668
1669     window.removeEventListener('mousemove', onMouseMove);
1670     window.removeEventListener('mouseup', onMouseUp);
1671     window.removeEventListener('touchmove', onMouseMove);
1672     window.removeEventListener('touchend', onMouseUp);
1673
1674     const currentH = parseInt(tableContainer.style.height, 1
0);
1675     if (!isNaN(currentH)) {
1676         try {
1677             localStorage.setItem('sys_inspector_net_height', Str
ing(currentH));
1678         } catch {}
1679     }
1680 }
1681
1682 function onMouseDown(e) {
1683     isDragging = true;
1684     startY = e.clientY ?? (e.touches && e.touches[0] ? e.tou
ches[0].clientY : 0);
1685     startHeight = tableContainer.getBoundingClientRect().hei
ght;
1686     resizer.classList.add('resizing');
1687     document.body.style.cursor = 'ns-resize';
1688     document.body.style.userSelect = 'none';
1689
1690     window.addEventListener('mousemove', onMouseMove, { pass
ive: false });
1691     window.addEventListener('mouseup', onMouseUp);
1692     window.addEventListener('touchmove', onMouseMove, { pass
ive: false });
1693     window.addEventListener('touchend', onMouseUp);
1694     e.preventDefault();
```

```
1695 |     }
1696 |
1697 |     resizer.addEventListener('mousedown', onMouseDown);
1698 |     resizer.addEventListener('touchstart', onMouseDown, { passive: false });
1699 |
1700 |     resizer.addEventListener('dblclick', () => {
1701 |         tableContainer.style.height = '300px';
1702 |         try {
1703 |             localStorage.removeItem('sys_inspector_net_height');
1704 |         } catch {}
1705 |     });
1706 | }
1707 |
1708 | function initWatcherTableResizer() {
1709 |     const resizer = document.getElementById('sys-watcher-table-resizer');
1710 |     const tableContainer = document.getElementById('sys-watcher-table-container');
1711 |     if (!resizer || !tableContainer) return;
1712 |
1713 |     if (resizer._resizerInitialized) return;
1714 |     resizer._resizerInitialized = true;
1715 |
1716 |     // Восстановление сохранённой пользователем высоты
1717 |     try {
1718 |         const savedHeight = localStorage.getItem('sys_inspector_watcher_height');
1719 |         if (savedHeight) {
1720 |             const parsed = parseInt(savedHeight, 10);
1721 |             if (!isNaN(parsed) && parsed >= 120 && parsed <= 1200)
1722 |             {
1723 |                 tableContainer.style.height = `${parsed}px`;
1724 |             }
1725 |         } catch {}
1726 |
1727 |         let startY = 0;
```

```
1728 |     let startHeight = 0;
1729 |     let isDragging = false;
1730 |
1731 |     function onMouseMove(e) {
1732 |         if (!isDragging) return;
1733 |         const clientY = e.clientY ?? (e.touches && e.touches[0]
1734 | ? e.touches[0].clientY : null);
1735 |         if (clientY === null) return;
1736 |
1737 |         const deltaY = clientY - startY;
1738 |         let newHeight = startHeight + deltaY;
1739 |         if (newHeight < 120) newHeight = 120;
1740 |         if (newHeight > 1200) newHeight = 1200;
1741 |
1742 |         tableContainer.style.height = `${newHeight}px`;
1743 |     }
1744 |
1745 |     function onMouseUp() {
1746 |         if (!isDragging) return;
1747 |         isDragging = false;
1748 |         resizer.classList.remove('resizing');
1749 |         document.body.style.cursor = '';
1750 |         document.body.style.userSelect = '';
1751 |
1752 |         window.removeEventListener('mousemove', onMouseMove);
1753 |         window.removeEventListener('mouseup', onMouseUp);
1754 |         window.removeEventListener('touchmove', onMouseMove);
1755 |         window.removeEventListener('touchend', onMouseUp);
1756 |
1757 |         const currentH = parseInt(tableContainer.style.height, 1
1758 | 0);
1759 |         if (!isNaN(currentH)) {
1760 |             try {
1761 |                 localStorage.setItem('sys_inspector_watcher_height',
1762 | String(currentH));
1763 |             } catch {}
1764 |         }
1765 |     }
```



```
1763
1764     function onMouseDown(e) {
1765         isDragging = true;
1766         startY = e.clientY ?? (e.touches && e.touches[0] ? e.tou
ches[0].clientY : 0);
1767         startHeight = tableContainer.getBoundingClientRect().hei
ght;
1768         resizer.classList.add('resizing');
1769         document.body.style.cursor = 'ns-resize';
1770         document.body.style.userSelect = 'none';
1771
1772         window.addEventListener('mousemove', onMouseMove, { pass
ive: false });
1773         window.addEventListener('mouseup', onMouseUp);
1774         window.addEventListener('touchmove', onMouseMove, { pass
ive: false });
1775         window.addEventListener('touchend', onMouseUp);
1776         e.preventDefault();
1777     }
1778
1779     resizer.addEventListener('mousedown', onMouseDown);
1780     resizer.addEventListener('touchstart', onMouseDown, { pass
ive: false });
1781
1782     // Сброс по двойному клику
1783     resizer.addEventListener('dblclick', () => {
1784         tableContainer.style.height = '280px';
1785         try {
1786             localStorage.removeItem('sys_inspector_watcher_height'
);
1787         } catch {}
1788     });
1789 }
1790
1791 // =====
=====
1792 // Telemetry & Polling Intervals Editor Controller (config.j
son)
```

```

1793 // =====
1794 =====
1795 const INTERVAL_PRESETS = [
1796   { value: '1 second', label: '1 сек' },
1797   { value: '2 seconds', label: '2 сек' },
1798   { value: '3 seconds', label: '3 сек' },
1799   { value: '5 seconds', label: '5 сек' },
1800   { value: '10 seconds', label: '10 сек' },
1801   { value: '30 seconds', label: '30 сек' },
1802   { value: '1 minute', label: '1 мин' },
1803   { value: '5 minutes', label: '5 мин' },
1804   { value: '10 minutes', label: '10 мин' },
1805   { value: '30 minutes', label: '30 мин' },
1806   { value: '1 hour', label: '1 час' },
1807   { value: '6 hours', label: '6 часов' },
1808   { value: '24 hours', label: '24 часа' },
1809 ];
1810
1811 const CORE_RESOURCE_LOGGERS = [
1812   'librehardwaremonitor',
1813   'system_inspector',
1814   'hardware_monitor'
1815 ];
1816
1817 const LOGGER_META = {
1818   librehardwaremonitor: {
1819     icon: '⚙',
1820     title: 'LibreHardwareMonitor',
1821     desc: 'Аппаратные сенсоры: температура CPU/GPU, вольтаж,
1822 обороты вентиляторов, частоты',
1823   },
1824   system_inspector: {
1825     icon: '⚙',
1826     title: 'System Inspector',
1827     desc: 'Комплексный снимок: загрузка CPU, RAM, дисковый в
1828 вод-вывод, процессы хоста',
1829   },
1830 };

```

```
1828     hardware_monitor: {
1829         icon: '□ ',
1830         title: 'Hardware Monitor',
1831         desc: 'Аппаратные датчики WMI, GPU SMI (NVIDIA/AMD/Intel
) и физические шины',
1832     },
1833     windows_sysadmin: {
1834         icon: '□ ',
1835         title: 'Windows SysAdmin',
1836         desc: 'Системные службы Windows, пользователи, локальные
группы и сетевые порты',
1837     },
1838     windows_defender: {
1839         icon: '□ ',
1840         title: 'Windows Defender',
1841         desc: 'Статус антивирусной защиты, сигнатуры, обнаруженн
ые угрозы и карантин',
1842     },
1843     windows_startup_auditor: {
1844         icon: '□ ',
1845         title: 'StartUp Auditor',
1846         desc: 'Автозагрузка программ, реестр Run/RunOnce, сервис
ы и планировщик задач',
1847     },
1848     windows_backup_manager: {
1849         icon: '□ ',
1850         title: 'Backup Manager',
1851         desc: 'Резервные копии баз данных SQLite, конфигураций и
снапшотов системы',
1852     },
1853     website_monitor: {
1854         icon: '□ ',
1855         title: 'Website Monitor',
1856         desc: 'Проверка доступности веб-сервисов, задержка ответ
ов HTTP/HTTPS',
1857     },
1858     gcloud_monitor: {
1859         icon: '□ ',
```

```
1860         title: 'GCloud Monitor',
1861         desc: 'Телеметрия виртуальных машин, квоты и мониторинг
Google Cloud',
1862     },
1863     cloudflared_monitor: {
1864         icon: '☐ ',
1865         title: 'Cloudflare Tunnel',
1866         desc: 'Статус защищенных туннелей cloudflared и исходящи
х подключений',
1867     },
1868     user_assistant: {
1869         icon: '☐ ',
1870         title: 'User Assistant',
1871         desc: 'Фоновые периодические напоминания, календарь и ув
едомления',
1872     },
1873     trading_terminal: {
1874         icon: '☐ ',
1875         title: 'Trading Terminal',
1876         desc: 'Торговые котировки, балансы криптобирж и открытые
ордера',
1877     },
1878     registry_viewer: {
1879         icon: '☐ ',
1880         title: 'Registry Viewer',
1881         desc: 'Мониторинг изменений системного реестра Windows',
1882     },
1883     software_audit: {
1884         icon: '☐ ',
1885         title: 'Software Audit',
1886         desc: 'Аудит неиспользуемого и редко запускаемого ПО (Us
erAssist/Prefetch)',
1887     },
1888     helpdesk: {
1889         icon: '☐ ',
1890         title: 'Helpdesk',
1891         desc: 'Мониторинг тикетов техподдержки и уведомлений',
1892     },
```

```
1893 | };
1894 |
1895 | let _sysIntervalsConfigData = null;
1896 |
1897 | function showAlertInModal(msg, type = 'success') {
1898 |     const alertEl = document.getElementById('modal-intervals-a
1899 | lert');
1900 |     if (!alertEl) return;
1901 |     alertEl.className = `alert alert-${type} py-2 px-3 small m
1902 | b-3`;
1903 |     alertEl.innerHTML = msg;
1904 |     alertEl.classList.remove('d-none');
1905 |     if (type === 'success') {
1906 |         setTimeout(() => {
1907 |             alertEl.classList.add('d-none');
1908 |         }, 4000);
1909 |     }
1910 | }
1911 |
1912 | async function loadSysIntervalsConfig(notify = false) {
1913 |     const btnRefresh = document.getElementById('btn-modal-inte
1914 | rvals-refresh');
1915 |     const icon = btnRefresh ? btnRefresh.querySelector('i') :
1916 | null;
1917 |     if (icon) icon.classList.add('spin-animation');
1918 |
1919 |     try {
1920 |         let res = await fetch('/api/autolog/config');
1921 |         if (!res.ok) {
1922 |             res = await fetch('/sysautologging/config');
1923 |         }
1924 |         if (!res.ok) {
1925 |             throw new Error(`HTTP ${res.status}`);
1926 |         }
1927 |
1928 |         _sysIntervalsConfigData = await res.json();
1929 |         renderSysIntervalsForm(_sysIntervalsConfigData);
1930 |     }
1931 | }
```

```
1927     if (notify) {
1928         showAlertInModal('<i class="bi bi-check2 me-1"></i> Ко
1929         нфигурация успешно загружена из файла', 'success');
1930     }
1931     } catch (e) {
1932         console.warn('[SystemInspectorTab] Failed to load interv
1933         als config:', e);
1934         showAlertInModal(`<i class="bi bi-exclamation-triangle m
1935         e-1"></i> Ошибка загрузки конфигурации: ${escapeHtml(e.message
1936         )}` , 'danger');
1937     } finally {
1938         if (icon) icon.classList.remove('spin-animation');
1939     }
1940 }
1941
1942 function buildIntervalSelectHtml(loggerName, currentInterval
1943 ) {
1944     const val = currentInterval || '1 minute';
1945     const isCustom = !INTERVAL_PRESETS.some(p => p.value === v
1946     al);
1947
1948     let html = `<select class="form-select form-select-sm bg-d
1949     ark text-white border-secondary sys-int-select" data-logger="$
1950     {loggerName}" style="min-width: 120px;">`;
1951     INTERVAL_PRESETS.forEach(p => {
1952         const sel = p.value === val ? 'selected' : '';
1953         html += `<option value="${p.value}" ${sel}>${p.label} ($
1954         {p.value})</option>`;
1955     });
1956     if (isCustom) {
1957         html += `<option value="${escapeHtml(val)}" selected>${e
1958         scapeHtml(val)} (пользовательский)</option>`;
1959     }
1960     html += `</select>`;
1961     return html;
1962 }
1963
1964 function renderSysIntervalsForm(data) {
```

```
1955 |     if (!data) return;
1956 |
1957 |     // 1. Config file badge
1958 |     const cfgBadge = document.getElementById('modal-intervals-
| cfg-file');
1959 |     if (cfgBadge && data.config_file) {
1960 |         cfgBadge.textContent = data.config_file;
1961 |     }
1962 |
1963 |     // 2. Engine status & global switch
1964 |     const engineStatusBadge = document.getElementById('modal-i
| ntervals-engine-status');
1965 |     if (engineStatusBadge) {
1966 |         if (data.is_running) {
1967 |             engineStatusBadge.textContent = 'AutoLog Active';
1968 |             engineStatusBadge.className = 'badge bg-success';
1969 |         } else {
1970 |             engineStatusBadge.textContent = 'AutoLog Paused';
1971 |             engineStatusBadge.className = 'badge bg-secondary';
1972 |         }
1973 |     }
1974 |
1975 |     const globalEnable = document.getElementById('modal-interv
| als-global-enable');
1976 |     if (globalEnable) {
1977 |         globalEnable.checked = data.enable_autolog !== false;
1978 |     }
1979 |
1980 |     // 3. Default fallback interval select
1981 |     const defSelect = document.getElementById('modal-intervals
| -default-select');
1982 |     if (defSelect && data.default_interval) {
1983 |         defSelect.value = data.default_interval;
1984 |     }
1985 |
1986 |     // 4. Populate Core resource loggers & Other loggers
1987 |     const coreContainer = document.getElementById('modal-core-
| loggers-container');
```

```

1988     const otherContainer = document.getElementById('modal-othe
1989     const otherCountEl = document.getElementById('modal-other-
1990
1991     if (!coreContainer || !otherContainer) return;
1992
1993     const loggers = data.loggers || {};
1994     let coreHtml = '';
1995     let otherHtml = '';
1996     let otherCount = 0;
1997
1998     // Process core loggers
1999     CORE_RESOURCE_LOGGERS.forEach(name => {
2000         const cfg = loggers[name] || { interval: '5 seconds', en
2001         const meta = LOGGER_META[name] || { icon: '□ ', title: na
2002         const isEnabled = cfg.enabled !== false;
2003         const intervalVal = cfg.interval || '5 seconds';
2004
2005         coreHtml += `
2006         <div class="p-2 rounded d-flex align-items-center just
2007         ify-content-between flex-wrap gap-2" style="background: var(--
2008         bg-color); border: 1px solid var(--border-color);">
2009         <div class="d-flex align-items-center gap-2" style="
2010         max-width: 58%;">
2011         <span class="fs-5">${meta.icon}</span>
2012         <div>
2013         <div class="fw-bold small" style="color: var(--t
2014         ext-color);">${meta.title}</div>
2015         <div class="small text-muted" style="font-size:
2016         0.72rem; line-height: 1.2;">${meta.desc}</div>
2017         </div>
2018         </div>
2019         <div class="d-flex align-items-center gap-2 ms-auto"
2020         >
2021         <div class="form-check form-switch m-0" title="Вкл

```



```

        ючить/отключить сбор метрик для этого сервиса">
2016         <input class="form-check-input sys-int-enable" t
        ype="checkbox" data-logger="${name}" ${isEnabled ? 'checked' :
        ''}>
2017     </div>
2018     ${buildIntervalSelectHtml(name, intervalVal)}
2019     <button class="btn btn-xs btn-outline-secondary ro
        unded px-1.5 py-0.5 sys-int-custom-btn" data-logger="${name}"
        title="Ввести произвольный интервал вручную">
2020         <i class="bi bi-pencil"></i>
2021     </button>
2022 </div>
2023 </div>
2024 `;
2025 });
2026
2027 // Process remaining loggers
2028 const allLoggerNames = Object.keys(loggers).length > 0
2029   ? Object.keys(loggers)
2030   : Object.keys(LOGGER_META);
2031
2032 allLoggerNames.forEach(name => {
2033   if (CORE_RESOURCE_LOGGERS.includes(name)) return;
2034   otherCount++;
2035   const cfg = loggers[name] || { interval: data.default_in
        terval || '1 minute', enabled: true };
2036   const meta = LOGGER_META[name] || { icon: '□ ', title: n
        ame, desc: 'Фоновый опрос службы' };
2037   const isEnabled = cfg.enabled !== false;
2038   const intervalVal = cfg.interval || '1 minute';
2039
2040   otherHtml += `
2041     <div class="p-2 rounded d-flex align-items-center just
        ify-content-between flex-wrap gap-2" style="background: var(--
        bg-color); border: 1px solid var(--border-color);">
2042       <div class="d-flex align-items-center gap-2" style="
        max-width: 58%;">
2043         <span class="fs-6">${meta.icon}</span>

```

```

2044         <div>
2045             <div class="fw-bold small" style="color: var(--t
ext-color); font-size: 0.78rem;">${meta.title}</div>
2046             <div class="small text-muted" style="font-size:
0.7rem; line-height: 1.2;">${meta.desc}</div>
2047         </div>
2048     </div>
2049     <div class="d-flex align-items-center gap-2 ms-auto"
>
2050         <div class="form-check form-switch m-0" title="Вкл
ючить/выключить">
2051             <input class="form-check-input sys-int-enable" t
ype="checkbox" data-logger="${name}" ${isEnabled ? 'checked' :
''}>
2052         </div>
2053         ${buildIntervalSelectHtml(name, intervalVal)}
2054         <button class="btn btn-xs btn-outline-secondary ro
unded px-1.5 py-0.5 sys-int-custom-btn" data-logger="${name}"
title="Ввести произвольный интервал вручную">
2055             <i class="bi bi-pencil"></i>
2056         </button>
2057     </div>
2058 </div>
2059 `;
2060 });
2061
2062 coreContainer.innerHTML = coreHtml;
2063 otherContainer.innerHTML = otherHtml;
2064 if (otherCountEl) otherCountEl.textContent = String(otherC
ount);
2065
2066 // Bind custom interval prompt buttons
2067 document.querySelectorAll('.sys-int-custom-btn').forEach(b
tn => {
2068     btn.onclick = () => {
2069         const lName = btn.getAttribute('data-logger');
2070         const sel = document.querySelector(`.sys-int-select[da
ta-logger="${lName}"]`);

```

```
2071 |         const currentVal = sel ? sel.value : '1 minute';
2072 |         const custom = prompt(`Введите интервал опроса для ${l
Name} (например: "2 seconds", "15 seconds", "10 minutes", "1 h
our"):`, currentVal);
2073 |         if (custom && custom.trim()) {
2074 |             const trimmed = custom.trim();
2075 |             let opt = sel.querySelector(`option[value="${trimmed
}"]`);
2076 |             if (!opt) {
2077 |                 opt = document.createElement('option');
2078 |                 opt.value = trimmed;
2079 |                 opt.textContent = `${trimmed} (пользовательский)`;
2080 |                 sel.appendChild(opt);
2081 |             }
2082 |             sel.value = trimmed;
2083 |         }
2084 |     };
2085 | });
2086 | }
2087 |
2088 | async function saveSysIntervalsConfig() {
2089 |     const btnSave = document.getElementById('btn-modal-interva
ls-save');
2090 |     if (btnSave) {
2091 |         btnSave.disabled = true;
2092 |         btnSave.innerHTML = '<span class="spinner-border spinner
-border-sm me-1" role="status"></span> Сохранение...';
2093 |     }
2094 |
2095 |     try {
2096 |         const globalEnable = document.getElementById('modal-inte
rvals-global-enable');
2097 |         const defSelect = document.getElementById('modal-interva
ls-default-select');
2098 |
2099 |         const payload = {
2100 |             enable_autolog: globalEnable ? globalEnable.checked :
true,
```

```

2101         default_interval: defSelect ? defSelect.value : '1 min
    ute',
2102         loggers: {}
2103     };
2104
2105     // Collect values from all interval selects & switches
2106     document.querySelectorAll('.sys-int-select').forEach(sel
    => {
2107         const lName = sel.getAttribute('data-logger');
2108         if (!lName) return;
2109         const sw = document.querySelector(`.sys-int-enable[dat
    a-logger="${lName}"]`);
2110         payload.loggers[lName] = {
2111             interval: sel.value,
2112             enabled: sw ? sw.checked : true
2113         };
2114     });
2115
2116     let saveRes = await fetch('/api/autolog/config', {
2117         method: 'POST',
2118         headers: { 'Content-Type': 'application/json' },
2119         body: JSON.stringify(payload)
2120     });
2121
2122     if (!saveRes.ok) {
2123         saveRes = await fetch('/sysautologging/config', {
2124             method: 'POST',
2125             headers: { 'Content-Type': 'application/json' },
2126             body: JSON.stringify(payload)
2127         });
2128     }
2129
2130     if (!saveRes.ok) {
2131         const errJson = await saveRes.json().catch(() => ({}))
    ;
2132         throw new Error(errJson.detail || `HTTP ${saveRes.stat
    us}`);
2133     }

```

```
2134 |
2135 |     const result = await saveRes.json();
2136 |     showAlertInModal(
2137 |         `<i class="bi bi-check-circle-fill text-success me-1">
    | </i> ${result.message || 'Интервалы успешно сохранены в config
    | .json и применены!'}` ,
2138 |         'success'
2139 |     );
2140 |
2141 |     // Refresh engine badge
2142 |     const engineStatusBadge = document.getElementById('modal
    | -intervals-engine-status');
2143 |     if (engineStatusBadge) {
2144 |         if (result.is_running) {
2145 |             engineStatusBadge.textContent = 'AutoLog Active';
2146 |             engineStatusBadge.className = 'badge bg-success';
2147 |         } else {
2148 |             engineStatusBadge.textContent = 'AutoLog Paused';
2149 |             engineStatusBadge.className = 'badge bg-secondary';
2150 |         }
2151 |     }
2152 | } catch (e) {
2153 |     console.error('[SystemInspectorTab] Save intervals confi
    | g error:', e);
2154 |     showAlertInModal(`<i class="bi bi-exclamation-triangle-f
    | ill text-danger me-1"></i> Не удалось сохранить интервалы: ${e
    | escapeHtml(e.message)}` , 'danger');
2155 | } finally {
2156 |     if (btnSave) {
2157 |         btnSave.disabled = false;
2158 |         btnSave.innerHTML = `<i class="bi bi-check2-circle me-
    | 1"></i> Сохранить в config.json';
2159 |     }
2160 | }
2161 | }
2162 |
2163 | let currentWatchDirs = [];
2164 | let currentWatchDir = '';
```

```
2165 | let stagedWatchDirs = [];
2166 | let currentBrowserPath = 'C:\\';
2167 | let cachedDrives = [];
2168 |
2169 | async function fetchLiveFileEvents() {
2170 |     try {
2171 |         const [resEvents, resTelem] = await Promise.all([
2172 |             fetch('/api/sysadmin/file-audit/live-events?limit=30')
2173 |             ,
2174 |             fetch('/api/sysadmin/file-audit/telemetry').catch(() => null)
2175 |         ]);
2176 |         if (resTelem && resTelem.ok) {
2177 |             const telem = await resTelem.json();
2178 |             const elRate = document.getElementById('sys-watcher-rate');
2179 |             const elCr = document.getElementById('sys-watcher-cr');
2180 |             ;
2181 |             const elMod = document.getElementById('sys-watcher-mod');
2182 |             const elDel = document.getElementById('sys-watcher-del');
2183 |             const elDrive = document.getElementById('sys-watcher-drive');
2184 |             const elDriveModel = document.getElementById('sys-watcher-drive-model');
2185 |             const elDriveTemp = document.getElementById('sys-watcher-temp');
2186 |             const elDiskR = document.getElementById('sys-watcher-r-kbs');
2187 |             const elDiskW = document.getElementById('sys-watcher-w-kbs');
2188 |             const elSecMatches = document.getElementById('sys-watcher-sec-matches');
2189 |             const elStatus = document.getElementById('sys-watcher-sensor-status');
```

```
2190 |         if (elRate) elRate.textContent = (telem.events_rate_per_sec || 0).toFixed(1);
2191 |         if (elCr) elCr.textContent = (telem.created_rate_per_sec || 0).toFixed(1);
2192 |         if (elMod) elMod.textContent = (telem.modified_rate_per_sec || 0).toFixed(1);
2193 |         if (elDel) elDel.textContent = (telem.deleted_rate_per_sec || 0).toFixed(1);
2194 |
2195 |         if (elDrive) {
2196 |             const drivesStr = (telem.drive_letters && telem.drive_letters.length)
2197 |                 ? telem.drive_letters.join(', ')
2198 |                 : (telem.drive_letter || 'C:');
2199 |             elDrive.textContent = drivesStr;
2200 |         }
2201 |         if (elDriveModel) elDriveModel.textContent = telem.drive_model || 'Storage';
2202 |         if (elDriveTemp) elDriveTemp.textContent = telem.drive_temperature_c !== null && telem.drive_temperature_c !== undefined ? `${telem.drive_temperature_c} °C` : '-- °C';
2203 |         if (elDiskR) elDiskR.textContent = Math.round(telem.disk_read_kbs || 0);
2204 |         if (elDiskW) elDiskW.textContent = Math.round(telem.disk_write_kbs || 0);
2205 |         if (elSecMatches) elSecMatches.textContent = telem.security_audit_matched_count || 0;
2206 |
2207 |         if (elStatus) {
2208 |             if (telem.burst_deletions_alert) {
2209 |                 elStatus.className = 'badge bg-danger-subtle text-danger border border-danger px-2 py-1';
2210 |                 elStatus.textContent = '⚠ Всплеск удалений';
2211 |             } else if (telem.high_activity_alert) {
2212 |                 elStatus.className = 'badge bg-warning-subtle text-warning border border-warning px-2 py-1';
2213 |                 elStatus.textContent = '⚠ Высокий I/O';
2214 |             } else {
```

```
2215 |         elStatus.className = 'badge bg-success-subtle text  
| -success border border-success px-2 py-1';  
2216 |         elStatus.textContent = 'Штатный режим';  
2217 |     }  
2218 | }  
2219 | }  
2220 |  
2221 | if (!resEvents || !resEvents.ok) return;  
2222 | const data = await resEvents.json();  
2223 | const events = data.events || [];  
2224 |  
2225 | if (data.filtered_count !== undefined) {  
2226 |     const filteredEl = document.getElementById('sys-watche  
| r-filtered-count');  
2227 |     if (filteredEl) filteredEl.textContent = data.filtered  
| _count;  
2228 | }  
2229 |  
2230 |     currentWatchDirs = data.watch_dirs || (data.watch_dir ?  
| [data.watch_dir] : []);  
2231 |     currentWatchDir = currentWatchDirs[0] || data.watch_dir  
| || '';  
2232 |  
2233 |     const dirPathEl = document.getElementById('sys-watch-dir  
| -path');  
2234 |     const badge = document.getElementById('sys-watch-dir-bad  
| ge');  
2235 |     if (dirPathEl) {  
2236 |         if (currentWatchDirs.length === 0) {  
2237 |             dirPathEl.innerText = 'Папки не выбраны';  
2238 |         } else if (currentWatchDirs.length === 1) {  
2239 |             const singleName = currentWatchDirs[0].split('\\').p  
| op() || currentWatchDirs[0];  
2240 |             dirPathEl.innerText = singleName;  
2241 |         } else {  
2242 |             const firstNames = currentWatchDirs.slice(0, 2).map(  
| p => p.split('\\').pop() || p).join(', ');  
2243 |             dirPathEl.innerText = `${currentWatchDirs.length} на
```



```

    пок: ${firstNames}${currentWatchDirs.length > 2 ? '...' : ''}`
    ;
2244     }
2245     }
2246     if (badge) {
2247         badge.title = `Отслеживаемые каталоги (${currentWatchD
irs.length}): \n${currentWatchDirs.join('\n')}\n\n(Нажмите для
настройки и выбора папок)`;
2248     }
2249
2250     const tbody = document.getElementById('sys-watcher-tbody
');
2251     if (tbody) {
2252         if (events.length === 0) {
2253             const labelDirs = currentWatchDirs.length > 0 ? curr
entWatchDirs.join(', ') : 'проекта';
2254             tbody.innerHTML = `|<td colspan="4" class="text-c
enter text-muted p-2">Ожидание изменений в папках <code>${labe
lDirs}</code>...</td></tr>`;
2255             return;
2256         }
2257         tbody.innerHTML = events.map((e, idx) => {
2258             const rootDirHint = e.watch_dir ? (e.watch_dir.split
('\').pop() || e.watch_dir) : '';
2259             const procDisplay = e.process_name
2260                 ? `

|  |

```

```

timestamp?.slice(11, 19) || ''}</td>
2265         <td>
2266             <span class="badge ${e.is_deletion ? 'bg-dange
r' : (e.action === 'Created' ? 'bg-success' : 'bg-secondary')}"
">${e.action}</span>
2267             ${rootDirHint && currentWatchDirs.length > 1 ?
`<span class="badge bg-dark border border-secondary text-mute
d ms-1" style="font-size: 0.65rem;" title="Корень: ${escapeHtm
l(e.watch_dir)}">${rootDirHint}</span>` : ''}
2268         </td>
2269         <td>${procDisplay}</td>
2270         <td class="font-monospace small text-light" styl
e="word-break: break-all;" title="${escapeHtml(e.path)}">${esc
apeHtml(e.path)}</td>
2271     </tr>
2272     `;
2273     }).join('');
2274
2275     tbody.querySelectorAll('.sys-live-row').forEach(row =>
{
2276         row.onclick = () => {
2277             const idx = parseInt(row.getAttribute('data-idx'),
10);
2278             const e = events[idx];
2279             if (!e) return;
2280
2281             if (window.AITableModal) {
2282                 window.AITableModal.show({
2283                     icon: '📁',
2284                     title: `Файловое событие: ${e.action}`,
2285                     subtitle: `${e.path} | ${e.timestamp}`,
2286                     tableType: 'file_event',
2287                     badges: [
2288                         { text: e.action, class: e.is_deletion ? 'ba
dge bg-danger' : (e.action === 'Created' ? 'badge bg-success'
: 'badge bg-info text-dark') },
2289                         { text: e.process_name ? `Программа: ${e.pro
cess_name}` : 'WinAPI ReadDirectoryChangesW', class: 'badge bg

```

```

    -dark border border-secondary text-info' },
2290         { text: 'WinAPI', class: 'badge bg-secondary
    ' }
2291     ],
2292     metadata: [
2293         { label: 'Действие', value: e.action },
2294         { label: 'Полный путь к файлу', value: e.pat
    h },
2295         { label: 'Программа / Процесс', value: e.pro
    cess_name ? `${e.process_name}${e.process_id ? ` (PID: ${e.pro
    cess_id})` : ''}` : 'Фоновый процесс / завершен' },
2296         { label: 'Время события', value: e.timestamp
    },
2297         { label: 'Признак удаления', value: e.is_del
    etion ? 'Да (Файл удален/переименован)' : 'Нет' },
2298         { label: 'Папка события', value: e.watch_dir
    || currentWatchDir },
2299         { label: 'Все отслеживаемые папки', value: c
    urrentWatchDirs.join('; ') }
2300     ],
2301     rawTitle: 'Детали события WinAPI & Process Inf
    o',
2302     rawContent: JSON.stringify(e, null, 2),
2303     requestData: e
    });
2304     });
2305     }
2306     };
2307     });
2308     }
2309     } catch (e) {
2310         console.error('[SystemInspectorTab] Failed to fetch live
    file events:', e);
2311     }
2312     }
2313
2314     // =====
    =====
2315     // Multi-Directory Watcher Manager & Folder Explorer Modal L

```

```

logic
2316 // =====
=====
2317
2318 function showWatchDirsAlert(msg, type = 'success') {
2319     const alertEl = document.getElementById('modal-watch-dirs-
alert');
2320     if (!alertEl) return;
2321     alertEl.className = `alert alert-${type} py-2 px-3 small m
b-3`;
2322     alertEl.innerHTML = msg;
2323     alertEl.classList.remove('d-none');
2324     if (type === 'success') {
2325         setTimeout(() => {
2326             alertEl.classList.add('d-none');
2327         }, 3500);
2328     }
2329 }
2330
2331 function renderModalActiveDirsList() {
2332     const container = document.getElementById('modal-watch-dir
s-active-list');
2333     const countBadge = document.getElementById('modal-watch-di
rs-count');
2334     const hintEl = document.getElementById('modal-watch-dirs-f
ooter-hint');
2335
2336     if (!container) return;
2337
2338     if (countBadge) countBadge.textContent = String(stagedWatc
hDirs.length);
2339     if (hintEl) hintEl.textContent = `Выбрано папок для монито
ринга: ${stagedWatchDirs.length}`;
2340
2341     if (stagedWatchDirs.length === 0) {
2342         container.innerHTML = `
2343             <div class="text-center text-muted small py-3">
2344                 <i class="bi bi-folder-x me-1"></i> Список пуст. Выб

```

ерите папки в проводнике ниже или воспользуйтесь быстрыми пресетами.

```
2345     </div>
2346     `;
2347     return;
2348   }
2349
2350   container.innerHTML = stagedWatchDirs.map((dirPath, idx) =
> {
2351     const folderName = dirPath.split('\\').pop() || dirPath;
2352     const driveLetter = (dirPath.slice(0, 2)).toUpperCase();
2353     return `
2354       <div class="p-1.5 px-2 rounded d-flex align-items-cent
er justify-content-between gap-2" style="background: var(--bg-
color); border: 1px solid var(--border-color);">
2355         <div class="d-flex align-items-center gap-2 text-tru
ncate" style="max-width: 82%;">
2356           <i class="bi bi-folder-check text-warning fs-6"></
i>
2357           <span class="badge bg-secondary font-monospace" st
yle="font-size: 0.68rem;">${driveLetter}</span>
2358           <div class="text-truncate">
2359             <span class="fw-bold small text-light">${escapeH
tml(folderName)}</span>
2360             <span class="small text-muted font-monospace d-b
lock text-truncate" style="font-size: 0.72rem;" title="${escap
eHtml(dirPath)}">${escapeHtml(dirPath)}</span>
2361           </div>
2362         </div>
2363         <button class="btn btn-xs btn-outline-danger rounded
-pill px-2 py-0.5" onclick="window._removeStagedWatchDir(${idx
})" title="Удалить из списка мониторинга">
2364           <i class="bi bi-trash3"></i>
2365         </button>
2366       </div>
2367     `;
2368   }).join('');
2369 }
```

```
2370 |
2371 | window._removeStagedWatchDir = function(index) {
2372 |     if (index >= 0 && index < stagedWatchDirs.length) {
2373 |         const removed = stagedWatchDirs.splice(index, 1)[0];
2374 |         renderModalActiveDirsList();
2375 |         showWatchDirsAlert(`Папка удалена из списка: <code>${escapeHtml(removed)}</code>`, 'info');
2376 |     }
2377 | };
2378 |
2379 | function addDirToStaged(pathToAdd) {
2380 |     if (!pathToAdd || !pathToAdd.trim()) return;
2381 |     const cleanPath = pathToAdd.trim();
2382 |     if (stagedWatchDirs.includes(cleanPath)) {
2383 |         showWatchDirsAlert(`Папка уже есть в списке: <code>${escapeHtml(cleanPath)}</code>`, 'warning');
2384 |         return;
2385 |     }
2386 |     stagedWatchDirs.push(cleanPath);
2387 |     renderModalActiveDirsList();
2388 |     showWatchDirsAlert(`Папка добавлена: <code>${escapeHtml(cleanPath)}</code>`, 'success');
2389 | }
2390 |
2391 | async function loadSystemDrives() {
2392 |     const drivesBar = document.getElementById('modal-watch-drives-bar');
2393 |     if (!drivesBar) return;
2394 |
2395 |     try {
2396 |         const res = await fetch('/api/sysadmin/filesystem/drives');
2397 |         if (res.ok) {
2398 |             const data = await res.json();
2399 |             cachedDrives = data.drives || [];
2400 |         }
2401 |     } catch (e) {
2402 |         console.warn('[SystemInspectorTab] Failed to fetch syste
```

```

    m drives:', e);
2403     cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
2404 }
2405
2406 if (cachedDrives.length === 0) {
2407     cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
2408 }
2409
2410 drivesBar.innerHTML = cachedDrives.map(d => {
2411     const label = d.mountpoint || 'C:\\';
2412     const freeTxt = d.free_gb ? `${d.free_gb} GB free` : '';
2413     return `
2414         <button class="btn btn-xs btn-outline-light rounded-pi
2415         ll px-2 py-0.5 sys-drive-btn font-monospace" data-drive="${esc
2416         apeHtml(label)}" title="${escapeHtml(label)} ${freeTxt}">
2417             <i class="bi bi-hdd me-1"></i>${escapeHtml(label)}
2418         </button>
2419     `;
2420     }).join('');
2421
2422 drivesBar.querySelectorAll('.sys-drive-btn').forEach(btn =
2423 > {
2424     btn.onclick = () => {
2425         const drv = btn.getAttribute('data-drive');
2426         if (drv) loadFolderBrowser(drv);
2427     };
2428 });
2429 }
2430
2431 function renderBreadcrumbs(currentPath) {
2432     const bcContainer = document.getElementById('modal-folder-
2433     breadcrumbs');
2434     if (!bcContainer) return;
2435
2436     const parts = currentPath.replace(/\\+$/, '').split('\\');
2437     let builtPath = '';
2438     const crumbsHtml = parts.map((part, idx) => {
2439         if (idx === 0) {

```

```

2436         builtPath = part + '\\';
2437     } else {
2438         builtPath = builtPath + (builtPath.endsWith('\\') ? ''
: '\\') + part;
2439     }
2440     const clickPath = builtPath;
2441     const isLast = idx === parts.length - 1;
2442     return `
2443         <span class="d-inline-flex align-items-center">
2444             ${idx > 0 ? '<span class="text-muted mx-1">/</span>'
: ''}
2445             <a href="#" class="sys-breadcrumb-link text-decorati
on-none ${isLast ? 'fw-bold text-info' : 'text-light'}" data-p
ath="${escapeHtml(clickPath)}">
2446                 ${escapeHtml(part || 'Корень')}
2447             </a>
2448         </span>
2449     `;
2450     }).join('');
2451
2452     bcContainer.innerHTML = crumbsHtml;
2453
2454     bcContainer.querySelectorAll('.sys-breadcrumb-link').forEa
ch(link => {
2455         link.onclick = (e) => {
2456             e.preventDefault();
2457             const p = link.getAttribute('data-path');
2458             if (p) loadFolderBrowser(p);
2459         };
2460     });
2461 }
2462
2463 async function loadFolderBrowser(targetPath) {
2464     const inputPath = document.getElementById('modal-folder-pa
th-input');
2465     const browserList = document.getElementById('modal-folder-
browser-list');
2466     if (!targetPath) targetPath = 'C:\\';

```



```

2467     currentBrowserPath = targetPath;
2468
2469     if (inputPath) inputPath.value = targetPath;
2470     renderBreadcrumbs(targetPath);
2471
2472     if (!browserList) return;
2473     browserList.innerHTML = `
```

```

runcate flex-grow-1 sys-nav-to-folder" data-path="${escapeHtml
(d.path)}" title="Нажмите для перехода в папку">
2495     <i class="bi ${d.has_subdirs ? 'bi-folder2 text-
warning' : 'bi-folder text-warning'}"></i>
2496     <span class="small text-light text-truncate">${e
scapeHtml(d.name)}</span>
2497     <span class="small text-muted font-monospace ms-
auto me-2" style="font-size: 0.68rem;">${d.modified || ''}</sp
an>
2498     </div>
2499     <button class="btn btn-xs ${isAlreadySelected ? 'b
tn-success' : 'btn-outline-info'} rounded-pill px-2 py-0.5 sys
-add-folder-btn" data-path="${escapeHtml(d.path)}" title="Доба
вить в отслеживаемые">
2500         <i class="bi ${isAlreadySelected ? 'bi-check2' :
'bi-plus-lg'} me-1"></i>${isAlreadySelected ? 'Выбрана' : 'Сл
едить'}
2501     </button>
2502 </div>
2503 `;
2504 }).join('');
2505
2506 // Bind folder navigation
2507 browserList.querySelectorAll('.sys-nav-to-folder').forEa
ch(el => {
2508     el.onclick = () => {
2509         const p = el.getAttribute('data-path');
2510         if (p) loadFolderBrowser(p);
2511     };
2512 });
2513
2514 // Bind Add buttons
2515 browserList.querySelectorAll('.sys-add-folder-btn').forE
ach(btn => {
2516     btn.onclick = (e) => {
2517         e.stopPropagation();
2518         const p = btn.getAttribute('data-path');
2519         if (p) {

```

```
2520         addDirToStaged(p);
2521         btn.className = 'btn btn-xs btn-success rounded-pi
|ll px-2 py-0.5 sys-add-folder-btn';
2522         btn.innerHTML = '<i class="bi bi-check2 me-1"></i>
|Выбрана';
2523     }
2524     };
2525     });
2526
2527     } catch (e) {
2528         console.warn('[SystemInspectorTab] Failed to browse fold
|er:', e);
2529         browserList.innerHTML = `<div class="text-danger small p
|y-2 px-2"><i class="bi bi-exclamation-triangle me-1"></i> Ошиб
|ка доступа: ${escapeHtml(e.message)}</div>`;
2530     }
2531 }
2532
2533 let currentExclusions = {
2534     enabled: true,
2535     paths: [],
2536     extensions: [],
2537     patterns: [],
2538     processes: [],
2539     filtered_count: 0
2540 };
2541
2542 function showExclusionsAlert(msg, type = 'success') {
2543     const alertEl = document.getElementById('modal-exclusions-
|alert');
2544     if (!alertEl) return;
2545     alertEl.className = `alert alert-${type} py-2 px-3 small m
|b-3`;
2546     alertEl.innerHTML = msg;
2547     alertEl.classList.remove('d-none');
2548     if (type === 'success' || type === 'info') {
2549         setTimeout(() => {
2550             alertEl.classList.add('d-none');
```

```
2551     }, 3500);
2552 }
2553 }
2554
2555 function renderExclusionsLists() {
2556     const totalCount = (currentExclusions.paths?.length || 0)
2557 +
2558     (currentExclusions.extensions?.length || 0) +
2559     (currentExclusions.patterns?.length || 0) +
2560     (currentExclusions.processes?.length || 0);
2561     const tabBadge = document.getElementById('modal-watch-excl
2562 usions-tab-count');
2563     const headerBadge = document.getElementById('sys-exclusion
2564 s-count-badge');
2565     if (tabBadge) tabBadge.textContent = String(totalCount);
2566     if (headerBadge) headerBadge.textContent = String(totalCou
2567 nt);
2568
2569     const switchEl = document.getElementById('switch-exclusion
2570 s-active');
2571     const statusBadge = document.getElementById('badge-exclusi
2572 ons-status');
2573     if (switchEl) switchEl.checked = Boolean(currentExclusions
2574 .enabled);
2575     if (statusBadge) {
2576         if (currentExclusions.enabled) {
2577             statusBadge.className = 'badge bg-success';
2578             statusBadge.textContent = 'ВКЛ';
2579         } else {
2580             statusBadge.className = 'badge bg-secondary';
2581             statusBadge.textContent = 'ВыКЛ';
2582         }
2583     }
2584
2585     const filteredTotalEl = document.getElementById('modal-exc
2586 lusions-filtered-total');
2587     if (filteredTotalEl) filteredTotalEl.textContent = String(
```

```

currentExclusions.filtered_count || 0);
2581
2582     // 1. Paths list
2583     const pathsContainer = document.getElementById('list-ex-pa
ths');
2584     const pathsBadge = document.getElementById('badge-ex-paths
-count');
2585     if (pathsBadge) pathsBadge.textContent = String(currentExc
lusions.paths?.length || 0);
2586     if (pathsContainer) {
2587         if (!currentExclusions.paths || currentExclusions.paths.
length === 0) {
2588             pathsContainer.innerHTML = `
```

```

        sions.extensions?.length || 0);
2603     if (extsContainer) {
2604         if (!currentExclusions.extensions || currentExclusions.e
xtensions.length === 0) {
2605             extsContainer.innerHTML = `
```

```

    ext-warning d-inline-flex align-items-center gap-1 font-monosp
    ace" style="font-size: 0.75rem;">
2626         ${escapeHtml(pat)}
2627         <button type="button" class="btn-close btn-close-w
    hite" style="font-size: 0.5rem;" onclick="window._removeExclus
    ionItem('patterns', '${escapeHtml(pat)}')" title="Удалить"></b
    utton>
2628     </span>
2629     `).join('') + `</div>`;
2630 }
2631 }
2632
2633 // 4. Processes list
2634 const procsContainer = document.getElementById('list-ex-pr
    ocesses');
2635 const procsBadge = document.getElementById('badge-ex-proce
    sses-count');
2636 if (procsBadge) procsBadge.textContent = String(currentExc
    lusions.processes?.length || 0);
2637 if (procsContainer) {
2638     if (!currentExclusions.processes || currentExclusions.pr
    ocesses.length === 0) {
2639         procsContainer.innerHTML = `<div class="text-muted sma
    ll text-center py-2">Нет исключенных программ</div>`;
2640     } else {
2641         procsContainer.innerHTML = `<div class="d-flex flex-wr
    ap gap-1">` + currentExclusions.processes.map(proc => `
2642         <span class="badge bg-dark border border-secondary t
    ext-danger d-inline-flex align-items-center gap-1 font-monospa
    ce" style="font-size: 0.75rem;">
2643             <i class="bi bi-cpu me-0.5"></i>${escapeHtml(proc)
    }
2644             <button type="button" class="btn-close btn-close-w
    hite" style="font-size: 0.5rem;" onclick="window._removeExclus
    ionItem('processes', '${escapeHtml(proc)}')" title="Удалить"><
    /button>
2645         </span>
2646         `).join('') + `</div>`;

```

```
2647     }
2648   }
2649 }
2650
2651 async function fetchExclusionsData() {
2652   try {
2653     const res = await fetch('/api/sysadmin/file-audit/exclusions');
2654     if (res.ok) {
2655       currentExclusions = await res.json();
2656       renderExclusionsLists();
2657     }
2658   } catch (e) {
2659     console.warn('[SystemInspectorTab] Failed to fetch exclusions:', e);
2660   }
2661 }
2662
2663 async function addExclusionItem(category, value) {
2664   if (!value || !value.trim()) return;
2665   const cleanVal = value.trim();
2666   try {
2667     const res = await fetch('/api/sysadmin/file-audit/exclusions/add', {
2668       method: 'POST',
2669       headers: { 'Content-Type': 'application/json' },
2670       body: JSON.stringify({ category, value: cleanVal })
2671     });
2672     const data = await res.json();
2673     if (res.ok && data.success) {
2674       currentExclusions = data.exclusions || currentExclusions;
2675       renderExclusionsLists();
2676       showExclusionsAlert(`Правило добавлено: <code>${escapeHtml(cleanVal)}</code>`, 'success');
2677       const inputVal = document.getElementById('input-exclusion-value');
2678       if (inputVal) inputVal.value = '';
```



```
2679         } else {
2680             showExclusionsAlert(data.message || 'Правило уже суцес
твует или невалидно', 'warning');
2681         }
2682     } catch (e) {
2683         showExclusionsAlert(`Ошибка добавления: ${e.message}`, '
danger');
2684     }
2685 }
2686
2687 window._removeExclusionItem = async function(category, value
) {
2688     if (!value) return;
2689     try {
2690         const res = await fetch('/api/sysadmin/file-audit/exclus
ions/remove', {
2691             method: 'POST',
2692             headers: { 'Content-Type': 'application/json' },
2693             body: JSON.stringify({ category, value })
2694         });
2695         const data = await res.json();
2696         if (res.ok && data.success) {
2697             currentExclusions = data.exclusions || currentExclusio
ns;
2698             renderExclusionsLists();
2699             showExclusionsAlert(`Правило удалено: <code>${escapeHt
ml(value)}</code>`, 'info');
2700         } else {
2701             showExclusionsAlert(data.message || 'Не удалось удалит
ь правило', 'warning');
2702         }
2703     } catch (e) {
2704         showExclusionsAlert(`Ошибка удаления: ${e.message}`, 'da
nger');
2705     }
2706 };
2707
2708 async function toggleExclusionsActive(enabled) {
```

```
2709     try {
2710         const res = await fetch('/api/sysadmin/file-audit/exclusions/toggle', {
2711             method: 'POST',
2712             headers: { 'Content-Type': 'application/json' },
2713             body: JSON.stringify({ enabled })
2714         });
2715         const data = await res.json();
2716         if (res.ok && data.success) {
2717             currentExclusions = data.exclusions || currentExclusions;
2718             renderExclusionsLists();
2719             showExclusionsAlert(`Фильтрация исключений ${data.enabled ? 'включена' : 'отключена'}`, 'info');
2720         }
2721     } catch (e) {
2722         showExclusionsAlert(`Ошибка переключения: ${e.message}`, 'danger');
2723     }
2724 }
2725
2726 async function applyExclusionPreset(preset) {
2727     if (preset === 'logs') {
2728         await addExclusionItem('paths', 'AppData\\Roaming\\AI-Boardboard\\apps\\windows\\telemetry\\logs');
2729         await addExclusionItem('extensions', '.log');
2730         await addExclusionItem('extensions', '.csv');
2731         await addExclusionItem('patterns', '*librehardwaremonitor_polls.csv*');
2732     } else if (preset === 'search') {
2733         await addExclusionItem('processes', 'SearchIndexer.exe');
2734
2735         await addExclusionItem('patterns', '*Windows.db*');
2736         await addExclusionItem('patterns', '*Windows.db-wal*');
2737         await addExclusionItem('patterns', '*Windows.db-shm*');
2738     } else if (preset === 'temp') {
2739         await addExclusionItem('paths', 'AppData\\Local\\Temp');
2740         await addExclusionItem('extensions', '.tmp');
```

```
2740     await addExclusionItem('patterns', '~$*');
2741   } else if (preset === 'dev') {
2742     await addExclusionItem('paths', '.git');
2743     await addExclusionItem('paths', 'node_modules');
2744     await addExclusionItem('paths', '__pycache__');
2745     await addExclusionItem('paths', '.pytest_cache');
2746   }
2747 }
2748
2749 async function openWatchFoldersModal(initialTab = 'folders')
2750 {
2751   const modalEl = document.getElementById('modal-sys-watch-f
2752   olders');
2753   if (!modalEl) {
2754     console.warn('[SystemInspectorTab] modal-sys-watch-folde
2755     rs element not found');
2756     return;
2757   }
2758   try {
2759     const res = await fetch('/api/sysadmin/file-audit/watch-
2760     dirs');
2761     if (res.ok) {
2762       const data = await res.json();
2763       stagedWatchDirs = data.watch_dirs && data.watch_dirs.l
2764       ength ? [...data.watch_dirs] : [currentWatchDir || 'C:\\'];
2765     } else {
2766       stagedWatchDirs = currentWatchDirs.length ? [...curren
2767       tWatchDirs] : [currentWatchDir || 'C:\\'];
2768     }
2769   } catch {
2770     stagedWatchDirs = currentWatchDirs.length ? [...currentW
2771     atchDirs] : [currentWatchDir || 'C:\\'];
2772   }
2773
2774   renderModalActiveDirsList();
2775   const tabFoldersBadge = document.getElementById('modal-wat
2776   ch-dirs-tab-count');
```

```
2770     if (tabFoldersBadge) tabFoldersBadge.textContent = String(
| stagedWatchDirs.length);
2771
2772     await loadSystemDrives();
2773     await fetchExclusionsData();
2774
2775     // Tab activation
2776     if (initialTab === 'exclusions') {
2777         const tabExBtn = document.getElementById('tab-btn-watch-
| exclusions');
2778         if (tabExBtn && window.bootstrap && window.bootstrap.Tab
| ) {
2779             const tab = window.bootstrap.Tab.getOrCreateInstance(t
| abExBtn);
2780             tab.show();
2781         } else if (tabExBtn) {
2782             tabExBtn.click();
2783         }
2784     } else {
2785         const tabFoldersBtn = document.getElementById('tab-btn-w
| atch-folders');
2786         if (tabFoldersBtn && window.bootstrap && window.bootstra
| p.Tab) {
2787             const tab = window.bootstrap.Tab.getOrCreateInstance(t
| abFoldersBtn);
2788             tab.show();
2789         } else if (tabFoldersBtn) {
2790             tabFoldersBtn.click();
2791         }
2792     }
2793
2794     const startPath = stagedWatchDirs[0] || currentBrowserPath
| || 'C:\\';
2795     loadFolderBrowser(startPath);
2796
2797     // Bind navigation buttons
2798     const btnUp = document.getElementById('btn-modal-folder-up
| ');
```

```
2799 |     if (btnUp) {
2800 |         btnUp.onclick = () => {
2801 |             const parts = currentBrowserPath.replace(/\\+$/, '').s
|plit('\\\\');
2802 |             if (parts.length > 1) {
2803 |                 parts.pop();
2804 |                 let parentPath = parts.join('\\\\');
2805 |                 if (parts.length === 1 && !parentPath.endsWith('\\\\')
|) parentPath += '\\\\';
2806 |                 loadFolderBrowser(parentPath);
2807 |             }
2808 |         };
2809 |     }
2810 |
2811 |     const btnGo = document.getElementById('btn-modal-folder-go
|');
2812 |     const inputPath = document.getElementById('modal-folder-pa
|th-input');
2813 |     if (btnGo && inputPath) {
2814 |         btnGo.onclick = () => {
2815 |             if (inputPath.value && inputPath.value.trim()) {
2816 |                 loadFolderBrowser(inputPath.value.trim());
2817 |             }
2818 |         };
2819 |         inputPath.onkeydown = (e) => {
2820 |             if (e.key === 'Enter') {
2821 |                 e.preventDefault();
2822 |                 btnGo.click();
2823 |             }
2824 |         };
2825 |     }
2826 |
2827 |     const btnAddCurrent = document.getElementById('btn-modal-f
|older-add-current');
2828 |     if (btnAddCurrent) {
2829 |         btnAddCurrent.onclick = () => {
2830 |             const val = inputPath ? inputPath.value.trim() : curre
|ntBrowserPath;
```

```
2831         if (val) addDirToStaged(val);
2832     };
2833 }
2834
2835 // Presets buttons (folders)
2836 document.querySelectorAll('.sys-preset-btn').forEach(btn =
> {
2837     btn.onclick = () => {
2838         const preset = btn.getAttribute('data-preset');
2839         let target = '';
2840         if (preset === 'project') target = 'C:\\Users\\onela\\
AppData\\Local\\AI-Breadboard';
2841         else if (preset === 'downloads') target = 'C:\\Users\\
onela\\Downloads';
2842         else if (preset === 'desktop') target = 'C:\\Users\\on
ela\\Desktop';
2843         else if (preset === 'temp') target = 'C:\\Users\\onela
\\AppData\\Local\\Temp';
2844
2845         if (target) {
2846             addDirToStaged(target);
2847             loadFolderBrowser(target);
2848         }
2849     };
2850 });
2851
2852 // Exclusions switch and add buttons binding
2853 const switchExActive = document.getElementById('switch-exc
lusions-active');
2854 if (switchExActive) {
2855     switchExActive.onchange = (e) => {
2856         toggleExclusionsActive(e.target.checked);
2857     };
2858 }
2859
2860 const btnAddEx = document.getElementById('btn-add-exclusio
n');
2861 const selectExCat = document.getElementById('select-exclus
```

```

ion-category');
2862     const inputExVal = document.getElementById('input-exclusio
n-value');
2863     if (btnAddEx && selectExCat && inputExVal) {
2864         btnAddEx.onclick = () => {
2865             addExclusionItem(selectExCat.value, inputExVal.value);
2866         };
2867         inputExVal.onkeydown = (e) => {
2868             if (e.key === 'Enter') {
2869                 e.preventDefault();
2870                 btnAddEx.click();
2871             }
2872         };
2873     }
2874
2875     // Exclusions preset buttons
2876     document.querySelectorAll('.sys-ex-preset-btn').forEach(bt
n => {
2877         btn.onclick = () => {
2878             const preset = btn.getAttribute('data-preset');
2879             if (preset) applyExclusionPreset(preset);
2880         };
2881     });
2882
2883     // Apply button
2884     const btnApply = document.getElementById('btn-modal-watch-
dirs-apply');
2885     if (btnApply) {
2886         btnApply.onclick = async () => {
2887             if (stagedWatchDirs.length === 0) {
2888                 showWatchDirsAlert('Выберите хотя бы одну папку для
мониторинга', 'warning');
2889                 return;
2890             }
2891
2892             btnApply.disabled = true;
2893             btnApply.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span> Применение...';

```

```
2894 |
2895 |     try {
2896 |         const res = await fetch('/api/sysadmin/file-audit/watch-dirs', {
2897 |             method: 'POST',
2898 |             headers: { 'Content-Type': 'application/json' },
2899 |             body: JSON.stringify({ paths: stagedWatchDirs })
2900 |         });
2901 |         const resData = await res.json();
2902 |         if (res.ok && resData.success) {
2903 |             if (window.bootstrap && window.bootstrap.Modal) {
2904 |                 const modal = window.bootstrap.Modal.getInstance
2905 | (modalEl);
2906 |                 if (modal) modal.hide();
2907 |             } else {
2908 |                 modalEl.classList.remove('show');
2909 |                 modalEl.style.display = 'none';
2910 |             }
2911 |             await fetchLiveFileEvents();
2912 |         } else {
2913 |             showWatchDirsAlert(`Ошибка: ${resData.detail || 'Н
2914 | е удалось применить список папок'}`, 'danger');
2915 |         }
2916 |     } catch (err) {
2917 |         showWatchDirsAlert(`Сетевая ошибка: ${err.message}`,
2918 | 'danger');
2919 |     } finally {
2920 |         btnApply.disabled = false;
2921 |         btnApply.innerHTML = '<i class="bi bi-check2-circle
2922 | me-1"></i> Применить и запустить';
2923 |     }
2924 | };
2925 | }

2926 |
2927 | if (window.bootstrap && window.bootstrap.Modal) {
2928 |     const modal = window.bootstrap.Modal.getOrCreateInstance
2929 | (modalEl);
2930 |     modal.show();
2931 | }
```



```
2926 |     } else {
2927 |         modalEl.classList.add('show');
2928 |         modalEl.style.display = 'block';
2929 |     }
2930 | }
2931 |
2932 | function showLiveWatcherHelpModal() {
2933 |     const dirsStr = currentWatchDirs.length ? currentWatchDirs
2934 | .join('\n- ') : (currentWatchDir || 'Рабочая папка');
2935 |     if (window.AITableModal) {
2936 |         window.AITableModal.show({
2937 |             icon: 'i',
2938 |             title: 'Справка:Изменения файлов в реальном времени (M
2939 |ulti-Directory)',
2940 |             subtitle: 'Низкоуровневый мониторинг файловой системы
2941 | Windows через WinAPI ReadDirectoryChangesW',
2942 |             tableType: 'help',
2943 |             badges: [
2944 |                 { text: 'WinAPI ReadDirectoryChangesW', class: 'badg
2945 |e bg-info text-dark' },
2946 |                 { text: 'Multi-Directory', class: 'badge bg-primary'
2947 | },
2948 |                 { text: 'Real-Time Streaming', class: 'badge bg-succ
2949 |ess' },
2950 |                 { text: 'Рекурсивно (bWatchSubtree = True)', class:
2951 | 'badge bg-warning text-dark' }
2952 |             ],
2953 |             metadata: [
2954 |                 { label: 'Технология', value: 'WinAPI ReadDirectoryC
2955 |hangesW (нативный вызов ядра Windows kernel32.dll)' },
2956 |                 { label: 'Режим работы', value: 'Множественные поток
2957 | и мониторинга с агрегацией в единый кольцевой буфер' },
2958 |                 { label: 'Отслеживаемые папки', value: currentWatchD
2959 |irs.join('; ') || 'Рабочая папка проекта' },
2960 |                 { label: 'Хранение настроек', value: 'apps/windows_s
2961 |ysadmin/config.json (ключ watch_directories)' }
2962 |             ],
2963 |             rawTitle: 'Подробное руководство по панели мониторинга
```

```

2953 |         rawContent: `# ПанельИзменения файлов в реальном време
2954 | ни
2955 | ### 1. Что это такое?
2956 | Компонент для мгновенного перехвата операций файловой системы
2957 | в режиме реального времени на базе WinAPI ReadDirectoryChanges
2958 | W с поддержкой одновременного наблюдения за произвольным колич
2959 | еством папок.
2960 |
2961 | ### 2. Типы отслеживаемых действий:
2962 | - Created: Создание нового файла или папки (FILE_ACTION_ADDED)
2963 | .
2964 | - Modified: Модификация содержимого, атрибутов или размера фай
2965 | ла (FILE_ACTION_MODIFIED).
2966 | - Deleted: Удаление файла или папки с диска (FILE_ACTION_REMOV
2967 | ED).
2968 | - Renamed: Переименование объекта (старое и новое имя).
2969 |
2970 | ### 3. Как выбрать или добавить папки?
2971 | 1. Нажмите кнопку «Папка» в заголовке панели.
2972 | 2. Откроется модальное окно с проводником файловой системы.
2973 | 3. Выберите логический диск (C:, D:, E:), перейдите в нужную п
2974 | апку или вставьте путь вручную.
2975 | 4. Нажмите «+ Добавить папку» или кнопку «Следить» напротив лю
2976 | бой подпапки.
2977 | 5. Нажмите «Применить и запустить». Список сохранится в config
2978 | .json и мониторинг стартует автоматически!`,
2979 |         requestData: {
2980 |             current_watch_dirs: currentWatchDirs,
2981 |             engine: 'ReadDirectoryChangesW (Multi-Directory)'
2982 |         }
2983 |     });
2984 | } else {
2985 |     alert('Мониторинг файловой системыИзменения файлов в реа
2986 | льном времени (Multi-Directory).\nОтслеживаемые папки:\n- ' +
2987 | dirsStr);
2988 | }

```

```
2978 | }
2979 |
2980 | function setupSysSensorInterval(seconds) {
2981 |     _currentUiRefreshSeconds = seconds || 5;
2982 |     const pollHandler = async () => {
2983 |         // Поэтапный независимый опрос подсистем
2984 |         fetchCoreMetrics();
2985 |         fetchProcessesFromDb();
2986 |         fetchLhmSensors();
2987 |         fetchLiveFileEvents();
2988 |         fetchNetworkActivity();
2989 |     };
2990 |     if (window.registerTabPoller) {
2991 |         window.registerTabPoller('tab-system-inspector', pollHan
2992 | dler, _currentUiRefreshSeconds * 1000, { immediate: false });
2993 |     } else {
2994 |         if (window._sysSensorInterval) {
2995 |             clearInterval(window._sysSensorInterval);
2996 |             window._sysSensorInterval = null;
2997 |         }
2998 |         window._sysSensorInterval = setInterval(() => {
2999 |             if (window.isTabActive ? window.isTabActive('tab-syste
3000 | m-inspector') : true) {
3001 |                 pollHandler();
3002 |             }, _currentUiRefreshSeconds * 1000);
3003 |     }
3004 |     console.log(`[SystemInspectorTab] UI sensor refresh interv
3005 | al set to ${_currentUiRefreshSeconds}s (SQLite processes & tel
3006 | emetry)`);
3007 | }
3008 |
3009 | function openSysIntervalsModal() {
3010 |     const modalEl = document.getElementById('sysIntervalsModal
3011 | ');
3012 |     if (!modalEl) {
3013 |         console.warn('[SystemInspectorTab] sysIntervalsModal ele
3014 | ment not found');
```

```
3010     return;
3011 }
3012
3013 // Set UI select to current seconds
3014 const selUi = document.getElementById('modal-ui-refresh-se
lect');
3015 if (selUi) {
3016     selUi.value = String(_currentUiRefreshSeconds);
3017 }
3018 const uiBadge = document.getElementById('modal-ui-refresh-
badge');
3019 if (uiBadge) {
3020     uiBadge.textContent = `${_currentUiRefreshSeconds} cek`;
3021 }
3022
3023 loadSysIntervalsConfig(false);
3024
3025 if (window.bootstrap && window.bootstrap.Modal) {
3026     const modal = window.bootstrap.Modal.getOrCreateInstance
(modalEl);
3027     modal.show();
3028 } else {
3029     modalEl.classList.add('show');
3030     modalEl.style.display = 'block';
3031 }
3032 }
3033
3034 function escapeHtml(str) {
3035     if (!str) return '';
3036     return String(str)
3037         .replace(/&/g, '&amp;')
3038         .replace(/</g, '&lt;')
3039         .replace(/>/g, '&gt;')
3040         .replace(/"/g, '&quot;')
3041         .replace(/'/g, '&#039;');
3042 }
3043
3044 async function initSystemInspectorTab() {
```

```
3045     console.log('[SystemInspectorTab] Initializing (progressiv
e telemetry & SQLite processes)...');
3046     bindTabEvents();
3047
3048     // 1. Поэтапный прогрессивный запуск: немедленно запрашива
ем быстрые базовые метрики
3049     fetchCoreMetrics();
3050
3051     // 2. Сразу же запрашиваем процессы из базы данных SQLite
3052     fetchProcessesFromDb();
3053
3054     // 3. Запрашиваем состояние аппаратных модулей (SMART диск
ов, планки RAM, порты)
3055     fetchHardwareQuick();
3056
3057     // 4. Опрашиваем аппаратные сенсоры хоста / LHM
3058     fetchLhmSensors();
3059
3060     // 5. Опрашиваем сетевую активность и события файлов
3061     fetchNetworkActivity();
3062     fetchLiveFileEvents();
3063
3064     // 6. Подключаем живой WebSocket-поток телеметрии
3065     connectSystemWebSocket();
3066
3067     // 7. Настраиваем периодический опрос (5 сек)
3068     if (!window._sysSensorInterval) {
3069         setupSysSensorInterval(_currentUiRefreshSeconds || 5);
3070     }
3071 }
3072
3073 function activateSystemInspectorTab() {
3074     if (window.isTabActive && !window.isTabActive('tab-system-
inspector')) return;
3075     console.log('[SystemInspectorTab] Tab activated, resuming
telemetry stream...');
3076     fetchCoreMetrics();
3077     fetchProcessesFromDb();
```

```
3078     fetchHardwareQuick();
3079     fetchLhmSensors();
3080     connectSystemWebSocket();
3081 }
3082
3083 function deactivateSystemInspectorTab() {
3084     console.log('[SystemInspectorTab] Tab deactivated, pausing
telemetry stream...');
3085     disconnectSystemWebSocket();
3086 }
3087
3088 window.initSystemInspectorTab = initSystemInspectorTab;
3089 window.activateSystemInspectorTab = activateSystemInspectorT
ab;
3090 window.deactivateSystemInspectorTab = deactivateSystemInspe
ctorTab;
3091 window.openSysIntervalsModal = openSysIntervalsModal;
3092 window.openProcStatsModal = openProcStatsModal;
3093 window.fetchProcessesFromDb = fetchProcessesFromDb;
3094 })();
3095
3096
```

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows System Log Center Layout
6 =====
7
8 Description:
9     Universal Windows OS Log Explorer, dynamic channel discovery, incident
10    correlation engine, timeline analysis, and AI diagnostic assistant.
11
12 File: index.html
13 Project: AI-Breadboard
14 Module: WebInterface.SystemLogsTab
15 Author: hypo69
16 Copyright: © 2026 hypo69
17 =====
18 =====
19 -->
20 <style>
21     .slc-container {
22         padding: 0.25rem 0.5rem;
23     }
24     .slc-compact-header {
25         background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%);
26         border: 1px solid #334155;
27         border-radius: 8px;
28         padding: 0.4rem 0.75rem;
29         margin-bottom: 0.5rem;
30     }
31     .slc-metric-pill {
32         background: rgba(15, 23, 42, 0.8);
33         border: 1px solid #334155;
34         border-radius: 6px;
```

```
31 | padding: 0.2rem 0.5rem;
32 | font-size: 0.75rem;
33 | display: inline-flex;
34 | align-items: center;
35 | gap: 0.35rem;
36 | }
37 | .slc-card {
38 |   background: #0f172a !important;
39 |   border: 1px solid #334155 !important;
40 |   border-radius: 8px;
41 |   padding: 0.5rem 0.75rem;
42 |   display: flex;
43 |   flex-direction: column;
44 |   gap: 0.1rem;
45 | }
46 | .slc-label {
47 |   font-size: 0.68rem;
48 |   text-transform: uppercase;
49 |   letter-spacing: 0.04em;
50 |   color: #94a3b8;
51 | }
52 | .slc-value {
53 |   font-size: 1.05rem;
54 |   font-weight: 700;
55 |   color: #f8fafc;
56 |   font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
57 | }
58 | .slc-panel {
59 |   background: #0f172a !important;
60 |   border: 1px solid #334155 !important;
61 |   border-radius: 8px;
62 |   overflow: hidden;
63 | }
64 | .slc-panel-header {
65 |   background: #1e293b !important;
66 |   border-bottom: 1px solid #334155 !important;
67 |   padding: 0.6rem 1rem;
```



```
68 |     font-weight: 600;
69 |     font-size: 0.85rem;
70 |     color: #38bdf8;
71 |     display: flex;
72 |     align-items: center;
73 |     justify-content: space-between;
74 | }
75 | .slc-tree-list {
76 |     max-height: 580px;
77 |     overflow-y: auto;
78 |     font-size: 0.8rem;
79 | }
80 | .slc-tree-item {
81 |     padding: 0.4rem 0.75rem;
82 |     cursor: pointer;
83 |     border-radius: 6px;
84 |     margin-bottom: 2px;
85 |     color: #cbd5e1;
86 |     display: flex;
87 |     justify-content: space-between;
88 |     align-items: center;
89 |     transition: background 0.15s ease;
90 | }
91 | .slc-tree-item:hover {
92 |     background: rgba(56, 189, 248, 0.1);
93 |     color: #38bdf8;
94 | }
95 | .slc-tree-item.active {
96 |     background: #0284c7;
97 |     color: #fff;
98 |     font-weight: 600;
99 | }
100 | .slc-table {
101 |     width: 100%;
102 |     margin-bottom: 0;
103 |     color: #e2e8f0;
104 |     font-size: 0.8rem;
105 | }
```

```
106 | .slc-table th {
107 |     background: #1e293b;
108 |     color: #94a3b8;
109 |     border-bottom: 1px solid #334155;
110 |     font-weight: 600;
111 |     text-transform: uppercase;
112 |     font-size: 0.7rem;
113 |     letter-spacing: 0.05em;
114 |     padding: 0.5rem 0.75rem;
115 | }
116 | .slc-table td {
117 |     border-bottom: 1px solid #1e293b;
118 |     padding: 0.45rem 0.75rem;
119 |     vertical-align: middle;
120 | }
121 | .slc-table tr:hover td {
122 |     background: rgba(56, 189, 248, 0.06);
123 |     cursor: pointer;
124 | }
125 | .slc-table tr.selected td,
126 | .slc-table tr.table-active td {
127 |     background: rgba(56, 189, 248, 0.2) !important;
128 |     color: #fff !important;
129 | }
130 | .badge-crit { background: #dc2626 !important; color: #fff !i
    mportant; }
131 | .badge-err { background: #ef4444 !important; color: #fff !im
    portant; }
132 | .badge-warn { background: #f59e0b !important; color: #000 !i
    mportant; }
133 | .badge-inf { background: #10b981 !important; color: #fff !im
    portant; }
134 | .badge-verb { background: #64748b !important; color: #fff !i
    mportant; }
135 |
136 | /* Interactive Channel Description Popover Tooltip */
137 | .slc-tooltip-popup {
138 |     position: fixed;
```

```
139 |     z-index: 99999;
140 |     max-width: 320px;
141 |     background: #0f172a;
142 |     border: 1px solid #38bdf8;
143 |     border-radius: 8px;
144 |     padding: 0.6rem 0.8rem;
145 |     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.7), 0 0 15px
    | rgba(56, 189, 248, 0.2);
146 |     pointer-events: none;
147 |     transition: opacity 0.15s ease, transform 0.15s ease;
148 |     opacity: 0;
149 |     transform: translateY(4px);
150 |     font-size: 0.75rem;
151 |     line-height: 1.35;
152 |     color: #e2e8f0;
153 | }
154 | .slc-tooltip-popup.show {
155 |     opacity: 1;
156 |     transform: translateY(0);
157 | }
158 | .slc-tooltip-title {
159 |     font-weight: 700;
160 |     color: #38bdf8;
161 |     margin-bottom: 0.25rem;
162 |     display: flex;
163 |     align-items: center;
164 |     gap: 0.4rem;
165 | }
166 | .slc-tooltip-meta {
167 |     color: #94a3b8;
168 |     font-size: 0.7rem;
169 |     border-top: 1px solid #334155;
170 |     padding-top: 0.35rem;
171 |     margin-top: 0.35rem;
172 |     display: flex;
173 |     justify-content: space-between;
174 | }
175 | </style>
```

```

176 |
177 | <div class="slc-container">
178 |   <!-- Compact Header Bar with Inline Stats & Actions -->
179 |   <div class="slc-compact-header d-flex justify-content-between align-items-center flex-wrap gap-2">
180 |     <!-- Title & Inline Metrics -->
181 |     <div class="d-flex align-items-center gap-2 flex-wrap">
182 |       <div class="d-flex align-items-center gap-1.5 me-1">
183 |         <span class="fs-6">□ </span>
184 |         <span class="fw-bold text-white small">Windows System
Log Center</span>
185 |       </div>
186 |
187 |       <!-- Quick Metrics Pills -->
188 |       <div class="slc-metric-pill" title="Discovered Log Sources">
189 |         <span class="text-secondary small"><i class="bi bi-diagram-3 me-1"></i>Sources:</span>
190 |         <span class="fw-bold text-info font-monospace" id="slc-stat-sources">1,247</span>
191 |       </div>
192 |       <div class="slc-metric-pill" title="Active Log Channel">
193 |         <span class="text-secondary small"><i class="bi bi-folder-fill me-1"></i>Channel:</span>
194 |         <span class="fw-bold text-warning font-monospace text-truncate" id="slc-stat-channel" style="max-width: 140px;">System</span>
195 |       </div>
196 |       <div class="slc-metric-pill" title="Events Loaded in View">
197 |         <span class="text-secondary small"><i class="bi bi-list-ol me-1"></i>Loaded:</span>
198 |         <span class="fw-bold text-light font-monospace" id="slc-stat-events">0</span>
199 |       </div>
200 |       <div class="slc-metric-pill" title="Errors and Incidents Detected">
201 |         <span class="text-secondary small"><i class="bi bi-exc

```

```

lamation-triangle-fill me-1"></i>Faults:</span>
202     <span class="fw-bold text-danger font-monospace" id="s
lc-stat-incidents">0</span>
203     </div>
204 </div>
205
206 <!-- Actions Toolbar -->
207 <div class="d-flex align-items-center gap-2 flex-wrap">
208     <button class="btn btn-outline-info rounded-pill px-2.5
py-1 small d-flex align-items-center gap-1" id="btn-slc-scan"
style="font-size: 0.78rem;">
209         <i class="bi bi-radar"></i> <span>Scan</span>
210     </button>
211     <div class="form-check form-switch text-light mb-0 d-inl
ine-flex align-items-center" style="font-size: 0.78rem; min-he
ight: auto;">
212         <input class="form-check-input me-1" type="checkbox" r
ole="switch" id="slc-live-switch" style="cursor: pointer;">
213         <label class="form-check-label text-secondary small us
er-select-none" for="slc-live-switch">Live</label>
214     </div>
215     <button class="btn btn-primary rounded-pill px-2.5 py-1
small d-flex align-items-center gap-1" id="btn-slc-refresh" st
yle="font-size: 0.78rem;">
216         <i class="bi bi-arrow-clockwise"></i> <span>Refresh</s
pan>
217     </button>
218     <div class="dropdown">
219         <button class="btn btn-xs btn-outline-secondary dropdo
wn-toggle rounded-pill px-2.5 py-1 text-light small d-flex ali
gn-items-center gap-1" type="button" data-bs-toggle="dropdown"
style="font-size: 0.78rem;">
220             <i class="bi bi-download"></i> <span>Export</span>
221         </button>
222         <ul class="dropdown-menu dropdown-menu-dark dropdown-m
enu-end shadow" style="font-size: 0.8rem;">
223             <li><a class="dropdown-item" href="#" id="btn-slc-ex
port-json"><i class="bi bi-filetype-json me-2"></i>JSON</a></li>

```

```

i>
224         <li><a class="dropdown-item" href="#" id="btn-slc-export-csv"><i class="bi bi-filetype-csv me-2"></i>CSV</a></li>
225     </ul>
226 </div>
227 </div>
228 </div>
229
230 <!-- Search & View Mode Navigation (Compact) -->
231 <div class="slc-panel mb-2 p-2">
232     <div class="d-flex justify-content-between align-items-center flex-wrap gap-1 mb-2 pb-1 border-bottom border-secondary">
233         <!-- Mode Tabs -->
234         <ul class="nav nav-pills gap-1" id="slc-mode-tabs">
235             <li class="nav-item">
236                 <button class="nav-link active py-0.5 px-2.5 small rounded-pill" data-mode="live" style="font-size: 0.78rem;">● Live Events</button>
237             </li>
238             <li class="nav-item">
239                 <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="audit" style="font-size: 0.78rem;">□ Аудит (Smart Digest)</button>
240             </li>
241             <li class="nav-item">
242                 <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="rag" style="font-size: 0.78rem;">□ □ RAG Поиск</button>
243             </li>
244             <li class="nav-item">
245                 <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="errors" style="font-size: 0.78rem;">□ □ Errors Only</button>
246             </li>
247             <li class="nav-item">
248                 <button class="nav-link py-0.5 px-2.5 small rounded-pill text-light" data-mode="incidents" style="font-size: 0.78rem;">□ □ Incidents</button>

```

```

249         </li>
250         <li class="nav-item">
251             <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="timeline" style="font-size: 0.78re
m;"> Timeline</button>
252         </li>
253         <li class="nav-item">
254             <button class="nav-link py-0.5 px-2.5 small rounded-
pill text-light" data-mode="raw" style="font-size: 0.78rem;">
Raw Record</button>
255         </li>
256     </ul>
257
258     <!-- Action AI Button -->
259     <div class="d-flex align-items-center gap-2">
260         <button class="btn btn-xs btn-outline-warning rounded-
pill px-2.5 py-1 small d-flex align-items-center gap-1" id="bt
n-slc-ask-ai" style="font-size: 0.78rem;">
261             <i class="bi bi-robot"></i> <span>Ask AI</span>
262         </button>
263     </div>
264 </div>
265
266 <!-- Filter Inputs Grid -->
267 <div class="row g-1.5 align-items-center">
268     <div class="col-lg-4 col-md-5">
269         <div class="input-group input-group-sm">
270             <span class="input-group-text bg-dark text-secondary
border-secondary py-0 px-2"><i class="bi bi-search" style="fo
nt-size: 0.75rem;"></i></span>
271             <input type="text" class="form-control form-control-
sm bg-dark text-light border-secondary py-1" id="slc-search-in
put" placeholder="Search message, provider..." style="font-siz
e: 0.8rem;">
272         </div>
273     </div>
274     <div class="col-lg-2 col-md-2 col-6">
275         <select class="form-select form-select-sm bg-dark text

```

```
    -light border-secondary py-1" id="slc-level-select" style="font-size: 0.8rem;">
276      <option value="">All Severities</option>
277      <option value="Critical">Critical</option>
278      <option value="Error">Error</option>
279      <option value="Warning">Warning</option>
280      <option value="Information">Information</option>
281    </select>
282  </div>
283  <div class="col-lg-2 col-md-2 col-6">
284    <select class="form-select form-select-sm bg-dark text-light border-secondary py-1" id="slc-hours-select" style="font-size: 0.8rem;">
285      <option value="1">Last 1 hour</option>
286      <option value="6">Last 6 hours</option>
287      <option value="24" selected>Last 24 hours</option>
288      <option value="168">Last 7 days</option>
289      <option value="0">All available</option>
290    </select>
291  </div>
292  <div class="col-lg-2 col-md-1.5 col-6">
293    <input type="number" class="form-control form-control-sm bg-dark text-light border-secondary py-1" id="slc-eventid-input" placeholder="Event ID" style="font-size: 0.8rem;">
294  </div>
295  <div class="col-lg-2 col-md-1.5 col-6">
296    <select class="form-select form-select-sm bg-dark text-light border-secondary py-1" id="slc-limit-select" style="font-size: 0.8rem;">
297      <option value="50">50 records</option>
298      <option value="100" selected>100 records</option>
299      <option value="250">250 records</option>
300      <option value="500">500 records</option>
301    </select>
302  </div>
303 </div>
304 </div>
305
```



```

306 | <!-- Main Multi-Column View -->
307 | <div class="row g-3">
308 |     <!-- Channel & Source Tree Sidebar -->
309 |     <div class="col-lg-3">
310 |         <div class="slc-panel h-100">
311 |             <div class="slc-panel-header">
312 |                 <span><i class="bi bi-folder2-open me-2"></i>Log Cha
nnels Tree</span>
313 |                 <span class="badge bg-secondary" id="slc-channel-cou
nt-badge">0</span>
314 |             </div>
315 |             <div class="p-2 border-bottom border-secondary">
316 |                 <input type="text" class="form-control form-control-
sm bg-dark text-light border-secondary" id="slc-channel-filter
-input" placeholder="Filter channels...">
317 |             </div>
318 |             <div class="p-2 slc-tree-list" id="slc-channel-tree-co
ntainer">
319 |                 <!-- Dynamic channels injected here -->
320 |             </div>
321 |         </div>
322 |     </div>
323 |
324 |     <!-- Events / Incidents Main Stream -->
325 |     <div class="col-lg-9">
326 |         <div class="slc-panel h-100">
327 |             <div class="slc-panel-header">
328 |                 <span id="slc-main-view-title"><i class="bi bi-list-
columns-reverse me-2"></i>Live Windows Event Stream</span>
329 |                 <div class="d-flex align-items-center gap-2">
330 |                     <button class="btn btn-sm btn-info rounded-pill px
-3 fw-bold text-dark" id="btn-slc-audit" title="Провести аудит
текущего журнала логов">
331 |                         <i class="bi bi-shield-check me-1"></i> Аудит
332 |                     </button>
333 |                     <span class="badge bg-secondary" id="slc-stream-co
unt-badge">0 entries</span>
334 |                 </div>

```

```

335 |         </div>
336 |
337 |         <!-- Table View Container -->
338 |         <div class="table-responsive" style="max-height: 580px
339 | ; overflow-y: auto;" id="slc-table-container">
340 |             <table class="table slc-table">
341 |                 <thead>
342 |                     <tr>
343 |                         <th style="width: 155px;">Timestamp</th>
344 |                         <th style="width: 90px;">Level</th>
345 |                         <th style="width: 65px;">ID</th>
346 |                         <th style="width: 160px;">Provider</th>
347 |                         <th style="width: 65px;">PID</th>
348 |                         <th>Message</th>
349 |                     </tr>
350 |                 </thead>
351 |                 <tbody id="slc-table-body">
352 |                     <tr>
353 |                         <td colspan="6" class="text-center text-muted
354 | py-4">
355 |                             <div class="spinner-border spinner-border-sm
356 | text-info me-2" role="status"></div>
357 |                             Loading system events...
358 |                         </td>
359 |                     </tr>
360 |                 </tbody>
361 |             </table>
362 |         </div>
363 |
364 |         <!-- Audit Digest View Container (Hidden by default) -
365 | -->
366 |         <div class="p-3 d-none" id="slc-audit-container" style
367 | ="max-height: 580px; overflow-y: auto;">
368 |             <div id="slc-audit-content"></div>
369 |         </div>
370 |
371 |         <!-- Dynamic Log RAG Search Container (Hidden by defau
372 | lt) -->

```

```
367 |         <div class="p-3 d-none" id="slc-rag-container" style="
| max-height: 580px; overflow-y: auto;">
368 |             <div class="card bg-dark border-info mb-3 shadow">
369 |                 <div class="card-header bg-black d-flex justify-co
| ntent-between align-items-center">
370 |                     <span class="text-info fw-bold"><i class="bi bi-
| search me-2"></i>Семантический RAG-поиск по логам</span>
371 |                     <button class="btn btn-sm btn-outline-info round
| ed-pill px-3" id="btn-slc-rag-rebuild">
372 |                         <i class="bi bi-arrow-repeat me-1"></i> Перест
| роить RAG-индекс
373 |                     </button>
374 |                 </div>
375 |                 <div class="card-body">
376 |                     <div class="input-group mb-2">
377 |                         <span class="input-group-text bg-black text-in
| fo border-secondary"><i class="bi bi-chat-left-dots"></i></spa
| n>
378 |                         <input type="text" class="form-control bg-dark
| text-light border-secondary" id="slc-rag-input" placeholder="
| Введите естественный запрос (например: 'почему завис BITS', 'о
| шибки DCOM 10016', 'сетевые сбои')...">
379 |                         <button class="btn btn-info px-4 fw-bold text-
| dark" id="btn-slc-rag-search">Найти</button>
380 |                     </div>
381 |                     <div class="small text-muted">
382 |                         Поиск использует динамический TF-IDF/BM25 инде
| кс кластеров, аномалий и сводок с нулевым расходом токенов LLM
| .
383 |                     </div>
384 |                 </div>
385 |             </div>
386 |
387 |             <div id="slc-rag-results">
388 |                 <div class="card bg-dark border-secondary p-4 text
| -center text-muted">
389 |                     Введите запрос для поиска по динамическому RAG-и
| ндексу логов или нажмите «Перестроить RAG-индекс».
```

```
390         </div>
391     </div>
392 </div>
393
394     <!-- Incidents View Container (Hidden by default) -->
395     <div class="p-3 d-none" id="slc-incidents-container" s
396 | tyle="max-height: 580px; overflow-y: auto;">
397         <div id="slc-incidents-list"></div>
398     </div>
399     <!-- Timeline Histogram View Container (Hidden by defa
400 | ult) -->
401     <div class="p-3 d-none" id="slc-timeline-container" st
402 | yle="max-height: 580px; overflow-y: auto;">
403         <div id="slc-timeline-list"></div>
404     </div>
405     <!-- Raw Record Container (Hidden by default) -->
406     <div class="p-3 d-none" id="slc-raw-container" style="
407 | max-height: 580px; overflow-y: auto;">
408         <pre class="bg-black text-info p-3 rounded font-mono
409 | space small" id="slc-raw-text">Select an event from the list t
410 | o view raw structured XML/JSON data.</pre>
411     </div>
412 </div>
413 <!-- Modal: Event Details & AI Diagnostics -->
414 <div class="modal fade" id="slc-event-detail-modal" tabindex="
415 | -1">
416     <div class="modal-dialog modal-lg modal-dialog-centered">
417         <div class="modal-content bg-dark text-light border-second
418 | ary">
419             <div class="modal-header border-secondary">
420                 <h5 class="modal-title fw-bold text-info" id="slc-moda
421 | l-title">Log Event Details</h5>
```

```

419         <button type="button" class="btn-close btn-close-white
" data-bs-dismiss="modal"></button>
420     </div>
421     <div class="modal-body">
422         <div class="row g-2 mb-3 small font-monospace">
423             <div class="col-sm-6"><strong class="text-secondary"
>Timestamp:</strong> <span id="slc-modal-time"></span></div>
424             <div class="col-sm-6"><strong class="text-secondary"
>Severity:</strong> <span id="slc-modal-level"></span></div>
425             <div class="col-sm-6"><strong class="text-secondary"
>Event ID:</strong> <span id="slc-modal-id"></span></div>
426             <div class="col-sm-6"><strong class="text-secondary"
>Provider:</strong> <span id="slc-modal-provider"></span></div>
>
427             <div class="col-sm-6"><strong class="text-secondary"
>Computer:</strong> <span id="slc-modal-computer"></span></div>
>
428             <div class="col-sm-6"><strong class="text-secondary"
>Process ID:</strong> <span id="slc-modal-pid"></span></div>
429             <div class="col-sm-12"><strong class="text-secondary"
">Channel:</strong> <span id="slc-modal-channel"></span></div>
430         </div>
431
432         <div class="card bg-black border-secondary p-3 mb-3">
433             <h6 class="small text-uppercase text-muted fw-bold m
b-2">Message Body</h6>
434             <div class="small font-monospace" id="slc-modal-mess
age" style="white-space: pre-wrap; word-break: break-word;"></
div>
435         </div>
436
437         <!-- AI Diagnostic Response Section -->
438         <div class="card border-info bg-dark p-3" id="slc-moda
l-ai-section">
439             <div class="d-flex justify-content-between align-ite
ms-center mb-2">
440                 <h6 class="fw-bold text-info mb-0"><i class="bi bi
-robot me-1"></i> AI Contextual Explanation</h6>

```

```
441 |         <button class="btn btn-sm btn-outline-info py-0 px
    | -2 rounded-pill" id="btn-slc-modal-diagnose">Analyze Context</
    | button>
442 |     </div>
443 |     <div class="small" id="slc-modal-ai-explanation">
444 |         Click "Analyze Context" to inspect surrounding eve
    | nts and identify the root cause of this incident.
445 |     </div>
446 | </div>
447 | </div>
448 | <div class="modal-footer border-secondary">
449 |     <button type="button" class="btn btn-sm btn-secondary"
    | data-bs-dismiss="modal">Close</button>
450 | </div>
451 | </div>
452 | </div>
453 | </div>
454 |
```

main.js

```
1 /**
2  * =====
3  * Process Name: Windows System Log Center Controller
4  * =====
5  * Description:
6  *   Frontend controller managing dynamic channel tree discovery, multi-mode
7  *   view rendering (Live, Errors, Incidents, Timeline, Raw), contextual AI
8  *   explanation, and multi-source log export.
9  *
10 * File: main.js
11 * Project: AI-Breadboard
12 * Module: WebInterface.SystemLogsTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
16 */
17
18 (function () {
19   'use strict';
20
21   let currentChannel = 'System';
22   let currentFilePath = '';
23   let currentMode = 'live';
24   let cachedEvents = [];
25   let cachedChannels = [];
26   let selectedEvent = null;
27   let liveIntervalTimer = null;
28   let isRequestInProgress = false;
29
30   /**
31    * Escape HTML entities to protect against XSS injection.
```

```

32 | *
33 | * @param {string} str - Raw input text.
34 | * @returns {string} Sanitized string.
35 | */
36 | function escapeHtml(str) {
37 |   if (!str) return '';
38 |   return String(str)
39 |     .replace(/&/g, '&amp;')
40 |     .replace(/</g, '&lt;')
41 |     .replace(/>/g, '&gt;')
42 |     .replace(/"/g, '&quot;')
43 |     .replace(/'/g, '&#039;');
44 | }
45 |
46 | /**
47 |  * Map log severity to badge styling class.
48 |  *
49 |  * @param {string} level - Severity level.
50 |  * @returns {string} CSS class name.
51 |  */
52 | function getSeverityClass(level) {
53 |   const lvl = (level || '').toLowerCase();
54 |   if (lvl.includes('crit')) return 'badge-crit';
55 |   if (lvl.includes('err')) return 'badge-err';
56 |   if (lvl.includes('warn')) return 'badge-warn';
57 |   if (lvl.includes('verb')) return 'badge-verb';
58 |   return 'badge-inf';
59 | }
60 |
61 | /**
62 |  * Perform dynamic system scan and populate channels tree.
63 |  *
64 |  * @param {boolean} force - Force refresh flag.
65 |  */
66 | async function performSystemScan(force = false) {
67 |   try {
68 |     const response = await fetch(`/api/v1/system_logs/scan?force=${force}`);

```



```

69 |         if (!response.ok) return;
70 |         const data = await response.json();
71 |
72 |         cachedChannels = data.channels || [];
73 |         const activeChannels = cachedChannels.filter((c) => (c.r
| record_count || 0) > 0);
74 |         const sourcesBadge = document.getElementById('slc-stat-s
| ources');
75 |         const countBadge = document.getElementById('slc-channel-
| count-badge');
76 |
77 |         if (sourcesBadge) {
78 |             sourcesBadge.textContent = `${activeChannels.length.to
| LocaleString()} active / ${data.total_sources.toLocaleString()
| } sources`;
79 |         }
80 |         if (countBadge) {
81 |             countBadge.textContent = String(activeChannels.length)
| ;
82 |         }
83 |
84 |         renderChannelsTree();
85 |     } catch (err) {
86 |         console.warn('Failed to perform system discovery scan:',
| err);
87 |     }
88 | }
89 |
90 | // Singleton global popup tooltip element
91 | let globalTooltipEl = null;
92 |
93 | function getOrCreateTooltipEl() {
94 |     if (!globalTooltipEl) {
95 |         globalTooltipEl = document.createElement('div');
96 |         globalTooltipEl.className = 'slc-tooltip-popup';
97 |         document.body.appendChild(globalTooltipEl);
98 |     }
99 |     return globalTooltipEl;

```

```
100 | }
101 |
102 | function showChannelTooltip(e, channel) {
103 |     const tip = getOrCreateTooltipEl();
104 |     const countFormatted = (channel.record_count || 0).toLocale
105 | eString();
106 |     const descText = channel.description || 'Служебный журнал
107 | операционной системы Windows.';
108 |     const typeLabel = channel.source_type === 'channel' ? 'Win
109 | dows Event Channel' : 'Log File';
110 |
111 |     tip.innerHTML = `
112 |         <div class="slc-tooltip-title">
113 |             <i class="bi bi-info-circle text-info"></i>
114 |             <span>${escapeHtml(channel.display_name)}</span>
115 |         </div>
116 |         <div style="color: #cbd5e1; margin-bottom: 0.35rem;">${e
117 | scapeHtml(descText)}</div>
118 |         <div class="slc-tooltip-meta">
119 |             <span><strong class="text-info">${countFormatted}</str
120 | ong> записей</span>
121 |             <span class="text-muted font-monospace">${typeLabel}</
122 | span>
123 |         </div>
124 |     `;
125 |
126 |     tip.classList.add('show');
127 |     positionTooltip(e, tip);
128 | }
129 |
130 | function hideChannelTooltip() {
131 |     if (globalTooltipEl) {
132 |         globalTooltipEl.classList.remove('show');
133 |     }
134 | }
135 |
136 | function positionTooltip(e, tip) {
137 |     const margin = 12;
```

```

132 |     let x = e.clientX + margin;
133 |     let y = e.clientY + margin;
134 |
135 |     const tipRect = tip.getBoundingClientRect();
136 |     if (x + tipRect.width > window.innerWidth) {
137 |         x = e.clientX - tipRect.width - margin;
138 |     }
139 |     if (y + tipRect.height > window.innerHeight) {
140 |         y = e.clientY - tipRect.height - margin;
141 |     }
142 |
143 |     tip.style.left = `${Math.max(10, x)}px`;
144 |     tip.style.top = `${Math.max(10, y)}px`;
145 | }
146 |
147 | /**
148 |  * Render channels tree list in sidebar with category groupi
ng, record count badges, and hover tooltips.
149 |  */
150 | function renderChannelsTree() {
151 |     const container = document.getElementById('slc-channel-tre
e-container');
152 |     const filterInput = document.getElementById('slc-channel-f
ilter-input');
153 |     const filterText = filterInput ? filterInput.value.trim().
toLowerCase() : '';
154 |
155 |     if (!container) return;
156 |     container.innerHTML = '';
157 |
158 |     const filtered = cachedChannels.filter((c) => {
159 |         // Не выводить каналы и источники, в которых 0 записей
160 |         if (c.record_count === 0) return false;
161 |
162 |         if (!filterText) return true;
163 |         return (
164 |             c.channel_name.toLowerCase().includes(filterText) ||
165 |             c.display_name.toLowerCase().includes(filterText) ||

```

```

166 |         (c.description && c.description.toLowerCase().includes
| (filterText)) ||
167 |         (c.category && c.category.toLowerCase().includes(filte
| rText))
168 |     );
169 | });
170 |
171 | // Group by category
172 | const grouped = {};
173 | filtered.forEach((c) => {
174 |     const cat = c.category || 'Windows Event Log';
175 |     if (!grouped[cat]) grouped[cat] = [];
176 |     grouped[cat].push(c);
177 | });
178 |
179 | const categoryIcons = {
180 |     'Windows Event Log': '📄 ',
181 |     'Windows OS Logs': '📄 ',
182 |     'Application Logs': '📄 ',
183 |     'AI-Breadboard Logs': '📄 ',
184 | };
185 |
186 | Object.keys(grouped).forEach((catName) => {
187 |     const catHeader = document.createElement('div');
188 |     catHeader.className = 'd-flex justify-content-between al
| ign-items-center px-2 py-1 mt-2 mb-1 rounded bg-dark border bo
| rder-secondary text-info small fw-bold';
189 |     const icon = categoryIcons[catName] || '📄 ';
190 |     catHeader.innerHTML = `
191 |         <span>${icon} ${escapeHtml(catName)}</span>
192 |         <span class="badge bg-secondary font-monospace">${grou
| ped[catName].length}</span>
193 |     `;
194 |     container.appendChild(catHeader);
195 |
196 |     grouped[catName].slice(0, 400).forEach((c) => {
197 |         const item = document.createElement('div');
198 |         const isActive = (c.channel_name === currentChannel &&

```

```

    !currentFilePath) || (currentFilePath && currentFilePath ===
c.location);
199     item.className = `slc-tree-item ${isActive ? 'active'
: ''}`;
200
201     let subBadge = '0';
202     let badgeStyle = 'bg-dark text-secondary';
203     if (c.record_count > 0) {
204         subBadge = c.record_count >= 1000 ? `${(c.record_cou
nt / 1000).toFixed(1)}k` : String(c.record_count);
205         badgeStyle = c.record_count > 1000 ? 'bg-primary tex
t-white' : 'bg-dark text-info border border-secondary';
206     } else if (c.source_type !== 'channel') {
207         subBadge = 'FILE';
208     }
209
210     item.innerHTML = `
211         <span class="text-truncate me-2" style="max-width: 1
75px;">${escapeHtml(c.display_name)}</span>
212         <span class="badge ${badgeStyle} font-monospace" sty
le="font-size: 0.72rem; min-width: 32px; text-align: right;">${
{subBadge}</span>
213     `;
214
215     // Tooltip events
216     item.onmouseenter = (e) => showChannelTooltip(e, c);
217     item.onmousemove = (e) => {
218         const tip = getOrCreateTooltipEl();
219         if (tip.classList.contains('show')) positionTooltip(
e, tip);
220     };
221     item.onmouseleave = () => hideChannelTooltip();
222
223     item.onclick = () => {
224         hideChannelTooltip();
225         if (c.source_type === 'channel') {
226             currentChannel = c.channel_name;
227             currentFilePath = '';

```

```

228         } else {
229             currentChannel = c.display_name;
230             currentFilePath = c.location;
231         }
232         const statChan = document.getElementById('slc-stat-c
hannel');
233         if (statChan) statChan.textContent = c.display_name;
234         renderChannelsTree();
235         loadEvents();
236     };
237     container.appendChild(item);
238 });
239 });
240 }
241
242 /**
243  * Fetch events matching current channel, filters, and activ
e view mode.
244  */
245 async function loadEvents() {
246     if (isRequestInProgress) return;
247     isRequestInProgress = true;
248
249     const levelSelect = document.getElementById('slc-level-sel
ect');
250     const hoursSelect = document.getElementById('slc-hours-sel
ect');
251     const searchInput = document.getElementById('slc-search-in
put');
252     const eventIdInput = document.getElementById('slc-eventid-
input');
253     const limitSelect = document.getElementById('slc-limit-sel
ect');
254     const tableBody = document.getElementById('slc-table-body'
);
255     const eventsBadge = document.getElementById('slc-stat-even
ts');
256     const faultsBadge = document.getElementById('slc-stat-inci

```

```

dents');
257     const streamBadge = document.getElementById('slc-stream-co
unt-badge');
258
259     const level = levelSelect ? levelSelect.value : '';
260     const hours = hoursSelect ? hoursSelect.value : '24';
261     const search = searchInput ? searchInput.value.trim() : ''
;
262     const eventId = eventIdInput && eventIdInput.value.trim()
? eventIdInput.value.trim() : '0';
263     const limit = limitSelect ? limitSelect.value : '100';
264
265     try {
266         const url = `/api/v1/system_logs/events?channel=${encode
URIComponent(currentChannel)}&limit=${limit}&level=${encodeURIComponent(level)}&search=${encodeURIComponent(search)}&hours=${
hours}&event_id=${eventId}&file_path=${encodeURIComponent(currentFilePath)}`;
267
268         const res = await fetch(url);
269         if (!res.ok) {
270             if (tableBody) tableBody.innerHTML = `<tr><td colspan=
"6" class="text-center text-danger py-3">Error reading channel
(HTTP ${res.status})</td></tr>`;
271             isRequestInProgress = false;
272             return;
273         }
274
275         const data = await res.json();
276         cachedEvents = data.entries || [];
277
278         if (eventsBadge) eventsBadge.textContent = String(cached
Events.length);
279         if (streamBadge) streamBadge.textContent = `${cachedEven
ts.length} records`;
280
281         let errCount = 0;
282         let critCount = 0;

```

```

283 |         cachedEvents.forEach((e) => {
284 |             const lvl = (e.level || '').toLowerCase();
285 |             if (lvl.includes('crit')) critCount++;
286 |             if (lvl.includes('err')) errCount++;
287 |         });
288 |
289 |         if (faultsBadge) {
290 |             faultsBadge.textContent = `${errCount + critCount} faults`;
291 |         }
292 |
293 |         renderCurrentModeView();
294 |     } catch (err) {
295 |         console.warn('Failed to load events:', err);
296 |         if (tableBody) tableBody.innerHTML = `<tr><td colspan="6"
| class="text-center text-danger py-3">Network error loading logs.</td></tr>`;
297 |     } finally {
298 |         isRequestInProgress = false;
299 |     }
300 | }
301 |
302 | /**
303 |  * Render view corresponding to active mode (Live, Errors, Incidents, Timeline, Raw).
304 |  */
305 | function renderCurrentModeView() {
306 |     const tableContainer = document.getElementById('slc-table-container');
307 |     const incidentsContainer = document.getElementById('slc-incidents-container');
308 |     const timelineContainer = document.getElementById('slc-timeline-container');
309 |     const rawContainer = document.getElementById('slc-raw-container');
310 |     const auditContainer = document.getElementById('slc-audit-container');
311 |     const tableBody = document.getElementById('slc-table-body')

```



```
);
312     const viewTitle = document.getElementById('slc-main-view-t
itle');
313
314     if (!tableContainer || !tableBody) return;
315
316     // Reset visibility
317     tableContainer.classList.add('d-none');
318     if (incidentsContainer) incidentsContainer.classList.add('
d-none');
319     if (timelineContainer) timelineContainer.classList.add('d-
none');
320     if (rawContainer) rawContainer.classList.add('d-none');
321     if (auditContainer) auditContainer.classList.add('d-none')
;
322
323     if (currentMode === 'audit') {
324         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-sh
ield-check text-info me-2"></i>Интеллектуальный Аудит Логирова
ния (Zero-LLM Overhead)`;
325         if (auditContainer) {
326             auditContainer.classList.remove('d-none');
327             loadAuditView();
328         }
329     } else if (currentMode === 'rag') {
330         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-br
ain text-info me-2"></i>Динамический RAG-поиск по логам`;
331         const ragContainer = document.getElementById('slc-rag-co
ntainer');
332         if (ragContainer) {
333             ragContainer.classList.remove('d-none');
334         }
335     } else if (currentMode === 'incidents') {
336         if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-di
agram-3 me-2"></i>Correlated Incident Cascades`;
337         if (incidentsContainer) {
338             incidentsContainer.classList.remove('d-none');
339             loadIncidentsView();
```

```

340     }
341     } else if (currentMode === 'timeline') {
342         if (viewTitle) viewTitle.innerHTML = `

```

```

includes('warn');
370     });
371 }
372
373     if (entriesToRender.length === 0) {
374         tableBody.innerHTML = `|
|  |

```

```

break: break-word;">${escapeHtml(e.message)}</td>
395     </tr>
396     `;
397     }).join('');
398
399     // Attach row click listeners for row selection and detail inspection
400     tableBody.querySelectorAll('tr').forEach((row) => {
401         row.onclick = () => {
402             const idx = parseInt(row.getAttribute('data-index')
|| '0', 10);
403             selectedEvent = entriesToRender[idx];
404             tableBody.querySelectorAll('tr').forEach(r => r.classList.remove('selected', 'table-active'));
405             row.classList.add('selected', 'table-active');
406             showEventModal(selectedEvent);
407         };
408     });
409 }
410 }
411
412 /**
413  * Fetch and render correlated incident clusters.
414  */
415 async function loadIncidentsView() {
416     const listEl = document.getElementById('slc-incidents-list');
417     if (!listEl) return;
418     listEl.innerHTML = `<div class="text-center py-4 text-muted"><div class="spinner-border spinner-border-sm text-info me-2"></div>Correlating incident cascades...</div>`;
419
420     try {
421         const res = await fetch('/api/v1/system_logs/incidents?window_seconds=25.0');
422         if (!res.ok) return;
423         const data = await res.json();
424         const incidents = data.incidents || [];

```

```

425
426     if (incidents.length === 0) {
427         listEl.innerHTML = `
```

```

    r retrieving incidents: ${err.message}</div>`;
450     }
451   }
452
453   /**
454    * Fetch and render timeline histogram.
455    */
456   async function loadTimelineView() {
457     const listEl = document.getElementById('slc-timeline-list'
458   );
459     if (!listEl) return;
460
461     try {
462       const res = await fetch('/api/v1/system_logs/timeline');
463       if (!res.ok) return;
464       const data = await res.json();
465       const histogram = data.histogram || [];
466
467       if (histogram.length === 0) {
468         listEl.innerHTML = `<div class="card bg-dark border-se
469   condary p-4 text-center text-muted">No timeline data available
470   .</div>`;
471         return;
472       }
473
474       listEl.innerHTML = histogram.map((h) => `
475         <div class="d-flex justify-content-between align-items
476         -center p-2 mb-2 bg-dark rounded border border-secondary">
477           <span class="font-monospace text-info">${escapeHtml(
478         h.time)}</span>
479           <div class="d-flex align-items-center gap-2">
480             <span class="badge bg-secondary">Total: ${h.total}
481           </span>
482             <span class="badge badge-err">Errors: ${h.errors}<
483           /span>
484             <span class="badge badge-warn">Warnings: ${h.warni
485         ngs}</span>
486             <span class="badge badge-inf">Info: ${h.info}</spa

```

```

n>
479         </div>
480     </div>
481     `).join('');
482 } catch (err) {
483     listEl.innerHTML = `<div class="alert alert-danger">Error
loading timeline: ${err.message}</div>`;
484 }
485 }
486
487 /**
488  * Fetch and render multi-level log audit digest (Noise filtering,
clustering, anomalies).
489  */
490 async function loadAuditView() {
491     const contentEl = document.getElementById('slc-audit-content');
492     if (!contentEl) return;
493
494     const hoursSelect = document.getElementById('slc-hours-select');
495     const limitSelect = document.getElementById('slc-limit-select');
496     const hours = hoursSelect ? hoursSelect.value : '24';
497     const limit = limitSelect ? limitSelect.value : '500';
498
499     contentEl.innerHTML = `
500         <div class="text-center py-5 text-muted">
501             <div class="spinner-border text-info mb-3" role="status"></div>
502             <div class="fw-bold text-light">Выполняется многоуровневый аудит массива логов...</div>
503             <div class="small text-secondary mt-1">Фильтрация шума
• Шаблонизация • Детекция аномалий • Сжатие данных</div>
504         </div>
505     `;
506
507     try {

```

```

508     const url = `/api/v1/system_logs/audit?channel=${encodeURIComponentComponent(currentChannel)}&limit=${limit}&hours=${hours}&file_path=${encodeURIComponentComponent(currentFilePath)}`;
509     const res = await fetch(url);
510     if (!res.ok) throw new Error(`HTTP ${res.status}`);
511     const data = await res.json();
512
513     const anomaliesHtml = (data.anomalies || []).map((a) =>
514     {
515         const badgeClass = getSeverityClass(a.level);
516         return `
517             <div class="p-2 mb-2 rounded bg-black border border-secondary d-flex justify-content-between align-items-center flex-wrap gap-2">
518                 <div>
519                     <span class="badge ${badgeClass} text-uppercase me-2">${escapeHtml(a.level)}</span>
520                     <strong class="text-info">${escapeHtml(a.provider)}</strong>
521                     <span class="text-muted font-monospace small">(ID ${a.event_id})</span>
522                     <div class="small text-light mt-1 font-monospace" style="word-break: break-word;">${escapeHtml(a.sample)}</div>
523                     <div class="small text-secondary mt-1"><i class="bi bi-clock me-1"></i>Окно: ${escapeHtml(a.time_window)}</div>
524                 </div>
525                 <div class="text-end">
526                     <span class="badge bg-danger rounded-pill px-3 py-2 fs-6">${a.count}x</span>
527                 </div>
528             `;
529         }).join('') || `<div class="text-success small p-2"><i class="bi bi-check-circle me-1"></i>Аномалий и критических сбоев не обнаружено.</div>`;
530

```



```

531     const clustersHtml = (data.top_clusters || []).map((cl,
|   idx) => {
532         const badgeClass = getSeverityClass(cl.level);
533         return `
534             <tr class="small">
535                 <td class="text-center font-monospace">${idx + 1}<
|   /td>
536                 <td><span class="badge ${badgeClass}">${escapeHtml
|   (cl.level)}</span></td>
537                 <td class="text-info text-truncate" style="max-wid
|   th: 140px;" title="${escapeHtml(cl.provider)}">${escapeHtml(cl
|   .provider)}</td>
538                 <td class="text-center font-monospace">${cl.event_
|   id || '-'}</td>
539                 <td class="font-monospace text-light" style="max-w
|   idth: 320px; word-break: break-word;" title="${escapeHtml(cl.t
|   emplate)}">${escapeHtml(cl.template)}</td>
540                 <td class="text-center"><span class="badge bg-seco
|   ndary font-monospace">${cl.count}</span></td>
541                 <td class="small text-secondary font-monospace tex
|   t-nowrap">${escapeHtml(cl.first_seen)}</td>
542             </tr>
543         `;
544     }).join('');
545
546     const totalScanned = data.total_analyzed ?? data.total_s
|   canned ?? 0;
547     const uniqueCount = data.unique_patterns_count ?? (data.
|   top_clusters ? data.top_clusters.length : 0);
548     const redundancyPct = data.redundancy_pct !== undefined
|   ? `${data.redundancy_pct.toFixed(1)}%` : (data.compression_rat
|   io || '0%');
549     const errSum = (data.error_count || 0) + (data.critical_
|   count || 0);
550     const warnSum = data.warning_count || 0;
551     const summaryText = data.strategy_rationale || data.exec
|   utive_summary || `Стратегия обработки: ${data.strategy || 'Ста
|   ндартная'}. Найдено ${uniqueCount} ключевых шаблонов.`;

```

```

552 |
553 |     contentEl.innerHTML = `
554 |         <!-- Executive Summary Card (Compact) -->
555 |         <div class="card bg-dark border-info mb-2 shadow-sm">
556 |             <div class="card-header bg-black text-info fw-bold p
557 | y-1 px-3 d-flex justify-content-between align-items-center sma
558 | ll">
559 |                 <span><i class="bi bi-cpu me-1"></i>Сводка аудита:
560 |                 <span class="badge bg-info text-dark font-monospac
561 | e">Дублирование: ${redundancyPct}</span>
562 |             </div>
563 |             <div class="card-body p-2">
564 |                 <div class="row g-2 text-center mb-2">
565 |                     <div class="col-sm-3 col-6">
566 |                         <div class="p-1 rounded bg-black border border
567 | -secondary">
568 |                             <div class="small text-muted" style="font-si
569 | ze: 0.72rem;">Просканировано</div>
570 |                             <div class="fs-5 fw-bold text-light font-mon
571 | ospace">${totalScanned}</div>
572 |                         </div>
573 |                         </div>
574 |                         <div class="col-sm-3 col-6">
575 |                             <div class="p-1 rounded bg-black border border
576 | -secondary">
577 |                                 <div class="small text-muted" style="font-si
578 | ze: 0.72rem;">Шаблонов</div>
579 |                                 <div class="fs-5 fw-bold text-info font-mono
580 | space">${uniqueCount}</div>
581 |                             </div>
582 |                             </div>
583 |                             <div class="col-sm-3 col-6">
584 |                                 <div class="p-1 rounded bg-black border border
585 | -secondary">
586 |                                     <div class="small text-muted" style="font-si
587 | ze: 0.72rem;">Ошибок / Сбоев</div>
588 |                                     <div class="fs-5 fw-bold text-danger font-mo

```

```

        nospace">${errSum}</div>
578         </div>
579     </div>
580     <div class="col-sm-3 col-6">
581         <div class="p-1 rounded bg-black border border
        -secondary">
582             <div class="small text-muted" style="font-si
        ze: 0.72rem;">Предупреждений</div>
583             <div class="fs-5 fw-bold text-warning font-m
        onospace">${warnSum}</div>
584         </div>
585     </div>
586 </div>
587     <p class="mb-0 text-light small"><i class="bi bi-i
        nfo-circle text-info me-1"></i>${escapeHtml(summaryText)}</p>
588     </div>
589 </div>
590
591     <!-- Detected Anomalies Section -->
592     <div class="card bg-dark border-secondary mb-2 shadow-
        sm">
593         <div class="card-header bg-black text-warning fw-bol
        d py-1 px-3 small">
594             <i class="bi bi-exclamation-triangle me-1"></i>Обн
        аруженные аномалии и всплески
595         </div>
596         <div class="card-body p-2">
597             ${anomaliesHtml}
598         </div>
599     </div>
600
601     <!-- Top Clustered Patterns Table -->
602     <div class="card bg-dark border-secondary">
603         <div class="card-header bg-black text-light fw-bold
        d-flex justify-content-between align-items-center">
604             <span><i class="bi bi-collection me-2"></i>Топ шаб
        лонов событий (Дедупликация)</span>
605             <span class="small text-muted">Топ 10 кластеров</s

```

```

        pan>
606         </div>
607         <div class="table-responsive">
608             <table class="table slc-table mb-0">
609                 <thead>
610                     <tr>
611                         <th style="width: 40px;">#</th>
612                         <th style="width: 80px;">Уровень</th>
613                         <th style="width: 140px;">Поставщик</th>
614                         <th style="width: 60px;">ID</th>
615                         <th>Шаблон сообщения (Masked)</th>
616                         <th style="width: 70px;" class="text-center"
        >Кол-во</th>
617                         <th style="width: 140px;">Впервые</th>
618                     </tr>
619                 </thead>
620                 <tbody>
621                     ${clustersHtml}
622                 </tbody>
623             </table>
624         </div>
625     </div>
626     `;
627     } catch (err) {
628         contentEl.innerHTML = `<div class="alert alert-danger">0
        ошибка проведения аудита: ${escapeHtml(err.message)}</div>`;
629     }
630 }
631
632 /**
633  * Execute dynamic RAG search over indexed patterns, anomali
        es, and digests.
634  */
635 async function performRAGSearch() {
636     const inputEl = document.getElementById('slc-rag-input');
637     const resultsEl = document.getElementById('slc-rag-results
        ');
638     if (!inputEl || !resultsEl) return;

```

```

639
640     const query = inputEl.value.trim();
641     if (!query) return;
642
643     resultsEl.innerHTML = `
```

```

w-sm">
669         <div class="card-header bg-black d-flex justify-co
        ntent-between align-items-center">
670             <span class="fw-bold text-info"><i class="bi bi-
        file-earmark-text me-2"></i>${escapeHtml(r.title)}</span>
671             <div class="d-flex align-items-center gap-2">
672                 <span class="badge ${badgeClass}">${escapeHtml
        (r.meta?.level || r.type)}</span>
673                 <span class="badge bg-secondary font-monospace
        ">Score: ${r.relevance_score}</span>
674             </div>
675         </div>
676         <div class="card-body">
677             <pre class="bg-black text-light p-3 rounded font
        -monospace small mb-0" style="white-space: pre-wrap; word-brea
        k: break-word;">${escapeHtml(r.text)}</pre>
678         </div>
679     </div>
680     `;
681     }).join('');
682     } catch (err) {
683         resultsEl.innerHTML = `<div class="alert alert-danger">0
       шибка поиска RAG: ${escapeHtml(err.message)}</div>`;
684     }
685 }
686
687 /**
688  * Trigger building or rebuilding dynamic RAG index for curr
        ent channel.
689  */
690 async function rebuildRAGIndex() {
691     const resultsEl = document.getElementById('slc-rag-results
        ');
692     if (!resultsEl) return;
693     resultsEl.innerHTML = `<div class="text-center py-4 text-m
        uted"><div class="spinner-border spinner-border-sm text-info m
        e-2"></div>Перестроение динамического RAG-индекса для «${escap
        eHtml(currentChannel)}»...</div>`;

```

```

694
695     try {
696         const res = await fetch(`/api/v1/system_logs/rag/build?c
channel=${encodeURIComponent(currentChannel)}&limit=500&hours=2
4&file_path=${encodeURIComponent(currentFilePath)}`, {
697             method: 'POST',
698         });
699         if (!res.ok) throw new Error(`HTTP ${res.status}`);
700         const data = await res.json();
701         resultsEl.innerHTML = `
702             <div class="alert alert-success d-flex align-items-cen
ter gap-2">
703                 <i class="bi bi-check-circle-fill fs-5"></i>
704                 <div>
705                     <strong>RAG-индекс успешно обновлен!</strong> Доба
влено ${data.chunks_added} чанков (Всего в индексе: ${data.tot
al_index_chunks}).
706                     <div class="small mt-1 text-light">${escapeHtml(da
ta.executive_summary)}</div>
707                 </div>
708             </div>
709         `;
710     } catch (err) {
711         resultsEl.innerHTML = `<div class="alert alert-danger">O
шибка перестроения RAG: ${escapeHtml(err.message)}</div>`;
712     }
713 }
714
715 /**
716  * Display event details modal with AI diagnostic action.
717  *
718  * @param {object} entry - Event object.
719  */
720 function showEventModal(entry) {
721     if (!entry) return;
722     selectedEvent = entry;
723
724     const modalEl = document.getElementById('slc-event-detail-

```

```

modal');
725     if (!modalEl) return;
726
727     document.getElementById('slc-modal-time').textContent = en
try.timestamp;
728     document.getElementById('slc-modal-level').innerHTML = `

```



```

    ', {
747         method: 'POST',
748         headers: { 'Content-Type': 'application/json' },
749         body: JSON.stringify({
750             target_timestamp: entry.timestamp,
751             window_minutes: 5,
752             channel: entry.channel || currentChannel,
753             event_id: entry.event_id || 0,
754             provider: entry.provider || entry.source || '',
755             level: entry.level || '',
756             message: entry.message || '',
757             target_entry: entry,
758             query_text: `Диагностика события ${entry.provider
r || entry.source} (ID ${entry.event_id || 0}): ${entry.message
e || ''}`,
759         }),
760     });
761     if (!res.ok) throw new Error(`HTTP ${res.status}`);
762     const report = await res.json();
763     aiExplanation.innerHTML = `
764         <div class="mb-2"><strong class="text-info">Сводка
: </strong> ${escapeHtml(report.summary)}</div>
765         <div class="mb-2"><strong class="text-warning">При
чина / Анализ:</strong> ${escapeHtml(report.root_cause)}</div>
766         <h6 class="small text-uppercase text-muted fw-bold
mb-1">Рекомендации:</h6>
767         <ul class="mb-0 ps-3">
768             ${((report.recommendations || []).map(r => `<li>${
escapeHtml(r)}</li>`).join(''))}
769         </ul>
770     `;
771     } catch (err) {
772         aiExplanation.innerHTML = `<div class="text-danger">
Ошибка диагностики: ${escapeHtml(err.message)}</div>`;
773     }
774 };
775 }
776

```

```
777     const bsModal = new bootstrap.Modal(modalEl);
778     bsModal.show();
779 }
780
781 /**
782  * Export logs as downloadable file.
783  *
784  * @param {string} format - "json" or "csv".
785  */
786 function exportLogs(format) {
787     const level = (document.getElementById('slc-level-select')
788     || {}).value || '';
789     const search = (document.getElementById('slc-search-input'
790     ) || {}).value || '';
791     const limit = (document.getElementById('slc-limit-select')
792     || {}).value || '200';
793     const hours = (document.getElementById('slc-hours-select')
794     || {}).value || '24';
795
796     const url = `/api/v1/system_logs/export?format=${format}&c
797     hannel=${encodeURIComponent(currentChannel)}&level=${encodeURIComponent(level)}&search=${encodeURIComponent(search)}&hours=${
798     {hours}&limit=${limit}`;
799     window.open(url, '_blank');
800 }
801
802 /**
803  * Main initializer for System Log Center tab.
804  */
805 function initSystemLogsTab() {
806     performSystemScan().then(() => {
807         loadEvents();
808     });
809
810     const scanBtn = document.getElementById('btn-slc-scan');
811     if (scanBtn) {
812         scanBtn.onclick = () => performSystemScan(true);
813     }
814 }
```

```
808
809     const refreshBtn = document.getElementById('btn-slc-refres
      h');
810     if (refreshBtn) {
811         refreshBtn.onclick = loadEvents;
812     }
813
814     const jsonExportBtn = document.getElementById('btn-slc-exp
      ort-json');
815     if (jsonExportBtn) {
816         jsonExportBtn.onclick = (e) => {
817             e.preventDefault();
818             exportLogs('json');
819         };
820     }
821
822     const csvExportBtn = document.getElementById('btn-slc-expo
      rt-csv');
823     if (csvExportBtn) {
824         csvExportBtn.onclick = (e) => {
825             e.preventDefault();
826             exportLogs('csv');
827         };
828     }
829
830     const modeTabs = document.querySelectorAll('#slc-mode-tabs
      .nav-link');
831     modeTabs.forEach((tab) => {
832         tab.onclick = () => {
833             modeTabs.forEach(t => t.classList.remove('active'));
834             tab.classList.add('active');
835             currentMode = tab.getAttribute('data-mode') || 'live';
836             renderCurrentModeView();
837         };
838     });
839
840     const auditBtn = document.getElementById('btn-slc-audit');
841     if (auditBtn) {
```

```

842 |     auditBtn.onclick = () => {
843 |         modeTabs.forEach(t => {
844 |             if (t.getAttribute('data-mode') === 'audit') {
845 |                 t.classList.add('active');
846 |             } else {
847 |                 t.classList.remove('active');
848 |             }
849 |         });
850 |         currentMode = 'audit';
851 |         renderCurrentModeView();
852 |     };
853 | }
854 |
855 |     const ragSearchBtn = document.getElementById('btn-slc-rag-
search');
856 |     if (ragSearchBtn) {
857 |         ragSearchBtn.onclick = performRAGSearch;
858 |     }
859 |
860 |     const ragRebuildBtn = document.getElementById('btn-slc-rag
-rebuild');
861 |     if (ragRebuildBtn) {
862 |         ragRebuildBtn.onclick = rebuildRAGIndex;
863 |     }
864 |
865 |     const ragInput = document.getElementById('slc-rag-input');
866 |     if (ragInput) {
867 |         ragInput.onkeydown = (e) => {
868 |             if (e.key === 'Enter') {
869 |                 e.preventDefault();
870 |                 performRAGSearch();
871 |             }
872 |         };
873 |     }
874 |
875 |     const askAiBtn = document.getElementById('btn-slc-ask-ai')
;
876 |     if (askAiBtn) {

```

```
877 |     askAiBtn.onclick = () => {
878 |         if (selectedEvent) {
879 |             showEventModal(selectedEvent);
880 |         } else if (cachedEvents.length > 0) {
881 |             selectedEvent = cachedEvents[0];
882 |             showEventModal(selectedEvent);
883 |         }
884 |     };
885 | }
886 |
887 | const levelSelect = document.getElementById('slc-level-select');
888 | if (levelSelect) levelSelect.onchange = loadEvents;
889 |
890 | const hoursSelect = document.getElementById('slc-hours-select');
891 | if (hoursSelect) hoursSelect.onchange = loadEvents;
892 |
893 | const limitSelect = document.getElementById('slc-limit-select');
894 | if (limitSelect) limitSelect.onchange = loadEvents;
895 |
896 | const eventIdInput = document.getElementById('slc-eventid-input');
897 | if (eventIdInput) eventIdInput.onchange = loadEvents;
898 |
899 | let searchTimer = null;
900 | const searchInput = document.getElementById('slc-search-input');
901 | if (searchInput) {
902 |     searchInput.oninput = () => {
903 |         clearTimeout(searchTimer);
904 |         searchTimer = setTimeout(loadEvents, 350);
905 |     };
906 | }
907 |
908 | const channelFilterInput = document.getElementById('slc-channel-filter-input');
```

```
909 |     if (channelFilterInput) {
910 |         channelFilterInput.oninput = renderChannelsTree;
911 |     }
912 |
913 |     const liveSwitch = document.getElementById('slc-live-switc
| h');
914 |     if (liveSwitch) {
915 |         liveSwitch.onchange = () => {
916 |             if (window.registerTabPoller && window.unregisterTabPo
| ller) {
917 |                 if (liveSwitch.checked) {
918 |                     window.registerTabPoller('tab-system-logs', loadEv
| ents, 4000, { immediate: true });
919 |                 } else {
920 |                     window.unregisterTabPoller('tab-system-logs_defaul
| t');
921 |                 }
922 |             } else {
923 |                 if (liveSwitch.checked) {
924 |                     liveIntervalTimer = setInterval(loadEvents, 4000);
925 |                 } else {
926 |                     clearInterval(liveIntervalTimer);
927 |                     liveIntervalTimer = null;
928 |                 }
929 |             }
930 |         };
931 |     }
932 | }
933 |
934 | // Export initializer for admin UI
935 | window.initSystemLogsTab = initSystemLogsTab;
936 | })();
937 |
```

index.html

```
1 <!-- Telegram Channel & Group RAG Search Tab -->
2 <div class="container-fluid py-2">
3   <!-- Top Banner / Header -->
4   <div class="alert alert-primary d-flex flex-wrap align-items
5   -center justify-content-between gap-2 mb-3 shadow-sm">
6     <div class="d-flex align-items-center gap-2">
7       <span class="fs-4">□ </span>
8       <div>
9         <h5 class="mb-0 fw-bold">Поиск по Telegram каналам и г
10        руппам (RAG)</h5>
11         <p class="mb-0 small text-muted">Индексация публичных
12        каналов, поиск по сообщениям с точными ссылками и мгновенным о
13        тветом</p>
14       </div>
15     </div>
16     <div class="d-flex align-items-center gap-2">
17       <span class="badge bg-primary-subtle text-primary border
18       border-primary-subtle px-3 py-2" id="tgram-stats-badge">
19         Загрузка каналов...
20       </span>
21       <button class="btn btn-sm btn-outline-primary" id="btn-r
22       efresh-tgram-channels" onclick="loadTelegramChannels()">
23         <i class="bi bi-arrow-clockwise"></i> Обновить
24       </button>
25     </div>
26   </div>
27
28   <div class="row g-3">
29     <!-- LEFT: Channel Management & Subscriptions -->
30     <div class="col-lg-4 col-md-5">
31       <!-- Add / Subscribe Card -->
32       <div class="card shadow-sm border-secondary-subtle mb-3"
33       >
34         <div class="card-header bg-body-tertiary py-2">
35           <h6 class="mb-0 fw-bold small text-uppercase d-flex
36           align-items-center gap-2">
```

```

29         <i class="bi bi-plus-circle-fill text-primary"></i
    > Добавить канал / группу
30     </h6>
31 </div>
32 <div class="card-body p-3">
33     <form id="tgram-add-channel-form" onsubmit="event.pr
eventDefault(); handleAddTelegramChannel();">
34         <div class="mb-2">
35             <label class="form-label small fw-semibold mb-1"
>Имя или ссылка на канал</label>
36             <div class="input-group input-group-sm">
37                 <span class="input-group-text bg-body-secondar
y">@</span>
38                 <input type="text" class="form-control" id="tg
ram-channel-input" placeholder="canozrimb или t.me/canozrimb"
required>
39             </div>
40             <div class="form-text text-muted small" style="f
ont-size: 0.75rem;">
41                 Поддерживаются публичные каналы и веб-превью T
elegram
42             </div>
43         </div>
44
45         <div class="mb-3">
46             <label class="form-label small fw-semibold mb-1"
>Лимит сообщений для сбора</label>
47             <select class="form-select form-select-sm" id="t
gram-max-msgs-select">
48                 <option value="500" selected>500 сообщений (бы
стро)</option>
49                 <option value="1000">1000 сообщений</option>
50                 <option value="3000">3000 сообщений</option>
51                 <option value="5000">5000 сообщений (глубокий
RAG)</option>
52             </select>
53         </div>
54

```



```

55 |         <button type="submit" class="btn btn-sm btn-primar
    | y w-100 d-flex align-items-center justify-content-center gap-2
    | shadow-sm" id="btn-tgram-subscribe">
56 |             <i class="bi bi-cloud-arrow-down-fill"></i>
57 |             <span>Подключить и индексировать</span>
58 |         </button>
59 |     </form>
60 | </div>
61 | </div>
62 |
63 | <!-- Channels List Card -->
64 | <div class="card shadow-sm border-secondary-subtle">
65 |     <div class="card-header bg-body-tertiary d-flex justif
    | y-content-between align-items-center py-2">
66 |         <span class="fw-semibold small text-uppercase">
67 |             Подключенные каналы (<span id="tgram-channels-coun
    | t">0</span>)
68 |         </span>
69 |         <span class="badge bg-secondary-subtle text-body" id
    | ="tgram-active-count-badge">0 в пуле</span>
70 |     </div>
71 |     <div class="card-body p-2" id="tgram-channels-list-con
    | tainer" style="max-height: 55vh; overflow-y: auto;">
72 |         <div class="text-center text-muted p-4">
73 |             <div class="spinner-border spinner-border-sm text-
    | primary mb-2" role="status"></div>
74 |             <div>Загрузка списка каналов...</div>
75 |         </div>
76 |     </div>
77 | </div>
78 | </div>
79 |
80 | <!-- RIGHT: Search Interface & Results -->
81 | <div class="col-lg-8 col-md-7">
82 |     <div class="card shadow-sm border-secondary-subtle h-100
    | d-flex flex-column">
83 |         <!-- Search Input Bar & Settings Header -->
84 |         <div class="card-header bg-body-tertiary p-3 border-bo

```

```

ttom">
85         <!-- Main Search Input -->
86         <form id="tgram-search-form" onsubmit="event.preventDefault(); executeTelegramSearch();" class="mb-3">
87             <div class="input-group shadow-sm">
88                 <span class="input-group-text bg-body text-primary"><i class="bi bi-search"></i></span>
89                 <input type="text" class="form-control form-control-lg fs-6" id="tgram-search-input" placeholder="Введите поисковый запрос, тему или вопрос..." required autofocus>
90                 <button class="btn btn-outline-secondary" type="button" title="Очистить" onclick="document.getElementById('tgram-search-input').value='';">
91                     <i class="bi bi-x-lg"></i>
92                 </button>
93                 <button class="btn btn-primary px-4 fw-semibold" type="submit" id="btn-tgram-do-search">
94                     <i class="bi bi-search me-1"></i> Найти
95                 </button>
96             </div>
97         </form>
98
99         <!-- Search Settings & Scope Filters -->
100        <div class="row g-2 align-items-center bg-body p-2 rounded border border-secondary-subtle small">
101            <!-- Channel Filter -->
102            <div class="col-md-5 col-12">
103                <label class="form-label text-muted mb-1 fw-semibold" style="font-size: 0.75rem;">
104                    <i class="bi bi-funnel-fill text-primary"></i>
105                    Область поиска:
106                </label>
107                <select class="form-select form-select-sm" id="tgram-filter-channel">
108                    <option value="all" selected>☐ Все подключенные каналы (Общий пул)</option>
109                </select>
110            </div>

```

```
110 |
111 |         <!-- Search Mode -->
112 |         <div class="col-md-4 col-12">
113 |             <label class="form-label text-muted mb-1 fw-semi
114 | bold" style="font-size: 0.75rem;">
115 |                 <i class="bi bi-diagram-2-fill text-info"></i>
116 | Режим индексов:
117 |             </label>
118 |             <select class="form-select form-select-sm" id="t
119 | gram-search-mode">
120 |                 <option value="unified" selected>Общий RAG пул
121 | (Unified)</option>
122 |                 <option value="individual">Раздельно по канала
123 | м</option>
124 |             </select>
125 |         </div>
126 |
127 |         <!-- Top K -->
128 |         <div class="col-md-3 col-12">
129 |             <label class="form-label text-muted mb-1 fw-semi
130 | bold" style="font-size: 0.75rem;">
131 |                 <i class="bi bi-list-ol text-warning"></i> Лим
132 | ит выдачи:
133 |             </label>
134 |             <select class="form-select form-select-sm" id="t
135 | gram-topk-select">
136 |                 <option value="5" selected>5 сообщений</option>
137 |                 <option value="10">10 сообщений</option>
138 |                 <option value="20">20 сообщений</option>
139 |                 <option value="50">50 сообщений</option>
140 |             </select>
141 |         </div>
142 |     </div>
143 |
144 |     <!-- Search Results Body -->
145 |     <div class="card-body p-3 flex-grow-1" style="max-heig
```

```
ht: 60vh; overflow-y: auto;" id="tgram-results-container">
139 |         <!-- Initial Placeholder -->
140 |         <div class="d-flex flex-column align-items-center ju
stify-content-center text-muted p-5 text-center" id="tgram-sea
rch-placeholder">
141 |             <span class="fs-1 mb-2">🔍 </span>
142 |             <h6 class="fw-bold">Поиск по базе знаний Telegram<
/h6>
143 |             <p class="small text-muted mb-0" style="max-width:
480px;">
144 |                 Введите ключевые слова или вопрос выше, чтобы на
йти точные сообщения с ссылками на первоисточники в Telegram.
145 |             </p>
146 |         </div>
147 |     </div>
148 | </div>
149 | </div>
150 | </div>
151 | </div>
152 |
```

main.js

```
1 // — TELEGRAM CHANNEL & GROUP RAG TAB MAIN.JS —————
2
3 let tgramSubscribedChannels = [];
4 let tgramAvailableChannels = [];
5
6 // Initialize Telegram RAG Tab
7 window.initTelegramRagTab = async function() {
8   console.log('[TelegramRAG] Initializing Telegram RAG Tab...')
9   await loadTelegramChannels();
10 };
11
12 // Notification helper
13 function showTgramNotification(message, type = 'info') {
14   if (typeof window.showNotification === 'function') {
15     window.showNotification(message, type);
16     return;
17   }
18   console.log(`[TelegramRAG Notification ${type}]: ${message}`
19 );
20
21 // Load channels from backend API
22 async function loadTelegramChannels() {
23   const container = document.getElementById('tgram-channels-li
24 st-container');
25   const statsBadge = document.getElementById('tgram-stats-badg
26 e');
27   const countEl = document.getElementById('tgram-channels-coun
28 t');
29   const activeCountBadge = document.getElementById('tgram-acti
30 ve-count-badge');
31   const filterSelect = document.getElementById('tgram-filter-c
32 hannel');
```

```

29 |   try {
30 |       const res = await fetch('/api/telegram_rag/channels');
31 |       if (!res.ok) throw new Error(`HTTP ${res.status}`);
32 |       const data = await res.json();
33 |
34 |       tgramSubscribedChannels = data.user_subscriptions || [];
35 |       tgramAvailableChannels = data.available_channels || [];
36 |
37 |       // Calculate total messages
38 |       const totalMsgs = tgramAvailableChannels.reduce((sum, c) =
> sum + (c.messages_count || 0), 0);
39 |
40 |       if (statsBadge) {
41 |         statsBadge.textContent = `${tgramSubscribedChannels.leng
th} подписок • ${totalMsgs} сообщений`;
42 |       }
43 |       if (countEl) countEl.textContent = tgramSubscribedChannels
.length;
44 |       if (activeCountBadge) activeCountBadge.textContent = `${tg
ramAvailableChannels.length} в пуле`;
45 |
46 |       // Render channels list
47 |       renderTelegramChannelsList(tgramSubscribedChannels, tgramA
vailableChannels);
48 |
49 |       // Update filter dropdown
50 |       if (filterSelect) {
51 |         const currentVal = filterSelect.value || 'all';
52 |         let optsHtml = '<option value="all">☐ Все подключенные
каналы (Общий пул)</option>';
53 |         tgramSubscribedChannels.forEach(ch => {
54 |           const isSel = ch === currentVal ? 'selected' : '';
55 |           const found = tgramAvailableChannels.find(a => a.chann
el === ch);
56 |           const countText = found ? ` (${found.messages_count} c
ообщ.)` : '';
57 |           optsHtml += `<option value="${ch}" ${isSel}>@${ch}${co
untText}</option>`;

```

```

58     });
59     filterSelect.innerHTML = optsHtml;
60 }
61 } catch (err) {
62     console.error('Error loading Telegram channels:', err);
63     if (container) {
64         container.innerHTML = `
```

```

91
92     const statusBadge = info.is_indexed
93     ? `<span class="badge bg-success-subtle text-success bor
    der border-success-subtle">${info.messages_count} сообщ.</span
    >`
94     : `<span class="badge bg-warning-subtle text-warning bor
    der">Не индексирован</span>`;
95
96     html += `
97         <div class="list-group-item p-2 rounded mb-1 border-0 bg
    -body-tertiary shadow-sm">
98             <div class="d-flex justify-content-between align-items
    -center mb-1">
99                 <div class="d-flex align-items-center gap-2 text-tru
    ncate">
100                     <span class="fs-5 text-primary">🔍 </span>
101                     <a href="https://t.me/${channelName}" target="_bla
    nk" rel="noopener noreferrer" class="fw-bold text-decoration-n
    one text-truncate" title="Открыть в Telegram">
102                         @${channelName}
103                         <i class="bi bi-box-arrow-up-right ms-1 text-mut
    ed" style="font-size: 0.75rem;"></i>
104                     </a>
105                 </div>
106                 <div>${statusBadge}</div>
107             </div>
108             <div class="d-flex justify-content-between align-items
    -center mt-2 pt-1 border-top border-secondary-subtle">
109                 <button class="btn btn-sm btn-outline-secondary py-0
    px-2 small" onclick="filterByChannel('${channelName}')" title
    ="Искать только в этом канале">
110                     <i class="bi bi-search"></i> Искать здесь
111                 </button>
112                 <div class="btn-group btn-group-sm">
113                     <button class="btn btn-outline-primary py-0 px-2"
    onclick="handleReindexTelegramChannel('${channelName}')" title
    ="Переиндексировать канал">
114                     <i class="bi bi-arrow-clockwise"></i>

```



```

115 |         </button>
116 |         <button class="btn btn-outline-danger py-0 px-2" o
nclick="handleUnsubscribeTelegramChannel('${channelName}')" ti
tle="Отключить канал">
117 |             <i class="bi bi-trash"></i>
118 |         </button>
119 |     </div>
120 | </div>
121 | </div>
122 | `;
123 | });
124 | html += '</div>';
125 | container.innerHTML = html;
126 | }
127 |
128 | // Quick filter helper from channel list
129 | function filterByChannel(channelName) {
130 |     const filterSelect = document.getElementById('tgram-filter-c
hannel');
131 |     if (filterSelect) {
132 |         filterSelect.value = channelName;
133 |     }
134 |     const searchInput = document.getElementById('tgram-search-in
put');
135 |     if (searchInput) {
136 |         searchInput.focus();
137 |     }
138 | }
139 |
140 | // Add / Subscribe Channel Handler
141 | async function handleAddTelegramChannel() {
142 |     const input = document.getElementById('tgram-channel-input')
;
143 |     const maxMsgsSelect = document.getElementById('tgram-max-msg
s-select');
144 |     const btn = document.getElementById('btn-tgram-subscribe');
145 |
146 |     if (!input || !input.value.trim()) return;

```

```

147 |   const channel = input.value.trim();
148 |   const maxMessages = parseInt(maxMsgsSelect?.value || '500',
    | 10);
149 |
150 |   try {
151 |     if (btn) {
152 |       btn.disabled = true;
153 |       btn.innerHTML = '<span class="spinner-border spinner-border-sm me-1" role="status"></span> Индексация...';
154 |     }
155 |
156 |     showTgramNotification(`Подключение канала @${channel}...`,
    | 'info');
157 |     const resp = await fetch('/api/telegram_rag/subscribe', {
158 |       method: 'POST',
159 |       headers: { 'Content-Type': 'application/json' },
160 |       body: JSON.stringify({ channel, max_messages: maxMessages
    | s })),
161 |     });
162 |
163 |     if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
164 |     const data = await resp.json();
165 |
166 |     showTgramNotification(data.message || `Канал @${channel} у
    | спешно подключен`, 'success');
167 |     input.value = '';
168 |     await loadTelegramChannels();
169 |   } catch (err) {
170 |     console.error('Error adding Telegram channel:', err);
171 |     showTgramNotification(`Ошибка: ${err.message}`, 'danger');
172 |   } finally {
173 |     if (btn) {
174 |       btn.disabled = false;
175 |       btn.innerHTML = '<i class="bi bi-cloud-arrow-down-fill me-1"></i> Подключить и индексировать';
176 |     }
177 |   }
178 | }

```

```

179
180 // Unsubscribe Channel Handler
181 async function handleUnsubscribeTelegramChannel(channel) {
182   if (!confirm(`Вы действительно хотите отключить канал @${channel}?`)) return;
183
184   try {
185     const resp = await fetch('/api/telegram_rag/unsubscribe',
186     {
187       method: 'POST',
188       headers: { 'Content-Type': 'application/json' },
189       body: JSON.stringify({ channel }),
190     });
191     if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
192     const data = await resp.json();
193     showTgramNotification(data.message || `Канал @${channel} отключен`, 'info');
194     await loadTelegramChannels();
195   } catch (err) {
196     console.error('Error unsubscribing channel:', err);
197     showTgramNotification(`Ошибка: ${err.message}`, 'danger');
198   }
199
200 // Reindex Channel Handler
201 async function handleReindexTelegramChannel(channel) {
202   const maxMsgs = parseInt(document.getElementById('tgram-max-msgs-select')?.value || '500', 10);
203   try {
204     showTgramNotification(`Обновление индекса @${channel}...`, 'info');
205     const resp = await fetch('/api/telegram_rag/reindex', {
206       method: 'POST',
207       headers: { 'Content-Type': 'application/json' },
208       body: JSON.stringify({ channel, max_messages: maxMsgs })
209     },
210     {
211       });
212     if (!resp.ok) throw new Error(`HTTP ${resp.status}`);

```

```
211     const data = await resp.json();
212     showTgramNotification(data.message || `Индекс @${channel}
успешно обновлен`, 'success');
213     await loadTelegramChannels();
214   } catch (err) {
215     console.error('Error reindexing channel:', err);
216     showTgramNotification(`Ошибка переиндексации: ${err.messag
e}`, 'danger');
217   }
218 }
219
220 // Highlight keywords in text
221 function highlightKeywords(text, query) {
222   if (!text || !query) return text || '';
223   const words = query.split(/\s+/).filter(w => w.length > 2);
224   if (words.length === 0) return text;
225
226   let escaped = text
227     .replace(/&/g, '&');
228     .replace(/</g, '<');
229     .replace(/>/g, '>');
230
231   words.forEach(w => {
232     const regex = new RegExp(`(${w.replace(/[\.*+?^${}()|[\]\$\`]/g, '\\$&')})`, 'gi');
233     escaped = escaped.replace(regex, '<mark class="bg-warning-
subtle text-body px-1 rounded">$1</mark>');
234   });
235   return escaped;
236 }
237
238 // Execute Telegram RAG Search
239 async function executeTelegramSearch() {
240   const searchInput = document.getElementById('tgram-search-in
put');
241   const filterSelect = document.getElementById('tgram-filter-c
hannel');
242   const modeSelect = document.getElementById('tgram-search-mod
```

```

    e');
243   const topkSelect = document.getElementById('tgram-topk-select');
244   const resultsContainer = document.getElementById('tgram-results-container');
245   const searchBtn = document.getElementById('btn-tgram-do-search');
246
247   if (!searchInput || !searchInput.value.trim()) return;
248   const query = searchInput.value.trim();
249   const channel = filterSelect ? filterSelect.value : 'all';
250   const searchMode = modeSelect ? modeSelect.value : 'unified'
    ;
251   const topK = parseInt(topkSelect?.value || '5', 10);
252
253   if (resultsContainer) {
254     resultsContainer.innerHTML = `
255       <div class="text-center text-muted p-5">
256         <div class="spinner-border text-primary mb-3" role="status"></div>
257         <h6>Поиск в базе знаний Telegram...</h6>
258         <div class="small text-muted">Запрос: «${query}»</div>
259       </div>
260     `;
261   }
262
263   if (searchBtn) searchBtn.disabled = true;
264
265   try {
266     const startTime = performance.now();
267     const resp = await fetch('/api/telegram_rag/search', {
268       method: 'POST',
269       headers: { 'Content-Type': 'application/json' },
270       body: JSON.stringify({
271         query,
272         channel: channel === 'all' ? null : channel,
273         search_mode: searchMode,
274         top_k: topK,

```

```

275     }},
276   });
277
278   if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
279   const data = await resp.json();
280   const duration = Math.round(performance.now() - startTime)
281   ;
282   renderTelegramSearchResults(data.results || [], query, duration);
283   } catch (err) {
284     console.error('Error executing Telegram search:', err);
285     if (resultsContainer) {
286       resultsContainer.innerHTML = `
287         <div class="alert alert-danger shadow-sm">
288           <h6><i class="bi bi-exclamation-triangle-fill me-2">
289             </i> Ошибка поиска</h6>
290           <p class="mb-0 small">${err.message}</p>
291         </div>
292       `;
293     }
294     finally {
295       if (searchBtn) searchBtn.disabled = false;
296     }
297   }
298 // Render Search Results Cards
299 function renderTelegramSearchResults(results, query, durationMs) {
300   const container = document.getElementById('tgram-results-container');
301   if (!container) return;
302
303   if (results.length === 0) {
304     container.innerHTML = `
305       <div class="text-center text-muted p-5">
306         <span class="fs-1 mb-2">⬜ ⬜ </span>
307         <h6 class="fw-bold">Ничего не найдено</h6>

```

```

308 |         <p class="small text-muted mb-0">По запросу «${query}»
      | в индексах Telegram сообщений не обнаружено. Попробуйте измен
      | ить формулировку или подключить дополнительные каналы.</p>
309 |     </div>
310 |     `;
311 |     return;
312 | }
313 |
314 | let html = `
315 |     <div class="d-flex justify-content-between align-items-cen
      | ter mb-3 pb-2 border-bottom">
316 |         <div class="small text-muted">
317 |             Найдено результатов: <strong class="text-primary">${re
      | sults.length}</strong> (${durationMs} мс)
318 |         </div>
319 |         <span class="badge bg-secondary-subtle text-body border"
      | >RAG Ranker</span>
320 |     </div>
321 |     <div class="d-flex flex-column gap-3">
322 |     `;
323 |
324 | results.forEach((r, idx) => {
325 |     const channelName = r.channel || 'unknown';
326 |     const author = r.author || 'Telegram';
327 |     const dateStr = r.date || '';
328 |     const msgUrl = r.url || `https://t.me/${channelName}/${r.i
      | d || ''}`;
329 |     const highlightedSnippet = highlightKeywords(r.snippet ||
      | r.text || '', query);
330 |     const scoreBadge = r.score !== undefined
331 |         ? `<span class="badge bg-info-subtle text-info border">P
      | елевантность: ${Math.round(r.score * 100)}%</span>`
332 |         : '';
333 |
334 |     html += `
335 |         <div class="card border shadow-sm rounded-3">
336 |             <div class="card-header bg-body-tertiary d-flex justif
      | y-content-between align-items-center py-2">

```

```

337         <div class="d-flex align-items-center gap-2">
338             <span class="badge bg-primary text-white">#{$idx +
1}</span>
339             <a href="https://t.me/${channelName}" target="_bla
nk" rel="noopener noreferrer" class="fw-bold text-decoration-n
one text-primary">
340                 @${channelName}
341             </a>
342             <span class="text-muted small">• ${author}</span>
343             ${dateStr ? `<span class="text-muted small">• ${da
teStr}</span>` : ''}
344         </div>
345         <div>${scoreBadge}</div>
346     </div>
347     <div class="card-body p-3">
348         <p class="mb-0 small text-body font-monospace" style
="white-space: pre-wrap; word-break: break-word;">
349             ${highlightedSnippet}
350         </p>
351     </div>
352     <div class="card-footer bg-body-secondary py-2 d-flex
justify-content-between align-items-center flex-wrap gap-2">
353         <a href="${msgUrl}" target="_blank" rel="noopener no
referrer" class="btn btn-sm btn-outline-primary d-inline-flex
align-items-center gap-1">
354             <i class="bi bi-box-arrow-up-right"></i>
355             <span>Открыть в Telegram</span>
356         </a>
357         <div class="btn-group btn-group-sm">
358             <button class="btn btn-outline-secondary" onclick=
"copyTelegramLink('${msgUrl}')" title="Скопировать ссылку на с
ообщение">
359                 <i class="bi bi-clipboard"></i> Ссылка
360             </button>
361             <button class="btn btn-outline-success" onclick="s
endTelegramResultToChat('${msgUrl}', '${encodeURIComponent(r.s
nippet || '')}')" title="Вставить цитату в чат">
362                 <i class="bi bi-chat-quote-fill"></i> В чат

```



```

363         </button>
364     </div>
365 </div>
366 </div>
367 `;
368 });
369
370 html += '</div>';
371 container.innerHTML = html;
372 }
373
374 // Copy link to clipboard
375 function copyTelegramLink(url) {
376     navigator.clipboard.writeText(url).then(() => {
377         showTgramNotification('Ссылка скопирована в буфер обмена',
378             'success');
379     }).catch(() => {
380         prompt('Скопируйте ссылку вручную:', url);
381     });
382 }
383
384 // Insert result snippet into main chat
385 function sendTelegramResultToChat(url, encodedSnippet) {
386     const snippet = decodeURIComponent(encodedSnippet);
387     const textToInsert = `[Сообщение Telegram](${url}):\n> ${snippet.replace(/\n/g, '\n> ')}`;
388
389     if (typeof window.switchTab === 'function') {
390         window.switchTab('tab-chat');
391     }
392
393     const msgInput = document.getElementById('message-input');
394     if (msgInput) {
395         msgInput.value = (msgInput.value ? msgInput.value + '\n\n' : '') + textToInsert;
396         msgInput.focus();
397         showTgramNotification('Цитата вставлена в поле ввода чата', 'success');

```

```
397 |     }
398 | }
399 |
400 | // Make globally accessible
401 | window.loadTelegramChannels = loadTelegramChannels;
402 | window.handleAddTelegramChannel = handleAddTelegramChannel;
403 | window.handleUnsubscribeTelegramChannel = handleUnsubscribeTel
    | egramChannel;
404 | window.handleReindexTelegramChannel = handleReindexTelegramCha
    | nnel;
405 | window.executeTelegramSearch = executeTelegramSearch;
406 | window.filterByChannel = filterByChannel;
407 | window.copyTelegramLink = copyTelegramLink;
408 | window.sendTelegramResultToChat = sendTelegramResultToChat;
409 |
```


CPU, RAM, GPU, Диск, Сеть, Температуры), аномалий и сводный Health Score</small>

27 </div>

28 </div>

29

30 <div class="d-flex align-items-center gap-2 flex-wrap">

31 <div class="input-group input-group-sm" style="width: auto;">

32 <i class="bi bi-folder2-open"></i>

33 <select class="form-select form-select-sm bg-dark text-white border-secondary" id="th-select-source" style="max-width: 260px;">

34 <option value="">Все обнаруженные логи</option>

35 </select>

36 </div>

37

38 <div class="form-check form-switch mb-0 small text-muted d-flex align-items-center gap-1">

39 <input class="form-check-input" type="checkbox" id="th-auto-refresh">

40 <label class="form-check-label small text-nowrap" for="th-auto-refresh">Авто (5с)</label>

41 </div>

42

43 <button id="th-btn-refresh" class="btn btn-sm btn-outline-info d-flex align-items-center gap-1 shadow-sm" title="Пересчитать аналитику и обновить графики">

44 <i class="bi bi-arrow-clockwise"></i>

45 Обновить

46 </button>

47

48 <button id="th-btn-export-html" class="btn btn-sm btn-outline-success d-flex align-items-center gap-1 shadow-sm" title="Открыть автономный HTML-отчет в новой вкладке">

49 <i class="bi bi-file-earmark-bar-graph"></i>

50 HTML-дашборд

```
51 |         </button>
52 |
53 |         <button id="th-btn-export-json" class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-1 shadow-sm"
54 |         title="Скачать полный JSON-отчет">
55 |             <i class="bi bi-download"></i>
56 |             <span>JSON</span>
57 |         </button>
58 |     </div>
59 |
60 |     <!-- KPI Metrics Cards -->
61 |     <div class="row g-2">
62 |         <!-- Health Score -->
63 |         <div class="col-6 col-md-3">
64 |             <div class="p-2.5 rounded border border-secondary d-
65 |             flex align-items-center justify-content-between" style="backgr
66 |             ound: rgba(15, 23, 42, 0.7); min-height: 70px;">
67 |                 <div>
68 |                     <div class="text-muted small text-uppercase" sty
69 |                     le="font-size: 0.68rem; letter-spacing: 0.04em;">Индекс здоров
70 |                     ья (Health)</div>
71 |                     <div class="fw-bold fs-4 mt-0.5 text-success" id
72 |                     ="th-kpi-health-score">--%</div>
73 |                 </div>
74 |                 <div class="fs-2 text-success opacity-75" id="th-k
75 |                 pi-health-icon"><i class="bi bi-heart-pulse-fill"></i></div>
76 |             </div>
77 |         </div>
78 |
79 |         <!-- Records Analyzed -->
80 |         <div class="col-6 col-md-3">
81 |             <div class="p-2.5 rounded border border-secondary d-
82 |             flex align-items-center justify-content-between" style="backgr
83 |             ound: rgba(15, 23, 42, 0.7); min-height: 70px;">
84 |                 <div>
85 |                     <div class="text-muted small text-uppercase" sty
86 |                     le="font-size: 0.68rem; letter-spacing: 0.04em;">Записей в выб
```

```

опке</div>
78         <div class="fw-bold fs-4 text-white mt-0.5" id="
th-kpi-records-count">0</div>
79     </div>
80     <div class="fs-2 text-info opacity-75"><i class="b
i bi-database-fill-check"></i></div>
81 </div>
82 </div>
83
84 <!-- Anomalies Count -->
85 <div class="col-6 col-md-3">
86     <div class="p-2.5 rounded border border-secondary d-
flex align-items-center justify-content-between" style="backgr
ound: rgba(15, 23, 42, 0.7); min-height: 70px;">
87         <div>
88             <div class="text-muted small text-uppercase" sty
le="font-size: 0.68rem; letter-spacing: 0.04em;">Аномалий / Сб
оев</div>
89             <div class="fw-bold fs-4 text-warning mt-0.5" id
="th-kpi-anomalies-count">0</div>
90         </div>
91         <div class="fs-2 text-warning opacity-75"><i class
="bi bi-exclamation-triangle-fill"></i></div>
92     </div>
93 </div>
94
95 <!-- Time Window -->
96 <div class="col-6 col-md-3">
97     <div class="p-2.5 rounded border border-secondary d-
flex align-items-center justify-content-between" style="backgr
ound: rgba(15, 23, 42, 0.7); min-height: 70px;">
98         <div class="text-truncate">
99             <div class="text-muted small text-uppercase" sty
le="font-size: 0.68rem; letter-spacing: 0.04em;">Временной диа
пазон</div>
100             <div class="fw-bold text-info small mt-1 text-tr
uncate" id="th-kpi-time-window" title="Диапазон дат">--</div>
101         </div>

```

```
102 |         <div class="fs-2 text-secondary opacity-75"><i cla
    | ss="bi bi-clock-history"></i></div>
103 |     </div>
104 | </div>
105 | </div>
106 |
107 | </div>
108 | </div>
109 |
110 | <!-- Main Tabs Navigation -->
111 | <ul class="nav nav-tabs border-secondary mb-3" id="telemetry
    | HistoryNavTabs" role="tablist">
112 |     <li class="nav-item" role="presentation">
113 |         <button class="nav-link active d-flex align-items-center
    | gap-1.5 py-1.5 px-3" id="th-tab-charts-btn" data-bs-toggle="t
    | ab" data-bs-target="#th-subtab-charts" type="button" role="tab
    | ">
114 |             <i class="bi bi-graph-up text-primary"></i>
115 |             <span>Интерактивные графики</span>
116 |         </button>
117 |     </li>
118 |     <li class="nav-item" role="presentation">
119 |         <button class="nav-link d-flex align-items-center gap-1.
    | 5 py-1.5 px-3" id="th-tab-records-btn" data-bs-toggle="tab" da
    | ta-bs-target="#th-subtab-records" type="button" role="tab">
120 |             <i class="bi bi-table text-info"></i>
121 |             <span>Таблица записей логов</span>
122 |             <span class="badge bg-secondary ms-1" id="th-badge-tab
    | le-records">0</span>
123 |         </button>
124 |     </li>
125 |     <li class="nav-item" role="presentation">
126 |         <button class="nav-link d-flex align-items-center gap-1.
    | 5 py-1.5 px-3" id="th-tab-anomalies-btn" data-bs-toggle="tab"
    | data-bs-target="#th-subtab-anomalies" type="button" role="tab"
    | >
127 |             <i class="bi bi-shield-exclamation text-warning"></i>
128 |             <span>Журнал аномалий</span>
```

```
129 |         <span class="badge bg-warning text-dark ms-1" id="th-b
| adge-table-anomalies">0</span>
130 |     </button>
131 | </li>
132 |     <li class="nav-item" role="presentation">
133 |         <button class="nav-link d-flex align-items-center gap-1.
| 5 py-1.5 px-3" id="th-tab-files-btn" data-bs-toggle="tab" data
| -bs-target="#th-subtab-files" type="button" role="tab">
134 |             <i class="bi bi-files text-success"></i>
135 |             <span>Файлы источников логов</span>
136 |             <span class="badge bg-secondary ms-1" id="th-badge-tab
| le-files">0</span>
137 |         </button>
138 |     </li>
139 | </ul>
140 |
141 | <!-- Tab Content -->
142 | <div class="tab-content" id="telemetryHistoryTabContent">
143 |
144 |     <!-- Subtab 1: Interactive Charts Grid -->
145 |     <div class="tab-pane fade show active" id="th-subtab-chart
| s" role="tabpanel">
146 |         <!-- AI Conclusions Summary Banner -->
147 |         <div class="card shadow-sm border-secondary mb-3 bg-dark
| " id="th-conclusions-card" style="display: none;">
148 |             <div class="card-body p-3">
149 |                 <h6 class="text-success fw-bold d-flex align-items-c
| enter gap-1.5 mb-2">
150 |                     <i class="bi bi-lightbulb-fill"></i>
151 |                     <span>Выводы и рекомендации аналитического движка<
| /span>
152 |                 </h6>
153 |                 <ul class="mb-0 text-white small ps-3" id="th-conclu
| sions-list"></ul>
154 |             </div>
155 |         </div>
156 |
157 |     <!-- Charts Grid -->
```



```

158 |         <div class="row g-3" id="th-charts-container">
159 |             <div class="col-12 text-center py-5 text-muted">
160 |                 <div class="spinner-border spinner-border-sm text-in
fo me-2" role="status"></div>
161 |                 Загрузка и построение графиков телеметрии...
162 |             </div>
163 |         </div>
164 |     </div>
165 |
166 |     <!-- Subtab 2: Logs Records Table -->
167 |     <div class="tab-pane fade" id="th-subtab-records" role="ta
bpanel">
168 |         <div class="card shadow-sm border-secondary-subtle">
169 |             <div class="card-header bg-dark d-flex flex-wrap justi
fy-content-between align-items-center gap-2 py-2 px-3 border-s
econdary">
170 |                 <div class="d-flex align-items-center gap-2">
171 |                     <span class="fw-semibold text-white small">Все заф
иксированные метрики и события</span>
172 |                 </div>
173 |
174 |                 <div class="d-flex align-items-center gap-2 flex-wra
p">
175 |                     <div class="input-group input-group-sm" style="max
-width: 280px;">
176 |                         <span class="input-group-text bg-dark text-muted
border-secondary"><i class="bi bi-search"></i></span>
177 |                         <input type="text" id="th-input-filter-records"
class="form-control bg-dark text-white border-secondary" place
holder="Поиск по метрикам...">
178 |                     </div>
179 |
180 |                     <select class="form-select form-select-sm bg-dark
text-white border-secondary" id="th-select-page-size" style="w
idth: 100px;">
181 |                         <option value="25">25 / стр</option>
182 |                         <option value="50" selected>50 / стр</option>
183 |                         <option value="100">100 / стр</option>

```

```

184         </select>
185
186         <button id="th-btn-reload-records" class="btn btn-
sm btn-outline-secondary" title="Обновить таблицу">
187             <i class="bi bi-arrow-clockwise"></i>
188         </button>
189         <button id="th-btn-log-windows-events" class="btn
btn-sm btn-outline-primary d-flex align-items-center gap-1 sha
dow-sm" title="Залогировать важные события Windows в базу">
190             <i class="bi bi-journal-text"></i>
191             <span>Логи Windows</span>
192         </button>
193     </div>
194 </div>
195
196     <div class="table-responsive" style="max-height: 540px
;">
197         <table class="table table-dark table-hover table-bor
dered align-middle mb-0 small" id="th-table-records">
198             <thead class="table-secondary text-nowrap sticky-t
op">
199                 <tr>
200                     <th style="width: 170px;">Временная метка</th>
201                     <th style="min-width: 160px;">Метрика / Катего
рия</th>
202                     <th style="width: 120px;" class="text-end">Зна
чение</th>
203                     <th style="width: 90px;" class="text-center">Е
д. изм.</th>
204                     <th style="min-width: 200px;">Детали / Атрибут
ы</th>
205                 </tr>
206             </thead>
207             <tbody id="th-tbody-records">
208                 <tr>
209                     <td colspan="5" class="text-center text-muted
py-4">
210                         <div class="spinner-border spinner-border-sm

```

```

    text-info me-2" role="status"></div>
211         Загрузка данных телеметрии...
212     </td>
213 </tr>
214 </tbody>
215 </table>
216 </div>
217
218 <div class="card-footer bg-dark border-secondary p-2 d
    -flex justify-content-between align-items-center flex-wrap gap
    -2">
219     <span class="small text-muted" id="th-records-pagina
    tion-info">Показано 0 из 0</span>
220     <div class="d-flex gap-1">
221         <button class="btn btn-sm btn-outline-secondary" i
    d="th-btn-prev-page" disabled>&laquo; Назад</button>
222         <span class="btn btn-sm btn-dark disabled border-0
    " id="th-page-indicator">1 / 1</span>
223         <button class="btn btn-sm btn-outline-secondary" i
    d="th-btn-next-page" disabled>Вперед &raquo;</button>
224     </div>
225 </div>
226 </div>
227 </div>
228
229 <!-- Subtab 3: Anomalies Journal -->
230 <div class="tab-pane fade" id="th-subtab-anomalies" role="
    tabpanel">
231     <div class="card shadow-sm border-secondary-subtle">
232         <div class="card-header bg-dark d-flex justify-content
    -between align-items-center py-2 px-3 border-secondary">
233             <span class="fw-semibold text-white small">Обнаружен
    ные отклонения, перегревы и троттлинг</span>
234             <span class="badge bg-secondary" id="th-anomalies-he
    ader-count">0</span>
235         </div>
236
237         <div class="table-responsive" style="max-height: 540px

```

```

; ">
238         <table class="table table-dark table-hover table-bordered align-middle mb-0 small" id="th-table-anomalies">
239             <thead class="table-secondary text-nowrap sticky-top">
240                 <tr>
241                     <th style="width: 110px;" class="text-center">
Уровень</th>
242                     <th style="width: 170px;">Время</th>
243                     <th style="min-width: 160px;">Метрика</th>
244                     <th style="width: 160px;">Значение / Порог</th>
>
245                     <th style="min-width: 250px;">Описание аномалии</th>
246                 </tr>
247             </thead>
248             <tbody id="th-tbody-anomalies">
249                 <tr>
250                     <td colspan="5" class="text-center text-muted py-4">Аномалий не зафиксировано</td>
251                 </tr>
252             </tbody>
253         </table>
254     </div>
255 </div>
256 </div>
257
258 <!-- Subtab 4: Log Files Storage -->
259 <div class="tab-pane fade" id="th-subtab-files" role="tabpanel">
260     <div class="card shadow-sm border-secondary-subtle">
261         <div class="card-header bg-dark d-flex justify-content-between align-items-center py-2 px-3 border-secondary">
262             <span class="fw-semibold text-white small">Доступные файлы логов телеметрии</span>
263             <button id="th-btn-refresh-files" class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-1">
264                 <i class="bi bi-arrow-clockwise"></i>

```

```

265         <span>Обновить</span>
266     </button>
267 </div>
268
269     <div class="table-responsive" style="max-height: 540px
    ;">
270         <table class="table table-dark table-hover table-bor
    dered align-middle mb-0 small" id="th-table-files">
271             <thead class="table-secondary text-nowrap sticky-t
    op">
272                 <tr>
273                     <th style="min-width: 220px;">Имя файла</th>
274                     <th style="min-width: 250px;">Полный путь</th>
275                     <th style="width: 100px;" class="text-end">Раз
    мер</th>
276                     <th style="width: 170px;">Дата изменения</th>
277                     <th style="width: 130px;" class="text-center">
    Действие</th>
278                 </tr>
279             </thead>
280             <tbody id="th-tbody-files">
281                 <tr>
282                     <td colspan="5" class="text-center text-muted
    py-4">
283                         <div class="spinner-border spinner-border-sm
    text-info me-2" role="status"></div>
284                         Поиск файлов логов...
285                     </td>
286                 </tr>
287             </tbody>
288         </table>
289     </div>
290 </div>
291 </div>
292
293 </div>
294
295 </div>

```


main.js

```
1 /**
2  * =====
3  * Process Name: Windows Telemetry History & Visualization Log
   ic
4  * =====
5  * Description:
6  *   Клиентский контроллер вкладки визуализации истории телеме
   трии:
7  *   построение графиков Chart.js, табличный рендеринг логов,
   пагинация,
8  *   детекция аномалий и интеграция с REST API.
9  *
10 * File: main.js
11 * Project: AI-Breadboard
12 * Module: WebInterface.TelemetryHistoryTab
13 * Author: hypo69
14 * Copyright: © 2026 hypo69
15 * =====
   =====
16 */
17
18 (function () {
19   'use strict';
20
21   let chartInstances = {};
22   let currentReport = null;
23   let autoRefreshTimer = null;
24   let currentPage = 1;
25   let pageSize = 50;
26   let currentSearchQuery = '';
27
28   /**
29    * Инициализация вкладки
30    */
```

```
31 | async function init() {
32 |     bindEvents();
33 |     await loadAvailableFiles();
34 |     await fetchReportAndRender();
35 |     await loadRecordsTable();
36 | }
37 |
38 | /**
39 |  * Привязка обработчиков событий
40 |  */
41 | function bindEvents() {
42 |     const btnRefresh = document.getElementById('th-btn-refresh
43 | ');
44 |     if (btnRefresh) {
45 |         btnRefresh.addEventListener('click', async () => {
46 |             btnRefresh.classList.add('disabled');
47 |             await Promise.all([fetchReportAndRender(), loadRecords
48 | Table(), loadAvailableFiles()]);
49 |             btnRefresh.classList.remove('disabled');
50 |         });
51 |     }
52 |     const selectSource = document.getElementById('th-select-so
53 | urce');
54 |     if (selectSource) {
55 |         selectSource.addEventListener('change', async () => {
56 |             currentPage = 1;
57 |             await Promise.all([fetchReportAndRender(), loadRecords
58 | Table()]);
59 |         });
60 |     }
61 |     const switchAuto = document.getElementById('th-auto-refres
62 | h');
63 |     if (switchAuto) {
64 |         switchAuto.addEventListener('change', (e) => {
65 |             if (e.target.checked) {
66 |                 if (autoRefreshTimer) clearInterval(autoRefreshTimer
```



```

    );
64     autoRefreshTimer = setInterval(async () => {
65         await fetchReportAndRender(false);
66     }, 5000);
67 } else {
68     if (autoRefreshTimer) {
69         clearInterval(autoRefreshTimer);
70         autoRefreshTimer = null;
71     }
72 }
73 });
74 }
75
76 const btnExportHtml = document.getElementById('th-btn-expo
rt-html');
77 if (btnExportHtml) {
78     btnExportHtml.addEventListener('click', () => {
79         const source = document.getElementById('th-select-sour
ce')?.value || '';
80         const url = `/api/windows/telemetry/research/dashboard
${source ? `?source_path=${encodeURIComponent(source)}` : ''}`
;
81         window.open(url, '_blank');
82     });
83 }
84
85 const btnExportJson = document.getElementById('th-btn-expo
rt-json');
86 if (btnExportJson) {
87     btnExportJson.addEventListener('click', () => {
88         if (!currentReport) return;
89         const dataStr = 'data:text/json;charset=utf-8,' + enco
encodeURIComponent(JSON.stringify(currentReport, null, 2));
90         const a = document.createElement('a');
91         a.setAttribute('href', dataStr);
92         a.setAttribute('download', `telemetry_report_${current
Report.report_id} || Date.now()}.json`);
93         document.body.appendChild(a);

```

```
94 |         a.click();
95 |         a.remove();
96 |     });
97 | }
98 |
99 |     const inputFilter = document.getElementById('th-input-filt
100 | er-records');
101 |     if (inputFilter) {
102 |         let debounceTimer = null;
103 |         inputFilter.addEventListener('input', (e) => {
104 |             clearTimeout(debounceTimer);
105 |             debounceTimer = setTimeout(() => {
106 |                 currentSearchQuery = e.target.value.trim();
107 |                 currentPage = 1;
108 |                 loadRecordsTable();
109 |             }, 300);
110 |         });
111 |     }
112 |     const selectPageSize = document.getElementById('th-select-
113 | page-size');
114 |     if (selectPageSize) {
115 |         selectPageSize.addEventListener('change', (e) => {
116 |             pageSize = parseInt(e.target.value, 10) || 50;
117 |             currentPage = 1;
118 |             loadRecordsTable();
119 |         });
120 |     }
121 |     const btnPrev = document.getElementById('th-btn-prev-page'
122 | );
123 |     if (btnPrev) {
124 |         btnPrev.addEventListener('click', () => {
125 |             if (currentPage > 1) {
126 |                 currentPage--;
127 |                 loadRecordsTable();
128 |             }
129 |         });
130 |     }
```

```
129 |     }
130 |
131 |     const btnNext = document.getElementById('th-btn-next-page'
    | );
132 |     if (btnNext) {
133 |         btnNext.addEventListener('click', () => {
134 |             currentPage++;
135 |             loadRecordsTable();
136 |         });
137 |     }
138 |
139 |     const btnReloadRecords = document.getElementById('th-btn-r
    | eload-records');
140 |     if (btnReloadRecords) {
141 |         btnReloadRecords.addEventListener('click', loadRecordsTa
    | ble);
142 |     }
143 |
144 |     const btnRefreshFiles = document.getElementById('th-btn-re
    | fresh-files');
145 |     if (btnRefreshFiles) {
146 |         btnRefreshFiles.addEventListener('click', loadAvailableF
    | iles);
147 |     }
148 | }
149 |
150 | /**
151 |  * Загрузка списка доступных файлов логов
152 |  */
153 | async function loadAvailableFiles() {
154 |     try {
155 |         const resp = await fetch('/api/windows/telemetry/researc
    | h/files');
156 |         if (!resp.ok) return;
157 |         const files = await resp.json();
158 |
159 |         const selectSource = document.getElementById('th-select-
    | source');
```

```

160     const badgeFiles = document.getElementById('th-badge-tab
le-files');
161     if (badgeFiles) badgeFiles.textContent = files.length;
162
163     if (selectSource) {
164         const curVal = selectSource.value;
165         selectSource.innerHTML = '<option value="">Все обнаруж
енные логи</option>';
166         files.forEach((f) => {
167             const opt = document.createElement('option');
168             opt.value = f.path;
169             opt.textContent = `${f.name} (${formatBytes(f.size_b
ytes)})`;
170             if (f.path === curVal) opt.selected = true;
171             selectSource.appendChild(opt);
172         });
173     }
174
175     const tbody = document.getElementById('th-tbody-files');
176     if (tbody) {
177         if (files.length === 0) {
178             tbody.innerHTML = '<tr><td colspan="5" class="text-c
enter text-muted py-4">Файлы логов не обнаружены</td></tr>';
179             return;
180         }
181         tbody.innerHTML = files
182             .map((f) => {
183                 const dateStr = new Date(f.modified * 1000).toLoca
leString('ru-RU');
184                 return `
185                 <tr>
186                     <td><span class="fw-semibold text-info">${f.name
}</span></td>
187                     <td><code class="small text-muted">${f.path}</co
de></td>
188                     <td class="text-end font-monospace">${formatByte
s(f.size_bytes)}</td>
189                     <td>${dateStr}</td>

```

```

190         <td class="text-center">
191             <button class="btn btn-xs btn-outline-primary
py-0 px-2 th-btn-select-file" data-path="{f.path}">
192                 Анализ
193             </button>
194         </td>
195     </tr>
196     `;
197 })
198 .join('');
199
200     tbody.querySelectorAll('.th-btn-select-file').forEach(
    (btn) => {
201         btn.addEventListener('click', (e) => {
202             const p = e.target.getAttribute('data-path');
203             if (selectSource) {
204                 selectSource.value = p;
205                 const evt = new Event('change');
206                 selectSource.dispatchEvent(evt);
207             }
208         });
209     });
210 }
211 } catch (err) {
212     console.warn('Ошибка загрузки списка файлов логов:', err
    );
213 }
214 }
215
216 /**
217  * Получение отчета исследования и рендеринг графиков
218  */
219     async function fetchReportAndRender(showLoading = true) {
220         const container = document.getElementById('th-charts-conta
    inner');
221         if (showLoading && container && Object.keys(chartInstances
    ).length === 0) {
222             container.innerHTML = `

```

```

223 |         <div class="col-12 text-center py-5 text-muted">
224 |             <div class="spinner-border spinner-border-sm text-in
fo me-2" role="status"></div>
225 |             Выполняется анализ телеметрии и построение графиков.
226 |             ..
227 |         </div>
228 |     `;
229 | }
230 |
231 | try {
232 |     const source = document.getElementById('th-select-source
233 |     ')?<div class="col-12 text-center py-4 text-warning">
234 |         <i class="bi bi-exclamation-triangle fs-4 d-bloc
235 |         k mb-1"></i>
236 |         Не удалось загрузить данные исследования (${resp
237 |         .status})
238 |     </div>
239 |     `;
240 | }
241 | return;
242 | }
243 |
244 | currentReport = await resp.json();
245 | renderKPIs(currentReport);
246 | renderConclusions(currentReport);
247 | renderAnomalies(currentReport);
248 | renderCharts(currentReport.charts || []);

```

```
255     } catch (err) {
256         console.error('Ошибка анализа телеметрии:', err);
257         if (container) {
258             container.innerHTML = `
259                 <div class="col-12 text-center py-4 text-danger">
260                     <i class="bi bi-bug fs-4 d-block mb-1"></i>
261                     Ошибка выполнения анализа: ${err.message}
262                 </div>
263             `;
264         }
265     }
266 }
267
268 /**
269  * Рендеринг KPI карточек
270  */
271 function renderKPIs(report) {
272     const elHealth = document.getElementById('th-kpi-health-score');
273     const elHealthIcon = document.getElementById('th-kpi-health-icon');
274     if (elHealth) {
275         const score = Math.round(report.health_score || 100);
276         elHealth.textContent = `${score}%`;
277         elHealth.className = score >= 85 ? 'fw-bold fs-4 mt-0.5 text-success' : score >= 60 ? 'fw-bold fs-4 mt-0.5 text-warning' : 'fw-bold fs-4 mt-0.5 text-danger';
278         if (elHealthIcon) {
279             elHealthIcon.className = score >= 85 ? 'fs-2 text-success opacity-75' : score >= 60 ? 'fs-2 text-warning opacity-75' : 'fs-2 text-danger opacity-75';
280         }
281     }
282
283     const elRecords = document.getElementById('th-kpi-records-count');
284     if (elRecords) elRecords.textContent = report.records_analyzed.toLocaleString('ru-RU');
```

```
285 |
286 |     const elAnomalies = document.getElementById('th-kpi-anomal
    ies-count');
287 |     if (elAnomalies) {
288 |         const anomCount = (report.anomalies || []).length;
289 |         elAnomalies.textContent = anomCount;
290 |         elAnomalies.className = anomCount > 0 ? 'fw-bold fs-4 te
    xt-warning mt-0.5' : 'fw-bold fs-4 text-success mt-0.5';
291 |     }
292 |
293 |     const elTimeWindow = document.getElementById('th-kpi-time-
    window');
294 |     if (elTimeWindow) {
295 |         const s = report.time_window_start ? report.time_window_
    start.split('T')[0] : '';
296 |         const e = report.time_window_end ? report.time_window_en
    d.split('T')[0] : '';
297 |         elTimeWindow.textContent = s && e ? `${s} – ${e}` : 'Акт
    уальный срез';
298 |     }
299 | }
300 |
301 | /**
302 |  * Рендеринг выводов аналитического движка
303 |  */
304 | function renderConclusions(report) {
305 |     const card = document.getElementById('th-conclusions-card'
    );
306 |     const list = document.getElementById('th-conclusions-list'
    );
307 |     if (!card || !list) return;
308 |
309 |     if (report.summary_conclusions && report.summary_conclusio
    ns.length > 0) {
310 |         list.innerHTML = report.summary_conclusions.map((c) => `
    <li>${c}</li>`).join('');
311 |         card.style.display = 'block';
312 |     } else {
```



```

313 |     card.style.display = 'none';
314 | }
315 | }
316 |
317 | /**
318 |  * Рендеринг таблицы аномалий
319 |  */
320 | function renderAnomalies(report) {
321 |     const tbody = document.getElementById('th-tbody-anomalies'
322 | );
323 |     const badge = document.getElementById('th-badge-table-anom
324 | alies');
325 |     const headerCount = document.getElementById('th-anomalies-
326 | header-count');
327 |     const anomalies = report.anomalies || [];
328 |
329 |     if (badge) badge.textContent = anomalies.length;
330 |     if (headerCount) headerCount.textContent = anomalies.lengt
331 | h;
332 |
333 |     if (!tbody) return;
334 |     if (anomalies.length === 0) {
335 |         tbody.innerHTML = '<tr><td colspan="5" class="text-cente
336 | r text-muted py-4">Аномалий и критических отклонений не зафикс
337 | ировано</td></tr>';
338 |         return;
339 |     }
340 |
341 |     tbody.innerHTML = anomalies
342 |         .map((a) => {
343 |             const isCrit = a.severity === 'critical';
344 |             const badgeCls = isCrit ? 'bg-danger text-white' : 'bg
345 | -warning text-dark';
346 |             return `
347 |                 <tr>
348 |                     <td class="text-center"><span class="badge ${badgeCl
349 | s}>${a.severity.toUpperCase()}</span></td>
350 |                     <td class="text-nowrap small font-monospace">${a.tim

```

```

    estamp || '--'}</td>
343     <td><strong class="text-info">${a.metric}</strong></
    td>
344     <td><span class="fw-bold">${a.value}</span> <span cl
    ass="text-muted small">(попор: ${a.threshold})</span></td>
345     <td>${a.description}</td>
346 </tr>
347 `;
348 })
349 .join('');
350 }
351
352 /**
353  * Построение графиков с помощью Chart.js
354  */
355 function renderCharts(charts) {
356     const container = document.getElementById('th-charts-conta
    inner');
357     if (!container) return;
358
359     // Уничтожаем старые инстансы Chart.js
360     Object.values(chartInstances).forEach((inst) => {
361         try {
362             inst.destroy();
363         } catch (e) {}
364     });
365     chartInstances = {};
366
367     if (!charts || charts.length === 0) {
368         container.innerHTML = `
369         <div class="col-12 text-center py-5 text-muted">
370             <i class="bi bi-info-circle fs-4 d-block mb-1"></i>
371             Нет достаточного количества данных для построения гр
    афиков
372         </div>
373     `;
374     return;
375 }

```

```

376
377 // Рендерим сетку карточек под графики
378 container.innerHTML = charts
379   .map((c) => {
380     const colClass = c.chart_type === 'doughnut' ? 'col-12
col-md-6' : 'col-12 col-xl-6';
381     return `
382       <div class="${colClass}">
383         <div class="card shadow-sm border-secondary h-100" s
style="background: #0f172a; border-radius: 8px;">
384           <div class="card-header bg-dark border-secondary p
y-2 px-3 d-flex justify-content-between align-items-center">
385             <div>
386               <h6 class="mb-0 fw-bold text-white small">${c.
title}</h6>
387               <small class="text-muted" style="font-size: 0.
7rem;">${c.description || ''}</small>
388             </div>
389             <span class="badge bg-secondary small">${c.chart
_type.toUpperCase()}</span>
390           </div>
391           <div class="card-body p-2" style="position: relati
ve; min-height: 250px; max-height: 320px;">
392             <canvas id="${c.id}"></canvas>
393           </div>
394         </div>
395       </div>
396     `;
397   })
398   .join('');
399
400 // Инициализируем Chart.js для каждого canvas
401 if (typeof Chart === 'undefined') {
402   loadChartJsScript(() => initChartObjects(charts));
403 } else {
404   initChartObjects(charts);
405 }
406 }

```



```

445 |         mode: isDoughnut ? 'nearest' : 'index',
446 |         intersect: false,
447 |     },
448 | },
449 |     scales: isDoughnut
450 |         ? {}
451 |         : {
452 |             x: {
453 |                 ticks: { color: '#94a3b8', font: { size: 10
| }, maxRotation: 45, maxTicksLimit: 12 },
454 |                 grid: { color: '#1e293b' },
455 |             },
456 |             y: {
457 |                 ticks: { color: '#94a3b8', font: { size: 10
| } },
458 |                 grid: { color: '#1e293b' },
459 |                 title: {
460 |                     display: !!cfg.y_axis_label,
461 |                     text: cfg.y_axis_label || '',
462 |                     color: '#94a3b8',
463 |                     font: { size: 11 },
464 |                 },
465 |             },
466 |         },
467 |     },
468 | });
469 |
470 |     chartInstances[cfg.id] = chartObj;
471 | });
472 | }
473 |
474 | /**
475 |  * Загрузка записей логов в таблицу
476 |  */
477 | async function loadRecordsTable() {
478 |     const tbody = document.getElementById('th-tbody-records');
479 |     const badge = document.getElementById('th-badge-table-reco
| rds');

```

```
480     const pageInfo = document.getElementById('th-records-pagin
ation-info');
481     const pageIndicator = document.getElementById('th-page-ind
icator');
482     const btnPrev = document.getElementById('th-btn-prev-page'
);
483     const btnNext = document.getElementById('th-btn-next-page'
);
484
485     const source = document.getElementById('th-select-source')
?.value || '';
486     const queryParams = new URLSearchParams({
487         page: currentPage.toString(),
488         page_size: pageSize.toString(),
489     });
490     if (source) queryParams.set('source_path', source);
491     if (currentSearchQuery) queryParams.set('query', currentSe
archQuery);
492
493     try {
494         const resp = await fetch(`/api/windows/telemetry/researc
h/records?${queryParams.toString()}`);
495         if (!resp.ok) return;
496         const data = await resp.json();
497
498         if (badge) badge.textContent = data.total.toLocaleString
('ru-RU');
499         if (pageInfo) {
500             const start = (data.page - 1) * data.page_size + 1;
501             const end = Math.min(data.page * data.page_size, data.
total);
502             pageInfo.textContent = `Показано ${data.total > 0 ? st
art : 0}-${end} из ${data.total}`;
503         }
504         if (pageIndicator) pageIndicator.textContent = `${data.p
age} / ${data.total_pages}`;
505         if (btnPrev) btnPrev.disabled = data.page <= 1;
506         if (btnNext) btnNext.disabled = data.page >= data.total_
```

```

pages;
507
508     if (!tbody) return;
509     if (data.items.length === 0) {
510         tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted py-4">Нет записей по заданному запросу</td></tr>';
511         return;
512     }
513
514     tbody.innerHTML = data.items
515         .map((r) => {
516             const ts = r.timestamp || r.time || r.datetime || '-';
517             const metric = r.metric_name || r.sensor_name || r.name || r.category || 'telemetry';
518             const val = r.value !== undefined ? r.value : r.val !== undefined ? r.val : '--';
519             const unit = r.unit || r.unit_symbol || '';
520
521             // Дополнительные детали
522             const details = Object.entries(r)
523                 .filter(([k]) => ![ 'timestamp', 'time', 'datetime', 'metric_name', 'sensor_name', 'name', 'value', 'val', 'unit', 'unit_symbol' ].includes(k))
524                 .map(([k, v]) => `${k}=${typeof v === 'object' ? JSON.stringify(v) : v}`)
525                 .join(', ');
526
527             return `
528             <tr>
529                 <td class="text-nowrap small font-monospace text-muted">${ts}</td>
530                 <td><strong class="text-info">${metric}</strong></td>
531                 <td class="text-end font-monospace fw-bold text-white">${val}</td>
532                 <td class="text-center small text-secondary">${uni

```

```

t}</td>
533         <td class="small text-truncate text-muted" style="
max-width: 320px;" title="{details}">{details || '-'}</td>
534     </tr>
535     `;
536     })
537     .join('');
538     } catch (err) {
539         console.error('Ошибка загрузки таблицы записей:', err);
540         if (tbody) tbody.innerHTML = '<tr><td colspan="5" class=
"text-center text-danger py-4">Ошибка загрузки записей</td></t
r>';
541     }
542 }
543
544 function formatBytes(bytes) {
545     if (bytes === 0 || !bytes) return '0 B';
546     const k = 1024;
547     const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
548     const i = Math.floor(Math.log(bytes) / Math.log(k));
549     return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
' + sizes[i];
550 }
551
552 // Запуск при загрузке
553 if (document.readyState === 'loading') {
554     document.addEventListener('DOMContentLoaded', init);
555 } else {
556     init();
557 }
558 })();
559

```


index.html

```
1 <!--
2 =====
3
4 Process Name: Telemetry Research Web GUI Tab
5 =====
6 Description:
7     Пользовательский интерфейс приложения глубокого исследован
8     ия телеметрии:
9     проверка системных гипотез, матрица корреляций, графики вр
10    еменных рядов,
11    детекция аномалий и аппаратных сбоев, генерация экспертных
12    заключений.
13
14 File: index.html
15 Project: AI-Breadboard
16 Module: WebInterface.TelemetryResearchTab
17 Author: hypo69
18 Copyright: © 2026 hypo69
19 =====
20 =====
21 -->
22 <div class="container-fluid p-2" id="telemetry-research-tab-co
23 ntainer">
24
25     <!-- Header & Control Toolbar -->
26     <div class="card shadow-sm border-secondary-subtle mb-3" sty
27 le="background: linear-gradient(135deg, #0b1329 0%, #1e1b4b 10
28 0%); border-radius: 10px;">
29         <div class="card-body p-3">
30             <div class="d-flex flex-wrap align-items-center justify-
31 content-between gap-2 mb-3">
32                 <div class="d-flex align-items-center gap-2">
33                     <div class="fs-2">❏ </div>
34                     <div>
35                         <h5 class="mb-0 fw-bold text-white">Исследования т
```

елеметрии и оборудования</h5>

```
27         <small class="text-light text-opacity-75">Глубокий
    анализ корреляций, проверка гипотез производительности, анома
    лии и Health Score</small>
28     </div>
29 </div>
30
31     <div class="d-flex align-items-center gap-2 flex-wrap"
    >
32         <div class="input-group input-group-sm" style="width
    : auto;">
33             <span class="input-group-text bg-dark text-muted b
    order-secondary"><i class="bi bi-folder2-open"></i></span>
34             <select class="form-select form-select-sm bg-dark
    text-white border-secondary" id="tr-select-source" style="max-
    width: 260px;">
35                 <option value="">Все доступные логи телеметрии</
    option>
36             </select>
37         </div>
38
39         <button id="tr-btn-run" class="btn btn-sm btn-primar
    y d-flex align-items-center gap-1 shadow-sm" title="Запустить
    полное исследование">
40             <i class="bi bi-play-circle-fill"></i>
41             <span>Исследовать</span>
42         </button>
43
44         <button id="tr-btn-dashboard" class="btn btn-sm btn-
    outline-success d-flex align-items-center gap-1 shadow-sm" tit
    le="Открыть автономный HTML-дашборд в новой вкладке">
45             <i class="bi bi-file-earmark-bar-graph"></i>
46             <span>HTML-дашборд</span>
47         </button>
48
49         <button id="tr-btn-json" class="btn btn-sm btn-outli
    ne-secondary d-flex align-items-center gap-1 shadow-sm" title=
    "Скачать JSON отчет">
```

```

50         <i class="bi bi-download"></i>
51         <span>JSON</span>
52     </button>
53 </div>
54 </div>
55
56 <!-- KPI Metrics Cards -->
57 <div class="row g-2 text-white">
58     <!-- Health Score -->
59     <div class="col-6 col-md-3">
60         <div class="p-2 rounded bg-dark bg-opacity-50 border
border-secondary d-flex align-items-center gap-3">
61             <div class="fs-3 text-info"><i class="bi bi-heart-
pulse-fill"></i></div>
62             <div class="flex-grow-1">
63                 <div class="text-muted small">Индекс здоровья</d
iv>
64                 <div class="d-flex align-items-baseline gap-2">
65                     <span class="h5 mb-0 fw-bold" id="tr-val-healt
h">--</span>
66                     <span class="small text-muted">/100</span>
67                 </div>
68                 <div class="progress mt-1" style="height: 4px;">
69                     <div id="tr-progress-health" class="progress-b
ar bg-success" style="width: 0%;"></div>
70                 </div>
71             </div>
72         </div>
73     </div>
74
75 <!-- Records Analyzed -->
76 <div class="col-6 col-md-3">
77     <div class="p-2 rounded bg-dark bg-opacity-50 border
border-secondary d-flex align-items-center gap-3">
78         <div class="fs-3 text-primary"><i class="bi bi-dat
abase-check"></i></div>
79         <div>
80             <div class="text-muted small">Записей исследован

```

```

o</div>
81         <div class="h5 mb-0 fw-bold" id="tr-val-records"
>0</div>
82         <div class="small text-muted" id="tr-val-timespa
n">Временной срез</div>
83     </div>
84 </div>
85 </div>
86
87 <!-- Hypotheses Confirmed -->
88 <div class="col-6 col-md-3">
89     <div class="p-2 rounded bg-dark bg-opacity-50 border
border-secondary d-flex align-items-center gap-3">
90         <div class="fs-3 text-warning"><i class="bi bi-lig
htbulb-fill"></i></div>
91         <div>
92             <div class="text-muted small">Проблемных гипотез
</div>
93             <div class="h5 mb-0 fw-bold" id="tr-val-hypothes
es">0</div>
94             <div class="small text-muted">Из проверенных</di
v>
95         </div>
96     </div>
97 </div>
98
99 <!-- Anomalies & Incidents -->
100 <div class="col-6 col-md-3">
101     <div class="p-2 rounded bg-dark bg-opacity-50 border
border-secondary d-flex align-items-center gap-3">
102         <div class="fs-3 text-danger"><i class="bi bi-excl
amation-triangle-fill"></i></div>
103         <div>
104             <div class="text-muted small">Аномалии и сбои</d
iv>
105             <div class="h5 mb-0 fw-bold" id="tr-val-anomalie
s">0</div>
106             <div class="small text-muted" id="tr-val-devices

```

```

    ">Устройства в норме</div>
107         </div>
108     </div>
109 </div>
110 </div>
111 </div>
112 </div>
113
114 <!-- Summary Alert Banner -->
115 <div id="tr-summary-banner" class="alert alert-info border-i
nfo d-flex align-items-center gap-2 mb-3 py-2 px-3 small d-non
e">
116     <i class="bi bi-info-circle-fill fs-5"></i>
117     <div id="tr-summary-text" class="flex-grow-1"></div>
118 </div>
119
120 <!-- Tabs Navigation -->
121 <ul class="nav nav-tabs border-secondary mb-3" id="trResearc
hTabs" role="tablist">
122     <li class="nav-item" role="presentation">
123         <button class="nav-link active fw-medium py-2 px-3" id="
tr-tab-hypotheses-btn" data-bs-toggle="tab" data-bs-target="#t
r-tab-hypotheses" type="button" role="tab">
124             □ Проверка гипотез
125         </button>
126     </li>
127     <li class="nav-item" role="presentation">
128         <button class="nav-link fw-medium py-2 px-3" id="tr-tab-
correlations-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-
correlations" type="button" role="tab">
129             □ Матрица корреляций
130         </button>
131     </li>
132     <li class="nav-item" role="presentation">
133         <button class="nav-link fw-medium py-2 px-3" id="tr-tab-
charts-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-chart
s" type="button" role="tab">
134             □ Графики метрик

```

```

135 |         </button>
136 |     </li>
137 |     <li class="nav-item" role="presentation">
138 |         <button class="nav-link fw-medium py-2 px-3" id="tr-tab-anomalies-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-anomalies" type="button" role="tab">
139 |             □ Аномалии и устройства
140 |         </button>
141 |     </li>
142 |     <li class="nav-item" role="presentation">
143 |         <button class="nav-link fw-medium py-2 px-3" id="tr-tab-records-btn" data-bs-toggle="tab" data-bs-target="#tr-tab-records" type="button" role="tab">
144 |             □ Записи телеметрии
145 |         </button>
146 |     </li>
147 | </ul>
148 |
149 | <!-- Tab Contents -->
150 | <div class="tab-content" id="trResearchTabContent">
151 |
152 |     <!-- 1. Hypotheses Tab -->
153 |     <div class="tab-pane fade show active" id="tr-tab-hypotheses" role="tabpanel">
154 |         <div class="row g-3" id="tr-hypotheses-list">
155 |             <div class="col-12 text-center text-muted py-4">Нажмите «Исследовать» для проверки системных гипотез...</div>
156 |         </div>
157 |     </div>
158 |
159 |     <!-- 2. Correlations Tab -->
160 |     <div class="tab-pane fade" id="tr-tab-correlations" role="tabpanel">
161 |         <div class="card bg-dark border-secondary">
162 |             <div class="card-header border-secondary d-flex align-items-center justify-content-between">
163 |                 <span class="fw-bold small text-light">Взаимосвязи между системными метриками (Пирсон)</span>

```

```

164 |         <span class="badge bg-secondary" id="tr-correlations
    | -count">0 на</span>
165 |     </div>
166 |     <div class="table-responsive">
167 |         <table class="table table-dark table-hover table-str
    | iped mb-0 align-middle small">
168 |             <thead>
169 |                 <tr class="text-muted border-secondary">
170 |                     <th>Метрика А</th>
171 |                     <th>Метрика Б</th>
172 |                     <th>Коэффициент (r)</th>
173 |                     <th>Выборка</th>
174 |                     <th>Интерпретация</th>
175 |                 </tr>
176 |             </thead>
177 |             <tbody id="tr-correlations-tbody">
178 |                 <tr><td colspan="5" class="text-center text-mute
    | d py-3">Нет данных корреляций</td></tr>
179 |             </tbody>
180 |         </table>
181 |     </div>
182 | </div>
183 | </div>
184 |
185 | <!-- 3. Charts Tab -->
186 | <div class="tab-pane fade" id="tr-tab-charts" role="tabpan
    | el">
187 |     <div class="row g-3" id="tr-charts-container">
188 |         <div class="col-12 text-center text-muted py-4">График
    | и появятся после запуска исследования...</div>
189 |     </div>
190 | </div>
191 |
192 | <!-- 4. Anomalies & Devices Tab -->
193 | <div class="tab-pane fade" id="tr-tab-anomalies" role="tab
    | panel">
194 |     <div class="row g-3">
195 |         <!-- Anomalies Table -->

```

```

196 |         <div class="col-12 col-lg-7">
197 |             <div class="card bg-dark border-secondary h-100">
198 |                 <div class="card-header border-secondary fw-bold s
mall text-light d-flex justify-content-between">
199 |                     <span>Зафиксированные аномалии</span>
200 |                     <span class="badge bg-danger" id="tr-anomalies-b
adge">0</span>
201 |                 </div>
202 |                 <div class="table-responsive">
203 |                     <table class="table table-dark table-hover table
-striped mb-0 align-middle small">
204 |                         <thead>
205 |                             <tr class="text-muted border-secondary">
206 |                                 <th>Время</th>
207 |                                 <th>Метрика</th>
208 |                                 <th>Уровень</th>
209 |                                 <th>Значение / Порог</th>
210 |                                 <th>Описание</th>
211 |                             </tr>
212 |                         </thead>
213 |                         <tbody id="tr-anomalies-tbody">
214 |                             <tr><td colspan="5" class="text-center text-
muted py-3">Аномалий не обнаружено</td></tr>
215 |                         </tbody>
216 |                     </table>
217 |                 </div>
218 |             </div>
219 |         </div>
220 |
221 |         <!-- Devices Summary -->
222 |         <div class="col-12 col-lg-5">
223 |             <div class="card bg-dark border-secondary h-100">
224 |                 <div class="card-header border-secondary fw-bold s
mall text-light d-flex justify-content-between">
225 |                     <span>Состояние оборудования</span>
226 |                     <span class="badge bg-info" id="tr-devices-badge
">OK</span>
227 |                 </div>

```



```

228         <div class="card-body p-3 small" id="tr-devices-co
| ntent">
229             <p class="text-muted">Данные об оборудовании заг
| ружаются...</p>
230         </div>
231     </div>
232 </div>
233 </div>
234 </div>
235
236 <!-- 5. Records Tab -->
237 <div class="tab-pane fade" id="tr-tab-records" role="tabpa
| nel">
238     <div class="card bg-dark border-secondary">
239         <div class="card-header border-secondary d-flex align-
| items-center justify-content-between gap-2 flex-wrap">
240             <div class="d-flex align-items-center gap-2">
241                 <input type="text" id="tr-records-search" class="f
| orm-control form-control-sm bg-dark text-white border-secondar
| y" placeholder="Поиск по записям..." style="width: 250px;">
242                 <button id="tr-records-search-btn" class="btn btn-
| sm btn-outline-secondary"><i class="bi bi-search"></i></button
| >
243             </div>
244             <div class="d-flex align-items-center gap-2">
245                 <span class="small text-muted" id="tr-records-pagi
| nation-info">Стр. 1 из 1</span>
246                 <div class="btn-group btn-group-sm">
247                     <button class="btn btn-outline-secondary" id="tr
| -records-prev-btn"><i class="bi bi-chevron-left"></i></button>
248                     <button class="btn btn-outline-secondary" id="tr
| -records-next-btn"><i class="bi bi-chevron-right"></i></button
| >
249                 </div>
250             </div>
251         </div>
252         <div class="table-responsive" style="max-height: 500px
| ;">

```

```

253 |         <table class="table table-dark table-hover table-str
      |iped mb-0 align-middle small font-monospace">
254 |             <thead>
255 |                 <tr class="text-muted border-secondary">
256 |                     <th>#</th>
257 |                     <th>Timestamp</th>
258 |                     <th>CPU %</th>
259 |                     <th>RAM %</th>
260 |                     <th>GPU %</th>
261 |                     <th>Диск (R/W)</th>
262 |                     <th>Событие / Детали</th>
263 |                 </tr>
264 |             </thead>
265 |             <tbody id="tr-records-tbody">
266 |                 <tr><td colspan="7" class="text-center text-mute
      |d py-3">Записи не загружены</td></tr>
267 |                 </tbody>
268 |             </table>
269 |         </div>
270 |     </div>
271 | </div>
272 |
273 | </div>
274 |
275 | </div>
276 |

```

main.js

```
1 /**
2  * =====
3  * Process Name: Telemetry Research Web GUI Controller
4  * =====
5  * Description:
6  *   Клиентский JavaScript контроллер для вкладки глубокого ис
7  *   следования телеметрии:
8  *   запуск исследовательских сценариев, проверка системных ги
9  *   потез, рендеринг
10  *   матрицы корреляций, графиков Chart.js, детекции аномалий
11  *   и записей.
12  *
13  * File: main.js
14  * Project: AI-Breadboard
15  * Module: WebInterface.TelemetryResearchTab
16  * Author: hypo69
17  * Copyright: © 2026 hypo69
18  * =====
19  */
20
21 let trCurrentReport = null;
22 let trChartsInstances = [];
23 let trCurrentPage = 1;
24 const trPageSize = 25;
25 let trTotalPages = 1;
26 let trSearchQuery = '';
27
28 /**
29  * Инициализация вкладки исследования телеметрии.
30  */
31 export async function initTelemetryResearchTab() {
32   bindEvents();
33   await loadSources();
34 }
```

```
31 |   await runResearch();
32 | }
33 |
34 | /**
35 |  * Привязка событий элементов управления.
36 |  */
37 | function bindEvents() {
38 |   const btnRun = document.getElementById('tr-btn-run');
39 |   if (btnRun) {
40 |     btnRun.onclick = () => runResearch();
41 |   }
42 |
43 |   const btnDashboard = document.getElementById('tr-btn-dashboa
  rd');
44 |   if (btnDashboard) {
45 |     btnDashboard.onclick = () => openHtmlDashboard();
46 |   }
47 |
48 |   const btnJson = document.getElementById('tr-btn-json');
49 |   if (btnJson) {
50 |     btnJson.onclick = () => downloadJsonReport();
51 |   }
52 |
53 |   const selectSource = document.getElementById('tr-select-sour
  ce');
54 |   if (selectSource) {
55 |     selectSource.onchange = () => runResearch();
56 |   }
57 |
58 |   const searchBtn = document.getElementById('tr-records-search
  -btn');
59 |   const searchInput = document.getElementById('tr-records-sear
  ch');
60 |   if (searchBtn && searchInput) {
61 |     searchBtn.onclick = () => {
62 |       trSearchQuery = searchInput.value.trim();
63 |       trCurrentPage = 1;
64 |       loadRecords();
```

```
65 |     };
66 |     searchInput.onkeydown = (e) => {
67 |         if (e.key === 'Enter') {
68 |             trSearchQuery = searchInput.value.trim();
69 |             trCurrentPage = 1;
70 |             loadRecords();
71 |         }
72 |     };
73 | }
74 |
75 | const prevBtn = document.getElementById('tr-records-prev-btn
    ');
76 | const nextBtn = document.getElementById('tr-records-next-btn
    ');
77 | if (prevBtn) {
78 |     prevBtn.onclick = () => {
79 |         if (trCurrentPage > 1) {
80 |             trCurrentPage--;
81 |             loadRecords();
82 |         }
83 |     };
84 | }
85 | if (nextBtn) {
86 |     nextBtn.onclick = () => {
87 |         if (trCurrentPage < trTotalPages) {
88 |             trCurrentPage++;
89 |             loadRecords();
90 |         }
91 |     };
92 | }
93 |
94 | const recordsTabBtn = document.getElementById('tr-tab-record
    s-btn');
95 | if (recordsTabBtn) {
96 |     recordsTabBtn.addEventListener('shown.bs.tab', () => {
97 |         loadRecords();
98 |     });
99 | }
```

```
100 | }
101 |
102 | /**
103 |  * Загрузка списка доступных источников телеметрии.
104 |  */
105 | async function loadSources() {
106 |     const select = document.getElementById('tr-select-source');
107 |     if (!select) return;
108 |
109 |     try {
110 |         const res = await fetch('/apps/telemetry_research/sources'
111 | );
112 |         if (!res.ok) return;
113 |         const files = await res.json();
114 |         select.innerHTML = '<option value="">Все доступные логи те
115 | леметрии</option>';
116 |         files.forEach((f) => {
117 |             const opt = document.createElement('option');
118 |             opt.value = f.path;
119 |             const sizeKb = (f.size_bytes / 1024).toFixed(1);
120 |             opt.textContent = `${f.name} (${sizeKb} KB)`;
121 |             select.appendChild(opt);
122 |         });
123 |     } catch (err) {
124 |         console.warn('[TelemetryResearch] Ошибка загрузки источник
125 | ов:', err);
126 |     }
127 | }
128 |
129 | /**
130 |  * Запуск глубокого исследования телеметрии.
131 |  */
132 | async function runResearch() {
133 |     const btnRun = document.getElementById('tr-btn-run');
134 |     const selectSource = document.getElementById('tr-select-sour
135 | ce');
```

```
134 |   if (btnRun) {
135 |       btnRun.disabled = true;
136 |       btnRun.innerHTML = '<span class="spinner-border spinner-bo
rder-sm" role="status"></span> Анализ...';
137 |   }
138 |
139 |   try {
140 |       const res = await fetch('/apps/telemetry_research/run-rese
arch', {
141 |           method: 'POST',
142 |           headers: { 'Content-Type': 'application/json' },
143 |           body: JSON.stringify({ source_path: sourcePath || null }
),
144 |       });
145 |
146 |       if (!res.ok) {
147 |           throw new Error(`Ошибка сервера: ${res.status}`);
148 |       }
149 |
150 |       const report = await res.json();
151 |       trCurrentReport = report;
152 |
153 |       renderKPI(report);
154 |       renderHypotheses(report.hypotheses);
155 |       renderCorrelations(report.correlations);
156 |       renderCharts(report.base_report.charts);
157 |       renderAnomalies(report.base_report.anomalies, report.base_
report.device_summary);
158 |
159 |       if (window.toast) {
160 |           window.toast.success('Исследование завершено', `Проанали
зировано ${report.base_report.records_analyzed} записей.`);
161 |       }
162 |   } catch (err) {
163 |       console.error('[TelemetryResearch] Ошибка анализа:', err);
164 |       if (window.toast) {
165 |           window.toast.error('Сбой исследования', err.message);
166 |       }
```

```
167 | } finally {
168 |     if (btnRun) {
169 |         btnRun.disabled = false;
170 |         btnRun.innerHTML = '<i class="bi bi-play-circle-fill"></i> <span>Исследовать</span>';
171 |     }
172 | }
173 | }
174 |
175 | /**
176 |  * Отрисовка сводных KPI карточек.
177 |  */
178 | function renderKPI(report) {
179 |     const base = report.base_report;
180 |     const valHealth = document.getElementById('tr-val-health');
181 |     const progHealth = document.getElementById('tr-progress-health');
182 |     const valRecords = document.getElementById('tr-val-records');
183 |     const valTimespan = document.getElementById('tr-val-timespan');
184 |     const valHypotheses = document.getElementById('tr-val-hypotheses');
185 |     const valAnomalies = document.getElementById('tr-val-anomalies');
186 |     const valDevices = document.getElementById('tr-val-devices');
187 |
188 |     if (valHealth) valHealth.textContent = `${base.health_score}`;
189 |     if (progHealth) {
190 |         progHealth.style.width = `${base.health_score}%`;
191 |         progHealth.className = `progress-bar ${
192 |             base.health_score >= 80 ? 'bg-success' : base.health_score >= 60 ? 'bg-warning' : 'bg-danger'
193 |         }`;
194 |     }
195 | }
```



```

196 |   if (valRecords) valRecords.textContent = `${base.records_ana
    | lyzed}`;
197 |   if (valTimespan && base.time_range) {
198 |     valTimespan.textContent = `${base.time_range.start ? base.
    | time_range.start.slice(11, 19) : ''} - ${
199 |       base.time_range.end ? base.time_range.end.slice(11, 19)
    | : ''
200 |     }`;
201 |   }
202 |
203 |   const confirmedCount = (report.hypotheses || []).filter((h)
    | => h.confirmed).length;
204 |   if (valHypotheses) valHypotheses.textContent = `${confirmedC
    | ount} / ${report.hypotheses ? report.hypotheses.length : 0}`;
205 |
206 |   if (valAnomalies) valAnomalies.textContent = `${(base.anomal
    | ies || []).length}`;
207 |   if (valDevices) {
208 |     valDevices.textContent = base.device_summary.error_count >
    | 0
209 |       ? `Ошибок устройств: ${base.device_summary.error_count}`
210 |       : 'Оборудование в норме';
211 |   }
212 |
213 |   // Summary alert banner
214 |   const banner = document.getElementById('tr-summary-banner');
215 |   const bannerText = document.getElementById('tr-summary-text'
    | );
216 |   if (banner && bannerText && report.investigation_summary) {
217 |     banner.classList.remove('d-none');
218 |     bannerText.textContent = report.investigation_summary;
219 |   }
220 | }
221 |
222 | /**
223 |  * Отрисовка списка гипотез.
224 |  */
225 | function renderHypotheses(hypotheses) {

```

```

226 |   const container = document.getElementById('tr-hypotheses-lis
    | t');
227 |   if (!container) return;
228 |
229 |   if (!hypotheses || hypotheses.length === 0) {
230 |     container.innerHTML = '<div class="col-12 text-center text
    | -muted py-4">Гипотезы не проверены</div>';
231 |     return;
232 |   }
233 |
234 |   container.innerHTML = '';
235 |   hypotheses.forEach((h) => {
236 |     const col = document.createElement('div');
237 |     col.className = 'col-12 col-md-6';
238 |
239 |     const cardBorder = h.confirmed ? 'border-warning' : 'borde
    | r-success-subtle';
240 |     const statusBadge = h.confirmed
241 |       ? '<span class="badge bg-warning text-dark"><i class="bi
    | bi-exclamation-triangle"></i> Подтверждена</span>'
242 |       : '<span class="badge bg-success"><i class="bi bi-check-
    | circle"></i> В норме</span>';
243 |
244 |     const evidenceItems = h.evidence.map((ev) => `<li>${ev}</l
    | i>`).join('');
245 |
246 |     col.innerHTML = `
247 |       <div class="card bg-dark ${cardBorder} h-100 shadow-sm">
248 |         <div class="card-header bg-transparent border-secondar
    | y d-flex justify-content-between align-items-center">
249 |           <span class="fw-bold text-light">${h.title}</span>
250 |           ${statusBadge}
251 |         </div>
252 |         <div class="card-body p-3 small">
253 |           <p class="text-muted mb-2">${h.description}</p>
254 |           <div class="mb-2">
255 |             <strong class="text-secondary">Факты телеметрии:</
    | strong>

```

```

256         <ul class="text-light text-opacity-75 ps-3 mb-1">$
| {evidenceItems}</ul>
257     </div>
258     ${
259         h.recommendation
260         ? `<div class="p-2 rounded bg-body-tertiary bg-o
| pacity-10 text-info mt-2">
261             <strong>□ Рекомендация:</strong> ${h.recom
| mendation}
262             </div>`
263         : ''
264     }
265 </div>
266 <div class="card-footer bg-transparent border-secondar
| y text-muted small d-flex justify-content-between">
267     <span>ID: <code>${h.hypothesis_id}</code></span>
268     <span>Уверенность: <strong>${Math.round(h.confidence
| * 100)}%</strong></span>
269 </div>
270 </div>
271 `;
272 container.appendChild(col);
273 });
274 }
275
276 /**
277  * Отрисовка матрицы корреляций.
278  */
279 function renderCorrelations(correlations) {
280     const tbody = document.getElementById('tr-correlations-tbody
| ');
281     const countBadge = document.getElementById('tr-correlations-
| count');
282     if (!tbody) return;
283
284     if (!correlations || correlations.length === 0) {
285         tbody.innerHTML = '<tr><td colspan="5" class="text-center
| text-muted py-3">Корреляции не рассчитаны</td></tr>';

```

```

286     if (countBadge) countBadge.textContent = '0 nap';
287     return;
288 }
289
290 if (countBadge) countBadge.textContent = `${correlations.length} nap`;
291
292 tbody.innerHTML = correlations
293     .map((c) => {
294         const absVal = Math.abs(c.coefficient);
295         let coeffBadgeClass = 'bg-secondary';
296         if (absVal >= 0.7) coeffBadgeClass = 'bg-danger';
297         else if (absVal >= 0.5) coeffBadgeClass = 'bg-warning text-dark';
298         else if (absVal >= 0.3) coeffBadgeClass = 'bg-info text-dark';
299
300         return `
301             <tr>
302                 <td><code class="text-info">${c.metric_a}</code></td>
303                 <td><code class="text-info">${c.metric_b}</code></td>
304                 <td><span class="badge ${coeffBadgeClass} font-monospace">${c.coefficient > 0 ? '+' : ''}${c.coefficient.toFixed(3)}</span></td>
305                 <td>${c.sample_size} точек</td>
306                 <td class="text-light text-opacity-75">${c.interpretation}</td>
307             </tr>
308         `;
309     })
310     .join('');
311 }
312
313 /**
314  * Отрисовка интерактивных графиков Chart.js.
315  */

```

```

316 function renderCharts(charts) {
317     const container = document.getElementById('tr-charts-contain
er');
318     if (!container) return;
319
320     // Очищаем старые инстансы графиков
321     trChartsInstances.forEach((inst) => {
322         try {
323             inst.destroy();
324         } catch {}
325     });
326     trChartsInstances = [];
327
328     if (!charts || charts.length === 0) {
329         container.innerHTML = '<div class="col-12 text-center text
-muted py-4">Нет доступных графиков</div>';
330         return;
331     }
332
333     container.innerHTML = '';
334
335     charts.forEach((ch, idx) => {
336         const col = document.createElement('div');
337         col.className = 'col-12 col-xl-6';
338
339         const canvasId = `tr-chart-canvas-${idx}`;
340         col.innerHTML = `
341             <div class="card bg-dark border-secondary h-100 shadow-s
m">
342                 <div class="card-header bg-transparent border-secondar
y d-flex justify-content-between align-items-center">
343                     <span class="fw-bold text-light">${ch.title}</span>
344                     <span class="badge bg-secondary small">${ch.chart_ty
pe}</span>
345                 </div>
346                 <div class="card-body p-2" style="position: relative;
height: 280px;">
347                     <canvas id="${canvasId}"></canvas>

```

```
348         </div>
349     </div>
350     `;
351     container.appendChild(col);
352
353     const ctx = document.getElementById(canvasId);
354     if (ctx && typeof Chart !== 'undefined') {
355         const labels = ch.labels.map((l) => (l.length > 19 ? l.slice(11, 19) : l));
356         const datasets = ch.datasets.map((ds) => ({
357             label: ds.label,
358             data: ds.data,
359             borderColor: ds.color || '#3b82f6',
360             backgroundColor: (ds.color || '#3b82f6') + '22',
361             borderWidth: 2,
362             fill: ch.chart_type === 'area',
363             tension: 0.25,
364             pointRadius: ds.data.length > 50 ? 0 : 3,
365         }));
366
367         const newInst = new Chart(ctx, {
368             type: ch.chart_type === 'bar' ? 'bar' : 'line',
369             data: { labels, datasets },
370             options: {
371                 responsive: true,
372                 maintainAspectRatio: false,
373                 animation: { duration: 400 },
374                 scales: {
375                     x: {
376                         grid: { color: 'rgba(255,255,255,0.06)' },
377                         ticks: { color: '#9ca3af', font: { size: 10 } },
378                     },
379                     y: {
380                         grid: { color: 'rgba(255,255,255,0.06)' },
381                         ticks: { color: '#9ca3af', font: { size: 10 } },
382                     },
383                 },
384                 plugins: {
```

```

385         legend: {
386             labels: { color: '#e5e7eb', boxWidth: 12, font:
387                 { size: 11 } },
388             },
389         },
390     });
391     trChartsInstances.push(newInst);
392 }
393 });
394 }
395
396 /**
397  * Отрисовка аномалий и состояния устройств.
398  */
399 function renderAnomalies(anomalies, deviceSummary) {
400     const tbody = document.getElementById('tr-anomalies-tbody');
401     const anomBadge = document.getElementById('tr-anomalies-badg
402         e');
403     const devBadge = document.getElementById('tr-devices-badge')
404         ;
405     const devContent = document.getElementById('tr-devices-conte
406         nt');
407
408     if (anomBadge) anomBadge.textContent = anomalies ? `${anomal
409         ies.length}` : '0';
410
411     if (tbody) {
412         if (!anomalies || anomalies.length === 0) {
413             tbody.innerHTML = '<tr><td colspan="5" class="text-cen
414                 ter text-success py-3"><i class="bi bi-check-circle"></i> Аномал
415                 ий в метриках не зафиксировано</td></tr>';
416         } else {
417             tbody.innerHTML = anomalies
418                 .map((a) => {
419                     const sevClass = a.severity === 'critical' ? 'bg-dan
420                         ger' : a.severity === 'high' ? 'bg-warning text-dark' : 'bg-in
421                             fo text-dark';

```

```

414         return `
415             <tr>
416                 <td class="text-muted">${a.timestamp ? a.timesta
mp.slice(11, 19) : '--'}</td>
417                 <td><strong>${a.metric}</strong></td>
418                 <td><span class="badge ${sevClass}">${a.severity
}</span></td>
419                 <td><code class="text-warning">${a.value}</code>
/ ${a.threshold}</td>
420                 <td class="text-light text-opacity-75">${a.messa
ge}</td>
421             </tr>
422         `;
423     })
424     .join('');
425 }
426 }
427
428 if (deviceSummary && devContent) {
429     const hasIssues = deviceSummary.error_count > 0 || deviceS
ummary.flapping_devices.length > 0;
430     if (devBadge) {
431         devBadge.className = `badge ${hasIssues ? 'bg-danger' :
'bg-success'}`;
432         devBadge.textContent = hasIssues ? 'Внимание' : 'Норма';
433     }
434
435     let flapHtml = '<span class="text-success">Нет</span>';
436     if (deviceSummary.flapping_devices && deviceSummary.flappi
ng_devices.length > 0) {
437         flapHtml = deviceSummary.flapping_devices.map((d) => `<s
pan class="badge bg-warning text-dark me-1">${d}</span>`).join
('');
438     }
439
440     devContent.innerHTML = `
441         <div class="mb-2"><strong>Всего инцидентов:</strong> ${d
eviceSummary.total_events}</div>

```



```
442 |     <div class="mb-2"><strong>Ошибок и сбоев:</strong> <span
    |     class="${deviceSummary.error_count > 0 ? 'text-danger fw-bold
    |     ' : 'text-success'}">${deviceSummary.error_count}</span></div>
443 |     <div class="mb-2"><strong>Флаппирующие устройства:</stro
    |     ng> ${flapHtml}</div>
444 |     `;
445 | }
446 | }
447 |
448 | /**
449 |  * Загрузка записей телеметрии с пагинацией.
450 |  */
451 | async function loadRecords() {
452 |     const tbody = document.getElementById('tr-records-tbody');
453 |     const infoEl = document.getElementById('tr-records-paginat
    |     ion-info');
454 |     const selectSource = document.getElementById('tr-select-sour
    |     ce');
455 |     const sourcePath = selectSource ? selectSource.value : null;
456 |
457 |     if (tbody) {
458 |         tbody.innerHTML = '<tr><td colspan="7" class="text-center
    |         text-muted py-3"><span class="spinner-border spinner-border-sm
    |         "></span> Загрузка...</td></tr>';
459 |     }
460 |
461 |     try {
462 |         const url = new URL('/apps/telemetry_research/records', wi
    |         ndow.location.origin);
463 |         if (sourcePath) url.searchParams.set('source_path', source
    |         Path);
464 |         if (trSearchQuery) url.searchParams.set('query', trSearchQ
    |         uery);
465 |         url.searchParams.set('page', trCurrentPage);
466 |         url.searchParams.set('page_size', trPageSize);
467 |
468 |         const res = await fetch(url.toString());
469 |         if (!res.ok) throw new Error(`HTTP ${res.status}`);
```

```
470     const data = await res.json();
471
472     trTotalPages = data.total_pages || 1;
473     if (infoEl) {
474         infoEl.textContent = `Стр. ${data.page} из ${trTotalPage
s} (всего ${data.total})`;
475     }
476
477     if (!data.items || data.items.length === 0) {
478         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
ter text-muted py-3">Записей не найдено</td></tr>';
479         return;
480     }
481
482     tbody.innerHTML = data.items
483         .map((r, i) => {
484             const num = (trCurrentPage - 1) * trPageSize + i + 1;
485             const ts = r.timestamp || r.time || '--';
486             const cpu = r.cpu ? `${r.cpu.total_percent || r.cpu.lo
ad_percent || '--}%` : '--';
487             const ram = r.memory ? `${r.memory.percent || '--}%`
: '--';
488             const gpu = r.gpu ? `${r.gpu.load_percent || '--}%` :
'--';
489             let disk = '--';
490             if (r.disk_io) {
491                 const rMb = ((r.disk_io.read_bytes_per_sec || 0) / 1
048576).toFixed(1);
492                 const wMb = ((r.disk_io.write_bytes_per_sec || 0) /
1048576).toFixed(1);
493                 disk = `${rMb} / ${wMb} MB/s`;
494             }
495             const eventInfo = r.event_type || r.friendly_name || J
SON.stringify(r).slice(0, 50);
496
497             return `
498                 <tr>
499                     <td>${num}</td>
```

```

500         <td class="text-muted">${ts}</td>
501         <td>${cpu}</td>
502         <td>${ram}</td>
503         <td>${gpu}</td>
504         <td>${disk}</td>
505         <td class="text-truncate" style="max-width: 250px;
" title="${eventInfo}">${eventInfo}</td>
506     </tr>
507     `;
508     })
509     .join('');
510 } catch (err) {
511     if (tbody) {
512         tbody.innerHTML = `<tr><td colspan="7" class="text-cen
r text-danger py-3">Ошибка: ${err.message}</td></tr>`;
513     }
514 }
515 }
516
517 /**
518  * Открытие автономного HTML дашборда.
519  */
520 function openHtmlDashboard() {
521     const selectSource = document.getElementById('tr-select-sour
ce');
522     const sourcePath = selectSource ? selectSource.value : '';
523     const url = sourcePath
524         ? `/apps/telemetry_research/dashboard?source_path=${encode
URIComponent(sourcePath)}`
525         : '/apps/telemetry_research/dashboard';
526     window.open(url, '_blank');
527 }
528
529 /**
530  * Экспорт JSON отчета.
531  */
532 function downloadJsonReport() {
533     if (!trCurrentReport) {

```

```
534     if (window.toast) window.toast.warning('Нет данных', 'Снач  
ала выполните исследование.');
```

535 return;

536 }

537 const blob = new Blob([JSON.stringify(trCurrentReport, null,
2)], { type: 'application/json' });

538 const url = URL.createObjectURL(blob);

539 const a = document.createElement('a');

540 a.href = url;

541 a.download = `telemetry\_research\_\${Date.now()}.json`;

542 a.click();

543 URL.revokeObjectURL(url);

544 }

545

546 // Автозапуск при подключении модуля

547 if (document.readyState === 'loading') {

548 document.addEventListener('DOMContentLoaded', initTelemetryR
esearchTab);

549 } else {

550 initTelemetryResearchTab();

551 }

552

index.html

```
1 <!--
2 =====
3
4 =====
5 Process Name: Windows Kernel Quality & Throttling Tab
6 =====
7
8 Description:
9     Вкладка телеметрии качества работы ядра Windows, задержек
10    DPC/ISR прерываний,
11    аппаратного троттлинга PROCHOT / Power Limit, режима PCIe
12    шины GPU и BSOD дампов.
13
14 File: index.html
15 Project: AI-Breadboard
16 Module: WebInterface.ThrottlingTab
17 Author: hypo69
18 Copyright: © 2026 hypo69
19 =====
20 =====
21 -->
22 <style>
23     .throttling-tab-wrapper {
24         padding: 0.75rem;
25         color: var(--text-color, #f8fafc);
26     }
27     .diag-meter-card {
28         background: var(--surface-2, #1e293b);
29         border: 1px solid var(--border-color, #334155);
30         border-radius: var(--radius-md, 8px);
31         padding: 0.75rem 1rem;
32     }
33     .diag-panel {
34         background: var(--surface-2, #1e293b);
35         border: 1px solid var(--border-color, #334155);
36         border-radius: 8px;
37         overflow: hidden;
```

```

32 | }
33 | .diag-panel-header {
34 |     background: rgba(15, 23, 42, 0.6);
35 |     padding: 8px 14px;
36 |     border-bottom: 1px solid var(--border-color, #334155);
37 |     font-weight: 600;
38 |     font-size: 0.85rem;
39 | }
40 | .about-sys-spec-table {
41 |     width: 100%;
42 |     border-collapse: separate;
43 |     border-spacing: 0;
44 | }
45 | .about-sys-spec-table td {
46 |     padding: 8px 12px;
47 |     border-bottom: 1px solid rgba(51, 65, 85, 0.35);
48 |     vertical-align: middle;
49 | }
50 | .about-sys-spec-key {
51 |     color: var(--text-muted, #94a3b8);
52 |     width: 220px;
53 |     font-size: 0.82rem;
54 |     font-weight: 500;
55 | }
56 | .about-sys-spec-val {
57 |     color: var(--text-color, #f8fafc);
58 |     font-weight: 500;
59 |     font-size: 0.82rem;
60 | }
61 | </style>
62 |
63 | <div class="throttling-tab-wrapper">
64 |     <!-- Top Header & Status -->
65 |     <div class="card shadow-sm border-secondary-subtle mb-3" sty
66 |         le="background: linear-gradient(135deg, #0f172a 0%, #1e293b 10
67 |         0%); border-radius: 10px;">
68 |         <div class="card-body p-3">
69 |             <div class="d-flex flex-wrap align-items-center justify-

```

```

content-between gap-2">
68     <div class="d-flex align-items-center gap-2">
69         <div class="fs-4">□ </div>
70     </div>
71     <h5 class="mb-0 fw-bold text-white">3. Качество яд
ра и Троттлинг (Kernel Quality & Hardware Health)</h5>
72     <small class="text-muted">DPC/ISR задержки драйвер
ов, аппаратный сброс частот PROCHOT/Power Limit, PCIe линии и
аварии BSOD</small>
73     </div>
74 </div>
75
76     <div class="d-flex align-items-center gap-2 flex-wrap"
>
77         <div class="form-check form-switch mb-0 small text-m
uted d-flex align-items-center gap-1">
78             <input class="form-check-input" type="checkbox" id
="diag-throttling-auto-refresh">
79             <label class="form-check-label small text-nowrap"
for="diag-throttling-auto-refresh">Авто (5с)</label>
80         </div>
81         <button class="btn btn-sm btn-outline-info rounded-p
ill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id=
"btn-diag-throttling-refresh">
82             <i class="bi bi-arrow-clockwise"></i> Обновить
83         </button>
84     </div>
85 </div>
86 </div>
87 </div>
88
89 <!-- KPI Latency and Status Meters -->
90 <div class="row g-2 mb-3">
91     <div class="col-12 col-sm-6 col-lg-3">
92         <div class="diag-meter-card">
93             <div class="d-flex justify-content-between align-items
-center small text-muted">
94                 <span>DPC Latency (Отложенные вызовы)</span>

```

```
95 |         <i class="bi bi-activity text-info"></i>
96 |     </div>
97 |     <div class="fs-4 fw-bold mt-1" id="diag-dpc-latency-val">0.0%</div>
98 |     <div class="small" id="diag-dpc-latency-sub"><span class="badge bg-success-subtle text-success border border-success">Optimal</span></div>
99 | </div>
100| </div>
101| <div class="col-12 col-sm-6 col-lg-3">
102|     <div class="diag-meter-card">
103|         <div class="d-flex justify-content-between align-items-center small text-muted">
104|             <span>ISR Latency (Прерывания драйверов)</span>
105|             <i class="bi bi-cpu text-warning"></i>
106|         </div>
107|         <div class="fs-4 fw-bold text-warning mt-1" id="diag-isr-latency-val">0.0%</div>
108|         <div class="small text-muted">Нагрузка драйверов на ЦП</div>
109|     </div>
110| </div>
111| <div class="col-12 col-sm-6 col-lg-3">
112|     <div class="diag-meter-card">
113|         <div class="d-flex justify-content-between align-items-center small text-muted">
114|             <span>Аппаратный троттлинг (PROCHOT)</span>
115|             <i class="bi bi-thermometer-sun text-danger"></i>
116|         </div>
117|         <div class="fs-4 fw-bold mt-1" id="diag-throttling-status-val">Inactive</div>
118|         <div class="small text-muted" id="diag-throttling-status-sub">Терморежим в норме</div>
119|     </div>
120| </div>
121| <div class="col-12 col-sm-6 col-lg-3">
122|     <div class="diag-meter-card">
123|         <div class="d-flex justify-content-between align-items
```



```

    -center small text-muted">
124         <span>Время непрерывной работы</span>
125         <i class="bi bi-clock text-primary"></i>
126     </div>
127     <div class="fs-4 fw-bold text-white mt-1" id="diag-uptime-val">--</div>
128     <div class="small text-muted">System Uptime</div>
129 </div>
130 </div>
131 </div>
132
133 <!-- GPU PCIe Bus & Kernel BSOD Crash Dumps -->
134 <div class="row g-2">
135     <div class="col-12 col-lg-6">
136         <div class="diag-panel h-100">
137             <div class="diag-panel-header d-flex justify-content-between align-items-center">
138                 <div class="d-flex align-items-center gap-2">
139                     <i class="bi bi-gpu-card text-success"></i>
140                     <span class="text-white">Режим шины PCIe графического ускорителя (GPU Link)</span>
141                 </div>
142                 <span class="badge bg-success-subtle text-success border border-success">PCIe Prober</span>
143             </div>
144             <div class="p-3">
145                 <table class="about-sys-spec-table">
146                     <tbody>
147                         <tr>
148                             <td class="about-sys-spec-key"><i class="bi bi-gpu-card me-1.5 text-success"></i>Видеокарта</td>
149                             <td class="about-sys-spec-val text-info" id="diag-gpu-pcie-name">GeForce GPU</td>
150                         </tr>
151                         <tr>
152                             <td class="about-sys-spec-key"><i class="bi bi-speedometer2 me-1.5 text-warning"></i>Скорость шины (Link Speed)</td>

```

```

153 |         <td class="about-sys-spec-val" id="diag-gpu-pc
| ie-speed">PCIe 3.0 / 4.0 (8.0 GT/s)</td>
154 |     </tr>
155 |     <tr>
156 |         <td class="about-sys-spec-key"><i class="bi bi
| -arrows-expand me-1.5 text-primary"></i>Ширина линий (Link Wid
| th)</td>
157 |         <td class="about-sys-spec-val" id="diag-gpu-pc
| ie-width">x16 Lanes (Full Width)</td>
158 |     </tr>
159 |     <tr>
160 |         <td class="about-sys-spec-key"><i class="bi bi
| -shield-check me-1.5 text-success"></i>Статус пропускной спосо
| бности</td>
161 |         <td class="about-sys-spec-val text-success" id
| ="diag-gpu-pcie-status">Штатный режим без деградации</td>
162 |     </tr>
163 | </tbody>
164 | </table>
165 | </div>
166 | </div>
167 | </div>
168 |
169 | <div class="col-12 col-lg-6">
170 |     <div class="diag-panel h-100">
171 |         <div class="diag-panel-header d-flex justify-content-b
| etween align-items-center">
172 |             <div class="d-flex align-items-center gap-2">
173 |                 <i class="bi bi-bug-fill text-danger"></i>
174 |                 <span class="text-white">Журнал аварийных дампов я
| дра (Minidump / BSOD)</span>
175 |             </div>
176 |             <span class="badge bg-secondary" id="diag-bsod-count
| ">0 сбоев</span>
177 |         </div>
178 |         <div class="p-2" id="diag-bsod-container" style="max-h
| eight: 260px; overflow-y: auto;">
179 |             <div class="text-center py-4 text-success small">

```

```
180 |         <i class="bi bi-check-circle-fill fs-5 d-block mb-
    | 1 text-success"></i>
181 |         Аварийных дампов падения ядра (BSOD) в Minidump не
    | обнаружено
182 |     </div>
183 | </div>
184 | </div>
185 | </div>
186 | </div>
187 </div>
188 |
```

main.js

```
1  /**
2   * =====
3   * Process Name: Windows Kernel Quality & Throttling Logic
4   * =====
5   * Description:
6   *   Клиентский контроллер вкладки качества ядра и троттлинга:
7   *   DPC/ISR задержки, PROCHOT троттлинг, PCIe шина GPU и дампы BSOD.
8   *
9   * File: main.js
10  * Project: AI-Breadboard
11  * Module: WebInterface.ThrottlingTab
12  * Author: hypo69
13  * Copyright: © 2026 hypo69
14  * =====
15  */
16
17 (function () {
18     'use strict';
19
20     let autoRefreshTimer = null;
21
22     function escapeHtml(str) {
23         if (str === null || str === undefined) return '';
24         return String(str)
25             .replace(/&/g, '&amp;')
26             .replace(/</g, '&lt;')
27             .replace(/>/g, '&gt;')
28             .replace(/"/g, '&quot;')
29             .replace(/'/g, '&#039;');
30     }
31
32     function setText(id, text) {
```

```

33     const el = document.getElementById(id);
34     if (el) el.textContent = text !== null && text !== undefined ? String(text) : '--';
35 }
36
37 async function apiFetch(url) {
38     if (window.api && typeof window.api.fetch === 'function')
39     {
40         return await window.api.fetch(url);
41     }
42     const res = await fetch(url);
43     if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.statusText}`);
44     return await res.json();
45 }
46
47 async function fetchKernelThrottling() {
48     try {
49         const data = await apiFetch('/api/v1/system/diagnostics/throttling');
50         if (!data) return;
51         setText('diag-dpc-latency-val', `${data.dpc_latency_pct.toFixed(2)}%`);
52         const dpcSub = document.getElementById('diag-dpc-latency-sub');
53         if (dpcSub) {
54             if (data.dpc_status === 'severe') {
55                 dpcSub.innerHTML = '<span class="badge bg-danger-subtle text-danger border border-danger">Критическая латентность DPC</span>';
56             } else if (data.dpc_status === 'elevated') {
57                 dpcSub.innerHTML = '<span class="badge bg-warning-subtle text-warning border border-warning">Повышенная задержка DPC</span>';
58             } else {
59                 dpcSub.innerHTML = '<span class="badge bg-success-subtle text-success border border-success">Оптимально (< 1.5%)</

```

```

    span>';
60     }
61     }
62
63     setText('diag-isr-latency-val', `${data.interrupt_latenc
y_pct.toFixed(2)}%`);
64     setText('diag-uptime-val', data.uptime_formatted || '--'
);
65
66     const thVal = document.getElementById('diag-throttling-s
tatus-val');
67     const thSub = document.getElementById('diag-throttling-s
tatus-sub');
68     if (thVal && thSub) {
69         if (data.thermal_throttling_detected) {
70             thVal.textContent = 'PROCHOT Active';
71             thVal.className = 'fs-4 fw-bold text-danger mt-1';
72             thSub.textContent = 'Обнаружен термический сброс час
тот CPU';
73         } else if (data.power_limit_throttling_detected) {
74             thVal.textContent = 'Power Limit PL1/PL2';
75             thVal.className = 'fs-4 fw-bold text-warning mt-1';
76             thSub.textContent = 'Ограничение по энергопотреблени
ю';
77         } else {
78             thVal.textContent = 'Inactive (Норма)';
79             thVal.className = 'fs-4 fw-bold text-success mt-1';
80             thSub.textContent = 'Терморезжим и питание в норме';
81         }
82     }
83
84     // GPU PCIe Link
85     if (data.gpu_pcie_link) {
86         setText('diag-gpu-pcie-name', data.gpu_pcie_link.gpu_n
ame || 'GPU Accelerator');
87         setText('diag-gpu-pcie-speed', data.gpu_pcie_link.curr
ent_link_speed || 'PCIe 3.0 / 4.0');
88         setText('diag-gpu-pcie-width', data.gpu_pcie_link.curr

```

```

ent_link_width || 'x16 Lanes');
89         setText('diag-gpu-pcie-status', data.gpu_pcie_link.sta
tus || 'Штатный режим без деградации');
90     }
91
92     // BSOD Minidump
93     const bsodContainer = document.getElementById('diag-bsod
-container');
94     if (bsodContainer && data.last_bsod_crashes) {
95         setText('diag-bsod-count', `${data.last_bsod_crashes.l
ength} сбоев`);
96         if (data.last_bsod_crashes.length === 0) {
97             bsodContainer.innerHTML = `
98                 <div class="text-center py-4 text-success small">
99                     <i class="bi bi-check-circle-fill fs-5 d-block m
b-1 text-success"></i>
100                     Аварийных дампов падения ядра (BSOD) в Minidump
не обнаружено
101                 </div>
102                 `;
103         } else {
104             bsodContainer.innerHTML = data.last_bsod_crashes.map
(c => `
105                 <div class="p-2 mb-1.5 rounded bg-black bg-opacity
-40 border border-danger-subtle d-flex align-items-center just
ify-content-between">
106                     <div>
107                         <strong class="text-danger"><i class="bi bi-fi
le-earmark-binary me-1"></i>${escapeHtml(c.file_name)}</strong>
108                         <div class="small text-muted font-monospace">${
escapeHtml(c.timestamp)} (${c.size_kb} KB)</div>
109                     </div>
110                     <span class="badge bg-danger">Crash Dump</span>
111                 </div>
112                 `).join('');
113         }
114     }

```

```
115     } catch (err) {
116         console.warn('[ThrottlingTab] fetchKernelThrottling erro
117         r:', err);
118     }
119 }
120 function bindEvents() {
121     const btnRefresh = document.getElementById('btn-diag-throt
122     tling-refresh');
123     if (btnRefresh) {
124         btnRefresh.onclick = () => fetchKernelThrottling();
125     }
126     const autoSwitch = document.getElementById('diag-throttlin
127     g-auto-refresh');
128     if (autoSwitch) {
129         autoSwitch.onchange = (e) => {
130             if (e.target.checked) {
131                 autoRefreshTimer = setInterval(fetchKernelThrottling
132                 , 5000);
133             } else if (autoRefreshTimer) {
134                 clearInterval(autoRefreshTimer);
135                 autoRefreshTimer = null;
136             }
137         };
138     }
139     async function init() {
140         bindEvents();
141         await fetchKernelThrottling();
142     }
143
144     if (document.readyState === 'loading') {
145         document.addEventListener('DOMContentLoaded', init);
146     } else {
147         setTimeout(init, 10);
148     }
```


149 | `})();`

150 |

index.html

```
1 <!-- Trading Terminal Application Tab -->
2 <style>
3   .trading-container {
4     padding: 1rem;
5   }
6   .trading-metric-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 10px;
10    padding: 1rem;
11    display: flex;
12    flex-direction: column;
13    gap: 0.25rem;
14  }
15  .trading-metric-label {
16    font-size: 0.75rem;
17    text-transform: uppercase;
18    letter-spacing: 0.05em;
19    color: #94a3b8;
20  }
21  .trading-metric-value {
22    font-size: 1.4rem;
23    font-weight: 700;
24    color: #f8fafc;
25    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
26    Consolas, monospace;
27  }
28  .trading-panel {
29    background: #0f172a !important;
30    border: 1px solid #334155 !important;
31    border-radius: 10px;
32    overflow: hidden;
33  }
34  .trading-panel-header {
35    background: #1e293b !important;
36    border-bottom: 1px solid #334155 !important;
```

```
36     padding: 0.75rem 1rem;
37     font-weight: 600;
38     font-size: 0.88rem;
39     color: #38bdf8;
40     display: flex;
41     align-items: center;
42     justify-content: space-between;
43 }
44 .trading-orderbook-table {
45     width: 100%;
46     font-size: 0.8rem;
47     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
48     border-collapse: collapse;
49 }
50 .trading-orderbook-table th {
51     background: #1e293b;
52     padding: 6px 12px;
53     color: #94a3b8;
54     font-weight: 600;
55     border-bottom: 1px solid #334155;
56 }
57 .trading-orderbook-table td {
58     padding: 5px 12px;
59     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
60 }
61 .trading-ask-row {
62     color: #f87171;
63     background: rgba(239, 68, 68, 0.05);
64 }
65 .trading-bid-row {
66     color: #4ade80;
67     background: rgba(34, 197, 94, 0.05);
68 }
69 .btn-kill-switch {
70     background: linear-gradient(135deg, #ef4444 0%, #b91c1c 10
0%) !important;
71     border: 1px solid #f87171 !important;
```

```

72     color: #ffffff !important;
73     font-weight: 700;
74     box-shadow: 0 0 12px rgba(239, 68, 68, 0.4);
75 }
76 .btn-kill-switch:hover {
77     background: linear-gradient(135deg, #dc2626 0%, #991b1b 10
0%) !important;
78     box-shadow: 0 0 18px rgba(239, 68, 68, 0.7);
79 }
80 </style>
81
82 <div class="trading-container">
83     <!-- Header Alert & Status -->
84     <div class="d-flex justify-content-between align-items-cente
r flex-wrap gap-3 mb-3 p-3 rounded-3 text-white" style="backgr
ound: linear-gradient(135deg, #0f172a 0%, #1e293b 100%); borde
r: 1px solid #334155;">
85         <div>
86             <h5 class="mb-1 fw-bold text-white d-flex align-items-ce
nter gap-2">
87                 <span class="fs-4">☐ </span> Торговый терминал (Exchan
ge Trading Desk)
88             </h5>
89             <p class="mb-0 small text-light" style="color: #cbd5e1 !
important;">
90                 Автономный торговый терминал с обновлением стакана цен
L2, расчётом PnL и кнопкой экстренной ликвидации (Kill-Switch
).
91             </p>
92         </div>
93         <div class="d-flex align-items-center gap-2">
94             <span class="badge rounded-pill bg-success-subtle text-s
uccess border border-success px-3 py-2" id="trading-stream-sta
tus">
95                 • Стрим активен
96             </span>
97             <button class="btn btn-sm btn-outline-info rounded-pill
px-3" id="btn-refresh-trading">

```

```

98         <i class="bi bi-arrow-clockwise me-1"></i> Обновить
99     </button>
100     <button class="btn btn-sm btn-outline-warning rounded-pi
101 ll px-3" id="btn-trading-config">
102         <i class="bi bi-gear-fill me-1"></i> Настройки (config
103 .json)
104     </button>
105     <button class="btn btn-sm btn-kill-switch rounded-pill p
106 x-3" id="btn-trigger-kill-switch">
107         ☐ Kill-Switch (Ликвидация)
108     </button>
109 </div>
110 </div>
111 <!-- Top Metrics Bar -->
112 <div class="row g-3 mb-3">
113     <div class="col-md-3 col-sm-6">
114         <div class="trading-metric-card">
115             <span class="trading-metric-label">Торговая пара</span
116 >
117             <span class="trading-metric-value" style="color: #38bd
118 f8;" id="trading-symbol">BTC/USDT</span>
119             <span class="small text-muted" id="trading-spread-info
120 ">Спред: --</span>
121         </div>
122     </div>
123     <div class="col-md-3 col-sm-6">
124         <div class="trading-metric-card">
125             <span class="trading-metric-label">Текущая цена</span>
126             <span class="trading-metric-value" id="trading-current
127 -price">$0.00</span>
128             <span class="small" id="trading-price-delta" style="co
129 lor: #4ade80;">Bid: $0 | Ask: $0</span>
130         </div>
131     </div>
132     <div class="col-md-3 col-sm-6">
133         <div class="trading-metric-card">
134             <span class="trading-metric-label">Баланс USDT / Эквит

```

```

    и</span>
128     <span class="trading-metric-value" id="trading-total-e
    quity">$0.00</span>
129     <span class="small text-muted" id="trading-cash-balanc
    e">Доступно: $0.00</span>
130     </div>
131 </div>
132 <div class="col-md-3 col-sm-6">
133     <div class="trading-metric-card">
134         <span class="trading-metric-label">Позиция / PnL</span
    >
135         <span class="trading-metric-value" id="trading-pnl-val
    " style="color: #4ade80;">+$0.00</span>
136         <span class="small text-muted" id="trading-pos-size">P
    азмep позиции: 0.0 BTC</span>
137     </div>
138 </div>
139 </div>
140
141 <!-- Main Trading Desk Grid -->
142 <div class="row g-3">
143     <!-- Left Column: Orderbook (L2 Depth) -->
144     <div class="col-lg-5">
145         <div class="trading-panel h-100">
146             <div class="trading-panel-header">
147                 <span><i class="bi bi-bar-chart-fill me-1"></i> Стак
    ан котировок (L2 Orderbook)</span>
148                 <span class="badge bg-secondary" id="trading-orderbo
    ok-depth">Глубина: 6</span>
149             </div>
150             <div class="p-0">
151                 <table class="trading-orderbook-table">
152                     <thead>
153                         <tr>
154                             <th>Цена (USDT)</th>
155                             <th style="text-align: right;">Объём</th>
156                             <th style="text-align: right;">Сумма</th>
157                         </tr>

```

```

158         </thead>
159         <tbody id="trading-asks-tbody">
160             <tr><td colspan="3" class="text-center py-2 text
-muted">Загрузка заявок на продажу...</td></tr>
161         </tbody>
162         <tbody style="border-top: 2px solid #334155; borde
r-bottom: 2px solid #334155;">
163             <tr style="background: #020617;">
164                 <td colspan="3" class="text-center py-2 fw-bol
d" id="trading-orderbook-midprice" style="font-size: 0.95rem;
color: #f8fafc;">
165                     $0.00
166                 </td>
167             </tr>
168         </tbody>
169         <tbody id="trading-bids-tbody">
170             <tr><td colspan="3" class="text-center py-2 text
-muted">Загрузка заявок на покупку...</td></tr>
171         </tbody>
172     </table>
173 </div>
174 </div>
175 </div>
176
177 <!-- Right Column: Order Placement & Recent Orders -->
178 <div class="col-lg-7 d-flex flex-column gap-3">
179     <!-- Order Form -->
180     <div class="trading-panel">
181         <div class="trading-panel-header">
182             <span><i class="bi bi-lightning-charge-fill me-1"></
i> Выставление ордера</span>
183             <span class="badge bg-primary" id="trading-mode-badg
e">Direct Execution</span>
184         </div>
185         <div class="p-3">
186             <form id="trading-order-form">
187                 <div class="row g-3">
188                     <div class="col-md-4">

```

```
189 |         <label class="form-label small fw-semibold tex  
| t-light">Тип ордера:</label>  
190 |         <select class="form-select form-select-sm bg-d  
| ark text-white border-secondary" id="trading-order-type">  
191 |             <option value="MARKET">Рыночный (Market)</op  
| tion>  
192 |             <option value="LIMIT">Лимитный (Limit)</opti  
| on>  
193 |         </select>  
194 |     </div>  
195 |     <div class="col-md-4">  
196 |         <label class="form-label small fw-semibold tex  
| t-light">Количество:</label>  
197 |         <div class="input-group input-group-sm">  
198 |             <input type="number" step="0.01" min="0.001"  
| class="form-control bg-dark text-white border-secondary" id="  
| trading-order-amount" value="0.1" required>  
199 |             <span class="input-group-text bg-secondary t  
| ext-white border-secondary">BTC</span>  
200 |         </div>  
201 |     </div>  
202 |     <div class="col-md-4">  
203 |         <label class="form-label small fw-semibold tex  
| t-light">Цена (для Limit):</label>  
204 |         <div class="input-group input-group-sm">  
205 |             <span class="input-group-text bg-secondary t  
| ext-white border-secondary">$</span>  
206 |             <input type="number" step="1" class="form-co  
| ntrol bg-dark text-white border-secondary" id="trading-order-p  
| rice" placeholder="По рынку">  
207 |         </div>  
208 |     </div>  
209 | </div>  
210 |  
211 |     <div class="d-flex justify-content-end gap-3 mt-3"  
| >  
212 |         <button type="button" class="btn btn-success fw-  
| bold px-4 rounded-pill" id="btn-order-buy">
```



```

213 |             <i class="bi bi-arrow-up-circle me-1"></i> Куп
      | ить (BUY)
214 |         </button>
215 |         <button type="button" class="btn btn-danger fw-b
      | old px-4 rounded-pill" id="btn-order-sell">
216 |             <i class="bi bi-arrow-down-circle me-1"></i> П
      | родать (SELL)
217 |         </button>
218 |     </div>
219 | </form>
220 | </div>
221 | </div>
222 |
223 | <!-- Recent Orders Table -->
224 | <div class="trading-panel flex-grow-1">
225 |     <div class="trading-panel-header">
226 |         <span><i class="bi bi-clock-history me-1"></i> Истоп
      | ия ордеров и исполнений</span>
227 |         <span class="badge bg-secondary" id="trading-orders-
      | count">0 ордеров</span>
228 |     </div>
229 |     <div class="table-responsive" style="max-height: 260px
      | ; overflow-y: auto;">
230 |         <table class="table table-dark table-hover table-sm
      | mb-0 small font-monospace">
231 |             <thead>
232 |                 <tr class="table-secondary text-dark">
233 |                     <th>Время</th>
234 |                     <th>ID</th>
235 |                     <th>Направление</th>
236 |                     <th>Тип</th>
237 |                     <th style="text-align: right;">Объём</th>
238 |                     <th style="text-align: right;">Цена исполнения
      | </th>
239 |                     <th style="text-align: right;">Статус</th>
240 |                 </tr>
241 |             </thead>
242 |             <tbody id="trading-orders-tbody">

```

```
243 |         <tr><td colspan="7" class="text-center py-3 text
    | -muted">История сделок пуста</td></tr>
244 |         </tbody>
245 |     </table>
246 | </div>
247 | </div>
248 | </div>
249 | </div>
250 | </div>
251 |
```

main.js

```
1 // Trading Terminal Tab JS Module
2 (function() {
3   let tradingWs = null;
4   let isTradingInitialized = false;
5
6   async function fetchTradingState() {
7     try {
8       const res = await fetch('/api/v1/trading/status');
9       if (!res.ok) return;
10      const data = await res.json();
11      updateTradingUI(data);
12    } catch (e) {
13      console.error('[TradingTab] Failed to fetch state:', e);
14    }
15  }
16
17  async function fetchOrderbook() {
18    try {
19      const res = await fetch('/api/v1/trading/orderbook?depth
20      =6');
21      if (!res.ok) return;
22      const data = await res.json();
23      renderOrderbook(data);
24    } catch (e) {
25      console.error('[TradingTab] Failed to fetch orderbook:',
26      e);
27    }
28  }
29
30  async function fetchOrdersList() {
31    try {
32      const res = await fetch('/api/v1/trading/orders');
33      if (!res.ok) return;
34      const orders = await res.json();
35      renderOrders(orders);
36    } catch (e) {
```

```

35 |     console.error('[TradingTab] Failed to fetch orders:', e)
    | ;
36 |   }
37 | }
38 |
39 | function updateTradingUI(state) {
40 |   if (!state) return;
41 |   const symEl = document.getElementById('trading-symbol');
42 |   const priceEl = document.getElementById('trading-current-p
rice');
43 |   const equityEl = document.getElementById('trading-total-eq
uity');
44 |   const balanceEl = document.getElementById('trading-cash-ba
lance');
45 |   const pnlEl = document.getElementById('trading-pnl-val');
46 |   const posEl = document.getElementById('trading-pos-size');
47 |
48 |   if (symEl) symEl.innerText = state.symbol || 'BTC/USDT';
49 |   if (priceEl) priceEl.innerText = `$$${Number(state.current_
price || 0).toLocaleString(undefined, { minimumFractionDigits:
2, maximumFractionDigits: 2 })}}`;
50 |   if (equityEl) equityEl.innerText = `$$${Number(state.total_
equity || 0).toLocaleString(undefined, { minimumFractionDigits
: 2, maximumFractionDigits: 2 })}}`;
51 |   if (balanceEl) balanceEl.innerText = `Доступно: $$${Number(
state.balance || 0).toLocaleString(undefined, { minimumFractio
nDigits: 2, maximumFractionDigits: 2 })}}`;
52 |
53 |   if (pnlEl) {
54 |     const pnl = Number(state.unrealized_pnl || 0);
55 |     const sign = pnl >= 0 ? '+' : '';
56 |     pnlEl.innerText = `${sign}$$${pnl.toLocaleString(undefine
d, { minimumFractionDigits: 2, maximumFractionDigits: 2 })}}`;
57 |     pnlEl.style.color = pnl >= 0 ? '#4ade80' : '#f87171';
58 |   }
59 |
60 |   if (posEl) {
61 |     posEl.innerText = `Размер позиции: ${state.position_size

```

```

    || 0} ${state.symbol?.split('/')[0] || 'BTC'}`;
62     }
63   }
64
65   function renderOrderbook(ob) {
66     if (!ob) return;
67     const asksTbody = document.getElementById('trading-asks-tb
ody');
68     const bidsTbody = document.getElementById('trading-bids-tb
ody');
69     const midEl = document.getElementById('trading-orderbook-m
idprice');
70     const spreadEl = document.getElementById('trading-spread-i
nfo');
71
72     if (midEl && ob.mid_price) {
73       midEl.innerText = `$$${Number(ob.mid_price).toLocaleStrin
g(undefined, { minimumFractionDigits: 2, maximumFractionDigits
: 2 })}`;
74     }
75     if (spreadEl && ob.spread !== undefined) {
76       spreadEl.innerText = `Спред: $$${Number(ob.spread).toFixe
d(2)}`;
77     }
78
79     if (asksTbody && Array.isArray(ob.asks)) {
80       // Show asks reversed (highest on top)
81       const reversedAsks = [...ob.asks].reverse();
82       asksTbody.innerHTML = reversedAsks.map(a => `
83         <tr class="trading-ask-row">
84           <td>$$${Number(a.price).toLocaleString(undefined, { m
inimumFractionDigits: 2, maximumFractionDigits: 2 })}</td>
85           <td style="text-align: right;">${Number(a.amount).to
Fixed(4)}</td>
86           <td style="text-align: right;">$$${Number(a.total).to
LocaleString(undefined, { minimumFractionDigits: 2, maximumFra
ctionDigits: 2 })}</td>
87         </tr>

```

```

88 |     `).join('');
89 | }
90 |
91 | if (bidsTbody && Array.isArray(ob.bids)) {
92 |     bidsTbody.innerHTML = ob.bids.map(b => `
93 |         <tr class="trading-bid-row">
94 |             <td>${Number(b.price).toLocaleString(undefined, { m
95 | inimumFractionDigits: 2, maximumFractionDigits: 2 })}</td>
96 |             <td style="text-align: right;">${Number(b.amount).to
97 | Fixed(4)}</td>
98 |             <td style="text-align: right;">${Number(b.total).to
99 | LocaleString(undefined, { minimumFractionDigits: 2, maximumFra
100 | ctionDigits: 2 })}</td>
101 |         </tr>
102 |     `).join('');
103 | }
104 | }
105 |
106 | function renderOrders(orders) {
107 |     const tbody = document.getElementById('trading-orders-tbod
108 | y');
109 |     const countEl = document.getElementById('trading-orders-co
110 | unt');
111 |     if (!tbody) return;
112 |
113 |     if (!Array.isArray(orders) || orders.length === 0) {
114 |         tbody.innerHTML = '<tr><td colspan="7" class="text-cen
115 | ter py-3 text-muted">История сделок пуста</td></tr>';
116 |         if (countEl) countEl.innerText = '0 ордеров';
117 |         return;
118 |     }
119 |
120 |     if (countEl) countEl.innerText = `${orders.length} ордеров
121 | `;
122 |
123 |     tbody.innerHTML = orders.map(o => {
124 |         const isBuy = (o.side || '').toUpperCase() === 'BUY';
125 |         const sideBadge = isBuy

```

```

118         ? '<span class="badge bg-success">BUY</span>'
119         : '<span class="badge bg-danger">SELL</span>';
120     return `
121         <tr>
122             <td>${new Date(o.timestamp * 1000).toLocaleTimeString()}</td>
123             <td class="text-muted">${o.order_id}</td>
124             <td>${sideBadge}</td>
125             <td>${o.order_type}</td>
126             <td style="text-align: right;">${Number(o.amount).toFixed(4)}</td>
127             <td style="text-align: right;">${Number(o.price).toLocaleString(
128                 undefined, { minimumFractionDigits: 2, maximumFractionDigits: 2 })}</td>
129             <td style="text-align: right;"><span class="badge bg-
130                 -secondary">${o.status}</span></td>
131         </tr>
132     `;
133     }).join('');
134 }
135
136 async function placeOrder(side) {
137     const amountInput = document.getElementById('trading-order-amount');
138     const priceInput = document.getElementById('trading-order-price');
139     const typeSelect = document.getElementById('trading-order-type');
140     const amount = parseFloat(amountInput?.value || '0');
141     const orderType = typeSelect?.value || 'MARKET';
142     const price = priceInput?.value ? parseFloat(priceInput.value) : null;
143
144     if (!amount || amount <= 0) {
145         alert('Укажите корректный объем ордера');
146         return;
147     }
148 }

```

```

147 |     try {
148 |         const payload = {
149 |             symbol: 'BTC/USDT',
150 |             side: side,
151 |             amount: amount,
152 |             order_type: orderType,
153 |             price: price
154 |         };
155 |
156 |         const res = await fetch('/api/v1/trading/order', {
157 |             method: 'POST',
158 |             headers: { 'Content-Type': 'application/json' },
159 |             body: JSON.stringify(payload)
160 |         });
161 |
162 |         if (!res.ok) {
163 |             const err = await res.json().catch(() => ({ detail: 'О
164 |шибка выполнения' }));
165 |             alert('Ошибка выставления ордера: ' + (err.detail || r
166 |es.statusText));
167 |             return;
168 |         }
169 |
170 |         await fetchTradingState();
171 |         await fetchOrdersList();
172 |         await fetchOrderbook();
173 |     } catch (e) {
174 |         console.error('[TradingTab] Order error:', e);
175 |         alert('Ошибка соединения при отправке ордера: ' + e.mess
176 |age);
177 |     }
178 | }
179 |
180 | async function triggerKillSwitch() {
181 |     if (!confirm('⚠ ВНИМАНИЕ: Вы действительно хотите немедлен
182 |но закрыть все позиции по рыночной цене (Kill-Switch)?')) {
183 |         return;
184 |     }

```



```
181
182     try {
183         const res = await fetch('/api/v1/trading/kill-switch', {
184             method: 'POST' });
185         if (!res.ok) throw new Error('Ошибка вызова Kill-Switch'
186         );
187         const data = await res.json();
188         alert(`Ликвидация завершена: ${data.message || 'Позиции
189         закрыты'}`);
190         await fetchTradingState();
191         await fetchOrdersList();
192     } catch (e) {
193         alert('Ошибка при вызове Kill-Switch: ' + e.message);
194     }
195
196     function isTradingTabActive() {
197         const el = document.getElementById('tab-trading');
198         if (!el) return false;
199         return el.classList.contains('active') || el.classList.con
200         tains('show') || (el.offsetWidth > 0 && el.offsetHeight > 0);
201     }
202
203     function connectTradingWebSocket() {
204         if (tradingWs) {
205             try { tradingWs.close(); } catch {}
206         }
207         const proto = window.location.protocol === 'https:' ? 'wss
208         :' : 'ws:';
209         const wsUrl = `${proto}://${window.location.host}/api/v1/tr
210         ading/ws/stream`;
211         try {
212             tradingWs = new WebSocket(wsUrl);
213             const statusBadge = document.getElementById('trading-str
214             eam-status');
215             tradingWs.onopen = () => {
```

```

212         if (statusBadge) {
213             statusBadge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2';
214             statusBadge.innerText = '● Стрим активен';
215         }
216     };
217
218     tradingWs.onmessage = (evt) => {
219         try {
220             const state = JSON.parse(evt.data);
221             updateTradingUI(state);
222             if (isTradingTabActive()) {
223                 fetchOrderbook();
224                 fetchOrdersList();
225             }
226         } catch (e) {
227             console.error('[TradingTab] WS message parse error:',
228                 , e);
229         }
230     };
231
232     tradingWs.onclose = () => {
233         if (statusBadge) {
234             statusBadge.className = 'badge rounded-pill bg-danger-subtle text-danger border border-danger px-3 py-2';
235             statusBadge.innerText = '● Отключено';
236         }
237     };
238     } catch (e) {
239         console.error('[TradingTab] WS connection error:', e);
240     }
241
242     function initTradingTab(force = false) {
243         if (!isTradingInitialized) {
244             const buyBtn = document.getElementById('btn-order-buy');
245             const sellBtn = document.getElementById('btn-order-sell'
246 );

```

```
246 |     const refreshBtn = document.getElementById('btn-refresh-
    | trading');
247 |     const killBtn = document.getElementById('btn-trigger-kill-
    | switch');
248 |
249 |     const configBtn = document.getElementById('btn-trading-c
    | onfig');
250 |
251 |     if (buyBtn) buyBtn.onclick = () => placeOrder('BUY');
252 |     if (sellBtn) sellBtn.onclick = () => placeOrder('SELL');
253 |     if (refreshBtn) refreshBtn.onclick = () => {
254 |         fetchTradingState();
255 |         fetchOrderbook();
256 |         fetchOrdersList();
257 |     };
258 |     if (killBtn) killBtn.onclick = triggerKillSwitch;
259 |     if (configBtn) configBtn.onclick = () => {
260 |         if (typeof window.openAppConfigModal === 'function') {
261 |             window.openAppConfigModal('trading_terminal', 'Excha
    | nge Trading Terminal');
262 |         }
263 |     };
264 |
265 |     isTradingInitialized = true;
266 | }
267 |
268 | if (!force && !isTradingTabActive()) {
269 |     console.log('[TradingTab] Tab is not active; skipping ba
    | ckground network polling and WebSocket connection.');
```

```
270 |     return;
271 | }
272 |
273 | console.log('[TradingTab] Initializing active Trading Term
    | inal network polling & WebSocket...');
274 | fetchTradingState();
275 | fetchOrderbook();
276 | fetchOrdersList();
277 | connectTradingWebSocket();
```

```
278 |     }  
279 |  
280 |     window.initTradingTab = initTradingTab;  
281 | })();  
282 |
```

index.html

```
1 <!-- TTS Settings Tab -->
2 <div class="container-fluid">
3   <div class="alert alert-info">
4     <h5 data-i18n="tts.title">□ Настройка TTS (Синтез речи)</
h5>
5     <p data-i18n="tts.description">Установите голос и движок п
о умолчанию для озвучки описания медиафайлов, а также протести
руйте их звучание.</p>
6   </div>
7
8   <div class="row g-3">
9     <!-- LEFT: Settings Configuration -->
10    <div class="col-5">
11      <div class="card mb-3">
12        <div class="card-header bg-dark text-white d-flex just
ify-content-between align-items-center">
13          <span>□ Параметры озвучки</span>
14        </div>
15        <div class="card-body">
16          <form id="admin-tts-form" onsubmit="event.preventDef
ault();">
17            <!-- Active Voice Badge -->
18            <div id="admin-active-voice-container" class="mb-3
" style="display: none;">
19              <span class="badge bg-success d-inline-flex alig
n-items-center gap-1 p-2">
20                <i class="bi bi-check-circle-fill"></i>
21                <span>Активный голос: <strong id="admin-active
-voice-name">...</strong></span>
22              </span>
23            </div>
24
25            <!-- Engine Selection -->
26            <div class="mb-3">
27              <label for="admin-tts-engine" class="form-label
small fw-bold">Система синтеза:</label>
```

```

28 |         <select id="admin-tts-engine" class="form-select
    | form-select-sm">
29 |             <option value="browser">Web Speech API (Браузе
    | рная)</option>
30 |             <option value="gtts">Google TTS (Базовая)</opt
    | ion>
31 |             <option value="edge-tts" selected>Microsoft Ed
    | ge TTS (Нейросетевая)</option>
32 |             <option value="silero">Silero TTS (Локальная о
    | ффлайн)</option>
33 |         </select>
34 |     </div>
35 |
36 |     <!-- Voice Selection -->
37 |     <div class="mb-3" id="admin-voice-select-container
    | ">
38 |         <label for="admin-tts-voice" class="form-label s
    | mall fw-bold">Голос / Спикер:</label>
39 |         <select id="admin-tts-voice" class="form-select
    | form-select-sm">
40 |             <!-- Dynamically populated -->
41 |         </select>
42 |     </div>
43 |
44 |     <!-- Save Default Button -->
45 |     <div class="d-grid">
46 |         <button type="button" class="btn btn-success btn
    | -sm" id="btn-admin-tts-save">
47 |             <i class="bi bi-floppy"></i> Применить по умол
    | чанию
48 |         </button>
49 |     </div>
50 | </form>
51 | </div>
52 | </div>
53 | </div>
54 |
55 | <!-- RIGHT: Text to Speech Playground -->

```

```

56 |     <div class="col-7">
57 |         <div class="card">
58 |             <div class="card-header bg-dark text-white">
59 |                 <span>□ Тестирование озвучки</span>
60 |             </div>
61 |             <div class="card-body">
62 |                 <div class="mb-3">
63 |                     <label for="admin-tts-text" class="form-label small fw-bold">Текст для проверки:</label>
64 |                     <textarea id="admin-tts-text" class="form-control form-control-sm" rows="5">Привет! Это проверка работы модульной системы синтеза речи. Выберите систему и голос слева, затем нажмите кнопку «Озвучить» для воспроизведения.</textarea>
65 |                 </div>
66 |
67 |                 <div class="d-flex align-items-center gap-3">
68 |                     <button type="button" class="btn btn-primary btn-sm px-4" id="btn-admin-tts-speak">
69 |                         <i class="bi bi-play-fill"></i> Озвучить текст
70 |                     </button>
71 |
72 |                     <!-- Loading Spinner -->
73 |                     <div class="status-badge small text-muted d-none" id="admin-tts-status">
74 |                         <div class="spinner-border spinner-border-sm text-primary" role="status"></div>
75 |                         <span id="admin-status-text" class="ms-1">Генерация аудио...</span>
76 |                     </div>
77 |                 </div>
78 |
79 |                 <!-- Audio Player Container -->
80 |                 <div class="mt-4 border-top pt-3 d-none" id="admin-tts-player-container">
81 |                     <label class="form-label small fw-bold">Результат генерации:</label>
82 |                     <audio id="admin-tts-audio-player" class="w-100" controls></audio>

```

```
83 |         </div>
84 |     </div>
85 | </div>
86 | </div>
87 | </div>
88 | </div>
89 |
```


main.js

```
1 // TTS Tab Configuration and Testing Logic
2 // Called via window.initTtsTab() when tab is loaded
3
4 'use strict';
5
6 function ttsFetch(url, options = {}) {
7     return window.api
8         ? window.api.fetch(url, options)
9         : fetch(url, options).then(r => {
10             if (!r.ok) throw new Error(`HTTP ${r.status}`);
11             return r.json();
12         });
13 }
14
15 function ttsNotify(msg, type = 'info') {
16     if (typeof showNotification === 'function') {
17         showNotification(msg, type);
18     } else {
19         console.log(`[${type}] ${msg}`);
20     }
21 }
22
23 // Available voices dictionary
24 const ADMIN_VOICE_MAP = {
25     'browser': [], // Populated dynamically from window.speechSynthesis
26     'gtts': [
27         { value: 'ru', label: 'Русский (Google TTS)' },
28         { value: 'en', label: 'English (Google TTS)' },
29         { value: 'he', label: 'עברית (Google TTS)' },
30         { value: 'de', label: 'Deutsch (Google TTS)' },
31         { value: 'fr', label: 'Français (Google TTS)' },
32         { value: 'es', label: 'Español (Google TTS)' }
33     ],
34     'edge-tts': [
35         { value: 'ru-RU-DmitryNeural', label: 'Дмитрий (RU - Мужск
```

```

ой)' },
36     { value: 'ru-RU-SvetlanaNeural', label: 'Светлана (RU - Же
НСКИЙ)' },
37     { value: 'en-US-JennyNeural', label: 'Jenny (US - Female)'
    },
38     { value: 'en-US-GuyNeural', label: 'Guy (US - Male)' },
39     { value: 'en-US-AriaNeural', label: 'Aria (US - Female)' }
    ,
40     { value: 'en-GB-SoniaNeural', label: 'Sonia (UK - Female)'
    },
41     { value: 'en-GB-RyanNeural', label: 'Ryan (UK - Male)' },
42     { value: 'he-IL-AvriNeural', label: 'Avri (HE - אברי)' },
43     { value: 'he-IL-HilaNeural', label: 'Hila (HE - הילה)' },
44     { value: 'de-DE-KatjaNeural', label: 'Katja (DE - Female)'
    },
45     { value: 'de-DE-ConradNeural', label: 'Conrad (DE - Male)'
    },
46     { value: 'fr-FR-DeniseNeural', label: 'Denise (FR - Female
)' },
47     { value: 'fr-FR-HenriNeural', label: 'Henri (FR - Male)' }
    ,
48     { value: 'es-ES-ElviraNeural', label: 'Elvira (ES - Female
)' },
49     { value: 'es-ES-AlvaroNeural', label: 'Alvaro (ES - Male)'
    }
50 ],
51 'silero': [
52     { value: 'eugene', label: 'Евгений (Silero RU)' },
53     { value: 'aidar', label: 'Айдар (Silero RU)' },
54     { value: 'baya', label: 'Бая (Silero RU)' },
55     { value: 'kseniya', label: 'Ксения (Silero RU)' },
56     { value: 'xenia', label: 'Ксения v2 (Silero RU)' },
57     { value: 'random', label: 'Рандомный (Silero RU)' }
58 ]
59 };
60
61 function getFriendlyVoiceName(system, voice) {
62     if (system === 'edge-tts') {

```

```
63     const found = ADMIN_VOICE_MAP['edge-tts'].find(v => v.value === voice);
64     return found ? `Edge: ${found.label}` : `Edge (${voice})`;
65 }
66 if (system === 'silero') {
67     const found = ADMIN_VOICE_MAP['silero'].find(v => v.value === voice);
68     return found ? found.label : `Silero (${voice})`;
69 }
70 if (system === 'gtts') {
71     const found = ADMIN_VOICE_MAP['gtts'].find(v => v.value ==
72     = voice);
73     return found ? found.label : `Google TTS (${voice})`;
74 }
75 if (system === 'browser') {
76     return `Браузерный (${voice} || 'По умолчанию')`;
77 }
78 return `${system} – ${voice}`;
79 }
80 async function initTtsTab() {
81     const engineSelect = document.getElementById('admin-tts-engine');
82     const voiceSelect = document.getElementById('admin-tts-voice');
83     const voiceContainer = document.getElementById('admin-voice-select-container');
84     const btnSpeak = document.getElementById('btn-admin-tts-speak');
85     const btnSave = document.getElementById('btn-admin-tts-save');
86     const ttsStatus = document.getElementById('admin-tts-status');
87     const statusText = document.getElementById('admin-status-text');
88     const playerContainer = document.getElementById('admin-tts-player-container');
89     const audioPlayer = document.getElementById('admin-tts-audio
```

```

    -player');
90   const activeVoiceContainer = document.getElementById('admin-
active-voice-container');
91   const activeVoiceName = document.getElementById('admin-activ
e-voice-name');
92
93   if (!engineSelect || !voiceSelect) {
94       console.error('TTS Settings Tab DOM elements not found.');
```

95 return;

96 }

97

98 function updateActiveBadge(system, voice) {

99 if (system && voice) {

100 activeVoiceName.textContent = getFriendlyVoiceName(system, voice);

101 activeVoiceContainer.style.display = 'block';

102 } else {

103 activeVoiceContainer.style.display = 'none';

104 }

105 }

106

107 // Populate voices based on selected engine

108 function populateVoices() {

109 const engine = engineSelect.value;

110 const voices = ADMIN_VOICE_MAP[engine] || [];

111

112 voiceSelect.innerHTML = '';

113

114 if (voices.length === 0) {

115 voiceContainer.style.display = 'none';

116 } else {

117 voiceContainer.style.display = 'block';

118 voices.forEach(voice => {

119 const opt = document.createElement('option');

120 opt.value = voice.value;

121 opt.textContent = voice.label;

122 voiceSelect.appendChild(opt);

123 });

```
124     }
125 }
126
127 // Load browser speech synthesis voices if available
128 function loadBrowserVoices() {
129     if ('speechSynthesis' in window) {
130         const voices = window.speechSynthesis.getVoices();
131         ADMIN_VOICE_MAP['browser'] = voices
132             .map(v => ({ value: v.name, label: `[${v.lang}] ${v.name}` }));
133
134         if (ADMIN_VOICE_MAP['browser'].length === 0) {
135             ADMIN_VOICE_MAP['browser'].push({ value: 'default', label: 'Браузер по умолчанию' });
136         }
137
138         if (engineSelect.value === 'browser') {
139             populateVoices();
140         }
141     }
142 }
143
144 if ('speechSynthesis' in window) {
145     window.speechSynthesis.onvoiceschanged = loadBrowserVoices;
146     loadBrowserVoices();
147 }
148
149 engineSelect.addEventListener('change', populateVoices);
150 populateVoices(); // Initial run
151
152 // Fetch current user settings
153 try {
154     const settings = await ttsFetch('/auth/settings');
155     if (settings) {
156         if (settings.tts_system) {
157             engineSelect.value = settings.tts_system;
158             populateVoices();
159         }
160     }
161 }
```

```
159 |     }
160 |     if (settings.tts_voice) {
161 |         voiceSelect.value = settings.tts_voice;
162 |     }
163 |     updateActiveBadge(settings.tts_system, settings.tts_voice);
164 | }
165 | } catch (e) {
166 |     console.error('Failed to load active TTS settings:', e);
167 | }
168 |
169 | // Speak button handler
170 | btnSpeak.addEventListener('click', async () => {
171 |     const text = document.getElementById('admin-tts-text').value.trim();
172 |     if (!text) {
173 |         ttsNotify('Пожалуйста, введите текст для озвучки', 'warning');
174 |         return;
175 |     }
176 |
177 |     const engine = engineSelect.value;
178 |     const voice = voiceSelect.value;
179 |
180 |     if (engine === 'browser') {
181 |         if ('speechSynthesis' in window) {
182 |             window.speechSynthesis.cancel();
183 |             const cleanText = text.replace(/<[^>]*>/g, '').replace(
184 | (/[*_`#]/g, ''));
185 |             const utterance = new SpeechSynthesisUtterance(cleanText);
186 |             const voices = window.speechSynthesis.getVoices();
187 |             const selectedVoice = voices.find(v => v.name === voice);
188 |             if (selectedVoice) {
189 |                 utterance.voice = selectedVoice;
190 |                 if (selectedVoice.lang) {
191 |                     utterance.lang = selectedVoice.lang;
192 |                 }
193 |                 window.speechSynthesis.speak(utterance);
194 |             }
195 |         }
196 |     }
197 | }
```

```

191         }
192     }
193
194     ttsStatus.classList.remove('d-none');
195     statusText.textContent = 'Воспроизведение браузером...';
196
197     utterance.onend = () => ttsStatus.classList.add('d-none');
198     utterance.onerror = () => ttsStatus.classList.add('d-none');
199
200     window.speechSynthesis.speak(utterance);
201     playerContainer.classList.add('d-none');
202 } else {
203     ttsNotify('Ваш браузер не поддерживает Web Speech API', 'danger');
204 }
205 return;
206 }
207
208 // Server-side synthesis
209 ttsStatus.classList.remove('d-none');
210 statusText.textContent = 'Синтез аудиофайла на сервере...';
211 ;
212 btnSpeak.disabled = true;
213 playerContainer.classList.add('d-none');
214
215 try {
216     const queryParams = new URLSearchParams({
217         text: text,
218         system: engine,
219         voice: voice
220     });
221
222     const audioUrl = `/api/tts/synthesize?${queryParams.toString}`;

```

```

223     audioPlayer.src = audioUrl;
224     audioPlayer.load();
225
226     audioPlayer.oncanplaythrough = () => {
227         ttsStatus.classList.add('d-none');
228         btnSpeak.disabled = false;
229         playerContainer.classList.remove('d-none');
230         audioPlayer.play().catch(err => console.log('Audio aut
    oplay blocked or failed:', err));
231     };
232
233     audioPlayer.onerror = (err) => {
234         console.error('Audio element error:', err);
235         statusText.textContent = 'Ошибка загрузки аудиофайла';
236         btnSpeak.disabled = false;
237         setTimeout(() => ttsStatus.classList.add('d-none'), 30
    00);
238     };
239
240     } catch (err) {
241         console.error(err);
242         statusText.textContent = 'Ошибка сети';
243         btnSpeak.disabled = false;
244         setTimeout(() => ttsStatus.classList.add('d-none'), 3000
    );
245     }
246 });
247
248 // Save Settings handler
249 btnSave.addEventListener('click', async () => {
250     const engine = engineSelect.value;
251     const voice = voiceSelect.value;
252
253     btnSave.disabled = true;
254     const originalText = btnSave.innerHTML;
255     btnSave.innerHTML = '<span class="spinner-border spinner-b
    order-sm" role="status" aria-hidden="true"></span> Сохранение.
    ..';

```



```
256 |
257 |     try {
258 |         const payload = {
259 |             tts_enabled: 1,
260 |             tts_system: engine,
261 |             tts_voice: voice
262 |         };
263 |
264 |         const response = await fetch('/auth/settings', {
265 |             method: 'POST',
266 |             headers: { 'Content-Type': 'application/json' },
267 |             body: JSON.stringify(payload)
268 |         });
269 |
270 |         if (response.ok) {
271 |             updateActiveBadge(engine, voice);
272 |             ttsNotify('Настройки TTS успешно применены по умолчанию!', 'success');
273 |         } else {
274 |             const errData = await response.json().catch(() => ({}))
275 |             );
276 |             throw new Error(errData.detail || 'Ошибка сохранения настроек');
277 |         }
278 |     } catch (err) {
279 |         console.error(err);
280 |         ttsNotify('Не удалось сохранить настройки: ' + err.message, 'danger');
281 |     } finally {
282 |         btnSave.disabled = false;
283 |         btnSave.innerHTML = originalText;
284 |     }
285 | });
286 |
287 | window.initTtsTab = initTtsTab;
288 |
```

google_signin.html

```
1 <!-- Google Sign-In Component -->
2 <div id="google-signin-container" class="d-flex align-items-center gap-2">
3   <button id="google-signin-btn" class="btn btn-outline-light btn-sm">
4     <i class="bi bi-google"></i>
5     <span data-i18n="auth.googleSignIn">Войти через Google</span>
6   </button>
7   <div id="user-profile" class="d-none flex-grow-1">
8     <div class="d-flex align-items-center gap-2">
9       <img id="user-avatar" src="" class="rounded-circle" width="32" height="32">
10      <span id="user-name" class="text-muted small"></span>
11      <button id="logout-btn" class="btn btn-sm btn-danger">
12        <i class="bi bi-box-arrow-right"></i> Выйти
13      </button>
14    </div>
15  </div>
16 </div>
17
18 <script>
19 // Google Sign-In Component Logic
20 (async () => {
21   const googleSignInBtn = document.getElementById('google-signin-btn');
22   const logoutBtn = document.getElementById('logout-btn');
23   const userProfile = document.getElementById('user-profile');
24   const userAvatar = document.getElementById('user-avatar');
25   const userName = document.getElementById('user-name');
26
27   // Check authentication status
28   async function checkAuthStatus() {
29     try {
30       const response = await fetch('/auth/check');
31       const data = await response.json();
```

```

32
33     if (data.authenticated) {
34         // User is authenticated
35         userProfile.classList.remove('d-none');
36         googleSignInBtn.classList.add('d-none');
37
38         if (data.name) {
39             userName.textContent = data.name;
40         }
41         if (data.email) {
42             userName.textContent = data.email.split('@')[0];
43         }
44         if (data.picture) {
45             userAvatar.src = data.picture;
46         }
47     } else {
48         // User is not authenticated
49         userProfile.classList.add('d-none');
50         googleSignInBtn.classList.remove('d-none');
51     }
52 } catch (error) {
53     console.error('Auth check failed:', error);
54     userProfile.classList.add('d-none');
55     googleSignInBtn.classList.remove('d-none');
56 }
57 }
58
59 // Handle Google Sign-In click
60 googleSignInBtn?.addEventListener('click', () => {
61     window.location.href = '/auth/google?next=' + encodeURIComponent(
window.location.pathname + window.location.search);
62 });
63
64 // Handle logout click
65 logoutBtn?.addEventListener('click', async () => {
66     try {
67         await fetch('/auth/logout', { method: 'POST' });
68         // Reload page to update UI

```

```
69 |         window.location.reload();
70 |     } catch (error) {
71 |         console.error('Logout failed:', error);
72 |     }
73 | });
74 |
75 | // Initial auth check
76 | checkAuthStatus();
77 | })();
78 | </script>
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
cale=1.0">
6   <title>AI Assistant</title>
7   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
t/css/bootstrap.min.css" rel="stylesheet">
8   <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.1
0.0/font/bootstrap-icons.css" rel="stylesheet">
9   <link href="/html/css/main.css?v=20260924_v1" rel="styleshee
t">
10  <script src="/html/js/toast.js?v=20260924_v1"></script>
11  <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i1
8next.min.js"></script>
12  <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
js"></script>
13 </head>
14 <body>
15 <div class="container-fluid mt-2" id="user-interface">
16
17   <div class="d-flex align-items-center justify-content-betwee
n mb-3 main-nav-container rounded px-2 py-1">
18     <ul class="nav nav-tabs flex-nowrap flex-grow-1 border-0 m
e-2 align-items-center" id="mainTabs">
19       <!-- Группа: Стандартные -->
20       <li class="nav-item dropdown">
21         <button class="nav-link dropdown-toggle active d-flex
align-items-center gap-1" type="button" id="standardTabsDropdo
wn">
22           <i class="bi bi-grid-fill text-primary me-1"></i>
23           <span data-i18n="tabs.groupStandard">Стандартные</sp
an>
24         </button>
25         <div class="dropdown-menu shadow border-secondary-subt
```

```

le">
26         <button class="dropdown-item active d-flex align-ite
ms-center gap-2" data-tab="tab-chat" data-bs-target="#tab-chat
" type="button" data-i18n="tabs.chat">
27             □ Чат
28         </button>
29         <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-voice" data-bs-target="#tab-voice" typ
e="button" data-i18n="tabs.voice">
30             □ Голос и аудио
31         </button>
32     </div>
33 </li>
34
35     <!-- Разделитель -->
36     <li class="nav-item d-flex align-items-center">
37         <div class="nav-group-divider"></div>
38     </li>
39
40     <!-- Группа: Расширенные -->
41     <li class="nav-item dropdown">
42         <button class="nav-link dropdown-toggle d-flex align-i
tems-center gap-1" type="button" id="extendedTabsDropdown">
43             <i class="bi bi-sliders2 text-info me-1"></i>
44             <span data-i18n="tabs.groupExtended">Расширенные</sp
an>
45         </button>
46         <div class="dropdown-menu shadow border-secondary-subt
le">
47             <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-rag" data-bs-target="#tab-rag" type="b
utton" data-i18n="tabs.rag">
48                 □ RAG
49             </button>
50             <button class="dropdown-item d-flex align-items-cent
er gap-2" data-tab="tab-tts" data-bs-target="#tab-tts" type="b
utton" data-i18n="tabs.tts">
51                 □ Настройка TTS

```

```

52 |         </button>
53 |         <button class="dropdown-item d-flex align-items-cent
    | er gap-2" data-tab="tab-skills" data-bs-target="#tab-skills" t
    | ype="button" data-i18n="tabs.skills">
54 |             □ Навыки
55 |         </button>
56 |         <button class="dropdown-item d-flex align-items-cent
    | er gap-2" data-tab="tab-news" data-bs-target="#tab-news" type=
    | "button" data-i18n="tabs.news">
57 |             □ Новости
58 |         </button>
59 |         <button class="dropdown-item d-flex align-items-cent
    | er gap-2" data-tab="tab-user-directories" data-bs-target="#tab
    | -user-directories" type="button" data-i18n="tabs.userDirectori
    | es">
60 |             □ Директории и файлы
61 |         </button>
62 |     </div>
63 | </li>
64 |
65 | <!-- Разделитель -->
66 | <li class="nav-item d-flex align-items-center">
67 |     <div class="nav-group-divider"></div>
68 | </li>
69 |
70 | <!-- Группа: Плагины -->
71 | <li class="nav-item dropdown" id="pluginsNavDropdownWrap
    | per">
72 |     <button class="nav-link dropdown-toggle d-flex align-i
    | tems-center gap-1" type="button" id="pluginsTabsDropdown">
73 |         <i class="bi bi-puzzle-fill text-warning me-1"></i>
74 |         <span data-i18n="tabs.plugins">Плагины</span>
75 |     </button>
76 |     <div class="dropdown-menu shadow border-secondary-subtl
    | e" id="nav-plugins-dropdown-menu">
77 |         <div id="nav-enabled-plugins-list" class="nav-enable
    | d-plugins-list">
78 |             <!-- Dynamic enabled plugins will be injected here

```

```

-->
79         </div>
80         <hr class="dropdown-divider my-1">
81         <button class="dropdown-item d-flex align-items-center gap-2" data-tab="tab-plugins" data-bs-target="#tab-plugins"
type="button" data-i18n="tabs.managePlugins">
82             □ Управление
83         </button>
84     </div>
85 </li>
86 <!-- Разделитель -->
87 <li class="nav-item d-flex align-items-center">
88     <div class="nav-group-divider"></div>
89 </li>
90
91 <!-- Группа: Сервисы -->
92 <li class="nav-item dropdown">
93     <button class="nav-link dropdown-toggle d-flex align-items-center gap-1" type="button" id="userServicesDropdown">
94         <i class="bi bi-grid-3x3-gap-fill text-success me-1"
></i>
95         <span data-i18n="tabs.groupUserServices">Сервисы</span>
96     </button>
97     <div class="dropdown-menu shadow border-secondary-subtle">
98         <button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-check-mail" type="button">
99             <i class="bi bi-envelope-check text-primary"></i>
100             <span>Почта</span>
101         </button>
102         <button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-search-gdocs" type="button">
103             <i class="bi bi-file-earmark-text text-info"></i>
104             <span>Google Docs</span>
105         </button>
106         <button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-gsheets" type="button">

```



```

107 |         <i class="bi bi-file-earmark-spreadsheet text-succ
    | ess"></i>
108 |         <span>Google Таблицы</span>
109 |     </button>
110 |     <button class="dropdown-item d-flex align-items-cent
    | er gap-2 py-1" id="nav-btn-news-digest" type="button">
111 |         <i class="bi bi-newspaper text-warning"></i>
112 |         <span>Новости</span>
113 |     </button>
114 |     <button class="dropdown-item d-flex align-items-cent
    | er gap-2 py-1" id="nav-btn-facebook-post" type="button">
115 |         <i class="bi bi-facebook text-primary"></i>
116 |         <span>Facebook</span>
117 |     </button>
118 | </div>
119 | </li>
120 | </ul>
121 |
122 | <div class="d-flex align-items-center gap-2 flex-shrink-0
    | ms-auto pb-1" id="nav-auth-container">
123 |     <!-- Google OAuth Register / Sign-in Button (Shown when
    | not logged in) -->
124 |     <button class="btn btn-sm btn-light text-dark fw-semibol
    | d d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-
    | pill btn-google-login" id="nav-google-btn" title="Войти или за
    | регистрироваться через Google" data-i18n-title="auth.googleSig
    | nInTitle" style="border: 1px solid rgba(0,0,0,0.15);">
125 |         <svg width="18" height="18" viewBox="0 0 24 24"><path
    | fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h
    | 5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.
    | 74 3.28-8.09z"/><path fill="#34A853" d="M12 23c2.97 0 5.46-.98
    | 7.28-2.66l-3.57-2.77c-.98-.66-2.23 1.06-3.71 1.06-2.86 0-5.29-
    | 1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/><path fill
    | ="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.
    | 35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.
    | 85-2.22.81-.63z"/><path fill="#EA4335" d="M12 5.38c1.62 0 3.06
    | .56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.4
    | 7 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/></svg>

```

```

126 |         <span data-i18n="auth.googleSignUp">Регистрация через
      | Google</span>
127 |     </button>
128 |
129 |     <!-- Authenticated User Menu Dropdown (Shown when logged
      | in) -->
130 |     <div class="dropdown d-none" id="user-menu-dropdown-wrap
      | per">
131 |         <button class="btn btn-sm btn-outline-secondary d-flex
      | align-items-center gap-2 px-2 py-1 rounded-pill dropdown-togg
      | le" id="user-settings-btn" type="button" data-bs-toggle="drop
      | down" aria-expanded="false" title="Пользовательское меню">
132 |             <img id="nav-user-avatar" src="" class="rounded-circ
      | le d-none" width="22" height="22" style="object-fit: cover;">
133 |             <i class="bi bi-person-circle" id="nav-user-icon"></
      | i>
134 |             <span id="nav-user-label" class="small fw-semibold"
      | data-i18n="userSettings.button">Пользователь</span>
135 |         </button>
136 |         <ul class="dropdown-menu dropdown-menu-end shadow bord
      | er-secondary-subtle py-1" aria-labelledby="user-settings-btn"
      | style="min-width: 220px;">
137 |             <li class="px-3 py-2 border-bottom border-secondary-
      | subtle">
138 |                 <div class="fw-bold text-truncate" id="user-menu-d
      | isplay-name">Пользователь</div>
139 |                 <div class="small text-muted text-truncate" id="us
      | er-menu-display-email" style="font-size: 0.75rem;">user@gmail.
      | com</div>
140 |             </li>
141 |             <li><h6 class="dropdown-header small text-uppercase
      | text-muted fw-semibold mt-1">Сервисы</h6></li>
142 |             <li>
143 |                 <button class="dropdown-item d-flex align-items-ce
      | nter gap-2 py-1" type="button" id="user-menu-btn-mail">
144 |                     <i class="bi bi-envelope-check text-primary"></i
      | >
145 |                     <span>Почта</span>

```

```
146         </button>
147     </li>
148     <li>
149         <button class="dropdown-item d-flex align-items-ce
150 nter gap-2 py-1" type="button" id="user-menu-btn-docs">
151             <i class="bi bi-file-earmark-text text-info"></i>
152             <span>Google Docs</span>
153         </button>
154     </li>
155     <li>
156         <button class="dropdown-item d-flex align-items-ce
157 nter gap-2 py-1" type="button" id="user-menu-btn-sheets">
158             <i class="bi bi-file-earmark-spreadsheet text-su
159 ccess"></i>
160             <span>Google Таблицы</span>
161         </button>
162     </li>
163     <li>
164         <button class="dropdown-item d-flex align-items-ce
165 nter gap-2 py-1" type="button" id="user-menu-btn-news">
166             <i class="bi bi-newspaper text-warning"></i>
167             <span>Новости</span>
168         </button>
169     </li>
170     <li>
171         <button class="dropdown-item d-flex align-items-ce
172 nter gap-2 py-1" type="button" id="user-menu-btn-facebook">
173             <i class="bi bi-facebook text-primary"></i>
174             <span>Facebook</span>
175         </button>
176     </li>
177     <li><hr class="dropdown-divider my-1"></li>
178     <li>
179         <button class="dropdown-item d-flex align-items-ce
180 nter gap-2 py-1" type="button" data-bs-toggle="modal" data-bs-
181 target="#userSettingsModal">
182             <i class="bi bi-gear-fill text-secondary"></i>
```

```

176 |         <span data-i18n="userSettings.title">Настройки п
    | рофиля</span>
177 |     </button>
178 | </li>
179 | <li>
180 |     <button class="dropdown-item d-flex align-items-ce
    | nter gap-2 py-1 text-danger btn-user-logout" type="button">
181 |         <i class="bi bi-box-arrow-right"></i>
182 |         <span>Выйти</span>
183 |     </button>
184 | </li>
185 | </ul>
186 | </div>
187 |
188 |
189 | <!-- Quick Settings Gear Button -->
190 | <button class="btn btn-sm btn-outline-secondary d-flex a
    | lign-items-center justify-content-center p-0 rounded-circle" i
    | d="user-settings-gear-btn" data-bs-toggle="modal" data-bs-targ
    | et="#userSettingsModal" title="Настройки" style="width: 32px;
    | height: 32px;" data-i18n-title="userSettings.title">
191 |     <i class="bi bi-gear-fill"></i>
192 | </button>
193 | </div>
194 | </div>
195 |
196 | <div class="tab-content" id="mainTabContent">
197 |     <!-- Chat Tab -->
198 |     <div id="tab-chat" class="tab-pane fade show active"></div
    | >
199 |
200 |     <!-- RAG Tab -->
201 |     <div id="tab-rag" class="tab-pane fade"></div>
202 |
203 |     <!-- Telegram RAG Tab -->
204 |     <div id="tab-telegram-rag" class="tab-pane fade"></div>
205 |
206 |     <!-- News Tab -->

```

```

207 |     <div id="tab-news" class="tab-pane fade"></div>
208 |
209 |     <!-- TTS Tab -->
210 |     <div id="tab-tts" class="tab-pane fade"></div>
211 |
212 |     <!-- Voice Tab -->
213 |     <div id="tab-voice" class="tab-pane fade"></div>
214 |
215 |     <!-- Skills Tab -->
216 |     <div id="tab-skills" class="tab-pane fade"></div>
217 |
218 |     <!-- Plugins Tab -->
219 |     <div id="tab-plugins" class="tab-pane fade"></div>
220 |
221 |     <!-- User Directories Tab -->
222 |     <div id="tab-user-directories" class="tab-pane fade"></div>
223 | >
224 | </div>
225 |
226 | <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    | /js/bootstrap.bundle.min.js"></script>
227 | <script src="/html/js/chatService.js?v=20260924_v1"></script>
228 | <script type="module" src="/html/user/main.js?v=20260924_v1"><
    | /script>
229 |
230 | <!-- Chat Logic & RAG Architecture Modal -->
231 | <div class="modal fade" id="chat-logic-modal" tabindex="-1" ar
    | ia-labelledby="chat-logic-modal-label" aria-hidden="true">
232 |     <div class="modal-dialog modal-xl modal-dialog-scrollable">
233 |         <div class="modal-content bg-dark text-white border-second
    | ary">
234 |             <div class="modal-header border-secondary d-flex justify
    | -content-between align-items-center">
235 |                 <h5 class="modal-title d-flex align-items-center gap-2
    | " id="chat-logic-modal-label">
236 |                     <i class="bi bi-diagram-3-fill text-info"></i> □ Ap
    | хитектура и алгоритм диспетчеризации ответов чата

```

```
237 |         </h5>
238 |         <button type="button" class="btn-close btn-close-white
| " data-bs-dismiss="modal" aria-label="Close"></button>
239 |     </div>
240 |     <div class="modal-body p-3">
241 |         <div class="p-2 border border-secondary rounded bg-bla
| ck d-flex justify-content-center align-items-center" style="ov
| erflow-x: auto; min-height: 520px;">
242 |             
243 |         </div>
244 |     </div>
245 |     <div class="modal-footer border-secondary">
246 |         <button type="button" class="btn btn-sm btn-secondary"
| data-bs-dismiss="modal">Закрыть</button>
247 |     </div>
248 | </div>
249 | </div>
250 | </div>
251 |
252 | <!-- User Settings Modal -->
253 | <div class="modal fade" id="userSettingsModal" tabindex="-1" a
| ria-labelledby="userSettingsModalLabel" aria-hidden="true">
254 |     <div class="modal-dialog modal-dialog-centered">
255 |         <div class="modal-content shadow border-secondary-subtle">
256 |             <div class="modal-header border-secondary-subtle">
257 |                 <h5 class="modal-title d-flex align-items-center gap-2
| " id="userSettingsModalLabel">
258 |                     <i class="bi bi-person-gear text-primary"></i>
259 |                     <span data-i18n="userSettings.title">Пользовательски
| е настройки</span>
260 |                 </h5>
261 |                 <button type="button" class="btn-close" data-bs-dismis
| s="modal" aria-label="Close"></button>
262 |             </div>
263 |             <div class="modal-body p-3">
264 |                 <!-- User Account & Google OAuth Card -->
```

```

265 |         <div class="card bg-body-tertiary border-secondary-sub
      | tle mb-3 p-3" id="user-settings-auth-card">
266 |             <div class="d-flex align-items-center gap-3">
267 |                 <img id="user-settings-avatar" src="" class="round
      | ed-circle border border-2 border-primary" width="48" height="4
      | 8" style="object-fit: cover;">
268 |                 <div class="flex-grow-1 min-w-0">
269 |                     <div class="d-flex align-items-center gap-2">
270 |                         <strong id="user-settings-name" class="text-tr
      |uncate">Пользователь</strong>
271 |                         <span id="user-settings-role-badge" class="bad
      |ge bg-primary">User</span>
272 |                     </div>
273 |                     <div id="user-settings-email" class="small text-
      |muted text-truncate">email@gmail.com</div>
274 |                     <div id="user-settings-google-badge" class="smal
      |l mt-1 text-muted">
275 |                         <i class="bi bi-google text-danger me-1"></i>G
      |oogle OAuth
276 |                     </div>
277 |                 </div>
278 |                 <button type="button" class="btn btn-outline-dange
      |r btn-sm btn-user-logout" title="Выйти">
279 |                     <i class="bi bi-box-arrow-right"></i>
280 |                 </button>
281 |             </div>
282 |         </div>
283 |
284 |         <!-- Guest / Connect Google OAuth Card -->
285 |         <div class="card bg-body-tertiary border-secondary-sub
      |tle mb-3 p-3 text-center d-none" id="user-settings-guest-card"
      |>
286 |             <div class="text-muted small mb-2">
287 |                 <i class="bi bi-shield-lock me-1"></i> Авторизация
      |и синхронизация аккаунта
288 |             </div>
289 |             <button type="button" class="btn btn-outline-danger
      |btn-sm w-100 d-flex align-items-center justify-content-center

```

```

gap-2 btn-google-login">
290     <i class="bi bi-google"></i>
291     <span>Войти через Google (OAuth)</span>
292 </button>
293 <div class="small text-muted mt-2" style="font-size:
    0.75rem;">
294     Синхронизация с Google Drive, Docs, Calendar и Con
tacts
295 </div>
296 </div>
297
298 <!-- Language Selection -->
299 <div class="mb-3">
300     <label class="form-label fw-semibold small d-flex al
    ign-items-center gap-2" for="modal-lang-selector">
301         <i class="bi bi-translate text-info"></i>
302         <span data-i18n="userSettings.language">Язык интер
фейса</span>
303     </label>
304     <select class="form-select lang-selector" id="modal-
    lang-selector">
305         <option value="ru">Русский</option>
306         <option value="en">English</option>
307         <option value="he">עברית</option>
308     </select>
309 </div>
310
311 <!-- Theme Selection -->
312 <div class="mb-2">
313     <label class="form-label fw-semibold small d-flex al
    ign-items-center gap-2" for="modal-theme-selector">
314         <i class="bi bi-palette text-warning"></i>
315         <span data-i18n="userSettings.theme">Тема оформлен
ия</span>
316     </label>
317     <select class="form-select theme-selector" id="modal-
    -theme-selector">
318         <option value="light" data-i18n="userSettings.them

```



```
    eLight">☐ Светлая</option>
319 |         <option value="dark" data-i18n="userSettings.theme
    Dark">☐ Тёмная</option>
320 |         <option value="system" data-i18n="userSettings.the
    meSystem">☐ Системная</option>
321 |     </select>
322 | </div>
323 | </div>
324 | <div class="modal-footer border-secondary-subtle py-2">
325 |     <button type="button" class="btn btn-secondary btn-sm"
    data-bs-dismiss="modal" data-i18n="userSettings.close">Закрыт
    ь</button>
326 | </div>
327 | </div>
328 | </div>
329 | </div>
330 |
331 | </body>
332 | </html>
```

main.js

```
1 // User Interface Main JS
2 import { initI18n, switchLang, applyTranslations } from '../js
  /i18n.js';
3 import { initTheme, setTheme, getThemeMode, getResolvedTheme }
  from '../js/theme.js';
4 import { initUserSettings, refreshUserProfile } from '../js/us
  erSettings.js';
5
6 // Make switchLang and theme functions available globally
7 window.switchLang = switchLang;
8 window.applyTranslations = applyTranslations;
9 window.setTheme = setTheme;
10 window.getThemeMode = getThemeMode;
11 window.getResolvedTheme = getResolvedTheme;
12 window.initUserSettings = initUserSettings;
13 window.refreshUserProfile = refreshUserProfile;
14
15 // Helper to open plugin from top navbar dropdown
16 window.openPluginFromDropdown = function(pluginName) {
17   if (!pluginName) return;
18   document.querySelectorAll('#mainTabs .dropdown-menu.show').f
  orEach((m) => {
19     m.classList.remove('show');
20     m.closest('.dropdown')?.querySelector('.dropdown-toggle')?
  .classList.remove('show');
21   });
22
23   const normalized = pluginName.toLowerCase().replace(/-/g, '_
  ');
24
25   if (normalized === 'telegram_channel_rag' || normalized ===
  'telegram_rag') {
26     const pane = document.getElementById('tab-telegram-rag') |
  | document.getElementById('tab-telegram_rag');
27     if (pane && typeof window.switchTab === 'function') {
28       window.switchTab('tab-telegram-rag');
```

```

29     return;
30 }
31 }
32
33 if (normalized === 'news_feed' || normalized === 'news' || normalized === 'smart_news' || normalized === 'smart_feed_news') {
34     const pane = document.getElementById('tab-news');
35     if (pane && typeof window.switchTab === 'function') {
36         window.switchTab('tab-news');
37         return;
38     }
39 }
40
41 if (typeof window.switchTab === 'function') {
42     window.switchTab('tab-plugins');
43 }
44
45 const select = () => {
46     if (typeof window.selectPlugin === 'function') {
47         window.selectPlugin(pluginName);
48     }
49 };
50
51 select();
52 setTimeout(select, 150);
53 setTimeout(select, 400);
54 };
55
56 // Global API module
57 window.api = {
58     async fetch(url, options = {}) {
59         const response = await fetch(url, options);
60         if (!response.ok) {
61             let msg = response.statusText;
62             try {
63                 const data = await response.json();
64                 if (data && data.detail) {

```

```

65 |         if (typeof data.detail === 'string') {
66 |             msg = data.detail;
67 |         } else if (Array.isArray(data.detail)) {
68 |             msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
69 |         } else {
70 |             msg = JSON.stringify(data.detail);
71 |         }
72 |     }
73 | } catch {}
74 |     throw new Error(`${response.status} ${msg}`);
75 | }
76 |     return response.json();
77 | }
78 | };
79 |
80 | // UI Notifications
81 | function showNotification(message, type = 'info') {
82 |     const notification = document.createElement('div');
83 |     notification.className = `alert alert-${type} position-fixed
top-0 end-0 m-3 shadow`;
84 |     notification.style.zIndex = '9999';
85 |     notification.style.maxWidth = '400px';
86 |     notification.textContent = message;
87 |     document.body.appendChild(notification);
88 |
89 |     setTimeout(() => {
90 |         notification.remove();
91 |     }, 5000);
92 | }
93 | window.showNotification = showNotification;
94 |
95 | // Tab Switch Handler
96 | function onTabSwitched(targetId) {
97 |     const cleanId = targetId.startsWith('#') ? targetId.slice(1)
: targetId;
98 |     if (cleanId === 'tab-chat') {
99 |         const msgInput = document.getElementById('message-input');

```

```
100     if (msgInput) msgInput.focus();
101   } else if (cleanId === 'tab-rag' && typeof window.initRagTab
    === 'function') {
102     console.log('[UserInterface] Switching to RAG tab...');
103     window.initRagTab();
104   } else if ((cleanId === 'tab-telegram-rag' || cleanId === 't
    ab-telegram_rag') && typeof window.initTelegramRagTab === 'fun
    ction') {
105     console.log('[UserInterface] Switching to Telegram RAG tab
    ...');
106     window.initTelegramRagTab();
107   } else if (cleanId === 'tab-tts' && typeof window.initTtsTab
    === 'function') {
108     console.log('[UserInterface] Switching to TTS tab...');
109     window.initTtsTab();
110   } else if (cleanId === 'tab-voice' && typeof window.initVoic
    eTab === 'function') {
111     console.log('[UserInterface] Switching to Voice tab...');
112     window.initVoiceTab();
113   } else if (cleanId === 'tab-skills' && typeof window.initSki
    llsTab === 'function') {
114     console.log('[UserInterface] Switching to Skills tab...');
115     window.initSkillsTab();
116   } else if (cleanId === 'tab-news' && typeof window.initNewsT
    ab === 'function') {
117     console.log('[UserInterface] Switching to News tab...');
118     window.initNewsTab();
119   } else if (cleanId === 'tab-plugins' && typeof window.initPl
    uuginsTab === 'function') {
120     console.log('[UserInterface] Switching to Plugins tab...')
    ;
121     window.initPluginsTab();
122   } else if (cleanId === 'tab-user-directories' && typeof wind
    ow.initUserDirectoriesTab === 'function') {
123     console.log('[UserInterface] Switching to User Directories
    tab...');
124     window.initUserDirectoriesTab();
125   }
```

```
126 }
127
128 // Карта конфигураций вкладок
129 const USER_TAB_DEFS = {
130   'chat': { html: '/html/chat/index.html', js: '/html/chat/main.js' },
131   'rag': { html: '/html/rag_tab/index.html', js: '/html/rag_tab/main.js' },
132   'telegram-rag': { html: '/html/telegram_rag_tab/index.html',
133     js: '/html/telegram_rag_tab/main.js' },
134   'news': { html: '/html/news_tab/index.html', js: '/html/news_tab/main.js' },
135   'tts': { html: '/html/tts_tab/index.html', js: '/html/tts_tab/main.js' },
136   'voice': { html: '/html/voice_tab/index.html', js: '/html/voice_tab/main.js' },
137   'skills': { html: '/html/skills_tab/index.html', js: '/html/skills_tab/main.js' },
138   'plugins': { html: '/html/plugins_tab/index.html', js: '/html/plugins_tab/main.js' },
139   'user-directories': { html: '/html/user_directories_tab/index.html',
140     js: '/html/user_directories_tab/main.js' },
141 };
142
143 const loadedUserTabs = new Set();
144 const loadingUserTabs = new Map();
145
146 async function ensureTabLoaded(tabName) {
147   if (loadedUserTabs.has(tabName)) return true;
148   if (loadingUserTabs.has(tabName)) return loadingUserTabs.get(tabName);
149
150   const def = USER_TAB_DEFS[tabName];
151   if (!def) return false;
152
153   const cb = Date.now();
154   const promise = (async () => {
155     try {
```

```

154     await loadTabContent(tabName, `${def.html}?v=${cb}`, `${
def.js}?v=${cb}`);
155     loadedUserTabs.add(tabName);
156     return true;
157   } finally {
158     loadingUserTabs.delete(tabName);
159   }
160   })();
161
162   loadingUserTabs.set(tabName, promise);
163   return promise;
164 }
165
166 // Programmatic tab switcher
167 async function switchTab(targetId) {
168   if (!targetId) return;
169   const cleanId = targetId.startsWith('#') ? targetId.slice(1)
: targetId;
170   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
cleanId}`;
171   const tabName = tabId.replace(/^tab-/, '');
172
173   // 1. Update active state on dropdown items & toggles
174   document.querySelectorAll('#mainTabs .dropdown-item').forEac
h((item) => {
175     const itemTarget = item.getAttribute('data-tab') || item.g
etAttribute('data-bs-target')?.replace('#', '');
176     if (itemTarget === tabId || itemTarget === cleanId) {
177       item.classList.add('active');
178     } else {
179       item.classList.remove('active');
180     }
181   });
182
183   document.querySelectorAll('#mainTabs .dropdown').forEach((dr
opdown) => {
184     const toggle = dropdown.querySelector('.dropdown-toggle');
185     const hasActiveChild = dropdown.querySelector('.dropdown-i

```

```
tem.active');
186     if (toggle) {
187         if (hasActiveChild) {
188             toggle.classList.add('active');
189         } else {
190             toggle.classList.remove('active');
191         }
192     }
193 });
194
195 // 2. Switch tab-pane (scoped to top-level container to preserve nested subtabs)
196 const mainTabContent = document.getElementById('mainTabContent');
197 if (mainTabContent) {
198     Array.from(mainTabContent.children).forEach((pane) => {
199         if (pane.classList.contains('tab-pane')) {
200             pane.classList.remove('show', 'active');
201         }
202     });
203 } else {
204     document.querySelectorAll('#mainTabContent > .tab-pane, body > .container-fluid > .tab-content > .tab-pane').forEach((pane) => {
205         pane.classList.remove('show', 'active');
206     });
207 }
208 const targetPane = document.getElementById(tabId) || document.getElementById(cleanId);
209 if (targetPane) {
210     targetPane.classList.add('show', 'active');
211 }
212
213 // 3. Загружаем содержимое вкладки по требованию, если ещё не загружено
214 await ensureTabLoaded(tabName);
215
216 // 4. Notify lifecycle callback
```



```

217 | onTabSwitched(tabId);
218 | }
219 | window.switchTab = switchTab;
220 | window.switchToTab = switchTab;
221 |
222 | // Setup dropdowns navigation
223 | function setupDropdownTabs() {
224 |     const mainTabs = document.getElementById('mainTabs');
225 |     if (!mainTabs) return;
226 |
227 |     // 1. Dropdown Toggle Buttons
228 |     mainTabs.querySelectorAll('.dropdown-toggle').forEach((btn)
    => {
229 |         btn.onclick = (e) => {
230 |             e.preventDefault();
231 |             e.stopPropagation();
232 |
233 |             const dropdown = btn.closest('.dropdown');
234 |             const menu = dropdown?.querySelector('.dropdown-menu');
235 |             const isAlreadyOpen = menu?.classList.contains('show');
236 |
237 |             // Close all dropdowns
238 |             document.querySelectorAll('#mainTabs .dropdown-menu.show
    ').forEach((m) => {
239 |                 m.classList.remove('show');
240 |                 m.closest('.dropdown')?.querySelector('.dropdown-toggl
    e')?.classList.remove('show');
241 |             });
242 |
243 |             // Toggle clicked dropdown
244 |             if (!isAlreadyOpen && menu) {
245 |                 menu.classList.add('show');
246 |                 btn.classList.add('show');
247 |             }
248 |         };
249 |     });
250 |
251 |     // 2. Dropdown Item Buttons (Tab Switchers & Action Buttons)

```

```

252 |   mainTabs.querySelectorAll('.dropdown-item').forEach((item) =
    | > {
253 |       item.onclick = async (e) => {
254 |           const targetId = item.getAttribute('data-tab') || item.g
    | etAttribute('data-bs-target')?.replace('#', '');
255 |           const pluginName = item.getAttribute('data-plugin');
256 |
257 |           // Close dropdown menu
258 |           item.closest('.dropdown-menu')?.classList.remove('show')
    | ;
259 |           item.closest('.dropdown')?.querySelector('.dropdown-togg
    | le')?.classList.remove('show');
260 |
261 |           if (pluginName && typeof window.openPluginFromDropdown =
    | == 'function') {
262 |               e.preventDefault();
263 |               e.stopPropagation();
264 |               window.openPluginFromDropdown(pluginName);
265 |           } else if (targetId) {
266 |               e.preventDefault();
267 |               e.stopPropagation();
268 |               await switchTab(targetId);
269 |           }
270 |       };
271 |   });
272 |
273 |   // 3. Document Click to Close Dropdowns
274 |   if (!document.body.dataset.dropdownOutsideBound) {
275 |       document.body.dataset.dropdownOutsideBound = 'true';
276 |       document.addEventListener('click', (e) => {
277 |           if (!e.target.closest('#mainTabs .dropdown')) {
278 |               document.querySelectorAll('#mainTabs .dropdown-menu.sh
    | ow').forEach((menu) => {
279 |                   menu.classList.remove('show');
280 |                   menu.closest('.dropdown')?.querySelector('.dropdown-
    | toggle')?.classList.remove('show');
281 |               });
282 |           }

```

```

283     });
284   }
285 }
286
287 async function startUserInterface() {
288   console.log('User interface initializing...');
289   try {
290     initTheme();
291     setupDropdownTabs();
292     const el = document.getElementById('user-interface');
293     if (el) el.style.display = 'block';
294     await initInterface();
295     console.log('User interface ready');
296   } catch (err) {
297     console.error('Error during startUserInterface initializat
ion:', err);
298     const el = document.getElementById('user-interface');
299     if (el) el.style.display = 'block';
300   }
301 }
302
303 if (document.readyState === 'loading') {
304   document.addEventListener('DOMContentLoaded', () => {
305     setupDropdownTabs();
306     startUserInterface();
307   });
308 } else {
309   setupDropdownTabs();
310   startUserInterface();
311 }
312
313 async function initInterface() {
314   // Initialize theme
315   initTheme();
316
317   // Initialize i18n
318   const savedLang = localStorage.getItem('app_language') || 'r
u';

```

```
319 | await initI18n(savedLang);
320 |
321 | // Setup language selector
322 | document.querySelectorAll('.lang-selector').forEach((sel) =>
    | {
323 |     sel.value = savedLang;
324 |     sel.addEventListener('change', (e) => {
325 |         switchLang(e.target.value);
326 |     });
327 | });
328 |
329 | // Initialize User Settings & Google OAuth
330 | await initUserSettings();
331 |
332 | // Определение и загрузка ТОЛЬКО активной вкладки
333 | const hash = location.hash.replace('#', '');
334 | const initialTabId = hash || 'tab-chat';
335 | const initialTabName = initialTabId.replace(/^tab-/, '');
336 |
337 | console.log(`[UserInterface] Loading initial active tab: ${i
    | nitialTabName}`);
338 | await ensureTabLoaded(initialTabName);
339 | await switchTab(initialTabId);
340 |
341 | // Apply translations
342 | applyTranslations();
343 |
344 | // Synchronize plugin tabs visibility
345 | try {
346 |     let pluginsResp = await fetch('/api/plugins?scope=user');
347 |     if (!pluginsResp.ok) {
348 |         pluginsResp = await fetch('/api/admin/plugins?scope=user
    | ');
349 |     }
350 |     if (pluginsResp.ok) {
351 |         const pluginsData = await pluginsResp.json();
352 |         if (window.syncPluginTabsVisibility && pluginsData && pl
    | uuginsData.plugins) {
```

```
353         window.syncPluginTabsVisibility(pluginsData.plugins);
354     }
355 }
356 } catch (err) {
357     console.error('Ошибка синхронизации видимости плагинов:',
358 err);
359 }
360 setupDropdownTabs();
361 }
362
363 async function loadTabContent(tabName, url, jsOverrideSrc) {
364     try {
365         const container = document.getElementById(`tab-${tabName}`
366 );
367         if (container && !container.innerHTML.trim()) {
368             container.innerHTML = `
369                 <div class="d-flex justify-content-center align-items-
370 center" style="height: 200px;">
371                     <div class="spinner-border text-primary" role="statu
372 s">
373                         <span class="visually-hidden">Загрузка...</span>
374                     </div>
375                 </div>
376             `;
377         }
378         const response = await fetch(url);
379         if (!response.ok) throw new Error(`HTTP ${response.status}
380 `);
381         const html = await response.text();
382         if (!container) return;
383         container.innerHTML = html;
384
385         // Load JS for the tab if provided or present
386         const cacheBuster = Date.now();
387         const scriptSrc = jsOverrideSrc || `/html/${tabName}/main.
388 js?v=${cacheBuster}`;
```

```

385
386     await new Promise((resolve) => {
387         const script = document.createElement('script');
388         if (tabName === 'instructions' || tabName === 'rag') {
389             script.type = 'module';
390         }
391         script.src = scriptSrc;
392         script.onload = () => {
393             console.log(`[User] Loaded JS for ${tabName}`);
394             resolve();
395         };
396         script.onerror = (err) => {
397             console.debug(`[User] Note: No JS loaded for ${tabName}
398             } from ${scriptSrc}`);
399             resolve();
400         };
401         document.body.appendChild(script);
402     });
403     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
404     .toUpperCase());
405     const initFuncName = 'init' + camelName.charAt(0).toUpperCase() + camelName.slice(1) + 'Tab';
406     if (typeof window[initFuncName] === 'function') {
407         console.log(`[User] Calling ${initFuncName} for ${tabName}
408         e`);
409         try {
410             await window[initFuncName]();
411         } catch (err) {
412             console.error(`[User] Error executing ${initFuncName}:
413             `, err);
414         }
415     }
416     applyTranslations();
417 } catch (e) {
418     console.error(`[User] Error loading tab ${tabName}:`, e);
419     const container = document.getElementById(`tab-${tabName}`
420     );

```

```
417 |     if (container) {
418 |         container.innerHTML = `<div class="alert alert-danger">0
    | ошибка загрузки: ${e.message}</div>`;
419 |     }
420 | }
421 | }
```

debounce-patch.js

```
1 /**
2  * Users Tab Debounce Patch
3  *
4  * Применяет улучшенный дебаунс к поиску пользователей,
5  * используя RequestManager для снижения нагрузки на сервер
6  */
7
8 import { createUserSearchDebounce } from '../js/debounce-integ
  ration.js';
9
10 // Ждем загрузки основного скрипта
11 if (window.initUsersTab) {
12   const originalInitUsersTab = window.initUsersTab;
13
14   window.initUsersTab = async function() {
15     // Вызываем оригинальную функцию
16     await originalInitUsersTab.apply(this, arguments);
17
18     console.log('[UsersDebouncePatch] Patching users tab searc
  h...');
19
20     // После инициализации, патчим поиск
21     setTimeout(() => {
22       const searchInput = document.getElementById('users-searc
  h-input');
23
24       if (searchInput && window.globalRequestManager) {
25         // Сохраняем оригинальный обработчик
26         const listeners = searchInput._eventListeners?.input |
  | [];
27
28         // Удаляем старые обработчики
29         searchInput.replaceWith(searchInput.cloneNode(true));
30         const newSearchInput = document.getElementById('users-
  search-input');
31
```



```
32 |         // Добавляем новый дебаунсированный обработчик
33 |         newSearchInput.addEventListener('input', (e) => {
34 |             const val = e.target.value;
35 |             const searchClear = document.getElementById('users-s
search-clear');
36 |
37 |             if (searchClear) {
38 |                 searchClear.classList.toggle('d-none', !val);
39 |             }
40 |
41 |             // Используем улучшенный дебаунс
42 |             window.globalRequestManager.debounce(
43 |                 'user-search-optimized',
44 |                 async () => {
45 |                     // Обновляем state поиска (симуляция)
46 |                     if (window.updateUserSearch) {
47 |                         window.updateUserSearch(val.trim());
48 |                     }
49 |                 },
50 |                 500
51 |             ).catch(e => console.error('Search debounce error:',
e));
52 |         });
53 |
54 |         // Обновляем кнопку очистки
55 |         const searchClear = document.getElementById('users-sea
rch-clear');
56 |         if (searchClear) {
57 |             searchClear.onclick = () => {
58 |                 newSearchInput.value = '';
59 |                 searchClear.classList.add('d-none');
60 |                 if (window.updateUserSearch) {
61 |                     window.updateUserSearch('');
62 |                 }
63 |             };
64 |         }
65 |
66 |         console.log('[UsersDebouncePatch] Users tab search pat
```

```
    ched with RequestManager debounce');
67     }
68     }, 500);
69 };
70 }
71
72 // Альтернативный подход: Если users tab уже загружена, патчим
    сразу
73 if (document.getElementById('users-search-input')) {
74     console.log('[UsersDebouncePatch] Users tab already loaded,
    applying patch immediately...');
75
76     const searchInput = document.getElementById('users-search-in
    put');
77     const searchClear = document.getElementById('users-search-cl
    ear');
78
79     // Создаем новый дебаунсированный обработчик
80     const debouncedSearch = createUserSearchDebounce(async (quer
    y) => {
81         console.log(`[UsersDebouncePatch] Executing debounced sear
    ch for: ${query}`);
82         // Обновляем UI через существующую функцию если она есть
83         if (window.updateUserSearch) {
84             await window.updateUserSearch(query);
85         }
86     });
87
88     // Заменяем обработчик
89     searchInput.replaceWith(searchInput.cloneNode(true));
90     const newSearchInput = document.getElementById('users-search
    -input');
91
92     newSearchInput.addEventListener('input', (e) => {
93         const val = e.target.value;
94
95         if (searchClear) {
96             searchClear.classList.toggle('d-none', !val);
```

```
97 |     }
98 |
99 |     debouncedSearch(val.trim());
100 | });
101 |
102 | if (searchClear) {
103 |     searchClear.addEventListener('click', () => {
104 |         newSearchInput.value = '';
105 |         searchClear.classList.add('d-none');
106 |         debouncedSearch('');
107 |     });
108 | }
109 |
110 | console.log('[UsersDebouncePatch] Search debouncing initiali
    | zed');
111 | }
112 |
113 | export {};
114 |
```

index.html

```
1 <!-- Users Management Tab -->
2 <div class="container-fluid py-2" id="users-tab-root">
3   <!-- Top Banner & Stats -->
4   <div class="d-flex flex-wrap align-items-center justify-content-between gap-3 mb-3 p-3 bg-body-tertiary border rounded shadow-sm">
5     <div>
6       <h5 class="mb-1 text-primary d-flex align-items-center gap-2">
7         <i class="bi bi-people-fill text-primary"></i>
8         <span>Управление пользователями</span>
9       </h5>
10      <p class="mb-0 text-muted small">Управление учетными записями, ролями, доступом к системе и сбросом паролей.</p>
11    </div>
12    <div class="d-flex flex-wrap gap-2">
13      <button class="btn btn-outline-warning btn-sm d-flex align-items-center gap-1" id="btn-check-orphaned-dirs" title="Проверить мертвые директории пользователей, которые уже удалены из системы">
14        <i class="bi bi-folder-x"></i> Проверить мертвые директории
15      </button>
16      <button class="btn btn-primary btn-sm d-flex align-items-center gap-1" id="btn-open-create-user">
17        <i class="bi bi-person-plus-fill"></i> Добавить пользователя
18      </button>
19      <button class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1" id="btn-refresh-users" title="Обновить список">
20        <i class="bi bi-arrow-clockwise"></i> Обновить
21      </button>
22    </div>
23  </div>
24
```

```
25 | <!-- Summary Metric Cards -->
26 | <div class="row g-3 mb-3">
27 |     <div class="col-6 col-md-3">
28 |         <div class="card bg-body-tertiary border h-100 shadow-sm
29 |         ">
30 |             <div class="card-body p-3 d-flex align-items-center ju
31 |             stify-content-between">
32 |                 <div>
33 |                     <div class="text-muted small">Всего пользователей<
34 |                     /div>
35 |                     <h4 class="mb-0 fw-bold" id="stat-total-users">0</
36 |                     h4>
37 |                 </div>
38 |                 <div class="bg-secondary bg-opacity-25 rounded p-2 t
39 |                 ext-primary">
40 |                     <i class="bi bi-people fs-4"></i>
41 |                 </div>
42 |             </div>
43 |             <div class="col-6 col-md-3">
44 |                 <div class="card bg-body-tertiary border h-100 shadow-sm
45 |                 ">
46 |                     <div class="card-body p-3 d-flex align-items-center ju
47 |                     stify-content-between">
48 |                         <div>
49 |                             <div class="text-muted small">Активные аккаунты</d
50 |                             iv>
51 |                             <h4 class="mb-0 text-success fw-bold" id="stat-act
52 |                             ive-users">0</h4>
53 |                         </div>
54 |                         <div class="bg-success bg-opacity-25 rounded p-2 tex
55 |                         t-success">
56 |                             <i class="bi bi-check-circle fs-4"></i>
57 |                         </div>
58 |                     </div>
59 |                 </div>
60 |             </div>
61 |         </div>
62 |     </div>
```

```
53 |     <div class="col-6 col-md-3">
54 |         <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
55 |             <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
56 |                 <div>
57 |                     <div class="text-muted small">Администраторы</div>
58 |                     <h4 class="mb-0 text-warning fw-bold" id="stat-adm
    | in-users">0</h4>
59 |                 </div>
60 |                 <div class="bg-warning bg-opacity-25 rounded p-2 tex
    | t-warning">
61 |                     <i class="bi bi-shield-lock fs-4"></i>
62 |                 </div>
63 |             </div>
64 |         </div>
65 |     </div>
66 |     <div class="col-6 col-md-3">
67 |         <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
68 |             <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
69 |                 <div>
70 |                     <div class="text-muted small">Telegram привязано</
    | div>
71 |                     <h4 class="mb-0 text-info fw-bold" id="stat-tg-use
    | rs">0</h4>
72 |                 </div>
73 |                 <div class="bg-info bg-opacity-25 rounded p-2 text-i
    | nfo">
74 |                     <i class="bi bi-telegram fs-4"></i>
75 |                 </div>
76 |             </div>
77 |         </div>
78 |     </div>
79 | </div>
80 |
81 | <!-- Search & Filter Controls -->
```

```
82 | <div class="card bg-body-tertiary border mb-3 shadow-sm">
83 |   <div class="card-body p-3">
84 |     <div class="row g-2 align-items-center">
85 |       <div class="col-12 col-md-6 col-lg-5">
86 |         <div class="input-group input-group-sm">
87 |           <span class="input-group-text"><i class="bi bi-sea
rch"></i></span>
88 |           <input type="text" class="form-control" id="users-
search-input" placeholder="Поиск по имени, email, Telegram..."
>
89 |           <button class="btn btn-outline-secondary d-none" t
ype="button" id="users-search-clear">⌫ </button>
90 |         </div>
91 |       </div>
92 |       <div class="col-6 col-md-3 col-lg-3">
93 |         <select class="form-select form-select-sm" id="users
-filter-role">
94 |           <option value="">Все роли</option>
95 |           <option value="admin">Администраторы (admin)</opti
on>
96 |           <option value="user">Пользователи (user)</option>
97 |           <option value="guest">Гости (guest)</option>
98 |         </select>
99 |       </div>
100 |      <div class="col-6 col-md-3 col-lg-2">
101 |        <select class="form-select form-select-sm" id="users
-filter-status">
102 |          <option value="">Все статусы</option>
103 |          <option value="active">Только активные</option>
104 |          <option value="inactive">Заблокированные</option>
105 |        </select>
106 |      </div>
107 |      <div class="col-12 col-lg-2 text-end text-muted small"
id="users-count-badge">
108 |        Загрузка...
109 |      </div>
110 |    </div>
111 |  </div>
```

```

112 | </div>
113 |
114 | <!-- User Alert/Status message -->
115 | <div id="users-alert-box" class="d-none alert alert-dismissi
ble fade show" role="alert">
116 |     <span id="users-alert-message"></span>
117 |     <button type="button" class="btn-close" aria-label="Close"
onClick="document.getElementById('users-alert-box').classList
.add('d-none')"></button>
118 | </div>
119 |
120 | <!-- Users Table Card -->
121 | <div class="card shadow-sm border">
122 |     <div class="card-body p-0">
123 |         <div class="table-responsive">
124 |             <table class="table table-hover mb-0 align-middle">
125 |                 <thead class="table-light small text-uppercase">
126 |                     <tr>
127 |                         <th style="width: 50px;" class="text-center">ID<
/th>
128 |                         <th>Пользователь</th>
129 |                         <th style="width: 140px;" class="text-center">Po
ль</th>
130 |                         <th style="width: 130px;" class="text-center">Ст
атус</th>
131 |                         <th style="width: 150px;">Telegram</th>
132 |                         <th style="width: 120px;" class="text-center">Па
роль</th>
133 |                         <th style="width: 160px;">Регистрация / Вход</th>
>
134 |                         <th style="width: 170px;" class="text-center">Де
йствия</th>
135 |                     </tr>
136 |                 </thead>
137 |                 <tbody id="users-table-body">
138 |                     <tr>
139 |                         <td colspan="8" class="text-center py-4 text-mut
ed">

```



```

140 |         <div class="spinner-border spinner-border-sm t
| ext-primary me-2" role="status"></div>
141 |         Загрузка списка пользователей...
142 |     </td>
143 | </tr>
144 | </tbody>
145 | </table>
146 | </div>
147 | </div>
148 | </div>
149 | </div>
150 |
151 | <!-- =====
| ===== -->
152 | <!-- MODAL: Create User -->
153 | <!-- =====
| ===== -->
154 | <div class="modal fade" id="modal-create-user" tabindex="-1" a
| ria-labelledby="modal-create-user-title" aria-hidden="true">
155 |     <div class="modal-dialog modal-dialog-centered">
156 |         <div class="modal-content shadow-lg">
157 |             <div class="modal-header">
158 |                 <h5 class="modal-title d-flex align-items-center gap-2
| " id="modal-create-user-title">
159 |                     <i class="bi bi-person-plus text-primary"></i> Добав
| ить нового пользователя
160 |                 </h5>
161 |                 <button type="button" class="btn-close" data-bs-dismis
| s="modal" aria-label="Close"></button>
162 |             </div>
163 |             <form id="form-create-user">
164 |                 <div class="modal-body p-3">
165 |                     <div class="mb-3">
166 |                         <label class="form-label small fw-bold">Email <spa
| n class="text-danger">*</span></label>
167 |                         <input type="email" class="form-control form-contr
| ol-sm" id="create-user-email" required placeholder="user@examp
| le.com">

```

```
168 |         </div>
169 |         <div class="mb-3">
170 |             <label class="form-label small fw-bold">Отображаем
| ое имя <span class="text-danger">*</span></label>
171 |             <input type="text" class="form-control form-control-sm" id="create-user-name" required placeholder="Иван Иванов"
| >
172 |         </div>
173 |         <div class="mb-3">
174 |             <label class="form-label small fw-bold">Пароль (оп
| ционально)</label>
175 |             <div class="input-group input-group-sm">
176 |                 <input type="text" class="form-control font-mono
| space" id="create-user-password" placeholder="Оставьте пустым
| или задайте пароль">
177 |                 <button class="btn btn-outline-secondary" type="
| button" id="btn-gen-create-password" title="Сгенерировать безо
| пасный пароль">
178 |                     <i class="bi bi-dice-5"></i>
179 |                 </button>
180 |             </div>
181 |             <div class="form-text text-muted small">Если парол
| ь не задан, пользователь сможет авторизоваться через Google ил
| и Telegram.</div>
182 |         </div>
183 |         <div class="mb-3 p-3 bg-body-tertiary border rounded
| ">
184 |             <div class="form-check form-switch d-flex align-it
| ems-center justify-content-between p-0 m-0">
185 |                 <label class="form-check-label fw-bold" for="cre
| ate-user-is-admin">
186 |                     <i class="bi bi-shield-lock-fill text-warning
| me-1"></i> Права Администратора
187 |                 </label>
188 |                 <input class="form-check-input ms-0 fs-5" type="
| checkbox" role="switch" id="create-user-is-admin">
189 |             </div>
190 |             <div class="small text-muted mt-1" id="create-role
```

```

    -hint">Роль: <strong class="text-info" id="create-role-label">
Пользователь (user)</strong></div>
191     </div>
192     <div class="row g-2 mb-2">
193         <div class="col-6">
194             <div class="form-check form-switch">
195                 <input class="form-check-input" type="checkbox"
" role="switch" id="create-user-is-active" checked>
196                 <label class="form-check-label small" for="cre
ate-user-is-active">Активен</label>
197             </div>
198         </div>
199         <div class="col-6">
200             <div class="form-check form-switch">
201                 <input class="form-check-input" type="checkbox"
" role="switch" id="create-user-verified" checked>
202                 <label class="form-check-label small" for="cre
ate-user-verified">Email подтвержден</label>
203             </div>
204         </div>
205     </div>
206 </div>
207 <div class="modal-footer">
208     <button type="button" class="btn btn-secondary btn-s
m rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
209     <button type="submit" class="btn btn-primary btn-sm
rounded-pill px-3" id="btn-submit-create-user">Создать пользо
ателя</button>
210 </div>
211 </form>
212 </div>
213 </div>
214 </div>
215
216 <!-- =====
===== -->
217 <!-- MODAL: Edit User -->
218 <!-- =====
===== -->

```

```

===== -->
219 <div class="modal fade" id="modal-edit-user" tabindex="-1" ari
    a-labelledby="modal-edit-user-title" aria-hidden="true">
220   <div class="modal-dialog modal-dialog-centered">
221     <div class="modal-content shadow-lg">
222       <div class="modal-header">
223         <h5 class="modal-title d-flex align-items-center gap-2
    " id="modal-edit-user-title">
224           <i class="bi bi-pencil-square text-warning"></i> Ред
    актирование пользователя #<span id="edit-user-id-badge"></span
    >
225         </h5>
226         <button type="button" class="btn-close" data-bs-dismis
    s="modal" aria-label="Close"></button>
227       </div>
228       <form id="form-edit-user">
229         <input type="hidden" id="edit-user-id">
230         <div class="modal-body p-3">
231           <div class="mb-3">
232             <label class="form-label small fw-bold">Email <spa
    n class="text-danger">*</span></label>
233             <input type="email" class="form-control form-contr
    ol-sm" id="edit-user-email" required>
234           </div>
235           <div class="mb-3">
236             <label class="form-label small fw-bold">Имя <span
    class="text-danger">*</span></label>
237             <input type="text" class="form-control form-contro
    l-sm" id="edit-user-name" required>
238           </div>
239           <div class="mb-3 p-3 bg-body-tertiary border rounded
    ">
240             <div class="form-check form-switch d-flex align-it
    ems-center justify-content-between p-0 m-0">
241               <label class="form-check-label fw-bold" for="edi
    t-user-is-admin">
242                 <i class="bi bi-shield-lock-fill text-warning
    me-1"></i> Права Администратора

```

```
243         </label>
244         <input class="form-check-input ms-0 fs-5" type="
checkbox" role="switch" id="edit-user-is-admin">
245     </div>
246     <div class="small text-muted mt-1" id="edit-role-h
int">Роль: <strong class="text-info" id="edit-role-label">Поль
зователь (user)</strong></div>
247 </div>
248 <div class="row g-2 mb-2">
249     <div class="col-6">
250         <div class="form-check form-switch">
251             <input class="form-check-input" type="checkbox
" role="switch" id="edit-user-is-active">
252             <label class="form-check-label small" for="edi
t-user-is-active">Активен</label>
253         </div>
254     </div>
255     <div class="col-6">
256         <div class="form-check form-switch">
257             <input class="form-check-input" type="checkbox
" role="switch" id="edit-user-verified">
258             <label class="form-check-label small" for="edi
t-user-verified">Email подтвержден</label>
259         </div>
260     </div>
261 </div>
262 </div>
263 <div class="modal-footer">
264     <button type="button" class="btn btn-secondary btn-s
m rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
265     <button type="submit" class="btn btn-warning btn-sm
rounded-pill px-3" id="btn-submit-edit-user">Сохранить изменен
ия</button>
266 </div>
267 </form>
268 </div>
269 </div>
270 </div>
```

```

271 |
272 | <!-- =====
    | ===== -->
273 | <!-- MODAL: Set / Reset Password -->
274 | <!-- =====
    | ===== -->
275 | <div class="modal fade" id="modal-password-user" tabindex="-1"
    |   aria-labelledby="modal-password-user-title" aria-hidden="true
    |   ">
276 |     <div class="modal-dialog modal-dialog-centered">
277 |       <div class="modal-content shadow-lg">
278 |         <div class="modal-header">
279 |           <h5 class="modal-title d-flex align-items-center gap-2
    |   " id="modal-password-user-title">
280 |             <i class="bi bi-key-fill text-info"></i> Смена пароля
    |   пользователя
281 |           </h5>
282 |           <button type="button" class="btn-close" data-bs-dismiss="modal"
    |   aria-label="Close"></button>
283 |         </div>
284 |         <form id="form-password-user">
285 |           <input type="hidden" id="pwd-user-id">
286 |           <div class="modal-body p-3">
287 |             <p class="small text-muted mb-3">
288 |               Установка нового пароля для <strong id="pwd-user-t
    |   arget"></strong>
289 |             </p>
290 |             <div class="mb-3">
291 |               <label class="form-label small fw-bold">Новый пароль
    |   <span class="text-danger">*</span></label>
292 |               <div class="input-group">
293 |                 <input type="text" class="form-control font-mono
    |   space" id="pwd-input-val" required minlength="4" placeholder="
    |   Введите пароль">
294 |                 <button class="btn btn-outline-info" type="button"
    |   id="btn-gen-pwd-val" title="Сгенерировать безопасный пароль"
    |   ">
295 |                   <i class="bi bi-dice-5"></i> Сгенерировать

```

```

296 |         </button>
297 |     </div>
298 | </div>
299 | </div>
300 | <div class="modal-footer">
301 |     <button type="button" class="btn btn-secondary btn-sm rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>
302 |     <button type="submit" class="btn btn-info btn-sm rounded-pill px-3" id="btn-submit-pwd-user">Установить пароль</button>
303 | </div>
304 | </form>
305 | </div>
306 | </div>
307 | </div>
308 |
309 | <!-- =====
310 | ===== -->
311 | <!-- MODAL: User Details & Profile Settings -->
312 | <!-- =====
313 | ===== -->
314 | <div class="modal fade" id="modal-details-user" tabindex="-1"
315 |     aria-labelledby="modal-details-user-title" aria-hidden="true">
316 |     <div class="modal-dialog modal-dialog-centered modal-lg">
317 |         <div class="modal-content shadow-lg">
318 |             <div class="modal-header">
319 |                 <h5 class="modal-title d-flex align-items-center gap-2" id="modal-details-user-title">
320 |                     <i class="bi bi-person-badge text-primary"></i> Профиль и настройки пользователя
321 |                 </h5>
322 |                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
323 |             </div>
324 |             <div class="modal-body p-3" id="details-user-content">
325 |                 <div class="text-center py-3 text-muted">Загрузка данных...</div>
326 |             </div>

```

```

324     <div class="modal-footer">
325         <button type="button" class="btn btn-secondary btn-sm
rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
326     </div>
327 </div>
328 </div>
329 </div>
330
331 <!-- =====
===== -->
332 <!-- MODAL: Delete Confirmation -->
333 <!-- =====
===== -->
334 <div class="modal fade" id="modal-delete-user" tabindex="-1" a
ria-labelledby="modal-delete-user-title" aria-hidden="true">
335     <div class="modal-dialog modal-dialog-centered">
336         <div class="modal-content border-danger shadow-lg">
337             <div class="modal-header border-danger">
338                 <h5 class="modal-title text-danger d-flex align-items-
center gap-2" id="modal-delete-user-title">
339                     <i class="bi bi-exclamation-triangle-fill"></i> Подт
верждение удаления
340                 </h5>
341                 <button type="button" class="btn-close" data-bs-dismis
s="modal" aria-label="Close"></button>
342             </div>
343             <div class="modal-body p-3">
344                 <p class="mb-2">Вы действительно хотите удалить пользо
вателя <strong class="text-danger" id="delete-user-name-target
"></strong>?</p>
345                 <p class="small text-muted mb-0">Это действие удалит у
четную запись, настройки и сессии пользователя. Операция необр
атима.</p>
346                 <input type="hidden" id="delete-user-id">
347             </div>
348             <div class="modal-footer">
349                 <button type="button" class="btn btn-secondary btn-sm
rounded-pill px-3" data-bs-dismiss="modal">Отмена</button>

```



```

350 |         <button type="button" class="btn btn-danger btn-sm rou
      | nded-pill px-3" id="btn-confirm-delete-user">Удалить пользоват
      | еля</button>
351 |     </div>
352 | </div>
353 | </div>
354 | </div>
355 |
356 | <!-- =====
      | ===== -->
357 | <!-- MODAL: Orphaned User Directories Management -->
358 | <!-- =====
      | ===== -->
359 | <div class="modal fade" id="modal-orphaned-dirs" tabindex="-1"
      | aria-labelledby="modal-orphaned-dirs-title" aria-hidden="true
      | ">
360 |     <div class="modal-dialog modal-dialog-centered modal-lg">
361 |         <div class="modal-content shadow-lg border-warning">
362 |             <div class="modal-header border-warning">
363 |                 <h5 class="modal-title text-warning d-flex align-items
      | -center gap-2" id="modal-orphaned-dirs-title">
364 |                     <i class="bi bi-folder-x"></i> Мертвые директории по
      | льзователей
365 |                 </h5>
366 |                 <button type="button" class="btn-close" data-bs-dismis
      | s="modal" aria-label="Close"></button>
367 |             </div>
368 |             <div class="modal-body p-3">
369 |                 <div class="alert alert-info border-0 small mb-3">
370 |                     <i class="bi bi-info-circle text-info me-1"></i>
371 |                     Обнаружены директории в хранилище <code>data/users/<
      | /code>, которые не привязаны ни к одному активному пользовател
      | ю в базе данных или остались после тестов.
372 |                 </div>
373 |
374 |                 <!-- Summary Stats Banner -->
375 |                 <div class="d-flex align-items-center justify-content-
      | between p-2 mb-3 bg-body-tertiary border rounded">

```

```

376         <div class="small">
377             <span class="text-muted">Найдено директорий:</span> <
378             <strong id="orphaned-dirs-count" class="text-warning">0</str
379             <span class="mx-2 text-secondary">|</span> <
380             <span class="text-muted">Занято на диске:</span> <
381             <strong id="orphaned-dirs-size" class="text-info">0 B</strong>
382             <span class="mx-2 text-secondary">|</span> <
383             <span class="text-muted">Файлов:</span> <strong id
384             ="orphaned-dirs-files">0</strong>
385         </div>
386         <button class="btn btn-outline-secondary btn-sm py-0
387         px-2 rounded-pill" id="btn-refresh-orphaned-dirs" title="Пере
388         сканировать">
389             <i class="bi bi-arrow-clockwise"></i> Обновить
390         </button>
391     </div>
392
393     <div id="orphaned-dirs-table-wrapper" class="table-res
394     ponsive" style="max-height: 380px; overflow-y: auto;">
395         <table class="table table-hover table-sm align-middl
396         e mb-0">
397             <thead class="table-light small text-uppercase sti
398             cky-top">
399                 <tr>
400                     <th>Папка / ID</th>
401                     <th>Причина</th>
402                     <th class="text-center" style="width: 80px;">Ф
403                     айлов</th>
404                     <th class="text-center" style="width: 100px;">
405                     Размер</th>
406                     <th style="width: 140px;">Изменено</th>
407                     <th class="text-center" style="width: 90px;">Д
408                     ействие</th>
409                 </tr>
410             </thead>
411             <tbody id="orphaned-dirs-tbody">
412                 <tr>

```

```

402 |         <td colspan="6" class="text-center py-4 text-m
| uted">
403 |             <div class="spinner-border spinner-border-sm
| text-warning me-2" role="status"></div>
404 |             Сканирование хранилища...
405 |         </td>
406 |     </tr>
407 | </tbody>
408 | </table>
409 | </div>
410 | </div>
411 | <div class="modal-footer justify-content-between">
412 |     <button type="button" class="btn btn-secondary btn-sm
| rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
413 |     <button type="button" class="btn btn-danger btn-sm rou
| nded-pill px-3 d-flex align-items-center gap-1" id="btn-cleanu
| p-all-orphaned">
414 |         <i class="bi bi-trash3-fill"></i> Удалить все мертвы
| е директории
415 |     </button>
416 | </div>
417 | </div>
418 | </div>
419 | </div>
420 |

```

main.js

```
1 // =====
  =====
2 // Webinterface: Users Tab Logic
3 // Module: webinterface/users_tab/main.js
4 // Author: hypo69
5 // Copyright: © 2026 hypo69
6 // =====
  =====
7
8 'use strict';
9
10 (function () {
11     const state = {
12         users: [],
13         stats: {
14             total: 0,
15             active: 0,
16             admins: 0,
17             telegram: 0
18         },
19         filterRole: '',
20         filterStatus: '',
21         searchQuery: '',
22         editingUserId: null,
23         savingUserId: null,
24         isLoading: false,
25         initialized: false
26     };
27
28     // Helper API fetch
29     async function apiFetch(url, options = {}) {
30         if (window.api && typeof window.api.fetch === 'function')
31         {
32             return window.api.fetch(url, options);
33         }
34         const res = await fetch(url, options);
```

```

34 |     if (!res.ok) {
35 |         let errMsg = `HTTP ${res.status}`;
36 |         try {
37 |             const errData = await res.json();
38 |             if (errData && errData.detail) {
39 |                 errMsg = typeof errData.detail === 'string' ? errData.detail : JSON.stringify(errData.detail);
40 |             }
41 |         } catch (_) {}
42 |         throw new Error(errMsg);
43 |     }
44 |     return res.json();
45 | }
46 |
47 | // Show notification
48 | function showStatusAlert(msg, type = 'info') {
49 |     const alertBox = document.getElementById('users-alert-box');
50 |     const alertMsg = document.getElementById('users-alert-message');
51 |     if (!alertBox || !alertMsg) return;
52 |
53 |     alertBox.className = `alert alert-${type} alert-dismissible fade show mb-3`;
54 |     alertMsg.innerHTML = msg;
55 |     alertBox.classList.remove('d-none');
56 |
57 |     setTimeout(() => {
58 |         alertBox.classList.add('d-none');
59 |     }, 5000);
60 | }
61 |
62 | // Generate secure password
63 | function generatePassword(length = 12) {
64 |     const charset = 'abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789!@#$%^&*';
65 |     let pwd = '';
66 |     const randomValues = new Uint32Array(length);

```

```

67 |     window.crypto.getRandomValues(randomValues);
68 |     for (let i = 0; i < length; i++) {
69 |         pwd += charset[randomValues[i] % charset.length];
70 |     }
71 |     return pwd;
72 | }
73 |
74 | // Escape HTML
75 | function escapeHtml(str) {
76 |     if (!str) return '';
77 |     return String(str)
78 |         .replace(/&/g, '&amp;')
79 |         .replace(/</g, '&lt;')
80 |         .replace(/>/g, '&gt;')
81 |         .replace(/"/g, '&quot;')
82 |         .replace(/'/g, '&#039;');
83 | }
84 |
85 | // Format datetime
86 | function formatDate(isoStr) {
87 |     if (!isoStr) return '<span class="text-muted">—</span>';
88 |     try {
89 |         const d = new Date(isoStr);
90 |         if (isNaN(d.getTime())) return escapeHtml(isoStr);
91 |         return `<span title="${escapeHtml(isoStr)}">${d.toLocale
DateString('ru-RU')} <small class="text-muted">${d.toLocaleTim
eString('ru-RU', { hour: '2-digit', minute: '2-digit' })}</sma
ll></span>`;
92 |     } catch (_) {
93 |         return escapeHtml(isoStr);
94 |     }
95 | }
96 |
97 | // Load Users List from Backend
98 | async function loadUsers() {
99 |     state.isLoading = true;
100 |     const tableBody = document.getElementById('users-table-bod
y');

```

```
101     const countBadge = document.getElementById('users-count-ba
102     dge');
103     if (tableBody && (!state.users || state.users.length === 0
104     )) {
105         tableBody.innerHTML = `
106             <tr>
107                 <td colspan="8" class="text-center py-4 text-muted">
108                     <div class="spinner-border spinner-border-sm text-
109                     primary me-2" role="status"></div>
110                     Загрузка списка пользователей...
111                 </td>
112             </tr>`;
113     }
114     try {
115         const params = new URLSearchParams();
116         if (state.searchQuery) params.append('q', state.searchQu
117         ery);
118         if (state.filterRole) params.append('role', state.filter
119         Role);
120         if (state.filterStatus) params.append('status', state.fi
121         lterStatus);
122         const url = `/api/admin/users?${params.toString()}`;
123         const data = await apiFetch(url);
124         state.users = data.users || [];
125         if (data.stats) {
126             state.stats = data.stats;
127         }
128         renderStats();
129         renderTable();
130     } catch (err) {
131         console.error('[UsersTab] Error loading users:', err);
132         showStatusAlert(`Ошибка загрузки пользователей: ${err.me
133         ssage}`, 'danger');
```

```

132         if (tableBody) {
133             tableBody.innerHTML = `
134                 <tr>
135                     <td colspan="8" class="text-center py-4 text-dange
r">
136                         <i class="bi bi-exclamation-octagon fs-4 d-block
mb-1"></i>
137                         Не удалось загрузить пользователей: ${escapeHtml
(err.message)}
138                     </td>
139                 </tr>`;
140         }
141     } finally {
142         state.isLoading = false;
143     }
144 }
145
146 // Render Metric Cards
147 function renderStats() {
148     const totalEl = document.getElementById('stat-total-users'
);
149     const activeEl = document.getElementById('stat-active-user
s');
150     const adminEl = document.getElementById('stat-admin-users'
);
151     const tgEl = document.getElementById('stat-tg-users');
152
153     if (totalEl) totalEl.textContent = state.stats.total || 0;
154     if (activeEl) activeEl.textContent = state.stats.active ||
0;
155     if (adminEl) adminEl.textContent = state.stats.admins || 0
;
156     if (tgEl) tgEl.textContent = state.stats.telegram || 0;
157 }
158
159 // Render Users Table
160 function renderTable() {
161     const tableBody = document.getElementById('users-table-bod

```



```

    y');
162     const countBadge = document.getElementById('users-count-ba
    dge');
163     if (!tableBody) return;
164
165     if (countBadge) {
166         countBadge.textContent = `Показано: ${state.users.length
    } из ${state.stats.total || state.users.length}`;
167     }
168
169     if (state.users.length === 0) {
170         tableBody.innerHTML = `
171         <tr>
172             <td colspan="8" class="text-center py-5 text-muted">
173                 <i class="bi bi-person-x fs-2 d-block mb-2 text-se
    condary"></i>
174                 Пользователи не найдены.
175             </td>
176         </tr>`;
177         return;
178     }
179
180     tableBody.innerHTML = state.users.map((user) => {
181         const isRoot = user.id === 1;
182         const isAdmin = Boolean(user.is_admin || user.role === '
    admin');
183         const isActive = Boolean(user.is_active);
184         const isEmailVerified = Boolean(user.is_email_verified);
185         const hasPassword = Boolean(user.has_password);
186         const isRowEditing = (state.editingUserId === user.id);
187         const isSaving = (state.savingUserId === user.id);
188
189         // User Initials or Avatar
190         let avatarHtml = '';
191         if (user.picture) {
192             avatarHtml = ``;

```

```

193     } else {
194         const initial = (user.name || user.email || 'U').charA
t(0).toUpperCase();
195         const bgClass = isAdmin ? 'bg-warning text-dark' : 'bg
-primary text-white';
196         avatarHtml = `

<i clas
s="bi bi-key-fill"></i> Задан</span>`
202         : `


```

```

218 |         <div class="input-group input-group-sm mb-1"
    | >
219 |             <span class="input-group-text text-muted">
    | <i class="bi bi-person"></i></span>
220 |             <input type="text" class="form-control inline-edit-name" value="{escapeHtml(user.name || '')}" placeholder="Имя пользователя">
    | </div>
221 |         <div class="input-group input-group-sm">
222 |             <span class="input-group-text text-muted">
    | <i class="bi bi-envelope"></i></span>
224 |             <input type="email" class="form-control inline-edit-email" value="{escapeHtml(user.email || '')}" placeholder="Email">
    | </div>
225 |         </div>
226 |     </div>
227 | </div>
228 | </td>
229 | <td class="text-center align-middle">
230 |     <select class="form-select form-select-sm inline-edit-role" ${isRoot ? 'disabled' : ''}>
231 |         <option value="user" ${user.role === 'user' ? 'selected' : ''}>☐ User</option>
232 |         <option value="admin" ${user.role === 'admin' ? 'selected' : ''}>☐ Admin</option>
233 |         <option value="guest" ${user.role === 'guest' ? 'selected' : ''}>☐ Guest</option>
234 |     </select>
235 | </td>
236 | <td class="text-center align-middle">
237 |     <select class="form-select form-select-sm inline-edit-status" ${isRoot ? 'disabled' : ''}>
238 |         <option value="1" ${isActive ? 'selected' : ''}>☐ Активен</option>
239 |         <option value="0" ${!isActive ? 'selected' : ''}>☐ Блок</option>
240 |     </select>
241 | </td>

```

```

242         <td class="align-middle">
243             <div class="input-group input-group-sm">
244                 <span class="input-group-text text-info"><i cl
class="bi bi-telegram"></i></span>
245                 <input type="text" class="form-control inline-
edit-tg" value="{escapeHtml(user.telegram_username || ' ')}" p
laceholder="Username">
246             </div>
247         </td>
248         <td class="text-center align-middle">{pwdHtml}</t
d>
249         <td class="small align-middle">
250             <div><span class="text-muted">Создан:</span> {f
ormatDate(user.created_at)}</div>
251         </td>
252         <td class="text-center align-middle">
253             <div class="d-flex justify-content-center gap-1"
>
254                 <button class="btn btn-success btn-sm btn-save
-inline" data-id="{user.id}" title="Сохранить изменения (Ente
r)" {isSaving ? 'disabled' : ''}>
255                     {isSaving ? '<span class="spinner-border sp
inner-border-sm"></span>' : '<i class="bi bi-check-lg"></i>'}
256                 </button>
257                 <button class="btn btn-outline-secondary btn-s
m btn-cancel-inline" data-id="{user.id}" title="Отмена (Esc)"
{isSaving ? 'disabled' : ''}>
258                     <i class="bi bi-x-lg"></i>
259                 </button>
260             </div>
261         </td>
262     </tr>
263     `;
264 }
265
266 // Normal view mode with DIRECT editable table fields:
267 const roleSelectHtml = `
268     <select class="form-select form-select-sm bg-dark bord

```

```

er-secondary user-table-role-select text-center ${isAdmin ? 't
ext-warning fw-bold' : 'text-info'}"
269     data-user-id="${user.id}"
270     ${isRoot ? 'disabled title="Нельзя изменить роль Roo
t"' : 'title="Изменить роль прямо в таблице"'}>
271     <option value="user" ${user.role === 'user' ? 'selec
ted' : ''} class="text-info">□ User</option>
272     <option value="admin" ${isAdmin ? 'selected' : ''} c
lass="text-warning">□ Admin</option>
273     <option value="guest" ${user.role === 'guest' ? 'sel
ected' : ''} class="text-secondary">□ Guest</option>
274     </select>
275     `;
276
277     const statusSelectHtml = `
278     <select class="form-select form-select-sm bg-dark bord
er-secondary user-table-status-select text-center ${isActive ?
'text-success' : 'text-danger'}"
279     data-user-id="${user.id}"
280     ${isRoot ? 'disabled title="Нельзя деактивировать Ro
ot"' : 'title="Изменить статус прямо в таблице"'}>
281     <option value="1" ${isActive ? 'selected' : ''} clas
s="text-success">□ Активен</option>
282     <option value="0" ${!isActive ? 'selected' : ''} cla
ss="text-danger">□ Блок</option>
283     </select>
284     `;
285
286     // Telegram column
287     let tgHtml = '<span class="text-muted small">—</span>';
288     if (user.telegram_username) {
289         tgHtml = `<a href="https://t.me/${escapeHtml(user.tele
gram_username)}" target="_blank" class="text-info text-decorat
ion-none small d-flex align-items-center gap-1" title="Открыть
профиль Telegram">
290         <i class="bi bi-telegram"></i> @${escapeHtml(user.te
legram_username)}
291         </a>`;

```

```

292         } else if (user.telegram_id) {
293             tgHtml = `<span class="text-muted small" title="Telegram ID"><i class="bi bi-telegram text-info"></i> ID: ${escapeHtml(user.telegram_id)}</span>`;
294         }
295
296         return `
297             <tr data-user-id="${user.id}">
298                 <td class="text-center text-muted fw-bold align-middle">${user.id}</td>
299                 <td class="align-middle">
300                     <div class="d-flex align-items-center justify-content-between">
301                         <div class="d-flex align-items-center">
302                             ${avatarHtml}
303                             <div>
304                                 <div class="fw-bold text-white user-cell-name" title="Дважды кликните для быстрого редактирования" style="cursor: pointer;">
305                                     ${escapeHtml(user.name || 'Без имени')}
306                                     <i class="bi bi-pencil-fill text-muted ms-1 opacity-25 hover-opacity-100" style="font-size:10px;"></i>
307                                 </div>
308                                 <div class="small text-muted d-flex align-items-center user-cell-email" title="Дважды кликните для быстрого редактирования" style="cursor: pointer;">
309                                     ${escapeHtml(user.email)} ${emailVerifiedBadge}
310                                 </div>
311                             </div>
312                         </div>
313                     </div>
314                 </td>
315                 <td class="text-center align-middle" style="min-width: 130px;">${roleSelectHtml}</td>
316                 <td class="text-center align-middle" style="min-width: 125px;">${statusSelectHtml}</td>
317                 <td class="align-middle" style="min-width: 140px;">

```

```

318         <div class="d-flex align-items-center justify-content-between">
319             <div class="user-cell-tg" title="Дважды кликните
                для быстрого редактирования" style="cursor: pointer;">${tgHtml}
            </div>
320         </div>
321     </td>
322     <td class="text-center align-middle">${pwdHtml}</td>
323     <td class="small align-middle">
324         <div><span class="text-muted">Создан:</span> ${formatDate(user.created_at)}</div>
325         ${user.last_login ? `<div><span class="text-muted">Вход:</span> ${formatDate(user.last_login)}</div>` : ''}
326     </td>
327     <td class="text-center align-middle">
328         <div class="btn-group btn-group-sm" role="group">
329             <button class="btn btn-outline-warning btn-inline-edit" data-id="${user.id}" title="Редактировать поля строки
                прямо в таблице">
330                 <i class="bi bi-pencil-square"></i>
331             </button>
332             <button class="btn btn-outline-info btn-pwd-user" data-id="${user.id}" data-name="${escapeHtml(user.name || user.email)}" title="Сменить пароль">
333                 <i class="bi bi-key-fill"></i>
334             </button>
335             <button class="btn btn-outline-light btn-details-user" data-id="${user.id}" title="Профиль и настройки">
336                 <i class="bi bi-info-circle-fill"></i>
337             </button>
338             <button class="btn btn-outline-secondary btn-edit-user-modal" data-id="${user.id}" title="Редактировать в модальном окне">
339                 <i class="bi bi-sliders"></i>
340             </button>
341             <button class="btn btn-outline-danger btn-delete-user" data-id="${user.id}" data-name="${escapeHtml(user.name || user.email)}" ${isRoot ? 'disabled' title="Нельзя удалить Ро

```

```

ot'' : 'title="Удалить"'>
342         <i class="bi bi-trash-fill"></i>
343     </button>
344 </div>
345 </td>
346 </tr>`;
347 }).join('');
348
349 attachTableEvents();
350 }
351
352 // Attach Table Action Button Events
353 function attachTableEvents() {
354     // Direct Table Role Selector Change
355     document.querySelectorAll('.user-table-role-select').forEa
ch(sel => {
356         sel.addEventListener('change', async (e) => {
357             const userId = parseInt(e.target.getAttribute('data-us
er-id'), 10);
358             const newRole = e.target.value;
359             const isAdmin = (newRole === 'admin') ? 1 : 0;
360             await handleUpdateField(userId, { role: newRole, is_ad
min: isAdmin }, `Роль изменена на "${newRole}"`);
361         });
362     });
363
364     // Direct Table Status Selector Change
365     document.querySelectorAll('.user-table-status-select').for
Each(sel => {
366         sel.addEventListener('change', async (e) => {
367             const userId = parseInt(e.target.getAttribute('data-us
er-id'), 10);
368             const newStatus = parseInt(e.target.value, 10);
369             await handleUpdateField(userId, { is_active: newStatus
}, `Статус изменен на "${newStatus ? 'Активен' : 'Заблокирова
н'}"`);
370         });
371     });

```



```

372
373     // Double-click row or name/email/tg to enter inline-edit
mode
374     document.querySelectorAll('#users-table-body tr').forEach(
row => {
375         const userId = parseInt(row.getAttribute('data-user-id')
, 10);
376         if (!userId) return;
377
378         const nameEl = row.querySelector('.user-cell-name');
379         const emailEl = row.querySelector('.user-cell-email');
380         const tgEl = row.querySelector('.user-cell-tg');
381
382         [nameEl, emailEl, tgEl].forEach(el => {
383             if (el) {
384                 el.addEventListener('dblclick', () => {
385                     state.editingUserId = userId;
386                     renderTable();
387                 });
388             }
389         });
390     });
391
392     // Inline Edit Button
393     document.querySelectorAll('.btn-inline-edit').forEach(btn
=> {
394         btn.addEventListener('click', (e) => {
395             const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
396             state.editingUserId = userId;
397             renderTable();
398         });
399     });
400
401     // Inline Cancel Button
402     document.querySelectorAll('.btn-cancel-inline').forEach(bt
n => {
403         btn.addEventListener('click', () => {

```

```

404         state.editingUserId = null;
405         renderTable();
406     });
407 });
408
409 // Inline Save Button
410 document.querySelectorAll('.btn-save-inline').forEach(btn
    => {
411     btn.addEventListener('click', async (e) => {
412         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
413         await saveInlineRow(userId);
414     });
415 });
416
417 // Enter & Escape key handling in inline inputs
418 document.querySelectorAll('.inline-edit-name, .inline-edit
-email, .inline-edit-tg').forEach(inp => {
419     inp.addEventListener('keydown', async (e) => {
420         const tr = inp.closest('tr');
421         const userId = tr ? parseInt(tr.getAttribute('data-use
r-id'), 10) : null;
422         if (!userId) return;
423
424         if (e.key === 'Enter') {
425             e.preventDefault();
426             await saveInlineRow(userId);
427         } else if (e.key === 'Escape') {
428             e.preventDefault();
429             state.editingUserId = null;
430             renderTable();
431         }
432     });
433 });
434
435 // Modal Edit User Button
436 document.querySelectorAll('.btn-edit-user-modal').forEach(
    btn => {

```

```

437         btn.addEventListener('click', (e) => {
438             const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
439             openEditModal(userId);
440         });
441     });
442
443     // Change Password Button
444     document.querySelectorAll('.btn-pwd-user').forEach(btn =>
{
445         btn.addEventListener('click', (e) => {
446             const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
447             const userName = e.currentTarget.getAttribute('data-na
me');
448             openPasswordModal(userId, userName);
449         });
450     });
451
452     // View Details Button
453     document.querySelectorAll('.btn-details-user').forEach(btn
=> {
454         btn.addEventListener('click', (e) => {
455             const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
456             openDetailsModal(userId);
457         });
458     });
459
460     // Delete User Button
461     document.querySelectorAll('.btn-delete-user').forEach(btn
=> {
462         btn.addEventListener('click', (e) => {
463             const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
464             const userName = e.currentTarget.getAttribute('data-na
me');
465             openDeleteModal(userId, userName);

```

```
466     });
467   });
468 }
469
470 // Save Inline Row Changes
471 async function saveInlineRow(userId) {
472   const row = document.querySelector(`tr[data-user-id="${userId}"]`);
473   if (!row) return;
474
475   const nameInput = row.querySelector('.inline-edit-name');
476   const emailInput = row.querySelector('.inline-edit-email');
477   ;
478   const roleSelect = row.querySelector('.inline-edit-role');
479   const statusSelect = row.querySelector('.inline-edit-status');
480
481   const tgInput = row.querySelector('.inline-edit-tg');
482
483   const name = nameInput ? nameInput.value.trim() : '';
484   const email = emailInput ? emailInput.value.trim() : '';
485   const role = roleSelect ? roleSelect.value : 'user';
486   const isActive = statusSelect ? parseInt(statusSelect.value, 10) : 1;
487   const tg = tgInput ? tgInput.value.trim().replace(/^@/, '') : '';
488
489   if (!email) {
490     showStatusAlert('Email не может быть пустым', 'warning');
491   };
492   return;
493 }
494
495 if (!name) {
496   showStatusAlert('Имя пользователя не может быть пустым', 'warning');
497   return;
498 }
499
500 const payload = {
```

```

497     name: name,
498     email: email,
499     role: role,
500     is_admin: (role === 'admin') ? 1 : 0,
501     is_active: isActive,
502     telegram_username: tg
503 };
504
505 state.savingUserId = userId;
506 const saveBtn = row.querySelector('.btn-save-inline');
507 if (saveBtn) saveBtn.disabled = true;
508
509 try {
510     const res = await apiFetch(`/api/admin/users/${userId}`,
511     {
512         method: 'PATCH',
513         headers: { 'Content-Type': 'application/json' },
514         body: JSON.stringify(payload)
515     });
516     state.editingUserId = null;
517     showStatusAlert(`Пользователь #${userId} (<strong>${escapeHtml(name)}</strong>) успешно сохранён`, 'success');
518     loadUsers();
519 } catch (err) {
520     showStatusAlert(`Ошибка сохранения: ${err.message}`, 'danger');
521 } finally {
522     state.savingUserId = null;
523     if (saveBtn) saveBtn.disabled = false;
524 }
525 }
526
527 // Update a Single Field or Subset of Fields
528 async function handleUpdateField(userId, payload, successMsg
529 = 'Данные сохранены') {
530     try {
531         const res = await apiFetch(`/api/admin/users/${userId}`,

```

```

    {
531         method: 'PATCH',
532         headers: { 'Content-Type': 'application/json' },
533         body: JSON.stringify(payload)
534     });
535
536     // Update local state user
537     const user = state.users.find(u => u.id === userId);
538     if (user && res.user) {
539         Object.assign(user, res.user);
540     }
541     showStatusAlert(`Пользователь #${userId}: ${successMsg}`
542 , 'success');
543     loadUsers();
544     } catch (err) {
545         showStatusAlert(`Ошибка обновления: ${err.message}`, 'da
546 nger');
547         loadUsers();
548     }
549 }
550 // Open Edit Modal
551 function openEditModal(userId) {
552     const user = state.users.find(u => u.id === userId);
553     if (!user) return;
554
555     const isEditAdmin = Boolean(user.is_admin || user.role ===
556 'admin');
557     document.getElementById('edit-user-id').value = user.id;
558     document.getElementById('edit-user-id-badge').textContent
559 = user.id;
560     document.getElementById('edit-user-email').value = user.em
561 ail || '';
562     document.getElementById('edit-user-name').value = user.nam
563 e || '';
564
565     const editAdminSw = document.getElementById('edit-user-is-
566 admin');

```

```
561     const editRoleLabel = document.getElementById('edit-role-label');
562     if (editAdminSw) {
563         editAdminSw.checked = isEditAdmin;
564         editAdminSw.disabled = (userId === 1);
565         if (editRoleLabel) {
566             editRoleLabel.textContent = isEditAdmin ? 'Администратор (admin)' : 'Пользователь (user)';
567             editRoleLabel.className = isEditAdmin ? 'text-warning' : 'text-info';
568         }
569     }
570
571     document.getElementById('edit-user-is-active').checked = Boolean(user.is_active);
572     document.getElementById('edit-user-verified').checked = Boolean(user.is_email_verified);
573
574     const modalEl = document.getElementById('modal-edit-user');
575     ;
576     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
577     ;
578     modal.show();
579 }
580
581 // Open Password Modal
582 function openPasswordModal(userId, userName) {
583     document.getElementById('pwd-user-id').value = userId;
584     document.getElementById('pwd-user-target').textContent = `${userName} (ID #${userId})`;
585     document.getElementById('pwd-input-val').value = '';
586
587     const modalEl = document.getElementById('modal-password-user');
588     const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
589     ;
590     modal.show();
591 }
```

```

589 |
590 | // Open Details Modal
591 | async function openDetailsModal(userId) {
592 |     const modalEl = document.getElementById('modal-details-use
r');
593 |     const contentEl = document.getElementById('details-user-co
ntent');
594 |     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
;
595 |     modal.show();
596 |
597 |     contentEl.innerHTML = `
598 |         <div class="text-center py-4 text-muted">
599 |             <div class="spinner-border spinner-border-sm text-prim
ary me-2" role="status"></div>
600 |             Загрузка профиля пользователя...
601 |         </div>`;
602 |
603 |     try {
604 |         const data = await apiFetch(`/api/admin/users/${userId}`
);
605 |         const u = data.user || {};
606 |         const s = data.settings || {};
607 |         const perms = data.permissions || [];
608 |
609 |         contentEl.innerHTML = `
610 |             <div class="row g-3">
611 |                 <div class="col-md-6">
612 |                     <h6 class="text-primary border-bottom pb-1 mb-2">□
□ Основные данные</h6>
613 |                     <table class="table table-sm table-borderless smal
l mb-0">
614 |                         <tr><td class="text-muted" style="width:120px;">
ID:</td><td class="fw-bold">${u.id}</td></tr>
615 |                         <tr><td class="text-muted">Email:</td><td>${esca
peHtml(u.email)} ${u.is_email_verified ? '<span class="badge b
g-success">Verified</span>' : '<span class="badge bg-secondary
">Unverified</span>'}</td></tr>

```



```

616 |         <tr><td class="text-muted">Имя:</td><td>${escape
|   <td>${escapeHtml(u.name || '-')}}</td></tr>
617 |         <tr><td class="text-muted">Роль:</td><td><span c
|   lass="badge bg-info text-dark">${escapeHtml(u.role || 'user')}}
|   </span> ${u.is_admin ? '<span class="badge bg-warning text-dar
|   k">Admin</span>' : ''}</td></tr>
618 |         <tr><td class="text-muted">Статус:</td><td>${u.i
|   s_active ? '<span class="text-success">Активен</span>' : '<spa
|   n class="text-danger">Заблокирован</span>'}</td></tr>
619 |         <tr><td class="text-muted">Пароль:</td><td>${u.h
|   as_password ? '<span class="text-success">Установлен</span>' :
|   '<span class="text-muted">Не задан</span>'}</td></tr>
620 |         <tr><td class="text-muted">Создан:</td><td>${esc
|   apeHtml(u.created_at || '-')}}</td></tr>
621 |         <tr><td class="text-muted">Последний вход:</td><
|   td>${escapeHtml(u.last_login || '-')}}</td></tr>
622 |         </table>
623 |     </div>
624 |     <div class="col-md-6">
625 |         <h6 class="text-info border-bottom pb-1 mb-2">Telegram & Настройки</h6>
626 |         <table class="table table-sm table-borderless smal
|   l mb-0">
627 |             <tr><td class="text-muted" style="width:130px;">
|   Telegram ID:</td><td>${u.telegram_id ? escapeHtml(u.telegram_i
|   d) : '<span class="text-muted">Не привязан</span>'}</td></tr>
628 |             <tr><td class="text-muted">TG Username:</td><td>
|   ${u.telegram_username ? `@${escapeHtml(u.telegram_username)}`
|   : '<span class="text-muted">—</span>'}</td></tr>
629 |             <tr><td class="text-muted">Тема UI:</td><td>${es
|   capeHtml(s.theme || 'dark')}}</td></tr>
630 |             <tr><td class="text-muted">Язык:</td><td>${escap
|   eHtml(s.language || 'ru')}}</td></tr>
631 |             <tr><td class="text-muted">TTS голос:</td><td>${
|   escapeHtml(s.tts_voice || 'ru-RU-DmitryNeural')}} (${escapeHtml
|   (s.tts_system || 'edge-tts')}}</td></tr>
632 |             <tr><td class="text-muted">Модель чата:</td><td>
|   ${escapeHtml(s.model || 'По умолчанию')}}</td></tr>

```

```

633         </table>
634     </div>
635     <div class="col-12 mt-3">
636         <h6 class="text-warning border-bottom pb-1 mb-2">□
637         □ Разрешения системы</h6>
638         <div class="d-flex flex-wrap gap-1">
639             ${perms.length > 0 ? perms.map(p => `<span class
640             ="badge bg-secondary">${escapeHtml(p)}</span>`).join('') : '<s
641             pan class="text-muted small">Нет явных разрешений</span>' }
642             </div>
643         </div>
644         ${s.system_instruction ? `
645         <div class="col-12 mt-2">
646             <h6 class="border-bottom pb-1 mb-1">□ Пользовател
647             ьская системная инструкция</h6>
648             <pre class="bg-body-tertiary p-2 rounded border sm
649             all font-monospace" style="max-height:120px;overflow-y:auto;">
650             ${escapeHtml(s.system_instruction)}</pre>
651             </div>` : ''}
652         </div>`;
653     } catch (err) {
654         contentEl.innerHTML = `<div class="text-danger py-3">Оши
655         бка загрузки данных: ${escapeHtml(err.message)}</div>`;
656     }
657 }
658
659 // Open Delete Modal
660 function openDeleteModal(userId, userName) {
661     document.getElementById('delete-user-id').value = userId;
662     document.getElementById('delete-user-name-target').textCon
663     tent = `${userName} (ID #${userId})`;
664
665     const modalEl = document.getElementById('modal-delete-user
666     ');
667     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
668     ;
669     modal.show();
670 }

```

```
661
662 // Initialize Event Listeners
663 function initListeners() {
664     // Open Create User Modal
665     const btnOpenCreate = document.getElementById('btn-open-cr
eate-user');
666     if (btnOpenCreate) {
667         btnOpenCreate.addEventListener('click', () => {
668             const form = document.getElementById('form-create-user
');
669             if (form) form.reset();
670             document.getElementById('create-user-is-admin').checke
d = false;
671             const createRoleLabel = document.getElementById('creat
e-role-label');
672             if (createRoleLabel) {
673                 createRoleLabel.textContent = 'Пользователь (user)';
674                 createRoleLabel.className = 'text-info';
675             }
676             document.getElementById('create-user-is-active').check
ed = true;
677             document.getElementById('create-user-verified').checke
d = true;
678             const modalEl = document.getElementById('modal-create-
user');
679             const modal = bootstrap.Modal.getOrCreateInstance(modal
El);
680             modal.show();
681         });
682     }
683
684     // Dynamic Switch Hints
685     const createAdminSw = document.getElementById('create-user
-is-admin');
686     const createRoleLabel = document.getElementById('create-ro
le-label');
687     if (createAdminSw && createRoleLabel) {
688         createAdminSw.addEventListener('change', (e) => {
```

```
689 |         createRoleLabel.textContent = e.target.checked ? 'Адми
| нистратор (admin)' : 'Пользователь (user)';
690 |         createRoleLabel.className = e.target.checked ? 'text-w
| arning' : 'text-info';
691 |     });
692 | }
693 |
694 |     const editAdminSw = document.getElementById('edit-user-is-
| admin');
695 |     const editRoleLabel = document.getElementById('edit-role-l
| abel');
696 |     if (editAdminSw && editRoleLabel) {
697 |         editAdminSw.addEventListener('change', (e) => {
698 |             editRoleLabel.textContent = e.target.checked ? 'Админи
| стратор (admin)' : 'Пользователь (user)';
699 |             editRoleLabel.className = e.target.checked ? 'text-war
| ning' : 'text-info';
700 |         });
701 |     }
702 |
703 |     // Refresh Users Button
704 |     const btnRefresh = document.getElementById('btn-refresh-us
| ers');
705 |     if (btnRefresh) {
706 |         btnRefresh.addEventListener('click', () => {
707 |             loadUsers();
708 |         });
709 |     }
710 |
711 |     // Search Input with Debounce
712 |     const searchInput = document.getElementById('users-search-
| input');
713 |     const searchClear = document.getElementById('users-search-
| clear');
714 |     let debounceTimeout = null;
715 |
716 |     if (searchInput) {
717 |         searchInput.addEventListener('input', (e) => {
```

```
718 |         const val = e.target.value;
719 |         if (searchClear) {
720 |             searchClear.classList.toggle('d-none', !val);
721 |         }
722 |         clearTimeout(debounceTimeout);
723 |         debounceTimeout = setTimeout(() => {
724 |             state.searchQuery = val.trim();
725 |             loadUsers();
726 |         }, 300);
727 |     });
728 | }
729 |
730 | if (searchClear) {
731 |     searchClear.addEventListener('click', () => {
732 |         if (searchInput) searchInput.value = '';
733 |         searchClear.classList.add('d-none');
734 |         state.searchQuery = '';
735 |         loadUsers();
736 |     });
737 | }
738 |
739 | // Role Filter
740 | const filterRole = document.getElementById('users-filter-r
741 | ole');
742 | if (filterRole) {
743 |     filterRole.addEventListener('change', (e) => {
744 |         state.filterRole = e.target.value;
745 |         loadUsers();
746 |     });
747 | }
748 |
749 | // Status Filter
750 | const filterStatus = document.getElementById('users-filter
751 | -status');
752 | if (filterStatus) {
753 |     filterStatus.addEventListener('change', (e) => {
754 |         state.filterStatus = e.target.value;
755 |         loadUsers();
756 |     });
757 | }
```

```

754     });
755 }
756
757 // Generator button for create modal
758 const btnGenCreatePwd = document.getElementById('btn-gen-c
reate-password');
759 if (btnGenCreatePwd) {
760     btnGenCreatePwd.addEventListener('click', () => {
761         const pwdInput = document.getElementById('create-user-
password');
762         if (pwdInput) pwdInput.value = generatePassword(12);
763     });
764 }
765
766 // Generator button for password change modal
767 const btnGenPwdVal = document.getElementById('btn-gen-pwd-
val');
768 if (btnGenPwdVal) {
769     btnGenPwdVal.addEventListener('click', () => {
770         const pwdInput = document.getElementById('pwd-input-va
l');
771         if (pwdInput) pwdInput.value = generatePassword(12);
772     });
773 }
774
775 // Create User Form Submit
776 const formCreate = document.getElementById('form-create-us
er');
777 if (formCreate) {
778     formCreate.addEventListener('submit', async (e) => {
779         e.preventDefault();
780         const submitBtn = document.getElementById('btn-submit-
create-user');
781         if (submitBtn) submitBtn.disabled = true;
782
783         const isAdmin = document.getElementById('create-user-i
s-admin').checked ? 1 : 0;
784         const payload = {

```

```
785 |         email: document.getElementById('create-user-email').
| value.trim(),
786 |         name: document.getElementById('create-user-name').va
| lue.trim(),
787 |         password: document.getElementById('create-user-passw
| ord').value.trim(),
788 |         role: isAdmin ? 'admin' : 'user',
789 |         is_admin: isAdmin,
790 |         is_active: document.getElementById('create-user-is-a
| ctive').checked ? 1 : 0,
791 |         is_email_verified: document.getElementById('create-u
| ser-verified').checked ? 1 : 0
792 |     };
793 |
794 |     try {
795 |         await apiFetch('/api/admin/users', {
796 |             method: 'POST',
797 |             headers: { 'Content-Type': 'application/json' },
798 |             body: JSON.stringify(payload)
799 |         });
800 |         const modalEl = document.getElementById('modal-creat
| e-user');
801 |         const modal = bootstrap.Modal.getInstance(modalEl);
802 |         if (modal) modal.hide();
803 |
804 |         showStatusAlert(`Пользователь ${escapeHtml(payload.n
| ame)} успешно создан`, 'success');
805 |         loadUsers();
806 |     } catch (err) {
807 |         showStatusAlert(`Ошибка создания пользователя: ${err
| .message}`, 'danger');
808 |     } finally {
809 |         if (submitBtn) submitBtn.disabled = false;
810 |     }
811 | });
812 | }
813 |
814 | // Edit User Form Submit (Modal)
```

```
815     const formEdit = document.getElementById('form-edit-user')
      ;
816     if (formEdit) {
817         formEdit.addEventListener('submit', async (e) => {
818             e.preventDefault();
819             const submitBtn = document.getElementById('btn-submit-
edit-user');
820             if (submitBtn) submitBtn.disabled = true;
821
822             const userId = parseInt(document.getElementById('edit-
user-id').value, 10);
823             const isAdmin = document.getElementById('edit-user-is-
admin').checked ? 1 : 0;
824             const payload = {
825                 email: document.getElementById('edit-user-email').va
lue.trim(),
826                 name: document.getElementById('edit-user-name').valu
e.trim(),
827                 role: isAdmin ? 'admin' : 'user',
828                 is_admin: isAdmin,
829                 is_active: document.getElementById('edit-user-is-act
ive').checked ? 1 : 0,
830                 is_email_verified: document.getElementById('edit-use
r-verified').checked ? 1 : 0
831             };
832
833             try {
834                 await apiFetch(`/api/admin/users/${userId}`, {
835                     method: 'PUT',
836                     headers: { 'Content-Type': 'application/json' },
837                     body: JSON.stringify(payload)
838                 });
839                 const modalEl = document.getElementById('modal-edit-
user');
840                 const modal = bootstrap.Modal.getInstance(modalEl);
841                 if (modal) modal.hide();
842
843                 showStatusAlert(`Данные пользователя #${userId} успе
```



```

        шно обновлены!`, 'success'));
844         loadUsers();
845     } catch (err) {
846         showStatusAlert(`Ошибка обновления: ${err.message}`,
        'danger');
847     } finally {
848         if (submitBtn) submitBtn.disabled = false;
849     }
850 });
851 }
852
853 // Change Password Form Submit
854 const formPwd = document.getElementById('form-password-use
r');
855 if (formPwd) {
856     formPwd.addEventListener('submit', async (e) => {
857         e.preventDefault();
858         const submitBtn = document.getElementById('btn-submit-
pwd-user');
859         if (submitBtn) submitBtn.disabled = true;
860
861         const userId = parseInt(document.getElementById('pwd-u
ser-id').value, 10);
862         const newPassword = document.getElementById('pwd-input
-val').value.trim();
863
864         try {
865             await apiFetch(`/api/admin/users/${userId}/password`
, {
866                 method: 'POST',
867                 headers: { 'Content-Type': 'application/json' },
868                 body: JSON.stringify({ password: newPassword })
869             });
870             const modalEl = document.getElementById('modal-passw
ord-user');
871             const modal = bootstrap.Modal.getInstance(modalEl);
872             if (modal) modal.hide();
873

```

```
874 |         showStatusAlert(`Пароль для пользователя #${userId}
      | успешно установлен!`, 'success');
875 |         loadUsers();
876 |     } catch (err) {
877 |         showStatusAlert(`Ошибка установки пароля: ${err.mess
      | age}`, 'danger');
878 |     } finally {
879 |         if (submitBtn) submitBtn.disabled = false;
880 |     }
881 | });
882 | }
883 |
884 | // Delete User Confirm Button
885 | const btnConfirmDelete = document.getElementById('btn-conf
      | irm-delete-user');
886 | if (btnConfirmDelete) {
887 |     btnConfirmDelete.addEventListener('click', async () => {
888 |         btnConfirmDelete.disabled = true;
889 |         const userId = parseInt(document.getElementById('delet
      | e-user-id').value, 10);
890 |
891 |         try {
892 |             await apiFetch(`/api/admin/users/${userId}`, { metho
      | d: 'DELETE' });
893 |             const modalEl = document.getElementById('modal-delet
      | e-user');
894 |             const modal = bootstrap.Modal.getInstance(modalEl);
895 |             if (modal) modal.hide();
896 |
897 |             showStatusAlert(`Пользователь #${userId} успешно уда
      | лён.`, 'success');
898 |             loadUsers();
899 |         } catch (err) {
900 |             showStatusAlert(`Ошибка удаления пользователя: ${err
      | .message}`, 'danger');
901 |         } finally {
902 |             btnConfirmDelete.disabled = false;
903 |         }
```

```
904 |     });
905 | }
906 |
907 | // Orphaned Directories Check Button
908 | const btnCheckOrphaned = document.getElementById('btn-check-orphaned-dirs');
909 | if (btnCheckOrphaned) {
910 |     btnCheckOrphaned.addEventListener('click', () => {
911 |         const modalEl = document.getElementById('modal-orphaned-dirs');
912 |         if (modalEl) {
913 |             const modal = new bootstrap.Modal(modalEl);
914 |             modal.show();
915 |             loadOrphanedDirs();
916 |         }
917 |     });
918 | }
919 |
920 | // Orphaned Directories Refresh Button
921 | const btnRefreshOrphaned = document.getElementById('btn-refresh-orphaned-dirs');
922 | if (btnRefreshOrphaned) {
923 |     btnRefreshOrphaned.addEventListener('click', () => {
924 |         loadOrphanedDirs();
925 |     });
926 | }
927 |
928 | // Cleanup All Orphaned Directories Button
929 | const btnCleanupAll = document.getElementById('btn-cleanup-all-orphaned');
930 | if (btnCleanupAll) {
931 |     btnCleanupAll.addEventListener('click', async () => {
932 |         if (!confirm('Вы уверены, что хотите удалить ВСЕ обнаруженные мертвые директории пользователей? Это действие необратимо.')) {
933 |             return;
934 |         }
935 |         btnCleanupAll.disabled = true;
```

```

936         try {
937             const res = await apiFetch('/api/admin/users/orphaned-dirs/clean', {
938                 method: 'POST',
939                 headers: { 'Content-Type': 'application/json' },
940                 body: JSON.stringify({})
941             });
942             showStatusAlert(`Успешно удалено мертвых директорий:
943             ${res.total_deleted} (освобождено: ${formatBytes(res.freed_bytes)})`, 'success');
944             loadOrphanedDirs();
945         } catch (err) {
946             showStatusAlert(`Ошибка очистки мертвых директорий:
947             ${err.message}`, 'danger');
948         } finally {
949             btnCleanupAll.disabled = false;
950         }
951     }
952
953     // Format bytes helper
954     function formatBytes(bytes, decimals = 1) {
955         if (!bytes || bytes <= 0) return '0 B';
956         const k = 1024;
957         const dm = decimals < 0 ? 0 : decimals;
958         const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
959         const i = Math.floor(Math.log(bytes) / Math.log(k));
960         return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
961         ' ' + sizes[i];
962     }
963
964     // Load Orphaned Directories from Backend
965     async function loadOrphanedDirs() {
966         const tbody = document.getElementById('orphaned-dirs-tbody');
967         const countEl = document.getElementById('orphaned-dirs-count');

```

```
967 |     const sizeEl = document.getElementById('orphaned-dirs-size
    | ');
968 |     const filesEl = document.getElementById('orphaned-dirs-fil
    | es');
969 |     const btnCleanupAll = document.getElementById('btn-cleanup
    | -all-orphaned');
970 |
971 |     if (tbody) {
972 |         tbody.innerHTML = `
973 |             <tr>
974 |                 <td colspan="6" class="text-center py-4 text-muted">
975 |                     <div class="spinner-border spinner-border-sm text-
    | warning me-2" role="status"></div>
976 |                     Сканирование хранилища data/users/...
977 |                 </td>
978 |             </tr>`;
979 |     }
980 |
981 |     try {
982 |         const data = await apiFetch('/api/admin/users/orphaned-d
    | irs');
983 |         const dirs = data.orphaned_dirs || [];
984 |
985 |         if (countEl) countEl.textContent = data.total || 0;
986 |         if (sizeEl) sizeEl.textContent = formatBytes(data.total_
    | size_bytes || 0);
987 |         if (filesEl) filesEl.textContent = data.total_files || 0
    | ;
988 |
989 |         if (btnCleanupAll) {
990 |             btnCleanupAll.disabled = (dirs.length === 0);
991 |         }
992 |
993 |         if (!tbody) return;
994 |
995 |         if (dirs.length === 0) {
996 |             tbody.innerHTML = `
997 |                 <tr>
```

```

998         <td colspan="6" class="text-center py-4 text-succe
| ss">
999         <i class="bi bi-check-circle-fill fs-4 d-block m
| b-1"></i>
1000         Мертвых директорий не обнаружено. Все папки соот
| ветствуют активным пользователям.
1001     </td>
1002 </tr>`;
1003     return;
1004 }
1005
1006     tbody.innerHTML = dirs.map(d => {
1007         const badgeClass = d.is_test_dir ? 'bg-secondary' : 'b
| g-warning text-dark';
1008         return `
1009         <tr data-dir-name="${escapeHtml(d.name)}">
1010             <td class="font-monospace fw-bold text-light">
1011                 <i class="bi bi-folder text-warning me-1"></i>${
| escapeHtml(d.name)}
1012             </td>
1013             <td>
1014                 <span class="badge ${badgeClass} small">${escape
| Html(d.reason)}</span>
1015             </td>
1016             <td class="text-center">${d.files_count}</td>
1017             <td class="text-center">${formatBytes(d.size_bytes
| )}</td>
1018             <td class="small">${formatDate(d.modified_at)}</td>
| >
1019             <td class="text-center">
1020                 <button class="btn btn-outline-danger btn-sm py-
| 0 px-2 btn-delete-single-orphaned" data-dir="${escapeHtml(d.na
| me)}" title="Удалить директорию">
1021                     <i class="bi bi-trash"></i>
1022                 </button>
1023             </td>
1024         </tr>`;
1025     }).join('');

```

```

1026
1027     // Add click listeners to single delete buttons
1028     tbody.querySelectorAll('.btn-delete-single-orphaned').fo
rEach(btn => {
1029         btn.addEventListener('click', async () => {
1030             const dirName = btn.getAttribute('data-dir');
1031             if (!dirName) return;
1032             if (!confirm(`Удалить мертвую директорию "${dirName}
"?`)) return;
1033
1034             btn.disabled = true;
1035             try {
1036                 const res = await apiFetch('/api/admin/users/orpha
ned-dirs/clean', {
1037                     method: 'POST',
1038                     headers: { 'Content-Type': 'application/json' },
1039                     body: JSON.stringify({ dirs: [dirName] })
1040                 });
1041                 showStatusAlert(`Директория "${dirName}" успешно у
далена.`, 'success');
1042                 loadOrphanedDirs();
1043             } catch (err) {
1044                 showStatusAlert(`Ошибка удаления директории ${dirN
ame}: ${err.message}`, 'danger');
1045                 btn.disabled = false;
1046             }
1047         });
1048     });
1049     } catch (err) {
1050         console.error('[UsersTab] Error loading orphaned dirs:',
err);
1051         if (tbody) {
1052             tbody.innerHTML = `
1053                 <tr>
1054                     <td colspan="6" class="text-center py-4 text-dange
r">
1055                         <i class="bi bi-exclamation-octagon fs-4 d-block
mb-1"></i>

```

```
1056 |             Ошибка сканирования: ${escapeHtml(err.message)}
1057 |         </td>
1058 |     </tr>`;
1059 |     }
1060 | }
1061 | }
1062 |
1063 | // Global Init Function
1064 | function initUsersTab() {
1065 |     console.log('[UsersTab] Initializing user management tab w
1066 | ith inline editing...');
1067 |     if (!state.initialized) {
1068 |         initListeners();
1069 |         state.initialized = true;
1070 |     }
1071 |     loadUsers();
1072 | }
1073 | // Register globally
1074 | window.initUsersTab = initUsersTab;
1075 | })();
1076 |
```


index.html

```
1 | <!-- User Assistant Application Tab -->
2 | <style>
3 |   .assistant-container {
4 |     padding: 1rem;
5 |   }
6 |   .assistant-card {
7 |     background: #0f172a !important;
8 |     border: 1px solid #334155 !important;
9 |     border-radius: 10px;
10 |     padding: 1rem;
11 |     display: flex;
12 |     flex-direction: column;
13 |     gap: 0.25rem;
14 |   }
15 |   .assistant-label {
16 |     font-size: 0.75rem;
17 |     text-transform: uppercase;
18 |     letter-spacing: 0.05em;
19 |     color: #94a3b8;
20 |   }
21 |   .assistant-value {
22 |     font-size: 1.35rem;
23 |     font-weight: 700;
24 |     color: #f8fafc;
25 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
26 |   }
27 |   .assistant-panel {
28 |     background: #0f172a !important;
29 |     border: 1px solid #334155 !important;
30 |     border-radius: 10px;
31 |     overflow: hidden;
32 |   }
33 |   .assistant-panel-header {
34 |     background: #1e293b !important;
35 |     border-bottom: 1px solid #334155 !important;
```

```

36     padding: 0.75rem 1rem;
37     font-weight: 600;
38     font-size: 0.88rem;
39     color: #38bdf8;
40     display: flex;
41     align-items: center;
42     justify-content: space-between;
43 }
44 .assistant-list-item {
45     padding: 0.75rem 1rem;
46     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
47     transition: background 0.15s ease;
48 }
49 .assistant-list-item:hover {
50     background: rgba(30, 41, 59, 0.5);
51 }
52 </style>
53
54 <div class="assistant-container">
55     <!-- Header Alert & Status -->
56     <div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 p-3 rounded-3 text-white" style="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%); border: 1px solid #334155;">
57         <div>
58             <h5 class="mb-1 fw-bold text-white d-flex align-items-center gap-2">
59                 <span class="fs-4">📧 </span> User Assistant (Agenda, Mail & Calendar)
60             </h5>
61             <p class="mb-0 small text-light" style="color: #cbd5e1 !important;">
62                 Персональный ассистент продуктивности: расписание дня, сортировка входящих писем, события календаря и поиск документов.
63             </p>
64         </div>
65         <div class="d-flex align-items-center gap-2">

```

```

66         <span class="badge rounded-pill bg-success-subtle text-s
| uccess border border-success px-3 py-2" id="assistant-status-b
| adge">
67             • Готов к работе
68         </span>
69         <button class="btn btn-sm btn-outline-info rounded-pill
| px-3" id="btn-assistant-refresh">
70             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
71         </button>
72         <button class="btn btn-sm btn-outline-warning rounded-pi
| ll px-3" id="btn-assistant-config">
73             <i class="bi bi-gear-fill me-1"></i> Настройки (config
| .json)
74         </button>
75     </div>
76 </div>
77
78 <!-- Top Metrics Bar -->
79 <div class="row g-3 mb-3">
80     <div class="col-md-3 col-sm-6">
81         <div class="assistant-card">
82             <span class="assistant-label">Событий на сегодня</span
| >
83             <span class="assistant-value text-info" id="assistant-
| events-count">0</span>
84         </div>
85     </div>
86     <div class="col-md-3 col-sm-6">
87         <div class="assistant-card">
88             <span class="assistant-label">Непрочитанных писем</spa
| n>
89             <span class="assistant-value text-warning" id="assista
| nt-unread-count">0</span>
90         </div>
91     </div>
92     <div class="col-md-3 col-sm-6">
93         <div class="assistant-card">
94             <span class="assistant-label">Недавних файлов</span>

```

```
95 |         <span class="assistant-value text-success" id="assista
| nt-files-count">0</span>
96 |     </div>
97 | </div>
98 | <div class="col-md-3 col-sm-6">
99 |     <div class="assistant-card">
100 |         <span class="assistant-label">Рабочее пространство</sp
| an>
101 |         <span class="assistant-value text-light" style="font-s
| ize: 1.05rem;" id="assistant-workspace-status">Активно</span>
102 |     </div>
103 | </div>
104 | </div>
105 |
106 | <!-- Main Content Panels Grid -->
107 | <div class="row g-3">
108 |     <!-- Calendar & Agenda Panel -->
109 |     <div class="col-lg-6">
110 |         <div class="assistant-panel h-100">
111 |             <div class="assistant-panel-header">
112 |                 <span>📅 Повестка дня и ближайшие события</span>
113 |                 <span class="badge bg-secondary" id="assistant-event
| s-badge">0 событий</span>
114 |             </div>
115 |             <div class="p-0" id="assistant-events-list" style="max
| -height: 450px; overflow-y: auto;">
116 |                 <div class="p-3 text-muted text-center small">Загруз
| ка повестки дня...</div>
117 |             </div>
118 |         </div>
119 |     </div>
120 |
121 |     <!-- Email Triage Panel -->
122 |     <div class="col-lg-6">
123 |         <div class="assistant-panel h-100">
124 |             <div class="assistant-panel-header">
125 |                 <span>✉ Непрочитанные сообщения</span>
126 |                 <span class="badge bg-secondary" id="assistant-mail-
```

```
    badge">0 писем</span>
127 |         </div>
128 |         <div class="p-0" id="assistant-mail-list" style="max-h
    | eight: 450px; overflow-y: auto;">
129 |             <div class="p-3 text-muted text-center small">Загруз
    | ка почты...</div>
130 |         </div>
131 |     </div>
132 | </div>
133 | </div>
134 | </div>
135 |
```

main.js

```
1 // =====
  =====
2 // Process Name: User Assistant Web Tab Module
3 // =====
  =====
4 // Description:
5 //   Client-side JavaScript controller for the User Assistant
6 //   administrative web interface tab.
7 //
8 // File: main.js
9 // Project: ai-breadboard
10 // Package: src.api.webinterface.user_assistant_tab
11 // Author: hypo69
12 // Copyright: © 2026 hypo69
13 // =====
  =====
14
15 (function() {
16   let isAssistantInitialized = false;
17
18   async function fetchAgenda() {
19     try {
20       const res = await fetch('/api/v1/assistant/agenda');
21       if (!res.ok) return;
22       const data = await res.json();
23
24       const eventsCountEl = document.getElementById('assistant-
25       -events-count');
26       const unreadCountEl = document.getElementById('assistant-
27       -unread-count');
28       const filesCountEl = document.getElementById('assistant-
29       files-count');
30       const eventsBadge = document.getElementById('assistant-e
31       vents-badge');
32       const mailBadge = document.getElementById('assistant-mai
33       l-badge');
```

```

29     const eventsList = document.getElementById('assistant-ev
ents-list');
30     const mailList = document.getElementById('assistant-mail
-list');
31
32     const events = data.events || [];
33     const emails = data.unread_emails || [];
34     const files = data.recent_files || [];
35
36     if (eventsCountEl) eventsCountEl.innerText = events.leng
th;
37     if (unreadCountEl) unreadCountEl.innerText = emails.leng
th;
38     if (filesCountEl) filesCountEl.innerText = files.length;
39     if (eventsBadge) eventsBadge.innerText = `${events.lengt
h} событий`;
40     if (mailBadge) mailBadge.innerText = `${emails.length} п
исем`;
41
42     if (eventsList) {
43         if (events.length === 0) {
44             eventsList.innerHTML = '<div class="p-3 text-muted t
ext-center small">Нет запланированных событий на сегодня.</div
>';
45         } else {
46             eventsList.innerHTML = events.map(e => `
47                 <div class="assistant-list-item d-flex justify-con
tent-between align-items-center">
48                     <div>
49                         <div class="fw-semibold text-light">${e.summar
y || 'Без названия'}</div>
50                         <div class="text-secondary small">📅 ${e.start
|| 'Весь день'} ${e.location ? '📍 ' + e.location : ''}</div>
51                     </div>
52                 </div>
53             `).join('');
54         }
55     }

```

```

56 |
57 |     if (maillist) {
58 |         if (emails.length === 0) {
59 |             maillist.innerHTML = '<div class="p-3 text-muted tex
60 | t-center small">Нет новых непрочитанных сообщений.</div>';
61 |         } else {
62 |             maillist.innerHTML = emails.map(m => `
63 |                 <div class="assistant-list-item d-flex justify-con
64 | tent-between align-items-center">
65 |                     <div>
66 |                         <div class="fw-semibold text-light">${m.subjec
67 | t || 'Без темы'}</div>
68 |                         <div class="text-secondary small">□ ${m.from
69 | || 'Неизвестный'} &bull; ${m.snippet || ''}</div>
70 |                     </div>
71 |                 </div>
72 |             `).join('');
73 |         }
74 |     } catch (e) {
75 |         console.error('[UserAssistantTab] Failed to fetch agenda
76 | :', e);
77 |     }
78 | }
79 |
80 | function initUserAssistantTab() {
81 |     fetchAgenda();
82 |     if (!isAssistantInitialized) {
83 |         const refreshBtn = document.getElementById('btn-assistan
84 | t-refresh');
85 |         if (refreshBtn) refreshBtn.addEventListener('click', fet
86 | chAgenda);
87 |
88 |         const configBtn = document.getElementById('btn-assistant
89 | -config');
90 |         if (configBtn) {
91 |             configBtn.addEventListener('click', () => {
92 |                 if (typeof window.openAppConfigModal === 'function')

```



```
86 |         window.openAppConfigModal('user_assistant');
87 |     }
88 | });
89 | }
90 |     isAssistantInitialized = true;
91 | }
92 | }
93 |
94 | window.initUserAssistantTab = initUserAssistantTab;
95 | })();
96 |
```

index.html

```
1 <!-- User Directories and Workspace Storage Tab -->
2 <div class="container-fluid py-2" id="user-directories-tab-root">
3   <!-- Top Banner & Quick Controls -->
4   <div class="d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3 p-3 bg-body-tertiary border rounded shadow-sm">
5     <div>
6       <h5 class="mb-1 text-primary d-flex align-items-center gap-2">
7         <i class="bi bi-folder2-open text-primary"></i>
8         <span>Пользовательские директории и хранилище</span>
9       </h5>
10      <p class="mb-0 text-muted small">Мониторинг файловых хранилищ пользователей (<code>data/users/&lt;user_id>;</code>), анализ дисковых квот, аудит деревьев файлов и очистка осиротевших каталогов.</p>
11    </div>
12    <div class="d-flex flex-wrap gap-2 align-items-center">
13      <button class="btn btn-outline-warning btn-sm d-flex align-items-center gap-1.5 rounded-pill px-3" id="btn-open-orphaned-dirs-modal" title="Проверить и удалить неиспользуемые папки">
14        <i class="bi bi-folder-x"></i> Мертвые директории (<span id="badge-orphaned-count">0</span>)
15      </button>
16      <button class="btn btn-primary btn-sm d-flex align-items-center gap-1.5 rounded-pill px-3" id="btn-refresh-user-dirs" title="Обновить данные хранилища">
17        <i class="bi bi-arrow-clockwise"></i> Обновить
18      </button>
19    </div>
20  </div>
21
22 <!-- Metric Cards -->
23 <div class="row g-3 mb-3">
```

```

24 |     <div class="col-6 col-md-3">
25 |         <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
26 |             <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
27 |                 <div>
28 |                     <div class="text-muted small">Пользователей с хран
    | илищем</div>
29 |                     <h4 class="mb-0 fw-bold text-primary" id="metric-t
    | otal-users">0</h4>
30 |                 </div>
31 |                 <div class="bg-primary bg-opacity-25 rounded p-2 tex
    | t-primary">
32 |                     <i class="bi bi-people-fill fs-4"></i>
33 |                 </div>
34 |             </div>
35 |         </div>
36 |     </div>
37 |     <div class="col-6 col-md-3">
38 |         <div class="card bg-body-tertiary border h-100 shadow-sm
    | ">
39 |             <div class="card-body p-3 d-flex align-items-center ju
    | stify-content-between">
40 |                 <div>
41 |                     <div class="text-muted small">Занято на диске</div
    | >
42 |                     <h4 class="mb-0 fw-bold text-info" id="metric-tota
    | l-storage">0 B</h4>
43 |                 </div>
44 |                 <div class="bg-info bg-opacity-25 rounded p-2 text-i
    | nfo">
45 |                     <i class="bi bi-hdd-fill fs-4"></i>
46 |                 </div>
47 |             </div>
48 |         </div>
49 |     </div>
50 |     <div class="col-6 col-md-3">
51 |         <div class="card bg-body-tertiary border h-100 shadow-sm

```

```

">
52         <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
53             <div>
54                 <div class="text-muted small">Всего файлов</div>
55                 <h4 class="mb-0 fw-bold text-success" id="metric-t
otal-files">0</h4>
56             </div>
57             <div class="bg-success bg-opacity-25 rounded p-2 tex
t-success">
58                 <i class="bi bi-file-earmark-text-fill fs-4"></i>
59             </div>
60         </div>
61     </div>
62 </div>
63 <div class="col-6 col-md-3">
64     <div class="card bg-body-tertiary border h-100 shadow-sm
">
65         <div class="card-body p-3 d-flex align-items-center ju
stify-content-between">
66             <div>
67                 <div class="text-muted small">Осиротевшие каталоги
</div>
68                 <h4 class="mb-0 fw-bold text-warning" id="metric-o
rphaned-size">0 B</h4>
69             </div>
70             <div class="bg-warning bg-opacity-25 rounded p-2 tex
t-warning">
71                 <i class="bi bi-trash3-fill fs-4"></i>
72             </div>
73         </div>
74     </div>
75 </div>
76 </div>
77
78 <!-- Search & Filter Controls -->
79 <div class="card bg-body-tertiary border mb-3 shadow-sm">
80     <div class="card-body p-2.5">

```

```

81 |         <div class="row g-2 align-items-center">
82 |             <div class="col-12 col-md-5">
83 |                 <div class="input-group input-group-sm">
84 |                     <span class="input-group-text"><i class="bi bi-sea
rch"></i></span>
85 |                     <input type="text" class="form-control" id="user-d
irs-search-input" placeholder="Поиск по имени пользователя, em
ail или имени файла...">
86 |                     <button class="btn btn-outline-secondary d-none" t
ype="button" id="user-dirs-search-clear">❌ </button>
87 |                 </div>
88 |             </div>
89 |             <div class="col-6 col-md-3">
90 |                 <select class="form-select form-select-sm" id="user-
dirs-ext-filter">
91 |                     <option value="">Все типы файлов</option>
92 |                     <option value="pdf">Документы PDF (.pdf)</option>
93 |                     <option value="txt">Текстовые (.txt)</option>
94 |                     <option value="md">Markdown (.md)</option>
95 |                     <option value="json">Данные JSON (.json)</option>
96 |                     <option value="csv">Таблицы CSV (.csv)</option>
97 |                     <option value="png">Изображения (.png, .jpg)</opti
on>
98 |                     <option value="mp3">Аудио (.mp3, .wav)</option>
99 |                 </select>
100 |             </div>
101 |             <div class="col-6 col-md-4 text-end text-muted small"
id="user-dirs-status-label">
102 |                 Загрузка...
103 |             </div>
104 |         </div>
105 |     </div>
106 | </div>
107 |
108 | <!-- Alert Message Box -->
109 | <div id="user-dirs-alert-box" class="d-none alert alert-dism
issible fade show" role="alert">
110 |     <span id="user-dirs-alert-message"></span>

```

```

111     <button type="button" class="btn-close" aria-label="Close"
      onclick="document.getElementById('user-dirs-alert-box').class
      List.add('d-none')"></button>
112 </div>
113
114 <!-- Two-Panel Split Layout (Users List & Files Explorer) --
      >
115 <div class="row g-3">
116     <!-- Left Column: Users List & Storage Quotas -->
117     <div class="col-12 col-lg-4">
118         <div class="card border shadow-sm h-100">
119             <div class="card-header bg-body-tertiary d-flex align-
      items-center justify-content-between py-2 px-3">
120                 <span class="fw-semibold small"><i class="bi bi-peop
      le me-1 text-primary"></i> Пользователи</span>
121                 <span class="badge bg-secondary rounded-pill" id="us
      ers-badge-count">0</span>
122             </div>
123             <div class="card-body p-0" style="max-height: 540px; o
      verflow-y: auto;">
124                 <div class="list-group list-group-flush" id="user-di
      rs-users-list">
125                     <div class="text-center py-4 text-muted small">
126                         <div class="spinner-border spinner-border-sm tex
      t-primary me-2" role="status"></div>
127                         Загрузка каталогов пользователей...
128                     </div>
129                 </div>
130             </div>
131         </div>
132     </div>
133
134 <!-- Right Column: Selected User File Tree & Inspector -->
135 <div class="col-12 col-lg-8">
136     <div class="card border shadow-sm h-100">
137         <div class="card-header bg-body-tertiary d-flex align-
      items-center justify-content-between py-2 px-3 flex-wrap gap-2
      ">

```

```

138         <div class="d-flex align-items-center gap-2">
139             <span class="fw-semibold text-truncate small" id="
explorerer-user-title">
140                 <i class="bi bi-folder-fill text-warning me-1"><
/i> Выберите пользователя слева
141             </span>
142             <span class="badge bg-info-subtle text-info border
border-info-subtle d-none" id="explorerer-user-path-badge"></sp
an>
143         </div>
144
145         <!-- Subfolder Filters Buttons -->
146         <div class="btn-group btn-group-sm" role="group" id=
"subfolder-filter-group">
147             <button type="button" class="btn btn-outline-prima
ry active" data-subfolder="">Все папки</button>
148             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="files">files</button>
149             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="rag">rag</button>
150             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="profile">profile</button>
151             <button type="button" class="btn btn-outline-prima
ry" data-subfolder="temp">temp</button>
152         </div>
153     </div>
154
155     <!-- Files Table -->
156     <div class="card-body p-0" style="max-height: 480px; o
verflow-y: auto;">
157         <div class="table-responsive">
158             <table class="table table-hover table-sm align-mid
dle mb-0" style="font-size: 0.84rem;">
159                 <thead class="table-light small text-uppercase s
ticky-top">
160                     <tr>
161                         <th style="width: 35px;"></th>
162                         <th>Имя файла / Путь</th>

```

```

163 |             <th style="width: 90px;" class="text-center"
    | >Раздел</th>
164 |             <th style="width: 95px;" class="text-center"
    | >Размер</th>
165 |             <th style="width: 140px;">Изменен</th>
166 |             <th style="width: 110px;" class="text-center
    | ">Действия</th>
167 |         </tr>
168 |     </thead>
169 |     <tbody id="user-files-table-body">
170 |     <tr>
171 |         <td colspan="6" class="text-center py-5 text
    | -muted">
172 |             <i class="bi bi-arrow-left-circle fs-3 d-b
    | lock mb-2 text-secondary"></i>
173 |             Выберите пользователя из списка слева для
    | просмотра файлов и директорий
174 |         </td>
175 |     </tr>
176 | </tbody>
177 | </table>
178 | </div>
179 | </div>
180 |
181 | <!-- Explorer Footer Stats -->
182 | <div class="card-footer bg-body-tertiary d-flex align-
    | items-center justify-content-between py-2 px-3 small text-mute
    | d">
183 |     <div id="explorer-footer-stats">
184 |         Файлов: <strong>0</strong> | Общий размер: <strong
    | >0 B</strong>
185 |     </div>
186 |     <div id="explorer-footer-quota">
187 |         Квота: <strong>-</strong>
188 |     </div>
189 | </div>
190 | </div>
191 | </div>

```



```

192 |     </div>
193 | </div>
194 |
195 | <!-- =====
196 | ===== -->
197 | <!-- MODAL: File Preview -->
198 | <!-- =====
199 | ===== -->
200 | <div class="modal fade" id="modal-file-preview" tabindex="-1"
201 |     aria-labelledby="modal-file-preview-title" aria-hidden="true">
202 |     <div class="modal-dialog modal-dialog-centered modal-lg">
203 |         <div class="modal-content shadow-lg border-primary">
204 |             <div class="modal-header border-primary py-2.5">
205 |                 <h6 class="modal-title d-flex align-items-center gap-2
206 |                     text-truncate" id="modal-file-preview-title">
207 |                     <i class="bi bi-file-earmark-text text-primary"></i>
208 |                     Предпросмотр файла
209 |                 </h6>
210 |                 <button type="button" class="btn-close" data-bs-dismiss="modal"
211 |                     aria-label="Close"></button>
212 |             </div>
213 |             <div class="modal-body p-3">
214 |                 <!-- File Info Bar -->
215 |                 <div class="d-flex align-items-center justify-content-between
216 |                     p-2 mb-2 bg-body-tertiary border rounded small">
217 |                     <div>
218 |                         <span class="text-muted">Путь:</span> <code id="pr
219 | eview-file-path"></code>
220 |                     </div>
221 |                     <div>
222 |                         <span class="badge bg-secondary me-1" id="preview-
223 | file-size"></span>
224 |                         <span class="badge bg-dark border" id="preview-fil
225 | e-type"></span>
226 |                     </div>
227 |                 </div>
228 |             </div>
229 |             <!-- File Content Container -->

```

```

220         <div id="preview-content-wrapper" class="bg-dark text-
white rounded p-3 font-monospace" style="max-height: 440px; ov
erflow-y: auto; font-size: 0.82rem; white-space: pre-wrap; wor
d-break: break-all;">
221             <div class="text-center py-4 text-muted">
222                 <div class="spinner-border spinner-border-sm text-
primary me-2"></div> Загрузка содержимого...
223             </div>
224         </div>
225     </div>
226     <div class="modal-footer justify-content-between py-2">
227         <button type="button" class="btn btn-outline-danger bt
n-sm rounded-pill px-3" id="btn-preview-delete-file">
228             <i class="bi bi-trash3 me-1"></i> Удалить файл
229         </button>
230         <div class="d-flex gap-2">
231             <button type="button" class="btn btn-outline-seconda
ry btn-sm rounded-pill px-3" data-bs-dismiss="modal">Закрыть</
button>
232             <a href="#" class="btn btn-primary btn-sm rounded-pi
ll px-3" id="btn-preview-download-file" target="_blank" downlo
ad>
233                 <i class="bi bi-download me-1"></i> Скачать
234             </a>
235         </div>
236     </div>
237 </div>
238 </div>
239 </div>
240
241 <!-- =====
===== -->
242 <!-- MODAL: Orphaned User Directories Management -->
243 <!-- =====
===== -->
244 <div class="modal fade" id="modal-orphaned-user-dirs" tabindex
="-1" aria-labelledby="modal-orphaned-user-dirs-title" aria-hi
dden="true">

```

```
245 | <div class="modal-dialog modal-dialog-centered modal-lg">
246 |   <div class="modal-content shadow-lg border-warning">
247 |     <div class="modal-header border-warning py-2.5">
248 |       <h6 class="modal-title text-warning d-flex align-items
- | center gap-2" id="modal-orphaned-user-dirs-title">
249 |         <i class="bi bi-folder-x"></i> Осиротевшие (мертвые)
 | директории пользователей
250 |       </h6>
251 |       <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
252 |     </div>
253 |     <div class="modal-body p-3">
254 |       <div class="alert alert-info border-0 small mb-3">
255 |         <i class="bi bi-info-circle text-info me-1"></i>
256 |         Директории в папке <code>data/users/</code>, которые
 | не принадлежат ни одной активной учетной записи в базе данных
 | .
257 |       </div>
258 |
259 |       <div class="d-flex align-items-center justify-content-between p-2 mb-3 bg-body-tertiary border rounded small">
260 |         <div>
261 |           Найдено папок: <strong id="modal-orphaned-count" class="text-warning">0</strong> |
262 |           Занято места: <strong id="modal-orphaned-size" class="text-info">0 В</strong> |
263 |           Файлов: <strong id="modal-orphaned-files">0</strong>
 | g>
264 |         </div>
265 |         <button class="btn btn-outline-secondary btn-xs rounded-pill px-2 py-0.5" id="btn-modal-refresh-orphaned">
266 |           <i class="bi bi-arrow-clockwise"></i> Обновить
267 |         </button>
268 |       </div>
269 |
270 |       <div class="table-responsive" style="max-height: 360px; overflow-y: auto;">
271 |         <table class="table table-hover table-sm align-middle
```

```

e mb-0" style="font-size: 0.84rem;">
272         <thead class="table-light small text-uppercase sti
cky-top">
273             <tr>
274                 <th>Каталог / ID</th>
275                 <th>Причина</th>
276                 <th class="text-center" style="width: 80px;">Ф
айлов</th>
277                 <th class="text-center" style="width: 95px;">Р
азмер</th>
278                 <th style="width: 140px;">Изменен</th>
279                 <th class="text-center" style="width: 85px;">Д
ействие</th>
280             </tr>
281         </thead>
282         <tbody id="modal-orphaned-dirs-tbody">
283             <tr>
284                 <td colspan="6" class="text-center py-4 text-m
uted">
285                     <div class="spinner-border spinner-border-sm
text-warning me-2"></div> Сканирование...
286                 </td>
287             </tr>
288         </tbody>
289     </table>
290 </div>
291 </div>
292     <div class="modal-footer justify-content-between py-2">
293         <button type="button" class="btn btn-secondary btn-sm
rounded-pill px-3" data-bs-dismiss="modal">Закрыть</button>
294         <button type="button" class="btn btn-danger btn-sm rou
nded-pill px-3 d-flex align-items-center gap-1.5" id="btn-moda
l-clean-all-orphaned">
295             <i class="bi bi-trash3-fill"></i> Удалить все мертвы
е директории
296         </button>
297     </div>
298 </div>

```

```
299 |   </div>
300 | </div>
301 |
```

main.js

```
1 /**
2  * user_directories_tab/main.js – Модуль управления пользоате
   льскими директориями и хранилищем
3  * AI Breadboard
4  */
5
6 let state = {
7   users: [],
8   selectedUserId: null,
9   selectedUserName: '',
10  currentSubfolder: '',
11  searchQuery: '',
12  extFilter: '',
13  files: [],
14  previewFile: null,
15  orphanedDirs: [],
16 };
17
18 /**
19  * Инициализация вкладки пользовательских директорий
20  */
21 export async function initUserDirectoriesTab() {
22   const root = document.getElementById('user-directories-tab-r
   oot');
23   if (!root) return;
24
25   setupEventListeners();
26   await refreshData();
27 }
28
29 /**
30  * Настройка обработчиков событий элементов управления
31  */
32 function setupEventListeners() {
33   // Кнопка обновления
34   const refreshBtn = document.getElementById('btn-refresh-user
```

```
-dirs');
35   if (refreshBtn) {
36     refreshBtn.onclick = () => refreshData();
37   }
38
39   // Поиск с дебаунсом
40   const searchInput = document.getElementById('user-dirs-search-input');
41   const searchClear = document.getElementById('user-dirs-search-clear');
42   if (searchInput) {
43     let timeout = null;
44     searchInput.oninput = () => {
45       clearTimeout(timeout);
46       state.searchQuery = searchInput.value.trim();
47       if (searchClear) {
48         searchClear.classList.toggle('d-none', !state.searchQuery);
49       }
50       timeout = setTimeout(() => {
51         applyFilters();
52       }, 300);
53     };
54   }
55
56   if (searchClear && searchInput) {
57     searchClear.onclick = () => {
58       searchInput.value = '';
59       state.searchQuery = '';
60       searchClear.classList.add('d-none');
61       applyFilters();
62     };
63   }
64
65   // Фильтр по расширениям
66   const extSelect = document.getElementById('user-dirs-ext-filter');
67   if (extSelect) {
```

```
68 |     extSelect.onChange = () => {
69 |         state.extFilter = extSelect.value;
70 |         if (state.selectedUserId) {
71 |             loadUserFiles(state.selectedUserId);
72 |         }
73 |     };
74 | }
75 |
76 | // Фильтры подпапок
77 | const subfolderGroup = document.getElementById('subfolder-filter-group');
78 | if (subfolderGroup) {
79 |     subfolderGroup.addEventListener('click', (e) => {
80 |         const btn = e.target.closest('button[data-subfolder]');
81 |         if (!btn) return;
82 |         subfolderGroup.querySelectorAll('button').forEach(b => b
83 | .classList.remove('active'));
84 |         btn.classList.add('active');
85 |         state.currentSubfolder = btn.dataset.subfolder || '';
86 |         if (state.selectedUserId) {
87 |             loadUserFiles(state.selectedUserId);
88 |         }
89 |     });
90 | }
91 |
92 | // Кнопки модального окна осиротевших каталогов
93 | const openOrphanedBtn = document.getElementById('btn-open-orphaned-dirs-modal');
94 | if (openOrphanedBtn) {
95 |     openOrphanedBtn.onclick = () => openOrphanedModal();
96 | }
97 |
98 | const refreshOrphanedBtn = document.getElementById('btn-modal-refresh-orphaned');
99 | if (refreshOrphanedBtn) {
100 |     refreshOrphanedBtn.onclick = () => loadOrphanedDirs();
101 | }
```



```
102 |   const cleanAllOrphanedBtn = document.getElementById('btn-mod
    |   al-clean-all-orphaned');
103 |   if (cleanAllOrphanedBtn) {
104 |     cleanAllOrphanedBtn.onclick = () => cleanOrphanedDirs();
105 |   }
106 |
107 |   // Кнопка удаления файла из предпросмотра
108 |   const previewDeleteBtn = document.getElementById('btn-previe
    |   w-delete-file');
109 |   if (previewDeleteBtn) {
110 |     previewDeleteBtn.onclick = () => {
111 |       if (state.previewFile && state.selectedUserId) {
112 |         deleteFile(state.selectedUserId, state.previewFile.rel
    |         ative_path);
113 |       }
114 |     };
115 |   }
116 | }
117 |
118 | /**
119 |  * Полное обновление данных: сводка и список пользователей
120 |  */
121 | async function refreshData() {
122 |   await Promise.all([
123 |     loadSummary(),
124 |     loadUsersList(),
125 |     loadOrphanedBadge(),
126 |   ]);
127 | }
128 |
129 | /**
130 |  * Загрузка сводной статистики хранилища
131 |  */
132 | async function loadSummary() {
133 |   try {
134 |     const res = await fetch('/api/admin/user-directories/summa
    |     ry');
135 |     if (!res.ok) return;
```

```
136 |     const data = await res.json();
137 |     if (data.status === 'ok' && data.summary) {
138 |         const s = data.summary;
139 |         const elUsers = document.getElementById('metric-total-us
    ers');
140 |         const elStorage = document.getElementById('metric-total-
    storage');
141 |         const elFiles = document.getElementById('metric-total-fi
    les');
142 |         const elOrphaned = document.getElementById('metric-orpha
    ned-size');
143 |
144 |         if (elUsers) elUsers.textContent = s.total_users || 0;
145 |         if (elStorage) elStorage.textContent = s.total_storage_f
    ormatted || '0 B';
146 |         if (elFiles) elFiles.textContent = s.total_files_count |
    | 0;
147 |         if (elOrphaned) elOrphaned.textContent = s.orphaned_form
    atted || '0 B';
148 |     }
149 | } catch (err) {
150 |     console.warn('Ошибка загрузки сводки хранилища:', err);
151 | }
152 | }
153 |
154 | /**
155 |  * Загрузка бейджа осиротевших каталогов
156 |  */
157 | async function loadOrphanedBadge() {
158 |     try {
159 |         const res = await fetch('/api/admin/user-directories/orpha
    ned');
160 |         if (!res.ok) return;
161 |         const data = await res.json();
162 |         if (data.status === 'ok') {
163 |             const badge = document.getElementById('badge-orphaned-co
    unt');
164 |             if (badge) badge.textContent = data.total || 0;
```

```
165 |     }
166 |   } catch (err) {
167 |     console.warn('Ошибка проверки осиротевших каталогов:', err
    |   );
168 |   }
169 | }
170 |
171 | /**
172 |  * Загрузка списка пользователей с метриками хранилища
173 |  */
174 | async function loadUsersList() {
175 |   const userListEl = document.getElementById('user-dirs-users
    | -list');
176 |   const countBadge = document.getElementById('users-badge-coun
    | t');
177 |   const statusLabel = document.getElementById('user-dirs-stat
    | us-label');
178 |
179 |   try {
180 |     if (statusLabel) statusLabel.textContent = 'Обновление спи
    | ска пользователей...';
181 |     const res = await fetch('/api/admin/user-directories/users
    | ');
182 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
183 |     const data = await res.json();
184 |
185 |     state.users = data.users || [];
186 |     if (countBadge) countBadge.textContent = state.users.lengt
    | h;
187 |     if (statusLabel) statusLabel.textContent = `Загружено поль
    | зователей: ${state.users.length}`;
188 |
189 |     renderUsersList();
190 |
191 |     // Если уже был выбран пользователь – перезагрузим его фай
    | лы, иначе выберем первого
192 |     if (state.selectedUserId && state.users.some(u => u.id ===
    | state.selectedUserId)) {
```

```

193     selectUser(state.selectedUserId);
194 } else if (state.users.length > 0) {
195     selectUser(state.users[0].id);
196 }
197 } catch (err) {
198     console.error('Ошибка загрузки пользователей:', err);
199     if (usersListEl) {
200         usersListEl.innerHTML = `
201             <div class="text-center py-4 text-danger small">
202                 <i class="bi bi-exclamation-triangle fs-4 d-block mb
-1"></i>
203                 Не удалось загрузить список пользователей: ${err.mes
sage}
204             </div>
205             `;
206         }
207     }
208 }
209
210 /**
211  * Рендеринг списка пользователей в левой колонке
212  */
213 function renderUsersList() {
214     const usersListEl = document.getElementById('user-dirs-users
-list');
215     if (!usersListEl) return;
216
217     const query = state.searchQuery.toLowerCase();
218     const filteredUsers = state.users.filter(u => {
219         if (!query) return true;
220         return (
221             (u.name && u.name.toLowerCase().includes(query)) ||
222             (u.email && u.email.toLowerCase().includes(query)) ||
223             String(u.id).includes(query)
224         );
225     });
226
227     if (filteredUsers.length === 0) {

```

```

228     usersListEl.innerHTML = `
229         <div class="text-center py-4 text-muted small">
230             Пользователи не найдены
231         </div>
232     `;
233     return;
234 }
235
236 usersListEl.innerHTML = filteredUsers.map(u => {
237     const isSelected = u.id === state.selectedUserId;
238     const activeClass = isSelected ? 'active border-primary' :
239     '';
240     const roleBadge = u.is_admin
241     ? '<span class="badge bg-warning text-dark me-1">admin</span>'
242     : '<span class="badge bg-secondary me-1">user</span>';
243     const diskWarning = u.quota_percent > 85 ? 'bg-danger' : u
244     .quota_percent > 60 ? 'bg-warning' : 'bg-primary';
245     return `
246         <a href="javascript:void(0)"
247             class="list-group-item list-group-item-action p-2.5 u
248             ser-select-item ${activeClass}"
249             data-user-id="${u.id}">
250             <div class="d-flex w-100 justify-content-between align
251             -items-center mb-1">
252                 <div class="d-flex align-items-center gap-1.5 text-t
253                 runcate">
254                     <span class="badge bg-dark border text-info" style
255                     ="font-size: 0.72rem;">#${u.id}</span>
256                     <strong class="text-truncate small ${isSelected ?
257                     'text-white' : ''}">${escapeHtml(u.name)}</strong>
258                 </div>
259                 <div>${roleBadge}</div>
260             </div>
261             <div class="text-truncate small text-muted mb-1.5" sty
262             le="font-size: 0.76rem;">

```

```

257         ${escapeHtml(u.email || '-')}
258     </div>
259     <div class="d-flex justify-content-between align-items
    -center small mb-1" style="font-size: 0.74rem;">
260         <span class="text-muted"><i class="bi bi-files me-1"
    ></i>${u.files_count} файлов</span>
261         <span class="fw-semibold text-info">${u.size_formatt
    ed}</span>
262     </div>
263     <div class="progress" style="height: 4px;" title="Испо
    льзовано квоты: ${u.quota_percent}%">
264         <div class="progress-bar ${diskWarning}" role="progr
    essbar" style="width: ${u.quota_percent}%;"></div>
265     </div>
266 </a>
267 `;
268 }).join('');
269
270 // Обработчики клика по элементам пользователей
271 userListEl.querySelectorAll('.user-select-item').forEach(it
    em => {
272     item.onclick = () => {
273         const uid = parseInt(item.dataset.userId, 10);
274         selectUser(uid);
275     };
276 });
277 }
278
279 /**
280  * Выбор пользователя и загрузка его файлового проводника
281  */
282 function selectUser(userId) {
283     state.selectedUserId = userId;
284     const user = state.users.find(u => u.id === userId);
285     state.selectedUserName = user ? user.name : `Пользователь #${
    {userId}`;
286
287     // Обновление подсветки в списке слева

```

```
288 |   const userListEl = document.getElementById('user-dirs-users
    | -list');
289 |   if (userListEl) {
290 |     userListEl.querySelectorAll('.user-select-item').forEach(
    | item => {
291 |       const uid = parseInt(item.dataset.userId, 10);
292 |       item.classList.toggle('active', uid === userId);
293 |       item.classList.toggle('border-primary', uid === userId);
294 |     });
295 |   }
296 |
297 |   // Обновление заголовка проводника
298 |   const titleEl = document.getElementById('explorer-user-title
    | ');
299 |   const pathBadge = document.getElementById('explorer-user-pat
    | h-badge');
300 |   if (titleEl) {
301 |     titleEl.innerHTML = `
```

```

    -quota');
318
319   if (tbody) {
320       tbody.innerHTML = `
321           <tr>
322               <td colspan="6" class="text-center py-4 text-muted">
323                   <div class="spinner-border spinner-border-sm text-pr
324                   imary me-2"></div>
325                   Загрузка файлов пользователя...
326               </td>
327           </tr>
328       `;
329   }
330   try {
331       let url = `/api/admin/user-directories/users/${userId}/tre
332       e?`;
333       const params = new URLSearchParams();
334       if (state.currentSubfolder) params.append('subfolder', sta
335       te.currentSubfolder);
336       if (state.searchQuery) params.append('q', state.searchQuer
337       y);
338       if (state.extFilter) params.append('extension', state.extF
339       ilter);
340
341       const res = await fetch(url + params.toString());
342       if (!res.ok) throw new Error(`HTTP ${res.status}`);
343       const data = await res.json();
344
345       state.files = data.items || [];
346
347       if (footerStats) {
348           footerStats.innerHTML = `Файлов: <strong>${data.total_it
349           ems || 0}</strong> | Общий размер: <strong>${data.total_size_f
350           ormatted || '0 B'}</strong>`;
351       }
352
353       const user = state.users.find(u => u.id === userId);

```



```

348     if (footerQuota && user) {
349         footerQuota.innerHTML = `Квота: <strong>${user.size_form
atted} / ${user.quota_formatted}</strong> (${user.quota_percen
t}%)`;
350     }
351
352     renderUserFilesTable();
353 } catch (err) {
354     console.error('Ошибка загрузки файлов:', err);
355     if (tbody) {
356         tbody.innerHTML = `
357             <tr>
358                 <td colspan="6" class="text-center py-4 text-danger
small">
359                     Ошибка загрузки файлов: ${err.message}
360                 </td>
361             </tr>
362         `;
363     }
364 }
365 }
366
367 /**
368  * Рендеринг таблицы файлов пользователя
369  */
370 function renderUserFilesTable() {
371     const tbody = document.getElementById('user-files-table-body
');
372     if (!tbody) return;
373
374     if (state.files.length === 0) {
375         tbody.innerHTML = `
376             <tr>
377                 <td colspan="6" class="text-center py-5 text-muted sma
ll">
378                     <i class="bi bi-inbox fs-3 d-block mb-1 text-seconda
ry"></i>
379                     В выбранном разделе нет файлов

```

```

380         </td>
381     </tr>
382     `;
383     return;
384 }
385
386 tbody.innerHTML = state.files.map(f => {
387     const isDir = f.is_dir;
388     const icon = isDir
389         ? '<i class="bi bi-folder-fill text-warning fs-6"></i>'
390         : getFileIcon(f.extension);
391
392     const subfolderBadge = f.subfolder === 'root'
393         ? '<span class="badge bg-secondary-subtle text-secondary border">root</span>'
394         : `<span class="badge bg-info-subtle text-info border bo
395         rder-info-subtle">${escapeHtml(f.subfolder)}</span>`;
396
397     return `
398         <tr>
399             <td class="text-center">${icon}</td>
400             <td>
401                 <div class="fw-semibold text-truncate text-white" st
402                 yle="max-width: 260px;" title="${escapeHtml(f.relative_path)}"
403                 >
404                     ${escapeHtml(f.name)}
405                 </div>
406                 <div class="text-muted small font-monospace" style="
407                 font-size: 0.72rem;">
408                     ${escapeHtml(f.relative_path)}
409                 </div>
410             </td>
411             <td class="text-center">${subfolderBadge}</td>
412             <td class="text-center font-monospace small">${f.size_
413             formatted}</td>
414             <td class="small text-muted font-monospace" style="fon
415             t-size: 0.76rem;">${f.modified_formatted}</td>
416             <td class="text-center">

```

```

411         <div class="btn-group btn-group-sm" role="group">
412             ${!isDir ? `
413                 <button class="btn btn-xs btn-outline-info btn-p
review-file"
414                     data-path="${escapeHtml(f.relative_path)
}"
415                         title="Предпросмотр файла">
416                             <i class="bi bi-eye"></i>
417                         </button>
418                         <a href="/api/admin/user-directories/users/${sta
te.selectedUserId}/file/download?path=${encodeURIComponent(f.r
elative_path)}"
419                             class="btn btn-xs btn-outline-success"
420                             title="Скачать файл"
421                             download>
422                                 <i class="bi bi-download"></i>
423                             </a>
424                         ` : ''}
425                 <button class="btn btn-xs btn-outline-danger btn-d
elete-file"
426                     data-path="${escapeHtml(f.relative_path)}"
427
428                     data-is-dir="${isDir}"
429                     title="Удалить">
430                         <i class="bi bi-trash3"></i>
431                     </button>
432                 </div>
433             </td>
434         </tr>
435     `;
436     }).join('');
437
438     // Навешивание событий предпросмотра
439     tbody.querySelectorAll('.btn-preview-file').forEach(btn => {
440         btn.onclick = () => {
441             const path = btn.dataset.path;
442             openFilePreview(state.selectedUserId, path);
443         };
444     });

```

```
443     });
444
445     // Навешивание событий удаления
446     tbody.querySelectorAll('.btn-delete-file').forEach(btn => {
447         btn.onclick = () => {
448             const path = btn.dataset.path;
449             const isDir = btn.dataset.isDir === 'true';
450             const promptText = isDir
451                 ? `Вы уверены, что хотите удалить директорию "${path}"
452                   со всем содержимым?`
453                 : `Вы уверены, что хотите удалить файл "${path}"?`;
454             if (confirm(promptText)) {
455                 deleteFile(state.selectedUserId, path);
456             }
457         };
458     });
459
460     /**
461     * Иконка по расширению файла
462     */
463     function getFileIcon(ext) {
464         const e = (ext || '').toLowerCase();
465         switch (e) {
466             case 'pdf':
467                 return '<i class="bi bi-file-earmark-pdf-fill text-dange
468                   r fs-6"></i>';
469             case 'txt':
470             case 'md':
471             case 'log':
472                 return '<i class="bi bi-file-earmark-text-fill text-info
473                   fs-6"></i>';
474             case 'json':
475             case 'xml':
476             case 'yaml':
477             case 'yml':
478                 return '<i class="bi bi-file-earmark-code-fill text-warn
479                   ing fs-6"></i>';
```

```

477 |     case 'csv':
478 |     case 'tsv':
479 |     case 'xlsx':
480 |         return '<i class="bi bi-file-earmark-spreadsheet-fill te
xt-success fs-6"></i>';
481 |     case 'png':
482 |     case 'jpg':
483 |     case 'jpeg':
484 |     case 'webp':
485 |     case 'gif':
486 |     case 'svg':
487 |         return '<i class="bi bi-file-earmark-image-fill text-pri
mary fs-6"></i>';
488 |     case 'mp3':
489 |     case 'wav':
490 |     case 'ogg':
491 |         return '<i class="bi bi-file-earmark-music-fill text-pur
ple fs-6"></i>';
492 |     case 'py':
493 |     case 'js':
494 |     case 'sh':
495 |     case 'ps1':
496 |         return '<i class="bi bi-filetype-py text-warning fs-6"><
/i>';
497 |     default:
498 |         return '<i class="bi bi-file-earmark-fill text-secondary
fs-6"></i>';
499 |     }
500 | }
501 |
502 | /**
503 |  * Открытие модального окна предпросмотра файла
504 |  */
505 | async function openFilePreview(userId, path) {
506 |     const modalEl = document.getElementById('modal-file-preview'
);
507 |     if (!modalEl) return;
508 |

```

```
509 |   const titleEl = document.getElementById('modal-file-preview-
    |   title');
510 |   const pathEl = document.getElementById('preview-file-path');
511 |   const sizeEl = document.getElementById('preview-file-size');
512 |   const typeEl = document.getElementById('preview-file-type');
513 |   const contentWrapper = document.getElementById('preview-cont
    |   ent-wrapper');
514 |   const downloadLink = document.getElementById('btn-preview-do
    |   wnload-file');
515 |
516 |   if (titleEl) titleEl.textContent = `Предпросмотр: ${path.spl
    |   it('/').pop()}`;
517 |   if (pathEl) pathEl.textContent = `data/users/${userId}/${pat
    |   h}`;
518 |   if (sizeEl) sizeEl.textContent = '...';
519 |   if (typeEl) typeEl.textContent = '...';
520 |   if (contentWrapper) {
521 |     contentWrapper.innerHTML = `
522 |       <div class="text-center py-4 text-muted">
523 |         <div class="spinner-border spinner-border-sm text-prim
    |   ary me-2"></div> Загрузка содержимого...
524 |       </div>
525 |     `;
526 |   }
527 |   if (downloadLink) {
528 |     downloadLink.href = `/api/admin/user-directories/users/${u
    |   serId}/file/download?path=${encodeURIComponent(path)}`;
529 |   }
530 |
531 |   const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
532 |   modal.show();
533 |
534 |   try {
535 |     const res = await fetch(`/api/admin/user-directories/users
    |   /${userId}/file/preview?path=${encodeURIComponent(path)}`);
536 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
537 |     const data = await res.json();
538 |     state.previewFile = data;
```

```

539
540     if (sizeEl) sizeEl.textContent = data.size_formatted;
541     if (typeEl) typeEl.textContent = data.mime_type || data.ex
tension;
542
543     if (contentWrapper) {
544         if (data.is_image) {
545             contentWrapper.innerHTML = `
546                 <div class="text-center p-2">
547                     
548                 </div>
549             `;
550         } else {
551             contentWrapper.textContent = data.content || '(Файл пу
ст)';
552         }
553     }
554 } catch (err) {
555     console.error('Ошибка предпросмотра файла:', err);
556     if (contentWrapper) {
557         contentWrapper.innerHTML = `<div class="text-danger p-3"
>Ошибка загрузки содержимого: ${err.message}</div>`;
558     }
559 }
560 }
561
562 /**
563  * Удаление файла или папки
564  */
565 async function deleteFile(userId, path) {
566     try {
567         const res = await fetch(`/api/admin/user-directories/users
/${userId}/file?path=${encodeURIComponent(path)}`, {
568             method: 'DELETE',
569         });
570         if (!res.ok) {

```

```
571     const errData = await res.json().catch(() => ({}));
572     throw new Error(errData.detail || `HTTP ${res.status}`);
573   }
574
575   showAlert(`Файл "${path}" успешно удален`, 'success');
576
577   // Закрываем модальное окно предпросмотра, если оно открыт
578   o
579   const previewModalEl = document.getElementById('modal-file
580   -preview');
581   if (previewModalEl) {
582     const modal = bootstrap.Modal.getInstance(previewModalEl
583     );
584     if (modal) modal.hide();
585   }
586
587   // Перезагружаем файлы и сводку
588   await loadUserFiles(userId);
589   await loadSummary();
590   await loadUsersList();
591   } catch (err) {
592     console.error('Ошибка удаления файла:', err);
593     showAlert(`Ошибка удаления файла: ${err.message}`, 'danger
594     ');
595   }
596 }
597
598 /**
599  * Открытие модального окна управления осиротевшими директория
600  ми
601  */
602 async function openOrphanedModal() {
603   const modalEl = document.getElementById('modal-orphaned-user
604   -dirs');
605   if (!modalEl) return;
606   const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
607   modal.show();
608   await loadOrphanedDirs();
609 }
```



```

603 }
604
605 /**
606  * Загрузка списка осиротевших директорий
607  */
608 async function loadOrphanedDirs() {
609     const tbody = document.getElementById('modal-orphaned-dirs-t
        body');
610     const countEl = document.getElementById('modal-orphaned-coun
        t');
611     const sizeEl = document.getElementById('modal-orphaned-size'
        );
612     const filesEl = document.getElementById('modal-orphaned-file
        s');
613
614     if (tbody) {
615         tbody.innerHTML = `
616             <tr>
617                 <td colspan="6" class="text-center py-4 text-muted">
618                     <div class="spinner-border spinner-border-sm text-wa
        rning me-2"></div> Сканирование...
619                 </td>
620             </tr>
621         `;
622     }
623
624     try {
625         const res = await fetch('/api/admin/user-directories/orpha
        ned');
626         if (!res.ok) throw new Error(`HTTP ${res.status}`);
627         const data = await res.json();
628
629         state.orphanedDirs = data.orphaned_dirs || [];
630
631         if (countEl) countEl.textContent = data.total || 0;
632         if (sizeEl) sizeEl.textContent = data.total_size_formatted
        || '0 B';
633         if (filesEl) filesEl.textContent = data.total_files || 0;

```

```

634
635     const badge = document.getElementById('badge-orphaned-coun
636     t');
637
638     if (badge) badge.textContent = data.total || 0;
639
640     renderOrphanedDirsTable();
641 } catch (err) {
642     console.error('Ошибка загрузки осиротевших директорий:', e
643     rr);
644     if (tbody) {
645         tbody.innerHTML = `
646         <tr>
647             <td colspan="6" class="text-center py-3 text-danger
648             small">
649                 Ошибка сканирования: ${err.message}
650             </td>
651         </tr>
652         `;
653     }
654 }
655
656 /**
657  * Рендеринг таблицы осиротевших каталогов
658  */
659
660 function renderOrphanedDirsTable() {
661     const tbody = document.getElementById('modal-orphaned-dirs-t
662     body');
663     if (!tbody) return;
664
665     if (state.orphanedDirs.length === 0) {
666         tbody.innerHTML = `
667         <tr>
668             <td colspan="6" class="text-center py-4 text-success s
669             mall">
670                 <i class="bi bi-check-circle fs-4 d-block mb-1"></i>
671                 Осиротевших каталогов не обнаружено. Все директории
672                 привязаны к активным пользователям!

```

```

666         </td>
667     </tr>
668     `;
669     return;
670 }
671
672 tbody.innerHTML = state.orphanedDirs.map(d => `
673     <tr>
674         <td>
675             <div class="fw-bold font-monospace text-warning small"
676             >${escapeHtml(d.dir_name)}</div>
677             <div class="text-muted small" style="font-size: 0.72re
678             m;">${escapeHtml(d.full_path)}</div>
679             </td>
680             <td><span class="badge bg-secondary-subtle text-secondar
681             y border">${escapeHtml(d.reason)}</span></td>
682             <td class="text-center font-monospace">${d.files_count}<
683             /td>
684             <td class="text-center font-monospace small">${d.size_fo
685             rmatted}</td>
686             <td class="small text-muted font-monospace" style="font-
687             size: 0.76rem;">${d.modified_formatted}</td>
688             <td class="text-center">
689                 <button class="btn btn-xs btn-outline-danger btn-clean
690                 -single-orphaned"
691                 data-dir="${escapeHtml(d.dir_name)}"
692                 title="Удалить этот каталог">
693                     <i class="bi bi-trash3"></i>
694                 </button>
695             </td>
696         </tr>
697     `).join('');
698
699 tbody.querySelectorAll('.btn-clean-single-orphaned').forEach
700 (btn => {
701     btn.onclick = () => {
702         const dirName = btn.dataset.dir;
703         if (confirm(`Удалить осиротевший каталог "${dirName}"?`))

```

```

    ) {
696         cleanOrphanedDirs([dirName]);
697     }
698 };
699 });
700 }
701
702 /**
703  * Очистка осиротевших каталогов (конкретных или всех)
704  */
705 async function cleanOrphanedDirs(dirsList = null) {
706     if (!dirsList && !confirm('Вы уверены, что хотите удалить ВС
Е осиротевшие каталоги пользователей? Это действие необратимо.
')) {
707         return;
708     }
709
710     try {
711         const payload = dirsList ? { dirs: dirsList } : {};
712         const res = await fetch('/api/admin/user-directories/orpha
ned/clean', {
713             method: 'POST',
714             headers: { 'Content-Type': 'application/json' },
715             body: JSON.stringify(payload),
716         });
717
718         if (!res.ok) throw new Error(`HTTP ${res.status}`);
719         const data = await res.json();
720
721         showAlert(`Удалено каталогов: ${data.deleted_count}, освоб
ождено места: ${data.freed_formatted}`, 'success');
722
723         await loadOrphanedDirs();
724         await loadSummary();
725     } catch (err) {
726         console.error('Ошибка очистки осиротевших каталогов:', err
);
727         showAlert(`Ошибка очистки каталогов: ${err.message}`, 'dan

```

```

    ger');
728     }
729 }
730
731 /**
732  * Применение клиентских фильтров поиска
733  */
734 function applyFilters() {
735     renderUsersList();
736     if (state.selectedUserId) {
737         loadUserFiles(state.selectedUserId);
738     }
739 }
740
741 /**
742  * Вывод сообщения alert
743  */
744 function showAlert(message, type = 'info') {
745     const alertBox = document.getElementById('user-dirs-alert-bo
746     x');
747     const alertMsg = document.getElementById('user-dirs-alert-me
748     ssage');
749     if (!alertBox || !alertMsg) return;
750     alertBox.className = `alert alert-${type} alert-dismissible
751     fade show mb-3 shadow-sm`;
752     alertMsg.textContent = message;
753     alertBox.classList.remove('d-none');
754     setTimeout(() => {
755         alertBox.classList.add('d-none');
756     }, 6000);
757 }
758 /**
759  * Экранирование HTML
760  */
761 function escapeHtml(text) {

```

```
762 |   if (!text) return '';
763 |   const map = {
764 |     '&': '&amp;',
765 |     '<': '&lt;',
766 |     '>': '&gt;',
767 |     '"': '&quot;',
768 |     "'": '&#039;',
769 |   };
770 |   return String(text).replace(/[&<>"']/g, m => map[m]);
771 | }
772 |
773 | // Экспорт по умолчанию
774 | export default initUserDirectoriesTab;
775 |
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title data-i18n="title">Эксперименты с TTS (Синтез речи)</t
8   itle>
9   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dis
10  t/css/bootstrap.min.css" rel="stylesheet">
11   <link href="https://cdn.jsdelivrivr.net/npm/bootstrap-icons@1.1
12   0.0/font/bootstrap-icons.css" rel="stylesheet">
13   <link href="/html/css/main.css?v=20260915_light_v2" rel="sty
14   lesheet">
15   <script src="https://cdn.jsdelivrivr.net/npm/i18next@23.11.5/i1
16   8next.min.js"></script>
17   <style>
18     /* Custom styles for TTS page */
19     :root {
20       --tts-nav-bg: #161b22;
21     }
22
23     body {
24       background: #0d1117;
25       color: #c9d1d9;
26       min-height: 100vh;
27     }
28
29     .tts-container {
30       max-width: 800px;
31       margin: 2rem auto;
32       padding: 0 1rem;
33     }
34
35     .tts-card {
36       background: #161b22;
```

```
31 |     border: 1px solid #30363d;
32 |     border-radius: 12px;
33 |     padding: 2rem;
34 |     box-shadow: 0 4px 12px rgba(0, 0, 0, 0.3);
35 | }
36 |
37 | h1 {
38 |     color: #c9d1d9;
39 |     font-weight: 600;
40 |     margin-bottom: 0.5rem;
41 | }
42 |
43 | .subtitle {
44 |     color: #8b949e;
45 |     font-size: 0.95rem;
46 |     margin-bottom: 2rem;
47 | }
48 |
49 | .form-label {
50 |     color: #8b949e;
51 |     font-weight: 500;
52 |     margin-bottom: 0.5rem;
53 | }
54 |
55 | .form-control, .form-select {
56 |     background: #0d1117;
57 |     border: 1px solid #30363d;
58 |     color: #c9d1d9;
59 |     border-radius: 8px;
60 |     padding: 0.75rem 1rem;
61 | }
62 |
63 | .form-control:focus, .form-select:focus {
64 |     background: #0d1117;
65 |     border-color: #58a6ff;
66 |     color: #c9d1d9;
67 |     box-shadow: 0 0 0 2px rgba(88, 166, 255, 0.25);
68 | }
```



```
69 |
70 | textarea.form-control {
71 |     resize: vertical;
72 |     min-height: 120px;
73 |     font-family: system-ui, -apple-system, sans-serif;
74 | }
75 |
76 | .btn-tts {
77 |     background: #1f6feb;
78 |     color: #fff;
79 |     border: none;
80 |     border-radius: 8px;
81 |     padding: 0.85rem 1.5rem;
82 |     font-weight: 500;
83 |     transition: background-color 0.2s;
84 | }
85 |
86 | .btn-tts:hover {
87 |     background: #388bfd;
88 |     color: #fff;
89 | }
90 |
91 | .btn-tts:disabled {
92 |     opacity: 0.6;
93 |     cursor: not-allowed;
94 | }
95 |
96 | .btn-apply {
97 |     background: #238636;
98 |     color: #fff;
99 |     border: none;
100 |     border-radius: 8px;
101 |     padding: 0.85rem 1.5rem;
102 |     font-weight: 500;
103 |     transition: background-color 0.2s;
104 | }
105 |
106 | .btn-apply:hover {
```

```
107 |     background: #2ea043;
108 |     color: #fff;
109 | }
110 |
111 | .btn-apply:disabled {
112 |     opacity: 0.6;
113 |     cursor: not-allowed;
114 | }
115 |
116 | .active-badge {
117 |     background: rgba(35, 134, 54, 0.15);
118 |     border: 1px solid rgba(35, 134, 54, 0.3);
119 |     color: #56d364;
120 |     padding: 0.5rem 1rem;
121 |     border-radius: 20px;
122 |     font-size: 0.85rem;
123 |     display: inline-flex;
124 |     align-items: center;
125 |     gap: 0.5rem;
126 |     margin-bottom: 1.5rem;
127 | }
128 |
129 | .audio-player-container {
130 |     margin-top: 2rem;
131 |     padding-top: 2rem;
132 |     border-top: 1px solid #30363d;
133 |     display: none;
134 | }
135 |
136 | audio {
137 |     width: 100%;
138 |     border-radius: 8px;
139 | }
140 |
141 | .status-badge {
142 |     display: inline-flex;
143 |     align-items: center;
144 |     gap: 0.5rem;
```

```
145 |     font-size: 0.9rem;
146 |     color: #8b949e;
147 |     margin-top: 1rem;
148 | }
149 |
150 | .spinner-border {
151 |     width: 1rem;
152 |     height: 1rem;
153 |     border-width: 0.15em;
154 | }
155 |
156 | .back-link {
157 |     display: inline-flex;
158 |     align-items: center;
159 |     gap: 0.5rem;
160 |     color: #8b949e;
161 |     text-decoration: none;
162 |     font-size: 0.95rem;
163 |     margin-top: 2rem;
164 |     transition: color 0.2s;
165 | }
166 |
167 | .back-link:hover {
168 |     color: #58a6ff;
169 | }
170 |
171 | /* Dark theme audio player controls */
172 | audio::-webkit-media-controls-panel {
173 |     background-color: rgba(13, 17, 23, 0.9);
174 | }
175 |
176 | audio::-webkit-media-controls-current-time-display,
177 | audio::-webkit-media-controls-time-remaining-display {
178 |     color: #c9d1d9;
179 | }
180 | </style>
181 | </head>
182 | <body>
```

```
183
184 <div class="tts-container">
185   <div class="tts-card">
186     <h1><i class="bi bi-mic"></i> <span data-i18n="title">Эксп
187     ерименты с TTS (Синтез речи)</span></h1>
188     <p class="subtitle" data-i18n="subtitle">Сравните качество
189     синтеза речи различных моделей: Web Speech, Google, Edge и Si
190     lero.</p>
191     <div id="active-voice-container" class="mb-3" style="display: none;">
192       <span class="active-badge">
193         <i class="bi bi-check-circle-fill"></i>
194         <span><span data-i18n="label_active_voice">Активный го
195         лос:</span> <strong id="active-voice-name">...</strong></span>
196       </span>
197     </div>
198     <form id="tts-form">
199       <div class="mb-4">
200         <label for="tts-text" class="form-label" data-i18n="fo
201         rm.label_text">Текст для озвучки</label>
202         <textarea id="tts-text" class="form-control" placeholder data-i18n-placeholder="form.placeholder">Привет! Это демонстрация работы различных систем синтеза речи. Какая из них звучит лучше всего?</textarea>
203       </div>
204       <div class="row">
205         <div class="col-md-6 mb-4">
206           <label for="tts-engine" class="form-label" data-i18n="form.label_engine">Система синтеза</label>
207           <select id="tts-engine" class="form-select">
208             <option value="browser" data-i18n="options.browser">Web Speech API (Браузерная)</option>
209             <option value="gtts" data-i18n="options.gtts">Google TTS (Базовая)</option>
210             <option value="edge-tts" selected data-i18n="optio
```

```
ns.edge-tts">Microsoft Edge TTS (Нейросетевая)</option>
209     <option value="silero" data-i18n="options.silero">
Silero TTS (Локальная оффлайн)</option>
210     </select>
211 </div>
212
213     <div class="col-md-6 mb-4" id="voice-select-container"
>
214         <label for="tts-voice" class="form-label" data-i18n=
"form.label_voice">Голос / Спикер</label>
215         <select id="tts-voice" class="form-select">
216             <!-- Populated dynamically via JS -->
217         </select>
218     </div>
219 </div>
220
221     <div class="d-flex gap-2">
222         <button type="submit" class="btn btn-tts flex-grow-1"
id="btn-speak" data-i18n="button_speak">Озвучить текст</button
>
223         <button type="button" class="btn btn-apply" id="btn-sa
ve-default" data-i18n="button_apply">Применить изменения</butt
on>
224     </div>
225
226     <div class="status-badge" id="tts-status" style="display
: none;">
227         <div class="spinner-border text-success" role="status"
></div>
228         <span id="status-text" data-i18n="status_generating">Г
енерация аудиофайла на сервере...</span>
229     </div>
230 </form>
231
232     <div class="audio-player-container" id="player-container">
233         <label class="form-label" data-i18n="label_result">Резул
ьтат генерации</label>
234         <audio id="audio-player" controls></audio>
```

```
235     </div>
236 </div>
237
238 <a href="/" class="back-link">
239     <i class="bi bi-arrow-left"></i>
240     <span data-i18n="back_link">Вернуться в медиатеку</span>
241 </a>
242 </div>
243
244 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    /js/bootstrap.bundle.min.js"></script>
245 <script type="module">
246     import { initI18n, switchLang, applyTranslations } from '/ht
    ml/js/i18n.js';
247
248     window.switchLang = switchLang;
249     window.applyTranslations = applyTranslations;
250
251     // Initialize i18n
252     const savedLang = localStorage.getItem('app_language') || 'r
    u';
253     await initI18n(savedLang);
254
255     const engineSelect = document.getElementById('tts-engine');
256     const voiceSelect = document.getElementById('tts-voice');
257     const voiceContainer = document.getElementById('voice-select
    -container');
258     const ttsForm = document.getElementById('tts-form');
259     const btnSpeak = document.getElementById('btn-speak');
260     const ttsStatus = document.getElementById('tts-status');
261     const statusText = document.getElementById('status-text');
262     const playerContainer = document.getElementById('player-cont
    ainer');
263     const audioPlayer = document.getElementById('audio-player');
264     const activeVoiceContainer = document.getElementById('active
    -voice-container');
265     const activeVoiceName = document.getElementById('active-voic
    e-name');
```

```
266
267 function getFriendlyVoiceName(system, voice) {
268   if (system === 'edge-tts') {
269     const found = voiceMap['edge-tts'].find(v => v.value ===
voice);
270     return found ? translateVoiceLabel(found.labelKey) : `Ed
ge (${voice})`;
271   }
272   if (system === 'silero') {
273     const found = voiceMap['silero'].find(v => v.value === v
oice);
274     return found ? translateVoiceLabel(found.labelKey) : `Si
lero (${voice})`;
275   }
276   if (system === 'gtts') {
277     const found = voiceMap['gtts'].find(v => v.value === voi
ce);
278     return found ? translateVoiceLabel(found.labelKey) : `Go
ogle TTS (${voice})`;
279   }
280   if (system === 'browser') {
281     return `Браузерный (${voice} || 'По умолчанию')`;
282   }
283   return `${system} – ${voice}`;
284 }
285
286 function updateActiveBadge(system, voice) {
287   if (system && voice) {
288     activeVoiceName.textContent = getFriendlyVoiceName(syste
m, voice);
289     activeVoiceContainer.style.display = 'block';
290   } else {
291     activeVoiceContainer.style.display = 'none';
292   }
293 }
294
295 // Available voices dictionary with translated labels
296 const voiceMap = {
```

```

297 |     'browser': [], // Populated dynamically from window.speech
    | Synthesis
298 |     'gtts': [
299 |         { value: 'ru', labelKey: 'voices.ru_standard' },
300 |         { value: 'en', labelKey: 'voices.en_standard' },
301 |         { value: 'he', labelKey: 'voices.he_standard' },
302 |         { value: 'de', labelKey: 'voices.de_standard' },
303 |         { value: 'fr', labelKey: 'voices.fr_standard' },
304 |         { value: 'es', labelKey: 'voices.es_standard' }
305 |     ],
306 |     'edge-tts': [
307 |         { value: 'ru-RU-DmitryNeural', labelKey: 'voices.dmitry'
    |     },
308 |         { value: 'ru-RU-SvetlanaNeural', labelKey: 'voices.svetl
    | ana' },
309 |         { value: 'en-US-JennyNeural', labelKey: 'voices.jenny' }
    |     ,
310 |         { value: 'en-US-GuyNeural', labelKey: 'voices.guy' },
311 |         { value: 'en-US-AriaNeural', labelKey: 'voices.aria' },
312 |         { value: 'en-GB-SoniaNeural', labelKey: 'voices.sonia' }
    |     ,
313 |         { value: 'en-GB-RyanNeural', labelKey: 'voices.ryan' },
314 |         { value: 'he-IL-AvriNeural', labelKey: 'voices.avri' },
315 |         { value: 'he-IL-HilaNeural', labelKey: 'voices.hila' },
316 |         { value: 'de-DE-KatjaNeural', labelKey: 'voices.katja' }
    |     ,
317 |         { value: 'de-DE-ConradNeural', labelKey: 'voices.conrad'
    |     },
318 |         { value: 'fr-FR-DeniseNeural', labelKey: 'voices.denise'
    |     },
319 |         { value: 'fr-FR-HenriNeural', labelKey: 'voices.henri' }
    |     ,
320 |         { value: 'es-ES-ElviraNeural', labelKey: 'voices.elvira'
    |     },
321 |         { value: 'es-ES-AlvaroNeural', labelKey: 'voices.alvaro'
    |     }
322 |     ],
323 |     'silero': [

```



```

324 |     { value: 'eugene', labelKey: 'voices.eugene' },
325 |     { value: 'aidar', labelKey: 'voices.aidar' },
326 |     { value: 'baya', labelKey: 'voices.baya' },
327 |     { value: 'kseniya', labelKey: 'voices.kseniya' },
328 |     { value: 'xenia', labelKey: 'voices.xenia' },
329 |     { value: 'random', labelKey: 'voices.random' }
330 |   ]
331 | };
332 |
333 | // Translate voice labels
334 | function translateVoiceLabel(labelKey) {
335 |   return i18next?.t(labelKey) || labelKey;
336 | }
337 |
338 | // Load browser native voices
339 | function loadBrowserVoices() {
340 |   if ('speechSynthesis' in window) {
341 |     const voices = window.speechSynthesis.getVoices();
342 |     voiceMap['browser'] = voices
343 |       .map(v => ({ value: v.name, labelKey: `[${v.lang}] ${v
344 |         .name}` }));
345 |
346 |     // Fallback if no voices found in browser
347 |     if (voiceMap['browser'].length === 0) {
348 |       voiceMap['browser'].push({ value: 'default', labelKey:
349 |         'voices.default' });
350 |     }
351 |
352 |     if (engineSelect.value === 'browser') {
353 |       updateVoiceSelect();
354 |     }
355 |   }
356 |
357 |   if ('speechSynthesis' in window) {
358 |     window.speechSynthesis.onvoiceschanged = loadBrowserVoices
;
loadBrowserVoices();

```

```

359 | }
360 |
361 | function updateVoiceSelect() {
362 |     const engine = engineSelect.value;
363 |     const voices = voiceMap[engine] || [];
364 |
365 |     voiceSelect.innerHTML = '';
366 |
367 |     if (voices.length === 0) {
368 |         voiceContainer.style.display = 'none';
369 |     } else {
370 |         voiceContainer.style.display = 'block';
371 |         voices.forEach(voice => {
372 |             const opt = document.createElement('option');
373 |             opt.value = voice.value;
374 |             opt.textContent = translateVoiceLabel(voice.labelKey);
375 |             voiceSelect.appendChild(opt);
376 |         });
377 |     }
378 | }
379 |
380 | engineSelect.addEventListener('change', updateVoiceSelect);
381 | updateVoiceSelect(); // Initial populate
382 |
383 | // On page load, fetch current settings to pre-select saved
    voice/engine
384 | try {
385 |     const settingsResponse = await fetch('/auth/settings');
386 |     if (settingsResponse.ok) {
387 |         const settings = await settingsResponse.json();
388 |         if (settings.tts_system) {
389 |             engineSelect.value = settings.tts_system;
390 |             updateVoiceSelect();
391 |         }
392 |         if (settings.tts_voice) {
393 |             voiceSelect.value = settings.tts_voice;
394 |         }
395 |         updateActiveBadge(settings.tts_system, settings.tts_voic

```

```

    e);
396     }
397   } catch (e) {
398     console.error('Failed to load user settings:', e);
399   }
400
401   // Update translations when language changes
402   const originalSwitchLang = window.switchLang;
403   window.switchLang = async function(lang) {
404     await originalSwitchLang(lang);
405     updateVoiceSelect();
406   };
407
408   ttsForm.addEventListener('submit', async (e) => {
409     e.preventDefault();
410     const text = document.getElementById('tts-text').value.trim();
411     if (!text) return;
412
413     const engine = engineSelect.value;
414     const voice = voiceSelect.value;
415
416     // 1. Browser Web Speech synthesis
417     if (engine === 'browser') {
418       if ('speechSynthesis' in window) {
419         window.speechSynthesis.cancel();
420         const cleanText = text.replace(/<[^>]*>/g, '').replace(
421           (/[*_`#]/g, '');
422         const utterance = new SpeechSynthesisUtterance(cleanText);
423         const voices = window.speechSynthesis.getVoices();
424         const selectedVoice = voices.find(v => v.name === voice);
425         if (selectedVoice) {
426           utterance.voice = selectedVoice;
427           if (selectedVoice.lang) {
428             utterance.lang = selectedVoice.lang;
429           }
430         }
431       }
432     }
433   });

```

```
429         }
430
431         ttsStatus.style.display = 'inline-flex';
432         statusText.textContent = i18next?.t('status_generating
433         ') || 'Воспроизведение речи браузером...';
434
435         utterance.onend = () => {
436             ttsStatus.style.display = 'none';
437         };
438         utterance.onerror = () => {
439             ttsStatus.style.display = 'none';
440         };
441
442         window.speechSynthesis.speak(utterance);
443         playerContainer.style.display = 'none';
444     } else {
445         const msg = i18next?.t('errors.no_api') || 'Ваш браузер
446         не поддерживает Web Speech API.';
447         window.showToast?.(msg, 'warning') || alert(msg);
448     }
449     return;
450 }
451
452 // 2. Server-side TTS synthesis (gtts, edge-tts, silero)
453 ttsStatus.style.display = 'inline-flex';
454 statusText.textContent = i18next?.t('status_generating') |
455 | 'Генерация аудиофайла на сервере...';
456 btnSpeak.disabled = true;
457 playerContainer.style.display = 'none';
458
459 try {
460     const queryParams = new URLSearchParams({
461         text: text,
462         system: engine,
463         voice: voice
464     });
465
466     // Load audio stream from server
```

```
464     const audioUrl = `/api/tts/synthesize?${queryParams.toString()} `;
465
466     // Force preload of the audio
467     audioPlayer.src = audioUrl;
468     audioPlayer.load();
469
470     audioPlayer.oncanplaythrough = () => {
471         ttsStatus.style.display = 'none';
472         btnSpeak.disabled = false;
473         playerContainer.style.display = 'block';
474         audioPlayer.play();
475     };
476
477     audioPlayer.onerror = (err) => {
478         console.error(err);
479         statusText.textContent = i18next?.t('errors.server_error') || 'Ошибка синтеза речи на сервере';
480         btnSpeak.disabled = false;
481         setTimeout(() => { ttsStatus.style.display = 'none'; },
482             3000);
483     };
484
485     } catch (err) {
486         console.error(err);
487         statusText.textContent = i18next?.t('errors.network_error') || 'Сбой сети при запросе к серверу';
488         btnSpeak.disabled = false;
489         setTimeout(() => { ttsStatus.style.display = 'none'; },
490             3000);
491     }
492
493     });
494
495     const btnSaveDefault = document.getElementById('btn-save-default');
496
497     btnSaveDefault.addEventListener('click', async () => {
498         const engine = engineSelect.value;
499         const voice = voiceSelect.value;
```

```
496 |
497 |     btnSaveDefault.disabled = true;
498 |     try {
499 |         const payload = {
500 |             tts_enabled: 1,
501 |             tts_system: engine,
502 |             tts_voice: voice
503 |         };
504 |
505 |         const response = await fetch('/auth/settings', {
506 |             method: 'POST',
507 |             headers: { 'Content-Type': 'application/json' },
508 |             body: JSON.stringify(payload)
509 |         });
510 |
511 |         if (response.ok) {
512 |             updateActiveBadge(engine, voice);
513 |             alert('Изменения успешно применены! Этот голос теперь
| установлен по умолчанию.');
```

514 | } else {

515 | alert('Не удалось сохранить настройки. Убедитесь, что

516 | вы вошли в систему.');

517 | } catch (err) {

518 | console.error(err);

519 | alert('Ошибка подключения к серверу при сохранении настр

520 | оек.');

521 | } finally {

522 | btnSaveDefault.disabled = false;

523 | });

524 | </script>

525 | </body>

526 | </html>

index.html

```
1 <!-- Voice Diarization & Audio Analyzer Tab -->
2 <div class="container-fluid py-2">
3   <div class="alert alert-danger d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 shadow-sm">
4     <div>
5       <h5 class="mb-1"><i class="bi bi-mic-fill me-2"></i><span data-i18n="voice.title">□ Разбор голосовых сообщений и диа-
изация</span></h5>
6       <p class="mb-0 small text-secondary" data-i18n="voice.subtitle">Запись с микрофона или загрузка аудиофайла: разделение
по собеседникам, стенограмма и сводка встреч.</p>
7     </div>
8     <div class="d-flex align-items-center gap-2">
9       <span class="badge bg-dark font-monospace p-2">Gemini Mu
ltimodal Audio</span>
10    </div>
11  </div>
12
13  <div class="row g-3">
14    <!-- LEFT COLUMN: Recording & Audio Upload Controls -->
15    <div class="col-lg-5 col-md-12">
16      <!-- Microphone Live Recording Card -->
17      <div class="card shadow-sm border-0 mb-3">
18        <div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
19          <span class="fw-semibold"><i class="bi bi-broadcast me-2 text-danger"></i><span data-i18n="voice.recordTitle">Запи-
сь с микрофона</span></span>
20          <span class="badge bg-danger font-monospace d-none"
id="vtab-record-timer">00:00</span>
21        </div>
22        <div class="card-body text-center p-4">
23          <div id="vtab-mic-visual" class="my-3">
24            <div class="d-inline-flex align-items-center justify-content-center rounded-circle p-4 border border-3 border-da
nger shadow" style="width: 100px; height: 100px; transition: a
```

```

    ll 0.3s;" id="vtab-mic-circle">
25         <i class="bi bi-mic-fill fs-1 text-danger" id="v
tab-mic-icon"></i>
26         </div>
27     </div>
28     <h6 class="fw-bold mb-1" id="vtab-record-status-labe
l" data-i18n="voice.clickToRecord">Нажмите для начала записи</
h6>
29     <p class="text-muted small mb-3">Говорите в микрофон
– система автоматически распознает разных спикеров.</p>
30
31     <div class="d-flex justify-content-center gap-2">
32         <button class="btn btn-danger px-4 py-2 rounded-pi
ll fw-bold shadow-sm" id="btn-vtab-record-toggle">
33             <i class="bi bi-record-fill me-1"></i> <span id=
"vtab-record-btn-text" data-i18n="voice.startRecord">Начать за
пись</span>
34         </button>
35         <button class="btn btn-outline-secondary px-3 py-2
rounded-pill d-none" id="btn-vtab-record-cancel">
36             Отмена
37         </button>
38     </div>
39 </div>
40 </div>
41
42 <!-- File Upload Card -->
43 <div class="card shadow-sm border-0 mb-3">
44     <div class="card-header bg-dark text-white d-flex just
ify-content-between align-items-center">
45         <span class="fw-semibold"><i class="bi bi-cloud-arro
w-up me-2 text-info"></i><span data-i18n="voice.uploadTitle">3
агрузка аудиофайла</span></span>
46     </div>
47     <div class="card-body">
48         <div id="vtab-dropzone" class="border border-2 borde
r-dashed rounded p-4 text-center cursor-pointer mb-2 bg-body-s
econdary bg-opacity-25" style="border-style: dashed !important

```



```

; ">
49         <i class="bi bi-file-earmark-music fs-1 text-info
mb-2 d-block"></i>
50         <h6 class="fw-bold mb-1" data-i18n="voice.dropzone
Text">Перетащите аудиофайл сюда или нажмите для выбора</h6>
51         <p class="text-muted small mb-2">Форматы: .mp3, .w
av, .ogg, .m4a, .webm, .aac</p>
52         <input type="file" id="vtab-file-input" class="d-n
one" accept="audio/*,.mp3,.wav,.ogg,.m4a,.webm,.aac">
53         <button class="btn btn-outline-info btn-sm px-3 ro
unded-pill mt-1" type="button" onclick="document.getElementByI
d('vtab-file-input').click();">
54         <i class="bi bi-folder2-open me-1"></i> <span da
ta-i18n="voice.browseAudio">Выбрать аудиофайл</span>
55         </button>
56     </div>
57 </div>
58 </div>
59
60 <!-- Audio Analysis Settings -->
61 <div class="card shadow-sm border-0">
62     <div class="card-header bg-dark text-white">
63         <span class="fw-semibold"><i class="bi bi-sliders me
-2"></i>Параметры распознавания</span>
64     </div>
65     <div class="card-body">
66         <div class="mb-2">
67             <label class="form-label small fw-bold" for="vtab-
model-select">Модель:</label>
68             <select class="form-select form-select-sm" id="vta
b-model-select">
69                 <option value="gemini-2.5-flash" selected>gemini
-2.5-flash (Рекомендуется для аудио)</option>
70                 <option value="gemini-2.0-flash">gemini-2.0-flas
h</option>
71                 <option value="gemini-1.5-flash">gemini-1.5-flas
h</option>
72                 <option value="gemini-1.5-pro">gemini-1.5-pro</o

```

```

ption>
73         </select>
74     </div>
75     <div class="mb-2">
76         <label class="form-label small fw-bold" for="vtab-
lang-select">Язык отчёта и сводки:</label>
77         <select class="form-select form-select-sm" id="vta
b-lang-select">
78             <option value="ru" selected>Русский (Russian)</o
ption>
79             <option value="en">English</option>
80             <option value="he">עברית (Hebrew)</option>
81         </select>
82     </div>
83     <div>
84         <label class="form-label small fw-bold" for="vtab-
api-key">Gemini API Key (опционально):</label>
85         <input type="password" class="form-control form-co
ntrol-sm" id="vtab-api-key" placeholder="Из .env по умолчанию"
autocomplete="new-password" data-lpignore="true" data-1p-igno
re="true">
86     </div>
87 </div>
88 </div>
89 </div>
90
91 <!-- RIGHT COLUMN: Audio Playback & Diarization Result -->
92 <div class="col-lg-7 col-md-12">
93     <div class="card shadow-sm border-0 mb-3" id="vtab-resul
t-card">
94         <div class="card-header bg-dark text-white d-flex just
ify-content-between align-items-center">
95             <span class="fw-semibold"><i class="bi bi-file-text
me-2 text-warning"></i><span data-i18n="voice.resultTitle">Рез
ультат анализа и стенограмма</span></span>
96             <div class="d-flex align-items-center gap-2">
97                 <button class="btn btn-success btn-sm d-none" id="
vtab-btn-save-rag">

```

```
98 |         <i class="bi bi-floppy me-1"></i> <span data-i18n="voice.saveToRag">Сохранить в RAG</span>
99 |         </button>
100 |     </div>
101 | </div>
102 |     <div class="card-body p-3" id="vtab-result-body">
103 |         <!-- Audio Player Preview -->
104 |         <div id="vtab-player-container" class="mb-3 d-none p-2 bg-body-tertiary border rounded">
105 |             <div class="small fw-semibold mb-1 text-muted"><i class="bi bi-play-circle me-1 text-danger"></i>Исходная аудиозапись: <span id="vtab-current-filename" class="text-body font-monospace"></span></div>
106 |             <audio id="vtab-audio-element" controls class="w-100" style="max-height: 38px;"></audio>
107 |         </div>
108 |
109 |         <!-- Loading State -->
110 |         <div id="vtab-loading" class="d-none text-center py-5">
111 |             <div class="spinner-border text-danger mb-3" role="status" style="width: 3rem; height: 3rem;"></div>
112 |             <h6 class="fw-bold text-primary">Идет распознавание и диаризация...</h6>
113 |             <p class="text-muted small">Gemini слушает аудио, разделяет собеседников и готовит сводку.</p>
114 |         </div>
115 |
116 |         <!-- Empty Initial State -->
117 |         <div id="vtab-empty-state" class="text-center py-5 text-muted">
118 |             <i class="bi bi-soundwave fs-1 d-block mb-2 opacity-50"></i>
119 |             <h6 class="fw-bold" data-i18n="voice.emptyStateTitle">Запишите голос или загрузите аудиофайл</h6>
120 |             <p class="small mb-0">Здесь появится краткая сводка, список ключевых решений и реплики каждого собеседника.</p>
121 |         </div>
```

```
122 |
123 |         <!-- Analysis Content Container -->
124 |         <div id="vtab-analysis-content" class="d-none"></div
    | >
125 |         </div>
126 |     </div>
127 | </div>
128 | </div>
129 | </div>
130 |
```

main.js

```
1 // — VOICE_TAB MAIN.JS —————  
2  
3 (function () {  
4   let isInitialized = false;  
5   let tabMediaRecorder = null;  
6   let tabAudioChunks = [];  
7   let tabRecordingStartTime = 0;  
8   let tabTimerInterval = null;  
9   let lastAnalysisData = null;  
10  let lastAudioFilename = 'voice_recording';  
11  
12  function initVoiceTab() {  
13    const recordBtn = document.getElementById('btn-vtab-record  
-toggle');  
14    if (!recordBtn) return;  
15    if (isInitialized) return;  
16    isInitialized = true;  
17    setupEventListeners();  
18  }  
19  
20  window.initVoiceTab = initVoiceTab;  
21  
22  function setupEventListeners() {  
23    const recordBtn = document.getElementById('btn-vtab-record  
-toggle');  
24    const cancelBtn = document.getElementById('btn-vtab-record  
-cancel');  
25    const fileInput = document.getElementById('vtab-file-input  
' );  
26    const dropzone = document.getElementById('vtab-dropzone');  
27    const saveRagBtn = document.getElementById('vtab-btn-save-  
rag');  
28  
29    if (recordBtn) {  
30      recordBtn.addEventListener('click', toggleRecording);
```

```
31 |     }
32 |     if (cancelBtn) {
33 |         cancelBtn.addEventListener('click', cancelRecording);
34 |     }
35 |     if (fileInput) {
36 |         fileInput.addEventListener('change', (e) => {
37 |             if (e.target.files && e.target.files.length > 0) {
38 |                 const file = e.target.files[0];
39 |                 analyzeAudioBlob(file, file.name);
40 |                 fileInput.value = '';
41 |             }
42 |         });
43 |     }
44 |     if (dropzone) {
45 |         ['dragenter', 'dragover'].forEach(eventName => {
46 |             dropzone.addEventListener(eventName, (e) => {
47 |                 e.preventDefault();
48 |                 e.stopPropagation();
49 |                 dropzone.classList.add('border-info', 'bg-info', 'bg
50 | -opacity-10');
51 |             }, false);
52 |         });
53 |         ['dragleave', 'drop'].forEach(eventName => {
54 |             dropzone.addEventListener(eventName, (e) => {
55 |                 e.preventDefault();
56 |                 e.stopPropagation();
57 |                 dropzone.classList.remove('border-info', 'bg-info',
58 | 'bg-opacity-10');
59 |             }, false);
60 |         });
61 |         dropzone.addEventListener('drop', (e) => {
62 |             const dt = e.dataTransfer;
63 |             if (dt && dt.files && dt.files.length > 0) {
64 |                 const file = dt.files[0];
65 |                 analyzeAudioBlob(file, file.name);
66 |             }

```

```
67 |     }, false);
68 | }
69 | if (saveRagBtn) {
70 |     saveRagBtn.addEventListener('click', saveResultToRag);
71 | }
72 | }
73 |
74 | async function toggleRecording() {
75 |     if (tabMediaRecorder && tabMediaRecorder.state === 'record
ing') {
76 |         stopRecording();
77 |     } else {
78 |         await startRecording();
79 |     }
80 | }
81 |
82 | async function startRecording() {
83 |     if (!navigator.mediaDevices || !navigator.mediaDevices.get
UserMedia) {
84 |         alert('Доступ к микрофону не поддерживается в вашем брау
зере. ');
85 |         return;
86 |     }
87 |
88 |     try {
89 |         const stream = await navigator.mediaDevices.getUserMedia
({ audio: true });
90 |         tabAudioChunks = [];
91 |
92 |         let options = {};
93 |         if (MediaRecorder.isTypeSupported('audio/webm;codecs=opus')) {
94 |             options = { mimeType: 'audio/webm;codecs=opus' };
95 |         } else if (MediaRecorder.isTypeSupported('audio/ogg;codecs=opus')) {
96 |             options = { mimeType: 'audio/ogg;codecs=opus' };
97 |         }
98 |
```

```

99 |         tabMediaRecorder = new MediaRecorder(stream, options);
100 |
101 |         tabMediaRecorder.ondataavailable = (e) => {
102 |             if (e.data && e.data.size > 0) {
103 |                 tabAudioChunks.push(e.data);
104 |             }
105 |         };
106 |
107 |         tabMediaRecorder.onstop = () => {
108 |             stream.getTracks().forEach(t => t.stop());
109 |             if (tabAudioChunks.length > 0) {
110 |                 const mime = tabMediaRecorder.mimeType || 'audio/web
111 | m';
112 |                 const blob = new Blob(tabAudioChunks, { type: mime }
113 | );
114 |                 const fname = `voice_${new Date().toISOString().slice(0, 19).replace(/[:T]/g, '_')}.webm`;
115 |                 analyzeAudioBlob(blob, fname);
116 |             }
117 |         };
118 |
119 |         tabMediaRecorder.start(250);
120 |         tabRecordingStartTime = Date.now();
121 |
122 |         // UI state update
123 |         const recordBtn = document.getElementById('btn-vtab-record-toggle');
124 |         const cancelBtn = document.getElementById('btn-vtab-record-cancel');
125 |         const btnText = document.getElementById('vtab-record-btn-text');
126 |         const statusLabel = document.getElementById('vtab-record-status-label');
127 |         const timerBadge = document.getElementById('vtab-record-timer');
128 |         const micCircle = document.getElementById('vtab-mic-circle');

```



```

128         if (recordBtn) {
129             recordBtn.className = 'btn btn-danger px-4 py-2 rounded-pill fw-bold shadow-sm';
130         }
131         if (btnText) btnText.textContent = 'Остановить и разобрать';
132         if (cancelBtn) cancelBtn.classList.remove('d-none');
133         if (statusLabel) statusLabel.textContent = 'Идет запись речи...';
134         if (timerBadge) {
135             timerBadge.classList.remove('d-none');
136             timerBadge.textContent = '00:00';
137         }
138         if (micCircle) {
139             micCircle.classList.add('bg-danger', 'bg-opacity-25');
140             micCircle.style.transform = 'scale(1.1)';
141         }
142
143         clearInterval(tabTimerInterval);
144         tabTimerInterval = setInterval(() => {
145             const elapsedSec = Math.floor((Date.now() - tabRecordingStartTime) / 1000);
146             const m = String(Math.floor(elapsedSec / 60)).padStart(2, '0');
147             const s = String(elapsedSec % 60).padStart(2, '0');
148             if (timerBadge) timerBadge.textContent = `${m}:${s}`;
149         }, 500);
150
151     } catch (err) {
152         console.error('Microphone error:', err);
153         alert(`Не удалось включить микрофон: ${err.message}`);
154     }
155 }
156
157 function stopRecording() {
158     resetRecordingUi();
159     if (tabMediaRecorder && tabMediaRecorder.state === 'recording') {

```

```
160     tabMediaRecorder.stop();
161 }
162 }
163
164 function cancelRecording() {
165     resetRecordingUi();
166     if (tabMediaRecorder) {
167         if (tabMediaRecorder.state === 'recording') {
168             tabMediaRecorder.onstop = null;
169             tabMediaRecorder.stop();
170         }
171         tabAudioChunks = [];
172     }
173 }
174
175 function resetRecordingUi() {
176     clearInterval(tabTimerInterval);
177     const cancelBtn = document.getElementById('btn-vtab-record
178 -cancel');
179     const btnText = document.getElementById('vtab-record-btn-t
180 ext');
181     const statusLabel = document.getElementById('vtab-record-s
182 tatus-label');
183     const timerBadge = document.getElementById('vtab-record-ti
184 mer');
185     const micCircle = document.getElementById('vtab-mic-circle
186 ');
187
188     if (cancelBtn) cancelBtn.classList.add('d-none');
189     if (btnText) btnText.textContent = 'Начать запись';
190     if (statusLabel) statusLabel.textContent = 'Нажмите для на
191 чала записи';
192     if (timerBadge) timerBadge.classList.add('d-none');
193     if (micCircle) {
194         micCircle.classList.remove('bg-danger', 'bg-opacity-25')
195     };
196     micCircle.style.transform = 'scale(1)';
197 }
```

```
191 | }
192 |
193 | async function analyzeAudioBlob(audioBlob, filename) {
194 |     lastAudioFilename = filename;
195 |     const playerContainer = document.getElementById('vtab-play
    er-container');
196 |     const audioElement = document.getElementById('vtab-audio-e
    lement');
197 |     const filenameLabel = document.getElementById('vtab-curren
    t-filename');
198 |     const loadingState = document.getElementById('vtab-loading
    ');
199 |     const emptyState = document.getElementById('vtab-empty-sta
    te');
200 |     const analysisContent = document.getElementById('vtab-anal
    ysis-content');
201 |     const saveRagBtn = document.getElementById('vtab-btn-save-
    rag');
202 |
203 |     // Show audio player
204 |     if (playerContainer && audioElement) {
205 |         const audioUrl = URL.createObjectURL(audioBlob);
206 |         audioElement.src = audioUrl;
207 |         if (filenameLabel) filenameLabel.textContent = filename;
208 |         playerContainer.classList.remove('d-none');
209 |     }
210 |
211 |     if (emptyState) emptyState.classList.add('d-none');
212 |     if (analysisContent) analysisContent.classList.add('d-none
    ');
213 |     if (loadingState) loadingState.classList.remove('d-none');
214 |     if (saveRagBtn) saveRagBtn.classList.add('d-none');
215 |
216 |     const model = document.getElementById('vtab-model-select')
    ?.value || 'gemini-2.5-flash';
217 |     const lang = document.getElementById('vtab-lang-select')?.
    value || 'ru';
218 |     const apiKey = document.getElementById('vtab-api-key')?.va
```

```

    lue || '';
219
220     const formData = new FormData();
221     formData.append('file', audioBlob, filename);
222     formData.append('model', model);
223     formData.append('language', lang);
224     if (apiKey) formData.append('api_key', apiKey);
225
226     try {
227         const res = await fetch('/api/audio/diarize', {
228             method: 'POST',
229             body: formData,
230         });
231
232         const json = await res.json();
233         if (!res.ok) throw new Error(json.detail || 'Analysis er
ror');
234
235         lastAnalysisData = json.data || {};
236         renderResults(lastAnalysisData);
237         if (saveRagBtn) saveRagBtn.classList.remove('d-none');
238     } catch (err) {
239         console.error('[VoiceTab] Diarization failed:', err);
240         if (analysisContent) {
241             analysisContent.innerHTML = `
242                 <div class="alert alert-danger p-3 mb-0">
243                     <h6 class="fw-bold mb-1"><i class="bi bi-exclamati
on-triangle-fill me-2"></i>Ошибка анализа аудио</h6>
244                     <p class="mb-0 small">${escapeHtml(err.message)}</
p>
245                 </div>
246             `;
247             analysisContent.classList.remove('d-none');
248         }
249     } finally {
250         if (loadingState) loadingState.classList.add('d-none');
251     }
252 }

```

```

253
254 function renderResults(data) {
255     const analysisContent = document.getElementById('vtab-anal
ysis-content');
256     if (!analysisContent) return;
257
258     const summary = data.summary || '';
259     const keyPoints = data.key_points || [];
260     const actionItems = data.action_items || [];
261     const transcript = data.transcript || [];
262     const speakers = data.speakers || [];
263
264     const speakerColors = [
265         '#3b82f6', '#10b981', '#f59e0b', '#ec4899', '#8b5cf6', '
#06b6d4'
266     ];
267
268     let keyPointsHtml = '';
269     if (keyPoints.length > 0) {
270         keyPointsHtml = `
271         <div class="card bg-body-tertiary border-0 mb-3 shadow
-sm">
272             <div class="card-body p-3">
273                 <h6 class="fw-bold text-info mb-2"><i class="bi bi
-key-fill me-2"></i>Ключевые темы и решения</h6>
274                 <ul class="mb-0 ps-3 small text-body-secondary">
275                     ${keyPoints.map(kp => `<li class="mb-1">${escape
Html(kp)}</li>`).join('')}
276                 </ul>
277             </div>
278         </div>
279         `;
280     }
281
282     let actionItemsHtml = '';
283     if (actionItems.length > 0) {
284         actionItemsHtml = `
285         <div class="card bg-body-tertiary border-0 mb-3 shadow

```

```

-sm">
286         <div class="card-body p-3">
287             <h6 class="fw-bold text-success mb-2"><i class="bi
bi-check2-square me-2"></i>Задачи и договоренности (Action It
ems)</h6>
288             <ul class="list-unstyled mb-0 small">
289                 ${actionItems.map(ai => `
290                     <li class="d-flex align-items-start gap-2 mb-1
">
291                         <span class="badge bg-success bg-opacity-25
text-success font-monospace">TODO</span>
292                         <span>${escapeHtml(ai)}</span>
293                     </li>
294                 `).join('')}
295             </ul>
296         </div>
297     </div>
298     `;
299 }
300
301 let transcriptHtml = '';
302 if (transcript.length > 0) {
303     transcriptHtml = `
304         <div class="card bg-body-tertiary border-0 shadow-sm">
305             <div class="card-header bg-dark text-white d-flex ju
stify-content-between align-items-center py-2">
306                 <span class="fw-semibold small"><i class="bi bi-ch
at-quote-fill me-2 text-warning"></i>Стенограмма по собеседник
ам (${speakers.length || 2} участников)</span>
307             </div>
308             <div class="card-body p-3 d-flex flex-column gap-2"
style="max-height: 400px; overflow-y: auto;">
309                 ${transcript.map((t, idx) => {
310                     const sp = t.speaker || `Собеседник ${idx % 2 +
1}`;
311                     const col = speakerColors[idx % speakerColors.le
ngth];
312                     const ts = t.timestamp ? `<span class="badge bg-

```

```

    dark text-muted font-monospace">${escapeHtml(t.timestamp)}</span>` : `';
313         return `
314         <div class="p-2 rounded bg-body border-start border-3 shadow-sm" style="border-left-color: ${col} !important; ">
315             <div class="d-flex align-items-center justify-content-between mb-1">
316                 <span class="fw-bold small" style="color: ${col}; "><i class="bi bi-person-fill me-1"></i>${escapeHtml(summary)}</span>
317                 ${ts}
318             </div>
319             <div class="small text-body">${escapeHtml(text)}</div>
320         </div>
321         `;
322         }).join('')}
323     </div>
324 </div>
325 `;
326 }
327
328 analysisContent.innerHTML = `
329     ${summary ? `
330         <div class="alert alert-primary mb-3 shadow-sm">
331             <h6 class="fw-bold mb-1"><i class="bi bi-card-text me-2"></i>Краткая сводка (Executive Summary)</h6>
332             <p class="mb-0 small">${escapeHtml(summary)}</p>
333         </div>
334         ` : ``}
335     ${keyPointsHtml}
336     ${actionItemsHtml}
337     ${transcriptHtml}
338 `;
339
340 analysisContent.classList.remove('d-none');
341 }

```

```

342
343 async function saveResultToRag() {
344     if (!lastAnalysisData) return;
345
346     try {
347         const res = await fetch('/api/audio/save-to-rag', {
348             method: 'POST',
349             headers: { 'Content-Type': 'application/json' },
350             body: JSON.stringify({
351                 title: `Голосовая запись: ${lastAudioFilename}`,
352                 summary: lastAnalysisData.summary || '',
353                 transcript: lastAnalysisData.transcript || [],
354                 key_points: lastAnalysisData.key_points || [],
355                 action_items: lastAnalysisData.action_items || [],
356                 markdown_report: lastAnalysisData.markdown_report ||
357                 '',
358                 }),
359             });
360
361         const json = await res.json();
362         if (!res.ok) throw new Error(json.detail || 'Save error'
363         );
364
365         alert(`👏 ${json.message || 'Разговор успешно сохранен в
366         базу знаний RAG!'}`);
367     } catch (err) {
368         console.error('[VoiceTab] Failed to save in RAG:', err);
369         alert(`❌ Ошибка сохранения в RAG: ${err.message}`);
370     }
371
372     function escapeHtml(text) {
373         if (!text) return '';
374         return String(text)
375             .replace(/&/g, '&')
376             .replace(/</g, '<')
377             .replace(/>/g, '>')
378             .replace(/"/g, '"');

```



```
377 |         .replace(/'/g, '&#039;');
378 |     }
379 |
380 |     if (document.readyState === 'loading') {
381 |         document.addEventListener('DOMContentLoaded', initVoiceTab
382 | );
383 |     } else {
384 |         initVoiceTab();
385 |     }
386 | }());
```

index.html

```
1 <!-- Website Intelligence Monitor Application Tab -->
2 <style>
3   .webmon-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .webmon-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    min-height: 58px;
15    gap: 0.2rem;
16  }
17  .webmon-label {
18    font-size: 0.68rem;
19    text-transform: uppercase;
20    letter-spacing: 0.05em;
21    color: #94a3b8;
22    line-height: 1.1;
23  }
24  .webmon-value {
25    font-size: 1.25rem;
26    font-weight: 700;
27    color: #f8fafc;
28    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
29    Consolas, monospace;
30    line-height: 1.15;
31  }
32  .webmon-panel {
33    background: #0f172a !important;
34    border: 1px solid #334155 !important;
35    border-radius: 8px;
36    overflow: hidden;
```

```
36 | }
37 | .webmon-panel-header {
38 |     background: #1e293b !important;
39 |     border-bottom: 1px solid #334155 !important;
40 |     padding: 0.45rem 0.85rem;
41 |     font-weight: 600;
42 |     font-size: 0.82rem;
43 |     color: #38bdf8;
44 |     display: flex;
45 |     align-items: center;
46 |     justify-content: space-between;
47 | }
48 | .webmon-table {
49 |     width: 100%;
50 |     font-size: 0.78rem;
51 |     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
    Consolas, monospace;
52 |     border-collapse: collapse;
53 | }
54 | .webmon-table th {
55 |     background: #1e293b;
56 |     padding: 5px 8px;
57 |     color: #94a3b8;
58 |     font-weight: 600;
59 |     border-bottom: 1px solid #334155;
60 | }
61 | .webmon-table td {
62 |     padding: 4px 8px;
63 |     border-bottom: 1px solid rgba(51, 65, 85, 0.4);
64 | }
65 | </style>
66 |
67 | <div class="webmon-container">
68 |     <!-- Header Alert & Status (Compact Strip) -->
69 |     <div class="d-flex justify-content-between align-items-cente
    r flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style
    ="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%
    ); border: 1px solid #334155;">
```

```

70     <div class="d-flex align-items-center gap-2 flex-wrap">
71         <span class="fs-5">□ </span>
72         <span class="fw-bold text-white small" style="font-size:
       0.92rem;">Website Intelligence</span>
73         <span class="badge bg-dark border border-secondary text-
       info fw-normal" style="font-size: 0.7rem;">Traffic Monitor</sp
       an>
74         <span class="text-secondary small d-none d-lg-inline" st
       yle="font-size: 0.76rem;">| GA4 Realtime, Google Search Consol
       e CTR & аномалии</span>
75     </div>
76     <div class="d-flex align-items-center gap-1.5 flex-wrap">
77         <span class="badge rounded-pill bg-success-subtle text-s
       uccess border border-success px-2 py-1" style="font-size: 0.72
       rem;" id="webmon-status-badge">
78             • АКТИВЕН
79         </span>
80         <button class="btn btn-xs btn-outline-info rounded-pill
       px-2.5 py-1" style="font-size: 0.75rem;" id="btn-webmon-refres
       h">
81             <i class="bi bi-arrow-clockwise me-1"></i> Обновить
82         </button>
83         <button class="btn btn-xs btn-outline-warning rounded-pi
       ll px-2 py-1" style="font-size: 0.75rem;" id="btn-webmon-confi
       g">
84             <i class="bi bi-gear-fill me-1"></i> Настройки
85         </button>
86     </div>
87 </div>
88
89 <!-- Top Metrics Bar (Compact Density Grid) -->
90 <div class="row g-2 mb-2">
91     <div class="col-6 col-md-3">
92         <div class="webmon-card">
93             <span class="webmon-label">Онлайн (Realtime)</span>
94             <div class="d-flex align-items-baseline justify-conten
       t-between mt-1">
95                 <span class="webmon-value text-success" id="webmon-r

```

```

    realtime-users">0</span>
96         <small class="text-muted" style="font-size: 0.68rem;
">Пользователей</small>
97     </div>
98 </div>
99 </div>
100 <div class="col-6 col-md-3">
101     <div class="webmon-card">
102         <span class="webmon-label">Доступность / Ping</span>
103         <div class="d-flex align-items-baseline justify-content-between mt-1">
104             <span class="webmon-value text-info" style="font-size: 0.95rem;" id="webmon-availability">100% (24ms)</span>
105         </div>
106     </div>
107 </div>
108 <div class="col-6 col-md-3">
109     <div class="webmon-card">
110         <span class="webmon-label">GSC CTR</span>
111         <div class="d-flex align-items-baseline justify-content-between mt-1">
112             <span class="webmon-value text-warning" id="webmon-ctr-score">4.2%</span>
113             <small class="text-muted" style="font-size: 0.68rem;
">Search Console</small>
114         </div>
115     </div>
116 </div>
117 <div class="col-6 col-md-3">
118     <div class="webmon-card">
119         <span class="webmon-label">Индекс здоровья</span>
120         <div class="d-flex align-items-baseline justify-content-between mt-1">
121             <span class="webmon-value text-light" id="webmon-health-score">98/100</span>
122         </div>
123     </div>
124 </div>

```

```

125 | </div>
126 |
127 | <!-- Main Content Panels Grid -->
128 | <div class="row g-3">
129 |     <!-- Top Pages Panel -->
130 |     <div class="col-lg-6">
131 |         <div class="webmon-panel h-100">
132 |             <div class="webmon-panel-header">
133 |                 <span>📄 Популярные страницы (GA4)</span>
134 |                 <span class="badge bg-secondary" id="webmon-pages-ba
135 | dge">Топ-10</span>
136 |             </div>
137 |             <div class="table-responsive">
138 |                 <table class="webmon-table">
139 |                     <thead>
140 |                         <tr>
141 |                             <th>Путь страницы</th>
142 |                             <th class="text-end">Просмотры</th>
143 |                             <th class="text-end">Пользователи</th>
144 |                         </tr>
145 |                     </thead>
146 |                     <tbody id="webmon-pages-table-body">
147 |                         <tr><td colspan="3" class="text-center text-mute
148 | d p-3">Загрузка популярных страниц...</td></tr>
149 |                     </tbody>
150 |                 </table>
151 |             </div>
152 |         </div>
153 |     </div>
154 | <!-- Traffic Sources & Diagnostics Panel -->
155 | <div class="col-lg-6">
156 |     <div class="webmon-panel h-100">
157 |         <div class="webmon-panel-header">
158 |             <span>📶 Каналы трафика и AI Диагностика</span>
159 |             <span class="badge bg-info text-dark">GA4 & Anomalie
160 | s</span>
161 |         </div>

```

```
160 |         <div class="p-3">
161 |             <div id="webmon-traffic-channels" class="mb-3">
162 |                 <div class="text-muted small text-center">Загрузка
| источников...</div>
163 |             </div>
164 |             <hr class="border-secondary my-2">
165 |             <h6 class="fw-bold text-info mb-1" id="webmon-diag-t
| itle">Анализ трендов трафика...</h6>
166 |             <p class="small text-light mb-0" id="webmon-diag-sum
| mary">Ожидание нормализованного отчёта...</p>
167 |         </div>
168 |     </div>
169 | </div>
170 | </div>
171 </div>
172 |
```

main.js

```
1 // =====
  =====
2 // Process Name: Website Intelligence Monitor Web Tab Module
3 // =====
  =====
4 // Description:
5 //   Client-side JavaScript controller for the Website Intelli
  gence Monitor
6 //   administrative web interface tab.
7 //
8 // File: main.js
9 // Project: ai-breadboard
10 // Package: src.api.webinterface.website_monitor_tab
11 // Author: hypo69
12 // Copyright: © 2026 hypo69
13 // =====
  =====
14
15 (function() {
16   let isWebmonInitialized = false;
17
18   async function fetchRealtime() {
19     try {
20       const res = await fetch('/api/v1/website-monitor/realtime');
21       if (!res.ok) return;
22       const data = await res.json();
23
24       const usersEl = document.getElementById('webmon-realtime
  -users');
25       if (usersEl) usersEl.innerText = data.active_users || 0;
26     } catch (e) {
27       console.error('[WebsiteMonitorTab] Failed to fetch realtime:', e);
28     }
29   }
```



```
30 |
31 |   async function fetchSummary() {
32 |     try {
33 |       const res = await fetch('/api/v1/website-monitor/summary
34 | ');
35 |       if (!res.ok) return;
36 |       const data = await res.json();
37 |
38 |       const availEl = document.getElementById('webmon-availabi
39 | lity');
40 |       const ctrEl = document.getElementById('webmon-ctr-score'
41 | );
42 |       const healthEl = document.getElementById('webmon-health-
43 | score');
44 |       const diagTitle = document.getElementById('webmon-diag-t
45 | itle');
46 |       const diagSummary = document.getElementById('webmon-diag
47 | -summary');
48 |
49 |       if (data.technical) {
50 |         if (availEl) availEl.innerText = `${data.technical.ava
51 | ilability_pct || 100}% (${data.technical.avg_latency_ms || 0}
52 | ms)`;
53 |       }
54 |       if (data.search_console) {
55 |         if (ctrEl) ctrEl.innerText = `${data.search_console.av
56 | g_ctr || 0}%`;
57 |       }
58 |       if (healthEl) healthEl.innerText = `${data.health_score
59 | || 100}/100`;
60 |
61 |       if (diagTitle) diagTitle.innerText = `Статус: ${data.sta
62 | tus || 'OK'}`;
63 |       if (diagSummary) diagSummary.innerText = data.summary ||
64 | 'Все веб-сервисы и каналы трафика функционируют стабильно.';
65 |     } catch (e) {
66 |       console.error('[WebsiteMonitorTab] Failed to fetch summa
67 | ry:', e);
68 |     }
69 |   }
70 | }
```

```

55 |     }
56 | }
57 |
58 | async function fetchPages() {
59 |     try {
60 |         const res = await fetch('/api/v1/website-monitor/pages?limit=8');
61 |         if (!res.ok) return;
62 |         const pages = await res.json();
63 |
64 |         const tbody = document.getElementById('webmon-pages-table-body');
65 |         if (tbody) {
66 |             if (!Array.isArray(pages) || pages.length === 0) {
67 |                 tbody.innerHTML = '<tr><td colspan="3" class="text-center text-muted p-3">Нет данных о просмотрах.</td></tr>';
68 |                 return;
69 |             }
70 |             tbody.innerHTML = pages.map((p, idx) => `
71 |                 <tr class="webmon-page-row" data-idx="${idx}" style="
72 |                 <td class="text-info text-truncate" style="max-width: 240px;" title="${p.page_path || '/'}">${p.page_path || '/'}
73 |                 <td class="text-end text-light">${p.screen_page_views || p.views || 0}</td>
74 |                 <td class="text-end text-secondary">${p.active_users || p.users || 0}</td>
75 |                 </tr>
76 |                 `).join('');
77 |
78 |             tbody.querySelectorAll('.webmon-page-row').forEach(row
79 | => {
80 |                 row.onclick = () => {
81 |                     const idx = parseInt(row.getAttribute('data-idx'),
82 | 10);
83 |
84 |                     const p = pages[idx];
85 |                     if (!p) return;

```

```

83 |
84 |         if (window.AITableModal) {
85 |             window.AITableModal.show({
86 |                 icon: '📄',
87 |                 title: `Веб-страница ${p.page_path || '/'}`,
88 |                 subtitle: `Просмотров: ${p.screen_page_views ||
| p.views || 0} | Пользователей: ${p.active_users || p.users ||
| 0}`,
89 |                 tableType: 'website',
90 |                 badges: [
91 |                     { text: `${p.screen_page_views || 0} views`,
| class: 'badge bg-info text-dark' },
92 |                     { text: `${p.active_users || 0} users`, clas
| s: 'badge bg-success' }
93 |                 ],
94 |                 metadata: [
95 |                     { label: 'URL / Путь страницы', value: p.pag
| e_path || '/' },
96 |                     { label: 'Просмотры страниц', value: String(
| p.screen_page_views || p.views || 0) },
97 |                     { label: 'Активные пользователи', value: Str
| ing(p.active_users || p.users || 0) },
98 |                     { label: 'Показатель отказов', value: p.boun
| ce_rate ? `${p.bounce_rate}%` : 'N/A' },
99 |                     { label: 'Средняя длительность', value: p.av
| g_session_duration ? `${p.avg_session_duration} сек` : 'N/A' }
100 |                 ],
101 |                 rawTitle: 'Аналитика веб-страницы',
102 |                 rawContent: JSON.stringify(p, null, 2),
103 |                 requestData: {
104 |                     url: p.page_path,
105 |                     views: p.screen_page_views || p.views,
106 |                     users: p.active_users || p.users
107 |                 }
108 |             });
109 |         }
110 |     };
111 | });

```

```

112     }
113   } catch (e) {
114     console.error('[WebsiteMonitorTab] Failed to fetch pages
: ', e);
115   }
116 }
117
118 async function fetchTrafficSources() {
119   try {
120     const res = await fetch('/api/v1/website-monitor/traffic
-sources');
121     if (!res.ok) return;
122     const channels = await res.json();
123
124     const container = document.getElementById('webmon-traffi
c-channels');
125     if (container && Array.isArray(channels)) {
126       container.innerHTML = channels.slice(0, 5).map(c => `
127         <div class="d-flex justify-content-between align-ite
ms-center py-1 border-bottom border-secondary-subtle small">
128           <span class="text-light">${c.channel_group} || c.na
me || 'Organic'}</span>
129           <span class="badge bg-secondary">${c.sessions} || c
.users || 0} сессий</span>
130         </div>
131       `).join('');
132     }
133   } catch (e) {
134     console.error('[WebsiteMonitorTab] Failed to fetch traff
ic sources:', e);
135   }
136 }
137
138 function initWebsiteMonitorTab() {
139   fetchRealtime();
140   fetchSummary();
141   fetchPages();
142   fetchTrafficSources();

```

```
143 |
144 |     if (!isWebmonInitialized) {
145 |         const refreshBtn = document.getElementById('btn-webmon-r
refresh');
146 |         if (refreshBtn) {
147 |             refreshBtn.addEventListener('click', () => {
148 |                 fetchRealtime();
149 |                 fetchSummary();
150 |                 fetchPages();
151 |                 fetchTrafficSources();
152 |             });
153 |         }
154 |
155 |         const configBtn = document.getElementById('btn-webmon-co
nfig');
156 |         if (configBtn) {
157 |             configBtn.addEventListener('click', () => {
158 |                 if (typeof window.openAppConfigModal === 'function')
{
159 |                     window.openAppConfigModal('website_monitor');
160 |                 }
161 |             });
162 |         }
163 |         isWebmonInitialized = true;
164 |     }
165 | }
166 |
167 | window.initWebsiteMonitorTab = initWebsiteMonitorTab;
168 | })();
169 |
```

index.html

```
1 <!-- Wikipedia Research & Multi-Model Benchmark Laboratory Tab
   -->
2 <style>
3   .wiki-lab-container {
4     padding: 1rem;
5   }
6   .wiki-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 10px;
10    padding: 1.25rem;
11    margin-bottom: 1rem;
12  }
13  .wiki-card-title {
14    font-size: 1rem;
15    font-weight: 700;
16    color: #38bdf8;
17    margin-bottom: 0.75rem;
18    display: flex;
19    align-items: center;
20    gap: 0.5rem;
21  }
22  .wiki-badge-pos {
23    background: rgba(34, 197, 94, 0.2);
24    color: #4ade80;
25    border: 1px solid rgba(34, 197, 94, 0.4);
26    padding: 2px 8px;
27    border-radius: 4px;
28    font-weight: 600;
29  }
30  .wiki-badge-neg {
31    background: rgba(239, 68, 68, 0.2);
32    color: #f87171;
33    border: 1px solid rgba(239, 68, 68, 0.4);
34    padding: 2px 8px;
35    border-radius: 4px;
```

```
36     font-weight: 600;
37 }
38 .wiki-badge-neu {
39     background: rgba(234, 179, 8, 0.2);
40     color: #facc15;
41     border: 1px solid rgba(234, 179, 8, 0.4);
42     padding: 2px 8px;
43     border-radius: 4px;
44     font-weight: 600;
45 }
46 .wiki-table {
47     width: 100%;
48     font-size: 0.85rem;
49     border-collapse: collapse;
50     color: #f1f5f9;
51 }
52 .wiki-table th {
53     background: #1e293b;
54     padding: 8px 12px;
55     color: #94a3b8;
56     font-weight: 600;
57     border-bottom: 1px solid #334155;
58     text-align: left;
59 }
60 .wiki-table td {
61     padding: 8px 12px;
62     border-bottom: 1px solid #1e293b;
63 }
64 .wiki-table tr:hover {
65     background: rgba(51, 65, 85, 0.3);
66 }
67 .wiki-checkbox-grid {
68     display: grid;
69     grid-template-columns: repeat(auto-fill, minmax(130px, 1fr
70 ));
71     gap: 0.5rem;
72 }
73 </style>
```

```

73 |
74 | <div class="wiki-lab-container">
75 |   <!-- Header -->
76 |   <div class="d-flex justify-content-between align-items-center mb-3">
77 |     <div>
78 |       <h4 class="fw-bold text-white mb-0">📖 Wikipedia Research & Model Laboratory</h4>
79 |       <small class="text-muted">Кросс-языковой сравнительный анализ контента Википедии и бенчмарк интерпретации моделей ИИ</small>
80 |     </div>
81 |     <div class="d-flex gap-2">
82 |       <button class="btn btn-outline-info btn-sm" id="btn-wiki-health" title="Проверить статус API">
83 |         <i class="bi bi-heart-pulse"></i> Статус
84 |       </button>
85 |     </div>
86 |   </div>
87 |
88 |   <!-- Experiment Configuration Panel -->
89 |   <div class="row g-3 mb-3">
90 |     <div class="col-lg-8">
91 |       <div class="wiki-card">
92 |         <div class="wiki-card-title">
93 |           <i class="bi bi-sliders"></i> Параметры исследования
94 |         </div>
95 |
96 |         <!-- Topic Search -->
97 |         <div class="mb-3">
98 |           <label class="form-label text-light small fw-bold">Тема для исследования (Wikipedia Topic):</label>
99 |           <div class="input-group">
100 |             <span class="input-group-text bg-dark border-secondary text-info"><i class="bi bi-search"></i></span>
101 |             <input type="text" class="form-control bg-dark text-white border-secondary" id="wiki-topic-input" placeholder="Например: Israel-Gaza war, Quantum computing, Artificial intell

```



```

102 |         ige" value="Israel-Gaza war">
103 |             <button class="btn btn-primary px-3" id="btn-run-e
104 | xp-a" title="Запустить кросс-языковое сравнение одной моделью"
105 | >
106 |                 <i class="bi bi-translate"></i> Exp A: Языки
107 |             </button>
108 |             <button class="btn btn-success px-3" id="btn-run-e
109 | xp-b" title="Запустить кросс-модельный бенчмарк">
110 |                 <i class="bi bi-cpu"></i> Exp B: Модели
111 |             </button>
112 |         </div>
113 |     </div>
114 |
115 |     <!-- Languages Selector -->
116 |     <div class="mb-3">
117 |         <label class="form-label text-light small fw-bold">Я
118 | зыковые разделы Википедии:</label>
119 |         <div class="wiki-checkbox-grid" id="wiki-languages-l
120 | ist">
121 |             <div class="form-check">
122 |                 <input class="form-check-input wiki-lang-check"
123 | type="checkbox" value="en" id="lang-en" checked>
124 |                 <label class="form-check-label text-white small"
125 | for="lang-en">English (en)</label>
126 |             </div>
127 |             <div class="form-check">
128 |                 <input class="form-check-input wiki-lang-check"
129 | type="checkbox" value="ru" id="lang-ru" checked>
130 |                 <label class="form-check-label text-white small"
131 | for="lang-ru">Русский (ru)</label>
132 |             </div>
133 |             <div class="form-check">
134 |                 <input class="form-check-input wiki-lang-check"
135 | type="checkbox" value="he" id="lang-he" checked>
136 |                 <label class="form-check-label text-white small"
137 | for="lang-he">עברית (he)</label>
138 |             </div>
139 |             <div class="form-check">

```

```
128 |         <input class="form-check-input wiki-lang-check"
| type="checkbox" value="de" id="lang-de" checked>
129 |         <label class="form-check-label text-white small"
| for="lang-de">Deutsch (de)</label>
130 |     </div>
131 |     <div class="form-check">
132 |         <input class="form-check-input wiki-lang-check"
| type="checkbox" value="fr" id="lang-fr" checked>
133 |         <label class="form-check-label text-white small"
| for="lang-fr">Français (fr)</label>
134 |     </div>
135 |     <div class="form-check">
136 |         <input class="form-check-input wiki-lang-check"
| type="checkbox" value="uk" id="lang-uk">
137 |         <label class="form-check-label text-white small"
| for="lang-uk">Українська (uk)</label>
138 |     </div>
139 | </div>
140 | </div>
141 |
142 | <!-- Models & Analysis Dimensions -->
143 | <div class="row">
144 |     <div class="col-md-6">
145 |         <label class="form-label text-light small fw-bold"
| >AI Модели (Experiment B):</label>
146 |         <div class="d-flex flex-wrap gap-2" id="wiki-model
| s-list">
147 |             <div class="form-check form-check-inline">
148 |                 <input class="form-check-input wiki-model-chec
| k" type="checkbox" value="gemini" id="mod-gemini" checked>
149 |                 <label class="form-check-label text-white smal
| l" for="mod-gemini">Gemini</label>
150 |             </div>
151 |             <div class="form-check form-check-inline">
152 |                 <input class="form-check-input wiki-model-chec
| k" type="checkbox" value="foundry" id="mod-foundry" checked>
153 |                 <label class="form-check-label text-white smal
| l" for="mod-foundry">Foundry Local</label>
```

```

154         </div>
155         <div class="form-check form-check-inline">
156             <input class="form-check-input wiki-model-chec
157 k" type="checkbox" value="ollama" id="mod-ollama" checked>
158             <label class="form-check-label text-white smal
159 l" for="mod-ollama">Ollama</label>
160         </div>
161         <div class="form-check form-check-inline">
162             <input class="form-check-input wiki-model-chec
163 k" type="checkbox" value="onnx" id="mod-onnx">
164             <label class="form-check-label text-white smal
165 l" for="mod-onnx">ONNX DirectML</label>
166         </div>
167     </div>
168 </div>
169 <div class="col-md-6">
170     <label class="form-label text-light small fw-bold"
171 >Базовая модель для Exp A:</label>
172     <select class="form-select form-select-sm bg-dark
173 text-white border-secondary" id="wiki-primary-model">
174         <option value="gemini" selected>Google Gemini</o
175 ption>
176         <option value="foundry">Microsoft Foundry Local<
177 /option>
178         <option value="ollama">Ollama Local LLM</option>
179         <option value="onnx">ONNX DirectML</option>
180     </select>
181 </div>
182 </div>
183 </div>
184 </div>
185 <!-- Quick Insights & Stats -->
186 <div class="col-lg-4">
187     <div class="wiki-card h-100">
188         <div class="wiki-card-title">
189             <i class="bi bi-lightbulb"></i> Аналитические инсайты

```

```
183 |         </div>
184 |         <div id="wiki-insights-container" class="text-secondary small">
185 |             <em>Запустите Experiment A или Experiment B для отображения выводов и кросс-сравнения...</em>
186 |         </div>
187 |     </div>
188 | </div>
189 </div>
190
191 <!-- Loading State -->
192 <div id="wiki-loading-spinner" class="text-center my-4 d-none">
193     <div class="spinner-border text-primary" role="status"></div>
194     <div class="text-light mt-2 small" id="wiki-loading-text">
195         Выполняется сбор статей и AI-анализ...</div>
196 </div>
197
198 <!-- Results Section -->
199 <div id="wiki-results-section" class="d-none">
200     <div class="wiki-card">
201         <div class="wiki-card-title" id="wiki-results-title">
202             <i class="bi bi-bar-chart-fill"></i> Результаты эксперимента
203         </div>
204         <div class="table-responsive">
205             <div id="wiki-results-table-container"></div>
206         </div>
207     </div>
208 </div>
209 </div>
210
```

main.js

```
1 // Wikipedia Research & Multi-Model Benchmark Laboratory WebGU
  I JS
2 (function () {
3   'use strict';
4
5   console.log('[WikiLab] Initializing Wikipedia Research Tab J
  S...');
6
7   async function checkHealth() {
8     try {
9       const res = await fetch('/api/v1/wikipedia-research/heal
  th');
10      if (res.ok) {
11        const data = await res.json();
12        alert(`Wikipedia Research API статус: ${data.status.to
  UpperCase()} (приложение: ${data.app})`);
13      } else {
14        alert(`Ошибка API: HTTP ${res.status}`);
15      }
16    } catch (e) {
17      alert(`Не удалось связаться с API: ${e.message}`);
18    }
19  }
20
21  function getSelectedLanguages() {
22    const checkboxes = document.querySelectorAll('.wiki-lang-c
  heck:checked');
23    const langs = [];
24    checkboxes.forEach((cb) => langs.push(cb.value));
25    return langs.length > 0 ? langs : ['en', 'ru'];
26  }
27
28  function getSelectedModels() {
29    const checkboxes = document.querySelectorAll('.wiki-model-
  check:checked');
30    const models = [];
```

```
31 |     checkboxes.forEach((cb) => models.push(cb.value));
32 |     return models.length > 0 ? models : ['gemini', 'foundry'];
33 | }
34 |
35 | function formatSentimentBadge(val) {
36 |     if (val === undefined || val === null) return '<span class
    ="text-muted">n/a</span>';
37 |     const num = parseFloat(val);
38 |     if (num > 0.05) {
39 |         return `<span class="wiki-badge-pos">+${num.toFixed(2)}<
    /span>`;
40 |     } else if (num < -0.05) {
41 |         return `<span class="wiki-badge-neg">${num.toFixed(2)}</
    span>`;
42 |     }
43 |     return `<span class="wiki-badge-neu">${num.toFixed(2)}</sp
    an>`;
44 | }
45 |
46 | function setLoading/loading, text = 'Выполняется сбор и анал
    из...') {
47 |     const spinner = document.getElementById('wiki-loading-spin
    ner');
48 |     const loadingText = document.getElementById('wiki-loading-
    text');
49 |     const resultsSec = document.getElementById('wiki-results-s
    ection');
50 |     if (spinner) {
51 |         if (loading) {
52 |             spinner.classList.remove('d-none');
53 |             if (loadingText) loadingText.textContent = text;
54 |             if (resultsSec) resultsSec.classList.add('d-none');
55 |         } else {
56 |             spinner.classList.add('d-none');
57 |             if (resultsSec) resultsSec.classList.remove('d-none');
58 |         }
59 |     }
60 | }
```

```
61 |
62 | function renderInsights(insights) {
63 |     const container = document.getElementById('wiki-insights-c
    ontainer');
64 |     if (!container) return;
65 |     if (!insights || insights.length === 0) {
66 |         container.innerHTML = '<div class="text-muted small">Нет
    выявленных значимых расхождений.</div>';
67 |         return;
68 |     }
69 |     let html = '<ul class="list-unstyled mb-0 d-flex flex-colu
    mn gap-2">';
70 |     insights.forEach((ins) => {
71 |         html += `<li class="d-flex align-items-start gap-2 text-
    light"><i class="bi bi-check-circle-fill text-info mt-1"></i>
    <span>${ins}</span></li>`;
72 |     });
73 |     html += '</ul>';
74 |     container.innerHTML = html;
75 | }
76 |
77 | async function runExperimentA() {
78 |     const topicInput = document.getElementById('wiki-topic-inp
    ut');
79 |     const topic = topicInput ? topicInput.value.trim() : 'Isra
    el-Gaza war';
80 |     const languages = getSelectedLanguages();
81 |     const model = document.getElementById('wiki-primary-model'
    )?.value || 'gemini';
82 |
83 |     setLoading(true, `Эксперимент A: сбор статей по теме "${to
    pic}" и анализ моделью ${model}...`);
84 |
85 |     try {
86 |         const response = await fetch('/api/v1/wikipedia-research
    /experiment/languages', {
87 |             method: 'POST',
88 |             headers: { 'Content-Type': 'application/json' },
```

```

89         body: JSON.stringify({ topic, languages, model }),
90     });
91
92     if (!response.ok) {
93         throw new Error(`HTTP ${response.status}: ${await response.text()}`);
94     }
95
96     const report = await response.json();
97     renderExperimentAResults(report);
98 } catch (err) {
99     alert(`Ошибка выполнения Эксперимента A: ${err.message}`
100 );
101 } finally {
102     setLoading(false);
103 }
104
105 function renderExperimentAResults(report) {
106     const titleEl = document.getElementById('wiki-results-title');
107     if (titleEl) {
108         titleEl.innerHTML = `<i class="bi bi-translate"></i> Experiment A: Кросс-языковое сравнение темы «${report.topic}» (Модель: ${report.model_used})`;
109     }
110
111     renderInsights(report.cross_language_insights);
112
113     const tableContainer = document.getElementById('wiki-results-table-container');
114     if (!tableContainer) return;
115
116     let html = `
117         <table class="wiki-table">
118             <thead>
119                 <tr>
120                     <th>Язык</th>

```



```

121         <th>Статья</th>
122         <th>Тональность</th>
123         <th>Субъективность</th>
124         <th>Критика</th>
125         <th>Похвала</th>
126         <th>Неопределенность</th>
127         <th>Спорные тезисы</th>
128         <th>Фрейминг</th>
129         <th>Слов</th>
130     </tr>
131 </thead>
132 <tbody>
133 `;
134
135     (report.metrics_summary_table || []).forEach((row) => {
136         html += `
137             <tr>
138                 <td><strong class="text-info">${row.lang.toUpperCase
139                 <td><a href="${row.url}" target="_blank" class="text
140                 <td>${formatSentimentBadge(row.sentiment)}</td>
141                 <td><span class="text-warning">${row.subjectivity.to
142                 <td><span class="text-danger">${row.criticism.toFixe
143                 <td><span class="text-success">${row.praise.toFixed(
144                 <td><span class="text-primary">${row.uncertainty.toF
145                 <td class="text-center"><span class="badge bg-second
146                 <td><span class="badge bg-dark border border-seconda
147                 <td class="text-muted small">${row.word_count.toLoca
148             </tr>
149         `;

```

```

150     });
151
152     html += `</tbody></table>`;
153     tableContainer.innerHTML = html;
154 }
155
156 async function runExperimentB() {
157     const topicInput = document.getElementById('wiki-topic-input');
158     const topic = topicInput ? topicInput.value.trim() : 'Israel-Gaza war';
159     const languages = getSelectedLanguages();
160     const models = getSelectedModels();
161
162     setLoading(true, `Эксперимент В: сопоставление интерпретаций моделями [${models.join(', ')}]...`);
163
164     try {
165         const response = await fetch('/api/v1/wikipedia-research/experiment/models', {
166             method: 'POST',
167             headers: { 'Content-Type': 'application/json' },
168             body: JSON.stringify({ topic, languages, models }),
169         });
170
171         if (!response.ok) {
172             throw new Error(`HTTP ${response.status}: ${await response.text()}`);
173         }
174
175         const report = await response.json();
176         renderExperimentBResults(report);
177     } catch (err) {
178         alert(`Ошибка выполнения Эксперимента В: ${err.message}`);
179     } finally {
180         setLoading(false);
181     }

```

```

182 | }
183 |
184 | function renderExperimentBResults(report) {
185 |     const titleEl = document.getElementById('wiki-results-titl
186 | e');
187 |     if (titleEl) {
188 |         titleEl.innerHTML = `<i class="bi bi-cpu"></i> Experimen
189 | t B: Мульти-модельный бенчмарк интерпретации «${report.topic}»
190 | `;
191 |     }
192 |     renderInsights(report.model_alignment_insights);
193 |
194 |     const tableContainer = document.getElementById('wiki-resul
195 | ts-table-container');
196 |     if (!tableContainer) return;
197 |
198 |     let html = `
199 |         <table class="wiki-table">
200 |             <thead>
201 |                 <tr>
202 |                     <th>Языковой раздел</th>
203 |                 </tr>
204 |             </thead>
205 |             <tbody>
206 |
207 |                 <tr>
208 |                     <td>
209 |                         <strong class="text-warning">${report.toU
210 | pperCase()}</strong>
211 |                     </td>
212 |                     <td>
213 |                         <strong class="text-warning">${report.toU
214 | pperCase()}</strong>

```

```

215 |         const sentiment = report.sentiment_grid?.[lang]?.[mod]
    | ;
216 |         const res = report.matrix?.[lang]?.[mod];
217 |         const framing = res?.metrics?.framing_tone || 'n/a';
218 |         html += `
219 |             <td>
220 |                 <div class="d-flex flex-column gap-1">
221 |                     <div>${formatSentimentBadge(sentiment)}</div>
222 |                     <small class="text-muted" style="font-size: 0.75
    | rem;">(${framing})</small>
223 |                 </div>
224 |             </td>
225 |         `;
226 |     });
227 |     html += `</tr>`;
228 | });
229 |
230 |     html += `</tbody></table>`;
231 |     tableContainer.innerHTML = html;
232 | }
233 |
234 | function bindEvents() {
235 |     const btnHealth = document.getElementById('btn-wiki-health
    | ');
236 |     if (btnHealth) btnHealth.onclick = checkHealth;
237 |
238 |     const btnExpA = document.getElementById('btn-run-exp-a');
239 |     if (btnExpA) btnExpA.onclick = runExperimentA;
240 |
241 |     const btnExpB = document.getElementById('btn-run-exp-b');
242 |     if (btnExpB) btnExpB.onclick = runExperimentB;
243 | }
244 |
245 | window.initWikipediaResearchTab = function () {
246 |     console.log('[WikiLab] initWikipediaResearchTab triggered'
    | );
247 |     bindEvents();
248 | };

```

```
249 |
250 |     if (document.readyState === 'loading') {
251 |         document.addEventListener('DOMContentLoaded', bindEvents);
252 |     } else {
253 |         bindEvents();
254 |     }
255 | })();
256 |
```

index.html

```
1 <!-- Windows System Administrator Application Tab -->
2 <style>
3   .winadmin-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .winadmin-table {
7     width: 100%;
8     font-size: 0.78rem;
9     font-family: var(--font-mono, monospace);
10    border-collapse: collapse;
11  }
12  .winadmin-table th {
13    background: var(--surface-2);
14    padding: 5px 8px;
15    color: var(--text-muted);
16    font-weight: 600;
17    border-bottom: 1px solid var(--border-color);
18  }
19  .winadmin-table td {
20    padding: 4px 8px;
21    border-bottom: 1px solid var(--border-subtle, rgba(51, 65,
22    85, 0.4));
23  }
24  .winadmin-filter-btn.active {
25    background-color: #3b82f6 !important;
26    color: #fff !important;
27    border-color: #3b82f6 !important;
28  }
29  .badge-hidden {
30    background-color: #6b21a8;
31    color: #e9d5ff;
32    border: 1px solid #9333ea;
33  }
34  .badge-admin {
35    background-color: #9a3412;
36    color: #ffedd5;
```

```

36     border: 1px solid #ea580c;
37 }
38 .badge-disabled {
39     background-color: #374151;
40     color: #9ca3af;
41 }
42 </style>
43
44 <div class="winadmin-container theme-sysadmin">
45     <!-- Header Alert & Status (Compact Strip) -->
46     <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 text-white" style="background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%); border: 1px solid #334155;">
47         <div class="d-flex align-items-center gap-2 flex-wrap">
48             <span class="fs-5">❏ </span>
49             <span class="fw-bold text-white small" style="font-size: 0.92rem;">Windows System Administrator</span>
50             <span class="badge bg-dark border border-secondary text-info fw-normal" style="font-size: 0.7rem;">AD & Security Auditing</span>
51             <span class="text-secondary small d-none d-lg-inline" style="font-size: 0.76rem;">| Учетные записи, Скрытые пользователи & Телеметрия</span>
52         </div>
53         <div class="d-flex align-items-center gap-1.5 flex-wrap">
54             <span class="badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1" style="font-size: 0.72rem;" id="winadmin-status-badge">
55                 • Готов
56             </span>
57             <button class="btn btn-xs btn-outline-info rounded-pill px-2.5 py-1" style="font-size: 0.75rem;" id="btn-winadmin-refresh">
58                 <i class="bi bi-arrow-clockwise me-1"></i> Обновить
59             </button>
60             <button class="btn btn-xs btn-outline-warning rounded-pill px-2 py-1" style="font-size: 0.75rem;" id="btn-winadmin-con

```

```

    fig">
61         <i class="bi bi-gear-fill me-1"></i> Настройки
62     </button>
63 </div>
64 </div>
65
66 <!-- Top Metrics Bar (Compact Density Grid) -->
67 <div class="row g-2 mb-2">
68     <div class="col-6 col-md-3">
69         <div class="winadmin-card">
70             <span class="winadmin-label">Хост / Домен</span>
71             <div class="d-flex align-items-baseline justify-content-between mt-1">
72                 <span class="winadmin-value text-truncate" id="winadmin-hostname" style="font-size: 0.95rem;">-</span>
73             </div>
74         </div>
75     </div>
76     <div class="col-6 col-md-3">
77         <div class="winadmin-card">
78             <span class="winadmin-label">Active Directory</span>
79             <div class="d-flex align-items-baseline justify-content-between mt-1">
80                 <span class="winadmin-value text-info text-truncate" id="winadmin-ad-status" style="font-size: 0.9rem;">Локальный хост</span>
81             </div>
82         </div>
83     </div>
84     <div class="col-6 col-md-3">
85         <div class="winadmin-card">
86             <span class="winadmin-label">Пользователи (Все / Онлайн)</span>
87             <div class="d-flex align-items-baseline justify-content-between mt-1">
88                 <span class="winadmin-value text-success" id="winadmin-user-count">0</span>
89                 <span class="badge bg-dark border border-secondary t

```



```

    ext-muted small" id="winadmin-user-breakdown" style="font-size
: 0.7rem;">0 скрытых</span>
90     </div>
91     </div>
92 </div>
93 <div class="col-6 col-md-3">
94     <div class="winadmin-card">
95         <span class="winadmin-label">События безопасности</spa
n>
96         <div class="d-flex align-items-baseline justify-conten
t-between mt-1">
97             <span class="winadmin-value text-warning" id="winadm
in-events-count">0</span>
98         </div>
99     </div>
100 </div>
101 </div>
102
103 <!-- Main Panels Row 1: Users & Security Log -->
104 <div class="row g-3 mb-3">
105     <!-- User Accounts & Sessions Panel -->
106     <div class="col-lg-7">
107         <div class="winadmin-panel h-100">
108             <div class="winadmin-panel-header d-flex justify-conte
nt-between align-items-center flex-wrap gap-2">
109                 <div class="d-flex align-items-center gap-2">
110                     <span><i class="bi bi-people-fill me-1"></i> Учетн
ые записи и сессии Windows</span>
111                     <span class="badge bg-secondary" id="winadmin-user
s-badge">0 аккаунтов</span>
112                 </div>
113                 <!-- Filter Buttons Toolbar -->
114                 <div class="btn-group btn-group-sm" role="group" id=
"winadmin-user-filters">
115                     <button type="button" class="btn btn-outline-secon
dary py-0 px-2 winadmin-filter-btn active" data-filter="all">В
сех</button>
116                     <button type="button" class="btn btn-outline-secon

```

```

    dary py-0 px-2 winadmin-filter-btn" data-filter="active">В сет
и</button>
117         <button type="button" class="btn btn-outline-secon
dary py-0 px-2 winadmin-filter-btn" data-filter="hidden">Скрыт
ые</button>
118         <button type="button" class="btn btn-outline-secon
dary py-0 px-2 winadmin-filter-btn" data-filter="admins">Админ
ы</button>
119     </div>
120 </div>
121     <div class="p-0 table-responsive" style="max-height: 3
80px; overflow-y: auto;">
122         <table class="winadmin-table">
123             <thead>
124                 <tr>
125                     <th>Учетная запись</th>
126                     <th>Группы / SID</th>
127                     <th>Профиль / Диск</th>
128                     <th>Ресурсы</th>
129                     <th>Статус</th>
130                     <th style="text-align: right;">Действие</th>
131                 </tr>
132             </thead>
133             <tbody id="winadmin-users-tbody">
134                 <tr><td colspan="6" class="text-center text-mute
d p-3">Загрузка пользователей и метрик...</td></tr>
135             </tbody>
136         </table>
137     </div>
138 </div>
139 </div>
140
141 <!-- Security Events Audit Log Panel -->
142 <div class="col-lg-5">
143     <div class="winadmin-panel h-100">
144         <div class="winadmin-panel-header">
145             <span><i class="bi bi-shield-exclamation me-1"></i>
Общий журнал безопасности</span>

```

```
146 |         <span class="badge bg-secondary" id="winadmin-events
    | -badge">0 записей</span>
147 |     </div>
148 |     <div class="p-0 table-responsive" style="max-height: 3
    | 80px; overflow-y: auto;">
149 |         <table class="winadmin-table">
150 |             <thead>
151 |                 <tr>
152 |                     <th>Время</th>
153 |                     <th>Event ID</th>
154 |                     <th>Уровень</th>
155 |                     <th>Источник / Описание</th>
156 |                 </tr>
157 |             </thead>
158 |             <tbody id="winadmin-events-tbody">
159 |                 <tr><td colspan="4" class="text-center text-mute
    | d p-3">Загрузка журнала событий...</td></tr>
160 |             </tbody>
161 |         </table>
162 |     </div>
163 | </div>
164 | </div>
165 | </div>
166 | </div>
167 |
```

main.js

```
1 // =====
  =====
2 // Process Name: Windows System Administrator Web Tab Module
3 // =====
  =====
4 // Description:
5 //   Клиентский контроллер вкладки Windows System Administrator:
6 //   - Отображение всех локальных и доменных учетных записей Windows
7 //   - Обнаружение скрытых пользователей (SpecialAccounts User List)
8 //   - Подробная телеметрия ресурсов (память, CPU, процессы, профили на диске)
9 //   - Фильтрация (Все, В сети, Скрытые, Администраторы)
10 //   - Интерактивное досье с поддержкой AI-аудита прав и безопасности
11 //
12 // File: main.js
13 // Project: ai-breadboard
14 // Package: src.api.webgui.windows_admin_tab
15 // Author: hypo69
16 // Copyright: © 2026 hypo69
17 // =====
  =====
18
19 (function() {
20   let isWinAdminInitialized = false;
21   let currentUserFilter = 'all';
22
23   async function fetchStatus() {
24     try {
25       const res = await fetch('/api/sysadmin/status');
26       if (!res.ok) return;
27       const data = await res.json();
28
```

```
29     const hostEl = document.getElementById('winadmin-hostnam
    e');
30     const userCountEl = document.getElementById('winadmin-us
    er-count');
31     const userBreakdownEl = document.getElementById('winadmi
    n-user-breakdown');
32     const adStatusEl = document.getElementById('winadmin-ad-
    status');
33     const eventsCountEl = document.getElementById('winadmin-
    events-count');
34
35     if (hostEl) hostEl.innerText = `${data.hostname || 'LOCA
    L'} / ${data.domain || 'WORKGROUP'}`;
36     if (userCountEl) {
37         userCountEl.innerText = `${data.total_accounts || data
    .user_count || 0} (${data.active_users || 0} онлайн)`;
38     }
39     if (userBreakdownEl) {
40         userBreakdownEl.innerText = `${data.hidden_users || 0}
    скрытых`;
41     }
42     if (adStatusEl) adStatusEl.innerText = data.ad_connected
    ? 'Подключен' : 'Локальный хост';
43     if (eventsCountEl) eventsCountEl.innerText = data.event_
    count || 0;
44 } catch (e) {
45     console.error('[WinAdminTab] Failed to fetch status:', e
    );
46 }
47 }
48
49 async function fetchAccounts(filterType = currentUserFilter)
    {
50     try {
51         const res = await fetch(`/api/sysadmin/accounts?filter_t
    ype=${encodeURIComponent(filterType)}`);
52         if (!res.ok) return;
53         const data = await res.json();
```

```

54 |     const accounts = data.accounts || [];
55 |
56 |     const tbody = document.getElementById('winadmin-users-tb
    ody');
57 |     const badge = document.getElementById('winadmin-users-ba
    dge');
58 |     if (badge) badge.innerText = `${accounts.length} аккаунт
    ов (${data.total || accounts.length} всего)`;
59 |
60 |     if (tbody) {
61 |         if (accounts.length === 0) {
62 |             tbody.innerHTML = '<tr><td colspan="6" class="text-c
    enter text-muted p-3">Учетные записи по выбранному фильтру не
    найдены</td></tr>';
63 |             return;
64 |         }
65 |
66 |         tbody.innerHTML = accounts.map((u, idx) => {
67 |             // Вычисление бейджей
68 |             const badges = [];
69 |             if (u.is_admin) badges.push('<span class="badge badg
    e-admin me-1">Admin</span>');
70 |             if (u.is_hidden) badges.push('<span class="badge bad
    ge-hidden me-1">Скрытый</span>');
71 |             if (!u.enabled) badges.push('<span class="badge badg
    e-disabled me-1">Отключен</span>');
72 |
73 |             // Статус
74 |             let statusBadge = '<span class="badge bg-secondary">
    Офлайн</span>';
75 |             if (u.is_logged_in) {
76 |                 statusBadge = '<span class="badge bg-success-subtl
    e text-success border border-success">● В сети</span>';
77 |             } else if (!u.enabled) {
78 |                 statusBadge = '<span class="badge bg-dark text-mut
    ed">Отключен</span>';
79 |             }
80 |

```

```

81 |         // Ресурсы
82 |         const resInfo = u.process_count > 0
83 |         ? `${u.process_count} проц
    | .</span> <span class="text-muted">(${u.memory_rss_mb} MB)</spa
    | n>`
84 |         : `>-</span>`;
85 |
86 |         // Профиль
87 |         const profInfo = u.profile_size_mb > 0
88 |         ? `

```

```

107 |         <div class="font-monospace text-muted small te
| xt-truncate" style="max-width: 130px;" title="\${u.sid}">\${u.si
| d.slice(-10) || '-'}</div>
108 |         </td>
109 |         <td class="small">
110 |             \${profInfo}
111 |         </td>
112 |         <td class="small font-monospace">
113 |             \${resInfo}
114 |         </td>
115 |         <td>\${statusBadge}</td>
116 |         <td style="text-align: right;">
117 |             <button class="btn btn-xs btn-outline-info py-
| 0 px-2 btn-inspect-user me-1" data-user="\${u.name}" title="Дос
| ье и метрики">
118 |                 Досье
119 |             </button>
120 |             \${u.is_logged_in ? `
121 |                 <button class="btn btn-xs btn-outline-danger
| py-0 px-1.5 btn-disconnect-user" data-user="\${u.name}" title=
| "Отключить сессию">
122 |                     Выход
123 |                 </button>
124 |             ` : ''}
125 |         </td>
126 |     </tr>
127 | `;
128 | }).join('');
129 |
130 | // Клик по строке для открытия досье
131 | tbody.querySelectorAll('.winadmin-user-row').forEach(r
| ow => {
132 |     row.onclick = (evt) => {
133 |         if (evt.target.closest('.btn-disconnect-user')) re
| turn;
134 |         const idx = parseInt(row.getAttribute('data-idx'),
| 10);
135 |         const u = accounts[idx];

```



```

136         if (!u) return;
137
138         openUserDossier(u);
139     };
140 });
141
142     // Кнопки "Досье"
143     tbody.querySelectorAll('.btn-inspect-user').forEach(bt
n => {
144         btn.onclick = (evt) => {
145             evt.stopPropagation();
146             const username = btn.getAttribute('data-user');
147             const u = accounts.find(a => a.name === username);
148             if (u) openUserDossier(u);
149         };
150     });
151
152     // Кнопки "Выход"
153     tbody.querySelectorAll('.btn-disconnect-user').forEach
(btn => {
154         btn.onclick = async (evt) => {
155             evt.stopPropagation();
156             const user = btn.getAttribute('data-user');
157             if (confirm(`Отключить сессию пользователя ${user}
?`)) {
158                 await fetch(`/api/sysadmin/users/${user}/disconn
ect`, { method: 'POST' });
159                 fetchAccounts();
160                 fetchStatus();
161             }
162         };
163     });
164 }
165 } catch (e) {
166     console.error('[WinAdminTab] Failed to fetch accounts:',
e);
167 }
168 }

```

```
169
170 function openUserDossier(u) {
171     if (!window.AITableModal) return;
172
173     const metadata = [
174         { label: 'Имя пользователя', value: u.name },
175         { label: 'Полное имя', value: u.full_name || 'Не указано' },
176         { label: 'Описание', value: u.description || 'Отсутствует' },
177         { label: 'SID пользователя', value: u.sid || 'N/A' },
178         { label: 'Тип аккаунта', value: `${u.account_type} (${u.is_admin ? 'Администратор' : 'Обычный'})` },
179         { label: 'Скрытая учетная запись', value: u.is_hidden ? 'Да (SpecialAccounts / Built-in)' : 'Нет' },
180         { label: 'Учетная запись включена', value: u.enabled ? 'Да' : 'Отключена' },
181         { label: 'Членство в группах', value: (u.groups && u.groups.length > 0) ? u.groups.join(', ') : 'Users' },
182         { label: 'Путь к профилю', value: u.profile_path || 'Не создан' },
183         { label: 'Размер профиля на диске', value: `${u.profile_size_mb} MB` },
184         { label: 'Последняя смена пароля', value: u.password_last_set || 'N/A' },
185         { label: 'Неверных попыток пароля', value: String(u.bad_password_count || 0) },
186         { label: 'Последний вход в систему', value: u.last_logon || 'Не зафиксирован' },
187         { label: 'Активных процессов', value: `${u.process_count} (RAM: ${u.memory_rss_mb} MB, CPU: ${u.cpu_percent}%)` }
188     ];
189
190     const badges = [
191         { text: u.is_logged_in ? 'Online' : 'Offline', class: u.is_logged_in ? 'badge bg-success' : 'badge bg-secondary' }
192     ];
193     if (u.is_admin) badges.push({ text: 'Администратор', class
```

```

: 'badge bg-warning text-dark' });
194     if (u.is_hidden) badges.push({ text: 'Скрытый', class: 'ba
dge bg-purple text-white' });
195     if (!u.enabled) badges.push({ text: 'Отключен', class: 'ba
dge bg-danger' });
196
197     window.AITableModal.show({
198         icon: '🔒',
199         title: `Учетная запись: ${u.name}`,
200         subtitle: `${u.full_name ? u.full_name + ' | ' : ''}SID:
${u.sid}`,
201         tableType: 'user_account',
202         badges: badges,
203         metadata: metadata,
204         rawTitle: 'Полная конфигурация и метрики пользователя',
205         rawContent: JSON.stringify(u, null, 2),
206         requestData: {
207             username: u.name,
208             is_admin: u.is_admin,
209             is_hidden: u.is_hidden,
210             groups: u.groups,
211             process_count: u.process_count,
212             memory_rss_mb: u.memory_rss_mb,
213             profile_path: u.profile_path
214         }
215     });
216 }
217
218 async function fetchEvents() {
219     try {
220         const res = await fetch('/api/sysadmin/events?hours=24')
221         ;
222         if (!res.ok) return;
223         const data = await res.json();
224         const events = data.events || [];
225
226         const tbody = document.getElementById('winadmin-events-t
body');

```

```

226 |     const badge = document.getElementById('winadmin-events-b
| adge');
227 |     if (badge) badge.innerText = `${events.length} записей`;
228 |
229 |     if (tbody) {
230 |         if (events.length === 0) {
231 |             tbody.innerHTML = '<tr><td colspan="4" class="text-c
| enter text-muted p-3">События не зафиксированы</td></tr>';
232 |             return;
233 |         }
234 |         const displayEvents = events.slice(0, 30);
235 |         tbody.innerHTML = displayEvents.map((e, idx) => `
236 |             <tr class="winadmin-event-row" data-idx="${idx}" sty
| le="cursor: pointer;" title="Нажмите для AI-диагностики событи
| я">
237 |                 <td class="font-monospace text-muted small">${e.ti
| mestamp?.slice(11, 19) || ''}</td>
238 |                 <td class="font-monospace fw-semibold text-info">$
| {e.event_id}</td>
239 |                 <td><span class="badge ${e.level === 'Warning' ? '
| bg-warning-subtle text-warning' : (e.level === 'Error' ? 'bg-d
| anger-subtle text-danger' : 'bg-info-subtle text-info')}">${e.
| level}</span></td>
240 |                 <td class="small text-light">${e.source}: ${e.desc
| ription}</td>
241 |             </tr>
242 |         `).join('');
243 |
244 |         tbody.querySelectorAll('.winadmin-event-row').forEach(
| row => {
245 |             row.onclick = () => {
246 |                 const idx = parseInt(row.getAttribute('data-idx'),
| 10);
247 |                 const e = displayEvents[idx];
248 |                 if (!e) return;
249 |
250 |                 if (window.AITableModal) {
251 |                     window.AITableModal.show({

```

```

252         icon: '□ ',
253         title: `Событие #${e.event_id} (${e.source})`,
254         subtitle: `Уровень: ${e.level} | ${e.timestamp}
    || ''}`,
255         tableType: 'generic',
256         badges: [
257             { text: e.level || 'Info', class: 'badge bg-
info text-dark' }
258         ],
259         metadata: [
260             { label: 'Event ID', value: String(e.event_id) },
261             { label: 'Источник / Provider', value: e.source },
262             { label: 'Уровень важности', value: e.level },
263             { label: 'Время регистрации', value: e.timestamp }
264         ],
265         rawTitle: 'Текст сообщения журнала',
266         rawContent: e.description || '',
267         requestData: {
268             event_id: e.event_id,
269             source: e.source,
270             level: e.level,
271             description: e.description
272         }
273     });
274 }
275 };
276 });
277 }
278 } catch (e) {
279     console.error('[WinAdminTab] Failed to fetch events:', e
);
280 }
281 }
282

```

```
283 | function initWindowsAdminTab() {
284 |     console.log('[WinAdminTab] Initializing Windows Sysadmin U
    | ser Metrics & Security tab...');
285 |     fetchStatus();
286 |     fetchAccounts('all');
287 |     fetchEvents();
288 |
289 |     if (!isWinAdminInitialized) {
290 |         const refreshBtn = document.getElementById('btn-winadmin-
    | -refresh');
291 |         const configBtn = document.getElementById('btn-winadmin-
    | config');
292 |
293 |         if (refreshBtn) {
294 |             refreshBtn.onclick = () => {
295 |                 fetchStatus();
296 |                 fetchAccounts(currentUserFilter);
297 |                 fetchEvents();
298 |             };
299 |         }
300 |
301 |         // Фильтры пользователей
302 |         const filterGroup = document.getElementById('winadmin-us
    | er-filters');
303 |         if (filterGroup) {
304 |             filterGroup.querySelectorAll('.winadmin-filter-btn').f
    | orEach(btn => {
305 |                 btn.onclick = () => {
306 |                     filterGroup.querySelectorAll('.winadmin-filter-btn
    | ').forEach(b => b.classList.remove('active'));
307 |                     btn.classList.add('active');
308 |                     currentUserFilter = btn.getAttribute('data-filter'
    | ) || 'all';
309 |                     fetchAccounts(currentUserFilter);
310 |                 };
311 |             });
312 |         }
313 |     }
```

```
314 |         if (configBtn) {
315 |             configBtn.onclick = () => {
316 |                 if (typeof window.openAppConfigModal === 'function')
317 |                 {
318 |                     window.openAppConfigModal('windows_sysadmin', 'Win
319 | dows System Administrator');
320 |                 }
321 |             };
322 |         }
323 |     }
324 | }
325 |
326 | window.initWindowsAdminTab = initWindowsAdminTab;
327 | })();
328 |
```

index.html

```
1 <!-- Windows Backup & Libraries Manager Tab -->
2 <style>
3   .wb-container {
4     padding: 0.5rem 0.75rem;
5   }
6   .wb-card {
7     background: #0f172a !important;
8     border: 1px solid #334155 !important;
9     border-radius: 8px;
10    padding: 0.45rem 0.75rem;
11    display: flex;
12    flex-direction: column;
13    justify-content: space-between;
14    height: 100%;
15    min-height: 58px;
16    transition: border-color 0.15s ease, box-shadow 0.15s ease
17    ;
18  }
19  .wb-card:hover {
20    border-color: #38bdf8 !important;
21    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
22  }
23  .wb-card-label {
24    font-size: 0.68rem;
25    text-transform: uppercase;
26    letter-spacing: 0.05em;
27    color: #94a3b8;
28    font-weight: 600;
29    display: flex;
30    justify-content: space-between;
31    align-items: center;
32    line-height: 1.1;
33  }
34  .wb-card-value {
35    font-size: 1.25rem;
36    font-weight: 700;
```



```
36     font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,  
    Consolas, monospace;  
37     line-height: 1.15;  
38 }  
39 .wb-card-sub {  
40     font-size: 0.7rem;  
41     color: #64748b;  
42     line-height: 1.1;  
43 }  
44  
45 .wb-panel {  
46     background: #0f172a !important;  
47     border: 1px solid #334155 !important;  
48     border-radius: 8px;  
49     overflow: hidden;  
50     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.2);  
51     margin-bottom: 0.75rem;  
52 }  
53 .wb-panel-header {  
54     background: #1e293b !important;  
55     border-bottom: 1px solid #334155 !important;  
56     padding: 0.45rem 0.85rem;  
57     font-weight: 600;  
58     font-size: 0.82rem;  
59     color: #38bdf8;  
60     display: flex;  
61     align-items: center;  
62     justify-content: space-between;  
63 }  
64 .wb-table {  
65     margin-bottom: 0;  
66     font-size: 0.8rem;  
67 }  
68 .wb-table th {  
69     background: #1e293b;  
70     color: #94a3b8;  
71     font-weight: 600;  
72     font-size: 0.72rem;
```

```

73     text-transform: uppercase;
74     border-bottom: 1px solid #334155;
75     padding: 0.4rem 0.6rem;
76 }
77 .wb-table td {
78     border-bottom: 1px solid #1e293b;
79     padding: 0.45rem 0.6rem;
80     vertical-align: middle;
81 }
82 </style>
83
84 <div class="wb-container">
85     <!-- Header Bar -->
86     <div class="d-flex align-items-center justify-content-between mb-2 flex-wrap gap-2">
87         <div>
88             <h6 class="mb-0 fw-bold text-white d-flex align-items-center gap-2">
89                 <span>⏏ </span> Резервное копирование и управление бэкапами Windows
90             </h6>
91             <div class="small text-muted" style="font-size: 0.72rem; ">
92                 Настройка бэкапов, аудит библиотек (.library-ms), служба File History (fhsvc), мониторинг изменений/удаления файлов и RAG-поиск по архивам
93             </div>
94         </div>
95         <div class="d-flex align-items-center gap-2">
96             <button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1" id="wb-btn-open-create-lib" title="Создать новую системную библиотеку Windows">
97                 <i class="bi bi-folder-plus"></i>
98                 <span>Создать библиотеку</span>
99             </button>
100             <button class="btn btn-sm btn-primary d-flex align-items-center gap-1" id="wb-btn-trigger-backup" title="Принудительный запуск архивации через fhexec">

```

```

101 |         <i class="bi bi-play-fill"></i>
102 |         <span>Запустить бэкап</span>
103 |     </button>
104 |     <button class="btn btn-sm btn-outline-secondary d-flex a
|     lign-items-center gap-1" id="wb-btn-refresh">
105 |         <i class="bi bi-arrow-clockwise"></i>
106 |         <span>Обновить</span>
107 |     </button>
108 | </div>
109 | </div>
110 |
111 | <!-- KPI Stats Row -->
112 | <div class="row g-2 mb-2">
113 |     <div class="col-6 col-md-3">
114 |         <div class="wb-card">
115 |             <div class="wb-card-label">
116 |                 <span>Health Score</span>
117 |                 <span class="badge bg-secondary" id="wb-badge-status
|                 ">-</span>
118 |             </div>
119 |             <div class="wb-card-value text-info" id="wb-val-health
|             ">- / 100</div>
120 |             <div class="wb-card-sub" id="wb-sub-health">Готовность
|             защиты данных</div>
121 |         </div>
122 |     </div>
123 |     <div class="col-6 col-md-3">
124 |         <div class="wb-card">
125 |             <div class="wb-card-label">
126 |                 <span>Служба File History</span>
127 |                 <span>□ </span>
128 |             </div>
129 |             <div class="wb-card-value text-white" id="wb-val-servi
|             ce">-</div>
130 |             <div class="wb-card-sub text-truncate" id="wb-sub-serv
|             ice">fhsvc статус</div>
131 |         </div>
132 |     </div>

```

```

133 |     <div class="col-6 col-md-3">
134 |         <div class="wb-card">
135 |             <div class="wb-card-label">
136 |                 <span>Библиотеки / VSS</span>
137 |                 <span>□ </span>
138 |             </div>
139 |             <div class="wb-card-value text-success" id="wb-val-lib
raries">-</div>
140 |             <div class="wb-card-sub" id="wb-sub-libraries">Библиот
ек / VSS снимков</div>
141 |         </div>
142 |     </div>
143 |     <div class="col-6 col-md-3">
144 |         <div class="wb-card">
145 |             <div class="wb-card-label">
146 |                 <span>Аудит удаления (24ч)</span>
147 |                 <button class="btn btn-xs btn-outline-primary py-0 p
x-1" id="wb-btn-enable-auditpol" title="Включить auditpol /sub
category:File System" style="font-size: 0.65rem;">Вкл</button>
148 |             </div>
149 |             <div class="wb-card-value text-danger" id="wb-val-dele
tions">0</div>
150 |             <div class="wb-card-sub" id="wb-sub-deletions">Event 4
663 / 4660</div>
151 |         </div>
152 |     </div>
153 | </div>
154 |
155 | <!-- Main Content Panels -->
156 | <div class="row g-2">
157 |     <!-- Left Column: Libraries & Folders -->
158 |     <div class="col-12 col-lg-7">
159 |         <div class="wb-panel">
160 |             <div class="wb-panel-header">
161 |                 <span class="d-flex align-items-center gap-1.5">
162 |                     <i class="bi bi-folder-symlink text-info"></i>
163 |                     <span>Системные библиотеки Windows (.library-ms)</
span>

```

```

164 |         </span>
165 |         <div class="d-flex align-items-center gap-2">
166 |             <button class="btn btn-xs btn-outline-info py-0 px
| -2" id="wb-btn-quick-add-folder" style="font-size: 0.72rem;" t
| itle="Добавить папку в существующую библиотеку">
167 |                 <i class="bi bi-plus-circle me-1"></i>Добавить п
| апку
168 |             </button>
169 |             <span class="badge bg-dark border border-secondary
| text-muted" id="wb-lib-count">0 шт.</span>
170 |         </div>
171 |     </div>
172 |     <div class="table-responsive" style="max-height: 240px
| ; overflow-y: auto;">
173 |         <table class="table table-dark table-hover wb-table"
| >
174 |             <thead>
175 |                 <tr>
176 |                     <th>Библиотека</th>
177 |                     <th>Папки / Источники</th>
178 |                     <th>Папка сохранения</th>
179 |                     <th>Закреплена</th>
180 |                 </tr>
181 |             </thead>
182 |             <tbody id="wb-libraries-body">
183 |                 <tr>
184 |                     <td colspan="4" class="text-center py-3 text-m
| uted">Загрузка библиотек...</td>
185 |                 </tr>
186 |             </tbody>
187 |         </table>
188 |     </div>
189 | </div>
190 |
191 | <!-- VSS Snapshots -->
192 | <div class="wb-panel">
193 |     <div class="wb-panel-header">
194 |         <span class="d-flex align-items-center gap-1.5">

```

```

195 |         <i class="bi bi-camera text-warning"></i>
196 |         <span>Теневые копии томов VSS (Volume Shadow Copie
    | s)</span>
197 |     </span>
198 |     <span class="badge bg-dark border border-secondary t
    | ext-muted" id="wb-vss-count">0 шт.</span>
199 | </div>
200 | <div class="table-responsive" style="max-height: 180px
    | ; overflow-y: auto;">
201 |     <table class="table table-dark table-hover wb-table"
    | >
202 |         <thead>
203 |             <tr>
204 |                 <th>ID снимка</th>
205 |                 <th>Исходный том</th>
206 |                 <th>Время создания</th>
207 |                 <th>Состояние</th>
208 |             </tr>
209 |         </thead>
210 |         <tbody id="wb-vss-body">
211 |             <tr>
212 |                 <td colspan="4" class="text-center py-3 text-m
    | uted">Загрузка VSS снимков...</td>
213 |             </tr>
214 |         </tbody>
215 |     </table>
216 | </div>
217 | </div>
218 | </div>
219 |
220 | <!-- Right Column: File History Config & Health & RAG Sear
    | ch -->
221 | <div class="col-12 col-lg-5">
222 |     <!-- File History Config & Target -->
223 |     <div class="wb-panel">
224 |         <div class="wb-panel-header">
225 |             <span class="d-flex align-items-center gap-1.5">
226 |                 <i class="bi bi-clock-history text-primary"></i>

```

```
227 |         <span>Настройка File History (История файлов)</spa
    | n>
228 |         </span>
229 |     </div>
230 |     <div class="p-2.5" style="font-size: 0.8rem;" id="wb-f
    | h-details">
231 |         <div class="mb-2">
232 |             <div class="text-muted small">Целевой каталог бэка
    | па:</div>
233 |             <div class="fw-semibold text-break font-monospace
    | small text-info" id="wb-fh-target">Не настроено</div>
234 |         </div>
235 |         <div class="mb-2">
236 |             <div class="text-muted small">Файл конфигурации:</
    | div>
237 |             <div class="text-break font-monospace small text-m
    | uted" id="wb-fh-config">-</div>
238 |         </div>
239 |         <div class="row g-2 mb-1">
240 |             <div class="col-6">
241 |                 <div class="text-muted small">Интервал архивации
    | :</div>
242 |                 <div class="fw-bold text-white" id="wb-fh-interv
    | al">-</div>
243 |             </div>
244 |             <div class="col-6">
245 |                 <div class="text-muted small">Срок хранения верс
    | ий:</div>
246 |                 <div class="fw-bold text-white" id="wb-fh-retent
    | ion">-</div>
247 |             </div>
248 |         </div>
249 |         <div class="mt-2 pt-2 border-top border-secondary">
250 |             <div class="text-muted small">Последняя успешная а
    | рхивация:</div>
251 |             <div class="fw-semibold text-warning" id="wb-fh-la
    | st-backup">-</div>
252 |         </div>
```

```

253         </div>
254     </div>
255
256     <!-- RAG Search in File History Archive -->
257     <div class="wb-panel">
258         <div class="wb-panel-header">
259             <span class="d-flex align-items-center gap-1.5">
260                 <i class="bi bi-search-heart text-info"></i>
261                 <span>RAG поиск по архивам бэкапа</span>
262             </span>
263             <button class="btn btn-xs btn-outline-primary py-0 p
x-2" id="wb-btn-rag-sync" title="Индексировать новые версии фа
йлов" style="font-size: 0.72rem;">
264                 <i class="bi bi-arrow-repeat me-1"></i>Синхронизац
ия
265             </button>
266         </div>
267         <div class="p-2" style="font-size: 0.8rem;">
268             <div class="input-group input-group-sm mb-2">
269                 <input type="text" class="form-control bg-dark tex
t-white border-secondary" id="wb-rag-query" placeholder="Поиск
документа или фрагмента текста..." />
270                 <button class="btn btn-primary" id="wb-btn-rag-sea
rch"><i class="bi bi-search"></i></button>
271             </div>
272             <div id="wb-rag-results" style="max-height: 140px; o
verflow-y: auto;">
273                 <div class="text-muted text-center py-2 small">Вве
дите запрос для поиска по архивным копиям</div>
274             </div>
275         </div>
276     </div>
277
278     <!-- Recommendations and Health Checklist -->
279     <div class="wb-panel">
280         <div class="wb-panel-header">
281             <span class="d-flex align-items-center gap-1.5">
282                 <i class="bi bi-shield-check text-success"></i>

```



```

283 |         <span>Рекомендации безопасности данных</span>
284 |     </span>
285 | </div>
286 |     <div class="p-2.5" style="max-height: 150px; overflow-
| y: auto;">
287 |         <ul class="list-unstyled mb-0" id="wb-recommendation
| s-list" style="font-size: 0.78rem;">
288 |             <li class="text-muted text-center py-2">Загрузка р
| екомендаций...</li>
289 |         </ul>
290 |     </div>
291 | </div>
292 | </div>
293 | </div>
294 |
295 | <!-- Row: User Shell Folders Storage & Relocation Management
| -->
296 | <div class="row g-2 mt-1">
297 |     <div class="col-12">
298 |         <div class="wb-panel">
299 |             <div class="wb-panel-header">
300 |                 <div class="d-flex align-items-center gap-2">
301 |                     <span class="d-flex align-items-center gap-1.5">
302 |                         <i class="bi bi-hdd-network-fill text-info"></i>
303 |                         <span>Пользовательские директории и перенос на в
| торичные диски</span>
304 |                     </span>
305 |                 </div>
306 |                 <div class="d-flex align-items-center gap-2">
307 |                     <span class="badge bg-primary-subtle text-primary
| border border-primary-subtle" id="wb-total-user-size">Объем: 0
| ГБ</span>
308 |                     <button class="btn btn-xs btn-outline-info py-0 px
| -2" id="wb-btn-refresh-user-folders" title="Пересчитать размер
| пользовательских папок" style="font-size: 0.72rem;">
309 |                         <i class="bi bi-arrow-repeat me-1"></i> Пересчит
| ать
310 |                     </button>

```

```

311         </div>
312     </div>
313
314     <div class="p-2">
315         <!-- Disks summary row -->
316         <div class="mb-2 p-2 rounded bg-dark border border-s
econdary" style="font-size: 0.76rem;" id="wb-drives-summary">
317             <div class="text-muted small mb-1 fw-bold">Обна
руженные физические накопители:</div>
318             <div class="d-flex flex-wrap gap-2" id="wb-drives-
list">
319                 <span class="text-muted">Загрузка информации о д
исках...</span>
320             </div>
321         </div>
322
323         <div class="table-responsive" style="max-height: 260
px; overflow-y: auto;">
324             <table class="table table-dark table-hover wb-tabl
e">
325                 <thead>
326                     <tr>
327                         <th>Директория</th>
328                         <th>Текущий путь</th>
329                         <th>Диск</th>
330                         <th>Размер</th>
331                         <th>Файлов</th>
332                         <th class="text-end">Действие</th>
333                     </tr>
334                 </thead>
335                 <tbody id="wb-user-folders-tbody">
336                     <tr>
337                         <td colspan="6" class="text-center text-mute
d p-3">Подсчет размера пользовательских папок...</td>
338                     </tr>
339                 </tbody>
340             </table>
341         </div>

```

```

342         </div>
343     </div>
344 </div>
345 </div>
346
347 <!-- Row 2: File Deletion Auditing (Event 4663 / 4660) -->
348 <div class="row g-2 mt-1">
349     <div class="col-12">
350         <div class="wb-panel mb-0">
351             <div class="wb-panel-header">
352                 <div class="d-flex align-items-center gap-2">
353                     <span class="d-flex align-items-center gap-1.5">
354                         <i class="bi bi-trash3-fill text-danger"></i>
355                         <span>Аудит изменения и удаления файлов (Event 4
356                         663 / 4660)</span>
357                     </span>
358                 </div>
359                 <div class="d-flex align-items-center gap-2">
360                     <button class="btn btn-xs btn-outline-info py-0 px
361                     -2" id="wb-btn-configure-sacl" title="Настроить SACL Delete дл
362                     я папки" style="font-size: 0.72rem;">
363                         <i class="bi bi-folder-check me-1"></i> Настроит
364                         ь SACL папки
365                     </button>
366                     <span class="badge bg-danger-subtle text-danger" i
367                     d="wb-deletions-badge">0 удалений</span>
368                 </div>
369             </div>
370             <div class="p-0 table-responsive" style="max-height: 2
371             80px; overflow-y: auto;">
372                 <table class="table table-dark table-hover wb-table"

```

```

373         <th>Пользователь</th>
374     </tr>
375 </thead>
376     <tbody id="wb-deletions-tbody">
377         <tr><td colspan="5" class="text-center text-mute
d p-3">Поиск событий удаления файлов...</td></tr>
378     </tbody>
379 </table>
380 </div>
381 </div>
382 </div>
383 </div>
384
385 <!-- Modal for Relocating Folder -->
386 <div class="modal fade" id="wb-relocate-modal" tabindex="-1"
    aria-hidden="true">
387     <div class="modal-dialog modal-dialog-centered">
388         <div class="modal-content bg-dark text-white border-seco
ndary">
389             <div class="modal-header border-secondary py-2">
390                 <h6 class="modal-title fw-bold text-info d-flex align-items-center gap-2">
391                     <i class="bi bi-box-arrow-right"></i>
392                     <span>Перенос пользовательской директории</span>
393                 </h6>
394                 <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
395             </div>
396             <div class="modal-body" style="font-size: 0.85rem;">
397                 <input type="hidden" id="wb-relocate-folder-id" />
398                 <div class="mb-2">
399                     <label class="text-muted small">Переносимая папка:
    </label>
400                     <div class="fw-bold text-white fs-6" id="wb-relocate-folder-name">-</div>
401                     <div class="font-monospace small text-muted text-break" id="wb-relocate-folder-path">-</div>
402                 </div>

```

```

403         <div class="mb-2">
404             <label class="text-muted small">Текущий размер:</l
abel>
405             <div class="fw-semibold text-warning" id="wb-reloc
ate-folder-size">-</div>
406         </div>
407         <div class="mb-2">
408             <label class="text-muted small mb-1" for="wb-reloc
ate-target-drive">Целевой диск для переноса:</label>
409             <select class="form-select form-select-sm bg-secon
dary text-white border-0" id="wb-relocate-target-drive">
410                 <option value="">Выберите диск с достаточным мес
том...</option>
411             </select>
412             <div class="form-text text-muted small" id="wb-rel
ocate-drive-hint"></div>
413         </div>
414         <div class="mb-3">
415             <label class="text-muted small mb-1" for="wb-reloc
ate-target-path">Целевая директория (папка):</label>
416             <div class="input-group input-group-sm">
417                 <input type="text" class="form-control form-cont
rol-sm bg-secondary text-white border-0 font-monospace" id="wb
-relocate-target-path" placeholder="D:\Users\Username\Pictures
" />
418                 <button class="btn btn-outline-info d-flex align
-items-center gap-1" type="button" id="wb-btn-browse-folder" t
itle="Выбрать целевую папку через системный проводник Windows"
>
419                     <i class="bi bi-folder2-open"></i>
420                     <span>Обзор...</span>
421                 </button>
422             </div>
423             <div class="form-text text-muted small">Можно выбр
ать готовую папку через проводник или отредактировать путь вру
чную.</div>
424         </div>
425         <div class="form-check mb-2">

```

```

426         <input class="form-check-input" type="checkbox" id
      | ="wb-relocate-delete-source" />
427         <label class="form-check-label text-white small" f
      | or="wb-relocate-delete-source">
428             Очистить исходную папку после успешного копирова
      | ния и перенастройки
429         </label>
430     </div>
431     <div class="alert alert-info py-2 px-3 small mb-0">
432         <i class="bi bi-info-circle me-1"></i>
433         Перенос скопирует файлы, обновит реестр Windows (U
      | ser Shell Folders) и системную библиотеку Windows, благодаря ч
      | ему служба File History и Проводник продолжат работать без сбо
      | ев.
434     </div>
435 </div>
436     <div class="modal-footer border-secondary py-1.5">
437         <button type="button" class="btn btn-sm btn-outline-
      | secondary" data-bs-dismiss="modal">Отмена</button>
438         <button type="button" class="btn btn-sm btn-primary
      | d-flex align-items-center gap-1" id="wb-btn-execute-relocate">
439             <i class="bi bi-check-circle"></i>
440             <span>Выполнить перенос</span>
441         </button>
442     </div>
443 </div>
444 </div>
445 </div>
446
447 <!-- Modal for Creating Library -->
448 <div class="modal fade" id="wb-create-lib-modal" tabindex="-
      | 1" aria-hidden="true">
449     <div class="modal-dialog modal-dialog-centered">
450         <div class="modal-content bg-dark text-white border-seco
      | ndary">
451             <div class="modal-header border-secondary py-2">
452                 <h6 class="modal-title fw-bold text-info d-flex alig
      | n-items-center gap-2">

```

```
453         <i class="bi bi-folder-plus"></i>
454         <span>Создать библиотеку Windows (.library-ms)</span>
455     </h6>
456     <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
457 </div>
458     <div class="modal-body" style="font-size: 0.85rem;">
459         <div class="mb-2">
460             <label class="text-muted small" for="wb-create-lib-name">Название библиотеки:</label>
461             <input type="text" class="form-control form-control-sm bg-secondary text-white border-0" id="wb-create-lib-name"
462                 placeholder="Например: Архивы Проектов" />
463         </div>
464         <div class="mb-2">
465             <label class="text-muted small" for="wb-create-lib-folders">Пути к папкам (по одной строке на папку):</label>
466             <textarea class="form-control form-control-sm bg-secondary text-white border-0 font-monospace" id="wb-create-lib-folders"
467                 rows="3" placeholder="C:\Data\Backups&#10;D:\Projects"></textarea>
468         </div>
469         <div class="form-check mb-2">
470             <input class="form-check-input" type="checkbox" id="wb-create-lib-pinned" checked />
471             <label class="form-check-label text-white small" for="wb-create-lib-pinned">
472                 Закрепить в панели навигации Проводника
473             </label>
474         </div>
475     </div>
476     <div class="modal-footer border-secondary py-1.5">
477         <button type="button" class="btn btn-sm btn-outline-secondary" data-bs-dismiss="modal">Отмена</button>
478         <button type="button" class="btn btn-sm btn-info d-flex align-items-center gap-1" id="wb-btn-execute-create-lib">
479             <i class="bi bi-check-lg"></i>
```

```

478         <span>Создать</span>
479     </button>
480 </div>
481 </div>
482 </div>
483 </div>
484
485 <!-- Modal for Adding Folder to Library -->
486 <div class="modal fade" id="wb-add-folder-modal" tabindex="-
1" aria-hidden="true">
487     <div class="modal-dialog modal-dialog-centered">
488         <div class="modal-content bg-dark text-white border-seco
ndary">
489             <div class="modal-header border-secondary py-2">
490                 <h6 class="modal-title fw-bold text-info d-flex align-items-center gap-2">
491                     <i class="bi bi-folder-plus"></i>
492                     <span>Добавить папку в библиотеку</span>
493                 </h6>
494                 <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal" aria-label="Close"></button>
495             </div>
496             <div class="modal-body" style="font-size: 0.85rem;">
497                 <div class="mb-2">
498                     <label class="text-muted small" for="wb-add-folder-lib-select">Выберите целевую библиотеку:</label>
499                     <select class="form-select form-select-sm bg-secondary text-white border-0" id="wb-add-folder-lib-select">
500                         <option value="">Загрузка библиотек...</option>
501                     </select>
502                 </div>
503                 <div class="mb-2">
504                     <label class="text-muted small" for="wb-add-folder-path">Полный путь к физической папке:</label>
505                     <input type="text" class="form-control form-control-sm bg-secondary text-white border-0 font-monospace" id="wb-add-folder-path" placeholder="D:\BackupFolder" />
506                 </div>

```



```
507 |         <div class="form-check mb-2">
508 |             <input class="form-check-input" type="checkbox" id
| ="wb-add-folder-default-save" />
509 |             <label class="form-check-label text-white small" f
| or="wb-add-folder-default-save">
510 |                 Сделать основной папкой сохранения по умолчанию
511 |             </label>
512 |         </div>
513 |     </div>
514 |     <div class="modal-footer border-secondary py-1.5">
515 |         <button type="button" class="btn btn-sm btn-outline-
| secondary" data-bs-dismiss="modal">Отмена</button>
516 |         <button type="button" class="btn btn-sm btn-primary
| d-flex align-items-center gap-1" id="wb-btn-execute-add-folder
| ">
517 |             <i class="bi bi-check-lg"></i>
518 |             <span>Добавить</span>
519 |         </button>
520 |     </div>
521 | </div>
522 | </div>
523 | </div>
524 | </div>
525 |
```

main.js

```
1 // Windows Backup & Libraries Manager Frontend Logic
2 (function () {
3   let _userFoldersData = null;
4   let _librariesData = [];
5
6   async function initWindowsBackupTab() {
7     console.log('[WindowsBackup] Initializing tab...');
8     bindEvents();
9     await loadBackupData();
10  }
11  window.initWindowsBackupTab = initWindowsBackupTab;
12
13  function bindEvents() {
14    const btnRefresh = document.getElementById('wb-btn-refresh
15  ');
16    if (btnRefresh) {
17      btnRefresh.onclick = () => loadBackupData();
18    }
19
20    const btnTrigger = document.getElementById('wb-btn-trigger
21  -backup');
22    if (btnTrigger) {
23      btnTrigger.onclick = async () => {
24        btnTrigger.disabled = true;
25        btnTrigger.innerHTML = '<span class="spinner-border sp
26  inner-border-sm me-1"></span> Запуск...';
27        try {
28          const res = await fetch('/api/v1/windows-backup/file
29  -history/trigger', { method: 'POST' });
30          const data = await res.json();
31          if (res.ok) {
32            alert('Архивация File History успешно запущена: '
33  + (data.message || 'OK'));
34          } else {
35            alert('Ошибка запуска: ' + (data.detail || JSON.st
36  ringify(data)));
37          }
38        } catch (error) {
39          console.error('Error during File History backup trigger:', error);
40        }
41      };
42    }
43  }
```

```
31 |         }
32 |     } catch (e) {
33 |         alert('Сетевая ошибка при запуске архивации: ' + e.m
message);
34 |     } finally {
35 |         btnTrigger.disabled = false;
36 |         btnTrigger.innerHTML = '<i class="bi bi-play-fill"><
/i> <span>Запустить бэкап</span>';
37 |         await loadBackupData();
38 |     }
39 | };
40 | }
41 |
42 | // Modal Create Library
43 | const btnOpenCreateLib = document.getElementById('wb-btn-o
pen-create-lib');
44 | if (btnOpenCreateLib) {
45 |     btnOpenCreateLib.onclick = () => {
46 |         const modalEl = document.getElementById('wb-create-lib
-modal');
47 |         if (modalEl) {
48 |             if (window.bootstrap?.Modal) {
49 |                 new window.bootstrap.Modal(modalEl).show();
50 |             } else {
51 |                 modalEl.classList.add('show');
52 |                 modalEl.style.display = 'block';
53 |             }
54 |         }
55 |     };
56 | }
57 |
58 | const btnExecCreateLib = document.getElementById('wb-btn-e
xecute-create-lib');
59 | if (btnExecCreateLib) {
60 |     btnExecCreateLib.onclick = async () => {
61 |         const name = document.getElementById('wb-create-lib-na
me')?.value?.trim();
62 |         const foldersRaw = document.getElementById('wb-create-
```

```

lib-folders')?.value || '';
63 |         const isPinned = document.getElementById('wb-create-li
lib-pinned')?.checked ?? true;
64 |
65 |         if (!name) {
66 |             alert('Укажите название библиотеки');
67 |             return;
68 |         }
69 |
70 |         const folders = foldersRaw
71 |             .split('\n')
72 |             .map(f => f.trim())
73 |             .filter(f => f.length > 0);
74 |
75 |         btnExecCreateLib.disabled = true;
76 |         btnExecCreateLib.innerHTML = '<span class="spinner-bor
der spinner-border-sm me-1"></span> Создание...';
77 |
78 |         try {
79 |             const res = await fetch('/api/v1/windows-backup/libr
aries', {
80 |                 method: 'POST',
81 |                 headers: { 'Content-Type': 'application/json' },
82 |                 body: JSON.stringify({ name, folders, is_pinned: i
sPinned })
83 |             });
84 |             const data = await res.json();
85 |             if (res.ok) {
86 |                 alert(`Библиотека "${name}" успешно создана!`);
87 |                 const modalEl = document.getElementById('wb-create
-lib-modal');
88 |                 if (modalEl && window.bootstrap?.Modal) {
89 |                     const modalInst = window.bootstrap.Modal.getInst
ance(modalEl);
90 |                     if (modalInst) modalInst.hide();
91 |                 }
92 |                 await loadLibraries();
93 |             } else {

```

```

94 |         alert('Ошибка: ' + (data.detail || JSON.stringify(
| data)));
95 |     }
96 |     } catch (e) {
97 |         alert('Ошибка сети: ' + e.message);
98 |     } finally {
99 |         btnExecCreateLib.disabled = false;
100 |         btnExecCreateLib.innerHTML = '<i class="bi bi-check-
| lg"></i> <span>Создать</span>';
101 |     }
102 | };
103 | }
104 |
105 | // Modal Add Folder to Library
106 | const btnQuickAddFolder = document.getElementById('wb-btn-
| quick-add-folder');
107 | if (btnQuickAddFolder) {
108 |     btnQuickAddFolder.onclick = () => {
109 |         const select = document.getElementById('wb-add-folder-
| lib-select');
110 |         if (select) {
111 |             select.innerHTML = _librariesData.length > 0
112 |                 ? _librariesData.map(l => `<option value="\${escape
| Html(l.name)}">\${escapeHtml(l.name)} (\${l.folder_count || 0} п
| апок)</option>`).join('')
113 |                 : '<option value="">Нет доступных библиотек</optio
| n>';
114 |         }
115 |         const modalEl = document.getElementById('wb-add-folder
| -modal');
116 |         if (modalEl) {
117 |             if (window.bootstrap?.Modal) {
118 |                 new window.bootstrap.Modal(modalEl).show();
119 |             } else {
120 |                 modalEl.classList.add('show');
121 |                 modalEl.style.display = 'block';
122 |             }
123 |         }

```

```
124     };
125   }
126
127   const btnExecAddFolder = document.getElementById('wb-btn-e
xecute-add-folder');
128   if (btnExecAddFolder) {
129     btnExecAddFolder.onclick = async () => {
130       const libName = document.getElementById('wb-add-folder
-lib-select')?.value;
131       const folderPath = document.getElementById('wb-add-fo
lder-path')?.value?.trim();
132       const isDefault = document.getElementById('wb-add-fold
er-default-save')?.checked ?? false;
133
134       if (!libName || !folderPath) {
135         alert('Выберите библиотеку и укажите путь к папке');
136         return;
137       }
138
139       btnExecAddFolder.disabled = true;
140       btnExecAddFolder.innerHTML = '<span class="spinner-bor
der spinner-border-sm me-1"></span> Добавление...';
141
142       try {
143         const res = await fetch(`/api/v1/windows-backup/libr
aries/${encodeURIComponent(libName)}/folders`, {
144           method: 'POST',
145           headers: { 'Content-Type': 'application/json' },
146           body: JSON.stringify({ folder_path: folderPath, is
_default_save: isDefault })
147         });
148         const data = await res.json();
149         if (res.ok) {
150           alert(`Папка успешно добавлена в библиотеку "${lib
Name}"!`);
151           const modalEl = document.getElementById('wb-add-fo
lder-modal');
152           if (modalEl && window.bootstrap?.Modal) {
```

```

153         const modalInst = window.bootstrap.Modal.getInstance(
modalEl);
154         if (modalInst) modalInst.hide();
155     }
156     await loadLibraries();
157 } else {
158     alert('Ошибка: ' + (data.detail || JSON.stringify(
data)));
159 }
160 } catch (e) {
161     alert('Ошибка сети: ' + e.message);
162 } finally {
163     btnExecAddFolder.disabled = false;
164     btnExecAddFolder.innerHTML = '<i class="bi bi-check-
lg"></i> <span>Добавить</span>';
165 }
166 };
167 }
168
169 // RAG Search & Sync
170 const btnRagSync = document.getElementById('wb-btn-rag-syn
c');
171 if (btnRagSync) {
172     btnRagSync.onclick = async () => {
173         btnRagSync.disabled = true;
174         btnRagSync.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span> Синхр...';
175         try {
176             const res = await fetch('/api/v1/windows-backup/file
-history/rag/sync', {
177                 method: 'POST',
178                 headers: { 'Content-Type': 'application/json' },
179                 body: JSON.stringify({ force_rebuild: false })
180             });
181             const data = await res.json();
182             if (res.ok) {
183                 alert(`Синхронизация RAG завершена!\nПроиндексиров
ано: ${data.total_indexed || 0} версий файлов.`);

```

```

184         } else {
185             alert('Ошибка синхронизации: ' + (data.detail || JSON.stringify(data)));
186         }
187     } catch (e) {
188         alert('Ошибка сети: ' + e.message);
189     } finally {
190         btnRagSync.disabled = false;
191         btnRagSync.innerHTML = '<i class="bi bi-arrow-repeat me-1"></i>Синхронизация';
192     }
193 };
194 }
195
196 const btnRagSearch = document.getElementById('wb-btn-rag-search');
197 const inputRagQuery = document.getElementById('wb-rag-query');
198 const performRagSearch = async () => {
199     const query = inputRagQuery?.value?.trim();
200     if (!query) return;
201
202     const resultsContainer = document.getElementById('wb-rag-results');
203     if (resultsContainer) {
204         resultsContainer.innerHTML = '<div class="text-center text-muted py-2"><span class="spinner-border spinner-border-sm me-1"></span>Поиск в архиве...</div>';
205     }
206
207     try {
208         const res = await fetch('/api/v1/windows-backup/file-history/rag/search', {
209             method: 'POST',
210             headers: { 'Content-Type': 'application/json' },
211             body: JSON.stringify({ query, top_k: 5 })
212         });
213         const items = await res.json();

```



```

214 |         if (resultsContainer) {
215 |             if (!items || items.length === 0) {
216 |                 resultsContainer.innerHTML = '<div class="text-mut
ed text-center py-2 small">Ничего не найдено по запросу</div>'
| ;
217 |                 return;
218 |             }
219 |             resultsContainer.innerHTML = items.map(item => `
220 |                 <div class="p-1.5 mb-1 rounded bg-dark border bord
er-secondary" style="font-size: 0.75rem;">
221 |                     <div class="d-flex justify-content-between align
-items-center mb-0.5">
222 |                         <span class="fw-bold text-info text-truncate"
title="${escapeHtml(item.file_name)}">${escapeHtml(item.file_n
ame)}</span>
223 |                         <span class="badge bg-primary-subtle text-prim
ary">${(item.score * 100).toFixed(0)}%</span>
224 |                     </div>
225 |                     <div class="text-muted font-monospace small text
-truncate" title="${escapeHtml(item.original_path)}">${escapeH
tml(item.original_path)}</div>
226 |                     ${item.snippet ? `<div class="text-white-50 mt-1
small text-truncate">${escapeHtml(item.snippet)}</div>` : ''}
227 |                     </div>
228 |                 `).join('');
229 |             }
230 |         } catch (e) {
231 |             if (resultsContainer) {
232 |                 resultsContainer.innerHTML = `<div class="text-dange
r small py-1">Ошибка поиска: ${escapeHtml(e.message)}</div>`;
233 |             }
234 |         }
235 |     };
236 |
237 |     if (btnRagSearch) btnRagSearch.onclick = performRagSearch;
238 |     if (inputRagQuery) {
239 |         inputRagQuery.onkeydown = (e) => {
240 |             if (e.key === 'Enter') {

```

```

241         e.preventDefault();
242         performRagSearch();
243     }
244 };
245 }
246
247 // Audit buttons
248 const btnEnableAudit = document.getElementById('wb-btn-enable-auditpol');
249 if (btnEnableAudit) {
250     btnEnableAudit.onclick = async () => {
251         if (confirm('Включить системный аудит файловой системы (auditpol /set /subcategory:File System)?')) {
252             try {
253                 const res = await fetch('/api/sysadmin/file-audit/policy', {
254                     method: 'POST',
255                     headers: { 'Content-Type': 'application/json' },
256                     body: JSON.stringify({ enable_success: true, enable_failure: true })
257                 });
258                 const data = await res.json();
259                 alert(data.success ? 'Аудит файловой системы успешно активирован!' : `Ошибка: ${data.error || 'Сбой'}`);
260                 await loadBackupData();
261             } catch (err) {
262                 alert(`Ошибка выполнения: ${err.message}`);
263             }
264         }
265     };
266 }
267
268 const btnConfigureSacl = document.getElementById('wb-btn-configure-sacl');
269 if (btnConfigureSacl) {
270     btnConfigureSacl.onclick = async () => {
271         const targetPath = prompt('Укажите путь к папке для настройки SACL аудита удаления:', 'c:\\Users\\onela\\AppData\\Lo

```

```

    cal\\AI-Breadboard');
272     if (targetPath) {
273         try {
274             const res = await fetch('/api/sysadmin/file-audit/
folder-sacl', {
275                 method: 'POST',
276                 headers: { 'Content-Type': 'application/json' },
277                 body: JSON.stringify({ path: targetPath, princip
al: 'Everyone', enable: true })
278             });
279             const data = await res.json();
280             alert(data.Success ? `SACL успешно настроен для ${
targetPath}!` : `Ошибка: ${data.Error || 'Сбой'}`);
281             await loadFileDeletions();
282         } catch (err) {
283             alert(`Ошибка: ${err.message}`);
284         }
285     }
286 };
287 }
288
289 const btnRefreshFolders = document.getElementById('wb-btn-
refresh-user-folders');
290 if (btnRefreshFolders) {
291     btnRefreshFolders.onclick = () => loadUserFoldersOvervie
w();
292 }
293
294 // Modal Relocate Target Drive Change
295 const selectRelocateDrive = document.getElementById('wb-re
locate-target-drive');
296 if (selectRelocateDrive) {
297     selectRelocateDrive.onchange = () => {
298         const driveVal = selectRelocateDrive.value;
299         const folderId = document.getElementById('wb-relocate-
folder-id')?.value;
300         const folder = (_userFoldersData?.folders || []).find(
f => f.folder_id === folderId);

```

```

301         const inputPath = document.getElementById('wb-relocate
302         -target-path');
303         if (driveVal && folder && inputPath) {
304             const d = driveVal.replace('/', '\\').replace(/\\+$/,
305             , '');
306             const parts = folder.current_path.split('\\');
307             const folderSubname = parts[parts.length - 1] || fol
308             der.name;
309             const userMatch = folder.current_path.match(/Users\\
310             ([^\\]+)/i);
311             const username = userMatch ? userMatch[1] : 'User';
312             inputPath.value = `${d}\\Users\\${username}\\${folde
313             rSubname}`;
314         }
315     };
316 }
317
318 // Modal Relocate Browse Folder Button
319 const btnBrowseFolder = document.getElementById('wb-btn-br
320 owse-folder');
321 if (btnBrowseFolder) {
322     btnBrowseFolder.onclick = async () => {
323         const currentPathVal = document.getElementById('wb-rel
324         ocate-target-path')?.value || '';
325         const currentDriveVal = document.getElementById('wb-re
326         locate-target-drive')?.value || '';
327         const initialPath = currentPathVal || (currentDriveVal
328         ? currentDriveVal + '\\ ' : '');
329
330         btnBrowseFolder.disabled = true;
331         btnBrowseFolder.innerHTML = '<span class="spinner-bord
332         er spinner-border-sm me-1"></span> Выбор...';
333
334         try {
335             const res = await fetch(`/api/v1/windows-backup/brow
336             se-folder?initial_path=${encodeURIComponent(initialPath)}`, {
337                 method: 'POST'
338             });
339         }
340     };
341 }

```

```

328         const data = await res.json();
329         if (res.ok && data.selected_path) {
330             const chosenPath = data.selected_path;
331             const inputPath = document.getElementById('wb-relo
cate-target-path');
332             if (inputPath) inputPath.value = chosenPath;
333
334             // Синхронизируем выпадающий список дисков
335             const driveLetter = chosenPath.match(/^[a-zA-Z]:/)
?.[0]?.toUpperCase();
336             if (driveLetter && selectRelocateDrive) {
337                 for (let opt of selectRelocateDrive.options) {
338                     if (opt.value && opt.value.toUpperCase().start
sWith(driveLetter)) {
339                         selectRelocateDrive.value = opt.value;
340                         break;
341                     }
342                 }
343             }
344         }
345     } catch (err) {
346         console.error('[WindowsBackup] Browse folder error:'
, err);
347     } finally {
348         btnBrowseFolder.disabled = false;
349         btnBrowseFolder.innerHTML = '<i class="bi bi-folder2
-open"></i> <span>Обзор...</span>';
350     }
351 };
352 }
353
354 // Modal Relocate Execution
355 const btnExecRelocate = document.getElementById('wb-btn-ex
ecute-relocate');
356 if (btnExecRelocate) {
357     btnExecRelocate.onclick = async () => {
358         const folderId = document.getElementById('wb-relocate-
folder-id')?.value;

```

```
359 |         const targetDrive = document.getElementById('wb-relocat
| te-target-drive')?.value;
360 |         const targetPath = document.getElementById('wb-relocat
| e-target-path')?.value?.trim();
361 |         const deleteSource = document.getElementById('wb-reloc
| ate-delete-source')?.checked || false;
362 |
363 |         if (!folderId || (!targetDrive && !targetPath)) {
364 |             alert('Пожалуйста, выберите целевой диск или укажите
| целевую папку для переноса.');
```

365 | return;

366 | }

367 |

368 | const destDisplay = targetPath || targetDrive;

369 | if (!confirm(`Подтвердите перенос папки в "\${destDispl
| ay}".\nВсе файлы будут скопированы, а системные пути и библиот
| еки перенастроены.`)) {

370 | return;

371 | }

372 |

373 | btnExecRelocate.disabled = true;

374 | btnExecRelocate.innerHTML = '<span class="spinner-bord
| er spinner-border-sm me-1"> Перенос...';

375 |

376 | try {

377 | const res = await fetch('/api/v1/windows-backup/user
| -folders/relocate', {

378 | method: 'POST',

379 | headers: { 'Content-Type': 'application/json' },

380 | body: JSON.stringify({

381 | folder_id: folderId,

382 | target_drive_letter: targetDrive || null,

383 | target_path: targetPath || null,

384 | delete_source_after: deleteSource

385 | })

386 | });

387 |

388 | const data = await res.json();

```

389         if (res.ok && data.success) {
390             alert(data.message || 'Перенос успешно завершён!')
391         };
392         const modalEl = document.getElementById('wb-relocate-modal');
393         if (modalEl && window.bootstrap?.Modal) {
394             const modalInstance = window.bootstrap.Modal.getInstance(modalEl);
395             if (modalInstance) modalInstance.hide();
396             await loadBackupData();
397         } else {
398             alert('Ошибка переноса: ' + (data.detail || data.message || JSON.stringify(data)));
399         }
400     } catch (e) {
401         alert('Сетевая ошибка при переносе: ' + e.message);
402     } finally {
403         btnExecRelocate.disabled = false;
404         btnExecRelocate.innerHTML = '<i class="bi bi-check-circle"></i> <span>Выполнить перенос</span>';
405     }
406 };
407 }
408 }
409
410 async function loadBackupData() {
411     await Promise.allSettled([
412         loadHealthReport(),
413         loadLibraries(),
414         loadFileHistoryStatus(),
415         loadUserFoldersOverview(),
416         loadVssSnapshots(),
417         loadFileDeletions()
418     ]);
419 }
420
421 async function loadHealthReport() {

```

```

422     try {
423         const res = await fetch('/api/v1/windows-backup/health')
424         ;
425         if (!res.ok) throw new Error(`HTTP ${res.status}`);
426         const data = await res.json();
427         const elHealth = document.getElementById('wb-val-health'
428         );
429         const elBadge = document.getElementById('wb-badge-status
430         ');
431         const elSub = document.getElementById('wb-sub-health');
432         const score = data.health_score ?? 0;
433         if (elHealth) elHealth.textContent = `${score} / 100`;
434         if (elBadge) {
435             elBadge.textContent = score >= 80 ? 'Готов' : score >=
436             50 ? 'Внимание' : 'Критично';
437             elBadge.className = `badge bg-${score >= 80 ? 'success
438             ' : score >= 50 ? 'warning' : 'danger'}`;
439         }
440         if (elSub) {
441             elSub.textContent = data.service_running ? 'Служба акт
442             ивна' : 'Служба остановлена';
443         }
444         const recList = document.getElementById('wb-recommendati
445         ons-list');
446         if (recList && Array.isArray(data.recommendations)) {
447             if (data.recommendations.length === 0) {
448                 recList.innerHTML = '<li class="text-success small">
449                 <i class="bi bi-check-circle me-1"></i>Все проверки безопаснос
450                 ти данных пройдены успешно.</li>';
451             } else {
452                 recList.innerHTML = data.recommendations.map(r => `
453                 <li class="mb-1.5 text-warning d-flex align-items-
454                 start gap-1">
455                     <i class="bi bi-exclamation-triangle-fill text-w
456                     arning flex-shrink-0 mt-0.5"></i>

```



```

449         <span>${escapeHtml(r)}</span>
450     </li>
451     `).join('');
452     }
453 }
454 } catch (e) {
455     console.warn('[WindowsBackup] Error loading health:', e)
456 };
457 }
458
459 async function loadLibraries() {
460     const tbody = document.getElementById('wb-libraries-body')
461     ;
462     const countBadge = document.getElementById('wb-lib-count')
463     ;
464     const valLibs = document.getElementById('wb-val-libraries'
465 );
466
467     try {
468         const res = await fetch('/api/v1/windows-backup/librarie
469 s');
470         if (!res.ok) throw new Error(`HTTP ${res.status}`);
471         const libs = await res.json();
472         _librariesData = libs || [];
473
474         if (countBadge) countBadge.textContent = `${libs.length}
475 шт.`;
476         if (valLibs) valLibs.textContent = `${libs.length}`;
477
478         if (!tbody) return;
479         if (!libs || libs.length === 0) {
480             tbody.innerHTML = '<tr><td colspan="4" class="text-cen
481 ter py-3 text-muted">Библиотеки не найдены</td></tr>';
482             return;
483         }
484
485         tbody.innerHTML = libs.map(lib => {

```

```

480 |         const foldersHtml = (lib.folders || []).map(f => {
481 |             const path = typeof f === 'string' ? f : f.folder_pa
th;
482 |             const exists = typeof f === 'object' ? f.exists : tr
ue;
483 |             return `
484 |                 <div class="text-break font-monospace small d-flex
align-items-center gap-1 ${exists ? 'text-white' : 'text-dang
er'}">
485 |                     <i class="bi ${exists ? 'bi-folder-check text-in
fo' : 'bi-folder-x text-danger'}"></i>
486 |                     <span>${escapeHtml(path)}</span>
487 |                     ${!exists ? '<span class="badge bg-danger-subtle
text-danger py-0 px-1" style="font-size:0.6rem;">Недоступна</
span>' : ''}
488 |                 </div>
489 |             `;
490 |         }).join('') || '<span class="text-muted small">Нет пап
ок</span>';
491 |
492 |         return `
493 |             <tr>
494 |                 <td class="fw-bold text-white">
495 |                     <i class="bi bi-collection me-1 text-primary"></
i>${escapeHtml(lib.name)}
496 |                 </td>
497 |                 <td>${foldersHtml}</td>
498 |                 <td class="text-break font-monospace small text-in
fo">
499 |                     ${lib.default_save_folder ? escapeHtml(lib.defau
lt_save_folder) : '<span class="text-muted">—</span>'}
500 |                 </td>
501 |                 <td>
502 |                     <span class="badge ${lib.is_pinned ? 'bg-primary
' : 'bg-secondary'}">${lib.is_pinned ? 'Да' : 'Нет'}</span>
503 |                 </td>
504 |             </tr>
505 |         `;

```

```
506 |     }).join('');
507 |   } catch (e) {
508 |     if (tbody) tbody.innerHTML = `<tr><td colspan="4" class=
    | "text-center py-3 text-danger">Ошибка: ${escapeHtml(e.message)
    | }</td></tr>`;
509 |   }
510 | }
511 |
512 | async function loadFileHistoryStatus() {
513 |   try {
514 |     const res = await fetch('/api/v1/windows-backup/file-his
    | tory/status');
515 |     if (!res.ok) throw new Error(`HTTP ${res.status}`);
516 |     const data = await res.json();
517 |
518 |     const elService = document.getElementById('wb-val-servic
    | e');
519 |     const elSubService = document.getElementById('wb-sub-ser
    | vice');
520 |     const elTarget = document.getElementById('wb-fh-target')
    | ;
521 |     const elConfig = document.getElementById('wb-fh-config')
    | ;
522 |     const elInterval = document.getElementById('wb-fh-interv
    | al');
523 |     const elRetention = document.getElementById('wb-fh-reten
    | tion');
524 |     const elLast = document.getElementById('wb-fh-last-backu
    | p');
525 |
526 |     const isRunning = data.is_service_running;
527 |     if (elService) {
528 |       elService.textContent = isRunning ? 'Работает' : 'Оста
    | новлена';
529 |       elService.className = `wb-card-value ${isRunning ? 'te
    | xt-success' : 'text-danger'}`;
530 |     }
531 |     if (elSubService) {
```

```
532 |         elSubService.textContent = `Заныск: ${data.service_sta  
| rt_type || 'Manual'}`;  
533 |     }  
534 |  
535 |     const cfg = data.config;  
536 |     if (cfg) {  
537 |         if (elTarget) elTarget.textContent = cfg.target_path |  
| cfg.target_url || 'Не настроено';  
538 |         if (elConfig) elConfig.textContent = cfg.config_file_p  
| ath || 'Стандартный профиль';  
539 |         if (elInterval) elInterval.textContent = cfg.backup_in  
| terval_minutes ? `${cfg.backup_interval_minutes} мин.` : 'По у  
| молчанию (1 ч)';  
540 |         if (elRetention) elRetention.textContent = cfg.retenti  
| on_policy || 'Всегда';  
541 |         if (elLast) elLast.textContent = cfg.last_backup_time  
| || 'Нет записей';  
542 |     } else {  
543 |         if (elTarget) elTarget.textContent = 'Конфигурация отс  
| утствует';  
544 |         if (elConfig) elConfig.textContent = '-';  
545 |         if (elInterval) elInterval.textContent = '-';  
546 |         if (elRetention) elRetention.textContent = '-';  
547 |         if (elLast) elLast.textContent = '-';  
548 |     }  
549 | } catch (e) {  
550 |     console.warn('[WindowsBackup] Error loading File History  
| status:', e);  
551 | }  
552 | }  
553 |  
554 | async function loadVssSnapshots() {  
555 |     const tbody = document.getElementById('wb-vss-body');  
556 |     const countBadge = document.getElementById('wb-vss-count')  
| ;  
557 |  
558 |     try {  
559 |         const res = await fetch('/api/v1/windows-backup/vss/snap
```

```

shots');
560     if (!res.ok) throw new Error(`HTTP ${res.status}`);
561     const snapshots = await res.json();
562
563     if (countBadge) countBadge.textContent = `${snapshots.le
length} шт.`;
564
565     if (!tbody) return;
566     if (!snapshots || snapshots.length === 0) {
567         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Теневые копии (VSS) не найдены</td></tr>'
;
568         return;
569     }
570
571     tbody.innerHTML = snapshots.map(s => `
572         <tr>
573             <td class="font-monospace small text-truncate" style
="max-width: 150px;" title="${escapeHtml(s.snapshot_id || '')}"
">
574                 ${escapeHtml(s.snapshot_id || '-')}
575             </td>
576             <td class="fw-semibold text-info">${escapeHtml(s.ori
ginal_volume || '-')}</td>
577             <td class="small text-muted">${escapeHtml(s.creation
_time || '-')}</td>
578             <td><span class="badge bg-success">${escapeHtml(s.st
atus || 'OK')}</span></td>
579         </tr>
580     `).join('');
581     } catch (e) {
582         if (tbody) tbody.innerHTML = `<tr><td colspan="4" class=
"text-center py-3 text-danger">Ошибка: ${escapeHtml(e.message)
}</td></tr>`;
583     }
584 }
585
586 async function loadFileDeletions() {

```

```

587     try {
588         const res = await fetch('/api/sysadmin/file-audit/deletions?hours=24&limit=100&deletions_only=true');
589         if (!res.ok) return;
590         const data = await res.json();
591         const events = data.events || [];
592
593         const tbody = document.getElementById('wb-deletions-tbody');
594         const badge = document.getElementById('wb-deletions-badge');
595         const valDeletions = document.getElementById('wb-val-deletions');
596
597         if (badge) badge.innerText = `${events.length} удалений`;
598         if (valDeletions) valDeletions.innerText = events.length;
599
600         if (tbody) {
601             if (events.length === 0) {
602                 tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted p-3">События удаления (4663/4660) не зафиксированы или аудит не включен</td></tr>';
603                 return;
604             }
605             tbody.innerHTML = events.slice(0, 30).map((e, idx) =>
606                 <tr class="wb-deletion-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для детального анализа удаления">
607                     <td class="font-monospace text-muted small">${e.timestamp?.slice(11, 19) || ''}</td>
608                     <td class="font-monospace fw-semibold ${e.event_id === 4660 ? 'text-danger' : 'text-warning'}">${e.event_id}</td>
609                     <td class="font-monospace text-truncate text-white" style="max-width: 260px;" title="${escapeHtml(e.object_name)}

```

```

    }">${escapeHtml(e.object_name.split('\\').pop() || e.object_name)}</td>
610 |         <td class="small text-info text-truncate" style="max-width: 140px;" title="${escapeHtml(e.process_name)}">${escapeHtml(e.process_name ? e.process_name.split('\\').pop() : 'N/A')}</td>
611 |         <td class="small text-muted">${escapeHtml(e.subject_user_name || 'SYSTEM')}</td>
612 |     </tr>
613 |     `).join('');
614 |
615 |     tbody.querySelectorAll('.wb-deletion-row').forEach(row
    => {
616 |         row.onclick = () => {
617 |             const idx = parseInt(row.getAttribute('data-idx'),
    10);
618 |             const e = events[idx];
619 |             if (!e) return;
620 |
621 |             if (window.AITableModal) {
622 |                 window.AITableModal.show({
623 |                     icon: '🗑',
624 |                     title: `Удаление: ${e.object_name.split('\\').pop() || e.object_name}`,
625 |                     subtitle: `Event ID: ${e.event_id} | ${e.timestamp}`,
626 |                     tableType: 'security_event',
627 |                     badges: [
628 |                         { text: `Event ${e.event_id}`, class: e.event_id === 4660 ? 'badge bg-danger' : 'badge bg-warning text-dark' },
629 |                         { text: e.status || 'Success', class: 'badge bg-success' }
630 |                     ],
631 |                     metadata: [
632 |                         { label: 'Целевой объект', value: e.object_name },
633 |                         { label: 'Тип объекта', value: e.object_type

```

```

    || 'File' },
634         { label: 'Процесс-инициатор', value: e.proce
ss_name || 'Не указан' },
635         { label: 'PID процесса', value: e.process_id
? String(e.process_id) : 'N/A' },
636         { label: 'Пользователь', value: `${e.subject
_domain_name}\\${e.subject_user_name}` },
637         { label: 'Дескриптор HandleId', value: e.han
dle_id || 'N/A' },
638         { label: 'Маска доступа', value: e.access_ma
sk || 'DELETE (0x10000)' }
639     ],
640     rawTitle: 'Полное сообщение Security Log',
641     rawContent: e.raw_message || JSON.stringify(e,
null, 2),
642     requestData: e
643     });
644 }
645 };
646 });
647 }
648 } catch (e) {
649     console.error('[WindowsBackup] Failed to fetch deletion
events:', e);
650 }
651 }
652
653 async function loadUserFoldersOverview() {
654     const tbody = document.getElementById('wb-user-folders-tbo
dy');
655     const totalSizeBadge = document.getElementById('wb-total-u
ser-size');
656     const drivesList = document.getElementById('wb-drives-list
');
657
658     try {
659         const res = await fetch('/api/v1/windows-backup/user-fol
ders/overview');

```



```

660         if (!res.ok) throw new Error(`HTTP ${res.status}`);
661         const data = await res.json();
662         _userFoldersData = data;
663
664         if (totalSizeBadge) {
665             totalSizeBadge.textContent = `Общий объем: ${data.total_user_size_gb} ГБ (${data.total_user_size_mb} МБ)`;
666         }
667
668         if (drivesList) {
669             if (!data.drives || data.drives.length === 0) {
670                 drivesList.innerHTML = '<span class="text-muted">Диски не обнаружены</span>';
671             } else {
672                 drivesList.innerHTML = data.drives.map(d => `
673                     <div class="px-2 py-1 rounded border ${d.is_system_drive ? 'border-info bg-dark' : 'border-secondary bg-dark'} d-flex align-items-center gap-1.5">
674                         <i class="bi ${d.is_system_drive ? 'bi-hdd-fill text-info' : 'bi-hdd text-success'}"></i>
675                         <span class="fw-bold">${escapeHtml(d.drive_letter)}</span>
676                         <span class="text-muted">(${escapeHtml(d.fstype)})</span>:
677                         <span class="text-success fw-semibold">${d.free_space_gb} ГБ своб.</span>
678                         <span class="text-muted small">из ${d.total_space_gb} ГБ</span>
679                         ${d.is_system_drive ? '<span class="badge bg-info-subtle text-info py-0 px-1" style="font-size: 0.65rem;">System</span>' : ''}
680                     </div>
681                 `).join('');
682             }
683         }
684
685         if (!tbody) return;
686         if (!data.folders || data.folders.length === 0) {

```

```

687         tbody.innerHTML = '<tr><td colspan="6" class="text-center py-3 text-muted">Пользовательские папки не найдены</td></tr>';
688         return;
689     }
690
691     tbody.innerHTML = data.folders.map(f => {
692         const suitableDrives = (data.available_target_drives |
693         | []).filter(d => {
694             const folderDrive = f.drive_letter.replace('\\', '')
695             ;
696             const targetDrive = d.drive_letter.replace('\\', '')
697             ;
698             return folderDrive !== targetDrive && d.free_space_gb
699             >= (f.size_gb + 1.0);
700         });
701
702         const canRelocate = suitableDrives.length > 0;
703
704         return `
705         <tr>
706             <td class="fw-bold text-white">
707                 <i class="bi bi-folder2-open me-1 text-primary">
708                 </i>${escapeHtml(f.name)}
709             </td>
710             <td class="font-monospace text-muted small text-truncate" style="max-width: 250px;" title="${escapeHtml(f.current_path)}">
711                 ${escapeHtml(f.current_path)}
712             </td>
713             <td>
714                 <span class="badge ${f.drive_letter.startsWith('C') ? 'bg-secondary' : 'bg-primary'}">${escapeHtml(f.drive_letter)}</span>
715             </td>
716             <td class="fw-semibold font-monospace ${f.size_gb > 1.0 ? 'text-warning' : 'text-info'}">
717                 ${f.size_gb >= 0.01 ? `${f.size_gb} ГБ` : `${f.s

```

```

    ize_mb} МБ`}
713         </td>
714         <td class="small text-muted font-monospace">${f.fi
le_count}</td>
715         <td class="text-end">
716             <button class="btn btn-xs ${canRelocate ? 'btn-o
utline-warning' : 'btn-outline-secondary'} py-0 px-2 wb-btn-re
locate-modal"
717                 data-folder-id="${escapeHtml(f.folder_id
)}"
718                 ${!canRelocate ? 'title="Нет подходящих
дисков с достаточным местом"' : 'title="Перенести на другой ди
ск"' }
719                 style="font-size: 0.72rem;">
720                 <i class="bi bi-box-arrow-right me-1"></i>Пере
нести
721             </button>
722         </td>
723     </tr>
724     `;
725     })).join('');
726
727     tbody.querySelectorAll('.wb-btn-relocate-modal').forEach
(btn => {
728         btn.onclick = () => {
729             const fid = btn.getAttribute('data-folder-id');
730             openRelocateModal(fid);
731         };
732     });
733
734     } catch (e) {
735         if (tbody) tbody.innerHTML = `<tr><td colspan="6" class=
"text-center py-3 text-danger">Ошибка загрузки: ${escapeHtml(e
.message)}</td></tr>`;
736     }
737 }
738
739 function openRelocateModal(folderId) {

```

```
740     if (!_userFoldersData) return;
741     const folder = (_userFoldersData.folders || []).find(f =>
742     f.folder_id === folderId);
743
744     const elId = document.getElementById('wb-relocate-folder-i
745     d');
746     const elName = document.getElementById('wb-relocate-folder
747     -name');
748     const elPath = document.getElementById('wb-relocate-folder
749     -path');
750     const elSize = document.getElementById('wb-relocate-folder
751     -size');
752     const selectDrive = document.getElementById('wb-relocate-t
753     arget-drive');
754     const elHint = document.getElementById('wb-relocate-drive-
755     hint');
756
757     if (elId) elId.value = folder.folder_id;
758     if (elName) elName.textContent = folder.name;
759     if (elPath) elPath.textContent = folder.current_path;
760     if (elSize) elSize.textContent = `${folder.size_gb >= 0.01
761     ? `${folder.size_gb} ГБ` : `${folder.size_mb} МБ`} (${folder.
762     file_count} файлов)`;
763
764     const inputPath = document.getElementById('wb-relocate-tar
765     get-path');
766     if (inputPath) inputPath.value = '';
767
768     if (selectDrive) {
769         selectDrive.innerHTML = '<option value="">-- Выберите ди
770         ск --</option>';
771         const folderDrive = folder.drive_letter.replace('\\', '
772         ').toUpperCase();
773         let firstSuitableDrive = null;
774
775         (_userFoldersData.drives || []).forEach(d => {
776             const dLetter = d.drive_letter.replace('\\', '
777             ').toUpp
```

```

erCase());
766     if (dLetter === folderDrive) return;
767
768     const isEnough = d.free_space_gb >= (folder.size_gb +
1.0);
769     const opt = document.createElement('option');
770     opt.value = d.drive_letter;
771     opt.textContent = `${d.drive_letter} (${d.fstype}) – С
вободно: ${d.free_space_gb} ГБ из ${d.total_space_gb} ГБ ${isE
nough ? '☐ Достаточно места' : '☐ Мало места'}`;
772     if (!isEnough) {
773         opt.disabled = true;
774     } else if (!firstSuitableDrive) {
775         firstSuitableDrive = d.drive_letter;
776     }
777     selectDrive.appendChild(opt);
778 });
779
780 if (firstSuitableDrive) {
781     selectDrive.value = firstSuitableDrive;
782     if (inputPath) {
783         const d = firstSuitableDrive.replace('/', '\\').repl
ace(/\\+$/, '');
784         const parts = folder.current_path.split('\\');
785         const folderSubname = parts[parts.length - 1] || fol
der.name;
786         const userMatch = folder.current_path.match(/Users\\
([^\]+)/i);
787         const username = userMatch ? userMatch[1] : 'User';
788         inputPath.value = `${d}\\Users\\${username}\\${folde
rSubname}`;
789     }
790 }
791 }
792
793 if (elHint) {
794     elHint.textContent = 'Требуется свободного места: миниму
м ' + (folder.size_gb + 1.0).toFixed(2) + ' ГБ (размер + 1 ГБ

```

```
резерва).';
795     }
796
797     const modalEl = document.getElementById('wb-relocate-modal
    ');
798     if (modalEl) {
799         if (window.bootstrap?.Modal) {
800             new window.bootstrap.Modal(modalEl).show();
801         } else {
802             modalEl.classList.add('show');
803             modalEl.style.display = 'block';
804         }
805     }
806 }
807
808 function escapeHtml(str) {
809     if (str === null || str === undefined) return '';
810     return String(str)
811         .replace(/&/g, '&amp;')
812         .replace(/</g, '&lt;')
813         .replace(/>/g, '&gt;')
814         .replace(/"/g, '&quot;')
815         .replace(/'/g, '&#039;');
816 }
817 }());
818
```

index.html

```
1 <!DOCTYPE html>
2 <html lang="ru">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-s
6   cale=1.0">
7   <title>Пользователи Windows</title>
8 </head>
9 <body>
10   <div class="container-fluid p-3">
11     <h4><i class="bi bi-people-fill text-primary"></i> Пользов
12     атели Windows</h4>
13     <p class="text-muted">Управление локальными учетными запис
14     ями операционной системы Windows.</p>
15
16     <div class="card border-secondary mb-3">
17       <div class="card-header bg-dark text-white">
18         <i class="bi bi-list-ul"></i> Список пользователей
19       </div>
20       <div class="card-body">
21         <div id="windows-users-list">
22           <p class="text-muted">Загрузка...</p>
23         </div>
24       </div>
25
26       <div class="card border-secondary">
27         <div class="card-header bg-dark text-white">
28           <i class="bi bi-people-fill"></i> Управление группами
29         </div>
30         <div class="card-body">
31           <div id="windows-groups-list">
32             <p class="text-muted">Загрузка...</p>
33           </div>
34         </div>
35       </div>
36     </div>
37   </div>
38 </body>
39 </html>
```

```
34 | </div>
35 |
36 | <script type="module">
37 |   import { initWindowsUsersTab } from '/html/windows_users_t
  ab/main.js?v=20260920_v1';
38 |   window.initWindowsUsersTab = initWindowsUsersTab;
39 | </script>
40 | </body>
41 | </html>
```


main.js

```
1 // Windows Users Tab Module
2 // =====
3
4 export async function initWindowsUsersTab() {
5   console.log('[WindowsUsersTab] Initializing...');
6
7   try {
8     // Load Windows users
9     const usersContainer = document.getElementById('windows-us
10    ers-list');
11     const groupsContainer = document.getElementById('windows-g
12    rroups-list');
13
14     if (usersContainer) {
15       usersContainer.innerHTML = '<div class="spinner-border t
16       ext-primary" role="status"><span class="visually-hidden">Loadi
17       ng...</span></div>';
18     }
19
20     if (groupsContainer) {
21       groupsContainer.innerHTML = '<div class="spinner-border
22       text-primary" role="status"><span class="visually-hidden">Load
23       ing...</span></div>';
24     }
25
26     // Fetch Windows users
27     try {
28       const response = await fetch('/api/windows-users');
29       if (response.ok) {
30         const data = await response.json();
31         if (data.status === 'ok' && data.users) {
32           if (usersContainer) {
33             usersContainer.innerHTML = `
34             <div class="table-responsive">
35             <table class="table table-dark table-sm table-
```

```

bordered">
30         <thead>
31             <tr>
32                 <th>Имя</th>
33                 <th>Enabled</th>
34                 <th>Описание</th>
35                 <th>SID</th>
36             </tr>
37         </thead>
38         <tbody>
39             ${data.users.map(u => `
40                 <tr>
41                     <td><strong>${escapeHtml(u.Name || '')}
42                 </strong></td>
43                     <td>${u.Enabled ? '<span class="badge
44                 bg-success">Да</span>' : '<span class="badge bg-danger">Нет</s
45                 pan>' }</td>
46                     <td>${escapeHtml(u.Description || '')}
47                 </td>
48                     <td class="text-muted small">${escapeH
49                 tml(u.SID || '')}</td>
50                 </tr>
51             `).join('')}
52         </tbody>
53     </table>
54 </div>
55 `;
56     }
57     } catch (err) {
58         console.error('[WindowsUsersTab] Error loading users:',
59         err);
60         if (usersContainer) {
61             usersContainer.innerHTML = `<div class="alert alert-da
62             nger">Ошибка загрузки: ${escapeHtml(err.message)}</div>`;
63         }
64     }
65 }

```

```

60
61 // Fetch Windows groups
62 try {
63     const response = await fetch('/api/windows-users/groups'
64 );
65     if (response.ok) {
66         const data = await response.json();
67         if (data.status === 'ok' && data.groups) {
68             if (groupsContainer) {
69                 groupsContainer.innerHTML = `
70                 <div class="table-responsive">
71                     <table class="table table-dark table-sm table-
72 bordered">
73                         <thead>
74                             <tr>
75                                 <th>Группа</th>
76                                 <th>SID</th>
77                                 <th>Описание</th>
78                             </tr>
79                         </thead>
80                         <tbody>
81                             ${data.groups.map(g => `
82                             <tr>
83                                 <td><strong>${escapeHtml(g.Name || '')}
84                                 </strong></td>
85                                 <td class="text-muted small">${escapeH
86 tml(g.SID || '')}</td>
87                                 <td>${escapeHtml(g.Description || '')}
88                                 </td>
89                             </tr>
90                             `).join('')}
91                         </tbody>
92                     </table>
93                 </div>
94                 `;
95             }
96         }
97     }
98 }

```

```
93 |     } catch (err) {
94 |         console.error('[WindowsUsersTab] Error loading groups:',
    |         err);
95 |         if (groupsContainer) {
96 |             groupsContainer.innerHTML = `
```